

GENERAL THETA FOUNDATIONS I: ROBUST COMPATIBILITY OF NUMERICAL WORD REALIZATIONS

QIAN QI

ABSTRACT. We study finite-horizon numerical experiments realized by stochastic registers with prescribed layer widths. For antisymmetric targets, two labels at every command cut admit an exact scalar tensor normal form: deleting affine offsets does not increase the optimal uniform error. A finite parity certificate detects failure of sign compatibility at arbitrary horizon. For constant-magnitude target entries it decides whether two labels can improve on the fair-coin error, and provides a matching constructive alternative. A planar one-command experiment has a complete two-cut error frontier: either cut alone can use two labels at error $\rho/4$, whereas using two labels at both forces error $\rho/2$. The obstruction survives arbitrary intervening hidden states and fixed perturbations. Compatible positive Hankel factors, finite all-word Gram certificates, convergent error bounds, arithmetic width laws and a separately priced finite-bit compiler are retained. The primary resource is nonuniform clocked atomic-row label width; we distinguish actual numerical error from average error and local compatibility certificates. Classical realization and tensor mechanisms are explicitly separated from the finite compatibility conclusions.

1. INTRODUCTION

A word response can have constant signed Hankel rank and constant nonnegative rank at every cut, yet require increasing positive state cardinality when its cut factorizations must use compatible stochastic updates. Positive realization, rather than ordinary linear realization, is therefore the relevant formalism. We study the controlled numerical response

$$(1.1) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, U_w x \rangle}{2}, \quad b \in \{-1, 1\},$$

for a finite orthogonal alphabet, a unit seed and one selected terminal coordinate query. Our basic resource is nonuniform clocked atomic-row positive-realization label width. The known horizon, row tables and exact real arithmetic are free in that model. A separate theorem below charges them in a uniform finite-bit command-stream model.

The principal new results concern an exact low-width classification, not another numerical test of an inherited asymptotic exponent. Theorem 3.1 identifies the actual optimum for every antisymmetric finite experiment with two labels at each cut. Theorem 3.2 gives a finite parity certificate for one side and a stochastic construction for the other. The certificate can be placed between any two two-label bottlenecks; widths elsewhere are unrestricted. Theorem 3.4 determines an entire one-command error frontier and provides all its rational upper witnesses. In particular the minimum feasible width at separate cuts need not form a jointly feasible profile even at positive error. The target probabilities are uniformly away from zero and one, and Corollary 3.6 retains a strict gap after perturbation.

These statements do not assert that tensor rank classifies arbitrary positive realizations. Their normal form is specific to two labels and antisymmetric target means. The sign dichotomy determines a strict improvement threshold rather than every optimum on its feasible side. Classical scalar stochastic updates and parity elimination are proved as needed and are not claimed as new

algebraic tools. The finite compatibility conclusions concern the prescribed numerical semantics, with every command and the held binary answer explicitly counted.

The finite-horizon question has two logically separate parts: whether a proposed machine realizes every word, and how close the best machine of a prescribed profile can be. Section 5 treats both with exact finite certificates. The all-word Gram identity checks numerical equality rather than average agreement; a backward recursion extracts a failed word when equality does not hold. It yields a polynomial-size existential real formulation of exact feasibility. The even-moment hierarchy and a rational grid theorem then bound the actual optimized terminal error from both sides. Their convergence is a theorem about all legal hidden machines, whereas the current verification program checks given rational realizations, sign systems and finite identities. No globally optimized grid minimum is reported by that program. These tasks must not be conflated.

The ingredients are deliberately distinguished from their use here. Polynomial-time equivalence of probabilistic automata is classical [20]; finite Gram and tensor manipulations are standard in weighted-automata theory [14]. Real quantifier elimination gives effectiveness, not a practical generic minimizer [1]. The categorical-output comparison imports the known nonnegative-rank complexity and field distinction [21]; it is not a hardness theorem for the fixed planar binary-query subclass. The finite certificate formulation adds neither an unpriced state nor an extra query to the numerical experiment.

Three previously established components are retained. First, Section 4 writes the exact finite-horizon object in the language of normalized positive Hankel factorizations and compatible future-response polytopes. This specializes the classical shift-invariant-cone principle; neither positivity nor compatibility is claimed as a new realization concept. For the planar numerical family, ordinary rank, nonnegative rank and normalized positive rank all equal three at every cut. This identifies precisely what separate factorization omits.

Second, write $\Gamma_{\mathcal{A}}(k, B)$ for optimized length- B word quantization loss and $\lambda_{\mathcal{A}}(k, a)$ for the least common dilation of a k -generator polytope containing a times the unit ball. Theorem 7.1 proves the universal comparison

$$(1.2) \quad \sup_{B \geq 1} \frac{-\log(1 - \Gamma_{\mathcal{A}}(k, B))}{B} \leq \log \lambda_{\mathcal{A}}(k, a).$$

It uses arbitrarily many independent packets to remove the initial inradius loss. Thus the paired profiles are related for every finite orthogonal alphabet, including noncommuting alphabets. Equality and optimality among all hidden realizations are not assumed.

Third, Theorem 12.3 gives a table-free finite-bit compiler for finitely many planar angles having fixed linear-space digit procedures. Its parameter search, dyadic probabilities, arithmetic scratch space, horizon counter and finite random words require $O(\log N + \log(1/\varepsilon))$ work bits. Every command is processed once; the output error is uniformly at most a prescribed $\varepsilon > 0$. For the explicit badly approximable algebraic-angle families the work-space order is exactly $\Theta(\log N)$, while the clean stochastic register retains the arithmetic label exponent. This is not an exact fair-bit implementation of arbitrary real rows, and the sharp coefficient of work space is not determined.

The packet, interval and enclosure theorems of the preceding development are reproduced with their proofs in Section 6. The packing-dilation balance there is a conditional meta-corollary, not a classification of finite alphabets. The arithmetic results below are retained: all of their analytic proofs remain in the current paper, with the branch of the finite-type minimum displayed explicitly.

Our arithmetic conclusions go beyond uniform bad approximation. For $\alpha \in \mathbb{R}^r$ with $1, \alpha_1, \dots, \alpha_r$ rationally independent, write

$$(1.3) \quad \omega^*(\alpha) = \limsup_{\|m\|_1 \rightarrow \infty} \frac{-\log \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}}{\log \|m\|_1}.$$

This is the ordinary dual exponent $\omega_{1,r}$ of the row vector α in [17, Definition 1], with an equivalent fixed-dimensional norm; it is not the uniform exponent $\hat{\omega}_{1,r}$ or the simultaneous exponent $\omega_{r,1}$. The alphabet consists of the identity and the rotations through $2\pi\alpha_i$. Throughout, r , the alphabet, $0 < \rho \leq 1/10$ and $0 \leq \varepsilon < \rho/(2\sqrt{2})$ are fixed.

Theorem 1.1 (Minimal Diophantine type). *If $\omega^*(\alpha) = r$, then*

$$(1.4) \quad \lim_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} = \frac{r}{2r+1}.$$

For Lebesgue-almost every α , and every fixed $\delta > 0$, there are positive constants c, C , depending on the fixed $\alpha, \delta, r, \rho, \varepsilon$, such that, for $N \geq 2$,

$$(1.5) \quad c \frac{N^{r/(2r+1)}}{(\log(N+2))^{2r(1+\delta)/(2r+1)}} \leq W_{N,\varepsilon}(\alpha) \leq CN^{r/(2r+1)} (\log(N+2))^{2r^2(1+\delta)/(2r+1)}.$$

The upper inequalities are achieved by exact common-row machines. No uniformity near rational relations or in growing r is asserted.

The exponent in (1.4) need not imply a two-sided constant-factor law. The earlier dual badly approximable theorem supplies the stronger Θ conclusion on its stated class, and is recovered in Corollary 9.4. Theorem 1.1 instead allows arbitrarily small Diophantine constants along subsequences. The quantitative arithmetic ingredients are classical; the memory conclusions concern their compatible realization and all-hidden-state converse.

There is also a genuine failure of a single exponent, even for one fixed irrational angle.

Theorem 1.2 (Irrational horizons and Liouville fluctuations). *For every irrational α there are horizons $M_n \rightarrow \infty$ such that $W_{M_n,\varepsilon}(\alpha) = \Theta(M_n^{1/3})$. If α is Liouville, then*

$$(1.6) \quad \liminf_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} = 0, \quad \frac{1}{3} \leq \limsup_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} \leq \frac{1}{2}.$$

Nevertheless $W_{N,\varepsilon}(\alpha) \rightarrow \infty$. All displayed upper bounds have exact realizations.

The upper limit in (1.6) is not determined here. The result establishes oscillation between subpower subsequences and a positive-power subsequence; it does not replace that distinction by a guessed exponent for all irrationals.

Provenance of the ingredients. The stochastic Hankel normal form and its finite separation dual are normalizations of established realization and convex-separation arguments [12, 16, 14]. Their role is to place the resource, not to claim priority for positive cones. The endpoint mechanism and arithmetic laws are inherited [7, 8, 19]. The distortion-rate comparison and the uniform finite-bit realization with its charged resource ledger were established in the preceding revision. The finite all-word and actual-error framework of Section 5 is retained from the readable revision-46 source. The new statements are the two-label normal form, balanced certificates and robust profile frontiers of Section 3. Dirichlet approximation, continued fractions, metric tail estimates and the algebraic examples remain classical. Section 13 compares the precise objectives and norms, rather than treating adjacent theories as absent.

2. THE NUMERICAL EXPERIMENT AND THE REGISTER

Let $V = \mathbb{R}^D$, let $\mathcal{A}$ be a nonempty finite alphabet, and fix orthogonal matrices U_a , $a \in \mathcal{A}$. Write $U_w = U_{a_N} \cdots U_{a_1}$ in the order of execution. Let $X \subset \mathbb{S}^{D-1}$ be a nonempty finite seed set and $0 < \rho < 1$. The required law is (1.1). Commands are externally supplied, not chosen by a policy inside the simulator. No distribution on command words is part of the specification.

Definition 2.1 (Clocked atomic-row realization). At horizon N , a machine consists of finite sets S_t with $|S_t| = K_t$ for $0 \leq t \leq N$, an initialization probability row E_x on S_0 , stochastic matrices $T_{t,a} : S_{t-1} \rightarrow S_t$, and terminal columns $d_j : S_N \rightarrow [-1, 1]$. Its numerical mean is

$$E_x T_{1,a_1} \cdots T_{N,a_N} d_j.$$

After the atomic query j , it draws and holds the binary answer with that mean. The final answer cut has two labels. Its peak is $\max(2, K_0, \dots, K_N)$. Available labels count even when unreachable; adding unused normalized rows is permitted.

All persistent randomness, input history and protocol distinctions must be in the current label. Only the external epoch, the prescribed horizon and the read-only row tables are free. Table construction, lookup, arithmetic and exact sampling of known real stochastic rows are uncharged. Matrices, rows and state sets may depend on N ; this is nonuniform positive-realization width, not uniform computational space. A common-row construction uses the same command matrices at all command epochs but may still have an N -dependent decoder. It need not answer at earlier stopping lengths.

Let $W_{N,\varepsilon}$ be the least peak among machines whose every binary row differs from (1.1) by at most ε in total variation. In particular $W_{N,\varepsilon} \geq 2$. For fixed finite $K_\bullet$ the optimal actual error $\mathcal{E}_N(K_\bullet)$ is attained: the row parameters lie in a finite product of compact sets and the error is a finite maximum of continuous functions. For planar results we use $X = \{\pm e_1, \pm e_2\}$ and include the identity letter. We keep $\rho \leq 1/10$ and $\varepsilon < \rho/(2\sqrt{2})$.

Lemma 2.2 (Calibration under any external test law). *Fix a seed $x \in X$ and any law on the complete command word, independent of private initialization. Let $Y_t = U_{a_t} \cdots U_{a_1} x$, $Z_t = \mathbb{E}[Y_t | S_t]$ and $q_t = \mathbb{E}\|Z_t\|$. Every machine with row error at most ε satisfies*

$$(2.1) \quad q_N \geq \kappa := \rho/\sqrt{D} - 2\varepsilon.$$

At a boundary preceding an independent complete packet, the next boundary admits the conditional-mean identity used in Lemma 6.2. No single-step independence inside a packet is asserted.

Proof. For a fixed word, every coordinate of $\mathbb{E}[d(S_N) | w] - \rho Y_N$ has modulus at most 2ε . Since $\|Y_N\| = 1$, averaging gives

$$\mathbb{E}\langle Y_N, d(S_N) \rangle \geq \rho - 2\varepsilon\sqrt{D}.$$

On the other hand this pairing equals $\mathbb{E}\langle Z_N, d(S_N) \rangle$ and is at most $\sqrt{D}q_N$. This proves (2.1). Combining terminal columns is a proof device, not a requirement to provide their joint output distribution. $\square$

3. EXACT TWO-LABEL GEOMETRY AND ROBUST COMPATIBILITY

The positive Hankel normal form below retains the whole future array. For two labels, a smaller exact normal form is available. It gives both a class of finite certificates and an explicit error frontier. The statements in this section concern the actual optimized numerical error, not a sum of local defects or an average over a test law.

3.1. Antisymmetric experiments. We first allow any finite response table, not necessarily one obtained from orthogonal matrices. Let the seeds be paired symbols $X = \{+1, -1, \dots, +n, -n\}$, let $w \in \mathcal{A}^N$, and let $j \in \{1, \dots, d\}$ be the selected query. Suppose the target means satisfy

$$(3.1) \quad f_{-i,w,j} = -f_{i,w,j}, \quad |f_{i,w,j}| \leq 1.$$

The binary law is $(1 + bf_{i,w,j})/2$. Write $\mathcal{E}_2(f)$ for the minimum worst-word binary-TV error of a machine with two available labels at *every* cut $0, \dots, N$. The final held answer also has two labels. Random initialization and arbitrary cut-dependent stochastic matrices are allowed. All extrema below are attained.

Theorem 3.1 (An exact scalar normal form). *For every table satisfying (3.1),*

$$(3.2) \quad 2\mathcal{E}_2(f) = \min_{|h_i| \leq 1, |\beta_{t,a}| \leq 1, |d_j| \leq 1} \max_{i,w,j} \left| f_{i,w,j} - h_i d_j \prod_{t=1}^N \beta_{t,at} \right|.$$

Thus the exact numerical optimization is the distance, in the uniform norm, to the set of bounded decomposable tensors with one factor for the positive seed, one for each command epoch, and one for the query. Every feasible tensor on the right has a two-label stochastic realization. For $N = 0$ the empty product is one.

Proof. Represent a probability row on two labels by its signed coordinate $r = 2p - 1 \in [-1, 1]$. Every stochastic update is an affine map $r \mapsto \alpha + \beta r$ with $|\alpha| + |\beta| \leq 1$; conversely this condition is sufficient. A bounded terminal mean has form $c_j + d_j r$ with $|c_j| + |d_j| \leq 1$. For a word w , the composite update has the form $A_w + B_w r$, where $B_w = \prod_t \beta_{t,at}$. If the two paired initialization coordinates are r_i, r_{-i} , the odd part of the realized mean is therefore

$$\frac{\hat{f}_{i,w,j} - \hat{f}_{-i,w,j}}{2} = h_i B_w d_j, \quad h_i = (r_i - r_{-i})/2.$$

Taking the odd part cannot increase uniform error against the odd target: each resulting residual is half the difference of two original residuals. This operation is realizable without mixing whole machines. Use initialization coordinates $\pm h_i$, replace every update by $r \mapsto \beta_{t,a} r$, and replace the terminal column by $d_j r$. All rows remain stochastic because $|\beta_{t,a}|, |d_j| \leq 1$. This proves that the right side is no larger than twice the original optimum, without retaining a symmetrization coin.

Conversely, initialize the positive label with probability $(1 \pm h_i)/2$, use the stochastic matrix

$$(3.3) \quad \frac{1}{2} \begin{pmatrix} 1 + \beta_{t,a} & 1 - \beta_{t,a} \\ 1 - \beta_{t,a} & 1 + \beta_{t,a} \end{pmatrix},$$

and terminal means $(d_j, -d_j)$. Induction gives the tensor in (3.2). Binary total variation is half the mean error. Compactness of the factor cubes and continuity prove attainment. $\square$

The scalar description of a two-state stochastic matrix is elementary. The content needed here is that deleting all affine offsets produces an admissible optimizer for an odd target, rather than merely a signed linear approximation. No averaging over an unrecorded global random choice is permitted. Tensor descriptions of stochastic processes are classical; Ohta's reachable/null-space reductions concern a different stationary observation model [16]. Equation (3.2) keeps the prescribed command order, bounded query columns and fixed numerical error objective simultaneously.

3.2. Balanced sign certificates. For an entry $e = (i, a_1, \dots, a_N, j)$, form its incidence vector $\iota(e) \in \mathbb{F}_2^M$, where

$$M = n + N|\mathcal{A}| + d.$$

There is one coordinate for each seed factor, each pair (t, a) , and each query factor; $\iota(e)$ marks the factors used by that entry. A set C of nonzero target entries is *balanced* if $\sum_{e \in C} \iota(e) = 0$. Equivalently each factor occurs an even number of times. It is *negative* if $\prod_{e \in C} f_e < 0$.

Theorem 3.2 (A finite sign obstruction and a gap dichotomy). *A negative balanced set C certifies*

$$(3.4) \quad \mathcal{E}_2(f) \geq \frac{1}{2} \min_{e \in C} |f_e|.$$

Suppose in addition that $f_e \in \{0, \rho, -\rho\}$ for a fixed $0 < \rho \leq 1$. Then exactly one of the following holds:

$$(3.5) \quad \begin{aligned} \mathcal{E}_2(f) &= \rho/2, & \text{the nonzero signs do not factor;} \\ \mathcal{E}_2(f) &\leq \rho/4, & \text{the nonzero signs factor.} \end{aligned}$$

Factorability, and hence the alternative in (3.5), is decided by a linear system over $\mathbb{F}_2$. In the first case there is a negative balanced certificate of at most $M + 1$ entries. In the second case the proof constructs a legal two-label machine. Complexity is polynomial in the explicit table size; the table can be exponential in the displayed horizon.

Proof. The product of any decomposable real tensor over a balanced set is nonnegative, since each factor has even exponent. If every tensor entry on C approximated the target within a mean error strictly smaller than $\min_C |f_e|$, it would be nonzero with the target sign. Its product would be negative, a contradiction. Apply Theorem 3.1 to obtain (3.4).

Let $s_e \in \mathbb{F}_2$ encode the nonzero target sign. The signs factor exactly when the system

$$(3.6) \quad \langle \iota(e), z \rangle = s_e \pmod{2} \quad (f_e \neq 0)$$

is consistent. If it is inconsistent, a linear combination of its rows has zero left side and right side one. The corresponding entries are negative and balanced. One may express a contradictory row in a basis of at most M earlier rows, so at most $M + 1$ entries suffice. Inequality (3.4) gives $\mathcal{E}_2(f) \geq \rho/2$; the fair-coin decoder attains the reverse inequality.

If (3.6) has a solution, take the seed and epoch factors to be the prescribed signs ± 1 , and take each query factor to be its prescribed sign times $\rho/2$. Every tensor entry then has magnitude $\rho/2$ and all nonzero targets have the correct sign. Both zero and nonzero entries have mean error at most $\rho/2$. The construction (3.3) realizes these factors with deterministic identity/swap command rows and bounded terminal means. Thus $\mathcal{E}_2(f) \leq \rho/4$. Ordinary finite-field elimination both decides consistency and records the indicated certificate. $\square$

The second alternative need not be an equality: a target which is already a bounded decomposable tensor may have zero error. The theorem classifies the strict-improvement threshold, not every smaller optimum. The certificate is an elementary parity identity, not a new nonnegative-rank theorem or a general polynomial-time minimization algorithm for stochastic automata. The difference is useful: an explicit witness certifies the error against *every* two-label machine, whereas an equivalence test checks a supplied machine.

3.3. Two bottlenecks and an exact finite frontier. Return to the orthogonal experiment. Put

$$(3.7) \quad X = \{\pm e_1, \pm e_2\}, \quad \mathcal{A} = \{I, R\}, \quad R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad 0 < \rho \leq 1/6.$$

The finite signal range in (3.7) is separate from the smaller sufficient calibration range of the arithmetic theorems. Every target answer probability is in $[(1 - \rho)/2, (1 + \rho)/2]$.

Theorem 3.3 (Two separated bottlenecks). *At any horizon $N \geq 1$, suppose a machine for (3.7) has at most two labels at two distinct cuts $s < t$ in $\{0, \dots, N\}$. Its actual worst-word binary-TV error is at least $\rho/2$, regardless of the widths at all other cuts. The fair-coin output attains this*

bound at every such prescribed profile. The same conclusion holds for any positive signal $\rho \leq 1$, independently of the upper construction used below.

Proof. Freeze all commands before s and after t to I . Between the two cuts retain the two words I^{t-s} and RI^{t-s-1} , with R in the first executed position. Integrating the intermediate computation gives two stochastic maps between the two endpoint alphabets. Pad a one-label endpoint if necessary. The initial probabilities may depend arbitrarily on the seed; the suffix gives one bounded column per query. Thus the odd part of this restricted experiment has the one-step scalar form of Theorem 3.1, even when intermediate alphabets are arbitrarily large.

Four entries of its positive-seed target are

$$f_{1,I,1} = \rho, \quad f_{2,I,2} = \rho, \quad f_{1,R,2} = \rho, \quad f_{2,R,1} = -\rho.$$

Each seed, intervening word and query occurs twice. These four entries form a negative balanced set. Taking the odd part of any putative approximation costs no extra error, so (3.4) gives $\rho/2$. The zero mean decoder proves sharpness. No reset occurs inside the interval and no command is replaced by a free input symbol: collapsing the interval is a converse argument only. $\square$

For $N = 1$ let $\mathcal{E}(k_0, k_1)$ denote the optimum at the two command-boundary widths. A positive minimum is attained even when the allowed profile is infeasible for exact realization.

Theorem 3.4 (A complete two-cut error frontier). *In (3.7) at $N = 1$, the exact frontier is*

$$(3.8) \quad \begin{array}{c|ccc} \mathcal{E}(k_0, k_1) & k_1 = 1 & k_1 = 2 & k_1 \geq 3 \\ \hline k_0 = 1 & \rho/2 & \rho/2 & \rho/2 \\ k_0 = 2 & \rho/2 & \rho/2 & \rho/4 \\ k_0 \geq 3 & \rho/2 & \rho/4 & 0 \end{array}$$

Every upper bound has an explicit rational stochastic realization when ρ is rational. In particular, minimizing either cut separately at tolerance $\rho/4$ permits two labels, but no single machine can attain both minima simultaneously.

Proof. A one-label bottleneck erases all seed dependence; for the identity word, opposite coordinate means force error at least $\rho/2$. Theorem 3.3 treats $(2, 2)$. If either cut has two labels, then, for the identity word, the realized four seed means lie on one affine line $n \cdot y = c$. If its uniform coordinate error is at most δ , then

$$\delta \|n\|_1 \geq \max_{y \in \{\pm \rho e_1, \pm \rho e_2\}} |n \cdot y - c| = \rho \|n\|_\infty + |c| \geq (\rho/2) \|n\|_1.$$

This proves the $\rho/4$ lower bounds after dividing mean error by two. A degenerate segment lies in such a line as well.

Here are matching machines. For $a > 0$, set

$$V_a = \{(a, 0), (0, a), (-a, -a)\}, \quad \beta_a(y) = \frac{1}{3} \left(1 + \frac{2y_1 - y_2}{a}, 1 + \frac{2y_2 - y_1}{a}, 1 - \frac{y_1 + y_2}{a} \right).$$

Whenever $\beta_a(y) \geq 0$, it sums to one and has barycenter y in V_a .

For $(3, 3)$ initialize with $\beta_{2\rho}(\rho x)$. For each vertex $v \in V_{2\rho}$ and command $U \in \{I, R\}$, use the row $\beta_{6\rho}(Uv)$ on $V_{6\rho}$. The six rows are nonnegative: substituting the vertices gives $2(Uv)_1 - (Uv)_2 \geq -6\rho$, the analogous second inequality, and $(Uv)_1 + (Uv)_2 \leq 6\rho$. Use the coordinates of $V_{6\rho}$ as terminal means. They lie in $[-1, 1]$ because $6\rho \leq 1$. The barycenter identity gives exactly ρUx for each word and seed.

For $(2, 3)$ let the two initial labels represent $\pm v$, where $v = (\rho/2, \rho/2)$; initialize e_1, e_2 at $+v$ and their negatives at $-v$. On command U , use the row $\beta_{2\rho}(\pm Uv)$ and decode with $V_{2\rho}$. All four rows are nonnegative. The output is $U[\rho(x_1 + x_2)(1, 1)/2]$, differing from ρUx in each coordinate by at most $\rho/2$ for both I and R .

For $(3, 2)$ initialize with $\beta_{2\rho}(\rho x)$ and decode the two terminal labels as $(1, 1)$ and $(-1, -1)$. For $v \in V_{2\rho}$ use positive-label probability $(1 + \ell(Uv))/2$, where $\ell(y) = (y_1 + y_2)/2$. Here $|\ell(Uv)| \leq 2\rho \leq 1$. The output mean is $\rho((Ux)_1 + (Ux)_2)(1, 1)/2$, again with coordinate error $\rho/2$. All remaining entries in the table follow by padding normalized unused rows, or by the zero decoder. Every coefficient just displayed is rational when ρ is rational. $\square$

For $\rho/4 \leq \varepsilon < \rho/2$, the feasible two-cut profiles are exactly the upward closure of $(2, 3)$ and $(3, 2)$. For $0 \leq \varepsilon < \rho/4$, both command cuts need at least three labels; for $\varepsilon \geq \rho/2$, one at each command cut suffices. The held answer is always charged separately as two labels. These are finite numerical frontiers, not asymptotic claims about an irrational alphabet. They complement rather than replace the growing-width laws.

Theorem 3.5 (An arbitrary-horizon profile frontier). *For the alphabet and seeds in (3.7), let $N \geq 1$ and suppose every command-cut width belongs to $\{1, 2\} \cup \{4, 5, \dots\}$. For any $0 < \rho \leq 1$, the exact optimized error is*

$$(3.9) \quad \mathcal{E}(K_\bullet) = \begin{cases} \rho/2, & \text{some } K_t = 1, \text{ or at least two cuts have } K_t = 2, \\ \rho/4, & \text{exactly one cut has } K_t = 2 \text{ and none has width one,} \\ 0, & \text{every } K_t \geq 4. \end{cases}$$

The constructive machines have deterministic initialization and command rows and rational decoder probabilities when ρ is rational. The excluded width-three profiles are not classified by this statement.

Proof. For a one-label cut freeze every word to the identity; opposite seeds force error $\rho/2$. Two two-label cuts are treated by Theorem 3.3. With one such cut, freeze the entire word to the identity. Its four output means lie in a segment, so the line argument in Theorem 3.4 gives error $\rho/4$.

For the matching construction without a bottleneck, store which of $\{\pm e_1, \pm e_2\}$ is the current vector. The command R permutes these four states, and the decoder is ρ times the vector. If the unique two-label cut is s , track this vector until that cut. At cut s replace the current vector y by $Py = (y_1 + y_2)(1, 1)/2$. There are only two possibilities, $\pm(1, 1)/2$, for these four axis vectors. This is a deterministic compression, with no retained old phase. Thereafter four labels track the quarter-rotation orbit of Py . The first transition out of the bottleneck embeds its two possibilities into these four labels and applies the actual command; subsequent transitions are permutations. At $s = 0$ perform the compression in initialization; at $s = N$ decode the two possibilities directly. The decoder multiplies by ρ . For each axis vector, $\|y - Py\|_\infty = 1/2$; every subsequent quarter rotation preserves this norm. The final mean error is exactly $\rho/2$, proving the second line of (3.9). A fair-coin decoder handles the first line. Available excess labels are padded by normalized unused rows. No random branch or clock history is stored for free. $\square$

Corollary 3.6 (Stability under arbitrary target perturbations). *If a finite binary response table is perturbed by at most η in uniform row TV, its optimal error at any fixed profile changes by at most η . In particular the two-bottleneck lower bound becomes $\max(0, \rho/2 - \eta)$ for any such perturbation of (3.7). The construction at $(2, 3)$ or $(3, 2)$ has error at most $\rho/4 + \eta$. Thus a strict compatibility gap persists whenever $\eta < \rho/8$, including full-support perturbations.*

Proof. For a fixed machine, the two target errors differ by at most η by the triangle inequality; minimizing in both directions proves the first claim. Apply it to (3.8); the gap between $\rho/2 - \eta$ and $\rho/4 + \eta$ is positive exactly when $\eta < \rho/8$. Antisymmetry need not survive the perturbation. $\square$

4. CONTROLLED HANKEL ARRAYS AND COMPATIBLE POSITIVE REALIZATIONS

For a horizon N , define the probability-valued prefix-suffix matrix

$$(4.1) \quad H_t[(x, u), (v, j, b)] = \frac{1 + b\rho\langle e_j, U_v U_u x \rangle}{2}, \quad |u| = t, \quad |v| = N - t.$$

The two columns indexed by $b = \pm 1$ sum to one for every fixed (v, j) . Let $\mathcal{C}_t$ be the product of these binary probability simplices. Thus a point of $\mathcal{C}_t$ is a legal collection of responses to all remaining words and queries, not a joint distribution of those answers. The restriction $L_{t,a} : \mathcal{C}_t \rightarrow \mathcal{C}_{t+1}$ reads the coordinates whose first remaining command is a .

A normalized positive factorization is $H_t = E_t B_t$, where E_t is row stochastic and every row of B_t belongs to $\mathcal{C}_t$. Its least inner dimension is denoted $\text{rank}_{\text{st}} H_t$. The usual nonnegative rank permits arbitrary nonnegative factors and need not impose these simultaneous normalizations. Consequently

$$(4.2) \quad \text{rank } H_t \leq \text{rank}_+ H_t \leq \text{rank}_{\text{st}} H_t \leq K_t$$

for every exact stochastic realization. The last inequality follows by conditioning the remaining experiment on the actual label at cut t . The factor E_t contains its state probabilities; B_t contains its complete future responses, including those of states not reached by a particular prefix.

Proposition 4.1 (Finite-horizon positive Hankel normal form). *For any finite table of binary responses, a profile $K_0, \dots, K_N$ is exactly realizable if and only if there are polytopes $P_t \subset \mathcal{C}_t$, generated by at most K_t points, such that*

$$(4.3) \quad H_0[x, \cdot] \in P_0 \quad (x \in X), \quad L_{t,a} P_t \subset P_{t+1} \quad (0 \leq t < N, a \in \mathcal{A}).$$

The held answer additionally costs two labels. The optimum error for a fixed profile is the least entrywise error from the specified table to one having such a compatible family.

Proof. Given a machine, let $b_{t,s} \in \mathcal{C}_t$ be the conditional response to every remaining word when its current label is s . Its stochastic update gives

$$L_{t,a} b_{t,s} = \sum_{s'} T_{t+1,a}(s, s') b_{t+1,s'}.$$

Taking convex hulls and using the initialization proves necessity. Conversely choose generators $b_{t,s}$, padding by repetitions when necessary. Express each $L_{t,a} b_{t,s}$ as a convex combination of the next generators, and use those coefficients as the stochastic row. Express each root row as a convex combination of the initial generators. At time N the generators themselves give the binary decoder rows. Backward induction on the remaining word proves equality of all responses. The construction is valid on every generator, not only on reachable mixtures. For approximation apply this equivalence to the actual response table of a machine, or to a candidate perturbed table; finite compactness of the stochastic parameter spaces gives attainment. $\square$

This is the finite, controlled, normalized version of the classical shift-invariant-cone realization principle, not a newly discovered characterization of positive systems. In particular, the cone generated by $(1, b)$ for $b \in P_t$ carries the positive restriction maps. Benvenuti-Farina's Theorems 2-3 give the stationary cone formulation; the multiple cuts here are essential [12]. The same compatibility issue was already explicit in the earlier lift formulation [18]. The normal form has exponentially many suffix coordinates. It is exact, but does not make arbitrary width minimization efficient or eliminate hidden directions by projecting them into the observed D -space.

Proposition 4.2 (A finite compatibility certificate). *For proposed generators B_t, B_{t+1} , put $c = L_{t,a}b_{t,s}$ and let $b'_1, \dots, b'_{K_{t+1}}$ be the next generators. The least rowwise uniform defect is*

$$(4.4) \quad \min_{\theta \in \Delta_{K_{t+1}}} \left\| c - \sum_i \theta_i b'_i \right\|_\infty = \max_{\|z\|_1 \leq 1} \left(\langle z, c \rangle - \max_i \langle z, b'_i \rangle \right).$$

It vanishes exactly when that row has a stochastic extension. If root rows are approximated within δ_0 and every cut restriction within δ_t , the resulting machine has worst-word binary-TV error at most $\delta_0 + \sum_{t=1}^N \delta_t$.

Proof. Write the infinity norm as the supremum of its pairing with the unit ℓ_1 ball. The simplex and this ball are compact and convex, so finite bilinear minimax interchanges the extrema. Minimizing the pairing over the simplex gives the maximum over its vertices with the displayed sign. For composition, restriction does not increase the infinity norm and stochastic averaging is a contraction. Induction on the suffix length therefore adds the successive errors. Each binary pair is normalized, so its total variation is exactly either coordinate error. Cancellation may reduce the true error; no equality with the local sum is asserted. $\square$

Proposition 4.3 (Constant separate ranks for the planar experiment). *For the four planar seeds, an identity command, and $0 < \rho \leq 1/10$, at every cut and every horizon,*

$$(4.5) \quad \text{rank } H_t = \text{rank}_+ H_t = \text{rank}_{\text{st}} H_t = 3.$$

Each factorization embeds in a complete machine when all other cuts are unrestricted. Nevertheless the compatible peak can grow as in Theorems 1.1 and 1.2.

Proof. The entries in (4.1) are affine functions of the two coordinates of $\rho U_u x$, so ordinary rank is at most three. Identity prefixes, identity suffixes, both coordinate queries and the four seeds give a rank-three submatrix. For the upper factorization, an equilateral triangle of circumradius 2ρ contains the disk of radius ρ . Encode $\rho U_u x$ in its three vertices v_i by barycentric coordinates. The continuation row at v_i is $(1 + b\langle e_j, U_v v_i \rangle)/2$, which is legal since $\|v_i\| = 2\rho < 1$. This is a normalized three-factor decomposition and proves the equalities. To embed it, retain the exact history until this cut, draw its factor label, retain that label and the remaining history afterwards, then use the factor decoder. Both retained histories are finite at a fixed horizon. They are not free registers in a simultaneous three-state realization. The growth assertions follow from the stated all-hidden converses, not from a claimed new separation of signed and positive realization in classical stationary theory. $\square$

5. FINITE CERTIFICATES FOR THE WHOLE NUMERICAL EXPERIMENT

The Hankel normal form uses all suffixes. We now give an equivalent finite certificate whose size is polynomial in a *displayed* horizon and proposed widths. We then treat the actual optimal error, not the sum of local compatibility defects. Throughout this section X and $\mathcal{A}$ are nonempty, $|\mathcal{A}| = m$, $D \geq 1$, and $K_0, \dots, K_N \geq 1$. The held binary answer is a separate two-label cut. A horizon is supplied in unary for statements about input size. Rational numbers have binary encodings; a real algebraic coefficient is supplied by a polynomial and a rational isolating interval. These conventions do not give an oracle for arbitrary real numbers.

In this section $T_{t,a}$ labels the edge from cut t to cut $t+1$, a reindexing of Definition 2.1. Write E_x for an initialization row, $T_{t,a}$ for the K_t -by- K_{t+1} row-stochastic transition, and $d_j \in [-1, 1]^{K_N}$ for the terminal mean column. Define the residual mean

$$(5.1) \quad R_{x,w,j} = E_x T_{0,a_1} \cdots T_{N-1,a_N} d_j - \rho \langle e_j, U_{a_N} \cdots U_{a_1} x \rangle.$$

Thus the actual worst-word binary-TV error is $e(\mathcal{M}) = \frac{1}{2} \max_{x,w,j} |R_{x,w,j}|$. In particular, centering the numerical response does not turn a bounded stochastic decoder into an unrestricted signed readout.

Set $n_t = K_t + D$ and form rectangular block matrices and vectors

$$(5.2) \quad A_{t,a} = \text{diag}(T_{t,a}, U_a^\top), \quad u_x = (E_x, -\rho x^\top), \quad v_j = \begin{pmatrix} d_j \\ e_j \end{pmatrix}.$$

The order and transpose in this definition are essential for noncommuting commands. Equation (5.1) equals $u_x A_w v_j$.

Theorem 5.1 (A finite all-word certificate). *Define*

$$(5.3) \quad G_0 = \sum_{x \in X} u_x^\top u_x, \quad G_{t+1} = \sum_{a \in \mathcal{A}} A_{t,a}^\top G_t A_{t,a}, \quad S_2 = \sum_{j=1}^D v_j^\top G_N v_j.$$

Then

$$S_2 = \sum_{x \in X} \sum_{w \in \mathcal{A}^N} \sum_{j=1}^D R_{x,w,j}^2.$$

Consequently a legal proposed machine is exact on every word if and only if $S_2 = 0$. For rational data this can be checked by exact arithmetic in polynomial time in the displayed data. If $S_2 > 0$, a seed, word and query with nonzero residual can also be found in polynomial time, without listing all words.

Exact feasibility of a prescribed profile for an algebraic orthogonal-word target has an existential real formula of polynomial size in $N, m, D, |X|$ and the explicitly displayed widths. Its equations may all be taken to have degree at most three after auxiliary arithmetic variables are introduced. Feasibility has a real algebraic witness whenever it has a real witness.

Proof. Induction in (5.3) gives $G_t = \sum_{x, |u|=t} (u_x A_u)^\top (u_x A_u)$. Pairing with v_j proves the identity. Every summand is a real square, so zero means equality of all numerical responses, not merely equality on average. No assumption about the ranks of reachable distributions or the observability of individual states is used.

For a witness, compute backwards

$$C_N = \sum_j v_j v_j^\top, \quad C_t = \sum_a A_{t,a} C_{t+1} A_{t,a}^\top.$$

Select x with $u_x C_0 u_x^\top > 0$. Given a current row u at cut t with $u C_t u^\top > 0$, at least one command satisfies $u A_{t,a} C_{t+1} A_{t,a}^\top u^\top > 0$. Append it and replace u by $u A_{t,a}$. At the last cut choose j with $u v_j \neq 0$. The matrices are positive semidefinite by their recursions, so this procedure cannot encounter a dead end. There are $O(Nm)$ matrix operations at dimensions at most $\max_t n_t$. For rational input, common-denominator bounds on these finite matrix products also give polynomial intermediate bit length. Exact algebraic sign testing gives the corresponding effective statement for algebraic inputs; no unspecified real-number comparison is invoked.

For existence, treat E, T, d, G as unknowns. Impose nonnegativity and unit row sums on E, T , the bounds on d , the recursions (5.3), and $S_2 = 0$. Each recursion entry has degree at most three, and its number of terms is polynomial in the matrix sizes. The Gram variables need no independent positivity constraint: the initial condition and recurrence already force the displayed sum of squares. If target coefficients are algebraic rather than rational, include their defining polynomials and isolating inequalities. Arithmetic circuit variables reduce larger degrees to quadratic equations. Elimination over real closed fields decides existence and permits algebraic sample points. This last conclusion is the usual real-algebraic sampling principle [1], not a polynomial-time optimizer. $\square$

The linear-algebraic equivalence mechanism is classical: compare Tzeng's equivalence algorithm [20] and the weighted-automata Hankel and Gram constructions of Balle–Panangaden–Precup [14]. The assertion here makes the finite clock, stochastic constraints, all seed initializations, and bounded query columns simultaneous. It must not be read as a new discovery of polynomial-time weighted-automata equivalence. Also, $S_2 \approx 0$ in floating point is *not* the exact certificate $S_2 = 0$.

5.1. A hierarchy for the actual optimal error. Let $Q = |X|Dm^N$. For each even integer $p \geq 2$, define

$$(5.4) \quad L_p(\mathcal{M}) = \frac{1}{2} \left(Q^{-1} \sum_{x,w,j} R_{x,w,j}^p \right)^{1/p}, \quad \ell_p(K_\bullet) = \min_{\mathcal{M}} L_p(\mathcal{M}),$$

where every minimization uses the *same* product of stochastic row simplices and bounded decoder cubes for the specified profile. Put $\mathcal{E}(K_\bullet) = \min_{\mathcal{M}} e(\mathcal{M})$.

Theorem 5.2 (Moment certificates and the error frontier). *All the preceding minima are attained. For every even $p \geq 2$,*

$$(5.5) \quad \ell_p(K_\bullet) \leq \mathcal{E}(K_\bullet) \leq Q^{1/p} \ell_p(K_\bullet).$$

The numbers ℓ_p increase along the even integers and converge to $\mathcal{E}$. The running minima of the right sides of (5.5) decrease to the same value. In particular, $\ell_2 = 0$ if and only if the profile admits an exact realization. For each fixed p , its moment sum is computable by a finite homogeneous polynomial recurrence without enumerating words. For fixed $\max_t(K_t + D)$, this recurrence has size polynomial in $p, N, m, D, |X|$.

For algebraic target data the actual optimum $\mathcal{E}$ and a minimizing machine are algebraic and effectively obtainable. For a finite prescribed width cap, the resulting threshold values give the exact finite-horizon error/width frontier. This is an effective finite optimization statement, not a closed formula or a polynomial-time algorithm for growing width.

Proof. The parameter set is nonempty and compact, and the objectives are continuous. The inequalities for a vector of Q numbers are

$$\left(Q^{-1} \sum |r_i|^p \right)^{1/p} \leq \max_i |r_i| \leq Q^{1/p} \left(Q^{-1} \sum |r_i|^p \right)^{1/p}.$$

The first inequality survives minimization. For the second, use a minimizer of L_p as a candidate for $\mathcal{E}$; no exchange of a maximum over words with a minimum over machines is made. Normalized finite-measure L^p norms increase with p , as do their minima. Since $Q^{1/p} \rightarrow 1$, the two sequences converge as asserted, including the case of a zero optimum.

The recurrence is particularly transparent in polynomial form. Let z be a row variable of the appropriate dimension and set

$$(5.6) \quad F_N(z) = \sum_j (zv_j)^p, \quad F_t(z) = \sum_a F_{t+1}(zA_{t,a}).$$

Then $\sum_x F_0(u_x)$ is the moment sum. A homogeneous degree- p polynomial in n_t variables has $\binom{n_t+p-1}{p}$ coefficients. Repeated polynomial multiplication and linear substitution compute these coefficients in polynomially many arithmetic operations when n_t is fixed. This is the symmetric-power form of the familiar tensor product of weighted representations. It does not imply a bound polynomial in both p and unbounded n_t .

For the last assertion one may instead introduce all finitely many word inequalities $-2e \leq R_{x,w,j} \leq 2e$ and minimize $e \in [0, 1]$. Projection of this compact semialgebraic feasible set gives its semialgebraic attainable-error set. Its left endpoint is algebraic, and real-algebraic sampling gives a machine above that endpoint. The horizon and word set are finite; the number of those

inequalities can be exponential. Standard real quantifier elimination proves the claim, with no missing infinite-horizon limit. Apply it successively to the finitely many allowed integer profiles or to padded constant caps. The held two-label output cut is then included in the reported peak. $\square$

For a *given* machine and tolerance factor $1 + \delta$, choosing an even $p \geq \log Q / \log(1 + \delta)$ gives lower and upper error certificates with ratio at most $1 + \delta$. This statement concerns checking the supplied machine, not finding a global moment minimizer. A certificate for a globally optimized bound must certify that optimization as well. A numerical local minimum has no such status. For fixed dimensions, exact rational coefficient growth in (5.6) is bounded by polynomially many input bit lengths, products and multinomial factors; certified root intervals can replace irrational p th roots in the output. The implementation below uses exact degree-two and degree-four recurrences, not an unreported large-degree optimization.

5.2. Finite precision without changing boundary widths. For an integer $L \geq 1$, let $\mathcal{G}_L$ contain all machines whose initial and transition probabilities, and positive-answer probabilities, are integral multiples of $1/L$. Let $e_L = \min_{\mathcal{M} \in \mathcal{G}_L} e(\mathcal{M})$ and set

$$(5.7) \quad C(K_\bullet) = (K_0 - 1) + \sum_{t=1}^N (K_t - 1) + 1.$$

These finite sets are nonempty even when some cuts have only one label.

Theorem 5.3 (A finite two-sided rational certificate). *For every finite binary target, without any orthogonality assumption,*

$$(5.8) \quad \max\{0, e_L - C(K_\bullet)/L\} \leq \mathcal{E}(K_\bullet) \leq e_L.$$

Consequently dyadic machines with the same available boundary profile attain every strictly larger error than $\mathcal{E}$. The theorem does not assert a dyadic or rational exact optimizer at the boundary. If two target probability tables differ by at most η uniformly, their optimal errors at the same profile differ by at most η . Thus a strict positive feasibility or infeasibility margin persists under sufficiently small target perturbations.

Proof. Round down the first $k - 1$ coordinates of any probability row of size k to multiples of $1/L$, and give the remaining mass to the last coordinate. The row stays stochastic and its TV change is at most $(k - 1)/L$. Round a terminal positive-answer probability to within $1/L$. Couple the two machines at initialization and then at each common command, or telescope the products by TV contraction. For every word the total output change is bounded by (5.7) divided by L . Applying this construction to a minimizing real machine gives $e_L \leq \mathcal{E} + C/L$; inclusion of the grid in the full parameter set gives $\mathcal{E} \leq e_L$. Both use the same labels; no scratch configuration is hidden in that assertion. Choosing $L = 2^b$ sufficiently large proves the strict-margin assertion. Finally, for any fixed machine its errors against two targets differ by at most η by the triangle inequality. Minimize in both directions. $\square$

An exhaustive grid search is a terminating certificate algorithm at fixed L , not an efficient proposal for large instances. If the target is supplied by certified intervals of radius η , the two endpoints in (5.8) can each be enlarged by η . Dyadic row sampling uses b fresh fair bits at denominator 2^b ; index, table and scratch storage still have to be charged in an implementation. The separate uniform compiler below supplies its own workspace accounting. The rational approximation statement must not be strengthened to exact rational factorization: such field restrictions genuinely matter in nonnegative realization [21].

Proposition 5.4 (An exact calibrated finite frontier). *Take $N = 0$, the four planar seeds $\{\pm e_1, \pm e_2\}$, two coordinate queries and $0 < \rho \leq 1/2$ (this wider signal range is used only for the*

present finite example). If k is the available initialization alphabet, the optimal binary-TV errors are

$$(5.9) \quad \mathcal{E}(1) = \rho/2, \quad \mathcal{E}(2) = \rho/4, \quad \mathcal{E}(k) = 0 \quad (k \geq 3).$$

The same formula holds for an arbitrary horizon consisting only of identity commands, with $k = \min_t K_t$. In all cases the final held answer still costs two labels.

Proof. A one-state decoder is one point of the mean square. Approximating $\pm \rho e_1, \pm \rho e_2$ by a common point has minimum uniform coordinate error ρ , attained at zero. For two states every seed mean lies on one affine line $n \cdot y = c$, with $n \neq 0$; a degenerate segment may be embedded in any such line. If all coordinate errors are at most d , duality of ℓ_∞ and ℓ_1 gives

$$d\|n\|_1 \geq \max_{y \in \{\pm \rho e_1, \pm \rho e_2\}} |n \cdot y - c| = \rho\|n\|_\infty + |c| \geq \frac{\rho}{2}\|n\|_1.$$

The diagonal segment with endpoints $\pm(\rho/2, \rho/2)$ attains coordinate error $\rho/2$: encode e_1, e_2 at its positive endpoint and their negatives at its negative endpoint. Divide mean errors by two to obtain binary TV. Three legal mean vectors $(2\rho, 0), (-\rho, 2\rho), (-\rho, -2\rho)$ enclose all four targets, so their barycentric initialization gives exact responses. For identity-only words, condition at a narrowest cut to obtain the same one-cut lower bound. Initialize the appropriate construction and carry its label through all cuts, padding unused labels as necessary, to obtain the upper bound. $\square$

5.3. A complexity boundary with a different output convention. The following comparison concerns one *categorical* terminal output, not a selected member of a family of binary queries. Its width is the pre-output memory alphabet; counting the held output would separately add its alphabet size.

Proposition 5.5 (Normalized factorization and the real field). *Given an explicitly listed rational row-stochastic matrix P , deciding whether it has a stochastic factorization $P = ED$ of inner dimension at most k is $\exists\mathbb{R}$ -complete. It is already a zero-command numerical realization problem. Rational target data need not have a rational exact optimizer at the minimum real inner dimension.*

Proof. Membership follows from the nonnegative variables, row-sum equations and quadratic equations $P = ED$. For hardness, remove the zero rows of a nonzero nonnegative rational matrix A and divide each remaining row by its positive row sum, giving P . Positive diagonal row scaling and removal of zero rows preserve nonnegative rank. If $P = WH$ is a nonnegative factorization, put $h_i = \sum_j H_{ij}$. For $h_i > 0$ set $D_{ij} = H_{ij}/h_i$ and $E_{xi} = W_{xi}h_i$. If $h_i = 0$, replace that unused decoder row by any probability row and set its E column to zero. Then $D\mathbf{1} = \mathbf{1}$ and $E\mathbf{1} = P\mathbf{1} = \mathbf{1}$. Thus stochastic inner dimension is exactly nonnegative rank, over either the real or rational field. Shitov's Theorem 2 gives $\exists\mathbb{R}$ -hardness and his Corollary 3 gives rational matrices with different minimum ranks over the rational and real algebraic fields [21]. Row normalization transfers both statements as just shown. Zero input matrices can be treated separately and do not affect the reduction. $\square$

This transfers a known complexity theorem; it is not a new hardness proof for nonnegative rank. Nor does it establish the same hardness for the fixed-dimensional orthogonal binary-query subclass. In that subclass Theorem 5.1 supplies the compact exact feasibility formula, while the inherited arithmetic theorems determine specific asymptotic widths. These are different assertions.

6. FINITE WORD PROFILES AND COMPATIBLE ENCLOSURES

The results in this section apply to every finite orthogonal alphabet, including noncommuting alphabets. No connectedness, arithmetic condition or spectral gap is assumed. Invariant directions may make a converse profile zero; this is a property of the stated profile, not an omitted hypothesis.

6.1. An optimized distortion. For $B, k \geq 1$ and a probability vector p on the actual words $\mathcal{A}^B$, define

$$(6.1) \quad D_k(p) = 1 - \max_{\substack{u \in \mathbb{S}^{D-1} \\ v_1, \dots, v_k \in \mathbb{S}^{D-1}}} \sum_{w \in \mathcal{A}^B} p_w \max_{1 \leq j \leq k} \langle v_j, U_w u \rangle,$$

$$(6.2) \quad \Gamma_{\mathcal{A}}(k, B) = \max_{p \in \Delta(\mathcal{A}^B)} D_k(p), \quad \chi_{\mathcal{A}}(k, B) = -\log(1 - \Gamma_{\mathcal{A}}(k, B)).$$

We allow $\chi = +\infty$. Repeated product matrices can be merged for the optimization: their probability masses add. Their representative words must still be executed one ordinary command at a time.

Proposition 6.1 (Finite-data properties). *The maxima in (6.1)–(6.2) are attained, $0 \leq \Gamma_{\mathcal{A}}(k, B) \leq 1$, and the profile is invariant under simultaneous orthogonal changes of coordinates. It is nonincreasing in k and unchanged by duplicating a letter. For two indexed orthogonal alphabets U, V ,*

$$(6.3) \quad |\Gamma_U(k, B) - \Gamma_V(k, B)| \leq B \max_a \|U_a - V_a\|_{\text{op}}.$$

For a fully listed finite alphabet of real-algebraic matrices and fixed integers k, B, D , $\Gamma_{\mathcal{A}}(k, B)$ and an optimizing word law are real-algebraic and can be determined by real quantifier elimination.

Proof. All parameter spaces are compact and the objective is continuous. Its maximum over directions is convex in p , so D_k is continuous and concave. The maximum scalar product is at most one. By choosing a single direction parallel to $\sum p_w U_w u$, the inner maximum is nonnegative; hence $D_k \in [0, 1]$. Conjugating all directions proves invariance. Additional centers only increase that inner maximum. Product telescoping gives $\|U_w - V_w\|_{\text{op}} \leq B \max_a \|U_a - V_a\|_{\text{op}}$, and maxima and minima preserve this uniform bound.

For the last assertion, enumerate the product matrices. Unit-sphere equations, simplex inequalities and finitely many maximum operations define a compact semialgebraic optimization with algebraic coefficients. Maximum operations can be expressed by finite alternatives or auxiliary variables. Quantifier elimination over the real closed field of real-algebraic numbers determines the endpoint of the feasible value interval and a witness [1]. The word table may have $|\mathcal{A}|^B$ entries; this is not a polynomial-time assertion or a decision procedure for an infinite spectral gap. No logarithm needs to be eliminated: finite budget comparisons can instead use products of $1 - \Gamma$. $\square$

Lemma 6.2 (Endpoint contraction; inherited mechanism). *Let a complete random word W with law p be independent of all information before its start. Internal letters of W may be correlated. If the endpoint register has at most k labels, then*

$$(6.4) \quad q' \leq (1 - D_k(p))q.$$

There is no bound on the intermediate registers.

Proof. Write $z_s = \mathbb{E}[Y \mid S = s]$ at the start. Composing all internal stochastic rows yields a common endpoint kernel $T(j \mid s, w)$. Since the entire word and the machine's fresh randomness are independent of the past,

$$z'_j = \mathbb{E}[U_W z_S \mid S' = j].$$

This is an endpoint identity; it would not generally hold after conditioning on a correlated next letter inside the packet. Choose $v_j = z'_j / \|z'_j\|$ for nonzero z'_j , and any unit vector otherwise. Then

$$\begin{aligned} q' &= \sum_s \mathbb{P}(S = s) \sum_w p_w \sum_j T(j \mid s, w) \langle v_j, U_w z_s \rangle \\ &\leq \sum_s \mathbb{P}(S = s) \|z_s\| \max_{\|u\|=1} \sum_w p_w \max_j \langle v_j, U_w u \rangle \leq (1 - D_k(p))q. \end{aligned}$$

Terms with $z_s = 0$ contribute zero. Replacing an executable assignment by the largest pairing is a relaxation in the correct direction. Arbitrary unobservable hidden coordinates are included in s ; no statewise observable-vector hypothesis is used. This is the conditional-centroid mechanism of [7, 8], now applied to the optimized finite word datum. $\square$

For an available profile $K_\bullet$, let $I_0 = 0$ and define

$$(6.5) \quad I_t = \max \left\{ I_{t-1}, \max_{1 \leq B \leq t} [I_{t-B} + \chi_{\mathcal{A}}(K_t, B)] \right\}, \quad \mathcal{I}_N(K_\bullet) = I_N.$$

If $\Gamma_{\mathcal{A}}(K_t, B) = 1$, the interval annihilates the conditional-centroid amplitude and is incompatible with any positive terminal calibration. This is the meaning of an infinite charge. The recurrence is the largest sum of charges of disjoint command intervals, with an interval of length B ending at t charged by $\chi_{\mathcal{A}}(K_t, B)$. The recurrence depends only on the alphabet and the available profile.

Theorem 6.3 (A nonuniform interval budget). *Every machine with $\kappa = \rho/\sqrt{D} - 2\varepsilon > 0$ satisfies*

$$(6.6) \quad \mathcal{I}_N(K_\bullet) \leq \log(1/\kappa).$$

More generally its optimal actual terminal error obeys

$$(6.7) \quad \mathcal{E}_N(K_\bullet) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{D}} - e^{-\mathcal{I}_N(K_\bullet)} \right]_+.$$

For a uniform peak bound K , every $B \leq N$ gives

$$(6.8) \quad \lfloor N/B \rfloor \chi_{\mathcal{A}}(K, B) \leq \log(1/\kappa).$$

For a fixed B, k with $\Gamma_{\mathcal{A}}(k, B) > 0$,

$$(6.9) \quad \#\{1 \leq t \leq N : K_t \leq k\} \leq B - 1 + \frac{B \log(1/\kappa)}{\chi_{\mathcal{A}}(k, B)}.$$

If $\chi = +\infty$, the last fraction is zero.

Proof. Choose any maximizing disjoint interval family in (6.5). On each selected interval put an independently sampled optimizing complete word from Proposition 6.1. All gaps and the final remainder contain fixed deterministic command words. A deterministic orthogonal command followed by a stochastic update cannot increase the conditional-mean norm, by conditional Jensen. Lemma 6.2 therefore gives

$$q_N \leq \prod_{(t-B, t]} (1 - \Gamma_{\mathcal{A}}(K_t, B)) = e^{-\mathcal{I}_N(K_\bullet)},$$

since $q_0 = 1$. The chosen law can depend on the available layer sizes, not on private states. Wordwise correctness requires the same machine to pass it. Lemma 2.2 proves (6.6) and (6.7); taking consecutive intervals proves (6.8).

For (6.9), discard candidate endpoints before B . Select the earliest remaining endpoint, delete all later candidates less than B away, and repeat. If M is the original number and J the number selected, the explicit count is

$$M - (B - 1) \leq BJ.$$

The associated length- B intervals are disjoint and each contributes at least $\chi_{\mathcal{A}}(k, B)$. Applying (6.6) gives the result. No restriction on the other registers was introduced. $\square$

The cap union estimate below is the geometric certificate already used in the predecessor packet method [7]. The finite incidence formulation specifies how to optimize that certificate on a fully listed word set.

Proposition 6.4 (Uniform cap certificates). *Suppose a law p on length- B words satisfies*

$$\sup_{u,v \in \mathbb{S}^{D-1}} p\{w : \|U_w u - v\| < \delta\} \leq b.$$

Then $\Gamma_{\mathcal{A}}(k, B) \geq \frac{1}{2}\delta^2(1 - kb)_+$. For algebraic product matrices and algebraic δ , the best such b is the value of a finite linear program once cap incidence types have been decided by real quantifier elimination.

Proof. The union of k open caps has probability at most kb . Outside it, $1 - \max_j \langle v_j, U_w u \rangle = \frac{1}{2} \min_j \|U_w u - v_j\|^2 \geq \delta^2/2$. For the algorithmic assertion, a subset of the finite word list either can or cannot be contained in one such cap for some u, v . This is a finite system of algebraic inequalities, including the sphere equations. For every feasible subset A impose $\sum_{w \in A} p_w \leq b$, along with the simplex constraints, and minimize b . Every actual incidence set is feasible; conversely a feasible subset is included in an actual incidence set and has no greater mass. Thus these constraints give exactly the desired supremum. The exponentially large enumeration is not concealed. $\square$

6.2. A separate exact-synthesis profile. Fix $0 < a < 1$. For k such that the following family is nonempty, let

$$(6.10) \quad \lambda_{\mathcal{A}}(k, a) = \min_P \inf \{\lambda \geq 1 : U_b P \subseteq \lambda P \text{ for all } b \in \mathcal{A}\},$$

where P ranges over convex hulls of at most k points in $\mathbb{B}^D$ containing $a\mathbb{B}^D$. An empty family has value $+\infty$.

Theorem 6.5 (Compatible enclosure synthesis). *The finite value in (6.10) is attained and lies in $[1, a^{-1}]$. If $\rho\lambda_{\mathcal{A}}(k, a)^N \leq a$, there is an exact machine with $\max(k, 2)$ labels, common command matrices and a horizon-dependent decoder. Each row has at most $D + 1$ successors. Algebraic matrices and algebraic a give an algebraically decidable profile value and algebraic row witnesses. For two indexed alphabets with a common feasible family,*

$$(6.11) \quad |\lambda_U(k, a) - \lambda_V(k, a)| \leq a^{-1} \max_b \|U_b - V_b\|_{\text{op}}.$$

Proof. Parameterize P by k points in the closed unit ball, allowing repetition. Containment of $a\mathbb{B}^D$ is closed in these coordinates. Every such P satisfies $U_b P \subseteq \mathbb{B}^D \subseteq a^{-1}P$. The containment conditions are closed on this compact set and $1 \leq \lambda \leq a^{-1}$; hence a minimum exists.

Let its vertices be v_s and λ its dilation. Initialize with a convex representation of ax . For each b, s , represent $\lambda^{-1}U_b v_s$ by the same vertices and use its convex coefficients as the row $T_b(s, \cdot)$. Induction gives

$$\mathbb{E}[v_{S_t} \mid w] = a\lambda^{-t}U_w x.$$

The decoder $d_j(s) = \rho\lambda^N \langle e_j, v_s \rangle / a$ lies in $[-1, 1]$ and gives (1.1). These are actual rows throughout the run, not independently chosen checkpoint factorizations. Carathéodory gives $D + 1$ successors; there are $|\mathcal{A}|k$ command rows, and their construction and storage remain uncharged.

The defining finite problem is semialgebraic: use convex coefficients for transformed vertices and a universal ball containment condition, then eliminate quantifiers. To prove (6.11), set $e = \max_b \|U_b - V_b\|_{\text{op}}$. For every feasible P , $V_b P \subseteq U_b P + e\mathbb{B}^D \subseteq (\lambda + e/a)P$. Interchanging U, V completes the proof. $\square$

The two profiles have different quantifiers. The converse tests every hidden machine; the enclosure theorem builds a particular positive realization. Neither theorem asserts that $\lambda_{\mathcal{A}}$ is the optimum over hidden machines. This is why finite-scale lower and upper bounds can be used even when they do not match.

Proposition 6.6 (Bounded word recoding). *Suppose every command of $\mathcal{A}$ is a product of exactly L commands of an alphabet $\mathcal{B}$ on the same space, with the same seeds and queries. Then*

$$\Gamma_{\mathcal{A}}(k, B) \leq \Gamma_{\mathcal{B}}(k, LB), \quad \lambda_{\mathcal{A}}(k, a) \leq \lambda_{\mathcal{B}}(k, a)^L, \quad W_{N, \varepsilon}(\mathcal{A}) \leq W_{LN, \varepsilon}(\mathcal{B}).$$

An identity letter permits words of length at most L to be padded. If both alphabets contain an identity and admit such bounded recodings into the other, the existence and value of a logarithmic width exponent agree.

Proof. Push forward a packet law along the word substitution to obtain the first inequality. Iterate the enclosure containments for the second. For the last inequality, compose the L stochastic rows of a $\mathcal{B}$ machine into one row for the corresponding $\mathcal{A}$ command, retaining only the block-end label. All rows may depend on the external epoch. This is an implementation of the atomic $\mathcal{A}$ interface, not a claim that its physical letters cost L letters. Applying the two comparisons and monotonicity under final identity padding proves the exponent statement. $\square$

Corollary 6.7 (Packing–dilation balance). *Fix a calibrated experiment, an inradius $a > \rho$, and positive numbers u, v . Suppose that for every sufficiently large k there are a length $B(k) \leq Ck^u$ and a word law for which*

$$\sup_{x, z \in \mathbb{S}^{D-1}} p\{w : \|U_w x - z\| < ck^{-v}\} \leq (2k)^{-1}.$$

Suppose also that $\lambda_{\mathcal{A}}(k, a) \leq \exp(C'k^{-u-2v})$. Then

$$W_{N, \varepsilon} = \Theta(N^{1/(u+2v)}).$$

The same conclusion holds with $u = 0$ when the lengths are uniformly bounded.

Proof. Proposition 6.4 gives $\Gamma(k, B(k)) \geq c^2 k^{-2v}/4$. Equation (6.8) implies $N \leq C_1 k^{u+2v}$ for a feasible peak k . Conversely choose k at a sufficiently large constant times $N^{1/(u+2v)}$ so that $C'Nk^{-u-2v} \leq \log(a/\rho)$, and apply Theorem 6.5. The hypotheses concern finite word orbits and compatible convex containments, and impose no commutativity on the matrices. $\square$

7. A DISTORTION–DILATION COMPARISON

The lower and upper profiles defined in Section 6 do not have identical quantifiers. They nevertheless obey a universal inequality. Define the optimized per-command distortion rate

$$(7.1) \quad \mathfrak{r}_{\mathcal{A}}(k) = \sup_{B \geq 1} \frac{-\log(1 - \Gamma_{\mathcal{A}}(k, B))}{B}.$$

All logarithms in this paper are natural. The value $+\infty$ has its usual extended meaning.

Theorem 7.1 (Distortion bounds every common enclosure). *For every finite orthogonal alphabet and every feasible k, a in (6.10),*

$$(7.2) \quad \mathfrak{r}_{\mathcal{A}}(k) \leq \log \lambda_{\mathcal{A}}(k, a).$$

In particular, every finite-word certificate gives the lower enclosure bound $\lambda_{\mathcal{A}}(k, a) \geq (1 - \Gamma_{\mathcal{A}}(k, B))^{-1/B}$. The assertion does not require commuting commands, a spectral gap, or any Diophantine hypothesis.

Proof. Take any feasible polytope $P = \text{conv}(v_1, \dots, v_k)$, contained in the unit ball and containing a times that ball, with common dilation λ . Use the stochastic rows representing $\lambda^{-1}U_b v_s$ by the same vertices. Fix a unit seed u and initialize the mean at au . This construction makes sense at every length, without imposing a terminal decoder. For each complete word w of length n it satisfies

$$\mathbb{E}[v_{S_n} \mid w] = a\lambda^{-n}U_w u.$$

Let $Y_n = U_w u$ and $q_n = \mathbb{E} \|\mathbb{E}[Y_n | S_n]\|$. Pairing with the bounded vector v_{S_n} gives, under every word law,

$$(7.3) \quad q_n \geq \mathbb{E} \langle Y_n, v_{S_n} \rangle = a\lambda^{-n}.$$

Fix B and independently repeat an optimizing length- B word law m times. The packet lemma applies at the m endpoints, each of which has at most k labels. Since $q_0 = 1$,

$$a\lambda^{-mB} \leq q_{mB} \leq (1 - \Gamma_{\mathcal{A}}(k, B))^m.$$

Taking m th roots and letting $m \rightarrow \infty$ removes the inradius factor. Hence $\lambda^{-B} \leq 1 - \Gamma_{\mathcal{A}}(k, B)$. Take the supremum over B and then the minimum over enclosures. In particular, $\Gamma = 1$ is incompatible with a finite feasible enclosure at this k , so no undefined logarithmic operation was used. $\square$

The removal of the initial inradius loss by repetition is important: a single-horizon application would leave an additive $\log(1/a)$. Theorem 7.1 is a comparison, not a duality asserting equality or a characterization of arbitrary hidden width. The Hankel normal form in Proposition 4.1 permits general future-response polytopes, whereas (6.10) uses one common polytope in the observed vector space.

Corollary 7.2 (Nonuniform enclosure chains). *Suppose $P_t \subset \mathbb{B}^D$ has at most K_t generators, $a\mathbb{B}^D \subset P_0$, and $U_b P_{t-1} \subset \lambda_t P_t$ for every command. Then*

$$\mathcal{I}_N(K_\bullet) \leq \log(1/a) + \sum_{t=1}^N \log \lambda_t.$$

Proof. Use the cut-dependent convex rows and the initialization with mean au . The same induction gives mean $a(\prod_t \lambda_t)^{-1} U_w u$. Choose the independent packet law attaining the interval recurrence. The pairing argument bounds the final amplitude below by this factor, while the packet theorem bounds it above by $\exp(-\mathcal{I}_N(K_\bullet))$. Taking logarithms proves the claim. $\square$

8. EXACT OPTIMIZATION OF PLANAR PACKETS

For a rotation alphabet, let $z_1, \dots, z_M \in \mathbb{S}^1 \subset \mathbb{C}$ be the distinct product phases of words of length B , ordered cyclically. A common rotation of all phases can be absorbed into the decoder centers. The initial direction in (6.1) therefore disappears. Any probability distribution on these phases is implemented by a distribution on representative words of the same length.

Let $\mathfrak{P}_k$ be the finite set of partitions of the cyclically ordered list into at most k nonempty consecutive arcs. For $p \in \Delta_M$, set

$$(8.1) \quad F_k(p) = \max_{\mathcal{P} \in \mathfrak{P}_k} \sum_{A \in \mathcal{P}} \left| \sum_{i \in A} p_i z_i \right|.$$

Theorem 8.1 (Optimal word laws as a cone program). *For the planar alphabet above,*

$$(8.2) \quad \Gamma_{\mathcal{A}}(k, B) = 1 - \min_{p \in \Delta_M} F_k(p).$$

Equivalently, minimize t subject to $p \in \Delta_M$ and, for every $\mathcal{P} \in \mathfrak{P}_k$,

$$(8.3) \quad \left\| \sum_{i \in A} p_i (\Re z_i, \Im z_i) \right\|_2 \leq r_{\mathcal{P}, A} \quad (A \in \mathcal{P}), \quad \sum_{A \in \mathcal{P}} r_{\mathcal{P}, A} \leq t.$$

This is a finite second-order cone program with attained optimum. A fixed law can be evaluated by cyclic segmentation in $O(kM^3)$ arithmetic operations, including computation of all interval sums. The optimization over laws may have exponentially many partition constraints.

Proof. The decoder centers are on the unit circle; maximizing their inner product is therefore exactly nearest chordal-distance assignment.

For fixed centers $v_1, \dots, v_k$ assign each phase to a nearest center in chordal distance; equivalently maximize $\Re(\bar{v}_j z_i)$. Distinct centers have connected cyclic Voronoi arcs, bounded by the angular midpoints of adjacent centers. Ties can be assigned consecutively; coincident centers can be merged. For the induced partition, the contribution of a block A is at most $|\sum_{i \in A} p_i z_i|$.

Conversely, for any cyclic partition choose the center of each block in the direction of its weighted complex sum, arbitrarily if that sum vanishes. The best assignment to those centers has value at least the sum in (8.1). Thus the maximum over centers equals $F_k(p)$. This proves (8.2); introducing the norm epigraphs proves (8.3). Zero weights follow either directly or by continuity from positive weights.

For a fixed cyclic cut, prefix sums give every block vector in constant time. The usual recurrence for the best j blocks among the first m points checks at most M previous split positions. It costs $O(kM^2)$ for that cut. Repeating for the M cuts costs $O(kM^3)$. This recurrence evaluates a given law; it is not a polynomial-size representation of all constraints in (8.3). Compactness of the simplex and continuity give attainment. $\square$

Nearest-center partitions and vector quantization are classical [5]. The additional optimization here chooses the distribution of actual command words, and Theorem 6.3 turns its value into a constraint on arbitrary hidden registers. There is no assertion that static quantization alone characterizes the least whole-run width.

Proposition 8.2 (A finite exact packet certificate). *For the alphabet consisting of identity and a quarter turn, at word length three the four distinct phases are $1, i, -1, -i$. Their uniform law is optimal and*

$$\Gamma(1, 3) = 1, \quad \Gamma(2, 3) = 1 - \frac{1}{\sqrt{2}}, \quad \Gamma(3, 3) = \frac{2 - \sqrt{2}}{4}, \quad \Gamma(k, 3) = 0 \quad (k \geq 4).$$

Proof. The distortion is concave in the phase law and invariant under its four cyclic rotations. Averaging any law over those rotations proves optimality of the uniform law. For two centers, adjacent pairs give total $\sqrt{2}/2$; a triple and a singleton give only $1/2$. For three centers, one pair and two singletons give $\sqrt{2}/4 + 1/2$. For one center the sum is zero; four centers fit all phases. Formula (8.2) gives the values. A uniform phase law is realized by the four ordinary words with zero, one, two or three initial quarter turns and then idle commands. $\square$

For algebraic rotation matrices, both the general quantifier-elimination formulation and the cone program have algebraic coefficients. No finite list of spherical harmonics is used. For merely computable-real entries, finite numerical approximations are diagnostics unless a rigorous error enclosure is supplied; (6.3) then quantifies how an entry enclosure propagates to a packet certificate.

Corollary 8.3 (Separated phases). *If $M \geq 2k$ distinct length- B word phases have pairwise circular distance at least $s \in (0, 1/2]$, then $\Gamma_{\mathcal{A}}(k, B) \geq s^2$.*

Proof. Use their uniform law. Each open arc of circular radius $s/2$ contains at most one phase. At least half the mass is at distance at least $s/2$ from every one of k centers. On $[0, 1/2]$, $1 - \cos(2\pi t) = 2\sin^2(\pi t) \geq 8t^2$. The average loss is at least s^2 . This is a lower certificate for (8.2), not necessarily its exact value. $\square$

9. ARITHMETIC PROFILES AT EVERY FINITE SCALE

Now $\mathcal{A}_\alpha = \{I, R_{2\pi\alpha_1}, \dots, R_{2\pi\alpha_r}\}$, with r fixed. Define

$$(9.1) \quad \psi_\alpha(H) = \min_{0 < \|m\|_1 \leq H} \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}, \quad \eta_\alpha(q) = \max_{1 \leq i \leq r} \|q\alpha_i\|_{\mathbb{R}/\mathbb{Z}}.$$

These are finite functions of the alphabet parameters. The first may vanish in the presence of a rational relation. The upper profile remains usable in that case.

Theorem 9.1 (Finite arithmetic bounds). *Let $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$. For a machine of peak K , put $L = \lceil (2K)^{1/r} \rceil$ and $B = r(L - 1)$. Then*

$$(9.2) \quad \lfloor N/B \rfloor \psi_\alpha(B)^2 \leq \log(1/\kappa).$$

For every integer $q \geq 4$,

$$(9.3) \quad N\eta_\alpha(q) \leq q^2/100 \implies W_{N,0}(\alpha) \leq q.$$

Consequently every finite planar rotation alphabet satisfies

$$(9.4) \quad W_{N,0}(\alpha) \leq \max(4, \lceil \sqrt{50N} \rceil).$$

All upper bounds use common command rows and hold also after adjoining reflection in the first coordinate axis.

Proof. Construct a word of length B in r consecutive segments of $L - 1$ positions. In segment i give J_i copies of the i th rotation followed by idle commands, independently and uniformly for $J_i \in \{0, \dots, L - 1\}$. Differences of count vectors have ℓ_1 norm at most B . If $\psi_\alpha(B) > 0$, the $L^r \geq 2K$ phases are distinct and separated by $\psi_\alpha(B)$. Corollary 8.3 and (6.8), with $-\log(1-x) \geq x$, give (9.2). If the minimum is zero the displayed inequality is automatic. All letters in a segment are actual commands, not a free count input.

For the upper bound let P_q be the regular q -gon inscribed in the unit circle. Choose nearest integers p_i to $q\alpha_i$, and put $\delta_i = 2\pi(\alpha_i - p_i/q)$, so $|\delta_i| \leq \pi/q$. Elementary chord geometry gives

$$(9.5) \quad R_{2\pi\alpha_i}P_q \subseteq \Lambda_i P_q, \quad \Lambda_i = \frac{\cos(\pi/q - |\delta_i|)}{\cos(\pi/q)}, \quad \log \Lambda_i \leq \frac{4\pi^2\eta_\alpha(q)}{q^2}.$$

Indeed the ray of a rotated vertex meets a neighboring edge at radial coordinate $\cos(\pi/q)/\cos(\pi/q - |\delta_i|)$. Integrating the derivative of $\log \cos(\pi/q - t)$ bounds the logarithm by $|\delta_i| \tan(\pi/q)$, and $\tan(\pi/q) \leq 2\pi/q$ for $q \geq 4$.

Take $\Lambda = \max_i \Lambda_i$, including one for the identity. Under (9.3), $\Lambda^N \leq e^{\pi^2/25} < 2$. Since P_q contains the half-unit disk, Theorem 6.5 with $a = 1/2$ applies: $\rho\Lambda^N \leq 1/5 < 1/2$. Explicitly, initialize at mean $x/2$, represent $\Lambda^{-1}R_{2\pi\alpha_i}v_s$ by the vertices for each row, and decode by $2\rho\Lambda^N \langle e_j, v_s \rangle$. This verifies every numerical word, including any deterministic remainder of a converse packet decomposition.

There are $(r + 1)q$ common command rows and two decoder columns, each row having at most three successors. These counts exclude row-description and arithmetic costs. Since $\eta_\alpha(q) \leq 1/2$, (9.4) follows. Reflection preserves P_q ; its contracted image uses the same vertex set. The reflected lower bound is obtained simply by restricting to rotation-only words. $\square$

The pair $(\psi_\alpha, \eta_\alpha)$ is a specialization of the word-distortion and enclosure profiles. It is not assumed to characterize exact finite width. In particular, (9.2) can be weak when long word packets have clusters; the optimized profile of Section 8 is available without that separation relaxation.

Lemma 9.2 (Denominators under a general power bound). *Suppose for $\tau \geq r$ and $b > 0$,*

$$(9.6) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b \|m\|_1^{-\tau} \quad (m \neq 0).$$

For every integer $Q \geq 2$ there is q with

$$(9.7) \quad cQ^{r/(\tau+1-r)} \leq q \leq Q^r, \quad \eta_\alpha(q) \leq Q^{-1}, \quad c = (b/r^{\tau+1})^{r/(\tau+1-r)}.$$

Proof. The $Q^r + 1$ points $j\alpha$ in Q^r equal torus boxes give integers p_i, q with $1 \leq q \leq Q^r$ and $|q\alpha_i - p_i| \leq Q^{-1}$. For $M = \lfloor q^{1/r} \rfloor$, the $(M+1)^r > q$ integer vectors in $\{0, \dots, M\}^r$ have a residue collision for $m \cdot p \pmod{q}$. Subtracting gives $m \neq 0$ with $H = \|m\|_1 \leq rq^{1/r}$ and $q \mid m \cdot p$. Thus

$$bH^{-\tau} \leq \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq H/(qQ), \quad bqQ \leq r^{\tau+1}q^{(\tau+1)/r}.$$

Rearranging yields (9.7). This direct residue argument supplies exactly the quantitative transference needed here, without an unspecified transference constant. $\square$

Corollary 9.3 (Power bounds at finite Diophantine type). *Under (9.6), for constants depending on the fixed data,*

$$(9.8) \quad c_1 N^{r/(2\tau+1)} \leq W_{N,\varepsilon}(\alpha) \leq C_1 N^{\min\{1/2, r(\tau+1-r)/(\tau+1+r)\}}.$$

No matching claim is made when $\tau > r$.

Proof. From (9.2), $N < B + b^{-2}B^{2\tau+1} \log(1/\kappa)$, while $B \leq r(2K)^{1/r}$. This gives the lower power after adjusting finitely many initial horizons. Lemma 9.2 bounds $N\eta_\alpha(q)/q^2 \leq c^{-2}NQ^{-1-2r/(\tau+1-r)}$. Choose Q at constant times $N^{(\tau+1-r)/(\tau+1+r)}$, large enough to satisfy (9.3) and $q \geq 4$. Then $q \leq Q^r$ proves the second candidate upper power; (9.4) proves the first. $\square$

Corollary 9.4 (The retained badly approximable law). *If (9.6) holds with $\tau = r$, then $W_{N,\varepsilon}(\alpha) = \Theta(N^{r/(2r+1)})$, also with the reflection extension. This is the previously established matched law [8].*

Proof. Substitute $\tau = r$ in (9.8); the two powers coincide. $\square$

For $1, \alpha_1, \dots, \alpha_r$ rationally independent, every optimized finite-block action gap on the circle is zero. Indeed simultaneous approximation supplies unbounded frequencies n with every $n\alpha_i$ close to an integer. Each fixed rotation word then acts almost identically on its Fourier mode. For the reflected alphabet use the real cosine mode, which reflection fixes. This is the established almost-invariance argument, not a claim that the finite packet distortion vanishes.

10. MINIMAL ORDINARY TYPE AND ALMOST-EVERYWHERE WIDTHS

Uniform bad approximation is substantially stronger than the condition needed to determine the logarithmic width exponent. We first deduce the minimal-type assertion without any measure-theoretic hypothesis.

Proof of (1.4) in Theorem 1.1. If $\omega^*(\alpha) = r$, then for every $h > 0$ there is $b_h > 0$ such that

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b_h \|m\|_1^{-r-h} \quad (m \neq 0).$$

Indeed the definition of the limsup gives this inequality for all sufficiently large $\|m\|_1$, and rational independence makes the finite remaining minimum positive. Apply Corollary 9.3 with $\tau = r + h$. It gives

$$\frac{r}{2r+2h+1} \leq \liminf_N \frac{\log W_{N,\varepsilon}}{\log N} \leq \limsup_N \frac{\log W_{N,\varepsilon}}{\log N} \leq \min \left\{ \frac{1}{2}, \frac{r(1+h)}{2r+1+h} \right\}.$$

For $0 < h < 1/(2r-1)$ the second branch is smaller. Let $h \downarrow 0$. This proves (1.4). No positive uniform constant at exponent r was used. $\square$

For completeness we prove a more quantitative metric statement, using only the summable part of metric Diophantine approximation. It does not claim a new metric theorem.

Lemma 10.1 (A summable logarithmic separation). *For Lebesgue-almost every $\alpha \in [0, 1]^r$, and every $s > 1$, there is $b_s > 0$ such that*

$$(10.1) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{b_s}{\|m\|_1^r (\log(2 + \|m\|_1))^s} \quad (m \neq 0).$$

Such a vector has $\omega^(\alpha) = r$.*

Proof. For a nonzero integer vector m , the map $\alpha \mapsto m \cdot \alpha \pmod{1}$ sends Haar measure on the torus to uniform circle measure. There are at most $C_r H^{r-1}$ integer vectors with $\|m\|_1 = H$. Hence the measures of the events

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} < [\|m\|_1^r (\log(2 + \|m\|_1))^s]^{-1}$$

have finite sum, bounded by a constant times $\sum_{H \geq 1} [H(\log(2 + H))^s]^{-1}$. The measure of their tail union tends to zero. Almost every vector violates the unscaled inequality only finitely often. Remove the measure-zero rational hyperplanes and reduce the constant to cover the finite exceptions. Intersect the full-measure sets for rational $s > 1$. For a real $s > 1$, choose a rational $1 < s_0 < s$; its stronger bound implies the bound with s , after a harmless change of constant. This gives the assertion on one full-measure set.

This bound implies $\omega^*(\alpha) \leq r$. For the reverse inequality, place the $(M + 1)^r$ values $m \cdot \alpha \pmod{1}$, $m \in \{0, \dots, M\}^r$, in fewer than $(M + 1)^r$ intervals of equal length. A collision gives a nonzero difference of norm at most rM and distance $O(M^{-r})$. As $M \rightarrow \infty$ the norms of some such differences must tend to infinity, since rational independence excludes an exact relation. Thus $\omega^*(\alpha) \geq r$. $\square$

Proposition 10.2 (Quantitative bounds under logarithmic separation). *Suppose (10.1) holds for some $s > 0$ and $b_s > 0$. Then there are $c, C > 0$ such that*

$$(10.2) \quad cN^{r/(2r+1)}(\log(N + 2))^{-2rs/(2r+1)} \leq W_{N,\varepsilon}(\alpha) \leq CN^{r/(2r+1)}(\log(N + 2))^{2r^2s/(2r+1)}.$$

Proof. The packet bound (9.2) gives

$$N < B + b_s^{-2} B^{2r+1} (\log(2 + B))^{2s} \log(1/\kappa).$$

With $B \leq r(2K)^{1/r}$ this implies $N \leq C_0 K^{(2r+1)/r} (\log(2 + K))^{2s}$. If $K > N$, the proposed lower bound is already true after reducing its constant. Otherwise $\log(2 + K) \leq \log(2 + N)$, and rearrangement proves the first inequality of (10.2).

For the upper inequality repeat the residue argument of Lemma 9.2. It produces $q \leq Q^r$, $\eta_\alpha(q) \leq Q^{-1}$ and $H \leq rq^{1/r}$ with

$$\frac{b_s}{H^r (\log(2 + H))^s} \leq \frac{H}{qQ}.$$

Since $q \leq Q^r$, it follows for $Q \geq 2$ that

$$(10.3) \quad q \geq c_0 \frac{Q^r}{(\log(Q + 2))^{rs}}.$$

Indeed $b_s q Q \leq H^{r+1} (\log(2 + H))^s \leq C_r q^{1+1/r} (\log(Q + 2))^s$, and raising the rearranged inequality to the r th power gives (10.3).

Choose an integer

$$Q = \left\lceil C_2 N^{1/(2r+1)} (\log(N + 2))^{2rs/(2r+1)} \right\rceil.$$

For sufficiently large fixed C_2 , (10.3) implies $N\eta_\alpha(q)/q^2 \leq 1/100$ and $q \geq 4$. To see the constant choice explicitly, for large N one has $\log(Q + 2) \leq C_3 \log(N + 2)$ with C_3 fixed; thus the ratio is bounded by a constant times $C_2^{-(2r+1)}$. Enlarge the final constant for the finitely many initial horizons. Theorem 9.1 gives an exact machine with $q \leq Q^r$ labels, proving the upper bound. $\square$

Completion of Theorem 1.1. Apply Lemma 10.1 with $s = 1 + \delta$ and then Proposition 10.2. This yields (1.5) on one full-measure set for all fixed $\delta > 0$. These statements concern a fixed alphabet before the horizon grows. None of their constants is asserted uniform in r or near a rational relation. $\square$

10.1. Rational dependencies without redundant exponents. The number of listed rotations need not be the relevant arithmetic rank. The following result covers finite rational reparametrizations rather than assigning one exponent to every redundant letter.

Corollary 10.3 (Finite rational extensions). *Suppose an alphabet contains I and rotations through angles $\beta_1, \dots, \beta_d$, where $1, \beta_1, \dots, \beta_d$ are rationally independent and $\omega^*(\beta) = d$. Suppose every additional angle belongs to $\mathbb{Q} + \sum_{j=1}^d \mathbb{Q}\beta_j$. Then*

$$\lim_N \frac{\log W_{N,\varepsilon}}{\log N} = \frac{d}{2d+1}.$$

The same assertion holds on a full-measure set of β with the logarithmic upper and lower powers of (1.5), replacing r by d . Reflection may be added. If all angles are rational, exact width is bounded independently of N .

Proof. Restriction to the displayed d -angle subalphabet gives the lower bound. Choose one common denominator h and integers t_i, n_{ij} so that every angle is $t_i/h + \sum_j n_{ij}\gamma_j$ with $\gamma_j = \beta_j/h$. Multiplication by h shows $\|m \cdot \gamma\|_{\mathbb{R}/\mathbb{Z}} \geq h^{-1} \|m \cdot \beta\|_{\mathbb{R}/\mathbb{Z}}$. Thus every separating power or logarithmic inequality for β holds for γ with a changed constant.

Take a simultaneous approximant p_j/q to γ from the proof above. A command's irrational part has residual angle at most $2\pi(\sum_j |n_{ij}|)/(qQ)$. The same polygon construction therefore works, after changing only fixed constants. Store in addition the rational phase $c \in \mathbb{Z}/h\mathbb{Z}$, updated by $c \mapsto c + t_i$. The decoder rotates the stored polygon vector by $2\pi c/h$. The peak, including the held answer, is $\max(2, hq)$, not an uncharged phase. Reflection sends c to $-c$ and conjugates the polygon phase; the regular polygon and the construction are reflection-invariant. The exact expectation identity is unchanged. The finite multiplier h does not change a logarithmic exponent or the displayed logarithmic powers.

For a purely rational alphabet, choose a common period h and retain the seed identifier and the accumulated phase. This uses at most $4h$ command labels for our four planar seeds, plus the held binary answer convention. No infinite-gap hypothesis is involved. $\square$

11. EVERY IRRATIONAL ANGLE HAS CUBIC SCALES

We keep the two-letter alphabet $\{I, R_{2\pi\alpha}\}$ and the planar resource and error conventions. No Diophantine lower bound is assumed in this section. The needed continued-fraction estimates are classical [3]; we include their short proof to specify the exact finite inequalities being used.

Lemma 11.1 (Convergent separation). *Let p_n/q_n be the convergents of an irrational α , ignoring the finitely many initial indices with repeated denominator. Then*

$$(11.1) \quad \frac{1}{q_{n+1} + q_n} < |q_n\alpha - p_n| < \frac{1}{q_{n+1}}, \quad \min_{1 \leq l < q_n} \|l\alpha\|_{\mathbb{R}/\mathbb{Z}} > \frac{1}{q_n + q_{n-1}} > \frac{1}{2q_n}.$$

Proof. The convergent recurrences imply $p_n q_{n-1} - p_{n-1} q_n = \pm 1$ and alternating signs of $e_n = q_n \alpha - p_n$. If ξ is the next complete quotient, then

$$\alpha = \frac{p_n \xi + p_{n-1}}{q_n \xi + q_{n-1}}, \quad a_{n+1} < \xi < a_{n+1} + 1.$$

It follows that $|e_n| = 1/(q_n \xi + q_{n-1})$, proving the first pair.

For $0 < l < q_n$ and an integer p , the determinant identity writes $(p, l) = a(p_n, q_n) + b(p_{n-1}, q_{n-1})$ with integers a, b . Here $b \neq 0$; a, b cannot both be positive, and cannot both be negative. If $a = 0$ the error has modulus at least $|e_{n-1}|$. Otherwise they have opposite signs, so ae_n and be_{n-1} have the same sign and again $|l\alpha - p| \geq |e_{n-1}|$. Minimize over p and apply the first pair with index $n - 1$. This proves the remaining inequalities. $\square$

Theorem 11.2 (Universal cubic subsequences). *Set $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$, $C_\kappa = 8(1 + \log(1/\kappa))$, and*

$$M_n = \lceil C_\kappa q_n^3 \rceil, \quad m_\kappa = \lceil 100(C_\kappa + 1) \rceil.$$

For all sufficiently large n ,

$$(11.2) \quad q_n/2 < W_{M_n, \varepsilon}(\alpha) \leq m_\kappa q_n.$$

The upper bound is exact, uses common command rows and requires no regularity of the partial quotients.

Proof. Use words of length $B = q_n - 1$ whose number of initial active letters ranges uniformly over $0, \dots, q_n - 1$. Lemma 11.1 gives q_n distinct phases with separation greater than $1/(2q_n)$. If the proposed peak K were at most $q_n/2$, Corollary 8.3 and the packet budget would give

$$\left\lfloor \frac{M_n}{q_n - 1} \right\rfloor \frac{1}{4q_n^2} \leq \log(1/\kappa).$$

The left side is at least $C_\kappa/4 - 1/(4q_n^2)$, strictly greater than the right side. This proves the lower bound against all hidden-state machines.

For the upper bound use the polygon denominator $Q_n = m_\kappa q_n$. Distance to the nearest integer is subadditive, so

$$\|Q_n \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq m_\kappa |q_n \alpha - p_n| < m_\kappa / q_{n+1} \leq m_\kappa / q_n.$$

Consequently $Q_n^2 / [100 \|Q_n \alpha\|_{\mathbb{R}/\mathbb{Z}}] > m_\kappa q_n^3 / 100 \geq (C_\kappa + 1) q_n^3 \geq M_n$. Equation (9.3) gives the exact upper construction. The constants have been chosen for transparency, not optimality. $\square$

Proposition 11.3 (Resonant long horizons). *For any irrational α and integer $q \geq 4$ with $N_q = \lfloor q^2 / (100 \|q\alpha\|_{\mathbb{R}/\mathbb{Z}}) \rfloor \geq 1$, one has $W_{N_q, 0}(\alpha) \leq q$. If α is Liouville, there is a sequence $N \rightarrow \infty$ on which $\log W_{N, \varepsilon}(\alpha) / \log N \rightarrow 0$.*

Proof. The first assertion is the finite enclosure implication (9.3). For a Liouville number choose arbitrarily large A and unbounded denominators q with $0 < \|q\alpha\|_{\mathbb{R}/\mathbb{Z}} < q^{-A}$. The first assertion then supplies horizons $N_q \geq q^{A+2}/200$ for large q , with width at most q . The logarithmic ratio is at most $1/(A+2) + o(1)$. A diagonal sequence with $A \rightarrow \infty$ proves the claim, since widths are at least two. $\square$

Proof of Theorem 1.2. Equation (11.2) and $M_n = \Theta(q_n^3)$ give the first assertion and a subsequence of logarithmic exponent $1/3$. Proposition 11.3 gives the zero lower limit for Liouville angles, and (9.4) gives the upper limit at most $1/2$.

It remains to verify unboundedness, which does not follow from a subsequence alone. Fix any proposed width K . The $2K$ phases of counts $0, \dots, 2K - 1$ are distinct for an irrational angle and have some positive minimum separation s_K . Independent words of length $2K - 1$ imply $\lfloor N/(2K - 1) \rfloor s_K^2 \leq \log(1/\kappa)$ for a width- K machine. This fails for all sufficiently large N . Thus $W_{N, \varepsilon} \rightarrow \infty$. $\square$

Corollary 11.4 (An explicit fluctuating alphabet). *Let $\alpha_L = \sum_{j=1}^{\infty} 10^{-j!}$. For $q_m = 10^{m!}$ and all large m , the horizon*

$$N_m = \lfloor q_m^{m+2} / 200 \rfloor$$

admits an exact machine with at most q_m labels. The same fixed two-letter alphabet also has the cubic subsequences of Theorem 11.2; hence no single logarithmic width exponent exists. Adding reflection preserves these conclusions.

Proof. The first m summands have denominator dividing q_m , and their remaining positive tail is less than $2 \cdot 10^{-(m+1)!}$. Thus $0 < \|q_m \alpha_L\|_{\mathbb{R}/\mathbb{Z}} < 2q_m^{-m}$ for large m , proving the horizon bound by (9.3). The same tail estimates preclude rationality and show the Liouville property. The general cubic theorem applies. Reflection leaves the upper polygons invariant, while the rotation-only test words give the same lower bounds. $\square$

These theorems describe different scales of one and the same experiment. A small label count on a long resonant horizon is not a constant-width simulator for all horizons, and it need not persist across the gaps between successive approximants. The exact finite optimum and the Liouville upper logarithmic limit remain separate quantitative questions. None of the proofs substitutes approximate output for the exact upper realizations.

12. FINITE RANDOM BITS AND UNIFORM SPACE

Exact real rows and finite-bit algorithms are different resources. We first give a finite perturbation statement and then remove the row table from a concrete uniform construction. Only this section uses a computational model. It does not change the exact atomic-row conclusions proved elsewhere.

Lemma 12.1 (Dyadic perturbation of a stochastic transducer). *Suppose initialization and every command row have at most s nonzero entries. They can be replaced by denominator- 2^b rows, on the same label sets, with row-TV error at most $(s-1)2^{-b}$. Rounding the terminal binary probabilities adds at most 2^{-b} . Consequently every numerical word changes by at most $((N+1)(s-1)+1)2^{-b}$ in binary total variation. Each rounded row can be sampled with exactly b independent fair bits.*

Proof. Order the s successors, round the first $s-1$ probabilities down to multiples of 2^{-b} , and give the remaining mass to the last successor. The total removed mass is at most $(s-1)2^{-b}$, which is exactly the row total variation. Couple the initializations and then the command updates, or telescope products using TV contraction, to sum their worst-row errors. A terminal Bernoulli parameter can be rounded down with error at most 2^{-b} . A uniform integer in $\{0, \dots, 2^b - 1\}$ selects one of the cumulative dyadic intervals. This uses a fixed number of bits, with no retained random tape or rejection-time side channel. $\square$

For a table implementation the storage is $O(|\mathcal{A}|Ks(\log K + b))$ bits for common rows of width K , in addition to initialization and decoder data. Cut-dependent rows can require a factor N more storage. Lemma 12.1 is the elementary stochastic-rounding principle; it is not claimed as a new finite-bit simulation method. The next construction uses no such table.

Definition 12.2 (Uniform command-stream model). A fixed finite program receives N in binary, one of the four planar seeds, exactly N one-way command symbols, and one final coordinate query. It has a work tape and fresh independent fair bits. The command input cannot be reread. All counters, arithmetic registers, stored parameters, random words and output storage count toward work space. An external source waits while a command is processed; waiting time is not an input or a persistent storage channel. The finite control is fixed independently of N . The program has no horizon-dependent advice or transition table.

An angle has an admissible digit procedure if a fixed program computes a dyadic approximation within 2^{-p} in polynomial time in p and $O(p)$ work bits. The same fixed r is used throughout. This explicit hypothesis excludes a free oracle for an arbitrary noncomputable real. Algebraic angles, in particular the examples below, meet it by rational bisection with fixed-degree integer arithmetic.

Theorem 12.3 (A table-free finite-bit compiler). *Fix r angles with admissible digit procedures, rational $0 < \rho \leq 1/10$, and rational $0 < \varepsilon < 1$. There is a uniform program in Definition 12.2 whose every binary output row has TV error at most ε from the rotation experiment. It uses*

$$(12.1) \quad O_{r,\alpha,\rho}(\log(N+2) + \log(1/\varepsilon)) \text{ work bits}, \quad (N+2)b \text{ fair bits},$$

where $b = \lceil \log_2(16(N+2)/\varepsilon) \rceil$. Its total running time is polynomial in $N + \log(1/\varepsilon)$ for fixed digit procedures. Its clean stochastic register has $q+1$ labels, including one central label, with $q = O(\sqrt{N})$.

If a denominator $q_0 \geq 4$ satisfies $N\eta_\alpha(q_0) \leq q_0^2/2000$, the search may choose $q \leq q_0$. Consequently the clean-register upper exponents and logarithmic bounds of Theorem 1.1 hold for this compiler as well, with changed constants. These clean labels do not include its scratch space or deterministic parameter registers; the full work bound is (12.1).

Proof. We give the parameter search, real reference rows, dyadic rounding, and finite-space evaluation separately.

Parameter search. Put $p = \lceil \log_2(16000(N+2)/\varepsilon) \rceil$ and obtain dyadic $\tilde{\alpha}_i$ with error at most 2^{-p} . For $q = 4, 5, \dots$ compute, by integer arithmetic,

$$e(q) = \max_i \|q\tilde{\alpha}_i\|_{\mathbb{R}/\mathbb{Z}} + q2^{-p}.$$

Stop at the first q for which $Ne(q) \leq q^2/1000$. The test is rational and certifies both $\eta_\alpha(q)$ and $\eta_{\tilde{\alpha}}(q)$ by the Lipschitz property of distance to $\mathbb{Z}$. It succeeds by $q = \max(4, \lceil \sqrt{2000(N+2)} \rceil)$: $\|q\tilde{\alpha}_i\| \leq 1/2$ and the added term is smaller than one at this bound. If q_0 satisfies the stronger displayed condition, then $e(q_0) \leq \eta_\alpha(q_0) + 2q_02^{-p}$ and $2Nq_02^{-p} \leq q_0^2/2000$. Hence that candidate succeeds. No exact comparison of an irrational angle with a rational sector boundary is performed.

Reference rows with slack. Use the regular q -gon $v_j = (\cos(2\pi j/q), \sin(2\pi j/q))$ and one additional label $*$ representing zero. Set $\Lambda = 1 + 1/(4N)$. For $N \geq 1$ the dilation estimate (9.5) and the search condition give

$$\log \Lambda \geq 1/(5N) > 4\pi^2\eta_{\tilde{\alpha}}(q)/q^2.$$

Thus $\Lambda^{-1}R_{2\pi\tilde{\alpha}_i}v_j$ lies strictly inside the polygon. The case $N = 0$ is handled by direct rounding of the seed response and a fixed amount of space. Write $q\tilde{\alpha}_i = n_i + t_i$, with integer n_i and $0 \leq t_i < 1$, and put $h = 2\pi/q$. The command row has successors $j + n_i$, $j + n_i + 1$ modulo q , and $*$, with respective weights

$$(12.2) \quad A_i = \frac{\sin((1-t_i)h)}{\Lambda \sin h}, \quad B_i = \frac{\sin(t_i h)}{\Lambda \sin h}, \quad 1 - A_i - B_i.$$

They are nonnegative by the dilation bound, and their vector mean is the required contracted rotation. The row at $*$ stays there. Initialize with vector mean $x/2$ using the same chord formula with radial factor $1/2$ and the exact seed angle $0, 1/4, 1/2, 3/4$. Decode with mean $2\rho\Lambda^N \langle e_j, v_s \rangle$, or zero at $*$. Since $\Lambda^N \leq e^{1/4}$, all decoder means are strictly inside $[-1, 1]$. The usual mean induction realizes every word for $\tilde{\alpha}$ exactly. These are reference real rows only.

Certified finite rows. Compute outward intervals for the first two weights in each row, each of width at most 2^{-b} . Clip their lower endpoints at zero and round down to denominator 2^b ; assign the residue to $*$. Relative to the reference row, at most $4 \cdot 2^{-b}$ mass is moved. Evaluate and round each terminal Bernoulli parameter within $2 \cdot 2^{-b}$. TV contraction therefore bounds initialization, N updates and final decoding by $(4N+6)2^{-b}$. Replacing the true angles by their dyadic approximants changes a product rotation by at most $2\pi N2^{-p}$ in operator norm, and changes a target binary row by at most $\pi\rho N2^{-p}$. The sum is less than ε with the chosen p, b . Use exactly b fair bits at each draw, including dummy bits when the central row is deterministic. The random-bit count is $(N+2)b$.

Work and uniform evaluation. Only q , the current vertex or central flag, the horizon and command counter, the r dyadic angles, and $O(r)$ offsets and dyadic weights are retained. The latter specify circulant rows, not a q -row table. The denominator search uses integers of $O(p + \log q)$ bits. Elementary sine and cosine on bounded intervals can be evaluated by fixed-point Taylor recurrences with outward rounding; π can be evaluated by Machin's arctangent identity. There are $O(b + \log q)$ terms and only a fixed number of live accumulators. Since $\sin(2\pi/q) \geq 4/q$ for $q \geq 4$, division loses only $O(\log q)$ bits. Repeated squaring evaluates Λ^N with $O(\log N)$ stages, all intermediate true values between 1 and $e^{1/4}$. An additional $O(\log(N+2) + \log(b+2))$ guard bits bounds accumulated rounding errors. Interval precision can instead be doubled until the stated widths are certified; these error estimates bound the final precision by the same order. No integer of unbounded exact denominator is stored. Every live register has $O(b + p + \log N)$ bits, and the number of registers is fixed for fixed r . This proves the work bound and a polynomial time bound, including parameter search. Finally the finite arithmetic upper proofs remain valid with $1/100$ replaced by $1/2000$, since only fixed constants change. Their candidate denominators therefore bound the search. $\square$

The proof describes a fixed-point implementation, not the numerical behavior of a floating-point library. The accompanying integer-interval reference code certifies each rounded row and uses fresh bounded-length random integers; its Python object overhead is not identified with the Turing work-space constant.

Corollary 12.4 (Sharp work-space order for explicit alphabets). *Let $\alpha_i = 2^{i/(r+1)}$, $1 \leq i \leq r$, and fix rational $0 < \varepsilon < \rho/(2\sqrt{2})$ and $0 < \rho \leq 1/10$. In Definition 12.2, numerical simulation requires and suffices with $\Theta(\log N)$ work bits. More precisely every fixed program using $s(N)$ work bits obeys*

$$(12.3) \quad s(N) \geq \frac{r}{2r+1} \log_2 N - O(\log \log(N+2)).$$

The compiler's clean register has $O(N^{r/(2r+1)})$ labels. This is not a claim that all its internal fair-bit states retain that label exponent or that the leading coefficient of work space is optimal.

Proof. Put $\theta = 2^{1/(r+1)}$. For a nonzero integer vector m with $H = \|m\|_1 \geq 1$, choose the integer p nearest to $-\sum_i m_i \theta^i$. The algebraic integer $P(\theta) = p + \sum_i m_i \theta^i$ is nonzero: the minimal polynomial $t^{r+1} - 2$ is Eisenstein and has degree $r+1$. Each of its other r conjugates has modulus at most $(9/2)H$, since $|p| \leq 2H + 1/2$ and $|\theta^i| < 2$. Its nonzero integer norm gives

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq (2/9)^r H^{-r}.$$

This classical algebraic example is credited in [4]; it is not new arithmetic. The inherited badly approximable theorem, Corollary 9.4, now supplies the all-hidden lower bound and the matching candidate denominators for the compiler. Bisection for θ and fixed-degree powers meet the digit hypothesis. A fixed program with $s(N)$ work bits has at most $C(s(N) + 1)^C 2^{s(N)}$ command-boundary configurations. Its private randomness since the last boundary is integrated into the next stochastic row, so it induces a legal atomic realization of at most that width. Apply the lower bound $W_{N,\varepsilon} \geq cN^{r/(2r+1)}$ and take logarithms. The polynomial factor accounts for the fixed number of work-tape head positions; it contributes $O(\log \log N)$ unless the displayed lower bound is already automatic. Theorem 12.3 supplies the upper work bound. $\square$

13. REALIZATION THEORY, ARITHMETIC AND COMPUTATIONAL SEMANTICS

Positive order and cone invariance. In the classical stationary problem one seeks a fixed nonnegative state-space realization of a transfer function. Minimal signed order need not equal minimal positive order. Benvenuti–Farina's Theorems 2–3 formulate positive existence by an invariant polyhedral

cone, either in a minimal linear realization or in a response-sequence space [12]; the 2022 survey concerns positive minimality [13]. Our enclosure construction is such a positive-cone construction after homogenization. The competing machines are different: their rows may vary with the cut and their entire realization may be redesigned for N . Proposition 4.1 makes the corresponding finite controlled normal form explicit. Classical stationary order cannot be substituted for its least simultaneous profile.

Hankel rank and signed weighted automata. For fixed seed and query the mean series is $w \mapsto \rho\langle e_j, U_w x \rangle$, with a signed linear representation of dimension at most D . Adding the constant probability offset requires at most $D + 1$ dimensions. Balle–Panangaden–Precup’s Theorem 2 identifies minimal weighted-automaton dimension with infinite Hankel rank [14]. That is the signed, all-length invariant, not stochastic label cardinality. Their Theorem 11 bounds an ℓ^2 approximation of absolutely summable rational series. Here the unweighted all-word probability series is not summable, and the objective is the maximum numerical error on a fixed length. A small weighted mean-square error under a chosen word law does not control rare-word conditional error. For an explicitly unweighted finite table, Frobenius error does bound entrywise error, but its optimizer still need not be stochastic or compatible. Equation (4.5) exhibits the precise failure of separate ranks to detect the present growing resource.

Stochastic and hidden Markov realization. Finesso–Grassi–Spreij use pseudo-Hankel nonnegative factorizations and an informational-divergence criterion to approximate a stationary output process at prescribed hidden order [15, Sections III–IV]. Their second realization step is itself evidence that a static factorization is not the complete dynamic specification. Our alphabet is externally controlled rather than generated as an observation process. Seed resets and selected query columns are separately specified; worst-word conditional probabilities replace a divergence between output laws. These are explicit changes of datum and objective, not a claim that Hankel positivity was absent from HMM theory.

Ohta’s reachable and null-space reductions, Theorems 6–9 and Remark 10, identify tensor and generalized-Hankel dimensions in the effective linear space of an HMM [16]. We use the same need to distinguish observable rank from hidden state count, but do not infer a positive stochastic basis from a signed quotient. His Section 4.4 also discusses rotational examples descended from earlier HMM realization work. Rotation-based rank/positive-order separations are therefore not claimed new here. Our quantitative theorems concern all horizon-specific stochastic rows and preserve bounded numerical responses on every controlled word.

Kronecker quasi-uniformity. Dick–Goda–Larcher–Pillichshammer–Suzuki [4], Theorem 3.9, characterize quasi-uniformity of the multidimensional sequence $n\alpha \pmod{\mathbb{Z}^r}$ by the simultaneous bound $\eta_\alpha(q) \geq cq^{-1/r}$. Their Lemma 3.11 bounds ratios of successive best simultaneous denominators; Section 3.3 also contains the vector $(2^{1/(r+1)}, \dots, 2^{r/(r+1)})$. These are direct arithmetic antecedents, not new examples of this paper.

Our packet points instead have the form $m \cdot \alpha \pmod{1}$. The relevant separation is dual. The residue argument of Lemma 9.2, with an arbitrary nearest-integer numerator and $\eta_\alpha(q)$ in place of Q^{-1} , proves directly that a dual bound with exponent r implies

$$\eta_\alpha(q) \geq br^{-r-1}q^{-1/r}.$$

Thus the cited quasi-uniformity theorem applies to the uniform badly approximable class used in the predecessor. Its denominator control supplies the arithmetic scales for a polygon upper bound. It supplies neither a compatible stochastic transition table nor a lower bound against discarded hidden history. The logarithmic separation of Lemma 10.1 is weaker than a uniform badly approximable inequality; it is proved by a summable tail estimate, not asserted to imply quasi-uniformity.

What is dynamic in the paired profiles. The geometry of nearest-center distortion and convex hull approximation is classical [5, 11]. A static k -center factorization does not specify how its encoder is calculated without retaining discarded inputs. Conversely, a convex codebook does not force arbitrary hidden states to possess coordinates in that codebook. The two implications in Section 6 avoid both identifications: the converse conditions on the actual state, and the upper theorem provides the same legal rows throughout the run. The packing–dilation balance in Corollary 6.7 is a conditional theorem on independently specified finite profiles, not a claimed equality of static and dynamic optima.

The circle program makes one of these profiles exactly finite, whereas the all-harmonic action-gap optimization of the predecessor remains infinite-dimensional. A finite word length can be exhaustively listed; an unbounded harmonic degree cannot be certified by a finite truncation without a tail theorem. No general action-gap algorithm follows from Proposition 6.1.

Branching programs and probability preservation. At fixed horizon our machine is a probabilistic ordered read-once layered program with arbitrary stochastic rows and bounded real output columns. We distinguish width from total size, uniform description and random-bit complexity. Fourier growth for regular, permutation and width-three branching programs concerns different hypotheses and objectives [9, 10, 6]. Circle-walk harmonic damping and its arithmetic behavior are also established subjects [2]. Our converse can use correlated letters inside independently sampled complete packets; it is not a theorem about average error under fair independent letters.

Preserving a strict cutpoint sign is weaker than preserving a numerical probability. The predecessor’s constant-width sign construction contracts the output amplitude with length and hence does not contradict the present width growth. Here signal and tolerance are fixed, one query is selected after the word, and neither joint answers nor anytime correctness is required. The exact arithmetic upper rows are atomic real kernels. Section 12 gives a separate finite-bit approximation theorem; it does not represent those exact rows by a finite number of fair bits. A row with three successors can still have costly or irrational coefficients.

Conclusions. The finite optimized word profile and the compatible enclosure profile retain quantitative information which qualitative gap data discard. They yield a minimal-type exponent and an almost-everywhere logarithmic estimate, and they exhibit fluctuating numerical width for every Liouville single angle. The reflected versions show that these conclusions survive a particular noncommuting finite extension, without asserting a classification of arbitrary noncommuting nongapped alphabets.

Closed-form finite widths beyond the explicitly solved frontier, the Liouville upper logarithmic limit, optimal enclosure shapes and equality in the distortion–dilation comparison remain distinct quantitative problems. The new profile comparison uses no external spectral gap. For fixed computable planar angles, storing all command counts already gives a classical $O(\log N)$ finite-accuracy work-space upper bound. We do not claim novelty for that upper order alone. The compiler proved here additionally maintains an arithmetic-scale clean stochastic register, generates circulant dyadic rows without a large advice table, and explicitly prices every arithmetic and random-bit register. The earlier expansion, calibration and representation theorems retain their original hypotheses in the unchanged supporting article.

Certificates and optimization. The Gram identity of Theorem 5.1 is a finite-length version of classical weighted-series equivalence, with stochastic rows and bounded numerical decoders imposed separately. In contrast to all-length ℓ_2 truncation, the moment sandwich in Theorem 5.2 targets the maximum error over the finite word set and optimizes over legal stochastic machines. Its global minimum is not supplied by an approximate matrix factorization. The rational grid certificate is complete at every fixed positive margin, but exact rational feasibility is different; the normalized nonnegative-rank example makes that distinction explicit. The new certificate framework does not change any earlier arithmetic exponent or imply a uniform solution of the infinite-gap problem.

REFERENCES

- [1] S. Basu, R. Pollack and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, 2nd ed., Springer, 2006.
- [2] I. Berkes and B. Borda, Random walks on the circle and Diophantine approximation, *J. London Math. Soc.* **108** (2023), 409–440.
- [3] J. W. S. Cassels, *An Introduction to Diophantine Approximation*, Cambridge Tracts in Mathematics and Mathematical Physics, vol. 45, Cambridge University Press, 1957.
- [4] J. Dick, T. Goda, G. Larcher, F. Pillichshammer and K. Suzuki, On the quasi-uniformity properties of quasi-Monte Carlo point sets and sequences—Part I: Lattices and Kronecker sequences, arXiv:2502.06202v2, version of 13 February 2026, Section 3.3.
- [5] S. Graf and H. Luschgy, *Foundations of Quantization for Probability Distributions*, Lecture Notes in Mathematics, vol. 1730, Springer, 2000.
- [6] C. H. Lee, E. Pyne and S. Vadhan, Fourier growth of regular branching programs, *APPROX/RANDOM 2022*, LIPIcs **245**, Art. 2, 2:1–2:21.
- [7] Q. Qi, *General Theta Foundations I: Sharp Width of Hidden Rotation Experiments*, manuscript, revision 35, 2026; repository publication [aabc0731a0c9](#).
- [8] Q. Qi, *General Theta Foundations I: Alphabet Dynamics and Calibrated Numerical Memory*, manuscript, revision 42, 2026; repository publication [b5f9874903a4](#).
- [9] O. Reingold, T. Steinke and S. Vadhan, Pseudorandomness for regular branching programs via Fourier analysis, *RANDOM 2013*, LNCS **8096**, 655–670.
- [10] T. Steinke, S. Vadhan and A. Wan, Pseudorandomness and Fourier growth bounds for width 3 branching programs, *Theory Comput.* **13** (2017), Art. 12, 1–50.
- [11] M. Vidyasagar, *Hidden Markov Processes: Theory and Applications to Biology*, Princeton University Press, 2014.
- [12] L. Benvenuti and L. Farina, A tutorial on the positive realization problem, *IEEE Trans. Automat. Control* **49** (2004), 651–664, Theorems 2–3 and Example 4.
- [13] L. Benvenuti, Minimal positive realizations: A survey, *Automatica* **143** (2022), Art. 110422.
- [14] B. Balle, P. Panangaden and D. Precup, A canonical form for weighted automata and applications to approximate minimization, arXiv:1501.06841v3, 2015, Theorems 2 and 11.
- [15] L. Finesso, A. Grassi and P. Spreij, Two-step nonnegative matrix factorization algorithm for the approximate realization of hidden Markov models, arXiv:1007.3435v1, 2010, Sections III–IV.
- [16] Y. Ohta, On the realization of hidden Markov models and tensor decomposition, arXiv:2008.11487v1, 2020, Theorems 6–9 and Remark 10.
- [17] Y. Bugeaud and M. Laurent, Exponents of Diophantine approximation, arXiv:math/0611354v1, 2006, Definition 1 and Section 2.
- [18] Q. Qi, *General Theta Foundations I: Compatible Polyhedral Lifts and Quantitative Causal Width*, manuscript, revision 33, 2026; repository publication [aab961081253](#).
- [19] Q. Qi, *General Theta Foundations I: Word Profiles and Arithmetic Fluctuations of Numerical Memory*, manuscript, revision 43, 2026; repository publication [153830f5dc8d](#).
- [20] W.-G. Tzeng, A polynomial-time algorithm for the equivalence of probabilistic automata, *SIAM J. Comput.* **21** (1992), 216–227, doi:10.1137/0221017.
- [21] Y. Shitov, A universality theorem for nonnegative matrix factorizations, arXiv:1606.09068v2, version of 5 April 2018, Theorem 2 and Corollary 3.

Preserved v44 published archive

The following predecessor pages are unchanged archival material.

GENERAL THETA FOUNDATIONS I: HANKEL COMPATIBILITY, DISTORTION RATES, AND FINITE-BIT MEMORY

QIAN QI

ABSTRACT. We place numerical orthogonal-word experiments in the finite-horizon positive Hankel realization framework and compare two previously separate complexity profiles. The optimized loss of any finite word packet, divided logarithmically by its length, is bounded by the logarithmic dilation of every compatible common polytope of the same cardinality. The proof allows arbitrary hidden stochastic states and removes initialization loss by repetition. For planar alphabets we construct a uniform table-free simulator using finite random bits and $O(\log N + \log(1/\varepsilon))$ total work bits, including parameter construction, counters and arithmetic. The explicit badly approximable families require $\Theta(\log N)$ work bits at fixed positive accuracy. The atomic label-width exponents, minimal-type metric bounds and Liouville fluctuations of the preceding development are retained with complete proofs. Atomic nonuniform width, clean register cardinality and the full finite-bit workspace are distinguished throughout. The profile comparison is one-sided; it is not a general solution of minimal positive realization or of arbitrary arithmetic width.

1. INTRODUCTION

A word response can have constant signed Hankel rank and constant nonnegative rank at every cut, yet require increasing positive state cardinality when its cut factorizations must use compatible stochastic updates. Positive realization, rather than ordinary linear realization, is therefore the relevant formalism. We study the controlled numerical response

$$(1.1) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, U_w x \rangle}{2}, \quad b \in \{-1, 1\},$$

for a finite orthogonal alphabet, a unit seed and one selected terminal coordinate query. Our basic resource is nonuniform clocked atomic-row positive-realization label width. The known horizon, row tables and exact real arithmetic are free in that model. A separate theorem below charges them in a uniform finite-bit command-stream model.

There are three contributions to the present treatment. First, Section 3 writes the exact finite-horizon object in the language of normalized positive Hankel factorizations and compatible future-response polytopes. This specializes the classical shift-invariant-cone principle; neither positivity nor compatibility is claimed as a new realization concept. For the planar numerical family, ordinary rank, nonnegative rank and normalized positive rank all equal three at every cut. This identifies precisely what separate factorization omits.

Second, write $\Gamma_{\mathcal{A}}(k, B)$ for optimized length- B word quantization loss and $\lambda_{\mathcal{A}}(k, a)$ for the least common dilation of a k -generator polytope containing a times the unit ball. Theorem 5.1 proves the universal comparison

$$(1.2) \quad \sup_{B \geq 1} \frac{-\log(1 - \Gamma_{\mathcal{A}}(k, B))}{B} \leq \log \lambda_{\mathcal{A}}(k, a).$$

It uses arbitrarily many independent packets to remove the initial inradius loss. Thus the paired profiles are related for every finite orthogonal alphabet, including noncommuting alphabets. Equality and optimality among all hidden realizations are not assumed.

Third, Theorem 10.3 gives a table-free finite-bit compiler for finitely many planar angles having fixed linear-space digit procedures. Its parameter search, dyadic probabilities, arithmetic scratch space, horizon counter and finite random words require $O(\log N + \log(1/\varepsilon))$ work bits. Every command is processed once; the output error is uniformly at most a prescribed $\varepsilon > 0$. For the explicit badly approximable algebraic-angle families the work-space order is exactly $\Theta(\log N)$, while the clean stochastic register retains the arithmetic label exponent. This is not an exact fair-bit implementation of arbitrary real rows, and the sharp coefficient of work space is not determined.

The packet, interval and enclosure theorems of the preceding development are reproduced with their proofs in Section 4. The packing–dilation balance there is a conditional meta-corollary, not a classification of finite alphabets. The arithmetic results below are retained: all of their analytic proofs remain in the current paper, with the branch of the finite-type minimum displayed explicitly.

Our arithmetic conclusions go beyond uniform bad approximation. For $\alpha \in \mathbb{R}^r$ with $1, \alpha_1, \dots, \alpha_r$ rationally independent, write

$$(1.3) \quad \omega^*(\alpha) = \limsup_{\|m\|_1 \rightarrow \infty} \frac{-\log \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}}{\log \|m\|_1}.$$

This is the ordinary dual exponent $\omega_{1,r}$ of the row vector α in [17, Definition 1], with an equivalent fixed-dimensional norm; it is not the uniform exponent $\widehat{\omega}_{1,r}$ or the simultaneous exponent $\omega_{r,1}$. The alphabet consists of the identity and the rotations through $2\pi\alpha_i$. Throughout, r , the alphabet, $0 < \rho \leq 1/10$ and $0 \leq \varepsilon < \rho/(2\sqrt{2})$ are fixed.

Theorem 1.1 (Minimal Diophantine type). *If $\omega^*(\alpha) = r$, then*

$$(1.4) \quad \lim_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} = \frac{r}{2r+1}.$$

For Lebesgue-almost every α , and every fixed $\delta > 0$, there are positive constants c, C , depending on the fixed $\alpha, \delta, r, \rho, \varepsilon$, such that, for $N \geq 2$,

$$(1.5) \quad c \frac{N^{r/(2r+1)}}{(\log(N+2))^{2r(1+\delta)/(2r+1)}} \leq W_{N,\varepsilon}(\alpha) \leq CN^{r/(2r+1)} (\log(N+2))^{2r^2(1+\delta)/(2r+1)}.$$

The upper inequalities are achieved by exact common-row machines. No uniformity near rational relations or in growing r is asserted.

The exponent in (1.4) need not imply a two-sided constant-factor law. The earlier dual badly approximable theorem supplies the stronger Θ conclusion on its stated class, and is recovered in Corollary 7.4. Theorem 1.1 instead allows arbitrarily small Diophantine constants along subsequences. The quantitative arithmetic ingredients are classical; the memory conclusions concern their compatible realization and all-hidden-state converse.

There is also a genuine failure of a single exponent, even for one fixed irrational angle.

Theorem 1.2 (Irrational horizons and Liouville fluctuations). *For every irrational α there are horizons $M_n \rightarrow \infty$ such that $W_{M_n,\varepsilon}(\alpha) = \Theta(M_n^{1/3})$. If α is Liouville, then*

$$(1.6) \quad \liminf_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} = 0, \quad \frac{1}{3} \leq \limsup_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} \leq \frac{1}{2}.$$

Nevertheless $W_{N,\varepsilon}(\alpha) \rightarrow \infty$. All displayed upper bounds have exact realizations.

The upper limit in (1.6) is not determined here. The result establishes oscillation between subpower subsequences and a positive-power subsequence; it does not replace that distinction by a guessed exponent for all irrationals.

Provenance of the ingredients. The stochastic Hankel normal form and its finite separation dual are normalizations of established realization and convex-separation arguments [12, 16, 14]. Their role is to place the resource, not to claim priority for positive cones. The endpoint mechanism and arithmetic laws are inherited [7, 8, 19]. The new statements are the distortion-rate comparison and the uniform finite-bit realization with its charged resource ledger. Dirichlet approximation, continued fractions, metric tail estimates and the algebraic examples remain classical. Section 11 compares the precise objectives and norms, rather than treating adjacent theories as absent.

2. THE NUMERICAL EXPERIMENT AND THE REGISTER

Let $V = \mathbb{R}^D$, let $\mathcal{A}$ be a nonempty finite alphabet, and fix orthogonal matrices U_a , $a \in \mathcal{A}$. Write $U_w = U_{a_N} \cdots U_{a_1}$ in the order of execution. Let $X \subset \mathbb{S}^{D-1}$ be a nonempty finite seed set and $0 < \rho < 1$. The required law is (1.1). Commands are externally supplied, not chosen by a policy inside the simulator. No distribution on command words is part of the specification.

Definition 2.1 (Clocked atomic-row realization). At horizon N , a machine consists of finite sets S_t with $|S_t| = K_t$ for $0 \leq t \leq N$, an initialization probability row E_x on S_0 , stochastic matrices $T_{t,a} : S_{t-1} \rightarrow S_t$, and terminal columns $d_j : S_N \rightarrow [-1, 1]$. Its numerical mean is

$$E_x T_{1,a_1} \cdots T_{N,a_N} d_j.$$

After the atomic query j , it draws and holds the binary answer with that mean. The final answer cut has two labels. Its peak is $\max(2, K_0, \dots, K_N)$. Available labels count even when unreachable; adding unused normalized rows is permitted.

All persistent randomness, input history and protocol distinctions must be in the current label. Only the external epoch, the prescribed horizon and the read-only row tables are free. Table construction, lookup, arithmetic and exact sampling of known real stochastic rows are uncharged. Matrices, rows and state sets may depend on N ; this is nonuniform positive-realization width, not uniform computational space. A common-row construction uses the same command matrices at all command epochs but may still have an N -dependent decoder. It need not answer at earlier stopping lengths.

Let $W_{N,\varepsilon}$ be the least peak among machines whose every binary row differs from (1.1) by at most ε in total variation. In particular $W_{N,\varepsilon} \geq 2$. For fixed finite $K_\bullet$ the optimal actual error $\mathcal{E}_N(K_\bullet)$ is attained: the row parameters lie in a finite product of compact sets and the error is a finite maximum of continuous functions. For planar results we use $X = \{\pm e_1, \pm e_2\}$ and include the identity letter. We keep $\rho \leq 1/10$ and $\varepsilon < \rho/(2\sqrt{2})$.

Lemma 2.2 (Calibration under any external test law). *Fix a seed $x \in X$ and any law on the complete command word, independent of private initialization. Let $Y_t = U_{a_t} \cdots U_{a_1} x$, $Z_t = \mathbb{E}[Y_t \mid S_t]$ and $q_t = \mathbb{E}\|Z_t\|$. Every machine with row error at most ε satisfies*

$$(2.1) \quad q_N \geq \kappa := \rho/\sqrt{D} - 2\varepsilon.$$

At a boundary preceding an independent complete packet, the next boundary admits the conditional-mean identity used in Lemma 4.2. No single-step independence inside a packet is asserted.

Proof. For a fixed word, every coordinate of $\mathbb{E}[d(S_N) \mid w] - \rho Y_N$ has modulus at most 2ε . Since $\|Y_N\| = 1$, averaging gives

$$\mathbb{E}\langle Y_N, d(S_N) \rangle \geq \rho - 2\varepsilon\sqrt{D}.$$

On the other hand this pairing equals $\mathbb{E}\langle Z_N, d(S_N) \rangle$ and is at most $\sqrt{D}q_N$. This proves (2.1). Combining terminal columns is a proof device, not a requirement to provide their joint output distribution. $\square$

3. CONTROLLED HANKEL ARRAYS AND COMPATIBLE POSITIVE REALIZATIONS

For a horizon N , define the probability-valued prefix-suffix matrix

$$(3.1) \quad H_t[(x, u), (v, j, b)] = \frac{1 + b\rho\langle e_j, U_v U_u x \rangle}{2}, \quad |u| = t, \quad |v| = N - t.$$

The two columns indexed by $b = \pm 1$ sum to one for every fixed (v, j) . Let $\mathcal{C}_t$ be the product of these binary probability simplices. Thus a point of $\mathcal{C}_t$ is a legal collection of responses to all remaining words and queries, not a joint distribution of those answers. The restriction $L_{t,a} : \mathcal{C}_t \rightarrow \mathcal{C}_{t+1}$ reads the coordinates whose first remaining command is a .

A normalized positive factorization is $H_t = E_t B_t$, where E_t is row stochastic and every row of B_t belongs to $\mathcal{C}_t$. Its least inner dimension is denoted $\text{rank}_{\text{st}} H_t$. The usual nonnegative rank permits arbitrary nonnegative factors and need not impose these simultaneous normalizations. Consequently

$$(3.2) \quad \text{rank } H_t \leq \text{rank}_+ H_t \leq \text{rank}_{\text{st}} H_t \leq K_t$$

for every exact stochastic realization. The last inequality follows by conditioning the remaining experiment on the actual label at cut t . The factor E_t contains its state probabilities; B_t contains its complete future responses, including those of states not reached by a particular prefix.

Proposition 3.1 (Finite-horizon positive Hankel normal form). *For any finite table of binary responses, a profile $K_0, \dots, K_N$ is exactly realizable if and only if there are polytopes $P_t \subset \mathcal{C}_t$, generated by at most K_t points, such that*

$$(3.3) \quad H_0[x, \cdot] \in P_0 \quad (x \in X), \quad L_{t,a} P_t \subset P_{t+1} \quad (0 \leq t < N, a \in \mathcal{A}).$$

The held answer additionally costs two labels. The optimum error for a fixed profile is the least entrywise error from the specified table to one having such a compatible family.

Proof. Given a machine, let $b_{t,s} \in \mathcal{C}_t$ be the conditional response to every remaining word when its current label is s . Its stochastic update gives

$$L_{t,a} b_{t,s} = \sum_{s'} T_{t+1,a}(s, s') b_{t+1,s'}.$$

Taking convex hulls and using the initialization proves necessity. Conversely choose generators $b_{t,s}$, padding by repetitions when necessary. Express each $L_{t,a} b_{t,s}$ as a convex combination of the next generators, and use those coefficients as the stochastic row. Express each root row as a convex combination of the initial generators. At time N the generators themselves give the binary decoder rows. Backward induction on the remaining word proves equality of all responses. The construction is valid on every generator, not only on reachable mixtures. For approximation apply this equivalence to the actual response table of a machine, or to a candidate perturbed table; finite compactness of the stochastic parameter spaces gives attainment. $\square$

This is the finite, controlled, normalized version of the classical shift-invariant-cone realization principle, not a newly discovered characterization of positive systems. In particular, the cone generated by $(1, b)$ for $b \in P_t$ carries the positive restriction maps. Benvenuti-Farina's Theorems 2-3 give the stationary cone formulation; the multiple cuts here are essential [12]. The same compatibility issue was already explicit in the earlier lift formulation [18]. The normal form has exponentially many suffix coordinates. It is exact, but does not make arbitrary width minimization efficient or eliminate hidden directions by projecting them into the observed D -space.

Proposition 3.2 (A finite compatibility certificate). *For proposed generators B_t, B_{t+1} , put $c = L_{t,a}b_{t,s}$ and let $b'_1, \dots, b'_{K_{t+1}}$ be the next generators. The least rowwise uniform defect is*

$$(3.4) \quad \min_{\theta \in \Delta_{K_{t+1}}} \left\| c - \sum_i \theta_i b'_i \right\|_\infty = \max_{\|z\|_1 \leq 1} \left(\langle z, c \rangle - \max_i \langle z, b'_i \rangle \right).$$

It vanishes exactly when that row has a stochastic extension. If root rows are approximated within δ_0 and every cut restriction within δ_t , the resulting machine has worst-word binary-TV error at most $\delta_0 + \sum_{t=1}^N \delta_t$.

Proof. Write the infinity norm as the supremum of its pairing with the unit ℓ_1 ball. The simplex and this ball are compact and convex, so finite bilinear minimax interchanges the extrema. Minimizing the pairing over the simplex gives the maximum over its vertices with the displayed sign. For composition, restriction does not increase the infinity norm and stochastic averaging is a contraction. Induction on the suffix length therefore adds the successive errors. Each binary pair is normalized, so its total variation is exactly either coordinate error. Cancellation may reduce the true error; no equality with the local sum is asserted. $\square$

Proposition 3.3 (Constant separate ranks for the planar experiment). *For the four planar seeds, an identity command, and $0 < \rho \leq 1/10$, at every cut and every horizon,*

$$(3.5) \quad \text{rank } H_t = \text{rank}_+ H_t = \text{rank}_{\text{st}} H_t = 3.$$

Each factorization embeds in a complete machine when all other cuts are unrestricted. Nevertheless the compatible peak can grow as in Theorems 1.1 and 1.2.

Proof. The entries in (3.1) are affine functions of the two coordinates of $\rho U_u x$, so ordinary rank is at most three. Identity prefixes, identity suffixes, both coordinate queries and the four seeds give a rank-three submatrix. For the upper factorization, an equilateral triangle of circumradius 2ρ contains the disk of radius ρ . Encode $\rho U_u x$ in its three vertices v_i by barycentric coordinates. The continuation row at v_i is $(1 + b\langle e_j, U_v v_i \rangle)/2$, which is legal since $\|v_i\| = 2\rho < 1$. This is a normalized three-factor decomposition and proves the equalities. To embed it, retain the exact history until this cut, draw its factor label, retain that label and the remaining history afterwards, then use the factor decoder. Both retained histories are finite at a fixed horizon. They are not free registers in a simultaneous three-state realization. The growth assertions follow from the stated all-hidden converses, not from a claimed new separation of signed and positive realization in classical stationary theory. $\square$

4. FINITE WORD PROFILES AND COMPATIBLE ENCLOSURES

The results in this section apply to every finite orthogonal alphabet, including noncommuting alphabets. No connectedness, arithmetic condition or spectral gap is assumed. Invariant directions may make a converse profile zero; this is a property of the stated profile, not an omitted hypothesis.

4.1. An optimized distortion. For $B, k \geq 1$ and a probability vector p on the actual words $\mathcal{A}^B$, define

$$(4.1) \quad D_k(p) = 1 - \max_{\substack{u \in \mathbb{S}^{D-1} \\ v_1, \dots, v_k \in \mathbb{S}^{D-1}}} \sum_{w \in \mathcal{A}^B} p_w \max_{1 \leq j \leq k} \langle v_j, U_w u \rangle,$$

$$(4.2) \quad \Gamma_{\mathcal{A}}(k, B) = \max_{p \in \Delta(\mathcal{A}^B)} D_k(p), \quad \chi_{\mathcal{A}}(k, B) = -\log(1 - \Gamma_{\mathcal{A}}(k, B)).$$

We allow $\chi = +\infty$. Repeated product matrices can be merged for the optimization: their probability masses add. Their representative words must still be executed one ordinary command at a time.

Proposition 4.1 (Finite-data properties). *The maxima in (4.1)–(4.2) are attained, $0 \leq \Gamma_{\mathcal{A}}(k, B) \leq 1$, and the profile is invariant under simultaneous orthogonal changes of coordinates. It is nonincreasing in k and unchanged by duplicating a letter. For two indexed orthogonal alphabets U, V ,*

$$(4.3) \quad |\Gamma_U(k, B) - \Gamma_V(k, B)| \leq B \max_a \|U_a - V_a\|_{\text{op}}.$$

For a fully listed finite alphabet of real-algebraic matrices and fixed integers k, B, D , $\Gamma_{\mathcal{A}}(k, B)$ and an optimizing word law are real-algebraic and can be determined by real quantifier elimination.

Proof. All parameter spaces are compact and the objective is continuous. Its maximum over directions is convex in p , so D_k is continuous and concave. The maximum scalar product is at most one. By choosing a single direction parallel to $\sum p_w U_w u$, the inner maximum is nonnegative; hence $D_k \in [0, 1]$. Conjugating all directions proves invariance. Additional centers only increase that inner maximum. Product telescoping gives $\|U_w - V_w\|_{\text{op}} \leq B \max_a \|U_a - V_a\|_{\text{op}}$, and maxima and minima preserve this uniform bound.

For the last assertion, enumerate the product matrices. Unit-sphere equations, simplex inequalities and finitely many maximum operations define a compact semialgebraic optimization with algebraic coefficients. Maximum operations can be expressed by finite alternatives or auxiliary variables. Quantifier elimination over the real closed field of real-algebraic numbers determines the endpoint of the feasible value interval and a witness [1]. The word table may have $|\mathcal{A}|^B$ entries; this is not a polynomial-time assertion or a decision procedure for an infinite spectral gap. No logarithm needs to be eliminated: finite budget comparisons can instead use products of $1 - \Gamma$. $\square$

Lemma 4.2 (Endpoint contraction; inherited mechanism). *Let a complete random word W with law p be independent of all information before its start. Internal letters of W may be correlated. If the endpoint register has at most k labels, then*

$$(4.4) \quad q' \leq (1 - D_k(p))q.$$

There is no bound on the intermediate registers.

Proof. Write $z_s = \mathbb{E}[Y \mid S = s]$ at the start. Composing all internal stochastic rows yields a common endpoint kernel $T(j \mid s, w)$. Since the entire word and the machine's fresh randomness are independent of the past,

$$z'_j = \mathbb{E}[U_W z_S \mid S' = j].$$

This is an endpoint identity; it would not generally hold after conditioning on a correlated next letter inside the packet. Choose $v_j = z'_j / \|z'_j\|$ for nonzero z'_j , and any unit vector otherwise. Then

$$\begin{aligned} q' &= \sum_s \mathbb{P}(S = s) \sum_w p_w \sum_j T(j \mid s, w) \langle v_j, U_w z_s \rangle \\ &\leq \sum_s \mathbb{P}(S = s) \|z_s\| \max_{\|u\|=1} \sum_w p_w \max_j \langle v_j, U_w u \rangle \leq (1 - D_k(p))q. \end{aligned}$$

Terms with $z_s = 0$ contribute zero. Replacing an executable assignment by the largest pairing is a relaxation in the correct direction. Arbitrary unobservable hidden coordinates are included in s ; no statewise observable-vector hypothesis is used. This is the conditional-centroid mechanism of [7, 8], now applied to the optimized finite word datum. $\square$

For an available profile $K_\bullet$, let $I_0 = 0$ and define

$$(4.5) \quad I_t = \max \left\{ I_{t-1}, \max_{1 \leq B \leq t} [I_{t-B} + \chi_{\mathcal{A}}(K_t, B)] \right\}, \quad \mathcal{I}_N(K_\bullet) = I_N.$$

If $\Gamma_{\mathcal{A}}(K_t, B) = 1$, the interval annihilates the conditional-centroid amplitude and is incompatible with any positive terminal calibration. This is the meaning of an infinite charge. The recurrence

is the largest sum of charges of disjoint command intervals, with an interval of length B ending at t charged by $\chi_{\mathcal{A}}(K_t, B)$. The recurrence depends only on the alphabet and the available profile.

Theorem 4.3 (A nonuniform interval budget). *Every machine with $\kappa = \rho/\sqrt{D} - 2\varepsilon > 0$ satisfies*

$$(4.6) \quad \mathcal{I}_N(K_{\bullet}) \leq \log(1/\kappa).$$

More generally its optimal actual terminal error obeys

$$(4.7) \quad \mathcal{E}_N(K_{\bullet}) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{D}} - e^{-\mathcal{I}_N(K_{\bullet})} \right]_+.$$

For a uniform peak bound K , every $B \leq N$ gives

$$(4.8) \quad \lfloor N/B \rfloor \chi_{\mathcal{A}}(K, B) \leq \log(1/\kappa).$$

For a fixed B, k with $\Gamma_{\mathcal{A}}(k, B) > 0$,

$$(4.9) \quad \#\{1 \leq t \leq N : K_t \leq k\} \leq B - 1 + \frac{B \log(1/\kappa)}{\chi_{\mathcal{A}}(k, B)}.$$

If $\chi = +\infty$, the last fraction is zero.

Proof. Choose any maximizing disjoint interval family in (4.5). On each selected interval put an independently sampled optimizing complete word from Proposition 4.1. All gaps and the final remainder contain fixed deterministic command words. A deterministic orthogonal command followed by a stochastic update cannot increase the conditional-mean norm, by conditional Jensen. Lemma 4.2 therefore gives

$$q_N \leq \prod_{(t-B, t]} (1 - \Gamma_{\mathcal{A}}(K_t, B)) = e^{-\mathcal{I}_N(K_{\bullet})},$$

since $q_0 = 1$. The chosen law can depend on the available layer sizes, not on private states. Wordwise correctness requires the same machine to pass it. Lemma 2.2 proves (4.6) and (4.7); taking consecutive intervals proves (4.8).

For (4.9), discard candidate endpoints before B . Select the earliest remaining endpoint, delete all later candidates less than B away, and repeat. If M is the original number and J the number selected, the explicit count is

$$M - (B - 1) \leq BJ.$$

The associated length- B intervals are disjoint and each contributes at least $\chi_{\mathcal{A}}(k, B)$. Applying (4.6) gives the result. No restriction on the other registers was introduced. $\square$

The cap union estimate below is the geometric certificate already used in the predecessor packet method [7]. The finite incidence formulation specifies how to optimize that certificate on a fully listed word set.

Proposition 4.4 (Uniform cap certificates). *Suppose a law p on length- B words satisfies*

$$\sup_{u, v \in \mathbb{S}^{D-1}} p\{w : \|U_w u - v\| < \delta\} \leq b.$$

Then $\Gamma_{\mathcal{A}}(k, B) \geq \frac{1}{2}\delta^2(1 - kb)_+$. For algebraic product matrices and algebraic δ , the best such b is the value of a finite linear program once cap incidence types have been decided by real quantifier elimination.

Proof. The union of k open caps has probability at most kb . Outside it, $1 - \max_j \langle v_j, U_w u \rangle = \frac{1}{2} \min_j \|U_w u - v_j\|^2 \geq \delta^2/2$. For the algorithmic assertion, a subset of the finite word list either can or cannot be contained in one such cap for some u, v . This is a finite system of algebraic inequalities, including the sphere equations. For every feasible subset A impose $\sum_{w \in A} p_w \leq b$, along with the simplex constraints, and minimize b . Every actual incidence set is feasible; conversely a feasible

subset is included in an actual incidence set and has no greater mass. Thus these constraints give exactly the desired supremum. The exponentially large enumeration is not concealed. $\square$

4.2. A separate exact-synthesis profile. Fix $0 < a < 1$. For k such that the following family is nonempty, let

$$(4.10) \quad \lambda_{\mathcal{A}}(k, a) = \min_P \inf \{ \lambda \geq 1 : U_b P \subseteq \lambda P \text{ for all } b \in \mathcal{A} \},$$

where P ranges over convex hulls of at most k points in $\mathbb{B}^D$ containing $a\mathbb{B}^D$. An empty family has value $+\infty$.

Theorem 4.5 (Compatible enclosure synthesis). *The finite value in (4.10) is attained and lies in $[1, a^{-1}]$. If $\rho \lambda_{\mathcal{A}}(k, a)^N \leq a$, there is an exact machine with $\max(k, 2)$ labels, common command matrices and a horizon-dependent decoder. Each row has at most $D + 1$ successors. Algebraic matrices and algebraic a give an algebraically decidable profile value and algebraic row witnesses. For two indexed alphabets with a common feasible family,*

$$(4.11) \quad |\lambda_U(k, a) - \lambda_V(k, a)| \leq a^{-1} \max_b \|U_b - V_b\|_{\text{op}}.$$

Proof. Parameterize P by k points in the closed unit ball, allowing repetition. Containment of $a\mathbb{B}^D$ is closed in these coordinates. Every such P satisfies $U_b P \subseteq \mathbb{B}^D \subseteq a^{-1}P$. The containment conditions are closed on this compact set and $1 \leq \lambda \leq a^{-1}$; hence a minimum exists.

Let its vertices be v_s and λ its dilation. Initialize with a convex representation of ax . For each b, s , represent $\lambda^{-1}U_b v_s$ by the same vertices and use its convex coefficients as the row $T_b(s, \cdot)$. Induction gives

$$\mathbb{E}[v_{S_t} \mid w] = a\lambda^{-t}U_w x.$$

The decoder $d_j(s) = \rho \lambda^N \langle e_j, v_s \rangle / a$ lies in $[-1, 1]$ and gives (1.1). These are actual rows throughout the run, not independently chosen checkpoint factorizations. Carathéodory gives $D + 1$ successors; there are $|\mathcal{A}|k$ command rows, and their construction and storage remain uncharged.

The defining finite problem is semialgebraic: use convex coefficients for transformed vertices and a universal ball containment condition, then eliminate quantifiers. To prove (4.11), set $e = \max_b \|U_b - V_b\|_{\text{op}}$. For every feasible P , $V_b P \subseteq U_b P + e\mathbb{B}^D \subseteq (\lambda + e/a)P$. Interchanging U, V completes the proof. $\square$

The two profiles have different quantifiers. The converse tests every hidden machine; the enclosure theorem builds a particular positive realization. Neither theorem asserts that $\lambda_{\mathcal{A}}$ is the optimum over hidden machines. This is why finite-scale lower and upper bounds can be used even when they do not match.

Proposition 4.6 (Bounded word recoding). *Suppose every command of $\mathcal{A}$ is a product of exactly L commands of an alphabet $\mathcal{B}$ on the same space, with the same seeds and queries. Then*

$$\Gamma_{\mathcal{A}}(k, B) \leq \Gamma_{\mathcal{B}}(k, LB), \quad \lambda_{\mathcal{A}}(k, a) \leq \lambda_{\mathcal{B}}(k, a)^L, \quad W_{N, \varepsilon}(\mathcal{A}) \leq W_{LN, \varepsilon}(\mathcal{B}).$$

An identity letter permits words of length at most L to be padded. If both alphabets contain an identity and admit such bounded recodings into the other, the existence and value of a logarithmic width exponent agree.

Proof. Push forward a packet law along the word substitution to obtain the first inequality. Iterate the enclosure containments for the second. For the last inequality, compose the L stochastic rows of a $\mathcal{B}$ machine into one row for the corresponding $\mathcal{A}$ command, retaining only the block-end label. All rows may depend on the external epoch. This is an implementation of the atomic $\mathcal{A}$ interface, not a claim that its physical letters cost L letters. Applying the two comparisons and monotonicity under final identity padding proves the exponent statement. $\square$

Corollary 4.7 (Packing–dilation balance). *Fix a calibrated experiment, an inradius $a > \rho$, and positive numbers u, v . Suppose that for every sufficiently large k there are a length $B(k) \leq Ck^u$ and a word law for which*

$$\sup_{x, z \in \mathbb{S}^{D-1}} p\{w : \|U_w x - z\| < ck^{-v}\} \leq (2k)^{-1}.$$

Suppose also that $\lambda_{\mathcal{A}}(k, a) \leq \exp(C'k^{-u-2v})$. Then

$$W_{N, \varepsilon} = \Theta(N^{1/(u+2v)}).$$

The same conclusion holds with $u = 0$ when the lengths are uniformly bounded.

Proof. Proposition 4.4 gives $\Gamma(k, B(k)) \geq c^2 k^{-2v}/4$. Equation (4.8) implies $N \leq C_1 k^{u+2v}$ for a feasible peak k . Conversely choose k at a sufficiently large constant times $N^{1/(u+2v)}$ so that $C'Nk^{-u-2v} \leq \log(a/\rho)$, and apply Theorem 4.5. The hypotheses concern finite word orbits and compatible convex containments, and impose no commutativity on the matrices. $\square$

5. A DISTORTION–DILATION COMPARISON

The lower and upper profiles defined in Section 4 do not have identical quantifiers. They nevertheless obey a universal inequality. Define the optimized per-command distortion rate

$$(5.1) \quad \mathfrak{r}_{\mathcal{A}}(k) = \sup_{B \geq 1} \frac{-\log(1 - \Gamma_{\mathcal{A}}(k, B))}{B}.$$

All logarithms in this paper are natural. The value $+\infty$ has its usual extended meaning.

Theorem 5.1 (Distortion bounds every common enclosure). *For every finite orthogonal alphabet and every feasible k, a in (4.10),*

$$(5.2) \quad \mathfrak{r}_{\mathcal{A}}(k) \leq \log \lambda_{\mathcal{A}}(k, a).$$

In particular, every finite-word certificate gives the lower enclosure bound $\lambda_{\mathcal{A}}(k, a) \geq (1 - \Gamma_{\mathcal{A}}(k, B))^{-1/B}$. The assertion does not require commuting commands, a spectral gap, or any Diophantine hypothesis.

Proof. Take any feasible polytope $P = \text{conv}(v_1, \dots, v_k)$, contained in the unit ball and containing a times that ball, with common dilation λ . Use the stochastic rows representing $\lambda^{-1}U_b v_s$ by the same vertices. Fix a unit seed u and initialize the mean at au . This construction makes sense at every length, without imposing a terminal decoder. For each complete word w of length n it satisfies

$$\mathbb{E}[v_{S_n} \mid w] = a\lambda^{-n}U_w u.$$

Let $Y_n = U_w u$ and $q_n = \mathbb{E}[\|Y_n\| \mid S_n]$. Pairing with the bounded vector v_{S_n} gives, under every word law,

$$(5.3) \quad q_n \geq \mathbb{E}\langle Y_n, v_{S_n} \rangle = a\lambda^{-n}.$$

Fix B and independently repeat an optimizing length- B word law m times. The packet lemma applies at the m endpoints, each of which has at most k labels. Since $q_0 = 1$,

$$a\lambda^{-mB} \leq q_{mB} \leq (1 - \Gamma_{\mathcal{A}}(k, B))^m.$$

Taking m th roots and letting $m \rightarrow \infty$ removes the inradius factor. Hence $\lambda^{-B} \leq 1 - \Gamma_{\mathcal{A}}(k, B)$. Take the supremum over B and then the minimum over enclosures. In particular, $\Gamma = 1$ is incompatible with a finite feasible enclosure at this k , so no undefined logarithmic operation was used. $\square$

The removal of the initial inradius loss by repetition is important: a single-horizon application would leave an additive $\log(1/a)$. Theorem 5.1 is a comparison, not a duality asserting equality or a characterization of arbitrary hidden width. The Hankel normal form in Proposition 3.1 permits general future-response polytopes, whereas (4.10) uses one common polytope in the observed vector space.

Corollary 5.2 (Nonuniform enclosure chains). *Suppose $P_t \subset \mathbb{B}^D$ has at most K_t generators, $a\mathbb{B}^D \subset P_0$, and $U_b P_{t-1} \subset \lambda_t P_t$ for every command. Then*

$$\mathcal{I}_N(K_\bullet) \leq \log(1/a) + \sum_{t=1}^N \log \lambda_t.$$

Proof. Use the cut-dependent convex rows and the initialization with mean au . The same induction gives mean $a(\prod_t \lambda_t)^{-1} U_w u$. Choose the independent packet law attaining the interval recurrence. The pairing argument bounds the final amplitude below by this factor, while the packet theorem bounds it above by $\exp(-\mathcal{I}_N(K_\bullet))$. Taking logarithms proves the claim. $\square$

6. EXACT OPTIMIZATION OF PLANAR PACKETS

For a rotation alphabet, let $z_1, \dots, z_M \in \mathbb{S}^1 \subset \mathbb{C}$ be the distinct product phases of words of length B , ordered cyclically. A common rotation of all phases can be absorbed into the decoder centers. The initial direction in (4.1) therefore disappears. Any probability distribution on these phases is implemented by a distribution on representative words of the same length.

Let $\mathfrak{P}_k$ be the finite set of partitions of the cyclically ordered list into at most k nonempty consecutive arcs. For $p \in \Delta_M$, set

$$(6.1) \quad F_k(p) = \max_{\mathcal{P} \in \mathfrak{P}_k} \sum_{A \in \mathcal{P}} \left| \sum_{i \in A} p_i z_i \right|.$$

Theorem 6.1 (Optimal word laws as a cone program). *For the planar alphabet above,*

$$(6.2) \quad \Gamma_A(k, B) = 1 - \min_{p \in \Delta_M} F_k(p).$$

Equivalently, minimize t subject to $p \in \Delta_M$ and, for every $\mathcal{P} \in \mathfrak{P}_k$,

$$(6.3) \quad \left\| \sum_{i \in A} p_i (\Re z_i, \Im z_i) \right\|_2 \leq r_{\mathcal{P}, A} \quad (A \in \mathcal{P}), \quad \sum_{A \in \mathcal{P}} r_{\mathcal{P}, A} \leq t.$$

This is a finite second-order cone program with attained optimum. A fixed law can be evaluated by cyclic segmentation in $O(kM^3)$ arithmetic operations, including computation of all interval sums. The optimization over laws may have exponentially many partition constraints.

Proof. The decoder centers are on the unit circle; maximizing their inner product is therefore exactly nearest chordal-distance assignment.

For fixed centers $v_1, \dots, v_k$ assign each phase to a nearest center in chordal distance; equivalently maximize $\Re(\bar{v}_j z_i)$. Distinct centers have connected cyclic Voronoi arcs, bounded by the angular midpoints of adjacent centers. Ties can be assigned consecutively; coincident centers can be merged. For the induced partition, the contribution of a block A is at most $|\sum_{i \in A} p_i z_i|$.

Conversely, for any cyclic partition choose the center of each block in the direction of its weighted complex sum, arbitrarily if that sum vanishes. The best assignment to those centers has value at least the sum in (6.1). Thus the maximum over centers equals $F_k(p)$. This proves (6.2); introducing the norm epigraphs proves (6.3). Zero weights follow either directly or by continuity from positive weights.

For a fixed cyclic cut, prefix sums give every block vector in constant time. The usual recurrence for the best j blocks among the first m points checks at most M previous split positions. It costs

$O(kM^2)$ for that cut. Repeating for the M cuts costs $O(kM^3)$. This recurrence evaluates a given law; it is not a polynomial-size representation of all constraints in (6.3). Compactness of the simplex and continuity give attainment. $\square$

Nearest-center partitions and vector quantization are classical [5]. The additional optimization here chooses the distribution of actual command words, and Theorem 4.3 turns its value into a constraint on arbitrary hidden registers. There is no assertion that static quantization alone characterizes the least whole-run width.

Proposition 6.2 (A finite exact packet certificate). *For the alphabet consisting of identity and a quarter turn, at word length three the four distinct phases are $1, i, -1, -i$. Their uniform law is optimal and*

$$\Gamma(1, 3) = 1, \quad \Gamma(2, 3) = 1 - \frac{1}{\sqrt{2}}, \quad \Gamma(3, 3) = \frac{2 - \sqrt{2}}{4}, \quad \Gamma(k, 3) = 0 \quad (k \geq 4).$$

Proof. The distortion is concave in the phase law and invariant under its four cyclic rotations. Averaging any law over those rotations proves optimality of the uniform law. For two centers, adjacent pairs give total $\sqrt{2}/2$; a triple and a singleton give only $1/2$. For three centers, one pair and two singletons give $\sqrt{2}/4 + 1/2$. For one center the sum is zero; four centers fit all phases. Formula (6.2) gives the values. A uniform phase law is realized by the four ordinary words with zero, one, two or three initial quarter turns and then idle commands. $\square$

For algebraic rotation matrices, both the general quantifier-elimination formulation and the cone program have algebraic coefficients. No finite list of spherical harmonics is used. For merely computable-real entries, finite numerical approximations are diagnostics unless a rigorous error enclosure is supplied; (4.3) then quantifies how an entry enclosure propagates to a packet certificate.

Corollary 6.3 (Separated phases). *If $M \geq 2k$ distinct length- B word phases have pairwise circular distance at least $s \in (0, 1/2]$, then $\Gamma_{\mathcal{A}}(k, B) \geq s^2$.*

Proof. Use their uniform law. Each open arc of circular radius $s/2$ contains at most one phase. At least half the mass is at distance at least $s/2$ from every one of k centers. On $[0, 1/2]$, $1 - \cos(2\pi t) = 2\sin^2(\pi t) \geq 8t^2$. The average loss is at least s^2 . This is a lower certificate for (6.2), not necessarily its exact value. $\square$

7. ARITHMETIC PROFILES AT EVERY FINITE SCALE

Now $\mathcal{A}_\alpha = \{I, R_{2\pi\alpha_1}, \dots, R_{2\pi\alpha_r}\}$, with r fixed. Define

$$(7.1) \quad \psi_\alpha(H) = \min_{0 < \|m\|_1 \leq H} \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}, \quad \eta_\alpha(q) = \max_{1 \leq i \leq r} \|q\alpha_i\|_{\mathbb{R}/\mathbb{Z}}.$$

These are finite functions of the alphabet parameters. The first may vanish in the presence of a rational relation. The upper profile remains usable in that case.

Theorem 7.1 (Finite arithmetic bounds). *Let $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$. For a machine of peak K , put $L = \lceil (2K)^{1/r} \rceil$ and $B = r(L - 1)$. Then*

$$(7.2) \quad \lfloor N/B \rfloor \psi_\alpha(B)^2 \leq \log(1/\kappa).$$

For every integer $q \geq 4$,

$$(7.3) \quad N\eta_\alpha(q) \leq q^2/100 \implies W_{N,0}(\alpha) \leq q.$$

Consequently every finite planar rotation alphabet satisfies

$$(7.4) \quad W_{N,0}(\alpha) \leq \max(4, \lceil \sqrt{50N} \rceil).$$

All upper bounds use common command rows and hold also after adjoining reflection in the first coordinate axis.

Proof. Construct a word of length B in r consecutive segments of $L - 1$ positions. In segment i give J_i copies of the i th rotation followed by idle commands, independently and uniformly for $J_i \in \{0, \dots, L - 1\}$. Differences of count vectors have ℓ_1 norm at most B . If $\psi_\alpha(B) > 0$, the $L^r \geq 2K$ phases are distinct and separated by $\psi_\alpha(B)$. Corollary 6.3 and (4.8), with $-\log(1-x) \geq x$, give (7.2). If the minimum is zero the displayed inequality is automatic. All letters in a segment are actual commands, not a free count input.

For the upper bound let P_q be the regular q -gon inscribed in the unit circle. Choose nearest integers p_i to $q\alpha_i$, and put $\delta_i = 2\pi(\alpha_i - p_i/q)$, so $|\delta_i| \leq \pi/q$. Elementary chord geometry gives

$$(7.5) \quad R_{2\pi\alpha_i}P_q \subseteq \Lambda_i P_q, \quad \Lambda_i = \frac{\cos(\pi/q - |\delta_i|)}{\cos(\pi/q)}, \quad \log \Lambda_i \leq \frac{4\pi^2 \eta_\alpha(q)}{q^2}.$$

Indeed the ray of a rotated vertex meets a neighboring edge at radial coordinate $\cos(\pi/q)/\cos(\pi/q - |\delta_i|)$. Integrating the derivative of $\log \cos(\pi/q - t)$ bounds the logarithm by $|\delta_i| \tan(\pi/q)$, and $\tan(\pi/q) \leq 2\pi/q$ for $q \geq 4$.

Take $\Lambda = \max_i \Lambda_i$, including one for the identity. Under (7.3), $\Lambda^N \leq e^{\pi^2/25} < 2$. Since P_q contains the half-unit disk, Theorem 4.5 with $a = 1/2$ applies: $\rho \Lambda^N \leq 1/5 < 1/2$. Explicitly, initialize at mean $x/2$, represent $\Lambda^{-1} R_{2\pi\alpha_i} v_s$ by the vertices for each row, and decode by $2\rho \Lambda^N \langle e_j, v_s \rangle$. This verifies every numerical word, including any deterministic remainder of a converse packet decomposition.

There are $(r+1)q$ common command rows and two decoder columns, each row having at most three successors. These counts exclude row-description and arithmetic costs. Since $\eta_\alpha(q) \leq 1/2$, (7.4) follows. Reflection preserves P_q ; its contracted image uses the same vertex set. The reflected lower bound is obtained simply by restricting to rotation-only words. $\square$

The pair $(\psi_\alpha, \eta_\alpha)$ is a specialization of the word-distortion and enclosure profiles. It is not assumed to characterize exact finite width. In particular, (7.2) can be weak when long word packets have clusters; the optimized profile of Section 6 is available without that separation relaxation.

Lemma 7.2 (Denominators under a general power bound). *Suppose for $\tau \geq r$ and $b > 0$,*

$$(7.6) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b \|m\|_1^{-\tau} \quad (m \neq 0).$$

For every integer $Q \geq 2$ there is q with

$$(7.7) \quad cQ^{r/(\tau+1-r)} \leq q \leq Q^r, \quad \eta_\alpha(q) \leq Q^{-1}, \quad c = (b/r^{\tau+1})^{r/(\tau+1-r)}.$$

Proof. The $Q^r + 1$ points $j\alpha$ in Q^r equal torus boxes give integers p_i, q with $1 \leq q \leq Q^r$ and $|q\alpha_i - p_i| \leq Q^{-1}$. For $M = \lfloor q^{1/r} \rfloor$, the $(M+1)^r > q$ integer vectors in $\{0, \dots, M\}^r$ have a residue collision for $m \cdot p \pmod{q}$. Subtracting gives $m \neq 0$ with $H = \|m\|_1 \leq rq^{1/r}$ and $q \mid m \cdot p$. Thus

$$bH^{-\tau} \leq \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq H/(qQ), \quad bqQ \leq r^{\tau+1} q^{(\tau+1)/r}.$$

Rearranging yields (7.7). This direct residue argument supplies exactly the quantitative transference needed here, without an unspecified transference constant. $\square$

Corollary 7.3 (Power bounds at finite Diophantine type). *Under (7.6), for constants depending on the fixed data,*

$$(7.8) \quad c_1 N^{r/(2\tau+1)} \leq W_{N,\varepsilon}(\alpha) \leq C_1 N^{\min\{1/2, r(\tau+1-r)/(\tau+1+r)\}}.$$

No matching claim is made when $\tau > r$.

Proof. From (7.2), $N < B + b^{-2} B^{2\tau+1} \log(1/\kappa)$, while $B \leq r(2K)^{1/r}$. This gives the lower power after adjusting finitely many initial horizons. Lemma 7.2 bounds $N\eta_\alpha(q)/q^2 \leq c^{-2} N Q^{-1-2r/(\tau+1-r)}$. Choose Q at constant times $N^{(\tau+1-r)/(\tau+1+r)}$, large enough to satisfy (7.3) and $q \geq 4$. Then $q \leq Q^r$ proves the second candidate upper power; (7.4) proves the first. $\square$

Corollary 7.4 (The retained badly approximable law). *If (7.6) holds with $\tau = r$, then $W_{N,\varepsilon}(\alpha) = \Theta(N^{r/(2r+1)})$, also with the reflection extension. This is the previously established matched law [8].*

Proof. Substitute $\tau = r$ in (7.8); the two powers coincide. $\square$

For $1, \alpha_1, \dots, \alpha_r$ rationally independent, every optimized finite-block action gap on the circle is zero. Indeed simultaneous approximation supplies unbounded frequencies n with every $n\alpha_i$ close to an integer. Each fixed rotation word then acts almost identically on its Fourier mode. For the reflected alphabet use the real cosine mode, which reflection fixes. This is the established almost-invariance argument, not a claim that the finite packet distortion vanishes.

8. MINIMAL ORDINARY TYPE AND ALMOST-EVERYWHERE WIDTHS

Uniform bad approximation is substantially stronger than the condition needed to determine the logarithmic width exponent. We first deduce the minimal-type assertion without any measure-theoretic hypothesis.

Proof of (1.4) in Theorem 1.1. If $\omega^*(\alpha) = r$, then for every $h > 0$ there is $b_h > 0$ such that

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b_h \|m\|_1^{-r-h} \quad (m \neq 0).$$

Indeed the definition of the limsup gives this inequality for all sufficiently large $\|m\|_1$, and rational independence makes the finite remaining minimum positive. Apply Corollary 7.3 with $\tau = r + h$. It gives

$$\frac{r}{2r+2h+1} \leq \liminf_N \frac{\log W_{N,\varepsilon}}{\log N} \leq \limsup_N \frac{\log W_{N,\varepsilon}}{\log N} \leq \min \left\{ \frac{1}{2}, \frac{r(1+h)}{2r+1+h} \right\}.$$

For $0 < h < 1/(2r-1)$ the second branch is smaller. Let $h \downarrow 0$. This proves (1.4). No positive uniform constant at exponent r was used. $\square$

For completeness we prove a more quantitative metric statement, using only the summable part of metric Diophantine approximation. It does not claim a new metric theorem.

Lemma 8.1 (A summable logarithmic separation). *For Lebesgue-almost every $\alpha \in [0, 1]^r$, and every $s > 1$, there is $b_s > 0$ such that*

$$(8.1) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{b_s}{\|m\|_1^r (\log(2 + \|m\|_1))^s} \quad (m \neq 0).$$

Such a vector has $\omega^(\alpha) = r$.*

Proof. For a nonzero integer vector m , the map $\alpha \mapsto m \cdot \alpha \pmod{1}$ sends Haar measure on the torus to uniform circle measure. There are at most $C_r H^{r-1}$ integer vectors with $\|m\|_1 = H$. Hence the measures of the events

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} < [\|m\|_1^r (\log(2 + \|m\|_1))^s]^{-1}$$

have finite sum, bounded by a constant times $\sum_{H \geq 1} [H(\log(2 + H))^s]^{-1}$. The measure of their tail union tends to zero. Almost every vector violates the unscaled inequality only finitely often. Remove the measure-zero rational hyperplanes and reduce the constant to cover the finite exceptions. Intersect the full-measure sets for rational $s > 1$. For a real $s > 1$, choose a rational $1 < s_0 < s$; its stronger bound implies the bound with s , after a harmless change of constant. This gives the assertion on one full-measure set.

This bound implies $\omega^*(\alpha) \leq r$. For the reverse inequality, place the $(M+1)^r$ values $m \cdot \alpha \pmod{1}$, $m \in \{0, \dots, M\}^r$, in fewer than $(M+1)^r$ intervals of equal length. A collision gives a nonzero difference of norm at most rM and distance $O(M^{-r})$. As $M \rightarrow \infty$ the norms of some such differences must tend to infinity, since rational independence excludes an exact relation. Thus $\omega^*(\alpha) \geq r$. $\square$

Proposition 8.2 (Quantitative bounds under logarithmic separation). *Suppose (8.1) holds for some $s > 0$ and $b_s > 0$. Then there are $c, C > 0$ such that*

$$(8.2) \quad cN^{r/(2r+1)}(\log(N+2))^{-2rs/(2r+1)} \leq W_{N,\varepsilon}(\alpha) \leq CN^{r/(2r+1)}(\log(N+2))^{2r^2s/(2r+1)}.$$

Proof. The packet bound (7.2) gives

$$N < B + b_s^{-2} B^{2r+1} (\log(2+B))^{2s} \log(1/\kappa).$$

With $B \leq r(2K)^{1/r}$ this implies $N \leq C_0 K^{(2r+1)/r} (\log(2+K))^{2s}$. If $K > N$, the proposed lower bound is already true after reducing its constant. Otherwise $\log(2+K) \leq \log(2+N)$, and rearrangement proves the first inequality of (8.2).

For the upper inequality repeat the residue argument of Lemma 7.2. It produces $q \leq Q^r$, $\eta_\alpha(q) \leq Q^{-1}$ and $H \leq rq^{1/r}$ with

$$\frac{b_s}{H^r (\log(2+H))^s} \leq \frac{H}{qQ}.$$

Since $q \leq Q^r$, it follows for $Q \geq 2$ that

$$(8.3) \quad q \geq c_0 \frac{Q^r}{(\log(Q+2))^{rs}}.$$

Indeed $b_s q Q \leq H^{r+1} (\log(2+H))^s \leq C_r q^{1+1/r} (\log(Q+2))^s$, and raising the rearranged inequality to the r th power gives (8.3).

Choose an integer

$$Q = \left\lceil C_2 N^{1/(2r+1)} (\log(N+2))^{2rs/(2r+1)} \right\rceil.$$

For sufficiently large fixed C_2 , (8.3) implies $N\eta_\alpha(q)/q^2 \leq 1/100$ and $q \geq 4$. To see the constant choice explicitly, for large N one has $\log(Q+2) \leq C_3 \log(N+2)$ with C_3 fixed; thus the ratio is bounded by a constant times $C_2^{-(2r+1)}$. Enlarge the final constant for the finitely many initial horizons. Theorem 7.1 gives an exact machine with $q \leq Q^r$ labels, proving the upper bound. $\square$

Completion of Theorem 1.1. Apply Lemma 8.1 with $s = 1 + \delta$ and then Proposition 8.2. This yields (1.5) on one full-measure set for all fixed $\delta > 0$. These statements concern a fixed alphabet before the horizon grows. None of their constants is asserted uniform in r or near a rational relation. $\square$

8.1. Rational dependencies without redundant exponents. The number of listed rotations need not be the relevant arithmetic rank. The following result covers finite rational reparametrizations rather than assigning one exponent to every redundant letter.

Corollary 8.3 (Finite rational extensions). *Suppose an alphabet contains I and rotations through angles $\beta_1, \dots, \beta_d$, where $1, \beta_1, \dots, \beta_d$ are rationally independent and $\omega^*(\beta) = d$. Suppose every additional angle belongs to $\mathbb{Q} + \sum_{j=1}^d \mathbb{Q}\beta_j$. Then*

$$\lim_N \frac{\log W_{N,\varepsilon}}{\log N} = \frac{d}{2d+1}.$$

The same assertion holds on a full-measure set of β with the logarithmic upper and lower powers of (1.5), replacing r by d . Reflection may be added. If all angles are rational, exact width is bounded independently of N .

Proof. Restriction to the displayed d -angle subalphabet gives the lower bound. Choose one common denominator h and integers t_i, n_{ij} so that every angle is $t_i/h + \sum_j n_{ij}\gamma_j$ with $\gamma_j = \beta_j/h$. Multiplication by h shows $\|m \cdot \gamma\|_{\mathbb{R}/\mathbb{Z}} \geq h^{-1} \|m \cdot \beta\|_{\mathbb{R}/\mathbb{Z}}$. Thus every separating power or logarithmic inequality for β holds for γ with a changed constant.

Take a simultaneous approximant p_j/q to γ from the proof above. A command's irrational part has residual angle at most $2\pi(\sum_j |n_{ij}|)/(qQ)$. The same polygon construction therefore works, after changing only fixed constants. Store in addition the rational phase $c \in \mathbb{Z}/h\mathbb{Z}$, updated by $c \mapsto c + t_i$. The decoder rotates the stored polygon vector by $2\pi c/h$. The peak, including the held answer, is $\max(2, hq)$, not an uncharged phase. Reflection sends c to $-c$ and conjugates the polygon phase; the regular polygon and the construction are reflection-invariant. The exact expectation identity is unchanged. The finite multiplier h does not change a logarithmic exponent or the displayed logarithmic powers.

For a purely rational alphabet, choose a common period h and retain the seed identifier and the accumulated phase. This uses at most $4h$ command labels for our four planar seeds, plus the held binary answer convention. No infinite-gap hypothesis is involved. $\square$

9. EVERY IRRATIONAL ANGLE HAS CUBIC SCALES

We keep the two-letter alphabet $\{I, R_{2\pi\alpha}\}$ and the planar resource and error conventions. No Diophantine lower bound is assumed in this section. The needed continued-fraction estimates are classical [3]; we include their short proof to specify the exact finite inequalities being used.

Lemma 9.1 (Convergent separation). *Let p_n/q_n be the convergents of an irrational α , ignoring the finitely many initial indices with repeated denominator. Then*

$$(9.1) \quad \frac{1}{q_{n+1} + q_n} < |q_n\alpha - p_n| < \frac{1}{q_{n+1}}, \quad \min_{1 \leq l < q_n} \|l\alpha\|_{\mathbb{R}/\mathbb{Z}} > \frac{1}{q_n + q_{n-1}} > \frac{1}{2q_n}.$$

Proof. The convergent recurrences imply $p_n q_{n-1} - p_{n-1} q_n = \pm 1$ and alternating signs of $e_n = q_n\alpha - p_n$. If ξ is the next complete quotient, then

$$\alpha = \frac{p_n \xi + p_{n-1}}{q_n \xi + q_{n-1}}, \quad a_{n+1} < \xi < a_{n+1} + 1.$$

It follows that $|e_n| = 1/(q_n \xi + q_{n-1})$, proving the first pair.

For $0 < l < q_n$ and an integer p , the determinant identity writes $(p, l) = a(p_n, q_n) + b(p_{n-1}, q_{n-1})$ with integers a, b . Here $b \neq 0$; a, b cannot both be positive, and cannot both be negative. If $a = 0$ the error has modulus at least $|e_{n-1}|$. Otherwise they have opposite signs, so ae_n and be_{n-1} have the same sign and again $|l\alpha - p| \geq |e_{n-1}|$. Minimize over p and apply the first pair with index $n - 1$. This proves the remaining inequalities. $\square$

Theorem 9.2 (Universal cubic subsequences). *Set $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$, $C_\kappa = 8(1 + \log(1/\kappa))$, and*

$$M_n = \lceil C_\kappa q_n^3 \rceil, \quad m_\kappa = \lceil 100(C_\kappa + 1) \rceil.$$

For all sufficiently large n ,

$$(9.2) \quad q_n/2 < W_{M_n, \varepsilon}(\alpha) \leq m_\kappa q_n.$$

The upper bound is exact, uses common command rows and requires no regularity of the partial quotients.

Proof. Use words of length $B = q_n - 1$ whose number of initial active letters ranges uniformly over $0, \dots, q_n - 1$. Lemma 9.1 gives q_n distinct phases with separation greater than $1/(2q_n)$. If the proposed peak K were at most $q_n/2$, Corollary 6.3 and the packet budget would give

$$\left\lfloor \frac{M_n}{q_n - 1} \right\rfloor \frac{1}{4q_n^2} \leq \log(1/\kappa).$$

The left side is at least $C_\kappa/4 - 1/(4q_n^2)$, strictly greater than the right side. This proves the lower bound against all hidden-state machines.

For the upper bound use the polygon denominator $Q_n = m_\kappa q_n$. Distance to the nearest integer is subadditive, so

$$\|Q_n \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq m_\kappa |q_n \alpha - p_n| < m_\kappa / q_{n+1} \leq m_\kappa / q_n.$$

Consequently $Q_n^2 / [100 \|Q_n \alpha\|_{\mathbb{R}/\mathbb{Z}}] > m_\kappa q_n^3 / 100 \geq (C_\kappa + 1) q_n^3 \geq M_n$. Equation (7.3) gives the exact upper construction. The constants have been chosen for transparency, not optimality. $\square$

Proposition 9.3 (Resonant long horizons). *For any irrational α and integer $q \geq 4$ with $N_q = \lfloor q^2 / (100 \|q\alpha\|_{\mathbb{R}/\mathbb{Z}}) \rfloor \geq 1$, one has $W_{N_q, 0}(\alpha) \leq q$. If α is Liouville, there is a sequence $N \rightarrow \infty$ on which $\log W_{N, \varepsilon}(\alpha) / \log N \rightarrow 0$.*

Proof. The first assertion is the finite enclosure implication (7.3). For a Liouville number choose arbitrarily large A and unbounded denominators q with $0 < \|q\alpha\|_{\mathbb{R}/\mathbb{Z}} < q^{-A}$. The first assertion then supplies horizons $N_q \geq q^{A+2}/200$ for large q , with width at most q . The logarithmic ratio is at most $1/(A+2) + o(1)$. A diagonal sequence with $A \rightarrow \infty$ proves the claim, since widths are at least two. $\square$

Proof of Theorem 1.2. Equation (9.2) and $M_n = \Theta(q_n^3)$ give the first assertion and a subsequence of logarithmic exponent $1/3$. Proposition 9.3 gives the zero lower limit for Liouville angles, and (7.4) gives the upper limit at most $1/2$.

It remains to verify unboundedness, which does not follow from a subsequence alone. Fix any proposed width K . The $2K$ phases of counts $0, \dots, 2K-1$ are distinct for an irrational angle and have some positive minimum separation s_K . Independent words of length $2K-1$ imply $\lfloor N/(2K-1) \rfloor s_K^2 \leq \log(1/\kappa)$ for a width- K machine. This fails for all sufficiently large N . Thus $W_{N, \varepsilon} \rightarrow \infty$. $\square$

Corollary 9.4 (An explicit fluctuating alphabet). *Let $\alpha_L = \sum_{j=1}^{\infty} 10^{-j!}$. For $q_m = 10^{m!}$ and all large m , the horizon*

$$N_m = \lfloor q_m^{m+2} / 200 \rfloor$$

admits an exact machine with at most q_m labels. The same fixed two-letter alphabet also has the cubic subsequences of Theorem 9.2; hence no single logarithmic width exponent exists. Adding reflection preserves these conclusions.

Proof. The first m summands have denominator dividing q_m , and their remaining positive tail is less than $2 \cdot 10^{-(m+1)!}$. Thus $0 < \|q_m \alpha_L\|_{\mathbb{R}/\mathbb{Z}} < 2q_m^{-m}$ for large m , proving the horizon bound by (7.3). The same tail estimates preclude rationality and show the Liouville property. The general cubic theorem applies. Reflection leaves the upper polygons invariant, while the rotation-only test words give the same lower bounds. $\square$

These theorems describe different scales of one and the same experiment. A small label count on a long resonant horizon is not a constant-width simulator for all horizons, and it need not persist across the gaps between successive approximants. The exact finite optimum and the Liouville upper logarithmic limit remain separate quantitative questions. None of the proofs substitutes approximate output for the exact upper realizations.

10. FINITE RANDOM BITS AND UNIFORM SPACE

Exact real rows and finite-bit algorithms are different resources. We first give a finite perturbation statement and then remove the row table from a concrete uniform construction. Only this section uses a computational model. It does not change the exact atomic-row conclusions proved elsewhere.

Lemma 10.1 (Dyadic perturbation of a stochastic transducer). *Suppose initialization and every command row have at most s nonzero entries. They can be replaced by denominator- 2^b rows, on the same label sets, with row-TV error at most $(s-1)2^{-b}$. Rounding the terminal binary probabilities adds at most 2^{-b} . Consequently every numerical word changes by at most $((N+1)(s-1)+1)2^{-b}$ in binary total variation. Each rounded row can be sampled with exactly b independent fair bits.*

Proof. Order the s successors, round the first $s-1$ probabilities down to multiples of 2^{-b} , and give the remaining mass to the last successor. The total removed mass is at most $(s-1)2^{-b}$, which is exactly the row total variation. Couple the initializations and then the command updates, or telescope products using TV contraction, to sum their worst-row errors. A terminal Bernoulli parameter can be rounded down with error at most 2^{-b} . A uniform integer in $\{0, \dots, 2^b - 1\}$ selects one of the cumulative dyadic intervals. This uses a fixed number of bits, with no retained random tape or rejection-time side channel. $\square$

For a table implementation the storage is $O(|\mathcal{A}|Ks(\log K + b))$ bits for common rows of width K , in addition to initialization and decoder data. Cut-dependent rows can require a factor N more storage. Lemma 10.1 is the elementary stochastic-rounding principle; it is not claimed as a new finite-bit simulation method. The next construction uses no such table.

Definition 10.2 (Uniform command-stream model). A fixed finite program receives N in binary, one of the four planar seeds, exactly N one-way command symbols, and one final coordinate query. It has a work tape and fresh independent fair bits. The command input cannot be reread. All counters, arithmetic registers, stored parameters, random words and output storage count toward work space. An external source waits while a command is processed; waiting time is not an input or a persistent storage channel. The finite control is fixed independently of N . The program has no horizon-dependent advice or transition table.

An angle has an admissible digit procedure if a fixed program computes a dyadic approximation within 2^{-p} in polynomial time in p and $O(p)$ work bits. The same fixed r is used throughout. This explicit hypothesis excludes a free oracle for an arbitrary noncomputable real. Algebraic angles, in particular the examples below, meet it by rational bisection with fixed-degree integer arithmetic.

Theorem 10.3 (A table-free finite-bit compiler). *Fix r angles with admissible digit procedures, rational $0 < \rho \leq 1/10$, and rational $0 < \varepsilon < 1$. There is a uniform program in Definition 10.2 whose every binary output row has TV error at most ε from the rotation experiment. It uses*

$$(10.1) \quad O_{r,\alpha,\rho}(\log(N+2) + \log(1/\varepsilon)) \text{ work bits}, \quad (N+2)b \text{ fair bits},$$

where $b = \lceil \log_2(16(N+2)/\varepsilon) \rceil$. Its total running time is polynomial in $N + \log(1/\varepsilon)$ for fixed digit procedures. Its clean stochastic register has $q+1$ labels, including one central label, with $q = O(\sqrt{N})$.

If a denominator $q_0 \geq 4$ satisfies $N\eta_\alpha(q_0) \leq q_0^2/2000$, the search may choose $q \leq q_0$. Consequently the clean-register upper exponents and logarithmic bounds of Theorem 1.1 hold for this compiler as well, with changed constants. These clean labels do not include its scratch space or deterministic parameter registers; the full work bound is (10.1).

Proof. We give the parameter search, real reference rows, dyadic rounding, and finite-space evaluation separately.

Parameter search. Put $p = \lceil \log_2(16000(N+2)/\varepsilon) \rceil$ and obtain dyadic $\tilde{\alpha}_i$ with error at most 2^{-p} . For $q = 4, 5, \dots$ compute, by integer arithmetic,

$$e(q) = \max_i \|q\tilde{\alpha}_i\|_{\mathbb{R}/\mathbb{Z}} + q2^{-p}.$$

Stop at the first q for which $Ne(q) \leq q^2/1000$. The test is rational and certifies both $\eta_\alpha(q)$ and $\eta_{\tilde{\alpha}}(q)$ by the Lipschitz property of distance to $\mathbb{Z}$. It succeeds by $q = \max(4, \lceil \sqrt{2000(N+2)} \rceil)$:

$\|q\tilde{\alpha}_i\| \leq 1/2$ and the added term is smaller than one at this bound. If q_0 satisfies the stronger displayed condition, then $e(q_0) \leq \eta_\alpha(q_0) + 2q_02^{-p}$ and $2Nq_02^{-p} \leq q_0^2/2000$. Hence that candidate succeeds. No exact comparison of an irrational angle with a rational sector boundary is performed.

Reference rows with slack. Use the regular q -gon $v_j = (\cos(2\pi j/q), \sin(2\pi j/q))$ and one additional label $*$ representing zero. Set $\Lambda = 1 + 1/(4N)$. For $N \geq 1$ the dilation estimate (7.5) and the search condition give

$$\log \Lambda \geq 1/(5N) > 4\pi^2\eta_\alpha(q)/q^2.$$

Thus $\Lambda^{-1}R_{2\pi\tilde{\alpha}_i}v_j$ lies strictly inside the polygon. The case $N = 0$ is handled by direct rounding of the seed response and a fixed amount of space. Write $q\tilde{\alpha}_i = n_i + t_i$, with integer n_i and $0 \leq t_i < 1$, and put $h = 2\pi/q$. The command row has successors $j + n_i$, $j + n_i + 1$ modulo q , and $*$, with respective weights

$$(10.2) \quad A_i = \frac{\sin((1 - t_i)h)}{\Lambda \sin h}, \quad B_i = \frac{\sin(t_i h)}{\Lambda \sin h}, \quad 1 - A_i - B_i.$$

They are nonnegative by the dilation bound, and their vector mean is the required contracted rotation. The row at $*$ stays there. Initialize with vector mean $x/2$ using the same chord formula with radial factor $1/2$ and the exact seed angle $0, 1/4, 1/2, 3/4$. Decode with mean $2\rho\Lambda^N\langle e_j, v_s \rangle$, or zero at $*$. Since $\Lambda^N \leq e^{1/4}$, all decoder means are strictly inside $[-1, 1]$. The usual mean induction realizes every word for $\tilde{\alpha}$ exactly. These are reference real rows only.

Certified finite rows. Compute outward intervals for the first two weights in each row, each of width at most 2^{-b} . Clip their lower endpoints at zero and round down to denominator 2^b ; assign the residue to $*$. Relative to the reference row, at most $4 \cdot 2^{-b}$ mass is moved. Evaluate and round each terminal Bernoulli parameter within $2 \cdot 2^{-b}$. TV contraction therefore bounds initialization, N updates and final decoding by $(4N + 6)2^{-b}$. Replacing the true angles by their dyadic approximants changes a product rotation by at most $2\pi N2^{-p}$ in operator norm, and changes a target binary row by at most $\pi\rho N2^{-p}$. The sum is less than ε with the chosen p, b . Use exactly b fair bits at each draw, including dummy bits when the central row is deterministic. The random-bit count is $(N + 2)b$.

Work and uniform evaluation. Only q , the current vertex or central flag, the horizon and command counter, the r dyadic angles, and $O(r)$ offsets and dyadic weights are retained. The latter specify circulant rows, not a q -row table. The denominator search uses integers of $O(p + \log q)$ bits. Elementary sine and cosine on bounded intervals can be evaluated by fixed-point Taylor recurrences with outward rounding; π can be evaluated by Machin's arctangent identity. There are $O(b + \log q)$ terms and only a fixed number of live accumulators. Since $\sin(2\pi/q) \geq 4/q$ for $q \geq 4$, division loses only $O(\log q)$ bits. Repeated squaring evaluates Λ^N with $O(\log N)$ stages, all intermediate true values between 1 and $e^{1/4}$. An additional $O(\log(N + 2) + \log(b + 2))$ guard bits bounds accumulated rounding errors. Interval precision can instead be doubled until the stated widths are certified; these error estimates bound the final precision by the same order. No integer of unbounded exact denominator is stored. Every live register has $O(b + p + \log N)$ bits, and the number of registers is fixed for fixed r . This proves the work bound and a polynomial time bound, including parameter search. Finally the finite arithmetic upper proofs remain valid with $1/100$ replaced by $1/2000$, since only fixed constants change. Their candidate denominators therefore bound the search. $\square$

The proof describes a fixed-point implementation, not the numerical behavior of a floating-point library. The accompanying integer-interval reference code certifies each rounded row and uses fresh bounded-length random integers; its Python object overhead is not identified with the Turing work-space constant.

Corollary 10.4 (Sharp work-space order for explicit alphabets). *Let $\alpha_i = 2^{i/(r+1)}$, $1 \leq i \leq r$, and fix rational $0 < \varepsilon < \rho/(2\sqrt{2})$ and $0 < \rho \leq 1/10$. In Definition 10.2, numerical simulation requires and suffices with $\Theta(\log N)$ work bits. More precisely every fixed program using $s(N)$ work bits obeys*

$$(10.3) \quad s(N) \geq \frac{r}{2r+1} \log_2 N - O(\log \log(N+2)).$$

The compiler's clean register has $O(N^{r/(2r+1)})$ labels. This is not a claim that all its internal fair-bit states retain that label exponent or that the leading coefficient of work space is optimal.

Proof. Put $\theta = 2^{1/(r+1)}$. For a nonzero integer vector m with $H = \|m\|_1 \geq 1$, choose the integer p nearest to $-\sum_i m_i \theta^i$. The algebraic integer $P(\theta) = p + \sum_i m_i \theta^i$ is nonzero: the minimal polynomial $t^{r+1} - 2$ is Eisenstein and has degree $r+1$. Each of its other r conjugates has modulus at most $(9/2)H$, since $|p| \leq 2H + 1/2$ and $|\theta^i| < 2$. Its nonzero integer norm gives

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq (2/9)^r H^{-r}.$$

This classical algebraic example is credited in [4]; it is not new arithmetic. The inherited badly approximable theorem, Corollary 7.4, now supplies the all-hidden lower bound and the matching candidate denominators for the compiler. Bisection for θ and fixed-degree powers meet the digit hypothesis. A fixed program with $s(N)$ work bits has at most $C(s(N) + 1)^{C2^{s(N)}}$ command-boundary configurations. Its private randomness since the last boundary is integrated into the next stochastic row, so it induces a legal atomic realization of at most that width. Apply the lower bound $W_{N,\varepsilon} \geq cN^{r/(2r+1)}$ and take logarithms. The polynomial factor accounts for the fixed number of work-tape head positions; it contributes $O(\log \log N)$ unless the displayed lower bound is already automatic. Theorem 10.3 supplies the upper work bound. $\square$

11. REALIZATION THEORY, ARITHMETIC AND COMPUTATIONAL SEMANTICS

Positive order and cone invariance. In the classical stationary problem one seeks a fixed nonnegative state-space realization of a transfer function. Minimal signed order need not equal minimal positive order. Benvenuti–Farina's Theorems 2–3 formulate positive existence by an invariant polyhedral cone, either in a minimal linear realization or in a response-sequence space [12]; the 2022 survey concerns positive minimality [13]. Our enclosure construction is such a positive-cone construction after homogenization. The competing machines are different: their rows may vary with the cut and their entire realization may be redesigned for N . Proposition 3.1 makes the corresponding finite controlled normal form explicit. Classical stationary order cannot be substituted for its least simultaneous profile.

Hankel rank and signed weighted automata. For fixed seed and query the mean series is $w \mapsto \rho\langle e_j, U_w x \rangle$, with a signed linear representation of dimension at most D . Adding the constant probability offset requires at most $D + 1$ dimensions. Balle–Panangaden–Precup's Theorem 2 identifies minimal weighted-automaton dimension with infinite Hankel rank [14]. That is the signed, all-length invariant, not stochastic label cardinality. Their Theorem 11 bounds an ℓ^2 approximation of absolutely summable rational series. Here the unweighted all-word probability series is not summable, and the objective is the maximum numerical error on a fixed length. A small weighted mean-square error under a chosen word law does not control rare-word conditional error. For an explicitly unweighted finite table, Frobenius error does bound entrywise error, but its optimizer still need not be stochastic or compatible. Equation (3.5) exhibits the precise failure of separate ranks to detect the present growing resource.

Stochastic and hidden Markov realization. Finesso–Grassi–Spreij use pseudo-Hankel nonnegative factorizations and an informational-divergence criterion to approximate a stationary output process at prescribed hidden order [15, Sections III–IV]. Their second realization step is itself evidence that a static factorization is not the complete dynamic specification. Our alphabet is externally controlled rather than generated as an observation process. Seed resets and selected query columns are separately specified; worst-word conditional probabilities replace a divergence between output laws. These are explicit changes of datum and objective, not a claim that Hankel positivity was absent from HMM theory.

Ohta’s reachable and null-space reductions, Theorems 6–9 and Remark 10, identify tensor and generalized-Hankel dimensions in the effective linear space of an HMM [16]. We use the same need to distinguish observable rank from hidden state count, but do not infer a positive stochastic basis from a signed quotient. His Section 4.4 also discusses rotational examples descended from earlier HMM realization work. Rotation-based rank/positive-order separations are therefore not claimed new here. Our quantitative theorems concern all horizon-specific stochastic rows and preserve bounded numerical responses on every controlled word.

Kronecker quasi-uniformity. Dick–Goda–Larcher–Pillichshammer–Suzuki [4], Theorem 3.9, characterize quasi-uniformity of the multidimensional sequence $n\alpha \pmod{\mathbb{Z}^r}$ by the simultaneous bound $\eta_\alpha(q) \geq cq^{-1/r}$. Their Lemma 3.11 bounds ratios of successive best simultaneous denominators; Section 3.3 also contains the vector $(2^{1/(r+1)}, \dots, 2^{r/(r+1)})$. These are direct arithmetic antecedents, not new examples of this paper.

Our packet points instead have the form $m \cdot \alpha \pmod{1}$. The relevant separation is dual. The residue argument of Lemma 7.2, with an arbitrary nearest-integer numerator and $\eta_\alpha(q)$ in place of Q^{-1} , proves directly that a dual bound with exponent r implies

$$\eta_\alpha(q) \geq br^{-r-1}q^{-1/r}.$$

Thus the cited quasi-uniformity theorem applies to the uniform badly approximable class used in the predecessor. Its denominator control supplies the arithmetic scales for a polygon upper bound. It supplies neither a compatible stochastic transition table nor a lower bound against discarded hidden history. The logarithmic separation of Lemma 8.1 is weaker than a uniform badly approximable inequality; it is proved by a summable tail estimate, not asserted to imply quasi-uniformity.

What is dynamic in the paired profiles. The geometry of nearest-center distortion and convex hull approximation is classical [5, 11]. A static k -center factorization does not specify how its encoder is calculated without retaining discarded inputs. Conversely, a convex codebook does not force arbitrary hidden states to possess coordinates in that codebook. The two implications in Section 4 avoid both identifications: the converse conditions on the actual state, and the upper theorem provides the same legal rows throughout the run. The packing–dilation balance in Corollary 4.7 is a conditional theorem on independently specified finite profiles, not a claimed equality of static and dynamic optima.

The circle program makes one of these profiles exactly finite, whereas the all-harmonic action-gap optimization of the predecessor remains infinite-dimensional. A finite word length can be exhaustively listed; an unbounded harmonic degree cannot be certified by a finite truncation without a tail theorem. No general action-gap algorithm follows from Proposition 4.1.

Branching programs and probability preservation. At fixed horizon our machine is a probabilistic ordered read-once layered program with arbitrary stochastic rows and bounded real output columns. We distinguish width from total size, uniform description and random-bit complexity. Fourier growth for regular, permutation and width-three branching programs concerns different hypotheses and objectives [9, 10, 6]. Circle-walk harmonic damping and its arithmetic behavior are also

established subjects [2]. Our converse can use correlated letters inside independently sampled complete packets; it is not a theorem about average error under fair independent letters.

Preserving a strict cutpoint sign is weaker than preserving a numerical probability. The predecessor’s constant-width sign construction contracts the output amplitude with length and hence does not contradict the present width growth. Here signal and tolerance are fixed, one query is selected after the word, and neither joint answers nor anytime correctness is required. The exact arithmetic upper rows are atomic real kernels. Section 10 gives a separate finite-bit approximation theorem; it does not represent those exact rows by a finite number of fair bits. A row with three successors can still have costly or irrational coefficients.

Conclusions. The finite optimized word profile and the compatible enclosure profile retain quantitative information which qualitative gap data discard. They yield a minimal-type exponent and an almost-everywhere logarithmic estimate, and they exhibit fluctuating numerical width for every Liouville single angle. The reflected versions show that these conclusions survive a particular noncommuting finite extension, without asserting a classification of arbitrary noncommuting nongapped alphabets.

Exact finite widths, the Liouville upper logarithmic limit, optimal enclosure shapes and equality in the distortion–dilation comparison remain distinct quantitative problems. The new profile comparison uses no external spectral gap. For fixed computable planar angles, storing all command counts already gives a classical $O(\log N)$ finite-accuracy work-space upper bound. We do not claim novelty for that upper order alone. The compiler proved here additionally maintains an arithmetic-scale clean stochastic register, generates circulant dyadic rows without a large advice table, and explicitly prices every arithmetic and random-bit register. The earlier expansion, calibration and representation theorems retain their original hypotheses in the unchanged supporting article.

REFERENCES

- [1] S. Basu, R. Pollack and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, 2nd ed., Springer, 2006.
- [2] I. Berkes and B. Borda, Random walks on the circle and Diophantine approximation, *J. London Math. Soc.* **108** (2023), 409–440.
- [3] J. W. S. Cassels, *An Introduction to Diophantine Approximation*, Cambridge Tracts in Mathematics and Mathematical Physics, vol. 45, Cambridge University Press, 1957.
- [4] J. Dick, T. Goda, G. Larcher, F. Pillichshammer and K. Suzuki, On the quasi-uniformity properties of quasi-Monte Carlo point sets and sequences—Part I: Lattices and Kronecker sequences, arXiv:2502.06202v2, version of 13 February 2026, Section 3.3.
- [5] S. Graf and H. Luschgy, *Foundations of Quantization for Probability Distributions*, Lecture Notes in Mathematics, vol. 1730, Springer, 2000.
- [6] C. H. Lee, E. Pyne and S. Vadhan, Fourier growth of regular branching programs, *APPROX/RANDOM 2022*, LIPIcs **245**, Art. 2, 2:1–2:21.
- [7] Q. Qi, *General Theta Foundations I: Sharp Width of Hidden Rotation Experiments*, manuscript, revision 35, 2026; repository publication [aabc0731a0c9](#).
- [8] Q. Qi, *General Theta Foundations I: Alphabet Dynamics and Calibrated Numerical Memory*, manuscript, revision 42, 2026; repository publication [b5f9874903a4](#).
- [9] O. Reingold, T. Steinke and S. Vadhan, Pseudorandomness for regular branching programs via Fourier analysis, *RANDOM 2013*, LNCS **8096**, 655–670.
- [10] T. Steinke, S. Vadhan and A. Wan, Pseudorandomness and Fourier growth bounds for width 3 branching programs, *Theory Comput.* **13** (2017), Art. 12, 1–50.
- [11] M. Vidyasagar, *Hidden Markov Processes: Theory and Applications to Biology*, Princeton University Press, 2014.
- [12] L. Benvenuti and L. Farina, A tutorial on the positive realization problem, *IEEE Trans. Automat. Control* **49** (2004), 651–664, Theorems 2–3 and Example 4.
- [13] L. Benvenuti, Minimal positive realizations: A survey, *Automatica* **143** (2022), Art. 110422.
- [14] B. Balle, P. Panangaden and D. Precup, A canonical form for weighted automata and applications to approximate minimization, arXiv:1501.06841v3, 2015, Theorems 2 and 11.

- [15] L. Finesso, A. Grassi and P. Spreij, Two-step nonnegative matrix factorization algorithm for the approximate realization of hidden Markov models, arXiv:1007.3435v1, 2010, Sections III–IV.
- [16] Y. Ohta, On the realization of hidden Markov models and tensor decomposition, arXiv:2008.11487v1, 2020, Theorems 6–9 and Remark 10.
- [17] Y. Bugeaud and M. Laurent, Exponents of Diophantine approximation, arXiv:math/0611354v1, 2006, Definition 1 and Section 2.
- [18] Q. Qi, *General Theta Foundations I: Compatible Polyhedral Lifts and Quantitative Causal Width*, manuscript, revision 33, 2026; repository publication **aab961081253**.
- [19] Q. Qi, *General Theta Foundations I: Word Profiles and Arithmetic Fluctuations of Numerical Memory*, manuscript, revision 43, 2026; repository publication **153830f5dc8d**.

PRESERVED MATHEMATICAL DEVELOPMENT

General Theta Foundations I - revision 43

The complete current article precedes this divider.

The following predecessor cumulative volume is reproduced without alteration.

Frozen predecessor publication:

153830f5dc8d13358f9103c307c619076c5ec80c

This archive is not part of the compact referee package. Independent historical analytic obligations retain their original hypotheses and proof status.

GENERAL THETA FOUNDATIONS I: WORD PROFILES AND ARITHMETIC FLUCTUATIONS OF NUMERICAL MEMORY

QIAN QI

ABSTRACT. We associate two finite-scale quantities to a finite orthogonal command alphabet: an optimized word-packet distortion and a compatible convex-enclosure contraction. The first bounds every hidden stochastic realization through a nonuniform interval budget; the second constructs exact realizations with common command rows. Neither requires a spectral gap. In the planar case, optimal packet distortion has a finite second-order cone formulation. These profiles yield a numerical-memory exponent $r/(2r + 1)$ for every r -angle alphabet of minimal ordinary dual Diophantine type, a class of full Lebesgue measure; explicit logarithmic bounds hold almost everywhere. For every irrational single angle we identify a sequence of horizons on which width has cubic-root order. For every Liouville angle the logarithmic width exponent has lower limit zero and upper limit at least one third, so a single power law need not exist. All lower bounds allow arbitrary hidden states and time-dependent stochastic rows, and the upper constructions preserve each numerical probability exactly. The resource is nonuniform clocked atomic-row label width, with a fixed signal, a fixed sufficiently small wordwise error, and one selected terminal query.

1. INTRODUCTION

A finite-dimensional orbit may admit a small positive factorization at each time and still require a growing register when those factorizations must be executed by one causal machine. For a fixed alphabet, that growth also depends on the distribution of its finite words. A qualitative assertion that all spectral gaps vanish loses precisely the arithmetic information which distinguishes such examples.

We study the numerical response

$$(1.1) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, U_w x \rangle}{2}, \quad b \in \{-1, 1\},$$

where the command word w has prescribed length N , the seed x is a unit vector, and only one final coordinate j is queried. The matrices in U_w are orthogonal. A classical register must preserve these probabilities, or approximate every one of them within a fixed binary total-variation error. Its stochastic rows may vary arbitrarily with the epoch. Section 2 defines the resource without a uniformity or finite-random-bit assumption.

The first main result is a pair of operational bounds in terms of finite data. For words of length B , let $\Gamma_{\mathcal{A}}(k, B)$ be the largest, over word distributions, of the smallest spherical distortion after replacing the word orbit by k centroid directions. This definition optimizes the converse law and permits every initial direction. Theorem 3.3 charges $-\log(1 - \Gamma_{\mathcal{A}}(K_t, B))$ to any interval of length B ending at time t , even when all other registers are unrestricted. A finite interval recurrence combines all such charges and bounds actual final error. In the opposite direction, Theorem 3.5 turns a finite polytope and its worst command contraction into common stochastic rows. These are distinct lower and upper profiles; we do not assume that an optimal hidden realization must be a vector-state polytope.

For circle actions, Theorem 4.1 removes the continuous search over centroid directions. The exact optimized packet value is a finite second-order cone problem indexed by cyclic partitions of the word phases. Thus a finite packet certificate is computationally different from an infinite harmonic-gap assertion. The theorem is not an efficiency assertion in a succinct word length.

Our arithmetic conclusions go beyond uniform bad approximation. For $\alpha \in \mathbb{R}^r$ with $1, \alpha_1, \dots, \alpha_r$ rationally independent, write

$$(1.2) \quad \omega^*(\alpha) = \limsup_{\|m\|_1 \rightarrow \infty} \frac{-\log \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}}{\log \|m\|_1}.$$

The alphabet consists of the identity and the rotations through $2\pi\alpha_i$. Throughout, r , the alphabet, $0 < \rho \leq 1/10$ and $0 \leq \varepsilon < \rho/(2\sqrt{2})$ are fixed.

Theorem 1.1 (Minimal Diophantine type). *If $\omega^*(\alpha) = r$, then*

$$(1.3) \quad \lim_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} = \frac{r}{2r+1}.$$

For Lebesgue-almost every α , and every fixed $\delta > 0$, there are positive constants c, C such that, for $N \geq 2$,

$$(1.4) \quad c \frac{N^{r/(2r+1)}}{(\log(N+2))^{2r(1+\delta)/(2r+1)}} \leq W_{N,\varepsilon}(\alpha) \leq CN^{r/(2r+1)} (\log(N+2))^{2r^2(1+\delta)/(2r+1)}.$$

The upper inequalities are achieved by exact common-row machines.

The exponent in (1.3) need not imply a two-sided constant-factor law. The earlier dual badly approximable theorem supplies the stronger Θ conclusion on its stated class, and is recovered in Corollary 5.4. Theorem 1.1 instead allows arbitrarily small Diophantine constants along subsequences. The quantitative arithmetic ingredients are classical; the memory conclusions concern their compatible realization and all-hidden-state converse.

There is also a genuine failure of a single exponent, even for one fixed irrational angle.

Theorem 1.2 (Irrational horizons and Liouville fluctuations). *For every irrational α there are horizons $M_n \rightarrow \infty$ such that $W_{M_n,\varepsilon}(\alpha) = \Theta(M_n^{1/3})$. If α is Liouville, then*

$$(1.5) \quad \liminf_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} = 0, \quad \frac{1}{3} \leq \limsup_{N \rightarrow \infty} \frac{\log W_{N,\varepsilon}(\alpha)}{\log N} \leq \frac{1}{2}.$$

Nevertheless $W_{N,\varepsilon}(\alpha) \rightarrow \infty$. All displayed upper bounds have exact realizations.

The upper limit in (1.5) is not determined here. The result establishes oscillation between subpower subsequences and a positive-power subsequence; it does not replace that distinction by a guessed exponent for all irrationals.

Relation to previous work. Finite codebooks, nearest-center partitions and positive convex realizations are classical tools [5, 11]. The endpoint conditional-centroid contraction used below was established in the preceding packet development [7, 8]; we give its complete proof and use it to define a finite alphabet profile. The general interval optimization, the circle word-law program, and the metric and Liouville memory conclusions are the new statements of this revision. Classical Dirichlet approximation, continued fractions and algebraic examples are not counted as new arithmetic. The direct comparison with Dick–Goda–Larcher–Pillichshammer–Suzuki [4], including their Theorem 3.9 and Lemma 3.11, is made in Section 8.

Quantity	Meaning
$K_t, W_{N,\varepsilon}$	Available labels at a cut; minimum whole-run peak.
$\Gamma_{\mathcal{A}}(k, B)$	Best finite-word distortion against k endpoint directions.
$\mathcal{I}_N(K_{\bullet})$	Maximum sum of distortion charges over disjoint intervals.
$\lambda_{\mathcal{A}}(k, a)$	Least common enclosure dilation with k vertices and inradius a .
$\psi_{\alpha}(H), \eta_{\alpha}(q)$	Dual separation and simultaneous rational-approximation errors.

2. THE NUMERICAL EXPERIMENT AND THE REGISTER

Let $V = \mathbb{R}^D$, let $\mathcal{A}$ be a nonempty finite alphabet, and fix orthogonal matrices U_a , $a \in \mathcal{A}$. Write $U_w = U_{a_N} \cdots U_{a_1}$ in the order of execution. Let $X \subset \mathbb{S}^{D-1}$ be a nonempty finite seed set and $0 < \rho < 1$. The required law is (1.1). Commands are externally supplied, not chosen by a policy inside the simulator. No distribution on command words is part of the specification.

Definition 2.1 (Clocked atomic-row realization). At horizon N , a machine consists of finite sets S_t with $|S_t| = K_t$ for $0 \leq t \leq N$, an initialization probability row E_x on S_0 , stochastic matrices $T_{t,a} : S_{t-1} \rightarrow S_t$, and terminal columns $d_j : S_N \rightarrow [-1, 1]$. Its numerical mean is

$$E_x T_{1,a_1} \cdots T_{N,a_N} d_j.$$

After the atomic query j , it draws and holds the binary answer with that mean. The final answer cut has two labels. Its peak is $\max(2, K_0, \dots, K_N)$. Available labels count even when unreachable; adding unused normalized rows is permitted.

All persistent randomness, input history and protocol distinctions must be in the current label. Only the external epoch, the prescribed horizon and the read-only row tables are free. Table construction, lookup, arithmetic and exact sampling of known real stochastic rows are uncharged. Matrices, rows and state sets may depend on N ; this is nonuniform positive-realization width, not uniform computational space. A common-row construction uses the same command matrices at all command epochs but may still have an N -dependent decoder. It need not answer at earlier stopping lengths.

Let $W_{N,\varepsilon}$ be the least peak among machines whose every binary row differs from (1.1) by at most ε in total variation. For fixed finite $K_{\bullet}$ the optimal actual error $\mathcal{E}_N(K_{\bullet})$ is attained: the row parameters lie in a finite product of compact sets and the error is a finite maximum of continuous functions. For planar results we use $X = \{\pm e_1, \pm e_2\}$ and include the identity letter. We keep $\rho \leq 1/10$ and $\varepsilon < \rho/(2\sqrt{2})$.

Lemma 2.2 (Calibration under any external test law). *Fix a seed $x \in X$ and any law on the complete command word, independent of private initialization. Let $Y_t = U_{a_t} \cdots U_{a_1} x$, $Z_t = \mathbb{E}[Y_t | S_t]$ and $q_t = \mathbb{E}\|Z_t\|$. Every machine with row error at most ε satisfies*

$$(2.1) \quad q_N \geq \kappa := \rho/\sqrt{D} - 2\varepsilon.$$

At a boundary preceding an independent complete packet, the next boundary admits the conditional-mean identity used in Lemma 3.2. No single-step independence inside a packet is asserted.

Proof. For a fixed word, every coordinate of $\mathbb{E}[d(S_N) | w] - \rho Y_N$ has modulus at most 2ε . Since $\|Y_N\| = 1$, averaging gives

$$\mathbb{E}\langle Y_N, d(S_N) \rangle \geq \rho - 2\varepsilon\sqrt{D}.$$

On the other hand this pairing equals $\mathbb{E}\langle Z_N, d(S_N) \rangle$ and is at most $\sqrt{D}q_N$. This proves (2.1). Combining terminal columns is a proof device, not a requirement to provide their joint output distribution. $\square$

3. FINITE WORD PROFILES AND COMPATIBLE ENCLOSURES

The results in this section apply to every finite orthogonal alphabet, including noncommuting alphabets. No connectedness, arithmetic condition or spectral gap is assumed. Invariant directions may make a converse profile zero; this is a property of the stated profile, not an omitted hypothesis.

3.1. An optimized distortion. For $B, k \geq 1$ and a probability vector p on the actual words $\mathcal{A}^B$, define

$$(3.1) \quad D_k(p) = 1 - \max_{\substack{u \in \mathbb{S}^{D-1} \\ v_1, \dots, v_k \in \mathbb{S}^{D-1}}} \sum_{w \in \mathcal{A}^B} p_w \max_{1 \leq j \leq k} \langle v_j, U_w u \rangle,$$

$$(3.2) \quad \Gamma_{\mathcal{A}}(k, B) = \max_{p \in \Delta(\mathcal{A}^B)} D_k(p), \quad \chi_{\mathcal{A}}(k, B) = -\log(1 - \Gamma_{\mathcal{A}}(k, B)).$$

We allow $\chi = +\infty$. Repeated product matrices can be merged for the optimization: their probability masses add. Their representative words must still be executed one ordinary command at a time.

Proposition 3.1 (Finite-data properties). *The maxima in (3.1)–(3.2) are attained, $0 \leq \Gamma_{\mathcal{A}}(k, B) \leq 1$, and the profile is invariant under simultaneous orthogonal changes of coordinates. It is nonincreasing in k and unchanged by duplicating a letter. For two indexed orthogonal alphabets U, V ,*

$$(3.3) \quad |\Gamma_U(k, B) - \Gamma_V(k, B)| \leq B \max_a \|U_a - V_a\|_{\text{op}}.$$

For a fully listed finite alphabet of real-algebraic matrices and fixed integers k, B, D , $\Gamma_{\mathcal{A}}(k, B)$ and an optimizing word law are real-algebraic and can be determined by real quantifier elimination.

Proof. All parameter spaces are compact and the objective is continuous. Its maximum over directions is convex in p , so D_k is continuous and concave. The maximum scalar product is at most one. By choosing a single direction parallel to $\sum p_w U_w u$, the inner maximum is nonnegative; hence $D_k \in [0, 1]$. Conjugating all directions proves invariance. Additional centers only increase that inner maximum. Product telescoping gives $\|U_w - V_w\|_{\text{op}} \leq B \max_a \|U_a - V_a\|_{\text{op}}$, and maxima and minima preserve this uniform bound.

For the last assertion, enumerate the product matrices. Unit-sphere equations, simplex inequalities and finitely many maximum operations define a compact semialgebraic optimization with algebraic coefficients. Maximum operations can be expressed by finite alternatives or auxiliary variables. Quantifier elimination over the real closed field of real-algebraic numbers determines the endpoint of the feasible value interval and a witness [1]. The word table may have $|\mathcal{A}|^B$ entries; this is not a polynomial-time assertion or a decision procedure for an infinite spectral gap. No logarithm needs to be eliminated: finite budget comparisons can instead use products of $1 - \Gamma$. $\square$

Lemma 3.2 (Endpoint contraction; inherited mechanism). *Let a complete random word W with law p be independent of all information before its start. Internal letters of W may be correlated. If the endpoint register has at most k labels, then*

$$(3.4) \quad q' \leq (1 - D_k(p))q.$$

There is no bound on the intermediate registers.

Proof. Write $z_s = \mathbb{E}[Y \mid S = s]$ at the start. Composing all internal stochastic rows yields a common endpoint kernel $T(j \mid s, w)$. Since the entire word and the machine's fresh randomness are independent of the past,

$$z'_j = \mathbb{E}[U_W z_S \mid S' = j].$$

This is an endpoint identity; it would not generally hold after conditioning on a correlated next letter inside the packet. Choose $v_j = z'_j / \|z'_j\|$ for nonzero z'_j , and any unit vector otherwise. Then

$$\begin{aligned} q' &= \sum_s \mathbb{P}(S = s) \sum_w p_w \sum_j T(j \mid s, w) \langle v_j, U_w z_s \rangle \\ &\leq \sum_s \mathbb{P}(S = s) \|z_s\| \max_{\|u\|=1} \sum_w p_w \max_j \langle v_j, U_w u \rangle \leq (1 - D_k(p))q. \end{aligned}$$

Terms with $z_s = 0$ contribute zero. Replacing an executable assignment by the largest pairing is a relaxation in the correct direction. Arbitrary unobservable hidden coordinates are included in s ; no statewise observable-vector hypothesis is used. This is the conditional-centroid mechanism of [7, 8], now applied to the optimized finite word datum. $\square$

For an available profile $K_\bullet$, let $I_0 = 0$ and define

$$(3.5) \quad I_t = \max \left\{ I_{t-1}, \max_{1 \leq B \leq t} [I_{t-B} + \chi_{\mathcal{A}}(K_t, B)] \right\}, \quad \mathcal{I}_N(K_\bullet) = I_N.$$

This is the largest sum of charges of disjoint command intervals, with an interval of length B ending at t charged by $\chi_{\mathcal{A}}(K_t, B)$. The recurrence depends only on the alphabet and the available profile.

Theorem 3.3 (A nonuniform interval budget). *Every machine with $\kappa = \rho/\sqrt{D} - 2\varepsilon > 0$ satisfies*

$$(3.6) \quad \mathcal{I}_N(K_\bullet) \leq \log(1/\kappa).$$

More generally its optimal actual terminal error obeys

$$(3.7) \quad \mathcal{E}_N(K_\bullet) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{D}} - e^{-\mathcal{I}_N(K_\bullet)} \right]_+.$$

For a uniform peak bound K , every $B \leq N$ gives

$$(3.8) \quad \lfloor N/B \rfloor \chi_{\mathcal{A}}(K, B) \leq \log(1/\kappa).$$

For a fixed B, k with $\Gamma_{\mathcal{A}}(k, B) > 0$,

$$(3.9) \quad \#\{1 \leq t \leq N : K_t \leq k\} \leq B - 1 + \frac{B \log(1/\kappa)}{\chi_{\mathcal{A}}(k, B)}.$$

If $\chi = +\infty$, the last fraction is zero.

Proof. Choose any maximizing disjoint interval family in (3.5). On each selected interval put an independently sampled optimizing complete word from Proposition 3.1. All gaps and the final remainder contain fixed deterministic command words. A deterministic orthogonal command followed by a stochastic update cannot increase the conditional-mean norm, by conditional Jensen. Lemma 3.2 therefore gives

$$q_N \leq \prod_{(t-B, t]} (1 - \Gamma_{\mathcal{A}}(K_t, B)) = e^{-\mathcal{I}_N(K_\bullet)},$$

since $q_0 = 1$. The chosen law can depend on the available layer sizes, not on private states. Wordwise correctness requires the same machine to pass it. Lemma 2.2 proves (3.6) and (3.7); taking consecutive intervals proves (3.8).

For (3.9), discard candidate endpoints before B . Select the earliest remaining endpoint, delete all later candidates less than B away, and repeat. If M is the original number and J the number selected, the explicit count is

$$M - (B - 1) \leq BJ.$$

The associated length- B intervals are disjoint and each contributes at least $\chi_{\mathcal{A}}(k, B)$. Applying (3.6) gives the result. No restriction on the other registers was introduced. $\square$

The cap union estimate below is the geometric certificate already used in the predecessor packet method [7]. The finite incidence formulation specifies how to optimize that certificate on a fully listed word set.

Proposition 3.4 (Uniform cap certificates). *Suppose a law p on length- B words satisfies*

$$\sup_{u,v \in \mathbb{S}^{D-1}} p\{w : \|U_w u - v\| < \delta\} \leq b.$$

Then $\Gamma_{\mathcal{A}}(k, B) \geq \frac{1}{2}\delta^2(1 - kb)_+$. For algebraic product matrices and algebraic δ , the best such b is the value of a finite linear program once cap incidence types have been decided by real quantifier elimination.

Proof. The union of k open caps has probability at most kb . Outside it, $1 - \max_j \langle v_j, U_w u \rangle = \frac{1}{2} \min_j \|U_w u - v_j\|^2 \geq \delta^2/2$. For the algorithmic assertion, a subset of the finite word list either can or cannot be contained in one such cap for some u, v . This is a finite system of algebraic inequalities, including the sphere equations. For every feasible subset A impose $\sum_{w \in A} p_w \leq b$, along with the simplex constraints, and minimize b . Every actual incidence set is feasible; conversely a feasible subset is included in an actual incidence set and has no greater mass. Thus these constraints give exactly the desired supremum. The exponentially large enumeration is not concealed. $\square$

3.2. A separate exact-synthesis profile. Fix $0 < a < 1$. For k such that the following family is nonempty, let

$$(3.10) \quad \lambda_{\mathcal{A}}(k, a) = \min_P \inf \{\lambda \geq 1 : U_b P \subseteq \lambda P \text{ for all } b \in \mathcal{A}\},$$

where P ranges over convex hulls of at most k points in $\mathbb{B}^D$ containing $a\mathbb{B}^D$. An empty family has value $+\infty$.

Theorem 3.5 (Compatible enclosure synthesis). *The finite value in (3.10) is attained and lies in $[1, a^{-1}]$. If $\rho\lambda_{\mathcal{A}}(k, a)^N \leq a$, there is an exact machine with $\max(k, 2)$ labels, common command matrices and a horizon-dependent decoder. Each row has at most $D + 1$ successors. Algebraic matrices and algebraic a give an algebraically decidable profile value and algebraic row witnesses. For two indexed alphabets with a common feasible family,*

$$(3.11) \quad |\lambda_U(k, a) - \lambda_V(k, a)| \leq a^{-1} \max_b \|U_b - V_b\|_{\text{op}}.$$

Proof. Parameterize P by k points in the closed unit ball, allowing repetition. Containment of $a\mathbb{B}^D$ is closed in these coordinates. Every such P satisfies $U_b P \subseteq \mathbb{B}^D \subseteq a^{-1}P$. The containment conditions are closed on this compact set and $1 \leq \lambda \leq a^{-1}$; hence a minimum exists.

Let its vertices be v_s and λ its dilation. Initialize with a convex representation of ax . For each b, s , represent $\lambda^{-1}U_b v_s$ by the same vertices and use its convex coefficients as the row $T_b(s, \cdot)$. Induction gives

$$\mathbb{E}[v_{S_t} \mid w] = a\lambda^{-t}U_w x.$$

The decoder $d_j(s) = \rho\lambda^N \langle e_j, v_s \rangle / a$ lies in $[-1, 1]$ and gives (1.1). These are actual rows throughout the run, not independently chosen checkpoint factorizations. Carathéodory gives $D + 1$ successors; there are $|\mathcal{A}|k$ command rows, and their construction and storage remain uncharged.

The defining finite problem is semialgebraic: use convex coefficients for transformed vertices and a universal ball containment condition, then eliminate quantifiers. To prove (3.11), set $e = \max_b \|U_b - V_b\|_{\text{op}}$. For every feasible P , $V_b P \subseteq U_b P + e\mathbb{B}^D \subseteq (\lambda + e/a)P$. Interchanging U, V completes the proof. $\square$

The two profiles have different quantifiers. The converse tests every hidden machine; the enclosure theorem builds a particular positive realization. Neither theorem asserts that $\lambda_{\mathcal{A}}$ is the

optimum over hidden machines. This is why finite-scale lower and upper bounds can be used even when they do not match.

Proposition 3.6 (Bounded word recoding). *Suppose every command of $\mathcal{A}$ is a product of exactly L commands of an alphabet $\mathcal{B}$ on the same space, with the same seeds and queries. Then*

$$\Gamma_{\mathcal{A}}(k, B) \leq \Gamma_{\mathcal{B}}(k, LB), \quad \lambda_{\mathcal{A}}(k, a) \leq \lambda_{\mathcal{B}}(k, a)^L, \quad W_{N,\varepsilon}(\mathcal{A}) \leq W_{LN,\varepsilon}(\mathcal{B}).$$

An identity letter permits words of length at most L to be padded. If both alphabets contain an identity and admit such bounded recodings into the other, the existence and value of a logarithmic width exponent agree.

Proof. Push forward a packet law along the word substitution to obtain the first inequality. Iterate the enclosure containments for the second. For the last inequality, compose the L stochastic rows of a $\mathcal{B}$ machine into one row for the corresponding $\mathcal{A}$ command, retaining only the block-end label. All rows may depend on the external epoch. This is an implementation of the atomic $\mathcal{A}$ interface, not a claim that its physical letters cost L letters. Applying the two comparisons and monotonicity under final identity padding proves the exponent statement. $\square$

Corollary 3.7 (Packing–dilation balance). *Fix a calibrated experiment, an inradius $a > \rho$, and positive numbers u, v . Suppose that for every sufficiently large k there are a length $B(k) \leq Ck^u$ and a word law for which*

$$\sup_{x,z \in \mathbb{S}^{D-1}} p\{w : \|U_w x - z\| < ck^{-v}\} \leq (2k)^{-1}.$$

Suppose also that $\lambda_{\mathcal{A}}(k, a) \leq \exp(C'k^{-u-2v})$. Then

$$W_{N,\varepsilon} = \Theta(N^{1/(u+2v)}).$$

The same conclusion holds with $u = 0$ when the lengths are uniformly bounded.

Proof. Proposition 3.4 gives $\Gamma(k, B(k)) \geq c^2 k^{-2v}/4$. Equation (3.8) implies $N \leq C_1 k^{u+2v}$ for a feasible peak k . Conversely choose k at a sufficiently large constant times $N^{1/(u+2v)}$ so that $C'Nk^{-u-2v} \leq \log(a/\rho)$, and apply Theorem 3.5. The hypotheses concern finite word orbits and compatible convex containments, and impose no commutativity on the matrices. $\square$

4. EXACT OPTIMIZATION OF PLANAR PACKETS

For a rotation alphabet, let $z_1, \dots, z_M \in \mathbb{S}^1 \subset \mathbb{C}$ be the distinct product phases of words of length B , ordered cyclically. A common rotation of all phases can be absorbed into the decoder centers. The initial direction in (3.1) therefore disappears. Any probability distribution on these phases is implemented by a distribution on representative words of the same length.

Let $\mathfrak{P}_k$ be the finite set of partitions of the cyclically ordered list into at most k nonempty consecutive arcs. For $p \in \Delta_M$, set

$$(4.1) \quad F_k(p) = \max_{\mathcal{P} \in \mathfrak{P}_k} \sum_{A \in \mathcal{P}} \left| \sum_{i \in A} p_i z_i \right|.$$

Theorem 4.1 (Optimal word laws as a cone program). *For the planar alphabet above,*

$$(4.2) \quad \Gamma_{\mathcal{A}}(k, B) = 1 - \min_{p \in \Delta_M} F_k(p).$$

Equivalently, minimize t subject to $p \in \Delta_M$ and, for every $\mathcal{P} \in \mathfrak{P}_k$,

$$(4.3) \quad \left\| \sum_{i \in A} p_i (\Re z_i, \Im z_i) \right\|_2 \leq r_{\mathcal{P},A} \quad (A \in \mathcal{P}), \quad \sum_{A \in \mathcal{P}} r_{\mathcal{P},A} \leq t.$$

This is a finite second-order cone program with attained optimum. A fixed law can be evaluated by cyclic segmentation in $O(kM^3)$ arithmetic operations, including computation of all interval sums. The optimization over laws may have exponentially many partition constraints.

Proof. For fixed centers $v_1, \dots, v_k$ assign each phase to a nearest center in chordal distance; equivalently maximize $\Re(\bar{v}_j z_i)$. Distinct centers have connected cyclic Voronoi arcs, bounded by the angular midpoints of adjacent centers. Ties can be assigned consecutively; coincident centers can be merged. For the induced partition, the contribution of a block A is at most $|\sum_{i \in A} p_i z_i|$.

Conversely, for any cyclic partition choose the center of each block in the direction of its weighted complex sum, arbitrarily if that sum vanishes. The best assignment to those centers has value at least the sum in (4.1). Thus the maximum over centers equals $F_k(p)$. This proves (4.2); introducing the norm epigraphs proves (4.3). Zero weights follow either directly or by continuity from positive weights.

For a fixed cyclic cut, prefix sums give every block vector in constant time. The usual recurrence for the best j blocks among the first m points checks at most M previous split positions. It costs $O(kM^2)$ for that cut. Repeating for the M cuts costs $O(kM^3)$. This recurrence evaluates a given law; it is not a polynomial-size representation of all constraints in (4.3). Compactness of the simplex and continuity give attainment. $\square$

Nearest-center partitions and vector quantization are classical [5]. The additional optimization here chooses the distribution of actual command words, and Theorem 3.3 turns its value into a constraint on arbitrary hidden registers. There is no assertion that static quantization alone characterizes the least whole-run width.

Proposition 4.2 (A finite exact packet certificate). *For the alphabet consisting of identity and a quarter turn, at word length three the four distinct phases are $1, i, -1, -i$. Their uniform law is optimal and*

$$\Gamma(1, 3) = 1, \quad \Gamma(2, 3) = 1 - \frac{1}{\sqrt{2}}, \quad \Gamma(3, 3) = \frac{2 - \sqrt{2}}{4}, \quad \Gamma(k, 3) = 0 \quad (k \geq 4).$$

Proof. The distortion is concave in the phase law and invariant under its four cyclic rotations. Averaging any law over those rotations proves optimality of the uniform law. For two centers, adjacent pairs give total $\sqrt{2}/2$; a triple and a singleton give only $1/2$. For three centers, one pair and two singletons give $\sqrt{2}/4 + 1/2$. For one center the sum is zero; four centers fit all phases. Formula (4.2) gives the values. A uniform phase law is realized by the four ordinary words with zero, one, two or three initial quarter turns and then idle commands. $\square$

For algebraic rotation matrices, both the general quantifier-elimination formulation and the cone program have algebraic coefficients. No finite list of spherical harmonics is used. For merely computable-real entries, finite numerical approximations are diagnostics unless a rigorous error enclosure is supplied; (3.3) then quantifies how an entry enclosure propagates to a packet certificate.

Corollary 4.3 (Separated phases). *If $M \geq 2k$ distinct length- B word phases have pairwise circular distance at least $s \in (0, 1/2]$, then $\Gamma_A(k, B) \geq s^2$.*

Proof. Use their uniform law. Each open arc of circular radius $s/2$ contains at most one phase. At least half the mass is at distance at least $s/2$ from every one of k centers. On $[0, 1/2]$, $1 - \cos(2\pi t) = 2\sin^2(\pi t) \geq 8t^2$. The average loss is at least s^2 . This is a lower certificate for (4.2), not necessarily its exact value. $\square$

5. ARITHMETIC PROFILES AT EVERY FINITE SCALE

Now $\mathcal{A}_\alpha = \{I, R_{2\pi\alpha_1}, \dots, R_{2\pi\alpha_r}\}$, with r fixed. Define

$$(5.1) \quad \psi_\alpha(H) = \min_{0 < \|m\|_1 \leq H} \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}, \quad \eta_\alpha(q) = \max_{1 \leq i \leq r} \|q\alpha_i\|_{\mathbb{R}/\mathbb{Z}}.$$

These are finite functions of the alphabet parameters. The first may vanish in the presence of a rational relation. The upper profile remains usable in that case.

Theorem 5.1 (Finite arithmetic bounds). *Let $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$. For a machine of peak K , put $L = \lceil (2K)^{1/r} \rceil$ and $B = r(L - 1)$. Then*

$$(5.2) \quad \lfloor N/B \rfloor \psi_\alpha(B)^2 \leq \log(1/\kappa).$$

For every integer $q \geq 4$,

$$(5.3) \quad N\eta_\alpha(q) \leq q^2/100 \implies W_{N,0}(\alpha) \leq q.$$

Consequently every finite planar rotation alphabet satisfies

$$(5.4) \quad W_{N,0}(\alpha) \leq \max(4, \lceil \sqrt{50N} \rceil).$$

All upper bounds use common command rows and hold also after adjoining reflection in the first coordinate axis.

Proof. Construct a word of length B in r consecutive segments of $L - 1$ positions. In segment i give J_i copies of the i th rotation followed by idle commands, independently and uniformly for $J_i \in \{0, \dots, L - 1\}$. Differences of count vectors have ℓ_1 norm at most B . If $\psi_\alpha(B) > 0$, the $L^r \geq 2K$ phases are distinct and separated by $\psi_\alpha(B)$. Corollary 4.3 and (3.8), with $-\log(1-x) \geq x$, give (5.2). If the minimum is zero the displayed inequality is automatic. All letters in a segment are actual commands, not a free count input.

For the upper bound let P_q be the regular q -gon inscribed in the unit circle. Choose nearest integers p_i to $q\alpha_i$, and put $\delta_i = 2\pi(\alpha_i - p_i/q)$, so $|\delta_i| \leq \pi/q$. Elementary chord geometry gives

$$(5.5) \quad R_{2\pi\alpha_i}P_q \subseteq \Lambda_i P_q, \quad \Lambda_i = \frac{\cos(\pi/q - |\delta_i|)}{\cos(\pi/q)}, \quad \log \Lambda_i \leq \frac{4\pi^2\eta_\alpha(q)}{q^2}.$$

Indeed the ray of a rotated vertex meets a neighboring edge at radial coordinate $\cos(\pi/q)/\cos(\pi/q - |\delta_i|)$. Integrating the derivative of $\log \cos(\pi/q - t)$ bounds the logarithm by $|\delta_i| \tan(\pi/q)$, and $\tan(\pi/q) \leq 2\pi/q$ for $q \geq 4$.

Take $\Lambda = \max_i \Lambda_i$, including one for the identity. Under (5.3), $\Lambda^N \leq e^{\pi^2/25} < 2$. Since P_q contains the half-unit disk, Theorem 3.5 with $a = 1/2$ applies: $\rho\Lambda^N \leq 1/5 < 1/2$. Explicitly, initialize at mean $x/2$, represent $\Lambda^{-1}R_{2\pi\alpha_i}v_s$ by the vertices for each row, and decode by $2\rho\Lambda^N \langle e_j, v_s \rangle$. This verifies every numerical word, including any deterministic remainder of a converse packet decomposition.

There are $(r + 1)q$ common command rows and two decoder columns, each row having at most three successors. These counts exclude row-description and arithmetic costs. Since $\eta_\alpha(q) \leq 1/2$, (5.4) follows. Reflection preserves P_q ; its contracted image uses the same vertex set. The reflected lower bound is obtained simply by restricting to rotation-only words. $\square$

The pair $(\psi_\alpha, \eta_\alpha)$ is a specialization of the word-distortion and enclosure profiles. It is not assumed to characterize exact finite width. In particular, (5.2) can be weak when long word packets have clusters; the optimized profile of Section 4 is available without that separation relaxation.

Lemma 5.2 (Denominators under a general power bound). *Suppose for $\tau \geq r$ and $b > 0$,*

$$(5.6) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b \|m\|_1^{-\tau} \quad (m \neq 0).$$

For every integer $Q \geq 2$ there is q with

$$(5.7) \quad cQ^{r/(\tau+1-r)} \leq q \leq Q^r, \quad \eta_\alpha(q) \leq Q^{-1}, \quad c = (b/r^{\tau+1})^{r/(\tau+1-r)}.$$

Proof. The $Q^r + 1$ points $j\alpha$ in Q^r equal torus boxes give integers p_i, q with $1 \leq q \leq Q^r$ and $|q\alpha_i - p_i| \leq Q^{-1}$. For $M = \lfloor q^{1/r} \rfloor$, the $(M+1)^r > q$ integer vectors in $\{0, \dots, M\}^r$ have a residue collision for $m \cdot p \pmod{q}$. Subtracting gives $m \neq 0$ with $H = \|m\|_1 \leq rq^{1/r}$ and $q \mid m \cdot p$. Thus

$$bH^{-\tau} \leq \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq H/(qQ), \quad bqQ \leq r^{\tau+1}q^{(\tau+1)/r}.$$

Rearranging yields (5.7). This direct residue argument supplies exactly the quantitative transference needed here, without an unspecified transference constant. $\square$

Corollary 5.3 (Power bounds at finite Diophantine type). *Under (5.6), for constants depending on the fixed data,*

$$(5.8) \quad c_1 N^{r/(2\tau+1)} \leq W_{N,\varepsilon}(\alpha) \leq C_1 N^{\min\{1/2, r(\tau+1-r)/(\tau+1+r)\}}.$$

No matching claim is made when $\tau > r$.

Proof. From (5.2), $N < B + b^{-2}B^{2\tau+1} \log(1/\kappa)$, while $B \leq r(2K)^{1/r}$. This gives the lower power after adjusting finitely many initial horizons. Lemma 5.2 bounds $N\eta_\alpha(q)/q^2 \leq c^{-2}NQ^{-1-2r/(\tau+1-r)}$. Choose Q at constant times $N^{(\tau+1-r)/(\tau+1+r)}$, large enough to satisfy (5.3) and $q \geq 4$. Then $q \leq Q^r$ proves the second candidate upper power; (5.4) proves the first. $\square$

Corollary 5.4 (The retained badly approximable law). *If (5.6) holds with $\tau = r$, then $W_{N,\varepsilon}(\alpha) = \Theta(N^{r/(2r+1)})$, also with the reflection extension. This is the previously established matched law [8].*

Proof. Substitute $\tau = r$ in (5.8); the two powers coincide. $\square$

For $1, \alpha_1, \dots, \alpha_r$ rationally independent, every optimized finite-block action gap on the circle is zero. Indeed simultaneous approximation supplies unbounded frequencies n with every $n\alpha_i$ close to an integer. Each fixed rotation word then acts almost identically on its Fourier mode. For the reflected alphabet use the real cosine mode, which reflection fixes. This is the established almost-invariance argument, not a claim that the finite packet distortion vanishes.

6. MINIMAL ORDINARY TYPE AND ALMOST-EVERYWHERE WIDTHS

Uniform bad approximation is substantially stronger than the condition needed to determine the logarithmic width exponent. We first deduce the minimal-type assertion without any measure-theoretic hypothesis.

Proof of (1.3) in Theorem 1.1. If $\omega^*(\alpha) = r$, then for every $h > 0$ there is $b_h > 0$ such that

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b_h \|m\|_1^{-r-h} \quad (m \neq 0).$$

Indeed the definition of the limsup gives this inequality for all sufficiently large $\|m\|_1$, and rational independence makes the finite remaining minimum positive. Apply Corollary 5.3 with $\tau = r + h$. It gives

$$\frac{r}{2r+2h+1} \leq \liminf_N \frac{\log W_{N,\varepsilon}}{\log N} \leq \limsup_N \frac{\log W_{N,\varepsilon}}{\log N} \leq \frac{r(1+h)}{2r+1+h}.$$

Let $h \downarrow 0$. This proves (1.3). No positive uniform constant at exponent r was used. $\square$

For completeness we prove a more quantitative metric statement, using only the summable part of metric Diophantine approximation. It does not claim a new metric theorem.

Lemma 6.1 (A summable logarithmic separation). *For Lebesgue-almost every $\alpha \in [0, 1]^r$, and every $s > 1$, there is $b_s > 0$ such that*

$$(6.1) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{b_s}{\|m\|_1^r (\log(2 + \|m\|_1))^s} \quad (m \neq 0).$$

Such a vector has $\omega^(\alpha) = r$.*

Proof. For a nonzero integer vector m , the map $\alpha \mapsto m \cdot \alpha \pmod{1}$ sends Haar measure on the torus to uniform circle measure. There are at most $C_r H^{r-1}$ integer vectors with $\|m\|_1 = H$. Hence the measures of the events

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} < [\|m\|_1^r (\log(2 + \|m\|_1))^s]^{-1}$$

have finite sum, bounded by a constant times $\sum_{H \geq 1} [H(\log(2 + H))^s]^{-1}$. The measure of their tail union tends to zero. Almost every vector violates the unscaled inequality only finitely often. Remove the measure-zero rational hyperplanes and reduce the constant to cover the finite exceptions. Intersecting the full-measure sets for rational $s > 1$ gives the assertion for every real $s > 1$.

This bound implies $\omega^*(\alpha) \leq r$. For the reverse inequality, place the $(M + 1)^r$ values $m \cdot \alpha \pmod{1}$, $m \in \{0, \dots, M\}^r$, in fewer than $(M + 1)^r$ intervals of equal length. A collision gives a nonzero difference of norm at most rM and distance $O(M^{-r})$. As $M \rightarrow \infty$ the norms of some such differences must tend to infinity, since rational independence excludes an exact relation. Thus $\omega^*(\alpha) \geq r$. $\square$

Proposition 6.2 (Quantitative bounds under logarithmic separation). *Suppose (6.1) holds for some $s > 0$ and $b_s > 0$. Then there are $c, C > 0$ such that*

$$(6.2) \quad cN^{r/(2r+1)}(\log(N+2))^{-2rs/(2r+1)} \leq W_{N,\varepsilon}(\alpha) \leq CN^{r/(2r+1)}(\log(N+2))^{2r^2s/(2r+1)}.$$

Proof. The packet bound (5.2) gives

$$N < B + b_s^{-2} B^{2r+1} (\log(2 + B))^{2s} \log(1/\kappa).$$

With $B \leq r(2K)^{1/r}$ this implies $N \leq C_0 K^{(2r+1)/r} (\log(2 + K))^{2s}$. If $K > N$, the proposed lower bound is already true after reducing its constant. Otherwise $\log(2 + K) \leq \log(2 + N)$, and rearrangement proves the first inequality of (6.2).

For the upper inequality repeat the residue argument of Lemma 5.2. It produces $q \leq Q^r$, $\eta_\alpha(q) \leq Q^{-1}$ and $H \leq rq^{1/r}$ with

$$\frac{b_s}{H^r (\log(2 + H))^s} \leq \frac{H}{qQ}.$$

Since $q \leq Q^r$, it follows for $Q \geq 2$ that

$$(6.3) \quad q \geq c_0 \frac{Q^r}{(\log(Q+2))^{rs}}.$$

Indeed $b_s q Q \leq H^{r+1} (\log(2 + H))^s \leq C_r q^{1+1/r} (\log(Q+2))^s$, and raising the rearranged inequality to the r th power gives (6.3).

Choose an integer

$$Q = \left\lceil C_2 N^{1/(2r+1)} (\log(N+2))^{2rs/(2r+1)} \right\rceil.$$

For sufficiently large fixed C_2 , (6.3) implies $N\eta_\alpha(q)/q^2 \leq 1/100$ and $q \geq 4$. To see the constant choice explicitly, for large N one has $\log(Q+2) \leq C_3 \log(N+2)$ with C_3 fixed; thus the ratio is bounded by a constant times $C_2^{-(2r+1)}$. Enlarge the final constant for the finitely many initial horizons. Theorem 5.1 gives an exact machine with $q \leq Q^r$ labels, proving the upper bound. $\square$

Completion of Theorem 1.1. Apply Lemma 6.1 with $s = 1 + \delta$ and then Proposition 6.2. This yields (1.4) on one full-measure set for all fixed $\delta > 0$. These statements concern a fixed alphabet before the horizon grows. None of their constants is asserted uniform in r or near a rational relation. $\square$

6.1. Rational dependencies without redundant exponents. The number of listed rotations need not be the relevant arithmetic rank. The following result covers finite rational reparametrizations rather than assigning one exponent to every redundant letter.

Corollary 6.3 (Finite rational extensions). *Suppose an alphabet contains I and rotations through angles $\beta_1, \dots, \beta_d$, where $1, \beta_1, \dots, \beta_d$ are rationally independent and $\omega^*(\beta) = d$. Suppose every additional angle belongs to $\mathbb{Q} + \sum_{j=1}^d \mathbb{Q}\beta_j$. Then*

$$\lim_N \frac{\log W_{N,\varepsilon}}{\log N} = \frac{d}{2d+1}.$$

The same assertion holds on a full-measure set of β with the logarithmic upper and lower powers of (1.4), replacing r by d . Reflection may be added. If all angles are rational, exact width is bounded independently of N .

Proof. Restriction to the displayed d -angle subalphabet gives the lower bound. Choose one common denominator h and integers t_i, n_{ij} so that every angle is $t_i/h + \sum_j n_{ij}\gamma_j$ with $\gamma_j = \beta_j/h$. Multiplication by h shows $\|m \cdot \gamma\|_{\mathbb{R}/\mathbb{Z}} \geq h^{-1} \|m \cdot \beta\|_{\mathbb{R}/\mathbb{Z}}$. Thus every separating power or logarithmic inequality for β holds for γ with a changed constant.

Take a simultaneous approximant p_j/q to γ from the proof above. A command's irrational part has residual angle at most $2\pi(\sum_j |n_{ij}|)/(qQ)$. The same polygon construction therefore works, after changing only fixed constants. Store in addition the rational phase $c \in \mathbb{Z}/h\mathbb{Z}$, updated by $c \mapsto c + t_i$. The decoder rotates the stored polygon vector by $2\pi c/h$. This uses hq labels, not an uncharged phase. Reflection sends c to $-c$ and conjugates the polygon phase; the regular polygon and the construction are reflection-invariant. The exact expectation identity is unchanged. The finite multiplier h does not change a logarithmic exponent or the displayed logarithmic powers.

For a purely rational alphabet, choose a common period h and retain the seed identifier and the accumulated phase. This uses at most $4h$ command labels for our four planar seeds, plus the held binary answer convention. No infinite-gap hypothesis is involved. $\square$

7. EVERY IRRATIONAL ANGLE HAS CUBIC SCALES

We keep the two-letter alphabet $\{I, R_{2\pi\alpha}\}$ and the planar resource and error conventions. No Diophantine lower bound is assumed in this section. The needed continued-fraction estimates are classical [3]; we include their short proof to specify the exact finite inequalities being used.

Lemma 7.1 (Convergent separation). *Let p_n/q_n be the convergents of an irrational α , ignoring the finitely many initial indices with repeated denominator. Then*

$$(7.1) \quad \frac{1}{q_{n+1} + q_n} < |q_n\alpha - p_n| < \frac{1}{q_{n+1}}, \quad \min_{1 \leq l < q_n} \|l\alpha\|_{\mathbb{R}/\mathbb{Z}} > \frac{1}{q_n + q_{n-1}} > \frac{1}{2q_n}.$$

Proof. The convergent recurrences imply $p_n q_{n-1} - p_{n-1} q_n = \pm 1$ and alternating signs of $e_n = q_n \alpha - p_n$. If ξ is the next complete quotient, then

$$\alpha = \frac{p_n \xi + p_{n-1}}{q_n \xi + q_{n-1}}, \quad a_{n+1} < \xi < a_{n+1} + 1.$$

It follows that $|e_n| = 1/(q_n \xi + q_{n-1})$, proving the first pair.

For $0 < l < q_n$ and an integer p , the determinant identity writes $(p, l) = a(p_n, q_n) + b(p_{n-1}, q_{n-1})$ with integers a, b . Here $b \neq 0$; a, b cannot both be positive, and cannot both be negative. If $a = 0$

the error has modulus at least $|e_{n-1}|$. Otherwise they have opposite signs, so ae_n and be_{n-1} have the same sign and again $|l\alpha - p| \geq |e_{n-1}|$. Minimize over p and apply the first pair with index $n - 1$. This proves the remaining inequalities. $\square$

Theorem 7.2 (Universal cubic subsequences). *Set $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$, $C_\kappa = 8(1 + \log(1/\kappa))$, and*

$$M_n = \lceil C_\kappa q_n^3 \rceil, \quad m_\kappa = \lceil 100(C_\kappa + 1) \rceil.$$

For all sufficiently large n ,

$$(7.2) \quad q_n/2 < W_{M_n, \varepsilon}(\alpha) \leq m_\kappa q_n.$$

The upper bound is exact, uses common command rows and requires no regularity of the partial quotients.

Proof. Use words of length $B = q_n - 1$ whose number of initial active letters ranges uniformly over $0, \dots, q_n - 1$. Lemma 7.1 gives q_n distinct phases with separation greater than $1/(2q_n)$. If the proposed peak K were at most $q_n/2$, Corollary 4.3 and the packet budget would give

$$\left\lfloor \frac{M_n}{q_n - 1} \right\rfloor \frac{1}{4q_n^2} \leq \log(1/\kappa).$$

The left side is at least $C_\kappa/4 - 1/(4q_n^2)$, strictly greater than the right side. This proves the lower bound against all hidden-state machines.

For the upper bound use the polygon denominator $Q_n = m_\kappa q_n$. Distance to the nearest integer is subadditive, so

$$\|Q_n \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq m_\kappa |q_n \alpha - p_n| < m_\kappa / q_{n+1} \leq m_\kappa / q_n.$$

Consequently $Q_n^2 / [100 \|Q_n \alpha\|_{\mathbb{R}/\mathbb{Z}}] > m_\kappa q_n^3 / 100 \geq (C_\kappa + 1) q_n^3 \geq M_n$. Equation (5.3) gives the exact upper construction. The constants have been chosen for transparency, not optimality. $\square$

Proposition 7.3 (Resonant long horizons). *For any irrational α and integer $q \geq 4$ with $N_q = \lfloor q^2 / (100 \|q\alpha\|_{\mathbb{R}/\mathbb{Z}}) \rfloor \geq 1$, one has $W_{N_q, 0}(\alpha) \leq q$. If α is Liouville, there is a sequence $N \rightarrow \infty$ on which $\log W_{N, \varepsilon}(\alpha) / \log N \rightarrow 0$.*

Proof. The first assertion is the finite enclosure implication (5.3). For a Liouville number choose arbitrarily large A and unbounded denominators q with $0 < \|q\alpha\|_{\mathbb{R}/\mathbb{Z}} < q^{-A}$. The first assertion then supplies horizons $N_q \geq q^{A+2}/200$ for large q , with width at most q . The logarithmic ratio is at most $1/(A+2) + o(1)$. A diagonal sequence with $A \rightarrow \infty$ proves the claim, since widths are at least two. $\square$

Proof of Theorem 1.2. Equation (7.2) and $M_n = \Theta(q_n^3)$ give the first assertion and a subsequence of logarithmic exponent $1/3$. Proposition 7.3 gives the zero lower limit for Liouville angles, and (5.4) gives the upper limit at most $1/2$.

It remains to verify unboundedness, which does not follow from a subsequence alone. Fix any proposed width K . The $2K$ phases of counts $0, \dots, 2K - 1$ are distinct for an irrational angle and have some positive minimum separation s_K . Independent words of length $2K - 1$ imply $\lfloor N/(2K - 1) \rfloor s_K^2 \leq \log(1/\kappa)$ for a width- K machine. This fails for all sufficiently large N . Thus $W_{N, \varepsilon} \rightarrow \infty$. $\square$

Corollary 7.4 (An explicit fluctuating alphabet). *Let $\alpha_L = \sum_{j=1}^{\infty} 10^{-j!}$. For $q_m = 10^{m!}$ and all large m , the horizon*

$$N_m = \lfloor q_m^{m+2} / 200 \rfloor$$

admits an exact machine with at most q_m labels. The same fixed two-letter alphabet also has the cubic subsequences of Theorem 7.2; hence no single logarithmic width exponent exists. Adding reflection preserves these conclusions.

Proof. The first m summands have denominator dividing q_m , and their remaining positive tail is less than $2 \cdot 10^{-(m+1)!}$. Thus $0 < \|q_m \alpha_L\|_{\mathbb{R}/\mathbb{Z}} < 2q_m^{-m}$ for large m , proving the horizon bound by (5.3). The same tail estimates preclude rationality and show the Liouville property. The general cubic theorem applies. Reflection leaves the upper polygons invariant, while the rotation-only test words give the same lower bounds. $\square$

These theorems describe different scales of one and the same experiment. A small label count on a long resonant horizon is not a constant-width simulator for all horizons, and it need not persist across the gaps between successive approximants. The exact finite optimum and the Liouville upper logarithmic limit remain separate quantitative questions. None of the proofs substitutes approximate output for the exact upper realizations.

8. ARITHMETIC, QUANTIZATION AND NUMERICAL SEMANTICS

Kronecker quasi-uniformity. Dick–Goda–Larcher–Pillichshammer–Suzuki [4], Theorem 3.9, characterize quasi-uniformity of the multidimensional sequence $n\alpha \pmod{\mathbb{Z}^r}$ by the simultaneous bound $\eta_\alpha(q) \geq cq^{-1/r}$. Their Lemma 3.11 bounds ratios of successive best simultaneous denominators; Section 3.3 also contains the vector $(2^{1/(r+1)}, \dots, 2^{r/(r+1)})$. These are direct arithmetic antecedents, not new examples of this paper.

Our packet points instead have the form $m \cdot \alpha \pmod{1}$. The relevant separation is dual. The residue argument of Lemma 5.2, with an arbitrary nearest-integer numerator and $\eta_\alpha(q)$ in place of Q^{-1} , proves directly that a dual bound with exponent r implies

$$\eta_\alpha(q) \geq br^{-r-1}q^{-1/r}.$$

Thus the cited quasi-uniformity theorem applies to the uniform badly approximable class used in the predecessor. Its denominator control supplies the arithmetic scales for a polygon upper bound. It supplies neither a compatible stochastic transition table nor a lower bound against discarded hidden history. The logarithmic separation of Lemma 6.1 is weaker than a uniform badly approximable inequality; it is proved by a summable tail estimate, not asserted to imply quasi-uniformity.

What is dynamic in the paired profiles. The geometry of nearest-center distortion and convex hull approximation is classical [5, 11]. A static k -center factorization does not specify how its encoder is calculated without retaining discarded inputs. Conversely, a convex codebook does not force arbitrary hidden states to possess coordinates in that codebook. The two implications in Section 3 avoid both identifications: the converse conditions on the actual state, and the upper theorem provides the same legal rows throughout the run. The packing–dilation balance in Corollary 3.7 is a conditional theorem on independently specified finite profiles, not a claimed equality of static and dynamic optima.

The circle program makes one of these profiles exactly finite, whereas the all-harmonic action-gap optimization of the predecessor remains infinite-dimensional. A finite word length can be exhaustively listed; an unbounded harmonic degree cannot be certified by a finite truncation without a tail theorem. No general action-gap algorithm follows from Proposition 3.1.

Branching programs and probability preservation. At fixed horizon our machine is a probabilistic ordered read-once layered program with arbitrary stochastic rows and bounded real output columns. We distinguish width from total size, uniform description and random-bit complexity. Fourier growth for regular, permutation and width-three branching programs concerns different hypotheses and objectives [9, 10, 6]. Circle-walk harmonic damping and its arithmetic behavior are also established subjects [2]. Our converse can use correlated letters inside independently sampled complete packets; it is not a theorem about average error under fair independent letters.

Preserving a strict cutpoint sign is weaker than preserving a numerical probability. The predecessor’s constant-width sign construction contracts the output amplitude with length and

hence does not contradict the present width growth. Here signal and tolerance are fixed, one query is selected after the word, and neither joint answers nor anytime correctness is required. Exact upper rows in this article are atomic real kernels, not finite-fair-bit algorithms. A row with three successors can still have costly or irrational coefficients.

Conclusions. The finite optimized word profile and the compatible enclosure profile retain quantitative information which qualitative gap data discard. They yield a minimal-type exponent and an almost-everywhere logarithmic estimate, and they exhibit fluctuating numerical width for every Liouville single angle. The reflected versions show that these conclusions survive a particular noncommuting finite extension, without asserting a classification of arbitrary noncommuting nongapped alphabets.

Exact finite widths, the Liouville upper logarithmic limit, optimal enclosure shapes and a general comparison between the two profiles remain distinct quantitative problems. The results here do not depend on a new external spectral-gap computation. The earlier expansion, calibration and representation theorems retain their original hypotheses in the unchanged supporting article.

REFERENCES

- [1] S. Basu, R. Pollack and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, 2nd ed., Springer, 2006.
- [2] I. Berkes and B. Borda, Random walks on the circle and Diophantine approximation, *J. London Math. Soc.* **108** (2023), 409–440.
- [3] J. W. S. Cassels, *An Introduction to Diophantine Approximation*, Cambridge Tracts in Mathematics and Mathematical Physics, vol. 45, Cambridge University Press, 1957.
- [4] J. Dick, T. Goda, G. Larcher, F. Pillichshammer and K. Suzuki, On the quasi-uniformity properties of quasi-Monte Carlo point sets and sequences—Part I: Lattices and Kronecker sequences, arXiv:2502.06202v2, version of 13 February 2026, Section 3.3.
- [5] S. Graf and H. Luschgy, *Foundations of Quantization for Probability Distributions*, Lecture Notes in Mathematics, vol. 1730, Springer, 2000.
- [6] C. H. Lee, E. Pyne and S. Vadhan, Fourier growth of regular branching programs, *APPROX/RANDOM 2022*, LIPIcs **245**, Art. 2, 2:1–2:21.
- [7] Q. Qi, *General Theta Foundations I: Sharp Width of Hidden Rotation Experiments*, manuscript, revision 35, 2026; repository publication [aabc0731a0c9](#).
- [8] Q. Qi, *General Theta Foundations I: Alphabet Dynamics and Calibrated Numerical Memory*, manuscript, revision 42, 2026; repository publication [b5f9874903a4](#).
- [9] O. Reingold, T. Steinke and S. Vadhan, Pseudorandomness for regular branching programs via Fourier analysis, *RANDOM 2013*, LNCS **8096**, 655–670.
- [10] T. Steinke, S. Vadhan and A. Wan, Pseudorandomness and Fourier growth bounds for width 3 branching programs, *Theory Comput.* **13** (2017), Art. 12, 1–50.
- [11] M. Vidyasagar, *Hidden Markov Processes: Theory and Applications to Biology*, Princeton University Press, 2014.

PRESERVED MATHEMATICAL DEVELOPMENT

General Theta Foundations I - revision 42

The complete current article precedes this divider.

The following predecessor cumulative volume is reproduced without alteration.

Frozen predecessor publication:

b5f9874903a429f9c1a9cb080547d50c891d050a

This archive is not part of the compact referee package. Independent historical analytic obligations retain their original hypotheses and proof status.

GENERAL THETA FOUNDATIONS I: ALPHABET DYNAMICS AND CALIBRATED NUMERICAL MEMORY

QIAN QI

ABSTRACT. We distinguish the geometry of a numerical experiment from the dynamics of its command alphabet. For a planar alphabet consisting of the identity and r rotations whose angles satisfy a dual badly approximable inequality, we prove that the minimum hidden stochastic width is $\Theta(N^{r/(2r+1)})$, including every fixed sufficiently small uniform error. Every finite-block optimized action gap is zero. Thus one circle action, one orbit space and one minimum orbit dimension support infinitely many distinct width exponents. The proof pairs a multivariate command-packet converse with an exact common-row polygon realization. For general orthogonal actions we optimize the command law, characterize finite-block expansion by the spectral radius of the uniform action average, and give a rational two-letter example with zero optimized one-step gap but positive two-step gap. We also turn the query-calibrated activation into a finite algebraic optimization, with a semidefinite upper certificate, explicit multiplicity examples and query-frame perturbation bounds. The resource throughout is nonuniform clocked atomic-row label width: tables and arithmetic are free, the horizon is fixed, and one terminal numerical query is answered.

1. INTRODUCTION

A finite numerical experiment prescribes probabilities, rather than only signs of probabilities relative to a threshold. Even when its linear response space has fixed dimension, a positive stochastic realization may need a growing number of persistent labels. Two features of the specification must be kept separate: the compact action generated by its commands, and the finite words that implement that action. The second feature is not determined by the first.

The experiment studied here has response

$$(1.1) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, A_w x \rangle}{2}, \quad A_w = A_{a_N} \cdots A_{a_1}.$$

The seed set is finite, the command alphabet is fixed, and exactly one coordinate query arrives after N commands. A simulator must satisfy the specification on every word. Its rows may vary with the epoch, and the simulator may be redesigned for each N . The minimum peak label count at binary total-variation error at most ε is denoted by $W_{N,\varepsilon}$.

For an expanding irreducible action, the earlier occupation and orbit-synthesis theorems give a width exponent equal to half the smallest nonzero orbit dimension. Their proofs are reproduced in the appendices. Expansion is sufficient, but neither a description of all alphabets nor a necessary condition for a growing width. Our first result addresses a class in which every finite-block norm-gap certificate vanishes.

Theorem 1.1 (A nongapped alphabet hierarchy). *Let $r \geq 1$ be fixed. Suppose $\alpha \in \mathbb{R}^r$ and $b_* > 0$ satisfy*

$$(1.2) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b_* \|m\|_1^{-r} \quad \text{for every } m \in \mathbb{Z}^r \setminus \{0\}.$$

Take the planar commands $I, R_{2\pi\alpha_1}, \dots, R_{2\pi\alpha_r}$, seeds $\{\pm e_1, \pm e_2\}$, and the two coordinate queries. For fixed $0 < \rho \leq 1/10$ and $0 \leq \varepsilon < \rho/(2\sqrt{2})$,

$$(1.3) \quad W_{N,\varepsilon} = \Theta_{\alpha,r,\rho,\varepsilon}(N^{r/(2r+1)}).$$

The lower bound allows arbitrary hidden stochastic states and time-dependent rows. An exact upper realization has common command rows and an N -dependent decoder. The same order holds after adjoining a reflection. Every finite-block optimized action gap of either alphabet is zero.

The hypotheses have explicit examples in every r : put $\xi = 2^{1/(r+1)}$ and $\alpha_i = \xi^i$. An algebraic-norm calculation proves (1.2) with $b_* = (2/9)^r$. The rotation matrices in this example have computable-real entries; algebraicity of the angles is not asserted to make those entries algebraic. The cases $r = 1, 2, 3$ give exponents $1/3, 2/5, 3/7$. The connected effective action is the standard $\text{SO}(2)$ action in all of them. Its minimum orbit dimension is always one. Section 6 proves the theorem, including every command in a packet and every persistent upper-state label.

A second distinction already occurs before the genuinely nongapped regime. Let V_a be the Koopman command operators on the complement of invariant functions. For a probability law p , write

$$g(p) = 1 - \|\sum_a p(a)V_a\|^2, \quad g_1^*(\mathcal{A}) = \max_{p \in \Delta(\mathcal{A})} g(p).$$

The optimized quantity depends only on the alphabet and action. Repeating this optimization over laws on actual length- L words gives g_L^* . We prove that some $g_L^* > 0$ if and only if the uniform one-step average has spectral radius less than one. This need not mean its norm is less than one. A two-letter rational $\text{SO}(3)$ alphabet has $g_1^* = 0$ but $g_2^* > 0$, and consequently linear numerical width. The proof of its positive block gap uses an explicitly identified qualitative algebraic spectral-gap theorem, not a numerical harmonic truncation.

The error tolerance is also part of the actual experiment. The canonical activation constant involves both an irreducible copy and the physical query basis. Section 4 gives an exact finite projection formulation, a semidefinite relaxation with a dual certificate, and a frame reconstruction problem with quantitative stability. It distinguishes these finite problems from infinite-dimensional gap positivity.

Notation	Meaning
$W_{N,\varepsilon}$	Peak available atomic labels; includes a held binary output.
g_L^*, r_L	Optimized length- L squared gap and complementary norm.
s_λ	Smallest orbit dimension in a nontrivial irreducible type.
c_λ	Seed activation divided by the norm of query reconstruction.
b_*, r	Dual Diophantine constant and number of rotation angles.

The new assertions are the multi-angle nongapped hierarchy, the alphabet/block formulation and its worked separation, and the finite calibration and stability results. The single-angle packet idea, entropy occupation estimate, uniform orbit bound and minimum-orbit compiler are inherited ingredients, identified as such. Convex spectral optimization, commutant structure, Diophantine approximation and transformation-group geometry are classical tools. No priority claim for those tools is made. The statement is about label width, not uniform computational space, total branching-program size or an exact finite integer optimum.

2. THE EXPERIMENT AND ITS CHARGED STATE

Fix orthogonal matrices $A_a \in O(\mathbb{R}^D)$ indexed by a finite alphabet $\mathcal{A}$, a nonempty finite set X of unit seeds, $0 < \rho \leq 1/10$, and a horizon $N \geq 1$. The target is (1.1).

Definition 2.1 (Clocked atomic-row realization). A realization consists of finite sets $S_0, \dots, S_N$, initial rows $E_x \in \Delta(S_0)$, stochastic matrices $T_{t,a} : S_t \rightarrow \Delta(S_{t+1})$, and columns $d_j : S_N \rightarrow [-1, 1]$. Its mean output on (x, w, j) is

$$E_x T_{0,a_1} \cdots T_{N-1,a_N} d_j.$$

It has error at most ε if this differs from $\rho\langle e_j, A_w x \rangle$ by at most 2ε for every triple. Its width is $\max(2, |S_0|, \dots, |S_N|)$. Available labels count, including unused padding. The factor two includes a charged held output after the final Bernoulli row.

The seed and query are atomic input symbols. Only the updated label persists: no seed, previous command, persistent random tape, protocol tag or acquisition history is free. A row may sample an arbitrary known real distribution. The epoch, horizon, row table, table construction and lookup, and real arithmetic are free. Common-row constructions below use the same T_a at every command epoch, but their final decoder may depend on N . They are not anytime simulators. A rational fair-bit implementation is a different resource, with sampler states and a self-paced interface; no equality with the atomic optimum is inferred.

Let $H = \langle A_a \rangle$ and let R be the invariant span of HX . Unless specified otherwise, geometric applications assume H connected. Its nontrivial real isotypic subspaces in R are denoted by R_λ , with irreducible models E_λ of dimension d_λ . Orthogonal projections and adjoints refer to the physical Euclidean structure. An isometric intertwiner $J : E_\lambda \rightarrow R_\lambda \subseteq \mathbb{R}^D$ is not an input to the machine.

2.1. Proof-side conditional centroids. A wordwise converse may choose a distribution on commands without changing the specification. If commands are fresh and independent under that test law, a unit orbit process Y_t and the actual register satisfy

$$(2.1) \quad Z_t = \mathbb{E}[Y_t | S_t], \quad q_t = \mathbb{E}\|Z_t\|^2, \quad v_t = \mathbb{E}\|Z_t\|^2, \quad Z_{t+1} = \mathbb{E}[A_a Z_t | S_{t+1}].$$

Indeed, conditional on (S_t, a) the future row uses no other past information. The conditional-expectation projection identity and Jensen's inequality give

$$(2.2) \quad q_{t+1} \leq q_t, \quad \Delta_t := \mathbb{E}\|A_a Z_t - Z_{t+1}\|^2 = v_t - v_{t+1}, \quad \sum_{t < N} \Delta_t \leq 1.$$

The same identities hold at boundaries of independent complete command words; their internal commands need not be independent. All internal updates are first integrated into their endpoint kernel. No identity requiring a fresh independent bit is invoked inside such a word.

Lemma 2.2 (Terminal calibration). *For a fixed unit seed, any realization with error ε satisfies*

$$(2.3) \quad q_N \geq \kappa := \rho/\sqrt{D} - 2\varepsilon.$$

*More generally, for an isometric intertwiner J and a seed with $a = \|J^*x\| > 0$, tracking $Y_t = J^*A_w x/a$ yields*

$$(2.4) \quad q_N \geq \rho a/B(J) - 2\varepsilon, \quad B(J) = \|J^*\|_{\ell_\infty^D \rightarrow \ell_2^{d_\lambda}}.$$

Proof. Let $d = (d_1, \dots, d_D)$. Wordwise coordinate error bounds the norm of the difference between its mean and the target by $2\varepsilon\sqrt{D}$. Thus $\mathbb{E}\langle Y_N, d(S_N) \rangle \geq \rho - 2\varepsilon\sqrt{D}$, whereas its absolute value is at most $\sqrt{D}q_N$. For the second assertion apply J^* before pairing with the normalized constituent orbit. The error is at most $2\varepsilon B(J)$ and $\|J^*d(s)\| \leq B(J)$. This proves (2.4). The pairings are analytical devices; only one query is answered. $\square$

For fixed sizes the optimum error is attained: parameters range over a finite product of compact simplices and intervals, and the objective is a finite maximum of continuous functions. Bounds below therefore concern actual optimal final error, irrespective of cancellation between intermediate errors.

3. CANONICAL ALPHABET AND BLOCK GAPS

Let E be a nontrivial orthogonal representation of the effective group H . On $L^2(\mathbb{S}(E))$ write $V_a f(u) = f(A_a^{-1}u)$ and let Π be Haar averaging. All operators below are restricted to $\mathcal{H} = (I - \Pi)L^2(\mathbb{S}(E))$, which each V_a preserves. The complexification is used for adjoints and spectra. Define

$$(3.1) \quad r_1(\mathcal{A}) = \min_{p \in \Delta(\mathcal{A})} \|P(p)\|, \quad P(p) = \sum_a p(a)V_a, \quad g_1^* = 1 - r_1^2.$$

Equivalent orthogonal actions conjugate these operators. Adding a redundant compact presenting factor does not change them. Repeating a letter does not change the set of admissible averages, although it changes the particular uniform law.

Proposition 3.1 (Optimization, full support and simultaneous laws). *The minimum in (3.1) is attained. The function $g(p) = 1 - \|P(p)\|^2$ is concave and 2-Lipschitz in ℓ_1 . Its supremum on full-support laws equals g_1^* . Every full-support law has positive gap if and only if $g_1^* > 0$. If $m = |\mathcal{A}|$ and u is uniform, then*

$$(3.2) \quad g(u) \geq g_1^*/m.$$

More generally, $q(a) \geq \tau p(a)$ for all letters implies $g(q) \geq \tau g(p)$. For finitely many represented types, individual positive optimized gaps can be certified by one common full-support law.

Proof. Unitary norms give $\|P(p) - P(q)\| \leq \|p - q\|_1$ and $\|P(p)\| \leq 1$. Continuity and compactness give attainment, and the square of a norm is convex. Squaring the preceding inequality gives the stated Lipschitz bound for gaps. Approximation of a maximizing law by its mixtures with the uniform law proves the supremum assertion, not attainment by a full-support optimizer.

If $q = \tau p + (1 - \tau)p'$ and $f \in \mathcal{H}$, convexity of the squared norm yields

$$\|P(q)f\|^2 \leq \tau \|P(p)f\|^2 + (1 - \tau) \|P(p')f\|^2 \leq (1 - \tau g(p)) \|f\|^2.$$

This proves domination. Uniform u dominates p/m for every p . Any full-support law dominates a positive multiple of a maximizing law. Finally a mixture $\sum_i w_i p_i$ dominates $w_i p_i$ for each type's chosen law; a further positive uniform mixture gives full support. Alternatively (3.2) applies to all types at once. Optimizers need not be unique. $\square$

Laziness can be priced precisely: if identity is already an allowed letter, adding it with weight $\eta < 1$ loses at most the factor $1 - \eta$ in this certificate. If it is not a letter, silently inserting idle steps changes the word interface. Nor does zero one-step gap imply zero gap after such a modification.

3.1. A dual problem and the limit of finite truncation. For unit $f \in \mathcal{H}$, let $Q(f)$ be the real Gram matrix

$$Q(f)_{ab} = \operatorname{Re}\langle V_a f, V_b f \rangle, \quad \mathcal{C} = \overline{\operatorname{conv}}\{Q(f) : \|f\| = 1\}.$$

It is a compact convex subset of real positive semidefinite m -by- m matrices with diagonal one.

Proposition 3.2 (Finite-dimensional Gram dual). *One has*

$$(3.3) \quad r_1^2 = \min_{p \in \Delta_m} \max_{Q \in \mathcal{C}} p^T Q p = \max_{Q \in \mathcal{C}} \min_{p \in \Delta_m} p^T Q p.$$

If $r_{1,L}$ is the same norm minimum restricted to spherical harmonics of degrees at most L , then $r_{1,L} \uparrow r_1$. These finite problems are convex semidefinite programs when their blocks are supplied. They give upper bounds on g_1^ , not finite certificates of positive full gap without a tail estimate.*

Proof. For fixed p , the squared norm is $\sup_f p^T Q(f)p$. Taking the closed convex hull does not change the supremum. The expression is convex in p and affine in Q ; the compact convex minimax theorem proves (3.3). This is the usual convex spectral-design mechanism, not a new minimax principle; compare the finite-chain optimization in [11].

The harmonic spaces are invariant under orthogonal substitutions and Haar averaging. Their finite orthogonal sums increase densely. Consequently the truncated norms increase pointwise to the full norm. Take a subsequence of truncated minimizing laws converging to p_* . For every fixed L_0 , continuity on its finite blocks gives $\|P(p_*)\|_{\leq L_0} \leq \lim_L r_{1,L}$. Taking the supremum over L_0 gives $r_1 \leq \lim_L r_{1,L}$; the reverse inequality is immediate. The block norm constraints are

$$\begin{pmatrix} tI & P_\ell(p) \\ P_\ell(p)^* & tI \end{pmatrix} \succeq 0 \quad (\ell \leq L),$$

with invariant subspaces removed. They describe a finite SDP. Convergence alone supplies no effective error bound for the omitted degrees. The set $\mathcal{C}$ is finite-dimensional but is specified by all unit vectors in an infinite-dimensional space; its dual is not an effective positivity algorithm. $\square$

3.2. The block invariant. For $L \geq 1$, give every word $w \in \mathcal{A}^L$ its actual operator V_w and put

$$(3.4) \quad r_L = \min_{\pi \in \Delta(\mathcal{A}^L)} \left\| \sum_w \pi(w) V_w \right\|, \quad g_L^* = 1 - r_L^2.$$

A block is L ordinary inputs. This definition supplies a test law, not a larger free input symbol or an extra runtime register.

Theorem 3.3 (Finite-block expansion). *Let $M = m^{-1} \sum_a V_a$ on $\mathcal{H}$. The following are equivalent:*

- (1) $g_L^* > 0$ for some finite L ;
- (2) $\|M^L\| < 1$ for some finite L ;
- (3) the spectral radius of M is less than one.

The equivalence remains true with M replaced by the average under any fixed full-support law. Also $r_{L+L'} \leq r_L r_{L'}$. For finitely many types with positive finite-block gaps, a common length and a common full-support word law work for all of them.

Proof. The uniform product law assigns m^{-L} to every word and hence dominates $m^{-L}\pi$ for any law π . Proposition 3.1, applied to word operators, gives

$$(3.5) \quad 1 - \|M^L\|^2 \geq m^{-L} g_L^*.$$

A fixed full-support law gives the same argument with the smallest product mass in place of m^{-L} . Thus (1) and (2) are equivalent. The spectral-radius formula $\text{spr}(M) = \lim_n \|M^n\|^{1/n}$ gives (2) $\Leftrightarrow$ (3). Concatenating independently chosen word laws proves submultiplicativity. A common multiple of the finitely many good lengths admits a positive gap for every type by repeating each type's good law; a full-support common word law dominates each one. All invariant complements refer to the original effective action. $\square$

This equivalence identifies a genuine difference between norm and spectral radius of a nonnormal action average. It is not a claim that either can be certified by finitely many unaccompanied harmonic calculations. The next sections make the finite numerical calibration explicit, then apply this distinction to a rational alphabet.

4. FINITE OPTIMIZATION OF THE ERROR CALIBRATION

Fix a nontrivial isotypic subspace R_λ of the reachable space and its orthogonal projection S_λ . All norms in this section use the actual coordinate queries of (1.1), not a freely rotated query

basis. Define

$$(4.1) \quad a_\lambda = \max_{x \in X, J} \|J^* x\|, \quad c_\lambda = \max_{x \in X, J} \frac{\|J^* x\|}{B(J)}, \quad J^* J = I,$$

where $J : E_\lambda \rightarrow R_\lambda$ ranges over isometric intertwiners. This is the canonical isotypic convention: a displayed splitting into equivalent copies is not canonical.

Theorem 4.1 (Projection formulation and algebraic evaluation). *Let $d = d_\lambda$ and let $\mathcal{P}_\lambda$ consist of the real matrices satisfying*

$$(4.2) \quad P = P^T = P^2, \quad \text{tr } P = d, \quad P = S_\lambda P S_\lambda, \quad P A_a = A_a P \quad (a \in \mathcal{A}).$$

Then

$$(4.3) \quad c_\lambda^2 = \max_{x \in X, P \in \mathcal{P}_\lambda} \frac{x^T P x}{\max_{\sigma \in \{-1, 1\}^D} \sigma^T P \sigma}, \quad \frac{a_\lambda}{\sqrt{D}} \leq c_\lambda \leq \frac{a_\lambda}{\sqrt{d}}.$$

If the finite matrices, seeds and the certified isotypic projection are real algebraic, this number and an optimizing projection are real algebraic and can be obtained by a finite quantifier-elimination procedure.

Proof. Norm duality and the extreme points of a cube give

$$(4.4) \quad B(J) = \max_{\|u\|=1} \|Ju\|_1 = \max_{\sigma \in \{-1, 1\}^D} \|J^* \sigma\|.$$

The projection $P = JJ^*$ satisfies (4.2). Conversely any such projection has an invariant image inside one isotypic component. Its real dimension d forces that image to be one irreducible copy, so an isometric intertwiner onto it exists. The value of both numerator and denominator depends only on P , proving the identity. Averaging $\sigma^T P \sigma$ over independent signs gives $\text{tr } P = d$, whereas $\sigma^T P \sigma \leq \|\sigma\|^2 = D$. Taking maxima yields the two inequalities.

The set of projections is compact and nonempty; the denominator is at least d . For any proposed squared value t , feasibility is the finite disjunction over seeds of (4.2) and

$$(4.5) \quad x^T P x \geq t \sigma^T P \sigma \quad \text{for every } \sigma \in \{-1, 1\}^D.$$

These are polynomial conditions of degree at most two in (t, P) . Real algebraic input is represented by defining polynomials and isolating intervals. Quantifier elimination over real closed fields, for example cylindrical algebraic decomposition [8], determines the compact feasible interval of t , its endpoint and a witness. Compact semialgebraic optimization over algebraic data has an algebraic endpoint and an algebraic witness. This is finite decidability, not a polynomial-time claim. There are exponentially many sign constraints. The certified projection is part of the input; no algorithm for discovering a compact-group spectral gap is hidden in this calculation. $\square$

4.1. The multiplicity models and a semidefinite certificate. The real commutant of an isotypic component is a matrix algebra over $\mathbb{F} = \mathbb{R}, \mathbb{C}$ or the quaternions. Its invariant real orthogonal projections of rank d correspond to rank-one Hermitian projections over $\mathbb{F}$. This follows by identifying the isotypic component with a multiplicity space tensored over the irreducible division-algebra commutant: invariant submodules correspond to $\mathbb{F}$ -linear multiplicity subspaces, and adjoints correspond to Hermitian adjoints. In particular the rank constraint is essential in every commutant type. It is not replaced by a trace condition alone.

For a matrix M , write

$$\mathcal{E}_\lambda(M) = \int_H h S_\lambda M S_\lambda h^T dm_H(h).$$

This is the orthogonal projection onto the self-adjoint commutant supported on R_λ . With algebraic command matrices it can equivalently be computed as the orthogonal projection onto the solution

of the finite linear commutation equations. The integral explains its positivity; an integration oracle is unnecessary for that finite linear problem.

Proposition 4.2 (A primal–dual upper certificate). *For each seed x , let v_x be the optimum of*

$$(4.6) \quad \begin{aligned} & \text{maximize } \text{tr}(xx^T Q), \\ & \text{subject to } Q \succeq 0, \quad Q = S_\lambda Q S_\lambda, \quad [Q, A_a] = 0, \quad \sigma^T Q \sigma \leq 1 \quad (\sigma \in \{-1, 1\}^D). \end{aligned}$$

Then $c_\lambda^2 \leq \max_x v_x$. The dual for fixed x is

$$(4.7) \quad \min_{y_\sigma \geq 0} \sum_{\sigma} y_{\sigma}, \quad \sum_{\sigma} y_{\sigma} \mathcal{E}_{\lambda}(\sigma \sigma^T) - \mathcal{E}_{\lambda}(xx^T) \succeq 0 \quad \text{on } R_{\lambda}.$$

Both optima are attained and equal. An optimal nonzero Q of real rank d certifies equality with the corresponding projection optimization. Higher-rank solutions are only upper bounds.

Proof. For an admissible P , set $Q = P / \max_{\sigma} \sigma^T P \sigma$. This is feasible and has objective equal to (4.3). The dual follows from the trace pairing in the self-adjoint commutant and positivity of Q . The feasible set is bounded: averaging its sign inequalities gives $\text{tr } Q \leq 1$. A sufficiently small positive multiple of S_{λ} is strictly feasible relative to the supported commutant. Taking all dual sign weights equal and large makes the dual constraint positive definite on R_{λ} , because the sign sum is a positive multiple of the identity. Finite-dimensional semidefinite strong duality gives equality and attainment. Finally a positive commuting matrix of rank d has one irreducible image; on that image its self-adjoint restriction is scalar. Hence $Q = tP$. Scaling to a saturated sign constraint recovers the original ratio. This last argument does not apply to higher commutant rank. $\square$

Proposition 4.3 (Computed calibrations). *For a multiplicity-one nontrivial representation filling the physical space of dimension D , with unit seeds, $c = 1/\sqrt{D}$. Thus the standard spherical, traceless complex Hermitian projective, and standard block $\mathbb{R}^3 \oplus \mathbb{R}^2$ examples have respectively*

$$c = 1/\sqrt{D}, \quad c = 1/\sqrt{q^2 - 1}, \quad c_3 = 1/\sqrt{3}, \quad c_2 = 1/\sqrt{2}$$

for unit coordinate seeds in their indicated blocks.

For m equivalent real-type copies in orthogonal physical coordinate blocks $(\mathbb{R}^d)^m$, assume the reachable isotypic space is the whole block space. Writing a seed as $x = (x_1, \dots, x_m)$, one has

$$(4.8) \quad c = \frac{1}{\sqrt{d}} \max_{x \in X, i} \|x_i\|.$$

For $d = 3, m = 2$ and the two seeds $(e_1, e_1)/\sqrt{2}$ and $(e_1, -e_1)/\sqrt{2}$, Euclidean activation is $a = 1$ but query calibration is $c = 1/\sqrt{6}$.

Proof. In multiplicity one the sole copy projection is I , and (4.4) gives $B = \sqrt{D}$. For a coordinate block replace I by its block projection. In the real-type multiplicity model, every isometric intertwiner is, up to an orthogonal change in the model, $Ju = (a_1 u, \dots, a_m u)$ with $\sum_i a_i^2 = 1$. Thus

$$B(J) = \sqrt{d} \sum_i |a_i|, \quad \|J^* x\| = \left\| \sum_i a_i x_i \right\| \leq \sum_i |a_i| \max_i \|x_i\|.$$

Taking one nonzero a_i attains the resulting bound. The two displayed seeds activate the full two-copy space, and each block has norm $1/\sqrt{2}$; a diagonal copy attains $a = 1$. $\square$

The last example admits a short dual certificate as well. For the positive diagonal seed use the sign vector $\sigma = (1, 1, 1, 1, 1, 1)$ and dual weight $1/6$. Twirling under the diagonal $\text{SO}(3)$ action gives $\mathcal{E}(xx^T) = \mathcal{E}(\sigma \sigma^T)/6$. A diagonal-copy projection scaled by $1/6$ is a matching feasible primal point. For the other seed change the signs of the last three coordinates. This certifies the value without a nonconvex numerical search.

4.2. Physical query frames. Let $F : \mathbb{R}^D \rightarrow \mathbb{R}^m$ have rows of Euclidean norm at most one, and replace the coordinate means by $\rho F A_w x$. For a copy J that can be reconstructed from these queries, define

$$(4.9) \quad \beta_F(J) = \min_{LF=J^*} \|L\|_{\ell_\infty^m \rightarrow \ell_2^d}, \quad c_{\lambda,F} = \max_{x,J} \|J^* x\| / \beta_F(J).$$

The minimum is finite exactly when $\ker F \subseteq \ker J^*$. Maxima range only over such copies. If there is none, no positive calibration is inferred. Full column rank is a sufficient condition.

Theorem 4.4 (Frame reconstruction and stability). *For fixed J , (4.9) is the finite second-order cone problem*

$$\min t \quad \text{subject to } LF = J^*, \quad \|L\sigma\| \leq t \quad (\sigma \in \{-1, 1\}^m).$$

Its minimum is attained, and terminal calibration becomes $q_N \geq \rho c_{\lambda,F} - 2\varepsilon$. Duplicating a query does not change β_F or $c_{\lambda,F}$. Adding query rows can only increase the calibrated activation; deleting rows can decrease it or destroy reconstructibility.

For full-column-rank frames F, F' with the same number of rows, set $\delta = \|F' - F\|_{2 \rightarrow 2}$. With the same seed set,

$$(4.10) \quad \frac{c_{\lambda,F}}{1 + \delta\sqrt{m}/\sigma_{\min}(F')} \leq c_{\lambda,F'} \leq (1 + \delta\sqrt{m}/\sigma_{\min}(F))c_{\lambda,F}.$$

If finite seed sets have Hausdorff distance at most h , their calibrated activations for fixed F differ by at most $h \max_j \|F_j\|$.

Proof. The sign formula for the operator norm gives the cone constraints. A bounded norm bounds every column of L , so minimizing sequences have convergent subsequences. Feasibility of $LF = J^*$ is precisely the stated kernel condition. Applying L to the vector of decoder means reproduces the target constituent; its norm is at most β_F and its error at most $2\varepsilon\beta_F$. The proof of Lemma 2.2 then applies.

An added query can be ignored. Conversely, if a row is duplicated, sum the columns of L assigned to its duplicates. The new reconstruction norm cannot exceed the old one, since giving equal signs to those duplicate entries is a restriction of the old cube. These two inequalities prove equality for duplication.

Write $\Delta = F' - F$ and use the Moore–Penrose left inverse F'^+ . If $LF = J^*$, then

$$L' = L - L\Delta F'^+ \quad \text{satisfies} \quad L'F' = J^*.$$

Furthermore,

$$\|\Delta F'^+\|_{\infty \rightarrow \infty} \leq \delta\sqrt{m}/\sigma_{\min}(F'),$$

after bounding each output coordinate by its Euclidean norm. Thus $\beta_{F'}(J) \leq \beta_F(J)(1 + \delta\sqrt{m}/\sigma_{\min}(F'))$. Swap the frames for the reverse comparison, take ratios and maxima, and obtain (4.10). Finally $\|J^*(x - x')\| \leq h$. Since $J^*Ju = u$, $1 \leq \beta_F(J)\|FJu\|_\infty \leq \beta_F(J)\max_j \|F_j\|$ for unit u . This proves the seed bound. No uniform stability is implied as a query frame loses rank. $\square$

These computations concern the sufficient terminal correlation used in a width converse. They do not assert that $\rho c/2$ is the exact critical error of the whole experiment. In particular, the algebraic calibration procedure is not an algorithm for deciding infinite-dimensional expansion.

5. A BLOCK-EXPANDED WIDTH THEOREM AND A RATIONAL EXAMPLE

For connected H , let $s_\lambda = \min_{\|u\|=1} \dim(Hu)$ in each active nontrivial type, and put $\sigma_X = \max_\lambda s_\lambda$. Here activation and reachability use the entire invariant seed span, not an arbitrary displayed decomposition into copies.

Theorem 5.1 (Dominant finite-block expansion). *Suppose a type λ with $s_\lambda = \sigma_X$ has $g_L^* > 0$ for a fixed finite L . If R has dimension D_R , $0 < \rho \leq \min(1/10, 1/(8D_R))$, and $0 \leq \varepsilon < \rho c_\lambda/2$, then*

$$(5.1) \quad W_{N,\varepsilon} = \Theta(N^{\sigma_X/2}).$$

The constants may depend on the fixed action, alphabet, block length, seeds, signal and error. Other active types need not have any block gap. The exact upper construction has common command rows and a horizon-dependent decoder.

Proof. Choose a maximizing seed and copy in (4.1), and choose a positive-gap length- L word law. Feed independent complete words with this law, followed by any fixed remainder of length less than L . The uncompressed constituent orbit stays of norm one. Compressing all intermediate computation gives a common stochastic endpoint row for each starting state and word. Conditional centroids at block boundaries satisfy (2.1)–(2.2); deterministic remaining commands cannot increase their mean norm. Terminal calibration therefore gives $q_t \geq \kappa = \rho c_\lambda - 2\varepsilon > 0$ at every block boundary.

The uniform orbit estimate of Theorem A.1 and the occupation theorem in Appendix B, now in dimension d_λ , imply, at peak width K ,

$$(5.2) \quad \lfloor N/L \rfloor g_L^* \leq B_{d_\lambda, A_\lambda, s_\lambda} \kappa^{-3-2/s_\lambda} K^{2/s_\lambda}.$$

Sphere and Lebesgue action norms agree by polar coordinates: both averages act trivially on the radial factor and identically on L^2 of the sphere. Thus no full presenting-group gap is substituted for the defined action gap. This proves the lower order.

For completeness the exact upper bound prices all branches. Split R orthogonally into irreducible copies of dimensions d_i , including trivial copies. Choose branch i initially with probability $w_i = d_i/D_R$ and store its identifier in a disjoint label set. In that branch synthesize the mean $(\rho/w_i)A_w\pi_i x$. The component seed has norm at most one, and $\rho/w_i \leq 1/(8d_i)$. The construction in Theorem C.2, whose initialization also accepts vectors in the unit ball, gives $O(N^{s_i/2})$ labels. A trivial copy needs a fixed two-label mixture to retain its initial scalar; its commands can be identities. The weighted sum of all branch outputs is exactly $\rho A_w x$. Total width is the sum of the branch widths, hence $O(N^{\sigma_X/2})$. Only one final query is answered, and its row is selected from the branch carried by the label. This proves the matching upper bound and prices the selector. $\square$

The full group norm-gap sufficient theorem and the earlier one-step dominant-action theorem are contained in this statement, not withdrawn. Its hypothesis is still sufficient rather than necessary. Section 6 treats a family outside every finite-block expansion certificate.

Theorem 5.2 (A rational alphabet whose first useful block has length two). *Let*

$$a = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3/5 & -4/5 \\ 0 & 4/5 & 3/5 \end{pmatrix}, \quad b = \begin{pmatrix} 3/5 & -4/5 & 0 \\ 4/5 & 3/5 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathcal{A} = \{a, ab\}.$$

For the standard $\text{SO}(3)$ action on $\mathbb{R}^3$,

$$g_1^*(\mathcal{A}) = 0, \quad g_2^*(\mathcal{A}) > 0.$$

The second inequality uses the algebraic spectral-gap theorem [2, Theorem 1.2]; it is a qualitative external certificate. With unit coordinate seeds, $0 < \rho \leq 1/24$ and $\varepsilon < \rho/(2\sqrt{3})$, the numerical width is $\Theta(N)$.

Proof. The common angle is irrational relative to 2π . Otherwise $2\cos\theta = 6/5$ would be a rational algebraic integer, hence an integer. Powers of a and b therefore contain the full rotations about their respective axes in their closures. Their infinitesimal rotations generate $\mathfrak{so}(3)$, so the closure is $\text{SO}(3)$. The given two-letter alphabet generates the same closed group since $a^{-1}(ab) = b$.

For any probability p , its average is $V_a((1-p)I + pV_b)$. There is a nonconstant mean-zero function fixed by V_b , for example the third coordinate on the sphere. This average preserves its norm. Every average is a contraction, so every one-step norm is exactly one.

Put $c = a^{-1}ba$. It rotates about an axis different from that of b , and b, c again generate a dense subgroup of $\text{SO}(3)$. They have rational matrices. For the symmetric average

$$Q = (V_b + V_{b^{-1}} + V_c + V_{c^{-1}})/4$$

the cited theorem, applied to the compact connected simple group $\text{SO}(3)$, gives a spectral gap. In particular there is $\delta > 0$ with $\langle (I - Q)f, f \rangle \geq \delta \|f\|^2$ on the mean-zero spherical subspace. The spherical action is a quotient of the regular group action; all entries of the adjoint matrices are algebraic. This is the sole deep input.

The uniform two-step average equals

$$V_{a^2} B, \quad B = (I + V_c)(I + V_b)/4.$$

For its four unitary summands $U_1 = I, U_2 = V_b, U_3 = V_c, U_4 = V_{cb}$, the variance identity gives

$$\begin{aligned} \|f\|^2 - \|Bf\|^2 &= \frac{1}{16} \sum_{i < j} \|U_i f - U_j f\|^2 \\ &\geq \frac{\|f - V_b f\|^2 + \|f - V_c f\|^2}{16} = \frac{1}{4} \langle (I - Q)f, f \rangle. \end{aligned}$$

Thus $g_2^* \geq \delta/4 > 0$. The minimum orbit dimension is two and Proposition 4.3 gives $c = 1/\sqrt{3}$. Apply Theorem 5.1. Finite matrix checks establish the displayed identities, not the external infinite-dimensional value of δ . $\square$

6. SHARP WIDTHS WITHOUT ANY FINITE-BLOCK GAP

We prove Theorem 1.1. Throughout this section $r, \alpha, b_*, \rho, \varepsilon$ are fixed as in that theorem. Distances on $\mathbb{R}/\mathbb{Z}$ use the shorter circular distance in $[0, 1/2]$, and logarithms are natural. We retain the entire wordwise specification; command laws appear only in the converse.

6.1. A packet can only lose conditional amplitude.

Lemma 6.1 (Endpoint quantization of a fresh packet). *Let a complete random command word W be independent of the past and of the old register's private randomness. Its net orthogonal action is U_W . If the register has at most k labels at its endpoint, and the product distribution obeys*

$$(6.1) \quad \sup_{\|u\|=1, v_1, \dots, v_k \in \mathbb{S}^1} \mathbb{E} \max_j \langle v_j, U_W u \rangle \leq 1 - \gamma,$$

then the conditional-mean norms at the two boundaries satisfy $q' \leq (1 - \gamma)q$.

Proof. Let $z_s = \mathbb{E}[Y \mid S = s]$ before the word. Integrate all internal rows and randomness into a stochastic endpoint kernel $T(j \mid s, W)$. Independence of the complete word and the update rule give

$$z'_j = \mathbb{E}[U_W z_s \mid S' = j].$$

Choose $v_j = z'_j / \|z'_j\|$ when $z'_j \neq 0$, and any unit direction otherwise. Then

$$\begin{aligned} q' &= \sum_s \mathbb{P}(S = s) \mathbb{E}_W \sum_j T(j \mid s, W) \langle v_j, U_W z_s \rangle \\ &\leq \sum_s \mathbb{P}(S = s) \|z_s\| \sup_{\|u\|=1} \mathbb{E}_W \max_j \langle v_j, U_W u \rangle \leq (1 - \gamma)q. \end{aligned}$$

No condition is imposed on the number of intermediate states. The maximizing assignment in the second line need not be executable; allowing it only increases an upper bound on surviving amplitude. Commands inside W may be correlated. $\square$

For an integer $k \geq 1$, put

$$(6.2) \quad L = \lceil (2k)^{1/r} \rceil, \quad B = r(L-1), \quad s = b_* B^{-r}.$$

A packet has r segments, each of length $L-1$. In segment i first give J_i copies of rotation i , then $L-1-J_i$ idle letters, where the J_i are independent uniform variables in $\{0, \dots, L-1\}$. Thus there are exactly B actual commands and $L^r \geq 2k$ equiprobable products.

Lemma 6.2 (Separated multivariate packets). *The products in this packet have pairwise circular angular distance at least s . Condition (6.1) holds with $\gamma = s^2$.*

Proof. A difference of distinct count vectors is nonzero with ℓ_1 norm at most B . Inequality (1.2) therefore gives the separation. A circular open cap of radius $s/2$ around one decoder direction contains at most one product point, even after an arbitrary common rotation by u . At most k of the L^r points lie in the union of the k caps. At least half the points have distance at least $s/2$ from every decoder direction. For $0 \leq d \leq 1/2$, $1 - \cos(2\pi d) = 2 \sin^2(\pi d) \geq 8d^2$. On that half the scalar-product loss is at least $2s^2$. Its average is at least s^2 , as claimed. Boundary points satisfy the same weak inequality. $\square$

Proposition 6.3 (Finite packet and occupation bounds). *Let $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$. For a simulator of peak width K , define B by (6.2) with $k = K$. Then*

$$(6.3) \quad \lfloor N/B \rfloor b_*^2 B^{-2r} \leq \log(1/\kappa), \quad N < B + b_*^{-2} B^{2r+1} \log(1/\kappa).$$

Consequently $K \geq cN^{r/(2r+1)}$ for a fixed positive c . More generally, if B is defined from a proposed small width k , any available profile satisfies

$$(6.4) \quad \#\{1 \leq t \leq N : K_t \leq k\} \leq B - 1 + b_*^{-2} B^{2r+1} \log(1/\kappa),$$

with no bound on other widths.

Proof. Feed independent packets up to time $B \lfloor N/B \rfloor$ and idle letters thereafter. The same realization must pass this law because its requirement is wordwise. Lemmas 6.1 and 6.2 give $q_N \leq (1-s^2)^{\lfloor N/B \rfloor}$. Terminal calibration gives $q_N \geq \kappa$. Taking logarithms and using $-\log(1-s^2) \geq s^2$ proves both inequalities. Since $B = r(\lceil (2K)^{1/r} \rceil - 1) \leq r(2K)^{1/r}$, the second inequality gives the stated lower power, after enlarging constants for finitely many small N .

For the profile statement discard small-width endpoints before B . Among the rest, greedily select the earliest available endpoint and then the next at least B later. At most B selected-or-discarded integer endpoints are accounted for by each choice. The chosen length- B intervals are disjoint. Put an independent packet in each and deterministic idle commands in all gaps. Only the chosen endpoints need have at most k labels. Repeating the loss argument bounds the number of chosen intervals by $s^{-2} \log(1/\kappa)$. Adding the at most $B-1$ initial endpoints proves (6.4). The test law is selected from available layer sizes, not by inspecting a machine's private state. $\square$

6.2. Simultaneous denominators at every scale. The following elementary argument records the arithmetic needed for an upper construction at every horizon, not merely on a subsequence.

Lemma 6.4 (Comparable simultaneous denominators). *For each integer $Q \geq 2$, there are integers $p_1, \dots, p_r, q$ such that*

$$(6.5) \quad c_* Q^r \leq q \leq Q^r, \quad \max_i |q\alpha_i - p_i| \leq Q^{-1}, \quad c_* = (b_*/r^{r+1})^r.$$

Proof. Place the $Q^r + 1$ vectors $j\alpha \bmod \mathbb{Z}^r$, $0 \leq j \leq Q^r$, in Q^r equal boxes. Two lie in the same box; subtraction gives $1 \leq q \leq Q^r$ and the stated errors. This is the usual simultaneous Dirichlet pigeonhole argument.

Put $M = \lfloor q^{1/r} \rfloor$. The $(M+1)^r > q$ integer vectors in $\{0, \dots, M\}^r$ cannot have distinct values of $m \cdot p \bmod q$. Their difference gives a nonzero m with $H = \|m\|_1 \leq rM$ and $q \mid m \cdot p$. Hence

$$b_* H^{-r} \leq \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{H}{qQ}.$$

It follows that $b_* qQ \leq H^{r+1} \leq r^{r+1} q^{(r+1)/r}$, which rearranges to the lower bound on q . No unexplained transference constant is used. $\square$

Proposition 6.5 (Exact common-row upper realization). *For every $N \geq 1$ there is an exact realization using $O(N^{r/(2r+1)})$ labels, with common command rows and an N -dependent decoder. It uses the original $r+1$ letters; a reflection may also be included at the same cost.*

Proof. Choose an integer Q so that $c_* Q^r \geq 4$ and

$$\frac{4\pi^2 N}{c_*^2 Q^{2r+1}} \leq \log 2.$$

It can be chosen $O(N^{1/(2r+1)})$, with a constant depending only on the fixed data. Take q, p_i from Lemma 6.4. Let P be the regular q -gon with vertices $u_j = (\cos(2\pi j/q), \sin(2\pi j/q))$. Write $\delta_i = 2\pi(\alpha_i - p_i/q)$, so $|\delta_i| \leq 2\pi/(qQ) \leq \pi/q$. The rotated vertices lie in $\Lambda_i P$, where

$$(6.6) \quad \Lambda_i = \frac{\cos(\pi/q - |\delta_i|)}{\cos(\pi/q)}, \quad \log \Lambda_i \leq |\delta_i| \tan(\pi/q) \leq \frac{4\pi^2}{q^2 Q}.$$

The first identity follows by intersecting the ray to a rotated vertex with the chord between the adjacent polygon vertices. The logarithmic inequality follows by integrating the derivative of $\log \cos(\pi/q - t)$, and $\tan(\pi/q) \leq 2\pi/q$ for $q \geq 4$.

Put $\Lambda = \max(1, \Lambda_1, \dots, \Lambda_r)$. Then $\Lambda^N \leq 2$, and $\Lambda^{-1} R_{2\pi\alpha_i} P \subseteq P$. The same containment holds for the identity and for reflection in the first coordinate axis, since the latter preserves P . For each label u_j and command choose convex coordinates in P of its image divided by Λ as the stochastic row. These choices are made once and do not depend on the epoch. Initially represent $x/2$ as a mixture of the vertices; this is possible since P contains the radius- $\cos(\pi/q)$ disk. At epoch t interpret label j as $2\Lambda^t u_j$. Induction using the common rows gives mean $A_w x$. The final decoder is

$$d_\ell(j) = 2\rho \Lambda^N \langle e_\ell, u_j \rangle,$$

whose modulus is at most $4\rho \leq 2/5$. Every conditional response is therefore exact. At most three successors per row suffice by planar Carathéodory; the vector interpretation is a description of fixed rows, not an extra real register. Finally $q \leq Q^r = O(N^{r/(2r+1)})$. Every partial command step uses the same q labels. $\square$

6.3. Vanishing gaps and explicit angles.

Proposition 6.6 (All block gaps vanish). *For any finite family of planar rotations generating a dense subgroup of $\text{SO}(2)$, all optimized finite-block gaps on its circle action are zero. This remains true after adjoining the reflection above when the effective group is $O(2)$.*

Proof. Simultaneous Dirichlet approximation gives integers $n_j \rightarrow \infty$ for which all $n_j \alpha_i$ approach integers. For an irrational generating family the denominators cannot remain bounded. On the mean-zero Fourier mode $e^{in_j \theta}$ every fixed rotation word then has eigenvalue approaching one. Every average on any fixed finite word set has norm one, so no law has positive gap. For the reflected alphabet use the normalized real function $\cos(n_j \theta)$. Reflection fixes it; rotations and any fixed words move it by a norm tending to zero. The same conclusion follows. The constants are invariant functions only at frequency zero because the effective actions are respectively $\text{SO}(2)$ and $O(2)$. $\square$

Proposition 6.7 (An explicit family in every alphabet size). *Let $\xi = 2^{1/(r+1)}$ and $\alpha_i = \xi^i$, $1 \leq i \leq r$. Then (1.2) holds with $b_* = (2/9)^r$.*

Proof. For nonzero m let $H = \|m\|_1 \geq 1$ and choose $m_0 \in \mathbb{Z}$ nearest to $-\sum_i m_i \xi^i$. The polynomial $P(T) = m_0 + \sum_i m_i T^i$ is nonzero of degree at most r . Since $T^{r+1} - 2$ is Eisenstein, $P(\xi) \neq 0$. It is an algebraic integer of nonzero integer norm. All conjugates of ξ have the same modulus and $|\xi^i| < 2$, so $|m_0| \leq 2H + 1/2 \leq 5H/2$. Each of the other r conjugates of $P(\xi)$ has modulus at most $9H/2$. The product of all $r+1$ moduli is at least one. Therefore

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} = |P(\xi)| \geq (2/9)^r H^{-r}.$$

The norm argument is classical algebraic Diophantine separation; its role here is to supply explicit input data for the memory theorem. $\square$

Proof of Theorem 1.1. Propositions 6.3 and 6.5 give the matching powers. Adding reflection retains the upper construction and the converse applies to its rotation-only subalphabet. Proposition 6.6 proves the gap assertion. The effective groups are the full indicated circle and orthogonal circle actions, since (1.2) makes each angle irrational. Proposition 6.7 supplies the announced examples. $\square$

At any one internal cut the exact separate positive minimum is still three. Indeed all actual residuals depend affinely on a planar vector, and the idle-prefix images of the four seeds give affine dimension two. Hence at least three normalized states are required. For the reverse bound choose an equilateral triangle of circumradius $2\rho \leq 1/5$. It contains the radius- ρ disk. Its three vertices define legal complete future continuations, since every future command is orthogonal and every final coordinate remains bounded by one. Their convex coordinates factor every actual residual at that cut. One can embed this factorization in a complete machine by retaining the full past before that cut and the chosen vertex plus the full remaining word thereafter. Other cuts are unrestricted in this separate-minimum assertion. Thus all exponents above measure causal compatibility beyond the same static value three.

7. FINITE COMPONENTS AND OBSERVABLE QUOTIENTS

Connectedness is not harmless bookkeeping. If H is a compact matrix group, its identity component H^0 has finite index, and other components can permute its isotypic subspaces. An uncharged component label would therefore change the realization problem.

Proposition 7.1 (A paid finite-component compiler). *Choose representatives r_c of the finite quotient H/H^0 , with $r_e = I$, and write*

$$(7.1) \quad A_a r_c = r_{c'} h_{a,c}, \quad h_{a,c} \in H^0.$$

Suppose the finite connected-component alphabet $\{h_{a,c}\}$ has an exact common-row vector realization of width K_N for the required seeds and signal, whose terminal represented vectors have norm at most one. Then the original alphabet has an exact realization of width at most $\max(2, [H : H^0] K_N)$. The terminal decoder may depend on N . If H itself is finite, the exact width is at most $\max(2, |HX|)$ at every horizon.

Proof. Use labels (c, s) . Given letter a and label (c, s) , update c to the uniquely determined c' in (7.1), and use the connected compiler's row for $h_{a,c}$ on s . The row depends only on the actual stored pair and the input. Inductively, if $A_w = r_c h_w$, the represented internal vector has mean $\rho h_w x$. At the final query use $\langle e_j, r_c v_s \rangle$. Its modulus is at most one and its overall mean is $\rho \langle e_j, A_w x \rangle$. The component identity is part of every label. No fixed choice of H^0 isotypic coordinates is assumed preserved by all components.

For finite H , retain the actual point in the finite orbit union HX , update it by the command permutation, and sample the prescribed bounded terminal row. This gives the stated bound. It is an upper bound, not a claim that every orbit point is a necessary label. $\square$

The minimum-orbit construction in Appendix C, followed by the paid branch construction in Theorem 5.1, supplies such vector compilers at the stated small signal even without expansion. A matching lower bound for an arbitrary finite extension requires information about actual return words and the action on the activated subspaces; it does not follow solely from the index. The reflection extension in Theorem 1.1 is a matched nongapped example: its upper polygon is reflection-invariant, while restriction to rotation words gives the full lower order.

Proposition 7.2 (Metric orbit quotients do not specify numerical memory). *The metric quotients of the standard actions $\mathrm{SO}(d) \curvearrowright \mathbb{R}^d$, $d \geq 2$, are all the half-line. Their minimum nonzero orbit dimensions are $d - 1$. The same half-line is the metric quotient of the finite reflection action $\{\pm 1\} \curvearrowright \mathbb{R}$. Moreover there are numerical alphabets with the identical effective standard $\mathrm{SO}(2)$ action, identical seeds and queries, and identical orbit geometry, but infinitely many distinct sharp width exponents.*

Proof. Two standard orthogonal orbits are spheres of radii r, s ; their distance is $|r - s|$. The reflection quotient has the same distance between the absolute values. Orbit dimensions in the standard connected actions are those of the spheres. Thus metric quotient-equivalence alone cannot preserve the minimum orbit dimension. The last assertion is Theorem 1.1 with the explicit angle families of Proposition 6.7: all act on the same plane and their exponents $r/(2r + 1)$ are distinct. $\square$

An ambient linear isometry taking every orbit onto an orbit does preserve the dimensions of those orbit submanifolds. To preserve the numerical experiment it must additionally intertwine the actual command letters and transform the seeds and queries. An isometry of the quotient, or a polar section with its finite Weyl group, need not do this. These are distinct equivalence relations, not competing proofs of the same invariant.

8. CLASSICAL BOUNDARIES AND SCOPE

8.1. Transformation groups. Gorodski–Lytschak [12, Section 1 and Proposition 1.1] distinguish metric quotient-equivalence from ambient orbit-equivalence and study reductions, boundary, cohomogeneity and copolarity. Their Proposition 1.1 imposes a no-boundary hypothesis when recovering the ambient dimension. The half-line example in Proposition 7.2 has boundary and makes no contrary assertion. Their principal-isotropy reductions and polar examples identify quotient metric spaces; they do not provide command intertwiners preserving specified numerical words. The full alphabet is additional dynamic data.

The sphere-orbifold classification in [13, Theorem 1.1] concerns the geometry of the quotient, not the wordwise positive width of a chosen generating set. Gorodski–Thorbergsson [14, Theorems 1.1 and 1.2] classify variationally complete and irreducible taut representations through orbit equivalence and osculating-space structure. Those results can identify useful orbit types in their classes; neither tautness nor polarity supplies a gap for a given alphabet. Our uniform orbital small-ball estimate is an inherited compact differential-chart argument. It is not presented as a replacement for the slice theorem or a new classification of compact representations. The algebraic evaluation of selected adjoint and exterior-power minima in the supporting manuscript remains intact; no general highest-weight classification is asserted here.

The support-net compiler uses the extrinsic convex hull of an orbit and quadratic loss at a support maximum. These are standard convex and Riemannian geometric facts [7, 10, 9]. Its role is to make one legal set of transition rows serve every layer. The nongapped hierarchy is more

sharply alphabet-dependent: even preserving the full orbit foliation in the same ambient plane does not preserve its width exponent.

8.2. Spectral design, circle walks and branching programs. Boyd–Diaconis–Xiao [11, Sections 2–4] formulate finite reversible Markov-chain design as convex spectral-norm minimization with semidefinite duality. Proposition 3.2 uses that classical convex structure, but its admissible operators act on an infinite spherical harmonic sum. Its truncated programs have the opposite certification direction from an upper tail estimate. We do not infer a positive infinite gap from a finite SDP.

The algebraic expansion theorem of Benoist–de Saxcé [2, Theorem 1.2 and Definition 2.1] is an external qualitative input. In Theorem 5.2 it is used only for the symmetric average of the explicitly verified dense generators b, c . Our four-word variance estimate then supplies the stated two-step norm loss. This separation avoids treating the norm of a nonnormal one-step average as its spectral radius.

Berkes–Borda [15] analyze circle random walks using Fourier multipliers and Diophantine data. The simultaneous pigeonhole argument and the algebraic-norm separation in Section 6 are elementary arithmetic tools. The additional assertion here concerns loss forced by compression into an arbitrary hidden register and a matching exact positive realization. Its test packets are independent as complete words, not independent-letter samples inside a packet. The $r = 1$ mechanism and the general conditional-centroid packet inequality are inherited from the earlier rotation development; the multivariate separation, every-scale denominator argument and their matched $r/(2r + 1)$ law are the extension proved here.

The clocked model is a nonuniform probabilistic ordered read-once layered program with real terminal columns. Regular and permutation branching-program Fourier bounds, such as [5, 6, 3], concern their stated transition assumptions and pseudorandomness or Fourier growth. This paper does not claim those objectives or count total program size. A Bernoulli row converts one bounded column into randomized Boolean acceptance, but preserving a cutpoint sign is weaker than preserving its numerical value. The constant-width attenuated sign simulator, the version-pinned quantum-automata comparison and the projective examples remain in the unaltered supporting manuscript; they are not new claims of this revision.

8.3. Resource ledger in the theorem. For a common-row upper realization with K labels there are $|\mathcal{A}|K$ command rows, in addition to the seed and decoder rows. A bound of $d + 1$ successors per row does not bound the complexity of constructing their weights or storing their real values. For a general clocked machine there can instead be $|\mathcal{A}| \sum_{t < N} K_t$ distinct command rows. None of these descriptions is charged as persistent label width.

The largest toral widths in Theorem 1.1 correspond to $\frac{r}{2r+1} \log_2 N + O(1)$ label bits, not polynomially many bits. Exact irrational stochastic rows are not finite random-bit algorithms. Output error, calibration, alphabet size and representation dimension are fixed before N grows. A simultaneous channel of all coordinate answers, repeated queries, an anytime decoder and adaptive experimental design are different tasks.

8.4. Conclusions. The alphabet can be expanding only after a finite word block, or nongapped at every finite block while still forcing a sharp growing width. Optimizing the proof law makes this distinction canonical. The explicit nongapped hierarchy shows why a representation exponent cannot classify all alphabets, and supplies matching laws in a substantial excluded family rather than merely restating the exclusion.

The numerical calibration is now a finite algebraic problem with certified convex upper bounds and quantitative frame stability. This does not make spectral-gap positivity decidable. The general dominant-nongapped problem beyond the dual badly approximable planar families, sharp finite integer widths, uniform varying-signal laws and optimal table or fair-bit costs remain distinct

questions. The optional arithmetic LPS application in the preserved manuscript retains its original explicitly stated external dependency and its original-page audit limitation. Neither these results nor their reproducible builds assert an independent priority certification or an editorial outcome.

APPENDIX A. THE SMALLEST ORBIT CONTROLS UNIFORM CONCENTRATION

In this section G is a compact connected Lie group, $U : G \rightarrow O(V)$, and $V^G = \{0\}$. Put

$$(A.1) \quad s(U) = \min_{u \in \mathbb{S}(V)} \dim(Gu).$$

The minimum is over *all* directions. It can be smaller than the dimension of a generic orbit. Since G is connected, $s(U) \geq 1$.

Theorem A.1 (Uniform concentration across orbit types). *There is a finite constant $A_U \geq 1$ such that*

$$(A.2) \quad \sup_{u, v \in \mathbb{S}(V)} m_G\{g : \|U_g u - v\| \leq r\} \leq A_U r^{s(U)} \quad (0 < r \leq 1).$$

This exponent is optimal among uniform power upper bounds: for a unit u_0 with $\dim(Gu_0) = s(U)$ there are $c, r_0 > 0$ such that

$$(A.3) \quad m_G\{g : \|U_g u_0 - u_0\| \leq r\} \geq c r^{s(U)} \quad (0 < r < r_0).$$

No regular-orbit, constant-stabilizer, or simple-spectrum hypothesis is needed in (A.2).

Proof. Write $m = \dim G$, $s = s(U)$ and let $\mathfrak{g}$ be the Lie algebra. For $u \in \mathbb{S}(V)$ the differential

$$A_u : \mathfrak{g} \longrightarrow V, \quad X \longmapsto dU(X)u$$

has rank $\dim(Gu) \geq s$. Its s th singular value is continuous in u and is strictly positive on the compact sphere. Its minimum is therefore positive. We give a chart argument that also records why variation of the stabilizer causes no loss in the exponent.

Fix $(g_0, u_0) \in G \times \mathbb{S}(V)$. In a smooth coordinate chart $g = \psi(z, w)$, where $z \in \mathbb{R}^s$ and $w \in \mathbb{R}^{m-s}$, select s coordinate directions and a linear projection $L : V \rightarrow \mathbb{R}^s$ for which

$$D_z[LU_{\psi(z, w)}u]_{(g_0, u_0)} = M$$

is invertible. This is possible by the rank bound. By shrinking to a product of bounded open rectangles $Z \times W$ and a neighborhood J of u_0 , arrange

$$\|D_z[LU_{\psi(z, w)}u] - M\|_{\text{op}} \leq \frac{1}{2}\sigma_{\min}(M) \quad (z, w, u) \in Z \times W \times J.$$

Take Z convex. For fixed (w, u) , integration along the segment from z to z' gives

$$(A.4) \quad \|LU_{\psi(z, w)}u - LU_{\psi(z', w)}u\| \geq \frac{1}{2}\sigma_{\min}(M)\|z - z'\|.$$

If both points lie in the inverse image of the radius- r ball around v , their projected distance is at most $2\|L\|r$. This inverse image, on the fixed w slice, consequently has diameter at most $4\|L\|r/\sigma_{\min}(M)$, and hence s -dimensional Lebesgue measure at most a constant times r^s . Haar measure has a bounded smooth density in the chosen relatively compact chart. Integration in the remaining $m - s$ coordinates yields the same bound there.

Choose finitely many of these product neighborhoods covering $G \times \mathbb{S}(V)$. For each fixed u , those whose J contains u cover G ; summing the preceding bounds proves (A.2). All constants are independent of the center v . An empty inverse image contributes zero. This use of a finite cover, rather than a separate density estimate on every orbit, is what controls neighborhoods of singular orbit types.

For the converse, $Gu_0 \simeq G/G_{u_0}$ is a compact smooth embedded s -dimensional submanifold. The pushforward of Haar probability is its normalized invariant Riemannian volume: the induced metric is G -invariant and the invariant probability on this homogeneous space is unique. A smooth

local graph at u_0 has volume comparable to r^s in a small ambient ball. This proves (A.3). If a uniform estimate with exponent $s' > s$ held, comparison at u_0 as $r \downarrow 0$ would contradict it. $\square$

The rank computation in (A.1) is intrinsic to the representation. For example, given algebraic skew matrices $B_1, \dots, B_m$ spanning its infinitesimal action, the condition $s(U) \leq r$ is the existence of a real vector u with $\sum u_i^2 = 1$ and all $(r+1)$ -minors of $[B_1 u \cdots B_m u]$ zero. Thus it is an explicit real-algebraic decision problem for fully specified algebraic data; quantifier elimination applies [8]. This is not a test for a spectral gap, and no efficient algorithm in a succinct representation is asserted.

Corollary A.2 (Intrinsic occupation exponent). *Suppose the representation satisfies (B.2). In the setting of Lemma 2.2, $q_N \geq \kappa > 0$ implies*

$$(A.5) \quad \sum_{\substack{0 \leq t \leq N \\ \bar{K}_t \leq k}} g_t \leq 2048 D (16 A_U)^{2/s(U)} \kappa^{-3-2/s(U)} k^{2/s(U)}.$$

The estimate allows arbitrary hidden continuations and imposes no bound on the other cuts.

Proof. Apply Theorem B.5 with the uniform estimate (A.2). In particular the conditional means may change orbit type at every step; their directions still satisfy the same estimate. $\square$

APPENDIX B. AN ORBITAL ENTROPY OCCUPATION THEOREM

Lebesgue measure is used on $V = \mathbb{R}^D$ and normalized Haar measure on G . For a function on V , set

$$(B.1) \quad \mathcal{U}_a f(z) = f(U_a^{-1} z), \quad P_t = \sum_a p_t(a) \mathcal{U}_a, \quad \Pi f(z) = \int_G f(U_g^{-1} z) dm_G(g).$$

Thus $\mathcal{U}_a^* = \mathcal{U}_{a^{-1}}$, Π is an orthogonal projection, and $P_t \Pi = \Pi P_t = \Pi$. The probability law p_t need not be symmetric. Its required squared norm gap is

$$(B.2) \quad \|P_t f\|_2^2 \leq \|\Pi f\|_2^2 + (1 - g_t) \|f - \Pi f\|_2^2, \quad 0 \leq g_t \leq 1.$$

Invariance in this formula means G -invariance, not necessarily radial symmetry. In a reducible representation invariant functions can carry many parameters.

Lemma B.1 (Transfer of a group norm gap). *If right convolution $R_{p_t} F(g) = \sum_a p_t(a) F(ga)$ on $L^2(G)$ has mean-zero norm at most $\sqrt{1 - g_t}$, then (B.2) holds in every orthogonal representation on V .*

Proof. For each z apply the group inequality to $F_z(g) = f(U_g^{-1} z)$. Its mean is $\Pi f(z)$ and $R_{p_t} F_z(g) = P_t f(U_g^{-1} z)$. Integrate the resulting inequality in z . Orthogonal changes of variables and Fubini identify the three integrals with those in (B.2). First take bounded compactly supported f and then use density. This proof does not replace the full group gap by a gap in the defining representation. $\square$

B.1. The conditional-mean cost. Let G_D be a standard independent real Gaussian vector and let γ_τ be the density of τG_D .

Lemma B.2 (A one-sided Gaussian coupling inequality). *For bounded X and a finite-valued variable J , put $M = \mathbb{E}[X \mid J]$. For every $\tau > 0$,*

$$(B.3) \quad h(X + \tau G_D) - h(M + \tau G_D) \leq \frac{\mathbb{E}\|X - M\|^2}{2\tau^2}.$$

Proof. Write $f = \text{Law}(M) * \gamma_\tau$. Differentiation of the Gaussian mixture and the conditional covariance identity give

$$(B.4) \quad \nabla^2 \log f(y) = -\tau^{-2}I + \tau^{-4} \text{Cov}(M \mid M + \tau G_D = y) \succeq -\tau^{-2}I.$$

Hence $\phi(x) = -\mathbb{E} \log f(x + \tau G_D)$ has Hessian at most $\tau^{-2}I$. Taylor's inequality at M yields $\phi(X) \leq \phi(M) + \langle \nabla \phi(M), X - M \rangle + \|X - M\|^2 / (2\tau^2)$. The conditional expectation of the linear term given J is zero. If $g = \text{Law}(X) * \gamma_\tau$, nonnegativity of $D(g \parallel f)$ gives $h(g) \leq -\int g \log f = \mathbb{E} \phi(X)$, whereas $\mathbb{E} \phi(M) = h(f)$. Boundedness of X, M provides Gaussian bounds with finite second moments and justifies differentiation and integration. These identities prove (B.3). $\square$

This elementary consequence of Gaussian posterior identities is used as an estimate, not as a claim of priority over the Gaussian estimation literature [1, 4]. The estimate is one-sided and can have a negative left side. Here J can be a finite register with no Gaussian observation model; the smoothing noise is added after conditioning.

Lemma B.3 (Entropy production away from invariant functions). *For any bounded-support Gaussian mixture density f ,*

$$(B.5) \quad h(P_t f) - h(f) \geq g_t \|\sqrt{f} - \Pi \sqrt{f}\|_2^2 \geq 0.$$

Proof. Put $f_a = \mathcal{U}_a f$ and $h = P_t f$. Orthogonality preserves entropy, so $h(h) - h(f) = \sum_a p_t(a) D(f_a \parallel h)$. For probability densities u, v , $D(u \parallel v) \geq \|\sqrt{u} - \sqrt{v}\|_2^2$; it follows, for example, from $-\log x \geq 2(1 - \sqrt{x})$ and integration. The mean is the least-squares minimizer, and thus

$$\sum_a p_t(a) \|\sqrt{f_a} - \sqrt{h}\|_2^2 \geq \sum_a p_t(a) \|\sqrt{f_a} - P_t \sqrt{f}\|_2^2 = 1 - \|P_t \sqrt{f}\|_2^2.$$

Apply (B.2) and $\|\sqrt{f}\|_2 = 1$. $\square$

No logarithmic Sobolev inequality or point-mass mixing assertion is used in this step.

B.2. Sparsity away from invariant densities. Suppose that the representation obeys the uniform orbital bound

$$(B.6) \quad m_G\{g : \|U_g u - v\| \leq r\} \leq A r^s \quad (u, v \in \mathbb{S}(V), \ 0 < r \leq 1),$$

where $A \geq 1$ and $s > 0$. In particular the representation has no nonzero fixed vector. It is essential that u range over all possible centroid directions, not merely the initial seed orbit.

Lemma B.4 (Finite centroids cannot be almost invariant). *Suppose Z has at most k values in the unit ball and $\mathbb{E}\|Z\| \geq \kappa$, where $0 < \kappa \leq 1$. Set*

$$(B.7) \quad r = \left(\frac{\kappa}{16Ak} \right)^{1/s}, \quad \tau = \frac{\kappa r}{8\sqrt{2D}}, \quad f = \text{Law}(Z) * \gamma_\tau.$$

Then $\|\sqrt{f} - \Pi \sqrt{f}\|_2^2 \geq \kappa/16$.

Proof. Atoms of norm at least $\kappa/2$ have total mass at least $\kappa/2$. Let Ω be the union of radius- r caps around their directions, and let C_Ω be its cone. By Markov's inequality, $\mathbb{P}(\|G_D\| \leq \sqrt{2D}) \geq 1/2$. On this event, perturbing a high-norm atom by τG_D changes its direction by at most $4\tau\sqrt{2D}/\kappa = r/2$. Therefore $\int_{C_\Omega} f \geq \kappa/4$.

For any G -invariant $u \in L^2(V)$, Haar averaging and (B.6) give

$$\int_{C_\Omega} |u(z)|^2 dz = \int |u(z)|^2 \int_G \mathbf{1}_{C_\Omega}(U_g z) dm_G(g) dz \leq k A r^s \|u\|_2^2.$$

The point $z = 0$ is Lebesgue-null. Taking $u = \Pi \sqrt{f}$, whose norm is at most one, bounds this integral by $\kappa/16$. The reverse triangle inequality in $L^2(C_\Omega)$ now gives a norm difference at least $\sqrt{\kappa/4} - \sqrt{\kappa/16} = \sqrt{\kappa}/4$. Repeated directions and zero centroids cause no change. $\square$

Theorem B.5 (Orbital occupation budget). *Under (B.2) and (B.6), let a finite hidden register track a unit-vector command process as in (2.1). If $q_N \geq \kappa > 0$, then, for every $k \geq 1$,*

$$(B.8) \quad \sum_{\substack{0 \leq t < N \\ K_t \leq k}} g_t \leq B_{D,A,s} \kappa^{-3-2/s} k^{2/s}, \quad B_{D,A,s} = 2048D(16A)^{2/s}.$$

The same inequality holds with K_t replaced by the number of positive-probability labels under the chosen test law. No bound is imposed on the other epochs.

Proof. Fix k and the single scale (B.7) for the entire run. Write $f_t = \text{Law}(Z_t) * \gamma_\tau$. Isotropy of the Gaussian makes the density of $U_a Z_t + \tau G_D$ equal to $P_t f_t$. By Lemmas B.2 and B.3,

$$h(f_{t+1}) - h(f_t) \geq g_t \|\sqrt{f_t} - \Pi \sqrt{f_t}\|_2^2 - \frac{\Delta_t}{2\tau^2}.$$

At an epoch with at most k labels, Lemma B.4 bounds the norm term below by $\kappa/16$. Elsewhere it is nonnegative. The initial density is a translate of γ_τ ; the terminal second moment is at most $1 + D\tau^2$. The Gaussian maximum-entropy bound and (2.2) yield

$$\frac{\kappa}{16} \sum_{K_t \leq k} g_t \leq \frac{D}{2} \log \left(1 + \frac{1}{D\tau^2} \right) + \frac{1}{2\tau^2} \leq \frac{1}{\tau^2}.$$

Substitute (B.7). The same Gaussian scale is retained through all epochs; no repeated mixing-time charge is incurred. $\square$

Corollary B.6 (Actual terminal error). *For a constant gap $g > 0$, let $\mathcal{E}_N(\mathbf{K})$ be the minimum actual wordwise binary-TV error over a prescribed available profile, and put $K = \max_t K_t$. Then*

$$(B.9) \quad \mathcal{E}_N(\mathbf{K}) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{D}} - \left(\frac{B_{D,A,s} K^{2/s}}{gN} \right)^{1/(3+2/s)} \right]_+.$$

In particular, $K = o(N^{s/2})$ implies $\liminf \mathcal{E}_N \geq \rho/(2\sqrt{D})$.

Proof. If the actual error is below the calibration threshold, combine (2.3) with (B.8) and solve for it. Otherwise the displayed lower bound already holds. Attainment or a minimizing sequence gives the same conclusion for the optimum. $\square$

Local simulation errors may cancel. Nothing in this corollary sums such errors or rules out cancellation; its objective is the actual final law.

APPENDIX C. EXACT SYNTHESIS FROM A MINIMAL ORBIT

Assume in this section that V is a nontrivial real irreducible orthogonal representation, of dimension d . Choose a unit vector v_0 with orbit dimension $s = s(U)$, and set

$$\mathcal{O} = Gv_0, \quad C = \text{conv } \mathcal{O}.$$

The orbit may be a singular orbit in the unit sphere. As a homogeneous space in its own right it is nevertheless a smooth compact manifold.

Lemma C.1 (Inradius and quadratic support approximation). *One has $d^{-1}\mathbb{B}(V) \subseteq C \subseteq \mathbb{B}(V)$. There are $c, C_0, h_0 > 0$, depending on U and v_0 , such that for $0 < h \leq h_0$ a set $\mathcal{V}_h \subset \mathcal{O}$ with $|\mathcal{V}_h| \leq C_0 h^{-s}$ satisfies*

$$(C.1) \quad (1 - ch^2)C \subseteq \text{conv } \mathcal{V}_h \subseteq C \quad \text{whenever } ch^2 < 1.$$

Proof. The mean of $U_g v_0$ is an invariant vector and is zero. Its covariance is I/d : the covariance is self-adjoint and commutes with the representation, so its eigenspaces are invariant; irreducibility and trace one give the claim. This is the real Schur orthogonality calculation, for example [7, Lemma 5]. For unit H , let $f(g) = \langle H, U_g v_0 \rangle$ and $M = \max_g f(g)$. Since $-1 \leq f \leq M$,

$$0 \leq \mathbb{E}[(M - f)(f + 1)] = M - \mathbb{E}f^2 = M - 1/d.$$

Every support function of C is thus at least $1/d$, which proves the inradius statement.

Equip $\mathcal{O}$ with its induced Riemannian metric. Finitely many bounded smooth coordinate patches, with their inverses uniformly Lipschitz on smaller patches, give an intrinsic h -net with at most $C_0 h^{-s}$ points. At a maximizer y of $z \mapsto \langle H, z \rangle$ on $\mathcal{O}$, its tangential first derivative is zero. The second fundamental form of the compact embedded orbit is bounded. Along a unit-speed minimizing geodesic of length at most h , Taylor's formula therefore gives

$$\langle H, y \rangle - \langle H, v \rangle \leq L \|H\| h^2$$

for a nearby net point v , with L independent of H, y . The inradius bound gives $\|H\| \leq d h_C(H)$. Hence $h_{\text{conv } \mathcal{V}_h}(H) \geq (1 - L d h^2) h_C(H)$ for every H . Taking $c \geq L d$, and using the support-function criterion, proves (C.1). $\square$

The covering exponent s and quadratic loss at a support maximum are geometric facts, not new quantization exponents. What is needed for realization is an enclosure compatible with *every command* and *every layer*; the following construction supplies it.

Theorem C.2 (A minimal-orbit positive realization). *For any finite command alphabet, any finite unit seed set, and $0 < \rho \leq 1/(8d)$, the experiment (1.1) has an exact realization with at most $C_U N^{s(U)/2}$ available labels, for $N \geq 1$. All command matrices can be chosen independent of the epoch; the final decoder may depend on N . Each row has at most $d + 1$ successors. If the seeds belong to $C \cap \mathbb{S}(V)$, the amplitude range improves to $0 < \rho \leq 1/10$.*

Proof. Choose

$$h_N = \min\{h_0, (8cN)^{-1/2}\}, \quad \eta = 1 - c h_N^2 \geq 1 - 1/(8N), \quad a_t = \eta^{-t}/4.$$

Let $v_1, \dots, v_M$ be the net points. The represented vector of a label i at layer t is $a_t v_i$. These vectors describe the fixed rows; they are not stored as real-valued runtime data.

The polytope $\text{conv}\{v_i\}$ contains ηC and hence a ball of radius η/d . For a unit seed x , $4\rho x \in \eta C$ if $\rho \leq 1/(8d)$, because $4\rho \leq 1/(2d) < \eta/d$. Choose its convex coordinates as the initial distribution. More generally, if $x \in C$ and $4\rho \leq 2/5 < \eta$, the same initialization is valid. For each command a and label i , invariance of C gives

$$\eta U_a v_i \in \eta C \subseteq \text{conv}\{v_j\}.$$

Choose any convex coordinates of this point for the row $T_a(i, \cdot)$. They do not depend on t . Since $a_{t+1}\eta = a_t$, conditional expectation updates the represented vector by exactly U_a . Induction gives mean $\rho U_w x$ after each complete prefix.

Use the terminal column $d_j(i) = a_N \langle e_j, v_i \rangle$. Bernoulli's inequality gives $\eta^N \geq 7/8$, so $|d_j(i)| \leq 2/7 < 1$. This proves every numerical conditional law exactly. Carathéodory's theorem gives $d + 1$ successors. Finally $M \leq C_0 h_N^{-s} \leq C_U N^{s/2}$, with the fixed two-state output cut absorbed into the constant. Neither convex feasibility nor row lookup is priced as computation in this model. $\square$

Corollary C.3 (The irreducible width law). *Suppose a probability law on the finite command alphabet has a full mean-zero group norm gap. For a nontrivial real irreducible V , any nonempty finite unit seed set, fixed $0 < \rho \leq 1/(8d)$, and $0 \leq \varepsilon < \rho/(2\sqrt{d})$,*

$$(C.2) \quad W_{N,\varepsilon} = \Theta_{U,p,\rho,\varepsilon}(N^{s(U)/2}).$$

The same order holds for exact vector-state realizations and for fixed-length realizations with common command rows. For seeds on a minimal orbit the amplitude range is $0 < \rho \leq 1/10$.

Proof. A full group norm gap transfers to (B.2) with a fixed $g > 0$. Apply Corollary A.2 at the peak k , using $\kappa = \rho/\sqrt{d} - 2\varepsilon > 0$. It gives $gN \leq B\kappa^{-3-2/s}k^{2/s}$ and therefore the lower order. The exact machine in Theorem C.2 is an upper bound for every allowed error, is vector-valued in its representation, and has common command rows. These subclasses lie inside the unrestricted clocked class, so the lower bound applies to them too. The last assertion uses the stronger initialization in that theorem. $\square$

REFERENCES

- [1] A. Atalik, A. Köse and M. Gastpar, Differential entropy of the conditional expectation under additive Gaussian noise, *IEEE Trans. Signal Process.* **70** (2022), 4851–4866; arXiv:2106.04677v3. doi:10.1109/TSP.2022.3211403.
- [2] Y. Benoist and N. de Saxcé, A spectral gap theorem in simple Lie groups, *Invent. Math.* **205** (2016), 337–361; arXiv:1405.1808. doi:10.1007/s00222-015-0636-2.
- [3] C. H. Lee, E. Pyne and S. Vadhan, Fourier growth of regular branching programs, *APPROX/RANDOM 2022*, LIPIcs **245**, Art. 2, 2:1–2:21.
- [4] Y. Polyanskiy and Y. Wu, Wasserstein continuity of entropy and outer bounds for interference channels, *IEEE Trans. Inform. Theory* **62** (2016), 3992–4002.
- [5] O. Reingold, T. Steinke and S. Vadhan, Pseudorandomness for regular branching programs via Fourier analysis, *RANDOM 2013*, LNCS **8096**, 655–670.
- [6] T. Steinke, S. Vadhan and A. Wan, Pseudorandomness and Fourier growth bounds for width 3 branching programs, *Theory Comput.* **13** (2017), Art. 12, 1–50.
- [7] A. Barvinok, Estimating L^∞ norms by L^{2^k} norms for functions on orbits, preprint (April 2002), Lemma 5. Author’s manuscript, 19 pp.
- [8] S. Basu, R. Pollack and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, 2nd ed., Springer, 2006.
- [9] M. Iacobelli, Asymptotic quantization for probability measures on Riemannian manifolds, arXiv:1412.4026v1 (2014), Theorem 1.4 and Section 2.
- [10] R. Sanyal, F. Sottile and B. Sturmfels, Orbitopes, *Mathematika* **57** (2011), 275–314; arXiv:0911.5436.
- [11] S. Boyd, P. Diaconis and L. Xiao, Fastest mixing Markov chain on a graph, *SIAM Review* **46** (2004), 667–689. doi:10.1137/S0036144503423264.
- [12] C. Gorodski and A. Lytchak, On orbit spaces of representations of compact Lie groups, *J. Reine Angew. Math.* **691** (2014), 61–100; arXiv:1109.1739v2.
- [13] C. Gorodski and A. Lytchak, Isometric actions on spheres with an orbifold quotient, *Math. Ann.* **365** (2016), 1041–1067; arXiv:1407.5863.
- [14] C. Gorodski and G. Thorbergsson, Representations of compact Lie groups and the osculating spaces of their orbits, *J. Reine Angew. Math.* **555** (2003), 187–235; arXiv:math/0203196, Theorems 1.1–1.2 in the inspected preprint.
- [15] I. Berkes and B. Borda, Random walks on the circle and Diophantine approximation, *J. London Math. Soc.* **108** (2023), 409–440.

PRESERVED MATHEMATICAL DEVELOPMENT

General Theta Foundations I - revision 41

The complete current article precedes this divider.

The following predecessor cumulative volume is reproduced without alteration.

Frozen predecessor publication:

08773b2705c5ac92716df5daf42a54f3b61bca34

This archive is not part of the compact referee package. Independent historical analytic obligations keep their original hypotheses and proof status.

GENERAL THETA FOUNDATIONS I: INTRINSIC ACTION GAPS AND NUMERICAL MEMORY

QIAN QI

ABSTRACT. We study the nonuniform clocked atomic-row label width of numerical experiments generated by a finite orthogonal command alphabet. We formulate expansion on the effective action rather than on a presenting group, and remove the dependence of calibration on a choice of irreducible copies. A commutant covariance formula computes an intrinsic isotypic activation magnitude; an optimized query norm gives a canonical error threshold. For arbitrary finite seeds, let R be their invariant span and let σ be the largest minimum unit-orbit dimension among its nontrivial irreducible types. If one type attaining σ has a positive action gap, then at the stated fixed signal and error the minimum hidden width is $\Theta(N^{\sigma/2})$. No expansion is required on the other types. We exhibit a faithful rational alphabet with an effective circle factor, no full group norm gap, and linear numerical width. We compute the minimum-orbit invariant for all compact simple adjoint representations and for classical defining and second exterior power modules, including their minimizing stabilizers. The lower bound retains the actual hidden register; the upper construction prices its branch labels and realizes every numerical response exactly. Epoch, horizon-specific design, tables, arithmetic and exact atomic sampling remain uncharged; this is not a uniform-space theorem.

1. INTRODUCTION

A numerical command experiment asks a finite register to retain enough information to answer one selected observable after a word of known length. The number of labels at a single cut is a static positive factorization problem. Using small factorizations at every cut requires one compatible chain of stochastic updates, and is a different problem. For orthogonal command dynamics, orbit geometry and a conditional-mean entropy budget quantify this difference.

The hypothesis should be a property of the experiment. A presentation by a group with an unused compact factor can change the full regular representation gap without changing a single response probability. Likewise, decomposing an isotypic space into equivalent irreducible copies can change individual seed projections without changing the experiment. We remove both choices: expansion is measured in the actual action, and activation is optimized over isometric intertwiners. The latter optimization has an exact commutant covariance formula for its unweighted magnitude. These changes also permit nonspanning seeds and experiments with nonexpanding, genuinely acting factors.

Fix orthogonal matrices A_a on $V = \mathbb{R}^D$, indexed by a finite alphabet, and a nonempty finite set $\mathcal{X}$ of unit seeds. For a word $w = a_1 \cdots a_N$ let $A_w = A_{a_N} \cdots A_{a_1}$. With a selected coordinate j and a binary answer B , the law is

$$(1) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, A_w x \rangle}{2}.$$

The error is the maximum binary total-variation error over all seeds, words and queries. The width $W_{N,\varepsilon}$ is the minimum of the maximum number of available persistent labels in a realizing

Date: September 26, 2026. Revision 41.

2020 Mathematics Subject Classification. Primary 93B15, 22D10; Secondary 94A17, 68Q10, 81P45.

Key words and phrases. Positive realization, compact group, orbit dimension, entropy, hidden memory, quantum automata, numerical simulation.

machine, including the final two-label answer cut. The horizon and epoch, nonuniform row tables, arithmetic and exact atomic sampling are free. A different machine may be used for every N . These are finite positive realizations, not uniform space-bounded algorithms.

Let $H = \overline{\langle A_a \rangle}$ be connected and put $R = \text{span}(H\mathcal{X})$, $r = \dim R$. Its canonical isotypic decomposition has invariant part R_0 and nontrivial types λ . For one irreducible model E_λ of each active type, define

$$s_\lambda = \min_{\|v\|=1} \dim(Hv), \quad \sigma_{\mathcal{X}} = \max_{\lambda} s_\lambda.$$

The maximum is zero if there is no nontrivial type. For an isometric intertwiner $J : E_\lambda \rightarrow R_\lambda$, let $B(J) = \|J^*\|_{\ell_D^\infty \rightarrow \ell^2}$, and define

$$c_\lambda = \max_{x \in \mathcal{X}, J} \frac{\|J^*x\|}{B(J)}.$$

All these quantities belong to the represented experiment and its specified query geometry. For a command probability p , the action gap on E_λ is $g_\lambda = 1 - \|P_\lambda - \Pi_\lambda\|^2$, where P_λ averages the commands on sphere functions and Π_λ is Haar projection onto the invariant functions.

Theorem 1.1 (Intrinsic dominant-action law). *Suppose $0 < \rho \leq \min\{1/10, 1/(8r)\}$. If an active type λ_* attaining $\sigma_{\mathcal{X}}$ has $g_{\lambda_*} > 0$ and $0 \leq \varepsilon < \rho c_{\lambda_*}/2$, then*

$$(2) \quad W_{N,\varepsilon} = \Theta(N^{\sigma_{\mathcal{X}}/2}).$$

No action gap is required on the other types. The lower bound permits arbitrary hidden states and time-dependent stochastic rows. An exact upper realization uses common command rows and a horizon-dependent terminal decoder, with its randomized branch identity charged to the register. If all active vectors are invariant, constant width suffices.

The action gap is identical on sphere functions and on Lebesgue-square-integrable functions on the representation space. It has an exact harmonic-block norm description. A full regular-representation gap of the effective group remains a sufficient certificate, but is no longer required by the headline. Proposition 8.1 makes the extension strict: ten rational orthogonal commands act faithfully on $\mathbb{R}^3 \oplus \mathbb{R}^2$, generate $\text{SO}(3) \times \text{SO}(2)$, and have no full group norm gap, while their numerical width is $\Theta(N)$. The three-dimensional constituent expands; the circle constituent does not.

We also compute rather than merely name the orbit invariant. Theorem 9.1 determines every compact simple adjoint minimum by a single-node root deletion, including all exceptional types and the minimizing centralizers. Theorem 9.3 determines the invariant throughout the second exterior-power fundamental family. In its $\text{SU}(6)$ example the minimizing orbit is $\text{SU}(6)/\text{Sp}(3)$ of dimension fourteen; the decomposable highest-weight orbit has dimension seventeen. This illustrates why projective highest-weight geometry cannot be substituted for the real-vector orbit required by the memory theorem.

Notation	Meaning
H, R	Effective compact matrix group and invariant span of the seeds.
R_λ, E_λ	Canonical isotypic part and one irreducible model.
$s_\lambda, \sigma_{\mathcal{X}}$	Minimum orbit dimension and dominant active minimum.
g_λ, c_λ	Intrinsic action gap and query-calibrated activation.
$K_t, W_{N,\varepsilon}$	Available labels at a cut and optimal whole-run peak.

The analytic foundation is the orbital entropy occupation theorem, whose complete proof is retained. Uniform orbit charts and contracted orbit hulls supply its geometric lower and constructive upper partners. Those results are inherited; the present extensions are the intrinsic action formulation, canonical activation on arbitrary seed spans, dominant-only expansion, and the

resulting faithful nongapped example. Root centralizers, skew congruence normal forms, covering geometry and positive attenuating embeddings are classical tools, not new names for discoveries. The original projective and arithmetic applications retain their separate signal ranges and external assumptions in the appendices. No main result relies on the optional numerical LPS input.

2. THE EXPERIMENT AND ITS POSITIVE REALIZATION

In the general notation below, G may be taken to be the effective matrix group H of Section 3; $U_a = A_a$. The main hypothesis is the constituent action gap, not the full regular gap of an arbitrary presenting group. Older sufficient full-group hypotheses are explicitly retained only where stated. Let G be a compact group, let $U : G \rightarrow O(V)$ be a continuous orthogonal representation on $V = \mathbb{R}^D$, and fix a finite command alphabet $\mathcal{A} \subset G$. A finite seed set $\mathcal{X} \subset \mathbb{S}(V)$ is given initially. For $w = (a_1, \dots, a_N)$ write $U_w = U_{a_N} \cdots U_{a_1}$. Relative to an orthonormal basis $e_1, \dots, e_D$, the prescribed numerical experiment is

$$(3) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, U_w x \rangle}{2}, \quad b \in \{-1, 1\}, \quad j \in [D], \quad 0 < \rho \leq 1/10.$$

All seeds, words and queries are required, not only almost-sure words under one source law. The probabilities are between $(1 - \rho)/2$ and $(1 + \rho)/2$.

Definition 2.1 (Clocked atomic-row realization). A horizon- N machine consists of finite sets $\mathcal{S}_t$ with $|\mathcal{S}_t| = K_t$, a stochastic initialization E_x on $\mathcal{S}_0$, stochastic matrices $T_{t,a} : \mathcal{S}_t \rightarrow \mathcal{S}_{t+1}$, and columns $d_j : \mathcal{S}_N \rightarrow [-1, 1]$. Its mean response is

$$(4) \quad E_x T_{0,a_1} \cdots T_{N-1,a_N} d_j.$$

It has error at most ε if this differs from $\rho\langle e_j, U_w x \rangle$ by at most 2ε for every (x, w, j) . Available labels, including padding, count. All rows, including unreachable ones, are normalized. Only the updated label persists. The seed, word prefix, old random choices and any data-dependent protocol information cannot be consulted again unless represented by that label. The external time, horizon, fixed program and exact sampling from its rows are free. The final binary answer may be sampled when j arrives; if retained, it uses a two-label final cut. Write $W_{N,\varepsilon}$ for the minimum of $\max(2, K_0, \dots, K_N)$.

The state alphabets and matrices may be chosen anew for each N , but the command symbols and their experimental meanings are fixed. Nothing requires one autonomous device or one decoder to serve all stopping lengths. No adaptive choice of the command is optimized in this model. A complex unitary representation can be treated by its realification; all inner products in the entropy argument are real.

Fix one seed x and an independent command law at each epoch for the converse. Let $Y_t = U_{a_t} \cdots U_{a_1} x$ and let S_t be the actual machine state under this test law. Define

$$(5) \quad Z_t = \mathbb{E}[Y_t \mid S_t], \quad q_t = \mathbb{E}\|Z_t\|, \quad v_t = \mathbb{E}\|Z_t\|^2.$$

These random vectors and their laws are analytical devices, not an additional runtime register or oracle.

Lemma 2.2 (Quadratic budget and terminal calibration). *For the fresh command a , put $X_t = U_a Z_t$. Then*

$$(6) \quad Z_{t+1} = \mathbb{E}[X_t \mid S_{t+1}], \quad \Delta_t := \mathbb{E}\|X_t - Z_{t+1}\|^2 = v_t - v_{t+1}, \quad \sum_{t < N} \Delta_t \leq 1.$$

The sequences q_t, v_t are nonincreasing and start at one. A machine with error at most ε satisfies

$$(7) \quad q_t \geq q_N \geq \kappa := \rho/\sqrt{D} - 2\varepsilon \quad (0 \leq t \leq N).$$

Proof. The current command is independent of the past, and the new label is generated from (S_t, a) by a common row. Conditioning first on (S_t, a, S_{t+1}) and then on S_{t+1} proves the first identity. The orthogonal projection identity for conditional expectation gives the second; Jensen and orthogonality give both monotonicities. The fixed seed has unit norm, even with randomized initialization. For calibration form $d = (d_1, \dots, d_D)$ only in the proof, so $\|d\| \leq \sqrt{D}$. Conditioning on the complete word and using (3)–(4) gives $\mathbb{E}\langle Y_N, d(S_N) \rangle \geq \rho - 2\sqrt{D}\varepsilon$. On conditioning instead on S_N , this is at most $\sqrt{D}q_N$. This proves (7). It does not require a joint measurement of the D coordinates. $\square$

For each fixed profile, the optimum actual error is attained: all stochastic rows and terminal columns form a finite product of compact sets and the objective is a maximum of finitely many continuous functions. Thus the subsequent lower bounds concern the actual optimization, not a sum of local error certificates.

3. EXPANSION OF THE EFFECTIVE ACTION

The stochastic experiment specifies matrices, not a presentation of a group. Let $\mathcal{A}$ be its finite command alphabet, let $A_a \in O(V)$ be its command matrices, and put

$$H = \overline{\langle A_a : a \in \mathcal{A} \rangle} \subset O(V).$$

We assume that this compact Lie group is connected. A probability p on $\mathcal{A}$ is used only in a converse; the required numerical law remains wordwise. All the quantities below are defined from these matrices and p . For a nonzero invariant subspace E , write σ_E for uniform sphere probability and define operators on $L^2(\mathbb{S}(E), \sigma_E)$ by

$$P_E f(u) = \sum_a p(a) f(A_a^{-1} u), \quad \Pi_E f(u) = \int_H f(h^{-1} u) dm_H(h).$$

The squared *action gap* is

$$(8) \quad g_E = 1 - \|P_E - \Pi_E\|_{2 \rightarrow 2}^2.$$

Thus $0 \leq g_E \leq 1$. When every function is invariant, the norm in this formula is zero. Such a subspace has no positive-dimensional orbit and is not used in an occupation lower bound.

Proposition 3.1 (Presentation invariance and the relevant spectrum). *The action gap has the following properties.*

- (1) *It is unchanged by an orthogonal intertwining of the action, by splitting a command into identical letters with the same total mass, or by replacing a presenting group by any compact extension with the same matrices and pushed-forward command law.*
- (2) *On $L^2(E, dz)$, with the same formulas for P_E and Π_E , the optimal inequality is*

$$(9) \quad \|P_E f\|_2^2 \leq \|\Pi_E f\|_2^2 + (1 - g_E) \|f - \Pi_E f\|_2^2.$$

In particular, the sphere and Lebesgue action gaps are identical.

- (3) *Let $\mathcal{H}_l(E)$ be the restrictions of homogeneous harmonic polynomials of degree l , and let $P_{E,l}, \Pi_{E,l}$ be the corresponding finite-dimensional restrictions. Then*

$$(10) \quad 1 - g_E = \sup_{l \geq 0} \|P_{E,l} - \Pi_{E,l}\|_{2 \rightarrow 2}^2.$$

No finite truncation of this supremum is an asserted gap certificate.

- (4) *A mean-zero regular-representation norm bound $\sqrt{1 - g}$ on H implies $g_E \geq g$. It is a sufficient certificate, not the definition of the hypothesis.*

Proof. The operators are determined by the probability measure $\sum_a p(a)\delta_{A_a|E}$ and the effective compact image on E . An orthogonal intertwiner induces a unitary conjugacy of both operators. Haar probability pushes forward to Haar probability on a compact image. This proves the first assertion, including extensions with a nontrivial kernel.

Polar coordinates identify $L^2(E, dz)$ with $L^2((0, \infty), r^{d-1}dr) \hat{\otimes} L^2(\mathbb{S}(E), \sigma_E)$, up to an immaterial fixed normalization. Both operators act as the identity on the radial factor. Consequently $\|P_E - \Pi_E\|$ is the same in the two spaces. Haar invariance gives $P_E \Pi_E = \Pi_E P_E = \Pi_E$, so the invariant and orthogonal components separate, proving (9). The usual orthogonal harmonic decomposition of sphere functions is invariant under every orthogonal matrix; the norm of a block-diagonal operator is the supremum of its block norms. This gives (10).

For the last assertion apply the group norm inequality, for fixed u , to $F_u(h) = f(h^{-1}u)$ and integrate in u . Orthogonal invariance of sphere measure identifies the terms with the corresponding action norms. One may first take continuous f , and then use density. This is the group-to-action transference argument, not an implication from a gap in the defining representation alone. $\square$

Corollary 3.2 (Constituent action hypotheses suffice). *In the orbital occupation theorem below, the full presenting-group assumption can be replaced by $g_E > 0$ on the particular irreducible space in which the conditional centroid is tracked. Its conclusion is*

$$g_E \# \{0 \leq t < N : K_t \leq k\} \leq B_{d,A,s} \kappa^{-3-2/s} k^{2/s}.$$

Other invariant summands need not have a positive action gap.

Proof. Equation (9) is exactly the analytic inequality used in entropy production. Apply Theorem 5.5 in E . The register is still the actual whole-machine register; no constituent identifier is supplied to it. The orbit estimate is that of Theorem 4.1 for the image on E . $\square$

The distinction is substantive even after removing trivially acting factors. Section 8 gives a faithful finite-alphabet experiment with an effective circle factor and no full group norm gap, yet with a matched numerical width law supplied by one dominant expanding constituent. The older full-group results remain valid as corollaries whenever their stronger hypotheses hold.

4. THE SMALLEST ORBIT CONTROLS UNIFORM CONCENTRATION

In this section G is a compact connected Lie group, $U : G \rightarrow O(V)$, and $V^G = \{0\}$. Put

$$(11) \quad s(U) = \min_{u \in \mathbb{S}(V)} \dim(Gu).$$

The minimum is over *all* directions. It can be smaller than the dimension of a generic orbit. Since G is connected, $s(U) \geq 1$.

Theorem 4.1 (Uniform concentration across orbit types). *There is a finite constant $A_U \geq 1$ such that*

$$(12) \quad \sup_{u,v \in \mathbb{S}(V)} m_G\{g : \|U_g u - v\| \leq r\} \leq A_U r^{s(U)} \quad (0 < r \leq 1).$$

This exponent is optimal among uniform power upper bounds: for a unit u_0 with $\dim(Gu_0) = s(U)$ there are $c, r_0 > 0$ such that

$$(13) \quad m_G\{g : \|U_g u_0 - u_0\| \leq r\} \geq c r^{s(U)} \quad (0 < r < r_0).$$

No regular-orbit, constant-stabilizer, or simple-spectrum hypothesis is needed in (12).

Proof. Write $m = \dim G$, $s = s(U)$ and let $\mathfrak{g}$ be the Lie algebra. For $u \in \mathbb{S}(V)$ the differential

$$A_u : \mathfrak{g} \longrightarrow V, \quad X \longmapsto dU(X)u$$

has rank $\dim(Gu) \geq s$. Its s th singular value is continuous in u and is strictly positive on the compact sphere. Its minimum is therefore positive. We give a chart argument that also records why variation of the stabilizer causes no loss in the exponent.

Fix $(g_0, u_0) \in G \times \mathbb{S}(V)$. In a smooth coordinate chart $g = \psi(z, w)$, where $z \in \mathbb{R}^s$ and $w \in \mathbb{R}^{m-s}$, select s coordinate directions and a linear projection $L : V \rightarrow \mathbb{R}^s$ for which

$$D_z[LU_{\psi(z,w)}u]_{(g_0, u_0)} = M$$

is invertible. This is possible by the rank bound. By shrinking to a product of bounded open rectangles $Z \times W$ and a neighborhood J of u_0 , arrange

$$\|D_z[LU_{\psi(z,w)}u] - M\|_{\text{op}} \leq \frac{1}{2}\sigma_{\min}(M) \quad (z, w, u) \in Z \times W \times J.$$

Take Z convex. For fixed (w, u) , integration along the segment from z to z' gives

$$(14) \quad \|LU_{\psi(z,w)}u - LU_{\psi(z',w)}u\| \geq \frac{1}{2}\sigma_{\min}(M)\|z - z'\|.$$

If both points lie in the inverse image of the radius- r ball around v , their projected distance is at most $2\|L\|r$. This inverse image, on the fixed w slice, consequently has diameter at most $4\|L\|r/\sigma_{\min}(M)$, and hence s -dimensional Lebesgue measure at most a constant times r^s . Haar measure has a bounded smooth density in the chosen relatively compact chart. Integration in the remaining $m - s$ coordinates yields the same bound there.

Choose finitely many of these product neighborhoods covering $G \times \mathbb{S}(V)$. For each fixed u , those whose J contains u cover G ; summing the preceding bounds proves (12). All constants are independent of the center v . An empty inverse image contributes zero. This use of a finite cover, rather than a separate density estimate on every orbit, is what controls neighborhoods of singular orbit types.

For the converse, $Gu_0 \simeq G/G_{u_0}$ is a compact smooth embedded s -dimensional submanifold. The pushforward of Haar probability is its normalized invariant Riemannian volume: the induced metric is G -invariant and the invariant probability on this homogeneous space is unique. A smooth local graph at u_0 has volume comparable to r^s in a small ambient ball. This proves (13). If a uniform estimate with exponent $s' > s$ held, comparison at u_0 as $r \downarrow 0$ would contradict it. $\square$

The rank computation in (11) is intrinsic to the representation. For example, given algebraic skew matrices $B_1, \dots, B_m$ spanning its infinitesimal action, the condition $s(U) \leq r$ is the existence of a real vector u with $\sum u_i^2 = 1$ and all $(r+1)$ -minors of $[B_1u \cdots B_mu]$ zero. Thus it is an explicit real-algebraic decision problem for fully specified algebraic data; quantifier elimination applies [22]. This is not a test for a spectral gap, and no efficient algorithm in a succinct representation is asserted.

Corollary 4.2 (Intrinsic occupation exponent). *Suppose the representation satisfies (17). In the setting of Lemma 2.2, $q_N \geq \kappa > 0$ implies*

$$(15) \quad \sum_{\substack{0 \leq t \leq N \\ \bar{K}_t \leq k}} g_t \leq 2048D(16A_U)^{2/s(U)} \kappa^{-3-2/s(U)} k^{2/s(U)}.$$

The estimate allows arbitrary hidden continuations and imposes no bound on the other cuts.

Proof. Apply Theorem 5.5 with the uniform estimate (12). In particular the conditional means may change orbit type at every step; their directions still satisfy the same estimate. $\square$

5. AN ORBITAL ENTROPY OCCUPATION THEOREM

Lebesgue measure is used on $V = \mathbb{R}^D$ and normalized Haar measure on G . For a function on V , set

$$(16) \quad \mathcal{U}_a f(z) = f(U_a^{-1}z), \quad P_t = \sum_a p_t(a) \mathcal{U}_a, \quad \Pi f(z) = \int_G f(U_g^{-1}z) dm_G(g).$$

Thus $\mathcal{U}_a^* = \mathcal{U}_{a^{-1}}$, Π is an orthogonal projection, and $P_t \Pi = \Pi P_t = \Pi$. The probability law p_t need not be symmetric. Its required squared norm gap is

$$(17) \quad \|P_t f\|_2^2 \leq \|\Pi f\|_2^2 + (1 - g_t) \|f - \Pi f\|_2^2, \quad 0 \leq g_t \leq 1.$$

Invariance in this formula means G -invariance, not necessarily radial symmetry. In a reducible representation invariant functions can carry many parameters.

Lemma 5.1 (Transfer of a group norm gap). *If right convolution $R_{p_t} F(g) = \sum_a p_t(a) F(ga)$ on $L^2(G)$ has mean-zero norm at most $\sqrt{1 - g_t}$, then (17) holds in every orthogonal representation on V .*

Proof. For each z apply the group inequality to $F_z(g) = f(U_g^{-1}z)$. Its mean is $\Pi f(z)$ and $R_{p_t} F_z(g) = P_t f(U_g^{-1}z)$. Integrate the resulting inequality in z . Orthogonal changes of variables and Fubini identify the three integrals with those in (17). First take bounded compactly supported f and then use density. This proof does not replace the full group gap by a gap in the defining representation. $\square$

5.1. The conditional-mean cost. Let G_D be a standard independent real Gaussian vector and let γ_τ be the density of τG_D .

Lemma 5.2 (A one-sided Gaussian coupling inequality). *For bounded X and a finite-valued variable J , put $M = \mathbb{E}[X \mid J]$. For every $\tau > 0$,*

$$(18) \quad h(X + \tau G_D) - h(M + \tau G_D) \leq \frac{\mathbb{E}\|X - M\|^2}{2\tau^2}.$$

Proof. Write $f = \text{Law}(M) * \gamma_\tau$. Differentiation of the Gaussian mixture and the conditional covariance identity give

$$(19) \quad \nabla^2 \log f(y) = -\tau^{-2} I + \tau^{-4} \text{Cov}(M \mid M + \tau G_D = y) \succeq -\tau^{-2} I.$$

Hence $\phi(x) = -\mathbb{E} \log f(x + \tau G_D)$ has Hessian at most $\tau^{-2} I$. Taylor's inequality at M yields $\phi(X) \leq \phi(M) + \langle \nabla \phi(M), X - M \rangle + \|X - M\|^2 / (2\tau^2)$. The conditional expectation of the linear term given J is zero. If $g = \text{Law}(X) * \gamma_\tau$, nonnegativity of $D(g \parallel f)$ gives $h(g) \leq -\int g \log f = \mathbb{E} \phi(X)$, whereas $\mathbb{E} \phi(M) = h(f)$. Boundedness of X, M provides Gaussian bounds with finite second moments and justifies differentiation and integration. These identities prove (18). $\square$

This elementary consequence of Gaussian posterior identities is used as an estimate, not as a claim of priority over the Gaussian estimation literature [1, 14]. The estimate is one-sided and can have a negative left side. Here J can be a finite register with no Gaussian observation model; the smoothing noise is added after conditioning.

Lemma 5.3 (Entropy production away from invariant functions). *For any bounded-support Gaussian mixture density f ,*

$$(20) \quad h(P_t f) - h(f) \geq g_t \|\sqrt{f} - \Pi \sqrt{f}\|_2^2 \geq 0.$$

Proof. Put $f_a = \mathcal{U}_a f$ and $h = P_t f$. Orthogonality preserves entropy, so $\mathbf{h}(h) - \mathbf{h}(f) = \sum_a p_t(a) D(f_a \| h)$. For probability densities u, v , $D(u \| v) \geq \|\sqrt{u} - \sqrt{v}\|_2^2$; it follows, for example, from $-\log x \geq 2(1 - \sqrt{x})$ and integration. The mean is the least-squares minimizer, and thus

$$\sum_a p_t(a) \|\sqrt{f_a} - \sqrt{h}\|_2^2 \geq \sum_a p_t(a) \|\sqrt{f_a} - P_t \sqrt{f}\|_2^2 = 1 - \|P_t \sqrt{f}\|_2^2.$$

Apply (17) and $\|\sqrt{f}\|_2 = 1$. □

No logarithmic Sobolev inequality or point-mass mixing assertion is used in this step.

5.2. Sparsity away from invariant densities. Suppose that the representation obeys the uniform orbital bound

$$(21) \quad m_G\{g : \|U_g u - v\| \leq r\} \leq A r^s \quad (u, v \in \mathbb{S}(V), 0 < r \leq 1),$$

where $A \geq 1$ and $s > 0$. In particular the representation has no nonzero fixed vector. It is essential that u range over all possible centroid directions, not merely the initial seed orbit.

Lemma 5.4 (Finite centroids cannot be almost invariant). *Suppose Z has at most k values in the unit ball and $\mathbb{E}\|Z\| \geq \kappa$, where $0 < \kappa \leq 1$. Set*

$$(22) \quad r = \left(\frac{\kappa}{16Ak} \right)^{1/s}, \quad \tau = \frac{\kappa r}{8\sqrt{2D}}, \quad f = \text{Law}(Z) * \gamma_\tau.$$

Then $\|\sqrt{f} - \Pi\sqrt{f}\|_2^2 \geq \kappa/16$.

Proof. Atoms of norm at least $\kappa/2$ have total mass at least $\kappa/2$. Let Ω be the union of radius- r caps around their directions, and let C_Ω be its cone. By Markov's inequality, $\mathbb{P}(\|G_D\| \leq \sqrt{2D}) \geq 1/2$. On this event, perturbing a high-norm atom by τG_D changes its direction by at most $4\tau\sqrt{2D}/\kappa = r/2$. Therefore $\int_{C_\Omega} f \geq \kappa/4$.

For any G -invariant $u \in L^2(V)$, Haar averaging and (21) give

$$\int_{C_\Omega} |u(z)|^2 dz = \int |u(z)|^2 \int_G \mathbf{1}_{C_\Omega}(U_g z) dm_G(g) dz \leq k A r^s \|u\|_2^2.$$

The point $z = 0$ is Lebesgue-null. Taking $u = \Pi\sqrt{f}$, whose norm is at most one, bounds this integral by $\kappa/16$. The reverse triangle inequality in $L^2(C_\Omega)$ now gives a norm difference at least $\sqrt{\kappa/4} - \sqrt{\kappa/16} = \sqrt{\kappa}/4$. Repeated directions and zero centroids cause no change. □

Theorem 5.5 (Orbital occupation budget). *Under (17) and (21), let a finite hidden register track a unit-vector command process as in (5). If $q_N \geq \kappa > 0$, then, for every $k \geq 1$,*

$$(23) \quad \sum_{\substack{0 \leq t < N \\ \bar{K}_t \leq k}} g_t \leq B_{D,A,s} \kappa^{-3-2/s} k^{2/s}, \quad B_{D,A,s} = 2048D(16A)^{2/s}.$$

The same inequality holds with K_t replaced by the number of positive-probability labels under the chosen test law. No bound is imposed on the other epochs.

Proof. Fix k and the single scale (22) for the entire run. Write $f_t = \text{Law}(Z_t) * \gamma_\tau$. Isotropy of the Gaussian makes the density of $U_a Z_t + \tau G_D$ equal to $P_t f_t$. By Lemmas 5.2 and 5.3,

$$\mathbf{h}(f_{t+1}) - \mathbf{h}(f_t) \geq g_t \|\sqrt{f_t} - \Pi\sqrt{f_t}\|_2^2 - \frac{\Delta_t}{2\tau^2}.$$

At an epoch with at most k labels, Lemma 5.4 bounds the norm term below by $\kappa/16$. Elsewhere it is nonnegative. The initial density is a translate of γ_τ ; the terminal second moment is at most

$1 + D\tau^2$. The Gaussian maximum-entropy bound and (6) yield

$$\frac{\kappa}{16} \sum_{K_t \leq k} g_t \leq \frac{D}{2} \log \left(1 + \frac{1}{D\tau^2} \right) + \frac{1}{2\tau^2} \leq \frac{1}{\tau^2}.$$

Substitute (22). The same Gaussian scale is retained through all epochs; no repeated mixing-time charge is incurred. $\square$

Corollary 5.6 (Actual terminal error). *For a constant gap $g > 0$, let $\mathcal{E}_N(\mathbf{K})$ be the minimum actual wordwise binary-TV error over a prescribed available profile, and put $K = \max_t K_t$. Then*

$$(24) \quad \mathcal{E}_N(\mathbf{K}) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{D}} - \left(\frac{B_{D,A,s} K^{2/s}}{gN} \right)^{1/(3+2/s)} \right]_+.$$

In particular, $K = o(N^{s/2})$ implies $\liminf \mathcal{E}_N \geq \rho/(2\sqrt{D})$.

Proof. If the actual error is below the calibration threshold, combine (7) with (23) and solve for it. Otherwise the displayed lower bound already holds. Attainment or a minimizing sequence gives the same conclusion for the optimum. $\square$

Local simulation errors may cancel. Nothing in this corollary sums such errors or rules out cancellation; its objective is the actual final law.

6. EXACT SYNTHESIS FROM A MINIMAL ORBIT

Assume in this section that V is a nontrivial real irreducible orthogonal representation, of dimension d . Choose a unit vector v_0 with orbit dimension $s = s(U)$, and set

$$\mathcal{O} = Gv_0, \quad C = \text{conv } \mathcal{O}.$$

The orbit may be a singular orbit in the unit sphere. As a homogeneous space in its own right it is nevertheless a smooth compact manifold.

Lemma 6.1 (Inradius and quadratic support approximation). *One has $d^{-1}\mathbb{B}(V) \subseteq C \subseteq \mathbb{B}(V)$. There are $c, C_0, h_0 > 0$, depending on U and v_0 , such that for $0 < h \leq h_0$ a set $\mathcal{V}_h \subset \mathcal{O}$ with $|\mathcal{V}_h| \leq C_0 h^{-s}$ satisfies*

$$(25) \quad (1 - ch^2)C \subseteq \text{conv } \mathcal{V}_h \subseteq C \quad \text{whenever } ch^2 < 1.$$

Proof. The mean of $U_g v_0$ is an invariant vector and is zero. Its covariance is I/d : the covariance is self-adjoint and commutes with the representation, so its eigenspaces are invariant; irreducibility and trace one give the claim. This is the real Schur orthogonality calculation, for example [21, Lemma 5]. For unit H , let $f(g) = \langle H, U_g v_0 \rangle$ and $M = \max_g f(g)$. Since $-1 \leq f \leq M$,

$$0 \leq \mathbb{E}[(M - f)(f + 1)] = M - \mathbb{E}f^2 = M - 1/d.$$

Every support function of C is thus at least $1/d$, which proves the inradius statement.

Equip $\mathcal{O}$ with its induced Riemannian metric. Finitely many bounded smooth coordinate patches, with their inverses uniformly Lipschitz on smaller patches, give an intrinsic h -net with at most $C_0 h^{-s}$ points. At a maximizer y of $z \mapsto \langle H, z \rangle$ on $\mathcal{O}$, its tangential first derivative is zero. The second fundamental form of the compact embedded orbit is bounded. Along a unit-speed minimizing geodesic of length at most h , Taylor's formula therefore gives

$$\langle H, y \rangle - \langle H, v \rangle \leq L \|H\| h^2$$

for a nearby net point v , with L independent of H, y . The inradius bound gives $\|H\| \leq dh_C(H)$. Hence $h_{\text{conv } \mathcal{V}_h}(H) \geq (1 - Ldh^2)h_C(H)$ for every H . Taking $c \geq Ld$, and using the support-function criterion, proves (25). $\square$

The covering exponent s and quadratic loss at a support maximum are geometric facts, not new quantization exponents. What is needed for realization is an enclosure compatible with *every command* and *every layer*; the following construction supplies it.

Theorem 6.2 (A minimal-orbit positive realization). *For any finite command alphabet, any finite unit seed set, and $0 < \rho \leq 1/(8d)$, the experiment (3) has an exact realization with at most $C_U N^{s(U)/2}$ available labels, for $N \geq 1$. All command matrices can be chosen independent of the epoch; the final decoder may depend on N . Each row has at most $d + 1$ successors. If the seeds belong to $C \cap \mathbb{S}(V)$, the amplitude range improves to $0 < \rho \leq 1/10$.*

Proof. Choose

$$h_N = \min\{h_0, (8cN)^{-1/2}\}, \quad \eta = 1 - ch_N^2 \geq 1 - 1/(8N), \quad a_t = \eta^{-t}/4.$$

Let $v_1, \dots, v_M$ be the net points. The represented vector of a label i at layer t is $a_t v_i$. These vectors describe the fixed rows; they are not stored as real-valued runtime data.

The polytope $\text{conv}\{v_i\}$ contains ηC and hence a ball of radius η/d . For a unit seed x , $4\rho x \in \eta C$ if $\rho \leq 1/(8d)$, because $4\rho \leq 1/(2d) < \eta/d$. Choose its convex coordinates as the initial distribution. More generally, if $x \in C$ and $4\rho \leq 2/5 < \eta$, the same initialization is valid. For each command a and label i , invariance of C gives

$$\eta U_a v_i \in \eta C \subseteq \text{conv}\{v_j\}.$$

Choose any convex coordinates of this point for the row $T_a(i, \cdot)$. They do not depend on t . Since $a_{t+1}\eta = a_t$, conditional expectation updates the represented vector by exactly U_a . Induction gives mean $\rho U_w x$ after each complete prefix.

Use the terminal column $d_j(i) = a_N \langle e_j, v_i \rangle$. Bernoulli's inequality gives $\eta^N \geq 7/8$, so $|d_j(i)| \leq 2/7 < 1$. This proves every numerical conditional law exactly. Carathéodory's theorem gives $d + 1$ successors. Finally $M \leq C_0 h_N^{-s} \leq C_U N^{s/2}$, with the fixed two-state output cut absorbed into the constant. Neither convex feasibility nor row lookup is priced as computation in this model. $\square$

Corollary 6.3 (The irreducible width law). *Suppose a probability law on the finite command alphabet has a full mean-zero group norm gap. For a nontrivial real irreducible V , any nonempty finite unit seed set, fixed $0 < \rho \leq 1/(8d)$, and $0 \leq \varepsilon < \rho/(2\sqrt{d})$,*

$$(26) \quad W_{N,\varepsilon} = \Theta_{U,p,\rho,\varepsilon}(N^{s(U)/2}).$$

The same order holds for exact vector-state realizations and for fixed-length realizations with common command rows. For seeds on a minimal orbit the amplitude range is $0 < \rho \leq 1/10$.

Proof. A full group norm gap transfers to (17) with a fixed $g > 0$. Apply Corollary 4.2 at the peak k , using $\kappa = \rho/\sqrt{d} - 2\varepsilon > 0$. It gives $gN \leq B\kappa^{-3-2/s}k^{2/s}$ and therefore the lower order. The exact machine in Theorem 6.2 is an upper bound for every allowed error, is vector-valued in its representation, and has common command rows. These subclasses lie inside the unrestricted clocked class, so the lower bound applies to them too. The last assertion uses the stronger initialization in that theorem. $\square$

7. CANONICAL ACTIVATION AND THE REACHABLE EXPERIMENT

No spanning hypothesis is needed on the given seeds. Their invariant reachable space is

$$(27) \quad R_{\mathcal{X}} = \text{span}\{hx : h \in H, x \in \mathcal{X}\}, \quad r = \dim R_{\mathcal{X}}.$$

Orthogonality makes its orthogonal complement invariant. Every response in (3) is unchanged when its query vector is projected to $R_{\mathcal{X}}$. Choose the *canonical isotypic decomposition*

$$R_{\mathcal{X}} = R_0 \oplus \bigoplus_{\lambda \in \Lambda} R_{\lambda}, \quad R_0 = R_{\mathcal{X}}^H,$$

where R_λ contains all copies of the nontrivial real irreducible type E_λ . Let $d_\lambda = \dim E_\lambda$, let m_λ be its real irreducible multiplicity, and put

$$s_\lambda = \min_{\|u\|=1, u \in E_\lambda} \dim(Hu), \quad \sigma_\lambda = \max_{\lambda \in \Lambda} s_\lambda.$$

For an empty set Λ we set $\sigma_\lambda = 0$. An isometric intertwiner $J : E_\lambda \rightarrow R_\lambda$ identifies a single irreducible copy, without choosing a decomposition into copies. The set $\mathcal{J}_\lambda$ of such intertwiners is nonempty and compact: it is the intersection of the linear intertwining equations with $J^*J = I$.

Definition 7.1 (Canonical seed and query calibration). Let P_λ be the isotypic orthogonal projection, and define

$$(28) \quad a_\lambda = \max_{x \in \mathcal{X}, J \in \mathcal{J}_\lambda} \|J^*x\|,$$

$$(29) \quad B(J) = \|J^*\|_{\ell_D^\infty \rightarrow \ell_{d_\lambda}^2} = \max_{\|u\|=1} \|Ju\|_1, \quad c_\lambda = \max_{x, J} \frac{\|J^*x\|}{B(J)}.$$

The ℓ^1 and ℓ^∞ norms refer to the actual coordinate queries e_j . Thus c_λ is invariant under changes of model coordinates when the queries are transformed with them, but a physical change of the query family is not declared costless. One always has $a_\lambda/\sqrt{D} \leq c_\lambda \leq a_\lambda$.

The equality defining B is norm duality. Since J is an isometry, $1 \leq B(J) \leq \sqrt{D}$, so maxima exist and denominators are nonzero.

Proposition 7.2 (A spectral formula in the commutant). *For $x \in \mathcal{X}$ set*

$$C_{\lambda,x} = \int_H (hP_\lambda x)(hP_\lambda x)^* dm_H(h) \quad \text{on } R_\lambda.$$

Then

$$(30) \quad a_\lambda^2 = d_\lambda \max_{x \in \mathcal{X}} \lambda_{\max}(C_{\lambda,x}), \quad \frac{\max_x \|P_\lambda x\|}{\sqrt{m_\lambda}} \leq a_\lambda \leq \max_x \|P_\lambda x\|.$$

Every active isotypic type has $a_\lambda > 0$. Both a_λ and c_λ are independent of a splitting into equivalent copies. For each fixed seed, $C_{\lambda,x}$ is the Hilbert–Schmidt orthogonal projection of $(P_\lambda x)(P_\lambda x)^$ onto the matrix commutant. Consequently the spectral formula is finite linear algebra once the isotypic projection is supplied.*

Proof. Haar averaging is a self-adjoint idempotent on the space of real matrices, with image the matrices commuting with H . This proves the last assertion. For an isometric intertwiner J , the self-adjoint operator $J^*C_{\lambda,x}J$ commutes with the irreducible action on E_λ . Its eigenspaces are invariant, so it is scalar, even in complex and quaternionic commutant types. Its trace is $\|J^*x\|^2$; therefore it equals $\|J^*x\|^2 I/d_\lambda$. This gives the upper bound by $\lambda_{\max}(C_{\lambda,x})$.

Conversely, the top eigenspace of $C_{\lambda,x}$ is invariant and contains an irreducible copy of type λ . Choose J with image in that eigenspace. The preceding scalar identity gives equality. The trace is $\|P_\lambda x\|^2$, the ambient dimension is $m_\lambda d_\lambda$, and the top eigenspace has dimension at least d_λ . These facts give both norm bounds in (30). If $P_\lambda x = 0$ for every seed, then it is zero on their invariant span, contrary to the presence of R_λ . The remaining assertions follow from the definitions. $\square$

This formula is not an appeal to a preferred basis of a multiplicity space. For instance, on two identical copies of a real-type d -dimensional irreducible, a seed $(v, v)/\sqrt{2}$ has projection norm $1/\sqrt{2}$ on each displayed copy and norm one on the diagonal copy. Its canonical a_λ is one. Taking a minimum over the displayed copies would produce a coordinate-dependent calibration. This seed need not span the ambient space; its reachable space is precisely the diagonal copy.

Theorem 7.3 (Intrinsic numerical width with active seeds). *Assume H is connected, $\mathcal{X}$ is a nonempty finite unit seed set, and $0 < \rho \leq \min\{1/10, 1/(8r)\}$. If $\Lambda \neq \emptyset$, suppose there is a type λ_* with*

$$(31) \quad s_{\lambda_*} = \sigma_{\mathcal{X}}, \quad g_{E_{\lambda_*}} > 0, \quad 0 \leq \varepsilon < \rho c_{\lambda_*}/2.$$

Then

$$(32) \quad W_{N,\varepsilon} = \Theta(N^{\sigma_{\mathcal{X}}/2}).$$

No action gap is required on the other types. The upper realization is exact and uses common command rows and an N -dependent decoder. If Λ is empty, an exact constant-width realization exists. All constants are for the fixed experiment, signal and error.

Proof. Fix a type with $g_{E_{\lambda}} > 0$ and choose x, J attaining c_{λ} , writing $a = \|J^*x\| > 0$ and $B = B(J)$. Track $Y_t = U_{w_t}|_{E_{\lambda}} J^*x/a$ under independent commands with law p , conditioned on the *actual* register of any candidate machine. For its decoder vector $d(S_N) \in [-1, 1]^D$, the wordwise error implies

$$\mathbb{E}\langle JY_N, d(S_N) \rangle \geq \rho a - 2\varepsilon B.$$

Indeed JY_N lies in the chosen invariant copy, its pairing with $U_w x$ is a , and its ℓ^1 norm is at most B for each word. On the other hand $\|J^*d(S_N)\| \leq B$, so conditioning on S_N bounds this correlation above by Bq_N . Thus

$$(33) \quad q_N \geq \kappa_{\lambda} := \rho c_{\lambda} - 2\varepsilon > 0.$$

Corollary 3.2 and the all-direction orbit bound now give $W_{N,\varepsilon} \geq cN^{s_{\lambda}/2}$. Apply this to λ_* . The proof charges no free information about J or the isotypic type: these are analytical choices for a converse.

For the upper bound choose, solely to construct a machine, any orthogonal irreducible decomposition of $R_{\mathcal{X}}$. Give a copy of dimension d_i the input-independent branch weight $w_i = d_i/r$. In that branch initialize the minimal-orbit construction of Theorem 6.2 at mean $(\rho/w_i)\pi_i x$. Its norm is at most $\rho r/d_i \leq 1/(8d_i)$, the sufficient initialization radius there. The same proof works for subunit initial vectors. Query j uses the restriction of the ambient functional e_j , of norm at most one. The disjoint union of branch labels stores the selected branch. Its weighted output mean is exactly $\rho U_w x$. A fixed subspace is implemented by the scaled cross-polytope construction in the proof of Theorem A.1; give it weight $\dim R_0/r$. The resulting total label count is

$$O(1) + \sum_{\lambda} m_{\lambda} C_{\lambda} N^{s_{\lambda}/2} = O(N^{\sigma_{\mathcal{X}}/2}).$$

All copies are fixed in number. Command rows are common in time; branch selection and the final two-label output cut are charged. If only fixed vectors are reachable, retain the finite seed identity or use the fixed polytope, and keep its state unchanged. $\square$

Corollary 7.4 (Canonical occupation and actual error). *For each type with $g_{E_{\lambda}} > 0$, set $B_{\lambda} = B_{d_{\lambda}, A_{\lambda}, s_{\lambda}}$ from the orbit theorem. Every machine of error ε with $\kappa_{\lambda} = \rho c_{\lambda} - 2\varepsilon > 0$ obeys*

$$g_{E_{\lambda}} \# \{t < N : K_t \leq k\} \leq B_{\lambda} \kappa_{\lambda}^{-3-2/s_{\lambda}} k^{2/s_{\lambda}}.$$

For a prescribed profile of peak K , the optimum actual error is at least

$$(34) \quad \frac{1}{2} \max_{\lambda: g_{E_{\lambda}} > 0} \left[\rho c_{\lambda} - \left(\frac{B_{\lambda} K^{2/s_{\lambda}}}{g_{E_{\lambda}} N} \right)^{1/(3+2/s_{\lambda})} \right]_+.$$

An empty maximum means zero. In particular only one calibrated expanding type of maximal s_{λ} is needed for the matched exponent.

Proof. Use (33) in the same occupation theorem, then solve its peak form for ε and maximize the resulting lower bounds. If the actual error exceeds a type's threshold, its displayed bound is already true. Compactness of all rows and decoders for a fixed profile gives the same inequality for the optimum. $\square$

Proof of Theorem 1.1. Apply Theorem 7.3 and Proposition 3.1. The definitions use only the effective command action, its reachable seed space, and the actual query norms. $\square$

8. A FAITHFUL ACTION WITHOUT A FULL GROUP GAP

The action formulation is not merely a way to remove a superfluous factor from the notation. Here the nonexpanding factor acts nontrivially and is observed by the experiment.

Proposition 8.1 (An expanding dominant summand is enough). *Let $R_x, R_z \in \text{SO}(3)$ be rotations about the first and third axes with cosine $3/5$ and sine $4/5$, and let*

$$\mu_3 = \frac{1}{5}(\delta_I + \delta_{R_x} + \delta_{R_x^{-1}} + \delta_{R_z} + \delta_{R_z^{-1}}).$$

Let $R \in \text{SO}(2)$ have the same cosine and sine, and let $\nu = \frac{1}{2}(\delta_I + \delta_R)$. The ten block-diagonal commands with product law $\mu_3 \otimes \nu$ act faithfully on $V = \mathbb{R}^3 \oplus \mathbb{R}^2$. For seeds $\{\pm e_1, \dots, \pm e_5\}$, $0 < \rho \leq 1/40$ and $0 \leq \varepsilon < \rho/(2\sqrt{3})$,

$$(35) \quad W_{N,\varepsilon} = \Theta(N).$$

The effective group is $\text{SO}(3) \times \text{SO}(2)$ and its mean-zero regular convolution norm is one. Thus (35) is outside the full-effective-group-gap hypothesis as well as outside the full presenting-group-gap hypothesis.

Proof. The complex number $(3 + 4i)/5$ is not a root of unity: a root of unity and its inverse are algebraic integers, whereas their sum $6/5$ is rational and not integral. The closure of its powers is therefore the circle. Powers of R_x and R_z give the two continuous axis rotation subgroups. Their infinitesimal generators and bracket span $\mathfrak{so}(3)$, so they generate a dense subgroup of $\text{SO}(3)$. The product alphabet includes the identities needed to generate the two factors separately, proving the assertion about the effective group.

The matrices are rational. Benoist–de Saxcé’s algebraic-generator theorem [2, Theorem 1.2] supplies a regular spectral gap on the compact simple factor $\text{SO}(3)$. Since μ_3 is symmetric, its operator is self-adjoint and spectral radius equals norm. (Its identity mass also bounds the negative spectral edge away from -1 .) Proposition 3.1 gives $g_{\mathbb{R}^3} > 0$. On the effective circle, put $R = e^{2\pi i \alpha}$ in complex notation. The functions $e^{2\pi i l \theta}$ have eigenvalues $(1 + e^{2\pi i l \alpha})/2$. Infinitely many l have $\|l\alpha\| \rightarrow 0$, by the pigeonhole approximation argument. Their moduli tend to one, proving the failure of a full product norm gap.

The two real irreducible summands have minimum orbit dimensions two and one. The first is dominant and has $c = 1/\sqrt{3}$: its standard basis seeds have projection norm one and its coordinate-query norm is $\sqrt{3}$. Theorem 7.3 now applies with $r = 5$. The exact upper compiler uses $O(N)$ sphere labels and $O(N^{1/2})$ circle labels. No Diophantine assumption on α and no numerical value of the qualitative $\text{SO}(3)$ gap is used. $\square$

The external input in this example is the algebraic spectral-gap theorem for the simple factor, not a newly proved expansion result. The conclusion shows why requiring expansion on every activated type would also be unnecessarily restrictive for a maximum-exponent law.

9. MINIMUM ORBITS FROM ROOT AND SINGULAR-VALUE DATA

The width theorem becomes a numerical exponent only after its orbit invariant is evaluated. This section evaluates it on every compact simple adjoint representation, on the defining classical representations, and on an infinite family of fundamental representations. The root and normal-form tools are classical; they are not asserted to originate in this paper. What they determine here is the dynamic width exponent under the action and calibration hypotheses of Theorem 7.3. We use the conventional root systems and highest-weight notation of [31, Sections 21–23, 27, 30–32].

9.1. All compact simple adjoint representations. Let $\mathfrak{h}$ be a compact simple real Lie algebra, with its invariant positive inner product, and let H be its connected adjoint group. Choose a maximal torus and a simple system Δ in the root system Φ of its complexification. For $I \subset \Delta$ let Φ_I be the root subsystem generated by I ; the empty subsystem has zero roots.

Theorem 9.1 (Adjoint minimum-orbit formula). *For the adjoint action on $\mathfrak{h}$,*

$$(36) \quad s(\text{Ad}) = |\Phi| - \max_{\alpha \in \Delta} |\Phi_{\Delta \setminus \{\alpha\}}|.$$

Every minimizing unit vector is conjugate to a point on a fundamental coweight ray corresponding to a maximizing deletion in (36). Conversely every such ray minimizes. The compact centralizer has Lie algebra consisting of the deleted-node Levi algebra and a one-dimensional center. The values are listed in Table 1.

Proof. Every element is conjugate into a closed Weyl chamber of the compact Cartan algebra. We identify that algebra with its real root-coordinate space. If X lies in this chamber, put $I(X) = \{\alpha \in \Delta : \alpha(X) = 0\}$. A positive root is a nonnegative integral combination of simple roots. Consequently its value on X vanishes exactly when it belongs to $\Phi_{I(X)}$. The real centralizer dimension is therefore $\text{rank } \Phi + |\Phi_{I(X)}|$, while $\dim \mathfrak{h} = \text{rank } \Phi + |\Phi|$. Hence

$$\dim(HX) = |\Phi| - |\Phi_{I(X)}|.$$

For $X \neq 0$, the simple roots span the dual Cartan space, so $I(X)$ is proper. Every proper subset is contained in a subset obtained by one deletion. If it is strictly contained, adjoining a missing simple root adds at least that root and its negative, and strictly increases the root count. Thus the largest possible centralizer occurs precisely at a maximizing single deletion. Points of the closed chamber with exactly that zero set are the positive fundamental coweight ray. Normalization chooses its unit point. This proves the formula and all the assertions about the minimizers and their centralizers.

For completeness the root counts entering the table can be obtained without an orbit-type classification. For A_ℓ , deletion of node i leaves $A_{i-1} + A_{\ell-i}$ and has $i(i-1) + (\ell-i)(\ell-i+1)$ roots. For B_ℓ, C_ℓ the analogous count is $i(i-1) + 2(\ell-i)^2$. These convex quadratic expressions are maximized at endpoints; they give the displayed counts, including the rank-two ties. For D_ℓ the chain deletions give $i(i-1) + 2(\ell-i)(\ell-i-1)$, while a spin-end deletion gives $\ell(\ell-1)$. Their maximum is $2(\ell-1)(\ell-2)$ for $\ell \geq 4$, with the familiar ties at D_4 . The exceptional diagrams have at most eight nodes: deleting them gives respectively a maximal A_1, B_3 or C_3, D_5, E_6 , and E_7 root subsystem. Their root counts are 2, 18, 40, 72, 126, whereas the original systems have 12, 48, 72, 126, 240 roots. These counts follow by reflection from their simple roots, or directly from the standard coordinate root lists. Substitution completes the table. $\square$

These are orbits of actual real vectors in the compact Lie algebra. They are not complex nilpotent orbits or projective highest-weight orbits. Confusing these objects changes the dimensions. The table is an evaluation of a classical centralizer calculation, not a claim of new classification of compact simple Lie algebras.

TABLE 1. Compact adjoint orbits and their conditional width exponents. The central one-dimensional torus in the centralizer is not a root.

Type	Largest deleted root subsystem	$s(\text{Ad})$	$s/2$
A_ℓ ($\ell \geq 1$)	$A_{\ell-1}$	2ℓ	ℓ
B_ℓ ($\ell \geq 2$)	$B_{\ell-1}$	$4\ell - 2$	$2\ell - 1$
C_ℓ ($\ell \geq 2$)	$C_{\ell-1}$	$4\ell - 2$	$2\ell - 1$
D_ℓ ($\ell \geq 4$)	$D_{\ell-1}$	$4\ell - 4$	$2\ell - 2$
G_2	A_1	10	5
F_4	B_3 or C_3	30	15
E_6	D_5	32	16
E_7	E_6	54	27
E_8	E_7	114	57

9.2. Defining and low-rank spin representations.

Proposition 9.2 (Transitive defining actions). *The defining real actions of $\text{SO}(n)$ on $\mathbb{R}^n$, $\text{SU}(n)$ on the underlying real space of $\mathbb{C}^n$, and $\text{Sp}(n)$ on $\mathbb{H}^n$ have minimum orbit dimensions*

$$n - 1, \quad 2n - 1, \quad 4n - 1,$$

respectively ($n \geq 3$, $n \geq 2$, and $n \geq 1$). Their unit-vector stabilizers are $\text{SO}(n - 1)$, $\text{SU}(n - 1)$, and $\text{Sp}(n - 1)$. The real spin representations of $\text{Spin}(3)$ and $\text{Spin}(5)$, and either complex half-spin representation of $\text{Spin}(6)$ viewed as a real representation, have minimum orbit dimensions 3, 7, 7.

Proof. Extend a unit vector to an orthonormal basis over the respective field. In the special orthogonal and special unitary cases, adjust the determinant on its orthogonal complement. This proves transitivity on the sphere. A matrix fixing the first basis vector acts freely by the indicated smaller group on its complement, giving the stabilizers and the dimensions. The standard low-rank identifications $\text{Spin}(3) = \text{Sp}(1)$, $\text{Spin}(5) = \text{Sp}(2)$ and $\text{Spin}(6) = \text{SU}(4)$ identify the stated spin modules with these defining modules (the two half-spin modules in the last case are dual). They follow also by matching the B_1 , $B_2 = C_2$, and $D_3 = A_3$ fundamental weights and dimensions [31, Sections 30–32]. Apply the preceding transitivity. $\square$

This proposition does not assert a formula for all higher spin modules. The adjoint table, including every exceptional type, and the next fundamental family have separate complete proofs.

9.3. A fundamental family whose minimum need not be a highest-weight orbit. Let $\text{SU}(n)$ act on complex skew-symmetric matrices by $A \mapsto gAg^T$. This is $\bigwedge^2 \mathbb{C}^n$, with its underlying real Euclidean structure. A fixed scalar normalization of the Hilbert–Schmidt norm does not affect orbit dimensions.

Theorem 9.3 (Second exterior powers). *For this action on its real unit sphere,*

$$(37) \quad s\left(\left(\bigwedge^2 \mathbb{C}^n\right)_{\mathbb{R}}\right) = \begin{cases} 5, & n = 3, 4, \\ 13, & n = 5, \\ 14, & n = 6, \\ 4n - 7, & n \geq 7. \end{cases}$$

For $n = 3$ and $n \geq 5$ the underlying real representation is irreducible. For $n = 4$ it is the sum of two equivalent real six-dimensional irreducibles, each with minimum orbit dimension five. At $n = 6$ a minimizing vector is a full-rank symplectic form, with stabilizer $\text{Sp}(3)$ and orbit $\text{SU}(6)/\text{Sp}(3)$; the decomposable highest-weight vector instead has orbit dimension seventeen.

Proof. We recall the skew-symmetric unitary congruence normal form and its argument. On $\mathbb{C}^n$ define the antilinear map $Fv = A\bar{v}$. Skew symmetry implies $F^2 = -AA^*$. On a positive eigenspace of AA^* with eigenvalue a^2 , F/a is antiunitary, squares to $-I$, and pairs a unit vector with an orthogonal unit vector. Inductively choosing such pairs gives blocks aJ , where $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, and a zero block. This is the classical unitary congruence construction [32]; the pairing argument supplies the needed special case. Although the basis change can have determinant different from one, conjugation by it preserves $SU(n)$ and therefore the dimension of the stabilizer inside $SU(n)$.

Suppose the distinct positive singular values have multiplicities $2k_1, \dots, 2k_b$, with $k = \sum_i k_i$ and $l = n - 2k$. A stabilizer preserves the eigenspaces of AA^* and commutes there with the associated quaternionic structure. Its unitary factors are therefore $Sp(k_i)$; on the kernel they form $U(l)$. The symplectic factors have determinant one, so imposing determinant one leaves $SU(l)$ on the kernel when $l > 0$. The stabilizer dimension is

$$\sum_i k_i(2k_i + 1) + \begin{cases} l^2 - 1, & l > 0, \\ 0, & l = 0. \end{cases}$$

At fixed rank, coalescing the positive singular values increases this quantity, because merging parts a, b increases it by $4ab$. Thus the smallest orbit at rank $2k < n$ has dimension

$$(38) \quad n^2 - 1 - k(2k + 1) - ((n - 2k)^2 - 1) = k(4n - 6k - 1).$$

At full rank $n = 2k$ it has dimension

$$(39) \quad n^2 - 1 - k(2k + 1) = n(n - 1)/2 - 1.$$

The quadratic in (38) is concave in k , so its minimum is at an endpoint. Comparing those endpoints with (39), when present, gives exactly (37). These minimizing strata are attained by normalized block forms, so this is not merely a dimension lower bound.

The complex module has highest weight ω_2 and dual weight ω_{n-2} ; for $n \neq 4$ these are different when $n \geq 3$. Consequently its underlying real module is irreducible of complex type. For $n = 4$ the conjugate-linear Hodge involution on two-forms has square one and commutes with $SU(4)$; its fixed real six-space and its imaginary multiple give the two equivalent real forms. The identification $SU(4) = Spin(6)$ makes each the standard real six-dimensional module, so its minimum orbit dimension is five. Alternatively the full-rank equality in (39) gives the same whole-space minimum. Finally $\dim SU(6) - \dim Sp(3) = 35 - 21 = 14$, whereas (38) at $k = 1$ gives seventeen. $\square$

Under the intrinsic action-gap and calibration hypotheses, divide the numbers in (37) by two to obtain the width exponents. In particular the $SU(6)$ fundamental module has order N^7 , not $N^{17/2}$; replacing the minimizing real-vector orbit by the highest-weight orbit would give the wrong answer. These results do not purport to classify every highest-weight representation. They compute the invariant, and its minimizing stabilizers, on the stated complete families rather than leaving those inputs as an existential rank test.

10. MATRIX AND HARMONIC REPRESENTATIONS

Theorem 4.1 identifies the correct geometric exponent without a separate estimate for every spectral type. We record its principal matrix consequences. The orbit dimensions below are classical; their role is to determine the dynamic width exponent.

Proposition 10.1 (Three conjugation families). *On the real space of traceless self-adjoint q by q matrices over $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{H}$, with inner product $\text{Re tr}(AB)$, let $\beta = \dim_{\mathbb{R}} \mathbb{F}$. The compact groups $SO(q), SU(q), Sp(q)$, respectively, act irreducibly for $q \geq 2$. The least nonzero unit-orbit dimension is $\beta(q - 1)$. Accordingly, under a full group gap, the width order in Corollary 6.3 is*

$$(40) \quad N^{(q-1)/2}, \quad N^{q-1}, \quad N^{2(q-1)},$$

respectively, in its fixed signal and fixed error ranges. The $\mathrm{SO}(2)$ case supplies no finitely supported full-gap example.

Proof. A self-adjoint matrix with distinct eigenvalue multiplicities $m_1, \dots, m_r$ has orbit dimension

$$\beta \sum_{i < j} m_i m_j = \frac{\beta}{2} \left(q^2 - \sum_i m_i^2 \right).$$

For real matrices subtract the dimensions of the orthogonal stabilizer blocks from $q(q-1)/2$; their disconnected components do not affect dimension. For complex matrices subtract $\sum m_i^2 - 1$ from $q^2 - 1$. For quaternionic matrices subtract $\sum m_i(2m_i + 1)$ from $q(2q + 1)$. These computations also identify the least orbits as real, complex and quaternionic projective spaces. A nonzero traceless matrix has at least two eigenvalues. The largest possible $\sum m_i^2$ under that restriction is $(q-1)^2 + 1$, attained by multiplicities $(1, q-1)$. This proves the minimum.

For completeness, the conjugation representation is irreducible over $\mathbb{R}$. Any nonzero invariant real subspace contains, after diagonalizing a nonzero member, a nonconstant traceless real diagonal matrix. Differences of its conjugates under coordinate interchanges give all traceless diagonal directions. Rotations in a pair of coordinate axes then give every self-adjoint off-diagonal direction; in the complex and quaternionic cases, diagonal unit phases give each real component of those entries. Signed interchanges and plane rotations can be chosen in the stated special groups. Thus the subspace is the entire traceless self-adjoint space. The general width law applies. Finally a finite measure on a torus has Fourier multipliers arbitrarily close to one by simultaneous approximation, so it has no full norm gap on its mean-zero L^2 space. $\square$

For complex matrices the projective seeds in (56) belong to a least-dimensional orbit. The signal need not be reduced to $1/(8(q^2 - 1))$ in this specialization.

Corollary 10.2 (The preserved projective width law). *Let $q \geq 2$, $0 < \rho \leq 1/10$, and $0 \leq \varepsilon < \rho/(2\sqrt{q^2 - 1})$ be fixed. For the finite projective seeds and coordinate queries of Section C, any finite command alphabet supporting a full group norm gap has*

$$W_{N,\varepsilon} = \Theta(N^{q-1}).$$

It has the fixed q -dimensional quantum realization of Proposition C.4. The explicit quantitative all-spectra estimate and the rational-projector net from the preceding revision are retained in Appendices B and C.

Proposition 10.3 (Every fixed spherical harmonic degree). *Let V_ℓ be the real irreducible $\mathrm{SO}(3)$ representation of harmonic homogeneous polynomials of degree $\ell \geq 1$, with $\dim V_\ell = 2\ell + 1$. Then $s(V_\ell) = 2$. Consequently every fixed ℓ has width $\Theta(N)$ under a full-gap command law, with the signal and query conventions of Corollary 6.3. The same six-command arithmetic alphabet in Appendix D supplies such a gap for all these representations.*

Proof. A nonzero vector has no three-dimensional stabilizer, since the representation is nontrivial irreducible. The Lie algebra $\mathfrak{so}(3)$ has no two-dimensional subalgebra: identifying its bracket with a cross product, two independent elements bracket to a third independent direction. The stabilizer dimension is therefore at most one, so every orbit has dimension at least two. A zonal spherical harmonic is fixed by the circle of rotations about its axis and has a two-dimensional orbit. This gives the minimum. The final assertion follows from Proposition E.1: the spherical norm bound controls all nontrivial $\mathrm{SO}(3)$ irreducibles, not just the defining three-dimensional action. $\square$

The degree is fixed before $N \rightarrow \infty$; no uniform-in- ℓ constants are claimed. Generic harmonic orbits can be three-dimensional, so their dimension would give the wrong exponent. In the real degree-two representation this is already the distinction between generic three-dimensional

matrix orbits and the two-dimensional projective orbit. These examples use the same general minimum-orbit theorem, not separate orbital cap computations.

11. NUMERICAL REALIZATION IS NOT CUTPOINT SIMULATION

A numerical simulator must preserve the values in (3), whereas a strict-cutpoint simulator need only preserve whether the answer probability is larger than $1/2$. The following construction makes the difference explicit in the present command model. Its stochastic embedding mechanism is classical; it is the quantitative width theorem that prevents numerical exactification at bounded width.

Proposition 11.1 (A constant-width sign simulator). *For any orthogonal action on $\mathbb{R}^D$, $D \geq 2$, there is a machine with $D + 1$ command-layer labels, common command rows, and one common query decoder for all lengths such that*

$$(41) \quad \tilde{\mathbb{P}}(B = 1 \mid x, w, j) - \frac{1}{2} = \frac{1}{2} \lambda^{N+1} \langle e_j, U_w x \rangle, \quad \lambda = 1/(2D).$$

It preserves every strict comparison of (3) with $1/2$, including equality. If $\lambda^{N+1} \leq \rho$, its worst numerical binary-TV error is at least $(\rho - \lambda^{N+1})/(2\sqrt{D})$.

Proof. Let $v_0, \dots, v_D$ be a regular simplex on the unit sphere of $\mathbb{R}^D$, centered at zero. Then

$$\sum_s v_s = 0, \quad \sum_s v_s v_s^T = \frac{D+1}{D} I, \quad b_s(z) = \frac{1 + D \langle v_s, z \rangle}{D+1}$$

are barycentric coordinates for every z in its convex hull. For $\|z\| \leq 1/(2D)$ they are all positive and sum to one. Initialize with weights $b_s(\lambda x)$. For command a use $T_a(s, r) = b_r(\lambda U_a v_s)$; these are stochastic rows, independent of N and of the epoch. Use the terminal mean $d_j(s) = \langle e_j, v_s \rangle$. The displayed barycentric identity shows inductively that the represented mean is $\lambda^{N+1} U_w x$. This proves (41) and its sign assertion. For every unit $U_w x$ some coordinate has modulus at least $1/\sqrt{D}$; binary total variation is half the difference of its means. This gives the numerical error bound. Only the label is stored. A held binary output uses a separate two-label cut. $\square$

Corollary 11.2 (Separation of numerical and sign-only width). *For every nontrivial expanding irreducible experiment in Theorem 7.3, numerical width is of order $N^{s(U)/2}$ at the stated fixed positive accuracy, but strict-cutpoint width in the same reset-command-query interface is at most $D + 1$ for all lengths simultaneously. For complex conjugation the numerical order is N^{q-1} , while the sign-only command width is at most q^2 .*

Proof. Combine Proposition 11.1 with the corresponding numerical width theorem. Here $D = q^2 - 1$ in the last assertion. The number counts command-layer labels, not the total state count of a language recognizer additionally parsing reset and query syntax. $\square$

11.1. The current quantum-automata comparison. We fix the version before comparing its theorem. The primary arXiv submission record inspected on September 26, 2026 lists version 1 dated April 8 and version 2 dated August 27, 2026. In version 1 [24], Proposition 3.1 is the q^2 -dimensional mixed-state linearization, Theorem 3.2 gives $2k + 6$ PFA states from a k -state GFA, and Corollary 3.3 gives $2q^2 + 6$. In the explicitly versioned version 2, the title and theorem numbering change. The PDF's printed pages 4 and 5 contain, respectively, Proposition 3.1 and Theorem 3.2 with the improved constants used below. These are distinct primary versions, not interchangeable theorem numbers. We make no inference about why an earlier access exposed only one version.

Chen–Wu’s version-2 generalized-automaton conversion [23, Proposition 3.1] maps a k -dimensional real zero-cutpoint generalized finite automaton to a probabilistic automaton with $k + 1$ states. For its parameters $c, \delta > 0$ the conversion has

$$(42) \quad f_{\text{PFA}}(w) - 1/(k + 1) = \delta c^{|w|} f_{\text{GFA}}(w).$$

Their Theorem 3.2 consequently gives $q^2 + 1$ states for strict-cutpoint simulation of a q -state general one-way QFA. The second version of that paper improves the larger bound in its original version. Mixed-state linearization is exact before the positive embedding; the factor in (42) is the distinction relevant here. The later paper [25, Proposition 4 and Theorems 11–12] uses the related conversion to compare hybrid quantum and classical language state costs. These are results about cutpoint languages, not fixed nonvanishing numerical accuracy.

For clarity, the algebra behind (42) is that a uniform-row matrix Q annihilates a zero-sum coordinate subspace, while a scaled matrix cK_a carries the desired signed dynamics there. Matrices $Q + cK_a$ are made stochastic by choosing c small. After N commands the signed deviation is multiplied by c^N . A length-dependent decoder attempting to divide by this factor need not lie in $[-1, 1]$ and is not a stochastic readout. More generally, a stochastic post-processing cannot increase total variation. Allowing redesign at every horizon does not overcome the numerical lower bound, which already permits that redesign.

A prefix–suffix sign or rank argument alone cannot substitute for the occupation proof. The matrix of centered target responses has entries $\langle \rho U_p x, U_s^* e_j \rangle$ and ordinary rank at most D at every split, independently of the horizon. Adding the constant probability column gives rank at most $D + 1$. The prepare–test sign-rank and shattering statements in [25] control a threshold relation in this fixed operator space; they do not force the numerical hidden-width growth proved here. Conversely the present argument gives no growing lower bound for the sign-only task, as Proposition 11.1 demonstrates.

Chen’s subsequent work [26], including its invariant-algebra simulation results and Section 7’s stabilizer-orbit capacity analysis, also uses strict-cutpoint behavioral equivalence. Its dimensions and symmetry constraints belong in this comparison. Orbit dimension as $\dim G - \dim G_v$, and quantum operator-space linearization, are not claimed new here. Our invariant first takes the *minimum over all nonzero directions in each irreducible representation*, then the *maximum across active constituents*; it governs a growing fixed-accuracy numerical width as the word length grows. These are different optimization and asymptotic variables. No assertion of exhaustive priority clearance is made from this comparison.

12. GEOMETRIC, PROBABILISTIC AND COMPUTATIONAL BOUNDARIES

The new theorem concerns a growing execution constrained by a single retained label. It neither changes the classical dimension of an orbit nor identifies numerical simulation with a threshold language.

12.1. What the covering exponent does and does not prove. For a smooth compact s -manifold, covering by $O(h^{-s})$ small balls is classical. Iacobelli’s manifold quantization theorem [27, Theorem 1.4] concerns the minimum integrated r th-power distance from a *fixed measure* to a codebook of k points, with its $k^{-r/s}$ asymptotics under the stated hypotheses. The codebook in that problem sees one sample from that measure; there are no stochastic command rows to be implemented at all layers. Theorem 6.2 uses the classical covering count, a quadratic support loss, and an explicit stochastic realization. Its new lower partner is not a static quantization lower bound. The conditional means of a hidden machine generally leave the seed orbit, and neither their distribution nor their orbit type is fixed. Theorem 4.1 controls all those directions at once; the entropy budget then charges every small-state epoch to one finite quadratic loss. This is the mechanism that prevents hidden lifts from outperforming the geometric synthesis order.

Orbit convex hulls and their representation-theoretic geometry are classical objects [21, 28]. The covariance identity and local covering calculation are reproved here to fix constants and hypotheses, not claimed as new geometric discoveries. Our minimum-orbit theorem does not give local volume asymptotics uniformly for each spectrum; only its uniform exponent is sharp. The elementary finite-rank chart argument also explains why no separate spectral projector calculation is needed for a new representation. The earlier projector argument is retained because it gives an explicit, albeit large, constant in its matrix family.

12.2. Positive rank and quantum dimension. At a fixed cut, a numerical classical realization factors the matrix of future response probabilities through K_t positive normalized state continuations. With one output alphabet this is a nonnegative factorization; with several selectable queries, each decoder block must normalize. Quantum prepare–measure probabilities instead admit positive-semidefinite factorizations. These static dimension relations are classical [29]. They do not by themselves supply common transition rows connecting independently chosen factorizations at successive cuts. In the projective experiment a fixed q -dimensional quantum register supplies all layers, while the compatible classical numerical width grows with N . No new quantum operation, seed reset, or quantum-memory model is introduced by that comparison.

Proposition 12.1 (Probabilistic ordered programs). *Definition 2.1 is equivalently a nonuniform ordered read-once layered program with stochastic command-labelled edges and real terminal columns. A selected column can be replaced by a Bernoulli acceptance row without changing any command-layer width; a retained final answer needs at most two labels.*

Proof. Identify $T_{t,a}(s, s')$ with the outgoing edge weight at layer t . Summing path probabilities gives (4), and conversely those edge weights define the stochastic rows. At terminal label s the selected query accepts with probability $(1 + d_j(s))/2$. $\square$

The Fourier-growth and pseudorandomness results in [15, 16, 9] have their stated deterministic, regular or permutation hypotheses and different objectives. The dominant action used by the converse has a norm gap; hidden reset rows and nonregular matrices remain allowed. Width is not total branching-program size, and real edge tables are not a uniform finite-bit implementation.

12.3. Expansion and implementation. Expansion in the present main theorem is a property of the actual command action on an activated irreducible space. It is unchanged by adding unused presenting factors. Theorem 7.3 requires it only on one dominant active type; Proposition 8.1 even allows a faithfully acting circle factor with no regular gap. Full effective-group expansion remains a sufficient certificate via Proposition 3.1. The algebraic alphabets in Proposition C.3 and in the mixed example use the deep external theorems [2, 3]; their constants are qualitative. Density or a finite harmonic calculation is not itself such a certificate. The optional numerical six-gate result uses exactly the separate external inequality (LPS5). Its original-source limitation is recorded in Appendix E.

Only one query is answered. A request for jointly independent answers to all queries changes the channel and need not satisfy the maximum law of Theorem 7.3. The model charges branch identifiers, but not the description length or construction of the row tables. Rational net coordinates do not make all transition weights rational when commands are merely algebraic. Carathéodory sparsity bounds successors, not table size. The separate fair-bit compilers in the preserved development incur further states and possibly error; they do not equate atomic width with bit complexity.

12.4. Conclusions. The effective action and the query-calibrated isotypic activation remove presentation and multiplicity choices from the width theorem. Arbitrary finite seeds first determine their invariant span. A single expanding type of maximal minimum-orbit dimension then determines

the matched exponent; other genuinely acting types may fail to expand. The upper realization charges its branch labels, while the lower bound continues to use the actual entire hidden register.

The root-deletion and singular-value calculations identify minimum orbits on the stated complete representation families. They do not classify all highest weights. Likewise the width law does not specify finite integer optima, sharp leading constants, arbitrary vanishing signals, nongapped dominant types, or uniform computational resources. Numerical probability preservation and strict-cutpoint recognition are separated by an explicit constant-width attenuating machine. The version-pinned comparison makes this semantic boundary independent of which Chen–Wu upper constant is quoted.

APPENDIX A. A MAXIMUM LAW FOR ORTHOGONAL SUMS

Let

$$(43) \quad V = V_0 \oplus V_1 \oplus \cdots \oplus V_m$$

be an orthogonal decomposition, where $V_0 = V^G$ and each V_i , $i \geq 1$, is a nontrivial real irreducible representation. Multiplicities are allowed. Put $d_i = \dim V_i$, $D = \dim V$ and

$$(44) \quad \sigma(U) = \max_{1 \leq i \leq m} s(U|_{V_i}), \quad \sigma(U) = 0 \quad \text{if } m = 0.$$

Equivalent irreducible orthogonal representations have isomorphic orbits. Thus $\sigma(U)$ does not depend on the chosen decomposition of an isotypic summand. In general it is *not* the smallest orbit dimension on the sphere of the whole reducible representation.

For the lower bound assume that the finite unit seeds span V and write

$$(45) \quad a_i = \max_{x \in \mathcal{X}} \|\pi_i x\| > 0, \quad a_* = \min_{1 \leq i \leq m} a_i,$$

where π_i is orthogonal projection; set $a_* = 1$ if $m = 0$. In a basis-adapted seed set $\{\pm e_j\}$ all a_i equal one. The query basis need not respect the decomposition.

Theorem A.1 (Irreducible maximum principle). *Suppose a law on the finite command alphabet has a full group norm gap. Let the seeds span V , let $0 < \rho \leq \min\{1/10, 1/(8D)\}$, and, when $m > 0$, let $0 \leq \varepsilon < \rho a_*/(2\sqrt{D})$. Then*

$$(46) \quad W_{N,\varepsilon} = \Theta(N^{\sigma(U)/2}).$$

The constants may depend on the representation, command law, seed set, signal and error, but not on N . If $m = 0$, an exact constant-width machine exists. The upper realization uses common command rows and only one selected terminal numerical query.

Proof. First fix $i \geq 1$ and a seed x with $\|\pi_i x\| = a_i$. Track the unit-vector process $Y_t^{(i)} = U_{a_t} \cdots U_{a_1} \pi_i x / a_i$ in V_i and its conditional mean given the *actual whole-machine state*. Form the ambient vector $d = (d_1, \dots, d_D)$ from the terminal columns in the proof. It has norm at most $\sqrt{D}$. The prescribed row means and the uniform error give

$$\mathbb{E}\langle Y_N^{(i)}, \pi_i d(S_N) \rangle \geq \rho a_i - 2\sqrt{D}\varepsilon.$$

Conditioning on S_N bounds the left side above by $\sqrt{D}\mathbb{E}\|\mathbb{E}[Y_N^{(i)} | S_N]\|$. Hence the terminal conditional amplitude is at least $\rho a_i / \sqrt{D} - 2\varepsilon > 0$. The inherited entropy occupation argument applies in the representation V_i to these conditional means, even though the hidden label also serves every other constituent. Theorem 4.1 supplies its exponent $s(U|_{V_i})$. It follows that $W_{N,\varepsilon} \geq c_i N^{s(U|_{V_i})/2}$. Choose a constituent attaining (44).

For the upper bound include only nonzero summands and set $w_i = d_i/D$. Choose a branch i at initialization with probability w_i , independently of the seed. The branch identifier is part

of the state label, not a free future input. In branch $i \geq 1$ use the minimal-orbit polytope of Theorem 6.2. Its initial represented mean is prescribed to be

$$(\rho/w_i)\pi_i x.$$

Its norm is at most $\rho D/d_i \leq 1/(8d_i)$, so the initialization proof there applies also to this possibly subunit vector. Subsequent rows are the common command rows from that construction. At a query j output the mean $\langle e_j, a_N v_{i,s} \rangle$. The conditional represented vector in the branch is $(\rho/w_i)U_w \pi_i x$; multiplying by w_i and summing gives exactly $\rho U_w x$. All terminal means are bounded by $2/7$.

In the invariant branch, if present, use the fixed vectors $\{\pm e : e \text{ in an orthonormal basis of } V_0\}$ scaled by $1/4$, with identity transitions. Their convex hull contains a ball of radius $1/(4\sqrt{d_0})$, which contains the required initial vector of norm at most $1/(8d_0)$. Its contribution is exact with $2d_0$ labels. The entire register is the disjoint union of these branch labels. Its cardinality is at most

$$2d_0 + \sum_{i=1}^m C_i N^{s(U|V_i)/2} = O(N^{\sigma(U)/2}).$$

No product of independently correct outputs is required. If $m = 0$ the invariant construction proves the asserted constant bound. The fixed two-label answer cut does not change the orders. $\square$

The original full-group maximum principle is retained here in its original sufficient range. Theorem 7.3 strengthens it by using the active span, canonical activation and only one dominant action gap; its proof of the current main theorem appears in Section 7.

The maximum in (44), rather than a sum, is substantive. For the direct product action of $\text{SO}(3) \times \text{SU}(3)$ on $\mathbb{R}^3 \oplus \text{Herm}_0(3)$, the two least orbit dimensions are two and four. With a full-gap product command law and the above signal conventions the label order is N^2 , not N^3 ; a generic product orbit and a joint-output product task describe different quantities. Adding a fixed subspace creates zero-dimensional whole-space orbits but does not remove this N^2 requirement. The proof explains both phenomena by paid branch labels and a constituent-wise converse.

Proposition A.2 (All separate cuts have constant positive minimum). *For any orthogonal command family on $\mathbb{R}^D$, seeds $\mathcal{X} = \{\pm e_1, \dots, \pm e_D\}$, and $0 < \rho \leq 1/D$, the normalized residual row rank and the separate positive minimum are both $D + 1$ at every internal cut. The separate minimum permits unrestricted memory at the other cuts. The assertion requires no group gap and does not assert simultaneous attainment.*

Proof. For a vector z , its remaining response on suffix v and query j has mean $\langle e_j, U_v z \rangle$. The resulting affine map from z to the normalized response array is injective: fix one suffix and vary the coordinate query; U_v is invertible. At a fixed cut, fix one preceding word w and vary the seeds. The vectors $\{\pm \rho U_w e_j\}$ affinely span $\mathbb{R}^D$. Thus the normalized residual arrays have row rank $D + 1$, giving the positive lower bound.

A regular unit simplex in $\mathbb{R}^D$ contains the ball of radius $1/D$. Every actual represented vector at that cut has norm ρ and is therefore a convex combination of its $D + 1$ vertices. A vertex v_s defines a legal normalized continuation, since $|\langle e_j, U_v v_s \rangle| \leq 1$ for every suffix and query. These are $D + 1$ positive generators of all cut rows. To embed the factorization in a whole machine, retain the complete finite prefix until the selected cut, sample its convex coordinates there, then retain the chosen label together with the remaining command prefix. There is no cap at those other cuts. The resulting finite machine attains $D + 1$ at the selected cut. This verifies the operational, not merely amputated-matrix, meaning of the separate minimum. $\square$

APPENDIX B. UNIFORM GEOMETRY OF HERMITIAN CONJUGATION

Fix $q \geq 2$ and let $V = \text{Herm}_0(q)$ be the real space of traceless Hermitian q by q matrices, with inner product $\text{tr}(AB)$. Its dimension is $D = q^2 - 1$. Unitary conjugation is orthogonal on this space. The pure-state orbit has dimension $2(q-1)$, but normalized hidden centroids can have arbitrary spectra. We therefore prove a bound on *every* unit sphere orbit.

B.1. A Grassmann small-ball estimate. Let P_0 be a rank- k orthogonal projector in $\mathbb{C}^q$, $1 \leq k < q$, and let P be a Haar-distributed rank- k projector. Put $m = k(q-k)$ and

$$(47) \quad b_{q,k} = \frac{1}{m!} \prod_{j=1}^k \frac{\Gamma(q-k+j)}{\Gamma(j)}.$$

Lemma B.1 (Projector balls). *For $0 < R \leq 1$,*

$$(48) \quad \mathbb{P}(\|P - P_0\|_{\text{HS}} \leq R) \leq b_{q,k} R^{2k(q-k)}.$$

Proof. By invariance take P_0 onto the first k coordinates. Apart from a null set, a Haar subspace is the graph of $Z \in \mathbb{C}^{(q-k) \times k}$. Its density with respect to complex Euclidean Lebesgue measure is

$$(49) \quad \pi^{-m} \prod_{j=1}^k \frac{\Gamma(q-k+j)}{\Gamma(j)} \det(I + Z^*Z)^{-q}.$$

For completeness, take a q by k matrix of independent standard complex Gaussian entries and split it into a square top block A and bottom block B . Its column space is Haar and $Z = BA^{-1}$ almost surely. The change $B = ZA$ has real Jacobian $|\det A|^{2(q-k)}$. Integrating A with weight $\exp[-\text{tr} A^*(I + Z^*Z)A]$ gives the determinant factor $\det(I + Z^*Z)^{-q}$. Complex Gaussian Gram-Schmidt gives $\mathbb{E}|\det A|^{2(q-k)} = \prod_{j=1}^k \Gamma(q-k+j)/\Gamma(j)$, since its squared orthogonal lengths have gamma shapes $1, \dots, k$. The powers of π give (49).

If θ_i are the principal angles, then $\|P - P_0\|_{\text{HS}}^2 = 2 \sum_i \sin^2 \theta_i$ and $\|Z\|_{\text{HS}}^2 = \sum_i \tan^2 \theta_i$. On the event in (48) each $\sin^2 \theta_i \leq R^2/2$, whence

$$\|Z\|_{\text{HS}}^2 \leq \frac{R^2}{2 - R^2} \leq R^2.$$

The density (49) is bounded by its value at zero. Multiply this value by the volume $\pi^m R^{2m}/m!$ of a real $2m$ -dimensional ball. This proves the assertion. $\square$

Define the explicit finite constant

$$(50) \quad A_q = \max_{1 \leq k < q} \left\{ \max(1, b_{q,k}) [9q(q-1)^2]^{k(q-k)} \right\}.$$

It is rational and at least one; no optimality is claimed.

Theorem B.2 (All-spectra orbital anti-concentration). *For all $A, B \in \text{Herm}_0(q)$ of Hilbert-Schmidt norm one and all $0 < r \leq 1$,*

$$(51) \quad m_{\text{SU}(q)}\{U : \|UAU^* - B\|_{\text{HS}} \leq r\} \leq A_q r^{2(q-1)}.$$

In particular, the estimate is uniform at repeated eigenvalues and over mixed-state centroid directions. No lower bound on each eigenvalue gap is assumed.

Proof. Let $\lambda_1 \geq \dots \geq \lambda_q$ be the eigenvalues of A . Trace zero and $\sum_i \lambda_i^2 = 1$ imply $\lambda_1 - \lambda_q \geq q^{-1/2}$. Hence some k has

$$(52) \quad \lambda_k - \lambda_{k+1} \geq \delta_q := \frac{1}{\sqrt{q}(q-1)}.$$

For two conjugates C, D of A , let P_C, P_D be their top- k spectral projectors. Eigenvectors u_i of C and v_j of D obey

$$(\lambda_i - \lambda_j)\langle u_i, v_j \rangle = \langle u_i, (C - D)v_j \rangle.$$

Summing squares for $i \leq k < j$ and using equal projector ranks gives

$$(53) \quad \|P_C - P_D\|_{\text{HS}}^2 = 2 \sum_{i \leq k < j} |\langle u_i, v_j \rangle|^2 \leq 2\delta_q^{-2} \|C - D\|_{\text{HS}}^2.$$

If the event in (51) is empty there is nothing to prove. Otherwise choose one conjugate D in that event. Every other C in the event has $\|C - D\|_{\text{HS}} \leq 2r$. The image P_C is a Haar rank- k projector and lies within $2\sqrt{2}\delta_q^{-1}r$ of P_D . Set $c_q = 3\sqrt{q}(q-1)$, which is larger than this coefficient. If $c_q r \leq 1$, Lemma B.1 gives a bound $b_{q,k}(c_q r)^{2k(q-k)}$. Since $k(q-k) \geq q-1$ and $r \leq 1$, this is at most the right side of (51). If $c_q r > 1$, then $A_q r^{2(q-1)} \geq c_q^{2(q-1)} r^{2(q-1)} > 1$, so the trivial probability bound suffices. Haar conjugation under $\text{SU}(q)$ and under $U(q)$ has the same law, completing the proof. $\square$

The proof isolates one uniformly separated spectral cut rather than requiring a simple spectrum. This is why pure-state and highly mixed centroids are handled by the same constant. The smallest possible nontrivial Grassmann exponent is $2(q-1)$.

Corollary B.3 (Conjugation occupation). *For conjugation commands satisfying a group gap g_t , any hidden register with $q_N \geq \kappa > 0$ obeys*

$$(54) \quad \sum_{K_t \leq k} g_t \leq C_q \kappa^{-3-1/(q-1)} k^{1/(q-1)}, \quad C_q = 2048(q^2 - 1)(16A_q)^{1/(q-1)}.$$

For a constant gap $g > 0$ and wordwise error ε with $\kappa = \rho/\sqrt{q^2 - 1} - 2\varepsilon > 0$,

$$(55) \quad W_{N,\varepsilon} \geq \left(\frac{g\kappa^{3+1/(q-1)}}{C_q} N \right)^{q-1}.$$

Proof. Apply Lemma 5.1, Theorem B.2 and Theorem 5.5, then use Lemma 2.2. $\square$

A group gap is stronger than a gap only on functions on the pure-state homogeneous quotient. The proof genuinely needs uniform control of the mixed-centroid directions, and does not silently replace this hypothesis by the weaker quotient gap.

APPENDIX C. SHARP MEMORY ON COMPLEX PROJECTIVE ORBITS

Write $r_q = \sqrt{1 - 1/q}$ and associate to a rank-one orthogonal projector P the unit vector

$$(56) \quad Y(P) = \frac{P - I/q}{r_q} \in \text{Herm}_0(q), \quad \mathcal{K}_q = \text{conv}\{Y(P) : \text{rank } P = 1\}.$$

The body $\mathcal{K}_q$ consists of normalized centered density matrices. It is invariant under conjugation and contains zero. Its extreme orbit is $\mathbb{CP}^{q-1}$. Fix an orthonormal real basis $F_1, \dots, F_D$ of $\text{Herm}_0(q)$ and use its coordinate queries in (3). One may take as seeds the q^2 rank-one projectors onto e_i , $(e_i + e_j)/\sqrt{2}$ and $(e_i + ie_j)/\sqrt{2}$, $i < j$. Their centered matrices affinely span $\text{Herm}_0(q)$. All seed and query alphabets are fixed when q is fixed.

C.1. A projective enclosure with quadratic loss. For rank-one projectors use the distance $d(P, Q) = \sqrt{1 - \text{tr}(PQ)}$, the sine of their principal angle.

Lemma C.1 (Finite projective synthesis). *For $0 < h \leq 1$, there is an h -net $\mathcal{V}_h$ of rank-one projectors with at most*

$$(57) \quad q(2\lceil 2\sqrt{2(q-1)/h} \rceil + 1)^{2(q-1)}$$

elements, and the entries of all these projectors can be chosen in $\mathbb{Q} + i\mathbb{Q}$. If $qh^2 < 1$, then

$$(58) \quad (1 - qh^2)\mathcal{K}_q \subseteq \text{conv}\{Y(P) : P \in \mathcal{V}_h\} \subseteq \mathcal{K}_q.$$

Proof. Every complex line has a representative with a largest coordinate scaled to one and the real and imaginary parts of all other coordinates in $[-1, 1]$. In each of the q charts round these $2(q-1)$ real coordinates to the grid of step $1/M$, where $M = \lceil 2\sqrt{2(q-1)/h} \rceil$. If v, w are the original and rounded representatives, their norms are at least one and $\|v - w\| \leq \sqrt{2(q-1)}/(2M)$. Normalization changes their distance by at most a factor two, and the sine distance is at most the distance of these normalized representatives. Thus the stated set is an h -net. The projector $ww^*/\|w\|^2$ has rational real and imaginary entries, and the grid count is (57).

For a nonzero traceless Hermitian H , let λ_1 and λ_q be its largest and smallest eigenvalues. A net projector within h of a top eigenline satisfies

$$\text{tr}(HP) \geq \lambda_1 - (\lambda_1 - \lambda_q)h^2 \geq (1 - qh^2)\lambda_1.$$

Indeed $-\lambda_q \leq (q-1)\lambda_1$ by trace zero. The support function of $\mathcal{K}_q$ at H is λ_1/r_q . The support-function criterion for convex containment proves (58). It applies to every direction, not just a finite set of tested queries. $\square$

Theorem C.2 (A finite-alphabet exact upper machine). *For every finite unitary command alphabet and $0 < \rho \leq 1/10$, the experiment on (56) has an exact machine of width $O_q(N^{q-1})$. For each N , all command transition matrices can be chosen independent of the epoch. Each row uses at most q^2 successors. The final decoder may depend on N .*

Proof. Take $h = (8qN)^{-1/2}$ and $\eta = 1 - 1/(8N)$ in Lemma C.1. Let V be the resulting finite set of centered vectors, and put $a_t = \eta^{-t}/4$. A label i represents $a_t Y(P_i)$ at epoch t . This is a proof description of a label; no matrix is stored in the register.

Since $\rho/a_0 = 4\rho \leq 2/5 < \eta$ and $0 \in \mathcal{K}_q$, any seed $\rho Y(P)$ is a convex combination of $a_0 V$. For each label i and each command U , the vector $\eta UY(P_i)U^*$ lies in $\text{conv } V$. Choose a convex representation and use its weights as the stochastic transition row. The choices depend on the command and label but not on t . Multiplying by a_{t+1} shows that the expected represented vector is updated by exact conjugation, since $a_{t+1}\eta = a_t$.

Induction gives expected vector $\rho Y(U_w P U_w^*)$. The terminal column for query j is $a_N \text{tr}(F_j Y(P_i))$. It is legal because $|\text{tr}(F_j Y(P_i))| \leq 1$ and Bernoulli's inequality gives $\eta^N \geq 7/8$, so $a_N \leq 2/7$. Sampling a binary answer with this mean proves every conditional probability in (3) exactly. Carathéodory's theorem in real dimension $D = q^2 - 1$ permits q^2 successors. The number in (57) is $O_q(h^{-2(q-1)}) = O_q(N^{q-1})$. $\square$

The upper argument uses a quadratic support loss along the projective orbit. Applying a net on the full ambient sphere would instead give the larger exponent $(q^2 - 2)/2$.

C.2. Fixed finite algebraic alphabets. For $1 \leq j < q$ let $R_j, D_j \in \text{SU}(q)$ act as the identity outside the adjacent coordinate pair $(j, j+1)$ and on that pair as

$$(59) \quad R = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}, \quad D = \begin{pmatrix} \zeta & 0 \\ 0 & \bar{\zeta} \end{pmatrix}, \quad \zeta = \frac{3 + 4i}{5}.$$

The alphabet has $4(q-1) + 1$ symbols: $\{I, R_j^{\pm 1}, D_j^{\pm 1} : 1 \leq j < q\}$. Assign probability $1/2$ to I and $1/[8(q-1)]$ to each other symbol.

Proposition C.3 (Qualitative group expansion). *This command law has a norm gap on $L_0^2(\text{SU}(q))$ for every fixed q . Its entries and probabilities are algebraic. No numerical gap value is asserted for this family.*

Proof. The number ζ is not a root of unity: $\zeta + \zeta^{-1} = 6/5$ is rational but not an algebraic integer. The powers of R_j and D_j therefore have closures containing two distinct one-parameter subgroups in the corresponding embedded $\text{SU}(2)$. Their Lie brackets generate that $\mathfrak{su}(2)$. The adjacent embedded Lie algebras generate $\mathfrak{su}(q)$: adjacent elementary off-diagonal directions produce all other off-diagonal directions by brackets, and their self-brackets produce the traceless diagonal directions. The closed generated subgroup is consequently $\text{SU}(q)$.

The adjoint entries are algebraic. The spectral-gap theorem of Benoist–de Saxcé [2, Theorem 1.2] applies to an adapted algebraic measure on this compact connected group with simple Lie algebra. Equivalently, one may use the algebraic unitary result of Bourgain–Gamburd [3]. The symmetric average has an upper gap; weight $1/2$ at the identity makes its spectrum nonnegative. Its mean-zero operator norm is therefore strictly below one. This is the external full-group theorem, not a calculation in the defining q -dimensional representation. $\square$

Proof of Corollary 10.2. Corollary B.3 gives the lower order, for every fixed positive calibration κ . Theorem C.2 gives the matching exact upper order. Proposition C.3 supplies fixed finite alphabets. The quantum assertion follows from the next proposition. Taking binary logarithms gives the label-bit formula. $\square$

Proposition C.4 (A fixed quantum realization). *The projective experiment has an exact quantum realization on $\mathbb{C}^q$ using fixed seed resets, the prescribed unitary commands and fixed query measurements. It is independent of N .*

Proof. On seed P reset the density matrix to

$$\sigma_P = I/q + \rho Y(P) = (1 - \rho/r_q)I/q + (\rho/r_q)P.$$

This is positive and has trace one, since $0 < \rho \leq 1/10 < r_q$. Commands act by conjugation. For query j , use effects $(I + F_j)/2, (I - F_j)/2$. Hilbert–Schmidt normalization implies $\|F_j\|_{\text{op}} \leq 1$, so these are positive effects summing to I . Their outcome mean is $\text{tr}(F_j \sigma_{U_w P U_w^*}) = \rho \text{tr}(F_j Y(U_w P U_w^*))$. This is precisely (3). The seed reset is a channel, not a unitary; only one query is measured. $\square$

The comparison is a fixed q -dimensional quantum register versus $(q-1) \log_2 N + O(1)$ classical label bits. Dimension or entropy advantages for quantum models are prior phenomena [6, 17, 5]. The assertion here is the proved classical width order for this complete worst-word numerical behavior. It does not request jointly independent answers to all queries or optimize arbitrary adaptive experiments.

APPENDIX D. AN EFFECTIVE SIX-COMMAND RATIONAL EXAMPLE

The preceding group-gap application is qualitative. We now choose a separate arithmetic alphabet for which a classical theorem supplies a numerical infinite-dimensional norm bound. This does not evaluate the gap of the earlier five-command example.

Let R_x, R_y, R_z be rotations about the coordinate axes with cosine $-3/5$ and sine $4/5$; explicitly,

$$(60) \quad R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3/5 & -4/5 \\ 0 & 4/5 & -3/5 \end{pmatrix}, \quad R_y = \begin{pmatrix} -3/5 & 0 & 4/5 \\ 0 & 1 & 0 \\ -4/5 & 0 & -3/5 \end{pmatrix}, \quad R_z = \begin{pmatrix} -3/5 & -4/5 & 0 \\ 4/5 & -3/5 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Use the fixed six-letter alphabet $\mathcal{A}_V = \{R_x^{\pm 1}, R_y^{\pm 1}, R_z^{\pm 1}\}$ and its uniform law. These are the adjoint rotations of the classical norm-five quaternions $1 \pm 2\mathbf{i}$, $1 \pm 2\mathbf{j}$, $1 \pm 2\mathbf{k}$, normalized by $\sqrt{5}$.

Proposition D.1 (External arithmetic gap certificate). *For $P_V f = (1/6) \sum_{R \in \mathcal{A}_V} f(R^{-1} \cdot)$ on $L_0^2(\mathbb{S}^2)$,*

$$(61) \quad \|P_V\| \leq \frac{2\sqrt{5}}{6} = \frac{\sqrt{5}}{3}.$$

Consequently $1 - \|P_V\|^2 \geq 4/9$.

Proof. This is the $p = 5$ specialization of the Lubotzky–Phillips–Sarnak spherical Hecke bound [10, 11]. In its unnormalized convention the $p + 1$ norm- p quaternion representatives act by a sum of rotations whose mean-zero norm is at most $2\sqrt{p}$. The representatives have positive odd real coordinate and even other coordinates when $p \equiv 1 \pmod{4}$; at $p = 5$ they are exactly the six displayed ones. Appendix E records the operator normalization and its source boundary. The coauthor account [12, Section 3.2, the V -gate example] spells out these representatives and their Hecke interpretation. The unnormalized spherical norm convention is also stated in [13, Theorem 3(ii)]. Dividing by six gives (61). Passing from quaternions to adjoint rotations identifies $\mathbb{S}^2$ as the homogeneous quotient; inverse signs do not affect this symmetric average. The arithmetic bound is invoked as an external theorem, not reproved by finite harmonic tests. $\square$

Use six seeds $\{\pm e_j : 1 \leq j \leq 3\}$, amplitude $\rho = 1/10$ and three final coordinate queries in (3).

Theorem D.2 (Numerical finite-horizon width certificate). *For $0 \leq \varepsilon < 1/(20\sqrt{3})$ and every integer $N \geq 1$,*

$$(62) \quad \frac{(1/(10\sqrt{3}) - 2\varepsilon)^4}{221184} N \leq W_{N,\varepsilon}(\mathcal{A}_V) \leq W_{N,0}(\mathcal{A}_V) < 150N.$$

The upper machine has exact rational command and decoder rows, common command matrices, and an N -dependent decoder. At zero error every internal selected cut has separate positive minimum four, while

$$(63) \quad W_{N,0} \geq \frac{N}{19\,906\,560\,000}.$$

The constants are proof constants, not sharp asymptotic constants.

Proof. For the standard $\text{SO}(3)$ action the orbit of every unit vector is uniform on $\mathbb{S}^2$. A chordal cap of radius $r \leq 1$ has probability $r^2/4 \leq r^2$, so take $D = 3, s = 2, A = 1$ in Theorem 5.5. The sphere gap transfers to $L^2(\mathbb{R}^3)$ by polar coordinates. Thus $B_{3,1,2} = 98304$, and the squared gap $4/9$ gives the lower bound in (62). At $\varepsilon = 0$ the fourth power of $1/(10\sqrt{3})$ is $1/90000$, which yields (63).

Here is an explicit cardinality estimate for the upper construction. Take both rational stereographic charts

$$t \mapsto \frac{(2t_1, 2t_2, \pm(1 - \|t\|^2))}{1 + \|t\|^2}, \quad t \in [-1, 1]^2,$$

with a grid of step $1/M$ and $M = \lceil 2\sqrt{2N} \rceil$. Every hemisphere is covered by its chart on the unit disk. Each chart is 2-Lipschitz, so its grid images form a sphere net of chordal radius at most $\sqrt{2}/M \leq 1/(2\sqrt{N})$. Its convex hull contains η times the unit ball, with $\eta = 1 - 1/(8N)$, by the support function identity $u \cdot v = 1 - \|u - v\|^2/2$. There are at most $2(2M + 1)^2 \leq (82 + 48\sqrt{2})N < 150N$ labels. Use scales $a_t = \eta^{-t}/4$, initialization representing ρx , rows representing $\eta R u$ in this hull, and decoder $a_N e_j \cdot u$. The verification is the induction in Theorem C.2. All data are rational; each nonempty rational convex system has rational weights, with at most four successors. The final decoder is bounded by $2/7$.

At a selected internal cut, fix any common command prefix. The six seed images affinely span $\mathbb{R}^3$, so the normalized continuation rank is four; future coordinate queries still separate these dimensions after any common suffix. For an upper factorization take a regular tetrahedron containing the radius- ρ ball and having circumradius $3\rho < 1$. Its four vectors give legal continuations under all future rotations. One embeds this selected-cut factorization in a complete machine by allowing uncapped raw histories elsewhere. Therefore the separate minimum is exactly four. This does not give a four-state machine at all cuts simultaneously. $\square$

For example, (63) exceeds four once $N \geq 79\,626\,240\,001$. The size of this number illustrates the conservative constants; the benefit is that no unspecified spectral constant is needed for this alphabet.

The associated qubit operations are $(I - 2i\sigma_j)/\sqrt{5}$ and their inverses, with seed Bloch vector $x/10$ and final Pauli query. They realize the numerical experiment exactly, independently of N . This quantum construction and the arithmetic gates are classical; (62) is the finite-register consequence. The earlier five-gate alphabet remains qualitatively gapped with a non-evaluated constant, and is retained unchanged in the supporting material.

APPENDIX E. THE ARITHMETIC OPERATOR AND ITS NORMALIZATION

This appendix fixes the exact external input used in Proposition D.1, then proves the remaining representation and normalization steps. Write $\mathbf{i}, \mathbf{j}, \mathbf{k}$ for quaternion units and

$$\mathcal{Q}_5 = \{1 \pm 2\mathbf{i}, 1 \pm 2\mathbf{j}, 1 \pm 2\mathbf{k}\}, \quad r_q(z) = qzq^{-1} \quad (z \in \text{Im } \mathbb{H}).$$

The required external LPS spherical Hecke inequality, specialized to this representative convention, is

$$\text{(LPS5)} \quad \left\| \sum_{q \in \mathcal{Q}_5} f(r_q^{-1} \cdot) \right\|_{L^2(\mathbb{S}^2)} \leq 2\sqrt{5} \|f\|_{L^2(\mathbb{S}^2)}, \quad \int f = 0.$$

This is an imported arithmetic theorem, not a finite-dimensional calculation. The original sources are [10, 11]; the unnormalized norm theorem is explicitly restated in [13, Theorem 3(ii), p. 8 of the preprint], while the quaternionic and Hecke representative conventions are discussed in [12, Section 3.2]. The restated existence theorem alone would not identify an arbitrary six rotations: the representative convention in (LPS5) is essential.

Proposition E.1 (Parity, quotient and full spherical spectrum). *With the external input (LPS5), the uniform average over the six rotations in (60) has mean-zero norm at most $\sqrt{5}/3$ both on $L^2(\mathbb{S}^2)$ and on $L^2(\text{SO}(3))$. Its squared gap is at least $4/9$. The conclusion does not assert the same numerical gap for all of $L_0^2(\text{SU}(2))$.*

Proof. An integral quaternion of norm five has one coordinate of magnitude one, one of magnitude two, and two zero coordinates. There are $4 \cdot 3 \cdot 4 = 48$ such quaternions. Multiplication by a Lipschitz unit in $\{\pm 1, \pm \mathbf{i}, \pm \mathbf{j}, \pm \mathbf{k}\}$ places the odd coordinate in the real position and makes it positive. This gives a unique representative with positive odd real part and even other coordinates, namely one of the six members of $\mathcal{Q}_5$. This specifies the parity class; it is not an average over all 48 quaternions or an extra quotient by their unit group.

Normalizing q to $q/\sqrt{5}$ does not change r_q . Quaternionic conjugation identifies the imaginary unit sphere with $\text{SO}(3)/\text{SO}(2)$, and q and $-q$ give the same rotation. For $(1 + 2\mathbf{i})/\sqrt{5}$ direct multiplication gives

$$\mathbf{i} \mapsto \mathbf{i}, \quad \mathbf{j} \mapsto (-3\mathbf{j} + 4\mathbf{k})/5, \quad \mathbf{k} \mapsto (-4\mathbf{j} - 3\mathbf{k})/5.$$

These are the columns of R_x in (60); cyclic permutation gives R_y, R_z . Conjugating a quaternion changes the sign of its imaginary coordinate and gives its normalized inverse. Thus the six

representatives already include inverse rotations, with one copy each. Division of the sum in (LPS5) by six gives $2\sqrt{5}/6 = \sqrt{5}/3$, and squaring gives $1 - 5/9 = 4/9$.

For the full-group statement, the complexified spherical space is the orthogonal sum of the harmonic spaces $\mathcal{H}_\ell$, $\ell \geq 0$. These are precisely all irreducible unitary representations of $\mathrm{SO}(3)$, one copy of each. The degree-zero space is the constants. The regular representation on $\mathrm{SO}(3)$ contains the same irreducibles with multiplicity $2\ell + 1$. Convolution acts on each copy by the same matrix, so its mean-zero operator norm is the supremum of the same matrix norms on both spaces. No extra invariant subspace is removed: on the sphere the constant is the only $\mathrm{SO}(3)$ -invariant function. The symmetric alphabet makes the inverse/right versus left convention immaterial for these norms. Half-integer-spin representations occur on $\mathrm{SU}(2)$ but not on its $\mathrm{SO}(3)$ quotient; this argument makes no claim about those extra blocks. $\square$

Source boundary. The original LPS publisher records were checked, but their complete 1986/1987 page images were not obtained in this revision. We therefore do not invent an original theorem number or label the secondary restatement as an inspected original proof. Equation (LPS5) states exactly the arithmetic line required, and the proposition proves every subsequent convention change. The effective theorem retains this explicit external dependency. The main width classification is conditional on an arbitrary full gap and does not depend on this numerical example or on an original-page attribution.

REFERENCES

- [1] A. Atalik, A. Köse and M. Gastpar, Differential entropy of the conditional expectation under additive Gaussian noise, *IEEE Trans. Signal Process.* **70** (2022), 4851–4866; arXiv:2106.04677v3. doi:10.1109/TSP.2022.3211403.
- [2] Y. Benoist and N. de Saxcé, A spectral gap theorem in simple Lie groups, *Invent. Math.* **205** (2016), 337–361; arXiv:1405.1808. doi:10.1007/s00222-015-0636-2.
- [3] J. Bourgain and A. Gamburd, A spectral gap theorem in $\mathrm{SU}(d)$, *J. Eur. Math. Soc.* **14** (2012), 1455–1511.
- [4] B. C. Geiger and C. Temmel, Lumpings of Markov chains, entropy rate preservation, and higher-order lumpability, *J. Appl. Probab.* **51** (2014), 1114–1132; extended version arXiv:1212.4375v6.
- [5] F. Ghafari et al., Dimensional quantum memory advantage in the simulation of stochastic processes, *Phys. Rev. X* **9** (2019), 041013.
- [6] M. Gu, K. Wiesner, E. Rieper and V. Vedral, Quantum mechanics can reduce the complexity of classical models, *Nature Commun.* **3** (2012), 762.
- [7] Z. He, C. D. Charalambous and P. A. Stavrou, A new finite-horizon dynamic programming analysis of nonanticipative rate-distortion function for Markov sources, arXiv:2411.11698v1 (2024).
- [8] A. Heller, On stochastic processes derived from Markov chains, *Ann. Math. Statist.* **36** (1965), 1286–1291.
- [9] C. H. Lee, E. Pyne and S. Vadhan, Fourier growth of regular branching programs, *APPROX/RANDOM 2022*, LIPIcs **245**, Art. 2, 2:1–2:21.
- [10] A. Lubotzky, R. Phillips and P. Sarnak, Hecke operators and distributing points on the sphere I, *Comm. Pure Appl. Math.* **39**, suppl. (1986), S149–S186.
- [11] A. Lubotzky, R. Phillips and P. Sarnak, Hecke operators and distributing points on S^2 . II, *Comm. Pure Appl. Math.* **40** (1987), 401–420.
- [12] O. Parzanchevski and P. Sarnak, Super-golden-gates for $\mathrm{PU}(2)$, *Adv. Math.* **327** (2018), 869–901; arXiv:1704.02106v1.
- [13] G. Pisier, Quadratic forms in unitary operators, *Linear Algebra Appl.* **267** (1997), 125–137; arXiv:math/9512207, Theorem 3(ii).
- [14] Y. Polyanskiy and Y. Wu, Wasserstein continuity of entropy and outer bounds for interference channels, *IEEE Trans. Inform. Theory* **62** (2016), 3992–4002.
- [15] O. Reingold, T. Steinke and S. Vadhan, Pseudorandomness for regular branching programs via Fourier analysis, *RANDOM 2013*, LNCS **8096**, 655–670.
- [16] T. Steinke, S. Vadhan and A. Wan, Pseudorandomness and Fourier growth bounds for width 3 branching programs, *Theory Comput.* **13** (2017), Art. 12, 1–50.
- [17] J. Thompson, A. J. P. Garner, V. Vedral and M. Gu, Using quantum theory to simplify input-output processes, *npj Quantum Inform.* **3** (2017), 6.
- [18] M. Vidyasagar, *Hidden Markov Processes: Theory and Applications to Biology*, Princeton University Press, 2014.

- [19] R. G. Wood, T. Linder and S. Yüksel, Optimal zero delay coding of Markov sources: stationary and finite memory codes, *IEEE Trans. Inform. Theory* (2017); arXiv:1606.09135, full revised version. doi:10.1109/TIT.2017.2692215.
- [20] Q. Qi, General Theta Foundations I: Entropy dissipation and sharp finite-alphabet memory, Revision 37 (2026), publication **c5b8015887f83833**. Preserved in the supporting article; cited only as a predecessor, not external validation.
- [21] A. Barvinok, Estimating L^∞ norms by L^{2^k} norms for functions on orbits, preprint (April 2002), Lemma 5. Author's manuscript, 19 pp.
- [22] S. Basu, R. Pollack and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, 2nd ed., Springer, 2006.
- [23] Z. Chen and J. Wu, The state cost of classical simulation of one-way general quantum finite automata, arXiv:2604.07058v2, submitted August 27, 2026, Proposition 3.1 and Theorem 3.2, printed pp. 4–5.
- [24] Z. Chen and J. Wu, The quadratic state cost of classical simulation of one-way quantum finite automata, arXiv:2604.07058v1, submitted April 8, 2026, Proposition 3.1, Theorem 3.2 and Corollary 3.3.
- [25] Z. Chen and J. Wu, On the simulation cost of quantum finite automata, arXiv:2605.10682v1 (2026).
- [26] Z. Chen, Behavioral memory under symmetry in one-way quantum automata, arXiv:2609.01451v1 (2026).
- [27] M. Iacobelli, Asymptotic quantization for probability measures on Riemannian manifolds, arXiv:1412.4026v1 (2014), Theorem 1.4 and Section 2.
- [28] R. Sanyal, F. Sottile and B. Sturmfels, Orbitopes, *Mathematika* **57** (2011), 275–314; arXiv:0911.5436.
- [29] H. Fawzi, J. Gouveia, P. A. Parrilo, R. Z. Robinson and R. R. Thomas, Positive semidefinite rank, *Math. Program.* **153** (2015), 133–177; arXiv:1407.4095.
- [30] Q. Qi, General Theta Foundations I: Orbit geometry and entropy budgets for finite memory, Revision 38 (2026), publication **8ec190c921656b98**. The unchanged supporting article is a predecessor, not external validation.
- [31] P. Etingof, *Lie Groups and Lie Algebras*, MIT OpenCourseWare, 18.755, Spring 2024, open textbook; Sections 21–23, 27, 30–32 and 35–36.
- [32] D. C. Youla, A normal form for a matrix under the unitary congruence group, *Canad. J. Math.* **13** (1961), 694–704. doi:10.4153/CJM-1961-059-8.

PRESERVED MATHEMATICAL DEVELOPMENT

General Theta Foundations I - revision 39

The complete current article precedes this divider.

The following predecessor cumulative volume is reproduced without alteration.

Frozen predecessor publication:

28890c62dd69f217bf2f1c205ff9542eb537c2ef

This archive is not part of the compact referee package. Independent historical analytic obligations keep their original hypotheses and proof status.

GENERAL THETA FOUNDATIONS I: MINIMAL ORBITS AND THE WIDTH OF NUMERICAL EXPERIMENTS

QIAN QI

ABSTRACT. For a compact connected Lie group acting orthogonally on a fixed finite-dimensional space, we identify the hidden-label width exponent of its numerical command experiment under a full group norm gap. In an irreducible representation, let s be the least dimension of a nonzero unit-vector orbit. We prove a uniform orbital small-ball bound of exponent s across every stabilizer type. Combined with an orbital entropy occupation budget and a compatible minimal-orbit construction, this gives sharp width $\Theta(N^{s/2})$ at fixed positive signal and fixed sufficiently small wordwise error. For reducible representations and spanning seeds at small fixed signal, the exponent is one half the maximum of the least orbit dimensions of the nontrivial irreducible constituents. The upper machine charges its randomized branch label and uses common command rows; the lower bound permits arbitrary hidden directions and time-dependent rows. We distinguish this numerical law from strict-cutpoint simulation by an explicit constant-width sign simulator with attenuated probabilities. The projective and arithmetic qubit results retain their original scope. The counted resource is nonuniform clocked atomic-row label width; tables, arithmetic and horizon-specific design are not charged.

1. INTRODUCTION

A low-dimensional linear statistic may have a small positive realization at every selected time and still require a large hidden register to follow a whole command sequence. The issue is compatibility: one stochastic update must serve all pasts that share a label. We study this constraint for numerical matrix coefficients of compact-group representations. The input alphabet is finite and fixed; the time horizon grows. Each numerical response is required on every input word, with one selected coordinate query at the end.

Our main result identifies a representation invariant rather than an analyst-chosen family of geometric witnesses. Let a compact connected Lie group act orthogonally on V , and decompose V as its invariant subspace and real irreducible constituents V_i . Define

$$s_i = \min_{v \in \mathbb{S}(V_i)} \dim(Gv), \quad \sigma(U) = \max_i s_i,$$

with $\sigma(U) = 0$ when there is no nontrivial constituent. Equivalent constituents have the same s_i , so this definition is independent of the orthogonal decomposition. It is the smallest orbit within each constituent, followed by a maximum across constituents, not the dimension of a generic orbit or of one chosen seed orbit.

Theorem 1.1 (Minimum orbits and numerical width). *Let G be a compact connected Lie group, $U : G \rightarrow O(\mathbb{R}^D)$ a fixed representation, and $\mathcal{A} \subset G$ a fixed finite alphabet supporting a probability law with full mean-zero $L^2(G)$ norm strictly below one. Let the finite unit seeds span V , and let $0 < \rho \leq \min\{1/10, 1/(8D)\}$. For every fixed $0 \leq \varepsilon < \rho a_*/(2\sqrt{D})$, with the positive activation*

Date: September 25, 2026. Revision 39.

2020 Mathematics Subject Classification. Primary 93B15, 22D10; Secondary 94A17, 68Q10, 81P45.

Key words and phrases. Positive realization, compact group, orbit dimension, entropy, hidden memory, quantum automata, numerical simulation.

constant a_* in (25) when a nontrivial constituent is present,

$$(1) \quad W_{N,\varepsilon} = \Theta_{G,U,\mathcal{A},p,\mathcal{X},\rho,\varepsilon}(N^{\sigma(U)/2}).$$

Here width means the maximum number of available labels in a nonuniform clocked atomic-row machine of Definition 2.1. A new machine may be chosen for every N . The lower bound allows arbitrary hidden states and cut-dependent rows. The upper bound is exact and uses common command rows with an N -dependent decoder. For a purely invariant representation an exact constant-width machine exists. In a nontrivial irreducible representation, any nonempty unit seed set suffices; if its seeds lie on a minimum-dimensional orbit, the amplitude range extends to $0 < \rho \leq 1/10$.

For conjugation on complex traceless Hermitian matrices, $s = 2(q - 1)$ and the theorem recovers N^{q-1} labels while preserving the earlier projective seed and signal conventions. It also gives $N^{(q-1)/2}$ and $N^{2(q-1)}$ for the real and quaternionic self-adjoint families under the corresponding full gap. On every fixed nontrivial spherical-harmonic representation of $\mathrm{SO}(3)$ the least orbit dimension is two, giving linear label growth regardless of the harmonic degree. The representation dimension is held fixed in each asymptotic statement. For a reducible action the exponent can remain unchanged after adding a large invariant component or many copies of an easier constituent.

1.1. The proof mechanism. The geometric input is now derived, rather than assumed separately for each application. A uniform rank bound for the differential of $(g, u) \mapsto U_g u$, followed by a finite chart covering, gives

$$\sup_{u,v} m_G \{g : \|U_g u - v\| \leq r\} \leq A r^{s(U)}.$$

This holds across all stabilizer types, including directions of mixed hidden conditional means. The orbital entropy occupation theorem then turns it into a cumulative bound on small-register epochs. That entropy theorem, proved in the preceding revision and reproduced here, is the dynamic converse input.

For the other direction choose a least-dimensional orbit, not a net on the ambient sphere. Its convex hull contains a fixed ball by irreducible orthogonality. At a support maximum the tangential first derivative vanishes, so an h -net loses only $O(h^2)$ in support. Taking $h \asymp N^{-1/2}$ gives exact compatible stochastic updates with $O(N^{s/2})$ labels. A randomized branch construction, with the branch identity charged inside the label, proves the maximum rule for reducible representations. One cannot replace this construction by a free selector or by a claim about joint-output products.

1.2. The numerical objective. Recent one-way quantum-automata simulation theorems use strict-cutpoint language equivalence [23, 24, 25]. Their positive embedding may attenuate a signed matrix coefficient exponentially with word length. Proposition 8.1 gives an explicit constant-label sign simulator for our own task, whereas Theorem 1.1 requires growing width to preserve its numerical probabilities at fixed accuracy. This comparison does not claim that exact mixed-state linearization or classical quantization geometry is new.

Symbol	Meaning
D, ρ, ε	Fixed real dimension, signal, binary-TV error
$K_t, W_{N,\varepsilon}$	Available labels at a cut; minimum peak
$s(U), \sigma(U)$	Least irreducible orbit dimension; constituent maximum
Z_t, q_t, Δ_t	Conditional centroid, amplitude, quadratic loss
g_t, Π	Squared norm gap; Haar-invariant projection
$C, \mathcal{V}_h$	Minimal-orbit convex hull; finite orbit net

The new conclusions are the uniform minimum-orbit concentration principle, the sharp representation width law and its reducible maximum rule. The entropy occupation argument and the quantitative projective and six-gate results are retained with their proofs. Classical orbit geometry, covering dimensions, stochastic positive embedding and external spectral gaps are explicitly attributed. For symmetric basis seeds, every separate internal cut still has positive minimum $D + 1$ (Proposition 6.2); the growing width is therefore a simultaneous compatibility cost. No program description, arithmetic cost or independent quantum query product is included in the label count.

2. THE EXPERIMENT AND ITS POSITIVE REALIZATION

Let G be a compact group, let $U : G \rightarrow O(V)$ be a continuous orthogonal representation on $V = \mathbb{R}^D$, and fix a finite command alphabet $\mathcal{A} \subset G$. A finite seed set $\mathcal{X} \subset \mathbb{S}(V)$ is given initially. For $w = (a_1, \dots, a_N)$ write $U_w = U_{a_N} \cdots U_{a_1}$. Relative to an orthonormal basis $e_1, \dots, e_D$, the prescribed numerical experiment is

$$(2) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, U_w x \rangle}{2}, \quad b \in \{-1, 1\}, \quad j \in [D], \quad 0 < \rho \leq 1/10.$$

All seeds, words and queries are required, not only almost-sure words under one source law. The probabilities are between $(1 - \rho)/2$ and $(1 + \rho)/2$.

Definition 2.1 (Clocked atomic-row realization). A horizon- N machine consists of finite sets $\mathcal{S}_t$ with $|\mathcal{S}_t| = K_t$, a stochastic initialization E_x on $\mathcal{S}_0$, stochastic matrices $T_{t,a} : \mathcal{S}_t \rightarrow \mathcal{S}_{t+1}$, and columns $d_j : \mathcal{S}_N \rightarrow [-1, 1]$. Its mean response is

$$(3) \quad E_x T_{0,a_1} \cdots T_{N-1,a_N} d_j.$$

It has error at most ε if this differs from $\rho\langle e_j, U_w x \rangle$ by at most 2ε for every (x, w, j) . Available labels, including padding, count. All rows, including unreachable ones, are normalized. Only the updated label persists. The seed, word prefix, old random choices and any data-dependent protocol information cannot be consulted again unless represented by that label. The external time, horizon, fixed program and exact sampling from its rows are free. The final binary answer may be sampled when j arrives; if retained, it uses a two-label final cut. Write $W_{N,\varepsilon}$ for the minimum of $\max(2, K_0, \dots, K_N)$.

The state alphabets and matrices may be chosen anew for each N , but the command symbols and their experimental meanings are fixed. Nothing requires one autonomous device or one decoder to serve all stopping lengths. No adaptive choice of the command is optimized in this model. A complex unitary representation can be treated by its realification; all inner products in the entropy argument are real.

Fix one seed x and an independent command law at each epoch for the converse. Let $Y_t = U_{a_t} \cdots U_{a_1} x$ and let S_t be the actual machine state under this test law. Define

$$(4) \quad Z_t = \mathbb{E}[Y_t \mid S_t], \quad q_t = \mathbb{E}\|Z_t\|, \quad v_t = \mathbb{E}\|Z_t\|^2.$$

These random vectors and their laws are analytical devices, not an additional runtime register or oracle.

Lemma 2.2 (Quadratic budget and terminal calibration). *For the fresh command a , put $X_t = U_a Z_t$. Then*

$$(5) \quad Z_{t+1} = \mathbb{E}[X_t \mid S_{t+1}], \quad \Delta_t := \mathbb{E}\|X_t - Z_{t+1}\|^2 = v_t - v_{t+1}, \quad \sum_{t < N} \Delta_t \leq 1.$$

The sequences q_t, v_t are nonincreasing and start at one. A machine with error at most ε satisfies

$$(6) \quad q_t \geq q_N \geq \kappa := \rho/\sqrt{D} - 2\varepsilon \quad (0 \leq t \leq N).$$

Proof. The current command is independent of the past, and the new label is generated from (S_t, a) by a common row. Conditioning first on (S_t, a, S_{t+1}) and then on S_{t+1} proves the first identity. The orthogonal projection identity for conditional expectation gives the second; Jensen and orthogonality give both monotonicities. The fixed seed has unit norm, even with randomized initialization. For calibration form $d = (d_1, \dots, d_D)$ only in the proof, so $\|d\| \leq \sqrt{D}$. Conditioning on the complete word and using (2)–(3) gives $\mathbb{E}\langle Y_N, d(S_N) \rangle \geq \rho - 2\sqrt{D}\varepsilon$. On conditioning instead on S_N , this is at most $\sqrt{D}q_N$. This proves (6). It does not require a joint measurement of the D coordinates. $\square$

For each fixed profile, the optimum actual error is attained: all stochastic rows and terminal columns form a finite product of compact sets and the objective is a maximum of finitely many continuous functions. Thus the subsequent lower bounds concern the actual optimization, not a sum of local error certificates.

3. THE SMALLEST ORBIT CONTROLS UNIFORM CONCENTRATION

In this section G is a compact connected Lie group, $U : G \rightarrow O(V)$, and $V^G = \{0\}$. Put

$$(7) \quad s(U) = \min_{u \in \mathbb{S}(V)} \dim(Gu).$$

The minimum is over *all* directions. It can be smaller than the dimension of a generic orbit. Since G is connected, $s(U) \geq 1$.

Theorem 3.1 (Uniform concentration across orbit types). *There is a finite constant $A_U \geq 1$ such that*

$$(8) \quad \sup_{u, v \in \mathbb{S}(V)} m_G\{g : \|U_g u - v\| \leq r\} \leq A_U r^{s(U)} \quad (0 < r \leq 1).$$

This exponent is optimal among uniform power upper bounds: for a unit u_0 with $\dim(Gu_0) = s(U)$ there are $c, r_0 > 0$ such that

$$(9) \quad m_G\{g : \|U_g u_0 - u_0\| \leq r\} \geq c r^{s(U)} \quad (0 < r < r_0).$$

No regular-orbit, constant-stabilizer, or simple-spectrum hypothesis is needed in (8).

Proof. Write $m = \dim G$, $s = s(U)$ and let $\mathfrak{g}$ be the Lie algebra. For $u \in \mathbb{S}(V)$ the differential

$$A_u : \mathfrak{g} \longrightarrow V, \quad X \longmapsto dU(X)u$$

has rank $\dim(Gu) \geq s$. Its s th singular value is continuous in u and is strictly positive on the compact sphere. Its minimum is therefore positive. We give a chart argument that also records why variation of the stabilizer causes no loss in the exponent.

Fix $(g_0, u_0) \in G \times \mathbb{S}(V)$. In a smooth coordinate chart $g = \psi(z, w)$, where $z \in \mathbb{R}^s$ and $w \in \mathbb{R}^{m-s}$, select s coordinate directions and a linear projection $L : V \rightarrow \mathbb{R}^s$ for which

$$D_z[LU_{\psi(z, w)}u]_{(g_0, u_0)} = M$$

is invertible. This is possible by the rank bound. By shrinking to a product of bounded open rectangles $Z \times W$ and a neighborhood J of u_0 , arrange

$$\|D_z[LU_{\psi(z, w)}u] - M\|_{\text{op}} \leq \frac{1}{2}\sigma_{\min}(M) \quad (z, w, u) \in Z \times W \times J.$$

Take Z convex. For fixed (w, u) , integration along the segment from z to z' gives

$$(10) \quad \|LU_{\psi(z, w)}u - LU_{\psi(z', w)}u\| \geq \frac{1}{2}\sigma_{\min}(M)\|z - z'\|.$$

If both points lie in the inverse image of the radius- r ball around v , their projected distance is at most $2\|L\|r$. This inverse image, on the fixed w slice, consequently has diameter at most $4\|L\|r/\sigma_{\min}(M)$, and hence s -dimensional Lebesgue measure at most a constant times r^s . Haar

measure has a bounded smooth density in the chosen relatively compact chart. Integration in the remaining $m - s$ coordinates yields the same bound there.

Choose finitely many of these product neighborhoods covering $G \times \mathbb{S}(V)$. For each fixed u , those whose J contains u cover G ; summing the preceding bounds proves (8). All constants are independent of the center v . An empty inverse image contributes zero. This use of a finite cover, rather than a separate density estimate on every orbit, is what controls neighborhoods of singular orbit types.

For the converse, $Gu_0 \simeq G/G_{u_0}$ is a compact smooth embedded s -dimensional submanifold. The pushforward of Haar probability is its normalized invariant Riemannian volume: the induced metric is G -invariant and the invariant probability on this homogeneous space is unique. A smooth local graph at u_0 has volume comparable to r^s in a small ambient ball. This proves (9). If a uniform estimate with exponent $s' > s$ held, comparison at u_0 as $r \downarrow 0$ would contradict it. $\square$

The rank computation in (7) is intrinsic to the representation. For example, given algebraic skew matrices $B_1, \dots, B_m$ spanning its infinitesimal action, the condition $s(U) \leq r$ is the existence of a real vector u with $\sum u_i^2 = 1$ and all $(r+1)$ -minors of $[B_1 u \cdots B_m u]$ zero. Thus it is an explicit real-algebraic decision problem for fully specified algebraic data; quantifier elimination applies [22]. This is not a test for a spectral gap, and no efficient algorithm in a succinct representation is asserted.

Corollary 3.2 (Intrinsic occupation exponent). *Suppose the representation satisfies (13). In the setting of Lemma 2.2, $q_N \geq \kappa > 0$ implies*

$$(11) \quad \sum_{\substack{0 \leq t \leq N \\ K_t \leq k}} g_t \leq 2048 D (16 A_U)^{2/s(U)} \kappa^{-3-2/s(U)} k^{2/s(U)}.$$

The estimate allows arbitrary hidden continuations and imposes no bound on the other cuts.

Proof. Apply Theorem 4.5 with the uniform estimate (8). In particular the conditional means may change orbit type at every step; their directions still satisfy the same estimate. $\square$

4. AN ORBITAL ENTROPY OCCUPATION THEOREM

Lebesgue measure is used on $V = \mathbb{R}^D$ and normalized Haar measure on G . For a function on V , set

$$(12) \quad \mathcal{U}_a f(z) = f(U_a^{-1} z), \quad P_t = \sum_a p_t(a) \mathcal{U}_a, \quad \Pi f(z) = \int_G f(U_g^{-1} z) dm_G(g).$$

Thus $\mathcal{U}_a^* = \mathcal{U}_{a^{-1}}$, Π is an orthogonal projection, and $P_t \Pi = \Pi P_t = \Pi$. The probability law p_t need not be symmetric. Its required squared norm gap is

$$(13) \quad \|P_t f\|_2^2 \leq \|\Pi f\|_2^2 + (1 - g_t) \|f - \Pi f\|_2^2, \quad 0 \leq g_t \leq 1.$$

Invariance in this formula means G -invariance, not necessarily radial symmetry. In a reducible representation invariant functions can carry many parameters.

Lemma 4.1 (Transfer of a group norm gap). *If right convolution $R_{p_t} F(g) = \sum_a p_t(a) F(ga)$ on $L^2(G)$ has mean-zero norm at most $\sqrt{1 - g_t}$, then (13) holds in every orthogonal representation on V .*

Proof. For each z apply the group inequality to $F_z(g) = f(U_g^{-1} z)$. Its mean is $\Pi f(z)$ and $R_{p_t} F_z(g) = P_t f(U_g^{-1} z)$. Integrate the resulting inequality in z . Orthogonal changes of variables and Fubini identify the three integrals with those in (13). First take bounded compactly supported f and then use density. This proof does not replace the full group gap by a gap in the defining representation. $\square$

4.1. The conditional-mean cost. Let G_D be a standard independent real Gaussian vector and let γ_τ be the density of τG_D .

Lemma 4.2 (A one-sided Gaussian coupling inequality). *For bounded X and a finite-valued variable J , put $M = \mathbb{E}[X \mid J]$. For every $\tau > 0$,*

$$(14) \quad \mathbf{h}(X + \tau G_D) - \mathbf{h}(M + \tau G_D) \leq \frac{\mathbb{E}\|X - M\|^2}{2\tau^2}.$$

Proof. Write $f = \text{Law}(M) * \gamma_\tau$. Differentiation of the Gaussian mixture and the conditional covariance identity give

$$(15) \quad \nabla^2 \log f(y) = -\tau^{-2}I + \tau^{-4} \text{Cov}(M \mid M + \tau G_D = y) \succeq -\tau^{-2}I.$$

Hence $\phi(x) = -\mathbb{E} \log f(x + \tau G_D)$ has Hessian at most $\tau^{-2}I$. Taylor's inequality at M yields $\phi(X) \leq \phi(M) + \langle \nabla \phi(M), X - M \rangle + \|X - M\|^2 / (2\tau^2)$. The conditional expectation of the linear term given J is zero. If $g = \text{Law}(X) * \gamma_\tau$, nonnegativity of $D(g \parallel f)$ gives $\mathbf{h}(g) \leq -\int g \log f = \mathbb{E} \phi(X)$, whereas $\mathbb{E} \phi(M) = \mathbf{h}(f)$. Boundedness of X, M provides Gaussian bounds with finite second moments and justifies differentiation and integration. These identities prove (14). $\square$

This elementary consequence of Gaussian posterior identities is used as an estimate, not as a claim of priority over the Gaussian estimation literature [1, 14]. The estimate is one-sided and can have a negative left side. Here J can be a finite register with no Gaussian observation model; the smoothing noise is added after conditioning.

Lemma 4.3 (Entropy production away from invariant functions). *For any bounded-support Gaussian mixture density f ,*

$$(16) \quad \mathbf{h}(P_t f) - \mathbf{h}(f) \geq g_t \|\sqrt{f} - \Pi \sqrt{f}\|_2^2 \geq 0.$$

Proof. Put $f_a = \mathcal{U}_a f$ and $h = P_t f$. Orthogonality preserves entropy, so $\mathbf{h}(h) - \mathbf{h}(f) = \sum_a p_t(a) D(f_a \parallel h)$. For probability densities u, v , $D(u \parallel v) \geq \|\sqrt{u} - \sqrt{v}\|_2^2$; it follows, for example, from $-\log x \geq 2(1 - \sqrt{x})$ and integration. The mean is the least-squares minimizer, and thus

$$\sum_a p_t(a) \|\sqrt{f_a} - \sqrt{h}\|_2^2 \geq \sum_a p_t(a) \|\sqrt{f_a} - P_t \sqrt{f}\|_2^2 = 1 - \|P_t \sqrt{f}\|_2^2.$$

Apply (13) and $\|\sqrt{f}\|_2 = 1$. $\square$

No logarithmic Sobolev inequality or point-mass mixing assertion is used in this step.

4.2. Sparsity away from invariant densities. Suppose that the representation obeys the uniform orbital bound

$$(17) \quad m_G\{g : \|U_g u - v\| \leq r\} \leq A r^s \quad (u, v \in \mathbb{S}(V), 0 < r \leq 1),$$

where $A \geq 1$ and $s > 0$. In particular the representation has no nonzero fixed vector. It is essential that u range over all possible centroid directions, not merely the initial seed orbit.

Lemma 4.4 (Finite centroids cannot be almost invariant). *Suppose Z has at most k values in the unit ball and $\mathbb{E}\|Z\| \geq \kappa$, where $0 < \kappa \leq 1$. Set*

$$(18) \quad r = \left(\frac{\kappa}{16Ak} \right)^{1/s}, \quad \tau = \frac{\kappa r}{8\sqrt{2D}}, \quad f = \text{Law}(Z) * \gamma_\tau.$$

Then $\|\sqrt{f} - \Pi \sqrt{f}\|_2^2 \geq \kappa/16$.

Proof. Atoms of norm at least $\kappa/2$ have total mass at least $\kappa/2$. Let Ω be the union of radius- r caps around their directions, and let C_Ω be its cone. By Markov's inequality, $\mathbb{P}(\|G_D\| \leq \sqrt{2D}) \geq 1/2$. On this event, perturbing a high-norm atom by τG_D changes its direction by at most $4\tau\sqrt{2D}/\kappa = r/2$. Therefore $\int_{C_\Omega} f \geq \kappa/4$.

For any G -invariant $u \in L^2(V)$, Haar averaging and (17) give

$$\int_{C_\Omega} |u(z)|^2 dz = \int |u(z)|^2 \int_G \mathbf{1}_{C_\Omega}(U_g z) dm_G(g) dz \leq k A r^s \|u\|_2^2.$$

The point $z = 0$ is Lebesgue-null. Taking $u = \Pi\sqrt{f}$, whose norm is at most one, bounds this integral by $\kappa/16$. The reverse triangle inequality in $L^2(C_\Omega)$ now gives a norm difference at least $\sqrt{\kappa/4} - \sqrt{\kappa/16} = \sqrt{\kappa}/4$. Repeated directions and zero centroids cause no change. $\square$

Theorem 4.5 (Orbital occupation budget). *Under (13) and (17), let a finite hidden register track a unit-vector command process as in (4). If $q_N \geq \kappa > 0$, then, for every $k \geq 1$,*

$$(19) \quad \sum_{\substack{0 \leq t < N \\ \bar{K}_t \leq k}} g_t \leq B_{D,A,s} \kappa^{-3-2/s} k^{2/s}, \quad B_{D,A,s} = 2048 D (16A)^{2/s}.$$

The same inequality holds with K_t replaced by the number of positive-probability labels under the chosen test law. No bound is imposed on the other epochs.

Proof. Fix k and the single scale (18) for the entire run. Write $f_t = \text{Law}(Z_t) * \gamma_\tau$. Isotropy of the Gaussian makes the density of $U_a Z_t + \tau G_D$ equal to $P_t f_t$. By Lemmas 4.2 and 4.3,

$$\mathbf{h}(f_{t+1}) - \mathbf{h}(f_t) \geq g_t \|\sqrt{f_t} - \Pi\sqrt{f_t}\|_2^2 - \frac{\Delta_t}{2\tau^2}.$$

At an epoch with at most k labels, Lemma 4.4 bounds the norm term below by $\kappa/16$. Elsewhere it is nonnegative. The initial density is a translate of γ_τ ; the terminal second moment is at most $1 + D\tau^2$. The Gaussian maximum-entropy bound and (5) yield

$$\frac{\kappa}{16} \sum_{K_t \leq k} g_t \leq \frac{D}{2} \log \left(1 + \frac{1}{D\tau^2} \right) + \frac{1}{2\tau^2} \leq \frac{1}{\tau^2}.$$

Substitute (18). The same Gaussian scale is retained through all epochs; no repeated mixing-time charge is incurred. $\square$

Corollary 4.6 (Actual terminal error). *For a constant gap $g > 0$, let $\mathcal{E}_N(\mathbf{K})$ be the minimum actual wordwise binary-TV error over a prescribed available profile, and put $K = \max_t K_t$. Then*

$$(20) \quad \mathcal{E}_N(\mathbf{K}) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{D}} - \left(\frac{B_{D,A,s} K^{2/s}}{gN} \right)^{1/(3+2/s)} \right]_+.$$

In particular, $K = o(N^{s/2})$ implies $\liminf \mathcal{E}_N \geq \rho/(2\sqrt{D})$.

Proof. If the actual error is below the calibration threshold, combine (6) with (19) and solve for it. Otherwise the displayed lower bound already holds. Attainment or a minimizing sequence gives the same conclusion for the optimum. $\square$

Local simulation errors may cancel. Nothing in this corollary sums such errors or rules out cancellation; its objective is the actual final law.

5. EXACT SYNTHESIS FROM A MINIMAL ORBIT

Assume in this section that V is a nontrivial real irreducible orthogonal representation, of dimension d . Choose a unit vector v_0 with orbit dimension $s = s(U)$, and set

$$\mathcal{O} = Gv_0, \quad C = \text{conv } \mathcal{O}.$$

The orbit may be a singular orbit in the unit sphere. As a homogeneous space in its own right it is nevertheless a smooth compact manifold.

Lemma 5.1 (Inradius and quadratic support approximation). *One has $d^{-1}\mathbb{B}(V) \subseteq C \subseteq \mathbb{B}(V)$. There are $c, C_0, h_0 > 0$, depending on U and v_0 , such that for $0 < h \leq h_0$ a set $\mathcal{V}_h \subset \mathcal{O}$ with $|\mathcal{V}_h| \leq C_0 h^{-s}$ satisfies*

$$(21) \quad (1 - ch^2)C \subseteq \text{conv } \mathcal{V}_h \subseteq C \quad \text{whenever } ch^2 < 1.$$

Proof. The mean of $U_g v_0$ is an invariant vector and is zero. Its covariance is I/d : the covariance is self-adjoint and commutes with the representation, so its eigenspaces are invariant; irreducibility and trace one give the claim. This is the real Schur orthogonality calculation, for example [21, Lemma 5]. For unit H , let $f(g) = \langle H, U_g v_0 \rangle$ and $M = \max_g f(g)$. Since $-1 \leq f \leq M$,

$$0 \leq \mathbb{E}[(M - f)(f + 1)] = M - \mathbb{E}f^2 = M - 1/d.$$

Every support function of C is thus at least $1/d$, which proves the inradius statement.

Equip $\mathcal{O}$ with its induced Riemannian metric. Finitely many bounded smooth coordinate patches, with their inverses uniformly Lipschitz on smaller patches, give an intrinsic h -net with at most $C_0 h^{-s}$ points. At a maximizer y of $z \mapsto \langle H, z \rangle$ on $\mathcal{O}$, its tangential first derivative is zero. The second fundamental form of the compact embedded orbit is bounded. Along a unit-speed minimizing geodesic of length at most h , Taylor's formula therefore gives

$$\langle H, y \rangle - \langle H, v \rangle \leq L \|H\| h^2$$

for a nearby net point v , with L independent of H, y . The inradius bound gives $\|H\| \leq dh_C(H)$. Hence $h_{\text{conv } \mathcal{V}_h}(H) \geq (1 - Ldh^2)h_C(H)$ for every H . Taking $c \geq Ld$, and using the support-function criterion, proves (21). $\square$

The covering exponent s and quadratic loss at a support maximum are geometric facts, not new quantization exponents. What is needed for realization is an enclosure compatible with *every command* and *every layer*; the following construction supplies it.

Theorem 5.2 (A minimal-orbit positive realization). *For any finite command alphabet, any finite unit seed set, and $0 < \rho \leq 1/(8d)$, the experiment (2) has an exact realization with at most $C_U N^{s(U)/2}$ available labels, for $N \geq 1$. All command matrices can be chosen independent of the epoch; the final decoder may depend on N . Each row has at most $d + 1$ successors. If the seeds belong to $C \cap \mathbb{S}(V)$, the amplitude range improves to $0 < \rho \leq 1/10$.*

Proof. Choose

$$h_N = \min\{h_0, (8cN)^{-1/2}\}, \quad \eta = 1 - ch_N^2 \geq 1 - 1/(8N), \quad a_t = \eta^{-t}/4.$$

Let $v_1, \dots, v_M$ be the net points. The represented vector of a label i at layer t is $a_t v_i$. These vectors describe the fixed rows; they are not stored as real-valued runtime data.

The polytope $\text{conv}\{v_i\}$ contains ηC and hence a ball of radius η/d . For a unit seed x , $4\rho x \in \eta C$ if $\rho \leq 1/(8d)$, because $4\rho \leq 1/(2d) < \eta/d$. Choose its convex coordinates as the initial distribution. More generally, if $x \in C$ and $4\rho \leq 2/5 < \eta$, the same initialization is valid. For each command a and label i , invariance of C gives

$$\eta U_a v_i \in \eta C \subseteq \text{conv}\{v_j\}.$$

Choose any convex coordinates of this point for the row $T_a(i, \cdot)$. They do not depend on t . Since $a_{t+1}\eta = a_t$, conditional expectation updates the represented vector by exactly U_a . Induction gives mean $\rho U_w x$ after each complete prefix.

Use the terminal column $d_j(i) = a_N \langle e_j, v_i \rangle$. Bernoulli's inequality gives $\eta^N \geq 7/8$, so $|d_j(i)| \leq 2/7 < 1$. This proves every numerical conditional law exactly. Carathéodory's theorem gives $d+1$ successors. Finally $M \leq C_0 h_N^{-s} \leq C_U N^{s/2}$, with the fixed two-state output cut absorbed into the constant. Neither convex feasibility nor row lookup is priced as computation in this model. $\square$

Corollary 5.3 (The irreducible width law). *Suppose a probability law on the finite command alphabet has a full mean-zero group norm gap. For a nontrivial real irreducible V , any nonempty finite unit seed set, fixed $0 < \rho \leq 1/(8d)$, and $0 \leq \varepsilon < \rho/(2\sqrt{d})$,*

$$(22) \quad W_{N,\varepsilon} = \Theta_{U,p,\rho,\varepsilon}(N^{s(U)/2}).$$

The same order holds for exact vector-state realizations and for fixed-length realizations with common command rows. For seeds on a minimal orbit the amplitude range is $0 < \rho \leq 1/10$.

Proof. A full group norm gap transfers to (13) with a fixed $g > 0$. Apply Corollary 3.2 at the peak k , using $\kappa = \rho/\sqrt{d} - 2\varepsilon > 0$. It gives $gN \leq B\kappa^{-3-2/s}k^{2/s}$ and therefore the lower order. The exact machine in Theorem 5.2 is an upper bound for every allowed error, is vector-valued in its representation, and has common command rows. These subclasses lie inside the unrestricted clocked class, so the lower bound applies to them too. The last assertion uses the stronger initialization in that theorem. $\square$

6. A MAXIMUM LAW FOR ORTHOGONAL SUMS

Let

$$(23) \quad V = V_0 \oplus V_1 \oplus \cdots \oplus V_m$$

be an orthogonal decomposition, where $V_0 = V^G$ and each V_i , $i \geq 1$, is a nontrivial real irreducible representation. Multiplicities are allowed. Put $d_i = \dim V_i$, $D = \dim V$ and

$$(24) \quad \sigma(U) = \max_{1 \leq i \leq m} s(U|_{V_i}), \quad \sigma(U) = 0 \quad \text{if } m = 0.$$

Equivalent irreducible orthogonal representations have isomorphic orbits. Thus $\sigma(U)$ does not depend on the chosen decomposition of an isotypic summand. In general it is *not* the smallest orbit dimension on the sphere of the whole reducible representation.

For the lower bound assume that the finite unit seeds span V and write

$$(25) \quad a_i = \max_{x \in \mathcal{X}} \|\pi_i x\| > 0, \quad a_* = \min_{1 \leq i \leq m} a_i,$$

where π_i is orthogonal projection; set $a_* = 1$ if $m = 0$. In a basis-adapted seed set $\{\pm e_j\}$ all a_i equal one. The query basis need not respect the decomposition.

Theorem 6.1 (Irreducible maximum principle). *Suppose a law on the finite command alphabet has a full group norm gap. Let the seeds span V , let $0 < \rho \leq \min\{1/10, 1/(8D)\}$, and, when $m > 0$, let $0 \leq \varepsilon < \rho a_*/(2\sqrt{D})$. Then*

$$(26) \quad W_{N,\varepsilon} = \Theta(N^{\sigma(U)/2}).$$

The constants may depend on the representation, command law, seed set, signal and error, but not on N . If $m = 0$, an exact constant-width machine exists. The upper realization uses common command rows and only one selected terminal numerical query.

Proof. First fix $i \geq 1$ and a seed x with $\|\pi_i x\| = a_i$. Track the unit-vector process $Y_t^{(i)} = U_{a_t} \cdots U_{a_1} \pi_i x / a_i$ in V_i and its conditional mean given the *actual whole-machine state*. Form the ambient vector $d = (d_1, \dots, d_D)$ from the terminal columns in the proof. It has norm at most $\sqrt{D}$. The prescribed row means and the uniform error give

$$\mathbb{E}\langle Y_N^{(i)}, \pi_i d(S_N) \rangle \geq \rho a_i - 2\sqrt{D}\varepsilon.$$

Conditioning on S_N bounds the left side above by $\sqrt{D}\mathbb{E}\|Y_N^{(i)} \mid S_N\|$. Hence the terminal conditional amplitude is at least $\rho a_i / \sqrt{D} - 2\varepsilon > 0$. The inherited entropy occupation argument applies in the representation V_i to these conditional means, even though the hidden label also serves every other constituent. Theorem 3.1 supplies its exponent $s(U|V_i)$. It follows that $W_{N,\varepsilon} \geq c_i N^{s(U|V_i)/2}$. Choose a constituent attaining (24).

For the upper bound include only nonzero summands and set $w_i = d_i/D$. Choose a branch i at initialization with probability w_i , independently of the seed. The branch identifier is part of the state label, not a free future input. In branch $i \geq 1$ use the minimal-orbit polytope of Theorem 5.2. Its initial represented mean is prescribed to be

$$(\rho/w_i)\pi_i x.$$

Its norm is at most $\rho D/d_i \leq 1/(8d_i)$, so the initialization proof there applies also to this possibly subunit vector. Subsequent rows are the common command rows from that construction. At a query j output the mean $\langle e_j, a_N v_{i,s} \rangle$. The conditional represented vector in the branch is $(\rho/w_i)U_w \pi_i x$; multiplying by w_i and summing gives exactly $\rho U_w x$. All terminal means are bounded by $2/7$.

In the invariant branch, if present, use the fixed vectors $\{\pm e : e \text{ in an orthonormal basis of } V_0\}$ scaled by $1/4$, with identity transitions. Their convex hull contains a ball of radius $1/(4\sqrt{d_0})$, which contains the required initial vector of norm at most $1/(8d_0)$. Its contribution is exact with $2d_0$ labels. The entire register is the disjoint union of these branch labels. Its cardinality is at most

$$2d_0 + \sum_{i=1}^m C_i N^{s(U|V_i)/2} = O(N^{\sigma(U)/2}).$$

No product of independently correct outputs is required. If $m = 0$ the invariant construction proves the asserted constant bound. The fixed two-label answer cut does not change the orders. $\square$

Proof of Theorem 1.1. Theorem 6.1 proves the stated general law. Corollary 5.3 gives its irreducible and minimal-orbit signal refinements. $\square$

The maximum in (24), rather than a sum, is substantive. For the direct product action of $\text{SO}(3) \times \text{SU}(3)$ on $\mathbb{R}^3 \oplus \text{Herm}_0(3)$, the two least orbit dimensions are two and four. With a full-gap product command law and the above signal conventions the label order is N^2 , not N^3 ; a generic product orbit and a joint-output product task describe different quantities. Adding a fixed subspace creates zero-dimensional whole-space orbits but does not remove this N^2 requirement. The proof explains both phenomena by paid branch labels and a constituent-wise converse.

Proposition 6.2 (All separate cuts have constant positive minimum). *For any orthogonal command family on $\mathbb{R}^D$, seeds $\mathcal{X} = \{\pm e_1, \dots, \pm e_D\}$, and $0 < \rho \leq 1/D$, the normalized residual row rank and the separate positive minimum are both $D + 1$ at every internal cut. The separate minimum permits unrestricted memory at the other cuts. The assertion requires no group gap and does not assert simultaneous attainment.*

Proof. For a vector z , its remaining response on suffix v and query j has mean $\langle e_j, U_v z \rangle$. The resulting affine map from z to the normalized response array is injective: fix one suffix and vary the coordinate query; U_v is invertible. At a fixed cut, fix one preceding word w and vary the

seeds. The vectors $\{\pm \rho U_w e_j\}$ affinely span $\mathbb{R}^D$. Thus the normalized residual arrays have row rank $D + 1$, giving the positive lower bound.

A regular unit simplex in $\mathbb{R}^D$ contains the ball of radius $1/D$. Every actual represented vector at that cut has norm ρ and is therefore a convex combination of its $D + 1$ vertices. A vertex v_s defines a legal normalized continuation, since $|\langle e_j, U_v v_s \rangle| \leq 1$ for every suffix and query. These are $D + 1$ positive generators of all cut rows. To embed the factorization in a whole machine, retain the complete finite prefix until the selected cut, sample its convex coordinates there, then retain the chosen label together with the remaining command prefix. There is no cap at those other cuts. The resulting finite machine attains $D + 1$ at the selected cut. This verifies the operational, not merely amputated-matrix, meaning of the separate minimum. $\square$

7. MATRIX AND HARMONIC REPRESENTATIONS

Theorem 3.1 identifies the correct geometric exponent without a separate estimate for every spectral type. We record its principal matrix consequences. The orbit dimensions below are classical; their role is to determine the dynamic width exponent.

Proposition 7.1 (Three conjugation families). *On the real space of traceless self-adjoint q by q matrices over $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{H}$, with inner product $\text{Re tr}(AB)$, let $\beta = \dim_{\mathbb{R}} \mathbb{F}$. The compact groups $\text{SO}(q), \text{SU}(q), \text{Sp}(q)$, respectively, act irreducibly for $q \geq 2$. The least nonzero unit-orbit dimension is $\beta(q - 1)$. Accordingly, under a full group gap, the width order in Corollary 5.3 is*

$$(27) \quad N^{(q-1)/2}, \quad N^{q-1}, \quad N^{2(q-1)},$$

respectively, in its fixed signal and fixed error ranges. The $\text{SO}(2)$ case supplies no finitely supported full-gap example.

Proof. A self-adjoint matrix with distinct eigenvalue multiplicities $m_1, \dots, m_r$ has orbit dimension

$$\beta \sum_{i < j} m_i m_j = \frac{\beta}{2} \left(q^2 - \sum_i m_i^2 \right).$$

For real matrices subtract the dimensions of the orthogonal stabilizer blocks from $q(q - 1)/2$; their disconnected components do not affect dimension. For complex matrices subtract $\sum m_i^2 - 1$ from $q^2 - 1$. For quaternionic matrices subtract $\sum m_i(2m_i + 1)$ from $q(2q + 1)$. These computations also identify the least orbits as real, complex and quaternionic projective spaces. A nonzero traceless matrix has at least two eigenvalues. The largest possible $\sum m_i^2$ under that restriction is $(q - 1)^2 + 1$, attained by multiplicities $(1, q - 1)$. This proves the minimum.

For completeness, the conjugation representation is irreducible over $\mathbb{R}$. Any nonzero invariant real subspace contains, after diagonalizing a nonzero member, a nonconstant traceless real diagonal matrix. Differences of its conjugates under coordinate interchanges give all traceless diagonal directions. Rotations in a pair of coordinate axes then give every self-adjoint off-diagonal direction; in the complex and quaternionic cases, diagonal unit phases give each real component of those entries. Signed interchanges and plane rotations can be chosen in the stated special groups. Thus the subspace is the entire traceless self-adjoint space. The general width law applies. Finally a finite measure on a torus has Fourier multipliers arbitrarily close to one by simultaneous approximation, so it has no full norm gap on its mean-zero L^2 space. $\square$

For complex matrices the projective seeds in (39) belong to a least-dimensional orbit. The signal need not be reduced to $1/(8(q^2 - 1))$ in this specialization.

Corollary 7.2 (The preserved projective width law). *Let $q \geq 2$, $0 < \rho \leq 1/10$, and $0 \leq \varepsilon < \rho/(2\sqrt{q^2 - 1})$ be fixed. For the finite projective seeds and coordinate queries of Section B, any*

finite command alphabet supporting a full group norm gap has

$$W_{N,\varepsilon} = \Theta(N^{q-1}).$$

It has the fixed q -dimensional quantum realization of Proposition B.4. The explicit quantitative all-spectra estimate and the rational-projector net from the preceding revision are retained in Appendices A and B.

Proposition 7.3 (Every fixed spherical harmonic degree). *Let V_ℓ be the real irreducible $\mathrm{SO}(3)$ representation of harmonic homogeneous polynomials of degree $\ell \geq 1$, with $\dim V_\ell = 2\ell + 1$. Then $s(V_\ell) = 2$. Consequently every fixed ℓ has width $\Theta(N)$ under a full-gap command law, with the signal and query conventions of Corollary 5.3. The same six-command arithmetic alphabet in Appendix C supplies such a gap for all these representations.*

Proof. A nonzero vector has no three-dimensional stabilizer, since the representation is nontrivial irreducible. The Lie algebra $\mathfrak{so}(3)$ has no two-dimensional subalgebra: identifying its bracket with a cross product, two independent elements bracket to a third independent direction. The stabilizer dimension is therefore at most one, so every orbit has dimension at least two. A zonal spherical harmonic is fixed by the circle of rotations about its axis and has a two-dimensional orbit. This gives the minimum. The final assertion follows from Proposition D.1: the spherical norm bound controls all nontrivial $\mathrm{SO}(3)$ irreducibles, not just the defining three-dimensional action. $\square$

The degree is fixed before $N \rightarrow \infty$; no uniform-in- ℓ constants are claimed. Generic harmonic orbits can be three-dimensional, so their dimension would give the wrong exponent. In the real degree-two representation this is already the distinction between generic three-dimensional matrix orbits and the two-dimensional projective orbit. These examples use the same general minimum-orbit theorem, not separate orbital cap computations.

8. NUMERICAL REALIZATION IS NOT CUTPOINT SIMULATION

A numerical simulator must preserve the values in (2), whereas a strict-cutpoint simulator need only preserve whether the answer probability is larger than $1/2$. The following construction makes the difference explicit in the present command model. Its stochastic embedding mechanism is classical; it is the quantitative width theorem that prevents numerical exactification at bounded width.

Proposition 8.1 (A constant-width sign simulator). *For any orthogonal action on $\mathbb{R}^D$, $D \geq 2$, there is a machine with $D + 1$ command-layer labels, common command rows, and one common query decoder for all lengths such that*

$$(28) \quad \tilde{\mathbb{P}}(B = 1 \mid x, w, j) - \frac{1}{2} = \frac{1}{2} \lambda^{N+1} \langle e_j, U_w x \rangle, \quad \lambda = 1/(2D).$$

It preserves every strict comparison of (2) with $1/2$, including equality. If $\lambda^{N+1} \leq \rho$, its worst numerical binary-TV error is at least $(\rho - \lambda^{N+1})/(2\sqrt{D})$.

Proof. Let $v_0, \dots, v_D$ be a regular simplex on the unit sphere of $\mathbb{R}^D$, centered at zero. Then

$$\sum_s v_s = 0, \quad \sum_s v_s v_s^T = \frac{D+1}{D} I, \quad b_s(z) = \frac{1 + D \langle v_s, z \rangle}{D+1}$$

are barycentric coordinates for every z in its convex hull. For $\|z\| \leq 1/(2D)$ they are all positive and sum to one. Initialize with weights $b_s(\lambda x)$. For command a use $T_a(s, r) = b_r(\lambda U_a v_s)$; these are stochastic rows, independent of N and of the epoch. Use the terminal mean $d_j(s) = \langle e_j, v_s \rangle$. The displayed barycentric identity shows inductively that the represented mean is $\lambda^{N+1} U_w x$. This proves (28) and its sign assertion. For every unit $U_w x$ some coordinate has modulus at least $1/\sqrt{D}$; binary total variation is half the difference of its means. This gives the numerical error bound. Only the label is stored. A held binary output uses a separate two-label cut. $\square$

Corollary 8.2 (Separation of numerical and sign-only width). *For every nontrivial expanding irreducible experiment in Corollary 5.3, numerical width is of order $N^{s(U)/2}$ at the stated fixed positive accuracy, but strict-cutpoint width in the same reset-command-query interface is at most $D + 1$ for all lengths simultaneously. For complex conjugation the numerical order is N^{q-1} , while the sign-only command width is at most q^2 .*

Proof. Combine Proposition 8.1 with the corresponding numerical width theorem. Here $D = q^2 - 1$ in the last assertion. The number counts command-layer labels, not the total state count of a language recognizer additionally parsing reset and query syntax. $\square$

8.1. The current quantum-automata comparison. Chen–Wu’s revised generalized-automaton conversion [23, Proposition 3.1] maps a k -dimensional real zero-cutpoint generalized finite automaton to a probabilistic automaton with $k + 1$ states. For its parameters $c, \delta > 0$ the conversion has

$$(29) \quad f_{\text{PFA}}(w) - 1/(k + 1) = \delta c^{|w|} f_{\text{GFA}}(w).$$

Their Theorem 3.2 consequently gives $q^2 + 1$ states for strict-cutpoint simulation of a q -state general one-way QFA. The second version of that paper improves the larger bound in its original version. Mixed-state linearization is exact before the positive embedding; the factor in (29) is the distinction relevant here. The later paper [24, Proposition 4 and Theorems 11–12] uses the related conversion to compare hybrid quantum and classical language state costs. These are results about cutpoint languages, not fixed nonvanishing numerical accuracy.

For clarity, the algebra behind (29) is that a uniform-row matrix Q annihilates a zero-sum coordinate subspace, while a scaled matrix cK_a carries the desired signed dynamics there. Matrices $Q + cK_a$ are made stochastic by choosing c small. After N commands the signed deviation is multiplied by c^N . A length-dependent decoder attempting to divide by this factor need not lie in $[-1, 1]$ and is not a stochastic readout. More generally, a stochastic post-processing cannot increase total variation. Allowing redesign at every horizon does not overcome the numerical lower bound, which already permits that redesign.

A prefix-suffix sign or rank argument alone cannot substitute for the occupation proof. The matrix of centered target responses has entries $\langle \rho U_p x, U_s^* e_j \rangle$ and ordinary rank at most D at every split, independently of the horizon. Adding the constant probability column gives rank at most $D + 1$. The prepare-test sign-rank and shattering statements in [24] control a threshold relation in this fixed operator space; they do not force the numerical hidden-width growth proved here. Conversely the present argument gives no growing lower bound for the sign-only task, as Proposition 8.1 demonstrates.

Chen’s subsequent work [25], including its invariant-algebra simulation results and Section 7’s stabilizer-orbit capacity analysis, also uses strict-cutpoint behavioral equivalence. Its dimensions and symmetry constraints belong in this comparison. Orbit dimension as $\dim G - \dim G_v$, and quantum operator-space linearization, are not claimed new here. Our invariant first takes the *minimum over all nonzero directions in each irreducible representation*, then the *maximum across active constituents*; it governs a growing fixed-accuracy numerical width as the word length grows. These are different optimization and asymptotic variables. No assertion of exhaustive priority clearance is made from this comparison.

9. GEOMETRIC, PROBABILISTIC AND COMPUTATIONAL BOUNDARIES

The new theorem concerns a growing execution constrained by a single retained label. It neither changes the classical dimension of an orbit nor identifies numerical simulation with a threshold language.

9.1. What the covering exponent does and does not prove. For a smooth compact s -manifold, covering by $O(h^{-s})$ small balls is classical. Iacobelli’s manifold quantization theorem [26, Theorem 1.4] concerns the minimum integrated r th-power distance from a *fixed measure* to a codebook of k points, with its $k^{-r/s}$ asymptotics under the stated hypotheses. The codebook in that problem sees one sample from that measure; there are no stochastic command rows to be implemented at all layers. Theorem 5.2 uses the classical covering count, a quadratic support loss, and an explicit stochastic realization. Its new lower partner is not a static quantization lower bound. The conditional means of a hidden machine generally leave the seed orbit, and neither their distribution nor their orbit type is fixed. Theorem 3.1 controls all those directions at once; the entropy budget then charges every small-state epoch to one finite quadratic loss. This is the mechanism that prevents hidden lifts from outperforming the geometric synthesis order.

Orbit convex hulls and their representation-theoretic geometry are classical objects [21, 27]. The covariance identity and local covering calculation are reproved here to fix constants and hypotheses, not claimed as new geometric discoveries. Our minimum-orbit theorem does not give local volume asymptotics uniformly for each spectrum; only its uniform exponent is sharp. The elementary finite-rank chart argument also explains why no separate spectral projector calculation is needed for a new representation. The earlier projector argument is retained because it gives an explicit, albeit large, constant in its matrix family.

9.2. Positive rank and quantum dimension. At a fixed cut, a numerical classical realization factors the matrix of future response probabilities through K_t positive normalized state continuations. With one output alphabet this is a nonnegative factorization; with several selectable queries, each decoder block must normalize. Quantum prepare–measure probabilities instead admit positive-semidefinite factorizations. These static dimension relations are classical [28]. They do not by themselves supply common transition rows connecting independently chosen factorizations at successive cuts. In the projective experiment a fixed q -dimensional quantum register supplies all layers, while the compatible classical numerical width grows with N . No new quantum operation, seed reset, or quantum-memory model is introduced by that comparison.

Proposition 9.1 (Probabilistic ordered programs). *Definition 2.1 is equivalently a nonuniform ordered read-once layered program with stochastic command-labelled edges and real terminal columns. A selected column can be replaced by a Bernoulli acceptance row without changing any command-layer width; a retained final answer needs at most two labels.*

Proof. Identify $T_{t,a}(s, s')$ with the outgoing edge weight at layer t . Summing path probabilities gives (3), and conversely those edge weights define the stochastic rows. At terminal label s the selected query accepts with probability $(1 + d_j(s))/2$. $\square$

The Fourier-growth and pseudorandomness results in [15, 16, 9] have their stated deterministic, regular or permutation hypotheses and different objectives. Our external group law has a norm gap; hidden reset rows and nonregular matrices remain allowed. Width is not total branching-program size, and real edge tables are not a uniform finite-bit implementation.

9.3. Expansion and implementation. Full group expansion is an explicit hypothesis, not a consequence of orbit dimension, density alone, or a calculation in the defining representation. The algebraic alphabets in Proposition B.3 use the deep external theorems [2, 3]; their gap constants remain qualitative. The numerical six-gate certificate uses the different external line (LPS5). It is not certified by a finite harmonic truncation. For example the circle has positive-dimensional orbits but admits no finite-command full group norm gap, and its earlier arithmetic width theorem is not an instance of the present hypothesis.

Only one query is answered. A request for jointly independent answers to all queries changes the channel and need not satisfy the maximum law of Theorem 6.1. The model charges branch

identifiers, but not the description length or construction of the row tables. Rational net coordinates do not make all transition weights rational when commands are merely algebraic. Carathéodory sparsity bounds successors, not table size. The separate fair-bit compilers in the preserved development incur further states and possibly error; they do not equate atomic width with bit complexity.

9.4. Conclusions. Under a full compact-group gap and the stated nonvanishing numerical calibration, the minimum nontrivial orbit dimension of each irreducible constituent determines the sharp hidden-width exponent. For spanning seed sets at small fixed signal, the largest of those exponents determines the whole representation, with an exact common-row upper realization. The uniform small-ball theorem and paid randomized-sum construction replace the former matrix-specific geometric verification by a representation-level conclusion.

The theorem is not an all-alphabet classification. It does not optimize leading constants, numerical gap values, uniform program size, variable-signal limits, or arbitrary experimental action selection. Strict-cutpoint simulation is a different task and has a constant-width implementation in the same interface. These boundaries are consequences of the definitions and explicit constructions, not an identification of several resource models.

APPENDIX A. UNIFORM GEOMETRY OF HERMITIAN CONJUGATION

Fix $q \geq 2$ and let $V = \text{Herm}_0(q)$ be the real space of traceless Hermitian q by q matrices, with inner product $\text{tr}(AB)$. Its dimension is $D = q^2 - 1$. Unitary conjugation is orthogonal on this space. The pure-state orbit has dimension $2(q - 1)$, but normalized hidden centroids can have arbitrary spectra. We therefore prove a bound on *every* unit sphere orbit.

A.1. A Grassmann small-ball estimate. Let P_0 be a rank- k orthogonal projector in $\mathbb{C}^q$, $1 \leq k < q$, and let P be a Haar-distributed rank- k projector. Put $m = k(q - k)$ and

$$(30) \quad b_{q,k} = \frac{1}{m!} \prod_{j=1}^k \frac{\Gamma(q - k + j)}{\Gamma(j)}.$$

Lemma A.1 (Projector balls). *For $0 < R \leq 1$,*

$$(31) \quad \mathbb{P}(\|P - P_0\|_{\text{HS}} \leq R) \leq b_{q,k} R^{2k(q-k)}.$$

Proof. By invariance take P_0 onto the first k coordinates. Apart from a null set, a Haar subspace is the graph of $Z \in \mathbb{C}^{(q-k) \times k}$. Its density with respect to complex Euclidean Lebesgue measure is

$$(32) \quad \pi^{-m} \prod_{j=1}^k \frac{\Gamma(q - k + j)}{\Gamma(j)} \det(I + Z^* Z)^{-q}.$$

For completeness, take a q by k matrix of independent standard complex Gaussian entries and split it into a square top block A and bottom block B . Its column space is Haar and $Z = BA^{-1}$ almost surely. The change $B = ZA$ has real Jacobian $|\det A|^{2(q-k)}$. Integrating A with weight $\exp[-\text{tr } A^*(I + Z^* Z)A]$ gives the determinant factor $\det(I + Z^* Z)^{-q}$. Complex Gaussian Gram-Schmidt gives $\mathbb{E}|\det A|^{2(q-k)} = \prod_{j=1}^k \Gamma(q - k + j)/\Gamma(j)$, since its squared orthogonal lengths have gamma shapes $1, \dots, k$. The powers of π give (32).

If θ_i are the principal angles, then $\|P - P_0\|_{\text{HS}}^2 = 2 \sum_i \sin^2 \theta_i$ and $\|Z\|_{\text{HS}}^2 = \sum_i \tan^2 \theta_i$. On the event in (31) each $\sin^2 \theta_i \leq R^2/2$, whence

$$\|Z\|_{\text{HS}}^2 \leq \frac{R^2}{2 - R^2} \leq R^2.$$

The density (32) is bounded by its value at zero. Multiply this value by the volume $\pi^m R^{2m}/m!$ of a real $2m$ -dimensional ball. This proves the assertion. $\square$

Define the explicit finite constant

$$(33) \quad A_q = \max_{1 \leq k < q} \left\{ \max(1, b_{q,k}) [9q(q-1)^2]^{k(q-k)} \right\}.$$

It is rational and at least one; no optimality is claimed.

Theorem A.2 (All-spectra orbital anti-concentration). *For all $A, B \in \text{Herm}_0(q)$ of Hilbert–Schmidt norm one and all $0 < r \leq 1$,*

$$(34) \quad m_{\text{SU}(q)}\{U : \|UAU^* - B\|_{\text{HS}} \leq r\} \leq A_q r^{2(q-1)}.$$

In particular, the estimate is uniform at repeated eigenvalues and over mixed-state centroid directions. No lower bound on each eigenvalue gap is assumed.

Proof. Let $\lambda_1 \geq \dots \geq \lambda_q$ be the eigenvalues of A . Trace zero and $\sum_i \lambda_i^2 = 1$ imply $\lambda_1 - \lambda_q \geq q^{-1/2}$. Hence some k has

$$(35) \quad \lambda_k - \lambda_{k+1} \geq \delta_q := \frac{1}{\sqrt{q}(q-1)}.$$

For two conjugates C, D of A , let P_C, P_D be their top- k spectral projectors. Eigenvectors u_i of C and v_j of D obey

$$(\lambda_i - \lambda_j) \langle u_i, v_j \rangle = \langle u_i, (C - D)v_j \rangle.$$

Summing squares for $i \leq k < j$ and using equal projector ranks gives

$$(36) \quad \|P_C - P_D\|_{\text{HS}}^2 = 2 \sum_{i \leq k < j} |\langle u_i, v_j \rangle|^2 \leq 2\delta_q^{-2} \|C - D\|_{\text{HS}}^2.$$

If the event in (34) is empty there is nothing to prove. Otherwise choose one conjugate D in that event. Every other C in the event has $\|C - D\|_{\text{HS}} \leq 2r$. The image P_C is a Haar rank- k projector and lies within $2\sqrt{2}\delta_q^{-1}r$ of P_D . Set $c_q = 3\sqrt{q}(q-1)$, which is larger than this coefficient. If $c_q r \leq 1$, Lemma A.1 gives a bound $b_{q,k}(c_q r)^{2k(q-k)}$. Since $k(q-k) \geq q-1$ and $r \leq 1$, this is at most the right side of (34). If $c_q r > 1$, then $A_q r^{2(q-1)} \geq c_q^{2(q-1)} r^{2(q-1)} > 1$, so the trivial probability bound suffices. Haar conjugation under $\text{SU}(q)$ and under $U(q)$ has the same law, completing the proof. $\square$

The proof isolates one uniformly separated spectral cut rather than requiring a simple spectrum. This is why pure-state and highly mixed centroids are handled by the same constant. The smallest possible nontrivial Grassmann exponent is $2(q-1)$.

Corollary A.3 (Conjugation occupation). *For conjugation commands satisfying a group gap g_t , any hidden register with $q_N \geq \kappa > 0$ obeys*

$$(37) \quad \sum_{K_t \leq k} g_t \leq C_q \kappa^{-3-1/(q-1)} k^{1/(q-1)}, \quad C_q = 2048(q^2 - 1)(16A_q)^{1/(q-1)}.$$

For a constant gap $g > 0$ and wordwise error ε with $\kappa = \rho/\sqrt{q^2 - 1} - 2\varepsilon > 0$,

$$(38) \quad W_{N,\varepsilon} \geq \left(\frac{g\kappa^{3+1/(q-1)}}{C_q} N \right)^{q-1}.$$

Proof. Apply Lemma 4.1, Theorem A.2 and Theorem 4.5, then use Lemma 2.2. $\square$

A group gap is stronger than a gap only on functions on the pure-state homogeneous quotient. The proof genuinely needs uniform control of the mixed-centroid directions, and does not silently replace this hypothesis by the weaker quotient gap.

APPENDIX B. SHARP MEMORY ON COMPLEX PROJECTIVE ORBITS

Write $r_q = \sqrt{1 - 1/q}$ and associate to a rank-one orthogonal projector P the unit vector

$$(39) \quad Y(P) = \frac{P - I/q}{r_q} \in \text{Herm}_0(q), \quad \mathcal{K}_q = \text{conv}\{Y(P) : \text{rank } P = 1\}.$$

The body $\mathcal{K}_q$ consists of normalized centered density matrices. It is invariant under conjugation and contains zero. Its extreme orbit is $\mathbb{CP}^{q-1}$. Fix an orthonormal real basis $F_1, \dots, F_D$ of $\text{Herm}_0(q)$ and use its coordinate queries in (2). One may take as seeds the q^2 rank-one projectors onto e_i , $(e_i + e_j)/\sqrt{2}$ and $(e_i + ie_j)/\sqrt{2}$, $i < j$. Their centered matrices affinely span $\text{Herm}_0(q)$. All seed and query alphabets are fixed when q is fixed.

B.1. A projective enclosure with quadratic loss. For rank-one projectors use the distance $d(P, Q) = \sqrt{1 - \text{tr}(PQ)}$, the sine of their principal angle.

Lemma B.1 (Finite projective synthesis). *For $0 < h \leq 1$, there is an h -net $\mathcal{V}_h$ of rank-one projectors with at most*

$$(40) \quad q(2\lceil 2\sqrt{2(q-1)}/h \rceil + 1)^{2(q-1)}$$

elements, and the entries of all these projectors can be chosen in $\mathbb{Q} + i\mathbb{Q}$. If $qh^2 < 1$, then

$$(41) \quad (1 - qh^2)\mathcal{K}_q \subseteq \text{conv}\{Y(P) : P \in \mathcal{V}_h\} \subseteq \mathcal{K}_q.$$

Proof. Every complex line has a representative with a largest coordinate scaled to one and the real and imaginary parts of all other coordinates in $[-1, 1]$. In each of the q charts round these $2(q-1)$ real coordinates to the grid of step $1/M$, where $M = \lceil 2\sqrt{2(q-1)}/h \rceil$. If v, w are the original and rounded representatives, their norms are at least one and $\|v - w\| \leq \sqrt{2(q-1)}/(2M)$. Normalization changes their distance by at most a factor two, and the sine distance is at most the distance of these normalized representatives. Thus the stated set is an h -net. The projector $ww^*/\|w\|^2$ has rational real and imaginary entries, and the grid count is (40).

For a nonzero traceless Hermitian H , let λ_1 and λ_q be its largest and smallest eigenvalues. A net projector within h of a top eigenline satisfies

$$\text{tr}(HP) \geq \lambda_1 - (\lambda_1 - \lambda_q)h^2 \geq (1 - qh^2)\lambda_1.$$

Indeed $-\lambda_q \leq (q-1)\lambda_1$ by trace zero. The support function of $\mathcal{K}_q$ at H is λ_1/r_q . The support-function criterion for convex containment proves (41). It applies to every direction, not just a finite set of tested queries. $\square$

Theorem B.2 (A finite-alphabet exact upper machine). *For every finite unitary command alphabet and $0 < \rho \leq 1/10$, the experiment on (39) has an exact machine of width $O_q(N^{q-1})$. For each N , all command transition matrices can be chosen independent of the epoch. Each row uses at most q^2 successors. The final decoder may depend on N .*

Proof. Take $h = (8qN)^{-1/2}$ and $\eta = 1 - 1/(8N)$ in Lemma B.1. Let V be the resulting finite set of centered vectors, and put $a_t = \eta^{-t}/4$. A label i represents $a_t Y(P_i)$ at epoch t . This is a proof description of a label; no matrix is stored in the register.

Since $\rho/a_0 = 4\rho \leq 2/5 < \eta$ and $0 \in \mathcal{K}_q$, any seed $\rho Y(P)$ is a convex combination of $a_0 V$. For each label i and each command U , the vector $\eta U Y(P_i) U^*$ lies in $\text{conv } V$. Choose a convex representation and use its weights as the stochastic transition row. The choices depend on the command and label but not on t . Multiplying by a_{t+1} shows that the expected represented vector is updated by exact conjugation, since $a_{t+1}\eta = a_t$.

Induction gives expected vector $\rho Y(U_w P U_w^*)$. The terminal column for query j is $a_N \text{tr}(F_j Y(P_i))$. It is legal because $|\text{tr}(F_j Y(P_i))| \leq 1$ and Bernoulli's inequality gives $\eta^N \geq 7/8$, so $a_N \leq 2/7$.

Sampling a binary answer with this mean proves every conditional probability in (2) exactly. Carathéodory's theorem in real dimension $D = q^2 - 1$ permits q^2 successors. The number in (40) is $O_q(h^{-2(q-1)}) = O_q(N^{q-1})$. $\square$

The upper argument uses a quadratic support loss along the projective orbit. Applying a net on the full ambient sphere would instead give the larger exponent $(q^2 - 2)/2$.

B.2. Fixed finite algebraic alphabets. For $1 \leq j < q$ let $R_j, D_j \in \text{SU}(q)$ act as the identity outside the adjacent coordinate pair $(j, j+1)$ and on that pair as

$$(42) \quad R = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}, \quad D = \begin{pmatrix} \zeta & 0 \\ 0 & \bar{\zeta} \end{pmatrix}, \quad \zeta = \frac{3+4i}{5}.$$

The alphabet has $4(q-1) + 1$ symbols: $\{I, R_j^{\pm 1}, D_j^{\pm 1} : 1 \leq j < q\}$. Assign probability $1/2$ to I and $1/[8(q-1)]$ to each other symbol.

Proposition B.3 (Qualitative group expansion). *This command law has a norm gap on $L_0^2(\text{SU}(q))$ for every fixed q . Its entries and probabilities are algebraic. No numerical gap value is asserted for this family.*

Proof. The number ζ is not a root of unity: $\zeta + \zeta^{-1} = 6/5$ is rational but not an algebraic integer. The powers of R_j and D_j therefore have closures containing two distinct one-parameter subgroups in the corresponding embedded $\text{SU}(2)$. Their Lie brackets generate that $\mathfrak{su}(2)$. The adjacent embedded Lie algebras generate $\mathfrak{su}(q)$: adjacent elementary off-diagonal directions produce all other off-diagonal directions by brackets, and their self-brackets produce the traceless diagonal directions. The closed generated subgroup is consequently $\text{SU}(q)$.

The adjoint entries are algebraic. The spectral-gap theorem of Benoist–de Saxcé [2, Theorem 1.2] applies to an adapted algebraic measure on this compact connected group with simple Lie algebra. Equivalently, one may use the algebraic unitary result of Bourgain–Gamburd [3]. The symmetric average has an upper gap; weight $1/2$ at the identity makes its spectrum nonnegative. Its mean-zero operator norm is therefore strictly below one. This is the external full-group theorem, not a calculation in the defining q -dimensional representation. $\square$

Proof of Corollary 7.2. Corollary A.3 gives the lower order, for every fixed positive calibration κ . Theorem B.2 gives the matching exact upper order. Proposition B.3 supplies fixed finite alphabets. The quantum assertion follows from the next proposition. Taking binary logarithms gives the label-bit formula. $\square$

Proposition B.4 (A fixed quantum realization). *The projective experiment has an exact quantum realization on $\mathbb{C}^q$ using fixed seed resets, the prescribed unitary commands and fixed query measurements. It is independent of N .*

Proof. On seed P reset the density matrix to

$$\sigma_P = I/q + \rho Y(P) = (1 - \rho/r_q)I/q + (\rho/r_q)P.$$

This is positive and has trace one, since $0 < \rho \leq 1/10 < r_q$. Commands act by conjugation. For query j , use effects $(I + F_j)/2, (I - F_j)/2$. Hilbert–Schmidt normalization implies $\|F_j\|_{\text{op}} \leq 1$, so these are positive effects summing to I . Their outcome mean is $\text{tr}(F_j \sigma_{U_w P U_w^*}) = \rho \text{tr}(F_j Y(U_w P U_w^*))$. This is precisely (2). The seed reset is a channel, not a unitary; only one query is measured. $\square$

The comparison is a fixed q -dimensional quantum register versus $(q-1) \log_2 N + O(1)$ classical label bits. Dimension or entropy advantages for quantum models are prior phenomena [6, 17, 5]. The assertion here is the proved classical width order for this complete worst-word numerical behavior. It does not request jointly independent answers to all queries or optimize arbitrary adaptive experiments.

APPENDIX C. AN EFFECTIVE SIX-COMMAND RATIONAL EXAMPLE

The preceding group-gap application is qualitative. We now choose a separate arithmetic alphabet for which a classical theorem supplies a numerical infinite-dimensional norm bound. This does not evaluate the gap of the earlier five-command example.

Let R_x, R_y, R_z be rotations about the coordinate axes with cosine $-3/5$ and sine $4/5$; explicitly,

$$(43) \quad R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3/5 & -4/5 \\ 0 & 4/5 & -3/5 \end{pmatrix}, \quad R_y = \begin{pmatrix} -3/5 & 0 & 4/5 \\ 0 & 1 & 0 \\ -4/5 & 0 & -3/5 \end{pmatrix}, \quad R_z = \begin{pmatrix} -3/5 & -4/5 & 0 \\ 4/5 & -3/5 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Use the fixed six-letter alphabet $\mathcal{A}_V = \{R_x^{\pm 1}, R_y^{\pm 1}, R_z^{\pm 1}\}$ and its uniform law. These are the adjoint rotations of the classical norm-five quaternions $1 \pm 2\mathbf{i}, 1 \pm 2\mathbf{j}, 1 \pm 2\mathbf{k}$, normalized by $\sqrt{5}$.

Proposition C.1 (External arithmetic gap certificate). *For $P_V f = (1/6) \sum_{R \in \mathcal{A}_V} f(R^{-1} \cdot)$ on $L_0^2(\mathbb{S}^2)$,*

$$(44) \quad \|P_V\| \leq \frac{2\sqrt{5}}{6} = \frac{\sqrt{5}}{3}.$$

Consequently $1 - \|P_V\|^2 \geq 4/9$.

Proof. This is the $p = 5$ specialization of the Lubotzky–Phillips–Sarnak spherical Hecke bound [10, 11]. In its unnormalized convention the $p + 1$ norm- p quaternion representatives act by a sum of rotations whose mean-zero norm is at most $2\sqrt{p}$. The representatives have positive odd real coordinate and even other coordinates when $p \equiv 1 \pmod{4}$; at $p = 5$ they are exactly the six displayed ones. Appendix D records the operator normalization and its source boundary. The coauthor account [12, Section 3.2, the V -gate example] spells out these representatives and their Hecke interpretation. The unnormalized spherical norm convention is also stated in [13, Theorem 3(ii)]. Dividing by six gives (44). Passing from quaternions to adjoint rotations identifies $\mathbb{S}^2$ as the homogeneous quotient; inverse signs do not affect this symmetric average. The arithmetic bound is invoked as an external theorem, not reproved by finite harmonic tests. $\square$

Use six seeds $\{\pm e_j : 1 \leq j \leq 3\}$, amplitude $\rho = 1/10$ and three final coordinate queries in (2).

Theorem C.2 (Numerical finite-horizon width certificate). *For $0 \leq \varepsilon < 1/(20\sqrt{3})$ and every integer $N \geq 1$,*

$$(45) \quad \frac{(1/(10\sqrt{3}) - 2\varepsilon)^4}{221184} N \leq W_{N,\varepsilon}(\mathcal{A}_V) \leq W_{N,0}(\mathcal{A}_V) < 150N.$$

The upper machine has exact rational command and decoder rows, common command matrices, and an N -dependent decoder. At zero error every internal selected cut has separate positive minimum four, while

$$(46) \quad W_{N,0} \geq \frac{N}{19\,906\,560\,000}.$$

The constants are proof constants, not sharp asymptotic constants.

Proof. For the standard $\mathrm{SO}(3)$ action the orbit of every unit vector is uniform on $\mathbb{S}^2$. A chordal cap of radius $r \leq 1$ has probability $r^2/4 \leq r^2$, so take $D = 3, s = 2, A = 1$ in Theorem 4.5. The sphere gap transfers to $L^2(\mathbb{R}^3)$ by polar coordinates. Thus $B_{3,1,2} = 98304$, and the squared gap $4/9$ gives the lower bound in (45). At $\varepsilon = 0$ the fourth power of $1/(10\sqrt{3})$ is $1/90000$, which yields (46).

Here is an explicit cardinality estimate for the upper construction. Take both rational stereographic charts

$$t \mapsto \frac{(2t_1, 2t_2, \pm(1 - \|t\|^2))}{1 + \|t\|^2}, \quad t \in [-1, 1]^2,$$

with a grid of step $1/M$ and $M = \lceil 2\sqrt{2N} \rceil$. Every hemisphere is covered by its chart on the unit disk. Each chart is 2-Lipschitz, so its grid images form a sphere net of chordal radius at most $\sqrt{2}/M \leq 1/(2\sqrt{N})$. Its convex hull contains η times the unit ball, with $\eta = 1 - 1/(8N)$, by the support function identity $u \cdot v = 1 - \|u - v\|^2/2$. There are at most $2(2M + 1)^2 \leq (82 + 48\sqrt{2})N < 150N$ labels. Use scales $a_t = \eta^{-t}/4$, initialization representing ρx , rows representing $\eta R u$ in this hull, and decoder $a_{Ne_j} \cdot u$. The verification is the induction in Theorem B.2. All data are rational; each nonempty rational convex system has rational weights, with at most four successors. The final decoder is bounded by $2/7$.

At a selected internal cut, fix any common command prefix. The six seed images affinely span $\mathbb{R}^3$, so the normalized continuation rank is four; future coordinate queries still separate these dimensions after any common suffix. For an upper factorization take a regular tetrahedron containing the radius- ρ ball and having circumradius $3\rho < 1$. Its four vectors give legal continuations under all future rotations. One embeds this selected-cut factorization in a complete machine by allowing uncapped raw histories elsewhere. Therefore the separate minimum is exactly four. This does not give a four-state machine at all cuts simultaneously. $\square$

For example, (46) exceeds four once $N \geq 79\,626\,240\,001$. The size of this number illustrates the conservative constants; the benefit is that no unspecified spectral constant is needed for this alphabet.

The associated qubit operations are $(I - 2i\sigma_j)/\sqrt{5}$ and their inverses, with seed Bloch vector $x/10$ and final Pauli query. They realize the numerical experiment exactly, independently of N . This quantum construction and the arithmetic gates are classical; (45) is the finite-register consequence. The earlier five-gate alphabet remains qualitatively gapped with a non-evaluated constant, and is retained unchanged in the supporting material.

APPENDIX D. THE ARITHMETIC OPERATOR AND ITS NORMALIZATION

This appendix fixes the exact external input used in Proposition C.1, then proves the remaining representation and normalization steps. Write $\mathbf{i}, \mathbf{j}, \mathbf{k}$ for quaternion units and

$$\mathcal{Q}_5 = \{1 \pm 2\mathbf{i}, 1 \pm 2\mathbf{j}, 1 \pm 2\mathbf{k}\}, \quad r_q(z) = qzq^{-1} \quad (z \in \text{Im } \mathbb{H}).$$

The required external LPS spherical Hecke inequality, specialized to this representative convention, is

$$(LPS5) \quad \left\| \sum_{q \in \mathcal{Q}_5} f(r_q^{-1} \cdot) \right\|_{L^2(\mathbb{S}^2)} \leq 2\sqrt{5} \|f\|_{L^2(\mathbb{S}^2)}, \quad \int f = 0.$$

This is an imported arithmetic theorem, not a finite-dimensional calculation. The original sources are [10, 11]; the unnormalized norm theorem is explicitly restated in [13, Theorem 3(ii), p. 8 of the preprint], while the quaternionic and Hecke representative conventions are discussed in [12, Section 3.2]. The restated existence theorem alone would not identify an arbitrary six rotations: the representative convention in (LPS5) is essential.

Proposition D.1 (Parity, quotient and full spherical spectrum). *With the external input (LPS5), the uniform average over the six rotations in (43) has mean-zero norm at most $\sqrt{5}/3$ both on $L^2(\mathbb{S}^2)$ and on $L^2(\text{SO}(3))$. Its squared gap is at least $4/9$. The conclusion does not assert the same numerical gap for all of $L_0^2(\text{SU}(2))$.*

Proof. An integral quaternion of norm five has one coordinate of magnitude one, one of magnitude two, and two zero coordinates. There are $4 \cdot 3 \cdot 4 = 48$ such quaternions. Multiplication by a Lipschitz unit in $\{\pm 1, \pm \mathbf{i}, \pm \mathbf{j}, \pm \mathbf{k}\}$ places the odd coordinate in the real position and makes it positive. This gives a unique representative with positive odd real part and even other coordinates, namely one of the six members of $\mathcal{Q}_5$. This specifies the parity class; it is not an average over all 48 quaternions or an extra quotient by their unit group.

Normalizing q to $q/\sqrt{5}$ does not change r_q . Quaternionic conjugation identifies the imaginary unit sphere with $\mathrm{SO}(3)/\mathrm{SO}(2)$, and q and $-q$ give the same rotation. For $(1 + 2\mathbf{i})/\sqrt{5}$ direct multiplication gives

$$\mathbf{i} \mapsto \mathbf{i}, \quad \mathbf{j} \mapsto (-3\mathbf{j} + 4\mathbf{k})/5, \quad \mathbf{k} \mapsto (-4\mathbf{j} - 3\mathbf{k})/5.$$

These are the columns of R_x in (43); cyclic permutation gives R_y, R_z . Conjugating a quaternion changes the sign of its imaginary coordinate and gives its normalized inverse. Thus the six representatives already include inverse rotations, with one copy each. Division of the sum in (LPS5) by six gives $2\sqrt{5}/6 = \sqrt{5}/3$, and squaring gives $1 - 5/9 = 4/9$.

For the full-group statement, the complexified spherical space is the orthogonal sum of the harmonic spaces $\mathcal{H}_\ell$, $\ell \geq 0$. These are precisely all irreducible unitary representations of $\mathrm{SO}(3)$, one copy of each. The degree-zero space is the constants. The regular representation on $\mathrm{SO}(3)$ contains the same irreducibles with multiplicity $2\ell + 1$. Convolution acts on each copy by the same matrix, so its mean-zero operator norm is the supremum of the same matrix norms on both spaces. No extra invariant subspace is removed: on the sphere the constant is the only $\mathrm{SO}(3)$ -invariant function. The symmetric alphabet makes the inverse/right versus left convention immaterial for these norms. Half-integer-spin representations occur on $\mathrm{SU}(2)$ but not on its $\mathrm{SO}(3)$ quotient; this argument makes no claim about those extra blocks. $\square$

Source boundary. The original LPS publisher records were checked, but their complete 1986/1987 page images were not obtained in this revision. We therefore do not invent an original theorem number or label the secondary restatement as an inspected original proof. Equation (LPS5) states exactly the arithmetic line required, and the proposition proves every subsequent convention change. The effective theorem retains this explicit external dependency. The main width classification is conditional on an arbitrary full gap and does not depend on this numerical example or on an original-page attribution.

REFERENCES

- [1] A. Atalik, A. Köse and M. Gastpar, Differential entropy of the conditional expectation under additive Gaussian noise, *IEEE Trans. Signal Process.* **70** (2022), 4851–4866; arXiv:2106.04677v3. doi:10.1109/TSP.2022.3211403.
- [2] Y. Benoist and N. de Saxcé, A spectral gap theorem in simple Lie groups, *Invent. Math.* **205** (2016), 337–361; arXiv:1405.1808. doi:10.1007/s00222-015-0636-2.
- [3] J. Bourgain and A. Gamburd, A spectral gap theorem in $\mathrm{SU}(d)$, *J. Eur. Math. Soc.* **14** (2012), 1455–1511.
- [4] B. C. Geiger and C. Temmel, Lumpings of Markov chains, entropy rate preservation, and higher-order lumpability, *J. Appl. Probab.* **51** (2014), 1114–1132; extended version arXiv:1212.4375v6.
- [5] F. Ghafari et al., Dimensional quantum memory advantage in the simulation of stochastic processes, *Phys. Rev. X* **9** (2019), 041013.
- [6] M. Gu, K. Wiesner, E. Rieper and V. Vedral, Quantum mechanics can reduce the complexity of classical models, *Nature Commun.* **3** (2012), 762.
- [7] Z. He, C. D. Charalambous and P. A. Stavrou, A new finite-horizon dynamic programming analysis of nonanticipative rate-distortion function for Markov sources, arXiv:2411.11698v1 (2024).
- [8] A. Heller, On stochastic processes derived from Markov chains, *Ann. Math. Statist.* **36** (1965), 1286–1291.
- [9] C. H. Lee, E. Pyne and S. Vadhan, Fourier growth of regular branching programs, *APPROX/RANDOM 2022*, LIPIcs **245**, Art. 2, 2:1–2:21.
- [10] A. Lubotzky, R. Phillips and P. Sarnak, Hecke operators and distributing points on the sphere I, *Comm. Pure Appl. Math.* **39**, suppl. (1986), S149–S186.

- [11] A. Lubotzky, R. Phillips and P. Sarnak, Hecke operators and distributing points on S^2 . II, *Comm. Pure Appl. Math.* **40** (1987), 401–420.
- [12] O. Parzanchevski and P. Sarnak, Super-golden-gates for $PU(2)$, *Adv. Math.* **327** (2018), 869–901; arXiv:1704.02106v1.
- [13] G. Pisier, Quadratic forms in unitary operators, *Linear Algebra Appl.* **267** (1997), 125–137; arXiv:math/9512207, Theorem 3(ii).
- [14] Y. Polyanskiy and Y. Wu, Wasserstein continuity of entropy and outer bounds for interference channels, *IEEE Trans. Inform. Theory* **62** (2016), 3992–4002.
- [15] O. Reingold, T. Steinke and S. Vadhan, Pseudorandomness for regular branching programs via Fourier analysis, *RANDOM 2013*, LNCS **8096**, 655–670.
- [16] T. Steinke, S. Vadhan and A. Wan, Pseudorandomness and Fourier growth bounds for width 3 branching programs, *Theory Comput.* **13** (2017), Art. 12, 1–50.
- [17] J. Thompson, A. J. P. Garner, V. Vedral and M. Gu, Using quantum theory to simplify input-output processes, *npj Quantum Inform.* **3** (2017), 6.
- [18] M. Vidyasagar, *Hidden Markov Processes: Theory and Applications to Biology*, Princeton University Press, 2014.
- [19] R. G. Wood, T. Linder and S. Yüksel, Optimal zero delay coding of Markov sources: stationary and finite memory codes, *IEEE Trans. Inform. Theory* (2017); arXiv:1606.09135, full revised version. doi:10.1109/TIT.2017.2692215.
- [20] Q. Qi, General Theta Foundations I: Entropy dissipation and sharp finite-alphabet memory, Revision 37 (2026), publication c5b8015887f83833. Preserved in the supporting article; cited only as a predecessor, not external validation.
- [21] A. Barvinok, Estimating L^∞ norms by L^{2^k} norms for functions on orbits, preprint (April 2002), Lemma 5. Author’s manuscript, 19 pp.
- [22] S. Basu, R. Pollack and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, 2nd ed., Springer, 2006.
- [23] Z. Chen and J. Wu, The state cost of classical simulation of one-way general quantum finite automata, arXiv:2604.07058v2 (2026), Proposition 3.1 and Theorem 3.2.
- [24] Z. Chen and J. Wu, On the simulation cost of quantum finite automata, arXiv:2605.10682v1 (2026).
- [25] Z. Chen, Behavioral memory under symmetry in one-way quantum automata, arXiv:2609.01451v1 (2026).
- [26] M. Iacobelli, Asymptotic quantization for probability measures on Riemannian manifolds, arXiv:1412.4026v1 (2014), Theorem 1.4 and Section 2.
- [27] R. Sanyal, F. Sottile and B. Sturmfels, Orbitopes, *Mathematika* **57** (2011), 275–314; arXiv:0911.5436.
- [28] H. Fawzi, J. Gouveia, P. A. Parrilo, R. Z. Robinson and R. R. Thomas, Positive semidefinite rank, *Math. Program.* **153** (2015), 133–177; arXiv:1407.4095.
- [29] Q. Qi, General Theta Foundations I: Orbit geometry and entropy budgets for finite memory, Revision 38 (2026), publication 8ec190c921656b98. The unchanged supporting article is a predecessor, not external validation.

PRESERVED MATHEMATICAL DEVELOPMENT

General Theta Foundations I - revision 38

The complete current article precedes this divider.

The following predecessor cumulative volume is reproduced without alteration.

Frozen predecessor publication:

8ec190c921656b9869f3520ec9bf7bf754622e97

This archive is not part of the compact referee package. Independent historical analytic obligations keep their original hypotheses and proof status.

GENERAL THETA FOUNDATIONS I: ORBIT GEOMETRY AND ENTROPY BUDGETS FOR FINITE MEMORY

QIAN QI

ABSTRACT. We prove a finite-memory occupation theorem for orthogonal representations of compact groups. A norm gap away from invariant functions and a uniform orbital small-ball bound of exponent s force the gap-weighted number of k -state epochs to be $O(k^{2/s})$. The competing transducer may have arbitrary hidden states and unrelated stochastic rows at every epoch. The small-ball hypothesis is required on the directions of all possible conditional means, not only on the original orbit. For conjugation on traceless Hermitian q by q matrices we establish it uniformly over all spectra with exponent $2(q-1)$. A projective-net construction then gives matching $\Theta(N^{q-1})$ classical labels for finite algebraic unitary alphabets with a full group spectral gap, at exact output and at every fixed sufficiently small wordwise error. The same numerical behavior has a fixed q -dimensional quantum realization. We also give a separate six-command rational three-dimensional example with an analytic numerical gap certificate from the classical Ramanujan bound, yielding an explicit finite-horizon lower constant. The resource is nonuniform clocked atomic-row width, not uniform computational space. Gaussian entropy identities, classical expansion and coding antecedents are distinguished from the orbital occupation and synthesis conclusions.

1. INTRODUCTION

A finite probabilistic device can preserve a linear statistic through a long sequence of transformations without retaining the underlying point. Consequently the dimension of the statistic does not determine the size of an exact positive realization. The distinction is particularly visible when the transformation group expands functions on its orbits. Mixing creates entropy, while replacement by finitely many conditional means removes it. An occupation inequality can compare these operations without assigning an observable vector to each hidden basis state.

The spherical form of this argument gives the order $N^{(d-1)/2}$ under a full spherical spectral gap [20]. Two issues arise beyond the sphere. First, invariant densities need not be radial. Second, a conditional mean of orbit points need not be another orbit point. The second issue is decisive for unitary conjugation: a conditional mean of pure states usually has several nontrivial eigenvalues. Applying a pure-state cap estimate directly to these means would be invalid.

We resolve both issues through orbital averaging on the whole ambient representation and a spectral-projector estimate uniform over all traceless Hermitian directions. The resulting exponent is intrinsic to the smallest nontrivial conjugation orbit, rather than the dimension of the ambient sphere. A construction on complex projective space matches this lower exponent. Thus the theorem is not obtained by substituting $q^2 - 1$ into a spherical formula.

1.1. Principal statements. Let G act orthogonally on $V = \mathbb{R}^D$. Write Π_G for Haar averaging and suppose that a finitely supported command law has squared norm gap $g > 0$ away from Π_G

Date: September 25, 2026. Revision 38.

2020 Mathematics Subject Classification. Primary 93B15, 22D10; Secondary 94A17, 68Q10, 81P45.

Key words and phrases. Positive realization, compact group, entropy, conditional expectation, projective space, probabilistic branching program, quantum memory.

on $L^2(V)$. Assume, for $s > 0$ and $A \geq 1$,

$$(1) \quad \sup_{u,v \in \mathbb{S}(V)} m_G\{a : \|au - v\| \leq r\} \leq Ar^s \quad (0 < r \leq 1).$$

The input word is universally quantified; the command law is chosen only for the lower proof. Theorem 3.5 proves

$$(2) \quad \sum_{t: K_t \leq k} g_t \leq 2048D(16A)^{2/s} \kappa^{-3-2/s} k^{2/s},$$

whenever a terminal correlation forces $\mathbb{E}\|\mathbb{E}[Y_N \mid S_N]\| \geq \kappa$. Widths at the other epochs are unrestricted. The gap may vary with time.

For $q \geq 2$, put $D = q^2 - 1$ and consider the normalized pure-state orbit in $\text{Herm}_0(q)$, equipped with its Hilbert–Schmidt inner product. A fixed finite unitary alphabet drives this orbit. One final query asks for a coordinate of the resulting vector, with fixed amplitude $0 < \rho \leq 1/10$.

Theorem 1.1 (Projective-orbit memory order). *Suppose the command alphabet supports a probability measure with a norm gap on $L_0^2(\text{SU}(q))$. For every fixed $0 \leq \varepsilon < \rho/(2\sqrt{q^2 - 1})$, the least classical width satisfies*

$$(3) \quad W_{N,\varepsilon} = \Theta_{q,\rho,\varepsilon,\mu}(N^{q-1}).$$

The lower bound permits arbitrary hidden states, time-dependent rows and horizon-dependent design. The exact upper construction uses common command rows and an N -dependent final decoder. There are explicit finite algebraic alphabets satisfying the hypothesis for every q . The behavior has a fixed quantum realization of dimension q ; its classical label bits are $(q - 1) \log_2 N + O(1)$.

The proof, including the alphabet and the fixed quantum device, is in Section 5. For $q > 2$ the exponent $q - 1$ differs from the ambient-sphere exponent $(q^2 - 2)/2$. The group-gap hypothesis here is qualitative; its existence follows from an external expansion theorem. It is not inferred merely from density of the generated subgroup.

A second application addresses effective constants. Section 6 uses the classical six V -gates, whose adjoint rotations have rational entries. The Lubotzky–Phillips–Sarnak bound gives a certified squared gap $4/9$ on the sphere. For amplitude $1/10$ the explicit consequence is

$$(4) \quad W_{N,\varepsilon} \geq \frac{(1/(10\sqrt{3}) - 2\varepsilon)^4}{221184} N.$$

This is a different alphabet from the earlier five-gate example, whose numerical gap is not evaluated here. The arithmetic spectral estimate is an external theorem, not a numerical truncation of spherical harmonics.

1.2. Conventions and attribution. All growth statements concern available persistent labels. The external epoch, the horizon, read-only row tables, their construction, arithmetic and exact atomic sampling are not charged. One query is selected after the command word; no independent joint answers are required. A finite random-bit implementation is a different resource problem.

Ingredient	Role in the argument
Gaussian score and conditional means	Classical identities; the one-sided quadratic entropy consequence is reproduced, not used as a priority claim.
Group spectral gaps	External inputs, with a separate numerical Ramanujan certificate for the six-gate example.
Orbital occupation	The representation-level extension, using uniform small balls for all centroid directions.
Hermitian conjugation	A uniform all-spectra estimate and matching projective synthesis give the $q - 1$ width exponent.
Quantum operations	Standard reset, conjugation and two-outcome measurement; the width lower bound concerns the classical competitor.

The comparison with causal coding, Gaussian conditional-expectation entropy, state aggregation and probabilistic branching programs is made at the level of the optimized objects in Section 7. **Notation.** N is the command horizon; K_t is the available label count; W is its maximum. D is the real ambient dimension and s is the orbital small-ball exponent. q denotes quantum dimension, while q_t below denotes a conditional-centroid norm. Entropy and log use natural units; $\log_2$ is reserved for label bits.

2. THE EXPERIMENT AND ITS POSITIVE REALIZATION

Let G be a compact group, let $U : G \rightarrow O(V)$ be a continuous orthogonal representation on $V = \mathbb{R}^D$, and fix a finite command alphabet $\mathcal{A} \subset G$. A finite seed set $\mathcal{X} \subset \mathbb{S}(V)$ is given initially. For $w = (a_1, \dots, a_N)$ write $U_w = U_{a_N} \cdots U_{a_1}$. Relative to an orthonormal basis $e_1, \dots, e_D$, the prescribed numerical experiment is

$$(5) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, U_w x \rangle}{2}, \quad b \in \{-1, 1\}, \quad j \in [D], \quad 0 < \rho \leq 1/10.$$

All seeds, words and queries are required, not only almost-sure words under one source law. The probabilities are between $(1 - \rho)/2$ and $(1 + \rho)/2$.

Definition 2.1 (Clocked atomic-row realization). A horizon- N machine consists of finite sets $\mathcal{S}_t$ with $|\mathcal{S}_t| = K_t$, a stochastic initialization E_x on $\mathcal{S}_0$, stochastic matrices $T_{t,a} : \mathcal{S}_t \rightarrow \mathcal{S}_{t+1}$, and columns $d_j : \mathcal{S}_N \rightarrow [-1, 1]$. Its mean response is

$$(6) \quad E_x T_{0,a_1} \cdots T_{N-1,a_N} d_j.$$

It has error at most ε if this differs from $\rho\langle e_j, U_w x \rangle$ by at most 2ε for every (x, w, j) . Available labels, including padding, count. All rows, including unreachable ones, are normalized. Only the updated label persists. The seed, word prefix, old random choices and any data-dependent protocol information cannot be consulted again unless represented by that label. The external time, horizon, fixed program and exact sampling from its rows are free. The final binary answer may be sampled when j arrives; if retained, it uses a two-label final cut. Write $W_{N,\varepsilon}$ for the minimum of $\max(2, K_0, \dots, K_N)$.

The state alphabets and matrices may be chosen anew for each N , but the command symbols and their experimental meanings are fixed. Nothing requires one autonomous device or one decoder to serve all stopping lengths. No adaptive choice of the command is optimized in this model. A complex unitary representation can be treated by its realification; all inner products in the entropy argument are real.

Fix one seed x and an independent command law at each epoch for the converse. Let $Y_t = U_{a_t} \cdots U_{a_1} x$ and let S_t be the actual machine state under this test law. Define

$$(7) \quad Z_t = \mathbb{E}[Y_t \mid S_t], \quad q_t = \mathbb{E}\|Z_t\|, \quad v_t = \mathbb{E}\|Z_t\|^2.$$

These random vectors and their laws are analytical devices, not an additional runtime register or oracle.

Lemma 2.2 (Quadratic budget and terminal calibration). *For the fresh command a , put $X_t = U_a Z_t$. Then*

$$(8) \quad Z_{t+1} = \mathbb{E}[X_t \mid S_{t+1}], \quad \Delta_t := \mathbb{E}\|X_t - Z_{t+1}\|^2 = v_t - v_{t+1}, \quad \sum_{t < N} \Delta_t \leq 1.$$

The sequences q_t, v_t are nonincreasing and start at one. A machine with error at most ε satisfies

$$(9) \quad q_t \geq q_N \geq \kappa := \rho/\sqrt{D} - 2\varepsilon \quad (0 \leq t \leq N).$$

Proof. The current command is independent of the past, and the new label is generated from (S_t, a) by a common row. Conditioning first on (S_t, a, S_{t+1}) and then on S_{t+1} proves the first identity. The orthogonal projection identity for conditional expectation gives the second; Jensen and orthogonality give both monotonicities. The fixed seed has unit norm, even with randomized initialization. For calibration form $d = (d_1, \dots, d_D)$ only in the proof, so $\|d\| \leq \sqrt{D}$. Conditioning on the complete word and using (5)–(6) gives $\mathbb{E}\langle Y_N, d(S_N) \rangle \geq \rho - 2\sqrt{D}\varepsilon$. On conditioning instead on S_N , this is at most $\sqrt{D}q_N$. This proves (9). It does not require a joint measurement of the D coordinates. $\square$

For each fixed profile, the optimum actual error is attained: all stochastic rows and terminal columns form a finite product of compact sets and the objective is a maximum of finitely many continuous functions. Thus the subsequent lower bounds concern the actual optimization, not a sum of local error certificates.

3. AN ORBITAL ENTROPY OCCUPATION THEOREM

Lebesgue measure is used on $V = \mathbb{R}^D$ and normalized Haar measure on G . For a function on V , set

$$(10) \quad \mathcal{U}_a f(z) = f(U_a^{-1}z), \quad P_t = \sum_a p_t(a) \mathcal{U}_a, \quad \Pi f(z) = \int_G f(U_g^{-1}z) dm_G(g).$$

Thus $\mathcal{U}_a^* = \mathcal{U}_{a^{-1}}$, Π is an orthogonal projection, and $P_t \Pi = \Pi P_t = \Pi$. The probability law p_t need not be symmetric. Its required squared norm gap is

$$(11) \quad \|P_t f\|_2^2 \leq \|\Pi f\|_2^2 + (1 - g_t) \|f - \Pi f\|_2^2, \quad 0 \leq g_t \leq 1.$$

Invariance in this formula means G -invariance, not necessarily radial symmetry. In a reducible representation invariant functions can carry many parameters.

Lemma 3.1 (Transfer of a group norm gap). *If right convolution $R_{p_t} F(g) = \sum_a p_t(a) F(ga)$ on $L^2(G)$ has mean-zero norm at most $\sqrt{1 - g_t}$, then (11) holds in every orthogonal representation on V .*

Proof. For each z apply the group inequality to $F_z(g) = f(U_g^{-1}z)$. Its mean is $\Pi f(z)$ and $R_{p_t} F_z(g) = P_t f(U_g^{-1}z)$. Integrate the resulting inequality in z . Orthogonal changes of variables and Fubini identify the three integrals with those in (11). First take bounded compactly supported f and then use density. This proof does not replace the full group gap by a gap in the defining representation. $\square$

3.1. The conditional-mean cost. Let G_D be a standard independent real Gaussian vector and let γ_τ be the density of τG_D .

Lemma 3.2 (A one-sided Gaussian coupling inequality). *For bounded X and a finite-valued variable J , put $M = \mathbb{E}[X \mid J]$. For every $\tau > 0$,*

$$(12) \quad \mathbf{h}(X + \tau G_D) - \mathbf{h}(M + \tau G_D) \leq \frac{\mathbb{E}\|X - M\|^2}{2\tau^2}.$$

Proof. Write $f = \text{Law}(M) * \gamma_\tau$. Differentiation of the Gaussian mixture and the conditional covariance identity give

$$(13) \quad \nabla^2 \log f(y) = -\tau^{-2}I + \tau^{-4} \text{Cov}(M \mid M + \tau G_D = y) \succeq -\tau^{-2}I.$$

Hence $\phi(x) = -\mathbb{E} \log f(x + \tau G_D)$ has Hessian at most $\tau^{-2}I$. Taylor's inequality at M yields $\phi(X) \leq \phi(M) + \langle \nabla \phi(M), X - M \rangle + \|X - M\|^2 / (2\tau^2)$. The conditional expectation of the linear term given J is zero. If $g = \text{Law}(X) * \gamma_\tau$, nonnegativity of $D(g \parallel f)$ gives $\mathbf{h}(g) \leq -\int g \log f = \mathbb{E} \phi(X)$, whereas $\mathbb{E} \phi(M) = \mathbf{h}(f)$. Boundedness of X, M provides Gaussian bounds with finite second moments and justifies differentiation and integration. These identities prove (12). $\square$

This elementary consequence of Gaussian posterior identities is used as an estimate, not as a claim of priority over the Gaussian estimation literature [1, 14]. The estimate is one-sided and can have a negative left side. Here J can be a finite register with no Gaussian observation model; the smoothing noise is added after conditioning.

Lemma 3.3 (Entropy production away from invariant functions). *For any bounded-support Gaussian mixture density f ,*

$$(14) \quad \mathbf{h}(P_t f) - \mathbf{h}(f) \geq g_t \|\sqrt{f} - \Pi \sqrt{f}\|_2^2 \geq 0.$$

Proof. Put $f_a = \mathcal{U}_a f$ and $h = P_t f$. Orthogonality preserves entropy, so $\mathbf{h}(h) - \mathbf{h}(f) = \sum_a p_t(a) D(f_a \parallel h)$. For probability densities u, v , $D(u \parallel v) \geq \|\sqrt{u} - \sqrt{v}\|_2^2$; it follows, for example, from $-\log x \geq 2(1 - \sqrt{x})$ and integration. The mean is the least-squares minimizer, and thus

$$\sum_a p_t(a) \|\sqrt{f_a} - \sqrt{h}\|_2^2 \geq \sum_a p_t(a) \|\sqrt{f_a} - P_t \sqrt{f}\|_2^2 = 1 - \|P_t \sqrt{f}\|_2^2.$$

Apply (11) and $\|\sqrt{f}\|_2 = 1$. $\square$

No logarithmic Sobolev inequality or point-mass mixing assertion is used in this step.

3.2. Sparsity away from invariant densities. Suppose that the representation obeys the uniform orbital bound

$$(15) \quad m_G\{g : \|U_g u - v\| \leq r\} \leq A r^s \quad (u, v \in \mathbb{S}(V), 0 < r \leq 1),$$

where $A \geq 1$ and $s > 0$. In particular the representation has no nonzero fixed vector. It is essential that u range over all possible centroid directions, not merely the initial seed orbit.

Lemma 3.4 (Finite centroids cannot be almost invariant). *Suppose Z has at most k values in the unit ball and $\mathbb{E}\|Z\| \geq \kappa$, where $0 < \kappa \leq 1$. Set*

$$(16) \quad r = \left(\frac{\kappa}{16Ak} \right)^{1/s}, \quad \tau = \frac{\kappa r}{8\sqrt{2D}}, \quad f = \text{Law}(Z) * \gamma_\tau.$$

Then $\|\sqrt{f} - \Pi \sqrt{f}\|_2^2 \geq \kappa/16$.

Proof. Atoms of norm at least $\kappa/2$ have total mass at least $\kappa/2$. Let Ω be the union of radius- r caps around their directions, and let C_Ω be its cone. By Markov's inequality, $\mathbb{P}(\|G_D\| \leq \sqrt{2D}) \geq 1/2$. On this event, perturbing a high-norm atom by τG_D changes its direction by at most $4\tau\sqrt{2D}/\kappa = r/2$. Therefore $\int_{C_\Omega} f \geq \kappa/4$.

For any G -invariant $u \in L^2(V)$, Haar averaging and (15) give

$$\int_{C_\Omega} |u(z)|^2 dz = \int |u(z)|^2 \int_G \mathbf{1}_{C_\Omega}(U_g z) dm_G(g) dz \leq k A r^s \|u\|_2^2.$$

The point $z = 0$ is Lebesgue-null. Taking $u = \Pi\sqrt{f}$, whose norm is at most one, bounds this integral by $\kappa/16$. The reverse triangle inequality in $L^2(C_\Omega)$ now gives a norm difference at least $\sqrt{\kappa/4} - \sqrt{\kappa/16} = \sqrt{\kappa}/4$. Repeated directions and zero centroids cause no change. $\square$

Theorem 3.5 (Orbital occupation budget). *Under (11) and (15), let a finite hidden register track a unit-vector command process as in (7). If $q_N \geq \kappa > 0$, then, for every $k \geq 1$,*

$$(17) \quad \sum_{\substack{0 \leq t < N \\ \bar{K}_t \leq k}} g_t \leq B_{D,A,s} \kappa^{-3-2/s} k^{2/s}, \quad B_{D,A,s} = 2048 D (16A)^{2/s}.$$

The same inequality holds with K_t replaced by the number of positive-probability labels under the chosen test law. No bound is imposed on the other epochs.

Proof. Fix k and the single scale (16) for the entire run. Write $f_t = \text{Law}(Z_t) * \gamma_\tau$. Isotropy of the Gaussian makes the density of $U_a Z_t + \tau G_D$ equal to $P_t f_t$. By Lemmas 3.2 and 3.3,

$$\mathbf{h}(f_{t+1}) - \mathbf{h}(f_t) \geq g_t \|\sqrt{f_t} - \Pi\sqrt{f_t}\|_2^2 - \frac{\Delta_t}{2\tau^2}.$$

At an epoch with at most k labels, Lemma 3.4 bounds the norm term below by $\kappa/16$. Elsewhere it is nonnegative. The initial density is a translate of γ_τ ; the terminal second moment is at most $1 + D\tau^2$. The Gaussian maximum-entropy bound and (8) yield

$$\frac{\kappa}{16} \sum_{K_t \leq k} g_t \leq \frac{D}{2} \log \left(1 + \frac{1}{D\tau^2} \right) + \frac{1}{2\tau^2} \leq \frac{1}{\tau^2}.$$

Substitute (16). The same Gaussian scale is retained through all epochs; no repeated mixing-time charge is incurred. $\square$

Corollary 3.6 (Actual terminal error). *For a constant gap $g > 0$, let $\mathcal{E}_N(\mathbf{K})$ be the minimum actual wordwise binary-TV error over a prescribed available profile, and put $K = \max_t K_t$. Then*

$$(18) \quad \mathcal{E}_N(\mathbf{K}) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{D}} - \left(\frac{B_{D,A,s} K^{2/s}}{gN} \right)^{1/(3+2/s)} \right]_+.$$

In particular, $K = o(N^{s/2})$ implies $\liminf \mathcal{E}_N \geq \rho/(2\sqrt{D})$.

Proof. If the actual error is below the calibration threshold, combine (9) with (17) and solve for it. Otherwise the displayed lower bound already holds. Attainment or a minimizing sequence gives the same conclusion for the optimum. $\square$

Local simulation errors may cancel. Nothing in this corollary sums such errors or rules out cancellation; its objective is the actual final law.

4. UNIFORM GEOMETRY OF HERMITIAN CONJUGATION

Fix $q \geq 2$ and let $V = \text{Herm}_0(q)$ be the real space of traceless Hermitian q by q matrices, with inner product $\text{tr}(AB)$. Its dimension is $D = q^2 - 1$. Unitary conjugation is orthogonal on this space. The pure-state orbit has dimension $2(q-1)$, but normalized hidden centroids can have arbitrary spectra. We therefore prove a bound on *every* unit sphere orbit.

4.1. A Grassmann small-ball estimate. Let P_0 be a rank- k orthogonal projector in $\mathbb{C}^q$, $1 \leq k < q$, and let P be a Haar-distributed rank- k projector. Put $m = k(q-k)$ and

$$(19) \quad b_{q,k} = \frac{1}{m!} \prod_{j=1}^k \frac{\Gamma(q-k+j)}{\Gamma(j)}.$$

Lemma 4.1 (Projector balls). *For $0 < R \leq 1$,*

$$(20) \quad \mathbb{P}(\|P - P_0\|_{\text{HS}} \leq R) \leq b_{q,k} R^{2k(q-k)}.$$

Proof. By invariance take P_0 onto the first k coordinates. Apart from a null set, a Haar subspace is the graph of $Z \in \mathbb{C}^{(q-k) \times k}$. Its density with respect to complex Euclidean Lebesgue measure is

$$(21) \quad \pi^{-m} \prod_{j=1}^k \frac{\Gamma(q-k+j)}{\Gamma(j)} \det(I + Z^*Z)^{-q}.$$

For completeness, take a q by k matrix of independent standard complex Gaussian entries and split it into a square top block A and bottom block B . Its column space is Haar and $Z = BA^{-1}$ almost surely. The change $B = ZA$ has real Jacobian $|\det A|^{2(q-k)}$. Integrating A with weight $\exp[-\text{tr} A^*(I + Z^*Z)A]$ gives the determinant factor $\det(I + Z^*Z)^{-q}$. Complex Gaussian Gram-Schmidt gives $\mathbb{E}|\det A|^{2(q-k)} = \prod_{j=1}^k \Gamma(q-k+j)/\Gamma(j)$, since its squared orthogonal lengths have gamma shapes $1, \dots, k$. The powers of π give (21).

If θ_i are the principal angles, then $\|P - P_0\|_{\text{HS}}^2 = 2 \sum_i \sin^2 \theta_i$ and $\|Z\|_{\text{HS}}^2 = \sum_i \tan^2 \theta_i$. On the event in (20) each $\sin^2 \theta_i \leq R^2/2$, whence

$$\|Z\|_{\text{HS}}^2 \leq \frac{R^2}{2 - R^2} \leq R^2.$$

The density (21) is bounded by its value at zero. Multiply this value by the volume $\pi^m R^{2m}/m!$ of a real $2m$ -dimensional ball. This proves the assertion. $\square$

Define the explicit finite constant

$$(22) \quad A_q = \max_{1 \leq k < q} \left\{ \max(1, b_{q,k}) [9q(q-1)^2]^{k(q-k)} \right\}.$$

It is rational and at least one; no optimality is claimed.

Theorem 4.2 (All-spectra orbital anti-concentration). *For all $A, B \in \text{Herm}_0(q)$ of Hilbert-Schmidt norm one and all $0 < r \leq 1$,*

$$(23) \quad m_{\text{SU}(q)}\{U : \|UAU^* - B\|_{\text{HS}} \leq r\} \leq A_q r^{2(q-1)}.$$

In particular, the estimate is uniform at repeated eigenvalues and over mixed-state centroid directions. No lower bound on each eigenvalue gap is assumed.

Proof. Let $\lambda_1 \geq \dots \geq \lambda_q$ be the eigenvalues of A . Trace zero and $\sum_i \lambda_i^2 = 1$ imply $\lambda_1 - \lambda_q \geq q^{-1/2}$. Hence some k has

$$(24) \quad \lambda_k - \lambda_{k+1} \geq \delta_q := \frac{1}{\sqrt{q}(q-1)}.$$

For two conjugates C, D of A , let P_C, P_D be their top- k spectral projectors. Eigenvectors u_i of C and v_j of D obey

$$(\lambda_i - \lambda_j)\langle u_i, v_j \rangle = \langle u_i, (C - D)v_j \rangle.$$

Summing squares for $i \leq k < j$ and using equal projector ranks gives

$$(25) \quad \|P_C - P_D\|_{\text{HS}}^2 = 2 \sum_{i \leq k < j} |\langle u_i, v_j \rangle|^2 \leq 2\delta_q^{-2} \|C - D\|_{\text{HS}}^2.$$

If the event in (23) is empty there is nothing to prove. Otherwise choose one conjugate D in that event. Every other C in the event has $\|C - D\|_{\text{HS}} \leq 2r$. The image P_C is a Haar rank- k projector and lies within $2\sqrt{2}\delta_q^{-1}r$ of P_D . Set $c_q = 3\sqrt{q}(q-1)$, which is larger than this coefficient. If $c_q r \leq 1$, Lemma 4.1 gives a bound $b_{q,k}(c_q r)^{2k(q-k)}$. Since $k(q-k) \geq q-1$ and $r \leq 1$, this is at most the right side of (23). If $c_q r > 1$, then $A_q r^{2(q-1)} \geq c_q^{2(q-1)} r^{2(q-1)} > 1$, so the trivial probability bound suffices. Haar conjugation under $\text{SU}(q)$ and under $U(q)$ has the same law, completing the proof. $\square$

The proof isolates one uniformly separated spectral cut rather than requiring a simple spectrum. This is why pure-state and highly mixed centroids are handled by the same constant. The smallest possible nontrivial Grassmann exponent is $2(q-1)$.

Corollary 4.3 (Conjugation occupation). *For conjugation commands satisfying a group gap g_t , any hidden register with $q_N \geq \kappa > 0$ obeys*

$$(26) \quad \sum_{K_i \leq k} g_t \leq C_q \kappa^{-3-1/(q-1)} k^{1/(q-1)}, \quad C_q = 2048(q^2 - 1)(16A_q)^{1/(q-1)}.$$

For a constant gap $g > 0$ and wordwise error ε with $\kappa = \rho/\sqrt{q^2 - 1} - 2\varepsilon > 0$,

$$(27) \quad W_{N,\varepsilon} \geq \left(\frac{g\kappa^{3+1/(q-1)}}{C_q} N \right)^{q-1}.$$

Proof. Apply Lemma 3.1, Theorem 4.2 and Theorem 3.5, then use Lemma 2.2. $\square$

A group gap is stronger than a gap only on functions on the pure-state homogeneous quotient. The proof genuinely needs uniform control of the mixed-centroid directions, and does not silently replace this hypothesis by the weaker quotient gap.

5. SHARP MEMORY ON COMPLEX PROJECTIVE ORBITS

Write $r_q = \sqrt{1 - 1/q}$ and associate to a rank-one orthogonal projector P the unit vector

$$(28) \quad Y(P) = \frac{P - I/q}{r_q} \in \text{Herm}_0(q), \quad \mathcal{K}_q = \text{conv}\{Y(P) : \text{rank } P = 1\}.$$

The body $\mathcal{K}_q$ consists of normalized centered density matrices. It is invariant under conjugation and contains zero. Its extreme orbit is $\mathbb{CP}^{q-1}$. Fix an orthonormal real basis $F_1, \dots, F_D$ of $\text{Herm}_0(q)$ and use its coordinate queries in (5). One may take as seeds the q^2 rank-one projectors onto e_i , $(e_i + e_j)/\sqrt{2}$ and $(e_i + ie_j)/\sqrt{2}$, $i < j$. Their centered matrices affinely span $\text{Herm}_0(q)$. All seed and query alphabets are fixed when q is fixed.

5.1. A projective enclosure with quadratic loss. For rank-one projectors use the distance $d(P, Q) = \sqrt{1 - \text{tr}(PQ)}$, the sine of their principal angle.

Lemma 5.1 (Finite projective synthesis). *For $0 < h \leq 1$, there is an h -net $\mathcal{V}_h$ of rank-one projectors with at most*

$$(29) \quad q(2\lceil 2\sqrt{2(q-1)/h} \rceil + 1)^{2(q-1)}$$

elements, and the entries of all these projectors can be chosen in $\mathbb{Q} + i\mathbb{Q}$. If $qh^2 < 1$, then

$$(30) \quad (1 - qh^2)\mathcal{K}_q \subseteq \text{conv}\{Y(P) : P \in \mathcal{V}_h\} \subseteq \mathcal{K}_q.$$

Proof. Every complex line has a representative with a largest coordinate scaled to one and the real and imaginary parts of all other coordinates in $[-1, 1]$. In each of the q charts round these $2(q-1)$ real coordinates to the grid of step $1/M$, where $M = \lceil 2\sqrt{2(q-1)/h} \rceil$. If v, w are the original and rounded representatives, their norms are at least one and $\|v - w\| \leq \sqrt{2(q-1)}/(2M)$. Normalization changes their distance by at most a factor two, and the sine distance is at most the distance of these normalized representatives. Thus the stated set is an h -net. The projector $w w^* / \|w\|^2$ has rational real and imaginary entries, and the grid count is (29).

For a nonzero traceless Hermitian H , let λ_1 and λ_q be its largest and smallest eigenvalues. A net projector within h of a top eigenline satisfies

$$\text{tr}(HP) \geq \lambda_1 - (\lambda_1 - \lambda_q)h^2 \geq (1 - qh^2)\lambda_1.$$

Indeed $-\lambda_q \leq (q-1)\lambda_1$ by trace zero. The support function of $\mathcal{K}_q$ at H is λ_1/r_q . The support-function criterion for convex containment proves (30). It applies to every direction, not just a finite set of tested queries. $\square$

Theorem 5.2 (A finite-alphabet exact upper machine). *For every finite unitary command alphabet and $0 < \rho \leq 1/10$, the experiment on (28) has an exact machine of width $O_q(N^{q-1})$. For each N , all command transition matrices can be chosen independent of the epoch. Each row uses at most q^2 successors. The final decoder may depend on N .*

Proof. Take $h = (8qN)^{-1/2}$ and $\eta = 1 - 1/(8N)$ in Lemma 5.1. Let V be the resulting finite set of centered vectors, and put $a_t = \eta^{-t}/4$. A label i represents $a_t Y(P_i)$ at epoch t . This is a proof description of a label; no matrix is stored in the register.

Since $\rho/a_0 = 4\rho \leq 2/5 < \eta$ and $0 \in \mathcal{K}_q$, any seed $\rho Y(P)$ is a convex combination of $a_0 V$. For each label i and each command U , the vector $\eta U Y(P_i) U^*$ lies in $\text{conv } V$. Choose a convex representation and use its weights as the stochastic transition row. The choices depend on the command and label but not on t . Multiplying by a_{t+1} shows that the expected represented vector is updated by exact conjugation, since $a_{t+1}\eta = a_t$.

Induction gives expected vector $\rho Y(U_w P U_w^*)$. The terminal column for query j is $a_N \text{tr}(F_j Y(P_i))$. It is legal because $|\text{tr}(F_j Y(P_i))| \leq 1$ and Bernoulli's inequality gives $\eta^N \geq 7/8$, so $a_N \leq 2/7$. Sampling a binary answer with this mean proves every conditional probability in (5) exactly. Carathéodory's theorem in real dimension $D = q^2 - 1$ permits q^2 successors. The number in (29) is $O_q(h^{-2(q-1)}) = O_q(N^{q-1})$. $\square$

The upper argument uses a quadratic support loss along the projective orbit. Applying a net on the full ambient sphere would instead give the larger exponent $(q^2 - 2)/2$.

5.2. Fixed finite algebraic alphabets. For $1 \leq j < q$ let $R_j, D_j \in \text{SU}(q)$ act as the identity outside the adjacent coordinate pair $(j, j+1)$ and on that pair as

$$(31) \quad R = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}, \quad D = \begin{pmatrix} \zeta & 0 \\ 0 & \bar{\zeta} \end{pmatrix}, \quad \zeta = \frac{3 + 4i}{5}.$$

The alphabet has $4(q-1) + 1$ symbols: $\{I, R_j^{\pm 1}, D_j^{\pm 1} : 1 \leq j < q\}$. Assign probability $1/2$ to I and $1/[8(q-1)]$ to each other symbol.

Proposition 5.3 (Qualitative group expansion). *This command law has a norm gap on $L_0^2(\text{SU}(q))$ for every fixed q . Its entries and probabilities are algebraic. No numerical gap value is asserted for this family.*

Proof. The number ζ is not a root of unity: $\zeta + \zeta^{-1} = 6/5$ is rational but not an algebraic integer. The powers of R_j and D_j therefore have closures containing two distinct one-parameter subgroups in the corresponding embedded $\text{SU}(2)$. Their Lie brackets generate that $\mathfrak{su}(2)$. The adjacent embedded Lie algebras generate $\mathfrak{su}(q)$: adjacent elementary off-diagonal directions produce all other off-diagonal directions by brackets, and their self-brackets produce the traceless diagonal directions. The closed generated subgroup is consequently $\text{SU}(q)$.

The adjoint entries are algebraic. The spectral-gap theorem of Benoist–de Saxcé [2, Theorem 1.2] applies to an adapted algebraic measure on this compact connected group with simple Lie algebra. Equivalently, one may use the algebraic unitary result of Bourgain–Gamburd [3]. The symmetric average has an upper gap; weight $1/2$ at the identity makes its spectrum nonnegative. Its mean-zero operator norm is therefore strictly below one. This is the external full-group theorem, not a calculation in the defining q -dimensional representation. $\square$

Proof of Theorem 1.1. Corollary 4.3 gives the lower order, for every fixed positive calibration κ . Theorem 5.2 gives the matching exact upper order. Proposition 5.3 supplies fixed finite alphabets. The quantum assertion follows from the next proposition. Taking binary logarithms gives the label-bit formula. $\square$

Proposition 5.4 (A fixed quantum realization). *The projective experiment has an exact quantum realization on $\mathbb{C}^q$ using fixed seed resets, the prescribed unitary commands and fixed query measurements. It is independent of N .*

Proof. On seed P reset the density matrix to

$$\sigma_P = I/q + \rho Y(P) = (1 - \rho/r_q)I/q + (\rho/r_q)P.$$

This is positive and has trace one, since $0 < \rho \leq 1/10 < r_q$. Commands act by conjugation. For query j , use effects $(I + F_j)/2, (I - F_j)/2$. Hilbert–Schmidt normalization implies $\|F_j\|_{\text{op}} \leq 1$, so these are positive effects summing to I . Their outcome mean is $\text{tr}(F_j \sigma_{U_w P U_w^*}) = \rho \text{tr}(F_j Y(U_w P U_w^*))$. This is precisely (5). The seed reset is a channel, not a unitary; only one query is measured. $\square$

The comparison is a fixed q -dimensional quantum register versus $(q-1) \log_2 N + O(1)$ classical label bits. Dimension or entropy advantages for quantum models are prior phenomena [6, 17, 5]. The assertion here is the proved classical width order for this complete worst-word numerical behavior. It does not request jointly independent answers to all queries or optimize arbitrary adaptive experiments.

6. AN EFFECTIVE SIX-COMMAND RATIONAL EXAMPLE

The preceding group-gap application is qualitative. We now choose a separate arithmetic alphabet for which a classical theorem supplies a numerical infinite-dimensional norm bound. This does not evaluate the gap of the earlier five-command example.

Let R_x, R_y, R_z be rotations about the coordinate axes with cosine $-3/5$ and sine $4/5$; explicitly,

$$(32) \quad R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3/5 & -4/5 \\ 0 & 4/5 & -3/5 \end{pmatrix}, \quad R_y = \begin{pmatrix} -3/5 & 0 & 4/5 \\ 0 & 1 & 0 \\ -4/5 & 0 & -3/5 \end{pmatrix}, \quad R_z = \begin{pmatrix} -3/5 & -4/5 & 0 \\ 4/5 & -3/5 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Use the fixed six-letter alphabet $\mathcal{A}_V = \{R_x^{\pm 1}, R_y^{\pm 1}, R_z^{\pm 1}\}$ and its uniform law. These are the adjoint rotations of the classical norm-five quaternions $1 \pm 2\mathbf{i}$, $1 \pm 2\mathbf{j}$, $1 \pm 2\mathbf{k}$, normalized by $\sqrt{5}$.

Proposition 6.1 (External arithmetic gap certificate). *For $P_V f = (1/6) \sum_{R \in \mathcal{A}_V} f(R^{-1} \cdot)$ on $L_0^2(\mathbb{S}^2)$,*

$$(33) \quad \|P_V\| \leq \frac{2\sqrt{5}}{6} = \frac{\sqrt{5}}{3}.$$

Consequently $1 - \|P_V\|^2 \geq 4/9$.

Proof. This is the $p = 5$ specialization of the Lubotzky–Phillips–Sarnak spherical Hecke bound [10, 11]. In its unnormalized convention the $p + 1$ norm- p quaternion representatives act by a sum of rotations whose mean-zero norm is at most $2\sqrt{p}$. The representatives have positive odd real coordinate and even other coordinates when $p \equiv 1 \pmod{4}$; at $p = 5$ they are exactly the six displayed ones. The coauthor account [12, Section 3.2, the V -gate example] spells out these representatives and their Hecke interpretation. The unnormalized spherical norm convention is also stated in [13, Theorem 3(ii)]. Dividing by six gives (33). Passing from quaternions to adjoint rotations identifies $\mathbb{S}^2$ as the homogeneous quotient; inverse signs do not affect this symmetric average. The arithmetic bound is invoked as an external theorem, not reproved by finite harmonic tests. $\square$

Use six seeds $\{\pm e_j : 1 \leq j \leq 3\}$, amplitude $\rho = 1/10$ and three final coordinate queries in (5).

Theorem 6.2 (Numerical finite-horizon width certificate). *For $0 \leq \varepsilon < 1/(20\sqrt{3})$ and every integer $N \geq 1$,*

$$(34) \quad \frac{(1/(10\sqrt{3}) - 2\varepsilon)^4}{221184} N \leq W_{N,\varepsilon}(\mathcal{A}_V) \leq W_{N,0}(\mathcal{A}_V) < 150N.$$

The upper machine has exact rational command and decoder rows, common command matrices, and an N -dependent decoder. At zero error every internal selected cut has separate positive minimum four, while

$$(35) \quad W_{N,0} \geq \frac{N}{19\,906\,560\,000}.$$

The constants are proof constants, not sharp asymptotic constants.

Proof. For the standard $\text{SO}(3)$ action the orbit of every unit vector is uniform on $\mathbb{S}^2$. A chordal cap of radius $r \leq 1$ has probability $r^2/4 \leq r^2$, so take $D = 3, s = 2, A = 1$ in Theorem 3.5. The sphere gap transfers to $L^2(\mathbb{R}^3)$ by polar coordinates. Thus $B_{3,1,2} = 98304$, and the squared gap $4/9$ gives the lower bound in (34). At $\varepsilon = 0$ the fourth power of $1/(10\sqrt{3})$ is $1/90000$, which yields (35).

Here is an explicit cardinality estimate for the upper construction. Take both rational stereographic charts

$$t \mapsto \frac{(2t_1, 2t_2, \pm(1 - \|t\|^2))}{1 + \|t\|^2}, \quad t \in [-1, 1]^2,$$

with a grid of step $1/M$ and $M = \lceil 2\sqrt{2N} \rceil$. Every hemisphere is covered by its chart on the unit disk. Each chart is 2-Lipschitz, so its grid images form a sphere net of chordal radius at most $\sqrt{2}/M \leq 1/(2\sqrt{N})$. Its convex hull contains η times the unit ball, with $\eta = 1 - 1/(8N)$, by the support function identity $u \cdot v = 1 - \|u - v\|^2/2$. There are at most $2(2M + 1)^2 \leq (82 + 48\sqrt{2})N < 150N$ labels. Use scales $a_t = \eta^{-t}/4$, initialization representing ρx , rows representing $\eta R u$ in this hull, and decoder $a_N e_j \cdot u$. The verification is the induction in Theorem 5.2. All data are rational; each nonempty rational convex system has rational weights, with at most four successors. The final decoder is bounded by $2/7$.

At a selected internal cut, fix any common command prefix. The six seed images affinely span $\mathbb{R}^3$, so the normalized continuation rank is four; future coordinate queries still separate these dimensions after any common suffix. For an upper factorization take a regular tetrahedron containing the radius- ρ ball and having circumradius $3\rho < 1$. Its four vectors give legal continuations under all future rotations. One embeds this selected-cut factorization in a complete machine by allowing uncapped raw histories elsewhere. Therefore the separate minimum is exactly four. This does not give a four-state machine at all cuts simultaneously. $\square$

For example, (35) exceeds four once $N \geq 79\,626\,240\,001$. The size of this number illustrates the conservative constants; the benefit is that no unspecified spectral constant is needed for this alphabet.

The associated qubit operations are $(I - 2i\sigma_j)/\sqrt{5}$ and their inverses, with seed Bloch vector $x/10$ and final Pauli query. They realize the numerical experiment exactly, independently of N . This quantum construction and the arithmetic gates are classical; (34) is the finite-register consequence. The earlier five-gate alphabet remains qualitatively gapped with a non-evaluated constant, and is retained unchanged in the supporting material.

7. OPERATIONAL AND THEOREM-LEVEL COMPARISONS

The comparisons below fix the minimized object before comparing results. They do not use a change of terminology as evidence of originality.

7.1. Ordered randomized programs.

Proposition 7.1 (The layered-program interpretation). *For a fixed command alphabet, Definition 2.1 is equivalent to a nonuniform probabilistic ordered read-once layered program with cut-dependent widths and real terminal columns. For one selected column, randomized Boolean acceptance has probability $(1 + m)/2$, where m is the column's mean, without changing any command-layer width. A separate charged terminal layer needs at most two labels. Maximum width is not total program size, and neither measure prices the row descriptions.*

Proof. At layer t , the outgoing edge weights labelled by command a are exactly the row $T_{t,a}(s, \cdot)$. Summing path probabilities gives (6). Conversely the edge weights of a probabilistic layered program form those stochastic matrices. At the terminal label s , toss a Bernoulli coin of parameter $(1 + d_j(s))/2$. This implements Boolean acceptance with the claimed probability. Choosing a query selects one column; it does not request independent outcomes for all columns. Padding a layer by unreachable labels and assigning normalized unused rows proves the same assertion for varying widths. $\square$

The equivalence permits arbitrary real edge probabilities and externally known layers. The Fourier-growth results for regular and width-three branching programs [15, 16, 9] have different structural assumptions and objectives. Here the external group average has a gap; the hidden transition matrices need not be regular or have a gap. In particular, a stochastic reset matrix is permitted. The all-spectra occupation inequality is a lower bound for the specified matrix-coefficient response, not a pseudorandom generator or a general lower bound for Boolean OBDDs.

7.2. Zero-delay coding and state aggregation. The structural coding results of Witsenhausen and Walrand–Varaiya, as formulated in [19, Theorems 1–2], reduce a Markov-source encoder to the current source and a posterior based on past transmitted messages. Wood–Linder–Yüksel prove stationary optimality and periodic finite-memory approximation for average distortion [19, Theorems 3–4]. A posterior is generally a point of a continuous simplex, not a finite alphabet. Those results optimize ongoing reproduction for a given stochastic source. Here a machine sees commands rather than the true orbit vector, and every command word must satisfy a final

numerical specification. Only its current label, not a free transcript of sent messages, is retained. Thus a sufficient belief state in the coding problem does not give a cardinality upper bound in the present problem. Conversely our width lower bound does not lower-bound the optimal average distortion of those coders.

Nonanticipative rate-distortion minimizes directed information over causal reproduction kernels under expected distortion constraints. For example, the finite-horizon Markov-source analysis in He–Charalambous–Stavrou [7, Sections II–III] derives a dynamic optimization of those kernels and an alternating computation. That objective is neither $\max_t K_t$ nor $\sum_t K_t$. In particular, mutual information can be small for a channel whose exact finite positive realization needs many labels. A source-law rate-distortion optimization cannot be inserted in place of the wordwise specification without a separate reduction.

Geiger–Tammel [4, Theorem 9] characterize higher-order strong lumpability by an equality of conditional entropies for a stationary finite Markov chain and a deterministic lumping. Their entropy-rate preservation results concern the whole lumped process. We allow randomized, epoch-dependent maps, retain a command-driven terminal observable, and do not require preservation of the trajectory entropy rate or of a Markov partition. State aggregation is therefore an adjacent comparison, not a theorem claimed to have been subsumed. The common use of conditional expectations does not identify the optimization tasks.

7.3. Conditional expectations and Gaussian entropy. Atalik–Köse–Gastpar [1, Theorem 1 and Proposition 2] study $M = \mathbb{E}[X \mid X + W]$ for Gaussian observation noise. In the identity observation case their inequality has the form $h(M) \geq 2h(X) - h(X + W)$, with a vector extension. Lemma 3.2 instead permits an arbitrary finite conditioning variable J , adds new independent noise *after* conditioning, and upper-bounds a difference of two smoothed entropies by the martingale coupling’s squared displacement. A discrete M is allowed here, whereas its unsmoothed differential entropy is not the quantity being compared. The two displayed inequalities are not direct substitutions of one another. The Gaussian score and posterior-covariance identities used in (13) are classical, and our short semiconcavity proof is not claimed as a new estimation identity. Polyanskiy–Wu’s Wasserstein entropy continuity [14] provides a further comparison; the quadratic cost here uses the particular conditional-mean coupling, not arbitrary transport. The argument needed for the occupation theorem is the amortization of that cost together with orbital sparsity.

7.4. Positive realization and orbit geometry. Positive realization and invariant convex sets have a long history [8, 18]. The occupation theorem does not require an internal state to be a point on an observed orbit or in the residual span. Such a restriction would miss hidden stochastic lifts. Classical sphere and projective covering exponents explain the net counts; the additional issue for conjugation is uniformity over spectra of all conditional centroids. Theorem 4.2 settles that issue through a separated spectral projector and the explicit Grassmann bound. Its exponent is matched by an exact projective, rather than ambient spherical, construction. This is the reason that $q - 1$, not $(q^2 - 2)/2$, appears in Theorem 1.1.

The deep group gaps are external inputs. Proposition 5.3 uses a qualitative theorem; Proposition 6.1 uses a numerical Hecke norm theorem for a different named alphabet. Finite diagonalization of a few harmonics is not a proof of either full gap. Neither density of a subgroup nor a gap in its defining finite-dimensional representation is silently promoted to a full L^2 norm gap.

7.5. Implementation and the remaining distinctions. All exact theorems here count atomic nonuniform labels. Replacing each row by a rational approximation and then using a fixed-duration padded fair-bit comparison sampler, as in the retained predecessor, incurs additional control labels and terminal approximation error. The padding removes dependence of duration on private coin outcomes; it does not price the read-only threshold tables or prove uniform space complexity. No exact finite-coin compiler for arbitrary irrational rows is asserted.

The proved results are the representation-valued occupation inequality, the all-spectra conjugation estimate, the matched projective cardinality order, and the effective six-command consequence. They do not classify all compact representations, all finite alphabets, or arbitrary adaptive experimental design. The sufficient calibration threshold is not shown optimal, and signal approaching zero with the horizon requires retaining the explicit constants rather than invoking a fixed-signal asymptotic. Exact finite integer optima and leading constants remain different questions. The standalone article contains all proofs used for its new claims; preservation records and computation are not mathematical or priority certificates.

REFERENCES

- [1] A. Atalik, A. Köse and M. Gastpar, Differential entropy of the conditional expectation under additive Gaussian noise, *IEEE Trans. Signal Process.* **70** (2022), 4851–4866; arXiv:2106.04677v3. doi:10.1109/TSP.2022.3211403.
- [2] Y. Benoist and N. de Saxcé, A spectral gap theorem in simple Lie groups, *Invent. Math.* **205** (2016), 337–361; arXiv:1405.1808. doi:10.1007/s00222-015-0636-2.
- [3] J. Bourgain and A. Gamburd, A spectral gap theorem in $SU(d)$, *J. Eur. Math. Soc.* **14** (2012), 1455–1511.
- [4] B. C. Geiger and C. Temmel, Lumpings of Markov chains, entropy rate preservation, and higher-order lumpability, *J. Appl. Probab.* **51** (2014), 1114–1132; extended version arXiv:1212.4375v6.
- [5] F. Ghafari et al., Dimensional quantum memory advantage in the simulation of stochastic processes, *Phys. Rev. X* **9** (2019), 041013.
- [6] M. Gu, K. Wiesner, E. Rieper and V. Vedral, Quantum mechanics can reduce the complexity of classical models, *Nature Commun.* **3** (2012), 762.
- [7] Z. He, C. D. Charalambous and P. A. Stavrou, A new finite-horizon dynamic programming analysis of nonanticipative rate-distortion function for Markov sources, arXiv:2411.11698v1 (2024).
- [8] A. Heller, On stochastic processes derived from Markov chains, *Ann. Math. Statist.* **36** (1965), 1286–1291.
- [9] C. H. Lee, E. Pyne and S. Vadhan, Fourier growth of regular branching programs, *APPROX/RANDOM 2022*, LIPIcs **245**, Art. 2, 2:1–2:21.
- [10] A. Lubotzky, R. Phillips and P. Sarnak, Hecke operators and distributing points on the sphere I, *Comm. Pure Appl. Math.* **39**, suppl. (1986), S149–S186.
- [11] A. Lubotzky, R. Phillips and P. Sarnak, Hecke operators and distributing points on S^2 . II, *Comm. Pure Appl. Math.* **40** (1987), 401–420.
- [12] O. Parzanchevski and P. Sarnak, Super-golden-gates for $PU(2)$, *Adv. Math.* **327** (2018), 869–901; arXiv:1704.02106v1.
- [13] G. Pisier, Quadratic forms in unitary operators, *Linear Algebra Appl.* **267** (1997), 125–137; arXiv:math/9512207, Theorem 3(ii).
- [14] Y. Polyanskiy and Y. Wu, Wasserstein continuity of entropy and outer bounds for interference channels, *IEEE Trans. Inform. Theory* **62** (2016), 3992–4002.
- [15] O. Reingold, T. Steinke and S. Vadhan, Pseudorandomness for regular branching programs via Fourier analysis, *RANDOM 2013*, LNCS **8096**, 655–670.
- [16] T. Steinke, S. Vadhan and A. Wan, Pseudorandomness and Fourier growth bounds for width 3 branching programs, *Theory Comput.* **13** (2017), Art. 12, 1–50.
- [17] J. Thompson, A. J. P. Garner, V. Vedral and M. Gu, Using quantum theory to simplify input-output processes, *npj Quantum Inform.* **3** (2017), 6.
- [18] M. Vidyasagar, *Hidden Markov Processes: Theory and Applications to Biology*, Princeton University Press, 2014.
- [19] R. G. Wood, T. Linder and S. Yüksel, Optimal zero delay coding of Markov sources: stationary and finite memory codes, *IEEE Trans. Inform. Theory* (2017); arXiv:1606.09135, full revised version. doi:10.1109/TIT.2017.2692215.
- [20] Q. Qi, General Theta Foundations I: Entropy dissipation and sharp finite-alphabet memory, Revision 37 (2026), publication c5b8015887f83833. Preserved in the supporting article; cited only as a predecessor, not external validation.

PRESERVED MATHEMATICAL DEVELOPMENT

General Theta Foundations I - revision 37

The complete current article precedes this divider.

The following predecessor cumulative volume is reproduced without alteration.

Frozen predecessor publication:

c5b8015887f83833832e393db69380afeb5a9a3b

This archive is not part of the compact referee package. Independent historical analytic obligations keep their original hypotheses and proof status.

GENERAL THETA FOUNDATIONS I: ENTROPY DISSIPATION AND SHARP FINITE-ALPHABET MEMORY

QIAN QI

ABSTRACT. Let a fixed finite family of orthogonal commands have an absolute spectral gap on the full mean-zero L^2 space of the sphere. We prove that its wordwise numerical behavior over N commands requires and admits $\Theta(N^{(d-1)/2})$ persistent classical labels, at zero error and at every fixed sufficiently small output error. The lower bound permits arbitrary hidden states, horizon-dependent design and unrelated rows at different epochs. It removes the logarithmic loss of a packet-mixing argument. The proof compares entropy production under random rotations with the entropy cost of conditional-mean compression after Gaussian smoothing; the latter is charged to one telescoping quadratic loss. A weighted occupation theorem also permits time-dependent command laws and unbounded widths at the other cuts. For five explicit rational three-dimensional commands, the classical width is consequently $\Theta(N)$, or $\log_2 N + O(1)$ persistent label bits, while a fixed qubit realizes the same numerical probabilities. The external spectral-gap input is qualitative, not a numerical gap certificate. Our resource is nonuniform clocked atomic-row width. A separate padded fair-bit compiler has no update-duration side channel and does not identify this width with uniform computational space.

1. INTRODUCTION

A positive factorization at one time does not specify a stochastic machine that remains valid at all later times. This distinction becomes quantitative when a finite register must preserve an orthogonal orbit under every command word. At each isolated cut a fixed-dimensional simplex can encode the observable vector, yet its successive positive realizations need not be compatible. The problem is to determine the number of labels required by one complete causal realization, with all hidden directions allowed.

Fix $U_a \in O(d)$ for a in a finite alphabet $\mathcal{A}$, seeds $x \in X = \{\pm e_1, \dots, \pm e_d\}$ and signal $0 < \rho \leq 1/10$. The last command acts on the left: $U_w = U_{a_N} \cdots U_{a_1}$. A query $j \in \{1, \dots, d\}$ is supplied only after the N commands. The required binary response is

$$(1) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho e_j^T U_w x}{2}, \quad b \in \{-1, 1\}.$$

All probabilities lie in $[9/20, 11/20]$. Equality, or the stated uniform error, is required on every word, not merely under a stationary input law. Let $W_{N,\varepsilon}$ be the least maximum number of persistent labels with binary row total-variation error at most ε ; the complete machine convention is Definition 2.1.

For a probability vector p on $\mathcal{A}$, write

$$(2) \quad Pf(v) = \sum_a p(a) f(U_a^{-1}v), \quad \|P|_{L_0^2(\sigma)}\| \leq \lambda < 1,$$

Date: September 25, 2026. Revision 37.

2020 Mathematics Subject Classification. Primary 93B15, 68Q10; Secondary 60J05, 22D10, 94A17, 81P45.

Key words and phrases. Positive realization, spectral gap, entropy dissipation, probabilistic branching programs, finite alphabets, quantum memory.

where σ is normalized surface measure on $\mathbb{S}^{d-1}$. The inverse convention makes P also the density transfer operator. The same norm condition holds for its adjoint. It is a hypothesis on *all* spherical harmonics, not the spectrum of the defining d -dimensional representation.

Theorem 1.1 (Sharp finite-alphabet width). *Suppose (2) holds and $0 \leq \varepsilon < \rho/(2\sqrt{d})$. Then, for constants depending only on $d, \lambda, \rho, \varepsilon$,*

$$(3) \quad cN^{(d-1)/2} \leq W_{N,\varepsilon} \leq W_{N,0} \leq C_d N^{(d-1)/2} \quad (N \geq 1).$$

The lower bound allows arbitrary hidden state continuations and cut-dependent stochastic rows. The upper realization has common command rows and an N -dependent terminal decoder. For rational commands, seeds and signal, its rows can all be rational and each update has at most $d+1$ successors. If the identity is available and $d\rho < 1$, every isolated cut nevertheless has positive minimum $d+1$.

The preceding finite-alphabet result [14] gave lower order $(N/\log(N+1))^{(d-1)/2}$ and the same upper construction. Its converse waited $O(\log k)$ command slots for the product distribution to become uniform to the resolution of a k -center quantizer. The proof here does not shorten that mixing time or assert one-step total-variation mixing of an atomic walk. It retains an entropy potential across *all* updates, so that repeated mixing need not be purchased afresh.

Put $Y_t = U_{a_t} \cdots U_{a_1} x$ and $Z_t = \mathbb{E}[Y_t \mid S_t]$. The auxiliary Gaussian density $f_t = \text{Law}(Z_t) * \gamma_\tau$ is used only in the proof. Randomizing a command increases its entropy by an angular Hellinger variance times the spectral gap. Replacing the resulting vector by a conditional mean decreases entropy by at most

$$\frac{\mathbb{E}\|Z_t\|^2 - \mathbb{E}\|Z_{t+1}\|^2}{2\tau^2}.$$

This cost telescopes. If Z_t has only k possible values but has nonvanishing mean norm, smoothing at scale $\tau \asymp k^{-1/(d-1)}$ still leaves a definite angular Hellinger variance. These statements combine to give an $O(k^{2/(d-1)})$ occupation bound, with no logarithmic factor. No noise is added to the actual command experiment.

The five rational gates of [14] consequently have exact and fixed-error classical width $\Theta(N)$. Their one-qubit realization is unchanged. This is a separation in worst-case numerical behavior under externally supplied words: the classical label bits are $\log_2 N + O(1)$, not linear bit space. The gap for these gates is supplied by Bourgain–Gamburd [1]; we do not give a numerical gap value.

Quantity	Meaning
$K_t, W_{N,\varepsilon}$	Available labels at a cut; smallest whole-run peak.
q_t, v_t	$\mathbb{E}\ Z_t\ $ and $\mathbb{E}\ Z_t\ ^2$, proof-side conditional moments.
$h(f), \Pi$	Differential entropy with natural logarithms; spherical averaging at each radius.
$g_t = 1 - \lambda_t^2$	The squared-norm gap of the command law at epoch t .
κ	A positive lower bound on retained terminal amplitude.

The new results are the one-step entropy comparison, the gap-weighted occupation law and its sharp width consequences. Rational spherical synthesis and the qubit operations are inherited and reproved. Spectral gaps, Gaussian score identities, Hellinger comparison and spherical quantization are classical inputs. Appendix A repairs the implementation timing convention; it is not counted as a new sampling algorithm. Section 7 gives the corresponding theorem comparisons. The prior packet, monomial and cyclic results remain intact in the supporting article rather than being silently weakened or used under altered hypotheses.

2. THE EXPERIMENT AND ITS REGISTERS

Definition 2.1 (Clocked stochastic realization). For a prescribed horizon N , a machine consists of finite alphabets $[K_t]$, $0 \leq t \leq N$, initial distributions E_x , row-stochastic matrices $T_{t,a}$ of size $K_t \times K_{t+1}$, and columns $d_j \in [-1, 1]^{K_N}$. Its response mean is $E_x T_{0,a_1} \cdots T_{N-1,a_N} d_j$. Its error is one half the maximum absolute discrepancy from $\rho e_j^T U_w x$ over the finite set of seeds, words and queries. Its peak is $\max_t K_t$, also including two labels if the final binary answer is retained at a charged cut. The minimum peak at error ε is $W_{N,\varepsilon}$.

The seed and query are each supplied once. The current command is consumed in its update. Only the updated register survives; a retained seed, input transcript, random seed, or data-dependent scheduling variable would be part of that register. Available, not merely reachable, labels are counted; padding therefore gives a larger admissible profile. Every unused row is normalized. The epoch, horizon, row tables, their construction, lookup and arithmetic, and exact atomic stochastic sampling are free. A machine may be redesigned for every N and need not use common rows. This is a nonuniform probabilistic ordered read-once width, not total program size, autonomous state count or uniform computational space.

All converse distributions below are imposed on the external word, independently of private randomness. Wordwise correctness licenses this choice; the test law is not additional runtime information. The commands in this paper's new converse are independent, with a possibly different law at each epoch. No independence of the hidden past given its current state is assumed.

Lemma 2.2 (Conditional moments and calibration). *Fix a unit seed x . Let the fresh command a_{t+1} be independent of the past and let private updates obey the stated Markov rule. Define*

$$Z_t = \mathbb{E}[Y_t | S_t], \quad q_t = \mathbb{E}\|Z_t\|, \quad v_t = \mathbb{E}\|Z_t\|^2, \quad Y_t = U_{a_t} \cdots U_{a_1} x.$$

Then $\|Z_t\| \leq 1$, q_t and v_t are nonincreasing, and

$$(4) \quad Z_{t+1} = \mathbb{E}[U_{a_{t+1}} Z_t | S_{t+1}], \quad \Delta_t := v_t - v_{t+1} = \mathbb{E}\|U_{a_{t+1}} Z_t - Z_{t+1}\|^2.$$

In particular $\sum_{t < N} \Delta_t \leq 1$. If the machine has wordwise error at most ε , then

$$(5) \quad q_t \geq q_N \geq \kappa := \rho/\sqrt{d} - 2\varepsilon.$$

Proof. Conditioning on S_t, a_{t+1} and its randomized successor does not change the conditional mean of Y_t : the new command is independent of the past and the update uses that past only through S_t . The tower property gives (4). Conditional Jensen and the orthogonality of U_a give both monotonicities. The orthogonal projection identity for conditional expectation gives the equality for Δ_t . Initially $Z_0 = x$, even when initialization randomizes the register.

In the proof only, combine the separate decoder columns as

$$D(S_N) = (d_1(S_N), \dots, d_d(S_N)), \quad \|D(S_N)\| \leq \sqrt{d}.$$

For every word its conditional mean is within $2\sqrt{d}\varepsilon$ of ρY_N . Hence

$$\rho - 2\sqrt{d}\varepsilon \leq \mathbb{E}\langle Y_N, D(S_N) \rangle = \mathbb{E}\langle Z_N, D(S_N) \rangle \leq \sqrt{d}q_N.$$

This does not require jointly sampling multiple answers. The vector combines different permitted final query columns only in the proof. $\square$

Proposition 2.3 (The isolated-cut minimum). *If $\text{Id} \in \mathcal{A}$ and $d\rho < 1$, the normalized residual rank and separate positive minimum at every internal cut equal $d + 1$.*

Proof. The residuals are affine functions of $\rho U_w x$. Idle prefixes from $\pm e_j$ span $\mathbb{R}^d$, and idle suffixes with coordinate queries separate these coordinates. Normalization adds an independent constant direction, so the rank is $d + 1$. A regular simplex with inradius ρ has circumradius $d\rho < 1$ and

contains every reachable predictive vector. Its vertices therefore define legal binary response channels after every orthogonal suffix. Decompose the selected-cut vector into those vertices. One may store the raw prefix before this cut and the raw suffix together with the simplex label after it. Those other state sets are finite and uncapped. Thus this is a factorization within a complete machine. Rank gives the matching lower bound. This proposition, inherited from [14], does not assert simultaneous attainment. $\square$

3. ENTROPY COST OF A CONDITIONAL CENTROID

For $\tau > 0$, let γ_τ be the centered Gaussian density on $\mathbb{R}^d$ with covariance $\tau^2 I$. If X is bounded, write $f_X = \text{Law}(X) * \gamma_\tau$ and $h(f) = -\int_{\mathbb{R}^d} f \log f$. All entropies and divergences in this section use natural logarithms. Gaussian mixtures with bounded centers have finite entropy and the Gaussian tail bounds needed below.

Lemma 3.1 (Quadratic cost of barycentric compression). *Let X be a bounded random vector and J a finite random variable on the same probability space. Set $M = \mathbb{E}[X \mid J]$. Then*

$$(6) \quad h(f_X) - h(f_M) \leq \frac{\mathbb{E}\|X - M\|^2}{2\tau^2}.$$

No sign assertion for the left side is required.

Proof. Put $f = f_M$. Direct differentiation of the finite Gaussian mixture gives

$$(7) \quad \nabla^2 \log f(y) = -\tau^{-2} I + \tau^{-4} \text{Cov}(M \mid M + \tau G = y) \succeq -\tau^{-2} I,$$

where G is an independent standard Gaussian. This identity also follows by differentiating the log-sum of exponentials after factoring out $\exp(-\|y\|^2/(2\tau^2))$. Define $\varphi(x) = -\mathbb{E}_G \log f(x + \tau G)$. Boundedness of the centers justifies differentiation and gives $\nabla^2 \varphi \preceq \tau^{-2} I$. Therefore

$$\varphi(X) - \varphi(M) \leq \langle \nabla \varphi(M), X - M \rangle + \frac{\|X - M\|^2}{2\tau^2}.$$

The first term averages to zero conditional on J . On the other hand, nonnegativity of $D(f_X \| f)$ gives

$$h(f_X) - h(f) \leq -\int (f_X - f) \log f = \mathbb{E}\varphi(X) - \mathbb{E}\varphi(M).$$

These inequalities prove the result. The Gaussian density used for convolution is a common mathematical kernel; no additional noisy state is assumed in the transducer. $\square$

The Gaussian score/Hessian mechanism is classical. For comparison, Polyanskiy–Wu [2, Proposition 1 and Corollary 4] control smoothed entropy differences by a Wasserstein distance. Here the specified martingale coupling $M = \mathbb{E}[X \mid J]$ cancels the linear Taylor term. The resulting one-sided *squared* error, rather than a square-root continuity estimate for unrelated measures, is what can be summed along a stochastic register. We use the explicit proof above, not an assumed entropy monotonicity under convex order.

3.1. An entropy increment from a spherical gap. Let Π be spherical averaging at each radius on $L^2(\mathbb{R}^d)$:

$$(\Pi u)(r\omega) = \int_{\mathbb{S}^{d-1}} u(r\theta) d\sigma(\theta).$$

It is the orthogonal projection onto radial functions. The angular gap (2), integrated in polar coordinates, gives

$$(8) \quad \|Pu\|_2^2 \leq \|\Pi u\|_2^2 + \lambda^2 \|u - \Pi u\|_2^2.$$

Here P acts on Euclidean functions by the same orthogonal pullbacks; it preserves radial functions and commutes with Π .

Lemma 3.2 (Entropy increment and angular Hellinger variance). *For a Gaussian mixture f with bounded centers,*

$$(9) \quad \mathbf{h}(Pf) - \mathbf{h}(f) \geq (1 - \lambda^2) \|\sqrt{f} - \Pi\sqrt{f}\|_2^2.$$

Without a strict gap the entropy increment is still nonnegative.

Proof. Set $f_a(y) = f(U_a^{-1}y)$ and $g = \sum_a p(a)f_a = Pf$. All f_a have the same entropy, and hence

$$\mathbf{h}(g) - \mathbf{h}(f) = \sum_a p(a) D(f_a \| g).$$

For probability densities u, v , Jensen's inequality gives $D(u \| v) \geq -2 \log \int \sqrt{uv} \geq 2(1 - \int \sqrt{uv}) = \|\sqrt{u} - \sqrt{v}\|_2^2$. Consequently the entropy increment is at least

$$\begin{aligned} \sum_a p(a) \|\sqrt{f_a} - \sqrt{g}\|_2^2 &= 1 - \|P\sqrt{f}\|_2^2 + \|P\sqrt{f} - \sqrt{g}\|_2^2 \\ &\geq 1 - \|P\sqrt{f}\|_2^2. \end{aligned}$$

As $\|\sqrt{f}\|_2^2 = 1$, (8) proves the assertion. The last statement also follows directly from entropy concavity. This argument uses the full L^2 gap only on square roots of smoothed probability densities; it assumes no logarithmic Sobolev inequality and no one-step mixing of point masses. $\square$

4. SPARSE CENTROIDS CANNOT BE ANGULARLY UNIFORM

Write $m = d - 1$. Fix a cap constant $A_d \geq 1$ such that

$$(10) \quad \sigma\{\theta : \|\theta - u\| \leq r\} \leq A_d r^m \quad (0 < r \leq 1, u \in \mathbb{S}^{d-1}).$$

For example, with $|\mathbb{S}^j|$ denoting surface area, one can use

$$A_d = \max \left\{ 1, \frac{|\mathbb{S}^{m-1}|}{m|\mathbb{S}^m|} \left(\frac{\pi}{2} \right)^m \right\}.$$

Indeed a chordal radius r has angular radius $2 \arcsin(r/2) \leq \pi r/2$, and polar integration with $\sin t \leq t$ proves the estimate. In dimension three, $A_3 = 1$ is a convenient nonoptimal choice, since the exact cap mass is $r^2/4$.

Lemma 4.1 (A scale-uniform sparse-mixture obstruction). *Suppose $\|Z\| \leq 1$, Z has at most k values, and $\mathbb{E}\|Z\| \geq \kappa$ for $0 < \kappa \leq 1$. Put*

$$(11) \quad r = \left(\frac{\kappa}{16A_d k} \right)^{1/m}, \quad \tau = \frac{\kappa r}{8\sqrt{2d}}, \quad f = \text{Law}(Z) * \gamma_\tau.$$

Then

$$(12) \quad \|\sqrt{f} - \Pi\sqrt{f}\|_2^2 \geq \frac{\kappa}{16}.$$

The locations and masses of the atoms are arbitrary.

Proof. Let $L = \{\|Z\| \geq \kappa/2\}$. Since $\mathbb{E}\|Z\| \leq \kappa/2 + \mathbb{P}(L)$, its probability is at least $\kappa/2$. Around each direction of a nonzero atom satisfying L place a spherical cap of chordal radius r . Their union Ω satisfies $\sigma(\Omega) \leq kA_d r^m = \kappa/16$.

Let G be independent standard Gaussian. Markov's inequality gives $\mathbb{P}(\|G\| \leq \sqrt{2d}) \geq 1/2$. On this event and L ,

$$\|\tau G\| \leq \kappa r/8, \quad \left\| \frac{Z + \tau G}{\|Z + \tau G\|} - \frac{Z}{\|Z\|} \right\| \leq \frac{2\|\tau G\|}{\|Z\|} \leq r/2.$$

The denominator is nonzero here because $r \leq 1$ and $\|\tau G\| \leq \|Z\|/4$. Thus the radial cone $C_\Omega = \{s\theta : s > 0, \theta \in \Omega\}$ has $\int_{C_\Omega} f \geq \kappa/4$.

Write $u = \Pi\sqrt{f}$. This is radial, and $\|u\|_2 \leq 1$. Polar integration yields $\int_{C_\Omega} u^2 = \sigma(\Omega)\|u\|_2^2 \leq \kappa/16$. The triangle inequality in $L^2(C_\Omega)$ now gives

$$\|\sqrt{f} - u\|_2 \geq \sqrt{\kappa/4} - \sqrt{\kappa/16} = \frac{\sqrt{\kappa}}{4}.$$

Squaring proves the assertion. Repeated atom directions and zero atoms only reduce the number of caps. No regularity or separation assumption on the centroid locations was used. $\square$

The cap scale in this lemma is the classical $k^{-1/m}$ quantization scale. To fix the comparison precisely, for a sphere probability measure ν its k -center cosine distortion is

$$\inf_{u_1, \dots, u_k \in \mathbb{S}^m} \int \min_i (1 - \langle v, u_i \rangle) d\nu(v) = \frac{1}{2} \inf_{u_1, \dots, u_k \in \mathbb{S}^m} \int \min_i \|v - u_i\|^2 d\nu(v).$$

Repeated centers are permitted. For surface measure its order is $k^{-2/m}$; the cap lower estimate and a covering upper estimate already prove that order. This is part of classical quadratic quantization, not a new geometric exponent; see Graf–Luschgy [3] and the manifold extension of Iacobelli [4, Theorem 1.4]. The quantity in (12) is different: it is an angular Hellinger obstruction for a *Gaussian-smoothed* finite register. Combined with the one-step entropy increment, it need not wait for a word product to approximate Haar measure at this scale.

5. A SHARP OCCUPATION LAW

The argument is useful in an inhomogeneous form. At epoch t , let the fresh command have a prescribed law p_t , independently across epochs, and let its angular operator satisfy

$$\|P_t|_{L_0^2(\sigma)}\| \leq \lambda_t \leq 1, \quad g_t = 1 - \lambda_t^2.$$

The alphabets and orthogonal matrices can also depend on t in this probabilistic statement. No strict gap is required at every epoch. Define the dimension-only constants

$$(13) \quad a_d = 3 + \frac{2}{d-1}, \quad B_d = 2048d(16A_d)^{2/(d-1)}.$$

Theorem 5.1 (Gap-weighted occupation without a mixing-time loss). *For the hidden process of Lemma 2.2, suppose $q_N \geq \kappa > 0$. For every integer $k \geq 1$,*

$$(14) \quad \sum_{\substack{0 \leq t < N \\ K_t \leq k}} g_t \leq B_d \kappa^{-a_d} k^{2/(d-1)}.$$

The other cuts may have arbitrarily many labels. The inequality also holds when K_t is replaced by the number of positive-probability states under this independent-command test law.

Proof. Fix k and choose one smoothing scale τ by (11); keep it fixed throughout the run. Set $f_t = \text{Law}(Z_t) * \gamma_\tau$ and $\tilde{Z}_t = U_{a_{t+1}} Z_t$. Orthogonal invariance of Gaussian noise and independence of the fresh command imply $f_{\tilde{Z}_t} = P_t f_t$. By (4), $Z_{t+1} = \mathbb{E}[\tilde{Z}_t | S_{t+1}]$. Lemma 3.1 therefore yields

$$(15) \quad h(f_{t+1}) - h(f_t) \geq h(P_t f_t) - h(f_t) - \frac{\Delta_t}{2\tau^2}.$$

Whenever $K_t \leq k$, Lemmas 3.2 and 4.1, together with $q_t \geq q_N \geq \kappa$, bound the entropy increment from randomizing the command below by $g_t \kappa/16$. At all other cuts that increment remains nonnegative.

Sum (15). The total quadratic loss is at most one. Furthermore, entropy concavity gives $h(f_0) \geq \frac{d}{2} \log(2\pi e \tau^2)$, while $\mathbb{E}\|Z_N + \tau G\|^2 \leq 1 + d\tau^2$ and Gaussian maximal entropy give

$$h(f_N) \leq \frac{d}{2} \log(2\pi e(\tau^2 + 1/d)).$$

The latter bound follows directly by comparing f_N with the centered Gaussian of that covariance and using nonnegative relative entropy. It does not require Z_N to be centered. Consequently

$$(16) \quad \frac{\kappa}{16} \sum_{K_t \leq k} g_t \leq \frac{d}{2} \log \left(1 + \frac{1}{d\tau^2} \right) + \frac{1}{2\tau^2} \leq \frac{1}{\tau^2}.$$

Since $\tau^{-2} = 128d\kappa^{-2}(16A_d k/\kappa)^{2/(d-1)}$, this is (14). Nothing in this summation restricts the intermediate hidden affine dimension, projected vertices, or the widths of the other registers. Under the test law, only positive-mass labels occur in the mixture f_t , proving the final assertion. $\square$

For a fixed gap $g = 1 - \lambda^2 > 0$, (14) gives $\#\{0 \leq t < N : K_t \leq k\} \leq B_d g^{-1} \kappa^{-a_d} k^{2/(d-1)}$. Counting endpoints $1, \dots, N$ instead changes this bound by at most one. Thus the earlier logarithmic loss in occupation, as well as in peak width, is removed. The proof does not assume a spectral gap for the stochastic state matrices; its gap belongs to the *external* orthogonal action on the sphere.

Lemma 5.2 (Exact spherical synthesis, inherited upper bound). *For every $N \geq 1$ and finite orthogonal command alphabet, the experiment (1) admits a common-row realization with at most $C_d N^{(d-1)/2}$ labels. Rational commands, seeds and signal admit rational rows with at most $d+1$ successors.*

Proof. This is the construction from [14]; we include it to make the matched theorem self-contained. Write $m = d - 1$ and $\eta = 1 - 1/(8N)$. The two rational stereographic charts

$$t \mapsto \left(\frac{2t}{1 + \|t\|^2}, \pm \frac{1 - \|t\|^2}{1 + \|t\|^2} \right), \quad t \in [-1, 1]^m,$$

are 2-Lipschitz. Every sphere point has a preimage in the unit ball in one chart. On a grid of mesh $1/M$, with $M = \lceil 2\sqrt{mN} \rceil$, their images form a rational set V of at most $2(2M+1)^m$ points. Every unit direction is within chordal distance $\sqrt{m}/M$ of V . Hence $\eta\mathbb{B}^d \subset P := \text{conv } V$ by the support-function criterion.

Associate label $u \in V$ at time t with $a_t u$, where $a_t = \eta^{-t}/4$. On command a choose a convex representation of $\eta U_a u$ in V . These are common command rows. Initialize the mean to ρx in $a_0 P$, which is possible because $\eta/4 \geq 7/32 > \rho$. Use terminal mean $a_N e_j^T u$. Bernoulli's inequality gives $\eta^N \geq 7/8$, so $a_N \leq 2/7 < 1$. The identity $a_{t+1} \mathbb{E}[u' \mid u, a] = a_t U_a u$ proves exactness on every word. For rational data, each convex feasibility problem has a rational basic solution with at most $d+1$ nonzero coefficients. The state holds only the label. A charged final bit uses two labels and is absorbed in C_d . No efficient table-generation algorithm is inferred from finite feasibility. $\square$

Proof of Theorem 1.1. Use the independent command law p from (2) and any fixed seed. Lemma 2.2 gives $\kappa = \rho/\sqrt{d} - 2\varepsilon > 0$. With $K = \max_t K_t$, Theorem 5.1 gives the explicit bound

$$(17) \quad K \geq \left(\frac{(1 - \lambda^2) \kappa^{a_d}}{B_d} N \right)^{(d-1)/2}.$$

The preceding lemma proves the exact upper bound. Proposition 2.3 proves the isolated-cut assertion. $\square$

Corollary 5.3 (An obstruction for actual terminal error). *For an available-label profile $\mathbf{K}$, let $\mathcal{E}_N(\mathbf{K})$ be the least actual worst-row terminal error among all machines with that profile. For every integer k with $I_k := \#\{0 \leq t < N : K_t \leq k\} > 0$,*

$$(18) \quad \mathcal{E}_N(\mathbf{K}) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{d}} - \left(\frac{B_d k^{2/(d-1)}}{(1 - \lambda^2) I_k} \right)^{1/a_d} \right]_+.$$

In particular, peak $o(N^{(d-1)/2})$ entails $\liminf \mathcal{E}_N \geq \rho/(2\sqrt{d})$.

Proof. The finite parameter domain of a fixed-profile machine is compact and the wordwise objective continuous, so the minimum exists. Apply (14) with $\kappa = q_N$ when $q_N > 0$, and use $q_N \geq \rho/\sqrt{d} - 2\mathcal{E}_N$. If $q_N = 0$, the same conclusion follows directly from calibration. The last assertion uses $I_K = N$. $\square$

This is an actual-error bound: it does not add local defects or rule out their cancellation. If a separately supplied diagram has an accumulated local deficiency certificate D , the classical composition argument produces a machine with final error at most D ; only then does (18) lower-bound that certificate. It need not equal the minimum terminal error.

6. A SHARP RATIONAL EXAMPLE AND ITS QUANTUM REALIZATION

Use the fixed five-letter alphabet $\mathcal{A}_* = \{\text{Id}, R_x, R_x^{-1}, R_z, R_z^{-1}\}$, where

$$(19) \quad R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -7/25 & -24/25 \\ 0 & 24/25 & -7/25 \end{pmatrix}, \quad R_z = \begin{pmatrix} -7/25 & -24/25 & 0 \\ 24/25 & -7/25 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

These are the rational commands from [14]. Take $\rho = 1/10$, six seeds $\{\pm e_j : 1 \leq j \leq 3\}$ and three final coordinate queries. No alphabet grows with the horizon.

Theorem 6.1 (Linear classical label complexity). *For every fixed $0 \leq \varepsilon < 1/(20\sqrt{3})$,*

$$(20) \quad W_{N,\varepsilon}(\mathcal{A}_*) = \Theta_\varepsilon(N), \quad \lceil \log_2 W_{N,\varepsilon}(\mathcal{A}_*) \rceil = \log_2 N + O_\varepsilon(1).$$

The exact upper machine may have rational common command rows. The separate positive minimum at each cut is four. If $\lambda_ < 1$ is the norm-gap constant for the walk with probabilities $(1/2, 1/8, 1/8, 1/8, 1/8)$, every such machine has*

$$(21) \quad K \geq \frac{(1 - \lambda_*^2)(1/(10\sqrt{3}) - 2\varepsilon)^4}{98304} N.$$

The existence of $\lambda_ < 1$ is an external spectral-gap input; its numerical value is not evaluated here.*

Proof. Let $\sigma_x, \sigma_y, \sigma_z$ be the Pauli matrices and define $A_x = (3I - 4i\sigma_x)/5$, $A_z = (3I - 4i\sigma_z)/5$ in $\text{SU}(2)$. Their adjoint actions are (19). Their eigenphases include $(3 + 4i)/5$, which is not a root of unity: its sum with its inverse is $6/5$, not an integer, whereas a rational algebraic integer is an integer. The closures of the respective cyclic subgroups therefore contain the two axial circles. The Lie algebra generated by $i\sigma_x, i\sigma_z$ also contains their bracket, a nonzero multiple of $i\sigma_y$. The closed generated subgroup is $\text{SU}(2)$.

The generators have algebraic entries. Bourgain–Gamburd [1, Theorem 1] supplies the upper spectral gap for their symmetric averaging operator on $L^2(\text{SU}(2))$. Adding weight $1/2$ at the identity makes its spectrum nonnegative and its mean-zero norm strictly less than one. Under the left action, lift a sphere function by $f(v) \mapsto f(\text{Ad}_g e_3)$. This is an isometry from the homogeneous quotient $L^2(\mathbb{S}^2)$ into $L^2(\text{SU}(2))$ and intertwines the left pullbacks. Thus the sphere operator has the same norm bound; inverse conventions are harmless for this symmetric walk.

Apply Theorem 1.1 and use $A_3 = 1$ in (13), giving $a_3 = 4$ and $B_3 = 98304$. This proves (21) and the matching upper order. Rational sparse rows follow from Lemma 5.2, and Proposition 2.3 gives four. Taking binary logarithms proves the bit statement, including the integer ceiling. $\square$

The rationality and alphabet are explicit, but a numerical finite-horizon lower threshold requires an additional certified bound on λ_* . We provide neither such a certificate nor a claim that finite-dimensional matrix eigenvalues establish the infinite-dimensional gap. Formula (21) gives the full dependence on any supplied gap certificate. The new conclusion here is removal of the logarithmic factor from the classical label count, not a new construction of expanding generators.

Proposition 6.2 (One qubit with reset, unitary commands and final query). *One qubit realizes (1) for (19) exactly, using a fixed finite collection of operations independent of N . Thus its quantum memory dimension is two, whereas its classical atomic label complexity and label bits satisfy (20) even at the stated fixed positive error. This is a numerical-behavior comparison, not a strict-cutpoint language-state separation.*

Proof. On seed x reset the qubit to $\varrho_x = (I + \rho \sum_j x_j \sigma_j)/2$. Its eigenvalues are positive because $\rho < 1$. Commands apply the unitaries $I, A_x^{\pm 1}, A_z^{\pm 1}$. The Bloch vector after the word is $\rho U_w x$. Measurement of σ_j has outcome b with probability $(1 + b \rho e_j^T U_w x)/2$. Only one final query is made. A seed reset is a completely positive operation; it is not being called a unitary measure-once transition. The classical statement is Theorem 6.1. $\square$

The same qubit operations appeared in the preceding revision. What is strengthened is the matching classical cost. In particular this is one qubit versus $\log_2 N + O(1)$ classical *label bits*, not one qubit versus linear bit space or an entropy lower bound for a chosen stationary input distribution.

7. COMPARISONS AND SCOPE

7.1. Quantization, smoothing and the new accumulated estimate. Classical quadratic quantization minimizes the expected squared distance to a finite codebook. The cosine loss on a sphere is exactly one half of the squared chordal distance. Graf–Luschgy [3] develops the general quantization framework, and Iacobelli [4, Theorem 1.4] proves the corresponding manifold asymptotics under explicit integrability hypotheses. On the compact sphere these give the familiar exponent $2/(d-1)$. Neither this geometric exponent nor the cap argument used here is a priority claim. Our lower bound concerns a sequence of *hidden conditional centroids*, not independently selected nearest neighbors in a static codebook.

Gaussian smoothing and entropy continuity are also established tools. The bound in [2, Corollary 4] applies to unrelated smoothed laws and is linear in an appropriate Wasserstein distance. Lemma 3.1 uses the particular conditional-mean coupling and a semiconvex Gaussian log density to remove the first-order term. We do not assert that this elementary Gaussian identity is absent from the literature. Its role here is to make the entire cost bounded by one quadratic variation budget, rather than by the sum of square roots of individual costs.

The entropy increment in Lemma 3.2 uses the standard relative-entropy/Hellinger comparison and an L^2 spectral bound, not a logarithmic Sobolev assumption. Lemma 4.1 gives a uniform positive angular defect after smoothing at the register’s resolution. The conclusion that combines these ingredients is Theorem 5.1: a finite horizon with arbitrarily varying hidden implementations has a gap-weighted small-register occupation of order at most $k^{2/(d-1)}$. That inequality, rather than a new quantization exponent or expansion theorem, is the main assertion.

7.2. Positive realization and branching programs. After padding layer alphabets, the model is a probabilistic ordered read-once layered program, with command-dependent row-stochastic matrices and a bounded real output. The scalar decoder for one query can be turned into randomized Boolean acceptance by one Bernoulli row. The total number of layer nodes and transition-table size are not the maximum width. In particular the result is not a lower bound in a uniform algorithmic-space model.

Fourier-growth and pseudorandomness results for regular, permutation and width-three branching programs [10, 11, 12] use their stated transition assumptions and control Boolean-input Fourier coefficients. Their regularity is not implied by row stochasticity: a reset matrix with every row $(1, 0, \dots, 0)$ does not preserve the uniform state law. Some bounds for unrestricted deterministic tables do extend to random tables by convexity, so randomness alone is not an

originality boundary. Here the assumed gap is on the external sphere action, whereas the arbitrary hidden matrices need have no gap or regularity. The converse uses the entropy of a smoothed conditional orbit mean, not an estimate on the Fourier mass of a Boolean acceptance function. We make no claim of an exhaustive branching-program priority classification.

Earlier positive-realization mechanisms and the packet theorem remain useful [13, 14]. In particular the full-word endpoint kernel must be allowed to distinguish different spellings of the same product. The present one-step proof uses actual independent command letters, not a product-sufficiency reduction. No previously corrected packet quantifier is relaxed.

7.3. Which quantum memory is compared. Quantum advantages for process simulation precede this work. Gu et al. [5] construct models with smaller stored entropy for classical stochastic processes. Thompson et al. [6] also treat *input-output* processes, not only autonomous outputs: their framework includes stationary input-dependent memory entropy and its worst-input counterpart. Our claim must not be distinguished from that work merely by the presence of input commands. The distinction here is finite-horizon worst-word numerical error, arbitrary horizon-specific classical tables, and maximum available label count; we do not minimize stationary memory entropy.

Ghafari et al. [7] demonstrate a memory-dimension advantage for stochastic process simulation. Thus dimension, as well as entropy, has established quantum-memory comparisons. Proposition 6.2 is an explicit driven family whose *classical* cardinality has a matched growth law in the horizon at fixed positive error. It is not a first claim of quantum dimension advantage. The recent world-model comparison [9] is also relevant; its process and update requirements are not replaced by our nonuniform one-query convention.

For strict cutpoints, Chen–Wu [8, Theorem 3.2] gives a classical PFA with at most $n^2 + 1$ states for a general n -dimensional one-way quantum automaton. The conversion preserves a threshold sign while allowing a length-dependent shrinking numerical margin [8, Proposition 3.1]. It therefore does not preserve the fixed terminal correlation used here. The result in (20) does not contradict or improve that theorem in its own language metric.

7.4. Conclusions. The full-sphere spectral-gap class has matched exact and fixed-error width $\Theta(N^{(d-1)/2})$. The explicit three-dimensional gate family has linear classical label count and logarithmic classical label bits. The proof allows hidden directions, cut-dependent rows, and fixed positive error, and it does not repeat a mixing-time charge.

The constants in the occupation theorem are not optimized. Neither an exact leading asymptotic constant nor every finite integer optimum is identified. A numerical lower constant for the displayed gates still requires a certified spectral-gap value. Arbitrary compact actions without the assumed sphere gap, all irrational cyclic arithmetic, multiple jointly constrained queries and adaptive experimental design are different problems. The retained monomial and cyclic theorems keep their own hypotheses in the supporting article.

Finally, nonuniform atomic-row width and its padded finite-coin implementation are different resources. The latter has the additional control factor proved below. Both leave table construction and storage outside the counted register, and neither is represented as a uniform space bound. The complete predecessor development is preserved separately; it is not needed to validate the new proof or part of the editorial claim for this article.

APPENDIX A. FIXED-DURATION FINITE-COIN IMPLEMENTATION

We make the timing repair explicit rather than assuming that duration is unobservable. The environment supplies a command at a ready boundary. Every completed update uses the same publicly specified number of microsteps, independent of the private bits, old label and command. No output or internal decision is visible during the update; only its fixed-time ready boundary is

visible. Threshold tables are nonuniform read-only data. The bit position, input latch, old label and control variables are counted. There is no uncharged microstep clock.

Theorem A.1 (Padded sparse-row compiler). *Let a finite transducer have peak K , common command rows and at most $L \geq 2$ successors per initialization, command or decoder row, with fixed seed, command and query alphabets. For $\delta > 0$, put*

$$b = \max \left\{ 1, \left\lceil \log_2 \frac{(N+2)(L-1)}{\delta} \right\rceil \right\}, \quad D = (L-1)b.$$

It admits a fair-bit implementation with exactly D private fair-bit microsteps per macro update, terminal error increased by at most δ , and at most $C_{A,X,J,L}K(b+1)$ persistent labels. Its duration discloses no information about the private trajectory.

Proof. For each categorical row, consider its conditional binary decision tree, with at most $L-1$ internal nodes. Replace each conditional probability by a lower dyadic approximation with b bits, keeping probabilities zero and one exact. Coupling decisions bounds the row total variation by $(L-1)2^{-b}$. Zero-mass nodes are given arbitrary legal rows.

To draw a threshold $m/2^b$, compare b private fair bits with its stored binary digits. Until the first mismatch the sampled prefix is equal to the table prefix, so no prefix integer is needed. Store the bit position and the relation (equal, below or above); after a mismatch, continue consuming the remaining bits of this comparison without changing the relation. After b steps, move to the selected child of the categorical tree. Once a leaf has been selected, hold its new label and continue consuming dummy bits until exactly D steps have elapsed. Deterministic thresholds and trees of shorter depth are padded in the same way.

Active states retain the old label, current input, tree node, relation and counted progress; their number is at most a constant times $K(b+1)$ for fixed alphabets and L . After leaf selection the old label is released; the sampler holds only the new label and counted progress. It does not hold two independent K -valued labels. Ready, initialization and query phases add only the same order. A final binary answer can be held inside the padding states. All timing information is the deterministic number D , so even a source observing ready times receives no private coin information from duration. This observation does not enlarge the main theorem into an adaptive-policy optimization result.

Each completed macro update has its rounded stochastic law regardless of padding. Stochastic contraction and a coupling of the at most $N+2$ updates bound final error by $(N+2)(L-1)2^{-b} \leq \delta$. The construction changes no mathematical target to a cutpoint language. $\square$

The comparison/tree construction is standard; the purpose is to price its control and remove the early-stopping side channel noted for the preceding self-paced compiler. It is not zero-error fair-bit sampling of arbitrary irrational probabilities.

Corollary A.2 (Padded label-bit rate). *For every fixed positive tolerance in Theorem 1.1, the exact spherical upper machine has a padded fair-bit approximation with $O(N^{(d-1)/2} \log(N+1))$ labels. In this separately defined nonuniform padded model, minimum persistent label bits obey*

$$\frac{d-1}{2} \log_2 N - O(1) \leq b_N^{\text{pad}} \leq \frac{d-1}{2} \log_2 N + O(\log \log(N+1)).$$

For the five rational gates the leading coefficient is one.

Proof. Lemma 5.2 gives common rows with $L = d+1$ (enlarge L for the binary output if necessary). Compile them with δ equal to the desired fixed tolerance, so $b = O(\log(N+1))$. This proves the upper label and bit bounds. At ready boundaries every padded machine induces an atomic transducer with no more labels than its total control alphabet. The sharp atomic lower bound therefore proves the lower bit bound. Tables, their generation, and arithmetic remain uncharged

in both nonuniform measures. No exact order for the padded label count is inferred from these leading-bit estimates. $\square$

REFERENCES

- [1] J. Bourgain and A. Gamburd, A spectral gap theorem in $SU(d)$, *J. Eur. Math. Soc.* **14** (2012), 1455–1511. Theorem 1; arXiv:1108.6264v1.
- [2] Y. Polyanskiy and Y. Wu, Wasserstein continuity of entropy and outer bounds for interference channels, *IEEE Trans. Inform. Theory* **62** (2016), 3992–4002.
- [3] S. Graf and H. Luschgy, *Foundations of Quantization for Probability Distributions*, Lecture Notes in Mathematics 1730, Springer, 2000.
- [4] M. Iacobelli, Asymptotic quantization for probability measures on Riemannian manifolds, *ESAIM Control Optim. Calc. Var.* **22** (2016), 770–785.
- [5] M. Gu, K. Wiesner, E. Rieper and V. Vedral, Quantum mechanics can reduce the complexity of classical models, *Nature Commun.* **3** (2012), Article 762.
- [6] J. Thompson, A. J. P. Garner, V. Vedral and M. Gu, Using quantum theory to simplify input–output processes, *npj Quantum Inform.* **3** (2017), Article 6.
- [7] F. Ghafari, N. Tischler, J. Thompson, M. Gu, L. K. Shalm, V. B. Verma, S. W. Nam, R. B. Patel, H. M. Wiseman and G. J. Pryde, Dimensional quantum memory advantage in the simulation of stochastic processes, *Phys. Rev. X* **9** (2019), 041013.
- [8] Z. Chen and J. Wu, The state cost of classical simulation of one-way general quantum finite automata, arXiv:2604.07058v2 (27 August 2026).
- [9] J. Lumbrellas, H. Ma, J. Thompson and M. Gu, An irreducible quantum advantage in aligning world models with reality, arXiv:2608.19779v1 (2026).
- [10] O. Reingold, T. Steinke and S. Vadhan, Pseudorandomness for regular branching programs via Fourier analysis, *APPROX/RANDOM 2013*, LNCS 8096, 655–670.
- [11] T. Steinke, S. Vadhan and A. Wan, Pseudorandomness and Fourier-growth bounds for width-3 branching programs, *Theory Comput.* **13** (2017), Article 12, 1–50.
- [12] C. H. Lee, E. Pyne and S. Vadhan, Fourier growth of regular branching programs, *APPROX/RANDOM 2022*, LIPIcs 245, Article 2.
- [13] Q. Qi, General Theta Foundations I: Sharp width of hidden rotation experiments, Revision 35 (2026), publication aabc0731a0c913f7.
- [14] Q. Qi, General Theta Foundations I: Finite alphabets, spectral gaps, and hidden memory, Revision 36 (2026), publication 1a08578710d55a23. Preserved without alteration in the supporting article.

PRESERVED MATHEMATICAL DEVELOPMENT

General Theta Foundations I - revision 36

The complete current article precedes this divider.

The following predecessor cumulative volume is reproduced without alteration.

Frozen predecessor publication:

1a08578710d55a2379cea56c43feb5eca56340a2

This archive is not part of the compact referee package. Independent historical analytic obligations keep their original hypotheses and proof status.

GENERAL THETA FOUNDATIONS I: FINITE ALPHABETS, SPECTRAL GAPS, AND HIDDEN MEMORY

QIAN QI

ABSTRACT. We study the width of finite stochastic transducers that reproduce every conditional probability of a prescribed orthogonal command experiment. For a fixed finite alphabet whose action on the sphere has an L^2 spectral gap, we prove lower and upper widths of orders $(N/\log(N+1))^{(d-1)/2}$ and $N^{(d-1)/2}$, respectively. The upper realization is exact; the lower bound permits arbitrary hidden states, unrelated transition rows at different epochs, and fixed small output error. Five explicit rational three-dimensional commands therefore have classical width between $cN/\log(N+1)$ and CN , although a single qubit realizes the same probabilities. This probability task is distinct from strict-cutpoint language equivalence. A second finite, noncommuting class, consisting of permutations and conjugations of resonant planar modes, has the matched order $N^{1/3}$. Its full group need not possess an irrational one-dimensional character. The common argument is a packet compression inequality formulated for the entire fresh command word, without assuming that its product is sufficient for the hidden machine. We also give a finite fair-bit compiler for sparse rows, with logarithmic rather than denominator-sized control overhead. The principal width is nonuniform clocked atomic-row width; the fair-bit interface is separately self-paced. The spectral-gap bounds determine a polynomial exponent, not a matched order up to constants.

1. INTRODUCTION

A small positive factorization at every observation cut does not, by itself, produce a small machine that realizes all cuts simultaneously. The additional requirement is causal compatibility: the same retained state must support each possible next input. We investigate this requirement for command experiments

$$(1) \quad \mathbb{P}(B = b \mid x, a_1 \cdots a_N, j) = \frac{1 + b\rho e_j^T U_{a_N} \cdots U_{a_1} x}{2}, \quad b \in \{-1, 1\}.$$

The command alphabet is fixed and finite. The seed is supplied once, and one coordinate query is supplied after the prescribed horizon. The command matrices act orthogonally; the realizing matrices act stochastically and may have arbitrary hidden directions.

The finite-alphabet restriction is substantive. An independent Haar rotation cannot be treated as one input when only a finite gate set is available. Nor can a whole word be replaced by its group product in the hidden transition law: equal products can have different intermediate computations. The present results retain each input slot and each full word throughout the converse.

Let $m = d - 1$. Suppose a probability distribution on the fixed alphabet induces an averaging operator P with

$$(2) \quad \|Pf\|_{L^2(\sigma)} \leq \lambda \|f\|_{L^2(\sigma)} \quad \text{when} \quad \int f d\sigma = 0, \quad \lambda < 1,$$

Date: September 25, 2026. Revision 36.

2020 Mathematics Subject Classification. Primary 93B15, 68Q10; Secondary 60J05, 22D10, 81P45.

Key words and phrases. Positive realization, probabilistic branching programs, finite alphabet, spectral gap, quantum automata, stochastic transducers.

where σ is normalized spherical measure. For fixed positive signal and sufficiently small uniform error, Theorem 4.3 proves

$$(3) \quad c \left(\frac{N}{\log(N+1)} \right)^{m/2} \leq W_{N,\varepsilon} \leq W_{N,0} \leq CN^{m/2}.$$

In particular $\log W_{N,\varepsilon} / \log N \rightarrow m/2$. The logarithmic gap in (3) is part of the statement, not absorbed into a Θ notation. The proof uses only a finite command distribution. A logarithmic number of actual commands makes its orbit distribution sufficiently close to spherical measure at the resolution imposed by the endpoint width.

Theorem 5.1 instantiates this criterion with five rational matrices in $\text{SO}(3)$. The spectral gap is an external theorem of Bourgain–Gamburd [1], not an assumption checked by numerics. Density and the algebraic hypotheses are verified for the stated gates. The experiment has separate positive minimum four at every internal cut, but its compatible hidden width grows almost linearly. A two-dimensional quantum memory realizes it exactly. Proposition 5.2 makes this comparison operational rather than terminological.

A different geometry gives a different order. In Theorem 6.1, finitely many orthogonal monomial commands permute planar modes, conjugate them, and rotate them by integer multiples of one badly approximable angle. Their exact and fixed-error widths are $\Theta(N^{1/3})$. The group generated by the displayed five-letter example is noncommuting, its real representation is irreducible, and no continuous one-dimensional character sees its irrational rotations. This is not an assertion that every finite noncommuting alphabet has the same exponent.

The methodological distinction is between mixing of the uncompressed orbit and irreversible loss at a finite register. The former supplies an orbit quantization scale; the latter converts that scale into a cumulative width constraint. All lower bounds allow new machines at each horizon and arbitrary changes of stochastic rows between epochs. The exact upper machines have common command rows for a fixed horizon, with a horizon-dependent decoder.

Notation	Meaning
K_t, W	available labels at cut t , and their maximum
$W_{N,\varepsilon}$	least peak at uniform binary row-TV error ε
$W, g(W)$ in §3	fresh packet word, and its ordered group product
$\Gamma_k(\mu)$	least k -center orbit distortion, including the starting direction
B_k	number of actual commands used by a test packet
q_t	norm of the conditional orbit mean, averaged over hidden states

The use of W for a word is local to the packet lemma; width elsewhere is scalar. Classical positive realization, spherical covering, spectral gap, and strict-cutpoint simulation are identified at their points of use. The new finite-alphabet consequences depend on these stated inputs. The sharp binary rotation result of [10] is retained and its packet and resonant arguments are reproduced within the larger monomial construction. Historical results not needed for the present proofs remain in the separately preserved supporting article.

2. FINITE EXPERIMENTS AND PERSISTENT REGISTERS

Fix a finite alphabet $\mathcal{A}$, matrices $U_a \in O(d)$, a finite seed set $X \subset \mathbb{S}^{d-1}$ and $0 < \rho \leq 1/10$. Unless otherwise stated, $X = \{\pm e_1, \dots, \pm e_d\}$ and $\text{Id} \in \mathcal{A}$. We use the full specification (1); no distribution or fairness assumption is imposed on the external word.

Definition 2.1 (Nonuniform clocked stochastic transducer). For horizon N , a transducer consists of finite state sets $[K_t]$, initial distributions E_x , stochastic matrices $T_{t,a} \in \mathbb{R}_+^{K_t \times K_{t+1}}$, and terminal columns $d_j \in [-1, 1]^{K_N}$. Its mean on a word is $E_x T_{0,a_1} \cdots T_{N-1,a_N} d_j$. Its error is half the

maximum absolute discrepancy from the mean in (1). The peak counts the available labels $\max_t K_t$, and also two terminal labels when the emitted answer is retained at a charged cut. The least peak with error at most ε is $W_{N,\varepsilon}$.

Only the updated register persists. The seed and the query appear once; the current command is consumed in its update. Private randomness does not persist outside the register. An input-dependent clock, a previous command, or an installation tag would be charged if retained. The epoch, N , complete row tables, their construction and arithmetic, and exact atomic sampling are free. Rows may vary with t and N . Unused rows are normalized, and padding counts as available labels. This is maximum layer width of a nonuniform probabilistic ordered read-once program, not uniform program space or total branching-program size. Section 7 specifies a different, finite-coin interface.

For comparison with the earlier compact-input theorem, a compact command space requires each matrix entry $a \mapsto T_{t,a}(s, s')$ to be Borel. Such a row function is nonuniform program data, not a finitely represented real-input algorithm. None of the finite-alphabet theorems below requires that access. Every input to their machines is a symbol from a fixed finite set.

Proposition 2.2 (Separate positive minimum). *If $\rho d < 1$ and the identity is available, the normalized residual rank and the separate positive minimum at every cut are $d + 1$. A separate minimum places no cap on the other registers.*

Proof. Residual rows are affine in $v = \rho U_w x$. Idle prefixes from $\pm e_j$ span $\mathbb{R}^d$, and an idle suffix followed by the coordinate queries separates the coordinates. Normalization adds the constant row, giving rank $d + 1$. A regular simplex centered at zero with circumradius $d\rho$ has inradius ρ . It contains every reachable vector and all its vertices have norm less than one. Each vertex consequently defines a valid future binary channel after every orthogonal command word. At the selected cut decompose v into these vertices. Before this cut one may store the raw prefix; afterwards one may store a simplex label and the raw suffix. At l later inputs this costs at most $(d + 1)|A|^l$ labels, and remains finite. It embeds the factorization in a complete machine rather than an amputated static matrix. Rank gives the converse. $\square$

Lemma 2.3 (Terminal correlation). *For a fixed unit seed x , put $Y_t = U_{a_t} \cdots U_{a_1} x$ and $q_t = \mathbb{E}[\|Y_t\| \mid S_t]$ under any external word law independent of the machine's private randomness. If the wordwise error is at most ε , then*

$$(4) \quad q_N \geq \kappa_d := \rho/\sqrt{d} - 2\varepsilon.$$

Proof. Combine the d decoder columns into $D(S_N) \in [-1, 1]^d$. Conditionally on a word, its mean differs from ρY_N by a vector of norm at most $2\sqrt{d}\varepsilon$. Thus $\mathbb{E}\langle Y_N, D(S_N) \rangle \geq \rho - 2\sqrt{d}\varepsilon$. Conditional expectation bounds the absolute value of this quantity by $\sqrt{d}q_N$. Two or more queries are not jointly sampled in this argument; their decoder columns are only combined in the proof. $\square$

3. COMPRESSION OF FULL COMMAND WORDS

Let a compact group act orthogonally on a finite-dimensional real Hilbert space H . Complex unitary representations are covered by the real inner product. For a group probability measure μ , define

$$(5) \quad \Gamma_k(\mu) = \inf_{v, u_1, \dots, u_k \in \mathbb{S}(H)} \int \min_{i \leq k} \{1 - \langle U_g v, u_i \rangle\} d\mu(g).$$

This continuous minimization has a minimum and lies in $[0, 1]$. The upper bound follows by including the direction of $\int U_g v d\mu(g)$ as a center. An invariant direction may make the coefficient zero. The conclusions below remain valid but then give no positive loss from that packet.

Lemma 3.1 (Full-word packet contraction). *Let $Y \in H$ be integrable and S finite. Let a random packet word W be independent of $\sigma(Y, S)$, and let $g(W)$ be its measurable ordered product. Suppose S' has at most k positive-mass values and, for every label r ,*

$$(6) \quad \mathbb{P}(S' = r \mid \sigma(Y, S, W)) = T_W(S, r) \quad a.s.,$$

where each T_w is a stochastic kernel. If $\mu = g_* \text{Law}(W)$, then

$$(7) \quad \mathbb{E} \|\mathbb{E}[U_{g(W)} Y \mid S']\| \leq (1 - \Gamma_k(\mu)) \mathbb{E} \|\mathbb{E}[Y \mid S]\|.$$

In particular T_w need not be constant on fibers of g .

Proof. Write $p_s = \mathbb{P}(S = s)$, $z_s = \mathbb{E}[Y \mid S = s]$, and let ν be the law of W . Independence and (6) imply

$$p'_r z'_r = \sum_s p_s \int T_w(s, r) U_{g(w)} z_s d\nu(w), \quad z'_r = \mathbb{E}[U_{g(W)} Y \mid S' = r].$$

For nonzero z'_r take $u_r = z'_r / \|z'_r\|$, with arbitrary unit choices for zero centroids. Summing inner products gives

$$\begin{aligned} \sum_r p'_r \|z'_r\| &= \sum_s p_s \int \sum_r T_w(s, r) \langle U_{g(w)} z_s, u_r \rangle d\nu(w) \\ &\leq \sum_{s: z_s \neq 0} p_s \|z_s\| \int \max_r \langle U_{g(w)} z_s / \|z_s\|, u_r \rangle d\nu(w) \\ &\leq (1 - \Gamma_k(\mu)) \sum_s p_s \|z_s\|. \end{aligned}$$

Only after removing the word-dependent assignment by a pointwise maximum do we push ν forward to μ . No product-sufficiency assertion is used. The old register and all internal packet cuts are unrestricted. $\square$

Remark 3.2 (Why the word matters). In a group with an involution r , the words ee and rr have the same product. A transducer can nevertheless record which first symbol occurred. Thus integration of the internal computation gives a kernel T_w , not necessarily a kernel $T_{g(w)}$. Also, independence of the product alone would be insufficient: an identity-product word correlated with Y could reveal Y through its spelling. The hypothesis concerns the entire fresh word. This repairs the word/product mismatch in the preceding formulation without restricting the class of machines or packet supports.

Theorem 3.3 (Packet budget). *Start with a fixed unit vector. On disjoint intervals use packet words whose full laws are fixed in advance, each independent of the earlier process and of all private random choices. Fixed commands may fill the gaps. If k_i bounds the register at packet endpoint i , and a terminal correlation forces $q_N \geq \kappa > 0$, then*

$$(8) \quad \kappa \leq \prod_i (1 - \Gamma_{k_i}(\mu_i)), \quad \sum_i \Gamma_{k_i}(\mu_i) \leq \log(1/\kappa).$$

The full words and their products may be noncommuting, and the product map may have collisions.

Proof. The ordered internal stochastic matrices of a whole packet multiply to T_w . Their Markov property gives (6); word freshness gives the required independence. Apply Lemma 3.1 at successive boundaries. A fixed command followed by stochastic compression cannot increase q , by conditional Jensen. Initially $q = 1$. Multiplication proves the product, and $1 - u \leq e^{-u}$ proves the sum. A factor zero precludes positive terminal correlation, as asserted. $\square$

For a wordwise specification the test law may be chosen after the available-label profile is known. It does not become information given to the machine. In the next section packets are made of independent letters. In Section 6 they have correlated letters; both satisfy the same full-word hypothesis.

4. FINITE SPECTRAL-GAP ALPHABETS

Let $d \geq 2$, $m = d - 1$, and σ be normalized measure on $\mathbb{S}^{d-1}$. For a distribution p on the finite alphabet, put $Pf(v) = \sum_a p(a)f(U_a v)$. Our hypothesis is the absolute L^2 gap

$$(9) \quad \|P|_{L^2_0(\sigma)}\| \leq \lambda < 1, \quad 0 < \lambda < 1.$$

Laziness makes an upper spectral gap absolute for symmetric walks. We assume (9) on the full mean-zero function space, not only on the defining d -dimensional representation. Density alone is not substituted for this analytic hypothesis.

Lemma 4.1 (Uniform mixing for Lipschitz tests). *There is C_d such that every real 1-Lipschitz f with values in $[-1, 1]$ satisfies*

$$(10) \quad \sup_v |P^B f(v) - \sigma f| \leq C_d \lambda^{2B/(m+2)}.$$

For $d = 3$ one can take $C_3 = 3$.

Proof. Orthogonal actions preserve distances, so $h = P^B f - \sigma f$ is 1-Lipschitz. Its L^2 norm is at most λ^B , since $\|f - \sigma f\|_2^2$ is a variance bounded by one. A chordal radius- r ball has mass at least $c_d r^m$ for $0 < r \leq 1$. Averaging $|h(v)| \leq |h(u)| + r$ over this ball and using Cauchy–Schwarz gives

$$|h(v)| \leq r + c_d^{-1/2} r^{-m/2} \lambda^B.$$

Take $r = \lambda^{2B/(m+2)}$. The ball estimate follows from polar integration and $\sin t \geq 2t/\pi$ on $[0, \pi/2]$. On $\mathbb{S}^2$ the chordal cap mass is exactly $r^2/4$; taking $r = \lambda^{B/2}$ gives $3\lambda^{B/2}$. $\square$

Lemma 4.2 (Finite-word orbit distortion). *There are constants $c_d > 0$ and integers $B_k = O_{d,\lambda}(\log(k+1))$ such that the product law μ_{B_k} of B_k independent command letters satisfies*

$$(11) \quad \Gamma_k(\mu_{B_k}) \geq c_d k^{-2/m}.$$

For $d = 3$, the explicit choices

$$(12) \quad B_k = \left\lceil \frac{2 \log(12k)}{-\log \lambda} \right\rceil, \quad \Gamma_k(\mu_{B_k}) \geq \frac{1}{4k}$$

are valid.

Proof. A chordal radius- r cap has mass at most $A_d r^m$, with $A_d \geq 1$. For any k unit centers, choose $r = (2A_d k)^{-1/m}$. Outside their union, of mass at least one half, the loss $1 - \max_i \langle v, u_i \rangle$ is at least $r^2/2$. Consequently the spherical average loss is at least $a_d k^{-2/m}$, where $a_d = (2A_d)^{-2/m}/4$. The function $f(v) = \max_i \langle v, u_i \rangle$ is 1-Lipschitz and lies in $[-1, 1]$. Lemma 4.1 gives, uniformly in the centers and starting direction,

$$P^B f(v) \leq 1 - a_d k^{-2/m} + C_d \lambda^{2B/(m+2)}.$$

Choose B_k so the last term is at most $a_d k^{-2/m}/2$. This proves the first assertion. Powers of P and chronologically ordered products give the same law here, since the independent letters have a common distribution and their joint law is invariant under reversal.

On $\mathbb{S}^2$ take $r = \sqrt{2/k} \leq \sqrt{2}$. The exact cap mass $r^2/4$ gives spherical loss at least $1/(2k)$. Equation (12) makes $3\lambda^{B_k/2} \leq 1/(4k)$, proving the sharper constants. Notice that B_k counts ordinary inputs, not one Haar input. $\square$

Theorem 4.3 (Finite-alphabet width and occupation). *Assume (9), $0 < \rho \leq 1/10$, and $0 \leq \varepsilon < \rho/(2\sqrt{d})$. Then*

$$(13) \quad c_{d,\lambda,\rho,\varepsilon} \left(\frac{N}{\log(N+1)} \right)^{m/2} \leq W_{N,\varepsilon} \leq W_{N,0} \leq C_d N^{m/2}.$$

In particular $\lim_{N \rightarrow \infty} \log W_{N,\varepsilon} / \log N = m/2$. Every available-label profile with this error satisfies

$$(14) \quad \#\{1 \leq t \leq N : K_t \leq k\} \leq B_k - 1 + c_d^{-1} B_k k^{2/m} \log(1/\kappa_d).$$

The other cuts may have arbitrarily large registers. For $d = 3$ one may replace the second term by $4B_k k \log(1/\kappa_3)$ using (12). The exact upper realization has common command rows and an N -dependent decoder.

Proof of the lower and occupation bounds. Under the independent-letter law, every consecutive length- B_k block is a fresh packet. Lemma 2.3 and Theorem 3.3 imply for a peak- k machine

$$(15) \quad \lfloor N/B_k \rfloor c_d k^{-2/m} \leq \log(1/\kappa_d).$$

Thus $N < B_k(1 + c_d^{-1} k^{2/m} \log(1/\kappa_d))$. As $B_k \leq C \log(k+1)$, this implies the first inequality in (13). Explicitly, if $k \geq N^{m/2}$ it is immediate; otherwise $\log(k+1) \leq C_m \log(N+1)$ and rearrangement suffices. Constants absorb the finitely many small N .

For occupation, discard at most $B_k - 1$ early endpoints and greedily select low-width endpoints at spacing at least B_k . Each selected interval removes at most B_k candidate endpoints. If l intervals are selected, their total loss is at least $l c_d k^{-2/m}$, irrespective of the internal widths. The packet budget bounds this by $\log(1/\kappa_d)$ and proves (14). The initialization cut is not in this count. The argument remains valid for positive-mass states under the selected independent-letter law. $\square$

Lemma 4.4 (Rational spherical synthesis). *For every $N \geq 1$ there is a rational finite set $V_N \subset \mathbb{S}^{d-1}$ of cardinality at most $C_d N^{m/2}$ such that*

$$\eta_N \mathbb{B}^d \subset \text{conv } V_N, \quad \eta_N = 1 - \frac{1}{8N}.$$

For finitely many orthogonal commands, this gives the exact upper bound in (13). If the commands, seeds and ρ are rational, all transition, initialization and decoder entries may be chosen rational, with at most $d+1$ successors in each command row.

Proof. Use the two stereographic charts

$$t \mapsto \left(\frac{2t}{1 + \|t\|^2}, \pm \frac{1 - \|t\|^2}{1 + \|t\|^2} \right), \quad t \in [-1, 1]^m.$$

Every sphere point has a preimage in the unit ball in one chart. Each chart is 2-Lipschitz. A grid of mesh $1/M$, with $M = \lceil 2\sqrt{mN} \rceil$, is therefore within chordal distance $\sqrt{m}/M$ of every sphere point. Its at most $2(2M+1)^m$ images are rational. In every unit direction the support function of their convex hull is at least $1 - m/(2M^2) \geq \eta_N$.

Write $P = \text{conv } V_N$, $a_t = \eta_N^{-t}/4$. From label $u \in V_N$ on command a , choose coefficients expressing $\eta_N U_a u$ as a convex combination of V_N . This is possible by ball containment. The coefficients depend on a, u, N , not on the epoch. Initialize to mean ρx in $a_0 P$, since $\rho \leq 1/10 < 7/32 \leq \eta_N/4$. At the final cut use mean decoder $a_N e_j^T u$. Bernoulli's inequality gives $\eta_N^N \geq 7/8$, so $a_N \leq 2/7 < 1$. The conditional mean update is $a_{t+1} \mathbb{E}[u' \mid u, a] = a_t U_a u$, proving exactness by induction. Only the vertex label is retained.

For rational data, convex feasibility is a rational linear system. A basic feasible solution has at most $d+1$ nonzero coefficients and is rational; apply this separately to the finitely many rows and

seeds. An extra binary-output cut has two labels and does not change the bound. This proves the upper assertion of Theorem 4.3. $\square$

The mixing and quantization inequalities are classical analytic and geometric mechanisms. Their use above is a finite-word, all-hidden-state compression bound. A logarithmic mixing time is explicitly paid. Neither an atomic Haar command nor a uniform-in- d constant is used.

5. FIVE RATIONAL COMMANDS AND A QUBIT

Consider the fixed matrices

$$(16) \quad R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -7/25 & -24/25 \\ 0 & 24/25 & -7/25 \end{pmatrix}, \quad R_z = \begin{pmatrix} -7/25 & -24/25 & 0 \\ 24/25 & -7/25 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and the five-letter alphabet $\mathcal{A}_* = \{\text{Id}, R_x, R_x^{-1}, R_z, R_z^{-1}\}$. Use the six seeds $\{\pm e_j : 1 \leq j \leq 3\}$, signal $\rho = 1/10$, and three terminal coordinate queries. Every specified probability lies in $[9/20, 11/20]$ at every horizon. Both the alphabet and the matrices are independent of N .

Theorem 5.1 (A finite noncommuting rational family). *For the experiment (16), every fixed $0 \leq \varepsilon < 1/(20\sqrt{3})$ has*

$$(17) \quad c_\varepsilon \frac{N}{\log(N+1)} \leq W_{N,\varepsilon}(\mathcal{A}_*) \leq W_{N,0}(\mathcal{A}_*) \leq CN.$$

The exact upper machine can have rational entries and common command rows. Every separate positive minimum is four. If $\lambda < 1$ is the sphere gap constant for the lazy walk with weights $1/2, 1/8, 1/8, 1/8, 1/8$, then a peak- K machine necessarily obeys

$$(18) \quad N < B_K \{1 + 4K \log(1/\kappa_3)\}, \quad B_K = \left\lceil \frac{2 \log(12K)}{-\log \lambda} \right\rceil, \quad \kappa_3 = \frac{1}{10\sqrt{3}} - 2\varepsilon.$$

No numerical value of the spectral gap is asserted.

Proof. Let $\sigma_x, \sigma_y, \sigma_z$ denote the Pauli matrices and put $A_x = (3I - 4i\sigma_x)/5$, $A_z = (3I - 4i\sigma_z)/5$ in $\text{SU}(2)$. Their adjoint actions on the traceless Hermitian matrices are precisely R_x, R_z . Their eigenvalues include $(3 + 4i)/5$. This number is not a root of unity: otherwise its sum with its inverse, $6/5$, would be a rational algebraic integer, hence an integer. Their powers are therefore dense in the respective one-parameter axial subgroups. The Lie algebra generated by $i\sigma_x, i\sigma_z$ contains their bracket, a nonzero multiple of $i\sigma_y$, so the closed subgroup they generate is all of $\text{SU}(2)$. In particular the generators do not commute.

The matrices have algebraic entries. Bourgain–Gamburd’s spectral-gap theorem [1, Theorem 1] applies to their symmetric averaging operator on $L^2(\text{SU}(2))$. Adding probability $1/2$ at the identity makes its spectrum nonnegative and its mean-zero norm strictly below one. The action on the sphere is a homogeneous quotient: lifting $f(v)$ to $f(gv_0)$ embeds $L^2(\mathbb{S}^2)$ isometrically into $L^2(\text{SU}(2))$ and intertwines the averaging operators. Hence the same gap holds on the sphere.

Apply Theorem 4.3, with the explicit constants in (12). Rational synthesis follows from Lemma 4.4. Proposition 2.2 gives four at each separate cut, since $3\rho < 1$. The spectral-gap theorem is the substantive external input; no finite collection of eigenvalue computations is used as its proof. $\square$

This theorem has a different arithmetic content from the binary example with rational rotation entries. One rotation may have exceptionally slow circle recurrences. Two noncommuting algebraic axes instead invoke a spectral gap on the entire sphere. We do not infer that either individual rotation angle is badly approximable.

Proposition 5.2 (Quantum probabilities versus threshold languages). *The experiment in Theorem 5.1 has an exact single-qubit realization with a finite, horizon-independent collection of operations. A general one-way quantum automaton of dimension two realizes its probabilities on valid prepare–command–query words. For any fixed strict cutpoint, its threshold language has a PFA realization with at most five states by [2, Theorem 3.2]. Nevertheless any classical clocked transducer approximating these numerical probabilities at the stated tolerance has width bounded below by (17).*

Proof. For seed x prepare

$$\varrho_x = \frac{1}{2}(I + \rho(x_1\sigma_x + x_2\sigma_y + x_3\sigma_z)).$$

It is a density matrix because $\|x\| = 1$ and $\rho < 1$. A command uses the corresponding unitary $I, A_x^{\pm 1}, A_z^{\pm 1}$. Their conjugations rotate the Bloch vector by (16). Measuring σ_j yields $\text{tr}(\varrho_w(I + b\sigma_j)/2) = (1 + b\rho e_j^T U_w x)/2$, exactly the required law. A query-dependent unitary before a fixed σ_z measurement implements all three queries; a seed symbol is a completely positive reset channel. Thus the operations define a dimension-two general quantum automaton on a finite alphabet, including all words outside the promised protocol with its usual automaton semantics.

The cited strict-cutpoint conversion concerns that entire language and therefore also its valid words. It preserves threshold signs, not numerical probabilities. In the conversion, the deviation from the classical cutpoint is multiplied by a positive length-dependent shrinking factor [2, Proposition 3.1]. Such attenuation is harmless for language equivalence and cannot supply the fixed-correlation requirement of Lemma 2.3. The last assertion is exactly Theorem 5.1; it even permits horizon-dependent and nonstationary classical simulators. $\square$

The variable in (17) is input length, with quantum dimension fixed. The quadratic state-cost results of Chen–Wu [2, 3] instead vary quantum dimension under strict-cutpoint equivalence. Neither result is contradicted or strengthened in its own metric by changing to the other objective.

6. A MATCHED NONCOMMUTING FINITE CLASS

The preceding finite spectral-gap class determines a growth exponent, with a logarithmic order gap. There is also a finite noncommuting class with a matched order. Its orbit structure is different from a sphere.

Identify $\mathbb{R}^{2r}$ with $\mathbb{C}^r$, fix an irrational α , and put $\zeta = e^{2\pi i\alpha}$. A *resonant monomial command* has the form

$$(19) \quad (U_a z)_j = \zeta^{n_{a,j}} \begin{cases} z_{\pi_a^{-1}(j)}, & \epsilon_{a,j} = 1, \\ \overline{z_{\pi_a^{-1}(j)}}, & \epsilon_{a,j} = -1, \end{cases}$$

where π_a is a permutation, $n_{a,j}$ is an integer, and $\epsilon_{a,j} \in \{-1, 1\}$. Fix finitely many such commands, and let $M = \max(1, \max_{a,j} |n_{a,j}|)$. Include the identity and the command A that multiplies the first coordinate by ζ and fixes the rest. Take the $4r$ seeds $(\pm 1 \pm i)e_j$, signal $0 < \rho \leq 1/(10r)$, and one of the $2r$ real coordinate queries. The specified mean is the corresponding coordinate of $\rho U_w x$.

Theorem 6.1 (Sharp monomial width). *If $b_\alpha := \inf_{l \geq 1} l \|l\alpha\|_{\mathbb{R}/\mathbb{Z}} > 0$, then for each fixed $0 \leq \varepsilon < \rho/2$,*

$$(20) \quad W_{N,\varepsilon}^{\text{mon}} = \Theta_{r,\mathcal{A},\alpha,\rho,\varepsilon}(N^{1/3}).$$

The exact upper machine has common command rows and at most four successors in any such row. Its final decoder may depend on N . Every separate exact positive minimum is $2r + 1$.

Lower bound. Restrict to seed $(1+i)e_1$, commands Id, A , and the first two real queries. The conditional target is the planar phase $\rho(1+i)\zeta^{L_N}$. Combining its two mean columns shows that $q_N \geq \kappa = \rho - 2\varepsilon$, as in Lemma 2.3 with the sharper planar normalization.

For a proposed peak k , put $B = 2k - 1$ and choose the length- B word $A^J \text{Id}^{B-J}$ with J uniform on $\{0, \dots, B\}$. Its $2k$ circle points are separated by $s_B = \min_{1 \leq l \leq B} \|l\alpha\|$. Each open arc of radius $s_B/2$ contains at most one of these points. Any k centers therefore leave at least half of them at distance at least $s_B/2$. The average loss is at least $\frac{1}{2}(1 - \cos(\pi s_B)) = \sin^2(\pi s_B/2) \geq s_B^2$. The full-word lemma applies although the packet's letters are correlated. Consecutive independent packets imply

$$\lfloor N/B \rfloor \frac{b_\alpha^2}{B^2} \leq \log(1/\kappa), \quad N < B + b_\alpha^{-2} B^3 \log(1/\kappa).$$

This proves the lower order against every hidden realization. No character of the full command group was used, only a sublanguage. $\square$

Sparse exact upper construction. Take a convergent p_0/q_0 of α , put $q = 2q_0$ and $\delta = 2\pi(\alpha - p_0/q_0)$. Choose the first convergent for which

$$(21) \quad q \geq \max\{8M, 8, (160MN)^{1/3}\}.$$

Then $|\delta| < 8\pi/q^2$ and $|n\delta| < \pi/q$ for $|n| \leq M$. Set $a = \pi/q$, and for each needed integer n put

$$\lambda_n = \frac{\cos(a - |n\delta|)}{\cos a}, \quad \omega_n = \frac{\sin |n\delta|}{\lambda_n \sin(2a)}, \quad \Lambda = \max_{|n| \leq M} \lambda_n.$$

These numbers obey $1 \leq \lambda_n \leq \Lambda$ and $0 \leq \omega_n \leq 1$. Let $u_l = e^{2\pi i l/q}$, indices modulo q . The two-point law on labels $l + 2np_0$ and $l + 2np_0 + \text{sgn}(n\delta)$, of weights $1 - \omega_n, \omega_n$, has mean $\zeta^n u_l / \lambda_n$. For $n = 0$ interpret it as the single label l . This is the intersection of a polygon edge with the rotated ray.

The register is a pair (j, l) , comprising one mode and one polygon label. Command a sends the mode to $j' = \pi_a(j)$ and, if needed, replaces l by $-l$ to conjugate. Draw the just described edge label for $n = n_{a,j'}$. With probability $(1 + \lambda_n/\Lambda)/2$ retain it; otherwise replace it by its antipode, label $l + q/2$. The expected complex label is exactly the commanded one divided by Λ . At most four labels occur. This antipodal step prices a common radial factor without a free zero state or a dense uniform redraw.

Associate $a_t u_l e_j$ with label (j, l) , where $a_t = \Lambda^t/4$. The conditional mean update is exactly U_a . Moreover

$$\log \Lambda \leq M|\delta| \tan a \leq 16\pi^2 M/q^3 < 160M/q^3, \quad a_N \leq e/4 < 1.$$

The seed occupies one mode. Its planar vector has radius $\rho\sqrt{2} \leq \sqrt{2}/10 < \cos(\pi/8)/4$, hence lies in the initial polygon and can be initialized using at most three vertices. Decoder means are the real coordinates of $a_N u_l e_j$, with zeros on other modes. Induction gives the exact specified law on all words, not just the restricted lower-bound words. Only (j, l) persists.

Bad approximability is equivalent to bounded partial quotients, which bound the ratio of consecutive convergent denominators. Thus the first choice in (21) has $q = O_{\alpha, M}(N^{1/3})$, proving the upper order with rq labels. $\square$

Separate minima. The seed vectors span $\mathbb{R}^{2r}$; identity suffixes separate their coordinates, so normalized residual rank is $2r + 1$. All predictive vectors have norm $\rho\sqrt{2}$. A regular simplex with inradius $\rho\sqrt{2}$ has circumradius $2r\rho\sqrt{2} \leq \sqrt{2}/5 < 1$. Its vertices define legal continuations under every orthogonal command. The selected-cut construction of Proposition 2.2 gives the matching positive minimum, with finite raw-word storage elsewhere. $\square$

Proposition 6.2 (A full group not seen by a character). *For every $r \geq 2$, at most five commands suffice to generate an instance of Theorem 6.1 whose compact closure is $O(2)^r \rtimes S_r$. Its real representation on $\mathbb{R}^{2r}$ is irreducible. Every continuous one-dimensional unitary character is trivial on the connected torus $SO(2)^r$, hence cannot see A 's irrational phase.*

Proof. Use Id, A , conjugation of the first mode, the cyclic permutation of modes, and the transposition of the first two modes. The last two permutations generate S_r ; for $r = 2$ duplicate commands are removed. Irrational powers of A fill one circle, and permutation conjugates fill all coordinate circles. Conjugations supply their reflections, proving the asserted closure.

An invariant real subspace under these circle actions decomposes into coordinate planes: averaging over all other circles projects onto a chosen plane. A nonzero projection contains that whole plane by rotation, and permutation invariance then supplies all planes. This proves irreducibility. Finally, conjugation of a coordinate circle element sends it to its inverse. In a one-dimensional character its value must therefore equal its inverse. Continuity on the connected circle forces that value to be one. Apply this in each coordinate. $\square$

This example is genuinely noncommuting and is not an application of a character of its entire compact group. Its lower bound still uses a cyclic sublanguage, and its upper bound uses the monomial structure. It does not classify arbitrary finite dense subsets of $SO(d)$. The case $r = 1$, $\rho = 1/10$ and commands Id, A recovers the original sharp rotation experiment of [10] without changing its signal, seed set, error tolerance or hidden-state quantifier.

7. FINITE COINS WITH CHARGED CONTROL

The atomic model permits arbitrary exact row probabilities. We now separate a finite random-bit implementation. The environment supplies an input only at a ready boundary and waits until the update has finished. Fair bits are private. The row table may contain precomputed finite binary thresholds, but the bit position, current input, old label and control state are all charged. There is no free microstep clock.

Theorem 7.1 (Sparse-row dyadic compiler). *Suppose a finite command transducer has peak K , at most L successors per row, fixed finite seed and query alphabets, and common command rows. For any $\delta > 0$ it has a self-paced fair-bit implementation whose uniform terminal row-TV error increases by at most δ , using*

$$(22) \quad C_{A,X,J,L}K(b+1) \text{ persistent labels,} \quad b = \max \left\{ 1, \left\lceil \log_2 \frac{C_L(N+2)}{\delta} \right\rceil \right\},$$

and at most $(L-1)b$ fair bits for a command update. The same bound holds for sparse initialization and the binary terminal decoder. Tables and the prescribed macroscopic final boundary remain nonuniform.

Proof. A categorical row with at most L outcomes is a binary decision tree with at most $L - 1$ internal nodes. At each node express the probability of choosing one subtree conditionally on reaching that node. A zero-mass node is unreachable and may use an arbitrary conditional row. Round each conditional probability down to a multiple of 2^{-b} . Coupling the binary decisions shows that the row-TV discrepancy is at most $(L - 1)2^{-b}$, without needing the original probabilities to be rational. Initialization and binary decoding are treated in the same way.

Thresholds zero and one are deterministic. For $0 < m < 2^b$, to sample the dyadic Bernoulli threshold $m/2^b$, compare independent fair bits with the b binary digits of m , from most to least significant. At the first mismatch the comparison is decided; if all digits agree, return the prescribed strict-comparison outcome. No prefix integer is stored. The comparison retains only its position $i \in \{1, \dots, b\}$; while all bits agree, their values are already the fixed threshold prefix.

A categorical tree additionally retains one of at most $L - 1$ nodes. It must retain the old label and the current input until a leaf gives the new label. These are all included in (22), along with ready, seed and query phases. There is no second uncharged output index. The common-row assumption eliminates a retained epoch counter inside the sampler. An external end query selects the final decoder after the prescribed number of completed inputs.

TV contraction under stochastic kernels and induction on the actual word give at most $C_L(N + 2)2^{-b}$ terminal discrepancy. This is at most δ by the chosen b . The argument couples legal completed updates and therefore prices every command even when the bit comparison stops early. Its duration is at most $(L - 1)b$ bits per update. $\square$

Corollary 7.2 (Leading label bits with finite coins). *For every fixed positive error below the thresholds of the preceding theorems, their explicit machines admit the following finite-coin peaks:*

$$\begin{aligned} O(N^{1/3} \log(N + 1)) & \quad \text{for the resonant monomial class,} \\ O(N^{(d-1)/2} \log(N + 1)) & \quad \text{for the spectral-gap class.} \end{aligned}$$

Consequently the least persistent bits in this self-paced, nonuniform fair-bit model have leading coefficients $1/3$ and $(d - 1)/2$ in $\log_2 N$, respectively, with additive $O(\log \log(N + 1))$ uncertainty.

Proof. The monomial upper has at most four successors; the spherical upper has at most $d + 1$. Their initialization is sparse and their decoder binary. Compile the exact machine with tolerance equal to the fixed desired error. Formula (22) gives the upper peaks. Every such sampler, when observed at ready boundaries, is itself an atomic transducer with at most its total number of labels. Hence the atomic lower bounds apply, irrespective of its internal timing. Taking binary logarithms and using (13) and (20) proves the stated leading coefficients. $\square$

This is not an exact finite-bit implementation of an irrational row at zero error. It is a fixed-error compiler with an explicitly charged input latch and bit counter. It does not make the row description or its construction free in an ordinary uniform-space model; those costs are outside both nonuniform measures. For the rational gates, exact rational atomic synthesis is additionally available, but exact rational sampling with unbounded rejection duration is a separate implementation choice.

8. OBJECTIVES, ANTECEDENTS, AND REMAINING DISTINCTIONS

8.1. Quantum and probabilistic automata. The prepare–rotate–query law is a numerical behavior, not just a language. Proposition 5.2 gives its exact quantum model and an explicit classical length-dependent obstruction. Chen–Wu’s revised paper [2] has a different title and sharper theorem than its April version: its stated cost is $n^2 + 1$ under strict cutpoints. Their companion work [3] treats the simulation cost of reversible quantum models through prepare–test and sign geometry. These compare threshold languages as quantum dimension varies. A shrinking positive acceptance margin can preserve such a language at all lengths and fail every fixed numerical tolerance in the present task.

The September symmetry study [4] likewise distinguishes reachable–observable rank from operational threshold memory. Its strict-cutpoint objective is not wordwise preservation of numerical probabilities. Our sphere example already has fixed real predictive rank four, so a rank theorem alone cannot yield its diverging compatible width. The present comparison does not claim that quantum/classical state separation itself is new.

Lumbreras–Ma–Thompson–Gu [5] analyze a stationary world model with an irrational phase, repeated histories, value error and decision loss, and supply an exact qutrit realization. Their Theorems 28–30 concern that stationary decision task. Here the classical machine may be redesigned for every finite horizon and may change its rows at every cut; one externally prescribed

terminal probability is tested. Neither a stationary peripheral-spectrum argument nor a strict-cutpoint conversion supplies our finite nonuniform width bounds.

8.2. Spectral and geometric inputs. Bourgain–Gamburd [1] supplies the full spectral-gap input for Theorem 5.1. We do not prove a new expansion theorem or a new sphere-covering law. The gap converts finite command length into a uniform Lipschitz mixing scale; the full-word contraction then converts that scale into a hidden-register cost. This distinguishes the finite alphabet from the all-rotation-input theorem retained in [10]. The gap constant is fixed but not evaluated numerically. Constants may deteriorate with dimension, the generating set and the error margin.

The lower bound uses a probability law on external commands, not an adaptive policy implemented by the simulator. In the spectral-gap class the test letters are independent. In the monomial class they are correlated within width-dependent packets. Berkes–Borda [6] study the harmonic and Diophantine behavior of the uncompressed circle walk; compression into an endpoint register is an additional operation.

The transducer is a probabilistic ordered read-once branching program with real readout. For a binary alphabet its usual Boolean-input matrix Fourier coefficient is the ordered product of the matrices $(T_{t,0} + (-1)^{s_t} T_{t,1})/2$. This algebraic identity is not the packet converse. The Fourier-growth results for regular and width-three programs [7, 8, 9] have their stated hypotheses and objectives. An arbitrary stochastic row need not preserve the uniform state law. Conversely, bounds valid for all deterministic tables can sometimes pass by convexity to randomized rows. Private row choices should not be mistaken for a persistent shared seed available to the decoder.

8.3. Inherited statements and preservation. The sharp bounded-type binary theorem, nonuniform occupation, actual terminal-error bounds, the angular coefficient $l^2 - 1$, and the separate compact-input result of [10] remain in the unaltered supporting article. The corrected full-word lemma supplies the advertised general packet implication without a product-sufficiency restriction. For a compact command table, a Borel convex choice can be made by triangulating a finite polytope and using the first containing simplex in a fixed order; affine barycentric coordinates on finitely many closed simplices are Borel. No such infinite table is used by the finite-alphabet results.

The accumulated local deficiency certificate from the preceding work is not the minimum actual terminal error. Given a fixed profile in the sphere case, the present proof directly yields

$$(23) \quad \mathcal{E}_N(K) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{d}} - \exp\{-c_d K^{-2/(d-1)} \lfloor N/B_K \rfloor\} \right]_+.$$

Indeed Lemma 2.3 holds for every candidate’s actual error, and the packet product bounds q_N above. This remains valid when intermediate approximation errors cancel. A sum of executable local garbling defects is only an upper certificate for a particular terminal error, by TV contraction; applying (23) to it does not turn that certificate into an intrinsic equality.

8.4. Conclusion. There are now two finite noncommuting classes with distinct quantitative laws. Spectral-gap actions on spheres have polynomial width exponent $(d-1)/2$, with the displayed logarithmic order gap. Resonant monomial actions have matched cube-root order and need not be detected by a character of the full group. Both conclusions allow arbitrary hidden states and layer-dependent stochastic rows. The explicit rational gates give a numerical-probability comparison with a single qubit, not a new strict-cutpoint state-cost theorem.

The logarithmic gap for spectral-gap alphabets, optimal finite constants, and classification of all irrational cyclic angles remain quantitative questions. General adaptive experimental design is a different task. The finite-coin corollary concerns a self-paced nonuniform sampler and does not

identify program-table space with label space. These distinctions leave the proved finite-alphabet results intact; they are not hypotheses silently replaced during implementation.

REFERENCES

- [1] J. Bourgain and A. Gamburd, A spectral gap theorem in $SU(d)$, *J. Eur. Math. Soc.* **14** (2012), 1455–1511. Theorem 1; arXiv:1108.6264v1.
- [2] Z. Chen and J. Wu, The state cost of classical simulation of one-way general quantum finite automata, arXiv:2604.07058v2 (27 August 2026).
- [3] Z. Chen and J. Wu, On the simulation cost of quantum finite automata, arXiv:2605.10682v1 (2026).
- [4] Z. Chen, Behavioral memory under symmetry in one-way quantum automata, arXiv:2609.01451v1 (2026).
- [5] J. Lumbreras, H. Ma, J. Thompson and M. Gu, An irreducible quantum advantage in aligning world models with reality, arXiv:2608.19779v1 (2026).
- [6] I. Berkes and B. Borda, Random walks on the circle and Diophantine approximation, *J. Lond. Math. Soc.* **108** (2023), 409–440.
- [7] O. Reingold, T. Steinke and S. Vadhan, Pseudorandomness for regular branching programs via Fourier analysis, *APPROX/RANDOM 2013*, LNCS 8096, 655–670.
- [8] T. Steinke, S. Vadhan and A. Wan, Pseudorandomness and Fourier growth bounds for width 3 branching programs, *Theory Comput.* **13** (2017), Article 12, 1–50.
- [9] C. H. Lee, E. Pyne and S. Vadhan, Fourier growth of regular branching programs, *APPROX/RANDOM 2022*, LIPIcs 245, Article 2.
- [10] Q. Qi, General Theta Foundations I: Sharp width of hidden rotation experiments, Revision 35 (2026), publication `aabc0731a0c913f7`. Preserved in the companion supporting article.

PRESERVED MATHEMATICAL DEVELOPMENT

General Theta Foundations I - revision 35

The complete current article precedes this divider.

The following predecessor cumulative volume is reproduced without alteration.

Frozen predecessor publication:

aabc0731a0c913f7b9e75e6efc08f7af665bdf23

This archive is not part of the compact referee package. Independent historical analytic obligations keep their original hypotheses and proof status.

GENERAL THETA FOUNDATIONS I: SHARP WIDTH OF HIDDEN ROTATION EXPERIMENTS

QIAN QI

ABSTRACT. We determine the order of the least hidden-state width of a uniformly positive binary-command rotation experiment. For every badly approximable angle and every fixed sufficiently small uniform output error, the minimum width is $\Theta(N^{1/3})$. The lower bound permits arbitrary stochastic transitions, cut-dependent state spaces, unobservable hidden directions, and a different machine for every horizon. It therefore matches the previous resonant-polygon construction without a vector-state restriction. The proof groups commands into independent packets whose rotation counts are separated. Compressing one packet into k labels loses a fixed multiple of k^{-2} of the conditional phase amplitude, while the packet uses $2k - 1$ commands. Multiplying these losses yields both the sharp order and a nonuniform occupation bound. A representation-valued version gives an orbit-quantization budget for general compact groups; for the separate model with all $\mathrm{SO}(d)$ rotations available as commands it yields the matched order $N^{(d-1)/2}$. We also bound the actual optimal terminal simulation error, independently of cancellation in local deficiency certificates. Throughout, width counts available persistent labels in a clocked atomic stochastic-row model; descriptions, arithmetic and exact real sampling are not charged. The fixed-alphabet rotation theorem and the compact-command extension have distinct input conventions.

1. INTRODUCTION

A small sufficient state at each separate experimental cut need not be compatible with small states at all cuts of one stochastic machine. The rotation experiment below makes this discrepancy quantitative while keeping the seed, command, query and answer alphabets fixed. Its separate positive minimum is three at every cut. Its simultaneous minimum grows with the number of commands, even when a fixed uniform approximation error is allowed.

Write R_α for planar rotation through $2\pi\alpha$. A seed $x \in \{-1, 1\}^2$ is followed by N binary commands and one coordinate query $j \in \{1, 2\}$. The answer $B \in \{-1, 1\}$ must have mean

$$(1) \quad \mathbb{E}[B \mid x, c_1, \dots, c_N, j] = \frac{1}{10} e_j^T R_\alpha^{c_1 + \dots + c_N} x.$$

All answer probabilities belong to $[(1 - \sqrt{2}/10)/2, (1 + \sqrt{2}/10)/2]$, independently of N . Let $W_{N,\varepsilon}(\alpha)$ denote the least peak number of persistent labels achieving total-variation error at most ε on every seed, word and query; Definition 2.1 is precise. Set

$$b_\alpha = \inf_{l \geq 1} l \|l\alpha\|_{\mathbb{R}/\mathbb{Z}}.$$

We call α *badly approximable* when $b_\alpha > 0$; equivalently its continued-fraction partial quotients are bounded.

Date: September 25, 2026. Revision 35.

2020 Mathematics Subject Classification. Primary 93B15, 68Q10; Secondary 60J10, 11J70, 62B15.

Key words and phrases. Positive realization, probabilistic branching programs, causal memory, rotation, quantization, comparison of experiments.

Theorem 1.1 (Sharp hidden width). *Let α be badly approximable and $0 \leq \varepsilon < 1/20$ be fixed. In the clocked atomic stochastic-row model,*

$$(2) \quad W_{N,\varepsilon}(\alpha) = \Theta_{\alpha,\varepsilon}(N^{1/3}).$$

More explicitly, put $\kappa = 1/10 - 2\varepsilon$. Every realization of peak at most K satisfies

$$(3) \quad N < (2K - 1) + \frac{(2K - 1)^3}{b_\alpha^2} \log \frac{1}{\kappa}.$$

The upper bound is exact, uses two command matrices independent of the epoch, and has a decoder depending on N . At the golden angle $\alpha = (\sqrt{5} - 1)/2$ and zero error,

$$(4) \quad \max\{3, (N/168)^{1/3}\} \leq W_{N,0}(\alpha), \quad W_{N,0}(\alpha) < 2 \max\{5, (40N)^{1/3}\}.$$

In fact $W_{N,0} \geq 3$ and $W_{N,0} > (N/168)^{1/3}$ separately.

None of the constants is optimized. The conclusion identifies the order of width, not the exact integer optimum or its leading constant. In persistent bits it gives $(\log_2 N)/3 + O_{\alpha,\varepsilon}(1)$. This is not a uniform space bound charging program description or exact real arithmetic.

The predecessor Fourier argument gives $\Omega(N^{1/5})$ for the same unrestricted hidden-state class [15]. Its conditional phase amplitude is the starting point here. The improvement does not come from changing a trigonometric-kernel constant. Instead we choose correlated commands *inside* fresh independent packets. A packet of $2k - 1$ commands produces $2k$ separated phase increments. Whatever hidden mechanism is used inside it, a k -label endpoint loses a proportion of amplitude of order k^{-2} . Since a wordwise simulator must pass this input distribution, multiplication over disjoint packets gives the cube-root converse. This also controls the number of low-width cuts without bounding the remaining widths.

The underlying model is a nonuniform probabilistic ordered read-once branching program, with two terminal affine readouts. Section 7 compares the result with the Fourier-growth theorems of Reingold–Steinke–Vadhan, Steinke–Vadhan–Wan and Lee–Pyne–Vadhan [1, 2, 3]. Our packet argument is a hidden-compression estimate, not a theorem about pseudorandom generators or Boolean Fourier growth. The circle-walk literature [5] describes the uncompressed walk; conditional averaging in a small register is an additional operation.

Notation	Meaning
$K_t, W_{N,\varepsilon}$	Available labels at cut t ; optimum peak.
q_t	Expected norm of a conditional past-orbit mean, used only in the proof.
$s_B(\alpha)$	$\min_{1 \leq l \leq B} \ l\alpha\ _{\mathbb{R}/\mathbb{Z}}$.
$\Gamma_k(\mu)$	Worst starting-direction k -center orbit distortion.
$\mathcal{E}_N(\mathbf{K})$	Minimum actual terminal row error for a profile.
$\mathcal{D}(E)$	Accumulated local deficiency certificate of a diagram.

Contribution and provenance. The packet contraction, sharp unrestricted order, nonuniform packet budget and compact-group applications are the new results. The conditional-amplitude variable is inherited from [15]; the resonant-polygon upper construction is inherited from [14] and is proved again here. Jensen’s inequality, sphere quantization, continued fractions and finite Blackwell garbling are classical tools. The earlier Fourier theorem is preserved, with its angular estimate sharpened in Appendix A. No accumulated archive is needed to verify the new proofs in this article.

2. THE EXPERIMENT AND ITS REALIZING MACHINES

State distributions are row vectors and $\Delta_k = \{e \in \mathbb{R}^k : e \geq 0, e\mathbf{1} = 1\}$.

Definition 2.1 (Clocked stochastic realization). For a specified N , a realizing machine has finite state sets $[K_t]$ for $0 \leq t \leq N$, initialization rows $E_x \in \Delta_{K_0}$, row-stochastic matrices $T_{t,c} \in \mathbb{R}_+^{K_t \times K_{t+1}}$ for $t < N$, $c \in \{0, 1\}$, and decoder means $d_j \in [-1, 1]^{K_N}$. Its conditional answer mean is

$$E_x T_{0,c_1} \cdots T_{N-1,c_N} d_j.$$

Its error is one half the maximum absolute difference from (1), over all seeds, command words and queries. The profile counts *available* labels, including padding. Its peak is $\max_{0 \leq t \leq N} K_t$, or the maximum of this number and two if a binary answer is retained at an additional output cut. $W_{N,\varepsilon}(\alpha)$ is the least such peak at error at most ε .

Only the updated register persists. An earlier random seed, a transcript, an input-dependent clock or any retained protocol variable must be in that register. The current input is consumed in its update; seed and query are supplied just once. The epoch is external and the complete row table may depend on N . All rows, including unused rows, are normalized. Exact atomic stochastic rows and read-only tables are allowed; row construction, arithmetic and table length are not part of this width. A self-paced finite fair-bit compiler for rational rows has additional workspace and is a different resource statement [12].

The commands are externally specified inputs. No independence or fairness is part of the simulation task. A lower bound may impose any distribution on command words, including one chosen after the available profile is known. That distribution is used in the proof and is not a persistent seed supplied to the machine. When counting states under a chosen test law, zero-probability labels may be removed; the conclusions in terms of available labels remain valid.

Proposition 2.2 (Separate positive minimum). *At every cut after initialization the exact normalized residual rank and the separate positive minimum of (1) are three. The latter permits arbitrary finite registers at the other cuts.*

Proof. The predictive vectors are $R_\alpha^a x/10$. Idle prefixes from the four seeds span a plane, and two coordinate queries after an idle suffix separate it. Normalization adds the constant row, giving rank three. The triangle with vertices $(1/2, 0)$ and $(-1/2, \pm 1/2)$ contains the seed disk of radius $\sqrt{2}/10$, since its side distances from the origin are $1/2$ and $1/(2\sqrt{5})$. Its vertices have norm at most $1/\sqrt{2}$, so all rotated continuations give valid binary laws. At the chosen cut decompose the actual predictive vector in this triangle. Before it retain the finite raw history; afterwards retain the triangle label and the remaining active-command count. This is a complete finite machine attaining three at that cut. Rank gives the lower bound. This separate-cut construction is inherited from [13, 14]. $\square$

The statement does not require a hidden basis state to possess a predictive vector. Indeed the converse below never projects a hidden simplex, or bounds the number of vertices in such a projection.

3. ORBIT QUANTIZATION UNDER STOCHASTIC COMPRESSION

We first state the probabilistic argument in a real finite-dimensional Hilbert space H . A complex unitary representation is covered by taking its real inner product. Let $\mathbb{S}(H)$ be the unit sphere and let μ be a probability measure on a compact group acting orthogonally on H , with action U_g .

Definition 3.1 (Orbit distortion). For $k \geq 1$ set

$$(5) \quad \Gamma_k(\mu) = \inf_{v, u_1, \dots, u_k \in \mathbb{S}(H)} \int \min_{i \leq k} (1 - \langle U_g v, u_i \rangle) d\mu(g).$$

It depends only on the representation and the law of a command packet, not on a proposed stochastic realization. It lies in $[0, 1]$.

The upper bound one follows by using the direction of $\int U_g v d\mu(g)$ as one of the centers, with any center if that integral vanishes. The infimum is attained by compactness and continuity. An invariant direction may make the coefficient zero; positivity is a geometric property to be established, not an automatic conclusion for all representations.

Lemma 3.2 (One-packet contraction). *Let $Y \in H$ be integrable, let S be a finite register, and let G be independent of (Y, S) with law μ . Suppose the new register S' has at most k positive-probability values and its conditional law, given (Y, S, G) , depends only on (S, G) . Then*

$$(6) \quad \mathbb{E} \|\mathbb{E}[U_G Y \mid S']\| \leq (1 - \Gamma_k(\mu)) \mathbb{E} \|\mathbb{E}[Y \mid S]\|.$$

Proof. Write $z_s = \mathbb{E}[Y \mid S = s]$, $p_s = \mathbb{P}(S = s)$ and $T_g(s, r) = \mathbb{P}(S' = r \mid S = s, G = g)$. Conditional independence gives

$$p'_r z'_r = \sum_s p_s \int T_g(s, r) U_g z_s d\mu(g), \quad z'_r = \mathbb{E}[U_G Y \mid S' = r].$$

Choose $u_r = z'_r / \|z'_r\|$ when the latter is nonzero and any unit vector otherwise. Thus

$$\begin{aligned} \sum_r p'_r \|z'_r\| &= \sum_s p_s \int \sum_r T_g(s, r) \langle U_g z_s, u_r \rangle d\mu(g) \\ &\leq \sum_{s: z_s \neq 0} p_s \|z_s\| \int \max_r \langle U_g(z_s / \|z_s\|), u_r \rangle d\mu(g) \\ &\leq (1 - \Gamma_k(\mu)) \sum_s p_s \|z_s\|. \end{aligned}$$

Zero centroids cause no loss of an identity: their inner product with any chosen direction is zero. The maximization only relaxes the actual possibly randomized assignment. In particular, neither the old register nor the computation inside the packet is required to have k states. $\square$

Theorem 3.3 (Representation-valued packet budget). *Let Y_0 be a fixed unit vector. Each command packet is sampled independently of the earlier process, with its full word law fixed in advance. Write G_i for its product and μ_i for the induced law. Between packets one may insert fixed commands. A finite stochastic register processes the commands in their prescribed order, with arbitrary cut-dependent rows. Let k_i bound its positive-probability states at the end of packet i . If a terminal function $d(S_N)$ satisfies $\|d\| \leq B$ and*

$$|\mathbb{E} \langle Y_N, d(S_N) \rangle| \geq B\kappa > 0,$$

then

$$(7) \quad \kappa \leq \prod_i (1 - \Gamma_{k_i}(\mu_i)), \quad \sum_i \Gamma_{k_i}(\mu_i) \leq \log(1/\kappa).$$

Here Y evolves by the same ordered representation products as the commands. The statement also applies to a complex scalar correlation with absolute value, using $q_N \geq \kappa$ as its terminal hypothesis.

Proof. At each boundary put $q = \mathbb{E}\|\mathbb{E}[Y \mid S]\|$. A whole packet has one stochastic input/output kernel conditional on its starting state and its command word. Its internal rows and all internal hidden states are integrated into that kernel. The fresh packet is independent of the past, so Lemma 3.2 applies. A fixed command can only decrease q , by conditional Jensen and preservation of norm by its action. Initially $q = 1$. At the terminal cut, $|\mathbb{E}\langle Y_N, d(S_N) \rangle| \leq Bq_N$. Multiplication proves the first assertion; $1 - u \leq e^{-u}$ proves the second. If a coefficient is one, positive terminal correlation is impossible, consistent with the first assertion. $\square$

The products G_i need not commute. The packet endpoints need not be consecutive cuts. Most importantly, the commands *inside* a packet need not be independent. Only its independence from the earlier process is used in the conditional-mean identity. This is what permits a stronger converse than a fair one-command test.

Proposition 3.4 (A small-ball criterion). *For $r > 0$ define*

$$A_\mu(r) = \sup_{v, u \in \mathbb{S}(H)} \mu\{g : \|U_g v - u\| < r\}.$$

Then

$$(8) \quad \Gamma_k(\mu) \geq \frac{r^2}{2} (1 - kA_\mu(r))_+.$$

If for every v the packet is uniform on m orbit points separated by Euclidean distance at least σ , then $\Gamma_k(\mu) \geq (1 - k/m)_+ \sigma^2/8$.

Proof. Outside the union of k radius- r balls the squared distance to each center is at least r^2 . Use $1 - \langle v, u \rangle = \|v - u\|^2/2$ and the union bound. For separated points, each open ball of radius $\sigma/2$ contains at most one, so $A_\mu(\sigma/2) \leq 1/m$. $\square$

4. THE SHARP BOUND FOR BINARY ROTATION COMMANDS

Put $\zeta = e^{2\pi i \alpha}$ and

$$(9) \quad s_B(\alpha) = \min_{1 \leq l \leq B} \|l\alpha\|_{\mathbb{R}/\mathbb{Z}}.$$

For irrational α this number is positive at every finite B . Given an integer $k \geq 1$, set $B = 2k - 1$ and choose J uniformly from $\{0, \dots, B\}$. The associated packet is the length- B command word

$$(10) \quad 1^J 0^{B-J}.$$

The command count is uniform on $2k$ possibilities; the physical packet still consists of B ordinary binary commands. Its random choice is not revealed as an additional input.

Lemma 4.1 (Separated packets). *The circle action of the packet (10) has*

$$(11) \quad \Gamma_k(\mu_B) \geq s_B(\alpha)^2.$$

Consequently it multiplies the conditional phase amplitude by at most $1 - s_B(\alpha)^2$ if its endpoint register has at most k states.

Proof. Write $s = s_B(\alpha)$. For any starting direction the $2k$ phase points differ pairwise by at least s in normalized circle distance. Each open arc of radius $s/2$ contains at most one of them. For any k center directions, at least k packet points are therefore at circle distance at least $s/2$ from every center. Their loss $1 - \cos(2\pi\theta)$ is at least $1 - \cos(\pi s)$. Since $0 \leq s \leq 1/2$ and $\sin(\pi s/2) \geq s$,

$$\frac{1}{2}(1 - \cos(\pi s)) = \sin^2(\pi s/2) \geq s^2.$$

Average over the $2k$ points and minimize over centers and initial phase. Lemma 3.2 proves the last assertion. This argument concerns actual hidden-state centroids, not their observable lift vertices. $\square$

Lemma 4.2 (Terminal calibration for every test law). *Fix seed $x = (1, 1)$. Under any command-word distribution independent of the machine's private randomness, a realization with error $\varepsilon < 1/20$ obeys*

$$(12) \quad q_N := \mathbb{E}|\mathbb{E}[\zeta^{L_N} \mid S_N]| \geq \kappa := 1/10 - 2\varepsilon.$$

Proof. Set $d = d_1 + id_2$, so $|d| \leq \sqrt{2}$. For each word the target conditional mean of $d(S_N)$ is $(1+i)\zeta^{L_N}/10$. A binary row-TV error ε changes each coordinate mean by at most 2ε , hence the complex mean by at most $2\sqrt{2}\varepsilon$. Multiply by ζ^{-L_N} and average to get

$$|\mathbb{E}[\zeta^{-L_N} d(S_N)]| \geq \sqrt{2}(1/10 - 2\varepsilon).$$

Conditioning on S_N bounds the left side by $\sqrt{2}q_N$. Only correctness at the final prescribed length was used. $\square$

Theorem 4.3 (Finite packet and occupation bounds). *For an arbitrary irrational α , a machine of error $\varepsilon < 1/20$ and available-label profile $(K_t)_{t=0}^N$ satisfies, for every $k \geq 1$ and $B = 2k - 1$,*

$$(13) \quad \#\{1 \leq t \leq N : K_t \leq k\} \leq B - 1 + \frac{B}{s_B(\alpha)^2} \log \frac{1}{\kappa}.$$

In particular, a peak- K machine has

$$(14) \quad \left\lfloor \frac{N}{2K-1} \right\rfloor s_{2K-1}(\alpha)^2 \leq \log \frac{1}{\kappa}.$$

More generally, for pairwise disjoint packets ending at cuts t_i , of lengths $B_i = 2k_i - 1$, with $K_{t_i} \leq k_i$, one has

$$(15) \quad \prod_i (1 - s_{B_i}(\alpha)^2) \geq \kappa, \quad \sum_i s_{B_i}(\alpha)^2 \leq \log \frac{1}{\kappa}.$$

Proof. On the chosen disjoint intervals use independent packets (10); set all other commands to zero. Fresh packet independence, Lemma 4.1 and Theorem 3.3 apply. Lemma 4.2 supplies the final correlation for this particular test law. This proves (15), and consecutive length- $2K - 1$ packets prove (14).

For (13), disregard low-width endpoints before B , of which there are at most $B - 1$. Select the first remaining such endpoint t , take the interval $(t - B, t]$, and repeat at the first low endpoint at least B epochs later. The selected intervals are disjoint. Each selected endpoint removes at most B low endpoints, since there are only B integer epochs in $[t, t + B - 1]$. If m intervals are selected, the total number of low endpoints is at most $B - 1 + Bm$. The second inequality in (15) gives $ms_B(\alpha)^2 \leq \log(1/\kappa)$.

The choice of intervals uses only the available profile, not the realized state or past command. It defines a legitimate external test distribution for this machine. No width bound is used inside a selected packet or at any unselected cut. $\square$

Proof of the lower bound in Theorem 1.1. For badly approximable α , $s_B(\alpha) \geq b_\alpha/B$. Equation (14) and $N < B(\lfloor N/B \rfloor + 1)$ prove (3). Since $2K - 1 \leq 2K$ and $K \leq K^3$,

$$(16) \quad W_{N,\varepsilon}(\alpha) > \left(\frac{N}{2 + 8b_\alpha^{-2} \log(1/\kappa)} \right)^{1/3}.$$

The corresponding occupation bound is $\#\{1 \leq t \leq N : K_t \leq k\} \leq 2k - 2 + 8b_\alpha^{-2} k^3 \log(1/\kappa)$.

For the golden angle, let $\alpha' = -(1 + \sqrt{5})/2$ and let p be nearest to $l\alpha$. Then

$$(l\alpha - p)(l\alpha' - p) = p^2 + lp - l^2 \in \mathbb{Z} \setminus \{0\}, \quad |l\alpha' - p| \leq \sqrt{5}l + 1/2 < 3l.$$

Thus $b_\alpha \geq 1/3$. At zero error, $2 + 72 \log 10 < 168$, proving the stated bound. The elementary inequality $\log 10 < 83/36$ can, for example, be checked by the positive exponential series. The rank-three bound follows from Proposition 2.2. $\square$

4.1. Matching realization. We reproduce the resonant upper construction from [14, 15] to make the match self-contained. Let p/q be a convergent of α with $q \geq \max\{5, (40N)^{1/3}\}$. Put

$$a = \pi/q, \quad \delta = 2\pi(\alpha - p/q), \quad \lambda = \frac{\cos(a - |\delta|)}{\cos a}, \quad \omega = \frac{\sin |\delta|}{\lambda \sin(2a)}, \quad u_s = (\cos(2\pi s/q), \sin(2\pi s/q)).$$

Subscripts on u_s are modulo q . Convergence gives $|\delta| < a$, so $\lambda \geq 1$ and $0 \leq \omega \leq 1$.

Proposition 4.4 (Exact resonant upper bound). *The experiment has an exact q -state realization. Its two command matrices are independent of the epoch, although its decoder depends on N . For badly approximable α , one may take $q = O_\alpha(N^{1/3})$.*

Proof. Associate to label s at time t the vector $r_t u_s$, $r_t = \lambda^t/4$. Define

$$T_0(s, \cdot) = \lambda^{-1} \delta_s + (1 - \lambda^{-1}) \text{Unif}[q], \quad T_1(s, \cdot) = (1 - \omega) \delta_{s+p} + \omega \delta_{s+p+\text{sgn } \delta}.$$

When $\delta = 0$ take $\omega = 0$. The chord between the two indicated active vertices meets the ray at angle $2\pi(s/q + \alpha)$ at radius $1/\lambda$. Its second coordinate gives exactly the formula for ω . Thus the expected next associated vector is $r_t u_s$ for an idle command and $R_\alpha r_t u_s$ for an active command. Furthermore

$$\log \lambda = \int_{a-|\delta|}^a \tan v \, dv \leq |\delta| \tan a < 4\pi^2/q^3 < 40/q^3,$$

since $q \geq 5$. Hence $r_N \leq e/4 < 1$. The initial polygon has inradius $\cos(\pi/q)/4 > \sqrt{2}/10$; initialize by any convex decomposition of $x/10$. The final decoder has means $r_N e_j^T u_s \in [-1, 1]$. Induction proves exactness for all words. Bounded partial quotients bound consecutive convergent-denominator ratios. For the golden angle the first denominator above a threshold $L \geq 5$ is less than $2L$, proving (4). $\square$

Corollary 4.5 (Hidden, vector and fixed-length common-row widths). *For a badly approximable angle, exact clocked hidden width, exact clocked vector-state width, and exact fixed-length width with common command matrices are all $\Theta_\alpha(N^{1/3})$.*

Proof. The first class contains the other two, while Proposition 4.4 belongs to their intersection. Apply the unrestricted lower bound. $\square$

An unchanged decoder correct after every earlier stopping length is a stronger, anytime specification; no such assertion is used here.

4.2. Other arithmetic. If $\|l\alpha\| \geq bl^{-\tau}$ for all $l \geq 1$, the same proof gives $N < B + b^{-2}B^{2\tau+1} \log(1/\kappa)$ for $B = 2K - 1$. No matching claim is made under this weaker arithmetic hypothesis. For the irrational angle whose rotation has rational entries $\zeta = (3 + 4i)/5$, the nonzero Gaussian integer $(3 + 4i)^l - 5^l$ gives $s_B(\alpha) \geq 5^{-B}/(2\pi)$. Consequently

$$N < B + 4\pi^2 B 25^B \log(1/\kappa), \quad B = 2K - 1.$$

The Gaussian integer is nonzero because otherwise $6/5 = \zeta + \zeta^{-1}$ would be an algebraic integer. This is an explicit logarithmic converse, not a proof that this angle is badly approximable. The earlier rational counter realization and its finite-coin accounting remain unchanged.

5. COMPACT COMMAND GROUPS

Theorem 3.3 applies to any orthogonal or unitary representation, including noncommuting commands. The following two consequences distinguish a finite command alphabet from a genuinely compact command alphabet.

Corollary 5.1 (Characters of compact groups). *Let a finite command alphabet in a compact group contain the identity and g , and let χ be a continuous one-dimensional unitary character with $\chi(g) = e^{2\pi i\alpha}$, α irrational. Any wordwise finite-register experiment whose terminal specification forces a conditional χ -amplitude at least $\kappa > 0$ satisfies (13) and (15) on these commands. In particular, a badly approximable character angle forces $\Omega(N^{1/3})$ hidden width.*

Proof. Use packets $g^J e^{B-J}$. Their character images are exactly those in Lemma 4.1. The other commands may be assigned probability zero by the test law. Apply Theorem 3.3 on $\mathbb{C}$. $\square$

This includes higher-dimensional tori through their characters. It is a lower bound for a selected observed character, not a multiplicative claim about simultaneous independent outputs of several characters. The next result uses a different orbit geometry rather than a character.

Fix an integer $d \geq 2$ and a unit vector $v \in \mathbb{R}^d$. Consider the separate experiment with commands $g_t \in \text{SO}(d)$, supplied as exact atomic matrices, and final coordinate query $j \in \{1, \dots, d\}$. Its specified mean is

$$(17) \quad \mathbb{E}[B \mid g_1, \dots, g_N, j] = \frac{1}{10} e_j^T g_N \cdots g_1 v.$$

A realizing finite-state machine has Borel stochastic rows $T_{t,g}$. Its width still counts only labels; access to this compact input and its row function is part of this model, not an implementation by finite binary input symbols. All terminal probabilities lie in $[9/20, 11/20]$.

Lemma 5.2 (Sphere quantization). *For normalized surface measure on $\mathbb{S}^{d-1}$ there are constants $c_d, C_d > 0$ such that the orbit distortion of Haar measure on $\text{SO}(d)$ satisfies*

$$\Gamma_k(\text{Haar}) \geq c_d k^{-2/(d-1)}.$$

For every $0 < \rho \leq 1/2$ there is a finite set $V \subset \mathbb{S}^{d-1}$ of at most $C_d \rho^{-(d-1)}$ points whose convex hull contains the Euclidean ball of radius $\cos \rho$.

Proof. Haar rotation sends every unit vector to normalized surface measure. A Euclidean radius- r cap, $0 < r \leq 1$, has angular radius $2 \arcsin(r/2) \leq \pi r/2$. Polar integration bounds its probability by $A_d r^{d-1}$ for some $A_d \geq 1$: its integral is proportional to $\int_0^{2 \arcsin(r/2)} \sin^{d-2} u \, du$. Take $r = (2A_d k)^{-1/(d-1)}$ in Proposition 3.4. The bound follows with $c_d = (2A_d)^{-2/(d-1)}/4$.

For the second assertion choose a maximal set with pairwise angular distance greater than ρ . The angular radius- $\rho/2$ caps are disjoint and each has surface mass at least $a_d \rho^{d-1}$, by the same polar integral and $\sin u \geq 2u/\pi$ on $[0, \pi/2]$. Thus the set has at most $a_d^{-1} \rho^{-(d-1)}$ points. Maximality makes it a ρ -net. Its support function in every unit direction is at least $\cos \rho$, proving the ball containment. These elementary estimates are the usual spherical quantization and covering mechanisms; their role here is sequential compression. $\square$

Theorem 5.3 (Sharp width for all rotation commands). *For fixed $d \geq 2$ and $0 \leq \varepsilon < 1/(20\sqrt{d})$, the least hidden-state width for (17), in its compact-command model, satisfies*

$$(18) \quad W_{N,\varepsilon}^{\text{SO}(d)} = \Theta_{d,\varepsilon}(N^{(d-1)/2}).$$

Writing $\kappa_d = 1/(10\sqrt{d}) - 2\varepsilon$, every profile obeys

$$(19) \quad c_d \sum_{t=1}^N K_t^{-2/(d-1)} \leq \log(1/\kappa_d).$$

The exact upper construction has common command rows and an N -dependent terminal decoder.

Proof. Use independent Haar commands as the test law and set $Y_t = g_t \cdots g_1 v$. Combine the coordinate decoder means into a vector $d(S_N)$ of norm at most $\sqrt{d}$. Wordwise approximation implies

$$\mathbb{E}\langle Y_N, d(S_N) \rangle \geq 1/10 - 2\sqrt{d}\varepsilon, \quad q_N \geq \kappa_d.$$

Theorem 3.3 and Lemma 5.2, one command per packet, give (19) and the lower bound.

For the upper bound take $\rho = 1/(2\sqrt{N})$ and a net V as in Lemma 5.2. Put $P = \text{conv } V$, $\lambda = (\cos \rho)^{-1}$, and $r_t = \lambda^t/4$. A label $u \in V$ has associated vector $r_t u$. For any command g , the vector gu/λ lies in $(\cos \rho)\mathbb{B}^d \subset P$. Fix a Borel convex decomposition of every point of P into V , and use the coefficients of gu/λ as the command row from label u . Such decompositions may be chosen by a finite triangulation of P using its vertices, with ties between simplices resolved in a fixed order. The resulting rows are Borel, normalized, and independent of t .

The expected next associated vector is exactly $g(r_t u)$. Since $-\log \cos \rho \leq \rho^2$ for $\rho \leq 1/2$, $r_N \leq e^{1/4}/4 < 1$. Also $v/10 \in r_0 P$, since $\cos(1/2)/4 > 1/10$. Initialize by its chosen convex decomposition and decode with means $r_N e_j^T u$. This realizes (17) exactly and uses at most $C_d(2\sqrt{N})^{d-1}$ labels (and two output labels, absorbed by the bound). No vectors are retained at runtime. $\square$

For $d \geq 3$ the commands here are noncommuting. For $d = 2$ the order $N^{1/2}$ concerns all circle rotations as inputs, while Theorem 1.1 has the smaller fixed alphabet $\{I, R_\alpha\}$ and order $N^{1/3}$. These statements concern different command sets. The compact theorem is not a claim of that exponent for an arbitrary finite generating set, or an uncharged approximation of a real command by a finite word.

6. ACTUAL TERMINAL ERROR AND LOCAL DIAGRAM CERTIFICATES

For a prescribed profile $\mathbf{K}$, define $\mathcal{E}_N(\mathbf{K})$ as the minimum uniform terminal row-TV error in Definition 2.1. The minimum exists: the row simplices, initialization rows and decoder intervals form a compact set, and the maximum of finitely many wordwise errors is continuous. Thus this is an intrinsic final simulation objective, not a sum of intermediate defects.

For $t \geq 1$ put $B_t = 2K_t - 1$. Call $(t - B_t, t]$ admissible when $B_t \leq t$. Let

$$(20) \quad \mathfrak{B}_\alpha(\mathbf{K}) = \max_{\substack{\text{pairwise disjoint} \\ \text{admissible intervals}}} \sum_i s_{B_{t_i}}(\alpha)^2.$$

This finite weighted-interval problem depends only on the profile and angle. Empty selections give zero. If $F_0 = 0$, its exact recurrence is

$$F_t = \begin{cases} \max\{F_{t-1}, F_{t-B_t} + s_{B_t}(\alpha)^2\}, & B_t \leq t, \\ F_{t-1}, & B_t > t. \end{cases} \quad \mathfrak{B}_\alpha(\mathbf{K}) = F_N.$$

The last chosen interval proves the recurrence. Computing the weights requires the stated angle representation; no bit-complexity claim for arbitrary exact real inputs is made.

Theorem 6.1 (Final-error obstruction). *For every irrational α and finite profile,*

$$(21) \quad \mathcal{E}_N(\mathbf{K}) \geq \frac{1}{2} \left[\frac{1}{10} - \exp\{-\mathfrak{B}_\alpha(\mathbf{K})\} \right]_+.$$

In particular, for peak at most K and $B = 2K - 1$,

$$(22) \quad \mathcal{E}_N(K) \geq \frac{1}{2} \left[\frac{1}{10} - \exp\{-\lfloor N/B \rfloor s_B(\alpha)^2\} \right]_+.$$

For a badly approximable angle, every sequence $K = o(N^{1/3})$ has $\liminf \mathcal{E}_N(K) \geq 1/20$.

Proof. Choose the disjoint intervals maximizing (20) and put independent separated packets on them, idle commands elsewhere. Every machine of that profile has terminal amplitude at most $\exp\{-\mathfrak{B}_\alpha(\mathbf{K})\}$, by Lemma 3.2 and Lemma 4.1. Terminal calibration gives $1/10 - 2\varepsilon \leq q_N$ whenever its left side is positive. Rearrange and then minimize over machines. Consecutive equal-length packets prove (22); the estimate $s_B \geq b_\alpha/B$ proves the asymptotic assertion. $\square$

This conclusion is about the actual terminal error, so possible cancellation of intermediate approximation errors cannot evade it. It does not give the exact value of $\mathcal{E}_N$ at every finite profile. In particular the limiting lower constant $1/20$ is not claimed optimal.

6.1. Transfer to executable diagrams. For finite row-stochastic experiments E, F indexed by the same finite set of prefixes, use the classical one-sided deficiency

$$(23) \quad \delta(E, F) = \min_{T \text{ stochastic}} \max_h \text{TV}((ET)_h, F_h).$$

The common row T cannot depend on h . The zero-defect case is Blackwell garbling [10, 11]; composition contracts total variation.

Let E_t specify a proposed state distribution for each actual input prefix at cut t . For every next command, choose a common update approximating the corresponding child rows, and likewise choose each terminal query decoder. Let ϵ_t be the worst row error of that update over prefixes and commands, and ϵ_{out} the worst terminal decoder error. If the initial rows are exact, put

$$\mathcal{D}(E) = \sum_{t < N} \epsilon_t + \epsilon_{\text{out}}.$$

Minimizing each one-step error gives (23) at that step. This is an *accumulated local deficiency certificate*, not a formula for the optimal final error.

Corollary 6.2 (Diagram obstruction). *Every such diagram of profile $\mathbf{K}$ satisfies the right side of (21) as a lower bound on $\mathcal{D}(E)$.*

Proof. Execute the selected common kernels. At each fixed prefix, induction and TV contraction bound the discrepancy from E_t by the sum of preceding ϵ_s . Append the common terminal decoder to obtain actual row error at most $\mathcal{D}(E)$. If this sum exceeds one, the trivial error bound one suffices. Thus $\mathcal{D}(E) \geq \mathcal{E}_N(\mathbf{K})$; apply Theorem 6.1. $\square$

The certificate can overestimate the achieved final error, since local errors may cancel. The corollary does not deny cancellation, minimize $\mathcal{D}$ over all diagrams, or assert equality of the two objectives. The new lower bound already applies before any diagram is introduced.

7. COMPARISON OF MODELS AND CONCLUSIONS

7.1. Probabilistic read-once branching programs. After fixing the seed and padding layers to a common width, the product in Definition 2.1 is a stochastic ordered read-once branching program: layer t reads command c_t once and applies T_{t-1, c_t} . This is the randomized-transition extension of the standard nonuniform model in [1, Section 1.1]; see also [4]. Width is the maximum layer size, whereas total size counts all layers. Our theorem is a width theorem, not a succinct uniform program-size or bit-operation theorem.

For a chosen query j , appending an accepting Bernoulli row with probability $(1 + d_j(s))/2$ turns the real readout into ordinary randomized acceptance with no larger width than $\max(K, 2)$. Combining the two means into $d_1 + id_2$ does not enlarge the retained alphabet. This combination is an identity between mean functions, not a requirement to sample both query answers jointly. Thus real readouts and the word “transducer” do not by themselves separate our problem from branching programs.

For a fixed initial row p and terminal column d , put $F(c) = pT_{0,c_1} \cdots T_{N-1,c_N}d$. Its Boolean-input Fourier coefficient at $A \subseteq [N]$ is exactly

$$(24) \quad \widehat{F}(A) = p \prod_{t=1}^N \frac{T_{t-1,0} + (-1)^{\mathbf{1}_{\{t \in A\}}} T_{t-1,1}}{2} d.$$

This follows by multiplying independent sums over input bits in their original order. It is a matrix-product Fourier coefficient, distinct from a harmonic of the conditional phase measure used in [15]. The present proof instead uses the nonlinear quantity $\mathbb{E}|\mathbb{E}[\zeta^{L_t} | S_t]|$ at packet boundaries.

Reingold–Steinke–Vadhan [1, Theorem 1.4] bound Boolean level- l Fourier mass by $(2w^2)^l$ for regular width- w programs. Steinke–Vadhan–Wan [2] establish width-three Fourier-growth bounds and use them for pseudorandomness. Lee–Pyne–Vadhan [3, Definitions 3–5 and Theorem 6] refine the regular-program bounds, including unbounded width and restrictions on accepting states; their Section 1.2.3 discusses generalized group products. These results concern Fourier growth or fooling classes of computations. They do not state the terminal-correlation packet inequality (6) or its nonuniform occupation consequence.

Regularity means preservation of a uniform layer distribution under a fair next input. It is not implied by row stochasticity: the matrix all of whose rows put mass one on a single label is legal here and destroys uniformity. Consequently a regular/permutation theorem cannot be invoked without its hypothesis. Conversely, some bounds for general deterministic programs extend to stochastic programs by resolving all private rows into a convex combination of deterministic tables and using convexity. We therefore make no claim that private randomness alone prevents use of the earlier literature. The extra statement proved here is the multiplicative loss forced by a separated command packet, for arbitrary starting conditional centroids and arbitrary endpoint assignments. It gives the matching growing-width law without a regularity assumption and without an estimate of the Fourier mass in (24).

7.2. Circle walks and the choice of test law. For independent Bernoulli commands of bias p , the uncompressed phase has harmonic multiplier

$$a_l = 1 - p + pe^{2\pi i l \alpha}, \quad |a_l|^2 = 1 - 4p(1-p)\sin^2(\pi l \alpha).$$

These are classical circle-walk quantities. Berkes–Borda [5] relate convergence and limit theorems for circle walks to rational approximation, including the transition along convergent denominators. That analysis concerns the uncompressed walk, not the information retained by an additional hidden register.

The earlier Fourier budget [15] combined this damping, common conditional-amplitude loss, and a positive finite-atom Fourier witness. It gave a fifth-root unrestricted converse. Our test law is different: its bits are correlated inside a packet so that the packet count is uniform, and different packets are independent. We are not asserting a cube-root occupation theorem for the fair-command law itself. Wordwise correctness allows the new law; Lemma 3.2 measures its compression directly. The packet costs $O(k)$ commands and loses $\Omega(k^{-2})$ amplitude, explaining the cube-root order. The angular averaging estimate remains valid with its sharp universal coefficient, proved in Appendix A; improving that coefficient’s quadratic order is not needed for the new result.

7.3. Positive realization and experiment comparison. Invariant convex sets, positive stochastic realizations, and irrational rotation obstructions for one repeated positive matrix belong to classical realization theory [6, 7, 8, 9]. The present lower bound permits unrelated matrices at every layer and a new machine for each horizon; it is not a peripheral-eigenvalue argument for a stationary realization. The upper construction is inherited from [14, 15] and has

been proved again in Proposition 4.4. What is new in the main theorem is its match against all hidden states, including under fixed row error.

Finite Blackwell garbling and one-sided deficiency are classical [10, 11]. Theorem 6.1 bounds the actual terminal optimization. Corollary 6.2 then bounds an accumulated local certificate; it does not identify that certificate with the optimized terminal distance. Orbit quantization, sphere-cap bounds and polytope nets in Section 5 are classical geometric mechanisms. Their representation-valued sequential loss is stated and proved explicitly in Theorem 3.3.

7.4. Conclusions and precise boundaries. The hidden and vector-state rotation widths have the same cube-root order for each badly approximable angle and fixed nonzero signal at sufficiently small fixed error. The finite constants are proof constants, not optimized thresholds or exact integer minima. The packet budget also controls selected low-width endpoints and actual terminal error. For compact rotation commands, a distinct matched sphere-dimension law follows from the same contraction theorem.

No sharp order is asserted for every irrational angle, for every finite noncommuting command set, or for a regime in which signal and permitted error vary with the horizon. Optimizing informative action choice is not the external-command problem studied here. Atomic computable-real rows are not finite fair-bit algorithms. No exact minimum dynamic Blackwell distance or independent analytic foundation for an unrelated model is inferred from these results. The original manuscript and its historical arguments remain separately available, with their hypotheses unchanged.

APPENDIX A. THE SHARP HOMOGENEOUS ANGULAR COEFFICIENT

The earlier Fourier proof remains useful when the external test law is required to have independent single commands. Its elementary angular estimate admits the following sharpened statement. This also resolves the coefficient issue in the preceding report without changing any previous published source.

Lemma A.1 (Homogeneous angular averaging). *Let $f_l(re^{i\theta}) = re^{il\theta}$ for $r > 0$ and $f_l(0) = 0$. For every integrable complex Z and integer $l \geq 1$,*

$$(25) \quad |\mathbb{E}f_l(Z) - f_l(\mathbb{E}Z)| \leq (l^2 - 1)(\mathbb{E}|Z| - |\mathbb{E}Z|).$$

For $l \geq 2$ the coefficient $l^2 - 1$ cannot be decreased uniformly over all such mixtures. At $l = 1$ both sides vanish.

Proof. The case $l = 1$ is linear. For $l \geq 2$, if $\mathbb{E}Z = 0$, the left side is at most $\mathbb{E}|Z|$ and $l^2 - 1 \geq 1$. Otherwise rotate the variable so $\mathbb{E}Z = \rho > 0$, and write $Z = re^{i\theta}$. For $u = e^{i\theta}$,

$$u^l - 1 - l(u - 1) = (u - 1) \sum_{j=0}^{l-1} (u^j - 1).$$

Since $|u^j - 1| \leq j|u - 1|$, the modulus of this remainder is at most $l(l - 1)(1 - \cos \theta)$. Retaining this coefficient gives

$$|u^l - \cos \theta - il \sin \theta| \leq \{l(l - 1) + (l - 1)\}(1 - \cos \theta) = (l^2 - 1)(1 - \cos \theta).$$

Multiply by r and average. The identities $\mathbb{E}(r \sin \theta) = 0$ and $\mathbb{E}(r \cos \theta) = \rho$ prove (25). Rotation covariance restores the general case. For sharpness take $Z = e^{i\theta}$ and $e^{-i\theta}$ with equal probabilities, and let $\theta \downarrow 0$. The ratio is

$$\frac{|\cos(l\theta) - \cos \theta|}{1 - \cos \theta} \rightarrow l^2 - 1.$$

□

Thus a uniformly linear-in- l replacement is impossible for unrestricted conditional mixtures. The improvement to cube-root width in this paper does not require such an estimate: it changes the test law to packets and measures the directly incurred geometric amplitude loss.

REFERENCES

- [1] O. Reingold, T. Steinke and S. Vadhan, *Pseudorandomness for regular branching programs via Fourier analysis*, APPROX/RANDOM 2013, LNCS 8096, 655–670. Full version: arXiv:1306.3004v2 (2013).
- [2] T. Steinke, S. Vadhan and A. Wan, *Pseudorandomness and Fourier-growth bounds for width-3 branching programs*, Theory of Computing **13** (2017), article 12, 1–50.
- [3] C. H. Lee, E. Pyne and S. Vadhan, *Fourier growth of regular branching programs*, APPROX/RANDOM 2022, LIPIcs **245**, article 2, 2:1–2:21.
- [4] I. Wegener, *Branching Programs and Binary Decision Diagrams*, SIAM, Philadelphia, 2000.
- [5] I. Berkes and B. Borda, *Random walks on the circle and Diophantine approximation*, J. London Math. Soc. **108** (2023), 409–440. Preprint: arXiv:2204.00274.
- [6] A. Heller, *On stochastic processes derived from Markov chains*, Ann. Math. Statist. **36** (1965), 1286–1291.
- [7] M. Vidyasagar, *The complete realization problem for hidden Markov models: a survey and some new results*, Math. Control Signals Systems **23** (2011), 1–65.
- [8] L. Benvenuti and L. Farina, *A tutorial on the positive realization problem*, IEEE Trans. Automat. Control **49** (2004), 651–664.
- [9] A. Monras and A. Winter, *Quantum learning of classical stochastic processes: the completely-positive realization problem*, J. Math. Phys. **57** (2016), 015219.
- [10] D. Blackwell, *Equivalent comparisons of experiments*, Ann. Math. Statist. **24** (1953), 265–272.
- [11] E. Torgersen, *Comparison of Statistical Experiments*, Cambridge University Press, Cambridge, 1991.
- [12] Q. Qi, *General Theta Foundations I: Compatibility of Positive Memory and Online Simplex Synthesis*, revision 31, repository manuscript, 2026.
- [13] Q. Qi, *General Theta Foundations I: Causal Width Beyond Cutwise Positive Rank*, revision 32, repository manuscript, 2026.
- [14] Q. Qi, *General Theta Foundations I*, revision 33, GTF-I-v33-compatible-lifts, repository manuscript, 2026. Frozen publication aab96108125317342e51e6a946efc4d778429338.
- [15] Q. Qi, *General Theta Foundations I: Fourier Budgets for Hidden Causal Memory*, revision 34, repository manuscript, 2026. Frozen publication 47bc47e5995e4b1b99a18aaa472a1309f1668f40.

PRESERVED MATHEMATICAL DEVELOPMENT

General Theta Foundations I - revision 34

The complete current article precedes this divider.

The following predecessor cumulative volume is reproduced without alteration.

Frozen predecessor publication:

47bc47e5995e4b1b99a18aaa472a1309f1668f40

This archive is not part of the compact referee package. Independent historical analytic obligations keep their original hypotheses and proof status.

GENERAL THETA FOUNDATIONS I: FOURIER BUDGETS FOR HIDDEN CAUSAL MEMORY

QIAN QI

ABSTRACT. We prove a polynomial memory lower bound for a uniformly positive rotation experiment whose separate positive minimum is three at every cut. The bound allows arbitrary hidden states, unrelated transition matrices at different epochs, and a new machine for each horizon. Its proof uses conditional past-phase moments rather than the number of vertices of a projected simplex section. A homogeneous angular inequality controls all Fourier moments by one cumulative loss of conditional phase amplitude; positivity of the Fejér kernel detects a register with few labels. For an angle satisfying $\inf_{m \geq 1} m \|m\alpha\| > 0$, the unrestricted width is at least a constant times $N^{1/5}$, improving the previous logarithmic lower bound, while an exact $O_\alpha(N^{1/3})$ construction is retained. The lower bound persists under any fixed sufficiently small uniform output error. For the rational rotation $(3 + 4i)/5$ it gives a logarithmic, rather than doubly logarithmic, lower bound. We formulate stochastic extension as finite Blackwell comparison and transfer the new obstruction to the accumulated deficiency of every proposed lift diagram. This rules out logarithmic-coordinate low-defect lift chains for the bounded-type family without imposing vector-state or equivariance restrictions. The exponents $1/5$ and $1/3$ are not identified, and the fixed-length and anytime autonomous specifications are kept distinct.

1. INTRODUCTION

An experiment can admit a small positive factorization at each separate cut while requiring substantially more states when the factorizations must be joined by common stochastic updates. For hidden realizations, a state need not carry an individually meaningful predictive vector. A projection of a K -coordinate simplex section can have exponentially many vertices. Consequently, a lower bound obtained only by counting the vertices of an observable polygon may lose an exponential factor in the memory parameter.

We study the following fixed-alphabet experiment. A seed $x \in \{-1, 1\}^2$ is followed by N commands $c_t \in \{0, 1\}$, a query $j \in \{1, 2\}$, and a binary answer B . For $\zeta = e^{2\pi i \alpha}$ and the corresponding planar rotation R_α , the prescribed answer law is

$$(1) \quad \mathbb{P}(B = b \mid x, c_1, \dots, c_N, j) = \frac{1 + b e_j^T R_\alpha^{c_1 + \dots + c_N} x / 10}{2}.$$

All these probabilities lie in the fixed interval $[(1 - \sqrt{2}/10)/2, (1 + \sqrt{2}/10)/2]$. The alphabets, amplitude and support margin do not change with N . Exact realization means equality for every input word, not merely after averaging over commands.

The separate positive minimum of this experiment is three. The previous mean-support argument gave $\Omega_\alpha(\log N)$ as an unrestricted hidden-state lower bound and $\Theta_\alpha(N^{1/3})$ within a vector-state subclass at bounded-type angles [17]. We retain its exact upper construction but replace the hidden lower argument. The new method does not associate a predictive vector with each basis state. Instead, it studies the conditional expectation of a random past phase *under a*

Date: September 25, 2026. Revision 34.

2020 Mathematics Subject Classification. Primary 93B15, 60J10; Secondary 42A16, 52B05, 62B15.

Key words and phrases. Positive realization, hidden memory, stochastic transducers, Fourier moments, comparison of experiments, causal lifts.

chosen test distribution on external words. These conditional expectations are proof objects, not information made available to the executing machine.

Write

$$(2) \quad \beta_m(\alpha) = \min_{1 \leq l \leq m} \sin^2(\pi l \alpha), \quad b_\alpha = \inf_{l \geq 1} l \|l \alpha\|_{\mathbb{R}/\mathbb{Z}}.$$

For irrational α , every β_m is positive. The assumption $b_\alpha > 0$ is the usual badly-approximable, equivalently bounded-type, condition. Let $W_{N,\varepsilon}(\alpha)$ be the least peak label count of a clocked stochastic realization with uniform row-total-variation error at most ε , and write $W_N = W_{N,0}$.

Theorem 1.1 (Polynomial obstruction for all hidden realizations). *Let $0 \leq \varepsilon < 1/20$ and put $\kappa = 1/10 - 2\varepsilon$. Every realizing profile $(K_t)_{t=0}^N$ satisfies, for every integer $k \geq 1$,*

$$(3) \quad \#\{0 \leq t < N : K_t \leq k\} \beta_{2k}(\alpha) \leq \frac{18k^3}{\kappa^2}.$$

In particular, if $b_\alpha > 0$, then

$$(4) \quad W_{N,\varepsilon}(\alpha) \geq \left(\frac{b_\alpha^2 \kappa^2 N}{18} \right)^{1/5}.$$

For $\alpha = (\sqrt{5} - 1)/2$, exact realization therefore obeys

$$(5) \quad \max\{3, (N/16200)^{1/5}\} \leq W_N(\alpha) < 2 \max\{5, (40N)^{1/3}\}.$$

More generally the same cube-root upper order holds whenever the partial quotients of α are bounded. For $\zeta = (3 + 4i)/5$,

$$(6) \quad N \leq 7200 W_N^3 25^{2W_N}, \quad W_N \geq \max\{3, \log_{5000}(N/7200)\}.$$

All lower bounds allow arbitrary cut-dependent hidden-state continuations.

The theorem excludes $O(\log N)$ hidden width at bounded-type angles, including at any fixed row error below $1/20$. It does not prove that hidden and vector-state widths have the same order: the current exact bounds are $\Omega_\alpha(N^{1/5})$ and $O_\alpha(N^{1/3})$. The cumulative version (3) also limits how often a small register may occur in an otherwise unrestricted profile.

The proof has three steps. Conditional phase amplitude decreases under a stochastic update. Its entire loss is at most one. A homogeneous angular inequality bounds the perturbation of the l th Fourier moment by $2l^2$ times this same loss. Finally, the positive Fejér kernel forces a nonnegligible Fourier potential whenever the conditional phase measure has few atoms. Randomizing the two external commands damps those moments at the rates $\sin^2(\pi l \alpha)$. The resulting potential telescopes for time-dependent as well as stationary matrices.

Section 5 interprets a supplied lift diagram as a sequence of finite experiment comparisons. The one-step garbling criterion and deficiency dual are classical [3, 12]. Their composition produces an executable approximate transducer. Applying Theorem 1.1 yields a new lower bound on the total defect of *every* such diagram for (1), rather than only a four-point warning example. Section 6 compares this notion with equivariant polyhedral lifts and distinguishes fixed-length autonomy from the stronger anytime query task.

Notation	Meaning
$K_t, W_{N,\varepsilon}$	Available labels at cut t ; minimum peak at row error ε .
$z_t(s), q_t$	Conditional past-phase expectation; its mean absolute value.
$\Phi_t^{(m)}$	Finite Fejér-weighted Fourier potential of the conditional phases.
$\delta(E, F)$	One-sided finite deficiency: simulate F by garbling E .
$\mathcal{D}(\mathcal{E})$	Sum of optimized one-step and terminal defects for a fixed diagram.
$A_N^{\text{fix}}, A_N^{\text{any}}$	Autonomous state counts for one fixed length, or all lengths up to N .

2. THE FINITE STOCHASTIC MODEL

State distributions are row vectors. Let $\Delta_k = \{e \in \mathbb{R}^k : e \geq 0, e\mathbf{1} = 1\}$.

Definition 2.1 (Clocked transducer and approximation). A transducer for a prescribed N has initialization rows $E_x \in \Delta_{K_0}$, row-stochastic matrices $T_{t,c} \in \mathbb{R}_+^{K_t \times K_{t+1}}$ for $0 \leq t < N$, $c \in \{0, 1\}$, and binary decoder means $d_j \in [-1, 1]^{K_N}$ for $j \in \{1, 2\}$. Its mean after a word is

$$E_x T_{0,c_1} \cdots T_{N-1,c_N} d_j.$$

Its uniform row error is the maximum total variation between this binary law and (1), over all seeds, command words and queries. The available-label profile is (K_t) and its peak is $\max_t K_t$. $W_{N,\varepsilon}$ minimizes this peak over transducers of row error at most ε . At error zero the defining product must equal the specified mean on every word.

Only the current state survives each update. All retained randomness, past inputs and protocol distinctions belong to that state. The seed and query are supplied once as atomic symbols. The command is an external input; it is not a hidden parameter chosen by the model. The epoch is an external clock, so different cuts can have different alphabets and rows. All rows, including unreachable rows, are normalized. Available labels are counted; padding is allowed. A retained final binary answer uses two labels at its own output cut. This does not enlarge any exact peak here, which is at least three.

Exact stochastic rows, their read-only descriptions, arithmetic, and row-construction time are not charged by this positive-realization width. A rational row can be compiled into fair-bit rejection sampling with extra phase, old-state, current-input and sampler-prefix storage, as in [15]. No equality between that implementation cost and the atomic width is asserted. Irrational rows below are exact mathematical kernels, not a finite fair-bit implementation claim.

Proposition 2.2 (The separate positive minimum remains three). *For exact realization of (1), every internal cut has normalized residual rank three and separate positive minimum three. The separate minimum allows arbitrary finite registers at all other cuts.*

Proof. A prefix has a predictive vector $R_\alpha^a x/10$. Its complete future array is affine in that vector. Four idle prefixes span an affine plane, and two idle-suffix coordinate queries separate it. Normalization adds the constant direction, so the residual rank is three. The triangle with vertices

$$(1/2, 0), \quad (-1/2, 1/2), \quad (-1/2, -1/2)$$

contains the disk of radius $\sqrt{2}/10$. Indeed its supporting-line distances from zero are $1/2, 1/(2\sqrt{5}), 1/(2\sqrt{5})$, all larger than that radius. Its vertices have norm at most $1/\sqrt{2}$, so every rotated continuation from a vertex gives legal binary answers. At the selected cut decompose the actual predictive vector in this triangle. Before that cut store the finite raw history; after it store the chosen vertex label and remaining command count. This is a complete finite realization with three states at the selected cut. Rank proves minimality. The construction is inherited from [16, 17]; the new result concerns simultaneous, not separate, width. $\square$

The lower argument below uses only the stochastic update rule and the correct terminal correlation. It does not project hidden basis states into this predictive plane. In particular, it covers the simplex-section lifts in [17], including lifts with unobservable directions.

3. A FOURIER BUDGET FOR A HIDDEN REGISTER

This section states the probabilistic argument independently of the particular seed amplitude. Let $C_1, \dots, C_N$ be independent fair bits, put $L_t = C_1 + \cdots + C_t$, and let S_t be an arbitrary

finite-state process with

$$(7) \quad \mathbb{P}(S_{t+1} = s' \mid S_0, C_1, \dots, C_t, S_t, C_{t+1}) = T_{t, C_{t+1}}(S_t, s').$$

The new command is independent of the past. Initialization may be random. The matrices and state spaces may depend on t and N . Set $Y_t = \zeta^{-L_t}$ and, on states of positive probability, define

$$(8) \quad p_t(s) = \mathbb{P}(S_t = s), \quad z_t(s) = \mathbb{E}[Y_t \mid S_t = s], \quad q_t = \sum_s p_t(s) |z_t(s)|.$$

Put $z_t = 0$ on a zero-mass state. These are conditional expectations used in the proof, not registers or extra inputs to the transducer.

Lemma 3.1 (One common amplitude loss). *The sequence q_t is nonincreasing, with $q_0 = 1$. If $Z_{t+1} = \zeta^{-C_{t+1}} z_t(S_t)$, then*

$$(9) \quad z_{t+1}(S_{t+1}) = \mathbb{E}[Z_{t+1} \mid S_{t+1}], \quad \ell_t := q_t - q_{t+1} \geq 0, \quad \sum_{t < N} \ell_t \leq 1.$$

If a complex function $d(S_N)$ obeys $|d| \leq B$ and $|\mathbb{E}[Y_N d(S_N)]| \geq B\kappa$, then $q_t \geq \kappa$ for every t .

Proof. The stochastic update and the independence of the new command imply $\mathbb{E}[Y_t \mid S_t, C_{t+1}, S_{t+1}] = z_t(S_t)$ on positive-mass events. The tower property gives the first identity. Conditional Jensen gives $q_{t+1} \leq \mathbb{E}|Z_{t+1}| = q_t$, hence the telescoping loss bound. Finally

$$|\mathbb{E}[Y_N d(S_N)]| = |\mathbb{E}[z_N(S_N) d(S_N)]| \leq Bq_N.$$

Monotonicity proves the last assertion. No conditional independence of past commands given the retained state is assumed. $\square$

For an integer $l \geq 1$ define the homogeneous angular function

$$(10) \quad f_l(re^{i\theta}) = re^{il\theta} \quad (r > 0), \quad f_l(0) = 0.$$

It is not necessary to differentiate this function at zero. The following inequality charges its failure to preserve averages to the *same* absolute-amplitude loss for every l .

Lemma 3.2 (Homogeneous angular averaging). *For any integrable complex random variable Z and integer $l \geq 1$,*

$$(11) \quad |\mathbb{E}f_l(Z) - f_l(\mathbb{E}Z)| \leq (l^2 + 1)(\mathbb{E}|Z| - |\mathbb{E}Z|).$$

Proof. If $\mathbb{E}Z = 0$, the left side is at most $\mathbb{E}|Z|$, and the result follows. Otherwise rotate so that $\mathbb{E}Z = \rho > 0$, and write $Z = re^{i\theta}$. Then $\mathbb{E}(r \sin \theta) = 0$ and $\mathbb{E}(r \cos \theta) = \rho$. For $u = e^{i\theta}$ the identity

$$u^l - 1 - l(u - 1) = (u - 1) \sum_{j=0}^{l-1} (u^j - 1)$$

and $|u^j - 1| \leq j|u - 1|$ give

$$|u^l - 1 - il \sin \theta| \leq l^2(1 - \cos \theta).$$

Consequently

$$|e^{il\theta} - \cos \theta - il \sin \theta| \leq (l^2 + 1)(1 - \cos \theta).$$

Multiply by r , average, and use the two identities for sine and cosine. Rotation covariance of f_l restores the general case. Values with $r = 0$ contribute zero. Thus the proof covers mixtures passing through zero. $\square$

Let

$$(12) \quad m_{t,l} = \sum_s p_t(s) f_l(z_t(s)), \quad a_l = \frac{1 + \zeta^{-l}}{2}.$$

By radial homogeneity, $|m_{t,l}| \leq q_t \leq 1$.

Lemma 3.3 (Moment recursion). *There are complex errors $e_{t,l}$ such that*

$$(13) \quad m_{t+1,l} = a_l m_{t,l} + e_{t,l}, \quad |e_{t,l}| \leq (l^2 + 1) \ell_t \leq 2l^2 \ell_t.$$

In particular

$$(14) \quad |m_{t+1,l}|^2 \leq (1 - \sin^2(\pi l \alpha)) |m_{t,l}|^2 + 4l^2 \ell_t.$$

Proof. Apply Lemma 3.2 conditionally on S_{t+1} to Z_{t+1} in Lemma 3.1, then average. The total conditional amplitude deficit is exactly ℓ_t . Rotation covariance and the fair command give $\mathbb{E} f_l(Z_{t+1}) = a_l m_{t,l}$. This proves the first line. Both $|m_{t+1,l}|$ and $|a_l m_{t,l}|$ are at most one, so the difference of their squares is at most $2|e_{t,l}|$. Use $|a_l|^2 = 1 - \sin^2(\pi l \alpha)$. $\square$

Associate to cut t the positive measure on the circle

$$(15) \quad \nu_t = \sum_{s: z_t(s) \neq 0} p_t(s) |z_t(s)| \delta_{\arg z_t(s)}.$$

Its mass is q_t , it has at most K_t atoms, and its l th Fourier moment is $m_{t,l}$. Equal directions may be combined; doing so only reduces the number of atoms.

Lemma 3.4 (A finite positive Fourier witness). *For a positive circle measure ν of mass q with at most k atoms, put*

$$\Phi^{(M)}(\nu) = \sum_{l=1}^M \left(1 - \frac{l}{M+1}\right) |\hat{\nu}(l)|^2.$$

Then

$$(16) \quad q^2 + 2\Phi^{(M)}(\nu) \geq \frac{M+1}{k} q^2.$$

In particular, $\Phi^{(2k)}(\nu) \geq q^2/2$.

Proof. The Fejér kernel

$$F_M(\theta) = \frac{1}{M+1} \left| \sum_{j=0}^M e^{ij\theta} \right|^2 = 1 + 2 \sum_{l=1}^M \left(1 - \frac{l}{M+1}\right) \cos(l\theta)$$

is nonnegative and has value $M+1$ at zero. If the atom weights are w_i , integrate $F_M(\theta - \phi)$ against $\nu \otimes \nu$. Dropping the nonnegative off-diagonal terms gives at least $(M+1) \sum_i w_i^2 \geq (M+1) q^2/k$. Expanding the same integral gives the left side of (16). This is the elementary positive-kernel identity; its use here is to detect a finite hidden register. $\square$

Theorem 3.5 (Spectral occupation budget). *In the process (7), suppose $q_N \geq \kappa > 0$. For every integer $k \geq 1$, even if other cuts have arbitrarily many states,*

$$(17) \quad \#\{0 \leq t < N : K_t \leq k\} \beta_{2k}(\alpha) \leq \frac{18k^3}{\kappa^2}.$$

The conclusion remains true with K_t replaced by the number of positive-probability states under this test distribution.

Proof. Fix k , set $M = 2k$, $w_l = 1 - l/(M + 1)$, and abbreviate $\Phi_t = \Phi^{(M)}(\nu_t)$. Multiply (14) by w_l and sum. This gives

$$\Phi_{t+1} \leq \Phi_t - \sum_{l=1}^M w_l \sin^2(\pi l \alpha) |m_{t,l}|^2 + 4\ell_t \sum_{l=1}^M w_l l^2.$$

Initially $z_0 = 1$ on every positive-mass state, so $m_{0,l} = 1$ and $\Phi_0 = M/2 = k$. Moreover

$$\sum_{l=1}^M w_l l^2 = \frac{M(M+1)(M+2)}{12} = \frac{k(k+1)(2k+1)}{3} \leq 2k^3.$$

Sum over all epochs, use $\Phi_N \geq 0$ and $\sum \ell_t \leq 1$, and obtain

$$(18) \quad \sum_{t < N} \sum_{l=1}^{2k} w_l \sin^2(\pi l \alpha) |m_{t,l}|^2 \leq k + 8k^3 \leq 9k^3.$$

At any cut with at most k positive-mass states, Lemma 3.4 and $q_t \geq q_N \geq \kappa$ imply $\Phi_t \geq \kappa^2/2$. The summand at such a cut is therefore at least $\beta_{2k}(\alpha)\kappa^2/2$. All other summands are nonnegative. Counting the small cuts proves (17). Nothing in the proof identifies the conditional past phase with a statewise future prediction, and no bound on vertices of a lift is used. $\square$

The test law uses fresh independent commands but makes no such assumption about the commands in the specification: a wordwise simulator must pass every chosen test distribution. The conditional objects in (8) need not be computable from the finite state by an available runtime table; they are used only in the converse.

4. CONSEQUENCES FOR THE ROTATION EXPERIMENT

We now apply the budget to the original wordwise specification, not to a statewise-intertwining subclass.

Proposition 4.1 (Terminal calibration of the past phase). *A realization of (1) with uniform row-TV error at most $\varepsilon < 1/20$ satisfies $q_N \geq 1/10 - 2\varepsilon$ under the fair-command test law with fixed seed $x = (1, 1)$.*

Proof. Let $d = d_1 + id_2$ be the complex conditional mean of the two possible answers from the final state. Its modulus is at most $\sqrt{2}$. For any fixed command word, exactness would give

$$\mathbb{E}[d(S_N) \mid C_1, \dots, C_N] = \frac{1+i}{10} \zeta^{L_N}.$$

A TV error ε for a binary answer changes its mean by at most 2ε . The complex error in this display is therefore at most $2\sqrt{2}\varepsilon$. Multiply by $Y_N = \zeta^{-L_N}$ and average. The reverse triangle inequality gives

$$|\mathbb{E}[Y_N d(S_N)]| \geq \sqrt{2}(1/10 - 2\varepsilon).$$

Apply Lemma 3.1. Only the final-length answer identities were used. There is no assumption of a correct answer at an earlier cut. $\square$

Proof of the lower assertions of Theorem 1.1. Proposition 4.1 and Theorem 3.5 give (3). If every cut has at most K states, then $N\beta_{2K}(\alpha) \leq 18K^3/\kappa^2$. For $b_\alpha > 0$,

$$|\sin(\pi l \alpha)| \geq 2\|l\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{2b_\alpha}{l} \geq \frac{b_\alpha}{K} \quad (1 \leq l \leq 2K).$$

This proves (4).

For the golden angle set $\alpha' = -(1 + \sqrt{5})/2$. If p is an integer nearest $l\alpha$, then

$$(l\alpha - p)(l\alpha' - p) = p^2 + lp - l^2 \in \mathbb{Z} \setminus \{0\}, \quad |l\alpha' - p| \leq \sqrt{5}l + 1/2 < 3l.$$

Hence $l\|\alpha\| > 1/3$ and $b_\alpha \geq 1/3$. With $\kappa = 1/10$ this gives $(N/16200)^{1/5}$. The exact rank-three bound is Proposition 2.2.

For $\zeta = (3 + 4i)/5$, no positive power equals one: otherwise $\zeta + \zeta^{-1} = 6/5$ would be a rational algebraic integer. Thus $(3 + 4i)^l - 5^l$ is a nonzero Gaussian integer and has modulus at least one. It follows that

$$\sin^2(\pi l \alpha) = \frac{|\zeta^l - 1|^2}{4} \geq \frac{25^{-l}}{4}, \quad \beta_{2K}(\alpha) \geq \frac{25^{-2K}}{4}.$$

The profile bound gives $N \leq 7200K^3 25^{2K}$ at zero error. Since $K^3 \leq 8^K$ for integer $K \geq 1$, it also gives $N \leq 7200 5000^K$. This proves (6). $\square$

The occupation conclusion contains more information than a peak bound. For example, at the golden angle and zero error there are at most $16200k^5$ command epochs with at most k available labels, irrespective of the widths at the other epochs. The analogous rational-angle count is at most $7200k^3 25^{2k}$. These are proof bounds, not exact finite complexity tables.

4.1. An exact upper construction retained from the preceding argument. For completeness we give the construction needed for the upper side of (5); its role here is inherited [17]. A convergent p/q of α satisfies $|\alpha - p/q| < q^{-2}$. Choose one with $q \geq \max\{5, (40N)^{1/3}\}$ and put

$$a = \pi/q, \quad \delta = 2\pi(\alpha - p/q), \quad \lambda = \frac{\cos(a - |\delta|)}{\cos a}, \quad \omega = \frac{\sin |\delta|}{\lambda \sin(2a)}.$$

Here $|\delta| < 2\pi/q^2 < a$, so $\lambda \geq 1$ and $0 \leq \omega \leq 1$. Write $u_s = (\cos(2\pi s/q), \sin(2\pi s/q))$ for the regular polygon vertices, with subscripts modulo q .

Proposition 4.2 (Resonant stochastic polygon). *There is a q -state exact realization of (1), with command matrices independent of the epoch and a horizon-dependent final decoder. For bounded partial quotients it gives $W_N = O_\alpha(N^{1/3})$. For the golden angle the upper bound in (5) holds.*

Proof. At cut t use the vector $r_t u_s$ for label s , where $r_t = \lambda^t/4$. The idle row is

$$T_0(s, \cdot) = \lambda^{-1} \delta_s + (1 - \lambda^{-1}) \text{Unif}[q].$$

The active row is

$$T_1(s, \cdot) = (1 - \omega) \delta_{s+p} + \omega \delta_{s+p+\text{sgn } \delta}.$$

If $\delta = 0$ use $\omega = 0$ and omit the second term. The line joining u_{s+p} to its indicated neighbor intersects the ray at angle $2\pi s/q + 2\pi\alpha$ at radius $1/\lambda$. Resolving its two coordinates, or using the sine rule, gives exactly the displayed ω . Consequently the expected next vector is $r_t u_s$ after an idle command and $R_\alpha r_t u_s$ after an active command.

Since $\tan(\pi/q) \leq 2\pi/q$ for $q \geq 5$,

$$\log \lambda = \int_{a-|\delta|}^a \tan v \, dv \leq |\delta| \tan a < 4\pi^2/q^3 < 40/q^3.$$

Thus $r_N \leq e/4 < 1$. The polygon at $t = 0$ has inradius $\cos(\pi/q)/4 > \sqrt{2}/10$ and therefore contains all four seed vectors $x/10$. Initialize by a convex representation of the seed in that polygon. The final decoder uses mean $r_N e_j^T u_s$, which lies in $[-1, 1]$. Induction on the commands proves exactness for every word.

Consecutive convergent denominators have bounded ratios at bounded-type angles, yielding the claimed order. At the golden angle the denominators are Fibonacci numbers, and the first one not smaller than a threshold $L \geq 5$ is strictly less than $2L$. This gives the stated numerical bound. $\square$

For the rational rotation an elementary exact rational alternative retains a triangle label and the number of active commands. Initialize in the triangle from Proposition 2.2, increment the count only on the active command, and decode the rotated triangle vertex. It uses at most $3(N+1)$ labels and rational rows. The preceding resonant construction uses exact real rows; a finite fair-bit implementation of those rows is not being asserted.

5. DEFICIENCY OF A STOCHASTIC LIFT DIAGRAM

We next turn the quantitative converse into an obstruction to approximate stochastic lifts of the main experiment. The local language is classical comparison of experiments [3, 12]; the new use is a uniform lower bound on a full diagram, derived from the Fourier budget.

5.1. Finite garbling and its normalization. Let $E \in \mathbb{R}_+^{H \times K}$ and $F \in \mathbb{R}_+^{H \times J}$ have stochastic rows, indexed by a common finite parameter set. With the convention $\text{TV}(p, q) = \frac{1}{2} \sum_j |p_j - q_j|$, define

$$(19) \quad \delta(E, F) = \min_{T \mathbf{1} = \mathbf{1}, T \geq 0} \max_{h \in H} \text{TV}((ET)_h, F_h).$$

The direction is fixed: the signal of E is to simulate the signal of F . This is a one-sided finite deficiency. It is not a distance unless both directions and appropriate equivalence classes are considered.

Proposition 5.1 (Finite comparison formula). *The minimum in (19) is attained, and*

$$(20) \quad \delta(E, F) = \max_{\sum_h \text{osc}(Z_h) \leq 1} \left\{ \langle Z, F \rangle - \sum_{i=1}^K \max_{1 \leq j \leq J} (E^T Z)_{ij} \right\}.$$

Here $\text{osc}(Z_h) = \max_j Z_{hj} - \min_j Z_{hj}$; adding a constant to a row of Z changes neither objective nor constraint. In particular, $\delta(E, F) = 0$ if and only if $F = ET$ for a stochastic T , equivalently if and only if the associated finite decision inequalities hold. For three finite experiments,

$$(21) \quad \delta(E, G) \leq \delta(E, F) + \delta(F, G).$$

Proof. The stochastic matrices form a compact polytope. If each row of A sums to zero, then

$$\max_h \frac{1}{2} \|A_h\|_1 = \max_{\sum_h \text{osc}(Z_h) \leq 1} \langle Z, A \rangle.$$

Indeed each row contributes at most $\text{osc}(Z_h) \|A_h\|_1 / 2$; equality follows by using an indicator of the positive coordinates on a maximizing row. Subtract the minimum of each row of Z . The permitted Z then form the compact convex set specified by auxiliary variables $u_h \geq 0$, $0 \leq Z_{hj} \leq u_h$, $\sum_h u_h \leq 1$. Apply finite convex minimax to $\langle Z, F - ET \rangle$. Each row of T maximizes a linear form independently, giving $\max_T \langle Z, ET \rangle = \sum_i \max_j (E^T Z)_{ij}$. This proves (20), including the normalization of the dual budget. At zero error it is exactly the finite Blackwell garbling criterion, not a new comparison theorem.

For the triangle bound choose minimizing post-processings T and U . For every row, contraction of TV under U and the triangle inequality give

$$\text{TV}((ETU)_h, G_h) \leq \text{TV}((ET)_h, F_h) + \text{TV}((FU)_h, G_h).$$

Taking maxima proves (21). □

5.2. One common transition for every prefix. Let H_t be all seed/command prefixes of length t in (1). A finite proposed diagram $\mathcal{E}$ assigns a row-stochastic matrix $E_t \in \mathbb{R}_+^{H_t \times K_t}$ at every cut. These are proposed ideal hidden-state distributions. For $c \in \{0, 1\}$ let $E_{t+1}^{[c]}$ have rows $E_{t+1}(hc, \cdot)$ indexed by H_t . Let Q_j have rows equal to the prescribed binary answer laws for query j and complete prefixes in H_N . Define

$$(22) \quad \begin{aligned} d_t(\mathcal{E}) &= \max_{c \in \{0,1\}} \delta(E_t, E_{t+1}^{[c]}), \\ d_{\text{out}}(\mathcal{E}) &= \max_{j \in \{1,2\}} \delta(E_N, Q_j), \\ \mathcal{D}(\mathcal{E}) &= \sum_{t < N} d_t(\mathcal{E}) + d_{\text{out}}(\mathcal{E}). \end{aligned}$$

A stochastic post-processing in each term is common to all prefixes; it cannot depend on an unrecorded history. Each fixed-diagram term is a finite linear program. The definition does not optimize over the unknown rows E_t and does not assert that local errors cannot cancel at the output.

Theorem 5.2 (Executable composition and main-family defect). *For every diagram $\mathcal{E}$ there is a clocked stochastic machine with its same profile and uniform output error at most $\min\{1, \mathcal{D}(\mathcal{E})\}$. If every $K_t \leq K$, then*

$$(23) \quad \mathcal{D}(\mathcal{E}) \geq \frac{1}{2} \left[\frac{1}{10} - \sqrt{\frac{18K^3}{N\beta_{2K}(\alpha)}} \right]_+.$$

More generally, for $J_k = \#\{t < N : K_t \leq k\} > 0$, the same inequality holds with K^3/N replaced by k^3/J_k .

Proof. Initialize using E_0 and use minimizing stochastic matrices in each term of (22). Denote the actual prefix-state law by $\hat{E}_t(h)$. Induction and TV contraction give

$$\max_{h \in H_t} \text{TV}(\hat{E}_t(h), E_t(h)) \leq \sum_{u < t} d_u(\mathcal{E}).$$

The selected terminal decoder adds at most d_{out} . This proves the constructive statement, including a common executable row at every step.

Write $D = \mathcal{D}(\mathcal{E})$. If $D \geq 1/20$, the lower bound is immediate. Otherwise the constructed machine has error at most D , so Theorem 1.1 applies with $\kappa = 1/10 - 2D$. It implies $N\beta_{2K}(\alpha) \leq 18K^3/(1/10 - 2D)^2$. Rearranging gives (23). The occupation form is identical, using (3) rather than its peak consequence. $\square$

For badly approximable α , the right side is at least

$$(24) \quad \frac{1}{2} \left[\frac{1}{10} - \sqrt{\frac{18K^5}{b_\alpha^2 N}} \right]_+.$$

Thus no family of diagrams with $K = o(N^{1/5})$ has total certified simulation defect tending to zero. In particular a static polygon lift using $O(\log N)$ coordinates cannot be turned into a vanishing-defect stochastic diagram for this task at the same width. This conclusion requires a specified initialization and bounded terminal decoder as in (22); it is not a lower bound on static extension complexity. It permits every hidden choice of section and projection, not merely a selected regular-polygon extended formulation.

The additive quantity $\mathcal{D}$ is a useful operational upper bound on terminal error, not a claimed exact formula for minimum terminal error. Approximation of intermediate lifts, terminal decoding and the complete wordwise task are thereby kept distinct. Theorem 5.2 provides the requested

connection between finite deficiency and the actual rotation family without relying on a separate toy example.

6. EQUIVARIANCE AND TWO AUTONOMOUS SPECIFICATIONS

An equivariant lift carries a group action on one extension, intertwining with the action on the base. In the orthant case this is represented by coordinate permutations after the usual normalization. Definitions 2.7–2.8 and Theorem 2.9 of the preprint of Gouveia–Parrilo–Thomas [7] relate symmetric cone lifts to symmetric slack factorizations. Symmetry can change extension size substantially [9]. In the regular-polygon case, equivariant LP lifts of prime-power order require linear size, whereas nonsymmetric LP lifts can have logarithmic size; the equivariant PSD theory has different bounds [4], including logarithmic-size lifts of regular 2^n -gons.

Our stochastic diagrams do not impose a group representation, inverses, one common section, or a permutation action on hidden coordinates. Their maps may be noninvertible and epoch dependent. Hence the cited equivariant lower bounds do not apply directly, and static small lifts do not provide the stochastic maps needed here. The following elementary comparison makes one exact point of contact. It is a consequence of classical stochastic matrix spectral theory, not a new symmetric-lift bound.

Proposition 6.1 (An exact cyclic stochastic lift). *Let $\xi = e^{2\pi i/m}$, $m \geq 2$, and suppose stochastic rows $E_0, \dots, E_{m-1} \in \Delta_K$, a stochastic matrix T , and a complex column d satisfy*

$$E_j T = E_{j+1 \bmod m}, \quad E_j d = r \xi^j \quad (r \neq 0).$$

Then $K \geq m$. The bound is attained by the m -cycle permutation matrix.

Proof. The complex row $w = m^{-1} \sum_{j=0}^{m-1} \xi^{-j} E_j$ has $wd = r$, so it is nonzero, and direct reindexing gives $wT = \xi w$. Thus ξ is an eigenvalue of T . Every unit-modulus eigenvalue of a finite stochastic matrix belongs to a recurrent irreducible class and is a root of unity of order dividing the period of that class. That period is at most the number of its states. For completeness, transient blocks have spectral radius less than one; in an irreducible recurrent block, equality in the triangle inequality for an eigenvector at its maximal modulus forces its phases to advance by the eigenvalue along every positive edge. Along every directed cycle the accumulated phase is one, so its order divides the greatest common divisor of the cycle lengths. The primitive m th root therefore requires at least m states. The permutation construction realizes equality. $\square$

Proposition 6.1 assumes exact cyclic closure, including its last-to-first transition. The finite irrational experiment has no such closure. Its nonstationary lower bound comes instead from the finite Fourier occupation theorem, which needs no eigenvalue shared by successive matrices. This is why replacing the latter proof by a peripheral-spectrum argument would be invalid.

6.1. Fixed length is not an anytime query. Define A_N^{fix} by requiring one common state set, two common command matrices T_0, T_1 , and one final decoder, but correctness only after exactly N commands. The entire machine may depend on N . Define A_N^{any} with the same data required to answer after every command length $0, 1, \dots, N$. A query is an atomic external input in both models. Counts refer to internal labels before emission; charging two additional absorbing answer labels increases the given upper bounds by at most two. No internal test of input length is required by either interface.

Corollary 6.2 (Bounds with the query-time quantifier fixed). *For bounded-type α ,*

$$\Omega_\alpha(N^{1/5}) \leq W_N(\alpha) \leq A_N^{\text{fix}}(\alpha) \leq O_\alpha(N^{1/3}).$$

For irrational α , the stronger anytime specification satisfies

$$N + 1 \leq A_N^{\text{any}}(\alpha) \leq 3(N + 1).$$

The anytime assertion is the inherited spectral comparison [17], not a linear lower bound for A_N^{fix} .

Proof. The unrestricted lower bound applies to stationary matrices in particular. Proposition 4.2 uses the same command matrices at all epochs, so it gives the fixed-length autonomous upper bound without phase tags. Its final radius and decoder depend on N , as permitted.

For the anytime lower bound fix seed $(1, 1)$, use only active commands, and combine the two decoders into d . If a machine with $K \leq N$ states were correct, then $ET_1^j d = c\zeta^j$ for $j = 0, \dots, K$, where $c = (1 + i)/10 \neq 0$. Cayley–Hamilton, applied to these $K + 1$ identities, forces the characteristic polynomial of T_1 to vanish at ζ . The unit-modulus eigenvalue property proved above contradicts irrationality. For the upper bound keep the triangle label of the seed and the number of active commands. The latter never exceeds N on a permitted word. Idle commands leave it unchanged; active commands increment it; the decoder rotates the stored triangle vertex by that count. Arbitrary normalized rows at count N complete the matrix on unused transitions. This machine answers correctly at every allowed length. $\square$

7. RELATION TO PRIOR MECHANISMS AND REMAINING QUESTIONS

Positive realization has long distinguished observable linear dimension from positive-state dimension. Benvenuti–Farina [1, 2] give fixed-degree examples requiring large positive order and explain invariant-cone constructions. Heller [8] and the realization survey of Vidyasagar [14] provide the stochastic-process background. Accessible spaces, observable quotients and positive intertwining are also explicit in Monràs–Winter [11]. None of these mechanisms is renamed here as a new normal form. The specific assertion proved in Theorem 3.5 is a finite-horizon lower bound for unrelated stochastic updates, using a final correlation rather than one repeated transition or a statewise predictive coordinate.

For comparison, the recent work of Lumbreras–Ma–Thompson–Gu [10] studies irreducible finite-classical prediction and decision errors in a stationary world-model setting. Its Section IV.B uses repeated application of the same update along a Wait–Tick trajectory and the peripheral behavior of that update. Here the matrices may change at every epoch, and even the complete machine may change with the horizon. The phase/Jensen/Fejér proof gives a quantitative finite budget under those quantifiers. This distinction does not claim that all neighboring approximate-realization results have been exhausted.

Static polyhedral lift theory identifies extension size through slack factorizations [7]. Regular polygons admit small nonsymmetric extensions [5, 13]; respecting a group action can change this substantially [9, 4]. Section 6 states the exact model distinction. Our converse does not assume equivariance or extract an exposed face for each state. A hidden register instead produces at most K atoms in a measure of conditional *past* phases, even when its reachable simplex section has exponentially many projected vertices. The common amplitude-loss estimate is what makes those atoms useful through time.

The one-step equation $ET = F$, its decision inequalities, and one-sided TV deficiency belong to classical comparison of experiments [3, 12]; see also the modern formulation of the Blackwell–Sherman–Stein theorem in [6]. Proposition 5.1 includes a proof to fix the direction and factor of two, not to claim a new one-step duality. The additive composition of stochastic errors is likewise standard. The main-family lower bound (23), which applies to every proposed small-width diagram, is a consequence of the new all-hidden-state budget.

Statement	Role in the present article
Conditional angular-loss and Fourier occupation theorem	New quantitative converse for arbitrary clocked hidden registers.
Bounded-type polynomial and rational-angle logarithmic lower bounds	Consequences for the original rotation family, including fixed row error.
Resonant cube-root exact realization	Inherited upper construction, reproduced with proof.
Finite Blackwell/deficiency dual and TV composition	Classical ingredients, with conventions and proof supplied.
Main-family diagram-defect inequality	New consequence combining the converse with executable composition.
Cyclic spectral and anytime comparisons	Classical spectral mechanism and inherited stronger-task comparison.

7.1. Conclusions. For the fixed-alphabet uniformly positive rotation experiment, a hidden register of bounded-type angle requires polynomial, not merely logarithmic, width. The proof permits arbitrary hidden states and arbitrary clocked updates. The same occupation theorem survives a fixed uniform row error below $1/20$ and obstructs every small-width stochastic lift diagram with vanishing total deficiency.

The exact hidden exponent is not determined: the present lower and upper orders are $N^{1/5}$ and $N^{1/3}$. The vector-state order remains $N^{1/3}$ under the inherited bounded-type hypotheses. Sharpening the homogeneous angular estimate, the finite Fourier witness, or the exact hidden construction are distinct possible routes to closing this gap. No optimal global lift-search algorithm is asserted by the finite comparison linear programs.

The model counts persistent classical labels, not construction time, read-only program size, or fair-bit sampling workspace. Fixed-length and anytime autonomous interfaces remain distinct. The historical geometric, analytic and operator-theoretic manuscripts in the broader program keep their independent proof obligations; no conclusion in this article asserts their closure. Earlier results are preserved separately without being used as evidence of priority or editorial significance.

REFERENCES

- [1] L. Benvenuti and L. Farina, An example of how positivity may force realizations of “large” dimension, *Systems Control Lett.* **36** (1999), 261–266, doi:10.1016/S0167-6911(98)00098-X.
- [2] L. Benvenuti and L. Farina, A tutorial on the positive realization problem, *IEEE Trans. Automat. Control* **49** (2004), 651–664.
- [3] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272, doi:10.1214/aoms/1177729032.
- [4] H. Fawzi, J. Saunderson and P. A. Parrilo, Equivariant semidefinite lifts of regular polygons, *Math. Oper. Res.* **42** (2017), 472–494, doi:10.1287/moor.2016.0813; arXiv:1409.4379.
- [5] S. Fiorini, T. Rothvoß and H. R. Tiwary, Extended formulations for polygons, *Discrete Comput. Geom.* **48** (2012), 658–668.
- [6] T. Fritz, T. Gonda, P. Perrone and E. F. Rischel, Representable Markov categories and comparison of statistical experiments in categorical probability, *Theoret. Comput. Sci.* **961** (2023), 113896; arXiv:2010.07416v3.
- [7] J. Gouveia, P. A. Parrilo and R. R. Thomas, Lifts of convex sets and cone factorizations, *Math. Oper. Res.* **38** (2013), 248–264, doi:10.1287/moor.1120.0575; arXiv:1111.3164.
- [8] A. Heller, On stochastic processes derived from Markov chains, *Ann. Math. Statist.* **36** (1965), 1286–1291.
- [9] V. Kaibel, K. Pashkovich and D. O. Theis, Symmetry matters for sizes of extended formulations, *SIAM J. Discrete Math.* **26** (2012), 1361–1382, doi:10.1137/110839813; arXiv:0911.3712.
- [10] J. Lumbresas, H. Ma, J. Thompson and M. Gu, An irreducible quantum advantage in aligning world models with reality, arXiv:2608.19779 (2026), Section IV.B.
- [11] A. Monràs and A. Winter, Quantum learning of classical stochastic processes: The completely-positive realization problem, *J. Math. Phys.* **57** (2016), 015219; arXiv:1412.3634.

- [12] E. Torgersen, *Comparison of Statistical Experiments*, Encyclopedia of Mathematics and its Applications, vol. 36, Cambridge University Press, 1991.
- [13] A. Vandaele, N. Gillis and F. Glineur, On the linear extension complexity of regular n -gons, *Linear Algebra Appl.* **521** (2017), 217–239.
- [14] M. Vidyasagar, The complete realization problem for hidden Markov models: a survey and some new results, *Math. Control Signals Systems* **23** (2011), 1–65.
- [15] Q. Qi, General Theta Foundations I: Compatibility of positive memory and online simplex synthesis, revision 31, September 25, 2026, repository manuscript, publication [90dc2c8f0c75e2cfbb95cc61b247ce4d5e216eb9](#).
- [16] Q. Qi, General Theta Foundations I: Causal width beyond cutwise positive rank, revision 32, September 25, 2026, repository manuscript, publication [d38fca741135d147dfdbdbd99cfd9a1a716a88a5](#).
- [17] Q. Qi, General Theta Foundations I: Compatible polyhedral lifts and quantitative causal width, revision 33, September 25, 2026, repository manuscript, publication [aab96108125317342e51e6a946efc4d778429338](#).

PRESERVED MATHEMATICAL DEVELOPMENT

General Theta Foundations I - revision 33

The complete current article precedes this divider.

The following predecessor cumulative volume is reproduced without alteration.

Frozen predecessor publication:

aab96108125317342e51e6a946efc4d778429338

This archive is not part of the compact referee package. Independent historical analytic obligations keep their original hypotheses and proof status.

GENERAL THETA FOUNDATIONS I: COMPATIBLE POLYHEDRAL LIFTS AND QUANTITATIVE CAUSAL WIDTH

QIAN QI

ABSTRACT. An exact finite-horizon stochastic transducer determines a sequence of simplex-section extensions joined by stochastic maps. Static extension size does not certify those maps. We formulate the compatibility condition, give its linear dual for fixed extensions, and exhibit an affine map whose best stochastic extension has uniform error exactly $1/4$. For a fixed-alphabet rotation experiment of separate positive rank three, a mean-support budget improves the all-hidden-state lower bound to $\sum_{t \leq N} 2^{-3Kt} \leq 320$ at the golden angle. A resonant polygon construction improves the exact upper width from order $N^{1/2}$ to order $N^{1/3}$. More generally, for every irrational angle with bounded continued-fraction coefficients, the minimum width among vector-state realizations is of order $N^{1/3}$; the lower bound covers all time-varying polygon chains, not only regular polygons. For general hidden realizations the resulting bounds are logarithmic and cube-root, and need not coincide. An elementary denominator argument gives an explicit doubly logarithmic lower bound for the rational rotation $(3 + 4i)/5$. Finally, requiring a single clock-free decoder to answer at any stopping length up to N gives linear autonomous state complexity. These results distinguish static lifts, compatible clocked lifts, vector-state realizations, and autonomous devices without identifying their different optimization problems.

1. INTRODUCTION

The state count of a positive realization is not determined by its linear rank. This is a classical fact. Benvenuti and Farina constructed transfer functions of fixed McMillan degree three with arbitrarily large minimum positive order [1]; subsequent work obtained matching realizations for their family [5, Example 2]. There is a second, different classical fact: a polygon with many vertices may have an extension with only logarithmically many facets [14, 6, 12]. The simultaneous finite-horizon problem lies between these observations. Its state register supplies an extension at each epoch, but also supplies stochastic maps joining the extensions for every allowed command. Neither independent positive factorizations nor independent small extensions impose this compatibility.

We study the complete binary-output experiment

$$(1) \quad C_{N,\alpha}(b \mid x, c_1, \dots, c_N, j) = \frac{1 + b e_j^T R_\alpha^{c_1 + \dots + c_N} x / 10}{2},$$

where $x \in \{-1, 1\}^2$, $c_t \in \{0, 1\}$, $j \in \{1, 2\}$ and $b \in \{-1, 1\}$. Here R_α is rotation through $2\pi\alpha$. All alphabets are fixed. All probabilities lie in the fixed interval

$$\left[\frac{1 - \sqrt{2}/10}{2}, \frac{1 + \sqrt{2}/10}{2} \right].$$

Date: September 25, 2026. Revision 33.

2020 Mathematics Subject Classification. Primary 93B15, 52B05; Secondary 68Q45, 11J70, 60J10.

Key words and phrases. Positive realization, stochastic transducers, compatible extensions, mean width, irrational rotation, finite-horizon realization.

Equality is required for every word, not just in expectation over an input prior. The separate positive minimum at every internal cut is three (Proposition 2.3). The question is the peak number of labels needed by one compatible machine.

1.1. The quantitative distinction. Write $W_N(\alpha)$ for unrestricted clocked positive-realization width. Write $V_N(\alpha)$ for its vector-state restriction, in which every state has a predictive vector and each stochastic command row intertwines those vectors, not merely their reachable affine combinations. These are defined in Section 2. Let $A_N(\alpha)$ be the number of states in a clock-free machine with one fixed decoder that answers a terminating query at any command length from zero through N . The last task has a stronger query-time specification, which is part of its definition.

Theorem 1.1 (Quantitative width distinctions). *Let α be irrational, and suppose all its continued-fraction partial quotients are at most an integer $A \geq 1$. Let q_N be the first convergent denominator at least $\max\{5, (40N)^{1/3}\}$. For $N \geq 1$,*

$$(2) \quad \max\left\{3, \frac{1}{3} \log_2 \frac{N}{160(A+1)}\right\} \leq W_N(\alpha) \leq V_N(\alpha) \leq q_N,$$

$$(3) \quad \max\left\{3, \left(\frac{N}{160(A+1)}\right)^{1/3}\right\} \leq V_N(\alpha) < (A+1) \max\{5, (40N)^{1/3}\}.$$

Thus $V_N(\alpha) = \Theta_\alpha(N^{1/3})$. In addition, for every infinite-order planar rotation, without the bounded-type assumption,

$$(4) \quad N+1 \leq A_N(\alpha) \leq 3(N+1)+2.$$

At the golden angle $\alpha = (\sqrt{5}-1)/2$, every exact hidden-state profile satisfies $\sum_{t < N} 2^{-3K_t} \leq 320$; every vector-state profile satisfies $\sum_{t < N} K_t^{-3} \leq 320$.

The hidden-state lower bound is not replaced by the stronger vector-state bound. A section of a K -coordinate simplex can project to many more than K vertices. This distinction is indispensable: regular polygons have small static extensions, and an unproved restriction to vector states would discard precisely the possible hidden savings. Theorem 3.5 specifies the additional stochastic compatibility required of a sequence of extensions. Theorem 3.2 checks it by a finite linear program when the extensions and the proposed lifted affine maps have been supplied. It does not optimize over all extensions efficiently.

For the rational rotation $R = (1/5) \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$, Theorem 6.2 gives the explicit necessary condition

$$N \leq 140 2^{2W_N} 5^{82^{W_N}}.$$

It requires neither quantifier elimination nor an unproved bounded-type property of $\arg(3+4i)/(2\pi)$. Its growth rate is deliberately separated from the sharper bounded-type statement.

1.2. Mechanism and scope. For an observable polygon P , use the mean support $m(P) = (2\pi)^{-1} \int h_P$, rather than its area. The one-step gain for $\text{conv}(P \cup R_\alpha P)$ accumulates linearly along a finite orbit. Comparing that orbit hull with the optimal perimeter of an inscribed polygon gives a gain of order M^{-3} for at most M vertices and bounded-type angles. No squaring of a Hausdorff displacement is needed. The matching vector-state upper bound aligns regular polygons with convergents of the angle; their required radial expansion per step is then of order M^{-3} rather than M^{-2} .

All exact clocked counts use atomic stochastic rows, an external epoch, and known read-only tables. They do not count arithmetic or table construction. The new autonomous comparison removes the external epoch and fixes the decoder, but still uses the same atomic-row convention.

The old rational fair-bit compiler remains a separately priced implementation theorem [15]. No table-free bit-space claim is made.

Item	Mathematical role
Reachable simplex sections	Inherited finite-horizon lift; classical accessible-space and extension principles.
Stochastic extension dual	Finite convex separation, used to distinguish static affine lifts from implementable maps.
Mean-support budget and resonance	New quantitative argument and exact clocked construction; matching order within the vector-state class.
Rational-angle rate	Explicit arithmetic consequence of the same budget.
Anytime autonomous comparison	Finite-horizon consequence of Cayley–Hamilton and the classical peripheral-spectrum constraint.

The complete earlier proofs, including the volume theorem and tagged profile composition, are preserved separately. They are not prerequisites for auditing the new argument, and their archive size is not part of its mathematical content.

2. MACHINES AND OBSERVABLE SECTIONS

We use row vectors for state distributions. Put $\Delta_k = \{e \in \mathbb{R}^k : e \geq 0, e\mathbf{1} = 1\}$, $S = [-1/10, 1/10]^2$, and $D = [-1, 1]^2$.

Definition 2.1 (Clocked realization). A realization of (1) consists of an initialization $E_x \in \Delta_{K_0}$, row-stochastic matrices $T_{t,c} \in \mathbb{R}_+^{K_t \times K_{t+1}}$ for $0 \leq t < N$, $c \in \{0, 1\}$, and decoder means $d_j \in [-1, 1]^{K_N}$, satisfying

$$(5) \quad E_x T_{0,c_1} \cdots T_{N-1,c_N} d_j = e_j^T R_\alpha^{\sum c_t} x / 10$$

for every input word and query. The decoder emits the binary law with this conditional mean. The available-label profile is $(K_t)_{t=0}^N$; its peak is $\max_t K_t$, and its minimum is $W_N(\alpha)$. A final retained binary output can be implemented in two further output labels at its own cut and does not change this peak, which is at least three.

Only the current register persists. The seed, old commands, random coins, and protocol history are not supplied again. The seed and terminal query are atomic inputs; commands are external, rather than selected by an experimental policy. A row may depend on the external epoch and current command, but not on a future symbol. Every row, including an unreachable one, is normalized. Available-label counting permits padding and makes feasible profiles upward closed. The number of program bits, exact row arithmetic, and sampling cost are not part of this width. For comparison, a separate minimum at cut t imposes no bound at any other cut. It is not the peak of a simultaneously minimal machine.

Definition 2.2 (Vector-state realization). A clocked realization is vector-state if there are vectors $z_{t,s} \in D$ satisfying

$$(6) \quad \sum_s E_x(s) z_{0,s} = x / 10, \quad \sum_{s'} T_{t,c}(s, s') z_{t+1,s'} = R_\alpha^c z_{t,s}, \quad d_j(s) = e_j^T z_{N,s}.$$

The middle equality is required at every available state, not only after averaging over reachable distributions. The least peak in this subclass is $V_N(\alpha)$.

Such a machine is equivalent to a sequence of polygons $P_t = \text{conv}\{z_{t,s}\}$ with

$$(7) \quad S \subset P_0, \quad P_t \subset D, \quad \text{conv}(P_t \cup R_\alpha P_t) \subset P_{t+1}.$$

The forward implication follows from (6); conversely, choose the vertices as states and the corresponding barycentric rows. Thus the minimum vertex count of a compatible chain is exactly the vector-state optimization. No such identification is imposed on W_N .

Proposition 2.3 (Separate minimum). *Every internal cut of (1) has normalized residual rank three and separate positive minimum three, for every N and rotation.*

Proof. The four prefixes with all commands idle have predictive vectors $x/10$ and span an affine plane. The two idle-suffix coordinate queries separate this plane. All residual arrays are affine functions of a two-dimensional predictive vector, so their normalized rank is exactly three. The vectors have Euclidean norm $\sqrt{2}/10$. The triangle with vertices

$$(1/2, 0), \quad (-1/2, 1/2), \quad (-1/2, -1/2)$$

contains the disk of this radius: its three supporting lines have distances $1/2, 1/(2\sqrt{5}), 1/(2\sqrt{5})$ from zero, all greater than $\sqrt{2}/10$. Its vertices have norm at most $1/\sqrt{2}$. Consequently every vertex defines a legal entire future continuation under all rotations and both queries. Barycentric initialization at the selected cut gives a three-generator factorization of all its residuals. To embed it into a complete machine, store the raw prefix until the selected cut, sample that three-valued label, and then retain the label and the remaining finite command count. Other cuts are unrestricted. Rank supplies the matching lower bound. This is the inherited separate minimum calculation of [16], included to fix its quantifier. $\square$

Lemma 2.4 (Reachable-section lift). *Every realization determines affine sections $Q_t = H_t \cap \Delta_{K_t}$ and linear maps π_t with $P_t = \pi_t(Q_t)$ satisfying (7) and*

$$(8) \quad |\text{ext } P_t| \leq 2^{K_t}.$$

For vector-state realizations one may instead use $|\text{ext } P_t| \leq K_t$.

Proof. Let E_h denote the state distribution following an actual prefix, and set $H_t = \text{aff}\{E_h\}$. Define $\pi_t(e)$ by the two terminal means obtained from e after idling all remaining commands. It maps the whole simplex into D . The identities

$$E_h T_{t,c} = E_{hc}, \quad \pi_{t+1}(E_h T_{t,c}) = R_\alpha^c \pi_t(E_h)$$

extend affinely across H_t . Stochasticity then gives $Q_t T_{t,c} \subset Q_{t+1}$ and the required observable inclusions. The idle command implies $P_t \subset P_{t+1}$, hence every P_t contains S . In its affine hull Q_t has at most K_t coordinate inequalities. Distinct vertices have distinct active zero-coordinate sets: otherwise their difference is a two-sided feasible direction at either vertex. There are at most 2^{K_t} such sets, and projection cannot increase the number of vertices. This is the standard face-count bound for extensions [6], not a special new combinatorial principle. For the vector-state class, use the polygon in (7). $\square$

The restriction to H_t is essential. A hidden basis state need not have the observable intertwining identity. The proof therefore retains the section before projection, as in accessible-space and observable-quotient constructions [3, 10]. It is this lift that separates the two bounds in Lemma 2.4.

3. STATIC EXTENSIONS AND STOCHASTIC COMPATIBILITY

A polyhedral extension is a polytope followed by an affine projection. Its size is the number of inequalities, not the number of projected vertices. The distinction is classical [14, 6]. We first record exactly what simplex normalization does, and does not, provide.

Proposition 3.1 (Normalization of bounded extensions). *Every positive-dimensional bounded polytope with f facets is affinely isomorphic to a section $H \cap \Delta_f$. Thus a bounded extension with f facets can be put in simplex-section form without increasing f . An affine map between two such sections need not extend to a stochastic map between their ambient simplices.*

Proof. Work in the affine hull and write an irredundant facet description $\ell_i(u) = b_i - a_i \cdot u \geq 0$, $1 \leq i \leq f$. Boundedness implies that the normals positively span the dual space. Consequently there are strictly positive w_i with $\sum_i w_i a_i = 0$. For example, express each $-a_i$ as a nonnegative combination of the normals and add these f dependences. The positive constant $c = \sum_i w_i b_i$ can be evaluated at a relative interior point, where it is a sum of positive slacks. The affine map

$$u \mapsto (w_i \ell_i(u)/c)_{i=1}^f$$

is injective, has coordinate sum one, and sends the polytope onto the intersection of its affine image with Δ_f . Injectivity follows because the normals span. A point polytope uses Δ_1 . The last assertion is proved quantitatively in Proposition 3.3 below. $\square$

The projection of a normalized extension is defined on its affine section. It need not map the whole ambient simplex into the observation domain. Nor does normalization produce transition kernels. These two additional requirements are not part of static extension complexity.

Theorem 3.2 (Stochastic extension criterion). *Let $Q \subset \Delta_k$ be a polytope, let $q^1, \dots, q^r$ affinely span Q , and let $F : Q \rightarrow \Delta_l$ be affine. Put $E_{hi} = q_i^h$ and $B_{hj} = F(q^h)_j$. A row-stochastic k by l matrix T satisfying $eT = F(e)$ on Q exists if and only if, for every real r by l matrix Z ,*

$$(9) \quad \langle Z, B \rangle \leq \sum_{i=1}^k \max_{1 \leq j \leq l} (E^T Z)_{ij}.$$

For fixed rational E, B , feasibility and a separating certificate are finite linear-programming problems.

Proof. The desired set of image matrices is the compact convex polytope $\mathcal{B} = \{ET : T \geq 0, T\mathbf{1} = \mathbf{1}\}$. Its support function at Z is the right side of (9), since the rows of T are independent probability vectors. Finite-dimensional separation says $B \in \mathcal{B}$ exactly when all the displayed inequalities hold. Equality on the spanning rows extends to Q by affinity. This is the ordinary support-function dual of a stochastic post-processing problem; it is not a new minimax interchange. The additional point is that the same test can be applied to each proposed inter-epoch lift. $\square$

Proposition 3.3 (An exact extension defect). *On*

$$Q = \{e \in \Delta_4 : e_1 + e_2 = e_3 + e_4 = 1/2\}$$

the affine map $F(e) = (2e_1, 2e_2)$ takes values in Δ_2 , but has no ambient stochastic extension. More precisely,

$$(10) \quad \min_{T \geq 0, T\mathbf{1} = \mathbf{1}} \max_{e \in Q} \text{TV}(eT, F(e)) = \frac{1}{4}.$$

Proof. Write $u_i = T_{i1} \in [0, 1]$. At the four vertices $e^{ij} = (e_i + e_j)/2$, $i \in \{1, 2\}$, $j \in \{3, 4\}$, the desired first output probability is one for $i = 1$ and zero for $i = 2$. For fixed j , the realized probabilities differ by $(u_1 - u_2)/2 \leq 1/2$. Their two errors must therefore sum to at least $1/2$, and one is at least $1/4$. Taking $(u_1, u_2, u_3, u_4) = (1, 0, 1/2, 1/2)$ attains error $1/4$ at the vertices. Convexity bounds the error everywhere else by the vertex maximum. This also gives a finite strict witness for (9). $\square$

Definition 3.4 (Compatible stochastic lift). For the mean dynamics in (1), a lift of profile (k_t) consists of nonempty sections $Q_t = H_t \cap \Delta_{k_t}$, affine observable maps $\pi_t : Q_t \rightarrow D$, seed lifts $E_x \in Q_0$ with $\pi_0(E_x) = x/10$, and affine maps $F_{t,c} : Q_t \rightarrow Q_{t+1}$ with

$$\pi_{t+1} F_{t,c} = R_\alpha^c \pi_t.$$

Each $F_{t,c}$ must extend to an ambient row-stochastic matrix. For each final query j , the map $e \mapsto (1 + e_j^T \pi_N(e))/2$ must extend to a $[0, 1]$ -valued affine map on the whole simplex, equivalently to a binary stochastic decoder. The least possible peak coordinate count is denoted $\text{cw}_N(\alpha)$.

Theorem 3.5 (Operational meaning of a compatible lift). *The complete feasible coordinate profiles of compatible stochastic lifts are exactly the feasible hidden-state profiles of (1). In particular, $\text{cw}_N(\alpha) = W_N(\alpha)$. For supplied finite sections and supplied affine lift maps, all stochastic extension requirements, including the final decoder requirements, can be tested by Theorem 3.2.*

Proof. A machine supplies the sections and idle-suffix maps of Lemma 2.4. Restrict its actual transition matrices to these sections; their observable identity holds there. The final decoder already extends to the whole simplex. Conversely choose the stochastic extensions and start at the specified seed lift. Induction keeps the running distribution in Q_t and its observable image equals the desired rotated seed. The final extended decoder then gives exactly (1). The stochastic maps at different cuts are used together, not selected afresh after seeing a hidden parameter. Affinely spanning points suffice for each extension test; convexity ensures legality throughout a section. $\square$

The theorem defines the dynamic object and gives a local feasibility criterion when its geometry is fixed. It is not an efficient algorithm for searching over all lifts. Proposition 3.3 does not prove that every logarithmic extension of a regular polygon is incompatible with rotation; it proves that compatibility cannot be inferred merely from affine positivity on the section. In particular, we neither assume nor exclude a logarithmic hidden-state construction for (1). Static $O(\log m)$ extensions of regular m -gons [6, 12] remain relevant candidates, but need the inter-epoch and readout extension tests.

4. MEAN-SUPPORT GROWTH ALONG AN ORBIT

For a planar convex body P , let $h_P(u) = \max_{z \in P} u \cdot z$ and

$$m(P) = \frac{1}{2\pi} \int_0^{2\pi} h_P(\cos \theta, \sin \theta) d\theta.$$

For polygons this is perimeter divided by 2π , by Cauchy's perimeter formula. Define

$$\mathcal{G}_\alpha(P) = \text{conv}(P \cup R_\alpha P), \quad D_\alpha(P) = m(\mathcal{G}_\alpha(P)) - m(P).$$

The functional is monotone under inclusion and invariant under rotation. For a chain (7),

$$(11) \quad \sum_{t < N} D_\alpha(P_t) \leq m(P_N) - m(P_0) \leq \sqrt{2}.$$

The last bound uses $P_N \subset [-1, 1]^2 \subset \sqrt{2}\mathbb{B}^2$. It is deliberately larger than the exact endpoint allowance.

Lemma 4.1 (Accumulation without squaring). *For any integer $q \geq 2$ and P ,*

$$(12) \quad m\left(\text{conv} \bigcup_{k=0}^{q-1} R_\alpha^k P\right) - m(P) \leq (q-1)D_\alpha(P).$$

Proof. Write $h_k(\theta) = h_P(\theta - 2\pi k\alpha)$. Pointwise, $\max_{k < q} h_k - h_0 \leq \sum_{k=1}^{q-1} (h_k - h_{k-1})_+$. Each summand has the same integral as $(h_1 - h_0)_+$, whose normalized integral is $D_\alpha(P)$. Integrate the inequality. $\square$

Lemma 4.2 (Inscribed-polygon deficit). *Suppose $a\mathbb{B}^2 \subset P$, and P has at most $M \geq 3$ vertices. Put $r = \max_{z \in P} \|z\|_2$. Then*

$$(13) \quad m(P) \leq r \frac{M}{\pi} \sin \frac{\pi}{M}.$$

Proof. Push each vertex radially to the circle of radius r . The resulting convex hull still contains zero and hence contains the old vertices and P . If its successive angular gaps are $\theta_1, \dots, \theta_m$, $m \leq M$, its perimeter is $2r \sum_i \sin(\theta_i/2)$. Concavity of sine on $[0, \pi]$, $\sum_i \theta_i = 2\pi$, and monotonicity of $m \sin(\pi/m)$ give at most $2rM \sin(\pi/M)$. Monotonicity of mean support and Cauchy's formula prove the assertion. The radial hull contains zero because any positive affine dependence of the old vertices giving zero can be reweighted by their radii. $\square$

Lemma 4.3 (A denominator gives an orbit net). *If $(p, q) = 1$ and $|\alpha - p/q| < q^{-2}$, the first q points of the rotation orbit of any unit vector form an angular $3\pi/q$ -net.*

Proof. The multiples of p/q give the uniform grid of mesh $1/q$ turns. The corresponding first q multiples of α differ by less than $1/q$ turns. Every point is within $1/(2q)$ turns of the grid, proving the bound after multiplying by 2π . $\square$

Theorem 4.4 (One-step quantitative mean-support gain). *Suppose $a\mathbb{B}^2 \subset P$, $|\text{ext } P| \leq M$, $M \geq 3$, and a reduced fraction p/q satisfies $q \geq 8M$ and $|\alpha - p/q| < q^{-2}$. Then*

$$(14) \quad D_\alpha(P) \geq \frac{3a}{4M^2q}.$$

Proof. Take a vertex of maximum radius r . Its first q rotations and Lemma 4.3 imply

$$m\left(\text{conv} \bigcup_{k < q} R_\alpha^k P\right) \geq r \cos(3\pi/q).$$

Subtract (13). Since $M \geq 3$ and $q \geq 8M$, Taylor's inequalities for sine and cosine give

$$\begin{aligned} \cos(3\pi/q) - \frac{M}{\pi} \sin(\pi/M) &\geq \frac{\pi^2}{M^2} \left(\frac{1}{6} - \frac{\pi^2}{120M^2} - \frac{9M^2}{2q^2} \right) \\ &\geq \frac{9}{M^2} \left(\frac{1}{6} - \frac{10}{1080} - \frac{9}{128} \right) = \frac{301}{384M^2} > \frac{3}{4M^2}. \end{aligned}$$

The inequalities use only $9 < \pi^2 < 10$. Apply Lemma 4.1, use $r \geq a$ and $q - 1 < q$. The gain is linear in the integrated support deficit; an area cap is not introduced. $\square$

For an irrational $\alpha \in (0, 1)$, let p_k/q_k be its continued-fraction convergents, ignoring the initial repeated denominator one, and define

$$(15) \quad Q_\alpha(M) = \min\{q_k : q_k \geq 8M\}.$$

We use the elementary relations

$$(16) \quad (p_k, q_k) = 1, \quad |\alpha - p_k/q_k| < \frac{1}{q_k q_{k+1}} \leq \frac{1}{q_k^2}, \quad q_{k+1} = a_{k+1} q_k + q_{k-1}.$$

They follow by writing α with its remaining continued-fraction complete quotient and using $p_k q_{k-1} - p_{k-1} q_k = (-1)^{k-1}$; see also [9]. In particular, if $a_k \leq A$ for all $k \geq 1$, $Q_\alpha(M) < 8(A+1)M$.

Theorem 4.5 (Profile budget and bounded-type width). *Every exact hidden-state realization of (1) satisfies*

$$(17) \quad \sum_{t < N} \frac{1}{2^{2K_t} Q_\alpha(2^{K_t})} \leq \frac{40\sqrt{2}}{3}.$$

If $a_k \leq A$, then

$$(18) \quad \sum_{t < N} 2^{-3K_t} \leq 160(A + 1).$$

For vector-state realizations the same statements hold with 2^{K_t} replaced by K_t . Consequently $W_N \geq \frac{1}{3} \log_2(N/[160(A + 1)])$ and $V_N \geq (N/[160(A + 1)])^{1/3}$.

Proof. Use Lemma 2.4 and Theorem 4.4 with $a = 1/10$, $M = 2^{K_t}$, $q = Q_\alpha(M)$ in (11). This gives (17). Under the bounded-type assumption each gain is at least $3/[320(A + 1)M^3]$. The endpoint allowance gives

$$\sum_{t < N} M^{-3} \leq \frac{320(A + 1)\sqrt{2}}{3} < 160(A + 1).$$

In the vector-state class, use the other vertex bound in Lemma 2.4. A peak bound supplies the stated consequences. $\square$

For the golden angle all partial quotients equal one. The explicit hidden lower bound exceeds three when $N > 320 \cdot 2^9 = 163840$. The earlier area estimate exceeded three only after $24000000 \cdot 2^{18} = 6291456000000$ commands [16]. The new constant is still not a practically sharp finite-horizon classification. Its significance is the improved exponent in the profile budget and its matching vector-state construction below.

The same mean-support proof works in any dimension for a specified orthogonal command family once one supplies an orbit-covering estimate and an inscribed-polytope mean-support deficit. No higher-dimensional numerical exponent is asserted here. Unlike a volume argument, the potential is not tied to a determinant formula. A strictly contractive rotation instead has a finite invariant polygon; the divergent conclusion cannot simply be retained under contraction.

5. RESONANT POLYGONS AND EXACT SYNTHESIS

A regular polygon need not be enlarged by the worst possible angular misalignment. Choosing its order from a convergent changes the cost from quadratic to cubic in its number of vertices.

Lemma 5.1 (Resonant containment and stochastic rows). *Let $q \geq 5$, $(p, q) = 1$, and $|\alpha - p/q| < q^{-2}$. Put*

$$a_q = \pi/q, \quad \delta = 2\pi(\alpha - p/q), \quad \lambda = \frac{\cos(a_q - |\delta|)}{\cos a_q}, \quad w = \frac{\sin |\delta|}{\lambda \sin(2a_q)}.$$

Then $1 \leq \lambda$, $0 \leq w \leq 1$, and

$$(19) \quad \log \lambda \leq 40/q^3.$$

For $v_j = (\cos(2\pi j/q), \sin(2\pi j/q))$ and $P_t = r_t \operatorname{conv}\{v_j : 0 \leq j < q\}$, $r_{t+1} = \lambda r_t$,

$$P_t \cup R_\alpha P_t \subset P_{t+1}.$$

The two stochastic rows realizing the inclusions are

$$(20) \quad T_0(j, \cdot) = \lambda^{-1} \delta_j + (1 - \lambda^{-1}) \operatorname{Unif}[q],$$

$$(21) \quad T_1(j, \cdot) = (1 - w) \delta_{j+p} + w \delta_{j+p+\operatorname{sgn} \delta},$$

with indices modulo q ; when $\delta = 0$, take the pure shift.

Proof. Here $|\delta| < 2\pi/q^2 \leq a_q$. The point $r_t v_{j+p}$ rotated by the residual angle δ lies on the edge of P_{t+1} joining the two displayed vertices. The imaginary coordinate gives the coefficient w ; the real coordinate identity is

$$\lambda = \cos |\delta| + \tan(a_q) \sin |\delta|.$$

This proves both barycentric equality and $0 \leq w \leq 1$ by edge containment. The idle row uses the zero barycenter of a regular polygon. Finally,

$$\log \lambda = \int_{a_q - |\delta|}^{a_q} \tan u \, du \leq |\delta| \tan a_q \leq (2\pi/q^2)(2\pi/q) < 40/q^3.$$

We used $a_q \leq \pi/5 < \pi/4$, where $\tan u \leq 2u$. $\square$

Theorem 5.2 (Cube-root synthesis). *Let q_N be the first convergent denominator of α at least $\max\{5, (40N)^{1/3}\}$. There is an exact vector-state realization of (1) with q_N states at every command cut. When the partial quotients are bounded by A ,*

$$q_N < (A + 1) \max\{5, (40N)^{1/3}\}, \quad V_N(\alpha) = \Theta_\alpha(N^{1/3}).$$

The command matrices can be the same at all epochs. The final decoder and the chosen polygon order are horizon-specific.

Proof. Take $q = q_N$, $r_0 = 1/4$ and $r_t = r_0 \lambda^t$. The inradius of P_0 is at least $\cos(\pi/5)/4 > \sqrt{2}/10$, so all seed vectors lie in P_0 . By (19), $r_N \leq e/4 < 3/4$, hence all polygons lie inside D . Initialize by any convex representation of $x/10$ in P_0 , and use (20)–(21). Given final label s and query j , use binary mean $r_N e_j^T v_s$. The row identities imply by induction that the distribution's barycenter in P_t is exactly the rotated seed. This proves every conditional row of (1), not an approximation to the angle. For an explicit initialization, the ray through x meets one edge of P_0 ; represent that boundary point by its two endpoints and mix with the uniform zero barycenter to obtain $x/10$. The continued-fraction recurrence bounds the first crossing denominator by $(A + 1)$ times the threshold. Theorem 4.5 supplies the matching vector-state lower order. $\square$

This proves (2)–(3). In the normalized label coordinates, both command matrices contract the observable barycenter by λ^{-1} at each command. Multiplying the final decoder by r_N compensates for exactly N contractions. Querying that same decoder earlier generally gives the wrong amplitude. Thus the stationary command rows in this construction are not a clock-free anytime realization, and do not contradict Section 7.

The construction uses direct vertex states; it does not exploit the logarithmic extensions of regular polygons. The matched order for V_N is a theorem about all vector-state chains, not evidence that $W_N = V_N$. The gap $\Omega_\alpha(\log N) \leq W_N \leq O_\alpha(N^{1/3})$ remains the quantitative hidden-lift question. The new upper bound improves the previous $O(N^{1/2})$ construction without asserting its own optimality in the unrestricted class.

Corollary 5.3 (A contractive comparison). *Replace the active mean map by ρR_α , $0 < \rho < 1$, and keep the idle map. A single horizon-independent vector-state device realizes all command lengths with any convergent order $q \geq \max\{5, (40/(-\log \rho))^{1/3}\}$. For bounded-type angles this uses $O_\alpha(1 + (-\log \rho)^{-1/3})$ processing labels.*

Proof. Use one fixed polygon of radius $1/4$. The resonant estimate implies $\rho \lambda \leq 1$, so $\rho R_\alpha P \subset P$ and the idle map preserves P . The active row is the boundary row of (21) mixed with uniform mass, with weight $\rho \lambda$ on that row. A fixed decoder reads the same polygon at all times. This is a quantitative instance of the classical invariant polytope construction [3], not a new existence principle. $\square$

6. AN EXPLICIT RATE FOR A RATIONAL ROTATION

Let $\zeta = (3 + 4i)/5 = e^{2\pi i\alpha}$, $0 < \alpha < 1/4$. This is not a root of unity: otherwise $\zeta + \zeta^{-1} = 6/5$ would be a rational algebraic integer and hence an integer. Thus α is irrational. No bounded-type assertion about this angle is needed.

Lemma 6.1 (Elementary denominator control). *For every integer $r \geq 1$,*

$$\|r\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{5^{-r}}{2\pi}.$$

Consequently the denominator function of (15) satisfies $Q_\alpha(M) < 7 \cdot 5^{8M}$ for every integer $M \geq 3$.

Proof. The Gaussian integer $(3 + 4i)^r - 5^r$ is nonzero, so its modulus is at least one. Therefore

$$5^{-r} \leq |\zeta^r - 1| \leq 2\pi \|r\alpha\|_{\mathbb{R}/\mathbb{Z}}.$$

If $q = Q_\alpha(M)$ and r is its preceding convergent denominator, then $r < 8M$ and (16) gives $\|r\alpha\| < 1/q$. Combine the two estimates and use $2\pi < 7$. The elementary bound is exponentially weak in M , but entirely explicit. $\square$

Theorem 6.2 (Rational-angle profile bounds). *For the rotation $R = (1/5) \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$, every exact hidden-state profile obeys*

$$(22) \quad \sum_{t < N} 2^{-2K_t} 5^{-8 \cdot 2^{K_t}} \leq 140.$$

In particular,

$$(23) \quad N \leq 140 \cdot 2^{2W_N} 5^{8 \cdot 2^{W_N}}.$$

For $N > 140$ this implies

$$W_N \geq \max \left\{ 3, \log_2 \left(\frac{1}{9} \log_5 \frac{N}{140} \right) \right\}.$$

In the vector-state class, $\sum_{t < N} K_t^{-2} 5^{-8K_t} \leq 140$ and, for $N > 140$, $V_N \geq \max\{3, \frac{1}{9} \log_5(N/140)\}$. Both classes have an exact rational realization with peak $3(N+1)$.

Proof. Insert Lemma 6.1 into (17). It gives

$$\sum_{t < N} M_t^{-2} 5^{-8M_t} \leq \frac{280\sqrt{2}}{3} < 140, \quad M_t = 2^{K_t}.$$

Each summand is decreasing in M_t , proving (23). For an integer $W \geq 1$, $2W \log_5 2 \leq 2^W$; hence $2 \log_5(2^W) + 8 \cdot 2^W \leq 9 \cdot 2^W$. Taking logarithms gives the stated hidden lower bound. For vector states use $M_t = K_t$ and $2 \log_5 V \leq V$ for integers $V \geq 1$.

For completeness, an exact rational upper machine stores (s, k) , where s is one of the three triangle vertices in Proposition 2.3 and k is the number of active commands. Initialize s by the rational barycentric coordinates of $x/10$. An idle command leaves (s, k) unchanged, and an active command increments k . At epoch t only $0 \leq k \leq t$ is needed. Assign predictive vector $R^k v_s$ to (s, k) and use its coordinate as the final binary mean. All assigned vectors have norm less than one, all transitions are rational, and the row identities hold on every state. The largest processing alphabet is $3(N+1)$; a held binary answer needs only two labels in its separate terminal cut. $\square$

The explicit inequalities replace the former quantifier-elimination-only exclusion for this angle. They do not assert its sharp asymptotic order. Sharper irrationality estimates for α would improve the denominator function in the same profile theorem, but none is silently assumed here. The rational upper machine admits the separately priced fair-bit implementation of [15]; its atomic label optimum and its sampler workspace are different quantities.

7. REMOVING THE CLOCK: ONE DECODER AT EVERY STOPPING LENGTH

We now change the query-time specification explicitly. The commands still come from the same external alphabet, and there is still only one terminal query. It may, however, arrive after any number $0 \leq n \leq N$ of commands, and the answer must have mean $e_j^T R_\alpha^{\sum c_t} x/10$ at that length. The machine does not receive n as a separate input.

Definition 7.1 (Anytime autonomous realization). An anytime autonomous machine has one finite set of available states, fixed command matrices T_0, T_1 , fixed initialization rows, and one fixed decoder for each query. The same rows work for every command length from zero through N . Let $A_N(\alpha)$ be the least total number of states, counting any states used to retain the emitted answer. The device may depend on the prescribed maximum N ; its rows may not depend on the actual stopping length. Tables and exact atomic sampling remain free, so this is not a succinct bit-space model.

Lemma 7.2 (Finite positive peripheral spectrum). *Every eigenvalue of modulus one of a finite row-stochastic matrix is a root of unity.*

Proof. Let $Tv = \lambda v$ and choose i with $|v_i|$ maximal and positive. Equality in the triangle inequality for $\lambda v_i = \sum_j T_{ij} v_j$ forces $v_j = \lambda v_i$ for every positive successor of i . All these successors have the same maximal modulus. Iteration follows a positive directed path, which eventually contains a cycle of length $l \geq 1$. Along that cycle, $v_j = \lambda^l v_j$ with $v_j \neq 0$. Thus $\lambda^l = 1$. This is the familiar peripheral-spectrum constraint for positive realizations; see [2, 3]. $\square$

Theorem 7.3 (Linear autonomous state complexity). *For any infinite-order planar rotation and every $N \geq 1$,*

$$(24) \quad N + 1 \leq A_N(\alpha) \leq 3(N + 1) + 2.$$

The upper machine has rational rows when R_α is rational. Thus $A_N(\alpha) = \Theta(N)$, whereas the fixed-length vector-state width is $\Theta_\alpha(N^{1/3})$ for bounded-type angles.

Proof. Suppose a machine has $K \leq N$ total states. Fix one nonzero seed and use only active commands. Let $d = d_1 + id_2$ be the complex column formed from its two fixed terminal means and let $z \neq 0$ be the initial complex predictive vector. With $T = T_1$ and $\zeta = e^{2\pi i \alpha}$, correctness says

$$ET^j d = z\zeta^j \quad (0 \leq j \leq N).$$

Write the characteristic polynomial of T as $\chi(X) = \sum_{j=0}^K a_j X^j$. Cayley–Hamilton and the first $K + 1$ required lengths give

$$0 = E\chi(T)d = z \sum_{j=0}^K a_j \zeta^j = z\chi(\zeta).$$

Thus ζ is an eigenvalue of T , contradicting Lemma 7.2. Hence $K \geq N + 1$. The argument needs only the finite length interval, not an infinite stationary specification or a minimality assumption.

For the upper bound use the labels (s, k) , $1 \leq s \leq 3$, $0 \leq k \leq N$, from the triangle-counter construction in Theorem 6.2, now all available in one state set. The idle matrix is the identity; the active matrix increments k , with arbitrary normalized rows after $k = N$. The decoder at (s, k) reads $R_\alpha^k v_s$ and has no epoch argument. Its norm is less than one. Initialization is the seed's triangle decomposition. The same matrices and decoder therefore answer at every permitted length. Add two absorbing answer states if the output must persist after its emission. This proves the count and rationality. $\square$

Equation (24) completes the proof of Theorem 1.1. It does not lower-bound an autonomous device required to answer only at one fixed length and allowed to be incorrect at all earlier lengths. That is another problem. In particular, the resonant construction in Section 5 already has stationary command rows, but its decoder compensates for exactly N normalizing contractions. The strengthened anytime requirement is what prevents that compensation from concealing the elapsed length. The finite-horizon comparison is a concrete use of classical spectral constraints, not a claim to discover the stationary rotation obstruction.

8. CLASSICAL COMPARISONS AND THE REMAINING LIFT PROBLEM

8.1. Fixed degree and positive order. The direct precedent for fixed degree three with unbounded positive order is Benvenuti–Farina [1], not merely the familiar stationary irrational-rotation obstruction. In the normalization reproduced by Czaja–Jaming–Matolcsi [5, Example 2, equation (3.1)], their family is

$$H_m(z) = \frac{1}{z-1} - \frac{4(5/2)^{m-2}}{z-2/5} + \frac{3 \cdot 5^{m-2}}{z-1/5}, \quad m \geq 4.$$

Its McMillan degree is three. The 1999 lower bound on positive order is m , and the later construction attains m [1, 5]. Those statements concern one repeated nonnegative matrix and an entire impulse response. They are not lower bounds against independently chosen stochastic matrices at every cut of a finite word experiment.

There is an equally direct precedent for the inadequacy of positive factorization without dynamics. Benvenuti–Farina [2, Section 3, Example 4 and Theorem 2] give a periodic Hankel matrix with a nonnegative factorization of inner size three but no three-dimensional positive realization; their minimality theorem includes a positive intertwining matrix. Thus neither the need for compatibility nor its failure in positive factorization is claimed as new here. For a finite prefix–suffix array, conditioning supplies a positive factorization at a cut, just as for a finite Hankel truncation. The issue in this article is one sequence of such factorizations under every command, allowing unrelated rows and a horizon-dependent machine. The profile budget proves a necessary inequality even in that enlarged class. It is not claimed to sharpen the known stationary lower bound for the family H_m .

Accessible spaces, observable quotients and invariant cones are classical [8, 3, 13, 10]. Lemma 2.4 is the finite-word version needed to avoid asserting an intertwining identity on hidden directions that the observations do not determine. The proof is included to expose that quantifier, not to claim invention of an accessible-space construction.

8.2. Static and dynamic extensions. Yannakakis’ factorization theorem relates extension size to nonnegative rank of a slack matrix [14]; see the explicit formulation in [6, Theorem 1]. The number of faces of an extension supplies the standard logarithmic lower estimate from projected vertices. Fiorini–Rothvoß–Tiwary [6, Theorem 2], building on Ben-Tal–Nemirovski [4], show that a regular m -gon has an $O(\log m)$ extension. Vandaele–Gillis–Glineur [12] refine regular-polygon bounds using slack factorizations.

These theorems minimize a static inequality count. They do not require the lifted command maps or the observable decoder to extend to stochastic maps on the whole ambient simplex. Proposition 3.1 shows that bounded polytope extensions can be normalized without increasing their facet count; Proposition 3.3 shows why this normalization alone does not settle dynamical implementation. The extension test (9) is elementary convex separation. Its role is to make the missing condition exact and checkable for fixed lift data, not to claim a new general linear-programming theorem.

The substantive quantitative conclusion is therefore divided in two. For arbitrary hidden realizations, the new potential yields a logarithmic lower bound, while the resonant construction yields a cube-root upper bound. For vector-state realizations, the same potential has no exponential facet-to-vertex conversion, and the order is exactly $N^{1/3}$. An independent small static lift does not bridge this gap. Conversely, the map-defect example does not rule out small compatible lifts that have a different geometry.

8.3. What is retained, and what is not inferred. The older volume obstruction, rank-three two-cut example, exact tagged profile identity, online query constructions and physical saddle results remain in their original sources and in the supporting archives [15, 16]. In particular the tagged identity still requires exact final tag recovery and a common suffix interface. Nothing in the present paper extends it to noisy tags or overlapping supports. The earlier additive replacement update is a weighted reservoir construction; its sampling mechanism is credited to that literature [11, Appendix A.2], not used as a new claim here.

The finite proofs above are self-contained apart from the named classical background identities whose short arguments are also given. The source record distinguishes the original 1999 reference from the full author survey and later matching-order proof used to inspect its family. The original 1999 article and the original Yannakakis article have not been claimed to receive an exhaustive proof-by-proof priority audit. Their results are expressly credited; the full accessible comparisons [2, 5, 6, 12] supply the concrete theorem statements used here. No exhaustive originality clearance is inferred from this bibliography.

The main unresolved issue is the optimal hidden width between logarithmic and cube-root growth. The vector-state result is not substituted for that question. The rational-angle bound is explicit but much weaker; its sharp denominator-controlled rate is likewise not identified.

The autonomous theorem concerns one fixed decoder at all stopping lengths, not just removal of a clock from the original single-length specification. All counts continue to price persistent labels rather than read-only programs, arithmetic, or physical realization of irrational coins. These are different optimization problems, and the comparisons depend on preserving their definitions.

The program title identifies this manuscript within the preserved research series. It does not assert that finite positive-realization arguments replace independent Fourier, large-deviation, semigroup, or operator-domain proof obligations elsewhere in that series, nor does it assert a journal decision. The theorem chain established here is the one displayed in Theorem 1.1 and its supporting lift and profile statements.

REFERENCES

- [1] L. Benvenuti and L. Farina, An example of how positivity may force realizations of “large” dimension, *Systems Control Lett.* **36** (1999), 261–266, doi:10.1016/S0167-6911(98)00098-X.
- [2] L. Benvenuti and L. Farina, Minimal positive realizations: A survey of recent results and open problems, *Kybernetika* **39** (2003), 217–228.
- [3] L. Benvenuti and L. Farina, A tutorial on the positive realization problem, *IEEE Trans. Automat. Control* **49** (2004), 651–664.
- [4] A. Ben-Tal and A. Nemirovski, On polyhedral approximations of the second-order cone, *Math. Oper. Res.* **26** (2001), 193–205.
- [5] W. Czaja, P. Jaming and M. Matolcsi, An efficient algorithm for positive realizations, *Systems Control Lett.* **57** (2008), 436–441; arXiv:math/0612551v2 (2007), doi:10.1016/j.sysconle.2007.11.001.
- [6] S. Fiorini, T. Rothvoß and H. R. Tiwary, Extended formulations for polygons, *Discrete Comput. Geom.* **48** (2012), 658–668; arXiv:1107.0371v2, doi:10.1007/s00454-012-9421-9.
- [7] N. Gillis and F. Glineur, On the geometric interpretation of the nonnegative rank, *Linear Algebra Appl.* **437** (2012), 2685–2712.
- [8] A. Heller, On stochastic processes derived from Markov chains, *Ann. Math. Statist.* **36** (1965), 1286–1291.
- [9] A. Ya. Khinchin, *Continued Fractions*, Dover, Mineola, NY, 1997, reprint.

- [10] A. Monràs and A. Winter, Quantum learning of classical stochastic processes: The completely-positive realization problem, *J. Math. Phys.* **57** (2016), 015219; arXiv:1412.3634.
- [11] M. Pharr, W. Jakob and G. Humphreys, *Physically Based Rendering: From Theory to Implementation*, 4th ed., MIT Press, 2023, Appendix A.2.
- [12] A. Vandaele, N. Gillis and F. Glineur, On the linear extension complexity of regular n -gons, *Linear Algebra Appl.* **521** (2017), 217–239; arXiv:1505.08031v2, doi:10.1016/j.laa.2016.12.023.
- [13] M. Vidyasagar, The complete realization problem for hidden Markov models: a survey and some new results, *Math. Control Signals Systems* **23** (2011), 1–65.
- [14] M. Yannakakis, Expressing combinatorial optimization problems by linear programs, *J. Comput. System Sci.* **43** (1991), 441–466, doi:10.1016/0022-0000(91)90024-Y.
- [15] Q. Qi, General Theta Foundations I: Compatibility of positive memory and online simplex synthesis, revision 31, September 25, 2026, repository manuscript, publication [90dc2c8f0c75e2cfbb95cc61b247ce4d5e216eb9](#).
- [16] Q. Qi, General Theta Foundations I: Causal width beyond cutwise positive rank, revision 32, September 25, 2026, repository manuscript, publication [d38fca741135d147dfdbdbd99cfd9a1a716a88a5](#).

PRESERVED MATHEMATICAL DEVELOPMENT

General Theta Foundations I - revision 32

The complete current article precedes this divider.

The following predecessor cumulative volume is reproduced without alteration.

Frozen predecessor publication:

d38fca741135d147dfdbdbd99cfd9a1a716a88a5

This archive is not part of the compact referee package. Independent historical analytic obligations keep their original hypotheses and proof status.

GENERAL THETA FOUNDATIONS I: CAUSAL WIDTH BEYOND CUTWISE POSITIVE RANK

QIAN QI

ABSTRACT. We compare the least memory at individual cuts with the least peak memory of one exact finite-horizon stochastic transducer. We construct a uniformly full-support family on fixed alphabets whose separate minimum is three at every internal cut, whereas its minimum simultaneous width tends to infinity. The lower bound allows arbitrary time-dependent stochastic rows and hidden continuations outside the observable affine space. Its mechanism is a volume budget for projections of reachable simplex sections: each command forces an enlargement, while bounded state number bounds the number of projected vertices. For a golden-angle rotation we obtain the explicit bound $W_N \geq \frac{1}{6} \log_2(N/24000000)$, alongside an exact $4\lceil\sqrt{N} + 1\rceil$ -state construction. A rational rotation gives the same unbounded incompatibility with effective algebraic horizon thresholds and rational finite realizations. A separate composition theorem identifies feasible profiles under a verifiable retained tag as Minkowski sums; applied to the two-cut obstruction, it gives the exact frontier $K_1, K_2 \geq 3m$, $K_1 + K_2 \geq 7m$. The finite-horizon volume inequality, rather than a stationary nonrealizability statement, is the central relation between local positive rank and causal memory.

1. INTRODUCTION

A stochastic response array can have small positive factorizations at every cut without admitting one machine that realizes those factorizations in succession. The issue is not only how many continuations enclose the observed rows. Every chosen continuation must also evolve by the same legal stochastic update as the other continuations at that time. We study the resulting difference between *separate* and *simultaneous* state complexity.

Our principal result gives an unbounded difference with fixed input and output alphabets. Fix $a = 1/10$, a planar rotation R , a seed $x \in \{-1, 1\}^2$, a word $c = (c_1, \dots, c_N) \in \{0, 1\}^N$, and a query $j \in \{1, 2\}$. The seed is read first, the commands are read in order, and the query is supplied last. The required binary response is

$$(1) \quad C_{N,R}(b \mid x, c, j) = \frac{1 + b e_j^T R^{c_1 + \dots + c_N} a x}{2}, \quad b \in \{-1, 1\}.$$

The same formula is required for every input word, not merely under a chosen distribution on words. Every entry lies between $(1 - \sqrt{2}/10)/2$ and $(1 + \sqrt{2}/10)/2$.

At any one internal cut, three normalized continuations suffice and are necessary. This remains true when the cut is required to belong to a complete machine and all its other registers may be enlarged. Write $W_N(R)$ for the least peak over machines satisfying all cuts at once.

Theorem 1.1 (Fixed local rank and growing causal width). *Let $\alpha = (\sqrt{5} - 1)/2$ and let R rotate by $2\pi\alpha$. Every internal cut of (1) has separate minimum three and ordinary normalized*

Date: September 25, 2026. Revision 32.

2020 Mathematics Subject Classification. Primary 93B15, 68Q45; Secondary 52A20, 60J10, 62B05.

Key words and phrases. Positive realization, finite stochastic transducers, causal memory, polyhedral lifts, rotation, compatible factorizations.

residual rank three. For $N \geq 1$,

$$(2) \quad \max \left\{ 3, \frac{1}{6} \log_2 \frac{N}{24000000} \right\} \leq W_N(R) \leq \min \left\{ 3(N+1), 4 \lceil \sqrt{N+1} \rceil \right\}.$$

All lower bounds allow arbitrary randomized encoders, arbitrary hidden state continuations, and cut-dependent stochastic transition matrices. The upper bounds realize every conditional probability exactly.

For the rational rotation

$$(3) \quad R_0 = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix},$$

the same separate minima and upper bounds hold and $W_N(R_0) \rightarrow \infty$. For each proposed state bound, an excluded horizon can be computed by real algebraic optimization. The $3(N+1)$ -state realization is rational.

The proof occupies Sections 3–5. The logarithmic lower bound and square-root upper bound are not asserted to be sharp. They show, with explicit constants in one family, that uniformly bounded separate positive ranks do not control simultaneous width even in the permissive clocked model. The rational version makes the qualitative phenomenon independent of transcendental input data.

The principal technical point is the treatment of nonminimal hidden representations. If a machine has K_t states, its hidden state vectors need not individually have an observable two-dimensional continuation. Projecting those individual states into a presumed invariant polygon would therefore restrict the competing class. Instead, let H_t be the affine span of the actually reachable state *distributions*, and intersect it with the probability simplex. Projecting $H_t \cap \Delta_{K_t}$ gives a valid observable polytope with at most 2^{K_t} vertices. Stochastic updates carry these sections forward. A command and an idle command then force a nested sequence of observable polytopes. A strictly positive volume increment, summed over time, provides the lower bound. This is a necessary argument for all hidden realizations, not a restriction to residual-state or affine-minimal machines.

A second result concerns exact finite profile composition. Theorem 6.1 proves that a verifiable retained tag makes feasible profiles add as sets. In particular, the earlier two-cut obstruction with frontier $(3, 4)$, $(4, 3)$ yields

$$K_1 \geq 3m, \quad K_2 \geq 3m, \quad K_1 + K_2 \geq 7m.$$

Its optimum peak is $\lceil 7m/2 \rceil$, whereas either cut separately has minimum $3m$. This closure law complements the temporal example: it identifies an exact growing Pareto frontier, while the rotation family keeps alphabets and all separate minima fixed. The tagged construction has structural output zeros; it is not called a full-support example.

1.1. Classical mechanisms and the new comparison. Invariant polyhedral cones and normalized convex sections are classical in positive realization [6, 1, 12]. Irrational rotation as an obstruction to a finite *stationary* positive realization is also classical: see [1, Example 4] and [8, Theorem 6, Example 2 and Lemma 7]. We do not claim either mechanism as a new discovery. Reachable subspaces and observable quotients are likewise established tools [8, Section 4].

The comparison proved here differs in its quantifiers. Every finite horizon has a finite exact realization; both the state spaces and the rows may change at every epoch. There is no common stationary matrix to which a peripheral-spectrum obstruction can be applied. Moreover, the least *positive* factorization at each cut, not just ordinary rank, is three. The cumulative volume bound and the explicit sequence of finite realizations concern this finite, time-dependent problem. Section 7 details the relation to positive realization, nonnegative factorization and residual

automata. The explicit bounds are claims of the present article; exhaustive priority clearance is not inferred from the bibliography.

Statement	Mathematical role
Reachable-section and volume budget	General multi-cut lower bound; Sections 3–5
Rotation family	Fixed-rank, fixed-alphabet unbounded incompatibility
Tagged profile sum	Exact composition of feasible profile sets
Two-cut primitive	Inherited from revision 31; proof retained in Appendix A
Invariant cones and weighted replacement	Classical mechanisms; not new claims

2. COMPLETE ARRAYS AND CHARGED STATE PROFILES

Definition 2.1 (Clocked stochastic transducer). Fix finite alphabets $X, Y_1, \dots, Y_N, J, B$. A complete terminal array is a probability distribution $C(\cdot \mid x, y_1, \dots, y_N, j)$ on B for every word in the full Cartesian product. A realization has finite registers $S_t = [K_t]$ for $0 \leq t \leq N$, an initialization row $E(s \mid x)$, transition rows $T_{t,y}(s, s')$ from S_t to S_{t+1} for $0 \leq t < N$, and terminal rows $D(b \mid s, j)$, all nonnegative and normalized. It realizes C when

$$(4) \quad C(b \mid x, y_1, \dots, y_N, j) = \sum_{s_0, \dots, s_N} E(s_0 \mid x) \prod_{t=0}^{N-1} T_{t, y_{t+1}}(s_t, s_{t+1}) D(b \mid s_N, j).$$

The current input is read atomically, and only the next register survives its update. There is an initial singleton before x and, when the output is retained, a final $|B|$ -label register. The full peak is $\max(1, |B|, K_0, \dots, K_N)$.

Inputs in this definition are externally supplied; they need not have a probability law. Equality is wordwise. A controller’s earlier randomness, input buffers, data-dependent scheduling information and previously emitted symbols cannot be read again unless retained in its register. The declared epoch t is a free external clock; the current query is consumed in the terminal update. All row tables are known program data, and sampling a stochastic row is atomic. Table size, arithmetic work, construction time and fair-bit sampling workspace are not included in K_t . A shared private seed is not an additional free input.

We count available labels. Unreachable labels may be removed, while padding by unused labels explains upward closure of feasible profiles. Every declared input row, including rows unused by the actual initial law, is a stochastic row. An autonomous implementation instead uses one state set and time-independent rows; phase tagging gives a sufficient sum-of-widths bound and is not identified with the clocked peak. These conventions separate the finite positive-realization problem from succinct streaming-program or bit-operation complexity.

Let $\mathcal{F}(C) \subset \mathbb{N}^{N+1}$ be the upward-closed set of realizable profiles. For a prefix $h = (x, y_1, \dots, y_t)$, let R_h be its complete conditional array on all suffixes and terminal queries. At this cut the ambient continuation polytope is a product of output simplices. Define $r_t = \text{rank}(R_h)_h$, with columns indexed by suffix, query and output. Normalized rows give $r_t = 1 + \dim \text{aff}\{R_h\}$.

Definition 2.2 (Separate minimum). The separate minimum $\kappa_t(C)$ is the least k such that every R_h is a convex combination of k normalized continuation arrays on the same future alphabet. There is no compatibility requirement between the dictionaries selected at distinct cuts. The minimum peak of a compatible realization is denoted $W(C)$.

For finite alphabets, $\kappa_t(C) = \min_{\mathbf{K} \in \mathcal{F}(C)} K_t$: store the whole prefix before the chosen cut, implement its encoder at that cut, and afterwards store the selected label and future inputs until the terminal update. Thus a separate factorization can always be embedded in a complete

machine, at possibly large costs elsewhere. Ordinary row rank is a lower bound on κ_t . For a single terminal output alphabet this minimum is ordinary nonnegative rank after normalization. For a queried continuation array, every suffix/query block of a generator must normalize separately; nonnegative rank of the concatenated matrix need only be a lower bound. These are not the same as residual-only state counts [5, 3].

Notation	Meaning
$K_t, W(C)$	available labels at a cut; least simultaneous peak
$r_t, \kappa_t(C)$	ordinary row rank; separate normalized positive minimum
E_h, H_t	reachable state distribution; its affine span
Q_t, P_t	reachable simplex section; observable projected section
S, D	compulsory seed hull; bounded observable domain
δ_M	minimum one-step volume expansion for at most M vertices

The input-only model is already a subclass of finite causal experiments: a prescribed action alphabet may consist of one action at each stage. A lower bound in this class does not solve optimization over informative experimental actions. The present advance concerns arbitrary-horizon compatibility of exact responses under a specified interface.

3. REACHABLE SECTIONS AND A CUMULATIVE VOLUME BOUND

We first treat a general finite collection of linear mean updates. Let $\mathcal{L}$ be a finite set of linear maps of $\mathbb{R}^d$, containing I_d . A finite seed alphabet has associated vectors $z(x)$, whose convex hull S has nonempty interior. Assume that every product of at most N allowed maps sends the seed vectors into $D = [-1, 1]^d$. After reading x and N commands $L_1, \dots, L_N \in \mathcal{L}$, a terminal query $j \in [d]$ requires a binary output of mean $e_j^T L_N \cdots L_1 z(x)$. Denote this complete array by C_N .

Lemma 3.1 (The reachable-section lift). *Every realization of C_N with profile $(K_t)_{t=0}^N$ determines full-dimensional polytopes $P_t \subset D$ such that*

$$(5) \quad S \subset P_t, \quad |\text{ext } P_t| \leq 2^{K_t}, \quad LP_t \subset P_{t+1} \quad (L \in \mathcal{L}, t < N).$$

The conclusion permits arbitrary hidden states and does not require individual state continuations to belong to the residual affine span.

Proof. For each actual prefix h at cut t , let $E_h \in \Delta_{K_t}$ be its hidden state distribution. Put

$$H_t = \text{aff}\{E_h : h \text{ a prefix at } t\}, \quad Q_t = H_t \cap \Delta_{K_t}.$$

For any state distribution $e \in \Delta_{K_t}$, run the remaining commands all equal to I_d . The vector of its d terminal means is a linear function $\pi_t(e)$ and belongs to D . Correctness on actual prefixes gives $\pi_t(E_h) = z_h$, the specified predictive vector. Set $P_t = \pi_t(Q_t)$.

Write $T_{t,L}$ for the realizing transition matrix of command L . The identity $E_h T_{t,L} = E_{hL}$ implies $H_t T_{t,L} \subset H_{t+1}$. Stochasticity also implies $\Delta_{K_t} T_{t,L} \subset \Delta_{K_{t+1}}$; consequently $Q_t T_{t,L} \subset Q_{t+1}$. On every actual prefix,

$$\pi_{t+1}(E_h T_{t,L}) = L \pi_t(E_h).$$

Both sides are affine in E_h , so this equality extends to all of H_t , including its points with some negative coordinates. Restricting back to Q_t yields $LP_t \subset P_{t+1}$. Taking the idle command shows $P_t \subset P_{t+1}$. The initial section contains all seeds, hence so does every P_t .

It remains to bound vertices without assuming a minimal hidden representation. In its affine hull, Q_t is a bounded polytope defined by the K_t coordinate inequalities $e_s \geq 0$. Two distinct vertices cannot have the same zero-coordinate set. Indeed, if they did, their difference would give a nonzero direction preserving all active equalities, and sufficiently small motion in both directions from either point would preserve the remaining strict inequalities. That point would

not be a vertex. There are at most 2^{K_t} zero sets. A linear image of a polytope is the convex hull of the images of its vertices, proving $|\text{ext } P_t| \leq |\text{ext } Q_t| \leq 2^{K_t}$. $\square$

The section Q_t , rather than the entire hidden simplex, is essential. For an arbitrary unit hidden state, the identity relating the two probe maps need not hold. It holds on the affine span of reachable *distributions*. This is the finite-horizon version of the accessible-space/observable-quotient precaution in classical realization [8, Section 4]; the vertex bound accounts for the resulting lift. In particular, replacing 2^{K_t} by K_t without further hypotheses would not be justified.

For a full-dimensional compact seed hull $S \subset D$, define

$$(6) \quad \delta_M(S, D, \mathcal{L}) = \min_{\substack{S \subset P \subset D \\ P \text{ a polytope, } |\text{ext } P| \leq M}} \left\{ \text{vol}_d \left(\text{conv} \bigcup_{L \in \mathcal{L}} LP \right) - \text{vol}_d(P) \right\}.$$

If the feasible set is empty, put $\delta_M = +\infty$. The maps in the union need not send all of D into D . Only the implementing sequence from Lemma 3.1 must obey that constraint. The quantity is a geometric optimization on observable polytopes, not an optimization over candidate transition tables.

Theorem 3.2 (Volume budget for a state profile). *Suppose $|\det L| = 1$ for every $L \in \mathcal{L}$ and $I_d \in \mathcal{L}$. If there is no full-dimensional polytope P with $S \subset P \subset D$ and $LP = P$ for all $L \in \mathcal{L}$, then every nonempty minimum in (6) is strictly positive. Every realization of C_N satisfies*

$$(7) \quad \sum_{t=0}^{N-1} \delta_{2^{K_t}}(S, D, \mathcal{L}) \leq \text{vol}_d(D) - \text{vol}_d(S).$$

In particular, a uniform state bound K implies $N\delta_{2^K} \leq \text{vol}_d(D) - \text{vol}_d(S)$.

Proof. The set of feasible polytopes with at most M vertices is compact in the Hausdorff metric: represent them by M points in the compact set D , permitting repetitions, and impose the closed condition that their convex hull contain S . Convex hull, finite union followed by convex hull, and volume of compact convex sets are continuous. Thus the minimum is attained whenever the set is nonempty.

Because the identity is allowed, the displayed volume difference is nonnegative. If it vanishes at P , the full-dimensional convex bodies P and $\text{conv} \bigcup_L LP$ have equal volume and one contains the other, so they are equal. Thus $LP \subset P$. The determinant hypothesis makes their volumes equal, hence $LP = P$ for every L , contrary to the assumption. Therefore $\delta_M > 0$.

For the polytopes in Lemma 3.1,

$$\text{vol}_d(P_{t+1}) - \text{vol}_d(P_t) \geq \text{vol}_d \left(\text{conv} \bigcup_L LP_t \right) - \text{vol}_d(P_t) \geq \delta_{2^{K_t}}.$$

Summing telescopes. The endpoint inclusions give (7). The minima are nonincreasing in M , which gives the assertion for a uniform bound. $\square$

The volume hypothesis is a property of the mean dynamics and the observation domain, not a small-ball or metric-dimension assumption. Under an invertible affine change of predictive coordinates, using the conjugate affine maps, both sides of (7) are multiplied by the same absolute determinant. Thus the normalized budget is independent of this coordinate choice. The proof also works for volume-preserving affine maps with an idle map.

Corollary 3.3 (Effective finite obstructions). *In dimension two, for explicitly given rational or real-algebraic polygons S, D and volume-preserving linear maps, every nonempty δ_M is a computable real-algebraic number. If the hypothesis of Theorem 3.2 holds, one can compute a*

rational $0 < c_M \leq \delta_M$ and hence a finite horizon excluding any specified uniform width K with $M = 2^K$.

Proof. Use M pairs of vertex coordinates in D as variables. Containment of each vertex of S in their hull is expressed with nonnegative barycentric variables. A finite partition by determinant signs and cyclic orders identifies each possible convex hull, including repetitions and collinear points. On every such piece the areas of P and $\text{conv} \bigcup_L LP$ are polynomial expressions, by the polygon area formula. Thus the graph of the objective, its feasible domain and the statement that a number is its minimum are first-order formulas over the real closed field of algebraic numbers. Quantifier elimination, followed by real-root isolation, computes the minimum and a positive rational lower bound when it is positive [2]. Algebraic inputs are represented by defining polynomials and isolating intervals. The procedure is finite; no polynomial complexity in K , the horizon, or a succinct circuit description is asserted. $\square$

4. A FIXED-ALPHABET FAMILY WITH UNBOUNDED COMPATIBILITY COST

We apply the volume bound with

$$a = 1/10, \quad S = a[-1, 1]^2, \quad D = [-1, 1]^2, \quad \mathcal{L} = \{I_2, R\},$$

where R is any planar rotation of infinite order. The terminal array is (1). The fixed alphabets have respectively four seed symbols, two command symbols, two queries and two outputs. Rotation preserves the norm of every actual predictive vector, so the uniform full-support bound stated in the introduction holds at every horizon.

Proposition 4.1 (All separate positive minima are three). *At every internal cut of $C_{N,R}$, ordinary normalized row rank and separate normalized positive minimum both equal three, for every rotation R and every $N \geq 0$.*

Proof. For a prefix at cut t , put $z = R^{c_1 + \dots + c_t} ax$. Its entire future response is affine in z . The all-idle prefixes contain the four seeds, which affinely span $\mathbb{R}^2$, and the all-idle suffix with the two coordinate queries distinguishes both coordinates. Normalization supplies an independent constant coordinate. Hence the residual row rank is exactly three.

Consider the triangle

$$(8) \quad T = \text{conv}\{v_0, v_1, v_2\}, \quad v_0 = (1/2, 0), \quad v_1 = (-1/2, 1/2), \quad v_2 = (-1/2, -1/2).$$

It contains the centered disk of radius $1/(2\sqrt{5})$, while every vertex has norm at most $1/\sqrt{2}$. All reachable z have norm $\sqrt{2}/10 < 1/(2\sqrt{5})$, and thus lie in T . Starting from any v_s , any future rotation word has both mean coordinates in $[-1, 1]$. It therefore defines a legal normalized future channel. Barycentric decomposition of z in T gives an exact three-state factorization of the whole suffix array. This proves $\kappa_t = 3$; Definition 2.2 embeds it in a complete machine with finite, unrestricted registers at other cuts. $\square$

Theorem 4.2 (Unbounded width with effective rational data). *For every infinite-order rotation R , $W_N(R) \rightarrow \infty$. More precisely, put $\delta_M = \delta_M(S, D, \{I, R\})$. These numbers are positive for $M \geq 3$, and every feasible profile satisfies*

$$(9) \quad \sum_{t=0}^{N-1} \delta_{2^{K_t}} \leq \frac{99}{25}.$$

For R_0 in (3), the target arrays are rational and the thresholds excluding a bounded width are effective in the sense of Corollary 3.3.

Proof. An invariant polygon containing S would have nonzero vertices whose orbits under R remain in its finite vertex set, since an invertible linear map taking a polygon onto itself permutes its vertices. An infinite-order planar rotation has no finite orbit away from zero. No such polygon exists. The triangle in (8) is feasible for the definition of δ_M when $M \geq 3$, so these minima are positive by Theorem 3.2. Its volume budget gives (9), since $4 - 4a^2 = 99/25$. For any K , horizons greater than $99/(25\delta_{2K})$ exclude all width- K machines. This proves the asserted divergence.

The eigenvalue $\zeta = (3 + 4i)/5$ of R_0 is not a root of unity. Otherwise $\zeta + \zeta^{-1} = 6/5$ would be a rational algebraic integer, hence an integer. The contradiction proves that R_0 has infinite order. Rationality of its powers and of the seeds proves rationality of every specified channel entry. Corollary 3.3 applies. $\square$

The divergence does not follow merely by invoking the classical absence of a stationary positive realization of an irrational rotation. The matrices of a proposed machine may change at every time, and depend on N . Inequality (9) excludes these machines as well. In particular there is no function of the maximum separate positive minimum alone that bounds simultaneous width on this class.

4.1. Constructive finite realization.

Theorem 4.3 (Exact polygon realization). *For any planar rotation R and any $N \geq 1$, the channel $C_{N,R}$ has an exact realization of peak at most $4\lceil\sqrt{N+1}\rceil$. It also has an exact realization of peak $3(N+1)$, with rational rows when R has rational entries. Both constructions store finite labels, not real predictive vectors.*

Proof. Put $M = 4\lceil\sqrt{N+1}\rceil$, $u_s = (\cos(2\pi s/M), \sin(2\pi s/M))$ for $0 \leq s < M$, and

$$\rho_t = \frac{\sqrt{2}}{10} \sec^{t+1}(\pi/M), \quad P_t = \text{conv}\{\rho_t u_s : 0 \leq s < M\} \quad (0 \leq t \leq N).$$

The inradius of P_0 is $\sqrt{2}/10$, so $S \subset P_0$. The inradius of P_{t+1} is ρ_t , and hence

$$(10) \quad P_t \cup RP_t \subset P_{t+1}.$$

All these polygons lie in D . Indeed, with $L = N + 1$,

$$\cos(\pi/M) \geq 1 - \frac{5}{16L}, \quad \cos^L(\pi/M) \geq 1 - \frac{5}{16} = \frac{11}{16}, \quad \rho_N \leq \frac{16\sqrt{2}}{110} < 1.$$

The first bound uses $M \geq 4\sqrt{L}$ and $\pi^2 < 10$; the second is Bernoulli's inequality.

Initialize by a convex decomposition of the observed seed ax in P_0 . At cut t , the machine stores $s \in [M]$. After command c , replace it by a random label whose mean vertex in P_{t+1} is $R^c \rho_t u_s$, using (10). Induction gives

$$\mathbb{E}[\rho_t u_{s_t} \mid x, c_1, \dots, c_t] = R^{c_1 + \dots + c_t} ax.$$

The terminal binary row has mean $e_j^T \rho_N u_s$. This proves the specified full conditional law. The final binary register fits within M states.

The update coefficients can be written explicitly. If the direction of $R^c u_s$ falls between u_l and u_{l+1} , let its offset be $0 \leq \theta \leq \omega = 2\pi/M$. Set

$$A = \frac{\rho_t \sin(\omega - \theta)}{\rho_{t+1} \sin \omega}, \quad B = \frac{\rho_t \sin \theta}{\rho_{t+1} \sin \omega}, \quad T_{t,c}(s, k) = A \mathbf{1}_{k=l} + B \mathbf{1}_{k=l+1} + \frac{1 - A - B}{M}.$$

Both coefficients are nonnegative and $A + B \leq (\rho_t/\rho_{t+1}) \sec(\pi/M) = 1$. Since $\sum_k u_k = 0$, these rows have the required mean. Boundary choices agree. Initialization uses the same triangular fan, replacing ρ_t by the norm of the seed. Only a vertex *index* is stored. In fact, the ratio ρ_t/ρ_{t+1}

is constant, so each command matrix is the same at every training epoch for this construction. The final decoder still depends on the chosen horizon.

For the rational alternative, decompose ax into the three vertices v_i of (8); its weights are

$$1/2 + ax_1, \quad 1/4 - ax_1/2 + ax_2, \quad 1/4 - ax_1/2 - ax_2.$$

Store the pair (i, q) , initially $q = 0$, and add c to q at each command. At cut t at most $3(t + 1)$ states are available. The decoder has mean $e_j^T R^q v_i$, which belongs to $[-1, 1]$ because $\|v_i\|_2 < 1$. This is an exact realization, with rational initialization, transitions and output rows when R is rational. $\square$

The polygon construction provides known exact stochastic rows. For a rational rotation, its trigonometric constants are algebraic; for the golden angle they are fixed computable real constants. Neither is being silently identified with a bounded fair-bit computation. The rational $3(N + 1)$ construction can use the separately charged rational-row sampler of the retained development [13]. Its extra workspace is not asserted to preserve the atomic peak. Similarly, an autonomous machine required to check the input length has an additional clock task; that task is not part of Definition 2.1.

5. AN EXPLICIT MULTI-CUT INEQUALITY

We now give a quantitative expansion estimate for the rotation in Theorem 1.1. Throughout this section, $\alpha = (\sqrt{5} - 1)/2$, R is rotation by $2\pi\alpha$, and $a = 1/10$. The Euclidean unit disk is denoted by $\mathbb{B}^2$.

Lemma 5.1 (Finite orbit covering). *For every integer $M \geq 3$, there is an integer q with $32M \leq q < 64M$ such that $\{R^k u : 0 \leq k < q\}$ is a $3\pi/q$ -net of the unit circle in angular distance, for every unit u .*

Proof. Let $F_0 = 0, F_1 = 1, F_{m+1} = F_m + F_{m-1}$ and take the first Fibonacci number $q = F_m$ at least $32M$. Consecutive positive Fibonacci numbers have ratio less than two, so $q < 64M$. Put $p = F_{m-1}$. The elementary identities

$$\gcd(p, q) = 1, \quad F_m \alpha - F_{m-1} = (-1)^{m-1} \alpha^m, \quad F_m \alpha^m < 1$$

follow by induction (or by the formula for F_m obtained from the two roots of $z^2 = z + 1$). Thus $|\alpha - p/q| < 1/q^2$. The first q multiples of p/q form the uniform q -grid modulo one. Each differs from the corresponding multiple of α by less than $1/q$ in circular distance. Every point of the circle is within $1/(2q)$ of a grid point and hence within $3/(2q)$ turns of an orbit point. Multiplying by 2π proves the assertion, also after a common rotation by u . $\square$

Lemma 5.2 (A polygon cannot be too nearly invariant). *If $a\mathbb{B}^2 \subset P \subset \sqrt{2}\mathbb{B}^2$ is a polygon with at most $M \geq 3$ vertices, then*

$$(11) \quad d_H(P, RP) \geq \frac{a}{64M^3}.$$

Proof. Write $\varepsilon = d_H(P, RP)$ and $r = \max_{z \in P} \|z\|_2$. Choose a vertex v of norm r . Rotational invariance of Hausdorff distance and the triangle inequality give $d_H(P, R^k P) \leq k\varepsilon$. By Lemma 5.1, for every unit direction u some $k < q$ puts $R^k v$ within angular distance $3\pi/q$ of u . For the support function h_P , it follows that

$$(12) \quad h_P(u) \geq r \cos(3\pi/q) - q\varepsilon, \quad r - h_P(u) \leq \frac{9\pi^2 \sqrt{2}}{2q^2} + q\varepsilon.$$

The directions of the at most M vertices have an empty circular arc of length at least $2\pi/M$. At its midpoint u_0 , every vertex has scalar product at most $r \cos(\pi/M)$. Consequently

$$r - h_P(u_0) \geq r(1 - \cos(\pi/M)) \geq \frac{2a}{M^2}.$$

Here $1 - \cos s \geq 2s^2/\pi^2$ for $0 \leq s \leq \pi$, and $r \geq a$. Using $q \geq 32M$, $\pi^2 < 10$ and $\sqrt{2} < 3/2$ shows that the first term on the right of (12) is less than a/M^2 . It follows that $q\varepsilon \geq a/M^2$. The bound $q < 64M$ proves (11). $\square$

Lemma 5.3 (Area pays for Hausdorff displacement). *If $a\mathbb{B}^2 \subset P \subset R_*\mathbb{B}^2$ and $U = \text{conv}(P \cup RP)$, for an arbitrary planar rotation R , then*

$$(13) \quad \text{area}(U) - \text{area}(P) \geq \frac{a}{R_*} d_H(P, RP)^2.$$

Proof. Let $d_+ = \max_{u \in \mathbb{S}^1} (h_{RP}(u) - h_P(u))_+$. When $d_+ > 0$, choose a maximizing direction u and $x \in RP$ with $u \cdot x = h_P(u) + d_+$. The triangle with vertices x and the two points $\pm av$, where $v \perp u$ is a unit vector, lies in U . Its part in $\{z : u \cdot z > h_P(u)\}$ lies outside P . Similar triangles give its area as

$$\frac{ad_+^2}{u \cdot x} \geq \frac{ad_+^2}{R_*}.$$

The zero case needs no argument. Repeating this with P and RP interchanged gives the same bound with d_- . Rotation preserves area, so the two excess areas coincide. For compact convex sets, $d_H(P, RP) = \max(d_+, d_-)$, proving the claim. $\square$

Theorem 5.4 (Quantitative temporal obstruction). *For the golden-angle rotation and S, D of Section 4,*

$$(14) \quad \delta_M \geq \frac{1}{6000000M^6} \quad (M \geq 3).$$

Every exact realization of $C_{N,R}$ therefore satisfies

$$(15) \quad \sum_{t=0}^{N-1} 2^{-6K_t} \leq 24000000.$$

In particular $W_N(R) \geq \frac{1}{6} \log_2(N/24000000)$.

Proof. A polygon feasible for (6) contains $a\mathbb{B}^2$ and lies in $\sqrt{2}\mathbb{B}^2$. The last two lemmas give

$$\text{area}(\text{conv}(P \cup RP)) - \text{area}(P) \geq \frac{a^3}{4096\sqrt{2}M^6} > \frac{1}{6000000M^6},$$

using $a = 1/10$ and $\sqrt{2} < 10/7$. Minimize over P to obtain (14). Apply Theorem 3.2 and bound $\text{area}(D) - \text{area}(S)$ by 4. If $K_t \leq W$ at all times, the left side of (15) is at least $N2^{-6W}$. $\square$

Combining Proposition 4.1, Theorems 4.2, 4.3 and 5.4 proves Theorem 1.1. The profile inequality gives more than a bound on one minimum peak. For example, at most $24000000 \cdot 2^{6k}$ of the command-input cuts can have width at most k , independently of the widths at the other cuts. The same statement follows directly by restricting the nonnegative sum.

Proposition 5.5 (Robustness at a fixed excluded profile). *Fix a finite horizon and a profile excluded for $C_{N,R}$ by either volume bound. There is a strictly positive uniform row-total-variation distance from $C_{N,R}$ to all arrays realized with that profile. For the rational family, a positive rational lower distance can be computed from the fully listed array and profile by real quantifier elimination.*

Proof. Initialization, update and decoder rows range over a finite product of compact simplices. The realized array is a polynomial, and hence continuous, function of those rows. Its image is compact. An excluded array has strictly positive distance to this image in the maximum of the finitely many row-TV distances. For rational target entries the minimum distance is a semialgebraic optimization, after expressing absolute values by auxiliary variables; real quantifier elimination and root isolation give the stated rational lower bound [2]. No uniform lower distance as the horizon or profile varies follows from this argument. $\square$

The exact golden-angle lower bound uses a specified computable real rotation; it is not presented as an algebraic-input computation. The rational family supplies algebraic excluded-horizon and fixed-profile robustness certificates instead. Neither the numerical values of its δ_M nor optimal rates are asserted by the finite regression tests.

6. EXACT COMPOSITION OF FEASIBLE PROFILES

The previous bounds accumulate a compatibility cost through time. There is also an exact composition operation across distinguished experiments. Its hypothesis is operational: the distinguishing label must be recovered at the end and is not externally available after its initial preparation.

Take m complete terminal arrays C_i , on a common horizon and common future input alphabets. Their initial alphabets may be identified by padding. Define the *verifiably tagged sum* $C^\oplus$ by receiving (i, x) initially and requiring terminal output (i, b) with law $C_i(b \mid x, \mathbf{y}, j)$. A wrong tag has probability zero. The tag is not repeated in later inputs, and previous outputs are not readable as memory. All branches have the same full suffix alphabet.

For two sets of integer profiles their sum below denotes coordinatewise Minkowski sum, including the already defined upward closure.

Theorem 6.1 (Profile addition under a retained tag). *For the internal registers of the tagged sum,*

$$(16) \quad \mathcal{F}(C^\oplus) = \mathcal{F}(C_1) + \cdots + \mathcal{F}(C_m).$$

The identity holds for arbitrary horizons and arbitrary stochastic realizations. It is an identity of complete feasible-profile sets, not just of separate minima or a scalar matrix rank.

Proof. Fix a realization and a cut t . Let $S_{t,i}$ consist of states having positive probability after some prefix from branch i . If one state belonged to $S_{t,i}$ and $S_{t,k}$ with $i \neq k$, run the same fixed future input word from that state. Exact tag recovery in branch i forces this conditional continuation to emit tag i with probability one: a positive mixture weight could not otherwise give zero wrong-tag probability. The same reasoning in branch k forces tag k . This is impossible. Thus the sets $S_{t,i}$ are disjoint.

For every allowed next input, a positive transition from $S_{t,i}$ leads to $S_{t+1,i}$, since its source is reachable after some branch- i prefix and the full product suffix is allowed. Restricting the machine to these state sets and stripping its terminal tag therefore gives a complete realization of C_i . In particular its profile $(|S_{t,i}|)_t$ is feasible. Their coordinatewise sum does not exceed the original profile. Padding any one of the restricted realizations absorbs each unused coordinate and gives membership in the right side of (16).

Conversely, use disjoint copies of the branch registers and transition rows. Initialization selects the declared branch, which is thereafter stored by membership in its disjoint register. The final decoder emits its tag and answer. This gives the sum of any feasible branch profiles and proves the reverse inclusion. No independent shared seed or free branch selector is supplied. $\square$

If the final tag were dropped, this proof would fail: states serving two branches could be identified. Likewise, branch-dependent legal suffixes must not be substituted for the common full alphabet without a separate argument. These are hypotheses of the operation, not conclusions about unlabelled mixtures.

Corollary 6.2 (An exact growing two-cut frontier). *Let every branch be the two-cut array (18), with a fixed $9/10 \leq h < 1$, and let $m \geq 1$. Its tagged sum has exactly the feasible internal profiles*

$$(17) \quad K_1 \geq 3m, \quad K_2 \geq 3m, \quad K_1 + K_2 \geq 7m.$$

The Pareto-minimal pairs are $(3m + j, 4m - j)$, $j = 0, \dots, m$. Both separate minima are $3m$, and the minimum full peak, including the $2m$ -symbol terminal output, is $\lceil 7m/2 \rceil$.

Proof. Appendix A proves the primitive feasible set as the upward closure of $(3, 4), (4, 3)$. Its m -fold Minkowski sum has the listed minimal pairs. To check that (17) has no additional holes, put $u = K_1 - 3m$, $v = K_2 - 3m$. If $u, v \geq 0$ and $u + v \geq m$, choose an integer j with $\max(0, m - v) \leq j \leq \min(m, u)$. Then $(K_1, K_2) \geq (3m + j, 4m - j)$. The peak is minimized by the nearest integer balance of j and $m - j$, giving $\lceil 7m/2 \rceil$. The output register has $2m \leq \lceil 7m/2 \rceil$ labels. $\square$

The primitive answer probabilities have full support, but the tagged sum has zeros at wrong tags. We do not use it as the full-support part of Theorem 1.1; that theorem already has a uniform full-support interval and fixed alphabets. The two results answer different questions: (17) is an exact finite Pareto law, whereas (15) excludes bounded simultaneous width while every separate positive minimum remains three.

7. COMPARISON, IMPLEMENTATION AND FURTHER QUESTIONS

7.1. Stationary positive realization and finite causal width. Benvenuti–Farina [1, Theorems 2 and 3] formulate existence of positive realizations by invariant polyhedral cones and shift-invariant positive sequences. Their Example 4 uses an irrational planar rotation to prevent a finite stationary realization. Monràs–Winter [8, Theorem 6 and Lemma 7] give the corresponding stochastic cone criterion and a rotation example. These are direct antecedents, not merely neighboring subjects.

Theorem 1.1 makes a different finite assertion. The matrices are allowed to be unrelated at different epochs, and a new machine may be selected at every horizon. Each entire finite array is positively realizable, while each of its cuts separately needs exactly three states. The lower bound is the accumulation of enlargement along $P_t \subset P_{t+1}$ and $RP_t \subset P_{t+1}$. An invariant-cone existence theorem or a peripheral-spectrum argument for one common matrix does not, by itself, give this quantitative inequality for changing rows. Conversely, our theorem does not re-prove or extend the full stationary positive-realization existence classification.

The reachable-section construction is deliberately compatible with classical quotient realizations. In [8, Section 4, Theorem 9], the accessible subspace and its unobservable kernel determine the minimal linear quotient; arbitrary positive realizations can contain more directions. That is why Lemma 3.1 uses a section of a simplex before projecting. It does not identify minimum hidden dimension with a minimum number of vertices of an unlifted observable polygon. The related distinction between restricted and unrestricted positive factorizations appears explicitly in [5, Definition 1, Theorem 1]. The exponential allowance 2^{K_t} is conservative, but makes the lower bound valid for all competing hidden representations.

The recent work of Lumbreras–Ma–Thompson–Gu [7] studies irreducible errors and decision failures of finite classical world models, with exact quantum realizations. Its Section IV.2 and ensuing proofs use repetition of one update, Perron–Frobenius limits and an irrational phase.

Our task has no quantum simulator or decision-loss claim: it compares exact *finite-horizon* cut-dependent classical realizations. The common rotation ancestry and different row-sharing quantifiers are both relevant. We do not infer a novelty boundary from the absence of a citation to this manuscript in that work.

7.2. Residual restrictions, direct sums and sampling. The finite-generation and canonical residual constructions of Esposito–Lemay–Denis–Dupont [4] and Denis–Esposito [3, Proposition 16] concern state languages constrained by the residual representation. The lower bound here places no such constraint on a state’s full continuation. Indeed all separate minima stay three in the rotation example, so a large list of actual residuals alone would not prove its causal width lower bound. Classical hidden-Markov and filtered-experiment comparison remain part of the surrounding theory [12, 9]; our wordwise finite profile statement is not claimed to be a new general comparison theorem for filtered spaces.

Nonnegative direct-sum additivity itself is familiar, including the tensor version in [11, Lemma 4.1]. Theorem 6.1 uses the same positivity principle to separate *reachable state sets at every time*; it identifies a sum of entire feasible profile sets. Taking only sums of separate ranks would lose the tradeoff in (17). Its tag and full common suffix alphabet are explicit and cannot be omitted.

The additive encoder construction in the preserved article [13] uses weighted replacement. The sampling step is weighted reservoir sampling, as described, for example, in [10, Appendix A.2]: replace the current sample with probability equal to the new weight divided by total weight. When total previous weight is zero, the dummy label is chosen from the existing alphabet. Neither that sampling algorithm nor the classical simplex inputs are novelty claims of the present paper. The earlier online consequences and rational sampler remain available without being needed for the rotation lower bound.

7.3. What is and is not charged. The finite-horizon bounds concern exact atomic stochastic transducers, not a bit-operation model. Both alphabet cardinality and all persistent hidden choices are charged. The row table and the prescribed external epoch are not. The rational family permits separately charged fair-bit sampling; the algebraic or computable-real polygon rows are not given a finite-denominator implementation claim. Sampling and verification costs therefore do not disappear into the phrase “exact realization.”

The new lower bound holds even with the external clock and arbitrary cut-dependent rows. The polygon upper construction has common command matrices across its training epochs, but its prescribed decoder depends on N . This does not define one finite machine that works at every horizon with an unchanged decoder. No automatic transfer from the clocked peak to autonomous length recognition is made.

7.4. Conclusions. The proved separation is between *separate positive minima* and *simultaneous positive width*, with fixed alphabets, uniformly positive response probabilities, and an arbitrarily long horizon. The projected reachable-section budget applies to a class of volume-preserving mean dynamics. The golden-angle family has an explicit multi-cut inequality and a constructive square-root upper bound; the rational family has effective excluded-horizon certificates. The tagged operation separately supplies exact growing Pareto frontiers.

The optimal growth of W_N , a sharp measure of lift complexity, quantitative stability uniform in the horizon, and general optimization of informative acquisition or validation actions remain distinct questions. The logarithmic lower bound is not matched by the square-root upper bound. Complete arbitrary-profile minimization is not claimed. These distinctions do not affect the unbounded separation or the exact tagged frontier.

The historical geometric, large-deviation and nonlinear-operator chains are preserved as separate mathematics. No finite-array argument here is claimed to discharge their analytic gates. The focused article contains the proofs required for its new claims; the cumulative volumes are archival records, not an additional burden of certification for its referee.

APPENDIX A. THE RETAINED TWO-CUT PRIMITIVE

The following proof is retained from revision 31 [13]; its role here is the input to Corollary 6.2, not a new claim.

The following experiment has only two internal memory cuts. The first input is $x = (x_1, x_2) \in \{-1, 1\}^2$, received as one symbol. A command $j \in \{1, 2, 3, 4, 5\}$ is then received, followed by a query $k \in \{1, 2\}$. The machine emits $B \in \{-1, 1\}$ in the update receiving k . Let K_1 count states after x and K_2 states after j ; a retained final answer has two labels. No further access to x or j is permitted. For $0 < h \leq 1$ prescribe the conditional mean vectors

$$(18) \quad \begin{array}{c|ccccc} j & 1 & 2 & 3 & 4 & 5 \\ \hline z_{x,j} & (h, h) & (h, -h) & (-h, 0) & (-h/2, hx_1/4) & (-h/2, hx_2/4) \end{array}.$$

The answer law is $\mathbb{P}(B = b \mid x, j, k) = (1 + bz_{x,j}(k))/2$.

The program is clocked and uses arbitrary stochastic rows. Thus every allowed implementation has common rows $E(s \mid x)$, $T(t \mid s, j)$ and $D(b \mid t, k)$; all past randomness is in s or t .

We first record the geometric estimate used by the converse.

Lemma A.1 (A quantitative forced triangle). *Put $h = 1 - \delta$, where $0 \leq \delta \leq 1/10$. If a triangle $P \subset [-1, 1]^2$ contains (h, h) , $(h, -h)$ and $(-h, 0)$, then every point $(-h/2, y) \in P$ satisfies*

$$(19) \quad |y| \leq \beta(\delta) := \frac{1}{4} + \frac{\delta}{2(1-\delta)} + \frac{\delta}{2-3\delta} < \frac{h}{2}.$$

Proof. A convex representation of (h, h) includes a vertex A for which $2 - A_1 - A_2 \leq 2\delta$, so $A_1, A_2 \geq 1 - 2\delta$. A representation of $(h, -h)$ similarly includes a vertex B with $B_1 \geq 1 - 2\delta$ and $B_2 \leq -1 + 2\delta$. These vertices are distinct. Call the third vertex D . In a representation of $(-h, 0)$, let u be the total weight on A, B . Since their first coordinates are at least $1 - 2\delta$ and $D_1 \geq -1$,

$$-1 + \delta \geq -1 + 2(1 - \delta)u, \quad u \leq \frac{\delta}{2(1 - \delta)}.$$

The zero second coordinate implies $|D_2| \leq u/(1 - u) \leq \delta/(2 - 3\delta)$. For a point of P with first coordinate $-h/2$, the total weight v on A, B satisfies

$$-h/2 \geq -1 + 2hv, \quad v \leq \frac{1 - h/2}{2h} = \frac{1}{4} + \frac{\delta}{2h}.$$

Its second coordinate has absolute value at most $v + (1 - v)|D_2| \leq \beta(\delta)$. Finally the two fractions in β increase with δ while $h/2$ decreases, and

$$\beta(1/10) = \frac{1}{4} + \frac{1}{18} + \frac{1}{17} = \frac{223}{612} < \frac{9}{20}.$$

This proves the strict bound throughout the interval, including zero. $\square$

Lemma A.2 (Centered square in an inscribed triangle). *If a triangle in $[-b, b]^2$ contains $[-a, a]^2$, with $a > 0$, then $b \geq 2a$.*

Proof. Write its three barycentric functions as $\lambda_s(u) = c_s + d_s \cdot u$. They are nonnegative on $[-a, a]^2$, so $c_s \geq a\|d_s\|_1$ and $\sum_s c_s = 1$. If the vertices are v_s , affine reconstruction gives $\sum_s v_s d_s^T = I_2$. Taking traces yields $2 \leq b \sum_s \|d_s\|_1 \leq b/a$. $\square$

This necessary bound is sufficient for our contradiction; no claim of its attainability in dimension two is used. The required two-dimensional trace argument has been proved here; its general-dimensional geometric background is recorded in [13].

Theorem A.3 (Exact incompatible minima and their Pareto frontier). *For every $9/10 \leq h \leq 1$, both cuts of (18) separately admit an exact three-state realization and have ordinary normalized residual rank three. No realization has $(K_1, K_2) = (3, 3)$. The entire feasible profile region is*

$$(20) \quad \{(K_1, K_2) : K_1 \geq 3, K_2 \geq 4\} \cup \{(K_1, K_2) : K_1 \geq 4, K_2 \geq 3\}.$$

Thus the exact Pareto-minimal profiles are $(3, 4)$ and $(4, 3)$, and the minimum full peak is four. For $h < 1$ the specified array has full support. At $h = 9/10$ all answer probabilities belong to $[1/20, 19/20]$.

Proof. The four rows at the first cut vary independently through the last two second coordinates of (18). They form a nondegenerate affine square, hence have normalized row rank three. At the second cut the response vectors lie in $\mathbb{R}^2$ and include three affinely independent points $(h, h), (h, -h), (-h, 0)$. Its normalized row rank is also three. Factoring through a stochastic register cannot increase rank, so both state counts are at least three.

Here are matching one-sided constructions. Put $a = h/4$. At the first cut, three continuation vectors with free coordinate pairs

$$(21) \quad (-a, -a), \quad (3a, -a), \quad (-a, 3a)$$

and the fixed coordinates from (18) enclose the four actual rows. The triangle is legal because $3a \leq 3/4$. The barycentric encoder weights for (ax_1, ax_2) are

$$(\lambda_0, \lambda_1, \lambda_2) = \left(\frac{2 - x_1 - x_2}{4}, \frac{1 + x_1}{4}, \frac{1 + x_2}{4} \right).$$

After j , represent the resulting two-dimensional mean in the convex hull of the four corners of $[-1, 1]^2$. For a mean (u, v) the weights on $(\varepsilon_1, \varepsilon_2)$ are $(1 + \varepsilon_1 u)(1 + \varepsilon_2 v)/4$. The decoder returns the queried corner coordinate. This is a single machine of profile $(3, 4)$.

For profile $(4, 3)$, first retain x . Every actual second-cut mean lies in the triangle with vertices $h(1, 1), h(1, -1), h(-1, 0)$. At a mean (u, v) its weights on those vertices are

$$(22) \quad \frac{h + u + 2v}{4h}, \quad \frac{h + u - 2v}{4h}, \quad \frac{h - u}{2h}.$$

For the last two columns they are respectively $(1/4, 0, 3/4)$ or $(0, 1/4, 3/4)$; for the first three columns they are the three unit vectors. These rows followed by the binary decoder prove the second construction. All tables are rational when h is rational.

Suppose now that a $(3, 3)$ machine exists. Let G_1, G_2, G_3 be its full first-cut continuation arrays. Since their convex hull contains an affine square of dimension two, their affine span equals the affine span of the four actual rows. Therefore each G_s has the first three columns of (18) unchanged and has last columns $(-h/2, u_s)$ and $(-h/2, v_s)$ for some u_s, v_s . The first encoder then forces

$$(23) \quad [-a, a]^2 \subseteq \text{conv}\{(u_s, v_s) : 1 \leq s \leq 3\}.$$

This affine-slice deduction allows arbitrary state continuations; it neither assumes that they are residuals nor restricts them beforehand to the residual hull.

The three states at the second cut have decoder mean vectors $D_1, D_2, D_3 \in [-1, 1]^2$. Their triangle P contains all actual second-cut rows, and hence the three points in Lemma A.1. Every shifted continuation of each G_s must also lie in P , because the update after j is a common stochastic row into these same three decoder states. In particular, $(-h/2, u_s), (-h/2, v_s) \in P$. Lemma A.1 gives $|u_s|, |v_s| \leq \beta(1 - h) < h/2 = 2a$. This contradicts Lemma A.2 applied to (23).

No $(3, 3)$ realization exists. Padding the two constructions proves (20). The full-support and probability bounds follow immediately from the displayed means. $\square$

REFERENCES

- [1] L. Benvenuti and L. Farina, A tutorial on the positive realization problem, *IEEE Trans. Automat. Control* **49** (2004), 651–664.
- [2] S. Basu, R. Pollack and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, 2nd ed., Algorithms and Computation in Mathematics 10, Springer, 2006.
- [3] F. Denis and Y. Esposito, On rational stochastic languages, *Fund. Inform.* **86** (2008), 41–77; full preprint arXiv:cs/0602093 (2006).
- [4] Y. Esposito, A. Lemay, F. Denis and P. Dupont, Learning probabilistic residual finite state automata, *Grammatical Inference: Algorithms and Applications*, LNCS 2484, Springer, 2002, 77–91, doi:10.1007/3-540-45790-9_7.
- [5] N. Gillis and F. Glineur, On the geometric interpretation of the nonnegative rank, *Linear Algebra Appl.* **437** (2012), 2685–2712.
- [6] A. Heller, On stochastic processes derived from Markov chains, *Ann. Math. Statist.* **36** (1965), 1286–1291.
- [7] J. Lumbreras, H. Ma, J. Thompson and M. Gu, An irreducible quantum advantage in aligning world models with reality, arXiv:2608.19779v1 (2026).
- [8] A. Monràs and A. Winter, Quantum learning of classical stochastic processes: The completely-positive realization problem, *J. Math. Phys.* **57** (2016), 015219; arXiv:1412.3634.
- [9] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28.
- [10] M. Pharr, W. Jakob and G. Humphreys, *Physically Based Rendering: From Theory to Implementation*, 4th ed., MIT Press, 2023, Appendix A.2, Reservoir sampling.
- [11] Y. Qi, P. Comon and L.-H. Lim, Semialgebraic geometry of nonnegative tensor rank, arXiv:1601.05351v3 (2016), Lemma 4.1.
- [12] M. Vidyasagar, The complete realization problem for hidden Markov models: a survey and some new results, *Math. Control Signals Systems* **23** (2011), 1–65.
- [13] Q. Qi, General Theta Foundations I: Compatibility of positive memory and online simplex synthesis, revision 31, September 25, 2026, repository manuscript, publication 90dc2c8f0c75e2cfbb95cc61b247ce4d5e216eb9.

PRESERVED MATHEMATICAL DEVELOPMENT

General Theta Foundations I - revision 31

The complete current article precedes this divider.

The following predecessor cumulative volume is reproduced without alteration.

Frozen predecessor publication:

90dc2c8f0c75e2cfbb95cc61b247ce4d5e216eb9

This archive is not part of the compact referee package. Independent historical analytic obligations keep their original hypotheses and proof status.

GENERAL THETA FOUNDATIONS I: COMPATIBILITY OF POSITIVE MEMORY AND ONLINE SIMPLEX SYNTHESIS

QIAN QI

ABSTRACT. We distinguish the separate positive realization of continuation arrays from their simultaneous realization by a finite causal machine. In the clocked atomic stochastic-row model, canonical affine sections yield a simultaneously rank-minimal machine whenever those sections are simplices; in particular, full-support arrays of cut rank at most two have no compatibility obstruction. We construct a full-support family of cut rank three whose two individual minimum state counts are both three, but whose exact Pareto-minimal profiles are $(3, 4)$ and $(4, 3)$. Thus the compatibility obstruction occurs already at rank three and persists away from support boundaries. In the opposite direction, every additive stochastic encoder admits an online realization with no increase in its label alphabet. Consequently every simplex enclosing a contracted cube gives an online query encoder with the same number of states. This transfers all classical critical cube-simplex inclusions, including the non-Hadamard dimensions five and nine, to optimal online realizations. We also identify exact conditional query synthesis with classical worst-case random-access geometry at a checkpoint, and separate that equivalence from the cost of computing an encoder. Explicit rational tables, a finite fair-bit sampling ledger, and internal stochastic-language conventions complete the realization statements.

1. INTRODUCTION

A finite statistical response law may have a small positive factorization at every cut and nevertheless require a larger state space when those factorizations must be executed in succession. The difficulty is not linear dimension alone: the generators selected at one cut determine continuations which must be available at the next. Conversely, some families have enough additive structure that a static factorization already specifies a lossless online implementation. This paper proves both statements in one precise model.

Our first main result is the family in Theorem 3.3. An input with four possible values is followed by a five-valued command and a binary query. Only the final binary answer is released. Each of the two internal cuts separately has minimum state count three. No machine has three states at both cuts, even though all prescribed answer probabilities can be bounded between $1/20$ and $19/20$. The profiles $(3, 4)$ and $(4, 3)$ are both realized explicitly. The proof excludes arbitrary stochastic encoders and decoders, not merely residual-state machines or a chosen architecture. It uses a forced affine slice at the first cut and a quantitative triangle constraint at the second.

The contrast is sharp in cut rank. Proposition 2.1 shows that the intersection of the normalized causal polytope with the affine span of the actual residuals is closed under normalized shifts. If these canonical sections are simplices, their vertices provide a simultaneously optimal machine (Theorem 2.2). All nonempty one-dimensional sections are intervals, so full-support cut ranks at

Date: September 25, 2026. Revision 31.

2020 Mathematics Subject Classification. Primary 68Q45, 52B05; Secondary 94A15, 62B05, 93E20.

Key words and phrases. Causal realization, stochastic automata, nonnegative rank, cube absorption, random access codes, compatible memory.

most two cannot exhibit the preceding failure. This criterion uses an intrinsic polytope, not unknown transition coefficients. The invariant-cone and residual-realization mechanisms have classical antecedents [5, 10, 2]; the explicit rank-three obstruction is stated separately from them.

Our second main result is Theorem 4.1. A stochastic encoder on a product input alphabet which is additive in its input coordinates can be computed by a finite-state replacement process using exactly the same label alphabet. There is no retained coordinate selector or common random seed. An enclosing simplex has affine barycentric coordinates, so every static simplex encoder for the binary-query channel

$$(1) \quad \mathbb{P}(B = b \mid x, j) = \frac{1 + b\eta x_j}{2}, \quad x \in \{-1, 1\}^n, \quad j \in [n], \quad b \in \{-1, 1\},$$

has an online realization of the same peak. In particular, in every dimension and at every nonzero signal,

$$(2) \quad K_n(\eta) = n + 1 \iff W_n^{\text{fo}}(\eta) = n + 1 \iff W_n^{\text{ad}}(\eta) = n + 1.$$

The quantities on the right count all acquisition cuts; K_n constrains only the cut preceding the query. The proof of the first implication works for every enclosing simplex, not only orthogonal or Hadamard ones.

The static critical inequality $\eta \leq 1/n$ is classical cube–simplex geometry. We state its relationship to the absorption index explicitly. Nevskii and Ukhalov’s critical simplices in dimensions five and nine [7] give, by the online theorem, six- and ten-state implementations at $\eta = 1/5$ and $\eta = 1/9$. We display the rational vertices and the replacement distributions, so this consequence does not rely on a hypothetical static-to-dynamic transfer. Critical simplex matrices are also closed under a normalized tensor construction, giving further explicit online families without invoking unproved Hadamard existence. The static inclusions are not claimed as new.

A further distinction can be settled exactly. The minimum checkpoint alphabet for (1) equals that for classical private-randomness random access with worst-case success at least $(1 + \eta)/2$. This follows already from the geometric characterization and cube-containment argument of Kondo et al. [6, Theorem 14, Lemma 15]. We give the short normalization and exactification proof in Section 6. No extra persistent symmetrization seed is required. Therefore exact conditional rows do not themselves establish a different checkpoint exponent from classical random-access coding [1]. The online implementation, not that exponent, is the object of the present synthesis theorem.

Notation	Meaning
$\mathcal{R}_t, \mathcal{P}_t$	actual normalized residuals, all legal future arrays
L_t, r_t	canonical affine section, affine dimension plus one
(K_1, K_2)	simultaneous register bounds at two designated cuts
$K_n(\eta)$	query-cut minimum, with unrestricted upstream memory
$W_n^{\text{fo}}, W_n^{\text{ad}}$	full acquisition peak: fixed or adaptive order
$\xi(S)$	cube absorption by homothety about the simplex centroid

A state count means the number of persistent labels. The primary model has exact atomic stochastic rows and an externally declared clock. Program descriptions and row-construction time are not part of that count. Section 7 gives a separate finite fair-bit implementation with its additional phase and sampling registers. A query is a current input, not a stored word; a model requiring a query buffer must charge it. These conventions are used consistently in both upper and lower bounds.

2. CANONICAL AFFINE SECTIONS OF A CAUSAL ARRAY

We use finite alphabets and a finite horizon. At a controller stage an action is chosen before its next report is received. A terminal answer is an action with a dummy following report. A complete normalized causal array $C(\mathbf{a} \mid \mathbf{y})$ prescribes the conditional law of all actions for every externally supplied report word. Nonanticipation means that the marginal of an action prefix depends only on reports preceding that prefix. Intermediate buffers that must survive an update are included as charged cuts. The rows may depend on the declared time, but not on an unrecorded history, schedule variable or persistent random seed.

At a cut t , let $\mathcal{P}_t$ be the polytope of normalized causal arrays on the remaining interface. For a controller prefix h whose probability, with past reports fixed, is positive, let $R_h \in \mathcal{P}_t$ be its normalized future array. Remove duplicate rows to form the finite residual set $\mathcal{R}_t$. This is a property of C , not of a chosen realization. We require the full array, including unsupported environmental inputs; an almost-sure partial specification is a different completion problem. Set

$$(3) \quad A_t = \text{aff } \mathcal{R}_t, \quad L_t = A_t \cap \mathcal{P}_t, \quad r_t = \dim A_t + 1.$$

The slice L_t is a compact polytope of dimension $r_t - 1$, since it contains $\mathcal{R}_t$. Normalization places A_t in a hyperplane not containing zero, so r_t is also the ordinary row rank of the normalized residuals.

If R is a future array, write $p_R(a)$ for its next-action probability. It is independent of the next report y . For $p_R(a) > 0$, denote by $R^{a,y}$ the normalized continuation after that action and report. For an actual residual this is another actual residual at the child cut.

Proposition 2.1 (Affine slices are closed under positive shifts). *If $G \in L_t$ and $p_G(a) > 0$, then $G^{a,y} \in L_{t+1}$ for every next report y .*

Proof. Write $G = \sum_h c_h R_h$ with real coefficients satisfying $\sum_h c_h = 1$. Taking an action marginal gives $p_G(a) = \sum_h c_h p_{R_h}(a)$. Restriction to (a, y) is linear in the unnormalized arrays, hence

$$G^{a,y} = \sum_{h: p_{R_h}(a) > 0} \frac{c_h p_{R_h}(a)}{p_G(a)} R_h^{a,y}.$$

The coefficients sum to one. They need not be positive: the display asserts membership in the affine span at the child, not a convex realization. Since $G \in \mathcal{P}_t$ and $p_G(a) > 0$, its normalized shift is a valid nonnegative causal array. Thus it lies in both factors of (3) at the child cut. $\square$

Theorem 2.2 (A canonical sufficient criterion and the rank-two case). *If every L_t is a simplex, a single stochastic clocked machine realizes C with exactly r_t labels at cut t , simultaneously. No realization can use fewer than r_t at that cut. In particular, every complete finite causal array with $r_t \leq 2$ at every cut has a simultaneously rank-minimal realization. This includes full-support arrays with arbitrary finite action and report alphabets.*

Proof. In any realization, conditioning on a prefix expresses R_h as a convex combination of the state continuations at cut t . Its affine dimension therefore forces at least r_t labels.

Choose the vertices of L_t as the states at cut t . Every actual residual is their convex combination. From vertex G , choose the next action with probability $p_G(a)$. After the next report, express $G^{a,y} \in L_{t+1}$ as a convex combination of the child vertices and use those weights as the update row. An action of probability zero may have an arbitrary unused update. Proposition 2.1 ensures that this construction is possible, and the action probability is independent of the unread report. Initialization uses the barycentric representation of C . Backward induction verifies the entire specified future law from every vertex state. A simplex of dimension $r_t - 1$ has exactly r_t vertices, proving the claim. A zero-dimensional slice is a point and a compact one-dimensional slice is a segment, which proves the last assertion. $\square$

This criterion differs from requiring the residual hull itself to be a simplex: internal vertices can lie outside that hull. It also differs from searching for unknown enclosing simplices and unknown shifts simultaneously. The polytope L_t is fixed by linear equations and inequalities from the input array; its normalized-shift closure is proved, not an additional feasibility hypothesis. For fully listed rational arrays, vertices and barycentric rows can be found by rational linear algebra and finite polyhedral enumeration. The explicit array may already be exponentially large in the horizon. No succinct-input polynomial bound, autonomous row-reuse theorem, or universal equality between rank and state complexity follows.

The next section shows that the rank-two conclusion is sharp as a uniform statement about the maximum cut rank. At rank three, separate rank-optimal enclosures need not be causally compatible even for a full-support family.

3. A FULL-SUPPORT OBSTRUCTION AT RANK THREE

The following experiment has only two internal memory cuts. The first input is $x = (x_1, x_2) \in \{-1, 1\}^2$, received as one symbol. A command $j \in \{1, 2, 3, 4, 5\}$ is then received, followed by a query $k \in \{1, 2\}$. The machine emits $B \in \{-1, 1\}$ in the update receiving k . Let K_1 count states after x and K_2 states after j ; a retained final answer has two labels. No further access to x or j is permitted. For $0 < h \leq 1$ prescribe the conditional mean vectors

$$(4) \quad \begin{array}{c|ccccc} j & 1 & 2 & 3 & 4 & 5 \\ \hline z_{x,j} & (h, h) & (h, -h) & (-h, 0) & (-h/2, hx_1/4) & (-h/2, hx_2/4) \end{array}.$$

The answer law is $\mathbb{P}(B = b \mid x, j, k) = (1 + bz_{x,j}(k))/2$.

The program is clocked and uses arbitrary stochastic rows. Thus every allowed implementation has common rows $E(s \mid x)$, $T(t \mid s, j)$ and $D(b \mid t, k)$; all past randomness is in s or t .

We first record the geometric estimate used by the converse.

Lemma 3.1 (A quantitative forced triangle). *Put $h = 1 - \delta$, where $0 \leq \delta \leq 1/10$. If a triangle $P \subset [-1, 1]^2$ contains (h, h) , $(h, -h)$ and $(-h, 0)$, then every point $(-h/2, y) \in P$ satisfies*

$$(5) \quad |y| \leq \beta(\delta) := \frac{1}{4} + \frac{\delta}{2(1-\delta)} + \frac{\delta}{2-3\delta} < \frac{h}{2}.$$

Proof. A convex representation of (h, h) includes a vertex A for which $2 - A_1 - A_2 \leq 2\delta$, so $A_1, A_2 \geq 1 - 2\delta$. A representation of $(h, -h)$ similarly includes a vertex B with $B_1 \geq 1 - 2\delta$ and $B_2 \leq -1 + 2\delta$. These vertices are distinct. Call the third vertex D . In a representation of $(-h, 0)$, let u be the total weight on A, B . Since their first coordinates are at least $1 - 2\delta$ and $D_1 \geq -1$,

$$-1 + \delta \geq -1 + 2(1 - \delta)u, \quad u \leq \frac{\delta}{2(1 - \delta)}.$$

The zero second coordinate implies $|D_2| \leq u/(1 - u) \leq \delta/(2 - 3\delta)$. For a point of P with first coordinate $-h/2$, the total weight v on A, B satisfies

$$-h/2 \geq -1 + 2hv, \quad v \leq \frac{1 - h/2}{2h} = \frac{1}{4} + \frac{\delta}{2h}.$$

Its second coordinate has absolute value at most $v + (1 - v)|D_2| \leq \beta(\delta)$. Finally the two fractions in β increase with δ while $h/2$ decreases, and

$$\beta(1/10) = \frac{1}{4} + \frac{1}{18} + \frac{1}{17} = \frac{223}{612} < \frac{9}{20}.$$

This proves the strict bound throughout the interval, including zero. $\square$

Lemma 3.2 (Centered square in an inscribed triangle). *If a triangle in $[-b, b]^2$ contains $[-a, a]^2$, with $a > 0$, then $b \geq 2a$.*

Proof. Write its three barycentric functions as $\lambda_s(u) = c_s + d_s \cdot u$. They are nonnegative on $[-a, a]^2$, so $c_s \geq a\|d_s\|_1$ and $\sum_s c_s = 1$. If the vertices are v_s , affine reconstruction gives $\sum_s v_s d_s^T = I_2$. Taking traces yields $2 \leq b \sum_s \|d_s\|_1 \leq b/a$. $\square$

This necessary bound is sufficient for our contradiction; no claim of its attainability in dimension two is used. Its general-dimensional version belongs to classical cube-absorption geometry; see Section 5.

Theorem 3.3 (Exact incompatible minima and their Pareto frontier). *For every $9/10 \leq h \leq 1$, both cuts of (4) separately admit an exact three-state realization and have ordinary normalized residual rank three. No realization has $(K_1, K_2) = (3, 3)$. The entire feasible profile region is*

$$(6) \quad \{(K_1, K_2) : K_1 \geq 3, K_2 \geq 4\} \cup \{(K_1, K_2) : K_1 \geq 4, K_2 \geq 3\}.$$

Thus the exact Pareto-minimal profiles are $(3, 4)$ and $(4, 3)$, and the minimum full peak is four. For $h < 1$ the specified array has full support. At $h = 9/10$ all answer probabilities belong to $[1/20, 19/20]$.

Proof. The four rows at the first cut vary independently through the last two second coordinates of (4). They form a nondegenerate affine square, hence have normalized row rank three. At the second cut the response vectors lie in $\mathbb{R}^2$ and include three affinely independent points $(h, h), (h, -h), (-h, 0)$. Its normalized row rank is also three. Factoring through a stochastic register cannot increase rank, so both state counts are at least three.

Here are matching one-sided constructions. Put $a = h/4$. At the first cut, three continuation vectors with free coordinate pairs

$$(7) \quad (-a, -a), \quad (3a, -a), \quad (-a, 3a)$$

and the fixed coordinates from (4) enclose the four actual rows. The triangle is legal because $3a \leq 3/4$. The barycentric encoder weights for (ax_1, ax_2) are

$$(\lambda_0, \lambda_1, \lambda_2) = \left(\frac{2 - x_1 - x_2}{4}, \frac{1 + x_1}{4}, \frac{1 + x_2}{4} \right).$$

After j , represent the resulting two-dimensional mean in the convex hull of the four corners of $[-1, 1]^2$. For a mean (u, v) the weights on $(\varepsilon_1, \varepsilon_2)$ are $(1 + \varepsilon_1 u)(1 + \varepsilon_2 v)/4$. The decoder returns the queried corner coordinate. This is a single machine of profile $(3, 4)$.

For profile $(4, 3)$, first retain x . Every actual second-cut mean lies in the triangle with vertices $h(1, 1), h(1, -1), h(-1, 0)$. At a mean (u, v) its weights on those vertices are

$$(8) \quad \frac{h + u + 2v}{4h}, \quad \frac{h + u - 2v}{4h}, \quad \frac{h - u}{2h}.$$

For the last two columns they are respectively $(1/4, 0, 3/4)$ or $(0, 1/4, 3/4)$; for the first three columns they are the three unit vectors. These rows followed by the binary decoder prove the second construction. All tables are rational when h is rational.

Suppose now that a $(3, 3)$ machine exists. Let G_1, G_2, G_3 be its full first-cut continuation arrays. Since their convex hull contains an affine square of dimension two, their affine span equals the affine span of the four actual rows. Therefore each G_s has the first three columns of (4) unchanged and has last columns $(-h/2, u_s)$ and $(-h/2, v_s)$ for some u_s, v_s . The first encoder then forces

$$(9) \quad [-a, a]^2 \subseteq \text{conv}\{(u_s, v_s) : 1 \leq s \leq 3\}.$$

This affine-slice deduction allows arbitrary state continuations; it neither assumes that they are residuals nor restricts them beforehand to the residual hull.

The three states at the second cut have decoder mean vectors $D_1, D_2, D_3 \in [-1, 1]^2$. Their triangle P contains all actual second-cut rows, and hence the three points in Lemma 3.1. Every shifted continuation of each G_s must also lie in P , because the update after j is a common stochastic row into these same three decoder states. In particular, $(-h/2, u_s), (-h/2, v_s) \in P$. Lemma 3.1 gives $|u_s|, |v_s| \leq \beta(1-h) < h/2 = 2a$. This contradicts Lemma 3.2 applied to (9). No (3,3) realization exists. Padding the two constructions proves (6). The full-support and probability bounds follow immediately from the displayed means. $\square$

For $h < 1$ all residual minimal faces at each cut coincide with the full terminal-channel face. Thus the support-component rank is just three at each cut; support separation is not the source of the failure. The obstruction is simultaneous positivity of the shifted generators. Theorem 2.2 and Theorem 3.3 together locate the first possible maximum cut rank for this phenomenon at three. They also distinguish an actual obstruction from a bilinear feasibility reformulation: the complete feasible profile set is determined here.

4. ADDITIVE ENCODERS ADMIT LOSSLESS ONLINE SYNTHESIS

The preceding obstruction involves two successive nonlinear choices of positive generators. Additivity provides a different situation in which the encoding stage can always be made causal without enlarging its alphabet.

Let $X_1, \dots, X_n$ be finite input sets and $S = [m]$ a label set. An encoder is a map $\lambda : \prod_i X_i \rightarrow \Delta_m$. Call it additive if there are vectors $a \in \mathbb{R}^m$ and $f_i(x_i) \in \mathbb{R}^m$ such that

$$(10) \quad \lambda(x) = a + \sum_{i=1}^n f_i(x_i).$$

The summands need not be probability vectors or have positive entries. All that is assumed is $\lambda(x) \in \Delta_m$ for every product input.

Theorem 4.1 (Additive encoding without a persistent selector). *Every encoder (10) has an exact fixed-order streaming realization with at most m states after each input. This holds for every externally prescribed permutation of the inputs. The construction uses only scalar replacement probabilities and fresh input-dependent draws on $[m]$; it never retains the index of the chosen summand. Rational encoder data give rational transition rows.*

Proof. For each label s , put

$$\ell_{i,s} = \min_{x_i \in X_i} f_{i,s}(x_i), \quad c_s = a_s + \sum_i \ell_{i,s}, \quad g_{i,s}(x_i) = f_{i,s}(x_i) - \ell_{i,s}.$$

Because the input is a full Cartesian product, each scalar minimum can be attained independently. Consequently $c_s = \min_x \lambda_s(x) \geq 0$, and $g_i(x_i)$ is nonnegative. Normalization of (10), with all other coordinates fixed, shows that $\sum_s f_{i,s}(x_i)$ is independent of x_i . Therefore

$$w_i = \sum_s g_{i,s}(x_i) \geq 0$$

is constant in x_i . With $w_0 = \sum_s c_s$, one has $w_0 + \sum_i w_i = 1$. Delete zero-weight summands. For the others define $q_0 = c/w_0$ and $q_i(x_i) = g_i(x_i)/w_i$, so

$$(11) \quad \lambda(x) = w_0 q_0 + \sum_i w_i q_i(x_i).$$

Let $W_t = w_0 + \sum_{i \leq t} w_i$. If $w_0 > 0$, initialize from q_0 . At the first positive-weight input when $W_{t-1} = 0$, simply draw from $q_t(x_t)$; before it a dummy label suffices. When $W_t > 0$, retain the

old label with probability W_{t-1}/W_t and otherwise replace it by a fresh draw from $q_t(x_t)$. A zero-weight input leaves the label unchanged. Induction gives the state law

$$\mathbb{P}(S_t = s \mid x_{1:t}) = \frac{w_0 q_0(s) + \sum_{i \leq t} w_i q_i(s \mid x_i)}{W_t}.$$

At $t = n$ the denominator is one and the law is exactly $\lambda(x)$. Equivalently, each update is the single atomic row

$$(12) \quad T_{t,x_t}(s, s') = \frac{W_{t-1}}{W_t} \mathbf{1}_{\{s=s'\}} + \frac{w_t}{W_t} q_t(s' \mid x_t).$$

Only s' persists. The weights and their prefix sums are known program data associated with the external clock. The same construction works in every prescribed permutation. Minima and arithmetic preserve rationality. $\square$

Corollary 4.2 (Rank-saturating additive channels). *Let C_x be a finite family of normalized terminal channels, additive in the product coordinates of x , with affine dimension $r - 1$. If C_x can be represented through r legal normalized generators at a checkpoint, then it can be acquired sequentially with at most r labels at every input cut, for any fixed order. Thus its checkpoint and acquisition-peak minima agree at r . If a final output must be retained, the complete peak is at most $\max(r, |B|)$ for output alphabet B .*

Proof. A convex representation through r generators enclosing a set of affine dimension $r - 1$ makes those generators affinely independent. Their barycentric coordinates are affine functions of C_x , hence additive in x . They are stochastic on all product inputs. Apply Theorem 4.1, then the unchanged terminal decoder. The affine-dimension lower bound holds for every acquisition method because the complete input must pass through the checkpoint label. $\square$

For (1), a decoder state has a mean vector $v_s \in [-1, 1]^n$. Conditioning on the state gives

$$(13) \quad \eta x = \sum_s E(s \mid x) v_s.$$

Consequently $K_n(\eta)$ is the smallest number of outer-cube points whose convex hull contains $\eta[-1, 1]^n$. This checkpoint formulation is classical random-access geometry in the normalization of Section 6. An adaptive acquisition schedule still produces one distribution $E(\cdot \mid x)$, with no past-dependent input supplied to the decoder, so it obeys the same lower bound.

Theorem 4.3 (Every simplex encoder has the same optimal online peak). *Let $n \geq 1$ and $\eta > 0$. If a simplex with vertices $v_0, \dots, v_n \in [-1, 1]^n$ contains $\eta[-1, 1]^n$, those very decoder vertices admit an exact online encoder with at most $n + 1$ states. Hence (2) holds for every n and η . No hypothesis concerning a Hadamard matrix is required.*

Proof. Write the simplex's affine barycentric coordinates as $\lambda_s(z) = a_s + b_s \cdot z$. Then $\lambda_s(\eta x) = a_s + \eta \sum_i b_{s,i} x_i$ is an additive stochastic encoder on the product cube. Theorem 4.1 computes it without expanding the label set. The unchanged binary decoder with mean $v_{s,j}$ gives (1). Conversely, $K_n(\eta) \leq W_n^{\text{ad}}(\eta) \leq W_n^{\text{fo}}(\eta)$ and $K_n(\eta) \geq n + 1$ by affine dimension. A final binary output requires no more than $n + 1$ labels. $\square$

The theorem concerns the peak, not simultaneous optimality of the individual $t + 1$ prefix bounds. Its sampler can occupy all $n + 1$ labels at an early cut. Nor does it assert online equality for arbitrary nonsimplicial checkpoint codebooks: their barycentric encoders need not be affine. The rank-three example in Section 3 shows why general static factorizations cannot be promoted to joint realizations without additional structure.

5. CRITICAL CUBE ABSORPTION AND NON-HADAMARD ONLINE ENCODERS

We separate the classical static geometry from the online consequence. Let $Q_n = [0, 1]^n$. For a nondegenerate simplex $S \subset Q_n$, let

$$\xi(S) = \inf\{\rho > 0 : Q_n \subset c(S) + \rho(S - c(S))\}, \quad \xi_n = \inf_{S \subset Q_n} \xi(S),$$

where $c(S)$ is its centroid. The universal lower bound $\xi_n \geq n$, the Hadamard equality construction, the axial-diameter formulas and the non-Hadamard equalities $\xi_5 = 5$, $\xi_9 = 9$ belong to the classical cube-absorption literature; Nevskii and Ukhlov give the needed results and antecedents in [7]. Their Theorem 2 proves that equality in $S \subset Q \subset nS$, with homothety about the simplex centroid, forces the centroids to coincide. Thus a critical simplex in their normalization gives, under $v = 2u - \mathbf{1}$, precisely

$$(14) \quad n^{-1}[-1, 1]^n \subset \text{conv}\{v_0, \dots, v_n\} \subset [-1, 1]^n.$$

Away from the critical value, centering at the origin and centering at the simplex centroid must not be silently identified.

For reference, the centered obstruction has a short proof. If the barycentric coordinates are $a_s + b_s \cdot z$, containment of $\eta[-1, 1]^n$ gives $a_s \geq \eta \|b_s\|_1$ and $\sum_s a_s = 1$. Reconstruction gives $\sum_s v_s b_s^T = I_n$, whence

$$(15) \quad n = \sum_{s,i} v_{s,i} b_{s,i} \leq \sum_s \|b_s\|_1 \leq \eta^{-1}.$$

This is a proof of a classical inequality in the present notation, not a new static threshold. The extra conclusion is that every feasible simplex, critical or not, has the same online peak by Theorem 4.3.

Proposition 5.1 (Explicit online rows for a critical simplex). *For any simplex satisfying (14), write its barycentric functions as $a_s + \sum_i b_{s,i} z_i$. Then*

$$(16) \quad a_s = \frac{1}{n+1} = \frac{1}{n} \sum_i |b_{s,i}|, \quad \sum_s |b_{s,i}| = 1, \quad \sum_s b_{s,i} = 0.$$

The distributions

$$(17) \quad q_i(s \mid x_i) = |b_{s,i}| + b_{s,i} x_i$$

and updates

$$(18) \quad T_{t,x_t}(s, s') = \frac{t-1}{t} \mathbf{1}_{\{s=s'\}} + \frac{1}{t} q_t(s' \mid x_t)$$

realize the critical query channel with $n+1$ states. Multiplying the final decoder means by $n\eta$ gives every $0 < \eta \leq 1/n$ at the same peak. The origin of a replacement draw is not retained.

Proof. Every inequality in (15) is an equality at $\eta = 1/n$. Thus $a_s = n^{-1} \|b_s\|_1$ for each s , and $v_{s,i} b_{s,i} = |b_{s,i}|$ for each pair (s, i) . Evaluating the s th barycentric function at its own vertex gives $1 = a_s + \|b_s\|_1 = (n+1)a_s$, so the centroid is zero. The i th diagonal entry of $\sum_s v_s b_s^T = I_n$ yields the second identity; barycentric normalization yields the third. Hence (17) is a probability vector. Induction in (18) gives

$$\mathbb{P}(S_n = s \mid x) = \frac{1}{n} \sum_i q_i(s \mid x_i) = a_s + \frac{1}{n} \sum_i b_{s,i} x_i.$$

This is exactly the barycentric distribution of x/n . Its decoder mean is x_j/n , and the claimed attenuation completes the proof. Unlike a Hadamard construction, an individual q_i need not have mean vector $x_i e_i$; the required cancellation holds after averaging all coordinates. $\square$

5.1. Two rational non-Hadamard implementations. The following vertices are taken from [7, Theorem 4 and Section 9], in its $[0, 1]^n$ normalization. They are reproduced to specify the actual online rows, not as new static constructions. In dimension five the six rows of U_5 are

$$(19) \quad U_5 = \begin{pmatrix} 1/2 & 1 & 1/3 & 1 & 1 \\ 1/2 & 0 & 1/3 & 1 & 1 \\ 1/2 & 1/2 & 1/3 & 0 & 1 \\ 1/2 & 1/2 & 0 & 1/3 & 0 \\ 0 & 1/2 & 1 & 1/3 & 0 \\ 1 & 1/2 & 1 & 1/3 & 0 \end{pmatrix}.$$

In dimension nine, choose the parameter $t = 1/2$ in their family:

$$(20) \quad U_9 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1/2 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1/2 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1/2 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1/2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1/2 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1/2 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1/2 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1/2 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Put $V_d = 2U_d - \mathbf{1}$ entrywise and $M_d = [\mathbf{1} \ V_d]$. The inverse gives $\lambda_s(z) = (1, z^T)(M_d^{-1})_{\cdot s}$; its first row is a_s and its other rows are $b_{s,i}$. Direct rational multiplication gives

$$(21) \quad \begin{array}{c|ccc} d & |\det M_d| & a_s & \sum_i |b_{s,i}| \\ \hline 5 & 32 & 1/6 & 5/6 \\ 9 & 12800 & 1/10 & 9/10 \end{array}$$

and $V_d^T(b_{s,i})_{s,i} = I_d$, $V_d^T a = 0$. Thus the displayed matrices also verify (14) directly: all barycentric coordinates are nonnegative on $d^{-1}[-1, 1]^d$, sum to one and reconstruct z . Equations (17)–(18) are full transition formulas requiring only these finite rational inputs.

Corollary 5.2 (Online optimality beyond Hadamard orders). *For $d = 5$ and $d = 9$ and every $\eta > 0$,*

$$K_d(\eta) = W_d^{\text{ad}}(\eta) = W_d^{\text{fo}}(\eta) = d + 1 \iff \eta \leq 1/d.$$

The matrices (19)–(20) and (18) give rational acquisition rows. No Hadamard matrix of order six or ten is used or needed.

Proof. Apply Proposition 5.1 to the two matrices, and use the dimension lower bound and (15) for the converse. $\square$

In the terminology of [7], the five-dimensional seed is perfect; their nine-dimensional seed is critical but is not perfect. Equality of the absorption factor, not the stronger condition that every cube vertex touches a dilated face, is what the online argument uses.

5.2. Closure of normalized critical matrices. This elementary closure gives an explicit growing family of such encoders. Let $M = [\mathbf{1} \ V]$ be the augmented matrix of a centered critical simplex of dimension d , and let $m = d + 1$. Its centroid condition and (16) say that the first row of M^{-1} is $\mathbf{1}^T/m$ and every column of M^{-1} has absolute sum one. Conversely, these two conditions together with $V \in [-1, 1]^{m \times d}$ and invertibility imply the critical inclusion by barycentric positivity.

Proposition 5.3 (Normalized tensor closure). *If M and N satisfy the preceding conditions, so does $M \otimes N$ with its all-ones column placed first. It gives a critical simplex of dimension $(d+1)(e+1) - 1$. Thus tensor products of the rational seeds of sizes 2, 6 and 10 give explicitly specified optimal online simplex encoders whenever*

$$n+1 = 2^a 6^b 10^c > 1, \quad a, b, c \text{ nonnegative integers.}$$

Proof. Entries of $M \otimes N$ are products of entries in $[-1, 1]$, so the new vertices are in the cube. The first inverse row is uniform on the product index set. Since $(M \otimes N)^{-1} = M^{-1} \otimes N^{-1}$, the absolute column sums multiply and remain one. Invertibility and the converse observation prove the critical inclusion; the online assertion follows from Proposition 5.1. The seed of size two is $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$. $\square$

No general multiplicativity of nonnegative rank is asserted. This is an explicit operation on the normalized critical matrices just defined; the static tensor algebra is not used as a priority claim.

6. EXACT QUERY SYNTHESIS AND CLASSICAL WORST-CASE CODING

Fix a finite decoder dictionary $D = \{v_1, \dots, v_m\} \subset [-1, 1]^n$, with private encoder and decoder randomness and no shared seed. A worst-case success guarantee $(1 + \eta)/2$ means that for every input x there is a probability vector E_x such that

$$x_i \sum_s E_x(s) v_{s,i} \geq \eta \quad \text{for all } i.$$

The same E_x must serve all future queries. Merely requiring average success over queries is a different condition.

Proposition 6.1 (Checkpoint exactification with no added states). *For $0 \leq \eta \leq 1$ the following are equivalent for the fixed dictionary:*

- (1) *it achieves worst-case random-access success at least $(1 + \eta)/2$;*
- (2) $\eta[-1, 1]^n \subseteq \text{conv } D$;
- (3) *it exactly realizes (1) after changing only the encoder.*

Thus the minimum checkpoint alphabet in the exact problem equals the minimum alphabet in the private-randomness worst-case coding problem. No runtime symmetrization or retained random partition is necessary.

Proof. Assume the first condition and fix $u \in \mathbb{R}^n$. Choose an input corner with $x_i = \text{sign}(u_i)$ when $u_i \neq 0$, and let $z_x = \sum_s E_x(s) v_s$. Then

$$\max_{v \in \text{conv } D} u \cdot v \geq u \cdot z_x \geq \eta \sum_i |u_i|.$$

The right side is the support function of $\eta[-1, 1]^n$. The separating hyperplane theorem gives the second condition. It gives an exact convex representation of every ηx using the same dictionary, which is the required new encoder in the third condition. The third condition implies the first immediately. $\square$

This is a translation of the geometry already present in [6, Theorem 14, Lemma 15 and equations (143)–(149)], not a new coding exponent. In their $[0, 1]^n$ normalization the dictionary is $S = (D + \mathbf{1})/2$, its directed ℓ^∞ Hausdorff error from the Boolean corners is $e(D)$, and the largest centered cube factor is

$$\eta(D) = 1 - 2e(D), \quad p_{\text{worst}}(D) = 1 - e(D) = \frac{1 + \eta(D)}{2}$$

whenever the factor is nonnegative. The factor two is essential when passing from errors to means. Their model has $m = 2^k$ messages; the proposition uses any integer m . Their Lemma 15 explicitly passes to cube containment. It is therefore not appropriate to call the exact checkpoint channel a strictly harder cardinality problem solely because it specifies equality rather than a lower bound on success.

Exactification can change a non-affine encoder into a different non-affine encoder. The proposition does not bound the storage needed to compute it online. Theorem 4.1 supplies that missing step for additive encoders, and Theorem 4.3 supplies it for every simplex dictionary. This is the precise division between the static coding theorem and the online theorem here.

The first-order entropy rate $1 - h_2((1 + \eta)/2)$ and logarithmic communication overhead are classical [1]. The preceding revision proved a fully charged blockwise exact realization with logarithmic state excess $O_\eta(\sqrt{n \log n})$, and uniform row-error results; those proofs and statements remain in the unchanged supporting article. Neither exactification nor the present online simplex theorem proves a matching second-order law for that block construction. The retained entropy theorem is not used as a new contribution of this revision.

Likewise, the optimal $(2^k - 1, k)$ classical construction and its static bound in [6, Theorems 16–17] overlap the Hadamard/Sylvester checkpoint case. The online synthesis applies to its simplex without increasing the number of messages, but does not change that existing optimality statement. Their conjecture-dependent results for other message sizes remain conjecture-dependent; none is used to infer an unconditional finite phase diagram here.

7. FINITE COINS AND STOCHASTIC-LANGUAGE CONVENTIONS

7.1. A separate fair-bit register ledger. The exact label counts above refer to atomic stochastic rows. They do not include the internal computation of a real-probability sampler. For the rational constructions a fully finite implementation is available, with an additional cost which can be stated without hiding a counter or the current input.

Proposition 7.1 (Finite fair-bit implementation of rational rows). *Consider T input stages, an at-most- m -label transducer with $m \geq 2$, and current input alphabet size at most d . Suppose every row has a common denominator for its entries of size at most $D \geq 2$. Put $b = \lceil \log_2 D \rceil$. In a self-paced protocol which supplies the next input only after the preceding update completes, the transducer has an exact fair-bit implementation using at most*

$$(22) \quad 2(T + 1)md2^b$$

states, including phase, current input, old label, sampler prefix and terminal output. Each stage uses fewer than $2b$ fair bits on average. Thus its persistent bit cost is at most

$$\lceil \log_2(T + 1) \rceil + \lceil \log_2 m \rceil + \lceil \log_2 d \rceil + b + 2.$$

This changes the implementation model and does not assert preservation of the atomic optimum in total states.

Proof. For a row of denominator $D_\rho \leq D$, let $b_\rho = \lceil \log_2 D_\rho \rceil$. Draw b_ρ fair bits to form an integer J between zero and $2^{b_\rho} - 1$. If $J \geq D_\rho$, repeat; otherwise map J through the row's integer cumulative sums to the new label. A denominator-one row is deterministic. The acceptance probability is $D_\rho/2^{b_\rho} > 1/2$, unless it is one, and the accepted integer is uniform. This proves exactness and the expectation bound.

Retain the phase, old label and current input throughout an attempted draw. The proper binary prefixes of length less than b_ρ form a tree with $2^{b_\rho} - 1 \leq 2^b - 1$ nodes. On the last bit, acceptance can move directly to the ready state containing the new label; rejection returns to the root. Hence a separate leaf or unbounded trial counter is unnecessary. There are at most

$(T + 1)m$ ready states, $Tmd(2^b - 1)$ sampler states and two final states. Their sum is at most (22). The input provider waits at an explicit ready boundary; the phase is stored, not supplied as a free data-dependent clock. $\square$

For (19) the distributions q_i have denominators dividing six, and for (20) they divide ten. Thus a row at acquisition stage t has denominator dividing $6t$ or $10t$, respectively. The critical decoders are also rational. More generally the normalized matrix tensors in Proposition 5.3 have rational denominators bounded by products of fixed seed denominators; their logarithms are $O(\log(n+1))$. The finite-coin implementation of those families therefore uses $O(\log(n+1))$ persistent bits, with explicit rational row data. This assertion concerns the critical polynomial-size matrix families, not efficient construction of a fixed-interior-signal exponential codebook.

7.2. Internal normalization of probabilistic automata.

Definition 7.2 (Proper stochastic-language automaton). On a finite alphabet Σ , a proper probabilistic automaton consists of a finite state set, an initial probability row α , nonnegative transition matrices T_a for $a \in \Sigma$, and stopping probabilities τ_s , with

$$\tau_s + \sum_{a,t} T_a(s, t) = 1.$$

Stopping emits no further symbol; a transition emits its label a . Termination is almost sure from every reachable state. The generated stochastic language and state languages are

$$L(w) = \alpha T_w \tau, \quad L_s(w) = e_s^T T_w \tau.$$

They are probability distributions on finite words. Unreachable states are removed. The automaton is residual if every reachable state language is $L_u(w) = L(uw)/L(u\Sigma^*)$ for some positive-mass prefix u .

These are the conventions used when comparing with probabilistic residual automata [3, 2]. If a nonnegative representation initially has state continuation masses $h_s \in (0, \infty)$, normalization replaces $T_a(s, t)$ by $T_a(s, t)h_t/h_s$, τ_s by τ_s/h_s , and α_s by $\alpha_s h_s$. The identity $h_s = \tau_s + \sum_{a,t} T_a(s, t)h_t$ gives stochastic rows. States with $h_s = 0$ have no contribution to any generated word and are deleted. Thus finite nonzero continuation masses can be normalized without adding states. Infinite continuation mass is not a stochastic language and is not included in this convention.

For completeness, we retain the operational separation with a proof in these internal terms. It is a result of the preceding revision, not a new succinctness-priority assertion.

Proposition 7.3 (A normalized residual/positive separation). *For $n \geq 2$ and rational $0 < \eta \leq (2n - 1)^{-1}$, define the finite language*

$$(23) \quad L(u_{x_1} \cdots u_{x_n} q_j o_b) = \frac{2^{-n}}{n} \frac{1 + b\eta x_j}{2},$$

using disjoint input, query and output symbols, and give zero mass to all other words. Every residual automaton has at least 2^n states. A proper rational positive automaton has at most $(n + 1)(n + 2)/2 + n(n + 1) + 1$ states, including phase and query storage.

Proof. The query-phase residuals are the 2^n vertices of an affine cube, with probabilities $(1 + b\eta x_j)/(2n)$ on the two-symbol words $q_j o_b$. Conditioning any proper automaton on a supported prefix expresses that residual as a convex combination of normalized state languages. Nonnegativity forces each used language to be supported on those two-symbol words. A residual of (23) after length t is concentrated on length $n + 2 - t$. Hence a residual state used here must itself be one of the query-phase cube vertices. Extremality forces a separate state for each of the 2^n vertices.

For the upper bound, generate the input bits independently and fairly. At phase t use labels $0, 1, \dots, t$. On the next input, retain the old label with probability $t/(t+1)$; otherwise replace it by $t+1$ for a positive bit and zero for a negative bit. After t inputs the probability of label $i > 0$ is $\mathbf{1}_{\{x_i=1\}}/t$ and the probability of zero is the fraction of negative bits. At phase n , associate means $v_0 = -\eta \mathbf{1}$ and $v_i = -\eta \mathbf{1} + 2n\eta e_i$. They are in the cube when $\eta \leq (2n-1)^{-1}$, and their mixture has mean ηx . Emit q_j uniformly, store (s, j) , emit o_b with mean $v_{s,j}$, and terminate. Phase tags require $\sum_{t=0}^n (t+1)$ states, query buffers require $n(n+1)$, and one state terminates. Every state language is normalized and every row is rational. Multiplication of the three stages yields (23). $\square$

8. CONSEQUENCES AND RETAINED RESULTS

The main distinction is now witnessed, rather than merely formulated. At full-support cut rank three, independent minimum positive factorizations need not fit one machine; the exact profile set (6) exhibits the cost of that incompatibility. At cut rank at most two, canonical affine sections always yield compatible minimum realizations. Additive encoders give a different structural class with no online alphabet penalty, regardless of the size or shape of the original residual hull.

For the binary-query problem, the minimal $n+1$ checkpoint alphabet has an online realization whenever it exists. The critical geometric threshold is therefore shared by checkpoint and full acquisition peak, not only in Hadamard dimensions. The static absorption results are classical; the transition formulas transfer each of them to an actual online machine. At larger message alphabets a generic encoder need not be additive, and the present theorem does not identify every finite state count or the second-order streaming excess. No new complete varying-signal phase diagram is inferred.

The full-support two-cut experiment has a fixed order of three atomic inputs. It establishes a compatibility obstruction, not a theorem optimizing all informative experimental actions or all report-dependent collision-validation schedules. The earlier controlled minimax, marked-saddle, positive-noise collision, fixed-clock scheduling, revelation, entropy-rate and acquired-transport results remain in the unchanged supporting article and cumulative mathematical volumes. Their statistical experiments and resource conventions are not replaced by the binary-query interface. Known-target exact stochastic rows and acquired-target approximate guarantees remain distinct.

The repository's historical A2/A3/A4 and B2/B1/B3/B4/C1/C2/D1 derivations have independent Fourier, large-deviation, operator-domain, semigroup and optional-projection hypotheses. Preserving those derivations does not make them consequences of a finite positive-realization theorem. The new proof dependencies are internal and explicit: affine slices locate the rank boundary; the forced-triangle obstruction proves a nonattainable profile; additive synthesis transfers critical simplices to online memory; exactification identifies the correct classical checkpoint comparison. No historical analytic gate is marked complete on the strength of that transfer.

The primary literature comparison also has precise limits. We use the cube-absorption statements and explicit simplices of [7], and the full v3 worst-case coding characterization of [6]. Positive realization and residual-state representations are older frameworks [5, 10, 3, 2]; their general existence language is not claimed as an invention. The support/affine arguments here must be distinguished from arbitrary nonnegative-rank minimization [4], and complete arrays from incomplete machine specifications [9]. General filtered-experiment comparison [8] is not a priority claim of the paper. The exact quantifiers and explicit constructions, rather than the size of the preserved development, are the mathematical assertions submitted for independent examination.

REFERENCES

- [1] A. Ambainis, A. Nayak, A. Ta-Shma and U. Vazirani, Dense quantum coding and quantum finite automata, *J. ACM* **49** (2002), 496–511, doi:10.1145/581771.581773; preprint arXiv:quant-ph/9804043v2.
- [2] F. Denis and Y. Esposito, Rational stochastic languages, arXiv:cs/0602093v1 (2006), especially Section 2 and Propositions 16 and 19.
- [3] Y. Esposito, A. Lemay, F. Denis and P. Dupont, Learning probabilistic residual finite state automata, in *Grammatical Inference: Algorithms and Applications (ICGI 2002)*, Lecture Notes in Computer Science **2484**, Springer, 2002, 77–91, doi:10.1007/3-540-45790-9_7.
- [4] N. Gillis and F. Glineur, On the geometric interpretation of the nonnegative rank, *Linear Algebra Appl.* **437** (2012), 2685–2712; arXiv:1009.0880.
- [5] A. Heller, On stochastic processes derived from Markov chains, *Ann. Math. Statist.* **36** (1965), 1286–1291, doi:10.1214/aoms/1177700000.
- [6] R. Kondo, Y. Sato, H. Yano, Y. Maeda, K. Ito and N. Yamamoto, Random access codes: explicit constructions, optimality, and classical–quantum gaps, arXiv:2604.21274v3 (July 16, 2026), Theorem 14, Lemma 15, and Theorems 16–17.
- [7] M. Nevskii and A. Ukhalov, Perfect simplices in $\mathbb{R}^5$, *Beitr. Algebra Geom.* **59** (2018), 501–521, doi:10.1007/s13366-018-0386-6. Inspected preprint: *Five-dimensional perfect simplices*, arXiv:1709.06068v3, Theorems 2–4 and Sections 6–9.
- [8] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28, doi:10.1524/strm.2002.20.14.1.
- [9] B. Reusch and W. Merzenich, Minimal coverings for incompletely specified sequential machines, *Acta Informatica* **22** (1986), 663–678, doi:10.1007/BF00263650.
- [10] M. Vidyasagar, The complete realization problem for hidden Markov models: a survey and some new results, *Math. Control Signals Systems* **23** (2011), 1–65, doi:10.1007/s00498-011-0066-7.

PRESERVED MATHEMATICAL DEVELOPMENT

General Theta Foundations I ? revision 30

The entire current focused article precedes this divider.

The following cumulative predecessor volume is reproduced without alteration.

Frozen predecessor publication:

a4abeb1c5e0323e5850e7f2446bf96fb1f7d4d8a

The independent geometric and analytic obligations of the historical program retain their stated hypotheses.

GENERAL THETA FOUNDATIONS I: POSITIVE REALIZATION AND EXACT STREAMING MEMORY

QIAN QI

ABSTRACT. We study the number of retained states needed to realize a specified finite causal experiment. A criterion for simultaneous attainment of the support-component rank is given by enclosing simplices and common positive shift equations. Unlike a residual-vertex criterion, it allows all internal continuations, including those outside the residual hull. For a binary-query channel on n acquired bits, we determine the exponential memory rate at every fixed signal strength $0 < \eta < 1$. The checkpoint, fixed-order streaming, and adaptive-acquisition minima all have rate $1 - h_2((1 + \eta)/2)$. The upper construction realizes every conditional probability exactly and has a fully charged sequential buffer; its excess logarithmic space is $O_\eta(\sqrt{n \log n})$. At row total-variation tolerance ε , the corresponding rate is obtained by replacing η by $(\eta - 2\varepsilon)_+$. We also construct an exact $(t + 1)$ -state online encoder for $\eta \leq (2n - 1)^{-1}$ and determine the sharp $n + 1$ -state threshold $\eta = 1/n$ whenever a Hadamard matrix of order $n + 1$ exists. The entropy exponent is classical in random-access coding; the conclusions here concern exact conditional channel synthesis and the complete retained-state cost. The relation to residual probabilistic automata is made explicit by a finite language with exponentially many residual states and a polynomial-size positive realization.

1. INTRODUCTION

A finite statistical memory is a collection of possible continuations, not a collection of numerical coordinates. A low-dimensional space of predictions can require many positive generators. Conversely, insisting that every internal state be an observed residual can exclude much smaller stochastic realizations. There is a third constraint: positive factorizations at successive cuts must arise from the same causal machine. The distinctions between dimension, positivity, and compatibility are the subject of this paper.

We work with complete finite response arrays, known primitive probabilities, and an explicit list of charged cuts. Arbitrary stochastic rows are atomic operations. Every retained selector, observation buffer, and data-dependent protocol state is included in the register. The program and the declared clock are fixed before observations. These conventions distinguish an exact state-count theorem from a finite-random-bit implementation or an acquired calibration theorem.

1.1. The realization question. At a cut t , normalized residual continuations form a finite set $\mathcal{R}_t$ in a causal polytope $\mathcal{P}_t$. Intersections of their minimal faces partition them into support components. If component j has affine dimension $r_{t,j} - 1$, every stochastic implementation needs at least $s_t = \sum_j r_{t,j}$ states. Residual vertices attain this bound when each component hull is a simplex. That sufficient condition does not describe all rank-saturating realizations.

Theorem 2.3 gives a necessary and sufficient condition for attaining the whole profile s_t . Each component must admit an enclosing simplex with $r_{t,j}$ normalized causal vertices in its affine slice. The vertices need not be residuals. Their one-step restrictions must satisfy common nonnegative

Date: September 24, 2026. Revision 30.

2020 Mathematics Subject Classification. Primary 68Q45, 52B05; Secondary 94A15, 62B05, 93E20.

Key words and phrases. Causal realization, stochastic automata, nonnegative rank, random access codes, streaming memory, exact channel synthesis.

shift equations with the next cut's vertices. Thus the condition separates static enclosure from simultaneous causal realization. For explicit algebraic arrays it becomes a finite existential system over the real numbers. This is an exact equality criterion, not a polynomial-time algorithm for a succinct controller or a solution of autonomous shared-row minimization.

1.2. The growing experiment. An input $x \in \{-1, 1\}^n$ is acquired one coordinate at a time. After all coordinates have been acquired, a query $j \in [n]$ is supplied. The required binary response is

$$(1) \quad \mathbb{P}(B = b \mid x, j) = \frac{1 + b\eta x_j}{2}, \quad b \in \{-1, 1\}.$$

The signal η is specified, and the law is required for every input and query. Write K_n for the smallest state count at the query barrier, W_n^{ad} for the smallest whole-schedule peak with adaptive acquisition, and W_n^{fo} for the fixed-order peak. In the latter two quantities all persistent choices during acquisition are charged. They satisfy $K_n \leq W_n^{\text{ad}} \leq W_n^{\text{fo}}$.

Let h_2 be binary entropy in bits and put

$$(2) \quad I(u) = 1 - h_2((1 + u)/2), \quad 0 \leq u \leq 1.$$

The main quantitative conclusion, Theorem 4.4, is

$$(3) \quad \lim_{n \rightarrow \infty} \frac{\log_2 K_n(\eta)}{n} = \lim_{n \rightarrow \infty} \frac{\log_2 W_n^{\text{ad}}(\eta)}{n} = \lim_{n \rightarrow \infty} \frac{\log_2 W_n^{\text{fo}}(\eta)}{n} = I(\eta) \quad (0 < \eta < 1).$$

The checkpoint upper error is $O_\eta(\log n)$ bits. The streaming error is $O_\eta(\sqrt{n \log n})$ bits. A fixed codebook is chosen in the existence proof, so there is no persistent public random seed hidden from the state count. For every rowwise total-variation tolerance ε , Theorem 5.1 gives the rate $I((\eta - 2\varepsilon)_+)$.

The exponent in (3) is not new as a random-access coding exponent. Ambainis, Nayak, Ta-Shma and Vazirani [1, 2] prove the matching first-order communication rate for success probability at least $(1 + \eta)/2$. We prove a version fixing *each* conditional response in (1), and then account for the largest retained register during a one-pass implementation. A success inequality by itself is not an exact conditional channel. Likewise, a short message by itself is not a space bound for computing that message from a stream. The proofs below keep both distinctions.

Finite-dimensional information is also obtained. An elementary online simplex construction gives the exact acquisition profile $t + 1$ for $0 < \eta \leq (2n - 1)^{-1}$, improving the smaller signal regime of the preceding development. Any $n + 1$ -state checkpoint factorization requires $\eta \leq 1/n$. If a Hadamard matrix of order $n + 1$ exists, a different online encoder attains that entire interval with peak $n + 1$ (Theorem 3.4). In particular, the threshold is sharp on the infinite sequence $n = 2^k - 1$. The latter construction has peak $n + 1$; it is not claimed to have only $t + 1$ states at every earlier cut.

1.3. One notation and resource convention.

Symbol	Meaning
$C, R_h, \mathcal{P}_t$	Complete causal law, normalized residual, and ambient causal polytope.
$r_{t,j}, s_t$	Rank of support component j , and their sum.
$B_{t,j,k}$	An internal normalized continuation, not necessarily a residual.
$K_n, W_n^{\text{fo}}, W_n^{\text{ad}}$	Query-cut, fixed-order peak, and adaptive-acquisition peak state counts.
η, ε, I	Specified signal, uniform row-TV tolerance, and entropy rate in bits.

State counts are cardinalities; their binary memory cost is the ceiling of their base-two logarithm. The query is an atomic current input and is answered without retaining an uncharged copy. Input-dependent scheduling information has to be retained. A terminal output can use two labels; this constant does not affect any rate. An externally scheduled overwrite may draw fresh independent randomness, but past randomness survives only through the retained label.

1.4. Relationship with the preceding foundations development. The causal-experiment, predictive-quotient, minimax-contact, and physical comparison results of the preceding development [10] are preserved in a separate complete mathematical companion. The present article proves its new claims without asking the reader to reconstruct a proof from that volume. Section 7 identifies the relations with positive realization, residual automata, and random-access coding. The old known-target physical memory theorems remain unchanged; neither a new fully adaptive collision theorem nor closure of the historical analytic pipeline is inferred from (3).

2. COHERENT POSITIVE REALIZATIONS AT THE RANK BOUND

2.1. Finite causal arrays. Fix a finite horizon and finite action and report alphabets. A complete causal array $C(\mathbf{a} \mid \mathbf{y})$ is nonnegative and normalized, and its action-prefix marginal depends only on preceding reports. A terminal output is an action with a dummy subsequent report. All storage cuts required by the interface are included. In particular, an action which must be remembered while awaiting a report has a charged buffer cut; it is not available later as an uncharged input. Legal operation and buffer-copy requirements can be specified as zeros and mandatory readouts in the complete array. Elementary phases may be split to expose them. The results below concern the resulting fixed, clocked interface.

For a controller prefix h of positive conditional mass, let R_h be its normalized array of future actions, with future reports supplied externally. Let $\mathcal{R}_t$ be all distinct such arrays at cut t and $\mathcal{P}_t$ the polytope of normalized causal arrays on its remaining interface. Neither object is a runtime oracle. At zero-mass prefixes no residual is defined; the complete array still specifies which continuations have zero mass. Partial almost-sure specifications require a choice, or an optimization, of complete extensions and are a different input to the theorem.

For $B \in \mathcal{P}_t$, let $p_B(a)$ be its next-action marginal, independent of the unread report. Let $S_{a,y}B$ be the unnormalized restriction obtained by fixing that action and the next report. If $p_B(a) > 0$, $S_{a,y}B/p_B(a) \in \mathcal{P}_{t+1}$; at zero probability the restriction is zero. For an actual residual, each positive normalized restriction is again an actual residual. These maps are linear before normalization.

Write $F_t(R)$ for the minimal face of $\mathcal{P}_t$ containing R . Join two members of $\mathcal{R}_t$ when their minimal faces intersect, and let $\mathcal{R}_{t,j}$, $1 \leq j \leq c_t$, be the connected components. Define

$$(4) \quad r_{t,j} = 1 + \dim \text{aff } \mathcal{R}_{t,j}, \quad s_t = \sum_{j=1}^{c_t} r_{t,j}.$$

Normalization places each component in an affine hyperplane not through zero, so $r_{t,j}$ is also its ordinary linear row rank.

Lemma 2.1 (Separated-generator bound). *Every stochastic realization has at least s_t states at cut t . If it has exactly s_t , the states used by component j are precisely $r_{t,j}$ affinely independent points of $\mathcal{P}_t \cap \text{aff } \mathcal{R}_{t,j}$.*

Proof. Starting the machine from a state s defines a normalized continuation $B_s \in \mathcal{P}_t$. Conditioning on a prefix gives

$$R_h = \sum_s E_h(s) B_s, \quad E_h(s) \geq 0, \quad \sum_s E_h(s) = 1.$$

Every positively used B_s belongs to $F_t(R_h)$, by the face property. Consequently the sets of states used by different support components are disjoint. The convex hull of the states used by component j contains $\mathcal{R}_{t,j}$, and hence has at least $r_{t,j}$ generators. This proves the lower bound. Equality in the sum leaves exactly that many generators in each component and no additional used states. Containment of an affine space of dimension $r_{t,j} - 1$ in their affine span forces independence and equality of the two affine spans. No residual-state restriction has been imposed. $\square$

Definition 2.2 (Coherent enclosing simplices). A coherent rank enclosure consists, at every cut and component, of $r_{t,j}$ points $B_{t,j,k} \in \mathcal{P}_t \cap \text{aff } \mathcal{R}_{t,j}$ whose convex hull contains $\mathcal{R}_{t,j}$, together with nonnegative numbers $w_{t,j,k,a,y,u}$, where u ranges over all vertices selected at the next cut, satisfying

$$(5) \quad S_{a,y} B_{t,j,k} = \sum_u w_{t,j,k,a,y,u} B_{t+1,u},$$

$$(6) \quad \sum_u w_{t,j,k,a,y,u} = p_{B_{t,j,k}}(a).$$

The initial array belongs to the initial selected hull. At the terminal cut there is the single empty continuation. Action-buffer phases obey the same equations on their explicitly split interface.

The enclosure points are automatically affinely independent: they contain a set of the full required affine dimension. In particular, “simplex” describes an enclosing object, not the residual hull.

Theorem 2.3 (Rank saturation and causal compatibility). *For a complete finite clocked causal array, the following are equivalent:*

- (1) *A stochastic realization has profile $K_t = s_t$ at every charged cut.*
- (2) *A coherent rank enclosure exists.*

If every component residual hull is a simplex, its own vertices give such an enclosure. The equivalence itself places no simpliciality condition on the residual hulls.

Proof. Suppose first that a rank-saturating machine exists. Delete unused states. Lemma 2.1 assigns its continuations to the indicated component affine slices and proves their independence and enclosure property. From a given state the actual next-action law is p_B . Conditional on action a and report y , the actual update probabilities $q(u \mid B, a, y)$ are common to all histories

giving that state. Take $w_u = p_B(a)q(u \mid B, a, y)$. The law of the future program gives (5) and (6). Thus the equations hold for one simultaneous collection, including buffer cuts.

Conversely, use the selected vertices as state labels. Initialize by an enclosing convex decomposition of C . From state B , draw the next action according to p_B . When $p_B(a) > 0$, update after report y using $w_u/p_B(a)$. At zero $p_B(a)$ the row is unused and can be assigned arbitrarily. Equation (6) makes each update stochastic, while p_B is independent of the unread report. Backward induction in (5) shows that the future array from state B is exactly B . The initialized law is therefore C . The number of labels is s_t at every cut.

For the final assertion, choose the vertices of each residual hull. They are actual residuals because the residual set is finite. Their positive normalized restrictions are residuals at the next cut and have convex decompositions in that cut's chosen vertices. Multiplying these decompositions by the action marginal supplies the required weights. This is the residual construction, now a special case of the equivalence. $\square$

Corollary 2.4 (Algebraic recognition). *For a fully listed rational or real-algebraic array, simultaneous attainment of s_t is decidable by a finite existential system of polynomial equalities and inequalities. A positive answer admits real-algebraic stochastic rows. Here algebraic input numbers are given by integer defining polynomials and isolating intervals.*

Proof. The finite residuals, component graphs, affine slices, and ranks are computed by exact linear algebra and linear feasibility. Write the unknown enclosure vertices in these affine slices and impose the linear causal-polytope constraints. Introduce convex coefficients for every residual enclosure, and weights for (5)–(6). All unknown products have degree at most two. Independence is forced by enclosure of the known full-dimensional component, so no extra determinant condition is needed. Real quantifier elimination, for example cylindrical algebraic decomposition [3, 4], decides this finite system. A nonempty semialgebraic set defined over the real algebraic numbers has a real algebraic point, which the same sample-point procedure supplies. The construction in the theorem then gives algebraic rows. $\square$

This is not a complexity bound polynomial in the horizon or in a succinct program. Nor does it replace equalities identifying a stochastic row at different autonomous times. Those equalities are extra constraints. The theorem characterizes rank saturation, not arbitrary minimum positive rank. Its essential additional condition over a list of static enclosures is the common shift equation.

2.2. What static rank does and does not measure. Let A be a nonnegative row-stochastic matrix with one output alphabet. Let $K(A)$ be the least number of normalized output rows whose convex hull contains its rows. Let $K_{\text{aff}}(A)$ impose the additional condition that generators lie in the affine hull of the rows, and let $K_{\text{res}}(A)$ require generators to be actual rows.

Proposition 2.5 (The normalization boundary). *For this single-output model,*

$$(7) \quad \text{rank } A \leq \text{rank}_+ A = K(A) \leq K_{\text{aff}}(A) \leq K_{\text{res}}(A) = |\text{ext conv}\{A_i\}|.$$

Writing $r = \text{rank } A$, equality $K(A) = r$ holds if and only if an $(r - 1)$ -simplex in the probability simplex contains every row and lies in their affine hull. For a family of queried output laws, the same convex characterization uses the product of the output simplices; ordinary nonnegative rank of the concatenated matrix is only a lower bound unless its factors satisfy all block-normalization equations.

Proof. A stochastic encoder-decoder factorization is a nonnegative factorization. Conversely, if $A = UV$ is nonnegative, discard zero rows of V and put $d_s = \sum_y V_{sy} > 0$. Then $B_{sy} = V_{sy}/d_s$ is stochastic and $E_{is} = U_{is}d_s$ has row sum one because A does. This proves $\text{rank}_+ A = K(A)$. The

remaining inequalities follow by restricting the generators. The extreme points of a finite hull are precisely the necessary actual-row generators, proving the last equality. If r generators enclose affine dimension $r - 1$, their affine hull must be the row affine hull and they are independent. This proves the simplex criterion and its converse.

With a query j , a physical state has a separate normalized decoder row for every j . Normalizing the total of a concatenated generator only fixes the sum over all blocks; it does not make every block sum one. The admissible ambient set is therefore the product of simplices, which proves the stated distinction. $\square$

The single-output affine-slice restriction is the transpose-normalized form of restricted non-negative rank [7]. At one cut, positive generator complexity also decomposes additively over the support components: a generator cannot serve two components by the proof of Lemma 2.1, and independent component decompositions can be united. For one output alphabet these components are the disjoint support blocks of the matrix. This static additivity is not asserted to solve the compatibility problem between different cuts.

Proposition 2.6 (A nonsimplicial equality and a strict inequality). *For the four binary-query mean rows $(\pm a, \pm a)$, ordinary normalized row rank is three. If $0 < a \leq 1/3$, their positive generator number is three although their residual hull is a square. If $1/2 < a < 1$, their positive generator number is four although all conditional probabilities are strictly positive.*

Proof. The rank assertion follows from affine dimension two. The triangle with vertices $(-a, -a), (3a, -a), (-a, 3a)$ lies in the outer square when $a \leq 1/3$ and contains the inner square; its defining inequalities are $z_i \geq -a$ and $z_1 + z_2 \leq 2a$. This proves the first assertion. For the second, put $d_x(z) = \frac{1}{2} \sum_{i=1}^2 (1 - x_i z_i)$ for $x \in \{-1, 1\}^2$. A convex representation of ax uses a generator with $d_x(z) < 1/2$, because its average is $1 - a < 1/2$. Such strict caps are disjoint for distinct x : a differing coordinate contributes one to $d_x + d_{x'}$. Four distinct generators are required, and the four rows themselves suffice. This proves a full-support rank gap without assuming residual-state realizations. $\square$

3. EXACT FINITE STREAMING AT THE RANK THRESHOLD

3.1. Checkpoint and streaming models. For $n \geq 1$ and $0 \leq \eta \leq 1$, the channel to be synthesized is (1). Every input coordinate is acquired exactly once before the query. In a fixed-order encoder the order is $1, \dots, n$. In an adaptive encoder the next unread coordinate can depend on the retained state and fresh randomness; any information required to make that choice legal must be retained. There is no readable transcript or unrecorded schedule. After acquisition, all input-dependent information passes through a single finite state before the external query arrives. One query is answered; a joint law for simultaneous answers to all queries is not required.

The checkpoint minimum $K_n(\eta)$ imposes no bound on upstream memory. The quantities $W_n^{\text{fo}}(\eta)$ and $W_n^{\text{ad}}(\eta)$ impose a bound on every cut, including intermediate compression and retained random selectors. The clock in the fixed-order convention is prescribed externally. A terminal binary label counts as at most two states. The query is a current atomic input; a model requiring its storage has to charge that storage separately.

Lemma 3.1 (Cube enclosure). *For $\eta > 0$, $K_n(\eta)$ is the smallest number of points $z_s \in [-1, 1]^n$ whose convex hull contains $[-\eta, \eta]^n$. In particular $K_n(\eta) \geq n + 1$. This lower bound and every lower bound obtained from such an enclosure also apply to adaptive acquisition.*

Proof. From state s , let $z_{s,j}$ be the mean binary output under query j . Conditioning on the complete input gives an encoder distribution E_x and hence

$$(8) \quad \eta x = \sum_s E_x(s) z_s.$$

Conversely, such a representation followed by a binary decoder with mean $z_{s,j}$ realizes the prescribed channel. Upstream memory is unrestricted for this converse construction. Convexity reduces enclosure of the inner cube to its corners. Its affine dimension gives $n + 1$. For adaptive acquisition, conditioning on the complete input still gives one distribution of the retained state; the subsequent decoder is common to all inputs. This uses the acquisition barrier, not an assertion about arbitrary interactive scheduling. $\square$

3.2. An online simplex with a linear signal window.

Theorem 3.2 (An exact minimal acquisition profile). *For every $n \geq 1$ and $0 < \eta \leq (2n - 1)^{-1}$,*

$$(9) \quad K_n(\eta) = W_n^{\text{ad}}(\eta) = W_n^{\text{fo}}(\eta) = n + 1.$$

One fixed-order encoder has exactly $t + 1$ states after t acquired bits, $0 \leq t \leq n$, and this profile is minimal at each acquisition cut for that order. Its stochastic acquisition rows are rational and independent of η .

Proof. For $t \geq 1$ use labels $0, 1, \dots, t$ with associated vectors

$$(10) \quad v_{t,0} = -\eta \mathbf{1}, \quad v_{t,i} = -\eta \mathbf{1} + 2t\eta e_i \quad (1 \leq i \leq t).$$

Their convex hull is

$$P_t = \{z \in \mathbb{R}^t : z_i \geq -\eta, \sum_i z_i \leq t\eta\}.$$

Every vertex lies in the outer cube because $(2t - 1)\eta \leq 1$. Moreover $P_t \times \{-\eta, \eta\} \subset P_{t+1}$. The barycentric coordinates of $w \in P_t$ are

$$\lambda_i(w) = \frac{w_i + \eta}{2t\eta}, \quad \lambda_0(w) = \frac{t\eta - \sum_i w_i}{2t\eta}.$$

They are nonnegative and sum to one.

Here is an explicit update avoiding any stored real vector. After t inputs, the state distribution is

$$(11) \quad \mathbb{P}(S_t = i \mid x_{1:t}) = \frac{\mathbf{1}_{\{x_i=1\}}}{t}, \quad \mathbb{P}(S_t = 0 \mid x_{1:t}) = \frac{|\{i \leq t : x_i = -1\}|}{t}.$$

To acquire bit x_{t+1} , keep the old label with probability $t/(t + 1)$. With probability $1/(t + 1)$ replace it by $t + 1$ when $x_{t+1} = 1$ and by zero when $x_{t+1} = -1$. For $t = 0$ only the replacement is used. Induction proves (11). The mean of v_{t,S_t} is exactly $\eta x_{1:t}$. At the query barrier use the binary decoder with mean $v_{n,S_n,j}$. This proves exact synthesis. Only the label survives an update. The last raw bit and the replacement coin do not persist in an auxiliary register.

The final lower bound follows from Lemma 3.1. At an earlier fixed-order cut, fix the unread bits identically for all prefixes. The future response vectors for queries among the first t coordinates contain the t -dimensional cube of means $\eta x_{1:t}$. Factoring these continuations through the current state requires at least $t + 1$ labels. Thus the entire acquisition profile is minimal. The final binary output uses no more labels than the peak $n + 1$. $\square$

For $t \geq 2$ the actual prefix residuals have a cube, not a simplex, as their hull. Equations (10) and (11) are therefore a growing nonsimplicial family of coherent rank enclosures. Theorem 2.3 applies without requiring that an internal vertex be a residual.

3.3. The sharp $n + 1$ -state boundary in Hadamard dimensions.

Lemma 3.3 (A simplex obstruction). *If $K_n(\eta) = n + 1$ with $\eta > 0$, then $\eta \leq 1/n$.*

Proof. The enclosing vertices $v_0, \dots, v_n \in [-1, 1]^n$ are affinely independent. Write their barycentric affine functions as $\lambda_s(z) = a_s + b_s \cdot z$. Nonnegativity on the inner cube gives $a_s \geq \eta \|b_s\|_1$, and $\sum_s a_s = 1$. The identity $z = \sum_s \lambda_s(z) v_s$ gives $\sum_s v_s b_s^T = I_n$. Taking its trace yields

$$n = \sum_s b_s \cdot v_s \leq \sum_s \|b_s\|_1 \leq \eta^{-1}.$$

This proves the assertion for arbitrary enclosing vertices, not just a chosen simplex construction. $\square$

Theorem 3.4 (A sharp threshold with an online realization). *Suppose that a Hadamard matrix of order $m = n + 1$ exists. For $\eta > 0$,*

$$(12) \quad K_n(\eta) = W_n^{\text{ad}}(\eta) = W_n^{\text{fo}}(\eta) = n + 1 \iff \eta \leq 1/n.$$

This applies in particular to every $n = 2^k - 1$, $k \geq 1$. More generally, a Hadamard matrix of order $m \geq n + 1$ gives an m -state streaming realization whenever $0 < \eta \leq 1/n$.

Proof. Normalize the first column of the Hadamard matrix to one, and select n other columns. Let $v_s \in \{-1, 1\}^n$, $1 \leq s \leq m$, be their rows. Orthogonality gives

$$(13) \quad \frac{1}{m} \sum_s v_{s,j} = 0, \quad \frac{1}{m} \sum_s v_{s,i} v_{s,j} = \mathbf{1}_{\{i=j\}}.$$

For each index i and observed bit x_i , define the probability law

$$q_{i,x_i}(s) = \frac{1 + x_i v_{s,i}}{m}.$$

Its mean vector is $x_i e_i$ by (13). At step i , retain the preceding state with probability $(i - 1)/i$; otherwise redraw a label from q_{i,x_i} . The first step always draws. After t bits its state law is $t^{-1} \sum_{i \leq t} q_{i,x_i}$. Consequently $\mathbb{E}[v_{S_n} | x] = x/n$. The final decoder with mean $n\eta v_{S_n,j}$ is legal for $\eta \leq 1/n$ and gives mean ηx_j . This is an online machine with m labels; the chosen index of a redraw is not separately retained. All acquisition probabilities are rational.

For $m = n + 1$, Lemma 3.1 proves optimality and Lemma 3.3 excludes larger signals. The Sylvester construction $H_1 = (1)$ and $H_{2m} = \begin{pmatrix} H_m & H_m \\ H_m & -H_m \end{pmatrix}$ gives the claimed infinite sequence directly. $\square$

The equality in (12) concerns the peak, not every intermediate width. The encoder in this proof may have $n + 1$ possible labels already after its first bit. In contrast, Theorem 3.2 attains the individual $t + 1$ bounds in its smaller signal interval. Choosing the next power of two for m also gives a less than $2(n + 1)$ -state upper bound at $\eta \leq 1/n$ for arbitrary n ; it does not decide the exact simplex threshold in every dimension.

4. EXACT SYNTHESIS THROUGHOUT THE FIXED-SIGNAL RANGE

All logarithms in rates are base two. Natural logarithms are denoted by $\ln$. Define I by (2); thus $I(0) = 0$, $I(1) = 1$, and I is increasing on $[0, 1]$.

4.1. A uniform corner-cap lower bound.

Lemma 4.1 (Entropy cap). *For $z \in [-1, 1]^n$ and $0 \leq u \leq 1$, the fraction of sign vectors x with $x \cdot z \geq nu$ is at most $2^{-nI(u)}$. Therefore*

$$(14) \quad K_n(\eta) \geq 2^{nI(\eta)}.$$

Proof. For a uniform sign vector X and $t \geq 0$,

$$\mathbb{E} e^{tX \cdot z} = \prod_{i=1}^n \cosh(tz_i) \leq \cosh(t)^n.$$

Exponential Markov inequality gives the bound $\exp\{-n \sup_{t \geq 0} (tu - \ln \cosh t)\}$. For $0 < u < 1$ the maximum occurs at $t = \operatorname{arctanh} u$. Substitution gives $\sup (tu - \ln \cosh t) = (\ln 2)I(u)$. The two endpoints follow by limits. If a set of generators encloses ηx , then some generator z_s has $x \cdot z_s \geq n\eta$. These caps therefore cover all 2^n sign vectors. The fraction bound and the union bound imply (14). $\square$

The exponent in this lemma is the classical random-access lower exponent [1, Theorem 2.1]. Its cap proof is useful here because it applies directly to exact and perturbed conditional means.

4.2. An enclosing codebook with exact, not one-sided, responses. For $0 < \eta < 1$ and $0 < \delta \leq (1 - \eta)/2$, put

$$(15) \quad k = \left\lceil \frac{n(1 + \eta + 2\delta)}{2} \right\rceil, \quad r = \frac{2k}{n} - 1,$$

$$(16) \quad M(n, \eta, \delta) = \left\lceil \frac{n+1}{\delta} \{n \ln(1 + 2/\delta) + 1\} 2^{nI(r)} \right\rceil.$$

Here $k \leq n$ and $\eta + 2\delta \leq r \leq \eta + 2\delta + 2/n$.

Lemma 4.2 (A Hamming-layer enclosure). *There is a set of at most $M(n, \eta, \delta)$ sign vertices whose convex hull contains $[-\eta, \eta]^n$. In particular,*

$$(17) \quad K_n(\eta) \leq \min\{2^n, M(n, \eta, \delta)\}.$$

For every fixed $0 < \eta < 1$ there is a constant C_η such that

$$(18) \quad nI(\eta) \leq \log_2 K_n(\eta) \leq nI(\eta) + C_\eta \log_2(n+1) \quad (n \geq 1).$$

Proof. Fix $a \in \mathbb{R}^n$ with $\|a\|_1 = 1$. Orient each coordinate by the sign of a_i , choosing either orientation at zeros. Condition a uniform sign vertex V on having exactly k coordinates aligned with these orientations. Under this conditional law $Z = a \cdot V \leq 1$ and $\mathbb{E}Z = r$. If $p = \mathbb{P}(Z \geq \eta + \delta \mid k)$, then

$$r \leq p + (1 - p)(\eta + \delta), \quad p \geq \frac{r - \eta - \delta}{1 - \eta - \delta} \geq \delta.$$

It follows that, for an unconditional uniform sign vertex,

$$(19) \quad \mathbb{P}(a \cdot V \geq \eta + \delta) \geq \delta \binom{n}{k} 2^{-n} \geq \frac{\delta}{n+1} 2^{-nI(r)} =: q.$$

For the second inequality, k is a mode of a binomial distribution with success probability k/n , so its probability is at least $1/(n+1)$. Rewriting that probability gives $\binom{n}{k} \geq 2^{nh_2(k/n)}/(n+1)$. The endpoints $k = 0, n$ have the same interpretation by continuity.

Choose a δ -net of the ℓ^1 unit sphere of size at most $(1 + 2/\delta)^n$. Such a net follows by taking a maximal separated set and comparing the volumes of disjoint ℓ^1 balls of radius $\delta/2$ with the ball of radius $1 + \delta/2$. Draw $M = M(n, \eta, \delta)$ independent uniform sign vertices. For any net

direction the probability that none has scalar product at least $\eta + \delta$ is at most e^{-Mq} . The probability that some net direction fails is at most $(1 + 2/\delta)^n e^{-Mq} \leq e^{-1} < 1$. Hence a fixed choice of vertices succeeds on every net direction. Their support function is one-Lipschitz in the ℓ^1 norm, so it is at least η on the entire unit sphere. Homogeneity and convex separation imply that their hull contains the inner cube. Repeated vertices can be discarded. Full retention gives the other upper bound.

For fixed η , take $\delta = 1/n$ once $n \geq 2/(1 - \eta)$. Then $r \leq \eta + 4/n$, and the derivative $I'(u) = \frac{1}{2} \log_2((1+u)/(1-u))$ is bounded on a fixed neighborhood of η lying below one. Thus $nI(r) \leq nI(\eta) + O_\eta(1)$ for sufficiently large n . The remaining factor in (16) is polynomial up to a logarithm and contributes $O(\log(n+1))$ bits. Enlarging C_η handles the finitely many smaller n . Combine this with Lemma 4.1. $\square$

The hull containment gives a convex representation of *each* ηx , and hence exactly the means in (1). It does not merely ensure that a coordinate is correct with probability at least $(1 + \eta)/2$. Once a successful vertex set has been chosen, all its probabilities are fixed in the program. No random choice of a codebook survives at runtime. For rational η , the enclosure linear systems admit rational encoder coefficients; for algebraic η they admit coefficients in the corresponding ordered field. The proof is an existence and finite-synthesis result, not a claim of fast codebook construction.

4.3. From a terminal codebook to a one-pass machine.

Lemma 4.3 (Charged block realization). *Suppose the ℓ -coordinate channel at signal η has a checkpoint codebook with M generators. Then, for $1 \leq \ell \leq n$,*

$$(20) \quad W_n^{\text{fo}}(\eta) \leq \max\{2, 2^\ell M^{\lfloor n/\ell \rfloor}\}.$$

All conditional output probabilities are realized exactly.

Proof. Split the fixed input order into $b = \lfloor n/\ell \rfloor$ full blocks and a final remainder of length less than ℓ . While acquiring a block, retain the previously chosen block labels and its current raw prefix. Upon completing a block, draw one label using that block's exact encoder, overwrite its raw buffer, and retain only the new label with the previous labels. The encoder table is a fixed row of the declared program. A remainder is retained raw.

At a cut the state is a tuple of at most b codebook labels and at most ℓ raw bits. During the overwrite the old buffer is replaced atomically, not retained alongside a new private selector. Thus the number of possible labels is bounded as in (20). The compression epochs are prescribed clock phases; they do not depend on observations.

For a query in a complete block, use the corresponding label's decoder coordinate. Its conditional mean is ηx_j independently of the other labels. For a query in the remainder, draw a binary response of mean ηx_j from the retained bit. At most two terminal labels are required. The label sampling uses fresh randomness, and all its persistent effect is in the tuple. No unread bit, chosen codebook, or previous random seed is a free register. $\square$

Theorem 4.4 (Exact fixed-signal streaming rate). *For every fixed $0 < \eta < 1$, there exists $C'_\eta < \infty$ such that*

$$(21) \quad nI(\eta) \leq \log_2 K_n(\eta) \leq \log_2 W_n^{\text{ad}}(\eta) \leq \log_2 W_n^{\text{fo}}(\eta),$$

$$(22) \quad \log_2 W_n^{\text{fo}}(\eta) \leq nI(\eta) + C'_\eta \sqrt{n \log_2(n+1)}.$$

Consequently all three normalized logarithmic complexities converge to $I(\eta)$, as in (3). The checkpoint has the stronger error bound (18).

Proof. The lower bounds have already been proved and survive adaptive acquisition. For sufficiently large n , choose $\ell = \lceil \sqrt{n \log_2(n+1)} \rceil \leq n$. Lemma 4.2 supplies a block codebook with $\log_2 M \leq \ell I(\eta) + C_\eta \log_2(\ell + 1)$. Lemma 4.3 gives, after increasing an absolute constant,

$$\log_2 W_n^{\text{fo}} \leq \ell + \frac{n}{\ell} \{ \ell I(\eta) + C_\eta \log_2(\ell + 1) \} \leq n I(\eta) + O_\eta(\sqrt{n \log(n+1)}).$$

Again the finitely many smaller n are absorbed into C'_η . Division by n proves the limits. $\square$

This theorem covers every fixed interior signal, rather than just two endpoint windows. It does not determine every finite integer value of K_n or W_n , their lower-order difference, or a uniform sharp law when η varies with n . Equations (16) and (20) remain explicit nonasymptotic bounds in those varying-signal problems. In the strong finite regime $\eta > 1 - 1/n$, the disjoint-cap argument gives $K_n = W_n^{\text{ad}} = W_n^{\text{fo}} = 2^n$: the encoder average of $\frac{1}{2} \sum_i (1 - x_i z_i)$ is below $1/2$, and its strict caps for distinct input corners are disjoint. Full retention matches this bound. That sharper endpoint statement is consistent with, but not needed for, Theorem 4.4.

5. UNIFORMLY APPROXIMATE EXPERIMENTS AND THEIR EXACT RATE

For two binary-output channels on the same input/query set, define

$$d_{\text{row}}(P, \tilde{P}) = \max_{x,j} \text{TV}(P(\cdot | x, j), \tilde{P}(\cdot | x, j)).$$

For $\varepsilon \geq 0$, this is a uniform comparison of experiments, not an error averaged over a selected input prior. Let $K_n^\varepsilon(\eta)$, $W_n^{\text{ad},\varepsilon}(\eta)$ and $W_n^{\text{fo},\varepsilon}(\eta)$ denote the corresponding minima allowing $d_{\text{row}} \leq \varepsilon$. Put $\kappa = (\eta - 2\varepsilon)_+$.

Theorem 5.1 (A sharp row-error memory rate). *If $0 < \kappa < 1$, then*

$$(23) \quad 2^{nI(\kappa)} \leq K_n^\varepsilon(\eta) \leq K_n(\kappa), \quad W_n^{\text{fo},\varepsilon}(\eta) \leq W_n^{\text{fo}}(\kappa).$$

In particular, for fixed $\eta \in (0, 1]$ and fixed $0 \leq \varepsilon < \eta/2$ with $\kappa < 1$, all three normalized logarithmic approximate complexities converge to $I(\kappa)$. For $\varepsilon \geq \eta/2$, one query state and at most two terminal states suffice, so the rate is zero. At $(\eta, \varepsilon) = (1, 0)$ the exact state counts equal 2^n .

Proof. Let $m_j(x)$ be the mean output of an admissible approximate channel. Binary total variation is half the absolute difference of means; hence $|m_j(x) - \eta x_j| \leq 2\varepsilon$. Factoring the approximate channel through states gives $m(x) = \sum_s E_x(s) z_s$, with $z_s \in [-1, 1]^n$. Therefore

$$x \cdot m(x) \geq n(\eta - 2\varepsilon) = n\kappa$$

when $\kappa > 0$. Some used generator has $x \cdot z_s \geq n\kappa$. The caps in Lemma 4.1 cover the input corners, giving the lower bound, also for adaptive acquisition.

Conversely, exactly synthesize the channel with mean κx_j . Its distance from the required one is $(\eta - \kappa)/2 = \varepsilon$. This proves the upper bounds and allows Theorem 4.4 to be applied at the fixed signal κ . If $\kappa = 0$, the fair-output channel is independent of input and query and has row defect $\eta/2$. It needs one state before output. Finally, an exact noiseless response to every query determines every input bit through the stored generator; the cap argument at $u = 1$, or disjoint supports, gives 2^n states, matched by retention. $\square$

5.1. A statistical consumer: reusable binary sensors. Consider a parameter $x \in \{-1, 1\}^n$ and a preparation interface that reveals its coordinates during acquisition. A later single coordinate measurement has conditional mean ηx_j . A simulator must be installed before the measurement index is chosen. Theorem 4.4 is an exact state-cost theorem for this simulator, not a theorem about a prior chosen to fit a codebook. Theorem 5.1 gives its uniform error-memory exponent, with the same lower bound against adaptive acquisition.

Corollary 5.2 (Comparison under downstream experiments). *A simulator with row defect ε changes the expectation of any common $[0, 1]$ -valued decision loss, after one queried report and arbitrary parameter-independent postprocessing, by at most ε , uniformly in x and in a randomized query distribution independent of the simulator's hidden state. If a sequence of fresh installations has conditional row defect at most ε at each round and is coupled to the same downstream history-dependent controller, the T -round path-law defect is at most $\min(1, T\varepsilon)$.*

Proof. At one round, mixture over the common query and contraction under postprocessing do not increase total variation; bounded losses then have expectation difference at most ε . For repeated installations, couple the two executions only while their public histories agree. The downstream action laws then agree and the chance of a first discrepant report in each active round is at most ε . A union bound proves the path statement. The conditional hypothesis is essential; it is ensured, for example, by fresh independent reinstallation and fresh sampling randomness at every round. $\square$

A single compressed label need not support a sequence of independent queries with the product output law. Observing an answer may change the conditional distribution of that label. Thus repeated use of the same unrefreshed code is not justified by the one-query theorem. If all n noisy answers are required jointly and independently, the specified product channel has full matrix rank 2^n for every $\eta > 0$, and needs at least 2^n query-barrier states. Indeed its transition matrix is the n -fold Kronecker power of the invertible binary symmetric channel. The distinction between a later chosen measurement and a joint vector of measurements is operational, not a change of notation.

The transport argument in the corollary is classical total-variation contraction and coupling. Its quantitative memory input is the new exact-synthesis law. It is a local statistical consumer, not a claim that a Fourier, large-deviation, or nonlinear-semigroup theorem in a different part of the foundations program has been proved by finite channel synthesis.

6. RESIDUAL AUTOMATA AND ARBITRARY POSITIVE STATES

For a stochastic language L on finite words and a prefix u with $L(u\Sigma^*) > 0$, its normalized residual is $L_u(v) = L(uv)/L(u\Sigma^*)$. A residual probabilistic automaton requires every state's generated language to be one of these residuals. An arbitrary probabilistic automaton need not satisfy that restriction. The distinction is explicit in the probabilistic residual line initiated by Esposito, Lemay, Denis and Dupont [6], and in [5, Section 2 and Propositions 16, 19]. Canonical minimality within a residual representation class is consequently not a minimum over all positive generators.

The following construction makes the distinction within the precise streaming family, without appealing to a conjectured difference between models. It concerns the total number of states of a generative automaton; its clock is explicitly incorporated in the states.

Theorem 6.1 (A finite-language separation). *Fix $n \geq 2$ and rational $0 < \eta \leq (2n - 1)^{-1}$. There is a finite stochastic language $L_{n,\eta}$ such that every residual probabilistic automaton generating it has at least 2^n states, whereas an unrestricted rational probabilistic automaton has at most*

$$(24) \quad \frac{(n+1)(n+2)}{2} + n(n+1) + 1$$

states. The lower bound holds already for the states used immediately before the query symbol.

Proof. Use distinct symbols u_-, u_+ for acquired bits, q_j for queries, and o_-, o_+ for answers. Give mass

$$(25) \quad L_{n,\eta}(u_{x_1} \cdots u_{x_n} q_j o_b) = \frac{2^{-n}}{n} \frac{1 + b\eta x_j}{2}$$

to those words and zero to every other word. The masses sum to one. After a length- n supported prefix, the residual has probabilities $(1 + b\eta x_j)/(2n)$ on $q_j o_b$. These 2^n residuals are an injective affine image of the corners of an n -cube. In particular, every one is an extreme point of their convex hull.

In a probabilistic realization, condition on such a prefix. The residual is a convex combination of the state languages then occupied. Since it is supported on words of length two of the displayed query-answer form, nonnegativity forces every positively used state language to have that same support restriction. In a residual automaton that state language must itself be a residual of $L_{n,\eta}$. A residual after a supported prefix of length t is concentrated on length $n + 2 - t$; hence a residual concentrated on length two must come from $t = n$. There are no other supported prefix types at that length. Therefore the used state languages belong to the 2^n cube-corner residuals above. Extremality forces each corner to occur as a state language. They are distinct, proving the lower bound.

For the upper bound, generate the u -symbols as independent fair bits. Simultaneously run the encoder of Theorem 3.2; a transition labelled u_x and updating to s' has probability $\frac{1}{2}T_{t,x}(s, s')$. Include the phase t in the state. The acquisition phases $t = 0, \dots, n$ have a total of $\sum_{t=0}^n (t+1) = (n+1)(n+2)/2$ states. At phase n , emit q_j uniformly and enter a state retaining (s, j) ; there are at most $n(n+1)$ such states. Emit o_b with the exact decoder probability and enter one common terminating state. Every transition and terminal probability is nonnegative rational, and the generated language is (25). This proves (24) with its clock and query buffer included. $\square$

The theorem does not assert that exponential succinctness differences between probabilistic models are new. Its role is to identify a concrete instance in which a residual-only minimum cannot be substituted for the all-machine minimum in this paper. In the present example the latter has a genuinely online positive realization. The state languages of that realization generally lie outside the residual hull.

One can equivalently treat the query as an external input, recovering the checkpoint and streaming models of Section 3. Turning it instead into a generated symbol in (25) is what permits a direct comparison with stochastic-language automata. The factor $1/n$ is essential: it normalizes the query choice rather than appending an unnormalized family of conditional laws.

7. THEOREM BOUNDARIES AND FURTHER QUESTIONS

7.1. What is inherited from realization theory. The probabilistic residual finite-state automata paper of Esposito-Lemay-Denis-Dupont [6] is an essential predecessor, not a peripheral citation. Its announced characterization uses finite generation by residual stochastic languages and canonical minimal forms. The later full treatment [5, Propositions 16, 19] provides a unique inclusion-minimal residual generating set and the finite-generation characterization by residual probabilistic automata. The relevant restriction is on the state languages. Our explicit comparison in Theorem 6.1 uses that restriction rather than inferring an all-machine minimum from the word “canonical.” The publisher record and abstract of [6], and the full cited sections of [5], are the bases of this comparison; we do not claim a completed line-by-line audit of the 2002 chapter.

Positive stochastic realization, including stable polyhedral sets of continuations, is classical [8, 12]. The stationary HMM existence question in [12] is not the same input as a fully listed finite controlled array. Nevertheless the principle that positive generators must be closed under appropriate shifts is shared. Theorem 2.3 should therefore not be read as an invention of positive realization. It makes rank saturation, affine-slice enclosure, and the simultaneous finite shift constraints explicit. Its use here is the non-residual streaming construction, the sharp

Hadamard-dimensional peak threshold, and the charged exact-synthesis bounds, not an assertion that a finite normal form alone settles the general realization problem.

Gillis–Glineur [7, Definition 1 and Theorem 1] distinguish restricted nonnegative rank from unrestricted nonnegative rank and relate the former to nested polytopes. Proposition 2.5 is the normalized single-output specialization, with the transpose convention made explicit. A queried law has block-normalization constraints, and coherent realization has constraints between cuts; these must not be removed when applying a static matrix theorem.

Compatible coverings for incompletely specified sequential machines [11] concern a different form of missing information. Our array theorem uses complete laws and allows support zeros in those laws; it does not choose a canonical completion of a partial law. Filtered-experiment comparison [9] is an adjacent comparison framework, not a result replaced by the constructions here. No general non-overlap assertion about that framework is made. The precise remaining source-access limits are recorded with the revision response.

7.2. What is inherited from random-access coding. The classical lower and upper communication rates in [1, Theorems 2.1 and 2.2] are $(1 - h_2(p))n$ and $(1 - h_2(p))n + O(\log n)$ for recovery success at least p . Their upper construction uses a finite family of masks and permutations to make the guarantee hold for every input and coordinate. Thus the entropy function and the polynomial-overhead communication scale are classical; neither is claimed as a new discovery here.

Our exact-channel proof uses a support-function enclosure of the entire inner cube. It supplies exact prescribed conditional probabilities, not only a success inequality. The block construction then prices all retained raw data and selected codewords. The limiting exponent remains the classical one, but the operational conclusion is about exact channel synthesis at every row and the peak of the whole streaming schedule. The proof of Theorem 5.1 shows precisely how a uniform row-error tolerance changes that exponent. It does not claim a result for arbitrary sequentially repeated queries on one installation.

7.3. Precision and implementation. Atomic stochastic rows can contain a specified real parameter. A rational row can be sampled exactly by a variable-length fair-bit procedure, but that statement alone does not give a bounded-time or bounded-workspace sampler. The present state counts concern atomic rows. The finite codebooks and rows may be enormous as program objects; program description length and computational time are not included in K_n or W_n . This is the same distinction as in the preceding known-target physical statements. Acquiring an unknown target and approximating its stochastic row requires calibration data, retained counters, and precision-dependent workspace. Those results are retained with their original costs in the mathematical companion [10].

7.4. Conclusions. The proved exact class is now described by the coherent-enclosure criterion, which is necessary and sufficient for the support-component rank profile to be attained. Residual-hull simpliciality is one sufficient subcase. The growing query family has nonsimplicial residual hulls, an explicit minimal acquisition profile in a linear signal window, and a sharp $n + 1$ -state threshold in Hadamard dimensions.

Unrestricted positive-state minimization, autonomous row sharing, and joint optimization over all controlled policies remain different problems. We have not supplied an example in which every static cut attains its rank bound but no simultaneous rank profile exists; the criterion states the exact compatibility test rather than claiming such a separation without proof. The historical geometric and unbounded-operator obligations of the larger program are likewise not consequences of a finite-state coding theorem.

The former fixed-interior-signal gap in the query family is resolved at the level of its exponential rate, both exactly and at a fixed uniform row-error tolerance. Finite-size checkpoint versus streaming gaps, optimal lower-order terms, the sharp simplex threshold in all dimensions, and critical regimes with signal varying with n are not identified by these limits. The explicit enclosure and block bounds provide quantitative statements in those regimes without replacing them by an unsupported exact phase diagram.

REFERENCES

- [1] A. Ambainis, A. Nayak, A. Ta-Shma and U. Vazirani, Dense quantum coding and a lower bound for 1-way quantum automata, arXiv:quant-ph/9804043v2 (1998), especially Theorems 2.1 and 2.2.
- [2] A. Ambainis, A. Nayak, A. Ta-Shma and U. Vazirani, Dense quantum coding and quantum finite automata, *J. ACM* **49** (2002), 496–511, doi:10.1145/581771.581773.
- [3] G. E. Collins, Quantifier elimination for real closed fields by cylindrical algebraic decomposition, in *Automata Theory and Formal Languages*, Lecture Notes in Computer Science **33**, Springer, 1975, 134–183, doi:10.1007/3-540-07407-4_17.
- [4] J. H. Davenport, Z. P. Tonks and A. K. Uncu, A poly-algorithmic approach to quantifier elimination, arXiv:2302.06814v2 (2023).
- [5] F. Denis and Y. Esposito, Rational stochastic languages, arXiv:cs/0602093v1 (2006), Sections 2, 3 and 4.3.
- [6] Y. Esposito, A. Lemay, F. Denis and P. Dupont, Learning probabilistic residual finite state automata, in *Grammatical Inference: Algorithms and Applications (ICGI 2002)*, Lecture Notes in Computer Science **2484**, Springer, 2002, 77–91, doi:10.1007/3-540-45790-9_7.
- [7] N. Gillis and F. Glineur, On the geometric interpretation of the nonnegative rank, *Linear Algebra Appl.* **437** (2012), 2685–2712; inspected preprint arXiv:1009.0880v1.
- [8] A. Heller, On stochastic processes derived from Markov chains, *Ann. Math. Statist.* **36** (1965), 1286–1291, doi:10.1214/aoms/1177700000.
- [9] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28, doi:10.1524/strm.2002.20.14.1.
- [10] Q. Qi, General Theta Foundations I: Intrinsic continuation geometry and causal memory, revision 29, September 24, 2026, repository `TrillioniumFoundation/theta-theory`, publication commit `b0bd13753456c15815b174db1106e2ce6ee7433b`. The complete mathematical manuscript and development are preserved with this revision as separate supporting volumes.
- [11] B. Reusch and W. Merzenich, Minimal coverings for incompletely specified sequential machines, *Acta Informatica* **22** (1986), 663–678, doi:10.1007/BF00263650.
- [12] M. Vidyasagar, The complete realization problem for hidden Markov models: a survey and some new results, *Math. Control Signals Systems* **23** (2011), 1–65, doi:10.1007/s00498-011-0066-7.

PRESERVED MATHEMATICAL DEVELOPMENT

General Theta Foundations I ? revision 29

The entire current focused article precedes this divider.

The following cumulative predecessor volume is reproduced without alteration.

Frozen predecessor publication:

b0bd13753456c15815b174db1106e2ce6ee7433b

The independent geometric and analytic obligations of the historical program retain their stated hypotheses.

GENERAL THETA FOUNDATIONS I: INTRINSIC CONTINUATION GEOMETRY AND CAUSAL MEMORY

QIAN QI

ABSTRACT. We study the number of persistent states needed to realize a causal statistical response. For a complete finite response array we define an intrinsic support-component rank from all normalized continuations. Every stochastic realization obeys this cutwise lower bound. If the convex hull in each component is a simplex, residual vertices attain all these bounds simultaneously through normalized causal updates. The criterion is finite and effective on rational data, and the complete class is closed under independent products of terminal channels, with multiplicative state counts. A growing binary random-access experiment then separates predictive dimension from positive memory: its exact minimum whole-schedule peak is $n + 1$ for $0 < \eta \leq n^{-2}$ and 2^n for $1 - n^{-1} < \eta \leq 1$, even when acquisition order is chosen adaptively. The weak-signal bound is attained by an explicit sequence of compatible enclosing simplices, not merely by a final encoder. Minimax contact connects the classification to the marked physical audit and its exact five-state decision law. The previously established positive-noise saddle, fixed-clock validation-order classification, support-boundary jump, causal foundations and transport proofs are retained as supporting mathematics. Exact physical counts assume a specified target and atomic stochastic rows; acquired calibration has separately priced approximation guarantees.

CONTENTS

1. Introduction	3
1.1. Resource and parameter conventions	4
1.2. Organization and the retained development	4
2. An intrinsic continuation invariant	4
2.1. Normalized causal residuals	4
2.2. Exact schedule fields without selected certificates	6
2.3. Why the completeness hypothesis has content	7
3. Products and intrinsic memory growth	7
4. A scalable full-support experiment with two exact memory laws	8
4.1. The intrinsic checkpoint geometry	9
4.2. A compatible sequence of enclosing simplices	9
4.3. A geometry stronger than rank	10
5. From saddle contact to intrinsic memory	11
5.1. The marked decision geometry as a complete instance	12
5.2. Known-target physical exactness and acquisition	12
6. Theorem-level relation to realization and coding	13
Supporting foundations and physical proofs	15
Appendix A.1. Causal experiments and their predictive quotients	15
A.1.1. The experimental object	15

Date: September 24, 2026. Revision 29.

2020 Mathematics Subject Classification. Primary 62B05, 62C20; Secondary 60J10, 68Q45, 93E20.

Key words and phrases. Statistical experiments, causal realization, predictive distributions, finite memory, minimax duality, physical certification.

A.1.2. Prepared predictive equivalence	16
Appendix A.2. Attainable resolution and resource-aware simulation	17
A.2.1. Checkpoint and online prediction	17
A.2.2. Causal morphisms and simultaneous error budgets	18
Appendix A.3. Experiments, architectures and attainable resources	20
Appendix A.4. A statistical dual that is uniform over architectures	21
Appendix A.5. Controlled experiments without a dominating action	22
A.5.1. An informative action choice	24
Appendix A.6. Controlled experiments at a prescribed memory profile	25
A.6.1. A parameterwise occupation state	25
A.6.2. An exact informative-control bottleneck	27
Appendix A.7. Saddle faces and causal realization	28
A.7.1. A normal form over a finite causal interface	28
A.7.2. The realization theorem	30
A.7.3. Faces as memory witnesses	31
Appendix A.8. Facial certificates for schedule-optimized memory	32
Appendix A.9. Exact minimax laws for a marked product family	34
A.9.1. Complementary slackness and the event alphabet	35
Appendix A.10. A self-contained collision interface	36
A.10.1. Preparation and elastic scattering	36
A.10.2. Positive noise and the actual mark	37
Appendix A.11. The physical experiment and its sharp preparation boundary	38
A.11.1. An exact rational certificate against one preparation	39
A.11.2. A marked five-event selector	40
A.11.3. Causal compilation and the full register profile	40
Appendix A.12. The exact two-preparation marked saddle	42
Appendix A.13. A continuous two-preparation marked family	45
Appendix A.14. Exact memory from a marked saddle	46
A.14.1. The sharp serial-validation peak	48
Appendix A.15. A full-support marked saddle	50
Appendix A.16. Exact statistical form of the Brownian collision sensor	52
Appendix A.17. Optimal validation orders and a positive-noise memory jump	54
Appendix A.18. Preparation limits and localization of least-favorable priors	57
Appendix A.19. An exact preparation–decision-memory law	59
Appendix A.20. Integer memory and the failure of a naive partition formula	61
Appendix A.21. Quantitative certificates for a finite compatible architecture	63
Appendix A.22. Autonomous noisy testing with a precision and time ledger	64
Appendix A.23. Physical consumers and the first-principles connection	66
Appendix A.24. A sharpened physical consumer and its simulation budget	67
Appendix A.25. Dependencies, retained results and remaining distinctions	68
References	68

1. INTRODUCTION

A finite stochastic experiment can have a low-dimensional predictive image and nevertheless require many persistent states. The reason is positivity: an internal state describes a normalized future experiment, not an arbitrary vector in a linear span. Causality imposes a second constraint. Positive representations at successive cuts must be joined by common stochastic updates, and a schedule chosen from observations cannot reappear as an uncharged clock.

We seek conditions under which those constraints admit an intrinsic and exact state count. A lower bound obtained from selected faces and an independently supplied upper program can certify a particular example. It does not yet tell us when the continuation geometry itself determines a complete answer. Theorem 2.2 supplies such a condition for a finite class of causal experiments. Its data are all normalized continuation rows of the response array, their minimal faces in the causal polytope, and the connected components of the face-intersection graph. The resulting sum of affine ranks is a lower bound against every stochastic realization. If the residual convex hull in each component is a simplex, a canonical residual-vertex construction attains those ranks at every cut in one machine.

The condition does not assume a dimension-growth law, a metric entropy exponent, or a proposed controller. It can be checked by exact linear feasibility and rank. It is also not universal: an explicitly given full-support square has rank three and needs four states. The failure example makes clear what the completeness hypothesis excludes. For independent terminal channels the complete class is closed under products, and its intrinsic capacities multiply (Theorem 3.1). At every declared schedule vertex this gives an exact field rather than a field of analyst-selected certificates. The schedule optimum follows by a bottleneck recurrence.

A second theorem gives a more substantial separation in a growing experiment. Read n bits, then answer a later coordinate query with specified bias η . Every input and every query must have exactly the required conditional output law. The checkpoint geometry is the containment of $[-\eta, \eta]^n$ in a convex hull inside $[-1, 1]^n$. For $0 < \eta \leq n^{-2}$, a sequence of enclosing simplices has $(t+1)$ vertices after t reports and is closed under appending the next bit. Its barycentric updates implement the channel with peak $n+1$, which the affine dimension forces. For $\eta > 1 - n^{-1}$, disjoint corner caps force 2^n states, attained by full retention. Both lower bounds hold after any adaptive acquisition schedule, because the whole acquired transcript must pass through the state before the query. Both regimes include full-support laws. Thus the same predictive rank $n+1$ can be attained by causal memory or be exponentially smaller than it.

The statistical application enters in the opposite direction: a minimax saddle first forces continuations, and the intrinsic geometry then prices them. Bayes indifference alone does not specify an optimal continuation. For the marked two-preparation family, contact and stationarity fix two coordinates left free by a least-favorable prior. The five forced rows have canonical support components of dimensions 1, 0, 1, recovering the exact count $2 + 1 + 2 = 5$. These rows remain forced in the full-support physical rectangle. Their independent query-channel products have capacity 5^m ; this is a statement about specified product continuations, not a new formula for the multi-block minimax value.

The proof structure is therefore

$$\begin{aligned}
 & \text{uniform saddle contact} \longrightarrow \text{forced normalized continuations} \\
 & \longrightarrow \text{intrinsic positive geometry} \longrightarrow \text{compatible simplex updates} \\
 & \longrightarrow \text{one causal realization with matching memory.}
 \end{aligned}$$

The general positive-realization and residual-automaton mechanisms have classical antecedents. Section 6 compares the precise claims with stochastic realization, residual automata, nested-polytope factorization, incomplete-machine minimization and random-access coding. We do not claim a new general minimax interchange or a solution of unrestricted positive-rank minimization.

1.1. Resource and parameter conventions. All persistent randomness belongs to the register. A fixed finite program and known row constants are read-only; an unknown target is not a free constant. The finite-state theorems use atomic stochastic rows, not free finite-length exact sampling of arbitrary real coins. The random-access query is one atomic input, explicitly distinct from a bit-serial query buffer. Input-dependent acquisition choices are allowed, but their retained protocol information is charged. The physical twelve-state optimum is over externally declared validation clocks; no adaptive collision optimum is inferred from it.

For a specified positive-noise Brownian target the exact physical saddle and five/twelve-state conclusions are retained. Acquired target calibration instead gives an approximate-score controller with finite counter and precision costs. The main theorems keep these quantifiers separate.

1.2. Organization and the retained development. Sections 2–6 contain the current theorem spine and its proofs. The supporting sections then retain the previous causal foundations, exact saddles, physical identification and validation-order proofs, revelation laws, finite-architecture certification, all-preparation localization, and transport consumers. Their stable theorem labels remain resolvable in this single manuscript. No predecessor proof is discarded to shorten the main narrative. The historical A2 geometric root and B4/C2 operator gates retain their own hypotheses; neither the present class theorem nor the growing query example replaces those independent derivations.

2. AN INTRINSIC CONTINUATION INVARIANT

We first give a finite realization theorem whose input is the complete response law, rather than a selected collection of lower certificates. The distinction between a law and an implementation is essential: states in a competing implementation need not themselves be residuals of the law. The lower bound below allows all such states.

2.1. Normalized causal residuals. Fix a finite horizon, finite alphabets, and a list of charged cuts. At an elementary stage the machine chooses an action a , then receives a report y . A terminal output is an action followed by a dummy report. Let $C(\mathbf{a} \mid \mathbf{y})$ be a nonnegative normalized array whose action-prefix marginal depends only on the reports preceding that prefix. Thus an action cannot read its own future report. Every necessary buffer cut is an elementary cut. Time-dependent rows are allowed; a persistent seed, data-dependent clock, or protocol register is not a free input.

These conventions describe exact realization of a *specified* causal array on the full declared alphabet. For a model imposing only almost-sure requirements, one must first specify a legal completion on unsupported words, or optimize over those completions. Changing the completion can change the answer. Additional autonomous row-identification constraints are not part of the clocked theorem below.

For a positive-mass controller prefix $h = (\mathbf{a}_{\leq t}, \mathbf{y}_{\leq t})$, let R_h be the normalized array of future actions conditional on that prefix and on externally supplied future reports. Positivity of the prefix refers to its marginal in C , with past reports fixed, and not to a prior on environmental parameters. Write $\mathcal{R}_t$ for the finite set of distinct such arrays. Duplicate arrays are removed. Let $\mathcal{P}_t$ be the polytope of all normalized causal arrays on the remaining interface. Its defining equations express normalization and nonanticipation; its inequalities express nonnegativity.

Legality restrictions are included in the response array's zero pattern. A mandatory retained output or buffer is included through its prescribed later readout, so that deleting it changes the continuation task.

If $R \in \mathcal{R}_t$ and its next action has probability $p_R(a) > 0$, then, for every next report y ,

$$(1) \quad R(a\mathbf{a} \mid y\mathbf{y}) = p_R(a)R^{a,y}(\mathbf{a} \mid \mathbf{y}), \quad R^{a,y} \in \mathcal{R}_{t+1}.$$

The scalar $p_R(a)$ is independent of y . If $p_R(a) = 0$, the entire left side is zero. These are normalized residuals of a controller, not posteriors in its runtime register.

Definition 2.1 (Support components and their rank). For $R \in \mathcal{R}_t$, denote its minimal face in $\mathcal{P}_t$ by $F_t(R)$. Join two residuals when their minimal faces intersect. Let $\mathcal{R}_{t,1}, \dots, \mathcal{R}_{t,c_t}$ be the connected components of this graph, and put

$$(2) \quad d_{t,j} = \dim \text{aff } \mathcal{R}_{t,j}, \quad s_t(C) = \sum_{j=1}^{c_t} (d_{t,j} + 1), \quad e_t(C) = \sum_{j=1}^{c_t} |\text{ext conv } \mathcal{R}_{t,j}|.$$

Call the residual geometry *componentwise simplicial* at t if each $\text{conv } \mathcal{R}_{t,j}$ is a simplex, including a point as a zero-dimensional simplex.

Everything in this definition is determined by C and the charged interface. For finite rational data, face intersection is a linear feasibility problem with the two zero patterns imposed, affine dimension is a matrix rank, and extreme-point membership is another linear feasibility problem. No witness subfamily or trial controller enters the definition. Since normalized residuals lie in an affine hyperplane not containing zero, $d_{t,j} + 1$ equals their ordinary row rank. For terminal-output channels the ambient polytope is a product of probability simplices; two minimal faces intersect precisely when their allowed-output sets intersect at every future input.

Theorem 2.2 (Intrinsic simultaneous realization). *Every stochastic clocked realization of C , with all persistent variables included in its register, has*

$$(3) \quad K_t \geq s_t(C)$$

at every charged cut. A single stochastic realization has profile $K_t = e_t(C)$ simultaneously at all cuts. If the residual geometry is componentwise simplicial at every cut, then

$$(4) \quad K_t^{\min} = s_t(C) = e_t(C) \quad \text{simultaneously,} \quad W^{\min}(C) = \max_t s_t(C).$$

A realizing machine is obtained directly from residual vertices and barycentric one-step updates. Thus the statement determines an exact profile, not just unrelated lower bounds at individual cuts.

Proof. Conditioning a realizing machine on a controller prefix gives

$$(5) \quad R_h = \sum_s E_h(s)B_{t,s}, \quad E_h(s) \geq 0, \quad \sum_s E_h(s) = 1, \quad B_{t,s} \in \mathcal{P}_t.$$

Here $B_{t,s}$ is the future program starting in the actual state s . It may lie outside the convex hull of the residuals. If $E_h(s) > 0$, the defining property of a face implies $B_{t,s} \in F_t(R_h)$. A state used for residuals in two different components would therefore belong to two intersecting minimal faces, which would join those components. This is impossible. The states used by distinct components are disjoint. Within component j , the affine span of its residuals is contained in the affine span of its used state continuations, and hence at least $d_{t,j} + 1$ states are required. Summing proves (3). This argument does not replace stochastic encoders by deterministic ones.

For the upper construction, at cut t choose as states all extreme points of the component convex hulls. Every such point is an actual residual, since the residual set is finite. Each residual is a convex combination of these chosen states. Given a state R , choose its next action with

probability $p_R(a)$. For positive $p_R(a)$, represent $R^{a,y}$ from (1) as a convex combination of the chosen states at cut $t + 1$, and use those coefficients as the update row after report y . At zero $p_R(a)$ choose an arbitrary update on the unused row. The action row remains independent of the unread report. Initialization is the decomposition of C at the first cut; the terminal continuation is realized by its prescribed output row.

Backward induction proves the entire future array from every chosen state. In particular, the updates at different cuts implement the same law; no separate factorization is being spliced without checking its shifts. This gives the profile $e_t(C)$. In the simplicial case each component has exactly $d_{t,j} + 1$ vertices, so the upper and lower profiles coincide. Within each component the barycentric coefficients are unique, although arbitrary minimum-size realizations need not be unique or have states equal to residuals. $\square$

Corollary 2.3 (A complete, finite test for the stated class). *For a complete finite rational response array, componentwise simpliciality and the numbers in (4) can be decided from linear systems, rational linear feasibility, and a finite graph. When the test succeeds, rational transition tables realizing the exact profile can be constructed. The same holds for real algebraic input using exact sign and linear-feasibility decisions over the specified ordered field.*

Proof. Enumerate the positive-prefix normalized residuals and their zero patterns. Solve the pairwise face-intersection problems in the stated causal polytope, find the connected components, and find the extreme residuals of each component hull. A component hull is a simplex exactly when its number of vertices equals its affine dimension plus one. Solve the barycentric systems in the proof of Theorem 2.2. All operations are finite and preserve the indicated field. $\square$

The input table can be exponentially large in the horizon. The corollary is a completeness and synthesis result for an explicit class, not a polynomial algorithm in a succinct program length. It does not say that ordinary rank solves unrestricted positive realization. Nor does it claim that residual-state minimization, by itself, minimizes over all probabilistic machines; that last conclusion uses the separated-component lower bound. The distinction is also present in residual automata [21] and nested-polytope formulations [19].

2.2. Exact schedule fields without selected certificates. Consider a terminal-output channel on finitely many tapes, internally ordered on each tape, with a full product input alphabet. At a vector v of consumed-prefix lengths let $\mathcal{R}_v$ consist of the channel rows for *all* prefixes at v as functions of the remaining suffixes. The rows and their minimal faces are independent of the path used to reach v . Define $s(v), e(v)$ by (2).

Corollary 2.4 (Intrinsic declared-order minimization). *If every component hull at every schedule vertex is a simplex, the least width on each declared schedule π is $\max_{v \in \pi} s(v)$. In particular,*

$$(6) \quad W_{\text{clk}}^{\min} = \min_{\pi} \max_{v \in \pi} s(v).$$

The minimum is computed by the bottleneck recurrence on the finite schedule graph. The same residual-vertex construction supplies the updates for every allowed next-tape restriction.

Proof. Restriction of an actual prefix row by any next tape symbol is an actual row at the corresponding child vertex. Apply the proof of Theorem 2.2 on the entire schedule graph, then on each path. The last-edge decomposition of paths gives the bottleneck recurrence. No lower certificate or upper residual family is an extra input to this corollary. $\square$

An observation-dependent choice of tape has to be included as a controller action in C . Theorem 2.2 also classifies such a specified causal array when its normalized residual geometry passes the same test. Equation (6) alone optimizes declared paths, not all adaptive arrays. A

scalable experiment for which the peak optimum is unchanged even by adaptive acquisition is solved in Section 4.

2.3. Why the completeness hypothesis has content. The condition is neither universal nor a restatement of the answer. Consider four binary-query channels with response vectors $(\epsilon, \epsilon), (\epsilon, 1 - \epsilon), (1 - \epsilon, \epsilon), (1 - \epsilon, 1 - \epsilon)$, where $0 < \epsilon < 1/4$. All output probabilities are positive, so there is one support component. Its affine dimension is two and $s = 3$, but four states are necessary. Indeed, for each corner of the square a convex representation of its displayed row needs a generator with total coordinate error at most $2\epsilon < 1/2$. Two distinct corner-error sums add to at least one, so no generator can serve two corners. Four deterministic corner generators suffice. The nonsimplicial square therefore has state count four rather than three. This is a strict failure of the lower bound outside its completeness class, not an unsupported claim that the class contains every experiment.

3. PRODUCTS AND INTRINSIC MEMORY GROWTH

A classification becomes more useful when it survives independent composition. For the following statement a continuation is a normalized channel from a finite future-query alphabet Y to a finite output alphabet O . Multiple components are allowed. The joint experiment requires the whole product output law, not merely the correct marginal means.

Theorem 3.1 (Multiplicativity on the complete class). *Let $\mathcal{R} \subset \prod_{y \in Y} \Delta(O)$ and $\mathcal{S} \subset \prod_{z \in Z} \Delta(A)$ be finite sets whose support-component hulls are simplices. Form the independent product rows*

$$(R \otimes S)(o, a \mid y, z) = R(o \mid y)S(a \mid z).$$

Their component hulls are again simplices, and their intrinsic capacities satisfy

$$(7) \quad s(\mathcal{R} \otimes \mathcal{S}) = s(\mathcal{R})s(\mathcal{S}).$$

Consequently every joint stochastic encoder and decoder for the product requires this many states; correlated internal implementations cannot reduce the number. For independent terminal channels whose residual sets pass the test at every individual cut, the product profile on any interleaving is exactly the product of the individual capacities at the corresponding vector of cuts.

Proof. Minimal faces of a query channel are given by the allowed outcomes at each query. Two product faces intersect exactly when the faces in each factor intersect: supports multiply, and a nonempty intersection must have nonempty projected supports in both factors for every query. The component graph of the product is the graph product with a loop at each vertex. Its connected components are precisely products of components, because one can move along a path in one factor while holding the other fixed.

In a component with k simplex vertices, the flattened normalized channel vectors of those vertices are linearly independent. Indeed, normalization turns any linear dependence into an affine dependence. For components with k and l vertices, their kl product vectors are linearly independent by the tensor-product rank identity. Every product of rows is a convex combination of these vectors, and every one of these product vertices is itself a product row. Hence the component hull is their $(kl - 1)$ -simplex. Summing kl over pairs of components gives (7). The cut lower bound allows arbitrary normalized joint generators, not just product generators. The upper construction uses products of the individual generators. Restriction of a product residual changes only the factor whose input is read, so compatibility holds simultaneously throughout the interleaving graph. $\square$

The independence in the specification of the output channel is indispensable. Independent inputs or additive payoffs alone do not force a product of optimal response channels. In particular, (7) is not a multiplicativity claim for arbitrary nonnegative rank or for the marked minimax values U_N .

Corollary 3.2 (A full-support family at a preparation barrier). *For $q \geq 2$ and $0 < \lambda < 1$, let*

$$T_\lambda(b | a) = \lambda \mathbf{1}_{a=b} + (1 - \lambda)/q, \quad a, b \in [q].$$

Acquire n input symbols before releasing any output; subsequently produce the specified independent product channel $T_\lambda^{\otimes n}$. The exact minimum peak is q^n , even if acquisition and output schedules depend on retained observations and internal randomization, provided all schedule memory is charged and no past transcript can be read back. For a declared schedule, at quiescent cuts between completed release operations, the exact width with m acquired but unreleased symbols is q^m . An intermediate held-output cut is charged separately using the active count immediately before that release. At $\lambda = 0$, one state suffices until an output-buffer cut.

Proof. T_λ has eigenvalues 1 and λ with multiplicity $q - 1$, so its q rows are independent and form a simplex. All entries are positive. Theorem 3.1 gives q^m at any fixed active cut. For an adaptive schedule at the barrier, condition on the complete input word x . Its state law is an encoder $E(s | x)$; the law of the entire subsequently produced labelled output vector is a decoder $D(\mathbf{b} | s)$. Any output-order choices are included in this law before taking the labelled-output projection. Thus $T_\lambda^{\otimes n} = ED$, whose rank q^n bounds the state count. This argument charges, rather than reveals, the acquisition transcript. Retaining the full input word gives the matching fixed-order upper bound. Releasing a symbol samples its output and discards that input; even if the output itself is held for one cut, there are at most $q q^{m-1} = q^m$ states. At $\lambda = 0$ the outputs are independent of all inputs, and can be sampled without retaining them. $\square$

This is a rank-boundary phenomenon entirely inside full support, unlike the physical support boundary in Theorem A.17.2. For every fixed n , the product law tends uniformly in total variation to the input-independent output law, with defect at most $n\lambda(1 - 1/q)$. Thus a one-state generator approximates it with that defect, while every nonzero λ has the stated exact memory. The estimate follows by coupling each output with an independent uniform symbol and summing the per-coordinate disagreements. Exact memory and approximate statistical performance are different quantities.

4. A SCALABLE FULL-SUPPORT EXPERIMENT WITH TWO EXACT MEMORY LAWS

We next keep the final answer alphabet binary while the preparation length grows. This distinguishes intrinsic positive memory from ordinary predictive dimension, and gives an exact causal construction rather than a checkpoint factorization alone.

An input $x = (x_1, \dots, x_n) \in \{-1, 1\}^n$ is fixed. Each coordinate can be read once during acquisition. After acquisition an external query $j \in [n]$ arrives as one atomic symbol, and a bit $B \in \{-1, 1\}$ is emitted. For a specified $0 \leq \eta \leq 1$, the required experiment is

$$(8) \quad \Pr(B = b | x, j) = \frac{1 + b\eta x_j}{2}.$$

Acquisition may choose the next unread coordinate using the retained state. At the query cut the entire acquisition transcript has been discarded except for that state. Neither a public persistent seed nor a transcript of the chosen order is given to the decoder. The external round number is a declared clock; any data-dependent schedule variables must be stored. The output is sampled in the update receiving j , so j need not be buffered after that update. Exact arbitrary stochastic rows are atomic, as in the preceding realization theorems.

Let $W_n(\eta)$ be the least whole-schedule peak over these controllers. Let $K_n(\eta)$ be the least number of states just before the query, without an upstream cap. For $0 < \eta < 1$ every required answer probability has full support. Equality with (8) is required for every input and every query; an average success criterion is different.

Theorem 4.1 (Linear and exponential exact memory regimes). *For every integer $n \geq 1$,*

$$(9) \quad K_n(\eta) = W_n(\eta) = n + 1, \quad 0 < \eta \leq n^{-2},$$

$$(10) \quad K_n(\eta) = W_n(\eta) = 2^n, \quad 1 - n^{-1} < \eta \leq 1.$$

In the weak-signal regime a fixed-order streaming controller has exactly $t + 1$ states after t input bits, simultaneously for $0 \leq t \leq n$. For that fixed order these counts are minimal. Both peak lower bounds hold for arbitrary adaptive acquisition schedules. The two displayed regimes are not asserted to exhaust the intermediate parameter range.

4.1. The intrinsic checkpoint geometry. For a query-state s , write $z_s(j) = \mathbb{E}[B \mid s, j] \in [-1, 1]$. An encoder gives (8) exactly if and only if

$$(11) \quad \eta x = \sum_s E(s \mid x) z_s \quad \text{for every } x \in \{-1, 1\}^n.$$

Thus $K_n(\eta)$ equals the smallest number of points in $[-1, 1]^n$ whose convex hull contains $[-\eta, \eta]^n$. Necessity is conditioning on the state. Conversely any such containment supplies a stochastic encoder and binary decoder. This is a normalized nested-polytope formulation, not an identification with unrestricted matrix rank [19]. It says nothing by itself about the memory needed to compute the encoder online.

For $\eta > 0$, the inner cube has affine dimension n , whence

$$(12) \quad K_n(\eta) \geq n + 1.$$

Equivalently, the matrix in (8), with input words as rows and pairs (j, b) as columns, has ordinary rank $n + 1$. Adaptive acquisition cannot evade (11): after the barrier there is still only one retained state law for each complete input, and the same decoder on every input.

4.2. A compatible sequence of enclosing simplices. We construct the encoder using only a label, never a stored real vector. For $t \geq 1$ put $b_t = t\eta$ and define

$$(13) \quad P_t = \{z \in \mathbb{R}^t : z_i \geq -b_t, \sum_{i=1}^t z_i \leq b_t\}.$$

Its vertices are $v_{t,0} = (-b_t, \dots, -b_t)$ and $v_{t,i} = v_{t,0} + (t+1)b_t e_i$, $1 \leq i \leq t$. The distinguished coordinate of $v_{t,i}$ is $t^2\eta$ and its other coordinates are $-t\eta$. Therefore $P_t \subset [-1, 1]^t$ when $\eta \leq n^{-2}$ and $t \leq n$. Set $P_0 = \{()\}$. The key causal inclusion is

$$(14) \quad P_t \times \{-\eta, \eta\} \subset P_{t+1}.$$

Indeed old coordinates are at least $-t\eta$, the new coordinate is at least $-\eta$, and the new sum is at most $t\eta + \eta = b_{t+1}$. For $w \in P_t$ its barycentric coordinates are

$$(15) \quad \lambda_0(w) = \frac{b_t - \sum_i w_i}{(t+1)b_t}, \quad \lambda_i(w) = \frac{w_i + b_t}{(t+1)b_t} \quad (1 \leq i \leq t).$$

They are nonnegative, sum to one, and reconstruct w .

Proof of (9). At time t the register is an index $s \in \{0, \dots, t\}$. After bit x_{t+1} , replace it by index s' sampled from the barycentric coordinates of $(v_{t,s}, \eta x_{t+1})$ in P_{t+1} . These are legal rows by (14). Induction gives

$$\sum_s \Pr(s \mid x_1, \dots, x_t) v_{t,s} = \eta(x_1, \dots, x_t).$$

Given the final state s and query j , draw the answer with mean $v_{n,s}(j)$. This is a valid binary row because $P_n \subset [-1, 1]^n$. The preceding identity proves (8) exactly. It also proves the entire online implementation, not just existence of a final stochastic matrix. The real vectors in the proof are read-only row descriptions; the machine stores only their indices. No past coin, real accumulator, or chosen-input transcript survives outside the register. Formula (15) shows that all acquisition rows are rational and independent of $\eta > 0$, because that common factor cancels. Only the final decoder depends on the specified signal strength.

The peak is $n + 1$; sampling and retaining the final binary answer uses two states. Bound (12) gives the converse even with adaptive acquisition. For a fixed acquisition order, after t bits fix all future input bits and consider queries among the first t coordinates. The resulting continuation rows have affine dimension t , forcing $t + 1$ states at that cut. This proves simultaneous minimality of the claimed streaming profile. $\square$

4.3. A geometry stronger than rank.

Proof of (10). For a corner x and a possible generator $z \in [-1, 1]^n$, set

$$d_x(z) = \frac{1}{2} \sum_{i=1}^n (1 - x_i z_i).$$

Equation (11) implies that its average under the encoder for x is $n(1 - \eta)/2 < 1/2$. Hence some used generator has $d_x(z) < 1/2$. For distinct corners x, x' , at a coordinate where they differ the two corresponding summands add to one; all other summands are nonnegative. Thus $d_x(z) + d_{x'}(z) \geq 1$, and no generator can serve both corners. All 2^n corners need distinct generators. Retaining the entire input word and then applying the single noisy query row gives the upper bound 2^n . The factorization argument is independent of the acquisition order or its internal randomization. $\square$

For $n \geq 2$, both exact regimes contain full-support laws. In the strong regime the ordinary rank is only $n + 1$, but the exact causal memory is 2^n . In the weak regime the nested simplices attain that rank with a compatible streaming implementation. Thus neither support alone nor linear dimension alone determines exact memory. In persistent bits these two regimes give respectively $\lceil \log_2(n + 1) \rceil$ and n . At $\eta = 0$ the query-state complexity is one; the peak is two if the emitted answer must be held for a charged output cut. Every nonzero weak signal has query-state complexity $n + 1$, despite full support and uniform convergence to the fair-output channel as $\eta \downarrow 0$.

Proposition 4.2 (Known rows, approximation, and a strict rank margin). *The input-independent fair-output channel has rowwise total-variation error $\eta/2$ in approximating (8). For $n \geq 2$, any approximation factoring through at most n query states has worst rowwise error at least $\eta/(2\sqrt{n})$. The exact state counts in Theorem 4.1 refer to known η and atomic stochastic rows, not to an unknown calibration or a fixed-length finite fair-bit implementation.*

Proof. The first assertion is the binary total-variation formula. For the second, form the 2^n by $2n$ probability matrix A . Its nonzero singular values are $\sqrt{n2^{n-1}}$ and $\eta\sqrt{2^{n-1}}$ with multiplicity n . This follows by using the orthogonal row functions $1, x_1, \dots, x_n$ and, in each query block, the orthogonal column vectors $(1, 1)$ and $(1, -1)$. A matrix of rank at most n differs from A

in Frobenius norm by at least its smallest nonzero singular value. If every binary-query row has total-variation error at most δ , each pair contributes at most $2\delta^2$, so the squared Frobenius error is at most $2n2^n\delta^2$. The claimed bound follows. This is only an approximation bound, not an assertion that the two sides match. $\square$

Random-access coding is a classical communication task [22, 23, 24]. Equation (8) fixes every conditional decoding probability, not merely a lower bound on worst-case success or an average over inputs and queries. The new causal assertion made here is the exact finite-horizon profile from (14), together with lower bounds that survive adaptive acquisition and all persistent-randomness accounting. No priority claim for the communication model or for convex coding is made.

5. FROM SADDLE CONTACT TO INTRINSIC MEMORY

The geometric theorems classify response laws. A statistical optimization must first determine which laws are forced. We make that logical order explicit, so that Bayes indifference is not mistaken for freedom to choose the cheapest response.

Let $\mathcal{C}$ be the finite polytope of full-history causal response arrays and $g_\theta(C)$ the continuous payoff against a fixed unknown parameter in a compact space. Suppose μ_* is a least-favorable prior with value U . The optimal set is

$$\mathcal{S}_* = \left\{ C : \int g_\theta(C) d\mu_* = U \right\} \cap \bigcap_{\theta} \{ C : g_\theta(C) \geq U \}.$$

For every $C \in \mathcal{S}_*$, nonnegativity and zero prior integral give $g_\theta(C) = U$ on $\text{supp } \mu_*$. Differentiability at an interior supporting parameter also gives $\nabla_\theta g_\theta(C) = 0$. These are the contact equations proved in Theorem A.7.3; they apply even when a response coordinate has zero Bayes coefficient.

Corollary 5.1 (A contact-rigid saddle classification). *Suppose the saddle conditions determine a unique complete causal response array C_* . If its normalized residual geometry is componentwise simplicial at every charged cut, the minimum profile attaining U is exactly $(s_t(C_*))_t$, simultaneously. Every smaller profile misses U by a strictly positive amount in a fixed finite interface. More generally, any finite family of common normalized continuation rows forced for every $C \in \mathcal{S}_*$ gives the support-component rank lower bound at its cut, even when those rows are obtained by parameter-independent averaging.*

Proof. Any value-attaining array belongs to $\mathcal{S}_*$, and hence equals C_* in the first assertion. Apply Theorem 2.2. The normalized finite-row realization space for a smaller fixed profile is compact. If its maximal worst-case score were U , it would contain an implementation of C_* , contrary to the lower bound. For the last assertion, each forced row factors through the same state continuations at that cut. Averaging encoder rows preserves normalization and this factorization. The separated-component lower argument therefore applies to the forced rows. This last lower bound does not, without completeness and compatibility, assert an exact whole-schedule profile. $\square$

This corollary states a checkable implication after contact has been solved; it does not assume that every saddle is response-rigid. The following application deliberately uses only the common forced rows, because the marked saddle does not determine every coupling of fresh validation coins.

5.1. The marked decision geometry as a complete instance. The full-support saddle of Theorem A.15.1, proved in the supporting development, concerns

$$(\gamma, \kappa) \in [6/25, 13/50] \times [0, 1/1000].$$

Its contact equations force the following five averaged response rows on the diagonal atoms 000, 001, 110, 111, with zero response at the four other atoms:

$$(16) \quad (t, 0, 1, 1), \quad (1, 0, 1, 1), \quad (1, 1, 1, 1), \quad (1, 1, 0, 1), \quad (1, 1, 0, t), \quad 0 < t < 1.$$

The inner three rows are individually forced by strict Bayes signs. The outer rows are averages over the independent training marks; contact and stationarity fix their formerly indifferent coordinates. Every actual state encoder still factors these averages through its same decoder. These facts are proved without a count-only restriction on competitors in Theorems A.14.1 and A.15.1.

Proposition 5.2 (Canonical five-state decision capacity). *For the forced response rows (16), the support graph has components of sizes 2, 1, 2, with affine dimensions 1, 0, 1. Their hulls are simplices. Thus their intrinsic capacity is*

$$(17) \quad 2 + 1 + 2 = 5.$$

This recovers the exact decision count against all the original stochastic encoders, training gates, stopping rules, and independent calibration, not just against a chosen five-event compiler.

Proof. Convert each response coordinate $h(y)$ into the two probabilities $(1 - h(y), h(y))$. This affine embedding identifies the response cube with its normalized binary-query channel polytope. The first two rows have intersecting minimal faces: the edge of the first contains the second. The last two rows have the reflected relation. Their faces are disjoint from one another and from the singleton face of the middle row. Hence these are exactly the three support components. The first and last components each contain two distinct points; the middle contains one. Corollary 5.1 gives the five-state lower bound. The retained optimal event selection attains it and its validation implementation is proved in the supporting physical development. No new physical hypothesis is inserted. $\square$

For m independent copies of this specified binary-query continuation, the exact joint checkpoint capacity is 5^m by Theorem 3.1. This requires the product output law for the independent copies. It is not a claim that the minimax value of an m -block audit is the product of U_2 , or that marginal optimality alone forces independent output randomization.

5.2. Known-target physical exactness and acquisition. The supporting proofs retain the earlier rational certificate and uniform two-preparation construction, with constants

$$(18) \quad u_1 = \frac{4035777}{10240000}, \quad m_2 = \frac{12 - 3\sqrt{3}}{16}, \quad \beta_0 = \frac{1}{300}.$$

These are not substituted for the later exact saddle value U_2 .

For every specified Brownian target $Q_{d,\sigma}$ in $9/10 \leq d \leq 11/10$, $0 < \sigma \leq 1/24$, the exact target-identification proof gives

$$\gamma = \frac{1}{4} - \frac{1}{2}\mathbb{E}e, \quad \kappa = \frac{1}{2}\mathbb{E}[e(1 - e)] > 0.$$

The conditional sign-error rate e is shared through the physical signal; replacing $\mathbb{E}e^2$ by $(\mathbb{E}e)^2$ is not justified. The retained full-support saddle and schedule theorems therefore give five decision states and minimum peak twelve over all externally declared, order-preserving validation clocks. Exactly the serial orders attain that minimum. The ideal reverse-order support limit has peak

ten. These are theorems about the specified target with its exact stochastic row available, and their proofs remain in full below.

An acquired target is different. Its finite counter and finite-bit construction in Proposition A.16.3 gives approximate score recovery with its stated calibration error and workspace. Neither that proposition nor Theorem 4.1 identifies an unknown algebraic row exactly from finitely many samples. The new intrinsic classifier does not change this distinction. Nor does its declared-path corollary convert the physical fixed-clock optimum into a lower bound for all report-dependent validation schedulers. The adaptive-acquisition quantifier of Theorem 4.1 is proved for that experiment by the barrier factorization, not imported from the collision example.

6. THEOREM-LEVEL RELATION TO REALIZATION AND CODING

There are three different minimization problems in the argument, and their classical precedents should not be conflated.

Positive realization versus an intrinsic complete class. Heller’s positive-realization viewpoint and the HMM realization survey of Vidyasagar [17, 18] concern the representation of stochastic laws by positive state systems. They motivate the normal form used here. Our main theorem does not replace that theory or solve its unrestricted minimization problem. Its finite input is a complete causal response array with declared cuts. From all its normalized residuals, it computes a face-intersection graph and a sum of affine ranks. Componentwise simpliciality is a finite geometric test, and under that test the lower sum equals a simultaneously realizable profile. No choice of a witness family remains in the answer. The general normal-form statement alone does not supply this all-realizations lower bound or its completeness test.

Residual minimality versus all-machine minimality. Denis and Esposito [21], Proposition 16 and their probabilistic residual-automaton results, establish canonical minimal generating subsets of residuals under their finite-generation hypotheses. The residual-vertex upper construction in our proof belongs to that line of ideas; its existence is not claimed as a new principle. But being smallest among residual-state representations need not mean being smallest among all positive representations. Our lower proof allows arbitrary state continuations outside the residual hull and rules out their sharing across distinct support components. It is this additional step, together with affine independence inside each component, that equates residual size with unrestricted stochastic size in the stated class. The square example after Corollary 2.4 exhibits failure of rank completeness outside that class.

Nested polytopes versus causally compatible covers. Gillis and Glineur [19], Definition 1 and Theorem 1, distinguish restricted nonnegative rank and give its nested-polytope formulation. Their distinction between ordinary and restricted rank is retained: we never force a competing generator into the residual affine span. For normalized binary-query channels the cube containment in (11) follows directly by conditioning. That checkpoint characterization is not a new identification of nonnegative factorization with convex geometry. The additional online question is whether covers at successive cuts admit stochastic shifts. The explicit inclusions (14) answer it for every preparation length and attain the lower rank at every acquisition cut. A final enclosing simplex without those inclusions would not establish Theorem 4.1.

Reusch and Merzenich [20] study compatible coverings for incompletely specified sequential machines. Their subject is particularly relevant when a target has support zeros. We do not infer a canonical completion from an almost-sure law. The present intrinsic theorem uses a complete array, and the retained physical support-boundary proof specifies its completion and supported tests explicitly.

Random-access codes versus exact online channel synthesis. Classical and quantum random-access coding, including the distinction between private and shared randomness, is established

in [22]. Recent work studies average and worst-case coding probabilities [23, 24]. The classical private-randomness factorization in [24], equation (4), is precisely the checkpoint conditioning mechanism underlying (11). Thus the random-access model and that factorization are not inventions of this article. The present statement fixes every conditional probability to (8) and prices the entire sequential encoder. The affine-dimension lower bound, the compatible $(t + 1)$ -state streaming construction, and the disjoint corner-cap converse give two exact regimes with different memory scales in one growing full-support family. We do not claim the optimal success probability for every message size or a complete intermediate-signal phase diagram. Statistical contact and the scope of originality. The new geometric classification applies to laws already specified or forced. A minimax problem supplies those laws only after Bayes optimality and uniform contact have both been imposed. The marked application makes this distinction concrete: the cheaper three-event Bayes tie rule is not uniformly minimax. Contact forces the two extra coordinates before intrinsic realization is considered. This comparison isolates the hypotheses and conclusions being used; it is not a claim that no reformulation of earlier theory could imply them. No complete priority or proof-level non-overlap audit of Norberg's filtered-experiment comparison [11] is asserted. The finite geometric and online results above are proved directly, with the inherited comparison, control, and testing mechanisms attributed in the supporting text.

SUPPORTING FOUNDATIONS AND PHYSICAL PROOFS

The following sections preserve the substantive mathematical body of revision 28. They supply the exact saddles, microscopic model, validation-order proofs, general causal foundations, and quantitative consumers used above. Earlier local scope statements describe the individual results in which they occur; the new classification and streaming theorems are stated in Sections 2–5. All predecessor theorem labels and all mathematical proofs remain available in this manuscript. The current introduction and theorem hierarchy replace repeated roadmap prose, not the supporting mathematics.

APPENDIX A.1. CAUSAL EXPERIMENTS AND THEIR PREDICTIVE QUOTIENTS

We begin with the object and morphism structure. Its elementary probability statements fix the meaning of later resource theorems; classical kernel construction, conditional expectation and quantization are not assigned a new priority. The substantive comparisons below retain their preparation, memory and approximation coordinates simultaneously.

A.1.1. The experimental object. An experiment has a fixed unknown parameter $\theta \in \Theta$, hidden states X_t , visible actions A_t , reports Y_{t+1} , and preparation laws μ_θ . We impose the following six conventions.

A0 (types and visibility). These spaces are standard Borel. Visible histories include actions, reports and publicly recorded resource increments. A legal-action set and its remaining budget are part of the visible protocol type. Two histories with different legal continuations are not identified.

A1 (positive kernels). There are jointly measurable normalized nonnegative kernels $K_t^\theta(dx', dy, dc \mid x, a)$, with c a nonnegative resource increment. An absorbing symbol represents termination or failure. Dirac kernels are allowed. Positivity here does not require every outcome to have strictly positive probability. A hidden state can contain the necessary physical past; this notation does not assume a finite-dimensional physical closure.

A2 (causality). A policy is a kernel $\alpha_t(da \mid h_t)$ depending only on visible past data and allowed independent randomization. It cannot read the unknown parameter. Stopping is with respect to the visible filtration.

A3 (executable tests). The specified bounded future tests include constants, required report events and their legal one-step continuations. For discrete reports the one-step instrument is

$$(\mathbb{T}_{t,a,y}f)(\theta, x) = \int f(\theta, x') K_t^\theta(dx', \{y\}, dc \mid x, a).$$

The test spaces are closed under the required instruments, finite linear combinations and the bounded monotone limits needed to recover the stated event laws. For general reports one uses event instruments and regular conditional distributions, not division by the probability of a continuous singleton.

A4 (actual preparation). Average risks and acquired state laws refer to the specified preparation and exploration protocol. Failed trials, calibration, resets and conditioning events retain their probabilities and implementation costs. A convenient measure on a reachable set does not replace its acquired law.

A5 (executable resources). Persistent labels, temporary buffers, program constants, calibration precision, preparations and clocks are distinguished. A retained selector is a state. A parameter-dependent table is not a known program constant. A simulator with R_t labels composed with a controller with K_t labels has the direct product bound $R_t K_t$, before any

justified quotient is taken. Atomic stochastic rows and fair-bit implementations are separately specified.

Proposition A.1.1 (Prepared path laws). *For every fixed parameter and causal policy, the preparation and kernels uniquely determine their finite-dimensional path laws. On a finite horizon they are the iterated kernel product*

$$\mu_\theta(dx_0) \prod_{t < T} \alpha_t(da_t | h_t) K_t^\theta(dx_{t+1}, dy_{t+1}, dc_t | x_t, a_t).$$

These laws are consistent under marginalization. For a countable horizon they determine the usual kernel extension. Deterministic dynamics give their original trajectory laws when represented by Dirac kernels.

Proof. Successive integration constructs each finite law. Normalization of the last two kernels gives its preceding marginal, and induction proves consistency and uniqueness on cylinder events. The countable kernel extension gives the full law; equivalently one applies the standard extension theorem to these consistent iterated kernels. Integration against a Dirac transition is evaluation along that transition, which proves the last assertion. $\square$

A.1.2. Prepared predictive equivalence. For Bayesian predictions additionally fix a prior on the parameter, and regard (θ, X_t) as the hidden variable. A prepared posterior π at a given protocol type evaluates the future tests. Define $\pi \sim_t \tilde{\pi}$ if their evaluations agree on every specified future test. This definition concerns one prepared experiment; it does not identify a prior-dependent posterior with a statistic sufficient simultaneously for all fixed parameters.

Theorem A.1.2 (Predictive quotient and causal closure). *For finite report alphabets, predictive equivalence is preserved by every legal positive-probability report update. Its class determines the laws of all specified future experiments. Every deterministic history statistic that determines those laws factors onto this quotient. For a measurable realization it is sufficient that a countable separating test family gives a measurable evaluation map with standard Borel image, and that the chosen statistic admits a measurable factor. In finite history spaces the latter conditions are automatic.*

Proof. For a legal action and report, Bayes' rule gives

$$\pi'(f) = \frac{\pi(\mathbb{T}_{t,a,y}f)}{\pi(\mathbb{T}_{t,a,y}1)}.$$

Both numerator and denominator depend only on the old class by A3. Equivalent posteriors assign the same probability to the report, so updates agree on the next test space whenever that probability is positive. Null reports do not affect the prepared law and may be assigned a fixed admissible version. Induction determines every finite future report word; the monotone-class extension gives the required future events. Conversely those event laws determine the tests. If two histories have the same sufficient statistic, they have the same future evaluations; the quotient is therefore constant on its fibers. This gives the factor set-theoretically. The stated measurable factor assumption supplies its measurable version; no claim that every abstract quotient of a standard Borel space is standard Borel is used. $\square$

Proposition A.1.3 (Barycenters, entropy and normalized response). *In a fixed preparation, suppose a hidden target Z , observation Y and postprocessing W form a Markov chain $Z \rightarrow Y \rightarrow W$. For any bounded vector of tests f ,*

$$p_W = \mathbb{E}[p_Y | W], \quad p_Y = \mathbb{E}[f(Z) | Y], \quad \mathbb{E}\varphi(p_W) \leq \mathbb{E}\varphi(p_Y)$$

for every convex φ for which these expectations exist. For a parameter-independent kernel G , $P \ll Q$, and finite $D(P\|Q)$, regular conditional laws give

$$D(P\|Q) = D(PG\|QG) + \int D(P(dx | w)\|Q(dx | w)) (PG)(dw).$$

Finally, if f, g are bounded and $\pi_u(dx) = e^{ug(x)}\pi(dx)/\pi(e^{ug})$, then

$$\left. \frac{d}{du} \pi_u(f) \right|_{u=0} = \text{Cov}_\pi(f, g).$$

Proof. The Markov property and the tower rule give the conditional barycenter; conditional Jensen gives convex order. The joint laws $P(dx)G(dw | x)$ and $Q(dx)G(dw | x)$ have relative entropy $D(P\|Q)$. Disintegrating by w and applying the entropy chain rule gives the identity, and hence data processing. The last equality follows by quotient differentiation; boundedness justifies differentiation under the integral. $\square$

The barycenter statement concerns a law of predictions, not a pointwise map between reachable sets. Likewise the covariance in the last formula is a normalized response, not an independently postulated Fisher matrix or a global sufficiency theorem.

APPENDIX A.2. ATTAINABLE RESOLUTION AND RESOURCE-AWARE SIMULATION

A.2.1. Checkpoint and online prediction. Fix an actual exploration protocol and its history law. Let $p(h)$ be a bounded d -dimensional conditional prediction vector, with law ν . For instance, a future test is sampled independently after encoding, and the coordinates include its sampling weights. Squared excess prediction loss is then $\mathbb{E}\|p(h) - \hat{p}\|^2$.

Proposition A.2.1 (Checkpoint resolution). *For at most M retained labels the least checkpoint excess loss is*

$$(19) \quad \inf_{c_1, \dots, c_M} \int \min_{1 \leq j \leq M} \|p - c_j\|^2 \nu(dp).$$

Independent randomized encoders or decoders cannot improve it. This formula concerns the stated preparation and does not provide a common codebook by separately optimizing at each unknown true parameter.

Proof. Conditional squared loss decomposes into its Bayes loss and squared distance from the conditional mean. For each retained label replace a random decoder by its mean. Random assignment to the resulting centers cannot do better than a nearest center. Conversely a nearest-center assignment attains the infimum over codebooks arbitrarily closely. Conditioning on an independent shared random seed, if such a seed is allowed, gives a codebook for each seed, each with risk at least (19). Averaging cannot improve the bound. The formula does not charge the computation of a checkpoint assignment as an online algorithm; online compatibility is a separate condition. $\square$

Theorem A.2.2 (A matched online resolution law). *Suppose a fixed prepared experiment has a predictive state s_t in a compact metric space S , a known initial state, and updates $s_{t+1} = F_{t,a,y}(s_t)$. Assume every permitted update is ρ -Lipschitz for one $0 \leq \rho < 1$, and the prediction map $p : S \rightarrow \mathbb{R}^d$ is L -Lipschitz. Suppose S has a δ -net of at most $C\delta^{-d}$ points for all sufficiently small δ . At a checkpoint T , assume the actual law ν of $p(s_T)$ satisfies the small-ball bound $\nu(B(z, a)) \leq Aa^d$ for every center z and all sufficiently small $a > 0$. Then for*

all sufficiently large M , the minimum online excess prediction loss with M retained predictive labels satisfies

$$(20) \quad c_0 M^{-2/d} \leq R_M^{\text{on}} \leq C_0 M^{-2/d}.$$

Here one may take

$$c_0 = \frac{1}{2}(2A)^{-2/d}, \quad C_0 = \frac{L^2 C^{2/d}}{(1-\rho)^2},$$

Any separate exploration controller, report buffer or clock is charged in addition to these M labels. The exploration protocol and the preparation are fixed before optimizing the encoder.

Proof. Any online M -label state is also a checkpoint encoder, so Proposition A.2.1 applies. For any M centers, the union of balls of radius $a = (2AM)^{-1/d}$ has ν -mass at most one half. On its complement squared distance is at least a^2 . Integration gives $c_0 M^{-2/d}$. This argument also permits randomized online encoders.

For the upper bound take a δ -net with $\delta = (C/M)^{1/d}$, and a fixed measurable nearest-net assignment (with a fixed tie rule). Quantize the known initial state and, after each legal update, quantize again. The state error obeys $e_{t+1} \leq \rho e_t + \delta$, $e_0 \leq \delta$, and hence $e_t \leq \delta/(1-\rho)$. The prediction error is at most Le_t . Squaring proves the upper bound, uniformly in time. Each update uses only the current net label and the current permitted input, so the construction is online. It does not grant the encoder free access to the original discarded history. If the exploration policy needs another register, that register is combined by a product construction. $\square$

For example, a Lebesgue density bounded above by B gives the small-ball hypothesis with $A = Bv_d$, where v_d is unit-ball volume. The lower hypothesis is a condition on the *acquired prediction law*. A reachable set of dimension d without a mass condition does not imply the lower bound. Similarly, the contraction assumption is additional dynamics, not a consequence of kernel positivity.

A.2.2. Causal morphisms and simultaneous error budgets. A causal simulator $\Gamma : \mathfrak{E} \rightarrow \mathfrak{F}$ is a parameter-independent executable transducer that uses source preparations to supply the target interface under every allowed target policy. Its defect ε is a uniform total-variation bound on the resulting path laws, including the fields required by the task. Its resource map ρ_Γ bounds the source resources of the transducer composed with a target controller.

Proposition A.2.3 (Composition and risk transfer). *Causal simulators compose. Their uniform defects add, capped at one, and their resource maps compose in the order $\rho_{\Gamma_1} \circ \rho_{\Gamma_2}$. For a terminal loss in $[0, L]$ and a common target task,*

$$\mathcal{R}_{\mathfrak{E}}(\rho_\Gamma(B)) \leq \mathcal{R}_{\mathfrak{F}}(B) + L\varepsilon.$$

For reward in $[-1, 1]$, the corresponding maximum worst-case rewards satisfy the reverse inequality with an error 2ε .

Proof. Interleave the two transducers; their pair of internal states is a causal implementation. Total variation contracts under a kernel and obeys the triangle inequality, proving the defect sum. If their state counts are R_t and M_t , the pair has at most $R_t M_t$ states. Other declared resource bounds compose by execution of the two protocols. A bounded loss changes expectation by at most its range times total variation. Apply this bound to every fixed parameter and target controller and take the appropriate infimum or supremum. Uniformity in the controller is essential to this step. $\square$

Theorem A.2.4 (Acquisition, quantization and model-error transport). *Suppose an exact source transducer has predictive state $s_t \in S_t$ and additional charged buffers of size R_t , supplies T target operations, and has uniform model defect at most ε_{mod} . Its initialization and update depend on a calibration quantity ξ . From N_{cal} source preparations acquire an estimate $\hat{\xi}$ with at most C_{cal} retained labels, such that*

$$\sup_{\theta} \mathbb{P}_{\theta}(d(\hat{\xi}, \xi) > a) \leq \eta.$$

Subsequent preparation noises are independent of calibration conditional on the fixed parameter. Let finite nets of sizes M_t have radii δ_t . Assume, on matching input/output histories and on the good calibration event, that the quantized transducer's state error satisfies

$$(21) \quad e_0 \leq d_0 + a_0 a + \delta_0, \quad e_{t+1} \leq \rho_t e_t + d_{t+1} + a_{t+1} a + \delta_{t+1}.$$

Assume its next output kernel and the exact transducer's output kernel differ in total variation by at most $\epsilon_t + L_t e_t + b_t a$, uniformly in the fixed parameter and legal policy. All constants and nets are declared independently of that unknown parameter. Then the quantized acquired transducer has defect at most

$$(22) \quad \Delta = \min \left\{ 1, \varepsilon_{\text{mod}} + \eta + \sum_{t < T} (\epsilon_t + L_t e_t + b_t a) \right\}.$$

Composing it with any target controller of cut widths K_t requires at most $C_{\text{cal}} R_t M_t K_t$ labels at aligned execution cuts and transfers its bounded reward with loss at most 2Δ . Its preparations are N_{cal} plus all source preparations specified by the transducer; no successful-trial conditioning removes the failures.

Proof. On the good calibration event, iterate (21). More explicitly, with $D_0 = d_0 + a_0 a + \delta_0$ and $D_j = d_j + a_j a + \delta_j$,

$$e_t \leq \sum_{j=0}^t D_j \prod_{k=j}^{t-1} \rho_k.$$

Couple each pair of output kernels maximally while their external histories agree. On agreement the target controller sees the same inputs, so it uses the same legal action and may use the same independent randomization. Its discontinuous action choices require no Lipschitz assumption: before a disagreement their arguments are identical. The chance of a first discrepancy is bounded by the sum in (22). Add the probability of bad calibration and the exact transducer's model defect. This proves the path bound, including stopping after padding by absorbing states.

The executable approximate transducer stores its calibration label, additional buffer, and net index. Combine these with the actual target register, not its full history or an uncharged belief. The resulting product gives the state count. If an operation contains further buffer cuts, apply the same product at each such cut; R_t was assumed to include them. Preparation costs add by execution. Finally apply Proposition A.2.3. $\square$

Corollary A.2.5 (A finite acquisition register). *Suppose r_0 calibration rows have at most d outcomes, and take n independent preparations from each row on a fixed schedule. Their empirical rows satisfy, with probability at least $1 - \eta$ uniformly in the row probabilities,*

$$\max \text{TV}(\hat{P}, P) \leq \frac{dz}{2}, \quad \eta = \min\{1, 2r_0 d \exp(-2nz^2)\}.$$

Their counts can be retained in at most $(n+1)^{r_0(d-1)}$ labels. Thus Theorem A.2.4 can use $N_{\text{cal}} = r_0 n$, $a = dz/2$ and this finite calibration register when its calibration metric is the maximum row total variation.

Proof. For each outcome the empirical frequency is the mean of independent Bernoulli indicators. Hoeffding's bound and a union bound over at most $r_0 d$ coordinates give the event stated. The total-variation estimate is half the sum of coordinate errors. Retain $d - 1$ counts per row; the last is reconstructed from the known number of draws at that stage of the clocked schedule. Each retained count takes at most $n + 1$ values. An autonomous implementation must additionally charge its schedule. $\square$

These results separate exact predictive equivalence, average checkpoint resolution, online realizability, and uniform simulation risk. In particular, neither an optimal checkpoint codebook nor the proof-state bundle of Section A.6 is automatically an executable finite register.

APPENDIX A.3. EXPERIMENTS, ARCHITECTURES AND ATTAINABLE RESOURCES

Throughout, total variation is $\text{TV}(P, Q) = \sup_{0 \leq h \leq 1} (P - Q)h = \frac{1}{2} \sum_x |P(x) - Q(x)|$. An experimental parameter is unknown and fixed throughout an execution. Controller randomization is independent of that parameter. A reset preparation is a fresh draw, independent of earlier preparations given the parameter. A controller may not recover discarded information from an uncharged tape, clock, selector or calibration table.

Definition A.3.1 (Finite architecture). A finite architecture A specifies a finite causal execution graph, its legal experimental operations, the observable inputs to each controller row, and a register alphabet at every cut. It also specifies which row occurrences share a stochastic table. A clocked architecture may use different rows at different scheduled times. An autonomous architecture may use time only through a charged state. For a parameter family Θ and terminal reward in $[-1, 1]$, let

$$V(A) = \sup_{q \in \prod_r \Delta_{s_r}} \inf_{\theta \in \Theta} g_\theta(A, q).$$

If $\mathcal{A}(R)$ is the class of architectures obeying a global resource budget R , its outer value is

$$V^{\text{out}}(R) = \sup_{A \in \mathcal{A}(R)} V(A).$$

Both the architecture and the simulator parameter are fixed before the controller's random choices. In particular, a mixture of complete controllers is not automatically an admissible controller at the original register profile.

For a finite hidden execution graph, compatibility can be stated without division. Let $x_{\theta ir}$ be the incoming mass of hidden history i at observable row r , and let $f_{\theta irj}$ be the outgoing mass through choice j . Include the usual fixed-kernel flow equations, as well as

$$(23) \quad f_{\theta irj} \geq 0, \quad \sum_j f_{\theta irj} = x_{\theta ir}, \quad f_{\theta irj} x_{\theta' i' r} = f_{\theta' i' rj} x_{\theta ir}.$$

For shared rows, the indices i, i' include all their occurrences.

Proposition A.3.2 (Compatible realization). *For a specified finite architecture, nonnegative flow arrays satisfying (23) are exactly those realized by a common stochastic row at every declared observable input, including at zero incoming mass.*

Proof. Necessity follows from $f_{\theta irj} = x_{\theta ir} q_r(j)$. For sufficiency, if some incoming mass at row r is positive, define $q_r(j)$ by dividing the corresponding outgoing mass by that incoming mass. Conservation makes q_r stochastic, and the cross-products force the same ratio at every other positive incoming mass. If an incoming mass vanishes, nonnegativity and conservation force all its outgoing masses to vanish. If the entire row is unreachable, choose any legal stochastic row.

Induction through the finite execution graph reconstructs the prescribed flows. The same ratio argument applies to identified occurrences of an autonomous row. $\square$

This formulation is inherited from the complete companion and is included to fix the controller class. Compact semialgebraic feasibility and an Archimedean positivity theorem provide a complete semialgebraic certificate hierarchy for a *fixed finite architecture*. They do not interchange the two optimizations in the definition. Section A.21 gives a quantitative alternative for the inner problem. The next section supplies a different certificate that survives the outer optimization.

APPENDIX A.4. A STATISTICAL DUAL THAT IS UNIFORM OVER ARCHITECTURES

Let Θ be compact, let Ω be finite, and let $\theta \mapsto P_\theta \in \mathcal{P}(\Omega)$ be continuous. Fix $Q \in \mathcal{P}(\Omega)$. An N -preparation reset selector observes $Z = (Z_1, \dots, Z_N) \sim P_\theta^{\otimes N}$ and chooses a response $h_Z \in [0, 1]^\Omega$. Its expected validation score is

$$(24) \quad G_\theta(h) = \sum_{z \in \Omega^N} P_\theta^{\otimes N}(z)(Q - P_\theta)h_z.$$

The response is fixed before two independent validation draws. Operationally it can be realized by choosing a random event with those coordinate marginals and retaining that event for both validations. No memory bound is imposed in the definition of $U_N(P, Q) = \max_h \min_\theta G_\theta(h)$.

Theorem A.4.1 (Posterior-predictive dual). *Both extrema in the following identity are attained:*

$$(25) \quad U_N(P, Q) = \min_{\mu \in \mathcal{P}(\Theta)} \sum_{z, \omega} \left[\int P_\theta^{\otimes N}(z) \{Q(\omega) - P_\theta(\omega)\} d\mu(\theta) \right]_+$$

$$(26) \quad = \min_{\mu \in \mathcal{P}(\Theta)} \sum_z m_\mu(z) \text{TV}(Q, \bar{P}_\mu(\cdot | z)).$$

Here $m_\mu(z) = \int P_\theta^{\otimes N}(z) d\mu(\theta)$ and, when $m_\mu(z) > 0$,

$$\bar{P}_\mu(\omega | z) = \frac{\int P_\theta^{\otimes N}(z) P_\theta(\omega) d\mu(\theta)}{m_\mu(z)}.$$

A zero-mass term in (26) is defined to be zero.

Proof. The response space is a finite-dimensional compact convex box. Probability measures on the compact space Θ form a weakly compact convex set. The integral of (24) is continuous and affine in each argument. The minimum over parameters equals the minimum over probability measures. The minimax theorem [3] therefore interchanges the maximum and minimum. For fixed μ , each response coordinate is independent in the maximization and its contribution is the positive part of its coefficient, proving (25). The coefficient vector at a fixed z has zero sum. For positive $m_\mu(z)$ it equals $m_\mu(z)(Q - \bar{P}_\mu(\cdot | z))$, proving (26). If $m_\mu(z) = 0$, every nonnegative likelihood integrand vanishes μ -almost surely and all coefficients are zero. Continuity on the two compact strategy spaces gives attainment. $\square$

Theorem A.4.2 (Dominating-row converse). *Suppose every permitted training row is a parameter-independent causal garbling of one fresh draw from P_θ . Suppose every permitted validation row is the same garbling of the target and candidate laws, and every selected row–event pair is fixed before validation. Then $U_N(P, Q)$ bounds the worst-case expected score of every auditor using at most N candidate training preparations, regardless of its register profile, feedback schedule, independent randomization, or stopping rule. Independent target-only calibration, whose law does not depend on θ , does not invalidate the bound. If the all-on row and all its events are permitted, the bound is attained in the class with no memory cap and N training preparations.*

Proof. Generate N independent all-on observations in advance. At the t th requested preparation, apply the declared garbling to observation t , using only previously simulated reports and independent randomization. This simulates the adaptive transcript with its exact law. A stopped execution ignores the unused observations; it does not reveal them to the auditor. Pull back its final row–event response to the all-on alphabet. Conditional on the entire generated training word and the independent random seed, this is a function in $[0, 1]^\Omega$. Averaging the seed, including any target-only calibration, produces one response h_z for each full word z . The score is consequently (24). This proves the upper bound for the union of all architectures. Conversely, retaining the entire word and then a selected event implements any such finite response kernel when memory is unrestricted. Fresh validation gives exactly (24). $\square$

Remark A.4.3 (What the dual does not convexify). The measure μ is an adversarial prior used to bound the minimum over *fixed* parameters. Its likelihood is $\int P_\theta^{\otimes N} d\mu(\theta)$, not $(\int P_\theta d\mu(\theta))^{\otimes N}$. It neither enlarges the private candidate class to mixtures nor gives the auditor a free persistent selector. Attainment of U_N without a memory cap does not assert attainment at any specified finite profile.

The state in (26) is generated by the actual preparation law, not by a geometric measure chosen independently of the experiment. For sequential continuation one retains the posterior μ_z or its future-test equivalence class; a single predictive marginal need not be closed under further observations. This is the distinction between a one-step response quotient and a future-predictive quotient.

Corollary A.4.4 (Finite moment support for the product family). *For two independent Bernoulli report bits with parameters (p, q) and an independent fair mark, an optimal prior in (25) can be chosen with at most $(N + 2)^2$ atoms. The two report counts (k, ℓ) are a causally updatable sufficient statistic for computing the dual.*

Proof. Conditional on the two counts, all arrangements of report bits and all mark words have parameter-independent probabilities. Summing their coefficients reduces (25) to the $(N + 1)^2$ count pairs. Their likelihoods are

$$\binom{N}{k} p^k (1 - p)^{N-k} \binom{N}{\ell} q^\ell (1 - q)^{N-\ell}.$$

After multiplication by $Q(\omega) - P_{p,q}(\omega)$, every integrand has degree at most $N + 1$ in each variable. Thus the objective depends only on the $(N + 2)^2$ rectangular moments. Their constant coordinate is one. Carathéodory's theorem in the remaining affine space replaces an optimal moment vector by a convex combination of at most $(N + 2)^2$ point evaluations. Counts update by addition of the current bits. No claim that this count register is optimal for the decision is made. $\square$

APPENDIX A.5. CONTROLLED EXPERIMENTS WITHOUT A DOMINATING ACTION

The reduction to one dominating row is not needed for a statistical dual, provided causal action choice is retained in the dual recursion. We give the finite-action theorem with a compact parameter space and an explicit protocol state. Its equality is an unrestricted-memory statement; its upper certificates apply to every smaller-memory class.

Let Θ be compact metric and E a finite set of protocol states. At $e \in E$ let $\mathcal{A}_e$ be a finite set of training actions. Action a produces x in a finite alphabet X_a with law P_θ^a and changes the protocol state to $T(e, a, x)$. The functions $\theta \mapsto P_\theta^a(x)$ are continuous. An unknown θ is fixed throughout the experiment. Conditional on it, each requested preparation uses a fresh

independent draw from the selected row. Legal validation designs form a nonempty finite set $\mathcal{B}_e$. Design b has a finite outcome alphabet Y_b , target law Q^b and continuous candidate law V_θ^b . A policy may stop after at most N training actions. It then freezes $b \in \mathcal{B}_e$ and an event on Y_b , before one candidate and one target validation. Randomization is independent of θ . No training action is assumed to dominate another.

For a finite nonnegative measure ν on Θ write $\nu_{a,x}(d\theta) = P_\theta^a(x)\nu(d\theta)$ and set

$$(27) \quad \Phi_e(\nu) = \max_{b \in \mathcal{B}_e} \sum_{y \in Y_b} \left[Q^b(y)\nu(\Theta) - \int V_\theta^b(y)d\nu \right]_+,$$

$$F_0(e, \nu) = \Phi_e(\nu),$$

$$(28) \quad F_n(e, \nu) = \max \left\{ \Phi_e(\nu), \max_{a \in \mathcal{A}_e} \sum_{x \in X_a} F_{n-1}(T(e, a, x), \nu_{a,x}) \right\}.$$

If no training action is legal, only the stopping term is retained. All the functions vanish at the zero measure, so no posterior division is required on null histories. For positive mass, (27) is the mass times the largest posterior-predictive total variation.

Theorem A.5.1 (Controlled posterior-predictive dual). *Let $\mathcal{U}_N(e_0)$ be the maximum worst-case expected validation score over all randomized causal policies in the declared protocol, with at most N training preparations and no memory cap. Then*

$$(29) \quad \mathcal{U}_N(e_0) = \min_{\mu \in \mathcal{P}(\Theta)} F_N(e_0, \mu).$$

Both extrema are attained. Each $F_n(e, \cdot)$ is the maximum of finitely many continuous linear functionals on the cone of measures; in particular it is convex, continuous and positively homogeneous. For every prior μ , $F_N(e_0, \mu)$ bounds the worst-case score of every finite-memory architecture implementing this protocol.

Proof. There are finitely many histories of length at most N and finitely many event masks at each stopping history. Consequently there are finitely many deterministic policy trees, denoted by $\Pi_N(e_0)$. For $\pi \in \Pi_N(e_0)$ its payoff $g_\pi(\theta)$ is a finite sum of finite products of continuous primitive probabilities and is continuous in θ .

A randomized causal policy induces the same outcome laws as a probability distribution on these trees: sample its finite random decision table in advance and condition on it. Conversely, with no memory cap, a sampled tree can be retained and followed. Thus the unrestricted payoff vectors are exactly the convex hull of the g_π . This step is not an assertion that mixing machines preserves a specified register width.

The simplex on $\Pi_N(e_0)$ and $\mathcal{P}(\Theta)$ are compact and convex. The expected payoff is continuous and bilinear. Minimax therefore gives

$$\max_{\lambda \in \Delta(\Pi_N)} \min_{\theta} \sum_{\pi} \lambda_{\pi} g_{\pi}(\theta) = \min_{\mu} \max_{\pi \in \Pi_N} \int g_{\pi} d\mu.$$

For a fixed measure, the terminal maximum is (27), because an event includes precisely the positive coordinates of its signed predictive measure. At a training node one selects an action and, after its observed outcome, selects a continuation tree separately on that branch. The unnormalized branch measure is exactly $\nu_{a,x}$. Backward induction proves (28) and identifies the right side with (29). It also proves the finite-linear-functional representation. Compactness gives an optimizing prior and an optimizing mixed tree. Finally, the minimum over fixed parameters of any policy is at most its average under μ , which is at most $F_N(e_0, \mu)$. This remains true after restricting its memory or architecture. $\square$

The minimization in (29) occurs once, outside the recursion. It cannot be replaced by fresh adversarial choices after observations. The posterior carries dependence on one fixed unknown parameter. Also, a Bayes-optimal tree for a minimizing prior need not by itself be a minimax policy; the saddle-point mixture is supplied by the finite game in the proof.

Corollary A.5.2 (Bellman upper certificates). *Suppose functions $\Psi_n(e, \nu)$ satisfy $\Psi_0 \geq \Phi$ and, for $n \geq 1$, satisfy*

$$\Psi_n(e, \nu) \geq \Phi_e(\nu), \quad \Psi_n(e, \nu) \geq \sum_x \Psi_{n-1}(T(e, a, x), \nu_{a,x}) \quad (a \in \mathcal{A}_e).$$

Then $\Psi_N(e_0, \mu)$ is an upper certificate for every admissible architecture and every prior μ . The family F_n is the least such family. An optimal prior can be chosen with at most $d + 1$ support points, where d is the affine dimension of $\{(g_\pi(\theta))_{\pi \in \Pi_N} : \theta \in \Theta\}$.

Proof. Induction in (28) gives $F_n \leq \Psi_n$. The equality case is the recursion itself. The objective depends on a prior only through its vector of integrals of the finitely many g_π . An optimal such vector belongs to the convex hull of their compact evaluation image. Carathéodory's theorem gives the stated support bound while preserving every pure-tree payoff. $\square$

Proposition A.5.3 (Causal stability). *Consider two models with the same protocol, alphabets and policy class. Suppose their training rows differ uniformly in total variation by at most ϵ , their candidate validation rows by at most η , and their target validation rows by at most β . Their unrestricted values, or their values in any common specified memory class, differ by at most $2N\epsilon + \eta + \beta$.*

Proof. Use the same policy randomization and couple corresponding training rows while the histories agree. The probability of a first discrepancy is at most $N\epsilon$, even with feedback and stopping. Conditional expected validation rewards lie in $[-1, 1]$, so differing training histories contribute at most $2N\epsilon$. On agreeing histories the same frozen design and event are used, and their conditional mean rewards differ by at most $\eta + \beta$. The bound is uniform in the fixed parameter and in the policy. Taking minima and suprema preserves it, with the reverse inequality obtained by interchanging the models. $\square$

A.5.1. An informative action choice. The following calculation distinguishes the controlled theorem from a common-row reduction. Let $\Theta = \{(i, j) : i, j \in \{0, 1\}\}$. Action R reports i . Action L reports j when $i = 0$ and otherwise reports an independent fair bit. Action H reports j when $i = 1$ and otherwise reports an independent fair bit. The candidate validation is $\delta_{(i,j)}$ and the target is uniform on Θ .

Proposition A.5.4 (Strict adaptive advantage). *With two training preparations, the unrestricted adaptive value is $3/4$ and the best nonadaptive value is $5/8$. Nonadaptive policies here include mixtures of schedules, with their random schedule retained. No available single training action dominates all the others.*

Proof. Take R first, and then L or H according to the observed i . The parameter is recovered exactly. The complement of its validation atom has score $3/4$. No policy can exceed $\text{TV}(Q, \delta_\theta) = 3/4$ even with full information.

For the converse for fixed schedules, use the uniform prior. For a pair of actions a, b its Bayes value is explicitly

$$(30) \quad \sum_{x,y,\theta} \left[\frac{1}{16} \sum_{\vartheta} P_{\vartheta}^a(x) P_{\vartheta}^b(y) - \frac{1}{4} P_{\theta}^a(x) P_{\theta}^b(y) \right]_+.$$

Substitution of the three displayed kernels gives

	R	L	H
R	$1/2$	$5/8$	$5/8$
L	$5/8$	$7/16$	$1/2$
H	$5/8$	$1/2$	$7/16$

This is an exact finite sum, not a parameter-grid estimate. It bounds any mixture of fixed schedules by $5/8$; an early stop can be padded with an ignored observation. To attain the bound uniformly in θ , choose with equal probabilities the schedules (R, L) and (R, H) . When the chosen branch-specific action matches i , the parameter is known and the score is $3/4$. Otherwise only i is known, and the complement of its two-atom fiber gives score $1/2$. Each fixed parameter therefore has mean score $5/8$.

Action R cannot distinguish parameters differing only in j . Action L cannot distinguish $(1, 0)$ and $(1, 1)$, and H cannot distinguish $(0, 0)$ and $(0, 1)$. An experiment which fails to distinguish such a pair cannot garble to one that distinguishes it. Hence no action is a common dominating row. $\square$

If every binary report is additionally passed through an independent binary symmetric channel of crossover ϵ , the primitive training rows change by at most ϵ and have full support for $0 < \epsilon < 1/2$. Proposition A.5.3, applied to both policy classes, leaves an adaptive advantage at least $1/8 - 8\epsilon$, which is positive for $0 < \epsilon < 1/64$. Thus the distinction is not confined to disjoint observation supports. The finite-horizon belief recursion and its piecewise-linear structure are classical [1]. The contribution here is their explicit integration with the frozen validation score, one fixed adversarial parameter, outer certificates, and the resource conventions; no priority for Bellman recursion itself is claimed.

APPENDIX A.6. CONTROLLED EXPERIMENTS AT A PRESCRIBED MEMORY PROFILE

An unrestricted posterior is a useful information state for a proof, but it is not a free runtime state. We first give a resource-exact dynamic formulation and then solve a controlled family in which replacing that formulation by a policy-tree mixture changes the value strictly.

A.6.1. A parameterwise occupation state. Fix a finite horizon T , a compact metric parameter space Θ , finite hidden execution spaces H_t , and register alphabets S_t of sizes at most K_t . The visible inputs to each stochastic row and all legal actions are specified. Any input that must survive a cut belongs to S_t : this includes a chosen action, an unread report buffer, a stop flag, a selected validation event and a validation accumulator. One may subdivide an operation into several cuts to expose these costs. The scheduled time is available only in the clocked convention.

For now rows at different scheduled times are independently prescribed. Let $\mathcal{Q}_t$ be the finite product of legal row simplices at stage t . A prescription $q \in \mathcal{Q}_t$ is a table, chosen at design time, whose runtime arguments are exactly the declared visible inputs. After summing the fixed environmental kernels its one-stage transition is a stochastic matrix $M_{t,\theta}^q$ on the hidden-state/register pairs. We assume this matrix is jointly continuous in (q, θ) . Both action selection and state updating occur in this matrix, in their specified causal order; it is not an arbitrary matrix chosen separately for each hidden state or each parameter.

For a continuous bundle of probability vectors $x = (x_\theta(h, s))_{\theta \in \Theta}$ define

$$(\mathcal{T}_{t,q}x)_\theta(h', s') = \sum_{h,s} x_\theta(h, s) M_{t,\theta}^q((h, s), (h', s')).$$

The initial bundle x_0 is fixed by preparation. Terminal rewards $r_\theta(h, s) \in [-1, 1]$ are continuous in θ . Absorbing stopped states and frozen validation phases are included in this execution, not added afterwards without charge. Put

$$(31) \quad B_T(x) = \min_{\theta \in \Theta} \sum_{h,s} x_\theta(h, s) r_\theta(h, s),$$

$$(32) \quad B_t(x) = \max_{q \in \mathcal{Q}_t} B_{t+1}(\mathcal{T}_{t,q}x), \quad 0 \leq t < T.$$

The argument x is an *offline occupation bundle*, not the realized history, not an observed parameter, and not an additional register made available to the machine.

Theorem A.6.1 (Resource-exact dynamic value and supersolutions). *For the specified clocked profile and observable-row interface, $B_0(x_0)$ is the exact maximum worst-case reward over all stochastic controllers. The maximum is attained by a sequence of legal common prescriptions. With the norm $\|x - x'\| = \sup_\theta \sum_{h,s} |x_\theta(h, s) - x'_\theta(h, s)|$, each B_t is 1-Lipschitz. Moreover,*

$$(33) \quad B_0(x_0) = \inf_H H_0(x_0),$$

where the infimum ranges over families of functions on occupation bundles satisfying

$$H_T \geq B_T, \quad H_t(x) \geq H_{t+1}(\mathcal{T}_{t,q}x) \quad (q \in \mathcal{Q}_t).$$

The infimum is attained by $H = B$. No convexification over retained controllers and no new adversarial choice after a report occurs in these identities.

Proof. For a fixed sequence $q_0, \dots, q_{T-1}$, iteration of the displayed transition gives its actual terminal law for every fixed θ . The terminal minimum is therefore exactly its worst-case reward. A runtime randomized controller in this interface is precisely such a sequence of common conditional rows; independence of fresh randomization and the declared register specify all its conditional choices. Persistent random correlations are allowed only through that register.

The occupation bundle at the next stage is determined by the current bundle and the prescription, without drawing a realized history in the design problem. Hence optimizing over the remaining prescription sequence gives (32) by ordinary backward induction. This does not select a different continuation for each unrecorded history. Each maximizing q is still a single legal table common to all hidden histories and parameters. At x_0 one follows the maximizing sequence of design prescriptions, computing their occupation bundles offline; the runtime controller only reads its allowed inputs.

Stochastic matrices contract the ℓ^1 norm. The terminal minimum is 1-Lipschitz in the displayed norm. Induction, using the same q when comparing two bundles and then taking the two maxima, proves the Lipschitz assertion. Joint continuity and compactness of $\mathcal{Q}_t$ give attainment at each stage, and compactness of Θ gives the terminal minimum. Any family H in (33) bounds the terminal reward along every prescription sequence. Backward induction therefore gives $H_t \geq B_t$. Equality is admissible, proving (33) and its attainment. $\square$

This is a nonlinear supersolution dual, not a least-favorable-prior saddle theorem at fixed width. Its optimization can be large and nonconvex. The principle of dynamic programming is classical; the point of this formulation is that the common-row and memory constraints remain in the maximizations. If rows are shared across scheduled times, as for an autonomous controller, choose the shared rows once and hold them fixed in all subsequent transitions; equivalently augment the *design* problem by that immutable prescription. Maximizing a shared row anew at each time would solve a different problem. A finite union of allowed interfaces can be handled by taking their maximum, or by a fully wired interface with the prescribed profile and its legal observable inputs. Neither operation enlarges a runtime register invisibly.

A.6.2. An exact informative-control bottleneck. Fix integers $m \geq 2$ and $1 \leq K \leq m$. The fixed parameter is $\theta = (i, j) \in [m] \times \{0, 1\}$. The first training experiment R reports i . Across the next cut the controller retains at most K labels, with an arbitrary stochastic encoder $E(s | i)$. It then chooses one of the m sensing actions with probabilities $D(a | s)$. The selected action is kept in a charged m -state action buffer and the old label is discarded. Action a reports j when $a = i$ and reports $\perp$ otherwise. The first report is no longer available.

The candidate validation law on $\{0, 1\}$ is δ_j ; the target is uniform. An event is frozen before one fresh candidate and one fresh target validation. Stopping before either training action is permitted. No independent persistent selector is given to the controller. The observation symbols are atomic; a bitwise input implementation would have additional cuts. Denote the maximum worst-case score by $V_{m,K}$.

Theorem A.6.2 (Exact finite-memory control and its convexification gap). *In this protocol,*

$$(34) \quad V_{m,K} = \frac{1}{2\lceil m/K \rceil}.$$

The value obtained by allowing an uncharged ex ante mixture of complete K -label controllers is instead

$$(35) \quad \bar{V}_{m,K} = \frac{K}{2m}.$$

The inequality is strict whenever K does not divide m . The matching controller for (34) has charged cut profile bounded by $(1, K, m, 3, 6, 3)$. The K -label cut controls both retention of the routing observation and the subsequent informative action choice.

Proof. For a fixed i , let

$$p_i = \sum_s E(s | i) D(i | s).$$

This is the probability of taking the informative action. On its complement, all retained information is independent of j . Averaging over the two values of j makes the expected validation score zero there, for any frozen event or randomized response. On the informative part the score is at most $\text{TV}(\text{Ber}(1/2), \delta_j) = 1/2$. Thus the minimum over j at a given i is at most $p_i/2$. A stop before sensing is also uninformative about j and obeys the same bound; it can be padded by an ignored sensing operation.

Set $L = \lceil m/K \rceil$. Since $p_i \leq \max_s D(i | s)$, a value $\min_i p_i > 1/L$ would require every i to have some row s with $D(i | s) > 1/L$. Each stochastic row has at most $L - 1$ such indices. But $K(L - 1) < m$. This contradiction proves the upper bound. Partition $[m]$ into at most K fibers of size at most L , retain the fiber label after R , and choose uniformly inside that fiber. On a successful report freeze $\{1\}$ when $j = 0$ and $\{0\}$ when $j = 1$; on $\perp$ freeze the empty event. The score for every j is $1/(2|\text{fiber}(i)|)$, proving equality. The event alphabet has three labels. Storing the event and its candidate indicator requires at most six labels, and the final difference has three. The chosen sensing action was separately charged, proving the displayed profile.

For the convexified upper bound use the uniform prior on (i, j) . For any one controller,

$$\frac{1}{m} \sum_i p_i = \frac{1}{m} \sum_{i,s} E(s | i) D(i | s) \leq \frac{1}{m} \sum_{i,s} D(i | s) = \frac{K}{m}.$$

The same upper bound holds for mixtures. For the matching mixture, choose uniformly one of the m cyclic intervals of K indices. For that interval hardcode a deterministic K -label encoder which routes each included index to its own sensing action, and assigns other indices arbitrarily. Each i belongs to exactly K intervals. A successful action is followed by the complementary

singleton event and failure by the empty event, as above. Every fixed (i, j) has mean score $K/(2m)$. This proves (35). $\square$

For example, $V_{3,2} = 1/4$ whereas $\bar{V}_{3,2} = 1/3$. The mixture just constructed can be implemented with a retained interval selector and a local label, using at most mK labels at the bottleneck. It cannot be declared a K -label realization: when the displayed gap is positive, (34) proves that no such realization exists. The action buffer of size m is not part of the K coordinate and has not been omitted from the profile. The theorem is a multicut resource law, not an assertion that its uniform peak equals K .

The actions are statistically incomparable. R cannot distinguish the two values of j at any fixed i , while action a does distinguish them at $i = a$. For $a \neq b$, action a does not distinguish the pair with $i = b$, while b does. A parameter-independent garbling cannot create a distinction absent from its input experiment.

Corollary A.6.3 (A strict gap with positive report probabilities). *Replace each of the two training kernels by a mixture with weight $\epsilon > 0$ of the uniform law on its report alphabet. Let $\Delta_{m,K} = K/(2m) - 1/(2\lceil m/K \rceil) > 0$. For $\epsilon < \Delta_{m,K}/8$ the resulting full-support model still has a strict gap between the genuine profile-constrained value and its free-selector converification.*

Proof. Each training row changes by at most ϵ in total variation. Couple corresponding rows until the first disagreement. For any fixed controller or mixed controller the probability of such a disagreement is at most 2ϵ , and its conditional reward lies in $[-1, 1]$. Each of the two optimal values changes by at most 4ϵ . Their difference is therefore at least $\Delta_{m,K} - 8\epsilon > 0$. All row arguments and register alphabets are unchanged. $\square$

The distinction between mixed and behavioral strategies without perfect recall is classical; it is not first asserted here. Contemporary polynomial formulations over behavioral simplices include [16]. Theorem A.6.2 evaluates the discrepancy in this specific frozen-validation experiment for every m, K , with a matching actual register and a noisy persistence statement. In particular, a single prior followed by an unrestricted policy-tree best response cannot be substituted for (32).

APPENDIX A.7. SADDLE FACES AND CAUSAL REALIZATION

The statistical saddle and its implementation are different objects. The former is convex; the latter need not be. The purpose of this section is to retain that distinction while extracting memory obstructions from an unrestricted saddle. In particular, a zero Bayes coefficient must not be discarded before the contact conditions at the worst parameters have been imposed.

A.7.1. A normal form over a finite causal interface. Fix a finite horizon and finite action and report alphabets. An action a_t is chosen before report y_t . A terminal output is included as one last action, followed by a dummy report. Illegal actions can be removed from each protocol type; equivalently, use a fixed padding of the finite protocol tree and impose its zero constraints. Define a *causal response array* $C(\mathbf{a} \mid \mathbf{y})$ by nonnegativity, normalization in $\mathbf{a}$ for every $\mathbf{y}$, and the requirement that its marginal on $a_1, \dots, a_t$ depends only on $y_1, \dots, y_{t-1}$. The resulting set $\mathcal{C}$ is a finite polytope. Conditioning successive marginals realizes every such array by a full-history randomized controller; arbitrary versions on zero marginals do not change the array.

For a fixed parameter θ , the environment has a history-dependent likelihood $L_\theta(\mathbf{a}, \mathbf{y})$, obtained by marginalizing its hidden states. The bounded reward has the linear form

$$(36) \quad g_\theta(C) = \sum_{\mathbf{a}, \mathbf{y}} L_\theta(\mathbf{a}, \mathbf{y}) r_\theta(\mathbf{a}, \mathbf{y}) C(\mathbf{a} \mid \mathbf{y}).$$

Assume its coefficient vector is continuous on a compact metric parameter space. The parameter is fixed once, not selected after a report. Protocol-specific equalities, including a frozen validation

choice, are represented by the corresponding finite protocol tree. A discarded protocol choice is not supplied again as a free input.

Here is an explicit positive realization of a response array. At cut t use k_t states, an initial distribution α , and nonnegative matrices $M_t^{a,y}$ of size k_{t-1} by k_t such that

$$(37) \quad M_t^{a,y} \mathbf{1} = p_t(a), \quad \sum_a p_t(a) = \mathbf{1}, \quad p_t(a) \text{ is independent of } y.$$

Then

$$(38) \quad C(\mathbf{a} \mid \mathbf{y}) = \alpha M_1^{a_1, y_1} \cdots M_T^{a_T, y_T} \mathbf{1}.$$

The independence in (37) is essential: otherwise an action could use its not-yet-observed report. If an action or an input must survive an intermediate cut, split this formula at that cut and include its buffer in the state alphabet. Likewise, impose any declared shared-row constraints before optimizing. An externally fixed clock permits time-dependent matrices; an autonomous implementation has to store its phase or use the prescribed shared rows.

Proposition A.7.1 (Outer normal form and compatible continuations). *For the finite interface and declared cuts, the response arrays of all controllers with profile at most $\mathbf{K}$ are exactly the arrays (38) with $k_t \leq K_t$, the normalization (37), and the declared interface constraints. This set, denoted $\mathcal{C}_{\mathbf{K}}$, is compact. The assertion is over all wirings of the allowed register, not one chosen transition graph.*

Equivalently, there are at most K_t normalized continuation arrays $B_{t,s}$ at each cut, with terminal continuation 1, and common nonnegative coefficients satisfying

$$(39) \quad B_{t-1,s}(a\mathbf{a} \mid y\mathbf{y}) = \sum_{s'} M_t^{a,y}(s, s') B_{t,s'}(\mathbf{a} \mid \mathbf{y}), \quad C = \sum_s \alpha(s) B_{0,s}.$$

Separate positive factorizations at the cuts are not sufficient without these common shift identities.

Proof. In a controller, let $p_t(a \mid s)$ be its action row and $q_t(s' \mid s, a, y)$ its update row. Their product defines $M_t^{a,y}$ and gives (37). Multiplication over a complete execution gives (38). Every finite wiring embeds into the fully connected register by assigning zero to unused transitions; smaller alphabets are padded by unused states. Thus no further maximization over graph presentations is necessary.

Conversely choose action a with probability $p_t(a \mid s)$, and, after observing y , use the row $M_t^{a,y}(s, s')/p_t(a \mid s)$ when the denominator is positive. At a zero denominator the entire corresponding nonnegative row vanishes; choose an arbitrary legal update there. This gives exactly the required runtime arguments. It does not expose a posterior, the parameter, or an offline occupation array to the machine. Apply the same construction at each declared buffer cut. A product of finitely many stochastic row simplices is compact; closed legality and sharing constraints preserve compactness. Its continuous image is $\mathcal{C}_{\mathbf{K}}$.

The future execution starting at a state s defines $B_{t,s}$ and satisfies (39). Conversely, induction from the terminal continuation and these identities gives the product formula. Thus the continuation description has exactly the same realizations. In particular, independently selected factorizations at different cuts cannot be spliced unless the shift equations hold. $\square$

This is a finite controlled positive-realization statement. The role of positivity and invariant generating cones in stochastic realization is classical [17, 18, 19]. Ordinary linear rank need not be a state count. Here the additional statistical question is which positive realizations meet an optimal saddle face. Proposition A.7.1 provides a normal form, not an efficient factorization algorithm. It also does not compactify an unbounded number of observable operations.

A.7.2. The realization theorem. Let

$$U = \max_{C \in \mathcal{C}} \min_{\theta} g_{\theta}(C), \quad V_{\mathbf{K}} = \max_{C \in \mathcal{C}_{\mathbf{K}}} \min_{\theta} g_{\theta}(C).$$

By compact convex minimax there is a least-favorable prior μ_* with $U = \max_{C \in \mathcal{C}} \int g_{\theta}(C) d\mu_*$. Put

$$\begin{aligned} \ell_*(C) &= \int g_{\theta}(C) d\mu_*, \\ F_* &= \{C \in \mathcal{C} : \ell_*(C) = U\}, \\ \mathcal{S}_* &= F_* \cap \bigcap_{\theta} \{C : g_{\theta}(C) \geq U\}. \end{aligned} \tag{40}$$

The set F_* is an exposed face. The second intersection is important: a Bayes best response to a least-favorable prior need not be minimax.

Here $\mathcal{C}$ is the full-history convex reference for the finite protocol. A nonconvex restriction, such as reuse of one stochastic table at several visits, is imposed on $\mathcal{C}_{\mathbf{K}}$, not inserted into $\mathcal{C}$ before invoking convex minimax. In that case U is a reference upper value unless exactness for the unrestricted physical class is separately known. In the reset applications below that exactness is supplied by Theorems A.4.1 and A.4.2.

Proposition A.7.2 (Bayes loss and worst-case loss). *Under the preceding assumptions,*

$$U - V_{\mathbf{K}} = \min_{C \in \mathcal{C}_{\mathbf{K}}} \left\{ U - \ell_*(C) + \ell_*(C) - \min_{\theta} g_{\theta}(C) \right\}. \tag{41}$$

Both summands are nonnegative and the minimum is attained.

Proof. The prior average is at most U and at least the minimum over fixed parameters. The sum in braces equals $U - \min_{\theta} g_{\theta}(C)$. Minimizing on the compact realization set proves the assertion. $\square$

Theorem A.7.3 (Saddle-face contact and compatible realization). *Under the preceding assumptions, $V_{\mathbf{K}} = U$ if and only if some array in $\mathcal{S}_*$ has compatible positive continuation generators of profile at most $\mathbf{K}$. If no such generators exist, $V_{\mathbf{K}} < U$ strictly.*

For $C \in \mathcal{S}_$ one has $g_{\theta}(C) = U$ on $\text{supp } \mu_*$. At an interior supporting parameter at which the payoff is differentiable, $\nabla_{\theta} g_{\theta}(C) = 0$. These equalities and stationarity conditions hold also in directions whose Bayes coefficient vanishes.*

At any reset continuation cut, let R_h denote the conditional channel of future outputs under each externally specified future input word. It factors as

$$R_h = \sum_{s=1}^{K_t} E(s \mid h) B_s, \quad E(\cdot \mid h) \in \Delta_{K_t}, \tag{42}$$

with one common family of continuation channels B_s . Any obstruction to this positive factorization for every member of $\mathcal{S}_$ is therefore a memory obstruction for every admissible architecture, regardless of its training wiring.*

Proof. By Proposition A.7.2, the two nonnegative losses vanish exactly when C belongs to $\mathcal{S}_*$. Proposition A.7.1 then proves the realization criterion. Compactness makes nonintersection a strictly positive value loss, not merely failure of one optimizer.

For an optimal array, $g_{\theta}(C) - U$ is continuous, nonnegative, and has zero μ_* integral. It is zero on the support of that measure. Every interior zero is a local minimum, so its gradient vanishes. No strict-sign hypothesis on individual coordinates is needed. Finally condition on the actual retained state at a reset cut. Given that state, the future program receives the same

specified inputs and has the same output law, irrespective of the discarded past. Averaging gives (42). All persistent random variables are part of that state. The assertion can equivalently be checked on controller input tapes, using the generators in (39); it does not assume that a hidden physical state resets at a cut where the interface has not reset it. $\square$

The identity (41) is an exact loss decomposition, not a new interchange of a minimum and maximum over nonconvex machines. Its use is that an unrestricted certificate identifies a finite face, while contact at its supporting parameters can determine the remaining coordinates. The resulting channels are then tested for causal realizability. This is stronger than choosing an arbitrary Bayes tie rule and counting the states of that one rule.

Corollary A.7.4 (Explicit coefficient form for frozen validation). *In the reset experiment of (24), write*

$$C_{z\omega}^* = \int P_{\theta}^{\otimes N}(z)(Q(\omega) - P_{\theta}(\omega)) d\mu_*(\theta).$$

The first summand of (41) is exactly

$$(43) \quad \sum_{C_{z\omega}^* > 0} C_{z\omega}^*(1 - h_z(\omega)) + \sum_{C_{z\omega}^* < 0} |C_{z\omega}^*| h_z(\omega).$$

Thus optimality fixes every strict-sign coordinate, while the support contact and stationarity equations must still be imposed at the zeros. For K labels at the post-training decision cut, an arbitrary upstream encoder gives the compact relaxation

$$(44) \quad h_z(\omega) = \sum_{s=1}^K E(s | z) D_s(\omega), \quad E \in (\Delta_K)^{\Omega^N}, \quad D_s \in [0, 1]^{\Omega}.$$

If this relaxation misses the saddle set, its strictly positive gap bounds every smaller architectural class.

Proof. Maximizing each coefficient separately gives the sum of its positive part. Subtraction gives (43). Conditioning on the decision state gives (44); the post-cut randomization is independent of the discarded word. Conversely an arbitrary encoder and a response list realize this factorization when other memory is unrestricted. The parameter spaces in (44) are compact, so the final assertion follows from Theorem A.7.3. $\square$

A.7.3. Faces as memory witnesses. For a finite input set $\mathcal{Y}$ and output alphabet $\mathcal{D}$, continuation channels form the polytope $\mathcal{P} = \prod_{y \in \mathcal{Y}} \Delta(\mathcal{D})$. Its vertices are deterministic output functions. The analogous polytope for scalar test responses is $[0, 1]^{\mathcal{Y}}$.

Lemma A.7.5 (Vertex and face packing). *Suppose channels to be factored through one common state alphabet include s distinct vertices $v_1, \dots, v_s$ of a polytope $\mathcal{P}$ and rows $r_1, \dots, r_j$. Suppose $r_i \in F_i$, the faces F_i are pairwise disjoint, and*

$$r_i \notin \text{conv}\{v_a : v_a \in F_i\}.$$

Every factorization of these rows as stochastic mixtures of generators in $\mathcal{P}$ needs at least $s + j$ generators.

Proof. A convex decomposition of a vertex uses that same vertex at every positive coefficient. Hence all s vertices must occur among the generators. If a convex combination belongs to a face, all generators with positive coefficients belong to that face. The stated exclusion therefore forces an additional generator in each F_i . Pairwise disjointness prevents any such generator from serving two faces. $\square$

The lemma does not assert that a stochastic state must itself be a history residual. Its generator can be any channel in the ambient polytope. This is why the argument also excludes randomized encoders and randomized continuations. It is a positive-factorization argument, not ordinary matrix rank or a restriction to a deterministic compiler.

Proposition A.7.6 (Resource equivalence and stable obstructions). *A causal, cut-preserving recoding with a causal inverse, both using only one transducer state and preserving the payoff, preserves every $V_{\mathbf{K}}$, U , and the set of profiles attaining U . More generally suppose simulations in both directions have defect at most ϵ and require at most R_t states at cut t . For rewards in $[-1, 1]$,*

$$V_{\epsilon}(\mathbf{RK}) \geq V_{\mathfrak{F}}(\mathbf{K}) - 2\epsilon, \quad V_{\mathfrak{F}}(\mathbf{RK}) \geq V_{\epsilon}(\mathbf{K}) - 2\epsilon.$$

For a common controller class and training-validation kernels whose values differ by at most η , a gap $U - V_{\mathbf{K}} = \delta$ remains at least $\delta - 2\eta$ after the perturbation.

Proof. Transport a controller through each simulator and retain the product of its state and the simulator state. Apply the uniform bounded-reward variation bound and optimize. With an exact one-state inverse the two inequalities are equalities at the same profile. In the last assertion both the unrestricted and the constrained optimum change by at most η ; subtract their two bounds. The state products must be applied at the actual buffer cuts, not only at the terminal decision. $\square$

These statements place the saddle face, the future-test quotient, and resource-aware causal morphisms in one theorem chain. The channel factorizations are presentation-independent under the stated causal recodings. The memory obstruction in the next section is derived from the experimental coefficients themselves, not from a metric dimension or a covering exponent supplied as an additional hypothesis.

APPENDIX A.8. FACIAL CERTIFICATES FOR SCHEDULE-OPTIMIZED MEMORY

A schedule is part of the resource problem. The following criterion extracts a whole family of schedule converses from a saddle, rather than minimizing the states of one selected program. Its hypotheses are finite identities of continuation functions and faces. Its conclusion applies to arbitrary stochastic state encoders.

Consider finitely many validation tapes, with a fixed internal order on each tape and finitely many symbols per slot. A *declared-clock schedule* is a word specifying which tape supplies each successive slot; it is fixed before training and is independent of all reports. Let $\mathcal{G}$ be the finite acyclic graph of consumed-prefix counts. Its directed paths from the initial vertex to the final vertex are the legal schedules. This includes all interleavings, not only serial ones. The lower bound below does not give an adaptive scheduler free storage: a report-dependent choice of the next tape belongs to a different controlled interface and requires its own charged rows.

At a vertex v of $\mathcal{G}$, a retained state supplies a conditional channel on the unread tape suffixes. Assume in this section that every prefix and suffix combination used in a witness has positive probability under a supporting experiment. One may instead restrict to a stated common suffix subalphabet with this property. The channels form a product of probability simplices, denoted by $\mathcal{P}_v$.

Definition A.8.1 (Contact-forced facial certificate). A certificate at v consists of distinct deterministic channels $d_1, \dots, d_s$ and channels $r_1, \dots, r_j$, such that every minimax-optimal auditor has these rows among its conditional continuations, allowing parameter-independent averaging of specified past rows. Each r_i lies in a face F_i of $\mathcal{P}_v$, the faces are pairwise disjoint, and

$$r_i \notin \text{conv}\{d_a : d_a \in F_i\}.$$

Its certified capacity is $s + j$. Define $c(v)$ to be the largest such capacity within a declared finite witness family. Conditional averaging must retain the same future-input coordinates and may not add the unknown parameter as an input. A channel of mean zero is not thereby a deterministic zero channel.

For an upper certificate choose a minimax-optimal finite randomization over frozen deterministic events, with the selected event retained at the decision cut. Let $\mathcal{D}_v$ be its distinct future deterministic output functions after all possible supported prefixes at v . When the input support is a full product, equality of these functions is closed under every next-symbol restriction. Put $r(v) = |\mathcal{D}_v|$. More general supported experiments may use any explicitly supplied shift-compatible deterministic completion; merely minimizing separate partial-function covers at different cuts is insufficient.

Theorem A.8.2 (Facial capacity and schedule realization). *Suppose contact with the unrestricted saddle supplies the preceding certificates. For each declared-clock path π and every stochastic auditor on that path attaining the unrestricted value,*

$$(45) \quad W(\pi) \geq \max_{v \in \pi} c(v).$$

A shift-compatible deterministic residual family realizes the selected optimal event rule simultaneously with widths $r(v)$ along any such path. If its training and selection peak is W_0 , then

$$(46) \quad \min_{\pi} \max_{v \in \pi} c(v) \leq W_{\text{clk}}^{\min} \leq \min_{\pi} \max \{W_0, \max_{v \in \pi} r(v)\}.$$

In particular, a certificate and a realizing path with the same peak identify the exact minimum over all declared-clock schedules and all stochastic architectures on those schedules. If $c(v) = r(v)$ at every vertex and W_0 does not exceed the path peak, the entire minimum-profile problem reduces to the intrinsic capacity field on $\mathcal{G}$. The peak optimum of that field is computed by

$$(47) \quad b(v_0) = c(v_0), \quad b(v) = \max\{c(v), \min_{u \rightarrow v} b(u)\}.$$

The result is invariant under bijective, causal, slot-preserving recodings of the tape alphabets.

Proof. Condition on a charged retained state at v . Every past row is a stochastic mixture of the same at most $W(\pi)$ future channels. Parameter-independent averaging preserves this factorization. The face-packing lemma therefore gives at least $s + j$ states for each certificate, proving (45). This step ranges over all state wirings and all stochastic encoders, not a specified graph.

For the upper bound retain the current residual output function. If two residuals agree on the future input product, their restrictions after each legal next symbol agree as well. Therefore the next retained label is determined by the old label and that symbol. At the last cut it is the output itself. This constructs all cuts compatibly in one machine. The initially sampled event and its training selector are charged in W_0 . Optimizing the two bounds over the finite path set proves (46). The recursion follows by classifying paths according to their last edge and taking the maximum with the cost of the final vertex. Induction proves it on the acyclic graph. Finally, a bijective recoding induces an affine isomorphism of every channel polytope, preserving vertices, faces, conditional averages and restriction maps. It preserves both certificates and realizing paths. $\square$

The general positive-realization normal form and residual quotient have classical antecedents [17, 18, 19]. The minimization of partially specified machines likewise requires compatible coverings, not unrelated cutwise minima [20]. The contribution used here is different: the witness rows are forced by *every member of the uniform saddle*, through contact on its Bayes-indifferent coordinates. This makes the lower field independent of a chosen optimal program.

The criterion is a certificate theorem for this class of experiments, not an assertion that arbitrary nonnegative-rank problems have become easy or that facial packing is always exact.

APPENDIX A.9. EXACT MINIMAX LAWS FOR A MARKED PRODUCT FAMILY

The posterior-predictive dual can be solved explicitly for a continuous family containing the collision target. Throughout this section, the training and candidate validation law is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad (p, q) \in [0, 1]^2.$$

In the order 000, 001, 110, 111, let Q_γ have masses

$$((1 + \gamma)/4, (1 - \gamma)/4, (1 - \gamma)/4, (1 + \gamma)/4),$$

and zero mass elsewhere. Write $U_1(\gamma) = U_1(b, Q_\gamma)$, with exactly the training-validation convention of (24). Set

$$(48) \quad \gamma_* = 1 - 1/\sqrt{2}, \quad s_\gamma = \sqrt{\gamma^2 - 2\gamma + 5}, \quad u_\gamma = \gamma - 2 + s_\gamma, \quad \tau_\gamma = \frac{2u_\gamma}{u_\gamma + 2 - \gamma}.$$

Theorem A.9.1 (Exact marked minimax family). *For every $0 \leq \gamma \leq \gamma_*$,*

$$(49) \quad U_1(\gamma) = \frac{3 + u_\gamma^2}{8}.$$

An optimal response depends on the training word only through the report count $K = X_1 + X_2$. On the four target atoms its rows are

$$(50) \quad h_0 = (\tau_\gamma, 0, 1, 1), \quad h_1 = (1, 1, 1, 1), \quad h_2 = (1, 1, 0, \tau_\gamma),$$

and all responses are zero off these atoms. A least-favorable prior is

$$(51) \quad \mu_\gamma = \frac{1}{2}\delta_{(t_\gamma, t_\gamma)} + \frac{1}{2}\delta_{(1-t_\gamma, 1-t_\gamma)}, \quad t_\gamma = \frac{1 - \sqrt{u_\gamma}}{2}.$$

The assertion optimizes over all stochastic responses to the full marked training word, not merely over count-based or symmetric responses.

Proof. First, u_γ is the positive root of

$$(52) \quad u^2 + (4 - 2\gamma)u - (1 + 2\gamma) = 0.$$

On the asserted interval, $0 < u_\gamma < 1$ and $0 < \tau_\gamma < 1$. Indeed the polynomial is negative at zero and positive at one; also $u_\gamma < 2 - \gamma$ because $\gamma < 1/2$. Substitution of $1 - 2\gamma$ into the polynomial gives $8(\gamma^2 - 2\gamma + 1/2)$, which is nonnegative precisely on the present interval. Monotonicity on $[0, \infty)$ therefore gives

$$(53) \quad u_\gamma \leq 1 - 2\gamma.$$

Put $a = (1 - p)(1 - q)$, $c = pq$, and $z = 1 - a - c$. The probabilities of the three training counts are a, z, c . The independent training mark may be discarded by the displayed construction. Let $G_\gamma(p, q)$ denote its score. With $u = (p + q - 1)^2$ and $v = (p - q)^2$, direct expansion gives

$$(54) \quad G_\gamma(p, q) - \frac{3 + u_\gamma^2}{8} = \frac{2 - \tau_\gamma}{16} \{(u - v - u_\gamma)^2 + 4v\}.$$

For clarity, this identity follows by substituting the three rows of (50) into

$$a(Q_\gamma - b)h_0 + z(Q_\gamma - b)h_1 + c(Q_\gamma - b)h_2,$$

using $a + c = (1 + u - v)/2$, $a - c = -(p + q - 1)$, and then eliminating u_γ^2 by (52). The coefficient of the resulting square is $(2 - \tau_\gamma)/16$ and that of v is four times this coefficient. Both

are nonnegative. Equality holds at both points of (51), since there $v = 0$ and $u = u_\gamma$. This proves the lower bound and identifies its equality set.

It remains to bound every response, including nonsymmetric ones. Use (51) in Theorem A.4.1. At $t = t_\gamma$ put $a = (1 - t)^2$, $c = t^2$, and $T = a + c = (1 + u_\gamma)/2$. For the training report pair 00, the four dual coefficients, apart from a common positive factor, are

$$(55) \quad \left(Q_H T - \frac{a^2 + c^2}{2}, \quad Q_L T - \frac{a^2 + c^2}{2}, \quad Q_L T - ac, \quad Q_H T - ac \right),$$

where $Q_H = (1 + \gamma)/4$ and $Q_L = (1 - \gamma)/4$. The first coordinate is zero by (52); the second is $-\gamma T/2 \leq 0$. The last two are positive: $T \geq 1/2$, $ac \leq 1/16$, and $\gamma < 1/2$ imply $Q_L T > 1/16 \geq ac$. Thus $(\tau_\gamma, 0, 1, 1)$ maximizes this linear functional. For the training pair 11, reversing all report bits and the mark reverses the four coordinates and gives the row h_2 .

For either training pair 01 or 10, the two prior support points have the same likelihood $t(1 - t)$. The posterior is still equally weighted, and the candidate predictive mass at each target atom is $T/4$. Every one of the four coefficients is therefore nonnegative by $Q_L - T/4 = (1 - 2\gamma - u_\gamma)/8 \geq 0$. The response h_1 is optimal. At off-diagonal validation atoms the target mass vanishes and the coefficient is nonpositive. Finally, conditioning on either training mark multiplies all these coefficients by $1/2$; it creates no further optimizing choice. We have therefore optimized every coordinate of the full marked-word response space.

The prior average of the displayed response equals its value at either support point, namely $(3 + u_\gamma^2)/8$, by (54). Every other response has prior average at most this number and hence has minimum over fixed (p, q) at most this number. Together with the lower bound this proves (49), optimality of the response, and least favorability of the prior. $\square$

Corollary A.9.2 (The exact one-preparation collision value). *For $Q_* = Q_{1/4}$,*

$$(56) \quad U_1(b, Q_*) = u_{\text{ex}} := \frac{85 - 7\sqrt{73}}{64}, \quad u_{1/4} = \frac{\sqrt{73} - 7}{4}, \quad \tau_{1/4} = 2 - \frac{14}{\sqrt{73}}.$$

If only the target changes to Q , then $|U_1(b, Q) - u_{\text{ex}}| \leq \text{TV}(Q, Q_)$.*

Proof. Substitute $\gamma = 1/4$ in Theorem A.9.1. For a fixed trained response the target perturbation is at most total variation, uniformly in (p, q) . Taking first the minimum and then the maximum preserves this bound in both directions. $\square$

A.9.1. Complementary slackness and the event alphabet. The tie in the first coordinate of (55) explains the randomization. Any value of that response coordinate is Bayes-optimal for the two-point prior. The minimax choice makes the uniform payoff stationary at its two support points; solving this condition gives τ_γ and the square identity (54). This derives the prior and response together, rather than postulating that a numerically effective prior is optimal.

Operationally, for $K = 0$ one draws once between the events with indicator vectors $(0, 0, 1, 1)$ and $(1, 0, 1, 1)$, with probabilities $1 - \tau_\gamma$ and τ_γ . For $K = 2$ one draws between $(1, 1, 0, 0)$ and $(1, 1, 0, 1)$. For $K = 1$ the event is $(1, 1, 1, 1)$. The resulting event identity is retained and used for both fresh validations. These are exactly the five events in the two-preparation construction below. The one-preparation theorem supplies their complementary-slackness interpretation; it does not prove that the two-preparation selector is optimal. The four-point rational prior retained below is a useful rational certificate, not a least-favorable prior.

The exact response uses an atomic stochastic row. It does not receive free persistent randomness. Replacing τ_γ by a dyadic number within 2^{-b} changes each conditional response, and hence the score, by at most 2^{-b} . A sequential fair-bit implementation must retain its bit

prefix and event identity. None of these implementation costs is needed by the deterministic two-preparation physical audit.

Proposition A.9.3 (The zero-preparation endpoint). *For $0 \leq \gamma \leq 1$, $U_0(b, Q_\gamma) = \gamma/2$.*

Proof. The fixed event $\{000, 111\}$ has score $(1 + \gamma)/2 - (a + c)/2 \geq \gamma/2$ because $a + c \leq 1$. Conversely, the equally weighted prior on $b(0, 0)$ and $b(1, 1)$ has predictive validation law uniform on the four target atoms. Its total variation from Q_γ is $\gamma/2$. Theorem A.4.1 with $N = 0$ gives the matching upper bound. $\square$

APPENDIX A.10. A SELF-CONTAINED COLLISION INTERFACE

We give the preparation, mechanical estimate, observation law and private simulator class needed for the physical specialization. The following argument does not appeal to an unpublished physical lemma. It retains the same positive-volume preparation and positive sensor noise as the earlier collision construction.

Definition A.10.1 (The reset row–event architecture). A preparation supplies two sequential report slots and, afterwards, an actual mark. The source reports in the two slots are constant. A private width-one simulator has no retained information between those slots and no persistent hidden seed. It may use fresh independent coins. Its first emission is $X_1 \sim \text{Ber}(p)$; after receiving the current gate $A \in \{0, 1\}$, its second emission has a fixed Bernoulli kernel with parameter q_A . The parameters (p, q_0, q_1) are fixed throughout the audit. The simulator does not observe the actual mark. All its coins are independent of that mark. Different preparations reset all unretained variables and are independent conditional on the parameters.

The auditor observes the first report, chooses a gate using its retained state and that report, observes the second report, and then observes the mark. During training it may choose rows and stop adaptively, using at most the declared number of candidate preparations. It next freezes one gate policy and one event, with any retained random choice included in its register. It uses that same row–event continuation on one fresh candidate preparation and one independent target preparation and returns the difference of their event indicators, target minus candidate. It may not reselect the event after either validation. Independent target-only calibration is allowed but all retained calibration and randomness are charged. Peak width counts every bit, mark, event and validation-accumulator cut. Clocked schedules are declared externally; an autonomous implementation must retain its phase. An unretained transcript, seed or stopping-time value is not available to a continuation.

In the all-on row the complete private law is $b(p, q_1)$, with an independent fair actual mark. For lower bounds it suffices to restrict the adversary to $q_0 = 0$, $q_1 = q$. In this subfamily every deterministic gate policy is the pushforward

$$T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$$

of $b(p, q)$. The gate, when included in the transcript, is another function of these same coordinates. Randomized gate policies are covered by averaging their independent random tables. This proves both the all-on product characterization for the upper construction and the common-garbling subfamily used for the outer converse. The width-one constraint is a simulator constraint; it is not an uncharged restriction on the auditor’s register.

A.10.1. Preparation and elastic scattering. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$, with no walls and contact distance d . Write $r = q_1 - q_2$, $g = v_1 - v_2$, $V = (v_1 + v_2)/2$, and let the center of mass position range over any fixed rational box of positive volume. Prepare

$(B, W) \in \{0, 1\}^2$ with probability $5/16$ at $B = W$ and $3/16$ at $B \neq W$. Conditional on (B, W) , use the uniform product preparation on the following boxes:

$$(57) \quad \begin{aligned} r &= (-a_0, (2B-1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & & & |V_z - (2W-1)| &\leq 1/100. \end{aligned}$$

Thus the actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$, preserved by momentum conservation. Set

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector, identically zero on the incoming transverse-velocity range.

Lemma A.10.2 (Uniform collision signal). *For every preparation in (57) and every $9/10 \leq d \leq 11/10$, exactly one collision occurs and its time lies in $(4/5, 3/2)$. Afterwards*

$$(2B-1)v_{1,y} > 1/2, \quad (2B-1)h(v_{1,y}) > 1/4.$$

For $0 < d \leq 2/5$ no collision occurs on $[0, 3]$ and the detector is zero throughout that interval.

Proof. Before contact, $r(t) = r + tg$. At $t \leq 4/5$ its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At $t = 3/2$ its absolute coordinate bounds are $1/4, 113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. The first contact therefore occurs in the claimed interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$. Hence the entry contact has $n_x < 0$, where $n = r(t_c)/d$. Its remaining coordinates give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B-1)n_y \geq 87/220.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The elastic rule gives

$$v_1^+ = V + g/2 - (g \cdot n)n, \quad g^+ = g - 2(g \cdot n)n.$$

Consequently

$$(2B-1)v_{1,y}^+ > -3/200 + (1443/1000)(87/220) > 1/2.$$

The definition of h implies the signal bound. Since $r(t_c) \cdot g^+ > 0$, squared separation increases strictly afterwards; there is no second collision. Finally, for $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. There is no contact, and the incoming $|v_{1,y}| \leq 3/200$ makes h zero. $\square$

A.10.2. Positive noise and the actual mark. Let $\mathcal{B}_t$ be Brownian motion independent of the preparation and set

$$Y_t = \int_0^t h(v_{1,y}(s)) ds + \sigma \mathcal{B}_t, \quad 0 < \sigma \leq 1/24.$$

The two potential all-on reports are

$$Z_1 = \mathbf{1}_{\{2(Y_2 - Y_{3/2}) > 0\}}, \quad Z_2 = \mathbf{1}_{\{2(Y_3 - Y_{5/2}) > 0\}}.$$

The actual second report is AZ_2 . The gate changes acquisition, not the mechanical path. The continuous path is implemented by the target sensor; it is not made available to the private simulator or supplied as an uncharged auditor buffer. Denote the law of (Z_1, Z_2, W) by $Q_{d,\sigma}$.

Theorem A.10.3 (Uniform physical presentation). *For $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$,*

$$(58) \quad \text{TV}(Q_{d,\sigma}, Q_*) < 1/300.$$

The same estimate holds after every declared feedback gate, including its observable gate value. The all-on private product family and the common-garbling subfamily in Definition A.10.1 apply to this very experiment.

Proof. Every collision occurs before $3/2$. Conditional on the preparation, the two normalized sensor windows are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. Lemma A.10.2 makes their means have sign $2B - 1$ and magnitude greater than $1/4$. Each potential report therefore differs from B with probability less than $\Phi(-3)$. The Gaussian tail estimate

$$\Phi(-x) \leq \frac{e^{-x^2/2}}{x\sqrt{2\pi}}$$

gives $\Phi(-3) < 1/600$: $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$ suffice. Couple the ideal and noisy reports using the same preparation, actual mark and gate random table. Except on a set of probability less than $2\Phi(-3) < 1/300$, both potential reports equal B and the entire gated transcript agrees. The ideal marked law is precisely Q_* , by the prepared (B, W) table. This proves (58) and the feedback-uniform version. Finally, the source is blind, its actual mark is unobserved by the simulator, and the gate is its only current input at the second emission. The product and subfamily assertions follow from the two independent private-emission kernels specified above. $\square$

For $d \leq 2/5$, the two windows instead contain only independent centered Gaussian noise. Their signs are independent fair bits, independent of W , and a private width-one simulator reproduces every gated row exactly. Thus the informative regime and its no-collision control use the same detector, preparation convention and actual mark.

APPENDIX A.11. THE PHYSICAL EXPERIMENT AND ITS SHARP PREPARATION BOUNDARY

Take $\Omega = \{0, 1\}^3$, with coordinates (x_1, x_2, w) . The all-on private candidate is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad 0 \leq p, q \leq 1.$$

The mark w is the actual mark, not an invented simulator bit. Put $a = (1-p)(1-q)$ and $c = pq$. On the ordered atoms

$$(59) \quad 000, \quad 001, \quad 110, \quad 111,$$

let Q_* have masses $5/16, 3/16, 3/16, 5/16$, and set it to zero elsewhere. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. A target Q is allowed whenever $\text{TV}(Q, Q_*) \leq \beta$.

The original feedback operation has the form

$$(60) \quad T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w), \quad g : \{0, 1\} \longrightarrow \{0, 1\}.$$

The selected gate is observable. It may depend on earlier preparations and on the first current report. The target rows are pushforwards by (60). For the converse, restrict the adversary to the admissible private subfamily whose second emission is AX_2 , where A is the current gate and $X_2 \sim \text{Ber}(q)$ is independent of the first emission and of the fair mark. Every row is then a causal garbling of $b(p, q)$. For the construction only the all-on row is used, and every private candidate in the original width-one class has the all-on product law just specified. These two facts, proved for the microscopic interface in Definition A.10.1 and Theorem A.10.3, explain why using a subfamily for the lower bound does not narrow the upper construction.

The admissible audit first trains, then selects one row–event pair, then uses one fresh candidate and one fresh target validation. It reports $D = \mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The event and row cannot be reselected after seeing a validation draw. All bit, mark, decision and validation cuts are charged. Define $N_c^{\text{phys}}(W, Q)$ to be the minimum number of candidate training preparations among these original-interface auditors, with peak register width at most W , whose score is uniformly strictly greater than c . Randomization, feedback and stopping are allowed. If no such audit exists, the value is $+\infty$.

Theorem A.11.1 (Two-preparation physical segment). *Let u_1, m_2 be as in (18). For every $\beta \geq 0$, $\text{TV}(Q, Q_*) \leq \beta$, and*

$$W \geq 12, \quad u_1 + \beta \leq c < m_2 - \beta,$$

one has $N_c^{\text{phys}}(W, Q) = 2$. The two-preparation audit is deterministic, uses no acquired calibration and no random bits, and has the full profile

$$(61) \quad (1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$$

from the initial cut through its twelve bit/mark updates. In particular, its decision alphabet has five labels and its peak has twelve. The lower bound remains valid with no memory cap.

The proof occupies the next three subsections. The interval in the statement can be empty for large β ; no assertion is then made. For $\beta = 1/300$, it contains $c = 2/5$ strictly between its endpoints.

A.11.1. An exact rational certificate against one preparation.

Lemma A.11.2. *Every original-interface auditor using at most one candidate training preparation has worst-case score at most $u_1 + \beta$, regardless of memory.*

Proof. By Theorem A.4.2, it suffices to bound $U_1(b, Q_*)$ and then perturb the target. In (25) use the four fixed parameters

$$(p, q) = (1/8, 1/8), (1/5, 1/5), (4/5, 4/5), (7/8, 7/8)$$

with weights $1/16, 7/16, 7/16, 1/16$, respectively. This measure is not asserted to be least favorable. Since $p = q$ on its support, the training count $K = x_1 + x_2$ is sufficient. The independent training mark and the order of unequal bits contain no further information about this prior.

For $k = 0, 1, 2$ and the four validation atoms in (59), the signed coefficients in (25), after grouping by K , are the entries of

$$(62) \quad \frac{1}{40960000} \begin{pmatrix} -1977 & -1775177 & 2170423 & 3943623 \\ 1765554 & 191954 & 191954 & 1765554 \\ 3943623 & 2170423 & -1775177 & -1977 \end{pmatrix}.$$

For example, the coefficient in column j is

$$\sum_{i=1}^4 \mu_i \binom{2}{k} p_i^k (1 - p_i)^{2-k} \left(Q_{*,j} - \frac{(1 - p_i)^2 \mathbf{1}_{j \leq 2} + p_i^2 \mathbf{1}_{j \geq 3}}{2} \right).$$

This identity proves every entry of (62) by rational arithmetic. All coefficients at off-diagonal validation atoms are nonpositive. The sum of the positive entries in (62) is $4035777/10240000 = u_1$. Each response coordinate is in $[0, 1]$, so this sum bounds its prior-averaged score and hence its minimum over the four fixed candidates.

Replacing Q_* by Q changes the score of any trained response by at most β . The training law is a candidate law and is unchanged. Consequently the same prior bounds every score by $u_1 + \beta$.

A procedure with no training, or one that stops before taking its permitted sample, is covered by ignoring the generated all-on observation. $\square$

A.11.2. A marked five-event selector. Take two independent all-on training preparations. Let S be the total number of ones among their four report bits; discard their marks. Choose the following events, represented by their indicators on (59) and with indicator zero at every off-diagonal atom:

$$(63) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & 0 \end{array}$$

Denote these indicators by $h_0, \dots, h_4$ and report $D = h_S(Y) - h_S(X)$ on independent validation draws.

Lemma A.11.3 (Exact uniform score of the selector). *Against the ideal target, the selector (63) has uniform minimum score $m_2 = (12 - 3\sqrt{3})/16$. Against Q , its score is at least $m_2 - \beta$.*

Proof. The generating polynomial of S is

$$((1-p) + pz)^2((1-q) + qz)^2.$$

Writing $g_s = (Q_* - b(p, q))h_s$, its expected score is $G(p, q) = \sum_{s=0}^4 [z^s]((1-p) + pz)^2((1-q) + qz)^2 g_s$. Substitution of the five rows in (63) and collection of terms with $x = p + q - 1$, $y = p - q$, $u = x^2$ and $v = y^2$ gives

$$(64) \quad \begin{aligned} G(p, q) = H(u, v) = & -\frac{u^3}{32} + \frac{3u^2v}{32} + \frac{3u^2}{16} - \frac{3uv^2}{32} - \frac{uv}{4} - \frac{3u}{32} \\ & + \frac{v^3}{32} + \frac{v^2}{16} + \frac{15v}{32} + \frac{7}{16}. \end{aligned}$$

This is a polynomial identity, not an interpolation over a grid. Since $0 \leq u \leq 1$ and $v \geq 0$ on the parameter square,

$$\partial_v H(u, v) = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}.$$

Thus $H(u, v) \geq H(u, 0) = (-u^3 + 6u^2 - 3u + 14)/32$. The derivative of the last expression is $(-3u^2 + 12u - 3)/32$. On $[0, 1]$ it changes sign from negative to positive at $u = 2 - \sqrt{3}$. Substitution gives $(12 - 3\sqrt{3})/16$. The minimizing values are feasible: take $p = q = (1 \pm \sqrt{2 - \sqrt{3}})/2$. This proves exactness of the minimum for this specified selector, not optimality of its score over all selectors with two preparations. Finally, for each s , the target expectation of $h_s \in [0, 1]^\Omega$ changes by at most β . Averaging over training proves the claimed robustness. $\square$

A.11.3. Causal compilation and the full register profile.

Lemma A.11.4. *The selector and its two validations admit the deterministic profile (61). No selector, first-bit buffer or validation accumulator is outside the charged register. An autonomous phase-tagged realization uses at most 75 states.*

Proof. After each training report bit retain the count accumulated so far. The two marks are read and discarded. The first six updates therefore have widths 2, 3, 3, 4, 5, 5 after the initial singleton. At the decision cut the five count values select the five events.

During candidate validation, after the first bit retain the event and that bit, using at most ten states. After the second bit retain the event and the function of the unread mark that gives its candidate-event indicator. For events 0, 1, 2, 3, 4, respectively, the possible functions are

$$\{0, 1\}, \quad \{0, 1 - w, 1\}, \quad \{0, 1\}, \quad \{0, w, 1\}, \quad \{0, 1\}.$$

There are twelve event–function pairs. After the candidate mark, retain the event and its now known candidate indicator $c_0 \in \{0, 1\}$, using at most ten states.

During target validation, after the first bit discard the event name and retain c_0 and its residual function of the second bit and mark. These residual functions belong to

$$\{0, x_2, (1 - x_2)(1 - w), 1 - x_2, x_2 w\}.$$

This uses at most ten states. After the second bit, the residual *difference* as a function of the unread mark belongs to

$$\{-1, 0, 1, w, 1 - w, w - 1, -w\},$$

so seven states suffice. The mark update gives the three possible outputs $-1, 0, 1$. Each transition depends only on its retained residual and the current report. Thus it is a legal common observable row, not a family of independently optimized hidden-history maps.

For completeness, equality of two residual functions implies equality of their restrictions after either next bit, so identifying equal residuals is closed under every subsequent update. The supplied transition tables are obtained by these restrictions and can be reconstructed directly from (63). Enumerating residuals is a convenience for reproducing the tables, not the proof of their sufficiency. Finally, disjointly tag the states at the thirteen cuts by their phase and use terminal self-loops. The sum of the displayed widths is 75. This implementation charges the clock; it does not assert that 75 is optimal. $\square$

Proof of Theorem A.11.1. Lemma A.11.2 excludes $N \leq 1$ for a strict score greater than $c \geq u_1 + \beta$, for every architecture and every memory budget. Lemmas A.11.3 and A.11.4 supply a two-preparation architecture of peak twelve with score at least $m_2 - \beta > c$. Larger width budgets can use the same machine. $\square$

Corollary A.11.5 (The positive-noise collision family). *For the original marked collision targets with $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the local microscopic estimate $\text{TV}(Q_{d,\sigma}, Q_*) < 1/300$ implies*

$$N_{2/5}^{\text{phys}}(W, Q_{d,\sigma}) = 2 \quad (W \geq 12).$$

There are four preparations in the complete audit: two candidate training, one candidate validation, and one target validation.

Proof. Apply Theorem A.11.1. The lower-end margin is the exact rational number

$$\frac{2}{5} - u_1 - \frac{1}{300} = \frac{78269}{30720000} > 0.$$

For the other endpoint, $\sqrt{3} < 7/4$ implies $m_2 > 27/64$ and $27/64 - 1/300 > 2/5$. The physical total-variation estimate, including the actual mark and positive sensor noise, is proved in Theorem A.10.3; the current deduction does not replace those hypotheses by a zero-noise limit. $\square$

Remark A.11.6 (The part of the frontier determined here). The deterministic theorem evaluates the minimum preparation coordinate for $W \geq 12$. Theorem A.12.1 and Corollary A.12.2 below additionally determine the ideal maximum score with two preparations on this width range. The least peak width over all validation orders and the full boundary for $3 \leq W \leq 11$ remain separate questions. Theorem A.14.3 below determines the peak for forward serial validation, and

Theorem A.14.1 determines the order-independent decision cut. The earlier exact decision-width result $\kappa_{2/5}^{\text{rows}} = 3$ is unchanged: it allows longer training, whereas this shortest audit uses five decision states. The older points (399, 40602), (320000, 1604) and $(12800 \cdot 2^{400}, 12)$ remain valid constructions with decision width three. They are dominated by (2, 12) in the projection onto training preparations and peak width, but not necessarily in a resource order that also constrains the decision alphabet separately.

Corollary A.11.7 (The enlarged physical preparation segment). *With the full architecture of Definition A.10.1,*

$$N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_{\text{ex}} + \beta \leq c < m_2 - \beta.$$

Here $u_{\text{ex}} = (85 - 7\sqrt{73})/64$ is the exact one-preparation all-on value, rather than the older rational upper bound.

Proof. Restrict the adversary to the common-garbling subfamily in Definition A.10.1. Theorem A.4.2 and Corollary A.9.2 bound every at-most-one-preparation auditor, even with unlimited memory, by $u_{\text{ex}} + \beta$. The same deterministic two-preparation machine in Lemmas A.11.3 and A.11.4 has score at least $m_2 - \beta$ against the full private candidate family. These inequalities prove the assertion. The four-preparation total, peak twelve and positive-noise specialization are unchanged. $\square$

APPENDIX A.12. THE EXACT TWO-PREPARATION MARKED SADDLE

The one-preparation saddle has an exact successor. A new zero coefficient of the least-favorable predictive measure makes it necessary to randomize at the two extreme training counts. This randomization changes the value, but does not enlarge the alphabet of frozen validation events.

Let $b(p, q)$ and Q_* be as in Section A.11. Let r be the unique root in $(1/3, 7/20)$ of

$$(65) \quad 2r^3 + 25r^2 - 3 = 0,$$

and define

$$(66) \quad \tau = \frac{6(-r^2 + 4r - 1)}{r(3r + 25)}, \quad v_2 = \frac{-r^3 + 6r^2 - 3r + 14}{32}, \quad p_{\pm} = \frac{1 \pm \sqrt{r}}{2}.$$

The polynomial in (65) is strictly increasing on the positive half-line and has opposite signs at the indicated endpoints. These observations prove the asserted existence, uniqueness and isolating interval. That interval also gives $0 < \tau < 1$.

Theorem A.12.1 (Exact two-preparation value). *For the full marked-word response class of (24),*

$$U_2(b, Q_*) = v_2.$$

A least-favorable prior gives mass 1/2 to each of (p_-, p_-) and (p_+, p_+) . An optimal response discards the two training marks, retains the total number S of ones in the four training report bits, and uses

$$(67) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & \tau & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & \tau \end{array}$$

with zero response at off-diagonal validation atoms. Numerically, $r = 0.3417694155\dots$, $\tau = 0.1688236447\dots$ and $v_2 = 0.4261127787\dots$. The exact value is determined by (65)–(66), not by these decimals.

Proof. Write $a = (1-p)(1-q)$, $c = pq$, $u = (p+q-1)^2$ and $v = (p-q)^2$. Both u and v belong to $[0, 1]$. The score of the deterministic five-event selector is the polynomial $H(u, v)$ in (64). Only counts zero and four are changed by (67). Their probabilities are a^2 and c^2 . Consequently the new score is

$$G(u, v) = H(u, v) + \tau J(u, v), \quad J = a^2(5/16 - a/2) + c^2(5/16 - c/2).$$

Expansion gives the polynomial identity

$$(68) \quad \begin{aligned} J(u, v) = & -\frac{u^3}{64} + \frac{3u^2v}{64} - \frac{25u^2}{128} - \frac{3uv^2}{64} + \frac{13uv}{64} \\ & + \frac{v^3}{64} - \frac{v^2}{128} - \frac{v}{32} + \frac{3}{128}. \end{aligned}$$

Thus

$$H(u, 0) = \frac{-u^3 + 6u^2 - 3u + 14}{32}, \quad J(u, 0) = \frac{3 - 25u^2 - 2u^3}{128}.$$

Equation (65) says $J(r, 0) = 0$, and the choice of τ says $\partial_u G(r, 0) = 0$. Direct factorization, using these two identities, yields

$$(69) \quad G(u, 0) - v_2 = \frac{(u-r)^2\{75 + 12r + 6u - 37r^2 - 74ru\}}{64r(3r + 25)}.$$

The expression in braces is strictly positive for $0 \leq u \leq 1$: it is at least $75 - 37(7/20)^2 - 74(7/20) > 0$. Moreover,

$$\partial_v H = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}, \quad \partial_v J = \frac{3(u-v)^2 + 13u - v - 2}{64} \geq -\frac{3}{64}.$$

Because $0 < \tau < 1$, $\partial_v G \geq 11/64$ on the unit square. It follows that $G(u, v) \geq G(u, 0) \geq v_2$, with equality precisely when $v = 0$ and $u = r$. Both equality points are feasible and are the two parameters in the asserted prior. This proves the global lower bound.

It remains to prove an upper bound against every response, not only those depending on S . Under the asserted prior the four training report bits are independent Bernoulli variables with common parameter p_- or p_+ . Conditional on $S = s$, their arrangements and the two independent training marks have parameter-independent distributions. The signed predictive coefficients can therefore be grouped by S without restricting any response coordinate. For the four displayed validation atoms put

$$C_{s,j} = \frac{1}{2} \sum_{p \in \{p_-, p_+\}} \binom{4}{s} p^s (1-p)^{4-s} \{Q_{*,j} - b_j(p, p)\}.$$

The first three rows of $256C$ are

$$\begin{aligned} s = 0 : & \quad (-(2r^3 + 25r^2 - 3), -(2r^3 + 27r^2 + 12r - 1), \\ & \quad -(2r^3 - 5r^2 - 20r - 1), -(2r^3 - 7r^2 - 32r - 3)), \\ s = 1 : & \quad 4(r-1)(2r^2 + 7r - 3, 2r^2 + 9r - 1, \\ & \quad 2r^2 - 7r - 1, 2r^2 - 9r - 3), \\ s = 2 : & \quad -6(r-1)^2(2r-3, 2r-1, 2r-1, 2r-3). \end{aligned}$$

Rows three and four are the reversals of rows one and zero, respectively. Their signs, using $1/3 < r < 7/20$, are

$$(0, -, +, +), \quad (+, -, +, +), \quad (+, +, +, +), \quad (+, +, -, +), \quad (+, +, -, 0).$$

Every off-diagonal validation coefficient is nonpositive because its target mass is zero. Hence (67) maximizes every signed coefficient: it is one at positive coefficients, zero at negative ones, and arbitrary at the two zeros. The lower-bound equality at both prior points shows that this Bayes maximum is exactly v_2 . The posterior-predictive dual, Theorem A.4.1, now gives the matching upper bound for the full response class. $\square$

Corollary A.12.2 (Exact physical score on the peak-twelve segment). *Use the atomic stochastic-row convention and the original marked interface of Definition A.10.1. At two candidate training preparations, every peak budget $W \geq 12$ has exact ideal worst-case score v_2 . For a target Q with $\text{TV}(Q, Q_*) \leq \beta$, the corresponding value lies in $[v_2 - \beta, v_2 + \beta]$. In particular, with $u_{\text{ex}} = (85 - 7\sqrt{73})/64$,*

$$(70) \quad N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad u_{\text{ex}} + \beta \leq c < v_2 - \beta.$$

Proof. Let $h_0, \dots, h_4$ be the five deterministic events in (63). At count zero select h_1 with probability τ and h_0 otherwise; at count four select h_3 with probability τ and h_4 otherwise. At the other counts retain the indicated event. Discard the count and retain the selected event for both validations. This realizes (67); the event alphabet remains five. The full charged profile, now including the separate event-selection update, is

$$(1, 2, 3, 3, 4, 5, 5, 5, 10, 12, 10, 10, 7, 3).$$

The common residual-function compiler in Lemma A.11.4 applies unchanged. In particular, no persistent random selector is outside the register. The all-on law for every private candidate is $b(p, q)$, so Theorem A.12.1 proves the constructive lower bound. For the converse restrict to the admissible common-garbling product subfamily and apply Theorem A.4.2 followed by Theorem A.12.1. This bounds all architectures, including those using feedback, stopping and more memory. Target perturbation changes each frozen-event expectation by at most β . Finally the exact one-preparation theorem excludes a strict score above c at $N \leq 1$, whereas the displayed machine achieves it at $N = 2$ under (70). $\square$

The root r is larger than $2 - \sqrt{3}$. The old diagonal polynomial $H(u, 0)$ is strictly increasing between these points, whence $v_2 > m_2$. The deterministic theorem, including its zero-random-bit implementation, remains valid; the new exact score uses an additional stochastic row.

Proposition A.12.3 (Finite precision for the new row). *Replacing τ by a dyadic $\tau_b = k/2^b \in [0, 1]$ with $|\tau_b - \tau| \leq 2^{-b}$ loses at most $(5/16)2^{-b}$ in uniform score. A clocked fair-bit implementation uses b extra random bits and at most $\max\{12, 6(b+1) + 5\}$ persistent states, including their phase, in addition to the unchanged preparation count.*

Proof. At each affected training word the response changes at a single atom; $|5/16 - a/2|$ and $|5/16 - c/2|$ are at most $5/16$. The two affected word probabilities sum to at most one. This gives the score estimate. To implement the dyadic coin, compare b fresh fair bits lexicographically with the fixed b -bit integer k . Retain the origin (count zero or four), one of the three comparison flags, and a phase in $\{0, \dots, b\}$. Once the comparison ends retain its chosen one of five events. Unaffected counts can wait in their event states. These states total at most $6(b+1) + 5$; the other cuts have peak twelve. The cases $k = 0, 2^b$ are deterministic. Exact atomic randomization and a finite fair-bit implementation are therefore distinct resource claims. $\square$

APPENDIX A.13. A CONTINUOUS TWO-PREPARATION MARKED FAMILY

The preceding saddle persists on an explicit interval of target laws, with exact rather than perturbative values. Use the target Q_γ from Section A.9 and write $U_2(\gamma) = U_2(b, Q_\gamma)$.

Theorem A.13.1 (A two-preparation saddle family). *For $6/25 \leq \gamma \leq 13/50$, let r_γ be the unique root in $(1/3, 7/20)$ of*

$$(71) \quad f_\gamma(r) = r^3 + (13 - 2\gamma)r^2 + (3 - 12\gamma)r - (1 + 2\gamma) = 0.$$

Set

$$D_\gamma(r) = 3r^2 + (26 - 4\gamma)r + 3 - 12\gamma, \quad t_\gamma = -\frac{2\{3r_\gamma^2 + (8\gamma - 14)r_\gamma + 3\}}{D_\gamma(r_\gamma)}.$$

Then $0 < t_\gamma < 1$, and

$$(72) \quad U_2(\gamma) = \frac{-r_\gamma^3 + (7 - 4\gamma)r_\gamma^2 - 3r_\gamma + 13 + 4\gamma}{32}.$$

The response (67), with $\tau = t_\gamma$, is optimal. A least-favorable prior gives equal masses to the two diagonal parameters $p = q = (1 \pm \sqrt{r_\gamma})/2$. The response has the same five-event, peak-twelve atomic realization. At $\gamma = 1/4$ these formulas reduce to Theorem A.12.1.

Proof. For brevity write $r = r_\gamma$, $D = D_\gamma(r)$ and $t = t_\gamma$. The derivative D_γ is positive on the rectangle $\gamma \in [6/25, 13/50]$, $r \in [1/3, 7/20]$. For example its lower bound is $1/3 + (26 - 52/50)/3 - 3/25 > 8$. The polynomial f_γ decreases with γ at a fixed positive argument. Direct rational evaluations give $f_{6/25}(1/3) < 0$ and $f_{13/50}(7/20) > 0$. Monotonicity therefore gives the root and its uniqueness in this interval. The numerator of t is $-16\gamma r - 6r^2 + 28r - 6$. It is bounded below by $28/3 - 6 - 16(13/50)(7/20) - 6(7/20)^2 > 0$, and above by $28(7/20) - 6 < 4 < D$. Thus $0 < t < 1$.

Let H, J be the polynomials from (64) and (68). Changing the target masses linearly in γ gives

$$\begin{aligned} H_\gamma(u, v) &= H(u, v) + (\gamma - 1/4) \frac{1 - (u - v)^2}{8}, \\ J_\gamma(u, v) &= J(u, v) + (\gamma - 1/4) \frac{(u - v)^2 + 6u - 2v + 1}{32}. \end{aligned}$$

The response's full-parameter score is $G_\gamma = H_\gamma + tJ_\gamma$. One has $J_\gamma(r, 0) = -f_\gamma(r)/64 = 0$ and $\partial_u G_\gamma(r, 0) = 0$. Algebraic division gives

$$(73) \quad G_\gamma(u, 0) - H_\gamma(r, 0) = \frac{(u - r)^2 A_\gamma(r, u)}{16D},$$

where

$$\begin{aligned} A_\gamma(r, u) &= 24\gamma^2 + (3\gamma - 10)r^2 + (6\gamma - 20)ru \\ &\quad + 12\gamma r + 6\gamma u - 51\gamma + 30. \end{aligned}$$

For $0 \leq u \leq 1$ this is at least $30 - 10(7/20)^2 - 20(7/20) - 51(13/50) > 0$. The transverse derivatives satisfy

$$\begin{aligned} 32\partial_v H_\gamma &= 3(u - v)^2 + (8\gamma - 10)u + (6 - 8\gamma)v + 15 \geq 5, \\ 64\partial_v J_\gamma &= 3(u - v)^2 + (14 - 4\gamma)u + (4\gamma - 2)v - 4\gamma - 1 \geq -3 \end{aligned}$$

on $0 \leq u, v \leq 1$. Consequently $\partial_v G_\gamma \geq 7/64$. Equation (73) proves a lower bound $H_\gamma(r, 0)$, with equality precisely at the proposed prior points. This quantity is the right side of (72).

For the upper bound group the full training words by S , as in the preceding theorem. Put $L(r) = r^2 + 6r + 1$. Using $f_\gamma(r) = 0$, the first three rows of $256C$ are

$$\begin{aligned} 0 : & (0, -8\gamma L(r), 32r(r+1) - 8\gamma L(r), 32r(r+1)), \\ 1 : & 8(r-1)(r^2 + 4r - 1 - 2\gamma(r+1), \\ & \quad r^2 + 4r - 1 + 2\gamma(r+1), \\ & \quad r^2 - 4r - 1 + 2\gamma(r+1), r^2 - 4r - 1 - 2\gamma(r+1)), \\ 2 : & -12(r-1)^2(r-1 - 2\gamma, r-1 + 2\gamma, r-1 + 2\gamma, r-1 - 2\gamma). \end{aligned}$$

The other rows are their reversals. The sign pattern is again

$$(0, -, +, +), \quad (+, -, +, +), \quad (+, +, +, +), \quad (+, +, -, +), \quad (+, +, -, 0).$$

Here are explicit interval checks for the potentially small margins. In row zero the third coordinate is bounded below by $32(1/3)(4/3) - 8(13/50)((7/20)^2 + 6(7/20) + 1) > 0$. The first parenthesis of row one is at most $(7/20)^2 + 4(7/20) - 1 - 2(6/25)(4/3) < 0$; the third is at most $(7/20)^2 - 4/3 - 1 + 2(13/50)(27/20) < 0$. The second parenthesis is positive already from $r > 1/3$ and $\gamma > 0$, and the fourth is negative. Finally $r - 1 + 2\gamma \leq -13/100 < 0$. All off-diagonal validation coefficients are nonpositive. Thus the stated response is a Bayes best response for this prior, including its two zero coefficients. Its score at both supporting parameters equals the lower bound. The full posterior-predictive dual proves (72). $\square$

This interval is a uniform certificate region for the displayed saddle, not a claim that the same masks remain optimal for every mark bias. The theorem supplies a continuous family at genuinely growing training history $N = 2$; it does not infer a recurrence for every N from the one-preparation formulas.

APPENDIX A.14. EXACT MEMORY FROM A MARKED SADDLE

The realization criterion is now applied to a saddle with genuine Bayes indifference. The zero coefficients in the two-preparation problem are not harmless: their values are fixed by contact with the parameter family. This supplies a converse for all optimal audits, rather than for a preselected response table.

Put $I = [6/25, 13/50]$. Throughout this section $r = r_\gamma$, $t = t_\gamma$, and $v = U_2(\gamma)$ are those of Theorem A.13.1. Denote the five deterministic events in (63) by $e_0, \dots, e_4$. Responses displayed below are ordered on 000, 001, 110, 111 and vanish at the other four atoms.

Theorem A.14.1 (Contact rigidity and exact decision memory). *For every $\gamma \in I$, every minimax optimal full-word response with two training preparations has, after averaging the independent training marks at the two extreme report counts, the five rows*

$$(74) \quad (t, 0, 1, 1), \quad (1, 0, 1, 1), \quad (1, 1, 1, 1), \quad (1, 1, 0, 1), \quad (1, 1, 0, t).$$

At the three interior counts the displayed response is forced at each individual training word, not merely after averaging.

In the original reset row-event interface, the least number of states at the post-training cut that permits the exact ideal score v is five. This converse allows arbitrary upstream finite memory, randomized encoding, adaptive training gates, stopping, independent target-only calibration, and all legal post-cut randomizations. Other cuts are unrestricted in this decision-memory assertion.

Proof. First consider a response to the full two-preparation word. Under the least-favorable prior of Theorem A.13.1, every training word has positive probability. The strict coefficient signs force e_1, e_2, e_3 at report counts 1, 2, 3. At count zero only the 000 coordinate is free; at count

four only the 111 coordinate is free. All off-diagonal validation coefficients are strictly negative: the prior parameters are interior, each word likelihood is positive, and the target assigns zero mass to those atoms. Thus those coordinates are zero for every optimal response.

Let a and b be the averages of the two free coordinates over the four equiprobable training mark words at their respective extreme counts. On the diagonal parameter (p, p) the score is

$$(75) \quad H_\gamma((2p-1)^2, 0) + aA_\gamma(p) + bB_\gamma(p),$$

where

$$A_\gamma(p) = (1-p)^4 \left(\frac{1+\gamma}{4} - \frac{(1-p)^2}{2} \right), \quad B_\gamma(p) = p^4 \left(\frac{1+\gamma}{4} - \frac{p^2}{2} \right).$$

Here H_γ is the deterministic five-event score in the proof of Theorem A.13.1. At $p_+ = (1+\sqrt{r})/2$, $A_\gamma(p_+) + B_\gamma(p_+) = J_\gamma(r, 0) = 0$ and $A_\gamma(p_+) > 0$. The latter follows from $p_+ > 1/2$ and $(1+\gamma)/4 > 1/4$. Reflection exchanges A and B at $p_- = 1 - p_+$. Also $H_\gamma(r, 0) = v$. Optimality forces contact at both prior points, by Theorem A.7.3. Subtracting v from (75) therefore gives $(a-b)A_\gamma(p_+) = 0$, hence $a = b$.

With this common value the diagonal score is $H_\gamma(u, 0) + aJ_\gamma(u, 0)$, where

$$H_\gamma(u, 0) = \frac{-u^3 + (7-4\gamma)u^2 - 3u + 13 + 4\gamma}{32}, \quad J_\gamma(u, 0) = -\frac{f_\gamma(u)}{64}.$$

The supporting value r lies in $(0, 1)$, so the derivative at r vanishes. Since $f'_\gamma(r) = D_\gamma(r) > 8$, the derivative equation has the unique solution

$$a = b = \frac{64 \partial_u H_\gamma(r, 0)}{D_\gamma(r)} = t_\gamma.$$

This proves (74). Individual extreme mark words may still use different tie values; no unnecessary uniqueness is asserted.

Now take an arbitrary audit with K states at the decision cut. Restrict the candidate to the common-garbling subfamily in Definition A.10.1. Generate two all-on words and simulate the adaptive training and its stopping with independent coins. The retained state is a parameter-independent stochastic function of those two words. Conditional on that state, the frozen row-event continuation pulls back to a response $D_s \in [0, 1]^8$ on the all-on alphabet. Any calibration is part of the same encoding and retained state. Thus (44) holds without increasing K . If the audit attains v against the full physical class, its pullback attains v against this subfamily, since the unrestricted upper bound is v . Averaging selected rows preserves their factorization.

The three interior rows in (74) are distinct vertices of $[0, 1]^8$, so the generators must include e_1, e_2, e_3 themselves. The first row lies in the face

$$F_0 = \{d : d(001) = 0, d(110) = d(111) = 1, d = 0 \text{ off the diagonal}\}.$$

Of those three obligatory vertices only e_1 lies in F_0 ; since $t < 1$, this row is not e_1 . It needs another generator in F_0 . Similarly the last row needs another generator in

$$F_4 = \{d : d(000) = d(001) = 1, d(110) = 0, d = 0 \text{ off the diagonal}\}.$$

Only e_3 among the obligatory vertices lies there. The two faces are disjoint. Lemma A.7.5 gives $K \geq 3 + 2 = 5$. Conversely the optimal response mixes e_0, e_1 at count zero and e_4, e_3 at count four and otherwise selects the indicated event. Only its five-valued event label survives the cut. The inherited compiler realizes its subsequent validation with finite memory and attains v . This proves exactness of the decision count. $\square$

The contact argument is essential. Setting both free coordinates equal to one gives a Bayes best response with only the three events e_1, e_2, e_3 , but its derivative at r is nonzero because $t < 1$. It is not minimax. Counting the states of a Bayes best response alone would give an incorrect resource conclusion.

Corollary A.14.2 (A strict gap and its physical persistence). *There is $\delta_* > 0$, uniform for $\gamma \in I$, such that every two-preparation audit with at most four states at the decision cut has ideal worst-case score at most $U_2(\gamma) - \delta_*$. In particular, this bound holds for every original-interface audit of whole-schedule peak at most four. If only the target changes by total variation at most $\beta < \delta_*/2$, the five-state decision construction still strictly outperforms this class.*

Proof. Let $W_4(\gamma)$ be the maximum worst-case score over the relaxed factorizations (44) with $K = 4$ and two full training words. Their encoder and decoder spaces are compact and fixed, independently of γ . The score is jointly continuous in these variables, the target bias, and (p, q) . Consequently W_4 is continuous. By Theorem A.14.1 it is strictly less than $U_2(\gamma)$ for each $\gamma \in I$. The minimum of this positive continuous gap on compact I is positive. This defines δ_* without pretending to have evaluated it numerically. The domination argument in that theorem makes it an upper bound for every original-interface audit in question. Changing only the target changes any fixed audit's mean score by at most β . The difference between the two classes therefore decreases by at most 2β . $\square$

For the microscopic family at $\gamma = 1/4$, the signal is larger than $1/4$ and the standard deviation of each normalized sensor noise is $\sqrt{2}\sigma$. The Gaussian tail bound used in Theorem A.10.3 gives the sharper, vanishing estimate

$$(76) \quad \text{TV}(Q_{d,\sigma}, Q_*) \leq \frac{8\sigma}{\sqrt{\pi}} \exp\{-1/(64\sigma^2)\}, \quad 9/10 \leq d \leq 11/10.$$

Thus Corollary A.14.2 holds uniformly in d for all sufficiently small positive σ . This is an existence statement with the displayed physical modulus. The present proof does not certify that every $\sigma \leq 1/24$ lies below the unknown gap threshold.

A.14.1. The sharp serial-validation peak. A decision cut and the whole execution have different resource costs. We now determine the latter in a declared validation schedule. In the *forward serial class*, both training preparations finish before validation; the candidate validation's two report slots and mark are read before any target validation report is read. The same frozen row-event continuation is used on both preparations, as in Definition A.10.1. Training gates, stochastic state updates and frozen gate policies are unrestricted within that interface. No intermediate output can be reread as external memory.

Theorem A.14.3 (Sharp forward serial peak). *For every $\gamma \in I$, the least whole-schedule peak width attaining $U_2(\gamma)$ in the forward serial class is twelve. In particular, a peak of at most eleven in this class incurs a strictly positive loss, which can be chosen uniformly for $\gamma \in I$.*

Proof. Use the common-garbling subfamily and the two raw all-on training words as in Theorem A.14.1. Any optimal audit must finish both training preparations almost surely under its least-favorable prior. Indeed a stopped branch after a first preparation would have a response independent of the unused second word. For each first report count 0, 1, 2, two choices of the unused word force two different interior vertices from e_1, e_2, e_3 . Each has positive conditional probability. A positive stopped component cannot be a convex component of two distinct vertices. The same argument excludes stopping before the first draw. Null branches may be completed arbitrarily.

Consider the cut after the candidate's second report slot and before its mark. On a canonical raw input tape its continuation is a channel from $(w, y) \in \{0, 1\} \times \{000, 001, 110, 111\}$ to $\{-1, 0, 1\}$, where w is the unread candidate mark and y the fresh target word. All these target words have positive probability. Each retained state supplies one such continuation channel, common to every past leading to it. Hence the rows indexed by selected pasts factor through at most the peak number of generators.

For the strict interior training counts, the following eight rows are forced deterministic channels:

$$(77) \quad e_1(y) - f(w), \quad f \in \{0, 1 - w, 1\}; \quad e_2(y) - f(w), \quad f \in \{0, 1\}; \quad e_3(y) - f(w), \quad f \in \{0, w, 1\}.$$

For example e_1 has candidate residual 0 after an off-diagonal report pair, $1 - w$ after 00, and 1 after 11. The other cases follow by restriction of their masks. Each listed past has positive probability under both supporting candidate parameters. The eight functions are distinct: subtracting two with the same mask compares their distinct w functions; for different masks their difference depends nontrivially on y and cannot equal a function of w alone. They are eight distinct vertices of the product of output simplices.

There are four additional required rows. Average the extreme training marks as in Theorem A.14.1, and take the candidate raw report pair to be, respectively, off diagonal and 11 at count zero, and off diagonal and 00 at count four. Independence of a fresh raw candidate preparation from the frozen continuation preserves the averaged tie probability t . On the four target words these rows have means

$$(78) \quad (t, 0, 1, 1), \quad (t - 1, -1, 0, 0), \quad (1, 1, 0, t), \quad (0, 0, -1, t - 1),$$

for both values of w . More precisely, each integer coordinate here is a point mass at that integer, and the remaining coordinate is the Bernoulli mixture between the two neighboring integers. This stronger channel statement, not a statement about a zero mean alone, follows because the candidate indicator is fixed at 0 or 1 on the selected raw pair, while the target indicator is a bit. Fresh gate randomness is already part of this channel; it supplies no additional past-dependent argument to a retained state.

The four rows lie in four pairwise disjoint faces of the product of output simplices. These faces fix, for both w , respectively,

	000	001	110	111
$F_{0,+}$	*	0	1	1
$F_{0,-}$	*	-1	0	0
$F_{4,+}$	1	1	0	*
$F_{4,-}$	0	0	-1	*

where a number specifies a point mass and $*$ is unrestricted. Among the eight vertices (77), each face contains exactly one: e_1 , $e_1 - 1$, e_3 , and $e_3 - 1$, respectively. Since $t < 1$, none of the four required rows is that vertex. Face packing forces four more generators, for a total of twelve. This argument uses conditional future-output laws and not the continued storage of an event name after a residual has replaced it.

The optimal five-event selection followed by the inherited forward compiler has peak twelve, including its stochastic selection cut. It proves the matching upper bound. To see strictness below twelve, use the finite-protocol normal form of Proposition A.7.1 with all logical cuts bounded by eleven, including selection and frozen-continuation buffers. Unobserved internal random computations can be collapsed to common transition kernels on the same register; an unretained random duration cannot be supplied as a later input. The normalized row spaces and closed frozen-interface restrictions are compact. If the constrained value equalled U_2 , the

preceding face obstruction would apply. Continuity and compactness in γ give a uniform strictly positive gap, just as in Corollary A.14.2. $\square$

Theorem A.14.3 is a minimum-width theorem for a specified serial order, not a claim that an arbitrary validation order has the same minimum. The ideal target has support restrictions that can change future residuals when the target is read first. Reverse-order and interleaved validations require their own continuation analysis. The decision-memory converse in Theorem A.14.1 does not make this serial-order assumption. Neither statement replaces the original physical interface by an uncharged atomic validation.

APPENDIX A.15. A FULL-SUPPORT MARKED SADDLE

The support restriction of the ideal target can be removed without replacing the private simulator or changing the training experiment. This section identifies an explicit region of positive targets on which both the statistical saddle and its decision-memory number are exact. The next section shows that the original Brownian sensor belongs to this region throughout its previously stated noise range.

Put

$$\mathcal{R} = [6/25, 13/50] \times [0, 1/1000].$$

For $(\gamma, \kappa) \in \mathcal{R}$, define $Q_{\gamma, \kappa}$ on $\{0, 1\}^3$ by the masses

$$(79) \quad \frac{\omega}{Q_{\gamma, \kappa}(\omega)} \left| \begin{array}{cccc|c} 000 & 001 & 110 & 111 & \text{each other atom} \\ a & b & b & a & \kappa \end{array} \right|, \quad a = \frac{1+\gamma}{4} - \kappa, \quad b = \frac{1-\gamma}{4} - \kappa.$$

These masses sum to one. For $\kappa > 0$ every atom has positive mass. The candidate is still $b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2)$; there is no additional candidate parameter.

Theorem A.15.1 (Full-support saddle and contact rigidity). *For every $(\gamma, \kappa) \in \mathcal{R}$, there is a unique $r = r_{\gamma, \kappa}$ in $(33/100, 7/20)$ satisfying*

$$(80) \quad f(r) = r^3 + (13 - 2\gamma + 8\kappa)r^2 + (3 - 12\gamma + 48\kappa)r - 1 - 2\gamma + 8\kappa = 0.$$

Set $D = f'(r)$ and

$$(81) \quad t = \frac{2\{-3r^2 + (14 - 8\gamma - 16\kappa)r - 3 + 48\kappa\}}{D},$$

$$(82) \quad v = \frac{-r^3 + (7 - 4\gamma - 8\kappa)r^2 + (-3 + 48\kappa)r + 13 + 4\gamma - 104\kappa}{32}.$$

Then $8 < D < 10$, $1/10 < t < 1/4$, and

$$U_2(b, Q_{\gamma, \kappa}) = v.$$

The optimal five response rows are (74), with this value of t , and zero at all off-diagonal atoms. The equal prior on $p = q = (1 \pm \sqrt{r})/2$ is least favorable. The same averaged rows are forced for every full-word minimax response. At the three interior counts the entire deterministic event is forced word by word.

The least post-training decision width attaining v is exactly five, including in the original gate interface with arbitrary stochastic training, stopping and independently acquired calibration.

Proof. On the indicated rectangle in (γ, κ, r) , direct monotonicity gives $8 < f'(r) < 10$. The polynomial decreases with γ and increases with κ . Its two endpoint evaluations are bounded by

$$f_{6/25, 1/1000}(33/100) = -81619/5000000 < 0, \quad f_{13/50, 0}(7/20) = 387/40000 > 0.$$

This proves existence and uniqueness in the stated interval. The numerator of t is $-16\gamma r - 32\kappa r + 96\kappa - 6r^2 + 28r - 6$. Bounding its negative and positive terms separately gives a lower

bound $10378/10000 > 1$ and an upper bound $19754/10000 < 2$. Together with $8 < D < 10$ these prove the asserted interval for t .

Let $u = (p + q - 1)^2$, $z = (p - q)^2$, and retain the polynomials H_γ, J_γ from Theorem A.13.1. The probabilities of the five training counts give

$$E(u, z) = \frac{(u - z)^2 - 6u + 2z + 13}{4},$$

$$L(u, z) = \frac{(u - z)^2 + 6u - 2z + 1}{8}.$$

Here E is the expected cardinality of the deterministic selected four-atom event, and L is the probability of an extreme training count. The score of the proposed response is therefore

$$G = H_{\gamma, \kappa} + tJ_{\gamma, \kappa}, \quad H_{\gamma, \kappa} = H_\gamma - \kappa E, \quad J_{\gamma, \kappa} = J_\gamma - \kappa L.$$

This is an identity in the actual training probabilities, not an estimate. On the diagonal, $J_{\gamma, \kappa}(u, 0) = -f(u)/64$ and $H_{\gamma, \kappa}(r, 0) = v$. Polynomial division, using $f(r) = 0$, gives

$$(83) \quad G(u, 0) - v = \frac{(u - r)^2 A(\gamma, \kappa, r, u)}{16D},$$

where

$$A = 24\gamma^2 + (3\gamma - 10)r^2 + (6\gamma - 20)ru + 12\gamma r + 6\gamma u - 51\gamma + 30$$

$$- 384\kappa^2 - 96\kappa r - 48\kappa u - 144\kappa.$$

For $0 \leq u \leq 1$ the negative terms alone give $A > 8$ throughout $\mathcal{R}$. Moreover, on $0 \leq u, z \leq 1$ the bounds in the preceding family theorem yield

$$\partial_z H_{\gamma, \kappa} \geq 5/32 - \kappa, \quad \partial_z J_{\gamma, \kappa} \geq -3/64.$$

Indeed the additional derivatives are respectively $\kappa(u - z - 1)/2$ and $\kappa(u - z + 1)/4$. Since $t < 1/4$, one has $\partial_z G > 1/8$. Thus (83) is a global lower certificate on the parameter square, with equality at exactly the two proposed prior points.

For the Bayes upper bound, group the four report bits by their total S ; the independent training marks and arrangements conditional on S are parameter independent under this prior. Put $L_r = r^2 + 6r + 1$. The first three rows of 256 times the signed coefficient table on 000, 001, 110, 111 are

$$S = 0 : \quad (0, -8\gamma L_r, 32r(r + 1) - 8\gamma L_r, 32r(r + 1)),$$

$$S = 1 : \quad 8(r - 1)(r^2 + 4r - 1 - 2\gamma(r + 1) + 8\kappa(r + 1),$$

$$r^2 + 4r - 1 + 2\gamma(r + 1) + 8\kappa(r + 1),$$

$$r^2 - 4r - 1 + 2\gamma(r + 1) + 8\kappa(r + 1),$$

$$r^2 - 4r - 1 - 2\gamma(r + 1) + 8\kappa(r + 1)),$$

$$S = 2 : \quad -12(r - 1)^2(r - 1 - 2\gamma + 8\kappa, r - 1 + 2\gamma + 8\kappa,$$

$$r - 1 + 2\gamma + 8\kappa, r - 1 - 2\gamma + 8\kappa).$$

The other rows are their reversals. For example the potentially small first parenthesis in row one is at most

$$(7/20)^2 + 4(7/20) - 1 - 2(6/25)(133/100) + 8(1/1000)(27/20) < -1/10.$$

The third parenthesis is negative by the same endpoint bounds; the second is positive and the fourth negative. In row two, $r - 1 + 2\gamma + 8\kappa \leq -61/500 < 0$. Row zero's third entry is positive since $32(33/100)(133/100) - 8(13/50)((7/20)^2 + 6(7/20) + 1) > 7$. Consequently the signs are

exactly those of the previous two-preparation saddle. At each off-diagonal atom the candidate mass on both prior points equals $(1-r)/8 > 13/160$, whereas $\kappa \leq 1/1000$. Every off-diagonal coefficient is strictly negative. Coordinatewise Bayes maximization therefore selects the proposed response, including its two indifferent coordinates. Its equal score at the prior support closes the minimax match against the full response class.

For completeness the contact argument persists, not just the sign pattern. If a_0, b_0 are the averaged extreme response coordinates, their extra diagonal contributions are

$$a_0(1-p)^4\{a - (1-p)^2/2\} + b_0p^4\{a - p^2/2\}.$$

At either support point their sum with $a_0 = b_0 = 1$ is zero by (80); the first term is positive at p_+ . Contact at both supports gives $a_0 = b_0$. Interior stationarity gives $a_0 = b_0 = 64\partial_u H_{\gamma,\kappa}(r, 0)/D = t$. All strict coordinates are forced individually. Thus the three vertices and two disjoint faces used in Theorem A.14.1 are unchanged. That theorem's positive-factorization argument gives five states for every admissible stochastic encoder, while the same five-event selection attains it. The common-garbbling reduction is unchanged because the candidate interface is unchanged. $\square$

The rectangle $\mathcal{R}$ is an explicit stability region for the *saddle and its exact decision-memory number*. It is not a claim that arbitrary nonsymmetric total-variation perturbations have the same optimizer. At $\kappa = 0$ every formula reduces to the retained ideal theorem. Positive off-diagonal target masses do not destroy the strict negative Bayes coefficients at those atoms: their magnitude relative to the candidate, rather than their being zero, is what is needed.

APPENDIX A.16. EXACT STATISTICAL FORM OF THE BROWNIAN COLLISION SENSOR

The full-support theorem applies to the original microscopic target, not only to an independently constructed contamination model. The preparation symmetries supply the required form exactly.

Theorem A.16.1 (Physical symmetry and a quantitative noise region). *Use exactly the preparation boxes, elastic collision, detector and Brownian windows of Section A.10. Conditional on the preparation put*

$$Z = |h(v_{1,y}^+)|, \quad e = \Phi\left(-\frac{Z}{\sqrt{2}\sigma}\right), \quad \mu_1 = \mathbb{E}e, \quad \mu_2 = \mathbb{E}e^2.$$

For every $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$ the exact all-on target law is

$$(84) \quad Q_{d,\sigma} = Q_{\gamma_{d,\sigma}, \kappa_{d,\sigma}}, \quad \gamma_{d,\sigma} = \frac{1}{4} - \frac{\mu_1}{2}, \quad \kappa_{d,\sigma} = \frac{\mu_1 - \mu_2}{2}.$$

In particular

$$\frac{1}{4} - \frac{1}{1200} < \gamma_{d,\sigma} < \frac{1}{4}, \quad 0 < \kappa_{d,\sigma} < \frac{1}{1200},$$

so the entire physical range lies inside $\mathcal{R}$, and every target atom has strictly positive probability. The physical two-preparation value in the unrestricted-memory original gate interface equals (82) at these parameters. Its exact decision-memory number is five.

Proof. The collision estimate gives $Z > 1/4$ for every prepared point. The conditional two window noises are independent, of equal variance, and the deterministic signal is constant after the collision. Thus, given (B, W, Z) , the two report bits are independent copies of B flipped with probability e .

We justify the distribution of Z rather than assume a noise symmetry. Reflection in the spatial y coordinate sends (r_y, g_y, V_y) to their negatives, interchanges the two B boxes, and preserves their uniform product laws and the elastic equations. It negates $v_{1,y}^+$ and leaves Z

unchanged. The only box that depends on W is the one for V_z ; changing this center-of-mass velocity cannot change the relative collision or the outgoing y velocity. All other box coordinates are independent of W . Consequently Z , and hence e , has the same conditional law for all four (B, W) labels. The Brownian increments are independent of these labels and of the preparation.

The prepared masses of (B, W) are $5/16$ for equality and $3/16$ otherwise. Direct averaging therefore gives

$$\begin{aligned} Q(000) &= Q(111) = 5/16 - 5\mu_1/8 + \mu_2/2, \\ Q(001) &= Q(110) = 3/16 - 3\mu_1/8 + \mu_2/2, \\ Q(010) &= Q(011) = Q(100) = Q(101) = (\mu_1 - \mu_2)/2. \end{aligned}$$

These are precisely (79) with (84). This calculation uses both the common signal and the actual mark; replacing the common signal by independent unconditional report errors would in general replace μ_2 incorrectly by μ_1^2 .

The Gaussian bound already proved in the physical bridge yields $0 < e < 1/600$ uniformly in this physical range. Thus $0 < \mu_2 < \mu_1 < 1/600$, proving the parameter bounds. Positivity follows also directly from $0 < e < 1$ and positive prepared label masses. The all-on construction works against every private candidate in Definition A.10.1; its score is the product-family score. Conversely the common-garbling private subfamily gives the architecture-uniform upper bound. Theorem A.15.1 therefore supplies both exact value and exact decision count for the original interface, not for an auxiliary sensor. $\square$

Remark A.16.2 (Known target and acquired calibration). The exact value and exact atomic stochastic realization are statements for a specified target law. Their event-selection probability depends on that target, not on the unknown candidate. When d, σ and the preparation are known, the expectations in (84) are defined integrals of that known physical law. They are not free information about the private parameter. An auditor that must acquire the target instead uses finite calibration and a precision allowance. The following explicit bound makes this separation operational.

Proposition A.16.3 (Acquisition and finite row precision). *Let the fixed physical target be in the preceding theorem. From n independent all-on target calibration words define*

$$\hat{\kappa} = \frac{1}{4n} \sum_{i=1}^n \mathbf{1}_{\{X_{1i} \neq X_{2i}\}}, \quad \hat{\gamma} = \frac{1}{n} \sum_{i=1}^n (\mathbf{1}_{\{000,111\}} - \mathbf{1}_{\{001,110\}})(X_i).$$

Clip these separately to the two intervals of $\mathcal{R}$. With probability at least $1 - 4\exp(-na^2/2)$, both coordinate errors are at most a . Choose the optimal response for this clipped target and replace its tie probability by a dyadic number with error at most 2^{-b} . Conditional on that calibration event, its worst-case score against the true target is at least

$$(85) \quad U_2(b, Q_{d,\sigma}) - 9a - 2^{-b}.$$

The calibration can be acquired with at most $(n+1)^3$ counter states and at most $4(n+1)^3$ states including the current report buffer; retaining the calibration output during the audit gives the conservative product bound $12(n+1)^3$ at logical report cuts. A sequential fair-bit realization of the tie coin, with its origin and bit phase retained, has a finite explicit bound $12(n+1)^3 + 8(b+2)(n+1)^3$.

Proof. The off-diagonal indicator has mean 4κ and range one. The signed diagonal indicator has mean $2a - 2b = \gamma$ and range two. Hoeffding's inequality and a union bound give the displayed failure probability; clipping to an interval containing the truth cannot increase coordinate error. The two target laws have total variation at most $|\Delta\gamma|/2 + 4|\Delta\kappa| \leq 9a/2$. Every fixed trained

response changes score by at most this amount when only the target changes, and so does the unrestricted optimal value. Using an optimizer for the estimated target therefore loses at most $9a$ against the true one. Only the two extreme-count event choices are rounded. Their score contribution changes by at most the coin error, because their total probability is at most one and the change in a single indicator expectation has magnitude at most one. This proves (85).

Three nonnegative category counts, each bounded by n , compute the two statistics; they also fix the clipped estimates and the subsequent program choice. Keeping this count triple with the twelve-state audit costs at most the stated product. At an extreme count, sample the dyadic coin using b fair bits and comparison with its binary threshold, retaining the two-valued origin, the comparison flag and the phase. At most $8(b+2)$ additional states suffice per calibration choice. All other counts select deterministic events. The bound includes the coin workspace instead of identifying an algebraic row with a finite random-bit primitive. The n target calibrations and the four audit preparations are all charged. $\square$

APPENDIX A.17. OPTIMAL VALIDATION ORDERS AND A POSITIVE-NOISE MEMORY JUMP

We now optimize the order in the original two-tape, bit-level validation interface. Each tape has slots $1, 2, w$ in that order. Write X for the candidate tape and Y for the target tape. There are twenty declared-clock interleavings. A fixed clock is common to all executions; its phase is not a report-dependent side channel. Gate choices, state updates, training and frozen-event selection remain stochastic and otherwise unrestricted as before.

Let $e_0, \dots, e_4$ be the five events (63). A full raw validation word has output $e_i(Y) - e_i(X)$. For the strict training counts, contact forces $i = 1, 2, 3$ respectively. At either extreme only one raw-event coordinate is indifferent; Theorem A.15.1 fixes its mark-averaged probability to $t \in (1/10, 1/4)$.

One precaution is needed. A freshly randomized gate need not couple its two validation coins as one stored event mask would. When both raw validation words equal the indifferent atom, their difference may be random with mean zero. None of the lower certificates below uses that coordinate as a forced deterministic zero. For every additional mixed witness, the known prefix excludes at least one validation word from the indifferent atom, so at least one of the two event indicators is already deterministic. Its future output law is then fixed by the remaining indicator's marginal. This covers both stored-mask and fresh gate implementations.

Theorem A.17.1 (The minimum over all declared-clock validation orders). *For every $(\gamma, \kappa) \in \mathcal{R}$ with $\kappa > 0$, the least whole-schedule peak attaining $U_2(b, Q_{\gamma, \kappa})$, minimized over all twenty declared-clock interleavings, is twelve. The only minimizing orders are the two serial orders*

$$XXXYYY, \quad YYYXXX.$$

Every nonserial declared order requires peak at least fourteen. The lower bounds permit the full stochastic gate and event interface, not merely an all-on deterministic compiler. These conclusions apply to $Q_{d, \sigma}$ for every $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$.

Proof. Every raw validation prefix has positive probability at either supporting candidate and at the target. Thus all cut witnesses may be compared on a common full future alphabet. Generate all-on words to simulate training gates as in Theorem A.14.1. The post-training response is an unrestricted optimizer on the supporting subfamily, hence has the contact-forced rows of Theorem A.15.1. Stopping before the two training preparations cannot occur with positive supporting probability: for each retained first-word branch, two possible unused second words force different strict vertices. A positive stopped component cannot belong to both vertices. The witnesses below therefore concern actual post-training validation cuts.

At a cut with two slots consumed from one tape and none from the other, the eight deterministic vertices and four disjoint additional faces from Theorem A.14.3 give twelve. That argument required only the four positive target diagonal words. It is still valid with full target support. Exchanging the two raw tapes negates the output and gives exactly the same channel-polytope certificate at the reversed cut. Thus either serial order needs twelve.

We give explicit certificates at the two possible nonserial entrance cuts. After one slot of each tape, write $x = X_1$, $y = Y_1$, and use suffix coordinates $(u, w, v, z) = (X_2, X_w, Y_2, Y_w)$. Define

i	$g_i^0(u, w)$	$g_i^1(u, w)$
1	$(1 - u)(1 - w)$	u
2	$1 - u$	u
3	$1 - u$	uw

and let $d_i^{xy} = g_i^y(v, z) - g_i^x(u, w)$. The twelve expressions with $i = 1, 2, 3$ and $x, y \in \{0, 1\}$ give ten distinct deterministic vertices. The only repetitions are $d_2^{11} = d_1^{11}$ and $d_3^{00} = d_2^{00}$, as is seen by substituting $u, w, v, z \in \{0, 1\}$ in the displayed functions.

There are four more required channels. Use count zero at prefixes $(x, y) = (0, 1), (1, 0)$ and count four at these same two prefixes. Their endpoints are respectively the continuations of (e_0, e_1) and (e_4, e_3) , with nontrivial weights $1 - t, t$. Each excludes simultaneous occurrence of the indifferent atom. The minimal face of each such channel consists of distributions supported, at each suffix, on the two endpoint output values. Among the ten mandatory vertices it contains only $d_1^{01}, d_1^{10}, d_3^{01}, d_3^{10}$, respectively. These four faces are pairwise disjoint. In their listed order, suffixes separating their six pairs are

pair	12	13	14	23	24	34
(u, w, v, z)	0110	0010	0000	0000	0001	0000

At each listed suffix the two sets of permitted output integers are disjoint; this follows directly from the displayed g_i^x and the masks e_0, e_4 . None of the required mixed rows is the sole mandatory vertex in its face, because $0 < t < 1$. Face packing gives $10 + 4 = 14$.

Suppose instead two slots of X and one of Y have been consumed. A deterministic continuation is indexed by (i, X_1X_2, Y_1) . The following twelve choices are distinct and include every strict-count continuation:

i	(X_1X_2, Y_1)
1	$(00, 0), (00, 1), (01, 0), (01, 1), (11, 0), (11, 1)$
2	$(00, 0), (01, 0)$
3	$(00, 1), (01, 1), (11, 0), (11, 1)$

They are verified by the formula $e_i(Y_1, Y_2, Y_w) - e_i(X_1, X_2, X_w)$ on the eight suffixes (X_w, Y_2, Y_w) . At training count zero take additionally prefixes $(01, 0)$ and $(11, 0)$. The corresponding two faces have endpoints $(0, (1 - Y_2)(1 - Y_w))$ and $(-1, -1 + (1 - Y_2)(1 - Y_w))$, respectively. Each contains just its listed $i = 1$ vertex, and they are disjoint at any suffix with $Y_2 = 1$. Both mixtures are nontrivial. Thus this cut also requires fourteen. Swapping X, Y proves the same result at the cut with one and two slots consumed.

Any nonserial interleaving either reads one slot from each tape first, or starts with two slots of one tape and then switches before its mark. It therefore passes one of the preceding cuts and needs fourteen. Every serial order has an upper realization of peak twelve: select the same five-valued frozen event and use its future residual functions. For full-support raw words the validation widths in either serial order are

$$(5, 10, 12, 10, 10, 7, 3).$$

These counts follow by restriction of the five masks, exactly as in Lemma A.11.4; for the reverse order negate all output residuals and exchange the tapes. Training and the charged stochastic selection cut have width at most five. Therefore both serial orders attain twelve, proving all assertions by Theorem A.8.2. The physical specialization follows from Theorem A.16.1. $\square$

This theorem optimizes the externally declared schedule; it does not optimize a report-dependent scheduler that changes which tape is read next. Such a scheduler has a different action interface, and its protocol state cannot be supplied for free. The phrase “all orders” in Theorem A.17.1 refers to all twenty declared interleavings, with this quantifier fixed explicitly.

Theorem A.17.2 (A singular support transition in reverse serial memory). *For every $\gamma \in [6/25, 13/50]$, the exact minimum peak in the reverse serial order $YYYXXX$ for the ideal target $Q_{\gamma,0}$ is ten. For every $0 < \kappa \leq 1/1000$ it is twelve. In particular the reverse-serial exact memory of the physical collision experiment jumps from ten at its zero-noise target to twelve at every positive $\sigma \leq 1/24$, although its target laws converge in total variation as $\sigma \downarrow 0$.*

Proof. For the upper bound at the ideal target, ignore its first report bit; its second bit equals the first almost surely, so this does not remove any needed information. After its second bit and mark, retain the residual $c - e_i(X)$, where $c = e_i(Y)$. Event e_2 has $c = 1$ surely; each other event permits two values of c . The compatible reverse validation profile is

$$(5, 5, 9, 9, 10, 7, 3).$$

After the second target bit the nine functions are, with the unread mark denoted by z ,

$$-e_0, 1 - e_0, 1 - z - e_1, 1 - e_1, 1 - e_2, 1 - e_3, z - e_3, 1 - e_4, -e_4.$$

Restricting after the target mark gives nine functions $c - e_i$. After the first candidate bit they give ten distinct functions of (X_2, X_w) ; after its second bit they give the seven scalar mark functions in Lemma A.11.4. Equality of residual functions is preserved by restriction, so these counts describe one compatible machine, not independent cutwise colorings. Training remains of peak five and the terminal score is correct almost surely for every candidate. On target words of zero probability, choose arbitrary legal transitions.

For the converse examine the cut after all three target slots and the first candidate bit. Write suffixes (X_2, X_w) in the order 00, 01, 10, 11. The three strict training events force the following eight distinct deterministic output channels:

$$\begin{aligned} (0, 1, 1, 1), & \quad (-1, 0, 0, 0), \quad (1, 1, 0, 0), \quad (0, 0, -1, -1), \\ (0, 0, 1, 1), & \quad (-1, -1, 0, 0), \quad (1, 1, 1, 0), \quad (0, 0, 0, -1). \end{aligned}$$

At training count zero take $X_1 = 0$ and target words 001 and 110. They force the channels with output means

$$(-t, 0, 0, 0), \quad (1 - t, 1, 1, 1).$$

Every integer coordinate is a point mass, and the remaining coordinate is the indicated mixture of neighboring output integers, because the target event indicator is fixed on these words. Their faces each contain exactly one of the eight mandatory vertices. They are disjoint at suffix 01, which requires output zero in the first and one in the second. Thus two additional generators are required, giving ten. All these prefixes have positive supporting probability. The full-support value twelve follows from the preceding theorem. Formula (84) gives the physical assertion and $\mu_1, \mu_2 \rightarrow 0$ by the uniform Gaussian bound. $\square$

The discontinuity concerns exact attainment. It neither contradicts continuity of the optimal score nor prevents a lower-memory device from approximating that score. It also explains why a

support-zero compiler cannot be transferred to the noisy physical target by simply keeping its state count unchanged.

APPENDIX A.18. PREPARATION LIMITS AND LOCALIZATION OF LEAST-FAVORABLE PRIORS

The contact structure has a second consequence, now for an arbitrary number of training preparations. We separate a general localization principle from the special geometry of the marked product family. An exact finite- N two-point prior is not assumed.

Theorem A.18.1 (Nearest-experiment limit and prior localization). *Let P_θ be a continuous family on a finite alphabet of size d , with compact metric parameter space, and let Q be fixed. Put*

$$d_Q(\theta) = \text{TV}(Q, P_\theta), \quad d_* = \min_{\theta} d_Q(\theta), \quad C_* = \{\theta : d_Q(\theta) = d_*\}.$$

For $N \geq 1$, the unrestricted frozen-validation values satisfy

$$(86) \quad d_* - \sqrt{(d-1)/N} \leq U_N(P, Q) \leq d_*, \quad U_N \uparrow d_*.$$

Every least-favorable prior μ_N in Theorem A.4.1 satisfies

$$(87) \quad \int (d_Q - d_*) d\mu_N \leq \sqrt{(d-1)/N}.$$

Consequently every weak limit of such priors is supported on C_ . For every $a > 0$ their mass on $\{d_Q - d_* \geq a\}$ is at most $\sqrt{(d-1)/N}/a$. More generally, an upper bound b_N on twice the expected total-variation error of a common empirical reconstruction replaces $\sqrt{(d-1)/N}$ in both conclusions.*

Proof. Let $\hat{P}$ be the empirical law of the N reports, and choose the indicator $h_{\hat{P}}$ maximizing $(Q - \hat{P})h$. If h_θ maximizes $(Q - P_\theta)h$, the two event inequalities give

$$(Q - P_\theta)h_{\hat{P}} \geq d_Q(\theta) - 2 \text{TV}(\hat{P}, P_\theta).$$

This inequality holds for the same parameter-independent response rule at every parameter. For any law P on d atoms,

$$\mathbb{E} \text{TV}(\hat{P}, P)^2 \leq \frac{d}{4N} \left(1 - \sum_x P(x)^2 \right) \leq \frac{d-1}{4N}.$$

Thus its expected score is at least $d_Q(\theta) - \sqrt{(d-1)/N}$. Every response has score at most $d_Q(\theta)$ at that parameter, which proves the upper bound at a minimizing parameter. Ignoring the last observation proves monotonicity and hence the limit.

Against a minimizing prior, the Bayes optimum is U_N . It is at least the prior average of the same empirical rule. Therefore

$$U_N \geq \int d_Q d\mu_N - \sqrt{(d-1)/N}.$$

Combine this with $U_N \leq d_*$ to get (87). Markov's inequality proves the mass estimate. The function $d_Q - d_*$ is nonnegative and continuous on the compact parameter space, so its integral under any weak limit is zero. Such a limit is supported on its zero set. The proof uses only the common reconstruction bound, which proves the last assertion. $\square$

The empirical estimate in (86) is the same finite-alphabet mechanism underlying the retained decision-spectrum classification. Its role here is different: it controls every least-favorable prior, not merely a constructive selector. In particular, finite-preparation saddle supports cannot remain far from the closest candidate experiments as the preparation number grows.

Theorem A.18.2 (Intrinsic marked contact geometry). *For every $\gamma \in I$, set $s_\gamma = \sqrt{(1+\gamma)/2}$ and*

$$d_\gamma = 2s_\gamma - 1 - \gamma/2, \quad C_\gamma = \{(s_\gamma, s_\gamma), (1 - s_\gamma, 1 - s_\gamma)\}.$$

Then

$$(88) \quad \min_{p,q} \text{TV}(Q_\gamma, b(p, q)) = d_\gamma, \quad \operatorname{argmin}_{p,q} \text{TV}(Q_\gamma, b(p, q)) = C_\gamma.$$

Moreover, for Euclidean distance in $[0, 1]^2$,

$$(89) \quad \text{TV}(Q_\gamma, b(p, q)) - d_\gamma \geq \frac{7}{2300} \operatorname{dist}((p, q), C_\gamma)^2.$$

The constant is uniform on I and is derived from the experiment's atom probabilities, not from a metric-entropy or small-ball assumption.

Proof. Put $a = (1-p)(1-q)$, $c = pq$, $l = (1-\gamma)/2$ and $h = (1+\gamma)/2$. The common mass of target and candidate is $\psi(a) + \psi(c)$, where

$$\psi(z) = \begin{cases} z, & 0 \leq z \leq l, \\ z/2 + (1-\gamma)/4, & l \leq z \leq h, \\ 1/2, & h \leq z \leq 1. \end{cases}$$

This follows by summing the smaller mass on each of the two marked atoms corresponding to the report pair. Reflection $(p, q) \mapsto (1-p, 1-q)$ exchanges a, c ; assume $c \geq a$. If $c < 1/4$, both terms are in the first segment of ψ , and their sum is at most $1/2$.

For $c \geq 1/4$, put $x = \sqrt{c} \geq 1/2$. By the arithmetic-geometric mean inequality,

$$a \leq (1-x)^2 \leq 1/4 < l.$$

Consequently the common mass is at most $F(x) = (1-x)^2 + \psi(x^2)$, with equality only when $p = q = x$. On the three consecutive intervals with endpoints $1/2, \sqrt{l}, \sqrt{h}, 1$, this function is respectively

$$(1-x)^2 + x^2, \quad (1-x)^2 + x^2/2 + (1-\gamma)/4, \quad (1-x)^2 + 1/2.$$

The first two are strictly convex and the last strictly decreases. The first increases away from $1/2$. Thus a maximum must occur at $\sqrt{l}$ or $\sqrt{h}$. Their difference is

$$(90) \quad F(\sqrt{h}) - F(\sqrt{l}) = \gamma \left(\frac{3}{2} - \frac{2}{\sqrt{h} + \sqrt{l}} \right) > \frac{7}{575}.$$

Indeed $\sqrt{h} > 39/50$, $\sqrt{l} > 3/5$, and $\gamma \geq 6/25$, so the right side is greater than $(6/25)(3/2 - 100/69) = 7/575$. The unique maximizing diagonal point on this side is $x = \sqrt{h}$. Its common mass is $(1 - \sqrt{h})^2 + 1/2 > 1/2$, also excluding $c < 1/4$. Subtracting it from one gives d_γ . Reflection proves (88) and the assertion about all minimizers.

We record a uniform quantitative form. Write $s = \sqrt{h}$ and $\Delta = 7/575$. For $1/2 \leq x \leq \sqrt{l}$, the common mass is at most $F(\sqrt{l})$, so its deficit from $F(s)$ is at least Δ . For $\sqrt{l} \leq x \leq s$, convexity puts F below the chord between its endpoint values. Since this interval has length at most one, its deficit is at least $\Delta(s-x)$. For $s \leq x \leq 1$ the deficit is

$$(1-s)^2 - (1-x)^2 = (x-s)(2-s-x) \geq (1-s)(x-s).$$

Since $s < 4/5$, this too is at least $\Delta|x-s|$. Together the three intervals give $F(s) - F(x) \geq \Delta|x-s|$.

Let $z = (p-q)^2$. At fixed $c = x^2$ the additional loss from the nondiagonal candidate is exactly

$$(1-x)^2 - a = p + q - 2x = \frac{z}{p+q+2x} \geq z/4.$$

The geometric mean x lies between p and q , and hence

$$(p - s)^2 + (q - s)^2 \leq 4z + 4(x - s)^2 \leq 4z + 4|x - s|.$$

It follows that the total deficit is at least $(\Delta/4)((p - s)^2 + (q - s)^2)$, proving (89) in this case. If $c < 1/4$, the total variation is at least $1/2$, whereas $d_\gamma < 12/25$ from $s < 4/5$ and $\gamma \geq 6/25$. The excess is therefore greater than $1/50$; Euclidean squared distance in the square is at most two, and $1/50 > 2(7/2300)$. This proves the remaining case. Reflection completes the proof. $\square$

Corollary A.18.3 (An all-preparation law and a limiting two-point prior). *For all $N \geq 1$ and $\gamma \in I$,*

$$(91) \quad d_\gamma - \sqrt{3/N} \leq U_N(b, Q_\gamma) \leq d_\gamma, \quad U_N(b, Q_\gamma) \uparrow d_\gamma.$$

Every least-favorable prior $\mu_{N,\gamma}$ satisfies

$$(92) \quad \int \text{dist}(\theta, C_\gamma)^2 d\mu_{N,\gamma}(\theta) \leq \frac{2300}{7} \sqrt{3/N}.$$

A least-favorable prior can be chosen invariant under simultaneous complementation of the two parameters. Every such symmetric choice satisfies, for the Euclidean quadratic transport distance,

$$(93) \quad W_2 \left(\mu_{N,\gamma}, \frac{1}{2} \delta_{(s_\gamma, s_\gamma)} + \frac{1}{2} \delta_{(1-s_\gamma, 1-s_\gamma)} \right) \leq \sqrt{2300/7} (3/N)^{1/4}.$$

In particular, the limiting symmetric least-favorable law is unique.

Proof. Estimate the four-atom report-pair law from training and tensor it with the known fair mark. Tensoring with a common probability preserves total variation. The proof of Theorem A.18.1 therefore uses $d = 4$ rather than eight, proving (91) and an excess bound $\sqrt{3/N}$. Combining the latter with (89) gives (92).

Simultaneously complement both report bits and the mark in every training and validation word. This leaves Q_γ invariant and sends $b(p, q)$ to $b(1 - p, 1 - q)$. The Bayes objective in (25) is invariant under this transformation and convex in the prior. Averaging a minimizing prior with its reflected image is therefore again minimizing and is symmetric.

Map each parameter to a closest point of C_γ , choosing the two points with equal probabilities at a tie. This is a Borel kernel, equivariant under reflection. A symmetric prior is pushed forward to the equal two-point law in (93); the induced coupling has squared transport cost exactly the left side of (92). Taking square roots proves the assertion. $\square$

These conclusions identify the endpoint and limiting contact geometry for every preparation number. They do not assert that the exact prior has two atoms at every finite N , or that the displayed convergence rates and constants are sharp. The finite- $N = 1, 2$ saddles are preserved as exact results. The new theorem explains their relationship to an intrinsically determined limiting contact set without extrapolating their particular event tables to unproved values of N .

APPENDIX A.19. AN EXACT PREPARATION–DECISION–MEMORY LAW

A second solvable family separates a genuine joint resource law from an implementation bound. Here the memory coordinate is the number of states at the training–validation cut. It is not identified with the bit-level peak width of the collision audit.

Fix $m = 2^d$, $0 < \alpha < 1$, and an unknown parameter $\theta \in \{1, \dots, m\}$. A fresh training preparation reports θ with probability α and an erasure $\perp$ otherwise. Conditional on θ , the preparations are independent. Candidate validation has law δ_θ and target validation is uniform on $\{1, \dots, m\}$. At the decision cut the auditor retains at most $K = 2^k$ states, $0 \leq k \leq d$. Its subsequent response may depend only on that state, current validation observations and fresh

independent randomness. In particular, a calibration table, remembered seed or stopping time outside the retained state is not available. Upstream and downstream memory are otherwise unrestricted for the converse. Let $V_N^{\text{er}}(K)$ be the best worst-case expected signed validation score with at most N training preparations.

Theorem A.19.1 (Exact joint law). *For every $N \geq 0$ and $K = 2^k \leq m$,*

$$(94) \quad V_N^{\text{er}}(K) = (1 - (1 - \alpha)^N) \left(1 - \frac{1}{K}\right).$$

The converse covers stochastic encoders and arbitrary legal stopping. The value is attained by a clocked machine with K training states. With atomic preparation and validation outcomes, the complete audit has peak at most $\max\{2K, 3\}$ and uses at most $N + 2$ preparations.

Proof. Write $q = (1 - \alpha)^N$ and pre-generate N potential training reports. A stopped auditor ignores the unused reports; this does not disclose extra information to it. On the event that all N reports are erasures, its retained-state law $r = (r_j)_{j=1}^K$ is independent of θ . After padding any smaller register with unused states, its channel to the decision register has the form

$$(95) \quad C(j \mid \theta) = qr_j + (1 - q)S(j \mid \theta)$$

for a stochastic channel S . This identity remains valid with randomized stopping and randomized state updates.

Use the uniform prior on parameters as an upper certificate. The part qr_j has zero expected score for every response: averaging δ_θ under this prior gives the target itself. For the other part, optimizing over all responses gives a convex function of S , a sum of positive parts of linear functions. Every stochastic channel is a convex combination of deterministic channels $j = g(\theta)$; therefore its value is at most the largest deterministic-channel value. This convexification is used only to bound a channel. It does not presuppose that a mixture of width- K machines is realizable at width K .

If the fibers of g have sizes $n_1, \dots, n_K$, a nonempty fiber has posterior uniform on that fiber, so its total variation from the uniform target is $1 - n_j/m$. The prior-averaged optimal score is

$$\sum_j \frac{n_j}{m} \left(1 - \frac{n_j}{m}\right) = 1 - \sum_j (n_j/m)^2 \leq 1 - 1/K.$$

Empty fibers contribute zero. Cauchy–Schwarz proves the inequality. This is an upper bound on the average score and hence on its minimum over fixed parameters. Multiplying by $1 - q$ proves the upper bound in (94).

For attainment, partition the parameter alphabet into K equal groups and let $g(\theta)$ denote its group. Initialize the register uniformly on the K group labels. On an erasure leave it unchanged; on a revealed parameter overwrite it by $g(\theta)$. No extra “revelation occurred” flag is needed. At the decision cut its law is $q \text{Unif}[K] + (1 - q)\delta_{g(\theta)}$. For register j , freeze the event which is the complement of group j . On a revelation its score is $1 - 1/K$. Under a uniformly random initial label, the mean score is zero for each fixed θ : one of the K labels has score $1 - 1/K$ and the others have score $-1/K$. This proves the exact worst-case value.

The k initialization bits can be accumulated in the K -state register; all earlier prefix counts are at most K . Thus no persistent seed lies outside it. Candidate validation retains the group label and its event indicator, using at most $2K$ states. Target validation produces one of the three values $-1, 0, 1$. This proves the atomic-outcome peak bound. The controller uses a declared external schedule; an autonomous stopping implementation must additionally retain its phase. A phase-tagged implementation is finite, but its cost is not included in K . $\square$

Corollary A.19.2 (Thresholds and matching resource scales). *For $K \geq 2$ and $0 \leq c < 1 - 1/K$, the least integer number of training preparations giving a score strictly greater than c is*

$$(96) \quad N_c^{\text{er}}(K) = \left\lfloor \frac{\log(1 - c/(1 - 1/K))}{\log(1 - \alpha)} \right\rfloor + 1.$$

If $c \geq 1 - 1/K$, no finite N suffices. For $0 < \varepsilon < 1$, achieving $V_N^{\text{er}}(K) \geq 1 - \varepsilon$ requires

$$K \geq \varepsilon^{-1}, \quad N \geq \frac{\log(1/\varepsilon)}{-\log(1 - \alpha)}.$$

Conversely, whenever an allowed $K \leq m$ satisfies $K \geq 2/\varepsilon$, the choice $N \geq \log(2/\varepsilon)/[-\log(1 - \alpha)]$ suffices.

Proof. The strict threshold is equivalent to $(1 - \alpha)^N < 1 - c/(1 - 1/K)$. Since the denominator logarithm is negative, the least integer satisfying this strict inequality is (96), including equality cases of the real quotient. Also

$$1 - V_N^{\text{er}}(K) = q + (1 - q)/K.$$

This is at least q and at least $1/K$, and is at most $q + 1/K$. These three inequalities prove the necessary and sufficient bounds. $\square$

The exact law concerns an entire (N, K, α, m) family, not a single finite diagnostic. Its mechanism is a parameter-independent erasure component followed by the convex geometry of the decision channel. It demonstrates how a preparation obstruction and a memory obstruction can be matched in one theorem. It does not identify the least physical peak width at two collision preparations. Encoding the m -symbol reports bit by bit also introduces cuts not present in this atomic model; the physical bit-level accounting is given separately.

APPENDIX A.20. INTEGER MEMORY AND THE FAILURE OF A NAIVE PARTITION FORMULA

The divisibility assumptions in Theorem A.19.1 can be removed, but a balanced-partition Bayes bound must not be mistaken for a minimax equality when a persistent random partition is not free. This distinction leads to another exact finite-resource transition.

Keep the revelation interface, but allow any integers $m \geq 2$ and $1 \leq K \leq m$. Put $\rho = 1 - (1 - \alpha)^N$, and denote the genuine K -label minimax value by $V_{N,m}^{\text{er}}(K)$. Write $m = Kq + s$, $0 \leq s < K$, and set

$$(97) \quad B_{m,K} = 1 - \frac{(K - s)q^2 + s(q + 1)^2}{m^2}.$$

Here and below all-erasure initialization randomness is a state initialization, not an unrecorded seed used again at validation.

Proposition A.20.1 (Arbitrary integer alphabet bounds and low-reveal equality). *The optimal uniform-prior Bayes score is $\rho B_{m,K}$. The minimax value satisfies*

$$(98) \quad V_{N,m}^{\text{er}}(K) \leq \min \left\{ \rho B_{m,K}, 1 - \frac{\lceil m/K \rceil}{m} \right\}.$$

If $\rho \leq m/(m + s)$, equality with the first term holds. At perfect revelation $\rho = 1$ the exact value is the second term. In particular, when $s > 0$ the uniform-prior Bayes optimum at perfect revelation is strictly larger than the minimax value.

Proof. The all-erasure decomposition (95) holds without any divisibility condition. Its parameter-independent part has zero uniform-prior score. Optimizing the other part over responses gives a convex function of its stochastic channel. A deterministic channel with fiber sizes n_j has Bayes score $1 - \sum_j (n_j/m)^2$. If two nonempty fiber sizes differ by at least two, moving one index from the larger to the smaller strictly decreases the sum of squares. Allowing empty fibers and repeating gives sizes q and $q + 1$, proving the Bayes bound. It is attained by this partition and its complementary events, with any parameter-independent initialization. This proves the first upper bound as well as the asserted Bayes optimum.

For perfect revelation, fix any K response vectors. For each known parameter choose a response maximizing its score; this deterministic assignment cannot decrease any parameter's payoff, even when the original encoder was randomized. Some assigned fiber has size at least $\lceil m/K \rceil$. Averaging its parameters, the score of their common response is at most the total variation between the uniform target and the uniform law on that fiber, namely $1 - |\text{fiber}|/m$. The minimum is no larger. Balanced fibers with their complementary events attain the resulting upper bound. A perfectly informed encoder can simulate any less informative encoder, so this is an upper bound at every ρ .

For the low-reveal equality let balanced fibers have weights $w_j = n_j/m$. When $\rho < 1$, initialize their label register with probabilities

$$(99) \quad r_j = \frac{\rho w_j + (1 - 2\rho)/K}{1 - \rho}.$$

These sum to one, and are nonnegative precisely throughout the stated range: the smallest numerator is $\rho q/m + (1 - 2\rho)/K$. On a revealed parameter overwrite the label by its fiber; on erasure do nothing. No revelation flag is retained. For a parameter in fiber i , the complementary-event score is

$$\begin{aligned} g_i &= \rho(1 - w_i) + (1 - \rho) \left(r_i - \sum_j r_j w_j \right) \\ &= \rho \left(1 - \sum_j w_j^2 \right) = \rho B_{m,K}. \end{aligned}$$

This matches the upper bound uniformly in the fixed parameter. If $\rho = 1$ and $s = 0$, the deterministic revealed label already attains that value, so the singular denominator in (99) is not used. Finally, with unequal balanced fibers the weighted average of their sizes is strictly smaller than the largest size; equivalently $B_{m,K} > 1 - \lceil m/K \rceil/m$. This proves the last assertion. $\square$

Theorem A.20.2 (The complete two-label revelation law). *For every $m \geq 2$, every $N \geq 0$, and $0 < \alpha < 1$, set $w = \lfloor m/2 \rfloor/m$ and $b = \lceil m/2 \rceil/m$. Then*

$$(100) \quad V_{N,m}^{\text{er}}(2) = \min\{2wb\rho, w\}, \quad \rho = 1 - (1 - \alpha)^N.$$

The same formula holds at the limiting endpoints $\rho = 0, 1$. For odd m it changes regime at $\rho = m/(m + 1)$; for even m it reduces to $\rho/2$. The attaining machine uses two training labels, a frozen two-event alphabet, and atomic peak at most four.

Proof. Proposition A.20.1 gives the upper bound and its attainment for $\rho \leq 1/(2b)$. Above this threshold, partition into fibers of weights w, b and initialize deterministically in the larger fiber's label. Continue to overwrite on a reveal and leave the state unchanged on an erasure. A parameter in the larger fiber has score w ; a parameter in the smaller fiber has score $(2\rho - 1)b$. The latter is at least w exactly when $\rho \geq 1/(2b)$. This attains the second upper bound. For even m the threshold is one, so the low-reveal construction suffices. The endpoint $\rho = 0$ may instead

freeze a constant event. Candidate validation retains the selected label and its indicator, with at most four states; target validation produces the difference. Initialization is atomic randomization when its probabilities are not dyadic. $\square$

For example, $m = 3, K = 2$ gives $V_{N,3}^{\text{er}}(2) = \min\{4\rho/9, 1/3\}$, rather than $4\rho/9$ for all ρ . The balanced-integer expression is therefore not the unqualified replacement of the divisible minimax formula. For $K \geq 3$ outside the low-reveal range, (98) remains a bound rather than an asserted equality. The compatible dynamic formulation of Section A.6 retains the exact optimization in that regime. These results strengthen the revelation example without identifying its decision-cut coordinate with the collision model's bit-level peak.

APPENDIX A.21. QUANTITATIVE CERTIFICATES FOR A FINITE COMPATIBLE ARCHITECTURE

We return to an arbitrary finite architecture, now with a finite parameter set. It can have hidden states, controlled experiments, more than two hypotheses and registers of any size. Enumerate its distinct stochastic rows by $r = 1, \dots, R$ and write $q_r \in \Delta_{s_r}$. A reused autonomous row appears only once in this product. Assume each execution uses at most H controller rows, and uses row r at most d_r times. Delete unused rows and put $S = \max_r s_r$. Fixed environmental kernels are not controller rows.

For an integer $m \geq 1$, intersect Δ_{s_r} with the coordinate boxes of side $1/m$. The resulting nonempty closed polytopes cover the simplex, with overlaps permitted at their boundaries. Write C for a product of such row cells and choose a vertex q_r^C in each factor. A row-cell vertex has at most one coordinate not fixed at a box boundary; consequently it has at most $v_r = s_r 2^{s_r-1}$ vertices, all on the $1/m$ grid. There are at most $m^{\sum_r s_r}$ product cells.

Write the vertices of row cell C_r as $v_{r,j}$ and parametrize $q_r = \sum_j \lambda_{r,j} v_{r,j}$ with λ_r in a simplex. A reward polynomial has degree at most d_r in block λ_r . Use the homogeneous simplex Bernstein basis of degree d_r in that block, obtaining

$$(101) \quad g_\theta(q(\lambda)) = \sum_\alpha b_{\theta,C,\alpha} B_\alpha(\lambda), \quad B_\alpha \geq 0, \quad \sum_\alpha B_\alpha = 1.$$

Degree elevation includes the factors $\sum_j \lambda_{r,j} = 1$. No independence between different visits of a shared row is imposed on its unknown row vector.

Lemma A.21.1 (Causal interpretation of the coefficients). *Every coefficient in (101) satisfies*

$$|b_{\theta,C,\alpha} - g_\theta(q^C)| \leq HS/m.$$

Proof. For each row r , independently prepare an ordered list of d_r vertex labels, sampled independently with law λ_r . At its k th visit, use the stochastic row given by the k th vertex label in that list. Before conditioning, the next label is independent of the previous execution, so this experiment has exactly the row q_r at every visit and expected reward $g_\theta(q(\lambda))$. Unused labels may be sampled after termination without altering the reward.

Condition on the counts of every vertex label in every list. The probability of those counts is $B_\alpha(\lambda)$. Conditional on the counts, each list is a uniformly ordered multiset. Its expected reward is independent of λ and is therefore the corresponding Bernstein coefficient. This also proves (101) directly.

Each row used by this conditioned experiment is a vertex of C_r or a mixture of its remaining vertices. Every two vectors in C_r have coordinate differences at most $1/m$, hence total variation at most $s_r/(2m) \leq S/(2m)$. Couple the conditioned experiment to the one using q^C , using the same environmental randomness while their executions agree. At each actual controller update the probability of a first row-level disagreement is at most $S/(2m)$. There are at most H such updates on the agreeing path. The terminal laws consequently differ in total variation by at

most $HS/(2m)$. A reward in $[-1, 1]$ has expectation difference at most twice that quantity. This proves the estimate. The conditioned urn is used to analyze a coefficient; it is not declared to be an admissible free-memory controller. $\square$

Theorem A.21.2 (Explicit subdivision certificate). *Define*

$$\ell_C = \min_{\theta} g_{\theta}(q^C), \quad u_C = \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha}, \quad L_m = \max_C \ell_C, \quad U_m = \max_C u_C.$$

For the exact all-stochastic value of the specified architecture,

$$(102) \quad L_m \leq V(A) \leq U_m \leq L_m + HS/m.$$

On each cell, a witnessing parameter θ_C supplies the nonnegative coefficient certificate

$$u_C - g_{\theta_C}(q(\lambda)) = \sum_{\alpha} (u_C - b_{\theta_C, C, \alpha}) B_{\alpha}(\lambda).$$

Its degree in row block r is d_r , and its total degree is at most $\sum_r d_r$. At most $\prod_r \binom{d_r + v_r - 1}{v_r - 1}$ coefficients per parameter and cell are needed. If the primitive data are rational, all coefficients and bounds can be checked by rational arithmetic. Taking $m \geq HS/\varepsilon$ gives a certified interval of length at most ε .

Proof. The point q^C is feasible, so ℓ_C is a lower bound on the value restricted to C . For any $q \in C$, choose a convex representation of its rows. Equation (101) implies

$$\min_{\theta} g_{\theta}(q) \leq \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha} = u_C.$$

Choose a parameter attaining the minimum defining ℓ_C and apply Lemma A.21.1; this gives $u_C \leq \ell_C + HS/m$. Taking the maximum over the finite cover proves (102). Choose θ_C attaining the definition of u_C ; all displayed coefficients of its certificate are nonnegative. The degree and count statements follow from the product Bernstein basis. Cell vertices are rational grid points, and a finite execution sums finite products, proving the rational statement. $\square$

Remark A.21.3. The number of cells and coefficients can be enormous. Neither polynomial running time nor a uniform degree bound for the original sum-of-squares hierarchy follows. Refinement changes the domain cover, not merely a single global SOS degree. For a finite collection of architectures obeying a specified budget, maxima of these intervals give outer intervals; an unbounded architecture class needs a separate uniform argument such as Theorem A.4.2. For compact parameter families, an effective net and a verified modulus for the primitive kernels are additional inputs, not consequences of compactness alone.

The resource estimate also makes row precision explicit. With $m = 2^b$, the lower witnesses have dyadic rows and lose at most $HS2^{-b}$ in value. Thus increasing row precision provides a quantified approximation for a fixed architecture, while its random-bit implementation still has to be charged. The physical theorem avoids this cost altogether because its rows are deterministic.

APPENDIX A.22. AUTONOMOUS NOISY TESTING WITH A PRECISION AND TIME LEDGER

This section supplies a substantive autonomous comparison rather than merely identifying shared rows. The stationary optimum is classical [6]; the proof and explicit finite-time implementation are recorded to make the resource conventions unambiguous. They are not used as novelty evidence for the physical theorem.

Let reports be independent, with law $\text{Ber}(p)$ under hypothesis $-$ and $\text{Ber}(q)$ under hypothesis $+$, where $0 < p < 1/2$, $q = 1 - p$, and $r = q/p$. A W -state autonomous machine has two time-independent stochastic transition matrices T_0, T_1 . Assume their induced chains $P_- = qT_0 + pT_1$ and $P_+ = pT_0 + qT_1$ are irreducible. A final decision uses only the state, with optional fresh terminal randomization.

Lemma A.22.1 (Likelihood-range bound). *If equivalent probability laws ν_+, ν_- on a finite set satisfy*

$$\frac{\max_i \nu_+(i)/\nu_-(i)}{\min_i \nu_+(i)/\nu_-(i)} \leq R,$$

then $\text{TV}(\nu_+, \nu_-) \leq (\sqrt{R} - 1)/(\sqrt{R} + 1)$.

Proof. Write $L = d\nu_+/d\nu_-$, $a = \min L$ and $b = \max L$. Since $\mathbb{E}_{\nu_-} L = 1$, one has $a \leq 1 \leq b$. The chord bound for $(L - 1)_+$ gives

$$\text{TV}(\nu_+, \nu_-) = \mathbb{E}_{\nu_-}(L - 1)_+ \leq \frac{(1 - a)(b - 1)}{b - a}.$$

For fixed b/a , maximizing the last expression gives $a = 1/\sqrt{b/a}$; substitution yields $(\sqrt{b/a} - 1)/(\sqrt{b/a} + 1)$. It is increasing in b/a , proving the assertion. The identical-law case follows directly. $\square$

Theorem A.22.2 (Autonomous stationary bound and an explicit approximation). *Put $D = W - 1$, $W \geq 2$. In the irreducible autonomous class the infimum stationary minimax error is*

$$(103) \quad e_W = \frac{1}{1 + r^D}.$$

For every $\epsilon = 2^{-b}$, $b \geq 0$, there is a machine whose rows have denominator dividing 2^{b+1} , and whose minimax error at report time n from any initial state is at most

$$(104) \quad e_W + \epsilon(W - 2) + \exp \left\{ - \left(\frac{p\epsilon}{2} \right)^D \left\lfloor \frac{n}{D} \right\rfloor \right\}.$$

The stationary lower bound is not a finite-horizon lower bound.

Proof. Every positive induced transition has its likelihood ratio in $[r^{-1}, r]$, and the supports of the two induced matrices are identical. The Markov chain tree formula writes each stationary weight as a sum of products over directed spanning trees rooted at that state. Each product has D transitions; its ratio lies in $[r^{-D}, r^D]$. The ratio of the two unnormalized stationary weights lies in the same interval. Normalizing multiplies all state likelihood ratios by a common number, so their maximum-to-minimum ratio is at most r^{2D} . Lemma A.22.1 therefore bounds stationary total variation by $(r^D - 1)/(r^D + 1)$. The equal-prior Bayes error is at least $1/(1 + r^D)$, and a minimax error cannot be smaller than that Bayes error. This is the equal-prior binary specialization of the Hellman–Cover bound.

For the upper approximation, take states $i = 0, \dots, D$, weights $c_0 = c_D = 1$, and $c_i = \epsilon$ at interior states. On report 1, move from i to $i + 1$ with probability $a_i = \frac{1}{2} \min(1, c_{i+1}/c_i)$, if that neighbor exists; otherwise stay. On report 0, move to $i - 1$ with probability $b_i = \frac{1}{2} \min(1, c_{i-1}/c_i)$, if it exists; otherwise stay. All other probability is assigned to staying. Detailed balance gives

$$\pi_+(i) = \frac{c_i r^i}{\sum_j c_j r^j}, \quad \pi_-(i) = \frac{c_i r^{-i}}{\sum_j c_j r^{-j}}.$$

Reflection $i \mapsto D - i$ exchanges these laws. Decide $+$ for $i > D/2$, $-$ for $i < D/2$, and use a fair terminal decision at $i = D/2$ when this state exists. The two errors agree. Conditional on being at an endpoint, the error is exactly e_W . The total interior stationary mass is at most $\epsilon(W - 2)$, since each $r^i \leq r^D$ and the denominator is at least r^D under $+$, with the reflected argument under $-$. This proves the first two terms of (104).

Every existing neighbor transition under either hypothesis has probability at least $p\epsilon/2$, and every self-loop has probability at least $1/2$. From any state, move down to zero and then stay until D steps have elapsed. This yields the common minorization $P_{\pm}^D(i, \{0\}) \geq \delta = (p\epsilon/2)^D$.

Subtracting that common atom gives contraction in total variation by $1 - \delta$ every D steps. Comparison with the stationary law yields $\text{TV}(P_{\pm}^n(i, \cdot), \pi_{\pm}) \leq (1 - \delta)^{\lfloor n/D \rfloor} \leq e^{-\delta \lfloor n/D \rfloor}$. A bounded decision changes error by at most this amount, proving (104). Finally, letting n tend to infinity and then ϵ to zero proves that the stationary infimum equals (103). All transition probabilities belong to $\{0, 1, 1/2, \epsilon/2, 1 - 1/2, 1 - \epsilon/2\}$. $\square$

The W states in this theorem refer to an atomic stochastic update after each report. A sequential fair-bit implementation must also retain the report, the current state, a bit counter and a flag. Each nontrivial move probability is $1/2$ or $2^{-(b+1)}$; test for a string of zero bits of the required length, and pad to $b + 1$ bits if a uniform report cycle is required. At most $4W(b + 2)$ states suffice for this implementation, including its bit-phase counter and a ready phase. This conservative bound is an implementation claim, not an optimal precision–state tradeoff. The time index n in (104) is an observation time at which an external analyst reads the autonomous machine, not a free input clock used by its transitions. The adversarial model of [10] is not covered by the independent-report proof above.

APPENDIX A.23. PHYSICAL CONSUMERS AND THE FIRST-PRINCIPLES CONNECTION

The physical preparation theorem needs no estimate of the target law as a stored input: the five events are fixed and their uniform validity uses a proved neighborhood of Q_* . This removes calibration memory from this particular consumer without asserting that target acquisition is unnecessary in an arbitrary comparison problem. The general acquired calibration theorem, with its full counter register, remains in the complete companion.

Corollary A.23.1 (A raw-interface confidence construction). *Let $m = m_2 - \beta > 0$ and $0 < c < m$. Repeat the full four-preparation audit independently R times and write $\overline{D}_R$ for the mean score. Under every private candidate,*

$$\mathbb{P}(\overline{D}_R \leq c) \leq \exp\{-R(m - c)^2/2\}.$$

Under the null in which the independent candidate and target validation laws agree, with the same event chosen before both validations,

$$\mathbb{P}(\overline{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

A clocked implementation uses $4R$ preparations and at most $12(2R + 1)$ persistent states, with no calibration and no random bits.

Proof. The independent block scores lie in $[-1, 1]$ and have mean at least m under a private candidate. Hoeffding’s inequality gives the first bound. Under the null, conditional on the selected event, the two independent validation indicators have the same mean. Each block therefore has mean zero, and the same inequality gives the second bound. Retain the integer sum in $\{-R, \dots, R\}$ along with the current twelve-state block. At a block boundary add its terminal score and reset its register. The deterministic external schedule is part of the clocked convention. $\square$

This is an upper construction on the raw marked interface. Its proof does not transfer a lower bound from a deliberately reduced Bernoulli interface, and no optimal confidence-memory frontier is claimed.

Proposition A.23.2 (Target transport without a training multiplier). *For the two-preparation audit, changing only the target from Q to Q' changes its expected score by at most $\text{TV}(Q, Q')$. Its candidate training law, register profile and event selector remain unchanged.*

Proof. Condition on the candidate training word. The expectation difference is $(Q - Q')h_S$, whose absolute value is at most $\text{TV}(Q, Q')$. Average over that unchanged training law. $\square$

In particular, every common-preparation transport estimate verified for the original positive-noise collision family can be inserted here without multiplying it by a long training horizon. If candidate kernels also change, their training and validation defects must be added by a causal coupling; the target-only statement does not remove those terms.

The first-principles route is now concrete. The positive experimental kernels and actual reset preparations give the likelihoods in (25); these generate posterior continuation states. The validation-test quotient gives the positive parts in the statistical dual. A prior then supplies an architecture-uniform impossibility certificate. On the constructive side, the five-event decision is compiled by taking its future residual functions, which are closed under observation shifts. These are compatible causal states and determine the charged profile. Thus the preparation law, predictive continuation and finite-resource realization are used in the same upper/lower theorem. This is not a deduction of geometry or nonlinear operator compactness from the existence of finite stochastic rows.

APPENDIX A.24. A SHARPENED PHYSICAL CONSUMER AND ITS SIMULATION BUDGET

The exact two-preparation theorem has a downstream use with explicit hypotheses, rather than a merely renamed dependency arrow. It sharpens the mean margin of a raw-interface confidence test, and the causal morphism theorem transfers that test, with its full memory and calibration costs, to another executable interface.

Theorem A.24.1 (Confidence and transported resource certificate). *Let Q be a physical marked target with $\text{TV}(Q, Q_*) \leq \beta$. Use the two-preparation event selector with a dyadic coin τ_b satisfying $|\tau_b - \tau| \leq 2^{-b}$, and put*

$$a_b = v_2 - \beta - \frac{5}{16}2^{-b}, \quad W_b = \max\{12, 6(b+1) + 5\}.$$

If $0 < c < a_b$, R independent four-preparation blocks, followed by the test $\overline{D}_R > c$, have error bounds

$$(105) \quad \sup_{\text{private}} \mathbb{P}(\overline{D}_R \leq c) \leq \exp\{-R(a_b - c)^2/2\},$$

$$(106) \quad \sup_{\text{equal}} \mathbb{P}(\overline{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

They use $4R$ preparations, at most bR fresh random bits, and at most $W_b(2R+1)$ persistent states under a clocked schedule. With an exact atomic coin, replace a_b, W_b by $v_2 - \beta, 12$.

Suppose a source interface simulates this entire repeated audit under Theorem A.2.4, uniformly over the private and equal-law families, with path defect Δ_R . Then the corresponding source error bounds are the right sides of (105)–(106) plus Δ_R , capped at one. At aligned cuts its state bound is $C_{\text{cal}}R_tM_tW_b(2R+1)$, and its preparation count is N_{cal} plus all source calls required to supply the $4R$ audit preparations. Calibration is acquired and retained, not supplied as a free real-valued table.

Proof. Corollary A.12.2 and Proposition A.12.3 give mean block score at least a_b under every fixed private parameter. Independent fresh preparations and independent coins give independent block scores in $[-1, 1]$. Hoeffding's inequality proves (105). Under the equal-law null, conditional on training and the frozen event, the two validation indicators have the same mean. Each block therefore has mean zero, proving (106) by the same inequality. Retain the integer sum in

$\{-R, \dots, R\}$ alongside the current block register. Updating the sum and resetting the block is a deterministic boundary operation. This proves the raw resource bounds.

Apply Theorem A.2.4 to the whole repeated experiment, including its stopping and validation fields. Its defect bound applies to each of the two terminal error events, whose indicators have range one; each probability therefore changes by at most Δ_R , not $2\Delta_R$. The same theorem gives the state product and preparation count. The source blocks need not be independent after a shared calibration stage: independence was used only for the reference audit, and path-law comparison transfers its final event probability. $\square$

The first part uses the exact saddle, the original microscopic bridge and the compiled residual machine. The second additionally needs the uniform acquired causal simulation theorem. A source which does not satisfy that theorem's stability and calibration hypotheses is not covered. Likewise this consumer supplies no reverse simulation and therefore does not transfer an optimal-width lower bound to an arbitrary source. It is an actual conditional theorem interface for subsequent models, not an assertion that the independent geometric A2 chain or the unbounded B4/C2 operator obligations have already been discharged.

APPENDIX A.25. DEPENDENCIES, RETAINED RESULTS AND REMAINING DISTINCTIONS

The exact three-report binary noisy frontier, arbitrary-horizon binary-register minimax recursion and compatible Bayesian interval recursion remain as proved in the complete companion. Its exact odd-horizon majority realization is also retained; it is not a new result of this revision. Section A.21 applies to registers of any finite width, controlled rows and any finite parameter set, but provides certified intervals rather than a closed minimax recursion for all such classes. The physical theorem supplies a matched preparation segment in a continuous two-parameter family with overlapping stochastic reports; its converse is not a finite parameter-grid check.

The stationary autonomous result in Section A.22 has an explicit formula, but its stationary lower bound must not be applied to a finite-horizon clocked machine. Its finite-time upper bound is purposely conservative and exposes the slowing caused by finite transition precision. An arbitrary compact-family presentation still requires additional effective information to construct a kernel net. A subdivided positive-basis degree bound must not be called a degree bound for a global SOS hierarchy.

The physical result is a genuine consumer of the experimental and continuation root. It does not make the independent A2 geometric proofs dependent on this root. The original B4 normalized-resolvent, range, compactness and control-transfer obligations, and the broad C2 strict-dual, form, rigidity and optional-projection obligations, are not proved by our finite-state theorem. Their statements, derivations and current status remain in the retained historical pipeline. No aggregate closure is inferred from preserving their source files.

Finally, the rational certificate (62), polynomial identity (64), and residual transitions are independently reproducible. The program in the accompanying source checks their exact identities and includes deliberately invalid controls. These checks are not substitutes for the proofs above. Neither compilation, preservation of earlier pages nor a successful finite test establishes novelty or an editorial decision.

REFERENCES

- [1] R. D. Smallwood and E. J. Sondik, The optimal control of partially observable Markov processes over a finite horizon, *Operations Research* **21** (1973), 1071–1088.
- [2] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272.
- [3] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176.

- [4] J. Müller and G. Montúfar, The geometry of memoryless stochastic policy optimization in infinite-horizon POMDPs, arXiv:2110.07409 (2021; revised versions).
- [5] M. Putinar, Positive polynomials on compact semi-algebraic sets, *Indiana Univ. Math. J.* **42** (1993), 969–984.
- [6] M. E. Hellman and T. M. Cover, Learning with finite memory, *Ann. Math. Statist.* **41** (1970), 765–782.
- [7] M. E. Hellman and T. M. Cover, On memory saved by randomization, *Ann. Math. Statist.* **42** (1971), 1075–1078.
- [8] R. A. Flower and M. E. Hellman, Hypothesis testing with finite memory in finite time, *IEEE Trans. Inform. Theory* **18** (1972), 429–431.
- [9] T. M. Cover, M. A. Freedman and M. E. Hellman, Optimal finite memory learning algorithms for the finite sample problem, *Information and Control* **30** (1976), 49–85.
- [10] M. A. Managoli and V. M. Prabhakaran, Memory constrained adversarial hypothesis testing, arXiv:2605.12063v1 (May 12, 2026).
- [11] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28, doi:10.1524/strm.2002.20.14.1.
- [12] Q. Qi, General Theta Foundations I: Compatible statistical experiments and resource frontiers, revision 23, September 24, 2026, complete mathematical companion, source commit `8a6043aca644043bd805668185eb227468f21170`; packaged commit `e926b783b643b3e7ff275a18105528927b76cc97b`. The 96-page article and 649-page development are reproduced without alteration after the present article in the two complete volumes.
- [13] H. Chernoff, Sequential design of experiments, *Ann. Math. Statist.* **30** (1959), 755–770.
- [14] M. Naghshvar and T. Javidi, Active sequential hypothesis testing, *Ann. Statist.* **41** (2013), 2703–2738, doi:10.1214/13-AOS1144.
- [15] S. Nitinawarat, G. K. Atia and V. V. Veeravalli, Controlled sensing for multihypothesis testing, *IEEE Trans. Automat. Control* **58** (2013), 2451–2464, arXiv:1205.0858v6.
- [16] R. Zheng, R. Sim and A. Varvitsiotis, Solving imperfect-recall games via sum-of-squares optimization, arXiv:2602.21722v1 (2026).
- [17] A. Heller, On stochastic processes derived from Markov chains, *Ann. Math. Statist.* **36** (1965), 1286–1291, doi:10.1214/aoms/1177700000.
- [18] M. Vidyasagar, The complete realization problem for hidden Markov models: a survey and some new results, *Math. Control Signals Systems* **23** (2011), 1–65, doi:10.1007/s00498-011-0066-7.
- [19] N. Gillis and F. Glineur, On the geometric interpretation of the nonnegative rank, *Linear Algebra Appl.* **437** (2012), 2685–2712, arXiv:1009.0880.
- [20] B. Reusch and W. Merzenich, Minimal coverings for incompletely specified sequential machines, *Acta Informatica* **22** (1986), 663–678, doi:10.1007/BF00263650.
- [21] F. Denis and Y. Esposito, Rational stochastic languages, arXiv:cs/0602093 (2006), version 1.
- [22] A. Ambainis, D. Leung, L. Mančinská and M. Ozols, Quantum random access codes with shared randomness, arXiv:0810.2937 (2008).
- [23] R. Kondo, Y. Sato, H. Yano, Y. Maeda, K. Ito and N. Yamamoto, Random access codes: explicit constructions, optimality, and classical–quantum gaps, arXiv:2604.21274v1 (April 23, 2026).
- [24] K. Liu, Classical codes violate the conjectured square-root bound for quantum random access codes, arXiv:2607.15617v1 (July 2026).

PRESERVED PREDECESSOR

General Theta Foundations I, revision 28

The following predecessor volume is reproduced without alteration.
The complete current article precedes this divider.

Frozen predecessor commit:

cdc749027c5f7a6f66c572ea498cd15e1b1a3391

GENERAL THETA FOUNDATIONS I: ROBUST SADDLES, CAUSAL MEMORY, AND VALIDATION ORDER

QIAN QI

ABSTRACT. We relate minimax contact geometry to the memory of causal statistical experiments. For a two-parameter family of full-support marked targets, we determine the exact two-preparation saddle and prove that five states are necessary and sufficient at the decision cut. The original positive-noise hard-sphere sensor is shown, by its preparation and reflection symmetries, to belong to this family throughout the explicit range $0 < \sigma \leq 1/24$. A facial scheduling criterion supplies converses uniform over stochastic encoders and over the declared validation order. Among all twenty clocked interleavings of the two three-slot validation tapes, the exact minimum peak is twelve; precisely the two serial orders attain it, and every nonserial order requires at least fourteen states. At zero noise, reverse serial validation instead has exact peak ten. Thus exact causal memory can jump at a support boundary while the statistical value varies continuously. We also retain the causal foundations, controlled minimax duality, exact ideal saddles, all-preparation localization, and resource-accounted transport results that enter this theorem chain.

CONTENTS

1. Introduction	2
1.1. The principal theorem chain	2
1.2. Foundations and retained results	3
1.3. Relation to earlier work	4
2. Causal experiments and their predictive quotients	4
2.1. The experimental object	4
2.2. Prepared predictive equivalence	5
3. Attainable resolution and resource-aware simulation	6
3.1. Checkpoint and online prediction	6
3.2. Causal morphisms and simultaneous error budgets	7
4. Experiments, architectures and attainable resources	9
5. A statistical dual that is uniform over architectures	10
6. Controlled experiments without a dominating action	12
6.1. An informative action choice	14
7. Controlled experiments at a prescribed memory profile	14
7.1. A parameterwise occupation state	15
7.2. An exact informative-control bottleneck	16
8. Saddle faces and causal realization	18
8.1. A normal form over a finite causal interface	18
8.2. The realization theorem	19
8.3. Faces as memory witnesses	21

Date: September 24, 2026. Revision 28.

2020 Mathematics Subject Classification. Primary 62B05, 62C20; Secondary 60J10, 68Q45, 93E20.

Key words and phrases. Statistical experiments, causal realization, predictive distributions, finite memory, minimax duality, physical certification.

9. Facial certificates for schedule-optimized memory	22
10. Exact minimax laws for a marked product family	23
10.1. Complementary slackness and the event alphabet	25
11. A self-contained collision interface	25
11.1. Preparation and elastic scattering	26
11.2. Positive noise and the actual mark	27
12. The physical experiment and its sharp preparation boundary	28
12.1. An exact rational certificate against one preparation	28
12.2. A marked five-event selector	29
12.3. Causal compilation and the full register profile	30
13. The exact two-preparation marked saddle	31
14. A continuous two-preparation marked family	34
15. Exact memory from a marked saddle	35
15.1. The sharp serial-validation peak	37
16. A full-support marked saddle	39
17. Exact statistical form of the Brownian collision sensor	41
18. Optimal validation orders and a positive-noise memory jump	43
19. Preparation limits and localization of least-favorable priors	46
20. An exact preparation–decision-memory law	49
21. Integer memory and the failure of a naive partition formula	51
22. Quantitative certificates for a finite compatible architecture	52
23. Autonomous noisy testing with a precision and time ledger	54
24. Physical consumers and the first-principles connection	55
25. A sharpened physical consumer and its simulation budget	56
26. Dependencies, retained results and remaining distinctions	57
References	58

1. INTRODUCTION

An optimal statistical decision and a minimum-memory realization of that decision are different objects. A least-favorable prior exposes a face of Bayes-optimal responses. Uniform minimax optimality imposes additional contact conditions, including on coordinates to which that prior assigns zero coefficient. Once those conditions have determined an optimal response, its realizability depends on common positive continuations across every charged cut. The order in which independent validation preparations are read is then a mathematical part of the memory problem, not a harmless implementation convention.

The present paper develops this connection from prepared causal experiments to an exact positive-noise physical theorem. We use finite registers, count all persistent selectors and report buffers, and freeze one row–event continuation before independent candidate and target validation. The unknown candidate parameter is fixed through the audit. The target is a specified law; acquisition of unknown target calibration has a separate resource charge. A clocked validation schedule is declared before data are observed. Report-dependent scheduling is a distinct controlled interface and is not treated as an uncharged clock.

1.1. The principal theorem chain. Let the private all-on experiment be $b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2)$. On the four diagonal marked atoms 000, 001, 110, 111, give the target masses

$$\frac{1+\gamma}{4} - \kappa, \quad \frac{1-\gamma}{4} - \kappa, \quad \frac{1-\gamma}{4} - \kappa, \quad \frac{1+\gamma}{4} - \kappa,$$

and give each other atom mass κ . For $(\gamma, \kappa) \in [6/25, 13/50] \times [0, 1/1000]$, the full-support saddle theorem determines the exact value by the unique root $r \in (33/100, 7/20)$ of

$$r^3 + (13 - 2\gamma + 8\kappa)r^2 + (3 - 12\gamma + 48\kappa)r - 1 - 2\gamma + 8\kappa = 0.$$

A global polynomial factorization proves the lower bound. Every signed posterior coefficient is checked for the matching two-point prior. Support contact and stationarity determine the two Bayes-indifferent response coordinates in an interval strictly inside $(0, 1)$. Consequently the minimum post-training decision width is exactly five, not merely at the ideal target but throughout this explicit rectangle. See Theorem 16.1.

The physical identification is exact. In the original collision model, the two Brownian report windows have one common random signal amplitude and conditionally independent noises. If e is their conditional sign error probability, put $\mu_1 = \mathbb{E}e$, $\mu_2 = \mathbb{E}e^2$. Preparation reflection and center-of-mass independence give

$$\gamma = \frac{1}{4} - \frac{\mu_1}{2}, \quad \kappa = \frac{\mu_1 - \mu_2}{2}.$$

The previously proved mechanical and Gaussian estimates place these parameters in the saddle rectangle for every contact distance $9/10 \leq d \leq 11/10$ and every $0 < \sigma \leq 1/24$. Theorem 17.1 thus proves exact physical memory on a quantified positive-noise range. It does not infer exact counts from an unevaluated compactness gap.

Theorem 9.2 gives a general schedule certificate. Contact-forced vertices and disjoint faces supply a capacity at each consumed-prefix configuration. Compatible residual functions supply an upper realization. A matching path in the finite schedule graph identifies the exact minimum across schedules and stochastic architectures. The distinction between expected responses and complete output channels is essential: two independent gate coins can produce a random output with mean zero. Our certificates do not replace such a channel by a deterministic zero.

For the full-support marked family the certificates imply

$$W_{\text{clk}}^{\min} = 12, \quad \text{argmin}_{\text{clock orders}} W = \{XXXXYYY, YYYXXX\}.$$

Here each letter reads the next slot of its tape, preserving the order of the two report bits and the mark. All other eighteen interleavings need at least fourteen states. At the ideal target the exact reverse serial peak is instead ten. Theorems 18.1 and 18.2 therefore show how a support change can alter exact memory even when the law and its minimax value change continuously. They do not identify a feedback scheduler with the fixed-clock class.

1.2. Foundations and retained results. The object-level route is the prepared causal experiment, its future-test quotient, an executable causal morphism, and the attainable resource profile. The first sections give A0–A5, conditional-mean and entropy identities, checkpoint resolution, online implementation, and joint calibration/model-error transport. The classical mechanisms are stated with their hypotheses rather than taken as a universal geometric law. The unrestricted controlled dual uses a single adversarial prior; the finite-memory occupation formulation does not convexify persistent selectors for free. The informative routing example remains a sharp multicut result.

The ideal marked values U_1 and U_2 , their continuous target family, and the self-contained microscopic bridge are included in full. For arbitrary preparation number the closest-experiment endpoint and least-favorable-prior localization remain as proved, including the explicit contact geometry of the ideal marked family. The present positive-noise and scheduling results do not assert a sharp large- N expansion or a finite- N support law. Revelation, finite-architecture certification, autonomous testing and transported confidence results are retained with their

distinct resource conventions. No result is removed to obtain the new memory theorem. For reference, the deterministic precursor constants retain their notation:

$$(1) \quad u_1 = \frac{4035777}{10240000}, \quad m_2 = \frac{12 - 3\sqrt{3}}{16}, \quad \beta_0 = \frac{1}{300}.$$

The sensor moments μ_1, μ_2 used above are different quantities.

1.3. Relation to earlier work. Classical experiment comparison and minimax supply the convex statistical ingredients [2, 3]. Controlled sensing, sequential experimental design and finite-horizon belief control are established subjects [13, 1, 14, 15]. Positive stochastic realization, invariant generating cones and nonnegative-rank geometry have a substantial literature [17, 18, 19]. Partial-input state minimization requires compatible coverings [20]. Finite-memory testing and the role of stochastic policies are likewise classical [6, 7, 8, 9], and the recent settings of [10, 16] have their own objectives and adversaries.

We do not claim the matrix-product normal form, the elementary loss identity, residual minimization, or dynamic programming as new. The loss identity is recorded separately as Proposition 8.2. The realization theorem retains its substantive contact and compatibility statements. The new consequences under consideration are the full-support saddle region containing the original Brownian model, the all-declared-order physical memory theorem, and the exact support-boundary jump. They use contact-forced channels common to every optimizer rather than the state count of an arbitrarily selected Bayes tie rule. The comparison with general filtered experiments [11] is not represented as a completed proof-level priority audit.

The historical geometric and unbounded-operator pipelines retain their own assumptions and derivations. In particular no finite-state argument here proves the Fourier/LLT, stopped-LDP, nonlinear semigroup or strict/form-response gates of A2, B4 or C2. The new physical memory converse is a genuine proof consumer of the foundations layer, not a replacement label for those independent analyses.

2. CAUSAL EXPERIMENTS AND THEIR PREDICTIVE QUOTIENTS

We begin with the object and morphism structure. Its elementary probability statements fix the meaning of later resource theorems; classical kernel construction, conditional expectation and quantization are not assigned a new priority. The substantive comparisons below retain their preparation, memory and approximation coordinates simultaneously.

2.1. The experimental object. An experiment has a fixed unknown parameter $\theta \in \Theta$, hidden states X_t , visible actions A_t , reports Y_{t+1} , and preparation laws μ_θ . We impose the following six conventions.

A0 (types and visibility). These spaces are standard Borel. Visible histories include actions, reports and publicly recorded resource increments. A legal-action set and its remaining budget are part of the visible protocol type. Two histories with different legal continuations are not identified.

A1 (positive kernels). There are jointly measurable normalized nonnegative kernels $K_t^\theta(dx', dy, dc | x, a)$, with c a nonnegative resource increment. An absorbing symbol represents termination or failure. Dirac kernels are allowed. Positivity here does not require every outcome to have strictly positive probability. A hidden state can contain the necessary physical past; this notation does not assume a finite-dimensional physical closure.

A2 (causality). A policy is a kernel $\alpha_t(da | h_t)$ depending only on visible past data and allowed independent randomization. It cannot read the unknown parameter. Stopping is with respect to the visible filtration.

A3 (executable tests). The specified bounded future tests include constants, required report events and their legal one-step continuations. For discrete reports the one-step instrument is

$$(\mathbb{T}_{t,a,y}f)(\theta, x) = \int f(\theta, x') K_t^\theta(dx', \{y\}, dc \mid x, a).$$

The test spaces are closed under the required instruments, finite linear combinations and the bounded monotone limits needed to recover the stated event laws. For general reports one uses event instruments and regular conditional distributions, not division by the probability of a continuous singleton.

A4 (actual preparation). Average risks and acquired state laws refer to the specified preparation and exploration protocol. Failed trials, calibration, resets and conditioning events retain their probabilities and implementation costs. A convenient measure on a reachable set does not replace its acquired law.

A5 (executable resources). Persistent labels, temporary buffers, program constants, calibration precision, preparations and clocks are distinguished. A retained selector is a state. A parameter-dependent table is not a known program constant. A simulator with R_t labels composed with a controller with K_t labels has the direct product bound $R_t K_t$, before any justified quotient is taken. Atomic stochastic rows and fair-bit implementations are separately specified.

Proposition 2.1 (Prepared path laws). *For every fixed parameter and causal policy, the preparation and kernels uniquely determine their finite-dimensional path laws. On a finite horizon they are the iterated kernel product*

$$\mu_\theta(dx_0) \prod_{t < T} \alpha_t(da_t \mid h_t) K_t^\theta(dx_{t+1}, dy_{t+1}, dc_t \mid x_t, a_t).$$

These laws are consistent under marginalization. For a countable horizon they determine the usual kernel extension. Deterministic dynamics give their original trajectory laws when represented by Dirac kernels.

Proof. Successive integration constructs each finite law. Normalization of the last two kernels gives its preceding marginal, and induction proves consistency and uniqueness on cylinder events. The countable kernel extension gives the full law; equivalently one applies the standard extension theorem to these consistent iterated kernels. Integration against a Dirac transition is evaluation along that transition, which proves the last assertion. $\square$

2.2. Prepared predictive equivalence. For Bayesian predictions additionally fix a prior on the parameter, and regard (θ, X_t) as the hidden variable. A prepared posterior π at a given protocol type evaluates the future tests. Define $\pi \sim_t \tilde{\pi}$ if their evaluations agree on every specified future test. This definition concerns one prepared experiment; it does not identify a prior-dependent posterior with a statistic sufficient simultaneously for all fixed parameters.

Theorem 2.2 (Predictive quotient and causal closure). *For finite report alphabets, predictive equivalence is preserved by every legal positive-probability report update. Its class determines the laws of all specified future experiments. Every deterministic history statistic that determines those laws factors onto this quotient. For a measurable realization it is sufficient that a countable separating test family gives a measurable evaluation map with standard Borel image, and that the chosen statistic admits a measurable factor. In finite history spaces the latter conditions are automatic.*

Proof. For a legal action and report, Bayes' rule gives

$$\pi'(f) = \frac{\pi(\mathbb{T}_{t,a,y}f)}{\pi(\mathbb{T}_{t,a,y}1)}.$$

Both numerator and denominator depend only on the old class by A3. Equivalent posteriors assign the same probability to the report, so updates agree on the next test space whenever that probability is positive. Null reports do not affect the prepared law and may be assigned a fixed admissible version. Induction determines every finite future report word; the monotone-class extension gives the required future events. Conversely those event laws determine the tests. If two histories have the same sufficient statistic, they have the same future evaluations; the quotient is therefore constant on its fibers. This gives the factor set-theoretically. The stated measurable factor assumption supplies its measurable version; no claim that every abstract quotient of a standard Borel space is standard Borel is used. $\square$

Proposition 2.3 (Barycenters, entropy and normalized response). *In a fixed preparation, suppose a hidden target Z , observation Y and postprocessing W form a Markov chain $Z \rightarrow Y \rightarrow W$. For any bounded vector of tests f ,*

$$p_W = \mathbb{E}[p_Y | W], \quad p_Y = \mathbb{E}[f(Z) | Y], \quad \mathbb{E}\varphi(p_W) \leq \mathbb{E}\varphi(p_Y)$$

for every convex φ for which these expectations exist. For a parameter-independent kernel G , $P \ll Q$, and finite $D(P\|Q)$, regular conditional laws give

$$\begin{aligned} D(P\|Q) &= D(PG\|QG) \\ &\quad + \int D(P(dx | w)\|Q(dx | w)) (PG)(dw). \end{aligned}$$

Finally, if f, g are bounded and $\pi_u(dx) = e^{ug(x)}\pi(dx)/\pi(e^{ug})$, then

$$\left. \frac{d}{du} \pi_u(f) \right|_{u=0} = \text{Cov}_\pi(f, g).$$

Proof. The Markov property and the tower rule give the conditional barycenter; conditional Jensen gives convex order. The joint laws $P(dx)G(dw | x)$ and $Q(dx)G(dw | x)$ have relative entropy $D(P\|Q)$. Disintegrating by w and applying the entropy chain rule gives the identity, and hence data processing. The last equality follows by quotient differentiation; boundedness justifies differentiation under the integral. $\square$

The barycenter statement concerns a law of predictions, not a pointwise map between reachable sets. Likewise the covariance in the last formula is a normalized response, not an independently postulated Fisher matrix or a global sufficiency theorem.

3. ATTAINABLE RESOLUTION AND RESOURCE-AWARE SIMULATION

3.1. Checkpoint and online prediction. Fix an actual exploration protocol and its history law. Let $p(h)$ be a bounded d -dimensional conditional prediction vector, with law ν . For instance, a future test is sampled independently after encoding, and the coordinates include its sampling weights. Squared excess prediction loss is then $\mathbb{E}\|p(h) - \hat{p}\|^2$.

Proposition 3.1 (Checkpoint resolution). *For at most M retained labels the least checkpoint excess loss is*

$$(2) \quad \inf_{c_1, \dots, c_M} \int \min_{1 \leq j \leq M} \|p - c_j\|^2 \nu(dp).$$

Independent randomized encoders or decoders cannot improve it. This formula concerns the stated preparation and does not provide a common codebook by separately optimizing at each unknown true parameter.

Proof. Conditional squared loss decomposes into its Bayes loss and squared distance from the conditional mean. For each retained label replace a random decoder by its mean. Random assignment to the resulting centers cannot do better than a nearest center. Conversely a nearest-center assignment attains the infimum over codebooks arbitrarily closely. Conditioning on an independent shared random seed, if such a seed is allowed, gives a codebook for each seed, each with risk at least (2). Averaging cannot improve the bound. The formula does not charge the computation of a checkpoint assignment as an online algorithm; online compatibility is a separate condition. $\square$

Theorem 3.2 (A matched online resolution law). *Suppose a fixed prepared experiment has a predictive state s_t in a compact metric space S , a known initial state, and updates $s_{t+1} = F_{t,a,y}(s_t)$. Assume every permitted update is ρ -Lipschitz for one $0 \leq \rho < 1$, and the prediction map $p : S \rightarrow \mathbb{R}^d$ is L -Lipschitz. Suppose S has a δ -net of at most $C\delta^{-d}$ points for all sufficiently small δ . At a checkpoint T , assume the actual law ν of $p(s_T)$ satisfies the small-ball bound $\nu(B(z, a)) \leq Aa^d$ for every center z and all sufficiently small $a > 0$. Then for all sufficiently large M , the minimum online excess prediction loss with M retained predictive labels satisfies*

$$(3) \quad c_0 M^{-2/d} \leq R_M^{\text{on}} \leq C_0 M^{-2/d}.$$

Here one may take

$$c_0 = \frac{1}{2}(2A)^{-2/d}, \quad C_0 = \frac{L^2 C^{2/d}}{(1-\rho)^2},$$

Any separate exploration controller, report buffer or clock is charged in addition to these M labels. The exploration protocol and the preparation are fixed before optimizing the encoder.

Proof. Any online M -label state is also a checkpoint encoder, so Proposition 3.1 applies. For any M centers, the union of balls of radius $a = (2AM)^{-1/d}$ has ν -mass at most one half. On its complement squared distance is at least a^2 . Integration gives $c_0 M^{-2/d}$. This argument also permits randomized online encoders.

For the upper bound take a δ -net with $\delta = (C/M)^{1/d}$, and a fixed measurable nearest-net assignment (with a fixed tie rule). Quantize the known initial state and, after each legal update, quantize again. The state error obeys $e_{t+1} \leq \rho e_t + \delta$, $e_0 \leq \delta$, and hence $e_t \leq \delta/(1-\rho)$. The prediction error is at most Le_t . Squaring proves the upper bound, uniformly in time. Each update uses only the current net label and the current permitted input, so the construction is online. It does not grant the encoder free access to the original discarded history. If the exploration policy needs another register, that register is combined by a product construction. $\square$

For example, a Lebesgue density bounded above by B gives the small-ball hypothesis with $A = Bv_d$, where v_d is unit-ball volume. The lower hypothesis is a condition on the *acquired prediction law*. A reachable set of dimension d without a mass condition does not imply the lower bound. Similarly, the contraction assumption is additional dynamics, not a consequence of kernel positivity.

3.2. Causal morphisms and simultaneous error budgets. A causal simulator $\Gamma : \mathfrak{E} \rightarrow \mathfrak{F}$ is a parameter-independent executable transducer that uses source preparations to supply the target interface under every allowed target policy. Its defect ε is a uniform total-variation bound on the resulting path laws, including the fields required by the task. Its resource map ρ_Γ bounds the source resources of the transducer composed with a target controller.

Proposition 3.3 (Composition and risk transfer). *Causal simulators compose. Their uniform defects add, capped at one, and their resource maps compose in the order $\rho_{\Gamma_1} \circ \rho_{\Gamma_2}$. For a terminal loss in $[0, L]$ and a common target task,*

$$\mathcal{R}_{\mathfrak{E}}(\rho_{\Gamma}(B)) \leq \mathcal{R}_{\mathfrak{F}}(B) + L\varepsilon.$$

For reward in $[-1, 1]$, the corresponding maximum worst-case rewards satisfy the reverse inequality with an error 2ε .

Proof. Interleave the two transducers; their pair of internal states is a causal implementation. Total variation contracts under a kernel and obeys the triangle inequality, proving the defect sum. If their state counts are R_t and M_t , the pair has at most $R_t M_t$ states. Other declared resource bounds compose by execution of the two protocols. A bounded loss changes expectation by at most its range times total variation. Apply this bound to every fixed parameter and target controller and take the appropriate infimum or supremum. Uniformity in the controller is essential to this step. $\square$

Theorem 3.4 (Acquisition, quantization and model-error transport). *Suppose an exact source transducer has predictive state $s_t \in S_t$ and additional charged buffers of size R_t , supplies T target operations, and has uniform model defect at most ε_{mod} . Its initialization and update depend on a calibration quantity ξ . From N_{cal} source preparations acquire an estimate $\hat{\xi}$ with at most C_{cal} retained labels, such that*

$$\sup_{\theta} \mathbb{P}_{\theta}(d(\hat{\xi}, \xi) > a) \leq \eta.$$

Subsequent preparation noises are independent of calibration conditional on the fixed parameter. Let finite nets of sizes M_t have radii δ_t . Assume, on matching input/output histories and on the good calibration event, that the quantized transducer's state error satisfies

$$(4) \quad e_0 \leq d_0 + a_0 a + \delta_0, \quad e_{t+1} \leq \rho_t e_t + d_{t+1} + a_{t+1} a + \delta_{t+1}.$$

Assume its next output kernel and the exact transducer's output kernel differ in total variation by at most $\epsilon_t + L_t e_t + b_t a$, uniformly in the fixed parameter and legal policy. All constants and nets are declared independently of that unknown parameter. Then the quantized acquired transducer has defect at most

$$(5) \quad \Delta = \min \left\{ 1, \varepsilon_{\text{mod}} + \eta + \sum_{t < T} (\epsilon_t + L_t e_t + b_t a) \right\}.$$

Composing it with any target controller of cut widths K_t requires at most $C_{\text{cal}} R_t M_t K_t$ labels at aligned execution cuts and transfers its bounded reward with loss at most 2Δ . Its preparations are N_{cal} plus all source preparations specified by the transducer; no successful-trial conditioning removes the failures.

Proof. On the good calibration event, iterate (4). More explicitly, with $D_0 = d_0 + a_0 a + \delta_0$ and $D_j = d_j + a_j a + \delta_j$,

$$e_t \leq \sum_{j=0}^t D_j \prod_{k=j}^{t-1} \rho_k.$$

Couple each pair of output kernels maximally while their external histories agree. On agreement the target controller sees the same inputs, so it uses the same legal action and may use the same independent randomization. Its discontinuous action choices require no Lipschitz assumption: before a disagreement their arguments are identical. The chance of a first discrepancy is bounded by the sum in (5). Add the probability of bad calibration and the exact transducer's model defect. This proves the path bound, including stopping after padding by absorbing states.

The executable approximate transducer stores its calibration label, additional buffer, and net index. Combine these with the actual target register, not its full history or an uncharged belief. The resulting product gives the state count. If an operation contains further buffer cuts, apply the same product at each such cut; R_t was assumed to include them. Preparation costs add by execution. Finally apply Proposition 3.3. $\square$

Corollary 3.5 (A finite acquisition register). *Suppose r_0 calibration rows have at most d outcomes, and take n independent preparations from each row on a fixed schedule. Their empirical rows satisfy, with probability at least $1 - \eta$ uniformly in the row probabilities,*

$$\max \text{TV}(\hat{P}, P) \leq \frac{dz}{2}, \quad \eta = \min\{1, 2r_0d \exp(-2nz^2)\}.$$

Their counts can be retained in at most $(n+1)^{r_0(d-1)}$ labels. Thus Theorem 3.4 can use $N_{\text{cal}} = r_0n$, $a = dz/2$ and this finite calibration register when its calibration metric is the maximum row total variation.

Proof. For each outcome the empirical frequency is the mean of independent Bernoulli indicators. Hoeffding's bound and a union bound over at most r_0d coordinates give the event stated. The total-variation estimate is half the sum of coordinate errors. Retain $d - 1$ counts per row; the last is reconstructed from the known number of draws at that stage of the clocked schedule. Each retained count takes at most $n + 1$ values. An autonomous implementation must additionally charge its schedule. $\square$

These results separate exact predictive equivalence, average checkpoint resolution, online realizability, and uniform simulation risk. In particular, neither an optimal checkpoint codebook nor the proof-state bundle of Section 7 is automatically an executable finite register.

4. EXPERIMENTS, ARCHITECTURES AND ATTAINABLE RESOURCES

Throughout, total variation is $\text{TV}(P, Q) = \sup_{0 \leq h \leq 1} (P - Q)h = \frac{1}{2} \sum_x |P(x) - Q(x)|$. An experimental parameter is unknown and fixed throughout an execution. Controller randomization is independent of that parameter. A reset preparation is a fresh draw, independent of earlier preparations given the parameter. A controller may not recover discarded information from an uncharged tape, clock, selector or calibration table.

Definition 4.1 (Finite architecture). A finite architecture A specifies a finite causal execution graph, its legal experimental operations, the observable inputs to each controller row, and a register alphabet at every cut. It also specifies which row occurrences share a stochastic table. A clocked architecture may use different rows at different scheduled times. An autonomous architecture may use time only through a charged state. For a parameter family Θ and terminal reward in $[-1, 1]$, let

$$V(A) = \sup_{q \in \prod_r \Delta_{s_r}} \inf_{\theta \in \Theta} g_\theta(A, q).$$

If $\mathcal{A}(R)$ is the class of architectures obeying a global resource budget R , its outer value is

$$V^{\text{out}}(R) = \sup_{A \in \mathcal{A}(R)} V(A).$$

Both the architecture and the simulator parameter are fixed before the controller's random choices. In particular, a mixture of complete controllers is not automatically an admissible controller at the original register profile.

For a finite hidden execution graph, compatibility can be stated without division. Let $x_{\theta ir}$ be the incoming mass of hidden history i at observable row r , and let $f_{\theta irj}$ be the outgoing mass through choice j . Include the usual fixed-kernel flow equations, as well as

$$(6) \quad f_{\theta irj} \geq 0, \quad \sum_j f_{\theta irj} = x_{\theta ir}, \quad f_{\theta irj} x_{\theta' i' r} = f_{\theta' i' rj} x_{\theta ir}.$$

For shared rows, the indices i, i' include all their occurrences.

Proposition 4.2 (Compatible realization). *For a specified finite architecture, nonnegative flow arrays satisfying (6) are exactly those realized by a common stochastic row at every declared observable input, including at zero incoming mass.*

Proof. Necessity follows from $f_{\theta irj} = x_{\theta ir} q_r(j)$. For sufficiency, if some incoming mass at row r is positive, define $q_r(j)$ by dividing the corresponding outgoing mass by that incoming mass. Conservation makes q_r stochastic, and the cross-products force the same ratio at every other positive incoming mass. If an incoming mass vanishes, nonnegativity and conservation force all its outgoing masses to vanish. If the entire row is unreachable, choose any legal stochastic row. Induction through the finite execution graph reconstructs the prescribed flows. The same ratio argument applies to identified occurrences of an autonomous row. $\square$

This formulation is inherited from the complete companion and is included to fix the controller class. Compact semialgebraic feasibility and an Archimedean positivity theorem provide a complete semialgebraic certificate hierarchy for a *fixed finite architecture*. They do not interchange the two optimizations in the definition. Section 22 gives a quantitative alternative for the inner problem. The next section supplies a different certificate that survives the outer optimization.

5. A STATISTICAL DUAL THAT IS UNIFORM OVER ARCHITECTURES

Let Θ be compact, let Ω be finite, and let $\theta \mapsto P_\theta \in \mathcal{P}(\Omega)$ be continuous. Fix $Q \in \mathcal{P}(\Omega)$. An N -preparation reset selector observes $Z = (Z_1, \dots, Z_N) \sim P_\theta^{\otimes N}$ and chooses a response $h_Z \in [0, 1]^\Omega$. Its expected validation score is

$$(7) \quad G_\theta(h) = \sum_{z \in \Omega^N} P_\theta^{\otimes N}(z) (Q - P_\theta) h_z.$$

The response is fixed before two independent validation draws. Operationally it can be realized by choosing a random event with those coordinate marginals and retaining that event for both validations. No memory bound is imposed in the definition of $U_N(P, Q) = \max_h \min_\theta G_\theta(h)$.

Theorem 5.1 (Posterior-predictive dual). *Both extrema in the following identity are attained:*

$$(8) \quad U_N(P, Q) = \min_{\mu \in \mathcal{P}(\Theta)} \sum_{z, \omega} \left[\int P_\theta^{\otimes N}(z) \{Q(\omega) - P_\theta(\omega)\} d\mu(\theta) \right]_+$$

$$(9) \quad = \min_{\mu \in \mathcal{P}(\Theta)} \sum_z m_\mu(z) \text{TV}(Q, \bar{P}_\mu(\cdot | z)).$$

Here $m_\mu(z) = \int P_\theta^{\otimes N}(z) d\mu(\theta)$ and, when $m_\mu(z) > 0$,

$$\bar{P}_\mu(\omega | z) = \frac{\int P_\theta^{\otimes N}(z) P_\theta(\omega) d\mu(\theta)}{m_\mu(z)}.$$

A zero-mass term in (9) is defined to be zero.

Proof. The response space is a finite-dimensional compact convex box. Probability measures on the compact space Θ form a weakly compact convex set. The integral of (7) is continuous and affine in each argument. The minimum over parameters equals the minimum over probability measures. The minimax theorem [3] therefore interchanges the maximum and minimum. For fixed μ , each response coordinate is independent in the maximization and its contribution is the positive part of its coefficient, proving (8). The coefficient vector at a fixed z has zero sum. For positive $m_\mu(z)$ it equals $m_\mu(z)(Q - \bar{P}_\mu(\cdot | z))$, proving (9). If $m_\mu(z) = 0$, every nonnegative likelihood integrand vanishes μ -almost surely and all coefficients are zero. Continuity on the two compact strategy spaces gives attainment. $\square$

Theorem 5.2 (Dominating-row converse). *Suppose every permitted training row is a parameter-independent causal garbling of one fresh draw from P_θ . Suppose every permitted validation row is the same garbling of the target and candidate laws, and every selected row–event pair is fixed before validation. Then $U_N(P, Q)$ bounds the worst-case expected score of every auditor using at most N candidate training preparations, regardless of its register profile, feedback schedule, independent randomization, or stopping rule. Independent target-only calibration, whose law does not depend on θ , does not invalidate the bound. If the all-on row and all its events are permitted, the bound is attained in the class with no memory cap and N training preparations.*

Proof. Generate N independent all-on observations in advance. At the t th requested preparation, apply the declared garbling to observation t , using only previously simulated reports and independent randomization. This simulates the adaptive transcript with its exact law. A stopped execution ignores the unused observations; it does not reveal them to the auditor. Pull back its final row–event response to the all-on alphabet. Conditional on the entire generated training word and the independent random seed, this is a function in $[0, 1]^\Omega$. Averaging the seed, including any target-only calibration, produces one response h_z for each full word z . The score is consequently (7). This proves the upper bound for the union of all architectures. Conversely, retaining the entire word and then a selected event implements any such finite response kernel when memory is unrestricted. Fresh validation gives exactly (7). $\square$

Remark 5.3 (What the dual does not convexify). The measure μ is an adversarial prior used to bound the minimum over *fixed* parameters. Its likelihood is $\int P_\theta^{\otimes N} d\mu(\theta)$, not $(\int P_\theta d\mu(\theta))^{\otimes N}$. It neither enlarges the private candidate class to mixtures nor gives the auditor a free persistent selector. Attainment of U_N without a memory cap does not assert attainment at any specified finite profile.

The state in (9) is generated by the actual preparation law, not by a geometric measure chosen independently of the experiment. For sequential continuation one retains the posterior μ_z or its future-test equivalence class; a single predictive marginal need not be closed under further observations. This is the distinction between a one-step response quotient and a future-predictive quotient.

Corollary 5.4 (Finite moment support for the product family). *For two independent Bernoulli report bits with parameters (p, q) and an independent fair mark, an optimal prior in (8) can be chosen with at most $(N + 2)^2$ atoms. The two report counts (k, ℓ) are a causally updatable sufficient statistic for computing the dual.*

Proof. Conditional on the two counts, all arrangements of report bits and all mark words have parameter-independent probabilities. Summing their coefficients reduces (8) to the $(N + 1)^2$ count pairs. Their likelihoods are

$$\binom{N}{k} p^k (1 - p)^{N-k} \binom{N}{\ell} q^\ell (1 - q)^{N-\ell}.$$

After multiplication by $Q(\omega) - P_{p,q}(\omega)$, every integrand has degree at most $N + 1$ in each variable. Thus the objective depends only on the $(N + 2)^2$ rectangular moments. Their constant coordinate is one. Carathéodory's theorem in the remaining affine space replaces an optimal moment vector by a convex combination of at most $(N + 2)^2$ point evaluations. Counts update by addition of the current bits. No claim that this count register is optimal for the decision is made. $\square$

6. CONTROLLED EXPERIMENTS WITHOUT A DOMINATING ACTION

The reduction to one dominating row is not needed for a statistical dual, provided causal action choice is retained in the dual recursion. We give the finite-action theorem with a compact parameter space and an explicit protocol state. Its equality is an unrestricted-memory statement; its upper certificates apply to every smaller-memory class.

Let Θ be compact metric and E a finite set of protocol states. At $e \in E$ let $\mathcal{A}_e$ be a finite set of training actions. Action a produces x in a finite alphabet X_a with law P_θ^a and changes the protocol state to $T(e, a, x)$. The functions $\theta \mapsto P_\theta^a(x)$ are continuous. An unknown θ is fixed throughout the experiment. Conditional on it, each requested preparation uses a fresh independent draw from the selected row. Legal validation designs form a nonempty finite set $\mathcal{B}_e$. Design b has a finite outcome alphabet Y_b , target law Q^b and continuous candidate law V_θ^b . A policy may stop after at most N training actions. It then freezes $b \in \mathcal{B}_e$ and an event on Y_b , before one candidate and one target validation. Randomization is independent of θ . No training action is assumed to dominate another.

For a finite nonnegative measure ν on Θ write $\nu_{a,x}(d\theta) = P_\theta^a(x)\nu(d\theta)$ and set

$$(10) \quad \begin{aligned} \Phi_e(\nu) &= \max_{b \in \mathcal{B}_e} \sum_{y \in Y_b} \left[Q^b(y)\nu(\Theta) - \int V_\theta^b(y)d\nu \right]_+, \\ F_0(e, \nu) &= \Phi_e(\nu), \end{aligned}$$

$$(11) \quad F_n(e, \nu) = \max \left\{ \Phi_e(\nu), \max_{a \in \mathcal{A}_e} \sum_{x \in X_a} F_{n-1}(T(e, a, x), \nu_{a,x}) \right\}.$$

If no training action is legal, only the stopping term is retained. All the functions vanish at the zero measure, so no posterior division is required on null histories. For positive mass, (10) is the mass times the largest posterior-predictive total variation.

Theorem 6.1 (Controlled posterior-predictive dual). *Let $\mathcal{U}_N(e_0)$ be the maximum worst-case expected validation score over all randomized causal policies in the declared protocol, with at most N training preparations and no memory cap. Then*

$$(12) \quad \mathcal{U}_N(e_0) = \min_{\mu \in \mathcal{P}(\Theta)} F_N(e_0, \mu).$$

Both extrema are attained. Each $F_n(e, \cdot)$ is the maximum of finitely many continuous linear functionals on the cone of measures; in particular it is convex, continuous and positively homogeneous. For every prior μ , $F_N(e_0, \mu)$ bounds the worst-case score of every finite-memory architecture implementing this protocol.

Proof. There are finitely many histories of length at most N and finitely many event masks at each stopping history. Consequently there are finitely many deterministic policy trees, denoted by $\Pi_N(e_0)$. For $\pi \in \Pi_N(e_0)$ its payoff $g_\pi(\theta)$ is a finite sum of finite products of continuous primitive probabilities and is continuous in θ .

A randomized causal policy induces the same outcome laws as a probability distribution on these trees: sample its finite random decision table in advance and condition on it. Conversely,

with no memory cap, a sampled tree can be retained and followed. Thus the unrestricted payoff vectors are exactly the convex hull of the g_π . This step is not an assertion that mixing machines preserves a specified register width.

The simplex on $\Pi_N(e_0)$ and $\mathcal{P}(\Theta)$ are compact and convex. The expected payoff is continuous and bilinear. Minimax therefore gives

$$\max_{\lambda \in \Delta(\Pi_N)} \min_{\theta} \sum_{\pi} \lambda_{\pi} g_{\pi}(\theta) = \min_{\mu} \max_{\pi \in \Pi_N} \int g_{\pi} d\mu.$$

For a fixed measure, the terminal maximum is (10), because an event includes precisely the positive coordinates of its signed predictive measure. At a training node one selects an action and, after its observed outcome, selects a continuation tree separately on that branch. The unnormalized branch measure is exactly $\nu_{a,x}$. Backward induction proves (11) and identifies the right side with (12). It also proves the finite-linear-functional representation. Compactness gives an optimizing prior and an optimizing mixed tree. Finally, the minimum over fixed parameters of any policy is at most its average under μ , which is at most $F_N(e_0, \mu)$. This remains true after restricting its memory or architecture. $\square$

The minimization in (12) occurs once, outside the recursion. It cannot be replaced by fresh adversarial choices after observations. The posterior carries dependence on one fixed unknown parameter. Also, a Bayes-optimal tree for a minimizing prior need not by itself be a minimax policy; the saddle-point mixture is supplied by the finite game in the proof.

Corollary 6.2 (Bellman upper certificates). *Suppose functions $\Psi_n(e, \nu)$ satisfy $\Psi_0 \geq \Phi$ and, for $n \geq 1$, satisfy*

$$\Psi_n(e, \nu) \geq \Phi_e(\nu), \quad \Psi_n(e, \nu) \geq \sum_x \Psi_{n-1}(T(e, a, x), \nu_{a,x}) \quad (a \in \mathcal{A}_e).$$

Then $\Psi_N(e_0, \mu)$ is an upper certificate for every admissible architecture and every prior μ . The family F_n is the least such family. An optimal prior can be chosen with at most $d + 1$ support points, where d is the affine dimension of $\{(g_{\pi}(\theta))_{\pi \in \Pi_N} : \theta \in \Theta\}$.

Proof. Induction in (11) gives $F_n \leq \Psi_n$. The equality case is the recursion itself. The objective depends on a prior only through its vector of integrals of the finitely many g_{π} . An optimal such vector belongs to the convex hull of their compact evaluation image. Carathéodory's theorem gives the stated support bound while preserving every pure-tree payoff. $\square$

Proposition 6.3 (Causal stability). *Consider two models with the same protocol, alphabets and policy class. Suppose their training rows differ uniformly in total variation by at most ϵ , their candidate validation rows by at most η , and their target validation rows by at most β . Their unrestricted values, or their values in any common specified memory class, differ by at most $2N\epsilon + \eta + \beta$.*

Proof. Use the same policy randomization and couple corresponding training rows while the histories agree. The probability of a first discrepancy is at most $N\epsilon$, even with feedback and stopping. Conditional expected validation rewards lie in $[-1, 1]$, so differing training histories contribute at most $2N\epsilon$. On agreeing histories the same frozen design and event are used, and their conditional mean rewards differ by at most $\eta + \beta$. The bound is uniform in the fixed parameter and in the policy. Taking minima and suprema preserves it, with the reverse inequality obtained by interchanging the models. $\square$

6.1. An informative action choice. The following calculation distinguishes the controlled theorem from a common-row reduction. Let $\Theta = \{(i, j) : i, j \in \{0, 1\}\}$. Action R reports i . Action L reports j when $i = 0$ and otherwise reports an independent fair bit. Action H reports j when $i = 1$ and otherwise reports an independent fair bit. The candidate validation is $\delta_{(i,j)}$ and the target is uniform on Θ .

Proposition 6.4 (Strict adaptive advantage). *With two training preparations, the unrestricted adaptive value is $3/4$ and the best nonadaptive value is $5/8$. Nonadaptive policies here include mixtures of schedules, with their random schedule retained. No available single training action dominates all the others.*

Proof. Take R first, and then L or H according to the observed i . The parameter is recovered exactly. The complement of its validation atom has score $3/4$. No policy can exceed $\text{TV}(Q, \delta_\theta) = 3/4$ even with full information.

For the converse for fixed schedules, use the uniform prior. For a pair of actions a, b its Bayes value is explicitly

$$(13) \quad \sum_{x,y,\theta} \left[\frac{1}{16} \sum_{\vartheta} P_{\vartheta}^a(x) P_{\vartheta}^b(y) - \frac{1}{4} P_{\theta}^a(x) P_{\theta}^b(y) \right]_+.$$

Substitution of the three displayed kernels gives

	R	L	H
R	$1/2$	$5/8$	$5/8$
L	$5/8$	$7/16$	$1/2$
H	$5/8$	$1/2$	$7/16$

This is an exact finite sum, not a parameter-grid estimate. It bounds any mixture of fixed schedules by $5/8$; an early stop can be padded with an ignored observation. To attain the bound uniformly in θ , choose with equal probabilities the schedules (R, L) and (R, H) . When the chosen branch-specific action matches i , the parameter is known and the score is $3/4$. Otherwise only i is known, and the complement of its two-atom fiber gives score $1/2$. Each fixed parameter therefore has mean score $5/8$.

Action R cannot distinguish parameters differing only in j . Action L cannot distinguish $(1, 0)$ and $(1, 1)$, and H cannot distinguish $(0, 0)$ and $(0, 1)$. An experiment which fails to distinguish such a pair cannot garble to one that distinguishes it. Hence no action is a common dominating row. $\square$

If every binary report is additionally passed through an independent binary symmetric channel of crossover ϵ , the primitive training rows change by at most ϵ and have full support for $0 < \epsilon < 1/2$. Proposition 6.3, applied to both policy classes, leaves an adaptive advantage at least $1/8 - 8\epsilon$, which is positive for $0 < \epsilon < 1/64$. Thus the distinction is not confined to disjoint observation supports. The finite-horizon belief recursion and its piecewise-linear structure are classical [1]. The contribution here is their explicit integration with the frozen validation score, one fixed adversarial parameter, outer certificates, and the resource conventions; no priority for Bellman recursion itself is claimed.

7. CONTROLLED EXPERIMENTS AT A PRESCRIBED MEMORY PROFILE

An unrestricted posterior is a useful information state for a proof, but it is not a free runtime state. We first give a resource-exact dynamic formulation and then solve a controlled family in which replacing that formulation by a policy-tree mixture changes the value strictly.

7.1. A parameterwise occupation state. Fix a finite horizon T , a compact metric parameter space Θ , finite hidden execution spaces H_t , and register alphabets S_t of sizes at most K_t . The visible inputs to each stochastic row and all legal actions are specified. Any input that must survive a cut belongs to S_t : this includes a chosen action, an unread report buffer, a stop flag, a selected validation event and a validation accumulator. One may subdivide an operation into several cuts to expose these costs. The scheduled time is available only in the clocked convention.

For now rows at different scheduled times are independently prescribed. Let $\mathcal{Q}_t$ be the finite product of legal row simplices at stage t . A prescription $q \in \mathcal{Q}_t$ is a table, chosen at design time, whose runtime arguments are exactly the declared visible inputs. After summing the fixed environmental kernels its one-stage transition is a stochastic matrix $M_{t,\theta}^q$ on the hidden-state/register pairs. We assume this matrix is jointly continuous in (q, θ) . Both action selection and state updating occur in this matrix, in their specified causal order; it is not an arbitrary matrix chosen separately for each hidden state or each parameter.

For a continuous bundle of probability vectors $x = (x_\theta(h, s))_{\theta \in \Theta}$ define

$$(\mathcal{T}_{t,q}x)_\theta(h', s') = \sum_{h,s} x_\theta(h, s) M_{t,\theta}^q((h, s), (h', s')).$$

The initial bundle x_0 is fixed by preparation. Terminal rewards $r_\theta(h, s) \in [-1, 1]$ are continuous in θ . Absorbing stopped states and frozen validation phases are included in this execution, not added afterwards without charge. Put

$$(14) \quad B_T(x) = \min_{\theta \in \Theta} \sum_{h,s} x_\theta(h, s) r_\theta(h, s),$$

$$(15) \quad B_t(x) = \max_{q \in \mathcal{Q}_t} B_{t+1}(\mathcal{T}_{t,q}x), \quad 0 \leq t < T.$$

The argument x is an *offline occupation bundle*, not the realized history, not an observed parameter, and not an additional register made available to the machine.

Theorem 7.1 (Resource-exact dynamic value and supersolutions). *For the specified clocked profile and observable-row interface, $B_0(x_0)$ is the exact maximum worst-case reward over all stochastic controllers. The maximum is attained by a sequence of legal common prescriptions. With the norm $\|x - x'\| = \sup_\theta \sum_{h,s} |x_\theta(h, s) - x'_\theta(h, s)|$, each B_t is 1-Lipschitz. Moreover,*

$$(16) \quad B_0(x_0) = \inf_H H_0(x_0),$$

where the infimum ranges over families of functions on occupation bundles satisfying

$$H_T \geq B_T, \quad H_t(x) \geq H_{t+1}(\mathcal{T}_{t,q}x) \quad (q \in \mathcal{Q}_t).$$

The infimum is attained by $H = B$. No convexification over retained controllers and no new adversarial choice after a report occurs in these identities.

Proof. For a fixed sequence $q_0, \dots, q_{T-1}$, iteration of the displayed transition gives its actual terminal law for every fixed θ . The terminal minimum is therefore exactly its worst-case reward. A runtime randomized controller in this interface is precisely such a sequence of common conditional rows; independence of fresh randomization and the declared register specify all its conditional choices. Persistent random correlations are allowed only through that register.

The occupation bundle at the next stage is determined by the current bundle and the prescription, without drawing a realized history in the design problem. Hence optimizing over the remaining prescription sequence gives (15) by ordinary backward induction. This does not select a different continuation for each unrecorded history. Each maximizing q is still a single legal table common to all hidden histories and parameters. At x_0 one follows the maximizing sequence

of design prescriptions, computing their occupation bundles offline; the runtime controller only reads its allowed inputs.

Stochastic matrices contract the ℓ^1 norm. The terminal minimum is 1-Lipschitz in the displayed norm. Induction, using the same q when comparing two bundles and then taking the two maxima, proves the Lipschitz assertion. Joint continuity and compactness of $\mathcal{Q}_t$ give attainment at each stage, and compactness of Θ gives the terminal minimum. Any family H in (16) bounds the terminal reward along every prescription sequence. Backward induction therefore gives $H_t \geq B_t$. Equality is admissible, proving (16) and its attainment. $\square$

This is a nonlinear supersolution dual, not a least-favorable-prior saddle theorem at fixed width. Its optimization can be large and nonconvex. The principle of dynamic programming is classical; the point of this formulation is that the common-row and memory constraints remain in the maximizations. If rows are shared across scheduled times, as for an autonomous controller, choose the shared rows once and hold them fixed in all subsequent transitions; equivalently augment the *design* problem by that immutable prescription. Maximizing a shared row anew at each time would solve a different problem. A finite union of allowed interfaces can be handled by taking their maximum, or by a fully wired interface with the prescribed profile and its legal observable inputs. Neither operation enlarges a runtime register invisibly.

7.2. An exact informative-control bottleneck. Fix integers $m \geq 2$ and $1 \leq K \leq m$. The fixed parameter is $\theta = (i, j) \in [m] \times \{0, 1\}$. The first training experiment R reports i . Across the next cut the controller retains at most K labels, with an arbitrary stochastic encoder $E(s \mid i)$. It then chooses one of the m sensing actions with probabilities $D(a \mid s)$. The selected action is kept in a charged m -state action buffer and the old label is discarded. Action a reports j when $a = i$ and reports $\perp$ otherwise. The first report is no longer available.

The candidate validation law on $\{0, 1\}$ is δ_j ; the target is uniform. An event is frozen before one fresh candidate and one fresh target validation. Stopping before either training action is permitted. No independent persistent selector is given to the controller. The observation symbols are atomic; a bitwise input implementation would have additional cuts. Denote the maximum worst-case score by $V_{m,K}$.

Theorem 7.2 (Exact finite-memory control and its convexification gap). *In this protocol,*

$$(17) \quad V_{m,K} = \frac{1}{2 \lceil m/K \rceil}.$$

The value obtained by allowing an uncharged ex ante mixture of complete K -label controllers is instead

$$(18) \quad \bar{V}_{m,K} = \frac{K}{2m}.$$

The inequality is strict whenever K does not divide m . The matching controller for (17) has charged cut profile bounded by $(1, K, m, 3, 6, 3)$. The K -label cut controls both retention of the routing observation and the subsequent informative action choice.

Proof. For a fixed i , let

$$p_i = \sum_s E(s \mid i) D(i \mid s).$$

This is the probability of taking the informative action. On its complement, all retained information is independent of j . Averaging over the two values of j makes the expected validation score zero there, for any frozen event or randomized response. On the informative part the score is at most $\text{TV}(\text{Ber}(1/2), \delta_j) = 1/2$. Thus the minimum over j at a given i is at

most $p_i/2$. A stop before sensing is also uninformative about j and obeys the same bound; it can be padded by an ignored sensing operation.

Set $L = \lceil m/K \rceil$. Since $p_i \leq \max_s D(i | s)$, a value $\min_i p_i > 1/L$ would require every i to have some row s with $D(i | s) > 1/L$. Each stochastic row has at most $L - 1$ such indices. But $K(L - 1) < m$. This contradiction proves the upper bound. Partition $[m]$ into at most K fibers of size at most L , retain the fiber label after R , and choose uniformly inside that fiber. On a successful report freeze $\{1\}$ when $j = 0$ and $\{0\}$ when $j = 1$; on $\perp$ freeze the empty event. The score for every j is $1/(2|\text{fiber}(i)|)$, proving equality. The event alphabet has three labels. Storing the event and its candidate indicator requires at most six labels, and the final difference has three. The chosen sensing action was separately charged, proving the displayed profile.

For the convexified upper bound use the uniform prior on (i, j) . For any one controller,

$$\frac{1}{m} \sum_i p_i = \frac{1}{m} \sum_{i,s} E(s | i) D(i | s) \leq \frac{1}{m} \sum_{i,s} D(i | s) = \frac{K}{m}.$$

The same upper bound holds for mixtures. For the matching mixture, choose uniformly one of the m cyclic intervals of K indices. For that interval hardcode a deterministic K -label encoder which routes each included index to its own sensing action, and assigns other indices arbitrarily. Each i belongs to exactly K intervals. A successful action is followed by the complementary singleton event and failure by the empty event, as above. Every fixed (i, j) has mean score $K/(2m)$. This proves (18). $\square$

For example, $V_{3,2} = 1/4$ whereas $\bar{V}_{3,2} = 1/3$. The mixture just constructed can be implemented with a retained interval selector and a local label, using at most mK labels at the bottleneck. It cannot be declared a K -label realization: when the displayed gap is positive, (17) proves that no such realization exists. The action buffer of size m is not part of the K coordinate and has not been omitted from the profile. The theorem is a multicut resource law, not an assertion that its uniform peak equals K .

The actions are statistically incomparable. R cannot distinguish the two values of j at any fixed i , while action a does distinguish them at $i = a$. For $a \neq b$, action a does not distinguish the pair with $i = b$, while b does. A parameter-independent garbling cannot create a distinction absent from its input experiment.

Corollary 7.3 (A strict gap with positive report probabilities). *Replace each of the two training kernels by a mixture with weight $\epsilon > 0$ of the uniform law on its report alphabet. Let $\Delta_{m,K} = K/(2m) - 1/(2\lceil m/K \rceil) > 0$. For $\epsilon < \Delta_{m,K}/8$ the resulting full-support model still has a strict gap between the genuine profile-constrained value and its free-selector convexification.*

Proof. Each training row changes by at most ϵ in total variation. Couple corresponding rows until the first disagreement. For any fixed controller or mixed controller the probability of such a disagreement is at most 2ϵ , and its conditional reward lies in $[-1, 1]$. Each of the two optimal values changes by at most 4ϵ . Their difference is therefore at least $\Delta_{m,K} - 8\epsilon > 0$. All row arguments and register alphabets are unchanged. $\square$

The distinction between mixed and behavioral strategies without perfect recall is classical; it is not first asserted here. Contemporary polynomial formulations over behavioral simplices include [16]. Theorem 7.2 evaluates the discrepancy in this specific frozen-validation experiment for every m, K , with a matching actual register and a noisy persistence statement. In particular, a single prior followed by an unrestricted policy-tree best response cannot be substituted for (15).

8. SADDLE FACES AND CAUSAL REALIZATION

The statistical saddle and its implementation are different objects. The former is convex; the latter need not be. The purpose of this section is to retain that distinction while extracting memory obstructions from an unrestricted saddle. In particular, a zero Bayes coefficient must not be discarded before the contact conditions at the worst parameters have been imposed.

8.1. A normal form over a finite causal interface. Fix a finite horizon and finite action and report alphabets. An action a_t is chosen before report y_t . A terminal output is included as one last action, followed by a dummy report. Illegal actions can be removed from each protocol type; equivalently, use a fixed padding of the finite protocol tree and impose its zero constraints. Define a *causal response array* $C(\mathbf{a} \mid \mathbf{y})$ by nonnegativity, normalization in $\mathbf{a}$ for every $\mathbf{y}$, and the requirement that its marginal on $a_1, \dots, a_t$ depends only on $y_1, \dots, y_{t-1}$. The resulting set $\mathcal{C}$ is a finite polytope. Conditioning successive marginals realizes every such array by a full-history randomized controller; arbitrary versions on zero marginals do not change the array.

For a fixed parameter θ , the environment has a history-dependent likelihood $L_\theta(\mathbf{a}, \mathbf{y})$, obtained by marginalizing its hidden states. The bounded reward has the linear form

$$(19) \quad g_\theta(C) = \sum_{\mathbf{a}, \mathbf{y}} L_\theta(\mathbf{a}, \mathbf{y}) r_\theta(\mathbf{a}, \mathbf{y}) C(\mathbf{a} \mid \mathbf{y}).$$

Assume its coefficient vector is continuous on a compact metric parameter space. The parameter is fixed once, not selected after a report. Protocol-specific equalities, including a frozen validation choice, are represented by the corresponding finite protocol tree. A discarded protocol choice is not supplied again as a free input.

Here is an explicit positive realization of a response array. At cut t use k_t states, an initial distribution α , and nonnegative matrices $M_t^{a,y}$ of size k_{t-1} by k_t such that

$$(20) \quad M_t^{a,y} \mathbf{1} = p_t(a), \quad \sum_a p_t(a) = \mathbf{1}, \quad p_t(a) \text{ is independent of } y.$$

Then

$$(21) \quad C(\mathbf{a} \mid \mathbf{y}) = \alpha M_1^{a_1, y_1} \dots M_T^{a_T, y_T} \mathbf{1}.$$

The independence in (20) is essential: otherwise an action could use its not-yet-observed report. If an action or an input must survive an intermediate cut, split this formula at that cut and include its buffer in the state alphabet. Likewise, impose any declared shared-row constraints before optimizing. An externally fixed clock permits time-dependent matrices; an autonomous implementation has to store its phase or use the prescribed shared rows.

Proposition 8.1 (Outer normal form and compatible continuations). *For the finite interface and declared cuts, the response arrays of all controllers with profile at most $\mathbf{K}$ are exactly the arrays (21) with $k_t \leq K_t$, the normalization (20), and the declared interface constraints. This set, denoted $\mathcal{C}_{\mathbf{K}}$, is compact. The assertion is over all wirings of the allowed register, not one chosen transition graph.*

Equivalently, there are at most K_t normalized continuation arrays $B_{t,s}$ at each cut, with terminal continuation 1, and common nonnegative coefficients satisfying

$$(22) \quad B_{t-1,s}(a\mathbf{a} \mid y\mathbf{y}) = \sum_{s'} M_t^{a,y}(s, s') B_{t,s'}(\mathbf{a} \mid \mathbf{y}), \quad C = \sum_s \alpha(s) B_{0,s}.$$

Separate positive factorizations at the cuts are not sufficient without these common shift identities.

Proof. In a controller, let $p_t(a \mid s)$ be its action row and $q_t(s' \mid s, a, y)$ its update row. Their product defines $M_t^{a,y}$ and gives (20). Multiplication over a complete execution gives (21). Every

finite wiring embeds into the fully connected register by assigning zero to unused transitions; smaller alphabets are padded by unused states. Thus no further maximization over graph presentations is necessary.

Conversely choose action a with probability $p_t(a \mid s)$, and, after observing y , use the row $M_t^{a,y}(s, s')/p_t(a \mid s)$ when the denominator is positive. At a zero denominator the entire corresponding nonnegative row vanishes; choose an arbitrary legal update there. This gives exactly the required runtime arguments. It does not expose a posterior, the parameter, or an offline occupation array to the machine. Apply the same construction at each declared buffer cut. A product of finitely many stochastic row simplices is compact; closed legality and sharing constraints preserve compactness. Its continuous image is $\mathcal{C}_{\mathbf{K}}$.

The future execution starting at a state s defines $B_{t,s}$ and satisfies (22). Conversely, induction from the terminal continuation and these identities gives the product formula. Thus the continuation description has exactly the same realizations. In particular, independently selected factorizations at different cuts cannot be spliced unless the shift equations hold. $\square$

This is a finite controlled positive-realization statement. The role of positivity and invariant generating cones in stochastic realization is classical [17, 18, 19]. Ordinary linear rank need not be a state count. Here the additional statistical question is which positive realizations meet an optimal saddle face. Proposition 8.1 provides a normal form, not an efficient factorization algorithm. It also does not compactify an unbounded number of observable operations.

8.2. The realization theorem. Let

$$U = \max_{C \in \mathcal{C}} \min_{\theta} g_{\theta}(C), \quad V_{\mathbf{K}} = \max_{C \in \mathcal{C}_{\mathbf{K}}} \min_{\theta} g_{\theta}(C).$$

By compact convex minimax there is a least-favorable prior μ_* with $U = \max_{C \in \mathcal{C}} \int g_{\theta}(C) d\mu_*$. Put

$$\begin{aligned} \ell_*(C) &= \int g_{\theta}(C) d\mu_*, \\ F_* &= \{C \in \mathcal{C} : \ell_*(C) = U\}, \\ \mathcal{S}_* &= F_* \cap \bigcap_{\theta} \{C : g_{\theta}(C) \geq U\}. \end{aligned} \tag{23}$$

The set F_* is an exposed face. The second intersection is important: a Bayes best response to a least-favorable prior need not be minimax.

Here $\mathcal{C}$ is the full-history convex reference for the finite protocol. A nonconvex restriction, such as reuse of one stochastic table at several visits, is imposed on $\mathcal{C}_{\mathbf{K}}$, not inserted into $\mathcal{C}$ before invoking convex minimax. In that case U is a reference upper value unless exactness for the unrestricted physical class is separately known. In the reset applications below that exactness is supplied by Theorems 5.1 and 5.2.

Proposition 8.2 (Bayes loss and worst-case loss). *Under the preceding assumptions,*

$$U - V_{\mathbf{K}} = \min_{C \in \mathcal{C}_{\mathbf{K}}} \left\{ U - \ell_*(C) + \ell_*(C) - \min_{\theta} g_{\theta}(C) \right\}. \tag{24}$$

Both summands are nonnegative and the minimum is attained.

Proof. The prior average is at most U and at least the minimum over fixed parameters. The sum in braces equals $U - \min_{\theta} g_{\theta}(C)$. Minimizing on the compact realization set proves the assertion. $\square$

Theorem 8.3 (Saddle-face contact and compatible realization). *Under the preceding assumptions, $V_{\mathbf{K}} = U$ if and only if some array in $\mathcal{S}_*$ has compatible positive continuation generators of profile at most $\mathbf{K}$. If no such generators exist, $V_{\mathbf{K}} < U$ strictly.*

For $C \in \mathcal{S}_$ one has $g_\theta(C) = U$ on $\text{supp } \mu_*$. At an interior supporting parameter at which the payoff is differentiable, $\nabla_\theta g_\theta(C) = 0$. These equalities and stationarity conditions hold also in directions whose Bayes coefficient vanishes.*

At any reset continuation cut, let R_h denote the conditional channel of future outputs under each externally specified future input word. It factors as

$$(25) \quad R_h = \sum_{s=1}^{K_t} E(s \mid h) B_s, \quad E(\cdot \mid h) \in \Delta_{K_t},$$

with one common family of continuation channels B_s . Any obstruction to this positive factorization for every member of $\mathcal{S}_$ is therefore a memory obstruction for every admissible architecture, regardless of its training wiring.*

Proof. By Proposition 8.2, the two nonnegative losses vanish exactly when C belongs to $\mathcal{S}_*$. Proposition 8.1 then proves the realization criterion. Compactness makes nonintersection a strictly positive value loss, not merely failure of one optimizer.

For an optimal array, $g_\theta(C) - U$ is continuous, nonnegative, and has zero μ_* integral. It is zero on the support of that measure. Every interior zero is a local minimum, so its gradient vanishes. No strict-sign hypothesis on individual coordinates is needed. Finally condition on the actual retained state at a reset cut. Given that state, the future program receives the same specified inputs and has the same output law, irrespective of the discarded past. Averaging gives (25). All persistent random variables are part of that state. The assertion can equivalently be checked on controller input tapes, using the generators in (22); it does not assume that a hidden physical state resets at a cut where the interface has not reset it. $\square$

The identity (24) is an exact loss decomposition, not a new interchange of a minimum and maximum over nonconvex machines. Its use is that an unrestricted certificate identifies a finite face, while contact at its supporting parameters can determine the remaining coordinates. The resulting channels are then tested for causal realizability. This is stronger than choosing an arbitrary Bayes tie rule and counting the states of that one rule.

Corollary 8.4 (Explicit coefficient form for frozen validation). *In the reset experiment of (7), write*

$$C_{z\omega}^* = \int P_\theta^{\otimes N}(z)(Q(\omega) - P_\theta(\omega)) d\mu_*(\theta).$$

The first summand of (24) is exactly

$$(26) \quad \sum_{C_{z\omega}^* > 0} C_{z\omega}^* (1 - h_z(\omega)) + \sum_{C_{z\omega}^* < 0} |C_{z\omega}^*| h_z(\omega).$$

Thus optimality fixes every strict-sign coordinate, while the support contact and stationarity equations must still be imposed at the zeros. For K labels at the post-training decision cut, an arbitrary upstream encoder gives the compact relaxation

$$(27) \quad h_z(\omega) = \sum_{s=1}^K E(s \mid z) D_s(\omega), \quad E \in (\Delta_K)^{\Omega^N}, \quad D_s \in [0, 1]^\Omega.$$

If this relaxation misses the saddle set, its strictly positive gap bounds every smaller architectural class.

Proof. Maximizing each coefficient separately gives the sum of its positive part. Subtraction gives (26). Conditioning on the decision state gives (27); the post-cut randomization is independent of the discarded word. Conversely an arbitrary encoder and a response list realize this factorization when other memory is unrestricted. The parameter spaces in (27) are compact, so the final assertion follows from Theorem 8.3. $\square$

8.3. Faces as memory witnesses. For a finite input set $\mathcal{Y}$ and output alphabet $\mathcal{D}$, continuation channels form the polytope $\mathcal{P} = \prod_{y \in \mathcal{Y}} \Delta(\mathcal{D})$. Its vertices are deterministic output functions. The analogous polytope for scalar test responses is $[0, 1]^{\mathcal{Y}}$.

Lemma 8.5 (Vertex and face packing). *Suppose channels to be factored through one common state alphabet include s distinct vertices $v_1, \dots, v_s$ of a polytope $\mathcal{P}$ and rows $r_1, \dots, r_j$. Suppose $r_i \in F_i$, the faces F_i are pairwise disjoint, and*

$$r_i \notin \text{conv}\{v_a : v_a \in F_i\}.$$

Every factorization of these rows as stochastic mixtures of generators in $\mathcal{P}$ needs at least $s + j$ generators.

Proof. A convex decomposition of a vertex uses that same vertex at every positive coefficient. Hence all s vertices must occur among the generators. If a convex combination belongs to a face, all generators with positive coefficients belong to that face. The stated exclusion therefore forces an additional generator in each F_i . Pairwise disjointness prevents any such generator from serving two faces. $\square$

The lemma does not assert that a stochastic state must itself be a history residual. Its generator can be any channel in the ambient polytope. This is why the argument also excludes randomized encoders and randomized continuations. It is a positive-factorization argument, not ordinary matrix rank or a restriction to a deterministic compiler.

Proposition 8.6 (Resource equivalence and stable obstructions). *A causal, cut-preserving recoding with a causal inverse, both using only one transducer state and preserving the payoff, preserves every $V_{\mathbf{K}}$, U , and the set of profiles attaining U . More generally suppose simulations in both directions have defect at most ϵ and require at most R_t states at cut t . For rewards in $[-1, 1]$,*

$$V_{\epsilon}(\mathbf{RK}) \geq V_{\mathfrak{F}}(\mathbf{K}) - 2\epsilon, \quad V_{\mathfrak{F}}(\mathbf{RK}) \geq V_{\epsilon}(\mathbf{K}) - 2\epsilon.$$

For a common controller class and training-validation kernels whose values differ by at most η , a gap $U - V_{\mathbf{K}} = \delta$ remains at least $\delta - 2\eta$ after the perturbation.

Proof. Transport a controller through each simulator and retain the product of its state and the simulator state. Apply the uniform bounded-reward variation bound and optimize. With an exact one-state inverse the two inequalities are equalities at the same profile. In the last assertion both the unrestricted and the constrained optimum change by at most η ; subtract their two bounds. The state products must be applied at the actual buffer cuts, not only at the terminal decision. $\square$

These statements place the saddle face, the future-test quotient, and resource-aware causal morphisms in one theorem chain. The channel factorizations are presentation-independent under the stated causal recodings. The memory obstruction in the next section is derived from the experimental coefficients themselves, not from a metric dimension or a covering exponent supplied as an additional hypothesis.

9. FACIAL CERTIFICATES FOR SCHEDULE-OPTIMIZED MEMORY

A schedule is part of the resource problem. The following criterion extracts a whole family of schedule converses from a saddle, rather than minimizing the states of one selected program. Its hypotheses are finite identities of continuation functions and faces. Its conclusion applies to arbitrary stochastic state encoders.

Consider finitely many validation tapes, with a fixed internal order on each tape and finitely many symbols per slot. A *declared-clock schedule* is a word specifying which tape supplies each successive slot; it is fixed before training and is independent of all reports. Let $\mathcal{G}$ be the finite acyclic graph of consumed-prefix counts. Its directed paths from the initial vertex to the final vertex are the legal schedules. This includes all interleavings, not only serial ones. The lower bound below does not give an adaptive scheduler free storage: a report-dependent choice of the next tape belongs to a different controlled interface and requires its own charged rows.

At a vertex v of $\mathcal{G}$, a retained state supplies a conditional channel on the unread tape suffixes. Assume in this section that every prefix and suffix combination used in a witness has positive probability under a supporting experiment. One may instead restrict to a stated common suffix subalphabet with this property. The channels form a product of probability simplices, denoted by $\mathcal{P}_v$.

Definition 9.1 (Contact-forced facial certificate). A certificate at v consists of distinct deterministic channels $d_1, \dots, d_s$ and channels $r_1, \dots, r_j$, such that every minimax-optimal auditor has these rows among its conditional continuations, allowing parameter-independent averaging of specified past rows. Each r_i lies in a face F_i of $\mathcal{P}_v$, the faces are pairwise disjoint, and

$$r_i \notin \text{conv}\{d_a : d_a \in F_i\}.$$

Its certified capacity is $s + j$. Define $c(v)$ to be the largest such capacity within a declared finite witness family. Conditional averaging must retain the same future-input coordinates and may not add the unknown parameter as an input. A channel of mean zero is not thereby a deterministic zero channel.

For an upper certificate choose a minimax-optimal finite randomization over frozen deterministic events, with the selected event retained at the decision cut. Let $\mathcal{D}_v$ be its distinct future deterministic output functions after all possible supported prefixes at v . When the input support is a full product, equality of these functions is closed under every next-symbol restriction. Put $r(v) = |\mathcal{D}_v|$. More general supported experiments may use any explicitly supplied shift-compatible deterministic completion; merely minimizing separate partial-function covers at different cuts is insufficient.

Theorem 9.2 (Facial capacity and schedule realization). *Suppose contact with the unrestricted saddle supplies the preceding certificates. For each declared-clock path π and every stochastic auditor on that path attaining the unrestricted value,*

$$(28) \quad W(\pi) \geq \max_{v \in \pi} c(v).$$

A shift-compatible deterministic residual family realizes the selected optimal event rule simultaneously with widths $r(v)$ along any such path. If its training and selection peak is W_0 , then

$$(29) \quad \min_{\pi} \max_{v \in \pi} c(v) \leq W_{\text{clk}}^{\min} \leq \min_{\pi} \max\{W_0, \max_{v \in \pi} r(v)\}.$$

In particular, a certificate and a realizing path with the same peak identify the exact minimum over all declared-clock schedules and all stochastic architectures on those schedules. If $c(v) = r(v)$ at every vertex and W_0 does not exceed the path peak, the entire minimum-profile problem reduces

to the intrinsic capacity field on $\mathcal{G}$. The peak optimum of that field is computed by

$$(30) \quad b(v_0) = c(v_0), \quad b(v) = \max\{c(v), \min_{u \rightarrow v} b(u)\}.$$

The result is invariant under bijective, causal, slot-preserving recodings of the tape alphabets.

Proof. Condition on a charged retained state at v . Every past row is a stochastic mixture of the same at most $W(\pi)$ future channels. Parameter-independent averaging preserves this factorization. The face-packing lemma therefore gives at least $s + j$ states for each certificate, proving (28). This step ranges over all state wirings and all stochastic encoders, not a specified graph.

For the upper bound retain the current residual output function. If two residuals agree on the future input product, their restrictions after each legal next symbol agree as well. Therefore the next retained label is determined by the old label and that symbol. At the last cut it is the output itself. This constructs all cuts compatibly in one machine. The initially sampled event and its training selector are charged in W_0 . Optimizing the two bounds over the finite path set proves (29). The recursion follows by classifying paths according to their last edge and taking the maximum with the cost of the final vertex. Induction proves it on the acyclic graph. Finally, a bijective recoding induces an affine isomorphism of every channel polytope, preserving vertices, faces, conditional averages and restriction maps. It preserves both certificates and realizing paths. $\square$

The general positive-realization normal form and residual quotient have classical antecedents [17, 18, 19]. The minimization of partially specified machines likewise requires compatible coverings, not unrelated cutwise minima [20]. The contribution used here is different: the witness rows are forced by *every member of the uniform saddle*, through contact on its Bayes-indifferent coordinates. This makes the lower field independent of a chosen optimal program. The criterion is a certificate theorem for this class of experiments, not an assertion that arbitrary nonnegative-rank problems have become easy or that facial packing is always exact.

10. EXACT MINIMAX LAWS FOR A MARKED PRODUCT FAMILY

The posterior-predictive dual can be solved explicitly for a continuous family containing the collision target. Throughout this section, the training and candidate validation law is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad (p, q) \in [0, 1]^2.$$

In the order 000, 001, 110, 111, let Q_γ have masses

$$((1 + \gamma)/4, (1 - \gamma)/4, (1 - \gamma)/4, (1 + \gamma)/4),$$

and zero mass elsewhere. Write $U_1(\gamma) = U_1(b, Q_\gamma)$, with exactly the training-validation convention of (7). Set

$$(31) \quad \gamma_* = 1 - 1/\sqrt{2}, \quad s_\gamma = \sqrt{\gamma^2 - 2\gamma + 5}, \quad u_\gamma = \gamma - 2 + s_\gamma, \quad \tau_\gamma = \frac{2u_\gamma}{u_\gamma + 2 - \gamma}.$$

Theorem 10.1 (Exact marked minimax family). *For every $0 \leq \gamma \leq \gamma_*$,*

$$(32) \quad U_1(\gamma) = \frac{3 + u_\gamma^2}{8}.$$

An optimal response depends on the training word only through the report count $K = X_1 + X_2$. On the four target atoms its rows are

$$(33) \quad h_0 = (\tau_\gamma, 0, 1, 1), \quad h_1 = (1, 1, 1, 1), \quad h_2 = (1, 1, 0, \tau_\gamma),$$

and all responses are zero off these atoms. A least-favorable prior is

$$(34) \quad \mu_\gamma = \frac{1}{2}\delta_{(t_\gamma, t_\gamma)} + \frac{1}{2}\delta_{(1-t_\gamma, 1-t_\gamma)}, \quad t_\gamma = \frac{1 - \sqrt{u_\gamma}}{2}.$$

The assertion optimizes over all stochastic responses to the full marked training word, not merely over count-based or symmetric responses.

Proof. First, u_γ is the positive root of

$$(35) \quad u^2 + (4 - 2\gamma)u - (1 + 2\gamma) = 0.$$

On the asserted interval, $0 < u_\gamma < 1$ and $0 < \tau_\gamma < 1$. Indeed the polynomial is negative at zero and positive at one; also $u_\gamma < 2 - \gamma$ because $\gamma < 1/2$. Substitution of $1 - 2\gamma$ into the polynomial gives $8(\gamma^2 - 2\gamma + 1/2)$, which is nonnegative precisely on the present interval. Monotonicity on $[0, \infty)$ therefore gives

$$(36) \quad u_\gamma \leq 1 - 2\gamma.$$

Put $a = (1 - p)(1 - q)$, $c = pq$, and $z = 1 - a - c$. The probabilities of the three training counts are a, z, c . The independent training mark may be discarded by the displayed construction. Let $G_\gamma(p, q)$ denote its score. With $u = (p + q - 1)^2$ and $v = (p - q)^2$, direct expansion gives

$$(37) \quad G_\gamma(p, q) - \frac{3 + u_\gamma^2}{8} = \frac{2 - \tau_\gamma}{16} \{(u - v - u_\gamma)^2 + 4v\}.$$

For clarity, this identity follows by substituting the three rows of (33) into

$$a(Q_\gamma - b)h_0 + z(Q_\gamma - b)h_1 + c(Q_\gamma - b)h_2,$$

using $a + c = (1 + u - v)/2$, $a - c = -(p + q - 1)$, and then eliminating u_γ^2 by (35). The coefficient of the resulting square is $(2 - \tau_\gamma)/16$ and that of v is four times this coefficient. Both are nonnegative. Equality holds at both points of (34), since there $v = 0$ and $u = u_\gamma$. This proves the lower bound and identifies its equality set.

It remains to bound every response, including nonsymmetric ones. Use (34) in Theorem 5.1. At $t = t_\gamma$ put $a = (1 - t)^2$, $c = t^2$, and $T = a + c = (1 + u_\gamma)/2$. For the training report pair 00, the four dual coefficients, apart from a common positive factor, are

$$(38) \quad \left(Q_H T - \frac{a^2 + c^2}{2}, \quad Q_L T - \frac{a^2 + c^2}{2}, \quad Q_L T - ac, \quad Q_H T - ac \right),$$

where $Q_H = (1 + \gamma)/4$ and $Q_L = (1 - \gamma)/4$. The first coordinate is zero by (35); the second is $-\gamma T/2 \leq 0$. The last two are positive: $T \geq 1/2$, $ac \leq 1/16$, and $\gamma < 1/2$ imply $Q_L T > 1/16 \geq ac$. Thus $(\tau_\gamma, 0, 1, 1)$ maximizes this linear functional. For the training pair 11, reversing all report bits and the mark reverses the four coordinates and gives the row h_2 .

For either training pair 01 or 10, the two prior support points have the same likelihood $t(1 - t)$. The posterior is still equally weighted, and the candidate predictive mass at each target atom is $T/4$. Every one of the four coefficients is therefore nonnegative by $Q_L - T/4 = (1 - 2\gamma - u_\gamma)/8 \geq 0$. The response h_1 is optimal. At off-diagonal validation atoms the target mass vanishes and the coefficient is nonpositive. Finally, conditioning on either training mark multiplies all these coefficients by $1/2$; it creates no further optimizing choice. We have therefore optimized every coordinate of the full marked-word response space.

The prior average of the displayed response equals its value at either support point, namely $(3 + u_\gamma^2)/8$, by (37). Every other response has prior average at most this number and hence has minimum over fixed (p, q) at most this number. Together with the lower bound this proves (32), optimality of the response, and least favorability of the prior. $\square$

Corollary 10.2 (The exact one-preparation collision value). *For $Q_* = Q_{1/4}$,*

$$(39) \quad U_1(b, Q_*) = u_{\text{ex}} := \frac{85 - 7\sqrt{73}}{64}, \quad u_{1/4} = \frac{\sqrt{73} - 7}{4}, \quad \tau_{1/4} = 2 - \frac{14}{\sqrt{73}}.$$

If only the target changes to Q , then $|U_1(b, Q) - u_{\text{ex}}| \leq \text{TV}(Q, Q_)$.*

Proof. Substitute $\gamma = 1/4$ in Theorem 10.1. For a fixed trained response the target perturbation is at most total variation, uniformly in (p, q) . Taking first the minimum and then the maximum preserves this bound in both directions. $\square$

10.1. Complementary slackness and the event alphabet. The tie in the first coordinate of (38) explains the randomization. Any value of that response coordinate is Bayes-optimal for the two-point prior. The minimax choice makes the uniform payoff stationary at its two support points; solving this condition gives τ_γ and the square identity (37). This derives the prior and response together, rather than postulating that a numerically effective prior is optimal.

Operationally, for $K = 0$ one draws once between the events with indicator vectors $(0, 0, 1, 1)$ and $(1, 0, 1, 1)$, with probabilities $1 - \tau_\gamma$ and τ_γ . For $K = 2$ one draws between $(1, 1, 0, 0)$ and $(1, 1, 0, 1)$. For $K = 1$ the event is $(1, 1, 1, 1)$. The resulting event identity is retained and used for both fresh validations. These are exactly the five events in the two-preparation construction below. The one-preparation theorem supplies their complementary-slackness interpretation; it does not prove that the two-preparation selector is optimal. The four-point rational prior retained below is a useful rational certificate, not a least-favorable prior.

The exact response uses an atomic stochastic row. It does not receive free persistent randomness. Replacing τ_γ by a dyadic number within 2^{-b} changes each conditional response, and hence the score, by at most 2^{-b} . A sequential fair-bit implementation must retain its bit prefix and event identity. None of these implementation costs is needed by the deterministic two-preparation physical audit.

Proposition 10.3 (The zero-preparation endpoint). *For $0 \leq \gamma \leq 1$, $U_0(b, Q_\gamma) = \gamma/2$.*

Proof. The fixed event $\{000, 111\}$ has score $(1 + \gamma)/2 - (a + c)/2 \geq \gamma/2$ because $a + c \leq 1$. Conversely, the equally weighted prior on $b(0, 0)$ and $b(1, 1)$ has predictive validation law uniform on the four target atoms. Its total variation from Q_γ is $\gamma/2$. Theorem 5.1 with $N = 0$ gives the matching upper bound. $\square$

11. A SELF-CONTAINED COLLISION INTERFACE

We give the preparation, mechanical estimate, observation law and private simulator class needed for the physical specialization. The following argument does not appeal to an unpublished physical lemma. It retains the same positive-volume preparation and positive sensor noise as the earlier collision construction.

Definition 11.1 (The reset row–event architecture). A preparation supplies two sequential report slots and, afterwards, an actual mark. The source reports in the two slots are constant. A private width-one simulator has no retained information between those slots and no persistent hidden seed. It may use fresh independent coins. Its first emission is $X_1 \sim \text{Ber}(p)$; after receiving the current gate $A \in \{0, 1\}$, its second emission has a fixed Bernoulli kernel with parameter q_A . The parameters (p, q_0, q_1) are fixed throughout the audit. The simulator does not observe the actual mark. All its coins are independent of that mark. Different preparations reset all unretained variables and are independent conditional on the parameters.

The auditor observes the first report, chooses a gate using its retained state and that report, observes the second report, and then observes the mark. During training it may choose rows and

stop adaptively, using at most the declared number of candidate preparations. It next freezes one gate policy and one event, with any retained random choice included in its register. It uses that same row–event continuation on one fresh candidate preparation and one independent target preparation and returns the difference of their event indicators, target minus candidate. It may not reselect the event after either validation. Independent target-only calibration is allowed but all retained calibration and randomness are charged. Peak width counts every bit, mark, event and validation-accumulator cut. Clocked schedules are declared externally; an autonomous implementation must retain its phase. An unretained transcript, seed or stopping-time value is not available to a continuation.

In the all-on row the complete private law is $b(p, q_1)$, with an independent fair actual mark. For lower bounds it suffices to restrict the adversary to $q_0 = 0$, $q_1 = q$. In this subfamily every deterministic gate policy is the pushforward

$$T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$$

of $b(p, q)$. The gate, when included in the transcript, is another function of these same coordinates. Randomized gate policies are covered by averaging their independent random tables. This proves both the all-on product characterization for the upper construction and the common-garbling subfamily used for the outer converse. The width-one constraint is a simulator constraint; it is not an uncharged restriction on the auditor's register.

11.1. Preparation and elastic scattering. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$, with no walls and contact distance d . Write $r = q_1 - q_2$, $g = v_1 - v_2$, $V = (v_1 + v_2)/2$, and let the center of mass position range over any fixed rational box of positive volume. Prepare $(B, W) \in \{0, 1\}^2$ with probability $5/16$ at $B = W$ and $3/16$ at $B \neq W$. Conditional on (B, W) , use the uniform product preparation on the following boxes:

$$(40) \quad \begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & & & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

Thus the actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$, preserved by momentum conservation. Set

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector, identically zero on the incoming transverse-velocity range.

Lemma 11.2 (Uniform collision signal). *For every preparation in (40) and every $9/10 \leq d \leq 11/10$, exactly one collision occurs and its time lies in $(4/5, 3/2)$. Afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4.$$

For $0 < d \leq 2/5$ no collision occurs on $[0, 3]$ and the detector is zero throughout that interval.

Proof. Before contact, $r(t) = r + tg$. At $t \leq 4/5$ its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At $t = 3/2$ its absolute coordinate bounds are $1/4$, $113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. The first contact therefore occurs in the claimed interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$. Hence the entry contact has $n_x < 0$, where $n = r(t_c)/d$. Its remaining coordinates give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq 87/220.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The elastic rule gives

$$v_1^+ = V + g/2 - (g \cdot n)n, \quad g^+ = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v_{1,y}^+ > -3/200 + (1443/1000)(87/220) > 1/2.$$

The definition of h implies the signal bound. Since $r(t_c) \cdot g^+ > 0$, squared separation increases strictly afterwards; there is no second collision. Finally, for $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. There is no contact, and the incoming $|v_{1,y}| \leq 3/200$ makes h zero. $\square$

11.2. Positive noise and the actual mark. Let $\mathcal{B}_t$ be Brownian motion independent of the preparation and set

$$Y_t = \int_0^t h(v_{1,y}(s)) ds + \sigma \mathcal{B}_t, \quad 0 < \sigma \leq 1/24.$$

The two potential all-on reports are

$$Z_1 = \mathbf{1}_{\{2(Y_2 - Y_{3/2}) > 0\}}, \quad Z_2 = \mathbf{1}_{\{2(Y_3 - Y_{5/2}) > 0\}}.$$

The actual second report is AZ_2 . The gate changes acquisition, not the mechanical path. The continuous path is implemented by the target sensor; it is not made available to the private simulator or supplied as an uncharged auditor buffer. Denote the law of (Z_1, Z_2, W) by $Q_{d,\sigma}$.

Theorem 11.3 (Uniform physical presentation). *For $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$,*

$$(41) \quad \text{TV}(Q_{d,\sigma}, Q_*) < 1/300.$$

The same estimate holds after every declared feedback gate, including its observable gate value. The all-on private product family and the common-garbling subfamily in Definition 11.1 apply to this very experiment.

Proof. Every collision occurs before $3/2$. Conditional on the preparation, the two normalized sensor windows are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. Lemma 11.2 makes their means have sign $2B - 1$ and magnitude greater than $1/4$. Each potential report therefore differs from B with probability less than $\Phi(-3)$. The Gaussian tail estimate

$$\Phi(-x) \leq \frac{e^{-x^2/2}}{x\sqrt{2\pi}}$$

gives $\Phi(-3) < 1/600$: $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$ suffice. Couple the ideal and noisy reports using the same preparation, actual mark and gate random table. Except on a set of probability less than $2\Phi(-3) < 1/300$, both potential reports equal B and the entire gated transcript agrees. The ideal marked law is precisely Q_* , by the prepared (B, W) table. This proves (41) and the feedback-uniform version. Finally, the source is blind, its actual mark is unobserved by the simulator, and the gate is its only current input at the second emission. The product and subfamily assertions follow from the two independent private-emission kernels specified above. $\square$

For $d \leq 2/5$, the two windows instead contain only independent centered Gaussian noise. Their signs are independent fair bits, independent of W , and a private width-one simulator reproduces every gated row exactly. Thus the informative regime and its no-collision control use the same detector, preparation convention and actual mark.

12. THE PHYSICAL EXPERIMENT AND ITS SHARP PREPARATION BOUNDARY

Take $\Omega = \{0, 1\}^3$, with coordinates (x_1, x_2, w) . The all-on private candidate is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad 0 \leq p, q \leq 1.$$

The mark w is the actual mark, not an invented simulator bit. Put $a = (1 - p)(1 - q)$ and $c = pq$. On the ordered atoms

$$(42) \quad 000, \quad 001, \quad 110, \quad 111,$$

let Q_* have masses $5/16, 3/16, 3/16, 5/16$, and set it to zero elsewhere. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. A target Q is allowed whenever $\text{TV}(Q, Q_*) \leq \beta$.

The original feedback operation has the form

$$(43) \quad T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w), \quad g : \{0, 1\} \longrightarrow \{0, 1\}.$$

The selected gate is observable. It may depend on earlier preparations and on the first current report. The target rows are pushforwards by (43). For the converse, restrict the adversary to the admissible private subfamily whose second emission is AX_2 , where A is the current gate and $X_2 \sim \text{Ber}(q)$ is independent of the first emission and of the fair mark. Every row is then a causal garbling of $b(p, q)$. For the construction only the all-on row is used, and every private candidate in the original width-one class has the all-on product law just specified. These two facts, proved for the microscopic interface in Definition 11.1 and Theorem 11.3, explain why using a subfamily for the lower bound does not narrow the upper construction.

The admissible audit first trains, then selects one row–event pair, then uses one fresh candidate and one fresh target validation. It reports $D = \mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The event and row cannot be reselected after seeing a validation draw. All bit, mark, decision and validation cuts are charged. Define $N_c^{\text{phys}}(W, Q)$ to be the minimum number of candidate training preparations among these original-interface auditors, with peak register width at most W , whose score is uniformly strictly greater than c . Randomization, feedback and stopping are allowed. If no such audit exists, the value is $+\infty$.

Theorem 12.1 (Two-preparation physical segment). *Let u_1, m_2 be as in (1). For every $\beta \geq 0$, $\text{TV}(Q, Q_*) \leq \beta$, and*

$$W \geq 12, \quad u_1 + \beta \leq c < m_2 - \beta,$$

one has $N_c^{\text{phys}}(W, Q) = 2$. The two-preparation audit is deterministic, uses no acquired calibration and no random bits, and has the full profile

$$(44) \quad (1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$$

from the initial cut through its twelve bit/mark updates. In particular, its decision alphabet has five labels and its peak has twelve. The lower bound remains valid with no memory cap.

The proof occupies the next three subsections. The interval in the statement can be empty for large β ; no assertion is then made. For $\beta = 1/300$, it contains $c = 2/5$ strictly between its endpoints.

12.1. An exact rational certificate against one preparation.

Lemma 12.2. *Every original-interface auditor using at most one candidate training preparation has worst-case score at most $u_1 + \beta$, regardless of memory.*

Proof. By Theorem 5.2, it suffices to bound $U_1(b, Q_*)$ and then perturb the target. In (8) use the four fixed parameters

$$(p, q) = (1/8, 1/8), (1/5, 1/5), (4/5, 4/5), (7/8, 7/8)$$

with weights $1/16, 7/16, 7/16, 1/16$, respectively. This measure is not asserted to be least favorable. Since $p = q$ on its support, the training count $K = x_1 + x_2$ is sufficient. The independent training mark and the order of unequal bits contain no further information about this prior.

For $k = 0, 1, 2$ and the four validation atoms in (42), the signed coefficients in (8), after grouping by K , are the entries of

$$(45) \quad \frac{1}{40960000} \begin{pmatrix} -1977 & -1775177 & 2170423 & 3943623 \\ 1765554 & 191954 & 191954 & 1765554 \\ 3943623 & 2170423 & -1775177 & -1977 \end{pmatrix}.$$

For example, the coefficient in column j is

$$\sum_{i=1}^4 \mu_i \binom{2}{k} p_i^k (1-p_i)^{2-k} \left(Q_{*,j} - \frac{(1-p_i)^2 \mathbf{1}_{j \leq 2} + p_i^2 \mathbf{1}_{j \geq 3}}{2} \right).$$

This identity proves every entry of (45) by rational arithmetic. All coefficients at off-diagonal validation atoms are nonpositive. The sum of the positive entries in (45) is $4035777/10240000 = u_1$. Each response coordinate is in $[0, 1]$, so this sum bounds its prior-averaged score and hence its minimum over the four fixed candidates.

Replacing Q_* by Q changes the score of any trained response by at most β . The training law is a candidate law and is unchanged. Consequently the same prior bounds every score by $u_1 + \beta$. A procedure with no training, or one that stops before taking its permitted sample, is covered by ignoring the generated all-on observation. $\square$

12.2. A marked five-event selector. Take two independent all-on training preparations. Let S be the total number of ones among their four report bits; discard their marks. Choose the following events, represented by their indicators on (42) and with indicator zero at every off-diagonal atom:

$$(46) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & 0 \end{array}$$

Denote these indicators by $h_0, \dots, h_4$ and report $D = h_S(Y) - h_S(X)$ on independent validation draws.

Lemma 12.3 (Exact uniform score of the selector). *Against the ideal target, the selector (46) has uniform minimum score $m_2 = (12 - 3\sqrt{3})/16$. Against Q , its score is at least $m_2 - \beta$.*

Proof. The generating polynomial of S is

$$((1-p) + pz)^2((1-q) + qz)^2.$$

Writing $g_s = (Q_* - b(p, q))h_s$, its expected score is $G(p, q) = \sum_{s=0}^4 [z^s]((1-p) + pz)^2((1-q) + qz)^2 g_s$. Substitution of the five rows in (46) and collection of terms with $x = p + q - 1$, $y = p - q$, $u = x^2$ and $v = y^2$ gives

$$(47) \quad G(p, q) = H(u, v) = -\frac{u^3}{32} + \frac{3u^2v}{32} + \frac{3u^2}{16} - \frac{3uv^2}{32} - \frac{uv}{4} - \frac{3u}{32} + \frac{v^3}{32} + \frac{v^2}{16} + \frac{15v}{32} + \frac{7}{16}.$$

This is a polynomial identity, not an interpolation over a grid. Since $0 \leq u \leq 1$ and $v \geq 0$ on the parameter square,

$$\partial_v H(u, v) = \frac{3(u - v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}.$$

Thus $H(u, v) \geq H(u, 0) = (-u^3 + 6u^2 - 3u + 14)/32$. The derivative of the last expression is $(-3u^2 + 12u - 3)/32$. On $[0, 1]$ it changes sign from negative to positive at $u = 2 - \sqrt{3}$. Substitution gives $(12 - 3\sqrt{3})/16$. The minimizing values are feasible: take $p = q = (1 \pm \sqrt{2 - \sqrt{3}})/2$. This proves exactness of the minimum for this specified selector, not optimality of its score over all selectors with two preparations. Finally, for each s , the target expectation of $h_s \in [0, 1]^\Omega$ changes by at most β . Averaging over training proves the claimed robustness. $\square$

12.3. Causal compilation and the full register profile.

Lemma 12.4. *The selector and its two validations admit the deterministic profile (44). No selector, first-bit buffer or validation accumulator is outside the charged register. An autonomous phase-tagged realization uses at most 75 states.*

Proof. After each training report bit retain the count accumulated so far. The two marks are read and discarded. The first six updates therefore have widths 2, 3, 3, 4, 5, 5 after the initial singleton. At the decision cut the five count values select the five events.

During candidate validation, after the first bit retain the event and that bit, using at most ten states. After the second bit retain the event and the function of the unread mark that gives its candidate-event indicator. For events 0, 1, 2, 3, 4, respectively, the possible functions are

$$\{0, 1\}, \quad \{0, 1 - w, 1\}, \quad \{0, 1\}, \quad \{0, w, 1\}, \quad \{0, 1\}.$$

There are twelve event–function pairs. After the candidate mark, retain the event and its now known candidate indicator $c_0 \in \{0, 1\}$, using at most ten states.

During target validation, after the first bit discard the event name and retain c_0 and its residual function of the second bit and mark. These residual functions belong to

$$\{0, x_2, (1 - x_2)(1 - w), 1 - x_2, x_2 w\}.$$

This uses at most ten states. After the second bit, the residual *difference* as a function of the unread mark belongs to

$$\{-1, 0, 1, w, 1 - w, w - 1, -w\},$$

so seven states suffice. The mark update gives the three possible outputs $-1, 0, 1$. Each transition depends only on its retained residual and the current report. Thus it is a legal common observable row, not a family of independently optimized hidden-history maps.

For completeness, equality of two residual functions implies equality of their restrictions after either next bit, so identifying equal residuals is closed under every subsequent update. The supplied transition tables are obtained by these restrictions and can be reconstructed directly from (46). Enumerating residuals is a convenience for reproducing the tables, not the proof of their sufficiency. Finally, disjointly tag the states at the thirteen cuts by their phase and use terminal self-loops. The sum of the displayed widths is 75. This implementation charges the clock; it does not assert that 75 is optimal. $\square$

Proof of Theorem 12.1. Lemma 12.2 excludes $N \leq 1$ for a strict score greater than $c \geq u_1 + \beta$, for every architecture and every memory budget. Lemmas 12.3 and 12.4 supply a two-preparation architecture of peak twelve with score at least $m_2 - \beta > c$. Larger width budgets can use the same machine. $\square$

Corollary 12.5 (The positive-noise collision family). *For the original marked collision targets with $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the local microscopic estimate $\text{TV}(Q_{d,\sigma}, Q_*) < 1/300$ implies*

$$N_{2/5}^{\text{phys}}(W, Q_{d,\sigma}) = 2 \quad (W \geq 12).$$

There are four preparations in the complete audit: two candidate training, one candidate validation, and one target validation.

Proof. Apply Theorem 12.1. The lower-end margin is the exact rational number

$$\frac{2}{5} - u_1 - \frac{1}{300} = \frac{78269}{30720000} > 0.$$

For the other endpoint, $\sqrt{3} < 7/4$ implies $m_2 > 27/64$ and $27/64 - 1/300 > 2/5$. The physical total-variation estimate, including the actual mark and positive sensor noise, is proved in Theorem 11.3; the current deduction does not replace those hypotheses by a zero-noise limit. $\square$

Remark 12.6 (The part of the frontier determined here). The deterministic theorem evaluates the minimum preparation coordinate for $W \geq 12$. Theorem 13.1 and Corollary 13.2 below additionally determine the ideal maximum score with two preparations on this width range. The least peak width over all validation orders and the full boundary for $3 \leq W \leq 11$ remain separate questions. Theorem 15.3 below determines the peak for forward serial validation, and Theorem 15.1 determines the order-independent decision cut. The earlier exact decision-width result $\kappa_{2/5}^{\text{rows}} = 3$ is unchanged: it allows longer training, whereas this shortest audit uses five decision states. The older points (399, 40602), (320000, 1604) and $(12800 \cdot 2^{400}, 12)$ remain valid constructions with decision width three. They are dominated by (2, 12) in the projection onto training preparations and peak width, but not necessarily in a resource order that also constrains the decision alphabet separately.

Corollary 12.7 (The enlarged physical preparation segment). *With the full architecture of Definition 11.1,*

$$N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_{\text{ex}} + \beta \leq c < m_2 - \beta.$$

Here $u_{\text{ex}} = (85 - 7\sqrt{73})/64$ is the exact one-preparation all-on value, rather than the older rational upper bound.

Proof. Restrict the adversary to the common-garbling subfamily in Definition 11.1. Theorem 5.2 and Corollary 10.2 bound every at-most-one-preparation auditor, even with unlimited memory, by $u_{\text{ex}} + \beta$. The same deterministic two-preparation machine in Lemmas 12.3 and 12.4 has score at least $m_2 - \beta$ against the full private candidate family. These inequalities prove the assertion. The four-preparation total, peak twelve and positive-noise specialization are unchanged. $\square$

13. THE EXACT TWO-PREPARATION MARKED SADDLE

The one-preparation saddle has an exact successor. A new zero coefficient of the least-favorable predictive measure makes it necessary to randomize at the two extreme training counts. This randomization changes the value, but does not enlarge the alphabet of frozen validation events.

Let $b(p, q)$ and Q_* be as in Section 12. Let r be the unique root in $(1/3, 7/20)$ of

$$(48) \quad 2r^3 + 25r^2 - 3 = 0,$$

and define

$$(49) \quad \tau = \frac{6(-r^2 + 4r - 1)}{r(3r + 25)}, \quad v_2 = \frac{-r^3 + 6r^2 - 3r + 14}{32}, \quad p_{\pm} = \frac{1 \pm \sqrt{r}}{2}.$$

The polynomial in (48) is strictly increasing on the positive half-line and has opposite signs at the indicated endpoints. These observations prove the asserted existence, uniqueness and isolating interval. That interval also gives $0 < \tau < 1$.

Theorem 13.1 (Exact two-preparation value). *For the full marked-word response class of (7),*

$$U_2(b, Q_*) = v_2.$$

A least-favorable prior gives mass $1/2$ to each of (p_-, p_-) and (p_+, p_+) . An optimal response discards the two training marks, retains the total number S of ones in the four training report bits, and uses

$$(50) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & \tau & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & \tau \end{array}$$

with zero response at off-diagonal validation atoms. Numerically, $r = 0.3417694155\dots$, $\tau = 0.1688236447\dots$ and $v_2 = 0.4261127787\dots$. The exact value is determined by (48)–(49), not by these decimals.

Proof. Write $a = (1 - p)(1 - q)$, $c = pq$, $u = (p + q - 1)^2$ and $v = (p - q)^2$. Both u and v belong to $[0, 1]$. The score of the deterministic five-event selector is the polynomial $H(u, v)$ in (47). Only counts zero and four are changed by (50). Their probabilities are a^2 and c^2 . Consequently the new score is

$$G(u, v) = H(u, v) + \tau J(u, v), \quad J = a^2(5/16 - a/2) + c^2(5/16 - c/2).$$

Expansion gives the polynomial identity

$$(51) \quad \begin{aligned} J(u, v) = & -\frac{u^3}{64} + \frac{3u^2v}{64} - \frac{25u^2}{128} - \frac{3uv^2}{64} + \frac{13uv}{64} \\ & + \frac{v^3}{64} - \frac{v^2}{128} - \frac{v}{32} + \frac{3}{128}. \end{aligned}$$

Thus

$$H(u, 0) = \frac{-u^3 + 6u^2 - 3u + 14}{32}, \quad J(u, 0) = \frac{3 - 25u^2 - 2u^3}{128}.$$

Equation (48) says $J(r, 0) = 0$, and the choice of τ says $\partial_u G(r, 0) = 0$. Direct factorization, using these two identities, yields

$$(52) \quad G(u, 0) - v_2 = \frac{(u - r)^2 \{75 + 12r + 6u - 37r^2 - 74ru\}}{64r(3r + 25)}.$$

The expression in braces is strictly positive for $0 \leq u \leq 1$: it is at least $75 - 37(7/20)^2 - 74(7/20) > 0$. Moreover,

$$\partial_v H = \frac{3(u - v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}, \quad \partial_v J = \frac{3(u - v)^2 + 13u - v - 2}{64} \geq -\frac{3}{64}.$$

Because $0 < \tau < 1$, $\partial_v G \geq 11/64$ on the unit square. It follows that $G(u, v) \geq G(u, 0) \geq v_2$, with equality precisely when $v = 0$ and $u = r$. Both equality points are feasible and are the two parameters in the asserted prior. This proves the global lower bound.

It remains to prove an upper bound against every response, not only those depending on S . Under the asserted prior the four training report bits are independent Bernoulli variables with common parameter p_- or p_+ . Conditional on $S = s$, their arrangements and the two

independent training marks have parameter-independent distributions. The signed predictive coefficients can therefore be grouped by S without restricting any response coordinate. For the four displayed validation atoms put

$$C_{s,j} = \frac{1}{2} \sum_{p \in \{p_-, p_+\}} \binom{4}{s} p^s (1-p)^{4-s} \{Q_{*,j} - b_j(p, p)\}.$$

The first three rows of $256C$ are

$$\begin{aligned} s = 0 : & \quad (-(2r^3 + 25r^2 - 3), -(2r^3 + 27r^2 + 12r - 1), \\ & \quad -(2r^3 - 5r^2 - 20r - 1), -(2r^3 - 7r^2 - 32r - 3)), \\ s = 1 : & \quad 4(r-1)(2r^2 + 7r - 3, 2r^2 + 9r - 1, \\ & \quad 2r^2 - 7r - 1, 2r^2 - 9r - 3), \\ s = 2 : & \quad -6(r-1)^2(2r-3, 2r-1, 2r-1, 2r-3). \end{aligned}$$

Rows three and four are the reversals of rows one and zero, respectively. Their signs, using $1/3 < r < 7/20$, are

$$(0, -, +, +), \quad (+, -, +, +), \quad (+, +, +, +), \quad (+, +, -, +), \quad (+, +, -, 0).$$

Every off-diagonal validation coefficient is nonpositive because its target mass is zero. Hence (50) maximizes every signed coefficient: it is one at positive coefficients, zero at negative ones, and arbitrary at the two zeros. The lower-bound equality at both prior points shows that this Bayes maximum is exactly v_2 . The posterior-predictive dual, Theorem 5.1, now gives the matching upper bound for the full response class. $\square$

Corollary 13.2 (Exact physical score on the peak-twelve segment). *Use the atomic stochastic-row convention and the original marked interface of Definition 11.1. At two candidate training preparations, every peak budget $W \geq 12$ has exact ideal worst-case score v_2 . For a target Q with $\text{TV}(Q, Q_*) \leq \beta$, the corresponding value lies in $[v_2 - \beta, v_2 + \beta]$. In particular, with $u_{\text{ex}} = (85 - 7\sqrt{73})/64$,*

$$(53) \quad N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad u_{\text{ex}} + \beta \leq c < v_2 - \beta.$$

Proof. Let $h_0, \dots, h_4$ be the five deterministic events in (46). At count zero select h_1 with probability τ and h_0 otherwise; at count four select h_3 with probability τ and h_4 otherwise. At the other counts retain the indicated event. Discard the count and retain the selected event for both validations. This realizes (50); the event alphabet remains five. The full charged profile, now including the separate event-selection update, is

$$(1, 2, 3, 3, 4, 5, 5, 5, 10, 12, 10, 10, 7, 3).$$

The common residual-function compiler in Lemma 12.4 applies unchanged. In particular, no persistent random selector is outside the register. The all-on law for every private candidate is $b(p, q)$, so Theorem 13.1 proves the constructive lower bound. For the converse restrict to the admissible common-garbling product subfamily and apply Theorem 5.2 followed by Theorem 13.1. This bounds all architectures, including those using feedback, stopping and more memory. Target perturbation changes each frozen-event expectation by at most β . Finally the exact one-preparation theorem excludes a strict score above c at $N \leq 1$, whereas the displayed machine achieves it at $N = 2$ under (53). $\square$

The root r is larger than $2 - \sqrt{3}$. The old diagonal polynomial $H(u, 0)$ is strictly increasing between these points, whence $v_2 > m_2$. The deterministic theorem, including its zero-random-bit implementation, remains valid; the new exact score uses an additional stochastic row.

Proposition 13.3 (Finite precision for the new row). *Replacing τ by a dyadic $\tau_b = k/2^b \in [0, 1]$ with $|\tau_b - \tau| \leq 2^{-b}$ loses at most $(5/16)2^{-b}$ in uniform score. A clocked fair-bit implementation uses b extra random bits and at most $\max\{12, 6(b+1) + 5\}$ persistent states, including their phase, in addition to the unchanged preparation count.*

Proof. At each affected training word the response changes at a single atom; $|5/16 - a/2|$ and $|5/16 - c/2|$ are at most $5/16$. The two affected word probabilities sum to at most one. This gives the score estimate. To implement the dyadic coin, compare b fresh fair bits lexicographically with the fixed b -bit integer k . Retain the origin (count zero or four), one of the three comparison flags, and a phase in $\{0, \dots, b\}$. Once the comparison ends retain its chosen one of five events. Unaffected counts can wait in their event states. These states total at most $6(b+1) + 5$; the other cuts have peak twelve. The cases $k = 0, 2^b$ are deterministic. Exact atomic randomization and a finite fair-bit implementation are therefore distinct resource claims. $\square$

14. A CONTINUOUS TWO-PREPARATION MARKED FAMILY

The preceding saddle persists on an explicit interval of target laws, with exact rather than perturbative values. Use the target Q_γ from Section 10 and write $U_2(\gamma) = U_2(b, Q_\gamma)$.

Theorem 14.1 (A two-preparation saddle family). *For $6/25 \leq \gamma \leq 13/50$, let r_γ be the unique root in $(1/3, 7/20)$ of*

$$(54) \quad f_\gamma(r) = r^3 + (13 - 2\gamma)r^2 + (3 - 12\gamma)r - (1 + 2\gamma) = 0.$$

Set

$$D_\gamma(r) = 3r^2 + (26 - 4\gamma)r + 3 - 12\gamma, \quad t_\gamma = -\frac{2\{3r_\gamma^2 + (8\gamma - 14)r_\gamma + 3\}}{D_\gamma(r_\gamma)}.$$

Then $0 < t_\gamma < 1$, and

$$(55) \quad U_2(\gamma) = \frac{-r_\gamma^3 + (7 - 4\gamma)r_\gamma^2 - 3r_\gamma + 13 + 4\gamma}{32}.$$

The response (50), with $\tau = t_\gamma$, is optimal. A least-favorable prior gives equal masses to the two diagonal parameters $p = q = (1 \pm \sqrt{r_\gamma})/2$. The response has the same five-event, peak-twelve atomic realization. At $\gamma = 1/4$ these formulas reduce to Theorem 13.1.

Proof. For brevity write $r = r_\gamma$, $D = D_\gamma(r)$ and $t = t_\gamma$. The derivative D_γ is positive on the rectangle $\gamma \in [6/25, 13/50]$, $r \in [1/3, 7/20]$. For example its lower bound is $1/3 + (26 - 52/50)/3 - 3/25 > 8$. The polynomial f_γ decreases with γ at a fixed positive argument. Direct rational evaluations give $f_{6/25}(1/3) < 0$ and $f_{13/50}(7/20) > 0$. Monotonicity therefore gives the root and its uniqueness in this interval. The numerator of t is $-16\gamma r - 6r^2 + 28r - 6$. It is bounded below by $28/3 - 6 - 16(13/50)(7/20) - 6(7/20)^2 > 0$, and above by $28(7/20) - 6 < 4 < D$. Thus $0 < t < 1$.

Let H, J be the polynomials from (47) and (51). Changing the target masses linearly in γ gives

$$\begin{aligned} H_\gamma(u, v) &= H(u, v) + (\gamma - 1/4) \frac{1 - (u - v)^2}{8}, \\ J_\gamma(u, v) &= J(u, v) + (\gamma - 1/4) \frac{(u - v)^2 + 6u - 2v + 1}{32}. \end{aligned}$$

The response's full-parameter score is $G_\gamma = H_\gamma + tJ_\gamma$. One has $J_\gamma(r, 0) = -f_\gamma(r)/64 = 0$ and $\partial_u G_\gamma(r, 0) = 0$. Algebraic division gives

$$(56) \quad G_\gamma(u, 0) - H_\gamma(r, 0) = \frac{(u - r)^2 A_\gamma(r, u)}{16D},$$

where

$$A_\gamma(r, u) = 24\gamma^2 + (3\gamma - 10)r^2 + (6\gamma - 20)ru \\ + 12\gamma r + 6\gamma u - 51\gamma + 30.$$

For $0 \leq u \leq 1$ this is at least $30 - 10(7/20)^2 - 20(7/20) - 51(13/50) > 0$. The transverse derivatives satisfy

$$32\partial_v H_\gamma = 3(u - v)^2 + (8\gamma - 10)u + (6 - 8\gamma)v + 15 \geq 5, \\ 64\partial_v J_\gamma = 3(u - v)^2 + (14 - 4\gamma)u + (4\gamma - 2)v - 4\gamma - 1 \geq -3$$

on $0 \leq u, v \leq 1$. Consequently $\partial_v G_\gamma \geq 7/64$. Equation (56) proves a lower bound $H_\gamma(r, 0)$, with equality precisely at the proposed prior points. This quantity is the right side of (55).

For the upper bound group the full training words by S , as in the preceding theorem. Put $L(r) = r^2 + 6r + 1$. Using $f_\gamma(r) = 0$, the first three rows of $256C$ are

$$\begin{aligned} 0: & (0, -8\gamma L(r), 32r(r+1) - 8\gamma L(r), 32r(r+1)), \\ 1: & 8(r-1)(r^2 + 4r - 1 - 2\gamma(r+1), \\ & r^2 + 4r - 1 + 2\gamma(r+1), \\ & r^2 - 4r - 1 + 2\gamma(r+1), r^2 - 4r - 1 - 2\gamma(r+1)), \\ 2: & -12(r-1)^2(r-1 - 2\gamma, r-1 + 2\gamma, r-1 + 2\gamma, r-1 - 2\gamma). \end{aligned}$$

The other rows are their reversals. The sign pattern is again

$$(0, -, +, +), \quad (+, -, +, +), \quad (+, +, +, +), \quad (+, +, -, +), \quad (+, +, -, 0).$$

Here are explicit interval checks for the potentially small margins. In row zero the third coordinate is bounded below by $32(1/3)(4/3) - 8(13/50)((7/20)^2 + 6(7/20) + 1) > 0$. The first parenthesis of row one is at most $(7/20)^2 + 4(7/20) - 1 - 2(6/25)(4/3) < 0$; the third is at most $(7/20)^2 - 4/3 - 1 + 2(13/50)(27/20) < 0$. The second parenthesis is positive already from $r > 1/3$ and $\gamma > 0$, and the fourth is negative. Finally $r - 1 + 2\gamma \leq -13/100 < 0$. All off-diagonal validation coefficients are nonpositive. Thus the stated response is a Bayes best response for this prior, including its two zero coefficients. Its score at both supporting parameters equals the lower bound. The full posterior-predictive dual proves (55). $\square$

This interval is a uniform certificate region for the displayed saddle, not a claim that the same masks remain optimal for every mark bias. The theorem supplies a continuous family at genuinely growing training history $N = 2$; it does not infer a recurrence for every N from the one-preparation formulas.

15. EXACT MEMORY FROM A MARKED SADDLE

The realization criterion is now applied to a saddle with genuine Bayes indifference. The zero coefficients in the two-preparation problem are not harmless: their values are fixed by contact with the parameter family. This supplies a converse for all optimal audits, rather than for a preselected response table.

Put $I = [6/25, 13/50]$. Throughout this section $r = r_\gamma$, $t = t_\gamma$, and $v = U_2(\gamma)$ are those of Theorem 14.1. Denote the five deterministic events in (46) by $e_0, \dots, e_4$. Responses displayed below are ordered on 000, 001, 110, 111 and vanish at the other four atoms.

Theorem 15.1 (Contact rigidity and exact decision memory). *For every $\gamma \in I$, every minimax optimal full-word response with two training preparations has, after averaging the independent training marks at the two extreme report counts, the five rows*

$$(57) \quad (t, 0, 1, 1), \quad (1, 0, 1, 1), \quad (1, 1, 1, 1), \quad (1, 1, 0, 1), \quad (1, 1, 0, t).$$

At the three interior counts the displayed response is forced at each individual training word, not merely after averaging.

In the original reset row–event interface, the least number of states at the post-training cut that permits the exact ideal score v is five. This converse allows arbitrary upstream finite memory, randomized encoding, adaptive training gates, stopping, independent target-only calibration, and all legal post-cut randomizations. Other cuts are unrestricted in this decision-memory assertion.

Proof. First consider a response to the full two-preparation word. Under the least-favorable prior of Theorem 14.1, every training word has positive probability. The strict coefficient signs force e_1, e_2, e_3 at report counts 1, 2, 3. At count zero only the 000 coordinate is free; at count four only the 111 coordinate is free. All off-diagonal validation coefficients are strictly negative: the prior parameters are interior, each word likelihood is positive, and the target assigns zero mass to those atoms. Thus those coordinates are zero for every optimal response.

Let a and b be the averages of the two free coordinates over the four equiprobable training mark words at their respective extreme counts. On the diagonal parameter (p, p) the score is

$$(58) \quad H_\gamma((2p-1)^2, 0) + aA_\gamma(p) + bB_\gamma(p),$$

where

$$A_\gamma(p) = (1-p)^4 \left(\frac{1+\gamma}{4} - \frac{(1-p)^2}{2} \right), \quad B_\gamma(p) = p^4 \left(\frac{1+\gamma}{4} - \frac{p^2}{2} \right).$$

Here H_γ is the deterministic five-event score in the proof of Theorem 14.1. At $p_+ = (1 + \sqrt{\gamma})/2$, $A_\gamma(p_+) + B_\gamma(p_+) = J_\gamma(r, 0) = 0$ and $A_\gamma(p_+) > 0$. The latter follows from $p_+ > 1/2$ and $(1 + \gamma)/4 > 1/4$. Reflection exchanges A and B at $p_- = 1 - p_+$. Also $H_\gamma(r, 0) = v$. Optimality forces contact at both prior points, by Theorem 8.3. Subtracting v from (58) therefore gives $(a - b)A_\gamma(p_+) = 0$, hence $a = b$.

With this common value the diagonal score is $H_\gamma(u, 0) + aJ_\gamma(u, 0)$, where

$$H_\gamma(u, 0) = \frac{-u^3 + (7 - 4\gamma)u^2 - 3u + 13 + 4\gamma}{32}, \quad J_\gamma(u, 0) = -\frac{f_\gamma(u)}{64}.$$

The supporting value r lies in $(0, 1)$, so the derivative at r vanishes. Since $f'_\gamma(r) = D_\gamma(r) > 8$, the derivative equation has the unique solution

$$a = b = \frac{64 \partial_u H_\gamma(r, 0)}{D_\gamma(r)} = t_\gamma.$$

This proves (57). Individual extreme mark words may still use different tie values; no unnecessary uniqueness is asserted.

Now take an arbitrary audit with K states at the decision cut. Restrict the candidate to the common-garbling subfamily in Definition 11.1. Generate two all-on words and simulate the adaptive training and its stopping with independent coins. The retained state is a parameter-independent stochastic function of those two words. Conditional on that state, the frozen row–event continuation pulls back to a response $D_s \in [0, 1]^8$ on the all-on alphabet. Any calibration is part of the same encoding and retained state. Thus (27) holds without increasing K . If the audit attains v against the full physical class, its pullback attains v against this subfamily, since the unrestricted upper bound is v . Averaging selected rows preserves their factorization.

The three interior rows in (57) are distinct vertices of $[0, 1]^8$, so the generators must include e_1, e_2, e_3 themselves. The first row lies in the face

$$F_0 = \{d : d(001) = 0, d(110) = d(111) = 1, d = 0 \text{ off the diagonal}\}.$$

Of those three obligatory vertices only e_1 lies in F_0 ; since $t < 1$, this row is not e_1 . It needs another generator in F_0 . Similarly the last row needs another generator in

$$F_4 = \{d : d(000) = d(001) = 1, d(110) = 0, d = 0 \text{ off the diagonal}\}.$$

Only e_3 among the obligatory vertices lies there. The two faces are disjoint. Lemma 8.5 gives $K \geq 3 + 2 = 5$. Conversely the optimal response mixes e_0, e_1 at count zero and e_4, e_3 at count four and otherwise selects the indicated event. Only its five-valued event label survives the cut. The inherited compiler realizes its subsequent validation with finite memory and attains v . This proves exactness of the decision count. $\square$

The contact argument is essential. Setting both free coordinates equal to one gives a Bayes best response with only the three events e_1, e_2, e_3 , but its derivative at r is nonzero because $t < 1$. It is not minimax. Counting the states of a Bayes best response alone would give an incorrect resource conclusion.

Corollary 15.2 (A strict gap and its physical persistence). *There is $\delta_* > 0$, uniform for $\gamma \in I$, such that every two-preparation audit with at most four states at the decision cut has ideal worst-case score at most $U_2(\gamma) - \delta_*$. In particular, this bound holds for every original-interface audit of whole-schedule peak at most four. If only the target changes by total variation at most $\beta < \delta_*/2$, the five-state decision construction still strictly outperforms this class.*

Proof. Let $W_4(\gamma)$ be the maximum worst-case score over the relaxed factorizations (27) with $K = 4$ and two full training words. Their encoder and decoder spaces are compact and fixed, independently of γ . The score is jointly continuous in these variables, the target bias, and (p, q) . Consequently W_4 is continuous. By Theorem 15.1 it is strictly less than $U_2(\gamma)$ for each $\gamma \in I$. The minimum of this positive continuous gap on compact I is positive. This defines δ_* without pretending to have evaluated it numerically. The domination argument in that theorem makes it an upper bound for every original-interface audit in question. Changing only the target changes any fixed audit's mean score by at most β . The difference between the two classes therefore decreases by at most 2β . $\square$

For the microscopic family at $\gamma = 1/4$, the signal is larger than $1/4$ and the standard deviation of each normalized sensor noise is $\sqrt{2}\sigma$. The Gaussian tail bound used in Theorem 11.3 gives the sharper, vanishing estimate

$$(59) \quad \text{TV}(Q_{d,\sigma}, Q_*) \leq \frac{8\sigma}{\sqrt{\pi}} \exp\{-1/(64\sigma^2)\}, \quad 9/10 \leq d \leq 11/10.$$

Thus Corollary 15.2 holds uniformly in d for all sufficiently small positive σ . This is an existence statement with the displayed physical modulus. The present proof does not certify that every $\sigma \leq 1/24$ lies below the unknown gap threshold.

15.1. The sharp serial-validation peak. A decision cut and the whole execution have different resource costs. We now determine the latter in a declared validation schedule. In the *forward serial class*, both training preparations finish before validation; the candidate validation's two report slots and mark are read before any target validation report is read. The same frozen row-event continuation is used on both preparations, as in Definition 11.1. Training gates, stochastic state updates and frozen gate policies are unrestricted within that interface. No intermediate output can be reread as external memory.

Theorem 15.3 (Sharp forward serial peak). *For every $\gamma \in I$, the least whole-schedule peak width attaining $U_2(\gamma)$ in the forward serial class is twelve. In particular, a peak of at most eleven in this class incurs a strictly positive loss, which can be chosen uniformly for $\gamma \in I$.*

Proof. Use the common-garbling subfamily and the two raw all-on training words as in Theorem 15.1. Any optimal audit must finish both training preparations almost surely under its least-favorable prior. Indeed a stopped branch after a first preparation would have a response independent of the unused second word. For each first report count 0, 1, 2, two choices of the unused word force two different interior vertices from e_1, e_2, e_3 . Each has positive conditional probability. A positive stopped component cannot be a convex component of two distinct vertices. The same argument excludes stopping before the first draw. Null branches may be completed arbitrarily.

Consider the cut after the candidate's second report slot and before its mark. On a canonical raw input tape its continuation is a channel from $(w, y) \in \{0, 1\} \times \{000, 001, 110, 111\}$ to $\{-1, 0, 1\}$, where w is the unread candidate mark and y the fresh target word. All these target words have positive probability. Each retained state supplies one such continuation channel, common to every past leading to it. Hence the rows indexed by selected pasts factor through at most the peak number of generators.

For the strict interior training counts, the following eight rows are forced deterministic channels:

$$(60) \quad e_1(y) - f(w), \quad f \in \{0, 1 - w, 1\}; \quad e_2(y) - f(w), \quad f \in \{0, 1\}; \quad e_3(y) - f(w), \quad f \in \{0, w, 1\}.$$

For example e_1 has candidate residual 0 after an off-diagonal report pair, $1 - w$ after 00, and 1 after 11. The other cases follow by restriction of their masks. Each listed past has positive probability under both supporting candidate parameters. The eight functions are distinct: subtracting two with the same mask compares their distinct w functions; for different masks their difference depends nontrivially on y and cannot equal a function of w alone. They are eight distinct vertices of the product of output simplices.

There are four additional required rows. Average the extreme training marks as in Theorem 15.1, and take the candidate raw report pair to be, respectively, off diagonal and 11 at count zero, and off diagonal and 00 at count four. Independence of a fresh raw candidate preparation from the frozen continuation preserves the averaged tie probability t . On the four target words these rows have means

$$(61) \quad (t, 0, 1, 1), \quad (t - 1, -1, 0, 0), \quad (1, 1, 0, t), \quad (0, 0, -1, t - 1),$$

for both values of w . More precisely, each integer coordinate here is a point mass at that integer, and the remaining coordinate is the Bernoulli mixture between the two neighboring integers. This stronger channel statement, not a statement about a zero mean alone, follows because the candidate indicator is fixed at 0 or 1 on the selected raw pair, while the target indicator is a bit. Fresh gate randomness is already part of this channel; it supplies no additional past-dependent argument to a retained state.

The four rows lie in four pairwise disjoint faces of the product of output simplices. These faces fix, for both w , respectively,

	000	001	110	111
$F_{0,+}$	*	0	1	1
$F_{0,-}$	*	-1	0	0
$F_{4,+}$	1	1	0	*
$F_{4,-}$	0	0	-1	*

where a number specifies a point mass and $*$ is unrestricted. Among the eight vertices (60), each face contains exactly one: e_1 , $e_1 - 1$, e_3 , and $e_3 - 1$, respectively. Since $t < 1$, none of the four required rows is that vertex. Face packing forces four more generators, for a total of twelve. This argument uses conditional future-output laws and not the continued storage of an event name after a residual has replaced it.

The optimal five-event selection followed by the inherited forward compiler has peak twelve, including its stochastic selection cut. It proves the matching upper bound. To see strictness below twelve, use the finite-protocol normal form of Proposition 8.1 with all logical cuts bounded by eleven, including selection and frozen-continuation buffers. Unobserved internal random computations can be collapsed to common transition kernels on the same register; an unretained random duration cannot be supplied as a later input. The normalized row spaces and closed frozen-interface restrictions are compact. If the constrained value equalled U_2 , the preceding face obstruction would apply. Continuity and compactness in γ give a uniform strictly positive gap, just as in Corollary 15.2. $\square$

Theorem 15.3 is a minimum-width theorem for a specified serial order, not a claim that an arbitrary validation order has the same minimum. The ideal target has support restrictions that can change future residuals when the target is read first. Reverse-order and interleaved validations require their own continuation analysis. The decision-memory converse in Theorem 15.1 does not make this serial-order assumption. Neither statement replaces the original physical interface by an uncharged atomic validation.

16. A FULL-SUPPORT MARKED SADDLE

The support restriction of the ideal target can be removed without replacing the private simulator or changing the training experiment. This section identifies an explicit region of positive targets on which both the statistical saddle and its decision-memory number are exact. The next section shows that the original Brownian sensor belongs to this region throughout its previously stated noise range.

Put

$$\mathcal{R} = [6/25, 13/50] \times [0, 1/1000].$$

For $(\gamma, \kappa) \in \mathcal{R}$, define $Q_{\gamma, \kappa}$ on $\{0, 1\}^3$ by the masses

$$(62) \quad \frac{\omega}{Q_{\gamma, \kappa}(\omega)} \left| \begin{array}{cccc|c} 000 & 001 & 110 & 111 & \text{each other atom} \\ a & b & b & a & \kappa \end{array} \right|, \quad a = \frac{1 + \gamma}{4} - \kappa, \quad b = \frac{1 - \gamma}{4} - \kappa.$$

These masses sum to one. For $\kappa > 0$ every atom has positive mass. The candidate is still $b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2)$; there is no additional candidate parameter.

Theorem 16.1 (Full-support saddle and contact rigidity). *For every $(\gamma, \kappa) \in \mathcal{R}$, there is a unique $r = r_{\gamma, \kappa}$ in $(33/100, 7/20)$ satisfying*

$$(63) \quad f(r) = r^3 + (13 - 2\gamma + 8\kappa)r^2 + (3 - 12\gamma + 48\kappa)r - 1 - 2\gamma + 8\kappa = 0.$$

Set $D = f'(r)$ and

$$(64) \quad t = \frac{2\{-3r^2 + (14 - 8\gamma - 16\kappa)r - 3 + 48\kappa\}}{D},$$

$$(65) \quad v = \frac{-r^3 + (7 - 4\gamma - 8\kappa)r^2 + (-3 + 48\kappa)r + 13 + 4\gamma - 104\kappa}{32}.$$

Then $8 < D < 10$, $1/10 < t < 1/4$, and

$$U_2(b, Q_{\gamma, \kappa}) = v.$$

The optimal five response rows are (57), with this value of t , and zero at all off-diagonal atoms. The equal prior on $p = q = (1 \pm \sqrt{r})/2$ is least favorable. The same averaged rows are forced for every full-word minimax response. At the three interior counts the entire deterministic event is forced word by word.

The least post-training decision width attaining v is exactly five, including in the original gate interface with arbitrary stochastic training, stopping and independently acquired calibration.

Proof. On the indicated rectangle in (γ, κ, r) , direct monotonicity gives $8 < f'(r) < 10$. The polynomial decreases with γ and increases with κ . Its two endpoint evaluations are bounded by

$$f_{6/25,1/1000}(33/100) = -81619/5000000 < 0, \quad f_{13/50,0}(7/20) = 387/40000 > 0.$$

This proves existence and uniqueness in the stated interval. The numerator of t is $-16\gamma r - 32\kappa r + 96\kappa - 6r^2 + 28r - 6$. Bounding its negative and positive terms separately gives a lower bound $10378/10000 > 1$ and an upper bound $19754/10000 < 2$. Together with $8 < D < 10$ these prove the asserted interval for t .

Let $u = (p + q - 1)^2$, $z = (p - q)^2$, and retain the polynomials H_γ, J_γ from Theorem 14.1. The probabilities of the five training counts give

$$E(u, z) = \frac{(u - z)^2 - 6u + 2z + 13}{4},$$

$$L(u, z) = \frac{(u - z)^2 + 6u - 2z + 1}{8}.$$

Here E is the expected cardinality of the deterministic selected four-atom event, and L is the probability of an extreme training count. The score of the proposed response is therefore

$$G = H_{\gamma,\kappa} + tJ_{\gamma,\kappa}, \quad H_{\gamma,\kappa} = H_\gamma - \kappa E, \quad J_{\gamma,\kappa} = J_\gamma - \kappa L.$$

This is an identity in the actual training probabilities, not an estimate. On the diagonal, $J_{\gamma,\kappa}(u, 0) = -f(u)/64$ and $H_{\gamma,\kappa}(r, 0) = v$. Polynomial division, using $f(r) = 0$, gives

$$(66) \quad G(u, 0) - v = \frac{(u - r)^2 A(\gamma, \kappa, r, u)}{16D},$$

where

$$A = 24\gamma^2 + (3\gamma - 10)r^2 + (6\gamma - 20)ru + 12\gamma r + 6\gamma u - 51\gamma + 30$$

$$- 384\kappa^2 - 96\kappa r - 48\kappa u - 144\kappa.$$

For $0 \leq u \leq 1$ the negative terms alone give $A > 8$ throughout $\mathcal{R}$. Moreover, on $0 \leq u, z \leq 1$ the bounds in the preceding family theorem yield

$$\partial_z H_{\gamma,\kappa} \geq 5/32 - \kappa, \quad \partial_z J_{\gamma,\kappa} \geq -3/64.$$

Indeed the additional derivatives are respectively $\kappa(u - z - 1)/2$ and $\kappa(u - z + 1)/4$. Since $t < 1/4$, one has $\partial_z G > 1/8$. Thus (66) is a global lower certificate on the parameter square, with equality at exactly the two proposed prior points.

For the Bayes upper bound, group the four report bits by their total S ; the independent training marks and arrangements conditional on S are parameter independent under this prior. Put $L_r = r^2 + 6r + 1$. The first three rows of 256 times the signed coefficient table on

000, 001, 110, 111 are

$$\begin{aligned}
S = 0 : & \quad (0, -8\gamma L_r, 32r(r+1) - 8\gamma L_r, 32r(r+1)), \\
S = 1 : & \quad 8(r-1)(r^2 + 4r - 1 - 2\gamma(r+1) + 8\kappa(r+1), \\
& \quad r^2 + 4r - 1 + 2\gamma(r+1) + 8\kappa(r+1), \\
& \quad r^2 - 4r - 1 + 2\gamma(r+1) + 8\kappa(r+1), \\
& \quad r^2 - 4r - 1 - 2\gamma(r+1) + 8\kappa(r+1)), \\
S = 2 : & \quad -12(r-1)^2(r-1-2\gamma+8\kappa, r-1+2\gamma+8\kappa, \\
& \quad r-1+2\gamma+8\kappa, r-1-2\gamma+8\kappa).
\end{aligned}$$

The other rows are their reversals. For example the potentially small first parenthesis in row one is at most

$$(7/20)^2 + 4(7/20) - 1 - 2(6/25)(133/100) + 8(1/1000)(27/20) < -1/10.$$

The third parenthesis is negative by the same endpoint bounds; the second is positive and the fourth negative. In row two, $r-1+2\gamma+8\kappa \leq -61/500 < 0$. Row zero's third entry is positive since $32(33/100)(133/100) - 8(13/50)((7/20)^2 + 6(7/20) + 1) > 7$. Consequently the signs are exactly those of the previous two-preparation saddle. At each off-diagonal atom the candidate mass on both prior points equals $(1-r)/8 > 13/160$, whereas $\kappa \leq 1/1000$. Every off-diagonal coefficient is strictly negative. Coordinatewise Bayes maximization therefore selects the proposed response, including its two indifferent coordinates. Its equal score at the prior support closes the minimax match against the full response class.

For completeness the contact argument persists, not just the sign pattern. If a_0, b_0 are the averaged extreme response coordinates, their extra diagonal contributions are

$$a_0(1-p)^4\{a - (1-p)^2/2\} + b_0p^4\{a - p^2/2\}.$$

At either support point their sum with $a_0 = b_0 = 1$ is zero by (63); the first term is positive at p_+ . Contact at both supports gives $a_0 = b_0$. Interior stationarity gives $a_0 = b_0 = 64\partial_u H_{\gamma,\kappa}(r, 0)/D = t$. All strict coordinates are forced individually. Thus the three vertices and two disjoint faces used in Theorem 15.1 are unchanged. That theorem's positive-factorization argument gives five states for every admissible stochastic encoder, while the same five-event selection attains it. The common-garbling reduction is unchanged because the candidate interface is unchanged. $\square$

The rectangle $\mathcal{R}$ is an explicit stability region for the *saddle and its exact decision-memory number*. It is not a claim that arbitrary nonsymmetric total-variation perturbations have the same optimizer. At $\kappa = 0$ every formula reduces to the retained ideal theorem. Positive off-diagonal target masses do not destroy the strict negative Bayes coefficients at those atoms: their magnitude relative to the candidate, rather than their being zero, is what is needed.

17. EXACT STATISTICAL FORM OF THE BROWNIAN COLLISION SENSOR

The full-support theorem applies to the original microscopic target, not only to an independently constructed contamination model. The preparation symmetries supply the required form exactly.

Theorem 17.1 (Physical symmetry and a quantitative noise region). *Use exactly the preparation boxes, elastic collision, detector and Brownian windows of Section 11. Conditional on the preparation put*

$$Z = |h(v_{1,y}^+)|, \quad e = \Phi\left(-\frac{Z}{\sqrt{2}\sigma}\right), \quad \mu_1 = \mathbb{E}e, \quad \mu_2 = \mathbb{E}e^2.$$

For every $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$ the exact all-on target law is

$$(67) \quad Q_{d,\sigma} = Q_{\gamma_{d,\sigma}, \kappa_{d,\sigma}}, \quad \gamma_{d,\sigma} = \frac{1}{4} - \frac{\mu_1}{2}, \quad \kappa_{d,\sigma} = \frac{\mu_1 - \mu_2}{2}.$$

In particular

$$\frac{1}{4} - \frac{1}{1200} < \gamma_{d,\sigma} < \frac{1}{4}, \quad 0 < \kappa_{d,\sigma} < \frac{1}{1200},$$

so the entire physical range lies inside $\mathcal{R}$, and every target atom has strictly positive probability. The physical two-preparation value in the unrestricted-memory original gate interface equals (65) at these parameters. Its exact decision-memory number is five.

Proof. The collision estimate gives $Z > 1/4$ for every prepared point. The conditional two window noises are independent, of equal variance, and the deterministic signal is constant after the collision. Thus, given (B, W, Z) , the two report bits are independent copies of B flipped with probability e .

We justify the distribution of Z rather than assume a noise symmetry. Reflection in the spatial y coordinate sends (r_y, g_y, V_y) to their negatives, interchanges the two B boxes, and preserves their uniform product laws and the elastic equations. It negates $v_{1,y}^+$ and leaves Z unchanged. The only box that depends on W is the one for V_z ; changing this center-of-mass velocity cannot change the relative collision or the outgoing y velocity. All other box coordinates are independent of W . Consequently Z , and hence e , has the same conditional law for all four (B, W) labels. The Brownian increments are independent of these labels and of the preparation.

The prepared masses of (B, W) are $5/16$ for equality and $3/16$ otherwise. Direct averaging therefore gives

$$\begin{aligned} Q(000) &= Q(111) = 5/16 - 5\mu_1/8 + \mu_2/2, \\ Q(001) &= Q(110) = 3/16 - 3\mu_1/8 + \mu_2/2, \\ Q(010) &= Q(011) = Q(100) = Q(101) = (\mu_1 - \mu_2)/2. \end{aligned}$$

These are precisely (62) with (67). This calculation uses both the common signal and the actual mark; replacing the common signal by independent unconditional report errors would in general replace μ_2 incorrectly by μ_1^2 .

The Gaussian bound already proved in the physical bridge yields $0 < e < 1/600$ uniformly in this physical range. Thus $0 < \mu_2 < \mu_1 < 1/600$, proving the parameter bounds. Positivity follows also directly from $0 < e < 1$ and positive prepared label masses. The all-on construction works against every private candidate in Definition 11.1; its score is the product-family score. Conversely the common-garbling private subfamily gives the architecture-uniform upper bound. Theorem 16.1 therefore supplies both exact value and exact decision count for the original interface, not for an auxiliary sensor. $\square$

Remark 17.2 (Known target and acquired calibration). The exact value and exact atomic stochastic realization are statements for a specified target law. Their event-selection probability depends on that target, not on the unknown candidate. When d, σ and the preparation are known, the expectations in (67) are defined integrals of that known physical law. They are not free information about the private parameter. An auditor that must acquire the target instead uses finite calibration and a precision allowance. The following explicit bound makes this separation operational.

Proposition 17.3 (Acquisition and finite row precision). *Let the fixed physical target be in the preceding theorem. From n independent all-on target calibration words define*

$$\hat{\kappa} = \frac{1}{4n} \sum_{i=1}^n \mathbf{1}_{\{X_{1i} \neq X_{2i}\}}, \quad \hat{\gamma} = \frac{1}{n} \sum_{i=1}^n (\mathbf{1}_{\{000,111\}} - \mathbf{1}_{\{001,110\}})(X_i).$$

Clip these separately to the two intervals of $\mathcal{R}$. With probability at least $1 - 4\exp(-na^2/2)$, both coordinate errors are at most a . Choose the optimal response for this clipped target and replace its tie probability by a dyadic number with error at most 2^{-b} . Conditional on that calibration event, its worst-case score against the true target is at least

$$(68) \quad U_2(b, Q_{d,\sigma}) - 9a - 2^{-b}.$$

The calibration can be acquired with at most $(n+1)^3$ counter states and at most $4(n+1)^3$ states including the current report buffer; retaining the calibration output during the audit gives the conservative product bound $12(n+1)^3$ at logical report cuts. A sequential fair-bit realization of the tie coin, with its origin and bit phase retained, has a finite explicit bound $12(n+1)^3 + 8(b+2)(n+1)^3$.

Proof. The off-diagonal indicator has mean 4κ and range one. The signed diagonal indicator has mean $2a - 2b = \gamma$ and range two. Hoeffding's inequality and a union bound give the displayed failure probability; clipping to an interval containing the truth cannot increase coordinate error. The two target laws have total variation at most $|\Delta\gamma|/2 + 4|\Delta\kappa| \leq 9a/2$. Every fixed trained response changes score by at most this amount when only the target changes, and so does the unrestricted optimal value. Using an optimizer for the estimated target therefore loses at most $9a$ against the true one. Only the two extreme-count event choices are rounded. Their score contribution changes by at most the coin error, because their total probability is at most one and the change in a single indicator expectation has magnitude at most one. This proves (68).

Three nonnegative category counts, each bounded by n , compute the two statistics; they also fix the clipped estimates and the subsequent program choice. Keeping this count triple with the twelve-state audit costs at most the stated product. At an extreme count, sample the dyadic coin using b fair bits and comparison with its binary threshold, retaining the two-valued origin, the comparison flag and the phase. At most $8(b+2)$ additional states suffice per calibration choice. All other counts select deterministic events. The bound includes the coin workspace instead of identifying an algebraic row with a finite random-bit primitive. The n target calibrations and the four audit preparations are all charged. $\square$

18. OPTIMAL VALIDATION ORDERS AND A POSITIVE-NOISE MEMORY JUMP

We now optimize the order in the original two-tape, bit-level validation interface. Each tape has slots $1, 2, w$ in that order. Write X for the candidate tape and Y for the target tape. There are twenty declared-clock interleavings. A fixed clock is common to all executions; its phase is not a report-dependent side channel. Gate choices, state updates, training and frozen-event selection remain stochastic and otherwise unrestricted as before.

Let $e_0, \dots, e_4$ be the five events (46). A full raw validation word has output $e_i(Y) - e_i(X)$. For the strict training counts, contact forces $i = 1, 2, 3$ respectively. At either extreme only one raw-event coordinate is indifferent; Theorem 16.1 fixes its mark-averaged probability to $t \in (1/10, 1/4)$.

One precaution is needed. A freshly randomized gate need not couple its two validation coins as one stored event mask would. When both raw validation words equal the indifferent atom, their difference may be random with mean zero. None of the lower certificates below uses that coordinate as a forced deterministic zero. For every additional mixed witness, the known

prefix excludes at least one validation word from the indifferent atom, so at least one of the two event indicators is already deterministic. Its future output law is then fixed by the remaining indicator's marginal. This covers both stored-mask and fresh gate implementations.

Theorem 18.1 (The minimum over all declared-clock validation orders). *For every $(\gamma, \kappa) \in \mathcal{R}$ with $\kappa > 0$, the least whole-schedule peak attaining $U_2(b, Q_{\gamma, \kappa})$, minimized over all twenty declared-clock interleavings, is twelve. The only minimizing orders are the two serial orders*

$$XXXYYY, \quad YYYXXX.$$

Every nonserial declared order requires peak at least fourteen. The lower bounds permit the full stochastic gate and event interface, not merely an all-on deterministic compiler. These conclusions apply to $Q_{d, \sigma}$ for every $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$.

Proof. Every raw validation prefix has positive probability at either supporting candidate and at the target. Thus all cut witnesses may be compared on a common full future alphabet. Generate all-on words to simulate training gates as in Theorem 15.1. The post-training response is an unrestricted optimizer on the supporting subfamily, hence has the contact-forced rows of Theorem 16.1. Stopping before the two training preparations cannot occur with positive supporting probability: for each retained first-word branch, two possible unused second words force different strict vertices. A positive stopped component cannot belong to both vertices. The witnesses below therefore concern actual post-training validation cuts.

At a cut with two slots consumed from one tape and none from the other, the eight deterministic vertices and four disjoint additional faces from Theorem 15.3 give twelve. That argument required only the four positive target diagonal words. It is still valid with full target support. Exchanging the two raw tapes negates the output and gives exactly the same channel-polytope certificate at the reversed cut. Thus either serial order needs twelve.

We give explicit certificates at the two possible nonserial entrance cuts. After one slot of each tape, write $x = X_1$, $y = Y_1$, and use suffix coordinates $(u, w, v, z) = (X_2, X_w, Y_2, Y_w)$. Define

i	$g_i^0(u, w)$	$g_i^1(u, w)$
1	$(1 - u)(1 - w)$	u
2	$1 - u$	u
3	$1 - u$	uw

and let $d_i^{xy} = g_i^y(v, z) - g_i^x(u, w)$. The twelve expressions with $i = 1, 2, 3$ and $x, y \in \{0, 1\}$ give ten distinct deterministic vertices. The only repetitions are $d_2^{11} = d_1^{11}$ and $d_3^{00} = d_2^{00}$, as is seen by substituting $u, w, v, z \in \{0, 1\}$ in the displayed functions.

There are four more required channels. Use count zero at prefixes $(x, y) = (0, 1), (1, 0)$ and count four at these same two prefixes. Their endpoints are respectively the continuations of (e_0, e_1) and (e_4, e_3) , with nontrivial weights $1 - t, t$. Each excludes simultaneous occurrence of the indifferent atom. The minimal face of each such channel consists of distributions supported, at each suffix, on the two endpoint output values. Among the ten mandatory vertices it contains only $d_1^{01}, d_1^{10}, d_3^{01}, d_3^{10}$, respectively. These four faces are pairwise disjoint. In their listed order, suffixes separating their six pairs are

pair	12	13	14	23	24	34
(u, w, v, z)	0110	0010	0000	0000	0001	0000

At each listed suffix the two sets of permitted output integers are disjoint; this follows directly from the displayed g_i^x and the masks e_0, e_4 . None of the required mixed rows is the sole mandatory vertex in its face, because $0 < t < 1$. Face packing gives $10 + 4 = 14$.

Suppose instead two slots of X and one of Y have been consumed. A deterministic continuation is indexed by (i, X_1X_2, Y_1) . The following twelve choices are distinct and include every strict-count continuation:

i	(X_1X_2, Y_1)
1	$(00, 0), (00, 1), (01, 0), (01, 1), (11, 0), (11, 1)$
2	$(00, 0), (01, 0)$
3	$(00, 1), (01, 1), (11, 0), (11, 1)$

They are verified by the formula $e_i(Y_1, Y_2, Y_w) - e_i(X_1, X_2, X_w)$ on the eight suffixes (X_w, Y_2, Y_w) . At training count zero take additionally prefixes $(01, 0)$ and $(11, 0)$. The corresponding two faces have endpoints $(0, (1 - Y_2)(1 - Y_w))$ and $(-1, -1 + (1 - Y_2)(1 - Y_w))$, respectively. Each contains just its listed $i = 1$ vertex, and they are disjoint at any suffix with $Y_2 = 1$. Both mixtures are nontrivial. Thus this cut also requires fourteen. Swapping X, Y proves the same result at the cut with one and two slots consumed.

Any nonserial interleaving either reads one slot from each tape first, or starts with two slots of one tape and then switches before its mark. It therefore passes one of the preceding cuts and needs fourteen. Every serial order has an upper realization of peak twelve: select the same five-valued frozen event and use its future residual functions. For full-support raw words the validation widths in either serial order are

$$(5, 10, 12, 10, 10, 7, 3).$$

These counts follow by restriction of the five masks, exactly as in Lemma 12.4; for the reverse order negate all output residuals and exchange the tapes. Training and the charged stochastic selection cut have width at most five. Therefore both serial orders attain twelve, proving all assertions by Theorem 9.2. The physical specialization follows from Theorem 17.1. $\square$

This theorem optimizes the externally declared schedule; it does not optimize a report-dependent scheduler that changes which tape is read next. Such a scheduler has a different action interface, and its protocol state cannot be supplied for free. The phrase “all orders” in Theorem 18.1 refers to all twenty declared interleavings, with this quantifier fixed explicitly.

Theorem 18.2 (A singular support transition in reverse serial memory). *For every $\gamma \in [6/25, 13/50]$, the exact minimum peak in the reverse serial order $YYYXXX$ for the ideal target $Q_{\gamma,0}$ is ten. For every $0 < \kappa \leq 1/1000$ it is twelve. In particular the reverse-serial exact memory of the physical collision experiment jumps from ten at its zero-noise target to twelve at every positive $\sigma \leq 1/24$, although its target laws converge in total variation as $\sigma \downarrow 0$.*

Proof. For the upper bound at the ideal target, ignore its first report bit; its second bit equals the first almost surely, so this does not remove any needed information. After its second bit and mark, retain the residual $c - e_i(X)$, where $c = e_i(Y)$. Event e_2 has $c = 1$ surely; each other event permits two values of c . The compatible reverse validation profile is

$$(5, 5, 9, 9, 10, 7, 3).$$

After the second target bit the nine functions are, with the unread mark denoted by z ,

$$-e_0, 1 - e_0, 1 - z - e_1, 1 - e_1, 1 - e_2, 1 - e_3, z - e_3, 1 - e_4, -e_4.$$

Restricting after the target mark gives nine functions $c - e_i$. After the first candidate bit they give ten distinct functions of (X_2, X_w) ; after its second bit they give the seven scalar mark functions in Lemma 12.4. Equality of residual functions is preserved by restriction, so these counts describe one compatible machine, not independent cutwise colorings. Training remains of peak five and the terminal score is correct almost surely for every candidate. On target words of zero probability, choose arbitrary legal transitions.

For the converse examine the cut after all three target slots and the first candidate bit. Write suffixes (X_2, X_w) in the order 00, 01, 10, 11. The three strict training events force the following eight distinct deterministic output channels:

$$\begin{aligned} (0, 1, 1, 1), & \quad (-1, 0, 0, 0), \quad (1, 1, 0, 0), \quad (0, 0, -1, -1), \\ (0, 0, 1, 1), & \quad (-1, -1, 0, 0), \quad (1, 1, 1, 0), \quad (0, 0, 0, -1). \end{aligned}$$

At training count zero take $X_1 = 0$ and target words 001 and 110. They force the channels with output means

$$(-t, 0, 0, 0), \quad (1 - t, 1, 1, 1).$$

Every integer coordinate is a point mass, and the remaining coordinate is the indicated mixture of neighboring output integers, because the target event indicator is fixed on these words. Their faces each contain exactly one of the eight mandatory vertices. They are disjoint at suffix 01, which requires output zero in the first and one in the second. Thus two additional generators are required, giving ten. All these prefixes have positive supporting probability. The full-support value twelve follows from the preceding theorem. Formula (67) gives the physical assertion and $\mu_1, \mu_2 \rightarrow 0$ by the uniform Gaussian bound. $\square$

The discontinuity concerns exact attainment. It neither contradicts continuity of the optimal score nor prevents a lower-memory device from approximating that score. It also explains why a support-zero compiler cannot be transferred to the noisy physical target by simply keeping its state count unchanged.

19. PREPARATION LIMITS AND LOCALIZATION OF LEAST-FAVORABLE PRIORS

The contact structure has a second consequence, now for an arbitrary number of training preparations. We separate a general localization principle from the special geometry of the marked product family. An exact finite- N two-point prior is not assumed.

Theorem 19.1 (Nearest-experiment limit and prior localization). *Let P_θ be a continuous family on a finite alphabet of size d , with compact metric parameter space, and let Q be fixed. Put*

$$d_Q(\theta) = \text{TV}(Q, P_\theta), \quad d_* = \min_\theta d_Q(\theta), \quad C_* = \{\theta : d_Q(\theta) = d_*\}.$$

For $N \geq 1$, the unrestricted frozen-validation values satisfy

$$(69) \quad d_* - \sqrt{(d-1)/N} \leq U_N(P, Q) \leq d_*, \quad U_N \uparrow d_*.$$

Every least-favorable prior μ_N in Theorem 5.1 satisfies

$$(70) \quad \int (d_Q - d_*) d\mu_N \leq \sqrt{(d-1)/N}.$$

Consequently every weak limit of such priors is supported on C_ . For every $a > 0$ their mass on $\{d_Q - d_* \geq a\}$ is at most $\sqrt{(d-1)/N}/a$. More generally, an upper bound b_N on twice the expected total-variation error of a common empirical reconstruction replaces $\sqrt{(d-1)/N}$ in both conclusions.*

Proof. Let $\hat{P}$ be the empirical law of the N reports, and choose the indicator $h_{\hat{P}}$ maximizing $(Q - \hat{P})h$. If h_θ maximizes $(Q - P_\theta)h$, the two event inequalities give

$$(Q - P_\theta)h_{\hat{P}} \geq d_Q(\theta) - 2\text{TV}(\hat{P}, P_\theta).$$

This inequality holds for the same parameter-independent response rule at every parameter. For any law P on d atoms,

$$\mathbb{E} \text{TV}(\hat{P}, P)^2 \leq \frac{d}{4N} \left(1 - \sum_x P(x)^2 \right) \leq \frac{d-1}{4N}.$$

Thus its expected score is at least $d_Q(\theta) - \sqrt{(d-1)/N}$. Every response has score at most $d_Q(\theta)$ at that parameter, which proves the upper bound at a minimizing parameter. Ignoring the last observation proves monotonicity and hence the limit.

Against a minimizing prior, the Bayes optimum is U_N . It is at least the prior average of the same empirical rule. Therefore

$$U_N \geq \int d_Q d\mu_N - \sqrt{(d-1)/N}.$$

Combine this with $U_N \leq d_*$ to get (70). Markov's inequality proves the mass estimate. The function $d_Q - d_*$ is nonnegative and continuous on the compact parameter space, so its integral under any weak limit is zero. Such a limit is supported on its zero set. The proof uses only the common reconstruction bound, which proves the last assertion. $\square$

The empirical estimate in (69) is the same finite-alphabet mechanism underlying the retained decision-spectrum classification. Its role here is different: it controls every least-favorable prior, not merely a constructive selector. In particular, finite-preparation saddle supports cannot remain far from the closest candidate experiments as the preparation number grows.

Theorem 19.2 (Intrinsic marked contact geometry). *For every $\gamma \in I$, set $s_\gamma = \sqrt{(1+\gamma)/2}$ and*

$$d_\gamma = 2s_\gamma - 1 - \gamma/2, \quad C_\gamma = \{(s_\gamma, s_\gamma), (1-s_\gamma, 1-s_\gamma)\}.$$

Then

$$(71) \quad \min_{p,q} \text{TV}(Q_\gamma, b(p, q)) = d_\gamma, \quad \operatorname{argmin}_{p,q} \text{TV}(Q_\gamma, b(p, q)) = C_\gamma.$$

Moreover, for Euclidean distance in $[0, 1]^2$,

$$(72) \quad \text{TV}(Q_\gamma, b(p, q)) - d_\gamma \geq \frac{7}{2300} \operatorname{dist}((p, q), C_\gamma)^2.$$

The constant is uniform on I and is derived from the experiment's atom probabilities, not from a metric-entropy or small-ball assumption.

Proof. Put $a = (1-p)(1-q)$, $c = pq$, $l = (1-\gamma)/2$ and $h = (1+\gamma)/2$. The common mass of target and candidate is $\psi(a) + \psi(c)$, where

$$\psi(z) = \begin{cases} z, & 0 \leq z \leq l, \\ z/2 + (1-\gamma)/4, & l \leq z \leq h, \\ 1/2, & h \leq z \leq 1. \end{cases}$$

This follows by summing the smaller mass on each of the two marked atoms corresponding to the report pair. Reflection $(p, q) \mapsto (1-p, 1-q)$ exchanges a, c ; assume $c \geq a$. If $c < 1/4$, both terms are in the first segment of ψ , and their sum is at most $1/2$.

For $c \geq 1/4$, put $x = \sqrt{c} \geq 1/2$. By the arithmetic-geometric mean inequality,

$$a \leq (1-x)^2 \leq 1/4 < l.$$

Consequently the common mass is at most $F(x) = (1-x)^2 + \psi(x^2)$, with equality only when $p = q = x$. On the three consecutive intervals with endpoints $1/2, \sqrt{l}, \sqrt{h}, 1$, this function is respectively

$$(1-x)^2 + x^2, \quad (1-x)^2 + x^2/2 + (1-\gamma)/4, \quad (1-x)^2 + 1/2.$$

The first two are strictly convex and the last strictly decreases. The first increases away from $1/2$. Thus a maximum must occur at $\sqrt{l}$ or $\sqrt{h}$. Their difference is

$$(73) \quad F(\sqrt{h}) - F(\sqrt{l}) = \gamma \left(\frac{3}{2} - \frac{2}{\sqrt{h} + \sqrt{l}} \right) > \frac{7}{575}.$$

Indeed $\sqrt{h} > 39/50$, $\sqrt{l} > 3/5$, and $\gamma \geq 6/25$, so the right side is greater than $(6/25)(3/2 - 100/69) = 7/575$. The unique maximizing diagonal point on this side is $x = \sqrt{h}$. Its common mass is $(1 - \sqrt{h})^2 + 1/2 > 1/2$, also excluding $c < 1/4$. Subtracting it from one gives d_γ . Reflection proves (71) and the assertion about all minimizers.

We record a uniform quantitative form. Write $s = \sqrt{h}$ and $\Delta = 7/575$. For $1/2 \leq x \leq \sqrt{l}$, the common mass is at most $F(\sqrt{l})$, so its deficit from $F(s)$ is at least Δ . For $\sqrt{l} \leq x \leq s$, convexity puts F below the chord between its endpoint values. Since this interval has length at most one, its deficit is at least $\Delta(s-x)$. For $s \leq x \leq 1$ the deficit is

$$(1-s)^2 - (1-x)^2 = (x-s)(2-s-x) \geq (1-s)(x-s).$$

Since $s < 4/5$, this too is at least $\Delta|x-s|$. Together the three intervals give $F(s) - F(x) \geq \Delta|x-s|$.

Let $z = (p-q)^2$. At fixed $c = x^2$ the additional loss from the nondiagonal candidate is exactly

$$(1-x)^2 - a = p + q - 2x = \frac{z}{p+q+2x} \geq z/4.$$

The geometric mean x lies between p and q , and hence

$$(p-s)^2 + (q-s)^2 \leq 4z + 4(x-s)^2 \leq 4z + 4|x-s|.$$

It follows that the total deficit is at least $(\Delta/4)((p-s)^2 + (q-s)^2)$, proving (72) in this case. If $c < 1/4$, the total variation is at least $1/2$, whereas $d_\gamma < 12/25$ from $s < 4/5$ and $\gamma \geq 6/25$. The excess is therefore greater than $1/50$; Euclidean squared distance in the square is at most two, and $1/50 > 2(7/2300)$. This proves the remaining case. Reflection completes the proof. $\square$

Corollary 19.3 (An all-preparation law and a limiting two-point prior). *For all $N \geq 1$ and $\gamma \in I$,*

$$(74) \quad d_\gamma - \sqrt{3/N} \leq U_N(b, Q_\gamma) \leq d_\gamma, \quad U_N(b, Q_\gamma) \uparrow d_\gamma.$$

Every least-favorable prior $\mu_{N,\gamma}$ satisfies

$$(75) \quad \int \text{dist}(\theta, C_\gamma)^2 d\mu_{N,\gamma}(\theta) \leq \frac{2300}{7} \sqrt{3/N}.$$

A least-favorable prior can be chosen invariant under simultaneous complementation of the two parameters. Every such symmetric choice satisfies, for the Euclidean quadratic transport distance,

$$(76) \quad W_2 \left(\mu_{N,\gamma}, \frac{1}{2} \delta_{(s_\gamma, s_\gamma)} + \frac{1}{2} \delta_{(1-s_\gamma, 1-s_\gamma)} \right) \leq \sqrt{2300/7} (3/N)^{1/4}.$$

In particular, the limiting symmetric least-favorable law is unique.

Proof. Estimate the four-atom report-pair law from training and tensor it with the known fair mark. Tensoring with a common probability preserves total variation. The proof of Theorem 19.1 therefore uses $d = 4$ rather than eight, proving (74) and an excess bound $\sqrt{3/N}$. Combining the latter with (72) gives (75).

Simultaneously complement both report bits and the mark in every training and validation word. This leaves Q_γ invariant and sends $b(p, q)$ to $b(1 - p, 1 - q)$. The Bayes objective in (8) is invariant under this transformation and convex in the prior. Averaging a minimizing prior with its reflected image is therefore again minimizing and is symmetric.

Map each parameter to a closest point of C_γ , choosing the two points with equal probabilities at a tie. This is a Borel kernel, equivariant under reflection. A symmetric prior is pushed forward to the equal two-point law in (76); the induced coupling has squared transport cost exactly the left side of (75). Taking square roots proves the assertion. $\square$

These conclusions identify the endpoint and limiting contact geometry for every preparation number. They do not assert that the exact prior has two atoms at every finite N , or that the displayed convergence rates and constants are sharp. The finite- $N = 1, 2$ saddles are preserved as exact results. The new theorem explains their relationship to an intrinsically determined limiting contact set without extrapolating their particular event tables to unproved values of N .

20. AN EXACT PREPARATION–DECISION-MEMORY LAW

A second solvable family separates a genuine joint resource law from an implementation bound. Here the memory coordinate is the number of states at the training–validation cut. It is not identified with the bit-level peak width of the collision audit.

Fix $m = 2^d$, $0 < \alpha < 1$, and an unknown parameter $\theta \in \{1, \dots, m\}$. A fresh training preparation reports θ with probability α and an erasure $\perp$ otherwise. Conditional on θ , the preparations are independent. Candidate validation has law δ_θ and target validation is uniform on $\{1, \dots, m\}$. At the decision cut the auditor retains at most $K = 2^k$ states, $0 \leq k \leq d$. Its subsequent response may depend only on that state, current validation observations and fresh independent randomness. In particular, a calibration table, remembered seed or stopping time outside the retained state is not available. Upstream and downstream memory are otherwise unrestricted for the converse. Let $V_N^{\text{er}}(K)$ be the best worst-case expected signed validation score with at most N training preparations.

Theorem 20.1 (Exact joint law). *For every $N \geq 0$ and $K = 2^k \leq m$,*

$$(77) \quad V_N^{\text{er}}(K) = (1 - (1 - \alpha)^N) \left(1 - \frac{1}{K}\right).$$

The converse covers stochastic encoders and arbitrary legal stopping. The value is attained by a clocked machine with K training states. With atomic preparation and validation outcomes, the complete audit has peak at most $\max\{2K, 3\}$ and uses at most $N + 2$ preparations.

Proof. Write $q = (1 - \alpha)^N$ and pre-generate N potential training reports. A stopped auditor ignores the unused reports; this does not disclose extra information to it. On the event that all N reports are erasures, its retained-state law $r = (r_j)_{j=1}^K$ is independent of θ . After padding any smaller register with unused states, its channel to the decision register has the form

$$(78) \quad C(j \mid \theta) = qr_j + (1 - q)S(j \mid \theta)$$

for a stochastic channel S . This identity remains valid with randomized stopping and randomized state updates.

Use the uniform prior on parameters as an upper certificate. The part qr_j has zero expected score for every response: averaging δ_θ under this prior gives the target itself. For the other

part, optimizing over all responses gives a convex function of S , a sum of positive parts of linear functions. Every stochastic channel is a convex combination of deterministic channels $j = g(\theta)$; therefore its value is at most the largest deterministic-channel value. This convexification is used only to bound a channel. It does not presuppose that a mixture of width- K machines is realizable at width K .

If the fibers of g have sizes $n_1, \dots, n_K$, a nonempty fiber has posterior uniform on that fiber, so its total variation from the uniform target is $1 - n_j/m$. The prior-averaged optimal score is

$$\sum_j \frac{n_j}{m} \left(1 - \frac{n_j}{m}\right) = 1 - \sum_j (n_j/m)^2 \leq 1 - 1/K.$$

Empty fibers contribute zero. Cauchy–Schwarz proves the inequality. This is an upper bound on the average score and hence on its minimum over fixed parameters. Multiplying by $1 - q$ proves the upper bound in (77).

For attainment, partition the parameter alphabet into K equal groups and let $g(\theta)$ denote its group. Initialize the register uniformly on the K group labels. On an erasure leave it unchanged; on a revealed parameter overwrite it by $g(\theta)$. No extra “revelation occurred” flag is needed. At the decision cut its law is $q \text{Unif}[K] + (1 - q)\delta_{g(\theta)}$. For register j , freeze the event which is the complement of group j . On a revelation its score is $1 - 1/K$. Under a uniformly random initial label, the mean score is zero for each fixed θ : one of the K labels has score $1 - 1/K$ and the others have score $-1/K$. This proves the exact worst-case value.

The k initialization bits can be accumulated in the K -state register; all earlier prefix counts are at most K . Thus no persistent seed lies outside it. Candidate validation retains the group label and its event indicator, using at most $2K$ states. Target validation produces one of the three values $-1, 0, 1$. This proves the atomic-outcome peak bound. The controller uses a declared external schedule; an autonomous stopping implementation must additionally retain its phase. A phase-tagged implementation is finite, but its cost is not included in K . $\square$

Corollary 20.2 (Thresholds and matching resource scales). *For $K \geq 2$ and $0 \leq c < 1 - 1/K$, the least integer number of training preparations giving a score strictly greater than c is*

$$(79) \quad N_c^{\text{er}}(K) = \left\lceil \frac{\log(1 - c/(1 - 1/K))}{\log(1 - \alpha)} \right\rceil + 1.$$

If $c \geq 1 - 1/K$, no finite N suffices. For $0 < \varepsilon < 1$, achieving $V_N^{\text{er}}(K) \geq 1 - \varepsilon$ requires

$$K \geq \varepsilon^{-1}, \quad N \geq \frac{\log(1/\varepsilon)}{-\log(1 - \alpha)}.$$

Conversely, whenever an allowed $K \leq m$ satisfies $K \geq 2/\varepsilon$, the choice $N \geq \log(2/\varepsilon)/[-\log(1 - \alpha)]$ suffices.

Proof. The strict threshold is equivalent to $(1 - \alpha)^N < 1 - c/(1 - 1/K)$. Since the denominator logarithm is negative, the least integer satisfying this strict inequality is (79), including equality cases of the real quotient. Also

$$1 - V_N^{\text{er}}(K) = q + (1 - q)/K.$$

This is at least q and at least $1/K$, and is at most $q + 1/K$. These three inequalities prove the necessary and sufficient bounds. $\square$

The exact law concerns an entire (N, K, α, m) family, not a single finite diagnostic. Its mechanism is a parameter-independent erasure component followed by the convex geometry of the decision channel. It demonstrates how a preparation obstruction and a memory obstruction can be matched in one theorem. It does not identify the least physical peak width at two collision

preparations. Encoding the m -symbol reports bit by bit also introduces cuts not present in this atomic model; the physical bit-level accounting is given separately.

21. INTEGER MEMORY AND THE FAILURE OF A NAIVE PARTITION FORMULA

The divisibility assumptions in Theorem 20.1 can be removed, but a balanced-partition Bayes bound must not be mistaken for a minimax equality when a persistent random partition is not free. This distinction leads to another exact finite-resource transition.

Keep the revelation interface, but allow any integers $m \geq 2$ and $1 \leq K \leq m$. Put $\rho = 1 - (1 - \alpha)^N$, and denote the genuine K -label minimax value by $V_{N,m}^{\text{er}}(K)$. Write $m = Kq + s$, $0 \leq s < K$, and set

$$(80) \quad B_{m,K} = 1 - \frac{(K-s)q^2 + s(q+1)^2}{m^2}.$$

Here and below all-erasure initialization randomness is a state initialization, not an unrecorded seed used again at validation.

Proposition 21.1 (Arbitrary integer alphabet bounds and low-reveal equality). *The optimal uniform-prior Bayes score is $\rho B_{m,K}$. The minimax value satisfies*

$$(81) \quad V_{N,m}^{\text{er}}(K) \leq \min \left\{ \rho B_{m,K}, 1 - \frac{\lceil m/K \rceil}{m} \right\}.$$

If $\rho \leq m/(m+s)$, equality with the first term holds. At perfect revelation $\rho = 1$ the exact value is the second term. In particular, when $s > 0$ the uniform-prior Bayes optimum at perfect revelation is strictly larger than the minimax value.

Proof. The all-erasure decomposition (78) holds without any divisibility condition. Its parameter-independent part has zero uniform-prior score. Optimizing the other part over responses gives a convex function of its stochastic channel. A deterministic channel with fiber sizes n_j has Bayes score $1 - \sum_j (n_j/m)^2$. If two nonempty fiber sizes differ by at least two, moving one index from the larger to the smaller strictly decreases the sum of squares. Allowing empty fibers and repeating gives sizes q and $q+1$, proving the Bayes bound. It is attained by this partition and its complementary events, with any parameter-independent initialization. This proves the first upper bound as well as the asserted Bayes optimum.

For perfect revelation, fix any K response vectors. For each known parameter choose a response maximizing its score; this deterministic assignment cannot decrease any parameter's payoff, even when the original encoder was randomized. Some assigned fiber has size at least $\lceil m/K \rceil$. Averaging its parameters, the score of their common response is at most the total variation between the uniform target and the uniform law on that fiber, namely $1 - |\text{fiber}|/m$. The minimum is no larger. Balanced fibers with their complementary events attain the resulting upper bound. A perfectly informed encoder can simulate any less informative encoder, so this is an upper bound at every ρ .

For the low-reveal equality let balanced fibers have weights $w_j = n_j/m$. When $\rho < 1$, initialize their label register with probabilities

$$(82) \quad r_j = \frac{\rho w_j + (1 - 2\rho)/K}{1 - \rho}.$$

These sum to one, and are nonnegative precisely throughout the stated range: the smallest numerator is $\rho q/m + (1 - 2\rho)/K$. On a revealed parameter overwrite the label by its fiber; on

erasure do nothing. No revelation flag is retained. For a parameter in fiber i , the complementary-event score is

$$\begin{aligned} g_i &= \rho(1 - w_i) + (1 - \rho) \left(r_i - \sum_j r_j w_j \right) \\ &= \rho \left(1 - \sum_j w_j^2 \right) = \rho B_{m,K}. \end{aligned}$$

This matches the upper bound uniformly in the fixed parameter. If $\rho = 1$ and $s = 0$, the deterministic revealed label already attains that value, so the singular denominator in (82) is not used. Finally, with unequal balanced fibers the weighted average of their sizes is strictly smaller than the largest size; equivalently $B_{m,K} > 1 - \lceil m/K \rceil / m$. This proves the last assertion. $\square$

Theorem 21.2 (The complete two-label revelation law). *For every $m \geq 2$, every $N \geq 0$, and $0 < \alpha < 1$, set $w = \lfloor m/2 \rfloor / m$ and $b = \lceil m/2 \rceil / m$. Then*

$$(83) \quad V_{N,m}^{\text{er}}(2) = \min\{2wb\rho, w\}, \quad \rho = 1 - (1 - \alpha)^N.$$

The same formula holds at the limiting endpoints $\rho = 0, 1$. For odd m it changes regime at $\rho = m/(m+1)$; for even m it reduces to $\rho/2$. The attaining machine uses two training labels, a frozen two-event alphabet, and atomic peak at most four.

Proof. Proposition 21.1 gives the upper bound and its attainment for $\rho \leq 1/(2b)$. Above this threshold, partition into fibers of weights w, b and initialize deterministically in the larger fiber's label. Continue to overwrite on a reveal and leave the state unchanged on an erasure. A parameter in the larger fiber has score w ; a parameter in the smaller fiber has score $(2\rho - 1)b$. The latter is at least w exactly when $\rho \geq 1/(2b)$. This attains the second upper bound. For even m the threshold is one, so the low-reveal construction suffices. The endpoint $\rho = 0$ may instead freeze a constant event. Candidate validation retains the selected label and its indicator, with at most four states; target validation produces the difference. Initialization is atomic randomization when its probabilities are not dyadic. $\square$

For example, $m = 3, K = 2$ gives $V_{N,3}^{\text{er}}(2) = \min\{4\rho/9, 1/3\}$, rather than $4\rho/9$ for all ρ . The balanced-integer expression is therefore not the unqualified replacement of the divisible minimax formula. For $K \geq 3$ outside the low-reveal range, (81) remains a bound rather than an asserted equality. The compatible dynamic formulation of Section 7 retains the exact optimization in that regime. These results strengthen the revelation example without identifying its decision-cut coordinate with the collision model's bit-level peak.

22. QUANTITATIVE CERTIFICATES FOR A FINITE COMPATIBLE ARCHITECTURE

We return to an arbitrary finite architecture, now with a finite parameter set. It can have hidden states, controlled experiments, more than two hypotheses and registers of any size. Enumerate its distinct stochastic rows by $r = 1, \dots, R$ and write $q_r \in \Delta_{s_r}$. A reused autonomous row appears only once in this product. Assume each execution uses at most H controller rows, and uses row r at most d_r times. Delete unused rows and put $S = \max_r s_r$. Fixed environmental kernels are not controller rows.

For an integer $m \geq 1$, intersect Δ_{s_r} with the coordinate boxes of side $1/m$. The resulting nonempty closed polytopes cover the simplex, with overlaps permitted at their boundaries. Write C for a product of such row cells and choose a vertex q_r^C in each factor. A row-cell vertex has at most one coordinate not fixed at a box boundary; consequently it has at most $v_r = s_r 2^{s_r - 1}$ vertices, all on the $1/m$ grid. There are at most $m^{\sum_r s_r}$ product cells.

Write the vertices of row cell C_r as $v_{r,j}$ and parametrize $q_r = \sum_j \lambda_{r,j} v_{r,j}$ with λ_r in a simplex. A reward polynomial has degree at most d_r in block λ_r . Use the homogeneous simplex Bernstein basis of degree d_r in that block, obtaining

$$(84) \quad g_\theta(q(\lambda)) = \sum_\alpha b_{\theta,C,\alpha} B_\alpha(\lambda), \quad B_\alpha \geq 0, \quad \sum_\alpha B_\alpha = 1.$$

Degree elevation includes the factors $\sum_j \lambda_{r,j} = 1$. No independence between different visits of a shared row is imposed on its unknown row vector.

Lemma 22.1 (Causal interpretation of the coefficients). *Every coefficient in (84) satisfies*

$$|b_{\theta,C,\alpha} - g_\theta(q^C)| \leq HS/m.$$

Proof. For each row r , independently prepare an ordered list of d_r vertex labels, sampled independently with law λ_r . At its k th visit, use the stochastic row given by the k th vertex label in that list. Before conditioning, the next label is independent of the previous execution, so this experiment has exactly the row q_r at every visit and expected reward $g_\theta(q(\lambda))$. Unused labels may be sampled after termination without altering the reward.

Condition on the counts of every vertex label in every list. The probability of those counts is $B_\alpha(\lambda)$. Conditional on the counts, each list is a uniformly ordered multiset. Its expected reward is independent of λ and is therefore the corresponding Bernstein coefficient. This also proves (84) directly.

Each row used by this conditioned experiment is a vertex of C_r or a mixture of its remaining vertices. Every two vectors in C_r have coordinate differences at most $1/m$, hence total variation at most $s_r/(2m) \leq S/(2m)$. Couple the conditioned experiment to the one using q^C , using the same environmental randomness while their executions agree. At each actual controller update the probability of a first row-level disagreement is at most $S/(2m)$. There are at most H such updates on the agreeing path. The terminal laws consequently differ in total variation by at most $HS/(2m)$. A reward in $[-1, 1]$ has expectation difference at most twice that quantity. This proves the estimate. The conditioned urn is used to analyze a coefficient; it is not declared to be an admissible free-memory controller. $\square$

Theorem 22.2 (Explicit subdivision certificate). *Define*

$$\ell_C = \min_\theta g_\theta(q^C), \quad u_C = \min_\theta \max_\alpha b_{\theta,C,\alpha}, \quad L_m = \max_C \ell_C, \quad U_m = \max_C u_C.$$

For the exact all-stochastic value of the specified architecture,

$$(85) \quad L_m \leq V(A) \leq U_m \leq L_m + HS/m.$$

On each cell, a witnessing parameter θ_C supplies the nonnegative coefficient certificate

$$u_C - g_{\theta_C}(q(\lambda)) = \sum_\alpha (u_C - b_{\theta_C,C,\alpha}) B_\alpha(\lambda).$$

Its degree in row block r is d_r , and its total degree is at most $\sum_r d_r$. At most $\prod_r \binom{d_r + v_r - 1}{v_r - 1}$ coefficients per parameter and cell are needed. If the primitive data are rational, all coefficients and bounds can be checked by rational arithmetic. Taking $m \geq HS/\varepsilon$ gives a certified interval of length at most ε .

Proof. The point q^C is feasible, so ℓ_C is a lower bound on the value restricted to C . For any $q \in C$, choose a convex representation of its rows. Equation (84) implies

$$\min_\theta g_\theta(q) \leq \min_\theta \max_\alpha b_{\theta,C,\alpha} = u_C.$$

Choose a parameter attaining the minimum defining ℓ_C and apply Lemma 22.1; this gives $u_C \leq \ell_C + HS/m$. Taking the maximum over the finite cover proves (85). Choose θ_C attaining

the definition of u_C ; all displayed coefficients of its certificate are nonnegative. The degree and count statements follow from the product Bernstein basis. Cell vertices are rational grid points, and a finite execution sums finite products, proving the rational statement. $\square$

Remark 22.3. The number of cells and coefficients can be enormous. Neither polynomial running time nor a uniform degree bound for the original sum-of-squares hierarchy follows. Refinement changes the domain cover, not merely a single global SOS degree. For a finite collection of architectures obeying a specified budget, maxima of these intervals give outer intervals; an unbounded architecture class needs a separate uniform argument such as Theorem 5.2. For compact parameter families, an effective net and a verified modulus for the primitive kernels are additional inputs, not consequences of compactness alone.

The resource estimate also makes row precision explicit. With $m = 2^b$, the lower witnesses have dyadic rows and lose at most $HS2^{-b}$ in value. Thus increasing row precision provides a quantified approximation for a fixed architecture, while its random-bit implementation still has to be charged. The physical theorem avoids this cost altogether because its rows are deterministic.

23. AUTONOMOUS NOISY TESTING WITH A PRECISION AND TIME LEDGER

This section supplies a substantive autonomous comparison rather than merely identifying shared rows. The stationary optimum is classical [6]; the proof and explicit finite-time implementation are recorded to make the resource conventions unambiguous. They are not used as novelty evidence for the physical theorem.

Let reports be independent, with law $\text{Ber}(p)$ under hypothesis $-$ and $\text{Ber}(q)$ under hypothesis $+$, where $0 < p < 1/2$, $q = 1 - p$, and $r = q/p$. A W -state autonomous machine has two time-independent stochastic transition matrices T_0, T_1 . Assume their induced chains $P_- = qT_0 + pT_1$ and $P_+ = pT_0 + qT_1$ are irreducible. A final decision uses only the state, with optional fresh terminal randomization.

Lemma 23.1 (Likelihood-range bound). *If equivalent probability laws ν_+, ν_- on a finite set satisfy*

$$\frac{\max_i \nu_+(i)/\nu_-(i)}{\min_i \nu_+(i)/\nu_-(i)} \leq R,$$

then $\text{TV}(\nu_+, \nu_-) \leq (\sqrt{R} - 1)/(\sqrt{R} + 1)$.

Proof. Write $L = d\nu_+/d\nu_-$, $a = \min L$ and $b = \max L$. Since $\mathbb{E}_{\nu_-} L = 1$, one has $a \leq 1 \leq b$. The chord bound for $(L - 1)_+$ gives

$$\text{TV}(\nu_+, \nu_-) = \mathbb{E}_{\nu_-} (L - 1)_+ \leq \frac{(1 - a)(b - 1)}{b - a}.$$

For fixed b/a , maximizing the last expression gives $a = 1/\sqrt{b/a}$; substitution yields $(\sqrt{b/a} - 1)/(\sqrt{b/a} + 1)$. It is increasing in b/a , proving the assertion. The identical-law case follows directly. $\square$

Theorem 23.2 (Autonomous stationary bound and an explicit approximation). *Put $D = W - 1$, $W \geq 2$. In the irreducible autonomous class the infimum stationary minimax error is*

$$(86) \quad e_W = \frac{1}{1 + r^D}.$$

For every $\epsilon = 2^{-b}$, $b \geq 0$, there is a machine whose rows have denominator dividing 2^{b+1} , and whose minimax error at report time n from any initial state is at most

$$(87) \quad e_W + \epsilon(W - 2) + \exp \left\{ - \left(\frac{p\epsilon}{2} \right)^D \left\lfloor \frac{n}{D} \right\rfloor \right\}.$$

The stationary lower bound is not a finite-horizon lower bound.

Proof. Every positive induced transition has its likelihood ratio in $[r^{-1}, r]$, and the supports of the two induced matrices are identical. The Markov chain tree formula writes each stationary weight as a sum of products over directed spanning trees rooted at that state. Each product has D transitions; its ratio lies in $[r^{-D}, r^D]$. The ratio of the two unnormalized stationary weights lies in the same interval. Normalizing multiplies all state likelihood ratios by a common number, so their maximum-to-minimum ratio is at most r^{2D} . Lemma 23.1 therefore bounds stationary total variation by $(r^D - 1)/(r^D + 1)$. The equal-prior Bayes error is at least $1/(1 + r^D)$, and a minimax error cannot be smaller than that Bayes error. This is the equal-prior binary specialization of the Hellman–Cover bound.

For the upper approximation, take states $i = 0, \dots, D$, weights $c_0 = c_D = 1$, and $c_i = \epsilon$ at interior states. On report 1, move from i to $i + 1$ with probability $a_i = \frac{1}{2} \min(1, c_{i+1}/c_i)$, if that neighbor exists; otherwise stay. On report 0, move to $i - 1$ with probability $b_i = \frac{1}{2} \min(1, c_{i-1}/c_i)$, if it exists; otherwise stay. All other probability is assigned to staying. Detailed balance gives

$$\pi_+(i) = \frac{c_i r^i}{\sum_j c_j r^j}, \quad \pi_-(i) = \frac{c_i r^{-i}}{\sum_j c_j r^{-j}}.$$

Reflection $i \mapsto D - i$ exchanges these laws. Decide $+$ for $i > D/2$, $-$ for $i < D/2$, and use a fair terminal decision at $i = D/2$ when this state exists. The two errors agree. Conditional on being at an endpoint, the error is exactly e_W . The total interior stationary mass is at most $\epsilon(W - 2)$, since each $r^i \leq r^D$ and the denominator is at least r^D under $+$, with the reflected argument under $-$. This proves the first two terms of (87).

Every existing neighbor transition under either hypothesis has probability at least $p\epsilon/2$, and every self-loop has probability at least $1/2$. From any state, move down to zero and then stay until D steps have elapsed. This yields the common minorization $P_\pm^D(i, \{0\}) \geq \delta = (p\epsilon/2)^D$. Subtracting that common atom gives contraction in total variation by $1 - \delta$ every D steps. Comparison with the stationary law yields $\text{TV}(P_\pm^n(i, \cdot), \pi_\pm) \leq (1 - \delta)^{\lfloor n/D \rfloor} \leq e^{-\delta \lfloor n/D \rfloor}$. A bounded decision changes error by at most this amount, proving (87). Finally, letting n tend to infinity and then ϵ to zero proves that the stationary infimum equals (86). All transition probabilities belong to $\{0, 1, 1/2, \epsilon/2, 1 - 1/2, 1 - \epsilon/2\}$. $\square$

The W states in this theorem refer to an atomic stochastic update after each report. A sequential fair-bit implementation must also retain the report, the current state, a bit counter and a flag. Each nontrivial move probability is $1/2$ or $2^{-(b+1)}$; test for a string of zero bits of the required length, and pad to $b + 1$ bits if a uniform report cycle is required. At most $4W(b + 2)$ states suffice for this implementation, including its bit-phase counter and a ready phase. This conservative bound is an implementation claim, not an optimal precision–state tradeoff. The time index n in (87) is an observation time at which an external analyst reads the autonomous machine, not a free input clock used by its transitions. The adversarial model of [10] is not covered by the independent-report proof above.

24. PHYSICAL CONSUMERS AND THE FIRST-PRINCIPLES CONNECTION

The physical preparation theorem needs no estimate of the target law as a stored input: the five events are fixed and their uniform validity uses a proved neighborhood of Q_* . This removes calibration memory from this particular consumer without asserting that target acquisition is unnecessary in an arbitrary comparison problem. The general acquired calibration theorem, with its full counter register, remains in the complete companion.

Corollary 24.1 (A raw-interface confidence construction). *Let $m = m_2 - \beta > 0$ and $0 < c < m$. Repeat the full four-preparation audit independently R times and write $\bar{D}_R$ for the mean score. Under every private candidate,*

$$\mathbb{P}(\bar{D}_R \leq c) \leq \exp\{-R(m - c)^2/2\}.$$

Under the null in which the independent candidate and target validation laws agree, with the same event chosen before both validations,

$$\mathbb{P}(\bar{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

A clocked implementation uses $4R$ preparations and at most $12(2R + 1)$ persistent states, with no calibration and no random bits.

Proof. The independent block scores lie in $[-1, 1]$ and have mean at least m under a private candidate. Hoeffding's inequality gives the first bound. Under the null, conditional on the selected event, the two independent validation indicators have the same mean. Each block therefore has mean zero, and the same inequality gives the second bound. Retain the integer sum in $\{-R, \dots, R\}$ along with the current twelve-state block. At a block boundary add its terminal score and reset its register. The deterministic external schedule is part of the clocked convention. $\square$

This is an upper construction on the raw marked interface. Its proof does not transfer a lower bound from a deliberately reduced Bernoulli interface, and no optimal confidence-memory frontier is claimed.

Proposition 24.2 (Target transport without a training multiplier). *For the two-preparation audit, changing only the target from Q to Q' changes its expected score by at most $\text{TV}(Q, Q')$. Its candidate training law, register profile and event selector remain unchanged.*

Proof. Condition on the candidate training word. The expectation difference is $(Q - Q')h_S$, whose absolute value is at most $\text{TV}(Q, Q')$. Average over that unchanged training law. $\square$

In particular, every common-preparation transport estimate verified for the original positive-noise collision family can be inserted here without multiplying it by a long training horizon. If candidate kernels also change, their training and validation defects must be added by a causal coupling; the target-only statement does not remove those terms.

The first-principles route is now concrete. The positive experimental kernels and actual reset preparations give the likelihoods in (8); these generate posterior continuation states. The validation-test quotient gives the positive parts in the statistical dual. A prior then supplies an architecture-uniform impossibility certificate. On the constructive side, the five-event decision is compiled by taking its future residual functions, which are closed under observation shifts. These are compatible causal states and determine the charged profile. Thus the preparation law, predictive continuation and finite-resource realization are used in the same upper/lower theorem. This is not a deduction of geometry or nonlinear operator compactness from the existence of finite stochastic rows.

25. A SHARPENED PHYSICAL CONSUMER AND ITS SIMULATION BUDGET

The exact two-preparation theorem has a downstream use with explicit hypotheses, rather than a merely renamed dependency arrow. It sharpens the mean margin of a raw-interface confidence test, and the causal morphism theorem transfers that test, with its full memory and calibration costs, to another executable interface.

Theorem 25.1 (Confidence and transported resource certificate). *Let Q be a physical marked target with $\text{TV}(Q, Q_*) \leq \beta$. Use the two-preparation event selector with a dyadic coin τ_b satisfying $|\tau_b - \tau| \leq 2^{-b}$, and put*

$$a_b = v_2 - \beta - \frac{5}{16}2^{-b}, \quad W_b = \max\{12, 6(b+1) + 5\}.$$

If $0 < c < a_b$, R independent four-preparation blocks, followed by the test $\overline{D}_R > c$, have error bounds

$$(88) \quad \sup_{\text{private}} \mathbb{P}(\overline{D}_R \leq c) \leq \exp\{-R(a_b - c)^2/2\},$$

$$(89) \quad \sup_{\text{equal}} \mathbb{P}(\overline{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

They use $4R$ preparations, at most bR fresh random bits, and at most $W_b(2R+1)$ persistent states under a clocked schedule. With an exact atomic coin, replace a_b, W_b by $v_2 - \beta, 12$.

Suppose a source interface simulates this entire repeated audit under Theorem 3.4, uniformly over the private and equal-law families, with path defect Δ_R . Then the corresponding source error bounds are the right sides of (88)–(89) plus Δ_R , capped at one. At aligned cuts its state bound is $C_{\text{cal}}R_tM_tW_b(2R+1)$, and its preparation count is N_{cal} plus all source calls required to supply the $4R$ audit preparations. Calibration is acquired and retained, not supplied as a free real-valued table.

Proof. Corollary 13.2 and Proposition 13.3 give mean block score at least a_b under every fixed private parameter. Independent fresh preparations and independent coins give independent block scores in $[-1, 1]$. Hoeffding’s inequality proves (88). Under the equal-law null, conditional on training and the frozen event, the two validation indicators have the same mean. Each block therefore has mean zero, proving (89) by the same inequality. Retain the integer sum in $\{-R, \dots, R\}$ alongside the current block register. Updating the sum and resetting the block is a deterministic boundary operation. This proves the raw resource bounds.

Apply Theorem 3.4 to the whole repeated experiment, including its stopping and validation fields. Its defect bound applies to each of the two terminal error events, whose indicators have range one; each probability therefore changes by at most Δ_R , not $2\Delta_R$. The same theorem gives the state product and preparation count. The source blocks need not be independent after a shared calibration stage: independence was used only for the reference audit, and path-law comparison transfers its final event probability. $\square$

The first part uses the exact saddle, the original microscopic bridge and the compiled residual machine. The second additionally needs the uniform acquired causal simulation theorem. A source which does not satisfy that theorem’s stability and calibration hypotheses is not covered. Likewise this consumer supplies no reverse simulation and therefore does not transfer an optimal-width lower bound to an arbitrary source. It is an actual conditional theorem interface for subsequent models, not an assertion that the independent geometric A2 chain or the unbounded B4/C2 operator obligations have already been discharged.

26. DEPENDENCIES, RETAINED RESULTS AND REMAINING DISTINCTIONS

The exact three-report binary noisy frontier, arbitrary-horizon binary-register minimax recursion and compatible Bayesian interval recursion remain as proved in the complete companion. Its exact odd-horizon majority realization is also retained; it is not a new result of this revision. Section 22 applies to registers of any finite width, controlled rows and any finite parameter set, but provides certified intervals rather than a closed minimax recursion for all such classes. The

physical theorem supplies a matched preparation segment in a continuous two-parameter family with overlapping stochastic reports; its converse is not a finite parameter-grid check.

The stationary autonomous result in Section 23 has an explicit formula, but its stationary lower bound must not be applied to a finite-horizon clocked machine. Its finite-time upper bound is purposely conservative and exposes the slowing caused by finite transition precision. An arbitrary compact-family presentation still requires additional effective information to construct a kernel net. A subdivided positive-basis degree bound must not be called a degree bound for a global SOS hierarchy.

The physical result is a genuine consumer of the experimental and continuation root. It does not make the independent A2 geometric proofs dependent on this root. The original B4 normalized-resolvent, range, compactness and control-transfer obligations, and the broad C2 strict-dual, form, rigidity and optional-projection obligations, are not proved by our finite-state theorem. Their statements, derivations and current status remain in the retained historical pipeline. No aggregate closure is inferred from preserving their source files.

Finally, the rational certificate (45), polynomial identity (47), and residual transitions are independently reproducible. The program in the accompanying source checks their exact identities and includes deliberately invalid controls. These checks are not substitutes for the proofs above. Neither compilation, preservation of earlier pages nor a successful finite test establishes novelty or an editorial decision.

REFERENCES

- [1] R. D. Smallwood and E. J. Sondik, The optimal control of partially observable Markov processes over a finite horizon, *Operations Research* **21** (1973), 1071–1088.
- [2] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272.
- [3] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176.
- [4] J. Müller and G. Montúfar, The geometry of memoryless stochastic policy optimization in infinite-horizon POMDPs, arXiv:2110.07409 (2021; revised versions).
- [5] M. Putinar, Positive polynomials on compact semi-algebraic sets, *Indiana Univ. Math. J.* **42** (1993), 969–984.
- [6] M. E. Hellman and T. M. Cover, Learning with finite memory, *Ann. Math. Statist.* **41** (1970), 765–782.
- [7] M. E. Hellman and T. M. Cover, On memory saved by randomization, *Ann. Math. Statist.* **42** (1971), 1075–1078.
- [8] R. A. Flower and M. E. Hellman, Hypothesis testing with finite memory in finite time, *IEEE Trans. Inform. Theory* **18** (1972), 429–431.
- [9] T. M. Cover, M. A. Freedman and M. E. Hellman, Optimal finite memory learning algorithms for the finite sample problem, *Information and Control* **30** (1976), 49–85.
- [10] M. A. Managoli and V. M. Prabhakaran, Memory constrained adversarial hypothesis testing, arXiv:2605.12063v1 (May 12, 2026).
- [11] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28, doi:10.1524/strm.2002.20.14.1.
- [12] Q. Qi, General Theta Foundations I: Compatible statistical experiments and resource frontiers, revision 23, September 24, 2026, complete mathematical companion, source commit 8a6043aca644043bd805668185eb227468f21170; packaged commit e926b783b643b3e7ff275a18105528927b76cc97b. The 96-page article and 649-page development are reproduced without alteration after the present article in the two complete volumes.
- [13] H. Chernoff, Sequential design of experiments, *Ann. Math. Statist.* **30** (1959), 755–770.
- [14] M. Naghshvar and T. Javidi, Active sequential hypothesis testing, *Ann. Statist.* **41** (2013), 2703–2738, doi:10.1214/13-AOS1144.
- [15] S. Nitinawarat, G. K. Atia and V. V. Veeravalli, Controlled sensing for multihypothesis testing, *IEEE Trans. Automat. Control* **58** (2013), 2451–2464, arXiv:1205.0858v6.
- [16] R. Zheng, R. Sim and A. Varvitsiotis, Solving imperfect-recall games via sum-of-squares optimization, arXiv:2602.21722v1 (2026).
- [17] A. Heller, On stochastic processes derived from Markov chains, *Ann. Math. Statist.* **36** (1965), 1286–1291, doi:10.1214/aoms/1177700000.

- [18] M. Vidyasagar, The complete realization problem for hidden Markov models: a survey and some new results, *Math. Control Signals Systems* **23** (2011), 1–65, doi:10.1007/s00498-011-0066-7.
- [19] N. Gillis and F. Glineur, On the geometric interpretation of the nonnegative rank, *Linear Algebra Appl.* **437** (2012), 2685–2712, arXiv:1009.0880.
- [20] B. Reusch and W. Merzenich, Minimal coverings for incompletely specified sequential machines, *Acta Informatica* **22** (1986), 663–678, doi:10.1007/BF00263650.

PRESERVED PREDECESSOR

General Theta Foundations I, revision 27

The following predecessor volume is reproduced without alteration.
The complete current article precedes this divider.

Frozen predecessor commit:

7a659e0a2a41cd91f1f6f4ddc3b01c50a77882e3

GENERAL THETA FOUNDATIONS I: SADDLE GEOMETRY AND THE MEMORY OF CAUSAL EXPERIMENTS

QIAN QI

ABSTRACT. We study the memory required to attain a statistical saddle in a causal experiment. For finite interfaces, admissible registers give normalized positive realizations of the experiment's continuation channels. An unrestricted minimax saddle determines an exposed Bayes face; optimal finite-memory realizations must also satisfy contact and stationarity at the supporting parameters. These additional conditions can fix responses on Bayes ties and thereby force memory that a Bayes best response alone would not require. For a continuous family of marked product experiments we obtain exactly five states at the decision cut of an optimal two-preparation audit. For serial validation with the candidate read first, the minimum whole-schedule peak is exactly twelve. Both converses include stochastic encoders and continuations. They are proved by positive face packing, not by minimizing a chosen deterministic machine. We also identify the all-preparation limiting value and derive the closest-product geometry directly from the experimental probabilities. Every least-favorable prior localizes near two intrinsic contact points; symmetric priors converge to their equal mixture with an explicit quadratic-transport bound. Prepared predictive quotients, causal simulation, informative finite-memory control, exact one- and two-preparation saddles, and positive-noise physical realization are developed within the same resource framework.

CONTENTS

1. Introduction	2
1.1. Saddle geometry and its realizations	2
1.2. Exact memory in a continuous marked experiment	3
1.3. A law for every preparation number	4
1.4. Prepared foundations and retained consequences	4
1.5. Relation to earlier work	4
1.6. Organization	5
2. Causal experiments and their predictive quotients	5
2.1. The experimental object	5
2.2. Prepared predictive equivalence	6
3. Attainable resolution and resource-aware simulation	7
3.1. Checkpoint and online prediction	7
3.2. Causal morphisms and simultaneous error budgets	8
4. Experiments, architectures and attainable resources	10
5. A statistical dual that is uniform over architectures	11
6. Controlled experiments without a dominating action	13
6.1. An informative action choice	14
7. Controlled experiments at a prescribed memory profile	15

Date: September 24, 2026. Revision 27.

2020 Mathematics Subject Classification. Primary 62B05, 62C20; Secondary 60J10, 68Q45, 93E20.

Key words and phrases. Statistical experiments, causal realization, predictive distributions, finite memory, minimax duality, physical certification.

7.1. A parameterwise occupation state	15
7.2. An exact informative-control bottleneck	17
8. Saddle faces and causal realization	18
8.1. A normal form over a finite causal interface	19
8.2. The realization theorem	20
8.3. Faces as memory witnesses	21
9. Exact minimax laws for a marked product family	22
9.1. Complementary slackness and the event alphabet	24
10. A self-contained collision interface	24
10.1. Preparation and elastic scattering	25
10.2. Positive noise and the actual mark	26
11. The physical experiment and its sharp preparation boundary	27
11.1. An exact rational certificate against one preparation	28
11.2. A marked five-event selector	28
11.3. Causal compilation and the full register profile	29
12. The exact two-preparation marked saddle	31
13. A continuous two-preparation marked family	33
14. Exact memory from a marked saddle	35
14.1. The sharp serial-validation peak	37
15. Preparation limits and localization of least-favorable priors	38
16. An exact preparation–decision-memory law	41
17. Integer memory and the failure of a naive partition formula	43
18. Quantitative certificates for a finite compatible architecture	44
19. Autonomous noisy testing with a precision and time ledger	46
20. Physical consumers and the first-principles connection	47
21. A sharpened physical consumer and its simulation budget	48
22. Dependencies, retained results and remaining distinctions	49
References	50

1. INTRODUCTION

A statistical saddle determines the best expected decision, but need not determine a machine that can make that decision. A Bayes response can lose information that is immaterial under its prior yet necessary for uniform optimality over the parameter family. In a causal experiment, this distinction has a resource cost: discarded information cannot be recovered by an uncharged random selector or a terminal decoder. The central question of this paper is how to read that cost from the saddle itself.

The underlying experiment consists of a preparation, positive normalized kernels, a causal action interface, executable future tests, and charged register cuts. The unknown parameter is fixed throughout an execution. The preparation determines path laws and posterior continuations; no geometric reference measure is supplied independently of the experiment. The predictive quotient identifies histories that admit identical legal future tests. Its stochastic realizations, rather than ordinary linear rank alone, govern finite memory.

1.1. Saddle geometry and its realizations. For a finite causal interface, all controller wirings within a given profile are represented by common normalized positive matrix products. The generators at successive cuts must be compatible under every one-step continuation. Independently factoring each cut does not suffice. Let $\mathcal{C}_{\mathbf{K}}$ be this compact, generally nonconvex response set,

and let U be the unrestricted minimax value. A least-favorable prior exposes the Bayes face F_* . The profiles attaining U are precisely those admitting compatible positive realizations of

$$\mathcal{S}_* = F_* \cap \{C : g_\theta(C) \geq U \text{ for all } \theta\}.$$

Theorem 8.2 gives the exact loss decomposition, attainment criterion, supporting-parameter contact equations, and the factorizations imposed at a retained-state cut. This is not a minimax interchange over nonconvex machines. Its operative distinction is between a Bayes exposed face and its uniformly optimal part.

This distinction is decisive when some Bayes coefficients vanish. Contact and stationarity at interior supporting parameters may determine these coordinates even though the Bayes objective does not. Once they are determined, a convex-face obstruction applies to every stochastic encoder and decoder. Lemma 8.4 gives a simple reusable form: obligatory vertices and additional rows in disjoint faces require separate positive generators. Exact causal recodings preserve the attainable profiles; approximate simulations transport the value with explicit state products and error allowances.

1.2. Exact memory in a continuous marked experiment. Our main application is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad (p, q) \in [0, 1]^2,$$

where the last coordinate is the actual preparation mark. The target Q_γ has masses $((1 + \gamma)/4, (1 - \gamma)/4, (1 - \gamma)/4, (1 + \gamma)/4)$ on 000, 001, 110, 111, and zero mass elsewhere. After candidate training, the auditor freezes a row and event and applies them to independent candidate and target validations. The score is target indicator minus candidate indicator. Write $U_N(\gamma)$ for its unrestricted minimax value with N training preparations.

The exact two-preparation saddle of Theorem 13.1 has a two-point least-favorable prior for $6/25 \leq \gamma \leq 13/50$. Its two extreme-count response coordinates are Bayes ties. Theorem 14.1 proves that every minimax response, not only the displayed symmetric optimizer, must average these coordinates to the same explicitly determined number $t_\gamma \in (0, 1)$. Three other count rows are forced cube vertices. Two disjoint exposed faces then require two additional generators. Consequently

minimum decision width at the exact two-preparation value = 5.

The result applies to the original adaptive row–event interface: training, gates, stopping, retained randomness, and independent calibration are included. A common-garbling subfamily supplies the converse; the construction works against the original private candidate class.

Applying the same argument to future *output channels* at a validation cut, rather than only to event responses, yields a stronger scheduled result. If the candidate's complete validation is read before any target validation report, eight deterministic continuation vertices and four additional disjoint faces force twelve generators. Theorem 14.3 therefore proves

minimum peak for exact forward serial validation = 12.

The matching implementation is the existing five-event residual machine. The lower bound does not assume that the event name remains stored after its residual is sufficient. This serial-order assertion is distinguished from the order-independent decision-width theorem. Reverse-order and interleaved validations are not assigned a peak optimum by this result.

For the collision bias $\gamma = 1/4$, the exact statistical values are

$$(1) \quad U_1 = \frac{85 - 7\sqrt{73}}{64}, \quad U_2 = \frac{-r^3 + 6r^2 - 3r + 14}{32}, \quad 2r^3 + 25r^2 - 3 = 0,$$

where $1/3 < r < 7/20$. Their full-parameter lower certificates and matching Bayes upper bounds are proved in the article. The new memory converses extract consequences from the entire optimal face of these calculations. They do not replace the exact values by a relaxation or a finite grid. Compactness also gives a strict four-state loss uniformly over the bias interval, and a strictly positive eleven-state loss for the forward serial class. The four-state loss survives sufficiently small positive sensor noise through an explicit vanishing Gaussian-tail modulus.

1.3. A law for every preparation number. The finite- N saddles also determine a natural limiting question. Theorem 15.1 proves that unrestricted values increase to the distance from the target to the nearest candidate experiment. More significantly, every least-favorable prior concentrates on that closest-experiment set. This follows from the same common empirical response used as a competitor in the saddle game; the prior is not replaced by a parameter-dependent algorithm.

For the marked family the closest candidates and a uniform quadratic growth inequality are derived directly from its four diagonal masses. With $s_\gamma = \sqrt{(1 + \gamma)}/2$, the exact limiting value is

$$d_\gamma = 2s_\gamma - 1 - \gamma/2,$$

and its contact set consists exactly of (s_γ, s_γ) and $(1 - s_\gamma, 1 - s_\gamma)$. For every $N \geq 1$,

$$d_\gamma - \sqrt{3/N} \leq U_N(\gamma) \leq d_\gamma.$$

Every symmetric least-favorable prior converges to the equal mixture of those points, with quadratic-transport distance at most $\sqrt{2300/7}(3/N)^{1/4}$. The intrinsic growth constant is derived, not assumed as a dimension or small-ball hypothesis. These are uniform bounds and a limiting classification, not a claim that a two-point prior is exact at every finite preparation number or that the rates are sharp.

1.4. Prepared foundations and retained consequences. The general foundations layer proves path construction, predictive quotient closure, checkpoint resolution, and acquired causal transport. The latter jointly charges calibration, preparation, quantization, numerical and model error, and the memory of a downstream controller. The informative routing family exhibits the exact values $1/(2\lceil m/K \rceil)$ with a real routing cut and $K/(2m)$ with a free persistent controller selector. The separate nonlinear occupation recursion retains all common-row constraints. These results distinguish an unrestricted saddle from its legal positive realizations.

The positive-noise collision construction realizes the marked experiment on the original report and actual-mark interface. The exact ideal score for $N = 2$, $W \geq 12$, the finite-precision implementation, and the transported confidence theorem remain part of the paper. So do the integer revelation laws, compatible subdivision certificates, and the explicitly attributed autonomous finite-memory comparison. For reference, the deterministic precursor constants are

$$(2) \quad u_1 = \frac{4035777}{10240000}, \quad m_2 = \frac{12 - 3\sqrt{3}}{16}, \quad \beta_0 = \frac{1}{300}.$$

Their earlier constructions and proofs are retained along with their later exact improvements.

1.5. Relation to earlier work. Positive stochastic realization and invariant cone methods have classical antecedents in Heller [17]; Vidyasagar [18] surveys the complete hidden-Markov realization problem. Nonnegative rank differs from ordinary linear rank, and its geometric interpretation is developed by Gillis and Glineur [19]. Our matrix normalization adds the specified causal action order and charged cuts. We do not claim the positive-realization mechanism itself as new. The additional argument links its generators to the uniformly optimal portion of a statistical saddle, including contact equations on Bayes-indifferent coordinates.

Blackwell comparison and convex minimax provide the statistical mechanisms [2, 3]. Belief-state dynamic programming is classical [1]. Controlled information acquisition and adaptive sensing have an established history [13, 14, 15]; neither causal action selection nor an adaptive advantage by itself is a priority claim here. Finite-memory testing and randomization are likewise classical [6, 7, 8, 9]. The autonomous stationary formula below is attributed to Hellman and Cover. Recent work on memory-constrained adversarial testing [10] and imperfect-recall optimization [16] concerns adjacent constraints with different adversarial or objective conventions. Polynomial policy optimization and positivity methods also have substantial antecedents [4, 5]. We retain the explicit distinctions in the literature crosswalk accompanying the source.

The empirical estimation bound used in the preparation limit is the same elementary finite-alphabet estimate appearing in the retained decision-spectrum theorem. The new use is localization of every least-favorable prior, followed by an explicit intrinsic contact calculation. Norberg's filtered-experiment comparison [11] remains an adjacent general theory; no proof-level non-overlap claim is made without a complete comparison of the original work.

1.6. Organization. The prepared foundations and finite interface are followed by statistical and controlled duality. Section 8 gives the central realization criterion. The exact marked saddles and physical bridge supply its hypotheses, and Section 14 extracts the decision and serial-peak converses. Section 15 gives the all-preparation law. The remaining resource and physical consumers follow. Complete predecessor volumes retain the development of the predictive, geometric, and operator interfaces. The independent historical A2 chain and the B4/C2 aggregate obligations are not turned into consequences of a finite-state theorem by a change of terminology.

2. CAUSAL EXPERIMENTS AND THEIR PREDICTIVE QUOTIENTS

We begin with the object and morphism structure. Its elementary probability statements fix the meaning of later resource theorems; classical kernel construction, conditional expectation and quantization are not assigned a new priority. The substantive comparisons below retain their preparation, memory and approximation coordinates simultaneously.

2.1. The experimental object. An experiment has a fixed unknown parameter $\theta \in \Theta$, hidden states X_t , visible actions A_t , reports Y_{t+1} , and preparation laws μ_θ . We impose the following six conventions.

A0 (types and visibility). These spaces are standard Borel. Visible histories include actions, reports and publicly recorded resource increments. A legal-action set and its remaining budget are part of the visible protocol type. Two histories with different legal continuations are not identified.

A1 (positive kernels). There are jointly measurable normalized nonnegative kernels $K_t^\theta(dx', dy, dc \mid x, a)$, with c a nonnegative resource increment. An absorbing symbol represents termination or failure. Dirac kernels are allowed. Positivity here does not require every outcome to have strictly positive probability. A hidden state can contain the necessary physical past; this notation does not assume a finite-dimensional physical closure.

A2 (causality). A policy is a kernel $\alpha_t(da \mid h_t)$ depending only on visible past data and allowed independent randomization. It cannot read the unknown parameter. Stopping is with respect to the visible filtration.

A3 (executable tests). The specified bounded future tests include constants, required report events and their legal one-step continuations. For discrete reports the one-step instrument is

$$(\mathbb{T}_{t,a,y}f)(\theta, x) = \int f(\theta, x') K_t^\theta(dx', \{y\}, dc \mid x, a).$$

The test spaces are closed under the required instruments, finite linear combinations and the bounded monotone limits needed to recover the stated event laws. For general reports one uses event instruments and regular conditional distributions, not division by the probability of a continuous singleton.

A4 (actual preparation). Average risks and acquired state laws refer to the specified preparation and exploration protocol. Failed trials, calibration, resets and conditioning events retain their probabilities and implementation costs. A convenient measure on a reachable set does not replace its acquired law.

A5 (executable resources). Persistent labels, temporary buffers, program constants, calibration precision, preparations and clocks are distinguished. A retained selector is a state. A parameter-dependent table is not a known program constant. A simulator with R_t labels composed with a controller with K_t labels has the direct product bound $R_t K_t$, before any justified quotient is taken. Atomic stochastic rows and fair-bit implementations are separately specified.

Proposition 2.1 (Prepared path laws). *For every fixed parameter and causal policy, the preparation and kernels uniquely determine their finite-dimensional path laws. On a finite horizon they are the iterated kernel product*

$$\mu_\theta(dx_0) \prod_{t < T} \alpha_t(da_t \mid h_t) K_t^\theta(dx_{t+1}, dy_{t+1}, dc_t \mid x_t, a_t).$$

These laws are consistent under marginalization. For a countable horizon they determine the usual kernel extension. Deterministic dynamics give their original trajectory laws when represented by Dirac kernels.

Proof. Successive integration constructs each finite law. Normalization of the last two kernels gives its preceding marginal, and induction proves consistency and uniqueness on cylinder events. The countable kernel extension gives the full law; equivalently one applies the standard extension theorem to these consistent iterated kernels. Integration against a Dirac transition is evaluation along that transition, which proves the last assertion. $\square$

2.2. Prepared predictive equivalence. For Bayesian predictions additionally fix a prior on the parameter, and regard (θ, X_t) as the hidden variable. A prepared posterior π at a given protocol type evaluates the future tests. Define $\pi \sim_t \tilde{\pi}$ if their evaluations agree on every specified future test. This definition concerns one prepared experiment; it does not identify a prior-dependent posterior with a statistic sufficient simultaneously for all fixed parameters.

Theorem 2.2 (Predictive quotient and causal closure). *For finite report alphabets, predictive equivalence is preserved by every legal positive-probability report update. Its class determines the laws of all specified future experiments. Every deterministic history statistic that determines those laws factors onto this quotient. For a measurable realization it is sufficient that a countable separating test family gives a measurable evaluation map with standard Borel image, and that the chosen statistic admits a measurable factor. In finite history spaces the latter conditions are automatic.*

Proof. For a legal action and report, Bayes' rule gives

$$\pi'(f) = \frac{\pi(\mathbb{T}_{t,a,y} f)}{\pi(\mathbb{T}_{t,a,y} 1)}.$$

Both numerator and denominator depend only on the old class by A3. Equivalent posteriors assign the same probability to the report, so updates agree on the next test space whenever that probability is positive. Null reports do not affect the prepared law and may be assigned a fixed

admissible version. Induction determines every finite future report word; the monotone-class extension gives the required future events. Conversely those event laws determine the tests. If two histories have the same sufficient statistic, they have the same future evaluations; the quotient is therefore constant on its fibers. This gives the factor set-theoretically. The stated measurable factor assumption supplies its measurable version; no claim that every abstract quotient of a standard Borel space is standard Borel is used. $\square$

Proposition 2.3 (Barycenters, entropy and normalized response). *In a fixed preparation, suppose a hidden target Z , observation Y and postprocessing W form a Markov chain $Z \rightarrow Y \rightarrow W$. For any bounded vector of tests f ,*

$$p_W = \mathbb{E}[p_Y | W], \quad p_Y = \mathbb{E}[f(Z) | Y], \quad \mathbb{E}\varphi(p_W) \leq \mathbb{E}\varphi(p_Y)$$

for every convex φ for which these expectations exist. For a parameter-independent kernel G , $P \ll Q$, and finite $D(P\|Q)$, regular conditional laws give

$$\begin{aligned} D(P\|Q) &= D(PG\|QG) \\ &\quad + \int D(P(dx | w)\|Q(dx | w)) (PG)(dw). \end{aligned}$$

Finally, if f, g are bounded and $\pi_u(dx) = e^{ug(x)}\pi(dx)/\pi(e^{ug})$, then

$$\left. \frac{d}{du} \pi_u(f) \right|_{u=0} = \text{Cov}_\pi(f, g).$$

Proof. The Markov property and the tower rule give the conditional barycenter; conditional Jensen gives convex order. The joint laws $P(dx)G(dw | x)$ and $Q(dx)G(dw | x)$ have relative entropy $D(P\|Q)$. Disintegrating by w and applying the entropy chain rule gives the identity, and hence data processing. The last equality follows by quotient differentiation; boundedness justifies differentiation under the integral. $\square$

The barycenter statement concerns a law of predictions, not a pointwise map between reachable sets. Likewise the covariance in the last formula is a normalized response, not an independently postulated Fisher matrix or a global sufficiency theorem.

3. ATTAINABLE RESOLUTION AND RESOURCE-AWARE SIMULATION

3.1. Checkpoint and online prediction. Fix an actual exploration protocol and its history law. Let $p(h)$ be a bounded d -dimensional conditional prediction vector, with law ν . For instance, a future test is sampled independently after encoding, and the coordinates include its sampling weights. Squared excess prediction loss is then $\mathbb{E}\|p(h) - \hat{p}\|^2$.

Proposition 3.1 (Checkpoint resolution). *For at most M retained labels the least checkpoint excess loss is*

$$(3) \quad \inf_{c_1, \dots, c_M} \int \min_{1 \leq j \leq M} \|p - c_j\|^2 \nu(dp).$$

Independent randomized encoders or decoders cannot improve it. This formula concerns the stated preparation and does not provide a common codebook by separately optimizing at each unknown true parameter.

Proof. Conditional squared loss decomposes into its Bayes loss and squared distance from the conditional mean. For each retained label replace a random decoder by its mean. Random assignment to the resulting centers cannot do better than a nearest center. Conversely a nearest-center assignment attains the infimum over codebooks arbitrarily closely. Conditioning on an independent shared random seed, if such a seed is allowed, gives a codebook for each seed,

each with risk at least (3). Averaging cannot improve the bound. The formula does not charge the computation of a checkpoint assignment as an online algorithm; online compatibility is a separate condition. $\square$

Theorem 3.2 (A matched online resolution law). *Suppose a fixed prepared experiment has a predictive state s_t in a compact metric space S , a known initial state, and updates $s_{t+1} = F_{t,a,y}(s_t)$. Assume every permitted update is ρ -Lipschitz for one $0 \leq \rho < 1$, and the prediction map $p : S \rightarrow \mathbb{R}^d$ is L -Lipschitz. Suppose S has a δ -net of at most $C\delta^{-d}$ points for all sufficiently small δ . At a checkpoint T , assume the actual law ν of $p(s_T)$ satisfies the small-ball bound $\nu(B(z,a)) \leq Aa^d$ for every center z and all sufficiently small $a > 0$. Then for all sufficiently large M , the minimum online excess prediction loss with M retained predictive labels satisfies*

$$(4) \quad c_0 M^{-2/d} \leq R_M^{\text{on}} \leq C_0 M^{-2/d}.$$

Here one may take

$$c_0 = \frac{1}{2}(2A)^{-2/d}, \quad C_0 = \frac{L^2 C^{2/d}}{(1-\rho)^2},$$

Any separate exploration controller, report buffer or clock is charged in addition to these M labels. The exploration protocol and the preparation are fixed before optimizing the encoder.

Proof. Any online M -label state is also a checkpoint encoder, so Proposition 3.1 applies. For any M centers, the union of balls of radius $a = (2AM)^{-1/d}$ has ν -mass at most one half. On its complement squared distance is at least a^2 . Integration gives $c_0 M^{-2/d}$. This argument also permits randomized online encoders.

For the upper bound take a δ -net with $\delta = (C/M)^{1/d}$, and a fixed measurable nearest-net assignment (with a fixed tie rule). Quantize the known initial state and, after each legal update, quantize again. The state error obeys $e_{t+1} \leq \rho e_t + \delta$, $e_0 \leq \delta$, and hence $e_t \leq \delta/(1-\rho)$. The prediction error is at most $L e_t$. Squaring proves the upper bound, uniformly in time. Each update uses only the current net label and the current permitted input, so the construction is online. It does not grant the encoder free access to the original discarded history. If the exploration policy needs another register, that register is combined by a product construction. $\square$

For example, a Lebesgue density bounded above by B gives the small-ball hypothesis with $A = Bv_d$, where v_d is unit-ball volume. The lower hypothesis is a condition on the *acquired prediction law*. A reachable set of dimension d without a mass condition does not imply the lower bound. Similarly, the contraction assumption is additional dynamics, not a consequence of kernel positivity.

3.2. Causal morphisms and simultaneous error budgets. A causal simulator $\Gamma : \mathfrak{E} \rightarrow \mathfrak{F}$ is a parameter-independent executable transducer that uses source preparations to supply the target interface under every allowed target policy. Its defect ε is a uniform total-variation bound on the resulting path laws, including the fields required by the task. Its resource map ρ_Γ bounds the source resources of the transducer composed with a target controller.

Proposition 3.3 (Composition and risk transfer). *Causal simulators compose. Their uniform defects add, capped at one, and their resource maps compose in the order $\rho_{\Gamma_1} \circ \rho_{\Gamma_2}$. For a terminal loss in $[0, L]$ and a common target task,*

$$\mathcal{R}_{\mathfrak{E}}(\rho_\Gamma(B)) \leq \mathcal{R}_{\mathfrak{F}}(B) + L\varepsilon.$$

For reward in $[-1, 1]$, the corresponding maximum worst-case rewards satisfy the reverse inequality with an error 2ε .

Proof. Interleave the two transducers; their pair of internal states is a causal implementation. Total variation contracts under a kernel and obeys the triangle inequality, proving the defect sum. If their state counts are R_t and M_t , the pair has at most $R_t M_t$ states. Other declared resource bounds compose by execution of the two protocols. A bounded loss changes expectation by at most its range times total variation. Apply this bound to every fixed parameter and target controller and take the appropriate infimum or supremum. Uniformity in the controller is essential to this step. $\square$

Theorem 3.4 (Acquisition, quantization and model-error transport). *Suppose an exact source transducer has predictive state $s_t \in S_t$ and additional charged buffers of size R_t , supplies T target operations, and has uniform model defect at most ε_{mod} . Its initialization and update depend on a calibration quantity ξ . From N_{cal} source preparations acquire an estimate $\hat{\xi}$ with at most C_{cal} retained labels, such that*

$$\sup_{\theta} \mathbb{P}_{\theta}(d(\hat{\xi}, \xi) > a) \leq \eta.$$

Subsequent preparation noises are independent of calibration conditional on the fixed parameter. Let finite nets of sizes M_t have radii δ_t . Assume, on matching input/output histories and on the good calibration event, that the quantized transducer's state error satisfies

$$(5) \quad e_0 \leq d_0 + a_0 a + \delta_0, \quad e_{t+1} \leq \rho_t e_t + d_{t+1} + a_{t+1} a + \delta_{t+1}.$$

Assume its next output kernel and the exact transducer's output kernel differ in total variation by at most $\epsilon_t + L_t e_t + b_t a$, uniformly in the fixed parameter and legal policy. All constants and nets are declared independently of that unknown parameter. Then the quantized acquired transducer has defect at most

$$(6) \quad \Delta = \min \left\{ 1, \varepsilon_{\text{mod}} + \eta + \sum_{t < T} (\epsilon_t + L_t e_t + b_t a) \right\}.$$

Composing it with any target controller of cut widths K_t requires at most $C_{\text{cal}} R_t M_t K_t$ labels at aligned execution cuts and transfers its bounded reward with loss at most 2Δ . Its preparations are N_{cal} plus all source preparations specified by the transducer; no successful-trial conditioning removes the failures.

Proof. On the good calibration event, iterate (5). More explicitly, with $D_0 = d_0 + a_0 a + \delta_0$ and $D_j = d_j + a_j a + \delta_j$,

$$e_t \leq \sum_{j=0}^t D_j \prod_{k=j}^{t-1} \rho_k.$$

Couple each pair of output kernels maximally while their external histories agree. On agreement the target controller sees the same inputs, so it uses the same legal action and may use the same independent randomization. Its discontinuous action choices require no Lipschitz assumption: before a disagreement their arguments are identical. The chance of a first discrepancy is bounded by the sum in (6). Add the probability of bad calibration and the exact transducer's model defect. This proves the path bound, including stopping after padding by absorbing states.

The executable approximate transducer stores its calibration label, additional buffer, and net index. Combine these with the actual target register, not its full history or an uncharged belief. The resulting product gives the state count. If an operation contains further buffer cuts, apply the same product at each such cut; R_t was assumed to include them. Preparation costs add by execution. Finally apply Proposition 3.3. $\square$

Corollary 3.5 (A finite acquisition register). *Suppose r_0 calibration rows have at most d outcomes, and take n independent preparations from each row on a fixed schedule. Their empirical rows satisfy, with probability at least $1 - \eta$ uniformly in the row probabilities,*

$$\max \text{TV}(\hat{P}, P) \leq \frac{dz}{2}, \quad \eta = \min\{1, 2r_0d \exp(-2nz^2)\}.$$

Their counts can be retained in at most $(n+1)^{r_0(d-1)}$ labels. Thus Theorem 3.4 can use $N_{\text{cal}} = r_0n$, $a = dz/2$ and this finite calibration register when its calibration metric is the maximum row total variation.

Proof. For each outcome the empirical frequency is the mean of independent Bernoulli indicators. Hoeffding's bound and a union bound over at most r_0d coordinates give the event stated. The total-variation estimate is half the sum of coordinate errors. Retain $d - 1$ counts per row; the last is reconstructed from the known number of draws at that stage of the clocked schedule. Each retained count takes at most $n + 1$ values. An autonomous implementation must additionally charge its schedule. $\square$

These results separate exact predictive equivalence, average checkpoint resolution, online realizability, and uniform simulation risk. In particular, neither an optimal checkpoint codebook nor the proof-state bundle of Section 7 is automatically an executable finite register.

4. EXPERIMENTS, ARCHITECTURES AND ATTAINABLE RESOURCES

Throughout, total variation is $\text{TV}(P, Q) = \sup_{0 \leq h \leq 1} (P - Q)h = \frac{1}{2} \sum_x |P(x) - Q(x)|$. An experimental parameter is unknown and fixed throughout an execution. Controller randomization is independent of that parameter. A reset preparation is a fresh draw, independent of earlier preparations given the parameter. A controller may not recover discarded information from an uncharged tape, clock, selector or calibration table.

Definition 4.1 (Finite architecture). A finite architecture A specifies a finite causal execution graph, its legal experimental operations, the observable inputs to each controller row, and a register alphabet at every cut. It also specifies which row occurrences share a stochastic table. A clocked architecture may use different rows at different scheduled times. An autonomous architecture may use time only through a charged state. For a parameter family Θ and terminal reward in $[-1, 1]$, let

$$V(A) = \sup_{q \in \prod_r \Delta_{s_r}} \inf_{\theta \in \Theta} g_\theta(A, q).$$

If $\mathcal{A}(R)$ is the class of architectures obeying a global resource budget R , its outer value is

$$V^{\text{out}}(R) = \sup_{A \in \mathcal{A}(R)} V(A).$$

Both the architecture and the simulator parameter are fixed before the controller's random choices. In particular, a mixture of complete controllers is not automatically an admissible controller at the original register profile.

For a finite hidden execution graph, compatibility can be stated without division. Let $x_{\theta ir}$ be the incoming mass of hidden history i at observable row r , and let $f_{\theta irj}$ be the outgoing mass through choice j . Include the usual fixed-kernel flow equations, as well as

$$(7) \quad f_{\theta irj} \geq 0, \quad \sum_j f_{\theta irj} = x_{\theta ir}, \quad f_{\theta irj} x_{\theta' i' r} = f_{\theta' i' rj} x_{\theta ir}.$$

For shared rows, the indices i, i' include all their occurrences.

Proposition 4.2 (Compatible realization). *For a specified finite architecture, nonnegative flow arrays satisfying (7) are exactly those realized by a common stochastic row at every declared observable input, including at zero incoming mass.*

Proof. Necessity follows from $f_{\theta irj} = x_{\theta ir} q_r(j)$. For sufficiency, if some incoming mass at row r is positive, define $q_r(j)$ by dividing the corresponding outgoing mass by that incoming mass. Conservation makes q_r stochastic, and the cross-products force the same ratio at every other positive incoming mass. If an incoming mass vanishes, nonnegativity and conservation force all its outgoing masses to vanish. If the entire row is unreachable, choose any legal stochastic row. Induction through the finite execution graph reconstructs the prescribed flows. The same ratio argument applies to identified occurrences of an autonomous row. $\square$

This formulation is inherited from the complete companion and is included to fix the controller class. Compact semialgebraic feasibility and an Archimedean positivity theorem provide a complete semialgebraic certificate hierarchy for a *fixed finite architecture*. They do not interchange the two optimizations in the definition. Section 18 gives a quantitative alternative for the inner problem. The next section supplies a different certificate that survives the outer optimization.

5. A STATISTICAL DUAL THAT IS UNIFORM OVER ARCHITECTURES

Let Θ be compact, let Ω be finite, and let $\theta \mapsto P_\theta \in \mathcal{P}(\Omega)$ be continuous. Fix $Q \in \mathcal{P}(\Omega)$. An N -preparation reset selector observes $Z = (Z_1, \dots, Z_N) \sim P_\theta^{\otimes N}$ and chooses a response $h_Z \in [0, 1]^\Omega$. Its expected validation score is

$$(8) \quad G_\theta(h) = \sum_{z \in \Omega^N} P_\theta^{\otimes N}(z) (Q - P_\theta) h_z.$$

The response is fixed before two independent validation draws. Operationally it can be realized by choosing a random event with those coordinate marginals and retaining that event for both validations. No memory bound is imposed in the definition of $U_N(P, Q) = \max_h \min_\theta G_\theta(h)$.

Theorem 5.1 (Posterior-predictive dual). *Both extrema in the following identity are attained:*

$$(9) \quad U_N(P, Q) = \min_{\mu \in \mathcal{P}(\Theta)} \sum_{z, \omega} \left[\int P_\theta^{\otimes N}(z) \{Q(\omega) - P_\theta(\omega)\} d\mu(\theta) \right]_+$$

$$(10) \quad = \min_{\mu \in \mathcal{P}(\Theta)} \sum_z m_\mu(z) \text{TV}(Q, \bar{P}_\mu(\cdot | z)).$$

Here $m_\mu(z) = \int P_\theta^{\otimes N}(z) d\mu(\theta)$ and, when $m_\mu(z) > 0$,

$$\bar{P}_\mu(\omega | z) = \frac{\int P_\theta^{\otimes N}(z) P_\theta(\omega) d\mu(\theta)}{m_\mu(z)}.$$

A zero-mass term in (10) is defined to be zero.

Proof. The response space is a finite-dimensional compact convex box. Probability measures on the compact space Θ form a weakly compact convex set. The integral of (8) is continuous and affine in each argument. The minimum over parameters equals the minimum over probability measures. The minimax theorem [3] therefore interchanges the maximum and minimum. For fixed μ , each response coordinate is independent in the maximization and its contribution is the positive part of its coefficient, proving (9). The coefficient vector at a fixed z has zero sum. For positive $m_\mu(z)$ it equals $m_\mu(z)(Q - \bar{P}_\mu(\cdot | z))$, proving (10). If $m_\mu(z) = 0$, every nonnegative likelihood integrand vanishes μ -almost surely and all coefficients are zero. Continuity on the two compact strategy spaces gives attainment. $\square$

Theorem 5.2 (Dominating-row converse). *Suppose every permitted training row is a parameter-independent causal garbling of one fresh draw from P_θ . Suppose every permitted validation row is the same garbling of the target and candidate laws, and every selected row–event pair is fixed before validation. Then $U_N(P, Q)$ bounds the worst-case expected score of every auditor using at most N candidate training preparations, regardless of its register profile, feedback schedule, independent randomization, or stopping rule. Independent target-only calibration, whose law does not depend on θ , does not invalidate the bound. If the all-on row and all its events are permitted, the bound is attained in the class with no memory cap and N training preparations.*

Proof. Generate N independent all-on observations in advance. At the t th requested preparation, apply the declared garbling to observation t , using only previously simulated reports and independent randomization. This simulates the adaptive transcript with its exact law. A stopped execution ignores the unused observations; it does not reveal them to the auditor. Pull back its final row–event response to the all-on alphabet. Conditional on the entire generated training word and the independent random seed, this is a function in $[0, 1]^\Omega$. Averaging the seed, including any target-only calibration, produces one response h_z for each full word z . The score is consequently (8). This proves the upper bound for the union of all architectures. Conversely, retaining the entire word and then a selected event implements any such finite response kernel when memory is unrestricted. Fresh validation gives exactly (8). $\square$

Remark 5.3 (What the dual does not convexify). The measure μ is an adversarial prior used to bound the minimum over *fixed* parameters. Its likelihood is $\int P_\theta^{\otimes N} d\mu(\theta)$, not $(\int P_\theta d\mu(\theta))^{\otimes N}$. It neither enlarges the private candidate class to mixtures nor gives the auditor a free persistent selector. Attainment of U_N without a memory cap does not assert attainment at any specified finite profile.

The state in (10) is generated by the actual preparation law, not by a geometric measure chosen independently of the experiment. For sequential continuation one retains the posterior μ_z or its future-test equivalence class; a single predictive marginal need not be closed under further observations. This is the distinction between a one-step response quotient and a future-predictive quotient.

Corollary 5.4 (Finite moment support for the product family). *For two independent Bernoulli report bits with parameters (p, q) and an independent fair mark, an optimal prior in (9) can be chosen with at most $(N + 2)^2$ atoms. The two report counts (k, ℓ) are a causally updatable sufficient statistic for computing the dual.*

Proof. Conditional on the two counts, all arrangements of report bits and all mark words have parameter-independent probabilities. Summing their coefficients reduces (9) to the $(N + 1)^2$ count pairs. Their likelihoods are

$$\binom{N}{k} p^k (1 - p)^{N-k} \binom{N}{\ell} q^\ell (1 - q)^{N-\ell}.$$

After multiplication by $Q(\omega) - P_{p,q}(\omega)$, every integrand has degree at most $N + 1$ in each variable. Thus the objective depends only on the $(N + 2)^2$ rectangular moments. Their constant coordinate is one. Carathéodory’s theorem in the remaining affine space replaces an optimal moment vector by a convex combination of at most $(N + 2)^2$ point evaluations. Counts update by addition of the current bits. No claim that this count register is optimal for the decision is made. $\square$

6. CONTROLLED EXPERIMENTS WITHOUT A DOMINATING ACTION

The reduction to one dominating row is not needed for a statistical dual, provided causal action choice is retained in the dual recursion. We give the finite-action theorem with a compact parameter space and an explicit protocol state. Its equality is an unrestricted-memory statement; its upper certificates apply to every smaller-memory class.

Let Θ be compact metric and E a finite set of protocol states. At $e \in E$ let $\mathcal{A}_e$ be a finite set of training actions. Action a produces x in a finite alphabet X_a with law P_θ^a and changes the protocol state to $T(e, a, x)$. The functions $\theta \mapsto P_\theta^a(x)$ are continuous. An unknown θ is fixed throughout the experiment. Conditional on it, each requested preparation uses a fresh independent draw from the selected row. Legal validation designs form a nonempty finite set $\mathcal{B}_e$. Design b has a finite outcome alphabet Y_b , target law Q^b and continuous candidate law V_θ^b . A policy may stop after at most N training actions. It then freezes $b \in \mathcal{B}_e$ and an event on Y_b , before one candidate and one target validation. Randomization is independent of θ . No training action is assumed to dominate another.

For a finite nonnegative measure ν on Θ write $\nu_{a,x}(d\theta) = P_\theta^a(x)\nu(d\theta)$ and set

$$(11) \quad \Phi_e(\nu) = \max_{b \in \mathcal{B}_e} \sum_{y \in Y_b} \left[Q^b(y)\nu(\Theta) - \int V_\theta^b(y) d\nu \right]_+,$$

$$F_0(e, \nu) = \Phi_e(\nu),$$

$$(12) \quad F_n(e, \nu) = \max \left\{ \Phi_e(\nu), \max_{a \in \mathcal{A}_e} \sum_{x \in X_a} F_{n-1}(T(e, a, x), \nu_{a,x}) \right\}.$$

If no training action is legal, only the stopping term is retained. All the functions vanish at the zero measure, so no posterior division is required on null histories. For positive mass, (11) is the mass times the largest posterior-predictive total variation.

Theorem 6.1 (Controlled posterior-predictive dual). *Let $\mathcal{U}_N(e_0)$ be the maximum worst-case expected validation score over all randomized causal policies in the declared protocol, with at most N training preparations and no memory cap. Then*

$$(13) \quad \mathcal{U}_N(e_0) = \min_{\mu \in \mathcal{P}(\Theta)} F_N(e_0, \mu).$$

Both extrema are attained. Each $F_n(e, \cdot)$ is the maximum of finitely many continuous linear functionals on the cone of measures; in particular it is convex, continuous and positively homogeneous. For every prior μ , $F_N(e_0, \mu)$ bounds the worst-case score of every finite-memory architecture implementing this protocol.

Proof. There are finitely many histories of length at most N and finitely many event masks at each stopping history. Consequently there are finitely many deterministic policy trees, denoted by $\Pi_N(e_0)$. For $\pi \in \Pi_N(e_0)$ its payoff $g_\pi(\theta)$ is a finite sum of finite products of continuous primitive probabilities and is continuous in θ .

A randomized causal policy induces the same outcome laws as a probability distribution on these trees: sample its finite random decision table in advance and condition on it. Conversely, with no memory cap, a sampled tree can be retained and followed. Thus the unrestricted payoff vectors are exactly the convex hull of the g_π . This step is not an assertion that mixing machines preserves a specified register width.

The simplex on $\Pi_N(e_0)$ and $\mathcal{P}(\Theta)$ are compact and convex. The expected payoff is continuous and bilinear. Minimax therefore gives

$$\max_{\lambda \in \Delta(\Pi_N)} \min_{\theta} \sum_{\pi} \lambda_{\pi} g_{\pi}(\theta) = \min_{\mu} \max_{\pi \in \Pi_N} \int g_{\pi} d\mu.$$

For a fixed measure, the terminal maximum is (11), because an event includes precisely the positive coordinates of its signed predictive measure. At a training node one selects an action and, after its observed outcome, selects a continuation tree separately on that branch. The unnormalized branch measure is exactly $\nu_{a,x}$. Backward induction proves (12) and identifies the right side with (13). It also proves the finite-linear-functional representation. Compactness gives an optimizing prior and an optimizing mixed tree. Finally, the minimum over fixed parameters of any policy is at most its average under μ , which is at most $F_N(e_0, \mu)$. This remains true after restricting its memory or architecture. $\square$

The minimization in (13) occurs once, outside the recursion. It cannot be replaced by fresh adversarial choices after observations. The posterior carries dependence on one fixed unknown parameter. Also, a Bayes-optimal tree for a minimizing prior need not by itself be a minimax policy; the saddle-point mixture is supplied by the finite game in the proof.

Corollary 6.2 (Bellman upper certificates). *Suppose functions $\Psi_n(e, \nu)$ satisfy $\Psi_0 \geq \Phi$ and, for $n \geq 1$, satisfy*

$$\Psi_n(e, \nu) \geq \Phi_e(\nu), \quad \Psi_n(e, \nu) \geq \sum_x \Psi_{n-1}(T(e, a, x), \nu_{a,x}) \quad (a \in \mathcal{A}_e).$$

Then $\Psi_N(e_0, \mu)$ is an upper certificate for every admissible architecture and every prior μ . The family F_n is the least such family. An optimal prior can be chosen with at most $d + 1$ support points, where d is the affine dimension of $\{(g_{\pi}(\theta))_{\pi \in \Pi_N} : \theta \in \Theta\}$.

Proof. Induction in (12) gives $F_n \leq \Psi_n$. The equality case is the recursion itself. The objective depends on a prior only through its vector of integrals of the finitely many g_{π} . An optimal such vector belongs to the convex hull of their compact evaluation image. Carathéodory's theorem gives the stated support bound while preserving every pure-tree payoff. $\square$

Proposition 6.3 (Causal stability). *Consider two models with the same protocol, alphabets and policy class. Suppose their training rows differ uniformly in total variation by at most ϵ , their candidate validation rows by at most η , and their target validation rows by at most β . Their unrestricted values, or their values in any common specified memory class, differ by at most $2N\epsilon + \eta + \beta$.*

Proof. Use the same policy randomization and couple corresponding training rows while the histories agree. The probability of a first discrepancy is at most $N\epsilon$, even with feedback and stopping. Conditional expected validation rewards lie in $[-1, 1]$, so differing training histories contribute at most $2N\epsilon$. On agreeing histories the same frozen design and event are used, and their conditional mean rewards differ by at most $\eta + \beta$. The bound is uniform in the fixed parameter and in the policy. Taking minima and suprema preserves it, with the reverse inequality obtained by interchanging the models. $\square$

6.1. An informative action choice. The following calculation distinguishes the controlled theorem from a common-row reduction. Let $\Theta = \{(i, j) : i, j \in \{0, 1\}\}$. Action R reports i . Action L reports j when $i = 0$ and otherwise reports an independent fair bit. Action H reports j when $i = 1$ and otherwise reports an independent fair bit. The candidate validation is $\delta_{(i,j)}$ and the target is uniform on Θ .

Proposition 6.4 (Strict adaptive advantage). *With two training preparations, the unrestricted adaptive value is $3/4$ and the best nonadaptive value is $5/8$. Nonadaptive policies here include mixtures of schedules, with their random schedule retained. No available single training action dominates all the others.*

Proof. Take R first, and then L or H according to the observed i . The parameter is recovered exactly. The complement of its validation atom has score $3/4$. No policy can exceed $\text{TV}(Q, \delta_\theta) = 3/4$ even with full information.

For the converse for fixed schedules, use the uniform prior. For a pair of actions a, b its Bayes value is explicitly

$$(14) \quad \sum_{x,y,\theta} \left[\frac{1}{16} \sum_{\vartheta} P_{\vartheta}^a(x) P_{\vartheta}^b(y) - \frac{1}{4} P_{\theta}^a(x) P_{\theta}^b(y) \right]_+.$$

Substitution of the three displayed kernels gives

	R	L	H
R	$1/2$	$5/8$	$5/8$
L	$5/8$	$7/16$	$1/2$
H	$5/8$	$1/2$	$7/16$

This is an exact finite sum, not a parameter-grid estimate. It bounds any mixture of fixed schedules by $5/8$; an early stop can be padded with an ignored observation. To attain the bound uniformly in θ , choose with equal probabilities the schedules (R, L) and (R, H) . When the chosen branch-specific action matches i , the parameter is known and the score is $3/4$. Otherwise only i is known, and the complement of its two-atom fiber gives score $1/2$. Each fixed parameter therefore has mean score $5/8$.

Action R cannot distinguish parameters differing only in j . Action L cannot distinguish $(1, 0)$ and $(1, 1)$, and H cannot distinguish $(0, 0)$ and $(0, 1)$. An experiment which fails to distinguish such a pair cannot garble to one that distinguishes it. Hence no action is a common dominating row. $\square$

If every binary report is additionally passed through an independent binary symmetric channel of crossover ϵ , the primitive training rows change by at most ϵ and have full support for $0 < \epsilon < 1/2$. Proposition 6.3, applied to both policy classes, leaves an adaptive advantage at least $1/8 - 8\epsilon$, which is positive for $0 < \epsilon < 1/64$. Thus the distinction is not confined to disjoint observation supports. The finite-horizon belief recursion and its piecewise-linear structure are classical [1]. The contribution here is their explicit integration with the frozen validation score, one fixed adversarial parameter, outer certificates, and the resource conventions; no priority for Bellman recursion itself is claimed.

7. CONTROLLED EXPERIMENTS AT A PRESCRIBED MEMORY PROFILE

An unrestricted posterior is a useful information state for a proof, but it is not a free runtime state. We first give a resource-exact dynamic formulation and then solve a controlled family in which replacing that formulation by a policy-tree mixture changes the value strictly.

7.1. A parameterwise occupation state. Fix a finite horizon T , a compact metric parameter space Θ , finite hidden execution spaces H_t , and register alphabets S_t of sizes at most K_t . The visible inputs to each stochastic row and all legal actions are specified. Any input that must survive a cut belongs to S_t : this includes a chosen action, an unread report buffer, a stop flag, a selected validation event and a validation accumulator. One may subdivide an operation into several cuts to expose these costs. The scheduled time is available only in the clocked convention.

For now rows at different scheduled times are independently prescribed. Let $\mathcal{Q}_t$ be the finite product of legal row simplices at stage t . A prescription $q \in \mathcal{Q}_t$ is a table, chosen at design time, whose runtime arguments are exactly the declared visible inputs. After summing the fixed environmental kernels its one-stage transition is a stochastic matrix $M_{t,\theta}^q$ on the hidden-state/register pairs. We assume this matrix is jointly continuous in (q, θ) . Both action selection and state updating occur in this matrix, in their specified causal order; it is not an arbitrary matrix chosen separately for each hidden state or each parameter.

For a continuous bundle of probability vectors $x = (x_\theta(h, s))_{\theta \in \Theta}$ define

$$(\mathcal{T}_{t,q}x)_\theta(h', s') = \sum_{h,s} x_\theta(h, s) M_{t,\theta}^q((h, s), (h', s')).$$

The initial bundle x_0 is fixed by preparation. Terminal rewards $r_\theta(h, s) \in [-1, 1]$ are continuous in θ . Absorbing stopped states and frozen validation phases are included in this execution, not added afterwards without charge. Put

$$(15) \quad B_T(x) = \min_{\theta \in \Theta} \sum_{h,s} x_\theta(h, s) r_\theta(h, s),$$

$$(16) \quad B_t(x) = \max_{q \in \mathcal{Q}_t} B_{t+1}(\mathcal{T}_{t,q}x), \quad 0 \leq t < T.$$

The argument x is an *offline occupation bundle*, not the realized history, not an observed parameter, and not an additional register made available to the machine.

Theorem 7.1 (Resource-exact dynamic value and supersolutions). *For the specified clocked profile and observable-row interface, $B_0(x_0)$ is the exact maximum worst-case reward over all stochastic controllers. The maximum is attained by a sequence of legal common prescriptions. With the norm $\|x - x'\| = \sup_\theta \sum_{h,s} |x_\theta(h, s) - x'_\theta(h, s)|$, each B_t is 1-Lipschitz. Moreover,*

$$(17) \quad B_0(x_0) = \inf_H H_0(x_0),$$

where the infimum ranges over families of functions on occupation bundles satisfying

$$H_T \geq B_T, \quad H_t(x) \geq H_{t+1}(\mathcal{T}_{t,q}x) \quad (q \in \mathcal{Q}_t).$$

The infimum is attained by $H = B$. No convexification over retained controllers and no new adversarial choice after a report occurs in these identities.

Proof. For a fixed sequence $q_0, \dots, q_{T-1}$, iteration of the displayed transition gives its actual terminal law for every fixed θ . The terminal minimum is therefore exactly its worst-case reward. A runtime randomized controller in this interface is precisely such a sequence of common conditional rows; independence of fresh randomization and the declared register specify all its conditional choices. Persistent random correlations are allowed only through that register.

The occupation bundle at the next stage is determined by the current bundle and the prescription, without drawing a realized history in the design problem. Hence optimizing over the remaining prescription sequence gives (16) by ordinary backward induction. This does not select a different continuation for each unrecorded history. Each maximizing q is still a single legal table common to all hidden histories and parameters. At x_0 one follows the maximizing sequence of design prescriptions, computing their occupation bundles offline; the runtime controller only reads its allowed inputs.

Stochastic matrices contract the ℓ^1 norm. The terminal minimum is 1-Lipschitz in the displayed norm. Induction, using the same q when comparing two bundles and then taking the two maxima, proves the Lipschitz assertion. Joint continuity and compactness of $\mathcal{Q}_t$ give attainment at each stage, and compactness of Θ gives the terminal minimum. Any family H

in (17) bounds the terminal reward along every prescription sequence. Backward induction therefore gives $H_t \geq B_t$. Equality is admissible, proving (17) and its attainment. $\square$

This is a nonlinear supersolution dual, not a least-favorable-prior saddle theorem at fixed width. Its optimization can be large and nonconvex. The principle of dynamic programming is classical; the point of this formulation is that the common-row and memory constraints remain in the maximizations. If rows are shared across scheduled times, as for an autonomous controller, choose the shared rows once and hold them fixed in all subsequent transitions; equivalently augment the *design* problem by that immutable prescription. Maximizing a shared row anew at each time would solve a different problem. A finite union of allowed interfaces can be handled by taking their maximum, or by a fully wired interface with the prescribed profile and its legal observable inputs. Neither operation enlarges a runtime register invisibly.

7.2. An exact informative-control bottleneck. Fix integers $m \geq 2$ and $1 \leq K \leq m$. The fixed parameter is $\theta = (i, j) \in [m] \times \{0, 1\}$. The first training experiment R reports i . Across the next cut the controller retains at most K labels, with an arbitrary stochastic encoder $E(s | i)$. It then chooses one of the m sensing actions with probabilities $D(a | s)$. The selected action is kept in a charged m -state action buffer and the old label is discarded. Action a reports j when $a = i$ and reports $\perp$ otherwise. The first report is no longer available.

The candidate validation law on $\{0, 1\}$ is δ_j ; the target is uniform. An event is frozen before one fresh candidate and one fresh target validation. Stopping before either training action is permitted. No independent persistent selector is given to the controller. The observation symbols are atomic; a bitwise input implementation would have additional cuts. Denote the maximum worst-case score by $V_{m,K}$.

Theorem 7.2 (Exact finite-memory control and its convexification gap). *In this protocol,*

$$(18) \quad V_{m,K} = \frac{1}{2 \lceil m/K \rceil}.$$

The value obtained by allowing an uncharged ex ante mixture of complete K -label controllers is instead

$$(19) \quad \bar{V}_{m,K} = \frac{K}{2m}.$$

The inequality is strict whenever K does not divide m . The matching controller for (18) has charged cut profile bounded by $(1, K, m, 3, 6, 3)$. The K -label cut controls both retention of the routing observation and the subsequent informative action choice.

Proof. For a fixed i , let

$$p_i = \sum_s E(s | i) D(i | s).$$

This is the probability of taking the informative action. On its complement, all retained information is independent of j . Averaging over the two values of j makes the expected validation score zero there, for any frozen event or randomized response. On the informative part the score is at most $\text{TV}(\text{Ber}(1/2), \delta_j) = 1/2$. Thus the minimum over j at a given i is at most $p_i/2$. A stop before sensing is also uninformative about j and obeys the same bound; it can be padded by an ignored sensing operation.

Set $L = \lceil m/K \rceil$. Since $p_i \leq \max_s D(i | s)$, a value $\min_i p_i > 1/L$ would require every i to have some row s with $D(i | s) > 1/L$. Each stochastic row has at most $L - 1$ such indices. But $K(L - 1) < m$. This contradiction proves the upper bound. Partition $[m]$ into at most K fibers of size at most L , retain the fiber label after R , and choose uniformly inside that fiber. On a successful report freeze $\{1\}$ when $j = 0$ and $\{0\}$ when $j = 1$; on $\perp$ freeze the empty event. The

score for every j is $1/(2|\text{fiber}(i)|)$, proving equality. The event alphabet has three labels. Storing the event and its candidate indicator requires at most six labels, and the final difference has three. The chosen sensing action was separately charged, proving the displayed profile.

For the convexified upper bound use the uniform prior on (i, j) . For any one controller,

$$\frac{1}{m} \sum_i p_i = \frac{1}{m} \sum_{i,s} E(s | i) D(i | s) \leq \frac{1}{m} \sum_{i,s} D(i | s) = \frac{K}{m}.$$

The same upper bound holds for mixtures. For the matching mixture, choose uniformly one of the m cyclic intervals of K indices. For that interval hardcode a deterministic K -label encoder which routes each included index to its own sensing action, and assigns other indices arbitrarily. Each i belongs to exactly K intervals. A successful action is followed by the complementary singleton event and failure by the empty event, as above. Every fixed (i, j) has mean score $K/(2m)$. This proves (19). $\square$

For example, $V_{3,2} = 1/4$ whereas $\bar{V}_{3,2} = 1/3$. The mixture just constructed can be implemented with a retained interval selector and a local label, using at most mK labels at the bottleneck. It cannot be declared a K -label realization: when the displayed gap is positive, (18) proves that no such realization exists. The action buffer of size m is not part of the K coordinate and has not been omitted from the profile. The theorem is a multicut resource law, not an assertion that its uniform peak equals K .

The actions are statistically incomparable. R cannot distinguish the two values of j at any fixed i , while action a does distinguish them at $i = a$. For $a \neq b$, action a does not distinguish the pair with $i = b$, while b does. A parameter-independent garbling cannot create a distinction absent from its input experiment.

Corollary 7.3 (A strict gap with positive report probabilities). *Replace each of the two training kernels by a mixture with weight $\epsilon > 0$ of the uniform law on its report alphabet. Let $\Delta_{m,K} = K/(2m) - 1/(2\lceil m/K \rceil) > 0$. For $\epsilon < \Delta_{m,K}/8$ the resulting full-support model still has a strict gap between the genuine profile-constrained value and its free-selector convexification.*

Proof. Each training row changes by at most ϵ in total variation. Couple corresponding rows until the first disagreement. For any fixed controller or mixed controller the probability of such a disagreement is at most 2ϵ , and its conditional reward lies in $[-1, 1]$. Each of the two optimal values changes by at most 4ϵ . Their difference is therefore at least $\Delta_{m,K} - 8\epsilon > 0$. All row arguments and register alphabets are unchanged. $\square$

The distinction between mixed and behavioral strategies without perfect recall is classical; it is not first asserted here. Contemporary polynomial formulations over behavioral simplices include [16]. Theorem 7.2 evaluates the discrepancy in this specific frozen-validation experiment for every m, K , with a matching actual register and a noisy persistence statement. In particular, a single prior followed by an unrestricted policy-tree best response cannot be substituted for (16).

8. SADDLE FACES AND CAUSAL REALIZATION

The statistical saddle and its implementation are different objects. The former is convex; the latter need not be. The purpose of this section is to retain that distinction while extracting memory obstructions from an unrestricted saddle. In particular, a zero Bayes coefficient must not be discarded before the contact conditions at the worst parameters have been imposed.

8.1. A normal form over a finite causal interface. Fix a finite horizon and finite action and report alphabets. An action a_t is chosen before report y_t . A terminal output is included as one last action, followed by a dummy report. Illegal actions can be removed from each protocol type; equivalently, use a fixed padding of the finite protocol tree and impose its zero constraints. Define a *causal response array* $C(\mathbf{a} \mid \mathbf{y})$ by nonnegativity, normalization in $\mathbf{a}$ for every $\mathbf{y}$, and the requirement that its marginal on $a_1, \dots, a_t$ depends only on $y_1, \dots, y_{t-1}$. The resulting set $\mathcal{C}$ is a finite polytope. Conditioning successive marginals realizes every such array by a full-history randomized controller; arbitrary versions on zero marginals do not change the array.

For a fixed parameter θ , the environment has a history-dependent likelihood $L_\theta(\mathbf{a}, \mathbf{y})$, obtained by marginalizing its hidden states. The bounded reward has the linear form

$$(20) \quad g_\theta(C) = \sum_{\mathbf{a}, \mathbf{y}} L_\theta(\mathbf{a}, \mathbf{y}) r_\theta(\mathbf{a}, \mathbf{y}) C(\mathbf{a} \mid \mathbf{y}).$$

Assume its coefficient vector is continuous on a compact metric parameter space. The parameter is fixed once, not selected after a report. Protocol-specific equalities, including a frozen validation choice, are represented by the corresponding finite protocol tree. A discarded protocol choice is not supplied again as a free input.

Here is an explicit positive realization of a response array. At cut t use k_t states, an initial distribution α , and nonnegative matrices $M_t^{a,y}$ of size k_{t-1} by k_t such that

$$(21) \quad M_t^{a,y} \mathbf{1} = p_t(a), \quad \sum_a p_t(a) = \mathbf{1}, \quad p_t(a) \text{ is independent of } y.$$

Then

$$(22) \quad C(\mathbf{a} \mid \mathbf{y}) = \alpha M_1^{a_1, y_1} \dots M_T^{a_T, y_T} \mathbf{1}.$$

The independence in (21) is essential: otherwise an action could use its not-yet-observed report. If an action or an input must survive an intermediate cut, split this formula at that cut and include its buffer in the state alphabet. Likewise, impose any declared shared-row constraints before optimizing. An externally fixed clock permits time-dependent matrices; an autonomous implementation has to store its phase or use the prescribed shared rows.

Proposition 8.1 (Outer normal form and compatible continuations). *For the finite interface and declared cuts, the response arrays of all controllers with profile at most $\mathbf{K}$ are exactly the arrays (22) with $k_t \leq K_t$, the normalization (21), and the declared interface constraints. This set, denoted $\mathcal{C}_{\mathbf{K}}$, is compact. The assertion is over all wirings of the allowed register, not one chosen transition graph.*

Equivalently, there are at most K_t normalized continuation arrays $B_{t,s}$ at each cut, with terminal continuation 1, and common nonnegative coefficients satisfying

$$(23) \quad B_{t-1,s}(a\mathbf{a} \mid y\mathbf{y}) = \sum_{s'} M_t^{a,y}(s, s') B_{t,s'}(\mathbf{a} \mid \mathbf{y}), \quad C = \sum_s \alpha(s) B_{0,s}.$$

Separate positive factorizations at the cuts are not sufficient without these common shift identities.

Proof. In a controller, let $p_t(a \mid s)$ be its action row and $q_t(s' \mid s, a, y)$ its update row. Their product defines $M_t^{a,y}$ and gives (21). Multiplication over a complete execution gives (22). Every finite wiring embeds into the fully connected register by assigning zero to unused transitions; smaller alphabets are padded by unused states. Thus no further maximization over graph presentations is necessary.

Conversely choose action a with probability $p_t(a \mid s)$, and, after observing y , use the row $M_t^{a,y}(s, s')/p_t(a \mid s)$ when the denominator is positive. At a zero denominator the entire corresponding nonnegative row vanishes; choose an arbitrary legal update there. This gives

exactly the required runtime arguments. It does not expose a posterior, the parameter, or an offline occupation array to the machine. Apply the same construction at each declared buffer cut. A product of finitely many stochastic row simplices is compact; closed legality and sharing constraints preserve compactness. Its continuous image is $\mathcal{C}_{\mathbf{K}}$.

The future execution starting at a state s defines $B_{t,s}$ and satisfies (23). Conversely, induction from the terminal continuation and these identities gives the product formula. Thus the continuation description has exactly the same realizations. In particular, independently selected factorizations at different cuts cannot be spliced unless the shift equations hold. $\square$

This is a finite controlled positive-realization statement. The role of positivity and invariant generating cones in stochastic realization is classical [17, 18, 19]. Ordinary linear rank need not be a state count. Here the additional statistical question is which positive realizations meet an optimal saddle face. Proposition 8.1 provides a normal form, not an efficient factorization algorithm. It also does not compactify an unbounded number of observable operations.

8.2. The realization theorem. Let

$$U = \max_{C \in \mathcal{C}} \min_{\theta} g_{\theta}(C), \quad V_{\mathbf{K}} = \max_{C \in \mathcal{C}_{\mathbf{K}}} \min_{\theta} g_{\theta}(C).$$

By compact convex minimax there is a least-favorable prior μ_* with $U = \max_{C \in \mathcal{C}} \int g_{\theta}(C) d\mu_*$. Put

$$\begin{aligned} \ell_*(C) &= \int g_{\theta}(C) d\mu_*, \\ F_* &= \{C \in \mathcal{C} : \ell_*(C) = U\}, \\ \mathcal{S}_* &= F_* \cap \bigcap_{\theta} \{C : g_{\theta}(C) \geq U\}. \end{aligned} \tag{24}$$

The set F_* is an exposed face. The second intersection is important: a Bayes best response to a least-favorable prior need not be minimax.

Here $\mathcal{C}$ is the full-history convex reference for the finite protocol. A nonconvex restriction, such as reuse of one stochastic table at several visits, is imposed on $\mathcal{C}_{\mathbf{K}}$, not inserted into $\mathcal{C}$ before invoking convex minimax. In that case U is a reference upper value unless exactness for the unrestricted physical class is separately known. In the reset applications below that exactness is supplied by Theorems 5.1 and 5.2.

Theorem 8.2 (Saddle-face realization and exact resource loss). *Under the preceding assumptions, the constrained outer value satisfies*

$$U - V_{\mathbf{K}} = \min_{C \in \mathcal{C}_{\mathbf{K}}} \left\{ U - \ell_*(C) + \ell_*(C) - \min_{\theta} g_{\theta}(C) \right\}. \tag{25}$$

Both summands are nonnegative. Consequently $V_{\mathbf{K}} = U$ if and only if some array in $\mathcal{S}_$ has compatible positive continuation generators of profile at most $\mathbf{K}$. If no such generators exist, $V_{\mathbf{K}} < U$ strictly.*

For $C \in \mathcal{S}_$ one has $g_{\theta}(C) = U$ on $\text{supp } \mu_*$. At an interior supporting parameter at which the payoff is differentiable, $\nabla_{\theta} g_{\theta}(C) = 0$. These equalities and stationarity conditions hold also in directions whose Bayes coefficient vanishes.*

At any reset continuation cut, let R_h denote the conditional channel of future outputs under each externally specified future input word. It factors as

$$R_h = \sum_{s=1}^{K_t} E(s \mid h) B_s, \quad E(\cdot \mid h) \in \Delta_{K_t}, \tag{26}$$

with one common family of continuation channels B_s . Any obstruction to this positive factorization for every member of $\mathcal{S}_*$ is therefore a memory obstruction for every admissible architecture, regardless of its training wiring.

Proof. For every C , its prior average is at most U and at least its minimum over fixed parameters. The expression in braces in (25) is identically $U - \min_{\theta} g_{\theta}(C)$. Taking its minimum over the compact set $\mathcal{C}_{\mathbf{K}}$ proves the identity and attainment. Its two nonnegative summands vanish exactly when C belongs to $\mathcal{S}_*$. Proposition 8.1 then proves the realization criterion. Compactness makes nonintersection a strictly positive value loss, not merely failure of one optimizer.

For an optimal array, $g_{\theta}(C) - U$ is continuous, nonnegative, and has zero μ_* integral. It is zero on the support of that measure. Every interior zero is a local minimum, so its gradient vanishes. No strict-sign hypothesis on individual coordinates is needed. Finally condition on the actual retained state at a reset cut. Given that state, the future program receives the same specified inputs and has the same output law, irrespective of the discarded past. Averaging gives (26). All persistent random variables are part of that state. The assertion can equivalently be checked on controller input tapes, using the generators in (23); it does not assume that a hidden physical state resets at a cut where the interface has not reset it. $\square$

The identity (25) is an exact loss decomposition, not a new interchange of a minimum and maximum over nonconvex machines. Its use is that an unrestricted certificate identifies a finite face, while contact at its supporting parameters can determine the remaining coordinates. The resulting channels are then tested for causal realizability. This is stronger than choosing an arbitrary Bayes tie rule and counting the states of that one rule.

Corollary 8.3 (Explicit coefficient form for frozen validation). *In the reset experiment of (8), write*

$$C_{z\omega}^* = \int P_{\theta}^{\otimes N}(z)(Q(\omega) - P_{\theta}(\omega)) d\mu_*(\theta).$$

The first summand of (25) is exactly

$$(27) \quad \sum_{C_{z\omega}^* > 0} C_{z\omega}^* (1 - h_z(\omega)) + \sum_{C_{z\omega}^* < 0} |C_{z\omega}^*| h_z(\omega).$$

Thus optimality fixes every strict-sign coordinate, while the support contact and stationarity equations must still be imposed at the zeros. For K labels at the post-training decision cut, an arbitrary upstream encoder gives the compact relaxation

$$(28) \quad h_z(\omega) = \sum_{s=1}^K E(s | z) D_s(\omega), \quad E \in (\Delta_K)^{\Omega^N}, \quad D_s \in [0, 1]^{\Omega}.$$

If this relaxation misses the saddle set, its strictly positive gap bounds every smaller architectural class.

Proof. Maximizing each coefficient separately gives the sum of its positive part. Subtraction gives (27). Conditioning on the decision state gives (28); the post-cut randomization is independent of the discarded word. Conversely an arbitrary encoder and a response list realize this factorization when other memory is unrestricted. The parameter spaces in (28) are compact, so the final assertion follows from Theorem 8.2. $\square$

8.3. Faces as memory witnesses. For a finite input set $\mathcal{Y}$ and output alphabet $\mathcal{D}$, continuation channels form the polytope $\mathcal{P} = \prod_{y \in \mathcal{Y}} \Delta(\mathcal{D})$. Its vertices are deterministic output functions. The analogous polytope for scalar test responses is $[0, 1]^{\mathcal{Y}}$.

Lemma 8.4 (Vertex and face packing). *Suppose channels to be factored through one common state alphabet include s distinct vertices $v_1, \dots, v_s$ of a polytope $\mathcal{P}$ and rows $r_1, \dots, r_j$. Suppose $r_i \in F_i$, the faces F_i are pairwise disjoint, and*

$$r_i \notin \text{conv}\{v_a : v_a \in F_i\}.$$

Every factorization of these rows as stochastic mixtures of generators in $\mathcal{P}$ needs at least $s + j$ generators.

Proof. A convex decomposition of a vertex uses that same vertex at every positive coefficient. Hence all s vertices must occur among the generators. If a convex combination belongs to a face, all generators with positive coefficients belong to that face. The stated exclusion therefore forces an additional generator in each F_i . Pairwise disjointness prevents any such generator from serving two faces. $\square$

The lemma does not assert that a stochastic state must itself be a history residual. Its generator can be any channel in the ambient polytope. This is why the argument also excludes randomized encoders and randomized continuations. It is a positive-factorization argument, not ordinary matrix rank or a restriction to a deterministic compiler.

Proposition 8.5 (Resource equivalence and stable obstructions). *A causal, cut-preserving recoding with a causal inverse, both using only one transducer state and preserving the payoff, preserves every $V_{\mathbf{K}}$, U , and the set of profiles attaining U . More generally suppose simulations in both directions have defect at most ϵ and require at most R_t states at cut t . For rewards in $[-1, 1]$,*

$$V_{\epsilon}(\mathbf{RK}) \geq V_{\mathfrak{F}}(\mathbf{K}) - 2\epsilon, \quad V_{\mathfrak{F}}(\mathbf{RK}) \geq V_{\epsilon}(\mathbf{K}) - 2\epsilon.$$

For a common controller class and training-validation kernels whose values differ by at most η , a gap $U - V_{\mathbf{K}} = \delta$ remains at least $\delta - 2\eta$ after the perturbation.

Proof. Transport a controller through each simulator and retain the product of its state and the simulator state. Apply the uniform bounded-reward variation bound and optimize. With an exact one-state inverse the two inequalities are equalities at the same profile. In the last assertion both the unrestricted and the constrained optimum change by at most η ; subtract their two bounds. The state products must be applied at the actual buffer cuts, not only at the terminal decision. $\square$

These statements place the saddle face, the future-test quotient, and resource-aware causal morphisms in one theorem chain. The channel factorizations are presentation-independent under the stated causal recodings. The memory obstruction in the next section is derived from the experimental coefficients themselves, not from a metric dimension or a covering exponent supplied as an additional hypothesis.

9. EXACT MINIMAX LAWS FOR A MARKED PRODUCT FAMILY

The posterior-predictive dual can be solved explicitly for a continuous family containing the collision target. Throughout this section, the training and candidate validation law is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad (p, q) \in [0, 1]^2.$$

In the order 000, 001, 110, 111, let Q_{γ} have masses

$$((1 + \gamma)/4, (1 - \gamma)/4, (1 - \gamma)/4, (1 + \gamma)/4),$$

and zero mass elsewhere. Write $U_1(\gamma) = U_1(b, Q_\gamma)$, with exactly the training-validation convention of (8). Set

$$(29) \quad \gamma_* = 1 - 1/\sqrt{2}, \quad s_\gamma = \sqrt{\gamma^2 - 2\gamma + 5}, \quad u_\gamma = \gamma - 2 + s_\gamma, \quad \tau_\gamma = \frac{2u_\gamma}{u_\gamma + 2 - \gamma}.$$

Theorem 9.1 (Exact marked minimax family). *For every $0 \leq \gamma \leq \gamma_*$,*

$$(30) \quad U_1(\gamma) = \frac{3 + u_\gamma^2}{8}.$$

An optimal response depends on the training word only through the report count $K = X_1 + X_2$. On the four target atoms its rows are

$$(31) \quad h_0 = (\tau_\gamma, 0, 1, 1), \quad h_1 = (1, 1, 1, 1), \quad h_2 = (1, 1, 0, \tau_\gamma),$$

and all responses are zero off these atoms. A least-favorable prior is

$$(32) \quad \mu_\gamma = \frac{1}{2}\delta_{(t_\gamma, t_\gamma)} + \frac{1}{2}\delta_{(1-t_\gamma, 1-t_\gamma)}, \quad t_\gamma = \frac{1 - \sqrt{u_\gamma}}{2}.$$

The assertion optimizes over all stochastic responses to the full marked training word, not merely over count-based or symmetric responses.

Proof. First, u_γ is the positive root of

$$(33) \quad u^2 + (4 - 2\gamma)u - (1 + 2\gamma) = 0.$$

On the asserted interval, $0 < u_\gamma < 1$ and $0 < \tau_\gamma < 1$. Indeed the polynomial is negative at zero and positive at one; also $u_\gamma < 2 - \gamma$ because $\gamma < 1/2$. Substitution of $1 - 2\gamma$ into the polynomial gives $8(\gamma^2 - 2\gamma + 1/2)$, which is nonnegative precisely on the present interval. Monotonicity on $[0, \infty)$ therefore gives

$$(34) \quad u_\gamma \leq 1 - 2\gamma.$$

Put $a = (1 - p)(1 - q)$, $c = pq$, and $z = 1 - a - c$. The probabilities of the three training counts are a, z, c . The independent training mark may be discarded by the displayed construction. Let $G_\gamma(p, q)$ denote its score. With $u = (p + q - 1)^2$ and $v = (p - q)^2$, direct expansion gives

$$(35) \quad G_\gamma(p, q) - \frac{3 + u_\gamma^2}{8} = \frac{2 - \tau_\gamma}{16} \{(u - v - u_\gamma)^2 + 4v\}.$$

For clarity, this identity follows by substituting the three rows of (31) into

$$a(Q_\gamma - b)h_0 + z(Q_\gamma - b)h_1 + c(Q_\gamma - b)h_2,$$

using $a + c = (1 + u - v)/2$, $a - c = -(p + q - 1)$, and then eliminating u_γ^2 by (33). The coefficient of the resulting square is $(2 - \tau_\gamma)/16$ and that of v is four times this coefficient. Both are nonnegative. Equality holds at both points of (32), since there $v = 0$ and $u = u_\gamma$. This proves the lower bound and identifies its equality set.

It remains to bound every response, including nonsymmetric ones. Use (32) in Theorem 5.1. At $t = t_\gamma$ put $a = (1 - t)^2$, $c = t^2$, and $T = a + c = (1 + u_\gamma)/2$. For the training report pair 00, the four dual coefficients, apart from a common positive factor, are

$$(36) \quad \left(Q_H T - \frac{a^2 + c^2}{2}, \quad Q_L T - \frac{a^2 + c^2}{2}, \quad Q_L T - ac, \quad Q_H T - ac \right),$$

where $Q_H = (1 + \gamma)/4$ and $Q_L = (1 - \gamma)/4$. The first coordinate is zero by (33); the second is $-\gamma T/2 \leq 0$. The last two are positive: $T \geq 1/2$, $ac \leq 1/16$, and $\gamma < 1/2$ imply $Q_L T > 1/16 \geq ac$. Thus $(\tau_\gamma, 0, 1, 1)$ maximizes this linear functional. For the training pair 11, reversing all report bits and the mark reverses the four coordinates and gives the row h_2 .

For either training pair 01 or 10, the two prior support points have the same likelihood $t(1-t)$. The posterior is still equally weighted, and the candidate predictive mass at each target atom is $T/4$. Every one of the four coefficients is therefore nonnegative by $Q_L - T/4 = (1 - 2\gamma - u_\gamma)/8 \geq 0$. The response h_1 is optimal. At off-diagonal validation atoms the target mass vanishes and the coefficient is nonpositive. Finally, conditioning on either training mark multiplies all these coefficients by $1/2$; it creates no further optimizing choice. We have therefore optimized every coordinate of the full marked-word response space.

The prior average of the displayed response equals its value at either support point, namely $(3 + u_\gamma^2)/8$, by (35). Every other response has prior average at most this number and hence has minimum over fixed (p, q) at most this number. Together with the lower bound this proves (30), optimality of the response, and least favorability of the prior. $\square$

Corollary 9.2 (The exact one-preparation collision value). *For $Q_* = Q_{1/4}$,*

$$(37) \quad U_1(b, Q_*) = u_{\text{ex}} := \frac{85 - 7\sqrt{73}}{64}, \quad u_{1/4} = \frac{\sqrt{73} - 7}{4}, \quad \tau_{1/4} = 2 - \frac{14}{\sqrt{73}}.$$

If only the target changes to Q , then $|U_1(b, Q) - u_{\text{ex}}| \leq \text{TV}(Q, Q_)$.*

Proof. Substitute $\gamma = 1/4$ in Theorem 9.1. For a fixed trained response the target perturbation is at most total variation, uniformly in (p, q) . Taking first the minimum and then the maximum preserves this bound in both directions. $\square$

9.1. Complementary slackness and the event alphabet. The tie in the first coordinate of (36) explains the randomization. Any value of that response coordinate is Bayes-optimal for the two-point prior. The minimax choice makes the uniform payoff stationary at its two support points; solving this condition gives τ_γ and the square identity (35). This derives the prior and response together, rather than postulating that a numerically effective prior is optimal.

Operationally, for $K = 0$ one draws once between the events with indicator vectors $(0, 0, 1, 1)$ and $(1, 0, 1, 1)$, with probabilities $1 - \tau_\gamma$ and τ_γ . For $K = 2$ one draws between $(1, 1, 0, 0)$ and $(1, 1, 0, 1)$. For $K = 1$ the event is $(1, 1, 1, 1)$. The resulting event identity is retained and used for both fresh validations. These are exactly the five events in the two-preparation construction below. The one-preparation theorem supplies their complementary-slackness interpretation; it does not prove that the two-preparation selector is optimal. The four-point rational prior retained below is a useful rational certificate, not a least-favorable prior.

The exact response uses an atomic stochastic row. It does not receive free persistent randomness. Replacing τ_γ by a dyadic number within 2^{-b} changes each conditional response, and hence the score, by at most 2^{-b} . A sequential fair-bit implementation must retain its bit prefix and event identity. None of these implementation costs is needed by the deterministic two-preparation physical audit.

Proposition 9.3 (The zero-preparation endpoint). *For $0 \leq \gamma \leq 1$, $U_0(b, Q_\gamma) = \gamma/2$.*

Proof. The fixed event $\{000, 111\}$ has score $(1 + \gamma)/2 - (a + c)/2 \geq \gamma/2$ because $a + c \leq 1$. Conversely, the equally weighted prior on $b(0, 0)$ and $b(1, 1)$ has predictive validation law uniform on the four target atoms. Its total variation from Q_γ is $\gamma/2$. Theorem 5.1 with $N = 0$ gives the matching upper bound. $\square$

10. A SELF-CONTAINED COLLISION INTERFACE

We give the preparation, mechanical estimate, observation law and private simulator class needed for the physical specialization. The following argument does not appeal to an unpublished

physical lemma. It retains the same positive-volume preparation and positive sensor noise as the earlier collision construction.

Definition 10.1 (The reset row–event architecture). A preparation supplies two sequential report slots and, afterwards, an actual mark. The source reports in the two slots are constant. A private width-one simulator has no retained information between those slots and no persistent hidden seed. It may use fresh independent coins. Its first emission is $X_1 \sim \text{Ber}(p)$; after receiving the current gate $A \in \{0, 1\}$, its second emission has a fixed Bernoulli kernel with parameter q_A . The parameters (p, q_0, q_1) are fixed throughout the audit. The simulator does not observe the actual mark. All its coins are independent of that mark. Different preparations reset all unretained variables and are independent conditional on the parameters.

The auditor observes the first report, chooses a gate using its retained state and that report, observes the second report, and then observes the mark. During training it may choose rows and stop adaptively, using at most the declared number of candidate preparations. It next freezes one gate policy and one event, with any retained random choice included in its register. It uses that same row–event continuation on one fresh candidate preparation and one independent target preparation and returns the difference of their event indicators, target minus candidate. It may not reselect the event after either validation. Independent target-only calibration is allowed but all retained calibration and randomness are charged. Peak width counts every bit, mark, event and validation-accumulator cut. Clocked schedules are declared externally; an autonomous implementation must retain its phase. An unretained transcript, seed or stopping-time value is not available to a continuation.

In the all-on row the complete private law is $b(p, q_1)$, with an independent fair actual mark. For lower bounds it suffices to restrict the adversary to $q_0 = 0$, $q_1 = q$. In this subfamily every deterministic gate policy is the pushforward

$$T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$$

of $b(p, q)$. The gate, when included in the transcript, is another function of these same coordinates. Randomized gate policies are covered by averaging their independent random tables. This proves both the all-on product characterization for the upper construction and the common-garbling subfamily used for the outer converse. The width-one constraint is a simulator constraint; it is not an uncharged restriction on the auditor’s register.

10.1. Preparation and elastic scattering. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$, with no walls and contact distance d . Write $r = q_1 - q_2$, $g = v_1 - v_2$, $V = (v_1 + v_2)/2$, and let the center of mass position range over any fixed rational box of positive volume. Prepare $(B, W) \in \{0, 1\}^2$ with probability $5/16$ at $B = W$ and $3/16$ at $B \neq W$. Conditional on (B, W) , use the uniform product preparation on the following boxes:

$$(38) \quad \begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & & & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

Thus the actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$, preserved by momentum conservation. Set

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector, identically zero on the incoming transverse-velocity range.

Lemma 10.2 (Uniform collision signal). *For every preparation in (38) and every $9/10 \leq d \leq 11/10$, exactly one collision occurs and its time lies in $(4/5, 3/2)$. Afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4.$$

For $0 < d \leq 2/5$ no collision occurs on $[0, 3]$ and the detector is zero throughout that interval.

Proof. Before contact, $r(t) = r + tg$. At $t \leq 4/5$ its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At $t = 3/2$ its absolute coordinate bounds are $1/4$, $113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. The first contact therefore occurs in the claimed interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$. Hence the entry contact has $n_x < 0$, where $n = r(t_c)/d$. Its remaining coordinates give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq 87/220.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The elastic rule gives

$$v_1^+ = V + g/2 - (g \cdot n)n, \quad g^+ = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v_{1,y}^+ > -3/200 + (1443/1000)(87/220) > 1/2.$$

The definition of h implies the signal bound. Since $r(t_c) \cdot g^+ > 0$, squared separation increases strictly afterwards; there is no second collision. Finally, for $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. There is no contact, and the incoming $|v_{1,y}| \leq 3/200$ makes h zero. $\square$

10.2. Positive noise and the actual mark. Let $\mathcal{B}_t$ be Brownian motion independent of the preparation and set

$$Y_t = \int_0^t h(v_{1,y}(s)) ds + \sigma \mathcal{B}_t, \quad 0 < \sigma \leq 1/24.$$

The two potential all-on reports are

$$Z_1 = \mathbf{1}_{\{2(Y_2 - Y_{3/2}) > 0\}}, \quad Z_2 = \mathbf{1}_{\{2(Y_3 - Y_{5/2}) > 0\}}.$$

The actual second report is AZ_2 . The gate changes acquisition, not the mechanical path. The continuous path is implemented by the target sensor; it is not made available to the private simulator or supplied as an uncharged auditor buffer. Denote the law of (Z_1, Z_2, W) by $Q_{d,\sigma}$.

Theorem 10.3 (Uniform physical presentation). *For $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$,*

$$(39) \quad \text{TV}(Q_{d,\sigma}, Q_*) < 1/300.$$

The same estimate holds after every declared feedback gate, including its observable gate value. The all-on private product family and the common-garbling subfamily in Definition 10.1 apply to this very experiment.

Proof. Every collision occurs before $3/2$. Conditional on the preparation, the two normalized sensor windows are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. Lemma 10.2 makes their means have sign $2B - 1$ and magnitude greater than $1/4$. Each potential report therefore differs from B with probability less than $\Phi(-3)$. The Gaussian tail estimate

$$\Phi(-x) \leq \frac{e^{-x^2/2}}{x\sqrt{2\pi}}$$

gives $\Phi(-3) < 1/600$: $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$ suffice. Couple the ideal and noisy reports using the same preparation, actual mark and gate random table. Except on a set of probability less than $2\Phi(-3) < 1/300$, both potential reports equal B and the entire gated transcript agrees. The ideal marked law is precisely Q_* , by the prepared (B, W) table. This proves (39) and the feedback-uniform version. Finally, the source is blind, its actual mark is unobserved by the simulator, and the gate is its only current input at the second emission. The product and subfamily assertions follow from the two independent private-emission kernels specified above. $\square$

For $d \leq 2/5$, the two windows instead contain only independent centered Gaussian noise. Their signs are independent fair bits, independent of W , and a private width-one simulator reproduces every gated row exactly. Thus the informative regime and its no-collision control use the same detector, preparation convention and actual mark.

11. THE PHYSICAL EXPERIMENT AND ITS SHARP PREPARATION BOUNDARY

Take $\Omega = \{0, 1\}^3$, with coordinates (x_1, x_2, w) . The all-on private candidate is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad 0 \leq p, q \leq 1.$$

The mark w is the actual mark, not an invented simulator bit. Put $a = (1-p)(1-q)$ and $c = pq$. On the ordered atoms

$$(40) \quad 000, \quad 001, \quad 110, \quad 111,$$

let Q_* have masses $5/16, 3/16, 3/16, 5/16$, and set it to zero elsewhere. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. A target Q is allowed whenever $\text{TV}(Q, Q_*) \leq \beta$.

The original feedback operation has the form

$$(41) \quad T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w), \quad g : \{0, 1\} \longrightarrow \{0, 1\}.$$

The selected gate is observable. It may depend on earlier preparations and on the first current report. The target rows are pushforwards by (41). For the converse, restrict the adversary to the admissible private subfamily whose second emission is AX_2 , where A is the current gate and $X_2 \sim \text{Ber}(q)$ is independent of the first emission and of the fair mark. Every row is then a causal garbling of $b(p, q)$. For the construction only the all-on row is used, and every private candidate in the original width-one class has the all-on product law just specified. These two facts, proved for the microscopic interface in Definition 10.1 and Theorem 10.3, explain why using a subfamily for the lower bound does not narrow the upper construction.

The admissible audit first trains, then selects one row–event pair, then uses one fresh candidate and one fresh target validation. It reports $D = \mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The event and row cannot be reselected after seeing a validation draw. All bit, mark, decision and validation cuts are charged. Define $N_c^{\text{phys}}(W, Q)$ to be the minimum number of candidate training preparations among these original-interface auditors, with peak register width at most W , whose score is uniformly strictly greater than c . Randomization, feedback and stopping are allowed. If no such audit exists, the value is $+\infty$.

Theorem 11.1 (Two-preparation physical segment). *Let u_1, m_2 be as in (2). For every $\beta \geq 0$, $\text{TV}(Q, Q_*) \leq \beta$, and*

$$W \geq 12, \quad u_1 + \beta \leq c < m_2 - \beta,$$

one has $N_c^{\text{phys}}(W, Q) = 2$. The two-preparation audit is deterministic, uses no acquired calibration and no random bits, and has the full profile

$$(42) \quad (1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$$

from the initial cut through its twelve bit/mark updates. In particular, its decision alphabet has five labels and its peak has twelve. The lower bound remains valid with no memory cap.

The proof occupies the next three subsections. The interval in the statement can be empty for large β ; no assertion is then made. For $\beta = 1/300$, it contains $c = 2/5$ strictly between its endpoints.

11.1. An exact rational certificate against one preparation.

Lemma 11.2. *Every original-interface auditor using at most one candidate training preparation has worst-case score at most $u_1 + \beta$, regardless of memory.*

Proof. By Theorem 5.2, it suffices to bound $U_1(b, Q_*)$ and then perturb the target. In (9) use the four fixed parameters

$$(p, q) = (1/8, 1/8), (1/5, 1/5), (4/5, 4/5), (7/8, 7/8)$$

with weights $1/16, 7/16, 7/16, 1/16$, respectively. This measure is not asserted to be least favorable. Since $p = q$ on its support, the training count $K = x_1 + x_2$ is sufficient. The independent training mark and the order of unequal bits contain no further information about this prior.

For $k = 0, 1, 2$ and the four validation atoms in (40), the signed coefficients in (9), after grouping by K , are the entries of

$$(43) \quad \frac{1}{40960000} \begin{pmatrix} -1977 & -1775177 & 2170423 & 3943623 \\ 1765554 & 191954 & 191954 & 1765554 \\ 3943623 & 2170423 & -1775177 & -1977 \end{pmatrix}.$$

For example, the coefficient in column j is

$$\sum_{i=1}^4 \mu_i \binom{2}{k} p_i^k (1-p_i)^{2-k} \left(Q_{*,j} - \frac{(1-p_i)^2 \mathbf{1}_{j \leq 2} + p_i^2 \mathbf{1}_{j \geq 3}}{2} \right).$$

This identity proves every entry of (43) by rational arithmetic. All coefficients at off-diagonal validation atoms are nonpositive. The sum of the positive entries in (43) is $4035777/10240000 = u_1$. Each response coordinate is in $[0, 1]$, so this sum bounds its prior-averaged score and hence its minimum over the four fixed candidates.

Replacing Q_* by Q changes the score of any trained response by at most β . The training law is a candidate law and is unchanged. Consequently the same prior bounds every score by $u_1 + \beta$. A procedure with no training, or one that stops before taking its permitted sample, is covered by ignoring the generated all-on observation. $\square$

11.2. A marked five-event selector. Take two independent all-on training preparations. Let S be the total number of ones among their four report bits; discard their marks. Choose the following events, represented by their indicators on (40) and with indicator zero at every off-diagonal atom:

$$(44) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & 0 \end{array}$$

Denote these indicators by $h_0, \dots, h_4$ and report $D = h_S(Y) - h_S(X)$ on independent validation draws.

Lemma 11.3 (Exact uniform score of the selector). *Against the ideal target, the selector (44) has uniform minimum score $m_2 = (12 - 3\sqrt{3})/16$. Against Q , its score is at least $m_2 - \beta$.*

Proof. The generating polynomial of S is

$$((1-p) + pz)^2((1-q) + qz)^2.$$

Writing $g_s = (Q_* - b(p, q))h_s$, its expected score is $G(p, q) = \sum_{s=0}^4 [z^s]((1-p) + pz)^2((1-q) + qz)^2 g_s$. Substitution of the five rows in (44) and collection of terms with $x = p + q - 1$, $y = p - q$, $u = x^2$ and $v = y^2$ gives

$$(45) \quad G(p, q) = H(u, v) = -\frac{u^3}{32} + \frac{3u^2v}{32} + \frac{3u^2}{16} - \frac{3uv^2}{32} - \frac{uv}{4} - \frac{3u}{32} + \frac{v^3}{32} + \frac{v^2}{16} + \frac{15v}{32} + \frac{7}{16}.$$

This is a polynomial identity, not an interpolation over a grid. Since $0 \leq u \leq 1$ and $v \geq 0$ on the parameter square,

$$\partial_v H(u, v) = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}.$$

Thus $H(u, v) \geq H(u, 0) = (-u^3 + 6u^2 - 3u + 14)/32$. The derivative of the last expression is $(-3u^2 + 12u - 3)/32$. On $[0, 1]$ it changes sign from negative to positive at $u = 2 - \sqrt{3}$. Substitution gives $(12 - 3\sqrt{3})/16$. The minimizing values are feasible: take $p = q = (1 \pm \sqrt{2 - \sqrt{3}})/2$. This proves exactness of the minimum for this specified selector, not optimality of its score over all selectors with two preparations. Finally, for each s , the target expectation of $h_s \in [0, 1]^\Omega$ changes by at most β . Averaging over training proves the claimed robustness. $\square$

11.3. Causal compilation and the full register profile.

Lemma 11.4. *The selector and its two validations admit the deterministic profile (42). No selector, first-bit buffer or validation accumulator is outside the charged register. An autonomous phase-tagged realization uses at most 75 states.*

Proof. After each training report bit retain the count accumulated so far. The two marks are read and discarded. The first six updates therefore have widths 2, 3, 3, 4, 5, 5 after the initial singleton. At the decision cut the five count values select the five events.

During candidate validation, after the first bit retain the event and that bit, using at most ten states. After the second bit retain the event and the function of the unread mark that gives its candidate-event indicator. For events 0, 1, 2, 3, 4, respectively, the possible functions are

$$\{0, 1\}, \quad \{0, 1 - w, 1\}, \quad \{0, 1\}, \quad \{0, w, 1\}, \quad \{0, 1\}.$$

There are twelve event–function pairs. After the candidate mark, retain the event and its now known candidate indicator $c_0 \in \{0, 1\}$, using at most ten states.

During target validation, after the first bit discard the event name and retain c_0 and its residual function of the second bit and mark. These residual functions belong to

$$\{0, x_2, (1 - x_2)(1 - w), 1 - x_2, x_2w\}.$$

This uses at most ten states. After the second bit, the residual *difference* as a function of the unread mark belongs to

$$\{-1, 0, 1, w, 1 - w, w - 1, -w\},$$

so seven states suffice. The mark update gives the three possible outputs $-1, 0, 1$. Each transition depends only on its retained residual and the current report. Thus it is a legal common observable row, not a family of independently optimized hidden-history maps.

For completeness, equality of two residual functions implies equality of their restrictions after either next bit, so identifying equal residuals is closed under every subsequent update. The supplied transition tables are obtained by these restrictions and can be reconstructed directly from (44). Enumerating residuals is a convenience for reproducing the tables, not the proof of their sufficiency. Finally, disjointly tag the states at the thirteen cuts by their phase and use terminal self-loops. The sum of the displayed widths is 75. This implementation charges the clock; it does not assert that 75 is optimal. $\square$

Proof of Theorem 11.1. Lemma 11.2 excludes $N \leq 1$ for a strict score greater than $c \geq u_1 + \beta$, for every architecture and every memory budget. Lemmas 11.3 and 11.4 supply a two-preparation architecture of peak twelve with score at least $m_2 - \beta > c$. Larger width budgets can use the same machine. $\square$

Corollary 11.5 (The positive-noise collision family). *For the original marked collision targets with $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the local microscopic estimate $\text{TV}(Q_{d,\sigma}, Q_*) < 1/300$ implies*

$$N_{2/5}^{\text{phys}}(W, Q_{d,\sigma}) = 2 \quad (W \geq 12).$$

There are four preparations in the complete audit: two candidate training, one candidate validation, and one target validation.

Proof. Apply Theorem 11.1. The lower-end margin is the exact rational number

$$\frac{2}{5} - u_1 - \frac{1}{300} = \frac{78269}{30720000} > 0.$$

For the other endpoint, $\sqrt{3} < 7/4$ implies $m_2 > 27/64$ and $27/64 - 1/300 > 2/5$. The physical total-variation estimate, including the actual mark and positive sensor noise, is proved in Theorem 10.3; the current deduction does not replace those hypotheses by a zero-noise limit. $\square$

Remark 11.6 (The part of the frontier determined here). The deterministic theorem evaluates the minimum preparation coordinate for $W \geq 12$. Theorem 12.1 and Corollary 12.2 below additionally determine the ideal maximum score with two preparations on this width range. The least peak width over all validation orders and the full boundary for $3 \leq W \leq 11$ remain separate questions. Theorem 14.3 below determines the peak for forward serial validation, and Theorem 14.1 determines the order-independent decision cut. The earlier exact decision-width result $\kappa_{2/5}^{\text{rows}} = 3$ is unchanged: it allows longer training, whereas this shortest audit uses five decision states. The older points (399, 40602), (320000, 1604) and $(12800 \cdot 2^{400}, 12)$ remain valid constructions with decision width three. They are dominated by (2, 12) in the projection onto training preparations and peak width, but not necessarily in a resource order that also constrains the decision alphabet separately.

Corollary 11.7 (The enlarged physical preparation segment). *With the full architecture of Definition 10.1,*

$$N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_{\text{ex}} + \beta \leq c < m_2 - \beta.$$

Here $u_{\text{ex}} = (85 - 7\sqrt{73})/64$ is the exact one-preparation all-on value, rather than the older rational upper bound.

Proof. Restrict the adversary to the common-garbling subfamily in Definition 10.1. Theorem 5.2 and Corollary 9.2 bound every at-most-one-preparation auditor, even with unlimited memory, by $u_{\text{ex}} + \beta$. The same deterministic two-preparation machine in Lemmas 11.3 and 11.4 has score at least $m_2 - \beta$ against the full private candidate family. These inequalities prove the assertion. The four-preparation total, peak twelve and positive-noise specialization are unchanged. $\square$

12. THE EXACT TWO-PREPARATION MARKED SADDLE

The one-preparation saddle has an exact successor. A new zero coefficient of the least-favorable predictive measure makes it necessary to randomize at the two extreme training counts. This randomization changes the value, but does not enlarge the alphabet of frozen validation events.

Let $b(p, q)$ and Q_* be as in Section 11. Let r be the unique root in $(1/3, 7/20)$ of

$$(46) \quad 2r^3 + 25r^2 - 3 = 0,$$

and define

$$(47) \quad \tau = \frac{6(-r^2 + 4r - 1)}{r(3r + 25)}, \quad v_2 = \frac{-r^3 + 6r^2 - 3r + 14}{32}, \quad p_{\pm} = \frac{1 \pm \sqrt{r}}{2}.$$

The polynomial in (46) is strictly increasing on the positive half-line and has opposite signs at the indicated endpoints. These observations prove the asserted existence, uniqueness and isolating interval. That interval also gives $0 < \tau < 1$.

Theorem 12.1 (Exact two-preparation value). *For the full marked-word response class of (8),*

$$U_2(b, Q_*) = v_2.$$

A least-favorable prior gives mass $1/2$ to each of (p_-, p_-) and (p_+, p_+) . An optimal response discards the two training marks, retains the total number S of ones in the four training report bits, and uses

$$(48) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & \tau & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & \tau \end{array}$$

with zero response at off-diagonal validation atoms. Numerically, $r = 0.3417694155\dots$, $\tau = 0.1688236447\dots$ and $v_2 = 0.4261127787\dots$. The exact value is determined by (46)–(47), not by these decimals.

Proof. Write $a = (1 - p)(1 - q)$, $c = pq$, $u = (p + q - 1)^2$ and $v = (p - q)^2$. Both u and v belong to $[0, 1]$. The score of the deterministic five-event selector is the polynomial $H(u, v)$ in (45). Only counts zero and four are changed by (48). Their probabilities are a^2 and c^2 . Consequently the new score is

$$G(u, v) = H(u, v) + \tau J(u, v), \quad J = a^2(5/16 - a/2) + c^2(5/16 - c/2).$$

Expansion gives the polynomial identity

$$(49) \quad \begin{aligned} J(u, v) = & -\frac{u^3}{64} + \frac{3u^2v}{64} - \frac{25u^2}{128} - \frac{3uv^2}{64} + \frac{13uv}{64} \\ & + \frac{v^3}{64} - \frac{v^2}{128} - \frac{v}{32} + \frac{3}{128}. \end{aligned}$$

Thus

$$H(u, 0) = \frac{-u^3 + 6u^2 - 3u + 14}{32}, \quad J(u, 0) = \frac{3 - 25u^2 - 2u^3}{128}.$$

Equation (46) says $J(r, 0) = 0$, and the choice of τ says $\partial_u G(r, 0) = 0$. Direct factorization, using these two identities, yields

$$(50) \quad G(u, 0) - v_2 = \frac{(u - r)^2 \{75 + 12r + 6u - 37r^2 - 74ru\}}{64r(3r + 25)}.$$

The expression in braces is strictly positive for $0 \leq u \leq 1$: it is at least $75 - 37(7/20)^2 - 74(7/20) > 0$. Moreover,

$$\partial_v H = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}, \quad \partial_v J = \frac{3(u-v)^2 + 13u - v - 2}{64} \geq -\frac{3}{64}.$$

Because $0 < \tau < 1$, $\partial_v G \geq 11/64$ on the unit square. It follows that $G(u, v) \geq G(u, 0) \geq v_2$, with equality precisely when $v = 0$ and $u = r$. Both equality points are feasible and are the two parameters in the asserted prior. This proves the global lower bound.

It remains to prove an upper bound against every response, not only those depending on S . Under the asserted prior the four training report bits are independent Bernoulli variables with common parameter p_- or p_+ . Conditional on $S = s$, their arrangements and the two independent training marks have parameter-independent distributions. The signed predictive coefficients can therefore be grouped by S without restricting any response coordinate. For the four displayed validation atoms put

$$C_{s,j} = \frac{1}{2} \sum_{p \in \{p_-, p_+\}} \binom{4}{s} p^s (1-p)^{4-s} \{Q_{*,j} - b_j(p, p)\}.$$

The first three rows of $256C$ are

$$\begin{aligned} s = 0 : & \quad (-(2r^3 + 25r^2 - 3), -(2r^3 + 27r^2 + 12r - 1), \\ & \quad -(2r^3 - 5r^2 - 20r - 1), -(2r^3 - 7r^2 - 32r - 3)), \\ s = 1 : & \quad 4(r-1)(2r^2 + 7r - 3, 2r^2 + 9r - 1, \\ & \quad 2r^2 - 7r - 1, 2r^2 - 9r - 3), \\ s = 2 : & \quad -6(r-1)^2(2r-3, 2r-1, 2r-1, 2r-3). \end{aligned}$$

Rows three and four are the reversals of rows one and zero, respectively. Their signs, using $1/3 < r < 7/20$, are

$$(0, -, +, +), \quad (+, -, +, +), \quad (+, +, +, +), \quad (+, +, -, +), \quad (+, +, -, 0).$$

Every off-diagonal validation coefficient is nonpositive because its target mass is zero. Hence (48) maximizes every signed coefficient: it is one at positive coefficients, zero at negative ones, and arbitrary at the two zeros. The lower-bound equality at both prior points shows that this Bayes maximum is exactly v_2 . The posterior-predictive dual, Theorem 5.1, now gives the matching upper bound for the full response class. $\square$

Corollary 12.2 (Exact physical score on the peak-twelve segment). *Use the atomic stochastic-row convention and the original marked interface of Definition 10.1. At two candidate training preparations, every peak budget $W \geq 12$ has exact ideal worst-case score v_2 . For a target Q with $\text{TV}(Q, Q_*) \leq \beta$, the corresponding value lies in $[v_2 - \beta, v_2 + \beta]$. In particular, with $u_{\text{ex}} = (85 - 7\sqrt{73})/64$,*

$$(51) \quad N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad u_{\text{ex}} + \beta \leq c < v_2 - \beta.$$

Proof. Let $h_0, \dots, h_4$ be the five deterministic events in (44). At count zero select h_1 with probability τ and h_0 otherwise; at count four select h_3 with probability τ and h_4 otherwise. At the other counts retain the indicated event. Discard the count and retain the selected event for both validations. This realizes (48); the event alphabet remains five. The full charged profile, now including the separate event-selection update, is

$$(1, 2, 3, 3, 4, 5, 5, 5, 10, 12, 10, 10, 7, 3).$$

The common residual-function compiler in Lemma 11.4 applies unchanged. In particular, no persistent random selector is outside the register. The all-on law for every private candidate is $b(p, q)$, so Theorem 12.1 proves the constructive lower bound. For the converse restrict to the admissible common-garbling product subfamily and apply Theorem 5.2 followed by Theorem 12.1. This bounds all architectures, including those using feedback, stopping and more memory. Target perturbation changes each frozen-event expectation by at most β . Finally the exact one-preparation theorem excludes a strict score above c at $N \leq 1$, whereas the displayed machine achieves it at $N = 2$ under (51). $\square$

The root r is larger than $2 - \sqrt{3}$. The old diagonal polynomial $H(u, 0)$ is strictly increasing between these points, whence $v_2 > m_2$. The deterministic theorem, including its zero-random-bit implementation, remains valid; the new exact score uses an additional stochastic row.

Proposition 12.3 (Finite precision for the new row). *Replacing τ by a dyadic $\tau_b = k/2^b \in [0, 1]$ with $|\tau_b - \tau| \leq 2^{-b}$ loses at most $(5/16)2^{-b}$ in uniform score. A clocked fair-bit implementation uses b extra random bits and at most $\max\{12, 6(b+1) + 5\}$ persistent states, including their phase, in addition to the unchanged preparation count.*

Proof. At each affected training word the response changes at a single atom; $|5/16 - a/2|$ and $|5/16 - c/2|$ are at most $5/16$. The two affected word probabilities sum to at most one. This gives the score estimate. To implement the dyadic coin, compare b fresh fair bits lexicographically with the fixed b -bit integer k . Retain the origin (count zero or four), one of the three comparison flags, and a phase in $\{0, \dots, b\}$. Once the comparison ends retain its chosen one of five events. Unaffected counts can wait in their event states. These states total at most $6(b+1) + 5$; the other cuts have peak twelve. The cases $k = 0, 2^b$ are deterministic. Exact atomic randomization and a finite fair-bit implementation are therefore distinct resource claims. $\square$

13. A CONTINUOUS TWO-PREPARATION MARKED FAMILY

The preceding saddle persists on an explicit interval of target laws, with exact rather than perturbative values. Use the target Q_γ from Section 9 and write $U_2(\gamma) = U_2(b, Q_\gamma)$.

Theorem 13.1 (A two-preparation saddle family). *For $6/25 \leq \gamma \leq 13/50$, let r_γ be the unique root in $(1/3, 7/20)$ of*

$$(52) \quad f_\gamma(r) = r^3 + (13 - 2\gamma)r^2 + (3 - 12\gamma)r - (1 + 2\gamma) = 0.$$

Set

$$D_\gamma(r) = 3r^2 + (26 - 4\gamma)r + 3 - 12\gamma, \quad t_\gamma = -\frac{2\{3r_\gamma^2 + (8\gamma - 14)r_\gamma + 3\}}{D_\gamma(r_\gamma)}.$$

Then $0 < t_\gamma < 1$, and

$$(53) \quad U_2(\gamma) = \frac{-r_\gamma^3 + (7 - 4\gamma)r_\gamma^2 - 3r_\gamma + 13 + 4\gamma}{32}.$$

The response (48), with $\tau = t_\gamma$, is optimal. A least-favorable prior gives equal masses to the two diagonal parameters $p = q = (1 \pm \sqrt{r_\gamma})/2$. The response has the same five-event, peak-twelve atomic realization. At $\gamma = 1/4$ these formulas reduce to Theorem 12.1.

Proof. For brevity write $r = r_\gamma$, $D = D_\gamma(r)$ and $t = t_\gamma$. The derivative D_γ is positive on the rectangle $\gamma \in [6/25, 13/50]$, $r \in [1/3, 7/20]$. For example its lower bound is $1/3 + (26 - 52/50)/3 - 3/25 > 8$. The polynomial f_γ decreases with γ at a fixed positive argument. Direct rational evaluations give $f_{6/25}(1/3) < 0$ and $f_{13/50}(7/20) > 0$. Monotonicity therefore gives the root and its uniqueness in this interval. The numerator of t is $-16\gamma r - 6r^2 + 28r - 6$. It is bounded

below by $28/3 - 6 - 16(13/50)(7/20) - 6(7/20)^2 > 0$, and above by $28(7/20) - 6 < 4 < D$. Thus $0 < t < 1$.

Let H, J be the polynomials from (45) and (49). Changing the target masses linearly in γ gives

$$\begin{aligned} H_\gamma(u, v) &= H(u, v) + (\gamma - 1/4) \frac{1 - (u - v)^2}{8}, \\ J_\gamma(u, v) &= J(u, v) + (\gamma - 1/4) \frac{(u - v)^2 + 6u - 2v + 1}{32}. \end{aligned}$$

The response's full-parameter score is $G_\gamma = H_\gamma + tJ_\gamma$. One has $J_\gamma(r, 0) = -f_\gamma(r)/64 = 0$ and $\partial_u G_\gamma(r, 0) = 0$. Algebraic division gives

$$(54) \quad G_\gamma(u, 0) - H_\gamma(r, 0) = \frac{(u - r)^2 A_\gamma(r, u)}{16D},$$

where

$$\begin{aligned} A_\gamma(r, u) &= 24\gamma^2 + (3\gamma - 10)r^2 + (6\gamma - 20)ru \\ &\quad + 12\gamma r + 6\gamma u - 51\gamma + 30. \end{aligned}$$

For $0 \leq u \leq 1$ this is at least $30 - 10(7/20)^2 - 20(7/20) - 51(13/50) > 0$. The transverse derivatives satisfy

$$\begin{aligned} 32\partial_v H_\gamma &= 3(u - v)^2 + (8\gamma - 10)u + (6 - 8\gamma)v + 15 \geq 5, \\ 64\partial_v J_\gamma &= 3(u - v)^2 + (14 - 4\gamma)u + (4\gamma - 2)v - 4\gamma - 1 \geq -3 \end{aligned}$$

on $0 \leq u, v \leq 1$. Consequently $\partial_v G_\gamma \geq 7/64$. Equation (54) proves a lower bound $H_\gamma(r, 0)$, with equality precisely at the proposed prior points. This quantity is the right side of (53).

For the upper bound group the full training words by S , as in the preceding theorem. Put $L(r) = r^2 + 6r + 1$. Using $f_\gamma(r) = 0$, the first three rows of $256C$ are

$$\begin{aligned} 0: & (0, -8\gamma L(r), 32r(r+1) - 8\gamma L(r), 32r(r+1)), \\ 1: & 8(r-1)(r^2 + 4r - 1 - 2\gamma(r+1), \\ & \quad r^2 + 4r - 1 + 2\gamma(r+1), \\ & \quad r^2 - 4r - 1 + 2\gamma(r+1), r^2 - 4r - 1 - 2\gamma(r+1)), \\ 2: & -12(r-1)^2(r-1-2\gamma, r-1+2\gamma, r-1+2\gamma, r-1-2\gamma). \end{aligned}$$

The other rows are their reversals. The sign pattern is again

$$(0, -, +, +), \quad (+, -, +, +), \quad (+, +, +, +), \quad (+, +, -, +), \quad (+, +, -, 0).$$

Here are explicit interval checks for the potentially small margins. In row zero the third coordinate is bounded below by $32(1/3)(4/3) - 8(13/50)((7/20)^2 + 6(7/20) + 1) > 0$. The first parenthesis of row one is at most $(7/20)^2 + 4(7/20) - 1 - 2(6/25)(4/3) < 0$; the third is at most $(7/20)^2 - 4/3 - 1 + 2(13/50)(27/20) < 0$. The second parenthesis is positive already from $r > 1/3$ and $\gamma > 0$, and the fourth is negative. Finally $r - 1 + 2\gamma \leq -13/100 < 0$. All off-diagonal validation coefficients are nonpositive. Thus the stated response is a Bayes best response for this prior, including its two zero coefficients. Its score at both supporting parameters equals the lower bound. The full posterior-predictive dual proves (53). $\square$

This interval is a uniform certificate region for the displayed saddle, not a claim that the same masks remain optimal for every mark bias. The theorem supplies a continuous family at genuinely growing training history $N = 2$; it does not infer a recurrence for every N from the one-preparation formulas.

14. EXACT MEMORY FROM A MARKED SADDLE

The realization criterion is now applied to a saddle with genuine Bayes indifference. The zero coefficients in the two-preparation problem are not harmless: their values are fixed by contact with the parameter family. This supplies a converse for all optimal audits, rather than for a preselected response table.

Put $I = [6/25, 13/50]$. Throughout this section $r = r_\gamma$, $t = t_\gamma$, and $v = U_2(\gamma)$ are those of Theorem 13.1. Denote the five deterministic events in (44) by $e_0, \dots, e_4$. Responses displayed below are ordered on 000, 001, 110, 111 and vanish at the other four atoms.

Theorem 14.1 (Contact rigidity and exact decision memory). *For every $\gamma \in I$, every minimax optimal full-word response with two training preparations has, after averaging the independent training marks at the two extreme report counts, the five rows*

$$(55) \quad (t, 0, 1, 1), \quad (1, 0, 1, 1), \quad (1, 1, 1, 1), \quad (1, 1, 0, 1), \quad (1, 1, 0, t).$$

At the three interior counts the displayed response is forced at each individual training word, not merely after averaging.

In the original reset row–event interface, the least number of states at the post-training cut that permits the exact ideal score v is five. This converse allows arbitrary upstream finite memory, randomized encoding, adaptive training gates, stopping, independent target-only calibration, and all legal post-cut randomizations. Other cuts are unrestricted in this decision-memory assertion.

Proof. First consider a response to the full two-preparation word. Under the least-favorable prior of Theorem 13.1, every training word has positive probability. The strict coefficient signs force e_1, e_2, e_3 at report counts 1, 2, 3. At count zero only the 000 coordinate is free; at count four only the 111 coordinate is free. All off-diagonal validation coefficients are strictly negative: the prior parameters are interior, each word likelihood is positive, and the target assigns zero mass to those atoms. Thus those coordinates are zero for every optimal response.

Let a and b be the averages of the two free coordinates over the four equiprobable training mark words at their respective extreme counts. On the diagonal parameter (p, p) the score is

$$(56) \quad H_\gamma((2p-1)^2, 0) + aA_\gamma(p) + bB_\gamma(p),$$

where

$$A_\gamma(p) = (1-p)^4 \left(\frac{1+\gamma}{4} - \frac{(1-p)^2}{2} \right), \quad B_\gamma(p) = p^4 \left(\frac{1+\gamma}{4} - \frac{p^2}{2} \right).$$

Here H_γ is the deterministic five-event score in the proof of Theorem 13.1. At $p_+ = (1 + \sqrt{r})/2$, $A_\gamma(p_+) + B_\gamma(p_+) = J_\gamma(r, 0) = 0$ and $A_\gamma(p_+) > 0$. The latter follows from $p_+ > 1/2$ and $(1 + \gamma)/4 > 1/4$. Reflection exchanges A and B at $p_- = 1 - p_+$. Also $H_\gamma(r, 0) = v$. Optimality forces contact at both prior points, by Theorem 8.2. Subtracting v from (56) therefore gives $(a - b)A_\gamma(p_+) = 0$, hence $a = b$.

With this common value the diagonal score is $H_\gamma(u, 0) + aJ_\gamma(u, 0)$, where

$$H_\gamma(u, 0) = \frac{-u^3 + (7 - 4\gamma)u^2 - 3u + 13 + 4\gamma}{32}, \quad J_\gamma(u, 0) = -\frac{f_\gamma(u)}{64}.$$

The supporting value r lies in $(0, 1)$, so the derivative at r vanishes. Since $f'_\gamma(r) = D_\gamma(r) > 8$, the derivative equation has the unique solution

$$a = b = \frac{64 \partial_u H_\gamma(r, 0)}{D_\gamma(r)} = t_\gamma.$$

This proves (55). Individual extreme mark words may still use different tie values; no unnecessary uniqueness is asserted.

Now take an arbitrary audit with K states at the decision cut. Restrict the candidate to the common-garbling subfamily in Definition 10.1. Generate two all-on words and simulate the adaptive training and its stopping with independent coins. The retained state is a parameter-independent stochastic function of those two words. Conditional on that state, the frozen row-event continuation pulls back to a response $D_s \in [0, 1]^8$ on the all-on alphabet. Any calibration is part of the same encoding and retained state. Thus (28) holds without increasing K . If the audit attains v against the full physical class, its pullback attains v against this subfamily, since the unrestricted upper bound is v . Averaging selected rows preserves their factorization.

The three interior rows in (55) are distinct vertices of $[0, 1]^8$, so the generators must include e_1, e_2, e_3 themselves. The first row lies in the face

$$F_0 = \{d : d(001) = 0, d(110) = d(111) = 1, d = 0 \text{ off the diagonal}\}.$$

Of those three obligatory vertices only e_1 lies in F_0 ; since $t < 1$, this row is not e_1 . It needs another generator in F_0 . Similarly the last row needs another generator in

$$F_4 = \{d : d(000) = d(001) = 1, d(110) = 0, d = 0 \text{ off the diagonal}\}.$$

Only e_3 among the obligatory vertices lies there. The two faces are disjoint. Lemma 8.4 gives $K \geq 3 + 2 = 5$. Conversely the optimal response mixes e_0, e_1 at count zero and e_4, e_3 at count four and otherwise selects the indicated event. Only its five-valued event label survives the cut. The inherited compiler realizes its subsequent validation with finite memory and attains v . This proves exactness of the decision count. $\square$

The contact argument is essential. Setting both free coordinates equal to one gives a Bayes best response with only the three events e_1, e_2, e_3 , but its derivative at r is nonzero because $t < 1$. It is not minimax. Counting the states of a Bayes best response alone would give an incorrect resource conclusion.

Corollary 14.2 (A strict gap and its physical persistence). *There is $\delta_* > 0$, uniform for $\gamma \in I$, such that every two-preparation audit with at most four states at the decision cut has ideal worst-case score at most $U_2(\gamma) - \delta_*$. In particular, this bound holds for every original-interface audit of whole-schedule peak at most four. If only the target changes by total variation at most $\beta < \delta_*/2$, the five-state decision construction still strictly outperforms this class.*

Proof. Let $W_4(\gamma)$ be the maximum worst-case score over the relaxed factorizations (28) with $K = 4$ and two full training words. Their encoder and decoder spaces are compact and fixed, independently of γ . The score is jointly continuous in these variables, the target bias, and (p, q) . Consequently W_4 is continuous. By Theorem 14.1 it is strictly less than $U_2(\gamma)$ for each $\gamma \in I$. The minimum of this positive continuous gap on compact I is positive. This defines δ_* without pretending to have evaluated it numerically. The domination argument in that theorem makes it an upper bound for every original-interface audit in question. Changing only the target changes any fixed audit's mean score by at most β . The difference between the two classes therefore decreases by at most 2β . $\square$

For the microscopic family at $\gamma = 1/4$, the signal is larger than $1/4$ and the standard deviation of each normalized sensor noise is $\sqrt{2}\sigma$. The Gaussian tail bound used in Theorem 10.3 gives the sharper, vanishing estimate

$$(57) \quad \text{TV}(Q_{d,\sigma}, Q_*) \leq \frac{8\sigma}{\sqrt{\pi}} \exp\{-1/(64\sigma^2)\}, \quad 9/10 \leq d \leq 11/10.$$

Thus Corollary 14.2 holds uniformly in d for all sufficiently small positive σ . This is an existence statement with the displayed physical modulus. The present proof does not certify that every $\sigma \leq 1/24$ lies below the unknown gap threshold.

14.1. The sharp serial-validation peak. A decision cut and the whole execution have different resource costs. We now determine the latter in a declared validation schedule. In the *forward serial class*, both training preparations finish before validation; the candidate validation's two report slots and mark are read before any target validation report is read. The same frozen row–event continuation is used on both preparations, as in Definition 10.1. Training gates, stochastic state updates and frozen gate policies are unrestricted within that interface. No intermediate output can be reread as external memory.

Theorem 14.3 (Sharp forward serial peak). *For every $\gamma \in I$, the least whole-schedule peak width attaining $U_2(\gamma)$ in the forward serial class is twelve. In particular, a peak of at most eleven in this class incurs a strictly positive loss, which can be chosen uniformly for $\gamma \in I$.*

Proof. Use the common-garbling subfamily and the two raw all-on training words as in Theorem 14.1. Any optimal audit must finish both training preparations almost surely under its least-favorable prior. Indeed a stopped branch after a first preparation would have a response independent of the unused second word. For each first report count 0, 1, 2, two choices of the unused word force two different interior vertices from e_1, e_2, e_3 . Each has positive conditional probability. A positive stopped component cannot be a convex component of two distinct vertices. The same argument excludes stopping before the first draw. Null branches may be completed arbitrarily.

Consider the cut after the candidate's second report slot and before its mark. On a canonical raw input tape its continuation is a channel from $(w, y) \in \{0, 1\} \times \{000, 001, 110, 111\}$ to $\{-1, 0, 1\}$, where w is the unread candidate mark and y the fresh target word. All these target words have positive probability. Each retained state supplies one such continuation channel, common to every past leading to it. Hence the rows indexed by selected pasts factor through at most the peak number of generators.

For the strict interior training counts, the following eight rows are forced deterministic channels:

$$(58) \quad e_1(y) - f(w), \quad f \in \{0, 1 - w, 1\}; \quad e_2(y) - f(w), \quad f \in \{0, 1\}; \quad e_3(y) - f(w), \quad f \in \{0, w, 1\}.$$

For example e_1 has candidate residual 0 after an off-diagonal report pair, $1 - w$ after 00, and 1 after 11. The other cases follow by restriction of their masks. Each listed past has positive probability under both supporting candidate parameters. The eight functions are distinct: subtracting two with the same mask compares their distinct w functions; for different masks their difference depends nontrivially on y and cannot equal a function of w alone. They are eight distinct vertices of the product of output simplices.

There are four additional required rows. Average the extreme training marks as in Theorem 14.1, and take the candidate raw report pair to be, respectively, off diagonal and 11 at count zero, and off diagonal and 00 at count four. Independence of a fresh raw candidate preparation from the frozen continuation preserves the averaged tie probability t . On the four target words these rows have means

$$(59) \quad (t, 0, 1, 1), \quad (t - 1, -1, 0, 0), \quad (1, 1, 0, t), \quad (0, 0, -1, t - 1),$$

for both values of w . More precisely, each integer coordinate here is a point mass at that integer, and the remaining coordinate is the Bernoulli mixture between the two neighboring integers. This stronger channel statement, not a statement about a zero mean alone, follows because the

candidate indicator is fixed at 0 or 1 on the selected raw pair, while the target indicator is a bit. Fresh gate randomness is already part of this channel; it supplies no additional past-dependent argument to a retained state.

The four rows lie in four pairwise disjoint faces of the product of output simplices. These faces fix, for both w , respectively,

	000	001	110	111
$F_{0,+}$	*	0	1	1
$F_{0,-}$	*	-1	0	0
$F_{4,+}$	1	1	0	*
$F_{4,-}$	0	0	-1	*

where a number specifies a point mass and $*$ is unrestricted. Among the eight vertices (58), each face contains exactly one: e_1 , $e_1 - 1$, e_3 , and $e_3 - 1$, respectively. Since $t < 1$, none of the four required rows is that vertex. Face packing forces four more generators, for a total of twelve. This argument uses conditional future-output laws and not the continued storage of an event name after a residual has replaced it.

The optimal five-event selection followed by the inherited forward compiler has peak twelve, including its stochastic selection cut. It proves the matching upper bound. To see strictness below twelve, use the finite-protocol normal form of Proposition 8.1 with all logical cuts bounded by eleven, including selection and frozen-continuation buffers. Unobserved internal random computations can be collapsed to common transition kernels on the same register; an unretained random duration cannot be supplied as a later input. The normalized row spaces and closed frozen-interface restrictions are compact. If the constrained value equalled U_2 , the preceding face obstruction would apply. Continuity and compactness in γ give a uniform strictly positive gap, just as in Corollary 14.2. $\square$

Theorem 14.3 is a minimum-width theorem for a specified serial order, not a claim that an arbitrary validation order has the same minimum. The ideal target has support restrictions that can change future residuals when the target is read first. Reverse-order and interleaved validations require their own continuation analysis. The decision-memory converse in Theorem 14.1 does not make this serial-order assumption. Neither statement replaces the original physical interface by an uncharged atomic validation.

15. PREPARATION LIMITS AND LOCALIZATION OF LEAST-FAVORABLE PRIORS

The contact structure has a second consequence, now for an arbitrary number of training preparations. We separate a general localization principle from the special geometry of the marked product family. An exact finite- N two-point prior is not assumed.

Theorem 15.1 (Nearest-experiment limit and prior localization). *Let P_θ be a continuous family on a finite alphabet of size d , with compact metric parameter space, and let Q be fixed. Put*

$$d_Q(\theta) = \text{TV}(Q, P_\theta), \quad d_* = \min_{\theta} d_Q(\theta), \quad C_* = \{\theta : d_Q(\theta) = d_*\}.$$

For $N \geq 1$, the unrestricted frozen-validation values satisfy

$$(60) \quad d_* - \sqrt{(d-1)/N} \leq U_N(P, Q) \leq d_*, \quad U_N \uparrow d_*.$$

Every least-favorable prior μ_N in Theorem 5.1 satisfies

$$(61) \quad \int (d_Q - d_*) d\mu_N \leq \sqrt{(d-1)/N}.$$

Consequently every weak limit of such priors is supported on C_* . For every $a > 0$ their mass on $\{d_Q - d_* \geq a\}$ is at most $\sqrt{(d-1)/N}/a$. More generally, an upper bound b_N on twice the expected total-variation error of a common empirical reconstruction replaces $\sqrt{(d-1)/N}$ in both conclusions.

Proof. Let $\hat{P}$ be the empirical law of the N reports, and choose the indicator $h_{\hat{P}}$ maximizing $(Q - \hat{P})h$. If h_θ maximizes $(Q - P_\theta)h$, the two event inequalities give

$$(Q - P_\theta)h_{\hat{P}} \geq d_Q(\theta) - 2 \text{TV}(\hat{P}, P_\theta).$$

This inequality holds for the same parameter-independent response rule at every parameter. For any law P on d atoms,

$$\mathbb{E} \text{TV}(\hat{P}, P)^2 \leq \frac{d}{4N} \left(1 - \sum_x P(x)^2\right) \leq \frac{d-1}{4N}.$$

Thus its expected score is at least $d_Q(\theta) - \sqrt{(d-1)/N}$. Every response has score at most $d_Q(\theta)$ at that parameter, which proves the upper bound at a minimizing parameter. Ignoring the last observation proves monotonicity and hence the limit.

Against a minimizing prior, the Bayes optimum is U_N . It is at least the prior average of the same empirical rule. Therefore

$$U_N \geq \int d_Q d\mu_N - \sqrt{(d-1)/N}.$$

Combine this with $U_N \leq d_*$ to get (61). Markov's inequality proves the mass estimate. The function $d_Q - d_*$ is nonnegative and continuous on the compact parameter space, so its integral under any weak limit is zero. Such a limit is supported on its zero set. The proof uses only the common reconstruction bound, which proves the last assertion. $\square$

The empirical estimate in (60) is the same finite-alphabet mechanism underlying the retained decision-spectrum classification. Its role here is different: it controls every least-favorable prior, not merely a constructive selector. In particular, finite-preparation saddle supports cannot remain far from the closest candidate experiments as the preparation number grows.

Theorem 15.2 (Intrinsic marked contact geometry). *For every $\gamma \in I$, set $s_\gamma = \sqrt{(1+\gamma)/2}$ and*

$$d_\gamma = 2s_\gamma - 1 - \gamma/2, \quad C_\gamma = \{(s_\gamma, s_\gamma), (1 - s_\gamma, 1 - s_\gamma)\}.$$

Then

$$(62) \quad \min_{p,q} \text{TV}(Q_\gamma, b(p, q)) = d_\gamma, \quad \operatorname{argmin}_{p,q} \text{TV}(Q_\gamma, b(p, q)) = C_\gamma.$$

Moreover, for Euclidean distance in $[0, 1]^2$,

$$(63) \quad \text{TV}(Q_\gamma, b(p, q)) - d_\gamma \geq \frac{7}{2300} \operatorname{dist}((p, q), C_\gamma)^2.$$

The constant is uniform on I and is derived from the experiment's atom probabilities, not from a metric-entropy or small-ball assumption.

Proof. Put $a = (1-p)(1-q)$, $c = pq$, $l = (1-\gamma)/2$ and $h = (1+\gamma)/2$. The common mass of target and candidate is $\psi(a) + \psi(c)$, where

$$\psi(z) = \begin{cases} z, & 0 \leq z \leq l, \\ z/2 + (1-\gamma)/4, & l \leq z \leq h, \\ 1/2, & h \leq z \leq 1. \end{cases}$$

This follows by summing the smaller mass on each of the two marked atoms corresponding to the report pair. Reflection $(p, q) \mapsto (1-p, 1-q)$ exchanges a, c ; assume $c \geq a$. If $c < 1/4$, both terms are in the first segment of ψ , and their sum is at most $1/2$.

For $c \geq 1/4$, put $x = \sqrt{c} \geq 1/2$. By the arithmetic-geometric mean inequality,

$$a \leq (1-x)^2 \leq 1/4 < l.$$

Consequently the common mass is at most $F(x) = (1-x)^2 + \psi(x^2)$, with equality only when $p = q = x$. On the three consecutive intervals with endpoints $1/2, \sqrt{l}, \sqrt{h}, 1$, this function is respectively

$$(1-x)^2 + x^2, \quad (1-x)^2 + x^2/2 + (1-\gamma)/4, \quad (1-x)^2 + 1/2.$$

The first two are strictly convex and the last strictly decreases. The first increases away from $1/2$. Thus a maximum must occur at $\sqrt{l}$ or $\sqrt{h}$. Their difference is

$$(64) \quad F(\sqrt{h}) - F(\sqrt{l}) = \gamma \left(\frac{3}{2} - \frac{2}{\sqrt{h} + \sqrt{l}} \right) > \frac{7}{575}.$$

Indeed $\sqrt{h} > 39/50$, $\sqrt{l} > 3/5$, and $\gamma \geq 6/25$, so the right side is greater than $(6/25)(3/2 - 100/69) = 7/575$. The unique maximizing diagonal point on this side is $x = \sqrt{h}$. Its common mass is $(1 - \sqrt{h})^2 + 1/2 > 1/2$, also excluding $c < 1/4$. Subtracting it from one gives d_γ . Reflection proves (62) and the assertion about all minimizers.

We record a uniform quantitative form. Write $s = \sqrt{h}$ and $\Delta = 7/575$. For $1/2 \leq x \leq \sqrt{l}$, the common mass is at most $F(\sqrt{l})$, so its deficit from $F(s)$ is at least Δ . For $\sqrt{l} \leq x \leq s$, convexity puts F below the chord between its endpoint values. Since this interval has length at most one, its deficit is at least $\Delta(s-x)$. For $s \leq x \leq 1$ the deficit is

$$(1-s)^2 - (1-x)^2 = (x-s)(2-s-x) \geq (1-s)(x-s).$$

Since $s < 4/5$, this too is at least $\Delta|x-s|$. Together the three intervals give $F(s) - F(x) \geq \Delta|x-s|$.

Let $z = (p-q)^2$. At fixed $c = x^2$ the additional loss from the nondiagonal candidate is exactly

$$(1-x)^2 - a = p + q - 2x = \frac{z}{p+q+2x} \geq z/4.$$

The geometric mean x lies between p and q , and hence

$$(p-s)^2 + (q-s)^2 \leq 4z + 4(x-s)^2 \leq 4z + 4|x-s|.$$

It follows that the total deficit is at least $(\Delta/4)((p-s)^2 + (q-s)^2)$, proving (63) in this case. If $c < 1/4$, the total variation is at least $1/2$, whereas $d_\gamma < 12/25$ from $s < 4/5$ and $\gamma \geq 6/25$. The excess is therefore greater than $1/50$; Euclidean squared distance in the square is at most two, and $1/50 > 2(7/2300)$. This proves the remaining case. Reflection completes the proof. $\square$

Corollary 15.3 (An all-preparation law and a limiting two-point prior). *For all $N \geq 1$ and $\gamma \in I$,*

$$(65) \quad d_\gamma - \sqrt{3/N} \leq U_N(b, Q_\gamma) \leq d_\gamma, \quad U_N(b, Q_\gamma) \uparrow d_\gamma.$$

Every least-favorable prior $\mu_{N,\gamma}$ satisfies

$$(66) \quad \int \text{dist}(\theta, C_\gamma)^2 d\mu_{N,\gamma}(\theta) \leq \frac{2300}{7} \sqrt{3/N}.$$

A least-favorable prior can be chosen invariant under simultaneous complementation of the two parameters. Every such symmetric choice satisfies, for the Euclidean quadratic transport distance,

$$(67) \quad W_2 \left(\mu_{N,\gamma}, \frac{1}{2} \delta_{(s_\gamma, s_\gamma)} + \frac{1}{2} \delta_{(1-s_\gamma, 1-s_\gamma)} \right) \leq \sqrt{2300/7} (3/N)^{1/4}.$$

In particular, the limiting symmetric least-favorable law is unique.

Proof. Estimate the four-atom report-pair law from training and tensor it with the known fair mark. Tensoring with a common probability preserves total variation. The proof of Theorem 15.1 therefore uses $d = 4$ rather than eight, proving (65) and an excess bound $\sqrt{3/N}$. Combining the latter with (63) gives (66).

Simultaneously complement both report bits and the mark in every training and validation word. This leaves Q_γ invariant and sends $b(p, q)$ to $b(1 - p, 1 - q)$. The Bayes objective in (9) is invariant under this transformation and convex in the prior. Averaging a minimizing prior with its reflected image is therefore again minimizing and is symmetric.

Map each parameter to a closest point of C_γ , choosing the two points with equal probabilities at a tie. This is a Borel kernel, equivariant under reflection. A symmetric prior is pushed forward to the equal two-point law in (67); the induced coupling has squared transport cost exactly the left side of (66). Taking square roots proves the assertion. $\square$

These conclusions identify the endpoint and limiting contact geometry for every preparation number. They do not assert that the exact prior has two atoms at every finite N , or that the displayed convergence rates and constants are sharp. The finite- $N = 1, 2$ saddles are preserved as exact results. The new theorem explains their relationship to an intrinsically determined limiting contact set without extrapolating their particular event tables to unproved values of N .

16. AN EXACT PREPARATION–DECISION-MEMORY LAW

A second solvable family separates a genuine joint resource law from an implementation bound. Here the memory coordinate is the number of states at the training–validation cut. It is not identified with the bit-level peak width of the collision audit.

Fix $m = 2^d$, $0 < \alpha < 1$, and an unknown parameter $\theta \in \{1, \dots, m\}$. A fresh training preparation reports θ with probability α and an erasure $\perp$ otherwise. Conditional on θ , the preparations are independent. Candidate validation has law δ_θ and target validation is uniform on $\{1, \dots, m\}$. At the decision cut the auditor retains at most $K = 2^k$ states, $0 \leq k \leq d$. Its subsequent response may depend only on that state, current validation observations and fresh independent randomness. In particular, a calibration table, remembered seed or stopping time outside the retained state is not available. Upstream and downstream memory are otherwise unrestricted for the converse. Let $V_N^{\text{er}}(K)$ be the best worst-case expected signed validation score with at most N training preparations.

Theorem 16.1 (Exact joint law). *For every $N \geq 0$ and $K = 2^k \leq m$,*

$$(68) \quad V_N^{\text{er}}(K) = (1 - (1 - \alpha)^N) \left(1 - \frac{1}{K}\right).$$

The converse covers stochastic encoders and arbitrary legal stopping. The value is attained by a clocked machine with K training states. With atomic preparation and validation outcomes, the complete audit has peak at most $\max\{2K, 3\}$ and uses at most $N + 2$ preparations.

Proof. Write $q = (1 - \alpha)^N$ and pre-generate N potential training reports. A stopped auditor ignores the unused reports; this does not disclose extra information to it. On the event that all N reports are erasures, its retained-state law $r = (r_j)_{j=1}^K$ is independent of θ . After padding any smaller register with unused states, its channel to the decision register has the form

$$(69) \quad C(j \mid \theta) = qr_j + (1 - q)S(j \mid \theta)$$

for a stochastic channel S . This identity remains valid with randomized stopping and randomized state updates.

Use the uniform prior on parameters as an upper certificate. The part qr_j has zero expected score for every response: averaging δ_θ under this prior gives the target itself. For the other part, optimizing over all responses gives a convex function of S , a sum of positive parts of linear functions. Every stochastic channel is a convex combination of deterministic channels $j = g(\theta)$; therefore its value is at most the largest deterministic-channel value. This convexification is used only to bound a channel. It does not presuppose that a mixture of width- K machines is realizable at width K .

If the fibers of g have sizes $n_1, \dots, n_K$, a nonempty fiber has posterior uniform on that fiber, so its total variation from the uniform target is $1 - n_j/m$. The prior-averaged optimal score is

$$\sum_j \frac{n_j}{m} \left(1 - \frac{n_j}{m}\right) = 1 - \sum_j (n_j/m)^2 \leq 1 - 1/K.$$

Empty fibers contribute zero. Cauchy–Schwarz proves the inequality. This is an upper bound on the average score and hence on its minimum over fixed parameters. Multiplying by $1 - q$ proves the upper bound in (68).

For attainment, partition the parameter alphabet into K equal groups and let $g(\theta)$ denote its group. Initialize the register uniformly on the K group labels. On an erasure leave it unchanged; on a revealed parameter overwrite it by $g(\theta)$. No extra “revelation occurred” flag is needed. At the decision cut its law is $q \text{Unif}[K] + (1 - q)\delta_{g(\theta)}$. For register j , freeze the event which is the complement of group j . On a revelation its score is $1 - 1/K$. Under a uniformly random initial label, the mean score is zero for each fixed θ : one of the K labels has score $1 - 1/K$ and the others have score $-1/K$. This proves the exact worst-case value.

The k initialization bits can be accumulated in the K -state register; all earlier prefix counts are at most K . Thus no persistent seed lies outside it. Candidate validation retains the group label and its event indicator, using at most $2K$ states. Target validation produces one of the three values $-1, 0, 1$. This proves the atomic-outcome peak bound. The controller uses a declared external schedule; an autonomous stopping implementation must additionally retain its phase. A phase-tagged implementation is finite, but its cost is not included in K . $\square$

Corollary 16.2 (Thresholds and matching resource scales). *For $K \geq 2$ and $0 \leq c < 1 - 1/K$, the least integer number of training preparations giving a score strictly greater than c is*

$$(70) \quad N_c^{\text{er}}(K) = \left\lfloor \frac{\log(1 - c/(1 - 1/K))}{\log(1 - \alpha)} \right\rfloor + 1.$$

If $c \geq 1 - 1/K$, no finite N suffices. For $0 < \varepsilon < 1$, achieving $V_N^{\text{er}}(K) \geq 1 - \varepsilon$ requires

$$K \geq \varepsilon^{-1}, \quad N \geq \frac{\log(1/\varepsilon)}{-\log(1 - \alpha)}.$$

Conversely, whenever an allowed $K \leq m$ satisfies $K \geq 2/\varepsilon$, the choice $N \geq \log(2/\varepsilon)/[-\log(1 - \alpha)]$ suffices.

Proof. The strict threshold is equivalent to $(1 - \alpha)^N < 1 - c/(1 - 1/K)$. Since the denominator logarithm is negative, the least integer satisfying this strict inequality is (70), including equality cases of the real quotient. Also

$$1 - V_N^{\text{er}}(K) = q + (1 - q)/K.$$

This is at least q and at least $1/K$, and is at most $q + 1/K$. These three inequalities prove the necessary and sufficient bounds. $\square$

The exact law concerns an entire (N, K, α, m) family, not a single finite diagnostic. Its mechanism is a parameter-independent erasure component followed by the convex geometry of

the decision channel. It demonstrates how a preparation obstruction and a memory obstruction can be matched in one theorem. It does not identify the least physical peak width at two collision preparations. Encoding the m -symbol reports bit by bit also introduces cuts not present in this atomic model; the physical bit-level accounting is given separately.

17. INTEGER MEMORY AND THE FAILURE OF A NAIVE PARTITION FORMULA

The divisibility assumptions in Theorem 16.1 can be removed, but a balanced-partition Bayes bound must not be mistaken for a minimax equality when a persistent random partition is not free. This distinction leads to another exact finite-resource transition.

Keep the revelation interface, but allow any integers $m \geq 2$ and $1 \leq K \leq m$. Put $\rho = 1 - (1 - \alpha)^N$, and denote the genuine K -label minimax value by $V_{N,m}^{\text{er}}(K)$. Write $m = Kq + s$, $0 \leq s < K$, and set

$$(71) \quad B_{m,K} = 1 - \frac{(K-s)q^2 + s(q+1)^2}{m^2}.$$

Here and below all-erasure initialization randomness is a state initialization, not an unrecorded seed used again at validation.

Proposition 17.1 (Arbitrary integer alphabet bounds and low-reveal equality). *The optimal uniform-prior Bayes score is $\rho B_{m,K}$. The minimax value satisfies*

$$(72) \quad V_{N,m}^{\text{er}}(K) \leq \min \left\{ \rho B_{m,K}, 1 - \frac{\lceil m/K \rceil}{m} \right\}.$$

If $\rho \leq m/(m+s)$, equality with the first term holds. At perfect revelation $\rho = 1$ the exact value is the second term. In particular, when $s > 0$ the uniform-prior Bayes optimum at perfect revelation is strictly larger than the minimax value.

Proof. The all-erasure decomposition (69) holds without any divisibility condition. Its parameter-independent part has zero uniform-prior score. Optimizing the other part over responses gives a convex function of its stochastic channel. A deterministic channel with fiber sizes n_j has Bayes score $1 - \sum_j (n_j/m)^2$. If two nonempty fiber sizes differ by at least two, moving one index from the larger to the smaller strictly decreases the sum of squares. Allowing empty fibers and repeating gives sizes q and $q+1$, proving the Bayes bound. It is attained by this partition and its complementary events, with any parameter-independent initialization. This proves the first upper bound as well as the asserted Bayes optimum.

For perfect revelation, fix any K response vectors. For each known parameter choose a response maximizing its score; this deterministic assignment cannot decrease any parameter's payoff, even when the original encoder was randomized. Some assigned fiber has size at least $\lceil m/K \rceil$. Averaging its parameters, the score of their common response is at most the total variation between the uniform target and the uniform law on that fiber, namely $1 - |\text{fiber}|/m$. The minimum is no larger. Balanced fibers with their complementary events attain the resulting upper bound. A perfectly informed encoder can simulate any less informative encoder, so this is an upper bound at every ρ .

For the low-reveal equality let balanced fibers have weights $w_j = n_j/m$. When $\rho < 1$, initialize their label register with probabilities

$$(73) \quad r_j = \frac{\rho w_j + (1 - 2\rho)/K}{1 - \rho}.$$

These sum to one, and are nonnegative precisely throughout the stated range: the smallest numerator is $\rho q/m + (1 - 2\rho)/K$. On a revealed parameter overwrite the label by its fiber; on

erasure do nothing. No revelation flag is retained. For a parameter in fiber i , the complementary-event score is

$$\begin{aligned} g_i &= \rho(1 - w_i) + (1 - \rho) \left(r_i - \sum_j r_j w_j \right) \\ &= \rho \left(1 - \sum_j w_j^2 \right) = \rho B_{m,K}. \end{aligned}$$

This matches the upper bound uniformly in the fixed parameter. If $\rho = 1$ and $s = 0$, the deterministic revealed label already attains that value, so the singular denominator in (73) is not used. Finally, with unequal balanced fibers the weighted average of their sizes is strictly smaller than the largest size; equivalently $B_{m,K} > 1 - \lceil m/K \rceil / m$. This proves the last assertion. $\square$

Theorem 17.2 (The complete two-label revelation law). *For every $m \geq 2$, every $N \geq 0$, and $0 < \alpha < 1$, set $w = \lfloor m/2 \rfloor / m$ and $b = \lceil m/2 \rceil / m$. Then*

$$(74) \quad V_{N,m}^{\text{er}}(2) = \min\{2wb\rho, w\}, \quad \rho = 1 - (1 - \alpha)^N.$$

The same formula holds at the limiting endpoints $\rho = 0, 1$. For odd m it changes regime at $\rho = m/(m+1)$; for even m it reduces to $\rho/2$. The attaining machine uses two training labels, a frozen two-event alphabet, and atomic peak at most four.

Proof. Proposition 17.1 gives the upper bound and its attainment for $\rho \leq 1/(2b)$. Above this threshold, partition into fibers of weights w, b and initialize deterministically in the larger fiber's label. Continue to overwrite on a reveal and leave the state unchanged on an erasure. A parameter in the larger fiber has score w ; a parameter in the smaller fiber has score $(2\rho - 1)b$. The latter is at least w exactly when $\rho \geq 1/(2b)$. This attains the second upper bound. For even m the threshold is one, so the low-reveal construction suffices. The endpoint $\rho = 0$ may instead freeze a constant event. Candidate validation retains the selected label and its indicator, with at most four states; target validation produces the difference. Initialization is atomic randomization when its probabilities are not dyadic. $\square$

For example, $m = 3, K = 2$ gives $V_{N,3}^{\text{er}}(2) = \min\{4\rho/9, 1/3\}$, rather than $4\rho/9$ for all ρ . The balanced-integer expression is therefore not the unqualified replacement of the divisible minimax formula. For $K \geq 3$ outside the low-reveal range, (72) remains a bound rather than an asserted equality. The compatible dynamic formulation of Section 7 retains the exact optimization in that regime. These results strengthen the revelation example without identifying its decision-cut coordinate with the collision model's bit-level peak.

18. QUANTITATIVE CERTIFICATES FOR A FINITE COMPATIBLE ARCHITECTURE

We return to an arbitrary finite architecture, now with a finite parameter set. It can have hidden states, controlled experiments, more than two hypotheses and registers of any size. Enumerate its distinct stochastic rows by $r = 1, \dots, R$ and write $q_r \in \Delta_{s_r}$. A reused autonomous row appears only once in this product. Assume each execution uses at most H controller rows, and uses row r at most d_r times. Delete unused rows and put $S = \max_r s_r$. Fixed environmental kernels are not controller rows.

For an integer $m \geq 1$, intersect Δ_{s_r} with the coordinate boxes of side $1/m$. The resulting nonempty closed polytopes cover the simplex, with overlaps permitted at their boundaries. Write C for a product of such row cells and choose a vertex q_r^C in each factor. A row-cell vertex has at most one coordinate not fixed at a box boundary; consequently it has at most $v_r = s_r 2^{s_r - 1}$ vertices, all on the $1/m$ grid. There are at most $m^{\sum_r s_r}$ product cells.

Write the vertices of row cell C_r as $v_{r,j}$ and parametrize $q_r = \sum_j \lambda_{r,j} v_{r,j}$ with λ_r in a simplex. A reward polynomial has degree at most d_r in block λ_r . Use the homogeneous simplex Bernstein basis of degree d_r in that block, obtaining

$$(75) \quad g_\theta(q(\lambda)) = \sum_\alpha b_{\theta,C,\alpha} B_\alpha(\lambda), \quad B_\alpha \geq 0, \quad \sum_\alpha B_\alpha = 1.$$

Degree elevation includes the factors $\sum_j \lambda_{r,j} = 1$. No independence between different visits of a shared row is imposed on its unknown row vector.

Lemma 18.1 (Causal interpretation of the coefficients). *Every coefficient in (75) satisfies*

$$|b_{\theta,C,\alpha} - g_\theta(q^C)| \leq HS/m.$$

Proof. For each row r , independently prepare an ordered list of d_r vertex labels, sampled independently with law λ_r . At its k th visit, use the stochastic row given by the k th vertex label in that list. Before conditioning, the next label is independent of the previous execution, so this experiment has exactly the row q_r at every visit and expected reward $g_\theta(q(\lambda))$. Unused labels may be sampled after termination without altering the reward.

Condition on the counts of every vertex label in every list. The probability of those counts is $B_\alpha(\lambda)$. Conditional on the counts, each list is a uniformly ordered multiset. Its expected reward is independent of λ and is therefore the corresponding Bernstein coefficient. This also proves (75) directly.

Each row used by this conditioned experiment is a vertex of C_r or a mixture of its remaining vertices. Every two vectors in C_r have coordinate differences at most $1/m$, hence total variation at most $s_r/(2m) \leq S/(2m)$. Couple the conditioned experiment to the one using q^C , using the same environmental randomness while their executions agree. At each actual controller update the probability of a first row-level disagreement is at most $S/(2m)$. There are at most H such updates on the agreeing path. The terminal laws consequently differ in total variation by at most $HS/(2m)$. A reward in $[-1, 1]$ has expectation difference at most twice that quantity. This proves the estimate. The conditioned urn is used to analyze a coefficient; it is not declared to be an admissible free-memory controller. $\square$

Theorem 18.2 (Explicit subdivision certificate). *Define*

$$\ell_C = \min_\theta g_\theta(q^C), \quad u_C = \min_\theta \max_\alpha b_{\theta,C,\alpha}, \quad L_m = \max_C \ell_C, \quad U_m = \max_C u_C.$$

For the exact all-stochastic value of the specified architecture,

$$(76) \quad L_m \leq V(A) \leq U_m \leq L_m + HS/m.$$

On each cell, a witnessing parameter θ_C supplies the nonnegative coefficient certificate

$$u_C - g_{\theta_C}(q(\lambda)) = \sum_\alpha (u_C - b_{\theta_C,C,\alpha}) B_\alpha(\lambda).$$

Its degree in row block r is d_r , and its total degree is at most $\sum_r d_r$. At most $\prod_r \binom{d_r + v_r - 1}{v_r - 1}$ coefficients per parameter and cell are needed. If the primitive data are rational, all coefficients and bounds can be checked by rational arithmetic. Taking $m \geq HS/\varepsilon$ gives a certified interval of length at most ε .

Proof. The point q^C is feasible, so ℓ_C is a lower bound on the value restricted to C . For any $q \in C$, choose a convex representation of its rows. Equation (75) implies

$$\min_\theta g_\theta(q) \leq \min_\theta \max_\alpha b_{\theta,C,\alpha} = u_C.$$

Choose a parameter attaining the minimum defining ℓ_C and apply Lemma 18.1; this gives $u_C \leq \ell_C + HS/m$. Taking the maximum over the finite cover proves (76). Choose θ_C attaining

the definition of u_C ; all displayed coefficients of its certificate are nonnegative. The degree and count statements follow from the product Bernstein basis. Cell vertices are rational grid points, and a finite execution sums finite products, proving the rational statement. $\square$

Remark 18.3. The number of cells and coefficients can be enormous. Neither polynomial running time nor a uniform degree bound for the original sum-of-squares hierarchy follows. Refinement changes the domain cover, not merely a single global SOS degree. For a finite collection of architectures obeying a specified budget, maxima of these intervals give outer intervals; an unbounded architecture class needs a separate uniform argument such as Theorem 5.2. For compact parameter families, an effective net and a verified modulus for the primitive kernels are additional inputs, not consequences of compactness alone.

The resource estimate also makes row precision explicit. With $m = 2^b$, the lower witnesses have dyadic rows and lose at most $HS2^{-b}$ in value. Thus increasing row precision provides a quantified approximation for a fixed architecture, while its random-bit implementation still has to be charged. The physical theorem avoids this cost altogether because its rows are deterministic.

19. AUTONOMOUS NOISY TESTING WITH A PRECISION AND TIME LEDGER

This section supplies a substantive autonomous comparison rather than merely identifying shared rows. The stationary optimum is classical [6]; the proof and explicit finite-time implementation are recorded to make the resource conventions unambiguous. They are not used as novelty evidence for the physical theorem.

Let reports be independent, with law $\text{Ber}(p)$ under hypothesis $-$ and $\text{Ber}(q)$ under hypothesis $+$, where $0 < p < 1/2$, $q = 1 - p$, and $r = q/p$. A W -state autonomous machine has two time-independent stochastic transition matrices T_0, T_1 . Assume their induced chains $P_- = qT_0 + pT_1$ and $P_+ = pT_0 + qT_1$ are irreducible. A final decision uses only the state, with optional fresh terminal randomization.

Lemma 19.1 (Likelihood-range bound). *If equivalent probability laws ν_+, ν_- on a finite set satisfy*

$$\frac{\max_i \nu_+(i)/\nu_-(i)}{\min_i \nu_+(i)/\nu_-(i)} \leq R,$$

then $\text{TV}(\nu_+, \nu_-) \leq (\sqrt{R} - 1)/(\sqrt{R} + 1)$.

Proof. Write $L = d\nu_+/d\nu_-$, $a = \min L$ and $b = \max L$. Since $\mathbb{E}_{\nu_-} L = 1$, one has $a \leq 1 \leq b$. The chord bound for $(L - 1)_+$ gives

$$\text{TV}(\nu_+, \nu_-) = \mathbb{E}_{\nu_-} (L - 1)_+ \leq \frac{(1 - a)(b - 1)}{b - a}.$$

For fixed b/a , maximizing the last expression gives $a = 1/\sqrt{b/a}$; substitution yields $(\sqrt{b/a} - 1)/(\sqrt{b/a} + 1)$. It is increasing in b/a , proving the assertion. The identical-law case follows directly. $\square$

Theorem 19.2 (Autonomous stationary bound and an explicit approximation). *Put $D = W - 1$, $W \geq 2$. In the irreducible autonomous class the infimum stationary minimax error is*

$$(77) \quad e_W = \frac{1}{1 + r^D}.$$

For every $\epsilon = 2^{-b}$, $b \geq 0$, there is a machine whose rows have denominator dividing 2^{b+1} , and whose minimax error at report time n from any initial state is at most

$$(78) \quad e_W + \epsilon(W - 2) + \exp \left\{ - \left(\frac{p\epsilon}{2} \right)^D \left\lfloor \frac{n}{D} \right\rfloor \right\}.$$

The stationary lower bound is not a finite-horizon lower bound.

Proof. Every positive induced transition has its likelihood ratio in $[r^{-1}, r]$, and the supports of the two induced matrices are identical. The Markov chain tree formula writes each stationary weight as a sum of products over directed spanning trees rooted at that state. Each product has D transitions; its ratio lies in $[r^{-D}, r^D]$. The ratio of the two unnormalized stationary weights lies in the same interval. Normalizing multiplies all state likelihood ratios by a common number, so their maximum-to-minimum ratio is at most r^{2D} . Lemma 19.1 therefore bounds stationary total variation by $(r^D - 1)/(r^D + 1)$. The equal-prior Bayes error is at least $1/(1 + r^D)$, and a minimax error cannot be smaller than that Bayes error. This is the equal-prior binary specialization of the Hellman–Cover bound.

For the upper approximation, take states $i = 0, \dots, D$, weights $c_0 = c_D = 1$, and $c_i = \epsilon$ at interior states. On report 1, move from i to $i + 1$ with probability $a_i = \frac{1}{2} \min(1, c_{i+1}/c_i)$, if that neighbor exists; otherwise stay. On report 0, move to $i - 1$ with probability $b_i = \frac{1}{2} \min(1, c_{i-1}/c_i)$, if it exists; otherwise stay. All other probability is assigned to staying. Detailed balance gives

$$\pi_+(i) = \frac{c_i r^i}{\sum_j c_j r^j}, \quad \pi_-(i) = \frac{c_i r^{-i}}{\sum_j c_j r^{-j}}.$$

Reflection $i \mapsto D - i$ exchanges these laws. Decide $+$ for $i > D/2$, $-$ for $i < D/2$, and use a fair terminal decision at $i = D/2$ when this state exists. The two errors agree. Conditional on being at an endpoint, the error is exactly e_W . The total interior stationary mass is at most $\epsilon(W - 2)$, since each $r^i \leq r^D$ and the denominator is at least r^D under $+$, with the reflected argument under $-$. This proves the first two terms of (78).

Every existing neighbor transition under either hypothesis has probability at least $p\epsilon/2$, and every self-loop has probability at least $1/2$. From any state, move down to zero and then stay until D steps have elapsed. This yields the common minorization $P_\pm^D(i, \{0\}) \geq \delta = (p\epsilon/2)^D$. Subtracting that common atom gives contraction in total variation by $1 - \delta$ every D steps. Comparison with the stationary law yields $\text{TV}(P_\pm^n(i, \cdot), \pi_\pm) \leq (1 - \delta)^{\lfloor n/D \rfloor} \leq e^{-\delta \lfloor n/D \rfloor}$. A bounded decision changes error by at most this amount, proving (78). Finally, letting n tend to infinity and then ϵ to zero proves that the stationary infimum equals (77). All transition probabilities belong to $\{0, 1, 1/2, \epsilon/2, 1 - 1/2, 1 - \epsilon/2\}$. $\square$

The W states in this theorem refer to an atomic stochastic update after each report. A sequential fair-bit implementation must also retain the report, the current state, a bit counter and a flag. Each nontrivial move probability is $1/2$ or $2^{-(b+1)}$; test for a string of zero bits of the required length, and pad to $b + 1$ bits if a uniform report cycle is required. At most $4W(b + 2)$ states suffice for this implementation, including its bit-phase counter and a ready phase. This conservative bound is an implementation claim, not an optimal precision–state tradeoff. The time index n in (78) is an observation time at which an external analyst reads the autonomous machine, not a free input clock used by its transitions. The adversarial model of [10] is not covered by the independent-report proof above.

20. PHYSICAL CONSUMERS AND THE FIRST-PRINCIPLES CONNECTION

The physical preparation theorem needs no estimate of the target law as a stored input: the five events are fixed and their uniform validity uses a proved neighborhood of Q_* . This removes calibration memory from this particular consumer without asserting that target acquisition is unnecessary in an arbitrary comparison problem. The general acquired calibration theorem, with its full counter register, remains in the complete companion.

Corollary 20.1 (A raw-interface confidence construction). *Let $m = m_2 - \beta > 0$ and $0 < c < m$. Repeat the full four-preparation audit independently R times and write $\bar{D}_R$ for the mean score. Under every private candidate,*

$$\mathbb{P}(\bar{D}_R \leq c) \leq \exp\{-R(m - c)^2/2\}.$$

Under the null in which the independent candidate and target validation laws agree, with the same event chosen before both validations,

$$\mathbb{P}(\bar{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

A clocked implementation uses $4R$ preparations and at most $12(2R + 1)$ persistent states, with no calibration and no random bits.

Proof. The independent block scores lie in $[-1, 1]$ and have mean at least m under a private candidate. Hoeffding's inequality gives the first bound. Under the null, conditional on the selected event, the two independent validation indicators have the same mean. Each block therefore has mean zero, and the same inequality gives the second bound. Retain the integer sum in $\{-R, \dots, R\}$ along with the current twelve-state block. At a block boundary add its terminal score and reset its register. The deterministic external schedule is part of the clocked convention. $\square$

This is an upper construction on the raw marked interface. Its proof does not transfer a lower bound from a deliberately reduced Bernoulli interface, and no optimal confidence-memory frontier is claimed.

Proposition 20.2 (Target transport without a training multiplier). *For the two-preparation audit, changing only the target from Q to Q' changes its expected score by at most $\text{TV}(Q, Q')$. Its candidate training law, register profile and event selector remain unchanged.*

Proof. Condition on the candidate training word. The expectation difference is $(Q - Q')h_S$, whose absolute value is at most $\text{TV}(Q, Q')$. Average over that unchanged training law. $\square$

In particular, every common-preparation transport estimate verified for the original positive-noise collision family can be inserted here without multiplying it by a long training horizon. If candidate kernels also change, their training and validation defects must be added by a causal coupling; the target-only statement does not remove those terms.

The first-principles route is now concrete. The positive experimental kernels and actual reset preparations give the likelihoods in (9); these generate posterior continuation states. The validation-test quotient gives the positive parts in the statistical dual. A prior then supplies an architecture-uniform impossibility certificate. On the constructive side, the five-event decision is compiled by taking its future residual functions, which are closed under observation shifts. These are compatible causal states and determine the charged profile. Thus the preparation law, predictive continuation and finite-resource realization are used in the same upper/lower theorem. This is not a deduction of geometry or nonlinear operator compactness from the existence of finite stochastic rows.

21. A SHARPENED PHYSICAL CONSUMER AND ITS SIMULATION BUDGET

The exact two-preparation theorem has a downstream use with explicit hypotheses, rather than a merely renamed dependency arrow. It sharpens the mean margin of a raw-interface confidence test, and the causal morphism theorem transfers that test, with its full memory and calibration costs, to another executable interface.

Theorem 21.1 (Confidence and transported resource certificate). *Let Q be a physical marked target with $\text{TV}(Q, Q_*) \leq \beta$. Use the two-preparation event selector with a dyadic coin τ_b satisfying $|\tau_b - \tau| \leq 2^{-b}$, and put*

$$a_b = v_2 - \beta - \frac{5}{16}2^{-b}, \quad W_b = \max\{12, 6(b+1) + 5\}.$$

If $0 < c < a_b$, R independent four-preparation blocks, followed by the test $\overline{D}_R > c$, have error bounds

$$(79) \quad \sup_{\text{private}} \mathbb{P}(\overline{D}_R \leq c) \leq \exp\{-R(a_b - c)^2/2\},$$

$$(80) \quad \sup_{\text{equal}} \mathbb{P}(\overline{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

They use $4R$ preparations, at most bR fresh random bits, and at most $W_b(2R+1)$ persistent states under a clocked schedule. With an exact atomic coin, replace a_b, W_b by $v_2 - \beta, 12$.

Suppose a source interface simulates this entire repeated audit under Theorem 3.4, uniformly over the private and equal-law families, with path defect Δ_R . Then the corresponding source error bounds are the right sides of (79)–(80) plus Δ_R , capped at one. At aligned cuts its state bound is $C_{\text{cal}}R_tM_tW_b(2R+1)$, and its preparation count is N_{cal} plus all source calls required to supply the $4R$ audit preparations. Calibration is acquired and retained, not supplied as a free real-valued table.

Proof. Corollary 12.2 and Proposition 12.3 give mean block score at least a_b under every fixed private parameter. Independent fresh preparations and independent coins give independent block scores in $[-1, 1]$. Hoeffding’s inequality proves (79). Under the equal-law null, conditional on training and the frozen event, the two validation indicators have the same mean. Each block therefore has mean zero, proving (80) by the same inequality. Retain the integer sum in $\{-R, \dots, R\}$ alongside the current block register. Updating the sum and resetting the block is a deterministic boundary operation. This proves the raw resource bounds.

Apply Theorem 3.4 to the whole repeated experiment, including its stopping and validation fields. Its defect bound applies to each of the two terminal error events, whose indicators have range one; each probability therefore changes by at most Δ_R , not $2\Delta_R$. The same theorem gives the state product and preparation count. The source blocks need not be independent after a shared calibration stage: independence was used only for the reference audit, and path-law comparison transfers its final event probability. $\square$

The first part uses the exact saddle, the original microscopic bridge and the compiled residual machine. The second additionally needs the uniform acquired causal simulation theorem. A source which does not satisfy that theorem’s stability and calibration hypotheses is not covered. Likewise this consumer supplies no reverse simulation and therefore does not transfer an optimal-width lower bound to an arbitrary source. It is an actual conditional theorem interface for subsequent models, not an assertion that the independent geometric A2 chain or the unbounded B4/C2 operator obligations have already been discharged.

22. DEPENDENCIES, RETAINED RESULTS AND REMAINING DISTINCTIONS

The exact three-report binary noisy frontier, arbitrary-horizon binary-register minimax recursion and compatible Bayesian interval recursion remain as proved in the complete companion. Its exact odd-horizon majority realization is also retained; it is not a new result of this revision. Section 18 applies to registers of any finite width, controlled rows and any finite parameter set, but provides certified intervals rather than a closed minimax recursion for all such classes. The

physical theorem supplies a matched preparation segment in a continuous two-parameter family with overlapping stochastic reports; its converse is not a finite parameter-grid check.

The stationary autonomous result in Section 19 has an explicit formula, but its stationary lower bound must not be applied to a finite-horizon clocked machine. Its finite-time upper bound is purposely conservative and exposes the slowing caused by finite transition precision. An arbitrary compact-family presentation still requires additional effective information to construct a kernel net. A subdivided positive-basis degree bound must not be called a degree bound for a global SOS hierarchy.

The physical result is a genuine consumer of the experimental and continuation root. It does not make the independent A2 geometric proofs dependent on this root. The original B4 normalized-resolvent, range, compactness and control-transfer obligations, and the broad C2 strict-dual, form, rigidity and optional-projection obligations, are not proved by our finite-state theorem. Their statements, derivations and current status remain in the retained historical pipeline. No aggregate closure is inferred from preserving their source files.

Finally, the rational certificate (43), polynomial identity (45), and residual transitions are independently reproducible. The program in the accompanying source checks their exact identities and includes deliberately invalid controls. These checks are not substitutes for the proofs above. Neither compilation, preservation of earlier pages nor a successful finite test establishes novelty or an editorial decision.

REFERENCES

- [1] R. D. Smallwood and E. J. Sondik, The optimal control of partially observable Markov processes over a finite horizon, *Operations Research* **21** (1973), 1071–1088.
- [2] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272.
- [3] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176.
- [4] J. Müller and G. Montúfar, The geometry of memoryless stochastic policy optimization in infinite-horizon POMDPs, arXiv:2110.07409 (2021; revised versions).
- [5] M. Putinar, Positive polynomials on compact semi-algebraic sets, *Indiana Univ. Math. J.* **42** (1993), 969–984.
- [6] M. E. Hellman and T. M. Cover, Learning with finite memory, *Ann. Math. Statist.* **41** (1970), 765–782.
- [7] M. E. Hellman and T. M. Cover, On memory saved by randomization, *Ann. Math. Statist.* **42** (1971), 1075–1078.
- [8] R. A. Flower and M. E. Hellman, Hypothesis testing with finite memory in finite time, *IEEE Trans. Inform. Theory* **18** (1972), 429–431.
- [9] T. M. Cover, M. A. Freedman and M. E. Hellman, Optimal finite memory learning algorithms for the finite sample problem, *Information and Control* **30** (1976), 49–85.
- [10] M. A. Managoli and V. M. Prabhakaran, Memory constrained adversarial hypothesis testing, arXiv:2605.12063v1 (May 12, 2026).
- [11] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28, doi:10.1524/strm.2002.20.14.1.
- [12] Q. Qi, General Theta Foundations I: Compatible statistical experiments and resource frontiers, revision 23, September 24, 2026, complete mathematical companion, source commit 8a6043aca644043bd805668185eb227468f21170; packaged commit e926b783b643b3e7ff275a18105528927b76cc97b. The 96-page article and 649-page development are reproduced without alteration after the present article in the two complete volumes.
- [13] H. Chernoff, Sequential design of experiments, *Ann. Math. Statist.* **30** (1959), 755–770.
- [14] M. Naghshvar and T. Javidi, Active sequential hypothesis testing, *Ann. Statist.* **41** (2013), 2703–2738, doi:10.1214/13-AOS1144.
- [15] S. Nitinawarat, G. K. Atia and V. V. Veeravalli, Controlled sensing for multihypothesis testing, *IEEE Trans. Automat. Control* **58** (2013), 2451–2464, arXiv:1205.0858v6.
- [16] R. Zheng, R. Sim and A. Varvitsiotis, Solving imperfect-recall games via sum-of-squares optimization, arXiv:2602.21722v1 (2026).
- [17] A. Heller, On stochastic processes derived from Markov chains, *Ann. Math. Statist.* **36** (1965), 1286–1291, doi:10.1214/aoms/1177700000.

- [18] M. Vidyasagar, The complete realization problem for hidden Markov models: a survey and some new results, *Math. Control Signals Systems* **23** (2011), 1–65, doi:10.1007/s00498-011-0066-7.
- [19] N. Gillis and F. Glineur, On the geometric interpretation of the nonnegative rank, *Linear Algebra Appl.* **437** (2012), 2685–2712, arXiv:1009.0880.

PRESERVED PREDECESSOR

General Theta Foundations I, revision 26

The following predecessor volume is reproduced without alteration.
The complete current article precedes this divider.

Frozen predecessor commit:

466bfcdcb8b7d594d3924dbfa82b4c6e29368858

GENERAL THETA FOUNDATIONS I: CAUSAL EXPERIMENTS, FINITE-MEMORY CONTROL, AND EXACT MINIMAX LAWS

QIAN QI

ABSTRACT. We study which statistical comparisons are attainable when both the choice of experiment and its retained continuation are resources. Prepared positive kernels generate a future-test quotient; causal simulators then transport prediction and decision guarantees with explicit acquisition, quantization and memory costs. At a prescribed finite register profile we give an occupation-state dynamic formulation and an exact controlled family whose value is $1/(2\lceil m/K \rceil)$, whereas a free mixture of complete controllers has value $K/(2m)$. This strict discrepancy persists under positive-noise perturbations. For a marked product experiment with two unknown Bernoulli parameters, we determine the exact two-preparation minimax value and its least-favorable two-point prior. The value is $(-r^3 + 6r^2 - 3r + 14)/32$, where $2r^3 + 25r^2 - 3 = 0$ and $1/3 < r < 7/20$. An explicit interval of mark biases has a corresponding exact two-preparation saddle family. A global polynomial certificate proves optimality over all responses to the full marked training word. The optimal response uses five frozen events and peak register width twelve, yielding an exact ideal physical score and an enlarged preparation boundary for positive-noise collision experiments. We also obtain arbitrary-integer revelation bounds, a complete two-label law, and a matched online prediction-resolution law under explicit dynamical and acquired-mass hypotheses.

CONTENTS

1. Introduction	2
1.1. Relation to comparison, control and finite memory	4
1.2. Organization	4
2. Causal experiments and their predictive quotients	4
2.1. The experimental object	5
2.2. Prepared predictive equivalence	6
3. Attainable resolution and resource-aware simulation	7
3.1. Checkpoint and online prediction	7
3.2. Causal morphisms and simultaneous error budgets	8
4. Experiments, architectures and attainable resources	9
5. A statistical dual that is uniform over architectures	10
6. Controlled experiments without a dominating action	12
6.1. An informative action choice	14
7. Controlled experiments at a prescribed memory profile	15
7.1. A parameterwise occupation state	15
7.2. An exact informative-control bottleneck	16
8. Exact minimax laws for a marked product family	18

Date: September 24, 2026. Revision 26.

2020 Mathematics Subject Classification. Primary 62B05, 62C20; Secondary 60J10, 68Q45, 93E20.

Key words and phrases. Statistical experiments, causal realization, predictive distributions, finite memory, minimax duality, physical certification.

8.1. Complementary slackness and the event alphabet	19
9. A self-contained collision interface	20
9.1. Preparation and elastic scattering	21
9.2. Positive noise and the actual mark	21
10. The physical experiment and its sharp preparation boundary	22
10.1. An exact rational certificate against one preparation	23
10.2. A marked five-event selector	24
10.3. Causal compilation and the full register profile	24
11. The exact two-preparation marked saddle	26
12. A continuous two-preparation marked family	28
13. An exact preparation–decision–memory law	30
14. Integer memory and the failure of a naive partition formula	32
15. Quantitative certificates for a finite compatible architecture	33
16. Autonomous noisy testing with a precision and time ledger	35
17. Physical consumers and the first-principles connection	37
18. A sharpened physical consumer and its simulation budget	38
19. Dependencies, retained results and remaining distinctions	39
References	39

1. INTRODUCTION

An experiment supplies observations, but a causal procedure must also retain enough information to choose its next experiment and interpret its eventual answer. These are different requirements. A posterior may summarize all relevant observations without fitting into the available register. Conversely, a small register sufficient for one terminal decision may not support a subsequent informative action. The purpose of this paper is to place these questions in one prepared causal model and to determine several of its exact resource boundaries.

The underlying object consists of a preparation, positive normalized experimental kernels, a causal action interface, executable future tests and a resource profile. No information metric or geometric measure is postulated independently of that object. The future-test quotient identifies prepared histories with identical legal continuations. Their actual probability law determines checkpoint resolution, whereas closure of the quotient under observations determines whether a quantization can be updated online. Sections 2 and 3 develop these distinctions. Under contractive updates and an acquired small-ball condition, online squared prediction loss is bounded above and below by constants times $M^{-2/d}$. A separate transport theorem simultaneously accounts for acquired calibration, state quantization, primitive model errors and the register used by a downstream controller.

The finite-memory question is not solved by unrestricted belief-state convexification. Section 7 retains the common observable rows in a parameterwise occupation-state recursion. The occupation state belongs to offline design, not to the executing controller. Its exact supersolution formulation contains neither an unobserved parameter oracle nor a free retained strategy tree. The distinction is quantitative. In a routing-and-sensing family with m incomparable branch-specific experiments and K labels at the routing cut, we prove

$$V_{m,K} = \frac{1}{2\lceil m/K \rceil}, \quad \bar{V}_{m,K} = \frac{K}{2m}.$$

The second value permits a persistent controller selector without charging it. The first does not. The values differ whenever K does not divide m , including $1/4$ versus $1/3$ at $(m, K) = (3, 2)$.

All other cuts, including the sensing-action buffer and the frozen validation event, are specified. A small full-support perturbation preserves the strict discrepancy.

Our principal explicit statistical calculation concerns

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad (p, q) \in [0, 1]^2.$$

The last bit is the mark of the actual preparation. On atoms 000, 001, 110, 111, the target Q_* has masses $5/16, 3/16, 3/16, 5/16$, and it vanishes elsewhere. Training is followed by a frozen event applied to independent candidate and target validations. If U_N is the unrestricted-memory worst-case mean difference using N candidate training preparations, the exact values at one and two are

$$(1) \quad U_1 = \frac{85 - 7\sqrt{73}}{64}, \quad U_2 = \frac{-r^3 + 6r^2 - 3r + 14}{32}, \quad 2r^3 + 25r^2 - 3 = 0,$$

where $r \in (1/3, 7/20)$. The first value belongs to the continuous marked-target family in Theorem 8.1. The second is proved in Theorem 11.1. Its optimal response depends on the total of four training bits, with a nontrivial randomization at the two extreme counts. A square factorization and strict transverse monotonicity prove its lower bound on the full parameter square. The upper bound uses a matching least-favorable prior and the signs of all posterior-predictive coefficients, including the real training marks. No restriction to symmetric responses or parameter grids is imposed on the converse. Theorem 12.1 extends this construction to every target mark bias $6/25 \leq \gamma \leq 13/50$, with a cubic-defined least-favorable prior and a matching exact $U_2(\gamma)$ formula.

The physical realization is part of the same argument. The five-event residual machine has peak twelve even after the new randomized selection step is charged. Its all-on law covers the original private candidate class; a common-garbling subfamily suffices for the converse. Thus the ideal two-preparation physical score is exactly U_2 at every $W \geq 12$. If $\text{TV}(Q, Q_*) \leq \beta$, then

$$N_c^{\text{phys}}(W, Q) = 2 \quad \text{whenever} \quad W \geq 12, \quad U_1 + \beta \leq c < U_2 - \beta.$$

One candidate and one target validation are charged in addition to the two training preparations. The local collision geometry and positive-noise estimate are proved in Section 9. The exact atomic stochastic row and its finite fair-bit approximations have distinct, explicit precision costs.

For comparison with the deterministic construction we retain the constants

$$(2) \quad u_1 = \frac{4035777}{10240000}, \quad m_2 = \frac{12 - 3\sqrt{3}}{16}, \quad \beta_0 = \frac{1}{300}.$$

The rational u_1 is a valid one-preparation certificate and m_2 is the exact minimum of the deterministic five-event selector. The exact values in (1) sharpen these two distinct bounds without discarding their proofs or implementations.

A revelation family illustrates another integer obstruction. For arbitrary alphabet size m , two decision labels have the complete law

$$V_{N,m}^{\text{er}}(2) = \min\{2wb[1 - (1 - \alpha)^N], w\}, \quad w = \lfloor m/2 \rfloor / m, \quad b = \lceil m/2 \rceil / m.$$

For odd m there is a saturation transition before perfect revelation. The corresponding balanced-partition Bayes formula is not the minimax value at all reveal probabilities. General integer bounds and an exact low-reveal range accompany this calculation. This example is a controlled comparison of decision-cut resources; it is not a surrogate for the collision model's bit-level width.

1.1. Relation to comparison, control and finite memory. Blackwell comparison and minimax theory provide the statistical mechanisms [2, 3]. The finite-horizon belief-state piecewise-linear representation is classical [1]. The unrestricted controlled dual in Section 6 specializes that machinery to a fixed adversarial parameter and a frozen validation score. It is not a new principle of dynamic programming. Its relation to the genuine constrained value is separated by Theorem 7.2.

Controlled information acquisition has a substantial sequential-design history, beginning with Chernoff [13]. Naghshvar and Javidi [14], Section 3 and equation (4), formulate a posterior-state stopping problem minimizing expected observation count plus a wrong-declaration penalty. Their continuation operator is the same Bayesian mechanism that underlies our unrestricted recursion. Our terminal functional instead compares fresh candidate and target draws, and our finite-profile recursion optimizes common row prescriptions without storing the full posterior. Their cost and asymptotic optimality results are not replaced by a finite-horizon validation formula.

Nitinawarat, Atia and Veeravalli [15], Theorem 2 and its controlled sensing example, already distinguish causal from open-loop experiments in a multihypothesis error-exponent problem. Thus informative adaptation or a strict adaptive advantage alone is not a novelty claim here. The exact ceiling law and its charged-selector discrepancy concern a different, fixed-horizon resource constraint; the marked saddle concerns a continuous two-parameter composite family and an exact finite- N validation value rather than an asymptotic error exponent.

Finite-memory testing and the role of randomization are treated in [6, 7, 8, 9]. Our autonomous stationary comparison explicitly retains the Hellman–Cover attribution. Managoli and Prabhakaran [10] study a time-invariant finite-state tester against adaptive sample-distribution choices. Here the unknown parameter is fixed throughout an execution. The lack of mixed/behavioral equivalence with imperfect recall is also established; Zheng, Sim and Varvitsiotis [16] develop sum-of-squares methods over behavioral strategies. We do not claim polynomial policy optimization, the mixed/behavioral distinction or a general positivity hierarchy as new [4, 5]. Our explicit finite-family values and full-parameter saddle certificates are the mathematical claims requiring comparison.

Norberg’s filtered comparison [11] remains relevant to the general object-level setting. No non-overlap theorem is asserted without a proof-level comparison of that original work. The foundational kernel, barycenter and quantization arguments here are proved to fix their resource and preparation meanings, not to establish priority for their classical ingredients.

1.2. Organization. The causal object, acquired resolution and simultaneous simulation budget come first. Compatible finite architectures and the posterior-predictive dual then lead to controlled finite memory, exact marked minimax laws and the physical realization. Integer revelation, quantitative finite-architecture certification and the autonomous comparison follow. A final consumer theorem transfers the sharpened physical audit to repeated confidence tests and causally simulated interfaces. The predecessor’s complete mathematical and development volumes are preserved separately. In particular, preservation of A2, B4 and C2 derivations is not a claim that their geometric or operator hypotheses follow from finite stochastic kernels.

2. CAUSAL EXPERIMENTS AND THEIR PREDICTIVE QUOTIENTS

We begin with the object and morphism structure. Its elementary probability statements fix the meaning of later resource theorems; classical kernel construction, conditional expectation and quantization are not assigned a new priority. The substantive comparisons below retain their preparation, memory and approximation coordinates simultaneously.

2.1. The experimental object. An experiment has a fixed unknown parameter $\theta \in \Theta$, hidden states X_t , visible actions A_t , reports Y_{t+1} , and preparation laws μ_θ . We impose the following six conventions.

A0 (types and visibility). These spaces are standard Borel. Visible histories include actions, reports and publicly recorded resource increments. A legal-action set and its remaining budget are part of the visible protocol type. Two histories with different legal continuations are not identified.

A1 (positive kernels). There are jointly measurable normalized nonnegative kernels $K_t^\theta(dx', dy, dc \mid x, a)$, with c a nonnegative resource increment. An absorbing symbol represents termination or failure. Dirac kernels are allowed. Positivity here does not require every outcome to have strictly positive probability. A hidden state can contain the necessary physical past; this notation does not assume a finite-dimensional physical closure.

A2 (causality). A policy is a kernel $\alpha_t(da \mid h_t)$ depending only on visible past data and allowed independent randomization. It cannot read the unknown parameter. Stopping is with respect to the visible filtration.

A3 (executable tests). The specified bounded future tests include constants, required report events and their legal one-step continuations. For discrete reports the one-step instrument is

$$(\mathbb{T}_{t,a,y}f)(\theta, x) = \int f(\theta, x') K_t^\theta(dx', \{y\}, dc \mid x, a).$$

The test spaces are closed under the required instruments, finite linear combinations and the bounded monotone limits needed to recover the stated event laws. For general reports one uses event instruments and regular conditional distributions, not division by the probability of a continuous singleton.

A4 (actual preparation). Average risks and acquired state laws refer to the specified preparation and exploration protocol. Failed trials, calibration, resets and conditioning events retain their probabilities and implementation costs. A convenient measure on a reachable set does not replace its acquired law.

A5 (executable resources). Persistent labels, temporary buffers, program constants, calibration precision, preparations and clocks are distinguished. A retained selector is a state. A parameter-dependent table is not a known program constant. A simulator with R_t labels composed with a controller with K_t labels has the direct product bound $R_t K_t$, before any justified quotient is taken. Atomic stochastic rows and fair-bit implementations are separately specified.

Proposition 2.1 (Prepared path laws). *For every fixed parameter and causal policy, the preparation and kernels uniquely determine their finite-dimensional path laws. On a finite horizon they are the iterated kernel product*

$$\mu_\theta(dx_0) \prod_{t < T} \alpha_t(da_t \mid h_t) K_t^\theta(dx_{t+1}, dy_{t+1}, dc_t \mid x_t, a_t).$$

These laws are consistent under marginalization. For a countable horizon they determine the usual kernel extension. Deterministic dynamics give their original trajectory laws when represented by Dirac kernels.

Proof. Successive integration constructs each finite law. Normalization of the last two kernels gives its preceding marginal, and induction proves consistency and uniqueness on cylinder events. The countable kernel extension gives the full law; equivalently one applies the standard extension theorem to these consistent iterated kernels. Integration against a Dirac transition is evaluation along that transition, which proves the last assertion. $\square$

2.2. Prepared predictive equivalence. For Bayesian predictions additionally fix a prior on the parameter, and regard (θ, X_t) as the hidden variable. A prepared posterior π at a given protocol type evaluates the future tests. Define $\pi \sim_t \tilde{\pi}$ if their evaluations agree on every specified future test. This definition concerns one prepared experiment; it does not identify a prior-dependent posterior with a statistic sufficient simultaneously for all fixed parameters.

Theorem 2.2 (Predictive quotient and causal closure). *For finite report alphabets, predictive equivalence is preserved by every legal positive-probability report update. Its class determines the laws of all specified future experiments. Every deterministic history statistic that determines those laws factors onto this quotient. For a measurable realization it is sufficient that a countable separating test family gives a measurable evaluation map with standard Borel image, and that the chosen statistic admits a measurable factor. In finite history spaces the latter conditions are automatic.*

Proof. For a legal action and report, Bayes' rule gives

$$\pi'(f) = \frac{\pi(\mathbb{T}_{t,a,y}f)}{\pi(\mathbb{T}_{t,a,y}1)}.$$

Both numerator and denominator depend only on the old class by A3. Equivalent posteriors assign the same probability to the report, so updates agree on the next test space whenever that probability is positive. Null reports do not affect the prepared law and may be assigned a fixed admissible version. Induction determines every finite future report word; the monotone-class extension gives the required future events. Conversely those event laws determine the tests. If two histories have the same sufficient statistic, they have the same future evaluations; the quotient is therefore constant on its fibers. This gives the factor set-theoretically. The stated measurable factor assumption supplies its measurable version; no claim that every abstract quotient of a standard Borel space is standard Borel is used. $\square$

Proposition 2.3 (Barycenters, entropy and normalized response). *In a fixed preparation, suppose a hidden target Z , observation Y and postprocessing W form a Markov chain $Z \rightarrow Y \rightarrow W$. For any bounded vector of tests f ,*

$$p_W = \mathbb{E}[p_Y | W], \quad p_Y = \mathbb{E}[f(Z) | Y], \quad \mathbb{E}\varphi(p_W) \leq \mathbb{E}\varphi(p_Y)$$

for every convex φ for which these expectations exist. For a parameter-independent kernel G , $P \ll Q$, and finite $D(P\|Q)$, regular conditional laws give

$$\begin{aligned} D(P\|Q) &= D(PG\|QG) \\ &\quad + \int D(P(dx | w)\|Q(dx | w)) (PG)(dw). \end{aligned}$$

Finally, if f, g are bounded and $\pi_u(dx) = e^{ug(x)}\pi(dx)/\pi(e^{ug})$, then

$$\left. \frac{d}{du} \pi_u(f) \right|_{u=0} = \text{Cov}_\pi(f, g).$$

Proof. The Markov property and the tower rule give the conditional barycenter; conditional Jensen gives convex order. The joint laws $P(dx)G(dw | x)$ and $Q(dx)G(dw | x)$ have relative entropy $D(P\|Q)$. Disintegrating by w and applying the entropy chain rule gives the identity, and hence data processing. The last equality follows by quotient differentiation; boundedness justifies differentiation under the integral. $\square$

The barycenter statement concerns a law of predictions, not a pointwise map between reachable sets. Likewise the covariance in the last formula is a normalized response, not an independently postulated Fisher matrix or a global sufficiency theorem.

3. ATTAINABLE RESOLUTION AND RESOURCE-AWARE SIMULATION

3.1. Checkpoint and online prediction. Fix an actual exploration protocol and its history law. Let $p(h)$ be a bounded d -dimensional conditional prediction vector, with law ν . For instance, a future test is sampled independently after encoding, and the coordinates include its sampling weights. Squared excess prediction loss is then $\mathbb{E}\|p(h) - \hat{p}\|^2$.

Proposition 3.1 (Checkpoint resolution). *For at most M retained labels the least checkpoint excess loss is*

$$(3) \quad \inf_{c_1, \dots, c_M} \int \min_{1 \leq j \leq M} \|p - c_j\|^2 \nu(dp).$$

Independent randomized encoders or decoders cannot improve it. This formula concerns the stated preparation and does not provide a common codebook by separately optimizing at each unknown true parameter.

Proof. Conditional squared loss decomposes into its Bayes loss and squared distance from the conditional mean. For each retained label replace a random decoder by its mean. Random assignment to the resulting centers cannot do better than a nearest center. Conversely a nearest-center assignment attains the infimum over codebooks arbitrarily closely. Conditioning on an independent shared random seed, if such a seed is allowed, gives a codebook for each seed, each with risk at least (3). Averaging cannot improve the bound. The formula does not charge the computation of a checkpoint assignment as an online algorithm; online compatibility is a separate condition. $\square$

Theorem 3.2 (A matched online resolution law). *Suppose a fixed prepared experiment has a predictive state s_t in a compact metric space S , a known initial state, and updates $s_{t+1} = F_{t,a,y}(s_t)$. Assume every permitted update is ρ -Lipschitz for one $0 \leq \rho < 1$, and the prediction map $p : S \rightarrow \mathbb{R}^d$ is L -Lipschitz. Suppose S has a δ -net of at most $C\delta^{-d}$ points for all sufficiently small δ . At a checkpoint T , assume the actual law ν of $p(s_T)$ satisfies the small-ball bound $\nu(B(z, a)) \leq Aa^d$ for every center z and all sufficiently small $a > 0$. Then for all sufficiently large M , the minimum online excess prediction loss with M retained predictive labels satisfies*

$$(4) \quad c_0 M^{-2/d} \leq R_M^{\text{on}} \leq C_0 M^{-2/d}.$$

Here one may take

$$c_0 = \frac{1}{2}(2A)^{-2/d}, \quad C_0 = \frac{L^2 C^{2/d}}{(1-\rho)^2},$$

Any separate exploration controller, report buffer or clock is charged in addition to these M labels. The exploration protocol and the preparation are fixed before optimizing the encoder.

Proof. Any online M -label state is also a checkpoint encoder, so Proposition 3.1 applies. For any M centers, the union of balls of radius $a = (2AM)^{-1/d}$ has ν -mass at most one half. On its complement squared distance is at least a^2 . Integration gives $c_0 M^{-2/d}$. This argument also permits randomized online encoders.

For the upper bound take a δ -net with $\delta = (C/M)^{1/d}$, and a fixed measurable nearest-net assignment (with a fixed tie rule). Quantize the known initial state and, after each legal update, quantize again. The state error obeys $e_{t+1} \leq \rho e_t + \delta$, $e_0 \leq \delta$, and hence $e_t \leq \delta/(1-\rho)$. The prediction error is at most Le_t . Squaring proves the upper bound, uniformly in time. Each update uses only the current net label and the current permitted input, so the construction is online. It does not grant the encoder free access to the original discarded history. If the exploration policy needs another register, that register is combined by a product construction. $\square$

For example, a Lebesgue density bounded above by B gives the small-ball hypothesis with $A = Bv_d$, where v_d is unit-ball volume. The lower hypothesis is a condition on the *acquired prediction law*. A reachable set of dimension d without a mass condition does not imply the lower bound. Similarly, the contraction assumption is additional dynamics, not a consequence of kernel positivity.

3.2. Causal morphisms and simultaneous error budgets. A causal simulator $\Gamma : \mathfrak{E} \rightarrow \mathfrak{F}$ is a parameter-independent executable transducer that uses source preparations to supply the target interface under every allowed target policy. Its defect ε is a uniform total-variation bound on the resulting path laws, including the fields required by the task. Its resource map ρ_Γ bounds the source resources of the transducer composed with a target controller.

Proposition 3.3 (Composition and risk transfer). *Causal simulators compose. Their uniform defects add, capped at one, and their resource maps compose in the order $\rho_{\Gamma_1} \circ \rho_{\Gamma_2}$. For a terminal loss in $[0, L]$ and a common target task,*

$$\mathcal{R}_{\mathfrak{E}}(\rho_\Gamma(B)) \leq \mathcal{R}_{\mathfrak{F}}(B) + L\varepsilon.$$

For reward in $[-1, 1]$, the corresponding maximum worst-case rewards satisfy the reverse inequality with an error 2ε .

Proof. Interleave the two transducers; their pair of internal states is a causal implementation. Total variation contracts under a kernel and obeys the triangle inequality, proving the defect sum. If their state counts are R_t and M_t , the pair has at most $R_t M_t$ states. Other declared resource bounds compose by execution of the two protocols. A bounded loss changes expectation by at most its range times total variation. Apply this bound to every fixed parameter and target controller and take the appropriate infimum or supremum. Uniformity in the controller is essential to this step. $\square$

Theorem 3.4 (Acquisition, quantization and model-error transport). *Suppose an exact source transducer has predictive state $s_t \in S_t$ and additional charged buffers of size R_t , supplies T target operations, and has uniform model defect at most ε_{mod} . Its initialization and update depend on a calibration quantity ξ . From N_{cal} source preparations acquire an estimate $\hat{\xi}$ with at most C_{cal} retained labels, such that*

$$\sup_{\theta} \mathbb{P}_{\theta}(d(\hat{\xi}, \xi) > a) \leq \eta.$$

Subsequent preparation noises are independent of calibration conditional on the fixed parameter. Let finite nets of sizes M_t have radii δ_t . Assume, on matching input/output histories and on the good calibration event, that the quantized transducer's state error satisfies

$$(5) \quad e_0 \leq d_0 + a_0 a + \delta_0, \quad e_{t+1} \leq \rho_t e_t + d_{t+1} + a_{t+1} a + \delta_{t+1}.$$

Assume its next output kernel and the exact transducer's output kernel differ in total variation by at most $\epsilon_t + L_t e_t + b_t a$, uniformly in the fixed parameter and legal policy. All constants and nets are declared independently of that unknown parameter. Then the quantized acquired transducer has defect at most

$$(6) \quad \Delta = \min \left\{ 1, \varepsilon_{\text{mod}} + \eta + \sum_{t < T} (\epsilon_t + L_t e_t + b_t a) \right\}.$$

Composing it with any target controller of cut widths K_t requires at most $C_{\text{cal}} R_t M_t K_t$ labels at aligned execution cuts and transfers its bounded reward with loss at most 2Δ . Its preparations are N_{cal} plus all source preparations specified by the transducer; no successful-trial conditioning removes the failures.

Proof. On the good calibration event, iterate (5). More explicitly, with $D_0 = d_0 + a_0a + \delta_0$ and $D_j = d_j + a_ja + \delta_j$,

$$e_t \leq \sum_{j=0}^t D_j \prod_{k=j}^{t-1} \rho_k.$$

Couple each pair of output kernels maximally while their external histories agree. On agreement the target controller sees the same inputs, so it uses the same legal action and may use the same independent randomization. Its discontinuous action choices require no Lipschitz assumption: before a disagreement their arguments are identical. The chance of a first discrepancy is bounded by the sum in (6). Add the probability of bad calibration and the exact transducer's model defect. This proves the path bound, including stopping after padding by absorbing states.

The executable approximate transducer stores its calibration label, additional buffer, and net index. Combine these with the actual target register, not its full history or an uncharged belief. The resulting product gives the state count. If an operation contains further buffer cuts, apply the same product at each such cut; R_t was assumed to include them. Preparation costs add by execution. Finally apply Proposition 3.3. $\square$

Corollary 3.5 (A finite acquisition register). *Suppose r_0 calibration rows have at most d outcomes, and take n independent preparations from each row on a fixed schedule. Their empirical rows satisfy, with probability at least $1 - \eta$ uniformly in the row probabilities,*

$$\max \text{TV}(\hat{P}, P) \leq \frac{dz}{2}, \quad \eta = \min\{1, 2r_0d \exp(-2nz^2)\}.$$

Their counts can be retained in at most $(n+1)^{r_0(d-1)}$ labels. Thus Theorem 3.4 can use $N_{\text{cal}} = r_0n$, $a = dz/2$ and this finite calibration register when its calibration metric is the maximum row total variation.

Proof. For each outcome the empirical frequency is the mean of independent Bernoulli indicators. Hoeffding's bound and a union bound over at most r_0d coordinates give the event stated. The total-variation estimate is half the sum of coordinate errors. Retain $d - 1$ counts per row; the last is reconstructed from the known number of draws at that stage of the clocked schedule. Each retained count takes at most $n + 1$ values. An autonomous implementation must additionally charge its schedule. $\square$

These results separate exact predictive equivalence, average checkpoint resolution, online realizability, and uniform simulation risk. In particular, neither an optimal checkpoint codebook nor the proof-state bundle of Section 7 is automatically an executable finite register.

4. EXPERIMENTS, ARCHITECTURES AND ATTAINABLE RESOURCES

Throughout, total variation is $\text{TV}(P, Q) = \sup_{0 \leq h \leq 1} (P - Q)h = \frac{1}{2} \sum_x |P(x) - Q(x)|$. An experimental parameter is unknown and fixed throughout an execution. Controller randomization is independent of that parameter. A reset preparation is a fresh draw, independent of earlier preparations given the parameter. A controller may not recover discarded information from an uncharged tape, clock, selector or calibration table.

Definition 4.1 (Finite architecture). A finite architecture A specifies a finite causal execution graph, its legal experimental operations, the observable inputs to each controller row, and a register alphabet at every cut. It also specifies which row occurrences share a stochastic table. A clocked architecture may use different rows at different scheduled times. An autonomous

architecture may use time only through a charged state. For a parameter family Θ and terminal reward in $[-1, 1]$, let

$$V(A) = \sup_{q \in \prod_r \Delta_{s_r}} \inf_{\theta \in \Theta} g_\theta(A, q).$$

If $\mathcal{A}(R)$ is the class of architectures obeying a global resource budget R , its outer value is

$$V^{\text{out}}(R) = \sup_{A \in \mathcal{A}(R)} V(A).$$

Both the architecture and the simulator parameter are fixed before the controller's random choices. In particular, a mixture of complete controllers is not automatically an admissible controller at the original register profile.

For a finite hidden execution graph, compatibility can be stated without division. Let $x_{\theta ir}$ be the incoming mass of hidden history i at observable row r , and let $f_{\theta irj}$ be the outgoing mass through choice j . Include the usual fixed-kernel flow equations, as well as

$$(7) \quad f_{\theta irj} \geq 0, \quad \sum_j f_{\theta irj} = x_{\theta ir}, \quad f_{\theta irj} x_{\theta' i' r} = f_{\theta' i' rj} x_{\theta ir}.$$

For shared rows, the indices i, i' include all their occurrences.

Proposition 4.2 (Compatible realization). *For a specified finite architecture, nonnegative flow arrays satisfying (7) are exactly those realized by a common stochastic row at every declared observable input, including at zero incoming mass.*

Proof. Necessity follows from $f_{\theta irj} = x_{\theta ir} q_r(j)$. For sufficiency, if some incoming mass at row r is positive, define $q_r(j)$ by dividing the corresponding outgoing mass by that incoming mass. Conservation makes q_r stochastic, and the cross-products force the same ratio at every other positive incoming mass. If an incoming mass vanishes, nonnegativity and conservation force all its outgoing masses to vanish. If the entire row is unreachable, choose any legal stochastic row. Induction through the finite execution graph reconstructs the prescribed flows. The same ratio argument applies to identified occurrences of an autonomous row. $\square$

This formulation is inherited from the complete companion and is included to fix the controller class. Compact semialgebraic feasibility and an Archimedean positivity theorem provide a complete semialgebraic certificate hierarchy for a *fixed finite architecture*. They do not interchange the two optimizations in the definition. Section 15 gives a quantitative alternative for the inner problem. The next section supplies a different certificate that survives the outer optimization.

5. A STATISTICAL DUAL THAT IS UNIFORM OVER ARCHITECTURES

Let Θ be compact, let Ω be finite, and let $\theta \mapsto P_\theta \in \mathcal{P}(\Omega)$ be continuous. Fix $Q \in \mathcal{P}(\Omega)$. An N -preparation reset selector observes $Z = (Z_1, \dots, Z_N) \sim P_\theta^{\otimes N}$ and chooses a response $h_Z \in [0, 1]^\Omega$. Its expected validation score is

$$(8) \quad G_\theta(h) = \sum_{z \in \Omega^N} P_\theta^{\otimes N}(z) (Q - P_\theta) h_z.$$

The response is fixed before two independent validation draws. Operationally it can be realized by choosing a random event with those coordinate marginals and retaining that event for both validations. No memory bound is imposed in the definition of $U_N(P, Q) = \max_h \min_\theta G_\theta(h)$.

Theorem 5.1 (Posterior-predictive dual). *Both extrema in the following identity are attained:*

$$(9) \quad U_N(P, Q) = \min_{\mu \in \mathcal{P}(\Theta)} \sum_{z, \omega} \left[\int P_{\theta}^{\otimes N}(z) \{Q(\omega) - P_{\theta}(\omega)\} d\mu(\theta) \right]_+$$

$$(10) \quad = \min_{\mu \in \mathcal{P}(\Theta)} \sum_z m_{\mu}(z) \text{TV}(Q, \bar{P}_{\mu}(\cdot | z)).$$

Here $m_{\mu}(z) = \int P_{\theta}^{\otimes N}(z) d\mu(\theta)$ and, when $m_{\mu}(z) > 0$,

$$\bar{P}_{\mu}(\omega | z) = \frac{\int P_{\theta}^{\otimes N}(z) P_{\theta}(\omega) d\mu(\theta)}{m_{\mu}(z)}.$$

A zero-mass term in (10) is defined to be zero.

Proof. The response space is a finite-dimensional compact convex box. Probability measures on the compact space Θ form a weakly compact convex set. The integral of (8) is continuous and affine in each argument. The minimum over parameters equals the minimum over probability measures. The minimax theorem [3] therefore interchanges the maximum and minimum. For fixed μ , each response coordinate is independent in the maximization and its contribution is the positive part of its coefficient, proving (9). The coefficient vector at a fixed z has zero sum. For positive $m_{\mu}(z)$ it equals $m_{\mu}(z)(Q - \bar{P}_{\mu}(\cdot | z))$, proving (10). If $m_{\mu}(z) = 0$, every nonnegative likelihood integrand vanishes μ -almost surely and all coefficients are zero. Continuity on the two compact strategy spaces gives attainment. $\square$

Theorem 5.2 (Dominating-row converse). *Suppose every permitted training row is a parameter-independent causal garbling of one fresh draw from P_{θ} . Suppose every permitted validation row is the same garbling of the target and candidate laws, and every selected row–event pair is fixed before validation. Then $U_N(P, Q)$ bounds the worst-case expected score of every auditor using at most N candidate training preparations, regardless of its register profile, feedback schedule, independent randomization, or stopping rule. Independent target-only calibration, whose law does not depend on θ , does not invalidate the bound. If the all-on row and all its events are permitted, the bound is attained in the class with no memory cap and N training preparations.*

Proof. Generate N independent all-on observations in advance. At the t th requested preparation, apply the declared garbling to observation t , using only previously simulated reports and independent randomization. This simulates the adaptive transcript with its exact law. A stopped execution ignores the unused observations; it does not reveal them to the auditor. Pull back its final row–event response to the all-on alphabet. Conditional on the entire generated training word and the independent random seed, this is a function in $[0, 1]^{\Omega}$. Averaging the seed, including any target-only calibration, produces one response h_z for each full word z . The score is consequently (8). This proves the upper bound for the union of all architectures. Conversely, retaining the entire word and then a selected event implements any such finite response kernel when memory is unrestricted. Fresh validation gives exactly (8). $\square$

Remark 5.3 (What the dual does not convexify). The measure μ is an adversarial prior used to bound the minimum over *fixed* parameters. Its likelihood is $\int P_{\theta}^{\otimes N} d\mu(\theta)$, not $(\int P_{\theta} d\mu(\theta))^{\otimes N}$. It neither enlarges the private candidate class to mixtures nor gives the auditor a free persistent selector. Attainment of U_N without a memory cap does not assert attainment at any specified finite profile.

The state in (10) is generated by the actual preparation law, not by a geometric measure chosen independently of the experiment. For sequential continuation one retains the posterior μ_z or its future-test equivalence class; a single predictive marginal need not be closed under further

observations. This is the distinction between a one-step response quotient and a future-predictive quotient.

Corollary 5.4 (Finite moment support for the product family). *For two independent Bernoulli report bits with parameters (p, q) and an independent fair mark, an optimal prior in (9) can be chosen with at most $(N + 2)^2$ atoms. The two report counts (k, ℓ) are a causally updatable sufficient statistic for computing the dual.*

Proof. Conditional on the two counts, all arrangements of report bits and all mark words have parameter-independent probabilities. Summing their coefficients reduces (9) to the $(N + 1)^2$ count pairs. Their likelihoods are

$$\binom{N}{k} p^k (1 - p)^{N-k} \binom{N}{\ell} q^\ell (1 - q)^{N-\ell}.$$

After multiplication by $Q(\omega) - P_{p,q}(\omega)$, every integrand has degree at most $N + 1$ in each variable. Thus the objective depends only on the $(N + 2)^2$ rectangular moments. Their constant coordinate is one. Carathéodory's theorem in the remaining affine space replaces an optimal moment vector by a convex combination of at most $(N + 2)^2$ point evaluations. Counts update by addition of the current bits. No claim that this count register is optimal for the decision is made. $\square$

6. CONTROLLED EXPERIMENTS WITHOUT A DOMINATING ACTION

The reduction to one dominating row is not needed for a statistical dual, provided causal action choice is retained in the dual recursion. We give the finite-action theorem with a compact parameter space and an explicit protocol state. Its equality is an unrestricted-memory statement; its upper certificates apply to every smaller-memory class.

Let Θ be compact metric and E a finite set of protocol states. At $e \in E$ let $\mathcal{A}_e$ be a finite set of training actions. Action a produces x in a finite alphabet X_a with law P_θ^a and changes the protocol state to $T(e, a, x)$. The functions $\theta \mapsto P_\theta^a(x)$ are continuous. An unknown θ is fixed throughout the experiment. Conditional on it, each requested preparation uses a fresh independent draw from the selected row. Legal validation designs form a nonempty finite set $\mathcal{B}_e$. Design b has a finite outcome alphabet Y_b , target law Q^b and continuous candidate law V_θ^b . A policy may stop after at most N training actions. It then freezes $b \in \mathcal{B}_e$ and an event on Y_b , before one candidate and one target validation. Randomization is independent of θ . No training action is assumed to dominate another.

For a finite nonnegative measure ν on Θ write $\nu_{a,x}(d\theta) = P_\theta^a(x)\nu(d\theta)$ and set

$$(11) \quad \Phi_e(\nu) = \max_{b \in \mathcal{B}_e} \sum_{y \in Y_b} \left[Q^b(y)\nu(\Theta) - \int V_\theta^b(y) d\nu \right]_+,$$

$$F_0(e, \nu) = \Phi_e(\nu),$$

$$(12) \quad F_n(e, \nu) = \max \left\{ \Phi_e(\nu), \max_{a \in \mathcal{A}_e} \sum_{x \in X_a} F_{n-1}(T(e, a, x), \nu_{a,x}) \right\}.$$

If no training action is legal, only the stopping term is retained. All the functions vanish at the zero measure, so no posterior division is required on null histories. For positive mass, (11) is the mass times the largest posterior-predictive total variation.

Theorem 6.1 (Controlled posterior-predictive dual). *Let $\mathcal{U}_N(e_0)$ be the maximum worst-case expected validation score over all randomized causal policies in the declared protocol, with at*

most N training preparations and no memory cap. Then

$$(13) \quad \mathcal{U}_N(e_0) = \min_{\mu \in \mathcal{P}(\Theta)} F_N(e_0, \mu).$$

Both extrema are attained. Each $F_n(e, \cdot)$ is the maximum of finitely many continuous linear functionals on the cone of measures; in particular it is convex, continuous and positively homogeneous. For every prior μ , $F_N(e_0, \mu)$ bounds the worst-case score of every finite-memory architecture implementing this protocol.

Proof. There are finitely many histories of length at most N and finitely many event masks at each stopping history. Consequently there are finitely many deterministic policy trees, denoted by $\Pi_N(e_0)$. For $\pi \in \Pi_N(e_0)$ its payoff $g_\pi(\theta)$ is a finite sum of finite products of continuous primitive probabilities and is continuous in θ .

A randomized causal policy induces the same outcome laws as a probability distribution on these trees: sample its finite random decision table in advance and condition on it. Conversely, with no memory cap, a sampled tree can be retained and followed. Thus the unrestricted payoff vectors are exactly the convex hull of the g_π . This step is not an assertion that mixing machines preserves a specified register width.

The simplex on $\Pi_N(e_0)$ and $\mathcal{P}(\Theta)$ are compact and convex. The expected payoff is continuous and bilinear. Minimax therefore gives

$$\max_{\lambda \in \Delta(\Pi_N)} \min_{\theta} \sum_{\pi} \lambda_{\pi} g_{\pi}(\theta) = \min_{\mu} \max_{\pi \in \Pi_N} \int g_{\pi} d\mu.$$

For a fixed measure, the terminal maximum is (11), because an event includes precisely the positive coordinates of its signed predictive measure. At a training node one selects an action and, after its observed outcome, selects a continuation tree separately on that branch. The unnormalized branch measure is exactly $\nu_{a,x}$. Backward induction proves (12) and identifies the right side with (13). It also proves the finite-linear-functional representation. Compactness gives an optimizing prior and an optimizing mixed tree. Finally, the minimum over fixed parameters of any policy is at most its average under μ , which is at most $F_N(e_0, \mu)$. This remains true after restricting its memory or architecture. $\square$

The minimization in (13) occurs once, outside the recursion. It cannot be replaced by fresh adversarial choices after observations. The posterior carries dependence on one fixed unknown parameter. Also, a Bayes-optimal tree for a minimizing prior need not by itself be a minimax policy; the saddle-point mixture is supplied by the finite game in the proof.

Corollary 6.2 (Bellman upper certificates). *Suppose functions $\Psi_n(e, \nu)$ satisfy $\Psi_0 \geq \Phi$ and, for $n \geq 1$, satisfy*

$$\Psi_n(e, \nu) \geq \Phi_e(\nu), \quad \Psi_n(e, \nu) \geq \sum_x \Psi_{n-1}(T(e, a, x), \nu_{a,x}) \quad (a \in \mathcal{A}_e).$$

Then $\Psi_N(e_0, \mu)$ is an upper certificate for every admissible architecture and every prior μ . The family F_n is the least such family. An optimal prior can be chosen with at most $d + 1$ support points, where d is the affine dimension of $\{(g_\pi(\theta))_{\pi \in \Pi_N} : \theta \in \Theta\}$.

Proof. Induction in (12) gives $F_n \leq \Psi_n$. The equality case is the recursion itself. The objective depends on a prior only through its vector of integrals of the finitely many g_π . An optimal such vector belongs to the convex hull of their compact evaluation image. Carathéodory's theorem gives the stated support bound while preserving every pure-tree payoff. $\square$

Proposition 6.3 (Causal stability). *Consider two models with the same protocol, alphabets and policy class. Suppose their training rows differ uniformly in total variation by at most ϵ , their*

candidate validation rows by at most η , and their target validation rows by at most β . Their unrestricted values, or their values in any common specified memory class, differ by at most $2N\epsilon + \eta + \beta$.

Proof. Use the same policy randomization and couple corresponding training rows while the histories agree. The probability of a first discrepancy is at most $N\epsilon$, even with feedback and stopping. Conditional expected validation rewards lie in $[-1, 1]$, so differing training histories contribute at most $2N\epsilon$. On agreeing histories the same frozen design and event are used, and their conditional mean rewards differ by at most $\eta + \beta$. The bound is uniform in the fixed parameter and in the policy. Taking minima and suprema preserves it, with the reverse inequality obtained by interchanging the models. $\square$

6.1. An informative action choice. The following calculation distinguishes the controlled theorem from a common-row reduction. Let $\Theta = \{(i, j) : i, j \in \{0, 1\}\}$. Action R reports i . Action L reports j when $i = 0$ and otherwise reports an independent fair bit. Action H reports j when $i = 1$ and otherwise reports an independent fair bit. The candidate validation is $\delta_{(i,j)}$ and the target is uniform on Θ .

Proposition 6.4 (Strict adaptive advantage). *With two training preparations, the unrestricted adaptive value is $3/4$ and the best nonadaptive value is $5/8$. Nonadaptive policies here include mixtures of schedules, with their random schedule retained. No available single training action dominates all the others.*

Proof. Take R first, and then L or H according to the observed i . The parameter is recovered exactly. The complement of its validation atom has score $3/4$. No policy can exceed $\text{TV}(Q, \delta_\theta) = 3/4$ even with full information.

For the converse for fixed schedules, use the uniform prior. For a pair of actions a, b its Bayes value is explicitly

$$(14) \quad \sum_{x,y,\theta} \left[\frac{1}{16} \sum_{\vartheta} P_{\vartheta}^a(x) P_{\vartheta}^b(y) - \frac{1}{4} P_{\theta}^a(x) P_{\theta}^b(y) \right]_+.$$

Substitution of the three displayed kernels gives

	R	L	H
R	$1/2$	$5/8$	$5/8$
L	$5/8$	$7/16$	$1/2$
H	$5/8$	$1/2$	$7/16$

This is an exact finite sum, not a parameter-grid estimate. It bounds any mixture of fixed schedules by $5/8$; an early stop can be padded with an ignored observation. To attain the bound uniformly in θ , choose with equal probabilities the schedules (R, L) and (R, H) . When the chosen branch-specific action matches i , the parameter is known and the score is $3/4$. Otherwise only i is known, and the complement of its two-atom fiber gives score $1/2$. Each fixed parameter therefore has mean score $5/8$.

Action R cannot distinguish parameters differing only in j . Action L cannot distinguish $(1, 0)$ and $(1, 1)$, and H cannot distinguish $(0, 0)$ and $(0, 1)$. An experiment which fails to distinguish such a pair cannot garble to one that distinguishes it. Hence no action is a common dominating row. $\square$

If every binary report is additionally passed through an independent binary symmetric channel of crossover ϵ , the primitive training rows change by at most ϵ and have full support for $0 < \epsilon < 1/2$. Proposition 6.3, applied to both policy classes, leaves an adaptive advantage at

least $1/8 - 8\epsilon$, which is positive for $0 < \epsilon < 1/64$. Thus the distinction is not confined to disjoint observation supports. The finite-horizon belief recursion and its piecewise-linear structure are classical [1]. The contribution here is their explicit integration with the frozen validation score, one fixed adversarial parameter, outer certificates, and the resource conventions; no priority for Bellman recursion itself is claimed.

7. CONTROLLED EXPERIMENTS AT A PRESCRIBED MEMORY PROFILE

An unrestricted posterior is a useful information state for a proof, but it is not a free runtime state. We first give a resource-exact dynamic formulation and then solve a controlled family in which replacing that formulation by a policy-tree mixture changes the value strictly.

7.1. A parameterwise occupation state. Fix a finite horizon T , a compact metric parameter space Θ , finite hidden execution spaces H_t , and register alphabets S_t of sizes at most K_t . The visible inputs to each stochastic row and all legal actions are specified. Any input that must survive a cut belongs to S_t : this includes a chosen action, an unread report buffer, a stop flag, a selected validation event and a validation accumulator. One may subdivide an operation into several cuts to expose these costs. The scheduled time is available only in the clocked convention.

For now rows at different scheduled times are independently prescribed. Let $\mathcal{Q}_t$ be the finite product of legal row simplices at stage t . A prescription $q \in \mathcal{Q}_t$ is a table, chosen at design time, whose runtime arguments are exactly the declared visible inputs. After summing the fixed environmental kernels its one-stage transition is a stochastic matrix $M_{t,\theta}^q$ on the hidden-state/register pairs. We assume this matrix is jointly continuous in (q, θ) . Both action selection and state updating occur in this matrix, in their specified causal order; it is not an arbitrary matrix chosen separately for each hidden state or each parameter.

For a continuous bundle of probability vectors $x = (x_\theta(h, s))_{\theta \in \Theta}$ define

$$(\mathcal{T}_{t,q}x)_\theta(h', s') = \sum_{h,s} x_\theta(h, s) M_{t,\theta}^q((h, s), (h', s')).$$

The initial bundle x_0 is fixed by preparation. Terminal rewards $r_\theta(h, s) \in [-1, 1]$ are continuous in θ . Absorbing stopped states and frozen validation phases are included in this execution, not added afterwards without charge. Put

$$(15) \quad B_T(x) = \min_{\theta \in \Theta} \sum_{h,s} x_\theta(h, s) r_\theta(h, s),$$

$$(16) \quad B_t(x) = \max_{q \in \mathcal{Q}_t} B_{t+1}(\mathcal{T}_{t,q}x), \quad 0 \leq t < T.$$

The argument x is an *offline occupation bundle*, not the realized history, not an observed parameter, and not an additional register made available to the machine.

Theorem 7.1 (Resource-exact dynamic value and supersolutions). *For the specified clocked profile and observable-row interface, $B_0(x_0)$ is the exact maximum worst-case reward over all stochastic controllers. The maximum is attained by a sequence of legal common prescriptions. With the norm $\|x - x'\| = \sup_\theta \sum_{h,s} |x_\theta(h, s) - x'_\theta(h, s)|$, each B_t is 1-Lipschitz. Moreover,*

$$(17) \quad B_0(x_0) = \inf_H H_0(x_0),$$

where the infimum ranges over families of functions on occupation bundles satisfying

$$H_T \geq B_T, \quad H_t(x) \geq H_{t+1}(\mathcal{T}_{t,q}x) \quad (q \in \mathcal{Q}_t).$$

The infimum is attained by $H = B$. No convexification over retained controllers and no new adversarial choice after a report occurs in these identities.

Proof. For a fixed sequence $q_0, \dots, q_{T-1}$, iteration of the displayed transition gives its actual terminal law for every fixed θ . The terminal minimum is therefore exactly its worst-case reward. A runtime randomized controller in this interface is precisely such a sequence of common conditional rows; independence of fresh randomization and the declared register specify all its conditional choices. Persistent random correlations are allowed only through that register.

The occupation bundle at the next stage is determined by the current bundle and the prescription, without drawing a realized history in the design problem. Hence optimizing over the remaining prescription sequence gives (16) by ordinary backward induction. This does not select a different continuation for each unrecorded history. Each maximizing q is still a single legal table common to all hidden histories and parameters. At x_0 one follows the maximizing sequence of design prescriptions, computing their occupation bundles offline; the runtime controller only reads its allowed inputs.

Stochastic matrices contract the ℓ^1 norm. The terminal minimum is 1-Lipschitz in the displayed norm. Induction, using the same q when comparing two bundles and then taking the two maxima, proves the Lipschitz assertion. Joint continuity and compactness of $\mathcal{Q}_t$ give attainment at each stage, and compactness of Θ gives the terminal minimum. Any family H in (17) bounds the terminal reward along every prescription sequence. Backward induction therefore gives $H_t \geq B_t$. Equality is admissible, proving (17) and its attainment. $\square$

This is a nonlinear supersolution dual, not a least-favorable-prior saddle theorem at fixed width. Its optimization can be large and nonconvex. The principle of dynamic programming is classical; the point of this formulation is that the common-row and memory constraints remain in the maximizations. If rows are shared across scheduled times, as for an autonomous controller, choose the shared rows once and hold them fixed in all subsequent transitions; equivalently augment the *design* problem by that immutable prescription. Maximizing a shared row anew at each time would solve a different problem. A finite union of allowed interfaces can be handled by taking their maximum, or by a fully wired interface with the prescribed profile and its legal observable inputs. Neither operation enlarges a runtime register invisibly.

7.2. An exact informative-control bottleneck. Fix integers $m \geq 2$ and $1 \leq K \leq m$. The fixed parameter is $\theta = (i, j) \in [m] \times \{0, 1\}$. The first training experiment R reports i . Across the next cut the controller retains at most K labels, with an arbitrary stochastic encoder $E(s | i)$. It then chooses one of the m sensing actions with probabilities $D(a | s)$. The selected action is kept in a charged m -state action buffer and the old label is discarded. Action a reports j when $a = i$ and reports $\perp$ otherwise. The first report is no longer available.

The candidate validation law on $\{0, 1\}$ is δ_j ; the target is uniform. An event is frozen before one fresh candidate and one fresh target validation. Stopping before either training action is permitted. No independent persistent selector is given to the controller. The observation symbols are atomic; a bitwise input implementation would have additional cuts. Denote the maximum worst-case score by $V_{m,K}$.

Theorem 7.2 (Exact finite-memory control and its convexification gap). *In this protocol,*

$$(18) \quad V_{m,K} = \frac{1}{2\lceil m/K \rceil}.$$

The value obtained by allowing an uncharged ex ante mixture of complete K -label controllers is instead

$$(19) \quad \bar{V}_{m,K} = \frac{K}{2m}.$$

The inequality is strict whenever K does not divide m . The matching controller for (18) has charged cut profile bounded by $(1, K, m, 3, 6, 3)$. The K -label cut controls both retention of the routing observation and the subsequent informative action choice.

Proof. For a fixed i , let

$$p_i = \sum_s E(s \mid i) D(i \mid s).$$

This is the probability of taking the informative action. On its complement, all retained information is independent of j . Averaging over the two values of j makes the expected validation score zero there, for any frozen event or randomized response. On the informative part the score is at most $\text{TV}(\text{Ber}(1/2), \delta_j) = 1/2$. Thus the minimum over j at a given i is at most $p_i/2$. A stop before sensing is also uninformative about j and obeys the same bound; it can be padded by an ignored sensing operation.

Set $L = \lceil m/K \rceil$. Since $p_i \leq \max_s D(i \mid s)$, a value $\min_i p_i > 1/L$ would require every i to have some row s with $D(i \mid s) > 1/L$. Each stochastic row has at most $L - 1$ such indices. But $K(L - 1) < m$. This contradiction proves the upper bound. Partition $[m]$ into at most K fibers of size at most L , retain the fiber label after R , and choose uniformly inside that fiber. On a successful report freeze $\{1\}$ when $j = 0$ and $\{0\}$ when $j = 1$; on $\perp$ freeze the empty event. The score for every j is $1/(2|\text{fiber}(i)|)$, proving equality. The event alphabet has three labels. Storing the event and its candidate indicator requires at most six labels, and the final difference has three. The chosen sensing action was separately charged, proving the displayed profile.

For the convexified upper bound use the uniform prior on (i, j) . For any one controller,

$$\frac{1}{m} \sum_i p_i = \frac{1}{m} \sum_{i,s} E(s \mid i) D(i \mid s) \leq \frac{1}{m} \sum_{i,s} D(i \mid s) = \frac{K}{m}.$$

The same upper bound holds for mixtures. For the matching mixture, choose uniformly one of the m cyclic intervals of K indices. For that interval hardcode a deterministic K -label encoder which routes each included index to its own sensing action, and assigns other indices arbitrarily. Each i belongs to exactly K intervals. A successful action is followed by the complementary singleton event and failure by the empty event, as above. Every fixed (i, j) has mean score $K/(2m)$. This proves (19). $\square$

For example, $V_{3,2} = 1/4$ whereas $\bar{V}_{3,2} = 1/3$. The mixture just constructed can be implemented with a retained interval selector and a local label, using at most mK labels at the bottleneck. It cannot be declared a K -label realization: when the displayed gap is positive, (18) proves that no such realization exists. The action buffer of size m is not part of the K coordinate and has not been omitted from the profile. The theorem is a multicut resource law, not an assertion that its uniform peak equals K .

The actions are statistically incomparable. R cannot distinguish the two values of j at any fixed i , while action a does distinguish them at $i = a$. For $a \neq b$, action a does not distinguish the pair with $i = b$, while b does. A parameter-independent garbling cannot create a distinction absent from its input experiment.

Corollary 7.3 (A strict gap with positive report probabilities). *Replace each of the two training kernels by a mixture with weight $\epsilon > 0$ of the uniform law on its report alphabet. Let $\Delta_{m,K} = K/(2m) - 1/(2\lceil m/K \rceil) > 0$. For $\epsilon < \Delta_{m,K}/8$ the resulting full-support model still has a strict gap between the genuine profile-constrained value and its free-selector convexification.*

Proof. Each training row changes by at most ϵ in total variation. Couple corresponding rows until the first disagreement. For any fixed controller or mixed controller the probability of such a disagreement is at most 2ϵ , and its conditional reward lies in $[-1, 1]$. Each of the two optimal

values changes by at most 4ϵ . Their difference is therefore at least $\Delta_{m,K} - 8\epsilon > 0$. All row arguments and register alphabets are unchanged. $\square$

The distinction between mixed and behavioral strategies without perfect recall is classical; it is not first asserted here. Contemporary polynomial formulations over behavioral simplices include [16]. Theorem 7.2 evaluates the discrepancy in this specific frozen-validation experiment for every m, K , with a matching actual register and a noisy persistence statement. In particular, a single prior followed by an unrestricted policy-tree best response cannot be substituted for (16).

8. EXACT MINIMAX LAWS FOR A MARKED PRODUCT FAMILY

The posterior-predictive dual can be solved explicitly for a continuous family containing the collision target. Throughout this section, the training and candidate validation law is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad (p, q) \in [0, 1]^2.$$

In the order 000, 001, 110, 111, let Q_γ have masses

$$((1 + \gamma)/4, (1 - \gamma)/4, (1 - \gamma)/4, (1 + \gamma)/4),$$

and zero mass elsewhere. Write $U_1(\gamma) = U_1(b, Q_\gamma)$, with exactly the training-validation convention of (8). Set

$$(20) \quad \gamma_* = 1 - 1/\sqrt{2}, \quad s_\gamma = \sqrt{\gamma^2 - 2\gamma + 5}, \quad u_\gamma = \gamma - 2 + s_\gamma, \quad \tau_\gamma = \frac{2u_\gamma}{u_\gamma + 2 - \gamma}.$$

Theorem 8.1 (Exact marked minimax family). *For every $0 \leq \gamma \leq \gamma_*$,*

$$(21) \quad U_1(\gamma) = \frac{3 + u_\gamma^2}{8}.$$

An optimal response depends on the training word only through the report count $K = X_1 + X_2$. On the four target atoms its rows are

$$(22) \quad h_0 = (\tau_\gamma, 0, 1, 1), \quad h_1 = (1, 1, 1, 1), \quad h_2 = (1, 1, 0, \tau_\gamma),$$

and all responses are zero off these atoms. A least-favorable prior is

$$(23) \quad \mu_\gamma = \frac{1}{2}\delta_{(t_\gamma, t_\gamma)} + \frac{1}{2}\delta_{(1-t_\gamma, 1-t_\gamma)}, \quad t_\gamma = \frac{1 - \sqrt{u_\gamma}}{2}.$$

The assertion optimizes over all stochastic responses to the full marked training word, not merely over count-based or symmetric responses.

Proof. First, u_γ is the positive root of

$$(24) \quad u^2 + (4 - 2\gamma)u - (1 + 2\gamma) = 0.$$

On the asserted interval, $0 < u_\gamma < 1$ and $0 < \tau_\gamma < 1$. Indeed the polynomial is negative at zero and positive at one; also $u_\gamma < 2 - \gamma$ because $\gamma < 1/2$. Substitution of $1 - 2\gamma$ into the polynomial gives $8(\gamma^2 - 2\gamma + 1/2)$, which is nonnegative precisely on the present interval. Monotonicity on $[0, \infty)$ therefore gives

$$(25) \quad u_\gamma \leq 1 - 2\gamma.$$

Put $a = (1 - p)(1 - q)$, $c = pq$, and $z = 1 - a - c$. The probabilities of the three training counts are a, z, c . The independent training mark may be discarded by the displayed construction. Let $G_\gamma(p, q)$ denote its score. With $u = (p + q - 1)^2$ and $v = (p - q)^2$, direct expansion gives

$$(26) \quad G_\gamma(p, q) - \frac{3 + u_\gamma^2}{8} = \frac{2 - \tau_\gamma}{16} \{(u - v - u_\gamma)^2 + 4v\}.$$

For clarity, this identity follows by substituting the three rows of (22) into

$$a(Q_\gamma - b)h_0 + z(Q_\gamma - b)h_1 + c(Q_\gamma - b)h_2,$$

using $a + c = (1 + u - v)/2$, $a - c = -(p + q - 1)$, and then eliminating u_γ^2 by (24). The coefficient of the resulting square is $(2 - \tau_\gamma)/16$ and that of v is four times this coefficient. Both are nonnegative. Equality holds at both points of (23), since there $v = 0$ and $u = u_\gamma$. This proves the lower bound and identifies its equality set.

It remains to bound every response, including nonsymmetric ones. Use (23) in Theorem 5.1. At $t = t_\gamma$ put $a = (1 - t)^2$, $c = t^2$, and $T = a + c = (1 + u_\gamma)/2$. For the training report pair 00, the four dual coefficients, apart from a common positive factor, are

$$(27) \quad \left(Q_H T - \frac{a^2 + c^2}{2}, \quad Q_L T - \frac{a^2 + c^2}{2}, \quad Q_L T - ac, \quad Q_H T - ac \right),$$

where $Q_H = (1 + \gamma)/4$ and $Q_L = (1 - \gamma)/4$. The first coordinate is zero by (24); the second is $-\gamma T/2 \leq 0$. The last two are positive: $T \geq 1/2$, $ac \leq 1/16$, and $\gamma < 1/2$ imply $Q_L T > 1/16 \geq ac$. Thus $(\tau_\gamma, 0, 1, 1)$ maximizes this linear functional. For the training pair 11, reversing all report bits and the mark reverses the four coordinates and gives the row h_2 .

For either training pair 01 or 10, the two prior support points have the same likelihood $t(1 - t)$. The posterior is still equally weighted, and the candidate predictive mass at each target atom is $T/4$. Every one of the four coefficients is therefore nonnegative by $Q_L - T/4 = (1 - 2\gamma - u_\gamma)/8 \geq 0$. The response h_1 is optimal. At off-diagonal validation atoms the target mass vanishes and the coefficient is nonpositive. Finally, conditioning on either training mark multiplies all these coefficients by $1/2$; it creates no further optimizing choice. We have therefore optimized every coordinate of the full marked-word response space.

The prior average of the displayed response equals its value at either support point, namely $(3 + u_\gamma^2)/8$, by (26). Every other response has prior average at most this number and hence has minimum over fixed (p, q) at most this number. Together with the lower bound this proves (21), optimality of the response, and least favorability of the prior. $\square$

Corollary 8.2 (The exact one-preparation collision value). *For $Q_* = Q_{1/4}$,*

$$(28) \quad U_1(b, Q_*) = u_{\text{ex}} := \frac{85 - 7\sqrt{73}}{64}, \quad u_{1/4} = \frac{\sqrt{73} - 7}{4}, \quad \tau_{1/4} = 2 - \frac{14}{\sqrt{73}}.$$

If only the target changes to Q , then $|U_1(b, Q) - u_{\text{ex}}| \leq \text{TV}(Q, Q_)$.*

Proof. Substitute $\gamma = 1/4$ in Theorem 8.1. For a fixed trained response the target perturbation is at most total variation, uniformly in (p, q) . Taking first the minimum and then the maximum preserves this bound in both directions. $\square$

8.1. Complementary slackness and the event alphabet. The tie in the first coordinate of (27) explains the randomization. Any value of that response coordinate is Bayes-optimal for the two-point prior. The minimax choice makes the uniform payoff stationary at its two support points; solving this condition gives τ_γ and the square identity (26). This derives the prior and response together, rather than postulating that a numerically effective prior is optimal.

Operationally, for $K = 0$ one draws once between the events with indicator vectors $(0, 0, 1, 1)$ and $(1, 0, 1, 1)$, with probabilities $1 - \tau_\gamma$ and τ_γ . For $K = 2$ one draws between $(1, 1, 0, 0)$ and $(1, 1, 0, 1)$. For $K = 1$ the event is $(1, 1, 1, 1)$. The resulting event identity is retained and used for both fresh validations. These are exactly the five events in the two-preparation construction below. The one-preparation theorem supplies their complementary-slackness interpretation;

it does not prove that the two-preparation selector is optimal. The four-point rational prior retained below is a useful rational certificate, not a least-favorable prior.

The exact response uses an atomic stochastic row. It does not receive free persistent randomness. Replacing τ_γ by a dyadic number within 2^{-b} changes each conditional response, and hence the score, by at most 2^{-b} . A sequential fair-bit implementation must retain its bit prefix and event identity. None of these implementation costs is needed by the deterministic two-preparation physical audit.

Proposition 8.3 (The zero-preparation endpoint). *For $0 \leq \gamma \leq 1$, $U_0(b, Q_\gamma) = \gamma/2$.*

Proof. The fixed event $\{000, 111\}$ has score $(1 + \gamma)/2 - (a + c)/2 \geq \gamma/2$ because $a + c \leq 1$. Conversely, the equally weighted prior on $b(0, 0)$ and $b(1, 1)$ has predictive validation law uniform on the four target atoms. Its total variation from Q_γ is $\gamma/2$. Theorem 5.1 with $N = 0$ gives the matching upper bound. $\square$

9. A SELF-CONTAINED COLLISION INTERFACE

We give the preparation, mechanical estimate, observation law and private simulator class needed for the physical specialization. The following argument does not appeal to an unpublished physical lemma. It retains the same positive-volume preparation and positive sensor noise as the earlier collision construction.

Definition 9.1 (The reset row–event architecture). A preparation supplies two sequential report slots and, afterwards, an actual mark. The source reports in the two slots are constant. A private width-one simulator has no retained information between those slots and no persistent hidden seed. It may use fresh independent coins. Its first emission is $X_1 \sim \text{Ber}(p)$; after receiving the current gate $A \in \{0, 1\}$, its second emission has a fixed Bernoulli kernel with parameter q_A . The parameters (p, q_0, q_1) are fixed throughout the audit. The simulator does not observe the actual mark. All its coins are independent of that mark. Different preparations reset all unretained variables and are independent conditional on the parameters.

The auditor observes the first report, chooses a gate using its retained state and that report, observes the second report, and then observes the mark. During training it may choose rows and stop adaptively, using at most the declared number of candidate preparations. It next freezes one gate policy and one event, with any retained random choice included in its register. It uses that same row–event continuation on one fresh candidate preparation and one independent target preparation and returns the difference of their event indicators, target minus candidate. It may not reselect the event after either validation. Independent target-only calibration is allowed but all retained calibration and randomness are charged. Peak width counts every bit, mark, event and validation-accumulator cut. Clocked schedules are declared externally; an autonomous implementation must retain its phase. An unretained transcript, seed or stopping-time value is not available to a continuation.

In the all-on row the complete private law is $b(p, q_1)$, with an independent fair actual mark. For lower bounds it suffices to restrict the adversary to $q_0 = 0$, $q_1 = q$. In this subfamily every deterministic gate policy is the pushforward

$$T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$$

of $b(p, q)$. The gate, when included in the transcript, is another function of these same coordinates. Randomized gate policies are covered by averaging their independent random tables. This proves both the all-on product characterization for the upper construction and the common-garbling subfamily used for the outer converse. The width-one constraint is a simulator constraint; it is not an uncharged restriction on the auditor’s register.

9.1. Preparation and elastic scattering. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$, with no walls and contact distance d . Write $r = q_1 - q_2$, $g = v_1 - v_2$, $V = (v_1 + v_2)/2$, and let the center of mass position range over any fixed rational box of positive volume. Prepare $(B, W) \in \{0, 1\}^2$ with probability $5/16$ at $B = W$ and $3/16$ at $B \neq W$. Conditional on (B, W) , use the uniform product preparation on the following boxes:

$$(29) \quad \begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & & & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

Thus the actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$, preserved by momentum conservation. Set

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector, identically zero on the incoming transverse-velocity range.

Lemma 9.2 (Uniform collision signal). *For every preparation in (29) and every $9/10 \leq d \leq 11/10$, exactly one collision occurs and its time lies in $(4/5, 3/2)$. Afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4.$$

For $0 < d \leq 2/5$ no collision occurs on $[0, 3]$ and the detector is zero throughout that interval.

Proof. Before contact, $r(t) = r + tg$. At $t \leq 4/5$ its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At $t = 3/2$ its absolute coordinate bounds are $1/4$, $113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. The first contact therefore occurs in the claimed interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$. Hence the entry contact has $n_x < 0$, where $n = r(t_c)/d$. Its remaining coordinates give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq 87/220.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The elastic rule gives

$$v_1^+ = V + g/2 - (g \cdot n)n, \quad g^+ = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v_{1,y}^+ > -3/200 + (1443/1000)(87/220) > 1/2.$$

The definition of h implies the signal bound. Since $r(t_c) \cdot g^+ > 0$, squared separation increases strictly afterwards; there is no second collision. Finally, for $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. There is no contact, and the incoming $|v_{1,y}| \leq 3/200$ makes h zero. $\square$

9.2. Positive noise and the actual mark. Let $\mathcal{B}_t$ be Brownian motion independent of the preparation and set

$$Y_t = \int_0^t h(v_{1,y}(s)) ds + \sigma \mathcal{B}_t, \quad 0 < \sigma \leq 1/24.$$

The two potential all-on reports are

$$Z_1 = \mathbf{1}_{\{2(Y_2 - Y_{3/2}) > 0\}}, \quad Z_2 = \mathbf{1}_{\{2(Y_3 - Y_{5/2}) > 0\}}.$$

The actual second report is AZ_2 . The gate changes acquisition, not the mechanical path. The continuous path is implemented by the target sensor; it is not made available to the private simulator or supplied as an uncharged auditor buffer. Denote the law of (Z_1, Z_2, W) by $Q_{d,\sigma}$.

Theorem 9.3 (Uniform physical presentation). *For $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$,*

$$(30) \quad \text{TV}(Q_{d,\sigma}, Q_*) < 1/300.$$

The same estimate holds after every declared feedback gate, including its observable gate value. The all-on private product family and the common-garbling subfamily in Definition 9.1 apply to this very experiment.

Proof. Every collision occurs before $3/2$. Conditional on the preparation, the two normalized sensor windows are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. Lemma 9.2 makes their means have sign $2B - 1$ and magnitude greater than $1/4$. Each potential report therefore differs from B with probability less than $\Phi(-3)$. The Gaussian tail estimate

$$\Phi(-x) \leq \frac{e^{-x^2/2}}{x\sqrt{2\pi}}$$

gives $\Phi(-3) < 1/600$: $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$ suffice. Couple the ideal and noisy reports using the same preparation, actual mark and gate random table. Except on a set of probability less than $2\Phi(-3) < 1/300$, both potential reports equal B and the entire gated transcript agrees. The ideal marked law is precisely Q_* , by the prepared (B, W) table. This proves (30) and the feedback-uniform version. Finally, the source is blind, its actual mark is unobserved by the simulator, and the gate is its only current input at the second emission. The product and subfamily assertions follow from the two independent private-emission kernels specified above. $\square$

For $d \leq 2/5$, the two windows instead contain only independent centered Gaussian noise. Their signs are independent fair bits, independent of W , and a private width-one simulator reproduces every gated row exactly. Thus the informative regime and its no-collision control use the same detector, preparation convention and actual mark.

10. THE PHYSICAL EXPERIMENT AND ITS SHARP PREPARATION BOUNDARY

Take $\Omega = \{0, 1\}^3$, with coordinates (x_1, x_2, w) . The all-on private candidate is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad 0 \leq p, q \leq 1.$$

The mark w is the actual mark, not an invented simulator bit. Put $a = (1 - p)(1 - q)$ and $c = pq$. On the ordered atoms

$$(31) \quad 000, \quad 001, \quad 110, \quad 111,$$

let Q_* have masses $5/16, 3/16, 3/16, 5/16$, and set it to zero elsewhere. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. A target Q is allowed whenever $\text{TV}(Q, Q_*) \leq \beta$.

The original feedback operation has the form

$$(32) \quad T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w), \quad g : \{0, 1\} \longrightarrow \{0, 1\}.$$

The selected gate is observable. It may depend on earlier preparations and on the first current report. The target rows are pushforwards by (32). For the converse, restrict the adversary to the admissible private subfamily whose second emission is AX_2 , where A is the current gate and $X_2 \sim \text{Ber}(q)$ is independent of the first emission and of the fair mark. Every row is then a causal garbling of $b(p, q)$. For the construction only the all-on row is used, and every private candidate in the original width-one class has the all-on product law just specified. These two facts, proved for the microscopic interface in Definition 9.1 and Theorem 9.3, explain why using a subfamily for the lower bound does not narrow the upper construction.

The admissible audit first trains, then selects one row–event pair, then uses one fresh candidate and one fresh target validation. It reports $D = \mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The event and row cannot be reselected after seeing a validation draw. All bit, mark, decision and validation cuts are charged. Define $N_c^{\text{phys}}(W, Q)$ to be the minimum number of candidate training preparations among these original-interface auditors, with peak register width at most W , whose score is uniformly strictly greater than c . Randomization, feedback and stopping are allowed. If no such audit exists, the value is $+\infty$.

Theorem 10.1 (Two-preparation physical segment). *Let u_1, m_2 be as in (2). For every $\beta \geq 0$, $\text{TV}(Q, Q_*) \leq \beta$, and*

$$W \geq 12, \quad u_1 + \beta \leq c < m_2 - \beta,$$

one has $N_c^{\text{phys}}(W, Q) = 2$. The two-preparation audit is deterministic, uses no acquired calibration and no random bits, and has the full profile

$$(33) \quad (1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$$

from the initial cut through its twelve bit/mark updates. In particular, its decision alphabet has five labels and its peak has twelve. The lower bound remains valid with no memory cap.

The proof occupies the next three subsections. The interval in the statement can be empty for large β ; no assertion is then made. For $\beta = 1/300$, it contains $c = 2/5$ strictly between its endpoints.

10.1. An exact rational certificate against one preparation.

Lemma 10.2. *Every original-interface auditor using at most one candidate training preparation has worst-case score at most $u_1 + \beta$, regardless of memory.*

Proof. By Theorem 5.2, it suffices to bound $U_1(b, Q_*)$ and then perturb the target. In (9) use the four fixed parameters

$$(p, q) = (1/8, 1/8), (1/5, 1/5), (4/5, 4/5), (7/8, 7/8)$$

with weights $1/16, 7/16, 7/16, 1/16$, respectively. This measure is not asserted to be least favorable. Since $p = q$ on its support, the training count $K = x_1 + x_2$ is sufficient. The independent training mark and the order of unequal bits contain no further information about this prior.

For $k = 0, 1, 2$ and the four validation atoms in (31), the signed coefficients in (9), after grouping by K , are the entries of

$$(34) \quad \frac{1}{40960000} \begin{pmatrix} -1977 & -1775177 & 2170423 & 3943623 \\ 1765554 & 191954 & 191954 & 1765554 \\ 3943623 & 2170423 & -1775177 & -1977 \end{pmatrix}.$$

For example, the coefficient in column j is

$$\sum_{i=1}^4 \mu_i \binom{2}{k} p_i^k (1 - p_i)^{2-k} \left(Q_{*,j} - \frac{(1 - p_i)^2 \mathbf{1}_{j \leq 2} + p_i^2 \mathbf{1}_{j \geq 3}}{2} \right).$$

This identity proves every entry of (34) by rational arithmetic. All coefficients at off-diagonal validation atoms are nonpositive. The sum of the positive entries in (34) is $4035777/10240000 = u_1$. Each response coordinate is in $[0, 1]$, so this sum bounds its prior-averaged score and hence its minimum over the four fixed candidates.

Replacing Q_* by Q changes the score of any trained response by at most β . The training law is a candidate law and is unchanged. Consequently the same prior bounds every score by $u_1 + \beta$.

A procedure with no training, or one that stops before taking its permitted sample, is covered by ignoring the generated all-on observation. $\square$

10.2. A marked five-event selector. Take two independent all-on training preparations. Let S be the total number of ones among their four report bits; discard their marks. Choose the following events, represented by their indicators on (31) and with indicator zero at every off-diagonal atom:

$$(35) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & 0 \end{array}$$

Denote these indicators by $h_0, \dots, h_4$ and report $D = h_S(Y) - h_S(X)$ on independent validation draws.

Lemma 10.3 (Exact uniform score of the selector). *Against the ideal target, the selector (35) has uniform minimum score $m_2 = (12 - 3\sqrt{3})/16$. Against Q , its score is at least $m_2 - \beta$.*

Proof. The generating polynomial of S is

$$((1-p) + pz)^2((1-q) + qz)^2.$$

Writing $g_s = (Q_* - b(p, q))h_s$, its expected score is $G(p, q) = \sum_{s=0}^4 [z^s]((1-p) + pz)^2((1-q) + qz)^2 g_s$. Substitution of the five rows in (35) and collection of terms with $x = p + q - 1$, $y = p - q$, $u = x^2$ and $v = y^2$ gives

$$(36) \quad \begin{aligned} G(p, q) = H(u, v) = & -\frac{u^3}{32} + \frac{3u^2v}{32} + \frac{3u^2}{16} - \frac{3uv^2}{32} - \frac{uv}{4} - \frac{3u}{32} \\ & + \frac{v^3}{32} + \frac{v^2}{16} + \frac{15v}{32} + \frac{7}{16}. \end{aligned}$$

This is a polynomial identity, not an interpolation over a grid. Since $0 \leq u \leq 1$ and $v \geq 0$ on the parameter square,

$$\partial_v H(u, v) = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}.$$

Thus $H(u, v) \geq H(u, 0) = (-u^3 + 6u^2 - 3u + 14)/32$. The derivative of the last expression is $(-3u^2 + 12u - 3)/32$. On $[0, 1]$ it changes sign from negative to positive at $u = 2 - \sqrt{3}$. Substitution gives $(12 - 3\sqrt{3})/16$. The minimizing values are feasible: take $p = q = (1 \pm \sqrt{2 - \sqrt{3}})/2$. This proves exactness of the minimum for this specified selector, not optimality of its score over all selectors with two preparations. Finally, for each s , the target expectation of $h_s \in [0, 1]^\Omega$ changes by at most β . Averaging over training proves the claimed robustness. $\square$

10.3. Causal compilation and the full register profile.

Lemma 10.4. *The selector and its two validations admit the deterministic profile (33). No selector, first-bit buffer or validation accumulator is outside the charged register. An autonomous phase-tagged realization uses at most 75 states.*

Proof. After each training report bit retain the count accumulated so far. The two marks are read and discarded. The first six updates therefore have widths 2, 3, 3, 4, 5, 5 after the initial singleton. At the decision cut the five count values select the five events.

During candidate validation, after the first bit retain the event and that bit, using at most ten states. After the second bit retain the event and the function of the unread mark that gives its candidate-event indicator. For events 0, 1, 2, 3, 4, respectively, the possible functions are

$$\{0, 1\}, \quad \{0, 1 - w, 1\}, \quad \{0, 1\}, \quad \{0, w, 1\}, \quad \{0, 1\}.$$

There are twelve event-function pairs. After the candidate mark, retain the event and its now known candidate indicator $c_0 \in \{0, 1\}$, using at most ten states.

During target validation, after the first bit discard the event name and retain c_0 and its residual function of the second bit and mark. These residual functions belong to

$$\{0, x_2, (1 - x_2)(1 - w), 1 - x_2, x_2 w\}.$$

This uses at most ten states. After the second bit, the residual *difference* as a function of the unread mark belongs to

$$\{-1, 0, 1, w, 1 - w, w - 1, -w\},$$

so seven states suffice. The mark update gives the three possible outputs $-1, 0, 1$. Each transition depends only on its retained residual and the current report. Thus it is a legal common observable row, not a family of independently optimized hidden-history maps.

For completeness, equality of two residual functions implies equality of their restrictions after either next bit, so identifying equal residuals is closed under every subsequent update. The supplied transition tables are obtained by these restrictions and can be reconstructed directly from (35). Enumerating residuals is a convenience for reproducing the tables, not the proof of their sufficiency. Finally, disjointly tag the states at the thirteen cuts by their phase and use terminal self-loops. The sum of the displayed widths is 75. This implementation charges the clock; it does not assert that 75 is optimal. $\square$

Proof of Theorem 10.1. Lemma 10.2 excludes $N \leq 1$ for a strict score greater than $c \geq u_1 + \beta$, for every architecture and every memory budget. Lemmas 10.3 and 10.4 supply a two-preparation architecture of peak twelve with score at least $m_2 - \beta > c$. Larger width budgets can use the same machine. $\square$

Corollary 10.5 (The positive-noise collision family). *For the original marked collision targets with $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the local microscopic estimate $\text{TV}(Q_{d,\sigma}, Q_*) < 1/300$ implies*

$$N_{2/5}^{\text{phys}}(W, Q_{d,\sigma}) = 2 \quad (W \geq 12).$$

There are four preparations in the complete audit: two candidate training, one candidate validation, and one target validation.

Proof. Apply Theorem 10.1. The lower-end margin is the exact rational number

$$\frac{2}{5} - u_1 - \frac{1}{300} = \frac{78269}{30720000} > 0.$$

For the other endpoint, $\sqrt{3} < 7/4$ implies $m_2 > 27/64$ and $27/64 - 1/300 > 2/5$. The physical total-variation estimate, including the actual mark and positive sensor noise, is proved in Theorem 9.3; the current deduction does not replace those hypotheses by a zero-noise limit. $\square$

Remark 10.6 (The part of the frontier determined here). The deterministic theorem evaluates the minimum preparation coordinate for $W \geq 12$. Theorem 11.1 and Corollary 11.2 below additionally determine the ideal maximum score with two preparations on this width range. The least peak width at $N = 2$ and the full boundary for $3 \leq W \leq 11$ remain separate questions. The earlier exact decision-width result $\kappa_{2/5}^{\text{rows}} = 3$ is unchanged: it allows longer training, whereas this shortest audit uses five decision states. The older points (399, 40602), (320000, 1604) and

$(12800 \cdot 2^{400}, 12)$ remain valid constructions with decision width three. They are dominated by $(2, 12)$ in the projection onto training preparations and peak width, but not necessarily in a resource order that also constrains the decision alphabet separately.

Corollary 10.7 (The enlarged physical preparation segment). *With the full architecture of Definition 9.1,*

$$N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_{\text{ex}} + \beta \leq c < m_2 - \beta.$$

Here $u_{\text{ex}} = (85 - 7\sqrt{73})/64$ is the exact one-preparation all-on value, rather than the older rational upper bound.

Proof. Restrict the adversary to the common-garbling subfamily in Definition 9.1. Theorem 5.2 and Corollary 8.2 bound every at-most-one-preparation auditor, even with unlimited memory, by $u_{\text{ex}} + \beta$. The same deterministic two-preparation machine in Lemmas 10.3 and 10.4 has score at least $m_2 - \beta$ against the full private candidate family. These inequalities prove the assertion. The four-preparation total, peak twelve and positive-noise specialization are unchanged. $\square$

11. THE EXACT TWO-PREPARATION MARKED SADDLE

The one-preparation saddle has an exact successor. A new zero coefficient of the least-favorable predictive measure makes it necessary to randomize at the two extreme training counts. This randomization changes the value, but does not enlarge the alphabet of frozen validation events.

Let $b(p, q)$ and Q_* be as in Section 10. Let r be the unique root in $(1/3, 7/20)$ of

$$(37) \quad 2r^3 + 25r^2 - 3 = 0,$$

and define

$$(38) \quad \tau = \frac{6(-r^2 + 4r - 1)}{r(3r + 25)}, \quad v_2 = \frac{-r^3 + 6r^2 - 3r + 14}{32}, \quad p_{\pm} = \frac{1 \pm \sqrt{r}}{2}.$$

The polynomial in (37) is strictly increasing on the positive half-line and has opposite signs at the indicated endpoints. These observations prove the asserted existence, uniqueness and isolating interval. That interval also gives $0 < \tau < 1$.

Theorem 11.1 (Exact two-preparation value). *For the full marked-word response class of (8),*

$$U_2(b, Q_*) = v_2.$$

A least-favorable prior gives mass $1/2$ to each of (p_-, p_-) and (p_+, p_+) . An optimal response discards the two training marks, retains the total number S of ones in the four training report bits, and uses

$$(39) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & \tau & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & \tau \end{array}$$

with zero response at off-diagonal validation atoms. Numerically, $r = 0.3417694155\dots$, $\tau = 0.1688236447\dots$ and $v_2 = 0.4261127787\dots$. The exact value is determined by (37)–(38), not by these decimals.

Proof. Write $a = (1 - p)(1 - q)$, $c = pq$, $u = (p + q - 1)^2$ and $v = (p - q)^2$. Both u and v belong to $[0, 1]$. The score of the deterministic five-event selector is the polynomial $H(u, v)$ in (36). Only counts zero and four are changed by (39). Their probabilities are a^2 and c^2 . Consequently the new score is

$$G(u, v) = H(u, v) + \tau J(u, v), \quad J = a^2(5/16 - a/2) + c^2(5/16 - c/2).$$

Expansion gives the polynomial identity

$$(40) \quad \begin{aligned} J(u, v) = & -\frac{u^3}{64} + \frac{3u^2v}{64} - \frac{25u^2}{128} - \frac{3uv^2}{64} + \frac{13uv}{64} \\ & + \frac{v^3}{64} - \frac{v^2}{128} - \frac{v}{32} + \frac{3}{128}. \end{aligned}$$

Thus

$$H(u, 0) = \frac{-u^3 + 6u^2 - 3u + 14}{32}, \quad J(u, 0) = \frac{3 - 25u^2 - 2u^3}{128}.$$

Equation (37) says $J(r, 0) = 0$, and the choice of τ says $\partial_u G(r, 0) = 0$. Direct factorization, using these two identities, yields

$$(41) \quad G(u, 0) - v_2 = \frac{(u - r)^2 \{75 + 12r + 6u - 37r^2 - 74ru\}}{64r(3r + 25)}.$$

The expression in braces is strictly positive for $0 \leq u \leq 1$: it is at least $75 - 37(7/20)^2 - 74(7/20) > 0$. Moreover,

$$\partial_v H = \frac{3(u - v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}, \quad \partial_v J = \frac{3(u - v)^2 + 13u - v - 2}{64} \geq -\frac{3}{64}.$$

Because $0 < \tau < 1$, $\partial_v G \geq 11/64$ on the unit square. It follows that $G(u, v) \geq G(u, 0) \geq v_2$, with equality precisely when $v = 0$ and $u = r$. Both equality points are feasible and are the two parameters in the asserted prior. This proves the global lower bound.

It remains to prove an upper bound against every response, not only those depending on S . Under the asserted prior the four training report bits are independent Bernoulli variables with common parameter p_- or p_+ . Conditional on $S = s$, their arrangements and the two independent training marks have parameter-independent distributions. The signed predictive coefficients can therefore be grouped by S without restricting any response coordinate. For the four displayed validation atoms put

$$C_{s,j} = \frac{1}{2} \sum_{p \in \{p_-, p_+\}} \binom{4}{s} p^s (1 - p)^{4-s} \{Q_{*,j} - b_j(p, p)\}.$$

The first three rows of $256C$ are

$$\begin{aligned} s = 0 : & \quad (-(2r^3 + 25r^2 - 3), -(2r^3 + 27r^2 + 12r - 1), \\ & \quad -(2r^3 - 5r^2 - 20r - 1), -(2r^3 - 7r^2 - 32r - 3)), \\ s = 1 : & \quad 4(r - 1)(2r^2 + 7r - 3, 2r^2 + 9r - 1, \\ & \quad 2r^2 - 7r - 1, 2r^2 - 9r - 3), \\ s = 2 : & \quad -6(r - 1)^2(2r - 3, 2r - 1, 2r - 1, 2r - 3). \end{aligned}$$

Rows three and four are the reversals of rows one and zero, respectively. Their signs, using $1/3 < r < 7/20$, are

$$(0, -, +, +), \quad (+, -, +, +), \quad (+, +, +, +), \quad (+, +, -, +), \quad (+, +, -, 0).$$

Every off-diagonal validation coefficient is nonpositive because its target mass is zero. Hence (39) maximizes every signed coefficient: it is one at positive coefficients, zero at negative ones, and arbitrary at the two zeros. The lower-bound equality at both prior points shows that this Bayes maximum is exactly v_2 . The posterior-predictive dual, Theorem 5.1, now gives the matching upper bound for the full response class. $\square$

Corollary 11.2 (Exact physical score on the peak-twelve segment). *Use the atomic stochastic-row convention and the original marked interface of Definition 9.1. At two candidate training preparations, every peak budget $W \geq 12$ has exact ideal worst-case score v_2 . For a target Q with $\text{TV}(Q, Q_*) \leq \beta$, the corresponding value lies in $[v_2 - \beta, v_2 + \beta]$. In particular, with $u_{\text{ex}} = (85 - 7\sqrt{73})/64$,*

$$(42) \quad N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad u_{\text{ex}} + \beta \leq c < v_2 - \beta.$$

Proof. Let $h_0, \dots, h_4$ be the five deterministic events in (35). At count zero select h_1 with probability τ and h_0 otherwise; at count four select h_3 with probability τ and h_4 otherwise. At the other counts retain the indicated event. Discard the count and retain the selected event for both validations. This realizes (39); the event alphabet remains five. The full charged profile, now including the separate event-selection update, is

$$(1, 2, 3, 3, 4, 5, 5, 5, 10, 12, 10, 10, 7, 3).$$

The common residual-function compiler in Lemma 10.4 applies unchanged. In particular, no persistent random selector is outside the register. The all-on law for every private candidate is $b(p, q)$, so Theorem 11.1 proves the constructive lower bound. For the converse restrict to the admissible common-garbling product subfamily and apply Theorem 5.2 followed by Theorem 11.1. This bounds all architectures, including those using feedback, stopping and more memory. Target perturbation changes each frozen-event expectation by at most β . Finally the exact one-preparation theorem excludes a strict score above c at $N \leq 1$, whereas the displayed machine achieves it at $N = 2$ under (42). $\square$

The root r is larger than $2 - \sqrt{3}$. The old diagonal polynomial $H(u, 0)$ is strictly increasing between these points, whence $v_2 > m_2$. The deterministic theorem, including its zero-random-bit implementation, remains valid; the new exact score uses an additional stochastic row.

Proposition 11.3 (Finite precision for the new row). *Replacing τ by a dyadic $\tau_b = k/2^b \in [0, 1]$ with $|\tau_b - \tau| \leq 2^{-b}$ loses at most $(5/16)2^{-b}$ in uniform score. A clocked fair-bit implementation uses b extra random bits and at most $\max\{12, 6(b+1) + 5\}$ persistent states, including their phase, in addition to the unchanged preparation count.*

Proof. At each affected training word the response changes at a single atom; $|5/16 - a/2|$ and $|5/16 - c/2|$ are at most $5/16$. The two affected word probabilities sum to at most one. This gives the score estimate. To implement the dyadic coin, compare b fresh fair bits lexicographically with the fixed b -bit integer k . Retain the origin (count zero or four), one of the three comparison flags, and a phase in $\{0, \dots, b\}$. Once the comparison ends retain its chosen one of five events. Unaffected counts can wait in their event states. These states total at most $6(b+1) + 5$; the other cuts have peak twelve. The cases $k = 0, 2^b$ are deterministic. Exact atomic randomization and a finite fair-bit implementation are therefore distinct resource claims. $\square$

12. A CONTINUOUS TWO-PREPARATION MARKED FAMILY

The preceding saddle persists on an explicit interval of target laws, with exact rather than perturbative values. Use the target Q_γ from Section 8 and write $U_2(\gamma) = U_2(b, Q_\gamma)$.

Theorem 12.1 (A two-preparation saddle family). *For $6/25 \leq \gamma \leq 13/50$, let r_γ be the unique root in $(1/3, 7/20)$ of*

$$(43) \quad f_\gamma(r) = r^3 + (13 - 2\gamma)r^2 + (3 - 12\gamma)r - (1 + 2\gamma) = 0.$$

Set

$$D_\gamma(r) = 3r^2 + (26 - 4\gamma)r + 3 - 12\gamma, \quad t_\gamma = -\frac{2\{3r_\gamma^2 + (8\gamma - 14)r_\gamma + 3\}}{D_\gamma(r_\gamma)}.$$

Then $0 < t_\gamma < 1$, and

$$(44) \quad U_2(\gamma) = \frac{-r_\gamma^3 + (7 - 4\gamma)r_\gamma^2 - 3r_\gamma + 13 + 4\gamma}{32}.$$

The response (39), with $\tau = t_\gamma$, is optimal. A least-favorable prior gives equal masses to the two diagonal parameters $p = q = (1 \pm \sqrt{r_\gamma})/2$. The response has the same five-event, peak-twelve atomic realization. At $\gamma = 1/4$ these formulas reduce to Theorem 11.1.

Proof. For brevity write $r = r_\gamma$, $D = D_\gamma(r)$ and $t = t_\gamma$. The derivative D_γ is positive on the rectangle $\gamma \in [6/25, 13/50]$, $r \in [1/3, 7/20]$. For example its lower bound is $1/3 + (26 - 52/50)/3 - 3/25 > 8$. The polynomial f_γ decreases with γ at a fixed positive argument. Direct rational evaluations give $f_{6/25}(1/3) < 0$ and $f_{13/50}(7/20) > 0$. Monotonicity therefore gives the root and its uniqueness in this interval. The numerator of t is $-16\gamma r - 6r^2 + 28r - 6$. It is bounded below by $28/3 - 6 - 16(13/50)(7/20) - 6(7/20)^2 > 0$, and above by $28(7/20) - 6 < 4 < D$. Thus $0 < t < 1$.

Let H, J be the polynomials from (36) and (40). Changing the target masses linearly in γ gives

$$\begin{aligned} H_\gamma(u, v) &= H(u, v) + (\gamma - 1/4) \frac{1 - (u - v)^2}{8}, \\ J_\gamma(u, v) &= J(u, v) + (\gamma - 1/4) \frac{(u - v)^2 + 6u - 2v + 1}{32}. \end{aligned}$$

The response's full-parameter score is $G_\gamma = H_\gamma + tJ_\gamma$. One has $J_\gamma(r, 0) = -f_\gamma(r)/64 = 0$ and $\partial_u G_\gamma(r, 0) = 0$. Algebraic division gives

$$(45) \quad G_\gamma(u, 0) - H_\gamma(r, 0) = \frac{(u - r)^2 A_\gamma(r, u)}{16D},$$

where

$$\begin{aligned} A_\gamma(r, u) &= 24\gamma^2 + (3\gamma - 10)r^2 + (6\gamma - 20)ru \\ &\quad + 12\gamma r + 6\gamma u - 51\gamma + 30. \end{aligned}$$

For $0 \leq u \leq 1$ this is at least $30 - 10(7/20)^2 - 20(7/20) - 51(13/50) > 0$. The transverse derivatives satisfy

$$\begin{aligned} 32\partial_v H_\gamma &= 3(u - v)^2 + (8\gamma - 10)u + (6 - 8\gamma)v + 15 \geq 5, \\ 64\partial_v J_\gamma &= 3(u - v)^2 + (14 - 4\gamma)u + (4\gamma - 2)v - 4\gamma - 1 \geq -3 \end{aligned}$$

on $0 \leq u, v \leq 1$. Consequently $\partial_v G_\gamma \geq 7/64$. Equation (45) proves a lower bound $H_\gamma(r, 0)$, with equality precisely at the proposed prior points. This quantity is the right side of (44).

For the upper bound group the full training words by S , as in the preceding theorem. Put $L(r) = r^2 + 6r + 1$. Using $f_\gamma(r) = 0$, the first three rows of $256C$ are

$$\begin{aligned} 0 : & (0, -8\gamma L(r), 32r(r+1) - 8\gamma L(r), 32r(r+1)), \\ 1 : & 8(r-1)(r^2 + 4r - 1 - 2\gamma(r+1), \\ & \quad r^2 + 4r - 1 + 2\gamma(r+1), \\ & \quad r^2 - 4r - 1 + 2\gamma(r+1), r^2 - 4r - 1 - 2\gamma(r+1)), \\ 2 : & -12(r-1)^2(r-1-2\gamma, r-1+2\gamma, r-1+2\gamma, r-1-2\gamma). \end{aligned}$$

The other rows are their reversals. The sign pattern is again

$$(0, -, +, +), \quad (+, -, +, +), \quad (+, +, +, +), \quad (+, +, -, +), \quad (+, +, -, 0).$$

Here are explicit interval checks for the potentially small margins. In row zero the third coordinate is bounded below by $32(1/3)(4/3) - 8(13/50)((7/20)^2 + 6(7/20) + 1) > 0$. The first parenthesis of row one is at most $(7/20)^2 + 4(7/20) - 1 - 2(6/25)(4/3) < 0$; the third is at most $(7/20)^2 - 4/3 - 1 + 2(13/50)(27/20) < 0$. The second parenthesis is positive already from $r > 1/3$ and $\gamma > 0$, and the fourth is negative. Finally $r - 1 + 2\gamma \leq -13/100 < 0$. All off-diagonal validation coefficients are nonpositive. Thus the stated response is a Bayes best response for this prior, including its two zero coefficients. Its score at both supporting parameters equals the lower bound. The full posterior-predictive dual proves (44). $\square$

This interval is a uniform certificate region for the displayed saddle, not a claim that the same masks remain optimal for every mark bias. The theorem supplies a continuous family at genuinely growing training history $N = 2$; it does not infer a recurrence for every N from the one-preparation formulas.

13. AN EXACT PREPARATION-DECISION-MEMORY LAW

A second solvable family separates a genuine joint resource law from an implementation bound. Here the memory coordinate is the number of states at the training-validation cut. It is not identified with the bit-level peak width of the collision audit.

Fix $m = 2^d$, $0 < \alpha < 1$, and an unknown parameter $\theta \in \{1, \dots, m\}$. A fresh training preparation reports θ with probability α and an erasure $\perp$ otherwise. Conditional on θ , the preparations are independent. Candidate validation has law δ_θ and target validation is uniform on $\{1, \dots, m\}$. At the decision cut the auditor retains at most $K = 2^k$ states, $0 \leq k \leq d$. Its subsequent response may depend only on that state, current validation observations and fresh independent randomness. In particular, a calibration table, remembered seed or stopping time outside the retained state is not available. Upstream and downstream memory are otherwise unrestricted for the converse. Let $V_N^{\text{er}}(K)$ be the best worst-case expected signed validation score with at most N training preparations.

Theorem 13.1 (Exact joint law). *For every $N \geq 0$ and $K = 2^k \leq m$,*

$$(46) \quad V_N^{\text{er}}(K) = (1 - (1 - \alpha)^N) \left(1 - \frac{1}{K}\right).$$

The converse covers stochastic encoders and arbitrary legal stopping. The value is attained by a clocked machine with K training states. With atomic preparation and validation outcomes, the complete audit has peak at most $\max\{2K, 3\}$ and uses at most $N + 2$ preparations.

Proof. Write $q = (1 - \alpha)^N$ and pre-generate N potential training reports. A stopped auditor ignores the unused reports; this does not disclose extra information to it. On the event that all

N reports are erasures, its retained-state law $r = (r_j)_{j=1}^K$ is independent of θ . After padding any smaller register with unused states, its channel to the decision register has the form

$$(47) \quad C(j \mid \theta) = qr_j + (1 - q)S(j \mid \theta)$$

for a stochastic channel S . This identity remains valid with randomized stopping and randomized state updates.

Use the uniform prior on parameters as an upper certificate. The part qr_j has zero expected score for every response: averaging δ_θ under this prior gives the target itself. For the other part, optimizing over all responses gives a convex function of S , a sum of positive parts of linear functions. Every stochastic channel is a convex combination of deterministic channels $j = g(\theta)$; therefore its value is at most the largest deterministic-channel value. This convexification is used only to bound a channel. It does not presuppose that a mixture of width- K machines is realizable at width K .

If the fibers of g have sizes $n_1, \dots, n_K$, a nonempty fiber has posterior uniform on that fiber, so its total variation from the uniform target is $1 - n_j/m$. The prior-averaged optimal score is

$$\sum_j \frac{n_j}{m} \left(1 - \frac{n_j}{m}\right) = 1 - \sum_j (n_j/m)^2 \leq 1 - 1/K.$$

Empty fibers contribute zero. Cauchy–Schwarz proves the inequality. This is an upper bound on the average score and hence on its minimum over fixed parameters. Multiplying by $1 - q$ proves the upper bound in (46).

For attainment, partition the parameter alphabet into K equal groups and let $g(\theta)$ denote its group. Initialize the register uniformly on the K group labels. On an erasure leave it unchanged; on a revealed parameter overwrite it by $g(\theta)$. No extra “revelation occurred” flag is needed. At the decision cut its law is $q \text{Unif}[K] + (1 - q)\delta_{g(\theta)}$. For register j , freeze the event which is the complement of group j . On a revelation its score is $1 - 1/K$. Under a uniformly random initial label, the mean score is zero for each fixed θ : one of the K labels has score $1 - 1/K$ and the others have score $-1/K$. This proves the exact worst-case value.

The k initialization bits can be accumulated in the K -state register; all earlier prefix counts are at most K . Thus no persistent seed lies outside it. Candidate validation retains the group label and its event indicator, using at most $2K$ states. Target validation produces one of the three values $-1, 0, 1$. This proves the atomic-outcome peak bound. The controller uses a declared external schedule; an autonomous stopping implementation must additionally retain its phase. A phase-tagged implementation is finite, but its cost is not included in K . $\square$

Corollary 13.2 (Thresholds and matching resource scales). *For $K \geq 2$ and $0 \leq c < 1 - 1/K$, the least integer number of training preparations giving a score strictly greater than c is*

$$(48) \quad N_c^{\text{er}}(K) = \left\lceil \frac{\log(1 - c/(1 - 1/K))}{\log(1 - \alpha)} \right\rceil + 1.$$

If $c \geq 1 - 1/K$, no finite N suffices. For $0 < \varepsilon < 1$, achieving $V_N^{\text{er}}(K) \geq 1 - \varepsilon$ requires

$$K \geq \varepsilon^{-1}, \quad N \geq \frac{\log(1/\varepsilon)}{-\log(1 - \alpha)}.$$

Conversely, whenever an allowed $K \leq m$ satisfies $K \geq 2/\varepsilon$, the choice $N \geq \log(2/\varepsilon)/[-\log(1 - \alpha)]$ suffices.

Proof. The strict threshold is equivalent to $(1 - \alpha)^N < 1 - c/(1 - 1/K)$. Since the denominator logarithm is negative, the least integer satisfying this strict inequality is (48), including equality

cases of the real quotient. Also

$$1 - V_N^{\text{er}}(K) = q + (1 - q)/K.$$

This is at least q and at least $1/K$, and is at most $q + 1/K$. These three inequalities prove the necessary and sufficient bounds. $\square$

The exact law concerns an entire (N, K, α, m) family, not a single finite diagnostic. Its mechanism is a parameter-independent erasure component followed by the convex geometry of the decision channel. It demonstrates how a preparation obstruction and a memory obstruction can be matched in one theorem. It does not identify the least physical peak width at two collision preparations. Encoding the m -symbol reports bit by bit also introduces cuts not present in this atomic model; the physical bit-level accounting is given separately.

14. INTEGER MEMORY AND THE FAILURE OF A NAIVE PARTITION FORMULA

The divisibility assumptions in Theorem 13.1 can be removed, but a balanced-partition Bayes bound must not be mistaken for a minimax equality when a persistent random partition is not free. This distinction leads to another exact finite-resource transition.

Keep the revelation interface, but allow any integers $m \geq 2$ and $1 \leq K \leq m$. Put $\rho = 1 - (1 - \alpha)^N$, and denote the genuine K -label minimax value by $V_{N,m}^{\text{er}}(K)$. Write $m = Kq + s$, $0 \leq s < K$, and set

$$(49) \quad B_{m,K} = 1 - \frac{(K - s)q^2 + s(q + 1)^2}{m^2}.$$

Here and below all-erasure initialization randomness is a state initialization, not an unrecorded seed used again at validation.

Proposition 14.1 (Arbitrary integer alphabet bounds and low-reveal equality). *The optimal uniform-prior Bayes score is $\rho B_{m,K}$. The minimax value satisfies*

$$(50) \quad V_{N,m}^{\text{er}}(K) \leq \min \left\{ \rho B_{m,K}, 1 - \frac{\lceil m/K \rceil}{m} \right\}.$$

If $\rho \leq m/(m + s)$, equality with the first term holds. At perfect revelation $\rho = 1$ the exact value is the second term. In particular, when $s > 0$ the uniform-prior Bayes optimum at perfect revelation is strictly larger than the minimax value.

Proof. The all-erasure decomposition (47) holds without any divisibility condition. Its parameter-independent part has zero uniform-prior score. Optimizing the other part over responses gives a convex function of its stochastic channel. A deterministic channel with fiber sizes n_j has Bayes score $1 - \sum_j (n_j/m)^2$. If two nonempty fiber sizes differ by at least two, moving one index from the larger to the smaller strictly decreases the sum of squares. Allowing empty fibers and repeating gives sizes q and $q + 1$, proving the Bayes bound. It is attained by this partition and its complementary events, with any parameter-independent initialization. This proves the first upper bound as well as the asserted Bayes optimum.

For perfect revelation, fix any K response vectors. For each known parameter choose a response maximizing its score; this deterministic assignment cannot decrease any parameter's payoff, even when the original encoder was randomized. Some assigned fiber has size at least $\lceil m/K \rceil$. Averaging its parameters, the score of their common response is at most the total variation between the uniform target and the uniform law on that fiber, namely $1 - \lceil m/K \rceil/m$. The minimum is no larger. Balanced fibers with their complementary events attain the resulting upper bound. A perfectly informed encoder can simulate any less informative encoder, so this is an upper bound at every ρ .

For the low-reveal equality let balanced fibers have weights $w_j = n_j/m$. When $\rho < 1$, initialize their label register with probabilities

$$(51) \quad r_j = \frac{\rho w_j + (1 - 2\rho)/K}{1 - \rho}.$$

These sum to one, and are nonnegative precisely throughout the stated range: the smallest numerator is $\rho q/m + (1 - 2\rho)/K$. On a revealed parameter overwrite the label by its fiber; on erasure do nothing. No revelation flag is retained. For a parameter in fiber i , the complementary-event score is

$$\begin{aligned} g_i &= \rho(1 - w_i) + (1 - \rho) \left(r_i - \sum_j r_j w_j \right) \\ &= \rho \left(1 - \sum_j w_j^2 \right) = \rho B_{m,K}. \end{aligned}$$

This matches the upper bound uniformly in the fixed parameter. If $\rho = 1$ and $s = 0$, the deterministic revealed label already attains that value, so the singular denominator in (51) is not used. Finally, with unequal balanced fibers the weighted average of their sizes is strictly smaller than the largest size; equivalently $B_{m,K} > 1 - \lceil m/K \rceil / m$. This proves the last assertion. $\square$

Theorem 14.2 (The complete two-label revelation law). *For every $m \geq 2$, every $N \geq 0$, and $0 < \alpha < 1$, set $w = \lfloor m/2 \rfloor / m$ and $b = \lceil m/2 \rceil / m$. Then*

$$(52) \quad V_{N,m}^{\text{er}}(2) = \min\{2wb\rho, w\}, \quad \rho = 1 - (1 - \alpha)^N.$$

The same formula holds at the limiting endpoints $\rho = 0, 1$. For odd m it changes regime at $\rho = m/(m + 1)$; for even m it reduces to $\rho/2$. The attaining machine uses two training labels, a frozen two-event alphabet, and atomic peak at most four.

Proof. Proposition 14.1 gives the upper bound and its attainment for $\rho \leq 1/(2b)$. Above this threshold, partition into fibers of weights w, b and initialize deterministically in the larger fiber's label. Continue to overwrite on a reveal and leave the state unchanged on an erasure. A parameter in the larger fiber has score w ; a parameter in the smaller fiber has score $(2\rho - 1)b$. The latter is at least w exactly when $\rho \geq 1/(2b)$. This attains the second upper bound. For even m the threshold is one, so the low-reveal construction suffices. The endpoint $\rho = 0$ may instead freeze a constant event. Candidate validation retains the selected label and its indicator, with at most four states; target validation produces the difference. Initialization is atomic randomization when its probabilities are not dyadic. $\square$

For example, $m = 3, K = 2$ gives $V_{N,3}^{\text{er}}(2) = \min\{4\rho/9, 1/3\}$, rather than $4\rho/9$ for all ρ . The balanced-integer expression is therefore not the unqualified replacement of the divisible minimax formula. For $K \geq 3$ outside the low-reveal range, (50) remains a bound rather than an asserted equality. The compatible dynamic formulation of Section 7 retains the exact optimization in that regime. These results strengthen the revelation example without identifying its decision-cut coordinate with the collision model's bit-level peak.

15. QUANTITATIVE CERTIFICATES FOR A FINITE COMPATIBLE ARCHITECTURE

We return to an arbitrary finite architecture, now with a finite parameter set. It can have hidden states, controlled experiments, more than two hypotheses and registers of any size. Enumerate its distinct stochastic rows by $r = 1, \dots, R$ and write $q_r \in \Delta_{s_r}$. A reused autonomous row appears only once in this product. Assume each execution uses at most H controller rows,

and uses row r at most d_r times. Delete unused rows and put $S = \max_r s_r$. Fixed environmental kernels are not controller rows.

For an integer $m \geq 1$, intersect Δ_{s_r} with the coordinate boxes of side $1/m$. The resulting nonempty closed polytopes cover the simplex, with overlaps permitted at their boundaries. Write C for a product of such row cells and choose a vertex q_r^C in each factor. A row-cell vertex has at most one coordinate not fixed at a box boundary; consequently it has at most $v_r = s_r 2^{s_r-1}$ vertices, all on the $1/m$ grid. There are at most $m^{\sum_r s_r}$ product cells.

Write the vertices of row cell C_r as $v_{r,j}$ and parametrize $q_r = \sum_j \lambda_{r,j} v_{r,j}$ with λ_r in a simplex. A reward polynomial has degree at most d_r in block λ_r . Use the homogeneous simplex Bernstein basis of degree d_r in that block, obtaining

$$(53) \quad g_\theta(q(\lambda)) = \sum_\alpha b_{\theta,C,\alpha} B_\alpha(\lambda), \quad B_\alpha \geq 0, \quad \sum_\alpha B_\alpha = 1.$$

Degree elevation includes the factors $\sum_j \lambda_{r,j} = 1$. No independence between different visits of a shared row is imposed on its unknown row vector.

Lemma 15.1 (Causal interpretation of the coefficients). *Every coefficient in (53) satisfies*

$$|b_{\theta,C,\alpha} - g_\theta(q^C)| \leq HS/m.$$

Proof. For each row r , independently prepare an ordered list of d_r vertex labels, sampled independently with law λ_r . At its k th visit, use the stochastic row given by the k th vertex label in that list. Before conditioning, the next label is independent of the previous execution, so this experiment has exactly the row q_r at every visit and expected reward $g_\theta(q(\lambda))$. Unused labels may be sampled after termination without altering the reward.

Condition on the counts of every vertex label in every list. The probability of those counts is $B_\alpha(\lambda)$. Conditional on the counts, each list is a uniformly ordered multiset. Its expected reward is independent of λ and is therefore the corresponding Bernstein coefficient. This also proves (53) directly.

Each row used by this conditioned experiment is a vertex of C_r or a mixture of its remaining vertices. Every two vectors in C_r have coordinate differences at most $1/m$, hence total variation at most $s_r/(2m) \leq S/(2m)$. Couple the conditioned experiment to the one using q^C , using the same environmental randomness while their executions agree. At each actual controller update the probability of a first row-level disagreement is at most $S/(2m)$. There are at most H such updates on the agreeing path. The terminal laws consequently differ in total variation by at most $HS/(2m)$. A reward in $[-1, 1]$ has expectation difference at most twice that quantity. This proves the estimate. The conditioned urn is used to analyze a coefficient; it is not declared to be an admissible free-memory controller. $\square$

Theorem 15.2 (Explicit subdivision certificate). *Define*

$$\ell_C = \min_\theta g_\theta(q^C), \quad u_C = \min_\theta \max_\alpha b_{\theta,C,\alpha}, \quad L_m = \max_C \ell_C, \quad U_m = \max_C u_C.$$

For the exact all-stochastic value of the specified architecture,

$$(54) \quad L_m \leq V(A) \leq U_m \leq L_m + HS/m.$$

On each cell, a witnessing parameter θ_C supplies the nonnegative coefficient certificate

$$u_C - g_{\theta_C}(q(\lambda)) = \sum_\alpha (u_C - b_{\theta_C,C,\alpha}) B_\alpha(\lambda).$$

Its degree in row block r is d_r , and its total degree is at most $\sum_r d_r$. At most $\prod_r \binom{d_r + v_r - 1}{v_r - 1}$ coefficients per parameter and cell are needed. If the primitive data are rational, all coefficients

and bounds can be checked by rational arithmetic. Taking $m \geq HS/\varepsilon$ gives a certified interval of length at most ε .

Proof. The point q^C is feasible, so ℓ_C is a lower bound on the value restricted to C . For any $q \in C$, choose a convex representation of its rows. Equation (53) implies

$$\min_{\theta} g_{\theta}(q) \leq \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha} = u_C.$$

Choose a parameter attaining the minimum defining ℓ_C and apply Lemma 15.1; this gives $u_C \leq \ell_C + HS/m$. Taking the maximum over the finite cover proves (54). Choose θ_C attaining the definition of u_C ; all displayed coefficients of its certificate are nonnegative. The degree and count statements follow from the product Bernstein basis. Cell vertices are rational grid points, and a finite execution sums finite products, proving the rational statement. $\square$

Remark 15.3. The number of cells and coefficients can be enormous. Neither polynomial running time nor a uniform degree bound for the original sum-of-squares hierarchy follows. Refinement changes the domain cover, not merely a single global SOS degree. For a finite collection of architectures obeying a specified budget, maxima of these intervals give outer intervals; an unbounded architecture class needs a separate uniform argument such as Theorem 5.2. For compact parameter families, an effective net and a verified modulus for the primitive kernels are additional inputs, not consequences of compactness alone.

The resource estimate also makes row precision explicit. With $m = 2^b$, the lower witnesses have dyadic rows and lose at most $HS2^{-b}$ in value. Thus increasing row precision provides a quantified approximation for a fixed architecture, while its random-bit implementation still has to be charged. The physical theorem avoids this cost altogether because its rows are deterministic.

16. AUTONOMOUS NOISY TESTING WITH A PRECISION AND TIME LEDGER

This section supplies a substantive autonomous comparison rather than merely identifying shared rows. The stationary optimum is classical [6]; the proof and explicit finite-time implementation are recorded to make the resource conventions unambiguous. They are not used as novelty evidence for the physical theorem.

Let reports be independent, with law $\text{Ber}(p)$ under hypothesis $-$ and $\text{Ber}(q)$ under hypothesis $+$, where $0 < p < 1/2$, $q = 1 - p$, and $r = q/p$. A W -state autonomous machine has two time-independent stochastic transition matrices T_0, T_1 . Assume their induced chains $P_- = qT_0 + pT_1$ and $P_+ = pT_0 + qT_1$ are irreducible. A final decision uses only the state, with optional fresh terminal randomization.

Lemma 16.1 (Likelihood-range bound). *If equivalent probability laws ν_+, ν_- on a finite set satisfy*

$$\frac{\max_i \nu_+(i)/\nu_-(i)}{\min_i \nu_+(i)/\nu_-(i)} \leq R,$$

then $\text{TV}(\nu_+, \nu_-) \leq (\sqrt{R} - 1)/(\sqrt{R} + 1)$.

Proof. Write $L = d\nu_+/d\nu_-$, $a = \min L$ and $b = \max L$. Since $\mathbb{E}_{\nu_-} L = 1$, one has $a \leq 1 \leq b$. The chord bound for $(L - 1)_+$ gives

$$\text{TV}(\nu_+, \nu_-) = \mathbb{E}_{\nu_-} (L - 1)_+ \leq \frac{(1 - a)(b - 1)}{b - a}.$$

For fixed b/a , maximizing the last expression gives $a = 1/\sqrt{b/a}$; substitution yields $(\sqrt{b/a} - 1)/(\sqrt{b/a} + 1)$. It is increasing in b/a , proving the assertion. The identical-law case follows directly. $\square$

Theorem 16.2 (Autonomous stationary bound and an explicit approximation). *Put $D = W - 1$, $W \geq 2$. In the irreducible autonomous class the infimum stationary minimax error is*

$$(55) \quad e_W = \frac{1}{1 + r^D}.$$

For every $\epsilon = 2^{-b}$, $b \geq 0$, there is a machine whose rows have denominator dividing 2^{b+1} , and whose minimax error at report time n from any initial state is at most

$$(56) \quad e_W + \epsilon(W - 2) + \exp \left\{ - \left(\frac{p\epsilon}{2} \right)^D \left\lfloor \frac{n}{D} \right\rfloor \right\}.$$

The stationary lower bound is not a finite-horizon lower bound.

Proof. Every positive induced transition has its likelihood ratio in $[r^{-1}, r]$, and the supports of the two induced matrices are identical. The Markov chain tree formula writes each stationary weight as a sum of products over directed spanning trees rooted at that state. Each product has D transitions; its ratio lies in $[r^{-D}, r^D]$. The ratio of the two unnormalized stationary weights lies in the same interval. Normalizing multiplies all state likelihood ratios by a common number, so their maximum-to-minimum ratio is at most r^{2D} . Lemma 16.1 therefore bounds stationary total variation by $(r^D - 1)/(r^D + 1)$. The equal-prior Bayes error is at least $1/(1 + r^D)$, and a minimax error cannot be smaller than that Bayes error. This is the equal-prior binary specialization of the Hellman–Cover bound.

For the upper approximation, take states $i = 0, \dots, D$, weights $c_0 = c_D = 1$, and $c_i = \epsilon$ at interior states. On report 1, move from i to $i + 1$ with probability $a_i = \frac{1}{2} \min(1, c_{i+1}/c_i)$, if that neighbor exists; otherwise stay. On report 0, move to $i - 1$ with probability $b_i = \frac{1}{2} \min(1, c_{i-1}/c_i)$, if it exists; otherwise stay. All other probability is assigned to staying. Detailed balance gives

$$\pi_+(i) = \frac{c_i r^i}{\sum_j c_j r^j}, \quad \pi_-(i) = \frac{c_i r^{-i}}{\sum_j c_j r^{-j}}.$$

Reflection $i \mapsto D - i$ exchanges these laws. Decide $+$ for $i > D/2$, $-$ for $i < D/2$, and use a fair terminal decision at $i = D/2$ when this state exists. The two errors agree. Conditional on being at an endpoint, the error is exactly e_W . The total interior stationary mass is at most $\epsilon(W - 2)$, since each $r^i \leq r^D$ and the denominator is at least r^D under $+$, with the reflected argument under $-$. This proves the first two terms of (56).

Every existing neighbor transition under either hypothesis has probability at least $p\epsilon/2$, and every self-loop has probability at least $1/2$. From any state, move down to zero and then stay until D steps have elapsed. This yields the common minorization $P_\pm^D(i, \{0\}) \geq \delta = (p\epsilon/2)^D$. Subtracting that common atom gives contraction in total variation by $1 - \delta$ every D steps. Comparison with the stationary law yields $\text{TV}(P_\pm^n(i, \cdot), \pi_\pm) \leq (1 - \delta)^{\lfloor n/D \rfloor} \leq e^{-\delta \lfloor n/D \rfloor}$. A bounded decision changes error by at most this amount, proving (56). Finally, letting n tend to infinity and then ϵ to zero proves that the stationary infimum equals (55). All transition probabilities belong to $\{0, 1, 1/2, \epsilon/2, 1 - 1/2, 1 - \epsilon/2\}$. $\square$

The W states in this theorem refer to an atomic stochastic update after each report. A sequential fair-bit implementation must also retain the report, the current state, a bit counter and a flag. Each nontrivial move probability is $1/2$ or $2^{-(b+1)}$; test for a string of zero bits of the required length, and pad to $b + 1$ bits if a uniform report cycle is required. At most $4W(b + 2)$ states suffice for this implementation, including its bit-phase counter and a ready phase. This conservative bound is an implementation claim, not an optimal precision–state tradeoff. The time index n in (56) is an observation time at which an external analyst reads the autonomous

machine, not a free input clock used by its transitions. The adversarial model of [10] is not covered by the independent-report proof above.

17. PHYSICAL CONSUMERS AND THE FIRST-PRINCIPLES CONNECTION

The physical preparation theorem needs no estimate of the target law as a stored input: the five events are fixed and their uniform validity uses a proved neighborhood of Q_* . This removes calibration memory from this particular consumer without asserting that target acquisition is unnecessary in an arbitrary comparison problem. The general acquired calibration theorem, with its full counter register, remains in the complete companion.

Corollary 17.1 (A raw-interface confidence construction). *Let $m = m_2 - \beta > 0$ and $0 < c < m$. Repeat the full four-preparation audit independently R times and write $\bar{D}_R$ for the mean score. Under every private candidate,*

$$\mathbb{P}(\bar{D}_R \leq c) \leq \exp\{-R(m - c)^2/2\}.$$

Under the null in which the independent candidate and target validation laws agree, with the same event chosen before both validations,

$$\mathbb{P}(\bar{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

A clocked implementation uses $4R$ preparations and at most $12(2R + 1)$ persistent states, with no calibration and no random bits.

Proof. The independent block scores lie in $[-1, 1]$ and have mean at least m under a private candidate. Hoeffding's inequality gives the first bound. Under the null, conditional on the selected event, the two independent validation indicators have the same mean. Each block therefore has mean zero, and the same inequality gives the second bound. Retain the integer sum in $\{-R, \dots, R\}$ along with the current twelve-state block. At a block boundary add its terminal score and reset its register. The deterministic external schedule is part of the clocked convention. $\square$

This is an upper construction on the raw marked interface. Its proof does not transfer a lower bound from a deliberately reduced Bernoulli interface, and no optimal confidence-memory frontier is claimed.

Proposition 17.2 (Target transport without a training multiplier). *For the two-preparation audit, changing only the target from Q to Q' changes its expected score by at most $\text{TV}(Q, Q')$. Its candidate training law, register profile and event selector remain unchanged.*

Proof. Condition on the candidate training word. The expectation difference is $(Q - Q')h_S$, whose absolute value is at most $\text{TV}(Q, Q')$. Average over that unchanged training law. $\square$

In particular, every common-preparation transport estimate verified for the original positive-noise collision family can be inserted here without multiplying it by a long training horizon. If candidate kernels also change, their training and validation defects must be added by a causal coupling; the target-only statement does not remove those terms.

The first-principles route is now concrete. The positive experimental kernels and actual reset preparations give the likelihoods in (9); these generate posterior continuation states. The validation-test quotient gives the positive parts in the statistical dual. A prior then supplies an architecture-uniform impossibility certificate. On the constructive side, the five-event decision is compiled by taking its future residual functions, which are closed under observation shifts. These are compatible causal states and determine the charged profile. Thus the preparation law, predictive continuation and finite-resource realization are used in the same upper/lower theorem.

This is not a deduction of geometry or nonlinear operator compactness from the existence of finite stochastic rows.

18. A SHARPENED PHYSICAL CONSUMER AND ITS SIMULATION BUDGET

The exact two-preparation theorem has a downstream use with explicit hypotheses, rather than a merely renamed dependency arrow. It sharpens the mean margin of a raw-interface confidence test, and the causal morphism theorem transfers that test, with its full memory and calibration costs, to another executable interface.

Theorem 18.1 (Confidence and transported resource certificate). *Let Q be a physical marked target with $\text{TV}(Q, Q_*) \leq \beta$. Use the two-preparation event selector with a dyadic coin τ_b satisfying $|\tau_b - \tau| \leq 2^{-b}$, and put*

$$a_b = v_2 - \beta - \frac{5}{16}2^{-b}, \quad W_b = \max\{12, 6(b+1) + 5\}.$$

If $0 < c < a_b$, R independent four-preparation blocks, followed by the test $\bar{D}_R > c$, have error bounds

$$(57) \quad \sup_{\text{private}} \mathbb{P}(\bar{D}_R \leq c) \leq \exp\{-R(a_b - c)^2/2\},$$

$$(58) \quad \sup_{\text{equal}} \mathbb{P}(\bar{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

They use $4R$ preparations, at most bR fresh random bits, and at most $W_b(2R+1)$ persistent states under a clocked schedule. With an exact atomic coin, replace a_b, W_b by $v_2 - \beta, 12$.

Suppose a source interface simulates this entire repeated audit under Theorem 3.4, uniformly over the private and equal-law families, with path defect Δ_R . Then the corresponding source error bounds are the right sides of (57)–(58) plus Δ_R , capped at one. At aligned cuts its state bound is $C_{\text{cal}}R_tM_tW_b(2R+1)$, and its preparation count is N_{cal} plus all source calls required to supply the $4R$ audit preparations. Calibration is acquired and retained, not supplied as a free real-valued table.

Proof. Corollary 11.2 and Proposition 11.3 give mean block score at least a_b under every fixed private parameter. Independent fresh preparations and independent coins give independent block scores in $[-1, 1]$. Hoeffding's inequality proves (57). Under the equal-law null, conditional on training and the frozen event, the two validation indicators have the same mean. Each block therefore has mean zero, proving (58) by the same inequality. Retain the integer sum in $\{-R, \dots, R\}$ alongside the current block register. Updating the sum and resetting the block is a deterministic boundary operation. This proves the raw resource bounds.

Apply Theorem 3.4 to the whole repeated experiment, including its stopping and validation fields. Its defect bound applies to each of the two terminal error events, whose indicators have range one; each probability therefore changes by at most Δ_R , not $2\Delta_R$. The same theorem gives the state product and preparation count. The source blocks need not be independent after a shared calibration stage: independence was used only for the reference audit, and path-law comparison transfers its final event probability. $\square$

The first part uses the exact saddle, the original microscopic bridge and the compiled residual machine. The second additionally needs the uniform acquired causal simulation theorem. A source which does not satisfy that theorem's stability and calibration hypotheses is not covered. Likewise this consumer supplies no reverse simulation and therefore does not transfer an optimal-width lower bound to an arbitrary source. It is an actual conditional theorem interface for

subsequent models, not an assertion that the independent geometric A2 chain or the unbounded B4/C2 operator obligations have already been discharged.

19. DEPENDENCIES, RETAINED RESULTS AND REMAINING DISTINCTIONS

The exact three-report binary noisy frontier, arbitrary-horizon binary-register minimax recursion and compatible Bayesian interval recursion remain as proved in the complete companion. Its exact odd-horizon majority realization is also retained; it is not a new result of this revision. Section 15 applies to registers of any finite width, controlled rows and any finite parameter set, but provides certified intervals rather than a closed minimax recursion for all such classes. The physical theorem supplies a matched preparation segment in a continuous two-parameter family with overlapping stochastic reports; its converse is not a finite parameter-grid check.

The stationary autonomous result in Section 16 has an explicit formula, but its stationary lower bound must not be applied to a finite-horizon clocked machine. Its finite-time upper bound is purposely conservative and exposes the slowing caused by finite transition precision. An arbitrary compact-family presentation still requires additional effective information to construct a kernel net. A subdivided positive-basis degree bound must not be called a degree bound for a global SOS hierarchy.

The physical result is a genuine consumer of the experimental and continuation root. It does not make the independent A2 geometric proofs dependent on this root. The original B4 normalized-resolvent, range, compactness and control-transfer obligations, and the broad C2 strict-dual, form, rigidity and optional-projection obligations, are not proved by our finite-state theorem. Their statements, derivations and current status remain in the retained historical pipeline. No aggregate closure is inferred from preserving their source files.

Finally, the rational certificate (34), polynomial identity (36), and residual transitions are independently reproducible. The program in the accompanying source checks their exact identities and includes deliberately invalid controls. These checks are not substitutes for the proofs above. Neither compilation, preservation of earlier pages nor a successful finite test establishes novelty or an editorial decision.

REFERENCES

- [1] R. D. Smallwood and E. J. Sondik, The optimal control of partially observable Markov processes over a finite horizon, *Operations Research* **21** (1973), 1071–1088.
- [2] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272.
- [3] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176.
- [4] J. Müller and G. Montúfar, The geometry of memoryless stochastic policy optimization in infinite-horizon POMDPs, arXiv:2110.07409 (2021; revised versions).
- [5] M. Putinar, Positive polynomials on compact semi-algebraic sets, *Indiana Univ. Math. J.* **42** (1993), 969–984.
- [6] M. E. Hellman and T. M. Cover, Learning with finite memory, *Ann. Math. Statist.* **41** (1970), 765–782.
- [7] M. E. Hellman and T. M. Cover, On memory saved by randomization, *Ann. Math. Statist.* **42** (1971), 1075–1078.
- [8] R. A. Flower and M. E. Hellman, Hypothesis testing with finite memory in finite time, *IEEE Trans. Inform. Theory* **18** (1972), 429–431.
- [9] T. M. Cover, M. A. Freedman and M. E. Hellman, Optimal finite memory learning algorithms for the finite sample problem, *Information and Control* **30** (1976), 49–85.
- [10] M. A. Managoli and V. M. Prabhakaran, Memory constrained adversarial hypothesis testing, arXiv:2605.12063v1 (May 12, 2026).
- [11] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28, doi:10.1524/strm.2002.20.14.1.
- [12] Q. Qi, General Theta Foundations I: Compatible statistical experiments and re-source frontiers, revision 23, September 24, 2026, complete mathematical companion, source commit 8a6043aca644043bd805668185eb227468f21170; packaged commit

- e926b783b643b3e7f275a18105528927b76cc97b. The 96-page article and 649-page development are reproduced without alteration after the present article in the two complete volumes.
- [13] H. Chernoff, Sequential design of experiments, *Ann. Math. Statist.* **30** (1959), 755–770.
 - [14] M. Naghshvar and T. Javidi, Active sequential hypothesis testing, *Ann. Statist.* **41** (2013), 2703–2738, doi:10.1214/13-AOS1144.
 - [15] S. Nitinawarat, G. K. Atia and V. V. Veeravalli, Controlled sensing for multihypothesis testing, *IEEE Trans. Automat. Control* **58** (2013), 2451–2464, arXiv:1205.0858v6.
 - [16] R. Zheng, R. Sim and A. Varvitsiotis, Solving imperfect-recall games via sum-of-squares optimization, arXiv:2602.21722v1 (2026).

PRESERVED PREDECESSOR

General Theta Foundations I, revision 25

The following predecessor volume is reproduced without alteration.
The complete current article precedes this divider.

Frozen predecessor commit:

818ca30bab506cfedaf6aee4c32bedff70c1e251

GENERAL THETA FOUNDATIONS I: CONTROLLED EXPERIMENT DUALITY AND EXACT MARKED MINIMAX LAWS

QIAN QI

ABSTRACT. We study finite-resource statistical comparison when training operations are chosen causally and a validation test is frozen before fresh candidate and target preparations. For finite controlled experiments with a compact parameter space, we prove an architecture-uniform posterior-predictive dual in which action choice remains inside a Bellman recursion and the adversarial prior is chosen only once. The actions need not be garblings of a common experiment. We solve the one-preparation minimax problem for a continuous family of marked product experiments, identifying both an optimal response and a two-point least-favorable prior by an explicit sum-of-squares identity. For the collision target the exact value is $(85 - 7\sqrt{73})/64$. Combined with a deterministic two-preparation audit, this gives an enlarged exact preparation segment at every peak width at least twelve. The microscopic collision and positive-noise bridge are proved within the article. A separate revelation family has the exact joint law $(1 - (1 - \alpha)^N)(1 - 1/K)$ for preparations and decision memory. Compatible finite-architecture certificates and a priced classical autonomous comparison complete the resource analysis.

CONTENTS

1. Introduction	2
1.1. Relation to earlier work	3
1.2. Organization and preserved development	3
2. Experiments, architectures and attainable resources	4
3. A statistical dual that is uniform over architectures	5
4. Controlled experiments without a dominating action	6
4.1. An informative action choice	8
5. Exact minimax laws for a marked product family	9
5.1. Complementary slackness and the event alphabet	11
6. A self-contained collision interface	11
6.1. Preparation and elastic scattering	12
6.2. Positive noise and the actual mark	13
7. The physical experiment and its sharp preparation boundary	13
7.1. An exact rational certificate against one preparation	14
7.2. A marked five-event selector	15
7.3. Causal compilation and the full register profile	16
8. An exact preparation–decision-memory law	17
9. Quantitative certificates for a finite compatible architecture	19
10. Autonomous noisy testing with a precision and time ledger	20
11. Physical consumers and the first-principles connection	22

Date: September 24, 2026. Revision 25.

2020 Mathematics Subject Classification. Primary 62B05, 62C20; Secondary 60J10, 68Q45, 93E20.

Key words and phrases. Statistical experiments, causal realization, predictive distributions, finite memory, minimax duality, physical certification.

12. Dependencies, retained results and remaining distinctions	23
References	24

1. INTRODUCTION

A statistical experiment determines not only which decisions can be made, but which observations and retained continuations are required to make them. In a controlled experiment, an action may change the information available at the next preparation. In a finite-memory experiment, retaining one continuation may preclude retaining another. A resource theorem must distinguish these two constraints rather than replace both by optimization within one chosen execution graph.

We consider an unknown parameter fixed throughout an audit. Training preparations are conditionally independent draws from causally selected experimental rows. The auditor then freezes a validation design and an event and compares fresh candidate and target draws through that event. All persistent randomness, report buffers and validation accumulators belong to the register. This model separates the simulator's resources from those of its auditor and keeps the mark of the actual preparation inside the statistical comparison.

The first result is a controlled posterior-predictive dual (Theorem 4.1). Its state is an unnormalized posterior measure together with the legal protocol state. At the terminal step, optimization over an event gives a positive-part functional, equivalently a posterior-predictive total variation. Earlier steps optimize the causal experimental action. One minimization over initial priors yields the exact unrestricted-memory minimax value. Every such prior, or a Bellman supersolution, gives a converse uniform over admissible finite-memory architectures. The common-dominating-row theorem becomes a specialization in which the experimental choice can be removed. The controlled formulation also applies when the available actions are statistically incomparable. A strict adaptive advantage persists under full-support perturbations (Proposition 4.4).

Our principal explicit minimax calculation concerns the product family $b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2)$ and targets Q_γ on the four marked diagonal atoms. For $0 \leq \gamma \leq 1 - 1/\sqrt{2}$, put

$$u_\gamma = \gamma - 2 + \sqrt{\gamma^2 - 2\gamma + 5}.$$

Theorem 5.1 gives

$$U_1(b, Q_\gamma) = (3 + u_\gamma^2)/8.$$

The optimal response and least-favorable prior are explicit. Their matching is certified over the entire parameter square by a square identity, not by an optimization grid. The endpoint of the displayed γ interval is the point where a posterior response coefficient changes sign. The zero-preparation value is also exact. At the physical mark bias $\gamma = 1/4$ the one-preparation value is

$$u_{\text{ex}} = (85 - 7\sqrt{73})/64.$$

This resolves the optimization problem hidden behind the earlier rational one-preparation certificate.

The physical consequence uses the same marked collision experiment, not a surrogate alphabet with a different cost convention. Define

$$(1) \quad u_1 = \frac{4035777}{10240000}, \quad m_2 = \frac{12 - 3\sqrt{3}}{16}, \quad \beta_0 = \frac{1}{300}.$$

The older rational bound u_1 remains a valid explicit certificate. The exact value u_{ex} improves it. We prove

$$(2) \quad N_c^{\text{phys}}(W, Q) = 2 \quad \text{for} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_{\text{ex}} + \beta \leq c < m_2 - \beta.$$

Here N counts candidate training preparations, and one candidate and one target validation are charged in addition. The upper construction is deterministic and has the complete bit-level profile $(1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$. Definition 6.1 and Theorem 6.3 prove locally every interface and physical approximation fact used in this deduction. Equation (2) is sharp in preparation count on its stated width range; it does not assert width optimality at two preparations or an exact value of U_2 .

A different theorem determines both coordinates in a revelation family. With reveal probability α , N preparations and K retained decision states, Theorem 8.1 proves

$$V_N^{\text{er}}(K) = (1 - (1 - \alpha)^N)(1 - 1/K).$$

The converse allows every stochastic encoder and legal stopping rule; the matching machine requires no uncharged revelation flag. The formula yields exact strict thresholds and matching sample and state scales. The memory coordinate here is a decision-cut width, not the collision machine's bit-level peak. Keeping those coordinates distinct is part of the resource theorem.

For a specified finite architecture, compatible occupation arrays retain all common-row constraints, including autonomous row reuse. The subdivision theorem gives an explicit HS/m certification error with controlled row-block degree. These are effective finite-architecture certificates, not a sharp universal complexity law. We also retain a resource-accounted autonomous binary comparison whose stationary formula is classical. None of these auxiliary results replaces the exact minimax or physical arguments.

1.1. Relation to earlier work. The comparison and minimax mechanisms are classical [2, 3]. Finite-horizon belief-state control and piecewise-linear value functions are established in [1]. Our controlled dual uses these mechanisms with a fixed adversarial parameter, frozen validation and explicit causal resource accounting. It is not a claim to have invented belief-state control or the minimax theorem. The exact marked-family saddle points, their equality certificate and their physical use are stated separately from these general mechanisms.

Polynomial stochastic-policy formulations and positivity certificates have substantial antecedents [4, 5]. Finite-memory testing, randomization and time-invariant rules are treated in [6, 7, 8, 9]. The stationary formula in Section 10 is explicitly attributed to Hellman and Cover. The May 2026 work of Managoli and Prabhakaran [10] treats memory-constrained adversarial binary hypothesis testing, including randomized time-invariant machines and adaptive distribution choices. Its objective and adversarial mechanism differ from the fixed-parameter, training-validation score here. No general priority for randomized state savings is asserted.

Norberg's filtered comparison [11] belongs to the surrounding comparison theory. We make no non-overlap or originality claim about that general theory without a proof-level comparison of the original paper. The novelty claims under consideration are the explicit theorems and resource realizations proved here, not an asserted absence of filtered-experiment predecessors.

1.2. Organization and preserved development. We first fix the architecture and common-row framework, then prove the controlled dual and exact marked minimax law. The physical bridge and charged two-preparation construction give (2). The joint revelation law, finite-architecture certification and autonomous comparison follow. Complete mathematical volumes retain the unaltered predecessor article and development, including the general continuation, Gaussian transport and historical pipeline proofs [12]. These remain mathematical content, not substitutes for a proof in the present article. The independent A2 geometric chain and

the original B4 and broad C2 obligations are not claimed to follow from the finite statistical theorems alone.

2. EXPERIMENTS, ARCHITECTURES AND ATTAINABLE RESOURCES

Throughout, total variation is $\text{TV}(P, Q) = \sup_{0 \leq h \leq 1} (P - Q)h = \frac{1}{2} \sum_x |P(x) - Q(x)|$. An experimental parameter is unknown and fixed throughout an execution. Controller randomization is independent of that parameter. A reset preparation is a fresh draw, independent of earlier preparations given the parameter. A controller may not recover discarded information from an uncharged tape, clock, selector or calibration table.

Definition 2.1 (Finite architecture). A finite architecture A specifies a finite causal execution graph, its legal experimental operations, the observable inputs to each controller row, and a register alphabet at every cut. It also specifies which row occurrences share a stochastic table. A clocked architecture may use different rows at different scheduled times. An autonomous architecture may use time only through a charged state. For a parameter family Θ and terminal reward in $[-1, 1]$, let

$$V(A) = \sup_{q \in \prod_r \Delta_{s_r}} \inf_{\theta \in \Theta} g_\theta(A, q).$$

If $\mathcal{A}(R)$ is the class of architectures obeying a global resource budget R , its outer value is

$$V^{\text{out}}(R) = \sup_{A \in \mathcal{A}(R)} V(A).$$

Both the architecture and the simulator parameter are fixed before the controller's random choices. In particular, a mixture of complete controllers is not automatically an admissible controller at the original register profile.

For a finite hidden execution graph, compatibility can be stated without division. Let $x_{\theta ir}$ be the incoming mass of hidden history i at observable row r , and let $f_{\theta irj}$ be the outgoing mass through choice j . Include the usual fixed-kernel flow equations, as well as

$$(3) \quad f_{\theta irj} \geq 0, \quad \sum_j f_{\theta irj} = x_{\theta ir}, \quad f_{\theta irj} x_{\theta' i' r} = f_{\theta' i' rj} x_{\theta ir}.$$

For shared rows, the indices i, i' include all their occurrences.

Proposition 2.2 (Compatible realization). *For a specified finite architecture, nonnegative flow arrays satisfying (3) are exactly those realized by a common stochastic row at every declared observable input, including at zero incoming mass.*

Proof. Necessity follows from $f_{\theta irj} = x_{\theta ir} q_r(j)$. For sufficiency, if some incoming mass at row r is positive, define $q_r(j)$ by dividing the corresponding outgoing mass by that incoming mass. Conservation makes q_r stochastic, and the cross-products force the same ratio at every other positive incoming mass. If an incoming mass vanishes, nonnegativity and conservation force all its outgoing masses to vanish. If the entire row is unreachable, choose any legal stochastic row. Induction through the finite execution graph reconstructs the prescribed flows. The same ratio argument applies to identified occurrences of an autonomous row. $\square$

This formulation is inherited from the complete companion and is included to fix the controller class. Compact semialgebraic feasibility and an Archimedean positivity theorem provide a complete semialgebraic certificate hierarchy for a *fixed finite architecture*. They do not interchange the two optimizations in the definition. Section 9 gives a quantitative alternative for the inner problem. The next section supplies a different certificate that survives the outer optimization.

3. A STATISTICAL DUAL THAT IS UNIFORM OVER ARCHITECTURES

Let Θ be compact, let Ω be finite, and let $\theta \mapsto P_\theta \in \mathcal{P}(\Omega)$ be continuous. Fix $Q \in \mathcal{P}(\Omega)$. An N -preparation reset selector observes $Z = (Z_1, \dots, Z_N) \sim P_\theta^{\otimes N}$ and chooses a response $h_Z \in [0, 1]^\Omega$. Its expected validation score is

$$(4) \quad G_\theta(h) = \sum_{z \in \Omega^N} P_\theta^{\otimes N}(z)(Q - P_\theta)h_z.$$

The response is fixed before two independent validation draws. Operationally it can be realized by choosing a random event with those coordinate marginals and retaining that event for both validations. No memory bound is imposed in the definition of $U_N(P, Q) = \max_h \min_\theta G_\theta(h)$.

Theorem 3.1 (Posterior-predictive dual). *Both extrema in the following identity are attained:*

$$(5) \quad U_N(P, Q) = \min_{\mu \in \mathcal{P}(\Theta)} \sum_{z, \omega} \left[\int P_\theta^{\otimes N}(z) \{Q(\omega) - P_\theta(\omega)\} d\mu(\theta) \right]_+$$

$$(6) \quad = \min_{\mu \in \mathcal{P}(\Theta)} \sum_z m_\mu(z) \text{TV}(Q, \bar{P}_\mu(\cdot | z)).$$

Here $m_\mu(z) = \int P_\theta^{\otimes N}(z) d\mu(\theta)$ and, when $m_\mu(z) > 0$,

$$\bar{P}_\mu(\omega | z) = \frac{\int P_\theta^{\otimes N}(z) P_\theta(\omega) d\mu(\theta)}{m_\mu(z)}.$$

A zero-mass term in (6) is defined to be zero.

Proof. The response space is a finite-dimensional compact convex box. Probability measures on the compact space Θ form a weakly compact convex set. The integral of (4) is continuous and affine in each argument. The minimum over parameters equals the minimum over probability measures. The minimax theorem [3] therefore interchanges the maximum and minimum. For fixed μ , each response coordinate is independent in the maximization and its contribution is the positive part of its coefficient, proving (5). The coefficient vector at a fixed z has zero sum. For positive $m_\mu(z)$ it equals $m_\mu(z)(Q - \bar{P}_\mu(\cdot | z))$, proving (6). If $m_\mu(z) = 0$, every nonnegative likelihood integrand vanishes μ -almost surely and all coefficients are zero. Continuity on the two compact strategy spaces gives attainment. $\square$

Theorem 3.2 (Dominating-row converse). *Suppose every permitted training row is a parameter-independent causal garbling of one fresh draw from P_θ . Suppose every permitted validation row is the same garbling of the target and candidate laws, and every selected row–event pair is fixed before validation. Then $U_N(P, Q)$ bounds the worst-case expected score of every auditor using at most N candidate training preparations, regardless of its register profile, feedback schedule, independent randomization, or stopping rule. Independent target-only calibration, whose law does not depend on θ , does not invalidate the bound. If the all-on row and all its events are permitted, the bound is attained in the class with no memory cap and N training preparations.*

Proof. Generate N independent all-on observations in advance. At the t th requested preparation, apply the declared garbling to observation t , using only previously simulated reports and independent randomization. This simulates the adaptive transcript with its exact law. A stopped execution ignores the unused observations; it does not reveal them to the auditor. Pull back its final row–event response to the all-on alphabet. Conditional on the entire generated training word and the independent random seed, this is a function in $[0, 1]^\Omega$. Averaging the seed, including any target-only calibration, produces one response h_z for each full word z . The score is consequently (4). This proves the upper bound for the union of all architectures. Conversely,

retaining the entire word and then a selected event implements any such finite response kernel when memory is unrestricted. Fresh validation gives exactly (4). $\square$

Remark 3.3 (What the dual does not convexify). The measure μ is an adversarial prior used to bound the minimum over *fixed* parameters. Its likelihood is $\int P_\theta^{\otimes N} d\mu(\theta)$, not $(\int P_\theta d\mu(\theta))^{\otimes N}$. It neither enlarges the private candidate class to mixtures nor gives the auditor a free persistent selector. Attainment of U_N without a memory cap does not assert attainment at any specified finite profile.

The state in (6) is generated by the actual preparation law, not by a geometric measure chosen independently of the experiment. For sequential continuation one retains the posterior μ_z or its future-test equivalence class; a single predictive marginal need not be closed under further observations. This is the distinction between a one-step response quotient and a future-predictive quotient.

Corollary 3.4 (Finite moment support for the product family). *For two independent Bernoulli report bits with parameters (p, q) and an independent fair mark, an optimal prior in (5) can be chosen with at most $(N + 2)^2$ atoms. The two report counts (k, ℓ) are a causally updatable sufficient statistic for computing the dual.*

Proof. Conditional on the two counts, all arrangements of report bits and all mark words have parameter-independent probabilities. Summing their coefficients reduces (5) to the $(N + 1)^2$ count pairs. Their likelihoods are

$$\binom{N}{k} p^k (1 - p)^{N-k} \binom{N}{\ell} q^\ell (1 - q)^{N-\ell}.$$

After multiplication by $Q(\omega) - P_{p,q}(\omega)$, every integrand has degree at most $N + 1$ in each variable. Thus the objective depends only on the $(N + 2)^2$ rectangular moments. Their constant coordinate is one. Carathéodory's theorem in the remaining affine space replaces an optimal moment vector by a convex combination of at most $(N + 2)^2$ point evaluations. Counts update by addition of the current bits. No claim that this count register is optimal for the decision is made. $\square$

4. CONTROLLED EXPERIMENTS WITHOUT A DOMINATING ACTION

The reduction to one dominating row is not needed for a statistical dual, provided causal action choice is retained in the dual recursion. We give the finite-action theorem with a compact parameter space and an explicit protocol state. Its equality is an unrestricted-memory statement; its upper certificates apply to every smaller-memory class.

Let Θ be compact metric and E a finite set of protocol states. At $e \in E$ let $\mathcal{A}_e$ be a finite set of training actions. Action a produces x in a finite alphabet X_a with law P_θ^a and changes the protocol state to $T(e, a, x)$. The functions $\theta \mapsto P_\theta^a(x)$ are continuous. An unknown θ is fixed throughout the experiment. Conditional on it, each requested preparation uses a fresh independent draw from the selected row. Legal validation designs form a nonempty finite set $\mathcal{B}_e$. Design b has a finite outcome alphabet Y_b , target law Q^b and continuous candidate law V_θ^b . A policy may stop after at most N training actions. It then freezes $b \in \mathcal{B}_e$ and an event on Y_b , before one candidate and one target validation. Randomization is independent of θ . No training action is assumed to dominate another.

For a finite nonnegative measure ν on Θ write $\nu_{a,x}(d\theta) = P_\theta^a(x)\nu(d\theta)$ and set

$$(7) \quad \Phi_e(\nu) = \max_{b \in \mathcal{B}_e} \sum_{y \in Y_b} \left[Q^b(y)\nu(\Theta) - \int V_\theta^b(y) d\nu \right]_+,$$

$$F_0(e, \nu) = \Phi_e(\nu),$$

$$(8) \quad F_n(e, \nu) = \max \left\{ \Phi_e(\nu), \max_{a \in \mathcal{A}_e} \sum_{x \in X_a} F_{n-1}(T(e, a, x), \nu_{a,x}) \right\}.$$

If no training action is legal, only the stopping term is retained. All the functions vanish at the zero measure, so no posterior division is required on null histories. For positive mass, (7) is the mass times the largest posterior-predictive total variation.

Theorem 4.1 (Controlled posterior-predictive dual). *Let $\mathcal{U}_N(e_0)$ be the maximum worst-case expected validation score over all randomized causal policies in the declared protocol, with at most N training preparations and no memory cap. Then*

$$(9) \quad \mathcal{U}_N(e_0) = \min_{\mu \in \mathcal{P}(\Theta)} F_N(e_0, \mu).$$

Both extrema are attained. Each $F_n(e, \cdot)$ is the maximum of finitely many continuous linear functionals on the cone of measures; in particular it is convex, continuous and positively homogeneous. For every prior μ , $F_N(e_0, \mu)$ bounds the worst-case score of every finite-memory architecture implementing this protocol.

Proof. There are finitely many histories of length at most N and finitely many event masks at each stopping history. Consequently there are finitely many deterministic policy trees, denoted by $\Pi_N(e_0)$. For $\pi \in \Pi_N(e_0)$ its payoff $g_\pi(\theta)$ is a finite sum of finite products of continuous primitive probabilities and is continuous in θ .

A randomized causal policy induces the same outcome laws as a probability distribution on these trees: sample its finite random decision table in advance and condition on it. Conversely, with no memory cap, a sampled tree can be retained and followed. Thus the unrestricted payoff vectors are exactly the convex hull of the g_π . This step is not an assertion that mixing machines preserves a specified register width.

The simplex on $\Pi_N(e_0)$ and $\mathcal{P}(\Theta)$ are compact and convex. The expected payoff is continuous and bilinear. Minimax therefore gives

$$\max_{\lambda \in \Delta(\Pi_N)} \min_{\theta} \sum_{\pi} \lambda_{\pi} g_{\pi}(\theta) = \min_{\mu} \max_{\pi \in \Pi_N} \int g_{\pi} d\mu.$$

For a fixed measure, the terminal maximum is (7), because an event includes precisely the positive coordinates of its signed predictive measure. At a training node one selects an action and, after its observed outcome, selects a continuation tree separately on that branch. The unnormalized branch measure is exactly $\nu_{a,x}$. Backward induction proves (8) and identifies the right side with (9). It also proves the finite-linear-functional representation. Compactness gives an optimizing prior and an optimizing mixed tree. Finally, the minimum over fixed parameters of any policy is at most its average under μ , which is at most $F_N(e_0, \mu)$. This remains true after restricting its memory or architecture. $\square$

The minimization in (9) occurs once, outside the recursion. It cannot be replaced by fresh adversarial choices after observations. The posterior carries dependence on one fixed unknown parameter. Also, a Bayes-optimal tree for a minimizing prior need not by itself be a minimax policy; the saddle-point mixture is supplied by the finite game in the proof.

Corollary 4.2 (Bellman upper certificates). *Suppose functions $\Psi_n(e, \nu)$ satisfy $\Psi_0 \geq \Phi$ and, for $n \geq 1$, satisfy*

$$\Psi_n(e, \nu) \geq \Phi_e(\nu), \quad \Psi_n(e, \nu) \geq \sum_x \Psi_{n-1}(T(e, a, x), \nu_{a,x}) \quad (a \in \mathcal{A}_e).$$

Then $\Psi_N(e_0, \mu)$ is an upper certificate for every admissible architecture and every prior μ . The family F_n is the least such family. An optimal prior can be chosen with at most $d + 1$ support points, where d is the affine dimension of $\{(g_\pi(\theta))_{\pi \in \Pi_N} : \theta \in \Theta\}$.

Proof. Induction in (8) gives $F_n \leq \Psi_n$. The equality case is the recursion itself. The objective depends on a prior only through its vector of integrals of the finitely many g_π . An optimal such vector belongs to the convex hull of their compact evaluation image. Carathéodory's theorem gives the stated support bound while preserving every pure-tree payoff. $\square$

Proposition 4.3 (Causal stability). *Consider two models with the same protocol, alphabets and policy class. Suppose their training rows differ uniformly in total variation by at most ϵ , their candidate validation rows by at most η , and their target validation rows by at most β . Their unrestricted values, or their values in any common specified memory class, differ by at most $2N\epsilon + \eta + \beta$.*

Proof. Use the same policy randomization and couple corresponding training rows while the histories agree. The probability of a first discrepancy is at most $N\epsilon$, even with feedback and stopping. Conditional expected validation rewards lie in $[-1, 1]$, so differing training histories contribute at most $2N\epsilon$. On agreeing histories the same frozen design and event are used, and their conditional mean rewards differ by at most $\eta + \beta$. The bound is uniform in the fixed parameter and in the policy. Taking minima and suprema preserves it, with the reverse inequality obtained by interchanging the models. $\square$

4.1. An informative action choice. The following calculation distinguishes the controlled theorem from a common-row reduction. Let $\Theta = \{(i, j) : i, j \in \{0, 1\}\}$. Action R reports i . Action L reports j when $i = 0$ and otherwise reports an independent fair bit. Action H reports j when $i = 1$ and otherwise reports an independent fair bit. The candidate validation is $\delta_{(i,j)}$ and the target is uniform on Θ .

Proposition 4.4 (Strict adaptive advantage). *With two training preparations, the unrestricted adaptive value is $3/4$ and the best nonadaptive value is $5/8$. Nonadaptive policies here include mixtures of schedules, with their random schedule retained. No available single training action dominates all the others.*

Proof. Take R first, and then L or H according to the observed i . The parameter is recovered exactly. The complement of its validation atom has score $3/4$. No policy can exceed $\text{TV}(Q, \delta_\theta) = 3/4$ even with full information.

For the converse for fixed schedules, use the uniform prior. For a pair of actions a, b its Bayes value is explicitly

$$(10) \quad \sum_{x,y,\theta} \left[\frac{1}{16} \sum_{\vartheta} P_{\vartheta}^a(x) P_{\vartheta}^b(y) - \frac{1}{4} P_{\theta}^a(x) P_{\theta}^b(y) \right]_+.$$

Substitution of the three displayed kernels gives

	R	L	H
R	$1/2$	$5/8$	$5/8$
L	$5/8$	$7/16$	$1/2$
H	$5/8$	$1/2$	$7/16$

This is an exact finite sum, not a parameter-grid estimate. It bounds any mixture of fixed schedules by $5/8$; an early stop can be padded with an ignored observation. To attain the bound uniformly in θ , choose with equal probabilities the schedules (R, L) and (R, H) . When the chosen branch-specific action matches i , the parameter is known and the score is $3/4$. Otherwise only i is known, and the complement of its two-atom fiber gives score $1/2$. Each fixed parameter therefore has mean score $5/8$.

Action R cannot distinguish parameters differing only in j . Action L cannot distinguish $(1, 0)$ and $(1, 1)$, and H cannot distinguish $(0, 0)$ and $(0, 1)$. An experiment which fails to distinguish such a pair cannot garble to one that distinguishes it. Hence no action is a common dominating row. $\square$

If every binary report is additionally passed through an independent binary symmetric channel of crossover ϵ , the primitive training rows change by at most ϵ and have full support for $0 < \epsilon < 1/2$. Proposition 4.3, applied to both policy classes, leaves an adaptive advantage at least $1/8 - 8\epsilon$, which is positive for $0 < \epsilon < 1/64$. Thus the distinction is not confined to disjoint observation supports. The finite-horizon belief recursion and its piecewise-linear structure are classical [1]. The contribution here is their explicit integration with the frozen validation score, one fixed adversarial parameter, outer certificates, and the resource conventions; no priority for Bellman recursion itself is claimed.

5. EXACT MINIMAX LAWS FOR A MARKED PRODUCT FAMILY

The posterior-predictive dual can be solved explicitly for a continuous family containing the collision target. Throughout this section, the training and candidate validation law is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad (p, q) \in [0, 1]^2.$$

In the order 000, 001, 110, 111, let Q_γ have masses

$$((1 + \gamma)/4, (1 - \gamma)/4, (1 - \gamma)/4, (1 + \gamma)/4),$$

and zero mass elsewhere. Write $U_1(\gamma) = U_1(b, Q_\gamma)$, with exactly the training-validation convention of (4). Set

$$(11) \quad \gamma_* = 1 - 1/\sqrt{2}, \quad s_\gamma = \sqrt{\gamma^2 - 2\gamma + 5}, \quad u_\gamma = \gamma - 2 + s_\gamma, \quad \tau_\gamma = \frac{2u_\gamma}{u_\gamma + 2 - \gamma}.$$

Theorem 5.1 (Exact marked minimax family). *For every $0 \leq \gamma \leq \gamma_*$,*

$$(12) \quad U_1(\gamma) = \frac{3 + u_\gamma^2}{8}.$$

An optimal response depends on the training word only through the report count $K = X_1 + X_2$. On the four target atoms its rows are

$$(13) \quad h_0 = (\tau_\gamma, 0, 1, 1), \quad h_1 = (1, 1, 1, 1), \quad h_2 = (1, 1, 0, \tau_\gamma),$$

and all responses are zero off these atoms. A least-favorable prior is

$$(14) \quad \mu_\gamma = \frac{1}{2}\delta_{(t_\gamma, t_\gamma)} + \frac{1}{2}\delta_{(1-t_\gamma, 1-t_\gamma)}, \quad t_\gamma = \frac{1 - \sqrt{u_\gamma}}{2}.$$

The assertion optimizes over all stochastic responses to the full marked training word, not merely over count-based or symmetric responses.

Proof. First, u_γ is the positive root of

$$(15) \quad u^2 + (4 - 2\gamma)u - (1 + 2\gamma) = 0.$$

On the asserted interval, $0 < u_\gamma < 1$ and $0 < \tau_\gamma < 1$. Indeed the polynomial is negative at zero and positive at one; also $u_\gamma < 2 - \gamma$ because $\gamma < 1/2$. Substitution of $1 - 2\gamma$ into the polynomial gives $8(\gamma^2 - 2\gamma + 1/2)$, which is nonnegative precisely on the present interval. Monotonicity on $[0, \infty)$ therefore gives

$$(16) \quad u_\gamma \leq 1 - 2\gamma.$$

Put $a = (1-p)(1-q)$, $c = pq$, and $z = 1 - a - c$. The probabilities of the three training counts are a, z, c . The independent training mark may be discarded by the displayed construction. Let $G_\gamma(p, q)$ denote its score. With $u = (p + q - 1)^2$ and $v = (p - q)^2$, direct expansion gives

$$(17) \quad G_\gamma(p, q) - \frac{3 + u_\gamma^2}{8} = \frac{2 - \tau_\gamma}{16} \{(u - v - u_\gamma)^2 + 4v\}.$$

For clarity, this identity follows by substituting the three rows of (13) into

$$a(Q_\gamma - b)h_0 + z(Q_\gamma - b)h_1 + c(Q_\gamma - b)h_2,$$

using $a + c = (1 + u - v)/2$, $a - c = -(p + q - 1)$, and then eliminating u_γ^2 by (15). The coefficient of the resulting square is $(2 - \tau_\gamma)/16$ and that of v is four times this coefficient. Both are nonnegative. Equality holds at both points of (14), since there $v = 0$ and $u = u_\gamma$. This proves the lower bound and identifies its equality set.

It remains to bound every response, including nonsymmetric ones. Use (14) in Theorem 3.1. At $t = t_\gamma$ put $a = (1 - t)^2$, $c = t^2$, and $T = a + c = (1 + u_\gamma)/2$. For the training report pair 00, the four dual coefficients, apart from a common positive factor, are

$$(18) \quad \left(Q_H T - \frac{a^2 + c^2}{2}, \quad Q_L T - \frac{a^2 + c^2}{2}, \quad Q_L T - ac, \quad Q_H T - ac \right),$$

where $Q_H = (1 + \gamma)/4$ and $Q_L = (1 - \gamma)/4$. The first coordinate is zero by (15); the second is $-\gamma T/2 \leq 0$. The last two are positive: $T \geq 1/2$, $ac \leq 1/16$, and $\gamma < 1/2$ imply $Q_L T > 1/16 \geq ac$. Thus $(\tau_\gamma, 0, 1, 1)$ maximizes this linear functional. For the training pair 11, reversing all report bits and the mark reverses the four coordinates and gives the row h_2 .

For either training pair 01 or 10, the two prior support points have the same likelihood $t(1 - t)$. The posterior is still equally weighted, and the candidate predictive mass at each target atom is $T/4$. Every one of the four coefficients is therefore nonnegative by $Q_L - T/4 = (1 - 2\gamma - u_\gamma)/8 \geq 0$. The response h_1 is optimal. At off-diagonal validation atoms the target mass vanishes and the coefficient is nonpositive. Finally, conditioning on either training mark multiplies all these coefficients by $1/2$; it creates no further optimizing choice. We have therefore optimized every coordinate of the full marked-word response space.

The prior average of the displayed response equals its value at either support point, namely $(3 + u_\gamma^2)/8$, by (17). Every other response has prior average at most this number and hence has minimum over fixed (p, q) at most this number. Together with the lower bound this proves (12), optimality of the response, and least favorability of the prior. $\square$

Corollary 5.2 (The exact one-preparation collision value). *For $Q_* = Q_{1/4}$,*

$$(19) \quad U_1(b, Q_*) = u_{\text{ex}} := \frac{85 - 7\sqrt{73}}{64}, \quad u_{1/4} = \frac{\sqrt{73} - 7}{4}, \quad \tau_{1/4} = 2 - \frac{14}{\sqrt{73}}.$$

If only the target changes to Q , then $|U_1(b, Q) - u_{\text{ex}}| \leq \text{TV}(Q, Q_)$.*

Proof. Substitute $\gamma = 1/4$ in Theorem 5.1. For a fixed trained response the target perturbation is at most total variation, uniformly in (p, q) . Taking first the minimum and then the maximum preserves this bound in both directions. $\square$

5.1. Complementary slackness and the event alphabet. The tie in the first coordinate of (18) explains the randomization. Any value of that response coordinate is Bayes-optimal for the two-point prior. The minimax choice makes the uniform payoff stationary at its two support points; solving this condition gives τ_γ and the square identity (17). This derives the prior and response together, rather than postulating that a numerically effective prior is optimal.

Operationally, for $K = 0$ one draws once between the events with indicator vectors $(0, 0, 1, 1)$ and $(1, 0, 1, 1)$, with probabilities $1 - \tau_\gamma$ and τ_γ . For $K = 2$ one draws between $(1, 1, 0, 0)$ and $(1, 1, 0, 1)$. For $K = 1$ the event is $(1, 1, 1, 1)$. The resulting event identity is retained and used for both fresh validations. These are exactly the five events in the two-preparation construction below. The one-preparation theorem supplies their complementary-slackness interpretation; it does not prove that the two-preparation selector is optimal. The four-point rational prior retained below is a useful rational certificate, not a least-favorable prior.

The exact response uses an atomic stochastic row. It does not receive free persistent randomness. Replacing τ_γ by a dyadic number within 2^{-b} changes each conditional response, and hence the score, by at most 2^{-b} . A sequential fair-bit implementation must retain its bit prefix and event identity. None of these implementation costs is needed by the deterministic two-preparation physical audit.

Proposition 5.3 (The zero-preparation endpoint). *For $0 \leq \gamma \leq 1$, $U_0(b, Q_\gamma) = \gamma/2$.*

Proof. The fixed event $\{000, 111\}$ has score $(1 + \gamma)/2 - (a + c)/2 \geq \gamma/2$ because $a + c \leq 1$. Conversely, the equally weighted prior on $b(0, 0)$ and $b(1, 1)$ has predictive validation law uniform on the four target atoms. Its total variation from Q_γ is $\gamma/2$. Theorem 3.1 with $N = 0$ gives the matching upper bound. $\square$

6. A SELF-CONTAINED COLLISION INTERFACE

We give the preparation, mechanical estimate, observation law and private simulator class needed for the physical specialization. The following argument does not appeal to an unpublished physical lemma. It retains the same positive-volume preparation and positive sensor noise as the earlier collision construction.

Definition 6.1 (The reset row–event architecture). A preparation supplies two sequential report slots and, afterwards, an actual mark. The source reports in the two slots are constant. A private width-one simulator has no retained information between those slots and no persistent hidden seed. It may use fresh independent coins. Its first emission is $X_1 \sim \text{Ber}(p)$; after receiving the current gate $A \in \{0, 1\}$, its second emission has a fixed Bernoulli kernel with parameter q_A . The parameters (p, q_0, q_1) are fixed throughout the audit. The simulator does not observe the actual mark. All its coins are independent of that mark. Different preparations reset all unretained variables and are independent conditional on the parameters.

The auditor observes the first report, chooses a gate using its retained state and that report, observes the second report, and then observes the mark. During training it may choose rows and stop adaptively, using at most the declared number of candidate preparations. It next freezes one gate policy and one event, with any retained random choice included in its register. It uses that same row–event continuation on one fresh candidate preparation and one independent target preparation and returns the difference of their event indicators, target minus candidate. It may not reselect the event after either validation. Independent target-only calibration is allowed but all retained calibration and randomness are charged. Peak width counts every bit, mark, event and validation-accumulator cut. Clocked schedules are declared externally; an autonomous implementation must retain its phase. An unretained transcript, seed or stopping-time value is not available to a continuation.

In the all-on row the complete private law is $b(p, q_1)$, with an independent fair actual mark. For lower bounds it suffices to restrict the adversary to $q_0 = 0$, $q_1 = q$. In this subfamily every deterministic gate policy is the pushforward

$$T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$$

of $b(p, q)$. The gate, when included in the transcript, is another function of these same coordinates. Randomized gate policies are covered by averaging their independent random tables. This proves both the all-on product characterization for the upper construction and the common-garbbling subfamily used for the outer converse. The width-one constraint is a simulator constraint; it is not an uncharged restriction on the auditor's register.

6.1. Preparation and elastic scattering. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$, with no walls and contact distance d . Write $r = q_1 - q_2$, $g = v_1 - v_2$, $V = (v_1 + v_2)/2$, and let the center of mass position range over any fixed rational box of positive volume. Prepare $(B, W) \in \{0, 1\}^2$ with probability $5/16$ at $B = W$ and $3/16$ at $B \neq W$. Conditional on (B, W) , use the uniform product preparation on the following boxes:

$$(20) \quad \begin{aligned} r &= (-a_0, (2B-1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & & & |V_z - (2W-1)| &\leq 1/100. \end{aligned}$$

Thus the actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$, preserved by momentum conservation. Set

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector, identically zero on the incoming transverse-velocity range.

Lemma 6.2 (Uniform collision signal). *For every preparation in (20) and every $9/10 \leq d \leq 11/10$, exactly one collision occurs and its time lies in $(4/5, 3/2)$. Afterwards*

$$(2B-1)v_{1,y} > 1/2, \quad (2B-1)h(v_{1,y}) > 1/4.$$

For $0 < d \leq 2/5$ no collision occurs on $[0, 3]$ and the detector is zero throughout that interval.

Proof. Before contact, $r(t) = r + tg$. At $t \leq 4/5$ its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At $t = 3/2$ its absolute coordinate bounds are $1/4$, $113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. The first contact therefore occurs in the claimed interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$. Hence the entry contact has $n_x < 0$, where $n = r(t_c)/d$. Its remaining coordinates give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B-1)n_y \geq 87/220.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The elastic rule gives

$$v_1^+ = V + g/2 - (g \cdot n)n, \quad g^+ = g - 2(g \cdot n)n.$$

Consequently

$$(2B-1)v_{1,y}^+ > -3/200 + (1443/1000)(87/220) > 1/2.$$

The definition of h implies the signal bound. Since $r(t_c) \cdot g^+ > 0$, squared separation increases strictly afterwards; there is no second collision. Finally, for $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. There is no contact, and the incoming $|v_{1,y}| \leq 3/200$ makes h zero. $\square$

6.2. Positive noise and the actual mark. Let $\mathcal{B}_t$ be Brownian motion independent of the preparation and set

$$Y_t = \int_0^t h(v_{1,y}(s)) ds + \sigma \mathcal{B}_t, \quad 0 < \sigma \leq 1/24.$$

The two potential all-on reports are

$$Z_1 = \mathbf{1}_{\{2(Y_2 - Y_{3/2}) > 0\}}, \quad Z_2 = \mathbf{1}_{\{2(Y_3 - Y_{5/2}) > 0\}}.$$

The actual second report is AZ_2 . The gate changes acquisition, not the mechanical path. The continuous path is implemented by the target sensor; it is not made available to the private simulator or supplied as an uncharged auditor buffer. Denote the law of (Z_1, Z_2, W) by $Q_{d,\sigma}$.

Theorem 6.3 (Uniform physical presentation). *For $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$,*

$$(21) \quad \text{TV}(Q_{d,\sigma}, Q_*) < 1/300.$$

The same estimate holds after every declared feedback gate, including its observable gate value. The all-on private product family and the common-garbling subfamily in Definition 6.1 apply to this very experiment.

Proof. Every collision occurs before $3/2$. Conditional on the preparation, the two normalized sensor windows are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. Lemma 6.2 makes their means have sign $2B - 1$ and magnitude greater than $1/4$. Each potential report therefore differs from B with probability less than $\Phi(-3)$. The Gaussian tail estimate

$$\Phi(-x) \leq \frac{e^{-x^2/2}}{x\sqrt{2\pi}}$$

gives $\Phi(-3) < 1/600$: $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$ suffice. Couple the ideal and noisy reports using the same preparation, actual mark and gate random table. Except on a set of probability less than $2\Phi(-3) < 1/300$, both potential reports equal B and the entire gated transcript agrees. The ideal marked law is precisely Q_* , by the prepared (B, W) table. This proves (21) and the feedback-uniform version. Finally, the source is blind, its actual mark is unobserved by the simulator, and the gate is its only current input at the second emission. The product and subfamily assertions follow from the two independent private-emission kernels specified above. $\square$

For $d \leq 2/5$, the two windows instead contain only independent centered Gaussian noise. Their signs are independent fair bits, independent of W , and a private width-one simulator reproduces every gated row exactly. Thus the informative regime and its no-collision control use the same detector, preparation convention and actual mark.

7. THE PHYSICAL EXPERIMENT AND ITS SHARP PREPARATION BOUNDARY

Take $\Omega = \{0, 1\}^3$, with coordinates (x_1, x_2, w) . The all-on private candidate is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad 0 \leq p, q \leq 1.$$

The mark w is the actual mark, not an invented simulator bit. Put $a = (1 - p)(1 - q)$ and $c = pq$. On the ordered atoms

$$(22) \quad 000, \quad 001, \quad 110, \quad 111,$$

let Q_* have masses $5/16, 3/16, 3/16, 5/16$, and set it to zero elsewhere. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. A target Q is allowed whenever $\text{TV}(Q, Q_*) \leq \beta$.

The original feedback operation has the form

$$(23) \quad T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w), \quad g : \{0, 1\} \longrightarrow \{0, 1\}.$$

The selected gate is observable. It may depend on earlier preparations and on the first current report. The target rows are pushforwards by (23). For the converse, restrict the adversary to the admissible private subfamily whose second emission is AX_2 , where A is the current gate and $X_2 \sim \text{Ber}(q)$ is independent of the first emission and of the fair mark. Every row is then a causal garbling of $b(p, q)$. For the construction only the all-on row is used, and every private candidate in the original width-one class has the all-on product law just specified. These two facts, proved for the microscopic interface in Definition 6.1 and Theorem 6.3, explain why using a subfamily for the lower bound does not narrow the upper construction.

The admissible audit first trains, then selects one row–event pair, then uses one fresh candidate and one fresh target validation. It reports $D = \mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The event and row cannot be reselected after seeing a validation draw. All bit, mark, decision and validation cuts are charged. Define $N_c^{\text{phys}}(W, Q)$ to be the minimum number of candidate training preparations among these original-interface auditors, with peak register width at most W , whose score is uniformly strictly greater than c . Randomization, feedback and stopping are allowed. If no such audit exists, the value is $+\infty$.

Theorem 7.1 (Two-preparation physical segment). *Let u_1, m_2 be as in (1). For every $\beta \geq 0$, $\text{TV}(Q, Q_*) \leq \beta$, and*

$$W \geq 12, \quad u_1 + \beta \leq c < m_2 - \beta,$$

one has $N_c^{\text{phys}}(W, Q) = 2$. The two-preparation audit is deterministic, uses no acquired calibration and no random bits, and has the full profile

$$(24) \quad (1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$$

from the initial cut through its twelve bit/mark updates. In particular, its decision alphabet has five labels and its peak has twelve. The lower bound remains valid with no memory cap.

The proof occupies the next three subsections. The interval in the statement can be empty for large β ; no assertion is then made. For $\beta = 1/300$, it contains $c = 2/5$ strictly between its endpoints.

7.1. An exact rational certificate against one preparation.

Lemma 7.2. *Every original-interface auditor using at most one candidate training preparation has worst-case score at most $u_1 + \beta$, regardless of memory.*

Proof. By Theorem 3.2, it suffices to bound $U_1(b, Q_*)$ and then perturb the target. In (5) use the four fixed parameters

$$(p, q) = (1/8, 1/8), (1/5, 1/5), (4/5, 4/5), (7/8, 7/8)$$

with weights $1/16, 7/16, 7/16, 1/16$, respectively. This measure is not asserted to be least favorable. Since $p = q$ on its support, the training count $K = x_1 + x_2$ is sufficient. The independent training mark and the order of unequal bits contain no further information about this prior.

For $k = 0, 1, 2$ and the four validation atoms in (22), the signed coefficients in (5), after grouping by K , are the entries of

$$(25) \quad \frac{1}{40960000} \begin{pmatrix} -1977 & -1775177 & 2170423 & 3943623 \\ 1765554 & 191954 & 191954 & 1765554 \\ 3943623 & 2170423 & -1775177 & -1977 \end{pmatrix}.$$

For example, the coefficient in column j is

$$\sum_{i=1}^4 \mu_i \binom{2}{k} p_i^k (1-p_i)^{2-k} \left(Q_{*,j} - \frac{(1-p_i)^2 \mathbf{1}_{j \leq 2} + p_i^2 \mathbf{1}_{j \geq 3}}{2} \right).$$

This identity proves every entry of (25) by rational arithmetic. All coefficients at off-diagonal validation atoms are nonpositive. The sum of the positive entries in (25) is $4035777/10240000 = u_1$. Each response coordinate is in $[0, 1]$, so this sum bounds its prior-averaged score and hence its minimum over the four fixed candidates.

Replacing Q_* by Q changes the score of any trained response by at most β . The training law is a candidate law and is unchanged. Consequently the same prior bounds every score by $u_1 + \beta$. A procedure with no training, or one that stops before taking its permitted sample, is covered by ignoring the generated all-on observation. $\square$

7.2. A marked five-event selector. Take two independent all-on training preparations. Let S be the total number of ones among their four report bits; discard their marks. Choose the following events, represented by their indicators on (22) and with indicator zero at every off-diagonal atom:

$$(26) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & 0 \end{array}$$

Denote these indicators by $h_0, \dots, h_4$ and report $D = h_S(Y) - h_S(X)$ on independent validation draws.

Lemma 7.3 (Exact uniform score of the selector). *Against the ideal target, the selector (26) has uniform minimum score $m_2 = (12 - 3\sqrt{3})/16$. Against Q , its score is at least $m_2 - \beta$.*

Proof. The generating polynomial of S is

$$((1-p) + pz)^2((1-q) + qz)^2.$$

Writing $g_s = (Q_* - b(p, q))h_s$, its expected score is $G(p, q) = \sum_{s=0}^4 [z^s]((1-p) + pz)^2((1-q) + qz)^2 g_s$. Substitution of the five rows in (26) and collection of terms with $x = p + q - 1$, $y = p - q$, $u = x^2$ and $v = y^2$ gives

$$(27) \quad \begin{aligned} G(p, q) = H(u, v) = & -\frac{u^3}{32} + \frac{3u^2v}{32} + \frac{3u^2}{16} - \frac{3uv^2}{32} - \frac{uv}{4} - \frac{3u}{32} \\ & + \frac{v^3}{32} + \frac{v^2}{16} + \frac{15v}{32} + \frac{7}{16}. \end{aligned}$$

This is a polynomial identity, not an interpolation over a grid. Since $0 \leq u \leq 1$ and $v \geq 0$ on the parameter square,

$$\partial_v H(u, v) = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}.$$

Thus $H(u, v) \geq H(u, 0) = (-u^3 + 6u^2 - 3u + 14)/32$. The derivative of the last expression is $(-3u^2 + 12u - 3)/32$. On $[0, 1]$ it changes sign from negative to positive at $u = 2 - \sqrt{3}$. Substitution gives $(12 - 3\sqrt{3})/16$. The minimizing values are feasible: take $p = q = (1 \pm \sqrt{2 - \sqrt{3}})/2$. This proves exactness of the minimum for this specified selector, not optimality of its score over all

selectors with two preparations. Finally, for each s , the target expectation of $h_s \in [0, 1]^\Omega$ changes by at most β . Averaging over training proves the claimed robustness. $\square$

7.3. Causal compilation and the full register profile.

Lemma 7.4. *The selector and its two validations admit the deterministic profile (24). No selector, first-bit buffer or validation accumulator is outside the charged register. An autonomous phase-tagged realization uses at most 75 states.*

Proof. After each training report bit retain the count accumulated so far. The two marks are read and discarded. The first six updates therefore have widths 2, 3, 3, 4, 5, 5 after the initial singleton. At the decision cut the five count values select the five events.

During candidate validation, after the first bit retain the event and that bit, using at most ten states. After the second bit retain the event and the function of the unread mark that gives its candidate-event indicator. For events 0, 1, 2, 3, 4, respectively, the possible functions are

$$\{0, 1\}, \quad \{0, 1 - w, 1\}, \quad \{0, 1\}, \quad \{0, w, 1\}, \quad \{0, 1\}.$$

There are twelve event–function pairs. After the candidate mark, retain the event and its now known candidate indicator $c_0 \in \{0, 1\}$, using at most ten states.

During target validation, after the first bit discard the event name and retain c_0 and its residual function of the second bit and mark. These residual functions belong to

$$\{0, x_2, (1 - x_2)(1 - w), 1 - x_2, x_2 w\}.$$

This uses at most ten states. After the second bit, the residual *difference* as a function of the unread mark belongs to

$$\{-1, 0, 1, w, 1 - w, w - 1, -w\},$$

so seven states suffice. The mark update gives the three possible outputs $-1, 0, 1$. Each transition depends only on its retained residual and the current report. Thus it is a legal common observable row, not a family of independently optimized hidden-history maps.

For completeness, equality of two residual functions implies equality of their restrictions after either next bit, so identifying equal residuals is closed under every subsequent update. The supplied transition tables are obtained by these restrictions and can be reconstructed directly from (26). Enumerating residuals is a convenience for reproducing the tables, not the proof of their sufficiency. Finally, disjointly tag the states at the thirteen cuts by their phase and use terminal self-loops. The sum of the displayed widths is 75. This implementation charges the clock; it does not assert that 75 is optimal. $\square$

Proof of Theorem 7.1. Lemma 7.2 excludes $N \leq 1$ for a strict score greater than $c \geq u_1 + \beta$, for every architecture and every memory budget. Lemmas 7.3 and 7.4 supply a two-preparation architecture of peak twelve with score at least $m_2 - \beta > c$. Larger width budgets can use the same machine. $\square$

Corollary 7.5 (The positive-noise collision family). *For the original marked collision targets with $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the local microscopic estimate $\text{TV}(Q_{d,\sigma}, Q_*) < 1/300$ implies*

$$N_{2/5}^{\text{phys}}(W, Q_{d,\sigma}) = 2 \quad (W \geq 12).$$

There are four preparations in the complete audit: two candidate training, one candidate validation, and one target validation.

Proof. Apply Theorem 7.1. The lower-end margin is the exact rational number

$$\frac{2}{5} - u_1 - \frac{1}{300} = \frac{78269}{30720000} > 0.$$

For the other endpoint, $\sqrt{3} < 7/4$ implies $m_2 > 27/64$ and $27/64 - 1/300 > 2/5$. The physical total-variation estimate, including the actual mark and positive sensor noise, is proved in Theorem 6.3; the current deduction does not replace those hypotheses by a zero-noise limit. $\square$

Remark 7.6 (The part of the frontier determined here). The theorem evaluates the minimum preparation coordinate for $W \geq 12$. It does not determine $N_c^{\text{phys}}(W, Q)$ for $3 \leq W \leq 11$, the maximum score with two preparations, or the least peak width at $N = 2$. The earlier exact decision-width result $\kappa_{2/5}^{\text{rows}} = 3$ is unchanged: it allows longer training, whereas this shortest audit uses five decision states. The older points (399, 40602), (320000, 1604) and $(12800 \cdot 2^{400}, 12)$ remain valid constructions with decision width three. They are dominated by (2, 12) in the projection onto training preparations and peak width, but not necessarily in a resource order that also constrains the decision alphabet separately.

Corollary 7.7 (The enlarged physical preparation segment). *With the full architecture of Definition 6.1,*

$$N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_{\text{ex}} + \beta \leq c < m_2 - \beta.$$

Here $u_{\text{ex}} = (85 - 7\sqrt{73})/64$ is the exact one-preparation all-on value, rather than the older rational upper bound.

Proof. Restrict the adversary to the common-garbling subfamily in Definition 6.1. Theorem 3.2 and Corollary 5.2 bound every at-most-one-preparation auditor, even with unlimited memory, by $u_{\text{ex}} + \beta$. The same deterministic two-preparation machine in Lemmas 7.3 and 7.4 has score at least $m_2 - \beta$ against the full private candidate family. These inequalities prove the assertion. The four-preparation total, peak twelve and positive-noise specialization are unchanged. $\square$

8. AN EXACT PREPARATION–DECISION-MEMORY LAW

A second solvable family separates a genuine joint resource law from an implementation bound. Here the memory coordinate is the number of states at the training-validation cut. It is not identified with the bit-level peak width of the collision audit.

Fix $m = 2^d$, $0 < \alpha < 1$, and an unknown parameter $\theta \in \{1, \dots, m\}$. A fresh training preparation reports θ with probability α and an erasure $\perp$ otherwise. Conditional on θ , the preparations are independent. Candidate validation has law δ_θ and target validation is uniform on $\{1, \dots, m\}$. At the decision cut the auditor retains at most $K = 2^k$ states, $0 \leq k \leq d$. Its subsequent response may depend only on that state, current validation observations and fresh independent randomness. In particular, a calibration table, remembered seed or stopping time outside the retained state is not available. Upstream and downstream memory are otherwise unrestricted for the converse. Let $V_N^{\text{er}}(K)$ be the best worst-case expected signed validation score with at most N training preparations.

Theorem 8.1 (Exact joint law). *For every $N \geq 0$ and $K = 2^k \leq m$,*

$$(28) \quad V_N^{\text{er}}(K) = (1 - (1 - \alpha)^N) \left(1 - \frac{1}{K}\right).$$

The converse covers stochastic encoders and arbitrary legal stopping. The value is attained by a clocked machine with K training states. With atomic preparation and validation outcomes, the complete audit has peak at most $\max\{2K, 3\}$ and uses at most $N + 2$ preparations.

Proof. Write $q = (1 - \alpha)^N$ and pre-generate N potential training reports. A stopped auditor ignores the unused reports; this does not disclose extra information to it. On the event that all N reports are erasures, its retained-state law $r = (r_j)_{j=1}^K$ is independent of θ . After padding any smaller register with unused states, its channel to the decision register has the form

$$(29) \quad C(j \mid \theta) = qr_j + (1 - q)S(j \mid \theta)$$

for a stochastic channel S . This identity remains valid with randomized stopping and randomized state updates.

Use the uniform prior on parameters as an upper certificate. The part qr_j has zero expected score for every response: averaging δ_θ under this prior gives the target itself. For the other part, optimizing over all responses gives a convex function of S , a sum of positive parts of linear functions. Every stochastic channel is a convex combination of deterministic channels $j = g(\theta)$; therefore its value is at most the largest deterministic-channel value. This convexification is used only to bound a channel. It does not presuppose that a mixture of width- K machines is realizable at width K .

If the fibers of g have sizes $n_1, \dots, n_K$, a nonempty fiber has posterior uniform on that fiber, so its total variation from the uniform target is $1 - n_j/m$. The prior-averaged optimal score is

$$\sum_j \frac{n_j}{m} \left(1 - \frac{n_j}{m}\right) = 1 - \sum_j (n_j/m)^2 \leq 1 - 1/K.$$

Empty fibers contribute zero. Cauchy–Schwarz proves the inequality. This is an upper bound on the average score and hence on its minimum over fixed parameters. Multiplying by $1 - q$ proves the upper bound in (28).

For attainment, partition the parameter alphabet into K equal groups and let $g(\theta)$ denote its group. Initialize the register uniformly on the K group labels. On an erasure leave it unchanged; on a revealed parameter overwrite it by $g(\theta)$. No extra “revelation occurred” flag is needed. At the decision cut its law is $q \text{Unif}[K] + (1 - q)\delta_{g(\theta)}$. For register j , freeze the event which is the complement of group j . On a revelation its score is $1 - 1/K$. Under a uniformly random initial label, the mean score is zero for each fixed θ : one of the K labels has score $1 - 1/K$ and the others have score $-1/K$. This proves the exact worst-case value.

The k initialization bits can be accumulated in the K -state register; all earlier prefix counts are at most K . Thus no persistent seed lies outside it. Candidate validation retains the group label and its event indicator, using at most $2K$ states. Target validation produces one of the three values $-1, 0, 1$. This proves the atomic-outcome peak bound. The controller uses a declared external schedule; an autonomous stopping implementation must additionally retain its phase. A phase-tagged implementation is finite, but its cost is not included in K . $\square$

Corollary 8.2 (Thresholds and matching resource scales). *For $K \geq 2$ and $0 \leq c < 1 - 1/K$, the least integer number of training preparations giving a score strictly greater than c is*

$$(30) \quad N_c^{\text{er}}(K) = \left\lceil \frac{\log(1 - c/(1 - 1/K))}{\log(1 - \alpha)} \right\rceil + 1.$$

If $c \geq 1 - 1/K$, no finite N suffices. For $0 < \varepsilon < 1$, achieving $V_N^{\text{er}}(K) \geq 1 - \varepsilon$ requires

$$K \geq \varepsilon^{-1}, \quad N \geq \frac{\log(1/\varepsilon)}{-\log(1 - \alpha)}.$$

Conversely, whenever an allowed $K \leq m$ satisfies $K \geq 2/\varepsilon$, the choice $N \geq \log(2/\varepsilon)/[-\log(1 - \alpha)]$ suffices.

Proof. The strict threshold is equivalent to $(1 - \alpha)^N < 1 - c/(1 - 1/K)$. Since the denominator logarithm is negative, the least integer satisfying this strict inequality is (30), including equality cases of the real quotient. Also

$$1 - V_N^{\text{er}}(K) = q + (1 - q)/K.$$

This is at least q and at least $1/K$, and is at most $q + 1/K$. These three inequalities prove the necessary and sufficient bounds. $\square$

The exact law concerns an entire (N, K, α, m) family, not a single finite diagnostic. Its mechanism is a parameter-independent erasure component followed by the convex geometry of the decision channel. It demonstrates how a preparation obstruction and a memory obstruction can be matched in one theorem. It does not identify the least physical peak width at two collision preparations. Encoding the m -symbol reports bit by bit also introduces cuts not present in this atomic model; the physical bit-level accounting is given separately.

9. QUANTITATIVE CERTIFICATES FOR A FINITE COMPATIBLE ARCHITECTURE

We return to an arbitrary finite architecture, now with a finite parameter set. It can have hidden states, controlled experiments, more than two hypotheses and registers of any size. Enumerate its distinct stochastic rows by $r = 1, \dots, R$ and write $q_r \in \Delta_{s_r}$. A reused autonomous row appears only once in this product. Assume each execution uses at most H controller rows, and uses row r at most d_r times. Delete unused rows and put $S = \max_r s_r$. Fixed environmental kernels are not controller rows.

For an integer $m \geq 1$, intersect Δ_{s_r} with the coordinate boxes of side $1/m$. The resulting nonempty closed polytopes cover the simplex, with overlaps permitted at their boundaries. Write C for a product of such row cells and choose a vertex q_r^C in each factor. A row-cell vertex has at most one coordinate not fixed at a box boundary; consequently it has at most $v_r = s_r 2^{s_r - 1}$ vertices, all on the $1/m$ grid. There are at most $m^{\sum_r s_r}$ product cells.

Write the vertices of row cell C_r as $v_{r,j}$ and parametrize $q_r = \sum_j \lambda_{r,j} v_{r,j}$ with λ_r in a simplex. A reward polynomial has degree at most d_r in block λ_r . Use the homogeneous simplex Bernstein basis of degree d_r in that block, obtaining

$$(31) \quad g_\theta(q(\lambda)) = \sum_\alpha b_{\theta,C,\alpha} B_\alpha(\lambda), \quad B_\alpha \geq 0, \quad \sum_\alpha B_\alpha = 1.$$

Degree elevation includes the factors $\sum_j \lambda_{r,j} = 1$. No independence between different visits of a shared row is imposed on its unknown row vector.

Lemma 9.1 (Causal interpretation of the coefficients). *Every coefficient in (31) satisfies*

$$|b_{\theta,C,\alpha} - g_\theta(q^C)| \leq HS/m.$$

Proof. For each row r , independently prepare an ordered list of d_r vertex labels, sampled independently with law λ_r . At its k th visit, use the stochastic row given by the k th vertex label in that list. Before conditioning, the next label is independent of the previous execution, so this experiment has exactly the row q_r at every visit and expected reward $g_\theta(q(\lambda))$. Unused labels may be sampled after termination without altering the reward.

Condition on the counts of every vertex label in every list. The probability of those counts is $B_\alpha(\lambda)$. Conditional on the counts, each list is a uniformly ordered multiset. Its expected reward is independent of λ and is therefore the corresponding Bernstein coefficient. This also proves (31) directly.

Each row used by this conditioned experiment is a vertex of C_r or a mixture of its remaining vertices. Every two vectors in C_r have coordinate differences at most $1/m$, hence total variation

at most $s_r/(2m) \leq S/(2m)$. Couple the conditioned experiment to the one using q^C , using the same environmental randomness while their executions agree. At each actual controller update the probability of a first row-level disagreement is at most $S/(2m)$. There are at most H such updates on the agreeing path. The terminal laws consequently differ in total variation by at most $HS/(2m)$. A reward in $[-1, 1]$ has expectation difference at most twice that quantity. This proves the estimate. The conditioned urn is used to analyze a coefficient; it is not declared to be an admissible free-memory controller. $\square$

Theorem 9.2 (Explicit subdivision certificate). *Define*

$$\ell_C = \min_{\theta} g_{\theta}(q^C), \quad u_C = \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha}, \quad L_m = \max_C \ell_C, \quad U_m = \max_C u_C.$$

For the exact all-stochastic value of the specified architecture,

$$(32) \quad L_m \leq V(A) \leq U_m \leq L_m + HS/m.$$

On each cell, a witnessing parameter θ_C supplies the nonnegative coefficient certificate

$$u_C - g_{\theta_C}(q(\lambda)) = \sum_{\alpha} (u_C - b_{\theta_C, C, \alpha}) B_{\alpha}(\lambda).$$

Its degree in row block r is d_r , and its total degree is at most $\sum_r d_r$. At most $\prod_r \binom{d_r + v_r - 1}{v_r - 1}$ coefficients per parameter and cell are needed. If the primitive data are rational, all coefficients and bounds can be checked by rational arithmetic. Taking $m \geq HS/\varepsilon$ gives a certified interval of length at most ε .

Proof. The point q^C is feasible, so ℓ_C is a lower bound on the value restricted to C . For any $q \in C$, choose a convex representation of its rows. Equation (31) implies

$$\min_{\theta} g_{\theta}(q) \leq \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha} = u_C.$$

Choose a parameter attaining the minimum defining ℓ_C and apply Lemma 9.1; this gives $u_C \leq \ell_C + HS/m$. Taking the maximum over the finite cover proves (32). Choose θ_C attaining the definition of u_C ; all displayed coefficients of its certificate are nonnegative. The degree and count statements follow from the product Bernstein basis. Cell vertices are rational grid points, and a finite execution sums finite products, proving the rational statement. $\square$

Remark 9.3. The number of cells and coefficients can be enormous. Neither polynomial running time nor a uniform degree bound for the original sum-of-squares hierarchy follows. Refinement changes the domain cover, not merely a single global SOS degree. For a finite collection of architectures obeying a specified budget, maxima of these intervals give outer intervals; an unbounded architecture class needs a separate uniform argument such as Theorem 3.2. For compact parameter families, an effective net and a verified modulus for the primitive kernels are additional inputs, not consequences of compactness alone.

The resource estimate also makes row precision explicit. With $m = 2^b$, the lower witnesses have dyadic rows and lose at most $HS2^{-b}$ in value. Thus increasing row precision provides a quantified approximation for a fixed architecture, while its random-bit implementation still has to be charged. The physical theorem avoids this cost altogether because its rows are deterministic.

10. AUTONOMOUS NOISY TESTING WITH A PRECISION AND TIME LEDGER

This section supplies a substantive autonomous comparison rather than merely identifying shared rows. The stationary optimum is classical [6]; the proof and explicit finite-time implementation are recorded to make the resource conventions unambiguous. They are not used as novelty evidence for the physical theorem.

Let reports be independent, with law $\text{Ber}(p)$ under hypothesis $-$ and $\text{Ber}(q)$ under hypothesis $+$, where $0 < p < 1/2$, $q = 1 - p$, and $r = q/p$. A W -state autonomous machine has two time-independent stochastic transition matrices T_0, T_1 . Assume their induced chains $P_- = qT_0 + pT_1$ and $P_+ = pT_0 + qT_1$ are irreducible. A final decision uses only the state, with optional fresh terminal randomization.

Lemma 10.1 (Likelihood-range bound). *If equivalent probability laws ν_+, ν_- on a finite set satisfy*

$$\frac{\max_i \nu_+(i)/\nu_-(i)}{\min_i \nu_+(i)/\nu_-(i)} \leq R,$$

then $\text{TV}(\nu_+, \nu_-) \leq (\sqrt{R} - 1)/(\sqrt{R} + 1)$.

Proof. Write $L = d\nu_+/d\nu_-$, $a = \min L$ and $b = \max L$. Since $\mathbb{E}_{\nu_-} L = 1$, one has $a \leq 1 \leq b$. The chord bound for $(L - 1)_+$ gives

$$\text{TV}(\nu_+, \nu_-) = \mathbb{E}_{\nu_-} (L - 1)_+ \leq \frac{(1 - a)(b - 1)}{b - a}.$$

For fixed b/a , maximizing the last expression gives $a = 1/\sqrt{b/a}$; substitution yields $(\sqrt{b/a} - 1)/(\sqrt{b/a} + 1)$. It is increasing in b/a , proving the assertion. The identical-law case follows directly. $\square$

Theorem 10.2 (Autonomous stationary bound and an explicit approximation). *Put $D = W - 1$, $W \geq 2$. In the irreducible autonomous class the infimum stationary minimax error is*

$$(33) \quad e_W = \frac{1}{1 + r^D}.$$

For every $\epsilon = 2^{-b}$, $b \geq 0$, there is a machine whose rows have denominator dividing 2^{b+1} , and whose minimax error at report time n from any initial state is at most

$$(34) \quad e_W + \epsilon(W - 2) + \exp \left\{ - \left(\frac{p\epsilon}{2} \right)^D \left\lfloor \frac{n}{D} \right\rfloor \right\}.$$

The stationary lower bound is not a finite-horizon lower bound.

Proof. Every positive induced transition has its likelihood ratio in $[r^{-1}, r]$, and the supports of the two induced matrices are identical. The Markov chain tree formula writes each stationary weight as a sum of products over directed spanning trees rooted at that state. Each product has D transitions; its ratio lies in $[r^{-D}, r^D]$. The ratio of the two unnormalized stationary weights lies in the same interval. Normalizing multiplies all state likelihood ratios by a common number, so their maximum-to-minimum ratio is at most r^{2D} . Lemma 10.1 therefore bounds stationary total variation by $(r^D - 1)/(r^D + 1)$. The equal-prior Bayes error is at least $1/(1 + r^D)$, and a minimax error cannot be smaller than that Bayes error. This is the equal-prior binary specialization of the Hellman–Cover bound.

For the upper approximation, take states $i = 0, \dots, D$, weights $c_0 = c_D = 1$, and $c_i = \epsilon$ at interior states. On report 1, move from i to $i + 1$ with probability $a_i = \frac{1}{2} \min(1, c_{i+1}/c_i)$, if that neighbor exists; otherwise stay. On report 0, move to $i - 1$ with probability $b_i = \frac{1}{2} \min(1, c_{i-1}/c_i)$, if it exists; otherwise stay. All other probability is assigned to staying. Detailed balance gives

$$\pi_+(i) = \frac{c_i r^i}{\sum_j c_j r^j}, \quad \pi_-(i) = \frac{c_i r^{-i}}{\sum_j c_j r^{-j}}.$$

Reflection $i \mapsto D - i$ exchanges these laws. Decide $+$ for $i > D/2$, $-$ for $i < D/2$, and use a fair terminal decision at $i = D/2$ when this state exists. The two errors agree. Conditional on being

at an endpoint, the error is exactly e_W . The total interior stationary mass is at most $\epsilon(W - 2)$, since each $r^i \leq r^D$ and the denominator is at least r^D under $+$, with the reflected argument under $-$. This proves the first two terms of (34).

Every existing neighbor transition under either hypothesis has probability at least $p\epsilon/2$, and every self-loop has probability at least $1/2$. From any state, move down to zero and then stay until D steps have elapsed. This yields the common minorization $P_{\pm}^D(i, \{0\}) \geq \delta = (p\epsilon/2)^D$. Subtracting that common atom gives contraction in total variation by $1 - \delta$ every D steps. Comparison with the stationary law yields $\text{TV}(P_{\pm}^n(i, \cdot), \pi_{\pm}) \leq (1 - \delta)^{\lfloor n/D \rfloor} \leq e^{-\delta \lfloor n/D \rfloor}$. A bounded decision changes error by at most this amount, proving (34). Finally, letting n tend to infinity and then ϵ to zero proves that the stationary infimum equals (33). All transition probabilities belong to $\{0, 1, 1/2, \epsilon/2, 1 - 1/2, 1 - \epsilon/2\}$. $\square$

The W states in this theorem refer to an atomic stochastic update after each report. A sequential fair-bit implementation must also retain the report, the current state, a bit counter and a flag. Each nontrivial move probability is $1/2$ or $2^{-(b+1)}$; test for a string of zero bits of the required length, and pad to $b + 1$ bits if a uniform report cycle is required. At most $4W(b + 2)$ states suffice for this implementation, including its bit-phase counter and a ready phase. This conservative bound is an implementation claim, not an optimal precision–state tradeoff. The time index n in (34) is an observation time at which an external analyst reads the autonomous machine, not a free input clock used by its transitions. The adversarial model of [10] is not covered by the independent-report proof above.

11. PHYSICAL CONSUMERS AND THE FIRST-PRINCIPLES CONNECTION

The physical preparation theorem needs no estimate of the target law as a stored input: the five events are fixed and their uniform validity uses a proved neighborhood of Q_* . This removes calibration memory from this particular consumer without asserting that target acquisition is unnecessary in an arbitrary comparison problem. The general acquired calibration theorem, with its full counter register, remains in the complete companion.

Corollary 11.1 (A raw-interface confidence construction). *Let $m = m_2 - \beta > 0$ and $0 < c < m$. Repeat the full four-preparation audit independently R times and write $\overline{D}_R$ for the mean score. Under every private candidate,*

$$\mathbb{P}(\overline{D}_R \leq c) \leq \exp\{-R(m - c)^2/2\}.$$

Under the null in which the independent candidate and target validation laws agree, with the same event chosen before both validations,

$$\mathbb{P}(\overline{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

A clocked implementation uses $4R$ preparations and at most $12(2R + 1)$ persistent states, with no calibration and no random bits.

Proof. The independent block scores lie in $[-1, 1]$ and have mean at least m under a private candidate. Hoeffding's inequality gives the first bound. Under the null, conditional on the selected event, the two independent validation indicators have the same mean. Each block therefore has mean zero, and the same inequality gives the second bound. Retain the integer sum in $\{-R, \dots, R\}$ along with the current twelve-state block. At a block boundary add its terminal score and reset its register. The deterministic external schedule is part of the clocked convention. $\square$

This is an upper construction on the raw marked interface. Its proof does not transfer a lower bound from a deliberately reduced Bernoulli interface, and no optimal confidence-memory frontier is claimed.

Proposition 11.2 (Target transport without a training multiplier). *For the two-preparation audit, changing only the target from Q to Q' changes its expected score by at most $\text{TV}(Q, Q')$. Its candidate training law, register profile and event selector remain unchanged.*

Proof. Condition on the candidate training word. The expectation difference is $(Q - Q')h_S$, whose absolute value is at most $\text{TV}(Q, Q')$. Average over that unchanged training law. $\square$

In particular, every common-preparation transport estimate verified for the original positive-noise collision family can be inserted here without multiplying it by a long training horizon. If candidate kernels also change, their training and validation defects must be added by a causal coupling; the target-only statement does not remove those terms.

The first-principles route is now concrete. The positive experimental kernels and actual reset preparations give the likelihoods in (5); these generate posterior continuation states. The validation-test quotient gives the positive parts in the statistical dual. A prior then supplies an architecture-uniform impossibility certificate. On the constructive side, the five-event decision is compiled by taking its future residual functions, which are closed under observation shifts. These are compatible causal states and determine the charged profile. Thus the preparation law, predictive continuation and finite-resource realization are used in the same upper/lower theorem. This is not a deduction of geometry or nonlinear operator compactness from the existence of finite stochastic rows.

12. DEPENDENCIES, RETAINED RESULTS AND REMAINING DISTINCTIONS

The exact three-report binary noisy frontier, arbitrary-horizon binary-register minimax recursion and compatible Bayesian interval recursion remain as proved in the complete companion. Its exact odd-horizon majority realization is also retained; it is not a new result of this revision. Section 9 applies to registers of any finite width, controlled rows and any finite parameter set, but provides certified intervals rather than a closed minimax recursion for all such classes. The physical theorem supplies a matched preparation segment in a continuous two-parameter family with overlapping stochastic reports; its converse is not a finite parameter-grid check.

The stationary autonomous result in Section 10 has an explicit formula, but its stationary lower bound must not be applied to a finite-horizon clocked machine. Its finite-time upper bound is purposely conservative and exposes the slowing caused by finite transition precision. An arbitrary compact-family presentation still requires additional effective information to construct a kernel net. A subdivided positive-basis degree bound must not be called a degree bound for a global SOS hierarchy.

The physical result is a genuine consumer of the experimental and continuation root. It does not make the independent A2 geometric proofs dependent on this root. The original B4 normalized-resolvent, range, compactness and control-transfer obligations, and the broad C2 strict-dual, form, rigidity and optional-projection obligations, are not proved by our finite-state theorem. Their statements, derivations and current status remain in the retained historical pipeline. No aggregate closure is inferred from preserving their source files.

Finally, the rational certificate (25), polynomial identity (27), and residual transitions are independently reproducible. The program in the accompanying source checks their exact identities and includes deliberately invalid controls. These checks are not substitutes for the proofs above. Neither compilation, preservation of earlier pages nor a successful finite test establishes novelty or an editorial decision.

REFERENCES

- [1] R. D. Smallwood and E. J. Sondik, The optimal control of partially observable Markov processes over a finite horizon, *Operations Research* **21** (1973), 1071–1088.
- [2] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272.
- [3] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176.
- [4] J. Müller and G. Montúfar, The geometry of memoryless stochastic policy optimization in infinite-horizon POMDPs, arXiv:2110.07409 (2021; revised versions).
- [5] M. Putinar, Positive polynomials on compact semi-algebraic sets, *Indiana Univ. Math. J.* **42** (1993), 969–984.
- [6] M. E. Hellman and T. M. Cover, Learning with finite memory, *Ann. Math. Statist.* **41** (1970), 765–782.
- [7] M. E. Hellman and T. M. Cover, On memory saved by randomization, *Ann. Math. Statist.* **42** (1971), 1075–1078.
- [8] R. A. Flower and M. E. Hellman, Hypothesis testing with finite memory in finite time, *IEEE Trans. Inform. Theory* **18** (1972), 429–431.
- [9] T. M. Cover, M. A. Freedman and M. E. Hellman, Optimal finite memory learning algorithms for the finite sample problem, *Information and Control* **30** (1976), 49–85.
- [10] M. A. Managoli and V. M. Prabhakaran, Memory constrained adversarial hypothesis testing, arXiv:2605.12063v1 (May 12, 2026).
- [11] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28, doi:10.1524/strm.2002.20.14.1.
- [12] Q. Qi, General Theta Foundations I: Compatible statistical experiments and resource frontiers, revision 23, September 24, 2026, complete mathematical companion, source commit 8a6043aca644043bd805668185eb227468f21170; packaged commit e926b783b643b3e7f275a18105528927b76cc97b. The 96-page article and 649-page development are reproduced without alteration after the present article in the two complete volumes.

PRESERVED PREDECESSOR

General Theta Foundations I, revision 24

The following volume is reproduced without alteration.

The current revision precedes this divider.

Predecessor commit:

ad1336e48e414b2c01e928b844c796ab4765178b

GENERAL THETA FOUNDATIONS I: STATISTICAL RESOURCE DUALITY AND TWO-PREPARATION PHYSICAL CERTIFICATION

QIAN QI

ABSTRACT. We study the resources needed to distinguish a marked experiment from a family of private causal simulators. We separate the optimization of one finite architecture from the optimization over all admissible architectures. For reset experiments whose feedback rows are garblings of a common row, we express the unrestricted-memory preparation value as a minimum of posterior-predictive total variations. The resulting dual bounds every adaptive finite-memory architecture, without selecting an architecture in advance. For the positive-noise marked collision experiment, a four-point rational dual certificate excludes all auditors with at most one training preparation. A deterministic auditor with two training preparations attains a strictly larger uniform score. Its complete bit-level register profile has peak twelve, proving that two training preparations are necessary and sufficient at every peak budget at least twelve on an explicit interval of score thresholds. For general finite architectures we give subdivided positive-basis certificates of explicitly bounded degree and error, including reused stochastic rows. We also give a fully priced autonomous binary-testing construction, with the stationary bound attributed to Hellman and Cover. The earlier compatible-occupation, noisy-testing, continuation and physical-comparison developments are retained in the accompanying complete mathematical volumes.

CONTENTS

1. Introduction	2
1.1. Relation to the literature	2
1.2. The mathematical volumes	3
2. Experiments, architectures and attainable resources	3
3. A statistical dual that is uniform over architectures	4
4. The physical experiment and its sharp preparation boundary	6
4.1. An exact rational certificate against one preparation	6
4.2. A marked five-event selector	7
4.3. Causal compilation and the full register profile	8
5. Quantitative certificates for a finite compatible architecture	9
6. Autonomous noisy testing with a precision and time ledger	11
7. Physical consumers and the first-principles connection	12
8. Dependencies, retained results and remaining distinctions	13
References	14

Date: September 24, 2026. Revision 24.

2020 Mathematics Subject Classification. Primary 62B05, 62C20; Secondary 60J10, 68Q45, 93E20.

Key words and phrases. Statistical experiments, causal realization, predictive distributions, finite memory, minimax duality, physical certification.

1. INTRODUCTION

An exact description of finite stochastic controllers does not by itself determine a resource frontier. There are two distinct questions. The first asks for the best controller within a specified architecture. The second asks whether any architecture satisfying a resource budget can perform a specified statistical task. A lower bound for the first question need not answer the second. We develop certificates that respect this distinction.

The principal application is a marked collision experiment. Its ideal all-on law is supported on four atoms, but neither the simulator nor the auditor is restricted to a four-atom transcript. The simulator produces two independent report bits and an independent actual mark; the target has a correlation between its reports and that mark. The auditor may use the original feedback rows, arbitrary stochastic transitions and stopping rules, subject to the reset and validation interface. Our upper certificate for the auditor is valid simultaneously for all such architectures. The matching construction uses the mark during validation and not during training. This distinction permits a much shorter preparation schedule than an audit based on unmarked diagonal events.

Put

$$(1) \quad u_1 = \frac{4035777}{10240000}, \quad m_2 = \frac{12 - 3\sqrt{3}}{16}, \quad \beta_0 = \frac{1}{300}.$$

The main resource conclusion is

$$(2) \quad N_c^{\text{phys}}(W, Q) = 2 \quad \text{if} \quad W \geq 12, \quad \text{TV}(Q, Q_*) \leq \beta, \quad u_1 + \beta \leq c < m_2 - \beta.$$

Here N counts candidate training preparations; two additional, fresh validation preparations are charged separately. The objective is a strict expected signed score greater than c . In particular, (2) applies to $c = 2/5$ and $\beta = \beta_0$. The lower bound permits unlimited memory, whereas the construction has peak width twelve. This is an exact preparation boundary on a high-width segment, not a claim that twelve is the least possible peak width.

Three mechanisms organize the proof. First, a dominating experimental row removes the outer choice of feedback schedules from the converse. Second, minimax duality turns a prior on fixed simulator parameters into a posterior-predictive upper certificate. Third, residual continuation functions compile the winning decision into a compatible small-memory machine. The general algebraic encoding is useful for checking that the compiled machine belongs to the intended class, but no generic Positivstellensatz is needed to prove (2).

For finite architectures we also supply quantitative information not contained in eventual sum-of-squares completeness. A partition of the stochastic-row simplices into cells of diameter $O(m^{-1})$ gives positive-basis certificates with error at most HS/m , where H bounds the number of row uses and S bounds the number of destinations in a row. The block degree is the maximum number of times the corresponding row can be used. The argument works with autonomous row reuse. It gives an effective, generally exponential, certification procedure; it is not a uniform degree bound for the unmodified Putinar hierarchy.

1.1. Relation to the literature. Comparison of experiments and minimax theory provide the classical background [1, 2]. Polynomial feasible-frequency formulations for partially observed stochastic control are treated by Müller and Montúfar [3]; the rank-one compatibility constraints below are not presented as a new general conditional-independence device. Likewise, strict positivity certificates use established real algebraic geometry [4]. Our quantitative subdivision theorem is a positive-basis construction with a causal error estimate, rather than a new general Positivstellensatz.

Finite-state testing has a substantial earlier theory. Hellman and Cover [5, 6], Flower and Hellman [7], and Cover, Freedman and Hellman [8] distinguish time-invariant rules, finite horizons

and randomization. We explicitly identify the stationary result in Section 6 as a specialization of that theory. Managoli and Prabhakaran [9], posted in May 2026, study randomized time-invariant finite-state binary testing against adaptive adversarial distribution choices and establish matching state exponents, with exact bounds for a class of problems. Our physical theorem has a different objective: a finite training–validation signed score against a fixed private simulator, with bit-level memory and preparations charged. We make no priority claim for the general phenomenon that randomization can save states.

Norberg’s filtered comparison of experiments [10] is pertinent to the broader causal comparison program. The original proof-level comparison remains incomplete: the accessible publisher record verifies the bibliographic identity but does not justify an assertion that its results exclude the present constructions. None of our proofs uses such an exclusion.

1.2. The mathematical volumes. The present article gives the new proof chain. The accompanying *Complete Mathematical Manuscript* contains this article followed by the unaltered 96-page revision 23. The *Complete Development* contains this article followed by the unaltered 649-page development. Thus the earlier general continuation, occupation, noisy binary, physical, Gaussian and operator interfaces remain available with their full hypotheses and proofs [11]. They are not silently replaced by (2). In particular, the independent A2 geometric chain and the unresolved historical B4 and broad C2 obligations remain distinct. The mathematical role of the present first foundation is an experimental and resource-comparison root; a universal derivation of every program branch is not assumed.

2. EXPERIMENTS, ARCHITECTURES AND ATTAINABLE RESOURCES

Throughout, total variation is $\text{TV}(P, Q) = \sup_{0 \leq h \leq 1} (P - Q)h = \frac{1}{2} \sum_x |P(x) - Q(x)|$. An experimental parameter is unknown and fixed throughout an execution. Controller randomization is independent of that parameter. A reset preparation is a fresh draw, independent of earlier preparations given the parameter. A controller may not recover discarded information from an uncharged tape, clock, selector or calibration table.

Definition 2.1 (Finite architecture). A finite architecture A specifies a finite causal execution graph, its legal experimental operations, the observable inputs to each controller row, and a register alphabet at every cut. It also specifies which row occurrences share a stochastic table. A clocked architecture may use different rows at different scheduled times. An autonomous architecture may use time only through a charged state. For a parameter family Θ and terminal reward in $[-1, 1]$, let

$$V(A) = \sup_{q \in \prod_r \Delta_{sr}} \inf_{\theta \in \Theta} g_\theta(A, q).$$

If $\mathcal{A}(R)$ is the class of architectures obeying a global resource budget R , its outer value is

$$V^{\text{out}}(R) = \sup_{A \in \mathcal{A}(R)} V(A).$$

Both the architecture and the simulator parameter are fixed before the controller’s random choices. In particular, a mixture of complete controllers is not automatically an admissible controller at the original register profile.

For a finite hidden execution graph, compatibility can be stated without division. Let $x_{\theta ir}$ be the incoming mass of hidden history i at observable row r , and let $f_{\theta irj}$ be the outgoing mass through choice j . Include the usual fixed-kernel flow equations, as well as

$$(3) \quad f_{\theta irj} \geq 0, \quad \sum_j f_{\theta irj} = x_{\theta ir}, \quad f_{\theta irj} x_{\theta' i' r} = f_{\theta' i' rj} x_{\theta ir}.$$

For shared rows, the indices i, i' include all their occurrences.

Proposition 2.2 (Compatible realization). *For a specified finite architecture, nonnegative flow arrays satisfying (3) are exactly those realized by a common stochastic row at every declared observable input, including at zero incoming mass.*

Proof. Necessity follows from $f_{\theta irj} = x_{\theta ir} q_r(j)$. For sufficiency, if some incoming mass at row r is positive, define $q_r(j)$ by dividing the corresponding outgoing mass by that incoming mass. Conservation makes q_r stochastic, and the cross-products force the same ratio at every other positive incoming mass. If an incoming mass vanishes, nonnegativity and conservation force all its outgoing masses to vanish. If the entire row is unreachable, choose any legal stochastic row. Induction through the finite execution graph reconstructs the prescribed flows. The same ratio argument applies to identified occurrences of an autonomous row. $\square$

This formulation is inherited from the complete companion and is included to fix the controller class. Compact semialgebraic feasibility and an Archimedean positivity theorem provide a complete semialgebraic certificate hierarchy for a *fixed finite architecture*. They do not interchange the two optimizations in the definition. Section 5 gives a quantitative alternative for the inner problem. The next section supplies a different certificate that survives the outer optimization.

3. A STATISTICAL DUAL THAT IS UNIFORM OVER ARCHITECTURES

Let Θ be compact, let Ω be finite, and let $\theta \mapsto P_\theta \in \mathcal{P}(\Omega)$ be continuous. Fix $Q \in \mathcal{P}(\Omega)$. An N -preparation reset selector observes $Z = (Z_1, \dots, Z_N) \sim P_\theta^{\otimes N}$ and chooses a response $h_Z \in [0, 1]^\Omega$. Its expected validation score is

$$(4) \quad G_\theta(h) = \sum_{z \in \Omega^N} P_\theta^{\otimes N}(z) (Q - P_\theta) h_z.$$

The response is fixed before two independent validation draws. Operationally it can be realized by choosing a random event with those coordinate marginals and retaining that event for both validations. No memory bound is imposed in the definition of $U_N(P, Q) = \max_h \min_\theta G_\theta(h)$.

Theorem 3.1 (Posterior-predictive dual). *Both extrema in the following identity are attained:*

$$(5) \quad U_N(P, Q) = \min_{\mu \in \mathcal{P}(\Theta)} \sum_{z, \omega} \left[\int P_\theta^{\otimes N}(z) \{Q(\omega) - P_\theta(\omega)\} d\mu(\theta) \right]_+$$

$$(6) \quad = \min_{\mu \in \mathcal{P}(\Theta)} \sum_z m_\mu(z) \text{TV}(Q, \bar{P}_\mu(\cdot | z)).$$

Here $m_\mu(z) = \int P_\theta^{\otimes N}(z) d\mu(\theta)$ and, when $m_\mu(z) > 0$,

$$\bar{P}_\mu(\omega | z) = \frac{\int P_\theta^{\otimes N}(z) P_\theta(\omega) d\mu(\theta)}{m_\mu(z)}.$$

A zero-mass term in (6) is defined to be zero.

Proof. The response space is a finite-dimensional compact convex box. Probability measures on the compact space Θ form a weakly compact convex set. The integral of (4) is continuous and affine in each argument. The minimum over parameters equals the minimum over probability measures. The minimax theorem [2] therefore interchanges the maximum and minimum. For fixed μ , each response coordinate is independent in the maximization and its contribution is the positive part of its coefficient, proving (5). The coefficient vector at a fixed z has zero sum. For positive $m_\mu(z)$ it equals $m_\mu(z)(Q - \bar{P}_\mu(\cdot | z))$, proving (6). If $m_\mu(z) = 0$, every nonnegative likelihood integrand vanishes μ -almost surely and all coefficients are zero. Continuity on the two compact strategy spaces gives attainment. $\square$

Theorem 3.2 (Dominating-row converse). *Suppose every permitted training row is a parameter-independent causal garbling of one fresh draw from P_θ . Suppose every permitted validation row is the same garbling of the target and candidate laws, and every selected row–event pair is fixed before validation. Then $U_N(P, Q)$ bounds the worst-case expected score of every auditor using at most N candidate training preparations, regardless of its register profile, feedback schedule, independent randomization, or stopping rule. Independent target-only calibration, whose law does not depend on θ , does not invalidate the bound. If the all-on row and all its events are permitted, the bound is attained in the class with no memory cap and N training preparations.*

Proof. Generate N independent all-on observations in advance. At the t th requested preparation, apply the declared garbling to observation t , using only previously simulated reports and independent randomization. This simulates the adaptive transcript with its exact law. A stopped execution ignores the unused observations; it does not reveal them to the auditor. Pull back its final row–event response to the all-on alphabet. Conditional on the entire generated training word and the independent random seed, this is a function in $[0, 1]^\Omega$. Averaging the seed, including any target-only calibration, produces one response h_z for each full word z . The score is consequently (4). This proves the upper bound for the union of all architectures. Conversely, retaining the entire word and then a selected event implements any such finite response kernel when memory is unrestricted. Fresh validation gives exactly (4). $\square$

Remark 3.3 (What the dual does not convexify). The measure μ is an adversarial prior used to bound the minimum over *fixed* parameters. Its likelihood is $\int P_\theta^{\otimes N} d\mu(\theta)$, not $(\int P_\theta d\mu(\theta))^{\otimes N}$. It neither enlarges the private candidate class to mixtures nor gives the auditor a free persistent selector. Attainment of U_N without a memory cap does not assert attainment at any specified finite profile.

The state in (6) is generated by the actual preparation law, not by a geometric measure chosen independently of the experiment. For sequential continuation one retains the posterior μ_z or its future-test equivalence class; a single predictive marginal need not be closed under further observations. This is the distinction between a one-step response quotient and a future-predictive quotient.

Corollary 3.4 (Finite moment support for the product family). *For two independent Bernoulli report bits with parameters (p, q) and an independent fair mark, an optimal prior in (5) can be chosen with at most $(N + 2)^2$ atoms. The two report counts (k, ℓ) are a causally updatable sufficient statistic for computing the dual.*

Proof. Conditional on the two counts, all arrangements of report bits and all mark words have parameter-independent probabilities. Summing their coefficients reduces (5) to the $(N + 1)^2$ count pairs. Their likelihoods are

$$\binom{N}{k} p^k (1 - p)^{N-k} \binom{N}{\ell} q^\ell (1 - q)^{N-\ell}.$$

After multiplication by $Q(\omega) - P_{p,q}(\omega)$, every integrand has degree at most $N + 1$ in each variable. Thus the objective depends only on the $(N + 2)^2$ rectangular moments. Their constant coordinate is one. Carathéodory’s theorem in the remaining affine space replaces an optimal moment vector by a convex combination of at most $(N + 2)^2$ point evaluations. Counts update by addition of the current bits. No claim that this count register is optimal for the decision is made. $\square$

4. THE PHYSICAL EXPERIMENT AND ITS SHARP PREPARATION BOUNDARY

Take $\Omega = \{0, 1\}^3$, with coordinates (x_1, x_2, w) . The all-on private candidate is

$$b(p, q) = \text{Ber}(p) \otimes \text{Ber}(q) \otimes \text{Ber}(1/2), \quad 0 \leq p, q \leq 1.$$

The mark w is the actual mark, not an invented simulator bit. Put $a = (1 - p)(1 - q)$ and $c = pq$. On the ordered atoms

$$(7) \quad 000, \quad 001, \quad 110, \quad 111,$$

let Q_* have masses $5/16, 3/16, 3/16, 5/16$, and set it to zero elsewhere. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. A target Q is allowed whenever $\text{TV}(Q, Q_*) \leq \beta$.

The original feedback operation has the form

$$(8) \quad T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w), \quad g : \{0, 1\} \longrightarrow \{0, 1\}.$$

The selected gate is observable. It may depend on earlier preparations and on the first current report. The target rows are pushforwards by (8). For the converse, restrict the adversary to the admissible private subfamily whose second emission is AX_2 , where A is the current gate and $X_2 \sim \text{Ber}(q)$ is independent of the first emission and of the fair mark. Every row is then a causal garbling of $b(p, q)$. For the construction only the all-on row is used, and every private candidate in the original width-one class has the all-on product law just specified. These two facts, proved for the microscopic interface in [11, physical-complexity, feedback-completion lemma], explain why using a subfamily for the lower bound does not narrow the upper construction.

The admissible audit first trains, then selects one row–event pair, then uses one fresh candidate and one fresh target validation. It reports $D = \mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The event and row cannot be reselected after seeing a validation draw. All bit, mark, decision and validation cuts are charged. Define $N_c^{\text{phys}}(W, Q)$ to be the minimum number of candidate training preparations among these original-interface auditors, with peak register width at most W , whose score is uniformly strictly greater than c . Randomization, feedback and stopping are allowed. If no such audit exists, the value is $+\infty$.

Theorem 4.1 (Two-preparation physical segment). *Let u_1, m_2 be as in (1). For every $\beta \geq 0$, $\text{TV}(Q, Q_*) \leq \beta$, and*

$$W \geq 12, \quad u_1 + \beta \leq c < m_2 - \beta,$$

one has $N_c^{\text{phys}}(W, Q) = 2$. The two-preparation audit is deterministic, uses no acquired calibration and no random bits, and has the full profile

$$(9) \quad (1, 2, 3, 3, 4, 5, 5, 10, 12, 10, 10, 7, 3)$$

from the initial cut through its twelve bit/mark updates. In particular, its decision alphabet has five labels and its peak has twelve. The lower bound remains valid with no memory cap.

The proof occupies the next three subsections. The interval in the statement can be empty for large β ; no assertion is then made. For $\beta = 1/300$, it contains $c = 2/5$ strictly between its endpoints.

4.1. An exact rational certificate against one preparation.

Lemma 4.2. *Every original-interface auditor using at most one candidate training preparation has worst-case score at most $u_1 + \beta$, regardless of memory.*

Proof. By Theorem 3.2, it suffices to bound $U_1(b, Q_*)$ and then perturb the target. In (5) use the four fixed parameters

$$(p, q) = (1/8, 1/8), (1/5, 1/5), (4/5, 4/5), (7/8, 7/8)$$

with weights $1/16, 7/16, 7/16, 1/16$, respectively. This measure is not asserted to be least favorable. Since $p = q$ on its support, the training count $K = x_1 + x_2$ is sufficient. The independent training mark and the order of unequal bits contain no further information about this prior.

For $k = 0, 1, 2$ and the four validation atoms in (7), the signed coefficients in (5), after grouping by K , are the entries of

$$(10) \quad \frac{1}{40960000} \begin{pmatrix} -1977 & -1775177 & 2170423 & 3943623 \\ 1765554 & 191954 & 191954 & 1765554 \\ 3943623 & 2170423 & -1775177 & -1977 \end{pmatrix}.$$

For example, the coefficient in column j is

$$\sum_{i=1}^4 \mu_i \binom{2}{k} p_i^k (1-p_i)^{2-k} \left(Q_{*,j} - \frac{(1-p_i)^2 \mathbf{1}_{j \leq 2} + p_i^2 \mathbf{1}_{j \geq 3}}{2} \right).$$

This identity proves every entry of (10) by rational arithmetic. All coefficients at off-diagonal validation atoms are nonpositive. The sum of the positive entries in (10) is $4035777/10240000 = u_1$. Each response coordinate is in $[0, 1]$, so this sum bounds its prior-averaged score and hence its minimum over the four fixed candidates.

Replacing Q_* by Q changes the score of any trained response by at most β . The training law is a candidate law and is unchanged. Consequently the same prior bounds every score by $u_1 + \beta$. A procedure with no training, or one that stops before taking its permitted sample, is covered by ignoring the generated all-on observation. $\square$

4.2. A marked five-event selector. Take two independent all-on training preparations. Let S be the total number of ones among their four report bits; discard their marks. Choose the following events, represented by their indicators on (7) and with indicator zero at every off-diagonal atom:

$$(11) \quad \begin{array}{c|cccc} S & 000 & 001 & 110 & 111 \\ \hline 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & 0 \end{array}$$

Denote these indicators by $h_0, \dots, h_4$ and report $D = h_S(Y) - h_S(X)$ on independent validation draws.

Lemma 4.3 (Exact uniform score of the selector). *Against the ideal target, the selector (11) has uniform minimum score $m_2 = (12 - 3\sqrt{3})/16$. Against Q , its score is at least $m_2 - \beta$.*

Proof. The generating polynomial of S is

$$((1-p) + pz)^2((1-q) + qz)^2.$$

Writing $g_s = (Q_* - b(p, q))h_s$, its expected score is $G(p, q) = \sum_{s=0}^4 [z^s]((1-p) + pz)^2((1-q) + qz)^2 g_s$. Substitution of the five rows in (11) and collection of terms with $x = p + q - 1$, $y = p - q$, $u = x^2$ and $v = y^2$ gives

$$(12) \quad \begin{aligned} G(p, q) = H(u, v) = & -\frac{u^3}{32} + \frac{3u^2v}{32} + \frac{3u^2}{16} - \frac{3uv^2}{32} - \frac{uv}{4} - \frac{3u}{32} \\ & + \frac{v^3}{32} + \frac{v^2}{16} + \frac{15v}{32} + \frac{7}{16}. \end{aligned}$$

This is a polynomial identity, not an interpolation over a grid. Since $0 \leq u \leq 1$ and $v \geq 0$ on the parameter square,

$$\partial_v H(u, v) = \frac{3(u-v)^2 - 8u + 4v + 15}{32} \geq \frac{7}{32}.$$

Thus $H(u, v) \geq H(u, 0) = (-u^3 + 6u^2 - 3u + 14)/32$. The derivative of the last expression is $(-3u^2 + 12u - 3)/32$. On $[0, 1]$ it changes sign from negative to positive at $u = 2 - \sqrt{3}$. Substitution gives $(12 - 3\sqrt{3})/16$. The minimizing values are feasible: take $p = q = (1 \pm \sqrt{2 - \sqrt{3}})/2$. This proves exactness of the minimum for this specified selector, not optimality of its score over all selectors with two preparations. Finally, for each s , the target expectation of $h_s \in [0, 1]^\Omega$ changes by at most β . Averaging over training proves the claimed robustness. $\square$

4.3. Causal compilation and the full register profile.

Lemma 4.4. *The selector and its two validations admit the deterministic profile (9). No selector, first-bit buffer or validation accumulator is outside the charged register. An autonomous phase-tagged realization uses at most 75 states.*

Proof. After each training report bit retain the count accumulated so far. The two marks are read and discarded. The first six updates therefore have widths 2, 3, 3, 4, 5, 5 after the initial singleton. At the decision cut the five count values select the five events.

During candidate validation, after the first bit retain the event and that bit, using at most ten states. After the second bit retain the event and the function of the unread mark that gives its candidate-event indicator. For events 0, 1, 2, 3, 4, respectively, the possible functions are

$$\{0, 1\}, \quad \{0, 1 - w, 1\}, \quad \{0, 1\}, \quad \{0, w, 1\}, \quad \{0, 1\}.$$

There are twelve event–function pairs. After the candidate mark, retain the event and its now known candidate indicator $c_0 \in \{0, 1\}$, using at most ten states.

During target validation, after the first bit discard the event name and retain c_0 and its residual function of the second bit and mark. These residual functions belong to

$$\{0, x_2, (1 - x_2)(1 - w), 1 - x_2, x_2 w\}.$$

This uses at most ten states. After the second bit, the residual *difference* as a function of the unread mark belongs to

$$\{-1, 0, 1, w, 1 - w, w - 1, -w\},$$

so seven states suffice. The mark update gives the three possible outputs $-1, 0, 1$. Each transition depends only on its retained residual and the current report. Thus it is a legal common observable row, not a family of independently optimized hidden-history maps.

For completeness, equality of two residual functions implies equality of their restrictions after either next bit, so identifying equal residuals is closed under every subsequent update. The supplied transition tables are obtained by these restrictions and can be reconstructed directly from (11). Enumerating residuals is a convenience for reproducing the tables, not the proof of their sufficiency. Finally, disjointly tag the states at the thirteen cuts by their phase and use terminal self-loops. The sum of the displayed widths is 75. This implementation charges the clock; it does not assert that 75 is optimal. $\square$

Proof of Theorem 4.1. Lemma 4.2 excludes $N \leq 1$ for a strict score greater than $c \geq u_1 + \beta$, for every architecture and every memory budget. Lemmas 4.3 and 4.4 supply a two-preparation architecture of peak twelve with score at least $m_2 - \beta > c$. Larger width budgets can use the same machine. $\square$

Corollary 4.5 (The positive-noise collision family). *For the original marked collision targets with $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the inherited microscopic estimate $\text{TV}(Q_{d,\sigma}, Q_*) < 1/300$ implies*

$$N_{2/5}^{\text{phys}}(W, Q_{d,\sigma}) = 2 \quad (W \geq 12).$$

There are four preparations in the complete audit: two candidate training, one candidate validation, and one target validation.

Proof. Apply Theorem 4.1. The lower-end margin is the exact rational number

$$\frac{2}{5} - u_1 - \frac{1}{300} = \frac{78269}{30720000} > 0.$$

For the other endpoint, $\sqrt{3} < 7/4$ implies $m_2 > 27/64$ and $27/64 - 1/300 > 2/5$. The physical total-variation estimate, including the actual mark and positive sensor noise, is proved in the retained microscopic-composition and physical-complexity modules [11]; the current deduction does not replace those hypotheses by a zero-noise limit. $\square$

Remark 4.6 (The part of the frontier determined here). The theorem evaluates the minimum preparation coordinate for $W \geq 12$. It does not determine $N_c^{\text{phys}}(W, Q)$ for $3 \leq W \leq 11$, the maximum score with two preparations, or the least peak width at $N = 2$. The earlier exact decision-width result $\kappa_{2/5}^{\text{rows}} = 3$ is unchanged: it allows longer training, whereas this shortest audit uses five decision states. The older points (399, 40602), (320000, 1604) and $(12800 \cdot 2^{400}, 12)$ remain valid constructions with decision width three. They are dominated by (2, 12) in the projection onto training preparations and peak width, but not necessarily in a resource order that also constrains the decision alphabet separately.

5. QUANTITATIVE CERTIFICATES FOR A FINITE COMPATIBLE ARCHITECTURE

We return to an arbitrary finite architecture, now with a finite parameter set. It can have hidden states, controlled experiments, more than two hypotheses and registers of any size. Enumerate its distinct stochastic rows by $r = 1, \dots, R$ and write $q_r \in \Delta_{s_r}$. A reused autonomous row appears only once in this product. Assume each execution uses at most H controller rows, and uses row r at most d_r times. Delete unused rows and put $S = \max_r s_r$. Fixed environmental kernels are not controller rows.

For an integer $m \geq 1$, intersect Δ_{s_r} with the coordinate boxes of side $1/m$. The resulting nonempty closed polytopes cover the simplex, with overlaps permitted at their boundaries. Write C for a product of such row cells and choose a vertex q_r^C in each factor. A row-cell vertex has at most one coordinate not fixed at a box boundary; consequently it has at most $v_r = s_r 2^{s_r - 1}$ vertices, all on the $1/m$ grid. There are at most $m^{\sum_r s_r}$ product cells.

Write the vertices of row cell C_r as $v_{r,j}$ and parametrize $q_r = \sum_j \lambda_{r,j} v_{r,j}$ with λ_r in a simplex. A reward polynomial has degree at most d_r in block λ_r . Use the homogeneous simplex Bernstein basis of degree d_r in that block, obtaining

$$(13) \quad g_\theta(q(\lambda)) = \sum_\alpha b_{\theta,C,\alpha} B_\alpha(\lambda), \quad B_\alpha \geq 0, \quad \sum_\alpha B_\alpha = 1.$$

Degree elevation includes the factors $\sum_j \lambda_{r,j} = 1$. No independence between different visits of a shared row is imposed on its unknown row vector.

Lemma 5.1 (Causal interpretation of the coefficients). *Every coefficient in (13) satisfies*

$$|b_{\theta,C,\alpha} - g_\theta(q^C)| \leq HS/m.$$

Proof. For each row r , independently prepare an ordered list of d_r vertex labels, sampled independently with law λ_r . At its k th visit, use the stochastic row given by the k th vertex label in that list. Before conditioning, the next label is independent of the previous execution, so this experiment has exactly the row q_r at every visit and expected reward $g_\theta(q(\lambda))$. Unused labels may be sampled after termination without altering the reward.

Condition on the counts of every vertex label in every list. The probability of those counts is $B_\alpha(\lambda)$. Conditional on the counts, each list is a uniformly ordered multiset. Its expected reward is independent of λ and is therefore the corresponding Bernstein coefficient. This also proves (13) directly.

Each row used by this conditioned experiment is a vertex of C_r or a mixture of its remaining vertices. Every two vectors in C_r have coordinate differences at most $1/m$, hence total variation at most $s_r/(2m) \leq S/(2m)$. Couple the conditioned experiment to the one using q^C , using the same environmental randomness while their executions agree. At each actual controller update the probability of a first row-level disagreement is at most $S/(2m)$. There are at most H such updates on the agreeing path. The terminal laws consequently differ in total variation by at most $HS/(2m)$. A reward in $[-1, 1]$ has expectation difference at most twice that quantity. This proves the estimate. The conditioned urn is used to analyze a coefficient; it is not declared to be an admissible free-memory controller. $\square$

Theorem 5.2 (Explicit subdivision certificate). *Define*

$$\ell_C = \min_{\theta} g_{\theta}(q^C), \quad u_C = \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha}, \quad L_m = \max_C \ell_C, \quad U_m = \max_C u_C.$$

For the exact all-stochastic value of the specified architecture,

$$(14) \quad L_m \leq V(A) \leq U_m \leq L_m + HS/m.$$

On each cell, a witnessing parameter θ_C supplies the nonnegative coefficient certificate

$$u_C - g_{\theta_C}(q(\lambda)) = \sum_{\alpha} (u_C - b_{\theta_C, C, \alpha}) B_{\alpha}(\lambda).$$

Its degree in row block r is d_r , and its total degree is at most $\sum_r d_r$. At most $\prod_r \binom{d_r + v_r - 1}{v_r - 1}$ coefficients per parameter and cell are needed. If the primitive data are rational, all coefficients and bounds can be checked by rational arithmetic. Taking $m \geq HS/\varepsilon$ gives a certified interval of length at most ε .

Proof. The point q^C is feasible, so ℓ_C is a lower bound on the value restricted to C . For any $q \in C$, choose a convex representation of its rows. Equation (13) implies

$$\min_{\theta} g_{\theta}(q) \leq \min_{\theta} \max_{\alpha} b_{\theta, C, \alpha} = u_C.$$

Choose a parameter attaining the minimum defining ℓ_C and apply Lemma 5.1; this gives $u_C \leq \ell_C + HS/m$. Taking the maximum over the finite cover proves (14). Choose θ_C attaining the definition of u_C ; all displayed coefficients of its certificate are nonnegative. The degree and count statements follow from the product Bernstein basis. Cell vertices are rational grid points, and a finite execution sums finite products, proving the rational statement. $\square$

Remark 5.3. The number of cells and coefficients can be enormous. Neither polynomial running time nor a uniform degree bound for the original sum-of-squares hierarchy follows. Refinement changes the domain cover, not merely a single global SOS degree. For a finite collection of architectures obeying a specified budget, maxima of these intervals give outer intervals; an unbounded architecture class needs a separate uniform argument such as Theorem 3.2. For compact parameter families, an effective net and a verified modulus for the primitive kernels are additional inputs, not consequences of compactness alone.

The resource estimate also makes row precision explicit. With $m = 2^b$, the lower witnesses have dyadic rows and lose at most $HS2^{-b}$ in value. Thus increasing row precision provides a quantified approximation for a fixed architecture, while its random-bit implementation still has to be charged. The physical theorem avoids this cost altogether because its rows are deterministic.

6. AUTONOMOUS NOISY TESTING WITH A PRECISION AND TIME LEDGER

This section supplies a substantive autonomous comparison rather than merely identifying shared rows. The stationary optimum is classical [5]; the proof and explicit finite-time implementation are recorded to make the resource conventions unambiguous. They are not used as novelty evidence for the physical theorem.

Let reports be independent, with law $\text{Ber}(p)$ under hypothesis $-$ and $\text{Ber}(q)$ under hypothesis $+$, where $0 < p < 1/2$, $q = 1 - p$, and $r = q/p$. A W -state autonomous machine has two time-independent stochastic transition matrices T_0, T_1 . Assume their induced chains $P_- = qT_0 + pT_1$ and $P_+ = pT_0 + qT_1$ are irreducible. A final decision uses only the state, with optional fresh terminal randomization.

Lemma 6.1 (Likelihood-range bound). *If equivalent probability laws ν_+, ν_- on a finite set satisfy*

$$\frac{\max_i \nu_+(i)/\nu_-(i)}{\min_i \nu_+(i)/\nu_-(i)} \leq R,$$

then $\text{TV}(\nu_+, \nu_-) \leq (\sqrt{R} - 1)/(\sqrt{R} + 1)$.

Proof. Write $L = d\nu_+/d\nu_-$, $a = \min L$ and $b = \max L$. Since $\mathbb{E}_{\nu_-} L = 1$, one has $a \leq 1 \leq b$. The chord bound for $(L - 1)_+$ gives

$$\text{TV}(\nu_+, \nu_-) = \mathbb{E}_{\nu_-} (L - 1)_+ \leq \frac{(1 - a)(b - 1)}{b - a}.$$

For fixed b/a , maximizing the last expression gives $a = 1/\sqrt{b/a}$; substitution yields $(\sqrt{b/a} - 1)/(\sqrt{b/a} + 1)$. It is increasing in b/a , proving the assertion. The identical-law case follows directly. $\square$

Theorem 6.2 (Autonomous stationary bound and an explicit approximation). *Put $D = W - 1$, $W \geq 2$. In the irreducible autonomous class the infimum stationary minimax error is*

$$(15) \quad e_W = \frac{1}{1 + r^D}.$$

For every $\epsilon = 2^{-b}$, $b \geq 0$, there is a machine whose rows have denominator dividing 2^{b+1} , and whose minimax error at report time n from any initial state is at most

$$(16) \quad e_W + \epsilon(W - 2) + \exp \left\{ - \left(\frac{p\epsilon}{2} \right)^D \left\lfloor \frac{n}{D} \right\rfloor \right\}.$$

The stationary lower bound is not a finite-horizon lower bound.

Proof. Every positive induced transition has its likelihood ratio in $[r^{-1}, r]$, and the supports of the two induced matrices are identical. The Markov chain tree formula writes each stationary weight as a sum of products over directed spanning trees rooted at that state. Each product has D transitions; its ratio lies in $[r^{-D}, r^D]$. The ratio of the two unnormalized stationary weights lies in the same interval. Normalizing multiplies all state likelihood ratios by a common number, so their maximum-to-minimum ratio is at most r^{2D} . Lemma 6.1 therefore bounds stationary total variation by $(r^D - 1)/(r^D + 1)$. The equal-prior Bayes error is at least $1/(1 + r^D)$,

and a minimax error cannot be smaller than that Bayes error. This is the equal-prior binary specialization of the Hellman–Cover bound.

For the upper approximation, take states $i = 0, \dots, D$, weights $c_0 = c_D = 1$, and $c_i = \epsilon$ at interior states. On report 1, move from i to $i + 1$ with probability $a_i = \frac{1}{2} \min(1, c_{i+1}/c_i)$, if that neighbor exists; otherwise stay. On report 0, move to $i - 1$ with probability $b_i = \frac{1}{2} \min(1, c_{i-1}/c_i)$, if it exists; otherwise stay. All other probability is assigned to staying. Detailed balance gives

$$\pi_+(i) = \frac{c_i r^i}{\sum_j c_j r^j}, \quad \pi_-(i) = \frac{c_i r^{-i}}{\sum_j c_j r^{-j}}.$$

Reflection $i \mapsto D - i$ exchanges these laws. Decide $+$ for $i > D/2$, $-$ for $i < D/2$, and use a fair terminal decision at $i = D/2$ when this state exists. The two errors agree. Conditional on being at an endpoint, the error is exactly e_W . The total interior stationary mass is at most $\epsilon(W - 2)$, since each $r^i \leq r^D$ and the denominator is at least r^D under $+$, with the reflected argument under $-$. This proves the first two terms of (16).

Every existing neighbor transition under either hypothesis has probability at least $p\epsilon/2$, and every self-loop has probability at least $1/2$. From any state, move down to zero and then stay until D steps have elapsed. This yields the common minorization $P_\pm^D(i, \{0\}) \geq \delta = (p\epsilon/2)^D$. Subtracting that common atom gives contraction in total variation by $1 - \delta$ every D steps. Comparison with the stationary law yields $\text{TV}(P_\pm^n(i, \cdot), \pi_\pm) \leq (1 - \delta)^{\lfloor n/D \rfloor} \leq e^{-\delta \lfloor n/D \rfloor}$. A bounded decision changes error by at most this amount, proving (16). Finally, letting n tend to infinity and then ϵ to zero proves that the stationary infimum equals (15). All transition probabilities belong to $\{0, 1, 1/2, \epsilon/2, 1 - 1/2, 1 - \epsilon/2\}$. $\square$

The W states in this theorem refer to an atomic stochastic update after each report. A sequential fair-bit implementation must also retain the report, the current state, a bit counter and a flag. Each nontrivial move probability is $1/2$ or $2^{-(b+1)}$; test for a string of zero bits of the required length, and pad to $b + 1$ bits if a uniform report cycle is required. At most $4W(b + 2)$ states suffice for this implementation, including its bit-phase counter and a ready phase. This conservative bound is an implementation claim, not an optimal precision–state tradeoff. The time index n in (16) is an observation time at which an external analyst reads the autonomous machine, not a free input clock used by its transitions. The adversarial model of [9] is not covered by the independent-report proof above.

7. PHYSICAL CONSUMERS AND THE FIRST-PRINCIPLES CONNECTION

The physical preparation theorem needs no estimate of the target law as a stored input: the five events are fixed and their uniform validity uses a proved neighborhood of Q_* . This removes calibration memory from this particular consumer without asserting that target acquisition is unnecessary in an arbitrary comparison problem. The general acquired calibration theorem, with its full counter register, remains in the complete companion.

Corollary 7.1 (A raw-interface confidence construction). *Let $m = m_2 - \beta > 0$ and $0 < c < m$. Repeat the full four-preparation audit independently R times and write $\bar{D}_R$ for the mean score. Under every private candidate,*

$$\mathbb{P}(\bar{D}_R \leq c) \leq \exp\{-R(m - c)^2/2\}.$$

Under the null in which the independent candidate and target validation laws agree, with the same event chosen before both validations,

$$\mathbb{P}(\bar{D}_R > c) \leq \exp\{-Rc^2/2\}.$$

A clocked implementation uses $4R$ preparations and at most $12(2R + 1)$ persistent states, with no calibration and no random bits.

Proof. The independent block scores lie in $[-1, 1]$ and have mean at least m under a private candidate. Hoeffding's inequality gives the first bound. Under the null, conditional on the selected event, the two independent validation indicators have the same mean. Each block therefore has mean zero, and the same inequality gives the second bound. Retain the integer sum in $\{-R, \dots, R\}$ along with the current twelve-state block. At a block boundary add its terminal score and reset its register. The deterministic external schedule is part of the clocked convention. $\square$

This is an upper construction on the raw marked interface. Its proof does not transfer a lower bound from a deliberately reduced Bernoulli interface, and no optimal confidence-memory frontier is claimed.

Proposition 7.2 (Target transport without a training multiplier). *For the two-preparation audit, changing only the target from Q to Q' changes its expected score by at most $\text{TV}(Q, Q')$. Its candidate training law, register profile and event selector remain unchanged.*

Proof. Condition on the candidate training word. The expectation difference is $(Q - Q')h_S$, whose absolute value is at most $\text{TV}(Q, Q')$. Average over that unchanged training law. $\square$

In particular, every common-preparation transport estimate verified for the original positive-noise collision family can be inserted here without multiplying it by a long training horizon. If candidate kernels also change, their training and validation defects must be added by a causal coupling; the target-only statement does not remove those terms.

The first-principles route is now concrete. The positive experimental kernels and actual reset preparations give the likelihoods in (5); these generate posterior continuation states. The validation-test quotient gives the positive parts in the statistical dual. A prior then supplies an architecture-uniform impossibility certificate. On the constructive side, the five-event decision is compiled by taking its future residual functions, which are closed under observation shifts. These are compatible causal states and determine the charged profile. Thus the preparation law, predictive continuation and finite-resource realization are used in the same upper/lower theorem. This is not a deduction of geometry or nonlinear operator compactness from the existence of finite stochastic rows.

8. DEPENDENCIES, RETAINED RESULTS AND REMAINING DISTINCTIONS

The exact three-report binary noisy frontier, arbitrary-horizon binary-register minimax recursion and compatible Bayesian interval recursion remain as proved in the complete companion. Its exact odd-horizon majority realization is also retained; it is not a new result of this revision. Section 5 applies to registers of any finite width, controlled rows and any finite parameter set, but provides certified intervals rather than a closed minimax recursion for all such classes. The physical theorem supplies a matched preparation segment in a continuous two-parameter family with overlapping stochastic reports; its converse is not a finite parameter-grid check.

The stationary autonomous result in Section 6 has an explicit formula, but its stationary lower bound must not be applied to a finite-horizon clocked machine. Its finite-time upper bound is purposely conservative and exposes the slowing caused by finite transition precision. An arbitrary compact-family presentation still requires additional effective information to construct a kernel net. A subdivided positive-basis degree bound must not be called a degree bound for a global SOS hierarchy.

The physical result is a genuine consumer of the experimental and continuation root. It does not make the independent A2 geometric proofs dependent on this root. The original B4 normalized-resolvent, range, compactness and control-transfer obligations, and the broad C2 strict-dual, form, rigidity and optional-projection obligations, are not proved by our finite-state theorem. Their statements, derivations and current status remain in the retained historical pipeline. No aggregate closure is inferred from preserving their source files.

Finally, the rational certificate (10), polynomial identity (12), and residual transitions are independently reproducible. The program in the accompanying source checks their exact identities and includes deliberately invalid controls. These checks are not substitutes for the proofs above. Neither compilation, preservation of earlier pages nor a successful finite test establishes novelty or an editorial decision.

REFERENCES

- [1] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272.
- [2] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176.
- [3] J. Müller and G. Montúfar, The geometry of memoryless stochastic policy optimization in infinite-horizon POMDPs, arXiv:2110.07409 (2021; revised versions).
- [4] M. Putinar, Positive polynomials on compact semi-algebraic sets, *Indiana Univ. Math. J.* **42** (1993), 969–984.
- [5] M. E. Hellman and T. M. Cover, Learning with finite memory, *Ann. Math. Statist.* **41** (1970), 765–782.
- [6] M. E. Hellman and T. M. Cover, On memory saved by randomization, *Ann. Math. Statist.* **42** (1971), 1075–1078.
- [7] R. A. Flower and M. E. Hellman, Hypothesis testing with finite memory in finite time, *IEEE Trans. Inform. Theory* **18** (1972), 429–431.
- [8] T. M. Cover, M. A. Freedman and M. E. Hellman, Optimal finite memory learning algorithms for the finite sample problem, *Information and Control* **30** (1976), 49–85.
- [9] M. A. Managoli and V. M. Prabhakaran, Memory constrained adversarial hypothesis testing, arXiv:2605.12063v1 (May 12, 2026).
- [10] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28, doi:10.1524/strm.2002.20.14.1.
- [11] Q. Qi, General Theta Foundations I: Compatible statistical experiments and resource frontiers, revision 23, September 24, 2026, complete mathematical companion, source commit 8a6043aca644043bd805668185eb227468f21170; packaged commit e926b783b643b3e7f275a18105528927b76cc97b. The 96-page article and 649-page development are reproduced without alteration after the present article in the two complete volumes.

General Theta Foundations I

Complete Development

Part II: preserved revision 23

The following 649 pages reproduce the predecessor volume without alteration.

Part I contains the new revision 24 proof chain. Earlier theorem numbers in Part II retain their original meaning. The source modules, proof qualifications and historical references remain available in the accompanying repository and source archive.

GENERAL THETA FOUNDATIONS I: COMPATIBLE STATISTICAL EXPERIMENTS AND RESOURCE FRONTIERS

QIAN QI

ABSTRACT. We study statistical experiments subject to a register constraint at each cut of a finite causal protocol. Their feasible occupation arrays are characterized by linear flow equations and quadratic compatibility equations, including at zero-probability histories. This yields constructive realizations and a complete polynomial upper-certificate hierarchy for the full-profile minimax value, without convexifying the class of machines. For binary experiments we obtain an exact posterior partition recursion for Bayesian decisions and an exact receiver-curve recursion for minimax decisions with binary registers. In a three-report experiment with overlapping binary noise, we compute the entire register-profile frontier for both objectives; the resulting strict gap shows that a free mixture of machines changes the minimax value. The characterization is connected to finite-precision response-cover auditors, acquired calibration, and positive-noise physical experiments, with preparations, intermediate registers and clocks explicitly charged. The marked continuation, residual, microscopic and Gaussian comparison results are developed with their proofs and with their distinct hypotheses retained.

CONTENTS

1. Introduction	3
1.1. The full-profile problem	3
1.2. Comparison with existing methods	4
1.3. Organization and scope of the applications	5
2. Compatible occupation arrays	5
2.1. The finite protocol and its observable rows	5
2.2. A complete polynomial converse	7
3. Continuation polytopes and noisy posterior partitions	8
3.1. One common continuation matrix	8
3.2. Binary posterior geometry	9
3.3. An exact minimax recursion for binary registers	11
4. An exact noisy frontier and the cost of hidden mixing	12
5. Acquisition, clocks, and the resource order	15
5.1. Comparable resource points	15
5.2. Acquiring a calibration rather than assuming it	16
6. Consumers of the compatible frontier	17
6.1. Reset scores and physical protocols	17
6.2. An exact confidence consumer	18
6.3. The foundation and the longer program	19
7. Intrinsic causal deficiency	19
7.1. Experiments and executable signatures	19

Date: September 24, 2026.

2020 Mathematics Subject Classification. Primary 62B05, 93E20; Secondary 60G44, 68Q45, 82C40.

Key words and phrases. Statistical experiments, finite memory, causal realization, minimax testing, likelihood-ratio quantization, polynomial certificates.

7.2.	The finite comparison theorem	20
7.3.	A delayed channel and a strict signature separation	22
7.4.	Optimal resource sets and decision certificates	24
8.	Continuation geometry and an exact memory obstruction	24
8.1.	Normalized continuation laws	24
8.2.	An erasure cut	26
9.	Decision continuation spectra	26
9.1.	A charged training-validation interface	27
9.2.	Several feedback rows	29
10.	Finite precision and charged randomization	29
11.	Approximate continuation profiles and resource transfer	30
11.1.	A causal comparison norm	30
11.2.	The full value and its effective approximation	31
12.	From a decision cover to an end-to-end machine	33
12.1.	The whole-schedule continuation value	33
13.	Finite-horizon sample-memory bounds	35
14.	An exact statistical frontier at all cuts	37
15.	Transport priced by active updates	39
16.	Weighted continuation obstructions	40
17.	Joint residual budgets and growing families	42
17.1.	Actual physical word weights	43
17.2.	A family with matching end-to-end growth	43
17.3.	A sharp simultaneous storage profile	44
18.	Finite-data testing on the original behavior image	44
18.1.	The reset interface	44
18.2.	A matching uniform sampling obstruction	46
18.3.	Transport of one fixed audit	47
19.	Continuation complexity of a finite-data test	47
19.1.	The width required to attain the binary reset value	48
20.	Polynomial testing of the original resource image	49
20.1.	A sharp two-time nonconvex example	51
21.	Adaptive testing on the observable behavior image	52
21.1.	The testing game	52
21.2.	A size bound in behavior dimension	53
21.3.	A three-weight certificate invisible to constant tests	55
21.4.	Strict certificates and effective physical input	56
22.	An exact algorithm for the observable chart	56
23.	A marked resource gap generated by a binary collision	57
23.1.	The exact marked two-report experiment	58
23.2.	A full-dimensional collision preparation	59
23.3.	Uniformly effective finite presentations	60
24.	Uniform stability of optional projections	61
24.1.	A microscopic sufficient condition	63
25.	Changing trajectories, likelihoods, and conditional consumers	64
25.1.	A common preparation, not a common flow	64
25.2.	The projected stochastic exponential	65
25.3.	A bounded entropic backward-equation consumer	66
25.4.	Resource transport is not the posterior coupling	67
26.	Observation densities and coupled process versions	68

27. Global transport of the backward equation	69
28. A uniform certificate through a moving collision	71
29. Sharp decision memory for the collision experiment	74
29.1. The marked family and its first two spectral values	74
29.2. A three-response construction	76
29.3. Simultaneous exact widths and the full acquisition schedule	77
29.4. What the two exact complexity statements measure	79
30. Full-profile certification of the collision family	79
30.1. The row-indexed decision invariant	79
31. A third resource regime for the same physical task	81
32. Confidence-level certification	82
33. Confidence tests with priced finite-state amplification	84
34. A finite-register certificate transported through the microscopic limit	85
35. Explicit phase transitions for the two-count certificate	87
Appendix A. The proof dependency chain	88
A.1. The full-profile resource chain	88
Appendix B. A namespaced map of the eleven-component program	90
Appendix C. Preserved organizing statements	90
C.1. Inherited baseline	92
Appendix D. Unrestricted-horizon multicut realization	93
Appendix E. Dependencies of the statistical resource frontier	94
Appendix F. The constructive resource theorem	94
References	95

1. INTRODUCTION

An experiment is used through a causal observer. Information acquired at one time may be lost at a later checkpoint, and a decision rule that is economical after training may be expensive to train. The statistical value of an observer therefore depends on its entire register profile, not only on the dimension of its last statistic. A further difficulty is that two feasible observers need not be mixable at the same profile: the variable that selects an observer has to be retained wherever their future actions differ.

We study this problem from positive experimental kernels, preparations, causal interventions and bounded executable tests. These objects do not presuppose a convex parameter image, an information metric, or a finite-dimensional closure of the underlying dynamics. The present paper develops their finite resource layer and its comparison maps. The word “finite” refers to a declared protocol and its observable alphabets, not to a removal of the microscopic or limiting questions in the larger theory.

1.1. The full-profile problem. Fix a finite schedule σ , a register profile $\mathbf{K}$, a family of environments B , and a terminal payoff $G \in [-1, 1]$. Let

$$V_{\sigma, \mathbf{K}}^{\text{clk}}(B, G) = \sup_A \inf_{b \in B} \mathbb{E}_b^A G,$$

where every persistent data-dependent variable of A passes through the specified registers. The clocked model allows a predetermined transition table at each epoch; its clock carries no discarded data. An autonomous controller instead reuses its declared transition rows and must retain any phase variable needed to select a different row. Refining an atomic operation into input-bit or random-bit epochs changes σ and its resource problem. These conventions are part of the theorem statements.

The central object is a compatible occupation array. Its coordinates record the probability mass entering and leaving each observable controller row, under every environment and hidden history. Ordinary flow conservation is necessary but not sufficient. All hidden histories that present the same observable row must use the same stochastic transition. This requirement is a collection of quadratic proportionality equations. Unlike a division-based parametrization, these equations also retain every zero-occupation boundary case.

Theorem 1.1 (Compatible statistical resource frontiers). *The following statements hold for the indicated finite protocols.*

- (i) *For a finite environment family, compatible occupation arrays are exactly the arrays of all stochastic machines of the full profile. Their minimax value is attained. It equals the decreasing limit of polynomial sum-of-squares upper certificates, and every feasible array reconstructs a machine. Shared-row equations give the corresponding autonomous characterization. Finite-precision controller witnesses and compact-family kernel nets give matching two-sided approximations.*
- (ii) *For binary, conditionally independent finite observations, possibly with a finite controlled choice of experiment, the optimal clocked Bayesian value at any finite register profile is given by a recursion of compatible interval partitions of incoming posterior atoms. For uncontrolled observations with one- or two-state registers, a receiver-curve recursion gives the exact clocked minimax value at every finite horizon. These are optimizations over all stochastic machines, not implementation bounds for a chosen statistic.*
- (iii) *With three independent binary reports of error probability $p \in (0, 1/2)$, put $d = p(1-p)(1-2p)$. At the atomic profile $(2, 2, 2)$ the exact fair-prior and minimax successes are respectively $1 - p + d/2$ and $1 - p + d/3$. At profile $(2, 3, 2)$ both equal $1 - p + d$. A free mixture of the two Bayesian optimal machines would raise the first profile's minimax value by $d/6$, so that convexification is strictly incorrect for that resource problem.*
- (iv) *A fixed K -response calibration can be acquired from $M = 2^s$ target preparations, with uniform error at most ρ except with probability β , whenever $M \geq (2\rho^2)^{-1} \log(2K/\beta)$. Its $(M + 1)^K$ -state counter vector is retained and priced through the subsequent auditor. The resulting response-cover constructions and their physical consumers belong to the same full-profile classes as the occupation converse.*

Proof. Part (i) is Theorems 2.1 and 2.2 and Corollary 2.3. Part (ii) is Theorems 3.2 and 3.3. Part (iii) is Theorem 4.1 and Corollary 4.3. Part (iv) is Theorem 5.2 and Corollary 6.1. $\square$

The certificate hierarchy in (i) uses a classical positivity theorem. Its role is to give an upper bound on every complete machine, paired with the reverse reconstruction from feasible occupations. It is not an efficiency or finite-termination assertion. The noisy recursions in (ii) and the closed formulas in (iii) supply structural information that a generic polynomial encoding alone would not provide. In particular, the distinction between Bayesian and minimax objectives survives in an experiment whose two laws have common support at every observation.

1.2. Comparison with existing methods. Comparison of experiments is classical [1, 2, 13]. Filtered randomization is also an existing subject. Weisshaupt's filtered criterion [10, Lemma 1 and Theorem 5] concerns compact convex classes of positive operators and a risk criterion for filtered randomization. Our feasible object is a specified finite register execution, whose behavior set need not be convex. We also cite Norberg [3]; the original proof text has not been obtained for a definitive theorem-level comparison, and no priority claim rests on that gap in access.

Finite-memory statistical learning and the role of time-dependent transition rules are treated by Cover and by Hellman and Cover [24, 26]. Likelihood-ratio quantizer optimality is a classical mechanism [27, Section III]. We use it both for compatible posterior partitions and for binary-channel domination. The receiver-curve and equal-error arguments are not introduced as new

statistical principles. Their role here is to solve one full-profile optimization, including its stochastic minimax part and the selector cost exposed by the explicit three-report frontier.

Polynomial descriptions of partially observed controller occupations are likewise established. Müller and Montúfar [28, Example 15, Theorem 16 and Remark 19] give semialgebraic feasible-frequency descriptions and polynomial planning for infinite-horizon memoryless POMDPs, with their stated positivity assumptions. The finite-protocol formulation here makes the observable row equivalences, changing register alphabets and zero-mass reconstruction explicit. Its upper-certificate completeness applies Putinar’s theorem and its rational version [29, 30]. The combined contribution is the exact compatible frontier, its noisy evaluations, and its connections to fully priced experimental consumers; neither polynomial optimization nor the vanishing-minor identity is claimed as a new invention.

1.3. Organization and scope of the applications. Sections 2–6 contain the principal proof chain. The marked continuation and decision spectra then identify the underlying experimental interfaces. The finite precision, response-cover tournaments, multicut revelation formula, weighted residuals, nonlinear tests and adaptive tests are developed with their proofs. The microscopic moving-collision construction and the bounded Gaussian comparison produce physical consumers of the same interfaces. The original short, polynomial-horizon and long-clock physical resource points are all retained.

Those three points are constructions, not an exact physical Pareto curve. The general converse now applies to each of their fixed protocols, but its numerical evaluation across all physical budgets is a separate task. Likewise a confidence lower bound at a specified retained-coin interface is not automatically a lower bound for an auditor that uses richer raw observations. These distinctions are mathematical typing conditions on the applications, not changes to the physical or operator targets.

The article retains the full development of its supporting results; versioned derivation and review records are kept outside the exposition. The dependency appendix records the namespaced eleven-component program. The primary A2 geometric chain remains independent, while B4 kinetic and broader C2 operator conclusions require their own stated hypotheses and proofs. No artificial dependence on a finite-state theorem is imposed on those branches of the program.

2. COMPATIBLE OCCUPATION ARRAYS

A fixed terminal statistic and an optimal statistic under a sequence of memory constraints are different objects. We begin with a characterization of the second object. All the spaces in this section are finite. No strict positivity assumption is imposed on a probability coordinate.

2.1. The finite protocol and its observable rows. Fix a finite schedule σ and a register profile $\mathbf{K} = (K_t)_{t \in \sigma}$. A cut is part of the specification of σ : subdividing an atomic observation or a randomization operation inserts additional register constraints. The clocked model permits a different transition table at each scheduled operation. It does not permit access to a discarded observation. We write $\mathcal{M}_\sigma^{\text{clk}}(\mathbf{K})$ for this class. Initial randomization, when allowed, is a transition of the same kind as all other randomizations.

Let Θ be a finite set of environments. Under θ , all preparation and experimental kernels are specified. One may include the complete physical and public past in a finite hidden index i . This index is a variable in the analysis, not an observer register. An elementary observer operation has an accessible row r and chooses j from a nonempty finite set J_r . The row contains exactly the current register and the current atomic input, if there is one. The choice j can specify a new register and a public action. A public action that is subsequently needed by the observer must be

retained in that register. Every permissible transition has the form

$$u(j \mid r), \quad u(j \mid r) \geq 0, \quad \sum_{j \in J_r} u(j \mid r) = 1. \quad (2.1)$$

Neither θ nor the unobserved part of i is an argument of u .

Known environment operations act linearly on conditional occupation masses. For example, if $z(\theta, h, m)$ is the mass before an observation, then its incoming mass at the observer is

$$x(\theta, h', (m, y)) = \sum_h z(\theta, h, m) L^\theta(h', y \mid h, m). \quad (2.2)$$

Dependence on a previously issued action is included in h or in the retained part of m . After an observer choice, the public history is updated and discarded indices are summed out, again by a known linear map. Alternating these two kinds of operation represents a finite controlled experiment, including feedback, calibration, validation, and a terminal readout. The cut alphabets and the allowed rows are fixed before optimization. In particular, this representation does not silently grant the observer the hidden history used to define the experimental kernels.

For each observer operation introduce outgoing masses $f(\theta, i, r, j)$. Impose the conservation and compatibility equations

$$f(\theta, i, r, j) \geq 0, \quad \sum_{j \in J_r} f(\theta, i, r, j) = x(\theta, i, r), \quad (2.3)$$

$$f(\theta, i, r, j) x(\theta', i', r) = f(\theta', i', r, j) x(\theta, i, r) \quad (\theta, i, \theta', i', r, j). \quad (2.4)$$

All incoming masses are nonnegative and are linked to the preceding operation by the known linear maps. Preparations are prescribed and have conditional mass one for each θ . We call the resulting set of arrays $\mathcal{O}_\sigma^{\text{clk}}(\mathbf{K})$.

Theorem 2.1 (Exact full-profile realization). *The occupation arrays of $\mathcal{M}_\sigma^{\text{clk}}(\mathbf{K})$ are exactly $\mathcal{O}_\sigma^{\text{clk}}(\mathbf{K})$. This set is compact and is defined by linear equations, nonnegativity, and the quadratic equations (2.4). Every feasible array determines an implementing machine, with arbitrary legal rows only where the incoming mass vanishes under every environment.*

More generally, suppose specified occurrences of observable rows must use the same transition distribution, with their choices identified. For each such class impose (2.4) also between different occurrences. The resulting arrays are exactly those realized by machines with these shared rows. In particular this gives an exact formulation for an autonomous controller with a fixed state alphabet and a single reused transition table.

Proof. A machine gives $f(\theta, i, r, j) = x(\theta, i, r) u(j \mid r)$, so conservation and every minor equation follow. Conversely, fix an occurrence of r . If some $x(\theta_0, i_0, r) > 0$, define

$$u(j \mid r) = \frac{f(\theta_0, i_0, r, j)}{x(\theta_0, i_0, r)}.$$

Conservation makes this a stochastic row. The minor with (θ_0, i_0) shows that the same formula reproduces f at every positive incoming mass. At a zero incoming mass, nonnegativity and conservation force each outgoing mass to be zero. If all incoming masses vanish, choose any legal stochastic row. Thus the recovered row is independent of both the environment and the hidden index.

Perform this construction at every observer operation. Induction along the schedule, using (2.2) and the corresponding linear outgoing maps, proves that the recovered machine has exactly the given occupation arrays. Its registers are the declared alphabets; the proof has not introduced a retained selector. For a shared row class, choose one positive incoming mass anywhere in that class and use the cross-occurrence equations. The same argument reconstructs one common row. A class with no positive incoming mass is treated arbitrarily.

Each coordinate is a conditional probability or a subprobability mass and is therefore at most one. Adding these redundant upper bounds makes boundedness explicit. The equations and inequalities are closed, and there are finitely many coordinates. Compactness follows. $\square$

The zero-mass provision is essential. Cancelling an occupation coordinate without a case distinction would remove valid boundary machines. Nor would separate low-rank descriptions at successive cuts imply (2.2)–(2.4). The latter equations refer to one execution, not to a list of independently selected encoders. For an autonomous machine, the phase cannot be supplied through a row name unless it is either an actual input or part of the charged state. A clocked machine can be embedded autonomously by retaining (t, m) ; Theorem 2.1 also permits direct optimization without making that possibly wasteful construction.

2.2. A complete polynomial converse. Let $G \in [-1, 1]$ be a terminal payoff and let $L_\theta(z)$ denote its expectation as a linear function of the complete occupation array z . The terminal decision is included among the observer operations. Set

$$V_{\sigma, \mathbf{K}}^{\text{clk}}(\Theta, G) = \max_{A \in \mathcal{M}_\sigma^{\text{clk}}(\mathbf{K})} \min_{\theta \in \Theta} \mathbb{E}_\theta^A G. \quad (2.5)$$

The admissible row sets are assumed nonempty. Introduce $v \in [-1, 1]$ and impose $L_\theta(z) - v \geq 0$ for every θ . Write $h_a = 0$ for all the linear and quadratic equalities and $g_b \geq 0$ for the remaining inequalities. Include the redundant ball inequality

$$R - \|(z, v)\|^2 \geq 0,$$

where the positive integer R is larger than the number of coordinates. Let $\mathcal{F}$ be this basic closed feasible set. For $d \geq 1$, define U_d as the infimum of $u \in \mathbb{R}$ for which the polynomial identity

$$u - v = \sigma_0 + \sum_b \sigma_b g_b + \sum_a q_a h_a \quad (2.6)$$

holds, with each σ_b a sum of squares, each q_a a polynomial, and every displayed summand of degree at most $2d$. An empty certificate class has infimum $+\infty$.

Theorem 2.2 (Complete full-profile certificates). *For the finite protocol above,*

$$V_{\sigma, \mathbf{K}}^{\text{clk}}(\Theta, G) = \max_{(z, v) \in \mathcal{F}} v = \inf_{d \geq 1} U_d, \quad U_{d+1} \leq U_d. \quad (2.7)$$

A feasible occupation array is a constructive lower certificate. An identity (2.6) is an upper certificate for every stochastic machine of the specified full profile. For rational experimental data, every rational $u > V_{\sigma, \mathbf{K}}^{\text{clk}}$ admits such a certificate with rational coefficients. The same conclusions hold with shared rows, including the autonomous case.

Let ℓ_r be the best worst-case payoff among the actual controller tables whose entries are dyadic with denominator 2^r . If transition t has at most s_t choices per row, then

$$\ell_r \leq V_{\sigma, \mathbf{K}}^{\text{clk}} \leq \min \left\{ U_d, \ell_r + 2 \sum_t (s_t - 1) 2^{-r} \right\}. \quad (2.8)$$

Here repeated uses of a shared row are counted repeatedly in the sum. Thus there are matching constructive and converse certificates to any positive accuracy, without a minimax interchange or a convexification of fixed-width machines.

Proof. The first equality follows from Theorem 2.1 and the inequalities $v \leq L_\theta(z)$. Compactness gives attainment. Evaluating (2.6) at any feasible point gives $v \leq u$. Increasing d only enlarges the certificate class.

For completeness, take $u > \max_{\mathcal{F}} v$. The polynomial $u - v$ is strictly positive on $\mathcal{F}$. Apply Putinar's positivity theorem [29] to the inequalities $g_b \geq 0$ and the pairs $h_a \geq 0, -h_a \geq 0$. The explicitly included ball makes the quadratic module Archimedean. The resulting identity has

finite degree. Collecting the two multipliers of h_a gives (2.6). Letting u decrease to the maximum proves the second equality. For rational data and rational u , the rational version with an explicit ball generator [30, Theorem 6] gives rational sums of squares and rational multipliers. This is an application of those positivity theorems, not a new Positivstellensatz.

Finally round each stochastic row to the dyadic simplex, preserving its causal zeros, at total-variation cost at most $(s_t - 1)2^{-r}$. Round a shared row once and use that rounded row at every occurrence. Coupling the original and rounded executions until their first discrepancy bounds the probability of a discrepancy by the sum in (2.8), capped at one. The payoff range is at most two. This estimate is uniform in θ , so rounding a maximizing machine proves the rightmost bound. The dyadic machines are actual machines, which proves the left bound. Their finite enumeration is a construction, even though its cost may be large. $\square$

A numerical semidefinite solution is not by itself an identity (2.6); coefficients and nonnegativity must be certified. The theorem proves convergence, not a uniform degree bound, finite termination at $u = V$, or an efficient algorithm. Its necessity statement is nevertheless different from a resource upper construction: every machine obeys every certified upper bound. Dropping compatibility constraints produces a relaxation. It does not preserve the exact value in general, as Theorem 4.1 will demonstrate explicitly.

Corollary 2.3 (Compact families and finite experimental adapters). *Suppose B is a compact family of finite experimental kernels on a fixed schedule with N environment calls. Let $B_j \subset B$ be a finite ω_j -net in the maximum conditional total-variation metric, with $\omega_j \rightarrow 0$. Use the same bounded terminal payoff throughout. Then*

$$\ell_r(B_j) - 2N\omega_j \leq V_{\sigma, \mathbf{K}}^{\text{clk}}(B, G) \leq U_d(B_j). \quad (2.9)$$

As j, r, d tend to infinity in a suitable order, the two sides have common limit $V_{\sigma, \mathbf{K}}^{\text{clk}}(B, G)$. A certified adapter changing call i by at most δ_i contributes $2\sum_i \delta_i$ to a payoff comparison. A simulator with profile $\mathbf{R}$ is charged by the product profile $(R_t K_t)$.

Proof. For any one controller, couple corresponding calls until the first unequal outcome. Its expected payoff varies by at most $2N\omega_j$ between a kernel and its net representative. Hence $V(B_j) - 2N\omega_j \leq V(B) \leq V(B_j)$. Apply Theorem 2.2 to B_j and choose the dyadic and certificate accuracies after choosing j . The same coupling with unequal δ_i gives the adapter bound. Retaining the ordered pair of simulator and controller states proves the product assertion. $\square$

An effective net is additional data; abstract compactness is not an algorithm for an arbitrary presentation. The finite-alphabet, compact parameter collision experiments considered below have this interface. Consequently their fixed-schedule, full-profile values admit the same all-auditor certificates. This implication does not evaluate those certificates numerically for every physical sample budget, nor does it identify a minimal autonomous implementation.

3. CONTINUATION POLYTOPES AND NOISY POSTERIOR PARTITIONS

We next obtain a finite exact recursion, and then an explicit solution, for noisy experiments. Unlike a revelation channel, these experiments need never disclose the parameter at any time.

3.1. One common continuation matrix. Let Θ and the terminal action set A be finite. Conditional on θ , report t has distribution p_t^θ on Y_t and reports are independent. A clocked observer starts in one state, retains at most K_t states after report t , and receives payoff $G(\theta, a)$. For the moment there are no prescribed zeros in its update tables. For a fixed prior π , denote its optimal Bayesian payoff by $B_{n, \mathbf{K}}^{\text{clk}}(\pi, G)$. The terminal readout has no further observation of a discarded report.

Define finite sets of continuation matrices, with one column per retained state, backwards by

$$\mathcal{C}_n = \{(G(\theta, a_s))_{\theta, s} : (a_s)_s \in A^{K_n}\}, \quad (3.1)$$

$$\mathcal{C}_{t-1} = \bigcup_{f: [K_{t-1}] \times Y_t \rightarrow [K_t]} T_{t,f}(\mathcal{C}_t), \quad (3.2)$$

$$(T_{t,f}C)_{\theta, s} = \sum_{y \in Y_t} p_t^\theta(y) C_{\theta, f(s, y)}, \quad K_0 = 1. \quad (3.3)$$

The same matrix C occurs in every term of (3.3). Different incoming states do not receive independently selected future machines. If there are finitely many controlled experiments, include an action choice a_s in the union and replace $p_t^\theta(y)$ by $p_t^\theta(y \mid a_s)$.

Proposition 3.1 (Exact Bayesian continuation recursion). *Every matrix in $\mathcal{C}_t$ is implemented by one deterministic suffix of the prescribed profile, and every such suffix has a matrix in $\mathcal{C}_t$. Moreover,*

$$B_{n, \mathbf{K}}^{\text{clk}}(\pi, G) = \max_{C \in \mathcal{C}_0} \sum_{\theta} \pi_{\theta} C_{\theta, 1}. \quad (3.4)$$

For Bayesian optimization the sets can be replaced by their joint convex hulls $\mathcal{P}_t = \text{conv } \mathcal{C}_t$, using $\mathcal{P}_{t-1} = \text{conv } \bigcup_f T_{t,f}(\mathcal{P}_t)$. This operation does not authorize a free shared random seed. Replacing $\mathcal{P}_t$ by the product of its separate column projections is not valid in general.

Proof. At the terminal time, each column records the selected terminal action. At an earlier time, a deterministic update f feeds the new state into one fixed suffix. Conditional expectation gives (3.3). Backward induction proves the realization and converse statements, including recovery of the update tables.

For a fixed prior the payoff is a multiaffine function of the finitely many time-indexed stochastic rows: an execution visits any such row at most once. A maximum exists. Holding all but one row fixed, replace that row by a maximizing vertex of its simplex. Doing this finitely many times gives a deterministic maximum without changing the budget. The same argument includes an experiment-selection row. This proves (3.4) for all stochastic machines. Linearity of $T_{t,f}$ gives the asserted convex-hull recursion. A maximizing vertex of the final finite hull has a deterministic witness; the hull is used only to calculate a support value.

For the last assertion, consider a suffix with two possible initial states, a compulsory next cut of width one, and no informative subsequent report. Its two continuation columns must be equal. Individually each column may implement either constant binary decision. The product of the two column projections also contains the matrix that makes opposite decisions in the two initial states, which the compulsory one-state cut cannot implement. Under a fair preparation where the two initial states reveal the binary parameter, this false matrix gives success one instead of the true value one half. $\square$

3.2. Binary posterior geometry. For $\Theta = \{0, 1\}$ write a current occupation list as $Z = (z_s)_s$, where

$$z_s = (\mathbb{P}(\theta = 0, M = s), \mathbb{P}(\theta = 1, M = s)) \in \mathbb{R}_+^2.$$

Zero columns can be discarded and unused labels can always be padded. After report t , its incoming atoms are

$$b_{s,y} = (z_{s,0} p_t^0(y), z_{s,1} p_t^1(y)). \quad (3.5)$$

Order the nonzero atoms by $q(b) = b_1/(b_0 + b_1) \in [0, 1]$. An interval partition divides that ordered list into at most K_t consecutive nonempty blocks; summing the atoms in each block gives a new list $\mathcal{I}_t(Z)$. Equal-posterior atoms may first be merged. No nestedness of the partitions at different times is required.

Set

$$\begin{aligned} F_n(Z) &= \sum_s \max_{a \in A} \{z_{s,0}G(0, a) + z_{s,1}G(1, a)\}, \\ F_{t-1}(Z) &= \max_{\mathcal{I}_t} F_t(\mathcal{I}_t(Z)), \quad t = n, \dots, 1. \end{aligned} \tag{3.6}$$

For controlled observations, first maximize over the finite vector of experiment choices $(a_s)_s$ and then form (3.5) with those choices. Only lists reachable from the initial list are needed in evaluating this recursion.

Theorem 3.2 (Exact noisy full-profile recursion). *For arbitrary binary finite experiments of the conditionally independent form above, any prior, any bounded terminal payoff, and any finite cardinality profile,*

$$B_{n, \mathbf{K}}^{\text{clk}}(\pi, G) = F_0((\pi_0, \pi_1)). \tag{3.7}$$

There is an optimal deterministic machine whose update at every time is an interval partition of the current incoming posterior atoms. The recursion produces such a machine and optimizes over all stochastic machines, not just over a preselected statistic. It also holds with a finite choice of controlled experiments at each state.

Proof. Deterministic optimality follows from Proposition 3.1. Fix a deterministic global maximum, and consider one of its update times. Hold its prefix, experiment choices at this time, and entire suffix fixed. If the suffix starting at label j has continuation rewards c_{0j}, c_{1j} , assigning incoming atom b to j contributes

$$b_0 c_{0j} + b_1 c_{1j} = (b_0 + b_1)((1 - q(b))c_{0j} + q(b)c_{1j}).$$

It is therefore optimal to select a maximizing affine function of q . The upper envelope of finitely many affine functions on $[0, 1]$ has interval regions. Choose tied affine functions consistently, ordering active slopes and discarding identical duplicates. Each selected label then occupies an interval. At an identical posterior, all atoms have the same maximizing set, so they can be assigned to the same label. Unused labels cause no difficulty. Relabel the active suffix states in interval order, including their future experiment choices.

Now perform these replacements in chronological order. A replacement does not alter any earlier incoming occupation list or earlier update. It cannot decrease the payoff and thus preserves global optimality. Later incoming lists may change; they are recomputed before their own replacement. Hence the final machine has the interval property at all times simultaneously. This forward argument is why independently optimizing a partition at each cut would not prove the theorem.

Conversely, every sequence of choices in (3.6) defines causal finite update tables and therefore one admissible machine. The terminal formula is exactly its optimal readout. The existence just proved shows that restricting the finite search to these choices loses nothing. This proves (3.7). With controlled observations, first keep the current deterministic experiment choices fixed in the replacement argument; the same affine-envelope reasoning applies to their incoming atoms. Maximizing over those finite choices completes the proof. $\square$

This theorem uses the likelihood-ratio quantizer principle [27]; the affine-envelope mechanism is not claimed as new. The point of the recursion is the common, changing occupation list and the simultaneous realizability of all its partitions. The posterior pairs are computed by the designer. The implemented machine stores only a finite label and uses a fixed finite lookup table; it does not store a real-valued posterior. The number of reachable lists can be large. No polynomial-time claim is part of the theorem.

The result is Bayesian and clocked. A maximum of a minimum over environments is not multiaffine, and a reused autonomous row can occur several times in one execution. Neither of those cases is covered by the purification argument. They remain covered by Theorems 2.1 and

2.2. The following example solves the minimax problem directly and shows why the distinction is substantive.

3.3. An exact minimax recursion for binary registers. There is also a structural recursion that does not purify the minimax objective. In this subsection observations are conditionally independent and uncontrolled, and every retained alphabet has size one or two. Represent a binary prefix experiment by its pair $r = (r_0, r_1)$, where $r_\theta = \mathbb{P}_\theta(M = 1)$. Its labels may be exchanged at no cost. Say that s dominates r if a single stochastic binary channel, independent of θ , maps s to r .

For a pair s and the next report law, let $E_t(s)$ be the experiment on $\{0, 1\} \times Y_t$ with masses

$$e_\theta(m, y) = s_\theta^m (1 - s_\theta)^{1-m} p_t^\theta(y).$$

At endpoints this formula has its direct Bernoulli meaning. For any finite binary experiment $E = (e_0, e_1)$ define its upper receiver curve by the finite linear program

$$R_E(a) = \max \left\{ \sum_x e_1(x) \phi(x) : 0 \leq \phi(x) \leq 1, \sum_x e_0(x) \phi(x) = a \right\}, \quad 0 \leq a \leq 1. \quad (3.8)$$

Ordering atoms by likelihood ratio solves this program: accept the largest ratios first, randomizing at at most one ratio level. Atoms with zero e_0 are treated by their direct contribution to the objective; no division by zero is needed.

Starting with $\mathcal{R}_0 = \{(0, 0)\}$, define

$$\mathcal{R}_t = \begin{cases} \{(0, 0)\}, & K_t = 1, \\ \{(a, R_{E_t(s)}(a)) : s \in \mathcal{R}_{t-1}, 0 \leq a \leq 1\}, & K_t = 2. \end{cases} \quad (3.9)$$

This is a recursion of attainable channels, not of their convex hulls. Its continuous parameters are design parameters of finite stochastic rows, not stored posterior coordinates. In particular, it need not have a polygonal reachable set after several stages.

Theorem 3.3 (Binary-register minimax frontier). *Every channel in $\mathcal{R}_t$ is attainable with the prefix profile, and every stochastic prefix experiment with that profile is dominated by some member of $\mathcal{R}_t$. For a bounded payoff G and finite terminal action set, let $\Gamma_G(r)$ be the optimal minimum of the two expected payoffs obtained by a terminal stochastic readout of the binary experiment r . Then*

$$V_{n, \mathbf{K}}^{\text{clk}}(\{0, 1\}, G) = \max_{r \in \mathcal{R}_n} \Gamma_G(r). \quad (3.10)$$

For correct-decision payoff this reduces to

$$V_{n, \mathbf{K}}^{\text{clk}} = \max_{r \in \mathcal{R}_n} \left\{ \frac{1}{2} + \frac{|r_1 - r_0|}{2(1 + |1 - r_0 - r_1|)} \right\}. \quad (3.11)$$

An optimizer can be implemented by likelihood-ratio quantizers with at most one randomized ratio level per update and a terminal readout. These assertions hold at every finite horizon, with overlapping noise and with arbitrary placement of one-state cuts.

Proof. First consider a binary summary of any finite binary experiment, with output probabilities (r_0, r_1) . Exchange its labels if necessary so that $r_1 \geq r_0$. The receiver-curve optimizer at $a = r_0$ has output probabilities (r_0, b) with $b \geq r_1$. It dominates (r_0, r_1) . Indeed, if $b > r_0$, put

$$\lambda = \frac{r_1 - r_0}{b - r_0} \in [0, 1].$$

The postprocessing probabilities $\mathbb{P}(M = 1 \mid S = 0) = r_0(1 - \lambda)$ and $\mathbb{P}(M = 1 \mid S = 1) = r_0(1 - \lambda) + \lambda$ are in $[0, 1]$ and give the required pair. If $b = r_0$, the desired summary is uninformative and can be generated by a constant stochastic row. Undoing the exchange of labels covers the original summary. The receiver program is solved by an ordered threshold because moving an equal amount

of e_0 -mass from a smaller likelihood ratio to a larger one cannot decrease its objective. Sorting the finite atoms and filling the prescribed mass a proves the stated one-level randomization rule, including zero-mass atoms.

Proceed by induction on t . Each member of (3.9) is attained by an attaining previous channel and a receiver-curve quantizer. Conversely, suppose a previous prefix channel r is dominated by an attainable s in $\mathcal{R}_{t-1}$. Conditional independence of the next observation implies that $E_t(s)$ dominates $E_t(r)$ by applying the same postprocessing to the state coordinate. Composing this postprocessing with any proposed new summary gives a binary quantizer of $E_t(s)$. The preceding argument dominates its output by a receiver-curve point, which belongs to $\mathcal{R}_t$. This composition can be incorporated in one update table and introduces no retained selector. At a one-state cut every prefix is uninformative and the reset to $(0, 0)$ is exact. This proves both inductive assertions.

A terminal readout can compose any dominating channel's postprocessing, so Γ_G is monotone under domination. The original finite machine class has a maximizer by compactness of its row simplices. The inductive domination statement supplies an attainable member of $\mathcal{R}_n$ with at least that value. No attainable member can exceed the optimum. This proves the maximum and attainment in (3.10), without needing a convexity claim about $\mathcal{R}_n$.

For the last formula, exchange labels to arrange $r_1 \geq r_0$. If $r_0 + r_1 \leq 1$, accept hypothesis one on state one and randomize on state zero with probability $(1 - r_0 - r_1)/(2 - r_0 - r_1)$. The two successes then equal $(1 - r_0)/(2 - r_0 - r_1)$. If $r_0 + r_1 \geq 1$, reject on state zero and accept on state one with probability $1/(r_0 + r_1)$; both successes equal $r_1/(r_0 + r_1)$. These are likelihood-ratio tests and are minimax by the same equal-error supporting-prior argument as for a finite receiver curve. Both expressions equal (3.11); uninformative endpoints give $1/2$ directly. Choosing the attaining prefix quantizers proves the final implementation assertion. $\square$

Equation (3.9) is a genuine minimax whole-schedule characterization for arbitrary finite horizons in this binary-register class. It uses the ordinary comparison order of binary experiments, not a minimax interchange for complete machines. The restriction to uncontrolled, conditionally independent reports is used exactly when the dominating prefix is tensored with the next report. General controlled experiments and larger minimax registers use the occupation theorem instead. The three-report result that follows is a closed-form evaluation, with a strict mixing gap, of this same noisy resource problem.

4. AN EXACT NOISY FRONTIER AND THE COST OF HIDDEN MIXING

Let $\theta \in \{0, 1\}$, and let Y_1, Y_2, Y_3 be conditionally independent binary reports with

$$\mathbb{P}_\theta(Y_t \neq \theta) = p, \quad 0 < p < 1/2.$$

Every report has overlapping support under the two hypotheses. A clocked machine has at most K_t labels after report t and makes a binary terminal decision. Updates are atomic stochastic operations; additional random-bit cuts are discussed below. Let $B_{\mathbf{K}}(p)$ be its optimal success under the fair prior, and let $V_{\mathbf{K}}(p)$ be its optimal minimum of the two conditional success probabilities. Both optimize over all stochastic machines of this profile. Put

$$d(p) = p(1 - p)(1 - 2p) > 0.$$

Theorem 4.1 (Complete three-report profile frontier). *For every positive integer profile (K_1, K_2, K_3) ,*

<i>profile condition</i>	$B_{\mathbf{K}}(p)$	$V_{\mathbf{K}}(p)$	
$K_3 = 1$	$1/2$	$1/2$	
$K_3 \geq 2, \min(K_1, K_2) = 1$	$1 - p$	$1 - p$	
$K_1, K_3 \geq 2, K_2 = 2$	$1 - p + \frac{1}{2}d(p)$	$1 - p + \frac{1}{3}d(p)$	
$K_1, K_3 \geq 2, K_2 \geq 3$	$1 - p + d(p)$	$1 - p + d(p)$	(4.1)

In the third line a minimax optimal machine retains Y_1 , then retains Y_1Y_2 (the Boolean AND), and after the third report decides one with probability

$$\frac{(M_2, Y_3)}{\mathbb{P}(\hat{\theta} = 1 \mid M_2, Y_3)} \mid \begin{array}{cccc} (0, 0) & (0, 1) & (1, 0) & (1, 1) \\ 0 & 2/3 & 1 & 1. \end{array} \quad (4.2)$$

The two conditional success probabilities are equal. In particular, for $p = 1/4$ and profile $(2, 2, 2)$ the exact Bayesian and minimax values are $51/64$ and $25/32$, whereas profile $(2, 3, 2)$ has value $27/32$ for both objectives.

We prove the minimax assertion without passing to a convex hull of machines. The following elementary binary-channel domination is useful.

Lemma 4.2 (Reduction of a two-report binary register). *Every stochastic binary summary of (Y_1, Y_2) is, up to exchanging its two output labels, a stochastic postprocessing of a summary with*

$$\mathbb{P}(M = 1 \mid 00) = 0, \quad \mathbb{P}(M = 1 \mid 01) = \mathbb{P}(M = 1 \mid 10) = w, \quad \mathbb{P}(M = 1 \mid 11) = 1 \quad (4.3)$$

for some $w \in [0, 1]$. This domination holds simultaneously under both hypotheses and continues to hold after appending an independent third report.

Proof. Write u, v for the summary probabilities on $00, 11$, and write b for the average of its probabilities on $01, 10$. Those two words have the same probability under either hypothesis, so their average contains all relevant information. Exchange output labels if necessary to arrange $v \geq u$, and put $\alpha = v - u \in [0, 1]$ and $c = 2p(1 - p) \in (0, 1)$. Its output-one probabilities satisfy

$$r_0 = u + c(b - u) + p^2\alpha, \quad r_1 - r_0 = (1 - 2p)\alpha.$$

For the summary (4.3) these probabilities are $s_0 = p^2 + cw$, $s_1 = (1 - p)^2 + cw$. Seek a postprocessing that sends input zero to one with probability a and input one to one with probability $a + \alpha$. It is stochastic precisely when $0 \leq a \leq 1 - \alpha$, and it reproduces r_0, r_1 if

$$a = u + c(b - u) - c\alpha w.$$

As w ranges over $[0, 1]$, the right side ranges from $u + c(b - v)$ to $u + c(b - u)$. The upper endpoint is nonnegative. The lower endpoint is at most $1 - \alpha$, since

$$1 - \alpha - u - c(b - v) = (1 - v) + c(v - b) \geq (1 - v)(1 - c) \geq 0.$$

Consequently this interval meets $[0, 1 - \alpha]$. This proves the claim, including $\alpha = 0$. Applying the postprocessing independently of the third report proves the last statement. $\square$

Proof of Theorem 4.1. If the last register has one state, its terminal decision is independent of the reports. The fair-prior value is one half, and a fair terminal randomization gives minimax value one half. If an earlier register has one state, at most the last two reports remain informative. The unrestricted optimal success from two equally reliable reports is $1 - p$ under either criterion: use their common value on agreement and a fair decision on disagreement. One report already gives $1 - p$. These rules respect the stated profiles and prove the second line.

Suppose henceforth that $K_1, K_3 \geq 2$. There is no loss in retaining Y_1 exactly. Indeed any stochastic first update followed by a second update induces one binary channel from (Y_1, Y_2) to M_2 , and that channel can be implemented directly when Y_1 is retained. Allowing more than two first-stage labels gives no larger set of such channels. At the last stage any randomized binary decision based on (M_2, Y_3) can be placed directly in a two-label register. Thus, when $K_2 = 2$, the unrestricted optimization is exactly over a binary summary of the first two reports and a randomized readout of it together with Y_3 .

For the Bayesian problem, Theorem 3.2 reduces the middle update to intervals of the three distinct likelihood-ratio classes $00, \{01, 10\}, 11$. A nonconstant two-interval update isolates 00 or

isolates 11. These are the Boolean OR and AND, up to labels. With AND, the Bayesian readout is $\hat{\theta} = Y_3 \vee (Y_1 \wedge Y_2)$. Its conditional successes are

$$c_0 = 1 - p - p^2 + p^3, \quad c_1 = 1 - 2p^2 + p^3.$$

Their average is $1 - p + d(p)/2$. The complementary rule interchanges c_0, c_1 . A constant middle update gives only $1 - p$, proving the Bayesian third line. This also covers updates that split the tied middle words: the theorem permits assigning the entire tied class to one maximizing suffix label without decreasing the Bayesian objective.

For the minimax problem, Lemma 4.2 reduces the search, without loss, to (4.3). Complementing all reports and output labels replaces w by $1 - w$ and interchanges the hypotheses, so it suffices to take $0 \leq w \leq 1/2$. Write

$$r_0 = p^2 + 2p(1 - p)w, \quad r_1 = (1 - p)^2 + 2p(1 - p)w.$$

In the experiment (M_2, Y_3) the likelihood ratios are ordered as 00, 01, 10, 11. Only the ordering of the middle two needs verification; after multiplying by their positive denominators it is equivalent to

$$r_1(1 - r_0)p^2 - r_0(1 - r_1)(1 - p)^2 = p^2(1 - p)^2(1 - 2p)(1 - 2w)^2 \geq 0.$$

Consider the test that accepts hypothesis one on 10, 11, rejects it on 00, and accepts it on 01 with probability

$$a(w) = \frac{1 - r_0 - r_1}{(1 - r_0)p + (1 - r_1)(1 - p)} = \frac{2(1 - 2w)}{3 - 2w}. \quad (4.4)$$

This number is in $[0, 1]$. Its two conditional success probabilities are equal, with common value

$$R(w) = 1 - r_0 - a(w)(1 - r_0)p. \quad (4.5)$$

It is a likelihood-ratio test, possibly randomized at one threshold. Choose the prior odds equal to that threshold. The test is then Bayesian optimal for those odds. Since its two successes are equal, no test can have a larger minimum: such a test would also have a larger payoff for this prior. Thus $R(w)$ is the exact minimax success for this summary, not merely an equal-error attainable value.

Direct differentiation gives

$$R'(w) = -\frac{2p(1 - p)(1 - 2p)(5 - 2w)(1 - 2w)}{(3 - 2w)^2} \leq 0. \quad (4.6)$$

Equivalently, an explicit nonnegative gap certificate is

$$R(0) - R(w) = \frac{2p(1 - p)(1 - 2p)w(5 - 6w)}{3(3 - 2w)} \geq 0, \quad 0 \leq w \leq 1/2. \quad (4.7)$$

The denominator is positive and every factor in the numerator is nonnegative on the stated domain. Therefore the best w is zero. Equations (4.4)–(4.5) give $a(0) = 2/3$ and

$$R(0) = 1 - \frac{2}{3}p - p^2 + \frac{2}{3}p^3 = 1 - p + \frac{1}{3}d(p).$$

The postprocessing in Lemma 4.2 can be composed with any final readout and does not need a retained selector. It follows that this upper bound applies to every original stochastic machine. The rule (4.2) attains it.

Finally, if $K_2 \geq 3$, retain the number of ones in the first two reports and output the majority of all three. Both successes are $1 - 3p^2 + 2p^3 = 1 - p + d(p)$. The likelihood-ratio rule for the full three reports is majority under the fair prior, so no machine can have a larger Bayesian value or a larger minimum. This proves the last line and completes all cases. $\square$

Corollary 4.3 (A strict fixed-profile mixing gap). *For profile $(2, 2, 2)$ a free initial selector that mixes the two complementary Bayesian optimal machines would give worst-case success $1 - p + d(p)/2$. Without that uncharged selector the exact value is $1 - p + d(p)/3$. The excess $d(p)/6$ is strictly positive.*

Proof. Mix the two payoff vectors (c_0, c_1) and (c_1, c_0) from the proof with equal weights. Each coordinate becomes their average. Theorem 4.1 gives the smaller all-stochastic value at the original profile. To implement the mixture literally, retain the selector jointly with the machine state; a profile $(4, 4, 2)$ suffices. The corollary does not claim that this product construction minimizes the expanded profile. $\square$

The gap identifies a specific failure of a convexified all-machine converse. A terminal output law can be a convex combination of laws while the machines producing those laws cannot be mixed within their original memory profile. In contrast, the quadratic compatibility constraints of Theorem 2.1 retain exactly the required coupling. The noisy loss $2d(p)/3$ between two and three middle states is also a strict full-schedule minimax loss; it is not an erasure probability or the cost of remembering a label that was already known.

Remark 4.4 (Atomic randomization and finite precision). The fraction $2/3$ in (4.2) is an exact rational row of the all-stochastic model. A fixed finite string of fair bits cannot realize this fraction exactly. Replacing it by a dyadic number within 2^{-r} gives a genuine finite-precision rule whose minimum success is at least $V_{(2,2,2)}(p) - 2^{-r}$. If each random-bit epoch is a charged cut, retain (M_2, Y_3) while executing the dyadic sampler; the three-status comparison sampler of Section 10 gives a conservative peak of 12 for this refined execution. The theorem’s exact two-state value refers to its stated atomic-update profile, not to this finer profile. The latter has its own exact occupation formulation. Neither an unbounded rejection sampler nor a free fair-bit clock is hidden in the two-state assertion.

5. ACQUISITION, CLOCKS, AND THE RESOURCE ORDER

A statistical bound is ambiguous unless the protocol and the resource coordinates being compared are specified. The superscript clk in the preceding theorems is therefore part of their notation, not a qualification confined to an implementation paragraph.

5.1. Comparable resource points. On one declared schedule, profiles are ordered coordinatewise. Enlarging a register alphabet preserves every previously feasible machine by padding unused states. Inserting observations is comparable only through an explicit embedding that says which new inputs are ignored and which registers survive the inserted cuts. Atomic and bit-serial schedules are different schedules, even when they have the same number of experimental preparations.

For a finite-precision implementation we record

$$\mathbf{r} = (N_B, N_Q, \mathbf{K}, r, D, \mathbf{C}). \quad (5.1)$$

Here N_B, N_Q count candidate and target preparations, r bounds fair bits per declared row sampler, D bounds a specified read-only description, and $\mathbf{C}$ specifies clocked or autonomous execution. A description is measured in a fixed code, not by an unspecified notion of compressibility. For example, with a known table layout, store the first $s - 1$ numerators of a 2^{-r} stochastic row in $(s - 1)(r + 1)$ bits; its last numerator is the remainder. The layout, response dictionary and any program used to expand repeated phases have their own fixed-code costs. Sample-dependent calibration is not part of that read-only description.

For the same code and schedule, increasing numerical resource caps or register alphabets gives an inclusion of machine classes. Changing $\mathbf{C}$ does not mean increasing one numerical coordinate. A clocked execution with T update epochs and profile $\mathbf{K}$ has an autonomous realization on the

disjoint union of its phase registers, using at most $\sum_{t=0}^T K_t$ states. This is an attainable conversion, not an equality of optimal frontiers. Shared-row optimization in Theorem 2.1 can be strictly smaller than that conversion.

Proposition 5.1 (Joint finite resource optimization). *Fix a finite experimental protocol and a finite table code. With finite caps on description and precision, the feasible machines form a finite set and their optimal worst-case payoff is exactly the maximum over that set. If the precision restriction is removed on a fixed table architecture, Theorems 2.1 and 2.2 characterize its exact value. A finite collection of permitted architectures is handled by taking the maximum of their values. Calibration calls can be included as an initial part of the same protocol, with all their retained information passed through the same profile constraints.*

Proof. There are finitely many descriptions of bounded length and finitely many allowed dyadic rows. Retain exactly the descriptions that decode to legal tables of the specified finite schedule and budgets. Each has a well-defined payoff. For unrestricted stochastic rows on a fixed architecture, use its observable row sets in the occupation formulation. Optimizing over a finite union is a maximum over its components. A calibration prefix uses known experimental kernels at the unknown environment in exactly the same way as any other prefix; its sample-dependent state is an ordinary retained register. Thus it introduces neither a parameter oracle nor a new exception to the compatibility constraints. $\square$

The proposition supplies a joint optimization model rather than merely a post hoc resource ledger. It does not claim a short closed formula for arbitrary description budgets. For compact continuous families the net condition in Corollary 2.3 is still required for effective finite certificates.

5.2. Acquiring a calibration rather than assuming it. The next result gives a fully priced constructive calibration prefix. Let $h_1, \dots, h_K : \Omega \rightarrow [0, 1]$ be a fixed known response list, chosen independently of the calibration observations, and suppose its values have denominator 2^r . Write $d = |\Omega|$ and $q_i = Qh_i$. The target Q need not be given as a table: independent preparations from Q are available. Let $M = 2^s$ and $0 < \beta < 1$.

Theorem 5.2 (Sampled calibration with persistent-register pricing). *For every $\rho > 0$ with*

$$M \geq \frac{1}{2\rho^2} \log \frac{2K}{\beta}, \quad (5.2)$$

there is a clocked finite-precision calibration protocol using M target preparations and MKr fair bits. It returns $\hat{q}_i \in \{0, 1/M, \dots, 1\}$ with

$$\mathbb{P} \left(\max_i |\hat{q}_i - q_i| \leq \rho \right) \geq 1 - \beta.$$

The complete counter register has at most $C = (M + 1)^K$ states; a word buffer and the three-status dyadic sampler give peak at most $3dC$. If a subsequent auditor has peak at most W for any exogenously fixed calibration vector, retaining the acquired calibration gives the bound $C \max(3d, W)$ for the combined execution.

Let $v \in [-1, 1]$ and $\epsilon \geq 0$. Suppose, uniformly for calibration vectors $\tilde{q}$, the auditor's true expected score is at least $v - \epsilon - 2\|\tilde{q} - q\|_\infty$, and its score is in $[-1, 1]$. With fresh training and validation after this prefix, the combined expected score is at least

$$v - \epsilon - 2\rho - 2\beta. \quad (5.3)$$

Proof. On target trial j , retain its word Y_j . For each i , generate a Bernoulli variable B_{ij} of parameter $h_i(Y_j)$ with an independent r -bit dyadic sampler, and increment counter i . After all

K updates, discard Y_j . For fixed i , the variables B_{ij} over different trials are independent, lie in $\{0, 1\}$, and have mean q_i . Their averages $\hat{q}_i$ therefore satisfy

$$\mathbb{P}(|\hat{q}_i - q_i| > \rho) \leq 2e^{-2M\rho^2}.$$

This bounded-variable exponential estimate follows from the exponential moment bound used in Theorem 9.2. A union bound proves the probability assertion. Independence between different counters is not required.

The K counters each have $M + 1$ possible values. The word buffer has d values, and the running comparison of a uniform r -bit word with a fixed dyadic threshold has three comparison statuses. These yield the stated state counts at every sampler cut. Trial, response and bit indices are in the declared clocked schedule, not in an unmentioned work tape. The persistent counter vector remains in the register of the subsequent auditor and multiplies its width by at most C . All its values are dyadic because $M = 2^s$.

Conditional on the calibration vector, later observations have their original independent laws. On the good calibration event the score bound is $v - \epsilon - 2\rho$. On the complementary event the score is at least -1 . Since $v \leq 1$ and $\epsilon, \rho \geq 0$, a loss of at most two times that event's probability suffices. This proves (5.3). $\square$

For the response-cover tournament of Theorem 13.2, one may take

$$N_B = n(K - 1), \quad W = 3dK(2m + 1), \quad \epsilon = 4(K - 1)\theta + 2(K - 1)(e^{-4\theta m} + e^{-2n\theta^2})$$

when v is the envelope of the already dyadic response list. Starting with an arbitrary list and rounding it to r bits incurs the additional 2^{1-r} term of that theorem. The new prefix prices target acquisition and its persistent calibration state. It is an explicit construction, not an optimal calibration-memory claim. Joint optimality at fixed finite budgets is the distinct problem in Proposition 5.1.

6. CONSUMERS OF THE COMPATIBLE FRONTIER

The preceding characterizations and the constructive continuation results below have the same object of optimization: one experiment, one schedule and one whole register profile. This section records the connections before developing the underlying marked and physical experiments in detail.

6.1. Reset scores and physical protocols. In the reset interface of Section 9, a fixed candidate supplies the training and candidate-validation kernels. The target supplies its own preparation and validation kernels. The final score is a function of the two retained validation bits. Include the response selection, all input-bit buffers, and all randomization cuts in σ . This produces precisely the finite protocol of Section 2. The image of the candidate family is not replaced by its convex hull.

Corollary 6.1 (Common constructive and converse interface). *At any fixed finite reset protocol and full profile, the best worst-case score has the occupation characterization of Theorem 2.1; for a finite family it has the complete dual of Theorem 2.2. For a compact family with effective finite kernel nets it has the converging upper and lower certificates of Corollary 2.3. The response-cover machines of Theorems 12.2 and 13.2 supply feasible lower certificates in these very classes. Any certified dual bound applies to those machines and to every other stochastic machine with the same protocol and profile.*

Proof. Augment the hidden index by the fixed candidate parameter and the finite public trial history. Selection and memory updates are parameter-independent observer rows. Experimental draws are the known linear environment operations. Terminal scoring is linear in the terminal

occupation array. Thus the realization theorem applies in both directions. The constructive machines specify legal rows at every cut and hence feasible occupation arrays. Compact-family and dual assertions are exactly the corresponding general theorems. $\square$

For the positive-noise marked collision experiment, the finite row-report alphabet is fixed and the private row family is compact. On compact subfamilies with the stated continuous kernel data, finite kernel nets give the preceding certificates. The microscopic and Gaussian comparison theorems below then transfer a certified payoff with their stated common-preparation error. If training kernels differ, their errors must also be charged; equality of the training source cannot be inferred from equality of the terminal validation task. The active-state estimate in Section 15 can replace a worst-case telescoping estimate for a fixed retained execution, but is not a bound on a full discarded transcript.

Thus the physical resource constructions (399, 40602), (320000, 1604) and $(12800 \cdot 2^{400}, 12)$ are now accompanied by a general all-auditor converse procedure on each fixed full profile. They remain attainable resource points, not a list of proved Pareto optima. In particular, the numerical statement $3 \leq W_{2/5}^{\text{phys}} \leq 12$ is not changed into an equality without an additional certificate or construction.

6.2. An exact confidence consumer. Let a reset audit return $D \in \{-1, 0, 1\}$. Suppose its null mean is zero and its alternative mean is at least g , where $0 < g \leq 1$ is known. On each independent repetition, use a fresh coin with

$$\mathbb{P}(Z = 1 \mid D) = \frac{1}{2} + \frac{D - g/2}{4}. \quad (6.1)$$

Under the null its success probability is $p = 1/2 - g/8$, and under the alternative it is at least $1 - p$. Consider the declared coin interface consisting of the null Bernoulli law p and all alternative Bernoulli laws with parameter in $[1 - p, 1]$, independently repeated.

Corollary 6.2 (Exact three-repetition confidence frontier). *At the atomic retained-coin profile (2, 2, 2), the minimum possible maximum of type-I and type-II error over this coin interface is*

$$p - \frac{1}{3}p(1 - p)(1 - 2p). \quad (6.2)$$

The rule (4.2) attains it. Increasing only the middle retained-coin alphabet to three changes this value to $p - p(1 - p)(1 - 2p)$. If one underlying audit uses b candidate training trials, two validation trials, and peak W , the three-repetition construction uses $3(b + 2)$ preparations. Its inner audit, coin sampler, and outer state are charged jointly, rather than identifying an outer two-state register with the entire physical register.

Proof. The alternative family contains the endpoint $1 - p$, so Theorem 4.1 gives the lower bound on the maximum error. For the attaining test, its probability of deciding one, as a function of the three input bits, is coordinatewise nondecreasing: the middle state is AND, and the four final probabilities are 0, $2/3$, 1, 1. Couple Bernoulli inputs with common independent uniforms. The power is then nondecreasing in their parameter, so the largest type-II error occurs at the endpoint. This proves attainment for the whole declared alternative interval. The majority test proves the assertion with three middle states in the same way. Preparation counts follow by repeating the base experiment independently three times. Product registers implement the outer transitions together with each inner repetition. $\square$

If g is dyadic, the coin in (6.1) has a finite dyadic sampler. At its internal cuts an explicit conservative bound is $2 \max(W, 9)$ for the outer state and inner audit/coin sampler; the additional $2/3$ terminal row is either an atomic rational operation or is replaced by the finite-precision realization

in Remark 4.4. A fully fair-bit-serial realization has the conservative peak $\max\{2\max(W, 9), 12\}$ and the indicated 2^{-r} error allowance. This does not erase its clock or its bits from the schedule.

The exact lower bound concerns the declared coin interface. A physical subfamily inherits that lower bound only if it contains its endpoint experiment (or an appropriate certified limiting experiment). A raw physical auditor that uses more of D or of the original report is not restricted to this interface. For such auditors the general occupation converse, with those richer inputs retained, is the relevant statement. The confidence result is therefore a proved consumer, not a relabeling of an expected-score lower bound as a hypothesis-testing lower bound.

6.3. The foundation and the longer program. The initial objects remain positive experimental kernels, preparations, causal interventions and executable continuations. Finite protocols are a resource-controlled subcategory, not a replacement for the standard Borel formulation or its limiting targets. The new occupation and continuation results supply a precise G3-type finite resource layer. The marked deficiency and Gaussian transport results supply the comparison maps used above. The finite resource theorems do not supply a missing kinetic generator domain, a compactness theorem for collision measures, or an optional-projection theorem on a general operator domain.

The full historical dependency graph is retained in Appendix B. The geometric A2 chain has its own premises. The B4 state-dependent control-transfer, range and compactness targets, and the broader C2 operator and rigidity targets, retain their original scope. Their relation to this paper is through explicit experiment and simulator interfaces when those have been established, not through a declaration that every manuscript is a consequence of a finite-state comparison theorem.

7. INTRINSIC CAUSAL DEFICIENCY

7.1. Experiments and executable signatures. Fix a finite parameter set Θ , a horizon T , and finite control alphabets $A_0, \dots, A_{T-1}$. An experiment has a report at checkpoints $0, \dots, T$ and an unobserved terminal mark W in a common finite set. It is specified by finite causal instruments: for each parameter and control history they give the joint law of the report path and the actual mark. Equivalently, a finite latent-state realization may be expanded into its history tree. The parameter and mark are unavailable to the encoder. The mark may depend on the physical controlled path. We compare experiments with the same control interface and mark space.

Write Y_t for the source report and Z_t for the target report. A private simulator of width K has stochastic rows

$$\begin{aligned} q_0(z, m \mid y), \quad m \in [K_0], \\ q_t(z, m \mid m', a, y), \quad m' \in [K_{t-1}], \quad m \in [K_t], \quad 1 \leq t \leq T. \end{aligned} \tag{7.1}$$

A row consumes the current source report, emits the target report, and retains the new state. The previous target reports are not available to that row unless encoded in m' . The clock and current control are part of the declared interface. Fresh private randomness is permitted, but no other persistent random variable is free. The product of all row simplexes is denoted $\mathcal{K}_K$.

A deterministic feedback tree g chooses a_t from $z_{0:t}$ and its own previous controls. It is the same controller on both sides of a comparison. Let Ω_g be the finite marked transcript space, including reports, controls, and W . Put

$$P_{\theta,g}^q = \text{Law}_\theta(E \xrightarrow{q} g), \quad Q_{\theta,g} = \text{Law}_\theta(F \rightarrow g).$$

Here W on the left is the source's actual mark. The total variation convention is $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. Private deficiency is

$$\delta_K^p(E, F) = \min_{q \in \mathcal{K}_K} \max_{\theta, g} \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}. \tag{7.2}$$

The minimum will be justified below. Randomized feedback need not be added to the maximum: conditioning on its independent random table expresses its law as a mixture of deterministic feedback laws. The same argument permits arbitrary independent side randomness in the tester.

A *hidden selector* $U \in [r]$, independent of the source, chooses one simulator q^u . It is not visible to the feedback controller or the final test. A *visible selector* is revealed before feedback begins. The two costs are

$$\delta_{K,r}^h(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, g} \left\| \sum_u \lambda_u P_{\theta, g}^{q^u} - Q_{\theta, g} \right\|_{\text{TV}}, \quad (7.3)$$

$$\delta_{K,r}^v(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, (g_u)} \left\| \sum_u \lambda_u \delta_u \otimes P_{\theta, g_u}^{q^u} - \sum_u \lambda_u \delta_u \otimes Q_{\theta, g_u} \right\|_{\text{TV}}. \quad (7.4)$$

In both expressions $\lambda \in \Delta_r$ and $q^u \in \mathcal{K}_K$. The second formula compares the joint seed–marked interface. Its target seed has the same distribution and is independent of the target experiment. A hidden selector can be stored privately by retaining (u, m) , at width rK_t . A visible selector is separate side information and is charged by its own alphabet size. Neither interpretation identifies an arbitrary real-valued design with a free finite register.

Definition 7.1 (Deficiency spectrum). The private resource spectrum of E relative to F is the upward-closed set of pairs (K, ε) satisfying $\delta_K^p(E, F) \leq \varepsilon$. The hidden spectrum consists of triples (K, r, ε) satisfying $\delta_{K,r}^h(E, F) \leq \varepsilon$, and the visible spectrum uses $\delta_{K,r}^v$. Width profiles are ordered coordinatewise; in the seeded spectra the seed bound and error are ordered as well. Unused states or selector values may be adjoined. The hidden convexification is $\delta_K^{\text{mix}} = \inf_{r \geq 1} \delta_{K,r}^h$.

The spectrum, not a presumed unique coordinatewise minimum profile, is the intrinsic object. For example one may minimize $\max_t K_t$, with a specified seed signature and tolerance, over its feasible profiles. Different checkpoints can have incomparable optimal tradeoffs.

7.2. The finite comparison theorem. Concatenate all probabilities $P_{\theta, g}^q(\omega)$ into $b(q) \in \mathbb{R}^N$, retaining their row indices; put $b_F = (Q_{\theta, g}(\omega))$. Let $\mathcal{D}_K$ be the finite set of deterministic choices of all rows in (7.1). Empty or unreachable rows remain harmless coordinates. Set

$$a_K = \dim \text{aff}\{b(q) : q \in \mathcal{K}_K\}, \quad \mathcal{I} = \{(\theta, g, B) : B \subseteq \Omega_g\}.$$

For $i = (\theta, g, B)$ and $d \in \mathcal{D}_K$, define

$$A_{i,d} = Q_{\theta, g}(B) - P_{\theta, g}^d(B). \quad (7.5)$$

Including both an event and its complement includes both signs.

Theorem 7.2 (Intrinsic deficiency, priced converse, and visibility). *For the finite marked causal experiments above the following assertions hold.*

(i) *Private deficiency and both fixed- r deficiencies are attained. The private problem has the exact polynomial formulation*

$$\min\{\varepsilon : q \in \mathcal{K}_K, 0 \leq \varepsilon \leq 1, Q_{\theta, g}(B) - P_{\theta, g}^q(B) \leq \varepsilon \text{ for all } (\theta, g, B) \in \mathcal{I}\}. \quad (7.6)$$

In particular its zero set is exactly the nonnegative causal factorization of the target marked behavior by the rows (7.1). At $K_t = 1$ for every t , this is precisely the same factorization with rows $q_t(z \mid a, y)$ and $q_0(z \mid y)$.

(ii) *Hidden convexification has the attained dual*

$$\delta_K^{\text{mix}}(E, F) = \min_{\lambda \in \Delta(\mathcal{D}_K)} \max_{i \in \mathcal{I}} \sum_d \lambda_d A_{i,d} = \max_{w \in \Delta(\mathcal{I})} \min_{d \in \mathcal{D}_K} \sum_i w_i A_{i,d}. \quad (7.7)$$

An optimal simulation uses at most $a_K + 1$ selector values. Thus every weighted causal test has optimal discrepancy at most ε if and only if a hidden mixture of K -state simulators has error at most ε ; it has a private realization of width at most $(a_K + 1)K$. More specifically,

$$\delta_{rK}^p \leq \delta_{K,r}^h, \quad \delta_K^{\text{mix}} = \delta_{K,a_K+1}^h \leq \delta_K^p. \quad (7.8)$$

(iii) Put $e_\theta(q) = \max_g \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}$. Then

$$\delta_{K,r}^v = \min_{\lambda, q^1, \dots, q^r} \max_{\theta} \sum_u \lambda_u e_\theta(q^u). \quad (7.9)$$

Writing $a_K^v = \dim \text{aff}\{(e_\theta(q))_\theta : q \in \mathcal{K}_K\}$, the infimum over all finite visible selectors is attained using $a_K^v + 1 \leq |\Theta| + 1$ values and equals

$$\max_{v \in \Delta(\Theta)} \min_{q \in \mathcal{K}_K} \sum_{\theta} v_\theta e_\theta(q). \quad (7.10)$$

For every $r \geq 1$, and also after optimizing over all r ,

$$\delta_{K,r}^v(E, F) = 0 \iff \delta_K^p(E, F) = 0. \quad (7.11)$$

No corresponding equivalence with $\delta_K^{\text{mix}} = 0$ holds in general.

(iv) For compatible experiments E, F, G and profiles K, L ,

$$\begin{aligned} \delta_{KL}^p(E, G) &\leq \delta_K^p(E, F) + \delta_L^p(F, G), \\ \delta_{KL,rs}^s(E, G) &\leq \delta_{K,r}^s(E, F) + \delta_{L,s}^s(F, G), \quad s \in \{h, v\}. \end{aligned} \quad (7.12)$$

Here KL is the pointwise product, and the second line uses independent selectors. If an F -consumer has widths S_t , composing it with a private simulator gives widths $K_t S_t$ and changes every $[0, 1]$ marked risk by at most $\delta_K^p(E, F)$. A hidden selector is stored at the additional price r , whereas visible selector alphabets multiply. The same bounds hold for optimized upper risk sets and their lower support functions with the indicated costs.

Proof. Expand a probability of a complete source path, simulator state path, and transcript. It is a product of fixed experiment probabilities and one chosen row entry per time. Summing latent and state paths gives a polynomial in the finitely many row entries. No row belonging to a fixed time can occur twice along one path. Consequently the polynomial is multiaffine in the groups of stochastic rows. Continuity and compactness of the product of closed finite simplexes prove attainment in (7.2); the same proof includes λ and the r policies in (7.3). For the visible problem there are only finitely many collections (g_u) and transcript events, so its objective is again continuous on a compact set. The finite-event formula for total variation proves (7.6). Setting $\varepsilon = 0$ forces equality of every transcript probability, in both directions of each event inequality. This proves the factorization and stateless claims, including their necessity.

For each stochastic row of q , sample its deterministic output once, independently of all other rows, before the experiment. This constructs a probability Λ_q on $\mathcal{D}_K$. On any fixed path the row variables encountered have different time indices, so the expectation of their product is the product of the row probabilities. Summing paths shows, simultaneously for every θ, g , that

$$b(q) = \sum_{d \in \mathcal{D}_K} \Lambda_q(d) b(d). \quad (7.13)$$

This is a behavior identity, not a same-width deterministic purification of minimax error. It implies $\text{conv } b(\mathcal{K}_K) = \text{conv}\{b(d) : d \in \mathcal{D}_K\}$. The latter is a compact polytope. Minimizing the continuous convex maximum of event discrepancies over this polytope gives the first expression in (7.7). Finite-dimensional minimax gives the second, with both optima attained. A point of this polytope is a convex combination of at most $a_K + 1$ of its generating points by Carathéodory's theorem. For completeness, if a representation uses more points, their affine dependence gives

coefficients c_d with $\sum c_d = 0$ and $\sum c_d b(d) = 0$. Subtract the largest permissible multiple of c from the positive mixing weights; a weight becomes zero without changing the represented point. Repeating gives the stated bound. Store its selector together with the machine state to obtain (7.8). The dual inequality for every w is equivalent to the maximum over w being at most ε , which proves the priced testing converse.

For disjoint seed fibers, total variation is the sum of the fiber total variations weighted by λ_u . For a fixed θ , the tester can choose a separate maximizing tree on each revealed fiber. Therefore its maximum is $\sum_u \lambda_u e_\theta(q^u)$, proving (7.9). The map $q \mapsto e(q)$ is continuous, being a finite maximum of continuous functions. Its image is compact. Convexifying it in $\mathbb{R}^{|\Theta|}$ and applying the preceding finite-dimensional separation and affine-dependence argument proves (7.10) and its selector bound. Notice that the inner minimum is over stochastic q , not over deterministic tables: the nonlinear function e_θ does not commute with (7.13). If the attained visible optimum is zero, then for every u of positive weight and every θ one has $e_\theta(q^u) = 0$, since every summand is nonnegative. Any such q^u is a private exact simulator. The converse uses a selector of one value. This also proves the assertion for the optimized visible selector, whose support is finite by the bound just established.

Finally, cascade a K -state E -to- F transducer and an L -state F -to- G transducer. Within a step, consume the intermediate report before discarding it. The retained pair has at most $K_t L_t$ values. For any terminal tester on G , the second transducer together with that tester is an admissible randomized feedback system on F . The first error bound applies to this system, and total variation contracts under the remaining channel. The second error bound applies to the genuine F system. The triangle inequality gives (7.12). With selectors, condition on the independent selectors in the appropriate places; in the visible case keep both selectors in the compared transcript. The same argument is valid because the target reference selectors are independent. Storing a hidden selector or exposing both visible selectors gives the stated costs. The minima established above may be used; equivalently one can use nearly optimal simulators and let the extra tolerance decrease.

A consumer is another causal transducer with task-specific readout and, where allowed, task-specific control, but with its required common encoder. The simulator is task blind, so its product with that encoder is still common. Condition on the consumer's independent random table to apply the feedback bound. Integration of any $[0, 1]$ marked loss changes by at most the total variation error. Taking the same paired consumer for every row proves the risk-set assertion, rather than a separate choice of simulator for each row. Infima of nonnegative weighted risks give the lower support inequalities. $\square$

Remark 7.3 (What the converse says). The dual (7.7) uses mixtures of indexed causal tests and optimizes over the entire common transducer. It is not a task-by-task choice of encoder. Its conclusion is an explicit hidden-selector simulation, with the selector subsequently charged if private storage is required. It does not replace the nonconvex private set by its convex hull at unchanged resource. The next example shows why that replacement would change the zero set.

7.3. A delayed channel and a strict signature separation. Let the source reveal $i \in [n]$ at checkpoint zero and reveal only a constant at checkpoint one. There is no control and the mark is constant. The target reports a constant at checkpoint zero and at checkpoint one draws $j \in [m]$ from a specified row-stochastic matrix $V = (V_{ij})$. The unknown parameter is i . The first target alphabet has one element: the simulator cannot use an early public output as unpriced storage. Only the intermediate register of size K matters.

Proposition 7.4 (The nonnegative-rank slice). *For this experiment,*

$$\delta_K^p(E, F_V) = \min_{A, B} \max_i \frac{1}{2} \sum_j |V_{ij} - (AB)_{ij}|, \quad (7.14)$$

where A is an n -by- K row-stochastic matrix and B is a K -by- m row-stochastic matrix. In particular

$$\delta_K^p(E, F_V) = 0 \iff \text{rank}_+(V) \leq K, \quad \delta_1^p(E, F_V) = 0 \iff V \text{ has identical rows.}$$

For $1 \leq K \leq n$ the identity channel has the sharp laws

$$\delta_K^p(E, F_{I_n}) = 1 - \frac{1}{\lceil n/K \rceil}, \quad \delta_K^{\text{mix}}(E, F_{I_n}) = 1 - \frac{K}{n}. \quad (7.15)$$

Thus, at tolerance $0 \leq \varepsilon < 1$, the least private width is $\lceil n/(1-\varepsilon)^{-1} \rceil$, whereas the least within-mode width with optimized hidden selection is $\max\{1, \lceil n(1-\varepsilon) \rceil\}$. The latter still pays for its selector. In particular, for $n = m = 3$ and $K = 2$,

$$\delta_2^p(E, F_{I_3}) = \frac{1}{2}, \quad \delta_2^{\text{mix}}(E, F_{I_3}) = \frac{1}{3}. \quad (7.16)$$

For $V_* = (1/2)I_3 + (1/6)\mathbf{1}\mathbf{1}^\top$,

$$\delta_{2,3}^h(E, F_{V_*}) = 0, \quad \delta_2^p(E, F_{V_*}) \geq \frac{1}{4\sqrt{3}} > 0, \quad \inf_r \delta_{2,r}^v(E, F_{V_*}) > 0. \quad (7.17)$$

Proof. The encoder maps the consumed i into its retained label by A ; the terminal decoder maps that label into j by B . This proves (7.14) including its converse implementation. If $V = CD$ is any nonnegative factorization, let $s_k = \sum_j D_{kj}$. Delete zero rows of D , define $B_{kj} = D_{kj}/s_k$ and $A_{ik} = C_{ik}s_k$. Since V is stochastic, the rows of A sum to one. Padding with unused states gives the claimed equivalence with nonnegative rank. The one-state statement follows immediately.

For $V = I_n$, row error is $1 - (AB)_{ii}$. If the least correct probability is c , for each i select a row $k(i)$ of B with $B_{k(i),i} \geq c$. Some row is selected at least $\lceil n/K \rceil$ times, so stochasticity gives $c \leq 1/\lceil n/K \rceil$. A partition into K groups of sizes at most $\lceil n/K \rceil$, each decoded uniformly, attains that bound. Also $\sum_i (AB)_{ii} \leq K$, by the calculation below with K in place of two. It remains valid under mixing, giving hidden error at least $1 - K/n$. Start with a deterministic channel that reproduces K input symbols exactly and sends all others to those K outputs. Average all simultaneous permutations of its input and output labels. Every diagonal entry is then K/n , proving the second formula. Inverting the two monotone formulas gives the stated least widths.

For $V = I_3$, total variation in row i is $1 - (AB)_{ii}$. If all three diagonal entries of AB were greater than $1/2$, then for each i some row of B would have its i th entry greater than $1/2$. A stochastic row has at most one such entry, so two rows cannot serve three indices. This proves the private lower bound $1/2$. Merge two input symbols into a retained label that decodes uniformly to that pair, and retain the third symbol exactly. This attains it. For every stochastic factorization with two intermediate states,

$$\sum_{i=1}^3 (AB)_{ii} = \sum_{k=1}^2 \sum_{i=1}^3 A_{ik} B_{ki} \leq \sum_{k=1}^2 \sum_{i=1}^3 B_{ki} = 2.$$

The same inequality holds under hidden mixtures, giving maximal row error at least $1/3$. Mix uniformly the three preceding pair-merge channels. The resulting diagonal entries are $2/3$ and the off-diagonal entries are $1/6$. This is exactly V_* and has error $1/3$ from I_3 . It proves the hidden claims in both displays.

The smallest singular value of V_* is $1/2$. Any AB with two intermediate states has a unit vector x in its right null space, so $\|V_* - AB\|_F \geq \|(V_* - AB)x\|_2 \geq 1/2$. If the maximal row total variation error is ε , each row has Euclidean error at most 2ε , and hence the Frobenius error is at most $2\sqrt{3}\varepsilon$. This proves the explicit private bound. The optimized visible value is attained by Theorem 7.2; if it were zero, its zero-set assertion would contradict that bound. This proves its strict positivity. $\square$

7.4. Optimal resource sets and decision certificates. For a class $\mathcal{C}_{S,q}(F)$ of common consumers, with all other information conventions fixed, let $\mathcal{R}_{S,q}(F)$ be its vector of attainable $[0, 1]$ risks. For a nonnegative row weight w summing to one, write

$$h_F(S, q; w) = \inf_{r \in \mathcal{R}_{S,q}(F)} \langle w, r \rangle.$$

If E privately simulates F with cost K and error ε , then

$$\mathcal{R}_{S,q}(F) \subseteq \mathcal{R}_{KS,q}(E) + [-\varepsilon, \varepsilon]^J, \quad h_E(KS, q; w) \leq h_F(S, q; w) + \varepsilon. \quad (7.18)$$

For nonfinite row sets the first expression means pointwise paired risk functions within uniform distance ε ; no finite coordinate box is needed. Upper sets give the equivalent one-sided order formulation. Hidden selectors multiply private width by r ; visible selectors multiply the separate public alphabet by r . These are direct consequences of the final part of Theorem 7.2, with the optimal intrinsic error in place of an externally estimated simulator error.

When two stateless comparison directions have error at most ε , paired risk sets and their support values at unchanged width are within ε . If each is centered by a paired unrestricted benchmark whose error is also at most ε , the centered values are within 2ε . Therefore a gap $g > 0$ between zero excess and infeasible widths is stable under $\varepsilon < g/4$: a value below $g/2$ in the approximation certifies feasibility in the exact finite problem, and a value above $g/2$ certifies infeasibility. The benchmark and the class being optimized must be specified; this assertion does not identify different information signatures.

Proposition 7.5 (Intrinsic finite-dimensional selector approximation). *Let $\mathcal{R}$ be a compact risk image in $C(J)$, where J is compact. Suppose a bounded linear operator $L : C(J) \rightarrow C(J)$ has rank at most m and $\sup_{r \in \mathcal{R}} \|r - Lr\|_\infty \leq \eta$. Every barycenter $\bar{r}$ of a probability on $\mathcal{R}$ has a barycenter r_* of at most $m + 1$ members of $\mathcal{R}$ with $\|\bar{r} - r_*\|_\infty \leq 2\eta$. If $\mathcal{R}$ lies in a finite affine space of dimension m , the error is zero with at most $m + 1$ members. Thus the selector price can be controlled by the dimension of attainable risks, rather than by a covering number of the row space.*

Proof. The compact set $L\mathcal{R}$ lies in an m -dimensional space. Its convex hull is compact, and $L\bar{r}$ belongs to it. The finite-dimensional affine-dependence argument in the proof of Theorem 7.2 gives $r_1, \dots, r_{m+1} \in \mathcal{R}$ and weights λ such that $L\bar{r} = \sum_i \lambda_i Lr_i$. Put $r_* = \sum_i \lambda_i r_i$. The two residual terms $\bar{r} - L\bar{r}$ and $r_* - Lr_*$ have norms at most η , proving the estimate. Apply the same argument after translation in the exact affine case. The implementation stores the selected design index whenever it is not provided as visible side information. $\square$

Remark 7.6 (Finite computation and general carriers). For rational finite instruments, (7.6) and the fixed- r formulations are finite polynomial feasibility problems; (7.7) is a finite linear program. These are exact variational characterizations, not assertions of polynomial running time in a succinct history description. The number of feedback trees, events, or deterministic tables can be large. On standard Borel carriers the same definitions use suprema over measurable feedback and tests and infima over executable kernels. Their product-state composition and decision transport remain valid by the same channel argument. Attainment and the finite testing dual are asserted here only for the stated finite class. The average-risk and physical results below give separate nonfinite theorems with their own hypotheses.

8. CONTINUATION GEOMETRY AND AN EXACT MEMORY OBSTRUCTION

8.1. Normalized continuation laws. We first separate the transducer from the source to which it is attached. There are L epochs with finite input alphabets X_t and output alphabets Z_t . A causal kernel $p(z_{1:L} \mid x_{1:L})$ is nonnegative, normalized for every input word, and has output-prefix marginals independent of later inputs. The current input can include both a source report and the

current control. It never includes an unavailable parameter, a terminal mark, or an earlier output not retained by the machine.

Let $\mathcal{C}_t$, $0 \leq t \leq L$, be the finite-dimensional polytope of normalized causal kernels from $x_{t+1:L}$ to $z_{t+1:L}$; put $\mathcal{C}_L = \{1\}$. For $v \in \mathcal{C}_t$ define the shift

$$(S_{x,z}v)(z_{t+2:L} \mid x_{t+2:L}) = v(z, z_{t+2:L} \mid x, x_{t+2:L}).$$

Its total mass $v(z \mid x)$ is independent of the future input word. A zero shift is allowed.

Definition 8.1 (Compatible continuation polytopes). A continuation cover of width $K = (K_0, \dots, K_L)$ consists of lists $V_t = (v_{t,m})_{m \leq K_t}$ in $\mathcal{C}_t$ such that, for every m, x, z ,

$$S_{x,z}v_{t,m} \in \text{cone } V_{t+1}. \quad (8.1)$$

It covers p when $p \in \text{conv } V_0$. Lists may repeat generators and may contain unreachable generators. Thus unused states can be adjoined.

Theorem 8.2 (Continuation realization with a width profile). *A causal kernel p has a stochastic transducer realization with an initial K_0 -valued register and K_t retained states after epoch t if and only if it has a compatible continuation cover of width K . Initial randomization is part of the charged initial register. With a deterministic initial state the condition is $p = v_{0,1}$ and $K_0 = 1$.*

Consequently the set of feasible width profiles depends only on the causal kernel and its declared input interface, not on a row presentation or a labeling of the hidden states. Every matrix of conditional continuation probabilities at cut t has nonnegative rank at most K_t . Individual rank bounds do not replace the simultaneous shift conditions (8.1).

Proof. Given a transducer, let $v_{t,m}$ be the suffix response when its register at time t equals m . The remaining rows define this response even for an unreachable state. If the next row is $q_{t+1}(z, m' \mid m, x)$, conditioning on its output and new state gives

$$S_{x,z}v_{t,m} = \sum_{m'} q_{t+1}(z, m' \mid m, x) v_{t+1,m'}.$$

The initial state distribution expresses p as a convex combination of V_0 .

Conversely choose nonnegative coefficients in every cone membership in (8.1). There are only finitely many choices to make. Since every generator on the right has mass one for each future input word, the sum of the coefficients for a fixed m, x, z is $v_{t,m}(z \mid x)$. Summing over z gives one. The coefficients are therefore stochastic rows $q_{t+1}(z, m' \mid m, x)$. Choose an initial distribution representing p in $\text{conv } V_0$. Backward induction from $\mathcal{C}_L = \{1\}$ proves that the constructed machine has exactly the response $v_{t,m}$ from each register state. In particular its initial response is p . Equality for open-loop input words implies equality under adaptive input selection: at each node of an input–output history tree, the same causal conditional probability is used on the two sides.

For the rank assertion, condition on any positive-probability observed history at the cut. Its continuation response is a convex combination of $v_{t,m}$ with weights equal to the posterior register distribution. Stack these responses as rows, and use all future input–output pairs as columns. This is a nonnegative factorization through K_t generators. The converse construction required the same generator families at adjacent cuts and all their shifts; an unrelated factorization of each cut matrix provides no such compatibility. The characterization, being stated entirely in terms of these response polytopes, also proves the asserted invariance. $\square$

Attach such a causal kernel to a marked source experiment. The output law is obtained by contraction with the source’s causal instruments, while its actual mark is retained. Hence zero private deficiency is equivalently the existence of a compatible continuation cover whose contraction equals the target marked behavior in every parameter and feedback row. This is the geometric form of the factorization in Theorem 7.2; no conditional dependence on the mark is introduced.

A hidden selector chooses one such cover independently of the source. It is a convex combination of complete realizations, not a convexification inside every retained-state simplex at the original width. Its private implementation uses the disjoint union of the covers and keeps the selector in the state. A visible selector keeps those fibers in the compared law. The nonnegative fiber argument in Theorem 7.2 therefore continues to give equality of the visible and private zero sets.

8.2. An erasure cut. A simple family makes the width obstruction exact, including its error. Let X take values in $[N]$ with law π . Before a specified cut the source reveals X . After the cut it has no reports correlated with X . The controller has no feedback channel carrying X across the cut, and all earlier emitted reports are prescribed constants. The only permitted cross-cut information of the simulator is its K -valued register and any independent selector explicitly included in its signature. The terminal mark is the actual $W = X$. The target's terminal report is X . All memory at other checkpoints is unrestricted.

Theorem 8.3 (Exact cut spectrum). *For this experiment, for private, hidden and visible signatures alike,*

$$\delta_K = 1 - \sum_{i=1}^{\min(K,N)} \pi_{(i)}. \quad (8.2)$$

For uniform X , $\delta_K = (1 - K/N)_+$. Thus an erasure cut for m unbiased bits requires $K \geq (1 - \epsilon)2^m$ whenever its error is at most ϵ . The bound holds for every surrounding width profile; equality is attained when the other checkpoints are unrestricted. If every $\pi_x > 0$, exact transmission has nonnegative cut rank N .

Proof. With all post-cut randomness averaged into the decoder, any private machine gives kernels $e(m | x)$ and $d(z | m)$. Its joint terminal law is $\pi_x \sum_m e(m | x) d(z | m)$. Since the target law is $\pi_x \mathbf{1}_{z=x}$, total variation is exactly one minus the correct-decoding probability. Put $u_x = \max_m d(x | m)$. Then

$$0 \leq u_x \leq 1, \quad \sum_x u_x \leq K, \quad \mathbb{P}(Z = X) \leq \sum_x \pi_x u_x \leq \sum_{i=1}^{\min(K,N)} \pi_{(i)}.$$

The last inequality follows by moving mass in u from a smaller weight to a larger one until the K largest weights are filled. To attain it, retain distinct labels for those letters and decode them deterministically; all other letters are mapped to any retained label. Before the cut one may use the unrestricted memory to compute this encoding.

For a hidden selector, condition on its value. It is independent of X , so every conditional correct-decoding probability obeys the same bound; averaging cannot improve it. For a visible selector, variation is the average of the fiber variations, with the same conclusion. The private construction works for both signatures. Finally, the exact joint cut matrix is diagonal with its N positive diagonal entries π_x . Its ordinary rank is N , so its nonnegative rank is at least N and its diagonal factorization attains N . $\square$

The absence of an auxiliary cross-cut feedback channel is essential. An unpriced controller storing X and returning it as a later control would change the experiment, not disprove the cut bound. Likewise a selector correlated with X is not an independent hidden selector. These distinctions are the same ones needed in the collision certificate below.

9. DECISION CONTINUATION SPECTRA

The memory at an experimental cut has a different role from the memory needed to reach that cut. We first identify its exact asymptotic value. This will give both an obstruction and

a construction for the same physical audit, rather than a lower bound for an auxiliary binary problem.

9.1. A charged training-validation interface. Let Ω be a finite marked alphabet, $d = |\Omega|$, let $B \subset \Delta(\Omega)$ be nonempty and compact, and fix $Q \in \Delta(\Omega)$. A candidate $b \in B$ is fixed throughout the experiment. Training consists of n independent observations from b . At the end of training the auditor retains a register $M \in [K]$. Everything depending on the training observations or on an earlier random seed must pass through this register. In particular, there is no external copy of the training history and no uncharged persistent selector.

After this cut, fresh independent randomness may select an event A using M . The event is chosen before independent candidate and target validation observations $X \sim b$ and $Y \sim Q$ are drawn. The score is $\mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The downstream register needed to keep A and to compute this score is distinct from the register at the specified cut. The upstream and downstream widths will be stated separately in concrete implementations. This is the reset interface of Section 18, with one additional charged checkpoint. The actual mark is part of a completed observation; it is not an input to the private candidate's online transition rows.

Write $\mathcal{H} = [0, 1]^\Omega$ and

$$f_h(b) = \sum_{w \in \Omega} (Q(w) - b(w))h(w), \quad h \in \mathcal{H}.$$

Every randomized event has a response $h(w) = \Pr(w \in A)$. Conversely, every $h \in \mathcal{H}$ is the response of a random event: include each atom independently with its specified probability. This is a finite probability distribution on 2^Ω . Only its mean response matters for expected signed score. The event itself must still be retained during validation.

Definition 9.1 (Decision continuation spectrum). The K -state decision continuation value is

$$\mathfrak{V}_K(B, Q) = \max_{h_1, \dots, h_K \in \mathcal{H}} \min_{b \in B} \max_{m \leq K} f_{h_m}(b). \quad (9.1)$$

Let $v_{n,K}(B, Q)$ be the supremum of the worst-case expected score of n -sample auditors at the preceding interface. Only the designated cut has width K in this definition; all other widths are finite but unrestricted. For $c \in \mathbb{R}$, let $\kappa_c(B, Q)$ be the least K for which $\mathfrak{V}_K(B, Q) > c$, with value $+\infty$ when no such K exists.

The definition uses only the marked behavior image and its target, not a parameterization of B or a labeling of candidate states. No convex hull of B is taken. A response list is a list of conditional validation continuations; it is not a mixture of feasible private candidates.

Theorem 9.2 (Operational classification at a decision cut). *All extrema in (9.1) are attained. For every $n \geq 1$,*

$$0 \leq \mathfrak{V}_K(B, Q) - v_{n,K}(B, Q) \leq \min \left\{ \sqrt{\frac{d-1}{n}}, \sqrt{\frac{2 \log(2K)}{n}} \right\}. \quad (9.2)$$

Consequently $v_{n,K} \rightarrow \mathfrak{V}_K$. Every auditor with K states at the designated cut, at any finite training length, has worst-case score at most $\mathfrak{V}_K$. If $\mathfrak{V}_K > c$, some finite-sample auditor with K states at that cut has worst-case score greater than c . Thus κ_c is exactly the minimum cut width permitting such a strict certificate with sufficiently many training samples.

Equivalently, κ_c is the least number of response half-spaces

$$\{b \in B : f_h(b) > c\}, \quad h \in \mathcal{H}, \quad (9.3)$$

that cover B . If two parameterized behavior families satisfy, uniformly in their common parameter q ,

$$\|b(q) - \tilde{b}(q)\|_{\text{TV}} \leq \alpha, \quad \|Q - \tilde{Q}\|_{\text{TV}} \leq \beta,$$

then

$$|\mathfrak{V}_K(B, Q) - \mathfrak{V}_K(\tilde{B}, \tilde{Q})| \leq \alpha + \beta. \quad (9.4)$$

Proof. For a fixed response list, the inner objective is continuous in b . Its variation when the list changes is bounded uniformly over b by $2 \max_m \|h_m - \tilde{h}_m\|_\infty$. Its minimum over compact B is therefore continuous in the list. Compactness of $\mathcal{H}^K$ proves attainment.

Condition on the retained state of any admissible auditor. Its conditional post-cut event law depends only on that state and fresh randomness, and has a response h_m . Independence of validation from training gives

$$g(b) = \sum_{m=1}^K \Pr_b(M = m) f_{h_m}(b) \leq \max_m f_{h_m}(b).$$

Taking the minimum over b , and then the supremum over auditors, proves the upper bound. This argument permits arbitrary stochastic training and arbitrary finite upstream memory. An extra training-dependent decoder would invalidate its first conditioning step and is excluded by the interface, not silently supplied.

For the reverse bound fix an optimal list. Let $\hat{b}$ be the empirical measure and choose the first index maximizing $f_{h_m}(\hat{b})$. Retain that index and discard the empirical table. If m_* maximizes the true score, then pointwise

$$\max_m f_{h_m}(b) - f_{h_{m_*}}(b) \leq 2 \max_{m \leq K} |(\hat{b} - b)h_m|. \quad (9.5)$$

The maximum is at most $\|\hat{b} - b\|_{\text{TV}}$, since $0 \leq h_m \leq 1$. Cauchy–Schwarz and the multinomial variances give

$$\mathbb{E} \|\hat{b} - b\|_{\text{TV}}^2 \leq \frac{d}{4n} \left(1 - \sum_w b(w)^2 \right) \leq \frac{d-1}{4n}.$$

This proves the first rate.

For completeness, if V takes values in an interval of length one, then $\log \mathbb{E} e^{s(V - \mathbb{E}V)} \leq s^2/8$. Indeed the second derivative of the logarithmic moment generating function is the variance under the exponentially tilted law, at most $1/4$; its value and first derivative vanish at zero. Integrating twice proves the inequality for both signs of s . Independence consequently gives $\mathbb{E} e^{s(b-b)h_m} \leq e^{s^2/(8n)}$. For $s > 0$,

$$\mathbb{E} \max_m |(\hat{b} - b)h_m| \leq \frac{\log(2K)}{s} + \frac{s}{8n}.$$

Minimizing in s gives $\sqrt{\log(2K)/(2n)}$ and proves the second rate in (9.2). The empirical table has finitely many values; the construction is therefore a finite-state auditor, with only the chosen index surviving the specified cut. Real response coefficients are allowed here; the physical construction below has deterministic rational transitions.

A finite cover in (9.3) says that the continuous function $b \mapsto \max_m f_{h_m}(b)$ is greater than c everywhere on compact B . Its attained minimum is then greater than c . This proves the covering characterization in both directions. It also proves the strict-threshold assertion; no claim about finite-sample attainment at the boundary $c = \mathfrak{V}_K$ is needed.

Finally $|(Q - b)h - (\tilde{Q} - \tilde{b})h| \leq \alpha + \beta$ for every response. Taking maxima, minima over the common parameter set, and maxima over lists preserves this uniform bound, proving (9.4). $\square$

Corollary 9.3 (Compression by a finite decision dictionary). *Suppose responses $h_1, \dots, h_K$ satisfy $\min_{b \in B} \max_m f_{h_m}(b) \geq c + \eta$ for $\eta > 0$. An empirical selector with $n > 2 \log(2K)/\eta^2$ achieves score greater than c and retains only K states at the decision cut. If these responses depend on a finite statistic $T : \Omega \rightarrow \Xi$, the training selector can count $T(X)$ instead of X . A sufficient*

upstream register has at most $\binom{n+|\Xi|-1}{|\Xi|-1}$ states at the last training checkpoint, and the corresponding time-dependent count registers suffice at earlier checkpoints.

Proof. Use the second bound in (9.2). Empirical expectations of h_m depend only on the histogram of $T(X)$. At time t , that histogram has $\binom{t+|\Xi|-1}{|\Xi|-1}$ possibilities. Its update and the final maximization are deterministic. No statistical sufficiency for estimating all of b is required; sufficiency for the specified responses is enough. $\square$

This corollary supplies an intermediate principle between a full marked histogram and a problem-specific compressed counter. The invariant is the number and geometry of validation continuations needed to cover the original candidate image at the desired score. It is not the algebraic dimension of a chosen parameterization. The empirical upper bound need not be sample-optimal, and a minimum decision-cut width is not a minimum of the largest width over the entire acquisition schedule.

9.2. Several feedback rows. The classification does not require that the declared experiment have only one feedback row. For the row-indexed interface of Section 18, let $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$ and define

$$f_\lambda(b) = \sum_{i \in \mathcal{I}} \lambda_i f_i(b), \quad \lambda \in \Delta(\mathcal{I}).$$

A post-cut continuation is now a probability distribution on row–event pairs; it is not an arbitrary test on a joint product of validation rows. Put

$$\mathfrak{V}_K^{\text{rows}}(B, Q) = \max_{\lambda_1, \dots, \lambda_K \in \Delta(\mathcal{I})} \min_{b \in B} \max_m f_{\lambda_m}(b).$$

Corollary 9.4 (Row-indexed classification). *With n training samples per row, the row-indexed decision-cut value converges to $\mathfrak{V}_K^{\text{rows}}$, and its deficit is at most*

$$\min\{\sqrt{D/n}, \sqrt{2 \log(2K)/n}\}, \quad D = \sum_j (|\Omega_j| - 1).$$

The strict half-space cover characterization holds with f_λ . The upper bound by $\mathfrak{V}_K^{\text{rows}}$ applies also to any other finite training schedule, including adaptive row selection.

Proof. Conditioning on the cut state gives the same convex-mixture upper bound as before; its proof places no restriction on how that state was reached. For empirical selection among an optimal list, $|\widehat{b} - b| \lambda_m \leq \max_j \|\widehat{b}_j - b_j\|_{\text{TV}}$. Sum the rowwise second moments to obtain the first rate. To obtain the second, align the r th independent training sample from each row into a tuple and write its response as $\sum_{j,A} \lambda_{m,j,A} \mathbf{1}_A(X_{j,r})$. This random variable belongs to $[0, 1]$, and its sample mean is the empirical candidate mean for that continuation. The logarithmic moment-generating argument in Theorem 9.2 therefore gives the same finite-list rate. Each chosen row–event distribution can be drawn using fresh randomness after the cut. Compactness and the covering argument are unchanged. $\square$

10. FINITE PRECISION AND CHARGED RANDOMIZATION

We distinguish the number of persistent labels from the description of a transition and from the clock that selects it. An observation in a finite alphabet is initially an atomic input. Every random bit used after that input is a separate, charged epoch. Section 11 gives the additional cost of splitting the observation itself.

Lemma 10.1 (Dyadic rows and finite randomizers). *A probability vector on s points has a dyadic approximation with denominator 2^r at total variation distance at most $(s-1)2^{-r}$. It can be sampled using exactly r independent fair bits and at most $2s-1$ persistent labels at each intermediate bit*

epoch. The label count uses the bit position supplied by the schedule. No partial random integer is retained outside these labels.

Proof. Round down the first $s - 1$ coordinates and assign the remaining mass to the last coordinate. The positive and negative parts of the change have equal mass, at most $(s - 1)2^{-r}$. For sampling, partition $\{0, \dots, 2^r - 1\}$ into consecutive intervals of the prescribed integer lengths. A binary prefix describes a dyadic interval. If that interval is contained in one output interval, keep the output label and ignore the remaining bits. Otherwise keep the prefix interval. At a fixed depth, every unresolved interval contains a partition boundary in its interior; there are at most $s - 1$ such intervals. Together with at most s resolved outputs this gives $2s - 1$ labels. The children and their resolved labels are fixed by the read-only partition and the depth. Thus the construction is an actual finite transition table, not a real-valued random seed. Zero-length intervals cause no difficulty. $\square$

Proposition 10.2 (Rational decision continuations). *Let $d = |\Omega|$. A list of K responses in $[0, 1]^\Omega$ can be replaced by dyadic responses of denominator 2^r , losing at most 2^{1-r} in its uniform score envelope. After a decision cut retaining K labels, the randomized event for the chosen dyadic response can be drawn using r fair bits, at most $K(2d + 1)$ labels during this draw, and at most $K(d + 1)$ labels afterwards. Keeping the event through one candidate and one target validation requires at most $2K(d + 1)$ labels at the intervening atomic-observation cuts. Its final signed score has three values.*

Proof. Round every coordinate down. The sup-norm error is at most 2^{-r} , and $\sum_w |Q(w) - b(w)| \leq 2$. This proves the envelope estimate. For a fixed rounded response h , draw a uniform dyadic integer U and choose $A_U = \{w : U < 2^r h(w)\}$. These events are nested and have at most $d + 1$ distinct values. Their mean response is exactly h . Apply Lemma 10.1 to their interval lengths, retaining the chosen response index during sampling. Afterwards the event label and the candidate-event indicator suffice; the target indicator emits their difference. All event masks are read-only data fixed before observing the candidate. $\square$

For K responses, their rounded coordinates require $Kd(r + 1)$ bits in a direct read-only representation. These bits are not charged as persistent sample-dependent labels, but are not declared costless precision. A construction using them records both quantities. In particular, the proposition turns every strict response-cover margin into a finite-randomness implementation; it says nothing about exact attainment of an irrational boundary value.

11. APPROXIMATE CONTINUATION PROFILES AND RESOURCE TRANSFER

A cut profile records the whole causal schedule, rather than its final compression. We give both a realization certificate and an operational value. These notions keep the compatibility condition of Theorem 8.2; separate nonnegative factorizations at individual cuts are not a substitute for that condition.

11.1. A causal comparison norm. For two normalized finite causal suffix kernels p, q at cut t , put

$$d_t^{\text{ca}}(p, q) = \sup_{\pi} \|P_p^\pi - P_q^\pi\|_{\text{TV}},$$

where π ranges over deterministic causal input policies on the remaining schedule and P_p^π is the law of its complete output transcript. Randomized policies give the same supremum, by conditioning on their independent random tables. In particular this distance includes feedback; the supremum over open-loop input words alone is not used.

For stochastic coefficients $q_t(z, j \mid m, x)$ and suffix kernels $v_{t+1,j}$, denote by $\mathcal{Q}_{t+1,m}$ the suffix kernel which, upon input x , emits z and chooses j with these coefficients, then uses $v_{t+1,j}$. The notation denotes one first-step operation, not a conditioning on an event of zero probability.

Theorem 11.1 (Approximate compatible realization). *Let $K_0, \dots, K_L$ be positive integers. Suppose normalized causal kernels $v_{t,m}$, $m \leq K_t$, and stochastic rows q_t satisfy*

$$d_t^{\text{ca}}(v_{t,m}, \mathcal{Q}_t v_{t+1,m}) \leq \eta_t \quad (t < L, m \leq K_t). \quad (11.1)$$

At the terminal cut the suffix kernels are the unit empty kernel. If $\rho \in \Delta_{K_0}$ and $d_0^{\text{ca}}(p, \sum_m \rho_m v_{0,m}) \leq \eta_{-1}$, then the machine with initial law ρ and transition rows q_t has the whole profile (K_t) and realizes a kernel $\hat{p}$ satisfying

$$d_0^{\text{ca}}(p, \hat{p}) \leq \min\{1, \eta_{-1} + \sum_{t=0}^{L-1} \eta_t\}. \quad (11.2)$$

Conversely, a machine with this profile within distance ε of p supplies such a certificate with $\eta_{-1} = \varepsilon$ and all local errors zero. If a deterministic initial state is required, use $K_0 = 1$ throughout both statements.

Proof. Let $\hat{v}_{t,m}$ be the actual continuation of the rows q from state m . Fix a future policy. Conditional on the first output and next state, its remaining actions form an admissible suffix policy. Replacing $v_{t+1,j}$ by $\hat{v}_{t+1,j}$ therefore changes its transcript law by at most $\max_j d_{t+1}^{\text{ca}}(v_{t+1,j}, \hat{v}_{t+1,j})$. Mixtures and pushforwards contract total variation. The triangle inequality and (11.1) imply, backwards from L ,

$$\max_m d_t^{\text{ca}}(v_{t,m}, \hat{v}_{t,m}) \leq \sum_{s=t}^{L-1} \eta_s.$$

Mixing with ρ proves (11.2). The converse takes the actual normalized continuations of the given machine, including arbitrary valid continuations for unreachable states. Their one-step equations are identities. There is no assertion that independently optimal approximate lists at successive cuts have compatible updates. $\square$

11.2. The full value and its effective approximation. Fix a finite schedule with finitely many observation and action symbols, N calls to an experiment, and a bounded terminal payoff $G \in [-1, 1]$. An environment $b \in B$ specifies the conditional observation kernels at all possible visible histories and actions. Assume B is a nonempty compact subset of the resulting finite product of probability simplexes. A controller A has a specified persistent profile $\mathbf{K}$; the clock has no access to sample history. All action choices and all retained randomness factor through the current register. Define

$$V_{N,\mathbf{K}}(B, G) = \max_{A \in \mathcal{M}(\mathbf{K})} \min_{b \in B} \mathbb{E}_b^A G. \quad (11.3)$$

Unused states pad smaller machines, so $\mathcal{M}(\mathbf{K})$ is a finite product of stochastic-row simplexes, with any prescribed causal zeros imposed. The terminal readout is included among its rows.

Theorem 11.2 (Full-profile resource transfer). *The maximum in (11.3) is attained. Its value is nondecreasing in each K_t . Removing a charged intermediate cut gives a relaxation, whereas inserting identity-holding cuts preserves every implementation with the same held register.*

Suppose two environments at each of their N calls differ in conditional total variation by at most δ_i , uniformly over corresponding histories, actions and parameters. Replace the controller rows at its L transitions by dyadic rows of denominator 2^r , with at most s_t possible next-state/output pairs at transition t . Any fixed controller loses at most

$$2 \min \left\{ 1, \sum_{i=1}^N \delta_i + \sum_{t=1}^L (s_t - 1) 2^{-r} \right\} \quad (11.4)$$

in expected payoff. The same bound compares optimal values on the same profile, with the appropriate dyadic approximation on one side. More generally an approximate continuation certificate contributes its error in (11.2) inside the braces.

If $B_0 \subset B$ is a finite ω -net in the maximum conditional-TV metric and v_0 is the maximum over the dyadic controller grid of the minimum over B_0 , then

$$v_0 - 2N\omega \leq V_{N,\mathbf{K}}(B, G) \leq v_0 + 2 \sum_{t=1}^L (s_t - 1)2^{-r}. \quad (11.5)$$

The bounds are certificates whenever the net and the finite evaluations are certified. Compactness alone supplies existence of nets, not an algorithm for an arbitrary presentation of B .

Proof. For fixed b , each path probability is a finite product of controller and environment coordinates. Summing the finitely many paths gives a jointly continuous payoff. Its minimum over compact B is continuous in the controller rows; the controller space is compact. This proves attainment. Padding proves monotonicity. Removing a constraint enlarges the feasible set, and identity transitions prove the insertion assertion.

For the perturbation bound, couple two executions until their first mismatch. On equal histories the conditional observation laws can be maximally coupled, with mismatch probability at most δ_i . Couple the controller transitions similarly, using Lemma 10.1. The probability of any mismatch is at most the sum in (11.4), capped at one. The payoff changes by at most two on that event. This coupling is recursive on finite histories and remains valid for feedback controls. Alternatively, approximate continuation kernels are compared in the causal distance and then composed, as in Theorem 11.1.

For a fixed grid controller, a minimizing point of B has a point of B_0 within ω ; the corresponding payoffs differ by at most $2N\omega$. Thus the grid optimum on all of B is at least $v_0 - 2N\omega$. Conversely approximate a maximizing unrestricted controller row by row. Its minimum over B decreases by at most the row-rounding bound, and its minimum over B_0 is no smaller. Taking the grid maximum proves the upper bound. No minimax interchange or convexification of the fixed-width behavior image occurs. $\square$

The formula records preparation count, persistent labels, calibration error, causal simulation error and transition precision in one execution. The training-count factor is present when training kernels change. It is absent only when those kernels are identical. Failure probabilities are also transported, with factor one rather than two, since their indicators have range $[0, 1]$.

Proposition 11.3 (Composition and schedule refinement). *A causal simulator of profile $\mathbf{R}$ followed on the same synchronized schedule by a controller of profile $\mathbf{K}$ has a realization with profile $(R_t K_t)$. Causal simulation errors add under successive composition. Two disjoint phases with a genuine reset and no retained bridge use the maximum of their peak widths. If a bridge with H labels must survive the second phase, its direct width bound is H times the second-phase width.*

If one atomic input x is encoded by ℓ symbols, let D_j be the number of possible length- j prefixes. Holding the pre-input state and the prefix realizes the refined acquisition cuts with widths $K_{\text{before}} D_j$, followed by the original update. A time-dependent machine running for T scheduled epochs becomes autonomous by retaining its clock, with at most $(T + 1) \max_t K_t$ states. These are implementation bounds, not claims of optimality for every refinement.

Proof. Keep the ordered pair of simulator and controller registers and compose their transition kernels. The simulator's emitted observation is the controller's current input. Conditional coupling until the first simulation discrepancy gives the error statement. A genuine reset permits reusing the alphabet, whereas a surviving bridge is another register. A partial encoded input is determined by its prefix, giving the stated refinement construction. Finally the augmented state (t, m) implements the time-indexed transition without an external clock. These constructions also show why neither hidden selectors nor an external transcript may be omitted from the register being counted. $\square$

12. FROM A DECISION COVER TO AN END-TO-END MACHINE

The unrestricted empirical table in Theorem 9.2 can be replaced by a bounded-state training procedure. This replacement trades preparation count against memory, rather than assuming that the training computation has already been performed. The elementary block mechanism belongs to the classical finite-statistics theory of Cover [24]; the result below tracks a finite horizon, a uniform indifference band, rational precision, the whole validation phase, and a compact nonconvex candidate image.

Lemma 12.1 (A finite-horizon four-state comparison). *Let Z_i be independent Bernoulli observations of unknown mean u . For $0 < \theta < 1/2$ and $0 < \delta < 1$, put*

$$L = \left\lceil \frac{\log(2/\delta)}{4\theta} \right\rceil, \quad R = \left\lceil 2^L \log(2/\delta) \right\rceil. \quad (12.1)$$

There is a deterministic clocked machine with four states at every observation cut, using exactly $2LR$ observations, that decides the sign of $u - 1/2$ with error at most δ whenever $|u - 1/2| \geq \theta$. No assertion is imposed within the indifference band.

Proof. Partition the observations into R pairs of blocks, each block of length L . A positive block succeeds when all its observations are one; the following negative block succeeds when all are zero. Stop updating the decision at the first successful block and retain its sign. If none succeeds, decide negative at the end. The prescribed remaining observations are still consumed, but ignored.

Two states are absorbing positive and negative decisions. The other two mean undecided with the current block still alive or already failed. At a block boundary the clock resets the undecided flag to alive. Its polarity and position are supplied by the clock; they are not inferred from an unrecorded sample. The last observation of each block either latches the sign or starts the next block. Thus four states suffice even at the block boundaries.

Write $x = u^L$, $y = (1 - u)^L$ and $s = x + (1 - x)y$. The probability of a positive decision is exactly

$$H_{L,R}(u) = \frac{x}{s} \{1 - (1 - s)^R\}. \quad (12.2)$$

Here $s > 0$, including at $u = 0, 1$. For $u \geq 1/2 + \theta$, the error is at most $y/x + (1 - s)^R$. For $u \leq 1/2 - \theta$, it is at most x/y ; the default negative decision is then correct. Since

$$\log \frac{1/2 + \theta}{1/2 - \theta} \geq 4\theta, \quad s \geq \max(x, y) \geq 2^{-L},$$

both errors are bounded by $e^{-4\theta L} + e^{-R2^{-L}} \leq \delta$. The logarithmic inequality follows by differentiation from its equality at zero. This also proves uniformity over the two composite hypotheses. $\square$

12.1. The whole-schedule continuation value. In the one-row reset interface of Section 9, let $W_c(B, Q)$ be the least peak width over all finite training lengths and all auditors whose worst-case expected signed score is strictly greater than c . Every training-observation cut, the decision cut, every fresh random-bit cut, and both validation cuts are charged. Observations in Ω are atomic for this definition. The event is selected before the two validations, exactly as in that interface. The schedule clock and read-only constants are recorded as separate resources. The class includes randomized machines; its rational subclass uses only finite fair-bit randomizers. Write W_c^{rat} for the corresponding minimum in the latter class. Empty feasible sets have value $+\infty$.

Theorem 12.2 (Uniform end-to-end realization of a decision cover). *Let $B \subset \Delta(\Omega)$ be nonempty and compact, Q fixed, and $d = |\Omega|$. For every strict threshold c ,*

$$\kappa_c(B, Q) \leq W_c(B, Q) \leq W_c^{\text{rat}}(B, Q) \leq 12d \kappa_c(B, Q). \quad (12.3)$$

The last inequality is interpreted only for finite κ_c ; when $\kappa_c = +\infty$ all three quantities are infinite.

More explicitly, fix $K \geq 2$ and a response list with uniform envelope at least v . Round its coordinates to denominator 2^r , and choose $q_i \in [0, 1] \cap 2^{-r}\mathbb{Z}$ with $|q_i - Qh_i^{(r)}| \leq \rho$. For θ, δ as in Lemma 12.1, a rational auditor uses

$$N = 2LR(K - 1) \quad (12.4)$$

candidate training observations and two validation observations. Its expected score is at least

$$v - 2^{1-r} - 2\rho - 4(K - 1)\theta - 2(K - 1)\delta. \quad (12.5)$$

Its completed-training-observation cuts have width at most $4K$, its decision cut has width K , and its entire random-bit-refined schedule has width at most $12Kd$. No empirical histogram is constructed. The response and comparison thresholds have a direct read-only representation of size $O(K^2d(r + 3) + \log L + \log R + \log K)$ bits.

Proof. Use a sequential tournament. Initially response 1 is incumbent. At stage $j = 2, \dots, K$, compare incumbent i with challenger j using fresh candidate observations. For one observation $w \sim b$, generate a Bernoulli variable with conditional success probability

$$\frac{1}{2} + \frac{q_j - q_i - h_j^{(r)}(w) + h_i^{(r)}(w)}{4}. \quad (12.6)$$

This lies in $[0, 1]$ and is dyadic of denominator 2^{r+2} . Its unconditional mean is $1/2 + (\tilde{f}_j(b) - \tilde{f}_i(b))/4$, where $\tilde{f}_i(b) = q_i - bh_i^{(r)}$. Conditional on the incumbent at the start of this stage, all these variables are independent and have this mean. The fixed candidate b is not changed by the comparison. Run Lemma 12.1 and replace i by j exactly when its decision is positive.

At each comparison, conditionally on all earlier stages, the probability of choosing the inferior response when their $\tilde{f}$ values differ by at least 4θ is at most δ . With probability at least $1 - (K - 1)\delta$, no such error occurs. On this event each stage loses at most 4θ relative to the better of its two responses. Induction on j gives

$$\tilde{f}_i(b) \geq \max_i \tilde{f}_i(b) - 4(K - 1)\theta.$$

Replacing the calibrated means by their true means loses at most 2ρ . Rounding the original list loses at most 2^{1-r} in its envelope, by Proposition 10.2. Any response payoff belongs to $[-1, 1]$, so failure of the good event loses at most $2(K - 1)\delta$ in expectation. This proves (12.5), uniformly over b .

During a comparison keep the incumbent index and one of the four block states. The challenger and block positions are fixed by the schedule. At the end of a comparison the sign updates the incumbent and is forgotten. This gives the $4K$ bound at completed-observation cuts. While drawing the coin in (12.6), keep this register and the current observation, and use the three-label binary sampler of Lemma 10.1. Thus at most $12Kd$ labels occur at any intervening random-bit cut. At the final decision cut only the incumbent remains. Proposition 10.2 gives at most $K(2d + 1)$ labels during event selection and $2K(d + 1)$ during validation; both are bounded by $12Kd$. The three-valued terminal output fits this bound as well. These are compatible transitions over the entire schedule.

The coin thresholds can be precomputed for all triples (i, j, w) and stored as $r + 3$ -bit integers. The event thresholds and L, R, K give the stated description bound; no arbitrary real coefficient is used by the executed machine. Formula (12.4) counts every training observation, including ignored observations after a block decision has been latched.

For (12.3), a width- W auditor has at most W labels at its decision cut. The operational upper bound of Theorem 9.2 implies $\kappa_c \leq W$. If $K = \kappa_c < \infty$, its optimal list has a positive margin over c . Take r large, $\rho \leq 2^{-r}$, and then θ, δ small enough that the loss in (12.5) is smaller than that margin. If $K = 1$, no training is needed: round the single response and use Proposition 10.2.

Rational auditors are a subclass of all auditors, proving the remaining inequality. Finally a finite-width successful machine would give finite κ_c , proving the assertion at infinity. $\square$

The dependence on the schedule is essential. The theorem is a constant-factor classification for fixed observation alphabet, not an optimal sample-complexity theorem. In particular R is exponential in L . For an autonomous implementation, a coarse bound on the number of epochs is $T \leq N(r+3) + r + 4$; Proposition 11.3 then gives at most $12Kd(T+1)$ states. The free-clock and autonomous statements are different resource bounds. Cover's theorem already exhibits the importance of this distinction [24, Theorem 2 and Section 4]; bounded-state learning itself is not a novelty claim of the present theorem.

Corollary 12.3 (Finite-precision, calibrated certificates). *Suppose the true training kernels differ from those used in an ideal auditor by at most α per training call, and the candidate and target validation laws differ by at most β_c and β_t . The right side of (12.5) is decreased by at most*

$$2 \min\{1, N\alpha\} + \beta_c + \beta_t$$

for the same implemented machine. A calibration error ρ in the known response means is already accounted for separately in that formula. When the training source is unchanged, $\alpha = 0$, regardless of how large N is.

Proof. Couple the training executions until their first mismatch, as in Theorem 11.2. The selected-event payoff has range $[-1, 1]$, giving the factor two. Conditional on a common selected event, its two validation means change by at most the two total variation errors. This uses no independence between a failed coupling and its terminal event. $\square$

13. FINITE-HORIZON SAMPLE-MEMORY BOUNDS

The minimum W_c in Theorem 12.2 permits arbitrarily long finite training. To retain the preparation constraint, let $V_{N,K}(B, Q)$ denote the supremum of the worst-case expected score over the same reset auditors, using at most N training trials and satisfying the specified register bound at each acquisition, random-bit, decision and validation cut. The observation order and the external clock are fixed parts of this interface. Write $V_{N,W}$ when every bound equals W . Padding an audit with ignored trials shows monotonicity in N . This value does not identify machines having only the same terminal dictionary.

A second realization of a response cover gives a polynomial preparation bound. Its counter is larger than the four-state block counter, and both constructions remain useful. The elementary random-walk argument below is classical; the purpose of retaining its proof is to price its use in the complete, calibrated continuation machine.

Lemma 13.1 (A truncated absorbing comparison). *Let Z_i be independent Bernoulli variables with mean u . Fix integers $m, n \geq 1$ and $0 < \theta < 1/2$. Start at $S_0 = 0$, add $2Z_i - 1$ while $|S_{i-1}| < m$, and retain the state forever after either boundary is reached. At time n declare positive exactly when $S_n \geq 0$. The machine has $2m + 1$ states. Whenever $|u - 1/2| \geq \theta$, its error probability is at most*

$$e_{m,n}(\theta) = e^{-4\theta m} + e^{-2n\theta^2}. \quad (13.1)$$

If τ is the boundary hitting time without truncation, then, with $x = 2u - 1$,

$$\mathbb{E}_u \tau = \begin{cases} m \tanh(m \operatorname{arctanh} x) / x, & 0 < |x| < 1, \\ m^2, & x = 0, \\ m, & |x| = 1. \end{cases} \quad (13.2)$$

In particular $\mathbb{E}_u(\tau \wedge n) \leq \min(n, m^2)$, and outside the indifference band it is at most $\min(n, m^2, m/(2\theta))$.

Proof. First suppose $1/2 < u < 1$ and put $r = (1 - u)/u$. For the unabsorbed walk $T_j = \sum_{i \leq j} (2Z_i - 1)$, the process r^{T_j} is a nonnegative martingale. Stopping on $\{-m, m\}$ gives

$$\mathbb{P}_u(T_\tau = -m) = \frac{r^m}{1 + r^m} \leq r^m.$$

For completeness, absorption has finite mean: in each block of $2m$ steps, a run of $2m$ positive increments has probability $u^{2m} > 0$ and forces absorption from every interior state. Thus the unabsorbed survival time is bounded by a geometric number of such blocks. Stopping first at $\tau \wedge j$, then using bounded convergence on the stopped exponential martingale, justifies the displayed identity. The estimate $\log((1 + 2\theta)/(1 - 2\theta)) \geq 4\theta$ gives $r^m \leq e^{-4\theta m}$ when $u \geq 1/2 + \theta$.

If the walk has not been absorbed, a negative final answer requires $T_n < 0$. Hoeffding's inequality [25] for variables in $[-1, 1]$ gives

$$\mathbb{P}_u(T_n \leq 0) \leq e^{-2n(u-1/2)^2} \leq e^{-2n\theta^2}.$$

Partitioning a wrong answer according to absorption proves (13.1). Reflection proves the lower-mean case; its nonabsorbed error includes $T_n = 0$ and satisfies the same bound. At $u = 0, 1$ the assertion follows directly.

The drift martingale $T_j - xj$ stopped at $\tau \wedge j$ gives $\mathbb{E}_u T_\tau = x\mathbb{E}_u \tau$, by integrability of τ and boundedness of the stopped walk. The hitting probabilities yield $\mathbb{E}_u T_\tau = m(1 - r^m)/(1 + r^m)$. Since $r = \exp(-2 \operatorname{arctanh} x)$, this is (13.2) when $x \neq 0$. For $x = 0$, stop $T_j^2 - j$ to obtain $\mathbb{E} \tau = m^2$. The endpoint formula is deterministic. Finally $\tanh(my) \leq m \tanh y$ for integer $m \geq 1$ and $y \geq 0$: induction follows from the addition formula for $\tanh$. This proves $\mathbb{E} \tau \leq m^2$; the bound $m/|x|$ follows from $|T_\tau| \leq m$. Truncation can only decrease the mean. $\square$

Theorem 13.2 (A preparation-width family of realizations). *Let $B \subset \Delta(\Omega)$ be compact, $d = |\Omega|$, and let $h_1, \dots, h_K \in [0, 1]^\Omega$, $K \geq 2$, have envelope $v = \min_{b \in B} \max_i (Q - b)h_i$. Round their coordinates to dyadic precision r , obtaining $h_i^{(r)}$, and choose known dyadic numbers $q_i \in [0, 1]$ with $|q_i - Qh_i^{(r)}| \leq \rho$. Suppose q_i also have denominator 2^r ; enlarging r accommodates any specified dyadic calibration. For $m, n \geq 1$ and $0 < \theta < 1/2$, an explicitly clocked fair-bit audit satisfies*

$$N = n(K - 1), \quad W \leq 3dK(2m + 1), \quad (13.3)$$

$$\inf_{b \in B} \mathbb{E}_b D \geq v - 2^{1-r} - 2\rho - 4(K - 1)\theta - 2(K - 1)e_{m,n}(\theta). \quad (13.4)$$

At completed training-trial cuts its width is at most $K(2m + 1)$; at the decision cut it is at most K . Every random-bit cut, the selected event, and both validations are included in W . In particular one can replace $e_{m,n}(\theta)$ by δ by taking

$$m = \left\lceil \frac{\log(2/\delta)}{4\theta} \right\rceil, \quad n = \left\lceil \frac{\log(2/\delta)}{2\theta^2} \right\rceil. \quad (13.5)$$

Proof. For a fixed unknown b , set $g_i(b) = q_i - bh_i^{(r)}$. Compare the current incumbent i with the next index j on fresh observations. Conditional on the observation w , generate a coin with probability

$$\frac{1}{2} + \frac{q_j - q_i - h_j^{(r)}(w) + h_i^{(r)}(w)}{4}. \quad (13.6)$$

This is in $[0, 1]$, is dyadic with denominator 2^{r+2} , and has mean $1/2 + (g_j(b) - g_i(b))/4$. Apply Lemma 13.1, selecting j for a positive answer and i otherwise. Conditional on all earlier stages, the observations and fair bits for this comparison are fresh. The probability of selecting the worse index when $|g_j - g_i| \geq 4\theta$ is at most $e_{m,n}(\theta)$.

Outside these exceptional comparisons each tournament stage loses at most 4θ relative to the better of its two contestants. Induction shows a total loss at most $4(K - 1)\theta$ relative to $\max_i g_i$.

The union probability of exceptional comparisons is at most $(K-1)e_{m,n}(\theta)$ even though the incumbent is random. All g_i lie in $[-1, 1]$, so their range is at most two. Consequently

$$\mathbb{E}_b g_i(b) \geq \max_i g_i(b) - 4(K-1)\theta - 2(K-1)e_{m,n}(\theta).$$

The calibration changes selection and evaluation by at most 2ρ . For each coordinate rounding, $\|h_i - h_i^{(r)}\|_\infty \leq 2^{-r}$, so $|(Q-b)(h_i - h_i^{(r)})| \leq 2^{1-r}$. A nested random event with mean $h_i^{(r)}$ has exactly that rounded payoff; hence (13.4).

At a completed trial retain the incumbent and the absorbing walk state. The stage number j is supplied by the predetermined clock, not by a sample-dependent register. During coin generation additionally retain w and at most three sampler labels, by Proposition 10.2. This gives $3dK(2m+1)$ labels. The terminal event sampler, held event, and validation storage use at most $K(2d+1)$, $K(d+1)$, and $2K(d+1)$ labels, respectively, as in Theorem 12.2; all are bounded by the stated W . The stage boundary discards the walk and initializes the next one. After absorption, remaining observations and bits in that stage are ignored, but are still charged to the fixed schedule. Thus the count N includes padding. Each active coin uses $r+2$ fair bits. A coarse bound for the full number of epochs is $T \leq N(r+3) + r + 4$.

The finite transition description consists of the dyadic coin table, the boundary and horizon integers, and the nested-event thresholds. It has $O(K^2d(r+3) + \log m + \log n + \log K)$ bits under binary encoding; the constants depend on a fixed encoding convention. No unknown b occurs in that description. The autonomous conversion stores the clock and has width at most $(T+1)W$, rather than W . Substitution of (13.5) proves the last assertion. $\square$

Corollary 13.3 (Two general preparation regimes). *For a strict K -response cover and a prescribed positive score margin, there are whole-schedule rational realizations of that margin with width at most $12dK$ and exponential block preparation count, and with width at most $3dK(2m+1)$ and preparation count $n(K-1)$ as in (13.5). Both constrain every training cut. Their resource pairs are upper bounds for $\mathbb{V}_{N,W}$, not a claim of an optimal frontier for every compact image.*

Proof. Choose rounding, calibration, band and error budgets whose sum is smaller than the strict margin. Apply Theorem 12.2 or Theorem 13.2. A one-response cover needs no training and is implemented directly by the rational event sampler. $\square$

14. AN EXACT STATISTICAL FRONTIER AT ALL CUTS

The preceding realizations are uniform upper bounds. A family with an exact answer illustrates what an all-algorithm statistical lower bound requires. Unlike exact computation of a prescribed word function, the objective here is the best decision risk among all rules, including stochastic rules, under a fixed preparation schedule and full profile.

Let $\Theta = \{1, \dots, M\}$ and fix a known prior $\pi_1 \geq \dots \geq \pi_M \geq 0$, $\sum_j \pi_j = 1$. At time t , an independent indicator E_t has known mean $q_t \in [0, 1]$. The report is $Y_t = \vartheta$ if $E_t = 1$, and $Y_t = *$ otherwise. All indicators are independent of the parameter and the machine's private randomization. The unknown parameter ϑ is fixed during the n observations. The machine starts in a parameter-independent state, has at most K_t labels after report t , and finally guesses ϑ . Its initial and final randomization is part of the same finite causal machine. The objective is Bayesian probability of a correct answer under π . Parameter-dependent loss is specified here and is not identified with the one-row physical signed-score problem.

Write $P_k = \sum_{j=1}^k \pi_j$ and put

$$k_t = \min\{M, K_t, K_{t+1}, \dots, K_n\}, \quad a = \prod_{s=1}^n (1 - q_s), \quad w_t = q_t \prod_{s=t+1}^n (1 - q_s). \quad (14.1)$$

Thus $a + \sum_t w_t = 1$ and k_t is nondecreasing in t .

Theorem 14.1 (Exact all-cut revelation frontier). *Over all stochastic causal machines satisfying the specified profile, the optimal Bayesian success probability is*

$$S_{\mathbf{q}, \mathbf{K}}(\pi) = a\pi_1 + \sum_{t=1}^n w_t P_{k_t}. \quad (14.2)$$

A single deterministic machine attains this value simultaneously for all $\mathbf{q} \in [0, 1]^n$. The upper bound therefore applies to all approximate statistical decision rules, not just to machines implementing a chosen selector exactly.

Proof. Let L be the last time with $E_L = 1$, with $L = 0$ when no revelation occurs. Then $\mathbb{P}(L = 0) = a$ and $\mathbb{P}(L = t) = w_t$. These events, and every particular pattern of indicators, are independent of ϑ .

First consider any stochastic machine. Realize all of its transition and output kernels by independent random variables indexed by time, state and input, and also realize its initial randomization. This is possible on a product space independent of ϑ and the indicators. Conditional on the entire random table the machine is deterministic. This conditioning is an analysis device, not a free persistent random seed in an implementation.

Fix that table and an indicator pattern with last revelation $t > 0$. The map from ϑ to the terminal guess factors through the state at every time $s \geq t$: all reports after t are the same symbol $*$, so the remaining transitions and output are fixed maps. The set of possible terminal guesses therefore has cardinality at most $\min_{s \geq t} K_s$, and at most M guesses can be correct. Its prior success probability is bounded by the mass of the k_t largest prior atoms, namely P_{k_t} . Earlier revelations and any parameter information encoded before t do not evade this future bottleneck. On the no-revelation pattern, the output is independent of ϑ and its success probability is at most π_1 . Integrate these bounds over the random table and indicator patterns to obtain the upper bound in (14.2). In particular no extremality claim about a nonconvex minimax set has been used.

For attainment, use the labels $\{1, \dots, k_t\}$ at time t , start with label one, and use the following deterministic updates. On $*$, hold the label. On a revelation $j \leq k_t$, replace it by j ; on a revelation $j > k_t$, replace it by one. The holding update is legal because $k_{t-1} \leq k_t$. At termination output the label. Conditional on $L = t > 0$, exactly the parameters $1, \dots, k_t$ are correct; on $L = 0$, exactly parameter one is correct. This proves equality. The updates use only the prescribed profile and prior ordering, not the probabilities q_t , establishing simultaneous attainment. $\square$

Corollary 14.2 (Sharp preparation–peak tradeoff). *For the uniform prior, a constant reveal probability $0 < q < 1$, and constant width $1 \leq K \leq M$, write $a_n = (1 - q)^n$. Then*

$$S_{n, K} = \frac{a_n + K(1 - a_n)}{M}. \quad (14.3)$$

For $n \geq 1$ and target success $s \in [1/M, 1]$, the least feasible width is

$$K_{n, s} = \max \left\{ 1, \left\lceil \frac{Ms - a_n}{1 - a_n} \right\rceil \right\}, \quad (14.4)$$

provided this integer is at most M ; otherwise the target is infeasible at that preparation count. If $s > 1/M$ and $K > Ms$, the least preparation count for width K is

$$n_{K, s} = \left\lceil \frac{\log((K - 1)/(K - Ms))}{-\log(1 - q)} \right\rceil. \quad (14.5)$$

If $K \leq Ms$ and $s > 1/M$, no finite n attains the target, except that equality can be attained when revelations are certain, a case excluded by $q < 1$. Width one gives success exactly $1/M$ for all n .

Proof. Here $P_k = k/M$ and $k_t = K$. Since $\sum_t w_t = 1 - a_n$, Theorem 14.1 gives (14.3). Solving $a_n + K(1 - a_n) \geq Ms$ first for K and then for n proves the formulas. For $K > 1$, the limiting value K/M is approached strictly from below when $0 < q < 1$, which proves the boundary assertion. $\square$

Corollary 14.3 (Exact cost of a changed checkpoint). *Replacing a profile $\mathbf{K}$ by a tighter profile $\widetilde{\mathbf{K}}$ loses exactly*

$$\sum_{t=1}^n w_t (P_{k_t} - P_{\widetilde{k}_t}).$$

Inserting a charged holding cut of width H immediately after time s replaces k_t by $\min(k_t, H)$ for $t \leq s$, and leaves the later terms unchanged. In particular a single narrow early cut does not constrain information acquired after that cut.

Proof. Subtract (14.2) for the two profiles. A holding cut is a further bottleneck exactly for revelations that precede it. The same deterministic construction realizes the refined schedule. $\square$

For example, with $M = 4$, $q_1 = q_2 = q_3 = 1/2$, and $(K_1, K_2, K_3) = (4, 1, 4)$, the effective profile is $(1, 1, 4)$ and the optimal uniform-prior success is $5/8$. Four labels at the terminal cut alone would instead give $29/32$. These are risks of complete experiments, not counts of residual classes of one word function. For a nonuniform prior the same theorem weights every surviving label by its actual mass. The nested optimal label sets are what makes all cut bounds jointly attainable. For a general loss or observation channel such nesting need not hold; compatible continuation updates, not independently optimized dictionaries, remain necessary.

The exact formula is Bayesian. Its upper bound also obstructs any uniform worst-case guarantee exceeding that Bayesian value, but the attaining prior-ordered machine is not asserted to be minimax. In particular an uncharged persistent random permutation cannot be used to turn its Bayesian success into a minimax statement. The prior, observation channel and all register cuts have been declared explicitly.

15. TRANSPORT PRICED BY ACTIVE UPDATES

A changed training source need not affect a latched computation at every subsequent scheduled preparation. The relevant quantity is the defect of its induced retained-state transition, not the number of raw reports that happen to be present. This distinction gives a sharper transfer bound without assuming equality of the training laws.

Theorem 15.1 (Occupation-weighted retained-law transfer). *Consider a fixed finite execution with common initial state law μ and retained-state transition kernels P_t, Q_t on the same finite state spaces, $1 \leq t \leq T$. The kernels may depend on a fixed unknown parameter; all inequalities below hold separately for each parameter. Let*

$$d_t(x) = \|P_t(x, \cdot) - Q_t(x, \cdot)\|_{\text{TV}}, \quad \mu_{t-1} = \mu P_1 \cdots P_{t-1}.$$

For the same terminal readout kernel F ,

$$\|\mu P_1 \cdots P_T F - \mu Q_1 \cdots Q_T F\|_{\text{TV}} \leq \min \left\{ 1, \sum_{t=1}^T \int d_t d\mu_{t-1} \right\}. \quad (15.1)$$

If $d_t(x) \leq \alpha_t \mathbf{1}_{A_t}(x) + \zeta_t(x)$, the right side can be replaced by the truncated sum $\sum_t \alpha_t \mathbb{P}_P(X_{t-1} \in A_t) + \sum_t \mathbb{E}_P \zeta_t(X_{t-1})$. For a readout in $[-1, 1]$ multiply the bound by two; for an indicator no factor two is needed. The reverse bound with P, Q exchanged is also valid.

Proof. For signed measures, expand the product difference in the order

$$\mu P_1 \cdots P_T - \mu Q_1 \cdots Q_T = \sum_{t=1}^T \mu P_1 \cdots P_{t-1} (P_t - Q_t) Q_{t+1} \cdots Q_T.$$

Integration of a row difference against μ_{t-1} has total variation at most $\int d_t d\mu_{t-1}$. All suffix kernels and F contract total variation. The triangle inequality and the bound one for probability-law

distance prove (15.1). Substitute the rowwise estimate for its occupation version. The expectation bounds follow from the oscillation of the readout. $\square$

Here the retained state must support the displayed transition kernels. For independent reset observations and a time-inhomogeneous finite machine this is automatic, even when the selected row depends on the state. For a non-reset physical process one must include the necessary analysis state or use its history kernels; ignoring a report need not remove its influence on subsequent physical dynamics. No Markov closure of an arbitrary compressed physical state is assumed.

Corollary 15.2 (Calibrated training with absorption). *Run the audit of Theorem 13.2 under an ideal reset source and a perturbed reset source whose row laws differ by at most α per active call. Use the same implemented coin and event randomizers. If A is the number of active comparison updates under the ideal execution, the change in its selected-response score is at most*

$$2 \min(1, \alpha \mathbb{E}A) + \beta_c + \beta_t, \quad \mathbb{E}A \leq (K - 1) \min(n, m^2), \quad (15.2)$$

where β_c, β_t bound candidate and target validation-law changes. A comparison with conditional coin mean u has the sharper active expectation $\mathbb{E}_u(\tau \wedge n)$ in (13.2). Uniform gaps $|u - 1/2| \geq \eta$ give the bound $\min(n, m^2, m/(2\eta))$ for that comparison.

Proof. At a completed comparison trial, marginalize the fresh coin bits into its transition kernel. Before absorption, data processing bounds its row defect by the source defect α . After absorption its retained transition is the same identity under both sources; its defect is zero. At predetermined stage boundaries the common reset of the walk is also defect-free. Theorem 15.1 therefore bounds the terminal incumbent-law distance by $\alpha \mathbb{E}A$. The score associated with an incumbent lies in $[-1, 1]$. Conditional on a common incumbent and common nested event, the two validation expectations change by at most $\beta_c + \beta_t$. This proves the score estimate. Condition on the incumbent at each stage and apply Lemma 13.1; summing proves the active-count bound. Marginalizing the coin here is a proof of a law estimate, not a waiver of its implementation states, which remain charged in (13.3). $\square$

The estimate is for the output or retained computation, not for the complete raw-observation transcript. A padded sequence of N changed reports can still have total variation of order $N\alpha$ even after the machine has stopped using them. Conversely, all N preparations remain in the resource ledger even when only A updates influence the answer. Finite row-rounding defects can be inserted as the terms ζ_t in Theorem 15.1; a changed initial law adds its total variation separately. Thus source-preserving transport is the case $\alpha = 0$, while the active bound also has content for $\alpha > 0$ and does not assert generic sample-count-free stability.

16. WEIGHTED CONTINUATION OBSTRUCTIONS

We next give a distribution-sensitive version of the wordwise obstruction. It applies to finite-valued terminal decisions and is useful even when a word of small probability should not be assigned the same weight as a likely one.

Let $f : \Sigma_1 \times \cdots \times \Sigma_L \rightarrow \mathcal{A}$ have finite range. The residual of a prefix x at cut t is the function $v \mapsto f(xv)$ on the suffix alphabet. Write $r_t(f)$ for the number of distinct residuals. A stochastic machine has a charged register at every cut; all persistent randomness is included in this register. Its wordwise error is $e(w) = \Pr(\hat{f}(w) \neq f(w))$.

Theorem 16.1 (Finite-valued realization and weighted cut bound). *Exact stochastic computation of f requires $K_t \geq r_t(f)$ at every cut. The whole residual profile is simultaneously attained by a deterministic machine.*

Fix a cut, prefixes $x_1, \dots, x_r$, and a graph on $[r]$. For each edge ij choose a suffix v_{ij} such that $f(x_i v_{ij}) \neq f(x_j v_{ij})$. Let $\mu(w) \geq 0$ be arbitrary word weights. Choose edge weights $\lambda_{ij} \geq 0$ satisfying the word capacity constraints

$$\sum_{ij: w=x_i v_{ij} \text{ or } w=x_j v_{ij}} \lambda_{ij} \leq \mu(w) \quad \text{for every word } w. \quad (16.1)$$

Every machine with K states at this cut satisfies

$$\sum_w \mu(w) e(w) \geq \min_{\phi: [r] \rightarrow [K]} \sum_{ij: \phi(i)=\phi(j)} \lambda_{ij}. \quad (16.2)$$

The right side is a weighted monochromatic-edge obstruction. Maximizing it over admissible witnesses and weights gives a finite, distribution-sensitive lower bound. The bound can be an equality with positive error.

Proof. Let $a_i(m)$ be the state distribution after prefix x_i . For a fixed edge and a common state m , the future output distribution on suffix v_{ij} is the same for both prefixes. Since their correct outputs are different, the sum of their two conditional error probabilities is at least one. Multiplying by the smaller state mass and summing gives

$$e(x_i v_{ij}) + e(x_j v_{ij}) \geq \sum_{m=1}^K \min\{a_i(m), a_j(m)\}. \quad (16.3)$$

Weight the inequalities by λ_{ij} and use (16.1). The resulting lower bound is

$$F(a_1, \dots, a_r) = \sum_{ij} \lambda_{ij} \sum_m \min\{a_i(m), a_j(m)\}.$$

This is a concave continuous function on $(\Delta_K)^r$. Every point of that product is a convex combination of its deterministic vertices; concavity implies that its value is at least the minimum over those vertices. A vertex is a labeling $\phi: [r] \rightarrow [K]$, and its value is exactly the right side of (16.2). This proves the bound without treating stochastic encodings as deterministic encodings.

For zero error, choose one distinguishing suffix for every two different residuals. Equation (16.3) forces the corresponding state supports to be disjoint. Hence at least $r_t(f)$ states are needed. For the converse, keep the residual class of the observed prefix. Equal residuals have equal shifts after every next input, so the next class is well defined. At the last epoch the residual is the output label itself. These quotient transitions realize all widths simultaneously.

For a positive-error equality, let the only relevant words be two one-letter prefixes with different required labels, weighted p and $1 - p$, where $0 < p < 1$. Put $K = 1$ and $\lambda_{12} = \min(p, 1 - p)$. The cut obstruction is $\min(p, 1 - p)$, attained by always emitting the more likely label. Thus the weighted constant is sometimes exact; universal sharpness for all graphs or input distributions is not asserted. With unit weights on the same two words the total-error lower bound is one, also attained by a constant output; this is equality in the earlier $(r - K)/(r - 1)$ bound at $r = 2$, $K = 1$. $\square$

Corollary 16.2 (A Bayes-regret obstruction for constrained audits). *Fix a finite action set $\mathcal{A}$, candidate class B , and a probability measure ν on B . For a training word w , let $p_w(b)$ be its likelihood and let $f_a(b)$ be the expected validation payoff. Put*

$$R_w(a) = \int_B p_w(b) f_a(b) d\nu(b), \quad S_\nu = \sum_w \max_a R_w(a).$$

Choose a maximizing label $f(w)$. Set $\mu(w) = R_w(f(w)) - \max_{a \neq f(w)} R_w(a)$, taking zero when a maximum is tied. Let $\Gamma_{t,K}(f, \mu)$ be any lower bound supplied by (16.2). Then every width- K audit

at cut t has worst-case expected payoff at most

$$S_\nu - \Gamma_{t,K}(f, \mu). \quad (16.4)$$

Proof. The conditional distribution of the auditor's output for a fixed word is independent of the unknown candidate. Its Bayes payoff is therefore $\sum_{w,a} R_w(a) \Pr(\hat{f}(w) = a)$. Subtracting this from S_ν gives at least $\sum_w \mu(w)e(w)$. Apply the theorem, and use that a minimum over candidates is at most the average under ν . $\square$

The corollary does not exchange a maximum and minimum over a nonconvex fixed-width machine class. The prior is a lower-bound device, not a randomization resource supplied to the candidate. With a unique maximizing label and strictly positive margins at every word, equality with S_ν forces exact wordwise computation. This makes explicit the uniqueness step used in the odd-sample binary theorem below. For even sample size, the balanced-count words have zero likelihood gap. A rule can differ on those ties without losing value; equality of values then does not force one prescribed total Boolean rule, and the odd-sample width profile cannot simply be asserted for that case.

17. JOINT RESIDUAL BUDGETS AND GROWING FAMILIES

We now put lower bounds on several cuts into one statistical budget. Summing separate cut bounds without controlling their common words would count the same regret more than once.

Theorem 17.1 (A shared-capacity multicut obstruction). *Let $f : \prod_{i=1}^L \Sigma_i \rightarrow \mathcal{A}$ and $\mu(w) \geq 0$ be as in Theorem 16.1. At each chosen cut t , choose a finite prefix graph with an edge $e = (u, v)$ carrying a distinguishing suffix s_e , and a weight $\lambda_{t,e} \geq 0$. Impose the single family of constraints*

$$\sum_{t,e: w=us_e \text{ or } w=vs_e} \lambda_{t,e} \leq \mu(w) \quad \text{for every complete word } w. \quad (17.1)$$

For a graph at cut t , write

$$\Gamma_{t,K_t}(\lambda) = \min_{\phi: \text{prefixes} \rightarrow [K_t]} \sum_{e=(u,v): \phi(u)=\phi(v)} \lambda_{t,e}.$$

Every stochastic machine of profile $\mathbf{K}$ satisfies

$$\sum_w \mu(w) \Pr(\hat{f}(w) \neq f(w)) \geq \sum_t \Gamma_{t,K_t}(\lambda). \quad (17.2)$$

Consequently, with R_w, S_ν, μ defined in Corollary 16.2, every audit of this profile and the fixed training schedule has worst-case payoff at most

$$S_\nu - \sup_{\lambda \text{ satisfying (17.1)}} \sum_t \Gamma_{t,K_t}(\lambda). \quad (17.3)$$

For fixed graphs the supremum is the value of a finite linear program.

Proof. For each edge use the overlap inequality (16.3). Multiply it by $\lambda_{t,e}$ and sum over all cuts and edges. The left side is bounded above by the left side of (17.2), by the shared capacity constraint. At each cut the resulting overlap expression is concave in its prefix state distributions, hence at least its minimum over deterministic labelings, as proved in Theorem 16.1. Adding these bounds proves (17.2). Although prefix distributions at different cuts are compatible, minimizing them separately only weakens the bound; it does not assume independent realizability. The Bayes-regret argument of Corollary 16.2 proves (17.3).

For the last assertion introduce variables z_t and impose, for each labeling ϕ of each finite graph,

$$z_t \leq \sum_{e=(u,v): \phi(u)=\phi(v)} \lambda_{t,e}.$$

Maximize $\sum_t z_t$ subject to these inequalities, nonnegative edge weights, and (17.1). There are finitely many constraints. This is a computable obstruction, not a claim that every multicut statistical optimum equals this relaxation. $\square$

17.1. Actual physical word weights. For n all-on trials of the private marked collision subfamily, a complete training word w specifies the two report bits and the mark of each trial. With N_{ab} its report-pair counts,

$$p_w(p, q) = 2^{-n} p^{N_{10}+N_{11}} (1-p)^{N_{00}+N_{01}} q^{N_{01}+N_{11}} (1-q)^{N_{00}+N_{10}}. \quad (17.4)$$

The formula has the continuous endpoint interpretation $0^0 = 1$. For any finitely supported prior on actual pairs $(p, q) \in [0, 1]^2$, use this likelihood in $R_w(A) = \int p_w(p, q)(Q - b(p, q))\mathbf{1}_A d\nu$, with all marked events $A \subseteq \Omega$ as actions. Randomized events are included by linearity. Equations (17.1) and (17.3) then give a finite all-algorithm obstruction for that full training profile, with the true physical word probabilities. Changing only Q by total variation β changes every audit's payoff by at most β .

This specialization bounds all stochastic selectors on the fixed all-on training schedule, not merely exact implementations of a majority rule. It does not assert a new numerical lower bound above three for every adaptive physical audit. Adaptive training has controller-dependent word likelihoods and must instead be included in the full experiment of (11.3); using (17.4) unchanged for such a likelihood would be incorrect. The feedback-uniform lower bound used below remains Lemma 29.2.

17.2. A family with matching end-to-end growth.

Theorem 17.2 (Point-source decision spectra). *Let $d \geq 2$, $B_d = \{\delta_1, \dots, \delta_d\}$ on $[d]$, and let Q_d be uniform. For every $K \geq 1$,*

$$\mathfrak{V}_K(B_d, Q_d) = 1 - \frac{\lceil d/K \rceil}{d}. \quad (17.5)$$

For $1 - 2/d < c < 1 - 1/d$, the decision complexity is exactly d , and the whole-schedule peak satisfies

$$d \leq W_c(B_d, Q_d) \leq W_c^{\text{rat}}(B_d, Q_d) \leq 2d.$$

The upper bound uses one training observation and two validations, with no randomization. Thus the end-to-end width grows linearly in this nontrivial sequence of compact behavior images.

Proof. Assign each point source to a response that maximizes its score. Some response is assigned a set S of at least $\lceil d/K \rceil$ indices. For $s = |S| < d$, the mean of its scores on S equals

$$\frac{d-s}{d} \left(\frac{1}{d-s} \sum_{i \notin S} h_i - \frac{1}{s} \sum_{i \in S} h_i \right) \leq 1 - \frac{s}{d}.$$

For $s = d$ the mean is zero. At least one assigned source has score no larger than this mean, proving the upper bound. Partition $[d]$ into at most K groups of size at most $\lceil d/K \rceil$ and use their complements as events. This attains the lower bound.

In the stated interval the formula forces $K \geq d$. A single training observation identifies the point source i exactly. Keep i and choose the complement of $\{i\}$. During validation retain i and the candidate indicator, at most $2d$ states, then emit the target indicator minus it. This computes the prescribed signed score on every input word, including words that have zero probability under the point-source family. Its expected score is $1 - 1/d$. The lower bound follows from the necessary decision-cut width. $\square$

17.3. A sharp simultaneous storage profile. A second class separates overlapping retention obligations. Independent uniform labels $X_j \in [q_j]$ are revealed at scheduled epochs a_j and must be reported at epochs $b_j > a_j$. No later input reveals an active label, and previous outputs cannot be read as external memory. Let $A_t = \{j : a_j \leq t < b_j\}$ and $D_t = \prod_{j \in A_t} q_j$.

Theorem 17.3 (Overlapping retention intervals). *Every stochastic machine whose probability of correctly reporting all labels is at least $1 - \varepsilon$ has*

$$K_t \geq (1 - \varepsilon)D_t \quad \text{at every cut } t. \quad (17.6)$$

The exact profile $K_t = D_t$ is attained simultaneously by a deterministic machine. Thus the minimum peak number of bits lies between $\log_2 \max_t D_t + \log_2(1 - \varepsilon)$ and $\lceil \log_2 \max_t D_t \rceil$ for $\varepsilon < 1$.

Proof. Fix a cut and give the decoder, as additional information, all labels outside A_t , whether past or future. They are independent of the uniform active vector. Conditional on them, every future prediction of that vector factors through the K_t -valued retained state and fresh randomness. If $a(m | x)$ is its conditional encoder and $d(x | m)$ the induced joint prediction law, its success probability is at most

$$\frac{1}{D_t} \sum_m \sum_x a(m | x) d(x | m) \leq \frac{1}{D_t} \sum_m \sum_x d(x | m) = \frac{K_t}{D_t}.$$

Correctness of all reports implies correctness of the active reports, so averaging over the independent additional information proves (17.6). For the upper bound, keep precisely the vector of active labels, append a label when revealed, and delete it when reported. Every update is deterministic and respects the schedule. The bit bounds follow by taking logarithms and rounding for an actual binary register. The lower bound is a counting argument, not an independence assumption about labels after conditioning on a compressed state. $\square$

18. FINITE-DATA TESTING ON THE ORIGINAL BEHAVIOR IMAGE

18.1. The reset interface. Let $B \subset \prod_{j=1}^J \Delta(\Omega_j)$ be compact, and let $Q = (Q_j)$ be fixed. Set $d_j = |\Omega_j|$, $D = \sum_j (d_j - 1)$, $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$, and $f_{j,A}(b) = Q_j(A) - b_j(A)$. Include the empty event. No convexity of B is assumed.

A candidate b is fixed for the whole audit. The auditor obtains n independent samples of each b_j , with independence between rows. In a causal experiment, row j means a specified parameter and deterministic feedback tree; the reset interface supplies that row. It is an additional experimental interface, not information available to the online simulator. A complete sample contains the terminal mark. Private randomness and any per-run hidden selector are reset between runs. The design of the candidate is not resampled.

Write $C_{j,w}$ for the counts, and $\mathcal{N}_n$ for all count arrays with $\sum_w C_{j,w} = n$. Their probability is the polynomial

$$p_C(b) = \prod_j \frac{n!}{\prod_w C_{j,w}!} \prod_w b_j(w)^{C_{j,w}}.$$

A randomized audit is an array $c = (c_{C,i})$ with $c_C \in \Delta(\mathcal{I})$. After training it chooses $i = (j, A)$ according to c_C . Independent candidate and target validation runs give the signed score $\mathbf{1}_A(\text{target}) - \mathbf{1}_A(\text{candidate})$. Its expected value is

$$g_c(b) = \sum_{C,i} p_C(b) c_{C,i} f_i(b), \quad v_n^{\text{aud}}(B, Q) = \max_c \min_{b \in B} g_c(b). \quad (18.1)$$

Thus every audit payoff is polynomial of total degree at most $Jn + 1$ in the behavior coordinates. The probability of a large observed score in a single audit is not asserted to equal its expectation; additional independent validation would be required for a specified confidence level.

Theorem 18.1 (Reset-audit duality and finite-sample approximation). *Let $\delta = \min_{b \in B} \max_i f_i(b)$. Then all the extrema below are attained, and*

$$v_n^{\text{aud}}(B, Q) = \min_{\mu \in \mathcal{P}(B)} \sum_{C \in \mathcal{N}_n} \max_{i \in \mathcal{I}} \int_B p_C(b) f_i(b) d\mu(b), \quad (18.2)$$

$$0 \leq \delta - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}. \quad (18.3)$$

The values are nondecreasing in n . If $a = \dim \text{aff } B$, a minimizing measure in (18.2) can be supported on at most $\binom{a+Jn+1}{a}$ points.

An explicit feasible audit attaining the lower bound $\delta - \sqrt{D/n}$ chooses an event maximizing $Q_j(A) - C_j(A)/n$, with a fixed deterministic tie rule. If the rational target probabilities are given, this is a rationally specified finite rule. With a free schedule clock, a conservative retained-state bound is

$$H(|\mathcal{I}| + 1)(n + 1) \sum_j d_j,$$

where H is the number of states needed to execute and record one causal transcript. This budget is charged to the auditor, not to the simulator.

Proof. The coefficient arrays form a compact convex product of finite simplexes. The integrand is continuous in b and affine in c . The probability measures on compact B form a compact convex set for weak convergence. The minimum over b equals the minimum over such measures for fixed c . Finite-dimensional minimax (or its compact measure version) therefore exchanges the two extrema. Maximization over each independent coefficient simplex selects its largest coordinate integral, giving (18.2). Continuity and compactness give attainment. This step uses measures on the original B , not feasible private realizations of their barycenters.

For any audit, $g_c(b) \leq F(b) = \max_i f_i(b)$. At a minimizer of F this gives $v_n^{\text{aud}} \leq \delta$. The zero event gives $v_n^{\text{aud}} \geq 0$. For the stated explicit rule put $\hat{b}_j = C_j/n$ and $\rho = \max_j \|\hat{b}_j - b_j\|_{\text{TV}}$. Comparing the empirical maximizer with a true maximizing event gives the pointwise regret bound

$$F(b) - f_{i(C)}(b) \leq 2\rho.$$

For a multinomial empirical measure,

$$\mathbb{E} \|\hat{b}_j - b_j\|_{\text{TV}}^2 \leq \frac{d_j}{4} \sum_w \mathbb{E} (\hat{b}_j(w) - b_j(w))^2 = \frac{d_j(1 - \sum_w b_j(w)^2)}{4n} \leq \frac{d_j - 1}{4n}.$$

Consequently $\mathbb{E} \rho \leq (\sum_j \mathbb{E} \|\hat{b}_j - b_j\|_{\text{TV}}^2)^{1/2} \leq \sqrt{D/n}/2$. Averaging the regret uniformly over B proves (18.3).

From $n + 1$ unordered samples in each row, discard one uniformly at random. The remaining counts have exactly the n -sample law, independently across rows. This realizes every n -sample audit as a randomized $(n + 1)$ -sample audit and proves monotonicity.

Choose affine coordinates on $\text{aff } B$. All the functions $p_C f_i$ lie in the span of monomials of total degree at most $Jn + 1$, a space of dimension at most $s = \binom{a+Jn+1}{a}$ including the constant. The moment vector of a probability measure is in the convex hull of the evaluation vectors; their constant coordinate is one, so their affine dimension is at most $s - 1$. Eliminating affine dependencies gives a representation by at most s points. It preserves every integral in (18.2), and hence preserves the objective.

Finally, each count is in $\{0, \dots, n\}$. The indicated product register stores all counts, the within-run state, and the selected event for validation. Only the latter event and the within-run buffer need survive once training ends, so the product bound is conservative. Randomizing an event uses fresh randomness. The schedule, reset access and terminal mark readout were declared before the theorem. No private simulator state is read by this implementation. $\square$

The dual measure chooses a candidate once, after which its samples are independent conditional on that candidate. Replacing it by $\bar{b} = \int b d\mu$ would replace $\int p_C(b) f_i(b) d\mu$ by $p_C(\bar{b}) f_i(\bar{b})$, generally a different quantity. It would also replace the joint replicated law by a product of its marginals. Neither operation is part of the theorem. This is the operational meaning of the moment side, rather than a same-budget convexification theorem.

Proposition 18.2 (Both sides sampled). *Suppose the target reference Q^b varies with the candidate, as it does when visible-selector weights are design variables. Supply independent training samples from both b and Q^b , and use their empirical difference. If their row dimensions are D_P, D_Q , the same construction has regret at most $\sqrt{D_P/n} + \sqrt{D_Q/n}$. Minimax and attainment hold on the compact image of pairs (b, Q^b) . In particular a variable reference law must not be treated as known fixed input to Theorem 18.1.*

Proof. The error in any event discrepancy is at most the sum of the maximal row variations of the two empirical tables. Apply the preceding second moment bound separately and then the empirical-maximizer regret inequality. Count probabilities are products for the two tables. They are continuous polynomials on the compact image of pairs, so the same minimax argument applies without changing the simulator signature. $\square$

18.2. A matching uniform sampling obstruction.

Theorem 18.3 (Binary reset lower bound). *Take one binary row with target $\text{Ber}(1/2)$ and candidate class $B_h = \{\text{Ber}(1/2 - h), \text{Ber}(1/2 + h)\}$, where $0 < h < 1/2$. Then*

$$\delta = h, \quad v_n^{\text{aud}} = h \|\text{Ber}(1/2 + h)^{\otimes n} - \text{Ber}(1/2 - h)^{\otimes n}\|_{\text{TV}}. \quad (18.4)$$

For every $n \geq 1$, choosing $h = 1/(8\sqrt{n})$ gives

$$\delta - v_n^{\text{aud}} \geq \frac{1}{16\sqrt{n}}.$$

Thus the exponent in (18.3) is sharp uniformly over compact candidate classes, already at affine dimension one. This does not assert a degree lower bound for revealed-behavior polynomials, nor that this two-point class equals an unrestricted private-row image.

Proof. Only the two complementary singleton events and the zero event matter. A randomized choice is a statistic d in $[-1, 1]$. The two expected payoffs are $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$. Their minimum is at most half their sum, which is at most $h\|P_+^n - P_-^n\|_{\text{TV}}$. Conversely take the sign of the likelihood difference, assigning zero at a tie. Complementing every sample interchanges P_+^n and P_-^n and changes that sign. The two payoffs are equal, and their average attains the variation bound. The statistic depends only on the count, so it is an admissible audit. The symmetric measure on the two candidates attains the dual value. Every Dirac measure has dual objective h , whereas for finite n the two product laws are equivalent and their variation is strictly less than one. Thus no dual minimizer is a Dirac measure. Their barycenter is the target law, but replacing the dual by that barycenter would incorrectly give value zero instead of the strictly positive value in (18.4).

For $h \leq 1/8$, elementary integration of $(1 - u^2)^{-1}$ gives

$$\text{KL}(\text{Ber}(1/2 + h) \parallel \text{Ber}(1/2 - h)) = 2h \log \frac{1 + 2h}{1 - 2h} \leq 16h^2.$$

The product relative entropy is at most $16nh^2$. Pinsker's inequality bounds its total variation by $\sqrt{8nh^2}$. At the specified h this is less than $1/2$. Substituting in (18.4) proves the claim. The inequality uses the classical two-point testing mechanism; the restriction being measured is the reset budget of the present game. $\square$

18.3. Transport of one fixed audit.

Proposition 18.4 (Reset cost under model perturbation). *Let q range over the same compact set of executable row designs in two models. Suppose uniformly in q, j that*

$$\|P_j^q - \tilde{P}_j^q\|_{\text{TV}} \leq \alpha, \quad \|Q_j - \tilde{Q}_j\|_{\text{TV}} \leq \beta.$$

For the same coefficient rule c , evaluated with each model's own target validation,

$$|g_c(q) - \tilde{g}_c(q)| \leq (2Jn + 1)\alpha + \beta. \quad (18.5)$$

The deficiencies themselves differ by at most $\alpha + \beta$, at the same resource budget. If the source behavior is unchanged, a fixed audit loses at most β , independently of its reset count.

Proof. A product coupling of the Jn training runs fails with probability at most $Jn\alpha$. Taking counts and choosing the event are contractions. A score with fixed target discrepancies lies in $[-1, 1]$, so the change in its expectation from this training-law perturbation is at most $2Jn\alpha$. At each event the discrepancy changes by at most $\alpha + \beta$. This proves (18.5). For deficiency, every row variation changes by at most $\alpha + \beta$; taking the maximum and then the minimum preserves that bound. No new simulator is inserted. $\square$

For a rational polynomial image, a rational rule and a rational threshold, strict positivity of $g_c(q)$ above that threshold is a first-order formula over the original row simplexes. Real quantifier elimination decides it. The same is true of an existential rule followed by this universal condition. This is an exact finite verification statement, not an efficient algorithm. The explicit empirical rule below will be verified by a uniform analytic inequality, rather than by storing its enormous complete table.

19. CONTINUATION COMPLEXITY OF A FINITE-DATA TEST

The reset budget does not by itself bound the auditor's retained memory. We now impose both constraints. At scheduled epoch t a streaming auditor reads one letter x_t from a finite alphabet X_t , updates a register, and emits no information to any auxiliary store. It makes its decision on reading the last letter. Transition rows may depend on the epoch; the clock is free. All other information crossing a cut belongs to the register. Fresh randomization is allowed. An independent persistent selector, when allowed, selects a complete auditor before any letters are read.

For a Boolean decision $\psi : X_1 \times \cdots \times X_n \rightarrow \{-1, 1\}$, let $u \sim_t v$ mean that

$$\psi(us) = \psi(vs) \quad \text{for every } s \in X_{t+1} \times \cdots \times X_n.$$

Write $r_t(\psi)$ for the number of equivalence classes of prefixes of length t , for $0 \leq t < n$. These are the deterministic continuation responses. The terminal decision is emitted on the last transition, so no terminal register is needed. In particular, $r_0 = 1$. The equivalence is a finite-horizon version of the classical residual-state construction [16]. The following quantitative statement will also measure the loss when a statistical auditor has fewer states than this construction.

Theorem 19.1 (Exact continuation width and a quantitative cut obstruction). *A stochastic streaming auditor computes ψ exactly on every word with width profile $(K_t)_{0 \leq t < n}$ if and only if $K_t \geq r_t(\psi)$ at every cut. The profiles are attained simultaneously by a deterministic auditor. Independent persistent randomization does not reduce these widths for exact computation.*

More generally, at a cut with $r = r_t(\psi) \geq 2$, every stochastic auditor of width K satisfies

$$\sum_{w \in X_1 \times \cdots \times X_n} e_\psi(w) \geq \frac{(r - K)_+}{r - 1}, \quad (19.1)$$

where $e_\psi(w)$ is its probability of an incorrect Boolean decision on input w . The inequality also holds after averaging over an independent selector whose conditional auditors each have width at most K .

Proof. From a prefix u let $e_u(m)$ be the distribution of the retained state. From state m and suffix s let $a_m(s)$ be the probability of terminal answer $+1$. Exact computation says

$$\sum_m e_u(m) a_m(s) = \frac{1 + \psi(us)}{2}.$$

The right side is zero or one. Every state of positive e_u -mass must therefore have precisely that zero-one continuation response for all s . Distinct prefix classes have disjoint positive supports. Hence $K_t \geq r_t$. Conversely, retain the prefix class. Appending the same letter to two equivalent prefixes gives equivalent new prefixes; thus every transition is well defined. At the last transition the class and the last letter determine ψ . This realizes every width r_t simultaneously. For a persistent independent selector, a zero average error on each of the finitely many words implies zero conditional error on all words outside a single null set of selector values. The preceding bound applies there.

For the quantitative claim choose representative prefixes $u_1, \dots, u_r$, one per class, and put $e_i(m) = e_{u_i}(m)$. Define

$$o_{ij} = \sum_{m=1}^K \min\{e_i(m), e_j(m)\}.$$

For nonnegative numbers $a_1, \dots, a_r$, ordering them shows that $\sum_{i < j} \min(a_i, a_j) \geq \sum_i a_i - \max_i a_i$. Apply this at each state. Since every e_i has mass one and each $\max_i e_i(m) \leq 1$, it follows that

$$\sum_{i < j} o_{ij} \geq r - K. \quad (19.2)$$

Choose a suffix s_{ij} distinguishing u_i, u_j . Conditional on a common state and this suffix, the two desired answers are opposite. Their error probabilities, weighted by $e_i(m), e_j(m)$, have sum at least $\min\{e_i(m), e_j(m)\}$. Thus

$$e_\psi(u_i s_{ij}) + e_\psi(u_j s_{ij}) \geq o_{ij}.$$

On summing, each complete word on the left occurs at most $r - 1$ times: its prefix identifies a single representative, which participates in $r - 1$ pairs. Nonnegativity of all other word errors and (19.2) prove (19.1). When $r \leq K$ the claim is just nonnegativity. Finally apply the bound conditional on a selector and integrate. Independence ensures that conditioning has not changed the input experiment. $\square$

The exact part is also the extremal zero-one case of Theorem 8.2. Its content is a simultaneous width profile, rather than separate unrelated factorizations at different cuts. The error bound does not assert that its constant is optimal for every Boolean function. Its use below is to convert a continuation obstruction into a strict loss of the attainable statistical value.

19.1. The width required to attain the binary reset value. Fix $0 < h < 1/2$ and the binary candidate class B_h of Theorem 18.3. A terminal decision $+1$ chooses the singleton event appropriate to the $+$ candidate, and -1 the complementary event. A zero-score event can equivalently be replaced by an independent fair choice of these decisions. The expected payoffs are therefore $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$, where $d \in [-1, 1]$ is the conditional mean decision and $P_\pm = \text{Ber}(1/2 \pm h)^{\otimes n}$. Let $v_{n,K}$ be the supremum of the smaller of these two payoffs over streaming auditors with the declared width profile. The final validation interface is unchanged; the widths here count training and selection only.

Theorem 19.2 (Exact memory threshold for optimal binary auditing). *Let $n = 2m + 1$ be odd. The unconstrained reset value is $v_n = h\|P_+ - P_-\|_{\text{TV}}$. It is attained at the profile*

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n, \quad (19.3)$$

and $v_{n,K} = v_n$ if and only if $K_t \geq r_t$ for all these cuts. For every cut with $r_t \geq 2$ one has the explicit deficit

$$v_n - v_{n,K} \geq 2h^2(1/4 - h^2)^m \frac{(r_t - K_t)_+}{r_t - 1}. \quad (19.4)$$

These conclusions remain valid when an independent persistent selector chooses auditors of the same conditional widths. For $n \geq 3$ the least peak training width attaining v_n is $m + 2$; for $n = 1$ it is one.

Proof. The sign of $P_+(w) - P_-(w)$ is the majority decision $\psi(w)$, with no zero difference since n is odd. Complementing every bit interchanges the two laws and reverses the decision. Consequently majority has equal payoffs and attains $h\|P_+ - P_-\|_{\text{TV}}$, as also proved in Theorem 18.3.

A prefix with k ones requires at least $m + 1 - k$ ones in its remaining $n - t$ places. Thresholds at most zero give the constant positive response; thresholds greater than $n - t$ give the constant negative response; the remaining thresholds give different nonconstant responses. Counting them for $0 \leq k \leq t$ gives (19.3). The exact realization in Theorem 19.1 implements majority at all these widths.

For an arbitrary randomized decision, its smaller payoff is bounded by the equal-prior average. The deficit of this average from the majority value is exactly

$$h \sum_w |P_+(w) - P_-(w)| e_\psi(w). \quad (19.5)$$

Put $a = 1/2 + h$ and $b = 1/2 - h$. A word with $k \geq m + 1$ ones has absolute likelihood difference $(ab)^{n/2} \{(a/b)^{k-n/2} - (a/b)^{-(k-n/2)}\}$, which increases with k . Symmetry treats the other half of the words. The smallest difference is therefore attained at $k = m, m + 1$ and equals

$$a^{m+1}b^m - b^{m+1}a^m = 2h(1/4 - h^2)^m.$$

Substitution of (19.1) into (19.5) gives (19.4). Any deficient cut gives a strictly positive loss; all sufficiently wide profiles implement majority. Maximizing (19.3) proves the peak-width assertion. The same average-payoff argument works conditional on an independent selector and hence after averaging. $\square$

The result identifies the exact memory needed to attain the optimal value, not the memory needed for a prescribed approximate value uniformly in h . The quantitative deficit can be small for large n . In particular it is not a sharp joint asymptotic tradeoff in sample size, state size and error. It complements, rather than replaces, the uniform sampling exponent in Theorem 18.3. No minimax interchange on the generally nonconvex set of fixed-width auditors has been used.

20. POLYNOMIAL TESTING OF THE ORIGINAL RESOURCE IMAGE

The private behavior image is usually nonconvex. Its testing criterion can nevertheless be written as a limit of finite algebraic lower certificates, without replacing that image by randomized designs. The testing weights in this section depend on the candidate simulator. Unlike Theorem 21.1, this identity is a certificate transform, not an independent variational dual. They are universal proof functions, not a hidden selector supplied to an executed experiment.

Use any one of the finite signatures of Theorem 7.2, with rational instrument data. Write its objective as $F(q) = \max_{1 \leq i \leq m} f_i(q)$, where f_i are all its marked event discrepancies, including the empty event $f_1 = 0$. For visible selectors the event and feedback rule include their selector

fibers, and the target selector law remains in f_i . Thus $-1 \leq f_i \leq 1$ for all three signatures; the polynomials may be different.

Parameterize a row with d outcomes by $d - 1$ variables in $[0, 1]$:

$$q_1 = x_1, \quad q_j = x_j \prod_{l < j} (1 - x_l) \quad (1 < j < d), \quad q_d = \prod_{l < d} (1 - x_l).$$

This map is onto the row simplex, including its boundary. Indeed, divide each desired probability by the remaining mass while that mass is positive; when it is zero choose the remaining variables arbitrarily. Taking the product of these maps, including a selector row when needed, gives an onto polynomial map $q(x)$ from $[0, 1]^D$ to the feasible row product. Put $h_i(x) = f_i(q(x))$ and $g_i = (1 + h_i)/2$. All h_i are rational polynomials, $0 \leq g_i \leq 1$, and $g_1 = 1/2$. The case $D = 0$ is a single feasible table and is interpreted directly.

Theorem 20.1 (Strict polynomial lower-certificate hierarchy). *For $p \geq 1$ define*

$$A_p(x) = \frac{\sum_{i=1}^m g_i(x)^p h_i(x)}{\sum_{i=1}^m g_i(x)^p}, \quad a_p = \min_{x \in [0, 1]^D} A_p(x), \quad \delta = \min_q F(q).$$

Then the denominators are positive, a_p is nondecreasing, and

$$\delta - 2(1 - m^{-1/(p+1)}) \leq a_p \leq \delta, \quad \delta = \sup_{p \geq 1} a_p. \quad (20.1)$$

For rational L let

$$P_{p,L}(x) = \sum_{i=1}^m g_i(x)^p (h_i(x) - L). \quad (20.2)$$

A nonnegative tensor Bernstein expansion of $P_{p,L}$ on $[0, 1]^D$ certifies $\delta \geq L$. If $L < \delta$, some p and some finite Bernstein degree have strictly positive coefficients. Conversely, a strictly positive such expansion certifies $L < \delta$. These are lower certificates at the original state and selector budgets. No convexification of the feasible behavior image occurs.

Proof. Fix x and put $b = \max_i g_i(x) \geq 1/2$ and $r_i = g_i(x)/b$. At least one r_i is one. Write $S_j = \sum_i r_i^j$. Hölder's inequality gives

$$S_p \leq m^{1/(p+1)} S_{p+1}^{p/(p+1)}, \quad S_{p+1} \geq 1.$$

Consequently $S_{p+1}/S_p \geq m^{-1/(p+1)}$. Since $A_p = 2bS_{p+1}/S_p - 1$ and $\max_i h_i = 2b - 1$,

$$0 \leq \max_i h_i - A_p \leq 2(1 - m^{-1/(p+1)}).$$

Cauchy-Schwarz gives $S_{p+2}S_p \geq S_{p+1}^2$, so A_p is nondecreasing in p . Its denominator is at least 2^{-p} and it is continuous on the cube. Taking minima proves (20.1). The bound is also at most $2 \log m/(p+1)$, by $1 - e^{-u} \leq u$.

Every tensor Bernstein basis function is nonnegative on the cube, and the basis functions sum to one. Nonnegative coefficients therefore imply $P_{p,L} \geq 0$. Dividing by the positive denominator gives $F(q(x)) \geq A_p(x) \geq L$ for every x , which proves soundness. For completeness choose p so that the error in (20.1) is less than $\delta - L$. Then

$$P_{p,L}(x) \geq 2^{-p} \{\delta - L - 2(1 - m^{-1/(p+1)})\} > 0.$$

Lemma 20.2 below gives a finite strictly positive Bernstein expansion. Conversely strictly positive coefficients have a positive minimum, and the denominator is at most m , so they give a strict positive uniform gap above L . $\square$

Positive Bernstein certificates and their degree elevation are classical; see de Klerk-Laurent [20]. The next elementary bound is retained for a self-contained exact verifier, not as a new general hypercube positivity theorem.

Lemma 20.2 (An explicit positivity degree). *Let $P(x) = \sum_k c_k x^k$ be a rational polynomial in D variables and $n \geq \max_{k,j} k_j$. In its tensor Bernstein expansion of degree n in each coordinate the coefficient at $\alpha \in \{0, \dots, n\}^D$ is*

$$b_{\alpha,n} = \sum_k c_k \prod_{j=1}^D \frac{\binom{\alpha_j}{k_j}}{\binom{n}{k_j}}, \quad |b_{\alpha,n} - P(\alpha/n)| \leq \frac{C(P)}{n}, \quad C(P) = \sum_k |c_k| \sum_j \binom{k_j}{2}. \quad (20.3)$$

In particular, $\min P > 0$ implies positivity of every coefficient whenever $n > C(P)/\min P$ and the degree restriction holds.

Proof. For a binomial random variable B with parameters (n, x) , $\mathbb{E} \binom{B}{k} = \binom{n}{k} x^k$. Expanding this expectation proves the univariate monomial formula, and products prove the displayed coefficient identity. The convention is $\binom{a}{k} = 0$ for $a < k$. To compare $\binom{a}{k}/\binom{n}{k}$ with $(a/n)^k$, draw k indices uniformly from $\{1, \dots, n\}$. Conditional on no repetition this is sampling without replacement. The probability of a repetition is at most $\binom{k}{2}/n$. Apply this total variation bound to the event that all drawn indices lie in a prescribed a -element subset. Hence the difference of the two factors is at most $\binom{k}{2}/n$. All factors lie in $[0, 1]$. A product telescoping bound followed by the triangle inequality proves (20.3). The final assertion follows by taking the minimum of P . $\square$

Corollary 20.3 (Rational lower witnesses without policy cells). *At every fixed finite rational signature, strict infeasibility above a rational level L is equivalent to the existence of the following finite rational certificate: an integer p , an integer n , and a strictly positive coefficient array for (20.2) whose identity is checked by (20.3). Dovetailing these certificates with rational original-budget upper witnesses produces nested intervals converging to δ . If a subfamily of the original tests gives a strict gap, the same construction applies to it after adjoining the empty test.*

Proof. Soundness and completeness are Theorem 20.1. Each finite candidate is checked by rational polynomial arithmetic. Enumerate all (p, n) and rational levels, and retain the largest verified lower level so far, together with zero. Rational row tables are dense in the feasible product, and their continuous objectives converge to the minimum, so enumeration and retention of the best upper table converge from above. This is a dovetailed enumeration, not a claim that any fixed (p, n) test search terminates. For a requested positive interval width, convergence of both endpoints proves termination. The reduced family conclusion uses only the same pointwise maximum argument. $\square$

Remark 20.4 (Meaning and size of the strict certificate transform). A constant distribution on tests generally detects only the convex hull of the behavior image. The functions $g_i^p / \sum_j g_j^p$ are instead functions of a proposed row table. Their universally quantified weighted inequality is an algebraic certificate; the tester need not observe a private state in an experiment. The displayed certificate identity is not an equality obtained by exchanging a private minimum with a fixed mixed test. Power means and Bernstein degree elevation are classical algebraic approximation devices, proved here in the precise form needed. The claim here is strict-level completeness for the original marked causal resource image, including visible selectors, and its physical transfer below. The coefficient count $(n+1)^D$, the number of original tests, and the algebraic degrees may be large. No polynomial complexity or intrinsic-dimension bound on this row-cube coefficient count is asserted. The independent adaptive test has the separate behavior-size bound of Theorem 21.3; its verification is a different cost.

20.1. A sharp two-time nonconvex example. The source has singleton reports, the control alphabet and actual mark are singleton, and the target emits $(0, 0)$ or $(1, 1)$ with probability $1/2$ each. At width one, a private simulator emits two independent bits with probabilities p, q . It cannot reread its first emitted bit.

Proposition 20.5 (A gap invisible to constant testing mixtures). *For this delayed experiment,*

$$\delta_{(1,1)}^p = \sqrt{2} - 1, \quad \delta_{(1,1),2}^h = 0, \quad \delta_{(1,1),r}^v = \sqrt{2} - 1 \quad (r \geq 1).$$

The private objective is the maximum of the four polynomials

$$0, \quad \frac{1}{2} - pq, \quad \frac{1}{2} - (1-p)(1-q), \quad p + q - 2pq.$$

Every constant mixture of these tests has minimum at most zero, whereas Corollary 20.3 certifies every rational $L < \sqrt{2} - 1$ at private width one.

Proof. Put $u = pq$ and $v = (1-p)(1-q)$. Direct total variation calculation gives $F = 1 - \min(u, 1/2) - \min(v, 1/2)$ and the displayed maximum. By Cauchy–Schwarz, $\sqrt{u} + \sqrt{v} \leq 1$. If $u \geq 1/2$, then $v \leq (1 - 1/\sqrt{2})^2$, so the sum of the two capped terms is at most $2 - \sqrt{2}$; the case $v \geq 1/2$ is symmetric. Otherwise set $a = \sqrt{u}$, $b = \sqrt{v}$. On $0 \leq a, b \leq 1/\sqrt{2}$, $a + b \leq 1$, the maximum of $a^2 + b^2$ is at a vertex of this polygon and equals $2 - \sqrt{2}$. Equality is attained at $p = q = 1/\sqrt{2}$. A hidden fair bit selects the two deterministic constant sequences and exactly reproduces the target. If it is exposed, variation is averaged over the selector fibers; each fiber has error at least the private minimum, which is itself attainable without a selector. Finally the target lies in the convex hull of the two private deterministic behaviors. The average discrepancy of any fixed test mixture at those behaviors is zero, so at least one has discrepancy at most zero. The strict polynomial certificates therefore detect information lost by constant convex separation. $\square$

21. ADAPTIVE TESTING ON THE OBSERVABLE BEHAVIOR IMAGE

The obstruction to linear randomization criteria at a fixed private register is the nonconvexity of the behavior image, not a failure of compactness. We give a variational formulation in which a test is an independent function of an observable candidate law. The construction does not give the simulator an extra random variable.

21.1. The testing game. Let B be a nonempty compact subset of a finite-dimensional affine space, and let $f_1, \dots, f_m$ be affine real functions, including the zero function. Put

$$F(b) = \max_{i \leq m} f_i(b), \quad \delta = \min_{b \in B} F(b).$$

For the finite comparison problem of Section 7, b is the array of discrepancies between the simulated and target marked laws, indexed by parameters, feedback trees and transcripts; f_i ranges over their signed event evaluations. In the visible case the array includes the selector and its target weights. Thus the target coordinate is part of the array when those weights vary. Private, fixed-hidden and fixed-visible signatures each give their own compact set B . No convex hull is taken.

An *adaptive test* is a continuous map $\kappa : B \rightarrow \Delta_m$, chosen before a candidate b . Its score against that candidate is

$$G(b, \kappa) = \sum_i \kappa_i(b) f_i(b).$$

This is a verification game: the selected candidate law is revealed to the evaluator of the already chosen function κ . It is not a one-observation statistical test, and does not make an unavailable parameter or mark observable to a physical controller.

Theorem 21.1 (Revealed-behavior adaptive representation). *For every such compact behavior image,*

$$\begin{aligned} \delta &= \sup_{\kappa \in C(B, \Delta_m)} \min_{b \in B} G(b, \kappa) \\ &= \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa \in C(B, \Delta_m)} \int_B G(b, \kappa) d\mu(b). \end{aligned} \tag{21.1}$$

The minimization over probability measures is attained. If Borel adaptive tests are allowed, the first supremum is attained as well. The continuous-test assertion is a supremum; continuous attainment is not needed or asserted.

Proof. Fix $\eta > 0$. The sets $U_i = \{b : f_i(b) > F(b) - \eta\}$ form an open cover of the compact metric space B . Choose a continuous partition of unity subordinate to this finite cover. The resulting test has $F(b) - \eta < G(b, \kappa) \leq F(b)$ for all b . Consequently the first supremum equals δ . The same uniform inequality shows, for every probability μ ,

$$\sup_{\kappa} \int G(b, \kappa) d\mu(b) = \int F d\mu.$$

Its minimum is δ , attained by a point mass at a minimizer of F . This also proves the displayed minimax exchange without an assumption that B is convex. Equivalently one can apply the usual compact-convex minimax theorem to $\mathcal{P}(B)$ and the convex class of continuous tests, with the pairing $\int G d\mu$. Finally choose the smallest index attaining $F(b)$. Its level sets are Borel, and its indicator vector is a Borel test with score exactly F . $\square$

The probability μ here describes a lottery over *revealed candidate laws*. The score is $\int G(b, \kappa) d\mu(b)$, not $G(\int b d\mu, \kappa)$. Hiding the candidate and replacing it by its barycenter is a different game; in general it lowers the value. This distinction is essential for a finite persistent-state budget.

21.2. A size bound in behavior dimension. Factor out directions annihilated by every $f_i - f_i(b_0)$, so that on the resulting difference space

$$\|v\|_* = \max_i |f_i(b_0 + v) - f_i(b_0)|$$

is a norm. Denote its dimension by a and let $R = \text{diam}_*(B)$. All conclusions below are unaffected by this observable quotient. The case $a = 0$ is a single constant comparison and requires one test. Suppose henceforth that $a \geq 1$, so $R > 0$.

Lemma 21.2 (A bounded behavior chart). *There exist $b_0 \in B$ and vectors $v_j \in B - b_0$, $1 \leq j \leq a$, forming a basis, such that $\|v_j\|_* \leq R$ and every $b \in B$ has coordinates $b = b_0 + \sum_j t_j v_j$ with $|t_j| \leq 2$. If B is a rational polynomial image of a product of simplexes, the points can be chosen rational and can be found effectively.*

Proof. Fix $b_0 \in B$. In any fixed coordinates on the difference space, let $D_0 > 0$ be the maximum absolute determinant of a vectors in $B - b_0$. Compactness and spanning give existence and positivity. Choose a tuple with determinant greater than $D_0/2$. Replacing its j th vector by $b - b_0$ multiplies the determinant by t_j ; maximality gives $|t_j| < 2$. The norm bound is the definition of R . For a rational polynomial image, take b_0 at rational rows. Rational row images are dense. An affine basis can be found by enumerating rational images and testing rational determinants; the true affine span and the maximum determinant are semialgebraic data. Quantifier elimination decides whether a proposed nonzero determinant exceeds half the maximum. Density therefore makes the search terminate. This is an effectiveness assertion, not a bit-complexity bound for finding the chart. $\square$

Set $b(x) = b_0 + \sum_j (4x_j - 2)v_j$ and $C = \{x \in [0, 1]^a : b(x) \in B\}$. The ambient cube is a coordinate box, not an enlargement of the feasible behavior set. Write $\ell_i(x) = f_i(b(x))$. Then

$$|\ell_i(x) - \ell_i(y)| \leq 4R \sum_j |x_j - y_j|. \quad (21.2)$$

For $n \geq 0$ let $\beta_{\alpha,n}$ be the tensor Bernstein basis of coordinatewise degree n . A feasible coefficient array has $c_{\alpha i} \geq 0$ and $\sum_i c_{\alpha i} = 1$ for every $\alpha \in \{0, \dots, n\}^a$. It specifies the independent test

$$\kappa_i^c(x) = \sum_{\alpha} c_{\alpha i} \beta_{\alpha,n}(x).$$

Define

$$d_n = \max_c \min_{x \in C} \sum_{\alpha, i} c_{\alpha i} \beta_{\alpha,n}(x) \ell_i(x). \quad (21.3)$$

Theorem 21.3 (Intrinsic adaptive degree and support bounds). *The maxima in (21.3) are attained and $d_n \leq d_{n+1} \leq \delta$. For $n \geq 1$,*

$$0 \leq \delta - d_n \leq \frac{4Ra}{\sqrt{n}}. \quad (21.4)$$

There is an array with at most $(n+1)^a$ nonzero coefficients, all equal to one, whose minimum score is at least $\delta - 4Ra/\sqrt{n}$. Its test has coordinatewise degree n , and its score degree at most $n+1$. Furthermore

$$d_n = \min_{\mu \in \mathcal{P}(C)} \sum_{\alpha} \max_i \int_C \beta_{\alpha,n}(x) \ell_i(x) d\mu(x), \quad (21.5)$$

and a minimizing measure can be supported on at most $(n+2)^a$ points. In particular a strict level $L < \delta$ admits an adaptive test of positive uniform margin with $n > (8Ra/(\delta - L))^2$ and at most $(n+1)^a$ occupied coefficient entries. These bounds depend on the observable image and its metric, not on the number of stochastic rows in a realization.

Proof. The coefficient space is a finite product of compact simplexes, and the minimum of a jointly continuous function over compact C is continuous in its coefficients. This proves attainment. Bernstein degree elevation replaces each coefficient by convex combinations of adjacent coefficients, preserving every simplex constraint and the represented function. It proves monotonicity. All scores are convex combinations of the $\ell_i(x)$, proving the upper bound by δ .

At each grid point α/n choose an index i_{α} maximizing ℓ_i there. The choice is made even at infeasible grid points; only the final score is minimized over C . Give that index coefficient one and the others zero. For fixed x let A_j be independent $\text{Bin}(n, x_j)$ variables. If i_* maximizes at x ,

$$\begin{aligned} \max_i \ell_i(x) - \mathbb{E} \ell_{i_A}(x) &\leq \mathbb{E} \{ \ell_{i_*}(x) - \ell_{i_*}(A/n) + \ell_{i_A}(A/n) - \ell_{i_A}(x) \} \\ &\leq 8R \sum_j \mathbb{E} |A_j/n - x_j| \leq \frac{4Ra}{\sqrt{n}}. \end{aligned}$$

Here the omitted middle difference is nonpositive, and the last bound uses $\text{Var}(A_j/n) \leq 1/(4n)$. This proves the rate and the sparse coefficient assertion.

For the last formula replace the minimum over C by a minimum over $\mathcal{P}(C)$. The pairing is bilinear and continuous, and both feasible sets are compact convex, so the compact minimax theorem [18] exchanges the extrema. Optimizing each coefficient simplex gives (21.5). The integrands are polynomials of coordinatewise degree at most $n+1$. Preserve all monomial moments of that degree by Carathéodory's theorem. There are $(n+2)^a$ monomials including the constant; their moment vectors lie in an affine hyperplane of dimension at most $(n+2)^a - 1$. A measure with at most $(n+2)^a$ atoms therefore has exactly the same objective. The strict-margin assertion follows from the rate with error less than $(\delta - L)/2$. $\square$

The score is a nonlinear function of behavior although each event functional is affine. This is the reason the measure in (21.5) cannot be replaced by its behavior barycenter. For $n = 0$ it can: d_0 is precisely the constant-test convex relaxation. In the two-time unmarked example of Proposition 20.5, the target belongs to the convex hull of private behaviors, so $d_0 = 0$, whereas $\lim_n d_n = \sqrt{2} - 1$. Thus the hierarchy recovers a gap which every constant test misses.

Remark 21.4 (What the size bound measures). The bound counts the nonzero entries of the independent adaptive test and controls its behavior-coordinate degree. It does not bound the cost of verifying its minimum on the generally nonconvex set C , the bit lengths of a rational chart, or the size of a description of C . Distinct row presentations of the same metric behavior image satisfy the same existence bound, although their coefficient arrays need not coincide. For the private fixed-target law array one may use $R \leq 1$ with the maximum total-variation metric. For the visible signed-discrepancy array the uniform bound $R \leq 2$ is sufficient. The old row-cube positivity certificate is a separate verification engine, with its own coordinate-dependent complexity.

21.3. A three-weight certificate invisible to constant tests.

Proposition 21.5 (A finite intrinsic nonlinear witness). *Consider a private width-one simulator producing two independent bits, with probabilities p, q , for the target which assigns probability $1/2$ to each of 00, 11. Put $u = pq$, $v = (1-p)(1-q)$ and $J(t) = 3t^2 - 2t^3$. The three behavior-dependent weights*

$$\kappa_1 = J(v), \quad \kappa_2 = J(u), \quad \kappa_3 = 1 - J(u) - J(v)$$

are nonnegative and sum to one. Against the event discrepancies $1/2 - u, 1/2 - v, 1 - u - v$, respectively, their score G satisfies

$$G \geq \frac{2}{5} + \frac{9}{8960}. \quad (21.6)$$

The certificate uses cubic weights in observable probabilities, a nine-term nonnegative derivative identity, and seventeen positive rational Bernstein coefficients in one variable. Every constant mixture of event discrepancies has minimum at most zero.

Proof. We have $u + v \leq 1$. Since J is increasing on $[0, 1]$ and $J(1-t) = 1 - J(t)$, the weights are feasible. Their score is

$$G = 1 - u - v + J(u)(u - 1/2) + J(v)(v - 1/2).$$

Set $a = (p + q - 1)^2$ and $b = (p - q)^2$. Then $a, b \geq 0$ and $a + b \leq 1$. Direct expansion gives

$$\begin{aligned} 64G &= 27 - 12a + 38a^2 - 20a^3 - a^4 + 36b - 60ab + 36a^2b + 4a^3b \\ &\quad + 6b^2 - 12ab^2 - 6a^2b^2 - 4b^3 + 4ab^3 - b^4. \end{aligned}$$

Let $c = 1 - a - b$. Another polynomial identity, valid when $a + b + c = 1$, is

$$\begin{aligned} 16\partial_b G &= 4a^3 + 6a^2c + 12ab^2 + 24abc + 12ac^2 \\ &\quad + 8b^3 + 30b^2c + 30bc^2 + 9c^3 \geq 0. \end{aligned}$$

Thus $G(a, b) \geq G(a, 0)$. The univariate polynomial

$$G(a, 0) - \frac{2}{5} = \frac{7}{320} - \frac{3}{16}a + \frac{19}{32}a^2 - \frac{5}{16}a^3 - \frac{1}{64}a^4$$

has degree-16 Bernstein coefficients $z_k/349440$, where

$$\begin{aligned} (z_0, \dots, z_{16}) &= (7644, 3549, 1183, 351, 855, 2494, 5064, 8358, \\ &\quad 12166, 16275, 20469, 24529, 28233, 31356, 33670, 34944, 34944). \end{aligned}$$

The Bernstein basis is nonnegative and sums to one. Its smallest coefficient is $351/349440 = 9/8960$, proving (21.6). All identities are rational polynomial identities; no numerical minimization is used. Finally the target is the barycenter of the two feasible deterministic laws 00 and 11. An affine mixture of signed event discrepancies has average zero at these two laws and hence minimum at most zero. The empty event supplies the reverse constant-test bound zero. $\square$

For comparison, the affine weights $v, u, 1 - u - v$ already give the smaller bound $3/8$, with the exact identity

$$G_{\text{affine}} - \frac{3}{8} = \frac{1}{2}(p - q)^2 + 2(p - 1/2)^2(q - 1/2)^2.$$

Neither certificate changes the private feasible set. The cubic certificate is an independent test chosen once for the whole behavior image, not a mixture chosen anew by an optimizer's unknown row values. Its evaluation does use the revealed behavior, as specified in the verification game. The 17-coefficient positivity check is classical; the distinction from constant tests is the behavior-dependent feasible weight family and its resource-sensitive minimum.

The infinite-dimensional representation above holds for continuous f_i on any compact metric space: its proof uses a partition of unity, not causal factorization. The measure minimum is attained at a Dirac mass at a minimizer of F . The finite-level polynomial theorem has a different content. It is an existence-size bound for revealed-law verification; the reset-audit theorem of Section 18 imposes an explicit finite-data restriction.

21.4. Strict certificates and effective physical input. Given rational finite instruments, a rational chart and the sparse array constructed above, its strict minimum can be certified by real quantifier elimination on the original row constraints. Alternatively, pull the score back through the original polynomial row map and certify its strict positivity using degree-elevated Bernstein coefficients. The pullback changes the verification degree and is not covered by the behavior-coordinate bound. Generic quantifier elimination also decides the exact finite semialgebraic boundary; this classical observation does not assert that a finite nonnegative Bernstein expansion exists at $L = \delta$.

Proposition 21.6 (Uniform effective presentation input). *Suppose a single algorithm, on input j , returns rational finite instruments E_j, F_j and nonnegative rational error enclosures a_j, c_j for two-sided stateless marked feedback presentations of E, F . Suppose $a_j + c_j \rightarrow 0$. All alphabets, widths, selector laws and finite subroutines must be computable uniformly from j and the requested signature. Then, for every positive rational η , an algorithm returns a rational interval of width below η containing $\delta_b(E, F)$ at the same signature.*

Proof. Enumerate j until $a_j + c_j < \eta/8$. This terminates without a known rate because the rational enclosures themselves tend to zero. For the selected finite problem, enumerate rational same-budget upper tables and strict rational lower certificates, dovetailing all finite checks; include the always valid lower bound zero. Compactness, density and strict completeness provide an interval of width below $\eta/2$. Enlarge it by $a_j + c_j$ at both ends using stateless comparison. The enlarged width is below $3\eta/4$. Every enumeration and comparison is uniform by the hypotheses. A mere sequence of individually computable presentations without a uniform enumeration is not an input satisfying this proposition. $\square$

22. AN EXACT ALGORITHM FOR THE OBSERVABLE CHART

The chart in Lemma 21.2 can be found without taking its dimension as an oracle. We state the input and output explicitly. Rational polynomials are encoded by finite lists of monomial exponent vectors and rational coefficients in reduced integer form. A product of simplexes is encoded by the number of entries in each row. Eliminating the last entry of every nontrivial row leaves a full-dimensional rational polytope $\Omega \subset \mathbb{R}^r$. Zero-dimensional factors are omitted.

Proposition 22.1 (Effective observable quotient and pullback). *The input consists of the preceding simplex description, a rational polynomial map $b : \Omega \rightarrow \mathbb{R}^N$, and finitely many rational affine functionals f_i on $\mathbb{R}^N$. There is a terminating algorithm which returns its observable affine*

dimension a , a rational basis and coordinate map for the observable quotient, and, when $a > 0$, rational parameter points $q_0, q_1, \dots, q_a$ with the following properties. With

$$z(q) = (f_i(b(q)) - f_i(b(q_0)))_i, \quad v_j = z(q_j),$$

the vectors v_j are a basis of the observable difference space, and all coordinates $t(q)$ in that basis satisfy $|t_j(q)| < 2$. The output also contains the rational polynomial cube map $x(q) = (t(q) + 2)/4$, a rational upper bound for the observable diameter, and the exact pullback of any subsequently supplied rational polynomial in x . The algorithm outputs the universal determinant formula it has decided. It makes no polynomial-time or coefficient-height claim.

Proof. Choose a rational relative interior point q_0 of the input simplexes. Expand the vector polynomial $z(q)$ in the free row coordinates. Let V be the rational span of all its coefficient vectors, including its constant coefficient. This equals $\text{span } z(\Omega)$: a linear functional vanishes on $z(\Omega)$ exactly when its polynomial vanishes on the nonempty interior of Ω , hence identically, hence on every coefficient. Gaussian elimination therefore gives $a = \dim V$ and a rational basis matrix U . Choosing independent rows gives a rational left inverse L with $LU = I_a$. Thus $y(q) = Lz(q)$ provides exact quotient coordinates. If $a = 0$, all discrepancies are constant and no chart is required.

For $a > 0$ let

$$D_0 = \max_{u_1, \dots, u_a \in \Omega} |\det(y(u_1), \dots, y(u_a))|.$$

Compactness gives a maximum, and spanning gives $D_0 > 0$. Enumerate rational tuples $(q_1, \dots, q_a)$ in Ω^a . Membership and their determinant d are decided by rational arithmetic. For $d \neq 0$, decide the first-order real formula

$$\forall u_1, \dots, u_a \in \Omega : \quad \det(y(u_1), \dots, y(u_a))^2 < 4d^2. \quad (22.1)$$

Quantifier elimination terminates for each formula. Density of rational points and continuity of determinant imply that some enumerated tuple has $|d| > D_0/2$. For such a tuple the strict formula is true, so the search terminates. No computation of an exact algebraic maximizer is needed.

Invert the rational matrix with columns $y(q_j)$ to obtain $t(q)$. Cramer's rule and (22.1) give $|t_j(q)| < 2$ for every q . In the observable norm $\|z\|_* = \max_i |z_i|$, each v_j has norm at most the diameter of $z(\Omega)$. An upper bound for that diameter can be produced, for example, by bounding each free coordinate in $[0, 1]$ and summing the absolute polynomial coefficients for $f_i(b(q)) - f_i(b(q'))$. This bound is rational and computable. Rational matrix multiplication and substitution give $x(q)$ and every requested pullback. They also give its full sparse coefficient encoding. All searches have now been specified by finite encodings and terminating operations. $\square$

There is a small but important distinction between the quotient and a projection onto a smaller set of simulator rows. The former identifies exactly the directions invisible to the chosen event family; the original row constraints remain present in every universal positivity test. The chart need not identify equivalent tasks after a change of marks or of information signature. Those are changes to the comparison problem.

23. A MARKED RESOURCE GAP GENERATED BY A BINARY COLLISION

We now realize a private–hidden–visible separation in a physical experiment whose information vanishes when the collision is removed. The controller chooses a later sensor gate after its first report. The gate controls acquisition, not the mechanical collision law. Both the source and the target carry the same actual, unobserved mark.

23.1. The exact marked two-report experiment. Let $B, W \in \{0, 1\}$ have the joint distribution

$$\mathbb{P}(B = b, W = w) = \begin{cases} 5/16, & b = w, \\ 3/16, & b \neq w. \end{cases} \quad (23.1)$$

The blind source E_* has two constant reports and actual mark W . The target F_* reports $Y_1 = B$, receives $A \in \{0, 1\}$ after that report, and reports $Y_2 = AB$. The feedback tester may choose A as any function of its earlier report. A width-one private simulator cannot reread its own first emission, except through the current gate A . Hidden selectors persist but are not revealed; visible selectors are included in the joint target law independently of (B, W) . Put

$$d_* = \sqrt{5/2} - 9/8.$$

Theorem 23.1 (An exact marked causal separation). *At the width profile $(1, 1)$,*

$$\delta^P(E_*, F_*) = d_*, \quad \delta_2^h(E_*, F_*) = 1/8, \quad \delta_r^v(E_*, F_*) = d_* \quad (r \geq 1). \quad (23.2)$$

The hidden upper bound is realized by two deterministic tables. The private upper bound is realized by two independent Bernoulli variables with parameter $\sqrt{5/8}$, followed by the gate. These bounds hold against every feedback rule, not just open-loop gates.

Proof. First choose the constant gate $A = 1$. The two private emissions are independent Bernoulli variables with parameters p, q and are independent of W . Let

$$u = pq, \quad v = (1 - p)(1 - q), \quad a = 3/8, \quad b = 5/8.$$

The overlap of the target and simulated marked laws is $H(u) + H(v)$, where

$$H(t) = \frac{1}{2} \{\min(t, a) + \min(t, b)\}.$$

Indeed the target is supported on equal report pairs, and its two W -masses at each equal pair are $3/16$ and $5/16$. Total variation is one minus this overlap.

Cauchy-Schwarz gives $\sqrt{u} + \sqrt{v} \leq 1$. By symmetry suppose $u \geq v$. If $u \leq 1/4$ the overlap is at most $1/2$. If $u \geq 1/4$, then $v \leq (1 - \sqrt{u})^2 \leq 1/4 < a$, so the overlap is at most

$$H(u) + (1 - \sqrt{u})^2.$$

This function is convex on each of $[1/4, a]$ and $[a, b]$ and nonincreasing on $[b, 1]$. Its maximum is therefore attained at $1/4, a$ or b . The values at a and b are $1 + 2a - 2\sqrt{a}$ and $3/2 + b - 2\sqrt{b}$ respectively. The second is larger, since

$$(3/2 + b - 2\sqrt{b}) - (1 + 2a - 2\sqrt{a}) = 3/8 - \frac{1/2}{\sqrt{a} + \sqrt{b}} > 0,$$

using $\sqrt{a} > 3/5$, $\sqrt{b} > 3/4$. It also exceeds $1/2$. Taking $p = q = \sqrt{b}$ attains this maximum, and hence the minimum error is $1 - (3/2 + b - 2\sqrt{b}) = d_*$.

For any hidden selector, the all-on report pair is still independent of W . If its probabilities of 11 and 00 are u, v , then $u + v \leq 1$. Concavity and monotonicity of H give

$$H(u) + H(v) \leq 2H((u + v)/2) \leq 2H(1/2) = 7/8.$$

A fair hidden bit J , emitting J twice, attains this overlap. This requires precisely two deterministic selector tables and gives error $1/8$. In particular hiding a seed does not erase the actual-mark obstruction.

These all-on bounds are lower bounds for the feedback problem. For the upper bounds first generate the corresponding all-on pair (X_1, X_2) , emit X_1 , and after receiving A emit AX_2 . For a deterministic feedback $A = g(X_1)$ the full gated marked transcript is the pushforward of the all-on marked pair. Applying the same map to the target pair cannot increase total variation. The construction is executable at width one: the second fresh Bernoulli variable is independent in the

private case, and the bit J is the priced persistent hidden selector in the hidden case. Randomized feedback follows by conditioning on its independent random table.

Finally, in the visible case the two compared laws are

$$\sum_j \lambda_j \delta_j \otimes P_{g_j}^{q_j} \quad \text{and} \quad \sum_j \lambda_j \delta_j \otimes Q_{g_j}.$$

The target selector is independent of (B, W) . The all-on gate in every fiber has error at least d_* in every private fiber. Joint variation is the weighted sum of these fiber errors. Ignoring the visible selector and using the private optimizer attains d_* . $\square$

23.2. A full-dimensional collision preparation. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$ with contact distance d . The center of mass $Q = (q_1 + q_2)/2$ ranges over any fixed rational position box. Write $r = q_1 - q_2$, $g = v_1 - v_2$, and $V = (v_1 + v_2)/2$. Conditional on (B, W) in (23.1), take the uniform product preparation on

$$\begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), \quad a_0 \in [29/10, 31/10], & b_0 &\in [9/20, 11/20], & (23.3) \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

The boxes have positive volume in the full phase space. The actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$; total momentum conservation retains it. The informative quantity will instead be the transverse velocity created at a collision. Put

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector which is identically zero on the incoming transverse-velocity range.

Lemma 23.2 (Uniform transverse collision signal). *For every preparation in (23.3) and every $d \in [9/10, 11/10]$, exactly one collision occurs before time 3. Its time t_c lies in $(4/5, 3/2)$, and afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4. \quad (23.4)$$

For $0 < d \leq 2/5$ no collision occurs before time 3, and the detector is zero throughout that interval.

Proof. Up to a collision $r(t) = r + tg$. For $t \leq 4/5$, its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At time $3/2$ the absolute coordinate bounds are $1/4, 113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. Thus a first contact occurs in the stated interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$ as well. The first contact therefore has $n_x < 0$, where $n = r(t_c)/d$. The entry contact is strictly incoming. The bounds above give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq \frac{87}{220}.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The equal-mass elastic collision rule is

$$v'_1 = V + g/2 - (g \cdot n)n, \quad g' = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v'_{1,y} > -3/200 + \frac{1443}{1000} \frac{87}{220} > 1/2.$$

The relative velocity is outgoing, $r(t_c) \cdot g' > 0$, so the squared separation increases strictly for all later times. There are only two spheres and no walls; no second collision is possible. This proves (23.4). For $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. Thus there is no collision. The incoming $|v_{1,y}| \leq 3/200$ then makes h zero. $\square$

At time 2 the physical target $F_{d,\sigma}$ reports

$$Y_1 = \mathbf{1}_{\{h(v_{1,y}(2)) + \sigma\xi_1 > 0\}}.$$

It then receives a gate $A \in \{0, 1\}$, and at time 3 reports

$$Y_2 = A\mathbf{1}_{\{h(v_{1,y}(3)) + \sigma\xi_2 > 0\}},$$

where the ξ_i are independent standard normal variables, independent of the preparation. The source E_d has the same preparation, evolution, control interface and actual mark, but emits two constant reports. The gates do not change the particle motion. This is a feedback acquisition experiment: taking $A = 0$ suppresses later information and taking $A = 1$ exposes another noisy observation of a collision-generated quantity.

Theorem 23.3 (Collision-sensitive physical resource separation). *Let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/12$. At retained width $(1, 1)$ the physical deficiencies obey, uniformly in d ,*

$$\begin{aligned} |\delta^P(E_d, F_{d,\sigma}) - d_*| &< 1/300, & |\delta_r^Y(E_d, F_{d,\sigma}) - d_*| &< 1/300 \quad (r \geq 1), \\ |\delta_2^h(E_d, F_{d,\sigma}) - 1/8| &< 1/300. \end{aligned} \quad (23.5)$$

In particular private and visible error exceed $9/20$, while hidden two-selector error is less than $13/100$. More generally the errors in these three comparisons are at most $2\Phi(-1/(4\sigma))$, which tends to zero as $\sigma \downarrow 0$. For $0 < d \leq 2/5$ all three deficiencies are zero. The interacting and noninteracting regimes use the same preparation and detector. The conclusions are unchanged by simultaneously scaling positions, diameter and observation times by any positive factor, with velocities and detector noise unchanged.

Proof. Couple the ideal and physical target by the same (B, W) and the same feedback random table. Lemma 23.2 bounds each potential bit error by $\Phi(-1/(4\sigma))$. With probability at least $1 - 2\Phi(-1/(4\sigma))$ both potential reports agree with their ideal bits; the gates and all emitted transcripts then agree as well. This gives a marked feedback variation bound uniform over all gates. The channels in both directions are the identity on the binary report alphabet and have one retained state. The blind source is already exactly E_* . Stateless comparison therefore changes each fixed-budget deficiency by at most the stated error, without adding memory or hiding an exposed selector.

For completeness, the elementary Gaussian tail bound $\Phi(-x) \leq e^{-x^2/2}/(x\sqrt{2\pi})$ at $x = 3$ gives $\Phi(-3) < 1/600$: use $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$. Hence the total error is below $1/300$. The inequalities $d_* - 1/300 > 9/20$ and $1/8 + 1/300 < 13/100$ follow by squaring positive rational bounds.

When $d \leq 2/5$, both potential reports are independent fair noise bits and are independent of W . A private width-one simulator draws those two bits and applies the current gate, exactly reproducing every feedback transcript. Hidden and visible versions can ignore their selectors. Finally elastic scattering is invariant under $(q, d, t) \mapsto (sq, sd, st)$ with v unchanged. This proves the scaling statement; it is geometric similarity, not a particle-number or Boltzmann–Grad limit. $\square$

The theorem combines three effects. Collisions create the sign of the informative transverse velocity. A gate chosen after the first report changes future information. A retained hidden bit, but not a visible one, permits temporal correlation without charging the width-one register. The actual mark remains partially inaccessible in every signature. The conserved longitudinal momentum labels that mark; it does not generate the observed transverse signal.

23.3. Uniformly effective finite presentations. The binary physical interface is already finite, but its probabilities are not generally rational. We make their effective approximation explicit rather than treating individual computability as a uniform algorithm.

Proposition 23.4 (Collision-chart rational presentations). *For computable $d \in [9/10, 11/10]$ and $\sigma > 0$, one algorithm produces rational marked causal instruments and certified two-sided stateless errors tending to zero for $(E_d, F_{d,\sigma})$. On a compact computable parameter interval with $\sigma \geq \sigma_0 > 0$, the errors and the construction are uniform. Thus Proposition 21.6 applies at every fixed finite state and selector signature, not only at $(1, 1)$.*

Proof. On every preparation box the entry time and outgoing velocity are

$$t_c = \frac{-r \cdot g - \sqrt{(r \cdot g)^2 - |g|^2(|r|^2 - d^2)}}{|g|^2}, \quad n = (r + t_c g)/d, \quad v'_1 = V + g/2 - (g \cdot n)n.$$

The positive square root is separated from zero: it equals $-d(g \cdot n) > (9/10)(1443/1000)$. All denominators are likewise bounded away from zero. These explicit bounds and the compact boxes allow interval differentiation of the displayed formulas, producing a rational common Lipschitz upper bound C_{geo} for $v'_{1,y}$ in the packet variables and d . This computation terminates because only addition, multiplication, division by separated positive quantities and a separated square root occur.

Conditional on the microscopic preparation the two potential all-on bits are independent Bernoulli with probability $p = \Phi(h(v'_{1,y})/\sigma)$. Their four probabilities are p^2 , $p(1-p)$, $p(1-p)$ and $(1-p)^2$. The normal distribution function is uniformly computable on bounded intervals, for example by integrating its convergent exponential series with a remainder bound. Its Lipschitz constant is less than one. Together with the 1-Lipschitz detector and the geometric bound, this yields a computable Lipschitz constant at most $2C_{\text{geo}}/\sigma_0$ for these four integrands on fixed-noise compact parameter sets. Equal-box Riemann sums of mesh h therefore have explicit error at most a computable constant times h , plus the chosen rational evaluation error. Parameter uncertainty is included by its interval enclosure.

Integrate the four all-on probabilities separately on each packet, with the exact rational weights (23.1). For each W , round its conditional probability vector to a rational simplex point on a sufficiently fine grid; the rounding error is bounded by the sum of the coordinate errors and the grid mesh. Keep $\mathbb{P}(W = 0) = \mathbb{P}(W = 1) = 1/2$ exact. Sample this finite joint law once at initialization, emit its first bit, and gate its second bit. This is a finite latent realization of the target instrument, not a simulator with extra free state. It makes every causal prefix consistent and never divides by a possibly small prefix probability. Every gated law is a pushforward of the all-on law, so the same marked variation bound holds uniformly under feedback. Identity report channels give both stateless directions. Choosing summable evaluation and mesh errors produces the asserted uniform sequence. $\square$

24. UNIFORM STABILITY OF OPTIONAL PROJECTIONS

Marked comparison controls more than terminal bounded risk when the compared laws share their latent variable. We prove a whole-process estimate for the resulting posterior martingales. The distinction between a probabilistic coupling and an executable causal channel is important in this section.

Let Z take values in a standard Borel space and let Y take values in a standard Borel canonical path space on $[0, T]$, with a right-continuous observation filtration. Consider equivalent joint probability measures P, Q on (Z, Y) , with the same Z -marginal ν . Use the usual completions, which agree for equivalent output laws. For a bounded measurable latent variable $f(Z)$ write

$$M_t^P = \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y], \quad M_t^Q = \mathbb{E}_Q[f(Z) \mid \mathcal{F}_t^Y].$$

Take their right-continuous martingale versions. All suprema below include T and can be evaluated on a countable dense time set.

Theorem 24.1 (Marked variation controls the whole posterior process). *Let $\varepsilon = \|P - Q\|_{\text{TV}}$ and $|f| \leq 1$. When both posterior functions are evaluated on the same canonical output path,*

$$\mathbb{E}_Q \sup_{t \leq T} |M_t^P(Y) - M_t^Q(Y)|^2 \leq 80\varepsilon. \quad (24.1)$$

There is a coupling of (Z, Y^P) and (Z, Y^Q) which retains exactly the same latent variable and satisfies

$$\mathbb{P}(Y^P \neq Y^Q) \leq \varepsilon, \quad \mathbb{E} \sup_{t \leq T} |M_t^P(Y^P) - M_t^Q(Y^Q)|^2 \leq 84\varepsilon. \quad (24.2)$$

The coupling is not asserted to be causal or to have finite state. The variation bound itself, not this coupling, is used for resource transport.

Proof. Let $D = dP/dQ$, $D_t = \mathbb{E}_Q[D \mid \mathcal{F}_t^Y]$, and set

$$A_t = \mathbb{E}_Q[(D - 1)f \mid \mathcal{F}_t^Y], \quad B_t = D_t - 1.$$

Bayes' formula gives

$$M_t^P - M_t^Q = \frac{A_t - B_t M_t^Q}{1 + B_t}.$$

Since $\mathbb{E}_Q|D - 1| = 2\varepsilon$, both martingales A, B have terminal absolute expectation at most 2ε . Doob's weak L^1 inequality gives, for $0 < u \leq 2$,

$$\begin{aligned} Q\{\sup_t |M_t^P - M_t^Q| > u\} &\leq Q\{\sup_t |B_t| > 1/2\} + Q\{\sup_t |A_t| > u/4\} + Q\{\sup_t |B_t| > u/4\} \\ &\leq 4\varepsilon + 16\varepsilon/u. \end{aligned}$$

Integrating $2u$ times this bound on $(0, 2)$ gives $16\varepsilon + 64\varepsilon = 80\varepsilon$, proving (24.1). No tightness conclusion is being inferred from finite-dimensional convergence.

Disintegrate both joint laws over their common latent marginal. Measurable maximal coupling of the two output kernels, conditional on Z , has disagreement probability

$$\int \|P(dy \mid z) - Q(dy \mid z)\|_{\text{TV}} d\nu(z) = \|P - Q\|_{\text{TV}}.$$

Such a coupling can be constructed by using the minimum of the two kernel densities with respect to their sum, then their residual kernels. It is measurable on standard Borel spaces. On the event of identical full output paths use (24.1) with the Q -marginal; on its complement the squared difference is at most four. This proves (24.2). The maximal coupling may depend on the full output path; it has no asserted online implementation. $\square$

Corollary 24.2 (Changing-filtration convergence from marked comparison). *Let P_n and P be equivalent joint laws with the same latent marginal ν , and $\|P_n - P\|_{\text{TV}} \rightarrow 0$. They admit a common probability-space realization $(Z, Y, Y^1, Y^2, \dots)$ such that, for every $f \in L^2(\nu)$,*

$$\mathbb{E} \sup_{t \leq T} \left| \mathbb{E}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}[f(Z) \mid \mathcal{F}_t^Y] \right|^2 \rightarrow 0.$$

The same realization gives convergence uniformly in time in probability for $f \in L^1(\nu)$. It also permits $f_n \rightarrow f$ in $L^2(\nu)$, respectively in $L^1(\nu)$, in the first conditional expectation.

Proof. Glue the conditional maximal couplings of (Z, Y^n) with the fixed reference (Z, Y) by taking their conditional kernels independently on a countable product. The coupling does not depend on f . For the truncation $f^{(K)} = (-K) \vee (f \wedge K)$, Theorem 24.1 bounds the middle difference in the L^2 norm of the supremum by $K\sqrt{84\varepsilon_n}$. Doob's L^2 inequality bounds each truncation tail by $2\|f - f^{(K)}\|_{L^2(\nu)}$. The triangle inequality therefore gives

$$\left\| \sup_t |M_t^{P_n, f} - M_t^{P, f}| \right\|_2 \leq 4\|f - f^{(K)}\|_2 + K\sqrt{84\varepsilon_n}.$$

First let $n \rightarrow \infty$, then $K \rightarrow \infty$. For L^1 use instead the weak maximal inequality for each tail and the bounded-case estimate for the middle term. An additional difference $f_n - f$ is handled by the same maximal inequalities under P_n , whose latent marginal remains ν . $\square$

24.1. A microscopic sufficient condition. Let $(\Phi_t)_{t \in \mathbb{R}}$ be an almost-everywhere regular hard-sphere flow on a labeled phase space, with invariant probability μ . Fix a prepared law ν with $d\nu/d\mu \leq C$, and bounded real observables h, h_n . These may be collision-sensitive tagged observables. Put $\eta_n = \|h_n - h\|_{L^2(\mu)}$. With a common latent initial state $Z \sim \nu$, consider the continuous observation laws

$$\begin{aligned} Y_t &= \int_0^t h(\Phi_s Z) ds + \sigma B_t, \\ Y_t^n &= \int_0^t h_n(\Phi_s Z) ds + \sigma B_t^n, \quad \sigma > 0, \end{aligned}$$

where the Brownian noise is independent of Z in each model. The notation does not prescribe a synchronous coupling.

Theorem 24.3 (Fixed-flow observation-approximation limit). *The joint latent-output laws are equivalent and satisfy*

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \eta_n. \quad (24.3)$$

For every $|f(Z)| \leq 1$ they admit a latent-preserving coupling with

$$\mathbb{E} \sup_{t \leq T} |M_t^{P_n, f} - M_t^{P, f}|^2 \leq \frac{42\sqrt{CT}}{\sigma} \eta_n. \quad (24.4)$$

In particular $\eta_n \rightarrow 0$ supplies whole-process optional-projection convergence, not merely convergence at finitely many times. This applies when h_n are orbit-mollified graph-core approximants to any bounded $h \in L^2(\mu)$.

Proof. Conditional on Z , the signals are deterministic bounded drifts. The exponential change-of-measure identity for Brownian motion gives

$$D_{\text{KL}}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |(h_n - h)(\Phi_s Z)|^2 ds \leq \frac{CT}{2\sigma^2} \eta_n^2.$$

The last inequality uses $\nu \leq C\mu$ and invariance of μ ; it does not require mixing, differentiability across a collision, or a claim that a tagged velocity belongs to the generator domain. Bounded drifts ensure equivalence and justify the exponential identity. Pinsker's inequality gives (24.3). Applying Theorem 24.1 proves (24.4).

For the last assertion choose a nonnegative smooth approximate identity ρ_n on the time line and set $h_n = \int \rho_n(s) h \circ \Phi_s ds$. Strong continuity of the Koopman group in $L^2(\mu)$ gives $\eta_n \rightarrow 0$ and $\|h_n\|_\infty \leq \|h\|_\infty$. Repeated integration by parts puts h_n in every power domain of the generator, with derivatives obtained from $\rho_n^{(k)}$. This is the orbit-core construction already used in the microscopic domain theory; the new conclusion is the uniform posterior estimate. $\square$

Sampling these observation paths at finitely many checkpoints and applying a common causal output postprocessing are contractions in marked variation. Thus the identity report comparison transfers every bounded finite-register risk at its original state and selector budgets with error bounded by (24.3). That is the resource-comparison consequence. The latent-preserving coupling in (24.4) has a different role: it realizes convergence of the changing observation filtrations and is not used as a finite-register simulator.

The theorem discharges a sufficient conditional-law hypothesis for optional projection in a genuine microscopic observation class. The constants depend on the fixed preparation, time

and noise level. No particle-uniform kinetic estimate, changing microscopic path limit, limiting stochastic-exponential identification or nonlinear BSDE limit is implied without its additional hypotheses. The whole-process estimate is what replaces an unsupported appeal to Doob's inequality as a tightness argument.

25. CHANGING TRAJECTORIES, LIKELIHOODS, AND CONDITIONAL CONSUMERS

25.1. A common preparation, not a common flow. Let $(\mathcal{Z}, \nu)$ be a standard Borel probability space, and let $H, H_n : [0, T] \times \mathcal{Z} \rightarrow \mathbb{R}$ be jointly measurable, with $|H|, |H_n| \leq M$. They may be observations of different flows on different phase spaces, pulled back by specified preparation maps from $\mathcal{Z}$. No common microscopic generator or invariant measure is assumed. Set

$$\eta_n^2 = \int_{\mathcal{Z}} \int_0^T |H_n(t, z) - H(t, z)|^2 dt d\nu(z).$$

On $\mathcal{Z} \times C([0, T])$ let R be the law of independent $Z \sim \nu$ and $Y = \sigma B$, where $\sigma > 0$. Define

$$L_t^n = \exp \left\{ \frac{1}{\sigma} \int_0^t H_n(s, Z) dB_s - \frac{1}{2\sigma^2} \int_0^t H_n(s, Z)^2 ds \right\}, \quad (25.1)$$

$$dP_n = L_T^n dR. \quad (25.2)$$

The reference model P, L uses H . Conditional on Z , these are Gaussian shifts, so under P_n the observed path has law $Y_t = \int_0^t H_n(s, Z) ds + \sigma B_t^n$. Boundedness gives true strictly positive density martingales. Write $\mathcal{F}_t$ for the usual canonical observation filtration and $\mathcal{H}_t = \sigma(Z, B_s : s \leq t)$ for the larger reference filtration. Equivalent laws have the same completed observation stopping times.

Theorem 25.1 (Likelihood and posterior transport for changing paths). *The measures P_n, P, R are equivalent and have common Z -marginal ν . With $A = M^2 T / \sigma^2$ and $\varepsilon_n = \min\{1, \eta_n / (2\sigma)\}$, one has*

$$\|P_n - P\|_{\text{TV}} \leq \varepsilon_n, \quad (25.3)$$

$$\mathbb{E}_R \sup_{t \leq T} |L_t^n - L_t|^2 \leq \frac{8}{\sigma^2} e^{3A} \eta_n^2. \quad (25.4)$$

For $|f(Z)| \leq K$ put

$$N_t^{n,f} = \mathbb{E}_R[L_T^n f(Z) \mid \mathcal{F}_t], \quad \ell_t^n = N_t^{n,1}.$$

Then

$$\mathbb{E}_R \sup_t |N_t^{n,f} - N_t^f|^2 \leq \frac{8K^2}{\sigma^2} e^{3A} \eta_n^2. \quad (25.5)$$

In particular this applies to the projected likelihood ℓ^n . For every $|f| \leq 1$, the posterior functions satisfy the same-path bound $80\varepsilon_n$ of Theorem 24.1, and there is one latent-preserving coupling, independent of f , on which

$$\mathbb{E} \sup_t |\mathbb{E}_{P_n}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y]|^2 \leq 84\varepsilon_n. \quad (25.6)$$

If $\eta_n \rightarrow 0$ and deterministic microscopic paths $X_n(Z) \rightarrow X(Z)$ in a specified Polish path metric in ν -probability, then on this coupling the microscopic paths, the full and projected likelihood processes evaluated on their own observation paths, and each bounded posterior process converge jointly in probability in that metric and uniformly in time for the last three coordinates.

Proof. Conditional Gaussian change of measure gives

$$\text{KL}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |H_n - H|^2 dt = \frac{\eta_n^2}{2\sigma^2}.$$

The same calculation proves conditional equivalence, and the conditional density has mean one, so the preparation marginal is unchanged. Integrating and applying Pinsker proves (25.3).

For the quantitative density estimate, condition first on $Z = z$ and write $D_t = L_t^n - L_t$. The stochastic differential equation is

$$dD_t = \sigma^{-1}\{D_t H_n(t, z) + L_t(H_n - H)(t, z)\} dB_t.$$

The conditional second moment of L_t is at most e^A . Itô's isometry and $(u + v)^2 \leq 2u^2 + 2v^2$ therefore imply

$$\mathbb{E}_R[D_t^2 | z] \leq \frac{2M^2}{\sigma^2} \int_0^t \mathbb{E}_R[D_s^2 | z] ds + \frac{2e^A}{\sigma^2} \int_0^t |H_n - H|^2(s, z) ds.$$

Gronwall, followed by integration in z , gives $\mathbb{E}_R D_T^2 \leq (2/\sigma^2) e^{3A} \eta_n^2$. The difference is a square-integrable $\mathcal{H}_t$ -martingale. Doob's L^2 inequality proves (25.4). Applying the same maximal inequality to the $\mathcal{F}_t$ -martingale $\mathbb{E}_R[(L_T^n - L_T)f | \mathcal{F}_t]$ proves (25.5). These are direct process estimates, not a claim that finite-dimensional convergence alone implies tightness.

The posterior assertions follow from Theorem 24.1 and its countable conditional-coupling construction. For the last claim, on the event $Y^n = Y$ the two likelihood functions are evaluated on the same (Z, Y) . Their uniform difference tends to zero in R -probability by the estimates just proved, and therefore in P -probability by absolute continuity of the fixed P with respect to R . The complementary event has probability at most ε_n . The same argument applies to ℓ^n , and (25.6) handles the posterior. The microscopic convergence holds on any coupling retaining the same Z . A finite union bound proves joint convergence. Stochastic integral versions are chosen measurably on the canonical product space; equivalence makes these common versions valid under every law in the countable sequence. $\square$

For $f \in L^2(\nu)$ the posterior conclusion follows by truncation exactly as in Corollary 24.2; it also permits $f_n \rightarrow f$ in $L^2(\nu)$. Boundedness of H_n is being used for the uniform likelihood second moments. It is not replaced by an unverified uniform-integrability assertion in a singular or zero-noise limit.

25.2. The projected stochastic exponential. Let $\hat{H}_t^n = \mathbb{E}_{P_n}[H_n(t, Z) | \mathcal{F}_t]$, using a predictable version for Lebesgue-almost every t . Put $\theta_t^n = \hat{H}_t^n / \sigma$.

Proposition 25.2 (Identification before passage to the limit). *The projected likelihood and the innovation satisfy*

$$\ell_t^n = \exp \left\{ \int_0^t \theta_s^n dB_s - \frac{1}{2} \int_0^t (\theta_s^n)^2 ds \right\}, \quad (25.7)$$

$$I_t^n = B_t - \int_0^t \theta_s^n ds. \quad (25.8)$$

Here I^n is a P_n -Brownian motion in $\mathcal{F}_t$. In particular the limit of the projected likelihoods in (25.5) is the projected stochastic exponential of the limiting model. Change of measure commutes with this limit on bounded measurable path tests.

Proof. Project $L_t^n = 1 + \int_0^t L_s^n H_n(s, Z) dB_s / \sigma$ onto the Brownian observation filtration. The integrands are square integrable; testing against bounded elementary predictable stochastic integrals and using Itô's isometry gives

$$d\ell_t^n = \sigma^{-1} \mathbb{E}_R[L_t^n H_n(t, Z) | \mathcal{F}_t] dB_t = \ell_t^n \theta_t^n dB_t.$$

The last equality is Bayes' formula. Since $|\theta_t^n| \leq M/\sigma$, the solution is the positive stochastic exponential (25.7); Girsanov gives the stated innovation. The same argument with H identifies the limiting exponential. Finally $|\mathbb{E}_{P_n} F - \mathbb{E}_P F| \leq 2\|F\|_\infty \varepsilon_n$ for bounded signed F . There is no interchange based only on a formal limit of logarithms. $\square$

25.3. A bounded entropic backward-equation consumer. Fix $\gamma > 0$ and a latent payoff f with $|f| \leq 1$. Define

$$U_t^n = \frac{1}{\gamma} \log \mathbb{E}_{P_n}[e^{\gamma f(Z)} \mid \mathcal{F}_t] = \frac{1}{\gamma} \log \frac{N_t^{n,e^{\gamma f}}}{\ell_t^n}. \quad (25.9)$$

The terminal condition here is the observable conditional certainty equivalent U_T^n , not the generally unobserved $f(Z)$ itself. This specification avoids imposing a latent terminal value on the smaller observation filtration.

Theorem 25.3 (Entropic values and localized backward integrands). *There is a unique bounded entropic value process with terminal value U_T^n ; for a predictable integrand Z^n it satisfies*

$$U_t^n = U_T^n + \int_t^T \frac{\gamma}{2} |Z_s^n|^2 ds - \int_t^T Z_s^n dI_s^n. \quad (25.10)$$

On the coupling of Theorem 25.1,

$$\mathbb{E} \sup_t |U_t^n(Y^n) - U_t(Y)|^2 \leq 84 \left(\frac{e^{2\gamma} - 1}{\gamma} \right)^2 \varepsilon_n. \quad (25.11)$$

For each $r > e^\gamma$ let τ_r^n be the first exit of any of $\ell^n, \ell, N^{n,e^{\gamma f}}, N^{e^{\gamma f}}$ from $(1/r, r)$, capped at T . If $\eta_n \rightarrow 0$, then

$$\mathbb{E}_R \int_0^{\tau_r^n} |Z_s^n - Z_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^{\tau_r^n} |\theta_s^n - \theta_s|^2 ds \rightarrow 0. \quad (25.12)$$

Furthermore $\lim_{r \rightarrow \infty} \sup_n R(\tau_r^n < T) = 0$. Thus the localizations exhaust the observation experiment in probability.

Proof. For a positive bounded latent g , write $N_t^{n,g} = \mathbb{E}_R[L_T^n g \mid \mathcal{F}_t]$ and $dN_t^{n,g} = a_t^{n,g} dB_t$. Projection of the full integral gives

$$a_t^{n,g} = \sigma^{-1} \mathbb{E}_R[L_t^n g H_n(t, Z) \mid \mathcal{F}_t], \quad |a_t^{n,g}| \leq (M/\sigma) N_t^{n,g}.$$

The same representation with $g = 1$ gives $d\ell_t^n = b_t^n dB_t$ and $b_t^n/\ell_t^n = \theta_t^n$. Put $N^n = N^{n,e^{\gamma f}}$ and $a^n = a^{n,e^{\gamma f}}$. Itô's formula in (25.9) gives the martingale coefficient

$$Z_t^n = \gamma^{-1} (a_t^n / N_t^n - b_t^n / \ell_t^n).$$

Replacing dB_t by $dI_t^n + \theta_t^n dt$ reduces its drift to $-\gamma |Z_t^n|^2/2$, which proves (25.10). The ratios are bounded by M/σ , so the integrals are well defined. Conversely, exponentiating any bounded solution of (25.10) produces a bounded P_n -martingale with terminal value $e^{\gamma U_T^n}$. Its conditional-expectation representation is unique, and martingale representation then identifies its integrand. This proves uniqueness in the bounded class.

Apply Theorem 24.1 to $g_0 = (e^{\gamma f} - e^{-\gamma})/(e^\gamma - e^{-\gamma}) \in [0, 1]$. The map taking its conditional expectation to $\gamma^{-1} \log \mathbb{E}[e^{\gamma f} \mid \mathcal{F}_t]$ has Lipschitz constant at most $(e^{2\gamma} - 1)/\gamma$. This proves (25.11).

The initial values of N^n and N agree, since Z has the same marginal. Itô's isometry and the terminal version of (25.5) imply

$$\mathbb{E}_R \int_0^T |a_s^n - a_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^T |b_s^n - b_s|^2 ds \rightarrow 0.$$

Before τ_r^n , all denominators lie in $[1/r, r]$, and the limiting numerators a, b are bounded in absolute value by $(M/\sigma)r$. For example,

$$\left| \frac{a^n}{N^n} - \frac{a}{N} \right| \leq r |a^n - a| + (M/\sigma) r^3 |N^n - N|.$$

The supremum estimates (25.5), followed by integration in time, prove (25.12). The same inequality applies to b^n/ℓ^n .

Finally $e^{-\gamma\ell^n} \leq N^n \leq e^{\gamma\ell^n}$. Each ℓ^n is a stochastic exponential with uniformly bounded integrand θ^n . Its positive and negative moments of the supremum are bounded uniformly in n : write any power as a stochastic exponential times its bounded compensator and apply Doob's inequality to a higher positive power. The same calculation applies to its inverse. Markov's inequality, the preceding comparison of N^n with ℓ^n , and a union bound prove the exhaustion assertion. This is a localized integrand theorem; it does not assert an unspecified global topology for backward equations with unbounded terminal data or different nonlinear drivers. $\square$

Corollary 25.4 (A changing-filtration stopping consumer). *Let $\psi(t, x) \in [0, 1]$ be measurable and L_ψ -Lipschitz in x , uniformly in t . For a bounded latent f with $|f| \leq 1$, put $M_t^n = \mathbb{E}_{P_n}[f(Z) \mid \mathcal{F}_t]$. Then*

$$\left| \sup_{\tau \leq T} \mathbb{E}_{P_n} \psi(\tau, M_\tau^n) - \sup_{\tau \leq T} \mathbb{E}_P \psi(\tau, M_\tau) \right| \leq \varepsilon_n + L_\psi \sqrt{80\varepsilon_n}.$$

The same statement for the entropic value U has the additional Lipschitz factor $(e^{2\gamma} - 1)/\gamma$ on the square-root term. The suprema range over the common canonical stopping times; attainment is not needed for the estimate.

Proof. For a fixed stopping time, first change the measure while keeping its $[0, 1]$ reward function, at cost at most ε_n . Then compare the two reward functions on the same canonical path under P , using Theorem 24.1, Cauchy-Schwarz and the Lipschitz bound. The estimate is uniform in the stopping time, so it survives both suprema. Equivalence of the observation laws is what allows the same stopping-time class. The entropic assertion follows from the Lipschitz calculation in the preceding theorem. $\square$

25.4. Resource transport is not the posterior coupling. Apply any common finite causal acquisition map to Y and retain a fixed finite mark $w(Z)$. Data processing makes its marked row variations at most ε_n , uniformly over a declared finite feedback family. If the source and target have respective bounds α_n, β_n , Theorem 7.2 gives same-budget deficiency perturbation at most $\alpha_n + \beta_n$, and Proposition 18.4 transports a fixed reset audit. A subsequent consumer of width S composed with a simulator of width K has width KS , with the stated selector price. None of these executable claims uses the latent-preserving maximal coupling. That coupling belongs to the proof of conditional-process convergence, not to the finite-state machine.

These results provide a direct changing-trajectory, likelihood, Girsanov and bounded entropic-value input for a conditional-law theory. They do not assert the strict-dual, rigidity, covariant-form or full kinetic conclusions of a larger program without their separate analytic proofs.

Example 25.5 (Why an amplitude bound is not a tightness argument). Let ξ_1, ξ_2 be independent symmetric signs, and, for $n \geq 5$, let $\mathcal{G}_t^n$ reveal ξ_1 at $1/2 - 2/n$ and ξ_2 at $1/2 - 1/n$. On $[0, 1]$ the bounded martingale

$$M_t^n = \mathbb{E}[\xi_1 + \xi_2 \mid \mathcal{G}_t^n]$$

converges at every finite collection of fixed times to the martingale $M_t = \mathbf{1}_{[1/2, 1]}(t)(\xi_1 + \xi_2)$. Nevertheless its laws are not tight in the Skorokhod J_1 topology. On the event $\xi_2 = -\xi_1$, of probability $1/2$, its path has an excursion of height one on an interval of length $1/n$ and is zero outside that interval. The three-time oscillation modulus of such a path is one at every scale greater than $2/n$. This violates the necessary J_1 tightness modulus criterion. All maximal moments are uniformly bounded by 2^p .

This example concerns the inference from finite-dimensional convergence and a maximal amplitude estimate alone. It does not claim that every possible stronger conditional-kernel hypothesis fails. The uniform estimates in Theorem 25.1 remove the need for that inference in the observation class treated here.

26. OBSERVATION DENSITIES AND COUPLED PROCESS VERSIONS

We record three measure-theoretic facts used in the global transport argument. Their purpose is to make explicit the passage from a tilted physical law to an observation law and from two marginal processes to one joint realization.

Lemma 26.1 (Observation-level return from a bounded tilt). *Let $dP^g/dP = g(Z)/c$, where $c = \mathbb{E}_P g$ and $e^{-\gamma} \leq g \leq e^\gamma$. For any observation sub- σ -field $\mathcal{F}$,*

$$\frac{d(P|_{\mathcal{F}})}{d(P^g|_{\mathcal{F}})} = \mathbb{E}_{P^g} \left[\frac{c}{g} \middle| \mathcal{F} \right] = \frac{c}{\mathbb{E}_P[g | \mathcal{F}]} \quad (26.1)$$

Both this density and its reciprocal lie in $[e^{-2\gamma}, e^{2\gamma}]$. In particular, for any nonnegative $\mathcal{F}$ -measurable V , $\mathbb{E}_P V \leq e^{2\gamma} \mathbb{E}_{P^g} V$.

Proof. For $A \in \mathcal{F}$, integrate c/g over A under P^g to obtain $P(A)$. Conditioning proves the first identity. In the other direction, $d(P^g|_{\mathcal{F}})/d(P|_{\mathcal{F}}) = \mathbb{E}_P[g | \mathcal{F}]/c$; equivalence and inversion prove the second. The bounds follow from the bounds on g , its conditional mean, and c . $\square$

Lemma 26.2 (Domination of the common-path measure). *Let (Y', Y) be a coupling on a standard Borel path space, with marginal laws P', P , and let $E = \{Y' = Y\}$. The subprobability measure $\kappa(A) = \Pr(E, Y \in A)$ is dominated by both P and P' . Consequently, for any nonnegative measurable path functional V ,*

$$\mathbb{E}[\mathbf{1}_E V(Y)] \leq \mathbb{E}_P V, \quad \mathbb{E}[\mathbf{1}_E V(Y')] \leq \mathbb{E}_{P'} V.$$

The assertion also holds on a common latent–observation space when the latent coordinate is shared by the coupling.

Proof. The event defining $\kappa(A)$ is contained in $\{Y \in A\}$ and, on E , in $\{Y' \in A\}$. This proves domination on sets. Integration of simple functions followed by monotone convergence proves the rest. Apply the same argument to the product coordinate for the last assertion. $\square$

Lemma 26.3 (Marginal stochastic integrals on an arbitrary coupling). *On a canonical standard Borel observation path space, suppose that under each of two marginal laws there are a continuous adapted process U , a predictable square-integrable process Z , and an innovation I such that, in that marginal's own filtration,*

$$\int_0^t Z_s dI_s = U_t - U_0 + \frac{\gamma}{2} \int_0^t |Z_s|^2 ds, \quad 0 \leq t \leq T. \quad (26.2)$$

Assume Z is bounded, as in the entropic model here. There are measurable versions of the two continuous integral processes which can be pulled back to any coupling of the observation laws. Each pulled-back version has the correct marginal law and satisfies its own identity (26.2). Neither integral thereby acquires a martingale or Itô-integral interpretation in the enlarged joint filtration.

Proof. A continuous process is determined by its values at rational times. Completed-measurable random variables on the canonical space have Borel versions. Choose such versions at the countably many rational times and extend by continuity on the measurable set where a continuous extension exists, using the zero path elsewhere. This gives a Borel $C([0, T])$ -valued version of U . Likewise take a jointly measurable version of Z modulo $dt \otimes P$; its bounded time integral has a continuous path, and its rational-time values give a measurable continuous version by the same construction. The right side of (26.2) is consequently a measurable continuous path functional. Marginal equality at all rational times, and continuity of the stochastic integral, make it indistinguishable from that integral. Repeat for the other marginal. Pullback along the two coordinate maps of any coupling preserves their respective almost-sure identities and laws. No enlargement theorem for innovations is used. $\square$

In Theorem 27.2, Lemma 26.1 is applied with $\mathcal{F} = \mathcal{F}_T$ to the integrated squared coefficient comparison. Lemma 26.2 justifies restricting a same-path estimate to the equality event of the latent-preserving coupling. Lemma 26.3 fixes the meaning of the coupled backward martingale integrals before their difference is estimated. Thus the coupled notation retains the marginal observation filtrations throughout.

27. GLOBAL TRANSPORT OF THE BACKWARD EQUATION

We retain exactly the common-preparation, bounded-signal and positive-noise hypotheses of Section 25. In particular the canonical joint laws P_n, P are equivalent and have the same latent marginal ν . All comparisons on a single observation path use common measurable versions of the conditional expectations. The following estimate avoids division by a small observation density in the final bound.

Lemma 27.1 (An integrated posterior comparison). *Let P, Q be equivalent laws on a standard Borel product carrying a latent variable Z and an observation Y , and let $\|P - Q\|_{\text{TV}} \leq \epsilon$. If $|u(Z)| \leq M$ and $\mathcal{F}$ is an observation sub- σ -field, then, under either $S = P$ or $S = Q$,*

$$\mathbb{E}_S |\mathbb{E}_Q[u \mid \mathcal{F}] - \mathbb{E}_P[u \mid \mathcal{F}]| \leq 4M\epsilon, \quad (27.1)$$

$$\mathbb{E}_S |\mathbb{E}_Q[u \mid \mathcal{F}] - \mathbb{E}_P[u \mid \mathcal{F}]|^2 \leq 8M^2\epsilon. \quad (27.2)$$

A common latent marginal is not needed for these two inequalities.

Proof. Use a dominating finite measure on $\mathcal{F}$. Write p, q for the densities of the restricted probability laws and a, b for the densities of the signed measures $\mathbb{E}_P[u \mathbf{1}_A], \mathbb{E}_Q[u \mathbf{1}_A]$ on $\mathcal{F}$. Their conditional means are $m_P = a/p$ and $m_Q = b/q$. Outside null sets, $|m_P|, |m_Q| \leq M$, and

$$p(m_Q - m_P) = m_Q(p - q) + (b - a).$$

The integral of $|p - q|$ is at most 2ϵ , by contraction of total variation. The integral of $|b - a|$ is at most $2M\epsilon$, since multiplication by u followed by restriction cannot increase signed variation by more than M . Integrating the displayed identity proves (27.1) under P . Interchanging the two laws proves it under Q . The squared bound follows from $|m_Q - m_P| \leq 2M$. $\square$

Let $|f| \leq 1$, $\gamma > 0$, and use the notation θ^n, U^n, Z^n, I^n of Proposition 25.2 and Theorem 25.3. Here Z^n denotes the scalar backward integrand, whereas the unadorned latent argument of H_n is the preparation variable. Define the deterministic bounds

$$A_n = \sigma^{-2} \{2\eta_n^2 + 16M^2T\epsilon_n\}, \quad (27.3)$$

$$C_n = \frac{2(1 + e^{4\gamma})}{\gamma^2} A_n, \quad (27.4)$$

where $\epsilon_n = \min\{1, \eta_n/(2\sigma)\}$ as before.

Theorem 27.2 (Global innovation and entropic-integrand transport). *Under the hypotheses of Theorem 25.1, under either physical law $S = P$ or $S = P_n$ on the same canonical path,*

$$\mathbb{E}_S \int_0^T |\theta_t^n - \theta_t|^2 dt \leq A_n, \quad \mathbb{E}_S \int_0^T |Z_t^n - Z_t|^2 dt \leq C_n. \quad (27.5)$$

No stopping at a density level is required. If $\eta_n \rightarrow 0$, both integrals also converge to zero in expectation under the reference law R .

On the single latent-preserving coupling of Theorem 25.1, evaluate each process on its own observation path, denoting these evaluations by a bar. Then

$$\mathbb{E} \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt \leq A_n + 4M^2 T \varepsilon_n / \sigma^2, \quad (27.6)$$

$$\mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt \leq C_n + 16M^2 T \varepsilon_n / (\gamma^2 \sigma^2). \quad (27.7)$$

If $\eta_n \rightarrow 0$, then $\bar{I}^n \rightarrow \bar{I}$ uniformly in probability. Writing $\bar{J}_t^n = \int_0^t \bar{Z}_s^n d\bar{I}_s^n$ for the stochastic integral in its own observation filtration, and $\bar{J}$ for its limit, one has

$$\mathbb{E} \sup_{t \leq T} |\bar{J}_t^n - \bar{J}_t|^2 \leq 2 \mathbb{E} \sup_{t \leq T} |\bar{U}_t^n - \bar{U}_t|^2 + \frac{8M^2 T}{\sigma^2} \mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt. \quad (27.8)$$

The first term on the right has the explicit bound in (25.11). Thus the full bounded entropic backward representation, including its martingale integral, is transported globally. The coupling need not make either innovation Brownian in the joined filtration; each integral is defined under its own marginal law.

Proof. For each deterministic t , under P split

$$\sigma(\theta_t^n - \theta_t) = \mathbb{E}_{P_n}[H_n(t, Z) | \mathcal{F}_t] - \mathbb{E}_P[H_n(t, Z) | \mathcal{F}_t] + \mathbb{E}_P[H_n(t, Z) - H(t, Z) | \mathcal{F}_t].$$

Lemma 27.1, conditional Jensen and $(x + y)^2 \leq 2x^2 + 2y^2$ bound its expected square by $16M^2 \varepsilon_n + 2\mathbb{E}_\nu |H_n(t, Z) - H(t, Z)|^2$. Integration gives the first bound in (27.5) under P . Under P_n use the same decomposition with the signal-difference term conditioned under P_n and the law-change term applied to H . The common latent marginal gives the identical bound. Jointly measurable conditional versions, and predictable versions equal for dt -almost every time, give the same integrated statements.

Put $g = e^{\gamma f}$ and $c = \mathbb{E}_\nu g$. Tilt the two full joint laws by $g(Z)/c$, writing P^g, P_n^g . They still have the same latent marginal $\nu_g = g\nu/c$, and their observation signals remain H, H_n . Since $e^{-\gamma} \leq g, c \leq e^\gamma$,

$$\|P_n^g - P^g\|_{\text{TV}} \leq e^{2\gamma} \varepsilon_n, \quad \eta_{n,g}^2 \leq e^{2\gamma} \eta_n^2.$$

Moreover, the reference martingale coefficient $a^{n,g}$ from the proof of Theorem 25.3 satisfies

$$\alpha_t^n := \frac{a_t^{n,g}}{N_t^{n,g}} = \sigma^{-1} \mathbb{E}_{P_n^g}[H_n(t, Z) | \mathcal{F}_t].$$

Apply the preceding innovation argument to these tilted laws. It bounds the squared integral of $\alpha^n - \alpha$ by $e^{2\gamma} A_n$ under either tilted physical law. For an observation-measurable nonnegative random variable, passage back to the corresponding original physical law costs at most $e^{2\gamma}$: the full density c/g is bounded by this constant, and so is its observation projection. Consequently

$$\mathbb{E}_S \int_0^T |\alpha_t^n - \alpha_t|^2 dt \leq e^{4\gamma} A_n.$$

The identity $Z^n = (\alpha^n - \theta^n)/\gamma$ proves the second bound in (27.5).

The pointwise bounds, valid for all models, are

$$|\theta^n|, |\alpha^n| \leq M/\sigma, \quad |Z^n| \leq 2M/(\gamma\sigma). \quad (27.9)$$

The squared time integrals are thus uniformly bounded random variables. Convergence in P -probability, followed by absolute continuity of the fixed R with respect to P , gives convergence in R -probability. Boundedness then gives convergence of their R -expectations. This argument establishes global reference convergence without claiming a new explicit reference rate.

For the coupling, let $E_n = \{Y^n = Y\}$, so that $\mathbb{P}(E_n^c) \leq \varepsilon_n$. On E_n both processes are functions of the same observation path. The restricted expectation of their squared difference is at most the

corresponding unrestricted expectation under the limiting marginal P . On E_n^c apply (27.9) and integrate for time T . This proves (27.6)–(27.7).

On E_n , the Brownian reference coordinates $Y^n/\sigma, Y/\sigma$ agree, so

$$\sup_{t \leq T} |\bar{I}_t^n - \bar{I}_t|^2 \leq T \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt.$$

Markov's inequality and $\mathbb{P}(E_n^c) \rightarrow 0$ prove uniform convergence in probability. This assertion does not require any moment estimate on the observation difference on the exceptional event.

Finally, the backward equation already proved gives, under each marginal,

$$\bar{J}_t^n = \bar{U}_t^n - U_0 + \frac{\gamma}{2} \int_0^t |\bar{Z}_s^n|^2 ds.$$

The common initial value is $U_0 = \gamma^{-1} \log \mathbb{E}_\nu g$. By (27.9),

$$||\bar{Z}_s^n|^2 - |\bar{Z}_s|^2| \leq \frac{4M}{\gamma\sigma} |\bar{Z}_s^n - \bar{Z}_s|.$$

Subtract the two integral identities, use Cauchy–Schwarz in time and $(x + y)^2 \leq 2x^2 + 2y^2$, and take expectations. This gives (27.8). The integral identity also defines compatible measurable versions on the coupling, without asserting a martingale property for its joint filtration. $\square$

The theorem strengthens the localized conclusion of Theorem 25.3 without changing its terminal condition or its driver. The terminal condition remains the observation-measurable conditional certainty equivalent, not the unobserved latent payoff. Positive fixed noise, bounded signals, and a common preparation are used in the proof. A singular zero-noise limit, an unbounded terminal condition, or a different nonlinear driver would require additional arguments.

28. A UNIFORM CERTIFICATE THROUGH A MOVING COLLISION

We now use all three parts of the argument on one family. The finite marked reference law and the uniform scattering geometry are those of Theorem 23.1 and Lemma 23.2. They are proved above, including the no-collision control. We use a continuous observation of the same detector so that finite reports and conditional processes are generated by one experiment.

Let z denote a preparation in the boxes of (23.3), and let ν be that prepared law. The sign of the impact orientation is $2B - 1$, and the conserved center-of-mass longitudinal sign is the actual mark W . Their correlation is prepared: $\mathbb{P}(B = b, W = w) = 5/16$ for $b = w$ and $3/16$ otherwise. The contact diameter d ranges over $I = [9/10, 11/10]$. Let $X_d(t, z)$ be the two-particle phase path, with outgoing velocities used at the collision, and define

$$H_d(t, z) = h(v_{1,y}^d(t, z)), \quad Y_t^d = \int_0^t H_d(s, z) ds + \sigma B_t^d, \quad 0 < \sigma \leq 1/24.$$

The Brownian noise is independent of the preparation. The source experiment has the same physical preparation and mark but constant reports. The target reports at times 2 and 3 are

$$Z_1^d = \mathbf{1}\{2(Y_2^d - Y_{3/2}^d) > 0\}, \quad Z_2^d = A\mathbf{1}\{2(Y_3^d - Y_{5/2}^d) > 0\}. \quad (28.1)$$

The acquisition gate $A \in \{0, 1\}$ is chosen after Z_1^d ; it does not act on the mechanical trajectory. The continuous path is the auxiliary observation experiment for conditional consumers; the finite simulator is not given this path or an uncharged buffer for computing its increments. The target sensor implements the declared window acquisition.

Lemma 28.1 (Quantitative change of the microscopic path). *For all preparations in the specified boxes, the collision time $t_c(d)$ and outgoing tagged velocities obey*

$$|t_c(d') - t_c(d)| \leq |d' - d|, \quad |v_i^+(d') - v_i^+(d)| \leq 30|d' - d| \quad (i = 1, 2).$$

Writing $\Delta = |d' - d|$, one has

$$\sup_z \int_0^3 |H_{d'}(t, z) - H_d(t, z)|^2 dt \leq \Delta + 2700\Delta^2. \quad (28.2)$$

The phase paths converge uniformly over the preparation in the Skorokhod J_1 metric as $d' \rightarrow d$. More precisely, for the product metric taking the maximum of the per-particle Euclidean position and velocity distances, a time change aligning the collision times gives J_1 distance at most 100Δ .

Proof. Write r and g for the incoming relative position and velocity, and $n(d) = (r + t_c(d)g)/d$ for the contact normal. The collision root satisfies $|r + t_c g| = d$. The nongrazing estimate from Lemma 23.2 is $-g \cdot n > 1443/1000$. Differentiating the root with respect to d gives

$$t'_c(d) = \frac{1}{g \cdot n(d)}, \quad |t'_c(d)| < 1.$$

The incoming bound is $|g| < 3$, so

$$|n'(d)| \leq \frac{|t'_c(d)| |g| + 1}{d} < 5.$$

Since $v_1^+ = V + g/2 - (g \cdot n)n$ and $v_2^+ = V - g/2 + (g \cdot n)n$, differentiation bounds either velocity derivative by $2|g| |n'| < 30$.

Before collision the detector is zero, and after collision it is a constant of absolute value at most one. Its Lipschitz constant is one. The two signals therefore differ on the interval between the two collision times by at most one and afterwards by at most 30Δ . The intervening interval has length at most Δ , and the whole observation lasts three units. This proves (28.2).

Choose the increasing piecewise affine time change fixing 0, 3 and sending $t_c(d)$ to $t_c(d')$. Its uniform displacement is at most Δ . Aligned velocities differ by at most 30Δ , with the same outgoing convention at the jump. Each particle's speed is below four and its velocity jump has norm below three. Without time change, the position difference is bounded by $3\Delta + 3(30\Delta) = 93\Delta$, by integrating the velocity difference before and after the two jump times. Aligning time adds at most 4Δ . The asserted bound follows, with a harmless rounding to 100Δ . $\square$

Theorem 28.2 (Microscopic-to-audit and conditional-law composition). *For the single family (28.1), all of the following hold.*

(a) *At the same width-one private register,*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \quad (r \geq 1), \quad \delta_2^h < 13/100,$$

uniformly for $d \in I$ and $0 < \sigma \leq 1/24$. When $d \leq 2/5$ the detector is zero on $[0, 3]$, and all three finite-experiment deficiencies are zero.

(b) *There is one explicit rational reset-audit rule with one all-on training row, eight marked outcomes and $n = 2800$ training runs, whose expected validation score exceeds $2/5$ against every private width-one simulator throughout that same parameter region. In detail, for counts C choose an event $A_C \subseteq \{0, 1\}^3$ maximizing*

$$Q_*(A_C) - C(A_C)/2800, \quad (28.3)$$

with lexicographic tie breaking. Here Q_ is the rational marked reference law of (B, B, W) . The independent validation uses the actual physical target, not Q_* . This finite rule has at most $\binom{2807}{7}$ training-count states at any scheduled epoch; a conservative complete auditor bound is $8(256 + 1)\binom{2807}{7}$. The state cost belongs to the auditor. The certificate is an expected-score inequality, not a confidence claim about one realized score.*

(c) Fix $\sigma > 0$ and let $d_n, d \in I$ with $d_n \rightarrow d$. Set

$$\epsilon_n = \min \left\{ 1, \frac{\sqrt{|d_n - d| + 2700|d_n - d|^2}}{2\sigma} \right\}. \quad (28.4)$$

The joint preparation–continuous-observation laws differ by at most ϵ_n . Every finite resource deficiency above changes by at most ϵ_n , and a fixed audit changes by at most ϵ_n because its source behavior is unchanged. On a latent-preserving coupling, $X_{d_n} \rightarrow X_d$ in J_1 and each bounded posterior converges in supremum L^2 with bound $84\epsilon_n$. The full and projected likelihoods converge in reference supremum L^2 , and jointly in probability with the microscopic paths on that coupling. The projected stochastic exponentials, bounded entropic backward values, localized integrands and stopping consumers have the conclusions of Proposition 25.2, Theorem 25.3, and Corollary 25.4.

(d) On compact parameter sets with $\sigma \geq \sigma_0 > 0$, the finite marked probabilities have uniform rational enclosures. The observable chart, strict same-budget polynomial verification and rational upper tables are therefore effective in the exact sense of Propositions 22.1 and 21.6.

Proof. Every collision occurs before time $3/2$. Conditional on preparation, the two normalized window increments in (28.1) are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. The outgoing detector has sign $2B - 1$ and absolute value greater than $1/4$. Coupling each noisy report to B gives total marked variation less than $1/300$ from the exact finite marked experiment, uniformly for every acquisition gate. The finite calculation of Theorem 23.1 gives

$$d_* = \sqrt{5/2} - 9/8, \quad \delta_*^p = \delta_*^v = d_*, \quad \delta_{*,2}^h = 1/8.$$

Thus private and visible values exceed $d_* - 1/300 > 9/20$, and the hidden value is less than $1/8 + 1/300 < 13/100$. The two-valued hidden witness stores its selector, not an observed mark. In the no-collision regime both window reports before gating are independent fair noise, independent of W . Fresh private coins reproduce them exactly, including feedback gates. This proves (a).

For (b), use only the all-on feedback row. Any private width-one simulator in this row emits independent bits with probabilities p, q , independently of the actual W . The calculation in Theorem 23.1 already gives minimum variation d_* for this row. Its alphabet (z_1, z_2, W) has eight elements, hence $D = 7$. The explicit empirical audit in Theorem 18.1, trained using Q_* , has expected ideal score at least

$$d_* - \sqrt{7/2800} = d_* - 1/20.$$

The candidate training and validation laws do not change when the target is replaced by the physical target. Conditional on any selected event, its target probability changes by less than $1/300$. Therefore the actual score is strictly greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (28.5)$$

For completeness, the last strict inequality is the positive-root comparison $\sqrt{5/2} > 947/600$; its squared margin is $5/2 - (947/600)^2 = 3191/360000 > 0$. All probabilities in Q_* are 0, $3/16$ or $5/16$, so (28.3) is a finite rational rule independent of d, σ, p, q . At training epoch k the number of eight-category count vectors is $\binom{k+7}{7} \leq \binom{2807}{7}$. A buffer of eight states records the current two reports and terminal mark; 256 event masks and an idle symbol suffice during event selection and validation. The conservative product bound follows. The clock and the source-reset interface are the declared resources. In particular the proof does not enumerate or claim to store a small complete coefficient table.

For (c), Lemma 28.1 gives the signal-distance hypothesis of Theorem 25.1, with $T = 3$ and $M = 1$. It also supplies its microscopic J_1 hypothesis. That theorem gives (28.4) and all the joint process assertions. The finite acquisition is a common measurable causal map of the continuous

observation, so marked variation contracts for every gate. The source is blind and independent of d . Consequently its executable candidate behaviors are identical across d , and Proposition 18.4 has $\alpha = 0$ and $\beta = \epsilon_n$. The same same-budget comparison applies in each visible fiber and in each hidden mixture. The exponential and backward claims now follow from the identified projected density and the explicitly bounded entropic consumer. This is where the finite certificate and the conditional-law theorem meet the same microscopic approximation.

Finally, the collision root is uniformly separated from grazing, its formula is algebraic on the rational preparation boxes, and the normalized window noise has variance $2\sigma^2$ bounded away from zero on the stated compact sets. The box subdivision and Gaussian-tail enclosure argument in Proposition 23.4 therefore applies with that variance. It returns rational enclosures with a computable uniform error. The rational quotient algorithm and same-budget strict certificate procedure then prove (d). Uniformity as $\sigma \downarrow 0$ is not asserted. $\square$

The all-on audit in part (b) is a single witness for the lower bound in the full feedback problem: a maximum over feedback trees includes this row. The visible lower bound is instead the exact fiber theorem; no claim is made that an auditor with a seed-independent state bound can store an arbitrarily large visible-seed transcript. This separates the two resource budgets rather than silently moving cost from one to the other.

The finite certificate is also preserved under an additional rational presentation error ρ : its lower score decreases by at most ρ when only the target is changed, and by the reset-cost estimate when the source is changed. Thus the certificate is an effective consumer of the physical presentation theorem, not an isolated exact finite example.

Theorem 28.2 and the preceding results prove Theorem C.3. The C2-type consequence is a joint changing-trajectory/likelihood/posterior theorem with an identified stochastic exponential and a fully specified bounded entropic backward consumer. Nothing in that conclusion identifies geometric similarity with kinetic scaling, or supplies particle-uniform recollision estimates by renaming this fixed-particle model.

29. SHARP DECISION MEMORY FOR THE COLLISION EXPERIMENT

We now apply the continuation spectrum to the same marked experiment that supplies the microscopic certificate. Both its impossibility statement and its construction concern the same score, the same private candidate family, and the same training-validation checkpoint.

29.1. The marked family and its first two spectral values. In the all-on row of Section 28, a private width-one candidate has two independent report bits with probabilities $p, q \in [0, 1]$, independently of the fair actual mark. Denote its law by $b(p, q)$ and put

$$a = (1 - p)(1 - q), \quad c = pq.$$

Let $B = \{b(p, q) : p, q \in [0, 1]\}$. The ideal target Q_* is supported on 000, 001, 110, 111, with masses $5/16, 3/16, 3/16, 5/16$, respectively. On these four atoms the candidate masses are $a/2, a/2, c/2, c/2$. The elementary continuation constraint implies

$$\sqrt{a} + \sqrt{c} \leq 1, \quad \min(a, c) \leq \frac{1}{4}. \quad (29.1)$$

Indeed the first inequality is Cauchy-Schwarz applied to $(\sqrt{1-p}, \sqrt{p})$ and $(\sqrt{1-q}, \sqrt{q})$; the second follows from the first. If $a \geq c$, then $c \leq 1/4$ and

$$a + 3c \leq (1 - \sqrt{c})^2 + 3c = 1 - 2\sqrt{c} + 4c \leq 1. \quad (29.2)$$

The symmetric inequality holds when $c \geq a$.

Theorem 29.1 (A sharp two-state obstruction). *For the ideal marked collision family,*

$$\mathfrak{V}_1(B, Q_*) = \frac{1}{8}, \quad \mathfrak{V}_2(B, Q_*) = \frac{3}{8}. \quad (29.3)$$

If $\|Q - Q_\|_{\text{TV}} \leq \beta$, every reset auditor retaining at most two states at the training-validation cut has worst-case expected score at most $3/8 + \beta$, regardless of its finite training length and its upstream or downstream memory. The assertion allows mark-dependent validation events and stochastic state transitions.*

Proof. For one state, use the event $\{000, 111\}$. Its score is $5/8 - (a + c)/2 \geq 1/8$. Conversely, average the two candidates with $(a, c) = (1, 0)$ and $(0, 1)$. Their average is uniform on the four supported atoms and is at total variation distance $1/8$ from Q_* . Every fixed response has average score at most $1/8$, so its minimum over the two candidates is at most $1/8$. This proves the first equality; the average is a bound on scores, not an allowed private candidate.

For the upper bound with two states, consider the three candidates $b_0 = b(0, 0)$, $b_1 = b(1, 1)$ and $b_* = b(1/2, 1/2)$. A response may be set to zero outside the support of Q_* without decreasing its payoff at any candidate. Write its four remaining coordinates as (x, y, z, t) . Its three payoffs are

$$\begin{aligned} f_0 &= (-3x - 5y + 3z + 5t)/16, \\ f_1 &= (5x + 3y - 5z - 3t)/16, \\ f_* &= (3x + y + z + 3t)/16. \end{aligned}$$

Since all four coordinates are in $[0, 1]$,

$$f_0 + f_1 \leq \frac{1}{4}, \quad f_0 + f_* \leq \frac{3}{4}, \quad f_1 + f_* \leq \frac{3}{4}. \quad (29.4)$$

Suppose two responses covered all three candidates with payoff strictly greater than $3/8$. The first inequality forces different responses to cover b_0 and b_1 . The other two inequalities then force both responses to have payoff less than $3/8$ at b_* . This is a contradiction. Hence $\mathfrak{V}_2 \leq 3/8$.

For attainment take the responses, in the same four-atom order,

$$h_0 = (2/3, 0, 1, 1), \quad h_1 = (1, 1, 0, 2/3),$$

with zero response elsewhere. Both have target mean $17/24$. Their payoffs are $17/24 - a/3 - c$ and $17/24 - a - c/3$. If $a \geq c$, the first is larger and (29.2) makes it at least $17/24 - 1/3 = 3/8$. The other case is symmetric. This proves equality. For the perturbed target, apply (9.4) and the operational upper bound of Theorem 9.2. $\square$

The obstruction has an immediate continuation interpretation. Each post-training state supplies one validation response. The three witness candidates in the proof cannot be covered above $3/8$ by two such responses. No increase of sample size can compensate for losing the third continuation at that cut. This argument does not use the unrelated Bernoulli majority calibration.

Lemma 29.2 (Feedback completion of the physical lower bound). *The upper bound $3/8 + \beta$ in Theorem 29.1 remains valid when the auditor may train on any of the original feedback rows and may select a row-event pair after the decision cut. Here every target row is the declared feedback pushforward of Q , with $\|Q - Q_*\|_{\text{TV}} \leq \beta$. In particular, two states cannot evade the bound by using a different sensor gate during training or validation.*

Proof. It suffices to restrict the adversary to an admissible subfamily. For $p, q \in [0, 1]$, let its first emission be $X_1 \sim \text{Ber}(p)$, and, after gate A , let its second emission be AX_2 , with an independent $X_2 \sim \text{Ber}(q)$ and the independent fair actual mark. These are genuine width-one private rows: the gate is a current input, and the second emission uses no retained copy of X_1 .

For any deterministic gate policy $g : \{0, 1\} \rightarrow \{0, 1\}$, the marked row is the pushforward of $b(p, q)$ under $T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$. The physical target row is the pushforward of Q by the same map, since acquisition does not act on the path. Including the gate in the transcript

changes nothing: it is also a deterministic function of the displayed coordinates. For a selected row event, the response pulls back to $\mathbf{1}_A \circ T_g$ on the all-on alphabet. A randomized row-event continuation pulls back to a function $h_m \in [0, 1]^\Omega$. Hence, conditional on cut state m , its expected score on this subfamily is exactly $(Q - b(p, q))h_m$.

Whatever training rows were used, their effect is solely the distribution of the retained state. The two-state score is thus a convex combination of two such responses and is bounded above by their envelope. Apply the three-candidate argument of Theorem 29.1, and then the uniform target perturbation bound. Since this is a subfamily of the full private candidate class, it is a valid lower-bound witness for that full class. Randomized feedback is covered by conditioning on its independent random table and then averaging the pulled-back response. $\square$

29.2. A three-response construction. Define three deterministic events, all ignoring the mark,

$$A_D = \{00, 11\} \times \{0, 1\}, \quad A_0 = \{00\} \times \{0, 1\}, \quad A_1 = \{11\} \times \{0, 1\}.$$

Their ideal scores are $1 - a - c$, $1/2 - a$ and $1/2 - c$.

Lemma 29.3 (The three-response envelope). *For the original private candidate image,*

$$\min_{p, q} \max\{1 - a - c, \tfrac{1}{2} - a, \tfrac{1}{2} - c\} = \sqrt{2} - 1. \quad (29.5)$$

In particular $\mathfrak{V}_3(B, Q_) \geq \sqrt{2} - 1$.*

Proof. By symmetry suppose $a \geq c$. Then $c \leq 1/4$ and the maximum is $1 - a - c$ for $a \leq 1/2$, and $1/2 - c$ for $a \geq 1/2$. If $a \leq 1/4$, the first expression is at least $1/2$. If $1/4 \leq a \leq 1/2$, (29.1) gives

$$a + c \leq 1 - 2\sqrt{a} + 2a \leq 2 - \sqrt{2},$$

since the last function is increasing on $[1/4, 1/2]$. If $a \geq 1/2$, then $c \leq (1 - \sqrt{a})^2 \leq (1 - 1/\sqrt{2})^2 = 3/2 - \sqrt{2}$. Both cases give the asserted lower bound. It is attained at $p = q = 1 - 1/\sqrt{2}$, where $a = 1/2$ and $c = 3/2 - \sqrt{2}$. The symmetric point gives the same value. $\square$

Lemma 29.4 (Odd-sample majority selection with a uniform regret bound). *Let $n = 2h - 1$ be odd. In n independent candidate trials, let C_0, C_1 count report pairs 00, 11. Choose A_1 if $C_0 \geq h$, choose A_0 if $C_1 \geq h$, and otherwise choose A_D . The first two cases are mutually exclusive. Its expected ideal score satisfies*

$$g_n(p, q) \geq \sqrt{2} - 1 - \frac{1}{4\sqrt{en}} - \frac{1}{4}e^{-n/8}. \quad (29.6)$$

Its event label has three possible values at the decision cut.

Proof. Because $a + c \leq 1$,

$$\max\{1 - a - c, \tfrac{1}{2} - a, \tfrac{1}{2} - c\} = 1 - a - c + (a - 1/2)_+ + (c - 1/2)_+.$$

The selected payoff is $1 - a - c + (a - 1/2)\mathbf{1}_{\{C_0 \geq h\}} + (c - 1/2)\mathbf{1}_{\{C_1 \geq h\}}$. Its expected loss from this envelope is exactly the sum, for $u = a, c$, of $|u - 1/2|$ times the probability that the binomial majority selects the wrong side of $u = 1/2$. At $u = 1/2$ the contribution is zero. No independence between C_0 and C_1 is used.

For $0 < u < 1/2$ and $k \geq h$, the ratio of the binomial mass at u to that at $1/2$ is

$$[4u(1 - u)]^{n/2} \left(\frac{u}{1 - u} \right)^{k - n/2} \leq [4u(1 - u)]^{n/2}.$$

The fair binomial upper tail is exactly $1/2$ because n is odd. Reflecting k treats $u > 1/2$. The endpoint cases follow directly or by continuity. Thus the wrong-side probability is at most

$$\tfrac{1}{2}[1 - 4(u - 1/2)^2]^{n/2} \leq \tfrac{1}{2}e^{-2n(u - 1/2)^2}.$$

For $x \geq 0$, the maximum of $xe^{-2nx^2}/2$ is $1/(4\sqrt{en})$, reached at $x = 1/(2\sqrt{n})$. At least one of a, c is at most $1/4$ by (29.1); that coordinate has $|u - 1/2| \in [1/4, 1/2]$ and contributes at most $e^{-n/8}/4$. Bound the other coordinate by the unrestricted maximum and use Lemma 29.3. $\square$

29.3. Simultaneous exact widths and the full acquisition schedule. The preceding construction admits more compression than storing its two raw counts. Write 0, 1, * for the three report-pair categories 00, 11 and the two off-diagonal pairs. For $n = 2h - 1$, let F_n be the terminal event-label function of this word. At a trial cut t , let $r = n - t$ and let (u, v) be the two prefix counts. Associate the residual key

$$\mathcal{R}_t(u, v) = \begin{cases} \text{L}, & u \geq h, \\ \text{R}, & v \geq h, \\ (\min(h - u, r + 1), \min(h - v, r + 1)), & u, v < h. \end{cases} \quad (29.7)$$

The symbols L, R mean the constant decisions A_1, A_0 . In a pair key, a coordinate $r + 1$ means that this majority is no longer reachable. The pair $(r + 1, r + 1)$ represents the constant decision A_D .

Theorem 29.5 (Exact profile of the physical audit rule). *The keys in (29.7) are exactly the distinct continuation residuals of F_n . Their number at trial cut t is*

$$r_t(F_n) = \begin{cases} \binom{t+2}{2}, & 0 \leq t \leq h - 1, \\ \binom{n-t+3}{2}, & h \leq t \leq n. \end{cases} \quad (29.8)$$

This is the minimum simultaneous width profile for exact computation of this rule, for deterministic and stochastic machines alike. Its maximum is $\binom{h+2}{2}$ at $t = h$.

For $n = 399$, the maximum trial-cut width is 20301, and the terminal training cut has exactly three states. Acquiring each report pair one bit at a time, discarding its training mark, and then executing independent candidate and target validations gives a deterministic machine with at most $40602 < 2^{16}$ states at every scheduled epoch. One additional ignored training trial may be inserted to give the same 400 training slots and 402 total trials as the earlier certificate. It changes neither the payoff nor these width bounds.

Proof. If a majority has already been reached, its label cannot change. Otherwise, the first coordinate of the pair key is exactly the number of additional 0 symbols needed, with the unreachable threshold clipped to $r + 1$; the second coordinate has the analogous meaning. These data determine the suffix function. Two different reachable pair keys can be distinguished by a suffix containing only one of the two symbols and enough * symbols to fill the remaining positions: use a count between the two different thresholds. The other majority cannot be reached by this suffix. The constant-majority keys are distinct from every pair key because an all-* suffix returns A_D for the latter. Thus the keys are exactly the residuals.

For $t < h$, no majority has been reached and every threshold is at most $r + 1$. The distinct pairs (u, v) with $u + v \leq t$ give $\binom{t+2}{2}$ keys. For $t \geq h$, the two constant-majority keys are reachable. The remaining keys are precisely

$$(i, j) \in \{1, \dots, r + 1\}^2, \quad i + j \geq r + 1.$$

To see surjectivity, use $u = h - i, v = h - j$; these counts are nonnegative, less than h , and sum to at most t because $2h - t = r + 1$. To see necessity before clipping, use the same inequality on $h - u, h - v$; if either coordinate is clipped, the inequality remains automatic. There are $(r + 1)^2 - (r - 1)r/2$ such pairs, with the expression valid also for $r = 0, 1$. Adding the two constant keys gives $(r + 2)(r + 3)/2$. This proves (29.8). The finite-valued realization theorem, Theorem 16.1, proves both the simultaneous realization and stochastic minimality for the specified rule. The two polynomial pieces of the profile give its maximum as stated.

For an explicit transition definition, order the reachable keys lexicographically, putting L, R first. For each key choose the lexicographically first prefix-count pair representing it. After the next category, increment the appropriate count and apply $\mathcal{R}_{t+1}$. Equality of residual functions implies equality of the shifted residuals, so this definition is independent of the chosen representative. The representative is part of the time-indexed transition table, not another stored register. The clock supplies t but no sample history. All these maps are deterministic and rational.

At a bit-level epoch, the state consists of a residual key and, only between the two report bits, the first bit. It therefore has at most $2 \max_t r_t$ values. On reading the second bit, determine the category, update the key and discard both bits. The mark is read and discarded; all-on controls and resets are prescribed by the free schedule. After training, keep just one of the three event labels. During validation, keep that label, the completed candidate-event indicator, and a first-bit buffer when needed. At most $3 \cdot 2 \cdot 2 = 12$ states suffice there; a completed target pair emits its event indicator minus the stored candidate indicator. The final score has three possible values. These phase-specific alphabets may be embedded in a single alphabet of size 40602 because the schedule clock identifies the phase.

The same trial-cut residuals remain distinct for the complete signed-score function, not merely for an externally emitted event name. Any two of A_0, A_1, A_D have different indicator functions. Take an off-diagonal candidate-validation pair, which belongs to none, and a target-validation pair distinguishing the two events. The resulting scores differ. Thus passing to the validation suffix does not merge the residuals already counted. $\square$

Theorem 29.6 (Matched physical decision complexity and stable certification). *For the two-sphere collision family of Theorem 28.2, let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/24$, and let $Q_{d,\sigma}$ be its all-on marked target. At the charged training-validation cut of the reset interface,*

$$\kappa_{2/5}(B, Q_{d,\sigma}) = 3. \quad (29.9)$$

More concretely, no auditor retaining at most two states at this cut can have expected score greater than $2/5$ uniformly against all private width-one candidates, at any finite training length, even with feedback-row selection as in Lemma 29.2. One rational auditor with three states at this cut does so with 399 informative training trials and two validation trials. With a redundant ignored training trial, it uses exactly 402 total trials, at most 40602 states at every scheduled epoch, and the exact trial-cut profile (29.8) on its informative training segment.

For fixed positive σ and $d_k \rightarrow d$, the same machine changes its expected score by at most

$$\epsilon_k = \min \left\{ 1, \frac{\sqrt{|d_k - d| + 2700|d_k - d|^2}}{2\sigma} \right\}. \quad (29.10)$$

The private, hidden and visible deficiencies, the no-collision control, and the joint microscopic, likelihood, posterior, innovation and global bounded entropic backward convergence retain all conclusions of Theorems 28.2 and 27.2.

Proof. Theorem 28.2 discharges the physical hypotheses and proves $\|Q_{d,\sigma} - Q_*\|_{\text{TV}} < 1/300$, uniformly on the stated parameters. Theorem 29.1 therefore gives the universal cut-width-two upper bound

$$\mathfrak{V}_2(B, Q_{d,\sigma}) < \frac{3}{8} + \frac{1}{300} < \frac{2}{5}.$$

Lemma 29.2 makes this a bound for the full original feedback-row interface, not only all-on training and not only the constructed rule. The upper construction uses all-on trials and works for every private candidate because only its two all-on parameters enter.

Use the deterministic selector of Lemma 29.4 with $n = 399$, but validate against the actual physical target. For each selected event, changing only the target from Q_* loses less than $1/300$.

Its physical expected score is consequently greater than

$$L_{399} := \sqrt{2} - 1 - \frac{1}{4\sqrt{399e}} - \frac{1}{4}e^{-399/8} - \frac{1}{300} > \frac{2}{5}. \quad (29.11)$$

Here is a rational verification of the last strict inequality, independent of decimal evaluation. The bounds $e > 2$, $\sqrt{798} > 28$, $\sqrt{2} > 707/500$, and $399/8 > 49$ imply

$$L_{399} > \frac{207}{500} - \frac{1}{112} - \frac{1}{300} - 2^{-51} > \frac{2}{5},$$

since $207/500 - 1/112 - 1/300 - 2/5 = 73/42000 > 2^{-51}$. The number L_{399} is approximately 0.4032891; the rational inequalities, not this approximation, establish the certificate. Its state realization was proved in Theorem 29.5. The general operational classification now yields (29.9) with both inequalities proved for this same physical task.

Changing d leaves the blind candidate source and all its training laws unchanged. The moving-collision estimate and the common-preparation Gaussian theorem bound the target variation by ϵ_k . For any fixed response its target mean changes by at most this amount; averaging over the unchanged training rule proves (29.10). There is no multiplier equal to the number of training trials. The same physical estimates verify the hypotheses of Theorem 27.2; the observation-level density and coupling versions used there are given in Section 26. This transports the same certificate and the bounded conditional-law consumer along the changing microscopic paths. The no-collision regime $d \leq 2/5$ has the vanishing deficiencies already proved in Theorem 28.2; it is not included in the positive-gap parameter interval. $\square$

29.4. What the two exact complexity statements measure. Equation (29.9) minimizes over every auditor at one specified physical decision cut. Equation (29.8) minimizes all trial-cut widths simultaneously for exact implementation of the stated 402-trial rule. These are different optimizations. The first is a task-level optimum, not an artifact of that rule. The second explains the whole online realization of the same rule, not a claim that every successful auditor must have 20301 states somewhere. The sixteen-bit bound includes bit acquisition and validation but does not bound the description length of time-dependent transition tables. It is an implementation upper bound for peak memory. Independent persistent randomization is charged; fresh randomness from the reset experiments is not a hidden cross-cut memory channel.

The former two-count 400-sample construction and its nineteen-bit bound remain in Section 34. They offer a different approximation to the full marked discrepancy. The new three-response rule does not reproduce that rule or attain its oracle envelope. It instead solves the specified physical certification task with an exactly minimal decision-cut alphabet and an explicit full continuation profile. The score is an expectation, not a confidence guarantee from a single realized validation pair.

30. FULL-PROFILE CERTIFICATION OF THE COLLISION FAMILY

The short audit of Theorem 29.6 is retained with its exact continuation profile. We now give a second audit of the same physical class whose entire acquisition-and-validation register is bounded independently of its accuracy parameters. Its cost is a much longer, explicitly counted, preparation schedule.

30.1. The row-indexed decision invariant. For the original row-event interface define

$$\kappa_c^{\text{rows}}(B, Q) = \min\{K : \mathfrak{R}_K^{\text{rows}}(B, Q) > c\}. \quad (30.1)$$

The minimum is $+\infty$ when the set is empty. This is not an implicit change of the one-row symbol κ_c . By Corollary 9.4, it is the exact minimum decision-cut alphabet for a strict score, allowing a finite training schedule and a post-cut distribution on row-event pairs.

Let W_c^{phys} be the minimum peak width among all finite auditors for the original physical row interface, now charging each report-bit, mark, decision and validation epoch. Its clock is externally scheduled, as in the earlier physical implementation. All persistent random selectors are charged.

Theorem 30.1 (A twelve-state physical audit). *In the physical parameter range of Theorem 29.6, put $\beta = 1/300$. For $0 < \theta < 1/4$ and $0 < \delta < 1$, let L, R be at least the respective quantities in (12.1). There is a deterministic audit with $N = 4LR$ candidate training trials and two validation trials whose expected score is strictly greater than*

$$\sqrt{2} - 1 - \beta - \theta - 2\delta. \quad (30.2)$$

It has at most five states at completed training-trial cuts, ten at intermediate training report-bit cuts, three at the decision cut, and twelve throughout validation. In particular

$$\kappa_{2/5}^{\text{rows}} = 3, \quad 3 \leq W_{2/5}^{\text{phys}} \leq 12. \quad (30.3)$$

One entirely integer schedule takes $\theta = 1/200$, $\delta = 1/1000$, $L = 400$ and $R = 8 \cdot 2^{400}$, so $N = 12800 \cdot 2^{400}$. Its lower bound exceeds $2/5$. The upper bound twelve is not asserted to be the exact minimum integer peak width.

Proof. Use the report-pair probabilities $a = (1-p)(1-q)$ and $c = pq$ from Section 29. In the first $2LR$ training trials run Lemma 12.1 on the indicator of the pair 00. If it returns positive, choose A_1 and ignore the remaining training trials. Otherwise, in the next $2LR$ trials run the same comparison on the indicator of 11, choosing A_0 if positive and A_D if negative. Both phases have fixed lengths, even after an absorbing decision.

A comparison whose parameter is outside its indifference band is correct with probability at least $1 - \delta$. For purposes of analysis, one may define the second comparison on its independent observations even when the implementation ignores them. A union bound gives probability at least $1 - 2\delta$ that neither comparison errs outside its band. Since $\min(a, c) \leq 1/4$ and $\theta < 1/4$, at most one of a, c lies in $[1/2 - \theta, 1/2 + \theta]$.

On this good event the chosen payoff is within θ of

$$E(a, c) = \max\{1 - a - c, 1/2 - a, 1/2 - c\}.$$

Indeed, if one coordinate exceeds $1/2 + \theta$, its associated event is selected and is optimal. If both are below $1/2 - \theta$, A_D is selected and is optimal. In the remaining case the low coordinate's comparison is negative, and choosing either of the two relevant events costs at most the distance of the near coordinate from $1/2$. The range of the three payoffs at any fixed (a, c) is at most one. The expected ideal score is therefore at least $E(a, c) - \theta - 2\delta$. Lemma 29.3 gives $E(a, c) \geq \sqrt{2} - 1$. Changing the target to the physical target loses strictly less than β , proving (30.2).

The first comparator has four states. The second phase has these four states and one extra absorbing label for a previously chosen A_1 , hence five. Retaining the first report bit doubles that count to ten. On the second bit, determine the relevant indicator, update the block state, and discard both bits. The training mark is read and discarded. At the phase boundary the fixed schedule maps the first decision to either the A_1 label or the initial state of the second comparator. At the end it maps the remaining decisions to A_0 or A_D . During validation keep the chosen event, the candidate-event indicator, and a first-bit buffer when needed. This requires at most $3 \cdot 2 \cdot 2 = 12$ states; all events ignore the mark. These are actual phase-compatible transitions, not a bound only on terminal labels.

For the integer schedule, the error estimate in the proof of Lemma 12.1 is $2e^{-8} < 1/1000$. To check it rationally, $e > 8/3$ and $(8/3)^8 > 2000$. Further $\sqrt{2} > 707/500$, and hence

$$\sqrt{2} - 1 - \frac{1}{300} - \frac{1}{200} - \frac{1}{500} - \frac{2}{5} > \frac{11}{3000} > 0.$$

This proves the strict score claim without numerical optimization. The training count is exactly $4LR$; early absorption does not delete trials from that count.

For the lower bounds in (30.3), every full-profile auditor has its decision cut among its charged cuts. The feedback pushforward subfamily of Lemma 29.2 bounds any width-two decision continuation by $3/8 + \beta < 2/5$, for every finite adaptive training schedule. The all-on construction just given is a member of the original row interface and has three decision labels. This proves the row-indexed equality and both peak inequalities. $\square$

Corollary 30.2 (Two physical resource regimes). *The same physical score threshold $2/5$ admits the two schedules*

<i>training trials</i>	<i>decision width</i>	<i>peak width</i>
399	3	40602
$12800 \cdot 2^{400}$	3	12

followed by two validations. The first schedule may be padded to 400 training trials as before. For either fixed machine, changing d to d_k at fixed positive noise level changes its expected score by at most ϵ_k in (29.10). If the candidate training kernels also change, their additional cost is the training term in Corollary 12.3.

Proof. The first line is Theorem 29.6; the second is Theorem 30.1. In the moving-collision comparison the candidate training source is identical. Conditional on its chosen event, only one target validation mean changes. The common-preparation Gaussian bound therefore contributes ϵ_k , independently of the length of the unchanged training. The general perturbation formula applies when this equality of sources is absent. $\square$

The long schedule is an existence bound on the resource frontier, not a practical substitute for the 399-trial rule. In an autonomous model the clock must also be stored; Proposition 11.3 gives its explicit additional factor. Both schedules use deterministic transitions, so no precision-dependent randomizer is hidden in the physical upper bound. The one-collision construction remains the verified physical realization. No many-particle or zero-noise theorem is inferred from the larger class of finite experiments.

31. A THIRD RESOURCE REGIME FOR THE SAME PHYSICAL TASK

The three-response obstruction, the short exact selector, and the constant-width block audit concern the same positive-noise marked collision family. A truncated absorbing comparison supplies a further resource pair without changing that family or its threshold.

Theorem 31.1 (Polynomial-horizon physical realization). *Use the original row-event interface and the parameter range of Theorem 30.1. Let $\beta = 1/300$, $0 < \theta < 1/4$, and $m, n \geq 1$. There is a deterministic audit with $2n$ candidate training trials and two validation trials whose expected signed score is strictly greater than*

$$\sqrt{2} - 1 - \beta - \theta - 2e_{m,n}(\theta). \quad (31.1)$$

It uses at most $2m + 2$ states at completed training-trial cuts, $4m + 4$ at intermediate report-bit cuts, three at the decision cut, and twelve throughout validation. In particular, $\theta = 1/200$, $m = 400$ and $n = 160000$ give score greater than $2/5$, with 320000 training trials and peak width at most 1604.

Proof. Write $a = (1-p)(1-q)$ and $c = pq$ for the actual blind product candidate report probabilities, as in Theorem 30.1. On the first n reports apply Lemma 13.1 to the indicator of 00. If the answer is positive retain A_1 and ignore the second training phase. Otherwise use the next n reports to compare the indicator of 11, retaining A_0 for a positive answer and A_D for a negative answer. Conceptually generate the second phase even when it is ignored. Independence of the

reset preparations gives an error probability at most $2e_{m,n}(\theta)$ for either comparison to be wrong outside its indifference band.

On the complementary event the selected ideal payoff is within θ of $\max(1-a-c, 1/2-a, 1/2-c)$. Indeed $\min(a, c) \leq 1/4$, so at most one coordinate is within θ of $1/2$. If neither is, the two comparisons select the maximizing event; if one is, switching the relevant events changes payoff by at most θ . The three payoffs have range at most one at each (a, c) . Their envelope is at least $\sqrt{2} - 1$ by Lemma 29.3. Replacing the ideal target by the physical target loses strictly less than β and does not change the candidate training law. This proves (31.1).

The walk has $2m + 1$ labels. The second phase adds the previously latched A_1 label, giving $2m + 2$. Retaining the first report bit doubles this number; after the second bit, determine the relevant indicator and discard the pair. The mark is read and discarded. A deterministic clock supplies the phase and the within-phase time. At its end map the remaining walk states to A_0 or A_D . The validation implementation from Theorem 30.1 uses at most twelve states, including the event, first bit, and candidate-event value. No real-valued randomization is needed by this construction.

For the stated integers, $4\theta m = 2n\theta^2 = 8$, so $e_{m,n}(\theta) = 2e^{-8} < 1/1000$. The rational estimates $e > 8/3$, $(8/3)^8 > 2000$, and $\sqrt{2} > 707/500$ give

$$\sqrt{2} - 1 - \frac{1}{300} - \frac{1}{200} - \frac{1}{500} - \frac{2}{5} > \frac{11}{3000} > 0.$$

This proves the strict physical margin, with the stated fixed training count and peak width. $\square$

Corollary 31.2 (Three compatible resource bounds). *The original physical task admits all three rows below, each followed by two validations and having decision width three:*

<i>training trials</i>	<i>peak width</i>	<i>training mechanism</i>
399	40602	<i>exact empirical selector</i>
320000	1604	<i>truncated absorbing comparisons</i>
$12800 \cdot 2^{400}$	12	<i>four-state block comparisons.</i>

The all-auditor row-indexed decision optimum remains $\kappa_{2/5}^{\text{rows}} = 3$ and the unrestricted finite-horizon physical peak still satisfies $3 \leq W_{2/5}^{\text{phys}} \leq 12$. The table gives attainable points, not an optimal physical fixed-sample frontier. In particular the 40602 bound is not an all-auditor lower bound for 399 samples.

Proof. Combine Theorems 29.6, 30.1 and 31.1. The feedback-uniform two-state obstruction in Theorem 30.1 continues to apply to every audit in this enlarged set of constructions. $\square$

For a perturbed blind training source with per-active-trial defect α , the two-phase construction also has the bound

$$2 \min\{1, 2\alpha \min(n, m^2)\} + \beta_c + \beta_t$$

on score change, by the proof of Corollary 15.2. The moving-collision parameter alone leaves that source unchanged, so its training contribution is zero; the previous target transport and bounded Gaussian/entropic consumer are retained without change. If one makes the machine autonomous, an explicit bound is $(6n + 12) \max(4m + 4, 12)$ states, since the $2n + 2$ trials each acquire two bits and a mark, with the phase/decision boundaries included. The clock multiplier is an implementation upper bound, not an optimized autonomous-memory claim.

32. CONFIDENCE-LEVEL CERTIFICATION

An expected signed score is an operational separation, not a confidence statement about one validation pair. We now distinguish these two tasks. The null hypothesis below is equality of the candidate and target validation laws; it need not belong to the private alternative family. All

preparations in the theorem are independent resets with one fixed candidate law. The test cannot be given a new adversarial candidate after each reset.

Theorem 32.1 (Confidence lift with a charged accumulator). *Suppose a base reset audit uses n training trials, one candidate and one target validation, and peak width W , with score $D \in \{-1, 0, 1\}$. Assume its expected score is at least $\gamma > 0$ for every alternative. Run J independent copies and reject equality when*

$$\sum_{j=1}^J D_j \geq J\gamma/2. \quad (32.1)$$

Then both the type-I error under equality and the worst-case type-II error are at most $e^{-J\gamma^2/8}$. The whole test has a realization with at most $W(2J+1)$ states and $J(n+2)$ trials. In particular $J \geq 8\gamma^{-2} \log(1/\alpha)$ suffices for both errors to be at most α .

Proof. Under equality, conditional on the training and selected event, the two independent validations have equal means; hence $\mathbb{E}D = 0$. Under each fixed alternative its mean is at least γ . The copies are independent and $D_j \in [-1, 1]$. The exponential-moment estimate for a variable in an interval of length two is $\mathbb{E}e^{s(D_j - \mathbb{E}D_j)} \leq e^{s^2/2}$, by the same tilted-variance argument used in Theorem 9.2. Markov's inequality and minimization in s give $\Pr(J^{-1} \sum (D_j - \mathbb{E}D_j) \geq a) \leq e^{-Ja^2/2}$, and the same lower-tail bound. Set $a = \gamma/2$.

Keep the running integer sum, with at most $2J+1$ values, beside the base machine's register. At each completion add its emitted score and reset only the base register. The sum is not stored in an external history. The clock supplies the repetition index and the fixed terminal threshold; all $J(n+2)$ trials are counted. $\square$

The 399-trial physical audit has mean strictly above $2/5$, so one may use the rational choice $\gamma = 2/5$ in this theorem. This gives the fully explicit sufficient count $J \geq 50 \log(1/\alpha)$ and $401J$ total trials (or $402J$ with the inherited ignored slot). The confidence guarantee uses repetition and its charged sum, not one pair of validation observations.

Theorem 32.2 (One training phase and repeated validation). *For the physical audit of Theorem 30.1, put $\gamma_0 = \sqrt{2} - 1 - 1/300 - \theta > 0$. Retain its chosen event and use m independent candidate-target validation pairs. Reject equality when their score sum is at least $m\gamma_0/2$. The type-I error is at most $e^{-m\gamma_0^2/8}$ and the worst-case type-II error is at most*

$$2\delta + e^{-m\gamma_0^2/8}.$$

The audit uses $4LR + 2m$ trials and at most $12(2m+1)$ states throughout training and validation. Thus $\delta \leq \alpha/4$ and $m \geq 8\gamma_0^{-2} \log(2/\alpha)$ suffice for error at most α .

Proof. The proof of Theorem 30.1 gives an event of training probability at least $1 - 2\delta$ on which the conditional mean score of the selected event is strictly greater than γ_0 . Conditional on that event and its selected label, the validation pairs are independent. Apply the tail bound in Theorem 32.1 conditionally and add the training failure probability. Under equality every selected event has zero mean, regardless of training, so no training failure term is needed for type-I error. Keep the three-valued event label, the within-pair indicator and first-bit buffer, and a sum with $2m+1$ values. Training itself needs no more than ten states, so the displayed bound covers every phase. $\square$

The tests in this section have a different output task from maximizing an expected witness score. The exact three-state decision-cut theorem is not asserted to be an optimal memory theorem for these binary tests. For perturbations of training and validation laws, the coupling argument of Theorem 11.2 bounds changes of each rejection probability by the sum of the per-call total variation errors. In a source-preserving physical transport, alternatively lower the conditional

mean margin by the single-row target variation and reapply the conditional tail estimate. This latter argument transports the confidence guarantee without pretending that an entire m -sample validation law has the same total variation distance as one validation observation.

33. CONFIDENCE TESTS WITH PRICED FINITE-STATE AMPLIFICATION

The signed-sum construction in Section 32 converts expectation into confidence. There is also a direct sample–memory choice for the amplifier itself. Its formulation keeps the binary testing objective distinct from the expected-score task.

Theorem 33.1 (Two finite-state confidence amplifiers). *Suppose a reset audit uses b training trials, two validations, peak width $W \geq 3$, and reports $D \in \{-1, 0, 1\}$. Under the equality null $\mathbb{E}D = 0$, and under every alternative in a fixed class $\mathbb{E}D \geq \gamma > 0$. Fix a known dyadic $0 < g \leq \gamma$, and put $\theta = g/8$.*

- (i) *For integers $m, n \geq 1$, there is a binary test with both errors at most $e_{m,n}(\theta)$, using $n(b+2)$ preparations and peak width at most $(2m+1)\max(W, 9)$. Thus both errors are at most α with the choices (13.5), with $\delta = \alpha$.*
- (ii) *With L, R as in (12.1), using $\delta = \alpha$ and the same θ , there is another test with both errors at most α , $2LR(b+2)$ preparations, and peak width at most $4\max(W, 9)$. Its very long external schedule is charged separately. The confidence error, not the input parameter, determines these read-only horizon constants.*

Proof. Run independent copies of the base audit with fresh preparations and fresh private bits. Conditional on each output D , generate an additional coin Z with probability

$$\mathbb{P}(Z = 1 \mid D) = \frac{1}{2} + \frac{D - g/2}{4}. \quad (33.1)$$

This is between zero and one, since $0 < g \leq \gamma \leq 1$ and $-1 \leq D \leq 1$. If g has denominator 2^s , the probability has denominator dividing 2^{s+3} and is sampled with $s+3$ fair bits. The coins are independent under each fixed parameter. Their mean is $1/2 - g/8$ under the null and at least $1/2 + g/8$ under the alternatives. Apply the walk of Lemma 13.1, rejecting the null for a positive result, to prove (i). Apply the block comparison of Lemma 12.1 to prove (ii).

At every base-audit epoch retain the amplifier state alongside its W labels. After obtaining D , retain that value and at most three Bernoulli-sampler labels while drawing (33.1). This requires at most nine labels per amplifier state. Update the amplifier and reset the base audit before the next repetition. The terminal rejection decision is a function of the amplifier state and requires no additional persistent random seed. This proves the stated full-cut peak bounds. Even after the amplifier is absorbed, padded base-audit repetitions remain charged to the preparation count. The clock supplying the repetition, base-audit epoch and fair-bit depth is external; storing it multiplies the displayed bound by at most one plus the total number of such epochs. $\square$

For the physical expected-score theorem one may take $g = 3/8$, since $3/8 < 2/5$. The additional coin then needs at most six fair bits. Thus confidence can be purchased either with a growing absorbing counter and polynomial repetition count, or with a constant register and a much longer clocked schedule. Neither statement identifies the minimum state count of the binary testing problem with the three conditional validation responses needed by the expected-score problem. The signed-accumulator and train-once confidence theorems remain available as different resource regimes.

The output-law transfer in Theorem 15.1 applies to a completed binary test. In particular, if perturbations change its output law by at most ε , its type-I and type-II bounds each increase by at most ε . Transporting all repeated raw data instead counts every affected preparation, as in Section 32. Independence of the base repetitions and uniformity over the fixed alternative class

are assumptions of the amplification theorem; adaptation of the alternative between repetitions is not being asserted.

34. A FINITE-REGISTER CERTIFICATE TRANSPORTED THROUGH THE MICROSCOPIC LIMIT

The general empirical audit in Theorem 18.1 treats every coordinate of a marked row as unknown. The private continuation constraint in the collision experiment forces a smaller family. Exploiting this constraint yields a much smaller executable auditor, without granting it knowledge of the candidate law.

In the all-on row of Section 28, every private width-one candidate emits two independent bits with parameters p, q , independently of the fair actual mark W . Put

$$a = (1 - p)(1 - q), \quad c = pq.$$

The ideal target Q_* is supported on 000, 001, 110, 111, with respective masses $5/16, 3/16, 3/16, 5/16$. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. Every other atom has target mass zero and cannot improve the maximizing event for $Q_* - b$.

Lemma 34.1 (Two-count audit and its exact register implementation). *In $n \geq 1$ independent all-on candidate trials let C_0, C_1 count report pairs 00, 11, respectively, ignoring the training mark. On the four supported atoms use the empirical masses $C_0/(2n), C_0/(2n), C_1/(2n), C_1/(2n)$, and select exactly those atoms whose target mass is strictly greater than this empirical mass. Call the resulting event A_C . For every $p, q \in [0, 1]$,*

$$\mathbb{E}\{Q_*(A_C) - b(A_C)\} \geq \|Q_* - b\|_{TV} - n^{-1/2}. \quad (34.1)$$

For $n = 400$, the rule is implemented by the two truncated counts $\min(C_0, 250), \min(C_1, 250)$, with comparisons to 150 and 250. Including execution of complete transcripts and the two independent validation trials, it has a deterministic streaming realization with at most 504008 retained states at any scheduled epoch. Thus a nineteen-bit register suffices. The schedule clock is free; no other history store is free. The table description and transition evaluation cost are not asserted to be bounded by nineteen bits.

Proof. On any supported atom let $x = Q_*(w) - b(w)$ and let $\hat{x}$ be the empirical discrepancy. If the sign rule selects a different event from an exact maximizing event, the lost contribution is at most $|x - \hat{x}|$. On unsupported atoms both rules can omit the atom. Summing gives regret at most

$$|C_0/n - a| + |C_1/n - c|.$$

The two counts need not be independent. Their marginal binomial variances and Cauchy–Schwarz give expected regret at most

$$\frac{\sqrt{a(1-a)} + \sqrt{c(1-c)}}{\sqrt{n}} \leq \frac{1}{\sqrt{n}}.$$

This proves (34.1).

For the 00 pair, select both marked atoms when $C_0 < 150$, only the atom of target mass $5/16$ when $150 \leq C_0 < 250$, and neither when $C_0 \geq 250$. The same rule applies to 11, with the other mark being the atom of larger target mass. At equality the zero empirical discrepancy is excluded. Increasing a count beyond 250 cannot change the final choice. Two registers with 251 possibilities each therefore implement the rule, with at most nine possible final event masks.

A buffer with eight possibilities is sufficient to execute and record a completed two-bit-plus-mark transcript. During training the pair of truncated counts and this buffer use at most $8 \cdot 251^2 = 504008$ states. After training, discard the counts and retain the event mask. During validation at most nine masks, one target-event indicator with two values, and the same eight-state transcript buffer are needed, a total of 144 states. The final signed score takes three values. Since the epoch is

known, labels can be reused between these phases; their state counts need not be multiplied. The update maps and threshold comparisons are parameter-independent deterministic maps. They are thus stochastic rows with entries zero or one, or equivalently a compatible continuation cover by Theorem 8.2. Fresh sampling randomness belongs to the specified reset experiments, not to an uncharged auditor register. Finally $504008 < 2^{19}$. $\square$

Theorem 34.2 (Finite-register microscopic certification and global transport). *Consider the marked hard-sphere family of Theorem 28.2, with $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/24$. There is one rational streaming audit, independent of d, σ, p, q , using 400 candidate training resets and one independent candidate and target validation trial, whose expected signed score is strictly greater than $2/5$ against every private width-one candidate. It uses at most 504008 retained auditor states. The candidate remains private width one; the auditor's register is a distinct, charged resource. The physical deficiencies still satisfy*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \ (r \geq 1), \quad \delta_2^h < 13/100.$$

For $d \leq 2/5$ all three finite-experiment deficiencies vanish.

Fix a positive σ and a sequence $d_k \rightarrow d$ within the collision interval. Put

$$\rho_k = |d_k - d| + 2700|d_k - d|^2, \quad \epsilon_k = \min\{1, \sqrt{\rho_k}/(2\sigma)\}.$$

The same 400-reset rule changes its expected score by at most ϵ_k under this perturbation. At the same time the microscopic paths, full and projected likelihoods, posteriors, and bounded entropic values have the joint convergence of Theorem 28.2. Their innovations, backward integrands and backward martingale integrals additionally have all the global conclusions of Theorem 27.2, with $T = 3$, $M = 1$, $\eta_k^2 \leq \rho_k$, and $\varepsilon_k \leq \epsilon_k$. On compact parameter sets bounded away from zero noise, the same audit has the effective probability-enclosure and rational-verification properties of Theorem 28.2(d).

Proof. The private width-one continuation restriction gives precisely the candidate family used in Lemma 34.1; the actual mark is not available to its transition rows. The exact finite separation in Theorem 23.1 gives

$$\min_{p,q} \|Q_* - b(p, q)\|_{TV} = d_* := \sqrt{5/2} - 9/8.$$

The scattering and observation estimates proved in Theorem 28.2 put the physical marked target within $1/300$ of Q_* , uniformly in all stated parameters. Train the two-count rule against the fixed rational Q_* , but use the actual physical target for validation. Each selected event's target probability changes by less than $1/300$. Hence its actual expected score is greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (34.2)$$

The last strict inequality is exact: it is equivalent to $\sqrt{5/2} > 947/600$, whose squared margin is $3191/360000$. The rule's executable memory bound was proved in Lemma 34.1. Its total trial count is 402, not 400 once validation is included. The private, hidden and visible deficiency bounds and the no-collision control are inherited from the same physical theorem, rather than inferred from an auditor whose signature was changed.

For changing d_k , Lemma 28.1 supplies the signal modulus ρ_k and the pathwise J_1 bound $100|d_k - d|$. Theorem 25.1 gives the joint marked variation bound ϵ_k and the likelihood/posterior convergence. The candidate source is blind and its behavior does not depend on d . Thus training laws and candidate validation laws are unchanged, and Proposition 18.4 has $\alpha = 0$ and $\beta = \epsilon_k$. There is no factor of 400 in this transport cost. The same preparation and signal bounds verify every hypothesis of Theorem 27.2, proving the global backward assertions. Finally the audit uses rational thresholds and finitely many deterministic row maps. The positive-noise enclosure

theorem therefore applies directly; an additional target-presentation error ζ reduces its score by at most ζ . $\square$

The two-count construction is specific to the private behavior family. A hidden mixture can destroy the independence of its two reports, and its mark relation must still be checked. The bound is therefore not asserted against the hidden class. The visible lower bound has its separate fiberwise proof; an arbitrarily large visible alphabet is not silently stored in the nineteen-bit auditor. The certificate concerns expected score, not a confidence statement about a single realized validation pair. Repeating validation for statistical confidence would incur an additional sample and storage budget.

The theorem uses a single mechanism: the continuation restriction fixes the feasible candidate laws and permits the compressed audit; scattering supplies its marked separation; common-preparation variation transports the same executable test and the global conditional-law consumer through a change in the microscopic trajectory. The collision transduces a prepared impact variable into a signal. It does not create the prepared correlation between that variable and the mark, and it is not a particle-number limit.

35. EXPLICIT PHASE TRANSITIONS FOR THE TWO-COUNT CERTIFICATE

For comparison with the new residual quotient, we give the complete phase-state specification of Lemma 34.1. It realizes the original 400-training-trial rule exactly, not the majority rule of Section 29. Let

$$C = (C_0, C_1) \in \{0, \dots, 250\}^2, \quad \tau(x) = \min(x, 250).$$

The free clock records the training index, the within-trial observation epoch, the validation phase, and deterministic processing epochs. It contains no data-dependent information. At the start, $C = (0, 0)$.

For a completed transcript (x_1, x_2, w) define the deterministic update

$$T(C, x_1, x_2, w) = (\tau(C_0 + \mathbf{1}_{\{(x_1, x_2) = (0, 0)\}}), \tau(C_1 + \mathbf{1}_{\{(x_1, x_2) = (1, 1)\}})).$$

The value of w does not enter this training update. The phase transitions are the following; x denotes the current input, not an earlier unretained report.

Phase	Retained data	Next transition
Training start	C	Read x_1 and keep (C, x_1) .
Second report	(C, x_1)	Read x_2 and keep (C, x_1, x_2) .
Terminal mark	(C, x_1, x_2)	Read w and keep (C, x_1, x_2, w) .
Count update	(C, x_1, x_2, w)	Replace by $T(C, x_1, x_2, w)$; discard the transcript.
Event formation	C	Replace by the event mask $A(C)$ defined below.
Candidate validation	A , partial transcript	Complete the transcript; replace it by $i = \mathbf{1}_A(x_1, x_2, w)$.
Target validation	(A, i) , partial transcript	Complete the transcript and emit $\mathbf{1}_A(y_1, y_2, v) - i$.
Score	A value in $\{-1, 0, 1\}$	Terminate.

The event formation map, including both threshold ties, is

$$\begin{aligned} 000 \in A(C) &\iff C_0 < 250, & 001 \in A(C) &\iff C_0 < 150, \\ 110 \in A(C) &\iff C_1 < 150, & 111 \in A(C) &\iff C_1 < 250. \end{aligned}$$

No other atom is included. For each report pair there are three choices: both marked atoms, only its larger-target-mass atom, or neither. Thus there are at most nine masks. Clipping at 250

preserves every comparison, including equality; no flag for a count exceeding the clipping threshold is required.

At a training epoch the possible states are bounded by $8 \cdot 251^2 = 504008$: the factor eight dominates every partial or complete transcript buffer. At validation epochs the largest alphabet has at most $9 \cdot 2 \cdot 8 = 144$ values, including the already computed candidate-event indicator. Event formation and final emission need at most nine and three states, respectively. Embed each phase alphabet into $[504008]$ by a fixed injective encoding. Because the phase is supplied by the free clock, these encodings need not have disjoint images. The next row decodes the current phase’s encoding and applies the displayed transition. This gives formal deterministic rows with entries zero or one at every epoch. The buffers, event mask, and validation indicator are all charged. No unpriced data store is introduced by label reuse.

APPENDIX A. THE PROOF DEPENDENCY CHAIN

An arrow below means an actual use of a theorem in the displayed proof, not an assertion of priority or an artificial dependency between papers. The preserved exact physical lower and upper bounds meet at the decision cut. The new full-profile theorem supplies an explicit training realization of general decision covers; its proof does not invoke the historical aggregate kinetic claims.

Input	Consequence using that input
Marked private rows and normalized continuations, Theorems 7.2, 8.2	The product behavior image and its validation continuation responses.
Decision continuation spectrum, Theorem 9.2	Universal two-state physical obstruction, Theorem 29.1; the strict-cut classification (29.9).
Product constraint (29.1) and three-response envelope, Lemma 29.3	Uniform finite-sample score, Lemma 29.4.
Finite-valued residual realization, Theorem 16.1	Exact full training profile and bit-level construction, Theorem 29.5.
Marked scattering and moving-collision estimate, Theorem 28.2	Both physical cut bounds and transport of the very same auditor, Theorem 29.6.
Common-preparation Gaussian comparison, Theorem 25.1; the three version lemmas in Section 26	The global conditional-law consumer, Theorem 27.2, on that moving-collision family.

The prior binary audit theorem remains a separate calibration, not a premise of the physical memory lower bound. The original two-count certificate remains a separate full-discrepancy approximation, not a claimed optimal implementation. The source-preserving build, rational checks and finite enumerations accompanying this article test the stated implementation and identities; they are not independent analytic proof certification.

A.1. The full-profile resource chain. The new central arrow is

strict response cover $\longrightarrow$ finite block tournament $\longrightarrow$ rational whole-schedule auditor.

Theorem 12.2 uses the decision spectrum, the finite block lemma and the dyadic sampler. It does not use a training oracle. Theorems 11.1 and 11.2 track approximation, calibration and precision on complete causal schedules. Theorem 17.1 uses one shared capacity budget for all cuts, and Theorem 32.1 retains the score accumulator. Together these establish a finite controlled-experiment instance of the program’s resource–precision–statistical-error objective. A universal singular-modulus rate or an infinite-horizon kinetic realization is not an input or a consequence of this finite theorem.

The current C2-scoped consumer remains the bounded, common-preparation, positive-noise Gaussian/entropic theorem. The original B4 normalized nonlinear resolvent formula cannot be inherited: its displayed linear difference identity fails on constant inputs. For a separately justified graph H with single-valued $R_\lambda = (I - H/\lambda)^{-1}$, the correctly typed identity is instead

$$R_\lambda h = R_\mu [(\lambda/\mu)h + (1 - \lambda/\mu)R_\lambda h].$$

Indeed both sides solve the same μ -resolvent inclusion. This conditional identity does not establish the graph's range or its kinetic corrector. Those historical targets remain in the program rather than being assigned proof credit from a finite resource theorem.

APPENDIX B. A NAMESPACE MAP OF THE ELEVEN-COMPONENT PROGRAM

The namespaces in this map separate historical paper targets from GTF statements proved here. Every historical component is retained. An interface means that the indicated consumer can be applied after its model-specific hypotheses are proved; it is not an automatic proof of the historical component’s aggregate theorem.

Paper	Retained historical role	GTF relation and precise proof credit
A1	Protocol and finite task reduction	Marked comparison, priced composition and decision-cut certification; no automatic all-order physical response conclusion.
A2	Independent primary geometric chain	Its protocol adapter is historical; the primary geometric chain is not a premise or consequence of a theorem in this article.
A3	Dynamical/potential-space targets	Conditional comparison interface; no new assertion of the historical dynamical closure.
A4	Conditional kernels and forgetting	Historical finite-model consumer; common-preparation posterior estimates feed the current C2-scoped consumer when their hypotheses hold.
B1	Microscopic realization targets	Declared marked experiment interface; the present scattering construction is fixed-particle.
B2	Dynamic action and kinetic targets	Historical action estimates are not used to prove the present decision-cut theorem.
B3	Balance, response and corrector targets	No current proof depends on the historical aggregate balance/kinetic claims.
B4	Nonlinear kinetic semigroup	The historical normalized-resolvent identity and its aggregate downstream claims are not premises here. Their independent repair remains a program obligation.
C1	Statistical acquisition/inference	Original behavior image, finite-reset experiments, and the present response-envelope and microscopic certification consumers.
C2	Duals, response, memory and projections	The current bounded Gaussian/entropic transport replaces the optional-projection step only in its explicitly proved class; it does not certify the historical strict-dual, form-response or rigidity aggregate.
D1	Phase and compact statistical targets	Historical finite-model consumer retained; no new complete phase-theory closure is inferred.

The machine-readable companion records the exact predecessor graph and source identities, including the original B4 and C2 statements inspected at their frozen commit. Names such as `gtf20:decision-cut` and `historical:B4:r17-main` denote different proof obligations. The organizing theorem is a general comparison-and-certification result with an essential physical consumer. It does not replace the entire historical pipeline by a declaration that every aggregate theorem follows from it. The broader program remains intact as a mathematical program.

APPENDIX C. PRESERVED ORGANIZING STATEMENTS

The following two statements retain the earlier organizing labels and claims for cross-reference. Their full proofs are included in the canonical modules above. The unchanged predecessor article and its complete historical development are appended in the separate companion, so that reorganization does not discard earlier derivations.

Theorem C.1 (Decision continuation complexity and physical certification). *The following assertions hold with all persistent data charged to the appropriate machine.*

(i) *The supremum $v_{n,K}$ of worst-case expected scores with n training samples and K states at the decision cut satisfies*

$$0 \leq \mathfrak{V}_K - v_{n,K} \leq \min \left\{ \sqrt{(|\Omega| - 1)/n}, \sqrt{2 \log(2K)/n} \right\}.$$

The least K permitting a strict score greater than c at some finite training length is the least size of a cover of B by sets $\{b : (Q - b)h > c\}$. This spectrum is Lipschitz under uniform total variation perturbations of candidate and target laws.

(ii) *For the ideal marked private product family B of the two-sphere experiment, $\mathfrak{V}_1 = 1/8$ and $\mathfrak{V}_2 = 3/8$. Throughout the collision interval $9/10 \leq d \leq 11/10$, $0 < \sigma \leq 1/24$, the minimum decision-cut width for uniform physical score greater than $2/5$ is exactly three. The lower bound applies to every finite-reset auditor with the specified interface, not merely a particular counter implementation; feedback-row selection does not evade it.*

(iii) *One deterministic rational rule has physical score greater than*

$$\sqrt{2} - 1 - \frac{1}{4\sqrt{399e}} - \frac{1}{4}e^{-399/8} - \frac{1}{300} > \frac{2}{5}.$$

It uses 399 informative training trials, an optional ignored training trial and two independent validations, hence fits the 402-trial schedule. It has exactly three decision-cut states. On its informative training segment, the minimum simultaneous widths for exact stochastic implementation are

$$r_t = \begin{cases} \binom{t+2}{2}, & 0 \leq t \leq 199, \\ \binom{402-t}{2}, & 200 \leq t \leq 399. \end{cases}$$

Their maximum is 20301. Including individual report-bit acquisition and the validations, at most $40602 < 2^{16}$ retained states suffice.

(iv) *At an arbitrary cut of a finite-valued decision, a weighted graph of distinguishing suffixes gives a distribution-sensitive error bound through its minimum monochromatic weight under K labels. Applied to Bayes margins, it yields a quantitative constrained-audit regret bound. At zero error it recovers the exact simultaneous residual profile, even for stochastic implementations.*

(v) *For fixed positive σ and changing d , the same physical rule has the target-variation modulus $\min\{1, \sqrt{|d' - d| + 2700|d' - d|^2/(2\sigma)}\}$, with no training-count factor. The microscopic paths, likelihoods, posterior processes, innovations, and bounded entropic backward values, integrands and martingale integrals have the joint transport conclusions proved below. The private and visible deficiencies exceed $9/20$, whereas a hidden two-valued selector has error less than $13/100$. In the no-collision regime $d \leq 2/5$ all three finite-experiment deficiencies vanish.*

Proof. Part (i) is Theorem 9.2. The exact ideal spectral values are Theorem 29.1; Lemmas 29.3 and 29.4 and Theorem 29.6 prove the matching physical construction. Theorem 29.5 proves (iii), including all acquisition and validation phases. Theorem 16.1 and Corollary 16.2 prove (iv). The microscopic estimates in Theorem 28.2, the explicit coupling-version lemmas in Section 26, and Theorem 27.2 prove (v). $\square$

Theorem C.2 (Continuation complexity and stable causal certification). *With these information conventions the following statements hold.*

(i) *A finite causal kernel is realizable at a width profile if and only if it admits a compatible normalized continuation cover of that profile. For a Boolean terminal test, the exact minimal profile is the number of distinct continuation responses at each cut, even when the implementing*

machine is stochastic. If a cut has $r \geq 2$ distinct responses and only K retained states, its total wordwise error is at least $(r - K)_+ / (r - 1)$.

(ii) For $0 < h < 1/2$, consider $n = 2m + 1$ independent binary samples from one of $\text{Ber}(1/2 - h)$ and $\text{Ber}(1/2 + h)$, and the signed event-discrepancy audit against $\text{Ber}(1/2)$. Its unrestricted value is $h\|P_+ - P_-\|_{\text{TV}}$. The exact training widths needed to attain it are

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n.$$

A smaller width at any cut has the strictly positive value deficit (19.4). Thus the continuation invariant controls both an implementation and a statistical obstruction.

(iii) In the prepared two-sphere experiment of Theorem 34.2, one rational audit using 400 candidate training resets, two validation trials, and at most 504008 retained auditor states has expected discrepancy greater than $2/5$ against every private width-one candidate. The same audit works for every contact diameter $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. The private and visible deficiencies exceed $9/20$, whereas a hidden two-valued selector has error less than $13/100$. Removing the collision, by taking $d \leq 2/5$, makes all three finite-experiment deficiencies zero.

(iv) With positive noise fixed, changing d changes both the collision time and the microscopic path. If $\rho = |d' - d| + 2700|d' - d|^2$, the same audit's expected score changes by at most $\min\{1, \sqrt{\rho}/(2\sigma)\}$. Along $d' \rightarrow d$, the microscopic paths, full and projected likelihoods, and bounded posterior processes converge jointly in the senses specified below. For the associated bounded entropic backward equation, values converge in supremum L^2 and integrands converge globally in time-integrated L^2 , including on the common-latent coupling. Innovations converge uniformly in probability, and backward martingale integrals converge in supremum L^2 .

C.1. Inherited baseline. The following summary is retained from the predecessor. The present article strengthens its memory and backward-integrand conclusions in Theorem C.2; the earlier 2800-trial audit remains valid.

Theorem C.3 (Finite-resource certification and microscopic transport). *The following assertions hold with the conventions just described.*

(i) *Finite causal kernels have a register realization with a prescribed width profile exactly when their normalized continuation laws admit compatible finitely generated polytopes of those widths. At an erasure cut carrying a latent letter of law π , the optimal error with K retained states is*

$$1 - \sum_{i=1}^{\min(K, N)} \pi_{(i)}, \quad \pi_{(1)} \geq \cdots \geq \pi_{(N)}.$$

Independent hidden or visible selectors do not improve this cut value.

(ii) *The reset-audit value is attained, has a finite-dimensional moment minimax representation on the original B , and satisfies*

$$0 \leq \delta(B, Q) - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}, \quad D = \sum_j (|\Omega_j| - 1).$$

The exponent $1/2$ cannot be improved uniformly over compact candidate classes. This assertion is about reset samples, not about the minimal degree of unrestricted polynomial tests.

(iii) *Suppose two prepared microscopic observation models have a common preparation variable, bounded signals H, H' on $[0, T]$, and the same positive Gaussian noise level σ . Set*

$$\eta^2 = \mathbb{E} \int_0^T |H - H'|^2 dt, \quad \varepsilon = \min\{1, \eta/(2\sigma)\}.$$

Their joint marked observation laws differ by at most ε . This estimate transports resource spectra at their stated budgets and transports each fixed finite audit with its explicitly charged reset cost. It

also gives whole-process posterior convergence. The unprojected and projected likelihoods converge in the supremum L^2 norm under the common Wiener reference when $\eta \rightarrow 0$. For bounded latent payoffs, the associated entropic backward equations converge in value, and their integrands converge after the stated denominator localization.

(iv) In the two-sphere model of Theorem 28.2, one rationally specified audit with 2,800 training runs has expected discrepancy strictly greater than $2/5$ against every private width-one simulator, uniformly for $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. A hidden two-valued selector has error less than $13/100$; every visible-selector width-one error exceeds $9/20$. For $d_n \rightarrow d$ the microscopic paths, likelihoods and posterior processes converge in the precise senses of that theorem, and the same audit remains valid. For $d \leq 2/5$ all three deficiencies vanish.

APPENDIX D. UNRESTRICTED-HORIZON MULTICUT REALIZATION

The following organizing statement retains the unrestricted finite-horizon comparison and its complete proof dependencies. The interfaces are those fixed in the introduction.

Theorem D.1 (Multicut realization and certification). *The following statements hold under the interfaces and resource conventions just described.*

(i) For $d = |\Omega|$, let W_c be the minimum peak width over all finite training schedules, including every random-bit and validation cut, and let W_c^{rat} restrict to finite fair-bit implementations. Then

$$\kappa_c(B, Q) \leq W_c(B, Q) \leq W_c^{\text{rat}}(B, Q) \leq 12d \kappa_c(B, Q).$$

For a K -response cover, a finite-horizon tournament realizes this bound with $2LR(K-1)$ training observations, where $L = \lceil \log(2/\delta)/(4\theta) \rceil$ and $R = \lceil 2^L \log(2/\delta) \rceil$. Its uniform score loss is at most $2^{1-r} + 2\rho + 4(K-1)\theta + 2(K-1)\delta$, with dyadic precision r and response-mean calibration error ρ .

(ii) Approximate normalized continuation lists with compatible one-step updates and local causal errors η_t realize their entire width profile with total error at most $\sum_t \eta_t$. Full-profile risk bounds also charge the sum of the per-call calibration errors and the row-rounding errors. Composition multiplies the simultaneous simulator and auditor profiles. A shared word-capacity constraint turns weighted residual obstructions at several cuts into a single Bayesian regret lower bound. A family of point-source experiments has exact decision spectra and matching linear peak-width growth; overlapping independent retention tasks have simultaneous sharp exact profiles and quantitative approximate lower bounds.

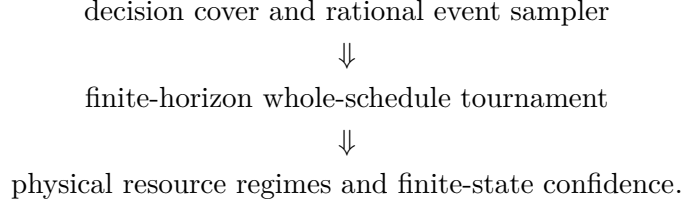
(iii) In the prepared two-sphere collision family, for $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the row-indexed decision complexity for score greater than $2/5$ is exactly three. A deterministic physical audit has peak width at most twelve throughout report-bit acquisition and validation, with an explicit finite preparation count. The earlier 399-training-trial audit is also retained, with peak width 40602 and its exact simultaneous residual profile. For the complete physical task these give $3 \leq W_{2/5}^{\text{phys}} \leq 12$; the upper endpoint is not claimed to be an exact integer optimum.

(iv) Independent repetition turns either positive expected-score certificate into a test with both errors at most α , charging the score accumulator and every preparation. The short physical audit needs $J \geq 50 \log(1/\alpha)$ repetitions. The physical certificate and the bounded common-preparation Gaussian/entropic conditional-law consumer are stable under the moving-collision perturbations specified below.

Proof. Part (i) is Theorem 12.2. Part (ii) follows from Theorems 11.1, 11.2, 17.1, 17.2, 17.3 and Proposition 11.3. Part (iii) is Theorem 30.1 together with the preserved Theorems 29.5 and 29.6. Part (iv) follows from Theorems 32.1 and 32.2, Corollary 30.2, and the common-preparation transport results in Theorems 28.2 and 27.2. $\square$

APPENDIX E. DEPENDENCIES OF THE STATISTICAL RESOURCE FRONTIER

The general preparation bound uses the decision spectrum only to supply a finite response list and its strict margin. Dyadic event realization supplies its executable randomizers. The absorbing-walk lemma then supplies both a finite comparison error and the expected number of active updates. Thus



The occupation-weighted transfer theorem acts on the induced retained kernels of each implemented machine. In the independent reset setting it consumes the walk's active-time estimate. Its proof does not invoke any claim about convergence of changing filtrations.

The exact revelation theorem is an independent, fully solved finite statistical model inside the same profile-constrained decision layer. Its necessary direction applies to every stochastic machine; its sufficient direction constructs compatible nested labels. It is not used as a physical collision lower bound. The lower bound three in that family still comes from the original feedback-row obstruction. Likewise, the complete finite-word residual profile of the short physical selector is retained as an exact implementation theorem and not relabeled as a statistical optimum.

These distinctions give an actual resource–precision–error consumer for the foundation blueprint: preparation and state budgets come from Theorem 13.2, calibration and rationalization from its explicit coin construction, perturbation from Theorem 15.1, and a confidence-level readout from Theorem 33.1. The exact frontier provides a sharp case of the same optimization with nonconstant alphabet and arbitrary state profiles. None of these arrows presupposes the historical B4 kinetic range theorem, the broader C2 rigidity aggregate, or an unproved dependency of the independent A2 chain.

APPENDIX F. THE CONSTRUCTIVE RESOURCE THEOREM

The constructive statement and its proof are retained here as a single reference point for the supporting development.

Theorem F.1 (Continuation profiles and statistical resources). *The following statements hold in the declared clocked finite-state model.*

(i) *Every strict K -response cover of envelope v has, for integers $m, n, r \geq 1$ and dyadic calibration error ρ , a rational whole-schedule realization with*

$$N = n(K - 1), \quad W \leq 3dK(2m + 1),$$

and worst-case score at least

$$v - 2^{1-r} - 2\rho - 4(K - 1)\theta - 2(K - 1)(e^{-4\theta m} + e^{-2n\theta^2}).$$

Here $K \geq 2$; one response needs no training. The previously available constant-width realization gives, in the same model, $\kappa_c \leq W_c \leq W_c^{\text{rat}} \leq 12d\kappa_c$.

(ii) *Suppose a parameter in $\{1, \dots, M\}$ has known ordered prior π , and report t independently reveals that parameter with probability q_t and otherwise reports $*$. Among all stochastic machines having at most K_t labels after each report, the exact Bayesian success probability is*

$$a\pi_1 + \sum_{t=1}^n w_t \sum_{j=1}^{k_t} \pi_j, \quad k_t = \min(M, K_t, \dots, K_n),$$

where $a = \prod_t (1 - q_t)$ and $w_t = q_t \prod_{s>t} (1 - q_s)$. One compatible deterministic machine attains all the cut bounds simultaneously. The formula yields exact sample–peak tradeoffs and exact losses under checkpoint insertion.

(iii) If P_t, Q_t are the induced retained-state transition kernels of a fixed execution with common initial law, its terminal-law change is bounded by

$$\min \left(1, \sum_t \mathbb{E}_P \|P_t(X_{t-1}, \cdot) - Q_t(X_{t-1}, \cdot)\|_{\text{TV}} \right).$$

For absorbed reset comparisons this prices active updates even when training laws differ. It does not remove ignored preparations from the schedule or from the full raw-data law.

(iv) In the marked positive-noise collision experiment, a 320000-training-trial audit with peak width at most 1604 has score greater than $2/5$. The 399-trial, 40602-state audit and the 12-state long-clock audit are also retained. The exact row-indexed decision width remains three. Separate finite-state amplifiers yield specified type-I and type-II errors, with a polynomial-horizon growing counter or a constant register and a longer clocked schedule.

Proof. Part (i) is Theorem 13.2 together with Theorem 12.2. Part (ii) is Theorem 14.1 and its corollaries. Part (iii) is Theorem 15.1 and Corollary 15.2. Part (iv) follows from Theorem 31.1, Corollary 31.2, and Theorem 33.1. $\square$

REFERENCES

- [1] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272. doi:10.1214/aoms/1177729032.
- [2] L. Le Cam, Sufficiency and approximate sufficiency, *Ann. Math. Statist.* **35** (1964), 1419–1455. doi:10.1214/aoms/1177700372.
- [3] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28. doi:10.1524/strm.2002.20.14.1.
- [4] A. Nayyar, A. Mahajan and D. Teneketzis, Decentralized stochastic control with partial history sharing: a common information approach, *IEEE Trans. Automat. Control* **58** (2013), 1644–1658. doi:10.1109/TAC.2013.2239000. The theorem numbers cited here refer to the author preprint, arXiv:1209.1695.
- [5] H. S. Witsenhausen, On the structure of real-time source coders, *Bell System Tech. J.* **58** (1979), 1437–1451. doi:10.1002/j.1538-7305.1979.tb02263.x.
- [6] I. Gallagher, L. Saint-Raymond and B. Texier, *From Newton to Boltzmann: Hard Spheres and Short-Range Potentials*, Zurich Lectures in Advanced Mathematics, European Mathematical Society, 2014. The flow and boundary discussion used here is in Sections 4.1–4.3 of arXiv:1208.5753v2.
- [7] C. Widder, J. Zimmer and T. Schilling, On the generalized Langevin equation and the Mori projection operator technique, preprint (2025), revised 2026. arXiv:2503.20457v6.
- [8] C. Widder and T. Schilling, Generalised Langevin dynamics: significance and limitations of the projection operator formalism, preprint (2026). arXiv:2604.20453v2.
- [9] H. F. Trotter, Approximation of semi-groups of operators, *Pacific J. Math.* **8** (1958), 887–919. doi:10.2140/pjm.1958.8.887. Theorem 5.2, p. 902, is the approximation theorem cited here.
- [10] H. Weisshaupt, LeCam’s randomization criterion in the setting of locally convex spaces without lattice structure, EURANDOM report 2002–017, 22 April 2002. Lemma 1 and Theorem 5, pp. 14–16. Author report.
- [11] S. Basu, Algorithms in real algebraic geometry: a survey, author preprint, arXiv:1409.1534 (2014). Published in *Panoramas et Synthèses* **51** (2017), 107–153. Theorem 2.1 and Section 2.1 in the author preprint are used here.
- [12] M. C. Paull and S. H. Unger, Minimizing the number of states in incompletely specified sequential switching functions, *IRE Trans. Electron. Comput.* **EC-8** (1959), 356–367. Publisher record.
- [13] E. Torgersen, *Comparison of Statistical Experiments*, Encyclopedia of Mathematics and its Applications 36, Cambridge University Press, Cambridge, 1991.
- [14] N. Saldi, A topology for team policies and existence of optimal team policies in stochastic team theory, preprint (2017), arXiv:1711.00639.
- [15] S. Yüksel and N. Saldi, Convex analysis in decentralized stochastic control, strategic measures and optimal solutions, author preprint arXiv:1609.07685v2.
- [16] A. Nerode, Linear automaton transformations, *Proc. Amer. Math. Soc.* **9** (1958), 541–544. doi:10.1090/S0002-9939-1958-0135681-9.

- [17] S. Kiefer, Notes on equivalence and minimization of weighted automata, preprint (2020), arXiv:2009.01217v1.
- [18] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176. doi:10.2140/pjm.1958.8.171.
- [19] H. Weisshaupt, A generalization of Le Cam’s randomization criterion, *Studia Sci. Math. Hungar.* **43** (2006), 443–486. doi:10.1556/SScMath.43.2006.4.3. Theorem numbers refer to the author’s 39-page preprint, Randomization.pdf.
- [20] E. de Klerk and M. Laurent, Error bounds for some semidefinite programming approaches to polynomial minimization on the hypercube, *SIAM J. Optim.* **20** (2010), 3104–3120. doi:10.1137/100790835.
- [21] M. Laurent and L. Slot, An effective version of Schmüdgen’s Positivstellensatz for the hypercube, arXiv:2109.09528v2, 2023.
- [22] E. de Klerk and J. C. Vera, The link between 1-norm approximation and effective Positivstellensätze for the hypercube, *Numer. Algebra Control Optim.* **16** (2026), 84–104. doi:10.3934/naco.2024060. Theorem numbers refer to arXiv:2404.04190v2.
- [23] M. Fliess, Séries rationnelles positives et processus stochastiques, *Ann. Inst. H. Poincaré Sect. B* **11** (1975), 1–21. Original article. Theorems 3.1 and 5.4 and Corollary 2, pp. 8–9 and 18–19.
- [24] T. M. Cover, Hypothesis testing with finite statistics, *Ann. Math. Statist.* **40** (1969), 828–835. doi:10.1214/aoms/1177697590. Theorem 2, pp. 833–834, and Section 4, pp. 834–835.
- [25] W. Hoeffding, Probability inequalities for sums of bounded random variables, *J. Amer. Statist. Assoc.* **58** (1963), 13–30. doi:10.1080/01621459.1963.10500830.
- [26] M. E. Hellman and T. M. Cover, Learning with finite memory, *Ann. Math. Statist.* **41** (1970), 765–782. doi:10.1214/aoms/1177696958.
- [27] J. N. Tsitsiklis, Extremal properties of likelihood-ratio quantizers, *IEEE Trans. Commun.* **41** (1993), 550–558. doi:10.1109/26.223779. The original article is available from the author’s MIT publication page.
- [28] J. Müller and G. Montúfar, The geometry of memoryless stochastic policy optimization in infinite-horizon POMDPs, *International Conference on Learning Representations*, 2022. arXiv:2110.07409. Theorem numbers refer to the 45-page author version.
- [29] M. Putinar, Positive polynomials on compact semi-algebraic sets, *Indiana Univ. Math. J.* **42** (1993), 969–984. doi:10.1512/iumj.1993.42.42045.
- [30] V. Powers, Rational certificates of positivity on compact semialgebraic sets, preprint (2009), arXiv:0911.1331. Theorem 6 is the explicit-ball rational positivity result used here.

GENERAL THETA FOUNDATIONS I: CAUSAL CONTINUATION AND STATISTICAL RESOURCE FRONTIERS

QIAN QI

ABSTRACT. We study statistical experiments in which memory is charged throughout acquisition, randomization and validation. For a compact, possibly nonconvex finite-alphabet behavior image, a finite response cover admits rational whole-schedule realizations with explicit preparation, state, precision and calibration bounds. Absorbing comparisons give a polynomial-horizon alternative to constant-width clocked learning. For independent erasure–revelation experiments we determine the exact Bayesian value under every prescribed multicut state profile, over all stochastic machines and for arbitrary priors. The value depends on the last revelation and each remaining memory bottleneck, yielding sharp sample–peak tradeoffs. Occupation-weighted transition defects transfer terminal guarantees under perturbed training laws without charging absorbed updates as active computation. For a marked positive-noise two-sphere collision family we retain the exact three-state decision optimum and give three distinct whole-schedule resource regimes. Finite-state amplification yields separate confidence guarantees. The underlying causal realization, residual obstructions, microscopic comparison and bounded Gaussian/entropic transport are developed with explicit interfaces and error budgets.

CONTENTS

1. Introduction	3
1.1. Statistical resource frontiers	3
1.2. Comparison, realization, and earlier work	4
1.3. Proof structure and the wider program	5
2. Intrinsic causal deficiency	5
2.1. Experiments and executable signatures	5
2.2. The finite comparison theorem	6
2.3. A delayed channel and a strict signature separation	8
2.4. Optimal resource sets and decision certificates	9
3. Continuation geometry and an exact memory obstruction	10
3.1. Normalized continuation laws	10
3.2. An erasure cut	12
4. Decision continuation spectra	12
4.1. A charged training–validation interface	12
4.2. Several feedback rows	15
5. Finite precision and charged randomization	15
6. Approximate continuation profiles and resource transfer	16
6.1. A causal comparison norm	16
6.2. The full value and its effective approximation	17
7. From a decision cover to an end-to-end machine	19
7.1. The whole-schedule continuation value	19

Date: September 24, 2026.

2020 Mathematics Subject Classification. Primary 62B05, 93E20; Secondary 60G44, 68Q45, 82C40.

Key words and phrases. causal comparison, continuation profile, finite-state decision, sample–memory tradeoff, statistical experiment, hard-sphere scattering.

8. Finite-horizon sample-memory bounds	21
9. An exact statistical frontier at all cuts	23
10. Transport priced by active updates	25
11. Weighted continuation obstructions	26
12. Joint residual budgets and growing families	28
12.1. Actual physical word weights	29
12.2. A family with matching end-to-end growth	29
12.3. A sharp simultaneous storage profile	30
13. Finite-data testing on the original behavior image	30
13.1. The reset interface	30
13.2. A matching uniform sampling obstruction	32
13.3. Transport of one fixed audit	33
14. Continuation complexity of a finite-data test	33
14.1. The width required to attain the binary reset value	34
15. Polynomial testing of the original resource image	35
15.1. A sharp two-time nonconvex example	37
16. Adaptive testing on the observable behavior image	38
16.1. The testing game	38
16.2. A size bound in behavior dimension	39
16.3. A three-weight certificate invisible to constant tests	41
16.4. Strict certificates and effective physical input	42
17. An exact algorithm for the observable chart	42
18. A marked resource gap generated by a binary collision	43
18.1. The exact marked two-report experiment	44
18.2. A full-dimensional collision preparation	45
18.3. Uniformly effective finite presentations	46
19. Uniform stability of optional projections	47
19.1. A microscopic sufficient condition	49
20. Changing trajectories, likelihoods, and conditional consumers	50
20.1. A common preparation, not a common flow	50
20.2. The projected stochastic exponential	51
20.3. A bounded entropic backward-equation consumer	52
20.4. Resource transport is not the posterior coupling	53
21. Observation densities and coupled process versions	54
22. Global transport of the backward equation	55
23. A uniform certificate through a moving collision	57
24. Sharp decision memory for the collision experiment	60
24.1. The marked family and its first two spectral values	60
24.2. A three-response construction	62
24.3. Simultaneous exact widths and the full acquisition schedule	63
24.4. What the two exact complexity statements measure	65
25. Full-profile certification of the collision family	65
25.1. The row-indexed decision invariant	65
26. A third resource regime for the same physical task	67
27. Confidence-level certification	68
28. Confidence tests with priced finite-state amplification	70
29. A finite-register certificate transported through the microscopic limit	71
30. Explicit phase transitions for the two-count certificate	73
Appendix A. The proof dependency chain	74

A.1. The full-profile resource chain	74
Appendix B. A namespaced map of the eleven-component program	76
Appendix C. Preserved organizing statements	76
C.1. Inherited baseline	78
Appendix D. Unrestricted-horizon multicut realization	79
Appendix E. Dependencies of the statistical resource frontier	80
References	80

1. INTRODUCTION

An observation can be informative and yet be unusable after a narrow causal checkpoint. Conversely, a large empirical statistic may be unnecessary when the observer is permitted a longer preparation schedule. A resource theory of statistical experiments must distinguish these effects. It must constrain the acquisition of a continuation, the information carried through each intervening cut, and the execution of its terminal decision. A bound on the last of these alone is not a bound on the complete experiment.

We study this question with positive experimental kernels, explicit preparations, and parameter-independent causal machines. Registers are charged at every declared observation and randomization cut. The schedule, calibration precision, read-only transition description and number of preparations are separate resources. An external clock supplies a predetermined transition, not a sample-dependent statistic; an autonomous implementation must store that clock. The parameter image is retained in its original form and need not be convex.

There are two complementary kinds of result. A realization theorem turns a finite response cover into a bounded-state, finite-precision training-and-validation machine. An exact risk theorem optimizes over all machines satisfying a given multicut profile, rather than over implementations of one chosen decision rule. Both can be transported under a controlled change of the experiment, with the simulator's register and the altered preparations priced explicitly.

1.1. Statistical resource frontiers. Let $B \subset \Delta(\Omega)$ be compact, $d = |\Omega|$, and let Q be a known target. A reset auditor trains on independent trials from one fixed unknown $b \in B$, selects a validation event, and reports the target-event indicator minus the candidate-event indicator on two fresh validations. A randomized event has a mean response $h \in [0, 1]^\Omega$. Its K -response envelope is

$$\mathfrak{V}_K(B, Q) = \max_{h_1, \dots, h_K \in [0, 1]^\Omega} \min_{b \in B} \max_{i \leq K} (Q - b)h_i.$$

Write $\kappa_c = \min\{K : \mathfrak{V}_K > c\}$. Let $V_{N,K}$ be the best worst-case score of a whole schedule with at most N training trials and state profile $\mathbf{K}$, and let W_c minimize peak width over all finite successful schedules. The distinction between κ_c and $V_{N,K}$ is the distinction between a decision interface and its complete statistical realization.

Theorem 1.1 (Continuation profiles and statistical resources). *The following statements hold in the declared clocked finite-state model.*

(i) *Every strict K -response cover of envelope v has, for integers $m, n, r \geq 1$ and dyadic calibration error ρ , a rational whole-schedule realization with*

$$N = n(K - 1), \quad W \leq 3dK(2m + 1),$$

and worst-case score at least

$$v - 2^{1-r} - 2\rho - 4(K - 1)\theta - 2(K - 1)(e^{-4\theta m} + e^{-2n\theta^2}).$$

Here $K \geq 2$; one response needs no training. The previously available constant-width realization gives, in the same model, $\kappa_c \leq W_c \leq W_c^{\text{rat}} \leq 12d\kappa_c$.

(ii) Suppose a parameter in $\{1, \dots, M\}$ has known ordered prior π , and report t independently reveals that parameter with probability q_t and otherwise reports $*$. Among all stochastic machines having at most K_t labels after each report, the exact Bayesian success probability is

$$a\pi_1 + \sum_{t=1}^n w_t \sum_{j=1}^{k_t} \pi_j, \quad k_t = \min(M, K_t, \dots, K_n),$$

where $a = \prod_t (1 - q_t)$ and $w_t = q_t \prod_{s>t} (1 - q_s)$. One compatible deterministic machine attains all the cut bounds simultaneously. The formula yields exact sample–peak tradeoffs and exact losses under checkpoint insertion.

(iii) If P_t, Q_t are the induced retained-state transition kernels of a fixed execution with common initial law, its terminal-law change is bounded by

$$\min \left(1, \sum_t \mathbb{E}_P \|P_t(X_{t-1}, \cdot) - Q_t(X_{t-1}, \cdot)\|_{\text{TV}} \right).$$

For absorbed reset comparisons this prices active updates even when training laws differ. It does not remove ignored preparations from the schedule or from the full raw-data law.

(iv) In the marked positive-noise collision experiment, a 320000-training-trial audit with peak width at most 1604 has score greater than $2/5$. The 399-trial, 40602-state audit and the 12-state long-clock audit are also retained. The exact row-indexed decision width remains three. Separate finite-state amplifiers yield specified type-I and type-II errors, with a polynomial-horizon growing counter or a constant register and a longer clocked schedule.

Proof. Part (i) is Theorem 8.2 together with Theorem 7.2. Part (ii) is Theorem 9.1 and its corollaries. Part (iii) is Theorem 10.1 and Corollary 10.2. Part (iv) follows from Theorem 26.1, Corollary 26.2, and Theorem 28.1. $\square$

The exact frontier in part (ii) is an all-algorithm statistical theorem for a growing family with arbitrary priors and profiles. Its optimum is Bayesian, whereas part (i) is uniform over the original candidate image. These objectives are not interchanged. The attaining revelation machine uses nested optimal label sets. Their compatibility is the reason that the separate bottleneck inequalities are jointly sharp; independently choosing an optimal dictionary at every cut would not prove such a statement for general experiments.

The general realization bounds in part (i) remain constructive upper bounds at fixed N . They do not purport to optimize every sample–memory frontier. In the physical family, the exact optimum at the decision cut and the full-schedule interval $3 \leq W_{2/5}^{\text{phys}} \leq 12$ are different assertions. The three displayed physical implementations exhibit distinct attainable resource regimes. An exact fixed-sample physical optimum is not inferred from the exact residual profile of its empirical selector.

1.2. Comparison, realization, and earlier work. The statistical comparison background is Blackwell–Le Cam theory and its filtered versions [1, 2, 13, 3, 10]. Weisshaupt’s filtered-operator definition and Lemma 1 on p. 14 and Theorem 5 on pp. 15–16 concern compact convex classes of positive operators, with risk criteria implying a filtered randomization. Here the constrained object is an executable finite-state machine with a priced register, precision and clock. Its behavior image need not be convex. A filtered existence theorem does not by itself give the full-profile budgets proved below. The original Norberg proof text has not been obtained for a definitive theorem-level crosswalk; no absence or priority claim is made on that basis.

Residual minimization, positive realization, and finite-state learning are established subjects [16, 23, 24, 26]. Cover’s four-state time-dependent threshold learner and explicit clock discussion are particularly relevant. The absorbing walk and the bounded-variable concentration estimate used

here are also classical [25]. The contribution is not a new name for those mechanisms. It is the finite-horizon passage from a general nonconvex response cover to a complete rational machine, the exact profile-constrained statistical optimization in the revelation class, and the compatible error and resource transfer to physical and confidence-level consumers.

1.3. Proof structure and the wider program. The underlying marked deficiency, normalized continuations and decision spectrum are followed by finite precision, compatible multicut realization, and bounded-state training. Sections 8 and 9 give the preparation–memory statements; Section 10 supplies their retained-law transport. The weighted and joint residual obstructions, reset minimax results, nonlinear and adaptive tests, and the entire microscopic and Gaussian transport chain are included with their proofs. Their physical and confidence consumers appear in Sections 26 and 28.

The bounded common-preparation Gaussian/entropic consumer is used only under its stated positive-noise hypotheses. Appendix B retains the namespaced eleven-component program and its real proof obligations. The primary A2 geometric chain is independent; the kinetic B4 and broader C2 operator targets are not replaced by fictitious arrows from finite-state comparison. The resource layer supplies preparation, memory, calibration and simulation bounds within that program, while its model-specific and singular-limit theorems retain their full mathematical targets.

2. INTRINSIC CAUSAL DEFICIENCY

2.1. Experiments and executable signatures. Fix a finite parameter set Θ , a horizon T , and finite control alphabets $A_0, \dots, A_{T-1}$. An experiment has a report at checkpoints $0, \dots, T$ and an unobserved terminal mark W in a common finite set. It is specified by finite causal instruments: for each parameter and control history they give the joint law of the report path and the actual mark. Equivalently, a finite latent-state realization may be expanded into its history tree. The parameter and mark are unavailable to the encoder. The mark may depend on the physical controlled path. We compare experiments with the same control interface and mark space.

Write Y_t for the source report and Z_t for the target report. A private simulator of width K has stochastic rows

$$\begin{aligned} q_0(z, m \mid y), \quad m &\in [K_0], \\ q_t(z, m \mid m', a, y), \quad m' &\in [K_{t-1}], \quad m \in [K_t], \quad 1 \leq t \leq T. \end{aligned} \tag{2.1}$$

A row consumes the current source report, emits the target report, and retains the new state. The previous target reports are not available to that row unless encoded in m' . The clock and current control are part of the declared interface. Fresh private randomness is permitted, but no other persistent random variable is free. The product of all row simplexes is denoted $\mathcal{K}_K$.

A deterministic feedback tree g chooses a_t from $z_{0:t}$ and its own previous controls. It is the same controller on both sides of a comparison. Let Ω_g be the finite marked transcript space, including reports, controls, and W . Put

$$P_{\theta,g}^q = \text{Law}_\theta(E \xrightarrow{q} g), \quad Q_{\theta,g} = \text{Law}_\theta(F \rightarrow g).$$

Here W on the left is the source's actual mark. The total variation convention is $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. Private deficiency is

$$\delta_K^p(E, F) = \min_{q \in \mathcal{K}_K} \max_{\theta, g} \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}. \tag{2.2}$$

The minimum will be justified below. Randomized feedback need not be added to the maximum: conditioning on its independent random table expresses its law as a mixture of deterministic feedback laws. The same argument permits arbitrary independent side randomness in the tester.

A *hidden selector* $U \in [r]$, independent of the source, chooses one simulator q^u . It is not visible to the feedback controller or the final test. A *visible selector* is revealed before feedback begins. The two costs are

$$\delta_{K,r}^h(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, g} \left\| \sum_u \lambda_u P_{\theta, g}^{q^u} - Q_{\theta, g} \right\|_{\text{TV}}, \quad (2.3)$$

$$\delta_{K,r}^v(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, (g_u)} \left\| \sum_u \lambda_u \delta_u \otimes P_{\theta, g_u}^{q^u} - \sum_u \lambda_u \delta_u \otimes Q_{\theta, g_u} \right\|_{\text{TV}}. \quad (2.4)$$

In both expressions $\lambda \in \Delta_r$ and $q^u \in \mathcal{K}_K$. The second formula compares the joint seed–marked interface. Its target seed has the same distribution and is independent of the target experiment. A hidden selector can be stored privately by retaining (u, m) , at width rK_t . A visible selector is separate side information and is charged by its own alphabet size. Neither interpretation identifies an arbitrary real-valued design with a free finite register.

Definition 2.1 (Deficiency spectrum). The private resource spectrum of E relative to F is the upward-closed set of pairs (K, ε) satisfying $\delta_K^p(E, F) \leq \varepsilon$. The hidden spectrum consists of triples (K, r, ε) satisfying $\delta_{K,r}^h(E, F) \leq \varepsilon$, and the visible spectrum uses $\delta_{K,r}^v$. Width profiles are ordered coordinatewise; in the seeded spectra the seed bound and error are ordered as well. Unused states or selector values may be adjoined. The hidden convexification is $\delta_K^{\text{mix}} = \inf_{r \geq 1} \delta_{K,r}^h$.

The spectrum, not a presumed unique coordinatewise minimum profile, is the intrinsic object. For example one may minimize $\max_t K_t$, with a specified seed signature and tolerance, over its feasible profiles. Different checkpoints can have incomparable optimal tradeoffs.

2.2. The finite comparison theorem. Concatenate all probabilities $P_{\theta, g}^q(\omega)$ into $b(q) \in \mathbb{R}^N$, retaining their row indices; put $b_F = (Q_{\theta, g}(\omega))$. Let $\mathcal{D}_K$ be the finite set of deterministic choices of all rows in (2.1). Empty or unreachable rows remain harmless coordinates. Set

$$a_K = \dim \text{aff}\{b(q) : q \in \mathcal{K}_K\}, \quad \mathcal{I} = \{(\theta, g, B) : B \subseteq \Omega_g\}.$$

For $i = (\theta, g, B)$ and $d \in \mathcal{D}_K$, define

$$A_{i,d} = Q_{\theta, g}(B) - P_{\theta, g}^d(B). \quad (2.5)$$

Including both an event and its complement includes both signs.

Theorem 2.2 (Intrinsic deficiency, priced converse, and visibility). *For the finite marked causal experiments above the following assertions hold.*

(i) *Private deficiency and both fixed- r deficiencies are attained. The private problem has the exact polynomial formulation*

$$\min\{\varepsilon : q \in \mathcal{K}_K, 0 \leq \varepsilon \leq 1, Q_{\theta, g}(B) - P_{\theta, g}^q(B) \leq \varepsilon \text{ for all } (\theta, g, B) \in \mathcal{I}\}. \quad (2.6)$$

In particular its zero set is exactly the nonnegative causal factorization of the target marked behavior by the rows (2.1). At $K_t = 1$ for every t , this is precisely the same factorization with rows $q_t(z \mid a, y)$ and $q_0(z \mid y)$.

(ii) *Hidden convexification has the attained dual*

$$\delta_K^{\text{mix}}(E, F) = \min_{\lambda \in \Delta(\mathcal{D}_K)} \max_{i \in \mathcal{I}} \sum_d \lambda_d A_{i,d} = \max_{w \in \Delta(\mathcal{I})} \min_{d \in \mathcal{D}_K} \sum_i w_i A_{i,d}. \quad (2.7)$$

An optimal simulation uses at most $a_K + 1$ selector values. Thus every weighted causal test has optimal discrepancy at most ε if and only if a hidden mixture of K -state simulators has error at most ε ; it has a private realization of width at most $(a_K + 1)K$. More specifically,

$$\delta_{rK}^p \leq \delta_{K,r}^h, \quad \delta_K^{\text{mix}} = \delta_{K, a_K+1}^h \leq \delta_K^p. \quad (2.8)$$

(iii) Put $e_\theta(q) = \max_g \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}$. Then

$$\delta_{K,r}^v = \min_{\lambda, q^1, \dots, q^r} \max_{\theta} \sum_u \lambda_u e_\theta(q^u). \quad (2.9)$$

Writing $a_K^v = \dim \text{aff}\{(e_\theta(q))_\theta : q \in \mathcal{K}_K\}$, the infimum over all finite visible selectors is attained using $a_K^v + 1 \leq |\Theta| + 1$ values and equals

$$\max_{v \in \Delta(\Theta)} \min_{q \in \mathcal{K}_K} \sum_{\theta} v_\theta e_\theta(q). \quad (2.10)$$

For every $r \geq 1$, and also after optimizing over all r ,

$$\delta_{K,r}^v(E, F) = 0 \iff \delta_K^p(E, F) = 0. \quad (2.11)$$

No corresponding equivalence with $\delta_K^{\text{mix}} = 0$ holds in general.

(iv) For compatible experiments E, F, G and profiles K, L ,

$$\begin{aligned} \delta_{KL}^p(E, G) &\leq \delta_K^p(E, F) + \delta_L^p(F, G), \\ \delta_{KL,rs}^s(E, G) &\leq \delta_{K,r}^s(E, F) + \delta_{L,s}^s(F, G), \quad s \in \{h, v\}. \end{aligned} \quad (2.12)$$

Here KL is the pointwise product, and the second line uses independent selectors. If an F -consumer has widths S_t , composing it with a private simulator gives widths $K_t S_t$ and changes every $[0, 1]$ marked risk by at most $\delta_K^p(E, F)$. A hidden selector is stored at the additional price r , whereas visible selector alphabets multiply. The same bounds hold for optimized upper risk sets and their lower support functions with the indicated costs.

Proof. Expand a probability of a complete source path, simulator state path, and transcript. It is a product of fixed experiment probabilities and one chosen row entry per time. Summing latent and state paths gives a polynomial in the finitely many row entries. No row belonging to a fixed time can occur twice along one path. Consequently the polynomial is multiaffine in the groups of stochastic rows. Continuity and compactness of the product of closed finite simplexes prove attainment in (2.2); the same proof includes λ and the r policies in (2.3). For the visible problem there are only finitely many collections (g_u) and transcript events, so its objective is again continuous on a compact set. The finite-event formula for total variation proves (2.6). Setting $\varepsilon = 0$ forces equality of every transcript probability, in both directions of each event inequality. This proves the factorization and stateless claims, including their necessity.

For each stochastic row of q , sample its deterministic output once, independently of all other rows, before the experiment. This constructs a probability Λ_q on $\mathcal{D}_K$. On any fixed path the row variables encountered have different time indices, so the expectation of their product is the product of the row probabilities. Summing paths shows, simultaneously for every θ, g , that

$$b(q) = \sum_{d \in \mathcal{D}_K} \Lambda_q(d) b(d). \quad (2.13)$$

This is a behavior identity, not a same-width deterministic purification of minimax error. It implies $\text{conv } b(\mathcal{K}_K) = \text{conv}\{b(d) : d \in \mathcal{D}_K\}$. The latter is a compact polytope. Minimizing the continuous convex maximum of event discrepancies over this polytope gives the first expression in (2.7). Finite-dimensional minimax gives the second, with both optima attained. A point of this polytope is a convex combination of at most $a_K + 1$ of its generating points by Carathéodory's theorem. For completeness, if a representation uses more points, their affine dependence gives coefficients c_d with $\sum c_d = 0$ and $\sum c_d b(d) = 0$. Subtract the largest permissible multiple of c from the positive mixing weights; a weight becomes zero without changing the represented point. Repeating gives the stated bound. Store its selector together with the machine state to obtain (2.8). The dual inequality for every w is equivalent to the maximum over w being at most ε , which proves the priced testing converse.

For disjoint seed fibers, total variation is the sum of the fiber total variations weighted by λ_u . For a fixed θ , the tester can choose a separate maximizing tree on each revealed fiber. Therefore its maximum is $\sum_u \lambda_u e_\theta(q^u)$, proving (2.9). The map $q \mapsto e(q)$ is continuous, being a finite maximum of continuous functions. Its image is compact. Convexifying it in $\mathbb{R}^{|\Theta|}$ and applying the preceding finite-dimensional separation and affine-dependence argument proves (2.10) and its selector bound. Notice that the inner minimum is over stochastic q , not over deterministic tables: the nonlinear function e_θ does not commute with (2.13). If the attained visible optimum is zero, then for every u of positive weight and every θ one has $e_\theta(q^u) = 0$, since every summand is nonnegative. Any such q^u is a private exact simulator. The converse uses a selector of one value. This also proves the assertion for the optimized visible selector, whose support is finite by the bound just established.

Finally, cascade a K -state E -to- F transducer and an L -state F -to- G transducer. Within a step, consume the intermediate report before discarding it. The retained pair has at most $K_t L_t$ values. For any terminal tester on G , the second transducer together with that tester is an admissible randomized feedback system on F . The first error bound applies to this system, and total variation contracts under the remaining channel. The second error bound applies to the genuine F system. The triangle inequality gives (2.12). With selectors, condition on the independent selectors in the appropriate places; in the visible case keep both selectors in the compared transcript. The same argument is valid because the target reference selectors are independent. Storing a hidden selector or exposing both visible selectors gives the stated costs. The minima established above may be used; equivalently one can use nearly optimal simulators and let the extra tolerance decrease.

A consumer is another causal transducer with task-specific readout and, where allowed, task-specific control, but with its required common encoder. The simulator is task blind, so its product with that encoder is still common. Condition on the consumer's independent random table to apply the feedback bound. Integration of any $[0, 1]$ marked loss changes by at most the total variation error. Taking the same paired consumer for every row proves the risk-set assertion, rather than a separate choice of simulator for each row. Infima of nonnegative weighted risks give the lower support inequalities. $\square$

Remark 2.3 (What the converse says). The dual (2.7) uses mixtures of indexed causal tests and optimizes over the entire common transducer. It is not a task-by-task choice of encoder. Its conclusion is an explicit hidden-selector simulation, with the selector subsequently charged if private storage is required. It does not replace the nonconvex private set by its convex hull at unchanged resource. The next example shows why that replacement would change the zero set.

2.3. A delayed channel and a strict signature separation. Let the source reveal $i \in [n]$ at checkpoint zero and reveal only a constant at checkpoint one. There is no control and the mark is constant. The target reports a constant at checkpoint zero and at checkpoint one draws $j \in [m]$ from a specified row-stochastic matrix $V = (V_{ij})$. The unknown parameter is i . The first target alphabet has one element: the simulator cannot use an early public output as unpriced storage. Only the intermediate register of size K matters.

Proposition 2.4 (The nonnegative-rank slice). *For this experiment,*

$$\delta_K^p(E, F_V) = \min_{A, B} \max_i \frac{1}{2} \sum_j |V_{ij} - (AB)_{ij}|, \quad (2.14)$$

where A is an n -by- K row-stochastic matrix and B is a K -by- m row-stochastic matrix. In particular

$$\delta_K^p(E, F_V) = 0 \iff \text{rank}_+(V) \leq K, \quad \delta_1^p(E, F_V) = 0 \iff V \text{ has identical rows.}$$

For $1 \leq K \leq n$ the identity channel has the sharp laws

$$\delta_K^p(E, F_{I_n}) = 1 - \frac{1}{\lceil n/K \rceil}, \quad \delta_K^{\text{mix}}(E, F_{I_n}) = 1 - \frac{K}{n}. \quad (2.15)$$

Thus, at tolerance $0 \leq \varepsilon < 1$, the least private width is $\lceil n/[(1 - \varepsilon)^{-1}] \rceil$, whereas the least within-mode width with optimized hidden selection is $\max\{1, \lceil n(1 - \varepsilon) \rceil\}$. The latter still pays for its selector. In particular, for $n = m = 3$ and $K = 2$,

$$\delta_2^p(E, F_{I_3}) = \frac{1}{2}, \quad \delta_2^{\text{mix}}(E, F_{I_3}) = \frac{1}{3}. \quad (2.16)$$

For $V_* = (1/2)I_3 + (1/6)\mathbf{1}\mathbf{1}^\top$,

$$\delta_{2,3}^h(E, F_{V_*}) = 0, \quad \delta_2^p(E, F_{V_*}) \geq \frac{1}{4\sqrt{3}} > 0, \quad \inf_r \delta_{2,r}^v(E, F_{V_*}) > 0. \quad (2.17)$$

Proof. The encoder maps the consumed i into its retained label by A ; the terminal decoder maps that label into j by B . This proves (2.14) including its converse implementation. If $V = CD$ is any nonnegative factorization, let $s_k = \sum_j D_{kj}$. Delete zero rows of D , define $B_{kj} = D_{kj}/s_k$ and $A_{ik} = C_{ik}s_k$. Since V is stochastic, the rows of A sum to one. Padding with unused states gives the claimed equivalence with nonnegative rank. The one-state statement follows immediately.

For $V = I_n$, row error is $1 - (AB)_{ii}$. If the least correct probability is c , for each i select a row $k(i)$ of B with $B_{k(i),i} \geq c$. Some row is selected at least $\lceil n/K \rceil$ times, so stochasticity gives $c \leq 1/\lceil n/K \rceil$. A partition into K groups of sizes at most $\lceil n/K \rceil$, each decoded uniformly, attains that bound. Also $\sum_i (AB)_{ii} \leq K$, by the calculation below with K in place of two. It remains valid under mixing, giving hidden error at least $1 - K/n$. Start with a deterministic channel that reproduces K input symbols exactly and sends all others to those K outputs. Average all simultaneous permutations of its input and output labels. Every diagonal entry is then K/n , proving the second formula. Inverting the two monotone formulas gives the stated least widths.

For $V = I_3$, total variation in row i is $1 - (AB)_{ii}$. If all three diagonal entries of AB were greater than $1/2$, then for each i some row of B would have its i th entry greater than $1/2$. A stochastic row has at most one such entry, so two rows cannot serve three indices. This proves the private lower bound $1/2$. Merge two input symbols into a retained label that decodes uniformly to that pair, and retain the third symbol exactly. This attains it. For every stochastic factorization with two intermediate states,

$$\sum_{i=1}^3 (AB)_{ii} = \sum_{k=1}^2 \sum_{i=1}^3 A_{ik} B_{ki} \leq \sum_{k=1}^2 \sum_{i=1}^3 B_{ki} = 2.$$

The same inequality holds under hidden mixtures, giving maximal row error at least $1/3$. Mix uniformly the three preceding pair-merge channels. The resulting diagonal entries are $2/3$ and the off-diagonal entries are $1/6$. This is exactly V_* and has error $1/3$ from I_3 . It proves the hidden claims in both displays.

The smallest singular value of V_* is $1/2$. Any AB with two intermediate states has a unit vector x in its right null space, so $\|V_* - AB\|_F \geq \|(V_* - AB)x\|_2 \geq 1/2$. If the maximal row total variation error is ε , each row has Euclidean error at most 2ε , and hence the Frobenius error is at most $2\sqrt{3}\varepsilon$. This proves the explicit private bound. The optimized visible value is attained by Theorem 2.2; if it were zero, its zero-set assertion would contradict that bound. This proves its strict positivity. $\square$

2.4. Optimal resource sets and decision certificates. For a class $\mathcal{C}_{S,q}(F)$ of common consumers, with all other information conventions fixed, let $\mathcal{R}_{S,q}(F)$ be its vector of attainable $[0, 1]$ risks. For a nonnegative row weight w summing to one, write

$$h_F(S, q; w) = \inf_{r \in \mathcal{R}_{S,q}(F)} \langle w, r \rangle.$$

If E privately simulates F with cost K and error ε , then

$$\mathcal{R}_{S,q}(F) \subseteq \mathcal{R}_{KS,q}(E) + [-\varepsilon, \varepsilon]^J, \quad h_E(KS, q; w) \leq h_F(S, q; w) + \varepsilon. \quad (2.18)$$

For nonfinite row sets the first expression means pointwise paired risk functions within uniform distance ε ; no finite coordinate box is needed. Upper sets give the equivalent one-sided order formulation. Hidden selectors multiply private width by r ; visible selectors multiply the separate public alphabet by r . These are direct consequences of the final part of Theorem 2.2, with the optimal intrinsic error in place of an externally estimated simulator error.

When two stateless comparison directions have error at most ε , paired risk sets and their support values at unchanged width are within ε . If each is centered by a paired unrestricted benchmark whose error is also at most ε , the centered values are within 2ε . Therefore a gap $g > 0$ between zero excess and infeasible widths is stable under $\varepsilon < g/4$: a value below $g/2$ in the approximation certifies feasibility in the exact finite problem, and a value above $g/2$ certifies infeasibility. The benchmark and the class being optimized must be specified; this assertion does not identify different information signatures.

Proposition 2.5 (Intrinsic finite-dimensional selector approximation). *Let $\mathcal{R}$ be a compact risk image in $C(J)$, where J is compact. Suppose a bounded linear operator $L : C(J) \rightarrow C(J)$ has rank at most m and $\sup_{r \in \mathcal{R}} \|r - Lr\|_\infty \leq \eta$. Every barycenter $\bar{r}$ of a probability on $\mathcal{R}$ has a barycenter r_* of at most $m + 1$ members of $\mathcal{R}$ with $\|\bar{r} - r_*\|_\infty \leq 2\eta$. If $\mathcal{R}$ lies in a finite affine space of dimension m , the error is zero with at most $m + 1$ members. Thus the selector price can be controlled by the dimension of attainable risks, rather than by a covering number of the row space.*

Proof. The compact set $L\mathcal{R}$ lies in an m -dimensional space. Its convex hull is compact, and $L\bar{r}$ belongs to it. The finite-dimensional affine-dependence argument in the proof of Theorem 2.2 gives $r_1, \dots, r_{m+1} \in \mathcal{R}$ and weights λ such that $L\bar{r} = \sum_i \lambda_i Lr_i$. Put $r_* = \sum_i \lambda_i r_i$. The two residual terms $\bar{r} - L\bar{r}$ and $r_* - Lr_*$ have norms at most η , proving the estimate. Apply the same argument after translation in the exact affine case. The implementation stores the selected design index whenever it is not provided as visible side information. $\square$

Remark 2.6 (Finite computation and general carriers). For rational finite instruments, (2.6) and the fixed- r formulations are finite polynomial feasibility problems; (2.7) is a finite linear program. These are exact variational characterizations, not assertions of polynomial running time in a succinct history description. The number of feedback trees, events, or deterministic tables can be large. On standard Borel carriers the same definitions use suprema over measurable feedback and tests and infima over executable kernels. Their product-state composition and decision transport remain valid by the same channel argument. Attainment and the finite testing dual are asserted here only for the stated finite class. The average-risk and physical results below give separate nonfinite theorems with their own hypotheses.

3. CONTINUATION GEOMETRY AND AN EXACT MEMORY OBSTRUCTION

3.1. Normalized continuation laws. We first separate the transducer from the source to which it is attached. There are L epochs with finite input alphabets X_t and output alphabets Z_t . A causal kernel $p(z_{1:L} \mid x_{1:L})$ is nonnegative, normalized for every input word, and has output-prefix marginals independent of later inputs. The current input can include both a source report and the current control. It never includes an unavailable parameter, a terminal mark, or an earlier output not retained by the machine.

Let $\mathcal{C}_t$, $0 \leq t \leq L$, be the finite-dimensional polytope of normalized causal kernels from $x_{t+1:L}$ to $z_{t+1:L}$; put $\mathcal{C}_L = \{1\}$. For $v \in \mathcal{C}_t$ define the shift

$$(S_{x,z}v)(z_{t+2:L} \mid x_{t+2:L}) = v(z, z_{t+2:L} \mid x, x_{t+2:L}).$$

Its total mass $v(z \mid x)$ is independent of the future input word. A zero shift is allowed.

Definition 3.1 (Compatible continuation polytopes). A continuation cover of width $K = (K_0, \dots, K_L)$ consists of lists $V_t = (v_{t,m})_{m \leq K_t}$ in $\mathcal{C}_t$ such that, for every m, x, z ,

$$S_{x,z}v_{t,m} \in \text{cone } V_{t+1}. \quad (3.1)$$

It covers p when $p \in \text{conv } V_0$. Lists may repeat generators and may contain unreachable generators. Thus unused states can be adjoined.

Theorem 3.2 (Continuation realization with a width profile). *A causal kernel p has a stochastic transducer realization with an initial K_0 -valued register and K_t retained states after epoch t if and only if it has a compatible continuation cover of width K . Initial randomization is part of the charged initial register. With a deterministic initial state the condition is $p = v_{0,1}$ and $K_0 = 1$.*

Consequently the set of feasible width profiles depends only on the causal kernel and its declared input interface, not on a row presentation or a labeling of the hidden states. Every matrix of conditional continuation probabilities at cut t has nonnegative rank at most K_t . Individual rank bounds do not replace the simultaneous shift conditions (3.1).

Proof. Given a transducer, let $v_{t,m}$ be the suffix response when its register at time t equals m . The remaining rows define this response even for an unreachable state. If the next row is $q_{t+1}(z, m' \mid m, x)$, conditioning on its output and new state gives

$$S_{x,z}v_{t,m} = \sum_{m'} q_{t+1}(z, m' \mid m, x) v_{t+1,m'}.$$

The initial state distribution expresses p as a convex combination of V_0 .

Conversely choose nonnegative coefficients in every cone membership in (3.1). There are only finitely many choices to make. Since every generator on the right has mass one for each future input word, the sum of the coefficients for a fixed m, x, z is $v_{t,m}(z \mid x)$. Summing over z gives one. The coefficients are therefore stochastic rows $q_{t+1}(z, m' \mid m, x)$. Choose an initial distribution representing p in $\text{conv } V_0$. Backward induction from $\mathcal{C}_L = \{1\}$ proves that the constructed machine has exactly the response $v_{t,m}$ from each register state. In particular its initial response is p . Equality for open-loop input words implies equality under adaptive input selection: at each node of an input–output history tree, the same causal conditional probability is used on the two sides.

For the rank assertion, condition on any positive-probability observed history at the cut. Its continuation response is a convex combination of $v_{t,m}$ with weights equal to the posterior register distribution. Stack these responses as rows, and use all future input–output pairs as columns. This is a nonnegative factorization through K_t generators. The converse construction required the same generator families at adjacent cuts and all their shifts; an unrelated factorization of each cut matrix provides no such compatibility. The characterization, being stated entirely in terms of these response polytopes, also proves the asserted invariance. $\square$

Attach such a causal kernel to a marked source experiment. The output law is obtained by contraction with the source’s causal instruments, while its actual mark is retained. Hence zero private deficiency is equivalently the existence of a compatible continuation cover whose contraction equals the target marked behavior in every parameter and feedback row. This is the geometric form of the factorization in Theorem 2.2; no conditional dependence on the mark is introduced.

A hidden selector chooses one such cover independently of the source. It is a convex combination of complete realizations, not a convexification inside every retained-state simplex at the original width. Its private implementation uses the disjoint union of the covers and keeps the selector in the state. A visible selector keeps those fibers in the compared law. The nonnegative fiber argument in Theorem 2.2 therefore continues to give equality of the visible and private zero sets.

3.2. An erasure cut. A simple family makes the width obstruction exact, including its error. Let X take values in $[N]$ with law π . Before a specified cut the source reveals X . After the cut it has no reports correlated with X . The controller has no feedback channel carrying X across the cut, and all earlier emitted reports are prescribed constants. The only permitted cross-cut information of the simulator is its K -valued register and any independent selector explicitly included in its signature. The terminal mark is the actual $W = X$. The target's terminal report is X . All memory at other checkpoints is unrestricted.

Theorem 3.3 (Exact cut spectrum). *For this experiment, for private, hidden and visible signatures alike,*

$$\delta_K = 1 - \sum_{i=1}^{\min(K,N)} \pi_{(i)}. \quad (3.2)$$

For uniform X , $\delta_K = (1 - K/N)_+$. Thus an erasure cut for m unbiased bits requires $K \geq (1 - \epsilon)2^m$ whenever its error is at most ϵ . The bound holds for every surrounding width profile; equality is attained when the other checkpoints are unrestricted. If every $\pi_x > 0$, exact transmission has nonnegative cut rank N .

Proof. With all post-cut randomness averaged into the decoder, any private machine gives kernels $e(m | x)$ and $d(z | m)$. Its joint terminal law is $\pi_x \sum_m e(m | x) d(z | m)$. Since the target law is $\pi_x \mathbf{1}_{z=x}$, total variation is exactly one minus the correct-decoding probability. Put $u_x = \max_m d(x | m)$. Then

$$0 \leq u_x \leq 1, \quad \sum_x u_x \leq K, \quad \mathbb{P}(Z = X) \leq \sum_x \pi_x u_x \leq \sum_{i=1}^{\min(K,N)} \pi_{(i)}.$$

The last inequality follows by moving mass in u from a smaller weight to a larger one until the K largest weights are filled. To attain it, retain distinct labels for those letters and decode them deterministically; all other letters are mapped to any retained label. Before the cut one may use the unrestricted memory to compute this encoding.

For a hidden selector, condition on its value. It is independent of X , so every conditional correct-decoding probability obeys the same bound; averaging cannot improve it. For a visible selector, variation is the average of the fiber variations, with the same conclusion. The private construction works for both signatures. Finally, the exact joint cut matrix is diagonal with its N positive diagonal entries π_x . Its ordinary rank is N , so its nonnegative rank is at least N and its diagonal factorization attains N . $\square$

The absence of an auxiliary cross-cut feedback channel is essential. An unpriced controller storing X and returning it as a later control would change the experiment, not disprove the cut bound. Likewise a selector correlated with X is not an independent hidden selector. These distinctions are the same ones needed in the collision certificate below.

4. DECISION CONTINUATION SPECTRA

The memory at an experimental cut has a different role from the memory needed to reach that cut. We first identify its exact asymptotic value. This will give both an obstruction and a construction for the same physical audit, rather than a lower bound for an auxiliary binary problem.

4.1. A charged training-validation interface. Let Ω be a finite marked alphabet, $d = |\Omega|$, let $B \subset \Delta(\Omega)$ be nonempty and compact, and fix $Q \in \Delta(\Omega)$. A candidate $b \in B$ is fixed throughout the experiment. Training consists of n independent observations from b . At the end of training the auditor retains a register $M \in [K]$. Everything depending on the training observations or on

an earlier random seed must pass through this register. In particular, there is no external copy of the training history and no uncharged persistent selector.

After this cut, fresh independent randomness may select an event A using M . The event is chosen before independent candidate and target validation observations $X \sim b$ and $Y \sim Q$ are drawn. The score is $\mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The downstream register needed to keep A and to compute this score is distinct from the register at the specified cut. The upstream and downstream widths will be stated separately in concrete implementations. This is the reset interface of Section 13, with one additional charged checkpoint. The actual mark is part of a completed observation; it is not an input to the private candidate's online transition rows.

Write $\mathcal{H} = [0, 1]^\Omega$ and

$$f_h(b) = \sum_{w \in \Omega} (Q(w) - b(w))h(w), \quad h \in \mathcal{H}.$$

Every randomized event has a response $h(w) = \Pr(w \in A)$. Conversely, every $h \in \mathcal{H}$ is the response of a random event: include each atom independently with its specified probability. This is a finite probability distribution on 2^Ω . Only its mean response matters for expected signed score. The event itself must still be retained during validation.

Definition 4.1 (Decision continuation spectrum). The K -state decision continuation value is

$$\mathfrak{V}_K(B, Q) = \max_{h_1, \dots, h_K \in \mathcal{H}} \min_{b \in B} \max_{m \leq K} f_{h_m}(b). \quad (4.1)$$

Let $v_{n,K}(B, Q)$ be the supremum of the worst-case expected score of n -sample auditors at the preceding interface. Only the designated cut has width K in this definition; all other widths are finite but unrestricted. For $c \in \mathbb{R}$, let $\kappa_c(B, Q)$ be the least K for which $\mathfrak{V}_K(B, Q) > c$, with value $+\infty$ when no such K exists.

The definition uses only the marked behavior image and its target, not a parameterization of B or a labeling of candidate states. No convex hull of B is taken. A response list is a list of conditional validation continuations; it is not a mixture of feasible private candidates.

Theorem 4.2 (Operational classification at a decision cut). *All extrema in (4.1) are attained. For every $n \geq 1$,*

$$0 \leq \mathfrak{V}_K(B, Q) - v_{n,K}(B, Q) \leq \min \left\{ \sqrt{\frac{d-1}{n}}, \sqrt{\frac{2 \log(2K)}{n}} \right\}. \quad (4.2)$$

Consequently $v_{n,K} \rightarrow \mathfrak{V}_K$. Every auditor with K states at the designated cut, at any finite training length, has worst-case score at most $\mathfrak{V}_K$. If $\mathfrak{V}_K > c$, some finite-sample auditor with K states at that cut has worst-case score greater than c . Thus κ_c is exactly the minimum cut width permitting such a strict certificate with sufficiently many training samples.

Equivalently, κ_c is the least number of response half-spaces

$$\{b \in B : f_h(b) > c\}, \quad h \in \mathcal{H}, \quad (4.3)$$

that cover B . If two parameterized behavior families satisfy, uniformly in their common parameter q ,

$$\|b(q) - \tilde{b}(q)\|_{\text{TV}} \leq \alpha, \quad \|Q - \tilde{Q}\|_{\text{TV}} \leq \beta,$$

then

$$|\mathfrak{V}_K(B, Q) - \mathfrak{V}_K(\tilde{B}, \tilde{Q})| \leq \alpha + \beta. \quad (4.4)$$

Proof. For a fixed response list, the inner objective is continuous in b . Its variation when the list changes is bounded uniformly over b by $2 \max_m \|h_m - \tilde{h}_m\|_\infty$. Its minimum over compact B is therefore continuous in the list. Compactness of $\mathcal{H}^K$ proves attainment.

Condition on the retained state of any admissible auditor. Its conditional post-cut event law depends only on that state and fresh randomness, and has a response h_m . Independence of validation from training gives

$$g(b) = \sum_{m=1}^K \Pr(M = m) f_{h_m}(b) \leq \max_m f_{h_m}(b).$$

Taking the minimum over b , and then the supremum over auditors, proves the upper bound. This argument permits arbitrary stochastic training and arbitrary finite upstream memory. An extra training-dependent decoder would invalidate its first conditioning step and is excluded by the interface, not silently supplied.

For the reverse bound fix an optimal list. Let $\hat{b}$ be the empirical measure and choose the first index maximizing $f_{h_m}(\hat{b})$. Retain that index and discard the empirical table. If m_* maximizes the true score, then pointwise

$$\max_m f_{h_m}(b) - f_{h_m}(\hat{b}) \leq 2 \max_{m \leq K} |(\hat{b} - b)h_m|. \quad (4.5)$$

The maximum is at most $\|\hat{b} - b\|_{\text{TV}}$, since $0 \leq h_m \leq 1$. Cauchy–Schwarz and the multinomial variances give

$$\mathbb{E}\|\hat{b} - b\|_{\text{TV}}^2 \leq \frac{d}{4n} \left(1 - \sum_w b(w)^2\right) \leq \frac{d-1}{4n}.$$

This proves the first rate.

For completeness, if V takes values in an interval of length one, then $\log \mathbb{E}e^{s(V - \mathbb{E}V)} \leq s^2/8$. Indeed the second derivative of the logarithmic moment generating function is the variance under the exponentially tilted law, at most $1/4$; its value and first derivative vanish at zero. Integrating twice proves the inequality for both signs of s . Independence consequently gives $\mathbb{E}e^{s(\hat{b}-b)h_m} \leq e^{s^2/(8n)}$. For $s > 0$,

$$\mathbb{E} \max_m |(\hat{b} - b)h_m| \leq \frac{\log(2K)}{s} + \frac{s}{8n}.$$

Minimizing in s gives $\sqrt{\log(2K)/(2n)}$ and proves the second rate in (4.2). The empirical table has finitely many values; the construction is therefore a finite-state auditor, with only the chosen index surviving the specified cut. Real response coefficients are allowed here; the physical construction below has deterministic rational transitions.

A finite cover in (4.3) says that the continuous function $b \mapsto \max_m f_{h_m}(b)$ is greater than c everywhere on compact B . Its attained minimum is then greater than c . This proves the covering characterization in both directions. It also proves the strict-threshold assertion; no claim about finite-sample attainment at the boundary $c = \mathfrak{V}_K$ is needed.

Finally $|(Q - b)h - (\tilde{Q} - \tilde{b})h| \leq \alpha + \beta$ for every response. Taking maxima, minima over the common parameter set, and maxima over lists preserves this uniform bound, proving (4.4). $\square$

Corollary 4.3 (Compression by a finite decision dictionary). *Suppose responses $h_1, \dots, h_K$ satisfy $\min_{b \in B} \max_m f_{h_m}(b) \geq c + \eta$ for $\eta > 0$. An empirical selector with $n > 2 \log(2K)/\eta^2$ achieves score greater than c and retains only K states at the decision cut. If these responses depend on a finite statistic $T : \Omega \rightarrow \Xi$, the training selector can count $T(X)$ instead of X . A sufficient upstream register has at most $\binom{n+|\Xi|-1}{|\Xi|-1}$ states at the last training checkpoint, and the corresponding time-dependent count registers suffice at earlier checkpoints.*

Proof. Use the second bound in (4.2). Empirical expectations of h_m depend only on the histogram of $T(X)$. At time t , that histogram has $\binom{t+|\Xi|-1}{|\Xi|-1}$ possibilities. Its update and the final maximization

are deterministic. No statistical sufficiency for estimating all of b is required; sufficiency for the specified responses is enough. $\square$

This corollary supplies an intermediate principle between a full marked histogram and a problem-specific compressed counter. The invariant is the number and geometry of validation continuations needed to cover the original candidate image at the desired score. It is not the algebraic dimension of a chosen parameterization. The empirical upper bound need not be sample-optimal, and a minimum decision-cut width is not a minimum of the largest width over the entire acquisition schedule.

4.2. Several feedback rows. The classification does not require that the declared experiment have only one feedback row. For the row-indexed interface of Section 13, let $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$ and define

$$f_\lambda(b) = \sum_{i \in \mathcal{I}} \lambda_i f_i(b), \quad \lambda \in \Delta(\mathcal{I}).$$

A post-cut continuation is now a probability distribution on row–event pairs; it is not an arbitrary test on a joint product of validation rows. Put

$$\mathfrak{V}_K^{\text{rows}}(B, Q) = \max_{\lambda_1, \dots, \lambda_K \in \Delta(\mathcal{I})} \min_{b \in B} \max_m f_{\lambda_m}(b).$$

Corollary 4.4 (Row-indexed classification). *With n training samples per row, the row-indexed decision-cut value converges to $\mathfrak{V}_K^{\text{rows}}$, and its deficit is at most*

$$\min\{\sqrt{D/n}, \sqrt{2 \log(2K)/n}\}, \quad D = \sum_j (|\Omega_j| - 1).$$

The strict half-space cover characterization holds with f_λ . The upper bound by $\mathfrak{V}_K^{\text{rows}}$ applies also to any other finite training schedule, including adaptive row selection.

Proof. Conditioning on the cut state gives the same convex-mixture upper bound as before; its proof places no restriction on how that state was reached. For empirical selection among an optimal list, $|(\hat{b} - b)\lambda_m| \leq \max_j \|\hat{b}_j - b_j\|_{\text{TV}}$. Sum the rowwise second moments to obtain the first rate. To obtain the second, align the r th independent training sample from each row into a tuple and write its response as $\sum_{j,A} \lambda_{m,j,A} \mathbf{1}_A(X_{j,r})$. This random variable belongs to $[0, 1]$, and its sample mean is the empirical candidate mean for that continuation. The logarithmic moment-generating argument in Theorem 4.2 therefore gives the same finite-list rate. Each chosen row–event distribution can be drawn using fresh randomness after the cut. Compactness and the covering argument are unchanged. $\square$

5. FINITE PRECISION AND CHARGED RANDOMIZATION

We distinguish the number of persistent labels from the description of a transition and from the clock that selects it. An observation in a finite alphabet is initially an atomic input. Every random bit used after that input is a separate, charged epoch. Section 6 gives the additional cost of splitting the observation itself.

Lemma 5.1 (Dyadic rows and finite randomizers). *A probability vector on s points has a dyadic approximation with denominator 2^r at total variation distance at most $(s-1)2^{-r}$. It can be sampled using exactly r independent fair bits and at most $2s-1$ persistent labels at each intermediate bit epoch. The label count uses the bit position supplied by the schedule. No partial random integer is retained outside these labels.*

Proof. Round down the first $s - 1$ coordinates and assign the remaining mass to the last coordinate. The positive and negative parts of the change have equal mass, at most $(s - 1)2^{-r}$. For sampling, partition $\{0, \dots, 2^r - 1\}$ into consecutive intervals of the prescribed integer lengths. A binary prefix describes a dyadic interval. If that interval is contained in one output interval, keep the output label and ignore the remaining bits. Otherwise keep the prefix interval. At a fixed depth, every unresolved interval contains a partition boundary in its interior; there are at most $s - 1$ such intervals. Together with at most s resolved outputs this gives $2s - 1$ labels. The children and their resolved labels are fixed by the read-only partition and the depth. Thus the construction is an actual finite transition table, not a real-valued random seed. Zero-length intervals cause no difficulty. $\square$

Proposition 5.2 (Rational decision continuations). *Let $d = |\Omega|$. A list of K responses in $[0, 1]^\Omega$ can be replaced by dyadic responses of denominator 2^r , losing at most 2^{1-r} in its uniform score envelope. After a decision cut retaining K labels, the randomized event for the chosen dyadic response can be drawn using r fair bits, at most $K(2d + 1)$ labels during this draw, and at most $K(d + 1)$ labels afterwards. Keeping the event through one candidate and one target validation requires at most $2K(d + 1)$ labels at the intervening atomic-observation cuts. Its final signed score has three values.*

Proof. Round every coordinate down. The sup-norm error is at most 2^{-r} , and $\sum_w |Q(w) - b(w)| \leq 2$. This proves the envelope estimate. For a fixed rounded response h , draw a uniform dyadic integer U and choose $A_U = \{w : U < 2^r h(w)\}$. These events are nested and have at most $d + 1$ distinct values. Their mean response is exactly h . Apply Lemma 5.1 to their interval lengths, retaining the chosen response index during sampling. Afterwards the event label and the candidate-event indicator suffice; the target indicator emits their difference. All event masks are read-only data fixed before observing the candidate. $\square$

For K responses, their rounded coordinates require $Kd(r + 1)$ bits in a direct read-only representation. These bits are not charged as persistent sample-dependent labels, but are not declared costless precision. A construction using them records both quantities. In particular, the proposition turns every strict response-cover margin into a finite-randomness implementation; it says nothing about exact attainment of an irrational boundary value.

6. APPROXIMATE CONTINUATION PROFILES AND RESOURCE TRANSFER

A cut profile records the whole causal schedule, rather than its final compression. We give both a realization certificate and an operational value. These notions keep the compatibility condition of Theorem 3.2; separate nonnegative factorizations at individual cuts are not a substitute for that condition.

6.1. A causal comparison norm. For two normalized finite causal suffix kernels p, q at cut t , put

$$d_t^{\text{ca}}(p, q) = \sup_{\pi} \|P_p^\pi - P_q^\pi\|_{\text{TV}},$$

where π ranges over deterministic causal input policies on the remaining schedule and P_p^π is the law of its complete output transcript. Randomized policies give the same supremum, by conditioning on their independent random tables. In particular this distance includes feedback; the supremum over open-loop input words alone is not used.

For stochastic coefficients $q_t(z, j \mid m, x)$ and suffix kernels $v_{t+1,j}$, denote by $\mathcal{Q}_t v_{t+1,m}$ the suffix kernel which, upon input x , emits z and chooses j with these coefficients, then uses $v_{t+1,j}$. The notation denotes one first-step operation, not a conditioning on an event of zero probability.

Theorem 6.1 (Approximate compatible realization). *Let $K_0, \dots, K_L$ be positive integers. Suppose normalized causal kernels $v_{t,m}$, $m \leq K_t$, and stochastic rows q_t satisfy*

$$d_t^{\text{ca}}(v_{t,m}, \mathcal{Q}_t v_{t+1,m}) \leq \eta_t \quad (t < L, m \leq K_t). \quad (6.1)$$

At the terminal cut the suffix kernels are the unit empty kernel. If $\rho \in \Delta_{K_0}$ and $d_0^{\text{ca}}(\rho, \sum_m \rho_m v_{0,m}) \leq \eta_{-1}$, then the machine with initial law ρ and transition rows q_t has the whole profile (K_t) and realizes a kernel $\hat{p}$ satisfying

$$d_0^{\text{ca}}(\rho, \hat{p}) \leq \min\{1, \eta_{-1} + \sum_{t=0}^{L-1} \eta_t\}. \quad (6.2)$$

Conversely, a machine with this profile within distance ε of p supplies such a certificate with $\eta_{-1} = \varepsilon$ and all local errors zero. If a deterministic initial state is required, use $K_0 = 1$ throughout both statements.

Proof. Let $\hat{v}_{t,m}$ be the actual continuation of the rows q from state m . Fix a future policy. Conditional on the first output and next state, its remaining actions form an admissible suffix policy. Replacing $v_{t+1,j}$ by $\hat{v}_{t+1,j}$ therefore changes its transcript law by at most $\max_j d_{t+1}^{\text{ca}}(v_{t+1,j}, \hat{v}_{t+1,j})$. Mixtures and pushforwards contract total variation. The triangle inequality and (6.1) imply, backwards from L ,

$$\max_m d_t^{\text{ca}}(v_{t,m}, \hat{v}_{t,m}) \leq \sum_{s=t}^{L-1} \eta_s.$$

Mixing with ρ proves (6.2). The converse takes the actual normalized continuations of the given machine, including arbitrary valid continuations for unreachable states. Their one-step equations are identities. There is no assertion that independently optimal approximate lists at successive cuts have compatible updates. $\square$

6.2. The full value and its effective approximation. Fix a finite schedule with finitely many observation and action symbols, N calls to an experiment, and a bounded terminal payoff $G \in [-1, 1]$. An environment $b \in B$ specifies the conditional observation kernels at all possible visible histories and actions. Assume B is a nonempty compact subset of the resulting finite product of probability simplexes. A controller A has a specified persistent profile $\mathbf{K}$; the clock has no access to sample history. All action choices and all retained randomness factor through the current register. Define

$$V_{N,\mathbf{K}}(B, G) = \max_{A \in \mathcal{M}(\mathbf{K})} \min_{b \in B} \mathbb{E}_b^A G. \quad (6.3)$$

Unused states pad smaller machines, so $\mathcal{M}(\mathbf{K})$ is a finite product of stochastic-row simplexes, with any prescribed causal zeros imposed. The terminal readout is included among its rows.

Theorem 6.2 (Full-profile resource transfer). *The maximum in (6.3) is attained. Its value is nondecreasing in each K_t . Removing a charged intermediate cut gives a relaxation, whereas inserting identity-holding cuts preserves every implementation with the same held register.*

Suppose two environments at each of their N calls differ in conditional total variation by at most δ_i , uniformly over corresponding histories, actions and parameters. Replace the controller rows at its L transitions by dyadic rows of denominator 2^r , with at most s_t possible next-state/output pairs at transition t . Any fixed controller loses at most

$$2 \min \left\{ 1, \sum_{i=1}^N \delta_i + \sum_{t=1}^L (s_t - 1) 2^{-r} \right\} \quad (6.4)$$

in expected payoff. The same bound compares optimal values on the same profile, with the appropriate dyadic approximation on one side. More generally an approximate continuation certificate contributes its error in (6.2) inside the braces.

If $B_0 \subset B$ is a finite ω -net in the maximum conditional-TV metric and v_0 is the maximum over the dyadic controller grid of the minimum over B_0 , then

$$v_0 - 2N\omega \leq V_{N,\mathbf{K}}(B, G) \leq v_0 + 2 \sum_{t=1}^L (s_t - 1) 2^{-r}. \quad (6.5)$$

The bounds are certificates whenever the net and the finite evaluations are certified. Compactness alone supplies existence of nets, not an algorithm for an arbitrary presentation of B .

Proof. For fixed b , each path probability is a finite product of controller and environment coordinates. Summing the finitely many paths gives a jointly continuous payoff. Its minimum over compact B is continuous in the controller rows; the controller space is compact. This proves attainment. Padding proves monotonicity. Removing a constraint enlarges the feasible set, and identity transitions prove the insertion assertion.

For the perturbation bound, couple two executions until their first mismatch. On equal histories the conditional observation laws can be maximally coupled, with mismatch probability at most δ_i . Couple the controller transitions similarly, using Lemma 5.1. The probability of any mismatch is at most the sum in (6.4), capped at one. The payoff changes by at most two on that event. This coupling is recursive on finite histories and remains valid for feedback controls. Alternatively, approximate continuation kernels are compared in the causal distance and then composed, as in Theorem 6.1.

For a fixed grid controller, a minimizing point of B has a point of B_0 within ω ; the corresponding payoffs differ by at most $2N\omega$. Thus the grid optimum on all of B is at least $v_0 - 2N\omega$. Conversely approximate a maximizing unrestricted controller row by row. Its minimum over B decreases by at most the row-rounding bound, and its minimum over B_0 is no smaller. Taking the grid maximum proves the upper bound. No minimax interchange or convexification of the fixed-width behavior image occurs. $\square$

The formula records preparation count, persistent labels, calibration error, causal simulation error and transition precision in one execution. The training-count factor is present when training kernels change. It is absent only when those kernels are identical. Failure probabilities are also transported, with factor one rather than two, since their indicators have range $[0, 1]$.

Proposition 6.3 (Composition and schedule refinement). *A causal simulator of profile $\mathbf{R}$ followed on the same synchronized schedule by a controller of profile $\mathbf{K}$ has a realization with profile $(R_t K_t)$. Causal simulation errors add under successive composition. Two disjoint phases with a genuine reset and no retained bridge use the maximum of their peak widths. If a bridge with H labels must survive the second phase, its direct width bound is H times the second-phase width.*

If one atomic input x is encoded by ℓ symbols, let D_j be the number of possible length- j prefixes. Holding the pre-input state and the prefix realizes the refined acquisition cuts with widths $K_{\text{before}} D_j$, followed by the original update. A time-dependent machine running for T scheduled epochs becomes autonomous by retaining its clock, with at most $(T + 1) \max_t K_t$ states. These are implementation bounds, not claims of optimality for every refinement.

Proof. Keep the ordered pair of simulator and controller registers and compose their transition kernels. The simulator's emitted observation is the controller's current input. Conditional coupling until the first simulation discrepancy gives the error statement. A genuine reset permits reusing the alphabet, whereas a surviving bridge is another register. A partial encoded input is determined by its prefix, giving the stated refinement construction. Finally the augmented state (t, m) implements the time-indexed transition without an external clock. These constructions also show why neither hidden selectors nor an external transcript may be omitted from the register being counted. $\square$

7. FROM A DECISION COVER TO AN END-TO-END MACHINE

The unrestricted empirical table in Theorem 4.2 can be replaced by a bounded-state training procedure. This replacement trades preparation count against memory, rather than assuming that the training computation has already been performed. The elementary block mechanism belongs to the classical finite-statistics theory of Cover [24]; the result below tracks a finite horizon, a uniform indifference band, rational precision, the whole validation phase, and a compact nonconvex candidate image.

Lemma 7.1 (A finite-horizon four-state comparison). *Let Z_i be independent Bernoulli observations of unknown mean u . For $0 < \theta < 1/2$ and $0 < \delta < 1$, put*

$$L = \left\lceil \frac{\log(2/\delta)}{4\theta} \right\rceil, \quad R = \left\lceil 2^L \log(2/\delta) \right\rceil. \quad (7.1)$$

There is a deterministic clocked machine with four states at every observation cut, using exactly $2LR$ observations, that decides the sign of $u - 1/2$ with error at most δ whenever $|u - 1/2| \geq \theta$. No assertion is imposed within the indifference band.

Proof. Partition the observations into R pairs of blocks, each block of length L . A positive block succeeds when all its observations are one; the following negative block succeeds when all are zero. Stop updating the decision at the first successful block and retain its sign. If none succeeds, decide negative at the end. The prescribed remaining observations are still consumed, but ignored.

Two states are absorbing positive and negative decisions. The other two mean undecided with the current block still alive or already failed. At a block boundary the clock resets the undecided flag to alive. Its polarity and position are supplied by the clock; they are not inferred from an unrecorded sample. The last observation of each block either latches the sign or starts the next block. Thus four states suffice even at the block boundaries.

Write $x = u^L$, $y = (1 - u)^L$ and $s = x + (1 - x)y$. The probability of a positive decision is exactly

$$H_{L,R}(u) = \frac{x}{s} \{1 - (1 - s)^R\}. \quad (7.2)$$

Here $s > 0$, including at $u = 0, 1$. For $u \geq 1/2 + \theta$, the error is at most $y/x + (1 - s)^R$. For $u \leq 1/2 - \theta$, it is at most x/y ; the default negative decision is then correct. Since

$$\log \frac{1/2 + \theta}{1/2 - \theta} \geq 4\theta, \quad s \geq \max(x, y) \geq 2^{-L},$$

both errors are bounded by $e^{-4\theta L} + e^{-R2^{-L}} \leq \delta$. The logarithmic inequality follows by differentiation from its equality at zero. This also proves uniformity over the two composite hypotheses. $\square$

7.1. The whole-schedule continuation value. In the one-row reset interface of Section 4, let $W_c(B, Q)$ be the least peak width over all finite training lengths and all auditors whose worst-case expected signed score is strictly greater than c . Every training-observation cut, the decision cut, every fresh random-bit cut, and both validation cuts are charged. Observations in Ω are atomic for this definition. The event is selected before the two validations, exactly as in that interface. The schedule clock and read-only constants are recorded as separate resources. The class includes randomized machines; its rational subclass uses only finite fair-bit randomizers. Write W_c^{rat} for the corresponding minimum in the latter class. Empty feasible sets have value $+\infty$.

Theorem 7.2 (Uniform end-to-end realization of a decision cover). *Let $B \subset \Delta(\Omega)$ be nonempty and compact, Q fixed, and $d = |\Omega|$. For every strict threshold c ,*

$$\kappa_c(B, Q) \leq W_c(B, Q) \leq W_c^{\text{rat}}(B, Q) \leq 12d \kappa_c(B, Q). \quad (7.3)$$

The last inequality is interpreted only for finite κ_c ; when $\kappa_c = +\infty$ all three quantities are infinite.

More explicitly, fix $K \geq 2$ and a response list with uniform envelope at least v . Round its coordinates to denominator 2^r , and choose $q_i \in [0, 1] \cap 2^{-r}\mathbb{Z}$ with $|q_i - Qh_i^{(r)}| \leq \rho$. For θ, δ as in Lemma 7.1, a rational auditor uses

$$N = 2LR(K - 1) \quad (7.4)$$

candidate training observations and two validation observations. Its expected score is at least

$$v - 2^{1-r} - 2\rho - 4(K - 1)\theta - 2(K - 1)\delta. \quad (7.5)$$

Its completed-training-observation cuts have width at most $4K$, its decision cut has width K , and its entire random-bit-refined schedule has width at most $12Kd$. No empirical histogram is constructed. The response and comparison thresholds have a direct read-only representation of size $O(K^2d(r + 3) + \log L + \log R + \log K)$ bits.

Proof. Use a sequential tournament. Initially response 1 is incumbent. At stage $j = 2, \dots, K$, compare incumbent i with challenger j using fresh candidate observations. For one observation $w \sim b$, generate a Bernoulli variable with conditional success probability

$$\frac{1}{2} + \frac{q_j - q_i - h_j^{(r)}(w) + h_i^{(r)}(w)}{4}. \quad (7.6)$$

This lies in $[0, 1]$ and is dyadic of denominator 2^{r+2} . Its unconditional mean is $1/2 + (\tilde{f}_j(b) - \tilde{f}_i(b))/4$, where $\tilde{f}_i(b) = q_i - bh_i^{(r)}$. Conditional on the incumbent at the start of this stage, all these variables are independent and have this mean. The fixed candidate b is not changed by the comparison. Run Lemma 7.1 and replace i by j exactly when its decision is positive.

At each comparison, conditionally on all earlier stages, the probability of choosing the inferior response when their $\tilde{f}$ values differ by at least 4θ is at most δ . With probability at least $1 - (K - 1)\delta$, no such error occurs. On this event each stage loses at most 4θ relative to the better of its two responses. Induction on j gives

$$\tilde{f}_i(b) \geq \max_i \tilde{f}_i(b) - 4(K - 1)\theta.$$

Replacing the calibrated means by their true means loses at most 2ρ . Rounding the original list loses at most 2^{1-r} in its envelope, by Proposition 5.2. Any response payoff belongs to $[-1, 1]$, so failure of the good event loses at most $2(K - 1)\delta$ in expectation. This proves (7.5), uniformly over b .

During a comparison keep the incumbent index and one of the four block states. The challenger and block positions are fixed by the schedule. At the end of a comparison the sign updates the incumbent and is forgotten. This gives the $4K$ bound at completed-observation cuts. While drawing the coin in (7.6), keep this register and the current observation, and use the three-label binary sampler of Lemma 5.1. Thus at most $12Kd$ labels occur at any intervening random-bit cut. At the final decision cut only the incumbent remains. Proposition 5.2 gives at most $K(2d + 1)$ labels during event selection and $2K(d + 1)$ during validation; both are bounded by $12Kd$. The three-valued terminal output fits this bound as well. These are compatible transitions over the entire schedule.

The coin thresholds can be precomputed for all triples (i, j, w) and stored as $r + 3$ -bit integers. The event thresholds and L, R, K give the stated description bound; no arbitrary real coefficient is used by the executed machine. Formula (7.4) counts every training observation, including ignored observations after a block decision has been latched.

For (7.3), a width- W auditor has at most W labels at its decision cut. The operational upper bound of Theorem 4.2 implies $\kappa_c \leq W$. If $K = \kappa_c < \infty$, its optimal list has a positive margin over c . Take r large, $\rho \leq 2^{-r}$, and then θ, δ small enough that the loss in (7.5) is smaller than that margin. If $K = 1$, no training is needed: round the single response and use Proposition 5.2. Rational

auditors are a subclass of all auditors, proving the remaining inequality. Finally a finite-width successful machine would give finite κ_c , proving the assertion at infinity. $\square$

The dependence on the schedule is essential. The theorem is a constant-factor classification for fixed observation alphabet, not an optimal sample-complexity theorem. In particular R is exponential in L . For an autonomous implementation, a coarse bound on the number of epochs is $T \leq N(r+3) + r+4$; Proposition 6.3 then gives at most $12Kd(T+1)$ states. The free-clock and autonomous statements are different resource bounds. Cover's theorem already exhibits the importance of this distinction [24, Theorem 2 and Section 4]; bounded-state learning itself is not a novelty claim of the present theorem.

Corollary 7.3 (Finite-precision, calibrated certificates). *Suppose the true training kernels differ from those used in an ideal auditor by at most α per training call, and the candidate and target validation laws differ by at most β_c and β_t . The right side of (7.5) is decreased by at most*

$$2 \min\{1, N\alpha\} + \beta_c + \beta_t$$

for the same implemented machine. A calibration error ρ in the known response means is already accounted for separately in that formula. When the training source is unchanged, $\alpha = 0$, regardless of how large N is.

Proof. Couple the training executions until their first mismatch, as in Theorem 6.2. The selected-event payoff has range $[-1, 1]$, giving the factor two. Conditional on a common selected event, its two validation means change by at most the two total variation errors. This uses no independence between a failed coupling and its terminal event. $\square$

8. FINITE-HORIZON SAMPLE-MEMORY BOUNDS

The minimum W_c in Theorem 7.2 permits arbitrarily long finite training. To retain the preparation constraint, let $V_{N,K}(B, Q)$ denote the supremum of the worst-case expected score over the same reset auditors, using at most N training trials and satisfying the specified register bound at each acquisition, random-bit, decision and validation cut. The observation order and the external clock are fixed parts of this interface. Write $V_{N,W}$ when every bound equals W . Padding an audit with ignored trials shows monotonicity in N . This value does not identify machines having only the same terminal dictionary.

A second realization of a response cover gives a polynomial preparation bound. Its counter is larger than the four-state block counter, and both constructions remain useful. The elementary random-walk argument below is classical; the purpose of retaining its proof is to price its use in the complete, calibrated continuation machine.

Lemma 8.1 (A truncated absorbing comparison). *Let Z_i be independent Bernoulli variables with mean u . Fix integers $m, n \geq 1$ and $0 < \theta < 1/2$. Start at $S_0 = 0$, add $2Z_i - 1$ while $|S_{i-1}| < m$, and retain the state forever after either boundary is reached. At time n declare positive exactly when $S_n \geq 0$. The machine has $2m+1$ states. Whenever $|u - 1/2| \geq \theta$, its error probability is at most*

$$e_{m,n}(\theta) = e^{-4\theta m} + e^{-2n\theta^2}. \quad (8.1)$$

If τ is the boundary hitting time without truncation, then, with $x = 2u - 1$,

$$\mathbb{E}_u \tau = \begin{cases} m \tanh(m \operatorname{arctanh} x) / x, & 0 < |x| < 1, \\ m^2, & x = 0, \\ m, & |x| = 1. \end{cases} \quad (8.2)$$

In particular $\mathbb{E}_u(\tau \wedge n) \leq \min(n, m^2)$, and outside the indifference band it is at most $\min(n, m^2, m/(2\theta))$.

Proof. First suppose $1/2 < u < 1$ and put $r = (1 - u)/u$. For the unabsorbed walk $T_j = \sum_{i \leq j} (2Z_i - 1)$, the process r^{T_j} is a nonnegative martingale. Stopping on $\{-m, m\}$ gives

$$\mathbb{P}_u(T_\tau = -m) = \frac{r^m}{1 + r^m} \leq r^m.$$

For completeness, absorption has finite mean: in each block of $2m$ steps, a run of $2m$ positive increments has probability $u^{2m} > 0$ and forces absorption from every interior state. Thus the unabsorbed survival time is bounded by a geometric number of such blocks. Stopping first at $\tau \wedge j$, then using bounded convergence on the stopped exponential martingale, justifies the displayed identity. The estimate $\log((1 + 2\theta)/(1 - 2\theta)) \geq 4\theta$ gives $r^m \leq e^{-4\theta m}$ when $u \geq 1/2 + \theta$.

If the walk has not been absorbed, a negative final answer requires $T_n < 0$. Hoeffding's inequality [25] for variables in $[-1, 1]$ gives

$$\mathbb{P}_u(T_n \leq 0) \leq e^{-2n(u-1/2)^2} \leq e^{-2n\theta^2}.$$

Partitioning a wrong answer according to absorption proves (8.1). Reflection proves the lower-mean case; its nonabsorbed error includes $T_n = 0$ and satisfies the same bound. At $u = 0, 1$ the assertion follows directly.

The drift martingale $T_j - xj$ stopped at $\tau \wedge j$ gives $\mathbb{E}_u T_\tau = x\mathbb{E}_u \tau$, by integrability of τ and boundedness of the stopped walk. The hitting probabilities yield $\mathbb{E}_u T_\tau = m(1 - r^m)/(1 + r^m)$. Since $r = \exp(-2 \operatorname{arctanh} x)$, this is (8.2) when $x \neq 0$. For $x = 0$, stop $T_j^2 - j$ to obtain $\mathbb{E} \tau = m^2$. The endpoint formula is deterministic. Finally $\tanh(my) \leq m \tanh y$ for integer $m \geq 1$ and $y \geq 0$: induction follows from the addition formula for $\tanh$. This proves $\mathbb{E} \tau \leq m^2$; the bound $m/|x|$ follows from $|T_\tau| \leq m$. Truncation can only decrease the mean. $\square$

Theorem 8.2 (A preparation-width family of realizations). *Let $B \subset \Delta(\Omega)$ be compact, $d = |\Omega|$, and let $h_1, \dots, h_K \in [0, 1]^\Omega$, $K \geq 2$, have envelope $v = \min_{b \in B} \max_i (Q - b)h_i$. Round their coordinates to dyadic precision r , obtaining $h_i^{(r)}$, and choose known dyadic numbers $q_i \in [0, 1]$ with $|q_i - Qh_i^{(r)}| \leq \rho$. Suppose q_i also have denominator 2^r ; enlarging r accommodates any specified dyadic calibration. For $m, n \geq 1$ and $0 < \theta < 1/2$, an explicitly clocked fair-bit audit satisfies*

$$N = n(K - 1), \quad W \leq 3dK(2m + 1), \quad (8.3)$$

$$\inf_{b \in B} \mathbb{E}_b D \geq v - 2^{1-r} - 2\rho - 4(K - 1)\theta - 2(K - 1)e_{m,n}(\theta). \quad (8.4)$$

At completed training-trial cuts its width is at most $K(2m + 1)$; at the decision cut it is at most K . Every random-bit cut, the selected event, and both validations are included in W . In particular one can replace $e_{m,n}(\theta)$ by δ by taking

$$m = \left\lceil \frac{\log(2/\delta)}{4\theta} \right\rceil, \quad n = \left\lceil \frac{\log(2/\delta)}{2\theta^2} \right\rceil. \quad (8.5)$$

Proof. For a fixed unknown b , set $g_i(b) = q_i - bh_i^{(r)}$. Compare the current incumbent i with the next index j on fresh observations. Conditional on the observation w , generate a coin with probability

$$\frac{1}{2} + \frac{q_j - q_i - h_j^{(r)}(w) + h_i^{(r)}(w)}{4}. \quad (8.6)$$

This is in $[0, 1]$, is dyadic with denominator 2^{r+2} , and has mean $1/2 + (g_j(b) - g_i(b))/4$. Apply Lemma 8.1, selecting j for a positive answer and i otherwise. Conditional on all earlier stages, the observations and fair bits for this comparison are fresh. The probability of selecting the worse index when $|g_j - g_i| \geq 4\theta$ is at most $e_{m,n}(\theta)$.

Outside these exceptional comparisons each tournament stage loses at most 4θ relative to the better of its two contestants. Induction shows a total loss at most $4(K - 1)\theta$ relative to $\max_i g_i$.

The union probability of exceptional comparisons is at most $(K - 1)e_{m,n}(\theta)$ even though the incumbent is random. All g_i lie in $[-1, 1]$, so their range is at most two. Consequently

$$\mathbb{E}_b g_i(b) \geq \max_i g_i(b) - 4(K - 1)\theta - 2(K - 1)e_{m,n}(\theta).$$

The calibration changes selection and evaluation by at most 2ρ . For each coordinate rounding, $\|h_i - h_i^{(r)}\|_\infty \leq 2^{-r}$, so $|(Q - b)(h_i - h_i^{(r)})| \leq 2^{1-r}$. A nested random event with mean $h_i^{(r)}$ has exactly that rounded payoff; hence (8.4).

At a completed trial retain the incumbent and the absorbing walk state. The stage number j is supplied by the predetermined clock, not by a sample-dependent register. During coin generation additionally retain w and at most three sampler labels, by Proposition 5.2. This gives $3dK(2m + 1)$ labels. The terminal event sampler, held event, and validation storage use at most $K(2d + 1)$, $K(d + 1)$, and $2K(d + 1)$ labels, respectively, as in Theorem 7.2; all are bounded by the stated W . The stage boundary discards the walk and initializes the next one. After absorption, remaining observations and bits in that stage are ignored, but are still charged to the fixed schedule. Thus the count N includes padding. Each active coin uses $r + 2$ fair bits. A coarse bound for the full number of epochs is $T \leq N(r + 3) + r + 4$.

The finite transition description consists of the dyadic coin table, the boundary and horizon integers, and the nested-event thresholds. It has $O(K^2 d(r + 3) + \log m + \log n + \log K)$ bits under binary encoding; the constants depend on a fixed encoding convention. No unknown b occurs in that description. The autonomous conversion stores the clock and has width at most $(T + 1)W$, rather than W . Substitution of (8.5) proves the last assertion. $\square$

Corollary 8.3 (Two general preparation regimes). *For a strict K -response cover and a prescribed positive score margin, there are whole-schedule rational realizations of that margin with width at most $12dK$ and exponential block preparation count, and with width at most $3dK(2m + 1)$ and preparation count $n(K - 1)$ as in (8.5). Both constrain every training cut. Their resource pairs are upper bounds for $\mathbb{V}_{N,W}$, not a claim of an optimal frontier for every compact image.*

Proof. Choose rounding, calibration, band and error budgets whose sum is smaller than the strict margin. Apply Theorem 7.2 or Theorem 8.2. A one-response cover needs no training and is implemented directly by the rational event sampler. $\square$

9. AN EXACT STATISTICAL FRONTIER AT ALL CUTS

The preceding realizations are uniform upper bounds. A family with an exact answer illustrates what an all-algorithm statistical lower bound requires. Unlike exact computation of a prescribed word function, the objective here is the best decision risk among all rules, including stochastic rules, under a fixed preparation schedule and full profile.

Let $\Theta = \{1, \dots, M\}$ and fix a known prior $\pi_1 \geq \dots \geq \pi_M \geq 0$, $\sum_j \pi_j = 1$. At time t , an independent indicator E_t has known mean $q_t \in [0, 1]$. The report is $Y_t = \vartheta$ if $E_t = 1$, and $Y_t = *$ otherwise. All indicators are independent of the parameter and the machine's private randomization. The unknown parameter ϑ is fixed during the n observations. The machine starts in a parameter-independent state, has at most K_t labels after report t , and finally guesses ϑ . Its initial and final randomization is part of the same finite causal machine. The objective is Bayesian probability of a correct answer under π . Parameter-dependent loss is specified here and is not identified with the one-row physical signed-score problem.

Write $P_k = \sum_{j=1}^k \pi_j$ and put

$$k_t = \min\{M, K_t, K_{t+1}, \dots, K_n\}, \quad a = \prod_{s=1}^n (1 - q_s), \quad w_t = q_t \prod_{s=t+1}^n (1 - q_s). \quad (9.1)$$

Thus $a + \sum_t w_t = 1$ and k_t is nondecreasing in t .

Theorem 9.1 (Exact all-cut revelation frontier). *Over all stochastic causal machines satisfying the specified profile, the optimal Bayesian success probability is*

$$S_{\mathbf{q}, \mathbf{K}}(\pi) = a\pi_1 + \sum_{t=1}^n w_t P_{k_t}. \quad (9.2)$$

A single deterministic machine attains this value simultaneously for all $\mathbf{q} \in [0, 1]^n$. The upper bound therefore applies to all approximate statistical decision rules, not just to machines implementing a chosen selector exactly.

Proof. Let L be the last time with $E_L = 1$, with $L = 0$ when no revelation occurs. Then $\mathbb{P}(L = 0) = a$ and $\mathbb{P}(L = t) = w_t$. These events, and every particular pattern of indicators, are independent of ϑ .

First consider any stochastic machine. Realize all of its transition and output kernels by independent random variables indexed by time, state and input, and also realize its initial randomization. This is possible on a product space independent of ϑ and the indicators. Conditional on the entire random table the machine is deterministic. This conditioning is an analysis device, not a free persistent random seed in an implementation.

Fix that table and an indicator pattern with last revelation $t > 0$. The map from ϑ to the terminal guess factors through the state at every time $s \geq t$: all reports after t are the same symbol $*$, so the remaining transitions and output are fixed maps. The set of possible terminal guesses therefore has cardinality at most $\min_{s \geq t} K_s$, and at most M guesses can be correct. Its prior success probability is bounded by the mass of the k_t largest prior atoms, namely P_{k_t} . Earlier revelations and any parameter information encoded before t do not evade this future bottleneck. On the no-revelation pattern, the output is independent of ϑ and its success probability is at most π_1 . Integrate these bounds over the random table and indicator patterns to obtain the upper bound in (9.2). In particular no extremality claim about a nonconvex minimax set has been used.

For attainment, use the labels $\{1, \dots, k_t\}$ at time t , start with label one, and use the following deterministic updates. On $*$, hold the label. On a revelation $j \leq k_t$, replace it by j ; on a revelation $j > k_t$, replace it by one. The holding update is legal because $k_{t-1} \leq k_t$. At termination output the label. Conditional on $L = t > 0$, exactly the parameters $1, \dots, k_t$ are correct; on $L = 0$, exactly parameter one is correct. This proves equality. The updates use only the prescribed profile and prior ordering, not the probabilities q_t , establishing simultaneous attainment. $\square$

Corollary 9.2 (Sharp preparation–peak tradeoff). *For the uniform prior, a constant reveal probability $0 < q < 1$, and constant width $1 \leq K \leq M$, write $a_n = (1 - q)^n$. Then*

$$S_{n, K} = \frac{a_n + K(1 - a_n)}{M}. \quad (9.3)$$

For $n \geq 1$ and target success $s \in [1/M, 1]$, the least feasible width is

$$K_{n, s} = \max \left\{ 1, \left\lceil \frac{Ms - a_n}{1 - a_n} \right\rceil \right\}, \quad (9.4)$$

provided this integer is at most M ; otherwise the target is infeasible at that preparation count. If $s > 1/M$ and $K > Ms$, the least preparation count for width K is

$$n_{K, s} = \left\lceil \frac{\log((K - 1)/(K - Ms))}{-\log(1 - q)} \right\rceil. \quad (9.5)$$

If $K \leq Ms$ and $s > 1/M$, no finite n attains the target, except that equality can be attained when revelations are certain, a case excluded by $q < 1$. Width one gives success exactly $1/M$ for all n .

Proof. Here $P_k = k/M$ and $k_t = K$. Since $\sum_t w_t = 1 - a_n$, Theorem 9.1 gives (9.3). Solving $a_n + K(1 - a_n) \geq Ms$ first for K and then for n proves the formulas. For $K > 1$, the limiting value K/M is approached strictly from below when $0 < q < 1$, which proves the boundary assertion. $\square$

Corollary 9.3 (Exact cost of a changed checkpoint). *Replacing a profile $\mathbf{K}$ by a tighter profile $\widetilde{\mathbf{K}}$ loses exactly*

$$\sum_{t=1}^n w_t (P_{k_t} - P_{\widetilde{k}_t}).$$

Inserting a charged holding cut of width H immediately after time s replaces k_t by $\min(k_t, H)$ for $t \leq s$, and leaves the later terms unchanged. In particular a single narrow early cut does not constrain information acquired after that cut.

Proof. Subtract (9.2) for the two profiles. A holding cut is a further bottleneck exactly for revelations that precede it. The same deterministic construction realizes the refined schedule. $\square$

For example, with $M = 4$, $q_1 = q_2 = q_3 = 1/2$, and $(K_1, K_2, K_3) = (4, 1, 4)$, the effective profile is $(1, 1, 4)$ and the optimal uniform-prior success is $5/8$. Four labels at the terminal cut alone would instead give $29/32$. These are risks of complete experiments, not counts of residual classes of one word function. For a nonuniform prior the same theorem weights every surviving label by its actual mass. The nested optimal label sets are what makes all cut bounds jointly attainable. For a general loss or observation channel such nesting need not hold; compatible continuation updates, not independently optimized dictionaries, remain necessary.

The exact formula is Bayesian. Its upper bound also obstructs any uniform worst-case guarantee exceeding that Bayesian value, but the attaining prior-ordered machine is not asserted to be minimax. In particular an uncharged persistent random permutation cannot be used to turn its Bayesian success into a minimax statement. The prior, observation channel and all register cuts have been declared explicitly.

10. TRANSPORT PRICED BY ACTIVE UPDATES

A changed training source need not affect a latched computation at every subsequent scheduled preparation. The relevant quantity is the defect of its induced retained-state transition, not the number of raw reports that happen to be present. This distinction gives a sharper transfer bound without assuming equality of the training laws.

Theorem 10.1 (Occupation-weighted retained-law transfer). *Consider a fixed finite execution with common initial state law μ and retained-state transition kernels P_t, Q_t on the same finite state spaces, $1 \leq t \leq T$. The kernels may depend on a fixed unknown parameter; all inequalities below hold separately for each parameter. Let*

$$d_t(x) = \|P_t(x, \cdot) - Q_t(x, \cdot)\|_{\text{TV}}, \quad \mu_{t-1} = \mu P_1 \cdots P_{t-1}.$$

For the same terminal readout kernel F ,

$$\|\mu P_1 \cdots P_T F - \mu Q_1 \cdots Q_T F\|_{\text{TV}} \leq \min \left\{ 1, \sum_{t=1}^T \int d_t d\mu_{t-1} \right\}. \quad (10.1)$$

If $d_t(x) \leq \alpha_t \mathbf{1}_{A_t}(x) + \zeta_t(x)$, the right side can be replaced by the truncated sum $\sum_t \alpha_t \mathbb{P}_P(X_{t-1} \in A_t) + \sum_t \mathbb{E}_P \zeta_t(X_{t-1})$. For a readout in $[-1, 1]$ multiply the bound by two; for an indicator no factor two is needed. The reverse bound with P, Q exchanged is also valid.

Proof. For signed measures, expand the product difference in the order

$$\mu P_1 \cdots P_T - \mu Q_1 \cdots Q_T = \sum_{t=1}^T \mu P_1 \cdots P_{t-1} (P_t - Q_t) Q_{t+1} \cdots Q_T.$$

Integration of a row difference against μ_{t-1} has total variation at most $\int d_t d\mu_{t-1}$. All suffix kernels and F contract total variation. The triangle inequality and the bound one for probability-law

distance prove (10.1). Substitute the rowwise estimate for its occupation version. The expectation bounds follow from the oscillation of the readout. $\square$

Here the retained state must support the displayed transition kernels. For independent reset observations and a time-inhomogeneous finite machine this is automatic, even when the selected row depends on the state. For a non-reset physical process one must include the necessary analysis state or use its history kernels; ignoring a report need not remove its influence on subsequent physical dynamics. No Markov closure of an arbitrary compressed physical state is assumed.

Corollary 10.2 (Calibrated training with absorption). *Run the audit of Theorem 8.2 under an ideal reset source and a perturbed reset source whose row laws differ by at most α per active call. Use the same implemented coin and event randomizers. If A is the number of active comparison updates under the ideal execution, the change in its selected-response score is at most*

$$2 \min(1, \alpha \mathbb{E}A) + \beta_c + \beta_t, \quad \mathbb{E}A \leq (K - 1) \min(n, m^2), \quad (10.2)$$

where β_c, β_t bound candidate and target validation-law changes. A comparison with conditional coin mean u has the sharper active expectation $\mathbb{E}_u(\tau \wedge n)$ in (8.2). Uniform gaps $|u - 1/2| \geq \eta$ give the bound $\min(n, m^2, m/(2\eta))$ for that comparison.

Proof. At a completed comparison trial, marginalize the fresh coin bits into its transition kernel. Before absorption, data processing bounds its row defect by the source defect α . After absorption its retained transition is the same identity under both sources; its defect is zero. At predetermined stage boundaries the common reset of the walk is also defect-free. Theorem 10.1 therefore bounds the terminal incumbent-law distance by $\alpha \mathbb{E}A$. The score associated with an incumbent lies in $[-1, 1]$. Conditional on a common incumbent and common nested event, the two validation expectations change by at most $\beta_c + \beta_t$. This proves the score estimate. Condition on the incumbent at each stage and apply Lemma 8.1; summing proves the active-count bound. Marginalizing the coin here is a proof of a law estimate, not a waiver of its implementation states, which remain charged in (8.3). $\square$

The estimate is for the output or retained computation, not for the complete raw-observation transcript. A padded sequence of N changed reports can still have total variation of order $N\alpha$ even after the machine has stopped using them. Conversely, all N preparations remain in the resource ledger even when only A updates influence the answer. Finite row-rounding defects can be inserted as the terms ζ_t in Theorem 10.1; a changed initial law adds its total variation separately. Thus source-preserving transport is the case $\alpha = 0$, while the active bound also has content for $\alpha > 0$ and does not assert generic sample-count-free stability.

11. WEIGHTED CONTINUATION OBSTRUCTIONS

We next give a distribution-sensitive version of the wordwise obstruction. It applies to finite-valued terminal decisions and is useful even when a word of small probability should not be assigned the same weight as a likely one.

Let $f : \Sigma_1 \times \cdots \times \Sigma_L \rightarrow \mathcal{A}$ have finite range. The residual of a prefix x at cut t is the function $v \mapsto f(xv)$ on the suffix alphabet. Write $r_t(f)$ for the number of distinct residuals. A stochastic machine has a charged register at every cut; all persistent randomness is included in this register. Its wordwise error is $e(w) = \Pr(\hat{f}(w) \neq f(w))$.

Theorem 11.1 (Finite-valued realization and weighted cut bound). *Exact stochastic computation of f requires $K_t \geq r_t(f)$ at every cut. The whole residual profile is simultaneously attained by a deterministic machine.*

Fix a cut, prefixes $x_1, \dots, x_r$, and a graph on $[r]$. For each edge ij choose a suffix v_{ij} such that $f(x_i v_{ij}) \neq f(x_j v_{ij})$. Let $\mu(w) \geq 0$ be arbitrary word weights. Choose edge weights $\lambda_{ij} \geq 0$ satisfying the word capacity constraints

$$\sum_{ij: w=x_i v_{ij} \text{ or } w=x_j v_{ij}} \lambda_{ij} \leq \mu(w) \quad \text{for every word } w. \quad (11.1)$$

Every machine with K states at this cut satisfies

$$\sum_w \mu(w) e(w) \geq \min_{\phi: [r] \rightarrow [K]} \sum_{ij: \phi(i)=\phi(j)} \lambda_{ij}. \quad (11.2)$$

The right side is a weighted monochromatic-edge obstruction. Maximizing it over admissible witnesses and weights gives a finite, distribution-sensitive lower bound. The bound can be an equality with positive error.

Proof. Let $a_i(m)$ be the state distribution after prefix x_i . For a fixed edge and a common state m , the future output distribution on suffix v_{ij} is the same for both prefixes. Since their correct outputs are different, the sum of their two conditional error probabilities is at least one. Multiplying by the smaller state mass and summing gives

$$e(x_i v_{ij}) + e(x_j v_{ij}) \geq \sum_{m=1}^K \min\{a_i(m), a_j(m)\}. \quad (11.3)$$

Weight the inequalities by λ_{ij} and use (11.1). The resulting lower bound is

$$F(a_1, \dots, a_r) = \sum_{ij} \lambda_{ij} \sum_m \min\{a_i(m), a_j(m)\}.$$

This is a concave continuous function on $(\Delta_K)^r$. Every point of that product is a convex combination of its deterministic vertices; concavity implies that its value is at least the minimum over those vertices. A vertex is a labeling $\phi: [r] \rightarrow [K]$, and its value is exactly the right side of (11.2). This proves the bound without treating stochastic encodings as deterministic encodings.

For zero error, choose one distinguishing suffix for every two different residuals. Equation (11.3) forces the corresponding state supports to be disjoint. Hence at least $r_t(f)$ states are needed. For the converse, keep the residual class of the observed prefix. Equal residuals have equal shifts after every next input, so the next class is well defined. At the last epoch the residual is the output label itself. These quotient transitions realize all widths simultaneously.

For a positive-error equality, let the only relevant words be two one-letter prefixes with different required labels, weighted p and $1 - p$, where $0 < p < 1$. Put $K = 1$ and $\lambda_{12} = \min(p, 1 - p)$. The cut obstruction is $\min(p, 1 - p)$, attained by always emitting the more likely label. Thus the weighted constant is sometimes exact; universal sharpness for all graphs or input distributions is not asserted. With unit weights on the same two words the total-error lower bound is one, also attained by a constant output; this is equality in the earlier $(r - K)/(r - 1)$ bound at $r = 2$, $K = 1$. $\square$

Corollary 11.2 (A Bayes-regret obstruction for constrained audits). *Fix a finite action set $\mathcal{A}$, candidate class B , and a probability measure ν on B . For a training word w , let $p_w(b)$ be its likelihood and let $f_a(b)$ be the expected validation payoff. Put*

$$R_w(a) = \int_B p_w(b) f_a(b) d\nu(b), \quad S_\nu = \sum_w \max_a R_w(a).$$

Choose a maximizing label $f(w)$. Set $\mu(w) = R_w(f(w)) - \max_{a \neq f(w)} R_w(a)$, taking zero when a maximum is tied. Let $\Gamma_{t,K}(f, \mu)$ be any lower bound supplied by (11.2). Then every width- K audit

at cut t has worst-case expected payoff at most

$$S_\nu - \Gamma_{t,K}(f, \mu). \quad (11.4)$$

Proof. The conditional distribution of the auditor's output for a fixed word is independent of the unknown candidate. Its Bayes payoff is therefore $\sum_{w,a} R_w(a) \Pr(\hat{f}(w) = a)$. Subtracting this from S_ν gives at least $\sum_w \mu(w)e(w)$. Apply the theorem, and use that a minimum over candidates is at most the average under ν . $\square$

The corollary does not exchange a maximum and minimum over a nonconvex fixed-width machine class. The prior is a lower-bound device, not a randomization resource supplied to the candidate. With a unique maximizing label and strictly positive margins at every word, equality with S_ν forces exact wordwise computation. This makes explicit the uniqueness step used in the odd-sample binary theorem below. For even sample size, the balanced-count words have zero likelihood gap. A rule can differ on those ties without losing value; equality of values then does not force one prescribed total Boolean rule, and the odd-sample width profile cannot simply be asserted for that case.

12. JOINT RESIDUAL BUDGETS AND GROWING FAMILIES

We now put lower bounds on several cuts into one statistical budget. Summing separate cut bounds without controlling their common words would count the same regret more than once.

Theorem 12.1 (A shared-capacity multicut obstruction). *Let $f : \prod_{i=1}^L \Sigma_i \rightarrow \mathcal{A}$ and $\mu(w) \geq 0$ be as in Theorem 11.1. At each chosen cut t , choose a finite prefix graph with an edge $e = (u, v)$ carrying a distinguishing suffix s_e , and a weight $\lambda_{t,e} \geq 0$. Impose the single family of constraints*

$$\sum_{t,e: w=us_e \text{ or } w=vs_e} \lambda_{t,e} \leq \mu(w) \quad \text{for every complete word } w. \quad (12.1)$$

For a graph at cut t , write

$$\Gamma_{t,K_t}(\lambda) = \min_{\phi: \text{prefixes} \rightarrow [K_t]} \sum_{e=(u,v): \phi(u)=\phi(v)} \lambda_{t,e}.$$

Every stochastic machine of profile $\mathbf{K}$ satisfies

$$\sum_w \mu(w) \Pr(\hat{f}(w) \neq f(w)) \geq \sum_t \Gamma_{t,K_t}(\lambda). \quad (12.2)$$

Consequently, with R_w, S_ν, μ defined in Corollary 11.2, every audit of this profile and the fixed training schedule has worst-case payoff at most

$$S_\nu - \sup_{\lambda \text{ satisfying (12.1)}} \sum_t \Gamma_{t,K_t}(\lambda). \quad (12.3)$$

For fixed graphs the supremum is the value of a finite linear program.

Proof. For each edge use the overlap inequality (11.3). Multiply it by $\lambda_{t,e}$ and sum over all cuts and edges. The left side is bounded above by the left side of (12.2), by the shared capacity constraint. At each cut the resulting overlap expression is concave in its prefix state distributions, hence at least its minimum over deterministic labelings, as proved in Theorem 11.1. Adding these bounds proves (12.2). Although prefix distributions at different cuts are compatible, minimizing them separately only weakens the bound; it does not assume independent realizability. The Bayes-regret argument of Corollary 11.2 proves (12.3).

For the last assertion introduce variables z_t and impose, for each labeling ϕ of each finite graph,

$$z_t \leq \sum_{e=(u,v): \phi(u)=\phi(v)} \lambda_{t,e}.$$

Maximize $\sum_t z_t$ subject to these inequalities, nonnegative edge weights, and (12.1). There are finitely many constraints. This is a computable obstruction, not a claim that every multicut statistical optimum equals this relaxation. $\square$

12.1. Actual physical word weights. For n all-on trials of the private marked collision subfamily, a complete training word w specifies the two report bits and the mark of each trial. With N_{ab} its report-pair counts,

$$p_w(p, q) = 2^{-n} p^{N_{10}+N_{11}} (1-p)^{N_{00}+N_{01}} q^{N_{01}+N_{11}} (1-q)^{N_{00}+N_{10}}. \quad (12.4)$$

The formula has the continuous endpoint interpretation $0^0 = 1$. For any finitely supported prior on actual pairs $(p, q) \in [0, 1]^2$, use this likelihood in $R_w(A) = \int p_w(p, q)(Q - b(p, q))\mathbf{1}_A d\nu$, with all marked events $A \subseteq \Omega$ as actions. Randomized events are included by linearity. Equations (12.1) and (12.3) then give a finite all-algorithm obstruction for that full training profile, with the true physical word probabilities. Changing only Q by total variation β changes every audit's payoff by at most β .

This specialization bounds all stochastic selectors on the fixed all-on training schedule, not merely exact implementations of a majority rule. It does not assert a new numerical lower bound above three for every adaptive physical audit. Adaptive training has controller-dependent word likelihoods and must instead be included in the full experiment of (6.3); using (12.4) unchanged for such a likelihood would be incorrect. The feedback-uniform lower bound used below remains Lemma 24.2.

12.2. A family with matching end-to-end growth.

Theorem 12.2 (Point-source decision spectra). *Let $d \geq 2$, $B_d = \{\delta_1, \dots, \delta_d\}$ on $[d]$, and let Q_d be uniform. For every $K \geq 1$,*

$$\mathfrak{V}_K(B_d, Q_d) = 1 - \frac{\lceil d/K \rceil}{d}. \quad (12.5)$$

For $1 - 2/d < c < 1 - 1/d$, the decision complexity is exactly d , and the whole-schedule peak satisfies

$$d \leq W_c(B_d, Q_d) \leq W_c^{\text{rat}}(B_d, Q_d) \leq 2d.$$

The upper bound uses one training observation and two validations, with no randomization. Thus the end-to-end width grows linearly in this nontrivial sequence of compact behavior images.

Proof. Assign each point source to a response that maximizes its score. Some response is assigned a set S of at least $\lceil d/K \rceil$ indices. For $s = |S| < d$, the mean of its scores on S equals

$$\frac{d-s}{d} \left(\frac{1}{d-s} \sum_{i \notin S} h_i - \frac{1}{s} \sum_{i \in S} h_i \right) \leq 1 - \frac{s}{d}.$$

For $s = d$ the mean is zero. At least one assigned source has score no larger than this mean, proving the upper bound. Partition $[d]$ into at most K groups of size at most $\lceil d/K \rceil$ and use their complements as events. This attains the lower bound.

In the stated interval the formula forces $K \geq d$. A single training observation identifies the point source i exactly. Keep i and choose the complement of $\{i\}$. During validation retain i and the candidate indicator, at most $2d$ states, then emit the target indicator minus it. This computes the prescribed signed score on every input word, including words that have zero probability under the point-source family. Its expected score is $1 - 1/d$. The lower bound follows from the necessary decision-cut width. $\square$

12.3. A sharp simultaneous storage profile. A second class separates overlapping retention obligations. Independent uniform labels $X_j \in [q_j]$ are revealed at scheduled epochs a_j and must be reported at epochs $b_j > a_j$. No later input reveals an active label, and previous outputs cannot be read as external memory. Let $A_t = \{j : a_j \leq t < b_j\}$ and $D_t = \prod_{j \in A_t} q_j$.

Theorem 12.3 (Overlapping retention intervals). *Every stochastic machine whose probability of correctly reporting all labels is at least $1 - \varepsilon$ has*

$$K_t \geq (1 - \varepsilon)D_t \quad \text{at every cut } t. \quad (12.6)$$

The exact profile $K_t = D_t$ is attained simultaneously by a deterministic machine. Thus the minimum peak number of bits lies between $\log_2 \max_t D_t + \log_2(1 - \varepsilon)$ and $\lceil \log_2 \max_t D_t \rceil$ for $\varepsilon < 1$.

Proof. Fix a cut and give the decoder, as additional information, all labels outside A_t , whether past or future. They are independent of the uniform active vector. Conditional on them, every future prediction of that vector factors through the K_t -valued retained state and fresh randomness. If $a(m | x)$ is its conditional encoder and $d(x | m)$ the induced joint prediction law, its success probability is at most

$$\frac{1}{D_t} \sum_m \sum_x a(m | x) d(x | m) \leq \frac{1}{D_t} \sum_m \sum_x d(x | m) = \frac{K_t}{D_t}.$$

Correctness of all reports implies correctness of the active reports, so averaging over the independent additional information proves (12.6). For the upper bound, keep precisely the vector of active labels, append a label when revealed, and delete it when reported. Every update is deterministic and respects the schedule. The bit bounds follow by taking logarithms and rounding for an actual binary register. The lower bound is a counting argument, not an independence assumption about labels after conditioning on a compressed state. $\square$

13. FINITE-DATA TESTING ON THE ORIGINAL BEHAVIOR IMAGE

13.1. The reset interface. Let $B \subset \prod_{j=1}^J \Delta(\Omega_j)$ be compact, and let $Q = (Q_j)$ be fixed. Set $d_j = |\Omega_j|$, $D = \sum_j (d_j - 1)$, $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$, and $f_{j,A}(b) = Q_j(A) - b_j(A)$. Include the empty event. No convexity of B is assumed.

A candidate b is fixed for the whole audit. The auditor obtains n independent samples of each b_j , with independence between rows. In a causal experiment, row j means a specified parameter and deterministic feedback tree; the reset interface supplies that row. It is an additional experimental interface, not information available to the online simulator. A complete sample contains the terminal mark. Private randomness and any per-run hidden selector are reset between runs. The design of the candidate is not resampled.

Write $C_{j,w}$ for the counts, and $\mathcal{N}_n$ for all count arrays with $\sum_w C_{j,w} = n$. Their probability is the polynomial

$$p_C(b) = \prod_j \frac{n!}{\prod_w C_{j,w}!} \prod_w b_j(w)^{C_{j,w}}.$$

A randomized audit is an array $c = (c_{C,i})$ with $c_C \in \Delta(\mathcal{I})$. After training it chooses $i = (j, A)$ according to c_C . Independent candidate and target validation runs give the signed score $\mathbf{1}_A(\text{target}) - \mathbf{1}_A(\text{candidate})$. Its expected value is

$$g_c(b) = \sum_{C,i} p_C(b) c_{C,i} f_i(b), \quad v_n^{\text{aud}}(B, Q) = \max_c \min_{b \in B} g_c(b). \quad (13.1)$$

Thus every audit payoff is polynomial of total degree at most $Jn + 1$ in the behavior coordinates. The probability of a large observed score in a single audit is not asserted to equal its expectation; additional independent validation would be required for a specified confidence level.

Theorem 13.1 (Reset-audit duality and finite-sample approximation). *Let $\delta = \min_{b \in B} \max_i f_i(b)$. Then all the extrema below are attained, and*

$$v_n^{\text{aud}}(B, Q) = \min_{\mu \in \mathcal{P}(B)} \sum_{C \in \mathcal{N}_n} \max_{i \in \mathcal{I}} \int_B p_C(b) f_i(b) d\mu(b), \quad (13.2)$$

$$0 \leq \delta - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}. \quad (13.3)$$

The values are nondecreasing in n . If $a = \dim \text{aff } B$, a minimizing measure in (13.2) can be supported on at most $\binom{a+Jn+1}{a}$ points.

An explicit feasible audit attaining the lower bound $\delta - \sqrt{D/n}$ chooses an event maximizing $Q_j(A) - C_j(A)/n$, with a fixed deterministic tie rule. If the rational target probabilities are given, this is a rationally specified finite rule. With a free schedule clock, a conservative retained-state bound is

$$H(|\mathcal{I}| + 1)(n + 1) \sum_j d_j,$$

where H is the number of states needed to execute and record one causal transcript. This budget is charged to the auditor, not to the simulator.

Proof. The coefficient arrays form a compact convex product of finite simplexes. The integrand is continuous in b and affine in c . The probability measures on compact B form a compact convex set for weak convergence. The minimum over b equals the minimum over such measures for fixed c . Finite-dimensional minimax (or its compact measure version) therefore exchanges the two extrema. Maximization over each independent coefficient simplex selects its largest coordinate integral, giving (13.2). Continuity and compactness give attainment. This step uses measures on the original B , not feasible private realizations of their barycenters.

For any audit, $g_c(b) \leq F(b) = \max_i f_i(b)$. At a minimizer of F this gives $v_n^{\text{aud}} \leq \delta$. The zero event gives $v_n^{\text{aud}} \geq 0$. For the stated explicit rule put $\hat{b}_j = C_j/n$ and $\rho = \max_j \|\hat{b}_j - b_j\|_{\text{TV}}$. Comparing the empirical maximizer with a true maximizing event gives the pointwise regret bound

$$F(b) - f_{i(C)}(b) \leq 2\rho.$$

For a multinomial empirical measure,

$$\mathbb{E} \|\hat{b}_j - b_j\|_{\text{TV}}^2 \leq \frac{d_j}{4} \sum_w \mathbb{E} (\hat{b}_j(w) - b_j(w))^2 = \frac{d_j(1 - \sum_w b_j(w)^2)}{4n} \leq \frac{d_j - 1}{4n}.$$

Consequently $\mathbb{E} \rho \leq (\sum_j \mathbb{E} \|\hat{b}_j - b_j\|_{\text{TV}}^2)^{1/2} \leq \sqrt{D/n}/2$. Averaging the regret uniformly over B proves (13.3).

From $n + 1$ unordered samples in each row, discard one uniformly at random. The remaining counts have exactly the n -sample law, independently across rows. This realizes every n -sample audit as a randomized $(n + 1)$ -sample audit and proves monotonicity.

Choose affine coordinates on $\text{aff } B$. All the functions $p_C f_i$ lie in the span of monomials of total degree at most $Jn + 1$, a space of dimension at most $s = \binom{a+Jn+1}{a}$ including the constant. The moment vector of a probability measure is in the convex hull of the evaluation vectors; their constant coordinate is one, so their affine dimension is at most $s - 1$. Eliminating affine dependencies gives a representation by at most s points. It preserves every integral in (13.2), and hence preserves the objective.

Finally, each count is in $\{0, \dots, n\}$. The indicated product register stores all counts, the within-run state, and the selected event for validation. Only the latter event and the within-run buffer need survive once training ends, so the product bound is conservative. Randomizing an event uses fresh randomness. The schedule, reset access and terminal mark readout were declared before the theorem. No private simulator state is read by this implementation. $\square$

The dual measure chooses a candidate once, after which its samples are independent conditional on that candidate. Replacing it by $\bar{b} = \int b d\mu$ would replace $\int p_C(b) f_i(b) d\mu$ by $p_C(\bar{b}) f_i(\bar{b})$, generally a different quantity. It would also replace the joint replicated law by a product of its marginals. Neither operation is part of the theorem. This is the operational meaning of the moment side, rather than a same-budget convexification theorem.

Proposition 13.2 (Both sides sampled). *Suppose the target reference Q^b varies with the candidate, as it does when visible-selector weights are design variables. Supply independent training samples from both b and Q^b , and use their empirical difference. If their row dimensions are D_P, D_Q , the same construction has regret at most $\sqrt{D_P/n} + \sqrt{D_Q/n}$. Minimax and attainment hold on the compact image of pairs (b, Q^b) . In particular a variable reference law must not be treated as known fixed input to Theorem 13.1.*

Proof. The error in any event discrepancy is at most the sum of the maximal row variations of the two empirical tables. Apply the preceding second moment bound separately and then the empirical-maximizer regret inequality. Count probabilities are products for the two tables. They are continuous polynomials on the compact image of pairs, so the same minimax argument applies without changing the simulator signature. $\square$

13.2. A matching uniform sampling obstruction.

Theorem 13.3 (Binary reset lower bound). *Take one binary row with target $\text{Ber}(1/2)$ and candidate class $B_h = \{\text{Ber}(1/2 - h), \text{Ber}(1/2 + h)\}$, where $0 < h < 1/2$. Then*

$$\delta = h, \quad v_n^{\text{aud}} = h \|\text{Ber}(1/2 + h)^{\otimes n} - \text{Ber}(1/2 - h)^{\otimes n}\|_{\text{TV}}. \quad (13.4)$$

For every $n \geq 1$, choosing $h = 1/(8\sqrt{n})$ gives

$$\delta - v_n^{\text{aud}} \geq \frac{1}{16\sqrt{n}}.$$

Thus the exponent in (13.3) is sharp uniformly over compact candidate classes, already at affine dimension one. This does not assert a degree lower bound for revealed-behavior polynomials, nor that this two-point class equals an unrestricted private-row image.

Proof. Only the two complementary singleton events and the zero event matter. A randomized choice is a statistic d in $[-1, 1]$. The two expected payoffs are $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$. Their minimum is at most half their sum, which is at most $h\|P_+^n - P_-^n\|_{\text{TV}}$. Conversely take the sign of the likelihood difference, assigning zero at a tie. Complementing every sample interchanges P_+^n and P_-^n and changes that sign. The two payoffs are equal, and their average attains the variation bound. The statistic depends only on the count, so it is an admissible audit. The symmetric measure on the two candidates attains the dual value. Every Dirac measure has dual objective h , whereas for finite n the two product laws are equivalent and their variation is strictly less than one. Thus no dual minimizer is a Dirac measure. Their barycenter is the target law, but replacing the dual by that barycenter would incorrectly give value zero instead of the strictly positive value in (13.4).

For $h \leq 1/8$, elementary integration of $(1 - u^2)^{-1}$ gives

$$\text{KL}(\text{Ber}(1/2 + h) \parallel \text{Ber}(1/2 - h)) = 2h \log \frac{1 + 2h}{1 - 2h} \leq 16h^2.$$

The product relative entropy is at most $16nh^2$. Pinsker's inequality bounds its total variation by $\sqrt{8nh^2}$. At the specified h this is less than $1/2$. Substituting in (13.4) proves the claim. The inequality uses the classical two-point testing mechanism; the restriction being measured is the reset budget of the present game. $\square$

13.3. Transport of one fixed audit.

Proposition 13.4 (Reset cost under model perturbation). *Let q range over the same compact set of executable row designs in two models. Suppose uniformly in q, j that*

$$\|P_j^q - \tilde{P}_j^q\|_{\text{TV}} \leq \alpha, \quad \|Q_j - \tilde{Q}_j\|_{\text{TV}} \leq \beta.$$

For the same coefficient rule c , evaluated with each model's own target validation,

$$|g_c(q) - \tilde{g}_c(q)| \leq (2Jn + 1)\alpha + \beta. \quad (13.5)$$

The deficiencies themselves differ by at most $\alpha + \beta$, at the same resource budget. If the source behavior is unchanged, a fixed audit loses at most β , independently of its reset count.

Proof. A product coupling of the Jn training runs fails with probability at most $Jn\alpha$. Taking counts and choosing the event are contractions. A score with fixed target discrepancies lies in $[-1, 1]$, so the change in its expectation from this training-law perturbation is at most $2Jn\alpha$. At each event the discrepancy changes by at most $\alpha + \beta$. This proves (13.5). For deficiency, every row variation changes by at most $\alpha + \beta$; taking the maximum and then the minimum preserves that bound. No new simulator is inserted. $\square$

For a rational polynomial image, a rational rule and a rational threshold, strict positivity of $g_c(q)$ above that threshold is a first-order formula over the original row simplexes. Real quantifier elimination decides it. The same is true of an existential rule followed by this universal condition. This is an exact finite verification statement, not an efficient algorithm. The explicit empirical rule below will be verified by a uniform analytic inequality, rather than by storing its enormous complete table.

14. CONTINUATION COMPLEXITY OF A FINITE-DATA TEST

The reset budget does not by itself bound the auditor's retained memory. We now impose both constraints. At scheduled epoch t a streaming auditor reads one letter x_t from a finite alphabet X_t , updates a register, and emits no information to any auxiliary store. It makes its decision on reading the last letter. Transition rows may depend on the epoch; the clock is free. All other information crossing a cut belongs to the register. Fresh randomization is allowed. An independent persistent selector, when allowed, selects a complete auditor before any letters are read.

For a Boolean decision $\psi : X_1 \times \cdots \times X_n \rightarrow \{-1, 1\}$, let $u \sim_t v$ mean that

$$\psi(us) = \psi(vs) \quad \text{for every } s \in X_{t+1} \times \cdots \times X_n.$$

Write $r_t(\psi)$ for the number of equivalence classes of prefixes of length t , for $0 \leq t < n$. These are the deterministic continuation responses. The terminal decision is emitted on the last transition, so no terminal register is needed. In particular, $r_0 = 1$. The equivalence is a finite-horizon version of the classical residual-state construction [16]. The following quantitative statement will also measure the loss when a statistical auditor has fewer states than this construction.

Theorem 14.1 (Exact continuation width and a quantitative cut obstruction). *A stochastic streaming auditor computes ψ exactly on every word with width profile $(K_t)_{0 \leq t < n}$ if and only if $K_t \geq r_t(\psi)$ at every cut. The profiles are attained simultaneously by a deterministic auditor. Independent persistent randomization does not reduce these widths for exact computation.*

More generally, at a cut with $r = r_t(\psi) \geq 2$, every stochastic auditor of width K satisfies

$$\sum_{w \in X_1 \times \cdots \times X_n} e_\psi(w) \geq \frac{(r - K)_+}{r - 1}, \quad (14.1)$$

where $e_\psi(w)$ is its probability of an incorrect Boolean decision on input w . The inequality also holds after averaging over an independent selector whose conditional auditors each have width at most K .

Proof. From a prefix u let $e_u(m)$ be the distribution of the retained state. From state m and suffix s let $a_m(s)$ be the probability of terminal answer $+1$. Exact computation says

$$\sum_m e_u(m) a_m(s) = \frac{1 + \psi(us)}{2}.$$

The right side is zero or one. Every state of positive e_u -mass must therefore have precisely that zero-one continuation response for all s . Distinct prefix classes have disjoint positive supports. Hence $K_t \geq r_t$. Conversely, retain the prefix class. Appending the same letter to two equivalent prefixes gives equivalent new prefixes; thus every transition is well defined. At the last transition the class and the last letter determine ψ . This realizes every width r_t simultaneously. For a persistent independent selector, a zero average error on each of the finitely many words implies zero conditional error on all words outside a single null set of selector values. The preceding bound applies there.

For the quantitative claim choose representative prefixes $u_1, \dots, u_r$, one per class, and put $e_i(m) = e_{u_i}(m)$. Define

$$o_{ij} = \sum_{m=1}^K \min\{e_i(m), e_j(m)\}.$$

For nonnegative numbers $a_1, \dots, a_r$, ordering them shows that $\sum_{i < j} \min(a_i, a_j) \geq \sum_i a_i - \max_i a_i$. Apply this at each state. Since every e_i has mass one and each $\max_i e_i(m) \leq 1$, it follows that

$$\sum_{i < j} o_{ij} \geq r - K. \quad (14.2)$$

Choose a suffix s_{ij} distinguishing u_i, u_j . Conditional on a common state and this suffix, the two desired answers are opposite. Their error probabilities, weighted by $e_i(m), e_j(m)$, have sum at least $\min\{e_i(m), e_j(m)\}$. Thus

$$e_\psi(u_i s_{ij}) + e_\psi(u_j s_{ij}) \geq o_{ij}.$$

On summing, each complete word on the left occurs at most $r - 1$ times: its prefix identifies a single representative, which participates in $r - 1$ pairs. Nonnegativity of all other word errors and (14.2) prove (14.1). When $r \leq K$ the claim is just nonnegativity. Finally apply the bound conditional on a selector and integrate. Independence ensures that conditioning has not changed the input experiment. $\square$

The exact part is also the extremal zero-one case of Theorem 3.2. Its content is a simultaneous width profile, rather than separate unrelated factorizations at different cuts. The error bound does not assert that its constant is optimal for every Boolean function. Its use below is to convert a continuation obstruction into a strict loss of the attainable statistical value.

14.1. The width required to attain the binary reset value. Fix $0 < h < 1/2$ and the binary candidate class B_h of Theorem 13.3. A terminal decision $+1$ chooses the singleton event appropriate to the $+$ candidate, and -1 the complementary event. A zero-score event can equivalently be replaced by an independent fair choice of these decisions. The expected payoffs are therefore $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$, where $d \in [-1, 1]$ is the conditional mean decision and $P_\pm = \text{Ber}(1/2 \pm h)^{\otimes n}$. Let $v_{n,K}$ be the supremum of the smaller of these two payoffs over streaming auditors with the declared width profile. The final validation interface is unchanged; the widths here count training and selection only.

Theorem 14.2 (Exact memory threshold for optimal binary auditing). *Let $n = 2m + 1$ be odd. The unconstrained reset value is $v_n = h\|P_+ - P_-\|_{\text{TV}}$. It is attained at the profile*

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n, \quad (14.3)$$

and $v_{n,K} = v_n$ if and only if $K_t \geq r_t$ for all these cuts. For every cut with $r_t \geq 2$ one has the explicit deficit

$$v_n - v_{n,K} \geq 2h^2(1/4 - h^2)^m \frac{(r_t - K_t)_+}{r_t - 1}. \quad (14.4)$$

These conclusions remain valid when an independent persistent selector chooses auditors of the same conditional widths. For $n \geq 3$ the least peak training width attaining v_n is $m + 2$; for $n = 1$ it is one.

Proof. The sign of $P_+(w) - P_-(w)$ is the majority decision $\psi(w)$, with no zero difference since n is odd. Complementing every bit interchanges the two laws and reverses the decision. Consequently majority has equal payoffs and attains $h\|P_+ - P_-\|_{\text{TV}}$, as also proved in Theorem 13.3.

A prefix with k ones requires at least $m + 1 - k$ ones in its remaining $n - t$ places. Thresholds at most zero give the constant positive response; thresholds greater than $n - t$ give the constant negative response; the remaining thresholds give different nonconstant responses. Counting them for $0 \leq k \leq t$ gives (14.3). The exact realization in Theorem 14.1 implements majority at all these widths.

For an arbitrary randomized decision, its smaller payoff is bounded by the equal-prior average. The deficit of this average from the majority value is exactly

$$h \sum_w |P_+(w) - P_-(w)| e_\psi(w). \quad (14.5)$$

Put $a = 1/2 + h$ and $b = 1/2 - h$. A word with $k \geq m + 1$ ones has absolute likelihood difference $(ab)^{n/2} \{(a/b)^{k-n/2} - (a/b)^{-(k-n/2)}\}$, which increases with k . Symmetry treats the other half of the words. The smallest difference is therefore attained at $k = m, m + 1$ and equals

$$a^{m+1}b^m - b^{m+1}a^m = 2h(1/4 - h^2)^m.$$

Substitution of (14.1) into (14.5) gives (14.4). Any deficient cut gives a strictly positive loss; all sufficiently wide profiles implement majority. Maximizing (14.3) proves the peak-width assertion. The same average-payoff argument works conditional on an independent selector and hence after averaging. $\square$

The result identifies the exact memory needed to attain the optimal value, not the memory needed for a prescribed approximate value uniformly in h . The quantitative deficit can be small for large n . In particular it is not a sharp joint asymptotic tradeoff in sample size, state size and error. It complements, rather than replaces, the uniform sampling exponent in Theorem 13.3. No minimax interchange on the generally nonconvex set of fixed-width auditors has been used.

15. POLYNOMIAL TESTING OF THE ORIGINAL RESOURCE IMAGE

The private behavior image is usually nonconvex. Its testing criterion can nevertheless be written as a limit of finite algebraic lower certificates, without replacing that image by randomized designs. The testing weights in this section depend on the candidate simulator. Unlike Theorem 16.1, this identity is a certificate transform, not an independent variational dual. They are universal proof functions, not a hidden selector supplied to an executed experiment.

Use any one of the finite signatures of Theorem 2.2, with rational instrument data. Write its objective as $F(q) = \max_{1 \leq i \leq m} f_i(q)$, where f_i are all its marked event discrepancies, including the empty event $f_1 = 0$. For visible selectors the event and feedback rule include their selector

fibers, and the target selector law remains in f_i . Thus $-1 \leq f_i \leq 1$ for all three signatures; the polynomials may be different.

Parameterize a row with d outcomes by $d - 1$ variables in $[0, 1]$:

$$q_1 = x_1, \quad q_j = x_j \prod_{l < j} (1 - x_l) \quad (1 < j < d), \quad q_d = \prod_{l < d} (1 - x_l).$$

This map is onto the row simplex, including its boundary. Indeed, divide each desired probability by the remaining mass while that mass is positive; when it is zero choose the remaining variables arbitrarily. Taking the product of these maps, including a selector row when needed, gives an onto polynomial map $q(x)$ from $[0, 1]^D$ to the feasible row product. Put $h_i(x) = f_i(q(x))$ and $g_i = (1 + h_i)/2$. All h_i are rational polynomials, $0 \leq g_i \leq 1$, and $g_1 = 1/2$. The case $D = 0$ is a single feasible table and is interpreted directly.

Theorem 15.1 (Strict polynomial lower-certificate hierarchy). *For $p \geq 1$ define*

$$A_p(x) = \frac{\sum_{i=1}^m g_i(x)^p h_i(x)}{\sum_{i=1}^m g_i(x)^p}, \quad a_p = \min_{x \in [0, 1]^D} A_p(x), \quad \delta = \min_q F(q).$$

Then the denominators are positive, a_p is nondecreasing, and

$$\delta - 2(1 - m^{-1/(p+1)}) \leq a_p \leq \delta, \quad \delta = \sup_{p \geq 1} a_p. \quad (15.1)$$

For rational L let

$$P_{p,L}(x) = \sum_{i=1}^m g_i(x)^p (h_i(x) - L). \quad (15.2)$$

A nonnegative tensor Bernstein expansion of $P_{p,L}$ on $[0, 1]^D$ certifies $\delta \geq L$. If $L < \delta$, some p and some finite Bernstein degree have strictly positive coefficients. Conversely, a strictly positive such expansion certifies $L < \delta$. These are lower certificates at the original state and selector budgets. No convexification of the feasible behavior image occurs.

Proof. Fix x and put $b = \max_i g_i(x) \geq 1/2$ and $r_i = g_i(x)/b$. At least one r_i is one. Write $S_j = \sum_i r_i^j$. Hölder's inequality gives

$$S_p \leq m^{1/(p+1)} S_{p+1}^{p/(p+1)}, \quad S_{p+1} \geq 1.$$

Consequently $S_{p+1}/S_p \geq m^{-1/(p+1)}$. Since $A_p = 2bS_{p+1}/S_p - 1$ and $\max_i h_i = 2b - 1$,

$$0 \leq \max_i h_i - A_p \leq 2(1 - m^{-1/(p+1)}).$$

Cauchy-Schwarz gives $S_{p+2}S_p \geq S_{p+1}^2$, so A_p is nondecreasing in p . Its denominator is at least 2^{-p} and it is continuous on the cube. Taking minima proves (15.1). The bound is also at most $2 \log m/(p+1)$, by $1 - e^{-u} \leq u$.

Every tensor Bernstein basis function is nonnegative on the cube, and the basis functions sum to one. Nonnegative coefficients therefore imply $P_{p,L} \geq 0$. Dividing by the positive denominator gives $F(q(x)) \geq A_p(x) \geq L$ for every x , which proves soundness. For completeness choose p so that the error in (15.1) is less than $\delta - L$. Then

$$P_{p,L}(x) \geq 2^{-p} \{\delta - L - 2(1 - m^{-1/(p+1)})\} > 0.$$

Lemma 15.2 below gives a finite strictly positive Bernstein expansion. Conversely strictly positive coefficients have a positive minimum, and the denominator is at most m , so they give a strict positive uniform gap above L . $\square$

Positive Bernstein certificates and their degree elevation are classical; see de Klerk-Laurent [20]. The next elementary bound is retained for a self-contained exact verifier, not as a new general hypercube positivity theorem.

Lemma 15.2 (An explicit positivity degree). *Let $P(x) = \sum_k c_k x^k$ be a rational polynomial in D variables and $n \geq \max_{k,j} k_j$. In its tensor Bernstein expansion of degree n in each coordinate the coefficient at $\alpha \in \{0, \dots, n\}^D$ is*

$$b_{\alpha,n} = \sum_k c_k \prod_{j=1}^D \frac{\binom{\alpha_j}{k_j}}{\binom{n}{k_j}}, \quad |b_{\alpha,n} - P(\alpha/n)| \leq \frac{C(P)}{n}, \quad C(P) = \sum_k |c_k| \sum_j \binom{k_j}{2}. \quad (15.3)$$

In particular, $\min P > 0$ implies positivity of every coefficient whenever $n > C(P)/\min P$ and the degree restriction holds.

Proof. For a binomial random variable B with parameters (n, x) , $\mathbb{E} \binom{B}{k} = \binom{n}{k} x^k$. Expanding this expectation proves the univariate monomial formula, and products prove the displayed coefficient identity. The convention is $\binom{a}{k} = 0$ for $a < k$. To compare $\binom{a}{k}/\binom{n}{k}$ with $(a/n)^k$, draw k indices uniformly from $\{1, \dots, n\}$. Conditional on no repetition this is sampling without replacement. The probability of a repetition is at most $\binom{k}{2}/n$. Apply this total variation bound to the event that all drawn indices lie in a prescribed a -element subset. Hence the difference of the two factors is at most $\binom{k}{2}/n$. All factors lie in $[0, 1]$. A product telescoping bound followed by the triangle inequality proves (15.3). The final assertion follows by taking the minimum of P . $\square$

Corollary 15.3 (Rational lower witnesses without policy cells). *At every fixed finite rational signature, strict infeasibility above a rational level L is equivalent to the existence of the following finite rational certificate: an integer p , an integer n , and a strictly positive coefficient array for (15.2) whose identity is checked by (15.3). Dovetailing these certificates with rational original-budget upper witnesses produces nested intervals converging to δ . If a subfamily of the original tests gives a strict gap, the same construction applies to it after adjoining the empty test.*

Proof. Soundness and completeness are Theorem 15.1. Each finite candidate is checked by rational polynomial arithmetic. Enumerate all (p, n) and rational levels, and retain the largest verified lower level so far, together with zero. Rational row tables are dense in the feasible product, and their continuous objectives converge to the minimum, so enumeration and retention of the best upper table converge from above. This is a dovetailed enumeration, not a claim that any fixed (p, n) test search terminates. For a requested positive interval width, convergence of both endpoints proves termination. The reduced family conclusion uses only the same pointwise maximum argument. $\square$

Remark 15.4 (Meaning and size of the strict certificate transform). A constant distribution on tests generally detects only the convex hull of the behavior image. The functions $g_i^p / \sum_j g_j^p$ are instead functions of a proposed row table. Their universally quantified weighted inequality is an algebraic certificate; the tester need not observe a private state in an experiment. The displayed certificate identity is not an equality obtained by exchanging a private minimum with a fixed mixed test. Power means and Bernstein degree elevation are classical algebraic approximation devices, proved here in the precise form needed. The claim here is strict-level completeness for the original marked causal resource image, including visible selectors, and its physical transfer below. The coefficient count $(n+1)^D$, the number of original tests, and the algebraic degrees may be large. No polynomial complexity or intrinsic-dimension bound on this row-cube coefficient count is asserted. The independent adaptive test has the separate behavior-size bound of Theorem 16.3; its verification is a different cost.

15.1. A sharp two-time nonconvex example. The source has singleton reports, the control alphabet and actual mark are singleton, and the target emits $(0, 0)$ or $(1, 1)$ with probability $1/2$ each. At width one, a private simulator emits two independent bits with probabilities p, q . It cannot reread its first emitted bit.

Proposition 15.5 (A gap invisible to constant testing mixtures). *For this delayed experiment,*

$$\delta_{(1,1)}^p = \sqrt{2} - 1, \quad \delta_{(1,1),2}^h = 0, \quad \delta_{(1,1),r}^v = \sqrt{2} - 1 \quad (r \geq 1).$$

The private objective is the maximum of the four polynomials

$$0, \quad \frac{1}{2} - pq, \quad \frac{1}{2} - (1-p)(1-q), \quad p + q - 2pq.$$

Every constant mixture of these tests has minimum at most zero, whereas Corollary 15.3 certifies every rational $L < \sqrt{2} - 1$ at private width one.

Proof. Put $u = pq$ and $v = (1-p)(1-q)$. Direct total variation calculation gives $F = 1 - \min(u, 1/2) - \min(v, 1/2)$ and the displayed maximum. By Cauchy–Schwarz, $\sqrt{u} + \sqrt{v} \leq 1$. If $u \geq 1/2$, then $v \leq (1 - 1/\sqrt{2})^2$, so the sum of the two capped terms is at most $2 - \sqrt{2}$; the case $v \geq 1/2$ is symmetric. Otherwise set $a = \sqrt{u}$, $b = \sqrt{v}$. On $0 \leq a, b \leq 1/\sqrt{2}$, $a + b \leq 1$, the maximum of $a^2 + b^2$ is at a vertex of this polygon and equals $2 - \sqrt{2}$. Equality is attained at $p = q = 1/\sqrt{2}$. A hidden fair bit selects the two deterministic constant sequences and exactly reproduces the target. If it is exposed, variation is averaged over the selector fibers; each fiber has error at least the private minimum, which is itself attainable without a selector. Finally the target lies in the convex hull of the two private deterministic behaviors. The average discrepancy of any fixed test mixture at those behaviors is zero, so at least one has discrepancy at most zero. The strict polynomial certificates therefore detect information lost by constant convex separation. $\square$

16. ADAPTIVE TESTING ON THE OBSERVABLE BEHAVIOR IMAGE

The obstruction to linear randomization criteria at a fixed private register is the nonconvexity of the behavior image, not a failure of compactness. We give a variational formulation in which a test is an independent function of an observable candidate law. The construction does not give the simulator an extra random variable.

16.1. The testing game. Let B be a nonempty compact subset of a finite-dimensional affine space, and let $f_1, \dots, f_m$ be affine real functions, including the zero function. Put

$$F(b) = \max_{i \leq m} f_i(b), \quad \delta = \min_{b \in B} F(b).$$

For the finite comparison problem of Section 2, b is the array of discrepancies between the simulated and target marked laws, indexed by parameters, feedback trees and transcripts; f_i ranges over their signed event evaluations. In the visible case the array includes the selector and its target weights. Thus the target coordinate is part of the array when those weights vary. Private, fixed-hidden and fixed-visible signatures each give their own compact set B . No convex hull is taken.

An *adaptive test* is a continuous map $\kappa : B \rightarrow \Delta_m$, chosen before a candidate b . Its score against that candidate is

$$G(b, \kappa) = \sum_i \kappa_i(b) f_i(b).$$

This is a verification game: the selected candidate law is revealed to the evaluator of the already chosen function κ . It is not a one-observation statistical test, and does not make an unavailable parameter or mark observable to a physical controller.

Theorem 16.1 (Revealed-behavior adaptive representation). *For every such compact behavior image,*

$$\begin{aligned} \delta &= \sup_{\kappa \in C(B, \Delta_m)} \min_{b \in B} G(b, \kappa) \\ &= \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa \in C(B, \Delta_m)} \int_B G(b, \kappa) d\mu(b). \end{aligned} \tag{16.1}$$

The minimization over probability measures is attained. If Borel adaptive tests are allowed, the first supremum is attained as well. The continuous-test assertion is a supremum; continuous attainment is not needed or asserted.

Proof. Fix $\eta > 0$. The sets $U_i = \{b : f_i(b) > F(b) - \eta\}$ form an open cover of the compact metric space B . Choose a continuous partition of unity subordinate to this finite cover. The resulting test has $F(b) - \eta < G(b, \kappa) \leq F(b)$ for all b . Consequently the first supremum equals δ . The same uniform inequality shows, for every probability μ ,

$$\sup_{\kappa} \int G(b, \kappa) d\mu(b) = \int F d\mu.$$

Its minimum is δ , attained by a point mass at a minimizer of F . This also proves the displayed minimax exchange without an assumption that B is convex. Equivalently one can apply the usual compact-convex minimax theorem to $\mathcal{P}(B)$ and the convex class of continuous tests, with the pairing $\int G d\mu$. Finally choose the smallest index attaining $F(b)$. Its level sets are Borel, and its indicator vector is a Borel test with score exactly F . $\square$

The probability μ here describes a lottery over *revealed candidate laws*. The score is $\int G(b, \kappa) d\mu(b)$, not $G(\int b d\mu, \kappa)$. Hiding the candidate and replacing it by its barycenter is a different game; in general it lowers the value. This distinction is essential for a finite persistent-state budget.

16.2. A size bound in behavior dimension. Factor out directions annihilated by every $f_i - f_i(b_0)$, so that on the resulting difference space

$$\|v\|_* = \max_i |f_i(b_0 + v) - f_i(b_0)|$$

is a norm. Denote its dimension by a and let $R = \text{diam}_*(B)$. All conclusions below are unaffected by this observable quotient. The case $a = 0$ is a single constant comparison and requires one test. Suppose henceforth that $a \geq 1$, so $R > 0$.

Lemma 16.2 (A bounded behavior chart). *There exist $b_0 \in B$ and vectors $v_j \in B - b_0$, $1 \leq j \leq a$, forming a basis, such that $\|v_j\|_* \leq R$ and every $b \in B$ has coordinates $b = b_0 + \sum_j t_j v_j$ with $|t_j| \leq 2$. If B is a rational polynomial image of a product of simplexes, the points can be chosen rational and can be found effectively.*

Proof. Fix $b_0 \in B$. In any fixed coordinates on the difference space, let $D_0 > 0$ be the maximum absolute determinant of a vectors in $B - b_0$. Compactness and spanning give existence and positivity. Choose a tuple with determinant greater than $D_0/2$. Replacing its j th vector by $b - b_0$ multiplies the determinant by t_j ; maximality gives $|t_j| < 2$. The norm bound is the definition of R . For a rational polynomial image, take b_0 at rational rows. Rational row images are dense. An affine basis can be found by enumerating rational images and testing rational determinants; the true affine span and the maximum determinant are semialgebraic data. Quantifier elimination decides whether a proposed nonzero determinant exceeds half the maximum. Density therefore makes the search terminate. This is an effectiveness assertion, not a bit-complexity bound for finding the chart. $\square$

Set $b(x) = b_0 + \sum_j (4x_j - 2)v_j$ and $C = \{x \in [0, 1]^a : b(x) \in B\}$. The ambient cube is a coordinate box, not an enlargement of the feasible behavior set. Write $\ell_i(x) = f_i(b(x))$. Then

$$|\ell_i(x) - \ell_i(y)| \leq 4R \sum_j |x_j - y_j|. \quad (16.2)$$

For $n \geq 0$ let $\beta_{\alpha,n}$ be the tensor Bernstein basis of coordinatewise degree n . A feasible coefficient array has $c_{\alpha i} \geq 0$ and $\sum_i c_{\alpha i} = 1$ for every $\alpha \in \{0, \dots, n\}^a$. It specifies the independent test

$$\kappa_i^c(x) = \sum_{\alpha} c_{\alpha i} \beta_{\alpha,n}(x).$$

Define

$$d_n = \max_c \min_{x \in C} \sum_{\alpha, i} c_{\alpha i} \beta_{\alpha,n}(x) \ell_i(x). \quad (16.3)$$

Theorem 16.3 (Intrinsic adaptive degree and support bounds). *The maxima in (16.3) are attained and $d_n \leq d_{n+1} \leq \delta$. For $n \geq 1$,*

$$0 \leq \delta - d_n \leq \frac{4Ra}{\sqrt{n}}. \quad (16.4)$$

There is an array with at most $(n+1)^a$ nonzero coefficients, all equal to one, whose minimum score is at least $\delta - 4Ra/\sqrt{n}$. Its test has coordinatewise degree n , and its score degree at most $n+1$. Furthermore

$$d_n = \min_{\mu \in \mathcal{P}(C)} \sum_{\alpha} \max_i \int_C \beta_{\alpha,n}(x) \ell_i(x) d\mu(x), \quad (16.5)$$

and a minimizing measure can be supported on at most $(n+2)^a$ points. In particular a strict level $L < \delta$ admits an adaptive test of positive uniform margin with $n > (8Ra/(\delta - L))^2$ and at most $(n+1)^a$ occupied coefficient entries. These bounds depend on the observable image and its metric, not on the number of stochastic rows in a realization.

Proof. The coefficient space is a finite product of compact simplexes, and the minimum of a jointly continuous function over compact C is continuous in its coefficients. This proves attainment. Bernstein degree elevation replaces each coefficient by convex combinations of adjacent coefficients, preserving every simplex constraint and the represented function. It proves monotonicity. All scores are convex combinations of the $\ell_i(x)$, proving the upper bound by δ .

At each grid point α/n choose an index i_{α} maximizing ℓ_i there. The choice is made even at infeasible grid points; only the final score is minimized over C . Give that index coefficient one and the others zero. For fixed x let A_j be independent $\text{Bin}(n, x_j)$ variables. If i_* maximizes at x ,

$$\begin{aligned} \max_i \ell_i(x) - \mathbb{E} \ell_{i_A}(x) &\leq \mathbb{E} \{ \ell_{i_*}(x) - \ell_{i_*}(A/n) + \ell_{i_A}(A/n) - \ell_{i_A}(x) \} \\ &\leq 8R \sum_j \mathbb{E} |A_j/n - x_j| \leq \frac{4Ra}{\sqrt{n}}. \end{aligned}$$

Here the omitted middle difference is nonpositive, and the last bound uses $\text{Var}(A_j/n) \leq 1/(4n)$. This proves the rate and the sparse coefficient assertion.

For the last formula replace the minimum over C by a minimum over $\mathcal{P}(C)$. The pairing is bilinear and continuous, and both feasible sets are compact convex, so the compact minimax theorem [18] exchanges the extrema. Optimizing each coefficient simplex gives (16.5). The integrands are polynomials of coordinatewise degree at most $n+1$. Preserve all monomial moments of that degree by Carathéodory's theorem. There are $(n+2)^a$ monomials including the constant; their moment vectors lie in an affine hyperplane of dimension at most $(n+2)^a - 1$. A measure with at most $(n+2)^a$ atoms therefore has exactly the same objective. The strict-margin assertion follows from the rate with error less than $(\delta - L)/2$. $\square$

The score is a nonlinear function of behavior although each event functional is affine. This is the reason the measure in (16.5) cannot be replaced by its behavior barycenter. For $n = 0$ it can: d_0 is precisely the constant-test convex relaxation. In the two-time unmarked example of Proposition 15.5, the target belongs to the convex hull of private behaviors, so $d_0 = 0$, whereas $\lim_n d_n = \sqrt{2} - 1$. Thus the hierarchy recovers a gap which every constant test misses.

Remark 16.4 (What the size bound measures). The bound counts the nonzero entries of the independent adaptive test and controls its behavior-coordinate degree. It does not bound the cost of verifying its minimum on the generally nonconvex set C , the bit lengths of a rational chart, or the size of a description of C . Distinct row presentations of the same metric behavior image satisfy the same existence bound, although their coefficient arrays need not coincide. For the private fixed-target law array one may use $R \leq 1$ with the maximum total-variation metric. For the visible signed-discrepancy array the uniform bound $R \leq 2$ is sufficient. The old row-cube positivity certificate is a separate verification engine, with its own coordinate-dependent complexity.

16.3. A three-weight certificate invisible to constant tests.

Proposition 16.5 (A finite intrinsic nonlinear witness). *Consider a private width-one simulator producing two independent bits, with probabilities p, q , for the target which assigns probability $1/2$ to each of 00, 11. Put $u = pq$, $v = (1-p)(1-q)$ and $J(t) = 3t^2 - 2t^3$. The three behavior-dependent weights*

$$\kappa_1 = J(v), \quad \kappa_2 = J(u), \quad \kappa_3 = 1 - J(u) - J(v)$$

are nonnegative and sum to one. Against the event discrepancies $1/2 - u, 1/2 - v, 1 - u - v$, respectively, their score G satisfies

$$G \geq \frac{2}{5} + \frac{9}{8960}. \quad (16.6)$$

The certificate uses cubic weights in observable probabilities, a nine-term nonnegative derivative identity, and seventeen positive rational Bernstein coefficients in one variable. Every constant mixture of event discrepancies has minimum at most zero.

Proof. We have $u + v \leq 1$. Since J is increasing on $[0, 1]$ and $J(1-t) = 1 - J(t)$, the weights are feasible. Their score is

$$G = 1 - u - v + J(u)(u - 1/2) + J(v)(v - 1/2).$$

Set $a = (p + q - 1)^2$ and $b = (p - q)^2$. Then $a, b \geq 0$ and $a + b \leq 1$. Direct expansion gives

$$\begin{aligned} 64G &= 27 - 12a + 38a^2 - 20a^3 - a^4 + 36b - 60ab + 36a^2b + 4a^3b \\ &\quad + 6b^2 - 12ab^2 - 6a^2b^2 - 4b^3 + 4ab^3 - b^4. \end{aligned}$$

Let $c = 1 - a - b$. Another polynomial identity, valid when $a + b + c = 1$, is

$$\begin{aligned} 16\partial_b G &= 4a^3 + 6a^2c + 12ab^2 + 24abc + 12ac^2 \\ &\quad + 8b^3 + 30b^2c + 30bc^2 + 9c^3 \geq 0. \end{aligned}$$

Thus $G(a, b) \geq G(a, 0)$. The univariate polynomial

$$G(a, 0) - \frac{2}{5} = \frac{7}{320} - \frac{3}{16}a + \frac{19}{32}a^2 - \frac{5}{16}a^3 - \frac{1}{64}a^4$$

has degree-16 Bernstein coefficients $z_k/349440$, where

$$\begin{aligned} (z_0, \dots, z_{16}) &= (7644, 3549, 1183, 351, 855, 2494, 5064, 8358, \\ &\quad 12166, 16275, 20469, 24529, 28233, 31356, 33670, 34944, 34944). \end{aligned}$$

The Bernstein basis is nonnegative and sums to one. Its smallest coefficient is $351/349440 = 9/8960$, proving (16.6). All identities are rational polynomial identities; no numerical minimization is used. Finally the target is the barycenter of the two feasible deterministic laws 00 and 11. An affine mixture of signed event discrepancies has average zero at these two laws and hence minimum at most zero. The empty event supplies the reverse constant-test bound zero. $\square$

For comparison, the affine weights $v, u, 1 - u - v$ already give the smaller bound $3/8$, with the exact identity

$$G_{\text{affine}} - \frac{3}{8} = \frac{1}{2}(p - q)^2 + 2(p - 1/2)^2(q - 1/2)^2.$$

Neither certificate changes the private feasible set. The cubic certificate is an independent test chosen once for the whole behavior image, not a mixture chosen anew by an optimizer's unknown row values. Its evaluation does use the revealed behavior, as specified in the verification game. The 17-coefficient positivity check is classical; the distinction from constant tests is the behavior-dependent feasible weight family and its resource-sensitive minimum.

The infinite-dimensional representation above holds for continuous f_i on any compact metric space: its proof uses a partition of unity, not causal factorization. The measure minimum is attained at a Dirac mass at a minimizer of F . The finite-level polynomial theorem has a different content. It is an existence-size bound for revealed-law verification; the reset-audit theorem of Section 13 imposes an explicit finite-data restriction.

16.4. Strict certificates and effective physical input. Given rational finite instruments, a rational chart and the sparse array constructed above, its strict minimum can be certified by real quantifier elimination on the original row constraints. Alternatively, pull the score back through the original polynomial row map and certify its strict positivity using degree-elevated Bernstein coefficients. The pullback changes the verification degree and is not covered by the behavior-coordinate bound. Generic quantifier elimination also decides the exact finite semialgebraic boundary; this classical observation does not assert that a finite nonnegative Bernstein expansion exists at $L = \delta$.

Proposition 16.6 (Uniform effective presentation input). *Suppose a single algorithm, on input j , returns rational finite instruments E_j, F_j and nonnegative rational error enclosures a_j, c_j for two-sided stateless marked feedback presentations of E, F . Suppose $a_j + c_j \rightarrow 0$. All alphabets, widths, selector laws and finite subroutines must be computable uniformly from j and the requested signature. Then, for every positive rational η , an algorithm returns a rational interval of width below η containing $\delta_b(E, F)$ at the same signature.*

Proof. Enumerate j until $a_j + c_j < \eta/8$. This terminates without a known rate because the rational enclosures themselves tend to zero. For the selected finite problem, enumerate rational same-budget upper tables and strict rational lower certificates, dovetailing all finite checks; include the always valid lower bound zero. Compactness, density and strict completeness provide an interval of width below $\eta/2$. Enlarge it by $a_j + c_j$ at both ends using stateless comparison. The enlarged width is below $3\eta/4$. Every enumeration and comparison is uniform by the hypotheses. A mere sequence of individually computable presentations without a uniform enumeration is not an input satisfying this proposition. $\square$

17. AN EXACT ALGORITHM FOR THE OBSERVABLE CHART

The chart in Lemma 16.2 can be found without taking its dimension as an oracle. We state the input and output explicitly. Rational polynomials are encoded by finite lists of monomial exponent vectors and rational coefficients in reduced integer form. A product of simplexes is encoded by the number of entries in each row. Eliminating the last entry of every nontrivial row leaves a full-dimensional rational polytope $\Omega \subset \mathbb{R}^r$. Zero-dimensional factors are omitted.

Proposition 17.1 (Effective observable quotient and pullback). *The input consists of the preceding simplex description, a rational polynomial map $b : \Omega \rightarrow \mathbb{R}^N$, and finitely many rational affine functionals f_i on $\mathbb{R}^N$. There is a terminating algorithm which returns its observable affine*

dimension a , a rational basis and coordinate map for the observable quotient, and, when $a > 0$, rational parameter points $q_0, q_1, \dots, q_a$ with the following properties. With

$$z(q) = (f_i(b(q)) - f_i(b(q_0)))_i, \quad v_j = z(q_j),$$

the vectors v_j are a basis of the observable difference space, and all coordinates $t(q)$ in that basis satisfy $|t_j(q)| < 2$. The output also contains the rational polynomial cube map $x(q) = (t(q) + 2)/4$, a rational upper bound for the observable diameter, and the exact pullback of any subsequently supplied rational polynomial in x . The algorithm outputs the universal determinant formula it has decided. It makes no polynomial-time or coefficient-height claim.

Proof. Choose a rational relative interior point q_0 of the input simplexes. Expand the vector polynomial $z(q)$ in the free row coordinates. Let V be the rational span of all its coefficient vectors, including its constant coefficient. This equals $\text{span } z(\Omega)$: a linear functional vanishes on $z(\Omega)$ exactly when its polynomial vanishes on the nonempty interior of Ω , hence identically, hence on every coefficient. Gaussian elimination therefore gives $a = \dim V$ and a rational basis matrix U . Choosing independent rows gives a rational left inverse L with $LU = I_a$. Thus $y(q) = Lz(q)$ provides exact quotient coordinates. If $a = 0$, all discrepancies are constant and no chart is required.

For $a > 0$ let

$$D_0 = \max_{u_1, \dots, u_a \in \Omega} |\det(y(u_1), \dots, y(u_a))|.$$

Compactness gives a maximum, and spanning gives $D_0 > 0$. Enumerate rational tuples $(q_1, \dots, q_a)$ in Ω^a . Membership and their determinant d are decided by rational arithmetic. For $d \neq 0$, decide the first-order real formula

$$\forall u_1, \dots, u_a \in \Omega : \quad \det(y(u_1), \dots, y(u_a))^2 < 4d^2. \quad (17.1)$$

Quantifier elimination terminates for each formula. Density of rational points and continuity of determinant imply that some enumerated tuple has $|d| > D_0/2$. For such a tuple the strict formula is true, so the search terminates. No computation of an exact algebraic maximizer is needed.

Invert the rational matrix with columns $y(q_j)$ to obtain $t(q)$. Cramer's rule and (17.1) give $|t_j(q)| < 2$ for every q . In the observable norm $\|z\|_* = \max_i |z_i|$, each v_j has norm at most the diameter of $z(\Omega)$. An upper bound for that diameter can be produced, for example, by bounding each free coordinate in $[0, 1]$ and summing the absolute polynomial coefficients for $f_i(b(q)) - f_i(b(q'))$. This bound is rational and computable. Rational matrix multiplication and substitution give $x(q)$ and every requested pullback. They also give its full sparse coefficient encoding. All searches have now been specified by finite encodings and terminating operations. $\square$

There is a small but important distinction between the quotient and a projection onto a smaller set of simulator rows. The former identifies exactly the directions invisible to the chosen event family; the original row constraints remain present in every universal positivity test. The chart need not identify equivalent tasks after a change of marks or of information signature. Those are changes to the comparison problem.

18. A MARKED RESOURCE GAP GENERATED BY A BINARY COLLISION

We now realize a private–hidden–visible separation in a physical experiment whose information vanishes when the collision is removed. The controller chooses a later sensor gate after its first report. The gate controls acquisition, not the mechanical collision law. Both the source and the target carry the same actual, unobserved mark.

18.1. The exact marked two-report experiment. Let $B, W \in \{0, 1\}$ have the joint distribution

$$\mathbb{P}(B = b, W = w) = \begin{cases} 5/16, & b = w, \\ 3/16, & b \neq w. \end{cases} \quad (18.1)$$

The blind source E_* has two constant reports and actual mark W . The target F_* reports $Y_1 = B$, receives $A \in \{0, 1\}$ after that report, and reports $Y_2 = AB$. The feedback tester may choose A as any function of its earlier report. A width-one private simulator cannot reread its own first emission, except through the current gate A . Hidden selectors persist but are not revealed; visible selectors are included in the joint target law independently of (B, W) . Put

$$d_* = \sqrt{5/2} - 9/8.$$

Theorem 18.1 (An exact marked causal separation). *At the width profile $(1, 1)$,*

$$\delta^P(E_*, F_*) = d_*, \quad \delta_2^h(E_*, F_*) = 1/8, \quad \delta_r^v(E_*, F_*) = d_* \quad (r \geq 1). \quad (18.2)$$

The hidden upper bound is realized by two deterministic tables. The private upper bound is realized by two independent Bernoulli variables with parameter $\sqrt{5/8}$, followed by the gate. These bounds hold against every feedback rule, not just open-loop gates.

Proof. First choose the constant gate $A = 1$. The two private emissions are independent Bernoulli variables with parameters p, q and are independent of W . Let

$$u = pq, \quad v = (1 - p)(1 - q), \quad a = 3/8, \quad b = 5/8.$$

The overlap of the target and simulated marked laws is $H(u) + H(v)$, where

$$H(t) = \frac{1}{2} \{\min(t, a) + \min(t, b)\}.$$

Indeed the target is supported on equal report pairs, and its two W -masses at each equal pair are $3/16$ and $5/16$. Total variation is one minus this overlap.

Cauchy-Schwarz gives $\sqrt{u} + \sqrt{v} \leq 1$. By symmetry suppose $u \geq v$. If $u \leq 1/4$ the overlap is at most $1/2$. If $u \geq 1/4$, then $v \leq (1 - \sqrt{u})^2 \leq 1/4 < a$, so the overlap is at most

$$H(u) + (1 - \sqrt{u})^2.$$

This function is convex on each of $[1/4, a]$ and $[a, b]$ and nonincreasing on $[b, 1]$. Its maximum is therefore attained at $1/4, a$ or b . The values at a and b are $1 + 2a - 2\sqrt{a}$ and $3/2 + b - 2\sqrt{b}$ respectively. The second is larger, since

$$(3/2 + b - 2\sqrt{b}) - (1 + 2a - 2\sqrt{a}) = 3/8 - \frac{1/2}{\sqrt{a} + \sqrt{b}} > 0,$$

using $\sqrt{a} > 3/5$, $\sqrt{b} > 3/4$. It also exceeds $1/2$. Taking $p = q = \sqrt{b}$ attains this maximum, and hence the minimum error is $1 - (3/2 + b - 2\sqrt{b}) = d_*$.

For any hidden selector, the all-on report pair is still independent of W . If its probabilities of 11 and 00 are u, v , then $u + v \leq 1$. Concavity and monotonicity of H give

$$H(u) + H(v) \leq 2H((u + v)/2) \leq 2H(1/2) = 7/8.$$

A fair hidden bit J , emitting J twice, attains this overlap. This requires precisely two deterministic selector tables and gives error $1/8$. In particular hiding a seed does not erase the actual-mark obstruction.

These all-on bounds are lower bounds for the feedback problem. For the upper bounds first generate the corresponding all-on pair (X_1, X_2) , emit X_1 , and after receiving A emit AX_2 . For a deterministic feedback $A = g(X_1)$ the full gated marked transcript is the pushforward of the all-on marked pair. Applying the same map to the target pair cannot increase total variation. The construction is executable at width one: the second fresh Bernoulli variable is independent in the

private case, and the bit J is the priced persistent hidden selector in the hidden case. Randomized feedback follows by conditioning on its independent random table.

Finally, in the visible case the two compared laws are

$$\sum_j \lambda_j \delta_j \otimes P_{g_j}^{q_j} \quad \text{and} \quad \sum_j \lambda_j \delta_j \otimes Q_{g_j}.$$

The target selector is independent of (B, W) . The all-on gate in every fiber has error at least d_* in every private fiber. Joint variation is the weighted sum of these fiber errors. Ignoring the visible selector and using the private optimizer attains d_* . $\square$

18.2. A full-dimensional collision preparation. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$ with contact distance d . The center of mass $Q = (q_1 + q_2)/2$ ranges over any fixed rational position box. Write $r = q_1 - q_2$, $g = v_1 - v_2$, and $V = (v_1 + v_2)/2$. Conditional on (B, W) in (18.1), take the uniform product preparation on

$$\begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), \quad a_0 \in [29/10, 31/10], & b_0 &\in [9/20, 11/20], \quad (18.3) \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

The boxes have positive volume in the full phase space. The actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$; total momentum conservation retains it. The informative quantity will instead be the transverse velocity created at a collision. Put

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector which is identically zero on the incoming transverse-velocity range.

Lemma 18.2 (Uniform transverse collision signal). *For every preparation in (18.3) and every $d \in [9/10, 11/10]$, exactly one collision occurs before time 3. Its time t_c lies in $(4/5, 3/2)$, and afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4. \quad (18.4)$$

For $0 < d \leq 2/5$ no collision occurs before time 3, and the detector is zero throughout that interval.

Proof. Up to a collision $r(t) = r + tg$. For $t \leq 4/5$, its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At time $3/2$ the absolute coordinate bounds are $1/4$, $113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. Thus a first contact occurs in the stated interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$ as well. The first contact therefore has $n_x < 0$, where $n = r(t_c)/d$. The entry contact is strictly incoming. The bounds above give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq \frac{87}{220}.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The equal-mass elastic collision rule is

$$v'_1 = V + g/2 - (g \cdot n)n, \quad g' = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v'_{1,y} > -3/200 + \frac{1443}{1000} \frac{87}{220} > 1/2.$$

The relative velocity is outgoing, $r(t_c) \cdot g' > 0$, so the squared separation increases strictly for all later times. There are only two spheres and no walls; no second collision is possible. This proves (18.4). For $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. Thus there is no collision. The incoming $|v_{1,y}| \leq 3/200$ then makes h zero. $\square$

At time 2 the physical target $F_{d,\sigma}$ reports

$$Y_1 = \mathbf{1}_{\{h(v_{1,y}(2)) + \sigma\xi_1 > 0\}}.$$

It then receives a gate $A \in \{0, 1\}$, and at time 3 reports

$$Y_2 = A\mathbf{1}_{\{h(v_{1,y}(3)) + \sigma\xi_2 > 0\}},$$

where the ξ_i are independent standard normal variables, independent of the preparation. The source E_d has the same preparation, evolution, control interface and actual mark, but emits two constant reports. The gates do not change the particle motion. This is a feedback acquisition experiment: taking $A = 0$ suppresses later information and taking $A = 1$ exposes another noisy observation of a collision-generated quantity.

Theorem 18.3 (Collision-sensitive physical resource separation). *Let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/12$. At retained width $(1, 1)$ the physical deficiencies obey, uniformly in d ,*

$$\begin{aligned} |\delta^p(E_d, F_{d,\sigma}) - d_*| &< 1/300, & |\delta_r^y(E_d, F_{d,\sigma}) - d_*| &< 1/300 \quad (r \geq 1), \\ |\delta_2^h(E_d, F_{d,\sigma}) - 1/8| &< 1/300. \end{aligned} \quad (18.5)$$

In particular private and visible error exceed $9/20$, while hidden two-selector error is less than $13/100$. More generally the errors in these three comparisons are at most $2\Phi(-1/(4\sigma))$, which tends to zero as $\sigma \downarrow 0$. For $0 < d \leq 2/5$ all three deficiencies are zero. The interacting and noninteracting regimes use the same preparation and detector. The conclusions are unchanged by simultaneously scaling positions, diameter and observation times by any positive factor, with velocities and detector noise unchanged.

Proof. Couple the ideal and physical target by the same (B, W) and the same feedback random table. Lemma 18.2 bounds each potential bit error by $\Phi(-1/(4\sigma))$. With probability at least $1 - 2\Phi(-1/(4\sigma))$ both potential reports agree with their ideal bits; the gates and all emitted transcripts then agree as well. This gives a marked feedback variation bound uniform over all gates. The channels in both directions are the identity on the binary report alphabet and have one retained state. The blind source is already exactly E_* . Stateless comparison therefore changes each fixed-budget deficiency by at most the stated error, without adding memory or hiding an exposed selector.

For completeness, the elementary Gaussian tail bound $\Phi(-x) \leq e^{-x^2/2}/(x\sqrt{2\pi})$ at $x = 3$ gives $\Phi(-3) < 1/600$: use $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$. Hence the total error is below $1/300$. The inequalities $d_* - 1/300 > 9/20$ and $1/8 + 1/300 < 13/100$ follow by squaring positive rational bounds.

When $d \leq 2/5$, both potential reports are independent fair noise bits and are independent of W . A private width-one simulator draws those two bits and applies the current gate, exactly reproducing every feedback transcript. Hidden and visible versions can ignore their selectors. Finally elastic scattering is invariant under $(q, d, t) \mapsto (sq, sd, st)$ with v unchanged. This proves the scaling statement; it is geometric similarity, not a particle-number or Boltzmann–Grad limit. $\square$

The theorem combines three effects. Collisions create the sign of the informative transverse velocity. A gate chosen after the first report changes future information. A retained hidden bit, but not a visible one, permits temporal correlation without charging the width-one register. The actual mark remains partially inaccessible in every signature. The conserved longitudinal momentum labels that mark; it does not generate the observed transverse signal.

18.3. Uniformly effective finite presentations. The binary physical interface is already finite, but its probabilities are not generally rational. We make their effective approximation explicit rather than treating individual computability as a uniform algorithm.

Proposition 18.4 (Collision-chart rational presentations). *For computable $d \in [9/10, 11/10]$ and $\sigma > 0$, one algorithm produces rational marked causal instruments and certified two-sided stateless errors tending to zero for $(E_d, F_{d,\sigma})$. On a compact computable parameter interval with $\sigma \geq \sigma_0 > 0$, the errors and the construction are uniform. Thus Proposition 16.6 applies at every fixed finite state and selector signature, not only at $(1, 1)$.*

Proof. On every preparation box the entry time and outgoing velocity are

$$t_c = \frac{-r \cdot g - \sqrt{(r \cdot g)^2 - |g|^2(|r|^2 - d^2)}}{|g|^2}, \quad n = (r + t_c g)/d, \quad v'_1 = V + g/2 - (g \cdot n)n.$$

The positive square root is separated from zero: it equals $-d(g \cdot n) > (9/10)(1443/1000)$. All denominators are likewise bounded away from zero. These explicit bounds and the compact boxes allow interval differentiation of the displayed formulas, producing a rational common Lipschitz upper bound C_{geo} for $v'_{1,y}$ in the packet variables and d . This computation terminates because only addition, multiplication, division by separated positive quantities and a separated square root occur.

Conditional on the microscopic preparation the two potential all-on bits are independent Bernoulli with probability $p = \Phi(h(v'_{1,y})/\sigma)$. Their four probabilities are p^2 , $p(1-p)$, $p(1-p)$ and $(1-p)^2$. The normal distribution function is uniformly computable on bounded intervals, for example by integrating its convergent exponential series with a remainder bound. Its Lipschitz constant is less than one. Together with the 1-Lipschitz detector and the geometric bound, this yields a computable Lipschitz constant at most $2C_{\text{geo}}/\sigma_0$ for these four integrands on fixed-noise compact parameter sets. Equal-box Riemann sums of mesh h therefore have explicit error at most a computable constant times h , plus the chosen rational evaluation error. Parameter uncertainty is included by its interval enclosure.

Integrate the four all-on probabilities separately on each packet, with the exact rational weights (18.1). For each W , round its conditional probability vector to a rational simplex point on a sufficiently fine grid; the rounding error is bounded by the sum of the coordinate errors and the grid mesh. Keep $\mathbb{P}(W = 0) = \mathbb{P}(W = 1) = 1/2$ exact. Sample this finite joint law once at initialization, emit its first bit, and gate its second bit. This is a finite latent realization of the target instrument, not a simulator with extra free state. It makes every causal prefix consistent and never divides by a possibly small prefix probability. Every gated law is a pushforward of the all-on law, so the same marked variation bound holds uniformly under feedback. Identity report channels give both stateless directions. Choosing summable evaluation and mesh errors produces the asserted uniform sequence. $\square$

19. UNIFORM STABILITY OF OPTIONAL PROJECTIONS

Marked comparison controls more than terminal bounded risk when the compared laws share their latent variable. We prove a whole-process estimate for the resulting posterior martingales. The distinction between a probabilistic coupling and an executable causal channel is important in this section.

Let Z take values in a standard Borel space and let Y take values in a standard Borel canonical path space on $[0, T]$, with a right-continuous observation filtration. Consider equivalent joint probability measures P, Q on (Z, Y) , with the same Z -marginal ν . Use the usual completions, which agree for equivalent output laws. For a bounded measurable latent variable $f(Z)$ write

$$M_t^P = \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y], \quad M_t^Q = \mathbb{E}_Q[f(Z) \mid \mathcal{F}_t^Y].$$

Take their right-continuous martingale versions. All suprema below include T and can be evaluated on a countable dense time set.

Theorem 19.1 (Marked variation controls the whole posterior process). *Let $\varepsilon = \|P - Q\|_{\text{TV}}$ and $|f| \leq 1$. When both posterior functions are evaluated on the same canonical output path,*

$$\mathbb{E}_Q \sup_{t \leq T} |M_t^P(Y) - M_t^Q(Y)|^2 \leq 80\varepsilon. \quad (19.1)$$

There is a coupling of (Z, Y^P) and (Z, Y^Q) which retains exactly the same latent variable and satisfies

$$\mathbb{P}(Y^P \neq Y^Q) \leq \varepsilon, \quad \mathbb{E} \sup_{t \leq T} |M_t^P(Y^P) - M_t^Q(Y^Q)|^2 \leq 84\varepsilon. \quad (19.2)$$

The coupling is not asserted to be causal or to have finite state. The variation bound itself, not this coupling, is used for resource transport.

Proof. Let $D = dP/dQ$, $D_t = \mathbb{E}_Q[D \mid \mathcal{F}_t^Y]$, and set

$$A_t = \mathbb{E}_Q[(D - 1)f \mid \mathcal{F}_t^Y], \quad B_t = D_t - 1.$$

Bayes' formula gives

$$M_t^P - M_t^Q = \frac{A_t - B_t M_t^Q}{1 + B_t}.$$

Since $\mathbb{E}_Q|D - 1| = 2\varepsilon$, both martingales A, B have terminal absolute expectation at most 2ε . Doob's weak L^1 inequality gives, for $0 < u \leq 2$,

$$\begin{aligned} Q\{\sup_t |M_t^P - M_t^Q| > u\} &\leq Q\{\sup_t |B_t| > 1/2\} + Q\{\sup_t |A_t| > u/4\} + Q\{\sup_t |B_t| > u/4\} \\ &\leq 4\varepsilon + 16\varepsilon/u. \end{aligned}$$

Integrating $2u$ times this bound on $(0, 2)$ gives $16\varepsilon + 64\varepsilon = 80\varepsilon$, proving (19.1). No tightness conclusion is being inferred from finite-dimensional convergence.

Disintegrate both joint laws over their common latent marginal. Measurable maximal coupling of the two output kernels, conditional on Z , has disagreement probability

$$\int \|P(dy \mid z) - Q(dy \mid z)\|_{\text{TV}} d\nu(z) = \|P - Q\|_{\text{TV}}.$$

Such a coupling can be constructed by using the minimum of the two kernel densities with respect to their sum, then their residual kernels. It is measurable on standard Borel spaces. On the event of identical full output paths use (19.1) with the Q -marginal; on its complement the squared difference is at most four. This proves (19.2). The maximal coupling may depend on the full output path; it has no asserted online implementation. $\square$

Corollary 19.2 (Changing-filtration convergence from marked comparison). *Let P_n and P be equivalent joint laws with the same latent marginal ν , and $\|P_n - P\|_{\text{TV}} \rightarrow 0$. They admit a common probability-space realization $(Z, Y, Y^1, Y^2, \dots)$ such that, for every $f \in L^2(\nu)$,*

$$\mathbb{E} \sup_{t \leq T} \left| \mathbb{E}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}[f(Z) \mid \mathcal{F}_t^Y] \right|^2 \rightarrow 0.$$

The same realization gives convergence uniformly in time in probability for $f \in L^1(\nu)$. It also permits $f_n \rightarrow f$ in $L^2(\nu)$, respectively in $L^1(\nu)$, in the first conditional expectation.

Proof. Glue the conditional maximal couplings of (Z, Y^n) with the fixed reference (Z, Y) by taking their conditional kernels independently on a countable product. The coupling does not depend on f . For the truncation $f^{(K)} = (-K) \vee (f \wedge K)$, Theorem 19.1 bounds the middle difference in the L^2 norm of the supremum by $K\sqrt{84\varepsilon_n}$. Doob's L^2 inequality bounds each truncation tail by $2\|f - f^{(K)}\|_{L^2(\nu)}$. The triangle inequality therefore gives

$$\left\| \sup_t |M_t^{P_n, f} - M_t^{P, f}| \right\|_2 \leq 4\|f - f^{(K)}\|_2 + K\sqrt{84\varepsilon_n}.$$

First let $n \rightarrow \infty$, then $K \rightarrow \infty$. For L^1 use instead the weak maximal inequality for each tail and the bounded-case estimate for the middle term. An additional difference $f_n - f$ is handled by the same maximal inequalities under P_n , whose latent marginal remains ν . $\square$

19.1. A microscopic sufficient condition. Let $(\Phi_t)_{t \in \mathbb{R}}$ be an almost-everywhere regular hard-sphere flow on a labeled phase space, with invariant probability μ . Fix a prepared law ν with $d\nu/d\mu \leq C$, and bounded real observables h, h_n . These may be collision-sensitive tagged observables. Put $\eta_n = \|h_n - h\|_{L^2(\mu)}$. With a common latent initial state $Z \sim \nu$, consider the continuous observation laws

$$\begin{aligned} Y_t &= \int_0^t h(\Phi_s Z) ds + \sigma B_t, \\ Y_t^n &= \int_0^t h_n(\Phi_s Z) ds + \sigma B_t^n, \quad \sigma > 0, \end{aligned}$$

where the Brownian noise is independent of Z in each model. The notation does not prescribe a synchronous coupling.

Theorem 19.3 (Fixed-flow observation-approximation limit). *The joint latent-output laws are equivalent and satisfy*

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \eta_n. \quad (19.3)$$

For every $|f(Z)| \leq 1$ they admit a latent-preserving coupling with

$$\mathbb{E} \sup_{t \leq T} |M_t^{P_n, f} - M_t^{P, f}|^2 \leq \frac{42\sqrt{CT}}{\sigma} \eta_n. \quad (19.4)$$

In particular $\eta_n \rightarrow 0$ supplies whole-process optional-projection convergence, not merely convergence at finitely many times. This applies when h_n are orbit-mollified graph-core approximants to any bounded $h \in L^2(\mu)$.

Proof. Conditional on Z , the signals are deterministic bounded drifts. The exponential change-of-measure identity for Brownian motion gives

$$D_{\text{KL}}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |(h_n - h)(\Phi_s Z)|^2 ds \leq \frac{CT}{2\sigma^2} \eta_n^2.$$

The last inequality uses $\nu \leq C\mu$ and invariance of μ ; it does not require mixing, differentiability across a collision, or a claim that a tagged velocity belongs to the generator domain. Bounded drifts ensure equivalence and justify the exponential identity. Pinsker's inequality gives (19.3). Applying Theorem 19.1 proves (19.4).

For the last assertion choose a nonnegative smooth approximate identity ρ_n on the time line and set $h_n = \int \rho_n(s) h \circ \Phi_s ds$. Strong continuity of the Koopman group in $L^2(\mu)$ gives $\eta_n \rightarrow 0$ and $\|h_n\|_\infty \leq \|h\|_\infty$. Repeated integration by parts puts h_n in every power domain of the generator, with derivatives obtained from $\rho_n^{(k)}$. This is the orbit-core construction already used in the microscopic domain theory; the new conclusion is the uniform posterior estimate. $\square$

Sampling these observation paths at finitely many checkpoints and applying a common causal output postprocessing are contractions in marked variation. Thus the identity report comparison transfers every bounded finite-register risk at its original state and selector budgets with error bounded by (19.3). That is the resource-comparison consequence. The latent-preserving coupling in (19.4) has a different role: it realizes convergence of the changing observation filtrations and is not used as a finite-register simulator.

The theorem discharges a sufficient conditional-law hypothesis for optional projection in a genuine microscopic observation class. The constants depend on the fixed preparation, time

and noise level. No particle-uniform kinetic estimate, changing microscopic path limit, limiting stochastic-exponential identification or nonlinear BSDE limit is implied without its additional hypotheses. The whole-process estimate is what replaces an unsupported appeal to Doob's inequality as a tightness argument.

20. CHANGING TRAJECTORIES, LIKELIHOODS, AND CONDITIONAL CONSUMERS

20.1. A common preparation, not a common flow. Let $(\mathcal{Z}, \nu)$ be a standard Borel probability space, and let $H, H_n : [0, T] \times \mathcal{Z} \rightarrow \mathbb{R}$ be jointly measurable, with $|H|, |H_n| \leq M$. They may be observations of different flows on different phase spaces, pulled back by specified preparation maps from $\mathcal{Z}$. No common microscopic generator or invariant measure is assumed. Set

$$\eta_n^2 = \int_{\mathcal{Z}} \int_0^T |H_n(t, z) - H(t, z)|^2 dt d\nu(z).$$

On $\mathcal{Z} \times C([0, T])$ let R be the law of independent $Z \sim \nu$ and $Y = \sigma B$, where $\sigma > 0$. Define

$$L_t^n = \exp \left\{ \frac{1}{\sigma} \int_0^t H_n(s, Z) dB_s - \frac{1}{2\sigma^2} \int_0^t H_n(s, Z)^2 ds \right\}, \quad (20.1)$$

$$dP_n = L_T^n dR. \quad (20.2)$$

The reference model P, L uses H . Conditional on Z , these are Gaussian shifts, so under P_n the observed path has law $Y_t = \int_0^t H_n(s, Z) ds + \sigma B_t^n$. Boundedness gives true strictly positive density martingales. Write $\mathcal{F}_t$ for the usual canonical observation filtration and $\mathcal{H}_t = \sigma(Z, B_s : s \leq t)$ for the larger reference filtration. Equivalent laws have the same completed observation stopping times.

Theorem 20.1 (Likelihood and posterior transport for changing paths). *The measures P_n, P, R are equivalent and have common Z -marginal ν . With $A = M^2 T / \sigma^2$ and $\varepsilon_n = \min\{1, \eta_n / (2\sigma)\}$, one has*

$$\|P_n - P\|_{\text{TV}} \leq \varepsilon_n, \quad (20.3)$$

$$\mathbb{E}_R \sup_{t \leq T} |L_t^n - L_t|^2 \leq \frac{8}{\sigma^2} e^{3A} \eta_n^2. \quad (20.4)$$

For $|f(Z)| \leq K$ put

$$N_t^{n,f} = \mathbb{E}_R[L_T^n f(Z) \mid \mathcal{F}_t], \quad \ell_t^n = N_t^{n,1}.$$

Then

$$\mathbb{E}_R \sup_t |N_t^{n,f} - N_t^f|^2 \leq \frac{8K^2}{\sigma^2} e^{3A} \eta_n^2. \quad (20.5)$$

In particular this applies to the projected likelihood ℓ^n . For every $|f| \leq 1$, the posterior functions satisfy the same-path bound $80\varepsilon_n$ of Theorem 19.1, and there is one latent-preserving coupling, independent of f , on which

$$\mathbb{E} \sup_t |\mathbb{E}_{P_n}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y]|^2 \leq 84\varepsilon_n. \quad (20.6)$$

If $\eta_n \rightarrow 0$ and deterministic microscopic paths $X_n(Z) \rightarrow X(Z)$ in a specified Polish path metric in ν -probability, then on this coupling the microscopic paths, the full and projected likelihood processes evaluated on their own observation paths, and each bounded posterior process converge jointly in probability in that metric and uniformly in time for the last three coordinates.

Proof. Conditional Gaussian change of measure gives

$$\text{KL}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |H_n - H|^2 dt = \frac{\eta_n^2}{2\sigma^2}.$$

The same calculation proves conditional equivalence, and the conditional density has mean one, so the preparation marginal is unchanged. Integrating and applying Pinsker proves (20.3).

For the quantitative density estimate, condition first on $Z = z$ and write $D_t = L_t^n - L_t$. The stochastic differential equation is

$$dD_t = \sigma^{-1}\{D_t H_n(t, z) + L_t(H_n - H)(t, z)\} dB_t.$$

The conditional second moment of L_t is at most e^A . Itô's isometry and $(u + v)^2 \leq 2u^2 + 2v^2$ therefore imply

$$\mathbb{E}_R[D_t^2 | z] \leq \frac{2M^2}{\sigma^2} \int_0^t \mathbb{E}_R[D_s^2 | z] ds + \frac{2e^A}{\sigma^2} \int_0^t |H_n - H|^2(s, z) ds.$$

Gronwall, followed by integration in z , gives $\mathbb{E}_R D_T^2 \leq (2/\sigma^2) e^{3A} \eta_n^2$. The difference is a square-integrable $\mathcal{H}_t$ -martingale. Doob's L^2 inequality proves (20.4). Applying the same maximal inequality to the $\mathcal{F}_t$ -martingale $\mathbb{E}_R[(L_T^n - L_T)f | \mathcal{F}_t]$ proves (20.5). These are direct process estimates, not a claim that finite-dimensional convergence alone implies tightness.

The posterior assertions follow from Theorem 19.1 and its countable conditional-coupling construction. For the last claim, on the event $Y^n = Y$ the two likelihood functions are evaluated on the same (Z, Y) . Their uniform difference tends to zero in R -probability by the estimates just proved, and therefore in P -probability by absolute continuity of the fixed P with respect to R . The complementary event has probability at most ε_n . The same argument applies to ℓ^n , and (20.6) handles the posterior. The microscopic convergence holds on any coupling retaining the same Z . A finite union bound proves joint convergence. Stochastic integral versions are chosen measurably on the canonical product space; equivalence makes these common versions valid under every law in the countable sequence. $\square$

For $f \in L^2(\nu)$ the posterior conclusion follows by truncation exactly as in Corollary 19.2; it also permits $f_n \rightarrow f$ in $L^2(\nu)$. Boundedness of H_n is being used for the uniform likelihood second moments. It is not replaced by an unverified uniform-integrability assertion in a singular or zero-noise limit.

20.2. The projected stochastic exponential. Let $\hat{H}_t^n = \mathbb{E}_{P_n}[H_n(t, Z) | \mathcal{F}_t]$, using a predictable version for Lebesgue-almost every t . Put $\theta_t^n = \hat{H}_t^n / \sigma$.

Proposition 20.2 (Identification before passage to the limit). *The projected likelihood and the innovation satisfy*

$$\ell_t^n = \exp \left\{ \int_0^t \theta_s^n dB_s - \frac{1}{2} \int_0^t (\theta_s^n)^2 ds \right\}, \quad (20.7)$$

$$I_t^n = B_t - \int_0^t \theta_s^n ds. \quad (20.8)$$

Here I^n is a P_n -Brownian motion in $\mathcal{F}_t$. In particular the limit of the projected likelihoods in (20.5) is the projected stochastic exponential of the limiting model. Change of measure commutes with this limit on bounded measurable path tests.

Proof. Project $L_t^n = 1 + \int_0^t L_s^n H_n(s, Z) dB_s / \sigma$ onto the Brownian observation filtration. The integrands are square integrable; testing against bounded elementary predictable stochastic integrals and using Itô's isometry gives

$$d\ell_t^n = \sigma^{-1} \mathbb{E}_R[L_t^n H_n(t, Z) | \mathcal{F}_t] dB_t = \ell_t^n \theta_t^n dB_t.$$

The last equality is Bayes' formula. Since $|\theta_t^n| \leq M/\sigma$, the solution is the positive stochastic exponential (20.7); Girsanov gives the stated innovation. The same argument with H identifies the limiting exponential. Finally $|\mathbb{E}_{P_n} F - \mathbb{E}_P F| \leq 2\|F\|_\infty \varepsilon_n$ for bounded signed F . There is no interchange based only on a formal limit of logarithms. $\square$

20.3. A bounded entropic backward-equation consumer. Fix $\gamma > 0$ and a latent payoff f with $|f| \leq 1$. Define

$$U_t^n = \frac{1}{\gamma} \log \mathbb{E}_{P_n}[e^{\gamma f(Z)} \mid \mathcal{F}_t] = \frac{1}{\gamma} \log \frac{N_t^{n,e^{\gamma f}}}{\ell_t^n}. \quad (20.9)$$

The terminal condition here is the observable conditional certainty equivalent U_T^n , not the generally unobserved $f(Z)$ itself. This specification avoids imposing a latent terminal value on the smaller observation filtration.

Theorem 20.3 (Entropic values and localized backward integrands). *There is a unique bounded entropic value process with terminal value U_T^n ; for a predictable integrand Z^n it satisfies*

$$U_t^n = U_T^n + \int_t^T \frac{\gamma}{2} |Z_s^n|^2 ds - \int_t^T Z_s^n dI_s^n. \quad (20.10)$$

On the coupling of Theorem 20.1,

$$\mathbb{E} \sup_t |U_t^n(Y^n) - U_t(Y)|^2 \leq 84 \left(\frac{e^{2\gamma} - 1}{\gamma} \right)^2 \varepsilon_n. \quad (20.11)$$

For each $r > e^\gamma$ let τ_r^n be the first exit of any of $\ell^n, \ell, N^{n,e^{\gamma f}}, N^{e^{\gamma f}}$ from $(1/r, r)$, capped at T . If $\eta_n \rightarrow 0$, then

$$\mathbb{E}_R \int_0^{\tau_r^n} |Z_s^n - Z_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^{\tau_r^n} |\theta_s^n - \theta_s|^2 ds \rightarrow 0. \quad (20.12)$$

Furthermore $\lim_{r \rightarrow \infty} \sup_n R(\tau_r^n < T) = 0$. Thus the localizations exhaust the observation experiment in probability.

Proof. For a positive bounded latent g , write $N_t^{n,g} = \mathbb{E}_R[L_T^n g \mid \mathcal{F}_t]$ and $dN_t^{n,g} = a_t^{n,g} dB_t$. Projection of the full integral gives

$$a_t^{n,g} = \sigma^{-1} \mathbb{E}_R[L_t^n g H_n(t, Z) \mid \mathcal{F}_t], \quad |a_t^{n,g}| \leq (M/\sigma) N_t^{n,g}.$$

The same representation with $g = 1$ gives $d\ell_t^n = b_t^n dB_t$ and $b_t^n/\ell_t^n = \theta_t^n$. Put $N^n = N^{n,e^{\gamma f}}$ and $a^n = a^{n,e^{\gamma f}}$. Itô's formula in (20.9) gives the martingale coefficient

$$Z_t^n = \gamma^{-1} (a_t^n/N_t^n - b_t^n/\ell_t^n).$$

Replacing dB_t by $dI_t^n + \theta_t^n dt$ reduces its drift to $-\gamma|Z_t^n|^2/2$, which proves (20.10). The ratios are bounded by M/σ , so the integrals are well defined. Conversely, exponentiating any bounded solution of (20.10) produces a bounded P_n -martingale with terminal value $e^{\gamma U_T^n}$. Its conditional-expectation representation is unique, and martingale representation then identifies its integrand. This proves uniqueness in the bounded class.

Apply Theorem 19.1 to $g_0 = (e^{\gamma f} - e^{-\gamma})/(e^\gamma - e^{-\gamma}) \in [0, 1]$. The map taking its conditional expectation to $\gamma^{-1} \log \mathbb{E}[e^{\gamma f} \mid \mathcal{F}_t]$ has Lipschitz constant at most $(e^{2\gamma} - 1)/\gamma$. This proves (20.11).

The initial values of N^n and N agree, since Z has the same marginal. Itô's isometry and the terminal version of (20.5) imply

$$\mathbb{E}_R \int_0^T |a_s^n - a_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^T |b_s^n - b_s|^2 ds \rightarrow 0.$$

Before τ_r^n , all denominators lie in $[1/r, r]$, and the limiting numerators a, b are bounded in absolute value by $(M/\sigma)r$. For example,

$$\left| \frac{a^n}{N^n} - \frac{a}{N} \right| \leq r|a^n - a| + (M/\sigma)r^3|N^n - N|.$$

The supremum estimates (20.5), followed by integration in time, prove (20.12). The same inequality applies to b^n/ℓ^n .

Finally $e^{-\gamma\ell^n} \leq N^n \leq e^{\gamma\ell^n}$. Each ℓ^n is a stochastic exponential with uniformly bounded integrand θ^n . Its positive and negative moments of the supremum are bounded uniformly in n : write any power as a stochastic exponential times its bounded compensator and apply Doob's inequality to a higher positive power. The same calculation applies to its inverse. Markov's inequality, the preceding comparison of N^n with ℓ^n , and a union bound prove the exhaustion assertion. This is a localized integrand theorem; it does not assert an unspecified global topology for backward equations with unbounded terminal data or different nonlinear drivers. $\square$

Corollary 20.4 (A changing-filtration stopping consumer). *Let $\psi(t, x) \in [0, 1]$ be measurable and L_ψ -Lipschitz in x , uniformly in t . For a bounded latent f with $|f| \leq 1$, put $M_t^n = \mathbb{E}_{P_n}[f(Z) \mid \mathcal{F}_t]$. Then*

$$\left| \sup_{\tau \leq T} \mathbb{E}_{P_n} \psi(\tau, M_\tau^n) - \sup_{\tau \leq T} \mathbb{E}_P \psi(\tau, M_\tau) \right| \leq \varepsilon_n + L_\psi \sqrt{80\varepsilon_n}.$$

The same statement for the entropic value U has the additional Lipschitz factor $(e^{2\gamma} - 1)/\gamma$ on the square-root term. The suprema range over the common canonical stopping times; attainment is not needed for the estimate.

Proof. For a fixed stopping time, first change the measure while keeping its $[0, 1]$ reward function, at cost at most ε_n . Then compare the two reward functions on the same canonical path under P , using Theorem 19.1, Cauchy-Schwarz and the Lipschitz bound. The estimate is uniform in the stopping time, so it survives both suprema. Equivalence of the observation laws is what allows the same stopping-time class. The entropic assertion follows from the Lipschitz calculation in the preceding theorem. $\square$

20.4. Resource transport is not the posterior coupling. Apply any common finite causal acquisition map to Y and retain a fixed finite mark $w(Z)$. Data processing makes its marked row variations at most ε_n , uniformly over a declared finite feedback family. If the source and target have respective bounds α_n, β_n , Theorem 2.2 gives same-budget deficiency perturbation at most $\alpha_n + \beta_n$, and Proposition 13.4 transports a fixed reset audit. A subsequent consumer of width S composed with a simulator of width K has width KS , with the stated selector price. None of these executable claims uses the latent-preserving maximal coupling. That coupling belongs to the proof of conditional-process convergence, not to the finite-state machine.

These results provide a direct changing-trajectory, likelihood, Girsanov and bounded entropic-value input for a conditional-law theory. They do not assert the strict-dual, rigidity, covariant-form or full kinetic conclusions of a larger program without their separate analytic proofs.

Example 20.5 (Why an amplitude bound is not a tightness argument). Let ξ_1, ξ_2 be independent symmetric signs, and, for $n \geq 5$, let $\mathcal{G}_t^n$ reveal ξ_1 at $1/2 - 2/n$ and ξ_2 at $1/2 - 1/n$. On $[0, 1]$ the bounded martingale

$$M_t^n = \mathbb{E}[\xi_1 + \xi_2 \mid \mathcal{G}_t^n]$$

converges at every finite collection of fixed times to the martingale $M_t = \mathbf{1}_{[1/2, 1]}(t)(\xi_1 + \xi_2)$. Nevertheless its laws are not tight in the Skorokhod J_1 topology. On the event $\xi_2 = -\xi_1$, of probability $1/2$, its path has an excursion of height one on an interval of length $1/n$ and is zero outside that interval. The three-time oscillation modulus of such a path is one at every scale greater than $2/n$. This violates the necessary J_1 tightness modulus criterion. All maximal moments are uniformly bounded by 2^p .

This example concerns the inference from finite-dimensional convergence and a maximal amplitude estimate alone. It does not claim that every possible stronger conditional-kernel hypothesis fails. The uniform estimates in Theorem 20.1 remove the need for that inference in the observation class treated here.

21. OBSERVATION DENSITIES AND COUPLED PROCESS VERSIONS

We record three measure-theoretic facts used in the global transport argument. Their purpose is to make explicit the passage from a tilted physical law to an observation law and from two marginal processes to one joint realization.

Lemma 21.1 (Observation-level return from a bounded tilt). *Let $dP^g/dP = g(Z)/c$, where $c = \mathbb{E}_P g$ and $e^{-\gamma} \leq g \leq e^\gamma$. For any observation sub- σ -field $\mathcal{F}$,*

$$\frac{d(P|_{\mathcal{F}})}{d(P^g|_{\mathcal{F}})} = \mathbb{E}_{P^g} \left[\frac{c}{g} \middle| \mathcal{F} \right] = \frac{c}{\mathbb{E}_P[g | \mathcal{F}]} \quad (21.1)$$

Both this density and its reciprocal lie in $[e^{-2\gamma}, e^{2\gamma}]$. In particular, for any nonnegative $\mathcal{F}$ -measurable V , $\mathbb{E}_P V \leq e^{2\gamma} \mathbb{E}_{P^g} V$.

Proof. For $A \in \mathcal{F}$, integrate c/g over A under P^g to obtain $P(A)$. Conditioning proves the first identity. In the other direction, $d(P^g|_{\mathcal{F}})/d(P|_{\mathcal{F}}) = \mathbb{E}_P[g | \mathcal{F}]/c$; equivalence and inversion prove the second. The bounds follow from the bounds on g , its conditional mean, and c . $\square$

Lemma 21.2 (Domination of the common-path measure). *Let (Y', Y) be a coupling on a standard Borel path space, with marginal laws P', P , and let $E = \{Y' = Y\}$. The subprobability measure $\kappa(A) = \Pr(E, Y \in A)$ is dominated by both P and P' . Consequently, for any nonnegative measurable path functional V ,*

$$\mathbb{E}[\mathbf{1}_E V(Y)] \leq \mathbb{E}_P V, \quad \mathbb{E}[\mathbf{1}_E V(Y')] \leq \mathbb{E}_{P'} V.$$

The assertion also holds on a common latent–observation space when the latent coordinate is shared by the coupling.

Proof. The event defining $\kappa(A)$ is contained in $\{Y \in A\}$ and, on E , in $\{Y' \in A\}$. This proves domination on sets. Integration of simple functions followed by monotone convergence proves the rest. Apply the same argument to the product coordinate for the last assertion. $\square$

Lemma 21.3 (Marginal stochastic integrals on an arbitrary coupling). *On a canonical standard Borel observation path space, suppose that under each of two marginal laws there are a continuous adapted process U , a predictable square-integrable process Z , and an innovation I such that, in that marginal's own filtration,*

$$\int_0^t Z_s dI_s = U_t - U_0 + \frac{\gamma}{2} \int_0^t |Z_s|^2 ds, \quad 0 \leq t \leq T. \quad (21.2)$$

Assume Z is bounded, as in the entropic model here. There are measurable versions of the two continuous integral processes which can be pulled back to any coupling of the observation laws. Each pulled-back version has the correct marginal law and satisfies its own identity (21.2). Neither integral thereby acquires a martingale or Itô-integral interpretation in the enlarged joint filtration.

Proof. A continuous process is determined by its values at rational times. Completed-measurable random variables on the canonical space have Borel versions. Choose such versions at the countably many rational times and extend by continuity on the measurable set where a continuous extension exists, using the zero path elsewhere. This gives a Borel $C([0, T])$ -valued version of U . Likewise take a jointly measurable version of Z modulo $dt \otimes P$; its bounded time integral has a continuous path, and its rational-time values give a measurable continuous version by the same construction. The right side of (21.2) is consequently a measurable continuous path functional. Marginal equality at all rational times, and continuity of the stochastic integral, make it indistinguishable from that integral. Repeat for the other marginal. Pullback along the two coordinate maps of any coupling preserves their respective almost-sure identities and laws. No enlargement theorem for innovations is used. $\square$

In Theorem 22.2, Lemma 21.1 is applied with $\mathcal{F} = \mathcal{F}_T$ to the integrated squared coefficient comparison. Lemma 21.2 justifies restricting a same-path estimate to the equality event of the latent-preserving coupling. Lemma 21.3 fixes the meaning of the coupled backward martingale integrals before their difference is estimated. Thus the coupled notation retains the marginal observation filtrations throughout.

22. GLOBAL TRANSPORT OF THE BACKWARD EQUATION

We retain exactly the common-preparation, bounded-signal and positive-noise hypotheses of Section 20. In particular the canonical joint laws P_n, P are equivalent and have the same latent marginal ν . All comparisons on a single observation path use common measurable versions of the conditional expectations. The following estimate avoids division by a small observation density in the final bound.

Lemma 22.1 (An integrated posterior comparison). *Let P, Q be equivalent laws on a standard Borel product carrying a latent variable Z and an observation Y , and let $\|P - Q\|_{\text{TV}} \leq \epsilon$. If $|u(Z)| \leq M$ and $\mathcal{F}$ is an observation sub- σ -field, then, under either $S = P$ or $S = Q$,*

$$\mathbb{E}_S |\mathbb{E}_Q[u | \mathcal{F}] - \mathbb{E}_P[u | \mathcal{F}]| \leq 4M\epsilon, \quad (22.1)$$

$$\mathbb{E}_S |\mathbb{E}_Q[u | \mathcal{F}] - \mathbb{E}_P[u | \mathcal{F}]|^2 \leq 8M^2\epsilon. \quad (22.2)$$

A common latent marginal is not needed for these two inequalities.

Proof. Use a dominating finite measure on $\mathcal{F}$. Write p, q for the densities of the restricted probability laws and a, b for the densities of the signed measures $\mathbb{E}_P[u \mathbf{1}_A], \mathbb{E}_Q[u \mathbf{1}_A]$ on $\mathcal{F}$. Their conditional means are $m_P = a/p$ and $m_Q = b/q$. Outside null sets, $|m_P|, |m_Q| \leq M$, and

$$p(m_Q - m_P) = m_Q(p - q) + (b - a).$$

The integral of $|p - q|$ is at most 2ϵ , by contraction of total variation. The integral of $|b - a|$ is at most $2M\epsilon$, since multiplication by u followed by restriction cannot increase signed variation by more than M . Integrating the displayed identity proves (22.1) under P . Interchanging the two laws proves it under Q . The squared bound follows from $|m_Q - m_P| \leq 2M$. $\square$

Let $|f| \leq 1$, $\gamma > 0$, and use the notation θ^n, U^n, Z^n, I^n of Proposition 20.2 and Theorem 20.3. Here Z^n denotes the scalar backward integrand, whereas the unadorned latent argument of H_n is the preparation variable. Define the deterministic bounds

$$A_n = \sigma^{-2} \{2\eta_n^2 + 16M^2T\epsilon_n\}, \quad (22.3)$$

$$C_n = \frac{2(1 + e^{4\gamma})}{\gamma^2} A_n, \quad (22.4)$$

where $\epsilon_n = \min\{1, \eta_n/(2\sigma)\}$ as before.

Theorem 22.2 (Global innovation and entropic-integrand transport). *Under the hypotheses of Theorem 20.1, under either physical law $S = P$ or $S = P_n$ on the same canonical path,*

$$\mathbb{E}_S \int_0^T |\theta_t^n - \theta_t|^2 dt \leq A_n, \quad \mathbb{E}_S \int_0^T |Z_t^n - Z_t|^2 dt \leq C_n. \quad (22.5)$$

No stopping at a density level is required. If $\eta_n \rightarrow 0$, both integrals also converge to zero in expectation under the reference law R .

On the single latent-preserving coupling of Theorem 20.1, evaluate each process on its own observation path, denoting these evaluations by a bar. Then

$$\mathbb{E} \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt \leq A_n + 4M^2 T \varepsilon_n / \sigma^2, \quad (22.6)$$

$$\mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt \leq C_n + 16M^2 T \varepsilon_n / (\gamma^2 \sigma^2). \quad (22.7)$$

If $\eta_n \rightarrow 0$, then $\bar{I}^n \rightarrow \bar{I}$ uniformly in probability. Writing $\bar{J}_t^n = \int_0^t \bar{Z}_s^n d\bar{I}_s^n$ for the stochastic integral in its own observation filtration, and $\bar{J}$ for its limit, one has

$$\mathbb{E} \sup_{t \leq T} |\bar{J}_t^n - \bar{J}_t|^2 \leq 2\mathbb{E} \sup_{t \leq T} |\bar{U}_t^n - \bar{U}_t|^2 + \frac{8M^2 T}{\sigma^2} \mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt. \quad (22.8)$$

The first term on the right has the explicit bound in (20.11). Thus the full bounded entropic backward representation, including its martingale integral, is transported globally. The coupling need not make either innovation Brownian in the joined filtration; each integral is defined under its own marginal law.

Proof. For each deterministic t , under P split

$$\sigma(\theta_t^n - \theta_t) = \mathbb{E}_{P_n}[H_n(t, Z) | \mathcal{F}_t] - \mathbb{E}_P[H_n(t, Z) | \mathcal{F}_t] + \mathbb{E}_P[H_n(t, Z) - H(t, Z) | \mathcal{F}_t].$$

Lemma 22.1, conditional Jensen and $(x + y)^2 \leq 2x^2 + 2y^2$ bound its expected square by $16M^2 \varepsilon_n + 2\mathbb{E}_\nu |H_n(t, Z) - H(t, Z)|^2$. Integration gives the first bound in (22.5) under P . Under P_n use the same decomposition with the signal-difference term conditioned under P_n and the law-change term applied to H . The common latent marginal gives the identical bound. Jointly measurable conditional versions, and predictable versions equal for dt -almost every time, give the same integrated statements.

Put $g = e^{\gamma f}$ and $c = \mathbb{E}_\nu g$. Tilt the two full joint laws by $g(Z)/c$, writing P^g, P_n^g . They still have the same latent marginal $\nu_g = g\nu/c$, and their observation signals remain H, H_n . Since $e^{-\gamma} \leq g, c \leq e^\gamma$,

$$\|P_n^g - P^g\|_{\text{TV}} \leq e^{2\gamma} \varepsilon_n, \quad \eta_{n,g}^2 \leq e^{2\gamma} \eta_n^2.$$

Moreover, the reference martingale coefficient $a^{n,g}$ from the proof of Theorem 20.3 satisfies

$$\alpha_t^n := \frac{a_t^{n,g}}{N_t^{n,g}} = \sigma^{-1} \mathbb{E}_{P_n^g}[H_n(t, Z) | \mathcal{F}_t].$$

Apply the preceding innovation argument to these tilted laws. It bounds the squared integral of $\alpha^n - \alpha$ by $e^{2\gamma} A_n$ under either tilted physical law. For an observation-measurable nonnegative random variable, passage back to the corresponding original physical law costs at most $e^{2\gamma}$: the full density c/g is bounded by this constant, and so is its observation projection. Consequently

$$\mathbb{E}_S \int_0^T |\alpha_t^n - \alpha_t|^2 dt \leq e^{4\gamma} A_n.$$

The identity $Z^n = (\alpha^n - \theta^n)/\gamma$ proves the second bound in (22.5).

The pointwise bounds, valid for all models, are

$$|\theta^n|, |\alpha^n| \leq M/\sigma, \quad |Z^n| \leq 2M/(\gamma\sigma). \quad (22.9)$$

The squared time integrals are thus uniformly bounded random variables. Convergence in P -probability, followed by absolute continuity of the fixed R with respect to P , gives convergence in R -probability. Boundedness then gives convergence of their R -expectations. This argument establishes global reference convergence without claiming a new explicit reference rate.

For the coupling, let $E_n = \{Y^n = Y\}$, so that $\mathbb{P}(E_n^c) \leq \varepsilon_n$. On E_n both processes are functions of the same observation path. The restricted expectation of their squared difference is at most the

corresponding unrestricted expectation under the limiting marginal P . On E_n^c apply (22.9) and integrate for time T . This proves (22.6)–(22.7).

On E_n , the Brownian reference coordinates $Y^n/\sigma, Y/\sigma$ agree, so

$$\sup_{t \leq T} |\bar{I}_t^n - \bar{I}_t|^2 \leq T \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt.$$

Markov's inequality and $\mathbb{P}(E_n^c) \rightarrow 0$ prove uniform convergence in probability. This assertion does not require any moment estimate on the observation difference on the exceptional event.

Finally, the backward equation already proved gives, under each marginal,

$$\bar{J}_t^n = \bar{U}_t^n - U_0 + \frac{\gamma}{2} \int_0^t |\bar{Z}_s^n|^2 ds.$$

The common initial value is $U_0 = \gamma^{-1} \log \mathbb{E}_\nu g$. By (22.9),

$$||\bar{Z}_s^n|^2 - |\bar{Z}_s|^2| \leq \frac{4M}{\gamma\sigma} |\bar{Z}_s^n - \bar{Z}_s|.$$

Subtract the two integral identities, use Cauchy–Schwarz in time and $(x + y)^2 \leq 2x^2 + 2y^2$, and take expectations. This gives (22.8). The integral identity also defines compatible measurable versions on the coupling, without asserting a martingale property for its joint filtration. $\square$

The theorem strengthens the localized conclusion of Theorem 20.3 without changing its terminal condition or its driver. The terminal condition remains the observation-measurable conditional certainty equivalent, not the unobserved latent payoff. Positive fixed noise, bounded signals, and a common preparation are used in the proof. A singular zero-noise limit, an unbounded terminal condition, or a different nonlinear driver would require additional arguments.

23. A UNIFORM CERTIFICATE THROUGH A MOVING COLLISION

We now use all three parts of the argument on one family. The finite marked reference law and the uniform scattering geometry are those of Theorem 18.1 and Lemma 18.2. They are proved above, including the no-collision control. We use a continuous observation of the same detector so that finite reports and conditional processes are generated by one experiment.

Let z denote a preparation in the boxes of (18.3), and let ν be that prepared law. The sign of the impact orientation is $2B - 1$, and the conserved center-of-mass longitudinal sign is the actual mark W . Their correlation is prepared: $\mathbb{P}(B = b, W = w) = 5/16$ for $b = w$ and $3/16$ otherwise. The contact diameter d ranges over $I = [9/10, 11/10]$. Let $X_d(t, z)$ be the two-particle phase path, with outgoing velocities used at the collision, and define

$$H_d(t, z) = h(v_{1,y}^d(t, z)), \quad Y_t^d = \int_0^t H_d(s, z) ds + \sigma B_t^d, \quad 0 < \sigma \leq 1/24.$$

The Brownian noise is independent of the preparation. The source experiment has the same physical preparation and mark but constant reports. The target reports at times 2 and 3 are

$$Z_1^d = \mathbf{1}\{2(Y_2^d - Y_{3/2}^d) > 0\}, \quad Z_2^d = A\mathbf{1}\{2(Y_3^d - Y_{5/2}^d) > 0\}. \quad (23.1)$$

The acquisition gate $A \in \{0, 1\}$ is chosen after Z_1^d ; it does not act on the mechanical trajectory. The continuous path is the auxiliary observation experiment for conditional consumers; the finite simulator is not given this path or an uncharged buffer for computing its increments. The target sensor implements the declared window acquisition.

Lemma 23.1 (Quantitative change of the microscopic path). *For all preparations in the specified boxes, the collision time $t_c(d)$ and outgoing tagged velocities obey*

$$|t_c(d') - t_c(d)| \leq |d' - d|, \quad |v_i^+(d') - v_i^+(d)| \leq 30|d' - d| \quad (i = 1, 2).$$

Writing $\Delta = |d' - d|$, one has

$$\sup_z \int_0^3 |H_{d'}(t, z) - H_d(t, z)|^2 dt \leq \Delta + 2700\Delta^2. \quad (23.2)$$

The phase paths converge uniformly over the preparation in the Skorokhod J_1 metric as $d' \rightarrow d$. More precisely, for the product metric taking the maximum of the per-particle Euclidean position and velocity distances, a time change aligning the collision times gives J_1 distance at most 100Δ .

Proof. Write r and g for the incoming relative position and velocity, and $n(d) = (r + t_c(d)g)/d$ for the contact normal. The collision root satisfies $|r + t_c g| = d$. The nongrazing estimate from Lemma 18.2 is $-g \cdot n > 1443/1000$. Differentiating the root with respect to d gives

$$t'_c(d) = \frac{1}{g \cdot n(d)}, \quad |t'_c(d)| < 1.$$

The incoming bound is $|g| < 3$, so

$$|n'(d)| \leq \frac{|t'_c(d)| |g| + 1}{d} < 5.$$

Since $v_1^+ = V + g/2 - (g \cdot n)n$ and $v_2^+ = V - g/2 + (g \cdot n)n$, differentiation bounds either velocity derivative by $2|g| |n'| < 30$.

Before collision the detector is zero, and after collision it is a constant of absolute value at most one. Its Lipschitz constant is one. The two signals therefore differ on the interval between the two collision times by at most one and afterwards by at most 30Δ . The intervening interval has length at most Δ , and the whole observation lasts three units. This proves (23.2).

Choose the increasing piecewise affine time change fixing 0, 3 and sending $t_c(d)$ to $t_c(d')$. Its uniform displacement is at most Δ . Aligned velocities differ by at most 30Δ , with the same outgoing convention at the jump. Each particle's speed is below four and its velocity jump has norm below three. Without time change, the position difference is bounded by $3\Delta + 3(30\Delta) = 93\Delta$, by integrating the velocity difference before and after the two jump times. Aligning time adds at most 4Δ . The asserted bound follows, with a harmless rounding to 100Δ . $\square$

Theorem 23.2 (Microscopic-to-audit and conditional-law composition). *For the single family (23.1), all of the following hold.*

(a) *At the same width-one private register,*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \quad (r \geq 1), \quad \delta_2^h < 13/100,$$

uniformly for $d \in I$ and $0 < \sigma \leq 1/24$. When $d \leq 2/5$ the detector is zero on $[0, 3]$, and all three finite-experiment deficiencies are zero.

(b) *There is one explicit rational reset-audit rule with one all-on training row, eight marked outcomes and $n = 2800$ training runs, whose expected validation score exceeds $2/5$ against every private width-one simulator throughout that same parameter region. In detail, for counts C choose an event $A_C \subseteq \{0, 1\}^3$ maximizing*

$$Q_*(A_C) - C(A_C)/2800, \quad (23.3)$$

with lexicographic tie breaking. Here Q_ is the rational marked reference law of (B, B, W) . The independent validation uses the actual physical target, not Q_* . This finite rule has at most $\binom{2807}{7}$ training-count states at any scheduled epoch; a conservative complete auditor bound is $8(256 + 1)\binom{2807}{7}$. The state cost belongs to the auditor. The certificate is an expected-score inequality, not a confidence claim about one realized score.*

(c) Fix $\sigma > 0$ and let $d_n, d \in I$ with $d_n \rightarrow d$. Set

$$\epsilon_n = \min \left\{ 1, \frac{\sqrt{|d_n - d| + 2700|d_n - d|^2}}{2\sigma} \right\}. \quad (23.4)$$

The joint preparation–continuous-observation laws differ by at most ϵ_n . Every finite resource deficiency above changes by at most ϵ_n , and a fixed audit changes by at most ϵ_n because its source behavior is unchanged. On a latent-preserving coupling, $X_{d_n} \rightarrow X_d$ in J_1 and each bounded posterior converges in supremum L^2 with bound $84\epsilon_n$. The full and projected likelihoods converge in reference supremum L^2 , and jointly in probability with the microscopic paths on that coupling. The projected stochastic exponentials, bounded entropic backward values, localized integrands and stopping consumers have the conclusions of Proposition 20.2, Theorem 20.3, and Corollary 20.4.

(d) On compact parameter sets with $\sigma \geq \sigma_0 > 0$, the finite marked probabilities have uniform rational enclosures. The observable chart, strict same-budget polynomial verification and rational upper tables are therefore effective in the exact sense of Propositions 17.1 and 16.6.

Proof. Every collision occurs before time $3/2$. Conditional on preparation, the two normalized window increments in (23.1) are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. The outgoing detector has sign $2B - 1$ and absolute value greater than $1/4$. Coupling each noisy report to B gives total marked variation less than $1/300$ from the exact finite marked experiment, uniformly for every acquisition gate. The finite calculation of Theorem 18.1 gives

$$d_* = \sqrt{5/2} - 9/8, \quad \delta_*^p = \delta_*^v = d_*, \quad \delta_{*,2}^h = 1/8.$$

Thus private and visible values exceed $d_* - 1/300 > 9/20$, and the hidden value is less than $1/8 + 1/300 < 13/100$. The two-valued hidden witness stores its selector, not an observed mark. In the no-collision regime both window reports before gating are independent fair noise, independent of W . Fresh private coins reproduce them exactly, including feedback gates. This proves (a).

For (b), use only the all-on feedback row. Any private width-one simulator in this row emits independent bits with probabilities p, q , independently of the actual W . The calculation in Theorem 18.1 already gives minimum variation d_* for this row. Its alphabet (z_1, z_2, W) has eight elements, hence $D = 7$. The explicit empirical audit in Theorem 13.1, trained using Q_* , has expected ideal score at least

$$d_* - \sqrt{7/2800} = d_* - 1/20.$$

The candidate training and validation laws do not change when the target is replaced by the physical target. Conditional on any selected event, its target probability changes by less than $1/300$. Therefore the actual score is strictly greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (23.5)$$

For completeness, the last strict inequality is the positive-root comparison $\sqrt{5/2} > 947/600$; its squared margin is $5/2 - (947/600)^2 = 3191/360000 > 0$. All probabilities in Q_* are 0, $3/16$ or $5/16$, so (23.3) is a finite rational rule independent of d, σ, p, q . At training epoch k the number of eight-category count vectors is $\binom{k+7}{7} \leq \binom{2807}{7}$. A buffer of eight states records the current two reports and terminal mark; 256 event masks and an idle symbol suffice during event selection and validation. The conservative product bound follows. The clock and the source-reset interface are the declared resources. In particular the proof does not enumerate or claim to store a small complete coefficient table.

For (c), Lemma 23.1 gives the signal-distance hypothesis of Theorem 20.1, with $T = 3$ and $M = 1$. It also supplies its microscopic J_1 hypothesis. That theorem gives (23.4) and all the joint process assertions. The finite acquisition is a common measurable causal map of the continuous

observation, so marked variation contracts for every gate. The source is blind and independent of d . Consequently its executable candidate behaviors are identical across d , and Proposition 13.4 has $\alpha = 0$ and $\beta = \epsilon_n$. The same same-budget comparison applies in each visible fiber and in each hidden mixture. The exponential and backward claims now follow from the identified projected density and the explicitly bounded entropic consumer. This is where the finite certificate and the conditional-law theorem meet the same microscopic approximation.

Finally, the collision root is uniformly separated from grazing, its formula is algebraic on the rational preparation boxes, and the normalized window noise has variance $2\sigma^2$ bounded away from zero on the stated compact sets. The box subdivision and Gaussian-tail enclosure argument in Proposition 18.4 therefore applies with that variance. It returns rational enclosures with a computable uniform error. The rational quotient algorithm and same-budget strict certificate procedure then prove (d). Uniformity as $\sigma \downarrow 0$ is not asserted. $\square$

The all-on audit in part (b) is a single witness for the lower bound in the full feedback problem: a maximum over feedback trees includes this row. The visible lower bound is instead the exact fiber theorem; no claim is made that an auditor with a seed-independent state bound can store an arbitrarily large visible-seed transcript. This separates the two resource budgets rather than silently moving cost from one to the other.

The finite certificate is also preserved under an additional rational presentation error ρ : its lower score decreases by at most ρ when only the target is changed, and by the reset-cost estimate when the source is changed. Thus the certificate is an effective consumer of the physical presentation theorem, not an isolated exact finite example.

Theorem 23.2 and the preceding results prove Theorem C.3. The C2-type consequence is a joint changing-trajectory/likelihood/posterior theorem with an identified stochastic exponential and a fully specified bounded entropic backward consumer. Nothing in that conclusion identifies geometric similarity with kinetic scaling, or supplies particle-uniform recollision estimates by renaming this fixed-particle model.

24. SHARP DECISION MEMORY FOR THE COLLISION EXPERIMENT

We now apply the continuation spectrum to the same marked experiment that supplies the microscopic certificate. Both its impossibility statement and its construction concern the same score, the same private candidate family, and the same training-validation checkpoint.

24.1. The marked family and its first two spectral values. In the all-on row of Section 23, a private width-one candidate has two independent report bits with probabilities $p, q \in [0, 1]$, independently of the fair actual mark. Denote its law by $b(p, q)$ and put

$$a = (1 - p)(1 - q), \quad c = pq.$$

Let $B = \{b(p, q) : p, q \in [0, 1]\}$. The ideal target Q_* is supported on 000, 001, 110, 111, with masses $5/16, 3/16, 3/16, 5/16$, respectively. On these four atoms the candidate masses are $a/2, a/2, c/2, c/2$. The elementary continuation constraint implies

$$\sqrt{a} + \sqrt{c} \leq 1, \quad \min(a, c) \leq \frac{1}{4}. \quad (24.1)$$

Indeed the first inequality is Cauchy-Schwarz applied to $(\sqrt{1-p}, \sqrt{p})$ and $(\sqrt{1-q}, \sqrt{q})$; the second follows from the first. If $a \geq c$, then $c \leq 1/4$ and

$$a + 3c \leq (1 - \sqrt{c})^2 + 3c = 1 - 2\sqrt{c} + 4c \leq 1. \quad (24.2)$$

The symmetric inequality holds when $c \geq a$.

Theorem 24.1 (A sharp two-state obstruction). *For the ideal marked collision family,*

$$\mathfrak{V}_1(B, Q_*) = \frac{1}{8}, \quad \mathfrak{V}_2(B, Q_*) = \frac{3}{8}. \quad (24.3)$$

If $\|Q - Q_\|_{\text{TV}} \leq \beta$, every reset auditor retaining at most two states at the training-validation cut has worst-case expected score at most $3/8 + \beta$, regardless of its finite training length and its upstream or downstream memory. The assertion allows mark-dependent validation events and stochastic state transitions.*

Proof. For one state, use the event $\{000, 111\}$. Its score is $5/8 - (a + c)/2 \geq 1/8$. Conversely, average the two candidates with $(a, c) = (1, 0)$ and $(0, 1)$. Their average is uniform on the four supported atoms and is at total variation distance $1/8$ from Q_* . Every fixed response has average score at most $1/8$, so its minimum over the two candidates is at most $1/8$. This proves the first equality; the average is a bound on scores, not an allowed private candidate.

For the upper bound with two states, consider the three candidates $b_0 = b(0, 0)$, $b_1 = b(1, 1)$ and $b_* = b(1/2, 1/2)$. A response may be set to zero outside the support of Q_* without decreasing its payoff at any candidate. Write its four remaining coordinates as (x, y, z, t) . Its three payoffs are

$$\begin{aligned} f_0 &= (-3x - 5y + 3z + 5t)/16, \\ f_1 &= (5x + 3y - 5z - 3t)/16, \\ f_* &= (3x + y + z + 3t)/16. \end{aligned}$$

Since all four coordinates are in $[0, 1]$,

$$f_0 + f_1 \leq \frac{1}{4}, \quad f_0 + f_* \leq \frac{3}{4}, \quad f_1 + f_* \leq \frac{3}{4}. \quad (24.4)$$

Suppose two responses covered all three candidates with payoff strictly greater than $3/8$. The first inequality forces different responses to cover b_0 and b_1 . The other two inequalities then force both responses to have payoff less than $3/8$ at b_* . This is a contradiction. Hence $\mathfrak{V}_2 \leq 3/8$.

For attainment take the responses, in the same four-atom order,

$$h_0 = (2/3, 0, 1, 1), \quad h_1 = (1, 1, 0, 2/3),$$

with zero response elsewhere. Both have target mean $17/24$. Their payoffs are $17/24 - a/3 - c$ and $17/24 - a - c/3$. If $a \geq c$, the first is larger and (24.2) makes it at least $17/24 - 1/3 = 3/8$. The other case is symmetric. This proves equality. For the perturbed target, apply (4.4) and the operational upper bound of Theorem 4.2. $\square$

The obstruction has an immediate continuation interpretation. Each post-training state supplies one validation response. The three witness candidates in the proof cannot be covered above $3/8$ by two such responses. No increase of sample size can compensate for losing the third continuation at that cut. This argument does not use the unrelated Bernoulli majority calibration.

Lemma 24.2 (Feedback completion of the physical lower bound). *The upper bound $3/8 + \beta$ in Theorem 24.1 remains valid when the auditor may train on any of the original feedback rows and may select a row-event pair after the decision cut. Here every target row is the declared feedback pushforward of Q , with $\|Q - Q_*\|_{\text{TV}} \leq \beta$. In particular, two states cannot evade the bound by using a different sensor gate during training or validation.*

Proof. It suffices to restrict the adversary to an admissible subfamily. For $p, q \in [0, 1]$, let its first emission be $X_1 \sim \text{Ber}(p)$, and, after gate A , let its second emission be AX_2 , with an independent $X_2 \sim \text{Ber}(q)$ and the independent fair actual mark. These are genuine width-one private rows: the gate is a current input, and the second emission uses no retained copy of X_1 .

For any deterministic gate policy $g : \{0, 1\} \rightarrow \{0, 1\}$, the marked row is the pushforward of $b(p, q)$ under $T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$. The physical target row is the pushforward of Q by the same map, since acquisition does not act on the path. Including the gate in the transcript

changes nothing: it is also a deterministic function of the displayed coordinates. For a selected row event, the response pulls back to $\mathbf{1}_A \circ T_g$ on the all-on alphabet. A randomized row-event continuation pulls back to a function $h_m \in [0, 1]^\Omega$. Hence, conditional on cut state m , its expected score on this subfamily is exactly $(Q - b(p, q))h_m$.

Whatever training rows were used, their effect is solely the distribution of the retained state. The two-state score is thus a convex combination of two such responses and is bounded above by their envelope. Apply the three-candidate argument of Theorem 24.1, and then the uniform target perturbation bound. Since this is a subfamily of the full private candidate class, it is a valid lower-bound witness for that full class. Randomized feedback is covered by conditioning on its independent random table and then averaging the pulled-back response. $\square$

24.2. A three-response construction. Define three deterministic events, all ignoring the mark,

$$A_D = \{00, 11\} \times \{0, 1\}, \quad A_0 = \{00\} \times \{0, 1\}, \quad A_1 = \{11\} \times \{0, 1\}.$$

Their ideal scores are $1 - a - c$, $1/2 - a$ and $1/2 - c$.

Lemma 24.3 (The three-response envelope). *For the original private candidate image,*

$$\min_{p, q} \max\{1 - a - c, \tfrac{1}{2} - a, \tfrac{1}{2} - c\} = \sqrt{2} - 1. \quad (24.5)$$

In particular $\mathfrak{V}_3(B, Q_) \geq \sqrt{2} - 1$.*

Proof. By symmetry suppose $a \geq c$. Then $c \leq 1/4$ and the maximum is $1 - a - c$ for $a \leq 1/2$, and $1/2 - c$ for $a \geq 1/2$. If $a \leq 1/4$, the first expression is at least $1/2$. If $1/4 \leq a \leq 1/2$, (24.1) gives

$$a + c \leq 1 - 2\sqrt{a} + 2a \leq 2 - \sqrt{2},$$

since the last function is increasing on $[1/4, 1/2]$. If $a \geq 1/2$, then $c \leq (1 - \sqrt{a})^2 \leq (1 - 1/\sqrt{2})^2 = 3/2 - \sqrt{2}$. Both cases give the asserted lower bound. It is attained at $p = q = 1 - 1/\sqrt{2}$, where $a = 1/2$ and $c = 3/2 - \sqrt{2}$. The symmetric point gives the same value. $\square$

Lemma 24.4 (Odd-sample majority selection with a uniform regret bound). *Let $n = 2h - 1$ be odd. In n independent candidate trials, let C_0, C_1 count report pairs 00, 11. Choose A_1 if $C_0 \geq h$, choose A_0 if $C_1 \geq h$, and otherwise choose A_D . The first two cases are mutually exclusive. Its expected ideal score satisfies*

$$g_n(p, q) \geq \sqrt{2} - 1 - \frac{1}{4\sqrt{en}} - \frac{1}{4}e^{-n/8}. \quad (24.6)$$

Its event label has three possible values at the decision cut.

Proof. Because $a + c \leq 1$,

$$\max\{1 - a - c, \tfrac{1}{2} - a, \tfrac{1}{2} - c\} = 1 - a - c + (a - 1/2)_+ + (c - 1/2)_+.$$

The selected payoff is $1 - a - c + (a - 1/2)\mathbf{1}_{\{C_0 \geq h\}} + (c - 1/2)\mathbf{1}_{\{C_1 \geq h\}}$. Its expected loss from this envelope is exactly the sum, for $u = a, c$, of $|u - 1/2|$ times the probability that the binomial majority selects the wrong side of $u = 1/2$. At $u = 1/2$ the contribution is zero. No independence between C_0 and C_1 is used.

For $0 < u < 1/2$ and $k \geq h$, the ratio of the binomial mass at u to that at $1/2$ is

$$[4u(1 - u)]^{n/2} \left(\frac{u}{1 - u} \right)^{k - n/2} \leq [4u(1 - u)]^{n/2}.$$

The fair binomial upper tail is exactly $1/2$ because n is odd. Reflecting k treats $u > 1/2$. The endpoint cases follow directly or by continuity. Thus the wrong-side probability is at most

$$\tfrac{1}{2}[1 - 4(u - 1/2)^2]^{n/2} \leq \tfrac{1}{2}e^{-2n(u - 1/2)^2}.$$

For $x \geq 0$, the maximum of $xe^{-2nx^2}/2$ is $1/(4\sqrt{en})$, reached at $x = 1/(2\sqrt{n})$. At least one of a, c is at most $1/4$ by (24.1); that coordinate has $|u - 1/2| \in [1/4, 1/2]$ and contributes at most $e^{-n/8}/4$. Bound the other coordinate by the unrestricted maximum and use Lemma 24.3. $\square$

24.3. Simultaneous exact widths and the full acquisition schedule. The preceding construction admits more compression than storing its two raw counts. Write 0, 1, * for the three report-pair categories 00, 11 and the two off-diagonal pairs. For $n = 2h - 1$, let F_n be the terminal event-label function of this word. At a trial cut t , let $r = n - t$ and let (u, v) be the two prefix counts. Associate the residual key

$$\mathcal{R}_t(u, v) = \begin{cases} \text{L}, & u \geq h, \\ \text{R}, & v \geq h, \\ (\min(h - u, r + 1), \min(h - v, r + 1)), & u, v < h. \end{cases} \quad (24.7)$$

The symbols L, R mean the constant decisions A_1, A_0 . In a pair key, a coordinate $r + 1$ means that this majority is no longer reachable. The pair $(r + 1, r + 1)$ represents the constant decision A_D .

Theorem 24.5 (Exact profile of the physical audit rule). *The keys in (24.7) are exactly the distinct continuation residuals of F_n . Their number at trial cut t is*

$$r_t(F_n) = \begin{cases} \binom{t+2}{2}, & 0 \leq t \leq h - 1, \\ \binom{n-t+3}{2}, & h \leq t \leq n. \end{cases} \quad (24.8)$$

This is the minimum simultaneous width profile for exact computation of this rule, for deterministic and stochastic machines alike. Its maximum is $\binom{h+2}{2}$ at $t = h$.

For $n = 399$, the maximum trial-cut width is 20301, and the terminal training cut has exactly three states. Acquiring each report pair one bit at a time, discarding its training mark, and then executing independent candidate and target validations gives a deterministic machine with at most $40602 < 2^{16}$ states at every scheduled epoch. One additional ignored training trial may be inserted to give the same 400 training slots and 402 total trials as the earlier certificate. It changes neither the payoff nor these width bounds.

Proof. If a majority has already been reached, its label cannot change. Otherwise, the first coordinate of the pair key is exactly the number of additional 0 symbols needed, with the unreachable threshold clipped to $r + 1$; the second coordinate has the analogous meaning. These data determine the suffix function. Two different reachable pair keys can be distinguished by a suffix containing only one of the two symbols and enough * symbols to fill the remaining positions: use a count between the two different thresholds. The other majority cannot be reached by this suffix. The constant-majority keys are distinct from every pair key because an all-* suffix returns A_D for the latter. Thus the keys are exactly the residuals.

For $t < h$, no majority has been reached and every threshold is at most $r + 1$. The distinct pairs (u, v) with $u + v \leq t$ give $\binom{t+2}{2}$ keys. For $t \geq h$, the two constant-majority keys are reachable. The remaining keys are precisely

$$(i, j) \in \{1, \dots, r + 1\}^2, \quad i + j \geq r + 1.$$

To see surjectivity, use $u = h - i, v = h - j$; these counts are nonnegative, less than h , and sum to at most t because $2h - t = r + 1$. To see necessity before clipping, use the same inequality on $h - u, h - v$; if either coordinate is clipped, the inequality remains automatic. There are $(r + 1)^2 - (r - 1)r/2$ such pairs, with the expression valid also for $r = 0, 1$. Adding the two constant keys gives $(r + 2)(r + 3)/2$. This proves (24.8). The finite-valued realization theorem, Theorem 11.1, proves both the simultaneous realization and stochastic minimality for the specified rule. The two polynomial pieces of the profile give its maximum as stated.

For an explicit transition definition, order the reachable keys lexicographically, putting L, R first. For each key choose the lexicographically first prefix-count pair representing it. After the next category, increment the appropriate count and apply $\mathcal{R}_{t+1}$. Equality of residual functions implies equality of the shifted residuals, so this definition is independent of the chosen representative. The representative is part of the time-indexed transition table, not another stored register. The clock supplies t but no sample history. All these maps are deterministic and rational.

At a bit-level epoch, the state consists of a residual key and, only between the two report bits, the first bit. It therefore has at most $2 \max_t r_t$ values. On reading the second bit, determine the category, update the key and discard both bits. The mark is read and discarded; all-on controls and resets are prescribed by the free schedule. After training, keep just one of the three event labels. During validation, keep that label, the completed candidate-event indicator, and a first-bit buffer when needed. At most $3 \cdot 2 \cdot 2 = 12$ states suffice there; a completed target pair emits its event indicator minus the stored candidate indicator. The final score has three possible values. These phase-specific alphabets may be embedded in a single alphabet of size 40602 because the schedule clock identifies the phase.

The same trial-cut residuals remain distinct for the complete signed-score function, not merely for an externally emitted event name. Any two of A_0, A_1, A_D have different indicator functions. Take an off-diagonal candidate-validation pair, which belongs to none, and a target-validation pair distinguishing the two events. The resulting scores differ. Thus passing to the validation suffix does not merge the residuals already counted. $\square$

Theorem 24.6 (Matched physical decision complexity and stable certification). *For the two-sphere collision family of Theorem 23.2, let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/24$, and let $Q_{d,\sigma}$ be its all-on marked target. At the charged training-validation cut of the reset interface,*

$$\kappa_{2/5}(B, Q_{d,\sigma}) = 3. \quad (24.9)$$

More concretely, no auditor retaining at most two states at this cut can have expected score greater than $2/5$ uniformly against all private width-one candidates, at any finite training length, even with feedback-row selection as in Lemma 24.2. One rational auditor with three states at this cut does so with 399 informative training trials and two validation trials. With a redundant ignored training trial, it uses exactly 402 total trials, at most 40602 states at every scheduled epoch, and the exact trial-cut profile (24.8) on its informative training segment.

For fixed positive σ and $d_k \rightarrow d$, the same machine changes its expected score by at most

$$\epsilon_k = \min \left\{ 1, \frac{\sqrt{|d_k - d| + 2700|d_k - d|^2}}{2\sigma} \right\}. \quad (24.10)$$

The private, hidden and visible deficiencies, the no-collision control, and the joint microscopic, likelihood, posterior, innovation and global bounded entropic backward convergence retain all conclusions of Theorems 23.2 and 22.2.

Proof. Theorem 23.2 discharges the physical hypotheses and proves $\|Q_{d,\sigma} - Q_*\|_{\text{TV}} < 1/300$, uniformly on the stated parameters. Theorem 24.1 therefore gives the universal cut-width-two upper bound

$$\mathfrak{V}_2(B, Q_{d,\sigma}) < \frac{3}{8} + \frac{1}{300} < \frac{2}{5}.$$

Lemma 24.2 makes this a bound for the full original feedback-row interface, not only all-on training and not only the constructed rule. The upper construction uses all-on trials and works for every private candidate because only its two all-on parameters enter.

Use the deterministic selector of Lemma 24.4 with $n = 399$, but validate against the actual physical target. For each selected event, changing only the target from Q_* loses less than $1/300$.

Its physical expected score is consequently greater than

$$L_{399} := \sqrt{2} - 1 - \frac{1}{4\sqrt{399e}} - \frac{1}{4}e^{-399/8} - \frac{1}{300} > \frac{2}{5}. \quad (24.11)$$

Here is a rational verification of the last strict inequality, independent of decimal evaluation. The bounds $e > 2$, $\sqrt{798} > 28$, $\sqrt{2} > 707/500$, and $399/8 > 49$ imply

$$L_{399} > \frac{207}{500} - \frac{1}{112} - \frac{1}{300} - 2^{-51} > \frac{2}{5},$$

since $207/500 - 1/112 - 1/300 - 2/5 = 73/42000 > 2^{-51}$. The number L_{399} is approximately 0.4032891; the rational inequalities, not this approximation, establish the certificate. Its state realization was proved in Theorem 24.5. The general operational classification now yields (24.9) with both inequalities proved for this same physical task.

Changing d leaves the blind candidate source and all its training laws unchanged. The moving-collision estimate and the common-preparation Gaussian theorem bound the target variation by ϵ_k . For any fixed response its target mean changes by at most this amount; averaging over the unchanged training rule proves (24.10). There is no multiplier equal to the number of training trials. The same physical estimates verify the hypotheses of Theorem 22.2; the observation-level density and coupling versions used there are given in Section 21. This transports the same certificate and the bounded conditional-law consumer along the changing microscopic paths. The no-collision regime $d \leq 2/5$ has the vanishing deficiencies already proved in Theorem 23.2; it is not included in the positive-gap parameter interval. $\square$

24.4. What the two exact complexity statements measure. Equation (24.9) minimizes over every auditor at one specified physical decision cut. Equation (24.8) minimizes all trial-cut widths simultaneously for exact implementation of the stated 402-trial rule. These are different optimizations. The first is a task-level optimum, not an artifact of that rule. The second explains the whole online realization of the same rule, not a claim that every successful auditor must have 20301 states somewhere. The sixteen-bit bound includes bit acquisition and validation but does not bound the description length of time-dependent transition tables. It is an implementation upper bound for peak memory. Independent persistent randomization is charged; fresh randomness from the reset experiments is not a hidden cross-cut memory channel.

The former two-count 400-sample construction and its nineteen-bit bound remain in Section 29. They offer a different approximation to the full marked discrepancy. The new three-response rule does not reproduce that rule or attain its oracle envelope. It instead solves the specified physical certification task with an exactly minimal decision-cut alphabet and an explicit full continuation profile. The score is an expectation, not a confidence guarantee from a single realized validation pair.

25. FULL-PROFILE CERTIFICATION OF THE COLLISION FAMILY

The short audit of Theorem 24.6 is retained with its exact continuation profile. We now give a second audit of the same physical class whose entire acquisition-and-validation register is bounded independently of its accuracy parameters. Its cost is a much longer, explicitly counted, preparation schedule.

25.1. The row-indexed decision invariant. For the original row–event interface define

$$\kappa_c^{\text{rows}}(B, Q) = \min\{K : \mathfrak{R}_K^{\text{rows}}(B, Q) > c\}. \quad (25.1)$$

The minimum is $+\infty$ when the set is empty. This is not an implicit change of the one-row symbol κ_c . By Corollary 4.4, it is the exact minimum decision-cut alphabet for a strict score, allowing a finite training schedule and a post-cut distribution on row–event pairs.

Let W_c^{phys} be the minimum peak width among all finite auditors for the original physical row interface, now charging each report-bit, mark, decision and validation epoch. Its clock is externally scheduled, as in the earlier physical implementation. All persistent random selectors are charged.

Theorem 25.1 (A twelve-state physical audit). *In the physical parameter range of Theorem 24.6, put $\beta = 1/300$. For $0 < \theta < 1/4$ and $0 < \delta < 1$, let L, R be at least the respective quantities in (7.1). There is a deterministic audit with $N = 4LR$ candidate training trials and two validation trials whose expected score is strictly greater than*

$$\sqrt{2} - 1 - \beta - \theta - 2\delta. \quad (25.2)$$

It has at most five states at completed training-trial cuts, ten at intermediate training report-bit cuts, three at the decision cut, and twelve throughout validation. In particular

$$\kappa_{2/5}^{\text{rows}} = 3, \quad 3 \leq W_{2/5}^{\text{phys}} \leq 12. \quad (25.3)$$

One entirely integer schedule takes $\theta = 1/200$, $\delta = 1/1000$, $L = 400$ and $R = 8 \cdot 2^{400}$, so $N = 12800 \cdot 2^{400}$. Its lower bound exceeds $2/5$. The upper bound twelve is not asserted to be the exact minimum integer peak width.

Proof. Use the report-pair probabilities $a = (1-p)(1-q)$ and $c = pq$ from Section 24. In the first $2LR$ training trials run Lemma 7.1 on the indicator of the pair 00. If it returns positive, choose A_1 and ignore the remaining training trials. Otherwise, in the next $2LR$ trials run the same comparison on the indicator of 11, choosing A_0 if positive and A_D if negative. Both phases have fixed lengths, even after an absorbing decision.

A comparison whose parameter is outside its indifference band is correct with probability at least $1 - \delta$. For purposes of analysis, one may define the second comparison on its independent observations even when the implementation ignores them. A union bound gives probability at least $1 - 2\delta$ that neither comparison errs outside its band. Since $\min(a, c) \leq 1/4$ and $\theta < 1/4$, at most one of a, c lies in $[1/2 - \theta, 1/2 + \theta]$.

On this good event the chosen payoff is within θ of

$$E(a, c) = \max\{1 - a - c, 1/2 - a, 1/2 - c\}.$$

Indeed, if one coordinate exceeds $1/2 + \theta$, its associated event is selected and is optimal. If both are below $1/2 - \theta$, A_D is selected and is optimal. In the remaining case the low coordinate's comparison is negative, and choosing either of the two relevant events costs at most the distance of the near coordinate from $1/2$. The range of the three payoffs at any fixed (a, c) is at most one. The expected ideal score is therefore at least $E(a, c) - \theta - 2\delta$. Lemma 24.3 gives $E(a, c) \geq \sqrt{2} - 1$. Changing the target to the physical target loses strictly less than β , proving (25.2).

The first comparator has four states. The second phase has these four states and one extra absorbing label for a previously chosen A_1 , hence five. Retaining the first report bit doubles that count to ten. On the second bit, determine the relevant indicator, update the block state, and discard both bits. The training mark is read and discarded. At the phase boundary the fixed schedule maps the first decision to either the A_1 label or the initial state of the second comparator. At the end it maps the remaining decisions to A_0 or A_D . During validation keep the chosen event, the candidate-event indicator, and a first-bit buffer when needed. This requires at most $3 \cdot 2 \cdot 2 = 12$ states; all events ignore the mark. These are actual phase-compatible transitions, not a bound only on terminal labels.

For the integer schedule, the error estimate in the proof of Lemma 7.1 is $2e^{-8} < 1/1000$. To check it rationally, $e > 8/3$ and $(8/3)^8 > 2000$. Further $\sqrt{2} > 707/500$, and hence

$$\sqrt{2} - 1 - \frac{1}{300} - \frac{1}{200} - \frac{1}{500} - \frac{2}{5} > \frac{11}{3000} > 0.$$

This proves the strict score claim without numerical optimization. The training count is exactly $4LR$; early absorption does not delete trials from that count.

For the lower bounds in (25.3), every full-profile auditor has its decision cut among its charged cuts. The feedback pushforward subfamily of Lemma 24.2 bounds any width-two decision continuation by $3/8 + \beta < 2/5$, for every finite adaptive training schedule. The all-on construction just given is a member of the original row interface and has three decision labels. This proves the row-indexed equality and both peak inequalities. $\square$

Corollary 25.2 (Two physical resource regimes). *The same physical score threshold $2/5$ admits the two schedules*

<i>training trials</i>	<i>decision width</i>	<i>peak width</i>
399	3	40602
$12800 \cdot 2^{400}$	3	12

followed by two validations. The first schedule may be padded to 400 training trials as before. For either fixed machine, changing d to d_k at fixed positive noise level changes its expected score by at most ϵ_k in (24.10). If the candidate training kernels also change, their additional cost is the training term in Corollary 7.3.

Proof. The first line is Theorem 24.6; the second is Theorem 25.1. In the moving-collision comparison the candidate training source is identical. Conditional on its chosen event, only one target validation mean changes. The common-preparation Gaussian bound therefore contributes ϵ_k , independently of the length of the unchanged training. The general perturbation formula applies when this equality of sources is absent. $\square$

The long schedule is an existence bound on the resource frontier, not a practical substitute for the 399-trial rule. In an autonomous model the clock must also be stored; Proposition 6.3 gives its explicit additional factor. Both schedules use deterministic transitions, so no precision-dependent randomizer is hidden in the physical upper bound. The one-collision construction remains the verified physical realization. No many-particle or zero-noise theorem is inferred from the larger class of finite experiments.

26. A THIRD RESOURCE REGIME FOR THE SAME PHYSICAL TASK

The three-response obstruction, the short exact selector, and the constant-width block audit concern the same positive-noise marked collision family. A truncated absorbing comparison supplies a further resource pair without changing that family or its threshold.

Theorem 26.1 (Polynomial-horizon physical realization). *Use the original row-event interface and the parameter range of Theorem 25.1. Let $\beta = 1/300$, $0 < \theta < 1/4$, and $m, n \geq 1$. There is a deterministic audit with $2n$ candidate training trials and two validation trials whose expected signed score is strictly greater than*

$$\sqrt{2} - 1 - \beta - \theta - 2e_{m,n}(\theta). \quad (26.1)$$

It uses at most $2m + 2$ states at completed training-trial cuts, $4m + 4$ at intermediate report-bit cuts, three at the decision cut, and twelve throughout validation. In particular, $\theta = 1/200$, $m = 400$ and $n = 160000$ give score greater than $2/5$, with 320000 training trials and peak width at most 1604.

Proof. Write $a = (1 - p)(1 - q)$ and $c = pq$ for the actual blind product candidate report probabilities, as in Theorem 25.1. On the first n reports apply Lemma 8.1 to the indicator of 00. If the answer is positive retain A_1 and ignore the second training phase. Otherwise use the next n reports to compare the indicator of 11, retaining A_0 for a positive answer and A_D for a negative answer. Conceptually generate the second phase even when it is ignored. Independence of the

reset preparations gives an error probability at most $2e_{m,n}(\theta)$ for either comparison to be wrong outside its indifference band.

On the complementary event the selected ideal payoff is within θ of $\max(1-a-c, 1/2-a, 1/2-c)$. Indeed $\min(a, c) \leq 1/4$, so at most one coordinate is within θ of $1/2$. If neither is, the two comparisons select the maximizing event; if one is, switching the relevant events changes payoff by at most θ . The three payoffs have range at most one at each (a, c) . Their envelope is at least $\sqrt{2} - 1$ by Lemma 24.3. Replacing the ideal target by the physical target loses strictly less than β and does not change the candidate training law. This proves (26.1).

The walk has $2m + 1$ labels. The second phase adds the previously latched A_1 label, giving $2m + 2$. Retaining the first report bit doubles this number; after the second bit, determine the relevant indicator and discard the pair. The mark is read and discarded. A deterministic clock supplies the phase and the within-phase time. At its end map the remaining walk states to A_0 or A_D . The validation implementation from Theorem 25.1 uses at most twelve states, including the event, first bit, and candidate-event value. No real-valued randomization is needed by this construction.

For the stated integers, $4\theta m = 2n\theta^2 = 8$, so $e_{m,n}(\theta) = 2e^{-8} < 1/1000$. The rational estimates $e > 8/3$, $(8/3)^8 > 2000$, and $\sqrt{2} > 707/500$ give

$$\sqrt{2} - 1 - \frac{1}{300} - \frac{1}{200} - \frac{1}{500} - \frac{2}{5} > \frac{11}{3000} > 0.$$

This proves the strict physical margin, with the stated fixed training count and peak width. $\square$

Corollary 26.2 (Three compatible resource bounds). *The original physical task admits all three rows below, each followed by two validations and having decision width three:*

<i>training trials</i>	<i>peak width</i>	<i>training mechanism</i>
399	40602	<i>exact empirical selector</i>
320000	1604	<i>truncated absorbing comparisons</i>
$12800 \cdot 2^{400}$	12	<i>four-state block comparisons.</i>

The all-auditor row-indexed decision optimum remains $\kappa_{2/5}^{\text{rows}} = 3$ and the unrestricted finite-horizon physical peak still satisfies $3 \leq W_{2/5}^{\text{phys}} \leq 12$. The table gives attainable points, not an optimal physical fixed-sample frontier. In particular the 40602 bound is not an all-auditor lower bound for 399 samples.

Proof. Combine Theorems 24.6, 25.1 and 26.1. The feedback-uniform two-state obstruction in Theorem 25.1 continues to apply to every audit in this enlarged set of constructions. $\square$

For a perturbed blind training source with per-active-trial defect α , the two-phase construction also has the bound

$$2 \min\{1, 2\alpha \min(n, m^2)\} + \beta_c + \beta_t$$

on score change, by the proof of Corollary 10.2. The moving-collision parameter alone leaves that source unchanged, so its training contribution is zero; the previous target transport and bounded Gaussian/entropic consumer are retained without change. If one makes the machine autonomous, an explicit bound is $(6n + 12) \max(4m + 4, 12)$ states, since the $2n + 2$ trials each acquire two bits and a mark, with the phase/decision boundaries included. The clock multiplier is an implementation upper bound, not an optimized autonomous-memory claim.

27. CONFIDENCE-LEVEL CERTIFICATION

An expected signed score is an operational separation, not a confidence statement about one validation pair. We now distinguish these two tasks. The null hypothesis below is equality of the candidate and target validation laws; it need not belong to the private alternative family. All

preparations in the theorem are independent resets with one fixed candidate law. The test cannot be given a new adversarial candidate after each reset.

Theorem 27.1 (Confidence lift with a charged accumulator). *Suppose a base reset audit uses n training trials, one candidate and one target validation, and peak width W , with score $D \in \{-1, 0, 1\}$. Assume its expected score is at least $\gamma > 0$ for every alternative. Run J independent copies and reject equality when*

$$\sum_{j=1}^J D_j \geq J\gamma/2. \quad (27.1)$$

Then both the type-I error under equality and the worst-case type-II error are at most $e^{-J\gamma^2/8}$. The whole test has a realization with at most $W(2J+1)$ states and $J(n+2)$ trials. In particular $J \geq 8\gamma^{-2} \log(1/\alpha)$ suffices for both errors to be at most α .

Proof. Under equality, conditional on the training and selected event, the two independent validations have equal means; hence $\mathbb{E}D = 0$. Under each fixed alternative its mean is at least γ . The copies are independent and $D_j \in [-1, 1]$. The exponential-moment estimate for a variable in an interval of length two is $\mathbb{E}e^{s(D_j - \mathbb{E}D_j)} \leq e^{s^2/2}$, by the same tilted-variance argument used in Theorem 4.2. Markov's inequality and minimization in s give $\Pr(J^{-1} \sum (D_j - \mathbb{E}D_j) \geq a) \leq e^{-Ja^2/2}$, and the same lower-tail bound. Set $a = \gamma/2$.

Keep the running integer sum, with at most $2J+1$ values, beside the base machine's register. At each completion add its emitted score and reset only the base register. The sum is not stored in an external history. The clock supplies the repetition index and the fixed terminal threshold; all $J(n+2)$ trials are counted. $\square$

The 399-trial physical audit has mean strictly above $2/5$, so one may use the rational choice $\gamma = 2/5$ in this theorem. This gives the fully explicit sufficient count $J \geq 50 \log(1/\alpha)$ and $401J$ total trials (or $402J$ with the inherited ignored slot). The confidence guarantee uses repetition and its charged sum, not one pair of validation observations.

Theorem 27.2 (One training phase and repeated validation). *For the physical audit of Theorem 25.1, put $\gamma_0 = \sqrt{2} - 1 - 1/300 - \theta > 0$. Retain its chosen event and use m independent candidate-target validation pairs. Reject equality when their score sum is at least $m\gamma_0/2$. The type-I error is at most $e^{-m\gamma_0^2/8}$ and the worst-case type-II error is at most*

$$2\delta + e^{-m\gamma_0^2/8}.$$

The audit uses $4LR + 2m$ trials and at most $12(2m+1)$ states throughout training and validation. Thus $\delta \leq \alpha/4$ and $m \geq 8\gamma_0^{-2} \log(2/\alpha)$ suffice for error at most α .

Proof. The proof of Theorem 25.1 gives an event of training probability at least $1 - 2\delta$ on which the conditional mean score of the selected event is strictly greater than γ_0 . Conditional on that event and its selected label, the validation pairs are independent. Apply the tail bound in Theorem 27.1 conditionally and add the training failure probability. Under equality every selected event has zero mean, regardless of training, so no training failure term is needed for type-I error. Keep the three-valued event label, the within-pair indicator and first-bit buffer, and a sum with $2m+1$ values. Training itself needs no more than ten states, so the displayed bound covers every phase. $\square$

The tests in this section have a different output task from maximizing an expected witness score. The exact three-state decision-cut theorem is not asserted to be an optimal memory theorem for these binary tests. For perturbations of training and validation laws, the coupling argument of Theorem 6.2 bounds changes of each rejection probability by the sum of the per-call total variation errors. In a source-preserving physical transport, alternatively lower the conditional mean margin

by the single-row target variation and reapply the conditional tail estimate. This latter argument transports the confidence guarantee without pretending that an entire m -sample validation law has the same total variation distance as one validation observation.

28. CONFIDENCE TESTS WITH PRICED FINITE-STATE AMPLIFICATION

The signed-sum construction in Section 27 converts expectation into confidence. There is also a direct sample–memory choice for the amplifier itself. Its formulation keeps the binary testing objective distinct from the expected-score task.

Theorem 28.1 (Two finite-state confidence amplifiers). *Suppose a reset audit uses b training trials, two validations, peak width $W \geq 3$, and reports $D \in \{-1, 0, 1\}$. Under the equality null $\mathbb{E}D = 0$, and under every alternative in a fixed class $\mathbb{E}D \geq \gamma > 0$. Fix a known dyadic $0 < g \leq \gamma$, and put $\theta = g/8$.*

- (i) *For integers $m, n \geq 1$, there is a binary test with both errors at most $e_{m,n}(\theta)$, using $n(b+2)$ preparations and peak width at most $(2m+1)\max(W, 9)$. Thus both errors are at most α with the choices (8.5), with $\delta = \alpha$.*
- (ii) *With L, R as in (7.1), using $\delta = \alpha$ and the same θ , there is another test with both errors at most α , $2LR(b+2)$ preparations, and peak width at most $4\max(W, 9)$. Its very long external schedule is charged separately. The confidence error, not the input parameter, determines these read-only horizon constants.*

Proof. Run independent copies of the base audit with fresh preparations and fresh private bits. Conditional on each output D , generate an additional coin Z with probability

$$\mathbb{P}(Z = 1 \mid D) = \frac{1}{2} + \frac{D - g/2}{4}. \quad (28.1)$$

This is between zero and one, since $0 < g \leq \gamma \leq 1$ and $-1 \leq D \leq 1$. If g has denominator 2^s , the probability has denominator dividing 2^{s+3} and is sampled with $s+3$ fair bits. The coins are independent under each fixed parameter. Their mean is $1/2 - g/8$ under the null and at least $1/2 + g/8$ under the alternatives. Apply the walk of Lemma 8.1, rejecting the null for a positive result, to prove (i). Apply the block comparison of Lemma 7.1 to prove (ii).

At every base-audit epoch retain the amplifier state alongside its W labels. After obtaining D , retain that value and at most three Bernoulli-sampler labels while drawing (28.1). This requires at most nine labels per amplifier state. Update the amplifier and reset the base audit before the next repetition. The terminal rejection decision is a function of the amplifier state and requires no additional persistent random seed. This proves the stated full-cut peak bounds. Even after the amplifier is absorbed, padded base-audit repetitions remain charged to the preparation count. The clock supplying the repetition, base-audit epoch and fair-bit depth is external; storing it multiplies the displayed bound by at most one plus the total number of such epochs. $\square$

For the physical expected-score theorem one may take $g = 3/8$, since $3/8 < 2/5$. The additional coin then needs at most six fair bits. Thus confidence can be purchased either with a growing absorbing counter and polynomial repetition count, or with a constant register and a much longer clocked schedule. Neither statement identifies the minimum state count of the binary testing problem with the three conditional validation responses needed by the expected-score problem. The signed-accumulator and train-once confidence theorems remain available as different resource regimes.

The output-law transfer in Theorem 10.1 applies to a completed binary test. In particular, if perturbations change its output law by at most ε , its type-I and type-II bounds each increase by at most ε . Transporting all repeated raw data instead counts every affected preparation, as in Section 27. Independence of the base repetitions and uniformity over the fixed alternative class

are assumptions of the amplification theorem; adaptation of the alternative between repetitions is not being asserted.

29. A FINITE-REGISTER CERTIFICATE TRANSPORTED THROUGH THE MICROSCOPIC LIMIT

The general empirical audit in Theorem 13.1 treats every coordinate of a marked row as unknown. The private continuation constraint in the collision experiment forces a smaller family. Exploiting this constraint yields a much smaller executable auditor, without granting it knowledge of the candidate law.

In the all-on row of Section 23, every private width-one candidate emits two independent bits with parameters p, q , independently of the fair actual mark W . Put

$$a = (1 - p)(1 - q), \quad c = pq.$$

The ideal target Q_* is supported on 000, 001, 110, 111, with respective masses $5/16, 3/16, 3/16, 5/16$. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. Every other atom has target mass zero and cannot improve the maximizing event for $Q_* - b$.

Lemma 29.1 (Two-count audit and its exact register implementation). *In $n \geq 1$ independent all-on candidate trials let C_0, C_1 count report pairs 00, 11, respectively, ignoring the training mark. On the four supported atoms use the empirical masses $C_0/(2n), C_0/(2n), C_1/(2n), C_1/(2n)$, and select exactly those atoms whose target mass is strictly greater than this empirical mass. Call the resulting event A_C . For every $p, q \in [0, 1]$,*

$$\mathbb{E}\{Q_*(A_C) - b(A_C)\} \geq \|Q_* - b\|_{TV} - n^{-1/2}. \quad (29.1)$$

For $n = 400$, the rule is implemented by the two truncated counts $\min(C_0, 250), \min(C_1, 250)$, with comparisons to 150 and 250. Including execution of complete transcripts and the two independent validation trials, it has a deterministic streaming realization with at most 504008 retained states at any scheduled epoch. Thus a nineteen-bit register suffices. The schedule clock is free; no other history store is free. The table description and transition evaluation cost are not asserted to be bounded by nineteen bits.

Proof. On any supported atom let $x = Q_*(w) - b(w)$ and let $\hat{x}$ be the empirical discrepancy. If the sign rule selects a different event from an exact maximizing event, the lost contribution is at most $|x - \hat{x}|$. On unsupported atoms both rules can omit the atom. Summing gives regret at most

$$|C_0/n - a| + |C_1/n - c|.$$

The two counts need not be independent. Their marginal binomial variances and Cauchy–Schwarz give expected regret at most

$$\frac{\sqrt{a(1-a)} + \sqrt{c(1-c)}}{\sqrt{n}} \leq \frac{1}{\sqrt{n}}.$$

This proves (29.1).

For the 00 pair, select both marked atoms when $C_0 < 150$, only the atom of target mass $5/16$ when $150 \leq C_0 < 250$, and neither when $C_0 \geq 250$. The same rule applies to 11, with the other mark being the atom of larger target mass. At equality the zero empirical discrepancy is excluded. Increasing a count beyond 250 cannot change the final choice. Two registers with 251 possibilities each therefore implement the rule, with at most nine possible final event masks.

A buffer with eight possibilities is sufficient to execute and record a completed two-bit-plus-mark transcript. During training the pair of truncated counts and this buffer use at most $8 \cdot 251^2 = 504008$ states. After training, discard the counts and retain the event mask. During validation at most nine masks, one target-event indicator with two values, and the same eight-state transcript buffer are needed, a total of 144 states. The final signed score takes three values. Since the epoch is

known, labels can be reused between these phases; their state counts need not be multiplied. The update maps and threshold comparisons are parameter-independent deterministic maps. They are thus stochastic rows with entries zero or one, or equivalently a compatible continuation cover by Theorem 3.2. Fresh sampling randomness belongs to the specified reset experiments, not to an uncharged auditor register. Finally $504008 < 2^{19}$. $\square$

Theorem 29.2 (Finite-register microscopic certification and global transport). *Consider the marked hard-sphere family of Theorem 23.2, with $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/24$. There is one rational streaming audit, independent of d, σ, p, q , using 400 candidate training resets and one independent candidate and target validation trial, whose expected signed score is strictly greater than $2/5$ against every private width-one candidate. It uses at most 504008 retained auditor states. The candidate remains private width one; the auditor's register is a distinct, charged resource. The physical deficiencies still satisfy*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \ (r \geq 1), \quad \delta_2^h < 13/100.$$

For $d \leq 2/5$ all three finite-experiment deficiencies vanish.

Fix a positive σ and a sequence $d_k \rightarrow d$ within the collision interval. Put

$$\rho_k = |d_k - d| + 2700|d_k - d|^2, \quad \epsilon_k = \min\{1, \sqrt{\rho_k}/(2\sigma)\}.$$

The same 400-reset rule changes its expected score by at most ϵ_k under this perturbation. At the same time the microscopic paths, full and projected likelihoods, posteriors, and bounded entropic values have the joint convergence of Theorem 23.2. Their innovations, backward integrands and backward martingale integrals additionally have all the global conclusions of Theorem 22.2, with $T = 3$, $M = 1$, $\eta_k^2 \leq \rho_k$, and $\varepsilon_k \leq \epsilon_k$. On compact parameter sets bounded away from zero noise, the same audit has the effective probability-enclosure and rational-verification properties of Theorem 23.2(d).

Proof. The private width-one continuation restriction gives precisely the candidate family used in Lemma 29.1; the actual mark is not available to its transition rows. The exact finite separation in Theorem 18.1 gives

$$\min_{p,q} \|Q_* - b(p, q)\|_{\text{TV}} = d_* := \sqrt{5/2} - 9/8.$$

The scattering and observation estimates proved in Theorem 23.2 put the physical marked target within $1/300$ of Q_* , uniformly in all stated parameters. Train the two-count rule against the fixed rational Q_* , but use the actual physical target for validation. Each selected event's target probability changes by less than $1/300$. Hence its actual expected score is greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (29.2)$$

The last strict inequality is exact: it is equivalent to $\sqrt{5/2} > 947/600$, whose squared margin is $3191/360000$. The rule's executable memory bound was proved in Lemma 29.1. Its total trial count is 402, not 400 once validation is included. The private, hidden and visible deficiency bounds and the no-collision control are inherited from the same physical theorem, rather than inferred from an auditor whose signature was changed.

For changing d_k , Lemma 23.1 supplies the signal modulus ρ_k and the pathwise J_1 bound $100|d_k - d|$. Theorem 20.1 gives the joint marked variation bound ϵ_k and the likelihood/posterior convergence. The candidate source is blind and its behavior does not depend on d . Thus training laws and candidate validation laws are unchanged, and Proposition 13.4 has $\alpha = 0$ and $\beta = \epsilon_k$. There is no factor of 400 in this transport cost. The same preparation and signal bounds verify every hypothesis of Theorem 22.2, proving the global backward assertions. Finally the audit uses rational thresholds and finitely many deterministic row maps. The positive-noise enclosure

theorem therefore applies directly; an additional target-presentation error ζ reduces its score by at most ζ . $\square$

The two-count construction is specific to the private behavior family. A hidden mixture can destroy the independence of its two reports, and its mark relation must still be checked. The bound is therefore not asserted against the hidden class. The visible lower bound has its separate fiberwise proof; an arbitrarily large visible alphabet is not silently stored in the nineteen-bit auditor. The certificate concerns expected score, not a confidence statement about a single realized validation pair. Repeating validation for statistical confidence would incur an additional sample and storage budget.

The theorem uses a single mechanism: the continuation restriction fixes the feasible candidate laws and permits the compressed audit; scattering supplies its marked separation; common-preparation variation transports the same executable test and the global conditional-law consumer through a change in the microscopic trajectory. The collision transduces a prepared impact variable into a signal. It does not create the prepared correlation between that variable and the mark, and it is not a particle-number limit.

30. EXPLICIT PHASE TRANSITIONS FOR THE TWO-COUNT CERTIFICATE

For comparison with the new residual quotient, we give the complete phase-state specification of Lemma 29.1. It realizes the original 400-training-trial rule exactly, not the majority rule of Section 24. Let

$$C = (C_0, C_1) \in \{0, \dots, 250\}^2, \quad \tau(x) = \min(x, 250).$$

The free clock records the training index, the within-trial observation epoch, the validation phase, and deterministic processing epochs. It contains no data-dependent information. At the start, $C = (0, 0)$.

For a completed transcript (x_1, x_2, w) define the deterministic update

$$T(C, x_1, x_2, w) = (\tau(C_0 + \mathbf{1}_{\{(x_1, x_2) = (0, 0)\}}), \tau(C_1 + \mathbf{1}_{\{(x_1, x_2) = (1, 1)\}})).$$

The value of w does not enter this training update. The phase transitions are the following; x denotes the current input, not an earlier unretained report.

Phase	Retained data	Next transition
Training start	C	Read x_1 and keep (C, x_1) .
Second report	(C, x_1)	Read x_2 and keep (C, x_1, x_2) .
Terminal mark	(C, x_1, x_2)	Read w and keep (C, x_1, x_2, w) .
Count update	(C, x_1, x_2, w)	Replace by $T(C, x_1, x_2, w)$; discard the transcript.
Event formation	C	Replace by the event mask $A(C)$ defined below.
Candidate validation	A , partial transcript	Complete the transcript; replace it by $i = \mathbf{1}_A(x_1, x_2, w)$.
Target validation	(A, i) , partial transcript	Complete the transcript and emit $\mathbf{1}_A(y_1, y_2, v) - i$.
Score	A value in $\{-1, 0, 1\}$	Terminate.

The event formation map, including both threshold ties, is

$$\begin{aligned} 000 \in A(C) &\iff C_0 < 250, & 001 \in A(C) &\iff C_0 < 150, \\ 110 \in A(C) &\iff C_1 < 150, & 111 \in A(C) &\iff C_1 < 250. \end{aligned}$$

No other atom is included. For each report pair there are three choices: both marked atoms, only its larger-target-mass atom, or neither. Thus there are at most nine masks. Clipping at 250

preserves every comparison, including equality; no flag for a count exceeding the clipping threshold is required.

At a training epoch the possible states are bounded by $8 \cdot 251^2 = 504008$: the factor eight dominates every partial or complete transcript buffer. At validation epochs the largest alphabet has at most $9 \cdot 2 \cdot 8 = 144$ values, including the already computed candidate-event indicator. Event formation and final emission need at most nine and three states, respectively. Embed each phase alphabet into $[504008]$ by a fixed injective encoding. Because the phase is supplied by the free clock, these encodings need not have disjoint images. The next row decodes the current phase’s encoding and applies the displayed transition. This gives formal deterministic rows with entries zero or one at every epoch. The buffers, event mask, and validation indicator are all charged. No unpriced data store is introduced by label reuse.

APPENDIX A. THE PROOF DEPENDENCY CHAIN

An arrow below means an actual use of a theorem in the displayed proof, not an assertion of priority or an artificial dependency between papers. The preserved exact physical lower and upper bounds meet at the decision cut. The new full-profile theorem supplies an explicit training realization of general decision covers; its proof does not invoke the historical aggregate kinetic claims.

Input	Consequence using that input
Marked private rows and normalized continuations, Theorems 2.2, 3.2	The product behavior image and its validation continuation responses.
Decision continuation spectrum, Theorem 4.2	Universal two-state physical obstruction, Theorem 24.1; the strict-cut classification (24.9).
Product constraint (24.1) and three-response envelope, Lemma 24.3	Uniform finite-sample score, Lemma 24.4.
Finite-valued residual realization, Theorem 11.1	Exact full training profile and bit-level construction, Theorem 24.5.
Marked scattering and moving-collision estimate, Theorem 23.2	Both physical cut bounds and transport of the very same auditor, Theorem 24.6.
Common-preparation Gaussian comparison, Theorem 20.1; the three version lemmas in Section 21	The global conditional-law consumer, Theorem 22.2, on that moving-collision family.

The prior binary audit theorem remains a separate calibration, not a premise of the physical memory lower bound. The original two-count certificate remains a separate full-discrepancy approximation, not a claimed optimal implementation. The source-preserving build, rational checks and finite enumerations accompanying this article test the stated implementation and identities; they are not independent analytic proof certification.

A.1. The full-profile resource chain. The new central arrow is

strict response cover $\longrightarrow$ finite block tournament $\longrightarrow$ rational whole-schedule auditor.

Theorem 7.2 uses the decision spectrum, the finite block lemma and the dyadic sampler. It does not use a training oracle. Theorems 6.1 and 6.2 track approximation, calibration and precision on complete causal schedules. Theorem 12.1 uses one shared capacity budget for all cuts, and Theorem 27.1 retains the score accumulator. Together these establish a finite controlled-experiment instance of the program’s resource–precision–statistical-error objective. A universal singular-modulus rate or an infinite-horizon kinetic realization is not an input or a consequence of this finite theorem.

The current C2-scoped consumer remains the bounded, common-preparation, positive-noise Gaussian/entropic theorem. The original B4 normalized nonlinear resolvent formula cannot be inherited: its displayed linear difference identity fails on constant inputs. For a separately justified graph H with single-valued $R_\lambda = (I - H/\lambda)^{-1}$, the correctly typed identity is instead

$$R_\lambda h = R_\mu [(\lambda/\mu)h + (1 - \lambda/\mu)R_\lambda h].$$

Indeed both sides solve the same μ -resolvent inclusion. This conditional identity does not establish the graph's range or its kinetic corrector. Those historical targets remain in the program rather than being assigned proof credit from a finite resource theorem.

APPENDIX B. A NAMESPACE MAP OF THE ELEVEN-COMPONENT PROGRAM

The namespaces in this map separate historical paper targets from GTF statements proved here. Every historical component is retained. An interface means that the indicated consumer can be applied after its model-specific hypotheses are proved; it is not an automatic proof of the historical component’s aggregate theorem.

Paper	Retained historical role	GTF relation and precise proof credit
A1	Protocol and finite task reduction	Marked comparison, priced composition and decision-cut certification; no automatic all-order physical response conclusion.
A2	Independent primary geometric chain	Its protocol adapter is historical; the primary geometric chain is not a premise or consequence of a theorem in this article.
A3	Dynamical/potential-space targets	Conditional comparison interface; no new assertion of the historical dynamical closure.
A4	Conditional kernels and forgetting	Historical finite-model consumer; common-preparation posterior estimates feed the current C2-scoped consumer when their hypotheses hold.
B1	Microscopic realization targets	Declared marked experiment interface; the present scattering construction is fixed-particle.
B2	Dynamic action and kinetic targets	Historical action estimates are not used to prove the present decision-cut theorem.
B3	Balance, response and corrector targets	No current proof depends on the historical aggregate balance/kinetic claims.
B4	Nonlinear kinetic semigroup	The historical normalized-resolvent identity and its aggregate downstream claims are not premises here. Their independent repair remains a program obligation.
C1	Statistical acquisition/inference	Original behavior image, finite-reset experiments, and the present response-envelope and microscopic certification consumers.
C2	Duals, response, memory and projections	The current bounded Gaussian/entropic transport replaces the optional-projection step only in its explicitly proved class; it does not certify the historical strict-dual, form-response or rigidity aggregate.
D1	Phase and compact statistical targets	Historical finite-model consumer retained; no new complete phase-theory closure is inferred.

The machine-readable companion records the exact predecessor graph and source identities, including the original B4 and C2 statements inspected at their frozen commit. Names such as `gtf20:decision-cut` and `historical:B4:r17-main` denote different proof obligations. The organizing theorem is a general comparison-and-certification result with an essential physical consumer. It does not replace the entire historical pipeline by a declaration that every aggregate theorem follows from it. The broader program remains intact as a mathematical program.

APPENDIX C. PRESERVED ORGANIZING STATEMENTS

The following two statements retain the earlier organizing labels and claims for cross-reference. Their full proofs are included in the canonical modules above. The unchanged predecessor article and its complete historical development are appended in the separate companion, so that reorganization does not discard earlier derivations.

Theorem C.1 (Decision continuation complexity and physical certification). *The following assertions hold with all persistent data charged to the appropriate machine.*

(i) *The supremum $v_{n,K}$ of worst-case expected scores with n training samples and K states at the decision cut satisfies*

$$0 \leq \mathfrak{V}_K - v_{n,K} \leq \min \left\{ \sqrt{(|\Omega| - 1)/n}, \sqrt{2 \log(2K)/n} \right\}.$$

The least K permitting a strict score greater than c at some finite training length is the least size of a cover of B by sets $\{b : (Q - b)h > c\}$. This spectrum is Lipschitz under uniform total variation perturbations of candidate and target laws.

(ii) *For the ideal marked private product family B of the two-sphere experiment, $\mathfrak{V}_1 = 1/8$ and $\mathfrak{V}_2 = 3/8$. Throughout the collision interval $9/10 \leq d \leq 11/10$, $0 < \sigma \leq 1/24$, the minimum decision-cut width for uniform physical score greater than $2/5$ is exactly three. The lower bound applies to every finite-reset auditor with the specified interface, not merely a particular counter implementation; feedback-row selection does not evade it.*

(iii) *One deterministic rational rule has physical score greater than*

$$\sqrt{2} - 1 - \frac{1}{4\sqrt{399e}} - \frac{1}{4}e^{-399/8} - \frac{1}{300} > \frac{2}{5}.$$

It uses 399 informative training trials, an optional ignored training trial and two independent validations, hence fits the 402-trial schedule. It has exactly three decision-cut states. On its informative training segment, the minimum simultaneous widths for exact stochastic implementation are

$$r_t = \begin{cases} \binom{t+2}{2}, & 0 \leq t \leq 199, \\ \binom{402-t}{2}, & 200 \leq t \leq 399. \end{cases}$$

Their maximum is 20301. Including individual report-bit acquisition and the validations, at most $40602 < 2^{16}$ retained states suffice.

(iv) *At an arbitrary cut of a finite-valued decision, a weighted graph of distinguishing suffixes gives a distribution-sensitive error bound through its minimum monochromatic weight under K labels. Applied to Bayes margins, it yields a quantitative constrained-audit regret bound. At zero error it recovers the exact simultaneous residual profile, even for stochastic implementations.*

(v) *For fixed positive σ and changing d , the same physical rule has the target-variation modulus $\min\{1, \sqrt{|d' - d| + 2700|d' - d|^2/(2\sigma)}\}$, with no training-count factor. The microscopic paths, likelihoods, posterior processes, innovations, and bounded entropic backward values, integrands and martingale integrals have the joint transport conclusions proved below. The private and visible deficiencies exceed $9/20$, whereas a hidden two-valued selector has error less than $13/100$. In the no-collision regime $d \leq 2/5$ all three finite-experiment deficiencies vanish.*

Proof. Part (i) is Theorem 4.2. The exact ideal spectral values are Theorem 24.1; Lemmas 24.3 and 24.4 and Theorem 24.6 prove the matching physical construction. Theorem 24.5 proves (iii), including all acquisition and validation phases. Theorem 11.1 and Corollary 11.2 prove (iv). The microscopic estimates in Theorem 23.2, the explicit coupling-version lemmas in Section 21, and Theorem 22.2 prove (v). $\square$

Theorem C.2 (Continuation complexity and stable causal certification). *With these information conventions the following statements hold.*

(i) *A finite causal kernel is realizable at a width profile if and only if it admits a compatible normalized continuation cover of that profile. For a Boolean terminal test, the exact minimal profile is the number of distinct continuation responses at each cut, even when the implementing*

machine is stochastic. If a cut has $r \geq 2$ distinct responses and only K retained states, its total wordwise error is at least $(r - K)_+/(r - 1)$.

(ii) For $0 < h < 1/2$, consider $n = 2m + 1$ independent binary samples from one of $\text{Ber}(1/2 - h)$ and $\text{Ber}(1/2 + h)$, and the signed event-discrepancy audit against $\text{Ber}(1/2)$. Its unrestricted value is $h\|P_+ - P_-\|_{\text{TV}}$. The exact training widths needed to attain it are

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n.$$

A smaller width at any cut has the strictly positive value deficit (14.4). Thus the continuation invariant controls both an implementation and a statistical obstruction.

(iii) In the prepared two-sphere experiment of Theorem 29.2, one rational audit using 400 candidate training resets, two validation trials, and at most 504008 retained auditor states has expected discrepancy greater than $2/5$ against every private width-one candidate. The same audit works for every contact diameter $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. The private and visible deficiencies exceed $9/20$, whereas a hidden two-valued selector has error less than $13/100$. Removing the collision, by taking $d \leq 2/5$, makes all three finite-experiment deficiencies zero.

(iv) With positive noise fixed, changing d changes both the collision time and the microscopic path. If $\rho = |d' - d| + 2700|d' - d|^2$, the same audit's expected score changes by at most $\min\{1, \sqrt{\rho}/(2\sigma)\}$. Along $d' \rightarrow d$, the microscopic paths, full and projected likelihoods, and bounded posterior processes converge jointly in the senses specified below. For the associated bounded entropic backward equation, values converge in supremum L^2 and integrands converge globally in time-integrated L^2 , including on the common-latent coupling. Innovations converge uniformly in probability, and backward martingale integrals converge in supremum L^2 .

C.1. Inherited baseline. The following summary is retained from the predecessor. The present article strengthens its memory and backward-integrand conclusions in Theorem C.2; the earlier 2800-trial audit remains valid.

Theorem C.3 (Finite-resource certification and microscopic transport). *The following assertions hold with the conventions just described.*

(i) *Finite causal kernels have a register realization with a prescribed width profile exactly when their normalized continuation laws admit compatible finitely generated polytopes of those widths. At an erasure cut carrying a latent letter of law π , the optimal error with K retained states is*

$$1 - \sum_{i=1}^{\min(K, N)} \pi_{(i)}, \quad \pi_{(1)} \geq \cdots \geq \pi_{(N)}.$$

Independent hidden or visible selectors do not improve this cut value.

(ii) *The reset-audit value is attained, has a finite-dimensional moment minimax representation on the original B , and satisfies*

$$0 \leq \delta(B, Q) - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}, \quad D = \sum_j (|\Omega_j| - 1).$$

The exponent $1/2$ cannot be improved uniformly over compact candidate classes. This assertion is about reset samples, not about the minimal degree of unrestricted polynomial tests.

(iii) *Suppose two prepared microscopic observation models have a common preparation variable, bounded signals H, H' on $[0, T]$, and the same positive Gaussian noise level σ . Set*

$$\eta^2 = \mathbb{E} \int_0^T |H - H'|^2 dt, \quad \varepsilon = \min\{1, \eta/(2\sigma)\}.$$

Their joint marked observation laws differ by at most ε . This estimate transports resource spectra at their stated budgets and transports each fixed finite audit with its explicitly charged reset cost. It

also gives whole-process posterior convergence. The unprojected and projected likelihoods converge in the supremum L^2 norm under the common Wiener reference when $\eta \rightarrow 0$. For bounded latent payoffs, the associated entropic backward equations converge in value, and their integrands converge after the stated denominator localization.

(iv) In the two-sphere model of Theorem 23.2, one rationally specified audit with 2,800 training runs has expected discrepancy strictly greater than $2/5$ against every private width-one simulator, uniformly for $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. A hidden two-valued selector has error less than $13/100$; every visible-selector width-one error exceeds $9/20$. For $d_n \rightarrow d$ the microscopic paths, likelihoods and posterior processes converge in the precise senses of that theorem, and the same audit remains valid. For $d \leq 2/5$ all three deficiencies vanish.

APPENDIX D. UNRESTRICTED-HORIZON MULTICUT REALIZATION

The following organizing statement retains the unrestricted finite-horizon comparison and its complete proof dependencies. The interfaces are those fixed in the introduction.

Theorem D.1 (Multicut realization and certification). *The following statements hold under the interfaces and resource conventions just described.*

(i) For $d = |\Omega|$, let W_c be the minimum peak width over all finite training schedules, including every random-bit and validation cut, and let W_c^{rat} restrict to finite fair-bit implementations. Then

$$\kappa_c(B, Q) \leq W_c(B, Q) \leq W_c^{\text{rat}}(B, Q) \leq 12d \kappa_c(B, Q).$$

For a K -response cover, a finite-horizon tournament realizes this bound with $2LR(K-1)$ training observations, where $L = \lceil \log(2/\delta)/(4\theta) \rceil$ and $R = \lceil 2^L \log(2/\delta) \rceil$. Its uniform score loss is at most $2^{1-r} + 2\rho + 4(K-1)\theta + 2(K-1)\delta$, with dyadic precision r and response-mean calibration error ρ .

(ii) Approximate normalized continuation lists with compatible one-step updates and local causal errors η_t realize their entire width profile with total error at most $\sum_t \eta_t$. Full-profile risk bounds also charge the sum of the per-call calibration errors and the row-rounding errors. Composition multiplies the simultaneous simulator and auditor profiles. A shared word-capacity constraint turns weighted residual obstructions at several cuts into a single Bayesian regret lower bound. A family of point-source experiments has exact decision spectra and matching linear peak-width growth; overlapping independent retention tasks have simultaneous sharp exact profiles and quantitative approximate lower bounds.

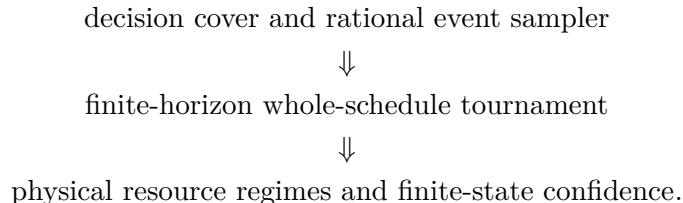
(iii) In the prepared two-sphere collision family, for $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the row-indexed decision complexity for score greater than $2/5$ is exactly three. A deterministic physical audit has peak width at most twelve throughout report-bit acquisition and validation, with an explicit finite preparation count. The earlier 399-training-trial audit is also retained, with peak width 40602 and its exact simultaneous residual profile. For the complete physical task these give $3 \leq W_{2/5}^{\text{phys}} \leq 12$; the upper endpoint is not claimed to be an exact integer optimum.

(iv) Independent repetition turns either positive expected-score certificate into a test with both errors at most α , charging the score accumulator and every preparation. The short physical audit needs $J \geq 50 \log(1/\alpha)$ repetitions. The physical certificate and the bounded common-preparation Gaussian/entropic conditional-law consumer are stable under the moving-collision perturbations specified below.

Proof. Part (i) is Theorem 7.2. Part (ii) follows from Theorems 6.1, 6.2, 12.1, 12.2, 12.3 and Proposition 6.3. Part (iii) is Theorem 25.1 together with the preserved Theorems 24.5 and 24.6. Part (iv) follows from Theorems 27.1 and 27.2, Corollary 25.2, and the common-preparation transport results in Theorems 23.2 and 22.2. $\square$

APPENDIX E. DEPENDENCIES OF THE STATISTICAL RESOURCE FRONTIER

The general preparation bound uses the decision spectrum only to supply a finite response list and its strict margin. Dyadic event realization supplies its executable randomizers. The absorbing-walk lemma then supplies both a finite comparison error and the expected number of active updates. Thus



The occupation-weighted transfer theorem acts on the induced retained kernels of each implemented machine. In the independent reset setting it consumes the walk’s active-time estimate. Its proof does not invoke any claim about convergence of changing filtrations.

The exact revelation theorem is an independent, fully solved finite statistical model inside the same profile-constrained decision layer. Its necessary direction applies to every stochastic machine; its sufficient direction constructs compatible nested labels. It is not used as a physical collision lower bound. The lower bound three in that family still comes from the original feedback-row obstruction. Likewise, the complete finite-word residual profile of the short physical selector is retained as an exact implementation theorem and not relabeled as a statistical optimum.

These distinctions give an actual resource–precision–error consumer for the foundation blueprint: preparation and state budgets come from Theorem 8.2, calibration and rationalization from its explicit coin construction, perturbation from Theorem 10.1, and a confidence-level readout from Theorem 28.1. The exact frontier provides a sharp case of the same optimization with nonconstant alphabet and arbitrary state profiles. None of these arrows presupposes the historical B4 kinetic range theorem, the broader C2 rigidity aggregate, or an unproved dependency of the independent A2 chain.

REFERENCES

- [1] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272. doi:10.1214/aoms/1177729032.
- [2] L. Le Cam, Sufficiency and approximate sufficiency, *Ann. Math. Statist.* **35** (1964), 1419–1455. doi:10.1214/aoms/1177700372.
- [3] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28. doi:10.1524/strm.2002.20.14.1.
- [4] A. Nayyar, A. Mahajan and D. Teneketzis, Decentralized stochastic control with partial history sharing: a common information approach, *IEEE Trans. Automat. Control* **58** (2013), 1644–1658. doi:10.1109/TAC.2013.2239000. The theorem numbers cited here refer to the author preprint, arXiv:1209.1695.
- [5] H. S. Witsenhausen, On the structure of real-time source coders, *Bell System Tech. J.* **58** (1979), 1437–1451. doi:10.1002/j.1538-7305.1979.tb02263.x.
- [6] I. Gallagher, L. Saint-Raymond and B. Texier, *From Newton to Boltzmann: Hard Spheres and Short-Range Potentials*, Zurich Lectures in Advanced Mathematics, European Mathematical Society, 2014. The flow and boundary discussion used here is in Sections 4.1–4.3 of arXiv:1208.5753v2.
- [7] C. Widder, J. Zimmer and T. Schilling, On the generalized Langevin equation and the Mori projection operator technique, preprint (2025), revised 2026. arXiv:2503.20457v6.
- [8] C. Widder and T. Schilling, Generalised Langevin dynamics: significance and limitations of the projection operator formalism, preprint (2026). arXiv:2604.20453v2.
- [9] H. F. Trotter, Approximation of semi-groups of operators, *Pacific J. Math.* **8** (1958), 887–919. doi:10.2140/pjm.1958.8.887. Theorem 5.2, p. 902, is the approximation theorem cited here.
- [10] H. Weisshaupt, LeCam’s randomization criterion in the setting of locally convex spaces without lattice structure, EURANDOM report 2002–017, 22 April 2002. Lemma 1 and Theorem 5, pp. 14–16. Author report.

- [11] S. Basu, Algorithms in real algebraic geometry: a survey, author preprint, arXiv:1409.1534 (2014). Published in *Panoramas et Synthèses* **51** (2017), 107–153. Theorem 2.1 and Section 2.1 in the author preprint are used here.
- [12] M. C. Paull and S. H. Unger, Minimizing the number of states in incompletely specified sequential switching functions, *IRE Trans. Electron. Comput.* **EC-8** (1959), 356–367. Publisher record.
- [13] E. Torgersen, *Comparison of Statistical Experiments*, Encyclopedia of Mathematics and its Applications 36, Cambridge University Press, Cambridge, 1991.
- [14] N. Saldi, A topology for team policies and existence of optimal team policies in stochastic team theory, preprint (2017), arXiv:1711.00639.
- [15] S. Yüksel and N. Saldi, Convex analysis in decentralized stochastic control, strategic measures and optimal solutions, author preprint arXiv:1609.07685v2.
- [16] A. Nerode, Linear automaton transformations, *Proc. Amer. Math. Soc.* **9** (1958), 541–544. doi:10.1090/S0002-9939-1958-0135681-9.
- [17] S. Kiefer, Notes on equivalence and minimization of weighted automata, preprint (2020), arXiv:2009.01217v1.
- [18] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176. doi:10.2140/pjm.1958.8.171.
- [19] H. Weisshaupt, A generalization of Le Cam’s randomization criterion, *Studia Sci. Math. Hungar.* **43** (2006), 443–486. doi:10.1556/SScMath.43.2006.4.3. Theorem numbers refer to the author’s 39-page preprint, Randomization.pdf.
- [20] E. de Klerk and M. Laurent, Error bounds for some semidefinite programming approaches to polynomial minimization on the hypercube, *SIAM J. Optim.* **20** (2010), 3104–3120. doi:10.1137/100790835.
- [21] M. Laurent and L. Slot, An effective version of Schmüdgen’s Positivstellensatz for the hypercube, arXiv:2109.09528v2, 2023.
- [22] E. de Klerk and J. C. Vera, The link between 1-norm approximation and effective Positivstellensätze for the hypercube, *Numer. Algebra Control Optim.* **16** (2026), 84–104. doi:10.3934/naco.2024060. Theorem numbers refer to arXiv:2404.04190v2.
- [23] M. Fliess, Séries rationnelles positives et processus stochastiques, *Ann. Inst. H. Poincaré Sect. B* **11** (1975), 1–21. Original article. Theorems 3.1 and 5.4 and Corollary 2, pp. 8–9 and 18–19.
- [24] T. M. Cover, Hypothesis testing with finite statistics, *Ann. Math. Statist.* **40** (1969), 828–835. doi:10.1214/aoms/1177697590. Theorem 2, pp. 833–834, and Section 4, pp. 834–835.
- [25] W. Hoeffding, Probability inequalities for sums of bounded random variables, *J. Amer. Statist. Assoc.* **58** (1963), 13–30. doi:10.1080/01621459.1963.10500830.
- [26] M. E. Hellman and T. M. Cover, Learning with finite memory, *Ann. Math. Statist.* **41** (1970), 765–782. doi:10.1214/aoms/1177696958.

GENERAL THETA FOUNDATIONS I: MULTICUT CONTINUATION COMPLEXITY AND STABLE CAUSAL CERTIFICATION

QIAN QI

ABSTRACT. We study causal certification when the auditor’s register is charged at every acquisition, randomization and validation cut. For a compact, possibly nonconvex, behavior image on a finite alphabet of size d , the strict decision-continuation complexity κ_c and the minimum whole-schedule peak width W_c satisfy $\kappa_c \leq W_c \leq 12d\kappa_c$ in the externally clocked model. A finite-horizon tournament gives a rational realization with explicit preparation, calibration and precision costs, without storing an empirical histogram. Compatible approximate continuations give full-profile error transfer, and a shared word-capacity obstruction converts multiple cut constraints into statistical regret. Growing point-source and overlapping-retention families provide quantitative lower bounds. For a marked two-sphere collision experiment we retain the exact three-state decision optimum and the short 399-trial construction, and give a second audit with at most twelve states throughout its bit-level schedule. Independent repetition yields confidence guarantees with a charged accumulator. The physical certificate and a bounded entropic conditional-law consumer remain stable under common-preparation Gaussian transport.

CONTENTS

1. Introduction	3
1.1. Classical ingredients and the contribution	4
1.2. Organization and the broader program	4
2. Intrinsic causal deficiency	4
2.1. Experiments and executable signatures	4
2.2. The finite comparison theorem	6
2.3. A delayed channel and a strict signature separation	8
2.4. Optimal resource sets and decision certificates	9
3. Continuation geometry and an exact memory obstruction	10
3.1. Normalized continuation laws	10
3.2. An erasure cut	11
4. Decision continuation spectra	12
4.1. A charged training-validation interface	12
4.2. Several feedback rows	14
5. Finite precision and charged randomization	15
6. Approximate continuation profiles and resource transfer	16
6.1. A causal comparison norm	16
6.2. The full value and its effective approximation	17
7. From a decision cover to an end-to-end machine	18
7.1. The whole-schedule continuation value	19
8. Weighted continuation obstructions	21

Date: September 24, 2026.

2020 Mathematics Subject Classification. Primary 62B05, 93E20; Secondary 60G44, 68Q45, 82C40.

Key words and phrases. causal comparison, continuation profile, finite memory, statistical experiments, hard-sphere scattering, nonlinear filtering.

9.	Joint residual budgets and growing families	22
9.1.	Actual physical word weights	23
9.2.	A family with matching end-to-end growth	23
9.3.	A sharp simultaneous storage profile	24
10.	Finite-data testing on the original behavior image	24
10.1.	The reset interface	24
10.2.	A matching uniform sampling obstruction	26
10.3.	Transport of one fixed audit	27
11.	Continuation complexity of a finite-data test	27
11.1.	The width required to attain the binary reset value	29
12.	Polynomial testing of the original resource image	30
12.1.	A sharp two-time nonconvex example	32
13.	Adaptive testing on the observable behavior image	32
13.1.	The testing game	32
13.2.	A size bound in behavior dimension	33
13.3.	A three-weight certificate invisible to constant tests	35
13.4.	Strict certificates and effective physical input	36
14.	An exact algorithm for the observable chart	37
15.	A marked resource gap generated by a binary collision	38
15.1.	The exact marked two-report experiment	38
15.2.	A full-dimensional collision preparation	39
15.3.	Uniformly effective finite presentations	41
16.	Uniform stability of optional projections	41
16.1.	A microscopic sufficient condition	43
17.	Changing trajectories, likelihoods, and conditional consumers	44
17.1.	A common preparation, not a common flow	44
17.2.	The projected stochastic exponential	45
17.3.	A bounded entropic backward-equation consumer	46
17.4.	Resource transport is not the posterior coupling	47
18.	Observation densities and coupled process versions	48
19.	Global transport of the backward equation	49
20.	A uniform certificate through a moving collision	52
21.	Sharp decision memory for the collision experiment	55
21.1.	The marked family and its first two spectral values	55
21.2.	A three-response construction	56
21.3.	Simultaneous exact widths and the full acquisition schedule	57
21.4.	What the two exact complexity statements measure	59
22.	Full-profile certification of the collision family	60
22.1.	The row-indexed decision invariant	60
23.	Confidence-level certification	61
24.	A finite-register certificate transported through the microscopic limit	63
25.	Explicit phase transitions for the two-count certificate	65
Appendix A.	The proof dependency chain	66
A.1.	The full-profile resource chain	66
Appendix B.	A namespaced map of the eleven-component program	68
Appendix C.	Preserved organizing statements	68
C.1.	Inherited baseline	70
References		71

1. INTRODUCTION

A finite causal experiment has two distinct memory problems. A simulator must retain enough of the past to produce the required future law. An auditor must retain enough of its observations to select and execute a statistically useful continuation. A restriction at the end of training does not, by itself, restrict the computation that produced that state. The question addressed here is how a decision continuation bound extends to a machine whose register is charged throughout acquisition, randomization and validation.

We work with declared causal interfaces and actual marked probability laws. Candidate behavior images need not be convex. Preparation count, retained labels, transition precision and the schedule clock are distinct resources. An external clock chooses a predetermined transition; it does not store sample history. An autonomous realization must retain that clock as part of its state. This distinction is essential even for a Bernoulli experiment.

For a compact candidate image $B \subset \Delta(\Omega)$ and target Q , a reset auditor observes independent trials from one fixed $b \in B$, then selects an event before one fresh candidate and one fresh target validation. Its signed score is the target indicator minus the candidate indicator. A randomized event has a response $h \in [0, 1]^\Omega$. Define

$$\mathfrak{V}_K(B, Q) = \max_{h_1, \dots, h_K} \min_{b \in B} \max_{i \leq K} (Q - b)h_i, \quad \kappa_c(B, Q) = \min\{K : \mathfrak{V}_K(B, Q) > c\}.$$

The decision-cut classification and its finite-sample approximation are proved in Section 4. The new point is that a response cover can be learned and implemented without an upstream histogram of unbounded width.

Theorem 1.1 (Multicut realization and certification). *The following statements hold under the interfaces and resource conventions just described.*

(i) *For $d = |\Omega|$, let W_c be the minimum peak width over all finite training schedules, including every random-bit and validation cut, and let W_c^{rat} restrict to finite fair-bit implementations. Then*

$$\kappa_c(B, Q) \leq W_c(B, Q) \leq W_c^{\text{rat}}(B, Q) \leq 12d \kappa_c(B, Q).$$

For a K -response cover, a finite-horizon tournament realizes this bound with $2LR(K-1)$ training observations, where $L = \lceil \log(2/\delta)/(4\theta) \rceil$ and $R = \lceil 2^L \log(2/\delta) \rceil$. Its uniform score loss is at most $2^{1-r} + 2\rho + 4(K-1)\theta + 2(K-1)\delta$, with dyadic precision r and response-mean calibration error ρ .

(ii) *Approximate normalized continuation lists with compatible one-step updates and local causal errors η_t realize their entire width profile with total error at most $\sum_t \eta_t$. Full-profile risk bounds also charge the sum of the per-call calibration errors and the row-rounding errors. Composition multiplies the simultaneous simulator and auditor profiles. A shared word-capacity constraint turns weighted residual obstructions at several cuts into a single Bayesian regret lower bound. A family of point-source experiments has exact decision spectra and matching linear peak-width growth; overlapping independent retention tasks have simultaneous sharp exact profiles and quantitative approximate lower bounds.*

(iii) *In the prepared two-sphere collision family, for $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$, the row-indexed decision complexity for score greater than $2/5$ is exactly three. A deterministic physical audit has peak width at most twelve throughout report-bit acquisition and validation, with an explicit finite preparation count. The earlier 399-training-trial audit is also retained, with peak width 40602 and its exact simultaneous residual profile. For the complete physical task these give $3 \leq W_{2/5}^{\text{phys}} \leq 12$; the upper endpoint is not claimed to be an exact integer optimum.*

(iv) *Independent repetition turns either positive expected-score certificate into a test with both errors at most α , charging the score accumulator and every preparation. The short physical audit needs $J \geq 50 \log(1/\alpha)$ repetitions. The physical certificate and the bounded common-preparation*

Gaussian/entropic conditional-law consumer are stable under the moving-collision perturbations specified below.

Proof. Part (i) is Theorem 7.2. Part (ii) follows from Theorems 6.1, 6.2, 9.1, 9.2, 9.3 and Proposition 6.3. Part (iii) is Theorem 22.1 together with the preserved Theorems 21.5 and 21.6. Part (iv) follows from Theorems 23.1 and 23.2, Corollary 22.2, and the common-preparation transport results in Theorems 20.2 and 19.2. $\square$

The central bound is about approximate statistical value over all finite training lengths. It is not the exact residual profile of a prescribed word function. Conversely, the exact profile of a particular selector cannot be used as an all-algorithm approximate-memory lower bound. The joint residual theorem explains when wordwise distinctions do carry statistical weight, through the actual likelihoods and Bayes margins. The growing point-source and retention families show that the full-profile statements have nonconstant resource consequences.

The physical application makes the training tradeoff especially visible. The short majority audit uses a moderate number of preparations and a large counter quotient. The second audit uses a finite rare-block comparison, a register of four bits, and a very long schedule. Neither is presented as optimizing preparation count and peak width simultaneously. Charging the clock changes the latter bound by the explicit autonomous simulation formula. Likewise, perturbing a source used in training generally introduces a training-count term; the source-preserving moving-collision estimate is identified separately.

1.1. Classical ingredients and the contribution. Blackwell–Le Cam comparison and earlier filtered comparison provide the statistical background [1, 2, 13, 3, 10]. In Weisshaupt’s report, Lemma 1 and Theorem 5 on pp. 14–16 use compact convex classes of filtered stochastic operators. Here the bounded register is a causal realization constraint; its behavior image is not replaced by a convex class of unpriced randomized machines. The full original Norberg article has not been obtained for a theorem-by-theorem priority audit. No claim that its theorems omit a particular feature is inferred from that limitation.

Positive-cone realization and residual minimization are classical mechanisms [23, 16]. Cover’s Theorem 2, pp. 833–834, already gives four-state clocked Bernoulli threshold learning, and his Section 4 explicitly discusses the external clock [24]. Our use of finite blocks is an implementation of that mechanism, not a claim to have discovered finite-state learning. The work here is the quantitative passage from a general nonconvex decision cover to a complete rational training-and-validation machine, its all-cut resource and perturbation accounting, and its integration with distribution-sensitive lower bounds and a microscopic marked experiment.

1.2. Organization and the broader program. Sections 5–9 develop the new realization and obstruction results after the underlying comparison and decision spectra. The finite-reset, streaming, nonlinear-testing, physical-scattering and conditional-transport arguments are retained in full. Sections 22 and 23 complete the physical resource and statistical-confidence consequences. Appendix B gives the actual proof dependencies for all eleven historical components. The independent A2 geometric chain, the nonlinear kinetic B4 targets and the broader C2 operator targets retain their own proof obligations. This finite controlled resource layer contributes to the program’s general resource–precision–error objective without replacing those model theorems by nominal arrows.

2. INTRINSIC CAUSAL DEFICIENCY

2.1. Experiments and executable signatures. Fix a finite parameter set Θ , a horizon T , and finite control alphabets $A_0, \dots, A_{T-1}$. An experiment has a report at checkpoints $0, \dots, T$ and an unobserved terminal mark W in a common finite set. It is specified by finite causal instruments:

for each parameter and control history they give the joint law of the report path and the actual mark. Equivalently, a finite latent-state realization may be expanded into its history tree. The parameter and mark are unavailable to the encoder. The mark may depend on the physical controlled path. We compare experiments with the same control interface and mark space.

Write Y_t for the source report and Z_t for the target report. A private simulator of width K has stochastic rows

$$\begin{aligned} q_0(z, m \mid y), \quad m &\in [K_0], \\ q_t(z, m \mid m', a, y), \quad m' &\in [K_{t-1}], \quad m \in [K_t], \quad 1 \leq t \leq T. \end{aligned} \quad (2.1)$$

A row consumes the current source report, emits the target report, and retains the new state. The previous target reports are not available to that row unless encoded in m' . The clock and current control are part of the declared interface. Fresh private randomness is permitted, but no other persistent random variable is free. The product of all row simplexes is denoted $\mathcal{K}_K$.

A deterministic feedback tree g chooses a_t from $z_{0:t}$ and its own previous controls. It is the same controller on both sides of a comparison. Let Ω_g be the finite marked transcript space, including reports, controls, and W . Put

$$P_{\theta,g}^q = \text{Law}_\theta(E \xrightarrow{q} g), \quad Q_{\theta,g} = \text{Law}_\theta(F \rightarrow g).$$

Here W on the left is the source's actual mark. The total variation convention is $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. Private deficiency is

$$\delta_K^p(E, F) = \min_{q \in \mathcal{K}_K} \max_{\theta, g} \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}. \quad (2.2)$$

The minimum will be justified below. Randomized feedback need not be added to the maximum: conditioning on its independent random table expresses its law as a mixture of deterministic feedback laws. The same argument permits arbitrary independent side randomness in the tester.

A *hidden selector* $U \in [r]$, independent of the source, chooses one simulator q^u . It is not visible to the feedback controller or the final test. A *visible selector* is revealed before feedback begins. The two costs are

$$\delta_{K,r}^h(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, g} \left\| \sum_u \lambda_u P_{\theta,g}^{q^u} - Q_{\theta,g} \right\|_{\text{TV}}, \quad (2.3)$$

$$\delta_{K,r}^v(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, (g_u)} \left\| \sum_u \lambda_u \delta_u \otimes P_{\theta, g_u}^{q^u} - \sum_u \lambda_u \delta_u \otimes Q_{\theta, g_u} \right\|_{\text{TV}}. \quad (2.4)$$

In both expressions $\lambda \in \Delta_r$ and $q^u \in \mathcal{K}_K$. The second formula compares the joint seed-marked interface. Its target seed has the same distribution and is independent of the target experiment. A hidden selector can be stored privately by retaining (u, m) , at width rK_t . A visible selector is separate side information and is charged by its own alphabet size. Neither interpretation identifies an arbitrary real-valued design with a free finite register.

Definition 2.1 (Deficiency spectrum). The private resource spectrum of E relative to F is the upward-closed set of pairs (K, ε) satisfying $\delta_K^p(E, F) \leq \varepsilon$. The hidden spectrum consists of triples (K, r, ε) satisfying $\delta_{K,r}^h(E, F) \leq \varepsilon$, and the visible spectrum uses $\delta_{K,r}^v$. Width profiles are ordered coordinatewise; in the seeded spectra the seed bound and error are ordered as well. Unused states or selector values may be adjoined. The hidden convexification is $\delta_K^{\text{mix}} = \inf_{r \geq 1} \delta_{K,r}^h$.

The spectrum, not a presumed unique coordinatewise minimum profile, is the intrinsic object. For example one may minimize $\max_t K_t$, with a specified seed signature and tolerance, over its feasible profiles. Different checkpoints can have incomparable optimal tradeoffs.

2.2. The finite comparison theorem. Concatenate all probabilities $P_{\theta,g}^q(\omega)$ into $b(q) \in \mathbb{R}^N$, retaining their row indices; put $b_F = (Q_{\theta,g}(\omega))$. Let $\mathcal{D}_K$ be the finite set of deterministic choices of all rows in (2.1). Empty or unreachable rows remain harmless coordinates. Set

$$a_K = \dim \text{aff}\{b(q) : q \in \mathcal{K}_K\}, \quad \mathcal{I} = \{(\theta, g, B) : B \subseteq \Omega_g\}.$$

For $i = (\theta, g, B)$ and $d \in \mathcal{D}_K$, define

$$A_{i,d} = Q_{\theta,g}(B) - P_{\theta,g}^d(B). \quad (2.5)$$

Including both an event and its complement includes both signs.

Theorem 2.2 (Intrinsic deficiency, priced converse, and visibility). *For the finite marked causal experiments above the following assertions hold.*

(i) *Private deficiency and both fixed- r deficiencies are attained. The private problem has the exact polynomial formulation*

$$\min\{\varepsilon : q \in \mathcal{K}_K, 0 \leq \varepsilon \leq 1, Q_{\theta,g}(B) - P_{\theta,g}^q(B) \leq \varepsilon \text{ for all } (\theta, g, B) \in \mathcal{I}\}. \quad (2.6)$$

In particular its zero set is exactly the nonnegative causal factorization of the target marked behavior by the rows (2.1). At $K_t = 1$ for every t , this is precisely the same factorization with rows $q_t(z \mid a, y)$ and $q_0(z \mid y)$.

(ii) *Hidden convexification has the attained dual*

$$\delta_K^{\text{mix}}(E, F) = \min_{\lambda \in \Delta(\mathcal{D}_K)} \max_{i \in \mathcal{I}} \sum_d \lambda_d A_{i,d} = \max_{w \in \Delta(\mathcal{I})} \min_{d \in \mathcal{D}_K} \sum_i w_i A_{i,d}. \quad (2.7)$$

An optimal simulation uses at most $a_K + 1$ selector values. Thus every weighted causal test has optimal discrepancy at most ε if and only if a hidden mixture of K -state simulators has error at most ε ; it has a private realization of width at most $(a_K + 1)K$. More specifically,

$$\delta_{rK}^p \leq \delta_{K,r}^h, \quad \delta_K^{\text{mix}} = \delta_{K,a_K+1}^h \leq \delta_K^p. \quad (2.8)$$

(iii) *Put $e_\theta(q) = \max_g \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}$. Then*

$$\delta_{K,r}^v = \min_{\lambda, q^1, \dots, q^r} \max_{\theta} \sum_u \lambda_u e_\theta(q^u). \quad (2.9)$$

Writing $a_K^v = \dim \text{aff}\{(e_\theta(q))_\theta : q \in \mathcal{K}_K\}$, the infimum over all finite visible selectors is attained using $a_K^v + 1 \leq |\Theta| + 1$ values and equals

$$\max_{v \in \Delta(\Theta)} \min_{q \in \mathcal{K}_K} \sum_{\theta} v_\theta e_\theta(q). \quad (2.10)$$

For every $r \geq 1$, and also after optimizing over all r ,

$$\delta_{K,r}^v(E, F) = 0 \iff \delta_K^p(E, F) = 0. \quad (2.11)$$

No corresponding equivalence with $\delta_K^{\text{mix}} = 0$ holds in general.

(iv) *For compatible experiments E, F, G and profiles K, L ,*

$$\begin{aligned} \delta_{KL}^p(E, G) &\leq \delta_K^p(E, F) + \delta_L^p(F, G), \\ \delta_{KL,rs}^s(E, G) &\leq \delta_{K,r}^s(E, F) + \delta_{L,s}^s(F, G), \quad s \in \{h, v\}. \end{aligned} \quad (2.12)$$

Here KL is the pointwise product, and the second line uses independent selectors. If an F -consumer has widths S_t , composing it with a private simulator gives widths $K_t S_t$ and changes every $[0, 1]$ marked risk by at most $\delta_K^p(E, F)$. A hidden selector is stored at the additional price r , whereas visible selector alphabets multiply. The same bounds hold for optimized upper risk sets and their lower support functions with the indicated costs.

Proof. Expand a probability of a complete source path, simulator state path, and transcript. It is a product of fixed experiment probabilities and one chosen row entry per time. Summing latent and state paths gives a polynomial in the finitely many row entries. No row belonging to a fixed time can occur twice along one path. Consequently the polynomial is multi-affine in the groups of stochastic rows. Continuity and compactness of the product of closed finite simplexes prove attainment in (2.2); the same proof includes λ and the r policies in (2.3). For the visible problem there are only finitely many collections (g_u) and transcript events, so its objective is again continuous on a compact set. The finite-event formula for total variation proves (2.6). Setting $\varepsilon = 0$ forces equality of every transcript probability, in both directions of each event inequality. This proves the factorization and stateless claims, including their necessity.

For each stochastic row of q , sample its deterministic output once, independently of all other rows, before the experiment. This constructs a probability Λ_q on $\mathcal{D}_K$. On any fixed path the row variables encountered have different time indices, so the expectation of their product is the product of the row probabilities. Summing paths shows, simultaneously for every θ, g , that

$$b(q) = \sum_{d \in \mathcal{D}_K} \Lambda_q(d) b(d). \quad (2.13)$$

This is a behavior identity, not a same-width deterministic purification of minimax error. It implies $\text{conv } b(\mathcal{K}_K) = \text{conv}\{b(d) : d \in \mathcal{D}_K\}$. The latter is a compact polytope. Minimizing the continuous convex maximum of event discrepancies over this polytope gives the first expression in (2.7). Finite-dimensional minimax gives the second, with both optima attained. A point of this polytope is a convex combination of at most $a_K + 1$ of its generating points by Carathéodory's theorem. For completeness, if a representation uses more points, their affine dependence gives coefficients c_d with $\sum c_d = 0$ and $\sum c_d b(d) = 0$. Subtract the largest permissible multiple of c from the positive mixing weights; a weight becomes zero without changing the represented point. Repeating gives the stated bound. Store its selector together with the machine state to obtain (2.8). The dual inequality for every w is equivalent to the maximum over w being at most ε , which proves the priced testing converse.

For disjoint seed fibers, total variation is the sum of the fiber total variations weighted by λ_u . For a fixed θ , the tester can choose a separate maximizing tree on each revealed fiber. Therefore its maximum is $\sum_u \lambda_u e_\theta(q^u)$, proving (2.9). The map $q \mapsto e(q)$ is continuous, being a finite maximum of continuous functions. Its image is compact. Convexifying it in $\mathbb{R}^{|\Theta|}$ and applying the preceding finite-dimensional separation and affine-dependence argument proves (2.10) and its selector bound. Notice that the inner minimum is over stochastic q , not over deterministic tables: the nonlinear function e_θ does not commute with (2.13). If the attained visible optimum is zero, then for every u of positive weight and every θ one has $e_\theta(q^u) = 0$, since every summand is nonnegative. Any such q^u is a private exact simulator. The converse uses a selector of one value. This also proves the assertion for the optimized visible selector, whose support is finite by the bound just established.

Finally, cascade a K -state E -to- F transducer and an L -state F -to- G transducer. Within a step, consume the intermediate report before discarding it. The retained pair has at most $K_t L_t$ values. For any terminal tester on G , the second transducer together with that tester is an admissible randomized feedback system on F . The first error bound applies to this system, and total variation contracts under the remaining channel. The second error bound applies to the genuine F system. The triangle inequality gives (2.12). With selectors, condition on the independent selectors in the appropriate places; in the visible case keep both selectors in the compared transcript. The same argument is valid because the target reference selectors are independent. Storing a hidden selector or exposing both visible selectors gives the stated costs. The minima established above may be used; equivalently one can use nearly optimal simulators and let the extra tolerance decrease.

A consumer is another causal transducer with task-specific readout and, where allowed, task-specific control, but with its required common encoder. The simulator is task blind, so its product

with that encoder is still common. Condition on the consumer's independent random table to apply the feedback bound. Integration of any $[0, 1]$ marked loss changes by at most the total variation error. Taking the same paired consumer for every row proves the risk-set assertion, rather than a separate choice of simulator for each row. Infima of nonnegative weighted risks give the lower support inequalities. $\square$

Remark 2.3 (What the converse says). The dual (2.7) uses mixtures of indexed causal tests and optimizes over the entire common transducer. It is not a task-by-task choice of encoder. Its conclusion is an explicit hidden-selector simulation, with the selector subsequently charged if private storage is required. It does not replace the nonconvex private set by its convex hull at unchanged resource. The next example shows why that replacement would change the zero set.

2.3. A delayed channel and a strict signature separation. Let the source reveal $i \in [n]$ at checkpoint zero and reveal only a constant at checkpoint one. There is no control and the mark is constant. The target reports a constant at checkpoint zero and at checkpoint one draws $j \in [m]$ from a specified row-stochastic matrix $V = (V_{ij})$. The unknown parameter is i . The first target alphabet has one element: the simulator cannot use an early public output as unpriced storage. Only the intermediate register of size K matters.

Proposition 2.4 (The nonnegative-rank slice). *For this experiment,*

$$\delta_K^p(E, F_V) = \min_{A, B} \max_i \frac{1}{2} \sum_j |V_{ij} - (AB)_{ij}|, \quad (2.14)$$

where A is an n -by- K row-stochastic matrix and B is a K -by- m row-stochastic matrix. In particular

$$\delta_K^p(E, F_V) = 0 \iff \text{rank}_+(V) \leq K, \quad \delta_1^p(E, F_V) = 0 \iff V \text{ has identical rows.}$$

For $1 \leq K \leq n$ the identity channel has the sharp laws

$$\delta_K^p(E, F_{I_n}) = 1 - \frac{1}{\lceil n/K \rceil}, \quad \delta_K^{\text{mix}}(E, F_{I_n}) = 1 - \frac{K}{n}. \quad (2.15)$$

Thus, at tolerance $0 \leq \varepsilon < 1$, the least private width is $\lceil n/(1 - \varepsilon)^{-1} \rceil$, whereas the least within-mode width with optimized hidden selection is $\max\{1, \lceil n(1 - \varepsilon) \rceil\}$. The latter still pays for its selector. In particular, for $n = m = 3$ and $K = 2$,

$$\delta_2^p(E, F_{I_3}) = \frac{1}{2}, \quad \delta_2^{\text{mix}}(E, F_{I_3}) = \frac{1}{3}. \quad (2.16)$$

For $V_* = (1/2)I_3 + (1/6)\mathbf{1}\mathbf{1}^\top$,

$$\delta_{2,3}^h(E, F_{V_*}) = 0, \quad \delta_2^p(E, F_{V_*}) \geq \frac{1}{4\sqrt{3}} > 0, \quad \inf_r \delta_{2,r}^v(E, F_{V_*}) > 0. \quad (2.17)$$

Proof. The encoder maps the consumed i into its retained label by A ; the terminal decoder maps that label into j by B . This proves (2.14) including its converse implementation. If $V = CD$ is any nonnegative factorization, let $s_k = \sum_j D_{kj}$. Delete zero rows of D , define $B_{kj} = D_{kj}/s_k$ and $A_{ik} = C_{ik}s_k$. Since V is stochastic, the rows of A sum to one. Padding with unused states gives the claimed equivalence with nonnegative rank. The one-state statement follows immediately.

For $V = I_n$, row error is $1 - (AB)_{ii}$. If the least correct probability is c , for each i select a row $k(i)$ of B with $B_{k(i),i} \geq c$. Some row is selected at least $\lceil n/K \rceil$ times, so stochasticity gives $c \leq 1/\lceil n/K \rceil$. A partition into K groups of sizes at most $\lceil n/K \rceil$, each decoded uniformly, attains that bound. Also $\sum_i (AB)_{ii} \leq K$, by the calculation below with K in place of two. It remains valid under mixing, giving hidden error at least $1 - K/n$. Start with a deterministic channel that reproduces K input symbols exactly and sends all others to those K outputs. Average all simultaneous permutations of its input and output labels. Every diagonal entry is then K/n , proving the second formula. Inverting the two monotone formulas gives the stated least widths.

For $V = I_3$, total variation in row i is $1 - (AB)_{ii}$. If all three diagonal entries of AB were greater than $1/2$, then for each i some row of B would have its i th entry greater than $1/2$. A stochastic row has at most one such entry, so two rows cannot serve three indices. This proves the private lower bound $1/2$. Merge two input symbols into a retained label that decodes uniformly to that pair, and retain the third symbol exactly. This attains it. For every stochastic factorization with two intermediate states,

$$\sum_{i=1}^3 (AB)_{ii} = \sum_{k=1}^2 \sum_{i=1}^3 A_{ik} B_{ki} \leq \sum_{k=1}^2 \sum_{i=1}^3 B_{ki} = 2.$$

The same inequality holds under hidden mixtures, giving maximal row error at least $1/3$. Mix uniformly the three preceding pair-merge channels. The resulting diagonal entries are $2/3$ and the off-diagonal entries are $1/6$. This is exactly V_* and has error $1/3$ from I_3 . It proves the hidden claims in both displays.

The smallest singular value of V_* is $1/2$. Any AB with two intermediate states has a unit vector x in its right null space, so $\|V_* - AB\|_F \geq \|(V_* - AB)x\|_2 \geq 1/2$. If the maximal row total variation error is ε , each row has Euclidean error at most 2ε , and hence the Frobenius error is at most $2\sqrt{3}\varepsilon$. This proves the explicit private bound. The optimized visible value is attained by Theorem 2.2; if it were zero, its zero-set assertion would contradict that bound. This proves its strict positivity. $\square$

2.4. Optimal resource sets and decision certificates. For a class $\mathcal{C}_{S,q}(F)$ of common consumers, with all other information conventions fixed, let $\mathcal{R}_{S,q}(F)$ be its vector of attainable $[0, 1]$ risks. For a nonnegative row weight w summing to one, write

$$h_F(S, q; w) = \inf_{r \in \mathcal{R}_{S,q}(F)} \langle w, r \rangle.$$

If E privately simulates F with cost K and error ε , then

$$\mathcal{R}_{S,q}(F) \subseteq \mathcal{R}_{KS,q}(E) + [-\varepsilon, \varepsilon]^J, \quad h_E(KS, q; w) \leq h_F(S, q; w) + \varepsilon. \quad (2.18)$$

For nonfinite row sets the first expression means pointwise paired risk functions within uniform distance ε ; no finite coordinate box is needed. Upper sets give the equivalent one-sided order formulation. Hidden selectors multiply private width by r ; visible selectors multiply the separate public alphabet by r . These are direct consequences of the final part of Theorem 2.2, with the optimal intrinsic error in place of an externally estimated simulator error.

When two stateless comparison directions have error at most ε , paired risk sets and their support values at unchanged width are within ε . If each is centered by a paired unrestricted benchmark whose error is also at most ε , the centered values are within 2ε . Therefore a gap $g > 0$ between zero excess and infeasible widths is stable under $\varepsilon < g/4$: a value below $g/2$ in the approximation certifies feasibility in the exact finite problem, and a value above $g/2$ certifies infeasibility. The benchmark and the class being optimized must be specified; this assertion does not identify different information signatures.

Proposition 2.5 (Intrinsic finite-dimensional selector approximation). *Let $\mathcal{R}$ be a compact risk image in $C(J)$, where J is compact. Suppose a bounded linear operator $L : C(J) \rightarrow C(J)$ has rank at most m and $\sup_{r \in \mathcal{R}} \|r - Lr\|_\infty \leq \eta$. Every barycenter $\bar{r}$ of a probability on $\mathcal{R}$ has a barycenter r_* of at most $m + 1$ members of $\mathcal{R}$ with $\|\bar{r} - r_*\|_\infty \leq 2\eta$. If $\mathcal{R}$ lies in a finite affine space of dimension m , the error is zero with at most $m + 1$ members. Thus the selector price can be controlled by the dimension of attainable risks, rather than by a covering number of the row space.*

Proof. The compact set $L\mathcal{R}$ lies in an m -dimensional space. Its convex hull is compact, and $L\bar{r}$ belongs to it. The finite-dimensional affine-dependence argument in the proof of Theorem 2.2 gives $r_1, \dots, r_{m+1} \in \mathcal{R}$ and weights λ such that $L\bar{r} = \sum_i \lambda_i Lr_i$. Put $r_* = \sum_i \lambda_i r_i$. The two residual

terms $\bar{r} - L\bar{r}$ and $r_* - Lr_*$ have norms at most η , proving the estimate. Apply the same argument after translation in the exact affine case. The implementation stores the selected design index whenever it is not provided as visible side information. $\square$

Remark 2.6 (Finite computation and general carriers). For rational finite instruments, (2.6) and the fixed- r formulations are finite polynomial feasibility problems; (2.7) is a finite linear program. These are exact variational characterizations, not assertions of polynomial running time in a succinct history description. The number of feedback trees, events, or deterministic tables can be large. On standard Borel carriers the same definitions use suprema over measurable feedback and tests and infima over executable kernels. Their product-state composition and decision transport remain valid by the same channel argument. Attainment and the finite testing dual are asserted here only for the stated finite class. The average-risk and physical results below give separate nonfinite theorems with their own hypotheses.

3. CONTINUATION GEOMETRY AND AN EXACT MEMORY OBSTRUCTION

3.1. Normalized continuation laws. We first separate the transducer from the source to which it is attached. There are L epochs with finite input alphabets X_t and output alphabets Z_t . A causal kernel $p(z_{1:L} \mid x_{1:L})$ is nonnegative, normalized for every input word, and has output-prefix marginals independent of later inputs. The current input can include both a source report and the current control. It never includes an unavailable parameter, a terminal mark, or an earlier output not retained by the machine.

Let $\mathcal{C}_t$, $0 \leq t \leq L$, be the finite-dimensional polytope of normalized causal kernels from $x_{t+1:L}$ to $z_{t+1:L}$; put $\mathcal{C}_L = \{1\}$. For $v \in \mathcal{C}_t$ define the shift

$$(S_{x,z}v)(z_{t+2:L} \mid x_{t+2:L}) = v(z, z_{t+2:L} \mid x, x_{t+2:L}).$$

Its total mass $v(z \mid x)$ is independent of the future input word. A zero shift is allowed.

Definition 3.1 (Compatible continuation polytopes). A continuation cover of width $K = (K_0, \dots, K_L)$ consists of lists $V_t = (v_{t,m})_{m \leq K_t}$ in $\mathcal{C}_t$ such that, for every m, x, z ,

$$S_{x,z}v_{t,m} \in \text{cone } V_{t+1}. \quad (3.1)$$

It covers p when $p \in \text{conv } V_0$. Lists may repeat generators and may contain unreachable generators. Thus unused states can be adjoined.

Theorem 3.2 (Continuation realization with a width profile). *A causal kernel p has a stochastic transducer realization with an initial K_0 -valued register and K_t retained states after epoch t if and only if it has a compatible continuation cover of width K . Initial randomization is part of the charged initial register. With a deterministic initial state the condition is $p = v_{0,1}$ and $K_0 = 1$.*

Consequently the set of feasible width profiles depends only on the causal kernel and its declared input interface, not on a row presentation or a labeling of the hidden states. Every matrix of conditional continuation probabilities at cut t has nonnegative rank at most K_t . Individual rank bounds do not replace the simultaneous shift conditions (3.1).

Proof. Given a transducer, let $v_{t,m}$ be the suffix response when its register at time t equals m . The remaining rows define this response even for an unreachable state. If the next row is $q_{t+1}(z, m' \mid m, x)$, conditioning on its output and new state gives

$$S_{x,z}v_{t,m} = \sum_{m'} q_{t+1}(z, m' \mid m, x) v_{t+1,m'}.$$

The initial state distribution expresses p as a convex combination of V_0 .

Conversely choose nonnegative coefficients in every cone membership in (3.1). There are only finitely many choices to make. Since every generator on the right has mass one for each future

input word, the sum of the coefficients for a fixed m, x, z is $v_{t,m}(z \mid x)$. Summing over z gives one. The coefficients are therefore stochastic rows $q_{t+1}(z, m' \mid m, x)$. Choose an initial distribution representing p in $\text{conv } V_0$. Backward induction from $\mathcal{C}_L = \{1\}$ proves that the constructed machine has exactly the response $v_{t,m}$ from each register state. In particular its initial response is p . Equality for open-loop input words implies equality under adaptive input selection: at each node of an input–output history tree, the same causal conditional probability is used on the two sides.

For the rank assertion, condition on any positive-probability observed history at the cut. Its continuation response is a convex combination of $v_{t,m}$ with weights equal to the posterior register distribution. Stack these responses as rows, and use all future input–output pairs as columns. This is a nonnegative factorization through K_t generators. The converse construction required the same generator families at adjacent cuts and all their shifts; an unrelated factorization of each cut matrix provides no such compatibility. The characterization, being stated entirely in terms of these response polytopes, also proves the asserted invariance. $\square$

Attach such a causal kernel to a marked source experiment. The output law is obtained by contraction with the source’s causal instruments, while its actual mark is retained. Hence zero private deficiency is equivalently the existence of a compatible continuation cover whose contraction equals the target marked behavior in every parameter and feedback row. This is the geometric form of the factorization in Theorem 2.2; no conditional dependence on the mark is introduced.

A hidden selector chooses one such cover independently of the source. It is a convex combination of complete realizations, not a convexification inside every retained-state simplex at the original width. Its private implementation uses the disjoint union of the covers and keeps the selector in the state. A visible selector keeps those fibers in the compared law. The nonnegative fiber argument in Theorem 2.2 therefore continues to give equality of the visible and private zero sets.

3.2. An erasure cut. A simple family makes the width obstruction exact, including its error. Let X take values in $[N]$ with law π . Before a specified cut the source reveals X . After the cut it has no reports correlated with X . The controller has no feedback channel carrying X across the cut, and all earlier emitted reports are prescribed constants. The only permitted cross-cut information of the simulator is its K -valued register and any independent selector explicitly included in its signature. The terminal mark is the actual $W = X$. The target’s terminal report is X . All memory at other checkpoints is unrestricted.

Theorem 3.3 (Exact cut spectrum). *For this experiment, for private, hidden and visible signatures alike,*

$$\delta_K = 1 - \sum_{i=1}^{\min(K,N)} \pi_{(i)}. \quad (3.2)$$

For uniform X , $\delta_K = (1 - K/N)_+$. Thus an erasure cut for m unbiased bits requires $K \geq (1 - \epsilon)2^m$ whenever its error is at most ϵ . The bound holds for every surrounding width profile; equality is attained when the other checkpoints are unrestricted. If every $\pi_x > 0$, exact transmission has nonnegative cut rank N .

Proof. With all post-cut randomness averaged into the decoder, any private machine gives kernels $e(m \mid x)$ and $d(z \mid m)$. Its joint terminal law is $\pi_x \sum_m e(m \mid x) d(z \mid m)$. Since the target law is $\pi_x \mathbf{1}_{z=x}$, total variation is exactly one minus the correct-decoding probability. Put $u_x = \max_m d(x \mid m)$. Then

$$0 \leq u_x \leq 1, \quad \sum_x u_x \leq K, \quad \mathbb{P}(Z = X) \leq \sum_x \pi_x u_x \leq \sum_{i=1}^{\min(K,N)} \pi_{(i)}.$$

The last inequality follows by moving mass in u from a smaller weight to a larger one until the K largest weights are filled. To attain it, retain distinct labels for those letters and decode them

deterministically; all other letters are mapped to any retained label. Before the cut one may use the unrestricted memory to compute this encoding.

For a hidden selector, condition on its value. It is independent of X , so every conditional correct-decoding probability obeys the same bound; averaging cannot improve it. For a visible selector, variation is the average of the fiber variations, with the same conclusion. The private construction works for both signatures. Finally, the exact joint cut matrix is diagonal with its N positive diagonal entries π_x . Its ordinary rank is N , so its nonnegative rank is at least N and its diagonal factorization attains N . $\square$

The absence of an auxiliary cross-cut feedback channel is essential. An unpriced controller storing X and returning it as a later control would change the experiment, not disprove the cut bound. Likewise a selector correlated with X is not an independent hidden selector. These distinctions are the same ones needed in the collision certificate below.

4. DECISION CONTINUATION SPECTRA

The memory at an experimental cut has a different role from the memory needed to reach that cut. We first identify its exact asymptotic value. This will give both an obstruction and a construction for the same physical audit, rather than a lower bound for an auxiliary binary problem.

4.1. A charged training-validation interface. Let Ω be a finite marked alphabet, $d = |\Omega|$, let $B \subset \Delta(\Omega)$ be nonempty and compact, and fix $Q \in \Delta(\Omega)$. A candidate $b \in B$ is fixed throughout the experiment. Training consists of n independent observations from b . At the end of training the auditor retains a register $M \in [K]$. Everything depending on the training observations or on an earlier random seed must pass through this register. In particular, there is no external copy of the training history and no uncharged persistent selector.

After this cut, fresh independent randomness may select an event A using M . The event is chosen before independent candidate and target validation observations $X \sim b$ and $Y \sim Q$ are drawn. The score is $\mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The downstream register needed to keep A and to compute this score is distinct from the register at the specified cut. The upstream and downstream widths will be stated separately in concrete implementations. This is the reset interface of Section 10, with one additional charged checkpoint. The actual mark is part of a completed observation; it is not an input to the private candidate's online transition rows.

Write $\mathcal{H} = [0, 1]^\Omega$ and

$$f_h(b) = \sum_{w \in \Omega} (Q(w) - b(w))h(w), \quad h \in \mathcal{H}.$$

Every randomized event has a response $h(w) = \Pr(w \in A)$. Conversely, every $h \in \mathcal{H}$ is the response of a random event: include each atom independently with its specified probability. This is a finite probability distribution on 2^Ω . Only its mean response matters for expected signed score. The event itself must still be retained during validation.

Definition 4.1 (Decision continuation spectrum). The K -state decision continuation value is

$$\mathfrak{V}_K(B, Q) = \max_{h_1, \dots, h_K \in \mathcal{H}} \min_{b \in B} \max_{m \leq K} f_{h_m}(b). \quad (4.1)$$

Let $v_{n,K}(B, Q)$ be the supremum of the worst-case expected score of n -sample auditors at the preceding interface. Only the designated cut has width K in this definition; all other widths are finite but unrestricted. For $c \in \mathbb{R}$, let $\kappa_c(B, Q)$ be the least K for which $\mathfrak{V}_K(B, Q) > c$, with value $+\infty$ when no such K exists.

The definition uses only the marked behavior image and its target, not a parameterization of B or a labeling of candidate states. No convex hull of B is taken. A response list is a list of conditional validation continuations; it is not a mixture of feasible private candidates.

Theorem 4.2 (Operational classification at a decision cut). *All extrema in (4.1) are attained. For every $n \geq 1$,*

$$0 \leq \mathfrak{V}_K(B, Q) - v_{n,K}(B, Q) \leq \min \left\{ \sqrt{\frac{d-1}{n}}, \sqrt{\frac{2 \log(2K)}{n}} \right\}. \quad (4.2)$$

Consequently $v_{n,K} \rightarrow \mathfrak{V}_K$. Every auditor with K states at the designated cut, at any finite training length, has worst-case score at most $\mathfrak{V}_K$. If $\mathfrak{V}_K > c$, some finite-sample auditor with K states at that cut has worst-case score greater than c . Thus κ_c is exactly the minimum cut width permitting such a strict certificate with sufficiently many training samples.

Equivalently, κ_c is the least number of response half-spaces

$$\{b \in B : f_h(b) > c\}, \quad h \in \mathcal{H}, \quad (4.3)$$

that cover B . If two parameterized behavior families satisfy, uniformly in their common parameter q ,

$$\|b(q) - \tilde{b}(q)\|_{\text{TV}} \leq \alpha, \quad \|Q - \tilde{Q}\|_{\text{TV}} \leq \beta,$$

then

$$|\mathfrak{V}_K(B, Q) - \mathfrak{V}_K(\tilde{B}, \tilde{Q})| \leq \alpha + \beta. \quad (4.4)$$

Proof. For a fixed response list, the inner objective is continuous in b . Its variation when the list changes is bounded uniformly over b by $2 \max_m \|h_m - \tilde{h}_m\|_\infty$. Its minimum over compact B is therefore continuous in the list. Compactness of $\mathcal{H}^K$ proves attainment.

Condition on the retained state of any admissible auditor. Its conditional post-cut event law depends only on that state and fresh randomness, and has a response h_m . Independence of validation from training gives

$$g(b) = \sum_{m=1}^K \Pr(M = m) f_{h_m}(b) \leq \max_m f_{h_m}(b).$$

Taking the minimum over b , and then the supremum over auditors, proves the upper bound. This argument permits arbitrary stochastic training and arbitrary finite upstream memory. An extra training-dependent decoder would invalidate its first conditioning step and is excluded by the interface, not silently supplied.

For the reverse bound fix an optimal list. Let $\hat{b}$ be the empirical measure and choose the first index maximizing $f_{h_m}(\hat{b})$. Retain that index and discard the empirical table. If m_* maximizes the true score, then pointwise

$$\max_m f_{h_m}(b) - f_{h_{m_*}}(b) \leq 2 \max_{m \leq K} |(\hat{b} - b)h_m|. \quad (4.5)$$

The maximum is at most $\|\hat{b} - b\|_{\text{TV}}$, since $0 \leq h_m \leq 1$. Cauchy–Schwarz and the multinomial variances give

$$\mathbb{E} \|\hat{b} - b\|_{\text{TV}}^2 \leq \frac{d}{4n} \left(1 - \sum_w b(w)^2 \right) \leq \frac{d-1}{4n}.$$

This proves the first rate.

For completeness, if V takes values in an interval of length one, then $\log \mathbb{E} e^{s(V - \mathbb{E}V)} \leq s^2/8$. Indeed the second derivative of the logarithmic moment generating function is the variance under the exponentially tilted law, at most $1/4$; its value and first derivative vanish at zero.

Integrating twice proves the inequality for both signs of s . Independence consequently gives $\mathbb{E}e^{s(\hat{b}-b)h_m} \leq e^{s^2/(8n)}$. For $s > 0$,

$$\mathbb{E} \max_m |(\hat{b} - b)h_m| \leq \frac{\log(2K)}{s} + \frac{s}{8n}.$$

Minimizing in s gives $\sqrt{\log(2K)/(2n)}$ and proves the second rate in (4.2). The empirical table has finitely many values; the construction is therefore a finite-state auditor, with only the chosen index surviving the specified cut. Real response coefficients are allowed here; the physical construction below has deterministic rational transitions.

A finite cover in (4.3) says that the continuous function $b \mapsto \max_m f_{h_m}(b)$ is greater than c everywhere on compact B . Its attained minimum is then greater than c . This proves the covering characterization in both directions. It also proves the strict-threshold assertion; no claim about finite-sample attainment at the boundary $c = \mathfrak{V}_K$ is needed.

Finally $|(Q - b)h - (\tilde{Q} - \tilde{b})h| \leq \alpha + \beta$ for every response. Taking maxima, minima over the common parameter set, and maxima over lists preserves this uniform bound, proving (4.4). $\square$

Corollary 4.3 (Compression by a finite decision dictionary). *Suppose responses $h_1, \dots, h_K$ satisfy $\min_{b \in B} \max_m f_{h_m}(b) \geq c + \eta$ for $\eta > 0$. An empirical selector with $n > 2 \log(2K)/\eta^2$ achieves score greater than c and retains only K states at the decision cut. If these responses depend on a finite statistic $T : \Omega \rightarrow \Xi$, the training selector can count $T(X)$ instead of X . A sufficient upstream register has at most $\binom{n+|\Xi|-1}{|\Xi|-1}$ states at the last training checkpoint, and the corresponding time-dependent count registers suffice at earlier checkpoints.*

Proof. Use the second bound in (4.2). Empirical expectations of h_m depend only on the histogram of $T(X)$. At time t , that histogram has $\binom{t+|\Xi|-1}{|\Xi|-1}$ possibilities. Its update and the final maximization are deterministic. No statistical sufficiency for estimating all of b is required; sufficiency for the specified responses is enough. $\square$

This corollary supplies an intermediate principle between a full marked histogram and a problem-specific compressed counter. The invariant is the number and geometry of validation continuations needed to cover the original candidate image at the desired score. It is not the algebraic dimension of a chosen parameterization. The empirical upper bound need not be sample-optimal, and a minimum decision-cut width is not a minimum of the largest width over the entire acquisition schedule.

4.2. Several feedback rows. The classification does not require that the declared experiment have only one feedback row. For the row-indexed interface of Section 10, let $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$ and define

$$f_\lambda(b) = \sum_{i \in \mathcal{I}} \lambda_i f_i(b), \quad \lambda \in \Delta(\mathcal{I}).$$

A post-cut continuation is now a probability distribution on row–event pairs; it is not an arbitrary test on a joint product of validation rows. Put

$$\mathfrak{V}_K^{\text{rows}}(B, Q) = \max_{\lambda_1, \dots, \lambda_K \in \Delta(\mathcal{I})} \min_{b \in B} \max_m f_{\lambda_m}(b).$$

Corollary 4.4 (Row-indexed classification). *With n training samples per row, the row-indexed decision-cut value converges to $\mathfrak{V}_K^{\text{rows}}$, and its deficit is at most*

$$\min\{\sqrt{D/n}, \sqrt{2 \log(2K)/n}\}, \quad D = \sum_j (|\Omega_j| - 1).$$

The strict half-space cover characterization holds with f_λ . The upper bound by $\mathfrak{V}_K^{\text{rows}}$ applies also to any other finite training schedule, including adaptive row selection.

Proof. Conditioning on the cut state gives the same convex-mixture upper bound as before; its proof places no restriction on how that state was reached. For empirical selection among an optimal list, $|(\hat{b} - b)\lambda_m| \leq \max_j \|\hat{b}_j - b_j\|_{TV}$. Sum the rowwise second moments to obtain the first rate. To obtain the second, align the r th independent training sample from each row into a tuple and write its response as $\sum_{j,A} \lambda_{m,j,A} \mathbf{1}_A(X_{j,r})$. This random variable belongs to $[0, 1]$, and its sample mean is the empirical candidate mean for that continuation. The logarithmic moment-generating argument in Theorem 4.2 therefore gives the same finite-list rate. Each chosen row–event distribution can be drawn using fresh randomness after the cut. Compactness and the covering argument are unchanged. $\square$

5. FINITE PRECISION AND CHARGED RANDOMIZATION

We distinguish the number of persistent labels from the description of a transition and from the clock that selects it. An observation in a finite alphabet is initially an atomic input. Every random bit used after that input is a separate, charged epoch. Section 6 gives the additional cost of splitting the observation itself.

Lemma 5.1 (Dyadic rows and finite randomizers). *A probability vector on s points has a dyadic approximation with denominator 2^r at total variation distance at most $(s-1)2^{-r}$. It can be sampled using exactly r independent fair bits and at most $2s-1$ persistent labels at each intermediate bit epoch. The label count uses the bit position supplied by the schedule. No partial random integer is retained outside these labels.*

Proof. Round down the first $s-1$ coordinates and assign the remaining mass to the last coordinate. The positive and negative parts of the change have equal mass, at most $(s-1)2^{-r}$. For sampling, partition $\{0, \dots, 2^r - 1\}$ into consecutive intervals of the prescribed integer lengths. A binary prefix describes a dyadic interval. If that interval is contained in one output interval, keep the output label and ignore the remaining bits. Otherwise keep the prefix interval. At a fixed depth, every unresolved interval contains a partition boundary in its interior; there are at most $s-1$ such intervals. Together with at most s resolved outputs this gives $2s-1$ labels. The children and their resolved labels are fixed by the read-only partition and the depth. Thus the construction is an actual finite transition table, not a real-valued random seed. Zero-length intervals cause no difficulty. $\square$

Proposition 5.2 (Rational decision continuations). *Let $d = |\Omega|$. A list of K responses in $[0, 1]^\Omega$ can be replaced by dyadic responses of denominator 2^r , losing at most 2^{1-r} in its uniform score envelope. After a decision cut retaining K labels, the randomized event for the chosen dyadic response can be drawn using r fair bits, at most $K(2d+1)$ labels during this draw, and at most $K(d+1)$ labels afterwards. Keeping the event through one candidate and one target validation requires at most $2K(d+1)$ labels at the intervening atomic-observation cuts. Its final signed score has three values.*

Proof. Round every coordinate down. The sup-norm error is at most 2^{-r} , and $\sum_w |Q(w) - b(w)| \leq 2$. This proves the envelope estimate. For a fixed rounded response h , draw a uniform dyadic integer U and choose $A_U = \{w : U < 2^r h(w)\}$. These events are nested and have at most $d+1$ distinct values. Their mean response is exactly h . Apply Lemma 5.1 to their interval lengths, retaining the chosen response index during sampling. Afterwards the event label and the candidate-event indicator suffice; the target indicator emits their difference. All event masks are read-only data fixed before observing the candidate. $\square$

For K responses, their rounded coordinates require $Kd(r+1)$ bits in a direct read-only representation. These bits are not charged as persistent sample-dependent labels, but are not declared costless precision. A construction using them records both quantities. In particular, the

proposition turns every strict response-cover margin into a finite-randomness implementation; it says nothing about exact attainment of an irrational boundary value.

6. APPROXIMATE CONTINUATION PROFILES AND RESOURCE TRANSFER

A cut profile records the whole causal schedule, rather than its final compression. We give both a realization certificate and an operational value. These notions keep the compatibility condition of Theorem 3.2; separate nonnegative factorizations at individual cuts are not a substitute for that condition.

6.1. A causal comparison norm. For two normalized finite causal suffix kernels p, q at cut t , put

$$d_t^{\text{ca}}(p, q) = \sup_{\pi} \|P_p^{\pi} - P_q^{\pi}\|_{\text{TV}},$$

where π ranges over deterministic causal input policies on the remaining schedule and P_p^{π} is the law of its complete output transcript. Randomized policies give the same supremum, by conditioning on their independent random tables. In particular this distance includes feedback; the supremum over open-loop input words alone is not used.

For stochastic coefficients $q_t(z, j \mid m, x)$ and suffix kernels $v_{t+1, j}$, denote by $\mathcal{Q}_t v_{t+1, m}$ the suffix kernel which, upon input x , emits z and chooses j with these coefficients, then uses $v_{t+1, j}$. The notation denotes one first-step operation, not a conditioning on an event of zero probability.

Theorem 6.1 (Approximate compatible realization). *Let $K_0, \dots, K_L$ be positive integers. Suppose normalized causal kernels $v_{t, m}$, $m \leq K_t$, and stochastic rows q_t satisfy*

$$d_t^{\text{ca}}(v_{t, m}, \mathcal{Q}_t v_{t+1, m}) \leq \eta_t \quad (t < L, m \leq K_t). \quad (6.1)$$

At the terminal cut the suffix kernels are the unit empty kernel. If $\rho \in \Delta_{K_0}$ and $d_0^{\text{ca}}(p, \sum_m \rho_m v_{0, m}) \leq \eta_{-1}$, then the machine with initial law ρ and transition rows q_t has the whole profile (K_t) and realizes a kernel $\hat{p}$ satisfying

$$d_0^{\text{ca}}(p, \hat{p}) \leq \min\{1, \eta_{-1} + \sum_{t=0}^{L-1} \eta_t\}. \quad (6.2)$$

Conversely, a machine with this profile within distance ε of p supplies such a certificate with $\eta_{-1} = \varepsilon$ and all local errors zero. If a deterministic initial state is required, use $K_0 = 1$ throughout both statements.

Proof. Let $\hat{v}_{t, m}$ be the actual continuation of the rows q from state m . Fix a future policy. Conditional on the first output and next state, its remaining actions form an admissible suffix policy. Replacing $v_{t+1, j}$ by $\hat{v}_{t+1, j}$ therefore changes its transcript law by at most $\max_j d_{t+1}^{\text{ca}}(v_{t+1, j}, \hat{v}_{t+1, j})$. Mixtures and pushforwards contract total variation. The triangle inequality and (6.1) imply, backwards from L ,

$$\max_m d_t^{\text{ca}}(v_{t, m}, \hat{v}_{t, m}) \leq \sum_{s=t}^{L-1} \eta_s.$$

Mixing with ρ proves (6.2). The converse takes the actual normalized continuations of the given machine, including arbitrary valid continuations for unreachable states. Their one-step equations are identities. There is no assertion that independently optimal approximate lists at successive cuts have compatible updates. $\square$

6.2. The full value and its effective approximation. Fix a finite schedule with finitely many observation and action symbols, N calls to an experiment, and a bounded terminal payoff $G \in [-1, 1]$. An environment $b \in B$ specifies the conditional observation kernels at all possible visible histories and actions. Assume B is a nonempty compact subset of the resulting finite product of probability simplexes. A controller A has a specified persistent profile $\mathbf{K}$; the clock has no access to sample history. All action choices and all retained randomness factor through the current register. Define

$$V_{N,\mathbf{K}}(B, G) = \max_{A \in \mathcal{M}(\mathbf{K})} \min_{b \in B} \mathbb{E}_b^A G. \quad (6.3)$$

Unused states pad smaller machines, so $\mathcal{M}(\mathbf{K})$ is a finite product of stochastic-row simplexes, with any prescribed causal zeros imposed. The terminal readout is included among its rows.

Theorem 6.2 (Full-profile resource transfer). *The maximum in (6.3) is attained. Its value is nondecreasing in each K_t . Removing a charged intermediate cut gives a relaxation, whereas inserting identity-holding cuts preserves every implementation with the same held register.*

Suppose two environments at each of their N calls differ in conditional total variation by at most δ_i , uniformly over corresponding histories, actions and parameters. Replace the controller rows at its L transitions by dyadic rows of denominator 2^r , with at most s_t possible next-state/output pairs at transition t . Any fixed controller loses at most

$$2 \min \left\{ 1, \sum_{i=1}^N \delta_i + \sum_{t=1}^L (s_t - 1) 2^{-r} \right\} \quad (6.4)$$

in expected payoff. The same bound compares optimal values on the same profile, with the appropriate dyadic approximation on one side. More generally an approximate continuation certificate contributes its error in (6.2) inside the braces.

If $B_0 \subset B$ is a finite ω -net in the maximum conditional-TV metric and v_0 is the maximum over the dyadic controller grid of the minimum over B_0 , then

$$v_0 - 2N\omega \leq V_{N,\mathbf{K}}(B, G) \leq v_0 + 2 \sum_{t=1}^L (s_t - 1) 2^{-r}. \quad (6.5)$$

The bounds are certificates whenever the net and the finite evaluations are certified. Compactness alone supplies existence of nets, not an algorithm for an arbitrary presentation of B .

Proof. For fixed b , each path probability is a finite product of controller and environment coordinates. Summing the finitely many paths gives a jointly continuous payoff. Its minimum over compact B is continuous in the controller rows; the controller space is compact. This proves attainment. Padding proves monotonicity. Removing a constraint enlarges the feasible set, and identity transitions prove the insertion assertion.

For the perturbation bound, couple two executions until their first mismatch. On equal histories the conditional observation laws can be maximally coupled, with mismatch probability at most δ_i . Couple the controller transitions similarly, using Lemma 5.1. The probability of any mismatch is at most the sum in (6.4), capped at one. The payoff changes by at most two on that event. This coupling is recursive on finite histories and remains valid for feedback controls. Alternatively, approximate continuation kernels are compared in the causal distance and then composed, as in Theorem 6.1.

For a fixed grid controller, a minimizing point of B has a point of B_0 within ω ; the corresponding payoffs differ by at most $2N\omega$. Thus the grid optimum on all of B is at least $v_0 - 2N\omega$. Conversely approximate a maximizing unrestricted controller row by row. Its minimum over B decreases by at most the row-rounding bound, and its minimum over B_0 is no smaller. Taking the grid maximum proves the upper bound. No minimax interchange or convexification of the fixed-width behavior image occurs. $\square$

The formula records preparation count, persistent labels, calibration error, causal simulation error and transition precision in one execution. The training-count factor is present when training kernels change. It is absent only when those kernels are identical. Failure probabilities are also transported, with factor one rather than two, since their indicators have range $[0, 1]$.

Proposition 6.3 (Composition and schedule refinement). *A causal simulator of profile $\mathbf{R}$ followed on the same synchronized schedule by a controller of profile $\mathbf{K}$ has a realization with profile $(R_t K_t)$. Causal simulation errors add under successive composition. Two disjoint phases with a genuine reset and no retained bridge use the maximum of their peak widths. If a bridge with H labels must survive the second phase, its direct width bound is H times the second-phase width.*

If one atomic input x is encoded by ℓ symbols, let D_j be the number of possible length- j prefixes. Holding the pre-input state and the prefix realizes the refined acquisition cuts with widths $K_{\text{before}} D_j$, followed by the original update. A time-dependent machine running for T scheduled epochs becomes autonomous by retaining its clock, with at most $(T + 1) \max_t K_t$ states. These are implementation bounds, not claims of optimality for every refinement.

Proof. Keep the ordered pair of simulator and controller registers and compose their transition kernels. The simulator's emitted observation is the controller's current input. Conditional coupling until the first simulation discrepancy gives the error statement. A genuine reset permits reusing the alphabet, whereas a surviving bridge is another register. A partial encoded input is determined by its prefix, giving the stated refinement construction. Finally the augmented state (t, m) implements the time-indexed transition without an external clock. These constructions also show why neither hidden selectors nor an external transcript may be omitted from the register being counted. $\square$

7. FROM A DECISION COVER TO AN END-TO-END MACHINE

The unrestricted empirical table in Theorem 4.2 can be replaced by a bounded-state training procedure. This replacement trades preparation count against memory, rather than assuming that the training computation has already been performed. The elementary block mechanism belongs to the classical finite-statistics theory of Cover [24]; the result below tracks a finite horizon, a uniform indifference band, rational precision, the whole validation phase, and a compact nonconvex candidate image.

Lemma 7.1 (A finite-horizon four-state comparison). *Let Z_i be independent Bernoulli observations of unknown mean u . For $0 < \theta < 1/2$ and $0 < \delta < 1$, put*

$$L = \left\lceil \frac{\log(2/\delta)}{4\theta} \right\rceil, \quad R = \left\lceil 2^L \log(2/\delta) \right\rceil. \quad (7.1)$$

There is a deterministic clocked machine with four states at every observation cut, using exactly $2LR$ observations, that decides the sign of $u - 1/2$ with error at most δ whenever $|u - 1/2| \geq \theta$. No assertion is imposed within the indifference band.

Proof. Partition the observations into R pairs of blocks, each block of length L . A positive block succeeds when all its observations are one; the following negative block succeeds when all are zero. Stop updating the decision at the first successful block and retain its sign. If none succeeds, decide negative at the end. The prescribed remaining observations are still consumed, but ignored.

Two states are absorbing positive and negative decisions. The other two mean undecided with the current block still alive or already failed. At a block boundary the clock resets the undecided flag to alive. Its polarity and position are supplied by the clock; they are not inferred from an unrecorded sample. The last observation of each block either latches the sign or starts the next block. Thus four states suffice even at the block boundaries.

Write $x = u^L$, $y = (1 - u)^L$ and $s = x + (1 - x)y$. The probability of a positive decision is exactly

$$H_{L,R}(u) = \frac{x}{s} \{1 - (1 - s)^R\}. \quad (7.2)$$

Here $s > 0$, including at $u = 0, 1$. For $u \geq 1/2 + \theta$, the error is at most $y/x + (1 - s)^R$. For $u \leq 1/2 - \theta$, it is at most x/y ; the default negative decision is then correct. Since

$$\log \frac{1/2 + \theta}{1/2 - \theta} \geq 4\theta, \quad s \geq \max(x, y) \geq 2^{-L},$$

both errors are bounded by $e^{-4\theta L} + e^{-R2^{-L}} \leq \delta$. The logarithmic inequality follows by differentiation from its equality at zero. This also proves uniformity over the two composite hypotheses. $\square$

7.1. The whole-schedule continuation value. In the one-row reset interface of Section 4, let $W_c(B, Q)$ be the least peak width over all finite training lengths and all auditors whose worst-case expected signed score is strictly greater than c . Every training-observation cut, the decision cut, every fresh random-bit cut, and both validation cuts are charged. Observations in Ω are atomic for this definition. The event is selected before the two validations, exactly as in that interface. The schedule clock and read-only constants are recorded as separate resources. The class includes randomized machines; its rational subclass uses only finite fair-bit randomizers. Write W_c^{rat} for the corresponding minimum in the latter class. Empty feasible sets have value $+\infty$.

Theorem 7.2 (Uniform end-to-end realization of a decision cover). *Let $B \subset \Delta(\Omega)$ be nonempty and compact, Q fixed, and $d = |\Omega|$. For every strict threshold c ,*

$$\kappa_c(B, Q) \leq W_c(B, Q) \leq W_c^{\text{rat}}(B, Q) \leq 12d \kappa_c(B, Q). \quad (7.3)$$

The last inequality is interpreted only for finite κ_c ; when $\kappa_c = +\infty$ all three quantities are infinite.

More explicitly, fix $K \geq 2$ and a response list with uniform envelope at least v . Round its coordinates to denominator 2^r , and choose $q_i \in [0, 1] \cap 2^{-r}\mathbb{Z}$ with $|q_i - Qh_i^{(r)}| \leq \rho$. For θ, δ as in Lemma 7.1, a rational auditor uses

$$N = 2LR(K - 1) \quad (7.4)$$

candidate training observations and two validation observations. Its expected score is at least

$$v - 2^{1-r} - 2\rho - 4(K - 1)\theta - 2(K - 1)\delta. \quad (7.5)$$

Its completed-training-observation cuts have width at most $4K$, its decision cut has width K , and its entire random-bit-refined schedule has width at most $12Kd$. No empirical histogram is constructed. The response and comparison thresholds have a direct read-only representation of size $O(K^2d(r + 3) + \log L + \log R + \log K)$ bits.

Proof. Use a sequential tournament. Initially response 1 is incumbent. At stage $j = 2, \dots, K$, compare incumbent i with challenger j using fresh candidate observations. For one observation $w \sim b$, generate a Bernoulli variable with conditional success probability

$$\frac{1}{2} + \frac{q_j - q_i - h_j^{(r)}(w) + h_i^{(r)}(w)}{4}. \quad (7.6)$$

This lies in $[0, 1]$ and is dyadic of denominator 2^{r+2} . Its unconditional mean is $1/2 + (\tilde{f}_j(b) - \tilde{f}_i(b))/4$, where $\tilde{f}_i(b) = q_i - bh_i^{(r)}$. Conditional on the incumbent at the start of this stage, all these variables are independent and have this mean. The fixed candidate b is not changed by the comparison. Run Lemma 7.1 and replace i by j exactly when its decision is positive.

At each comparison, conditionally on all earlier stages, the probability of choosing the inferior response when their $\tilde{f}$ values differ by at least 4θ is at most δ . With probability at least $1 - (K - 1)\delta$,

no such error occurs. On this event each stage loses at most 4θ relative to the better of its two responses. Induction on j gives

$$\tilde{f}_i(b) \geq \max_i \tilde{f}_i(b) - 4(K-1)\theta.$$

Replacing the calibrated means by their true means loses at most 2ρ . Rounding the original list loses at most 2^{1-r} in its envelope, by Proposition 5.2. Any response payoff belongs to $[-1, 1]$, so failure of the good event loses at most $2(K-1)\delta$ in expectation. This proves (7.5), uniformly over b .

During a comparison keep the incumbent index and one of the four block states. The challenger and block positions are fixed by the schedule. At the end of a comparison the sign updates the incumbent and is forgotten. This gives the $4K$ bound at completed-observation cuts. While drawing the coin in (7.6), keep this register and the current observation, and use the three-label binary sampler of Lemma 5.1. Thus at most $12Kd$ labels occur at any intervening random-bit cut. At the final decision cut only the incumbent remains. Proposition 5.2 gives at most $K(2d+1)$ labels during event selection and $2K(d+1)$ during validation; both are bounded by $12Kd$. The three-valued terminal output fits this bound as well. These are compatible transitions over the entire schedule.

The coin thresholds can be precomputed for all triples (i, j, w) and stored as $r+3$ -bit integers. The event thresholds and L, R, K give the stated description bound; no arbitrary real coefficient is used by the executed machine. Formula (7.4) counts every training observation, including ignored observations after a block decision has been latched.

For (7.3), a width- W auditor has at most W labels at its decision cut. The operational upper bound of Theorem 4.2 implies $\kappa_c \leq W$. If $K = \kappa_c < \infty$, its optimal list has a positive margin over c . Take r large, $\rho \leq 2^{-r}$, and then θ, δ small enough that the loss in (7.5) is smaller than that margin. If $K = 1$, no training is needed: round the single response and use Proposition 5.2. Rational auditors are a subclass of all auditors, proving the remaining inequality. Finally a finite-width successful machine would give finite κ_c , proving the assertion at infinity. $\square$

The dependence on the schedule is essential. The theorem is a constant-factor classification for fixed observation alphabet, not an optimal sample-complexity theorem. In particular R is exponential in L . For an autonomous implementation, a coarse bound on the number of epochs is $T \leq N(r+3) + r + 4$; Proposition 6.3 then gives at most $12Kd(T+1)$ states. The free-clock and autonomous statements are different resource bounds. Cover's theorem already exhibits the importance of this distinction [24, Theorem 2 and Section 4]; bounded-state learning itself is not a novelty claim of the present theorem.

Corollary 7.3 (Finite-precision, calibrated certificates). *Suppose the true training kernels differ from those used in an ideal auditor by at most α per training call, and the candidate and target validation laws differ by at most β_c and β_t . The right side of (7.5) is decreased by at most*

$$2 \min\{1, N\alpha\} + \beta_c + \beta_t$$

for the same implemented machine. A calibration error ρ in the known response means is already accounted for separately in that formula. When the training source is unchanged, $\alpha = 0$, regardless of how large N is.

Proof. Couple the training executions until their first mismatch, as in Theorem 6.2. The selected-event payoff has range $[-1, 1]$, giving the factor two. Conditional on a common selected event, its two validation means change by at most the two total variation errors. This uses no independence between a failed coupling and its terminal event. $\square$

8. WEIGHTED CONTINUATION OBSTRUCTIONS

We next give a distribution-sensitive version of the wordwise obstruction. It applies to finite-valued terminal decisions and is useful even when a word of small probability should not be assigned the same weight as a likely one.

Let $f : \Sigma_1 \times \cdots \times \Sigma_L \rightarrow \mathcal{A}$ have finite range. The residual of a prefix x at cut t is the function $v \mapsto f(xv)$ on the suffix alphabet. Write $r_t(f)$ for the number of distinct residuals. A stochastic machine has a charged register at every cut; all persistent randomness is included in this register. Its wordwise error is $e(w) = \Pr(\hat{f}(w) \neq f(w))$.

Theorem 8.1 (Finite-valued realization and weighted cut bound). *Exact stochastic computation of f requires $K_t \geq r_t(f)$ at every cut. The whole residual profile is simultaneously attained by a deterministic machine.*

Fix a cut, prefixes $x_1, \dots, x_r$, and a graph on $[r]$. For each edge ij choose a suffix v_{ij} such that $f(x_i v_{ij}) \neq f(x_j v_{ij})$. Let $\mu(w) \geq 0$ be arbitrary word weights. Choose edge weights $\lambda_{ij} \geq 0$ satisfying the word capacity constraints

$$\sum_{ij: w=x_i v_{ij} \text{ or } w=x_j v_{ij}} \lambda_{ij} \leq \mu(w) \quad \text{for every word } w. \quad (8.1)$$

Every machine with K states at this cut satisfies

$$\sum_w \mu(w) e(w) \geq \min_{\phi: [r] \rightarrow [K]} \sum_{ij: \phi(i)=\phi(j)} \lambda_{ij}. \quad (8.2)$$

The right side is a weighted monochromatic-edge obstruction. Maximizing it over admissible witnesses and weights gives a finite, distribution-sensitive lower bound. The bound can be an equality with positive error.

Proof. Let $a_i(m)$ be the state distribution after prefix x_i . For a fixed edge and a common state m , the future output distribution on suffix v_{ij} is the same for both prefixes. Since their correct outputs are different, the sum of their two conditional error probabilities is at least one. Multiplying by the smaller state mass and summing gives

$$e(x_i v_{ij}) + e(x_j v_{ij}) \geq \sum_{m=1}^K \min\{a_i(m), a_j(m)\}. \quad (8.3)$$

Weight the inequalities by λ_{ij} and use (8.1). The resulting lower bound is

$$F(a_1, \dots, a_r) = \sum_{ij} \lambda_{ij} \sum_m \min\{a_i(m), a_j(m)\}.$$

This is a concave continuous function on $(\Delta_K)^r$. Every point of that product is a convex combination of its deterministic vertices; concavity implies that its value is at least the minimum over those vertices. A vertex is a labeling $\phi : [r] \rightarrow [K]$, and its value is exactly the right side of (8.2). This proves the bound without treating stochastic encodings as deterministic encodings.

For zero error, choose one distinguishing suffix for every two different residuals. Equation (8.3) forces the corresponding state supports to be disjoint. Hence at least $r_t(f)$ states are needed. For the converse, keep the residual class of the observed prefix. Equal residuals have equal shifts after every next input, so the next class is well defined. At the last epoch the residual is the output label itself. These quotient transitions realize all widths simultaneously.

For a positive-error equality, let the only relevant words be two one-letter prefixes with different required labels, weighted p and $1-p$, where $0 < p < 1$. Put $K = 1$ and $\lambda_{12} = \min(p, 1-p)$. The cut obstruction is $\min(p, 1-p)$, attained by always emitting the more likely label. Thus the weighted constant is sometimes exact; universal sharpness for all graphs or input distributions is not asserted. With unit weights on the same two words the total-error lower bound is one,

also attained by a constant output; this is equality in the earlier $(r - K)/(r - 1)$ bound at $r = 2$, $K = 1$. $\square$

Corollary 8.2 (A Bayes-regret obstruction for constrained audits). *Fix a finite action set $\mathcal{A}$, candidate class B , and a probability measure ν on B . For a training word w , let $p_w(b)$ be its likelihood and let $f_a(b)$ be the expected validation payoff. Put*

$$R_w(a) = \int_B p_w(b) f_a(b) d\nu(b), \quad S_\nu = \sum_w \max_a R_w(a).$$

Choose a maximizing label $f(w)$. Set $\mu(w) = R_w(f(w)) - \max_{a \neq f(w)} R_w(a)$, taking zero when a maximum is tied. Let $\Gamma_{t,K}(f, \mu)$ be any lower bound supplied by (8.2). Then every width- K audit at cut t has worst-case expected payoff at most

$$S_\nu - \Gamma_{t,K}(f, \mu). \quad (8.4)$$

Proof. The conditional distribution of the auditor's output for a fixed word is independent of the unknown candidate. Its Bayes payoff is therefore $\sum_{w,a} R_w(a) \Pr(\hat{f}(w) = a)$. Subtracting this from S_ν gives at least $\sum_w \mu(w) e(w)$. Apply the theorem, and use that a minimum over candidates is at most the average under ν . $\square$

The corollary does not exchange a maximum and minimum over a nonconvex fixed-width machine class. The prior is a lower-bound device, not a randomization resource supplied to the candidate. With a unique maximizing label and strictly positive margins at every word, equality with S_ν forces exact wordwise computation. This makes explicit the uniqueness step used in the odd-sample binary theorem below. For even sample size, the balanced-count words have zero likelihood gap. A rule can differ on those ties without losing value; equality of values then does not force one prescribed total Boolean rule, and the odd-sample width profile cannot simply be asserted for that case.

9. JOINT RESIDUAL BUDGETS AND GROWING FAMILIES

We now put lower bounds on several cuts into one statistical budget. Summing separate cut bounds without controlling their common words would count the same regret more than once.

Theorem 9.1 (A shared-capacity multicut obstruction). *Let $f : \prod_{i=1}^L \Sigma_i \rightarrow \mathcal{A}$ and $\mu(w) \geq 0$ be as in Theorem 8.1. At each chosen cut t , choose a finite prefix graph with an edge $e = (u, v)$ carrying a distinguishing suffix s_e , and a weight $\lambda_{t,e} \geq 0$. Impose the single family of constraints*

$$\sum_{t,e: w=us_e \text{ or } w=vs_e} \lambda_{t,e} \leq \mu(w) \quad \text{for every complete word } w. \quad (9.1)$$

For a graph at cut t , write

$$\Gamma_{t,K_t}(\lambda) = \min_{\phi: \text{prefixes} \rightarrow [K_t]} \sum_{e=(u,v): \phi(u)=\phi(v)} \lambda_{t,e}.$$

Every stochastic machine of profile $\mathbf{K}$ satisfies

$$\sum_w \mu(w) \Pr(\hat{f}(w) \neq f(w)) \geq \sum_t \Gamma_{t,K_t}(\lambda). \quad (9.2)$$

Consequently, with R_w, S_ν, μ defined in Corollary 8.2, every audit of this profile and the fixed training schedule has worst-case payoff at most

$$S_\nu - \sup_{\lambda \text{ satisfying (9.1)}} \sum_t \Gamma_{t,K_t}(\lambda). \quad (9.3)$$

For fixed graphs the supremum is the value of a finite linear program.

Proof. For each edge use the overlap inequality (8.3). Multiply it by $\lambda_{t,e}$ and sum over all cuts and edges. The left side is bounded above by the left side of (9.2), by the shared capacity constraint. At each cut the resulting overlap expression is concave in its prefix state distributions, hence at least its minimum over deterministic labelings, as proved in Theorem 8.1. Adding these bounds proves (9.2). Although prefix distributions at different cuts are compatible, minimizing them separately only weakens the bound; it does not assume independent realizability. The Bayes-regret argument of Corollary 8.2 proves (9.3).

For the last assertion introduce variables z_t and impose, for each labeling ϕ of each finite graph,

$$z_t \leq \sum_{e=(u,v): \phi(u)=\phi(v)} \lambda_{t,e}.$$

Maximize $\sum_t z_t$ subject to these inequalities, nonnegative edge weights, and (9.1). There are finitely many constraints. This is a computable obstruction, not a claim that every multicut statistical optimum equals this relaxation. $\square$

9.1. Actual physical word weights. For n all-on trials of the private marked collision subfamily, a complete training word w specifies the two report bits and the mark of each trial. With N_{ab} its report-pair counts,

$$p_w(p, q) = 2^{-n} p^{N_{10}+N_{11}} (1-p)^{N_{00}+N_{01}} q^{N_{01}+N_{11}} (1-q)^{N_{00}+N_{10}}. \quad (9.4)$$

The formula has the continuous endpoint interpretation $0^0 = 1$. For any finitely supported prior on actual pairs $(p, q) \in [0, 1]^2$, use this likelihood in $R_w(A) = \int p_w(p, q)(Q - b(p, q))\mathbf{1}_A d\nu$, with all marked events $A \subseteq \Omega$ as actions. Randomized events are included by linearity. Equations (9.1) and (9.3) then give a finite all-algorithm obstruction for that full training profile, with the true physical word probabilities. Changing only Q by total variation β changes every audit's payoff by at most β .

This specialization bounds all stochastic selectors on the fixed all-on training schedule, not merely exact implementations of a majority rule. It does not assert a new numerical lower bound above three for every adaptive physical audit. Adaptive training has controller-dependent word likelihoods and must instead be included in the full experiment of (6.3); using (9.4) unchanged for such a likelihood would be incorrect. The feedback-uniform lower bound used below remains Lemma 21.2.

9.2. A family with matching end-to-end growth.

Theorem 9.2 (Point-source decision spectra). *Let $d \geq 2$, $B_d = \{\delta_1, \dots, \delta_d\}$ on $[d]$, and let Q_d be uniform. For every $K \geq 1$,*

$$\mathfrak{V}_K(B_d, Q_d) = 1 - \frac{\lceil d/K \rceil}{d}. \quad (9.5)$$

For $1 - 2/d < c < 1 - 1/d$, the decision complexity is exactly d , and the whole-schedule peak satisfies

$$d \leq W_c(B_d, Q_d) \leq W_c^{\text{rat}}(B_d, Q_d) \leq 2d.$$

The upper bound uses one training observation and two validations, with no randomization. Thus the end-to-end width grows linearly in this nontrivial sequence of compact behavior images.

Proof. Assign each point source to a response that maximizes its score. Some response is assigned a set S of at least $\lceil d/K \rceil$ indices. For $s = |S| < d$, the mean of its scores on S equals

$$\frac{d-s}{d} \left(\frac{1}{d-s} \sum_{i \notin S} h_i - \frac{1}{s} \sum_{i \in S} h_i \right) \leq 1 - \frac{s}{d}.$$

For $s = d$ the mean is zero. At least one assigned source has score no larger than this mean, proving the upper bound. Partition $[d]$ into at most K groups of size at most $\lceil d/K \rceil$ and use their complements as events. This attains the lower bound.

In the stated interval the formula forces $K \geq d$. A single training observation identifies the point source i exactly. Keep i and choose the complement of $\{i\}$. During validation retain i and the candidate indicator, at most $2d$ states, then emit the target indicator minus it. This computes the prescribed signed score on every input word, including words that have zero probability under the point-source family. Its expected score is $1 - 1/d$. The lower bound follows from the necessary decision-cut width. $\square$

9.3. A sharp simultaneous storage profile. A second class separates overlapping retention obligations. Independent uniform labels $X_j \in [q_j]$ are revealed at scheduled epochs a_j and must be reported at epochs $b_j > a_j$. No later input reveals an active label, and previous outputs cannot be read as external memory. Let $A_t = \{j : a_j \leq t < b_j\}$ and $D_t = \prod_{j \in A_t} q_j$.

Theorem 9.3 (Overlapping retention intervals). *Every stochastic machine whose probability of correctly reporting all labels is at least $1 - \varepsilon$ has*

$$K_t \geq (1 - \varepsilon)D_t \quad \text{at every cut } t. \quad (9.6)$$

The exact profile $K_t = D_t$ is attained simultaneously by a deterministic machine. Thus the minimum peak number of bits lies between $\log_2 \max_t D_t + \log_2(1 - \varepsilon)$ and $\lceil \log_2 \max_t D_t \rceil$ for $\varepsilon < 1$.

Proof. Fix a cut and give the decoder, as additional information, all labels outside A_t , whether past or future. They are independent of the uniform active vector. Conditional on them, every future prediction of that vector factors through the K_t -valued retained state and fresh randomness. If $a(m \mid x)$ is its conditional encoder and $d(x \mid m)$ the induced joint prediction law, its success probability is at most

$$\frac{1}{D_t} \sum_m \sum_x a(m \mid x) d(x \mid m) \leq \frac{1}{D_t} \sum_m \sum_x d(x \mid m) = \frac{K_t}{D_t}.$$

Correctness of all reports implies correctness of the active reports, so averaging over the independent additional information proves (9.6). For the upper bound, keep precisely the vector of active labels, append a label when revealed, and delete it when reported. Every update is deterministic and respects the schedule. The bit bounds follow by taking logarithms and rounding for an actual binary register. The lower bound is a counting argument, not an independence assumption about labels after conditioning on a compressed state. $\square$

10. FINITE-DATA TESTING ON THE ORIGINAL BEHAVIOR IMAGE

10.1. The reset interface. Let $B \subset \prod_{j=1}^J \Delta(\Omega_j)$ be compact, and let $Q = (Q_j)$ be fixed. Set $d_j = |\Omega_j|$, $D = \sum_j (d_j - 1)$, $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$, and $f_{j,A}(b) = Q_j(A) - b_j(A)$. Include the empty event. No convexity of B is assumed.

A candidate b is fixed for the whole audit. The auditor obtains n independent samples of each b_j , with independence between rows. In a causal experiment, row j means a specified parameter and deterministic feedback tree; the reset interface supplies that row. It is an additional experimental interface, not information available to the online simulator. A complete sample contains the terminal mark. Private randomness and any per-run hidden selector are reset between runs. The design of the candidate is not resampled.

Write $C_{j,w}$ for the counts, and $\mathcal{N}_n$ for all count arrays with $\sum_w C_{j,w} = n$. Their probability is the polynomial

$$p_C(b) = \prod_j \frac{n!}{\prod_w C_{j,w}!} \prod_w b_j(w)^{C_{j,w}}.$$

A randomized audit is an array $c = (c_{C,i})$ with $c_C \in \Delta(\mathcal{I})$. After training it chooses $i = (j, A)$ according to c_C . Independent candidate and target validation runs give the signed score $\mathbf{1}_A(\text{target}) - \mathbf{1}_A(\text{candidate})$. Its expected value is

$$g_c(b) = \sum_{C,i} p_C(b) c_{C,i} f_i(b), \quad v_n^{\text{aud}}(B, Q) = \max_c \min_{b \in B} g_c(b). \quad (10.1)$$

Thus every audit payoff is polynomial of total degree at most $Jn + 1$ in the behavior coordinates. The probability of a large observed score in a single audit is not asserted to equal its expectation; additional independent validation would be required for a specified confidence level.

Theorem 10.1 (Reset-audit duality and finite-sample approximation). *Let $\delta = \min_{b \in B} \max_i f_i(b)$. Then all the extrema below are attained, and*

$$v_n^{\text{aud}}(B, Q) = \min_{\mu \in \mathcal{P}(B)} \sum_{C \in \mathcal{N}_n} \max_{i \in \mathcal{I}} \int_B p_C(b) f_i(b) d\mu(b), \quad (10.2)$$

$$0 \leq \delta - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}. \quad (10.3)$$

The values are nondecreasing in n . If $a = \dim \text{aff } B$, a minimizing measure in (10.2) can be supported on at most $\binom{a+Jn+1}{a}$ points.

An explicit feasible audit attaining the lower bound $\delta - \sqrt{D/n}$ chooses an event maximizing $Q_j(A) - C_j(A)/n$, with a fixed deterministic tie rule. If the rational target probabilities are given, this is a rationally specified finite rule. With a free schedule clock, a conservative retained-state bound is

$$H(|\mathcal{I}| + 1)(n + 1) \sum_j d_j,$$

where H is the number of states needed to execute and record one causal transcript. This budget is charged to the auditor, not to the simulator.

Proof. The coefficient arrays form a compact convex product of finite simplexes. The integrand is continuous in b and affine in c . The probability measures on compact B form a compact convex set for weak convergence. The minimum over b equals the minimum over such measures for fixed c . Finite-dimensional minimax (or its compact measure version) therefore exchanges the two extrema. Maximization over each independent coefficient simplex selects its largest coordinate integral, giving (10.2). Continuity and compactness give attainment. This step uses measures on the original B , not feasible private realizations of their barycenters.

For any audit, $g_c(b) \leq F(b) = \max_i f_i(b)$. At a minimizer of F this gives $v_n^{\text{aud}} \leq \delta$. The zero event gives $v_n^{\text{aud}} \geq 0$. For the stated explicit rule put $\hat{b}_j = C_j/n$ and $\rho = \max_j \|\hat{b}_j - b_j\|_{\text{TV}}$. Comparing the empirical maximizer with a true maximizing event gives the pointwise regret bound

$$F(b) - f_{i(C)}(b) \leq 2\rho.$$

For a multinomial empirical measure,

$$\mathbb{E} \|\hat{b}_j - b_j\|_{\text{TV}}^2 \leq \frac{d_j}{4} \sum_w \mathbb{E} (\hat{b}_j(w) - b_j(w))^2 = \frac{d_j(1 - \sum_w b_j(w)^2)}{4n} \leq \frac{d_j - 1}{4n}.$$

Consequently $\mathbb{E} \rho \leq (\sum_j \mathbb{E} \|\hat{b}_j - b_j\|_{\text{TV}}^2)^{1/2} \leq \sqrt{D/n}/2$. Averaging the regret uniformly over B proves (10.3).

From $n + 1$ unordered samples in each row, discard one uniformly at random. The remaining counts have exactly the n -sample law, independently across rows. This realizes every n -sample audit as a randomized $(n + 1)$ -sample audit and proves monotonicity.

Choose affine coordinates on $\text{aff } B$. All the functions $p_C f_i$ lie in the span of monomials of total degree at most $Jn + 1$, a space of dimension at most $s = \binom{a+Jn+1}{a}$ including the constant. The moment vector of a probability measure is in the convex hull of the evaluation vectors;

their constant coordinate is one, so their affine dimension is at most $s - 1$. Eliminating affine dependencies gives a representation by at most s points. It preserves every integral in (10.2), and hence preserves the objective.

Finally, each count is in $\{0, \dots, n\}$. The indicated product register stores all counts, the within-run state, and the selected event for validation. Only the latter event and the within-run buffer need survive once training ends, so the product bound is conservative. Randomizing an event uses fresh randomness. The schedule, reset access and terminal mark readout were declared before the theorem. No private simulator state is read by this implementation. $\square$

The dual measure chooses a candidate once, after which its samples are independent conditional on that candidate. Replacing it by $\bar{b} = \int b d\mu$ would replace $\int p_C(b) f_i(b) d\mu$ by $p_C(\bar{b}) f_i(\bar{b})$, generally a different quantity. It would also replace the joint replicated law by a product of its marginals. Neither operation is part of the theorem. This is the operational meaning of the moment side, rather than a same-budget convexification theorem.

Proposition 10.2 (Both sides sampled). *Suppose the target reference Q^b varies with the candidate, as it does when visible-selector weights are design variables. Supply independent training samples from both b and Q^b , and use their empirical difference. If their row dimensions are D_P, D_Q , the same construction has regret at most $\sqrt{D_P/n} + \sqrt{D_Q/n}$. Minimax and attainment hold on the compact image of pairs (b, Q^b) . In particular a variable reference law must not be treated as known fixed input to Theorem 10.1.*

Proof. The error in any event discrepancy is at most the sum of the maximal row variations of the two empirical tables. Apply the preceding second moment bound separately and then the empirical-maximizer regret inequality. Count probabilities are products for the two tables. They are continuous polynomials on the compact image of pairs, so the same minimax argument applies without changing the simulator signature. $\square$

10.2. A matching uniform sampling obstruction.

Theorem 10.3 (Binary reset lower bound). *Take one binary row with target $\text{Ber}(1/2)$ and candidate class $B_h = \{\text{Ber}(1/2 - h), \text{Ber}(1/2 + h)\}$, where $0 < h < 1/2$. Then*

$$\delta = h, \quad v_n^{\text{aud}} = h \|\text{Ber}(1/2 + h)^{\otimes n} - \text{Ber}(1/2 - h)^{\otimes n}\|_{\text{TV}}. \quad (10.4)$$

For every $n \geq 1$, choosing $h = 1/(8\sqrt{n})$ gives

$$\delta - v_n^{\text{aud}} \geq \frac{1}{16\sqrt{n}}.$$

Thus the exponent in (10.3) is sharp uniformly over compact candidate classes, already at affine dimension one. This does not assert a degree lower bound for revealed-behavior polynomials, nor that this two-point class equals an unrestricted private-row image.

Proof. Only the two complementary singleton events and the zero event matter. A randomized choice is a statistic d in $[-1, 1]$. The two expected payoffs are $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$. Their minimum is at most half their sum, which is at most $h\|P_+^n - P_-^n\|_{\text{TV}}$. Conversely take the sign of the likelihood difference, assigning zero at a tie. Complementing every sample interchanges P_+^n and P_-^n and changes that sign. The two payoffs are equal, and their average attains the variation bound. The statistic depends only on the count, so it is an admissible audit. The symmetric measure on the two candidates attains the dual value. Every Dirac measure has dual objective h , whereas for finite n the two product laws are equivalent and their variation is strictly less than one. Thus no dual minimizer is a Dirac measure. Their barycenter is the target law, but replacing the dual by that barycenter would incorrectly give value zero instead of the strictly positive value in (10.4).

For $h \leq 1/8$, elementary integration of $(1 - u^2)^{-1}$ gives

$$\text{KL}(\text{Ber}(1/2 + h) \parallel \text{Ber}(1/2 - h)) = 2h \log \frac{1 + 2h}{1 - 2h} \leq 16h^2.$$

The product relative entropy is at most $16nh^2$. Pinsker's inequality bounds its total variation by $\sqrt{8nh^2}$. At the specified h this is less than $1/2$. Substituting in (10.4) proves the claim. The inequality uses the classical two-point testing mechanism; the restriction being measured is the reset budget of the present game. $\square$

10.3. Transport of one fixed audit.

Proposition 10.4 (Reset cost under model perturbation). *Let q range over the same compact set of executable row designs in two models. Suppose uniformly in q, j that*

$$\|P_j^q - \tilde{P}_j^q\|_{\text{TV}} \leq \alpha, \quad \|Q_j - \tilde{Q}_j\|_{\text{TV}} \leq \beta.$$

For the same coefficient rule c , evaluated with each model's own target validation,

$$|g_c(q) - \tilde{g}_c(q)| \leq (2Jn + 1)\alpha + \beta. \quad (10.5)$$

The deficiencies themselves differ by at most $\alpha + \beta$, at the same resource budget. If the source behavior is unchanged, a fixed audit loses at most β , independently of its reset count.

Proof. A product coupling of the Jn training runs fails with probability at most $Jn\alpha$. Taking counts and choosing the event are contractions. A score with fixed target discrepancies lies in $[-1, 1]$, so the change in its expectation from this training-law perturbation is at most $2Jn\alpha$. At each event the discrepancy changes by at most $\alpha + \beta$. This proves (10.5). For deficiency, every row variation changes by at most $\alpha + \beta$; taking the maximum and then the minimum preserves that bound. No new simulator is inserted. $\square$

For a rational polynomial image, a rational rule and a rational threshold, strict positivity of $g_c(q)$ above that threshold is a first-order formula over the original row simplexes. Real quantifier elimination decides it. The same is true of an existential rule followed by this universal condition. This is an exact finite verification statement, not an efficient algorithm. The explicit empirical rule below will be verified by a uniform analytic inequality, rather than by storing its enormous complete table.

11. CONTINUATION COMPLEXITY OF A FINITE-DATA TEST

The reset budget does not by itself bound the auditor's retained memory. We now impose both constraints. At scheduled epoch t a streaming auditor reads one letter x_t from a finite alphabet X_t , updates a register, and emits no information to any auxiliary store. It makes its decision on reading the last letter. Transition rows may depend on the epoch; the clock is free. All other information crossing a cut belongs to the register. Fresh randomization is allowed. An independent persistent selector, when allowed, selects a complete auditor before any letters are read.

For a Boolean decision $\psi : X_1 \times \cdots \times X_n \rightarrow \{-1, 1\}$, let $u \sim_t v$ mean that

$$\psi(us) = \psi(vs) \quad \text{for every } s \in X_{t+1} \times \cdots \times X_n.$$

Write $r_t(\psi)$ for the number of equivalence classes of prefixes of length t , for $0 \leq t < n$. These are the deterministic continuation responses. The terminal decision is emitted on the last transition, so no terminal register is needed. In particular, $r_0 = 1$. The equivalence is a finite-horizon version of the classical residual-state construction [16]. The following quantitative statement will also measure the loss when a statistical auditor has fewer states than this construction.

Theorem 11.1 (Exact continuation width and a quantitative cut obstruction). *A stochastic streaming auditor computes ψ exactly on every word with width profile $(K_t)_{0 \leq t < n}$ if and only if $K_t \geq r_t(\psi)$ at every cut. The profiles are attained simultaneously by a deterministic auditor. Independent persistent randomization does not reduce these widths for exact computation.*

More generally, at a cut with $r = r_t(\psi) \geq 2$, every stochastic auditor of width K satisfies

$$\sum_{w \in X_1 \times \cdots \times X_n} e_\psi(w) \geq \frac{(r - K)_+}{r - 1}, \quad (11.1)$$

where $e_\psi(w)$ is its probability of an incorrect Boolean decision on input w . The inequality also holds after averaging over an independent selector whose conditional auditors each have width at most K .

Proof. From a prefix u let $e_u(m)$ be the distribution of the retained state. From state m and suffix s let $a_m(s)$ be the probability of terminal answer $+1$. Exact computation says

$$\sum_m e_u(m) a_m(s) = \frac{1 + \psi(us)}{2}.$$

The right side is zero or one. Every state of positive e_u -mass must therefore have precisely that zero-one continuation response for all s . Distinct prefix classes have disjoint positive supports. Hence $K_t \geq r_t$. Conversely, retain the prefix class. Appending the same letter to two equivalent prefixes gives equivalent new prefixes; thus every transition is well defined. At the last transition the class and the last letter determine ψ . This realizes every width r_t simultaneously. For a persistent independent selector, a zero average error on each of the finitely many words implies zero conditional error on all words outside a single null set of selector values. The preceding bound applies there.

For the quantitative claim choose representative prefixes $u_1, \dots, u_r$, one per class, and put $e_i(m) = e_{u_i}(m)$. Define

$$o_{ij} = \sum_{m=1}^K \min\{e_i(m), e_j(m)\}.$$

For nonnegative numbers $a_1, \dots, a_r$, ordering them shows that $\sum_{i < j} \min(a_i, a_j) \geq \sum_i a_i - \max_i a_i$. Apply this at each state. Since every e_i has mass one and each $\max_i e_i(m) \leq 1$, it follows that

$$\sum_{i < j} o_{ij} \geq r - K. \quad (11.2)$$

Choose a suffix s_{ij} distinguishing u_i, u_j . Conditional on a common state and this suffix, the two desired answers are opposite. Their error probabilities, weighted by $e_i(m), e_j(m)$, have sum at least $\min\{e_i(m), e_j(m)\}$. Thus

$$e_\psi(u_i s_{ij}) + e_\psi(u_j s_{ij}) \geq o_{ij}.$$

On summing, each complete word on the left occurs at most $r - 1$ times: its prefix identifies a single representative, which participates in $r - 1$ pairs. Nonnegativity of all other word errors and (11.2) prove (11.1). When $r \leq K$ the claim is just nonnegativity. Finally apply the bound conditional on a selector and integrate. Independence ensures that conditioning has not changed the input experiment. $\square$

The exact part is also the extremal zero-one case of Theorem 3.2. Its content is a simultaneous width profile, rather than separate unrelated factorizations at different cuts. The error bound does not assert that its constant is optimal for every Boolean function. Its use below is to convert a continuation obstruction into a strict loss of the attainable statistical value.

11.1. The width required to attain the binary reset value. Fix $0 < h < 1/2$ and the binary candidate class B_h of Theorem 10.3. A terminal decision $+1$ chooses the singleton event appropriate to the $+$ candidate, and -1 the complementary event. A zero-score event can equivalently be replaced by an independent fair choice of these decisions. The expected payoffs are therefore $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$, where $d \in [-1, 1]$ is the conditional mean decision and $P_\pm = \text{Ber}(1/2 \pm h)^{\otimes n}$. Let $v_{n,K}$ be the supremum of the smaller of these two payoffs over streaming auditors with the declared width profile. The final validation interface is unchanged; the widths here count training and selection only.

Theorem 11.2 (Exact memory threshold for optimal binary auditing). *Let $n = 2m + 1$ be odd. The unconstrained reset value is $v_n = h\|P_+ - P_-\|_{\text{TV}}$. It is attained at the profile*

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n, \quad (11.3)$$

and $v_{n,K} = v_n$ if and only if $K_t \geq r_t$ for all these cuts. For every cut with $r_t \geq 2$ one has the explicit deficit

$$v_n - v_{n,K} \geq 2h^2(1/4 - h^2)^m \frac{(r_t - K_t)_+}{r_t - 1}. \quad (11.4)$$

These conclusions remain valid when an independent persistent selector chooses auditors of the same conditional widths. For $n \geq 3$ the least peak training width attaining v_n is $m + 2$; for $n = 1$ it is one.

Proof. The sign of $P_+(w) - P_-(w)$ is the majority decision $\psi(w)$, with no zero difference since n is odd. Complementing every bit interchanges the two laws and reverses the decision. Consequently majority has equal payoffs and attains $h\|P_+ - P_-\|_{\text{TV}}$, as also proved in Theorem 10.3.

A prefix with k ones requires at least $m + 1 - k$ ones in its remaining $n - t$ places. Thresholds at most zero give the constant positive response; thresholds greater than $n - t$ give the constant negative response; the remaining thresholds give different nonconstant responses. Counting them for $0 \leq k \leq t$ gives (11.3). The exact realization in Theorem 11.1 implements majority at all these widths.

For an arbitrary randomized decision, its smaller payoff is bounded by the equal-prior average. The deficit of this average from the majority value is exactly

$$h \sum_w |P_+(w) - P_-(w)| e_\psi(w). \quad (11.5)$$

Put $a = 1/2 + h$ and $b = 1/2 - h$. A word with $k \geq m + 1$ ones has absolute likelihood difference $(ab)^{n/2} \{(a/b)^{k-n/2} - (a/b)^{-(k-n/2)}\}$, which increases with k . Symmetry treats the other half of the words. The smallest difference is therefore attained at $k = m, m + 1$ and equals

$$a^{m+1}b^m - b^{m+1}a^m = 2h(1/4 - h^2)^m.$$

Substitution of (11.1) into (11.5) gives (11.4). Any deficient cut gives a strictly positive loss; all sufficiently wide profiles implement majority. Maximizing (11.3) proves the peak-width assertion. The same average-payoff argument works conditional on an independent selector and hence after averaging. $\square$

The result identifies the exact memory needed to attain the optimal value, not the memory needed for a prescribed approximate value uniformly in h . The quantitative deficit can be small for large n . In particular it is not a sharp joint asymptotic tradeoff in sample size, state size and error. It complements, rather than replaces, the uniform sampling exponent in Theorem 10.3. No minimax interchange on the generally nonconvex set of fixed-width auditors has been used.

12. POLYNOMIAL TESTING OF THE ORIGINAL RESOURCE IMAGE

The private behavior image is usually nonconvex. Its testing criterion can nevertheless be written as a limit of finite algebraic lower certificates, without replacing that image by randomized designs. The testing weights in this section depend on the candidate simulator. Unlike Theorem 13.1, this identity is a certificate transform, not an independent variational dual. They are universal proof functions, not a hidden selector supplied to an executed experiment.

Use any one of the finite signatures of Theorem 2.2, with rational instrument data. Write its objective as $F(q) = \max_{1 \leq i \leq m} f_i(q)$, where f_i are all its marked event discrepancies, including the empty event $f_1 = 0$. For visible selectors the event and feedback rule include their selector fibers, and the target selector law remains in f_i . Thus $-1 \leq f_i \leq 1$ for all three signatures; the polynomials may be different.

Parameterize a row with d outcomes by $d - 1$ variables in $[0, 1]$:

$$q_1 = x_1, \quad q_j = x_j \prod_{l < j} (1 - x_l) \quad (1 < j < d), \quad q_d = \prod_{l < d} (1 - x_l).$$

This map is onto the row simplex, including its boundary. Indeed, divide each desired probability by the remaining mass while that mass is positive; when it is zero choose the remaining variables arbitrarily. Taking the product of these maps, including a selector row when needed, gives an onto polynomial map $q(x)$ from $[0, 1]^D$ to the feasible row product. Put $h_i(x) = f_i(q(x))$ and $g_i = (1 + h_i)/2$. All h_i are rational polynomials, $0 \leq g_i \leq 1$, and $g_1 = 1/2$. The case $D = 0$ is a single feasible table and is interpreted directly.

Theorem 12.1 (Strict polynomial lower-certificate hierarchy). *For $p \geq 1$ define*

$$A_p(x) = \frac{\sum_{i=1}^m g_i(x)^p h_i(x)}{\sum_{i=1}^m g_i(x)^p}, \quad a_p = \min_{x \in [0, 1]^D} A_p(x), \quad \delta = \min_q F(q).$$

Then the denominators are positive, a_p is nondecreasing, and

$$\delta - 2(1 - m^{-1/(p+1)}) \leq a_p \leq \delta, \quad \delta = \sup_{p \geq 1} a_p. \quad (12.1)$$

For rational L let

$$P_{p,L}(x) = \sum_{i=1}^m g_i(x)^p (h_i(x) - L). \quad (12.2)$$

A nonnegative tensor Bernstein expansion of $P_{p,L}$ on $[0, 1]^D$ certifies $\delta \geq L$. If $L < \delta$, some p and some finite Bernstein degree have strictly positive coefficients. Conversely, a strictly positive such expansion certifies $L < \delta$. These are lower certificates at the original state and selector budgets. No convexification of the feasible behavior image occurs.

Proof. Fix x and put $b = \max_i g_i(x) \geq 1/2$ and $r_i = g_i(x)/b$. At least one r_i is one. Write $S_j = \sum_i r_i^j$. Hölder's inequality gives

$$S_p \leq m^{1/(p+1)} S_{p+1}^{p/(p+1)}, \quad S_{p+1} \geq 1.$$

Consequently $S_{p+1}/S_p \geq m^{-1/(p+1)}$. Since $A_p = 2bS_{p+1}/S_p - 1$ and $\max_i h_i = 2b - 1$,

$$0 \leq \max_i h_i - A_p \leq 2(1 - m^{-1/(p+1)}).$$

Cauchy-Schwarz gives $S_{p+2}S_p \geq S_{p+1}^2$, so A_p is nondecreasing in p . Its denominator is at least 2^{-p} and it is continuous on the cube. Taking minima proves (12.1). The bound is also at most $2 \log m/(p+1)$, by $1 - e^{-u} \leq u$.

Every tensor Bernstein basis function is nonnegative on the cube, and the basis functions sum to one. Nonnegative coefficients therefore imply $P_{p,L} \geq 0$. Dividing by the positive denominator

gives $F(q(x)) \geq A_p(x) \geq L$ for every x , which proves soundness. For completeness choose p so that the error in (12.1) is less than $\delta - L$. Then

$$P_{p,L}(x) \geq 2^{-p}\{\delta - L - 2(1 - m^{-1/(p+1)})\} > 0.$$

Lemma 12.2 below gives a finite strictly positive Bernstein expansion. Conversely strictly positive coefficients have a positive minimum, and the denominator is at most m , so they give a strict positive uniform gap above L . $\square$

Positive Bernstein certificates and their degree elevation are classical; see de Klerk–Laurent [20]. The next elementary bound is retained for a self-contained exact verifier, not as a new general hypercube positivity theorem.

Lemma 12.2 (An explicit positivity degree). *Let $P(x) = \sum_k c_k x^k$ be a rational polynomial in D variables and $n \geq \max_{k,j} k_j$. In its tensor Bernstein expansion of degree n in each coordinate the coefficient at $\alpha \in \{0, \dots, n\}^D$ is*

$$b_{\alpha,n} = \sum_k c_k \prod_{j=1}^D \frac{\binom{\alpha_j}{k_j}}{\binom{n}{k_j}}, \quad |b_{\alpha,n} - P(\alpha/n)| \leq \frac{C(P)}{n}, \quad C(P) = \sum_k |c_k| \sum_j \binom{k_j}{2}. \quad (12.3)$$

In particular, $\min P > 0$ implies positivity of every coefficient whenever $n > C(P)/\min P$ and the degree restriction holds.

Proof. For a binomial random variable B with parameters (n, x) , $\mathbb{E}\binom{B}{k} = \binom{n}{k} x^k$. Expanding this expectation proves the univariate monomial formula, and products prove the displayed coefficient identity. The convention is $\binom{a}{k} = 0$ for $a < k$. To compare $\binom{a}{k}/\binom{n}{k}$ with $(a/n)^k$, draw k indices uniformly from $\{1, \dots, n\}$. Conditional on no repetition this is sampling without replacement. The probability of a repetition is at most $\binom{k}{2}/n$. Apply this total variation bound to the event that all drawn indices lie in a prescribed a -element subset. Hence the difference of the two factors is at most $\binom{k}{2}/n$. All factors lie in $[0, 1]$. A product telescoping bound followed by the triangle inequality proves (12.3). The final assertion follows by taking the minimum of P . $\square$

Corollary 12.3 (Rational lower witnesses without policy cells). *At every fixed finite rational signature, strict infeasibility above a rational level L is equivalent to the existence of the following finite rational certificate: an integer p , an integer n , and a strictly positive coefficient array for (12.2) whose identity is checked by (12.3). Dovetailing these certificates with rational original-budget upper witnesses produces nested intervals converging to δ . If a subfamily of the original tests gives a strict gap, the same construction applies to it after adjoining the empty test.*

Proof. Soundness and completeness are Theorem 12.1. Each finite candidate is checked by rational polynomial arithmetic. Enumerate all (p, n) and rational levels, and retain the largest verified lower level so far, together with zero. Rational row tables are dense in the feasible product, and their continuous objectives converge to the minimum, so enumeration and retention of the best upper table converge from above. This is a dovetailed enumeration, not a claim that any fixed (p, n) test search terminates. For a requested positive interval width, convergence of both endpoints proves termination. The reduced family conclusion uses only the same pointwise maximum argument. $\square$

Remark 12.4 (Meaning and size of the strict certificate transform). A constant distribution on tests generally detects only the convex hull of the behavior image. The functions $g_i^p / \sum_j g_j^p$ are instead functions of a proposed row table. Their universally quantified weighted inequality is an algebraic certificate; the tester need not observe a private state in an experiment. The displayed certificate identity is not an equality obtained by exchanging a private minimum with a fixed mixed test. Power means and Bernstein degree elevation are classical algebraic approximation devices, proved here in the precise form needed. The claim here is strict-level completeness for the

original marked causal resource image, including visible selectors, and its physical transfer below. The coefficient count $(n+1)^D$, the number of original tests, and the algebraic degrees may be large. No polynomial complexity or intrinsic-dimension bound on this row-cube coefficient count is asserted. The independent adaptive test has the separate behavior-size bound of Theorem 13.3; its verification is a different cost.

12.1. A sharp two-time nonconvex example. The source has singleton reports, the control alphabet and actual mark are singleton, and the target emits $(0,0)$ or $(1,1)$ with probability $1/2$ each. At width one, a private simulator emits two independent bits with probabilities p, q . It cannot reread its first emitted bit.

Proposition 12.5 (A gap invisible to constant testing mixtures). *For this delayed experiment,*

$$\delta_{(1,1)}^p = \sqrt{2} - 1, \quad \delta_{(1,1),2}^h = 0, \quad \delta_{(1,1),r}^v = \sqrt{2} - 1 \quad (r \geq 1).$$

The private objective is the maximum of the four polynomials

$$0, \quad \frac{1}{2} - pq, \quad \frac{1}{2} - (1-p)(1-q), \quad p + q - 2pq.$$

Every constant mixture of these tests has minimum at most zero, whereas Corollary 12.3 certifies every rational $L < \sqrt{2} - 1$ at private width one.

Proof. Put $u = pq$ and $v = (1-p)(1-q)$. Direct total variation calculation gives $F = 1 - \min(u, 1/2) - \min(v, 1/2)$ and the displayed maximum. By Cauchy–Schwarz, $\sqrt{u} + \sqrt{v} \leq 1$. If $u \geq 1/2$, then $v \leq (1 - 1/\sqrt{2})^2$, so the sum of the two capped terms is at most $2 - \sqrt{2}$; the case $v \geq 1/2$ is symmetric. Otherwise set $a = \sqrt{u}$, $b = \sqrt{v}$. On $0 \leq a, b \leq 1/\sqrt{2}$, $a + b \leq 1$, the maximum of $a^2 + b^2$ is at a vertex of this polygon and equals $2 - \sqrt{2}$. Equality is attained at $p = q = 1/\sqrt{2}$. A hidden fair bit selects the two deterministic constant sequences and exactly reproduces the target. If it is exposed, variation is averaged over the selector fibers; each fiber has error at least the private minimum, which is itself attainable without a selector. Finally the target lies in the convex hull of the two private deterministic behaviors. The average discrepancy of any fixed test mixture at those behaviors is zero, so at least one has discrepancy at most zero. The strict polynomial certificates therefore detect information lost by constant convex separation. $\square$

13. ADAPTIVE TESTING ON THE OBSERVABLE BEHAVIOR IMAGE

The obstruction to linear randomization criteria at a fixed private register is the nonconvexity of the behavior image, not a failure of compactness. We give a variational formulation in which a test is an independent function of an observable candidate law. The construction does not give the simulator an extra random variable.

13.1. The testing game. Let B be a nonempty compact subset of a finite-dimensional affine space, and let $f_1, \dots, f_m$ be affine real functions, including the zero function. Put

$$F(b) = \max_{i \leq m} f_i(b), \quad \delta = \min_{b \in B} F(b).$$

For the finite comparison problem of Section 2, b is the array of discrepancies between the simulated and target marked laws, indexed by parameters, feedback trees and transcripts; f_i ranges over their signed event evaluations. In the visible case the array includes the selector and its target weights. Thus the target coordinate is part of the array when those weights vary. Private, fixed-hidden and fixed-visible signatures each give their own compact set B . No convex hull is taken.

An *adaptive test* is a continuous map $\kappa : B \rightarrow \Delta_m$, chosen before a candidate b . Its score against that candidate is

$$G(b, \kappa) = \sum_i \kappa_i(b) f_i(b).$$

This is a verification game: the selected candidate law is revealed to the evaluator of the already chosen function κ . It is not a one-observation statistical test, and does not make an unavailable parameter or mark observable to a physical controller.

Theorem 13.1 (Revealed-behavior adaptive representation). *For every such compact behavior image,*

$$\begin{aligned} \delta &= \sup_{\kappa \in C(B, \Delta_m)} \min_{b \in B} G(b, \kappa) \\ &= \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa \in C(B, \Delta_m)} \int_B G(b, \kappa) d\mu(b). \end{aligned} \quad (13.1)$$

The minimization over probability measures is attained. If Borel adaptive tests are allowed, the first supremum is attained as well. The continuous-test assertion is a supremum; continuous attainment is not needed or asserted.

Proof. Fix $\eta > 0$. The sets $U_i = \{b : f_i(b) > F(b) - \eta\}$ form an open cover of the compact metric space B . Choose a continuous partition of unity subordinate to this finite cover. The resulting test has $F(b) - \eta < G(b, \kappa) \leq F(b)$ for all b . Consequently the first supremum equals δ . The same uniform inequality shows, for every probability μ ,

$$\sup_{\kappa} \int G(b, \kappa) d\mu(b) = \int F d\mu.$$

Its minimum is δ , attained by a point mass at a minimizer of F . This also proves the displayed minimax exchange without an assumption that B is convex. Equivalently one can apply the usual compact-convex minimax theorem to $\mathcal{P}(B)$ and the convex class of continuous tests, with the pairing $\int G d\mu$. Finally choose the smallest index attaining $F(b)$. Its level sets are Borel, and its indicator vector is a Borel test with score exactly F . $\square$

The probability μ here describes a lottery over *revealed candidate laws*. The score is $\int G(b, \kappa) d\mu(b)$, not $G(\int b d\mu, \kappa)$. Hiding the candidate and replacing it by its barycenter is a different game; in general it lowers the value. This distinction is essential for a finite persistent-state budget.

13.2. A size bound in behavior dimension. Factor out directions annihilated by every $f_i - f_i(b_0)$, so that on the resulting difference space

$$\|v\|_* = \max_i |f_i(b_0 + v) - f_i(b_0)|$$

is a norm. Denote its dimension by a and let $R = \text{diam}_*(B)$. All conclusions below are unaffected by this observable quotient. The case $a = 0$ is a single constant comparison and requires one test. Suppose henceforth that $a \geq 1$, so $R > 0$.

Lemma 13.2 (A bounded behavior chart). *There exist $b_0 \in B$ and vectors $v_j \in B - b_0$, $1 \leq j \leq a$, forming a basis, such that $\|v_j\|_* \leq R$ and every $b \in B$ has coordinates $b = b_0 + \sum_j t_j v_j$ with $|t_j| \leq 2$. If B is a rational polynomial image of a product of simplexes, the points can be chosen rational and can be found effectively.*

Proof. Fix $b_0 \in B$. In any fixed coordinates on the difference space, let $D_0 > 0$ be the maximum absolute determinant of a vectors in $B - b_0$. Compactness and spanning give existence and positivity. Choose a tuple with determinant greater than $D_0/2$. Replacing its j th vector by $b - b_0$ multiplies the determinant by t_j ; maximality gives $|t_j| < 2$. The norm bound is the definition of R . For a rational polynomial image, take b_0 at rational rows. Rational row images are dense. An affine basis can be found by enumerating rational images and testing rational determinants; the true affine span and the maximum determinant are semialgebraic data. Quantifier elimination

decides whether a proposed nonzero determinant exceeds half the maximum. Density therefore makes the search terminate. This is an effectiveness assertion, not a bit-complexity bound for finding the chart. $\square$

Set $b(x) = b_0 + \sum_j (4x_j - 2)v_j$ and $C = \{x \in [0, 1]^a : b(x) \in B\}$. The ambient cube is a coordinate box, not an enlargement of the feasible behavior set. Write $\ell_i(x) = f_i(b(x))$. Then

$$|\ell_i(x) - \ell_i(y)| \leq 4R \sum_j |x_j - y_j|. \quad (13.2)$$

For $n \geq 0$ let $\beta_{\alpha,n}$ be the tensor Bernstein basis of coordinatewise degree n . A feasible coefficient array has $c_{\alpha i} \geq 0$ and $\sum_i c_{\alpha i} = 1$ for every $\alpha \in \{0, \dots, n\}^a$. It specifies the independent test

$$\kappa_i^c(x) = \sum_{\alpha} c_{\alpha i} \beta_{\alpha,n}(x).$$

Define

$$d_n = \max_c \min_{x \in C} \sum_{\alpha, i} c_{\alpha i} \beta_{\alpha,n}(x) \ell_i(x). \quad (13.3)$$

Theorem 13.3 (Intrinsic adaptive degree and support bounds). *The maxima in (13.3) are attained and $d_n \leq d_{n+1} \leq \delta$. For $n \geq 1$,*

$$0 \leq \delta - d_n \leq \frac{4Ra}{\sqrt{n}}. \quad (13.4)$$

There is an array with at most $(n+1)^a$ nonzero coefficients, all equal to one, whose minimum score is at least $\delta - 4Ra/\sqrt{n}$. Its test has coordinatewise degree n , and its score degree at most $n+1$. Furthermore

$$d_n = \min_{\mu \in \mathcal{P}(C)} \sum_{\alpha} \max_i \int_C \beta_{\alpha,n}(x) \ell_i(x) d\mu(x), \quad (13.5)$$

and a minimizing measure can be supported on at most $(n+2)^a$ points. In particular a strict level $L < \delta$ admits an adaptive test of positive uniform margin with $n > (8Ra/(\delta - L))^2$ and at most $(n+1)^a$ occupied coefficient entries. These bounds depend on the observable image and its metric, not on the number of stochastic rows in a realization.

Proof. The coefficient space is a finite product of compact simplexes, and the minimum of a jointly continuous function over compact C is continuous in its coefficients. This proves attainment. Bernstein degree elevation replaces each coefficient by convex combinations of adjacent coefficients, preserving every simplex constraint and the represented function. It proves monotonicity. All scores are convex combinations of the $\ell_i(x)$, proving the upper bound by δ .

At each grid point α/n choose an index i_{α} maximizing ℓ_i there. The choice is made even at infeasible grid points; only the final score is minimized over C . Give that index coefficient one and the others zero. For fixed x let A_j be independent $\text{Bin}(n, x_j)$ variables. If i_* maximizes at x ,

$$\begin{aligned} \max_i \ell_i(x) - \mathbb{E} \ell_{i_A}(x) &\leq \mathbb{E} \{ \ell_{i_*}(x) - \ell_{i_*}(A/n) + \ell_{i_A}(A/n) - \ell_{i_A}(x) \} \\ &\leq 8R \sum_j \mathbb{E} |A_j/n - x_j| \leq \frac{4Ra}{\sqrt{n}}. \end{aligned}$$

Here the omitted middle difference is nonpositive, and the last bound uses $\text{Var}(A_j/n) \leq 1/(4n)$. This proves the rate and the sparse coefficient assertion.

For the last formula replace the minimum over C by a minimum over $\mathcal{P}(C)$. The pairing is bilinear and continuous, and both feasible sets are compact convex, so the compact minimax theorem [18] exchanges the extrema. Optimizing each coefficient simplex gives (13.5). The integrands are polynomials of coordinatewise degree at most $n+1$. Preserve all monomial moments of that degree by Carathéodory's theorem. There are $(n+2)^a$ monomials including the constant;

their moment vectors lie in an affine hyperplane of dimension at most $(n+2)^a - 1$. A measure with at most $(n+2)^a$ atoms therefore has exactly the same objective. The strict-margin assertion follows from the rate with error less than $(\delta - L)/2$. $\square$

The score is a nonlinear function of behavior although each event functional is affine. This is the reason the measure in (13.5) cannot be replaced by its behavior barycenter. For $n = 0$ it can: d_0 is precisely the constant-test convex relaxation. In the two-time unmarked example of Proposition 12.5, the target belongs to the convex hull of private behaviors, so $d_0 = 0$, whereas $\lim_n d_n = \sqrt{2} - 1$. Thus the hierarchy recovers a gap which every constant test misses.

Remark 13.4 (What the size bound measures). The bound counts the nonzero entries of the independent adaptive test and controls its behavior-coordinate degree. It does not bound the cost of verifying its minimum on the generally nonconvex set C , the bit lengths of a rational chart, or the size of a description of C . Distinct row presentations of the same metric behavior image satisfy the same existence bound, although their coefficient arrays need not coincide. For the private fixed-target law array one may use $R \leq 1$ with the maximum total-variation metric. For the visible signed-discrepancy array the uniform bound $R \leq 2$ is sufficient. The old row-cube positivity certificate is a separate verification engine, with its own coordinate-dependent complexity.

13.3. A three-weight certificate invisible to constant tests.

Proposition 13.5 (A finite intrinsic nonlinear witness). *Consider a private width-one simulator producing two independent bits, with probabilities p, q , for the target which assigns probability $1/2$ to each of $00, 11$. Put $u = pq$, $v = (1-p)(1-q)$ and $J(t) = 3t^2 - 2t^3$. The three behavior-dependent weights*

$$\kappa_1 = J(v), \quad \kappa_2 = J(u), \quad \kappa_3 = 1 - J(u) - J(v)$$

are nonnegative and sum to one. Against the event discrepancies $1/2 - u, 1/2 - v, 1 - u - v$, respectively, their score G satisfies

$$G \geq \frac{2}{5} + \frac{9}{8960}. \quad (13.6)$$

The certificate uses cubic weights in observable probabilities, a nine-term nonnegative derivative identity, and seventeen positive rational Bernstein coefficients in one variable. Every constant mixture of event discrepancies has minimum at most zero.

Proof. We have $u + v \leq 1$. Since J is increasing on $[0, 1]$ and $J(1-t) = 1 - J(t)$, the weights are feasible. Their score is

$$G = 1 - u - v + J(u)(u - 1/2) + J(v)(v - 1/2).$$

Set $a = (p + q - 1)^2$ and $b = (p - q)^2$. Then $a, b \geq 0$ and $a + b \leq 1$. Direct expansion gives

$$\begin{aligned} 64G &= 27 - 12a + 38a^2 - 20a^3 - a^4 + 36b - 60ab + 36a^2b + 4a^3b \\ &\quad + 6b^2 - 12ab^2 - 6a^2b^2 - 4b^3 + 4ab^3 - b^4. \end{aligned}$$

Let $c = 1 - a - b$. Another polynomial identity, valid when $a + b + c = 1$, is

$$\begin{aligned} 16\partial_b G &= 4a^3 + 6a^2c + 12ab^2 + 24abc + 12ac^2 \\ &\quad + 8b^3 + 30b^2c + 30bc^2 + 9c^3 \geq 0. \end{aligned}$$

Thus $G(a, b) \geq G(a, 0)$. The univariate polynomial

$$G(a, 0) - \frac{2}{5} = \frac{7}{320} - \frac{3}{16}a + \frac{19}{32}a^2 - \frac{5}{16}a^3 - \frac{1}{64}a^4$$

has degree-16 Bernstein coefficients $z_k/349440$, where

$$(z_0, \dots, z_{16}) = (7644, 3549, 1183, 351, 855, 2494, 5064, 8358, \\ 12166, 16275, 20469, 24529, 28233, 31356, 33670, 34944, 34944).$$

The Bernstein basis is nonnegative and sums to one. Its smallest coefficient is $351/349440 = 9/8960$, proving (13.6). All identities are rational polynomial identities; no numerical minimization is used. Finally the target is the barycenter of the two feasible deterministic laws 00 and 11. An affine mixture of signed event discrepancies has average zero at these two laws and hence minimum at most zero. The empty event supplies the reverse constant-test bound zero. $\square$

For comparison, the affine weights $v, u, 1 - u - v$ already give the smaller bound $3/8$, with the exact identity

$$G_{\text{affine}} - \frac{3}{8} = \frac{1}{2}(p - q)^2 + 2(p - 1/2)^2(q - 1/2)^2.$$

Neither certificate changes the private feasible set. The cubic certificate is an independent test chosen once for the whole behavior image, not a mixture chosen anew by an optimizer's unknown row values. Its evaluation does use the revealed behavior, as specified in the verification game. The 17-coefficient positivity check is classical; the distinction from constant tests is the behavior-dependent feasible weight family and its resource-sensitive minimum.

The infinite-dimensional representation above holds for continuous f_i on any compact metric space: its proof uses a partition of unity, not causal factorization. The measure minimum is attained at a Dirac mass at a minimizer of F . The finite-level polynomial theorem has a different content. It is an existence-size bound for revealed-law verification; the reset-audit theorem of Section 10 imposes an explicit finite-data restriction.

13.4. Strict certificates and effective physical input. Given rational finite instruments, a rational chart and the sparse array constructed above, its strict minimum can be certified by real quantifier elimination on the original row constraints. Alternatively, pull the score back through the original polynomial row map and certify its strict positivity using degree-elevated Bernstein coefficients. The pullback changes the verification degree and is not covered by the behavior-coordinate bound. Generic quantifier elimination also decides the exact finite semialgebraic boundary; this classical observation does not assert that a finite nonnegative Bernstein expansion exists at $L = \delta$.

Proposition 13.6 (Uniform effective presentation input). *Suppose a single algorithm, on input j , returns rational finite instruments E_j, F_j and nonnegative rational error enclosures a_j, c_j for two-sided stateless marked feedback presentations of E, F . Suppose $a_j + c_j \rightarrow 0$. All alphabets, widths, selector laws and finite subroutines must be computable uniformly from j and the requested signature. Then, for every positive rational η , an algorithm returns a rational interval of width below η containing $\delta_b(E, F)$ at the same signature.*

Proof. Enumerate j until $a_j + c_j < \eta/8$. This terminates without a known rate because the rational enclosures themselves tend to zero. For the selected finite problem, enumerate rational same-budget upper tables and strict rational lower certificates, dovetailing all finite checks; include the always valid lower bound zero. Compactness, density and strict completeness provide an interval of width below $\eta/2$. Enlarge it by $a_j + c_j$ at both ends using stateless comparison. The enlarged width is below $3\eta/4$. Every enumeration and comparison is uniform by the hypotheses. A mere sequence of individually computable presentations without a uniform enumeration is not an input satisfying this proposition. $\square$

14. AN EXACT ALGORITHM FOR THE OBSERVABLE CHART

The chart in Lemma 13.2 can be found without taking its dimension as an oracle. We state the input and output explicitly. Rational polynomials are encoded by finite lists of monomial exponent vectors and rational coefficients in reduced integer form. A product of simplexes is encoded by the number of entries in each row. Eliminating the last entry of every nontrivial row leaves a full-dimensional rational polytope $\Omega \subset \mathbb{R}^r$. Zero-dimensional factors are omitted.

Proposition 14.1 (Effective observable quotient and pullback). *The input consists of the preceding simplex description, a rational polynomial map $b : \Omega \rightarrow \mathbb{R}^N$, and finitely many rational affine functionals f_i on $\mathbb{R}^N$. There is a terminating algorithm which returns its observable affine dimension a , a rational basis and coordinate map for the observable quotient, and, when $a > 0$, rational parameter points $q_0, q_1, \dots, q_a$ with the following properties. With*

$$z(q) = (f_i(b(q)) - f_i(b(q_0)))_i, \quad v_j = z(q_j),$$

the vectors v_j are a basis of the observable difference space, and all coordinates $t(q)$ in that basis satisfy $|t_j(q)| < 2$. The output also contains the rational polynomial cube map $x(q) = (t(q) + 2)/4$, a rational upper bound for the observable diameter, and the exact pullback of any subsequently supplied rational polynomial in x . The algorithm outputs the universal determinant formula it has decided. It makes no polynomial-time or coefficient-height claim.

Proof. Choose a rational relative interior point q_0 of the input simplexes. Expand the vector polynomial $z(q)$ in the free row coordinates. Let V be the rational span of all its coefficient vectors, including its constant coefficient. This equals $\text{span } z(\Omega)$: a linear functional vanishes on $z(\Omega)$ exactly when its polynomial vanishes on the nonempty interior of Ω , hence identically, hence on every coefficient. Gaussian elimination therefore gives $a = \dim V$ and a rational basis matrix U . Choosing independent rows gives a rational left inverse L with $LU = I_a$. Thus $y(q) = Lz(q)$ provides exact quotient coordinates. If $a = 0$, all discrepancies are constant and no chart is required.

For $a > 0$ let

$$D_0 = \max_{u_1, \dots, u_a \in \Omega} |\det(y(u_1), \dots, y(u_a))|.$$

Compactness gives a maximum, and spanning gives $D_0 > 0$. Enumerate rational tuples $(q_1, \dots, q_a)$ in Ω^a . Membership and their determinant d are decided by rational arithmetic. For $d \neq 0$, decide the first-order real formula

$$\forall u_1, \dots, u_a \in \Omega : \quad \det(y(u_1), \dots, y(u_a))^2 < 4d^2. \quad (14.1)$$

Quantifier elimination terminates for each formula. Density of rational points and continuity of determinant imply that some enumerated tuple has $|d| > D_0/2$. For such a tuple the strict formula is true, so the search terminates. No computation of an exact algebraic maximizer is needed.

Invert the rational matrix with columns $y(q_j)$ to obtain $t(q)$. Cramer's rule and (14.1) give $|t_j(q)| < 2$ for every q . In the observable norm $\|z\|_* = \max_i |z_i|$, each v_j has norm at most the diameter of $z(\Omega)$. An upper bound for that diameter can be produced, for example, by bounding each free coordinate in $[0, 1]$ and summing the absolute polynomial coefficients for $f_i(b(q)) - f_i(b(q'))$. This bound is rational and computable. Rational matrix multiplication and substitution give $x(q)$ and every requested pullback. They also give its full sparse coefficient encoding. All searches have now been specified by finite encodings and terminating operations. $\square$

There is a small but important distinction between the quotient and a projection onto a smaller set of simulator rows. The former identifies exactly the directions invisible to the chosen event family; the original row constraints remain present in every universal positivity test. The chart need not identify equivalent tasks after a change of marks or of information signature. Those are changes to the comparison problem.

15. A MARKED RESOURCE GAP GENERATED BY A BINARY COLLISION

We now realize a private–hidden–visible separation in a physical experiment whose information vanishes when the collision is removed. The controller chooses a later sensor gate after its first report. The gate controls acquisition, not the mechanical collision law. Both the source and the target carry the same actual, unobserved mark.

15.1. The exact marked two-report experiment. Let $B, W \in \{0, 1\}$ have the joint distribution

$$\mathbb{P}(B = b, W = w) = \begin{cases} 5/16, & b = w, \\ 3/16, & b \neq w. \end{cases} \quad (15.1)$$

The blind source E_* has two constant reports and actual mark W . The target F_* reports $Y_1 = B$, receives $A \in \{0, 1\}$ after that report, and reports $Y_2 = AB$. The feedback tester may choose A as any function of its earlier report. A width-one private simulator cannot reread its own first emission, except through the current gate A . Hidden selectors persist but are not revealed; visible selectors are included in the joint target law independently of (B, W) . Put

$$d_* = \sqrt{5/2} - 9/8.$$

Theorem 15.1 (An exact marked causal separation). *At the width profile $(1, 1)$,*

$$\delta^P(E_*, F_*) = d_*, \quad \delta_2^h(E_*, F_*) = 1/8, \quad \delta_r^v(E_*, F_*) = d_* \quad (r \geq 1). \quad (15.2)$$

The hidden upper bound is realized by two deterministic tables. The private upper bound is realized by two independent Bernoulli variables with parameter $\sqrt{5/8}$, followed by the gate. These bounds hold against every feedback rule, not just open-loop gates.

Proof. First choose the constant gate $A = 1$. The two private emissions are independent Bernoulli variables with parameters p, q and are independent of W . Let

$$u = pq, \quad v = (1 - p)(1 - q), \quad a = 3/8, \quad b = 5/8.$$

The overlap of the target and simulated marked laws is $H(u) + H(v)$, where

$$H(t) = \frac{1}{2} \{ \min(t, a) + \min(t, b) \}.$$

Indeed the target is supported on equal report pairs, and its two W -masses at each equal pair are $3/16$ and $5/16$. Total variation is one minus this overlap.

Cauchy–Schwarz gives $\sqrt{u} + \sqrt{v} \leq 1$. By symmetry suppose $u \geq v$. If $u \leq 1/4$ the overlap is at most $1/2$. If $u \geq 1/4$, then $v \leq (1 - \sqrt{u})^2 \leq 1/4 < a$, so the overlap is at most

$$H(u) + (1 - \sqrt{u})^2.$$

This function is convex on each of $[1/4, a]$ and $[a, b]$ and nonincreasing on $[b, 1]$. Its maximum is therefore attained at $1/4$, a or b . The values at a and b are $1 + 2a - 2\sqrt{a}$ and $3/2 + b - 2\sqrt{b}$ respectively. The second is larger, since

$$(3/2 + b - 2\sqrt{b}) - (1 + 2a - 2\sqrt{a}) = 3/8 - \frac{1/2}{\sqrt{a} + \sqrt{b}} > 0,$$

using $\sqrt{a} > 3/5$, $\sqrt{b} > 3/4$. It also exceeds $1/2$. Taking $p = q = \sqrt{b}$ attains this maximum, and hence the minimum error is $1 - (3/2 + b - 2\sqrt{b}) = d_*$.

For any hidden selector, the all-on report pair is still independent of W . If its probabilities of 11 and 00 are u, v , then $u + v \leq 1$. Concavity and monotonicity of H give

$$H(u) + H(v) \leq 2H((u + v)/2) \leq 2H(1/2) = 7/8.$$

A fair hidden bit J , emitting J twice, attains this overlap. This requires precisely two deterministic selector tables and gives error $1/8$. In particular hiding a seed does not erase the actual-mark obstruction.

These all-on bounds are lower bounds for the feedback problem. For the upper bounds first generate the corresponding all-on pair (X_1, X_2) , emit X_1 , and after receiving A emit AX_2 . For a deterministic feedback $A = g(X_1)$ the full gated marked transcript is the pushforward of the all-on marked pair. Applying the same map to the target pair cannot increase total variation. The construction is executable at width one: the second fresh Bernoulli variable is independent in the private case, and the bit J is the priced persistent hidden selector in the hidden case. Randomized feedback follows by conditioning on its independent random table.

Finally, in the visible case the two compared laws are

$$\sum_j \lambda_j \delta_j \otimes P_{g_j}^{q_j} \quad \text{and} \quad \sum_j \lambda_j \delta_j \otimes Q_{g_j}.$$

The target selector is independent of (B, W) . The all-on gate in every fiber has error at least d_* in every private fiber. Joint variation is the weighted sum of these fiber errors. Ignoring the visible selector and using the private optimizer attains d_* . $\square$

15.2. A full-dimensional collision preparation. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$ with contact distance d . The center of mass $Q = (q_1 + q_2)/2$ ranges over any fixed rational position box. Write $r = q_1 - q_2$, $g = v_1 - v_2$, and $V = (v_1 + v_2)/2$. Conditional on (B, W) in (15.1), take the uniform product preparation on

$$\begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), \quad a_0 \in [29/10, 31/10], & b_0 &\in [9/20, 11/20], \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & |V_z - (2W - 1)| &\leq 1/100. \end{aligned} \quad (15.3)$$

The boxes have positive volume in the full phase space. The actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$; total momentum conservation retains it. The informative quantity will instead be the transverse velocity created at a collision. Put

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector which is identically zero on the incoming transverse-velocity range.

Lemma 15.2 (Uniform transverse collision signal). *For every preparation in (15.3) and every $d \in [9/10, 11/10]$, exactly one collision occurs before time 3. Its time t_c lies in $(4/5, 3/2)$, and afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4. \quad (15.4)$$

For $0 < d \leq 2/5$ no collision occurs before time 3, and the detector is zero throughout that interval.

Proof. Up to a collision $r(t) = r + tg$. For $t \leq 4/5$, its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At time $3/2$ the absolute coordinate bounds are $1/4, 113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. Thus a first contact occurs in the stated interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$ as well. The first contact therefore has $n_x < 0$, where $n = r(t_c)/d$. The entry contact is strictly incoming. The bounds above give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq \frac{87}{220}.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The equal-mass elastic collision rule is

$$v'_1 = V + g/2 - (g \cdot n)n, \quad g' = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v'_{1,y} > -3/200 + \frac{1443}{1000} \frac{87}{220} > 1/2.$$

The relative velocity is outgoing, $r(t_c) \cdot g' > 0$, so the squared separation increases strictly for all later times. There are only two spheres and no walls; no second collision is possible. This proves (15.4). For $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. Thus there is no collision. The incoming $|v_{1,y}| \leq 3/200$ then makes h zero. $\square$

At time 2 the physical target $F_{d,\sigma}$ reports

$$Y_1 = \mathbf{1}_{\{h(v_{1,y}(2)) + \sigma \xi_1 > 0\}}.$$

It then receives a gate $A \in \{0, 1\}$, and at time 3 reports

$$Y_2 = A \mathbf{1}_{\{h(v_{1,y}(3)) + \sigma \xi_2 > 0\}},$$

where the ξ_i are independent standard normal variables, independent of the preparation. The source E_d has the same preparation, evolution, control interface and actual mark, but emits two constant reports. The gates do not change the particle motion. This is a feedback acquisition experiment: taking $A = 0$ suppresses later information and taking $A = 1$ exposes another noisy observation of a collision-generated quantity.

Theorem 15.3 (Collision-sensitive physical resource separation). *Let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/12$. At retained width $(1, 1)$ the physical deficiencies obey, uniformly in d ,*

$$\begin{aligned} |\delta^p(E_d, F_{d,\sigma}) - d_*| &< 1/300, & |\delta_r^y(E_d, F_{d,\sigma}) - d_*| &< 1/300 \quad (r \geq 1), \\ |\delta_2^h(E_d, F_{d,\sigma}) - 1/8| &< 1/300. \end{aligned} \quad (15.5)$$

In particular private and visible error exceed $9/20$, while hidden two-selector error is less than $13/100$. More generally the errors in these three comparisons are at most $2\Phi(-1/(4\sigma))$, which tends to zero as $\sigma \downarrow 0$. For $0 < d \leq 2/5$ all three deficiencies are zero. The interacting and noninteracting regimes use the same preparation and detector. The conclusions are unchanged by simultaneously scaling positions, diameter and observation times by any positive factor, with velocities and detector noise unchanged.

Proof. Couple the ideal and physical target by the same (B, W) and the same feedback random table. Lemma 15.2 bounds each potential bit error by $\Phi(-1/(4\sigma))$. With probability at least $1 - 2\Phi(-1/(4\sigma))$ both potential reports agree with their ideal bits; the gates and all emitted transcripts then agree as well. This gives a marked feedback variation bound uniform over all gates. The channels in both directions are the identity on the binary report alphabet and have one retained state. The blind source is already exactly E_* . Stateless comparison therefore changes each fixed-budget deficiency by at most the stated error, without adding memory or hiding an exposed selector.

For completeness, the elementary Gaussian tail bound $\Phi(-x) \leq e^{-x^2/2}/(x\sqrt{2\pi})$ at $x = 3$ gives $\Phi(-3) < 1/600$: use $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$. Hence the total error is below $1/300$. The inequalities $d_* - 1/300 > 9/20$ and $1/8 + 1/300 < 13/100$ follow by squaring positive rational bounds.

When $d \leq 2/5$, both potential reports are independent fair noise bits and are independent of W . A private width-one simulator draws those two bits and applies the current gate, exactly reproducing every feedback transcript. Hidden and visible versions can ignore their selectors. Finally elastic scattering is invariant under $(q, d, t) \mapsto (sq, sd, st)$ with v unchanged. This proves the scaling statement; it is geometric similarity, not a particle-number or Boltzmann–Grad limit. $\square$

The theorem combines three effects. Collisions create the sign of the informative transverse velocity. A gate chosen after the first report changes future information. A retained hidden bit, but not a visible one, permits temporal correlation without charging the width-one register. The actual mark remains partially inaccessible in every signature. The conserved longitudinal momentum labels that mark; it does not generate the observed transverse signal.

15.3. Uniformly effective finite presentations. The binary physical interface is already finite, but its probabilities are not generally rational. We make their effective approximation explicit rather than treating individual computability as a uniform algorithm.

Proposition 15.4 (Collision-chart rational presentations). *For computable $d \in [9/10, 11/10]$ and $\sigma > 0$, one algorithm produces rational marked causal instruments and certified two-sided stateless errors tending to zero for $(E_d, F_{d,\sigma})$. On a compact computable parameter interval with $\sigma \geq \sigma_0 > 0$, the errors and the construction are uniform. Thus Proposition 13.6 applies at every fixed finite state and selector signature, not only at $(1, 1)$.*

Proof. On every preparation box the entry time and outgoing velocity are

$$t_c = \frac{-r \cdot g - \sqrt{(r \cdot g)^2 - |g|^2(|r|^2 - d^2)}}{|g|^2}, \quad n = (r + t_c g)/d, \quad v'_1 = V + g/2 - (g \cdot n)n.$$

The positive square root is separated from zero: it equals $-d(g \cdot n) > (9/10)(1443/1000)$. All denominators are likewise bounded away from zero. These explicit bounds and the compact boxes allow interval differentiation of the displayed formulas, producing a rational common Lipschitz upper bound C_{geo} for $v'_{1,y}$ in the packet variables and d . This computation terminates because only addition, multiplication, division by separated positive quantities and a separated square root occur.

Conditional on the microscopic preparation the two potential all-on bits are independent Bernoulli with probability $p = \Phi(h(v'_{1,y})/\sigma)$. Their four probabilities are p^2 , $p(1-p)$, $p(1-p)$ and $(1-p)^2$. The normal distribution function is uniformly computable on bounded intervals, for example by integrating its convergent exponential series with a remainder bound. Its Lipschitz constant is less than one. Together with the 1-Lipschitz detector and the geometric bound, this yields a computable Lipschitz constant at most $2C_{\text{geo}}/\sigma_0$ for these four integrands on fixed-noise compact parameter sets. Equal-box Riemann sums of mesh h therefore have explicit error at most a computable constant times h , plus the chosen rational evaluation error. Parameter uncertainty is included by its interval enclosure.

Integrate the four all-on probabilities separately on each packet, with the exact rational weights (15.1). For each W , round its conditional probability vector to a rational simplex point on a sufficiently fine grid; the rounding error is bounded by the sum of the coordinate errors and the grid mesh. Keep $\mathbb{P}(W = 0) = \mathbb{P}(W = 1) = 1/2$ exact. Sample this finite joint law once at initialization, emit its first bit, and gate its second bit. This is a finite latent realization of the target instrument, not a simulator with extra free state. It makes every causal prefix consistent and never divides by a possibly small prefix probability. Every gated law is a pushforward of the all-on law, so the same marked variation bound holds uniformly under feedback. Identity report channels give both stateless directions. Choosing summable evaluation and mesh errors produces the asserted uniform sequence. $\square$

16. UNIFORM STABILITY OF OPTIONAL PROJECTIONS

Marked comparison controls more than terminal bounded risk when the compared laws share their latent variable. We prove a whole-process estimate for the resulting posterior martingales. The distinction between a probabilistic coupling and an executable causal channel is important in this section.

Let Z take values in a standard Borel space and let Y take values in a standard Borel canonical path space on $[0, T]$, with a right-continuous observation filtration. Consider equivalent joint probability measures P, Q on (Z, Y) , with the same Z -marginal ν . Use the usual completions, which agree for equivalent output laws. For a bounded measurable latent variable $f(Z)$ write

$$M_t^P = \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y], \quad M_t^Q = \mathbb{E}_Q[f(Z) \mid \mathcal{F}_t^Y].$$

Take their right-continuous martingale versions. All suprema below include T and can be evaluated on a countable dense time set.

Theorem 16.1 (Marked variation controls the whole posterior process). *Let $\varepsilon = \|P - Q\|_{\text{TV}}$ and $|f| \leq 1$. When both posterior functions are evaluated on the same canonical output path,*

$$\mathbb{E}_Q \sup_{t \leq T} |M_t^P(Y) - M_t^Q(Y)|^2 \leq 80\varepsilon. \quad (16.1)$$

There is a coupling of (Z, Y^P) and (Z, Y^Q) which retains exactly the same latent variable and satisfies

$$\mathbb{P}(Y^P \neq Y^Q) \leq \varepsilon, \quad \mathbb{E} \sup_{t \leq T} |M_t^P(Y^P) - M_t^Q(Y^Q)|^2 \leq 84\varepsilon. \quad (16.2)$$

The coupling is not asserted to be causal or to have finite state. The variation bound itself, not this coupling, is used for resource transport.

Proof. Let $D = dP/dQ$, $D_t = \mathbb{E}_Q[D \mid \mathcal{F}_t^Y]$, and set

$$A_t = \mathbb{E}_Q[(D - 1)f \mid \mathcal{F}_t^Y], \quad B_t = D_t - 1.$$

Bayes' formula gives

$$M_t^P - M_t^Q = \frac{A_t - B_t M_t^Q}{1 + B_t}.$$

Since $\mathbb{E}_Q|D - 1| = 2\varepsilon$, both martingales A, B have terminal absolute expectation at most 2ε . Doob's weak L^1 inequality gives, for $0 < u \leq 2$,

$$\begin{aligned} Q\{\sup_t |M_t^P - M_t^Q| > u\} &\leq Q\{\sup_t |B_t| > 1/2\} + Q\{\sup_t |A_t| > u/4\} + Q\{\sup_t |B_t| > u/4\} \\ &\leq 4\varepsilon + 16\varepsilon/u. \end{aligned}$$

Integrating $2u$ times this bound on $(0, 2)$ gives $16\varepsilon + 64\varepsilon = 80\varepsilon$, proving (16.1). No tightness conclusion is being inferred from finite-dimensional convergence.

Disintegrate both joint laws over their common latent marginal. Measurable maximal coupling of the two output kernels, conditional on Z , has disagreement probability

$$\int \|P(dy \mid z) - Q(dy \mid z)\|_{\text{TV}} d\nu(z) = \|P - Q\|_{\text{TV}}.$$

Such a coupling can be constructed by using the minimum of the two kernel densities with respect to their sum, then their residual kernels. It is measurable on standard Borel spaces. On the event of identical full output paths use (16.1) with the Q -marginal; on its complement the squared difference is at most four. This proves (16.2). The maximal coupling may depend on the full output path; it has no asserted online implementation. $\square$

Corollary 16.2 (Changing-filtration convergence from marked comparison). *Let P_n and P be equivalent joint laws with the same latent marginal ν , and $\|P_n - P\|_{\text{TV}} \rightarrow 0$. They admit a common probability-space realization $(Z, Y, Y^1, Y^2, \dots)$ such that, for every $f \in L^2(\nu)$,*

$$\mathbb{E} \sup_{t \leq T} |\mathbb{E}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}[f(Z) \mid \mathcal{F}_t^Y]|^2 \rightarrow 0.$$

The same realization gives convergence uniformly in time in probability for $f \in L^1(\nu)$. It also permits $f_n \rightarrow f$ in $L^2(\nu)$, respectively in $L^1(\nu)$, in the first conditional expectation.

Proof. Glue the conditional maximal couplings of (Z, Y^n) with the fixed reference (Z, Y) by taking their conditional kernels independently on a countable product. The coupling does not depend on f . For the truncation $f^{(K)} = (-K) \vee (f \wedge K)$, Theorem 16.1 bounds the middle difference in the L^2 norm of the supremum by $K\sqrt{84\epsilon_n}$. Doob's L^2 inequality bounds each truncation tail by $2\|f - f^{(K)}\|_{L^2(\nu)}$. The triangle inequality therefore gives

$$\left\| \sup_t |M_t^{P_n, f} - M_t^{P, f}| \right\|_2 \leq 4\|f - f^{(K)}\|_2 + K\sqrt{84\epsilon_n}.$$

First let $n \rightarrow \infty$, then $K \rightarrow \infty$. For L^1 use instead the weak maximal inequality for each tail and the bounded-case estimate for the middle term. An additional difference $f_n - f$ is handled by the same maximal inequalities under P_n , whose latent marginal remains ν . $\square$

16.1. A microscopic sufficient condition. Let $(\Phi_t)_{t \in \mathbb{R}}$ be an almost-everywhere regular hard-sphere flow on a labeled phase space, with invariant probability μ . Fix a prepared law ν with $d\nu/d\mu \leq C$, and bounded real observables h, h_n . These may be collision-sensitive tagged observables. Put $\eta_n = \|h_n - h\|_{L^2(\mu)}$. With a common latent initial state $Z \sim \nu$, consider the continuous observation laws

$$\begin{aligned} Y_t &= \int_0^t h(\Phi_s Z) ds + \sigma B_t, \\ Y_t^n &= \int_0^t h_n(\Phi_s Z) ds + \sigma B_t^n, \quad \sigma > 0, \end{aligned}$$

where the Brownian noise is independent of Z in each model. The notation does not prescribe a synchronous coupling.

Theorem 16.3 (Fixed-flow observation-approximation limit). *The joint latent-output laws are equivalent and satisfy*

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \eta_n. \quad (16.3)$$

For every $|f(Z)| \leq 1$ they admit a latent-preserving coupling with

$$\mathbb{E} \sup_{t \leq T} |M_t^{P_n, f} - M_t^{P, f}|^2 \leq \frac{42\sqrt{CT}}{\sigma} \eta_n. \quad (16.4)$$

In particular $\eta_n \rightarrow 0$ supplies whole-process optional-projection convergence, not merely convergence at finitely many times. This applies when h_n are orbit-mollified graph-core approximants to any bounded $h \in L^2(\mu)$.

Proof. Conditional on Z , the signals are deterministic bounded drifts. The exponential change-of-measure identity for Brownian motion gives

$$D_{\text{KL}}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |(h_n - h)(\Phi_s Z)|^2 ds \leq \frac{CT}{2\sigma^2} \eta_n^2.$$

The last inequality uses $\nu \leq C\mu$ and invariance of μ ; it does not require mixing, differentiability across a collision, or a claim that a tagged velocity belongs to the generator domain. Bounded drifts ensure equivalence and justify the exponential identity. Pinsker's inequality gives (16.3). Applying Theorem 16.1 proves (16.4).

For the last assertion choose a nonnegative smooth approximate identity ρ_n on the time line and set $h_n = \int \rho_n(s) h \circ \Phi_s ds$. Strong continuity of the Koopman group in $L^2(\mu)$ gives $\eta_n \rightarrow 0$ and $\|h_n\|_\infty \leq \|h\|_\infty$. Repeated integration by parts puts h_n in every power domain of the generator,

with derivatives obtained from $\rho_n^{(k)}$. This is the orbit-core construction already used in the microscopic domain theory; the new conclusion is the uniform posterior estimate. $\square$

Sampling these observation paths at finitely many checkpoints and applying a common causal output postprocessing are contractions in marked variation. Thus the identity report comparison transfers every bounded finite-register risk at its original state and selector budgets with error bounded by (16.3). That is the resource-comparison consequence. The latent-preserving coupling in (16.4) has a different role: it realizes convergence of the changing observation filtrations and is not used as a finite-register simulator.

The theorem discharges a sufficient conditional-law hypothesis for optional projection in a genuine microscopic observation class. The constants depend on the fixed preparation, time and noise level. No particle-uniform kinetic estimate, changing microscopic path limit, limiting stochastic-exponential identification or nonlinear BSDE limit is implied without its additional hypotheses. The whole-process estimate is what replaces an unsupported appeal to Doob's inequality as a tightness argument.

17. CHANGING TRAJECTORIES, LIKELIHOODS, AND CONDITIONAL CONSUMERS

17.1. A common preparation, not a common flow. Let $(\mathcal{Z}, \nu)$ be a standard Borel probability space, and let $H, H_n : [0, T] \times \mathcal{Z} \rightarrow \mathbb{R}$ be jointly measurable, with $|H|, |H_n| \leq M$. They may be observations of different flows on different phase spaces, pulled back by specified preparation maps from $\mathcal{Z}$. No common microscopic generator or invariant measure is assumed. Set

$$\eta_n^2 = \int_{\mathcal{Z}} \int_0^T |H_n(t, z) - H(t, z)|^2 dt d\nu(z).$$

On $\mathcal{Z} \times C([0, T])$ let R be the law of independent $Z \sim \nu$ and $Y = \sigma B$, where $\sigma > 0$. Define

$$L_t^n = \exp \left\{ \frac{1}{\sigma} \int_0^t H_n(s, Z) dB_s - \frac{1}{2\sigma^2} \int_0^t H_n(s, Z)^2 ds \right\}, \quad (17.1)$$

$$dP_n = L_T^n dR. \quad (17.2)$$

The reference model P, L uses H . Conditional on Z , these are Gaussian shifts, so under P_n the observed path has law $Y_t = \int_0^t H_n(s, Z) ds + \sigma B_t^n$. Boundedness gives true strictly positive density martingales. Write $\mathcal{F}_t$ for the usual canonical observation filtration and $\mathcal{H}_t = \sigma(Z, B_s : s \leq t)$ for the larger reference filtration. Equivalent laws have the same completed observation stopping times.

Theorem 17.1 (Likelihood and posterior transport for changing paths). *The measures P_n, P, R are equivalent and have common Z -marginal ν . With $A = M^2 T / \sigma^2$ and $\varepsilon_n = \min\{1, \eta_n / (2\sigma)\}$, one has*

$$\|P_n - P\|_{\text{TV}} \leq \varepsilon_n, \quad (17.3)$$

$$\mathbb{E}_R \sup_{t \leq T} |L_t^n - L_t|^2 \leq \frac{8}{\sigma^2} e^{3A} \eta_n^2. \quad (17.4)$$

For $|f(Z)| \leq K$ put

$$N_t^{n,f} = \mathbb{E}_R[L_T^n f(Z) \mid \mathcal{F}_t], \quad \ell_t^n = N_t^{n,1}.$$

Then

$$\mathbb{E}_R \sup_t |N_t^{n,f} - N_t^f|^2 \leq \frac{8K^2}{\sigma^2} e^{3A} \eta_n^2. \quad (17.5)$$

In particular this applies to the projected likelihood ℓ^n . For every $|f| \leq 1$, the posterior functions satisfy the same-path bound $80\varepsilon_n$ of Theorem 16.1, and there is one latent-preserving coupling, independent of f , on which

$$\mathbb{E} \sup_t |\mathbb{E}_{P_n}[f(Z) | \mathcal{F}_t^{Y^n}] - \mathbb{E}_P[f(Z) | \mathcal{F}_t^Y]|^2 \leq 84\varepsilon_n. \quad (17.6)$$

If $\eta_n \rightarrow 0$ and deterministic microscopic paths $X_n(Z) \rightarrow X(Z)$ in a specified Polish path metric in ν -probability, then on this coupling the microscopic paths, the full and projected likelihood processes evaluated on their own observation paths, and each bounded posterior process converge jointly in probability in that metric and uniformly in time for the last three coordinates.

Proof. Conditional Gaussian change of measure gives

$$\text{KL}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |H_n - H|^2 dt = \frac{\eta_n^2}{2\sigma^2}.$$

The same calculation proves conditional equivalence, and the conditional density has mean one, so the preparation marginal is unchanged. Integrating and applying Pinsker proves (17.3).

For the quantitative density estimate, condition first on $Z = z$ and write $D_t = L_t^n - L_t$. The stochastic differential equation is

$$dD_t = \sigma^{-1} \{D_t H_n(t, z) + L_t(H_n - H)(t, z)\} dB_t.$$

The conditional second moment of L_t is at most e^A . Itô's isometry and $(u + v)^2 \leq 2u^2 + 2v^2$ therefore imply

$$\mathbb{E}_R[D_t^2 | z] \leq \frac{2M^2}{\sigma^2} \int_0^t \mathbb{E}_R[D_s^2 | z] ds + \frac{2e^A}{\sigma^2} \int_0^t |H_n - H|^2(s, z) ds.$$

Gronwall, followed by integration in z , gives $\mathbb{E}_R D_T^2 \leq (2/\sigma^2) e^{3A} \eta_n^2$. The difference is a square-integrable $\mathcal{H}_t$ -martingale. Doob's L^2 inequality proves (17.4). Applying the same maximal inequality to the $\mathcal{F}_t$ -martingale $\mathbb{E}_R[(L_T^n - L_T)f | \mathcal{F}_t]$ proves (17.5). These are direct process estimates, not a claim that finite-dimensional convergence alone implies tightness.

The posterior assertions follow from Theorem 16.1 and its countable conditional-coupling construction. For the last claim, on the event $Y^n = Y$ the two likelihood functions are evaluated on the same (Z, Y) . Their uniform difference tends to zero in R -probability by the estimates just proved, and therefore in P -probability by absolute continuity of the fixed P with respect to R . The complementary event has probability at most ε_n . The same argument applies to ℓ^n , and (17.6) handles the posterior. The microscopic convergence holds on any coupling retaining the same Z . A finite union bound proves joint convergence. Stochastic integral versions are chosen measurably on the canonical product space; equivalence makes these common versions valid under every law in the countable sequence. $\square$

For $f \in L^2(\nu)$ the posterior conclusion follows by truncation exactly as in Corollary 16.2; it also permits $f_n \rightarrow f$ in $L^2(\nu)$. Boundedness of H_n is being used for the uniform likelihood second moments. It is not replaced by an unverified uniform-integrability assertion in a singular or zero-noise limit.

17.2. The projected stochastic exponential. Let $\hat{H}_t^n = \mathbb{E}_{P_n}[H_n(t, Z) | \mathcal{F}_t]$, using a predictable version for Lebesgue-almost every t . Put $\theta_t^n = \hat{H}_t^n / \sigma$.

Proposition 17.2 (Identification before passage to the limit). *The projected likelihood and the innovation satisfy*

$$\ell_t^n = \exp \left\{ \int_0^t \theta_s^n dB_s - \frac{1}{2} \int_0^t (\theta_s^n)^2 ds \right\}, \quad (17.7)$$

$$I_t^n = B_t - \int_0^t \theta_s^n ds. \quad (17.8)$$

Here I^n is a P_n -Brownian motion in $\mathcal{F}_t$. In particular the limit of the projected likelihoods in (17.5) is the projected stochastic exponential of the limiting model. Change of measure commutes with this limit on bounded measurable path tests.

Proof. Project $L_t^n = 1 + \int_0^t L_s^n H_n(s, Z) dB_s / \sigma$ onto the Brownian observation filtration. The integrands are square integrable; testing against bounded elementary predictable stochastic integrals and using Itô's isometry gives

$$d\ell_t^n = \sigma^{-1} \mathbb{E}_R[L_t^n H_n(t, Z) | \mathcal{F}_t] dB_t = \ell_t^n \theta_t^n dB_t.$$

The last equality is Bayes' formula. Since $|\theta_t^n| \leq M/\sigma$, the solution is the positive stochastic exponential (17.7); Girsanov gives the stated innovation. The same argument with H identifies the limiting exponential. Finally $|\mathbb{E}_{P_n} F - \mathbb{E}_P F| \leq 2\|F\|_\infty \varepsilon_n$ for bounded signed F . There is no interchange based only on a formal limit of logarithms. $\square$

17.3. A bounded entropic backward-equation consumer. Fix $\gamma > 0$ and a latent payoff f with $|f| \leq 1$. Define

$$U_t^n = \frac{1}{\gamma} \log \mathbb{E}_{P_n}[e^{\gamma f(Z)} | \mathcal{F}_t] = \frac{1}{\gamma} \log \frac{N_t^{n, e^{\gamma f}}}{\ell_t^n}. \quad (17.9)$$

The terminal condition here is the observable conditional certainty equivalent U_T^n , not the generally unobserved $f(Z)$ itself. This specification avoids imposing a latent terminal value on the smaller observation filtration.

Theorem 17.3 (Entropic values and localized backward integrands). *There is a unique bounded entropic value process with terminal value U_T^n ; for a predictable integrand Z^n it satisfies*

$$U_t^n = U_T^n + \int_t^T \frac{\gamma}{2} |Z_s^n|^2 ds - \int_t^T Z_s^n dI_s^n. \quad (17.10)$$

On the coupling of Theorem 17.1,

$$\mathbb{E} \sup_t |U_t^n(Y^n) - U_t(Y)|^2 \leq 84 \left(\frac{e^{2\gamma} - 1}{\gamma} \right)^2 \varepsilon_n. \quad (17.11)$$

For each $r > e^\gamma$ let τ_r^n be the first exit of any of $\ell^n, \ell, N^{n, e^{\gamma f}}, N^{e^{\gamma f}}$ from $(1/r, r)$, capped at T . If $\eta_n \rightarrow 0$, then

$$\mathbb{E}_R \int_0^{\tau_r^n} |Z_s^n - Z_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^{\tau_r^n} |\theta_s^n - \theta_s|^2 ds \rightarrow 0. \quad (17.12)$$

Furthermore $\lim_{r \rightarrow \infty} \sup_n R(\tau_r^n < T) = 0$. Thus the localizations exhaust the observation experiment in probability.

Proof. For a positive bounded latent g , write $N_t^{n, g} = \mathbb{E}_R[L_T^n g | \mathcal{F}_t]$ and $dN_t^{n, g} = a_t^{n, g} dB_t$. Projection of the full integral gives

$$a_t^{n, g} = \sigma^{-1} \mathbb{E}_R[L_t^n g H_n(t, Z) | \mathcal{F}_t], \quad |a_t^{n, g}| \leq (M/\sigma) N_t^{n, g}.$$

The same representation with $g = 1$ gives $d\ell_t^n = b_t^n dB_t$ and $b_t^n/\ell_t^n = \theta_t^n$. Put $N^n = N^{n,e^{\gamma f}}$ and $a^n = a^{n,e^{\gamma f}}$. Itô's formula in (17.9) gives the martingale coefficient

$$Z_t^n = \gamma^{-1} (a_t^n/N_t^n - b_t^n/\ell_t^n).$$

Replacing dB_t by $dI_t^n + \theta_t^n dt$ reduces its drift to $-\gamma|Z_t^n|^2/2$, which proves (17.10). The ratios are bounded by M/σ , so the integrals are well defined. Conversely, exponentiating any bounded solution of (17.10) produces a bounded P_n -martingale with terminal value $e^{\gamma U_T^n}$. Its conditional-expectation representation is unique, and martingale representation then identifies its integrand. This proves uniqueness in the bounded class.

Apply Theorem 16.1 to $g_0 = (e^{\gamma f} - e^{-\gamma})/(e^{\gamma} - e^{-\gamma}) \in [0, 1]$. The map taking its conditional expectation to $\gamma^{-1} \log \mathbb{E}[e^{\gamma f} | \mathcal{F}_t]$ has Lipschitz constant at most $(e^{2\gamma} - 1)/\gamma$. This proves (17.11).

The initial values of N^n and N agree, since Z has the same marginal. Itô's isometry and the terminal version of (17.5) imply

$$\mathbb{E}_R \int_0^T |a_s^n - a_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^T |b_s^n - b_s|^2 ds \rightarrow 0.$$

Before τ_r^n , all denominators lie in $[1/r, r]$, and the limiting numerators a, b are bounded in absolute value by $(M/\sigma)r$. For example,

$$\left| \frac{a^n}{N^n} - \frac{a}{N} \right| \leq r|a^n - a| + (M/\sigma)r^3|N^n - N|.$$

The supremum estimates (17.5), followed by integration in time, prove (17.12). The same inequality applies to b^n/ℓ^n .

Finally $e^{-\gamma \ell^n} \leq N^n \leq e^{\gamma \ell^n}$. Each ℓ^n is a stochastic exponential with uniformly bounded integrand θ^n . Its positive and negative moments of the supremum are bounded uniformly in n : write any power as a stochastic exponential times its bounded compensator and apply Doob's inequality to a higher positive power. The same calculation applies to its inverse. Markov's inequality, the preceding comparison of N^n with ℓ^n , and a union bound prove the exhaustion assertion. This is a localized integrand theorem; it does not assert an unspecified global topology for backward equations with unbounded terminal data or different nonlinear drivers. $\square$

Corollary 17.4 (A changing-filtration stopping consumer). *Let $\psi(t, x) \in [0, 1]$ be measurable and L_ψ -Lipschitz in x , uniformly in t . For a bounded latent f with $|f| \leq 1$, put $M_t^n = \mathbb{E}_{P_n}[f(Z) | \mathcal{F}_t]$. Then*

$$\left| \sup_{\tau \leq T} \mathbb{E}_{P_n} \psi(\tau, M_\tau^n) - \sup_{\tau \leq T} \mathbb{E}_P \psi(\tau, M_\tau) \right| \leq \varepsilon_n + L_\psi \sqrt{80\varepsilon_n}.$$

The same statement for the entropic value U has the additional Lipschitz factor $(e^{2\gamma} - 1)/\gamma$ on the square-root term. The suprema range over the common canonical stopping times; attainment is not needed for the estimate.

Proof. For a fixed stopping time, first change the measure while keeping its $[0, 1]$ reward function, at cost at most ε_n . Then compare the two reward functions on the same canonical path under P , using Theorem 16.1, Cauchy-Schwarz and the Lipschitz bound. The estimate is uniform in the stopping time, so it survives both suprema. Equivalence of the observation laws is what allows the same stopping-time class. The entropic assertion follows from the Lipschitz calculation in the preceding theorem. $\square$

17.4. Resource transport is not the posterior coupling. Apply any common finite causal acquisition map to Y and retain a fixed finite mark $w(Z)$. Data processing makes its marked row variations at most ε_n , uniformly over a declared finite feedback family. If the source and target have respective bounds α_n, β_n , Theorem 2.2 gives same-budget deficiency perturbation at most

$\alpha_n + \beta_n$, and Proposition 10.4 transports a fixed reset audit. A subsequent consumer of width S composed with a simulator of width K has width KS , with the stated selector price. None of these executable claims uses the latent-preserving maximal coupling. That coupling belongs to the proof of conditional-process convergence, not to the finite-state machine.

These results provide a direct changing-trajectory, likelihood, Girsanov and bounded entropic-value input for a conditional-law theory. They do not assert the strict-dual, rigidity, covariant-form or full kinetic conclusions of a larger program without their separate analytic proofs.

Example 17.5 (Why an amplitude bound is not a tightness argument). Let ξ_1, ξ_2 be independent symmetric signs, and, for $n \geq 5$, let $\mathcal{G}_t^n$ reveal ξ_1 at $1/2 - 2/n$ and ξ_2 at $1/2 - 1/n$. On $[0, 1]$ the bounded martingale

$$M_t^n = \mathbb{E}[\xi_1 + \xi_2 \mid \mathcal{G}_t^n]$$

converges at every finite collection of fixed times to the martingale $M_t = \mathbf{1}_{[1/2, 1]}(t)(\xi_1 + \xi_2)$. Nevertheless its laws are not tight in the Skorokhod J_1 topology. On the event $\xi_2 = -\xi_1$, of probability $1/2$, its path has an excursion of height one on an interval of length $1/n$ and is zero outside that interval. The three-time oscillation modulus of such a path is one at every scale greater than $2/n$. This violates the necessary J_1 tightness modulus criterion. All maximal moments are uniformly bounded by 2^p .

This example concerns the inference from finite-dimensional convergence and a maximal amplitude estimate alone. It does not claim that every possible stronger conditional-kernel hypothesis fails. The uniform estimates in Theorem 17.1 remove the need for that inference in the observation class treated here.

18. OBSERVATION DENSITIES AND COUPLED PROCESS VERSIONS

We record three measure-theoretic facts used in the global transport argument. Their purpose is to make explicit the passage from a tilted physical law to an observation law and from two marginal processes to one joint realization.

Lemma 18.1 (Observation-level return from a bounded tilt). *Let $dP^g/dP = g(Z)/c$, where $c = \mathbb{E}_P g$ and $e^{-\gamma} \leq g \leq e^\gamma$. For any observation sub- σ -field $\mathcal{F}$,*

$$\frac{d(P|\mathcal{F})}{d(P^g|\mathcal{F})} = \mathbb{E}_{P^g} \left[\frac{c}{g} \mid \mathcal{F} \right] = \frac{c}{\mathbb{E}_P[g \mid \mathcal{F}]} \quad (18.1)$$

Both this density and its reciprocal lie in $[e^{-2\gamma}, e^{2\gamma}]$. In particular, for any nonnegative $\mathcal{F}$ -measurable V , $\mathbb{E}_P V \leq e^{2\gamma} \mathbb{E}_{P^g} V$.

Proof. For $A \in \mathcal{F}$, integrate c/g over A under P^g to obtain $P(A)$. Conditioning proves the first identity. In the other direction, $d(P^g|\mathcal{F})/d(P|\mathcal{F}) = \mathbb{E}_P[g \mid \mathcal{F}]/c$; equivalence and inversion prove the second. The bounds follow from the bounds on g , its conditional mean, and c . $\square$

Lemma 18.2 (Domination of the common-path measure). *Let (Y', Y) be a coupling on a standard Borel path space, with marginal laws P', P , and let $E = \{Y' = Y\}$. The subprobability measure $\kappa(A) = \Pr(E, Y \in A)$ is dominated by both P and P' . Consequently, for any nonnegative measurable path functional V ,*

$$\mathbb{E}[\mathbf{1}_E V(Y)] \leq \mathbb{E}_P V, \quad \mathbb{E}[\mathbf{1}_E V(Y')] \leq \mathbb{E}_{P'} V.$$

The assertion also holds on a common latent-observation space when the latent coordinate is shared by the coupling.

Proof. The event defining $\kappa(A)$ is contained in $\{Y \in A\}$ and, on E , in $\{Y' \in A\}$. This proves domination on sets. Integration of simple functions followed by monotone convergence proves the rest. Apply the same argument to the product coordinate for the last assertion. $\square$

Lemma 18.3 (Marginal stochastic integrals on an arbitrary coupling). *On a canonical standard Borel observation path space, suppose that under each of two marginal laws there are a continuous adapted process U , a predictable square-integrable process Z , and an innovation I such that, in that marginal's own filtration,*

$$\int_0^t Z_s dI_s = U_t - U_0 + \frac{\gamma}{2} \int_0^t |Z_s|^2 ds, \quad 0 \leq t \leq T. \quad (18.2)$$

Assume Z is bounded, as in the entropic model here. There are measurable versions of the two continuous integral processes which can be pulled back to any coupling of the observation laws. Each pulled-back version has the correct marginal law and satisfies its own identity (18.2). Neither integral thereby acquires a martingale or Itô-integral interpretation in the enlarged joint filtration.

Proof. A continuous process is determined by its values at rational times. Completed-measurable random variables on the canonical space have Borel versions. Choose such versions at the countably many rational times and extend by continuity on the measurable set where a continuous extension exists, using the zero path elsewhere. This gives a Borel $C([0, T])$ -valued version of U . Likewise take a jointly measurable version of Z modulo $dt \otimes P$; its bounded time integral has a continuous path, and its rational-time values give a measurable continuous version by the same construction. The right side of (18.2) is consequently a measurable continuous path functional. Marginal equality at all rational times, and continuity of the stochastic integral, make it indistinguishable from that integral. Repeat for the other marginal. Pullback along the two coordinate maps of any coupling preserves their respective almost-sure identities and laws. No enlargement theorem for innovations is used. $\square$

In Theorem 19.2, Lemma 18.1 is applied with $\mathcal{F} = \mathcal{F}_T$ to the integrated squared coefficient comparison. Lemma 18.2 justifies restricting a same-path estimate to the equality event of the latent-preserving coupling. Lemma 18.3 fixes the meaning of the coupled backward martingale integrals before their difference is estimated. Thus the coupled notation retains the marginal observation filtrations throughout.

19. GLOBAL TRANSPORT OF THE BACKWARD EQUATION

We retain exactly the common-preparation, bounded-signal and positive-noise hypotheses of Section 17. In particular the canonical joint laws P_n, P are equivalent and have the same latent marginal ν . All comparisons on a single observation path use common measurable versions of the conditional expectations. The following estimate avoids division by a small observation density in the final bound.

Lemma 19.1 (An integrated posterior comparison). *Let P, Q be equivalent laws on a standard Borel product carrying a latent variable Z and an observation Y , and let $\|P - Q\|_{\text{TV}} \leq \epsilon$. If $|u(Z)| \leq M$ and $\mathcal{F}$ is an observation sub- σ -field, then, under either $S = P$ or $S = Q$,*

$$\mathbb{E}_S |\mathbb{E}_Q[u | \mathcal{F}] - \mathbb{E}_P[u | \mathcal{F}]| \leq 4M\epsilon, \quad (19.1)$$

$$\mathbb{E}_S |\mathbb{E}_Q[u | \mathcal{F}] - \mathbb{E}_P[u | \mathcal{F}]|^2 \leq 8M^2\epsilon. \quad (19.2)$$

A common latent marginal is not needed for these two inequalities.

Proof. Use a dominating finite measure on $\mathcal{F}$. Write p, q for the densities of the restricted probability laws and a, b for the densities of the signed measures $\mathbb{E}_P[u \mathbf{1}_A], \mathbb{E}_Q[u \mathbf{1}_A]$ on $\mathcal{F}$. Their conditional means are $m_P = a/p$ and $m_Q = b/q$. Outside null sets, $|m_P|, |m_Q| \leq M$, and

$$p(m_Q - m_P) = m_Q(p - q) + (b - a).$$

The integral of $|p - q|$ is at most 2ϵ , by contraction of total variation. The integral of $|b - a|$ is at most $2M\epsilon$, since multiplication by u followed by restriction cannot increase signed variation by

more than M . Integrating the displayed identity proves (19.1) under P . Interchanging the two laws proves it under Q . The squared bound follows from $|m_Q - m_P| \leq 2M$. $\square$

Let $|f| \leq 1$, $\gamma > 0$, and use the notation θ^n, U^n, Z^n, I^n of Proposition 17.2 and Theorem 17.3. Here Z^n denotes the scalar backward integrand, whereas the unadorned latent argument of H_n is the preparation variable. Define the deterministic bounds

$$A_n = \sigma^{-2} \{2\eta_n^2 + 16M^2 T \varepsilon_n\}, \quad (19.3)$$

$$C_n = \frac{2(1 + e^{4\gamma})}{\gamma^2} A_n, \quad (19.4)$$

where $\varepsilon_n = \min\{1, \eta_n/(2\sigma)\}$ as before.

Theorem 19.2 (Global innovation and entropic-integrand transport). *Under the hypotheses of Theorem 17.1, under either physical law $S = P$ or $S = P_n$ on the same canonical path,*

$$\mathbb{E}_S \int_0^T |\theta_t^n - \theta_t|^2 dt \leq A_n, \quad \mathbb{E}_S \int_0^T |Z_t^n - Z_t|^2 dt \leq C_n. \quad (19.5)$$

No stopping at a density level is required. If $\eta_n \rightarrow 0$, both integrals also converge to zero in expectation under the reference law R .

On the single latent-preserving coupling of Theorem 17.1, evaluate each process on its own observation path, denoting these evaluations by a bar. Then

$$\mathbb{E} \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt \leq A_n + 4M^2 T \varepsilon_n / \sigma^2, \quad (19.6)$$

$$\mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt \leq C_n + 16M^2 T \varepsilon_n / (\gamma^2 \sigma^2). \quad (19.7)$$

If $\eta_n \rightarrow 0$, then $\bar{I}^n \rightarrow \bar{I}$ uniformly in probability. Writing $\bar{J}_t^n = \int_0^t \bar{Z}_s^n d\bar{I}_s^n$ for the stochastic integral in its own observation filtration, and $\bar{J}$ for its limit, one has

$$\mathbb{E} \sup_{t \leq T} |\bar{J}_t^n - \bar{J}_t|^2 \leq 2\mathbb{E} \sup_{t \leq T} |\bar{U}_t^n - \bar{U}_t|^2 + \frac{8M^2 T}{\sigma^2} \mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt. \quad (19.8)$$

The first term on the right has the explicit bound in (17.11). Thus the full bounded entropic backward representation, including its martingale integral, is transported globally. The coupling need not make either innovation Brownian in the joined filtration; each integral is defined under its own marginal law.

Proof. For each deterministic t , under P split

$$\sigma(\theta_t^n - \theta_t) = \mathbb{E}_{P_n}[H_n(t, Z) \mid \mathcal{F}_t] - \mathbb{E}_P[H_n(t, Z) \mid \mathcal{F}_t] + \mathbb{E}_P[H_n(t, Z) - H(t, Z) \mid \mathcal{F}_t].$$

Lemma 19.1, conditional Jensen and $(x + y)^2 \leq 2x^2 + 2y^2$ bound its expected square by $16M^2 \varepsilon_n + 2\mathbb{E}_\nu |H_n(t, Z) - H(t, Z)|^2$. Integration gives the first bound in (19.5) under P . Under P_n use the same decomposition with the signal-difference term conditioned under P_n and the law-change term applied to H . The common latent marginal gives the identical bound. Jointly measurable conditional versions, and predictable versions equal for dt -almost every time, give the same integrated statements.

Put $g = e^{\gamma f}$ and $c = \mathbb{E}_\nu g$. Tilt the two full joint laws by $g(Z)/c$, writing P^g, P_n^g . They still have the same latent marginal $\nu_g = g\nu/c$, and their observation signals remain H, H_n . Since $e^{-\gamma} \leq g, c \leq e^\gamma$,

$$\|P_n^g - P^g\|_{\text{TV}} \leq e^{2\gamma} \varepsilon_n, \quad \eta_{n,g}^2 \leq e^{2\gamma} \eta_n^2.$$

Moreover, the reference martingale coefficient $a^{n,g}$ from the proof of Theorem 17.3 satisfies

$$\alpha_t^n := \frac{a_t^{n,g}}{N_t^{n,g}} = \sigma^{-1} \mathbb{E}_{P_n^g}[H_n(t, Z) \mid \mathcal{F}_t].$$

Apply the preceding innovation argument to these tilted laws. It bounds the squared integral of $\alpha^n - \alpha$ by $e^{2\gamma} A_n$ under either tilted physical law. For an observation-measurable nonnegative random variable, passage back to the corresponding original physical law costs at most $e^{2\gamma}$: the full density c/g is bounded by this constant, and so is its observation projection. Consequently

$$\mathbb{E}_S \int_0^T |\alpha_t^n - \alpha_t|^2 dt \leq e^{4\gamma} A_n.$$

The identity $Z^n = (\alpha^n - \theta^n)/\gamma$ proves the second bound in (19.5).

The pointwise bounds, valid for all models, are

$$|\theta^n|, |\alpha^n| \leq M/\sigma, \quad |Z^n| \leq 2M/(\gamma\sigma). \quad (19.9)$$

The squared time integrals are thus uniformly bounded random variables. Convergence in P -probability, followed by absolute continuity of the fixed R with respect to P , gives convergence in R -probability. Boundedness then gives convergence of their R -expectations. This argument establishes global reference convergence without claiming a new explicit reference rate.

For the coupling, let $E_n = \{Y^n = Y\}$, so that $\mathbb{P}(E_n^c) \leq \varepsilon_n$. On E_n both processes are functions of the same observation path. The restricted expectation of their squared difference is at most the corresponding unrestricted expectation under the limiting marginal P . On E_n^c apply (19.9) and integrate for time T . This proves (19.6)–(19.7).

On E_n , the Brownian reference coordinates $Y^n/\sigma, Y/\sigma$ agree, so

$$\sup_{t \leq T} |\bar{I}_t^n - \bar{I}_t|^2 \leq T \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt.$$

Markov's inequality and $\mathbb{P}(E_n^c) \rightarrow 0$ prove uniform convergence in probability. This assertion does not require any moment estimate on the observation difference on the exceptional event.

Finally, the backward equation already proved gives, under each marginal,

$$\bar{J}_t^n = \bar{U}_t^n - U_0 + \frac{\gamma}{2} \int_0^t |\bar{Z}_s^n|^2 ds.$$

The common initial value is $U_0 = \gamma^{-1} \log \mathbb{E}_\nu g$. By (19.9),

$$||\bar{Z}_s^n|^2 - |\bar{Z}_s|^2| \leq \frac{4M}{\gamma\sigma} |\bar{Z}_s^n - \bar{Z}_s|.$$

Subtract the two integral identities, use Cauchy–Schwarz in time and $(x + y)^2 \leq 2x^2 + 2y^2$, and take expectations. This gives (19.8). The integral identity also defines compatible measurable versions on the coupling, without asserting a martingale property for its joint filtration. $\square$

The theorem strengthens the localized conclusion of Theorem 17.3 without changing its terminal condition or its driver. The terminal condition remains the observation-measurable conditional certainty equivalent, not the unobserved latent payoff. Positive fixed noise, bounded signals, and a common preparation are used in the proof. A singular zero-noise limit, an unbounded terminal condition, or a different nonlinear driver would require additional arguments.

20. A UNIFORM CERTIFICATE THROUGH A MOVING COLLISION

We now use all three parts of the argument on one family. The finite marked reference law and the uniform scattering geometry are those of Theorem 15.1 and Lemma 15.2. They are proved above, including the no-collision control. We use a continuous observation of the same detector so that finite reports and conditional processes are generated by one experiment.

Let z denote a preparation in the boxes of (15.3), and let ν be that prepared law. The sign of the impact orientation is $2B - 1$, and the conserved center-of-mass longitudinal sign is the actual mark W . Their correlation is prepared: $\mathbb{P}(B = b, W = w) = 5/16$ for $b = w$ and $3/16$ otherwise. The contact diameter d ranges over $I = [9/10, 11/10]$. Let $X_d(t, z)$ be the two-particle phase path, with outgoing velocities used at the collision, and define

$$H_d(t, z) = h(v_{1,y}^d(t, z)), \quad Y_t^d = \int_0^t H_d(s, z) ds + \sigma B_t^d, \quad 0 < \sigma \leq 1/24.$$

The Brownian noise is independent of the preparation. The source experiment has the same physical preparation and mark but constant reports. The target reports at times 2 and 3 are

$$Z_1^d = \mathbf{1}\{2(Y_2^d - Y_{3/2}^d) > 0\}, \quad Z_2^d = A\mathbf{1}\{2(Y_3^d - Y_{5/2}^d) > 0\}. \quad (20.1)$$

The acquisition gate $A \in \{0, 1\}$ is chosen after Z_1^d ; it does not act on the mechanical trajectory. The continuous path is the auxiliary observation experiment for conditional consumers; the finite simulator is not given this path or an uncharged buffer for computing its increments. The target sensor implements the declared window acquisition.

Lemma 20.1 (Quantitative change of the microscopic path). *For all preparations in the specified boxes, the collision time $t_c(d)$ and outgoing tagged velocities obey*

$$|t_c(d') - t_c(d)| \leq |d' - d|, \quad |v_i^+(d') - v_i^+(d)| \leq 30|d' - d| \quad (i = 1, 2).$$

Writing $\Delta = |d' - d|$, one has

$$\sup_z \int_0^3 |H_{d'}(t, z) - H_d(t, z)|^2 dt \leq \Delta + 2700\Delta^2. \quad (20.2)$$

The phase paths converge uniformly over the preparation in the Skorokhod J_1 metric as $d' \rightarrow d$. More precisely, for the product metric taking the maximum of the per-particle Euclidean position and velocity distances, a time change aligning the collision times gives J_1 distance at most 100Δ .

Proof. Write r and g for the incoming relative position and velocity, and $n(d) = (r + t_c(d)g)/d$ for the contact normal. The collision root satisfies $|r + t_c g| = d$. The nongrazing estimate from Lemma 15.2 is $-g \cdot n > 1443/1000$. Differentiating the root with respect to d gives

$$t'_c(d) = \frac{1}{g \cdot n(d)}, \quad |t'_c(d)| < 1.$$

The incoming bound is $|g| < 3$, so

$$|n'(d)| \leq \frac{|t'_c(d)| |g| + 1}{d} < 5.$$

Since $v_1^+ = V + g/2 - (g \cdot n)n$ and $v_2^+ = V - g/2 + (g \cdot n)n$, differentiation bounds either velocity derivative by $2|g| |n'| < 30$.

Before collision the detector is zero, and after collision it is a constant of absolute value at most one. Its Lipschitz constant is one. The two signals therefore differ on the interval between the two collision times by at most one and afterwards by at most 30Δ . The intervening interval has length at most Δ , and the whole observation lasts three units. This proves (20.2).

Choose the increasing piecewise affine time change fixing 0, 3 and sending $t_c(d)$ to $t_c(d')$. Its uniform displacement is at most Δ . Aligned velocities differ by at most 30Δ , with the same

outgoing convention at the jump. Each particle's speed is below four and its velocity jump has norm below three. Without time change, the position difference is bounded by $3\Delta + 3(30\Delta) = 93\Delta$, by integrating the velocity difference before and after the two jump times. Aligning time adds at most 4Δ . The asserted bound follows, with a harmless rounding to 100Δ . $\square$

Theorem 20.2 (Microscopic-to-audit and conditional-law composition). *For the single family (20.1), all of the following hold.*

(a) *At the same width-one private register,*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \quad (r \geq 1), \quad \delta_2^h < 13/100,$$

uniformly for $d \in I$ and $0 < \sigma \leq 1/24$. When $d \leq 2/5$ the detector is zero on $[0, 3]$, and all three finite-experiment deficiencies are zero.

(b) *There is one explicit rational reset-audit rule with one all-on training row, eight marked outcomes and $n = 2800$ training runs, whose expected validation score exceeds $2/5$ against every private width-one simulator throughout that same parameter region. In detail, for counts C choose an event $A_C \subseteq \{0, 1\}^3$ maximizing*

$$Q_*(A_C) - C(A_C)/2800, \tag{20.3}$$

with lexicographic tie breaking. Here Q_ is the rational marked reference law of (B, B, W) . The independent validation uses the actual physical target, not Q_* . This finite rule has at most $\binom{2807}{7}$ training-count states at any scheduled epoch; a conservative complete auditor bound is $8(256 + 1)\binom{2807}{7}$. The state cost belongs to the auditor. The certificate is an expected-score inequality, not a confidence claim about one realized score.*

(c) *Fix $\sigma > 0$ and let $d_n, d \in I$ with $d_n \rightarrow d$. Set*

$$\epsilon_n = \min \left\{ 1, \frac{\sqrt{|d_n - d| + 2700|d_n - d|^2}}{2\sigma} \right\}. \tag{20.4}$$

The joint preparation–continuous-observation laws differ by at most ϵ_n . Every finite resource deficiency above changes by at most ϵ_n , and a fixed audit changes by at most ϵ_n because its source behavior is unchanged. On a latent-preserving coupling, $X_{d_n} \rightarrow X_d$ in J_1 and each bounded posterior converges in supremum L^2 with bound $84\epsilon_n$. The full and projected likelihoods converge in reference supremum L^2 , and jointly in probability with the microscopic paths on that coupling. The projected stochastic exponentials, bounded entropic backward values, localized integrands and stopping consumers have the conclusions of Proposition 17.2, Theorem 17.3, and Corollary 17.4.

(d) *On compact parameter sets with $\sigma \geq \sigma_0 > 0$, the finite marked probabilities have uniform rational enclosures. The observable chart, strict same-budget polynomial verification and rational upper tables are therefore effective in the exact sense of Propositions 14.1 and 13.6.*

Proof. Every collision occurs before time $3/2$. Conditional on preparation, the two normalized window increments in (20.1) are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. The outgoing detector has sign $2B - 1$ and absolute value greater than $1/4$. Coupling each noisy report to B gives total marked variation less than $1/300$ from the exact finite marked experiment, uniformly for every acquisition gate. The finite calculation of Theorem 15.1 gives

$$d_* = \sqrt{5/2} - 9/8, \quad \delta_*^p = \delta_*^v = d_*, \quad \delta_{*,2}^h = 1/8.$$

Thus private and visible values exceed $d_* - 1/300 > 9/20$, and the hidden value is less than $1/8 + 1/300 < 13/100$. The two-valued hidden witness stores its selector, not an observed mark. In the no-collision regime both window reports before gating are independent fair noise, independent of W . Fresh private coins reproduce them exactly, including feedback gates. This proves (a).

For (b), use only the all-on feedback row. Any private width-one simulator in this row emits independent bits with probabilities p, q , independently of the actual W . The calculation in Theorem 15.1 already gives minimum variation d_* for this row. Its alphabet (z_1, z_2, W) has eight elements, hence $D = 7$. The explicit empirical audit in Theorem 10.1, trained using Q_* , has expected ideal score at least

$$d_* - \sqrt{7/2800} = d_* - 1/20.$$

The candidate training and validation laws do not change when the target is replaced by the physical target. Conditional on any selected event, its target probability changes by less than $1/300$. Therefore the actual score is strictly greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (20.5)$$

For completeness, the last strict inequality is the positive-root comparison $\sqrt{5/2} > 947/600$; its squared margin is $5/2 - (947/600)^2 = 3191/360000 > 0$. All probabilities in Q_* are 0, $3/16$ or $5/16$, so (20.3) is a finite rational rule independent of d, σ, p, q . At training epoch k the number of eight-category count vectors is $\binom{k+7}{7} \leq \binom{2807}{7}$. A buffer of eight states records the current two reports and terminal mark; 256 event masks and an idle symbol suffice during event selection and validation. The conservative product bound follows. The clock and the source-reset interface are the declared resources. In particular the proof does not enumerate or claim to store a small complete coefficient table.

For (c), Lemma 20.1 gives the signal-distance hypothesis of Theorem 17.1, with $T = 3$ and $M = 1$. It also supplies its microscopic J_1 hypothesis. That theorem gives (20.4) and all the joint process assertions. The finite acquisition is a common measurable causal map of the continuous observation, so marked variation contracts for every gate. The source is blind and independent of d . Consequently its executable candidate behaviors are identical across d , and Proposition 10.4 has $\alpha = 0$ and $\beta = \epsilon_n$. The same same-budget comparison applies in each visible fiber and in each hidden mixture. The exponential and backward claims now follow from the identified projected density and the explicitly bounded entropic consumer. This is where the finite certificate and the conditional-law theorem meet the same microscopic approximation.

Finally, the collision root is uniformly separated from grazing, its formula is algebraic on the rational preparation boxes, and the normalized window noise has variance $2\sigma^2$ bounded away from zero on the stated compact sets. The box subdivision and Gaussian-tail enclosure argument in Proposition 15.4 therefore applies with that variance. It returns rational enclosures with a computable uniform error. The rational quotient algorithm and same-budget strict certificate procedure then prove (d). Uniformity as $\sigma \downarrow 0$ is not asserted. $\square$

The all-on audit in part (b) is a single witness for the lower bound in the full feedback problem: a maximum over feedback trees includes this row. The visible lower bound is instead the exact fiber theorem; no claim is made that an auditor with a seed-independent state bound can store an arbitrarily large visible-seed transcript. This separates the two resource budgets rather than silently moving cost from one to the other.

The finite certificate is also preserved under an additional rational presentation error ρ : its lower score decreases by at most ρ when only the target is changed, and by the reset-cost estimate when the source is changed. Thus the certificate is an effective consumer of the physical presentation theorem, not an isolated exact finite example.

Theorem 20.2 and the preceding results prove Theorem C.3. The C2-type consequence is a joint changing-trajectory/likelihood/posterior theorem with an identified stochastic exponential and a fully specified bounded entropic backward consumer. Nothing in that conclusion identifies geometric similarity with kinetic scaling, or supplies particle-uniform recollision estimates by renaming this fixed-particle model.

21. SHARP DECISION MEMORY FOR THE COLLISION EXPERIMENT

We now apply the continuation spectrum to the same marked experiment that supplies the microscopic certificate. Both its impossibility statement and its construction concern the same score, the same private candidate family, and the same training–validation checkpoint.

21.1. The marked family and its first two spectral values. In the all-on row of Section 20, a private width-one candidate has two independent report bits with probabilities $p, q \in [0, 1]$, independently of the fair actual mark. Denote its law by $b(p, q)$ and put

$$a = (1 - p)(1 - q), \quad c = pq.$$

Let $B = \{b(p, q) : p, q \in [0, 1]\}$. The ideal target Q_* is supported on 000, 001, 110, 111, with masses $5/16, 3/16, 3/16, 5/16$, respectively. On these four atoms the candidate masses are $a/2, a/2, c/2, c/2$. The elementary continuation constraint implies

$$\sqrt{a} + \sqrt{c} \leq 1, \quad \min(a, c) \leq \frac{1}{4}. \quad (21.1)$$

Indeed the first inequality is Cauchy–Schwarz applied to $(\sqrt{1-p}, \sqrt{p})$ and $(\sqrt{1-q}, \sqrt{q})$; the second follows from the first. If $a \geq c$, then $c \leq 1/4$ and

$$a + 3c \leq (1 - \sqrt{c})^2 + 3c = 1 - 2\sqrt{c} + 4c \leq 1. \quad (21.2)$$

The symmetric inequality holds when $c \geq a$.

Theorem 21.1 (A sharp two-state obstruction). *For the ideal marked collision family,*

$$\mathfrak{V}_1(B, Q_*) = \frac{1}{8}, \quad \mathfrak{V}_2(B, Q_*) = \frac{3}{8}. \quad (21.3)$$

If $\|Q - Q_\|_{\text{TV}} \leq \beta$, every reset auditor retaining at most two states at the training–validation cut has worst-case expected score at most $3/8 + \beta$, regardless of its finite training length and its upstream or downstream memory. The assertion allows mark-dependent validation events and stochastic state transitions.*

Proof. For one state, use the event $\{000, 111\}$. Its score is $5/8 - (a + c)/2 \geq 1/8$. Conversely, average the two candidates with $(a, c) = (1, 0)$ and $(0, 1)$. Their average is uniform on the four supported atoms and is at total variation distance $1/8$ from Q_* . Every fixed response has average score at most $1/8$, so its minimum over the two candidates is at most $1/8$. This proves the first equality; the average is a bound on scores, not an allowed private candidate.

For the upper bound with two states, consider the three candidates $b_0 = b(0, 0)$, $b_1 = b(1, 1)$ and $b_* = b(1/2, 1/2)$. A response may be set to zero outside the support of Q_* without decreasing its payoff at any candidate. Write its four remaining coordinates as (x, y, z, t) . Its three payoffs are

$$\begin{aligned} f_0 &= (-3x - 5y + 3z + 5t)/16, \\ f_1 &= (5x + 3y - 5z - 3t)/16, \\ f_* &= (3x + y + z + 3t)/16. \end{aligned}$$

Since all four coordinates are in $[0, 1]$,

$$f_0 + f_1 \leq \frac{1}{4}, \quad f_0 + f_* \leq \frac{3}{4}, \quad f_1 + f_* \leq \frac{3}{4}. \quad (21.4)$$

Suppose two responses covered all three candidates with payoff strictly greater than $3/8$. The first inequality forces different responses to cover b_0 and b_1 . The other two inequalities then force both responses to have payoff less than $3/8$ at b_* . This is a contradiction. Hence $\mathfrak{V}_2 \leq 3/8$.

For attainment take the responses, in the same four-atom order,

$$h_0 = (2/3, 0, 1, 1), \quad h_1 = (1, 1, 0, 2/3),$$

with zero response elsewhere. Both have target mean $17/24$. Their payoffs are $17/24 - a/3 - c$ and $17/24 - a - c/3$. If $a \geq c$, the first is larger and (21.2) makes it at least $17/24 - 1/3 = 3/8$.

The other case is symmetric. This proves equality. For the perturbed target, apply (4.4) and the operational upper bound of Theorem 4.2. $\square$

The obstruction has an immediate continuation interpretation. Each post-training state supplies one validation response. The three witness candidates in the proof cannot be covered above $3/8$ by two such responses. No increase of sample size can compensate for losing the third continuation at that cut. This argument does not use the unrelated Bernoulli majority calibration.

Lemma 21.2 (Feedback completion of the physical lower bound). *The upper bound $3/8 + \beta$ in Theorem 21.1 remains valid when the auditor may train on any of the original feedback rows and may select a row–event pair after the decision cut. Here every target row is the declared feedback pushforward of Q , with $\|Q - Q_*\|_{\text{TV}} \leq \beta$. In particular, two states cannot evade the bound by using a different sensor gate during training or validation.*

Proof. It suffices to restrict the adversary to an admissible subfamily. For $p, q \in [0, 1]$, let its first emission be $X_1 \sim \text{Ber}(p)$, and, after gate A , let its second emission be AX_2 , with an independent $X_2 \sim \text{Ber}(q)$ and the independent fair actual mark. These are genuine width-one private rows: the gate is a current input, and the second emission uses no retained copy of X_1 .

For any deterministic gate policy $g : \{0, 1\} \rightarrow \{0, 1\}$, the marked row is the pushforward of $b(p, q)$ under $T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$. The physical target row is the pushforward of Q by the same map, since acquisition does not act on the path. Including the gate in the transcript changes nothing: it is also a deterministic function of the displayed coordinates. For a selected row event, the response pulls back to $\mathbf{1}_A \circ T_g$ on the all-on alphabet. A randomized row–event continuation pulls back to a function $h_m \in [0, 1]^\Omega$. Hence, conditional on cut state m , its expected score on this subfamily is exactly $(Q - b(p, q))h_m$.

Whatever training rows were used, their effect is solely the distribution of the retained state. The two-state score is thus a convex combination of two such responses and is bounded above by their envelope. Apply the three-candidate argument of Theorem 21.1, and then the uniform target perturbation bound. Since this is a subfamily of the full private candidate class, it is a valid lower-bound witness for that full class. Randomized feedback is covered by conditioning on its independent random table and then averaging the pulled-back response. $\square$

21.2. A three-response construction. Define three deterministic events, all ignoring the mark,

$$A_D = \{00, 11\} \times \{0, 1\}, \quad A_0 = \{00\} \times \{0, 1\}, \quad A_1 = \{11\} \times \{0, 1\}.$$

Their ideal scores are $1 - a - c$, $1/2 - a$ and $1/2 - c$.

Lemma 21.3 (The three-response envelope). *For the original private candidate image,*

$$\min_{p, q} \max\{1 - a - c, \tfrac{1}{2} - a, \tfrac{1}{2} - c\} = \sqrt{2} - 1. \quad (21.5)$$

In particular $\mathfrak{V}_3(B, Q_) \geq \sqrt{2} - 1$.*

Proof. By symmetry suppose $a \geq c$. Then $c \leq 1/4$ and the maximum is $1 - a - c$ for $a \leq 1/2$, and $1/2 - c$ for $a \geq 1/2$. If $a \leq 1/4$, the first expression is at least $1/2$. If $1/4 \leq a \leq 1/2$, (21.1) gives

$$a + c \leq 1 - 2\sqrt{a} + 2a \leq 2 - \sqrt{2},$$

since the last function is increasing on $[1/4, 1/2]$. If $a \geq 1/2$, then $c \leq (1 - \sqrt{a})^2 \leq (1 - 1/\sqrt{2})^2 = 3/2 - \sqrt{2}$. Both cases give the asserted lower bound. It is attained at $p = q = 1 - 1/\sqrt{2}$, where $a = 1/2$ and $c = 3/2 - \sqrt{2}$. The symmetric point gives the same value. $\square$

Lemma 21.4 (Odd-sample majority selection with a uniform regret bound). *Let $n = 2h - 1$ be odd. In n independent candidate trials, let C_0, C_1 count report pairs 00, 11. Choose A_1 if $C_0 \geq h$,*

choose A_0 if $C_1 \geq h$, and otherwise choose A_D . The first two cases are mutually exclusive. Its expected ideal score satisfies

$$g_n(p, q) \geq \sqrt{2} - 1 - \frac{1}{4\sqrt{en}} - \frac{1}{4}e^{-n/8}. \quad (21.6)$$

Its event label has three possible values at the decision cut.

Proof. Because $a + c \leq 1$,

$$\max\{1 - a - c, \frac{1}{2} - a, \frac{1}{2} - c\} = 1 - a - c + (a - 1/2)_+ + (c - 1/2)_+.$$

The selected payoff is $1 - a - c + (a - 1/2)\mathbf{1}_{\{C_0 \geq h\}} + (c - 1/2)\mathbf{1}_{\{C_1 \geq h\}}$. Its expected loss from this envelope is exactly the sum, for $u = a, c$, of $|u - 1/2|$ times the probability that the binomial majority selects the wrong side of $u = 1/2$. At $u = 1/2$ the contribution is zero. No independence between C_0 and C_1 is used.

For $0 < u < 1/2$ and $k \geq h$, the ratio of the binomial mass at u to that at $1/2$ is

$$[4u(1 - u)]^{n/2} \left(\frac{u}{1 - u} \right)^{k - n/2} \leq [4u(1 - u)]^{n/2}.$$

The fair binomial upper tail is exactly $1/2$ because n is odd. Reflecting k treats $u > 1/2$. The endpoint cases follow directly or by continuity. Thus the wrong-side probability is at most

$$\frac{1}{2}[1 - 4(u - 1/2)^2]^{n/2} \leq \frac{1}{2}e^{-2n(u - 1/2)^2}.$$

For $x \geq 0$, the maximum of $xe^{-2nx^2}/2$ is $1/(4\sqrt{en})$, reached at $x = 1/(2\sqrt{n})$. At least one of a, c is at most $1/4$ by (21.1); that coordinate has $|u - 1/2| \in [1/4, 1/2]$ and contributes at most $e^{-n/8}/4$. Bound the other coordinate by the unrestricted maximum and use Lemma 21.3. $\square$

21.3. Simultaneous exact widths and the full acquisition schedule. The preceding construction admits more compression than storing its two raw counts. Write $0, 1, *$ for the three report-pair categories $00, 11$ and the two off-diagonal pairs. For $n = 2h - 1$, let F_n be the terminal event-label function of this word. At a trial cut t , let $r = n - t$ and let (u, v) be the two prefix counts. Associate the residual key

$$\mathcal{R}_t(u, v) = \begin{cases} \text{L}, & u \geq h, \\ \text{R}, & v \geq h, \\ (\min(h - u, r + 1), \min(h - v, r + 1)), & u, v < h. \end{cases} \quad (21.7)$$

The symbols L, R mean the constant decisions A_1, A_0 . In a pair key, a coordinate $r + 1$ means that this majority is no longer reachable. The pair $(r + 1, r + 1)$ represents the constant decision A_D .

Theorem 21.5 (Exact profile of the physical audit rule). *The keys in (21.7) are exactly the distinct continuation residuals of F_n . Their number at trial cut t is*

$$r_t(F_n) = \begin{cases} \binom{t+2}{2}, & 0 \leq t \leq h - 1, \\ \binom{n-t+3}{2}, & h \leq t \leq n. \end{cases} \quad (21.8)$$

This is the minimum simultaneous width profile for exact computation of this rule, for deterministic and stochastic machines alike. Its maximum is $\binom{h+2}{2}$ at $t = h$.

For $n = 399$, the maximum trial-cut width is 20301, and the terminal training cut has exactly three states. Acquiring each report pair one bit at a time, discarding its training mark, and then executing independent candidate and target validations gives a deterministic machine with at most $40602 < 2^{16}$ states at every scheduled epoch. One additional ignored training trial may be inserted to give the same 400 training slots and 402 total trials as the earlier certificate. It changes neither the payoff nor these width bounds.

Proof. If a majority has already been reached, its label cannot change. Otherwise, the first coordinate of the pair key is exactly the number of additional 0 symbols needed, with the unreachable threshold clipped to $r + 1$; the second coordinate has the analogous meaning. These data determine the suffix function. Two different reachable pair keys can be distinguished by a suffix containing only one of the two symbols and enough $*$ symbols to fill the remaining positions: use a count between the two different thresholds. The other majority cannot be reached by this suffix. The constant-majority keys are distinct from every pair key because an all- $*$ suffix returns A_D for the latter. Thus the keys are exactly the residuals.

For $t < h$, no majority has been reached and every threshold is at most $r + 1$. The distinct pairs (u, v) with $u + v \leq t$ give $\binom{t+2}{2}$ keys. For $t \geq h$, the two constant-majority keys are reachable. The remaining keys are precisely

$$(i, j) \in \{1, \dots, r + 1\}^2, \quad i + j \geq r + 1.$$

To see surjectivity, use $u = h - i$, $v = h - j$; these counts are nonnegative, less than h , and sum to at most t because $2h - t = r + 1$. To see necessity before clipping, use the same inequality on $h - u, h - v$; if either coordinate is clipped, the inequality remains automatic. There are $(r + 1)^2 - (r - 1)r/2$ such pairs, with the expression valid also for $r = 0, 1$. Adding the two constant keys gives $(r + 2)(r + 3)/2$. This proves (21.8). The finite-valued realization theorem, Theorem 8.1, proves both the simultaneous realization and stochastic minimality for the specified rule. The two polynomial pieces of the profile give its maximum as stated.

For an explicit transition definition, order the reachable keys lexicographically, putting L, R first. For each key choose the lexicographically first prefix-count pair representing it. After the next category, increment the appropriate count and apply $\mathcal{R}_{t+1}$. Equality of residual functions implies equality of the shifted residuals, so this definition is independent of the chosen representative. The representative is part of the time-indexed transition table, not another stored register. The clock supplies t but no sample history. All these maps are deterministic and rational.

At a bit-level epoch, the state consists of a residual key and, only between the two report bits, the first bit. It therefore has at most $2 \max_t r_t$ values. On reading the second bit, determine the category, update the key and discard both bits. The mark is read and discarded; all-on controls and resets are prescribed by the free schedule. After training, keep just one of the three event labels. During validation, keep that label, the completed candidate-event indicator, and a first-bit buffer when needed. At most $3 \cdot 2 \cdot 2 = 12$ states suffice there; a completed target pair emits its event indicator minus the stored candidate indicator. The final score has three possible values. These phase-specific alphabets may be embedded in a single alphabet of size 40602 because the schedule clock identifies the phase.

The same trial-cut residuals remain distinct for the complete signed-score function, not merely for an externally emitted event name. Any two of A_0, A_1, A_D have different indicator functions. Take an off-diagonal candidate-validation pair, which belongs to none, and a target-validation pair distinguishing the two events. The resulting scores differ. Thus passing to the validation suffix does not merge the residuals already counted. $\square$

Theorem 21.6 (Matched physical decision complexity and stable certification). *For the two-sphere collision family of Theorem 20.2, let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/24$, and let $Q_{d,\sigma}$ be its all-on marked target. At the charged training-validation cut of the reset interface,*

$$\kappa_{2/5}(B, Q_{d,\sigma}) = 3. \tag{21.9}$$

More concretely, no auditor retaining at most two states at this cut can have expected score greater than $2/5$ uniformly against all private width-one candidates, at any finite training length, even with feedback-row selection as in Lemma 21.2. One rational auditor with three states at this cut does so with 399 informative training trials and two validation trials. With a redundant ignored

training trial, it uses exactly 402 total trials, at most 40602 states at every scheduled epoch, and the exact trial-cut profile (21.8) on its informative training segment.

For fixed positive σ and $d_k \rightarrow d$, the same machine changes its expected score by at most

$$\epsilon_k = \min \left\{ 1, \frac{\sqrt{|d_k - d| + 2700|d_k - d|^2}}{2\sigma} \right\}. \quad (21.10)$$

The private, hidden and visible deficiencies, the no-collision control, and the joint microscopic, likelihood, posterior, innovation and global bounded entropic backward convergence retain all conclusions of Theorems 20.2 and 19.2.

Proof. Theorem 20.2 discharges the physical hypotheses and proves $\|Q_{d,\sigma} - Q_*\|_{\text{TV}} < 1/300$, uniformly on the stated parameters. Theorem 21.1 therefore gives the universal cut-width-two upper bound

$$\mathfrak{V}_2(B, Q_{d,\sigma}) < \frac{3}{8} + \frac{1}{300} < \frac{2}{5}.$$

Lemma 21.2 makes this a bound for the full original feedback-row interface, not only all-on training and not only the constructed rule. The upper construction uses all-on trials and works for every private candidate because only its two all-on parameters enter.

Use the deterministic selector of Lemma 21.4 with $n = 399$, but validate against the actual physical target. For each selected event, changing only the target from Q_* loses less than $1/300$. Its physical expected score is consequently greater than

$$L_{399} := \sqrt{2} - 1 - \frac{1}{4\sqrt{399e}} - \frac{1}{4}e^{-399/8} - \frac{1}{300} > \frac{2}{5}. \quad (21.11)$$

Here is a rational verification of the last strict inequality, independent of decimal evaluation. The bounds $e > 2$, $\sqrt{798} > 28$, $\sqrt{2} > 707/500$, and $399/8 > 49$ imply

$$L_{399} > \frac{207}{500} - \frac{1}{112} - \frac{1}{300} - 2^{-51} > \frac{2}{5},$$

since $207/500 - 1/112 - 1/300 - 2/5 = 73/42000 > 2^{-51}$. The number L_{399} is approximately 0.4032891; the rational inequalities, not this approximation, establish the certificate. Its state realization was proved in Theorem 21.5. The general operational classification now yields (21.9) with both inequalities proved for this same physical task.

Changing d leaves the blind candidate source and all its training laws unchanged. The moving-collision estimate and the common-preparation Gaussian theorem bound the target variation by ϵ_k . For any fixed response its target mean changes by at most this amount; averaging over the unchanged training rule proves (21.10). There is no multiplier equal to the number of training trials. The same physical estimates verify the hypotheses of Theorem 19.2; the observation-level density and coupling versions used there are given in Section 18. This transports the same certificate and the bounded conditional-law consumer along the changing microscopic paths. The no-collision regime $d \leq 2/5$ has the vanishing deficiencies already proved in Theorem 20.2; it is not included in the positive-gap parameter interval. $\square$

21.4. What the two exact complexity statements measure. Equation (21.9) minimizes over every auditor at one specified physical decision cut. Equation (21.8) minimizes all trial-cut widths simultaneously for exact implementation of the stated 402-trial rule. These are different optimizations. The first is a task-level optimum, not an artifact of that rule. The second explains the whole online realization of the same rule, not a claim that every successful auditor must have 20301 states somewhere. The sixteen-bit bound includes bit acquisition and validation but does not bound the description length of time-dependent transition tables. It is an implementation upper bound for peak memory. Independent persistent randomization is charged; fresh randomness from the reset experiments is not a hidden cross-cut memory channel.

The former two-count 400-sample construction and its nineteen-bit bound remain in Section 24. They offer a different approximation to the full marked discrepancy. The new three-response rule does not reproduce that rule or attain its oracle envelope. It instead solves the specified physical certification task with an exactly minimal decision-cut alphabet and an explicit full continuation profile. The score is an expectation, not a confidence guarantee from a single realized validation pair.

22. FULL-PROFILE CERTIFICATION OF THE COLLISION FAMILY

The short audit of Theorem 21.6 is retained with its exact continuation profile. We now give a second audit of the same physical class whose entire acquisition-and-validation register is bounded independently of its accuracy parameters. Its cost is a much longer, explicitly counted, preparation schedule.

22.1. The row-indexed decision invariant. For the original row-event interface define

$$\kappa_c^{\text{rows}}(B, Q) = \min\{K : \mathfrak{V}_K^{\text{rows}}(B, Q) > c\}. \quad (22.1)$$

The minimum is $+\infty$ when the set is empty. This is not an implicit change of the one-row symbol κ_c . By Corollary 4.4, it is the exact minimum decision-cut alphabet for a strict score, allowing a finite training schedule and a post-cut distribution on row-event pairs.

Let W_c^{phys} be the minimum peak width among all finite auditors for the original physical row interface, now charging each report-bit, mark, decision and validation epoch. Its clock is externally scheduled, as in the earlier physical implementation. All persistent random selectors are charged.

Theorem 22.1 (A twelve-state physical audit). *In the physical parameter range of Theorem 21.6, put $\beta = 1/300$. For $0 < \theta < 1/4$ and $0 < \delta < 1$, let L, R be at least the respective quantities in (7.1). There is a deterministic audit with $N = 4LR$ candidate training trials and two validation trials whose expected score is strictly greater than*

$$\sqrt{2} - 1 - \beta - \theta - 2\delta. \quad (22.2)$$

It has at most five states at completed training-trial cuts, ten at intermediate training report-bit cuts, three at the decision cut, and twelve throughout validation. In particular

$$\kappa_{2/5}^{\text{rows}} = 3, \quad 3 \leq W_{2/5}^{\text{phys}} \leq 12. \quad (22.3)$$

One entirely integer schedule takes $\theta = 1/200$, $\delta = 1/1000$, $L = 400$ and $R = 8 \cdot 2^{400}$, so $N = 12800 \cdot 2^{400}$. Its lower bound exceeds $2/5$. The upper bound twelve is not asserted to be the exact minimum integer peak width.

Proof. Use the report-pair probabilities $a = (1-p)(1-q)$ and $c = pq$ from Section 21. In the first $2LR$ training trials run Lemma 7.1 on the indicator of the pair 00. If it returns positive, choose A_1 and ignore the remaining training trials. Otherwise, in the next $2LR$ trials run the same comparison on the indicator of 11, choosing A_0 if positive and A_D if negative. Both phases have fixed lengths, even after an absorbing decision.

A comparison whose parameter is outside its indifference band is correct with probability at least $1 - \delta$. For purposes of analysis, one may define the second comparison on its independent observations even when the implementation ignores them. A union bound gives probability at least $1 - 2\delta$ that neither comparison errs outside its band. Since $\min(a, c) \leq 1/4$ and $\theta < 1/4$, at most one of a, c lies in $[1/2 - \theta, 1/2 + \theta]$.

On this good event the chosen payoff is within θ of

$$E(a, c) = \max\{1 - a - c, 1/2 - a, 1/2 - c\}.$$

Indeed, if one coordinate exceeds $1/2 + \theta$, its associated event is selected and is optimal. If both are below $1/2 - \theta$, A_D is selected and is optimal. In the remaining case the low coordinate's

comparison is negative, and choosing either of the two relevant events costs at most the distance of the near coordinate from $1/2$. The range of the three payoffs at any fixed (a, c) is at most one. The expected ideal score is therefore at least $E(a, c) - \theta - 2\delta$. Lemma 21.3 gives $E(a, c) \geq \sqrt{2} - 1$. Changing the target to the physical target loses strictly less than β , proving (22.2).

The first comparator has four states. The second phase has these four states and one extra absorbing label for a previously chosen A_1 , hence five. Retaining the first report bit doubles that count to ten. On the second bit, determine the relevant indicator, update the block state, and discard both bits. The training mark is read and discarded. At the phase boundary the fixed schedule maps the first decision to either the A_1 label or the initial state of the second comparator. At the end it maps the remaining decisions to A_0 or A_D . During validation keep the chosen event, the candidate-event indicator, and a first-bit buffer when needed. This requires at most $3 \cdot 2 \cdot 2 = 12$ states; all events ignore the mark. These are actual phase-compatible transitions, not a bound only on terminal labels.

For the integer schedule, the error estimate in the proof of Lemma 7.1 is $2e^{-8} < 1/1000$. To check it rationally, $e > 8/3$ and $(8/3)^8 > 2000$. Further $\sqrt{2} > 707/500$, and hence

$$\sqrt{2} - 1 - \frac{1}{300} - \frac{1}{200} - \frac{1}{500} - \frac{2}{5} > \frac{11}{3000} > 0.$$

This proves the strict score claim without numerical optimization. The training count is exactly $4LR$; early absorption does not delete trials from that count.

For the lower bounds in (22.3), every full-profile auditor has its decision cut among its charged cuts. The feedback pushforward subfamily of Lemma 21.2 bounds any width-two decision continuation by $3/8 + \beta < 2/5$, for every finite adaptive training schedule. The all-on construction just given is a member of the original row interface and has three decision labels. This proves the row-indexed equality and both peak inequalities. $\square$

Corollary 22.2 (Two physical resource regimes). *The same physical score threshold $2/5$ admits the two schedules*

<i>training trials</i>	<i>decision width</i>	<i>peak width</i>
399	3	40602
$12800 \cdot 2^{400}$	3	12

followed by two validations. The first schedule may be padded to 400 training trials as before. For either fixed machine, changing d to d_k at fixed positive noise level changes its expected score by at most ϵ_k in (21.10). If the candidate training kernels also change, their additional cost is the training term in Corollary 7.3.

Proof. The first line is Theorem 21.6; the second is Theorem 22.1. In the moving-collision comparison the candidate training source is identical. Conditional on its chosen event, only one target validation mean changes. The common-preparation Gaussian bound therefore contributes ϵ_k , independently of the length of the unchanged training. The general perturbation formula applies when this equality of sources is absent. $\square$

The long schedule is an existence bound on the resource frontier, not a practical substitute for the 399-trial rule. In an autonomous model the clock must also be stored; Proposition 6.3 gives its explicit additional factor. Both schedules use deterministic transitions, so no precision-dependent randomizer is hidden in the physical upper bound. The one-collision construction remains the verified physical realization. No many-particle or zero-noise theorem is inferred from the larger class of finite experiments.

23. CONFIDENCE-LEVEL CERTIFICATION

An expected signed score is an operational separation, not a confidence statement about one validation pair. We now distinguish these two tasks. The null hypothesis below is equality of the

candidate and target validation laws; it need not belong to the private alternative family. All preparations in the theorem are independent resets with one fixed candidate law. The test cannot be given a new adversarial candidate after each reset.

Theorem 23.1 (Confidence lift with a charged accumulator). *Suppose a base reset audit uses n training trials, one candidate and one target validation, and peak width W , with score $D \in \{-1, 0, 1\}$. Assume its expected score is at least $\gamma > 0$ for every alternative. Run J independent copies and reject equality when*

$$\sum_{j=1}^J D_j \geq J\gamma/2. \quad (23.1)$$

Then both the type-I error under equality and the worst-case type-II error are at most $e^{-J\gamma^2/8}$. The whole test has a realization with at most $W(2J+1)$ states and $J(n+2)$ trials. In particular $J \geq 8\gamma^{-2} \log(1/\alpha)$ suffices for both errors to be at most α .

Proof. Under equality, conditional on the training and selected event, the two independent validations have equal means; hence $\mathbb{E}D = 0$. Under each fixed alternative its mean is at least γ . The copies are independent and $D_j \in [-1, 1]$. The exponential-moment estimate for a variable in an interval of length two is $\mathbb{E}e^{s(D_j - \mathbb{E}D_j)} \leq e^{s^2/2}$, by the same tilted-variance argument used in Theorem 4.2. Markov's inequality and minimization in s give $\Pr(J^{-1} \sum (D_j - \mathbb{E}D_j) \geq a) \leq e^{-Ja^2/2}$, and the same lower-tail bound. Set $a = \gamma/2$.

Keep the running integer sum, with at most $2J+1$ values, beside the base machine's register. At each completion add its emitted score and reset only the base register. The sum is not stored in an external history. The clock supplies the repetition index and the fixed terminal threshold; all $J(n+2)$ trials are counted. $\square$

The 399-trial physical audit has mean strictly above $2/5$, so one may use the rational choice $\gamma = 2/5$ in this theorem. This gives the fully explicit sufficient count $J \geq 50 \log(1/\alpha)$ and $401J$ total trials (or $402J$ with the inherited ignored slot). The confidence guarantee uses repetition and its charged sum, not one pair of validation observations.

Theorem 23.2 (One training phase and repeated validation). *For the physical audit of Theorem 22.1, put $\gamma_0 = \sqrt{2} - 1 - 1/300 - \theta > 0$. Retain its chosen event and use m independent candidate–target validation pairs. Reject equality when their score sum is at least $m\gamma_0/2$. The type-I error is at most $e^{-m\gamma_0^2/8}$ and the worst-case type-II error is at most*

$$2\delta + e^{-m\gamma_0^2/8}.$$

The audit uses $4LR + 2m$ trials and at most $12(2m+1)$ states throughout training and validation. Thus $\delta \leq \alpha/4$ and $m \geq 8\gamma_0^{-2} \log(2/\alpha)$ suffice for error at most α .

Proof. The proof of Theorem 22.1 gives an event of training probability at least $1 - 2\delta$ on which the conditional mean score of the selected event is strictly greater than γ_0 . Conditional on that event and its selected label, the validation pairs are independent. Apply the tail bound in Theorem 23.1 conditionally and add the training failure probability. Under equality every selected event has zero mean, regardless of training, so no training failure term is needed for type-I error. Keep the three-valued event label, the within-pair indicator and first-bit buffer, and a sum with $2m+1$ values. Training itself needs no more than ten states, so the displayed bound covers every phase. $\square$

The tests in this section have a different output task from maximizing an expected witness score. The exact three-state decision-cut theorem is not asserted to be an optimal memory theorem for these binary tests. For perturbations of training and validation laws, the coupling argument of Theorem 6.2 bounds changes of each rejection probability by the sum of the per-call total variation

errors. In a source-preserving physical transport, alternatively lower the conditional mean margin by the single-row target variation and reapply the conditional tail estimate. This latter argument transports the confidence guarantee without pretending that an entire m -sample validation law has the same total variation distance as one validation observation.

24. A FINITE-REGISTER CERTIFICATE TRANSPORTED THROUGH THE MICROSCOPIC LIMIT

The general empirical audit in Theorem 10.1 treats every coordinate of a marked row as unknown. The private continuation constraint in the collision experiment forces a smaller family. Exploiting this constraint yields a much smaller executable auditor, without granting it knowledge of the candidate law.

In the all-on row of Section 20, every private width-one candidate emits two independent bits with parameters p, q , independently of the fair actual mark W . Put

$$a = (1 - p)(1 - q), \quad c = pq.$$

The ideal target Q_* is supported on 000, 001, 110, 111, with respective masses $5/16, 3/16, 3/16, 5/16$. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. Every other atom has target mass zero and cannot improve the maximizing event for $Q_* - b$.

Lemma 24.1 (Two-count audit and its exact register implementation). *In $n \geq 1$ independent all-on candidate trials let C_0, C_1 count report pairs 00, 11, respectively, ignoring the training mark. On the four supported atoms use the empirical masses $C_0/(2n), C_0/(2n), C_1/(2n), C_1/(2n)$, and select exactly those atoms whose target mass is strictly greater than this empirical mass. Call the resulting event A_C . For every $p, q \in [0, 1]$,*

$$\mathbb{E}\{Q_*(A_C) - b(A_C)\} \geq \|Q_* - b\|_{\text{TV}} - n^{-1/2}. \quad (24.1)$$

For $n = 400$, the rule is implemented by the two truncated counts $\min(C_0, 250), \min(C_1, 250)$, with comparisons to 150 and 250. Including execution of complete transcripts and the two independent validation trials, it has a deterministic streaming realization with at most 504008 retained states at any scheduled epoch. Thus a nineteen-bit register suffices. The schedule clock is free; no other history store is free. The table description and transition evaluation cost are not asserted to be bounded by nineteen bits.

Proof. On any supported atom let $x = Q_*(w) - b(w)$ and let $\hat{x}$ be the empirical discrepancy. If the sign rule selects a different event from an exact maximizing event, the lost contribution is at most $|x - \hat{x}|$. On unsupported atoms both rules can omit the atom. Summing gives regret at most

$$|C_0/n - a| + |C_1/n - c|.$$

The two counts need not be independent. Their marginal binomial variances and Cauchy–Schwarz give expected regret at most

$$\frac{\sqrt{a(1-a)} + \sqrt{c(1-c)}}{\sqrt{n}} \leq \frac{1}{\sqrt{n}}.$$

This proves (24.1).

For the 00 pair, select both marked atoms when $C_0 < 150$, only the atom of target mass $5/16$ when $150 \leq C_0 < 250$, and neither when $C_0 \geq 250$. The same rule applies to 11, with the other mark being the atom of larger target mass. At equality the zero empirical discrepancy is excluded. Increasing a count beyond 250 cannot change the final choice. Two registers with 251 possibilities each therefore implement the rule, with at most nine possible final event masks.

A buffer with eight possibilities is sufficient to execute and record a completed two-bit-plus-mark transcript. During training the pair of truncated counts and this buffer use at most $8 \cdot 251^2 = 504008$ states. After training, discard the counts and retain the event mask. During validation at most

nine masks, one target-event indicator with two values, and the same eight-state transcript buffer are needed, a total of 144 states. The final signed score takes three values. Since the epoch is known, labels can be reused between these phases; their state counts need not be multiplied. The update maps and threshold comparisons are parameter-independent deterministic maps. They are thus stochastic rows with entries zero or one, or equivalently a compatible continuation cover by Theorem 3.2. Fresh sampling randomness belongs to the specified reset experiments, not to an uncharged auditor register. Finally $504008 < 2^{19}$. $\square$

Theorem 24.2 (Finite-register microscopic certification and global transport). *Consider the marked hard-sphere family of Theorem 20.2, with $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/24$. There is one rational streaming audit, independent of d, σ, p, q , using 400 candidate training resets and one independent candidate and target validation trial, whose expected signed score is strictly greater than $2/5$ against every private width-one candidate. It uses at most 504008 retained auditor states. The candidate remains private width one; the auditor's register is a distinct, charged resource. The physical deficiencies still satisfy*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \ (r \geq 1), \quad \delta_2^h < 13/100.$$

For $d \leq 2/5$ all three finite-experiment deficiencies vanish.

Fix a positive σ and a sequence $d_k \rightarrow d$ within the collision interval. Put

$$\rho_k = |d_k - d| + 2700|d_k - d|^2, \quad \epsilon_k = \min\{1, \sqrt{\rho_k}/(2\sigma)\}.$$

The same 400-reset rule changes its expected score by at most ϵ_k under this perturbation. At the same time the microscopic paths, full and projected likelihoods, posteriors, and bounded entropic values have the joint convergence of Theorem 20.2. Their innovations, backward integrands and backward martingale integrals additionally have all the global conclusions of Theorem 19.2, with $T = 3$, $M = 1$, $\eta_k^2 \leq \rho_k$, and $\epsilon_k \leq \epsilon_k$. On compact parameter sets bounded away from zero noise, the same audit has the effective probability-enclosure and rational-verification properties of Theorem 20.2(d).

Proof. The private width-one continuation restriction gives precisely the candidate family used in Lemma 24.1; the actual mark is not available to its transition rows. The exact finite separation in Theorem 15.1 gives

$$\min_{p,q} \|Q_* - b(p,q)\|_{TV} = d_* := \sqrt{5/2} - 9/8.$$

The scattering and observation estimates proved in Theorem 20.2 put the physical marked target within $1/300$ of Q_* , uniformly in all stated parameters. Train the two-count rule against the fixed rational Q_* , but use the actual physical target for validation. Each selected event's target probability changes by less than $1/300$. Hence its actual expected score is greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (24.2)$$

The last strict inequality is exact: it is equivalent to $\sqrt{5/2} > 947/600$, whose squared margin is $3191/360000$. The rule's executable memory bound was proved in Lemma 24.1. Its total trial count is 402, not 400 once validation is included. The private, hidden and visible deficiency bounds and the no-collision control are inherited from the same physical theorem, rather than inferred from an auditor whose signature was changed.

For changing d_k , Lemma 20.1 supplies the signal modulus ρ_k and the pathwise J_1 bound $100|d_k - d|$. Theorem 17.1 gives the joint marked variation bound ϵ_k and the likelihood/posterior convergence. The candidate source is blind and its behavior does not depend on d . Thus training laws and candidate validation laws are unchanged, and Proposition 10.4 has $\alpha = 0$ and $\beta = \epsilon_k$. There is no factor of 400 in this transport cost. The same preparation and signal bounds verify every hypothesis of Theorem 19.2, proving the global backward assertions. Finally the audit

uses rational thresholds and finitely many deterministic row maps. The positive-noise enclosure theorem therefore applies directly; an additional target-presentation error ζ reduces its score by at most ζ . $\square$

The two-count construction is specific to the private behavior family. A hidden mixture can destroy the independence of its two reports, and its mark relation must still be checked. The bound is therefore not asserted against the hidden class. The visible lower bound has its separate fiberwise proof; an arbitrarily large visible alphabet is not silently stored in the nineteen-bit auditor. The certificate concerns expected score, not a confidence statement about a single realized validation pair. Repeating validation for statistical confidence would incur an additional sample and storage budget.

The theorem uses a single mechanism: the continuation restriction fixes the feasible candidate laws and permits the compressed audit; scattering supplies its marked separation; common-preparation variation transports the same executable test and the global conditional-law consumer through a change in the microscopic trajectory. The collision transduces a prepared impact variable into a signal. It does not create the prepared correlation between that variable and the mark, and it is not a particle-number limit.

25. EXPLICIT PHASE TRANSITIONS FOR THE TWO-COUNT CERTIFICATE

For comparison with the new residual quotient, we give the complete phase-state specification of Lemma 24.1. It realizes the original 400-training-trial rule exactly, not the majority rule of Section 21. Let

$$C = (C_0, C_1) \in \{0, \dots, 250\}^2, \quad \tau(x) = \min(x, 250).$$

The free clock records the training index, the within-trial observation epoch, the validation phase, and deterministic processing epochs. It contains no data-dependent information. At the start, $C = (0, 0)$.

For a completed transcript (x_1, x_2, w) define the deterministic update

$$T(C, x_1, x_2, w) = (\tau(C_0 + \mathbf{1}_{\{(x_1, x_2) = (0, 0)\}}), \tau(C_1 + \mathbf{1}_{\{(x_1, x_2) = (1, 1)\}})).$$

The value of w does not enter this training update. The phase transitions are the following; x denotes the current input, not an earlier unretained report.

Phase	Retained data	Next transition
Training start	C	Read x_1 and keep (C, x_1) .
Second report	(C, x_1)	Read x_2 and keep (C, x_1, x_2) .
Terminal mark	(C, x_1, x_2)	Read w and keep (C, x_1, x_2, w) .
Count update	(C, x_1, x_2, w)	Replace by $T(C, x_1, x_2, w)$; discard the transcript.
Event formation	C	Replace by the event mask $A(C)$ defined below.
Candidate validation	A , partial transcript	Complete the transcript; replace it by $i = \mathbf{1}_A(x_1, x_2, w)$.
Target validation	(A, i) , partial transcript	Complete the transcript and emit $\mathbf{1}_A(y_1, y_2, v) - i$.
Score	A value in $\{-1, 0, 1\}$	Terminate.

The event formation map, including both threshold ties, is

$$\begin{aligned} 000 \in A(C) &\iff C_0 < 250, & 001 \in A(C) &\iff C_0 < 150, \\ 110 \in A(C) &\iff C_1 < 150, & 111 \in A(C) &\iff C_1 < 250. \end{aligned}$$

No other atom is included. For each report pair there are three choices: both marked atoms, only its larger-target-mass atom, or neither. Thus there are at most nine masks. Clipping at 250 preserves every comparison, including equality; no flag for a count exceeding the clipping threshold is required.

At a training epoch the possible states are bounded by $8 \cdot 251^2 = 504008$: the factor eight dominates every partial or complete transcript buffer. At validation epochs the largest alphabet has at most $9 \cdot 2 \cdot 8 = 144$ values, including the already computed candidate-event indicator. Event formation and final emission need at most nine and three states, respectively. Embed each phase alphabet into $[504008]$ by a fixed injective encoding. Because the phase is supplied by the free clock, these encodings need not have disjoint images. The next row decodes the current phase’s encoding and applies the displayed transition. This gives formal deterministic rows with entries zero or one at every epoch. The buffers, event mask, and validation indicator are all charged. No unpriced data store is introduced by label reuse.

APPENDIX A. THE PROOF DEPENDENCY CHAIN

An arrow below means an actual use of a theorem in the displayed proof, not an assertion of priority or an artificial dependency between papers. The preserved exact physical lower and upper bounds meet at the decision cut. The new full-profile theorem supplies an explicit training realization of general decision covers; its proof does not invoke the historical aggregate kinetic claims.

Input	Consequence using that input
Marked private rows and normalized continuations, Theorems 2.2, 3.2	The product behavior image and its validation continuation responses.
Decision continuation spectrum, Theorem 4.2	Universal two-state physical obstruction, Theorem 21.1; the strict-cut classification (21.9).
Product constraint (21.1) and three-response envelope, Lemma 21.3	Uniform finite-sample score, Lemma 21.4.
Finite-valued residual realization, Theorem 8.1	Exact full training profile and bit-level construction, Theorem 21.5.
Marked scattering and moving-collision estimate, Theorem 20.2	Both physical cut bounds and transport of the very same auditor, Theorem 21.6.
Common-preparation Gaussian comparison, Theorem 17.1; the three version lemmas in Section 18	The global conditional-law consumer, Theorem 19.2, on that moving-collision family.

The prior binary audit theorem remains a separate calibration, not a premise of the physical memory lower bound. The original two-count certificate remains a separate full-discrepancy approximation, not a claimed optimal implementation. The source-preserving build, rational checks and finite enumerations accompanying this article test the stated implementation and identities; they are not independent analytic proof certification.

A.1. The full-profile resource chain. The new central arrow is

strict response cover $\longrightarrow$ finite block tournament $\longrightarrow$ rational whole-schedule auditor.

Theorem 7.2 uses the decision spectrum, the finite block lemma and the dyadic sampler. It does not use a training oracle. Theorems 6.1 and 6.2 track approximation, calibration and precision on complete causal schedules. Theorem 9.1 uses one shared capacity budget for all cuts, and Theorem 23.1 retains the score accumulator. Together these establish a finite controlled-experiment instance of the program’s resource–precision–statistical-error objective. A universal

singular-modulus rate or an infinite-horizon kinetic realization is not an input or a consequence of this finite theorem.

The current C2-scoped consumer remains the bounded, common-preparation, positive-noise Gaussian/entropic theorem. The original B4 normalized nonlinear resolvent formula cannot be inherited: its displayed linear difference identity fails on constant inputs. For a separately justified graph H with single-valued $R_\lambda = (I - H/\lambda)^{-1}$, the correctly typed identity is instead

$$R_\lambda h = R_\mu [(\lambda/\mu)h + (1 - \lambda/\mu)R_\lambda h].$$

Indeed both sides solve the same μ -resolvent inclusion. This conditional identity does not establish the graph's range or its kinetic corrector. Those historical targets remain in the program rather than being assigned proof credit from a finite resource theorem.

APPENDIX B. A NAMESPACE MAP OF THE ELEVEN-COMPONENT PROGRAM

The namespaces in this map separate historical paper targets from GTF statements proved here. Every historical component is retained. An interface means that the indicated consumer can be applied after its model-specific hypotheses are proved; it is not an automatic proof of the historical component’s aggregate theorem.

Paper	Retained historical role	GTF relation and precise proof credit
A1	Protocol and finite task reduction	Marked comparison, priced composition and decision-cut certification; no automatic all-order physical response conclusion.
A2	Independent primary geometric chain	Its protocol adapter is historical; the primary geometric chain is not a premise or consequence of a theorem in this article.
A3	Dynamical/potential-space targets	Conditional comparison interface; no new assertion of the historical dynamical closure.
A4	Conditional kernels and forgetting	Historical finite-model consumer; common-preparation posterior estimates feed the current C2-scoped consumer when their hypotheses hold.
B1	Microscopic realization targets	Declared marked experiment interface; the present scattering construction is fixed-particle.
B2	Dynamic action and kinetic targets	Historical action estimates are not used to prove the present decision-cut theorem.
B3	Balance, response and corrector targets	No current proof depends on the historical aggregate balance/kinetic claims.
B4	Nonlinear kinetic semigroup	The historical normalized-resolvent identity and its aggregate downstream claims are not premises here. Their independent repair remains a program obligation.
C1	Statistical acquisition/inference	Original behavior image, finite-reset experiments, and the present response-envelope and microscopic certification consumers.
C2	Duals, response, memory and projections	The current bounded Gaussian/entropic transport replaces the optional-projection step only in its explicitly proved class; it does not certify the historical strict-dual, form-response or rigidity aggregate.
D1	Phase and compact statistical targets	Historical finite-model consumer retained; no new complete phase-theory closure is inferred.

The machine-readable companion records the exact predecessor graph and source identities, including the original B4 and C2 statements inspected at their frozen commit. Names such as `gtf20:decision-cut` and `historical:B4:r17-main` denote different proof obligations. The organizing theorem is a general comparison-and-certification result with an essential physical consumer. It does not replace the entire historical pipeline by a declaration that every aggregate theorem follows from it. The broader program remains intact as a mathematical program.

APPENDIX C. PRESERVED ORGANIZING STATEMENTS

The following two statements retain the earlier organizing labels and claims for cross-reference. Their full proofs are included in the canonical modules above. The unchanged predecessor article and its complete historical development are appended in the separate companion, so that reorganization does not discard earlier derivations.

Theorem C.1 (Decision continuation complexity and physical certification). *The following assertions hold with all persistent data charged to the appropriate machine.*

(i) *The supremum $v_{n,K}$ of worst-case expected scores with n training samples and K states at the decision cut satisfies*

$$0 \leq \mathfrak{V}_K - v_{n,K} \leq \min \left\{ \sqrt{(|\Omega| - 1)/n}, \sqrt{2 \log(2K)/n} \right\}.$$

The least K permitting a strict score greater than c at some finite training length is the least size of a cover of B by sets $\{b : (Q - b)h > c\}$. This spectrum is Lipschitz under uniform total variation perturbations of candidate and target laws.

(ii) *For the ideal marked private product family B of the two-sphere experiment, $\mathfrak{V}_1 = 1/8$ and $\mathfrak{V}_2 = 3/8$. Throughout the collision interval $9/10 \leq d \leq 11/10$, $0 < \sigma \leq 1/24$, the minimum decision-cut width for uniform physical score greater than $2/5$ is exactly three. The lower bound applies to every finite-reset auditor with the specified interface, not merely a particular counter implementation; feedback-row selection does not evade it.*

(iii) *One deterministic rational rule has physical score greater than*

$$\sqrt{2} - 1 - \frac{1}{4\sqrt{399e}} - \frac{1}{4}e^{-399/8} - \frac{1}{300} > \frac{2}{5}.$$

It uses 399 informative training trials, an optional ignored training trial and two independent validations, hence fits the 402-trial schedule. It has exactly three decision-cut states. On its informative training segment, the minimum simultaneous widths for exact stochastic implementation are

$$r_t = \begin{cases} \binom{t+2}{2}, & 0 \leq t \leq 199, \\ \binom{402-t}{2}, & 200 \leq t \leq 399. \end{cases}$$

Their maximum is 20301. Including individual report-bit acquisition and the validations, at most $40602 < 2^{16}$ retained states suffice.

(iv) *At an arbitrary cut of a finite-valued decision, a weighted graph of distinguishing suffixes gives a distribution-sensitive error bound through its minimum monochromatic weight under K labels. Applied to Bayes margins, it yields a quantitative constrained-audit regret bound. At zero error it recovers the exact simultaneous residual profile, even for stochastic implementations.*

(v) *For fixed positive σ and changing d , the same physical rule has the target-variation modulus $\min\{1, \sqrt{|d' - d| + 2700|d' - d|^2/(2\sigma)}\}$, with no training-count factor. The microscopic paths, likelihoods, posterior processes, innovations, and bounded entropic backward values, integrands and martingale integrals have the joint transport conclusions proved below. The private and visible deficiencies exceed $9/20$, whereas a hidden two-valued selector has error less than $13/100$. In the no-collision regime $d \leq 2/5$ all three finite-experiment deficiencies vanish.*

Proof. Part (i) is Theorem 4.2. The exact ideal spectral values are Theorem 21.1; Lemmas 21.3 and 21.4 and Theorem 21.6 prove the matching physical construction. Theorem 21.5 proves (iii), including all acquisition and validation phases. Theorem 8.1 and Corollary 8.2 prove (iv). The microscopic estimates in Theorem 20.2, the explicit coupling-version lemmas in Section 18, and Theorem 19.2 prove (v). $\square$

Theorem C.2 (Continuation complexity and stable causal certification). *With these information conventions the following statements hold.*

(i) *A finite causal kernel is realizable at a width profile if and only if it admits a compatible normalized continuation cover of that profile. For a Boolean terminal test, the exact minimal profile is the number of distinct continuation responses at each cut, even when the implementing*

machine is stochastic. If a cut has $r \geq 2$ distinct responses and only K retained states, its total wordwise error is at least $(r - K)_+ / (r - 1)$.

(ii) For $0 < h < 1/2$, consider $n = 2m + 1$ independent binary samples from one of $\text{Ber}(1/2 - h)$ and $\text{Ber}(1/2 + h)$, and the signed event-discrepancy audit against $\text{Ber}(1/2)$. Its unrestricted value is $h\|P_+ - P_-\|_{\text{TV}}$. The exact training widths needed to attain it are

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n.$$

A smaller width at any cut has the strictly positive value deficit (11.4). Thus the continuation invariant controls both an implementation and a statistical obstruction.

(iii) In the prepared two-sphere experiment of Theorem 24.2, one rational audit using 400 candidate training resets, two validation trials, and at most 504008 retained auditor states has expected discrepancy greater than $2/5$ against every private width-one candidate. The same audit works for every contact diameter $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. The private and visible deficiencies exceed $9/20$, whereas a hidden two-valued selector has error less than $13/100$. Removing the collision, by taking $d \leq 2/5$, makes all three finite-experiment deficiencies zero.

(iv) With positive noise fixed, changing d changes both the collision time and the microscopic path. If $\rho = |d' - d| + 2700|d' - d|^2$, the same audit's expected score changes by at most $\min\{1, \sqrt{\rho}/(2\sigma)\}$. Along $d' \rightarrow d$, the microscopic paths, full and projected likelihoods, and bounded posterior processes converge jointly in the senses specified below. For the associated bounded entropic backward equation, values converge in supremum L^2 and integrands converge globally in time-integrated L^2 , including on the common-latent coupling. Innovations converge uniformly in probability, and backward martingale integrals converge in supremum L^2 .

C.1. Inherited baseline. The following summary is retained from the predecessor. The present article strengthens its memory and backward-integrand conclusions in Theorem C.2; the earlier 2800-trial audit remains valid.

Theorem C.3 (Finite-resource certification and microscopic transport). *The following assertions hold with the conventions just described.*

(i) *Finite causal kernels have a register realization with a prescribed width profile exactly when their normalized continuation laws admit compatible finitely generated polytopes of those widths. At an erasure cut carrying a latent letter of law π , the optimal error with K retained states is*

$$1 - \sum_{i=1}^{\min(K, N)} \pi_{(i)}, \quad \pi_{(1)} \geq \cdots \geq \pi_{(N)}.$$

Independent hidden or visible selectors do not improve this cut value.

(ii) *The reset-audit value is attained, has a finite-dimensional moment minimax representation on the original B , and satisfies*

$$0 \leq \delta(B, Q) - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}, \quad D = \sum_j (|\Omega_j| - 1).$$

The exponent $1/2$ cannot be improved uniformly over compact candidate classes. This assertion is about reset samples, not about the minimal degree of unrestricted polynomial tests.

(iii) *Suppose two prepared microscopic observation models have a common preparation variable, bounded signals H, H' on $[0, T]$, and the same positive Gaussian noise level σ . Set*

$$\eta^2 = \mathbb{E} \int_0^T |H - H'|^2 dt, \quad \varepsilon = \min\{1, \eta/(2\sigma)\}.$$

Their joint marked observation laws differ by at most ε . This estimate transports resource spectra at their stated budgets and transports each fixed finite audit with its explicitly charged reset cost. It

also gives whole-process posterior convergence. The unprojected and projected likelihoods converge in the supremum L^2 norm under the common Wiener reference when $\eta \rightarrow 0$. For bounded latent payoffs, the associated entropic backward equations converge in value, and their integrands converge after the stated denominator localization.

(iv) In the two-sphere model of Theorem 20.2, one rationally specified audit with 2,800 training runs has expected discrepancy strictly greater than $2/5$ against every private width-one simulator, uniformly for $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. A hidden two-valued selector has error less than $13/100$; every visible-selector width-one error exceeds $9/20$. For $d_n \rightarrow d$ the microscopic paths, likelihoods and posterior processes converge in the precise senses of that theorem, and the same audit remains valid. For $d \leq 2/5$ all three deficiencies vanish.

REFERENCES

- [1] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272. doi:10.1214/aoms/1177729032.
- [2] L. Le Cam, Sufficiency and approximate sufficiency, *Ann. Math. Statist.* **35** (1964), 1419–1455. doi:10.1214/aoms/1177700372.
- [3] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28. doi:10.1524/strm.2002.20.14.1.
- [4] A. Nayyar, A. Mahajan and D. Teneketzis, Decentralized stochastic control with partial history sharing: a common information approach, *IEEE Trans. Automat. Control* **58** (2013), 1644–1658. doi:10.1109/TAC.2013.2239000. The theorem numbers cited here refer to the author preprint, arXiv:1209.1695.
- [5] H. S. Witsenhausen, On the structure of real-time source coders, *Bell System Tech. J.* **58** (1979), 1437–1451. doi:10.1002/j.1538-7305.1979.tb02263.x.
- [6] I. Gallagher, L. Saint-Raymond and B. Texier, *From Newton to Boltzmann: Hard Spheres and Short-Range Potentials*, Zurich Lectures in Advanced Mathematics, European Mathematical Society, 2014. The flow and boundary discussion used here is in Sections 4.1–4.3 of arXiv:1208.5753v2.
- [7] C. Widder, J. Zimmer and T. Schilling, On the generalized Langevin equation and the Mori projection operator technique, preprint (2025), revised 2026. arXiv:2503.20457v6.
- [8] C. Widder and T. Schilling, Generalised Langevin dynamics: significance and limitations of the projection operator formalism, preprint (2026). arXiv:2604.20453v2.
- [9] H. F. Trotter, Approximation of semi-groups of operators, *Pacific J. Math.* **8** (1958), 887–919. doi:10.2140/pjm.1958.8.887. Theorem 5.2, p. 902, is the approximation theorem cited here.
- [10] H. Weisshaupt, LeCam’s randomization criterion in the setting of locally convex spaces without lattice structure, EURANDOM report 2002–017, 22 April 2002. Lemma 1 and Theorem 5, pp. 14–16. Author report.
- [11] S. Basu, Algorithms in real algebraic geometry: a survey, author preprint, arXiv:1409.1534 (2014). Published in *Panoramas et Synthèses* **51** (2017), 107–153. Theorem 2.1 and Section 2.1 in the author preprint are used here.
- [12] M. C. Paull and S. H. Unger, Minimizing the number of states in incompletely specified sequential switching functions, *IRE Trans. Electron. Comput.* **EC-8** (1959), 356–367. Publisher record.
- [13] E. Torgersen, *Comparison of Statistical Experiments*, Encyclopedia of Mathematics and its Applications 36, Cambridge University Press, Cambridge, 1991.
- [14] N. Saldi, A topology for team policies and existence of optimal team policies in stochastic team theory, preprint (2017), arXiv:1711.00639.
- [15] S. Yüksel and N. Saldi, Convex analysis in decentralized stochastic control, strategic measures and optimal solutions, author preprint arXiv:1609.07685v2.
- [16] A. Nerode, Linear automaton transformations, *Proc. Amer. Math. Soc.* **9** (1958), 541–544. doi:10.1090/S0002-9939-1958-0135681-9.
- [17] S. Kiefer, Notes on equivalence and minimization of weighted automata, preprint (2020), arXiv:2009.01217v1.
- [18] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176. doi:10.2140/pjm.1958.8.171.
- [19] H. Weisshaupt, A generalization of Le Cam’s randomization criterion, *Studia Sci. Math. Hungar.* **43** (2006), 443–486. doi:10.1556/SScMath.43.2006.4.3. Theorem numbers refer to the author’s 39-page preprint, Randomization.pdf.
- [20] E. de Klerk and M. Laurent, Error bounds for some semidefinite programming approaches to polynomial minimization on the hypercube, *SIAM J. Optim.* **20** (2010), 3104–3120. doi:10.1137/100790835.
- [21] M. Laurent and L. Slot, An effective version of Schmüdgen’s Positivstellensatz for the hypercube, arXiv:2109.09528v2, 2023.

- [22] E. de Klerk and J. C. Vera, The link between 1-norm approximation and effective Positivstellensätze for the hypercube, *Numer. Algebra Control Optim.* **16** (2026), 84–104. doi:10.3934/naco.2024060. Theorem numbers refer to arXiv:2404.04190v2.
- [23] M. Fliess, Séries rationnelles positives et processus stochastiques, *Ann. Inst. H. Poincaré Sect. B* **11** (1975), 1–21. Original article. Theorems 3.1 and 5.4 and Corollary 2, pp. 8–9 and 18–19.
- [24] T. M. Cover, Hypothesis testing with finite statistics, *Ann. Math. Statist.* **40** (1969), 828–835. doi:10.1214/aoms/1177697590. Theorem 2, pp. 833–834, and Section 4, pp. 834–835.

GENERAL THETA FOUNDATIONS I: CONTINUATION COMPLEXITY AND STABLE CAUSAL CERTIFICATION

QIAN QI

ABSTRACT. We study finite marked causal experiments with separately charged simulator and auditor memory. At a training-validation cut, a decision continuation spectrum characterizes the minimum retained alphabet for a strict expected score, with a uniform finite-sample approximation. For the marked private product family arising from a prepared two-sphere collision, its one-state and two-state values are exactly $1/8$ and $3/8$. Three retained states are necessary and sufficient to certify score greater than $2/5$ throughout the stated physical parameter range. A single rational 402-trial audit attains this score; its informative training segment has an exact simultaneous stochastic continuation profile and its complete scheduled implementation uses fewer than 2^{16} states. A weighted separation obstruction converts missing continuation states into distribution-sensitive regret for finite-valued decisions. The same physical certificate and the bounded entropic backward representation are transported along moving collision paths by a common-preparation Gaussian comparison. The full comparison, realization, sampling, effectiveness and conditional-transport proofs are retained, with private, hidden and visible randomization accounted for separately.

CONTENTS

1. Introduction	2
1.1. The decision cut and the principal theorem	3
1.2. The full comparison and transport framework	4
1.3. Relation to classical comparison and realization	4
2. Intrinsic causal deficiency	5
2.1. Experiments and executable signatures	5
2.2. The finite comparison theorem	6
2.3. A delayed channel and a strict signature separation	8
2.4. Optimal resource sets and decision certificates	9
3. Continuation geometry and an exact memory obstruction	10
3.1. Normalized continuation laws	10
3.2. An erasure cut	12
4. Decision continuation spectra	12
4.1. A charged training-validation interface	12
4.2. Several feedback rows	15
5. Weighted continuation obstructions	15
6. Finite-data testing on the original behavior image	17
6.1. The reset interface	17
6.2. A matching uniform sampling obstruction	19
6.3. Transport of one fixed audit	19
7. Continuation complexity of a finite-data test	20
7.1. The width required to attain the binary reset value	21

Date: September 23, 2026.

2020 Mathematics Subject Classification. Primary 62B05, 93E20; Secondary 60G44, 68Q45, 82C40.

Key words and phrases. causal comparison, continuation cone, finite memory, statistical experiments, hard-sphere scattering, nonlinear filtering.

8. Polynomial testing of the original resource image	22
8.1. A sharp two-time nonconvex example	24
9. Adaptive testing on the observable behavior image	25
9.1. The testing game	25
9.2. A size bound in behavior dimension	26
9.3. A three-weight certificate invisible to constant tests	27
9.4. Strict certificates and effective physical input	29
10. An exact algorithm for the observable chart	29
11. A marked resource gap generated by a binary collision	30
11.1. The exact marked two-report experiment	30
11.2. A full-dimensional collision preparation	32
11.3. Uniformly effective finite presentations	33
12. Uniform stability of optional projections	34
12.1. A microscopic sufficient condition	36
13. Changing trajectories, likelihoods, and conditional consumers	37
13.1. A common preparation, not a common flow	37
13.2. The projected stochastic exponential	38
13.3. A bounded entropic backward-equation consumer	38
13.4. Resource transport is not the posterior coupling	40
14. Observation densities and coupled process versions	40
15. Global transport of the backward equation	42
16. A uniform certificate through a moving collision	44
17. Sharp decision memory for the collision experiment	47
17.1. The marked family and its first two spectral values	47
17.2. A three-response construction	49
17.3. Simultaneous exact widths and the full acquisition schedule	50
17.4. What the two exact complexity statements measure	52
18. A finite-register certificate transported through the microscopic limit	52
19. Explicit phase transitions for the two-count certificate	55
Appendix A. The proof dependency chain	55
Appendix B. A namespaced map of the eleven-component program	57
Appendix C. Preserved organizing statements	57
C.1. Inherited baseline	58
References	59

1. INTRODUCTION

A causal experiment supplies observations, admits specified controls and carries an actual terminal mark. A reduction of that experiment consumes its observations through a retained register. Its resource restriction is not determined by the set of its output laws alone: the permitted state at each cut and the visibility of persistent randomization must also be specified. A finite-data auditor has its own such restrictions. In particular, a test given an exact candidate law and a test learning from repeated trials solve different mathematical problems.

The object organizing this paper is a family of continuation responses. At a causal cut, those responses must be compatible with the one-step shifts at the next cut. At a training-validation cut, they are the conditional validation responses available from each retained state. The first viewpoint characterizes exact realization; the second gives a geometric spectrum for approximate statistical certification. We show that these viewpoints control the same physical audit: a lower bound on its

possible continuation responses, an attaining finite-sample rule, and the exact continuation profile of that rule.

1.1. The decision cut and the principal theorem. Let B be a compact set of laws on a finite marked alphabet Ω , and let Q be the target law. A reset auditor trains on independent samples from one candidate $b \in B$, held fixed for the whole audit. Only its charged register survives the end of training. It then selects an event before independent target and candidate validations and emits the difference of their event indicators. Its score means the expectation of this signed difference. A response $h \in [0, 1]^\Omega$ is the mean indicator of a randomized event. Put

$$\mathfrak{V}_K(B, Q) = \max_{h_1, \dots, h_K} \min_{b \in B} \max_{m \leq K} (Q - b)h_m.$$

This is a continuation invariant of the marked decision interface. It does not convexify the candidate image and does not grant the auditor an external copy of its training record.

Theorem 1.1 (Decision continuation complexity and physical certification). *The following assertions hold with all persistent data charged to the appropriate machine.*

(i) *The supremum $v_{n,K}$ of worst-case expected scores with n training samples and K states at the decision cut satisfies*

$$0 \leq \mathfrak{V}_K - v_{n,K} \leq \min \left\{ \sqrt{(|\Omega| - 1)/n}, \sqrt{2 \log(2K)/n} \right\}.$$

The least K permitting a strict score greater than c at some finite training length is the least size of a cover of B by sets $\{b : (Q - b)h > c\}$. This spectrum is Lipschitz under uniform total variation perturbations of candidate and target laws.

(ii) *For the ideal marked private product family B of the two-sphere experiment, $\mathfrak{V}_1 = 1/8$ and $\mathfrak{V}_2 = 3/8$. Throughout the collision interval $9/10 \leq d \leq 11/10$, $0 < \sigma \leq 1/24$, the minimum decision-cut width for uniform physical score greater than $2/5$ is exactly three. The lower bound applies to every finite-reset auditor with the specified interface, not merely a particular counter implementation; feedback-row selection does not evade it.*

(iii) *One deterministic rational rule has physical score greater than*

$$\sqrt{2} - 1 - \frac{1}{4\sqrt{399e}} - \frac{1}{4}e^{-399/8} - \frac{1}{300} > \frac{2}{5}.$$

It uses 399 informative training trials, an optional ignored training trial and two independent validations, hence fits the 402-trial schedule. It has exactly three decision-cut states. On its informative training segment, the minimum simultaneous widths for exact stochastic implementation are

$$r_t = \begin{cases} \binom{t+2}{2}, & 0 \leq t \leq 199, \\ \binom{402-t}{2}, & 200 \leq t \leq 399. \end{cases}$$

Their maximum is 20301. Including individual report-bit acquisition and the validations, at most $40602 < 2^{16}$ retained states suffice.

(iv) *At an arbitrary cut of a finite-valued decision, a weighted graph of distinguishing suffixes gives a distribution-sensitive error bound through its minimum monochromatic weight under K labels. Applied to Bayes margins, it yields a quantitative constrained-audit regret bound. At zero error it recovers the exact simultaneous residual profile, even for stochastic implementations.*

(v) *For fixed positive σ and changing d , the same physical rule has the target-variation modulus $\min\{1, \sqrt{|d' - d| + 2700|d' - d|^2/(2\sigma)}\}$, with no training-count factor. The microscopic paths, likelihoods, posterior processes, innovations, and bounded entropic backward values, integrands and martingale integrals have the joint transport conclusions proved below. The private and visible*

deficiencies exceed 9/20, whereas a hidden two-valued selector has error less than 13/100. In the no-collision regime $d \leq 2/5$ all three finite-experiment deficiencies vanish.

Proof. Part (i) is Theorem 4.2. The exact ideal spectral values are Theorem 17.1; Lemmas 17.3 and 17.4 and Theorem 17.6 prove the matching physical construction. Theorem 17.5 proves (iii), including all acquisition and validation phases. Theorem 5.1 and Corollary 5.2 prove (iv). The microscopic estimates in Theorem 16.2, the explicit coupling-version lemmas in Section 14, and Theorem 15.2 prove (v). $\square$

The key point is that the two-state obstruction and the three-state construction are for the same physical score. Three witness candidates force three validation continuations above a prescribed level; a three-event dictionary then covers the entire original nonconvex product image. Its empirical majority selector has a residual quotient that can be computed exactly at every training cut. Thus the continuation invariant both limits the physical certificate and constructs it. The larger unrestricted marked discrepancy is a different optimization. Likewise, minimum width at one specified cut and minimum peak width over all successful algorithms are different quantities. The theorem gives an exact answer to the former, and an exact implementation profile for the stated successful rule.

1.2. The full comparison and transport framework. The general finite experiment has a private register profile K , a possible independent hidden selector, or a visible selector included in the compared law. Compatible normalized continuation covers characterize its exact causal realization. Compactness, nonnegative factorization, priced composition and hidden/visible comparison are proved in Theorems 2.2 and 3.2. The actual mark cannot be read by the online candidate. The reset auditor's completed marked trials are a separate interface with a separate paid register.

The finite-reset minimax theorem and its moment representation operate on the original compact behavior image, without replacing a sampled candidate by the barycenter of a prior. The sampling error is uniformly of order $n^{-1/2}$, with a matching binary sampling obstruction. The Boolean streaming theorem determines the widths needed to attain that binary testing value. The weighted finite-valued result extends its error mechanism to nonuniform word weights and general finite decisions. The nonlinear revealed-behavior witness, Bernstein hierarchy and rational chart retain their full proofs. Their evaluator sees an exact behavior law, unlike a reset auditor; their privileges are not interchanged.

The physical model consists of two labeled hard spheres with a prepared impact-mark correlation and one nongrazing collision. The collision transduces the impact variable into an observed transverse velocity; the acquisition gate does not change the mechanics. Varying the contact diameter changes the collision time and trajectory. A common-preparation Gaussian observation estimate transports both the finite certificate and its conditional-law consumer. The latter is a bounded scalar entropic backward equation with fixed positive observation noise. Its terminal condition is the observation-measurable conditional certainty equivalent, not an unobserved latent payoff. The bounded tilt yields global integrand transport without a density stopping time. The coupled stochastic integrals retain their own marginal observation filtrations.

1.3. Relation to classical comparison and realization. The classical comparison theory of Blackwell and Le Cam [1, 2, 13] and earlier filtered comparison [3, 19] provide essential context. Introducing a filtration or a randomization criterion is not itself a novelty claim. Stable positive-cone realization [23] and residual-state minimization [16] are classical mechanisms. Their finite controlled normalization is proved here so that the resource profile and the input interface remain explicit. We do not claim a new deterministic automata minimization theorem.

The contribution is the link from compatible continuation responses to statistical certification at a charged physical decision cut. The response-envelope classification has a uniform finite-sample

realization; its first two values on the physical product image can be computed exactly; and a three-response cover attains the prescribed microscopic score with a completely specified streaming quotient. The weighted obstruction uses state-distribution overlap and concavity rather than an unsupported deterministic reduction of randomized memory. Scattering, Bayes' formula, stochastic exponentials and martingale inequalities are used as tools, not claimed as discoveries.

The original full text of Norberg's paper was not obtained for a proof-level comparison. The literature crosswalk therefore separates verified bibliographic information, inherited original-source comparisons, and unresolved priority questions. In particular, no absence claim about Norberg's theorems is inferred from this access limitation. The results here are presented by their precise statements and proofs, not an unverified assertion of priority for filtered comparison.

The larger theta-theory program retains all eleven historical components. The program map in Appendix B distinguishes theorem inputs, scoped replacements, historical aggregate obligations and the independent A2 geometric chain. The present foundational layer organizes the causal comparison and certification chain; independence of another chain is not replaced by a nominal dependency arrow. All predecessor proofs and program targets remain available in the unchanged historical companion to this article.

2. INTRINSIC CAUSAL DEFICIENCY

2.1. Experiments and executable signatures. Fix a finite parameter set Θ , a horizon T , and finite control alphabets $A_0, \dots, A_{T-1}$. An experiment has a report at checkpoints $0, \dots, T$ and an unobserved terminal mark W in a common finite set. It is specified by finite causal instruments: for each parameter and control history they give the joint law of the report path and the actual mark. Equivalently, a finite latent-state realization may be expanded into its history tree. The parameter and mark are unavailable to the encoder. The mark may depend on the physical controlled path. We compare experiments with the same control interface and mark space.

Write Y_t for the source report and Z_t for the target report. A private simulator of width K has stochastic rows

$$\begin{aligned} q_0(z, m \mid y), \quad m \in [K_0], \\ q_t(z, m \mid m', a, y), \quad m' \in [K_{t-1}], \quad m \in [K_t], \quad 1 \leq t \leq T. \end{aligned} \tag{2.1}$$

A row consumes the current source report, emits the target report, and retains the new state. The previous target reports are not available to that row unless encoded in m' . The clock and current control are part of the declared interface. Fresh private randomness is permitted, but no other persistent random variable is free. The product of all row simplexes is denoted $\mathcal{K}_K$.

A deterministic feedback tree g chooses a_t from $z_{0:t}$ and its own previous controls. It is the same controller on both sides of a comparison. Let Ω_g be the finite marked transcript space, including reports, controls, and W . Put

$$P_{\theta,g}^q = \text{Law}_\theta(E \xrightarrow{q} g), \quad Q_{\theta,g} = \text{Law}_\theta(F \rightarrow g).$$

Here W on the left is the source's actual mark. The total variation convention is $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. Private deficiency is

$$\delta_K^p(E, F) = \min_{q \in \mathcal{K}_K} \max_{\theta, g} \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}. \tag{2.2}$$

The minimum will be justified below. Randomized feedback need not be added to the maximum: conditioning on its independent random table expresses its law as a mixture of deterministic feedback laws. The same argument permits arbitrary independent side randomness in the tester.

A *hidden selector* $U \in [r]$, independent of the source, chooses one simulator q^u . It is not visible to the feedback controller or the final test. A *visible selector* is revealed before feedback begins.

The two costs are

$$\delta_{K,r}^h(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, g} \left\| \sum_u \lambda_u P_{\theta, g}^{q^u} - Q_{\theta, g} \right\|_{\text{TV}}, \quad (2.3)$$

$$\delta_{K,r}^v(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, (g_u)} \left\| \sum_u \lambda_u \delta_u \otimes P_{\theta, g_u}^{q^u} - \sum_u \lambda_u \delta_u \otimes Q_{\theta, g_u} \right\|_{\text{TV}}. \quad (2.4)$$

In both expressions $\lambda \in \Delta_r$ and $q^u \in \mathcal{K}_K$. The second formula compares the joint seed–marked interface. Its target seed has the same distribution and is independent of the target experiment. A hidden selector can be stored privately by retaining (u, m) , at width rK_t . A visible selector is separate side information and is charged by its own alphabet size. Neither interpretation identifies an arbitrary real-valued design with a free finite register.

Definition 2.1 (Deficiency spectrum). The private resource spectrum of E relative to F is the upward-closed set of pairs (K, ε) satisfying $\delta_K^p(E, F) \leq \varepsilon$. The hidden spectrum consists of triples (K, r, ε) satisfying $\delta_{K,r}^h(E, F) \leq \varepsilon$, and the visible spectrum uses $\delta_{K,r}^v$. Width profiles are ordered coordinatewise; in the seeded spectra the seed bound and error are ordered as well. Unused states or selector values may be adjoined. The hidden convexification is $\delta_K^{\text{mix}} = \inf_{r \geq 1} \delta_{K,r}^h$.

The spectrum, not a presumed unique coordinatewise minimum profile, is the intrinsic object. For example one may minimize $\max_t K_t$, with a specified seed signature and tolerance, over its feasible profiles. Different checkpoints can have incomparable optimal tradeoffs.

2.2. The finite comparison theorem. Concatenate all probabilities $P_{\theta, g}^q(\omega)$ into $b(q) \in \mathbb{R}^N$, retaining their row indices; put $b_F = (Q_{\theta, g}(\omega))$. Let $\mathcal{D}_K$ be the finite set of deterministic choices of all rows in (2.1). Empty or unreachable rows remain harmless coordinates. Set

$$a_K = \dim \text{aff}\{b(q) : q \in \mathcal{K}_K\}, \quad \mathcal{I} = \{(\theta, g, B) : B \subseteq \Omega_g\}.$$

For $i = (\theta, g, B)$ and $d \in \mathcal{D}_K$, define

$$A_{i,d} = Q_{\theta, g}(B) - P_{\theta, g}^d(B). \quad (2.5)$$

Including both an event and its complement includes both signs.

Theorem 2.2 (Intrinsic deficiency, priced converse, and visibility). *For the finite marked causal experiments above the following assertions hold.*

(i) *Private deficiency and both fixed- r deficiencies are attained. The private problem has the exact polynomial formulation*

$$\min\{\varepsilon : q \in \mathcal{K}_K, 0 \leq \varepsilon \leq 1, Q_{\theta, g}(B) - P_{\theta, g}^q(B) \leq \varepsilon \text{ for all } (\theta, g, B) \in \mathcal{I}\}. \quad (2.6)$$

In particular its zero set is exactly the nonnegative causal factorization of the target marked behavior by the rows (2.1). At $K_t = 1$ for every t , this is precisely the same factorization with rows $q_t(z \mid a, y)$ and $q_0(z \mid y)$.

(ii) *Hidden convexification has the attained dual*

$$\delta_K^{\text{mix}}(E, F) = \min_{\lambda \in \Delta(\mathcal{D}_K)} \max_{i \in \mathcal{I}} \sum_d \lambda_d A_{i,d} = \max_{w \in \Delta(\mathcal{I})} \min_{d \in \mathcal{D}_K} \sum_i w_i A_{i,d}. \quad (2.7)$$

An optimal simulation uses at most $a_K + 1$ selector values. Thus every weighted causal test has optimal discrepancy at most ε if and only if a hidden mixture of K -state simulators has error at most ε ; it has a private realization of width at most $(a_K + 1)K$. More specifically,

$$\delta_{rK}^p \leq \delta_{K,r}^h, \quad \delta_K^{\text{mix}} = \delta_{K, a_K+1}^h \leq \delta_K^p. \quad (2.8)$$

(iii) Put $e_\theta(q) = \max_g \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}$. Then

$$\delta_{K,r}^v = \min_{\lambda, q^1, \dots, q^r} \max_{\theta} \sum_u \lambda_u e_\theta(q^u). \quad (2.9)$$

Writing $a_K^v = \dim \text{aff}\{(e_\theta(q))_\theta : q \in \mathcal{K}_K\}$, the infimum over all finite visible selectors is attained using $a_K^v + 1 \leq |\Theta| + 1$ values and equals

$$\max_{v \in \Delta(\Theta)} \min_{q \in \mathcal{K}_K} \sum_{\theta} v_\theta e_\theta(q). \quad (2.10)$$

For every $r \geq 1$, and also after optimizing over all r ,

$$\delta_{K,r}^v(E, F) = 0 \iff \delta_K^p(E, F) = 0. \quad (2.11)$$

No corresponding equivalence with $\delta_K^{\text{mix}} = 0$ holds in general.

(iv) For compatible experiments E, F, G and profiles K, L ,

$$\begin{aligned} \delta_{KL}^p(E, G) &\leq \delta_K^p(E, F) + \delta_L^p(F, G), \\ \delta_{KL,rs}^s(E, G) &\leq \delta_{K,r}^s(E, F) + \delta_{L,s}^s(F, G), \quad s \in \{h, v\}. \end{aligned} \quad (2.12)$$

Here KL is the pointwise product, and the second line uses independent selectors. If an F -consumer has widths S_t , composing it with a private simulator gives widths $K_t S_t$ and changes every $[0, 1]$ marked risk by at most $\delta_K^p(E, F)$. A hidden selector is stored at the additional price r , whereas visible selector alphabets multiply. The same bounds hold for optimized upper risk sets and their lower support functions with the indicated costs.

Proof. Expand a probability of a complete source path, simulator state path, and transcript. It is a product of fixed experiment probabilities and one chosen row entry per time. Summing latent and state paths gives a polynomial in the finitely many row entries. No row belonging to a fixed time can occur twice along one path. Consequently the polynomial is multiaffine in the groups of stochastic rows. Continuity and compactness of the product of closed finite simplexes prove attainment in (2.2); the same proof includes λ and the r policies in (2.3). For the visible problem there are only finitely many collections (g_u) and transcript events, so its objective is again continuous on a compact set. The finite-event formula for total variation proves (2.6). Setting $\varepsilon = 0$ forces equality of every transcript probability, in both directions of each event inequality. This proves the factorization and stateless claims, including their necessity.

For each stochastic row of q , sample its deterministic output once, independently of all other rows, before the experiment. This constructs a probability Λ_q on $\mathcal{D}_K$. On any fixed path the row variables encountered have different time indices, so the expectation of their product is the product of the row probabilities. Summing paths shows, simultaneously for every θ, g , that

$$b(q) = \sum_{d \in \mathcal{D}_K} \Lambda_q(d) b(d). \quad (2.13)$$

This is a behavior identity, not a same-width deterministic purification of minimax error. It implies $\text{conv } b(\mathcal{K}_K) = \text{conv}\{b(d) : d \in \mathcal{D}_K\}$. The latter is a compact polytope. Minimizing the continuous convex maximum of event discrepancies over this polytope gives the first expression in (2.7). Finite-dimensional minimax gives the second, with both optima attained. A point of this polytope is a convex combination of at most $a_K + 1$ of its generating points by Carathéodory's theorem. For completeness, if a representation uses more points, their affine dependence gives coefficients c_d with $\sum c_d = 0$ and $\sum c_d b(d) = 0$. Subtract the largest permissible multiple of c from the positive mixing weights; a weight becomes zero without changing the represented point. Repeating gives the stated bound. Store its selector together with the machine state to obtain (2.8). The dual inequality for every w is equivalent to the maximum over w being at most ε , which proves the priced testing converse.

For disjoint seed fibers, total variation is the sum of the fiber total variations weighted by λ_u . For a fixed θ , the tester can choose a separate maximizing tree on each revealed fiber. Therefore its maximum is $\sum_u \lambda_u e_\theta(q^u)$, proving (2.9). The map $q \mapsto e(q)$ is continuous, being a finite maximum of continuous functions. Its image is compact. Convexifying it in $\mathbb{R}^{|\Theta|}$ and applying the preceding finite-dimensional separation and affine-dependence argument proves (2.10) and its selector bound. Notice that the inner minimum is over stochastic q , not over deterministic tables: the nonlinear function e_θ does not commute with (2.13). If the attained visible optimum is zero, then for every u of positive weight and every θ one has $e_\theta(q^u) = 0$, since every summand is nonnegative. Any such q^u is a private exact simulator. The converse uses a selector of one value. This also proves the assertion for the optimized visible selector, whose support is finite by the bound just established.

Finally, cascade a K -state E -to- F transducer and an L -state F -to- G transducer. Within a step, consume the intermediate report before discarding it. The retained pair has at most $K_t L_t$ values. For any terminal tester on G , the second transducer together with that tester is an admissible randomized feedback system on F . The first error bound applies to this system, and total variation contracts under the remaining channel. The second error bound applies to the genuine F system. The triangle inequality gives (2.12). With selectors, condition on the independent selectors in the appropriate places; in the visible case keep both selectors in the compared transcript. The same argument is valid because the target reference selectors are independent. Storing a hidden selector or exposing both visible selectors gives the stated costs. The minima established above may be used; equivalently one can use nearly optimal simulators and let the extra tolerance decrease.

A consumer is another causal transducer with task-specific readout and, where allowed, task-specific control, but with its required common encoder. The simulator is task blind, so its product with that encoder is still common. Condition on the consumer's independent random table to apply the feedback bound. Integration of any $[0, 1]$ marked loss changes by at most the total variation error. Taking the same paired consumer for every row proves the risk-set assertion, rather than a separate choice of simulator for each row. Infima of nonnegative weighted risks give the lower support inequalities. $\square$

Remark 2.3 (What the converse says). The dual (2.7) uses mixtures of indexed causal tests and optimizes over the entire common transducer. It is not a task-by-task choice of encoder. Its conclusion is an explicit hidden-selector simulation, with the selector subsequently charged if private storage is required. It does not replace the nonconvex private set by its convex hull at unchanged resource. The next example shows why that replacement would change the zero set.

2.3. A delayed channel and a strict signature separation. Let the source reveal $i \in [n]$ at checkpoint zero and reveal only a constant at checkpoint one. There is no control and the mark is constant. The target reports a constant at checkpoint zero and at checkpoint one draws $j \in [m]$ from a specified row-stochastic matrix $V = (V_{ij})$. The unknown parameter is i . The first target alphabet has one element: the simulator cannot use an early public output as unpriced storage. Only the intermediate register of size K matters.

Proposition 2.4 (The nonnegative-rank slice). *For this experiment,*

$$\delta_K^p(E, F_V) = \min_{A, B} \max_i \frac{1}{2} \sum_j |V_{ij} - (AB)_{ij}|, \quad (2.14)$$

where A is an n -by- K row-stochastic matrix and B is a K -by- m row-stochastic matrix. In particular

$$\delta_K^p(E, F_V) = 0 \iff \text{rank}_+(V) \leq K, \quad \delta_1^p(E, F_V) = 0 \iff V \text{ has identical rows.}$$

For $1 \leq K \leq n$ the identity channel has the sharp laws

$$\delta_K^p(E, F_{I_n}) = 1 - \frac{1}{\lceil n/K \rceil}, \quad \delta_K^{\text{mix}}(E, F_{I_n}) = 1 - \frac{K}{n}. \quad (2.15)$$

Thus, at tolerance $0 \leq \varepsilon < 1$, the least private width is $\lceil n/[(1 - \varepsilon)^{-1}] \rceil$, whereas the least within-mode width with optimized hidden selection is $\max\{1, \lceil n(1 - \varepsilon) \rceil\}$. The latter still pays for its selector. In particular, for $n = m = 3$ and $K = 2$,

$$\delta_2^p(E, F_{I_3}) = \frac{1}{2}, \quad \delta_2^{\text{mix}}(E, F_{I_3}) = \frac{1}{3}. \quad (2.16)$$

For $V_* = (1/2)I_3 + (1/6)\mathbf{1}\mathbf{1}^\top$,

$$\delta_{2,3}^h(E, F_{V_*}) = 0, \quad \delta_2^p(E, F_{V_*}) \geq \frac{1}{4\sqrt{3}} > 0, \quad \inf_r \delta_{2,r}^v(E, F_{V_*}) > 0. \quad (2.17)$$

Proof. The encoder maps the consumed i into its retained label by A ; the terminal decoder maps that label into j by B . This proves (2.14) including its converse implementation. If $V = CD$ is any nonnegative factorization, let $s_k = \sum_j D_{kj}$. Delete zero rows of D , define $B_{kj} = D_{kj}/s_k$ and $A_{ik} = C_{ik}s_k$. Since V is stochastic, the rows of A sum to one. Padding with unused states gives the claimed equivalence with nonnegative rank. The one-state statement follows immediately.

For $V = I_n$, row error is $1 - (AB)_{ii}$. If the least correct probability is c , for each i select a row $k(i)$ of B with $B_{k(i),i} \geq c$. Some row is selected at least $\lceil n/K \rceil$ times, so stochasticity gives $c \leq 1/\lceil n/K \rceil$. A partition into K groups of sizes at most $\lceil n/K \rceil$, each decoded uniformly, attains that bound. Also $\sum_i (AB)_{ii} \leq K$, by the calculation below with K in place of two. It remains valid under mixing, giving hidden error at least $1 - K/n$. Start with a deterministic channel that reproduces K input symbols exactly and sends all others to those K outputs. Average all simultaneous permutations of its input and output labels. Every diagonal entry is then K/n , proving the second formula. Inverting the two monotone formulas gives the stated least widths.

For $V = I_3$, total variation in row i is $1 - (AB)_{ii}$. If all three diagonal entries of AB were greater than $1/2$, then for each i some row of B would have its i th entry greater than $1/2$. A stochastic row has at most one such entry, so two rows cannot serve three indices. This proves the private lower bound $1/2$. Merge two input symbols into a retained label that decodes uniformly to that pair, and retain the third symbol exactly. This attains it. For every stochastic factorization with two intermediate states,

$$\sum_{i=1}^3 (AB)_{ii} = \sum_{k=1}^2 \sum_{i=1}^3 A_{ik} B_{ki} \leq \sum_{k=1}^2 \sum_{i=1}^3 B_{ki} = 2.$$

The same inequality holds under hidden mixtures, giving maximal row error at least $1/3$. Mix uniformly the three preceding pair-merge channels. The resulting diagonal entries are $2/3$ and the off-diagonal entries are $1/6$. This is exactly V_* and has error $1/3$ from I_3 . It proves the hidden claims in both displays.

The smallest singular value of V_* is $1/2$. Any AB with two intermediate states has a unit vector x in its right null space, so $\|V_* - AB\|_F \geq \|(V_* - AB)x\|_2 \geq 1/2$. If the maximal row total variation error is ε , each row has Euclidean error at most 2ε , and hence the Frobenius error is at most $2\sqrt{3}\varepsilon$. This proves the explicit private bound. The optimized visible value is attained by Theorem 2.2; if it were zero, its zero-set assertion would contradict that bound. This proves its strict positivity. $\square$

2.4. Optimal resource sets and decision certificates. For a class $\mathcal{C}_{S,q}(F)$ of common consumers, with all other information conventions fixed, let $\mathcal{R}_{S,q}(F)$ be its vector of attainable $[0, 1]$ risks. For a nonnegative row weight w summing to one, write

$$h_F(S, q; w) = \inf_{r \in \mathcal{R}_{S,q}(F)} \langle w, r \rangle.$$

If E privately simulates F with cost K and error ε , then

$$\mathcal{R}_{S,q}(F) \subseteq \mathcal{R}_{KS,q}(E) + [-\varepsilon, \varepsilon]^J, \quad h_E(KS, q; w) \leq h_F(S, q; w) + \varepsilon. \quad (2.18)$$

For nonfinite row sets the first expression means pointwise paired risk functions within uniform distance ε ; no finite coordinate box is needed. Upper sets give the equivalent one-sided order formulation. Hidden selectors multiply private width by r ; visible selectors multiply the separate public alphabet by r . These are direct consequences of the final part of Theorem 2.2, with the optimal intrinsic error in place of an externally estimated simulator error.

When two stateless comparison directions have error at most ε , paired risk sets and their support values at unchanged width are within ε . If each is centered by a paired unrestricted benchmark whose error is also at most ε , the centered values are within 2ε . Therefore a gap $g > 0$ between zero excess and infeasible widths is stable under $\varepsilon < g/4$: a value below $g/2$ in the approximation certifies feasibility in the exact finite problem, and a value above $g/2$ certifies infeasibility. The benchmark and the class being optimized must be specified; this assertion does not identify different information signatures.

Proposition 2.5 (Intrinsic finite-dimensional selector approximation). *Let $\mathcal{R}$ be a compact risk image in $C(J)$, where J is compact. Suppose a bounded linear operator $L : C(J) \rightarrow C(J)$ has rank at most m and $\sup_{r \in \mathcal{R}} \|r - Lr\|_\infty \leq \eta$. Every barycenter $\bar{r}$ of a probability on $\mathcal{R}$ has a barycenter r_* of at most $m + 1$ members of $\mathcal{R}$ with $\|\bar{r} - r_*\|_\infty \leq 2\eta$. If $\mathcal{R}$ lies in a finite affine space of dimension m , the error is zero with at most $m + 1$ members. Thus the selector price can be controlled by the dimension of attainable risks, rather than by a covering number of the row space.*

Proof. The compact set $L\mathcal{R}$ lies in an m -dimensional space. Its convex hull is compact, and $L\bar{r}$ belongs to it. The finite-dimensional affine-dependence argument in the proof of Theorem 2.2 gives $r_1, \dots, r_{m+1} \in \mathcal{R}$ and weights λ such that $L\bar{r} = \sum_i \lambda_i Lr_i$. Put $r_* = \sum_i \lambda_i r_i$. The two residual terms $\bar{r} - L\bar{r}$ and $r_* - Lr_*$ have norms at most η , proving the estimate. Apply the same argument after translation in the exact affine case. The implementation stores the selected design index whenever it is not provided as visible side information. $\square$

Remark 2.6 (Finite computation and general carriers). For rational finite instruments, (2.6) and the fixed- r formulations are finite polynomial feasibility problems; (2.7) is a finite linear program. These are exact variational characterizations, not assertions of polynomial running time in a succinct history description. The number of feedback trees, events, or deterministic tables can be large. On standard Borel carriers the same definitions use suprema over measurable feedback and tests and infima over executable kernels. Their product-state composition and decision transport remain valid by the same channel argument. Attainment and the finite testing dual are asserted here only for the stated finite class. The average-risk and physical results below give separate nonfinite theorems with their own hypotheses.

3. CONTINUATION GEOMETRY AND AN EXACT MEMORY OBSTRUCTION

3.1. Normalized continuation laws. We first separate the transducer from the source to which it is attached. There are L epochs with finite input alphabets X_t and output alphabets Z_t . A causal kernel $p(z_{1:L} \mid x_{1:L})$ is nonnegative, normalized for every input word, and has output-prefix marginals independent of later inputs. The current input can include both a source report and the current control. It never includes an unavailable parameter, a terminal mark, or an earlier output not retained by the machine.

Let $\mathcal{C}_t$, $0 \leq t \leq L$, be the finite-dimensional polytope of normalized causal kernels from $x_{t+1:L}$ to $z_{t+1:L}$; put $\mathcal{C}_L = \{1\}$. For $v \in \mathcal{C}_t$ define the shift

$$(S_{x,z}v)(z_{t+2:L} \mid x_{t+2:L}) = v(z, z_{t+2:L} \mid x, x_{t+2:L}).$$

Its total mass $v(z \mid x)$ is independent of the future input word. A zero shift is allowed.

Definition 3.1 (Compatible continuation polytopes). A continuation cover of width $K = (K_0, \dots, K_L)$ consists of lists $V_t = (v_{t,m})_{m \leq K_t}$ in $\mathcal{C}_t$ such that, for every m, x, z ,

$$S_{x,z}v_{t,m} \in \text{cone } V_{t+1}. \quad (3.1)$$

It covers p when $p \in \text{conv } V_0$. Lists may repeat generators and may contain unreachable generators. Thus unused states can be adjoined.

Theorem 3.2 (Continuation realization with a width profile). *A causal kernel p has a stochastic transducer realization with an initial K_0 -valued register and K_t retained states after epoch t if and only if it has a compatible continuation cover of width K . Initial randomization is part of the charged initial register. With a deterministic initial state the condition is $p = v_{0,1}$ and $K_0 = 1$.*

Consequently the set of feasible width profiles depends only on the causal kernel and its declared input interface, not on a row presentation or a labeling of the hidden states. Every matrix of conditional continuation probabilities at cut t has nonnegative rank at most K_t . Individual rank bounds do not replace the simultaneous shift conditions (3.1).

Proof. Given a transducer, let $v_{t,m}$ be the suffix response when its register at time t equals m . The remaining rows define this response even for an unreachable state. If the next row is $q_{t+1}(z, m' \mid m, x)$, conditioning on its output and new state gives

$$S_{x,z}v_{t,m} = \sum_{m'} q_{t+1}(z, m' \mid m, x) v_{t+1,m'}.$$

The initial state distribution expresses p as a convex combination of V_0 .

Conversely choose nonnegative coefficients in every cone membership in (3.1). There are only finitely many choices to make. Since every generator on the right has mass one for each future input word, the sum of the coefficients for a fixed m, x, z is $v_{t,m}(z \mid x)$. Summing over z gives one. The coefficients are therefore stochastic rows $q_{t+1}(z, m' \mid m, x)$. Choose an initial distribution representing p in $\text{conv } V_0$. Backward induction from $\mathcal{C}_L = \{1\}$ proves that the constructed machine has exactly the response $v_{t,m}$ from each register state. In particular its initial response is p . Equality for open-loop input words implies equality under adaptive input selection: at each node of an input–output history tree, the same causal conditional probability is used on the two sides.

For the rank assertion, condition on any positive-probability observed history at the cut. Its continuation response is a convex combination of $v_{t,m}$ with weights equal to the posterior register distribution. Stack these responses as rows, and use all future input–output pairs as columns. This is a nonnegative factorization through K_t generators. The converse construction required the same generator families at adjacent cuts and all their shifts; an unrelated factorization of each cut matrix provides no such compatibility. The characterization, being stated entirely in terms of these response polytopes, also proves the asserted invariance. $\square$

Attach such a causal kernel to a marked source experiment. The output law is obtained by contraction with the source’s causal instruments, while its actual mark is retained. Hence zero private deficiency is equivalently the existence of a compatible continuation cover whose contraction equals the target marked behavior in every parameter and feedback row. This is the geometric form of the factorization in Theorem 2.2; no conditional dependence on the mark is introduced.

A hidden selector chooses one such cover independently of the source. It is a convex combination of complete realizations, not a convexification inside every retained-state simplex at the original width. Its private implementation uses the disjoint union of the covers and keeps the selector in the state. A visible selector keeps those fibers in the compared law. The nonnegative fiber argument in Theorem 2.2 therefore continues to give equality of the visible and private zero sets.

3.2. An erasure cut. A simple family makes the width obstruction exact, including its error. Let X take values in $[N]$ with law π . Before a specified cut the source reveals X . After the cut it has no reports correlated with X . The controller has no feedback channel carrying X across the cut, and all earlier emitted reports are prescribed constants. The only permitted cross-cut information of the simulator is its K -valued register and any independent selector explicitly included in its signature. The terminal mark is the actual $W = X$. The target's terminal report is X . All memory at other checkpoints is unrestricted.

Theorem 3.3 (Exact cut spectrum). *For this experiment, for private, hidden and visible signatures alike,*

$$\delta_K = 1 - \sum_{i=1}^{\min(K,N)} \pi_{(i)}. \quad (3.2)$$

For uniform X , $\delta_K = (1 - K/N)_+$. Thus an erasure cut for m unbiased bits requires $K \geq (1 - \epsilon)2^m$ whenever its error is at most ϵ . The bound holds for every surrounding width profile; equality is attained when the other checkpoints are unrestricted. If every $\pi_x > 0$, exact transmission has nonnegative cut rank N .

Proof. With all post-cut randomness averaged into the decoder, any private machine gives kernels $e(m | x)$ and $d(z | m)$. Its joint terminal law is $\pi_x \sum_m e(m | x) d(z | m)$. Since the target law is $\pi_x \mathbf{1}_{z=x}$, total variation is exactly one minus the correct-decoding probability. Put $u_x = \max_m d(x | m)$. Then

$$0 \leq u_x \leq 1, \quad \sum_x u_x \leq K, \quad \mathbb{P}(Z = X) \leq \sum_x \pi_x u_x \leq \sum_{i=1}^{\min(K,N)} \pi_{(i)}.$$

The last inequality follows by moving mass in u from a smaller weight to a larger one until the K largest weights are filled. To attain it, retain distinct labels for those letters and decode them deterministically; all other letters are mapped to any retained label. Before the cut one may use the unrestricted memory to compute this encoding.

For a hidden selector, condition on its value. It is independent of X , so every conditional correct-decoding probability obeys the same bound; averaging cannot improve it. For a visible selector, variation is the average of the fiber variations, with the same conclusion. The private construction works for both signatures. Finally, the exact joint cut matrix is diagonal with its N positive diagonal entries π_x . Its ordinary rank is N , so its nonnegative rank is at least N and its diagonal factorization attains N . $\square$

The absence of an auxiliary cross-cut feedback channel is essential. An unpriced controller storing X and returning it as a later control would change the experiment, not disprove the cut bound. Likewise a selector correlated with X is not an independent hidden selector. These distinctions are the same ones needed in the collision certificate below.

4. DECISION CONTINUATION SPECTRA

The memory at an experimental cut has a different role from the memory needed to reach that cut. We first identify its exact asymptotic value. This will give both an obstruction and a construction for the same physical audit, rather than a lower bound for an auxiliary binary problem.

4.1. A charged training-validation interface. Let Ω be a finite marked alphabet, $d = |\Omega|$, let $B \subset \Delta(\Omega)$ be nonempty and compact, and fix $Q \in \Delta(\Omega)$. A candidate $b \in B$ is fixed throughout the experiment. Training consists of n independent observations from b . At the end of training the auditor retains a register $M \in [K]$. Everything depending on the training observations or on

an earlier random seed must pass through this register. In particular, there is no external copy of the training history and no uncharged persistent selector.

After this cut, fresh independent randomness may select an event A using M . The event is chosen before independent candidate and target validation observations $X \sim b$ and $Y \sim Q$ are drawn. The score is $\mathbf{1}_A(Y) - \mathbf{1}_A(X)$. The downstream register needed to keep A and to compute this score is distinct from the register at the specified cut. The upstream and downstream widths will be stated separately in concrete implementations. This is the reset interface of Section 6, with one additional charged checkpoint. The actual mark is part of a completed observation; it is not an input to the private candidate's online transition rows.

Write $\mathcal{H} = [0, 1]^\Omega$ and

$$f_h(b) = \sum_{w \in \Omega} (Q(w) - b(w))h(w), \quad h \in \mathcal{H}.$$

Every randomized event has a response $h(w) = \Pr(w \in A)$. Conversely, every $h \in \mathcal{H}$ is the response of a random event: include each atom independently with its specified probability. This is a finite probability distribution on 2^Ω . Only its mean response matters for expected signed score. The event itself must still be retained during validation.

Definition 4.1 (Decision continuation spectrum). The K -state decision continuation value is

$$\mathfrak{V}_K(B, Q) = \max_{h_1, \dots, h_K \in \mathcal{H}} \min_{b \in B} \max_{m \leq K} f_{h_m}(b). \quad (4.1)$$

Let $v_{n,K}(B, Q)$ be the supremum of the worst-case expected score of n -sample auditors at the preceding interface. Only the designated cut has width K in this definition; all other widths are finite but unrestricted. For $c \in \mathbb{R}$, let $\kappa_c(B, Q)$ be the least K for which $\mathfrak{V}_K(B, Q) > c$, with value $+\infty$ when no such K exists.

The definition uses only the marked behavior image and its target, not a parameterization of B or a labeling of candidate states. No convex hull of B is taken. A response list is a list of conditional validation continuations; it is not a mixture of feasible private candidates.

Theorem 4.2 (Operational classification at a decision cut). *All extrema in (4.1) are attained. For every $n \geq 1$,*

$$0 \leq \mathfrak{V}_K(B, Q) - v_{n,K}(B, Q) \leq \min \left\{ \sqrt{\frac{d-1}{n}}, \sqrt{\frac{2 \log(2K)}{n}} \right\}. \quad (4.2)$$

Consequently $v_{n,K} \rightarrow \mathfrak{V}_K$. Every auditor with K states at the designated cut, at any finite training length, has worst-case score at most $\mathfrak{V}_K$. If $\mathfrak{V}_K > c$, some finite-sample auditor with K states at that cut has worst-case score greater than c . Thus κ_c is exactly the minimum cut width permitting such a strict certificate with sufficiently many training samples.

Equivalently, κ_c is the least number of response half-spaces

$$\{b \in B : f_h(b) > c\}, \quad h \in \mathcal{H}, \quad (4.3)$$

that cover B . If two parameterized behavior families satisfy, uniformly in their common parameter q ,

$$\|b(q) - \tilde{b}(q)\|_{\text{TV}} \leq \alpha, \quad \|Q - \tilde{Q}\|_{\text{TV}} \leq \beta,$$

then

$$|\mathfrak{V}_K(B, Q) - \mathfrak{V}_K(\tilde{B}, \tilde{Q})| \leq \alpha + \beta. \quad (4.4)$$

Proof. For a fixed response list, the inner objective is continuous in b . Its variation when the list changes is bounded uniformly over b by $2 \max_m \|h_m - \tilde{h}_m\|_\infty$. Its minimum over compact B is therefore continuous in the list. Compactness of $\mathcal{H}^K$ proves attainment.

Condition on the retained state of any admissible auditor. Its conditional post-cut event law depends only on that state and fresh randomness, and has a response h_m . Independence of validation from training gives

$$g(b) = \sum_{m=1}^K \Pr(M = m) f_{h_m}(b) \leq \max_m f_{h_m}(b).$$

Taking the minimum over b , and then the supremum over auditors, proves the upper bound. This argument permits arbitrary stochastic training and arbitrary finite upstream memory. An extra training-dependent decoder would invalidate its first conditioning step and is excluded by the interface, not silently supplied.

For the reverse bound fix an optimal list. Let $\hat{b}$ be the empirical measure and choose the first index maximizing $f_{h_m}(\hat{b})$. Retain that index and discard the empirical table. If m_* maximizes the true score, then pointwise

$$\max_m f_{h_m}(b) - f_{h_m}(\hat{b}) \leq 2 \max_{m \leq K} |(\hat{b} - b)h_m|. \quad (4.5)$$

The maximum is at most $\|\hat{b} - b\|_{\text{TV}}$, since $0 \leq h_m \leq 1$. Cauchy–Schwarz and the multinomial variances give

$$\mathbb{E}\|\hat{b} - b\|_{\text{TV}}^2 \leq \frac{d}{4n} \left(1 - \sum_w b(w)^2\right) \leq \frac{d-1}{4n}.$$

This proves the first rate.

For completeness, if V takes values in an interval of length one, then $\log \mathbb{E}e^{s(V - \mathbb{E}V)} \leq s^2/8$. Indeed the second derivative of the logarithmic moment generating function is the variance under the exponentially tilted law, at most $1/4$; its value and first derivative vanish at zero. Integrating twice proves the inequality for both signs of s . Independence consequently gives $\mathbb{E}e^{s(\hat{b}-b)h_m} \leq e^{s^2/(8n)}$. For $s > 0$,

$$\mathbb{E} \max_m |(\hat{b} - b)h_m| \leq \frac{\log(2K)}{s} + \frac{s}{8n}.$$

Minimizing in s gives $\sqrt{\log(2K)/(2n)}$ and proves the second rate in (4.2). The empirical table has finitely many values; the construction is therefore a finite-state auditor, with only the chosen index surviving the specified cut. Real response coefficients are allowed here; the physical construction below has deterministic rational transitions.

A finite cover in (4.3) says that the continuous function $b \mapsto \max_m f_{h_m}(b)$ is greater than c everywhere on compact B . Its attained minimum is then greater than c . This proves the covering characterization in both directions. It also proves the strict-threshold assertion; no claim about finite-sample attainment at the boundary $c = \mathfrak{V}_K$ is needed.

Finally $|(Q - b)h - (\tilde{Q} - \tilde{b})h| \leq \alpha + \beta$ for every response. Taking maxima, minima over the common parameter set, and maxima over lists preserves this uniform bound, proving (4.4). $\square$

Corollary 4.3 (Compression by a finite decision dictionary). *Suppose responses $h_1, \dots, h_K$ satisfy $\min_{b \in B} \max_m f_{h_m}(b) \geq c + \eta$ for $\eta > 0$. An empirical selector with $n > 2 \log(2K)/\eta^2$ achieves score greater than c and retains only K states at the decision cut. If these responses depend on a finite statistic $T : \Omega \rightarrow \Xi$, the training selector can count $T(X)$ instead of X . A sufficient upstream register has at most $\binom{n+|\Xi|-1}{|\Xi|-1}$ states at the last training checkpoint, and the corresponding time-dependent count registers suffice at earlier checkpoints.*

Proof. Use the second bound in (4.2). Empirical expectations of h_m depend only on the histogram of $T(X)$. At time t , that histogram has $\binom{t+|\Xi|-1}{|\Xi|-1}$ possibilities. Its update and the final maximization

are deterministic. No statistical sufficiency for estimating all of b is required; sufficiency for the specified responses is enough. $\square$

This corollary supplies an intermediate principle between a full marked histogram and a problem-specific compressed counter. The invariant is the number and geometry of validation continuations needed to cover the original candidate image at the desired score. It is not the algebraic dimension of a chosen parameterization. The empirical upper bound need not be sample-optimal, and a minimum decision-cut width is not a minimum of the largest width over the entire acquisition schedule.

4.2. Several feedback rows. The classification does not require that the declared experiment have only one feedback row. For the row-indexed interface of Section 6, let $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$ and define

$$f_\lambda(b) = \sum_{i \in \mathcal{I}} \lambda_i f_i(b), \quad \lambda \in \Delta(\mathcal{I}).$$

A post-cut continuation is now a probability distribution on row–event pairs; it is not an arbitrary test on a joint product of validation rows. Put

$$\mathfrak{V}_K^{\text{rows}}(B, Q) = \max_{\lambda_1, \dots, \lambda_K \in \Delta(\mathcal{I})} \min_{b \in B} \max_m f_{\lambda_m}(b).$$

Corollary 4.4 (Row-indexed classification). *With n training samples per row, the row-indexed decision-cut value converges to $\mathfrak{V}_K^{\text{rows}}$, and its deficit is at most*

$$\min\{\sqrt{D/n}, \sqrt{2 \log(2K)/n}\}, \quad D = \sum_j (|\Omega_j| - 1).$$

The strict half-space cover characterization holds with f_λ . The upper bound by $\mathfrak{V}_K^{\text{rows}}$ applies also to any other finite training schedule, including adaptive row selection.

Proof. Conditioning on the cut state gives the same convex-mixture upper bound as before; its proof places no restriction on how that state was reached. For empirical selection among an optimal list, $|(\hat{b} - b)\lambda_m| \leq \max_j \|\hat{b}_j - b_j\|_{\text{TV}}$. Sum the rowwise second moments to obtain the first rate. To obtain the second, align the r th independent training sample from each row into a tuple and write its response as $\sum_{j,A} \lambda_{m,j,A} \mathbf{1}_A(X_{j,r})$. This random variable belongs to $[0, 1]$, and its sample mean is the empirical candidate mean for that continuation. The logarithmic moment-generating argument in Theorem 4.2 therefore gives the same finite-list rate. Each chosen row–event distribution can be drawn using fresh randomness after the cut. Compactness and the covering argument are unchanged. $\square$

5. WEIGHTED CONTINUATION OBSTRUCTIONS

We next give a distribution-sensitive version of the wordwise obstruction. It applies to finite-valued terminal decisions and is useful even when a word of small probability should not be assigned the same weight as a likely one.

Let $f : \Sigma_1 \times \dots \times \Sigma_L \rightarrow \mathcal{A}$ have finite range. The residual of a prefix x at cut t is the function $v \mapsto f(xv)$ on the suffix alphabet. Write $r_t(f)$ for the number of distinct residuals. A stochastic machine has a charged register at every cut; all persistent randomness is included in this register. Its wordwise error is $e(w) = \Pr(\hat{f}(w) \neq f(w))$.

Theorem 5.1 (Finite-valued realization and weighted cut bound). *Exact stochastic computation of f requires $K_t \geq r_t(f)$ at every cut. The whole residual profile is simultaneously attained by a deterministic machine.*

Fix a cut, prefixes $x_1, \dots, x_r$, and a graph on $[r]$. For each edge ij choose a suffix v_{ij} such that $f(x_i v_{ij}) \neq f(x_j v_{ij})$. Let $\mu(w) \geq 0$ be arbitrary word weights. Choose edge weights $\lambda_{ij} \geq 0$ satisfying the word capacity constraints

$$\sum_{ij: w=x_i v_{ij} \text{ or } w=x_j v_{ij}} \lambda_{ij} \leq \mu(w) \quad \text{for every word } w. \quad (5.1)$$

Every machine with K states at this cut satisfies

$$\sum_w \mu(w) e(w) \geq \min_{\phi: [r] \rightarrow [K]} \sum_{ij: \phi(i)=\phi(j)} \lambda_{ij}. \quad (5.2)$$

The right side is a weighted monochromatic-edge obstruction. Maximizing it over admissible witnesses and weights gives a finite, distribution-sensitive lower bound. The bound can be an equality with positive error.

Proof. Let $a_i(m)$ be the state distribution after prefix x_i . For a fixed edge and a common state m , the future output distribution on suffix v_{ij} is the same for both prefixes. Since their correct outputs are different, the sum of their two conditional error probabilities is at least one. Multiplying by the smaller state mass and summing gives

$$e(x_i v_{ij}) + e(x_j v_{ij}) \geq \sum_{m=1}^K \min\{a_i(m), a_j(m)\}. \quad (5.3)$$

Weight the inequalities by λ_{ij} and use (5.1). The resulting lower bound is

$$F(a_1, \dots, a_r) = \sum_{ij} \lambda_{ij} \sum_m \min\{a_i(m), a_j(m)\}.$$

This is a concave continuous function on $(\Delta_K)^r$. Every point of that product is a convex combination of its deterministic vertices; concavity implies that its value is at least the minimum over those vertices. A vertex is a labeling $\phi: [r] \rightarrow [K]$, and its value is exactly the right side of (5.2). This proves the bound without treating stochastic encodings as deterministic encodings.

For zero error, choose one distinguishing suffix for every two different residuals. Equation (5.3) forces the corresponding state supports to be disjoint. Hence at least $r_t(f)$ states are needed. For the converse, keep the residual class of the observed prefix. Equal residuals have equal shifts after every next input, so the next class is well defined. At the last epoch the residual is the output label itself. These quotient transitions realize all widths simultaneously.

For a positive-error equality, let the only relevant words be two one-letter prefixes with different required labels, weighted p and $1 - p$, where $0 < p < 1$. Put $K = 1$ and $\lambda_{12} = \min(p, 1 - p)$. The cut obstruction is $\min(p, 1 - p)$, attained by always emitting the more likely label. Thus the weighted constant is sometimes exact; universal sharpness for all graphs or input distributions is not asserted. With unit weights on the same two words the total-error lower bound is one, also attained by a constant output; this is equality in the earlier $(r - K)/(r - 1)$ bound at $r = 2$, $K = 1$. $\square$

Corollary 5.2 (A Bayes-regret obstruction for constrained audits). *Fix a finite action set $\mathcal{A}$, candidate class B , and a probability measure ν on B . For a training word w , let $p_w(b)$ be its likelihood and let $f_a(b)$ be the expected validation payoff. Put*

$$R_w(a) = \int_B p_w(b) f_a(b) d\nu(b), \quad S_\nu = \sum_w \max_a R_w(a).$$

Choose a maximizing label $f(w)$. Set $\mu(w) = R_w(f(w)) - \max_{a \neq f(w)} R_w(a)$, taking zero when a maximum is tied. Let $\Gamma_{t,K}(f, \mu)$ be any lower bound supplied by (5.2). Then every width- K audit

at cut t has worst-case expected payoff at most

$$S_\nu - \Gamma_{t,K}(f, \mu). \quad (5.4)$$

Proof. The conditional distribution of the auditor's output for a fixed word is independent of the unknown candidate. Its Bayes payoff is therefore $\sum_{w,a} R_w(a) \Pr(\hat{f}(w) = a)$. Subtracting this from S_ν gives at least $\sum_w \mu(w)e(w)$. Apply the theorem, and use that a minimum over candidates is at most the average under ν . $\square$

The corollary does not exchange a maximum and minimum over a nonconvex fixed-width machine class. The prior is a lower-bound device, not a randomization resource supplied to the candidate. With a unique maximizing label and strictly positive margins at every word, equality with S_ν forces exact wordwise computation. This makes explicit the uniqueness step used in the odd-sample binary theorem below. For even sample size, the balanced-count words have zero likelihood gap. A rule can differ on those ties without losing value; equality of values then does not force one prescribed total Boolean rule, and the odd-sample width profile cannot simply be asserted for that case.

6. FINITE-DATA TESTING ON THE ORIGINAL BEHAVIOR IMAGE

6.1. The reset interface. Let $B \subset \prod_{j=1}^J \Delta(\Omega_j)$ be compact, and let $Q = (Q_j)$ be fixed. Set $d_j = |\Omega_j|$, $D = \sum_j (d_j - 1)$, $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$, and $f_{j,A}(b) = Q_j(A) - b_j(A)$. Include the empty event. No convexity of B is assumed.

A candidate b is fixed for the whole audit. The auditor obtains n independent samples of each b_j , with independence between rows. In a causal experiment, row j means a specified parameter and deterministic feedback tree; the reset interface supplies that row. It is an additional experimental interface, not information available to the online simulator. A complete sample contains the terminal mark. Private randomness and any per-run hidden selector are reset between runs. The design of the candidate is not resampled.

Write $C_{j,w}$ for the counts, and $\mathcal{N}_n$ for all count arrays with $\sum_w C_{j,w} = n$. Their probability is the polynomial

$$p_C(b) = \prod_j \frac{n!}{\prod_w C_{j,w}!} \prod_w b_j(w)^{C_{j,w}}.$$

A randomized audit is an array $c = (c_{C,i})$ with $c_C \in \Delta(\mathcal{I})$. After training it chooses $i = (j, A)$ according to c_C . Independent candidate and target validation runs give the signed score $\mathbf{1}_A(\text{target}) - \mathbf{1}_A(\text{candidate})$. Its expected value is

$$g_c(b) = \sum_{C,i} p_C(b) c_{C,i} f_i(b), \quad v_n^{\text{aud}}(B, Q) = \max_c \min_{b \in B} g_c(b). \quad (6.1)$$

Thus every audit payoff is polynomial of total degree at most $Jn + 1$ in the behavior coordinates. The probability of a large observed score in a single audit is not asserted to equal its expectation; additional independent validation would be required for a specified confidence level.

Theorem 6.1 (Reset-audit duality and finite-sample approximation). *Let $\delta = \min_{b \in B} \max_i f_i(b)$. Then all the extrema below are attained, and*

$$v_n^{\text{aud}}(B, Q) = \min_{\mu \in \mathcal{P}(B)} \sum_{C \in \mathcal{N}_n} \max_{i \in \mathcal{I}} \int_B p_C(b) f_i(b) d\mu(b), \quad (6.2)$$

$$0 \leq \delta - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}. \quad (6.3)$$

The values are nondecreasing in n . If $a = \dim \text{aff } B$, a minimizing measure in (6.2) can be supported on at most $\binom{a+Jn+1}{a}$ points.

An explicit feasible audit attaining the lower bound $\delta - \sqrt{D/n}$ chooses an event maximizing $Q_j(A) - C_j(A)/n$, with a fixed deterministic tie rule. If the rational target probabilities are given, this is a rationally specified finite rule. With a free schedule clock, a conservative retained-state bound is

$$H(|\mathcal{I}| + 1)(n + 1) \sum_j d_j,$$

where H is the number of states needed to execute and record one causal transcript. This budget is charged to the auditor, not to the simulator.

Proof. The coefficient arrays form a compact convex product of finite simplexes. The integrand is continuous in b and affine in c . The probability measures on compact B form a compact convex set for weak convergence. The minimum over b equals the minimum over such measures for fixed c . Finite-dimensional minimax (or its compact measure version) therefore exchanges the two extrema. Maximization over each independent coefficient simplex selects its largest coordinate integral, giving (6.2). Continuity and compactness give attainment. This step uses measures on the original B , not feasible private realizations of their barycenters.

For any audit, $g_c(b) \leq F(b) = \max_i f_i(b)$. At a minimizer of F this gives $v_n^{\text{aud}} \leq \delta$. The zero event gives $v_n^{\text{aud}} \geq 0$. For the stated explicit rule put $\hat{b}_j = C_j/n$ and $\rho = \max_j \|\hat{b}_j - b_j\|_{\text{TV}}$. Comparing the empirical maximizer with a true maximizing event gives the pointwise regret bound

$$F(b) - f_{i(C)}(b) \leq 2\rho.$$

For a multinomial empirical measure,

$$\mathbb{E} \|\hat{b}_j - b_j\|_{\text{TV}}^2 \leq \frac{d_j}{4} \sum_w \mathbb{E} (\hat{b}_j(w) - b_j(w))^2 = \frac{d_j(1 - \sum_w b_j(w)^2)}{4n} \leq \frac{d_j - 1}{4n}.$$

Consequently $\mathbb{E}\rho \leq (\sum_j \mathbb{E} \|\hat{b}_j - b_j\|_{\text{TV}}^2)^{1/2} \leq \sqrt{D/n}/2$. Averaging the regret uniformly over B proves (6.3).

From $n + 1$ unordered samples in each row, discard one uniformly at random. The remaining counts have exactly the n -sample law, independently across rows. This realizes every n -sample audit as a randomized $(n + 1)$ -sample audit and proves monotonicity.

Choose affine coordinates on $\text{aff } B$. All the functions $p_C f_i$ lie in the span of monomials of total degree at most $Jn + 1$, a space of dimension at most $s = \binom{a + Jn + 1}{a}$ including the constant. The moment vector of a probability measure is in the convex hull of the evaluation vectors; their constant coordinate is one, so their affine dimension is at most $s - 1$. Eliminating affine dependencies gives a representation by at most s points. It preserves every integral in (6.2), and hence preserves the objective.

Finally, each count is in $\{0, \dots, n\}$. The indicated product register stores all counts, the within-run state, and the selected event for validation. Only the latter event and the within-run buffer need survive once training ends, so the product bound is conservative. Randomizing an event uses fresh randomness. The schedule, reset access and terminal mark readout were declared before the theorem. No private simulator state is read by this implementation. $\square$

The dual measure chooses a candidate once, after which its samples are independent conditional on that candidate. Replacing it by $\bar{b} = \int b d\mu$ would replace $\int p_C(b) f_i(b) d\mu$ by $p_C(\bar{b}) f_i(\bar{b})$, generally a different quantity. It would also replace the joint replicated law by a product of its marginals. Neither operation is part of the theorem. This is the operational meaning of the moment side, rather than a same-budget convexification theorem.

Proposition 6.2 (Both sides sampled). *Suppose the target reference Q^b varies with the candidate, as it does when visible-selector weights are design variables. Supply independent training samples from both b and Q^b , and use their empirical difference. If their row dimensions are D_P, D_Q , the*

same construction has regret at most $\sqrt{D_P/n} + \sqrt{D_Q/n}$. Minimax and attainment hold on the compact image of pairs (b, Q^b) . In particular a variable reference law must not be treated as known fixed input to Theorem 6.1.

Proof. The error in any event discrepancy is at most the sum of the maximal row variations of the two empirical tables. Apply the preceding second moment bound separately and then the empirical-maximizer regret inequality. Count probabilities are products for the two tables. They are continuous polynomials on the compact image of pairs, so the same minimax argument applies without changing the simulator signature. $\square$

6.2. A matching uniform sampling obstruction.

Theorem 6.3 (Binary reset lower bound). *Take one binary row with target $\text{Ber}(1/2)$ and candidate class $B_h = \{\text{Ber}(1/2 - h), \text{Ber}(1/2 + h)\}$, where $0 < h < 1/2$. Then*

$$\delta = h, \quad v_n^{\text{aud}} = h \|\text{Ber}(1/2 + h)^{\otimes n} - \text{Ber}(1/2 - h)^{\otimes n}\|_{\text{TV}}. \quad (6.4)$$

For every $n \geq 1$, choosing $h = 1/(8\sqrt{n})$ gives

$$\delta - v_n^{\text{aud}} \geq \frac{1}{16\sqrt{n}}.$$

Thus the exponent in (6.3) is sharp uniformly over compact candidate classes, already at affine dimension one. This does not assert a degree lower bound for revealed-behavior polynomials, nor that this two-point class equals an unrestricted private-row image.

Proof. Only the two complementary singleton events and the zero event matter. A randomized choice is a statistic d in $[-1, 1]$. The two expected payoffs are $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$. Their minimum is at most half their sum, which is at most $h\|P_+^n - P_-^n\|_{\text{TV}}$. Conversely take the sign of the likelihood difference, assigning zero at a tie. Complementing every sample interchanges P_+^n and P_-^n and changes that sign. The two payoffs are equal, and their average attains the variation bound. The statistic depends only on the count, so it is an admissible audit. The symmetric measure on the two candidates attains the dual value. Every Dirac measure has dual objective h , whereas for finite n the two product laws are equivalent and their variation is strictly less than one. Thus no dual minimizer is a Dirac measure. Their barycenter is the target law, but replacing the dual by that barycenter would incorrectly give value zero instead of the strictly positive value in (6.4).

For $h \leq 1/8$, elementary integration of $(1 - u^2)^{-1}$ gives

$$\text{KL}(\text{Ber}(1/2 + h) \parallel \text{Ber}(1/2 - h)) = 2h \log \frac{1 + 2h}{1 - 2h} \leq 16h^2.$$

The product relative entropy is at most $16nh^2$. Pinsker's inequality bounds its total variation by $\sqrt{8nh^2}$. At the specified h this is less than $1/2$. Substituting in (6.4) proves the claim. The inequality uses the classical two-point testing mechanism; the restriction being measured is the reset budget of the present game. $\square$

6.3. Transport of one fixed audit.

Proposition 6.4 (Reset cost under model perturbation). *Let q range over the same compact set of executable row designs in two models. Suppose uniformly in q, j that*

$$\|P_j^q - \tilde{P}_j^q\|_{\text{TV}} \leq \alpha, \quad \|Q_j - \tilde{Q}_j\|_{\text{TV}} \leq \beta.$$

For the same coefficient rule c , evaluated with each model's own target validation,

$$|g_c(q) - \tilde{g}_c(q)| \leq (2Jn + 1)\alpha + \beta. \quad (6.5)$$

The deficiencies themselves differ by at most $\alpha + \beta$, at the same resource budget. If the source behavior is unchanged, a fixed audit loses at most β , independently of its reset count.

Proof. A product coupling of the Jn training runs fails with probability at most $Jn\alpha$. Taking counts and choosing the event are contractions. A score with fixed target discrepancies lies in $[-1, 1]$, so the change in its expectation from this training-law perturbation is at most $2Jn\alpha$. At each event the discrepancy changes by at most $\alpha + \beta$. This proves (6.5). For deficiency, every row variation changes by at most $\alpha + \beta$; taking the maximum and then the minimum preserves that bound. No new simulator is inserted. $\square$

For a rational polynomial image, a rational rule and a rational threshold, strict positivity of $g_c(q)$ above that threshold is a first-order formula over the original row simplexes. Real quantifier elimination decides it. The same is true of an existential rule followed by this universal condition. This is an exact finite verification statement, not an efficient algorithm. The explicit empirical rule below will be verified by a uniform analytic inequality, rather than by storing its enormous complete table.

7. CONTINUATION COMPLEXITY OF A FINITE-DATA TEST

The reset budget does not by itself bound the auditor's retained memory. We now impose both constraints. At scheduled epoch t a streaming auditor reads one letter x_t from a finite alphabet X_t , updates a register, and emits no information to any auxiliary store. It makes its decision on reading the last letter. Transition rows may depend on the epoch; the clock is free. All other information crossing a cut belongs to the register. Fresh randomization is allowed. An independent persistent selector, when allowed, selects a complete auditor before any letters are read.

For a Boolean decision $\psi : X_1 \times \cdots \times X_n \rightarrow \{-1, 1\}$, let $u \sim_t v$ mean that

$$\psi(us) = \psi(vs) \quad \text{for every } s \in X_{t+1} \times \cdots \times X_n.$$

Write $r_t(\psi)$ for the number of equivalence classes of prefixes of length t , for $0 \leq t < n$. These are the deterministic continuation responses. The terminal decision is emitted on the last transition, so no terminal register is needed. In particular, $r_0 = 1$. The equivalence is a finite-horizon version of the classical residual-state construction [16]. The following quantitative statement will also measure the loss when a statistical auditor has fewer states than this construction.

Theorem 7.1 (Exact continuation width and a quantitative cut obstruction). *A stochastic streaming auditor computes ψ exactly on every word with width profile $(K_t)_{0 \leq t < n}$ if and only if $K_t \geq r_t(\psi)$ at every cut. The profiles are attained simultaneously by a deterministic auditor. Independent persistent randomization does not reduce these widths for exact computation.*

More generally, at a cut with $r = r_t(\psi) \geq 2$, every stochastic auditor of width K satisfies

$$\sum_{w \in X_1 \times \cdots \times X_n} e_\psi(w) \geq \frac{(r - K)_+}{r - 1}, \quad (7.1)$$

where $e_\psi(w)$ is its probability of an incorrect Boolean decision on input w . The inequality also holds after averaging over an independent selector whose conditional auditors each have width at most K .

Proof. From a prefix u let $e_u(m)$ be the distribution of the retained state. From state m and suffix s let $a_m(s)$ be the probability of terminal answer $+1$. Exact computation says

$$\sum_m e_u(m) a_m(s) = \frac{1 + \psi(us)}{2}.$$

The right side is zero or one. Every state of positive e_u -mass must therefore have precisely that zero-one continuation response for all s . Distinct prefix classes have disjoint positive supports. Hence $K_t \geq r_t$. Conversely, retain the prefix class. Appending the same letter to two equivalent prefixes gives equivalent new prefixes; thus every transition is well defined. At the last transition

the class and the last letter determine ψ . This realizes every width r_t simultaneously. For a persistent independent selector, a zero average error on each of the finitely many words implies zero conditional error on all words outside a single null set of selector values. The preceding bound applies there.

For the quantitative claim choose representative prefixes $u_1, \dots, u_r$, one per class, and put $e_i(m) = e_{u_i}(m)$. Define

$$o_{ij} = \sum_{m=1}^K \min\{e_i(m), e_j(m)\}.$$

For nonnegative numbers $a_1, \dots, a_r$, ordering them shows that $\sum_{i < j} \min(a_i, a_j) \geq \sum_i a_i - \max_i a_i$. Apply this at each state. Since every e_i has mass one and each $\max_i e_i(m) \leq 1$, it follows that

$$\sum_{i < j} o_{ij} \geq r - K. \quad (7.2)$$

Choose a suffix s_{ij} distinguishing u_i, u_j . Conditional on a common state and this suffix, the two desired answers are opposite. Their error probabilities, weighted by $e_i(m), e_j(m)$, have sum at least $\min\{e_i(m), e_j(m)\}$. Thus

$$e_\psi(u_i s_{ij}) + e_\psi(u_j s_{ij}) \geq o_{ij}.$$

On summing, each complete word on the left occurs at most $r - 1$ times: its prefix identifies a single representative, which participates in $r - 1$ pairs. Nonnegativity of all other word errors and (7.2) prove (7.1). When $r \leq K$ the claim is just nonnegativity. Finally apply the bound conditional on a selector and integrate. Independence ensures that conditioning has not changed the input experiment. $\square$

The exact part is also the extremal zero-one case of Theorem 3.2. Its content is a simultaneous width profile, rather than separate unrelated factorizations at different cuts. The error bound does not assert that its constant is optimal for every Boolean function. Its use below is to convert a continuation obstruction into a strict loss of the attainable statistical value.

7.1. The width required to attain the binary reset value. Fix $0 < h < 1/2$ and the binary candidate class B_h of Theorem 6.3. A terminal decision $+1$ chooses the singleton event appropriate to the $+$ candidate, and -1 the complementary event. A zero-score event can equivalently be replaced by an independent fair choice of these decisions. The expected payoffs are therefore $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$, where $d \in [-1, 1]$ is the conditional mean decision and $P_\pm = \text{Ber}(1/2 \pm h)^{\otimes n}$. Let $v_{n,K}$ be the supremum of the smaller of these two payoffs over streaming auditors with the declared width profile. The final validation interface is unchanged; the widths here count training and selection only.

Theorem 7.2 (Exact memory threshold for optimal binary auditing). *Let $n = 2m + 1$ be odd. The unconstrained reset value is $v_n = h\|P_+ - P_-\|_{\text{TV}}$. It is attained at the profile*

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n, \quad (7.3)$$

and $v_{n,K} = v_n$ if and only if $K_t \geq r_t$ for all these cuts. For every cut with $r_t \geq 2$ one has the explicit deficit

$$v_n - v_{n,K} \geq 2h^2(1/4 - h^2)^m \frac{(r_t - K_t)_+}{r_t - 1}. \quad (7.4)$$

These conclusions remain valid when an independent persistent selector chooses auditors of the same conditional widths. For $n \geq 3$ the least peak training width attaining v_n is $m + 2$; for $n = 1$ it is one.

Proof. The sign of $P_+(w) - P_-(w)$ is the majority decision $\psi(w)$, with no zero difference since n is odd. Complementing every bit interchanges the two laws and reverses the decision. Consequently majority has equal payoffs and attains $h\|P_+ - P_-\|_{\text{TV}}$, as also proved in Theorem 6.3.

A prefix with k ones requires at least $m + 1 - k$ ones in its remaining $n - t$ places. Thresholds at most zero give the constant positive response; thresholds greater than $n - t$ give the constant negative response; the remaining thresholds give different nonconstant responses. Counting them for $0 \leq k \leq t$ gives (7.3). The exact realization in Theorem 7.1 implements majority at all these widths.

For an arbitrary randomized decision, its smaller payoff is bounded by the equal-prior average. The deficit of this average from the majority value is exactly

$$h \sum_w |P_+(w) - P_-(w)| e_\psi(w). \quad (7.5)$$

Put $a = 1/2 + h$ and $b = 1/2 - h$. A word with $k \geq m + 1$ ones has absolute likelihood difference $(ab)^{n/2} \{(a/b)^{k-n/2} - (a/b)^{-(k-n/2)}\}$, which increases with k . Symmetry treats the other half of the words. The smallest difference is therefore attained at $k = m, m + 1$ and equals

$$a^{m+1}b^m - b^{m+1}a^m = 2h(1/4 - h^2)^m.$$

Substitution of (7.1) into (7.5) gives (7.4). Any deficient cut gives a strictly positive loss; all sufficiently wide profiles implement majority. Maximizing (7.3) proves the peak-width assertion. The same average-payoff argument works conditional on an independent selector and hence after averaging. $\square$

The result identifies the exact memory needed to attain the optimal value, not the memory needed for a prescribed approximate value uniformly in h . The quantitative deficit can be small for large n . In particular it is not a sharp joint asymptotic tradeoff in sample size, state size and error. It complements, rather than replaces, the uniform sampling exponent in Theorem 6.3. No minimax interchange on the generally nonconvex set of fixed-width auditors has been used.

8. POLYNOMIAL TESTING OF THE ORIGINAL RESOURCE IMAGE

The private behavior image is usually nonconvex. Its testing criterion can nevertheless be written as a limit of finite algebraic lower certificates, without replacing that image by randomized designs. The testing weights in this section depend on the candidate simulator. Unlike Theorem 9.1, this identity is a certificate transform, not an independent variational dual. They are universal proof functions, not a hidden selector supplied to an executed experiment.

Use any one of the finite signatures of Theorem 2.2, with rational instrument data. Write its objective as $F(q) = \max_{1 \leq i \leq m} f_i(q)$, where f_i are all its marked event discrepancies, including the empty event $f_1 = 0$. For visible selectors the event and feedback rule include their selector fibers, and the target selector law remains in f_i . Thus $-1 \leq f_i \leq 1$ for all three signatures; the polynomials may be different.

Parameterize a row with d outcomes by $d - 1$ variables in $[0, 1]$:

$$q_1 = x_1, \quad q_j = x_j \prod_{l < j} (1 - x_l) \quad (1 < j < d), \quad q_d = \prod_{l < d} (1 - x_l).$$

This map is onto the row simplex, including its boundary. Indeed, divide each desired probability by the remaining mass while that mass is positive; when it is zero choose the remaining variables arbitrarily. Taking the product of these maps, including a selector row when needed, gives an onto polynomial map $q(x)$ from $[0, 1]^D$ to the feasible row product. Put $h_i(x) = f_i(q(x))$ and $g_i = (1 + h_i)/2$. All h_i are rational polynomials, $0 \leq g_i \leq 1$, and $g_1 = 1/2$. The case $D = 0$ is a single feasible table and is interpreted directly.

Theorem 8.1 (Strict polynomial lower-certificate hierarchy). *For $p \geq 1$ define*

$$A_p(x) = \frac{\sum_{i=1}^m g_i(x)^p h_i(x)}{\sum_{i=1}^m g_i(x)^p}, \quad a_p = \min_{x \in [0,1]^D} A_p(x), \quad \delta = \min_q F(q).$$

Then the denominators are positive, a_p is nondecreasing, and

$$\delta - 2(1 - m^{-1/(p+1)}) \leq a_p \leq \delta, \quad \delta = \sup_{p \geq 1} a_p. \quad (8.1)$$

For rational L let

$$P_{p,L}(x) = \sum_{i=1}^m g_i(x)^p (h_i(x) - L). \quad (8.2)$$

A nonnegative tensor Bernstein expansion of $P_{p,L}$ on $[0,1]^D$ certifies $\delta \geq L$. If $L < \delta$, some p and some finite Bernstein degree have strictly positive coefficients. Conversely, a strictly positive such expansion certifies $L < \delta$. These are lower certificates at the original state and selector budgets. No convexification of the feasible behavior image occurs.

Proof. Fix x and put $b = \max_i g_i(x) \geq 1/2$ and $r_i = g_i(x)/b$. At least one r_i is one. Write $S_j = \sum_i r_i^j$. Hölder's inequality gives

$$S_p \leq m^{1/(p+1)} S_{p+1}^{p/(p+1)}, \quad S_{p+1} \geq 1.$$

Consequently $S_{p+1}/S_p \geq m^{-1/(p+1)}$. Since $A_p = 2bS_{p+1}/S_p - 1$ and $\max_i h_i = 2b - 1$,

$$0 \leq \max_i h_i - A_p \leq 2(1 - m^{-1/(p+1)}).$$

Cauchy–Schwarz gives $S_{p+2}S_p \geq S_{p+1}^2$, so A_p is nondecreasing in p . Its denominator is at least 2^{-p} and it is continuous on the cube. Taking minima proves (8.1). The bound is also at most $2 \log m/(p+1)$, by $1 - e^{-u} \leq u$.

Every tensor Bernstein basis function is nonnegative on the cube, and the basis functions sum to one. Nonnegative coefficients therefore imply $P_{p,L} \geq 0$. Dividing by the positive denominator gives $F(q(x)) \geq A_p(x) \geq L$ for every x , which proves soundness. For completeness choose p so that the error in (8.1) is less than $\delta - L$. Then

$$P_{p,L}(x) \geq 2^{-p} \{\delta - L - 2(1 - m^{-1/(p+1)})\} > 0.$$

Lemma 8.2 below gives a finite strictly positive Bernstein expansion. Conversely strictly positive coefficients have a positive minimum, and the denominator is at most m , so they give a strict positive uniform gap above L . $\square$

Positive Bernstein certificates and their degree elevation are classical; see de Klerk–Laurent [20]. The next elementary bound is retained for a self-contained exact verifier, not as a new general hypercube positivity theorem.

Lemma 8.2 (An explicit positivity degree). *Let $P(x) = \sum_k c_k x^k$ be a rational polynomial in D variables and $n \geq \max_{k,j} k_j$. In its tensor Bernstein expansion of degree n in each coordinate the coefficient at $\alpha \in \{0, \dots, n\}^D$ is*

$$b_{\alpha,n} = \sum_k c_k \prod_{j=1}^D \frac{\binom{\alpha_j}{k_j}}{\binom{n}{k_j}}, \quad |b_{\alpha,n} - P(\alpha/n)| \leq \frac{C(P)}{n}, \quad C(P) = \sum_k |c_k| \sum_j \binom{k_j}{2}. \quad (8.3)$$

In particular, $\min P > 0$ implies positivity of every coefficient whenever $n > C(P)/\min P$ and the degree restriction holds.

Proof. For a binomial random variable B with parameters (n, x) , $\mathbb{E}\binom{B}{k} = \binom{n}{k}x^k$. Expanding this expectation proves the univariate monomial formula, and products prove the displayed coefficient identity. The convention is $\binom{a}{k} = 0$ for $a < k$. To compare $\binom{a}{k}/\binom{n}{k}$ with $(a/n)^k$, draw k indices uniformly from $\{1, \dots, n\}$. Conditional on no repetition this is sampling without replacement. The probability of a repetition is at most $\binom{k}{2}/n$. Apply this total variation bound to the event that all drawn indices lie in a prescribed a -element subset. Hence the difference of the two factors is at most $\binom{k}{2}/n$. All factors lie in $[0, 1]$. A product telescoping bound followed by the triangle inequality proves (8.3). The final assertion follows by taking the minimum of P . $\square$

Corollary 8.3 (Rational lower witnesses without policy cells). *At every fixed finite rational signature, strict infeasibility above a rational level L is equivalent to the existence of the following finite rational certificate: an integer p , an integer n , and a strictly positive coefficient array for (8.2) whose identity is checked by (8.3). Dovetailing these certificates with rational original-budget upper witnesses produces nested intervals converging to δ . If a subfamily of the original tests gives a strict gap, the same construction applies to it after adjoining the empty test.*

Proof. Soundness and completeness are Theorem 8.1. Each finite candidate is checked by rational polynomial arithmetic. Enumerate all (p, n) and rational levels, and retain the largest verified lower level so far, together with zero. Rational row tables are dense in the feasible product, and their continuous objectives converge to the minimum, so enumeration and retention of the best upper table converge from above. This is a dovetailed enumeration, not a claim that any fixed (p, n) test search terminates. For a requested positive interval width, convergence of both endpoints proves termination. The reduced family conclusion uses only the same pointwise maximum argument. $\square$

Remark 8.4 (Meaning and size of the strict certificate transform). A constant distribution on tests generally detects only the convex hull of the behavior image. The functions $g_i^p / \sum_j g_j^p$ are instead functions of a proposed row table. Their universally quantified weighted inequality is an algebraic certificate; the tester need not observe a private state in an experiment. The displayed certificate identity is not an equality obtained by exchanging a private minimum with a fixed mixed test. Power means and Bernstein degree elevation are classical algebraic approximation devices, proved here in the precise form needed. The claim here is strict-level completeness for the original marked causal resource image, including visible selectors, and its physical transfer below. The coefficient count $(n+1)^D$, the number of original tests, and the algebraic degrees may be large. No polynomial complexity or intrinsic-dimension bound on this row-cube coefficient count is asserted. The independent adaptive test has the separate behavior-size bound of Theorem 9.3; its verification is a different cost.

8.1. A sharp two-time nonconvex example. The source has singleton reports, the control alphabet and actual mark are singleton, and the target emits $(0, 0)$ or $(1, 1)$ with probability $1/2$ each. At width one, a private simulator emits two independent bits with probabilities p, q . It cannot reread its first emitted bit.

Proposition 8.5 (A gap invisible to constant testing mixtures). *For this delayed experiment,*

$$\delta_{(1,1)}^p = \sqrt{2} - 1, \quad \delta_{(1,1),2}^h = 0, \quad \delta_{(1,1),r}^v = \sqrt{2} - 1 \quad (r \geq 1).$$

The private objective is the maximum of the four polynomials

$$0, \quad \frac{1}{2} - pq, \quad \frac{1}{2} - (1-p)(1-q), \quad p + q - 2pq.$$

Every constant mixture of these tests has minimum at most zero, whereas Corollary 8.3 certifies every rational $L < \sqrt{2} - 1$ at private width one.

Proof. Put $u = pq$ and $v = (1-p)(1-q)$. Direct total variation calculation gives $F = 1 - \min(u, 1/2) - \min(v, 1/2)$ and the displayed maximum. By Cauchy–Schwarz, $\sqrt{u} + \sqrt{v} \leq 1$. If

$u \geq 1/2$, then $v \leq (1 - 1/\sqrt{2})^2$, so the sum of the two capped terms is at most $2 - \sqrt{2}$; the case $v \geq 1/2$ is symmetric. Otherwise set $a = \sqrt{u}$, $b = \sqrt{v}$. On $0 \leq a, b \leq 1/\sqrt{2}$, $a + b \leq 1$, the maximum of $a^2 + b^2$ is at a vertex of this polygon and equals $2 - \sqrt{2}$. Equality is attained at $p = q = 1/\sqrt{2}$. A hidden fair bit selects the two deterministic constant sequences and exactly reproduces the target. If it is exposed, variation is averaged over the selector fibers; each fiber has error at least the private minimum, which is itself attainable without a selector. Finally the target lies in the convex hull of the two private deterministic behaviors. The average discrepancy of any fixed test mixture at those behaviors is zero, so at least one has discrepancy at most zero. The strict polynomial certificates therefore detect information lost by constant convex separation. $\square$

9. ADAPTIVE TESTING ON THE OBSERVABLE BEHAVIOR IMAGE

The obstruction to linear randomization criteria at a fixed private register is the nonconvexity of the behavior image, not a failure of compactness. We give a variational formulation in which a test is an independent function of an observable candidate law. The construction does not give the simulator an extra random variable.

9.1. The testing game. Let B be a nonempty compact subset of a finite-dimensional affine space, and let $f_1, \dots, f_m$ be affine real functions, including the zero function. Put

$$F(b) = \max_{i \leq m} f_i(b), \quad \delta = \min_{b \in B} F(b).$$

For the finite comparison problem of Section 2, b is the array of discrepancies between the simulated and target marked laws, indexed by parameters, feedback trees and transcripts; f_i ranges over their signed event evaluations. In the visible case the array includes the selector and its target weights. Thus the target coordinate is part of the array when those weights vary. Private, fixed-hidden and fixed-visible signatures each give their own compact set B . No convex hull is taken.

An *adaptive test* is a continuous map $\kappa : B \rightarrow \Delta_m$, chosen before a candidate b . Its score against that candidate is

$$G(b, \kappa) = \sum_i \kappa_i(b) f_i(b).$$

This is a verification game: the selected candidate law is revealed to the evaluator of the already chosen function κ . It is not a one-observation statistical test, and does not make an unavailable parameter or mark observable to a physical controller.

Theorem 9.1 (Revealed-behavior adaptive representation). *For every such compact behavior image,*

$$\begin{aligned} \delta &= \sup_{\kappa \in C(B, \Delta_m)} \min_{b \in B} G(b, \kappa) \\ &= \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa \in C(B, \Delta_m)} \int_B G(b, \kappa) d\mu(b). \end{aligned} \tag{9.1}$$

The minimization over probability measures is attained. If Borel adaptive tests are allowed, the first supremum is attained as well. The continuous-test assertion is a supremum; continuous attainment is not needed or asserted.

Proof. Fix $\eta > 0$. The sets $U_i = \{b : f_i(b) > F(b) - \eta\}$ form an open cover of the compact metric space B . Choose a continuous partition of unity subordinate to this finite cover. The resulting test has $F(b) - \eta < G(b, \kappa) \leq F(b)$ for all b . Consequently the first supremum equals δ . The same uniform inequality shows, for every probability μ ,

$$\sup_{\kappa} \int G(b, \kappa) d\mu(b) = \int F d\mu.$$

Its minimum is δ , attained by a point mass at a minimizer of F . This also proves the displayed minimax exchange without an assumption that B is convex. Equivalently one can apply the usual compact-convex minimax theorem to $\mathcal{P}(B)$ and the convex class of continuous tests, with the pairing $\int G d\mu$. Finally choose the smallest index attaining $F(b)$. Its level sets are Borel, and its indicator vector is a Borel test with score exactly F . $\square$

The probability μ here describes a lottery over *revealed candidate laws*. The score is $\int G(b, \kappa) d\mu(b)$, not $G(\int b d\mu, \kappa)$. Hiding the candidate and replacing it by its barycenter is a different game; in general it lowers the value. This distinction is essential for a finite persistent-state budget.

9.2. A size bound in behavior dimension. Factor out directions annihilated by every $f_i - f_i(b_0)$, so that on the resulting difference space

$$\|v\|_* = \max_i |f_i(b_0 + v) - f_i(b_0)|$$

is a norm. Denote its dimension by a and let $R = \text{diam}_*(B)$. All conclusions below are unaffected by this observable quotient. The case $a = 0$ is a single constant comparison and requires one test. Suppose henceforth that $a \geq 1$, so $R > 0$.

Lemma 9.2 (A bounded behavior chart). *There exist $b_0 \in B$ and vectors $v_j \in B - b_0$, $1 \leq j \leq a$, forming a basis, such that $\|v_j\|_* \leq R$ and every $b \in B$ has coordinates $b = b_0 + \sum_j t_j v_j$ with $|t_j| \leq 2$. If B is a rational polynomial image of a product of simplexes, the points can be chosen rational and can be found effectively.*

Proof. Fix $b_0 \in B$. In any fixed coordinates on the difference space, let $D_0 > 0$ be the maximum absolute determinant of a vectors in $B - b_0$. Compactness and spanning give existence and positivity. Choose a tuple with determinant greater than $D_0/2$. Replacing its j th vector by $b - b_0$ multiplies the determinant by t_j ; maximality gives $|t_j| < 2$. The norm bound is the definition of R . For a rational polynomial image, take b_0 at rational rows. Rational row images are dense. An affine basis can be found by enumerating rational images and testing rational determinants; the true affine span and the maximum determinant are semialgebraic data. Quantifier elimination decides whether a proposed nonzero determinant exceeds half the maximum. Density therefore makes the search terminate. This is an effectiveness assertion, not a bit-complexity bound for finding the chart. $\square$

Set $b(x) = b_0 + \sum_j (4x_j - 2)v_j$ and $C = \{x \in [0, 1]^a : b(x) \in B\}$. The ambient cube is a coordinate box, not an enlargement of the feasible behavior set. Write $\ell_i(x) = f_i(b(x))$. Then

$$|\ell_i(x) - \ell_i(y)| \leq 4R \sum_j |x_j - y_j|. \quad (9.2)$$

For $n \geq 0$ let $\beta_{\alpha, n}$ be the tensor Bernstein basis of coordinatewise degree n . A feasible coefficient array has $c_{\alpha i} \geq 0$ and $\sum_i c_{\alpha i} = 1$ for every $\alpha \in \{0, \dots, n\}^a$. It specifies the independent test

$$\kappa_i^c(x) = \sum_{\alpha} c_{\alpha i} \beta_{\alpha, n}(x).$$

Define

$$d_n = \max_c \min_{x \in C} \sum_{\alpha, i} c_{\alpha i} \beta_{\alpha, n}(x) \ell_i(x). \quad (9.3)$$

Theorem 9.3 (Intrinsic adaptive degree and support bounds). *The maxima in (9.3) are attained and $d_n \leq d_{n+1} \leq \delta$. For $n \geq 1$,*

$$0 \leq \delta - d_n \leq \frac{4Ra}{\sqrt{n}}. \quad (9.4)$$

There is an array with at most $(n+1)^a$ nonzero coefficients, all equal to one, whose minimum score is at least $\delta - 4Ra/\sqrt{n}$. Its test has coordinatewise degree n , and its score degree at most $n+1$. Furthermore

$$d_n = \min_{\mu \in \mathcal{P}(C)} \sum_{\alpha} \max_i \int_C \beta_{\alpha,n}(x) \ell_i(x) d\mu(x), \quad (9.5)$$

and a minimizing measure can be supported on at most $(n+2)^a$ points. In particular a strict level $L < \delta$ admits an adaptive test of positive uniform margin with $n > (8Ra/(\delta - L))^2$ and at most $(n+1)^a$ occupied coefficient entries. These bounds depend on the observable image and its metric, not on the number of stochastic rows in a realization.

Proof. The coefficient space is a finite product of compact simplexes, and the minimum of a jointly continuous function over compact C is continuous in its coefficients. This proves attainment. Bernstein degree elevation replaces each coefficient by convex combinations of adjacent coefficients, preserving every simplex constraint and the represented function. It proves monotonicity. All scores are convex combinations of the $\ell_i(x)$, proving the upper bound by δ .

At each grid point α/n choose an index i_{α} maximizing ℓ_i there. The choice is made even at infeasible grid points; only the final score is minimized over C . Give that index coefficient one and the others zero. For fixed x let A_j be independent $\text{Bin}(n, x_j)$ variables. If i_* maximizes at x ,

$$\begin{aligned} \max_i \ell_i(x) - \mathbb{E} \ell_{i_A}(x) &\leq \mathbb{E} \{ \ell_{i_*}(x) - \ell_{i_*}(A/n) + \ell_{i_A}(A/n) - \ell_{i_A}(x) \} \\ &\leq 8R \sum_j \mathbb{E} |A_j/n - x_j| \leq \frac{4Ra}{\sqrt{n}}. \end{aligned}$$

Here the omitted middle difference is nonpositive, and the last bound uses $\text{Var}(A_j/n) \leq 1/(4n)$. This proves the rate and the sparse coefficient assertion.

For the last formula replace the minimum over C by a minimum over $\mathcal{P}(C)$. The pairing is bilinear and continuous, and both feasible sets are compact convex, so the compact minimax theorem [18] exchanges the extrema. Optimizing each coefficient simplex gives (9.5). The integrands are polynomials of coordinatewise degree at most $n+1$. Preserve all monomial moments of that degree by Carathéodory's theorem. There are $(n+2)^a$ monomials including the constant; their moment vectors lie in an affine hyperplane of dimension at most $(n+2)^a - 1$. A measure with at most $(n+2)^a$ atoms therefore has exactly the same objective. The strict-margin assertion follows from the rate with error less than $(\delta - L)/2$. $\square$

The score is a nonlinear function of behavior although each event functional is affine. This is the reason the measure in (9.5) cannot be replaced by its behavior barycenter. For $n=0$ it can: d_0 is precisely the constant-test convex relaxation. In the two-time unmarked example of Proposition 8.5, the target belongs to the convex hull of private behaviors, so $d_0 = 0$, whereas $\lim_n d_n = \sqrt{2} - 1$. Thus the hierarchy recovers a gap which every constant test misses.

Remark 9.4 (What the size bound measures). The bound counts the nonzero entries of the independent adaptive test and controls its behavior-coordinate degree. It does not bound the cost of verifying its minimum on the generally nonconvex set C , the bit lengths of a rational chart, or the size of a description of C . Distinct row presentations of the same metric behavior image satisfy the same existence bound, although their coefficient arrays need not coincide. For the private fixed-target law array one may use $R \leq 1$ with the maximum total-variation metric. For the visible signed-discrepancy array the uniform bound $R \leq 2$ is sufficient. The old row-cube positivity certificate is a separate verification engine, with its own coordinate-dependent complexity.

9.3. A three-weight certificate invisible to constant tests.

Proposition 9.5 (A finite intrinsic nonlinear witness). *Consider a private width-one simulator producing two independent bits, with probabilities p, q , for the target which assigns probability $1/2$ to each of 00, 11. Put $u = pq$, $v = (1-p)(1-q)$ and $J(t) = 3t^2 - 2t^3$. The three behavior-dependent weights*

$$\kappa_1 = J(v), \quad \kappa_2 = J(u), \quad \kappa_3 = 1 - J(u) - J(v)$$

are nonnegative and sum to one. Against the event discrepancies $1/2 - u, 1/2 - v, 1 - u - v$, respectively, their score G satisfies

$$G \geq \frac{2}{5} + \frac{9}{8960}. \quad (9.6)$$

The certificate uses cubic weights in observable probabilities, a nine-term nonnegative derivative identity, and seventeen positive rational Bernstein coefficients in one variable. Every constant mixture of event discrepancies has minimum at most zero.

Proof. We have $u + v \leq 1$. Since J is increasing on $[0, 1]$ and $J(1-t) = 1 - J(t)$, the weights are feasible. Their score is

$$G = 1 - u - v + J(u)(u - 1/2) + J(v)(v - 1/2).$$

Set $a = (p + q - 1)^2$ and $b = (p - q)^2$. Then $a, b \geq 0$ and $a + b \leq 1$. Direct expansion gives

$$\begin{aligned} 64G &= 27 - 12a + 38a^2 - 20a^3 - a^4 + 36b - 60ab + 36a^2b + 4a^3b \\ &\quad + 6b^2 - 12ab^2 - 6a^2b^2 - 4b^3 + 4ab^3 - b^4. \end{aligned}$$

Let $c = 1 - a - b$. Another polynomial identity, valid when $a + b + c = 1$, is

$$\begin{aligned} 16\partial_b G &= 4a^3 + 6a^2c + 12ab^2 + 24abc + 12ac^2 \\ &\quad + 8b^3 + 30b^2c + 30bc^2 + 9c^3 \geq 0. \end{aligned}$$

Thus $G(a, b) \geq G(a, 0)$. The univariate polynomial

$$G(a, 0) - \frac{2}{5} = \frac{7}{320} - \frac{3}{16}a + \frac{19}{32}a^2 - \frac{5}{16}a^3 - \frac{1}{64}a^4$$

has degree-16 Bernstein coefficients $z_k/349440$, where

$$\begin{aligned} (z_0, \dots, z_{16}) &= (7644, 3549, 1183, 351, 855, 2494, 5064, 8358, \\ &\quad 12166, 16275, 20469, 24529, 28233, 31356, 33670, 34944, 34944). \end{aligned}$$

The Bernstein basis is nonnegative and sums to one. Its smallest coefficient is $351/349440 = 9/8960$, proving (9.6). All identities are rational polynomial identities; no numerical minimization is used. Finally the target is the barycenter of the two feasible deterministic laws 00 and 11. An affine mixture of signed event discrepancies has average zero at these two laws and hence minimum at most zero. The empty event supplies the reverse constant-test bound zero. $\square$

For comparison, the affine weights $v, u, 1 - u - v$ already give the smaller bound $3/8$, with the exact identity

$$G_{\text{affine}} - \frac{3}{8} = \frac{1}{2}(p - q)^2 + 2(p - 1/2)^2(q - 1/2)^2.$$

Neither certificate changes the private feasible set. The cubic certificate is an independent test chosen once for the whole behavior image, not a mixture chosen anew by an optimizer's unknown row values. Its evaluation does use the revealed behavior, as specified in the verification game. The 17-coefficient positivity check is classical; the distinction from constant tests is the behavior-dependent feasible weight family and its resource-sensitive minimum.

The infinite-dimensional representation above holds for continuous f_i on any compact metric space: its proof uses a partition of unity, not causal factorization. The measure minimum is attained at a Dirac mass at a minimizer of F . The finite-level polynomial theorem has a different

content. It is an existence-size bound for revealed-law verification; the reset-audit theorem of Section 6 imposes an explicit finite-data restriction.

9.4. Strict certificates and effective physical input. Given rational finite instruments, a rational chart and the sparse array constructed above, its strict minimum can be certified by real quantifier elimination on the original row constraints. Alternatively, pull the score back through the original polynomial row map and certify its strict positivity using degree-elevated Bernstein coefficients. The pullback changes the verification degree and is not covered by the behavior-coordinate bound. Generic quantifier elimination also decides the exact finite semialgebraic boundary; this classical observation does not assert that a finite nonnegative Bernstein expansion exists at $L = \delta$.

Proposition 9.6 (Uniform effective presentation input). *Suppose a single algorithm, on input j , returns rational finite instruments E_j, F_j and nonnegative rational error enclosures a_j, c_j for two-sided stateless marked feedback presentations of E, F . Suppose $a_j + c_j \rightarrow 0$. All alphabets, widths, selector laws and finite subroutines must be computable uniformly from j and the requested signature. Then, for every positive rational η , an algorithm returns a rational interval of width below η containing $\delta_b(E, F)$ at the same signature.*

Proof. Enumerate j until $a_j + c_j < \eta/8$. This terminates without a known rate because the rational enclosures themselves tend to zero. For the selected finite problem, enumerate rational same-budget upper tables and strict rational lower certificates, dovetailing all finite checks; include the always valid lower bound zero. Compactness, density and strict completeness provide an interval of width below $\eta/2$. Enlarge it by $a_j + c_j$ at both ends using stateless comparison. The enlarged width is below $3\eta/4$. Every enumeration and comparison is uniform by the hypotheses. A mere sequence of individually computable presentations without a uniform enumeration is not an input satisfying this proposition. $\square$

10. AN EXACT ALGORITHM FOR THE OBSERVABLE CHART

The chart in Lemma 9.2 can be found without taking its dimension as an oracle. We state the input and output explicitly. Rational polynomials are encoded by finite lists of monomial exponent vectors and rational coefficients in reduced integer form. A product of simplexes is encoded by the number of entries in each row. Eliminating the last entry of every nontrivial row leaves a full-dimensional rational polytope $\Omega \subset \mathbb{R}^r$. Zero-dimensional factors are omitted.

Proposition 10.1 (Effective observable quotient and pullback). *The input consists of the preceding simplex description, a rational polynomial map $b : \Omega \rightarrow \mathbb{R}^N$, and finitely many rational affine functionals f_i on $\mathbb{R}^N$. There is a terminating algorithm which returns its observable affine dimension a , a rational basis and coordinate map for the observable quotient, and, when $a > 0$, rational parameter points $q_0, q_1, \dots, q_a$ with the following properties. With*

$$z(q) = (f_i(b(q)) - f_i(b(q_0)))_i, \quad v_j = z(q_j),$$

the vectors v_j are a basis of the observable difference space, and all coordinates $t(q)$ in that basis satisfy $|t_j(q)| < 2$. The output also contains the rational polynomial cube map $x(q) = (t(q) + 2)/4$, a rational upper bound for the observable diameter, and the exact pullback of any subsequently supplied rational polynomial in x . The algorithm outputs the universal determinant formula it has decided. It makes no polynomial-time or coefficient-height claim.

Proof. Choose a rational relative interior point q_0 of the input simplexes. Expand the vector polynomial $z(q)$ in the free row coordinates. Let V be the rational span of all its coefficient vectors, including its constant coefficient. This equals $\text{span } z(\Omega)$: a linear functional vanishes on $z(\Omega)$ exactly when its polynomial vanishes on the nonempty interior of Ω , hence identically, hence on

every coefficient. Gaussian elimination therefore gives $a = \dim V$ and a rational basis matrix U . Choosing independent rows gives a rational left inverse L with $LU = I_a$. Thus $y(q) = Lz(q)$ provides exact quotient coordinates. If $a = 0$, all discrepancies are constant and no chart is required.

For $a > 0$ let

$$D_0 = \max_{u_1, \dots, u_a \in \Omega} |\det(y(u_1), \dots, y(u_a))|.$$

Compactness gives a maximum, and spanning gives $D_0 > 0$. Enumerate rational tuples $(q_1, \dots, q_a)$ in Ω^a . Membership and their determinant d are decided by rational arithmetic. For $d \neq 0$, decide the first-order real formula

$$\forall u_1, \dots, u_a \in \Omega : \quad \det(y(u_1), \dots, y(u_a))^2 < 4d^2. \quad (10.1)$$

Quantifier elimination terminates for each formula. Density of rational points and continuity of determinant imply that some enumerated tuple has $|d| > D_0/2$. For such a tuple the strict formula is true, so the search terminates. No computation of an exact algebraic maximizer is needed.

Invert the rational matrix with columns $y(q_j)$ to obtain $t(q)$. Cramer's rule and (10.1) give $|t_j(q)| < 2$ for every q . In the observable norm $\|z\|_* = \max_i |z_i|$, each v_j has norm at most the diameter of $z(\Omega)$. An upper bound for that diameter can be produced, for example, by bounding each free coordinate in $[0, 1]$ and summing the absolute polynomial coefficients for $f_i(b(q)) - f_i(b(q'))$. This bound is rational and computable. Rational matrix multiplication and substitution give $x(q)$ and every requested pullback. They also give its full sparse coefficient encoding. All searches have now been specified by finite encodings and terminating operations. $\square$

There is a small but important distinction between the quotient and a projection onto a smaller set of simulator rows. The former identifies exactly the directions invisible to the chosen event family; the original row constraints remain present in every universal positivity test. The chart need not identify equivalent tasks after a change of marks or of information signature. Those are changes to the comparison problem.

11. A MARKED RESOURCE GAP GENERATED BY A BINARY COLLISION

We now realize a private–hidden–visible separation in a physical experiment whose information vanishes when the collision is removed. The controller chooses a later sensor gate after its first report. The gate controls acquisition, not the mechanical collision law. Both the source and the target carry the same actual, unobserved mark.

11.1. The exact marked two-report experiment. Let $B, W \in \{0, 1\}$ have the joint distribution

$$\mathbb{P}(B = b, W = w) = \begin{cases} 5/16, & b = w, \\ 3/16, & b \neq w. \end{cases} \quad (11.1)$$

The blind source E_* has two constant reports and actual mark W . The target F_* reports $Y_1 = B$, receives $A \in \{0, 1\}$ after that report, and reports $Y_2 = AB$. The feedback tester may choose A as any function of its earlier report. A width-one private simulator cannot reread its own first emission, except through the current gate A . Hidden selectors persist but are not revealed; visible selectors are included in the joint target law independently of (B, W) . Put

$$d_* = \sqrt{5/2} - 9/8.$$

Theorem 11.1 (An exact marked causal separation). *At the width profile $(1, 1)$,*

$$\delta^p(E_*, F_*) = d_*, \quad \delta_2^h(E_*, F_*) = 1/8, \quad \delta_r^v(E_*, F_*) = d_* \quad (r \geq 1). \quad (11.2)$$

The hidden upper bound is realized by two deterministic tables. The private upper bound is realized by two independent Bernoulli variables with parameter $\sqrt{5/8}$, followed by the gate. These bounds hold against every feedback rule, not just open-loop gates.

Proof. First choose the constant gate $A = 1$. The two private emissions are independent Bernoulli variables with parameters p, q and are independent of W . Let

$$u = pq, \quad v = (1 - p)(1 - q), \quad a = 3/8, \quad b = 5/8.$$

The overlap of the target and simulated marked laws is $H(u) + H(v)$, where

$$H(t) = \frac{1}{2} \{\min(t, a) + \min(t, b)\}.$$

Indeed the target is supported on equal report pairs, and its two W -masses at each equal pair are $3/16$ and $5/16$. Total variation is one minus this overlap.

Cauchy-Schwarz gives $\sqrt{u} + \sqrt{v} \leq 1$. By symmetry suppose $u \geq v$. If $u \leq 1/4$ the overlap is at most $1/2$. If $u \geq 1/4$, then $v \leq (1 - \sqrt{u})^2 \leq 1/4 < a$, so the overlap is at most

$$H(u) + (1 - \sqrt{u})^2.$$

This function is convex on each of $[1/4, a]$ and $[a, b]$ and nonincreasing on $[b, 1]$. Its maximum is therefore attained at $1/4$, a or b . The values at a and b are $1 + 2a - 2\sqrt{a}$ and $3/2 + b - 2\sqrt{b}$ respectively. The second is larger, since

$$(3/2 + b - 2\sqrt{b}) - (1 + 2a - 2\sqrt{a}) = 3/8 - \frac{1/2}{\sqrt{a} + \sqrt{b}} > 0,$$

using $\sqrt{a} > 3/5$, $\sqrt{b} > 3/4$. It also exceeds $1/2$. Taking $p = q = \sqrt{b}$ attains this maximum, and hence the minimum error is $1 - (3/2 + b - 2\sqrt{b}) = d_*$.

For any hidden selector, the all-on report pair is still independent of W . If its probabilities of 11 and 00 are u, v , then $u + v \leq 1$. Concavity and monotonicity of H give

$$H(u) + H(v) \leq 2H((u + v)/2) \leq 2H(1/2) = 7/8.$$

A fair hidden bit J , emitting J twice, attains this overlap. This requires precisely two deterministic selector tables and gives error $1/8$. In particular hiding a seed does not erase the actual-mark obstruction.

These all-on bounds are lower bounds for the feedback problem. For the upper bounds first generate the corresponding all-on pair (X_1, X_2) , emit X_1 , and after receiving A emit AX_2 . For a deterministic feedback $A = g(X_1)$ the full gated marked transcript is the pushforward of the all-on marked pair. Applying the same map to the target pair cannot increase total variation. The construction is executable at width one: the second fresh Bernoulli variable is independent in the private case, and the bit J is the priced persistent hidden selector in the hidden case. Randomized feedback follows by conditioning on its independent random table.

Finally, in the visible case the two compared laws are

$$\sum_j \lambda_j \delta_j \otimes P_{g_j}^{q_j} \quad \text{and} \quad \sum_j \lambda_j \delta_j \otimes Q_{g_j}.$$

The target selector is independent of (B, W) . The all-on gate in every fiber has error at least d_* in every private fiber. Joint variation is the weighted sum of these fiber errors. Ignoring the visible selector and using the private optimizer attains d_* . $\square$

11.2. A full-dimensional collision preparation. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$ with contact distance d . The center of mass $Q = (q_1 + q_2)/2$ ranges over any fixed rational position box. Write $r = q_1 - q_2$, $g = v_1 - v_2$, and $V = (v_1 + v_2)/2$. Conditional on (B, W) in (11.1), take the uniform product preparation on

$$\begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), \quad a_0 \in [29/10, 31/10], & b_0 &\in [9/20, 11/20], \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & |V_z - (2W - 1)| &\leq 1/100. \end{aligned} \quad (11.3)$$

The boxes have positive volume in the full phase space. The actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$; total momentum conservation retains it. The informative quantity will instead be the transverse velocity created at a collision. Put

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector which is identically zero on the incoming transverse-velocity range.

Lemma 11.2 (Uniform transverse collision signal). *For every preparation in (11.3) and every $d \in [9/10, 11/10]$, exactly one collision occurs before time 3. Its time t_c lies in $(4/5, 3/2)$, and afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4. \quad (11.4)$$

For $0 < d \leq 2/5$ no collision occurs before time 3, and the detector is zero throughout that interval.

Proof. Up to a collision $r(t) = r + tg$. For $t \leq 4/5$, its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At time $3/2$ the absolute coordinate bounds are $1/4$, $113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. Thus a first contact occurs in the stated interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$ as well. The first contact therefore has $n_x < 0$, where $n = r(t_c)/d$. The entry contact is strictly incoming. The bounds above give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq \frac{87}{220}.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The equal-mass elastic collision rule is

$$v'_1 = V + g/2 - (g \cdot n)n, \quad g' = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v'_{1,y} > -3/200 + \frac{1443}{1000} \frac{87}{220} > 1/2.$$

The relative velocity is outgoing, $r(t_c) \cdot g' > 0$, so the squared separation increases strictly for all later times. There are only two spheres and no walls; no second collision is possible. This proves (11.4). For $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. Thus there is no collision. The incoming $|v_{1,y}| \leq 3/200$ then makes h zero. $\square$

At time 2 the physical target $F_{d,\sigma}$ reports

$$Y_1 = \mathbf{1}_{\{h(v_{1,y}(2)) + \sigma \xi_1 > 0\}}.$$

It then receives a gate $A \in \{0, 1\}$, and at time 3 reports

$$Y_2 = A \mathbf{1}_{\{h(v_{1,y}(3)) + \sigma \xi_2 > 0\}},$$

where the ξ_i are independent standard normal variables, independent of the preparation. The source E_d has the same preparation, evolution, control interface and actual mark, but emits two constant reports. The gates do not change the particle motion. This is a feedback acquisition

experiment: taking $A = 0$ suppresses later information and taking $A = 1$ exposes another noisy observation of a collision-generated quantity.

Theorem 11.3 (Collision-sensitive physical resource separation). *Let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/12$. At retained width $(1, 1)$ the physical deficiencies obey, uniformly in d ,*

$$\begin{aligned} |\delta^p(E_d, F_{d,\sigma}) - d_*| &< 1/300, & |\delta_r^y(E_d, F_{d,\sigma}) - d_*| &< 1/300 \quad (r \geq 1), \\ |\delta_2^h(E_d, F_{d,\sigma}) - 1/8| &< 1/300. \end{aligned} \quad (11.5)$$

In particular private and visible error exceed $9/20$, while hidden two-selector error is less than $13/100$. More generally the errors in these three comparisons are at most $2\Phi(-1/(4\sigma))$, which tends to zero as $\sigma \downarrow 0$. For $0 < d \leq 2/5$ all three deficiencies are zero. The interacting and noninteracting regimes use the same preparation and detector. The conclusions are unchanged by simultaneously scaling positions, diameter and observation times by any positive factor, with velocities and detector noise unchanged.

Proof. Couple the ideal and physical target by the same (B, W) and the same feedback random table. Lemma 11.2 bounds each potential bit error by $\Phi(-1/(4\sigma))$. With probability at least $1 - 2\Phi(-1/(4\sigma))$ both potential reports agree with their ideal bits; the gates and all emitted transcripts then agree as well. This gives a marked feedback variation bound uniform over all gates. The channels in both directions are the identity on the binary report alphabet and have one retained state. The blind source is already exactly E_* . Stateless comparison therefore changes each fixed-budget deficiency by at most the stated error, without adding memory or hiding an exposed selector.

For completeness, the elementary Gaussian tail bound $\Phi(-x) \leq e^{-x^2/2}/(x\sqrt{2\pi})$ at $x = 3$ gives $\Phi(-3) < 1/600$: use $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$. Hence the total error is below $1/300$. The inequalities $d_* - 1/300 > 9/20$ and $1/8 + 1/300 < 13/100$ follow by squaring positive rational bounds.

When $d \leq 2/5$, both potential reports are independent fair noise bits and are independent of W . A private width-one simulator draws those two bits and applies the current gate, exactly reproducing every feedback transcript. Hidden and visible versions can ignore their selectors. Finally elastic scattering is invariant under $(q, d, t) \mapsto (sq, sd, st)$ with v unchanged. This proves the scaling statement; it is geometric similarity, not a particle-number or Boltzmann–Grad limit. $\square$

The theorem combines three effects. Collisions create the sign of the informative transverse velocity. A gate chosen after the first report changes future information. A retained hidden bit, but not a visible one, permits temporal correlation without charging the width-one register. The actual mark remains partially inaccessible in every signature. The conserved longitudinal momentum labels that mark; it does not generate the observed transverse signal.

11.3. Uniformly effective finite presentations. The binary physical interface is already finite, but its probabilities are not generally rational. We make their effective approximation explicit rather than treating individual computability as a uniform algorithm.

Proposition 11.4 (Collision-chart rational presentations). *For computable $d \in [9/10, 11/10]$ and $\sigma > 0$, one algorithm produces rational marked causal instruments and certified two-sided stateless errors tending to zero for $(E_d, F_{d,\sigma})$. On a compact computable parameter interval with $\sigma \geq \sigma_0 > 0$, the errors and the construction are uniform. Thus Proposition 9.6 applies at every fixed finite state and selector signature, not only at $(1, 1)$.*

Proof. On every preparation box the entry time and outgoing velocity are

$$t_c = \frac{-r \cdot g - \sqrt{(r \cdot g)^2 - |g|^2(|r|^2 - d^2)}}{|g|^2}, \quad n = (r + t_c g)/d, \quad v'_1 = V + g/2 - (g \cdot n)n.$$

The positive square root is separated from zero: it equals $-d(g \cdot n) > (9/10)(1443/1000)$. All denominators are likewise bounded away from zero. These explicit bounds and the compact boxes allow interval differentiation of the displayed formulas, producing a rational common Lipschitz upper bound C_{geo} for $v'_{1,y}$ in the packet variables and d . This computation terminates because only addition, multiplication, division by separated positive quantities and a separated square root occur.

Conditional on the microscopic preparation the two potential all-on bits are independent Bernoulli with probability $p = \Phi(h(v'_{1,y})/\sigma)$. Their four probabilities are p^2 , $p(1-p)$, $p(1-p)$ and $(1-p)^2$. The normal distribution function is uniformly computable on bounded intervals, for example by integrating its convergent exponential series with a remainder bound. Its Lipschitz constant is less than one. Together with the 1-Lipschitz detector and the geometric bound, this yields a computable Lipschitz constant at most $2C_{\text{geo}}/\sigma_0$ for these four integrands on fixed-noise compact parameter sets. Equal-box Riemann sums of mesh h therefore have explicit error at most a computable constant times h , plus the chosen rational evaluation error. Parameter uncertainty is included by its interval enclosure.

Integrate the four all-on probabilities separately on each packet, with the exact rational weights (11.1). For each W , round its conditional probability vector to a rational simplex point on a sufficiently fine grid; the rounding error is bounded by the sum of the coordinate errors and the grid mesh. Keep $\mathbb{P}(W = 0) = \mathbb{P}(W = 1) = 1/2$ exact. Sample this finite joint law once at initialization, emit its first bit, and gate its second bit. This is a finite latent realization of the target instrument, not a simulator with extra free state. It makes every causal prefix consistent and never divides by a possibly small prefix probability. Every gated law is a pushforward of the all-on law, so the same marked variation bound holds uniformly under feedback. Identity report channels give both stateless directions. Choosing summable evaluation and mesh errors produces the asserted uniform sequence. $\square$

12. UNIFORM STABILITY OF OPTIONAL PROJECTIONS

Marked comparison controls more than terminal bounded risk when the compared laws share their latent variable. We prove a whole-process estimate for the resulting posterior martingales. The distinction between a probabilistic coupling and an executable causal channel is important in this section.

Let Z take values in a standard Borel space and let Y take values in a standard Borel canonical path space on $[0, T]$, with a right-continuous observation filtration. Consider equivalent joint probability measures P, Q on (Z, Y) , with the same Z -marginal ν . Use the usual completions, which agree for equivalent output laws. For a bounded measurable latent variable $f(Z)$ write

$$M_t^P = \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y], \quad M_t^Q = \mathbb{E}_Q[f(Z) \mid \mathcal{F}_t^Y].$$

Take their right-continuous martingale versions. All suprema below include T and can be evaluated on a countable dense time set.

Theorem 12.1 (Marked variation controls the whole posterior process). *Let $\varepsilon = \|P - Q\|_{\text{TV}}$ and $|f| \leq 1$. When both posterior functions are evaluated on the same canonical output path,*

$$\mathbb{E}_Q \sup_{t \leq T} |M_t^P(Y) - M_t^Q(Y)|^2 \leq 80\varepsilon. \quad (12.1)$$

There is a coupling of (Z, Y^P) and (Z, Y^Q) which retains exactly the same latent variable and satisfies

$$\mathbb{P}(Y^P \neq Y^Q) \leq \varepsilon, \quad \mathbb{E} \sup_{t \leq T} |M_t^P(Y^P) - M_t^Q(Y^Q)|^2 \leq 84\varepsilon. \quad (12.2)$$

The coupling is not asserted to be causal or to have finite state. The variation bound itself, not this coupling, is used for resource transport.

Proof. Let $D = dP/dQ$, $D_t = \mathbb{E}_Q[D \mid \mathcal{F}_t^Y]$, and set

$$A_t = \mathbb{E}_Q[(D - 1)f \mid \mathcal{F}_t^Y], \quad B_t = D_t - 1.$$

Bayes' formula gives

$$M_t^P - M_t^Q = \frac{A_t - B_t M_t^Q}{1 + B_t}.$$

Since $\mathbb{E}_Q|D - 1| = 2\varepsilon$, both martingales A, B have terminal absolute expectation at most 2ε . Doob's weak L^1 inequality gives, for $0 < u \leq 2$,

$$\begin{aligned} Q\{\sup_t |M_t^P - M_t^Q| > u\} &\leq Q\{\sup_t |B_t| > 1/2\} + Q\{\sup_t |A_t| > u/4\} + Q\{\sup_t |B_t| > u/4\} \\ &\leq 4\varepsilon + 16\varepsilon/u. \end{aligned}$$

Integrating $2u$ times this bound on $(0, 2)$ gives $16\varepsilon + 64\varepsilon = 80\varepsilon$, proving (12.1). No tightness conclusion is being inferred from finite-dimensional convergence.

Disintegrate both joint laws over their common latent marginal. Measurable maximal coupling of the two output kernels, conditional on Z , has disagreement probability

$$\int \|P(dy \mid z) - Q(dy \mid z)\|_{\text{TV}} d\nu(z) = \|P - Q\|_{\text{TV}}.$$

Such a coupling can be constructed by using the minimum of the two kernel densities with respect to their sum, then their residual kernels. It is measurable on standard Borel spaces. On the event of identical full output paths use (12.1) with the Q -marginal; on its complement the squared difference is at most four. This proves (12.2). The maximal coupling may depend on the full output path; it has no asserted online implementation. $\square$

Corollary 12.2 (Changing-filtration convergence from marked comparison). *Let P_n and P be equivalent joint laws with the same latent marginal ν , and $\|P_n - P\|_{\text{TV}} \rightarrow 0$. They admit a common probability-space realization $(Z, Y, Y^1, Y^2, \dots)$ such that, for every $f \in L^2(\nu)$,*

$$\mathbb{E} \sup_{t \leq T} \left| \mathbb{E}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}[f(Z) \mid \mathcal{F}_t^Y] \right|^2 \rightarrow 0.$$

The same realization gives convergence uniformly in time in probability for $f \in L^1(\nu)$. It also permits $f_n \rightarrow f$ in $L^2(\nu)$, respectively in $L^1(\nu)$, in the first conditional expectation.

Proof. Glue the conditional maximal couplings of (Z, Y^n) with the fixed reference (Z, Y) by taking their conditional kernels independently on a countable product. The coupling does not depend on f . For the truncation $f^{(K)} = (-K) \vee (f \wedge K)$, Theorem 12.1 bounds the middle difference in the L^2 norm of the supremum by $K\sqrt{84\varepsilon_n}$. Doob's L^2 inequality bounds each truncation tail by $2\|f - f^{(K)}\|_{L^2(\nu)}$. The triangle inequality therefore gives

$$\left\| \sup_t |M_t^{P_n, f} - M_t^{P, f}| \right\|_2 \leq 4\|f - f^{(K)}\|_2 + K\sqrt{84\varepsilon_n}.$$

First let $n \rightarrow \infty$, then $K \rightarrow \infty$. For L^1 use instead the weak maximal inequality for each tail and the bounded-case estimate for the middle term. An additional difference $f_n - f$ is handled by the same maximal inequalities under P_n , whose latent marginal remains ν . $\square$

12.1. A microscopic sufficient condition. Let $(\Phi_t)_{t \in \mathbb{R}}$ be an almost-everywhere regular hard-sphere flow on a labeled phase space, with invariant probability μ . Fix a prepared law ν with $d\nu/d\mu \leq C$, and bounded real observables h, h_n . These may be collision-sensitive tagged observables. Put $\eta_n = \|h_n - h\|_{L^2(\mu)}$. With a common latent initial state $Z \sim \nu$, consider the continuous observation laws

$$\begin{aligned} Y_t &= \int_0^t h(\Phi_s Z) ds + \sigma B_t, \\ Y_t^n &= \int_0^t h_n(\Phi_s Z) ds + \sigma B_t^n, \quad \sigma > 0, \end{aligned}$$

where the Brownian noise is independent of Z in each model. The notation does not prescribe a synchronous coupling.

Theorem 12.3 (Fixed-flow observation-approximation limit). *The joint latent-output laws are equivalent and satisfy*

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \eta_n. \quad (12.3)$$

For every $|f(Z)| \leq 1$ they admit a latent-preserving coupling with

$$\mathbb{E} \sup_{t \leq T} |M_t^{P_n, f} - M_t^{P, f}|^2 \leq \frac{42\sqrt{CT}}{\sigma} \eta_n^2. \quad (12.4)$$

In particular $\eta_n \rightarrow 0$ supplies whole-process optional-projection convergence, not merely convergence at finitely many times. This applies when h_n are orbit-mollified graph-core approximants to any bounded $h \in L^2(\mu)$.

Proof. Conditional on Z , the signals are deterministic bounded drifts. The exponential change-of-measure identity for Brownian motion gives

$$D_{\text{KL}}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |(h_n - h)(\Phi_s Z)|^2 ds \leq \frac{CT}{2\sigma^2} \eta_n^2.$$

The last inequality uses $\nu \leq C\mu$ and invariance of μ ; it does not require mixing, differentiability across a collision, or a claim that a tagged velocity belongs to the generator domain. Bounded drifts ensure equivalence and justify the exponential identity. Pinsker's inequality gives (12.3). Applying Theorem 12.1 proves (12.4).

For the last assertion choose a nonnegative smooth approximate identity ρ_n on the time line and set $h_n = \int \rho_n(s) h \circ \Phi_s ds$. Strong continuity of the Koopman group in $L^2(\mu)$ gives $\eta_n \rightarrow 0$ and $\|h_n\|_\infty \leq \|h\|_\infty$. Repeated integration by parts puts h_n in every power domain of the generator, with derivatives obtained from $\rho_n^{(k)}$. This is the orbit-core construction already used in the microscopic domain theory; the new conclusion is the uniform posterior estimate. $\square$

Sampling these observation paths at finitely many checkpoints and applying a common causal output postprocessing are contractions in marked variation. Thus the identity report comparison transfers every bounded finite-register risk at its original state and selector budgets with error bounded by (12.3). That is the resource-comparison consequence. The latent-preserving coupling in (12.4) has a different role: it realizes convergence of the changing observation filtrations and is not used as a finite-register simulator.

The theorem discharges a sufficient conditional-law hypothesis for optional projection in a genuine microscopic observation class. The constants depend on the fixed preparation, time and noise level. No particle-uniform kinetic estimate, changing microscopic path limit, limiting stochastic-exponential identification or nonlinear BSDE limit is implied without its additional hypotheses. The whole-process estimate is what replaces an unsupported appeal to Doob's inequality as a tightness argument.

13. CHANGING TRAJECTORIES, LIKELIHOODS, AND CONDITIONAL CONSUMERS

13.1. A common preparation, not a common flow. Let $(\mathcal{Z}, \nu)$ be a standard Borel probability space, and let $H, H_n : [0, T] \times \mathcal{Z} \rightarrow \mathbb{R}$ be jointly measurable, with $|H|, |H_n| \leq M$. They may be observations of different flows on different phase spaces, pulled back by specified preparation maps from $\mathcal{Z}$. No common microscopic generator or invariant measure is assumed. Set

$$\eta_n^2 = \int_{\mathcal{Z}} \int_0^T |H_n(t, z) - H(t, z)|^2 dt d\nu(z).$$

On $\mathcal{Z} \times C([0, T])$ let R be the law of independent $Z \sim \nu$ and $Y = \sigma B$, where $\sigma > 0$. Define

$$L_t^n = \exp \left\{ \frac{1}{\sigma} \int_0^t H_n(s, Z) dB_s - \frac{1}{2\sigma^2} \int_0^t H_n(s, Z)^2 ds \right\}, \quad (13.1)$$

$$dP_n = L_T^n dR. \quad (13.2)$$

The reference model P, L uses H . Conditional on Z , these are Gaussian shifts, so under P_n the observed path has law $Y_t = \int_0^t H_n(s, Z) ds + \sigma B_t^n$. Boundedness gives true strictly positive density martingales. Write $\mathcal{F}_t$ for the usual canonical observation filtration and $\mathcal{H}_t = \sigma(Z, B_s : s \leq t)$ for the larger reference filtration. Equivalent laws have the same completed observation stopping times.

Theorem 13.1 (Likelihood and posterior transport for changing paths). *The measures P_n, P, R are equivalent and have common Z -marginal ν . With $A = M^2 T / \sigma^2$ and $\varepsilon_n = \min\{1, \eta_n / (2\sigma)\}$, one has*

$$\|P_n - P\|_{\text{TV}} \leq \varepsilon_n, \quad (13.3)$$

$$\mathbb{E}_R \sup_{t \leq T} |L_t^n - L_t|^2 \leq \frac{8}{\sigma^2} e^{3A} \eta_n^2. \quad (13.4)$$

For $|f(Z)| \leq K$ put

$$N_t^{n,f} = \mathbb{E}_R[L_T^n f(Z) | \mathcal{F}_t], \quad \ell_t^n = N_t^{n,1}.$$

Then

$$\mathbb{E}_R \sup_t |N_t^{n,f} - N_t^f|^2 \leq \frac{8K^2}{\sigma^2} e^{3A} \eta_n^2. \quad (13.5)$$

In particular this applies to the projected likelihood ℓ^n . For every $|f| \leq 1$, the posterior functions satisfy the same-path bound $80\varepsilon_n$ of Theorem 12.1, and there is one latent-preserving coupling, independent of f , on which

$$\mathbb{E} \sup_t |\mathbb{E}_{P_n}[f(Z) | \mathcal{F}_t^{Y^n}] - \mathbb{E}_P[f(Z) | \mathcal{F}_t^Y]|^2 \leq 84\varepsilon_n. \quad (13.6)$$

If $\eta_n \rightarrow 0$ and deterministic microscopic paths $X_n(Z) \rightarrow X(Z)$ in a specified Polish path metric in ν -probability, then on this coupling the microscopic paths, the full and projected likelihood processes evaluated on their own observation paths, and each bounded posterior process converge jointly in probability in that metric and uniformly in time for the last three coordinates.

Proof. Conditional Gaussian change of measure gives

$$\text{KL}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |H_n - H|^2 dt = \frac{\eta_n^2}{2\sigma^2}.$$

The same calculation proves conditional equivalence, and the conditional density has mean one, so the preparation marginal is unchanged. Integrating and applying Pinsker proves (13.3).

For the quantitative density estimate, condition first on $Z = z$ and write $D_t = L_t^n - L_t$. The stochastic differential equation is

$$dD_t = \sigma^{-1} \{D_t H_n(t, z) + L_t (H_n - H)(t, z)\} dB_t.$$

The conditional second moment of L_t is at most e^A . Itô's isometry and $(u + v)^2 \leq 2u^2 + 2v^2$ therefore imply

$$\mathbb{E}_R[D_t^2 \mid z] \leq \frac{2M^2}{\sigma^2} \int_0^t \mathbb{E}_R[D_s^2 \mid z] ds + \frac{2e^A}{\sigma^2} \int_0^t |H_n - H|^2(s, z) ds.$$

Gronwall, followed by integration in z , gives $\mathbb{E}_R D_T^2 \leq (2/\sigma^2) e^{3A} \eta_n^2$. The difference is a square-integrable $\mathcal{H}_t$ -martingale. Doob's L^2 inequality proves (13.4). Applying the same maximal inequality to the $\mathcal{F}_t$ -martingale $\mathbb{E}_R[(L_T^n - L_T)f \mid \mathcal{F}_t]$ proves (13.5). These are direct process estimates, not a claim that finite-dimensional convergence alone implies tightness.

The posterior assertions follow from Theorem 12.1 and its countable conditional-coupling construction. For the last claim, on the event $Y^n = Y$ the two likelihood functions are evaluated on the same (Z, Y) . Their uniform difference tends to zero in R -probability by the estimates just proved, and therefore in P -probability by absolute continuity of the fixed P with respect to R . The complementary event has probability at most ε_n . The same argument applies to ℓ^n , and (13.6) handles the posterior. The microscopic convergence holds on any coupling retaining the same Z . A finite union bound proves joint convergence. Stochastic integral versions are chosen measurably on the canonical product space; equivalence makes these common versions valid under every law in the countable sequence. $\square$

For $f \in L^2(\nu)$ the posterior conclusion follows by truncation exactly as in Corollary 12.2; it also permits $f_n \rightarrow f$ in $L^2(\nu)$. Boundedness of H_n is being used for the uniform likelihood second moments. It is not replaced by an unverified uniform-integrability assertion in a singular or zero-noise limit.

13.2. The projected stochastic exponential. Let $\hat{H}_t^n = \mathbb{E}_{P_n}[H_n(t, Z) \mid \mathcal{F}_t]$, using a predictable version for Lebesgue-almost every t . Put $\theta_t^n = \hat{H}_t^n / \sigma$.

Proposition 13.2 (Identification before passage to the limit). *The projected likelihood and the innovation satisfy*

$$\ell_t^n = \exp \left\{ \int_0^t \theta_s^n dB_s - \frac{1}{2} \int_0^t (\theta_s^n)^2 ds \right\}, \quad (13.7)$$

$$I_t^n = B_t - \int_0^t \theta_s^n ds. \quad (13.8)$$

Here I^n is a P_n -Brownian motion in $\mathcal{F}_t$. In particular the limit of the projected likelihoods in (13.5) is the projected stochastic exponential of the limiting model. Change of measure commutes with this limit on bounded measurable path tests.

Proof. Project $L_t^n = 1 + \int_0^t L_s^n H_n(s, Z) dB_s / \sigma$ onto the Brownian observation filtration. The integrands are square integrable; testing against bounded elementary predictable stochastic integrals and using Itô's isometry gives

$$d\ell_t^n = \sigma^{-1} \mathbb{E}_R[L_t^n H_n(t, Z) \mid \mathcal{F}_t] dB_t = \ell_t^n \theta_t^n dB_t.$$

The last equality is Bayes' formula. Since $|\theta_t^n| \leq M/\sigma$, the solution is the positive stochastic exponential (13.7); Girsanov gives the stated innovation. The same argument with H identifies the limiting exponential. Finally $|\mathbb{E}_{P_n} F - \mathbb{E}_P F| \leq 2\|F\|_\infty \varepsilon_n$ for bounded signed F . There is no interchange based only on a formal limit of logarithms. $\square$

13.3. A bounded entropic backward-equation consumer. Fix $\gamma > 0$ and a latent payoff f with $|f| \leq 1$. Define

$$U_t^n = \frac{1}{\gamma} \log \mathbb{E}_{P_n}[e^{\gamma f(Z)} \mid \mathcal{F}_t] = \frac{1}{\gamma} \log \frac{N_t^{n, e^{\gamma f}}}{\ell_t^n}. \quad (13.9)$$

The terminal condition here is the observable conditional certainty equivalent U_T^n , not the generally unobserved $f(Z)$ itself. This specification avoids imposing a latent terminal value on the smaller observation filtration.

Theorem 13.3 (Entropic values and localized backward integrands). *There is a unique bounded entropic value process with terminal value U_T^n ; for a predictable integrand Z^n it satisfies*

$$U_t^n = U_T^n + \int_t^T \frac{\gamma}{2} |Z_s^n|^2 ds - \int_t^T Z_s^n dI_s^n. \quad (13.10)$$

On the coupling of Theorem 13.1,

$$\mathbb{E} \sup_t |U_t^n(Y^n) - U_t(Y)|^2 \leq 84 \left(\frac{e^{2\gamma} - 1}{\gamma} \right)^2 \varepsilon_n. \quad (13.11)$$

For each $r > e^\gamma$ let τ_r^n be the first exit of any of $\ell^n, \ell, N^{n,e^{\gamma f}}, N^{e^{\gamma f}}$ from $(1/r, r)$, capped at T . If $\eta_n \rightarrow 0$, then

$$\mathbb{E}_R \int_0^{\tau_r^n} |Z_s^n - Z_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^{\tau_r^n} |\theta_s^n - \theta_s|^2 ds \rightarrow 0. \quad (13.12)$$

Furthermore $\lim_{r \rightarrow \infty} \sup_n R(\tau_r^n < T) = 0$. Thus the localizations exhaust the observation experiment in probability.

Proof. For a positive bounded latent g , write $N_t^{n,g} = \mathbb{E}_R[L_T^n g \mid \mathcal{F}_t]$ and $dN_t^{n,g} = a_t^{n,g} dB_t$. Projection of the full integral gives

$$a_t^{n,g} = \sigma^{-1} \mathbb{E}_R[L_t^n g H_n(t, Z) \mid \mathcal{F}_t], \quad |a_t^{n,g}| \leq (M/\sigma) N_t^{n,g}.$$

The same representation with $g = 1$ gives $d\ell_t^n = b_t^n dB_t$ and $b_t^n / \ell_t^n = \theta_t^n$. Put $N^n = N^{n,e^{\gamma f}}$ and $a^n = a^{n,e^{\gamma f}}$. Itô's formula in (13.9) gives the martingale coefficient

$$Z_t^n = \gamma^{-1} (a_t^n / N_t^n - b_t^n / \ell_t^n).$$

Replacing dB_t by $dI_t^n + \theta_t^n dt$ reduces its drift to $-\gamma |Z_t^n|^2 / 2$, which proves (13.10). The ratios are bounded by M/σ , so the integrals are well defined. Conversely, exponentiating any bounded solution of (13.10) produces a bounded P_n -martingale with terminal value $e^{\gamma U_T^n}$. Its conditional-expectation representation is unique, and martingale representation then identifies its integrand. This proves uniqueness in the bounded class.

Apply Theorem 12.1 to $g_0 = (e^{\gamma f} - e^{-\gamma}) / (e^\gamma - e^{-\gamma}) \in [0, 1]$. The map taking its conditional expectation to $\gamma^{-1} \log \mathbb{E}[e^{\gamma f} \mid \mathcal{F}_t]$ has Lipschitz constant at most $(e^{2\gamma} - 1)/\gamma$. This proves (13.11).

The initial values of N^n and N agree, since Z has the same marginal. Itô's isometry and the terminal version of (13.5) imply

$$\mathbb{E}_R \int_0^T |a_s^n - a_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^T |b_s^n - b_s|^2 ds \rightarrow 0.$$

Before τ_r^n , all denominators lie in $[1/r, r]$, and the limiting numerators a, b are bounded in absolute value by $(M/\sigma)r$. For example,

$$\left| \frac{a^n}{N^n} - \frac{a}{N} \right| \leq r |a^n - a| + (M/\sigma) r^3 |N^n - N|.$$

The supremum estimates (13.5), followed by integration in time, prove (13.12). The same inequality applies to b^n / ℓ^n .

Finally $e^{-\gamma} \ell^n \leq N^n \leq e^\gamma \ell^n$. Each ℓ^n is a stochastic exponential with uniformly bounded integrand θ^n . Its positive and negative moments of the supremum are bounded uniformly in n : write any power as a stochastic exponential times its bounded compensator and apply Doob's inequality to a higher positive power. The same calculation applies to its inverse. Markov's

inequality, the preceding comparison of N^n with ℓ^n , and a union bound prove the exhaustion assertion. This is a localized integrand theorem; it does not assert an unspecified global topology for backward equations with unbounded terminal data or different nonlinear drivers. $\square$

Corollary 13.4 (A changing-filtration stopping consumer). *Let $\psi(t, x) \in [0, 1]$ be measurable and L_ψ -Lipschitz in x , uniformly in t . For a bounded latent f with $|f| \leq 1$, put $M_t^n = \mathbb{E}_{P_n}[f(Z) \mid \mathcal{F}_t]$. Then*

$$\left| \sup_{\tau \leq T} \mathbb{E}_{P_n} \psi(\tau, M_\tau^n) - \sup_{\tau \leq T} \mathbb{E}_P \psi(\tau, M_\tau) \right| \leq \varepsilon_n + L_\psi \sqrt{80\varepsilon_n}.$$

The same statement for the entropic value U has the additional Lipschitz factor $(e^{2\gamma} - 1)/\gamma$ on the square-root term. The suprema range over the common canonical stopping times; attainment is not needed for the estimate.

Proof. For a fixed stopping time, first change the measure while keeping its $[0, 1]$ reward function, at cost at most ε_n . Then compare the two reward functions on the same canonical path under P , using Theorem 12.1, Cauchy–Schwarz and the Lipschitz bound. The estimate is uniform in the stopping time, so it survives both suprema. Equivalence of the observation laws is what allows the same stopping-time class. The entropic assertion follows from the Lipschitz calculation in the preceding theorem. $\square$

13.4. Resource transport is not the posterior coupling. Apply any common finite causal acquisition map to Y and retain a fixed finite mark $w(Z)$. Data processing makes its marked row variations at most ε_n , uniformly over a declared finite feedback family. If the source and target have respective bounds α_n, β_n , Theorem 2.2 gives same-budget deficiency perturbation at most $\alpha_n + \beta_n$, and Proposition 6.4 transports a fixed reset audit. A subsequent consumer of width S composed with a simulator of width K has width KS , with the stated selector price. None of these executable claims uses the latent-preserving maximal coupling. That coupling belongs to the proof of conditional-process convergence, not to the finite-state machine.

These results provide a direct changing-trajectory, likelihood, Girsanov and bounded entropic-value input for a conditional-law theory. They do not assert the strict-dual, rigidity, covariant-form or full kinetic conclusions of a larger program without their separate analytic proofs.

Example 13.5 (Why an amplitude bound is not a tightness argument). Let ξ_1, ξ_2 be independent symmetric signs, and, for $n \geq 5$, let $\mathcal{G}_t^n$ reveal ξ_1 at $1/2 - 2/n$ and ξ_2 at $1/2 - 1/n$. On $[0, 1]$ the bounded martingale

$$M_t^n = \mathbb{E}[\xi_1 + \xi_2 \mid \mathcal{G}_t^n]$$

converges at every finite collection of fixed times to the martingale $M_t = \mathbf{1}_{[1/2, 1]}(t)(\xi_1 + \xi_2)$. Nevertheless its laws are not tight in the Skorokhod J_1 topology. On the event $\xi_2 = -\xi_1$, of probability $1/2$, its path has an excursion of height one on an interval of length $1/n$ and is zero outside that interval. The three-time oscillation modulus of such a path is one at every scale greater than $2/n$. This violates the necessary J_1 tightness modulus criterion. All maximal moments are uniformly bounded by 2^p .

This example concerns the inference from finite-dimensional convergence and a maximal amplitude estimate alone. It does not claim that every possible stronger conditional-kernel hypothesis fails. The uniform estimates in Theorem 13.1 remove the need for that inference in the observation class treated here.

14. OBSERVATION DENSITIES AND COUPLED PROCESS VERSIONS

We record three measure-theoretic facts used in the global transport argument. Their purpose is to make explicit the passage from a tilted physical law to an observation law and from two marginal processes to one joint realization.

Lemma 14.1 (Observation-level return from a bounded tilt). *Let $dP^g/dP = g(Z)/c$, where $c = \mathbb{E}_P g$ and $e^{-\gamma} \leq g \leq e^\gamma$. For any observation sub- σ -field $\mathcal{F}$,*

$$\frac{d(P|_{\mathcal{F}})}{d(P^g|_{\mathcal{F}})} = \mathbb{E}_{P^g} \left[\frac{c}{g} \middle| \mathcal{F} \right] = \frac{c}{\mathbb{E}_P[g | \mathcal{F}]}. \quad (14.1)$$

Both this density and its reciprocal lie in $[e^{-2\gamma}, e^{2\gamma}]$. In particular, for any nonnegative $\mathcal{F}$ -measurable V , $\mathbb{E}_P V \leq e^{2\gamma} \mathbb{E}_{P^g} V$.

Proof. For $A \in \mathcal{F}$, integrate c/g over A under P^g to obtain $P(A)$. Conditioning proves the first identity. In the other direction, $d(P^g|_{\mathcal{F}})/d(P|_{\mathcal{F}}) = \mathbb{E}_P[g | \mathcal{F}]/c$; equivalence and inversion prove the second. The bounds follow from the bounds on g , its conditional mean, and c . $\square$

Lemma 14.2 (Domination of the common-path measure). *Let (Y', Y) be a coupling on a standard Borel path space, with marginal laws P', P , and let $E = \{Y' = Y\}$. The subprobability measure $\kappa(A) = \Pr(E, Y \in A)$ is dominated by both P and P' . Consequently, for any nonnegative measurable path functional V ,*

$$\mathbb{E}[\mathbf{1}_E V(Y)] \leq \mathbb{E}_P V, \quad \mathbb{E}[\mathbf{1}_E V(Y')] \leq \mathbb{E}_{P'} V.$$

The assertion also holds on a common latent–observation space when the latent coordinate is shared by the coupling.

Proof. The event defining $\kappa(A)$ is contained in $\{Y \in A\}$ and, on E , in $\{Y' \in A\}$. This proves domination on sets. Integration of simple functions followed by monotone convergence proves the rest. Apply the same argument to the product coordinate for the last assertion. $\square$

Lemma 14.3 (Marginal stochastic integrals on an arbitrary coupling). *On a canonical standard Borel observation path space, suppose that under each of two marginal laws there are a continuous adapted process U , a predictable square-integrable process Z , and an innovation I such that, in that marginal's own filtration,*

$$\int_0^t Z_s dI_s = U_t - U_0 + \frac{\gamma}{2} \int_0^t |Z_s|^2 ds, \quad 0 \leq t \leq T. \quad (14.2)$$

Assume Z is bounded, as in the entropic model here. There are measurable versions of the two continuous integral processes which can be pulled back to any coupling of the observation laws. Each pulled-back version has the correct marginal law and satisfies its own identity (14.2). Neither integral thereby acquires a martingale or Itô-integral interpretation in the enlarged joint filtration.

Proof. A continuous process is determined by its values at rational times. Completed-measurable random variables on the canonical space have Borel versions. Choose such versions at the countably many rational times and extend by continuity on the measurable set where a continuous extension exists, using the zero path elsewhere. This gives a Borel $C([0, T])$ -valued version of U . Likewise take a jointly measurable version of Z modulo $dt \otimes P$; its bounded time integral has a continuous path, and its rational-time values give a measurable continuous version by the same construction. The right side of (14.2) is consequently a measurable continuous path functional. Marginal equality at all rational times, and continuity of the stochastic integral, make it indistinguishable from that integral. Repeat for the other marginal. Pullback along the two coordinate maps of any coupling preserves their respective almost-sure identities and laws. No enlargement theorem for innovations is used. $\square$

In Theorem 15.2, Lemma 14.1 is applied with $\mathcal{F} = \mathcal{F}_T$ to the integrated squared coefficient comparison. Lemma 14.2 justifies restricting a same-path estimate to the equality event of the latent-preserving coupling. Lemma 14.3 fixes the meaning of the coupled backward martingale integrals before their difference is estimated. Thus the coupled notation retains the marginal observation filtrations throughout.

15. GLOBAL TRANSPORT OF THE BACKWARD EQUATION

We retain exactly the common-preparation, bounded-signal and positive-noise hypotheses of Section 13. In particular the canonical joint laws P_n, P are equivalent and have the same latent marginal ν . All comparisons on a single observation path use common measurable versions of the conditional expectations. The following estimate avoids division by a small observation density in the final bound.

Lemma 15.1 (An integrated posterior comparison). *Let P, Q be equivalent laws on a standard Borel product carrying a latent variable Z and an observation Y , and let $\|P - Q\|_{\text{TV}} \leq \epsilon$. If $|u(Z)| \leq M$ and $\mathcal{F}$ is an observation sub- σ -field, then, under either $S = P$ or $S = Q$,*

$$\mathbb{E}_S |\mathbb{E}_Q[u | \mathcal{F}] - \mathbb{E}_P[u | \mathcal{F}]| \leq 4M\epsilon, \quad (15.1)$$

$$\mathbb{E}_S |\mathbb{E}_Q[u | \mathcal{F}] - \mathbb{E}_P[u | \mathcal{F}]|^2 \leq 8M^2\epsilon. \quad (15.2)$$

A common latent marginal is not needed for these two inequalities.

Proof. Use a dominating finite measure on $\mathcal{F}$. Write p, q for the densities of the restricted probability laws and a, b for the densities of the signed measures $\mathbb{E}_P[u \mathbf{1}_A], \mathbb{E}_Q[u \mathbf{1}_A]$ on $\mathcal{F}$. Their conditional means are $m_P = a/p$ and $m_Q = b/q$. Outside null sets, $|m_P|, |m_Q| \leq M$, and

$$p(m_Q - m_P) = m_Q(p - q) + (b - a).$$

The integral of $|p - q|$ is at most 2ϵ , by contraction of total variation. The integral of $|b - a|$ is at most $2M\epsilon$, since multiplication by u followed by restriction cannot increase signed variation by more than M . Integrating the displayed identity proves (15.1) under P . Interchanging the two laws proves it under Q . The squared bound follows from $|m_Q - m_P| \leq 2M$. $\square$

Let $|f| \leq 1$, $\gamma > 0$, and use the notation θ^n, U^n, Z^n, I^n of Proposition 13.2 and Theorem 13.3. Here Z^n denotes the scalar backward integrand, whereas the unadorned latent argument of H_n is the preparation variable. Define the deterministic bounds

$$A_n = \sigma^{-2} \{2\eta_n^2 + 16M^2 T \varepsilon_n\}, \quad (15.3)$$

$$C_n = \frac{2(1 + e^{4\gamma})}{\gamma^2} A_n, \quad (15.4)$$

where $\varepsilon_n = \min\{1, \eta_n/(2\sigma)\}$ as before.

Theorem 15.2 (Global innovation and entropic-integrand transport). *Under the hypotheses of Theorem 13.1, under either physical law $S = P$ or $S = P_n$ on the same canonical path,*

$$\mathbb{E}_S \int_0^T |\theta_t^n - \theta_t|^2 dt \leq A_n, \quad \mathbb{E}_S \int_0^T |Z_t^n - Z_t|^2 dt \leq C_n. \quad (15.5)$$

No stopping at a density level is required. If $\eta_n \rightarrow 0$, both integrals also converge to zero in expectation under the reference law R .

On the single latent-preserving coupling of Theorem 13.1, evaluate each process on its own observation path, denoting these evaluations by a bar. Then

$$\mathbb{E} \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt \leq A_n + 4M^2 T \varepsilon_n / \sigma^2, \quad (15.6)$$

$$\mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt \leq C_n + 16M^2 T \varepsilon_n / (\gamma^2 \sigma^2). \quad (15.7)$$

If $\eta_n \rightarrow 0$, then $\bar{I}^n \rightarrow \bar{I}$ uniformly in probability. Writing $\bar{J}_t^n = \int_0^t \bar{Z}_s^n d\bar{I}_s^n$ for the stochastic integral in its own observation filtration, and $\bar{J}$ for its limit, one has

$$\mathbb{E} \sup_{t \leq T} |\bar{J}_t^n - \bar{J}_t|^2 \leq 2 \mathbb{E} \sup_{t \leq T} |\bar{U}_t^n - \bar{U}_t|^2 + \frac{8M^2 T}{\sigma^2} \mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt. \quad (15.8)$$

The first term on the right has the explicit bound in (13.11). Thus the full bounded entropic backward representation, including its martingale integral, is transported globally. The coupling need not make either innovation Brownian in the joined filtration; each integral is defined under its own marginal law.

Proof. For each deterministic t , under P split

$$\sigma(\theta_t^n - \theta_t) = \mathbb{E}_{P_n}[H_n(t, Z) \mid \mathcal{F}_t] - \mathbb{E}_P[H_n(t, Z) \mid \mathcal{F}_t] + \mathbb{E}_P[H_n(t, Z) - H(t, Z) \mid \mathcal{F}_t].$$

Lemma 15.1, conditional Jensen and $(x + y)^2 \leq 2x^2 + 2y^2$ bound its expected square by $16M^2\varepsilon_n + 2\mathbb{E}_\nu|H_n(t, Z) - H(t, Z)|^2$. Integration gives the first bound in (15.5) under P . Under P_n use the same decomposition with the signal-difference term conditioned under P_n and the law-change term applied to H . The common latent marginal gives the identical bound. Jointly measurable conditional versions, and predictable versions equal for dt -almost every time, give the same integrated statements.

Put $g = e^{\gamma f}$ and $c = \mathbb{E}_\nu g$. Tilt the two full joint laws by $g(Z)/c$, writing P^g, P_n^g . They still have the same latent marginal $\nu_g = g\nu/c$, and their observation signals remain H, H_n . Since $e^{-\gamma} \leq g, c \leq e^\gamma$,

$$\|P_n^g - P^g\|_{\text{TV}} \leq e^{2\gamma}\varepsilon_n, \quad \eta_{n,g}^2 \leq e^{2\gamma}\eta_n^2.$$

Moreover, the reference martingale coefficient $a^{n,g}$ from the proof of Theorem 13.3 satisfies

$$\alpha_t^n := \frac{a_t^{n,g}}{N_t^{n,g}} = \sigma^{-1} \mathbb{E}_{P_n^g}[H_n(t, Z) \mid \mathcal{F}_t].$$

Apply the preceding innovation argument to these tilted laws. It bounds the squared integral of $\alpha^n - \alpha$ by $e^{2\gamma}A_n$ under either tilted physical law. For an observation-measurable nonnegative random variable, passage back to the corresponding original physical law costs at most $e^{2\gamma}$: the full density c/g is bounded by this constant, and so is its observation projection. Consequently

$$\mathbb{E}_S \int_0^T |\alpha_t^n - \alpha_t|^2 dt \leq e^{4\gamma} A_n.$$

The identity $Z^n = (\alpha^n - \theta^n)/\gamma$ proves the second bound in (15.5).

The pointwise bounds, valid for all models, are

$$|\theta^n|, |\alpha^n| \leq M/\sigma, \quad |Z^n| \leq 2M/(\gamma\sigma). \quad (15.9)$$

The squared time integrals are thus uniformly bounded random variables. Convergence in P -probability, followed by absolute continuity of the fixed R with respect to P , gives convergence in R -probability. Boundedness then gives convergence of their R -expectations. This argument establishes global reference convergence without claiming a new explicit reference rate.

For the coupling, let $E_n = \{Y^n = Y\}$, so that $\mathbb{P}(E_n^c) \leq \varepsilon_n$. On E_n both processes are functions of the same observation path. The restricted expectation of their squared difference is at most the corresponding unrestricted expectation under the limiting marginal P . On E_n^c apply (15.9) and integrate for time T . This proves (15.6)–(15.7).

On E_n , the Brownian reference coordinates $Y^n/\sigma, Y/\sigma$ agree, so

$$\sup_{t \leq T} |\bar{I}_t^n - \bar{I}_t|^2 \leq T \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt.$$

Markov's inequality and $\mathbb{P}(E_n^c) \rightarrow 0$ prove uniform convergence in probability. This assertion does not require any moment estimate on the observation difference on the exceptional event.

Finally, the backward equation already proved gives, under each marginal,

$$\bar{J}_t^n = \bar{U}_t^n - U_0 + \frac{\gamma}{2} \int_0^t |\bar{Z}_s^n|^2 ds.$$

The common initial value is $U_0 = \gamma^{-1} \log \mathbb{E}_\nu g$. By (15.9),

$$||\bar{Z}_s^n|^2 - |\bar{Z}_s|^2| \leq \frac{4M}{\gamma\sigma} |\bar{Z}_s^n - \bar{Z}_s|.$$

Subtract the two integral identities, use Cauchy–Schwarz in time and $(x + y)^2 \leq 2x^2 + 2y^2$, and take expectations. This gives (15.8). The integral identity also defines compatible measurable versions on the coupling, without asserting a martingale property for its joint filtration. $\square$

The theorem strengthens the localized conclusion of Theorem 13.3 without changing its terminal condition or its driver. The terminal condition remains the observation-measurable conditional certainty equivalent, not the unobserved latent payoff. Positive fixed noise, bounded signals, and a common preparation are used in the proof. A singular zero-noise limit, an unbounded terminal condition, or a different nonlinear driver would require additional arguments.

16. A UNIFORM CERTIFICATE THROUGH A MOVING COLLISION

We now use all three parts of the argument on one family. The finite marked reference law and the uniform scattering geometry are those of Theorem 11.1 and Lemma 11.2. They are proved above, including the no-collision control. We use a continuous observation of the same detector so that finite reports and conditional processes are generated by one experiment.

Let z denote a preparation in the boxes of (11.3), and let ν be that prepared law. The sign of the impact orientation is $2B - 1$, and the conserved center-of-mass longitudinal sign is the actual mark W . Their correlation is prepared: $\mathbb{P}(B = b, W = w) = 5/16$ for $b = w$ and $3/16$ otherwise. The contact diameter d ranges over $I = [9/10, 11/10]$. Let $X_d(t, z)$ be the two-particle phase path, with outgoing velocities used at the collision, and define

$$H_d(t, z) = h(v_{1,y}^d(t, z)), \quad Y_t^d = \int_0^t H_d(s, z) ds + \sigma B_t^d, \quad 0 < \sigma \leq 1/24.$$

The Brownian noise is independent of the preparation. The source experiment has the same physical preparation and mark but constant reports. The target reports at times 2 and 3 are

$$Z_1^d = \mathbf{1}\{2(Y_2^d - Y_{3/2}^d) > 0\}, \quad Z_2^d = A\mathbf{1}\{2(Y_3^d - Y_{5/2}^d) > 0\}. \quad (16.1)$$

The acquisition gate $A \in \{0, 1\}$ is chosen after Z_1^d ; it does not act on the mechanical trajectory. The continuous path is the auxiliary observation experiment for conditional consumers; the finite simulator is not given this path or an uncharged buffer for computing its increments. The target sensor implements the declared window acquisition.

Lemma 16.1 (Quantitative change of the microscopic path). *For all preparations in the specified boxes, the collision time $t_c(d)$ and outgoing tagged velocities obey*

$$|t_c(d') - t_c(d)| \leq |d' - d|, \quad |v_i^+(d') - v_i^+(d)| \leq 30|d' - d| \quad (i = 1, 2).$$

Writing $\Delta = |d' - d|$, one has

$$\sup_z \int_0^3 |H_{d'}(t, z) - H_d(t, z)|^2 dt \leq \Delta + 2700\Delta^2. \quad (16.2)$$

The phase paths converge uniformly over the preparation in the Skorokhod J_1 metric as $d' \rightarrow d$. More precisely, for the product metric taking the maximum of the per-particle Euclidean position and velocity distances, a time change aligning the collision times gives J_1 distance at most 100Δ .

Proof. Write r and g for the incoming relative position and velocity, and $n(d) = (r + t_c(d)g)/d$ for the contact normal. The collision root satisfies $|r + t_c g| = d$. The nongrazing estimate from Lemma 11.2 is $-g \cdot n > 1443/1000$. Differentiating the root with respect to d gives

$$t'_c(d) = \frac{1}{g \cdot n(d)}, \quad |t'_c(d)| < 1.$$

The incoming bound is $|g| < 3$, so

$$|n'(d)| \leq \frac{|t'_c(d)| |g| + 1}{d} < 5.$$

Since $v_1^+ = V + g/2 - (g \cdot n)n$ and $v_2^+ = V - g/2 + (g \cdot n)n$, differentiation bounds either velocity derivative by $2|g| |n'| < 30$.

Before collision the detector is zero, and after collision it is a constant of absolute value at most one. Its Lipschitz constant is one. The two signals therefore differ on the interval between the two collision times by at most one and afterwards by at most 30Δ . The intervening interval has length at most Δ , and the whole observation lasts three units. This proves (16.2).

Choose the increasing piecewise affine time change fixing 0, 3 and sending $t_c(d)$ to $t_c(d')$. Its uniform displacement is at most Δ . Aligned velocities differ by at most 30Δ , with the same outgoing convention at the jump. Each particle's speed is below four and its velocity jump has norm below three. Without time change, the position difference is bounded by $3\Delta + 3(30\Delta) = 93\Delta$, by integrating the velocity difference before and after the two jump times. Aligning time adds at most 4Δ . The asserted bound follows, with a harmless rounding to 100Δ . $\square$

Theorem 16.2 (Microscopic-to-audit and conditional-law composition). *For the single family (16.1), all of the following hold.*

(a) *At the same width-one private register,*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \quad (r \geq 1), \quad \delta_2^h < 13/100,$$

uniformly for $d \in I$ and $0 < \sigma \leq 1/24$. When $d \leq 2/5$ the detector is zero on $[0, 3]$, and all three finite-experiment deficiencies are zero.

(b) *There is one explicit rational reset-audit rule with one all-on training row, eight marked outcomes and $n = 2800$ training runs, whose expected validation score exceeds $2/5$ against every private width-one simulator throughout that same parameter region. In detail, for counts C choose an event $A_C \subseteq \{0, 1\}^3$ maximizing*

$$Q_*(A_C) - C(A_C)/2800, \tag{16.3}$$

with lexicographic tie breaking. Here Q_ is the rational marked reference law of (B, B, W) . The independent validation uses the actual physical target, not Q_* . This finite rule has at most $\binom{2807}{7}$ training-count states at any scheduled epoch; a conservative complete auditor bound is $8(256 + 1)\binom{2807}{7}$. The state cost belongs to the auditor. The certificate is an expected-score inequality, not a confidence claim about one realized score.*

(c) *Fix $\sigma > 0$ and let $d_n, d \in I$ with $d_n \rightarrow d$. Set*

$$\epsilon_n = \min \left\{ 1, \frac{\sqrt{|d_n - d| + 2700|d_n - d|^2}}{2\sigma} \right\}. \tag{16.4}$$

The joint preparation-continuous-observation laws differ by at most ϵ_n . Every finite resource deficiency above changes by at most ϵ_n , and a fixed audit changes by at most ϵ_n because its source

behavior is unchanged. On a latent-preserving coupling, $X_{d_n} \rightarrow X_d$ in J_1 and each bounded posterior converges in supremum L^2 with bound $84\epsilon_n$. The full and projected likelihoods converge in reference supremum L^2 , and jointly in probability with the microscopic paths on that coupling. The projected stochastic exponentials, bounded entropic backward values, localized integrands and stopping consumers have the conclusions of Proposition 13.2, Theorem 13.3, and Corollary 13.4.

(d) On compact parameter sets with $\sigma \geq \sigma_0 > 0$, the finite marked probabilities have uniform rational enclosures. The observable chart, strict same-budget polynomial verification and rational upper tables are therefore effective in the exact sense of Propositions 10.1 and 9.6.

Proof. Every collision occurs before time $3/2$. Conditional on preparation, the two normalized window increments in (16.1) are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. The outgoing detector has sign $2B - 1$ and absolute value greater than $1/4$. Coupling each noisy report to B gives total marked variation less than $1/300$ from the exact finite marked experiment, uniformly for every acquisition gate. The finite calculation of Theorem 11.1 gives

$$d_* = \sqrt{5/2} - 9/8, \quad \delta_*^p = \delta_*^v = d_*, \quad \delta_{*,2}^h = 1/8.$$

Thus private and visible values exceed $d_* - 1/300 > 9/20$, and the hidden value is less than $1/8 + 1/300 < 13/100$. The two-valued hidden witness stores its selector, not an observed mark. In the no-collision regime both window reports before gating are independent fair noise, independent of W . Fresh private coins reproduce them exactly, including feedback gates. This proves (a).

For (b), use only the all-on feedback row. Any private width-one simulator in this row emits independent bits with probabilities p, q , independently of the actual W . The calculation in Theorem 11.1 already gives minimum variation d_* for this row. Its alphabet (z_1, z_2, W) has eight elements, hence $D = 7$. The explicit empirical audit in Theorem 6.1, trained using Q_* , has expected ideal score at least

$$d_* - \sqrt{7/2800} = d_* - 1/20.$$

The candidate training and validation laws do not change when the target is replaced by the physical target. Conditional on any selected event, its target probability changes by less than $1/300$. Therefore the actual score is strictly greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (16.5)$$

For completeness, the last strict inequality is the positive-root comparison $\sqrt{5/2} > 947/600$; its squared margin is $5/2 - (947/600)^2 = 3191/360000 > 0$. All probabilities in Q_* are 0, $3/16$ or $5/16$, so (16.3) is a finite rational rule independent of d, σ, p, q . At training epoch k the number of eight-category count vectors is $\binom{k+7}{7} \leq \binom{2807}{7}$. A buffer of eight states records the current two reports and terminal mark; 256 event masks and an idle symbol suffice during event selection and validation. The conservative product bound follows. The clock and the source-reset interface are the declared resources. In particular the proof does not enumerate or claim to store a small complete coefficient table.

For (c), Lemma 16.1 gives the signal-distance hypothesis of Theorem 13.1, with $T = 3$ and $M = 1$. It also supplies its microscopic J_1 hypothesis. That theorem gives (16.4) and all the joint process assertions. The finite acquisition is a common measurable causal map of the continuous observation, so marked variation contracts for every gate. The source is blind and independent of d . Consequently its executable candidate behaviors are identical across d , and Proposition 6.4 has $\alpha = 0$ and $\beta = \epsilon_n$. The same same-budget comparison applies in each visible fiber and in each hidden mixture. The exponential and backward claims now follow from the identified projected density and the explicitly bounded entropic consumer. This is where the finite certificate and the conditional-law theorem meet the same microscopic approximation.

Finally, the collision root is uniformly separated from grazing, its formula is algebraic on the rational preparation boxes, and the normalized window noise has variance $2\sigma^2$ bounded away from zero on the stated compact sets. The box subdivision and Gaussian-tail enclosure argument in Proposition 11.4 therefore applies with that variance. It returns rational enclosures with a computable uniform error. The rational quotient algorithm and same-budget strict certificate procedure then prove (d). Uniformity as $\sigma \downarrow 0$ is not asserted. $\square$

The all-on audit in part (b) is a single witness for the lower bound in the full feedback problem: a maximum over feedback trees includes this row. The visible lower bound is instead the exact fiber theorem; no claim is made that an auditor with a seed-independent state bound can store an arbitrarily large visible-seed transcript. This separates the two resource budgets rather than silently moving cost from one to the other.

The finite certificate is also preserved under an additional rational presentation error ρ : its lower score decreases by at most ρ when only the target is changed, and by the reset-cost estimate when the source is changed. Thus the certificate is an effective consumer of the physical presentation theorem, not an isolated exact finite example.

Theorem 16.2 and the preceding results prove Theorem C.2. The C2-type consequence is a joint changing-trajectory/likelihood/posterior theorem with an identified stochastic exponential and a fully specified bounded entropic backward consumer. Nothing in that conclusion identifies geometric similarity with kinetic scaling, or supplies particle-uniform recollision estimates by renaming this fixed-particle model.

17. SHARP DECISION MEMORY FOR THE COLLISION EXPERIMENT

We now apply the continuation spectrum to the same marked experiment that supplies the microscopic certificate. Both its impossibility statement and its construction concern the same score, the same private candidate family, and the same training-validation checkpoint.

17.1. The marked family and its first two spectral values. In the all-on row of Section 16, a private width-one candidate has two independent report bits with probabilities $p, q \in [0, 1]$, independently of the fair actual mark. Denote its law by $b(p, q)$ and put

$$a = (1 - p)(1 - q), \quad c = pq.$$

Let $B = \{b(p, q) : p, q \in [0, 1]\}$. The ideal target Q_* is supported on 000, 001, 110, 111, with masses $5/16, 3/16, 3/16, 5/16$, respectively. On these four atoms the candidate masses are $a/2, a/2, c/2, c/2$. The elementary continuation constraint implies

$$\sqrt{a} + \sqrt{c} \leq 1, \quad \min(a, c) \leq \frac{1}{4}. \quad (17.1)$$

Indeed the first inequality is Cauchy-Schwarz applied to $(\sqrt{1-p}, \sqrt{p})$ and $(\sqrt{1-q}, \sqrt{q})$; the second follows from the first. If $a \geq c$, then $c \leq 1/4$ and

$$a + 3c \leq (1 - \sqrt{c})^2 + 3c = 1 - 2\sqrt{c} + 4c \leq 1. \quad (17.2)$$

The symmetric inequality holds when $c \geq a$.

Theorem 17.1 (A sharp two-state obstruction). *For the ideal marked collision family,*

$$\mathfrak{V}_1(B, Q_*) = \frac{1}{8}, \quad \mathfrak{V}_2(B, Q_*) = \frac{3}{8}. \quad (17.3)$$

If $\|Q - Q_\|_{\text{TV}} \leq \beta$, every reset auditor retaining at most two states at the training-validation cut has worst-case expected score at most $3/8 + \beta$, regardless of its finite training length and its upstream or downstream memory. The assertion allows mark-dependent validation events and stochastic state transitions.*

Proof. For one state, use the event $\{000, 111\}$. Its score is $5/8 - (a + c)/2 \geq 1/8$. Conversely, average the two candidates with $(a, c) = (1, 0)$ and $(0, 1)$. Their average is uniform on the four supported atoms and is at total variation distance $1/8$ from Q_* . Every fixed response has average score at most $1/8$, so its minimum over the two candidates is at most $1/8$. This proves the first equality; the average is a bound on scores, not an allowed private candidate.

For the upper bound with two states, consider the three candidates $b_0 = b(0, 0)$, $b_1 = b(1, 1)$ and $b_* = b(1/2, 1/2)$. A response may be set to zero outside the support of Q_* without decreasing its payoff at any candidate. Write its four remaining coordinates as (x, y, z, t) . Its three payoffs are

$$\begin{aligned} f_0 &= (-3x - 5y + 3z + 5t)/16, \\ f_1 &= (5x + 3y - 5z - 3t)/16, \\ f_* &= (3x + y + z + 3t)/16. \end{aligned}$$

Since all four coordinates are in $[0, 1]$,

$$f_0 + f_1 \leq \frac{1}{4}, \quad f_0 + f_* \leq \frac{3}{4}, \quad f_1 + f_* \leq \frac{3}{4}. \quad (17.4)$$

Suppose two responses covered all three candidates with payoff strictly greater than $3/8$. The first inequality forces different responses to cover b_0 and b_1 . The other two inequalities then force both responses to have payoff less than $3/8$ at b_* . This is a contradiction. Hence $\mathfrak{V}_2 \leq 3/8$.

For attainment take the responses, in the same four-atom order,

$$h_0 = (2/3, 0, 1, 1), \quad h_1 = (1, 1, 0, 2/3),$$

with zero response elsewhere. Both have target mean $17/24$. Their payoffs are $17/24 - a/3 - c$ and $17/24 - a - c/3$. If $a \geq c$, the first is larger and (17.2) makes it at least $17/24 - 1/3 = 3/8$. The other case is symmetric. This proves equality. For the perturbed target, apply (4.4) and the operational upper bound of Theorem 4.2. $\square$

The obstruction has an immediate continuation interpretation. Each post-training state supplies one validation response. The three witness candidates in the proof cannot be covered above $3/8$ by two such responses. No increase of sample size can compensate for losing the third continuation at that cut. This argument does not use the unrelated Bernoulli majority calibration.

Lemma 17.2 (Feedback completion of the physical lower bound). *The upper bound $3/8 + \beta$ in Theorem 17.1 remains valid when the auditor may train on any of the original feedback rows and may select a row–event pair after the decision cut. Here every target row is the declared feedback pushforward of Q , with $\|Q - Q_*\|_{\text{TV}} \leq \beta$. In particular, two states cannot evade the bound by using a different sensor gate during training or validation.*

Proof. It suffices to restrict the adversary to an admissible subfamily. For $p, q \in [0, 1]$, let its first emission be $X_1 \sim \text{Ber}(p)$, and, after gate A , let its second emission be AX_2 , with an independent $X_2 \sim \text{Ber}(q)$ and the independent fair actual mark. These are genuine width-one private rows: the gate is a current input, and the second emission uses no retained copy of X_1 .

For any deterministic gate policy $g : \{0, 1\} \rightarrow \{0, 1\}$, the marked row is the pushforward of $b(p, q)$ under $T_g(x_1, x_2, w) = (x_1, g(x_1)x_2, w)$. The physical target row is the pushforward of Q by the same map, since acquisition does not act on the path. Including the gate in the transcript changes nothing: it is also a deterministic function of the displayed coordinates. For a selected row event, the response pulls back to $\mathbf{1}_A \circ T_g$ on the all-on alphabet. A randomized row–event continuation pulls back to a function $h_m \in [0, 1]^\Omega$. Hence, conditional on cut state m , its expected score on this subfamily is exactly $(Q - b(p, q))h_m$.

Whatever training rows were used, their effect is solely the distribution of the retained state. The two-state score is thus a convex combination of two such responses and is bounded above by their envelope. Apply the three-candidate argument of Theorem 17.1, and then the uniform

target perturbation bound. Since this is a subfamily of the full private candidate class, it is a valid lower-bound witness for that full class. Randomized feedback is covered by conditioning on its independent random table and then averaging the pulled-back response. $\square$

17.2. A three-response construction. Define three deterministic events, all ignoring the mark,

$$A_D = \{00, 11\} \times \{0, 1\}, \quad A_0 = \{00\} \times \{0, 1\}, \quad A_1 = \{11\} \times \{0, 1\}.$$

Their ideal scores are $1 - a - c$, $1/2 - a$ and $1/2 - c$.

Lemma 17.3 (The three-response envelope). *For the original private candidate image,*

$$\min_{p,q} \max\{1 - a - c, \frac{1}{2} - a, \frac{1}{2} - c\} = \sqrt{2} - 1. \quad (17.5)$$

In particular $\mathfrak{V}_3(B, Q_) \geq \sqrt{2} - 1$.*

Proof. By symmetry suppose $a \geq c$. Then $c \leq 1/4$ and the maximum is $1 - a - c$ for $a \leq 1/2$, and $1/2 - c$ for $a \geq 1/2$. If $a \leq 1/4$, the first expression is at least $1/2$. If $1/4 \leq a \leq 1/2$, (17.1) gives

$$a + c \leq 1 - 2\sqrt{a} + 2a \leq 2 - \sqrt{2},$$

since the last function is increasing on $[1/4, 1/2]$. If $a \geq 1/2$, then $c \leq (1 - \sqrt{a})^2 \leq (1 - 1/\sqrt{2})^2 = 3/2 - \sqrt{2}$. Both cases give the asserted lower bound. It is attained at $p = q = 1 - 1/\sqrt{2}$, where $a = 1/2$ and $c = 3/2 - \sqrt{2}$. The symmetric point gives the same value. $\square$

Lemma 17.4 (Odd-sample majority selection with a uniform regret bound). *Let $n = 2h - 1$ be odd. In n independent candidate trials, let C_0, C_1 count report pairs 00, 11. Choose A_1 if $C_0 \geq h$, choose A_0 if $C_1 \geq h$, and otherwise choose A_D . The first two cases are mutually exclusive. Its expected ideal score satisfies*

$$g_n(p, q) \geq \sqrt{2} - 1 - \frac{1}{4\sqrt{en}} - \frac{1}{4}e^{-n/8}. \quad (17.6)$$

Its event label has three possible values at the decision cut.

Proof. Because $a + c \leq 1$,

$$\max\{1 - a - c, \frac{1}{2} - a, \frac{1}{2} - c\} = 1 - a - c + (a - 1/2)_+ + (c - 1/2)_+.$$

The selected payoff is $1 - a - c + (a - 1/2)\mathbf{1}_{\{C_0 \geq h\}} + (c - 1/2)\mathbf{1}_{\{C_1 \geq h\}}$. Its expected loss from this envelope is exactly the sum, for $u = a, c$, of $|u - 1/2|$ times the probability that the binomial majority selects the wrong side of $u = 1/2$. At $u = 1/2$ the contribution is zero. No independence between C_0 and C_1 is used.

For $0 < u < 1/2$ and $k \geq h$, the ratio of the binomial mass at u to that at $1/2$ is

$$[4u(1 - u)]^{n/2} \left(\frac{u}{1 - u} \right)^{k - n/2} \leq [4u(1 - u)]^{n/2}.$$

The fair binomial upper tail is exactly $1/2$ because n is odd. Reflecting k treats $u > 1/2$. The endpoint cases follow directly or by continuity. Thus the wrong-side probability is at most

$$\frac{1}{2}[1 - 4(u - 1/2)^2]^{n/2} \leq \frac{1}{2}e^{-2n(u - 1/2)^2}.$$

For $x \geq 0$, the maximum of $xe^{-2nx^2}/2$ is $1/(4\sqrt{en})$, reached at $x = 1/(2\sqrt{n})$. At least one of a, c is at most $1/4$ by (17.1); that coordinate has $|u - 1/2| \in [1/4, 1/2]$ and contributes at most $e^{-n/8}/4$. Bound the other coordinate by the unrestricted maximum and use Lemma 17.3. $\square$

17.3. Simultaneous exact widths and the full acquisition schedule. The preceding construction admits more compression than storing its two raw counts. Write 0, 1, * for the three report-pair categories 00, 11 and the two off-diagonal pairs. For $n = 2h - 1$, let F_n be the terminal event-label function of this word. At a trial cut t , let $r = n - t$ and let (u, v) be the two prefix counts. Associate the residual key

$$\mathcal{R}_t(u, v) = \begin{cases} \text{L}, & u \geq h, \\ \text{R}, & v \geq h, \\ (\min(h - u, r + 1), \min(h - v, r + 1)), & u, v < h. \end{cases} \quad (17.7)$$

The symbols L, R mean the constant decisions A_1, A_0 . In a pair key, a coordinate $r + 1$ means that this majority is no longer reachable. The pair $(r + 1, r + 1)$ represents the constant decision A_D .

Theorem 17.5 (Exact profile of the physical audit rule). *The keys in (17.7) are exactly the distinct continuation residuals of F_n . Their number at trial cut t is*

$$r_t(F_n) = \begin{cases} \binom{t+2}{2}, & 0 \leq t \leq h - 1, \\ \binom{n-t+3}{2}, & h \leq t \leq n. \end{cases} \quad (17.8)$$

This is the minimum simultaneous width profile for exact computation of this rule, for deterministic and stochastic machines alike. Its maximum is $\binom{h+2}{2}$ at $t = h$.

For $n = 399$, the maximum trial-cut width is 20301, and the terminal training cut has exactly three states. Acquiring each report pair one bit at a time, discarding its training mark, and then executing independent candidate and target validations gives a deterministic machine with at most $40602 < 2^{16}$ states at every scheduled epoch. One additional ignored training trial may be inserted to give the same 400 training slots and 402 total trials as the earlier certificate. It changes neither the payoff nor these width bounds.

Proof. If a majority has already been reached, its label cannot change. Otherwise, the first coordinate of the pair key is exactly the number of additional 0 symbols needed, with the unreachable threshold clipped to $r + 1$; the second coordinate has the analogous meaning. These data determine the suffix function. Two different reachable pair keys can be distinguished by a suffix containing only one of the two symbols and enough * symbols to fill the remaining positions: use a count between the two different thresholds. The other majority cannot be reached by this suffix. The constant-majority keys are distinct from every pair key because an all-* suffix returns A_D for the latter. Thus the keys are exactly the residuals.

For $t < h$, no majority has been reached and every threshold is at most $r + 1$. The distinct pairs (u, v) with $u + v \leq t$ give $\binom{t+2}{2}$ keys. For $t \geq h$, the two constant-majority keys are reachable. The remaining keys are precisely

$$(i, j) \in \{1, \dots, r + 1\}^2, \quad i + j \geq r + 1.$$

To see surjectivity, use $u = h - i$, $v = h - j$; these counts are nonnegative, less than h , and sum to at most t because $2h - t = r + 1$. To see necessity before clipping, use the same inequality on $h - u, h - v$; if either coordinate is clipped, the inequality remains automatic. There are $(r + 1)^2 - (r - 1)r/2$ such pairs, with the expression valid also for $r = 0, 1$. Adding the two constant keys gives $(r + 2)(r + 3)/2$. This proves (17.8). The finite-valued realization theorem, Theorem 5.1, proves both the simultaneous realization and stochastic minimality for the specified rule. The two polynomial pieces of the profile give its maximum as stated.

For an explicit transition definition, order the reachable keys lexicographically, putting L, R first. For each key choose the lexicographically first prefix-count pair representing it. After the next category, increment the appropriate count and apply $\mathcal{R}_{t+1}$. Equality of residual functions implies equality of the shifted residuals, so this definition is independent of the chosen representative. The

representative is part of the time-indexed transition table, not another stored register. The clock supplies t but no sample history. All these maps are deterministic and rational.

At a bit-level epoch, the state consists of a residual key and, only between the two report bits, the first bit. It therefore has at most $2 \max_t r_t$ values. On reading the second bit, determine the category, update the key and discard both bits. The mark is read and discarded; all-on controls and resets are prescribed by the free schedule. After training, keep just one of the three event labels. During validation, keep that label, the completed candidate-event indicator, and a first-bit buffer when needed. At most $3 \cdot 2 \cdot 2 = 12$ states suffice there; a completed target pair emits its event indicator minus the stored candidate indicator. The final score has three possible values. These phase-specific alphabets may be embedded in a single alphabet of size 40602 because the schedule clock identifies the phase.

The same trial-cut residuals remain distinct for the complete signed-score function, not merely for an externally emitted event name. Any two of A_0, A_1, A_D have different indicator functions. Take an off-diagonal candidate-validation pair, which belongs to none, and a target-validation pair distinguishing the two events. The resulting scores differ. Thus passing to the validation suffix does not merge the residuals already counted. $\square$

Theorem 17.6 (Matched physical decision complexity and stable certification). *For the two-sphere collision family of Theorem 16.2, let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/24$, and let $Q_{d,\sigma}$ be its all-on marked target. At the charged training-validation cut of the reset interface,*

$$\kappa_{2/5}(B, Q_{d,\sigma}) = 3. \quad (17.9)$$

More concretely, no auditor retaining at most two states at this cut can have expected score greater than $2/5$ uniformly against all private width-one candidates, at any finite training length, even with feedback-row selection as in Lemma 17.2. One rational auditor with three states at this cut does so with 399 informative training trials and two validation trials. With a redundant ignored training trial, it uses exactly 402 total trials, at most 40602 states at every scheduled epoch, and the exact trial-cut profile (17.8) on its informative training segment.

For fixed positive σ and $d_k \rightarrow d$, the same machine changes its expected score by at most

$$\epsilon_k = \min \left\{ 1, \frac{\sqrt{|d_k - d| + 2700|d_k - d|^2}}{2\sigma} \right\}. \quad (17.10)$$

The private, hidden and visible deficiencies, the no-collision control, and the joint microscopic, likelihood, posterior, innovation and global bounded entropic backward convergence retain all conclusions of Theorems 16.2 and 15.2.

Proof. Theorem 16.2 discharges the physical hypotheses and proves $\|Q_{d,\sigma} - Q_*\|_{\text{TV}} < 1/300$, uniformly on the stated parameters. Theorem 17.1 therefore gives the universal cut-width-two upper bound

$$\mathfrak{V}_2(B, Q_{d,\sigma}) < \frac{3}{8} + \frac{1}{300} < \frac{2}{5}.$$

Lemma 17.2 makes this a bound for the full original feedback-row interface, not only all-on training and not only the constructed rule. The upper construction uses all-on trials and works for every private candidate because only its two all-on parameters enter.

Use the deterministic selector of Lemma 17.4 with $n = 399$, but validate against the actual physical target. For each selected event, changing only the target from Q_* loses less than $1/300$. Its physical expected score is consequently greater than

$$L_{399} := \sqrt{2} - 1 - \frac{1}{4\sqrt{399}e} - \frac{1}{4}e^{-399/8} - \frac{1}{300} > \frac{2}{5}. \quad (17.11)$$

Here is a rational verification of the last strict inequality, independent of decimal evaluation. The bounds $e > 2$, $\sqrt{798} > 28$, $\sqrt{2} > 707/500$, and $399/8 > 49$ imply

$$L_{399} > \frac{207}{500} - \frac{1}{112} - \frac{1}{300} - 2^{-51} > \frac{2}{5},$$

since $207/500 - 1/112 - 1/300 - 2/5 = 73/42000 > 2^{-51}$. The number L_{399} is approximately 0.4032891; the rational inequalities, not this approximation, establish the certificate. Its state realization was proved in Theorem 17.5. The general operational classification now yields (17.9) with both inequalities proved for this same physical task.

Changing d leaves the blind candidate source and all its training laws unchanged. The moving-collision estimate and the common-preparation Gaussian theorem bound the target variation by ϵ_k . For any fixed response its target mean changes by at most this amount; averaging over the unchanged training rule proves (17.10). There is no multiplier equal to the number of training trials. The same physical estimates verify the hypotheses of Theorem 15.2; the observation-level density and coupling versions used there are given in Section 14. This transports the same certificate and the bounded conditional-law consumer along the changing microscopic paths. The no-collision regime $d \leq 2/5$ has the vanishing deficiencies already proved in Theorem 16.2; it is not included in the positive-gap parameter interval. $\square$

17.4. What the two exact complexity statements measure. Equation (17.9) minimizes over every auditor at one specified physical decision cut. Equation (17.8) minimizes all trial-cut widths simultaneously for exact implementation of the stated 402-trial rule. These are different optimizations. The first is a task-level optimum, not an artifact of that rule. The second explains the whole online realization of the same rule, not a claim that every successful auditor must have 20301 states somewhere. The sixteen-bit bound includes bit acquisition and validation but does not bound the description length of time-dependent transition tables. It is an implementation upper bound for peak memory. Independent persistent randomization is charged; fresh randomness from the reset experiments is not a hidden cross-cut memory channel.

The former two-count 400-sample construction and its nineteen-bit bound remain in Section 18. They offer a different approximation to the full marked discrepancy. The new three-response rule does not reproduce that rule or attain its oracle envelope. It instead solves the specified physical certification task with an exactly minimal decision-cut alphabet and an explicit full continuation profile. The score is an expectation, not a confidence guarantee from a single realized validation pair.

18. A FINITE-REGISTER CERTIFICATE TRANSPORTED THROUGH THE MICROSCOPIC LIMIT

The general empirical audit in Theorem 6.1 treats every coordinate of a marked row as unknown. The private continuation constraint in the collision experiment forces a smaller family. Exploiting this constraint yields a much smaller executable auditor, without granting it knowledge of the candidate law.

In the all-on row of Section 16, every private width-one candidate emits two independent bits with parameters p, q , independently of the fair actual mark W . Put

$$a = (1 - p)(1 - q), \quad c = pq.$$

The ideal target Q_* is supported on 000, 001, 110, 111, with respective masses $5/16, 3/16, 3/16, 5/16$. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. Every other atom has target mass zero and cannot improve the maximizing event for $Q_* - b$.

Lemma 18.1 (Two-count audit and its exact register implementation). *In $n \geq 1$ independent all-on candidate trials let C_0, C_1 count report pairs 00, 11, respectively, ignoring the training mark. On the four supported atoms use the empirical masses $C_0/(2n), C_0/(2n), C_1/(2n), C_1/(2n)$, and*

select exactly those atoms whose target mass is strictly greater than this empirical mass. Call the resulting event A_C . For every $p, q \in [0, 1]$,

$$\mathbb{E}\{Q_*(A_C) - b(A_C)\} \geq \|Q_* - b\|_{\text{TV}} - n^{-1/2}. \quad (18.1)$$

For $n = 400$, the rule is implemented by the two truncated counts $\min(C_0, 250)$, $\min(C_1, 250)$, with comparisons to 150 and 250. Including execution of complete transcripts and the two independent validation trials, it has a deterministic streaming realization with at most 504008 retained states at any scheduled epoch. Thus a nineteen-bit register suffices. The schedule clock is free; no other history store is free. The table description and transition evaluation cost are not asserted to be bounded by nineteen bits.

Proof. On any supported atom let $x = Q_*(w) - b(w)$ and let $\hat{x}$ be the empirical discrepancy. If the sign rule selects a different event from an exact maximizing event, the lost contribution is at most $|x - \hat{x}|$. On unsupported atoms both rules can omit the atom. Summing gives regret at most

$$|C_0/n - a| + |C_1/n - c|.$$

The two counts need not be independent. Their marginal binomial variances and Cauchy–Schwarz give expected regret at most

$$\frac{\sqrt{a(1-a)} + \sqrt{c(1-c)}}{\sqrt{n}} \leq \frac{1}{\sqrt{n}}.$$

This proves (18.1).

For the 00 pair, select both marked atoms when $C_0 < 150$, only the atom of target mass 5/16 when $150 \leq C_0 < 250$, and neither when $C_0 \geq 250$. The same rule applies to 11, with the other mark being the atom of larger target mass. At equality the zero empirical discrepancy is excluded. Increasing a count beyond 250 cannot change the final choice. Two registers with 251 possibilities each therefore implement the rule, with at most nine possible final event masks.

A buffer with eight possibilities is sufficient to execute and record a completed two-bit-plus-mark transcript. During training the pair of truncated counts and this buffer use at most $8 \cdot 251^2 = 504008$ states. After training, discard the counts and retain the event mask. During validation at most nine masks, one target-event indicator with two values, and the same eight-state transcript buffer are needed, a total of 144 states. The final signed score takes three values. Since the epoch is known, labels can be reused between these phases; their state counts need not be multiplied. The update maps and threshold comparisons are parameter-independent deterministic maps. They are thus stochastic rows with entries zero or one, or equivalently a compatible continuation cover by Theorem 3.2. Fresh sampling randomness belongs to the specified reset experiments, not to an uncharged auditor register. Finally $504008 < 2^{19}$. $\square$

Theorem 18.2 (Finite-register microscopic certification and global transport). *Consider the marked hard-sphere family of Theorem 16.2, with $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/24$. There is one rational streaming audit, independent of d, σ, p, q , using 400 candidate training resets and one independent candidate and target validation trial, whose expected signed score is strictly greater than $2/5$ against every private width-one candidate. It uses at most 504008 retained auditor states. The candidate remains private width one; the auditor’s register is a distinct, charged resource. The physical deficiencies still satisfy*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \ (r \geq 1), \quad \delta_2^h < 13/100.$$

For $d \leq 2/5$ all three finite-experiment deficiencies vanish.

Fix a positive σ and a sequence $d_k \rightarrow d$ within the collision interval. Put

$$\rho_k = |d_k - d| + 2700|d_k - d|^2, \quad \epsilon_k = \min\{1, \sqrt{\rho_k}/(2\sigma)\}.$$

The same 400-reset rule changes its expected score by at most ϵ_k under this perturbation. At the same time the microscopic paths, full and projected likelihoods, posteriors, and bounded entropic values have the joint convergence of Theorem 16.2. Their innovations, backward integrands and backward martingale integrals additionally have all the global conclusions of Theorem 15.2, with $T = 3$, $M = 1$, $\eta_k^2 \leq \rho_k$, and $\varepsilon_k \leq \epsilon_k$. On compact parameter sets bounded away from zero noise, the same audit has the effective probability-enclosure and rational-verification properties of Theorem 16.2(d).

Proof. The private width-one continuation restriction gives precisely the candidate family used in Lemma 18.1; the actual mark is not available to its transition rows. The exact finite separation in Theorem 11.1 gives

$$\min_{p,q} \|Q_* - b(p,q)\|_{\text{TV}} = d_* := \sqrt{5/2} - 9/8.$$

The scattering and observation estimates proved in Theorem 16.2 put the physical marked target within $1/300$ of Q_* , uniformly in all stated parameters. Train the two-count rule against the fixed rational Q_* , but use the actual physical target for validation. Each selected event's target probability changes by less than $1/300$. Hence its actual expected score is greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (18.2)$$

The last strict inequality is exact: it is equivalent to $\sqrt{5/2} > 947/600$, whose squared margin is $3191/360000$. The rule's executable memory bound was proved in Lemma 18.1. Its total trial count is 402, not 400 once validation is included. The private, hidden and visible deficiency bounds and the no-collision control are inherited from the same physical theorem, rather than inferred from an auditor whose signature was changed.

For changing d_k , Lemma 16.1 supplies the signal modulus ρ_k and the pathwise J_1 bound $100|d_k - d|$. Theorem 13.1 gives the joint marked variation bound ϵ_k and the likelihood/posterior convergence. The candidate source is blind and its behavior does not depend on d . Thus training laws and candidate validation laws are unchanged, and Proposition 6.4 has $\alpha = 0$ and $\beta = \epsilon_k$. There is no factor of 400 in this transport cost. The same preparation and signal bounds verify every hypothesis of Theorem 15.2, proving the global backward assertions. Finally the audit uses rational thresholds and finitely many deterministic row maps. The positive-noise enclosure theorem therefore applies directly; an additional target-presentation error ζ reduces its score by at most ζ . $\square$

The two-count construction is specific to the private behavior family. A hidden mixture can destroy the independence of its two reports, and its mark relation must still be checked. The bound is therefore not asserted against the hidden class. The visible lower bound has its separate fiberwise proof; an arbitrarily large visible alphabet is not silently stored in the nineteen-bit auditor. The certificate concerns expected score, not a confidence statement about a single realized validation pair. Repeating validation for statistical confidence would incur an additional sample and storage budget.

The theorem uses a single mechanism: the continuation restriction fixes the feasible candidate laws and permits the compressed audit; scattering supplies its marked separation; common-preparation variation transports the same executable test and the global conditional-law consumer through a change in the microscopic trajectory. The collision transduces a prepared impact variable into a signal. It does not create the prepared correlation between that variable and the mark, and it is not a particle-number limit.

19. EXPLICIT PHASE TRANSITIONS FOR THE TWO-COUNT CERTIFICATE

For comparison with the new residual quotient, we give the complete phase-state specification of Lemma 18.1. It realizes the original 400-training-trial rule exactly, not the majority rule of Section 17. Let

$$C = (C_0, C_1) \in \{0, \dots, 250\}^2, \quad \tau(x) = \min(x, 250).$$

The free clock records the training index, the within-trial observation epoch, the validation phase, and deterministic processing epochs. It contains no data-dependent information. At the start, $C = (0, 0)$.

For a completed transcript (x_1, x_2, w) define the deterministic update

$$T(C, x_1, x_2, w) = (\tau(C_0 + \mathbf{1}_{\{(x_1, x_2) = (0, 0)\}}), \tau(C_1 + \mathbf{1}_{\{(x_1, x_2) = (1, 1)\}})).$$

The value of w does not enter this training update. The phase transitions are the following; x denotes the current input, not an earlier unretained report.

Phase	Retained data	Next transition
Training start	C	Read x_1 and keep (C, x_1) .
Second report	(C, x_1)	Read x_2 and keep (C, x_1, x_2) .
Terminal mark	(C, x_1, x_2)	Read w and keep (C, x_1, x_2, w) .
Count update	(C, x_1, x_2, w)	Replace by $T(C, x_1, x_2, w)$; discard the transcript.
Event formation	C	Replace by the event mask $A(C)$ defined below.
Candidate validation	A , partial transcript	Complete the transcript; replace it by $i = \mathbf{1}_A(x_1, x_2, w)$.
Target validation	(A, i) , partial transcript	Complete the transcript and emit $\mathbf{1}_A(y_1, y_2, v) - i$.
Score	A value in $\{-1, 0, 1\}$	Terminate.

The event formation map, including both threshold ties, is

$$\begin{aligned} 000 \in A(C) &\iff C_0 < 250, & 001 \in A(C) &\iff C_0 < 150, \\ 110 \in A(C) &\iff C_1 < 150, & 111 \in A(C) &\iff C_1 < 250. \end{aligned}$$

No other atom is included. For each report pair there are three choices: both marked atoms, only its larger-target-mass atom, or neither. Thus there are at most nine masks. Clipping at 250 preserves every comparison, including equality; no flag for a count exceeding the clipping threshold is required.

At a training epoch the possible states are bounded by $8 \cdot 251^2 = 504008$: the factor eight dominates every partial or complete transcript buffer. At validation epochs the largest alphabet has at most $9 \cdot 2 \cdot 8 = 144$ values, including the already computed candidate-event indicator. Event formation and final emission need at most nine and three states, respectively. Embed each phase alphabet into $[504008]$ by a fixed injective encoding. Because the phase is supplied by the free clock, these encodings need not have disjoint images. The next row decodes the current phase's encoding and applies the displayed transition. This gives formal deterministic rows with entries zero or one at every epoch. The buffers, event mask, and validation indicator are all charged. No unpriced data store is introduced by label reuse.

APPENDIX A. THE PROOF DEPENDENCY CHAIN

An arrow below means an actual use of a theorem in the displayed proof, not an assertion of priority or an artificial dependency between papers. The current physical lower and upper bounds meet at one decision cut.

Input	Consequence using that input
Marked private rows and normalized continuations, Theorems 2.2, 3.2	The product behavior image and its validation continuation responses.
Decision continuation spectrum, Theorem 4.2	Universal two-state physical obstruction, Theorem 17.1; the strict-cut classification (17.9).
Product constraint (17.1) and three-response envelope, Lemma 17.3	Uniform finite-sample score, Lemma 17.4.
Finite-valued residual realization, Theorem 5.1	Exact full training profile and bit-level construction, Theorem 17.5.
Marked scattering and moving-collision estimate, Theorem 16.2	Both physical cut bounds and transport of the very same auditor, Theorem 17.6.
Common-preparation Gaussian comparison, Theorem 13.1; the three version lemmas in Section 14	The global conditional-law consumer, Theorem 15.2, on that moving-collision family.

The prior binary audit theorem remains a separate calibration, not a premise of the physical memory lower bound. The original two-count certificate remains a separate full-discrepancy approximation, not a claimed optimal implementation. The source-preserving build, rational checks and finite enumerations accompanying this article test the stated implementation and identities; they are not independent analytic proof certification.

APPENDIX B. A NAMESPACE MAP OF THE ELEVEN-COMPONENT PROGRAM

The namespaces in this map separate historical paper targets from GTF statements proved here. Every historical component is retained. An interface means that the indicated consumer can be applied after its model-specific hypotheses are proved; it is not an automatic proof of the historical component’s aggregate theorem.

Paper	Retained historical role	GTF relation and precise proof credit
A1	Protocol and finite task reduction	Marked comparison, priced composition and decision-cut certification; no automatic all-order physical response conclusion.
A2	Independent primary geometric chain	Its protocol adapter is historical; the primary geometric chain is not a premise or consequence of a theorem in this article.
A3	Dynamical/potential-space targets	Conditional comparison interface; no new assertion of the historical dynamical closure.
A4	Conditional kernels and forgetting	Historical finite-model consumer; common-preparation posterior estimates feed the current C2-scoped consumer when their hypotheses hold.
B1	Microscopic realization targets	Declared marked experiment interface; the present scattering construction is fixed-particle.
B2	Dynamic action and kinetic targets	Historical action estimates are not used to prove the present decision-cut theorem.
B3	Balance, response and corrector targets	No current proof depends on the historical aggregate balance/kinetic claims.
B4	Nonlinear kinetic semigroup	The historical normalized-resolvent identity and its aggregate downstream claims are not premises here. Their independent repair remains a program obligation.
C1	Statistical acquisition/inference	Original behavior image, finite-reset experiments, and the present response-envelope and microscopic certification consumers.
C2	Duals, response, memory and projections	The current bounded Gaussian/entropic transport replaces the optional-projection step only in its explicitly proved class; it does not certify the historical strict-dual, form-response or rigidity aggregate.
D1	Phase and compact statistical targets	Historical finite-model consumer retained; no new complete phase-theory closure is inferred.

The machine-readable companion records the exact predecessor graph and source identities, including the original B4 and C2 statements inspected at their frozen commit. Names such as `gtf20:decision-cut` and `historical:B4:r17-main` denote different proof obligations. The organizing theorem is a general comparison-and-certification result with an essential physical consumer. It does not replace the entire historical pipeline by a declaration that every aggregate theorem follows from it. The broader program remains intact as a mathematical program.

APPENDIX C. PRESERVED ORGANIZING STATEMENTS

The following two statements retain the earlier organizing labels and claims for cross-reference. Their full proofs are included in the canonical modules above. The unchanged predecessor article and its complete historical development are appended in the separate companion, so that reorganization does not discard earlier derivations.

Theorem C.1 (Continuation complexity and stable causal certification). *With these information conventions the following statements hold.*

- (i) *A finite causal kernel is realizable at a width profile if and only if it admits a compatible normalized continuation cover of that profile. For a Boolean terminal test, the exact minimal profile is the number of distinct continuation responses at each cut, even when the implementing machine is stochastic. If a cut has $r \geq 2$ distinct responses and only K retained states, its total wordwise error is at least $(r - K)_+/(r - 1)$.*
- (ii) *For $0 < h < 1/2$, consider $n = 2m + 1$ independent binary samples from one of $\text{Ber}(1/2 - h)$ and $\text{Ber}(1/2 + h)$, and the signed event-discrepancy audit against $\text{Ber}(1/2)$. Its unrestricted value is $h\|P_+ - P_-\|_{\text{TV}}$. The exact training widths needed to attain it are*

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n.$$

A smaller width at any cut has the strictly positive value deficit (7.4). Thus the continuation invariant controls both an implementation and a statistical obstruction.

- (iii) *In the prepared two-sphere experiment of Theorem 18.2, one rational audit using 400 candidate training resets, two validation trials, and at most 504008 retained auditor states has expected discrepancy greater than $2/5$ against every private width-one candidate. The same audit works for every contact diameter $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. The private and visible deficiencies exceed $9/20$, whereas a hidden two-valued selector has error less than $13/100$. Removing the collision, by taking $d \leq 2/5$, makes all three finite-experiment deficiencies zero.*
- (iv) *With positive noise fixed, changing d changes both the collision time and the microscopic path. If $\rho = |d' - d| + 2700|d' - d|^2$, the same audit's expected score changes by at most $\min\{1, \sqrt{\rho}/(2\sigma)\}$. Along $d' \rightarrow d$, the microscopic paths, full and projected likelihoods, and bounded posterior processes converge jointly in the senses specified below. For the associated bounded entropic backward equation, values converge in supremum L^2 and integrands converge globally in time-integrated L^2 , including on the common-latent coupling. Innovations converge uniformly in probability, and backward martingale integrals converge in supremum L^2 .*

C.1. Inherited baseline. The following summary is retained from the predecessor. The present article strengthens its memory and backward-integrand conclusions in Theorem C.1; the earlier 2800-trial audit remains valid.

Theorem C.2 (Finite-resource certification and microscopic transport). *The following assertions hold with the conventions just described.*

- (i) *Finite causal kernels have a register realization with a prescribed width profile exactly when their normalized continuation laws admit compatible finitely generated polytopes of those widths. At an erasure cut carrying a latent letter of law π , the optimal error with K retained states is*

$$1 - \sum_{i=1}^{\min(K, N)} \pi_{(i)}, \quad \pi_{(1)} \geq \cdots \geq \pi_{(N)}.$$

Independent hidden or visible selectors do not improve this cut value.

- (ii) *The reset-audit value is attained, has a finite-dimensional moment minimax representation on the original B , and satisfies*

$$0 \leq \delta(B, Q) - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}, \quad D = \sum_j (|\Omega_j| - 1).$$

The exponent $1/2$ cannot be improved uniformly over compact candidate classes. This assertion is about reset samples, not about the minimal degree of unrestricted polynomial tests.

(iii) Suppose two prepared microscopic observation models have a common preparation variable, bounded signals H, H' on $[0, T]$, and the same positive Gaussian noise level σ . Set

$$\eta^2 = \mathbb{E} \int_0^T |H - H'|^2 dt, \quad \varepsilon = \min\{1, \eta/(2\sigma)\}.$$

Their joint marked observation laws differ by at most ε . This estimate transports resource spectra at their stated budgets and transports each fixed finite audit with its explicitly charged reset cost. It also gives whole-process posterior convergence. The unprojected and projected likelihoods converge in the supremum L^2 norm under the common Wiener reference when $\eta \rightarrow 0$. For bounded latent payoffs, the associated entropic backward equations converge in value, and their integrands converge after the stated denominator localization.

(iv) In the two-sphere model of Theorem 16.2, one rationally specified audit with 2,800 training runs has expected discrepancy strictly greater than $2/5$ against every private width-one simulator, uniformly for $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. A hidden two-valued selector has error less than $13/100$; every visible-selector width-one error exceeds $9/20$. For $d_n \rightarrow d$ the microscopic paths, likelihoods and posterior processes converge in the precise senses of that theorem, and the same audit remains valid. For $d \leq 2/5$ all three deficiencies vanish.

REFERENCES

- [1] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272. doi:10.1214/aoms/1177729032.
- [2] L. Le Cam, Sufficiency and approximate sufficiency, *Ann. Math. Statist.* **35** (1964), 1419–1455. doi:10.1214/aoms/1177700372.
- [3] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28. doi:10.1524/strm.2002.20.14.1.
- [4] A. Nayyar, A. Mahajan and D. Teneketzis, Decentralized stochastic control with partial history sharing: a common information approach, *IEEE Trans. Automat. Control* **58** (2013), 1644–1658. doi:10.1109/TAC.2013.2239000. The theorem numbers cited here refer to the author preprint, arXiv:1209.1695.
- [5] H. S. Witsenhausen, On the structure of real-time source coders, *Bell System Tech. J.* **58** (1979), 1437–1451. doi:10.1002/j.1538-7305.1979.tb02263.x.
- [6] I. Gallagher, L. Saint-Raymond and B. Texier, *From Newton to Boltzmann: Hard Spheres and Short-Range Potentials*, Zurich Lectures in Advanced Mathematics, European Mathematical Society, 2014. The flow and boundary discussion used here is in Sections 4.1–4.3 of arXiv:1208.5753v2.
- [7] C. Widder, J. Zimmer and T. Schilling, On the generalized Langevin equation and the Mori projection operator technique, preprint (2025), revised 2026. arXiv:2503.20457v6.
- [8] C. Widder and T. Schilling, Generalised Langevin dynamics: significance and limitations of the projection operator formalism, preprint (2026). arXiv:2604.20453v2.
- [9] H. F. Trotter, Approximation of semi-groups of operators, *Pacific J. Math.* **8** (1958), 887–919. doi:10.2140/pjm.1958.8.887. Theorem 5.2, p. 902, is the approximation theorem cited here.
- [10] H. Weisshaupt, LeCam’s randomization criterion in the setting of locally convex spaces without lattice structure, EURANDOM report 2002–017, 22 April 2002. Lemma 1 and Theorem 5, pp. 14–16. Author report.
- [11] S. Basu, Algorithms in real algebraic geometry: a survey, author preprint, arXiv:1409.1534 (2014). Published in *Panoramas et Synthèses* **51** (2017), 107–153. Theorem 2.1 and Section 2.1 in the author preprint are used here.
- [12] M. C. Paull and S. H. Unger, Minimizing the number of states in incompletely specified sequential switching functions, *IRE Trans. Electron. Comput.* **EC-8** (1959), 356–367. Publisher record.
- [13] E. Torgersen, *Comparison of Statistical Experiments*, Encyclopedia of Mathematics and its Applications 36, Cambridge University Press, Cambridge, 1991.
- [14] N. Saldi, A topology for team policies and existence of optimal team policies in stochastic team theory, preprint (2017), arXiv:1711.00639.
- [15] S. Yüksel and N. Saldi, Convex analysis in decentralized stochastic control, strategic measures and optimal solutions, author preprint arXiv:1609.07685v2.
- [16] A. Nerode, Linear automaton transformations, *Proc. Amer. Math. Soc.* **9** (1958), 541–544. doi:10.1090/S0002-9939-1958-0135681-9.
- [17] S. Kiefer, Notes on equivalence and minimization of weighted automata, preprint (2020), arXiv:2009.01217v1.
- [18] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176. doi:10.2140/pjm.1958.8.171.

- [19] H. Weisshaupt, A generalization of Le Cam's randomization criterion, *Studia Sci. Math. Hungar.* **43** (2006), 443–486. doi:10.1556/SScMath.43.2006.4.3. Theorem numbers refer to the author's 39-page preprint, Randomization.pdf.
- [20] E. de Klerk and M. Laurent, Error bounds for some semidefinite programming approaches to polynomial minimization on the hypercube, *SIAM J. Optim.* **20** (2010), 3104–3120. doi:10.1137/100790835.
- [21] M. Laurent and L. Slot, An effective version of Schmüdgen's Positivstellensatz for the hypercube, arXiv:2109.09528v2, 2023.
- [22] E. de Klerk and J. C. Vera, The link between 1-norm approximation and effective Positivstellensätze for the hypercube, *Numer. Algebra Control Optim.* **16** (2026), 84–104. doi:10.3934/naco.2024060. Theorem numbers refer to arXiv:2404.04190v2.
- [23] M. Fliess, Séries rationnelles positives et processus stochastiques, *Ann. Inst. H. Poincaré Sect. B* **11** (1975), 1–21. Original article. Theorems 3.1 and 5.4 and Corollary 2, pp. 8–9 and 18–19.

GENERAL THETA FOUNDATIONS I: CONTINUATION COMPLEXITY AND STABLE CAUSAL CERTIFICATION

QIAN QI

ABSTRACT. We study marked causal experiments with separately charged simulator and verifier memory. Compatible continuation responses characterize realizability at a prescribed width profile. For Boolean streaming tests we turn a deficit of continuation states into a quantitative statistical loss, and determine exactly the widths attaining the optimal odd-sample binary reset value. On a genuinely interacting two-sphere family, the private continuation restriction yields one rational audit with 400 training trials and a nineteen-bit retained register whose expected discrepancy exceeds $2/5$. The same audit remains valid while contact distance changes the collision time and microscopic path. A common-preparation Gaussian comparison transports likelihoods and posterior processes; a bounded terminal tilt then gives global, rather than localized, convergence of the associated entropic backward integrands and their martingale integrals. The certificate and this conditional-law consumer are consequences of one microscopic transport estimate. Hidden and visible selectors retain distinct costs, and removing the collision makes the finite resource gaps vanish.

CONTENTS

1. Introduction	2
1.1. Main results	2
1.2. The general comparison framework	3
1.3. Classical inputs and the contribution	4
1.4. Microscopic and programmatic scope	4
2. Intrinsic causal deficiency	4
2.1. Experiments and executable signatures	4
2.2. The finite comparison theorem	6
2.3. A delayed channel and a strict signature separation	8
2.4. Optimal resource sets and decision certificates	9
3. Continuation geometry and an exact memory obstruction	10
3.1. Normalized continuation laws	10
3.2. An erasure cut	11
4. Finite-data testing on the original behavior image	12
4.1. The reset interface	12
4.2. A matching uniform sampling obstruction	14
4.3. Transport of one fixed audit	14
5. Continuation complexity of a finite-data test	15
5.1. The width required to attain the binary reset value	16
6. Polynomial testing of the original resource image	17
6.1. A sharp two-time nonconvex example	19
7. Adaptive testing on the observable behavior image	20

Date: September 23, 2026.

2020 Mathematics Subject Classification. Primary 62B05, 93E20; Secondary 60G44, 68Q45, 82C40.

Key words and phrases. causal comparison, continuation cone, finite memory, statistical experiments, hard-sphere scattering, nonlinear filtering.

7.1. The testing game	20
7.2. A size bound in behavior dimension	21
7.3. A three-weight certificate invisible to constant tests	22
7.4. Strict certificates and effective physical input	24
8. An exact algorithm for the observable chart	24
9. A marked resource gap generated by a binary collision	25
9.1. The exact marked two-report experiment	25
9.2. A full-dimensional collision preparation	27
9.3. Uniformly effective finite presentations	28
10. Uniform stability of optional projections	29
10.1. A microscopic sufficient condition	31
11. Changing trajectories, likelihoods, and conditional consumers	32
11.1. A common preparation, not a common flow	32
11.2. The projected stochastic exponential	33
11.3. A bounded entropic backward-equation consumer	33
11.4. Resource transport is not the posterior coupling	35
12. Global transport of the backward equation	35
13. A uniform certificate through a moving collision	38
14. A finite-register certificate transported through the microscopic limit	41
Appendix A. The canonical proof chain	43
A.1. Inherited baseline	44
References	45

1. INTRODUCTION

A causal reduction consumes observations, emits reports and retains a state. The dimension of that state and the visibility of persistent randomization are part of the mathematical problem. The same is true of a test intended to distinguish a reduction from its target: a test given the complete candidate law has a different information budget from a test which reads finitely many trials through a finite register. We study these two sides of causal comparison together, without convexifying an unpriced private resource class.

The organizing object is a family of normalized continuation responses, with one compatible list at each cut. Compatibility means that every one-step input–output shift is a positive combination of the next list. It characterizes realizability at a prescribed register profile. For zero–one terminal decisions, the lists reduce to residual equivalence classes. This allows an exact calculation of the memory needed to attain a statistical testing value, as opposed to an upper bound for the size of an unconstrained certificate. We then use a private continuation restriction to construct a small streaming certificate for an interacting microscopic experiment. The same certificate and its conditional-law consumers are transported through a change in the microscopic trajectory.

1.1. Main results. The resource signature distinguishes fresh private randomness from an independent persistent hidden selector and from a visible selector. An actual terminal mark belongs to the experiment; the online simulator cannot read it, but the terminal comparison can. A reset auditor is a separate machine. It reads specified independent completed trials of a candidate held fixed during the audit. Its schedule is known and its retained register is charged. It has no access to the candidate law or to the candidate’s private state.

Theorem 1.1 (Continuation complexity and stable causal certification). *With these information conventions the following statements hold.*

(i) A finite causal kernel is realizable at a width profile if and only if it admits a compatible normalized continuation cover of that profile. For a Boolean terminal test, the exact minimal profile is the number of distinct continuation responses at each cut, even when the implementing machine is stochastic. If a cut has $r \geq 2$ distinct responses and only K retained states, its total wordwise error is at least $(r - K)_+/(r - 1)$.

(ii) For $0 < h < 1/2$, consider $n = 2m + 1$ independent binary samples from one of $\text{Ber}(1/2 - h)$ and $\text{Ber}(1/2 + h)$, and the signed event-discrepancy audit against $\text{Ber}(1/2)$. Its unrestricted value is $h\|P_+ - P_-\|_{\text{TV}}$. The exact training widths needed to attain it are

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n.$$

A smaller width at any cut has the strictly positive value deficit (5.4). Thus the continuation invariant controls both an implementation and a statistical obstruction.

(iii) In the prepared two-sphere experiment of Theorem 14.2, one rational audit using 400 candidate training resets, two validation trials, and at most 504008 retained auditor states has expected discrepancy greater than $2/5$ against every private width-one candidate. The same audit works for every contact diameter $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. The private and visible deficiencies exceed $9/20$, whereas a hidden two-valued selector has error less than $13/100$. Removing the collision, by taking $d \leq 2/5$, makes all three finite-experiment deficiencies zero.

(iv) With positive noise fixed, changing d changes both the collision time and the microscopic path. If $\rho = |d' - d| + 2700|d' - d|^2$, the same audit's expected score changes by at most $\min\{1, \sqrt{\rho}/(2\sigma)\}$. Along $d' \rightarrow d$, the microscopic paths, full and projected likelihoods, and bounded posterior processes converge jointly in the senses specified below. For the associated bounded entropic backward equation, values converge in supremum L^2 and integrands converge globally in time-integrated L^2 , including on the common-latent coupling. Innovations converge uniformly in probability, and backward martingale integrals converge in supremum L^2 .

Part (i) follows from Theorems 3.2 and 5.1; part (ii) is Theorem 5.2. Parts (iii)–(iv) are the single composition theorem proved in Section 14, using the global transport theorem of Section 12. The two registers in part (iii) have different roles: the candidate remains private width one, while the auditor's nineteen-bit implementation has its own explicit budget. This is not a statement that the candidate can borrow the auditor's memory.

The finite-data construction is materially tied to the resource geometry. The private width-one candidate has independent reports and a fair independent mark. Only two report-pair probabilities determine the four relevant marked coordinates. Two truncated empirical counts therefore replace a full multinomial table. A direct regret inequality gives the 400-sample rule, and its deterministic continuation states give the register bound. The collision supplies the separating marked target. The common-preparation variation estimate then transports the very same rule, not a reoptimized rule, together with the global backward representation.

1.2. The general comparison framework. For a compact marked behavior image B and fixed target rows Q_j , set

$$\delta(B, Q) = \min_{b \in B} \max_{j, A} \{Q_j(A) - b_j(A)\}.$$

Complements supply both signs. Theorem 2.2 supplies compactness, exact factorization, priced composition and perturbation at the declared signature. Theorem 4.1 gives an attained finite-reset moment dual on the original, possibly nonconvex B and $0 \leq \delta - v_n^{\text{aud}} \leq \sqrt{D/n}$. Its dual chooses one candidate for the entire replicated experiment; its barycenter cannot replace that candidate mixture. Theorem 4.3 proves a matching uniform sampling exponent. The new memory obstruction is different: it identifies exactly when a streaming auditor attains the unconstrained value, and gives a strict deficit otherwise. We do not exchange minimax on the nonconvex set of fixed-width auditors.

The revealed-behavior representation and its intrinsic Bernstein hierarchy remain separate tools, with their full proofs and exact nonlinear example. Their evaluator sees an exact law, unlike the reset auditor. The formal rational chart algorithm and physical probability enclosures retain this distinction. Effective here means an exact terminating construction under specified rational inputs and positive-noise hypotheses, not polynomial bit complexity.

1.3. Classical inputs and the contribution. Statistical comparison and randomization have a classical foundation in Blackwell and Le Cam [1, 2, 13]; filtered comparison predates this work [3, 19]. We do not claim priority for introducing a filtration or a randomization criterion. The original proof-level crosswalk with Norberg remains unverified; no assertion about absence of the present results from that work is made.

Stable positive-cone realization is classical [23], as is residual-state minimization [16]. The continuation theorem is a finite controlled normalization with a simultaneous width profile. The zero-one residual characterization is proved here in a stochastic implementation model, not claimed as a new deterministic automata theorem. The added step is the overlap inequality (5.2), which converts insufficient continuation width into statistical regret and an exact optimal-value threshold for the binary reset game. The quantitative penalty is explicit, but not claimed to be a sharp joint asymptotic law in sample size, memory and error.

The minimax theorem, moment support reduction, empirical second moments, two-point testing, and polynomial positivity are classical tools [18, 20, 21, 22]. Likewise the scattering formula, Bayes' formula, Girsanov's theorem and martingale inequalities are inputs, not discoveries. The global backward estimate uses a denominator-free integrated posterior comparison and a bounded terminal tilt. It improves the localized transport estimate while retaining the same actual terminal condition and driver. The contribution offered here is the information-budgeted theorem chain: compatible continuations, constrained testing value, an executable microscopic certificate, and global transport of its conditional-law consumer.

1.4. Microscopic and programmatic scope. The physical family has two labeled spheres and one nongrazing collision. The correlation between the impact variable and the actual mark is prepared; the collision creates an observable transverse signal of that variable. The feedback is an acquisition gate, not a mechanical force. Varying contact diameter is a genuinely changing microscopic path, not a Boltzmann-Grad limit. The global transport theorem assumes a common preparation, bounded signals and fixed positive noise. Its bounded scalar entropic equation has an observation-measurable conditional certainty equivalent at terminal time, not an unobserved latent terminal payoff.

This gives an explicit conditional-law consumer in the larger theta-theory program, including changing paths, likelihoods, projected stochastic exponentials and the complete bounded backward representation. It does not manufacture a dependence of the independent A2 geometric chain, nor use an unverified kinetic resolvent or rigidity theorem as an input. The wider program and its historical targets are retained in full. Their source assertions, current proof credit and independent-review status are recorded separately in the accompanying statement ledger.

The canonical dependency diagram is in Appendix A. Every mathematical module used here has a full proof in this article. The complete predecessor development is preserved as a companion; its historical introductions are not substitutes for the present theorem statements.

2. INTRINSIC CAUSAL DEFICIENCY

2.1. Experiments and executable signatures. Fix a finite parameter set Θ , a horizon T , and finite control alphabets $A_0, \dots, A_{T-1}$. An experiment has a report at checkpoints $0, \dots, T$ and an unobserved terminal mark W in a common finite set. It is specified by finite causal instruments: for each parameter and control history they give the joint law of the report path and the actual

mark. Equivalently, a finite latent-state realization may be expanded into its history tree. The parameter and mark are unavailable to the encoder. The mark may depend on the physical controlled path. We compare experiments with the same control interface and mark space.

Write Y_t for the source report and Z_t for the target report. A private simulator of width K has stochastic rows

$$\begin{aligned} q_0(z, m \mid y), \quad m &\in [K_0], \\ q_t(z, m \mid m', a, y), \quad m' &\in [K_{t-1}], \quad m \in [K_t], \quad 1 \leq t \leq T. \end{aligned} \quad (2.1)$$

A row consumes the current source report, emits the target report, and retains the new state. The previous target reports are not available to that row unless encoded in m' . The clock and current control are part of the declared interface. Fresh private randomness is permitted, but no other persistent random variable is free. The product of all row simplexes is denoted $\mathcal{K}_K$.

A deterministic feedback tree g chooses a_t from $z_{0:t}$ and its own previous controls. It is the same controller on both sides of a comparison. Let Ω_g be the finite marked transcript space, including reports, controls, and W . Put

$$P_{\theta,g}^q = \text{Law}_\theta(E \xrightarrow{q} g), \quad Q_{\theta,g} = \text{Law}_\theta(F \rightarrow g).$$

Here W on the left is the source's actual mark. The total variation convention is $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. Private deficiency is

$$\delta_K^p(E, F) = \min_{q \in \mathcal{K}_K} \max_{\theta, g} \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}. \quad (2.2)$$

The minimum will be justified below. Randomized feedback need not be added to the maximum: conditioning on its independent random table expresses its law as a mixture of deterministic feedback laws. The same argument permits arbitrary independent side randomness in the tester.

A *hidden selector* $U \in [r]$, independent of the source, chooses one simulator q^u . It is not visible to the feedback controller or the final test. A *visible selector* is revealed before feedback begins. The two costs are

$$\delta_{K,r}^h(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, g} \left\| \sum_u \lambda_u P_{\theta,g}^{q^u} - Q_{\theta,g} \right\|_{\text{TV}}, \quad (2.3)$$

$$\delta_{K,r}^v(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, (g_u)} \left\| \sum_u \lambda_u \delta_u \otimes P_{\theta, g_u}^{q^u} - \sum_u \lambda_u \delta_u \otimes Q_{\theta, g_u} \right\|_{\text{TV}}. \quad (2.4)$$

In both expressions $\lambda \in \Delta_r$ and $q^u \in \mathcal{K}_K$. The second formula compares the joint seed-marked interface. Its target seed has the same distribution and is independent of the target experiment. A hidden selector can be stored privately by retaining (u, m) , at width rK_t . A visible selector is separate side information and is charged by its own alphabet size. Neither interpretation identifies an arbitrary real-valued design with a free finite register.

Definition 2.1 (Deficiency spectrum). The private resource spectrum of E relative to F is the upward-closed set of pairs (K, ε) satisfying $\delta_K^p(E, F) \leq \varepsilon$. The hidden spectrum consists of triples (K, r, ε) satisfying $\delta_{K,r}^h(E, F) \leq \varepsilon$, and the visible spectrum uses $\delta_{K,r}^v$. Width profiles are ordered coordinatewise; in the seeded spectra the seed bound and error are ordered as well. Unused states or selector values may be adjoined. The hidden convexification is $\delta_K^{\text{mix}} = \inf_{r \geq 1} \delta_{K,r}^h$.

The spectrum, not a presumed unique coordinatewise minimum profile, is the intrinsic object. For example one may minimize $\max_t K_t$, with a specified seed signature and tolerance, over its feasible profiles. Different checkpoints can have incomparable optimal tradeoffs.

2.2. The finite comparison theorem. Concatenate all probabilities $P_{\theta,g}^q(\omega)$ into $b(q) \in \mathbb{R}^N$, retaining their row indices; put $b_F = (Q_{\theta,g}(\omega))$. Let $\mathcal{D}_K$ be the finite set of deterministic choices of all rows in (2.1). Empty or unreachable rows remain harmless coordinates. Set

$$a_K = \dim \text{aff}\{b(q) : q \in \mathcal{K}_K\}, \quad \mathcal{I} = \{(\theta, g, B) : B \subseteq \Omega_g\}.$$

For $i = (\theta, g, B)$ and $d \in \mathcal{D}_K$, define

$$A_{i,d} = Q_{\theta,g}(B) - P_{\theta,g}^d(B). \quad (2.5)$$

Including both an event and its complement includes both signs.

Theorem 2.2 (Intrinsic deficiency, priced converse, and visibility). *For the finite marked causal experiments above the following assertions hold.*

(i) *Private deficiency and both fixed- r deficiencies are attained. The private problem has the exact polynomial formulation*

$$\min\{\varepsilon : q \in \mathcal{K}_K, 0 \leq \varepsilon \leq 1, Q_{\theta,g}(B) - P_{\theta,g}^q(B) \leq \varepsilon \text{ for all } (\theta, g, B) \in \mathcal{I}\}. \quad (2.6)$$

In particular its zero set is exactly the nonnegative causal factorization of the target marked behavior by the rows (2.1). At $K_t = 1$ for every t , this is precisely the same factorization with rows $q_t(z \mid a, y)$ and $q_0(z \mid y)$.

(ii) *Hidden convexification has the attained dual*

$$\delta_K^{\text{mix}}(E, F) = \min_{\lambda \in \Delta(\mathcal{D}_K)} \max_{i \in \mathcal{I}} \sum_d \lambda_d A_{i,d} = \max_{w \in \Delta(\mathcal{I})} \min_{d \in \mathcal{D}_K} \sum_i w_i A_{i,d}. \quad (2.7)$$

An optimal simulation uses at most $a_K + 1$ selector values. Thus every weighted causal test has optimal discrepancy at most ε if and only if a hidden mixture of K -state simulators has error at most ε ; it has a private realization of width at most $(a_K + 1)K$. More specifically,

$$\delta_{rK}^p \leq \delta_{K,r}^h, \quad \delta_K^{\text{mix}} = \delta_{K,a_K+1}^h \leq \delta_K^p. \quad (2.8)$$

(iii) *Put $e_\theta(q) = \max_g \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}$. Then*

$$\delta_{K,r}^v = \min_{\lambda, q^1, \dots, q^r} \max_{\theta} \sum_u \lambda_u e_\theta(q^u). \quad (2.9)$$

Writing $a_K^v = \dim \text{aff}\{(e_\theta(q))_\theta : q \in \mathcal{K}_K\}$, the infimum over all finite visible selectors is attained using $a_K^v + 1 \leq |\Theta| + 1$ values and equals

$$\max_{v \in \Delta(\Theta)} \min_{q \in \mathcal{K}_K} \sum_{\theta} v_\theta e_\theta(q). \quad (2.10)$$

For every $r \geq 1$, and also after optimizing over all r ,

$$\delta_{K,r}^v(E, F) = 0 \iff \delta_K^p(E, F) = 0. \quad (2.11)$$

No corresponding equivalence with $\delta_K^{\text{mix}} = 0$ holds in general.

(iv) *For compatible experiments E, F, G and profiles K, L ,*

$$\begin{aligned} \delta_{KL}^p(E, G) &\leq \delta_K^p(E, F) + \delta_L^p(F, G), \\ \delta_{KL,rs}^s(E, G) &\leq \delta_{K,r}^s(E, F) + \delta_{L,s}^s(F, G), \quad s \in \{h, v\}. \end{aligned} \quad (2.12)$$

Here KL is the pointwise product, and the second line uses independent selectors. If an F -consumer has widths S_t , composing it with a private simulator gives widths $K_t S_t$ and changes every $[0, 1]$ marked risk by at most $\delta_K^p(E, F)$. A hidden selector is stored at the additional price r , whereas visible selector alphabets multiply. The same bounds hold for optimized upper risk sets and their lower support functions with the indicated costs.

Proof. Expand a probability of a complete source path, simulator state path, and transcript. It is a product of fixed experiment probabilities and one chosen row entry per time. Summing latent and state paths gives a polynomial in the finitely many row entries. No row belonging to a fixed time can occur twice along one path. Consequently the polynomial is multi-affine in the groups of stochastic rows. Continuity and compactness of the product of closed finite simplexes prove attainment in (2.2); the same proof includes λ and the r policies in (2.3). For the visible problem there are only finitely many collections (g_u) and transcript events, so its objective is again continuous on a compact set. The finite-event formula for total variation proves (2.6). Setting $\varepsilon = 0$ forces equality of every transcript probability, in both directions of each event inequality. This proves the factorization and stateless claims, including their necessity.

For each stochastic row of q , sample its deterministic output once, independently of all other rows, before the experiment. This constructs a probability Λ_q on $\mathcal{D}_K$. On any fixed path the row variables encountered have different time indices, so the expectation of their product is the product of the row probabilities. Summing paths shows, simultaneously for every θ, g , that

$$b(q) = \sum_{d \in \mathcal{D}_K} \Lambda_q(d) b(d). \quad (2.13)$$

This is a behavior identity, not a same-width deterministic purification of minimax error. It implies $\text{conv } b(\mathcal{K}_K) = \text{conv}\{b(d) : d \in \mathcal{D}_K\}$. The latter is a compact polytope. Minimizing the continuous convex maximum of event discrepancies over this polytope gives the first expression in (2.7). Finite-dimensional minimax gives the second, with both optima attained. A point of this polytope is a convex combination of at most $a_K + 1$ of its generating points by Carathéodory's theorem. For completeness, if a representation uses more points, their affine dependence gives coefficients c_d with $\sum c_d = 0$ and $\sum c_d b(d) = 0$. Subtract the largest permissible multiple of c from the positive mixing weights; a weight becomes zero without changing the represented point. Repeating gives the stated bound. Store its selector together with the machine state to obtain (2.8). The dual inequality for every w is equivalent to the maximum over w being at most ε , which proves the priced testing converse.

For disjoint seed fibers, total variation is the sum of the fiber total variations weighted by λ_u . For a fixed θ , the tester can choose a separate maximizing tree on each revealed fiber. Therefore its maximum is $\sum_u \lambda_u e_\theta(q^u)$, proving (2.9). The map $q \mapsto e(q)$ is continuous, being a finite maximum of continuous functions. Its image is compact. Convexifying it in $\mathbb{R}^{|\Theta|}$ and applying the preceding finite-dimensional separation and affine-dependence argument proves (2.10) and its selector bound. Notice that the inner minimum is over stochastic q , not over deterministic tables: the nonlinear function e_θ does not commute with (2.13). If the attained visible optimum is zero, then for every u of positive weight and every θ one has $e_\theta(q^u) = 0$, since every summand is nonnegative. Any such q^u is a private exact simulator. The converse uses a selector of one value. This also proves the assertion for the optimized visible selector, whose support is finite by the bound just established.

Finally, cascade a K -state E -to- F transducer and an L -state F -to- G transducer. Within a step, consume the intermediate report before discarding it. The retained pair has at most $K_t L_t$ values. For any terminal tester on G , the second transducer together with that tester is an admissible randomized feedback system on F . The first error bound applies to this system, and total variation contracts under the remaining channel. The second error bound applies to the genuine F system. The triangle inequality gives (2.12). With selectors, condition on the independent selectors in the appropriate places; in the visible case keep both selectors in the compared transcript. The same argument is valid because the target reference selectors are independent. Storing a hidden selector or exposing both visible selectors gives the stated costs. The minima established above may be used; equivalently one can use nearly optimal simulators and let the extra tolerance decrease.

A consumer is another causal transducer with task-specific readout and, where allowed, task-specific control, but with its required common encoder. The simulator is task blind, so its product

with that encoder is still common. Condition on the consumer's independent random table to apply the feedback bound. Integration of any $[0, 1]$ marked loss changes by at most the total variation error. Taking the same paired consumer for every row proves the risk-set assertion, rather than a separate choice of simulator for each row. Infima of nonnegative weighted risks give the lower support inequalities. $\square$

Remark 2.3 (What the converse says). The dual (2.7) uses mixtures of indexed causal tests and optimizes over the entire common transducer. It is not a task-by-task choice of encoder. Its conclusion is an explicit hidden-selector simulation, with the selector subsequently charged if private storage is required. It does not replace the nonconvex private set by its convex hull at unchanged resource. The next example shows why that replacement would change the zero set.

2.3. A delayed channel and a strict signature separation. Let the source reveal $i \in [n]$ at checkpoint zero and reveal only a constant at checkpoint one. There is no control and the mark is constant. The target reports a constant at checkpoint zero and at checkpoint one draws $j \in [m]$ from a specified row-stochastic matrix $V = (V_{ij})$. The unknown parameter is i . The first target alphabet has one element: the simulator cannot use an early public output as unpriced storage. Only the intermediate register of size K matters.

Proposition 2.4 (The nonnegative-rank slice). *For this experiment,*

$$\delta_K^p(E, F_V) = \min_{A, B} \max_i \frac{1}{2} \sum_j |V_{ij} - (AB)_{ij}|, \quad (2.14)$$

where A is an n -by- K row-stochastic matrix and B is a K -by- m row-stochastic matrix. In particular

$$\delta_K^p(E, F_V) = 0 \iff \text{rank}_+(V) \leq K, \quad \delta_1^p(E, F_V) = 0 \iff V \text{ has identical rows.}$$

For $1 \leq K \leq n$ the identity channel has the sharp laws

$$\delta_K^p(E, F_{I_n}) = 1 - \frac{1}{\lceil n/K \rceil}, \quad \delta_K^{\text{mix}}(E, F_{I_n}) = 1 - \frac{K}{n}. \quad (2.15)$$

Thus, at tolerance $0 \leq \varepsilon < 1$, the least private width is $\lceil n/(1-\varepsilon)^{-1} \rceil$, whereas the least within-mode width with optimized hidden selection is $\max\{1, \lceil n(1-\varepsilon) \rceil\}$. The latter still pays for its selector. In particular, for $n = m = 3$ and $K = 2$,

$$\delta_2^p(E, F_{I_3}) = \frac{1}{2}, \quad \delta_2^{\text{mix}}(E, F_{I_3}) = \frac{1}{3}. \quad (2.16)$$

For $V_* = (1/2)I_3 + (1/6)\mathbf{1}\mathbf{1}^\top$,

$$\delta_{2,3}^h(E, F_{V_*}) = 0, \quad \delta_2^p(E, F_{V_*}) \geq \frac{1}{4\sqrt{3}} > 0, \quad \inf_r \delta_{2,r}^v(E, F_{V_*}) > 0. \quad (2.17)$$

Proof. The encoder maps the consumed i into its retained label by A ; the terminal decoder maps that label into j by B . This proves (2.14) including its converse implementation. If $V = CD$ is any nonnegative factorization, let $s_k = \sum_j D_{kj}$. Delete zero rows of D , define $B_{kj} = D_{kj}/s_k$ and $A_{ik} = C_{ik}s_k$. Since V is stochastic, the rows of A sum to one. Padding with unused states gives the claimed equivalence with nonnegative rank. The one-state statement follows immediately.

For $V = I_n$, row error is $1 - (AB)_{ii}$. If the least correct probability is c , for each i select a row $k(i)$ of B with $B_{k(i),i} \geq c$. Some row is selected at least $\lceil n/K \rceil$ times, so stochasticity gives $c \leq 1/\lceil n/K \rceil$. A partition into K groups of sizes at most $\lceil n/K \rceil$, each decoded uniformly, attains that bound. Also $\sum_i (AB)_{ii} \leq K$, by the calculation below with K in place of two. It remains valid under mixing, giving hidden error at least $1 - K/n$. Start with a deterministic channel that reproduces K input symbols exactly and sends all others to those K outputs. Average all simultaneous permutations of its input and output labels. Every diagonal entry is then K/n , proving the second formula. Inverting the two monotone formulas gives the stated least widths.

For $V = I_3$, total variation in row i is $1 - (AB)_{ii}$. If all three diagonal entries of AB were greater than $1/2$, then for each i some row of B would have its i th entry greater than $1/2$. A stochastic row has at most one such entry, so two rows cannot serve three indices. This proves the private lower bound $1/2$. Merge two input symbols into a retained label that decodes uniformly to that pair, and retain the third symbol exactly. This attains it. For every stochastic factorization with two intermediate states,

$$\sum_{i=1}^3 (AB)_{ii} = \sum_{k=1}^2 \sum_{i=1}^3 A_{ik} B_{ki} \leq \sum_{k=1}^2 \sum_{i=1}^3 B_{ki} = 2.$$

The same inequality holds under hidden mixtures, giving maximal row error at least $1/3$. Mix uniformly the three preceding pair-merge channels. The resulting diagonal entries are $2/3$ and the off-diagonal entries are $1/6$. This is exactly V_* and has error $1/3$ from I_3 . It proves the hidden claims in both displays.

The smallest singular value of V_* is $1/2$. Any AB with two intermediate states has a unit vector x in its right null space, so $\|V_* - AB\|_F \geq \|(V_* - AB)x\|_2 \geq 1/2$. If the maximal row total variation error is ε , each row has Euclidean error at most 2ε , and hence the Frobenius error is at most $2\sqrt{3}\varepsilon$. This proves the explicit private bound. The optimized visible value is attained by Theorem 2.2; if it were zero, its zero-set assertion would contradict that bound. This proves its strict positivity. $\square$

2.4. Optimal resource sets and decision certificates. For a class $\mathcal{C}_{S,q}(F)$ of common consumers, with all other information conventions fixed, let $\mathcal{R}_{S,q}(F)$ be its vector of attainable $[0, 1]$ risks. For a nonnegative row weight w summing to one, write

$$h_F(S, q; w) = \inf_{r \in \mathcal{R}_{S,q}(F)} \langle w, r \rangle.$$

If E privately simulates F with cost K and error ε , then

$$\mathcal{R}_{S,q}(F) \subseteq \mathcal{R}_{KS,q}(E) + [-\varepsilon, \varepsilon]^J, \quad h_E(KS, q; w) \leq h_F(S, q; w) + \varepsilon. \quad (2.18)$$

For nonfinite row sets the first expression means pointwise paired risk functions within uniform distance ε ; no finite coordinate box is needed. Upper sets give the equivalent one-sided order formulation. Hidden selectors multiply private width by r ; visible selectors multiply the separate public alphabet by r . These are direct consequences of the final part of Theorem 2.2, with the optimal intrinsic error in place of an externally estimated simulator error.

When two stateless comparison directions have error at most ε , paired risk sets and their support values at unchanged width are within ε . If each is centered by a paired unrestricted benchmark whose error is also at most ε , the centered values are within 2ε . Therefore a gap $g > 0$ between zero excess and infeasible widths is stable under $\varepsilon < g/4$: a value below $g/2$ in the approximation certifies feasibility in the exact finite problem, and a value above $g/2$ certifies infeasibility. The benchmark and the class being optimized must be specified; this assertion does not identify different information signatures.

Proposition 2.5 (Intrinsic finite-dimensional selector approximation). *Let $\mathcal{R}$ be a compact risk image in $C(J)$, where J is compact. Suppose a bounded linear operator $L : C(J) \rightarrow C(J)$ has rank at most m and $\sup_{r \in \mathcal{R}} \|r - Lr\|_\infty \leq \eta$. Every barycenter $\bar{r}$ of a probability on $\mathcal{R}$ has a barycenter r_* of at most $m + 1$ members of $\mathcal{R}$ with $\|\bar{r} - r_*\|_\infty \leq 2\eta$. If $\mathcal{R}$ lies in a finite affine space of dimension m , the error is zero with at most $m + 1$ members. Thus the selector price can be controlled by the dimension of attainable risks, rather than by a covering number of the row space.*

Proof. The compact set $L\mathcal{R}$ lies in an m -dimensional space. Its convex hull is compact, and $L\bar{r}$ belongs to it. The finite-dimensional affine-dependence argument in the proof of Theorem 2.2 gives $r_1, \dots, r_{m+1} \in \mathcal{R}$ and weights λ such that $L\bar{r} = \sum_i \lambda_i Lr_i$. Put $r_* = \sum_i \lambda_i r_i$. The two residual

terms $\bar{r} - L\bar{r}$ and $r_* - Lr_*$ have norms at most η , proving the estimate. Apply the same argument after translation in the exact affine case. The implementation stores the selected design index whenever it is not provided as visible side information. $\square$

Remark 2.6 (Finite computation and general carriers). For rational finite instruments, (2.6) and the fixed- r formulations are finite polynomial feasibility problems; (2.7) is a finite linear program. These are exact variational characterizations, not assertions of polynomial running time in a succinct history description. The number of feedback trees, events, or deterministic tables can be large. On standard Borel carriers the same definitions use suprema over measurable feedback and tests and infima over executable kernels. Their product-state composition and decision transport remain valid by the same channel argument. Attainment and the finite testing dual are asserted here only for the stated finite class. The average-risk and physical results below give separate nonfinite theorems with their own hypotheses.

3. CONTINUATION GEOMETRY AND AN EXACT MEMORY OBSTRUCTION

3.1. Normalized continuation laws. We first separate the transducer from the source to which it is attached. There are L epochs with finite input alphabets X_t and output alphabets Z_t . A causal kernel $p(z_{1:L} \mid x_{1:L})$ is nonnegative, normalized for every input word, and has output-prefix marginals independent of later inputs. The current input can include both a source report and the current control. It never includes an unavailable parameter, a terminal mark, or an earlier output not retained by the machine.

Let $\mathcal{C}_t$, $0 \leq t \leq L$, be the finite-dimensional polytope of normalized causal kernels from $x_{t+1:L}$ to $z_{t+1:L}$; put $\mathcal{C}_L = \{1\}$. For $v \in \mathcal{C}_t$ define the shift

$$(S_{x,z}v)(z_{t+2:L} \mid x_{t+2:L}) = v(z, z_{t+2:L} \mid x, x_{t+2:L}).$$

Its total mass $v(z \mid x)$ is independent of the future input word. A zero shift is allowed.

Definition 3.1 (Compatible continuation polytopes). A continuation cover of width $K = (K_0, \dots, K_L)$ consists of lists $V_t = (v_{t,m})_{m \leq K_t}$ in $\mathcal{C}_t$ such that, for every m, x, z ,

$$S_{x,z}v_{t,m} \in \text{cone } V_{t+1}. \quad (3.1)$$

It covers p when $p \in \text{conv } V_0$. Lists may repeat generators and may contain unreachable generators. Thus unused states can be adjoined.

Theorem 3.2 (Continuation realization with a width profile). *A causal kernel p has a stochastic transducer realization with an initial K_0 -valued register and K_t retained states after epoch t if and only if it has a compatible continuation cover of width K . Initial randomization is part of the charged initial register. With a deterministic initial state the condition is $p = v_{0,1}$ and $K_0 = 1$.*

Consequently the set of feasible width profiles depends only on the causal kernel and its declared input interface, not on a row presentation or a labeling of the hidden states. Every matrix of conditional continuation probabilities at cut t has nonnegative rank at most K_t . Individual rank bounds do not replace the simultaneous shift conditions (3.1).

Proof. Given a transducer, let $v_{t,m}$ be the suffix response when its register at time t equals m . The remaining rows define this response even for an unreachable state. If the next row is $q_{t+1}(z, m' \mid m, x)$, conditioning on its output and new state gives

$$S_{x,z}v_{t,m} = \sum_{m'} q_{t+1}(z, m' \mid m, x) v_{t+1,m'}.$$

The initial state distribution expresses p as a convex combination of V_0 .

Conversely choose nonnegative coefficients in every cone membership in (3.1). There are only finitely many choices to make. Since every generator on the right has mass one for each future

input word, the sum of the coefficients for a fixed m, x, z is $v_{t,m}(z \mid x)$. Summing over z gives one. The coefficients are therefore stochastic rows $q_{t+1}(z, m' \mid m, x)$. Choose an initial distribution representing p in $\text{conv } V_0$. Backward induction from $\mathcal{C}_L = \{1\}$ proves that the constructed machine has exactly the response $v_{t,m}$ from each register state. In particular its initial response is p . Equality for open-loop input words implies equality under adaptive input selection: at each node of an input–output history tree, the same causal conditional probability is used on the two sides.

For the rank assertion, condition on any positive-probability observed history at the cut. Its continuation response is a convex combination of $v_{t,m}$ with weights equal to the posterior register distribution. Stack these responses as rows, and use all future input–output pairs as columns. This is a nonnegative factorization through K_t generators. The converse construction required the same generator families at adjacent cuts and all their shifts; an unrelated factorization of each cut matrix provides no such compatibility. The characterization, being stated entirely in terms of these response polytopes, also proves the asserted invariance. $\square$

Attach such a causal kernel to a marked source experiment. The output law is obtained by contraction with the source’s causal instruments, while its actual mark is retained. Hence zero private deficiency is equivalently the existence of a compatible continuation cover whose contraction equals the target marked behavior in every parameter and feedback row. This is the geometric form of the factorization in Theorem 2.2; no conditional dependence on the mark is introduced.

A hidden selector chooses one such cover independently of the source. It is a convex combination of complete realizations, not a convexification inside every retained-state simplex at the original width. Its private implementation uses the disjoint union of the covers and keeps the selector in the state. A visible selector keeps those fibers in the compared law. The nonnegative fiber argument in Theorem 2.2 therefore continues to give equality of the visible and private zero sets.

3.2. An erasure cut. A simple family makes the width obstruction exact, including its error. Let X take values in $[N]$ with law π . Before a specified cut the source reveals X . After the cut it has no reports correlated with X . The controller has no feedback channel carrying X across the cut, and all earlier emitted reports are prescribed constants. The only permitted cross-cut information of the simulator is its K -valued register and any independent selector explicitly included in its signature. The terminal mark is the actual $W = X$. The target’s terminal report is X . All memory at other checkpoints is unrestricted.

Theorem 3.3 (Exact cut spectrum). *For this experiment, for private, hidden and visible signatures alike,*

$$\delta_K = 1 - \sum_{i=1}^{\min(K,N)} \pi_{(i)}. \quad (3.2)$$

For uniform X , $\delta_K = (1 - K/N)_+$. Thus an erasure cut for m unbiased bits requires $K \geq (1 - \epsilon)2^m$ whenever its error is at most ϵ . The bound holds for every surrounding width profile; equality is attained when the other checkpoints are unrestricted. If every $\pi_x > 0$, exact transmission has nonnegative cut rank N .

Proof. With all post-cut randomness averaged into the decoder, any private machine gives kernels $e(m \mid x)$ and $d(z \mid m)$. Its joint terminal law is $\pi_x \sum_m e(m \mid x) d(z \mid m)$. Since the target law is $\pi_x \mathbf{1}_{z=x}$, total variation is exactly one minus the correct-decoding probability. Put $u_x = \max_m d(x \mid m)$. Then

$$0 \leq u_x \leq 1, \quad \sum_x u_x \leq K, \quad \mathbb{P}(Z = X) \leq \sum_x \pi_x u_x \leq \sum_{i=1}^{\min(K,N)} \pi_{(i)}.$$

The last inequality follows by moving mass in u from a smaller weight to a larger one until the K largest weights are filled. To attain it, retain distinct labels for those letters and decode them

deterministically; all other letters are mapped to any retained label. Before the cut one may use the unrestricted memory to compute this encoding.

For a hidden selector, condition on its value. It is independent of X , so every conditional correct-decoding probability obeys the same bound; averaging cannot improve it. For a visible selector, variation is the average of the fiber variations, with the same conclusion. The private construction works for both signatures. Finally, the exact joint cut matrix is diagonal with its N positive diagonal entries π_x . Its ordinary rank is N , so its nonnegative rank is at least N and its diagonal factorization attains N . $\square$

The absence of an auxiliary cross-cut feedback channel is essential. An unpriced controller storing X and returning it as a later control would change the experiment, not disprove the cut bound. Likewise a selector correlated with X is not an independent hidden selector. These distinctions are the same ones needed in the collision certificate below.

4. FINITE-DATA TESTING ON THE ORIGINAL BEHAVIOR IMAGE

4.1. The reset interface. Let $B \subset \prod_{j=1}^J \Delta(\Omega_j)$ be compact, and let $Q = (Q_j)$ be fixed. Set $d_j = |\Omega_j|$, $D = \sum_j (d_j - 1)$, $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$, and $f_{j,A}(b) = Q_j(A) - b_j(A)$. Include the empty event. No convexity of B is assumed.

A candidate b is fixed for the whole audit. The auditor obtains n independent samples of each b_j , with independence between rows. In a causal experiment, row j means a specified parameter and deterministic feedback tree; the reset interface supplies that row. It is an additional experimental interface, not information available to the online simulator. A complete sample contains the terminal mark. Private randomness and any per-run hidden selector are reset between runs. The design of the candidate is not resampled.

Write $C_{j,w}$ for the counts, and $\mathcal{N}_n$ for all count arrays with $\sum_w C_{j,w} = n$. Their probability is the polynomial

$$p_C(b) = \prod_j \frac{n!}{\prod_w C_{j,w}!} \prod_w b_j(w)^{C_{j,w}}.$$

A randomized audit is an array $c = (c_{C,i})$ with $c_C \in \Delta(\mathcal{I})$. After training it chooses $i = (j, A)$ according to c_C . Independent candidate and target validation runs give the signed score $\mathbf{1}_A(\text{target}) - \mathbf{1}_A(\text{candidate})$. Its expected value is

$$g_c(b) = \sum_{C,i} p_C(b) c_{C,i} f_i(b), \quad v_n^{\text{aud}}(B, Q) = \max_c \min_{b \in B} g_c(b). \quad (4.1)$$

Thus every audit payoff is polynomial of total degree at most $Jn + 1$ in the behavior coordinates. The probability of a large observed score in a single audit is not asserted to equal its expectation; additional independent validation would be required for a specified confidence level.

Theorem 4.1 (Reset-audit duality and finite-sample approximation). *Let $\delta = \min_{b \in B} \max_i f_i(b)$. Then all the extrema below are attained, and*

$$v_n^{\text{aud}}(B, Q) = \min_{\mu \in \mathcal{P}(B)} \sum_{C \in \mathcal{N}_n} \max_{i \in \mathcal{I}} \int_B p_C(b) f_i(b) d\mu(b), \quad (4.2)$$

$$0 \leq \delta - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}. \quad (4.3)$$

The values are nondecreasing in n . If $a = \dim \text{aff } B$, a minimizing measure in (4.2) can be supported on at most $\binom{a+Jn+1}{a}$ points.

An explicit feasible audit attaining the lower bound $\delta - \sqrt{D/n}$ chooses an event maximizing $Q_j(A) - C_j(A)/n$, with a fixed deterministic tie rule. If the rational target probabilities are given,

this is a rationally specified finite rule. With a free schedule clock, a conservative retained-state bound is

$$H(|\mathcal{I}| + 1)(n + 1)^{\sum_j d_j},$$

where H is the number of states needed to execute and record one causal transcript. This budget is charged to the auditor, not to the simulator.

Proof. The coefficient arrays form a compact convex product of finite simplexes. The integrand is continuous in b and affine in c . The probability measures on compact B form a compact convex set for weak convergence. The minimum over b equals the minimum over such measures for fixed c . Finite-dimensional minimax (or its compact measure version) therefore exchanges the two extrema. Maximization over each independent coefficient simplex selects its largest coordinate integral, giving (4.2). Continuity and compactness give attainment. This step uses measures on the original B , not feasible private realizations of their barycenters.

For any audit, $g_c(b) \leq F(b) = \max_i f_i(b)$. At a minimizer of F this gives $v_n^{\text{aud}} \leq \delta$. The zero event gives $v_n^{\text{aud}} \geq 0$. For the stated explicit rule put $\hat{b}_j = C_j/n$ and $\rho = \max_j \|\hat{b}_j - b_j\|_{\text{TV}}$. Comparing the empirical maximizer with a true maximizing event gives the pointwise regret bound

$$F(b) - f_{i(C)}(b) \leq 2\rho.$$

For a multinomial empirical measure,

$$\mathbb{E}\|\hat{b}_j - b_j\|_{\text{TV}}^2 \leq \frac{d_j}{4} \sum_w \mathbb{E}(\hat{b}_j(w) - b_j(w))^2 = \frac{d_j(1 - \sum_w b_j(w)^2)}{4n} \leq \frac{d_j - 1}{4n}.$$

Consequently $\mathbb{E}\rho \leq (\sum_j \mathbb{E}\|\hat{b}_j - b_j\|_{\text{TV}}^2)^{1/2} \leq \sqrt{D/n}/2$. Averaging the regret uniformly over B proves (4.3).

From $n + 1$ unordered samples in each row, discard one uniformly at random. The remaining counts have exactly the n -sample law, independently across rows. This realizes every n -sample audit as a randomized $(n + 1)$ -sample audit and proves monotonicity.

Choose affine coordinates on $\text{aff } B$. All the functions $p_C f_i$ lie in the span of monomials of total degree at most $Jn + 1$, a space of dimension at most $s = \binom{a + Jn + 1}{a}$ including the constant. The moment vector of a probability measure is in the convex hull of the evaluation vectors; their constant coordinate is one, so their affine dimension is at most $s - 1$. Eliminating affine dependencies gives a representation by at most s points. It preserves every integral in (4.2), and hence preserves the objective.

Finally, each count is in $\{0, \dots, n\}$. The indicated product register stores all counts, the within-run state, and the selected event for validation. Only the latter event and the within-run buffer need survive once training ends, so the product bound is conservative. Randomizing an event uses fresh randomness. The schedule, reset access and terminal mark readout were declared before the theorem. No private simulator state is read by this implementation. $\square$

The dual measure chooses a candidate once, after which its samples are independent conditional on that candidate. Replacing it by $\bar{b} = \int b d\mu$ would replace $\int p_C(b) f_i(b) d\mu$ by $p_C(\bar{b}) f_i(\bar{b})$, generally a different quantity. It would also replace the joint replicated law by a product of its marginals. Neither operation is part of the theorem. This is the operational meaning of the moment side, rather than a same-budget convexification theorem.

Proposition 4.2 (Both sides sampled). *Suppose the target reference Q^b varies with the candidate, as it does when visible-selector weights are design variables. Supply independent training samples from both b and Q^b , and use their empirical difference. If their row dimensions are D_P, D_Q , the same construction has regret at most $\sqrt{D_P/n} + \sqrt{D_Q/n}$. Minimax and attainment hold on the*

compact image of pairs (b, Q^b) . In particular a variable reference law must not be treated as known fixed input to Theorem 4.1.

Proof. The error in any event discrepancy is at most the sum of the maximal row variations of the two empirical tables. Apply the preceding second moment bound separately and then the empirical-maximizer regret inequality. Count probabilities are products for the two tables. They are continuous polynomials on the compact image of pairs, so the same minimax argument applies without changing the simulator signature. $\square$

4.2. A matching uniform sampling obstruction.

Theorem 4.3 (Binary reset lower bound). *Take one binary row with target $\text{Ber}(1/2)$ and candidate class $B_h = \{\text{Ber}(1/2 - h), \text{Ber}(1/2 + h)\}$, where $0 < h < 1/2$. Then*

$$\delta = h, \quad v_n^{\text{aud}} = h \|\text{Ber}(1/2 + h)^{\otimes n} - \text{Ber}(1/2 - h)^{\otimes n}\|_{\text{TV}}. \quad (4.4)$$

For every $n \geq 1$, choosing $h = 1/(8\sqrt{n})$ gives

$$\delta - v_n^{\text{aud}} \geq \frac{1}{16\sqrt{n}}.$$

Thus the exponent in (4.3) is sharp uniformly over compact candidate classes, already at affine dimension one. This does not assert a degree lower bound for revealed-behavior polynomials, nor that this two-point class equals an unrestricted private-row image.

Proof. Only the two complementary singleton events and the zero event matter. A randomized choice is a statistic d in $[-1, 1]$. The two expected payoffs are $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$. Their minimum is at most half their sum, which is at most $h\|P_+^n - P_-^n\|_{\text{TV}}$. Conversely take the sign of the likelihood difference, assigning zero at a tie. Complementing every sample interchanges P_+^n and P_-^n and changes that sign. The two payoffs are equal, and their average attains the variation bound. The statistic depends only on the count, so it is an admissible audit. The symmetric measure on the two candidates attains the dual value. Every Dirac measure has dual objective h , whereas for finite n the two product laws are equivalent and their variation is strictly less than one. Thus no dual minimizer is a Dirac measure. Their barycenter is the target law, but replacing the dual by that barycenter would incorrectly give value zero instead of the strictly positive value in (4.4).

For $h \leq 1/8$, elementary integration of $(1 - u^2)^{-1}$ gives

$$\text{KL}(\text{Ber}(1/2 + h) \parallel \text{Ber}(1/2 - h)) = 2h \log \frac{1 + 2h}{1 - 2h} \leq 16h^2.$$

The product relative entropy is at most $16nh^2$. Pinsker's inequality bounds its total variation by $\sqrt{8nh^2}$. At the specified h this is less than $1/2$. Substituting in (4.4) proves the claim. The inequality uses the classical two-point testing mechanism; the restriction being measured is the reset budget of the present game. $\square$

4.3. Transport of one fixed audit.

Proposition 4.4 (Reset cost under model perturbation). *Let q range over the same compact set of executable row designs in two models. Suppose uniformly in q, j that*

$$\|P_j^q - \tilde{P}_j^q\|_{\text{TV}} \leq \alpha, \quad \|Q_j - \tilde{Q}_j\|_{\text{TV}} \leq \beta.$$

For the same coefficient rule c , evaluated with each model's own target validation,

$$|g_c(q) - \tilde{g}_c(q)| \leq (2Jn + 1)\alpha + \beta. \quad (4.5)$$

The deficiencies themselves differ by at most $\alpha + \beta$, at the same resource budget. If the source behavior is unchanged, a fixed audit loses at most β , independently of its reset count.

Proof. A product coupling of the Jn training runs fails with probability at most $Jn\alpha$. Taking counts and choosing the event are contractions. A score with fixed target discrepancies lies in $[-1, 1]$, so the change in its expectation from this training-law perturbation is at most $2Jn\alpha$. At each event the discrepancy changes by at most $\alpha + \beta$. This proves (4.5). For deficiency, every row variation changes by at most $\alpha + \beta$; taking the maximum and then the minimum preserves that bound. No new simulator is inserted. $\square$

For a rational polynomial image, a rational rule and a rational threshold, strict positivity of $g_c(q)$ above that threshold is a first-order formula over the original row simplexes. Real quantifier elimination decides it. The same is true of an existential rule followed by this universal condition. This is an exact finite verification statement, not an efficient algorithm. The explicit empirical rule below will be verified by a uniform analytic inequality, rather than by storing its enormous complete table.

5. CONTINUATION COMPLEXITY OF A FINITE-DATA TEST

The reset budget does not by itself bound the auditor's retained memory. We now impose both constraints. At scheduled epoch t a streaming auditor reads one letter x_t from a finite alphabet X_t , updates a register, and emits no information to any auxiliary store. It makes its decision on reading the last letter. Transition rows may depend on the epoch; the clock is free. All other information crossing a cut belongs to the register. Fresh randomization is allowed. An independent persistent selector, when allowed, selects a complete auditor before any letters are read.

For a Boolean decision $\psi : X_1 \times \cdots \times X_n \rightarrow \{-1, 1\}$, let $u \sim_t v$ mean that

$$\psi(us) = \psi(vs) \quad \text{for every } s \in X_{t+1} \times \cdots \times X_n.$$

Write $r_t(\psi)$ for the number of equivalence classes of prefixes of length t , for $0 \leq t < n$. These are the deterministic continuation responses. The terminal decision is emitted on the last transition, so no terminal register is needed. In particular, $r_0 = 1$. The equivalence is a finite-horizon version of the classical residual-state construction [16]. The following quantitative statement will also measure the loss when a statistical auditor has fewer states than this construction.

Theorem 5.1 (Exact continuation width and a quantitative cut obstruction). *A stochastic streaming auditor computes ψ exactly on every word with width profile $(K_t)_{0 \leq t < n}$ if and only if $K_t \geq r_t(\psi)$ at every cut. The profiles are attained simultaneously by a deterministic auditor. Independent persistent randomization does not reduce these widths for exact computation.*

More generally, at a cut with $r = r_t(\psi) \geq 2$, every stochastic auditor of width K satisfies

$$\sum_{w \in X_1 \times \cdots \times X_n} e_\psi(w) \geq \frac{(r - K)_+}{r - 1}, \quad (5.1)$$

where $e_\psi(w)$ is its probability of an incorrect Boolean decision on input w . The inequality also holds after averaging over an independent selector whose conditional auditors each have width at most K .

Proof. From a prefix u let $e_u(m)$ be the distribution of the retained state. From state m and suffix s let $a_m(s)$ be the probability of terminal answer $+1$. Exact computation says

$$\sum_m e_u(m) a_m(s) = \frac{1 + \psi(us)}{2}.$$

The right side is zero or one. Every state of positive e_u -mass must therefore have precisely that zero-one continuation response for all s . Distinct prefix classes have disjoint positive supports. Hence $K_t \geq r_t$. Conversely, retain the prefix class. Appending the same letter to two equivalent prefixes gives equivalent new prefixes; thus every transition is well defined. At the last transition

the class and the last letter determine ψ . This realizes every width r_t simultaneously. For a persistent independent selector, a zero average error on each of the finitely many words implies zero conditional error on all words outside a single null set of selector values. The preceding bound applies there.

For the quantitative claim choose representative prefixes $u_1, \dots, u_r$, one per class, and put $e_i(m) = e_{u_i}(m)$. Define

$$o_{ij} = \sum_{m=1}^K \min\{e_i(m), e_j(m)\}.$$

For nonnegative numbers $a_1, \dots, a_r$, ordering them shows that $\sum_{i < j} \min(a_i, a_j) \geq \sum_i a_i - \max_i a_i$. Apply this at each state. Since every e_i has mass one and each $\max_i e_i(m) \leq 1$, it follows that

$$\sum_{i < j} o_{ij} \geq r - K. \quad (5.2)$$

Choose a suffix s_{ij} distinguishing u_i, u_j . Conditional on a common state and this suffix, the two desired answers are opposite. Their error probabilities, weighted by $e_i(m), e_j(m)$, have sum at least $\min\{e_i(m), e_j(m)\}$. Thus

$$e_\psi(u_i s_{ij}) + e_\psi(u_j s_{ij}) \geq o_{ij}.$$

On summing, each complete word on the left occurs at most $r - 1$ times: its prefix identifies a single representative, which participates in $r - 1$ pairs. Nonnegativity of all other word errors and (5.2) prove (5.1). When $r \leq K$ the claim is just nonnegativity. Finally apply the bound conditional on a selector and integrate. Independence ensures that conditioning has not changed the input experiment. $\square$

The exact part is also the extremal zero-one case of Theorem 3.2. Its content is a simultaneous width profile, rather than separate unrelated factorizations at different cuts. The error bound does not assert that its constant is optimal for every Boolean function. Its use below is to convert a continuation obstruction into a strict loss of the attainable statistical value.

5.1. The width required to attain the binary reset value. Fix $0 < h < 1/2$ and the binary candidate class B_h of Theorem 4.3. A terminal decision $+1$ chooses the singleton event appropriate to the $+$ candidate, and -1 the complementary event. A zero-score event can equivalently be replaced by an independent fair choice of these decisions. The expected payoffs are therefore $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$, where $d \in [-1, 1]$ is the conditional mean decision and $P_\pm = \text{Ber}(1/2 \pm h)^{\otimes n}$. Let $v_{n,K}$ be the supremum of the smaller of these two payoffs over streaming auditors with the declared width profile. The final validation interface is unchanged; the widths here count training and selection only.

Theorem 5.2 (Exact memory threshold for optimal binary auditing). *Let $n = 2m + 1$ be odd. The unconstrained reset value is $v_n = h\|P_+ - P_-\|_{\text{TV}}$. It is attained at the profile*

$$r_t = \min\{t + 1, n - t + 2\}, \quad 0 \leq t < n, \quad (5.3)$$

and $v_{n,K} = v_n$ if and only if $K_t \geq r_t$ for all these cuts. For every cut with $r_t \geq 2$ one has the explicit deficit

$$v_n - v_{n,K} \geq 2h^2(1/4 - h^2)^m \frac{(r_t - K_t)_+}{r_t - 1}. \quad (5.4)$$

These conclusions remain valid when an independent persistent selector chooses auditors of the same conditional widths. For $n \geq 3$ the least peak training width attaining v_n is $m + 2$; for $n = 1$ it is one.

Proof. The sign of $P_+(w) - P_-(w)$ is the majority decision $\psi(w)$, with no zero difference since n is odd. Complementing every bit interchanges the two laws and reverses the decision. Consequently majority has equal payoffs and attains $h\|P_+ - P_-\|_{\text{TV}}$, as also proved in Theorem 4.3.

A prefix with k ones requires at least $m + 1 - k$ ones in its remaining $n - t$ places. Thresholds at most zero give the constant positive response; thresholds greater than $n - t$ give the constant negative response; the remaining thresholds give different nonconstant responses. Counting them for $0 \leq k \leq t$ gives (5.3). The exact realization in Theorem 5.1 implements majority at all these widths.

For an arbitrary randomized decision, its smaller payoff is bounded by the equal-prior average. The deficit of this average from the majority value is exactly

$$h \sum_w |P_+(w) - P_-(w)| e_\psi(w). \quad (5.5)$$

Put $a = 1/2 + h$ and $b = 1/2 - h$. A word with $k \geq m + 1$ ones has absolute likelihood difference $(ab)^{n/2} \{(a/b)^{k-n/2} - (a/b)^{-(k-n/2)}\}$, which increases with k . Symmetry treats the other half of the words. The smallest difference is therefore attained at $k = m, m + 1$ and equals

$$a^{m+1}b^m - b^{m+1}a^m = 2h(1/4 - h^2)^m.$$

Substitution of (5.1) into (5.5) gives (5.4). Any deficient cut gives a strictly positive loss; all sufficiently wide profiles implement majority. Maximizing (5.3) proves the peak-width assertion. The same average-payoff argument works conditional on an independent selector and hence after averaging. $\square$

The result identifies the exact memory needed to attain the optimal value, not the memory needed for a prescribed approximate value uniformly in h . The quantitative deficit can be small for large n . In particular it is not a sharp joint asymptotic tradeoff in sample size, state size and error. It complements, rather than replaces, the uniform sampling exponent in Theorem 4.3. No minimax interchange on the generally nonconvex set of fixed-width auditors has been used.

6. POLYNOMIAL TESTING OF THE ORIGINAL RESOURCE IMAGE

The private behavior image is usually nonconvex. Its testing criterion can nevertheless be written as a limit of finite algebraic lower certificates, without replacing that image by randomized designs. The testing weights in this section depend on the candidate simulator. Unlike Theorem 7.1, this identity is a certificate transform, not an independent variational dual. They are universal proof functions, not a hidden selector supplied to an executed experiment.

Use any one of the finite signatures of Theorem 2.2, with rational instrument data. Write its objective as $F(q) = \max_{1 \leq i \leq m} f_i(q)$, where f_i are all its marked event discrepancies, including the empty event $f_1 = 0$. For visible selectors the event and feedback rule include their selector fibers, and the target selector law remains in f_i . Thus $-1 \leq f_i \leq 1$ for all three signatures; the polynomials may be different.

Parameterize a row with d outcomes by $d - 1$ variables in $[0, 1]$:

$$q_1 = x_1, \quad q_j = x_j \prod_{l < j} (1 - x_l) \quad (1 < j < d), \quad q_d = \prod_{l < d} (1 - x_l).$$

This map is onto the row simplex, including its boundary. Indeed, divide each desired probability by the remaining mass while that mass is positive; when it is zero choose the remaining variables arbitrarily. Taking the product of these maps, including a selector row when needed, gives an onto polynomial map $q(x)$ from $[0, 1]^D$ to the feasible row product. Put $h_i(x) = f_i(q(x))$ and $g_i = (1 + h_i)/2$. All h_i are rational polynomials, $0 \leq g_i \leq 1$, and $g_1 = 1/2$. The case $D = 0$ is a single feasible table and is interpreted directly.

Theorem 6.1 (Strict polynomial lower-certificate hierarchy). *For $p \geq 1$ define*

$$A_p(x) = \frac{\sum_{i=1}^m g_i(x)^p h_i(x)}{\sum_{i=1}^m g_i(x)^p}, \quad a_p = \min_{x \in [0,1]^D} A_p(x), \quad \delta = \min_q F(q).$$

Then the denominators are positive, a_p is nondecreasing, and

$$\delta - 2(1 - m^{-1/(p+1)}) \leq a_p \leq \delta, \quad \delta = \sup_{p \geq 1} a_p. \quad (6.1)$$

For rational L let

$$P_{p,L}(x) = \sum_{i=1}^m g_i(x)^p (h_i(x) - L). \quad (6.2)$$

A nonnegative tensor Bernstein expansion of $P_{p,L}$ on $[0,1]^D$ certifies $\delta \geq L$. If $L < \delta$, some p and some finite Bernstein degree have strictly positive coefficients. Conversely, a strictly positive such expansion certifies $L < \delta$. These are lower certificates at the original state and selector budgets. No convexification of the feasible behavior image occurs.

Proof. Fix x and put $b = \max_i g_i(x) \geq 1/2$ and $r_i = g_i(x)/b$. At least one r_i is one. Write $S_j = \sum_i r_i^j$. Hölder's inequality gives

$$S_p \leq m^{1/(p+1)} S_{p+1}^{p/(p+1)}, \quad S_{p+1} \geq 1.$$

Consequently $S_{p+1}/S_p \geq m^{-1/(p+1)}$. Since $A_p = 2bS_{p+1}/S_p - 1$ and $\max_i h_i = 2b - 1$,

$$0 \leq \max_i h_i - A_p \leq 2(1 - m^{-1/(p+1)}).$$

Cauchy-Schwarz gives $S_{p+2}S_p \geq S_{p+1}^2$, so A_p is nondecreasing in p . Its denominator is at least 2^{-p} and it is continuous on the cube. Taking minima proves (6.1). The bound is also at most $2 \log m/(p+1)$, by $1 - e^{-u} \leq u$.

Every tensor Bernstein basis function is nonnegative on the cube, and the basis functions sum to one. Nonnegative coefficients therefore imply $P_{p,L} \geq 0$. Dividing by the positive denominator gives $F(q(x)) \geq A_p(x) \geq L$ for every x , which proves soundness. For completeness choose p so that the error in (6.1) is less than $\delta - L$. Then

$$P_{p,L}(x) \geq 2^{-p} \{\delta - L - 2(1 - m^{-1/(p+1)})\} > 0.$$

Lemma 6.2 below gives a finite strictly positive Bernstein expansion. Conversely strictly positive coefficients have a positive minimum, and the denominator is at most m , so they give a strict positive uniform gap above L . $\square$

Positive Bernstein certificates and their degree elevation are classical; see de Klerk-Laurent [20]. The next elementary bound is retained for a self-contained exact verifier, not as a new general hypercube positivity theorem.

Lemma 6.2 (An explicit positivity degree). *Let $P(x) = \sum_k c_k x^k$ be a rational polynomial in D variables and $n \geq \max_{k,j} k_j$. In its tensor Bernstein expansion of degree n in each coordinate the coefficient at $\alpha \in \{0, \dots, n\}^D$ is*

$$b_{\alpha,n} = \sum_k c_k \prod_{j=1}^D \frac{\binom{\alpha_j}{k_j}}{\binom{n}{k_j}}, \quad |b_{\alpha,n} - P(\alpha/n)| \leq \frac{C(P)}{n}, \quad C(P) = \sum_k |c_k| \sum_j \binom{k_j}{2}. \quad (6.3)$$

In particular, $\min P > 0$ implies positivity of every coefficient whenever $n > C(P)/\min P$ and the degree restriction holds.

Proof. For a binomial random variable B with parameters (n, x) , $\mathbb{E}\binom{B}{k} = \binom{n}{k}x^k$. Expanding this expectation proves the univariate monomial formula, and products prove the displayed coefficient identity. The convention is $\binom{a}{k} = 0$ for $a < k$. To compare $\binom{a}{k}/\binom{n}{k}$ with $(a/n)^k$, draw k indices uniformly from $\{1, \dots, n\}$. Conditional on no repetition this is sampling without replacement. The probability of a repetition is at most $\binom{k}{2}/n$. Apply this total variation bound to the event that all drawn indices lie in a prescribed a -element subset. Hence the difference of the two factors is at most $\binom{k}{2}/n$. All factors lie in $[0, 1]$. A product telescoping bound followed by the triangle inequality proves (6.3). The final assertion follows by taking the minimum of P . $\square$

Corollary 6.3 (Rational lower witnesses without policy cells). *At every fixed finite rational signature, strict infeasibility above a rational level L is equivalent to the existence of the following finite rational certificate: an integer p , an integer n , and a strictly positive coefficient array for (6.2) whose identity is checked by (6.3). Dovetailing these certificates with rational original-budget upper witnesses produces nested intervals converging to δ . If a subfamily of the original tests gives a strict gap, the same construction applies to it after adjoining the empty test.*

Proof. Soundness and completeness are Theorem 6.1. Each finite candidate is checked by rational polynomial arithmetic. Enumerate all (p, n) and rational levels, and retain the largest verified lower level so far, together with zero. Rational row tables are dense in the feasible product, and their continuous objectives converge to the minimum, so enumeration and retention of the best upper table converge from above. This is a dovetailed enumeration, not a claim that any fixed (p, n) test search terminates. For a requested positive interval width, convergence of both endpoints proves termination. The reduced family conclusion uses only the same pointwise maximum argument. $\square$

Remark 6.4 (Meaning and size of the strict certificate transform). A constant distribution on tests generally detects only the convex hull of the behavior image. The functions $g_i^p / \sum_j g_j^p$ are instead functions of a proposed row table. Their universally quantified weighted inequality is an algebraic certificate; the tester need not observe a private state in an experiment. The displayed certificate identity is not an equality obtained by exchanging a private minimum with a fixed mixed test. Power means and Bernstein degree elevation are classical algebraic approximation devices, proved here in the precise form needed. The claim here is strict-level completeness for the original marked causal resource image, including visible selectors, and its physical transfer below. The coefficient count $(n+1)^D$, the number of original tests, and the algebraic degrees may be large. No polynomial complexity or intrinsic-dimension bound on this row-cube coefficient count is asserted. The independent adaptive test has the separate behavior-size bound of Theorem 7.3; its verification is a different cost.

6.1. A sharp two-time nonconvex example. The source has singleton reports, the control alphabet and actual mark are singleton, and the target emits $(0, 0)$ or $(1, 1)$ with probability $1/2$ each. At width one, a private simulator emits two independent bits with probabilities p, q . It cannot reread its first emitted bit.

Proposition 6.5 (A gap invisible to constant testing mixtures). *For this delayed experiment,*

$$\delta_{(1,1)}^p = \sqrt{2} - 1, \quad \delta_{(1,1),2}^h = 0, \quad \delta_{(1,1),r}^v = \sqrt{2} - 1 \quad (r \geq 1).$$

The private objective is the maximum of the four polynomials

$$0, \quad \frac{1}{2} - pq, \quad \frac{1}{2} - (1-p)(1-q), \quad p + q - 2pq.$$

Every constant mixture of these tests has minimum at most zero, whereas Corollary 6.3 certifies every rational $L < \sqrt{2} - 1$ at private width one.

Proof. Put $u = pq$ and $v = (1-p)(1-q)$. Direct total variation calculation gives $F = 1 - \min(u, 1/2) - \min(v, 1/2)$ and the displayed maximum. By Cauchy–Schwarz, $\sqrt{u} + \sqrt{v} \leq 1$. If

$u \geq 1/2$, then $v \leq (1 - 1/\sqrt{2})^2$, so the sum of the two capped terms is at most $2 - \sqrt{2}$; the case $v \geq 1/2$ is symmetric. Otherwise set $a = \sqrt{u}$, $b = \sqrt{v}$. On $0 \leq a, b \leq 1/\sqrt{2}$, $a + b \leq 1$, the maximum of $a^2 + b^2$ is at a vertex of this polygon and equals $2 - \sqrt{2}$. Equality is attained at $p = q = 1/\sqrt{2}$. A hidden fair bit selects the two deterministic constant sequences and exactly reproduces the target. If it is exposed, variation is averaged over the selector fibers; each fiber has error at least the private minimum, which is itself attainable without a selector. Finally the target lies in the convex hull of the two private deterministic behaviors. The average discrepancy of any fixed test mixture at those behaviors is zero, so at least one has discrepancy at most zero. The strict polynomial certificates therefore detect information lost by constant convex separation. $\square$

7. ADAPTIVE TESTING ON THE OBSERVABLE BEHAVIOR IMAGE

The obstruction to linear randomization criteria at a fixed private register is the nonconvexity of the behavior image, not a failure of compactness. We give a variational formulation in which a test is an independent function of an observable candidate law. The construction does not give the simulator an extra random variable.

7.1. The testing game. Let B be a nonempty compact subset of a finite-dimensional affine space, and let $f_1, \dots, f_m$ be affine real functions, including the zero function. Put

$$F(b) = \max_{i \leq m} f_i(b), \quad \delta = \min_{b \in B} F(b).$$

For the finite comparison problem of Section 2, b is the array of discrepancies between the simulated and target marked laws, indexed by parameters, feedback trees and transcripts; f_i ranges over their signed event evaluations. In the visible case the array includes the selector and its target weights. Thus the target coordinate is part of the array when those weights vary. Private, fixed-hidden and fixed-visible signatures each give their own compact set B . No convex hull is taken.

An *adaptive test* is a continuous map $\kappa : B \rightarrow \Delta_m$, chosen before a candidate b . Its score against that candidate is

$$G(b, \kappa) = \sum_i \kappa_i(b) f_i(b).$$

This is a verification game: the selected candidate law is revealed to the evaluator of the already chosen function κ . It is not a one-observation statistical test, and does not make an unavailable parameter or mark observable to a physical controller.

Theorem 7.1 (Revealed-behavior adaptive representation). *For every such compact behavior image,*

$$\begin{aligned} \delta &= \sup_{\kappa \in C(B, \Delta_m)} \min_{b \in B} G(b, \kappa) \\ &= \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa \in C(B, \Delta_m)} \int_B G(b, \kappa) d\mu(b). \end{aligned} \tag{7.1}$$

The minimization over probability measures is attained. If Borel adaptive tests are allowed, the first supremum is attained as well. The continuous-test assertion is a supremum; continuous attainment is not needed or asserted.

Proof. Fix $\eta > 0$. The sets $U_i = \{b : f_i(b) > F(b) - \eta\}$ form an open cover of the compact metric space B . Choose a continuous partition of unity subordinate to this finite cover. The resulting test has $F(b) - \eta < G(b, \kappa) \leq F(b)$ for all b . Consequently the first supremum equals δ . The same uniform inequality shows, for every probability μ ,

$$\sup_{\kappa} \int G(b, \kappa) d\mu(b) = \int F d\mu.$$

Its minimum is δ , attained by a point mass at a minimizer of F . This also proves the displayed minimax exchange without an assumption that B is convex. Equivalently one can apply the usual compact-convex minimax theorem to $\mathcal{P}(B)$ and the convex class of continuous tests, with the pairing $\int G d\mu$. Finally choose the smallest index attaining $F(b)$. Its level sets are Borel, and its indicator vector is a Borel test with score exactly F . $\square$

The probability μ here describes a lottery over *revealed candidate laws*. The score is $\int G(b, \kappa) d\mu(b)$, not $G(\int b d\mu, \kappa)$. Hiding the candidate and replacing it by its barycenter is a different game; in general it lowers the value. This distinction is essential for a finite persistent-state budget.

7.2. A size bound in behavior dimension. Factor out directions annihilated by every $f_i - f_i(b_0)$, so that on the resulting difference space

$$\|v\|_* = \max_i |f_i(b_0 + v) - f_i(b_0)|$$

is a norm. Denote its dimension by a and let $R = \text{diam}_*(B)$. All conclusions below are unaffected by this observable quotient. The case $a = 0$ is a single constant comparison and requires one test. Suppose henceforth that $a \geq 1$, so $R > 0$.

Lemma 7.2 (A bounded behavior chart). *There exist $b_0 \in B$ and vectors $v_j \in B - b_0$, $1 \leq j \leq a$, forming a basis, such that $\|v_j\|_* \leq R$ and every $b \in B$ has coordinates $b = b_0 + \sum_j t_j v_j$ with $|t_j| \leq 2$. If B is a rational polynomial image of a product of simplexes, the points can be chosen rational and can be found effectively.*

Proof. Fix $b_0 \in B$. In any fixed coordinates on the difference space, let $D_0 > 0$ be the maximum absolute determinant of a vectors in $B - b_0$. Compactness and spanning give existence and positivity. Choose a tuple with determinant greater than $D_0/2$. Replacing its j th vector by $b - b_0$ multiplies the determinant by t_j ; maximality gives $|t_j| < 2$. The norm bound is the definition of R . For a rational polynomial image, take b_0 at rational rows. Rational row images are dense. An affine basis can be found by enumerating rational images and testing rational determinants; the true affine span and the maximum determinant are semialgebraic data. Quantifier elimination decides whether a proposed nonzero determinant exceeds half the maximum. Density therefore makes the search terminate. This is an effectiveness assertion, not a bit-complexity bound for finding the chart. $\square$

Set $b(x) = b_0 + \sum_j (4x_j - 2)v_j$ and $C = \{x \in [0, 1]^a : b(x) \in B\}$. The ambient cube is a coordinate box, not an enlargement of the feasible behavior set. Write $\ell_i(x) = f_i(b(x))$. Then

$$|\ell_i(x) - \ell_i(y)| \leq 4R \sum_j |x_j - y_j|. \quad (7.2)$$

For $n \geq 0$ let $\beta_{\alpha, n}$ be the tensor Bernstein basis of coordinatewise degree n . A feasible coefficient array has $c_{\alpha i} \geq 0$ and $\sum_i c_{\alpha i} = 1$ for every $\alpha \in \{0, \dots, n\}^a$. It specifies the independent test

$$\kappa_i^c(x) = \sum_{\alpha} c_{\alpha i} \beta_{\alpha, n}(x).$$

Define

$$d_n = \max_c \min_{x \in C} \sum_{\alpha, i} c_{\alpha i} \beta_{\alpha, n}(x) \ell_i(x). \quad (7.3)$$

Theorem 7.3 (Intrinsic adaptive degree and support bounds). *The maxima in (7.3) are attained and $d_n \leq d_{n+1} \leq \delta$. For $n \geq 1$,*

$$0 \leq \delta - d_n \leq \frac{4Ra}{\sqrt{n}}. \quad (7.4)$$

There is an array with at most $(n+1)^a$ nonzero coefficients, all equal to one, whose minimum score is at least $\delta - 4Ra/\sqrt{n}$. Its test has coordinatewise degree n , and its score degree at most $n+1$. Furthermore

$$d_n = \min_{\mu \in \mathcal{P}(C)} \sum_{\alpha} \max_i \int_C \beta_{\alpha,n}(x) \ell_i(x) d\mu(x), \quad (7.5)$$

and a minimizing measure can be supported on at most $(n+2)^a$ points. In particular a strict level $L < \delta$ admits an adaptive test of positive uniform margin with $n > (8Ra/(\delta - L))^2$ and at most $(n+1)^a$ occupied coefficient entries. These bounds depend on the observable image and its metric, not on the number of stochastic rows in a realization.

Proof. The coefficient space is a finite product of compact simplexes, and the minimum of a jointly continuous function over compact C is continuous in its coefficients. This proves attainment. Bernstein degree elevation replaces each coefficient by convex combinations of adjacent coefficients, preserving every simplex constraint and the represented function. It proves monotonicity. All scores are convex combinations of the $\ell_i(x)$, proving the upper bound by δ .

At each grid point α/n choose an index i_{α} maximizing ℓ_i there. The choice is made even at infeasible grid points; only the final score is minimized over C . Give that index coefficient one and the others zero. For fixed x let A_j be independent $\text{Bin}(n, x_j)$ variables. If i_* maximizes at x ,

$$\begin{aligned} \max_i \ell_i(x) - \mathbb{E} \ell_{i_A}(x) &\leq \mathbb{E} \{ \ell_{i_*}(x) - \ell_{i_*}(A/n) + \ell_{i_A}(A/n) - \ell_{i_A}(x) \} \\ &\leq 8R \sum_j \mathbb{E} |A_j/n - x_j| \leq \frac{4Ra}{\sqrt{n}}. \end{aligned}$$

Here the omitted middle difference is nonpositive, and the last bound uses $\text{Var}(A_j/n) \leq 1/(4n)$. This proves the rate and the sparse coefficient assertion.

For the last formula replace the minimum over C by a minimum over $\mathcal{P}(C)$. The pairing is bilinear and continuous, and both feasible sets are compact convex, so the compact minimax theorem [18] exchanges the extrema. Optimizing each coefficient simplex gives (7.5). The integrands are polynomials of coordinatewise degree at most $n+1$. Preserve all monomial moments of that degree by Carathéodory's theorem. There are $(n+2)^a$ monomials including the constant; their moment vectors lie in an affine hyperplane of dimension at most $(n+2)^a - 1$. A measure with at most $(n+2)^a$ atoms therefore has exactly the same objective. The strict-margin assertion follows from the rate with error less than $(\delta - L)/2$. $\square$

The score is a nonlinear function of behavior although each event functional is affine. This is the reason the measure in (7.5) cannot be replaced by its behavior barycenter. For $n=0$ it can: d_0 is precisely the constant-test convex relaxation. In the two-time unmarked example of Proposition 6.5, the target belongs to the convex hull of private behaviors, so $d_0 = 0$, whereas $\lim_n d_n = \sqrt{2} - 1$. Thus the hierarchy recovers a gap which every constant test misses.

Remark 7.4 (What the size bound measures). The bound counts the nonzero entries of the independent adaptive test and controls its behavior-coordinate degree. It does not bound the cost of verifying its minimum on the generally nonconvex set C , the bit lengths of a rational chart, or the size of a description of C . Distinct row presentations of the same metric behavior image satisfy the same existence bound, although their coefficient arrays need not coincide. For the private fixed-target law array one may use $R \leq 1$ with the maximum total-variation metric. For the visible signed-discrepancy array the uniform bound $R \leq 2$ is sufficient. The old row-cube positivity certificate is a separate verification engine, with its own coordinate-dependent complexity.

7.3. A three-weight certificate invisible to constant tests.

Proposition 7.5 (A finite intrinsic nonlinear witness). *Consider a private width-one simulator producing two independent bits, with probabilities p, q , for the target which assigns probability $1/2$ to each of 00, 11. Put $u = pq$, $v = (1-p)(1-q)$ and $J(t) = 3t^2 - 2t^3$. The three behavior-dependent weights*

$$\kappa_1 = J(v), \quad \kappa_2 = J(u), \quad \kappa_3 = 1 - J(u) - J(v)$$

are nonnegative and sum to one. Against the event discrepancies $1/2 - u, 1/2 - v, 1 - u - v$, respectively, their score G satisfies

$$G \geq \frac{2}{5} + \frac{9}{8960}. \quad (7.6)$$

The certificate uses cubic weights in observable probabilities, a nine-term nonnegative derivative identity, and seventeen positive rational Bernstein coefficients in one variable. Every constant mixture of event discrepancies has minimum at most zero.

Proof. We have $u + v \leq 1$. Since J is increasing on $[0, 1]$ and $J(1-t) = 1 - J(t)$, the weights are feasible. Their score is

$$G = 1 - u - v + J(u)(u - 1/2) + J(v)(v - 1/2).$$

Set $a = (p + q - 1)^2$ and $b = (p - q)^2$. Then $a, b \geq 0$ and $a + b \leq 1$. Direct expansion gives

$$\begin{aligned} 64G &= 27 - 12a + 38a^2 - 20a^3 - a^4 + 36b - 60ab + 36a^2b + 4a^3b \\ &\quad + 6b^2 - 12ab^2 - 6a^2b^2 - 4b^3 + 4ab^3 - b^4. \end{aligned}$$

Let $c = 1 - a - b$. Another polynomial identity, valid when $a + b + c = 1$, is

$$\begin{aligned} 16\partial_b G &= 4a^3 + 6a^2c + 12ab^2 + 24abc + 12ac^2 \\ &\quad + 8b^3 + 30b^2c + 30bc^2 + 9c^3 \geq 0. \end{aligned}$$

Thus $G(a, b) \geq G(a, 0)$. The univariate polynomial

$$G(a, 0) - \frac{2}{5} = \frac{7}{320} - \frac{3}{16}a + \frac{19}{32}a^2 - \frac{5}{16}a^3 - \frac{1}{64}a^4$$

has degree-16 Bernstein coefficients $z_k/349440$, where

$$\begin{aligned} (z_0, \dots, z_{16}) &= (7644, 3549, 1183, 351, 855, 2494, 5064, 8358, \\ &\quad 12166, 16275, 20469, 24529, 28233, 31356, 33670, 34944, 34944). \end{aligned}$$

The Bernstein basis is nonnegative and sums to one. Its smallest coefficient is $351/349440 = 9/8960$, proving (7.6). All identities are rational polynomial identities; no numerical minimization is used. Finally the target is the barycenter of the two feasible deterministic laws 00 and 11. An affine mixture of signed event discrepancies has average zero at these two laws and hence minimum at most zero. The empty event supplies the reverse constant-test bound zero. $\square$

For comparison, the affine weights $v, u, 1 - u - v$ already give the smaller bound $3/8$, with the exact identity

$$G_{\text{affine}} - \frac{3}{8} = \frac{1}{2}(p - q)^2 + 2(p - 1/2)^2(q - 1/2)^2.$$

Neither certificate changes the private feasible set. The cubic certificate is an independent test chosen once for the whole behavior image, not a mixture chosen anew by an optimizer's unknown row values. Its evaluation does use the revealed behavior, as specified in the verification game. The 17-coefficient positivity check is classical; the distinction from constant tests is the behavior-dependent feasible weight family and its resource-sensitive minimum.

The infinite-dimensional representation above holds for continuous f_i on any compact metric space: its proof uses a partition of unity, not causal factorization. The measure minimum is attained at a Dirac mass at a minimizer of F . The finite-level polynomial theorem has a different

content. It is an existence-size bound for revealed-law verification; the reset-audit theorem of Section 4 imposes an explicit finite-data restriction.

7.4. Strict certificates and effective physical input. Given rational finite instruments, a rational chart and the sparse array constructed above, its strict minimum can be certified by real quantifier elimination on the original row constraints. Alternatively, pull the score back through the original polynomial row map and certify its strict positivity using degree-elevated Bernstein coefficients. The pullback changes the verification degree and is not covered by the behavior-coordinate bound. Generic quantifier elimination also decides the exact finite semialgebraic boundary; this classical observation does not assert that a finite nonnegative Bernstein expansion exists at $L = \delta$.

Proposition 7.6 (Uniform effective presentation input). *Suppose a single algorithm, on input j , returns rational finite instruments E_j, F_j and nonnegative rational error enclosures a_j, c_j for two-sided stateless marked feedback presentations of E, F . Suppose $a_j + c_j \rightarrow 0$. All alphabets, widths, selector laws and finite subroutines must be computable uniformly from j and the requested signature. Then, for every positive rational η , an algorithm returns a rational interval of width below η containing $\delta_b(E, F)$ at the same signature.*

Proof. Enumerate j until $a_j + c_j < \eta/8$. This terminates without a known rate because the rational enclosures themselves tend to zero. For the selected finite problem, enumerate rational same-budget upper tables and strict rational lower certificates, dovetailing all finite checks; include the always valid lower bound zero. Compactness, density and strict completeness provide an interval of width below $\eta/2$. Enlarge it by $a_j + c_j$ at both ends using stateless comparison. The enlarged width is below $3\eta/4$. Every enumeration and comparison is uniform by the hypotheses. A mere sequence of individually computable presentations without a uniform enumeration is not an input satisfying this proposition. $\square$

8. AN EXACT ALGORITHM FOR THE OBSERVABLE CHART

The chart in Lemma 7.2 can be found without taking its dimension as an oracle. We state the input and output explicitly. Rational polynomials are encoded by finite lists of monomial exponent vectors and rational coefficients in reduced integer form. A product of simplexes is encoded by the number of entries in each row. Eliminating the last entry of every nontrivial row leaves a full-dimensional rational polytope $\Omega \subset \mathbb{R}^r$. Zero-dimensional factors are omitted.

Proposition 8.1 (Effective observable quotient and pullback). *The input consists of the preceding simplex description, a rational polynomial map $b : \Omega \rightarrow \mathbb{R}^N$, and finitely many rational affine functionals f_i on $\mathbb{R}^N$. There is a terminating algorithm which returns its observable affine dimension a , a rational basis and coordinate map for the observable quotient, and, when $a > 0$, rational parameter points $q_0, q_1, \dots, q_a$ with the following properties. With*

$$z(q) = (f_i(b(q)) - f_i(b(q_0)))_i, \quad v_j = z(q_j),$$

the vectors v_j are a basis of the observable difference space, and all coordinates $t(q)$ in that basis satisfy $|t_j(q)| < 2$. The output also contains the rational polynomial cube map $x(q) = (t(q) + 2)/4$, a rational upper bound for the observable diameter, and the exact pullback of any subsequently supplied rational polynomial in x . The algorithm outputs the universal determinant formula it has decided. It makes no polynomial-time or coefficient-height claim.

Proof. Choose a rational relative interior point q_0 of the input simplexes. Expand the vector polynomial $z(q)$ in the free row coordinates. Let V be the rational span of all its coefficient vectors, including its constant coefficient. This equals $\text{span } z(\Omega)$: a linear functional vanishes on $z(\Omega)$ exactly when its polynomial vanishes on the nonempty interior of Ω , hence identically, hence on

every coefficient. Gaussian elimination therefore gives $a = \dim V$ and a rational basis matrix U . Choosing independent rows gives a rational left inverse L with $LU = I_a$. Thus $y(q) = Lz(q)$ provides exact quotient coordinates. If $a = 0$, all discrepancies are constant and no chart is required.

For $a > 0$ let

$$D_0 = \max_{u_1, \dots, u_a \in \Omega} |\det(y(u_1), \dots, y(u_a))|.$$

Compactness gives a maximum, and spanning gives $D_0 > 0$. Enumerate rational tuples $(q_1, \dots, q_a)$ in Ω^a . Membership and their determinant d are decided by rational arithmetic. For $d \neq 0$, decide the first-order real formula

$$\forall u_1, \dots, u_a \in \Omega : \quad \det(y(u_1), \dots, y(u_a))^2 < 4d^2. \quad (8.1)$$

Quantifier elimination terminates for each formula. Density of rational points and continuity of determinant imply that some enumerated tuple has $|d| > D_0/2$. For such a tuple the strict formula is true, so the search terminates. No computation of an exact algebraic maximizer is needed.

Invert the rational matrix with columns $y(q_j)$ to obtain $t(q)$. Cramer's rule and (8.1) give $|t_j(q)| < 2$ for every q . In the observable norm $\|z\|_* = \max_i |z_i|$, each v_j has norm at most the diameter of $z(\Omega)$. An upper bound for that diameter can be produced, for example, by bounding each free coordinate in $[0, 1]$ and summing the absolute polynomial coefficients for $f_i(b(q)) - f_i(b(q'))$. This bound is rational and computable. Rational matrix multiplication and substitution give $x(q)$ and every requested pullback. They also give its full sparse coefficient encoding. All searches have now been specified by finite encodings and terminating operations. $\square$

There is a small but important distinction between the quotient and a projection onto a smaller set of simulator rows. The former identifies exactly the directions invisible to the chosen event family; the original row constraints remain present in every universal positivity test. The chart need not identify equivalent tasks after a change of marks or of information signature. Those are changes to the comparison problem.

9. A MARKED RESOURCE GAP GENERATED BY A BINARY COLLISION

We now realize a private–hidden–visible separation in a physical experiment whose information vanishes when the collision is removed. The controller chooses a later sensor gate after its first report. The gate controls acquisition, not the mechanical collision law. Both the source and the target carry the same actual, unobserved mark.

9.1. The exact marked two-report experiment. Let $B, W \in \{0, 1\}$ have the joint distribution

$$\mathbb{P}(B = b, W = w) = \begin{cases} 5/16, & b = w, \\ 3/16, & b \neq w. \end{cases} \quad (9.1)$$

The blind source E_* has two constant reports and actual mark W . The target F_* reports $Y_1 = B$, receives $A \in \{0, 1\}$ after that report, and reports $Y_2 = AB$. The feedback tester may choose A as any function of its earlier report. A width-one private simulator cannot reread its own first emission, except through the current gate A . Hidden selectors persist but are not revealed; visible selectors are included in the joint target law independently of (B, W) . Put

$$d_* = \sqrt{5/2} - 9/8.$$

Theorem 9.1 (An exact marked causal separation). *At the width profile $(1, 1)$,*

$$\delta^p(E_*, F_*) = d_*, \quad \delta_2^h(E_*, F_*) = 1/8, \quad \delta_r^v(E_*, F_*) = d_* \quad (r \geq 1). \quad (9.2)$$

The hidden upper bound is realized by two deterministic tables. The private upper bound is realized by two independent Bernoulli variables with parameter $\sqrt{5/8}$, followed by the gate. These bounds hold against every feedback rule, not just open-loop gates.

Proof. First choose the constant gate $A = 1$. The two private emissions are independent Bernoulli variables with parameters p, q and are independent of W . Let

$$u = pq, \quad v = (1-p)(1-q), \quad a = 3/8, \quad b = 5/8.$$

The overlap of the target and simulated marked laws is $H(u) + H(v)$, where

$$H(t) = \frac{1}{2} \{\min(t, a) + \min(t, b)\}.$$

Indeed the target is supported on equal report pairs, and its two W -masses at each equal pair are $3/16$ and $5/16$. Total variation is one minus this overlap.

Cauchy-Schwarz gives $\sqrt{u} + \sqrt{v} \leq 1$. By symmetry suppose $u \geq v$. If $u \leq 1/4$ the overlap is at most $1/2$. If $u \geq 1/4$, then $v \leq (1 - \sqrt{u})^2 \leq 1/4 < a$, so the overlap is at most

$$H(u) + (1 - \sqrt{u})^2.$$

This function is convex on each of $[1/4, a]$ and $[a, b]$ and nonincreasing on $[b, 1]$. Its maximum is therefore attained at $1/4$, a or b . The values at a and b are $1 + 2a - 2\sqrt{a}$ and $3/2 + b - 2\sqrt{b}$ respectively. The second is larger, since

$$(3/2 + b - 2\sqrt{b}) - (1 + 2a - 2\sqrt{a}) = 3/8 - \frac{1/2}{\sqrt{a} + \sqrt{b}} > 0,$$

using $\sqrt{a} > 3/5$, $\sqrt{b} > 3/4$. It also exceeds $1/2$. Taking $p = q = \sqrt{b}$ attains this maximum, and hence the minimum error is $1 - (3/2 + b - 2\sqrt{b}) = d_*$.

For any hidden selector, the all-on report pair is still independent of W . If its probabilities of 11 and 00 are u, v , then $u + v \leq 1$. Concavity and monotonicity of H give

$$H(u) + H(v) \leq 2H((u+v)/2) \leq 2H(1/2) = 7/8.$$

A fair hidden bit J , emitting J twice, attains this overlap. This requires precisely two deterministic selector tables and gives error $1/8$. In particular hiding a seed does not erase the actual-mark obstruction.

These all-on bounds are lower bounds for the feedback problem. For the upper bounds first generate the corresponding all-on pair (X_1, X_2) , emit X_1 , and after receiving A emit AX_2 . For a deterministic feedback $A = g(X_1)$ the full gated marked transcript is the pushforward of the all-on marked pair. Applying the same map to the target pair cannot increase total variation. The construction is executable at width one: the second fresh Bernoulli variable is independent in the private case, and the bit J is the priced persistent hidden selector in the hidden case. Randomized feedback follows by conditioning on its independent random table.

Finally, in the visible case the two compared laws are

$$\sum_j \lambda_j \delta_j \otimes P_{g_j}^{q_j} \quad \text{and} \quad \sum_j \lambda_j \delta_j \otimes Q_{g_j}.$$

The target selector is independent of (B, W) . The all-on gate in every fiber has error at least d_* in every private fiber. Joint variation is the weighted sum of these fiber errors. Ignoring the visible selector and using the private optimizer attains d_* . $\square$

9.2. A full-dimensional collision preparation. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$ with contact distance d . The center of mass $Q = (q_1 + q_2)/2$ ranges over any fixed rational position box. Write $r = q_1 - q_2$, $g = v_1 - v_2$, and $V = (v_1 + v_2)/2$. Conditional on (B, W) in (9.1), take the uniform product preparation on

$$\begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], & (9.3) \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

The boxes have positive volume in the full phase space. The actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$; total momentum conservation retains it. The informative quantity will instead be the transverse velocity created at a collision. Put

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector which is identically zero on the incoming transverse-velocity range.

Lemma 9.2 (Uniform transverse collision signal). *For every preparation in (9.3) and every $d \in [9/10, 11/10]$, exactly one collision occurs before time 3. Its time t_c lies in $(4/5, 3/2)$, and afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4. \quad (9.4)$$

For $0 < d \leq 2/5$ no collision occurs before time 3, and the detector is zero throughout that interval.

Proof. Up to a collision $r(t) = r + tg$. For $t \leq 4/5$, its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At time $3/2$ the absolute coordinate bounds are $1/4$, $113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. Thus a first contact occurs in the stated interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$ as well. The first contact therefore has $n_x < 0$, where $n = r(t_c)/d$. The entry contact is strictly incoming. The bounds above give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq \frac{87}{220}.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The equal-mass elastic collision rule is

$$v'_1 = V + g/2 - (g \cdot n)n, \quad g' = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v'_{1,y} > -3/200 + \frac{1443}{1000} \frac{87}{220} > 1/2.$$

The relative velocity is outgoing, $r(t_c) \cdot g' > 0$, so the squared separation increases strictly for all later times. There are only two spheres and no walls; no second collision is possible. This proves (9.4). For $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. Thus there is no collision. The incoming $|v_{1,y}| \leq 3/200$ then makes h zero. $\square$

At time 2 the physical target $F_{d,\sigma}$ reports

$$Y_1 = \mathbf{1}_{\{h(v_{1,y}(2)) + \sigma \xi_1 > 0\}}.$$

It then receives a gate $A \in \{0, 1\}$, and at time 3 reports

$$Y_2 = A \mathbf{1}_{\{h(v_{1,y}(3)) + \sigma \xi_2 > 0\}},$$

where the ξ_i are independent standard normal variables, independent of the preparation. The source E_d has the same preparation, evolution, control interface and actual mark, but emits two constant reports. The gates do not change the particle motion. This is a feedback acquisition

experiment: taking $A = 0$ suppresses later information and taking $A = 1$ exposes another noisy observation of a collision-generated quantity.

Theorem 9.3 (Collision-sensitive physical resource separation). *Let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/12$. At retained width $(1, 1)$ the physical deficiencies obey, uniformly in d ,*

$$\begin{aligned} |\delta^p(E_d, F_{d,\sigma}) - d_*| &< 1/300, & |\delta_r^v(E_d, F_{d,\sigma}) - d_*| &< 1/300 \quad (r \geq 1), \\ |\delta_2^h(E_d, F_{d,\sigma}) - 1/8| &< 1/300. \end{aligned} \quad (9.5)$$

In particular private and visible error exceed $9/20$, while hidden two-selector error is less than $13/100$. More generally the errors in these three comparisons are at most $2\Phi(-1/(4\sigma))$, which tends to zero as $\sigma \downarrow 0$. For $0 < d \leq 2/5$ all three deficiencies are zero. The interacting and noninteracting regimes use the same preparation and detector. The conclusions are unchanged by simultaneously scaling positions, diameter and observation times by any positive factor, with velocities and detector noise unchanged.

Proof. Couple the ideal and physical target by the same (B, W) and the same feedback random table. Lemma 9.2 bounds each potential bit error by $\Phi(-1/(4\sigma))$. With probability at least $1 - 2\Phi(-1/(4\sigma))$ both potential reports agree with their ideal bits; the gates and all emitted transcripts then agree as well. This gives a marked feedback variation bound uniform over all gates. The channels in both directions are the identity on the binary report alphabet and have one retained state. The blind source is already exactly E_* . Stateless comparison therefore changes each fixed-budget deficiency by at most the stated error, without adding memory or hiding an exposed selector.

For completeness, the elementary Gaussian tail bound $\Phi(-x) \leq e^{-x^2/2}/(x\sqrt{2\pi})$ at $x = 3$ gives $\Phi(-3) < 1/600$: use $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$. Hence the total error is below $1/300$. The inequalities $d_* - 1/300 > 9/20$ and $1/8 + 1/300 < 13/100$ follow by squaring positive rational bounds.

When $d \leq 2/5$, both potential reports are independent fair noise bits and are independent of W . A private width-one simulator draws those two bits and applies the current gate, exactly reproducing every feedback transcript. Hidden and visible versions can ignore their selectors. Finally elastic scattering is invariant under $(q, d, t) \mapsto (sq, sd, st)$ with v unchanged. This proves the scaling statement; it is geometric similarity, not a particle-number or Boltzmann–Grad limit. $\square$

The theorem combines three effects. Collisions create the sign of the informative transverse velocity. A gate chosen after the first report changes future information. A retained hidden bit, but not a visible one, permits temporal correlation without charging the width-one register. The actual mark remains partially inaccessible in every signature. The conserved longitudinal momentum labels that mark; it does not generate the observed transverse signal.

9.3. Uniformly effective finite presentations. The binary physical interface is already finite, but its probabilities are not generally rational. We make their effective approximation explicit rather than treating individual computability as a uniform algorithm.

Proposition 9.4 (Collision-chart rational presentations). *For computable $d \in [9/10, 11/10]$ and $\sigma > 0$, one algorithm produces rational marked causal instruments and certified two-sided stateless errors tending to zero for $(E_d, F_{d,\sigma})$. On a compact computable parameter interval with $\sigma \geq \sigma_0 > 0$, the errors and the construction are uniform. Thus Proposition 7.6 applies at every fixed finite state and selector signature, not only at $(1, 1)$.*

Proof. On every preparation box the entry time and outgoing velocity are

$$t_c = \frac{-r \cdot g - \sqrt{(r \cdot g)^2 - |g|^2(|r|^2 - d^2)}}{|g|^2}, \quad n = (r + t_c g)/d, \quad v'_1 = V + g/2 - (g \cdot n)n.$$

The positive square root is separated from zero: it equals $-d(g \cdot n) > (9/10)(1443/1000)$. All denominators are likewise bounded away from zero. These explicit bounds and the compact boxes allow interval differentiation of the displayed formulas, producing a rational common Lipschitz upper bound C_{geo} for $v'_{1,y}$ in the packet variables and d . This computation terminates because only addition, multiplication, division by separated positive quantities and a separated square root occur.

Conditional on the microscopic preparation the two potential all-on bits are independent Bernoulli with probability $p = \Phi(h(v'_{1,y})/\sigma)$. Their four probabilities are p^2 , $p(1-p)$, $p(1-p)$ and $(1-p)^2$. The normal distribution function is uniformly computable on bounded intervals, for example by integrating its convergent exponential series with a remainder bound. Its Lipschitz constant is less than one. Together with the 1-Lipschitz detector and the geometric bound, this yields a computable Lipschitz constant at most $2C_{\text{geo}}/\sigma_0$ for these four integrands on fixed-noise compact parameter sets. Equal-box Riemann sums of mesh h therefore have explicit error at most a computable constant times h , plus the chosen rational evaluation error. Parameter uncertainty is included by its interval enclosure.

Integrate the four all-on probabilities separately on each packet, with the exact rational weights (9.1). For each W , round its conditional probability vector to a rational simplex point on a sufficiently fine grid; the rounding error is bounded by the sum of the coordinate errors and the grid mesh. Keep $\mathbb{P}(W = 0) = \mathbb{P}(W = 1) = 1/2$ exact. Sample this finite joint law once at initialization, emit its first bit, and gate its second bit. This is a finite latent realization of the target instrument, not a simulator with extra free state. It makes every causal prefix consistent and never divides by a possibly small prefix probability. Every gated law is a pushforward of the all-on law, so the same marked variation bound holds uniformly under feedback. Identity report channels give both stateless directions. Choosing summable evaluation and mesh errors produces the asserted uniform sequence. $\square$

10. UNIFORM STABILITY OF OPTIONAL PROJECTIONS

Marked comparison controls more than terminal bounded risk when the compared laws share their latent variable. We prove a whole-process estimate for the resulting posterior martingales. The distinction between a probabilistic coupling and an executable causal channel is important in this section.

Let Z take values in a standard Borel space and let Y take values in a standard Borel canonical path space on $[0, T]$, with a right-continuous observation filtration. Consider equivalent joint probability measures P, Q on (Z, Y) , with the same Z -marginal ν . Use the usual completions, which agree for equivalent output laws. For a bounded measurable latent variable $f(Z)$ write

$$M_t^P = \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y], \quad M_t^Q = \mathbb{E}_Q[f(Z) \mid \mathcal{F}_t^Y].$$

Take their right-continuous martingale versions. All suprema below include T and can be evaluated on a countable dense time set.

Theorem 10.1 (Marked variation controls the whole posterior process). *Let $\varepsilon = \|P - Q\|_{\text{TV}}$ and $|f| \leq 1$. When both posterior functions are evaluated on the same canonical output path,*

$$\mathbb{E}_Q \sup_{t \leq T} |M_t^P(Y) - M_t^Q(Y)|^2 \leq 80\varepsilon. \quad (10.1)$$

There is a coupling of (Z, Y^P) and (Z, Y^Q) which retains exactly the same latent variable and satisfies

$$\mathbb{P}(Y^P \neq Y^Q) \leq \varepsilon, \quad \mathbb{E} \sup_{t \leq T} |M_t^P(Y^P) - M_t^Q(Y^Q)|^2 \leq 84\varepsilon. \quad (10.2)$$

The coupling is not asserted to be causal or to have finite state. The variation bound itself, not this coupling, is used for resource transport.

Proof. Let $D = dP/dQ$, $D_t = \mathbb{E}_Q[D \mid \mathcal{F}_t^Y]$, and set

$$A_t = \mathbb{E}_Q[(D - 1)f \mid \mathcal{F}_t^Y], \quad B_t = D_t - 1.$$

Bayes' formula gives

$$M_t^P - M_t^Q = \frac{A_t - B_t M_t^Q}{1 + B_t}.$$

Since $\mathbb{E}_Q|D - 1| = 2\varepsilon$, both martingales A, B have terminal absolute expectation at most 2ε . Doob's weak L^1 inequality gives, for $0 < u \leq 2$,

$$\begin{aligned} Q\{\sup_t |M_t^P - M_t^Q| > u\} &\leq Q\{\sup_t |B_t| > 1/2\} + Q\{\sup_t |A_t| > u/4\} + Q\{\sup_t |B_t| > u/4\} \\ &\leq 4\varepsilon + 16\varepsilon/u. \end{aligned}$$

Integrating $2u$ times this bound on $(0, 2)$ gives $16\varepsilon + 64\varepsilon = 80\varepsilon$, proving (10.1). No tightness conclusion is being inferred from finite-dimensional convergence.

Disintegrate both joint laws over their common latent marginal. Measurable maximal coupling of the two output kernels, conditional on Z , has disagreement probability

$$\int \|P(dy \mid z) - Q(dy \mid z)\|_{\text{TV}} d\nu(z) = \|P - Q\|_{\text{TV}}.$$

Such a coupling can be constructed by using the minimum of the two kernel densities with respect to their sum, then their residual kernels. It is measurable on standard Borel spaces. On the event of identical full output paths use (10.1) with the Q -marginal; on its complement the squared difference is at most four. This proves (10.2). The maximal coupling may depend on the full output path; it has no asserted online implementation. $\square$

Corollary 10.2 (Changing-filtration convergence from marked comparison). *Let P_n and P be equivalent joint laws with the same latent marginal ν , and $\|P_n - P\|_{\text{TV}} \rightarrow 0$. They admit a common probability-space realization $(Z, Y, Y^1, Y^2, \dots)$ such that, for every $f \in L^2(\nu)$,*

$$\mathbb{E} \sup_{t \leq T} \left| \mathbb{E}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}[f(Z) \mid \mathcal{F}_t^Y] \right|^2 \rightarrow 0.$$

The same realization gives convergence uniformly in time in probability for $f \in L^1(\nu)$. It also permits $f_n \rightarrow f$ in $L^2(\nu)$, respectively in $L^1(\nu)$, in the first conditional expectation.

Proof. Glue the conditional maximal couplings of (Z, Y^n) with the fixed reference (Z, Y) by taking their conditional kernels independently on a countable product. The coupling does not depend on f . For the truncation $f^{(K)} = (-K) \vee (f \wedge K)$, Theorem 10.1 bounds the middle difference in the L^2 norm of the supremum by $K\sqrt{84\varepsilon_n}$. Doob's L^2 inequality bounds each truncation tail by $2\|f - f^{(K)}\|_{L^2(\nu)}$. The triangle inequality therefore gives

$$\left\| \sup_t |M_t^{P_n, f} - M_t^{P, f}| \right\|_2 \leq 4\|f - f^{(K)}\|_2 + K\sqrt{84\varepsilon_n}.$$

First let $n \rightarrow \infty$, then $K \rightarrow \infty$. For L^1 use instead the weak maximal inequality for each tail and the bounded-case estimate for the middle term. An additional difference $f_n - f$ is handled by the same maximal inequalities under P_n , whose latent marginal remains ν . $\square$

10.1. A microscopic sufficient condition. Let $(\Phi_t)_{t \in \mathbb{R}}$ be an almost-everywhere regular hard-sphere flow on a labeled phase space, with invariant probability μ . Fix a prepared law ν with $d\nu/d\mu \leq C$, and bounded real observables h, h_n . These may be collision-sensitive tagged observables. Put $\eta_n = \|h_n - h\|_{L^2(\mu)}$. With a common latent initial state $Z \sim \nu$, consider the continuous observation laws

$$Y_t = \int_0^t h(\Phi_s Z) ds + \sigma B_t,$$

$$Y_t^n = \int_0^t h_n(\Phi_s Z) ds + \sigma B_t^n, \quad \sigma > 0,$$

where the Brownian noise is independent of Z in each model. The notation does not prescribe a synchronous coupling.

Theorem 10.3 (Fixed-flow observation-approximation limit). *The joint latent-output laws are equivalent and satisfy*

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \eta_n. \quad (10.3)$$

For every $|f(Z)| \leq 1$ they admit a latent-preserving coupling with

$$\mathbb{E} \sup_{t \leq T} |M_t^{P_n, f} - M_t^{P, f}|^2 \leq \frac{42\sqrt{CT}}{\sigma} \eta_n^2. \quad (10.4)$$

In particular $\eta_n \rightarrow 0$ supplies whole-process optional-projection convergence, not merely convergence at finitely many times. This applies when h_n are orbit-mollified graph-core approximants to any bounded $h \in L^2(\mu)$.

Proof. Conditional on Z , the signals are deterministic bounded drifts. The exponential change-of-measure identity for Brownian motion gives

$$D_{\text{KL}}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |(h_n - h)(\Phi_s Z)|^2 ds \leq \frac{CT}{2\sigma^2} \eta_n^2.$$

The last inequality uses $\nu \leq C\mu$ and invariance of μ ; it does not require mixing, differentiability across a collision, or a claim that a tagged velocity belongs to the generator domain. Bounded drifts ensure equivalence and justify the exponential identity. Pinsker's inequality gives (10.3). Applying Theorem 10.1 proves (10.4).

For the last assertion choose a nonnegative smooth approximate identity ρ_n on the time line and set $h_n = \int \rho_n(s) h \circ \Phi_s ds$. Strong continuity of the Koopman group in $L^2(\mu)$ gives $\eta_n \rightarrow 0$ and $\|h_n\|_\infty \leq \|h\|_\infty$. Repeated integration by parts puts h_n in every power domain of the generator, with derivatives obtained from $\rho_n^{(k)}$. This is the orbit-core construction already used in the microscopic domain theory; the new conclusion is the uniform posterior estimate. $\square$

Sampling these observation paths at finitely many checkpoints and applying a common causal output postprocessing are contractions in marked variation. Thus the identity report comparison transfers every bounded finite-register risk at its original state and selector budgets with error bounded by (10.3). That is the resource-comparison consequence. The latent-preserving coupling in (10.4) has a different role: it realizes convergence of the changing observation filtrations and is not used as a finite-register simulator.

The theorem discharges a sufficient conditional-law hypothesis for optional projection in a genuine microscopic observation class. The constants depend on the fixed preparation, time and noise level. No particle-uniform kinetic estimate, changing microscopic path limit, limiting stochastic-exponential identification or nonlinear BSDE limit is implied without its additional hypotheses. The whole-process estimate is what replaces an unsupported appeal to Doob's inequality as a tightness argument.

11. CHANGING TRAJECTORIES, LIKELIHOODS, AND CONDITIONAL CONSUMERS

11.1. A common preparation, not a common flow. Let $(\mathcal{Z}, \nu)$ be a standard Borel probability space, and let $H, H_n : [0, T] \times \mathcal{Z} \rightarrow \mathbb{R}$ be jointly measurable, with $|H|, |H_n| \leq M$. They may be observations of different flows on different phase spaces, pulled back by specified preparation maps from $\mathcal{Z}$. No common microscopic generator or invariant measure is assumed. Set

$$\eta_n^2 = \int_{\mathcal{Z}} \int_0^T |H_n(t, z) - H(t, z)|^2 dt d\nu(z).$$

On $\mathcal{Z} \times C([0, T])$ let R be the law of independent $Z \sim \nu$ and $Y = \sigma B$, where $\sigma > 0$. Define

$$L_t^n = \exp \left\{ \frac{1}{\sigma} \int_0^t H_n(s, Z) dB_s - \frac{1}{2\sigma^2} \int_0^t H_n(s, Z)^2 ds \right\}, \quad (11.1)$$

$$dP_n = L_T^n dR. \quad (11.2)$$

The reference model P, L uses H . Conditional on Z , these are Gaussian shifts, so under P_n the observed path has law $Y_t = \int_0^t H_n(s, Z) ds + \sigma B_t^n$. Boundedness gives true strictly positive density martingales. Write $\mathcal{F}_t$ for the usual canonical observation filtration and $\mathcal{H}_t = \sigma(Z, B_s : s \leq t)$ for the larger reference filtration. Equivalent laws have the same completed observation stopping times.

Theorem 11.1 (Likelihood and posterior transport for changing paths). *The measures P_n, P, R are equivalent and have common Z -marginal ν . With $A = M^2 T / \sigma^2$ and $\varepsilon_n = \min\{1, \eta_n / (2\sigma)\}$, one has*

$$\|P_n - P\|_{\text{TV}} \leq \varepsilon_n, \quad (11.3)$$

$$\mathbb{E}_R \sup_{t \leq T} |L_t^n - L_t|^2 \leq \frac{8}{\sigma^2} e^{3A} \eta_n^2. \quad (11.4)$$

For $|f(Z)| \leq K$ put

$$N_t^{n,f} = \mathbb{E}_R[L_T^n f(Z) | \mathcal{F}_t], \quad \ell_t^n = N_t^{n,1}.$$

Then

$$\mathbb{E}_R \sup_t |N_t^{n,f} - N_t^f|^2 \leq \frac{8K^2}{\sigma^2} e^{3A} \eta_n^2. \quad (11.5)$$

In particular this applies to the projected likelihood ℓ^n . For every $|f| \leq 1$, the posterior functions satisfy the same-path bound $80\varepsilon_n$ of Theorem 10.1, and there is one latent-preserving coupling, independent of f , on which

$$\mathbb{E} \sup_t |\mathbb{E}_{P_n}[f(Z) | \mathcal{F}_t^{Y^n}] - \mathbb{E}_P[f(Z) | \mathcal{F}_t^Y]|^2 \leq 84\varepsilon_n. \quad (11.6)$$

If $\eta_n \rightarrow 0$ and deterministic microscopic paths $X_n(Z) \rightarrow X(Z)$ in a specified Polish path metric in ν -probability, then on this coupling the microscopic paths, the full and projected likelihood processes evaluated on their own observation paths, and each bounded posterior process converge jointly in probability in that metric and uniformly in time for the last three coordinates.

Proof. Conditional Gaussian change of measure gives

$$\text{KL}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |H_n - H|^2 dt = \frac{\eta_n^2}{2\sigma^2}.$$

The same calculation proves conditional equivalence, and the conditional density has mean one, so the preparation marginal is unchanged. Integrating and applying Pinsker proves (11.3).

For the quantitative density estimate, condition first on $Z = z$ and write $D_t = L_t^n - L_t$. The stochastic differential equation is

$$dD_t = \sigma^{-1} \{D_t H_n(t, z) + L_t (H_n - H)(t, z)\} dB_t.$$

The conditional second moment of L_t is at most e^A . Itô's isometry and $(u + v)^2 \leq 2u^2 + 2v^2$ therefore imply

$$\mathbb{E}_R[D_t^2 \mid z] \leq \frac{2M^2}{\sigma^2} \int_0^t \mathbb{E}_R[D_s^2 \mid z] ds + \frac{2e^A}{\sigma^2} \int_0^t |H_n - H|^2(s, z) ds.$$

Gronwall, followed by integration in z , gives $\mathbb{E}_R D_T^2 \leq (2/\sigma^2) e^{3A} \eta_n^2$. The difference is a square-integrable $\mathcal{H}_t$ -martingale. Doob's L^2 inequality proves (11.4). Applying the same maximal inequality to the $\mathcal{F}_t$ -martingale $\mathbb{E}_R[(L_T^n - L_T)f \mid \mathcal{F}_t]$ proves (11.5). These are direct process estimates, not a claim that finite-dimensional convergence alone implies tightness.

The posterior assertions follow from Theorem 10.1 and its countable conditional-coupling construction. For the last claim, on the event $Y^n = Y$ the two likelihood functions are evaluated on the same (Z, Y) . Their uniform difference tends to zero in R -probability by the estimates just proved, and therefore in P -probability by absolute continuity of the fixed P with respect to R . The complementary event has probability at most ε_n . The same argument applies to ℓ^n , and (11.6) handles the posterior. The microscopic convergence holds on any coupling retaining the same Z . A finite union bound proves joint convergence. Stochastic integral versions are chosen measurably on the canonical product space; equivalence makes these common versions valid under every law in the countable sequence. $\square$

For $f \in L^2(\nu)$ the posterior conclusion follows by truncation exactly as in Corollary 10.2; it also permits $f_n \rightarrow f$ in $L^2(\nu)$. Boundedness of H_n is being used for the uniform likelihood second moments. It is not replaced by an unverified uniform-integrability assertion in a singular or zero-noise limit.

11.2. The projected stochastic exponential. Let $\hat{H}_t^n = \mathbb{E}_{P_n}[H_n(t, Z) \mid \mathcal{F}_t]$, using a predictable version for Lebesgue-almost every t . Put $\theta_t^n = \hat{H}_t^n / \sigma$.

Proposition 11.2 (Identification before passage to the limit). *The projected likelihood and the innovation satisfy*

$$\ell_t^n = \exp \left\{ \int_0^t \theta_s^n dB_s - \frac{1}{2} \int_0^t (\theta_s^n)^2 ds \right\}, \quad (11.7)$$

$$I_t^n = B_t - \int_0^t \theta_s^n ds. \quad (11.8)$$

Here I^n is a P_n -Brownian motion in $\mathcal{F}_t$. In particular the limit of the projected likelihoods in (11.5) is the projected stochastic exponential of the limiting model. Change of measure commutes with this limit on bounded measurable path tests.

Proof. Project $L_t^n = 1 + \int_0^t L_s^n H_n(s, Z) dB_s / \sigma$ onto the Brownian observation filtration. The integrands are square integrable; testing against bounded elementary predictable stochastic integrals and using Itô's isometry gives

$$d\ell_t^n = \sigma^{-1} \mathbb{E}_R[L_t^n H_n(t, Z) \mid \mathcal{F}_t] dB_t = \ell_t^n \theta_t^n dB_t.$$

The last equality is Bayes' formula. Since $|\theta_t^n| \leq M/\sigma$, the solution is the positive stochastic exponential (11.7); Girsanov gives the stated innovation. The same argument with H identifies the limiting exponential. Finally $|\mathbb{E}_{P_n} F - \mathbb{E}_P F| \leq 2\|F\|_\infty \varepsilon_n$ for bounded signed F . There is no interchange based only on a formal limit of logarithms. $\square$

11.3. A bounded entropic backward-equation consumer. Fix $\gamma > 0$ and a latent payoff f with $|f| \leq 1$. Define

$$U_t^n = \frac{1}{\gamma} \log \mathbb{E}_{P_n}[e^{\gamma f(Z)} \mid \mathcal{F}_t] = \frac{1}{\gamma} \log \frac{N_t^{n, e^{\gamma f}}}{\ell_t^n}. \quad (11.9)$$

The terminal condition here is the observable conditional certainty equivalent U_T^n , not the generally unobserved $f(Z)$ itself. This specification avoids imposing a latent terminal value on the smaller observation filtration.

Theorem 11.3 (Entropic values and localized backward integrands). *There is a unique bounded entropic value process with terminal value U_T^n ; for a predictable integrand Z^n it satisfies*

$$U_t^n = U_T^n + \int_t^T \frac{\gamma}{2} |Z_s^n|^2 ds - \int_t^T Z_s^n dI_s^n. \quad (11.10)$$

On the coupling of Theorem 11.1,

$$\mathbb{E} \sup_t |U_t^n(Y^n) - U_t(Y)|^2 \leq 84 \left(\frac{e^{2\gamma} - 1}{\gamma} \right)^2 \varepsilon_n. \quad (11.11)$$

For each $r > e^\gamma$ let τ_r^n be the first exit of any of $\ell^n, \ell, N^n, e^{\gamma f}, N e^{\gamma f}$ from $(1/r, r)$, capped at T . If $\eta_n \rightarrow 0$, then

$$\mathbb{E}_R \int_0^{\tau_r^n} |Z_s^n - Z_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^{\tau_r^n} |\theta_s^n - \theta_s|^2 ds \rightarrow 0. \quad (11.12)$$

Furthermore $\lim_{r \rightarrow \infty} \sup_n R(\tau_r^n < T) = 0$. Thus the localizations exhaust the observation experiment in probability.

Proof. For a positive bounded latent g , write $N_t^{n,g} = \mathbb{E}_R[L_T^n g \mid \mathcal{F}_t]$ and $dN_t^{n,g} = a_t^{n,g} dB_t$. Projection of the full integral gives

$$a_t^{n,g} = \sigma^{-1} \mathbb{E}_R[L_t^n g H_n(t, Z) \mid \mathcal{F}_t], \quad |a_t^{n,g}| \leq (M/\sigma) N_t^{n,g}.$$

The same representation with $g = 1$ gives $d\ell_t^n = b_t^n dB_t$ and $b_t^n / \ell_t^n = \theta_t^n$. Put $N^n = N^n, e^{\gamma f}$ and $a^n = a^n, e^{\gamma f}$. Itô's formula in (11.9) gives the martingale coefficient

$$Z_t^n = \gamma^{-1} (a_t^n / N_t^n - b_t^n / \ell_t^n).$$

Replacing dB_t by $dI_t^n + \theta_t^n dt$ reduces its drift to $-\gamma |Z_t^n|^2 / 2$, which proves (11.10). The ratios are bounded by M/σ , so the integrals are well defined. Conversely, exponentiating any bounded solution of (11.10) produces a bounded P_n -martingale with terminal value $e^{\gamma U_T^n}$. Its conditional-expectation representation is unique, and martingale representation then identifies its integrand. This proves uniqueness in the bounded class.

Apply Theorem 10.1 to $g_0 = (e^{\gamma f} - e^{-\gamma}) / (e^\gamma - e^{-\gamma}) \in [0, 1]$. The map taking its conditional expectation to $\gamma^{-1} \log \mathbb{E}[e^{\gamma f} \mid \mathcal{F}_t]$ has Lipschitz constant at most $(e^{2\gamma} - 1)/\gamma$. This proves (11.11).

The initial values of N^n and N agree, since Z has the same marginal. Itô's isometry and the terminal version of (11.5) imply

$$\mathbb{E}_R \int_0^T |a_s^n - a_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^T |b_s^n - b_s|^2 ds \rightarrow 0.$$

Before τ_r^n , all denominators lie in $[1/r, r]$, and the limiting numerators a, b are bounded in absolute value by $(M/\sigma)r$. For example,

$$\left| \frac{a^n}{N^n} - \frac{a}{N} \right| \leq r |a^n - a| + (M/\sigma) r^3 |N^n - N|.$$

The supremum estimates (11.5), followed by integration in time, prove (11.12). The same inequality applies to b^n / ℓ^n .

Finally $e^{-\gamma} \ell^n \leq N^n \leq e^{\gamma} \ell^n$. Each ℓ^n is a stochastic exponential with uniformly bounded integrand θ^n . Its positive and negative moments of the supremum are bounded uniformly in n : write any power as a stochastic exponential times its bounded compensator and apply Doob's inequality to a higher positive power. The same calculation applies to its inverse. Markov's

inequality, the preceding comparison of N^n with ℓ^n , and a union bound prove the exhaustion assertion. This is a localized integrand theorem; it does not assert an unspecified global topology for backward equations with unbounded terminal data or different nonlinear drivers. $\square$

Corollary 11.4 (A changing-filtration stopping consumer). *Let $\psi(t, x) \in [0, 1]$ be measurable and L_ψ -Lipschitz in x , uniformly in t . For a bounded latent f with $|f| \leq 1$, put $M_t^n = \mathbb{E}_{P_n}[f(Z) \mid \mathcal{F}_t]$. Then*

$$\left| \sup_{\tau \leq T} \mathbb{E}_{P_n} \psi(\tau, M_\tau^n) - \sup_{\tau \leq T} \mathbb{E}_P \psi(\tau, M_\tau) \right| \leq \varepsilon_n + L_\psi \sqrt{80\varepsilon_n}.$$

The same statement for the entropic value U has the additional Lipschitz factor $(e^{2\gamma} - 1)/\gamma$ on the square-root term. The suprema range over the common canonical stopping times; attainment is not needed for the estimate.

Proof. For a fixed stopping time, first change the measure while keeping its $[0, 1]$ reward function, at cost at most ε_n . Then compare the two reward functions on the same canonical path under P , using Theorem 10.1, Cauchy–Schwarz and the Lipschitz bound. The estimate is uniform in the stopping time, so it survives both suprema. Equivalence of the observation laws is what allows the same stopping-time class. The entropic assertion follows from the Lipschitz calculation in the preceding theorem. $\square$

11.4. Resource transport is not the posterior coupling. Apply any common finite causal acquisition map to Y and retain a fixed finite mark $w(Z)$. Data processing makes its marked row variations at most ε_n , uniformly over a declared finite feedback family. If the source and target have respective bounds α_n, β_n , Theorem 2.2 gives same-budget deficiency perturbation at most $\alpha_n + \beta_n$, and Proposition 4.4 transports a fixed reset audit. A subsequent consumer of width S composed with a simulator of width K has width KS , with the stated selector price. None of these executable claims uses the latent-preserving maximal coupling. That coupling belongs to the proof of conditional-process convergence, not to the finite-state machine.

These results provide a direct changing-trajectory, likelihood, Girsanov and bounded entropic-value input for a conditional-law theory. They do not assert the strict-dual, rigidity, covariant-form or full kinetic conclusions of a larger program without their separate analytic proofs.

Example 11.5 (Why an amplitude bound is not a tightness argument). Let ξ_1, ξ_2 be independent symmetric signs, and, for $n \geq 5$, let $\mathcal{G}_t^n$ reveal ξ_1 at $1/2 - 2/n$ and ξ_2 at $1/2 - 1/n$. On $[0, 1]$ the bounded martingale

$$M_t^n = \mathbb{E}[\xi_1 + \xi_2 \mid \mathcal{G}_t^n]$$

converges at every finite collection of fixed times to the martingale $M_t = \mathbf{1}_{[1/2, 1]}(t)(\xi_1 + \xi_2)$. Nevertheless its laws are not tight in the Skorokhod J_1 topology. On the event $\xi_2 = -\xi_1$, of probability $1/2$, its path has an excursion of height one on an interval of length $1/n$ and is zero outside that interval. The three-time oscillation modulus of such a path is one at every scale greater than $2/n$. This violates the necessary J_1 tightness modulus criterion. All maximal moments are uniformly bounded by 2^p .

This example concerns the inference from finite-dimensional convergence and a maximal amplitude estimate alone. It does not claim that every possible stronger conditional-kernel hypothesis fails. The uniform estimates in Theorem 11.1 remove the need for that inference in the observation class treated here.

12. GLOBAL TRANSPORT OF THE BACKWARD EQUATION

We retain exactly the common-preparation, bounded-signal and positive-noise hypotheses of Section 11. In particular the canonical joint laws P_n, P are equivalent and have the same latent marginal ν . All comparisons on a single observation path use common measurable versions of the

conditional expectations. The following estimate avoids division by a small observation density in the final bound.

Lemma 12.1 (An integrated posterior comparison). *Let P, Q be equivalent laws on a standard Borel product carrying a latent variable Z and an observation Y , and let $\|P - Q\|_{\text{TV}} \leq \epsilon$. If $|u(Z)| \leq M$ and $\mathcal{F}$ is an observation sub- σ -field, then, under either $S = P$ or $S = Q$,*

$$\mathbb{E}_S[\mathbb{E}_Q[u \mid \mathcal{F}] - \mathbb{E}_P[u \mid \mathcal{F}]] \leq 4M\epsilon, \quad (12.1)$$

$$\mathbb{E}_S[\mathbb{E}_Q[u \mid \mathcal{F}] - \mathbb{E}_P[u \mid \mathcal{F}]]^2 \leq 8M^2\epsilon. \quad (12.2)$$

A common latent marginal is not needed for these two inequalities.

Proof. Use a dominating finite measure on $\mathcal{F}$. Write p, q for the densities of the restricted probability laws and a, b for the densities of the signed measures $\mathbb{E}_P[u\mathbf{1}_A], \mathbb{E}_Q[u\mathbf{1}_A]$ on $\mathcal{F}$. Their conditional means are $m_P = a/p$ and $m_Q = b/q$. Outside null sets, $|m_P|, |m_Q| \leq M$, and

$$p(m_Q - m_P) = m_Q(p - q) + (b - a).$$

The integral of $|p - q|$ is at most 2ϵ , by contraction of total variation. The integral of $|b - a|$ is at most $2M\epsilon$, since multiplication by u followed by restriction cannot increase signed variation by more than M . Integrating the displayed identity proves (12.1) under P . Interchanging the two laws proves it under Q . The squared bound follows from $|m_Q - m_P| \leq 2M$. $\square$

Let $|f| \leq 1$, $\gamma > 0$, and use the notation θ^n, U^n, Z^n, I^n of Proposition 11.2 and Theorem 11.3. Here Z^n denotes the scalar backward integrand, whereas the unadorned latent argument of H_n is the preparation variable. Define the deterministic bounds

$$A_n = \sigma^{-2}\{2\eta_n^2 + 16M^2T\epsilon_n\}, \quad (12.3)$$

$$C_n = \frac{2(1 + e^{4\gamma})}{\gamma^2} A_n, \quad (12.4)$$

where $\epsilon_n = \min\{1, \eta_n/(2\sigma)\}$ as before.

Theorem 12.2 (Global innovation and entropic-integrand transport). *Under the hypotheses of Theorem 11.1, under either physical law $S = P$ or $S = P_n$ on the same canonical path,*

$$\mathbb{E}_S \int_0^T |\theta_t^n - \theta_t|^2 dt \leq A_n, \quad \mathbb{E}_S \int_0^T |Z_t^n - Z_t|^2 dt \leq C_n. \quad (12.5)$$

No stopping at a density level is required. If $\eta_n \rightarrow 0$, both integrals also converge to zero in expectation under the reference law R .

On the single latent-preserving coupling of Theorem 11.1, evaluate each process on its own observation path, denoting these evaluations by a bar. Then

$$\mathbb{E} \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt \leq A_n + 4M^2T\epsilon_n/\sigma^2, \quad (12.6)$$

$$\mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt \leq C_n + 16M^2T\epsilon_n/(\gamma^2\sigma^2). \quad (12.7)$$

If $\eta_n \rightarrow 0$, then $\bar{I}^n \rightarrow \bar{I}$ uniformly in probability. Writing $\bar{J}_t^n = \int_0^t \bar{Z}_s^n d\bar{I}_s^n$ for the stochastic integral in its own observation filtration, and $\bar{J}$ for its limit, one has

$$\mathbb{E} \sup_{t \leq T} |\bar{J}_t^n - \bar{J}_t|^2 \leq 2\mathbb{E} \sup_{t \leq T} |\bar{U}_t^n - \bar{U}_t|^2 + \frac{8M^2T}{\sigma^2} \mathbb{E} \int_0^T |\bar{Z}_t^n - \bar{Z}_t|^2 dt. \quad (12.8)$$

The first term on the right has the explicit bound in (11.11). Thus the full bounded entropic backward representation, including its martingale integral, is transported globally. The coupling

need not make either innovation Brownian in the joined filtration; each integral is defined under its own marginal law.

Proof. For each deterministic t , under P split

$$\sigma(\theta_t^n - \theta_t) = \mathbb{E}_{P_n}[H_n(t, Z) \mid \mathcal{F}_t] - \mathbb{E}_P[H_n(t, Z) \mid \mathcal{F}_t] + \mathbb{E}_P[H_n(t, Z) - H(t, Z) \mid \mathcal{F}_t].$$

Lemma 12.1, conditional Jensen and $(x + y)^2 \leq 2x^2 + 2y^2$ bound its expected square by $16M^2\varepsilon_n + 2\mathbb{E}_\nu|H_n(t, Z) - H(t, Z)|^2$. Integration gives the first bound in (12.5) under P . Under P_n use the same decomposition with the signal-difference term conditioned under P_n and the law-change term applied to H . The common latent marginal gives the identical bound. Jointly measurable conditional versions, and predictable versions equal for dt -almost every time, give the same integrated statements.

Put $g = e^{\gamma f}$ and $c = \mathbb{E}_\nu g$. Tilt the two full joint laws by $g(Z)/c$, writing P^g, P_n^g . They still have the same latent marginal $\nu_g = g\nu/c$, and their observation signals remain H, H_n . Since $e^{-\gamma} \leq g, c \leq e^\gamma$,

$$\|P_n^g - P^g\|_{\text{TV}} \leq e^{2\gamma}\varepsilon_n, \quad \eta_{n,g}^2 \leq e^{2\gamma}\eta_n^2.$$

Moreover, the reference martingale coefficient $a^{n,g}$ from the proof of Theorem 11.3 satisfies

$$\alpha_t^n := \frac{a_t^{n,g}}{N_t^{n,g}} = \sigma^{-1} \mathbb{E}_{P_n^g}[H_n(t, Z) \mid \mathcal{F}_t].$$

Apply the preceding innovation argument to these tilted laws. It bounds the squared integral of $\alpha^n - \alpha$ by $e^{2\gamma}A_n$ under either tilted physical law. For an observation-measurable nonnegative random variable, passage back to the corresponding original physical law costs at most $e^{2\gamma}$: the full density c/g is bounded by this constant, and so is its observation projection. Consequently

$$\mathbb{E}_S \int_0^T |\alpha_t^n - \alpha_t|^2 dt \leq e^{4\gamma}A_n.$$

The identity $Z^n = (\alpha^n - \theta^n)/\gamma$ proves the second bound in (12.5).

The pointwise bounds, valid for all models, are

$$|\theta^n|, |\alpha^n| \leq M/\sigma, \quad |Z^n| \leq 2M/(\gamma\sigma). \quad (12.9)$$

The squared time integrals are thus uniformly bounded random variables. Convergence in P -probability, followed by absolute continuity of the fixed R with respect to P , gives convergence in R -probability. Boundedness then gives convergence of their R -expectations. This argument establishes global reference convergence without claiming a new explicit reference rate.

For the coupling, let $E_n = \{Y^n = Y\}$, so that $\mathbb{P}(E_n^c) \leq \varepsilon_n$. On E_n both processes are functions of the same observation path. The restricted expectation of their squared difference is at most the corresponding unrestricted expectation under the limiting marginal P . On E_n^c apply (12.9) and integrate for time T . This proves (12.6)–(12.7).

On E_n , the Brownian reference coordinates $Y^n/\sigma, Y/\sigma$ agree, so

$$\sup_{t \leq T} |\bar{I}_t^n - \bar{I}_t|^2 \leq T \int_0^T |\bar{\theta}_t^n - \bar{\theta}_t|^2 dt.$$

Markov's inequality and $\mathbb{P}(E_n^c) \rightarrow 0$ prove uniform convergence in probability. This assertion does not require any moment estimate on the observation difference on the exceptional event.

Finally, the backward equation already proved gives, under each marginal,

$$\bar{J}_t^n = \bar{U}_t^n - U_0 + \frac{\gamma}{2} \int_0^t |\bar{Z}_s^n|^2 ds.$$

The common initial value is $U_0 = \gamma^{-1} \log \mathbb{E}_\nu g$. By (12.9),

$$||\bar{Z}_s^n|^2 - |\bar{Z}_s|^2| \leq \frac{4M}{\gamma\sigma} |\bar{Z}_s^n - \bar{Z}_s|.$$

Subtract the two integral identities, use Cauchy–Schwarz in time and $(x + y)^2 \leq 2x^2 + 2y^2$, and take expectations. This gives (12.8). The integral identity also defines compatible measurable versions on the coupling, without asserting a martingale property for its joint filtration. $\square$

The theorem strengthens the localized conclusion of Theorem 11.3 without changing its terminal condition or its driver. The terminal condition remains the observation-measurable conditional certainty equivalent, not the unobserved latent payoff. Positive fixed noise, bounded signals, and a common preparation are used in the proof. A singular zero-noise limit, an unbounded terminal condition, or a different nonlinear driver would require additional arguments.

13. A UNIFORM CERTIFICATE THROUGH A MOVING COLLISION

We now use all three parts of the argument on one family. The finite marked reference law and the uniform scattering geometry are those of Theorem 9.1 and Lemma 9.2. They are proved above, including the no-collision control. We use a continuous observation of the same detector so that finite reports and conditional processes are generated by one experiment.

Let z denote a preparation in the boxes of (9.3), and let ν be that prepared law. The sign of the impact orientation is $2B - 1$, and the conserved center-of-mass longitudinal sign is the actual mark W . Their correlation is prepared: $\mathbb{P}(B = b, W = w) = 5/16$ for $b = w$ and $3/16$ otherwise. The contact diameter d ranges over $I = [9/10, 11/10]$. Let $X_d(t, z)$ be the two-particle phase path, with outgoing velocities used at the collision, and define

$$H_d(t, z) = h(v_{1,y}^d(t, z)), \quad Y_t^d = \int_0^t H_d(s, z) ds + \sigma B_t^d, \quad 0 < \sigma \leq 1/24.$$

The Brownian noise is independent of the preparation. The source experiment has the same physical preparation and mark but constant reports. The target reports at times 2 and 3 are

$$Z_1^d = \mathbf{1}\{2(Y_2^d - Y_{3/2}^d) > 0\}, \quad Z_2^d = A\mathbf{1}\{2(Y_3^d - Y_{5/2}^d) > 0\}. \quad (13.1)$$

The acquisition gate $A \in \{0, 1\}$ is chosen after Z_1^d ; it does not act on the mechanical trajectory. The continuous path is the auxiliary observation experiment for conditional consumers; the finite simulator is not given this path or an uncharged buffer for computing its increments. The target sensor implements the declared window acquisition.

Lemma 13.1 (Quantitative change of the microscopic path). *For all preparations in the specified boxes, the collision time $t_c(d)$ and outgoing tagged velocities obey*

$$|t_c(d') - t_c(d)| \leq |d' - d|, \quad |v_i^+(d') - v_i^+(d)| \leq 30|d' - d| \quad (i = 1, 2).$$

Writing $\Delta = |d' - d|$, one has

$$\sup_z \int_0^3 |H_{d'}(t, z) - H_d(t, z)|^2 dt \leq \Delta + 2700\Delta^2. \quad (13.2)$$

The phase paths converge uniformly over the preparation in the Skorokhod J_1 metric as $d' \rightarrow d$. More precisely, for the product metric taking the maximum of the per-particle Euclidean position and velocity distances, a time change aligning the collision times gives J_1 distance at most 100Δ .

Proof. Write r and g for the incoming relative position and velocity, and $n(d) = (r + t_c(d)g)/d$ for the contact normal. The collision root satisfies $|r + t_c g| = d$. The nongrazing estimate from Lemma 9.2 is $-g \cdot n > 1443/1000$. Differentiating the root with respect to d gives

$$t'_c(d) = \frac{1}{g \cdot n(d)}, \quad |t'_c(d)| < 1.$$

The incoming bound is $|g| < 3$, so

$$|n'(d)| \leq \frac{|t'_c(d)| |g| + 1}{d} < 5.$$

Since $v_1^+ = V + g/2 - (g \cdot n)n$ and $v_2^+ = V - g/2 + (g \cdot n)n$, differentiation bounds either velocity derivative by $2|g| |n'| < 30$.

Before collision the detector is zero, and after collision it is a constant of absolute value at most one. Its Lipschitz constant is one. The two signals therefore differ on the interval between the two collision times by at most one and afterwards by at most 30Δ . The intervening interval has length at most Δ , and the whole observation lasts three units. This proves (13.2).

Choose the increasing piecewise affine time change fixing 0, 3 and sending $t_c(d)$ to $t_c(d')$. Its uniform displacement is at most Δ . Aligned velocities differ by at most 30Δ , with the same outgoing convention at the jump. Each particle's speed is below four and its velocity jump has norm below three. Without time change, the position difference is bounded by $3\Delta + 3(30\Delta) = 93\Delta$, by integrating the velocity difference before and after the two jump times. Aligning time adds at most 4Δ . The asserted bound follows, with a harmless rounding to 100Δ . $\square$

Theorem 13.2 (Microscopic-to-audit and conditional-law composition). *For the single family (13.1), all of the following hold.*

(a) *At the same width-one private register,*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \quad (r \geq 1), \quad \delta_2^h < 13/100,$$

uniformly for $d \in I$ and $0 < \sigma \leq 1/24$. When $d \leq 2/5$ the detector is zero on $[0, 3]$, and all three finite-experiment deficiencies are zero.

(b) *There is one explicit rational reset-audit rule with one all-on training row, eight marked outcomes and $n = 2800$ training runs, whose expected validation score exceeds $2/5$ against every private width-one simulator throughout that same parameter region. In detail, for counts C choose an event $A_C \subseteq \{0, 1\}^3$ maximizing*

$$Q_*(A_C) - C(A_C)/2800, \tag{13.3}$$

with lexicographic tie breaking. Here Q_ is the rational marked reference law of (B, B, W) . The independent validation uses the actual physical target, not Q_* . This finite rule has at most $\binom{2807}{7}$ training-count states at any scheduled epoch; a conservative complete auditor bound is $8(256 + 1)\binom{2807}{7}$. The state cost belongs to the auditor. The certificate is an expected-score inequality, not a confidence claim about one realized score.*

(c) *Fix $\sigma > 0$ and let $d_n, d \in I$ with $d_n \rightarrow d$. Set*

$$\epsilon_n = \min \left\{ 1, \frac{\sqrt{|d_n - d| + 2700|d_n - d|^2}}{2\sigma} \right\}. \tag{13.4}$$

The joint preparation-continuous-observation laws differ by at most ϵ_n . Every finite resource deficiency above changes by at most ϵ_n , and a fixed audit changes by at most ϵ_n because its source behavior is unchanged. On a latent-preserving coupling, $X_{d_n} \rightarrow X_d$ in J_1 and each bounded posterior converges in supremum L^2 with bound $84\epsilon_n$. The full and projected likelihoods converge in reference supremum L^2 , and jointly in probability with the microscopic paths on that coupling.

The projected stochastic exponentials, bounded entropic backward values, localized integrands and stopping consumers have the conclusions of Proposition 11.2, Theorem 11.3, and Corollary 11.4.

(d) *On compact parameter sets with $\sigma \geq \sigma_0 > 0$, the finite marked probabilities have uniform rational enclosures. The observable chart, strict same-budget polynomial verification and rational upper tables are therefore effective in the exact sense of Propositions 8.1 and 7.6.*

Proof. Every collision occurs before time $3/2$. Conditional on preparation, the two normalized window increments in (13.1) are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. The outgoing detector has sign $2B - 1$ and absolute value greater than $1/4$. Coupling each noisy report to B gives total marked variation less than $1/300$ from the exact finite marked experiment, uniformly for every acquisition gate. The finite calculation of Theorem 9.1 gives

$$d_* = \sqrt{5/2} - 9/8, \quad \delta_*^p = \delta_*^v = d_*, \quad \delta_{*,2}^h = 1/8.$$

Thus private and visible values exceed $d_* - 1/300 > 9/20$, and the hidden value is less than $1/8 + 1/300 < 13/100$. The two-valued hidden witness stores its selector, not an observed mark. In the no-collision regime both window reports before gating are independent fair noise, independent of W . Fresh private coins reproduce them exactly, including feedback gates. This proves (a).

For (b), use only the all-on feedback row. Any private width-one simulator in this row emits independent bits with probabilities p, q , independently of the actual W . The calculation in Theorem 9.1 already gives minimum variation d_* for this row. Its alphabet (z_1, z_2, W) has eight elements, hence $D = 7$. The explicit empirical audit in Theorem 4.1, trained using Q_* , has expected ideal score at least

$$d_* - \sqrt{7/2800} = d_* - 1/20.$$

The candidate training and validation laws do not change when the target is replaced by the physical target. Conditional on any selected event, its target probability changes by less than $1/300$. Therefore the actual score is strictly greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (13.5)$$

For completeness, the last strict inequality is the positive-root comparison $\sqrt{5/2} > 947/600$; its squared margin is $5/2 - (947/600)^2 = 3191/360000 > 0$. All probabilities in Q_* are 0, $3/16$ or $5/16$, so (13.3) is a finite rational rule independent of d, σ, p, q . At training epoch k the number of eight-category count vectors is $\binom{k+7}{7} \leq \binom{2807}{7}$. A buffer of eight states records the current two reports and terminal mark; 256 event masks and an idle symbol suffice during event selection and validation. The conservative product bound follows. The clock and the source-reset interface are the declared resources. In particular the proof does not enumerate or claim to store a small complete coefficient table.

For (c), Lemma 13.1 gives the signal-distance hypothesis of Theorem 11.1, with $T = 3$ and $M = 1$. It also supplies its microscopic J_1 hypothesis. That theorem gives (13.4) and all the joint process assertions. The finite acquisition is a common measurable causal map of the continuous observation, so marked variation contracts for every gate. The source is blind and independent of d . Consequently its executable candidate behaviors are identical across d , and Proposition 4.4 has $\alpha = 0$ and $\beta = \epsilon_n$. The same same-budget comparison applies in each visible fiber and in each hidden mixture. The exponential and backward claims now follow from the identified projected density and the explicitly bounded entropic consumer. This is where the finite certificate and the conditional-law theorem meet the same microscopic approximation.

Finally, the collision root is uniformly separated from grazing, its formula is algebraic on the rational preparation boxes, and the normalized window noise has variance $2\sigma^2$ bounded away from zero on the stated compact sets. The box subdivision and Gaussian-tail enclosure argument

in Proposition 9.4 therefore applies with that variance. It returns rational enclosures with a computable uniform error. The rational quotient algorithm and same-budget strict certificate procedure then prove (d). Uniformity as $\sigma \downarrow 0$ is not asserted. $\square$

The all-on audit in part (b) is a single witness for the lower bound in the full feedback problem: a maximum over feedback trees includes this row. The visible lower bound is instead the exact fiber theorem; no claim is made that an auditor with a seed-independent state bound can store an arbitrarily large visible-seed transcript. This separates the two resource budgets rather than silently moving cost from one to the other.

The finite certificate is also preserved under an additional rational presentation error ρ : its lower score decreases by at most ρ when only the target is changed, and by the reset-cost estimate when the source is changed. Thus the certificate is an effective consumer of the physical presentation theorem, not an isolated exact finite example.

Theorem 13.2 and the preceding results prove Theorem A.1. The C2-type consequence is a joint changing-trajectory/likelihood/posterior theorem with an identified stochastic exponential and a fully specified bounded entropic backward consumer. Nothing in that conclusion identifies geometric similarity with kinetic scaling, or supplies particle-uniform recollision estimates by renaming this fixed-particle model.

14. A FINITE-REGISTER CERTIFICATE TRANSPORTED THROUGH THE MICROSCOPIC LIMIT

The general empirical audit in Theorem 4.1 treats every coordinate of a marked row as unknown. The private continuation constraint in the collision experiment forces a smaller family. Exploiting this constraint yields a much smaller executable auditor, without granting it knowledge of the candidate law.

In the all-on row of Section 13, every private width-one candidate emits two independent bits with parameters p, q , independently of the fair actual mark W . Put

$$a = (1 - p)(1 - q), \quad c = pq.$$

The ideal target Q_* is supported on 000, 001, 110, 111, with respective masses $5/16, 3/16, 3/16, 5/16$. The candidate masses on these atoms are $a/2, a/2, c/2, c/2$. Every other atom has target mass zero and cannot improve the maximizing event for $Q_* - b$.

Lemma 14.1 (Two-count audit and its exact register implementation). *In $n \geq 1$ independent all-on candidate trials let C_0, C_1 count report pairs 00, 11, respectively, ignoring the training mark. On the four supported atoms use the empirical masses $C_0/(2n), C_0/(2n), C_1/(2n), C_1/(2n)$, and select exactly those atoms whose target mass is strictly greater than this empirical mass. Call the resulting event A_C . For every $p, q \in [0, 1]$,*

$$\mathbb{E}\{Q_*(A_C) - b(A_C)\} \geq \|Q_* - b\|_{TV} - n^{-1/2}. \quad (14.1)$$

For $n = 400$, the rule is implemented by the two truncated counts $\min(C_0, 250), \min(C_1, 250)$, with comparisons to 150 and 250. Including execution of complete transcripts and the two independent validation trials, it has a deterministic streaming realization with at most 504008 retained states at any scheduled epoch. Thus a nineteen-bit register suffices. The schedule clock is free; no other history store is free. The table description and transition evaluation cost are not asserted to be bounded by nineteen bits.

Proof. On any supported atom let $x = Q_*(w) - b(w)$ and let $\hat{x}$ be the empirical discrepancy. If the sign rule selects a different event from an exact maximizing event, the lost contribution is at most $|x - \hat{x}|$. On unsupported atoms both rules can omit the atom. Summing gives regret at most

$$|C_0/n - a| + |C_1/n - c|.$$

The two counts need not be independent. Their marginal binomial variances and Cauchy–Schwarz give expected regret at most

$$\frac{\sqrt{a(1-a)} + \sqrt{c(1-c)}}{\sqrt{n}} \leq \frac{1}{\sqrt{n}}.$$

This proves (14.1).

For the 00 pair, select both marked atoms when $C_0 < 150$, only the atom of target mass 5/16 when $150 \leq C_0 < 250$, and neither when $C_0 \geq 250$. The same rule applies to 11, with the other mark being the atom of larger target mass. At equality the zero empirical discrepancy is excluded. Increasing a count beyond 250 cannot change the final choice. Two registers with 251 possibilities each therefore implement the rule, with at most nine possible final event masks.

A buffer with eight possibilities is sufficient to execute and record a completed two-bit-plus-mark transcript. During training the pair of truncated counts and this buffer use at most $8 \cdot 251^2 = 504008$ states. After training, discard the counts and retain the event mask. During validation at most nine masks, one target-event indicator with two values, and the same eight-state transcript buffer are needed, a total of 144 states. The final signed score takes three values. Since the epoch is known, labels can be reused between these phases; their state counts need not be multiplied. The update maps and threshold comparisons are parameter-independent deterministic maps. They are thus stochastic rows with entries zero or one, or equivalently a compatible continuation cover by Theorem 3.2. Fresh sampling randomness belongs to the specified reset experiments, not to an uncharged auditor register. Finally $504008 < 2^{19}$. $\square$

Theorem 14.2 (Finite-register microscopic certification and global transport). *Consider the marked hard-sphere family of Theorem 13.2, with $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/24$. There is one rational streaming audit, independent of d, σ, p, q , using 400 candidate training resets and one independent candidate and target validation trial, whose expected signed score is strictly greater than $2/5$ against every private width-one candidate. It uses at most 504008 retained auditor states. The candidate remains private width one; the auditor’s register is a distinct, charged resource. The physical deficiencies still satisfy*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \ (r \geq 1), \quad \delta_2^h < 13/100.$$

For $d \leq 2/5$ all three finite-experiment deficiencies vanish.

Fix a positive σ and a sequence $d_k \rightarrow d$ within the collision interval. Put

$$\rho_k = |d_k - d| + 2700|d_k - d|^2, \quad \epsilon_k = \min\{1, \sqrt{\rho_k}/(2\sigma)\}.$$

The same 400-reset rule changes its expected score by at most ϵ_k under this perturbation. At the same time the microscopic paths, full and projected likelihoods, posteriors, and bounded entropic values have the joint convergence of Theorem 13.2. Their innovations, backward integrands and backward martingale integrals additionally have all the global conclusions of Theorem 12.2, with $T = 3$, $M = 1$, $\eta_k^2 \leq \rho_k$, and $\epsilon_k \leq \epsilon_k$. On compact parameter sets bounded away from zero noise, the same audit has the effective probability-enclosure and rational-verification properties of Theorem 13.2(d).

Proof. The private width-one continuation restriction gives precisely the candidate family used in Lemma 14.1; the actual mark is not available to its transition rows. The exact finite separation in Theorem 9.1 gives

$$\min_{p,q} \|Q_* - b(p, q)\|_{\text{TV}} = d_* := \sqrt{5/2} - 9/8.$$

The scattering and observation estimates proved in Theorem 13.2 put the physical marked target within $1/300$ of Q_* , uniformly in all stated parameters. Train the two-count rule against the

fixed rational Q_* , but use the actual physical target for validation. Each selected event's target probability changes by less than $1/300$. Hence its actual expected score is greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (14.2)$$

The last strict inequality is exact: it is equivalent to $\sqrt{5/2} > 947/600$, whose squared margin is $3191/360000$. The rule's executable memory bound was proved in Lemma 14.1. Its total trial count is 402, not 400 once validation is included. The private, hidden and visible deficiency bounds and the no-collision control are inherited from the same physical theorem, rather than inferred from an auditor whose signature was changed.

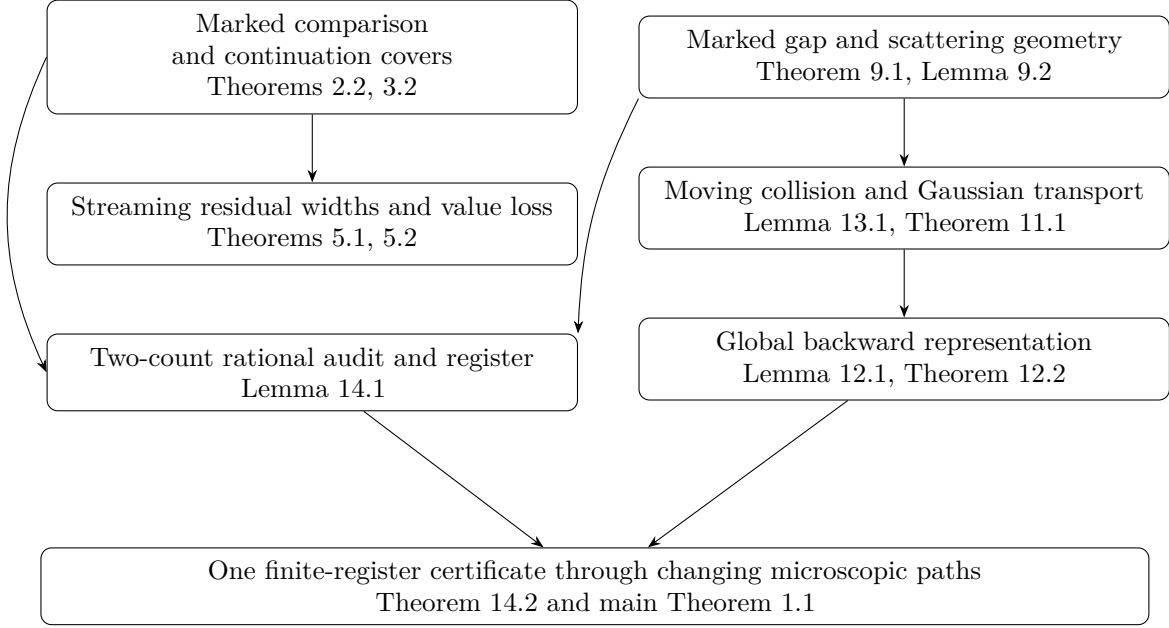
For changing d_k , Lemma 13.1 supplies the signal modulus ρ_k and the pathwise J_1 bound $100|d_k - d|$. Theorem 11.1 gives the joint marked variation bound ϵ_k and the likelihood/posterior convergence. The candidate source is blind and its behavior does not depend on d . Thus training laws and candidate validation laws are unchanged, and Proposition 4.4 has $\alpha = 0$ and $\beta = \epsilon_k$. There is no factor of 400 in this transport cost. The same preparation and signal bounds verify every hypothesis of Theorem 12.2, proving the global backward assertions. Finally the audit uses rational thresholds and finitely many deterministic row maps. The positive-noise enclosure theorem therefore applies directly; an additional target-presentation error ζ reduces its score by at most ζ . $\square$

The two-count construction is specific to the private behavior family. A hidden mixture can destroy the independence of its two reports, and its mark relation must still be checked. The bound is therefore not asserted against the hidden class. The visible lower bound has its separate fiberwise proof; an arbitrarily large visible alphabet is not silently stored in the nineteen-bit auditor. The certificate concerns expected score, not a confidence statement about a single realized validation pair. Repeating validation for statistical confidence would incur an additional sample and storage budget.

The theorem uses a single mechanism: the continuation restriction fixes the feasible candidate laws and permits the compressed audit; scattering supplies its marked separation; common-preparation variation transports the same executable test and the global conditional-law consumer through a change in the microscopic trajectory. The collision transduces a prepared impact variable into a signal. It does not create the prepared correlation between that variable and the mark, and it is not a particle-number limit.

APPENDIX A. THE CANONICAL PROOF CHAIN

The diagram separates exact realization and testing complexity from the physical verification of the common-preparation hypotheses. Both feed the same final certification theorem. A dependency arrow means that the stated proof uses the source result, not that the source is the only possible method of proof.



The binary optimal-value theorem is the classification and lower-bound branch of the main result; it is not needed to prove the two-count upper bound. The latter uses the deterministic continuation realization and the specific private behavior geometry directly. The finite-reset minimax and its moment support bound describe the more general audit framework. The exact nonlinear witness, revealed-behavior hierarchy and rational chart retain their separate information conventions and full proofs.

The comparison theorem, positive continuation realization, erasure-cut law, reset minimax and sampling obstruction, chart, nonlinear witness, marked scattering gap, posterior estimate, moving-collision estimate and localized entropic theorem are inherited inputs. The stochastic residual error inequality, exact binary audit-width threshold, compressed certificate and global backward-integrand bounds are the additions of this revision. The preceding introduction states which proof mechanisms are classical. None of the label, source, build or diagnostic checks accompanying this article is an independent verification of the analytic proofs.

The actual mark is never exposed to the online simulator. The reset auditor has a separate completed-trial interface. A revealed-behavior test sees an exact law. The common-latent maximal coupling may use an entire observation path and is not an executable causal simulator. These privileges are not interchangeable.

A.1. Inherited baseline. The following summary is retained from the predecessor. The present article strengthens its memory and backward-integrand conclusions in Theorem 1.1; the earlier 2800-trial audit remains valid.

Theorem A.1 (Finite-resource certification and microscopic transport). *The following assertions hold with the conventions just described.*

(i) *Finite causal kernels have a register realization with a prescribed width profile exactly when their normalized continuation laws admit compatible finitely generated polytopes of those widths. At an erasure cut carrying a latent letter of law π , the optimal error with K retained states is*

$$1 - \sum_{i=1}^{\min(K,N)} \pi_{(i)}, \quad \pi_{(1)} \geq \cdots \geq \pi_{(N)}.$$

Independent hidden or visible selectors do not improve this cut value.

(ii) *The reset-audit value is attained, has a finite-dimensional moment minimax representation on the original B , and satisfies*

$$0 \leq \delta(B, Q) - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}, \quad D = \sum_j (|\Omega_j| - 1).$$

The exponent $1/2$ cannot be improved uniformly over compact candidate classes. This assertion is about reset samples, not about the minimal degree of unrestricted polynomial tests.

(iii) *Suppose two prepared microscopic observation models have a common preparation variable, bounded signals H, H' on $[0, T]$, and the same positive Gaussian noise level σ . Set*

$$\eta^2 = \mathbb{E} \int_0^T |H - H'|^2 dt, \quad \varepsilon = \min\{1, \eta/(2\sigma)\}.$$

Their joint marked observation laws differ by at most ε . This estimate transports resource spectra at their stated budgets and transports each fixed finite audit with its explicitly charged reset cost. It also gives whole-process posterior convergence. The unprojected and projected likelihoods converge in the supremum L^2 norm under the common Wiener reference when $\eta \rightarrow 0$. For bounded latent payoffs, the associated entropic backward equations converge in value, and their integrands converge after the stated denominator localization.

(iv) *In the two-sphere model of Theorem 13.2, one rationally specified audit with 2,800 training runs has expected discrepancy strictly greater than $2/5$ against every private width-one simulator, uniformly for $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. A hidden two-valued selector has error less than $13/100$; every visible-selector width-one error exceeds $9/20$. For $d_n \rightarrow d$ the microscopic paths, likelihoods and posterior processes converge in the precise senses of that theorem, and the same audit remains valid. For $d \leq 2/5$ all three deficiencies vanish.*

REFERENCES

- [1] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272. doi:10.1214/aoms/1177729032.
- [2] L. Le Cam, Sufficiency and approximate sufficiency, *Ann. Math. Statist.* **35** (1964), 1419–1455. doi:10.1214/aoms/1177700372.
- [3] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28. doi:10.1524/strm.2002.20.14.1.
- [4] A. Nayyar, A. Mahajan and D. Teneketzis, Decentralized stochastic control with partial history sharing: a common information approach, *IEEE Trans. Automat. Control* **58** (2013), 1644–1658. doi:10.1109/TAC.2013.2239000. The theorem numbers cited here refer to the author preprint, arXiv:1209.1695.
- [5] H. S. Witsenhausen, On the structure of real-time source coders, *Bell System Tech. J.* **58** (1979), 1437–1451. doi:10.1002/j.1538-7305.1979.tb02263.x.
- [6] I. Gallagher, L. Saint-Raymond and B. Texier, *From Newton to Boltzmann: Hard Spheres and Short-Range Potentials*, Zurich Lectures in Advanced Mathematics, European Mathematical Society, 2014. The flow and boundary discussion used here is in Sections 4.1–4.3 of arXiv:1208.5753v2.
- [7] C. Widder, J. Zimmer and T. Schilling, On the generalized Langevin equation and the Mori projection operator technique, preprint (2025), revised 2026. arXiv:2503.20457v6.
- [8] C. Widder and T. Schilling, Generalised Langevin dynamics: significance and limitations of the projection operator formalism, preprint (2026). arXiv:2604.20453v2.
- [9] H. F. Trotter, Approximation of semi-groups of operators, *Pacific J. Math.* **8** (1958), 887–919. doi:10.2140/pjm.1958.8.887. Theorem 5.2, p. 902, is the approximation theorem cited here.
- [10] H. Weisshaupt, LeCam’s randomization criterion in the setting of locally convex spaces without lattice structure, EURANDOM report 2002–017, 22 April 2002. Lemma 1 and Theorem 5, pp. 14–16. Author report.
- [11] S. Basu, Algorithms in real algebraic geometry: a survey, author preprint, arXiv:1409.1534 (2014). Published in *Panoramas et Synthèses* **51** (2017), 107–153. Theorem 2.1 and Section 2.1 in the author preprint are used here.
- [12] M. C. Paull and S. H. Unger, Minimizing the number of states in incompletely specified sequential switching functions, *IRE Trans. Electron. Comput.* **EC-8** (1959), 356–367. Publisher record.
- [13] E. Torgersen, *Comparison of Statistical Experiments*, Encyclopedia of Mathematics and its Applications 36, Cambridge University Press, Cambridge, 1991.

- [14] N. Saldi, A topology for team policies and existence of optimal team policies in stochastic team theory, preprint (2017), arXiv:1711.00639.
- [15] S. Yüksel and N. Saldi, Convex analysis in decentralized stochastic control, strategic measures and optimal solutions, author preprint arXiv:1609.07685v2.
- [16] A. Nerode, Linear automaton transformations, *Proc. Amer. Math. Soc.* **9** (1958), 541–544. doi:10.1090/S0002-9939-1958-0135681-9.
- [17] S. Kiefer, Notes on equivalence and minimization of weighted automata, preprint (2020), arXiv:2009.01217v1.
- [18] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176. doi:10.2140/pjm.1958.8.171.
- [19] H. Weisshaupt, A generalization of Le Cam’s randomization criterion, *Studia Sci. Math. Hungar.* **43** (2006), 443–486. doi:10.1556/SScMath.43.2006.4.3. Theorem numbers refer to the author’s 39-page preprint, Randomization.pdf.
- [20] E. de Klerk and M. Laurent, Error bounds for some semidefinite programming approaches to polynomial minimization on the hypercube, *SIAM J. Optim.* **20** (2010), 3104–3120. doi:10.1137/100790835.
- [21] M. Laurent and L. Slot, An effective version of Schmüdgen’s Positivstellensatz for the hypercube, arXiv:2109.09528v2, 2023.
- [22] E. de Klerk and J. C. Vera, The link between 1-norm approximation and effective Positivstellensätze for the hypercube, *Numer. Algebra Control Optim.* **16** (2026), 84–104. doi:10.3934/naco.2024060. Theorem numbers refer to arXiv:2404.04190v2.
- [23] M. Fliess, Séries rationnelles positives et processus stochastiques, *Ann. Inst. H. Poincaré Sect. B* **11** (1975), 1–21. Original article. Theorems 3.1 and 5.4 and Corollary 2, pp. 8–9 and 18–19.

GENERAL THETA FOUNDATIONS I: CONTINUATION, TESTING, AND MICROSCOPIC TRANSPORT

QIAN QI

ABSTRACT. We study marked causal experiments with a fixed persistent register and separately charged hidden or visible randomization. We characterize finite registers by compatible continuation polytopes and determine an exact memory–error law at an erasure cut. On the original nonconvex behavior image, a finite reset-audit game has an attained minimax representation and error at most $\sqrt{D/n}$, where D counts independent observable probabilities. A binary family gives a matching uniform sampling exponent. We then prove transport of these certificates together with likelihoods, posterior processes, and bounded entropic backward equations under Gaussian observation of changing microscopic trajectories. For two hard spheres this yields one explicit finite-data certificate, uniform on a contact-diameter interval, separating private and visible width one from a hidden two-valued selector. The collision creates the observed signal from a prepared impact variable; removing the collision makes all three deficiencies zero. Varying the contact diameter changes the collision time and the microscopic path, while the certified resource gap and the conditional-law consequences persist. The effective constructions retain the original resource constraints.

CONTENTS

1. Introduction	2
1.1. The principal mechanism	2
1.2. What the information restrictions do	3
1.3. Relation to earlier comparison and realization theory	4
1.4. Scope and organization	4
2. Intrinsic causal deficiency	4
2.1. Experiments and executable signatures	4
2.2. The finite comparison theorem	6
2.3. A delayed channel and a strict signature separation	8
2.4. Optimal resource sets and decision certificates	9
3. Continuation geometry and an exact memory obstruction	10
3.1. Normalized continuation laws	10
3.2. An erasure cut	11
4. Finite-data testing on the original behavior image	12
4.1. The reset interface	12
4.2. A matching uniform sampling obstruction	14
4.3. Transport of one fixed audit	14
5. Polynomial testing of the original resource image	15
5.1. A sharp two-time nonconvex example	17
6. Adaptive testing on the observable behavior image	17
6.1. The testing game	18

Date: September 23, 2026.

2020 Mathematics Subject Classification. Primary 62B05, 93E20; Secondary 60G44, 68Q45, 82C40.

Key words and phrases. causal comparison, continuation cone, finite memory, statistical experiments, hard-sphere scattering, nonlinear filtering.

6.2.	A size bound in behavior dimension	18
6.3.	A three-weight certificate invisible to constant tests	20
6.4.	Strict certificates and effective physical input	21
7.	An exact algorithm for the observable chart	22
8.	A marked resource gap generated by a binary collision	23
8.1.	The exact marked two-report experiment	23
8.2.	A full-dimensional collision preparation	24
8.3.	Uniformly effective finite presentations	26
9.	Uniform stability of optional projections	27
9.1.	A microscopic sufficient condition	28
10.	Changing trajectories, likelihoods, and conditional consumers	29
10.1.	A common preparation, not a common flow	29
10.2.	The projected stochastic exponential	31
10.3.	A bounded entropic backward-equation consumer	31
10.4.	Resource transport is not the posterior coupling	33
11.	A uniform certificate through a moving collision	33
	Appendix A. The canonical theorem chain	37
	References	37

1. INTRODUCTION

A causal reduction consumes observations, emits reports, and carries some information into the future. The size and visibility of that carried information are mathematical data. In particular, a persistent hidden selector is not equivalent to fresh private randomization, and revealing the selector can change a comparison even when the retained register is unchanged. A theory of reduction must therefore compare marked laws before convexifying its feasible set.

There are two further requirements. A separating test should specify what it observes about a proposed reduction; an exact probability law is a much stronger observation than finitely many independent runs. Moreover, a certificate obtained in a microscopic model should survive an approximation of the dynamics, not only an approximation of a formula written after the dynamics have been fixed. This paper joins these requirements.

Our basic object is the resource spectrum of a marked causal experiment. A point of that spectrum records a register-width profile, an error, and, where applicable, a hidden or visible selector alphabet. The mark is an actual system variable. It is not available to the online encoder. The comparison tests include its terminal value. Theorem 2.2 gives the elementary compactness, factorization and costed composition properties. The subsequent argument does not replace the private spectrum by its convex hull.

1.1. The principal mechanism. Let B be the compact behavior image of a fixed resource signature. A behavior $b = (b_j)_{j=1}^J$ lists the laws of finitely many marked causal transcripts. Fix target laws Q_j and put

$$F(b) = \max_{j,A} \{Q_j(A) - b_j(A)\}, \quad \delta(B, Q) = \min_{b \in B} F(b).$$

Complements of events supply both signs. An auditor is allowed n independent, reset runs of each row, followed by independent validation. The candidate is held fixed during the whole audit, but its realized private randomness is reset between runs. The auditor sees completed marked transcripts, not the candidate's probabilities or its retained states. This information pattern is different both from an ordinary single-run comparison and from a revealed-law verification game.

Theorem 1.1 (Finite-resource certification and microscopic transport). *The following assertions hold with the conventions just described.*

(i) *Finite causal kernels have a register realization with a prescribed width profile exactly when their normalized continuation laws admit compatible finitely generated polytopes of those widths. At an erasure cut carrying a latent letter of law π , the optimal error with K retained states is*

$$1 - \sum_{i=1}^{\min(K,N)} \pi(i), \quad \pi_{(1)} \geq \cdots \geq \pi_{(N)}.$$

Independent hidden or visible selectors do not improve this cut value.

(ii) *The reset-audit value is attained, has a finite-dimensional moment minimax representation on the original B , and satisfies*

$$0 \leq \delta(B, Q) - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}, \quad D = \sum_j (|\Omega_j| - 1).$$

The exponent $1/2$ cannot be improved uniformly over compact candidate classes. This assertion is about reset samples, not about the minimal degree of unrestricted polynomial tests.

(iii) *Suppose two prepared microscopic observation models have a common preparation variable, bounded signals H, H' on $[0, T]$, and the same positive Gaussian noise level σ . Set*

$$\eta^2 = \mathbb{E} \int_0^T |H - H'|^2 dt, \quad \varepsilon = \min\{1, \eta/(2\sigma)\}.$$

Their joint marked observation laws differ by at most ε . This estimate transports resource spectra at their stated budgets and transports each fixed finite audit with its explicitly charged reset cost. It also gives whole-process posterior convergence. The unprojected and projected likelihoods converge in the supremum L^2 norm under the common Wiener reference when $\eta \rightarrow 0$. For bounded latent payoffs, the associated entropic backward equations converge in value, and their integrands converge after the stated denominator localization.

(iv) *In the two-sphere model of Theorem 11.2, one rationally specified audit with 2,800 training runs has expected discrepancy strictly greater than $2/5$ against every private width-one simulator, uniformly for $9/10 \leq d \leq 11/10$ and $0 < \sigma \leq 1/24$. A hidden two-valued selector has error less than $13/100$; every visible-selector width-one error exceeds $9/20$. For $d_n \rightarrow d$ the microscopic paths, likelihoods and posterior processes converge in the precise senses of that theorem, and the same audit remains valid. For $d \leq 2/5$ all three deficiencies vanish.*

Part (iv) is a composition, not a juxtaposition: the scattering calculation provides a marked gap, the reset-audit theorem turns it into one finite-data certificate on the nonconvex private image, and microscopic transport carries this certificate together with a conditional-law consumer through a changing collision geometry. The proofs are given in Theorems 3.2, 3.3, 4.1, 4.3, 10.1, 10.3, and 11.2.

1.2. What the information restrictions do. The continuation criterion is a geometric realization criterion, not a claim that ordinary Hankel rank equals nonnegative state complexity. Its generators must be stable under all one-step input–output shifts. The cut theorem gives a sharp obstruction even when all other checkpoints have unrestricted memory. In particular, m erased unbiased bits require at least $(1 - \epsilon)2^m$ retained states to attain error ϵ .

The reset-audit dual randomizes over a candidate chosen once for an entire audit. Its moment measure cannot be replaced by the barycenter behavior: the replicated samples are conditionally independent given that candidate, not after forgetting it. This is why minimax on the auditor’s finite coefficient simplex does not supply free persistent randomness to the simulator. The auditor’s

count register, within-run buffer, validation choice, and reset interface are all explicit. No sample-complexity statement here is a polynomial-time statement about minimizing on a nonconvex image.

We also retain revealed-behavior testing and its behavior-dimensional Bernstein bound. The infinite-dimensional representation is a general partition-of-unity identity. Its finite polynomial hierarchy and its small exact nonlinear witness remain useful, but they have different information requirements from the reset audit. Proposition 7.1 gives an exact input–output algorithm for the rational observable quotient and chart used by that hierarchy.

1.3. Relation to earlier comparison and realization theory. Blackwell and Le Cam supply the classical comparison viewpoint [1, 2, 13]. Filtered comparison has a substantial earlier literature, including Norberg and Weisshaupt [3, 19]. We do not assert that inserting a filtration into an experiment is new. The extra data here are fixed persistent width, the visibility of separately charged selectors, and the finite reset interface of the testing problem. An original-text proof-by-proof crosswalk with Norberg is not established in this revision; no claim of priority over an uninspected theorem is used in a proof.

Positive realization by stable finitely generated cones is classical. Fliess’s Theorem 3.1 and the stochastic-process specialization in Theorem 5.4 and its second corollary make this particularly clear [23]. Our continuation statement is a finite-horizon, input-indexed normalization of that mechanism, with a width at each cut. The new use is its connection to the marked resource spectrum and to an exact cut obstruction; the existence of a positive-cone realization is not claimed as a discovery.

The minimax exchange is classical [18]. The statistical lower bound is a two-point testing argument, and the empirical upper bound uses multinomial second moments. Bernstein and positivity methods are also classical [20, 21, 22]. What is proved here is the compatibility of these constructions with the unconvexified resource object, including their information costs and their transport through an interacting microscopic limit. Bayes’ formula, Gaussian change of measure, martingale representation, and maximal inequalities are the analytic inputs to the transport step; the moving-collision estimate verifies that step without replacing the microscopic trajectory by a fixed flow.

1.4. Scope and organization. The physical theorem is at fixed particle number and fixed positive noise. The B – W correlation belongs to the preparation. Collision transduces B into the measured transverse signal; it does not create that correlation. The feedback is an acquisition gate, not a mechanical force. Changing the contact diameter changes a genuine microscopic path, but is not a Boltzmann–Grad or particle-number limit.

The foundations developed here remain applicable to the larger dynamical program. A concrete downstream consequence is the bounded Gaussian changing-path likelihood, posterior and entropic-value theorem proved in Section 10; it replaces a finite-dimensional-to-tightness shortcut by uniform estimates. Algebraic geometry, kinetic action compactness and particle-uniform hierarchy recovery retain their own hypotheses. No independence of an algebraic-geometric theorem is converted into a dependency merely by the order of papers.

The canonical proof chain follows the diagram in Appendix A. The full preceding comparison, physical-domain, task, and historical development is preserved in the companion volume. It is not necessary to reconstruct that archive to read the proofs in this article.

2. INTRINSIC CAUSAL DEFICIENCY

2.1. Experiments and executable signatures. Fix a finite parameter set Θ , a horizon T , and finite control alphabets $A_0, \dots, A_{T-1}$. An experiment has a report at checkpoints $0, \dots, T$ and an unobserved terminal mark W in a common finite set. It is specified by finite causal instruments:

for each parameter and control history they give the joint law of the report path and the actual mark. Equivalently, a finite latent-state realization may be expanded into its history tree. The parameter and mark are unavailable to the encoder. The mark may depend on the physical controlled path. We compare experiments with the same control interface and mark space.

Write Y_t for the source report and Z_t for the target report. A private simulator of width K has stochastic rows

$$\begin{aligned} q_0(z, m \mid y), \quad m \in [K_0], \\ q_t(z, m \mid m', a, y), \quad m' \in [K_{t-1}], \quad m \in [K_t], \quad 1 \leq t \leq T. \end{aligned} \quad (2.1)$$

A row consumes the current source report, emits the target report, and retains the new state. The previous target reports are not available to that row unless encoded in m' . The clock and current control are part of the declared interface. Fresh private randomness is permitted, but no other persistent random variable is free. The product of all row simplexes is denoted $\mathcal{K}_K$.

A deterministic feedback tree g chooses a_t from $z_{0:t}$ and its own previous controls. It is the same controller on both sides of a comparison. Let Ω_g be the finite marked transcript space, including reports, controls, and W . Put

$$P_{\theta,g}^q = \text{Law}_\theta(E \xrightarrow{q} g), \quad Q_{\theta,g} = \text{Law}_\theta(F \rightarrow g).$$

Here W on the left is the source's actual mark. The total variation convention is $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. Private deficiency is

$$\delta_K^p(E, F) = \min_{q \in \mathcal{K}_K} \max_{\theta, g} \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}. \quad (2.2)$$

The minimum will be justified below. Randomized feedback need not be added to the maximum: conditioning on its independent random table expresses its law as a mixture of deterministic feedback laws. The same argument permits arbitrary independent side randomness in the tester.

A *hidden selector* $U \in [r]$, independent of the source, chooses one simulator q^u . It is not visible to the feedback controller or the final test. A *visible selector* is revealed before feedback begins. The two costs are

$$\delta_{K,r}^h(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, g} \left\| \sum_u \lambda_u P_{\theta,g}^{q^u} - Q_{\theta,g} \right\|_{\text{TV}}, \quad (2.3)$$

$$\delta_{K,r}^v(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, (g_u)} \left\| \sum_u \lambda_u \delta_u \otimes P_{\theta, g_u}^{q^u} - \sum_u \lambda_u \delta_u \otimes Q_{\theta, g_u} \right\|_{\text{TV}}. \quad (2.4)$$

In both expressions $\lambda \in \Delta_r$ and $q^u \in \mathcal{K}_K$. The second formula compares the joint seed-marked interface. Its target seed has the same distribution and is independent of the target experiment. A hidden selector can be stored privately by retaining (u, m) , at width rK_t . A visible selector is separate side information and is charged by its own alphabet size. Neither interpretation identifies an arbitrary real-valued design with a free finite register.

Definition 2.1 (Deficiency spectrum). The private resource spectrum of E relative to F is the upward-closed set of pairs (K, ε) satisfying $\delta_K^p(E, F) \leq \varepsilon$. The hidden spectrum consists of triples (K, r, ε) satisfying $\delta_{K,r}^h(E, F) \leq \varepsilon$, and the visible spectrum uses $\delta_{K,r}^v$. Width profiles are ordered coordinatewise; in the seeded spectra the seed bound and error are ordered as well. Unused states or selector values may be adjoined. The hidden convexification is $\delta_K^{\text{mix}} = \inf_{r \geq 1} \delta_{K,r}^h$.

The spectrum, not a presumed unique coordinatewise minimum profile, is the intrinsic object. For example one may minimize $\max_t K_t$, with a specified seed signature and tolerance, over its feasible profiles. Different checkpoints can have incomparable optimal tradeoffs.

2.2. The finite comparison theorem. Concatenate all probabilities $P_{\theta,g}^q(\omega)$ into $b(q) \in \mathbb{R}^N$, retaining their row indices; put $b_F = (Q_{\theta,g}(\omega))$. Let $\mathcal{D}_K$ be the finite set of deterministic choices of all rows in (2.1). Empty or unreachable rows remain harmless coordinates. Set

$$a_K = \dim \text{aff}\{b(q) : q \in \mathcal{K}_K\}, \quad \mathcal{I} = \{(\theta, g, B) : B \subseteq \Omega_g\}.$$

For $i = (\theta, g, B)$ and $d \in \mathcal{D}_K$, define

$$A_{i,d} = Q_{\theta,g}(B) - P_{\theta,g}^d(B). \quad (2.5)$$

Including both an event and its complement includes both signs.

Theorem 2.2 (Intrinsic deficiency, priced converse, and visibility). *For the finite marked causal experiments above the following assertions hold.*

(i) *Private deficiency and both fixed- r deficiencies are attained. The private problem has the exact polynomial formulation*

$$\min\{\varepsilon : q \in \mathcal{K}_K, 0 \leq \varepsilon \leq 1, Q_{\theta,g}(B) - P_{\theta,g}^q(B) \leq \varepsilon \text{ for all } (\theta, g, B) \in \mathcal{I}\}. \quad (2.6)$$

In particular its zero set is exactly the nonnegative causal factorization of the target marked behavior by the rows (2.1). At $K_t = 1$ for every t , this is precisely the same factorization with rows $q_t(z \mid a, y)$ and $q_0(z \mid y)$.

(ii) *Hidden convexification has the attained dual*

$$\delta_K^{\text{mix}}(E, F) = \min_{\lambda \in \Delta(\mathcal{D}_K)} \max_{i \in \mathcal{I}} \sum_d \lambda_d A_{i,d} = \max_{w \in \Delta(\mathcal{I})} \min_{d \in \mathcal{D}_K} \sum_i w_i A_{i,d}. \quad (2.7)$$

An optimal simulation uses at most $a_K + 1$ selector values. Thus every weighted causal test has optimal discrepancy at most ε if and only if a hidden mixture of K -state simulators has error at most ε ; it has a private realization of width at most $(a_K + 1)K$. More specifically,

$$\delta_{rK}^p \leq \delta_{K,r}^h, \quad \delta_K^{\text{mix}} = \delta_{K,a_K+1}^h \leq \delta_K^p. \quad (2.8)$$

(iii) *Put $e_\theta(q) = \max_g \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}$. Then*

$$\delta_{K,r}^v = \min_{\lambda, q^1, \dots, q^r} \max_{\theta} \sum_u \lambda_u e_\theta(q^u). \quad (2.9)$$

Writing $a_K^v = \dim \text{aff}\{(e_\theta(q))_\theta : q \in \mathcal{K}_K\}$, the infimum over all finite visible selectors is attained using $a_K^v + 1 \leq |\Theta| + 1$ values and equals

$$\max_{v \in \Delta(\Theta)} \min_{q \in \mathcal{K}_K} \sum_{\theta} v_\theta e_\theta(q). \quad (2.10)$$

For every $r \geq 1$, and also after optimizing over all r ,

$$\delta_{K,r}^v(E, F) = 0 \iff \delta_K^p(E, F) = 0. \quad (2.11)$$

No corresponding equivalence with $\delta_K^{\text{mix}} = 0$ holds in general.

(iv) *For compatible experiments E, F, G and profiles K, L ,*

$$\begin{aligned} \delta_{KL}^p(E, G) &\leq \delta_K^p(E, F) + \delta_L^p(F, G), \\ \delta_{KL,rs}^s(E, G) &\leq \delta_{K,r}^s(E, F) + \delta_{L,s}^s(F, G), \quad s \in \{h, v\}. \end{aligned} \quad (2.12)$$

Here KL is the pointwise product, and the second line uses independent selectors. If an F -consumer has widths S_t , composing it with a private simulator gives widths $K_t S_t$ and changes every $[0, 1]$ marked risk by at most $\delta_K^p(E, F)$. A hidden selector is stored at the additional price r , whereas visible selector alphabets multiply. The same bounds hold for optimized upper risk sets and their lower support functions with the indicated costs.

Proof. Expand a probability of a complete source path, simulator state path, and transcript. It is a product of fixed experiment probabilities and one chosen row entry per time. Summing latent and state paths gives a polynomial in the finitely many row entries. No row belonging to a fixed time can occur twice along one path. Consequently the polynomial is multi-affine in the groups of stochastic rows. Continuity and compactness of the product of closed finite simplexes prove attainment in (2.2); the same proof includes λ and the r policies in (2.3). For the visible problem there are only finitely many collections (g_u) and transcript events, so its objective is again continuous on a compact set. The finite-event formula for total variation proves (2.6). Setting $\varepsilon = 0$ forces equality of every transcript probability, in both directions of each event inequality. This proves the factorization and stateless claims, including their necessity.

For each stochastic row of q , sample its deterministic output once, independently of all other rows, before the experiment. This constructs a probability Λ_q on $\mathcal{D}_K$. On any fixed path the row variables encountered have different time indices, so the expectation of their product is the product of the row probabilities. Summing paths shows, simultaneously for every θ, g , that

$$b(q) = \sum_{d \in \mathcal{D}_K} \Lambda_q(d) b(d). \quad (2.13)$$

This is a behavior identity, not a same-width deterministic purification of minimax error. It implies $\text{conv } b(\mathcal{K}_K) = \text{conv}\{b(d) : d \in \mathcal{D}_K\}$. The latter is a compact polytope. Minimizing the continuous convex maximum of event discrepancies over this polytope gives the first expression in (2.7). Finite-dimensional minimax gives the second, with both optima attained. A point of this polytope is a convex combination of at most $a_K + 1$ of its generating points by Carathéodory's theorem. For completeness, if a representation uses more points, their affine dependence gives coefficients c_d with $\sum c_d = 0$ and $\sum c_d b(d) = 0$. Subtract the largest permissible multiple of c from the positive mixing weights; a weight becomes zero without changing the represented point. Repeating gives the stated bound. Store its selector together with the machine state to obtain (2.8). The dual inequality for every w is equivalent to the maximum over w being at most ε , which proves the priced testing converse.

For disjoint seed fibers, total variation is the sum of the fiber total variations weighted by λ_u . For a fixed θ , the tester can choose a separate maximizing tree on each revealed fiber. Therefore its maximum is $\sum_u \lambda_u e_\theta(q^u)$, proving (2.9). The map $q \mapsto e(q)$ is continuous, being a finite maximum of continuous functions. Its image is compact. Convexifying it in $\mathbb{R}^{|\Theta|}$ and applying the preceding finite-dimensional separation and affine-dependence argument proves (2.10) and its selector bound. Notice that the inner minimum is over stochastic q , not over deterministic tables: the nonlinear function e_θ does not commute with (2.13). If the attained visible optimum is zero, then for every u of positive weight and every θ one has $e_\theta(q^u) = 0$, since every summand is nonnegative. Any such q^u is a private exact simulator. The converse uses a selector of one value. This also proves the assertion for the optimized visible selector, whose support is finite by the bound just established.

Finally, cascade a K -state E -to- F transducer and an L -state F -to- G transducer. Within a step, consume the intermediate report before discarding it. The retained pair has at most $K_t L_t$ values. For any terminal tester on G , the second transducer together with that tester is an admissible randomized feedback system on F . The first error bound applies to this system, and total variation contracts under the remaining channel. The second error bound applies to the genuine F system. The triangle inequality gives (2.12). With selectors, condition on the independent selectors in the appropriate places; in the visible case keep both selectors in the compared transcript. The same argument is valid because the target reference selectors are independent. Storing a hidden selector or exposing both visible selectors gives the stated costs. The minima established above may be used; equivalently one can use nearly optimal simulators and let the extra tolerance decrease.

A consumer is another causal transducer with task-specific readout and, where allowed, task-specific control, but with its required common encoder. The simulator is task blind, so its product

with that encoder is still common. Condition on the consumer's independent random table to apply the feedback bound. Integration of any $[0, 1]$ marked loss changes by at most the total variation error. Taking the same paired consumer for every row proves the risk-set assertion, rather than a separate choice of simulator for each row. Infima of nonnegative weighted risks give the lower support inequalities. $\square$

Remark 2.3 (What the converse says). The dual (2.7) uses mixtures of indexed causal tests and optimizes over the entire common transducer. It is not a task-by-task choice of encoder. Its conclusion is an explicit hidden-selector simulation, with the selector subsequently charged if private storage is required. It does not replace the nonconvex private set by its convex hull at unchanged resource. The next example shows why that replacement would change the zero set.

2.3. A delayed channel and a strict signature separation. Let the source reveal $i \in [n]$ at checkpoint zero and reveal only a constant at checkpoint one. There is no control and the mark is constant. The target reports a constant at checkpoint zero and at checkpoint one draws $j \in [m]$ from a specified row-stochastic matrix $V = (V_{ij})$. The unknown parameter is i . The first target alphabet has one element: the simulator cannot use an early public output as unpriced storage. Only the intermediate register of size K matters.

Proposition 2.4 (The nonnegative-rank slice). *For this experiment,*

$$\delta_K^p(E, F_V) = \min_{A, B} \max_i \frac{1}{2} \sum_j |V_{ij} - (AB)_{ij}|, \quad (2.14)$$

where A is an n -by- K row-stochastic matrix and B is a K -by- m row-stochastic matrix. In particular

$$\delta_K^p(E, F_V) = 0 \iff \text{rank}_+(V) \leq K, \quad \delta_1^p(E, F_V) = 0 \iff V \text{ has identical rows.}$$

For $1 \leq K \leq n$ the identity channel has the sharp laws

$$\delta_K^p(E, F_{I_n}) = 1 - \frac{1}{\lceil n/K \rceil}, \quad \delta_K^{\text{mix}}(E, F_{I_n}) = 1 - \frac{K}{n}. \quad (2.15)$$

Thus, at tolerance $0 \leq \varepsilon < 1$, the least private width is $\lceil n/(1 - \varepsilon)^{-1} \rceil$, whereas the least within-mode width with optimized hidden selection is $\max\{1, \lceil n(1 - \varepsilon) \rceil\}$. The latter still pays for its selector. In particular, for $n = m = 3$ and $K = 2$,

$$\delta_2^p(E, F_{I_3}) = \frac{1}{2}, \quad \delta_2^{\text{mix}}(E, F_{I_3}) = \frac{1}{3}. \quad (2.16)$$

For $V_* = (1/2)I_3 + (1/6)\mathbf{1}\mathbf{1}^\top$,

$$\delta_{2,3}^h(E, F_{V_*}) = 0, \quad \delta_2^p(E, F_{V_*}) \geq \frac{1}{4\sqrt{3}} > 0, \quad \inf_r \delta_{2,r}^v(E, F_{V_*}) > 0. \quad (2.17)$$

Proof. The encoder maps the consumed i into its retained label by A ; the terminal decoder maps that label into j by B . This proves (2.14) including its converse implementation. If $V = CD$ is any nonnegative factorization, let $s_k = \sum_j D_{kj}$. Delete zero rows of D , define $B_{kj} = D_{kj}/s_k$ and $A_{ik} = C_{ik}s_k$. Since V is stochastic, the rows of A sum to one. Padding with unused states gives the claimed equivalence with nonnegative rank. The one-state statement follows immediately.

For $V = I_n$, row error is $1 - (AB)_{ii}$. If the least correct probability is c , for each i select a row $k(i)$ of B with $B_{k(i),i} \geq c$. Some row is selected at least $\lceil n/K \rceil$ times, so stochasticity gives $c \leq 1/\lceil n/K \rceil$. A partition into K groups of sizes at most $\lceil n/K \rceil$, each decoded uniformly, attains that bound. Also $\sum_i (AB)_{ii} \leq K$, by the calculation below with K in place of two. It remains valid under mixing, giving hidden error at least $1 - K/n$. Start with a deterministic channel that reproduces K input symbols exactly and sends all others to those K outputs. Average all simultaneous permutations of its input and output labels. Every diagonal entry is then K/n , proving the second formula. Inverting the two monotone formulas gives the stated least widths.

For $V = I_3$, total variation in row i is $1 - (AB)_{ii}$. If all three diagonal entries of AB were greater than $1/2$, then for each i some row of B would have its i th entry greater than $1/2$. A stochastic row has at most one such entry, so two rows cannot serve three indices. This proves the private lower bound $1/2$. Merge two input symbols into a retained label that decodes uniformly to that pair, and retain the third symbol exactly. This attains it. For every stochastic factorization with two intermediate states,

$$\sum_{i=1}^3 (AB)_{ii} = \sum_{k=1}^2 \sum_{i=1}^3 A_{ik} B_{ki} \leq \sum_{k=1}^2 \sum_{i=1}^3 B_{ki} = 2.$$

The same inequality holds under hidden mixtures, giving maximal row error at least $1/3$. Mix uniformly the three preceding pair-merge channels. The resulting diagonal entries are $2/3$ and the off-diagonal entries are $1/6$. This is exactly V_* and has error $1/3$ from I_3 . It proves the hidden claims in both displays.

The smallest singular value of V_* is $1/2$. Any AB with two intermediate states has a unit vector x in its right null space, so $\|V_* - AB\|_F \geq \|(V_* - AB)x\|_2 \geq 1/2$. If the maximal row total variation error is ε , each row has Euclidean error at most 2ε , and hence the Frobenius error is at most $2\sqrt{3}\varepsilon$. This proves the explicit private bound. The optimized visible value is attained by Theorem 2.2; if it were zero, its zero-set assertion would contradict that bound. This proves its strict positivity. $\square$

2.4. Optimal resource sets and decision certificates. For a class $\mathcal{C}_{S,q}(F)$ of common consumers, with all other information conventions fixed, let $\mathcal{R}_{S,q}(F)$ be its vector of attainable $[0, 1]$ risks. For a nonnegative row weight w summing to one, write

$$h_F(S, q; w) = \inf_{r \in \mathcal{R}_{S,q}(F)} \langle w, r \rangle.$$

If E privately simulates F with cost K and error ε , then

$$\mathcal{R}_{S,q}(F) \subseteq \mathcal{R}_{KS,q}(E) + [-\varepsilon, \varepsilon]^J, \quad h_E(KS, q; w) \leq h_F(S, q; w) + \varepsilon. \quad (2.18)$$

For nonfinite row sets the first expression means pointwise paired risk functions within uniform distance ε ; no finite coordinate box is needed. Upper sets give the equivalent one-sided order formulation. Hidden selectors multiply private width by r ; visible selectors multiply the separate public alphabet by r . These are direct consequences of the final part of Theorem 2.2, with the optimal intrinsic error in place of an externally estimated simulator error.

When two stateless comparison directions have error at most ε , paired risk sets and their support values at unchanged width are within ε . If each is centered by a paired unrestricted benchmark whose error is also at most ε , the centered values are within 2ε . Therefore a gap $g > 0$ between zero excess and infeasible widths is stable under $\varepsilon < g/4$: a value below $g/2$ in the approximation certifies feasibility in the exact finite problem, and a value above $g/2$ certifies infeasibility. The benchmark and the class being optimized must be specified; this assertion does not identify different information signatures.

Proposition 2.5 (Intrinsic finite-dimensional selector approximation). *Let $\mathcal{R}$ be a compact risk image in $C(J)$, where J is compact. Suppose a bounded linear operator $L : C(J) \rightarrow C(J)$ has rank at most m and $\sup_{r \in \mathcal{R}} \|r - Lr\|_\infty \leq \eta$. Every barycenter $\bar{r}$ of a probability on $\mathcal{R}$ has a barycenter r_* of at most $m + 1$ members of $\mathcal{R}$ with $\|\bar{r} - r_*\|_\infty \leq 2\eta$. If $\mathcal{R}$ lies in a finite affine space of dimension m , the error is zero with at most $m + 1$ members. Thus the selector price can be controlled by the dimension of attainable risks, rather than by a covering number of the row space.*

Proof. The compact set $L\mathcal{R}$ lies in an m -dimensional space. Its convex hull is compact, and $L\bar{r}$ belongs to it. The finite-dimensional affine-dependence argument in the proof of Theorem 2.2 gives $r_1, \dots, r_{m+1} \in \mathcal{R}$ and weights λ such that $L\bar{r} = \sum_i \lambda_i Lr_i$. Put $r_* = \sum_i \lambda_i r_i$. The two residual

terms $\bar{r} - L\bar{r}$ and $r_* - Lr_*$ have norms at most η , proving the estimate. Apply the same argument after translation in the exact affine case. The implementation stores the selected design index whenever it is not provided as visible side information. $\square$

Remark 2.6 (Finite computation and general carriers). For rational finite instruments, (2.6) and the fixed- r formulations are finite polynomial feasibility problems; (2.7) is a finite linear program. These are exact variational characterizations, not assertions of polynomial running time in a succinct history description. The number of feedback trees, events, or deterministic tables can be large. On standard Borel carriers the same definitions use suprema over measurable feedback and tests and infima over executable kernels. Their product-state composition and decision transport remain valid by the same channel argument. Attainment and the finite testing dual are asserted here only for the stated finite class. The average-risk and physical results below give separate nonfinite theorems with their own hypotheses.

3. CONTINUATION GEOMETRY AND AN EXACT MEMORY OBSTRUCTION

3.1. Normalized continuation laws. We first separate the transducer from the source to which it is attached. There are L epochs with finite input alphabets X_t and output alphabets Z_t . A causal kernel $p(z_{1:L} \mid x_{1:L})$ is nonnegative, normalized for every input word, and has output-prefix marginals independent of later inputs. The current input can include both a source report and the current control. It never includes an unavailable parameter, a terminal mark, or an earlier output not retained by the machine.

Let $\mathcal{C}_t$, $0 \leq t \leq L$, be the finite-dimensional polytope of normalized causal kernels from $x_{t+1:L}$ to $z_{t+1:L}$; put $\mathcal{C}_L = \{1\}$. For $v \in \mathcal{C}_t$ define the shift

$$(S_{x,z}v)(z_{t+2:L} \mid x_{t+2:L}) = v(z, z_{t+2:L} \mid x, x_{t+2:L}).$$

Its total mass $v(z \mid x)$ is independent of the future input word. A zero shift is allowed.

Definition 3.1 (Compatible continuation polytopes). A continuation cover of width $K = (K_0, \dots, K_L)$ consists of lists $V_t = (v_{t,m})_{m \leq K_t}$ in $\mathcal{C}_t$ such that, for every m, x, z ,

$$S_{x,z}v_{t,m} \in \text{cone } V_{t+1}. \quad (3.1)$$

It covers p when $p \in \text{conv } V_0$. Lists may repeat generators and may contain unreachable generators. Thus unused states can be adjoined.

Theorem 3.2 (Continuation realization with a width profile). *A causal kernel p has a stochastic transducer realization with an initial K_0 -valued register and K_t retained states after epoch t if and only if it has a compatible continuation cover of width K . Initial randomization is part of the charged initial register. With a deterministic initial state the condition is $p = v_{0,1}$ and $K_0 = 1$.*

Consequently the set of feasible width profiles depends only on the causal kernel and its declared input interface, not on a row presentation or a labeling of the hidden states. Every matrix of conditional continuation probabilities at cut t has nonnegative rank at most K_t . Individual rank bounds do not replace the simultaneous shift conditions (3.1).

Proof. Given a transducer, let $v_{t,m}$ be the suffix response when its register at time t equals m . The remaining rows define this response even for an unreachable state. If the next row is $q_{t+1}(z, m' \mid m, x)$, conditioning on its output and new state gives

$$S_{x,z}v_{t,m} = \sum_{m'} q_{t+1}(z, m' \mid m, x) v_{t+1,m'}.$$

The initial state distribution expresses p as a convex combination of V_0 .

Conversely choose nonnegative coefficients in every cone membership in (3.1). There are only finitely many choices to make. Since every generator on the right has mass one for each future

input word, the sum of the coefficients for a fixed m, x, z is $v_{t,m}(z \mid x)$. Summing over z gives one. The coefficients are therefore stochastic rows $q_{t+1}(z, m' \mid m, x)$. Choose an initial distribution representing p in $\text{conv } V_0$. Backward induction from $\mathcal{C}_L = \{1\}$ proves that the constructed machine has exactly the response $v_{t,m}$ from each register state. In particular its initial response is p . Equality for open-loop input words implies equality under adaptive input selection: at each node of an input–output history tree, the same causal conditional probability is used on the two sides.

For the rank assertion, condition on any positive-probability observed history at the cut. Its continuation response is a convex combination of $v_{t,m}$ with weights equal to the posterior register distribution. Stack these responses as rows, and use all future input–output pairs as columns. This is a nonnegative factorization through K_t generators. The converse construction required the same generator families at adjacent cuts and all their shifts; an unrelated factorization of each cut matrix provides no such compatibility. The characterization, being stated entirely in terms of these response polytopes, also proves the asserted invariance. $\square$

Attach such a causal kernel to a marked source experiment. The output law is obtained by contraction with the source’s causal instruments, while its actual mark is retained. Hence zero private deficiency is equivalently the existence of a compatible continuation cover whose contraction equals the target marked behavior in every parameter and feedback row. This is the geometric form of the factorization in Theorem 2.2; no conditional dependence on the mark is introduced.

A hidden selector chooses one such cover independently of the source. It is a convex combination of complete realizations, not a convexification inside every retained-state simplex at the original width. Its private implementation uses the disjoint union of the covers and keeps the selector in the state. A visible selector keeps those fibers in the compared law. The nonnegative fiber argument in Theorem 2.2 therefore continues to give equality of the visible and private zero sets.

3.2. An erasure cut. A simple family makes the width obstruction exact, including its error. Let X take values in $[N]$ with law π . Before a specified cut the source reveals X . After the cut it has no reports correlated with X . The controller has no feedback channel carrying X across the cut, and all earlier emitted reports are prescribed constants. The only permitted cross-cut information of the simulator is its K -valued register and any independent selector explicitly included in its signature. The terminal mark is the actual $W = X$. The target’s terminal report is X . All memory at other checkpoints is unrestricted.

Theorem 3.3 (Exact cut spectrum). *For this experiment, for private, hidden and visible signatures alike,*

$$\delta_K = 1 - \sum_{i=1}^{\min(K,N)} \pi_{(i)}. \quad (3.2)$$

For uniform X , $\delta_K = (1 - K/N)_+$. Thus an erasure cut for m unbiased bits requires $K \geq (1 - \epsilon)2^m$ whenever its error is at most ϵ . The bound holds for every surrounding width profile; equality is attained when the other checkpoints are unrestricted. If every $\pi_x > 0$, exact transmission has nonnegative cut rank N .

Proof. With all post-cut randomness averaged into the decoder, any private machine gives kernels $e(m \mid x)$ and $d(z \mid m)$. Its joint terminal law is $\pi_x \sum_m e(m \mid x) d(z \mid m)$. Since the target law is $\pi_x \mathbf{1}_{z=x}$, total variation is exactly one minus the correct-decoding probability. Put $u_x = \max_m d(x \mid m)$. Then

$$0 \leq u_x \leq 1, \quad \sum_x u_x \leq K, \quad \mathbb{P}(Z = X) \leq \sum_x \pi_x u_x \leq \sum_{i=1}^{\min(K,N)} \pi_{(i)}.$$

The last inequality follows by moving mass in u from a smaller weight to a larger one until the K largest weights are filled. To attain it, retain distinct labels for those letters and decode them

deterministically; all other letters are mapped to any retained label. Before the cut one may use the unrestricted memory to compute this encoding.

For a hidden selector, condition on its value. It is independent of X , so every conditional correct-decoding probability obeys the same bound; averaging cannot improve it. For a visible selector, variation is the average of the fiber variations, with the same conclusion. The private construction works for both signatures. Finally, the exact joint cut matrix is diagonal with its N positive diagonal entries π_x . Its ordinary rank is N , so its nonnegative rank is at least N and its diagonal factorization attains N . $\square$

The absence of an auxiliary cross-cut feedback channel is essential. An unpriced controller storing X and returning it as a later control would change the experiment, not disprove the cut bound. Likewise a selector correlated with X is not an independent hidden selector. These distinctions are the same ones needed in the collision certificate below.

4. FINITE-DATA TESTING ON THE ORIGINAL BEHAVIOR IMAGE

4.1. The reset interface. Let $B \subset \prod_{j=1}^J \Delta(\Omega_j)$ be compact, and let $Q = (Q_j)$ be fixed. Set $d_j = |\Omega_j|$, $D = \sum_j (d_j - 1)$, $\mathcal{I} = \{(j, A) : A \subseteq \Omega_j\}$, and $f_{j,A}(b) = Q_j(A) - b_j(A)$. Include the empty event. No convexity of B is assumed.

A candidate b is fixed for the whole audit. The auditor obtains n independent samples of each b_j , with independence between rows. In a causal experiment, row j means a specified parameter and deterministic feedback tree; the reset interface supplies that row. It is an additional experimental interface, not information available to the online simulator. A complete sample contains the terminal mark. Private randomness and any per-run hidden selector are reset between runs. The design of the candidate is not resampled.

Write $C_{j,w}$ for the counts, and $\mathcal{N}_n$ for all count arrays with $\sum_w C_{j,w} = n$. Their probability is the polynomial

$$p_C(b) = \prod_j \frac{n!}{\prod_w C_{j,w}!} \prod_w b_j(w)^{C_{j,w}}.$$

A randomized audit is an array $c = (c_{C,i})$ with $c_C \in \Delta(\mathcal{I})$. After training it chooses $i = (j, A)$ according to c_C . Independent candidate and target validation runs give the signed score $\mathbf{1}_A(\text{target}) - \mathbf{1}_A(\text{candidate})$. Its expected value is

$$g_c(b) = \sum_{C,i} p_C(b) c_{C,i} f_i(b), \quad v_n^{\text{aud}}(B, Q) = \max_c \min_{b \in B} g_c(b). \quad (4.1)$$

Thus every audit payoff is polynomial of total degree at most $Jn + 1$ in the behavior coordinates. The probability of a large observed score in a single audit is not asserted to equal its expectation; additional independent validation would be required for a specified confidence level.

Theorem 4.1 (Reset-audit duality and finite-sample approximation). *Let $\delta = \min_{b \in B} \max_i f_i(b)$. Then all the extrema below are attained, and*

$$v_n^{\text{aud}}(B, Q) = \min_{\mu \in \mathcal{P}(B)} \sum_{C \in \mathcal{N}_n} \max_{i \in \mathcal{I}} \int_B p_C(b) f_i(b) d\mu(b), \quad (4.2)$$

$$0 \leq \delta - v_n^{\text{aud}}(B, Q) \leq \sqrt{D/n}. \quad (4.3)$$

The values are nondecreasing in n . If $a = \dim \text{aff } B$, a minimizing measure in (4.2) can be supported on at most $\binom{a+Jn+1}{a}$ points.

An explicit feasible audit attaining the lower bound $\delta - \sqrt{D/n}$ chooses an event maximizing $Q_j(A) - C_j(A)/n$, with a fixed deterministic tie rule. If the rational target probabilities are given,

this is a rationally specified finite rule. With a free schedule clock, a conservative retained-state bound is

$$H(|\mathcal{I}| + 1)(n + 1)^{\sum_j d_j},$$

where H is the number of states needed to execute and record one causal transcript. This budget is charged to the auditor, not to the simulator.

Proof. The coefficient arrays form a compact convex product of finite simplexes. The integrand is continuous in b and affine in c . The probability measures on compact B form a compact convex set for weak convergence. The minimum over b equals the minimum over such measures for fixed c . Finite-dimensional minimax (or its compact measure version) therefore exchanges the two extrema. Maximization over each independent coefficient simplex selects its largest coordinate integral, giving (4.2). Continuity and compactness give attainment. This step uses measures on the original B , not feasible private realizations of their barycenters.

For any audit, $g_c(b) \leq F(b) = \max_i f_i(b)$. At a minimizer of F this gives $v_n^{\text{aud}} \leq \delta$. The zero event gives $v_n^{\text{aud}} \geq 0$. For the stated explicit rule put $\hat{b}_j = C_j/n$ and $\rho = \max_j \|\hat{b}_j - b_j\|_{\text{TV}}$. Comparing the empirical maximizer with a true maximizing event gives the pointwise regret bound

$$F(b) - f_{i(C)}(b) \leq 2\rho.$$

For a multinomial empirical measure,

$$\mathbb{E}\|\hat{b}_j - b_j\|_{\text{TV}}^2 \leq \frac{d_j}{4} \sum_w \mathbb{E}(\hat{b}_j(w) - b_j(w))^2 = \frac{d_j(1 - \sum_w b_j(w)^2)}{4n} \leq \frac{d_j - 1}{4n}.$$

Consequently $\mathbb{E}\rho \leq (\sum_j \mathbb{E}\|\hat{b}_j - b_j\|_{\text{TV}}^2)^{1/2} \leq \sqrt{D/n}/2$. Averaging the regret uniformly over B proves (4.3).

From $n + 1$ unordered samples in each row, discard one uniformly at random. The remaining counts have exactly the n -sample law, independently across rows. This realizes every n -sample audit as a randomized $(n + 1)$ -sample audit and proves monotonicity.

Choose affine coordinates on $\text{aff } B$. All the functions $p_C f_i$ lie in the span of monomials of total degree at most $Jn + 1$, a space of dimension at most $s = \binom{a + Jn + 1}{a}$ including the constant. The moment vector of a probability measure is in the convex hull of the evaluation vectors; their constant coordinate is one, so their affine dimension is at most $s - 1$. Eliminating affine dependencies gives a representation by at most s points. It preserves every integral in (4.2), and hence preserves the objective.

Finally, each count is in $\{0, \dots, n\}$. The indicated product register stores all counts, the within-run state, and the selected event for validation. Only the latter event and the within-run buffer need survive once training ends, so the product bound is conservative. Randomizing an event uses fresh randomness. The schedule, reset access and terminal mark readout were declared before the theorem. No private simulator state is read by this implementation. $\square$

The dual measure chooses a candidate once, after which its samples are independent conditional on that candidate. Replacing it by $\bar{b} = \int b d\mu$ would replace $\int p_C(b) f_i(b) d\mu$ by $p_C(\bar{b}) f_i(\bar{b})$, generally a different quantity. It would also replace the joint replicated law by a product of its marginals. Neither operation is part of the theorem. This is the operational meaning of the moment side, rather than a same-budget convexification theorem.

Proposition 4.2 (Both sides sampled). *Suppose the target reference Q^b varies with the candidate, as it does when visible-selector weights are design variables. Supply independent training samples from both b and Q^b , and use their empirical difference. If their row dimensions are D_P, D_Q , the same construction has regret at most $\sqrt{D_P/n} + \sqrt{D_Q/n}$. Minimax and attainment hold on the*

compact image of pairs (b, Q^b) . In particular a variable reference law must not be treated as known fixed input to Theorem 4.1.

Proof. The error in any event discrepancy is at most the sum of the maximal row variations of the two empirical tables. Apply the preceding second moment bound separately and then the empirical-maximizer regret inequality. Count probabilities are products for the two tables. They are continuous polynomials on the compact image of pairs, so the same minimax argument applies without changing the simulator signature. $\square$

4.2. A matching uniform sampling obstruction.

Theorem 4.3 (Binary reset lower bound). *Take one binary row with target $\text{Ber}(1/2)$ and candidate class $B_h = \{\text{Ber}(1/2 - h), \text{Ber}(1/2 + h)\}$, where $0 < h < 1/2$. Then*

$$\delta = h, \quad v_n^{\text{aud}} = h \|\text{Ber}(1/2 + h)^{\otimes n} - \text{Ber}(1/2 - h)^{\otimes n}\|_{\text{TV}}. \quad (4.4)$$

For every $n \geq 1$, choosing $h = 1/(8\sqrt{n})$ gives

$$\delta - v_n^{\text{aud}} \geq \frac{1}{16\sqrt{n}}.$$

Thus the exponent in (4.3) is sharp uniformly over compact candidate classes, already at affine dimension one. This does not assert a degree lower bound for revealed-behavior polynomials, nor that this two-point class equals an unrestricted private-row image.

Proof. Only the two complementary singleton events and the zero event matter. A randomized choice is a statistic d in $[-1, 1]$. The two expected payoffs are $h\mathbb{E}_+d$ and $-h\mathbb{E}_-d$. Their minimum is at most half their sum, which is at most $h\|P_+^n - P_-^n\|_{\text{TV}}$. Conversely take the sign of the likelihood difference, assigning zero at a tie. Complementing every sample interchanges P_+^n and P_-^n and changes that sign. The two payoffs are equal, and their average attains the variation bound. The statistic depends only on the count, so it is an admissible audit. The symmetric measure on the two candidates attains the dual value. Every Dirac measure has dual objective h , whereas for finite n the two product laws are equivalent and their variation is strictly less than one. Thus no dual minimizer is a Dirac measure. Their barycenter is the target law, but replacing the dual by that barycenter would incorrectly give value zero instead of the strictly positive value in (4.4).

For $h \leq 1/8$, elementary integration of $(1 - u^2)^{-1}$ gives

$$\text{KL}(\text{Ber}(1/2 + h) \parallel \text{Ber}(1/2 - h)) = 2h \log \frac{1 + 2h}{1 - 2h} \leq 16h^2.$$

The product relative entropy is at most $16nh^2$. Pinsker's inequality bounds its total variation by $\sqrt{8nh^2}$. At the specified h this is less than $1/2$. Substituting in (4.4) proves the claim. The inequality uses the classical two-point testing mechanism; the restriction being measured is the reset budget of the present game. $\square$

4.3. Transport of one fixed audit.

Proposition 4.4 (Reset cost under model perturbation). *Let q range over the same compact set of executable row designs in two models. Suppose uniformly in q, j that*

$$\|P_j^q - \tilde{P}_j^q\|_{\text{TV}} \leq \alpha, \quad \|Q_j - \tilde{Q}_j\|_{\text{TV}} \leq \beta.$$

For the same coefficient rule c , evaluated with each model's own target validation,

$$|g_c(q) - \tilde{g}_c(q)| \leq (2Jn + 1)\alpha + \beta. \quad (4.5)$$

The deficiencies themselves differ by at most $\alpha + \beta$, at the same resource budget. If the source behavior is unchanged, a fixed audit loses at most β , independently of its reset count.

Proof. A product coupling of the Jn training runs fails with probability at most $Jn\alpha$. Taking counts and choosing the event are contractions. A score with fixed target discrepancies lies in $[-1, 1]$, so the change in its expectation from this training-law perturbation is at most $2Jn\alpha$. At each event the discrepancy changes by at most $\alpha + \beta$. This proves (4.5). For deficiency, every row variation changes by at most $\alpha + \beta$; taking the maximum and then the minimum preserves that bound. No new simulator is inserted. $\square$

For a rational polynomial image, a rational rule and a rational threshold, strict positivity of $g_c(q)$ above that threshold is a first-order formula over the original row simplexes. Real quantifier elimination decides it. The same is true of an existential rule followed by this universal condition. This is an exact finite verification statement, not an efficient algorithm. The explicit empirical rule below will be verified by a uniform analytic inequality, rather than by storing its enormous complete table.

5. POLYNOMIAL TESTING OF THE ORIGINAL RESOURCE IMAGE

The private behavior image is usually nonconvex. Its testing criterion can nevertheless be written as a limit of finite algebraic lower certificates, without replacing that image by randomized designs. The testing weights in this section depend on the candidate simulator. Unlike Theorem 6.1, this identity is a certificate transform, not an independent variational dual. They are universal proof functions, not a hidden selector supplied to an executed experiment.

Use any one of the finite signatures of Theorem 2.2, with rational instrument data. Write its objective as $F(q) = \max_{1 \leq i \leq m} f_i(q)$, where f_i are all its marked event discrepancies, including the empty event $f_1 = 0$. For visible selectors the event and feedback rule include their selector fibers, and the target selector law remains in f_i . Thus $-1 \leq f_i \leq 1$ for all three signatures; the polynomials may be different.

Parameterize a row with d outcomes by $d - 1$ variables in $[0, 1]$:

$$q_1 = x_1, \quad q_j = x_j \prod_{l < j} (1 - x_l) \quad (1 < j < d), \quad q_d = \prod_{l < d} (1 - x_l).$$

This map is onto the row simplex, including its boundary. Indeed, divide each desired probability by the remaining mass while that mass is positive; when it is zero choose the remaining variables arbitrarily. Taking the product of these maps, including a selector row when needed, gives an onto polynomial map $q(x)$ from $[0, 1]^D$ to the feasible row product. Put $h_i(x) = f_i(q(x))$ and $g_i = (1 + h_i)/2$. All h_i are rational polynomials, $0 \leq g_i \leq 1$, and $g_1 = 1/2$. The case $D = 0$ is a single feasible table and is interpreted directly.

Theorem 5.1 (Strict polynomial lower-certificate hierarchy). *For $p \geq 1$ define*

$$A_p(x) = \frac{\sum_{i=1}^m g_i(x)^p h_i(x)}{\sum_{i=1}^m g_i(x)^p}, \quad a_p = \min_{x \in [0, 1]^D} A_p(x), \quad \delta = \min_q F(q).$$

Then the denominators are positive, a_p is nondecreasing, and

$$\delta - 2(1 - m^{-1/(p+1)}) \leq a_p \leq \delta, \quad \delta = \sup_{p \geq 1} a_p. \quad (5.1)$$

For rational L let

$$P_{p,L}(x) = \sum_{i=1}^m g_i(x)^p (h_i(x) - L). \quad (5.2)$$

A nonnegative tensor Bernstein expansion of $P_{p,L}$ on $[0, 1]^D$ certifies $\delta \geq L$. If $L < \delta$, some p and some finite Bernstein degree have strictly positive coefficients. Conversely, a strictly positive such expansion certifies $L < \delta$. These are lower certificates at the original state and selector budgets. No convexification of the feasible behavior image occurs.

Proof. Fix x and put $b = \max_i g_i(x) \geq 1/2$ and $r_i = g_i(x)/b$. At least one r_i is one. Write $S_j = \sum_i r_i^j$. Hölder's inequality gives

$$S_p \leq m^{1/(p+1)} S_{p+1}^{p/(p+1)}, \quad S_{p+1} \geq 1.$$

Consequently $S_{p+1}/S_p \geq m^{-1/(p+1)}$. Since $A_p = 2bS_{p+1}/S_p - 1$ and $\max_i h_i = 2b - 1$,

$$0 \leq \max_i h_i - A_p \leq 2(1 - m^{-1/(p+1)}).$$

Cauchy–Schwarz gives $S_{p+2}S_p \geq S_{p+1}^2$, so A_p is nondecreasing in p . Its denominator is at least 2^{-p} and it is continuous on the cube. Taking minima proves (5.1). The bound is also at most $2 \log m/(p+1)$, by $1 - e^{-u} \leq u$.

Every tensor Bernstein basis function is nonnegative on the cube, and the basis functions sum to one. Nonnegative coefficients therefore imply $P_{p,L} \geq 0$. Dividing by the positive denominator gives $F(q(x)) \geq A_p(x) \geq L$ for every x , which proves soundness. For completeness choose p so that the error in (5.1) is less than $\delta - L$. Then

$$P_{p,L}(x) \geq 2^{-p} \{\delta - L - 2(1 - m^{-1/(p+1)})\} > 0.$$

Lemma 5.2 below gives a finite strictly positive Bernstein expansion. Conversely strictly positive coefficients have a positive minimum, and the denominator is at most m , so they give a strict positive uniform gap above L . $\square$

Positive Bernstein certificates and their degree elevation are classical; see de Klerk–Laurent [20]. The next elementary bound is retained for a self-contained exact verifier, not as a new general hypercube positivity theorem.

Lemma 5.2 (An explicit positivity degree). *Let $P(x) = \sum_k c_k x^k$ be a rational polynomial in D variables and $n \geq \max_{k,j} k_j$. In its tensor Bernstein expansion of degree n in each coordinate the coefficient at $\alpha \in \{0, \dots, n\}^D$ is*

$$b_{\alpha,n} = \sum_k c_k \prod_{j=1}^D \frac{\binom{\alpha_j}{k_j}}{\binom{n}{k_j}}, \quad |b_{\alpha,n} - P(\alpha/n)| \leq \frac{C(P)}{n}, \quad C(P) = \sum_k |c_k| \sum_j \binom{k_j}{2}. \quad (5.3)$$

In particular, $\min P > 0$ implies positivity of every coefficient whenever $n > C(P)/\min P$ and the degree restriction holds.

Proof. For a binomial random variable B with parameters (n, x) , $\mathbb{E} \binom{B}{k} = \binom{n}{k} x^k$. Expanding this expectation proves the univariate monomial formula, and products prove the displayed coefficient identity. The convention is $\binom{a}{k} = 0$ for $a < k$. To compare $\binom{a}{k}/\binom{n}{k}$ with $(a/n)^k$, draw k indices uniformly from $\{1, \dots, n\}$. Conditional on no repetition this is sampling without replacement. The probability of a repetition is at most $\binom{k}{2}/n$. Apply this total variation bound to the event that all drawn indices lie in a prescribed a -element subset. Hence the difference of the two factors is at most $\binom{k}{2}/n$. All factors lie in $[0, 1]$. A product telescoping bound followed by the triangle inequality proves (5.3). The final assertion follows by taking the minimum of P . $\square$

Corollary 5.3 (Rational lower witnesses without policy cells). *At every fixed finite rational signature, strict infeasibility above a rational level L is equivalent to the existence of the following finite rational certificate: an integer p , an integer n , and a strictly positive coefficient array for (5.2) whose identity is checked by (5.3). Dovetailing these certificates with rational original-budget upper witnesses produces nested intervals converging to δ . If a subfamily of the original tests gives a strict gap, the same construction applies to it after adjoining the empty test.*

Proof. Soundness and completeness are Theorem 5.1. Each finite candidate is checked by rational polynomial arithmetic. Enumerate all (p, n) and rational levels, and retain the largest verified lower level so far, together with zero. Rational row tables are dense in the feasible product, and their continuous objectives converge to the minimum, so enumeration and retention of the best upper table converge from above. This is a dovetailed enumeration, not a claim that any fixed (p, n) test search terminates. For a requested positive interval width, convergence of both endpoints proves termination. The reduced family conclusion uses only the same pointwise maximum argument. $\square$

Remark 5.4 (Meaning and size of the strict certificate transform). A constant distribution on tests generally detects only the convex hull of the behavior image. The functions $g_i^p / \sum_j g_j^p$ are instead functions of a proposed row table. Their universally quantified weighted inequality is an algebraic certificate; the tester need not observe a private state in an experiment. The displayed certificate identity is not an equality obtained by exchanging a private minimum with a fixed mixed test. Power means and Bernstein degree elevation are classical algebraic approximation devices, proved here in the precise form needed. The claim here is strict-level completeness for the original marked causal resource image, including visible selectors, and its physical transfer below. The coefficient count $(n + 1)^D$, the number of original tests, and the algebraic degrees may be large. No polynomial complexity or intrinsic-dimension bound on this row-cube coefficient count is asserted. The independent adaptive test has the separate behavior-size bound of Theorem 6.3; its verification is a different cost.

5.1. A sharp two-time nonconvex example. The source has singleton reports, the control alphabet and actual mark are singleton, and the target emits $(0, 0)$ or $(1, 1)$ with probability $1/2$ each. At width one, a private simulator emits two independent bits with probabilities p, q . It cannot reread its first emitted bit.

Proposition 5.5 (A gap invisible to constant testing mixtures). *For this delayed experiment,*

$$\delta_{(1,1)}^p = \sqrt{2} - 1, \quad \delta_{(1,1),2}^h = 0, \quad \delta_{(1,1),r}^v = \sqrt{2} - 1 \quad (r \geq 1).$$

The private objective is the maximum of the four polynomials

$$0, \quad \frac{1}{2} - pq, \quad \frac{1}{2} - (1 - p)(1 - q), \quad p + q - 2pq.$$

Every constant mixture of these tests has minimum at most zero, whereas Corollary 5.3 certifies every rational $L < \sqrt{2} - 1$ at private width one.

Proof. Put $u = pq$ and $v = (1 - p)(1 - q)$. Direct total variation calculation gives $F = 1 - \min(u, 1/2) - \min(v, 1/2)$ and the displayed maximum. By Cauchy–Schwarz, $\sqrt{u} + \sqrt{v} \leq 1$. If $u \geq 1/2$, then $v \leq (1 - 1/\sqrt{2})^2$, so the sum of the two capped terms is at most $2 - \sqrt{2}$; the case $v \geq 1/2$ is symmetric. Otherwise set $a = \sqrt{u}$, $b = \sqrt{v}$. On $0 \leq a, b \leq 1/\sqrt{2}$, $a + b \leq 1$, the maximum of $a^2 + b^2$ is at a vertex of this polygon and equals $2 - \sqrt{2}$. Equality is attained at $p = q = 1/\sqrt{2}$. A hidden fair bit selects the two deterministic constant sequences and exactly reproduces the target. If it is exposed, variation is averaged over the selector fibers; each fiber has error at least the private minimum, which is itself attainable without a selector. Finally the target lies in the convex hull of the two private deterministic behaviors. The average discrepancy of any fixed test mixture at those behaviors is zero, so at least one has discrepancy at most zero. The strict polynomial certificates therefore detect information lost by constant convex separation. $\square$

6. ADAPTIVE TESTING ON THE OBSERVABLE BEHAVIOR IMAGE

The obstruction to linear randomization criteria at a fixed private register is the nonconvexity of the behavior image, not a failure of compactness. We give a variational formulation in which a test is an independent function of an observable candidate law. The construction does not give the simulator an extra random variable.

6.1. The testing game. Let B be a nonempty compact subset of a finite-dimensional affine space, and let $f_1, \dots, f_m$ be affine real functions, including the zero function. Put

$$F(b) = \max_{i \leq m} f_i(b), \quad \delta = \min_{b \in B} F(b).$$

For the finite comparison problem of Section 2, b is the array of discrepancies between the simulated and target marked laws, indexed by parameters, feedback trees and transcripts; f_i ranges over their signed event evaluations. In the visible case the array includes the selector and its target weights. Thus the target coordinate is part of the array when those weights vary. Private, fixed-hidden and fixed-visible signatures each give their own compact set B . No convex hull is taken.

An *adaptive test* is a continuous map $\kappa : B \rightarrow \Delta_m$, chosen before a candidate b . Its score against that candidate is

$$G(b, \kappa) = \sum_i \kappa_i(b) f_i(b).$$

This is a verification game: the selected candidate law is revealed to the evaluator of the already chosen function κ . It is not a one-observation statistical test, and does not make an unavailable parameter or mark observable to a physical controller.

Theorem 6.1 (Revealed-behavior adaptive representation). *For every such compact behavior image,*

$$\begin{aligned} \delta &= \sup_{\kappa \in C(B, \Delta_m)} \min_{b \in B} G(b, \kappa) \\ &= \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa \in C(B, \Delta_m)} \int_B G(b, \kappa) d\mu(b). \end{aligned} \tag{6.1}$$

The minimization over probability measures is attained. If Borel adaptive tests are allowed, the first supremum is attained as well. The continuous-test assertion is a supremum; continuous attainment is not needed or asserted.

Proof. Fix $\eta > 0$. The sets $U_i = \{b : f_i(b) > F(b) - \eta\}$ form an open cover of the compact metric space B . Choose a continuous partition of unity subordinate to this finite cover. The resulting test has $F(b) - \eta < G(b, \kappa) \leq F(b)$ for all b . Consequently the first supremum equals δ . The same uniform inequality shows, for every probability μ ,

$$\sup_{\kappa} \int G(b, \kappa) d\mu(b) = \int F d\mu.$$

Its minimum is δ , attained by a point mass at a minimizer of F . This also proves the displayed minimax exchange without an assumption that B is convex. Equivalently one can apply the usual compact-convex minimax theorem to $\mathcal{P}(B)$ and the convex class of continuous tests, with the pairing $\int G d\mu$. Finally choose the smallest index attaining $F(b)$. Its level sets are Borel, and its indicator vector is a Borel test with score exactly F . $\square$

The probability μ here describes a lottery over *revealed candidate laws*. The score is $\int G(b, \kappa) d\mu(b)$, not $G(\int b d\mu, \kappa)$. Hiding the candidate and replacing it by its barycenter is a different game; in general it lowers the value. This distinction is essential for a finite persistent-state budget.

6.2. A size bound in behavior dimension. Factor out directions annihilated by every $f_i - f_i(b_0)$, so that on the resulting difference space

$$\|v\|_* = \max_i |f_i(b_0 + v) - f_i(b_0)|$$

is a norm. Denote its dimension by a and let $R = \text{diam}_*(B)$. All conclusions below are unaffected by this observable quotient. The case $a = 0$ is a single constant comparison and requires one test. Suppose henceforth that $a \geq 1$, so $R > 0$.

Lemma 6.2 (A bounded behavior chart). *There exist $b_0 \in B$ and vectors $v_j \in B - b_0$, $1 \leq j \leq a$, forming a basis, such that $\|v_j\|_* \leq R$ and every $b \in B$ has coordinates $b = b_0 + \sum_j t_j v_j$ with $|t_j| \leq 2$. If B is a rational polynomial image of a product of simplexes, the points can be chosen rational and can be found effectively.*

Proof. Fix $b_0 \in B$. In any fixed coordinates on the difference space, let $D_0 > 0$ be the maximum absolute determinant of a vectors in $B - b_0$. Compactness and spanning give existence and positivity. Choose a tuple with determinant greater than $D_0/2$. Replacing its j th vector by $b - b_0$ multiplies the determinant by t_j ; maximality gives $|t_j| < 2$. The norm bound is the definition of R . For a rational polynomial image, take b_0 at rational rows. Rational row images are dense. An affine basis can be found by enumerating rational images and testing rational determinants; the true affine span and the maximum determinant are semialgebraic data. Quantifier elimination decides whether a proposed nonzero determinant exceeds half the maximum. Density therefore makes the search terminate. This is an effectiveness assertion, not a bit-complexity bound for finding the chart. $\square$

Set $b(x) = b_0 + \sum_j (4x_j - 2)v_j$ and $C = \{x \in [0, 1]^a : b(x) \in B\}$. The ambient cube is a coordinate box, not an enlargement of the feasible behavior set. Write $\ell_i(x) = f_i(b(x))$. Then

$$|\ell_i(x) - \ell_i(y)| \leq 4R \sum_j |x_j - y_j|. \quad (6.2)$$

For $n \geq 0$ let $\beta_{\alpha,n}$ be the tensor Bernstein basis of coordinatewise degree n . A feasible coefficient array has $c_{\alpha i} \geq 0$ and $\sum_i c_{\alpha i} = 1$ for every $\alpha \in \{0, \dots, n\}^a$. It specifies the independent test

$$\kappa_i^c(x) = \sum_{\alpha} c_{\alpha i} \beta_{\alpha,n}(x).$$

Define

$$d_n = \max_c \min_{x \in C} \sum_{\alpha, i} c_{\alpha i} \beta_{\alpha,n}(x) \ell_i(x). \quad (6.3)$$

Theorem 6.3 (Intrinsic adaptive degree and support bounds). *The maxima in (6.3) are attained and $d_n \leq d_{n+1} \leq \delta$. For $n \geq 1$,*

$$0 \leq \delta - d_n \leq \frac{4Ra}{\sqrt{n}}. \quad (6.4)$$

There is an array with at most $(n+1)^a$ nonzero coefficients, all equal to one, whose minimum score is at least $\delta - 4Ra/\sqrt{n}$. Its test has coordinatewise degree n , and its score degree at most $n+1$. Furthermore

$$d_n = \min_{\mu \in \mathcal{P}(C)} \sum_{\alpha} \max_i \int_C \beta_{\alpha,n}(x) \ell_i(x) d\mu(x), \quad (6.5)$$

and a minimizing measure can be supported on at most $(n+2)^a$ points. In particular a strict level $L < \delta$ admits an adaptive test of positive uniform margin with $n > (8Ra/(\delta - L))^2$ and at most $(n+1)^a$ occupied coefficient entries. These bounds depend on the observable image and its metric, not on the number of stochastic rows in a realization.

Proof. The coefficient space is a finite product of compact simplexes, and the minimum of a jointly continuous function over compact C is continuous in its coefficients. This proves attainment. Bernstein degree elevation replaces each coefficient by convex combinations of adjacent coefficients, preserving every simplex constraint and the represented function. It proves monotonicity. All scores are convex combinations of the $\ell_i(x)$, proving the upper bound by δ .

At each grid point α/n choose an index i_{α} maximizing ℓ_i there. The choice is made even at infeasible grid points; only the final score is minimized over C . Give that index coefficient one and

the others zero. For fixed x let A_j be independent $\text{Bin}(n, x_j)$ variables. If i_* maximizes at x ,

$$\begin{aligned} \max_i \ell_i(x) - \mathbb{E}\ell_{i_A}(x) &\leq \mathbb{E}\{\ell_{i_*}(x) - \ell_{i_*}(A/n) + \ell_{i_A}(A/n) - \ell_{i_A}(x)\} \\ &\leq 8R \sum_j \mathbb{E}|A_j/n - x_j| \leq \frac{4Ra}{\sqrt{n}}. \end{aligned}$$

Here the omitted middle difference is nonpositive, and the last bound uses $\text{Var}(A_j/n) \leq 1/(4n)$. This proves the rate and the sparse coefficient assertion.

For the last formula replace the minimum over C by a minimum over $\mathcal{P}(C)$. The pairing is bilinear and continuous, and both feasible sets are compact convex, so the compact minimax theorem [18] exchanges the extrema. Optimizing each coefficient simplex gives (6.5). The integrands are polynomials of coordinatewise degree at most $n+1$. Preserve all monomial moments of that degree by Carathéodory's theorem. There are $(n+2)^a$ monomials including the constant; their moment vectors lie in an affine hyperplane of dimension at most $(n+2)^a - 1$. A measure with at most $(n+2)^a$ atoms therefore has exactly the same objective. The strict-margin assertion follows from the rate with error less than $(\delta - L)/2$. $\square$

The score is a nonlinear function of behavior although each event functional is affine. This is the reason the measure in (6.5) cannot be replaced by its behavior barycenter. For $n=0$ it can: d_0 is precisely the constant-test convex relaxation. In the two-time unmarked example of Proposition 5.5, the target belongs to the convex hull of private behaviors, so $d_0 = 0$, whereas $\lim_n d_n = \sqrt{2} - 1$. Thus the hierarchy recovers a gap which every constant test misses.

Remark 6.4 (What the size bound measures). The bound counts the nonzero entries of the independent adaptive test and controls its behavior-coordinate degree. It does not bound the cost of verifying its minimum on the generally nonconvex set C , the bit lengths of a rational chart, or the size of a description of C . Distinct row presentations of the same metric behavior image satisfy the same existence bound, although their coefficient arrays need not coincide. For the private fixed-target law array one may use $R \leq 1$ with the maximum total-variation metric. For the visible signed-discrepancy array the uniform bound $R \leq 2$ is sufficient. The old row-cube positivity certificate is a separate verification engine, with its own coordinate-dependent complexity.

6.3. A three-weight certificate invisible to constant tests.

Proposition 6.5 (A finite intrinsic nonlinear witness). *Consider a private width-one simulator producing two independent bits, with probabilities p, q , for the target which assigns probability $1/2$ to each of 00, 11. Put $u = pq$, $v = (1-p)(1-q)$ and $J(t) = 3t^2 - 2t^3$. The three behavior-dependent weights*

$$\kappa_1 = J(v), \quad \kappa_2 = J(u), \quad \kappa_3 = 1 - J(u) - J(v)$$

are nonnegative and sum to one. Against the event discrepancies $1/2 - u, 1/2 - v, 1 - u - v$, respectively, their score G satisfies

$$G \geq \frac{2}{5} + \frac{9}{8960}. \quad (6.6)$$

The certificate uses cubic weights in observable probabilities, a nine-term nonnegative derivative identity, and seventeen positive rational Bernstein coefficients in one variable. Every constant mixture of event discrepancies has minimum at most zero.

Proof. We have $u + v \leq 1$. Since J is increasing on $[0, 1]$ and $J(1-t) = 1 - J(t)$, the weights are feasible. Their score is

$$G = 1 - u - v + J(u)(u - 1/2) + J(v)(v - 1/2).$$

Set $a = (p + q - 1)^2$ and $b = (p - q)^2$. Then $a, b \geq 0$ and $a + b \leq 1$. Direct expansion gives

$$\begin{aligned} 64G &= 27 - 12a + 38a^2 - 20a^3 - a^4 + 36b - 60ab + 36a^2b + 4a^3b \\ &\quad + 6b^2 - 12ab^2 - 6a^2b^2 - 4b^3 + 4ab^3 - b^4. \end{aligned}$$

Let $c = 1 - a - b$. Another polynomial identity, valid when $a + b + c = 1$, is

$$\begin{aligned} 16\partial_b G &= 4a^3 + 6a^2c + 12ab^2 + 24abc + 12ac^2 \\ &\quad + 8b^3 + 30b^2c + 30bc^2 + 9c^3 \geq 0. \end{aligned}$$

Thus $G(a, b) \geq G(a, 0)$. The univariate polynomial

$$G(a, 0) - \frac{2}{5} = \frac{7}{320} - \frac{3}{16}a + \frac{19}{32}a^2 - \frac{5}{16}a^3 - \frac{1}{64}a^4$$

has degree-16 Bernstein coefficients $z_k/349440$, where

$$\begin{aligned} (z_0, \dots, z_{16}) &= (7644, 3549, 1183, 351, 855, 2494, 5064, 8358, \\ &\quad 12166, 16275, 20469, 24529, 28233, 31356, 33670, 34944, 34944). \end{aligned}$$

The Bernstein basis is nonnegative and sums to one. Its smallest coefficient is $351/349440 = 9/8960$, proving (6.6). All identities are rational polynomial identities; no numerical minimization is used. Finally the target is the barycenter of the two feasible deterministic laws 00 and 11. An affine mixture of signed event discrepancies has average zero at these two laws and hence minimum at most zero. The empty event supplies the reverse constant-test bound zero. $\square$

For comparison, the affine weights $v, u, 1 - u - v$ already give the smaller bound $3/8$, with the exact identity

$$G_{\text{affine}} - \frac{3}{8} = \frac{1}{2}(p - q)^2 + 2(p - 1/2)^2(q - 1/2)^2.$$

Neither certificate changes the private feasible set. The cubic certificate is an independent test chosen once for the whole behavior image, not a mixture chosen anew by an optimizer's unknown row values. Its evaluation does use the revealed behavior, as specified in the verification game. The 17-coefficient positivity check is classical; the distinction from constant tests is the behavior-dependent feasible weight family and its resource-sensitive minimum.

The infinite-dimensional representation above holds for continuous f_i on any compact metric space: its proof uses a partition of unity, not causal factorization. The measure minimum is attained at a Dirac mass at a minimizer of F . The finite-level polynomial theorem has a different content. It is an existence-size bound for revealed-law verification; the reset-audit theorem of Section 4 imposes an explicit finite-data restriction.

6.4. Strict certificates and effective physical input. Given rational finite instruments, a rational chart and the sparse array constructed above, its strict minimum can be certified by real quantifier elimination on the original row constraints. Alternatively, pull the score back through the original polynomial row map and certify its strict positivity using degree-elevated Bernstein coefficients. The pullback changes the verification degree and is not covered by the behavior-coordinate bound. Generic quantifier elimination also decides the exact finite semialgebraic boundary; this classical observation does not assert that a finite nonnegative Bernstein expansion exists at $L = \delta$.

Proposition 6.6 (Uniform effective presentation input). *Suppose a single algorithm, on input j , returns rational finite instruments E_j, F_j and nonnegative rational error enclosures a_j, c_j for two-sided stateless marked feedback presentations of E, F . Suppose $a_j + c_j \rightarrow 0$. All alphabets, widths, selector laws and finite subroutines must be computable uniformly from j and the requested*

signature. Then, for every positive rational η , an algorithm returns a rational interval of width below η containing $\delta_b(E, F)$ at the same signature.

Proof. Enumerate j until $a_j + c_j < \eta/8$. This terminates without a known rate because the rational enclosures themselves tend to zero. For the selected finite problem, enumerate rational same-budget upper tables and strict rational lower certificates, dovetailing all finite checks; include the always valid lower bound zero. Compactness, density and strict completeness provide an interval of width below $\eta/2$. Enlarge it by $a_j + c_j$ at both ends using stateless comparison. The enlarged width is below $3\eta/4$. Every enumeration and comparison is uniform by the hypotheses. A mere sequence of individually computable presentations without a uniform enumeration is not an input satisfying this proposition. $\square$

7. AN EXACT ALGORITHM FOR THE OBSERVABLE CHART

The chart in Lemma 6.2 can be found without taking its dimension as an oracle. We state the input and output explicitly. Rational polynomials are encoded by finite lists of monomial exponent vectors and rational coefficients in reduced integer form. A product of simplexes is encoded by the number of entries in each row. Eliminating the last entry of every nontrivial row leaves a full-dimensional rational polytope $\Omega \subset \mathbb{R}^r$. Zero-dimensional factors are omitted.

Proposition 7.1 (Effective observable quotient and pullback). *The input consists of the preceding simplex description, a rational polynomial map $b : \Omega \rightarrow \mathbb{R}^N$, and finitely many rational affine functionals f_i on $\mathbb{R}^N$. There is a terminating algorithm which returns its observable affine dimension a , a rational basis and coordinate map for the observable quotient, and, when $a > 0$, rational parameter points $q_0, q_1, \dots, q_a$ with the following properties. With*

$$z(q) = (f_i(b(q)) - f_i(b(q_0)))_i, \quad v_j = z(q_j),$$

the vectors v_j are a basis of the observable difference space, and all coordinates $t(q)$ in that basis satisfy $|t_j(q)| < 2$. The output also contains the rational polynomial cube map $x(q) = (t(q) + 2)/4$, a rational upper bound for the observable diameter, and the exact pullback of any subsequently supplied rational polynomial in x . The algorithm outputs the universal determinant formula it has decided. It makes no polynomial-time or coefficient-height claim.

Proof. Choose a rational relative interior point q_0 of the input simplexes. Expand the vector polynomial $z(q)$ in the free row coordinates. Let V be the rational span of all its coefficient vectors, including its constant coefficient. This equals $\text{span } z(\Omega)$: a linear functional vanishes on $z(\Omega)$ exactly when its polynomial vanishes on the nonempty interior of Ω , hence identically, hence on every coefficient. Gaussian elimination therefore gives $a = \dim V$ and a rational basis matrix U . Choosing independent rows gives a rational left inverse L with $LU = I_a$. Thus $y(q) = Lz(q)$ provides exact quotient coordinates. If $a = 0$, all discrepancies are constant and no chart is required.

For $a > 0$ let

$$D_0 = \max_{u_1, \dots, u_a \in \Omega} |\det(y(u_1), \dots, y(u_a))|.$$

Compactness gives a maximum, and spanning gives $D_0 > 0$. Enumerate rational tuples $(q_1, \dots, q_a)$ in Ω^a . Membership and their determinant d are decided by rational arithmetic. For $d \neq 0$, decide the first-order real formula

$$\forall u_1, \dots, u_a \in \Omega : \quad \det(y(u_1), \dots, y(u_a))^2 < 4d^2. \quad (7.1)$$

Quantifier elimination terminates for each formula. Density of rational points and continuity of determinant imply that some enumerated tuple has $|d| > D_0/2$. For such a tuple the strict formula is true, so the search terminates. No computation of an exact algebraic maximizer is needed.

Invert the rational matrix with columns $y(q_j)$ to obtain $t(q)$. Cramer's rule and (7.1) give $|t_j(q)| < 2$ for every q . In the observable norm $\|z\|_* = \max_i |z_i|$, each v_j has norm at most the diameter of $z(\Omega)$. An upper bound for that diameter can be produced, for example, by bounding each free coordinate in $[0, 1]$ and summing the absolute polynomial coefficients for $f_i(b(q)) - f_i(b(q'))$. This bound is rational and computable. Rational matrix multiplication and substitution give $x(q)$ and every requested pullback. They also give its full sparse coefficient encoding. All searches have now been specified by finite encodings and terminating operations. $\square$

There is a small but important distinction between the quotient and a projection onto a smaller set of simulator rows. The former identifies exactly the directions invisible to the chosen event family; the original row constraints remain present in every universal positivity test. The chart need not identify equivalent tasks after a change of marks or of information signature. Those are changes to the comparison problem.

8. A MARKED RESOURCE GAP GENERATED BY A BINARY COLLISION

We now realize a private–hidden–visible separation in a physical experiment whose information vanishes when the collision is removed. The controller chooses a later sensor gate after its first report. The gate controls acquisition, not the mechanical collision law. Both the source and the target carry the same actual, unobserved mark.

8.1. The exact marked two-report experiment. Let $B, W \in \{0, 1\}$ have the joint distribution

$$\mathbb{P}(B = b, W = w) = \begin{cases} 5/16, & b = w, \\ 3/16, & b \neq w. \end{cases} \quad (8.1)$$

The blind source E_* has two constant reports and actual mark W . The target F_* reports $Y_1 = B$, receives $A \in \{0, 1\}$ after that report, and reports $Y_2 = AB$. The feedback tester may choose A as any function of its earlier report. A width-one private simulator cannot reread its own first emission, except through the current gate A . Hidden selectors persist but are not revealed; visible selectors are included in the joint target law independently of (B, W) . Put

$$d_* = \sqrt{5/2} - 9/8.$$

Theorem 8.1 (An exact marked causal separation). *At the width profile $(1, 1)$,*

$$\delta^p(E_*, F_*) = d_*, \quad \delta_2^h(E_*, F_*) = 1/8, \quad \delta_r^v(E_*, F_*) = d_* \quad (r \geq 1). \quad (8.2)$$

The hidden upper bound is realized by two deterministic tables. The private upper bound is realized by two independent Bernoulli variables with parameter $\sqrt{5/8}$, followed by the gate. These bounds hold against every feedback rule, not just open-loop gates.

Proof. First choose the constant gate $A = 1$. The two private emissions are independent Bernoulli variables with parameters p, q and are independent of W . Let

$$u = pq, \quad v = (1 - p)(1 - q), \quad a = 3/8, \quad b = 5/8.$$

The overlap of the target and simulated marked laws is $H(u) + H(v)$, where

$$H(t) = \frac{1}{2} \{\min(t, a) + \min(t, b)\}.$$

Indeed the target is supported on equal report pairs, and its two W -masses at each equal pair are $3/16$ and $5/16$. Total variation is one minus this overlap.

Cauchy–Schwarz gives $\sqrt{u} + \sqrt{v} \leq 1$. By symmetry suppose $u \geq v$. If $u \leq 1/4$ the overlap is at most $1/2$. If $u \geq 1/4$, then $v \leq (1 - \sqrt{u})^2 \leq 1/4 < a$, so the overlap is at most

$$H(u) + (1 - \sqrt{u})^2.$$

This function is convex on each of $[1/4, a]$ and $[a, b]$ and nonincreasing on $[b, 1]$. Its maximum is therefore attained at $1/4$, a or b . The values at a and b are $1 + 2a - 2\sqrt{a}$ and $3/2 + b - 2\sqrt{b}$ respectively. The second is larger, since

$$(3/2 + b - 2\sqrt{b}) - (1 + 2a - 2\sqrt{a}) = 3/8 - \frac{1/2}{\sqrt{a} + \sqrt{b}} > 0,$$

using $\sqrt{a} > 3/5$, $\sqrt{b} > 3/4$. It also exceeds $1/2$. Taking $p = q = \sqrt{b}$ attains this maximum, and hence the minimum error is $1 - (3/2 + b - 2\sqrt{b}) = d_*$.

For any hidden selector, the all-on report pair is still independent of W . If its probabilities of 11 and 00 are u, v , then $u + v \leq 1$. Concavity and monotonicity of H give

$$H(u) + H(v) \leq 2H((u + v)/2) \leq 2H(1/2) = 7/8.$$

A fair hidden bit J , emitting J twice, attains this overlap. This requires precisely two deterministic selector tables and gives error $1/8$. In particular hiding a seed does not erase the actual-mark obstruction.

These all-on bounds are lower bounds for the feedback problem. For the upper bounds first generate the corresponding all-on pair (X_1, X_2) , emit X_1 , and after receiving A emit AX_2 . For a deterministic feedback $A = g(X_1)$ the full gated marked transcript is the pushforward of the all-on marked pair. Applying the same map to the target pair cannot increase total variation. The construction is executable at width one: the second fresh Bernoulli variable is independent in the private case, and the bit J is the priced persistent hidden selector in the hidden case. Randomized feedback follows by conditioning on its independent random table.

Finally, in the visible case the two compared laws are

$$\sum_j \lambda_j \delta_j \otimes P_{g_j}^{q_j} \quad \text{and} \quad \sum_j \lambda_j \delta_j \otimes Q_{g_j}.$$

The target selector is independent of (B, W) . The all-on gate in every fiber has error at least d_* in every private fiber. Joint variation is the weighted sum of these fiber errors. Ignoring the visible selector and using the private optimizer attains d_* . $\square$

8.2. A full-dimensional collision preparation. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$ with contact distance d . The center of mass $Q = (q_1 + q_2)/2$ ranges over any fixed rational position box. Write $r = q_1 - q_2$, $g = v_1 - v_2$, and $V = (v_1 + v_2)/2$. Conditional on (B, W) in (8.1), take the uniform product preparation on

$$\begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], & (8.3) \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

The boxes have positive volume in the full phase space. The actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$; total momentum conservation retains it. The informative quantity will instead be the transverse velocity created at a collision. Put

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector which is identically zero on the incoming transverse-velocity range.

Lemma 8.2 (Uniform transverse collision signal). *For every preparation in (8.3) and every $d \in [9/10, 11/10]$, exactly one collision occurs before time 3. Its time t_c lies in $(4/5, 3/2)$, and afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4. \quad (8.4)$$

For $0 < d \leq 2/5$ no collision occurs before time 3, and the detector is zero throughout that interval.

Proof. Up to a collision $r(t) = r + tg$. For $t \leq 4/5$, its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At time $3/2$ the absolute coordinate bounds are $1/4$, $113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. Thus a first contact occurs in the stated interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$ as well. The first contact therefore has $n_x < 0$, where $n = r(t_c)/d$. The entry contact is strictly incoming. The bounds above give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq \frac{87}{220}.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The equal-mass elastic collision rule is

$$v'_1 = V + g/2 - (g \cdot n)n, \quad g' = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v'_{1,y} > -3/200 + \frac{1443}{1000} \frac{87}{220} > 1/2.$$

The relative velocity is outgoing, $r(t_c) \cdot g' > 0$, so the squared separation increases strictly for all later times. There are only two spheres and no walls; no second collision is possible. This proves (8.4). For $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. Thus there is no collision. The incoming $|v_{1,y}| \leq 3/200$ then makes h zero. $\square$

At time 2 the physical target $F_{d,\sigma}$ reports

$$Y_1 = \mathbf{1}_{\{h(v_{1,y}(2)) + \sigma \xi_1 > 0\}}.$$

It then receives a gate $A \in \{0, 1\}$, and at time 3 reports

$$Y_2 = A \mathbf{1}_{\{h(v_{1,y}(3)) + \sigma \xi_2 > 0\}},$$

where the ξ_i are independent standard normal variables, independent of the preparation. The source E_d has the same preparation, evolution, control interface and actual mark, but emits two constant reports. The gates do not change the particle motion. This is a feedback acquisition experiment: taking $A = 0$ suppresses later information and taking $A = 1$ exposes another noisy observation of a collision-generated quantity.

Theorem 8.3 (Collision-sensitive physical resource separation). *Let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/12$. At retained width $(1, 1)$ the physical deficiencies obey, uniformly in d ,*

$$\begin{aligned} |\delta^p(E_d, F_{d,\sigma}) - d_*| &< 1/300, & |\delta_r^y(E_d, F_{d,\sigma}) - d_*| &< 1/300 \quad (r \geq 1), \\ |\delta_2^h(E_d, F_{d,\sigma}) - 1/8| &< 1/300. \end{aligned} \quad (8.5)$$

In particular private and visible error exceed $9/20$, while hidden two-selector error is less than $13/100$. More generally the errors in these three comparisons are at most $2\Phi(-1/(4\sigma))$, which tends to zero as $\sigma \downarrow 0$. For $0 < d \leq 2/5$ all three deficiencies are zero. The interacting and noninteracting regimes use the same preparation and detector. The conclusions are unchanged by simultaneously scaling positions, diameter and observation times by any positive factor, with velocities and detector noise unchanged.

Proof. Couple the ideal and physical target by the same (B, W) and the same feedback random table. Lemma 8.2 bounds each potential bit error by $\Phi(-1/(4\sigma))$. With probability at least $1 - 2\Phi(-1/(4\sigma))$ both potential reports agree with their ideal bits; the gates and all emitted transcripts then agree as well. This gives a marked feedback variation bound uniform over all gates. The channels in both directions are the identity on the binary report alphabet and have one retained state. The blind source is already exactly E_* . Stateless comparison therefore changes each fixed-budget deficiency by at most the stated error, without adding memory or hiding an exposed selector.

For completeness, the elementary Gaussian tail bound $\Phi(-x) \leq e^{-x^2/2}/(x\sqrt{2\pi})$ at $x = 3$ gives $\Phi(-3) < 1/600$: use $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$. Hence the total error is below $1/300$. The inequalities $d_* - 1/300 > 9/20$ and $1/8 + 1/300 < 13/100$ follow by squaring positive rational bounds.

When $d \leq 2/5$, both potential reports are independent fair noise bits and are independent of W . A private width-one simulator draws those two bits and applies the current gate, exactly reproducing every feedback transcript. Hidden and visible versions can ignore their selectors. Finally elastic scattering is invariant under $(q, d, t) \mapsto (sq, sd, st)$ with v unchanged. This proves the scaling statement; it is geometric similarity, not a particle-number or Boltzmann–Grad limit. $\square$

The theorem combines three effects. Collisions create the sign of the informative transverse velocity. A gate chosen after the first report changes future information. A retained hidden bit, but not a visible one, permits temporal correlation without charging the width-one register. The actual mark remains partially inaccessible in every signature. The conserved longitudinal momentum labels that mark; it does not generate the observed transverse signal.

8.3. Uniformly effective finite presentations. The binary physical interface is already finite, but its probabilities are not generally rational. We make their effective approximation explicit rather than treating individual computability as a uniform algorithm.

Proposition 8.4 (Collision-chart rational presentations). *For computable $d \in [9/10, 11/10]$ and $\sigma > 0$, one algorithm produces rational marked causal instruments and certified two-sided stateless errors tending to zero for $(E_d, F_{d,\sigma})$. On a compact computable parameter interval with $\sigma \geq \sigma_0 > 0$, the errors and the construction are uniform. Thus Proposition 6.6 applies at every fixed finite state and selector signature, not only at $(1, 1)$.*

Proof. On every preparation box the entry time and outgoing velocity are

$$t_c = \frac{-r \cdot g - \sqrt{(r \cdot g)^2 - |g|^2(|r|^2 - d^2)}}{|g|^2}, \quad n = (r + t_c g)/d, \quad v'_1 = V + g/2 - (g \cdot n)n.$$

The positive square root is separated from zero: it equals $-d(g \cdot n) > (9/10)(1443/1000)$. All denominators are likewise bounded away from zero. These explicit bounds and the compact boxes allow interval differentiation of the displayed formulas, producing a rational common Lipschitz upper bound C_{geo} for $v'_{1,y}$ in the packet variables and d . This computation terminates because only addition, multiplication, division by separated positive quantities and a separated square root occur.

Conditional on the microscopic preparation the two potential all-on bits are independent Bernoulli with probability $p = \Phi(h(v'_{1,y})/\sigma)$. Their four probabilities are p^2 , $p(1-p)$, $p(1-p)$ and $(1-p)^2$. The normal distribution function is uniformly computable on bounded intervals, for example by integrating its convergent exponential series with a remainder bound. Its Lipschitz constant is less than one. Together with the 1-Lipschitz detector and the geometric bound, this yields a computable Lipschitz constant at most $2C_{\text{geo}}/\sigma_0$ for these four integrands on fixed-noise compact parameter sets. Equal-box Riemann sums of mesh h therefore have explicit error at most a computable constant times h , plus the chosen rational evaluation error. Parameter uncertainty is included by its interval enclosure.

Integrate the four all-on probabilities separately on each packet, with the exact rational weights (8.1). For each W , round its conditional probability vector to a rational simplex point on a sufficiently fine grid; the rounding error is bounded by the sum of the coordinate errors and the grid mesh. Keep $\mathbb{P}(W = 0) = \mathbb{P}(W = 1) = 1/2$ exact. Sample this finite joint law once at initialization, emit its first bit, and gate its second bit. This is a finite latent realization of the target instrument, not a simulator with extra free state. It makes every causal prefix consistent

and never divides by a possibly small prefix probability. Every gated law is a pushforward of the all-on law, so the same marked variation bound holds uniformly under feedback. Identity report channels give both stateless directions. Choosing summable evaluation and mesh errors produces the asserted uniform sequence. $\square$

9. UNIFORM STABILITY OF OPTIONAL PROJECTIONS

Marked comparison controls more than terminal bounded risk when the compared laws share their latent variable. We prove a whole-process estimate for the resulting posterior martingales. The distinction between a probabilistic coupling and an executable causal channel is important in this section.

Let Z take values in a standard Borel space and let Y take values in a standard Borel canonical path space on $[0, T]$, with a right-continuous observation filtration. Consider equivalent joint probability measures P, Q on (Z, Y) , with the same Z -marginal ν . Use the usual completions, which agree for equivalent output laws. For a bounded measurable latent variable $f(Z)$ write

$$M_t^P = \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y], \quad M_t^Q = \mathbb{E}_Q[f(Z) \mid \mathcal{F}_t^Y].$$

Take their right-continuous martingale versions. All suprema below include T and can be evaluated on a countable dense time set.

Theorem 9.1 (Marked variation controls the whole posterior process). *Let $\varepsilon = \|P - Q\|_{\text{TV}}$ and $|f| \leq 1$. When both posterior functions are evaluated on the same canonical output path,*

$$\mathbb{E}_Q \sup_{t \leq T} |M_t^P(Y) - M_t^Q(Y)|^2 \leq 80\varepsilon. \quad (9.1)$$

There is a coupling of (Z, Y^P) and (Z, Y^Q) which retains exactly the same latent variable and satisfies

$$\mathbb{P}(Y^P \neq Y^Q) \leq \varepsilon, \quad \mathbb{E} \sup_{t \leq T} |M_t^P(Y^P) - M_t^Q(Y^Q)|^2 \leq 84\varepsilon. \quad (9.2)$$

The coupling is not asserted to be causal or to have finite state. The variation bound itself, not this coupling, is used for resource transport.

Proof. Let $D = dP/dQ$, $D_t = \mathbb{E}_Q[D \mid \mathcal{F}_t^Y]$, and set

$$A_t = \mathbb{E}_Q[(D - 1)f \mid \mathcal{F}_t^Y], \quad B_t = D_t - 1.$$

Bayes' formula gives

$$M_t^P - M_t^Q = \frac{A_t - B_t M_t^Q}{1 + B_t}.$$

Since $\mathbb{E}_Q|D - 1| = 2\varepsilon$, both martingales A, B have terminal absolute expectation at most 2ε . Doob's weak L^1 inequality gives, for $0 < u \leq 2$,

$$\begin{aligned} Q\{\sup_t |M_t^P - M_t^Q| > u\} &\leq Q\{\sup_t |B_t| > 1/2\} + Q\{\sup_t |A_t| > u/4\} + Q\{\sup_t |B_t| > u/4\} \\ &\leq 4\varepsilon + 16\varepsilon/u. \end{aligned}$$

Integrating $2u$ times this bound on $(0, 2)$ gives $16\varepsilon + 64\varepsilon = 80\varepsilon$, proving (9.1). No tightness conclusion is being inferred from finite-dimensional convergence.

Disintegrate both joint laws over their common latent marginal. Measurable maximal coupling of the two output kernels, conditional on Z , has disagreement probability

$$\int \|P(dy \mid z) - Q(dy \mid z)\|_{\text{TV}} d\nu(z) = \|P - Q\|_{\text{TV}}.$$

Such a coupling can be constructed by using the minimum of the two kernel densities with respect to their sum, then their residual kernels. It is measurable on standard Borel spaces. On the

event of identical full output paths use (9.1) with the Q -marginal; on its complement the squared difference is at most four. This proves (9.2). The maximal coupling may depend on the full output path; it has no asserted online implementation. $\square$

Corollary 9.2 (Changing-filtration convergence from marked comparison). *Let P_n and P be equivalent joint laws with the same latent marginal ν , and $\|P_n - P\|_{\text{TV}} \rightarrow 0$. They admit a common probability-space realization $(Z, Y, Y^1, Y^2, \dots)$ such that, for every $f \in L^2(\nu)$,*

$$\mathbb{E} \sup_{t \leq T} \left| \mathbb{E}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}[f(Z) \mid \mathcal{F}_t^Y] \right|^2 \rightarrow 0.$$

The same realization gives convergence uniformly in time in probability for $f \in L^1(\nu)$. It also permits $f_n \rightarrow f$ in $L^2(\nu)$, respectively in $L^1(\nu)$, in the first conditional expectation.

Proof. Glue the conditional maximal couplings of (Z, Y^n) with the fixed reference (Z, Y) by taking their conditional kernels independently on a countable product. The coupling does not depend on f . For the truncation $f^{(K)} = (-K) \vee (f \wedge K)$, Theorem 9.1 bounds the middle difference in the L^2 norm of the supremum by $K\sqrt{84\varepsilon_n}$. Doob's L^2 inequality bounds each truncation tail by $2\|f - f^{(K)}\|_{L^2(\nu)}$. The triangle inequality therefore gives

$$\left\| \sup_t |M_t^{P_n, f} - M_t^{P, f}| \right\|_2 \leq 4\|f - f^{(K)}\|_2 + K\sqrt{84\varepsilon_n}.$$

First let $n \rightarrow \infty$, then $K \rightarrow \infty$. For L^1 use instead the weak maximal inequality for each tail and the bounded-case estimate for the middle term. An additional difference $f_n - f$ is handled by the same maximal inequalities under P_n , whose latent marginal remains ν . $\square$

9.1. A microscopic sufficient condition. Let $(\Phi_t)_{t \in \mathbb{R}}$ be an almost-everywhere regular hard-sphere flow on a labeled phase space, with invariant probability μ . Fix a prepared law ν with $d\nu/d\mu \leq C$, and bounded real observables h, h_n . These may be collision-sensitive tagged observables. Put $\eta_n = \|h_n - h\|_{L^2(\mu)}$. With a common latent initial state $Z \sim \nu$, consider the continuous observation laws

$$\begin{aligned} Y_t &= \int_0^t h(\Phi_s Z) ds + \sigma B_t, \\ Y_t^n &= \int_0^t h_n(\Phi_s Z) ds + \sigma B_t^n, \quad \sigma > 0, \end{aligned}$$

where the Brownian noise is independent of Z in each model. The notation does not prescribe a synchronous coupling.

Theorem 9.3 (Fixed-flow observation-approximation limit). *The joint latent-output laws are equivalent and satisfy*

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \eta_n. \quad (9.3)$$

For every $|f(Z)| \leq 1$ they admit a latent-preserving coupling with

$$\mathbb{E} \sup_{t \leq T} |M_t^{P_n, f} - M_t^{P, f}|^2 \leq \frac{42\sqrt{CT}}{\sigma} \eta_n. \quad (9.4)$$

In particular $\eta_n \rightarrow 0$ supplies whole-process optional-projection convergence, not merely convergence at finitely many times. This applies when h_n are orbit-mollified graph-core approximants to any bounded $h \in L^2(\mu)$.

Proof. Conditional on Z , the signals are deterministic bounded drifts. The exponential change-of-measure identity for Brownian motion gives

$$D_{\text{KL}}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |(h_n - h)(\Phi_s Z)|^2 ds \leq \frac{CT}{2\sigma^2} \eta_n^2.$$

The last inequality uses $\nu \leq C\mu$ and invariance of μ ; it does not require mixing, differentiability across a collision, or a claim that a tagged velocity belongs to the generator domain. Bounded drifts ensure equivalence and justify the exponential identity. Pinsker's inequality gives (9.3). Applying Theorem 9.1 proves (9.4).

For the last assertion choose a nonnegative smooth approximate identity ρ_n on the time line and set $h_n = \int \rho_n(s) h \circ \Phi_s ds$. Strong continuity of the Koopman group in $L^2(\mu)$ gives $\eta_n \rightarrow 0$ and $\|h_n\|_\infty \leq \|h\|_\infty$. Repeated integration by parts puts h_n in every power domain of the generator, with derivatives obtained from $\rho_n^{(k)}$. This is the orbit-core construction already used in the microscopic domain theory; the new conclusion is the uniform posterior estimate. $\square$

Sampling these observation paths at finitely many checkpoints and applying a common causal output postprocessing are contractions in marked variation. Thus the identity report comparison transfers every bounded finite-register risk at its original state and selector budgets with error bounded by (9.3). That is the resource-comparison consequence. The latent-preserving coupling in (9.4) has a different role: it realizes convergence of the changing observation filtrations and is not used as a finite-register simulator.

The theorem discharges a sufficient conditional-law hypothesis for optional projection in a genuine microscopic observation class. The constants depend on the fixed preparation, time and noise level. No particle-uniform kinetic estimate, changing microscopic path limit, limiting stochastic-exponential identification or nonlinear BSDE limit is implied without its additional hypotheses. The whole-process estimate is what replaces an unsupported appeal to Doob's inequality as a tightness argument.

10. CHANGING TRAJECTORIES, LIKELIHOODS, AND CONDITIONAL CONSUMERS

10.1. A common preparation, not a common flow. Let $(\mathcal{Z}, \nu)$ be a standard Borel probability space, and let $H, H_n : [0, T] \times \mathcal{Z} \rightarrow \mathbb{R}$ be jointly measurable, with $|H|, |H_n| \leq M$. They may be observations of different flows on different phase spaces, pulled back by specified preparation maps from $\mathcal{Z}$. No common microscopic generator or invariant measure is assumed. Set

$$\eta_n^2 = \int_{\mathcal{Z}} \int_0^T |H_n(t, z) - H(t, z)|^2 dt d\nu(z).$$

On $\mathcal{Z} \times C([0, T])$ let R be the law of independent $Z \sim \nu$ and $Y = \sigma B$, where $\sigma > 0$. Define

$$L_t^n = \exp \left\{ \frac{1}{\sigma} \int_0^t H_n(s, Z) dB_s - \frac{1}{2\sigma^2} \int_0^t H_n(s, Z)^2 ds \right\}, \quad (10.1)$$

$$dP_n = L_T^n dR. \quad (10.2)$$

The reference model P, L uses H . Conditional on Z , these are Gaussian shifts, so under P_n the observed path has law $Y_t = \int_0^t H_n(s, Z) ds + \sigma B_t^n$. Boundedness gives true strictly positive density martingales. Write $\mathcal{F}_t$ for the usual canonical observation filtration and $\mathcal{H}_t = \sigma(Z, B_s : s \leq t)$ for the larger reference filtration. Equivalent laws have the same completed observation stopping times.

Theorem 10.1 (Likelihood and posterior transport for changing paths). *The measures P_n, P, R are equivalent and have common Z -marginal ν . With $A = M^2 T / \sigma^2$ and $\varepsilon_n = \min\{1, \eta_n / (2\sigma)\}$,*

one has

$$\|P_n - P\|_{\text{TV}} \leq \varepsilon_n, \quad (10.3)$$

$$\mathbb{E}_R \sup_{t \leq T} |L_t^n - L_t|^2 \leq \frac{8}{\sigma^2} e^{3A} \eta_n^2. \quad (10.4)$$

For $|f(Z)| \leq K$ put

$$N_t^{n,f} = \mathbb{E}_R[L_T^n f(Z) \mid \mathcal{F}_t], \quad \ell_t^n = N_t^{n,1}.$$

Then

$$\mathbb{E}_R \sup_t |N_t^{n,f} - N_t^f|^2 \leq \frac{8K^2}{\sigma^2} e^{3A} \eta_n^2. \quad (10.5)$$

In particular this applies to the projected likelihood ℓ^n . For every $|f| \leq 1$, the posterior functions satisfy the same-path bound $80\varepsilon_n$ of Theorem 9.1, and there is one latent-preserving coupling, independent of f , on which

$$\mathbb{E} \sup_t |\mathbb{E}_{P_n}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y]|^2 \leq 84\varepsilon_n. \quad (10.6)$$

If $\eta_n \rightarrow 0$ and deterministic microscopic paths $X_n(Z) \rightarrow X(Z)$ in a specified Polish path metric in ν -probability, then on this coupling the microscopic paths, the full and projected likelihood processes evaluated on their own observation paths, and each bounded posterior process converge jointly in probability in that metric and uniformly in time for the last three coordinates.

Proof. Conditional Gaussian change of measure gives

$$\text{KL}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |H_n - H|^2 dt = \frac{\eta_n^2}{2\sigma^2}.$$

The same calculation proves conditional equivalence, and the conditional density has mean one, so the preparation marginal is unchanged. Integrating and applying Pinsker proves (10.3).

For the quantitative density estimate, condition first on $Z = z$ and write $D_t = L_t^n - L_t$. The stochastic differential equation is

$$dD_t = \sigma^{-1} \{D_t H_n(t, z) + L_t(H_n - H)(t, z)\} dB_t.$$

The conditional second moment of L_t is at most e^A . Itô's isometry and $(u + v)^2 \leq 2u^2 + 2v^2$ therefore imply

$$\mathbb{E}_R[D_t^2 \mid z] \leq \frac{2M^2}{\sigma^2} \int_0^t \mathbb{E}_R[D_s^2 \mid z] ds + \frac{2e^A}{\sigma^2} \int_0^t |H_n - H|^2(s, z) ds.$$

Gronwall, followed by integration in z , gives $\mathbb{E}_R D_T^2 \leq (2/\sigma^2) e^{3A} \eta_n^2$. The difference is a square-integrable $\mathcal{H}_t$ -martingale. Doob's L^2 inequality proves (10.4). Applying the same maximal inequality to the $\mathcal{F}_t$ -martingale $\mathbb{E}_R[(L_T^n - L_T)f \mid \mathcal{F}_t]$ proves (10.5). These are direct process estimates, not a claim that finite-dimensional convergence alone implies tightness.

The posterior assertions follow from Theorem 9.1 and its countable conditional-coupling construction. For the last claim, on the event $Y^n = Y$ the two likelihood functions are evaluated on the same (Z, Y) . Their uniform difference tends to zero in R -probability by the estimates just proved, and therefore in P -probability by absolute continuity of the fixed P with respect to R . The complementary event has probability at most ε_n . The same argument applies to ℓ^n , and (10.6) handles the posterior. The microscopic convergence holds on any coupling retaining the same Z . A finite union bound proves joint convergence. Stochastic integral versions are chosen measurably on the canonical product space; equivalence makes these common versions valid under every law in the countable sequence. $\square$

For $f \in L^2(\nu)$ the posterior conclusion follows by truncation exactly as in Corollary 9.2; it also permits $f_n \rightarrow f$ in $L^2(\nu)$. Boundedness of H_n is being used for the uniform likelihood second moments. It is not replaced by an unverified uniform-integrability assertion in a singular or zero-noise limit.

10.2. The projected stochastic exponential. Let $\hat{H}_t^n = \mathbb{E}_{P_n}[H_n(t, Z) \mid \mathcal{F}_t]$, using a predictable version for Lebesgue-almost every t . Put $\theta_t^n = \hat{H}_t^n / \sigma$.

Proposition 10.2 (Identification before passage to the limit). *The projected likelihood and the innovation satisfy*

$$\ell_t^n = \exp \left\{ \int_0^t \theta_s^n dB_s - \frac{1}{2} \int_0^t (\theta_s^n)^2 ds \right\}, \quad (10.7)$$

$$I_t^n = B_t - \int_0^t \theta_s^n ds. \quad (10.8)$$

Here I^n is a P_n -Brownian motion in $\mathcal{F}_t$. In particular the limit of the projected likelihoods in (10.5) is the projected stochastic exponential of the limiting model. Change of measure commutes with this limit on bounded measurable path tests.

Proof. Project $L_t^n = 1 + \int_0^t L_s^n H_n(s, Z) dB_s / \sigma$ onto the Brownian observation filtration. The integrands are square integrable; testing against bounded elementary predictable stochastic integrals and using Itô's isometry gives

$$d\ell_t^n = \sigma^{-1} \mathbb{E}_R[L_t^n H_n(t, Z) \mid \mathcal{F}_t] dB_t = \ell_t^n \theta_t^n dB_t.$$

The last equality is Bayes' formula. Since $|\theta_t^n| \leq M/\sigma$, the solution is the positive stochastic exponential (10.7); Girsanov gives the stated innovation. The same argument with H identifies the limiting exponential. Finally $|\mathbb{E}_{P_n} F - \mathbb{E}_P F| \leq 2\|F\|_\infty \varepsilon_n$ for bounded signed F . There is no interchange based only on a formal limit of logarithms. $\square$

10.3. A bounded entropic backward-equation consumer. Fix $\gamma > 0$ and a latent payoff f with $|f| \leq 1$. Define

$$U_t^n = \frac{1}{\gamma} \log \mathbb{E}_{P_n}[e^{\gamma f(Z)} \mid \mathcal{F}_t] = \frac{1}{\gamma} \log \frac{N_t^{n, e^{\gamma f}}}{\ell_t^n}. \quad (10.9)$$

The terminal condition here is the observable conditional certainty equivalent U_T^n , not the generally unobserved $f(Z)$ itself. This specification avoids imposing a latent terminal value on the smaller observation filtration.

Theorem 10.3 (Entropic values and localized backward integrands). *There is a unique bounded entropic value process with terminal value U_T^n ; for a predictable integrand Z^n it satisfies*

$$U_t^n = U_T^n + \int_t^T \frac{\gamma}{2} |Z_s^n|^2 ds - \int_t^T Z_s^n dI_s^n. \quad (10.10)$$

On the coupling of Theorem 10.1,

$$\mathbb{E} \sup_t |U_t^n(Y^n) - U_t(Y)|^2 \leq 84 \left(\frac{e^{2\gamma} - 1}{\gamma} \right)^2 \varepsilon_n. \quad (10.11)$$

For each $r > e^\gamma$ let τ_r^n be the first exit of any of $\ell^n, \ell, N^{n, e^{\gamma f}}, N^{e^{\gamma f}}$ from $(1/r, r)$, capped at T . If $\eta_n \rightarrow 0$, then

$$\mathbb{E}_R \int_0^{\tau_r^n} |Z_s^n - Z_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^{\tau_r^n} |\theta_s^n - \theta_s|^2 ds \rightarrow 0. \quad (10.12)$$

Furthermore $\lim_{r \rightarrow \infty} \sup_n R(\tau_r^n < T) = 0$. Thus the localizations exhaust the observation experiment in probability.

Proof. For a positive bounded latent g , write $N_t^{n,g} = \mathbb{E}_R[L_T^n g \mid \mathcal{F}_t]$ and $dN_t^{n,g} = a_t^{n,g} dB_t$. Projection of the full integral gives

$$a_t^{n,g} = \sigma^{-1} \mathbb{E}_R[L_t^n g H_n(t, Z) \mid \mathcal{F}_t], \quad |a_t^{n,g}| \leq (M/\sigma) N_t^{n,g}.$$

The same representation with $g = 1$ gives $d\ell_t^n = b_t^n dB_t$ and $b_t^n/\ell_t^n = \theta_t^n$. Put $N^n = N^{n,e^{\gamma f}}$ and $a^n = a^{n,e^{\gamma f}}$. Itô's formula in (10.9) gives the martingale coefficient

$$Z_t^n = \gamma^{-1} (a_t^n/N_t^n - b_t^n/\ell_t^n).$$

Replacing dB_t by $dI_t^n + \theta_t^n dt$ reduces its drift to $-\gamma|Z_t^n|^2/2$, which proves (10.10). The ratios are bounded by M/σ , so the integrals are well defined. Conversely, exponentiating any bounded solution of (10.10) produces a bounded P_n -martingale with terminal value $e^{\gamma U_T^n}$. Its conditional-expectation representation is unique, and martingale representation then identifies its integrand. This proves uniqueness in the bounded class.

Apply Theorem 9.1 to $g_0 = (e^{\gamma f} - e^{-\gamma})/(e^\gamma - e^{-\gamma}) \in [0, 1]$. The map taking its conditional expectation to $\gamma^{-1} \log \mathbb{E}[e^{\gamma f} \mid \mathcal{F}_t]$ has Lipschitz constant at most $(e^{2\gamma} - 1)/\gamma$. This proves (10.11).

The initial values of N^n and N agree, since Z has the same marginal. Itô's isometry and the terminal version of (10.5) imply

$$\mathbb{E}_R \int_0^T |a_s^n - a_s|^2 ds \rightarrow 0, \quad \mathbb{E}_R \int_0^T |b_s^n - b_s|^2 ds \rightarrow 0.$$

Before τ_r^n , all denominators lie in $[1/r, r]$, and the limiting numerators a, b are bounded in absolute value by $(M/\sigma)r$. For example,

$$\left| \frac{a^n}{N^n} - \frac{a}{N} \right| \leq r|a^n - a| + (M/\sigma)r^3|N^n - N|.$$

The supremum estimates (10.5), followed by integration in time, prove (10.12). The same inequality applies to b^n/ℓ^n .

Finally $e^{-\gamma} \ell^n \leq N^n \leq e^\gamma \ell^n$. Each ℓ^n is a stochastic exponential with uniformly bounded integrand θ^n . Its positive and negative moments of the supremum are bounded uniformly in n : write any power as a stochastic exponential times its bounded compensator and apply Doob's inequality to a higher positive power. The same calculation applies to its inverse. Markov's inequality, the preceding comparison of N^n with ℓ^n , and a union bound prove the exhaustion assertion. This is a localized integrand theorem; it does not assert an unspecified global topology for backward equations with unbounded terminal data or different nonlinear drivers. $\square$

Corollary 10.4 (A changing-filtration stopping consumer). *Let $\psi(t, x) \in [0, 1]$ be measurable and L_ψ -Lipschitz in x , uniformly in t . For a bounded latent f with $|f| \leq 1$, put $M_t^n = \mathbb{E}_{P_n}[f(Z) \mid \mathcal{F}_t]$. Then*

$$\left| \sup_{\tau \leq T} \mathbb{E}_{P_n} \psi(\tau, M_\tau^n) - \sup_{\tau \leq T} \mathbb{E}_P \psi(\tau, M_\tau) \right| \leq \varepsilon_n + L_\psi \sqrt{80\varepsilon_n}.$$

The same statement for the entropic value U has the additional Lipschitz factor $(e^{2\gamma} - 1)/\gamma$ on the square-root term. The suprema range over the common canonical stopping times; attainment is not needed for the estimate.

Proof. For a fixed stopping time, first change the measure while keeping its $[0, 1]$ reward function, at cost at most ε_n . Then compare the two reward functions on the same canonical path under P , using Theorem 9.1, Cauchy-Schwarz and the Lipschitz bound. The estimate is uniform in the stopping time, so it survives both suprema. Equivalence of the observation laws is what allows the same stopping-time class. The entropic assertion follows from the Lipschitz calculation in the preceding theorem. $\square$

10.4. Resource transport is not the posterior coupling. Apply any common finite causal acquisition map to Y and retain a fixed finite mark $w(Z)$. Data processing makes its marked row variations at most ε_n , uniformly over a declared finite feedback family. If the source and target have respective bounds α_n, β_n , Theorem 2.2 gives same-budget deficiency perturbation at most $\alpha_n + \beta_n$, and Proposition 4.4 transports a fixed reset audit. A subsequent consumer of width S composed with a simulator of width K has width KS , with the stated selector price. None of these executable claims uses the latent-preserving maximal coupling. That coupling belongs to the proof of conditional-process convergence, not to the finite-state machine.

These results provide a direct changing-trajectory, likelihood, Girsanov and bounded entropic-value input for a conditional-law theory. They do not assert the strict-dual, rigidity, covariant-form or full kinetic conclusions of a larger program without their separate analytic proofs.

Example 10.5 (Why an amplitude bound is not a tightness argument). Let ξ_1, ξ_2 be independent symmetric signs, and, for $n \geq 5$, let $\mathcal{G}_t^n$ reveal ξ_1 at $1/2 - 2/n$ and ξ_2 at $1/2 - 1/n$. On $[0, 1]$ the bounded martingale

$$M_t^n = \mathbb{E}[\xi_1 + \xi_2 \mid \mathcal{G}_t^n]$$

converges at every finite collection of fixed times to the martingale $M_t = \mathbf{1}_{[1/2, 1]}(t)(\xi_1 + \xi_2)$. Nevertheless its laws are not tight in the Skorokhod J_1 topology. On the event $\xi_2 = -\xi_1$, of probability $1/2$, its path has an excursion of height one on an interval of length $1/n$ and is zero outside that interval. The three-time oscillation modulus of such a path is one at every scale greater than $2/n$. This violates the necessary J_1 tightness modulus criterion. All maximal moments are uniformly bounded by 2^p .

This example concerns the inference from finite-dimensional convergence and a maximal amplitude estimate alone. It does not claim that every possible stronger conditional-kernel hypothesis fails. The uniform estimates in Theorem 10.1 remove the need for that inference in the observation class treated here.

11. A UNIFORM CERTIFICATE THROUGH A MOVING COLLISION

We now use all three parts of the argument on one family. The finite marked reference law and the uniform scattering geometry are those of Theorem 8.1 and Lemma 8.2. They are proved above, including the no-collision control. We use a continuous observation of the same detector so that finite reports and conditional processes are generated by one experiment.

Let z denote a preparation in the boxes of (8.3), and let ν be that prepared law. The sign of the impact orientation is $2B - 1$, and the conserved center-of-mass longitudinal sign is the actual mark W . Their correlation is prepared: $\mathbb{P}(B = b, W = w) = 5/16$ for $b = w$ and $3/16$ otherwise. The contact diameter d ranges over $I = [9/10, 11/10]$. Let $X_d(t, z)$ be the two-particle phase path, with outgoing velocities used at the collision, and define

$$H_d(t, z) = h(v_{1,y}^d(t, z)), \quad Y_t^d = \int_0^t H_d(s, z) ds + \sigma B_t^d, \quad 0 < \sigma \leq 1/24.$$

The Brownian noise is independent of the preparation. The source experiment has the same physical preparation and mark but constant reports. The target reports at times 2 and 3 are

$$Z_1^d = \mathbf{1}\{2(Y_2^d - Y_{3/2}^d) > 0\}, \quad Z_2^d = A\mathbf{1}\{2(Y_3^d - Y_{5/2}^d) > 0\}. \quad (11.1)$$

The acquisition gate $A \in \{0, 1\}$ is chosen after Z_1^d ; it does not act on the mechanical trajectory. The continuous path is the auxiliary observation experiment for conditional consumers; the finite simulator is not given this path or an uncharged buffer for computing its increments. The target sensor implements the declared window acquisition.

Lemma 11.1 (Quantitative change of the microscopic path). *For all preparations in the specified boxes, the collision time $t_c(d)$ and outgoing tagged velocities obey*

$$|t_c(d') - t_c(d)| \leq |d' - d|, \quad |v_i^+(d') - v_i^+(d)| \leq 30|d' - d| \quad (i = 1, 2).$$

Writing $\Delta = |d' - d|$, one has

$$\sup_z \int_0^3 |H_{d'}(t, z) - H_d(t, z)|^2 dt \leq \Delta + 2700\Delta^2. \quad (11.2)$$

The phase paths converge uniformly over the preparation in the Skorokhod J_1 metric as $d' \rightarrow d$. More precisely, for the product metric taking the maximum of the per-particle Euclidean position and velocity distances, a time change aligning the collision times gives J_1 distance at most 100Δ .

Proof. Write r and g for the incoming relative position and velocity, and $n(d) = (r + t_c(d)g)/d$ for the contact normal. The collision root satisfies $|r + t_cg| = d$. The nongrazing estimate from Lemma 8.2 is $-g \cdot n > 1443/1000$. Differentiating the root with respect to d gives

$$t'_c(d) = \frac{1}{g \cdot n(d)}, \quad |t'_c(d)| < 1.$$

The incoming bound is $|g| < 3$, so

$$|n'(d)| \leq \frac{|t'_c(d)| |g| + 1}{d} < 5.$$

Since $v_1^+ = V + g/2 - (g \cdot n)n$ and $v_2^+ = V - g/2 + (g \cdot n)n$, differentiation bounds either velocity derivative by $2|g| |n'| < 30$.

Before collision the detector is zero, and after collision it is a constant of absolute value at most one. Its Lipschitz constant is one. The two signals therefore differ on the interval between the two collision times by at most one and afterwards by at most 30Δ . The intervening interval has length at most Δ , and the whole observation lasts three units. This proves (11.2).

Choose the increasing piecewise affine time change fixing 0, 3 and sending $t_c(d)$ to $t_c(d')$. Its uniform displacement is at most Δ . Aligned velocities differ by at most 30Δ , with the same outgoing convention at the jump. Each particle's speed is below four and its velocity jump has norm below three. Without time change, the position difference is bounded by $3\Delta + 3(30\Delta) = 93\Delta$, by integrating the velocity difference before and after the two jump times. Aligning time adds at most 4Δ . The asserted bound follows, with a harmless rounding to 100Δ . $\square$

Theorem 11.2 (Microscopic-to-audit and conditional-law composition). *For the single family (11.1), all of the following hold.*

(a) *At the same width-one private register,*

$$\delta^p > 9/20, \quad \delta_r^v > 9/20 \quad (r \geq 1), \quad \delta_2^h < 13/100,$$

uniformly for $d \in I$ and $0 < \sigma \leq 1/24$. When $d \leq 2/5$ the detector is zero on $[0, 3]$, and all three finite-experiment deficiencies are zero.

(b) *There is one explicit rational reset-audit rule with one all-on training row, eight marked outcomes and $n = 2800$ training runs, whose expected validation score exceeds $2/5$ against every private width-one simulator throughout that same parameter region. In detail, for counts C choose an event $A_C \subseteq \{0, 1\}^3$ maximizing*

$$Q_*(A_C) - C(A_C)/2800, \quad (11.3)$$

with lexicographic tie breaking. Here Q_ is the rational marked reference law of (B, B, W) . The independent validation uses the actual physical target, not Q_* . This finite rule has at most $\binom{2807}{7}$ training-count states at any scheduled epoch; a conservative complete auditor bound is*

$8(256 + 1)\binom{2807}{7}$. The state cost belongs to the auditor. The certificate is an expected-score inequality, not a confidence claim about one realized score.

(c) Fix $\sigma > 0$ and let $d_n, d \in I$ with $d_n \rightarrow d$. Set

$$\epsilon_n = \min \left\{ 1, \frac{\sqrt{|d_n - d| + 2700|d_n - d|^2}}{2\sigma} \right\}. \quad (11.4)$$

The joint preparation–continuous-observation laws differ by at most ϵ_n . Every finite resource deficiency above changes by at most ϵ_n , and a fixed audit changes by at most ϵ_n because its source behavior is unchanged. On a latent-preserving coupling, $X_{d_n} \rightarrow X_d$ in J_1 and each bounded posterior converges in supremum L^2 with bound $84\epsilon_n$. The full and projected likelihoods converge in reference supremum L^2 , and jointly in probability with the microscopic paths on that coupling. The projected stochastic exponentials, bounded entropic backward values, localized integrands and stopping consumers have the conclusions of Proposition 10.2, Theorem 10.3, and Corollary 10.4.

(d) On compact parameter sets with $\sigma \geq \sigma_0 > 0$, the finite marked probabilities have uniform rational enclosures. The observable chart, strict same-budget polynomial verification and rational upper tables are therefore effective in the exact sense of Propositions 7.1 and 6.6.

Proof. Every collision occurs before time $3/2$. Conditional on preparation, the two normalized window increments in (11.1) are independent Gaussians with mean $h(v_{1,y}^+)$ and standard deviation $\sqrt{2}\sigma \leq 1/12$. The outgoing detector has sign $2B - 1$ and absolute value greater than $1/4$. Coupling each noisy report to B gives total marked variation less than $1/300$ from the exact finite marked experiment, uniformly for every acquisition gate. The finite calculation of Theorem 8.1 gives

$$d_* = \sqrt{5/2} - 9/8, \quad \delta_*^p = \delta_*^v = d_*, \quad \delta_{*,2}^h = 1/8.$$

Thus private and visible values exceed $d_* - 1/300 > 9/20$, and the hidden value is less than $1/8 + 1/300 < 13/100$. The two-valued hidden witness stores its selector, not an observed mark. In the no-collision regime both window reports before gating are independent fair noise, independent of W . Fresh private coins reproduce them exactly, including feedback gates. This proves (a).

For (b), use only the all-on feedback row. Any private width-one simulator in this row emits independent bits with probabilities p, q , independently of the actual W . The calculation in Theorem 8.1 already gives minimum variation d_* for this row. Its alphabet (z_1, z_2, W) has eight elements, hence $D = 7$. The explicit empirical audit in Theorem 4.1, trained using Q_* , has expected ideal score at least

$$d_* - \sqrt{7/2800} = d_* - 1/20.$$

The candidate training and validation laws do not change when the target is replaced by the physical target. Conditional on any selected event, its target probability changes by less than $1/300$. Therefore the actual score is strictly greater than

$$d_* - \frac{1}{20} - \frac{1}{300} > \frac{2}{5}. \quad (11.5)$$

For completeness, the last strict inequality is the positive-root comparison $\sqrt{5/2} > 947/600$; its squared margin is $5/2 - (947/600)^2 = 3191/360000 > 0$. All probabilities in Q_* are 0, $3/16$ or $5/16$, so (11.3) is a finite rational rule independent of d, σ, p, q . At training epoch k the number of eight-category count vectors is $\binom{k+7}{7} \leq \binom{2807}{7}$. A buffer of eight states records the current two reports and terminal mark; 256 event masks and an idle symbol suffice during event selection and validation. The conservative product bound follows. The clock and the source-reset interface are the declared resources. In particular the proof does not enumerate or claim to store a small complete coefficient table.

For (c), Lemma 11.1 gives the signal-distance hypothesis of Theorem 10.1, with $T = 3$ and $M = 1$. It also supplies its microscopic J_1 hypothesis. That theorem gives (11.4) and all the joint process assertions. The finite acquisition is a common measurable causal map of the continuous observation, so marked variation contracts for every gate. The source is blind and independent of d . Consequently its executable candidate behaviors are identical across d , and Proposition 4.4 has $\alpha = 0$ and $\beta = \epsilon_n$. The same same-budget comparison applies in each visible fiber and in each hidden mixture. The exponential and backward claims now follow from the identified projected density and the explicitly bounded entropic consumer. This is where the finite certificate and the conditional-law theorem meet the same microscopic approximation.

Finally, the collision root is uniformly separated from grazing, its formula is algebraic on the rational preparation boxes, and the normalized window noise has variance $2\sigma^2$ bounded away from zero on the stated compact sets. The box subdivision and Gaussian-tail enclosure argument in Proposition 8.4 therefore applies with that variance. It returns rational enclosures with a computable uniform error. The rational quotient algorithm and same-budget strict certificate procedure then prove (d). Uniformity as $\sigma \downarrow 0$ is not asserted. $\square$

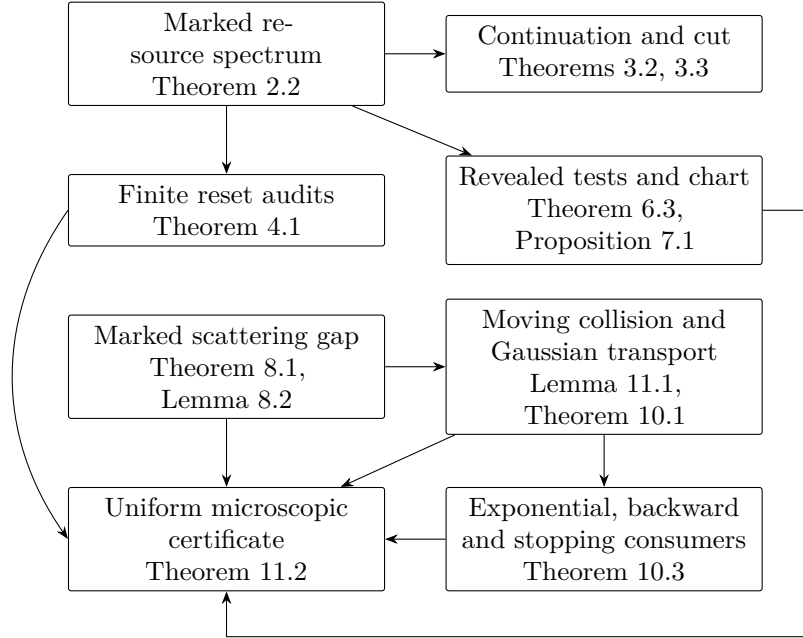
The all-on audit in part (b) is a single witness for the lower bound in the full feedback problem: a maximum over feedback trees includes this row. The visible lower bound is instead the exact fiber theorem; no claim is made that an auditor with a seed-independent state bound can store an arbitrarily large visible-seed transcript. This separates the two resource budgets rather than silently moving cost from one to the other.

The finite certificate is also preserved under an additional rational presentation error ρ : its lower score decreases by at most ρ when only the target is changed, and by the reset-cost estimate when the source is changed. Thus the certificate is an effective consumer of the physical presentation theorem, not an isolated exact finite example.

Theorem 11.2 and the preceding results prove Theorem 1.1. The C2-type consequence is a joint changing-trajectory/likelihood/posterior theorem with an identified stochastic exponential and a fully specified bounded entropic backward consumer. Nothing in that conclusion identifies geometric similarity with kinetic scaling, or supplies particle-uniform recollision estimates by renaming this fixed-particle model.

APPENDIX A. THE CANONICAL THEOREM CHAIN

The diagram records dependencies within this article. A solid arrow means that the conclusion at its tail is used in a proof at its head. The continuation characterization identifies the original finite-resource object; the exact cut theorem supplies its independent sharp obstruction. The two testing constructions have different information conventions.



In detail, the proof of Theorem 11.2(b) uses the audit theorem and the exact marked gap, while part (c) uses the moving-collision lemma, the TV-to-posterior estimate, the likelihood theorem and its conditional consumers. Part (d) consumes the chart and physical effective-presentation constructions. No testing theorem is an input to the elementary elastic-scattering calculation.

The comparison theorem, strict polynomial testing, revealed-behavior hierarchy, exact marked gap, scattering bounds, and common-latent TV-to-posterior estimate are inherited mathematical inputs with complete proofs reproduced here. The continuation normalization, exact cut spectrum, reset-audit game and lower bound, formal chart algorithm, changing-trajectory likelihood estimates, entropic consumer, and uniform moving-collision composition are additions of the present revision. The positive-cone, minimax, two-point testing, polynomial positivity, and stochastic-calculus mechanisms are classical inputs, not priority claims.

The parameter and the actual mark remain unavailable to the online simulator. The reset auditor sees completed marked trials through its separate interface. A revealed-behavior test sees an exact proposed law. The maximal coupling of observation paths may use their entire future and is not an executable transducer. These three information privileges are never substituted for one another.

REFERENCES

- [1] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272. doi:10.1214/aoms/1177729032.
- [2] L. Le Cam, Sufficiency and approximate sufficiency, *Ann. Math. Statist.* **35** (1964), 1419–1455. doi:10.1214/aoms/1177700372.
- [3] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28. doi:10.1524/strm.2002.20.14.1.

- [4] A. Nayyar, A. Mahajan and D. Teneketzis, Decentralized stochastic control with partial history sharing: a common information approach, *IEEE Trans. Automat. Control* **58** (2013), 1644–1658. doi:10.1109/TAC.2013.2239000. The theorem numbers cited here refer to the author preprint, arXiv:1209.1695.
- [5] H. S. Witsenhausen, On the structure of real-time source coders, *Bell System Tech. J.* **58** (1979), 1437–1451. doi:10.1002/j.1538-7305.1979.tb02263.x.
- [6] I. Gallagher, L. Saint-Raymond and B. Texier, *From Newton to Boltzmann: Hard Spheres and Short-Range Potentials*, Zurich Lectures in Advanced Mathematics, European Mathematical Society, 2014. The flow and boundary discussion used here is in Sections 4.1–4.3 of arXiv:1208.5753v2.
- [7] C. Widder, J. Zimmer and T. Schilling, On the generalized Langevin equation and the Mori projection operator technique, preprint (2025), revised 2026. arXiv:2503.20457v6.
- [8] C. Widder and T. Schilling, Generalised Langevin dynamics: significance and limitations of the projection operator formalism, preprint (2026). arXiv:2604.20453v2.
- [9] H. F. Trotter, Approximation of semi-groups of operators, *Pacific J. Math.* **8** (1958), 887–919. doi:10.2140/pjm.1958.8.887. Theorem 5.2, p. 902, is the approximation theorem cited here.
- [10] H. Weisshaupt, LeCam’s randomization criterion in the setting of locally convex spaces without lattice structure, EURANDOM report 2002–017, 22 April 2002. Lemma 1 and Theorem 5, pp. 14–16. Author report.
- [11] S. Basu, Algorithms in real algebraic geometry: a survey, author preprint, arXiv:1409.1534 (2014). Published in *Panoramas et Synthèses* **51** (2017), 107–153. Theorem 2.1 and Section 2.1 in the author preprint are used here.
- [12] M. C. Paull and S. H. Unger, Minimizing the number of states in incompletely specified sequential switching functions, *IRE Trans. Electron. Comput.* **EC-8** (1959), 356–367. Publisher record.
- [13] E. Torgersen, *Comparison of Statistical Experiments*, Encyclopedia of Mathematics and its Applications 36, Cambridge University Press, Cambridge, 1991.
- [14] N. Saldi, A topology for team policies and existence of optimal team policies in stochastic team theory, preprint (2017), arXiv:1711.00639.
- [15] S. Yüksel and N. Saldi, Convex analysis in decentralized stochastic control, strategic measures and optimal solutions, author preprint arXiv:1609.07685v2.
- [16] A. Nerode, Linear automaton transformations, *Proc. Amer. Math. Soc.* **9** (1958), 541–544. doi:10.1090/S0002-9939-1958-0135681-9.
- [17] S. Kiefer, Notes on equivalence and minimization of weighted automata, preprint (2020), arXiv:2009.01217v1.
- [18] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176. doi:10.2140/pjm.1958.8.171.
- [19] H. Weisshaupt, A generalization of Le Cam’s randomization criterion, *Studia Sci. Math. Hungar.* **43** (2006), 443–486. doi:10.1556/SScMath.43.2006.4.3. Theorem numbers refer to the author’s 39-page preprint, Randomization.pdf.
- [20] E. de Klerk and M. Laurent, Error bounds for some semidefinite programming approaches to polynomial minimization on the hypercube, *SIAM J. Optim.* **20** (2010), 3104–3120. doi:10.1137/100790835.
- [21] M. Laurent and L. Slot, An effective version of Schmüdgen’s Positivstellensatz for the hypercube, arXiv:2109.09528v2, 2023.
- [22] E. de Klerk and J. C. Vera, The link between 1-norm approximation and effective Positivstellensätze for the hypercube, *Numer. Algebra Control Optim.* **16** (2026), 84–104. doi:10.3934/naco.2024060. Theorem numbers refer to arXiv:2404.04190v2.
- [23] M. Fliess, Séries rationnelles positives et processus stochastiques, *Ann. Inst. H. Poincaré Sect. B* **11** (1975), 1–21. Original article. Theorems 3.1 and 5.4 and Corollary 2, pp. 8–9 and 18–19.

GENERAL THETA FOUNDATIONS I: ADAPTIVE TESTING AND COLLISION-SENSITIVE COMPARISON

QIAN QI

ABSTRACT. We study comparison of marked causal experiments with fixed persistent state and separately priced hidden or visible randomness. On the nonconvex observable behavior image we construct an independent adaptive testing game and polynomial test classes with attained finite-level duality. In behavior dimension a and diameter R , their error is at most $4Ra/\sqrt{n}$, using at most $(n+1)^a$ occupied test coefficients; this does not charge redundant simulator rows. We then construct a two-sphere scattering experiment with an actual correlated mark and a gate chosen after the first noisy report. Its private and visible deficiencies exceed $9/20$, whereas a hidden two-valued selector gives error below $13/100$, at the same width-one register. Removing the collision makes every deficiency zero. Exact finite values, uniform collision bounds and computable rational presentations prove this separation without convexifying the private image. Finally, marked variation controls the entire optional-projection process; Gaussian observation of graph-core approximants yields a quantitative microscopic changing-filtration limit. The costed comparison, strict-certificate, task and domain theory is developed with its separate information and analytic hypotheses.

COMPLETE DEVELOPMENT

The canonical article and the preserved historical mathematical development.

CONTENTS

1. Introduction	7
1.1. Main results	8
1.2. Relation to classical theory	9
1.3. Organization and analytic scope	9
2. Intrinsic causal deficiency	10
2.1. Experiments and executable signatures	10
2.2. The finite comparison theorem	11
2.3. A delayed channel and a strict signature separation	13
2.4. Optimal resource sets and decision certificates	14
2.5. Local testing at the prescribed private budget	15
2.6. Transferring intrinsic certificates across presentations	18
2.7. Witness size in the intrinsic behavior dimension	18
3. Adaptive testing on the observable behavior image	19
3.1. The testing game	19
3.2. A size bound in behavior dimension	20
3.3. A three-weight certificate invisible to constant tests	22
3.4. Strict certificates and effective physical input	23
4. A marked resource gap generated by a binary collision	24
4.1. The exact marked two-report experiment	24

Date: September 23, 2026.

2020 Mathematics Subject Classification. Primary 62B05, 93E20; Secondary 60G44, 68Q45, 47D06, 82C40.

Key words and phrases. causal deficiency, adaptive testing, behavior dimension, finite memory, hard-sphere scattering, optional projections.

4.2.	A full-dimensional collision preparation	25
4.3.	Uniformly effective finite presentations	27
5.	Uniform stability of optional projections	27
5.1.	A microscopic sufficient condition	29
6.	Polynomial testing of the original resource image	30
6.1.	A sharp two-time nonconvex example	32
7.	Endogenous common decisions	32
7.1.	A shared encoder on a controlled tree	32
7.2.	Lossy common design and intrinsic comparison	34
7.3.	Task resolution, positive error, and the complete state	35
8.	Regenerative average risks	37
8.1.	The reset signature	37
8.2.	Cycle comparison before optimization	38
9.	An active microscopic realization	39
9.1.	Interventions and the unbounded flow	39
9.2.	A graph-core approximation and controlled words	40
9.3.	Actual marked acquisition under feedback	41
9.4.	Active regenerative average decisions	42
9.5.	A posteriori bounds from graph and intervention residuals	43
10.	Finite rational presentations of Gaussian acquisitions	45
10.1.	A stateless report channel	45
10.2.	Rationalization without a small-prefix denominator	46
11.	Enriched defect certificates and their convergence	48
12.	Effective graph data from regular collision charts	50
13.	Physical intrinsic resource certification	53
14.	Certified data for active hard spheres	54
14.1.	A conserved mode with an exact marked deficiency	54
14.2.	A nonzero collision-compatible graph residual	56
15.	A nonconserved collision-compatible acquisition	57
15.1.	One actual mark, one nonzero residual, and a positive certificate	59
16.	A joint physical and nonconvex certificate theorem	61
16.1.	Exponential acquisition values	62
	Appendix A. Theorem dependencies and information conventions	63
	Preserved development: twelfth article and earlier mathematics	64
	Preserved development: twelfth article and earlier mathematics	64
	Appendix 22. Introduction	64
22.1.	The structural results	64
22.2.	Antecedents and the comparison boundary	65
22.3.	Scope and organization	65
	Appendix 23. The marked decision spectrum	66
23.1.	Experiments, tasks, and signatures	66
23.2.	Typed simulations and invariance	67
23.3.	The terminal decision-complete slice	68
	Appendix 24. State-costed comparison and complete-state fusion	69
24.1.	The information interface and priced randomness	69
24.2.	Deficiency frontiers and spectra	71
24.3.	The marked quotient as a zero-overhead object	72
	Appendix 25. Causal task completion on the actual support	73
25.1.	Supported trees and simultaneous decisions	73

25.2.	A separating risk gap and prediction completion	75
25.3.	Coherent exact machines on an unbounded time interval	76
Appendix 26.	Compact duality for dominated causal experiments	77
26.1.	The experiment and its information pattern	77
26.2.	The topology and an executable realization	77
26.3.	Finite observation approximations, uniformly in memory	80
Appendix 27.	A singular observation class and conditional compact duality	81
27.1.	Conditional innovation models	81
Appendix 28.	Infinite-horizon risk bodies and coherent comparison	83
28.1.	Finite-prefix hypotheses and executable policies	84
28.2.	Costed comparison at all times	86
Appendix 29.	The minimal marked continuation quotient	87
29.1.	An operational necessity theorem	88
29.2.	Two exact marked causal channels	89
Appendix 30.	Task-indexed exact compression and its completion	90
30.1.	Finite decision families	90
30.2.	Prediction tasks and a canonical completion	91
30.3.	A task-indexed causal quotient	91
Appendix 31.	Synthesis from continuation spaces	92
Appendix 32.	Positive-state growth and task-synthesis complexity	94
Appendix 33.	A common-encoder obstruction with an interior dual witness	95
Appendix 34.	Controlled common encoders and the minimax risk body	96
34.1.	Finite instruments and the information pattern	96
34.2.	The variational theorem	97
34.3.	Controlled continuation partitions	98
34.4.	Private randomization is a different finite problem	99
Appendix 35.	One variational principle: occupancies, Bayes envelopes and pressure	100
35.1.	An exact support-function recursion	100
35.2.	The common quadratic task family	101
35.3.	Exponential criteria and hidden-phase control	102
Appendix 36.	Minimal posterior-and-report-predictive coordinates	103
Appendix 37.	Finite compilation and several charged resources	105
Appendix 38.	A Gaussian experiment with a genuine temporal penalty	107
Appendix 39.	Marked projections and finite numerical acquisition	109
39.1.	A finite table for the delayed Gaussian problem	110
Appendix 40.	Positive observation models and history reduction	111
40.1.	A proved forgetting estimate	112
Appendix 41.	Nonlinear finite-state semigroups	113
Appendix 42.	Hidden phases and risk-sensitive continuation	115
Appendix 43.	A microscopic graph core and width-uniform acquisition transfer	116
43.1.	The microscopic flow and invariant spaces	116
43.2.	Noisy acquisitions of physical observables	118
43.3.	The precise historical implication	119
Appendix 44.	Unbounded generators, forced memory, and physical decision limits	119
44.1.	A common domain for resolved and orthogonal dynamics	120
44.2.	A stability estimate which retains the full observable	121
44.3.	Galerkin convergence from the microscopic graph core	122
44.4.	The common-encoder limit for actual physical observations	123
Appendix 45.	Stable certificates for exact resource profiles	124

Appendix 46. A prior-independent experiment behind predictive coordinates	126
46.1. The causal factorization	127
Appendix 47. A necessity theorem for causal realization	128
Appendix 48. Finite-prefix, finite-description realization	130
Appendix 49. The measure-valued slice: sharp presentation entropy	132
Appendix 50. A dynamic finite-experiment consequence	134
50.1. A theorem-level downstream arrow	135
Supporting theory and model verification	136
Supporting theory and model verification	136
Appendix 51. Positive experiments and predictive quotients	136
51.1. Actual histories	136
51.2. The future-test quotient	136
51.3. Attainable Bayes geometry	137
51.4. Normalized response and observation modules	138
Appendix 52. Causal morphisms and charged approximation	138
Appendix 53. A structural theorem for predictive trees	140
53.1. Local hypotheses and the finite profile	140
53.2. Balance, dilation and exact closure	141
53.3. Static lower bound and the causal converse	142
Appendix 54. Two model verifications	143
54.1. Markov-renewal orbit geometry	143
54.2. An infinite hidden shift with noisy continual reports	144
Appendix 55. Finite presentations of measure-valued filters	145
55.1. A nonlinear physical realization	146
Appendix 56. Sharp finite-experiment budgets	147
56.1. The cardinality obstruction	147
56.2. Positive second-order reconstruction	148
Appendix 57. A sharp budget for the physical count-to-contact experiment	149
57.1. The A2 matrices and the scope of the consequence	151
Preserved historical introductions and supporting bodies	152
Preserved historical introductions and supporting bodies	152
Appendix 58. Introduction	152
58.1. The main theorem and its components	152
58.2. Classical ingredients and the comparison of scope	153
58.3. Organization	153
Appendix 59. Introduction	154
59.1. The principal theorem	154
59.2. Classical antecedents and the distinction of resources	155
59.3. Organization and the program of applications	155
Appendix 60. Introduction	156
60.1. The main statement	156
60.2. Comparison with classical results	157
60.3. Organization and scope	157
Appendix 61. Introduction	158
61.1. The main theorem	158
61.2. Endogenous decisions and nonsummable risks	158
61.3. Active microscopic realization	159
61.4. Relation to earlier theories	159
61.5. Organization	159

Appendix 62. Introduction	160
62.1. The structural result	160
62.2. Analytic realization and computational distinctions	160
62.3. A physical consequence and the place of the other results	161
62.4. Antecedents and the claim of distinction	161
Appendix 63. Introduction	162
63.1. The main results	162
63.2. Related work and the distinction of scope	163
63.3. Consequences and organization	163
Appendix 64. Introduction	164
64.1. Relation to existing theory	165
64.2. Organization and scope	166
Appendix 65. Introduction	166
65.1. Relation to previous theory	167
65.2. Organization and quantitative conventions	168
Appendix 66. Introduction	168
66.1. Relation to previous work	169
66.2. Reading convention	170
Appendix 67. Introduction	170
67.1. The intrinsic law	170
67.2. A nonregenerative realization principle	171
67.3. The resource and its invariances	171
67.4. Relation to previous work	171
67.5. Organization and dependencies	172
Appendix 68. Markov geometry and the finite-state law	172
68.1. Observables and acquisition times	173
68.2. A computable geometric profile	173
Appendix 69. Renewal energies and admissible suffix trees	174
Appendix 70. Two pressures on an admissible language	177
Appendix 71. Critical corrections and a uniform renewal window	178
Appendix 72. Finite registers without a common regeneration law	180
Appendix 73. Operational changes of the finite-memory interface	182
Appendix 74. Posterior orbits and finite-state realization	182
Appendix 75. A finite-register refinement of the current contact experiment	186
Appendix 76. Introduction	187
76.1. The principal results	187
76.2. Resource convention	188
76.3. Relation to the literature	188
76.4. Organization and dependencies	189
Appendix 77. An intrinsic law on Gibbs repellers	190
77.1. A finite cylinder algorithm	190
77.2. The pressure law	191
Appendix 78. Cylinder geometry and causal realization	192
Appendix 79. Pressure, distortion and survival	194
79.1. Markov branch weights and unequal multipliers	196
Appendix 80. Acquisitions with persistent hidden state	197
Appendix 81. Noisy expanding experiments and a uniform critical window	199
81.1. Geometric renewals and the critical window	202
81.2. Nonconstant slopes and other readouts	203

Appendix 82. Calibrated moment observations at root collisions	204
Complementary quantitative development	207
Appendix 83. Regeneration and the geometry of future predictions	207
83.1. A retained-label experiment	207
83.2. The future-orbit converse	208
Appendix 84. A sharp causal-memory transition for an expanding experiment	209
84.1. An explicit finite machine	209
84.2. The lower bound without a restriction on encoders	210
Appendix 85. An exact calibration quotient and singular resolution	211
85.1. Correlated paired measurements	211
85.2. Contact geometry in the noise metric	212
Appendix 86. Acquired geometry and nonuniform causal resolution	214
86.1. The experiment, the metric, and the resource	214
86.2. A resolution theorem	215
86.3. Intermittent contraction	216
Appendix 87. Positive and intermittent hidden Markov experiments	217
87.1. Projective stability	217
87.2. An experiment with unbounded noncontracting intervals	219
87.3. Discrete and singular acquisition	220
Appendix 88. Singular statistical resolution with measured calibration	220
Appendix 89. Optimization under a retained-label constraint	222
Appendix 90. Finite descriptions and numerical implementation	223
Appendix 91. Comparison of results and mathematical dependencies	224
Foundational development and boundary examples	225
Appendix A. Positive causal experiments	225
A.1. Types and primitive data	225
A.2. Strategies, paths, and stopping	226
A.3. Prepared and parameter-uniform questions	227
A.4. Instruments and evidence	227
A.5. Preparing a conditional experiment	228
Appendix B. Future tests and predictive quotients	228
B.1. Operator words and adaptive protocols	228
B.2. Safe measurable realizations	229
B.3. Finite remaining-test coordinates	229
B.4. Three different fibres	230
Appendix C. Executable reductions and charged resources	230
C.1. Finite-state implementations	230
C.2. The path-kernel estimate	231
C.3. Checking an exact state reduction locally	232
C.4. Terminal and causal equivalence	232
Appendix D. Posterior geometry and information loss	232
D.1. Posterior barycentres	232
D.2. Relative entropy and recovery	233
D.3. Scores and Fisher information	233
D.4. Sequential information in finite positive models	234
Appendix E. Attained prediction and checkpoint quantization	235
E.1. The physical loss	235
E.2. A small-ball lower bound	235
E.3. Whole-image covers and all integer budgets	236

E.4. Causal lower bounds use one exploration law	236
E.5. Calibrated anisotropy	236
Appendix F. Causal quantization with feedback and stopping	236
F.1. Admissible state domains	236
F.2. What feedback does and does not require	238
F.3. Uniform prediction and finite-horizon path comparison	238
F.4. Model or calibration perturbations	239
Appendix G. A dynamic model with a uniform memory law	239
G.1. The experiment and its resource convention	239
G.2. Explicit posterior recursion	240
G.3. The actual acquisition density	241
G.4. A fixed codebook and the proof of the theorem	242
G.5. Feedback-uniform emulation in this model	243
G.6. Calibration and finite precision	243
Appendix H. Normalized response and observation modules	243
H.1. The covariance pairing	243
H.2. Finite positive functionals and exact product fibres	244
H.3. The contact calculation	244
H.4. Why general dynamic tests need not form a finite algebra	245
Appendix I. Exponential transforms and conditional limit interfaces	245
I.1. Random channels have nonlinear exponential pullbacks	245
I.2. A finite risk-sensitive recursion	246
I.3. Deterministic LDP contractions	247
I.4. Regular quadratic and labelled-phase interfaces	247
I.5. Closure and finite-dimensional memory	247
Appendix J. Boundary examples and non-implications	248
Appendix K. Interfaces with singular geometry and dynamics	249
K.1. What is established in this installment	250
K.2. The role of the two geometric manuscripts	250
K.3. A precise next target	250
K.4. The physical realization obligations remain	250
K.5. Conclusion	251
Appendix L. Measurability, finite covers, and a scalar benchmark	251
L.1. Finite-label implementation	251
L.2. Conditional distributions and common versions	251
L.3. Quantization after a readout change	251
L.4. Predictable error budgets and random horizons	252
Appendix M. Explicit finite checks and the limits of computation	252
M.1. A two-state posterior calculation	252
M.2. A direct resource calculation	253
M.3. A finite precision requirement rather than a free real tape	253
M.4. Role of the accompanying diagnostics	253
References	254

1. INTRODUCTION

Comparison of experiments asks whether the information supplied by one observation system can replace that supplied by another. A causal comparison must also respect the order in which

observations and actions become available. A resource-constrained comparison asks a further question: how much persistent state and shared randomness does the replacement require? These questions differ even for two binary reports. A simulator that cannot retain or reread its first emission cannot in general reproduce a correlated second emission. Revealing a persistent random selector to the tester is also different from hiding it. The distinction concerns the joint laws being compared, not merely the implementation of a convex combination.

We study marked experiments. A mark is an actual latent quantity of the source or target system, not a fresh variable sampled after a comparison. The comparison tests the joint law of the mark and the controlled report history. Private simulation uses finite retained registers and fresh private randomization. A hidden selector is an additional priced persistent variable; a visible selector is exposed and must occur with the same independent reference law on the target side. With these conventions the attainable behavior image is compact but usually nonconvex. The central issue is to compare this original image rather than a convex relaxation with a different information pattern.

1.1. Main results. For fixed finite instruments and a resource signature, let B denote the observable discrepancy image and $F(b) = \max_i f_i(b)$ its marked feedback error. The f_i are affine event tests, including zero. The deficiency is $\delta = \min_B F$. An adaptive test is an independently chosen function $\kappa : B \rightarrow \Delta_m$, scored at b by $\sum_i \kappa_i(b) f_i(b)$. It observes a proposed law for purposes of verification; it is not an added physical information channel.

Theorems 3.1 and 3.3 give the independent variational identity

$$\delta = \sup_{\kappa} \min_{b \in B} \sum_i \kappa_i(b) f_i(b) = \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa} \int \sum_i \kappa_i(b) f_i(b) d\mu(b),$$

and a quantitative polynomial hierarchy. In a bounded chart of the observable affine space, of dimension a and metric diameter R , its attained finite-level values satisfy

$$0 \leq \delta - d_n \leq 4Ra/\sqrt{n}.$$

An approximating test needs at most $(n+1)^a$ occupied Bernstein coefficients, and the corresponding moment dual needs at most $(n+2)^a$ support points. The dimension counts behavior, not row variables. The nonlinear score prevents replacement of a lottery over revealed candidate laws by its barycenter. At degree zero such a replacement is possible, and the resulting convex relaxation can miss a strictly positive private deficiency.

Proposition 3.5 gives a concrete nonlinear witness with three cubic behavior weights, nine nonnegative derivative terms and seventeen positive rational coefficients. It certifies a private lower bound above $2/5$, while constant tests give zero.

The independent game and its size theorem complement the strict polynomial certificate hierarchy proved in Theorem 6.1. The latter remains an exact verifier on a chosen row cube; it is not itself called an independent dual. Strict lower levels are complete, and their verification complexity is kept separate from the behavior-coordinate degree bound. With uniformly effective stateless finite presentations this produces certified intervals at the original state and randomization budgets.

The second main result makes collision mechanics essential. We prepare two spheres in full-dimensional incoming phase-space boxes and observe a bounded function of the tagged transverse velocity after scattering. An actual binary mark, correlated with the incoming impact orientation, is retained but never revealed. A binary gate is chosen after the first noisy report. Theorem 4.1 first solves the limiting finite experiment exactly:

$$\delta^p = \delta_r^v = \sqrt{5/2} - 9/8, \quad \delta_2^h = 1/8 \quad \text{at width } (1, 1).$$

Uniform nongrazing collision estimates and Gaussian tails then give, for contact distance $9/10 \leq d \leq 11/10$ and noise $0 < \sigma \leq 1/12$,

$$\delta^p, \delta_r^v > 9/20, \quad \delta_2^h < 13/100.$$

The same detector and preparation have zero deficiency for $0 < d \leq 2/5$, when no collision reaches the detector during acquisition. Thus conserved collective motion cannot account for the gap. The estimates are uniform on an interval of contact distances and under geometric similarity. Explicit collision charts also produce rational marked presentations with certified errors tending to zero.

The third result strengthens the conditional-law content of marked comparison. For equivalent joint laws with a common latent marginal, Theorem 5.1 constructs a latent-preserving coupling under which the squared uniform distance between bounded posterior martingales is at most 84 times the joint total variation. The coupling is a convergence device, not an executable simulator. Gaussian observation of an invariant microscopic flow gives

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \|h_n - h\|_{L^2(\mu)},$$

so graph-core approximation yields convergence of entire optional projection processes. The maximum-over-time estimate, rather than a finite-dimensional argument, supplies the changing-filtration input. Finite-checkpoint risk transport still uses the stateless identity comparison and keeps its original budget.

1.2. Relation to classical theory. Classical statistical comparison and randomization originate with Blackwell and Le Cam [1, 47, 46]. Filtered comparison has important antecedents, including Norberg [48]. We do not claim priority for introducing causal deficiency. In the adapted setting Weisshaupt’s Definition 5.1 and Theorems 5.1–5.2 [72] use a compact convex closure of adapted stochastic operators, allowing finitely additive outputs. A fixed finite-state private image need not have that convexity. The two-report examples identify the precise failed step in inserting the state constraint into such a separation proof. Our testing game retains the nonconvex image by changing the verifier’s strategy space, not by asserting a new unrestricted randomization criterion. Its minimax ingredient is classical [71].

Bernstein positivity is likewise not new here. Theorem 1.4 of de Klerk–Laurent [73] gives constructive hypercube degree and error bounds through Bernstein and Handelman representations. Laurent–Slot [74], Theorem 2 and Corollary 3, obtain stronger Schmüdgen degree bounds using Jackson kernels. Theorem 9 and Proposition 2 of de Klerk–Vera [75] connect Chebyshev coefficient norms to effective hypercube certificates. Our bound concerns an independent simplex-valued adaptive test on a nonconvex behavior image; it is not an improvement to those general positivity bounds. Pulling that test back to row coordinates still incurs a separate positivity-verification cost.

The finite-state, common-information and automata antecedents [54, 53, 70, 58] remain relevant to the task and synthesis theory. The physical flow and collision rule are classical [63]. The contribution of the scattering construction is the marked, feedback-sensitive resource comparison with uniform quantitative certificates in that flow, not a new mechanics of elastic collision.

1.3. Organization and analytic scope. After the exact comparison conventions, Sections 3 and 4 prove the adaptive hierarchy and its collision-sensitive realization. Section 5 proves the posterior-process estimate and its microscopic sufficient condition. The remaining sections develop the complete finite certificate, task, regeneration, graph-domain and physical-presentation machinery used for broader acquisitions, including the explicitly solvable collective benchmark. Appendix A records the theorem dependencies in mathematical terms.

All physical conclusions have their stated finite-time and preparation hypotheses. A binary scattering chart and a fixed-particle optional-projection estimate are different from particle-uniform

recollision estimates or a nonlinear kinetic limit. In particular a large-deviation action, a hierarchy corrector uniform on action sublevels, and a limiting nonlinear resolvent require their own analytic inputs. The comparison theorem transports consequences of such inputs; it does not create them by naming a dependency.

2. INTRINSIC CAUSAL DEFICIENCY

2.1. Experiments and executable signatures. Fix a finite parameter set Θ , a horizon T , and finite control alphabets $A_0, \dots, A_{T-1}$. An experiment has a report at checkpoints $0, \dots, T$ and an unobserved terminal mark W in a common finite set. It is specified by finite causal instruments: for each parameter and control history they give the joint law of the report path and the actual mark. Equivalently, a finite latent-state realization may be expanded into its history tree. The parameter and mark are unavailable to the encoder. The mark may depend on the physical controlled path. We compare experiments with the same control interface and mark space.

Write Y_t for the source report and Z_t for the target report. A private simulator of width K has stochastic rows

$$\begin{aligned} q_0(z, m \mid y), \quad m \in [K_0], \\ q_t(z, m \mid m', a, y), \quad m' \in [K_{t-1}], \quad m \in [K_t], \quad 1 \leq t \leq T. \end{aligned} \quad (2.1)$$

A row consumes the current source report, emits the target report, and retains the new state. The previous target reports are not available to that row unless encoded in m' . The clock and current control are part of the declared interface. Fresh private randomness is permitted, but no other persistent random variable is free. The product of all row simplexes is denoted $\mathcal{K}_K$.

A deterministic feedback tree g chooses a_t from $z_{0:t}$ and its own previous controls. It is the same controller on both sides of a comparison. Let Ω_g be the finite marked transcript space, including reports, controls, and W . Put

$$P_{\theta,g}^q = \text{Law}_\theta(E \xrightarrow{q} g), \quad Q_{\theta,g} = \text{Law}_\theta(F \rightarrow g).$$

Here W on the left is the source's actual mark. The total variation convention is $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. Private deficiency is

$$\delta_K^p(E, F) = \min_{q \in \mathcal{K}_K} \max_{\theta, g} \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}. \quad (2.2)$$

The minimum will be justified below. Randomized feedback need not be added to the maximum: conditioning on its independent random table expresses its law as a mixture of deterministic feedback laws. The same argument permits arbitrary independent side randomness in the tester.

A *hidden selector* $U \in [r]$, independent of the source, chooses one simulator q^u . It is not visible to the feedback controller or the final test. A *visible selector* is revealed before feedback begins. The two costs are

$$\delta_{K,r}^h(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, g} \left\| \sum_u \lambda_u P_{\theta,g}^{q^u} - Q_{\theta,g} \right\|_{\text{TV}}, \quad (2.3)$$

$$\delta_{K,r}^v(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, (g_u)} \left\| \sum_u \lambda_u \delta_u \otimes P_{\theta,g_u}^{q^u} - \sum_u \lambda_u \delta_u \otimes Q_{\theta,g_u} \right\|_{\text{TV}}. \quad (2.4)$$

In both expressions $\lambda \in \Delta_r$ and $q^u \in \mathcal{K}_K$. The second formula compares the joint seed-marked interface. Its target seed has the same distribution and is independent of the target experiment. A hidden selector can be stored privately by retaining (u, m) , at width rK_t . A visible selector is separate side information and is charged by its own alphabet size. Neither interpretation identifies an arbitrary real-valued design with a free finite register.

Definition 2.1 (Deficiency spectrum). The private resource spectrum of E relative to F is the upward-closed set of pairs (K, ε) satisfying $\delta_K^p(E, F) \leq \varepsilon$. The hidden spectrum consists of triples (K, r, ε) satisfying $\delta_{K,r}^h(E, F) \leq \varepsilon$, and the visible spectrum uses $\delta_{K,r}^v$. Width profiles are ordered coordinatewise; in the seeded spectra the seed bound and error are ordered as well. Unused states or selector values may be adjoined. The hidden convexification is $\delta_K^{\text{mix}} = \inf_{r \geq 1} \delta_{K,r}^h$.

The spectrum, not a presumed unique coordinatewise minimum profile, is the intrinsic object. For example one may minimize $\max_t K_t$, with a specified seed signature and tolerance, over its feasible profiles. Different checkpoints can have incomparable optimal tradeoffs.

2.2. The finite comparison theorem. Concatenate all probabilities $P_{\theta,g}^q(\omega)$ into $b(q) \in \mathbb{R}^N$, retaining their row indices; put $b_F = (Q_{\theta,g}(\omega))$. Let $\mathcal{D}_K$ be the finite set of deterministic choices of all rows in (2.1). Empty or unreachable rows remain harmless coordinates. Set

$$a_K = \dim \text{aff}\{b(q) : q \in \mathcal{K}_K\}, \quad \mathcal{I} = \{(\theta, g, B) : B \subseteq \Omega_g\}.$$

For $i = (\theta, g, B)$ and $d \in \mathcal{D}_K$, define

$$A_{i,d} = Q_{\theta,g}(B) - P_{\theta,g}^d(B). \quad (2.5)$$

Including both an event and its complement includes both signs.

Theorem 2.2 (Intrinsic deficiency, priced converse, and visibility). *For the finite marked causal experiments above the following assertions hold.*

(i) *Private deficiency and both fixed- r deficiencies are attained. The private problem has the exact polynomial formulation*

$$\min\{\varepsilon : q \in \mathcal{K}_K, 0 \leq \varepsilon \leq 1, Q_{\theta,g}(B) - P_{\theta,g}^q(B) \leq \varepsilon \text{ for all } (\theta, g, B) \in \mathcal{I}\}. \quad (2.6)$$

In particular its zero set is exactly the nonnegative causal factorization of the target marked behavior by the rows (2.1). At $K_t = 1$ for every t , this is precisely the same factorization with rows $q_t(z \mid a, y)$ and $q_0(z \mid y)$.

(ii) *Hidden convexification has the attained dual*

$$\delta_K^{\text{mix}}(E, F) = \min_{\lambda \in \Delta(\mathcal{D}_K)} \max_{i \in \mathcal{I}} \sum_d \lambda_d A_{i,d} = \max_{w \in \Delta(\mathcal{I})} \min_{d \in \mathcal{D}_K} \sum_i w_i A_{i,d}. \quad (2.7)$$

An optimal simulation uses at most $a_K + 1$ selector values. Thus every weighted causal test has optimal discrepancy at most ε if and only if a hidden mixture of K -state simulators has error at most ε ; it has a private realization of width at most $(a_K + 1)K$. More specifically,

$$\delta_{rK}^p \leq \delta_{K,r}^h, \quad \delta_K^{\text{mix}} = \delta_{K,a_K+1}^h \leq \delta_K^p. \quad (2.8)$$

(iii) *Put $e_\theta(q) = \max_g \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}$. Then*

$$\delta_{K,r}^v = \min_{\lambda, q^1, \dots, q^r} \max_{\theta} \sum_u \lambda_u e_\theta(q^u). \quad (2.9)$$

Writing $a_K^v = \dim \text{aff}\{(e_\theta(q))_\theta : q \in \mathcal{K}_K\}$, the infimum over all finite visible selectors is attained using $a_K^v + 1 \leq |\Theta| + 1$ values and equals

$$\max_{v \in \Delta(\Theta)} \min_{q \in \mathcal{K}_K} \sum_{\theta} v_\theta e_\theta(q). \quad (2.10)$$

For every $r \geq 1$, and also after optimizing over all r ,

$$\delta_{K,r}^v(E, F) = 0 \iff \delta_K^p(E, F) = 0. \quad (2.11)$$

No corresponding equivalence with $\delta_K^{\text{mix}} = 0$ holds in general.

(iv) For compatible experiments E, F, G and profiles K, L ,

$$\begin{aligned}\delta_{KL}^p(E, G) &\leq \delta_K^p(E, F) + \delta_L^p(F, G), \\ \delta_{KL,rs}^s(E, G) &\leq \delta_{K,r}^s(E, F) + \delta_{L,s}^s(F, G), \quad s \in \{h, v\}.\end{aligned}\tag{2.12}$$

Here KL is the pointwise product, and the second line uses independent selectors. If an F -consumer has widths S_t , composing it with a private simulator gives widths $K_t S_t$ and changes every $[0, 1]$ marked risk by at most $\delta_K^p(E, F)$. A hidden selector is stored at the additional price r , whereas visible selector alphabets multiply. The same bounds hold for optimized upper risk sets and their lower support functions with the indicated costs.

Proof. Expand a probability of a complete source path, simulator state path, and transcript. It is a product of fixed experiment probabilities and one chosen row entry per time. Summing latent and state paths gives a polynomial in the finitely many row entries. No row belonging to a fixed time can occur twice along one path. Consequently the polynomial is multi-affine in the groups of stochastic rows. Continuity and compactness of the product of closed finite simplexes prove attainment in (2.2); the same proof includes λ and the r policies in (2.3). For the visible problem there are only finitely many collections (g_u) and transcript events, so its objective is again continuous on a compact set. The finite-event formula for total variation proves (2.6). Setting $\varepsilon = 0$ forces equality of every transcript probability, in both directions of each event inequality. This proves the factorization and stateless claims, including their necessity.

For each stochastic row of q , sample its deterministic output once, independently of all other rows, before the experiment. This constructs a probability Λ_q on $\mathcal{D}_K$. On any fixed path the row variables encountered have different time indices, so the expectation of their product is the product of the row probabilities. Summing paths shows, simultaneously for every θ, g , that

$$b(q) = \sum_{d \in \mathcal{D}_K} \Lambda_q(d) b(d).\tag{2.13}$$

This is a behavior identity, not a same-width deterministic purification of minimax error. It implies $\text{conv } b(\mathcal{K}_K) = \text{conv } \{b(d) : d \in \mathcal{D}_K\}$. The latter is a compact polytope. Minimizing the continuous convex maximum of event discrepancies over this polytope gives the first expression in (2.7). Finite-dimensional minimax gives the second, with both optima attained. A point of this polytope is a convex combination of at most $a_K + 1$ of its generating points by Carathéodory's theorem. For completeness, if a representation uses more points, their affine dependence gives coefficients c_d with $\sum c_d = 0$ and $\sum c_d b(d) = 0$. Subtract the largest permissible multiple of c from the positive mixing weights; a weight becomes zero without changing the represented point. Repeating gives the stated bound. Store its selector together with the machine state to obtain (2.8). The dual inequality for every w is equivalent to the maximum over w being at most ε , which proves the priced testing converse.

For disjoint seed fibers, total variation is the sum of the fiber total variations weighted by λ_u . For a fixed θ , the tester can choose a separate maximizing tree on each revealed fiber. Therefore its maximum is $\sum_u \lambda_u e_\theta(q^u)$, proving (2.9). The map $q \mapsto e(q)$ is continuous, being a finite maximum of continuous functions. Its image is compact. Convexifying it in $\mathbb{R}^{|\Theta|}$ and applying the preceding finite-dimensional separation and affine-dependence argument proves (2.10) and its selector bound. Notice that the inner minimum is over stochastic q , not over deterministic tables: the nonlinear function e_θ does not commute with (2.13). If the attained visible optimum is zero, then for every u of positive weight and every θ one has $e_\theta(q^u) = 0$, since every summand is nonnegative. Any such q^u is a private exact simulator. The converse uses a selector of one value. This also proves the assertion for the optimized visible selector, whose support is finite by the bound just established.

Finally, cascade a K -state E -to- F transducer and an L -state F -to- G transducer. Within a step, consume the intermediate report before discarding it. The retained pair has at most $K_t L_t$ values.

For any terminal tester on G , the second transducer together with that tester is an admissible randomized feedback system on F . The first error bound applies to this system, and total variation contracts under the remaining channel. The second error bound applies to the genuine F system. The triangle inequality gives (2.12). With selectors, condition on the independent selectors in the appropriate places; in the visible case keep both selectors in the compared transcript. The same argument is valid because the target reference selectors are independent. Storing a hidden selector or exposing both visible selectors gives the stated costs. The minima established above may be used; equivalently one can use nearly optimal simulators and let the extra tolerance decrease.

A consumer is another causal transducer with task-specific readout and, where allowed, task-specific control, but with its required common encoder. The simulator is task blind, so its product with that encoder is still common. Condition on the consumer's independent random table to apply the feedback bound. Integration of any $[0, 1]$ marked loss changes by at most the total variation error. Taking the same paired consumer for every row proves the risk-set assertion, rather than a separate choice of simulator for each row. Infima of nonnegative weighted risks give the lower support inequalities. $\square$

Remark 2.3 (What the converse says). The dual (2.7) uses mixtures of indexed causal tests and optimizes over the entire common transducer. It is not a task-by-task choice of encoder. Its conclusion is an explicit hidden-selector simulation, with the selector subsequently charged if private storage is required. It does not replace the nonconvex private set by its convex hull at unchanged resource. The next example shows why that replacement would change the zero set.

2.3. A delayed channel and a strict signature separation. Let the source reveal $i \in [n]$ at checkpoint zero and reveal only a constant at checkpoint one. There is no control and the mark is constant. The target reports a constant at checkpoint zero and at checkpoint one draws $j \in [m]$ from a specified row-stochastic matrix $V = (V_{ij})$. The unknown parameter is i . The first target alphabet has one element: the simulator cannot use an early public output as unpriced storage. Only the intermediate register of size K matters.

Proposition 2.4 (The nonnegative-rank slice). *For this experiment,*

$$\delta_K^p(E, F_V) = \min_{A, B} \max_i \frac{1}{2} \sum_j |V_{ij} - (AB)_{ij}|, \quad (2.14)$$

where A is an n -by- K row-stochastic matrix and B is a K -by- m row-stochastic matrix. In particular

$$\delta_K^p(E, F_V) = 0 \iff \text{rank}_+(V) \leq K, \quad \delta_1^p(E, F_V) = 0 \iff V \text{ has identical rows.}$$

For $1 \leq K \leq n$ the identity channel has the sharp laws

$$\delta_K^p(E, F_{I_n}) = 1 - \frac{1}{\lceil n/K \rceil}, \quad \delta_K^{\text{mix}}(E, F_{I_n}) = 1 - \frac{K}{n}. \quad (2.15)$$

Thus, at tolerance $0 \leq \varepsilon < 1$, the least private width is $\lceil n/(1 - \varepsilon)^{-1} \rceil$, whereas the least within-mode width with optimized hidden selection is $\max\{1, \lceil n(1 - \varepsilon) \rceil\}$. The latter still pays for its selector. In particular, for $n = m = 3$ and $K = 2$,

$$\delta_2^p(E, F_{I_3}) = \frac{1}{2}, \quad \delta_2^{\text{mix}}(E, F_{I_3}) = \frac{1}{3}. \quad (2.16)$$

For $V_* = (1/2)I_3 + (1/6)\mathbf{1}\mathbf{1}^\top$,

$$\delta_{2,3}^h(E, F_{V_*}) = 0, \quad \delta_2^p(E, F_{V_*}) \geq \frac{1}{4\sqrt{3}} > 0, \quad \inf_r \delta_{2,r}^v(E, F_{V_*}) > 0. \quad (2.17)$$

Proof. The encoder maps the consumed i into its retained label by A ; the terminal decoder maps that label into j by B . This proves (2.14) including its converse implementation. If $V = CD$ is any nonnegative factorization, let $s_k = \sum_j D_{kj}$. Delete zero rows of D , define $B_{kj} = D_{kj}/s_k$ and

$A_{ik} = C_{ik}s_k$. Since V is stochastic, the rows of A sum to one. Padding with unused states gives the claimed equivalence with nonnegative rank. The one-state statement follows immediately.

For $V = I_n$, row error is $1 - (AB)_{ii}$. If the least correct probability is c , for each i select a row $k(i)$ of B with $B_{k(i),i} \geq c$. Some row is selected at least $\lceil n/K \rceil$ times, so stochasticity gives $c \leq 1/\lceil n/K \rceil$. A partition into K groups of sizes at most $\lceil n/K \rceil$, each decoded uniformly, attains that bound. Also $\sum_i (AB)_{ii} \leq K$, by the calculation below with K in place of two. It remains valid under mixing, giving hidden error at least $1 - K/n$. Start with a deterministic channel that reproduces K input symbols exactly and sends all others to those K outputs. Average all simultaneous permutations of its input and output labels. Every diagonal entry is then K/n , proving the second formula. Inverting the two monotone formulas gives the stated least widths.

For $V = I_3$, total variation in row i is $1 - (AB)_{ii}$. If all three diagonal entries of AB were greater than $1/2$, then for each i some row of B would have its i th entry greater than $1/2$. A stochastic row has at most one such entry, so two rows cannot serve three indices. This proves the private lower bound $1/2$. Merge two input symbols into a retained label that decodes uniformly to that pair, and retain the third symbol exactly. This attains it. For every stochastic factorization with two intermediate states,

$$\sum_{i=1}^3 (AB)_{ii} = \sum_{k=1}^2 \sum_{i=1}^3 A_{ik} B_{ki} \leq \sum_{k=1}^2 \sum_{i=1}^3 B_{ki} = 2.$$

The same inequality holds under hidden mixtures, giving maximal row error at least $1/3$. Mix uniformly the three preceding pair-merge channels. The resulting diagonal entries are $2/3$ and the off-diagonal entries are $1/6$. This is exactly V_* and has error $1/3$ from I_3 . It proves the hidden claims in both displays.

The smallest singular value of V_* is $1/2$. Any AB with two intermediate states has a unit vector x in its right null space, so $\|V_* - AB\|_F \geq \|(V_* - AB)x\|_2 \geq 1/2$. If the maximal row total variation error is ε , each row has Euclidean error at most 2ε , and hence the Frobenius error is at most $2\sqrt{3}\varepsilon$. This proves the explicit private bound. The optimized visible value is attained by Theorem 2.2; if it were zero, its zero-set assertion would contradict that bound. This proves its strict positivity. $\square$

2.4. Optimal resource sets and decision certificates. For a class $\mathcal{C}_{S,q}(F)$ of common consumers, with all other information conventions fixed, let $\mathcal{R}_{S,q}(F)$ be its vector of attainable $[0, 1]$ risks. For a nonnegative row weight w summing to one, write

$$h_F(S, q; w) = \inf_{r \in \mathcal{R}_{S,q}(F)} \langle w, r \rangle.$$

If E privately simulates F with cost K and error ε , then

$$\mathcal{R}_{S,q}(F) \subseteq \mathcal{R}_{KS,q}(E) + [-\varepsilon, \varepsilon]^J, \quad h_E(KS, q; w) \leq h_F(S, q; w) + \varepsilon. \quad (2.18)$$

For nonfinite row sets the first expression means pointwise paired risk functions within uniform distance ε ; no finite coordinate box is needed. Upper sets give the equivalent one-sided order formulation. Hidden selectors multiply private width by r ; visible selectors multiply the separate public alphabet by r . These are direct consequences of the final part of Theorem 2.2, with the optimal intrinsic error in place of an externally estimated simulator error.

When two stateless comparison directions have error at most ε , paired risk sets and their support values at unchanged width are within ε . If each is centered by a paired unrestricted benchmark whose error is also at most ε , the centered values are within 2ε . Therefore a gap $g > 0$ between zero excess and infeasible widths is stable under $\varepsilon < g/4$: a value below $g/2$ in the approximation certifies feasibility in the exact finite problem, and a value above $g/2$ certifies infeasibility. The benchmark and the class being optimized must be specified; this assertion does not identify different information signatures.

Proposition 2.5 (Intrinsic finite-dimensional selector approximation). *Let $\mathcal{R}$ be a compact risk image in $C(J)$, where J is compact. Suppose a bounded linear operator $L : C(J) \rightarrow C(J)$ has rank at most m and $\sup_{r \in \mathcal{R}} \|r - Lr\|_\infty \leq \eta$. Every barycenter $\bar{r}$ of a probability on $\mathcal{R}$ has a barycenter r_* of at most $m + 1$ members of $\mathcal{R}$ with $\|\bar{r} - r_*\|_\infty \leq 2\eta$. If $\mathcal{R}$ lies in a finite affine space of dimension m , the error is zero with at most $m + 1$ members. Thus the selector price can be controlled by the dimension of attainable risks, rather than by a covering number of the row space.*

Proof. The compact set $L\mathcal{R}$ lies in an m -dimensional space. Its convex hull is compact, and $L\bar{r}$ belongs to it. The finite-dimensional affine-dependence argument in the proof of Theorem 2.2 gives $r_1, \dots, r_{m+1} \in \mathcal{R}$ and weights λ such that $L\bar{r} = \sum_i \lambda_i Lr_i$. Put $r_* = \sum_i \lambda_i r_i$. The two residual terms $\bar{r} - L\bar{r}$ and $r_* - Lr_*$ have norms at most η , proving the estimate. Apply the same argument after translation in the exact affine case. The implementation stores the selected design index whenever it is not provided as visible side information. $\square$

Remark 2.6 (Finite computation and general carriers). For rational finite instruments, (2.6) and the fixed- r formulations are finite polynomial feasibility problems; (2.7) is a finite linear program. These are exact variational characterizations, not assertions of polynomial running time in a succinct history description. The number of feedback trees, events, or deterministic tables can be large. On standard Borel carriers the same definitions use suprema over measurable feedback and tests and infima over executable kernels. Their product-state composition and decision transport remain valid by the same channel argument. Attainment and the finite testing dual are asserted here only for the stated finite class. The average-risk and physical results below give separate nonfinite theorems with their own hypotheses.

2.5. Local testing at the prescribed private budget. The convex converse in Theorem 2.2 has an explicit selector price. We now give a different converse that leaves the private register unchanged. Its witnesses are finite covers of the policy space by local testing inequalities, rather than one separating functional on a convex hull. Throughout this subsection the experiments, horizon, and alphabets are finite; for effective statements their instrument probabilities are rational. The prescribed widths and, when present, selector cardinality are fixed.

Let $\mathcal{P}$ be the product of the row simplexes of the private transducer. For a hidden or visible selector with at most r values, include its simplex and r copies of the transducer rows in $\mathcal{P}$. Zero selector weights are permitted. The relevant deficiency has the form

$$\delta = \min_{q \in \mathcal{P}} F(q), \quad F(q) = \max_{i \in I} f_i(q), \quad 0 \leq F \leq 1. \quad (2.19)$$

Here I consists of all parameter, feedback-tree, and marked-event tests. The empty event is included. For a visible selector a test specifies a feedback tree and event in each selector fiber. Thus

$$f_i(\lambda, q) = \sum_u \lambda_u \{Q_{\theta, g_u}(B_u) - P_{\theta, g_u}^{q_u}(B_u)\}.$$

For a hidden selector the target term is independent of λ . Every f_i is affine in each row separately, with rational coefficients. This follows by expanding complete controlled paths: a time-indexed row is used at most once along each path. It holds simultaneously for all feedback tests, not just for open-loop words.

For an integer $M \geq 1$, intersect a row simplex Δ_{d-1} with each closed coordinate box

$$\prod_{j=1}^d [b_j/M, (b_j + 1)/M], \quad b_j \in \{0, \dots, M - 1\}.$$

Discard empty intersections. Products of these row polytopes form a finite closed cover $\mathcal{C}_M$ of $\mathcal{P}$. Overlapping faces cause no ambiguity. Write $V(C)$ for the products of row vertices of a cell C . These are exactly its vertices. They are rational and are obtained by setting all but at most one

coordinate of each row to a box endpoint and imposing that the coordinates sum to one. This prescription also includes lower-dimensional cells and one-point rows. Define

$$\ell(C) = \max_{i \in I} \min_{v \in V(C)} f_i(v), \quad u(C) = \min_{v \in V(C)} F(v), \quad (2.20)$$

$$L_M = \min_{C \in \mathcal{C}_M} \ell(C), \quad U_M = \min_{C \in \mathcal{C}_M} u(C). \quad (2.21)$$

For time t , let d_t be the largest row alphabet cardinality, including both emitted symbol and next retained label. Put

$$\rho_t(M) = \begin{cases} 0, & d_t = 1, \\ \min\{1, d_t/(2M)\}, & d_t > 1, \end{cases} \quad b_M = 1 - \prod_{t=0}^T (1 - \rho_t(M)).$$

For a selector row set $s_M = 0$ if $r = 1$, and $s_M = \min\{1, r/(2M)\}$ otherwise. The three bounds used below are

$$\omega_M^p = b_M, \quad \omega_M^h = s_M + (1 - s_M)b_M, \quad \omega_M^v = \min\{1, 2s_M + (1 - s_M)b_M\}. \quad (2.22)$$

They do not count the number of histories or the number of policy rows. The visible-selector bound includes a second selector term because the reference joint experiment changes when its selector law changes.

Theorem 2.7 (Complete local testing certificates). *For any of the three fixed resource signatures in (2.19), the rational quantities (2.21) satisfy*

$$0 \leq L_M \leq \delta \leq U_M \leq 1, \quad U_M - L_M \leq \omega_M. \quad (2.23)$$

The upper witness is an executable rational transducer at the prescribed widths and selector cardinality. A lower witness consists of one test i_C for each cell and its inequalities $f_{i_C}(v) \geq L_M$ at all $v \in V(C)$.

For $M = 2^n$, the lower bounds increase and the upper bounds decrease to δ . In particular,

$$\delta = \sup_n \max_{(i_C)_{C \in \mathcal{C}_{2^n}}} \min_{C \in \mathcal{C}_{2^n}} \min_{v \in V(C)} f_{i_C}(v) = \inf_n U_{2^n}. \quad (2.24)$$

If $\delta > \varepsilon$, a finite local testing certificate eventually proves this strict inequality. If $\delta < \varepsilon$, a rational original-budget simulator eventually proves feasibility below ε . For rational ε , non-strict feasibility, including equality, is decidable by real quantifier elimination. The optimum is algebraic, and an optimal row table can be chosen with real algebraic entries.

Proof. Express each row of $q \in C$ as a convex combination of its row vertices. Expand a separately affine f_i in those combinations, one row at a time. The resulting weights on $V(C)$ are nonnegative and sum to one. Consequently

$$\min_{q \in C} f_i(q) = \min_{v \in V(C)} f_i(v). \quad (2.25)$$

It follows that $\ell(C) \leq \min_C F \leq u(C)$ and hence $L_M \leq \delta \leq U_M$. No assertion that F itself is separately affine has been made. This distinction is essential at a randomized optimum. The empty event gives $\ell(C) \geq 0$.

We next prove the oscillation estimate used for the gap. Two probability rows in the same M -cell differ in total variation by at most $\rho_t(M)$. Run the two transducers against the same tester and source. As long as all earlier emitted symbols and retained labels coincide, the controls and source laws agree. Maximally couple the next joint output–state row. The conditional chance of a mismatch is at most $\rho_t(M)$. Thus the chance of any mismatch through time T is at most b_M . On the event of no mismatch the source can be coupled identically, including its actual mark. This bounds the difference of the probability of every marked feedback event by b_M , uniformly in the parameter and tester. A finite causal instrument provides precisely the conditional kernels needed for this coupling.

For a hidden selector first maximally couple its two laws. The selector mismatch has probability at most s_M ; on matched fibers the preceding bound is b_M . The difference of any fixed test is at most $s_M + (1 - s_M)b_M$. There is only one executed fiber, so this estimate has no extra factor r . For a visible selector write its fixed test as the expectation of $a_u(q) = Q_{\theta, g_u}(B_u) - P_{\theta, g_u}^{q_u}(B_u)$. These numbers lie in $[-1, 1]$. On mismatched selector fibers their difference is at most two; on matched fibers it is at most b_M . Thus its oscillation is at most $2s_M + (1 - s_M)b_M$. This estimate does not forget the selector in either compared experiment.

Choose a vertex v_* minimizing F on $V(C)$ and an active test i_* there. For every other vertex,

$$f_{i_*}(v) \geq F(v_*) - \omega'_M = u(C) - \omega'_M,$$

where ω'_M is the untruncated corresponding oscillation bound. It follows that $u(C) - \ell(C) \leq \omega'_M$. Since both bounds lie in $[0, 1]$, truncation by one is valid. Choose a cell attaining L_M to obtain $U_M - L_M \leq \omega_M$. A vertex attaining U_M is an original row table; it is not a mixture of different designs.

Dyadic refinement only replaces each cell by subsets. By (2.25), its descendants cannot lower the minimum of a fixed test and hence cannot lower L_M . Every coarse vertex belongs to a finer cell and remains an extreme point of that subset; therefore the set of available vertices can only increase, and U_M cannot increase. The bounds in (2.22) tend to zero. Independent choices of the maximizing test in each of the finitely many cells give (2.24) and the two strict-decision assertions.

Finally $q \in \mathcal{P}$ and $f_i(q) \leq \varepsilon$ are finitely many polynomial conditions over the rationals. Quantifier elimination over real closed fields decides their existential conjunction; this is the classical effective Tarski–Seidenberg theorem, not a new elimination algorithm [69, §2.1, Theorem 2.1]. The set of feasible objective values is a semialgebraic ray with included endpoint δ . Its endpoint is algebraic over $\mathbb{Q}$. The nonempty optimum fiber has a point over the real algebraic numbers by the same real-closed-field transfer, proving the final assertion. $\square$

Corollary 2.8 (Private feasibility is locally test complete). *At fixed private widths, $\delta_K^P(E, F) \leq \varepsilon$ if and only if no finite dyadic local testing certificate has value greater than ε . For any finite prescribed list of resource profiles, all strictly feasible and strictly infeasible profiles can be separated by rational original-budget witnesses and finite local certificates. With rational tolerance, exact feasibility and the minimal elements in that finite list are decidable. These assertions require neither a free public selector nor a convex private behavior set.*

Proof. The first claim is (2.24). Apply the strict parts of Theorem 2.7 to each profile in the finite list and its quantifier-elimination clause to profiles on the boundary. The partial order is coordinatewise state and selector cost. Enumeration of feasible profiles then determines the minimal elements. It need not produce a single least profile, and no polynomial running time is asserted. $\square$

For the three-symbol channel V_* in Proposition 2.4, $\delta_2^{\text{mix}} = 0$ whereas $\delta_2^P \geq 1/(4\sqrt{3}) > 1/8$. Theorem 2.7 therefore eventually gives a finite local certificate exceeding $1/8$ at private width two. A single convex testing separator cannot have that conclusion, since the hidden convex hull already contains the target. This example identifies the logical work done by localization; it does not claim a polynomial bound on the size of the certificate.

The exact use of row vertices and real algebraic decision procedures is classical. So is the unrestricted filtered randomization criterion: The precise filtered antecedent is [68, Lemma 1 and Theorem 5], discussed in the introduction. The conclusion here is instead a complete system of local tests with executable witnesses at that fixed register, together with an explicit feedback modulus. These are different conclusions even on the finite delayed-channel slice. The original-source audit records separately the still-unverified comparison with Norberg’s full proof and with the original incomplete-machine minimization papers.

2.6. Transferring intrinsic certificates across presentations. For compatible marked experiments write $E \simeq_a \tilde{E}$ when two stateless, seed-free, marked causal channels, one in each direction, have feedback error at most a . Their report timing and action interfaces are the same. On a visible signature they compare the complete joint interface with the independent selector. These are assumptions about executable channels, not about unseeded marginal distributions.

Theorem 2.9 (Original-budget stability of intrinsic certificates). *Suppose $E \simeq_a \tilde{E}$ and $F \simeq_b \tilde{F}$. For private, fixed hidden-selector, and fixed visible-selector deficiencies at the same budget,*

$$|\delta_K(E, F) - \delta_K(\tilde{E}, \tilde{F})| \leq a + b. \quad (2.26)$$

If the approximating experiments are finite and rational, their local certificates consequently give the rigorous interval

$$\max\{0, L_M - a - b\} \leq \delta_K(E, F) \leq \min\{1, U_M + a + b\}. \quad (2.27)$$

An upper witness consists of the rational transducer and the two stateless bridge channels and has the original persistent resource budget. A lower certificate excludes every simulator at that budget for the original experiments whenever $L_M > a + b + \varepsilon$.

Proof. Compose $E \rightarrow \tilde{E}$, a near-optimal $\tilde{E} \rightarrow \tilde{F}$ transducer, and $\tilde{F} \rightarrow F$. The optimized composition inequality in Theorem 2.2, or its identical channel proof on measurable carriers, adds the three errors and multiplies the widths by $1, K, 1$. The bridges introduce no selector, so a prescribed hidden or visible cardinality remains unchanged. Reverse the bridges for the opposite inequality and take the infima. Equation (2.27) follows from Theorem 2.7. For the last assertion any original simulator would, after reverse bridging, contradict the finite lower certificate. Thus the lower statement does not assume the existence of an optimal simulator on a nonfinite carrier. $\square$

2.7. Witness size in the intrinsic behavior dimension. The local theorem can be complemented by a bound on how many different tests are needed. This bound uses the dimension of attainable behaviors, not the number of stochastic-row coordinates in a chosen realization. It does not confuse the number of tests with the cost of verifying that their violation regions cover a nonconvex set.

Let $\mathcal{B}_K = \{b(q) : q \in \mathcal{K}_K\}$ be the private behavior image from Theorem 2.2. Differences of behaviors have zero mass in each parameter–feedback row. On their linear span put

$$\|b - b'\|_* = \max_{\theta, g} \frac{1}{2} \sum_{\omega \in \Omega_g} |b_{\theta, g}(\omega) - b'_{\theta, g}(\omega)|.$$

This is a norm, $\text{diam } \mathcal{B}_K \leq 1$, and every event discrepancy $f_i(b) = Q_{\theta, g}(B) - b_{\theta, g}(B)$ is 1-Lipschitz. Let $a = \dim \text{aff } \mathcal{B}_K$; redundant realization coordinates do not change a .

Theorem 2.10 (Intrinsic-dimension testing witnesses). *Suppose $\delta_K^p(E, F) \geq L + \gamma$ with $\gamma > 0$. There is a subfamily $\mathcal{I}_0$ of the original marked feedback tests, of size at most*

$$|\mathcal{I}_0| \leq \lceil (1 + 4/\gamma)^a \rceil, \quad (2.28)$$

such that

$$\min_{q \in \mathcal{K}_K} \max_{i \in \mathcal{I}_0} f_i(b(q)) \geq L + \gamma/2. \quad (2.29)$$

Thus the same-budget private lower certificate can use this fixed subfamily on every policy cell. With rational finite input, rational L, γ , and classical real quantifier elimination, such a subfamily can be extracted effectively. No convex mixture of simulators, state increase or selector is introduced.

Proof. In the a -dimensional affine behavior space choose a maximal set $b_1, \dots, b_s \in \mathcal{B}_K$ with pairwise distances strictly greater than $r = \gamma/2$. It is an r -net by maximality. Balls of radius $r/2$ about these points have disjoint interiors and lie in a ball of radius $1 + r/2$ about b_1 . Lebesgue

volume in this affine space, and its homogeneity under dilation, give $s \leq (1 + 2/r)^a$. For $a = 0$ one point suffices. At each center choose a test i_j attaining its maximal discrepancy. This discrepancy is at least $L + \gamma$. Every behavior within r of that center therefore satisfies $f_{i_j} \geq L + \gamma/2$. Take the union of these tests and remove repetitions. This proves both assertions.

For effective extraction use rational row tables as centers. Their behavior images are dense, since rational points are dense in products of simplexes and b is continuous. Given finitely many centers, the assertion that some feasible q lies at distance strictly greater than r from all centers is a finite existential semialgebraic assertion: absolute values and row maxima can be expressed by finitely many polynomial inequalities and disjunctions. Real quantifier elimination decides it [69]. When true, rational row enumeration finds such a point because the strict inequality defines a relatively open nonempty subset of the row simplexes. When false, the chosen points form an r -net. The volume bound forces termination. All discrepancies at rational centers are rational and can be compared exactly.

Finally apply Theorem 2.7 to the reduced finite family. Its maximal discrepancy has the strict gap in (2.29); sufficiently fine policy cells certify L using only these tests. The extraction bound is on the number of distinct tests, not on the number of cells or the complexity of real quantifier elimination. $\square$

Proposition 2.11 (Dimension dependence cannot be discarded). *For the delayed identity experiment on $n \geq 2$ symbols at private width one, $a = n - 1$. If*

$$0 < \gamma < \frac{1}{n(n-1)}, \quad L = 1 - \frac{1}{n} - \gamma,$$

any subfamily of original event tests that proves $\min_q \max_i f_i(q) > L$ contains at least $n = a + 1$ tests.

Proof. A private one-state behavior is a probability vector p independent of the parameter. A positive discrepancy in parameter row i has form $1 - p(B)$ with $i \in B$, and is bounded above by $1 - p_i$. If fewer than n tests are used, let S be the parameter indices of the positive-discrepancy tests. It has size at most $n - 1$. When it is nonempty choose p uniform on S . Every chosen positive discrepancy is at most $1 - 1/|S| \leq 1 - 1/(n - 1) < L$; tests with target probability zero are nonpositive. If S is empty all tests are nonpositive. Hence no such subfamily certifies the stated bound. The full behavior set is the $(n - 1)$ -simplex, giving its dimension. $\square$

The volumetric packing step and real-algebraic elimination are classical. The conclusion here is a structural bound for the nonconvex original resource image, invariant under the addition of unobservable row coordinates. It supplies an alternative to retaining every possible event at every policy cell. It is neither a polynomial-time algorithm nor a claim that one constant convex separator characterizes the private feasible set.

3. ADAPTIVE TESTING ON THE OBSERVABLE BEHAVIOR IMAGE

The obstruction to linear randomization criteria at a fixed private register is the nonconvexity of the behavior image, not a failure of compactness. We give a variational formulation in which a test is an independent function of an observable candidate law. The construction does not give the simulator an extra random variable.

3.1. The testing game. Let B be a nonempty compact subset of a finite-dimensional affine space, and let $f_1, \dots, f_m$ be affine real functions, including the zero function. Put

$$F(b) = \max_{i \leq m} f_i(b), \quad \delta = \min_{b \in B} F(b).$$

For the finite comparison problem of Section 2, b is the array of discrepancies between the simulated and target marked laws, indexed by parameters, feedback trees and transcripts; f_i ranges over their

signed event evaluations. In the visible case the array includes the selector and its target weights. Thus the target coordinate is part of the array when those weights vary. Private, fixed-hidden and fixed-visible signatures each give their own compact set B . No convex hull is taken.

An *adaptive test* is a continuous map $\kappa : B \rightarrow \Delta_m$, chosen before a candidate b . Its score against that candidate is

$$G(b, \kappa) = \sum_i \kappa_i(b) f_i(b).$$

This is a verification game: the selected candidate law is revealed to the evaluator of the already chosen function κ . It is not a one-observation statistical test, and does not make an unavailable parameter or mark observable to a physical controller.

Theorem 3.1 (Independent adaptive testing duality). *For every such compact behavior image,*

$$\begin{aligned} \delta &= \sup_{\kappa \in C(B, \Delta_m)} \min_{b \in B} G(b, \kappa) \\ &= \min_{\mu \in \mathcal{P}(B)} \sup_{\kappa \in C(B, \Delta_m)} \int_B G(b, \kappa) d\mu(b). \end{aligned} \tag{3.1}$$

The minimization over probability measures is attained. If Borel adaptive tests are allowed, the first supremum is attained as well. The continuous-test assertion is a supremum; continuous attainment is not needed or asserted.

Proof. Fix $\eta > 0$. The sets $U_i = \{b : f_i(b) > F(b) - \eta\}$ form an open cover of the compact metric space B . Choose a continuous partition of unity subordinate to this finite cover. The resulting test has $F(b) - \eta < G(b, \kappa) \leq F(b)$ for all b . Consequently the first supremum equals δ . The same uniform inequality shows, for every probability μ ,

$$\sup_{\kappa} \int G(b, \kappa) d\mu(b) = \int F d\mu.$$

Its minimum is δ , attained by a point mass at a minimizer of F . This also proves the displayed minimax exchange without an assumption that B is convex. Equivalently one can apply the usual compact-convex minimax theorem to $\mathcal{P}(B)$ and the convex class of continuous tests, with the pairing $\int G d\mu$. Finally choose the smallest index attaining $F(b)$. Its level sets are Borel, and its indicator vector is a Borel test with score exactly F . $\square$

The probability μ here describes a lottery over *revealed candidate laws*. The score is $\int G(b, \kappa) d\mu(b)$, not $G(\int b d\mu, \kappa)$. Hiding the candidate and replacing it by its barycenter is a different game; in general it lowers the value. This distinction is essential for a finite persistent-state budget.

3.2. A size bound in behavior dimension. Factor out directions annihilated by every $f_i - f_i(b_0)$, so that on the resulting difference space

$$\|v\|_* = \max_i |f_i(b_0 + v) - f_i(b_0)|$$

is a norm. Denote its dimension by a and let $R = \text{diam}_*(B)$. All conclusions below are unaffected by this observable quotient. The case $a = 0$ is a single constant comparison and requires one test. Suppose henceforth that $a \geq 1$, so $R > 0$.

Lemma 3.2 (A bounded behavior chart). *There exist $b_0 \in B$ and vectors $v_j \in B - b_0$, $1 \leq j \leq a$, forming a basis, such that $\|v_j\|_* \leq R$ and every $b \in B$ has coordinates $b = b_0 + \sum_j t_j v_j$ with $|t_j| \leq 2$. If B is a rational polynomial image of a product of simplexes, the points can be chosen rational and can be found effectively.*

Proof. Fix $b_0 \in B$. In any fixed coordinates on the difference space, let $D_0 > 0$ be the maximum absolute determinant of a vectors in $B - b_0$. Compactness and spanning give existence and positivity. Choose a tuple with determinant greater than $D_0/2$. Replacing its j th vector by $b - b_0$ multiplies the determinant by t_j ; maximality gives $|t_j| < 2$. The norm bound is the definition of R . For a rational polynomial image, take b_0 at rational rows. Rational row images are dense. An affine basis can be found by enumerating rational images and testing rational determinants; the true affine span and the maximum determinant are semialgebraic data. Quantifier elimination decides whether a proposed nonzero determinant exceeds half the maximum. Density therefore makes the search terminate. This is an effectiveness assertion, not a bit-complexity bound for finding the chart. $\square$

Set $b(x) = b_0 + \sum_j (4x_j - 2)v_j$ and $C = \{x \in [0, 1]^a : b(x) \in B\}$. The ambient cube is a coordinate box, not an enlargement of the feasible behavior set. Write $\ell_i(x) = f_i(b(x))$. Then

$$|\ell_i(x) - \ell_i(y)| \leq 4R \sum_j |x_j - y_j|. \quad (3.2)$$

For $n \geq 0$ let $\beta_{\alpha,n}$ be the tensor Bernstein basis of coordinatewise degree n . A feasible coefficient array has $c_{\alpha i} \geq 0$ and $\sum_i c_{\alpha i} = 1$ for every $\alpha \in \{0, \dots, n\}^a$. It specifies the independent test

$$\kappa_i^c(x) = \sum_{\alpha} c_{\alpha i} \beta_{\alpha,n}(x).$$

Define

$$d_n = \max_c \min_{x \in C} \sum_{\alpha, i} c_{\alpha i} \beta_{\alpha,n}(x) \ell_i(x). \quad (3.3)$$

Theorem 3.3 (Intrinsic adaptive degree and support bounds). *The maxima in (3.3) are attained and $d_n \leq d_{n+1} \leq \delta$. For $n \geq 1$,*

$$0 \leq \delta - d_n \leq \frac{4Ra}{\sqrt{n}}. \quad (3.4)$$

There is an array with at most $(n+1)^a$ nonzero coefficients, all equal to one, whose minimum score is at least $\delta - 4Ra/\sqrt{n}$. Its test has coordinatewise degree n , and its score degree at most $n+1$. Furthermore

$$d_n = \min_{\mu \in \mathcal{P}(C)} \sum_{\alpha} \max_i \int_C \beta_{\alpha,n}(x) \ell_i(x) d\mu(x), \quad (3.5)$$

and a minimizing measure can be supported on at most $(n+2)^a$ points. In particular a strict level $L < \delta$ admits an adaptive test of positive uniform margin with $n > (8Ra/(\delta - L))^2$ and at most $(n+1)^a$ occupied coefficient entries. These bounds depend on the observable image and its metric, not on the number of stochastic rows in a realization.

Proof. The coefficient space is a finite product of compact simplexes, and the minimum of a jointly continuous function over compact C is continuous in its coefficients. This proves attainment. Bernstein degree elevation replaces each coefficient by convex combinations of adjacent coefficients, preserving every simplex constraint and the represented function. It proves monotonicity. All scores are convex combinations of the $\ell_i(x)$, proving the upper bound by δ .

At each grid point α/n choose an index i_{α} maximizing ℓ_i there. The choice is made even at infeasible grid points; only the final score is minimized over C . Give that index coefficient one and the others zero. For fixed x let A_j be independent $\text{Bin}(n, x_j)$ variables. If i_* maximizes at x ,

$$\begin{aligned} \max_i \ell_i(x) - \mathbb{E} \ell_{i_A}(x) &\leq \mathbb{E} \{ \ell_{i_*}(x) - \ell_{i_*}(A/n) + \ell_{i_A}(A/n) - \ell_{i_A}(x) \} \\ &\leq 8R \sum_j \mathbb{E} |A_j/n - x_j| \leq \frac{4Ra}{\sqrt{n}}. \end{aligned}$$

Here the omitted middle difference is nonpositive, and the last bound uses $\text{Var}(A_j/n) \leq 1/(4n)$. This proves the rate and the sparse coefficient assertion.

For the last formula replace the minimum over C by a minimum over $\mathcal{P}(C)$. The pairing is bilinear and continuous, and both feasible sets are compact convex, so the compact minimax theorem [71] exchanges the extrema. Optimizing each coefficient simplex gives (3.5). The integrands are polynomials of coordinatewise degree at most $n + 1$. Preserve all monomial moments of that degree by Carathéodory's theorem. There are $(n + 2)^a$ monomials including the constant; their moment vectors lie in an affine hyperplane of dimension at most $(n + 2)^a - 1$. A measure with at most $(n + 2)^a$ atoms therefore has exactly the same objective. The strict-margin assertion follows from the rate with error less than $(\delta - L)/2$. $\square$

The score is a nonlinear function of behavior although each event functional is affine. This is the reason the measure in (3.5) cannot be replaced by its behavior barycenter. For $n = 0$ it can: d_0 is precisely the constant-test convex relaxation. In the two-time unmarked example of Proposition 6.5, the target belongs to the convex hull of private behaviors, so $d_0 = 0$, whereas $\lim_n d_n = \sqrt{2} - 1$. Thus the hierarchy recovers a gap which every constant test misses.

Remark 3.4 (What the size bound measures). The bound counts the nonzero entries of the independent adaptive test and controls its behavior-coordinate degree. It does not bound the cost of verifying its minimum on the generally nonconvex set C , the bit lengths of a rational chart, or the size of a description of C . Distinct row presentations of the same metric behavior image satisfy the same existence bound, although their coefficient arrays need not coincide. For the private fixed-target law array one may use $R \leq 1$ with the maximum total-variation metric. For the visible signed-discrepancy array the uniform bound $R \leq 2$ is sufficient. The old row-cube positivity certificate is a separate verification engine, with its own coordinate-dependent complexity.

3.3. A three-weight certificate invisible to constant tests.

Proposition 3.5 (A finite intrinsic nonlinear witness). *Consider a private width-one simulator producing two independent bits, with probabilities p, q , for the target which assigns probability $1/2$ to each of 00, 11. Put $u = pq$, $v = (1 - p)(1 - q)$ and $J(t) = 3t^2 - 2t^3$. The three behavior-dependent weights*

$$\kappa_1 = J(v), \quad \kappa_2 = J(u), \quad \kappa_3 = 1 - J(u) - J(v)$$

are nonnegative and sum to one. Against the event discrepancies $1/2 - u, 1/2 - v, 1 - u - v$, respectively, their score G satisfies

$$G \geq \frac{2}{5} + \frac{9}{8960}. \quad (3.6)$$

The certificate uses cubic weights in observable probabilities, a nine-term nonnegative derivative identity, and seventeen positive rational Bernstein coefficients in one variable. Every constant mixture of event discrepancies has minimum at most zero.

Proof. We have $u + v \leq 1$. Since J is increasing on $[0, 1]$ and $J(1 - t) = 1 - J(t)$, the weights are feasible. Their score is

$$G = 1 - u - v + J(u)(u - 1/2) + J(v)(v - 1/2).$$

Set $a = (p + q - 1)^2$ and $b = (p - q)^2$. Then $a, b \geq 0$ and $a + b \leq 1$. Direct expansion gives

$$\begin{aligned} 64G &= 27 - 12a + 38a^2 - 20a^3 - a^4 + 36b - 60ab + 36a^2b + 4a^3b \\ &\quad + 6b^2 - 12ab^2 - 6a^2b^2 - 4b^3 + 4ab^3 - b^4. \end{aligned}$$

Let $c = 1 - a - b$. Another polynomial identity, valid when $a + b + c = 1$, is

$$\begin{aligned} 16\partial_b G &= 4a^3 + 6a^2c + 12ab^2 + 24abc + 12ac^2 \\ &\quad + 8b^3 + 30b^2c + 30bc^2 + 9c^3 \geq 0. \end{aligned}$$

Thus $G(a, b) \geq G(a, 0)$. The univariate polynomial

$$G(a, 0) - \frac{2}{5} = \frac{7}{320} - \frac{3}{16}a + \frac{19}{32}a^2 - \frac{5}{16}a^3 - \frac{1}{64}a^4$$

has degree-16 Bernstein coefficients $z_k/349440$, where

$$(z_0, \dots, z_{16}) = (7644, 3549, 1183, 351, 855, 2494, 5064, 8358, \\ 12166, 16275, 20469, 24529, 28233, 31356, 33670, 34944, 34944).$$

The Bernstein basis is nonnegative and sums to one. Its smallest coefficient is $351/349440 = 9/8960$, proving (3.6). All identities are rational polynomial identities; no numerical minimization is used. Finally the target is the barycenter of the two feasible deterministic laws 00 and 11. An affine mixture of signed event discrepancies has average zero at these two laws and hence minimum at most zero. The empty event supplies the reverse constant-test bound zero. $\square$

For comparison, the affine weights $v, u, 1 - u - v$ already give the smaller bound $3/8$, with the exact identity

$$G_{\text{affine}} - \frac{3}{8} = \frac{1}{2}(p - q)^2 + 2(p - 1/2)^2(q - 1/2)^2.$$

Neither certificate changes the private feasible set. The cubic certificate is an independent test chosen once for the whole behavior image, not a mixture chosen anew by an optimizer's unknown row values. Its evaluation does use the revealed behavior, as specified in the verification game. The 17-coefficient positivity check is classical; the distinction from constant tests is the behavior-dependent feasible weight family and its resource-sensitive minimum.

3.4. Strict certificates and effective physical input. Given rational finite instruments, a rational chart and the sparse array constructed above, its strict minimum can be certified by real quantifier elimination on the original row constraints. Alternatively, pull the score back through the original polynomial row map and certify its strict positivity using degree-elevated Bernstein coefficients. The pullback changes the verification degree and is not covered by the behavior-coordinate bound. Generic quantifier elimination also decides the exact finite semialgebraic boundary; this classical observation does not assert that a finite nonnegative Bernstein expansion exists at $L = \delta$.

Proposition 3.6 (Uniform effective presentation input). *Suppose a single algorithm, on input j , returns rational finite instruments E_j, F_j and nonnegative rational error enclosures a_j, c_j for two-sided stateless marked feedback presentations of E, F . Suppose $a_j + c_j \rightarrow 0$. All alphabets, widths, selector laws and finite subroutines must be computable uniformly from j and the requested signature. Then, for every positive rational η , an algorithm returns a rational interval of width below η containing $\delta_b(E, F)$ at the same signature.*

Proof. Enumerate j until $a_j + c_j < \eta/8$. This terminates without a known rate because the rational enclosures themselves tend to zero. For the selected finite problem, enumerate rational same-budget upper tables and strict rational lower certificates, dovetailing all finite checks; include the always valid lower bound zero. Compactness, density and strict completeness provide an interval of width below $\eta/2$. Enlarge it by $a_j + c_j$ at both ends using stateless comparison. The enlarged width is below $3\eta/4$. Every enumeration and comparison is uniform by the hypotheses. A mere sequence of individually computable presentations without a uniform enumeration is not an input satisfying this proposition. $\square$

4. A MARKED RESOURCE GAP GENERATED BY A BINARY COLLISION

We now realize a private–hidden–visible separation in a physical experiment whose information vanishes when the collision is removed. The controller chooses a later sensor gate after its first report. The gate controls acquisition, not the mechanical collision law. Both the source and the target carry the same actual, unobserved mark.

4.1. The exact marked two-report experiment. Let $B, W \in \{0, 1\}$ have the joint distribution

$$\mathbb{P}(B = b, W = w) = \begin{cases} 5/16, & b = w, \\ 3/16, & b \neq w. \end{cases} \quad (4.1)$$

The blind source E_* has two constant reports and actual mark W . The target F_* reports $Y_1 = B$, receives $A \in \{0, 1\}$ after that report, and reports $Y_2 = AB$. The feedback tester may choose A as any function of its earlier report. A width-one private simulator cannot reread its own first emission, except through the current gate A . Hidden selectors persist but are not revealed; visible selectors are included in the joint target law independently of (B, W) . Put

$$d_* = \sqrt{5/2} - 9/8.$$

Theorem 4.1 (An exact marked causal separation). *At the width profile $(1, 1)$,*

$$\delta^P(E_*, F_*) = d_*, \quad \delta_2^h(E_*, F_*) = 1/8, \quad \delta_r^v(E_*, F_*) = d_* \quad (r \geq 1). \quad (4.2)$$

The hidden upper bound is realized by two deterministic tables. The private upper bound is realized by two independent Bernoulli variables with parameter $\sqrt{5/8}$, followed by the gate. These bounds hold against every feedback rule, not just open-loop gates.

Proof. First choose the constant gate $A = 1$. The two private emissions are independent Bernoulli variables with parameters p, q and are independent of W . Let

$$u = pq, \quad v = (1 - p)(1 - q), \quad a = 3/8, \quad b = 5/8.$$

The overlap of the target and simulated marked laws is $H(u) + H(v)$, where

$$H(t) = \frac{1}{2} \{ \min(t, a) + \min(t, b) \}.$$

Indeed the target is supported on equal report pairs, and its two W -masses at each equal pair are $3/16$ and $5/16$. Total variation is one minus this overlap.

Cauchy–Schwarz gives $\sqrt{u} + \sqrt{v} \leq 1$. By symmetry suppose $u \geq v$. If $u \leq 1/4$ the overlap is at most $1/2$. If $u \geq 1/4$, then $v \leq (1 - \sqrt{u})^2 \leq 1/4 < a$, so the overlap is at most

$$H(u) + (1 - \sqrt{u})^2.$$

This function is convex on each of $[1/4, a]$ and $[a, b]$ and nonincreasing on $[b, 1]$. Its maximum is therefore attained at $1/4$, a or b . The values at a and b are $1 + 2a - 2\sqrt{a}$ and $3/2 + b - 2\sqrt{b}$ respectively. The second is larger, since

$$(3/2 + b - 2\sqrt{b}) - (1 + 2a - 2\sqrt{a}) = 3/8 - \frac{1/2}{\sqrt{a} + \sqrt{b}} > 0,$$

using $\sqrt{a} > 3/5$, $\sqrt{b} > 3/4$. It also exceeds $1/2$. Taking $p = q = \sqrt{b}$ attains this maximum, and hence the minimum error is $1 - (3/2 + b - 2\sqrt{b}) = d_*$.

For any hidden selector, the all-on report pair is still independent of W . If its probabilities of 11 and 00 are u, v , then $u + v \leq 1$. Concavity and monotonicity of H give

$$H(u) + H(v) \leq 2H((u + v)/2) \leq 2H(1/2) = 7/8.$$

A fair hidden bit J , emitting J twice, attains this overlap. This requires precisely two deterministic selector tables and gives error $1/8$. In particular hiding a seed does not erase the actual-mark obstruction.

These all-on bounds are lower bounds for the feedback problem. For the upper bounds first generate the corresponding all-on pair (X_1, X_2) , emit X_1 , and after receiving A emit AX_2 . For a deterministic feedback $A = g(X_1)$ the full gated marked transcript is the pushforward of the all-on marked pair. Applying the same map to the target pair cannot increase total variation. The construction is executable at width one: the second fresh Bernoulli variable is independent in the private case, and the bit J is the priced persistent hidden selector in the hidden case. Randomized feedback follows by conditioning on its independent random table.

Finally, in the visible case the two compared laws are

$$\sum_j \lambda_j \delta_j \otimes P_{g_j}^{q_j} \quad \text{and} \quad \sum_j \lambda_j \delta_j \otimes Q_{g_j}.$$

The target selector is independent of (B, W) . The all-on gate in every fiber has error at least d_* in every private fiber. Joint variation is the weighted sum of these fiber errors. Ignoring the visible selector and using the private optimizer attains d_* . $\square$

4.2. A full-dimensional collision preparation. Consider two labeled equal-mass hard spheres in $\mathbb{R}^3$ with contact distance d . The center of mass $Q = (q_1 + q_2)/2$ ranges over any fixed rational position box. Write $r = q_1 - q_2$, $g = v_1 - v_2$, and $V = (v_1 + v_2)/2$. Conditional on (B, W) in (4.1), take the uniform product preparation on

$$\begin{aligned} r &= (-a_0, (2B - 1)b_0, c_0), & a_0 &\in [29/10, 31/10], & b_0 &\in [9/20, 11/20], & (4.3) \\ |c_0| &\leq 1/100, & g_x &\in [19/10, 21/10], & |g_y|, |g_z| &\leq 1/100, \\ |V_x|, |V_y| &\leq 1/100, & |V_z - (2W - 1)| &\leq 1/100. \end{aligned}$$

The boxes have positive volume in the full phase space. The actual mark is $W = \mathbf{1}_{\{V_z > 0\}}$; total momentum conservation retains it. The informative quantity will instead be the transverse velocity created at a collision. Put

$$h(z) = \text{sgn}(z) \min\{1, (|z| - 1/4)_+\}.$$

This is a bounded 1-Lipschitz detector which is identically zero on the incoming transverse-velocity range.

Lemma 4.2 (Uniform transverse collision signal). *For every preparation in (4.3) and every $d \in [9/10, 11/10]$, exactly one collision occurs before time 3. Its time t_c lies in $(4/5, 3/2)$, and afterwards*

$$(2B - 1)v_{1,y} > 1/2, \quad (2B - 1)h(v_{1,y}) > 1/4. \quad (4.4)$$

For $0 < d \leq 2/5$ no collision occurs before time 3, and the detector is zero throughout that interval.

Proof. Up to a collision $r(t) = r + tg$. For $t \leq 4/5$, its x coordinate is at most $-29/10 + (4/5)(21/10) = -61/50$, so its norm exceeds $11/10$. At time $3/2$ the absolute coordinate bounds are $1/4, 113/200$ and $1/40$, whose squared sum is less than $(9/10)^2$. Thus a first contact occurs in the stated interval. At the free zero of the x coordinate, whose time is at most $31/19$, the transverse norm is less than $9/10$ as well. The first contact therefore has $n_x < 0$, where $n = r(t_c)/d$. The entry contact is strictly incoming. The bounds above give

$$|n_x|^2 \geq 1 - \frac{(113/200)^2 + (1/40)^2}{(9/10)^2} > (77/100)^2, \quad (2B - 1)n_y \geq \frac{87}{220}.$$

It follows that $-g \cdot n > (19/10)(77/100) - 2/100 = 1443/1000$. The equal-mass elastic collision rule is

$$v'_1 = V + g/2 - (g \cdot n)n, \quad g' = g - 2(g \cdot n)n.$$

Consequently

$$(2B - 1)v'_{1,y} > -3/200 + \frac{1443}{1000} \frac{87}{220} > 1/2.$$

The relative velocity is outgoing, $r(t_c) \cdot g' > 0$, so the squared separation increases strictly for all later times. There are only two spheres and no walls; no second collision is possible. This proves (4.4). For $d \leq 2/5$ and $0 \leq t \leq 3$, $|(r + tg)_y| \geq 9/20 - 3/100 = 21/50 > d$. Thus there is no collision. The incoming $|v_{1,y}| \leq 3/200$ then makes h zero. $\square$

At time 2 the physical target $F_{d,\sigma}$ reports

$$Y_1 = \mathbf{1}_{\{h(v_{1,y}(2)) + \sigma \xi_1 > 0\}}.$$

It then receives a gate $A \in \{0, 1\}$, and at time 3 reports

$$Y_2 = A \mathbf{1}_{\{h(v_{1,y}(3)) + \sigma \xi_2 > 0\}},$$

where the ξ_i are independent standard normal variables, independent of the preparation. The source E_d has the same preparation, evolution, control interface and actual mark, but emits two constant reports. The gates do not change the particle motion. This is a feedback acquisition experiment: taking $A = 0$ suppresses later information and taking $A = 1$ exposes another noisy observation of a collision-generated quantity.

Theorem 4.3 (Collision-sensitive physical resource separation). *Let $d \in [9/10, 11/10]$ and $0 < \sigma \leq 1/12$. At retained width $(1, 1)$ the physical deficiencies obey, uniformly in d ,*

$$\begin{aligned} |\delta^p(E_d, F_{d,\sigma}) - d_*| &< 1/300, & |\delta_r^v(E_d, F_{d,\sigma}) - d_*| &< 1/300 \quad (r \geq 1), \\ |\delta_2^h(E_d, F_{d,\sigma}) - 1/8| &< 1/300. \end{aligned} \quad (4.5)$$

In particular private and visible error exceed $9/20$, while hidden two-selector error is less than $13/100$. More generally the errors in these three comparisons are at most $2\Phi(-1/(4\sigma))$, which tends to zero as $\sigma \downarrow 0$. For $0 < d \leq 2/5$ all three deficiencies are zero. The interacting and noninteracting regimes use the same preparation and detector. The conclusions are unchanged by simultaneously scaling positions, diameter and observation times by any positive factor, with velocities and detector noise unchanged.

Proof. Couple the ideal and physical target by the same (B, W) and the same feedback random table. Lemma 4.2 bounds each potential bit error by $\Phi(-1/(4\sigma))$. With probability at least $1 - 2\Phi(-1/(4\sigma))$ both potential reports agree with their ideal bits; the gates and all emitted transcripts then agree as well. This gives a marked feedback variation bound uniform over all gates. The channels in both directions are the identity on the binary report alphabet and have one retained state. The blind source is already exactly E_* . Stateless comparison therefore changes each fixed-budget deficiency by at most the stated error, without adding memory or hiding an exposed selector.

For completeness, the elementary Gaussian tail bound $\Phi(-x) \leq e^{-x^2/2}/(x\sqrt{2\pi})$ at $x = 3$ gives $\Phi(-3) < 1/600$: use $e^{9/2} > 80$ and $\sqrt{2\pi} > 5/2$. Hence the total error is below $1/300$. The inequalities $d_* - 1/300 > 9/20$ and $1/8 + 1/300 < 13/100$ follow by squaring positive rational bounds.

When $d \leq 2/5$, both potential reports are independent fair noise bits and are independent of W . A private width-one simulator draws those two bits and applies the current gate, exactly reproducing every feedback transcript. Hidden and visible versions can ignore their selectors. Finally elastic scattering is invariant under $(q, d, t) \mapsto (sq, sd, st)$ with v unchanged. This proves the scaling statement; it is geometric similarity, not a particle-number or Boltzmann–Grad limit. $\square$

The theorem combines three effects. Collisions create the sign of the informative transverse velocity. A gate chosen after the first report changes future information. A retained hidden bit, but not a visible one, permits temporal correlation without charging the width-one register. The actual mark remains partially inaccessible in every signature. The conserved longitudinal momentum labels that mark; it does not generate the observed transverse signal.

4.3. Uniformly effective finite presentations. The binary physical interface is already finite, but its probabilities are not generally rational. We make their effective approximation explicit rather than treating individual computability as a uniform algorithm.

Proposition 4.4 (Collision-chart rational presentations). *For computable $d \in [9/10, 11/10]$ and $\sigma > 0$, one algorithm produces rational marked causal instruments and certified two-sided stateless errors tending to zero for $(E_d, F_{d,\sigma})$. On a compact computable parameter interval with $\sigma \geq \sigma_0 > 0$, the errors and the construction are uniform. Thus Proposition 3.6 applies at every fixed finite state and selector signature, not only at $(1, 1)$.*

Proof. On every preparation box the entry time and outgoing velocity are

$$t_c = \frac{-r \cdot g - \sqrt{(r \cdot g)^2 - |g|^2(|r|^2 - d^2)}}{|g|^2}, \quad n = (r + t_c g)/d, \quad v'_1 = V + g/2 - (g \cdot n)n.$$

The positive square root is separated from zero: it equals $-d(g \cdot n) > (9/10)(1443/1000)$. All denominators are likewise bounded away from zero. These explicit bounds and the compact boxes allow interval differentiation of the displayed formulas, producing a rational common Lipschitz upper bound C_{geo} for $v'_{1,y}$ in the packet variables and d . This computation terminates because only addition, multiplication, division by separated positive quantities and a separated square root occur.

Conditional on the microscopic preparation the two potential all-on bits are independent Bernoulli with probability $p = \Phi(h(v'_{1,y})/\sigma)$. Their four probabilities are p^2 , $p(1-p)$, $p(1-p)$ and $(1-p)^2$. The normal distribution function is uniformly computable on bounded intervals, for example by integrating its convergent exponential series with a remainder bound. Its Lipschitz constant is less than one. Together with the 1-Lipschitz detector and the geometric bound, this yields a computable Lipschitz constant at most $2C_{\text{geo}}/\sigma_0$ for these four integrands on fixed-noise compact parameter sets. Equal-box Riemann sums of mesh h therefore have explicit error at most a computable constant times h , plus the chosen rational evaluation error. Parameter uncertainty is included by its interval enclosure.

Integrate the four all-on probabilities separately on each packet, with the exact rational weights (4.1). For each W , round its conditional probability vector to a rational simplex point on a sufficiently fine grid; the rounding error is bounded by the sum of the coordinate errors and the grid mesh. Keep $\mathbb{P}(W = 0) = \mathbb{P}(W = 1) = 1/2$ exact. Sample this finite joint law once at initialization, emit its first bit, and gate its second bit. This is a finite latent realization of the target instrument, not a simulator with extra free state. It makes every causal prefix consistent and never divides by a possibly small prefix probability. Every gated law is a pushforward of the all-on law, so the same marked variation bound holds uniformly under feedback. Identity report channels give both stateless directions. Choosing summable evaluation and mesh errors produces the asserted uniform sequence. $\square$

5. UNIFORM STABILITY OF OPTIONAL PROJECTIONS

Marked comparison controls more than terminal bounded risk when the compared laws share their latent variable. We prove a whole-process estimate for the resulting posterior martingales. The distinction between a probabilistic coupling and an executable causal channel is important in this section.

Let Z take values in a standard Borel space and let Y take values in a standard Borel canonical path space on $[0, T]$, with a right-continuous observation filtration. Consider equivalent joint probability measures P, Q on (Z, Y) , with the same Z -marginal ν . Use the usual completions, which agree for equivalent output laws. For a bounded measurable latent variable $f(Z)$ write

$$M_t^P = \mathbb{E}_P[f(Z) \mid \mathcal{F}_t^Y], \quad M_t^Q = \mathbb{E}_Q[f(Z) \mid \mathcal{F}_t^Y].$$

Take their right-continuous martingale versions. All suprema below include T and can be evaluated on a countable dense time set.

Theorem 5.1 (Marked variation controls the whole posterior process). *Let $\varepsilon = \|P - Q\|_{\text{TV}}$ and $|f| \leq 1$. When both posterior functions are evaluated on the same canonical output path,*

$$\mathbb{E}_Q \sup_{t \leq T} |M_t^P(Y) - M_t^Q(Y)|^2 \leq 80\varepsilon. \quad (5.1)$$

There is a coupling of (Z, Y^P) and (Z, Y^Q) which retains exactly the same latent variable and satisfies

$$\mathbb{P}(Y^P \neq Y^Q) \leq \varepsilon, \quad \mathbb{E} \sup_{t \leq T} |M_t^P(Y^P) - M_t^Q(Y^Q)|^2 \leq 84\varepsilon. \quad (5.2)$$

The coupling is not asserted to be causal or to have finite state. The variation bound itself, not this coupling, is used for resource transport.

Proof. Let $D = dP/dQ$, $D_t = \mathbb{E}_Q[D \mid \mathcal{F}_t^Y]$, and set

$$A_t = \mathbb{E}_Q[(D - 1)f \mid \mathcal{F}_t^Y], \quad B_t = D_t - 1.$$

Bayes' formula gives

$$M_t^P - M_t^Q = \frac{A_t - B_t M_t^Q}{1 + B_t}.$$

Since $\mathbb{E}_Q|D - 1| = 2\varepsilon$, both martingales A, B have terminal absolute expectation at most 2ε . Doob's weak L^1 inequality gives, for $0 < u \leq 2$,

$$\begin{aligned} Q\{\sup_t |M_t^P - M_t^Q| > u\} &\leq Q\{\sup_t |B_t| > 1/2\} + Q\{\sup_t |A_t| > u/4\} + Q\{\sup_t |B_t| > u/4\} \\ &\leq 4\varepsilon + 16\varepsilon/u. \end{aligned}$$

Integrating $2u$ times this bound on $(0, 2)$ gives $16\varepsilon + 64\varepsilon = 80\varepsilon$, proving (5.1). No tightness conclusion is being inferred from finite-dimensional convergence.

Disintegrate both joint laws over their common latent marginal. Measurable maximal coupling of the two output kernels, conditional on Z , has disagreement probability

$$\int \|P(dy \mid z) - Q(dy \mid z)\|_{\text{TV}} d\nu(z) = \|P - Q\|_{\text{TV}}.$$

Such a coupling can be constructed by using the minimum of the two kernel densities with respect to their sum, then their residual kernels. It is measurable on standard Borel spaces. On the event of identical full output paths use (5.1) with the Q -marginal; on its complement the squared difference is at most four. This proves (5.2). The maximal coupling may depend on the full output path; it has no asserted online implementation. $\square$

Corollary 5.2 (Changing-filtration convergence from marked comparison). *Let P_n and P be equivalent joint laws with the same latent marginal ν , and $\|P_n - P\|_{\text{TV}} \rightarrow 0$. They admit a common probability-space realization $(Z, Y, Y^1, Y^2, \dots)$ such that, for every $f \in L^2(\nu)$,*

$$\mathbb{E} \sup_{t \leq T} \left| \mathbb{E}[f(Z) \mid \mathcal{F}_t^{Y^n}] - \mathbb{E}[f(Z) \mid \mathcal{F}_t^Y] \right|^2 \rightarrow 0.$$

The same realization gives convergence uniformly in time in probability for $f \in L^1(\nu)$. It also permits $f_n \rightarrow f$ in $L^2(\nu)$, respectively in $L^1(\nu)$, in the first conditional expectation.

Proof. Glue the conditional maximal couplings of (Z, Y^n) with the fixed reference (Z, Y) by taking their conditional kernels independently on a countable product. The coupling does not depend on f . For the truncation $f^{(K)} = (-K) \vee (f \wedge K)$, Theorem 5.1 bounds the middle difference in the L^2 norm of the supremum by $K\sqrt{84\epsilon_n}$. Doob's L^2 inequality bounds each truncation tail by $2\|f - f^{(K)}\|_{L^2(\nu)}$. The triangle inequality therefore gives

$$\left\| \sup_t |M_t^{P_n, f} - M_t^{P, f}| \right\|_2 \leq 4\|f - f^{(K)}\|_2 + K\sqrt{84\epsilon_n}.$$

First let $n \rightarrow \infty$, then $K \rightarrow \infty$. For L^1 use instead the weak maximal inequality for each tail and the bounded-case estimate for the middle term. An additional difference $f_n - f$ is handled by the same maximal inequalities under P_n , whose latent marginal remains ν . $\square$

5.1. A microscopic sufficient condition. Let $(\Phi_t)_{t \in \mathbb{R}}$ be an almost-everywhere regular hard-sphere flow on a labeled phase space, with invariant probability μ . Fix a prepared law ν with $d\nu/d\mu \leq C$, and bounded real observables h, h_n . These may be collision-sensitive tagged observables. Put $\eta_n = \|h_n - h\|_{L^2(\mu)}$. With a common latent initial state $Z \sim \nu$, consider the continuous observation laws

$$\begin{aligned} Y_t &= \int_0^t h(\Phi_s Z) ds + \sigma B_t, \\ Y_t^n &= \int_0^t h_n(\Phi_s Z) ds + \sigma B_t^n, \quad \sigma > 0, \end{aligned}$$

where the Brownian noise is independent of Z in each model. The notation does not prescribe a synchronous coupling.

Theorem 5.3 (Collision-observation optional-projection limit). *The joint latent-output laws are equivalent and satisfy*

$$\|P_n - P\|_{\text{TV}} \leq \frac{\sqrt{CT}}{2\sigma} \eta_n. \quad (5.3)$$

For every $|f(Z)| \leq 1$ they admit a latent-preserving coupling with

$$\mathbb{E} \sup_{t \leq T} |M_t^{P_n, f} - M_t^{P, f}|^2 \leq \frac{42\sqrt{CT}}{\sigma} \eta_n. \quad (5.4)$$

In particular $\eta_n \rightarrow 0$ supplies whole-process optional-projection convergence, not merely convergence at finitely many times. This applies when h_n are orbit-mollified graph-core approximants to any bounded $h \in L^2(\mu)$.

Proof. Conditional on Z , the signals are deterministic bounded drifts. The exponential change-of-measure identity for Brownian motion gives

$$D_{\text{KL}}(P_n \| P) = \frac{1}{2\sigma^2} \mathbb{E}_\nu \int_0^T |(h_n - h)(\Phi_s Z)|^2 ds \leq \frac{CT}{2\sigma^2} \eta_n^2.$$

The last inequality uses $\nu \leq C\mu$ and invariance of μ ; it does not require mixing, differentiability across a collision, or a claim that a tagged velocity belongs to the generator domain. Bounded drifts ensure equivalence and justify the exponential identity. Pinsker's inequality gives (5.3). Applying Theorem 5.1 proves (5.4).

For the last assertion choose a nonnegative smooth approximate identity ρ_n on the time line and set $h_n = \int \rho_n(s) h \circ \Phi_s ds$. Strong continuity of the Koopman group in $L^2(\mu)$ gives $\eta_n \rightarrow 0$ and $\|h_n\|_\infty \leq \|h\|_\infty$. Repeated integration by parts puts h_n in every power domain of the generator,

with derivatives obtained from $\rho_n^{(k)}$. This is the orbit-core construction already used in the microscopic domain theory; the new conclusion is the uniform posterior estimate. $\square$

Sampling these observation paths at finitely many checkpoints and applying a common causal output postprocessing are contractions in marked variation. Thus the identity report comparison transfers every bounded finite-register risk at its original state and selector budgets with error bounded by (5.3). That is the resource-comparison consequence. The latent-preserving coupling in (5.4) has a different role: it realizes convergence of the changing observation filtrations and is not used as a finite-register simulator.

The theorem discharges a sufficient conditional-law hypothesis for optional projection in a genuine microscopic observation class. The constants depend on the fixed preparation, time and noise level. No particle-uniform kinetic estimate, changing microscopic path limit, limiting stochastic-exponential identification or nonlinear BSDE limit is implied without its additional hypotheses. The whole-process estimate is what replaces an unsupported appeal to Doob's inequality as a tightness argument.

6. POLYNOMIAL TESTING OF THE ORIGINAL RESOURCE IMAGE

The private behavior image is usually nonconvex. Its testing criterion can nevertheless be written as a limit of finite algebraic lower certificates, without replacing that image by randomized designs. The testing weights in this section depend on the candidate simulator. Unlike Theorem 3.1, this identity is a certificate transform, not an independent variational dual. They are universal proof functions, not a hidden selector supplied to an executed experiment.

Use any one of the finite signatures of Theorem 2.2, with rational instrument data. Write its objective as $F(q) = \max_{1 \leq i \leq m} f_i(q)$, where f_i are all its marked event discrepancies, including the empty event $f_1 = 0$. For visible selectors the event and feedback rule include their selector fibers, and the target selector law remains in f_i . Thus $-1 \leq f_i \leq 1$ for all three signatures; the polynomials may be different.

Parameterize a row with d outcomes by $d - 1$ variables in $[0, 1]$:

$$q_1 = x_1, \quad q_j = x_j \prod_{l < j} (1 - x_l) \quad (1 < j < d), \quad q_d = \prod_{l < d} (1 - x_l).$$

This map is onto the row simplex, including its boundary. Indeed, divide each desired probability by the remaining mass while that mass is positive; when it is zero choose the remaining variables arbitrarily. Taking the product of these maps, including a selector row when needed, gives an onto polynomial map $q(x)$ from $[0, 1]^D$ to the feasible row product. Put $h_i(x) = f_i(q(x))$ and $g_i = (1 + h_i)/2$. All h_i are rational polynomials, $0 \leq g_i \leq 1$, and $g_1 = 1/2$. The case $D = 0$ is a single feasible table and is interpreted directly.

Theorem 6.1 (Strict polynomial lower-certificate hierarchy). *For $p \geq 1$ define*

$$A_p(x) = \frac{\sum_{i=1}^m g_i(x)^p h_i(x)}{\sum_{i=1}^m g_i(x)^p}, \quad a_p = \min_{x \in [0, 1]^D} A_p(x), \quad \delta = \min_q F(q).$$

Then the denominators are positive, a_p is nondecreasing, and

$$\delta - 2(1 - m^{-1/(p+1)}) \leq a_p \leq \delta, \quad \delta = \sup_{p \geq 1} a_p. \quad (6.1)$$

For rational L let

$$P_{p,L}(x) = \sum_{i=1}^m g_i(x)^p (h_i(x) - L). \quad (6.2)$$

A nonnegative tensor Bernstein expansion of $P_{p,L}$ on $[0, 1]^D$ certifies $\delta \geq L$. If $L < \delta$, some p and some finite Bernstein degree have strictly positive coefficients. Conversely, a strictly positive such

expansion certifies $L < \delta$. These are lower certificates at the original state and selector budgets. No convergence of the feasible behavior image occurs.

Proof. Fix x and put $b = \max_i g_i(x) \geq 1/2$ and $r_i = g_i(x)/b$. At least one r_i is one. Write $S_j = \sum_i r_i^j$. Hölder's inequality gives

$$S_p \leq m^{1/(p+1)} S_{p+1}^{p/(p+1)}, \quad S_{p+1} \geq 1.$$

Consequently $S_{p+1}/S_p \geq m^{-1/(p+1)}$. Since $A_p = 2bS_{p+1}/S_p - 1$ and $\max_i h_i = 2b - 1$,

$$0 \leq \max_i h_i - A_p \leq 2(1 - m^{-1/(p+1)}).$$

Cauchy–Schwarz gives $S_{p+2}S_p \geq S_{p+1}^2$, so A_p is nondecreasing in p . Its denominator is at least 2^{-p} and it is continuous on the cube. Taking minima proves (6.1). The bound is also at most $2 \log m/(p+1)$, by $1 - e^{-u} \leq u$.

Every tensor Bernstein basis function is nonnegative on the cube, and the basis functions sum to one. Nonnegative coefficients therefore imply $P_{p,L} \geq 0$. Dividing by the positive denominator gives $F(q(x)) \geq A_p(x) \geq L$ for every x , which proves soundness. For completeness choose p so that the error in (6.1) is less than $\delta - L$. Then

$$P_{p,L}(x) \geq 2^{-p} \{\delta - L - 2(1 - m^{-1/(p+1)})\} > 0.$$

Lemma 6.2 below gives a finite strictly positive Bernstein expansion. Conversely strictly positive coefficients have a positive minimum, and the denominator is at most m , so they give a strict positive uniform gap above L . $\square$

Positive Bernstein certificates and their degree elevation are classical; see de Klerk–Laurent [73]. The next elementary bound is retained for a self-contained exact verifier, not as a new general hypercube positivity theorem.

Lemma 6.2 (An explicit positivity degree). *Let $P(x) = \sum_k c_k x^k$ be a rational polynomial in D variables and $n \geq \max_{k,j} k_j$. In its tensor Bernstein expansion of degree n in each coordinate the coefficient at $\alpha \in \{0, \dots, n\}^D$ is*

$$b_{\alpha,n} = \sum_k c_k \prod_{j=1}^D \frac{\binom{\alpha_j}{k_j}}{\binom{n}{k_j}}, \quad |b_{\alpha,n} - P(\alpha/n)| \leq \frac{C(P)}{n}, \quad C(P) = \sum_k |c_k| \sum_j \binom{k_j}{2}. \quad (6.3)$$

In particular, $\min P > 0$ implies positivity of every coefficient whenever $n > C(P)/\min P$ and the degree restriction holds.

Proof. For a binomial random variable B with parameters (n, x) , $\mathbb{E} \binom{B}{k} = \binom{n}{k} x^k$. Expanding this expectation proves the univariate monomial formula, and products prove the displayed coefficient identity. The convention is $\binom{a}{k} = 0$ for $a < k$. To compare $\binom{a}{k}/\binom{n}{k}$ with $(a/n)^k$, draw k indices uniformly from $\{1, \dots, n\}$. Conditional on no repetition this is sampling without replacement. The probability of a repetition is at most $\binom{k}{2}/n$. Apply this total variation bound to the event that all drawn indices lie in a prescribed a -element subset. Hence the difference of the two factors is at most $\binom{k}{2}/n$. All factors lie in $[0, 1]$. A product telescoping bound followed by the triangle inequality proves (6.3). The final assertion follows by taking the minimum of P . $\square$

Corollary 6.3 (Rational lower witnesses without policy cells). *At every fixed finite rational signature, strict infeasibility above a rational level L is equivalent to the existence of the following finite rational certificate: an integer p , an integer n , and a strictly positive coefficient array for (6.2) whose identity is checked by (6.3). Dovetailing these certificates with rational original-budget upper witnesses produces nested intervals converging to δ . If a subfamily of the original tests gives a strict gap, the same construction applies to it after adjoining the empty test.*

Proof. Soundness and completeness are Theorem 6.1. Each finite candidate is checked by rational polynomial arithmetic. Enumerate all (p, n) and rational levels, and retain the largest verified lower level so far, together with zero. Rational row tables are dense in the feasible product, and their continuous objectives converge to the minimum, so enumeration and retention of the best upper table converge from above. This is a dovetailed enumeration, not a claim that any fixed (p, n) test search terminates. For a requested positive interval width, convergence of both endpoints proves termination. The reduced family conclusion uses only the same pointwise maximum argument. $\square$

Remark 6.4 (Meaning and size of the strict certificate transform). A constant distribution on tests generally detects only the convex hull of the behavior image. The functions $g_i^p / \sum_j g_j^p$ are instead functions of a proposed row table. Their universally quantified weighted inequality is an algebraic certificate; the tester need not observe a private state in an experiment. The displayed certificate identity is not an equality obtained by exchanging a private minimum with a fixed mixed test. Power means and Bernstein degree elevation are classical algebraic approximation devices, proved here in the precise form needed. The claim here is strict-level completeness for the original marked causal resource image, including visible selectors, and its physical transfer below. The coefficient count $(n+1)^D$, the number of original tests, and the algebraic degrees may be large. No polynomial complexity or intrinsic-dimension bound on this row-cube coefficient count is asserted. The independent adaptive test has the separate behavior-size bound of Theorem 3.3; its verification is a different cost.

6.1. A sharp two-time nonconvex example. The source has singleton reports, the control alphabet and actual mark are singleton, and the target emits $(0, 0)$ or $(1, 1)$ with probability $1/2$ each. At width one, a private simulator emits two independent bits with probabilities p, q . It cannot reread its first emitted bit.

Proposition 6.5 (A gap invisible to constant testing mixtures). *For this delayed experiment,*

$$\delta_{(1,1)}^p = \sqrt{2} - 1, \quad \delta_{(1,1),2}^h = 0, \quad \delta_{(1,1),r}^v = \sqrt{2} - 1 \quad (r \geq 1).$$

The private objective is the maximum of the four polynomials

$$0, \quad \frac{1}{2} - pq, \quad \frac{1}{2} - (1-p)(1-q), \quad p + q - 2pq.$$

Every constant mixture of these tests has minimum at most zero, whereas Corollary 6.3 certifies every rational $L < \sqrt{2} - 1$ at private width one.

Proof. Put $u = pq$ and $v = (1-p)(1-q)$. Direct total variation calculation gives $F = 1 - \min(u, 1/2) - \min(v, 1/2)$ and the displayed maximum. By Cauchy–Schwarz, $\sqrt{u} + \sqrt{v} \leq 1$. If $u \geq 1/2$, then $v \leq (1 - 1/\sqrt{2})^2$, so the sum of the two capped terms is at most $2 - \sqrt{2}$; the case $v \geq 1/2$ is symmetric. Otherwise set $a = \sqrt{u}$, $b = \sqrt{v}$. On $0 \leq a, b \leq 1/\sqrt{2}$, $a + b \leq 1$, the maximum of $a^2 + b^2$ is at a vertex of this polygon and equals $2 - \sqrt{2}$. Equality is attained at $p = q = 1/\sqrt{2}$. A hidden fair bit selects the two deterministic constant sequences and exactly reproduces the target. If it is exposed, variation is averaged over the selector fibers; each fiber has error at least the private minimum, which is itself attainable without a selector. Finally the target lies in the convex hull of the two private deterministic behaviors. The average discrepancy of any fixed test mixture at those behaviors is zero, so at least one has discrepancy at most zero. The strict polynomial certificates therefore detect information lost by constant convex separation. $\square$

7. ENDOGENOUS COMMON DECISIONS

7.1. A shared encoder on a controlled tree. Let $\mathcal{H}_t$ be the finite supported observation–control histories of a known controlled experiment. The initial history has distribution μ , positive on $\mathcal{H}_0$. For $h \in \mathcal{H}_t$ and $a \in A_t$, write $p_t(y \mid h, a)$ for the next-report law and hay for the successor

when this probability is positive. Zero-probability successors impose no condition. The finite task set is D . Task d has conditional expected stage costs $c_{t,d}(h, a)$ and terminal cost $c_{T,d}(h)$. These may be obtained by conditioning bounded losses of the actual hidden mark; the task does not alter the physical transition law. Controls do alter that law through a .

The common source-consuming encoder is

$$q_0(m \mid h), \quad q_{t+1}(m' \mid m, a, y), \quad m \in [S_t], \quad m' \in [S_{t+1}].$$

Controller d chooses its action by $\alpha_{t,d}(a \mid m)$. It cannot query the discarded history. The clock is available. The encoder does not know d ; the update sees the actual selected action, as allowed by the interface. All kernels can be stochastic. The same encoder must serve every task, even though different tasks induce different laws on histories.

Define the full-history Bellman functions and regrets by

$$\begin{aligned} V_{T,d}(h) &= c_{T,d}(h), \\ V_{t,d}(h) &= \min_{a \in A_t} \left(c_{t,d}(h, a) + \sum_y p_t(y \mid h, a) V_{t+1,d}(hay) \right), \\ G_{t,d}(h, a) &= c_{t,d}(h, a) + \sum_y p_t(y \mid h, a) V_{t+1,d}(hay) - V_{t,d}(h) \geq 0. \end{aligned} \tag{7.1}$$

The benchmark for task d is $v_d = \sum_h \mu(h) V_{0,d}(h)$. It is an ordinary full-history control value, not a posterior quantity conditioned on a fixed open-loop preparation.

Lemma 7.1 (Occupation regret). *For every admissible common encoder and task controller, the task excess satisfies the exact identity*

$$R_d - v_d = \mathbb{E}_d \sum_{t=0}^{T-1} G_{t,d}(H_t, A_t). \tag{7.2}$$

Moreover the entire vector of risks of a randomized finite-horizon design is a convex combination of risk vectors of deterministic designs with the same widths and one common encoder.

Proof. Take conditional expectations in (7.1) along the actual controlled path. Summing the resulting expressions for $G_{t,d}$ telescopes the terms $V_{t,d}(H_t)$, leaving the sum of stage and terminal costs minus $\mathbb{E} V_{0,d}(H_0)$. This proves (7.2) without assuming the control is open loop.

Sample in advance a deterministic outcome for every time-indexed row of the common encoder and every time-indexed row of every task controller. Row choices are independent with their prescribed probabilities; the sampled encoder table is shared across tasks. Along each trajectory each time-indexed row is encountered at most once. The product of its row probabilities is therefore the probability of that path under the sampled-table mixture. This identity holds for all tasks simultaneously after summing paths and costs. An additional independent visible or hidden seed is first conditioned on and then integrated. The conclusion is a mixture identity; the mixing variable is not claimed to be executable privately without retaining its index. $\square$

Definition 7.2 (Endogenous compatible cover). A cover of profile S consists of an initial map $i : \mathcal{H}_0 \rightarrow [S_0]$, common maps $\delta_t : [S_t] \times A_t \times Y_{t+1} \rightarrow [S_{t+1}]$, task action maps $a_{t,d} : [S_t] \rightarrow A_t$, and cells $C_{t,m}^d \subseteq \mathcal{H}_t$. Empty cells and overlapping cells are allowed. The requirements are

$$\begin{aligned} h &\in C_{0,i(h)}^d & (h \in \mathcal{H}_0, d \in D), \\ G_{t,d}(h, a_{t,d}(m)) &= 0 & (h \in C_{t,m}^d), \\ h \in C_{t,m}^d, p_t(y \mid h, a_{t,d}(m)) > 0 &\implies ha_{t,d}(m)y \in C_{t+1,\delta_t(m,a_{t,d}(m),y)}^d. \end{aligned} \tag{7.3}$$

Thus the cells cover the subtree generated by the selected actions. They need not cover histories generated by actions the controller never selects. Crucially, i and δ_t cannot depend on d .

Theorem 7.3 (Exact endogenous compression). *For a prescribed finite width profile S , the following are equivalent: a deterministic common design attains $R_d = v_d$ for every task; a privately randomized common design does so; a common design with an independent visible or hidden seed does so; and an endogenous compatible cover of profile S exists.*

Let $\beta_d > 0$ sum to one, and put $\mathcal{G} = \sum_d \beta_d (R_d - v_d)$. If the profile is infeasible, then every such design satisfies

$$\mathcal{G} \geq g_S := \beta_* \mu_* p_*^T \gamma_* > 0, \quad (7.4)$$

where $\beta_* = \min_d \beta_d$, μ_* is the least positive initial mass, p_* is the least positive transition probability, and γ_* is the least positive Bellman regret among supported entries. If there are no positive regrets, every profile with positive widths is feasible. The convention $p_* = 1$ applies when there are no transitions.

More generally, replacing the zero in (7.3) by the requirement $G_{t,d}(h, a_{t,d}(m)) \leq \eta_{t,d}$ gives a common design with $R_d - v_d \leq \sum_t \eta_{t,d}$. The sharper value is exactly the occupation-weighted sum in (7.2).

Proof. A cover defines the deterministic encoder $M_0 = i(H_0)$ and $M_{t+1} = \delta_t(M_t, A_t, Y_{t+1})$, and the task controller uses $a_{t,d}(M_t)$. Induction in (7.3) shows that every reached history lies in its indicated cell. Each incurred Bellman regret is zero, so Lemma 7.1 proves exactness. The same induction and identity prove the approximate-cover statement.

Conversely, an exact deterministic design defines $C_{t,m}^d$ to be the set of histories reached with positive probability under task d and state m . Initial coverage and common successor closure follow from its encoder. Since the nonnegative sum in (7.2) has zero expectation, every positively reached history has zero regret for its selected action. This proves all cover requirements. It also explains why adding closure for every unused action would impose a strictly different problem.

For a randomized design, Lemma 7.1 expresses the vector of excesses as an average of deterministic excess vectors. Each coordinate is nonnegative. If all averaged coordinates are zero, then, for almost every sampled design, all coordinates are zero, since the task set is finite. Choose one such deterministic table. This applies after conditioning on an arbitrary independent seed, with recovery from the seed and retained state when the seed is visible. It does not use a seed correlated with the initial history. The deterministic cover already constructed then proves necessity.

If no exact deterministic design exists, each deterministic design has some task, stage, and positively reached history with positive selected regret. That history has probability at least $\mu_* p_*^t$, and its regret is at least γ_* . Since $0 < p_* \leq 1$ and $t \leq T$, its contribution to $\mathcal{G}$ is at least the right side of (7.4). Every randomized design is an average of these vectors, so the same bound follows. When there is no positive regret, every action is Bellman optimal wherever supported, and a one-state encoder with arbitrary task actions is exact. $\square$

Example 7.4 (Why unused controls cannot define the cover). At the first time there is no report. Action **rest** leads to a deterministic hidden bit zero and a constant report, at zero cost. Action **probe** costs one, draws a fair bit, and reports it. After this report is consumed there is one blank checkpoint, followed by a binary guess with zero-one loss. The full-history optimum chooses **rest** and guesses zero; one retained state is exact. A restart family that insists on optimal guessing after the unused **probe** branch needs two retained states. The endogenous criterion and the all-controls restart criterion are both meaningful, but they have different feasible sets. The cover in Definition 7.2 implements the former, not the latter.

7.2. Lossy common design and intrinsic comparison. Let $\mathcal{D}_S^{\text{ctl}}$ be the finite set of deterministic common encoders and deterministic task controllers in this section. Put $r_d(\pi) = R_d(\pi) - v_d$. Then the hidden-selector lossy minimax value is

$$\min_{\lambda \in \Delta(\mathcal{D}_S^{\text{ctl}})} \max_d \sum_{\pi} \lambda_{\pi} r_d(\pi) = \max_{\beta \in \Delta(D)} \min_{\pi \in \mathcal{D}_S^{\text{ctl}}} \sum_d \beta_d r_d(\pi). \quad (7.5)$$

Here a selector chooses the entire design, including its task-controller tables; its mode is internal to that design. This follows from the simultaneous mixture identity in Lemma 7.1 and finite minimax. A mixture using at most $|D| + 1$ designs suffices and can be stored with at most $(|D| + 1)S_t$ private labels. Private finite-width optimization is instead the polynomial optimization in the actual encoder and controller rows. The exact zero sets agree by Theorem 7.3, but positive minimax values need not agree. Thus the cover theorem supplies a distinguished zero face inside a quantitative lossy theory, rather than purporting to solve every lossy design problem by equivalence classes.

Corollary 7.5 (Intrinsic certification of endogenous widths). *Suppose the losses of this finite problem are normalized so that every total task loss lies in $[0, 1]$. Let an approximation F have the same control and task interfaces, with each task represented by the same bounded marked-transcript loss on both sides and the full-history benchmark included in the paired decision class. If*

$$\max\{\delta_1^P(E, F), \delta_1^P(F, E)\} \leq \varepsilon < g_S/4,$$

then the optimized β -weighted excess at profile S in F is less than $g_S/2$ exactly when the endogenous cover exists in E . If only a cost- K simulation from E to F is known and an exact S -state F design transports with zero error and the same benchmark, then an endogenous cover of profile KS exists in E . A hidden selector multiplies that profile by its charged selector size.

Proof. Full-history controllers also transport through a stateless simulator, so their optimized benchmarks differ by at most ε . Equation (2.18) bounds the optimized constrained values by ε in each direction. The optimized excesses therefore differ by at most 2ε . In E the value is zero on the cover face and at least g_S otherwise, by Theorem 7.3. The strict threshold follows. For the one-sided statement, the product construction transports the exact design at the stated width and benchmark. The same theorem then converts exactness into a cover. $\square$

Remark 7.6 (Dependence of the finite witness). The number g_S depends on the least positive probability and regret of the specified finite model. It can tend to zero as a transition or a cost is perturbed. Neither (7.4) nor the certification corollary asserts a model-uniform robustness radius. Likewise an inverse-limit assertion that every finite prefix has a feasible cover would give existence of an infinite consistent design, not a computable bound on the first infeasible prefix. No such bound is used here.

7.3. Task resolution, positive error, and the complete state. We give a quantitative state law at a source-consuming restart. It complements the endogenous zero-face theorem; it does not replace the causal update constraint by a terminal partition.

Let $\mathcal{H}$ be a finite set of supported histories at a fixed checkpoint. Choose d actual marked continuation-event prediction tasks, each including its declared open-loop continuation control word. For $h \in \mathcal{H}$ let $v(h) \in [0, 1]^d$ be the vector of their conditional probabilities. A source-consuming statistic has rows $Q(m | h)$, $m \in [K]$. For every preparation λ on $\mathcal{H}$, the task is announced after storage and its forecast is scored by squared error against the actual event. Define

$$e(Q) = \sup_{\lambda} \max_{1 \leq j \leq d} \mathbb{E}_{\lambda} \text{Var}(v_j(H) | M), \quad e_K = \inf_Q e(Q).$$

The expression is precisely excess Brier risk over the full-history forecast: conditional on H , event variance is an unavoidable common term, and the law of total variance gives the displayed residual. Decoders may use the announced preparation and task, but the same statistic Q must work for all of them. Let $N(\eta)$ be the maximum size of a subset of $v(\mathcal{H})$ whose Euclidean pair distances are at least $\eta > 0$. Let $C(r)$ be the minimum number of sets covering $v(\mathcal{H})$ with coordinatewise diameter at most r .

Theorem 7.7 (Positive-error state bounds). *For all positive K and η ,*

$$e_K \geq \frac{\eta^2}{2d} \left(1 - \frac{K}{N(\eta)}\right)_+. \quad (7.6)$$

For $0 \leq \varepsilon \leq 1/4$ there is a deterministic statistic of width $C(2\sqrt{\varepsilon})$ with $e(Q) \leq \varepsilon$. Consequently, writing $K(\varepsilon) = \min\{K : e_K \leq \varepsilon\}$,

$$N(\eta) \left(1 - \frac{2d\varepsilon}{\eta^2}\right) \leq K(\varepsilon) \leq C(2\sqrt{\varepsilon}). \quad (7.7)$$

At zero error the exact width is $|v(\mathcal{H})|$. It is at most the universal marked continuation width, and it equals that width when the event family separates all distinct marked continuation laws. The lower bound also holds when an independent exposed selector is available and each selector fiber has at most K private labels; in that case the conditional variance is taken given (U, M) .

Proof. Choose $n = N(\eta)$ separated histories and prepare them uniformly. For an output m , write $\pi_h(m) = \mathbb{P}(H = h \mid M = m)$. The variance identity gives

$$\begin{aligned} \sum_{j=1}^d \mathbb{E} \text{Var}(v_j(H) \mid M) &= \frac{1}{2} \sum_m \mathbb{P}(M = m) \sum_{h, h'} \pi_h(m) \pi_{h'}(m) \|v(h) - v(h')\|_2^2 \\ &\geq \frac{\eta^2}{2} \left(1 - \sum_m \mathbb{P}(M = m) \sum_h \pi_h(m)^2\right). \end{aligned}$$

Furthermore

$$\sum_m \mathbb{P}(M = m) \sum_h \pi_h(m)^2 \leq \sum_m \mathbb{P}(M = m) \max_h \pi_h(m) = \frac{1}{n} \sum_m \max_h Q(m \mid h) \leq \frac{K}{n}.$$

At least one coordinate variance is the average of these d terms or larger. Nonnegativity proves (7.6). Conditioning on an independent exposed selector gives the identical bound on the sum of coordinate variances at every fiber, with the same uniform preparation. Average this sum over the selector and only then choose a coordinate at least as large as its average. Thus the task need not depend on the selector, and the same lower bound follows.

For the upper bound assign each history to a covering set containing its feature vector and retain its set index. Every conditional coordinate range then has length at most $2\sqrt{\varepsilon}$. A real random variable in $[a, b]$ has variance at most $(b - a)^2/4$: $\mathbb{E}(X - a)(b - X) \geq 0$ implies $\text{Var } X \leq (\mathbb{E}X - a)(b - \mathbb{E}X) \leq (b - a)^2/4$. This proves the upper bound uniformly over preparations.

If two histories with different feature vectors share a retained symbol with positive probability, a preparation on just those two histories has a strictly positive posterior variance at that symbol in a differing coordinate. Therefore zero excess forces disjoint retained supports for distinct feature vectors. Conversely the feature vector itself is an exact statistic. This proves the zero law. Every feature is determined by the universal marked continuation law; a finite separating event family distinguishes all its finitely many classes. The final assertion follows. $\square$

Corollary 7.8 (A positive-error obstruction for marked simulation). *Suppose a prescribed source-consuming reverse simulator of width K compares the stored-state restart to the full-history restart, preserving the actual marked continuation experiment with error at most δ uniformly over preparations and announced continuation tasks. Then*

$$\delta \geq e_K \geq \frac{\eta^2}{2d} \left(1 - \frac{K}{N(\eta)}\right)_+.$$

Proof. Compose the reverse simulator with the full-history optimal Brier forecast for each announced task. Squared error belongs to $[0, 1]$; marked comparison changes its expectation by at

most δ . The optimal stored-state forecast is no worse than this composed forecast. Hence its excess is at most δ for every preparation and task. Apply Theorem 7.7. $\square$

For several checkpoints these are necessary width bounds at every restart. Their simultaneous sufficiency additionally requires that the chosen covers admit causal updates on the reached support, exactly as in the preceding endogenous theorem. A feature cover alone does not construct a reverse channel preserving an arbitrary actual mark. For a full separating family the zero-error envelope recovers the universal continuation quotient; before completion, the quantitative bounds depend on the declared tasks rather than on irrelevant marks or unused future controls.

8. REGENERATIVE AVERAGE RISKS

8.1. The reset signature. Let the physical state space X be standard Borel, with any actual persistent hidden mark included in X . Reports and actions are finite. A stationary common finite-register design consists of stochastic encoder rows and task-controller rows, independent of time; their product of simplexes is a compact set Π_S . A fixed finite task set is part of the row index $j \in J$, where J is compact metrizable. For each design and row, the joint physical–register transition has form

$$P_j^\pi = \eta N_j + (1 - \eta) Q_j^\pi, \quad 0 < \eta \leq 1, \quad N_j(x, m; \cdot) = \nu_j \otimes \delta_{m_0}. \quad (8.1)$$

The independent reset flag is physically supplied; it refreshes the physical preparation and sets the within-policy memory to m_0 . Thus N_j is a constant kernel independent of π . Actions between resets may change the physical state. No domination of Q_j^π by one reference measure is assumed.

Let $c_j^\pi : X \times [S] \rightarrow [0, 1]$ be the expected one-stage marked cost. We assume the following uniform continuity conditions, using the supremum over starting states for kernel total variation:

$$(j, \pi) \mapsto Q_j^\pi \text{ is continuous in } \sup_{x, m} \|\cdot\|_{\text{TV}}, \quad j \mapsto \nu_j \text{ in TV}, \quad (j, \pi) \mapsto c_j^\pi \text{ in } \|\cdot\|_\infty. \quad (8.2)$$

For fixed physical kernels with finite action and report alphabets, continuity in π follows directly by expanding the finite sums of encoder and controller probabilities; the physical kernels can be singular. Row dependence must independently satisfy (8.2). In particular changing a deterministic flow with the row does not automatically satisfy a total-variation continuity hypothesis.

Theorem 8.1 (Nonsummable average-risk duality). *Under (8.1)–(8.2), every (j, π) has a unique invariant probability ζ_j^π . Its average risk*

$$r_\pi(j) = \zeta_j^\pi c_j^\pi$$

is the limit of expected Cesàro average cost from every initial law. For $N \geq 1$ and $0 < \beta < 1$ the following errors are uniform in j, π , and the initial law ξ :

$$\left| \frac{1}{N} \sum_{t=0}^{N-1} \xi(P_j^\pi)^t c_j^\pi - r_\pi(j) \right| \leq \frac{1}{N\eta}, \quad (8.3)$$

$$\left| (1 - \beta) \sum_{t=0}^{\infty} \beta^t \xi(P_j^\pi)^t c_j^\pi - r_\pi(j) \right| \leq \frac{1 - \beta}{1 - \beta(1 - \eta)}. \quad (8.4)$$

The map $\pi \mapsto r_\pi$ is continuous into $C(J)$ and has compact image. The attainable average-risk body under independent selection of a stationary design is compact and convex in $C(J)$. For continuous $b : J \rightarrow \mathbb{R}$ its common-design minimax value satisfies

$$\min_{\Lambda \in \mathcal{P}(\Pi_S)} \max_{j \in J} \left(\int r_\pi(j) d\Lambda(\pi) - b(j) \right) = \max_{w \in \mathcal{P}(J)} \min_{\pi \in \Pi_S} \int_J (r_\pi(j) - b(j)) dw(j), \quad (8.5)$$

and all displayed extrema are attained. For finite J a selector of at most $|J| + 1$ values realizes every required optimum. More generally Proposition 2.5 gives an intrinsic finite-dimensional selector approximation.

Proof. A probability kernel contracts total variation. Subtracting two probabilities in (8.1) cancels the common reset part, giving

$$\|\xi P_j^\pi - \xi' P_j^\pi\|_{\text{TV}} \leq (1 - \eta) \|\xi - \xi'\|_{\text{TV}}.$$

The explicit probability

$$\zeta_j^\pi = \eta \sum_{k=0}^{\infty} (1 - \eta)^k (\nu_j \otimes \delta_{m_0}) (Q_j^\pi)^k \quad (8.6)$$

is invariant by direct substitution, with convergence in total variation. Contraction proves uniqueness and $\|\xi(P_j^\pi)^t - \zeta_j^\pi\|_{\text{TV}} \leq (1 - \eta)^t$. Integrating the bounded cost and summing the geometric series proves (8.3) and (8.4), including their uniformity.

For two parameter-policy pairs let P, P' denote their kernels and ζ, ζ' their invariant laws. Invariance, the preceding contraction, and a one-step difference give

$$\|\zeta - \zeta'\|_{\text{TV}} \leq (1 - \eta) \|\zeta - \zeta'\|_{\text{TV}} + \sup_{x, m} \|P(x, m; \cdot) - P'(x, m; \cdot)\|_{\text{TV}}.$$

Thus the invariant-law difference is at most the last supremum divided by η . Together with the cost difference, this proves joint continuity of $(j, \pi) \mapsto r_\pi(j)$. Compactness of $J \times \Pi_S$ upgrades it to continuity of $\pi \mapsto r_\pi$ in the uniform norm. Its image is compact.

Probabilities on compact Π_S form a weakly compact set. The barycenter map into $C(J)$ is norm continuous: approximate the compact risk image uniformly by finitely many continuous coordinates, or use uniform equicontinuity of $r_\pi(j)$ and a finite net in J . The image is therefore compact and convex. The payoff $\int (r_\pi - b) d\Lambda dw$ is continuous and affine in both probabilities on compact spaces. Convex minimax gives (8.5). The inner minimum over a probability on Π_S equals the minimum over π , and compactness attains it. A finite-dimensional barycenter is reduced by the affine-dependence argument already proved in Theorem 2.2. This proves the finite selector assertion. $\square$

Remark 8.2 (Which policies are optimized). The theorem optimizes stationary finite-register common policies with the reset signature (8.1). It does not assert that a stationary policy is optimal among arbitrary nonstationary policies without that signature. A selector choosing a stationary policy is sampled once; its mode persists through resets while that mode's within-policy register is reset. If the selector must be private, store the mode and the register, paying their product. Conditioned on the mode, (8.3)–(8.4) hold with the same constants, so they also hold after averaging. The joint chain including a persistent mode need not have a unique invariant law across modes, and no such claim is needed.

8.2. Cycle comparison before optimization. Consider two reset experiments with the same reset probability and reset flag. A comparison within a fresh cycle starts with the specified physical preparation and initial register. Suppose, for every feedback design and every independent visible side seed, the joint laws of the marked information needed to evaluate the age- k stage cost differ by at most $\varepsilon_k \leq 1$. Here age zero means the first stage following a reset; its law includes any reports used at that stage. The same control law and encoder rows are used on both sides. This is a uniform channel comparison, not a comparison obtained after separately optimizing policies.

Theorem 8.3 (Regenerative resource transport). *The resulting paired average risks satisfy, for every consumer width,*

$$\|r_E - r_F\|_\infty \leq \mathfrak{e} := \eta \sum_{k=0}^{\infty} (1 - \eta)^k \varepsilon_k. \quad (8.7)$$

The bound holds with a priced simulator by replacing the consumer width S by its product with the simulator width and by charging the selector as in Theorem 2.2. For a sequence of stateless two-sided comparisons with $\varepsilon_{n,k} \rightarrow 0$ for each fixed k , $\mathfrak{e}_n \rightarrow 0$, uniformly over all consumer widths. At every width, the corresponding average-risk bodies and optimized lower support values have the same error bound. The minimax values in (8.5) with the same centering b differ by at most $\mathfrak{e}_n$.

Proof. Formula (8.6) expresses a stationary average risk as the geometric mixture of costs at each cycle age. The expectation of a $[0, 1]$ marked stage cost changes by at most ε_k in that age's joint law, yielding (8.7). The proof is uniform in the consumer because its error hypothesis is uniform before integration. For a priced transducer, include its retained state in the joint register and reset it to its specified initial state at the cycle boundary. The same product construction and conditioning on independent mode choices applies.

The geometric weights sum to one. Given $\delta > 0$, first choose L so their tail beyond L is less than $\delta/2$, and then choose n so the finite weighted sum through L is less than $\delta/2$. This proves $\mathfrak{e}_n \rightarrow 0$ without a rate uniform in k . Pairing each design with the same design on the other side gives the risk-body comparison. Taking weighted infima or the supremum of centered row risks proves the optimized statements; taking mixtures preserves the same bound. $\square$

Example 8.4 (A nondominated physical kernel). On $X = [0, 1]$, let $Q^a(x, \cdot) = \delta_{(x+a) \bmod 1}$ for finitely many rotations a , and let ν be Lebesgue probability. The transition $\eta\nu + (1 - \eta)Q^a(x, \cdot)$, as x ranges over X , is not dominated by a common σ -finite measure when $\eta < 1$: such a measure would have to give positive mass to uncountably many singleton points. A σ -finite measure has at most countably many positive-mass atoms, by decomposing into finite-measure sets and positive lower bounds on atomic mass. Finite report partitions and any common finite encoder fit Theorem 8.1. This is a nonsummable controlled risk theorem with a genuinely nondominated physical transition family, not an application of full-path density compactness.

Remark 8.5 (Why the reset is explicit). A hidden physical process may mix while a retained controller bit never changes. Physical mixing alone then gives no uniform contraction on the joint physical–register space. Equation (8.1) requires an actual clearing of that register. Conversely, without a cycle tail or another uniform long-time condition, finite-prefix comparison errors need not control average cost: a sequence of processes that agrees with a zero-cost process for its first n stages and incurs unit cost forever afterward has vanishing error at each fixed prefix but average cost one. The positive theorem uses the explicit geometric cycle law to control this precise obstruction.

9. AN ACTIVE MICROSCOPIC REALIZATION

9.1. Interventions and the unbounded flow. Fix a nonempty finite system of $N \geq 2$ identical hard spheres of positive diameter on a flat torus, with positive configuration volume. Let Γ_N be the admissible phase space and let μ be the normalized measure with density proportional to $\mathbf{1}_{\Gamma_N}(q, v) \exp(-b \sum_i |v_i|^2)$, $b > 0$. We use the classical almost-everywhere hard-sphere flow Φ_t , which is an invertible measure-preserving flow off a null exceptional set; see [63] and the detailed microscopic construction retained in the companion. No claim about particle-number uniformity is needed. On $H = L^2(\mu)$,

$$U(t)f = f \circ \Phi_t$$

is a strongly continuous unitary group, with skew-adjoint generator L . Indeed invariance gives isometry and invertibility; for bounded continuous phase functions, almost-everywhere continuity of the orbit at zero and dominated convergence give strong continuity. Density and isometry extend it to H . The generator domain includes the collision identifications of the flow; it is not the unrestricted formal transport derivative on arbitrary boundary data.

Let A be finite and choose orthogonal matrices O_a on velocity space. The intervention

$$R_a(q_1, \dots, q_N, v_1, \dots, v_N) = (q_1, \dots, q_N, O_a v_1, \dots, O_a v_N)$$

is an invertible μ -preserving map. Put $C_a f = f \circ R_a$. For a fixed step length $\Delta > 0$, the controlled physical step is $F_a = \Phi_\Delta \circ R_a$, and its Koopman operator is

$$T_a = C_a U(\Delta). \quad (9.1)$$

The order matters. If both $O_a = I$ and $O_a = -I$ are allowed, the two steps give different subsequent positions on a positive-measure set of configurations with no collision in the next Δ units of time and nonzero sufficiently small velocities. Such a set exists inside any nonempty interior admissible configuration region by choosing small velocities and then a positive interval away from zero. The action therefore changes the dynamics, rather than merely choosing which observable to read. We require no assertion that $C_a D(L)$ is contained in $D(L)$.

For all finite control words, the flow and intervention compositions are well defined on a common full-measure subset: the preimage of a null exceptional flow set under each finite measure-preserving composition is null, and there are countably many finite words. Every preparation absolutely continuous with respect to μ is supported there, including under a feedback choice of the word.

9.2. A graph-core approximation and controlled words. Choose an increasing family of finite-dimensional real subspaces $V_n \subset D(L)$ whose union is a graph core, and let P_n be the orthogonal projections. Such a family can be constructed without postulating smooth collision-boundary coordinates. Start with a countable dense family (f_j) in H and a smooth compactly supported approximate identity χ_ε on $\mathbb{R}$. Define

$$f_{j,\varepsilon} = \int_{\mathbb{R}} \chi_\varepsilon(s) U(s) f_j ds.$$

A difference quotient followed by translation of χ_ε shows that $f_{j,\varepsilon} \in D(L^k)$ for every k and

$$L^k f_{j,\varepsilon} = (-1)^k \int \chi_\varepsilon^{(k)}(s) U(s) f_j ds, \quad \|L^k f_{j,\varepsilon}\|_2 \leq \|\chi_\varepsilon^{(k)}\|_1 \|f_j\|_2. \quad (9.2)$$

For $f \in D(L)$ its smoothed versions converge to f in graph norm: commuting L on its domain reduces this to strong continuity for f and Lf . At each fixed smoothing scale, approximation of f by the dense family controls both the function and its first derivative by (9.2). A countable sequence of scales and finite linear spans consequently gives a graph core. Enumerate those vectors and let V_n be their successive spans. This also shows that their collision compatibility comes from the actual flow domain, not from an imposed incompatible trace.

Set $L_n = P_n L P_n$ on V_n , $U_n(t) = \exp(tL_n)$ on V_n , and, as an operator on H extended by zero on $V_n^\perp$, define

$$T_{a,n} = P_n C_a P_n U_n(\Delta) P_n. \quad (9.3)$$

These operators are contractions. Their finite-dimensional matrices need not be Markov matrices: they approximate *observables*; the observation experiment constructed below is nevertheless a genuine probability experiment.

Lemma 9.1 (Graph-core and controlled-word convergence). *For every $f \in H$ and $T_0 < \infty$,*

$$\sup_{0 \leq t \leq T_0} \|U_n(t) P_n f - U(t) f\|_2 \longrightarrow 0. \quad (9.4)$$

For every fixed word length k ,

$$\max_{(a_1, \dots, a_k) \in A^k} \|T_{a_1, n} \cdots T_{a_k, n} f - T_{a_1} \cdots T_{a_k} f\|_2 \longrightarrow 0. \quad (9.5)$$

Neither statement needs domain invariance under the interventions.

Proof. The matrix L_n is skew-adjoint on V_n because L is skew-adjoint and $V_n \subset D(L)$. Thus its group is unitary and $\|(z - L_n)^{-1}\| \leq 1/\Re z$ when $\Re z > 0$. If $u \in V_m$ and $n \geq m$, then

$$(z - L_n)^{-1} P_n (z - L) u = u.$$

The set $(z - L) \bigcup_m V_m$ is dense in H , since the union is a graph core and $z - L$ maps $D(L)$ onto H . The uniform resolvent bounds and $P_n \rightarrow I$ strongly therefore give

$$(z - L_n)^{-1} P_n f \rightarrow (z - L)^{-1} f \quad (f \in H).$$

This is the graph-core consistency and stable-resolvent setting of Trotter's theorem [67, Theorem 5.2]. We give the contraction argument to specify the hypotheses used here. For any contraction generator A and $u \in D(A)$, the Euler resolvent formula can be written as a Gamma integral:

$$(I - tA/m)^{-m} u = \int_0^\infty e^{sA} u d\gamma_{m,t}(s), \quad \mathbb{E}_{\gamma_{m,t}} s = t, \quad \text{Var}_{\gamma_{m,t}} s = t^2/m.$$

This follows by multiplying the Laplace resolvent integrals and convolving their exponential densities. Since $\|e^{sA} u - e^{tA} u\| \leq |s - t| \|Au\|$, its error from $e^{tA} u$ is at most $t \|Au\|/\sqrt{m}$. For $u \in V_l$ one has $\|L_n u\| \leq \|Lu\|$ for all $n \geq l$. At each fixed $t > 0$ and Euler order m , the resolvent convergence already proved gives convergence of its m th power. First letting $n \rightarrow \infty$ and then $m \rightarrow \infty$ proves orbit convergence on the core. The orbit Lipschitz bound with constant $\|Lu\|$, uniform in n , upgrades it to uniformity on $[0, T_0]$ by a finite time mesh. Density and contraction extend it to all $f \in H$. No derivative of $U(t)f$ for $f \notin D(L)$ was taken.

Now $P_n C_a P_n \rightarrow C_a$ strongly by boundedness of C_a and strong convergence of P_n . Equation (9.4) gives $T_{a,n} \rightarrow T_a$ strongly. For a fixed finite word, subtract the two products one factor at a time. Each difference acts on a fixed vector formed from the limiting factors, while every factor to its left is a contraction. Each term tends to zero. There are finitely many words of length k , which proves the maximum in (9.5). At no step was an expression such as $LC_a f$ required. $\square$

9.3. Actual marked acquisition under feedback. Fix real observables $f_k \in H$ and noises $\sigma_k > 0$. At step k , a controller chooses a_k from its previously available reports and retained state, the action is applied to the physical state, and the new report is

$$Y_k = (T_{a_1} \cdots T_{a_k} f_k)(x_0) + \sigma_k \xi_k, \quad \xi_k \text{ independent standard Gaussians.} \quad (9.6)$$

The initial law is $p_\theta d\mu$, with $\sup_\theta \|p_\theta\|_2 \leq D < \infty$. An unobserved initial physical mark W is drawn jointly with x_0 through a fixed channel $\gamma(dw | x_0)$, independently of the acquisition noises. Losses may use that actual mark and the observed control-report transcript. This specification does not allow an unproved replacement of a controlled terminal physical quantity by an independent variable of the same marginal law.

The approximation replaces the product in (9.6) by $T_{a_1,n} \cdots T_{a_k,n} f_k$. It has a finite-dimensional physical realization. In a real orthonormal basis $(\varphi_1, \dots, \varphi_{d_n})$ of V_n , write the matrix of $T_{a,n}$ as $M_{a,n}$ and the coefficient vector of $P_n f_k$ as $c_{k,n}$. The initial row vector is $v_0 = (\varphi_1(x_0), \dots, \varphi_{d_n}(x_0))$, and

$$v_k = v_{k-1} M_{a_k,n}, \quad Y_k^{(n)} = v_k c_{k,n} + \sigma_k \xi_k.$$

Initialize (v_0, W) with the actual push-forward joint law of (x_0, W) . This is a valid controlled probability model with a real vector state; no positivity of $M_{a,n}$ is needed. Its real dimension d_n is not counted as d_n finite retained labels. The consumer register remains a separate constrained object. Measurable representatives of the finitely many functions may be chosen once; their null ambiguities have zero probability under the specified preparations.

Theorem 9.2 (Active marked deficiency bound). *For each fixed horizon T , define*

$$\epsilon_{n,T} = 1 \wedge \frac{D}{\sqrt{2\pi}} \sum_{k=1}^T \frac{1}{\sigma_k} \sum_{a \in A^k} \|T_{a_1} \cdots T_{a_k} f_k - T_{a_1,n} \cdots T_{a_k,n} f_k\|_2. \quad (9.7)$$

The exact and approximate acquisitions have two-sided stateless marked causal deficiency at most $\epsilon_{n,T}$, using the identity report channel. The bound includes every independent visible seed in the joint comparison, holds uniformly over every finite-register consumer and feedback design, and tends to zero for each fixed T . Consequently all paired $[0, 1]$ task risks and optimized lower support values have error at most $\epsilon_{n,T}$ at unchanged widths; paired centered values have error at most $2\epsilon_{n,T}$ when their unrestricted benchmarks are also paired.

Proof. Couple the two experiments with the same x_0, W and every independent visible seed. Couple their private controller randomness so that the same available history produces the same action. At each step before the first report mismatch, use a maximal coupling of the two Gaussian reports. Gaussian densities satisfy

$$\|N(u, \sigma^2) - N(v, \sigma^2)\|_{\text{TV}} = 2\Phi(|u - v|/(2\sigma)) - 1 \leq \frac{|u - v|}{\sigma\sqrt{2\pi}}.$$

The equality follows by separating the densities at their midpoint; the inequality follows by bounding the standard normal density by $1/\sqrt{2\pi}$. Maximal couplings can be chosen measurably from these densities by their common part and residual parts.

For a fixed x_0 , the probability of a first mismatch at time k is bounded by the sum, over the possible matched action words $a \in A^k$, of their matched-history probabilities times the displayed Gaussian bound for that word. Discarding these probabilities gives an upper bound by the sum over all words. Integrate under the *original* preparation $p_\theta d\mu$. Cauchy–Schwarz bounds the integral of each absolute mean difference by D times its L^2 norm. Summing first-mismatch probabilities yields (9.7). In particular no bound on a feedback-conditioned posterior density has been assumed. When no mismatch occurs the entire transcript and actual mark, including the visible seed, coincide. The coupling inequality gives their joint total variation bound.

The same argument is symmetric in the two report models; the stateless identity channel is executable in both directions, so it bounds the intrinsic infima. Lemma 9.1 makes every term in the fixed finite sum tend to zero. The comparison precedes any consumer optimization, hence its bound does not depend on the consumer’s number of labels. The remaining conclusions are the channel and support inequalities in Theorem 2.2 and (2.18), which remain valid on these standard Borel carriers. $\square$

9.4. Active regenerative average decisions. Take a fixed observable f , fixed noise $\sigma > 0$, and a fixed finite measurable report partition applied to each Gaussian acquisition. At each cycle boundary the physical system is re-prepared from $p_\theta d\mu$ with its actual mark channel γ , and the within-policy register is reset to m_0 . Between boundaries the orthogonal interventions and hard-sphere flow continue as above. The independent reset probability is $\eta > 0$. In the approximate system reset the real vector and mark by the same joint push-forward preparation. A stationary consumer uses the partitioned reports, actions, and its finite register. Its $[0, 1]$ one-stage losses depend on the actual cycle mark, control, and current finite report; reset stage costs are fixed and identically paired. A finite task index is available to the controller but not to the common encoder.

Assume $j \mapsto p_j$ is continuous in $L^1(\mu)$ on a compact row set, the L^2 bounds are uniform, and the bounded row losses vary uniformly continuously. The flow, intervention maps, observable, noise law, report partition, and mark channel are fixed across rows. Then the exact and every approximate model satisfy (8.2): only the reset law and losses vary with the row, and the dependence on finite policy rows is a finite polynomial expansion in kernels of norm one. The underlying physical transitions are allowed to be deterministic between resets.

Corollary 9.3 (Active average-risk realization). *Both models have the compact average-risk bodies and attained duality of Theorem 8.1. At every finite consumer width their paired average risks,*

lower supports, and equally centered minimax values differ by at most

$$\epsilon_n = \eta \sum_{k=0}^{\infty} (1 - \eta)^k \epsilon_{n,k+1} \longrightarrow 0. \quad (9.8)$$

The convergence is uniform in the consumer width and in any independent priced mixture of stationary designs. It does not require a controlled word approximation rate uniform in word length.

Proof. A deterministic report partition contracts total variation. The age- k stage is evaluated after at most $k + 1$ controlled acquisitions, so Theorem 9.2 supplies the cycle comparison with $\epsilon_{n,k} = \epsilon_{n,k+1}$. Apply Theorem 8.3. Its geometric tail proof gives the limit although the finite-word bound in (9.7) can grow with k . The reset and row-continuity assertions just verified give Theorem 8.1. Conditioning on each persistent mode and charging its index proves the mixture statement. $\square$

Remark 9.4 (Fixed physical scope). The interventions above are active changes to microscopic dynamics, and the average theorem uses actual repeated physical preparation. Neither observation control alone nor an assumed posterior L^2 bound is substituted for them. The assertions are for fixed N , sphere diameter, A , Δ , and positive noise, and for the declared initial-mark losses. No estimate is asserted uniform in particle number, a growing action alphabet, vanishing noise, or a kinetic scaling. The historical nonlinear kinetic corrector and the Sinai left-strip spectral problem remain different analytic statements. They are preserved in the development with their original hypotheses, not inferred from the present operator approximation.

9.5. A posteriori bounds from graph and intervention residuals. Strong controlled-word convergence supplies an asymptotic comparison. For a fixed approximation space one can also bound its actual comparison error using finite Gram data. We retain the physical setup of Theorem 9.2: the particle number, interventions, noise, step length, and actual initial-mark channel are fixed. These statements do not require the interventions to map $D(L)$ into itself.

Suppress n and let $V \subset D(L)$ have orthonormal basis $\phi_1, \dots, \phi_d$. Identify V with $\mathbb{R}^d$ by an isometry ι ; complexification gives the same formulas with adjoints. Write

$$P = \iota^* \iota, \quad J = \iota^* L \iota, \quad B_a = \iota^* C_a \iota, \quad M_a = B_a e^{\Delta J}.$$

Because L is skew-adjoint and C_a is unitary, J is skew-adjoint, e^{sJ} is unitary and M_a is a contraction. The two residual Gram matrices are

$$R = (L \iota)^* L \iota - J^* J = ((I - P) L \iota)^* ((I - P) L \iota), \quad (9.9)$$

$$S_a = I - B_a^* B_a = ((I - P) C_a \iota)^* ((I - P) C_a \iota). \quad (9.10)$$

In particular both are positive semidefinite. For a coefficient vector c , set

$$d_a(c) = \int_0^{\Delta} \sqrt{(e^{sJ} c)^* R (e^{sJ} c)} ds + \sqrt{(e^{\Delta J} c)^* S_a (e^{\Delta J} c)}. \quad (9.11)$$

All quantities in this formula are finite-dimensional. For a real observable $f \in H$ put

$$c_f = \iota^* f, \quad \tau_f = (\|f\|_H^2 - \|c_f\|^2)^{1/2}.$$

If $w = (a_1, \dots, a_k)$ and $c_j = M_{a_{j+1}} \cdots M_{a_k} c_f$ (empty product equal to the identity), define

$$E_w(f) = \tau_f + \sum_{j=1}^k d_{a_j}(c_j). \quad (9.12)$$

The suffix order agrees with the Koopman product for a physically chronological word; its reversal would approximate a different dynamics.

Theorem 9.5 (Residual control of the active marked experiment). *For every such space V and every controlled word,*

$$\|T_{a_1} \cdots T_{a_k} f - \iota M_{a_1} \cdots M_{a_k} c_f\|_H \leq E_w(f). \quad (9.13)$$

For the preparations p_θ with $\sup_\theta \|p_\theta\|_2 \leq D$ and Gaussian acquisitions of Theorem 9.2, the two stateless marked feedback deficiencies are bounded by

$$\hat{\epsilon}_{V,T} = \min \left\{ 1, \frac{D}{\sqrt{2\pi}} \sum_{k=1}^T \frac{1}{\sigma_k} \left(\sum_{w \in A^k} E_w(f_k)^2 \right)^{1/2} \right\}. \quad (9.14)$$

The comparison includes every independent exposed seed and is uniform over all finite consumer widths. Thus it bounds raw paired risks and lower supports at unchanged widths. When unrestricted benchmarks are also paired, the centered risk bound is $2\hat{\epsilon}_{V,T}$.

Proof. The orthogonal splitting of $L\iota c$ into its V component ιJc and its orthogonal component proves (9.9). Unitarity of C_a similarly proves (9.10). For $v = \iota c$, the path $s \mapsto \iota e^{sJ} c$ lies in the finite space $V \subset D(L)$ and is differentiable in the graph norm. Hence variation of constants is legitimate and gives

$$U(\Delta)v - \iota e^{\Delta J} c = \int_0^\Delta U(\Delta - s)(L\iota - \iota J)e^{sJ} c ds.$$

Taking norms and applying (9.9) bounds this difference by the integral in (9.11). Apply the unitary C_a and then add the orthogonal projection error $\|(I - P)C_a \iota e^{\Delta J} c\|$ to obtain

$$\|T_a \iota c - \iota M_a c\| \leq d_a(c).$$

No derivative of $C_a \iota c$ has occurred. First remove $(I - P)f$ at cost τ_f . Telescope the difference of the two word products with exact prefixes and approximate suffixes. All prefixes are isometries and all approximate suffixes are contractions. The j th term is bounded by $d_{a_j}(c_j)$, proving (9.13).

For the probabilistic assertion use the common initial x_0 , its actual conditional mark W , and the same visible seed, as in Theorem 9.2. Before the first report mismatch the selected actions agree. Conditional on x_0 , the weights of the matched histories leading to the possible words of length k sum to at most one. If $e_w(x)$ is the corresponding exact-minus-approximate mean, the first-mismatch contribution at time k is therefore at most

$$\frac{1}{\sigma_k \sqrt{2\pi}} \int p_\theta(x) \max_{w \in A^k} |e_w(x)| d\mu(x).$$

This retains the history weights until after taking their maximum, instead of replacing each by one. Cauchy–Schwarz and $\max_w |e_w|^2 \leq \sum_w |e_w|^2$ bound the integral by $D(\sum_w \|e_w\|_2^2)^{1/2}$. Apply (9.13) and sum the first-mismatch probabilities. The coupling is symmetric and preserves the whole actual marked transcript on its matching event, so it proves both stateless inequalities in (9.14). It conditions only on the original preparation, not on a feedback posterior with a supposed L^2 bound. Conditioning on an independent exposed seed gives the same uniform estimate. Risk and support statements follow by composition with any consumer and by taking the appropriate infima. $\square$

Corollary 9.6 (Finite-data average-risk and presentation certificates). *In the active regenerative model of Corollary 9.3, a finite age cutoff H gives an error bound for paired average risks and lower supports, uniform in consumer width,*

$$\hat{\epsilon}_{V,H} = \eta \sum_{k=0}^H (1 - \eta)^k \hat{\epsilon}_{V,k+1} + (1 - \eta)^{H+1}. \quad (9.15)$$

The same bound applies to identically centered values; separately paired unrestricted benchmarks give twice the bound. Finite-horizon bounds (9.14), combined with any other verified stateless presentation bridges, can be inserted into (2.27) without changing the intrinsic resource budget.

Proof. At age k there are at most $k+1$ acquisitions. Use Theorem 9.5 at those horizons and bound each remaining age error by one. The remaining geometric mass is exactly $(1-\eta)^{H+1}$. The cycle formula in Theorem 8.3 proves (9.15). The final assertion is Theorem 2.9. A finite-dimensional vector model is not automatically a finite report experiment; a further finite-report bridge must actually be proved before invoking the rational local testing algorithm. No such bridge is implicit in a matrix representation. $\square$

The data required in (9.9)–(9.12) are inner products of $f, \phi_i, L\phi_i, C_a\phi_i$. Once these data have rigorous enclosures, the formulas give a posteriori bounds without knowing the exact controlled orbit. For example, certified numbers $r_0 \geq \|R\|$ and $s_{a,0} \geq \|S_a\|$ imply the simpler bound

$$d_a(c) \leq (\Delta\sqrt{r_0} + \sqrt{s_{a,0}})\|c\|.$$

With matrix-entry enclosures, interval arithmetic and this inequality avoid an uncertified numerical integral. Floating-point Gram entries alone are not an analytic certificate. Nor is it asserted that these residual majorants converge for every graph-core sequence: cancellation may make the actual orbit error small while its residual bound is large. The convergence theorem remains Lemma 9.1; Theorem 9.5 is a complementary finite-space bound. This distinction separates a proved limiting approximation from a checkable error estimate for a particular space.

10. FINITE RATIONAL PRESENTATIONS OF GAUSSIAN ACQUISITIONS

We now join finite resource certification to the actual marked physical experiment. The intermediate model is allowed to have a nonfinite latent state. Its reports, not merely its observable coordinates, must eventually be finite. Throughout this section the parameter, action and mark alphabets are finite, the horizon is finite, and the mark is drawn with the initial state. An initial report, when present, is treated as one additional acquisition. All channels below are parameter blind and task blind.

10.1. A stateless report channel. Let (X, μ) be a probability space, let the preparations be $p_\theta\mu$ with $\|p_\theta\|_2 \leq D$, and let $\gamma(dw | x)$ be the actual initial-mark channel. For each action word $w \in A^k$, let $m_{k,w} \in L^2(\mu)$ be a real mean function. The report at time k is

$$Y_k = m_{k,a_{1:k}}(x) + \sigma_k \xi_k, \quad \sigma_k > 0,$$

where the noises are independent standard Gaussians, independent of (x, W) . Actions may be selected from all previously emitted reports. The finite-dimensional models in the preceding section have this form; so do the enriched approximations in Section 11. Put

$$B_k = \left(\sum_{w \in A^k} \|m_{k,w}\|_2^2 \right)^{1/2}.$$

Choose $R > 0$ and a finite interval partition of $[-R, R]$ with maximum cell length h . Adjoin the two tail cells. Let Q return the cell of a real report. Let H reconstruct an interior cell uniformly and reconstruct the tail cells at $-R$ and R , respectively. The endpoints may be assigned arbitrarily to adjacent cells. These maps use no persistent memory, seed, parameter or mark. Fresh randomness in H is transient. Denote by $F^{R,h}$ the physical acquisition with the report $Q(Y_k)$; its latent state and actual mark are unchanged.

Lemma 10.1 (Uniform marked report quantization). *The channel Q simulates $F^{R,h}$ from F exactly. The reverse channel H has marked feedback error at most*

$$\beta(R, h) = 1 \wedge \sum_{k=1}^T \left\{ \frac{h}{\sigma_k \sqrt{2\pi}} + \frac{2DB_k}{R} + \frac{4\sigma_k^2}{R^2} \right\}. \quad (10.1)$$

The comparison includes the joint law of every independent exposed selector and is uniform over every finite consumer width. In particular $h \downarrow 0$ and $R \uparrow \infty$ give two-sided stateless marked comparison with error tending to zero.

Proof. For an absolutely continuous density g and an interval I of length at most h , write $g_I = |I|^{-1} \int_I g$. The fundamental theorem of calculus and Fubini give

$$\int_I |g - g_I| \leq |I|^{-1} \int_{I^2} |g(u) - g(v)| du dv \leq h \int_I |g'|.$$

Consequently the variation error of the interior histogram is at most $\frac{h}{2} \int |g'|$. For a Gaussian of variance σ^2 the latter integral is $\sqrt{2/\pi}/\sigma$. Replacement of the tails costs at most their total probability, even though the reconstruction is atomic. For any mean m , this probability is at most

$$\mathbf{1}_{\{|m| > R/2\}} + \mathbb{P}\{|\sigma\xi| > R/2\} \leq \frac{2|m|}{R} + \frac{4\sigma^2}{R^2}.$$

Thus the conditional report error, given the original x and the selected action word, has the three terms in (10.1) before integration.

Couple the exact and reconstructed systems with the same original (x, W) and independent selector. Before their first emitted-report mismatch use the same controller randomness and maximally couple the next reports. Matched-history weights of different words at time k sum to at most one. Therefore the mean term is bounded by

$$\frac{2}{R} \int p_\theta(x) \max_{w \in A^k} |m_{k,w}(x)| d\mu(x) \leq \frac{2D}{R} \left(\sum_w \|m_{k,w}\|_2^2 \right)^{1/2}.$$

This uses the original preparation, not an assumed bound for a feedback-conditioned posterior. Sum the first-mismatch bounds. On the matching event the complete emitted transcript, controls, mark and selector agree. This proves the reverse comparison. Applying Q to each original report is the exact forward simulation, also under feedback. Conditioning on the exposed selector throughout proves the joint assertion. Composition never enlarges a retained register. $\square$

The bound is deliberately an elementary certified bound; Gaussian tail integrals or higher moments may improve it. No uniform bound on the pointwise means is needed. The quantity B_k can grow with the number of action words. Uniformity in memory width is not uniformity in the size of the finite instrument being constructed.

10.2. Rationalization without a small-prefix denominator. Let G now be a finite-report acquisition of this form. For each θ , write

$$\pi_\theta(w) = \mathbb{P}_\theta(W = w), \quad p_\theta(w, h; y | a) = \mathbb{P}_\theta(W = w, Y_{1:k-1} = h_y, Y_k = y | a_{1:k-1} = h_a, a_k = a)$$

for its joint marked prefix masses under the indicated open-loop control word. Here $h = (h_y, h_a)$, and the conditioning on controls means intervention, not conditioning on the probability of choosing a control. Causality and the initial-mark convention imply

$$\sum_y p_\theta(w, h; y | a) = p_\theta(w, h)$$

independently of the current action a . A parameterized latent history-tree instrument may use these masses; its latent dependence on θ is part of the experiment, not information given to a simulator.

Suppose nonnegative rational arrays $t_\theta(w)$ and $t_\theta(w, h; y | a)$ approximate the indicated joint masses. Normalize $t_\theta(w)$ to obtain a rational initial-mark law. At each node (w, h, a) normalize its vector of children to obtain a rational report kernel. If a normalizing sum is zero, use a fixed rational probability vector, for example the uniform vector. This defines a coherent finite rational

marked causal experiment $G_{\mathbb{Q}}$. Its prefix probabilities need not equal the originally rounded joint masses; they are the probabilities generated by these normalized kernels. That distinction is essential.

Lemma 10.2 (Weighted normalization). *If $u, v \in [0, \infty)^b$, $s = \sum u_j$, and $\bar{u} = u/s$ when $s > 0$, then, for $\bar{v} = v/\sum v_j$ with an arbitrary probability vector chosen when its denominator vanishes,*

$$s\|\bar{u} - \bar{v}\|_{\text{TV}} \leq \sum_j |u_j - v_j|. \quad (10.2)$$

For $s = 0$ the left side is defined to be zero.

Proof. If both sums are positive, the triangle inequality gives

$$\sum_j |u_j - s\bar{v}_j| \leq \sum_j |u_j - v_j| + \left| \sum_j v_j - s \right| \leq 2 \sum_j |u_j - v_j|.$$

If $\sum v_j = 0$, the left side of (10.2) is at most $s = \sum |u_j - v_j|$. The remaining case is immediate. $\square$

Theorem 10.3 (Rational causal tree bridge). *The identity report channels compare G and $G_{\mathbb{Q}}$ in both directions, including the actual marks, with feedback error at most*

$$\rho = 1 \wedge \max_{\theta} \left[\sum_w |\pi_{\theta}(w) - t_{\theta}(w)| + \sum_{k=1}^T \sum_{w,h,a,y} |p_{\theta}(w, h; y | a) - t_{\theta}(w, h; y | a)| \right]. \quad (10.3)$$

Here the finite sum includes all open-loop histories at the specified time, not only the histories reached by a particular controller. The bound is independent of simulator and consumer memory. It remains valid with a visible selector and its joint reference law.

If every joint mass has computable rational enclosures of arbitrarily small width, the construction is effective and produces $\rho \rightarrow 0$. No lower bound for any nonzero prefix probability, and no decision whether a prefix has exactly zero probability, is required.

Proof. Couple the initial marks by their normalized laws. Lemma 10.2 bounds the initial failure probability by the first sum in (10.3). Use the same controller and selector on both sides. Conditional on a matched marked history, maximally couple the next reports. Under the original instrument, the probability of reaching a particular marked history under any deterministic controller is either zero or the corresponding open-loop prefix mass; for a randomized controller it is at most that mass. The probability of reaching it with the coupling still matched is smaller still. Thus its first-mismatch contribution is bounded by the original prefix mass times the conditional variation distance. Applying (10.2) at that node bounds this product by the sum of joint child-mass errors. Sum over all nodes and times. The coupling inequality gives the asserted joint variation distance. The estimate is symmetric as a comparison of the two emitted laws; identity channels require no state in either direction.

For effectiveness, take a nonnegative rational lower endpoint of a certified enclosure for every mass, clipping a negative lower endpoint at zero. If there are M mass entries for a parameter and every interval has width at most η/M , the displayed bound is at most η . Normalization uses only rational arithmetic. A zero rounded sum invokes the declared fallback. The weighted bound, rather than a bound on a ratio at an individual rare history, proves convergence. Conditioning on an independent visible selector leaves every bound unchanged. The mark is part of the finite tree and is never redrawn independently of its reports. $\square$

For the Gaussian acquisition its required masses are explicitly

$$\int_X p_\theta(x) \gamma(w \mid x) \prod_{j=1}^k \mathbb{P}\{m_{j,a_{1,j}}(x) + \sigma_j \xi_j \in I_{y_j}\} d\mu(x). \quad (10.4)$$

This is an integration problem for a specified preparation and trial family. The theorem does not infer its effective solvability from the mere words “standard Borel”. Rigorous enclosures for (10.4) are part of the effective data specification in Theorem 13.1; explicit physical families are treated in Section 14.

11. ENRICHED DEFECT CERTIFICATES AND THEIR CONVERGENCE

A small error of a Galerkin orbit need not force a chosen residual majorant to be small. We replace that implication by an estimator whose own convergence is proved. The approximation is allowed to be a finite tree of trial observables rather than powers of one compressed matrix. This enrichment changes the numerical presentation, not the physical experiment or the charged consumer register.

Let H be a separable real Hilbert space, $U(t)$ a strongly continuous orthogonal group with generator L , and C_a finitely many orthogonal operators. Write $T_a = C_a U(\Delta)$, with fixed rational $\Delta > 0$. Let $\mathcal{D}$ be a countable rational vector space in $D(L)$ whose real span is graph dense in $D(L)$, and suppose its rational span is also graph dense. It suffices to start with a countable graph-core family and all rational linear combinations.

Definition 11.1 (Effective graph data). The data are effective if vectors of $\mathcal{D}$ can be enumerated and rational intervals of any requested width can be computed for inner products among finitely many vectors of the forms

$$v, \quad Lv, \quad C_a v, \quad f_1, \dots, f_T, \quad v \in \mathcal{D}.$$

Only inner products actually used in the following norm expressions are necessary. The initial observables have specified H names. For physical probability certification we require, in addition, computable enclosures for (10.4) for finite trial families selected by this procedure and certified preparation bounds. These are finite analytic input operations; no deficiency optimizer or exact controlled orbit is an input operation.

For nodes $v_0, \dots, v_l \in \mathcal{D}$ at rational times $0 = s_0 < \dots < s_l = \Delta$, let $z(s)$ be their piecewise affine interpolant, and set

$$\begin{aligned} \mathcal{D}(z) &= \frac{1}{2} \sum_{j=0}^{l-1} (s_{j+1} - s_j) (\|d_j - Lv_j\| + \|d_j - Lv_{j+1}\|), \\ d_j &= (v_{j+1} - v_j) / (s_{j+1} - s_j). \end{aligned} \quad (11.1)$$

The interpolation lies in $D(L)$ at every time and is absolutely continuous in its graph norm. For a trial input $g \in \mathcal{D}$ and trial output $v \in \mathcal{D}$, define the one-step certificate

$$\mathcal{E}_a(g, z, v) = \|g - z(0)\| + \mathcal{D}(z) + \|C_a z(\Delta) - v\|. \quad (11.2)$$

No derivative of an intervened vector appears.

Lemma 11.2 (Certified path defect). *For every such triple,*

$$\|T_a g - v\| \leq \mathcal{E}_a(g, z, v).$$

Given $g \in \mathcal{D}$, an action a , and $\eta > 0$, there are rational nodes with $z(0) = g$ and a rational-core output v such that $\mathcal{E}_a(g, z, v) < \eta$.

Proof. On each interpolation interval, variation of constants gives

$$U(\Delta)z(0) - z(\Delta) = \int_0^\Delta U(\Delta - s)(Lz(s) - z'(s)) ds.$$

The identity follows first on each graph-differentiable interval and then by concatenation. The integrand's norm is bounded by the affine interpolation of its endpoint norms, since z' is constant on that interval and Lz is affine. Thus its integral is at most $\mathcal{D}(z)$. Orthogonality of C_a and the two endpoint errors give (11.2).

For existence put $u(s) = U(s)g$. It is continuously differentiable in H , continuous in the graph norm, and $u' = Lu$. Let a uniform rational mesh have size h . Its exact-node interpolant has derivative error bounded by the modulus of continuity of u' at scale h , and its L -image differs from $Lu(s)$ by at most the modulus of continuity of Lu at that scale. Both moduli tend to zero. Approximate each noninitial node by a member of $\mathcal{D}$ within h^2 in graph norm; keep the initial node equal to g . The change in the interpolant derivative is at most $2h$ and the change in its L -image at most h^2 . The endpoint norms in (11.1) obey these same modulus bounds, so $\mathcal{D}(z) \rightarrow 0$, not only the orbit error. Finally use density of $\mathcal{D}$ in H to approximate $C_a z(\Delta)$ by v . No assertion that $C_a z(\Delta) \in D(L)$ is needed. $\square$

Theorem 11.3 (Effective enriched controlled approximation). *Under the effective graph-data hypothesis there is an algorithm which, for a fixed finite horizon and rational tolerance $\eta > 0$, returns finite-core mean functions $\hat{m}_{k,w}$ and rational upper bounds $e_{k,w}$ such that*

$$\|T_{a_1} \cdots T_{a_k} f_k - \hat{m}_{k,w}\| \leq e_{k,w} < \eta, \quad w = (a_1, \dots, a_k), \quad 1 \leq k \leq T. \quad (11.3)$$

Each bound is a sum of an initial approximation error and certified one-step path defects. The algorithm terminates; no rate or polynomial complexity is asserted. Its certified physical marked error is

$$\alpha = 1 \wedge \frac{D}{\sqrt{2\pi}} \sum_{k=1}^T \frac{1}{\sigma_k} \left(\sum_{w \in A^k} e_{k,w}^2 \right)^{1/2}, \quad (11.4)$$

and tends to zero along requested tolerances tending to zero. The source and approximation are compared by stateless identity report channels, including all independent exposed selectors.

Proof. Enumerate $\mathcal{D}$ and increasingly accurate rational norm upper bounds until a trial vector for f_k has error strictly below a prescribed small tolerance. Density ensures termination. For each step enumerate all finite rational time partitions and all rational core nodes and outputs, with initial node the trial input. Dovetail these candidates and interval precisions. Lemma 11.2 provides a candidate whose true defect is strictly smaller than the requested tolerance, so eventually its rational upper enclosure is also smaller. This is a terminating semidecision search with a proved strict witness. It does not test an undecidable equality of real numbers.

For each final acquisition k , build a suffix tree starting at the trial f_k . At the next level approximate T_a applied to each current trial vector using the preceding search. After k levels, the chronological word $a_1, \dots, a_k$ corresponds to applying a_k first to the observable and a_1 last. This gives exactly $T_{a_1} \cdots T_{a_k} f_k$, not its reversed physical dynamics. Orthogonality propagates the accumulated error additively. Allocate less than $\eta/(2(k+1))$ to the initial approximation and each step, and refine finite norm enclosures until their rational sum is below η . There are finitely many nodes and every search terminates.

All trial vectors used at finitely many nodes span a finite-dimensional subspace of H . Initialize its evaluation vector jointly with the actual mark and use the stored action-word table to read the appropriate linear functional at each time. This is a causal probability model. Its physical lookup table is not declared to be a finite consumer register, and its coefficient matrices are not called Markov matrices. The first-mismatch proof of Theorem 9.5, with the verified errors in (11.3), gives

(11.4). Every term tends to zero at fixed horizon. All numerical square roots and constants in an implementation are replaced by rational upper enclosures; their extra tolerances may also be made arbitrarily small. $\square$

Remark 11.4 (Existence, effectiveness, and the old Galerkin estimator). Graph-core density alone proves existence of arbitrarily small enriched defects. Effective graph data make the search an algorithm. Neither assertion says that the Galerkin majorants (9.12) converge along every graph-core sequence. Those estimates remain valid for every selected space and are kept in the article. The new estimator has a different convergence proof: its piecewise affine residual itself is made small. Effective integration of the resulting finite report probabilities is an additional operation, explicit in Definition 11.1, and is not inferred from Hilbert-space density.

12. EFFECTIVE GRAPH DATA FROM REGULAR COLLISION CHARTS

The effective-data hypothesis can be discharged for a class of genuine colliding systems; it need not remain an oracle for their exact flow. Here the particle number is fixed. No bound uniform in particle number, number of collisions, or kinetic scaling is used or concluded.

Fix $N \geq 2$, a flat square or cubic torus of positive rational side length, and a positive rational sphere diameter smaller than half that length. Assume the allowed configuration region has nonempty interior. Let μ be the normalized equilibrium law with independent standard Gaussian velocity coordinates, and let the finite intervention family consist of algebraic orthogonal transformations applied identically to every particle velocity. Flight times are rational. Noise scales are computable positive reals with supplied rational positive lower bounds. We use the classical almost-everywhere regular hard-sphere flow: on each bounded time interval almost every phase point has a finite itinerary of simple, nongrazing binary collisions, and the flow preserves equilibrium measure [63]. The following effective statements are consequences of that flow theorem and the explicit collision formulas, rather than additional collision-count estimates.

A bounded observable or preparation is called elementary-computable here if it has a supplied rational bound and admits uniformly computable rational interval enclosures on compact rational boxes of continuity, with mesh errors tending to zero. We allow piecewise descriptions whose exceptional boundaries are a specified finite union of nonzero polynomial zero sets. Globally computable continuous functions with known local moduli, and bounded piecewise polynomial functions, are examples. Preparations are nonnegative normalized densities of this class. A finite actual-mark channel has elementary-computable probabilities $\gamma(w \mid x)$ summing to one. These specifications exclude arbitrary noncomputable measurable inputs.

Lemma 12.1 (Effective full-measure chart exhaustion). *For any fixed finite collection of controlled words and bounded time windows, regular collision charts can be enumerated with rational box certificates. Their union has full equilibrium measure. Integrals of bounded functions which have computable interval enclosures on these charts have rational enclosures of arbitrarily small width, provided a uniform bound for the integrand is supplied. The assertion also holds with finitely many compact time-integration variables.*

Proof. On a velocity-cutoff box and a bounded time window, energy conservation gives a computable speed bound. Only finitely many periodic images can participate in a collision; enumerate all images within that displacement bound. The candidate time for a pair is a root of

$$|q_i - q_j + n\ell + t(v_i - v_j)|^2 = d_0^2,$$

where ℓ is the torus side and d_0 the diameter. On a simple nongrazing itinerary, the relevant discriminant is positive, the selected future root is separated from every competing root, and the normal relative velocity does not vanish. Reflection of the velocities is an explicit algebraic function of the positions and velocities at that root. An already realized contact has its zero

root removed; the outgoing sign specifies the next free flight. In unfolded coordinates there is no coordinate-wrap event to treat as a physical collision. A pair with zero relative velocity is handled by a strict positive separation bound on its whole flight interval, not by division by a vanishing quadratic coefficient. The same no-contact interval test handles sufficiently small perturbations when no root is present.

Enumerate finite collision lists, rational velocity and position boxes, and rational strict margins. Interval arithmetic for algebraic operations and square roots verifies the selected roots, their order, nongrazing margins, and absence of another collision before the final time. On sufficiently small boxes about any regular itinerary with a noncollision final time, all these strict inequalities hold. This is because there are finitely many collisions and every relevant gap and nongrazing angle is positive there. Thus enumeration produces a neighborhood of every such point. Endpoint-collision sets have measure zero. Reverse time uses the same construction. Algebraic kicks have computable inverses and preserve μ , so the finite intersections needed for controlled words and transformed chart inputs again have full measure. This proves effective enumerability and exhaustion; it does not assert a computable uniform collision-count bound.

The equilibrium measure itself is computable on boxes. The configuration normalizer is the volume of a bounded region defined by finitely many algebraic separation inequalities, with boundary of Lebesgue measure zero. Inside and outside rational-box approximations converge to its volume. A rational interior box provides a strictly positive lower bound for the normalizer; it can be found by enumeration because the interior is nonempty. Velocity-box masses are products of computable Gaussian interval probabilities. Gaussian tails provide effective cutoffs. These facts give rational upper and lower enclosures for μ and for the mass of every box used in the construction.

Replace finitely many accepted boxes by a disjoint dyadic refinement. Continue enumeration and refinement until the certified lower mass of their union exceeds $1 - \varepsilon$. This search terminates since the countable union has full measure. On the accepted compact boxes use interval enclosures and subdivision to bound the integrand's integral; on the complement use its known absolute bound M , at cost at most $M\varepsilon$. Refining the mass and integration enclosures gives any specified absolute error. Exceptional input-function boundaries are also null and can be excluded by strict interval tests for their nonzero defining polynomials. Compact time variables are handled by product-box integration; their collision-time sets are null by the finite-itinerary property and Fubini's theorem. At no point is an uncertified sample estimate for the omitted set used. $\square$

Let (ψ_j) be a countable uniformly described dense family of bounded, compact-velocity-supported continuous piecewise polynomial functions on phase space. For instance, use rational tent functions on successively finer coordinate boxes. Restrictions of this family are dense in $L^2(\mu)$. For rational $\delta > 0$ set

$$\kappa_\delta(s) = \frac{15}{16\delta}(1 - s^2/\delta^2)^2 \mathbf{1}_{[-\delta, \delta]}(s), \quad v_{j,\delta} = \int \kappa_\delta(s) U(s) \psi_j ds.$$

Let $\mathcal{D}$ be all rational linear combinations of these vectors.

Theorem 12.2 (Constructive microscopic graph data). *For the fixed-particle equilibrium system just specified, $\mathcal{D}$ is an enumerable graph core for L . Each vector has the exact representation*

$$Lv_{j,\delta} = - \int \kappa'_\delta(s) U(s) \psi_j ds. \tag{12.1}$$

All inner products required in Definition 11.1, for bounded elementary-computable acquisition observables, have arbitrarily accurate rational enclosures. The marked Gaussian prefix masses of every finite trial family in $\mathcal{D}$ also have such enclosures. Thus the complete effective-data hypothesis of Theorem 13.1 holds in this class, including its probability integration clause.

Proof. The kernel is continuously differentiable, compactly supported, has integral one, and vanishes at its support endpoints. The strong group identity and integration by parts prove domain membership and (12.1). If $u \in D(L)$, its mollified orbit converges to u in graph norm as $\delta \downarrow 0$. At fixed δ , approximating u in H by rational combinations of the ψ_j approximates its mollification in graph norm, since

$$\|A_\delta(u - \psi)\| + \|LA_\delta(u - \psi)\| \leq (\|\kappa_\delta\|_1 + \|\kappa'_\delta\|_1)\|u - \psi\|.$$

A diagonal choice proves graph density. No decision of linear independence between formal core vectors is needed by the enriched search.

For inner products insert the time-integral representations on each side and use Fubini. The resulting integrands are bounded products of known kernels and of observables evaluated along finitely many exact collision itineraries, possibly after an algebraic kick. Known sup-norm bounds for ψ_j , κ_δ and κ'_δ bound these products. Lemma 12.1 therefore computes their integrals, including cross terms involving $C_a v$ and the given bounded f_k . The signs of κ' present no difficulty because absolute bounds are available.

For completeness, the time-integral value of a core function itself admits uniform enclosures on each compact regular itinerary chart. Enclose all collision times there. Away from small strips containing those times, the algebraic flow formulas and input observables have uniform interval approximations. The strips contribute at most their total length times the known kernel and observable bounds. Strict itinerary margins allow these strips to shrink under subdivision. Thus both $v(x)$ and $Lv(x)$ are computable continuous functions on that chart, even if a velocity-dependent ψ_j jumps at a collision. The integral across its moving jump is controlled by the strip bound, not by pretending the jump is absent.

Every trial mean is a finite rational combination of such bounded core functions. Gaussian cell probabilities are computable continuous functions of that mean, including the two tails, and their derivatives are bounded by the positive noise scales. The integrand in (10.4), multiplied by a bounded preparation and actual-mark probability, consequently satisfies the hypotheses of Lemma 12.1. The lemma computes its mass enclosures. It also supplies L^2 preparation and trial-norm enclosures. This verifies all finite operations required by Definition 11.1. The exhaustive searches may have no useful uniform rate, but each requested enclosure terminates by the full-measure exhaustion proved above. $\square$

Corollary 12.3 (Certified approximation for a colliding physical class). *For two systems in Theorem 12.2 with a common finite control interface, finite parameter and mark alphabets, bounded elementary-computable preparations and observables, rational positive noise variances and rational observation times, the procedure of Theorem 13.1 produces convergent same-budget intervals for their physical intrinsic deficiency. It uses certified collision charts rather than an exact-flow integration oracle. It does not require the acquisition observables to be conserved quantities.*

Proof. The theorem supplies the effective graph data and marked prefix integrals; Theorem 11.3 supplies convergent analytic certificates; the finite report and rational-tree bridges preserve the resource signature. Apply Theorem 13.1. One may work in the full labeled L^2 space. For symmetric input data the construction also restricts to its symmetric subspace by symmetrizing the tent family; the flow and identical-particle kicks commute with that finite symmetrization. $\square$

The use of a null exceptional set is constructive here because its regular complement has an effective increasing chart enumeration and because the probability of each accepted finite cover is certified. A bare assertion of almost-everywhere computability without those two properties would not justify the conclusion. Conversely, the argument is an exhaustion algorithm, not a practical complexity bound or a uniform estimate in the Boltzmann–Grad regime.

13. PHYSICAL INTRINSIC RESOURCE CERTIFICATION

We can now state a single conclusion joining the analytic and finite parts of the theory. The original experiments, not their approximations, are the objects whose intrinsic deficiency is bounded. Fix one of the private, hidden-selector, or visible-selector signatures, denoted b ; its state widths and, where relevant, selector cardinality are fixed. Write $\delta_b(E, F)$ for the corresponding intrinsic deficiency, with source E simulating target F . The marked feedback comparison always uses the actual initial mark. For a visible selector its reference law is the joint law specified in Section 2.

Theorem 13.1 (Same-budget physical certification). *Let E, F be finite-horizon active Gaussian acquisitions on Hilbert probability models of the form considered above, with finite parameter, action and actual-mark alphabets, positive noise variances, and certified L^2 preparation bounds. For each model choose a finite trial presentation with rigorous marked error α_E, α_F , supplied either by Theorem 9.5 or by Theorem 11.3. Choose finite report channels as in Lemma 10.1 and rational marked trees as in Theorem 10.3, with certified errors β_E, β_F and ρ_E, ρ_F . Put*

$$a = 1 \wedge (\alpha_E + \beta_E + \rho_E), \quad c = 1 \wedge (\alpha_F + \beta_F + \rho_F).$$

Let $[L, U]$ be any verified local-certificate interval for $\delta_b(E_{\mathbb{Q}}, F_{\mathbb{Q}})$ at the original signature b . Then the original physical deficiency belongs to

$$[\max\{0, L - a - c\}, \min\{1, U + a + c\}]. \quad (13.1)$$

An upper witness is an executable simulator of the original physical experiment with exactly the declared persistent-state and selector budget. A strict lower endpoint is an impossibility certificate at that same budget; it does not refer to a convexified private feasible set.

For the effective assertion, the noise scales are computable reals with supplied rational positive lower bounds. Under effective graph data, including the marked cell-integral operation in Definition 11.1, there is a terminating procedure producing intervals of the form (13.1) of arbitrarily small prescribed width. Their nested intersections converge to $\delta_b(E, F)$. The procedure uses only finite rational certificates and certified analytic input operations. It does not assume attainment of the original nonfinite optimization.

Proof. Each of the three presentation changes is implemented by a stateless, seed-free, parameter-blind and task-blind report channel. The physical trial change uses the identity report channel; the quantization change uses Q in one direction and transiently randomized H in the other; the rationalization change uses the identity finite-report channel. Their compositions give two-sided marked feedback comparisons $E \leftrightarrow E_{\mathbb{Q}}$ with error a and $F \leftrightarrow F_{\mathbb{Q}}$ with error c . Fresh per-report randomness in a presentation channel is integrated into a simulator row; it is not a persistent public or hidden mode. All comparisons include the joint reference of a visible selector.

For an executable finite upper witness, first pass a report of E through the forward presentation, apply the finite simulator, and pass its output through the reverse presentation for F . The stored state and persistent selector are exactly those of the finite witness. The marked feedback triangle inequality bounds its error by $U + a + c$. Conversely, any physical simulator with error smaller than $L - a - c$ could be composed with the opposite stateless presentations to give a finite simulator of the same signature with error smaller than L . This contradicts the verified finite lower certificate. Equivalently, Theorem 2.9 gives

$$|\delta_b(E, F) - \delta_b(E_{\mathbb{Q}}, F_{\mathbb{Q}})| \leq a + c.$$

Taking infima in this argument requires no nonfinite minimizer.

For effectiveness, first choose enriched tolerances so that the two rational upper bounds for α are as small as requested. The terminating search of Theorem 11.3 supplies the trial means and their norm bounds. Choose rational R large and h small until the explicit rational upper bounds for β are small. There are then finitely many marked prefix masses. Request their enclosures at

a precision making the finite sum in (10.3) small, and normalize rational lower endpoints. This gives actual finite rational instruments, not floating-point transition tables.

For these fixed finite instruments, Theorem 2.7 supplies convergent dyadic local lower certificates and executable rational upper witnesses. Refine until $U - L$ is below the requested remaining budget; convergence, or the explicit causal oscillation modulus, ensures termination. Although the finite alphabet and row dimension may grow as the physical tolerance decreases, at each stage this is a finite problem and the mesh is selected for that problem. Thus the total interval width is at most $U - L + 2(a + c)$ and can be made arbitrarily small. Intersect each interval with its predecessors. They all contain the same true value, so the intersections are nonempty and nested, and their lengths tend to zero. $\square$

Remark 13.2 (What is constructive in the theorem). The quantitative implication (13.1) holds for any rigorously supplied analytic enclosures. Its effective-convergence assertion has an explicit analytic data hypothesis. A graph core known only to exist is not, by itself, a numerical integration algorithm. Neither arbitrary real preparations nor arbitrary measurable marks are silently declared computable. Section 14 gives genuine hard-sphere data for which the required calculations can be certified, including an exact continuous-report deficiency. The private certificate search remains nonconvex and may be expensive; the witness-dimension theorem reduces the number of distinct tests, not the number of cells or the cost of real quantifier elimination.

Corollary 13.3 (Regenerative risk certificates). *Retain the actual synchronous reset and stationary-policy signature of Section 8. For losses in $[0, 1]$, certified marked presentation errors ϵ_j for ages $0 \leq j \leq N$ give a uniform stationary-risk error at most*

$$\eta \sum_{j=0}^N (1 - \eta)^j \epsilon_j + (1 - \eta)^{N+1}.$$

The same bound holds for each lower support function of the stationary risk body. Finite-age rational local certificates may be transferred at their declared budgets by Theorem 13.1. Under its effective hypotheses the presentation part of the displayed risk bound can be made arbitrarily small, and then N can be increased.

Proof. Condition the stationary regenerative law on its age. The law of age j has weight $\eta(1 - \eta)^j$, and the omitted ages have total weight $(1 - \eta)^{N+1}$. Apply the marked bounded-loss inequality to each retained age and then sum. The bound precedes optimization, so it also bounds lower support functions. Finite-age certificates concern those finite-age experiments; this corollary does not identify a finite-age optimized deficiency with an unrestricted infinite-history deficiency. The register is actually cleared at reset and any persistent selector remains charged, exactly as in the inherited theorem. $\square$

14. CERTIFIED DATA FOR ACTIVE HARD SPHERES

The preceding theorems require analytic data, not only matrix regressions. We give two explicit families. The first has an exactly computable continuous-report intrinsic deficiency. The second has a nonzero graph residual obtained from a collision-compatible position observable. The hard-sphere flow and its almost-everywhere realization are the same as in the inherited active microscopic construction; no new assumption on a kicked vector's generator domain is introduced.

14.1. A conserved mode with an exact marked deficiency. Take $N \geq 2$ identical hard spheres on a flat torus with a nonempty configuration region. Normalize the equilibrium law to have independent standard Gaussian velocity coordinates. Its configuration factor is uniform on

the allowed configurations. For a fixed coordinate put

$$X = N^{-1/2} \sum_{i=1}^N v_{i,1}, \quad W = \operatorname{sgn} X.$$

Then X is standard Gaussian and W is a fair actual initial mark. Elastic collisions preserve total momentum, hence $U(t)X = X$ and $X \in \bigcap_{j \geq 1} D(L^j)$ with $LX = 0$. The actions $a \in \{-1, 1\}$ reverse all velocities or leave them fixed, so $\bar{C}_a X = aX$. They are physical interventions, even though the selected observable lies in a conserved sector. In the normalized space $V = \operatorname{span}\{X\}$, the exact Gram data are

$$G = [1], \quad J = [0], \quad B_a = [a], \quad R = [0], \quad S_a = [0]. \quad (14.1)$$

All entries are rational. These identities hold for the full colliding system, not a collision-free restriction of its equilibrium law.

Consider one acquisition after the chosen action and any fixed flight time. Source E reports a fresh standard Gaussian independent of the state, while target F reports

$$Y = aX + \xi, \quad \xi \sim N(0, 1).$$

There is no informative earlier report. Source and target carry the same actual mark W . The simulator knows the current action, not X or W ; every persistent selector is independent of the physical state.

Theorem 14.1 (An exact continuous physical certificate). *For every nonempty finite private width profile, and every finite hidden or visible independent-selector budget,*

$$\delta_b(E, F) = \frac{1}{4}. \quad (14.2)$$

The upper witness is the stateless simulator that emits an independent $N(0, 2)$ report. A lower witness is the marked event $\{W = \operatorname{sgn}(aY)\}$. The report-sign coarsening of F has the exact rational joint table

$$\mathbb{P}_F(W = w, Z = z \mid a) = \begin{cases} 3/8, & z = aw, \\ 1/8, & z = -aw, \end{cases} \quad Z = \operatorname{sgn} Y. \quad (14.3)$$

It also has deficiency $1/4$ from a source independent of W .

Proof. For any source simulator its emitted report is independent of W , conditional on an exposed selector and the externally fixed action. The indicated event therefore has probability $1/2$. At the target, $aY = X + a\xi$. The plane vector $(X, a\xi)$ is rotationally invariant; the two sign-disagreement wedges between X and $X + a\xi$ have total angle $\pi/2$. Consequently agreement has probability $3/4$ and the variation discrepancy is at least $1/4$. The action may be fixed in this test, so taking the feedback supremum cannot weaken the bound. This argument holds in each exposed-selector fibre and also after hiding a selector.

For the upper witness write $\varphi_{\sqrt{2}}$ for the $N(0, 2)$ density. The conditional Gaussian law $X \mid Y = y$ has mean $ay/2$ and variance $1/2$. Thus the target joint densities with respect to counting measure on w and Lebesgue measure on y are

$$p(w, y) = \varphi_{\sqrt{2}}(y) \Phi(way/\sqrt{2}).$$

The source witness has joint density $q(w, y) = \frac{1}{2} \varphi_{\sqrt{2}}(y)$. The sign of $p - q$ is positive exactly on $w = \operatorname{sgn}(ay)$ and negative on its complement. Hence that event attains the total variation, which is $3/4 - 1/2 = 1/4$. There are only the two possible actions; the same calculation holds for both, and for mixtures determined by any independent selector. The witness uses no persistent state. Integrating over the two report signs and using symmetry proves (14.3). Its variation

discrepancy from the independent fair-sign witness is again $1/4$, and the same event proves the lower bound. $\square$

The sign table is an exact finite lower-witness presentation, not a two-sided zero-error reconstruction of the continuous Gaussian report. The continuous upper bound above is proved directly. For an arbitrary small two-sided bridge one must instead use the partitions and error budgets in Section 10. Confusing these two uses of a coarsening would give an incorrect certificate.

Proposition 14.2 (Effective cell data in the conserved sector). *For finitely many repeated acquisitions of X , with the two actions above, rational noise variances bounded away from zero, and the mark $\text{sgn } X$, every marked Gaussian prefix mass in (10.4) admits rational enclosures of arbitrarily small prescribed width. The exact trial space has zero graph and intervention errors for every horizon. Consequently Theorem 13.1 gives a convergent same-budget finite rational certificate procedure for this active physical family.*

Proof. Every controlled mean is X or $-X$. The mass is a one-dimensional integral on one of the half-lines of the standard Gaussian density times finitely many Gaussian interval probabilities. Its tail outside $[-M, M]$ is at most M^{-2} , by the second moment. On the remaining compact interval, each interval probability and its derivative has a computable uniform bound: its derivative is a difference of two Gaussian densities divided by its positive noise scale. Products therefore have a computable Lipschitz bound. Uniform Riemann sums have an explicit mesh error given by this bound times the interval length and mesh size. Gaussian density and distribution values at algebraic arguments have rational enclosures: integrate the power series of $\exp(-u^2/2)$ on a bounded interval with a Taylor remainder bound, using symmetry and a Gaussian tail bound for unbounded endpoints. Square roots and the normalizing constant have elementary rational interval algorithms. Combining these enclosures with the mesh and tail errors gives the requested rational mass interval. All quantities are scalar; the configuration-volume normalizer cancels. Equation (14.1) removes the analytic trial error for every controlled word. Apply the finite-report and rational-tree constructions and then the finite local-certificate theorem. This proves an actual procedure for this family, not computability of arbitrary preparations. $\square$

14.2. A nonzero collision-compatible graph residual. Consider two spheres of diameter one on the two-dimensional square torus of side ten. The allowed ordered configuration volume is $100(100 - \pi)$; velocities remain independent standard Gaussians. Let $b(u) = (1 - u^2)^2$ on $|u| \leq 1$ and zero elsewhere, and let $h = 1/4$. Place squares of half-side h at $c_1 = (2, 5)$ and $c_2 = (8, 5)$. Let $\chi_1(q_1, q_2)$ be the product of the four rescaled functions b when particle one is near c_1 and particle two near c_2 , and put $f = \chi_1(q_1, q_2) + \chi_1(q_2, q_1)$. The two supports are disjoint, are separated from the collision boundary, and define a symmetric C^1 position observable on the torus.

Proposition 14.3 (Exact nonzero physical Gram data). *The vector f belongs to $D(L)$ and $Lf = v_1 \cdot \nabla_{q_1} f + v_2 \cdot \nabla_{q_2} f$. With $V = \text{span}\{\phi\}$, $\phi = f/\|f\|_2$, its exact data are*

$$J = 0, \quad R = [192], \quad B_{\pm} = 1, \quad S_{\pm} = 0, \quad \|f\|_2^2 = \frac{2h^4(256/315)^4}{100(100 - \pi)}. \quad (14.4)$$

Here the actions still reverse all velocities or leave them fixed. For one acquisition of f with unit Gaussian noise after $\Delta = 1/1000$, under equilibrium preparation and with any initial finite mark channel, the static trial mean f and the true active physical acquisition have two-sided marked feedback error less than $1/100000$. All constants in this last assertion follow from rational Gram enclosures and $3 < \pi < 4$.

Proof. The position support is at positive distance from contact, including periodic images; its minimal separation is greater than one. On a regular orbit f is continuously differentiable between collisions, and its transport derivative vanishes near each collision configuration. Thus the integral

transport identity holds across collisions without a jump. Its difference quotients are bounded in absolute value by a constant times the conserved total speed. This is square integrable under the Gaussian velocity law. Dominated convergence gives the strong L^2 generator derivative claimed. This also proves the required collision-domain membership rather than presuming a smooth-boundary extension of a kicked function.

Direct polynomial integration gives

$$I_0 = \int_{-1}^1 b(u)^2 du = 256/315, \quad I_1 = \int_{-1}^1 b'(u)^2 du = 256/105.$$

The two disjoint product supports give $\|f\|^2 = 2h^4 I_0^4/Z$, $Z = 100(100 - \pi)$. Independence, centering and unit variance of velocity coordinates give

$$\langle f, Lf \rangle = 0, \quad \|Lf\|^2 = \frac{4I_1}{h^2 I_0} \|f\|^2 = 192 \|f\|^2.$$

Velocity reversal fixes the position vector f , so $B_{\pm} = 1$ and $S_{\pm} = 0$. The displayed Gram residual follows. Since $f \in V$, there is no projection error and the one-step residual is $\Delta\sqrt{192}\|f\|$. The two-action bound in Theorem 9.5 is therefore

$$\alpha \leq \frac{\Delta\sqrt{384}\|f\|}{\sqrt{2\pi}}.$$

Using $I_0 < 1$, $h = 1/4$ and $Z > 9600$ gives $\|f\|^2 < 1/1228800$. Together with $2\pi > 6$ this yields $\alpha^2 < 64/(10^6 \cdot 1228800) < 10^{-10}$. Every inequality is rational except the explicitly enclosed π ; no floating-point Gram estimate is used. A mark coupled to the original state stays coupled in the first-mismatch argument. $\square$

These two examples answer different questions. A conserved sector can carry nontrivial hidden information and an exact positive deficiency; a localized transport sector supplies genuinely nonzero graph residuals and certified error bounds. The velocity interventions change the physical dynamics in both cases. Neither example is a proof of the particle-number-uniform nonlinear kinetic corrector in the historical B4 program. Their downstream edge is the explicit marked physical-to-finite certification theorem, with its own stated analytic inputs and error topology.

15. A NONCONSERVED COLLISION-COMPATIBLE ACQUISITION

The following example lies on the full equilibrium phase space of an interacting hard-sphere system. It is neither a collision-free restriction nor an observable constant along the flow. Its explicitly solvable collective mode permits simultaneous evaluation of a nonzero generator residual and of a marked decision deficiency. The orbit factors through total momentum, so collisions cancel from this particular observable. The residual measures projection error, not interaction strength. The collision-sensitive comparison is proved in Section 4; the solvable collective calculation is retained here as a separate exact benchmark.

Fix $N \geq 2$ equal-mass hard spheres on the square flat torus of side $\ell > 0$, with positive admissible configuration volume. Velocities have independent standard Gaussian coordinates under equilibrium μ . Let $U(t)$ be the full hard-sphere Koopman group, and allow the two physical interventions $v \mapsto av$, $a \in \{-1, 1\}$, before each flight of length Δ . Define the circle-valued coordinate and the normalized total momentum

$$\vartheta = \frac{2\pi}{\ell} \sum_{i=1}^N q_{i,1} \pmod{2\pi}, \quad X = N^{-1/2} \sum_{i=1}^N v_{i,1}, \quad \omega = \frac{2\pi\sqrt{N}}{\ell}, \quad f = \cos \vartheta.$$

The exponential of the displayed coordinate is well defined on the torus; no choice of lifted particle positions enters f .

Theorem 15.1 (Collective domain and controlled-word reduction). *The coordinate ϑ is Haar uniform and independent of the standard normal X . On the full-measure domain of the hard-sphere flow,*

$$U(t)f = \cos(\vartheta + \omega Xt), \quad L^j f = \omega^j X^j \cos(\vartheta + j\pi/2), \quad \|L^j f\|_2^2 = \frac{\omega^{2j}(2j-1)!!}{2}. \quad (15.1)$$

Here $f \in \bigcap_{j \geq 1} D(L^j)$ and $(-1)!! = 1$. In the normalized one-dimensional trial space spanned by $\varphi = \sqrt{2}f$, the matrices of the preceding residual theorem are

$$G = 1, \quad J = 0, \quad R = \omega^2 > 0, \quad B_{-1} = B_1 = 1, \quad S_{-1} = S_1 = 0.$$

For a word $w = (a_1, \dots, a_k)$ put

$$s(w) = \Delta \sum_{j=1}^k \prod_{i=1}^j a_i.$$

Then $T_{a_1} \cdots T_{a_k} f = \cos(\vartheta + \omega X s(w))$. Exactly $k+1$ distinct observable functions occur among the 2^k words, indexed by $s/\Delta \in \{-k, -k+2, \dots, k\}$.

Proof. Simultaneous translation of every particle by r in the first torus coordinate preserves the admissible configuration measure and shifts ϑ by $2\pi Nr/\ell$. These shifts exhaust the circle, so its push-forward measure is Haar. Configuration and velocity are independent, and Gaussian addition proves the assertion about X . Elastic collisions conserve total momentum and do not jump particle positions. Between collisions the derivative of ϑ is ωX . This proves the orbit formula, including the periodic identifications, on the classical almost-everywhere flow domain. Difference quotients of that formula, dominated successively by polynomials in $|X|$, converge in $L^2(\mu)$ since all Gaussian moments are finite. Induction gives membership in every generator power domain and the derivative formula. Independence and the uniform circle integral of squared sine or cosine give the norm formula. This also establishes collision compatibility in the actual generator domain, rather than by applying a formal derivative to arbitrary boundary data.

The inner product of $\cos \vartheta$ and $X \sin \vartheta$ vanishes. Thus $J = 0$ and the squared residual norm of $L\varphi$ is ω^2 . A velocity reversal fixes φ , giving $B_a = 1$ and $S_a = 0$. After the j th intervention the total momentum is $(a_1 \cdots a_j)X$, so summing flight displacements proves the word formula in the order $T_a = C_a U(\Delta)$. Every sequence of k signs can be the successive cumulative products, which gives all $k+1$ clock values. They are distinct as observable functions because

$$\|\cos(\vartheta + \omega X s) - \cos(\vartheta + \omega X t)\|_2^2 = 1 - \exp\{-\omega^2(s-t)^2/2\} > 0 \quad (s \neq t).$$

□

Theorem 15.2 (Explicit converging finite physical presentations). *Let $m \geq 1$ and replace the mean at clock s by*

$$f_{m,s} = \sum_{j=0}^{m-1} \frac{s^j}{j!} L^j f.$$

For $|s| \leq k\Delta$ the mean error is bounded by

$$E_{m,k} = \frac{(k\Delta\omega)^m}{m!} \sqrt{\frac{(2m-1)!!}{2}}. \quad (15.2)$$

For Gaussian report noises $\sigma_k > 0$ and preparations satisfying $\sup_\theta \|p_\theta\|_2 \leq D$, the exact and polynomial acquisitions have two-sided stateless marked feedback comparison error at most

$$\alpha_{m,T} = 1 \wedge \frac{D}{\sqrt{2\pi}} \sum_{k=1}^T \frac{\sqrt{k+1}}{\sigma_k} E_{m,k}. \quad (15.3)$$

The same actual initial mark and every exposed independent selector are preserved. The estimate is uniform over all consumer register widths, and $\alpha_{m,T} \rightarrow 0$ at every fixed finite horizon. The physical trial functions and all their generator and intervention Gram entries are explicitly integrable against equilibrium.

Proof. The integral Taylor remainder for the unitary orbit on $D(L^m)$ has norm at most $|s|^m \|L^m f\|_2 / m!$, also for negative s . Theorem 15.1 gives (15.2). Its successive ratio is

$$\frac{E_{m+1,k}}{E_{m,k}} = k\Delta\omega \frac{\sqrt{2m+1}}{m+1} \rightarrow 0,$$

which is an effective bound for the convergence claimed here.

Couple the two acquisitions from the same initial x, W and the same independent selector. Until the first unequal report, their controls agree. Two unit-dimensional Gaussians with the same variance have variation distance at most their mean difference divided by $\sigma_k \sqrt{2\pi}$; this follows by integrating the derivative of the translated Gaussian density. At fixed x , the probabilities of all matched prefixes sum to at most one. Keep those probabilities until bounding the mismatch by the largest mean error at that x . There are only $k+1$ different errors at time k , even though there are 2^k words. Cauchy–Schwarz gives

$$\int p_\theta(x) \max_s |e_{m,s}(x)| d\mu(x) \leq D \left(\sum_s \|e_{m,s}\|_2^2 \right)^{1/2} \leq D\sqrt{k+1} E_{m,k}.$$

Sum first-mismatch probabilities over k . The construction in both directions uses the identity report interface, has no added persistent state, and couples the actual mark, proving (15.3). No bound on feedback-conditioned preparation densities was used.

The trial family consists of $X^j \cos \vartheta$ and $X^j \sin \vartheta$ for $j < m$. Its inner products are products of circle integrals and Gaussian moments. Generator entries multiply by ωX and rotate sine and cosine; the interventions replace X by $-X$. Every required finite entry is thus a polynomial in ω with rational coefficients before normalization. This supplies exact symbolic expressions and arbitrarily tight rational interval enclosures. The polynomial means may be unbounded in X ; that is harmless for (15.3). For finite-report integration one first truncates $|X|$ with its explicit Gaussian tail, then integrates bounded Gaussian report probabilities over the resulting compact cylinder. For the equilibrium and mark used below, the only mark boundaries are $\vartheta = \pi/2, 3\pi/2$, and can be split explicitly. On each piece trigonometric, polynomial and Gaussian distribution functions have computable uniform moduli. Rational upper and lower Riemann enclosures therefore converge. This verifies the effective marked integration hypothesis of the finite-presentation theorem for this class, without an exact hard-sphere flow integration oracle. $\square$

Remark 15.3. The signed clock is an index of physical mean functions, not an unpriced extra register offered to a consumer. The width-uniform proof uses equality of mean functions to reduce a bound; it does not allow a finite-state simulator to remember its entire control word. Neither the polynomial coefficient vector nor its real dimension is a finite consumer-label count. The underlying microscopic system still has collisions; solvability here comes from the conserved total momentum that transports a *nonconserved* position observable.

15.1. One actual mark, one nonzero residual, and a positive certificate. Let $W = \text{sgn} \cos \vartheta$ in the initial equilibrium state; the zero set has measure zero. Both experiments have a singleton initial report and allow $a \in \{-1, 1\}$ before their single acquisition. The source reports an independent standard Gaussian. The target reports

$$Y = \cos(\vartheta + a\tau X) + \xi, \quad \tau = \omega\Delta, \quad \xi \sim N(0, 1)$$

with noise independent of the actual initial mark and microscopic state. Let Φ and ϕ denote the standard normal distribution and density.

Theorem 15.4 (Exact marked physical deficiency). *Here the action precedes any informative report and its reversal symmetry leaves the relevant marked law unchanged. In the visible comparison the independent selector is retained jointly with the same selector law on the target side. For every nonempty finite private, hidden-selector, or visible-selector signature in this one-acquisition model, its intrinsic deficiency equals*

$$d(\tau) = \mathbb{E}[W\{\Phi(\cos(\vartheta + \tau X)) - 1/2\}] > 0. \quad (15.4)$$

It is attained by the stateless simulator that draws the target's unconditional report marginal. If

$$c_k(\tau) = 2^{-(2k+1)} \sum_{j=0}^{2k+1} \binom{2k+1}{j} \frac{2(-1)^{(|2k+1-2j|-1)/2}}{\pi|2k+1-2j|} e^{-(2k+1-2j)^2\tau^2/2},$$

then

$$d(\tau) = \frac{1}{\sqrt{2\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k c_k(\tau)}{2^k k! (2k+1)}. \quad (15.5)$$

The error after $k = K$ is at most $[\sqrt{2\pi} 2^{K+1} (K+1)! (2K+3)]^{-1}$.

Proof. The action is chosen before any informative report. Reversal of the independent symmetric X does not change the joint law of (W, Y) , including conditionally on any exposed independent seed. Set $H_\tau(\theta) = \mathbb{E} \operatorname{sgn} \cos(\theta + \tau X)$. This is the circle heat evolution of $\operatorname{sgn} \cos \theta$ at time $\tau^2/2$. It is strictly positive on $(-\pi/2, \pi/2)$ and strictly negative on the opposite arc. To see the sign without assuming it from a Fourier expansion, reflect the heat solution oddly across both endpoints of the positive arc. Its restriction solves the heat equation on that interval with zero boundary values and initial value one. Equivalently it is the survival probability of Brownian motion killed at those endpoints. This is positive in the interior at every finite positive time. The odd reflected solution agrees with the circle solution by uniqueness (the difference has zero initial data and its squared integral has nonpositive derivative). At $\tau = 0$ the same sign assertion is immediate.

The final angle is uniform. Reversibility of the Gaussian circle convolution gives $\mathbb{E}[W \mid \vartheta + \tau X = \theta] = H_\tau(\theta)$. Let ν be the unconditional density of Y . The density of (W, Y) , minus $\frac{1}{2}\nu(y)$ in its w th row, is

$$\frac{w}{4\pi} \int_0^{2\pi} H_\tau(\theta) \phi(y - \cos \theta) d\theta.$$

Pair points θ and $\theta + \pi$. On the positive arc both $H_\tau(\theta)$ and $\cos \theta$ are positive, and $\phi(y - \cos \theta) - \phi(y + \cos \theta)$ has the sign of y . Thus the density difference has the sign of wy , and is not zero. Its positive event is $\{wy > 0\}$. The excess probability of that event is (15.4), proving positivity and giving a stateless upper witness of that variation distance.

Every source simulator at the declared signature, conditionally on its exposed seed, produces a report independent of the fair W . For any such report law the event $\{wy > 0\}$ has probability at most $1/2$, including possible mass at zero. The target probability is $1/2 + d(\tau)$. This supplies the matching lower bound for private and hidden randomization; the same bound in each visible fiber supplies it for visible randomization. Mixtures of the two action choices do not change this conclusion.

Finally, uniformly for $|z| \leq 1$,

$$\Phi(z) - 1/2 = \frac{1}{\sqrt{2\pi}} \sum_{k \geq 0} \frac{(-1)^k z^{2k+1}}{2^k k! (2k+1)}.$$

The absolute remainder is bounded by the next term, by the alternating Taylor expansion of $e^{-u^2/2}$ followed by integration. Expand $\cos^{2k+1}$ into its finite complex exponential sum. For odd n ,

$$\mathbb{E}[W \cos(n\vartheta)] = \frac{2(-1)^{(|n|-1)/2}}{\pi|n|}, \quad \mathbb{E}e^{in\tau X} = e^{-n^2\tau^2/2}.$$

The sine terms vanish. These identities give c_k and the claimed series and remainder, with no unproved exchange of a nonuniform sum. $\square$

Corollary 15.5 (A rational, nonvacuous collision certificate). *For $N = 2$, $\ell = 10$, any rational diameter in $(0, 5)$ giving nonempty interior, and $\Delta = 1/2$, the same physical acquisition satisfies*

$$\frac{205809122887}{10^{12}} < d(\tau) < \frac{205809122889}{10^{12}}, \quad R = \frac{2\pi^2}{25} > 0.$$

The degree-fifteen physical Taylor presentation has marked comparison error less than $3 \cdot 10^{-11}$ for the single report.

Proof. Here $\tau^2/2 = \pi^2/100$. Evaluate (15.5) through $K = 10$ with outward rational arithmetic and add its stated remainder. For reproducibility the source supplies every arithmetic operation: $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ is bounded by alternating series; square roots are enclosed by rational bisection; exponentials are range-reduced to arguments in $[0, 1]$, enclosed by alternating series, and squared back with outward rounding. The resulting enclosure has lower endpoint greater than the displayed lower rational and upper endpoint smaller than the displayed upper rational. All comparisons are integer comparisons. Applying (15.2) with $m = 16$, and replacing $\sqrt{2}$ in (15.3) by the larger rational 2, yields the final strict bound. The generator residual follows exactly from Theorem 15.1. The supplied checker redoes these finite rational computations; the analytic justification is the preceding domain, coupling and series proofs, not the execution of that checker. $\square$

16. A JOINT PHYSICAL AND NONCONVEX CERTIFICATE THEOREM

The finite testing and microscopic approximation results now give a single lower–upper certificate for the original resource problem. The stateless transfer inequality and physical-to-rational bridge are those of Theorem 13.1; the polynomial lower module is a separate strengthening, not a new physical transfer principle. Unlike a constant mixture of tests, the polynomial lower certificate can separate a target from a private behavior image even when the target belongs to its convex hull.

Theorem 16.1 (Constructive same-budget physical testing). *Fix a finite horizon and one of the private, hidden-selector, or visible-selector signatures. Let E, F be marked physical acquisitions with finite compatible controls and marks. Suppose that rational finite presentations E_j, F_j have executable two-sided stateless marked feedback errors $a_j, c_j \rightarrow 0$, with computable upper bounds. For the algorithmic assertion require the uniform enumeration and rational error enclosures of Proposition 3.6, not merely individual computability of each presentation. Let $P_{p,L}$ be the polynomial (6.2) of the finite intrinsic deficiency problem for E_j, F_j , and let a nonnegative rational Bernstein array certify it. Let an actual rational simulator table at the same signature have finite error U . Then*

$$\max\{0, L - a_j - c_j\} \leq \delta_b(E, F) \leq \min\{1, U + a_j + c_j\}. \quad (16.1)$$

There is a terminating dovetailed search for such an interval of any prescribed positive rational width. Every lower certificate is checked by a finite polynomial identity and sign comparisons; every upper certificate is an executable table at the original budget. The visible case retains the selector in both compared marked laws.

The hypotheses hold for the effective fixed-particle collision class of Theorem 12.2, and in particular for the nonconserved collective acquisitions of Theorem 15.2 with equilibrium preparation and a finite computable marked channel with the piecewise effective integration property proved there. For the sign mark of Theorem 15.4, that integration property is explicit.

Proof. The finite polynomial identity gives $L \leq \delta_b(E_j, F_j)$ by Theorem 6.1; direct evaluation of the rational table gives the reverse bound by U . Compose it with the two stateless physical

presentation channels. Their identity state costs preserve all retained widths and selector budgets. The marked triangle and presentation-stability inequalities give

$$|\delta_b(E, F) - \delta_b(E_j, F_j)| \leq a_j + c_j.$$

This proves (16.1); no attainment assertion for the original nonfinite minimum is required. The visible comparison is joint with the same selector, not its hidden marginal.

Given a target width $\eta > 0$, enumerate presentations until their certified sum of errors is less than $\eta/4$. For that presentation, Corollary 6.3 produces lower and upper endpoints with gap less than $\eta/2$. Retain maxima of all verified lower endpoints and minima of all verified upper endpoints. The resulting nested physical intervals have width less than η and converge. Zero lower bounds are always available, so this argument also covers zero deficiency and does not ask a strict-positivity certificate to prove equality at a boundary.

The cited constructive collision theorem provides effective graph approximation and marked integration; the finite-report and joint-prefix rationalization theorems then give the presentations and their error bounds. In the collective class, (15.2) replaces the unquantified graph approximation by an explicit rate. Its compact cylinder integration and Gaussian tail bounds give every finite marked-prefix probability. The retained finite-presentation theorem rationalizes these joint prefixes without dividing by a small conditional probability. Thus all links in this specialization are supplied either by an explicitly cited retained theorem or by a proof above, not by assuming a rational physical transition oracle. $\square$

16.1. Exponential acquisition values. The distinction between small total variation and accuracy at an exponential scale is important for the nonlinear-observable direction of the program. The following estimate gives its precise price. It is a theorem about bounded marked acquisition functionals, not a construction of a kinetic large-deviation action.

Theorem 16.2 (Sharp exponential transfer). *Let P, Q be two laws of the same marked transcript with $\|P - Q\|_{TV} \leq \varepsilon$. If h is measurable and $A \leq h \leq B$, then for every $\beta > 0$,*

$$\left| \frac{1}{\beta} \log \mathbb{E}_P e^{\beta h} - \frac{1}{\beta} \log \mathbb{E}_Q e^{\beta h} \right| \leq \frac{1}{\beta} \log \{1 + \varepsilon(e^{\beta(B-A)} - 1)\}. \quad (16.2)$$

The bound is sharp. If a pair of stateless marked presentation channels has error ε uniformly over a declared resource signature, the payoff is evaluated on the same compared marked output interface. Under this convention the same bound holds between optimal infimum values, and between the optimal supremum values, over that signature. The policies are transported through the channels at unchanged budgets.

For the effective physical class of Theorem 16.1, let $\beta_n > 0$ and an upper bound $M_n \geq B_n - A_n$ be computable. One can construct finite rational presentations with certified errors $\varepsilon_n \leq e^{-\beta_n M_n}/n$. Their optimized normalized logarithmic acquisition values then differ from the physical ones by at most $\log(1 + 1/n)/\beta_n$.

Proof. For any bounded g , integration of its level sets gives $|\mathbb{E}_P g - \mathbb{E}_Q g| \leq \varepsilon(\sup g - \inf g)$. Take $g = e^{\beta(h-A)} \in [1, e^{\beta(B-A)}]$. Both expectations are at least one. Their larger-to-smaller ratio is at most $1 + \varepsilon(e^{\beta(B-A)} - 1)$, which proves the estimate. Equality holds when Q is concentrated where $h = A$ and P moves mass ε to a point where $h = B$.

For any admissible physical policy transport it to the presentation and apply the uniform inequality; this gives one inequality between optimal values. Transport in the other direction for the reverse inequality. Approximate optimizing policies suffice, so attainment is unnecessary. The same reasoning works for infima and suprema, but does not replace a common policy by separate phasewise optimizers.

Finally choose a strictly positive rational tolerance below $e^{-\beta_n M_n}/n$, using computable exponential enclosures. The terminating presentation search of Theorem 16.1 attains that tolerance.

Substitution gives the final claim. This is an accuracy-versus-scale construction; it asserts neither an efficient complexity rate nor a large-deviation limit of the microscopic model. $\square$

Remark 16.3 (The downstream implication). Theorems 15.1–16.2 provide a verified fixed-particle link from a nonconserved collision-domain observable to finite-resource marked and exponential acquisition values, with rigorously computable errors. They do not assert particle-number-uniform bounds, a Boltzmann–Grad limit, a nonlinear BBGKY graph corrector on action sublevels, or a range theorem for the historical kinetic Hamiltonian. Those are distinct analytic assertions in the retained program. In particular (16.2) explains why plain unscaled total variation alone does not supply exponential-scale control. All eleven historical components remain in the dependency graph; neither component order nor the existence of an adapter is used as a proof of a missing kinetic theorem.

APPENDIX A. THEOREM DEPENDENCIES AND INFORMATION CONVENTIONS

The comparison conventions of Section 2 supply a compact original behavior image for each fixed signature. Theorem 3.1 uses only that image and its affine tests. Lemma 3.2 and the binomial estimate then give Theorem 3.3. The strict row-cube verifier of Theorem 6.1 is a separate route for checking lower bounds. Proposition 3.6 combines finite strict certificates, executable upper tables and stateless presentation errors; it does not require an inverse constrained randomization theorem.

The collision path has a second, independent finite calculation: Theorem 4.1 solves its marked gated reference experiment, and Lemma 4.2 proves the physical separation of incoming and outgoing detector values. The stateless marked variation estimate joins these two arguments in Theorem 4.3. Proposition 4.4 verifies the effective presentation input using a separated collision root. None of these steps assumes the collective benchmark’s conserved-mode formula.

For optional projections, Bayes’ formula and maximal inequalities first prove Theorem 5.1. Conditional Gaussian change of measure then verifies its hypotheses in Theorem 5.3. The same-latent maximal coupling is used only for process convergence. Finite-register risk transport is instead a data-processing consequence of the original marked variation estimate.

The graph-domain, residual and finite-prefix sections supply the broader physical presentation machinery. Their input is an actual flow, a specified Hilbert realization, a prepared law and an observable with the declared approximation properties. The bound in Theorem 13.1 already expresses the same-budget transfer principle. Theorem 16.1 retains it with the strict polynomial lower module. The present adaptive testing and scattering results add independent-test size control and a collision-generated resource separation; they do not reclassify the transfer principle as a new result each time a certificate module changes.

Throughout the article fresh private randomization has no memory. A hidden persistent selector and a visible persistent selector are separate signatures. A visible comparison is joint with its independent target selector; marginalizing it changes the problem. The computational chart, the revealed-candidate testing game and the latent-preserving convergence coupling are proof constructions, not unpriced channels available to the physical decision maker.

PRESERVED DEVELOPMENT: TWELFTH ARTICLE AND EARLIER MATHEMATICS

The preceding sections constitute the seventeenth canonical article. The following archival body retains the earlier mathematical development, including the twelfth canonical body, historical introductions and the annotations already made in the fourteenth development. Original files are unchanged. The seventeenth local copies only clarify the certificate terminology, collective benchmark and effective transfer hypotheses. Historical scope statements are read as historical, not as competing claims for the current article. The preservation map identifies every local change and preserves all predecessor theorem labels.

APPENDIX 22. INTRODUCTION

A causal experiment supplies observations at specified times. A decision procedure with a finite retained register cannot in general replace this interface by an equivalent full-history experiment at no cost. Information which is supplied again by one presentation may have to be carried across time by the other. Moreover, a single encoder serving several tasks cannot be optimized separately for each task. These two constraints make the attainable risk body depend on both the experiment and its resource signature.

This paper studies that dependence for marked experiments. A mark is the actual target variable appearing in a loss, including its coupling with the observations and any latent parameter. The observation controller may know the task; the common encoder does not. Private retained labels, public seed information, and side observations supplied afresh by the physical interface are specified separately. This distinction is needed both for positive approximation results and for exact minimum-state statements.

22.1. The structural results. The first comparison theorem charges a simulator's persistent state. If a simulator of widths K_t and seed size r approximates the target experiment with marked feedback error ε , then an S_t -state, q -seed target consumer has a source implementation of widths $K_t S_t$ and seed size $r q$. The construction retains reachable pairs and gives risk-body inclusion before optimization over consumers. Composition multiplies the state costs and adds the errors. When a consumer's state already determines the marked continuation class, the class register can instead be fused with the consumer's register. This gives the exact same-width complete-state comparison in Theorem 24.7. General consumers need not admit this cancellation.

Our main exact completion result, Theorem 25.2, identifies which width profiles preserve a declared family of decisions throughout a supported history tree. Its object is a causally closed compatible cover. Each cell must admit one optimal action for every task, and all its actual successors under a common control and report must fit in one next cell. Histories with no such successor impose no fictitious constraint. The existence of these covers is equivalent to an exact causal common encoder, whether deterministic, privately randomized, or publicly randomized. Thus neither a coarsest partition nor a positive next-report hypothesis is needed. Corollary 25.5 places this combinatorial characterization on the zero face of the minimax risk body: infeasibility has a positive decision witness $\alpha\gamma$, explicitly specified by the finite task table. Completing Brier prediction tests recovers the universal marked continuation quotient.

The nonfinite theory has two complementary extensions. The dominated finite-horizon compactness theorem uses weak-star policy rows and L^1 -continuous loss effects. A conditional-kernel version permits a physically redisplayed nonatomic covariate, and hence singular full report laws. Theorem 28.2 then treats an unbounded time interval under coherent dominated finite-prefix effects and a summable criterion. With $r_{j,t} \in [0, 1]$ and $\sum a_t = 1$, it proves compact attainable risk

bodies and the attained identity

$$\min_{\lambda} \max_j \left\{ \int \sum_{t \geq 0} a_t r_{j,t}(\pi) \lambda(d\pi) - b_j \right\} = \max_w \min_{\pi} \int \left\{ \sum_{t \geq 0} a_t r_{j,t}(\pi) - b_j \right\} w(dj).$$

No measure dominating the entire infinite path is assumed. For a fixed accuracy the finite public-seed approximation depends on row covering, not the truncation horizon. Its index is priced explicitly when stored in the private register. Prefix-coherent comparison has error $\sum a_t \varepsilon_t$, preserving the same product resource law.

A physical consumer connects these assertions to unbounded microscopic transport. For a contraction generator L and a finite resolved space in $D(L) \cap D(L^*)$, we construct the orthogonal generator on its actual domain and prove the forced-memory and Schur-complement formulas, including the contribution of the unresolved initial observable. A collision-compatible hard-sphere graph core gives strong, locally time-uniform Galerkin approximation of the full Koopman orbit. For Gaussian noisy physical acquisitions with uniformly L^2 -bounded initial densities, Theorem 44.4 converts this analytic approximation into one marked comparison bound uniform over every finite register profile and feedback design. The same bound passes to the infinite-horizon dual values. Crucially, neither the Galerkin coefficients nor a resampled mark are supplied to the encoder.

Finally Theorem 45.1 joins the two sides. If an approximate experiment has resource-preserving risk and benchmark errors at most $\varepsilon < \alpha\gamma/4$, its minimax value separates exact causal cover feasibility from infeasibility by the threshold $\alpha\gamma/2$. The comparison is performed before optimization and its cost remains visible in the one-sided version. This is a precise sense in which analytic experiment approximation can certify an exact common-encoder resource profile.

22.2. Antecedents and the comparison boundary. Classical experiment comparison and randomization criteria are due to Blackwell, Le Cam and their successors [1, 47, 46]. Filtered statistical experiment comparison is itself an established subject [48]; the observation timing is not introduced here as a new concept. The weak-star policy topology and strategic-measure compactification used in the finite-prefix theory have close antecedents in stochastic teams [55, 56]. Common-information dynamic programming and finite-memory causal coding are also established tools [53, 54, 57]. Our encoder remains coupled across decision rows, and every memory or seed conversion has an explicit implementation cost.

The continuation-space equivalence engine is classical weighted-automaton linear algebra [58, 61]. Positive reachable state cardinality, in contrast, need not equal linear rank. The synthesis results below include polynomial count-vector families, exponential positive width at rank two, and a finite-task graph-coloring reduction. The compatible-cover device likewise belongs to partial-machine minimization. Here its support propagation is proved directly for randomized common encoders, and the resulting exact profiles are related to positive risk witnesses and marked comparison. We make no claim that the cover device or graph coloring by itself is new.

The operator section uses the classical finite-rank projection method, not a new general resolution of orthogonal dynamics. The semigroup and domain assumptions in [65, 66] are particularly relevant: a finite-rank projection does not make PL bounded unless its range lies in the adjoint domain. We impose that condition and provide the finite-block proof, the forcing and reconstruction estimates, and the width-uniform physical decision consequence. The graph-core resolvent argument, product compactness, Carathéodory's theorem and convex minimax are classical ingredients. The result proposed for assessment is their resource-sensitive connection, not a priority claim for any of these ingredients.

22.3. Scope and organization. All hypotheses remain attached to their statements. The infinite-horizon compactness result concerns summable criteria and dominated finite prefixes, not arbitrary average-cost control or arbitrary filtered strategic-measure closure. The exact cover theorem

concerns supported restart decision families; an arbitrary task-dependent control law is not silently replaced by an open-loop test. Public seed information is part of the compared joint interface whenever a consumer may read it. A marginal simulation bound which discards that seed is insufficient. The physical approximation proves a microscopic L^2 decision limit, not a nonlinear kinetic action-sublevel or Boltzmann–Grad limit.

The theory also retains the finite common-membership separation, delayed Gaussian resource exponents, positive filtering, finite nonlinear semigroups, hidden-phase decisions, predictive trees and measure-valued consequences. The historical program is a dependency graph: a proved model consumer, an acquisition adapter and a primary theorem dependency are different relationships. In particular the primary algebraic-geometric A2 line is independent of the present comparison theory. The complete development preserves the earlier mathematical bodies with their local hypotheses. Reproducibility and source-identity records are separate from the mathematical arguments and are not used as proof premises.

APPENDIX 23. THE MARKED DECISION SPECTRUM

We first fix the object whose finite presentations are to be compared. This requires specifying both the decision tasks and the information available to the decoder. These specifications cannot be recovered from a dimension or a cardinality alone.

23.1. Experiments, tasks, and signatures. An experiment is a standard Borel causal instrument, with an initial law and transition–observation kernels indexed by a parameter θ . Actions, when present, are chosen before the corresponding transition. The same observation-based policy is used at every value of θ . A *mark* d specifies a horizon, decision times, action spaces, a loss in $[0, 1]$, and a risk functional. The latter is either expectation under a declared prior or the supremum over a declared parameter set, taken after selecting one decoder. A minimax decoder does not know the unknown parameter. A mark may instead specify a fixed Bayesian prediction problem with an integrable squared loss; assertions involving total variation below apply only after a uniform loss bound is supplied. When a loss concerns a latent target, the marked experiment includes that target, and a simulation must preserve its joint law with the observations. Equality of report laws alone does not suffice for such a mark.

An encoder Q is an initialized causal transducer

$$I_t \sim q_t(\cdot \mid I_{t-1}, Y_t, A_{t-1}), \quad I_t \in \{1, \dots, S\}. \quad (23.1)$$

After an update, a decoder sees only (t, I_t) and any explicitly declared current side input. It does not see the discarded observation, an output transcript, old actions, or a data-dependent random seed. All persistent decoder state belongs to I_t . Fresh randomization is allowed. A public seed, when allowed, is independent of the experiment; a data-dependent position or modification of that seed is charged. A terminal reconstruction kernel may have a continuous output even when the physical statistic has a finite alphabet.

We use the resource signature

$$b = (S, A, L, P, p, W, \tau; R). \quad (23.2)$$

Here S counts persistent states; A counts physical terminal labels; L bounds raw symbols read per acquisition; P bounds fixed description bits; p is numerical precision; W is transient work space; τ is update time; and R records the permitted independent randomization. A coordinate ∞ means that no bound is asserted. For a precision constraint, the order is the order on permitted implementations, not the numerical order on an accuracy tolerance. The *cardinality signature* constrains only S (and A when explicitly stated). In particular, exact real constants in a fixed program are free only under that signature. All theorems below state which signature they use.

For an encoder admissible at b , let $r_E(Q, d)$ be the infimal value of the declared risk functional over its allowed decoders. Let $\beta_E(d)$ be the infimal risk using the full observation history under the same decision timing. The encoder is selected before the mark; its decoder may depend on the mark. For a set $\mathcal{D}$ of marks define

$$\mathcal{A}_E(b) = \{u \in [0, 1]^{\mathcal{D}} : \text{some admissible } Q \text{ has } r_E(Q, d) \leq u_d \ (d \in \mathcal{D})\}, \quad (23.3)$$

$$\Theta_E(b; \mathcal{D}) = \inf_Q \sup_{d \in \mathcal{D}} \{r_E(Q, d) - \beta_E(d)\}. \quad (23.4)$$

Infima need not be attained. We use the displayed attainable sets, not their product-topology closure: for an infinite task family, taking that closure before a uniform task supremum can change the value. The common encoder and the order of quantifiers in (23.4) are part of the invariant. The collection of attainable sets, baselines, and resource signatures is the *marked decision spectrum*, denoted $\Theta(E)$. This is the meaning of the symbol Θ in the present paper. It is a profile, not an additional physical parameter. Unbounded squared marks are treated separately without restricting the vectors to $[0, 1]$.

Example 23.1 (The encoder cannot follow the task). Let $X = (X_1, X_2)$ consist of two independent fair bits. A terminal statistic has two values. The two marks ask, respectively, for X_1 and X_2 , with zero-one loss. Each mark separately admits risk zero, but one common encoder has worst risk at least $1/4$. Indeed, for each fixed independent coding seed and each encoder label, the two optimal answers form a word in $\{0, 1\}^2$. Two reconstruction words have average Hamming distance at least $1/2$ from the four equally likely input words: two words may be exact, and each remaining word is at distance at least one. Randomization does not lower the average of the two risks. Hence their maximum is at least $1/4$. Conversely, with a public fair coin, reveal one of the two coordinates and guess the other. Thus the value is $1/4$ when that coin is permitted, whereas interchanging $\inf_Q$ and the task supremum gives zero. No such interchange is made in this paper.

23.2. Typed simulations and invariance. A certified simulation $S : E \Rightarrow F$ consists of initialized causal kernels, a monotone resource transformer ρ_S , and an error bound $\epsilon_S(T)$. The kernels do not know θ or the latent variables. For every target-side admissible policy and every parameter, the simulated marked transcript through horizon T has total variation distance at most $\epsilon_S(T)$ from its law in F . Moreover, composing the simulation with any target encoder of signature b produces a source encoder of signature at most $\rho_S(b)$. These are uniform requirements on one simulator, not permission to choose a different simulator for each task or policy. When marks have different horizons, use the corresponding error for each coordinate; a common ϵ below denotes their supremum.

Two certified simulators represent the same morphism when they have the same transformer and error certificate and induce the same marked execution laws under every admissible continuation. Internal register names are immaterial. We use standard Borel kernels modulo this execution equivalence. The exact morphisms have error zero. A K -state cardinality simulator has the transformer $S \mapsto KS$; its own output is not a second free memory register.

Theorem 23.2 (Spectrum transport and resource equivalence). *For bounded marks and signatures as in (23.2), certified simulations have identities and associative composition up to execution equivalence. If $S : E \Rightarrow F$ and $T : F \Rightarrow G$, the composite has transformer $\rho_S \circ \rho_T$ and error at most $\min\{1, \epsilon_S + \epsilon_T\}$. For every target-admissible Q there is a source-admissible Q' with*

$$r_E(Q', d) \leq r_F(Q, d) + \epsilon_S, \quad \beta_E(d) \leq \beta_F(d) + \epsilon_S. \quad (23.5)$$

If a reverse simulation $R : F \Rightarrow E$ is also supplied, then

$$\Theta_E(\rho_S(b); \mathcal{D}) \leq \Theta_F(b; \mathcal{D}) + \epsilon_S + \epsilon_R. \quad (23.6)$$

In particular, mutually exact finite-state simulations give equal ideal baselines and comparable spectra under the two budget dilations. Whenever a positive power exponent of either cardinality profile exists, the other has the same exponent. These assertions also hold coordinatewise for the attainable risk sets, before taking the task supremum.

Proof. Compose the kernels on the product of the two internal states. Fix one policy for G , regard its composition with T as a policy and continuation for F , and apply the certificate for S . Data processing followed by the certificate for T gives the sum of the two errors. The composition of transformers is forced by the same execution. Associativity follows from iterated kernel integration and the canonical reassociation of the triple product of registers; execution equivalence is preserved under composition because its quantifier includes all continuations. The identity kernel, identity transformer, and zero error give identities. This establishes a category for exact simulations and the corresponding error calculus for approximate ones; it is not an assertion of a new general randomization theorem.

Run Q on the simulated reports and use its decoder. Every loss in $[0, 1]$ changes in expectation by at most the variation error. Taking infima over decoders proves the first inequality in (23.5); an arbitrarily good full-history policy proves the second. The reverse simulator implies $\beta_F(d) - \beta_E(d) \leq \epsilon_R$. Subtract the baselines, take the common task supremum, and only then take the infimum over Q . This proves (23.6). No compactness or attainment of an optimal encoder is used.

For exact cardinality equivalence with factors K_1, K_2 , the inequalities are $\Theta_E(K_1 M) \leq \Theta_F(M)$ and $\Theta_F(K_2 M) \leq \Theta_E(M)$. Monotonicity in M and $\log(K_i M)/\log M \rightarrow 1$ sandwich any existing limit of $-\log \Theta(M)/\log M$. The same reasoning applies to a single bounded squared prediction mark; multiplication by its loss bound changes no positive power exponent. $\square$

The theorem is an invariance statement for a *typed* equivalence. Ordinary statistical equivalence without a resource certificate need not preserve a causal implementation budget. Conversely, retaining every emitted label in an external transcript changes the signature and thus changes the spectrum. Products of independent observations and products of memories are different operations: the former concern the experiment, the latter the implementation.

Proposition 23.3 (Forgetting causal and resource structure). *Fix a finite horizon and restrict to uncontrolled instruments and parameter-decision marks. Sending an instrument to its full transcript experiment, and an exact causal simulator to its transcript kernel, defines a functor to parameterized statistical experiments with parameter-independent kernels modulo equality on every parameter law. It preserves identities and composition. It forgets the resource certificate; the resulting statistical equivalence alone does not supply an equality of causal-register spectra.*

Proof. Iterating a simulator's initialized kernels at a fixed input transcript gives a parameter-independent Markov kernel from source transcripts to target transcripts. Iterated integration identifies the transcript kernel of a composite with the composite of these kernels. Execution equivalence implies equality on every source parameter law, so this assignment is well defined on the stated classes. The identity gives the identity transcript kernel. A general inverse transcript kernel need not be causal, and a causal inverse may store a nonfinite state; neither datum is reconstructed by the forgetful operation. $\square$

23.3. The terminal decision-complete slice. In this subsection $E = \{P_\theta : \theta \in K\}$ has standard Borel observation space X , K is compact metric, and $\theta \mapsto P_\theta$ is continuous in total variation. A terminal encoder is any Markov kernel $Q : X \rightarrow [S]$. Only its physical alphabet $A \leq S$ is constrained; transient computation and the reconstruction kernel are unrestricted. Let $\mathcal{D}_{\text{all}}$ consist of all finitely supported priors on K and all finite-action losses in $[0, 1]$. For these families we use $\delta(F, E) = \inf_R \sup_{\theta \in K} \|RF_\theta - P_\theta\|_{\text{TV}}$, where R is a parameter-independent reconstruction kernel and $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$.

Theorem 23.4 (Decision completeness at a fixed common encoder). *For every such Q ,*

$$\sup_{d \in \mathcal{D}_{\text{all}}} \{r_E(Q, d) - \beta_E(d)\} = \delta(QE, E). \quad (23.7)$$

Consequently

$$\Theta_E(S; \mathcal{D}_{\text{all}}) = \inf_{Q: X \rightarrow [S]} \delta(QE, E). \quad (23.8)$$

This is the Blackwell–Le Cam randomization criterion in a compact setting, with the common finite encoder kept outside the decision supremum. In particular, no new priority is claimed for the unconstrained criterion [1, 47, 46].

Proof. A reconstruction R with variation error ϵ transfers each bounded decision rule, proving that the left side is at most the right side. First suppose that both K and X are finite. Put $F = QE$. Finite-dimensional convex minimax applied to the row-stochastic matrices R and the total-variation dual gives

$$\delta(F, E) = \sup_{\pi, h} \left[\sum_{\theta, x} \pi_{\theta} h_{\theta x} P_{\theta}(x) - \sup_R \sum_{\theta, i, x} \pi_{\theta} F_{\theta}(i) R(i, x) h_{\theta x} \right],$$

where π is a prior and $0 \leq h_{\theta x} \leq 1$. To justify convexity of the dual domain, write its variables as $u_{\theta x} = \pi_{\theta} h_{\theta x}$ with $0 \leq u_{\theta x} \leq \pi_{\theta}$ and $\sum \pi_{\theta} = 1$. Regard h as a reward with action set X . The first term is the reward of the identity decision in E , bounded above by its optimal reward; the second is the optimal reward in F . Thus every dual expression is at most one of the optimal reward differences. Changing reward to loss proves equality.

Here is the approximation needed for general X, K . A dense sequence (θ_j) and positive weights with sum one give $m = \sum_j w_j P_{\theta_j}$ dominating every P_{θ} ; domination follows by total-variation continuity. The set of densities $\{dP_{\theta}/dm : \theta \in K\}$ is compact in $L^1(m)$. Given $\eta > 0$, choose a finite L^1 net and approximate its elements by simple functions on a common finite partition $\mathcal{C}$ of X . Conditional expectation onto $\mathcal{C}$ is an L^1 contraction, so the cell reconstruction $R_{\mathcal{C}}(\cdot | C) = m(\cdot | C)$ satisfies

$$\sup_{\theta} \|R_{\mathcal{C}} P_{\theta} - P_{\theta}\|_{\text{TV}} \leq \eta$$

after choosing the net and approximations sufficiently accurately. Zero-mass cells are irrelevant. Hence $\delta(F, E) \leq \delta(F, \mathcal{C}E) + \eta$. Now choose a finite parameter net on which both F and $\mathcal{C}E$ vary by at most η in total variation. The supremum reconstruction error on K differs from that on the net by at most 2η , uniformly in the reconstruction. Apply the finite result to this net. Its priors are allowed priors on K , and observing X is at least as informative as observing $\mathcal{C}(X)$, so the resulting decision differences for E dominate those for $\mathcal{C}E$. It follows that the left side of (23.7) is at least $\delta(F, E) - 3\eta$. Let $\eta \downarrow 0$. Equation (23.8) now follows by taking the infimum over Q , with no exchange of quantifiers. $\square$

APPENDIX 24. STATE-COSTED COMPARISON AND COMPLETE-STATE FUSION

An experiment is not identified with an implementation of its observation channel. Two equivalent experiments can expose the same information at different times. The cost of carrying that information between those times is the subject of this section. All losses in this section take values in $[0, 1]$; a bound L multiplies every error term by L .

24.1. The information interface and priced randomness. Fix a finite horizon T , finite control and terminal decision alphabets, a parameter set, a common marked target W , and a family J of decision rows. A row specifies a parameter and a task. The task is available to its controller and terminal decision rule, but not to the common encoder. The parameter is unavailable to all three. The timing is the timing of Section 23: a report is consumed once, and only the

retained register is passed to the next checkpoint. Any current side observation supplied afresh by the instrument is part of the interface and is specified separately. It is not silently included in the register.

Write $S = (S_0, \dots, S_T)$ for the private register bounds. A *public seed of size q* is an independent random variable with at most q values, retained and readable by the encoder and the controllers. Its law is chosen as part of the design. Fresh private randomization remains available. The notation $q = \infty$ permits one independent uniform random variable on $[0, 1]$, with no bound on its information content. A seed is never allowed to depend on the source or parameter. Let

$$\mathcal{R}_E(S, q) = \{(r_j(\pi))_{j \in J} : \pi \text{ is an admissible common-encoder design on } E\}.$$

For continuous risk functions on compact J , put

$$\mathcal{U}_E(S, q) = \mathcal{R}_E(S, q) + C(J)_+, \quad h_E(S, q; w) = \inf_{r \in \mathcal{R}_E(S, q)} \int_J r_j w(dj), \quad w \in \mathcal{P}(J).$$

The lower support value h does not assert convexity of the private risk set. In particular, $q = 1$ is not replaced by $q = \infty$ in this notation.

Definition 24.1 (Primitive marked simulator). A simulator $\Gamma : E \rightarrow F$ consumes the reports of E and produces the reports of F with the same timing and control interface. At checkpoint t it retains $J_t \in \{1, \dots, K_t\}$, and it uses a public seed with at most r values. Its update and report kernels depend only on its old state, the current input, the currently supplied side observations, and the issued control. They do not depend on the task or parameter. The simulator has error ε when, for every parameter and every admissible randomized feedback strategy at the output interface, the total variation distance between the simulated and true joint laws of the output transcript and the *actual* mark W is at most ε . The same requirement applies after adjoining independent randomization used by the output strategy. The simulator neither observes an unavailable mark nor replaces it by an independent draw.

This definition is a constraint on a causal channel, not a claim that a channel exists whenever certain decision inequalities hold. It is often verified by a parameter-uniform coupling or by factoring the actual marked law. The abstract simulation hypothesis of Theorem 23.2 is thereby supplied with an explicit machine and a charged state space.

Theorem 24.2 (Resource-costed comparison). *Suppose $\Gamma : E \rightarrow F$ is a primitive simulator of cost (K, r) and error ε . Then, for every target width S and seed bound q , every target design has a source implementation retaining only reachable pairs (J_t, I_t) . In particular its width is at most $K_t S_t$ and its public seed has at most $r q$ values. Its risk differs from the target risk by at most ε in every row. Consequently*

$$\mathcal{U}_F(S, q) + \varepsilon \mathbf{1} \subseteq \mathcal{U}_E(KS, rq), \quad (24.1)$$

$$h_E(KS, rq; w) \leq h_F(S, q; w) + \varepsilon. \quad (24.2)$$

Products involving ∞ have their natural meaning. Two composable simulators of costs (K, r) and (L, s) and errors ε, η have a composite of cost (KL, rs) and error at most $\min(1, \varepsilon + \eta)$. Identity simulators have cost $(1, 1)$ and error zero.

Proof. Keep (J_{t-1}, I_{t-1}) and the two independent seed indices. The target controller uses I_{t-1} to issue the next control. The simulator consumes the source input, draws its synthetic report and new internal state, and passes that report to the target encoder. The encoder draws I_t and then discards the report. This is one transition on the finite product state space; no earlier input is retained. The common encoder remains common because the simulator does not use the task and the only task information entering its transition is the legitimately issued control. Unreachable

pairs may be removed, which proves the stated sharper implementation bound as well as the product bound.

Definition 24.1 applies to this feedback strategy. Applying a bounded terminal loss, or a bounded measurable loss of the specified marked transcript, changes its expectation by at most total variation. The same simulated design works for all rows; it is not chosen after the row is known. This proves the risk assertion. If r_E and r_F are the resulting risks, then $r_F + \varepsilon \mathbf{1} - r_E \geq 0$, which gives (24.1); integration and an infimum give (24.2).

For composition retain both simulator states and both seed indices. Compare the composite first with the second simulator fed by the exact intermediate experiment. Data processing does not increase the first error. The second comparison costs at most η . This argument is valid under the feedback of the final consumer, since that consumer and the second simulator together are an admissible feedback strategy at the intermediate interface. The triangle inequality proves the error bound. $\square$

Remark 24.3 (Local error certificates). If the two conditional transition laws can be coupled at step t with failure probability at most e_t , conditional on every previously matched history, and the terminal conditional mark can be coupled with failure probability at most e_W , then the simulator error is at most $1 - (1 - e_W) \prod_t (1 - e_t)$. Indeed the probability that all couplings succeed is at least the product. A bound only for uncontrolled one-step marginals is not such a certificate: the conditional and feedback quantifiers are necessary.

24.2. Deficiency frontiers and spectra. Define $\delta_{K,r}(E, F)$ as the infimum of the errors of the simulators in Definition 24.1 with the indicated cost bounds. Set the value to $+\infty$ when there is no such simulator. No attainment is presupposed.

Corollary 24.4 (Composition and decision witnesses). *For compatible interfaces,*

$$\delta_{KL,rs}(E, G) \leq \delta_{K,r}(E, F) + \delta_{L,s}(F, G).$$

Moreover each choice of S, q, w gives the lower bound

$$\delta_{K,r}(E, F) \geq [h_E(KS, rq; w) - h_F(S, q; w)]_+.$$

If the two upper risk sets under consideration are closed and convex, (24.1) is equivalent to (24.2) for every $w \in \mathcal{P}(J)$. This last equivalence concerns risk sets, and is not an inverse theorem constructing a simulator.

Proof. Use simulators within an arbitrary positive tolerance of each finite infimum, apply Theorem 24.2, and let the tolerance decrease to zero. For the risk-set assertion, separate a point outside the closed convex upper set from that set. Any separating continuous linear functional must be nonnegative on $C(J)_+$, since the set is upper. It is therefore integration against a finite positive measure. Normalize its mass to one. The resulting separation contradicts the asserted lower support inequalities. The other implication is immediate. $\square$

For a fixed bounded loss family, let $b_{E,j}$ be the full-history Bayes risk with the same observation timing and parameter-blind decision rules, and define the centered minimax spectrum

$$\Theta_E(S, q) = \inf_{r \in \mathcal{R}_E(S, q)} \sup_{j \in J} (r_j - b_{E,j}).$$

When the unrestricted baseline is defined using a prior rather than a single parameter row, the same argument applies to that prior throughout. If exact full-history simulation is possible in both directions, these baselines agree. More generally, forward and reverse simulation errors ε, η imply $b_{F,j} - b_{E,j} \leq \eta$, and hence

$$\Theta_E(KS, rq) \leq \Theta_F(S, q) + \varepsilon + \eta. \quad (24.3)$$

Here the baseline inequality follows by simulating any source full-history rule on the target and taking its infimum. Thus neither a parameter oracle nor a change of timing enters the centering. For a fixed horizon and two exact simulations with fixed finite costs, a scalar budget B controlling all S_t and the charged finite seed index is changed by at most constant factors. Whenever a positive spectrum has a power exponent, the inequalities in both directions preserve that exponent. A growing horizon with growing K_t is not covered by this constant-factor assertion.

Proposition 24.5 (Public-to-private conversion and finite seed cost). *A design with a q -valued public seed has an exactly equivalent privately randomized implementation of width qS_t : store the sampled index with the private state. Suppose, in addition, that a family of risks satisfies $|r_j(\pi) - r_{j'}(\pi)| \leq \omega(d(j, j'))$ for all private designs, where J is compact. For an ideal public design and any $h, \zeta > 0$, there is a public mixture of at most $N(J, h) + 1$ private designs whose uniform risk error is at most $\zeta + 2\omega(h)$. Charging this index gives private width at most $(N(J, h) + 1)S_t$.*

Proof. The exact conversion samples the seed before observing the source and makes its index part of every register. Since the controller reads that register, all the original kernels can be executed. This is not a claim that qS_t is always necessary.

Choose an h -net of $N = N(J, h)$ rows. At these rows the risk vector of the ideal design is an expectation of vectors in $[0, 1]^N$, and hence belongs to the closure of their convex hull. Approximate it within ζ by a finite convex combination. Caratheodory's theorem reduces this combination to at most $N + 1$ actual private designs without changing those N coordinates. Move from an arbitrary row to a net row in the target and the approximation; each move costs at most $\omega(h)$. This proves the claim without assuming that the private risk set itself is closed. $\square$

24.3. The marked quotient as a zero-overhead object. For this subsection use the finite marked instrument of Section 29. Let $c_t(h)$ be its marked continuation class, K_t the number of supported classes, and $Q(E)$ the class experiment of Theorem 29.5. It has class update $F_t(c, a, y)$ and the parameter-independent reverse report kernel $L_t(y \mid c, a, c')$ constructed there.

Definition 24.6 (Complete retained design). A design is complete when, at every checkpoint, there is a map ϕ_t such that $c_t(H_t) = \phi_t(U, I_t)$ almost surely for every supported preparation and open-loop control word. With a finite instrument the same maps can be chosen outside one seed-null set, simultaneously for all supported histories. On $Q(E)$ the corresponding condition is $C_t = \phi_t(U, I_t)$. Let $\mathcal{R}^c$ denote the risk body restricted to these designs.

Completeness is equivalent to preserving all the restart Brier risks of Theorem 29.3. It is not imposed on an arbitrary finite list of decision tasks.

Theorem 24.7 (Complete-state fusion). *For the finite marked instrument, every width vector and seed signature satisfy*

$$\mathcal{R}_E^c(S, q) = \mathcal{R}_{Q(E)}^c(S, q). \quad (24.4)$$

Both sides are empty if $S_t < K_t$ at any required checkpoint. For general, not necessarily complete, designs the two exact simulations give instead

$$\begin{aligned} \mathcal{U}_{Q(E)}(S, q) &\subseteq \mathcal{U}_E(KS, q), \\ \mathcal{U}_E(S, q) &\subseteq \mathcal{U}_{Q(E)}(KS, q). \end{aligned}$$

Thus exact marked equivalence is resource-neutral on complete retained designs, while on arbitrary task designs it has an explicitly charged state distortion.

Proof. Consider first a complete design on E to be run on $Q(E)$. At the start of a step recover the old class c from (U, I_{t-1}) . When the class instrument supplies c' , sample y from $L_t(\cdot \mid c, a, c')$ and perform the original encoder update with y . The factorization in Theorem 29.5 shows that this report has the correct conditional law, jointly with the actual future marked experiment. In

particular every positive-probability updated register symbol has class c' , by completeness of the original design. The new register therefore again recovers the new class. The sampled report and the old class need not be retained separately. Induction gives an implementation with precisely the original private labels and the original seed.

Conversely, start with a complete design on $Q(E)$ and an actual report y from E . Recover c from the old register, compute $c' = F_t(c, a, y)$, and apply the class encoder with that input. Its new register recovers c' . The class update and terminal marked channel are exactly those of the class experiment, so induction again proves equality of all marked risks. At the initial checkpoint use the stipulated initial class (or the same class construction after the initial report). Neither construction inserts an additional persistent state.

The lower bound is Theorem 29.3, applied conditionally on the independent seed; the finite supported-history set permits a common null set. Finally, without completeness one retains c separately. The forward class tracker and the reverse report transducer each need at most K_t states, and Theorem 24.2 gives the last two inclusions. $\square$

Example 24.8 (A sharp temporal state charge). Let X be uniform on $\{1, \dots, k\}$ and let $W = X$. Experiment E reports X at time one and a constant at time two; experiment F reports X at both times. These marked experiments simulate each other exactly. The forward simulator $E \rightarrow F$ stores X between reports; the reverse one is stateless. At widths $(S_1, S_2) = (s, k)$, the terminal error for identifying W is zero on F , whereas on E it is at least $1 - s/k$ when $s \leq k$, with equality attainable. Indeed, for a private message channel $Q(m | x)$, its maximal guessing success is

$$\frac{1}{k} \sum_m \max_x Q(m | x) \leq \frac{s}{k}.$$

Independent public randomization averages this bound and cannot improve it. Partitioning the k inputs into s nonempty cells attains it. The same argument applied to a simulator which must reproduce the second report shows that its exact intermediate state count is at least k . The marked continuation quotient has k states. A complete consumer already carries them, and Theorem 24.7 eliminates the additional factor for exactly that reason.

Theorem 24.2, the lower support characterization in Corollary 24.4, and Theorem 24.7 are the resource-aware comparison statement of this article. The compactness results below supply cases in which its dual risk invariant is attained; the task-completion results identify which exact state requirement a particular decision family imposes.

APPENDIX 25. CAUSAL TASK COMPLETION ON THE ACTUAL SUPPORT

A finite task family need not induce an equivalence relation on histories. Moreover, a report which is impossible after one history must not be confused with a report having a prescribed continuation there. We give a support-sensitive characterization which addresses both issues at once. The underlying device is the compatible-cover construction for partial machines. The point here is its exact relation to a common encoder, randomized resource signatures, and the zero face of the decision spectrum.

25.1. Supported trees and simultaneous decisions. Let $\mathcal{H}_t$, $0 \leq t \leq T$, be finite sets of observable histories. The initial history $h \in \mathcal{H}_0$ is consumed by the initial encoder. Controls $a \in \mathcal{A}_t$ and reports $y \in \mathcal{Y}_{t+1}$ have finite alphabets. For each h and a , a nonempty set of successors hay is specified in $\mathcal{H}_{t+1}$. These are precisely the positive-probability successors in a fixed reference preparation of the controlled instrument. The reference preparation charges every history under consideration. Zero-probability successors are absent; their probabilities are not changed. Controls in the restart tests are fixed open-loop words. This ensures that the encoder's private randomization

remains independent of the source conditional on the supplied history. Conditional marked laws, not a parameter oracle, define all the losses below.

At time t there is a finite task set D_t . For $d \in D_t$ let B_{td} be a finite decision alphabet and let $\ell_{td}(h, b) \in [0, 1]$ be the conditional loss, obtained from the actual marked continuation and any continuation protocol specified by the task. Put

$$m_{td}(h) = \min_b \ell_{td}(h, b), \quad A_{td}(h) = \operatorname{argmin}_b \ell_{td}(h, b).$$

Choose a preparation $\eta_t(h) > 0$ on every $\mathcal{H}_t$. The full-history restart benchmark is $\sum_h \eta_t(h) m_{td}(h)$. At every checkpoint the same source-consuming encoder is used for all tasks. Its initial kernel is $q_0(m | h)$, its updates are $q_{t+1}(m' | m, a, y)$, and its task readouts are $v_{td}(b | m)$. Time dependence is allowed and is not persistent memory. Write $Q_t(m | h)$ for the message distribution obtained by running these kernels along the supplied history. An independent public seed, if present, is an additional named input to these kernels.

A set $C \subseteq \mathcal{H}_t$ is *task compatible* if

$$\bigcap_{h \in C} A_{td}(h) \neq \emptyset \quad \text{for every } d \in D_t. \quad (25.1)$$

Empty sets are permitted. Define its supported successor set by

$$\operatorname{Succ}_{a,y}(C) = \{hay \in \mathcal{H}_{t+1} : h \in C\}.$$

Definition 25.1 (Causally closed compatible cover). A width profile $S = (S_0, \dots, S_T)$ admits a causally closed compatible cover if there are compatible sets $C_{t,m}$, $1 \leq m \leq S_t$, which cover $\mathcal{H}_t$ and satisfy the following condition: for every $t < T$, m, a, y there is one index $\delta_t(m, a, y)$ such that

$$\operatorname{Succ}_{a,y}(C_{t,m}) \subseteq C_{t+1, \delta_t(m, a, y)}. \quad (25.2)$$

The sets may overlap. The single index in (25.2) must work for all histories in the cell for which this successor exists.

Theorem 25.2 (Exact causal task spectrum). *For a finite supported instrument and the finite restart tasks above, the following conditions on S are equivalent:*

- (1) *a deterministic common encoder with width profile at most S attains every full-history restart benchmark at every checkpoint;*
- (2) *a privately randomized common encoder does so, with or without an independent public seed of any prescribed positive size, including an ideal uniform seed;*
- (3) *S admits a causally closed compatible cover.*

The equivalence is independent of the strictly positive preparations η_t . In particular the exact feasible width profiles are not changed by private or independent public randomization. No positivity assumption is imposed on a next-report probability which is absent from the actual support.

Proof. For a fixed task the excess risk is

$$\sum_{h, m, b} \eta_t(h) Q_t(m | h) v_{td}(b | m) [\ell_{td}(h, b) - m_{td}(h)]. \quad (25.3)$$

It is a sum of nonnegative terms. If it vanishes for all tasks, then $C_{t,m} = \{h : Q_t(m | h) > 0\}$ is compatible: every decision in the nonempty support of $v_{td}(\cdot | m)$ belongs to every $A_{td}(h)$ with $h \in C_{t,m}$. The cells cover the supported histories.

For fixed t, m, a, y , select any m' with $q_{t+1}(m' | m, a, y) > 0$. If hay is a supported successor of some $h \in C_{t,m}$, the recursion for Q includes the positive product $Q_t(m | h) q_{t+1}(m' | m, a, y)$. Consequently $hay \in C_{t+1, m'}$, proving (25.2). There is no inference about a history hay outside the support.

Conversely, choose an initial containing cell for each $h \in \mathcal{H}_0$. Use the maps δ_t as deterministic updates. Induction on time shows that the retained cell always contains the actual history. For each cell and task choose an action from the intersection in (25.1). All terms in (25.3) are zero. Empty cells and empty successor sets have arbitrary decisions and updates. This constructs one machine attaining the entire width profile, rather than unrelated checkpoint statistics.

For a public seed U , condition the nonnegative expression (25.3) on U . There are finitely many checkpoints and tasks. Outside one seed-null set all their conditional excesses vanish. Because the restart preparation and the open-loop controls are independent of U , they still charge the same supported histories. Fixing one seed in this set gives a private machine to which the preceding argument applies. The converse may ignore the seed. Only positivity, not the numerical values, of η_t has been used. $\square$

The theorem is a zero-excess purification statement, not purification of the entire private risk body. Positive excess can benefit from private or public randomization, as Theorem 33.1 demonstrates. Nor does a family of arbitrary checkpoint-optimal covers automatically satisfy (25.2).

Example 25.3 (Support carries information without an extra register). Let the first report be a bit X , and let the next report again be X . The initial task has a constant optimal action and the terminal task is to predict X exactly. A one-cell initial cover and two terminal cells are causally closed: report y selects its own terminal cell. Thus $(S_0, S_1) = (1, 2)$ is feasible. Replacing the second report by a constant leaves the separate task widths equal to 1 and 2, but the profile $(1, 2)$ is no longer feasible. Its initial cell has both supported successors under the same report, and these require different decisions. The information timing, not merely a list of checkpoint cover numbers, determines the causal answer.

Example 25.4 (Overlapping cells and no coarsest partition). There are three initial histories u, v, w with constant initial tasks. History u has only successor report 0, with terminal target 0; v has only report 1, with terminal target 0; and w has only report 0, with terminal target 1. The initial cover $\{u, v\}, \{v, w\}$, together with terminal target cells, is causally closed. Both partitions $\{u, v\} \mid \{w\}$ and $\{u\} \mid \{v, w\}$ are feasible, whereas their common coarsening is not. Hence no single coarsest feasible partition describes this problem. A fictitious signature for an unsupported successor would impose a condition absent from the experiment.

25.2. A separating risk gap and prediction completion. Choose positive numbers β_{td} with $\sum_{t,d} \beta_{td} = 1$ and form the aggregate excess

$$G(\pi) = \sum_{t,d} \beta_{td} [r_{td}(\pi) - \sum_h \eta_t(h) m_{td}(h)].$$

This is a positive decision-row witness for the common risk body. Let

$$\alpha = \min_{t,d,h} \beta_{td} \eta_t(h) > 0.$$

If a strictly positive entry regret exists, let γ be the least such regret among $\ell_{td}(h, b) - m_{td}(h)$.

Corollary 25.5 (Zero face and a finite decision witness). *If the cover condition holds, the minimum of G and the minimax restart excess are zero. If it fails, a positive entry regret exists and, for all private or independently public-randomized machines of width S ,*

$$G(\pi) \geq \alpha\gamma, \quad \max_{t,d} [r_{td}(\pi) - \sum_h \eta_t(h) m_{td}(h)] \geq \alpha\gamma. \quad (25.4)$$

Thus causally closed covers characterize exactly the zero face of the common-encoder spectrum, with a strictly positive finite witness off that face.

Proof. Fix all entries of a deterministic time-indexed encoder and all task readouts. Infeasibility means that at least one supported history at one checkpoint has a nonoptimal decision for some task. Its contribution to G is at least $\alpha\gamma$. If every entry regret were zero, the one-cell cover at every time would work, so γ exists when needed.

A behavioral private design is a mixture, for the purpose of computing its risks, of deterministic tables: sample an independent value for each initial-encoder row, each time-indexed update row, and each time/task readout row according to the design's specified kernels. Along any fixed history each such time-indexed row is used at most once, so averaging these tables reproduces every expression (25.3). A public seed can first be conditioned upon and then integrated out. Every deterministic table has the lower bound, and hence so does their mixture. This use of a table mixture is a proof of an inequality; it does not grant its table index to an admissible private encoder. The second inequality follows because G is a convex combination of task excesses. $\square$

Corollary 25.6 (Brier tasks and universal marked completion). *Theorem 25.2 also holds for finitely many Brier tasks whose conditional probabilities are $p_{td}(h)$ and whose actions range over $[0, 1]$. A cell is then compatible precisely when each probability is constant on it. If the cover condition fails, let $\delta > 0$ be the least positive difference between probabilities for a common time and task at two histories. Then every randomized design satisfies*

$$G(\pi) \geq \alpha\delta^2/2.$$

For a finite marked instrument, completing the tests by a finite pair-separating family of all marked continuation-event predictions recovers the universal marked quotient and its least width profile. With strictly positive report probabilities this specializes to the signature construction in Theorem 30.4.

Proof. The conditional Brier excess is $(b - p_{td}(h))^2$. Zero excess forces the readout distribution to be concentrated at $p_{td}(h)$ on each message support. The support and transition proof is otherwise unchanged. For a fixed deterministic encoder, if all its cells had constant probabilities, they would form a causally closed compatible cover. In the infeasible case some cell contains two histories with probabilities p, p' separated by at least δ for one task. For any readout b , their weighted contribution is at least $\alpha[(b - p)^2 + (b - p')^2] \geq \alpha\delta^2/2$. Integration over randomized readouts and the finite encoder-table mixture proves the bound. Finally, equality of all marked continuation-event probabilities is precisely equality of the continuation law. Its supported class transitions are well defined by Lemma 29.2; different classes cannot share a cell. The marked quotient tracker attains these class counts. A finite pair-separating subfamily suffices because there are finitely many classes. $\square$

25.3. Coherent exact machines on an unbounded time interval.

Corollary 25.7 (Finite-prefix criterion). *Let the supported tree have infinitely many levels but finitely many histories, controls, reports, and finite-action tasks at each level. Fix finite widths S_t in advance. One deterministic infinite machine attains all restart benchmarks if and only if every finite prefix admits a causally closed compatible cover with those widths. Independent public or private randomization does not change this criterion.*

Proof. The deterministic initial table and every time-indexed update and readout table take values in finite sets. Their countable product is compact. Exactness through time N defines a closed cylinder set of tables. These sets are nested and are nonempty by Theorem 25.2 and arbitrary extension of a finite table. Compactness gives a point in their intersection. The reverse implication is restriction. For a randomized infinite machine, restriction to each prefix gives the covers by the finite theorem. The argument is an existence theorem; it asserts neither a finite stopping rule for an infeasible infinite problem nor an effective bound on the prefix which witnesses infeasibility. $\square$

APPENDIX 26. COMPACT DUALITY FOR DOMINATED CAUSAL EXPERIMENTS

We begin with a nonfinite experiment. The observation spaces in this section need not be locally compact, and the encoder kernels need not be continuous in the observation. The hypothesis used in place of such regularity is domination of the complete marked observation law.

26.1. The experiment and its information pattern. Fix a horizon T , finite control sets A_t , nonempty finite registers $I_t = [S_t]$, and a finite task set D . Task d has a finite decision set B_d . Let Y_t and W be standard Borel spaces, with probability reference measures μ_t and ν , and write $\mu = \bigotimes_{t=1}^T \mu_t$. The mark W is the actual terminal quantity on the source probability space; it is not a reconstructed sample with the same marginal distribution.

A causal experiment is specified by a consistent family of laws of $(Y_1, \dots, Y_T, W)$ under open-loop action words $a = (a_0, \dots, a_{T-1})$, indexed by a parameter $\theta \in \Theta$. Consistency means that the marginal law of $Y_{1:t}$ is independent of $a_t, \dots, a_{T-1}$, and that feedback laws are obtained by inserting their conditional action probabilities in the open-loop laws. Suppose that these laws have densities

$$p_{\theta,a}(y, w) \quad \text{with respect to } \mu(dy)\nu(dw). \quad (26.1)$$

This hypothesis is satisfied by dominated hidden Markov instruments, including continuous observation models. It excludes, for example, an unnoised repeated real observation supported on a diagonal of product Lebesgue measure. We do not replace domination by an unproved closure claim for arbitrary strategic measures.

The encoder is a single family

$$q_0 \in \mathcal{P}(I_0), \quad q_t(i' \mid i, a, y) \in \mathcal{P}(I_{t+1}), \quad 0 \leq t < T,$$

where $y \in Y_{t+1}$. It is common to every task and every parameter. The task controller and terminal rule are $g_{td}(a \mid i)$ and $v_d(b \mid i)$; neither depends on θ . The task is known to the controller, not supplied as an additional argument to q_t . The current control and observation are transient inputs. Between updates, the only private retained state is $i \in I_t$. Time and the prescribed tables are part of the fixed program. All private draws are fresh and conditionally independent given their arguments. A retained independent public seed, when allowed, is a separate resource. These conventions rule out both a hidden controller transcript and an uncharged retained random tape.

Let $J = D \times \Theta$ be compact metrizable, with D discrete and Θ compact metrizable. More generally the argument works for a compact row set with a continuous map into the finite task set. For $j = (d, \theta)$ and $b \in B_d$, let $0 \leq \ell_j(w, b) \leq 1$ be measurable. Our regularity condition is

$$f_{j,a,b}(y) := \int \ell_j(w, b) p_{\theta,a}(y, w) \nu(dw), \quad j \mapsto f_{j,a,b} \text{ continuous in } L^1(\mu) \quad (26.2)$$

on each task component, for every a, b . A continuous baseline $b_0 \in C(J)$ is fixed. A full-history minimax baseline for task d is constant on its parameter rows, and is one permitted choice; a parameterwise oracle is a different baseline and is not substituted for it.

Write K_S for the space of all such private designs $k = (q_0, q, g, v)$. The risk of a design is

$$R_j(k) = \sum_{i_0, \dots, i_T} \sum_{a,b} q_0(i_0) v_d(b \mid i_T) \prod_{t=0}^{T-1} g_{td}(a_t \mid i_t) \int f_{j,a,b}(y) \prod_{t=0}^{T-1} q_t(i_{t+1} \mid i_t, a_t, y_{t+1}) \mu(dy). \quad (26.3)$$

The sums over b are in B_d . This expression follows by first fixing the control and register paths in the causal law. In particular it does not optimize the parameter rows separately.

26.2. The topology and an executable realization. Each encoder row is a simplex-valued element of a finite product of $L^\infty(\mu_{t+1})$, endowed with its weak star topology. Positivity and the constraint that its coordinates sum to one are weak star closed: test them against nonnegative L^1 functions. Banach–Alaoglu therefore makes the row space compact. Since the probability space is

standard Borel, L^1 is separable, and this compact space is metrizable. Give the finite controller, initialization and decoder simplices their usual topologies, and $K_{\mathbf{S}}$ the product topology.

Lemma 26.1 (Tensor testing). *Suppose $u_{tn}, u_t \in L^\infty(\mu_t)$ have absolute value at most one and $u_{tn} \rightarrow u_t$ weak star, for each t . Then*

$$\int F(y) \prod_t u_{tn}(y_t) \mu(dy) \longrightarrow \int F(y) \prod_t u_t(y_t) \mu(dy)$$

for every $F \in L^1(\mu)$. Convergence is uniform over every compact subset of $L^1(\mu)$.

Proof. For $F(y) = \prod_t F_t(y_t)$ with $F_t \in L^1(\mu_t)$ the integrals factor, and each factor converges by the definition of the topology. Finite sums of these tensors are dense in $L^1(\mu)$: rectangular simple functions generate the product sigma field, and a monotone class argument followed by truncation gives the density statement. An L^1 error of ϵ in the test function changes either integral by at most ϵ . This proves convergence for one F . A finite ϵ -net of a compact family proves the uniform assertion. $\square$

Lemma 26.2 (Jointly measurable representatives). *The encoder row space has a choice of representatives $(q, y) \mapsto \hat{q}(y)$ that is jointly Borel, is simplex valued at every y , and represents each equivalence class q almost everywhere. Consequently a Borel probability on $K_{\mathbf{S}}$ can be sampled as one independent public seed and implemented by measurable causal update rules.*

Proof. Choose increasing finite measurable partitions generating the report sigma field modulo completion. The coordinates of $\mathbb{E}[q \mid \mathcal{F}_n](y)$ are finite sums of functions of the form

$$\mathbf{1}_C(y) \frac{\int_C q d\mu_t}{\mu_t(C)}, \quad \mu_t(C) > 0.$$

On each zero-measure partition cell use one fixed probability vector. Their coefficients are weak star continuous in q , so these functions are jointly Borel. For fixed q , the bounded martingale convergence theorem gives coordinatewise convergence to a representative of q almost everywhere. On the jointly Borel set where all coordinates converge use their limit, and elsewhere use one fixed probability vector. Every finite-stage vector is in the simplex, so is the limit. There are only finitely many rows. A sampled k is thus executable by inverse transforms with fresh independent uniform variables. Domination ensures that exceptional report sets have zero probability for every actual action path and parameter. Compact metrizability also ensures that every probability on $K_{\mathbf{S}}$ is the law of a Borel function of a uniform seed. The sampled design is retained publicly; this last retention is not free private memory. $\square$

Theorem 26.3 (Nonfinite common-encoder duality). *Under (26.1)–(26.2), the map $R : K_{\mathbf{S}} \rightarrow C(J)$ is continuous in the uniform norm. The attainable shared-seed risk set and its upper extension are*

$$\mathcal{A}_{\mathbf{S}} = \left\{ \int R(k) \lambda(dk) : \lambda \in \mathcal{P}(K_{\mathbf{S}}) \right\}, \quad \mathcal{U}_{\mathbf{S}} = \mathcal{A}_{\mathbf{S}} + C(J)_+.$$

The first is compact and convex and the second is closed and convex. Set

$$h_{\mathbf{S}}(w) = \min_{k \in K_{\mathbf{S}}} \int_J R_j(k) w(dj), \quad w \in \mathcal{P}(J).$$

Then $u \in \mathcal{U}_{\mathbf{S}}$ if and only if $\int u dw \geq h_{\mathbf{S}}(w)$ for every $w \in \mathcal{P}(J)$, and

$$\min_{\lambda \in \mathcal{P}(K_{\mathbf{S}})} \max_{j \in J} \left\{ \int R_j(k) \lambda(dk) - b_0(j) \right\} = \max_{w \in \mathcal{P}(J)} \left\{ h_{\mathbf{S}}(w) - \int b_0 dw \right\}. \quad (26.4)$$

Both extrema are attained. The private minimax optimum is also attained, but its attainable risk set is $R(K_{\mathbf{S}})$, not its convex hull.

Proof. Compactness of $K_{\mathbf{S}}$ was proved above. Apply Lemma 26.1 to each term of (26.3). The effect functions form finitely many compact subsets of L^1 , by (26.2). Thus convergence of designs gives uniform convergence of all row risks. This proves the first assertion without enumerating deterministic encoders.

For completeness, the barycenter map is continuous from weak convergence of λ to norm convergence in $C(J)$. Approximate the continuous map R uniformly on compact $K_{\mathbf{S}}$ by a finite-valued continuous partition-of-unity interpolation. Integrals of the finitely many continuous coefficients converge; the uniform approximation controls the remainder. The image of compact $\mathcal{P}(K_{\mathbf{S}})$ is therefore compact. Lemma 26.2 identifies this image with executable public mixtures, rather than only a formal relaxed set. Convexity follows by mixing the public seed.

If $u_n = a_n + c_n \rightarrow u$, $a_n \in \mathcal{A}_{\mathbf{S}}$ and $c_n \geq 0$, pass to a convergent subsequence of a_n . Its limit a satisfies $u - a \geq 0$, proving closedness of $\mathcal{U}_{\mathbf{S}}$. A separating continuous functional for a point outside this closed convex upper set must be nonnegative: otherwise translating an attainable risk in a nonnegative direction contradicts its lower separating bound. By the Riesz representation theorem this functional is integration against a finite positive measure, which may be normalized to a probability. Minimizing it over the upper set gives $h_{\mathbf{S}}$. This proves the support characterization.

The bilinear payoff

$$F(\lambda, w) = \int_J \int_{K_{\mathbf{S}}} R_j(k) \lambda(dk) w(dj) - \int_J b_0(j) w(dj)$$

is continuous separately on the two compact convex probability spaces. The classical convex minimax theorem gives equality of the two extrema. For fixed w , a linear functional on probabilities has minimum at a Dirac probability, which gives the inner minimum in (26.4). The left objective is continuous and the right objective, as an infimum of continuous functions of w , is upper semicontinuous. Compactness gives attainment. Finally the private objective is continuous on compact $K_{\mathbf{S}}$. No convexification was used for that assertion. $\square$

The use of policy topologies and static reduction belongs to stochastic team theory; see Saldi [55], especially the weak star construction and continuity/existence results, and Yüksel–Saldi [56], who explain why conditional independence is not generally closed under weak convergence. Here the product domination and finite action-path expansion supply the needed continuity, while the same encoder coordinates remain coupled across all tasks and unknown-parameter rows. Convex separation and minimax are classical tools, not additional novelty claims.

Corollary 26.4 (Effect formulation with arbitrary marks). *Theorem 26.3 still holds when there is no common reference measure for W , provided each open-loop report marginal is dominated by μ and the signed effect measures*

$$C \mapsto \int_{C \times W} \ell_j(w, b) P_{\theta, a}(dy, dw)$$

have $L^1(\mu)$ densities satisfying (26.2). The actual joint marked laws, not arbitrarily selected couplings, are used to define these effects.

Proof. Boundedness of the loss and report domination give absolute continuity of each effect measure. Disintegration of the actual marked law, or simply integration first in that measure, gives (26.3). All compactness and separation arguments in the theorem use only these effects. Report domination still makes the jointly measurable representatives executable. No common measure on W is needed in any of these steps. By contrast, the marked total-variation approximation theorem below retains its stronger joint-density hypothesis. $\square$

Corollary 26.5 (Countably many tasks). *Suppose that the task-row set is countable, every row has the dominated representation above, and every task has finitely many controller and decoder*

entries. No uniform continuity across rows is required. For a bounded baseline sequence, the shared optimum is attained and

$$\min_{\lambda} \sup_j \{\mathbb{E}_{\lambda} R_j - b_0(j)\} = \sup_{w \in \mathcal{P}_f(J)} \left\{ \min_k \sum_j w_j R_j(k) - \sum_j w_j b_0(j) \right\},$$

where $\mathcal{P}_f(J)$ denotes the finitely supported probabilities. The right side is a supremum, not an asserted maximum.

Proof. The countable product of the task-controller and decoder simplices with the common encoder space is compact metrizable. Each risk is continuous by the tensor lemma; the representative construction is unchanged. On $\mathcal{P}(K_{\mathbf{S}})$, the objective is a supremum of continuous functions, hence lower semicontinuous and attains its minimum. If every finite subset of row constraints $\mathbb{E} R_j - b_0(j) \leq r$ is feasible, compactness and the finite intersection property make all constraints feasible simultaneously. Finite-row convex minimax therefore gives the displayed identity, by taking r strictly above its right side and then decreasing r to the infimum. This argument makes no uniform finite-seed assertion. $\square$

Corollary 26.6 (The price of a finite mixture). *In Theorem 26.3, put*

$$\omega(\delta) = \sup_{k, d_J(j, j') \leq \delta} |R_j(k) - R_{j'}(k)|.$$

Then $\omega(\delta) \rightarrow 0$. If J has a δ -net of size N , every attainable shared risk can be approximated within $2\omega(\delta)$ in uniform norm by a mixture of at most $N + 1$ private designs. Such a mixture has a private implementation with registers $(N + 1)S_t$, including initialization. If its weights are approximated by a probability vector with denominator 2^p , an additional error at most $(N + 1)2^{-p}$ suffices.

Proof. The compact subset $R(K_{\mathbf{S}})$ of $C(J)$ is equicontinuous. Carathéodory's theorem applied to its evaluations on the net gives $N + 1$ designs whose mixture agrees there with the prescribed risk. Move an arbitrary row to the net; each of the two risks changes by at most $\omega(\delta)$. Store the chosen design index with the register label for the private implementation. Approximate weights by rounding down the first $L - 1$ coordinates and putting the remainder in the last, where $L \leq N + 1$. Their total variation distance is at most $L2^{-p}$, which bounds the change of every $[0, 1]$ -valued risk. This approximation can be sampled with p fair bits; exact non-dyadic weights do not have an asserted fixed worst-case fair-bit cost. $\square$

26.3. Finite observation approximations, uniformly in memory. Let $\mathcal{F}_{tn}$ be increasing finite partitions generating Y_t , and retain the actual mark W . The finite-report experiment E_n reports only the cell of each observation. Its reverse report channel samples independently from μ_t conditioned on that cell. Cells of zero reference measure are assigned arbitrary outputs and have zero probability in the experiment. Write

$$p_{\theta, a}^{(n)} = \mathbb{E}_{\mu \otimes \nu} [p_{\theta, a} \mid \bigotimes_t \mathcal{F}_{tn} \otimes \mathcal{B}(W)], \quad \delta_n = \frac{1}{2} \sum_a \sup_{\theta} \|p_{\theta, a} - p_{\theta, a}^{(n)}\|_1.$$

Theorem 26.7 (Width-uniform approximation). *If each family $\{p_{\theta, a} : \theta \in \Theta\}$ is compact in $L^1(\mu \otimes \nu)$, then $\delta_n \rightarrow 0$. For every width vector, private or shared randomness, and every task family with losses in $[0, 1]$, the raw minimax risks of E and E_n differ by at most δ_n . The same comparison of excess risks costs at most $2\delta_n$ when the full-history baselines are recomputed. The simulations neither add persistent states nor reveal the mark.*

Proof. Conditional expectations on the generating partitions converge in L^1 for each density. Their norms are at most one as operators on L^1 , so a finite-net argument gives uniform convergence on each compact family. There are finitely many action words.

Reporting the cell is an exact causal channel from E to E_n . After the reverse channel, the open-loop joint density of synthetic reports and the actual W is $p_{\theta,a}^{(n)}$. Under an arbitrary adaptive policy and encoder, expand the joint law by its action and register paths. The conditional rule multiplying each density is between zero and one. Summing the absolute differences over action words, and integrating over the register randomization, bounds the total variation of the marked output laws by δ_n . This bound is uniform over policies, tasks, seeds and widths. Composition of any code with either report channel needs only the original register: cell conversion or within-cell sampling is a current-report operation. Thus every code transfers with the stated risk error in the reverse direction, and with zero error in the forward direction. Taking minima and worst rows proves the raw comparison. The same argument without a width bound compares full-history baselines, and the triangle inequality gives the excess-risk bound. $\square$

APPENDIX 27. A SINGULAR OBSERVATION CLASS AND CONDITIONAL COMPACT DUALITY

Product domination is not invariant under repeatedly displaying a continuous observation. A physical instrument may display the same covariate at every checkpoint, without asking the private register to store it. We prove compact duality for this non-product observation class. The distinction between a repeatedly supplied observation and a privately remembered one is essential, as Example 24.8 shows.

27.1. Conditional innovation models. Let X take values in a standard Borel space $\mathbf{X}$ and be supplied *currently* to the encoder, controller, and terminal decision rule at each of their operations. The distribution of X may depend on the parameter; X is not a public random seed. The successive innovation reports belong to standard Borel spaces $\mathbf{Z}_t$. At every fixed x , all private registers, controls, and terminal action sets are finite. Let ν and μ_t be reference probabilities, and set $\mu = \bigotimes_{t=1}^T \mu_t$. As in Section 26, for each open-loop control word a , terminal action b , and row j , define the nonnegative loss effect $f_{j,a,b}(x, z)$. It is the density with respect to $\nu \otimes \mu$ of the measure obtained by integrating the bounded loss against the *actual* open-loop joint law of (X, Z, W) . We require

$$j \mapsto f_{j,a,b} \text{ continuous in } L^1(\nu \otimes \mu), \quad J \text{ compact metric.} \quad (27.1)$$

The finite collection of these effects includes the initial probabilities and all parameter-dependent weights. It need not factor in x, z , and there is no hypothesis of domination by a product measure on the full *displayed* report path $(X, (X, Z_1), \dots, (X, Z_T))$.

Let $\mathcal{K}_S$ be the compact metrizable space of private innovation policies at a fixed value of x , using the weak star simplex topology of Theorem 26.3. This includes initial-state, controller, and terminal decision simplices and the common encoder kernels. A private policy with the displayed covariate is a measurable section $x \mapsto k_x \in \mathcal{K}_S$. A public policy is a measurable $(x, u) \mapsto k_{x,u}$, where u is uniform and independent of the source. The jointly measurable representatives constructed in the dominated section turn these sections into actual kernels on (x, u, z_t) .

For a fixed x , let $R_j(x, k)$ be the finite sum over control, register, and terminal paths of their policy factors integrated against $f_{j,a,b}(x, \cdot)$. It is an unnormalized conditional contribution:

$$r_j(k) = \int_{\mathbf{X}} R_j(x, k_x) \nu(dx).$$

This notation avoids dividing by a possibly vanishing density of X .

Lemma 27.1 (Conditional risk integrands). *Each R_j represents an element of $\mathcal{B} = L^1(\nu; C(\mathcal{K}_S))$, and the map $j \mapsto R_j$ is continuous into $\mathcal{B}$. After choosing measurable versions, the integrands are measurable in x and continuous in k for almost every x (for each fixed integrand). Every measurable section of $\mathcal{K}_S$ has an executable common-encoder representative.*

Proof. For almost every x , the slices $f_{j,a,b}(x, \cdot)$ are integrable. The tensor-testing lemma in the dominated section implies continuity in k of each path integral. Its absolute value is bounded by the sum of the $L^1(\mu)$ norms of its finitely many sliced effects, multiplied by the finite number of state paths. This is an integrable envelope. For two rows, the same estimate applied to the difference gives

$$\|R_j - R_{j'}\|_{\mathcal{B}} \leq C_S \sum_{a,b} \|f_{j,a,b} - f_{j',a,b}\|_{L^1(\nu \otimes \mu)}.$$

Measurability first follows on the countable dense family of simple effects and policies and then by the displayed bound and tensor testing. Since $C(\mathcal{K}_S)$ is separable, this gives strong measurability in $\mathcal{B}$. Continuity follows from (27.1).

For executability, use the Borel representative of a weak star simplex coordinate furnished by successive conditional expectations on finite partitions, as in the dominated section. This representative is jointly Borel in the coordinate and the innovation. Composing it with the measurable map $x \mapsto k_x$, or $(x, u) \mapsto k_{x,u}$, gives measurable encoder kernels. The finite-dimensional policy coordinates require no additional choice. A section measurable for the completion of ν has a Borel version outside a ν -null set, since the target is standard Borel; the reference domination makes this replacement harmless for every parameter. Conversely, a measurable implemented kernel gives a measurable section by integration against a countable dense family of $L^1(\mu_t)$ tests. Thus this parametrization neither loses policies nor adds nonexecutable ones. $\square$

Lemma 27.2 (Compact conditional randomization). *The set $\mathcal{Y}$ of probability kernels $\Lambda_x(dk)$ from $(\mathbf{X}, \nu)$ to $\mathcal{K}_S$, modulo ν -null sets, is compact metrizable for integration against $\mathcal{B}$. Its elements are precisely the positive continuous linear functionals L on $\mathcal{B}$ satisfying*

$$L(g(x)\mathbf{1}(k)) = \int g \, d\nu \quad (g \in L^1(\nu)).$$

Every such kernel has a realization $k = F(x, U)$ with F measurable and U uniform independent of the source.

Proof. Positivity and normalization imply $|L(V)| \leq \int \|V(x, \cdot)\|_{\infty} \nu(dx)$, so all such functionals lie in the closed unit ball of $\mathcal{B}^*$. Their defining equalities and inequalities are weak star closed. Banach–Alaoglu therefore gives compactness; separability of $\mathcal{B}$ gives metrizability on this ball.

For completeness, the representation does not assume a density in the policy coordinate. For each $\varphi \in C(\mathcal{K}_S)$, the bounded functional $g \mapsto L(g\varphi)$ on $L^1(\nu)$ has a Radon–Nikodym representative $v_{\varphi} \in L^{\infty}(\nu)$. Choose these representatives on a countable dense rational vector lattice of continuous functions containing constants. Outside a single null set, they are linear, positive, normalized, and bounded by $\|\varphi\|_{\infty}$. At each remaining x extend by continuity to $C(\mathcal{K}_S)$. The Riesz representation theorem gives a unique probability Λ_x . The integrals of a countable determining family are measurable, hence $x \mapsto \Lambda_x$ is a measurable probability kernel. Simple $C(\mathcal{K}_S)$ -valued functions and L^1 approximation now prove $L(V) = \int V(x, k) \Lambda_x(dk) \nu(dx)$ for every $V \in \mathcal{B}$.

Finally a probability kernel to a standard Borel space is realized by a measurable function of its input and one independent uniform variable (the kernel randomization lemma; see [4]). One may also construct it by successively splitting $[0, 1]$ along refining countable partitions of the compact policy space. This uses an independent seed; it does not treat X as independent of the source. $\square$

Theorem 27.3 (Singular conditional common-encoder duality). *Under (27.1) and the displayed-covariate interface, the ideal-public attainable risk body is the compact convex subset of $C(J)$*

$$\mathcal{A}_S = \left\{ \left(\int R_j(x, k) \Lambda_x(dk) \nu(dx) \right)_{j \in J} : \Lambda \in \mathcal{Y} \right\}.$$

Every point is executed with the stated finite private register and one independent public seed. Its upper set is closed, and, for $w \in \mathcal{P}(J)$, its lower support value is

$$h_S(w) = \int_X \min_{k \in \mathcal{K}_S} \left\{ \int_J R_j(x, k) w(dj) \right\} \nu(dx). \quad (27.2)$$

In particular

$$\min_{r \in \mathcal{A}_S} \max_{j \in J} r_j = \max_{w \in \mathcal{P}(J)} h_S(w),$$

and both extrema are attained. For each fixed w the support value is attained by one measurable private section. Finite-seed approximation and its state charge are those of Proposition 24.5.

Proof. By Lemma 27.1, the image of J in $\mathcal{B}$ is compact. If $\Lambda^n \rightarrow \Lambda$ in the topology of Lemma 27.2, the risk integrals converge on each fixed integrand. Approximate the compact set of row integrands by a finite $\mathcal{B}$ -net and use the uniform norm-one bound on the functionals. Convergence is then uniform in j . Thus the risk map is continuous into $C(J)$, and its image is compact and convex. The kernel realization and Lemma 27.1 prove executability. A sum of a compact set and the closed positive cone is closed.

The Bochner integral $V_w = \int_J R_j w(dj)$ belongs to $\mathcal{B}$. It has a Caratheodory version $V_w(x, k)$. The measurable minimum theorem on a compact metric space gives a measurable minimizing section k_x ; more explicitly, its nonempty compact argmin correspondence is measurable, as can be tested by infima over a countable dense set of policies, and admits a measurable selector. Its minimum is integrable by the envelope bound. Any kernel averages values no smaller than that minimum, while the Dirac kernel at k_x attains it. This proves (27.2). The choice depends on x and w , not on the realized task or parameter; all task-dependent controls remain coupled to a single encoder inside k_x .

The payoff $(r, w) \mapsto \int r dw$ is continuous and affine on two compact convex sets. The compact minimax theorem therefore proves the identity and attainment; equivalently use separation of the compact upper risk body. The common modulus needed for finite seeds is supplied by the compact family of integrands and its $\mathcal{B}$ -continuity. $\square$

Example 27.4 (A genuinely non-product report law). Take a nonatomic X on $[0, 1]$, display it at times zero and one, and adjoin any finite-horizon innovation model satisfying (27.1). The full displayed law is concentrated on the diagonal $x_0 = x_1$. It is not dominated by any product of probability reference measures on the two display coordinates. Indeed a product measure charges the diagonal only at common atoms; there are at most countably many such atoms. Outside those atoms it assigns zero mass to the diagonal, whereas the nonatomic displayed law assigns mass one. The same conclusion holds for sigma-finite product references, by restriction to countably many finite-measure sets. Nevertheless Theorem 27.3 applies with a single reference for X .

Remark 27.5 (What has and has not been compactified). The theorem treats arbitrary measurable dependence on the repeated covariate, not a finite quantization of it. It does not assert weak closure of all strategic measures with arbitrary information structures. Nor does it assert compactness of the entire private risk body in this conditional model. The conditional Young kernels are exactly the ideal-public relaxation, and finite retained randomness is charged by Proposition 24.5. If the instrument ceases to display X , a policy that still accesses X is no longer admissible. These are information-interface distinctions, not exceptional null-set conventions.

APPENDIX 28. INFINITE-HORIZON RISK BODIES AND COHERENT COMPARISON

The resource profile and the comparison channel can be defined for an unbounded time interval without a single reference measure dominating all infinite trajectories. What is needed for the compactness argument below is a coherent family of dominated finite-prefix loss effects and a

uniform tail bound for the criterion. This is a different assertion from weak closedness of arbitrary filtered strategic measures.

28.1. Finite-prefix hypotheses and executable policies. Time is indexed by $t \in \mathbb{N}_0$. Fix finite private widths S_t , finite control and decision alphabets at each time, and a finite set of task labels available to controllers but not to the common encoder. Reports range over standard Borel spaces (Y_t, μ_t) . The parameter and task rows form a compact metric space J . The controlled instrument and its actual marks are specified by coherent stochastic kernels. A mark used in the stage- t loss is part of the actual prefix experiment at that checkpoint. No independently regenerated target is substituted for it.

For each N , assume the finite-prefix loss effects satisfy the hypotheses of Theorem 26.3: after integrating out the actual mark, each effect for a fixed control, register, and decision string is an $L^1(\mu_0 \otimes \cdots \otimes \mu_N)$ function, continuous in the row j , and the losses take values in $[0, 1]$. These hypotheses are local in N ; no uniform density bound in N is required. Effects for shorter prefixes agree with those obtained from the specified instrument. The policy information pattern is the finite-register pattern in Section 24.

Let K_t be the compact metrizable space of stage- t weak-star encoder rows and finite controller/readout simplexes. The initial stage has its corresponding initial-encoder rows. Put

$$K = \prod_{t \geq 0} K_t.$$

This notation fixes time-dependent policies, not stationary policies. For $\pi \in K$, denote the expected stage loss by $r_{j,t}(\pi)$. It depends only on the first $t + 1$ blocks and is a continuous map into $C(J)$ by the finite-prefix tensor-testing argument.

Choose $a_t \geq 0$ with $\sum_{t \geq 0} a_t = 1$, and define

$$r_j(\pi) = \sum_{t \geq 0} a_t r_{j,t}(\pi), \quad A_N = \sum_{t > N} a_t. \quad (28.1)$$

Geometric discounting is the case $a_t = (1 - \beta)\beta^t$, $0 < \beta < 1$. More generally all arguments apply to a row family with a common summable dominating weight sequence.

Lemma 28.1 (Coherent policy realization). *Every point of K has executable Borel representatives, and the representatives can be chosen jointly Borel in the policy and its report arguments. One independent uniform seed executes any Borel probability on K as a publicly randomized policy. For a fixed parameter and feedback design, changing the chosen representatives on the reference null sets changes none of its finite-prefix or infinite path laws.*

Proof. Apply Lemma 26.2 to each of the countably many encoder rows. Conditional averages on increasing finite report partitions and the Borel convergence set give jointly measurable simplex-valued representatives; on the exceptional set choose a fixed simplex point. For each prefix there are finitely many relevant rows. Product domination makes their reference-null exceptional sets null for every actual open-loop law; the finite control-string expansion gives the same fact under feedback. A countable union over prefixes remains null for the fixed parameter and design. The resulting Borel kernels define the infinite law by successive kernel extension. Equality on every prefix implies equality of the infinite path laws.

The space K is compact metrizable, hence standard Borel. A Borel map from a uniform variable to K realizes any probability on K . Evaluating the jointly Borel representatives at the sampled policy supplies all successive encoder rows. The seed is sampled independently of the source and is available as specified at each time. This is an ideal public resource; no finite storage bound is inferred from this realization. $\square$

Theorem 28.2 (Summable infinite-horizon common-encoder duality). *Under the finite-prefix hypotheses above, the following statements hold. The private risk body $\mathcal{R}(S, 1) = \{r(\pi) : \pi \in K\}$ is compact in $C(J)$. For each finite public seed bound q , the body $\mathcal{R}(S, q)$ is compact. The ideal public body is the compact convex set*

$$\mathcal{R}(S, \infty) = \left\{ \int_K r(\pi) \lambda(d\pi) : \lambda \in \mathcal{P}(K) \right\}. \quad (28.2)$$

For $b \in C(J)$ its minimax excess satisfies the attained duality

$$\min_{\lambda \in \mathcal{P}(K)} \max_{j \in J} \left[\int r_j d\lambda - b_j \right] = \max_{w \in \mathcal{P}(J)} \min_{\pi \in K} \int_J [r_j(\pi) - b_j] w(dj). \quad (28.3)$$

There is a saddle pair of mixed policies and row probabilities. The truncated and full risk bodies, at the same width and public seed signature, have Hausdorff distance at most A_N in $C(J)$, after truncation is identified with arbitrary continuation of its policy. Consequently their minimax excesses differ by at most A_N for a common baseline, and by at most $A_N + \|b - b^{(N)}\|_\infty$ for different baselines.

Proof. The countable product K is compact metrizable. Its finite-prefix risk maps are continuous into $C(J)$. Equation (28.1) gives the uniform bound

$$\left\| r(\pi) - \sum_{t=0}^N a_t r_{\cdot, t}(\pi) \right\|_\infty \leq A_N,$$

independently of π and of the widths. Thus $\pi \mapsto r(\pi)$ is continuous into $C(J)$, proving compactness of the private body. For finite q , the map from $\Delta_q \times K^q$ which takes weights and policies to their weighted risk is continuous and has exactly the q -seed body as its image.

For completeness, norm continuity of the barycentre map in (28.2) does not follow from scalar weak convergence alone. Cover K by finitely many sets on which r oscillates by at most ε in the $C(J)$ norm. A continuous partition of unity subordinate to the cover approximates r uniformly by a finite sum $\sum_i \psi_i(\pi) r(\pi_i)$. Its barycentre depends continuously in norm on λ for weak convergence of probabilities. The error is at most ε . Letting $\varepsilon \downarrow 0$ proves norm continuity, so the image of $\mathcal{P}(K)$ is compact and convex. Executability is Lemma 28.1.

The payoff obtained by integrating $r_j - b_j$ against λ and w is continuous and affine in each probability on the two compact convex spaces. The compact minimax theorem gives equality and attainment. Minimizing a linear functional over $\mathcal{P}(K)$ is the same as minimizing it over K , which gives (28.3). A primal minimizer and a dual maximizer form a saddle pair by their two defining inequalities. Equivalently the compact upper risk body is separated by positive row functionals, as in the finite-prefix proof.

Every infinite policy restricts to a prefix policy and every prefix policy has an arbitrary admissible continuation. The uniform tail bound therefore gives both Hausdorff inclusions for the private bodies and, after integrating, for every public signature. Taking suprema, infima, and subtracting the respective baselines proves the last assertion. $\square$

Corollary 28.3 (Finite seed cost independent of truncation time). *The family $\{r(\pi) : \pi \in K\}$ has a common modulus of continuity ω on J . If $N(J, h)$ is the size of a finite h -net of J , every ideal public risk can be approximated, in supremum norm, within $2\omega(h)$ by a mixture of at most $N(J, h) + 1$ private policies. Its mixture index can instead be stored privately by enlarging every register to at most $(N(J, h) + 1)S_t$. The number of policies does not grow with the number of time checkpoints at this fixed accuracy.*

Proof. A compact subset of $C(J)$ is uniformly equicontinuous: take a finite supremum-norm approximation and use the uniform continuity of its finitely many members. This supplies $\omega(h) \downarrow 0$. For a net of size n , project the private risk body into $\mathbb{R}^n$. Its convex hull is compact and

Carathéodory's theorem expresses any projected public barycentre using at most $n + 1$ projected private risks. The resulting mixture agrees exactly at the net points. The modulus for the original risk and for the finite mixture bounds their difference at an arbitrary row by $2\omega(h)$. To replace the public index by private storage, initialize the joint state (i, m_0) , keep i unchanged, and use its selected policy at all later times. The task readouts use the same index. This is the explicit product-state construction, and has no finite-horizon restriction. $\square$

Example 28.4 (No dominating measure for the complete path). Let the reports be independent Bernoulli variables with parameter $p \in [1/4, 3/4]$. With finite decision alphabets and bounded row-continuous stage losses, every finite prefix satisfies the effect hypotheses. Yet the full path laws have no common σ -finite dominating measure. Indeed the disjoint Borel sets $E_p = \{y : \lim_n n^{-1} \sum_{i < n} y_i = p\}$ have probability one under their respective laws. If a σ -finite measure dominated all these laws, it would give positive measure to uncountably many disjoint sets. Intersecting with its countable finite-measure cover and then with positive lower measure thresholds shows this is impossible. The convergence of the empirical mean here is the ordinary strong law, not a new ingredient. Theorem 28.2 therefore has genuinely infinite-path applications without assuming a dominating measure on the complete path.

28.2. Costed comparison at all times. An infinite simulator is *prefix coherent* if one set of causal kernels defines all its restrictions. At time t it has state cost K_t . A public simulator seed, if used, is explicitly part of the output interface: the comparison tests its joint law with the simulated transcript and actual marks against the target experiment supplied with an independent copy of that seed. The same applies to an independent consumer seed. This joint-law convention prevents a small marginal simulation error from hiding dependence revealed by a public seed. For deterministic or privately randomized stateless observation channels there is no additional public interface to test.

Example 28.5 (Why the simulator seed belongs to the interface). Let the source observe a fair bit W , let U be an independent public fair bit, and output $Y = W \oplus U$. As a law of (Y, W) this is exactly the uninformative experiment in which Y is independent of W . But a consumer reading U can use $Y \oplus U$ to recover W from the simulated interface, whereas that decision has error $1/2$ in the true uninformative experiment supplied with an independent U . The total variation distance of the two joint (Y, W, U) laws is $1/2$, not zero. Thus a zero marginal error cannot justify a zero error for this seed-aware composition. Testing the joint interface, as required here, excludes this ambiguity without forbidding public randomization.

Suppose the prefix- t error, uniform in parameter and output feedback strategy in this seed-aware sense, is at most $\varepsilon_t \leq 1$.

Theorem 28.6 (Coherent resource comparison). *An infinite target consumer with width S_t and seed bound q , composed with such a simulator of width K_t and seed bound r , has an executable source realization with widths $K_t S_t$ and seed bound rq . For the criterion (28.1), every row-risk difference is at most*

$$E_a = \sum_{t \geq 0} a_t \varepsilon_t. \quad (28.4)$$

The upper-risk inclusion and lower-support inequality of Theorem 24.2 hold with error E_a . Two prefix-coherent simulations compose with multiplied state and seed costs and prefix errors at most $\min\{1, \varepsilon_t + \varepsilon'_t\}$. For two-sided cost-one simulations all private and public risk bodies have Hausdorff distance at most E_a ; their support values differ by at most E_a . If the respective full-history baselines also differ by at most E_a , their centered minimax spectra differ by at most $2E_a$.

Proof. At each checkpoint retain the reachable simulator/consumer state pair. This single infinite machine has every finite-prefix product realization in Theorem 24.2. The joint seed convention

makes its feedback rule admissible for the stated prefix comparison. A stage loss is a $[0, 1]$ -valued measurable function of the prefix, the actual mark, and the terminal randomization. The total variation bound therefore bounds its expectation difference by ε_t . Sum through N and bound the two remaining weighted tails by their common summable majorant; letting $N \rightarrow \infty$ gives (28.4). The risk and support statements follow by the same explicit implementation, before any optimization over consumers. Composition is prefixwise contraction of total variation and its triangle inequality. Two-sided comparison gives both inclusions. Subtracting a baseline with the stated bound adds at most one further E_a . $\square$

Corollary 28.7 (Geometric error accumulation). *If a conditional coupling at each step fails with probability at most e , uniformly on previously matched prefixes, then one may take $\varepsilon_t \leq 1 - (1 - e)^{t+1}$. Under geometric discounting,*

$$E_a \leq \frac{e}{1 - \beta + \beta e}.$$

This statement includes any stage mark in the coupled prefix. A bound only on its marginal, with no coupling to the actual reports, does not meet the hypothesis.

Proof. The conditional success bound iterates to probability at least $(1 - e)^{t+1}$ of matching through time t . Summing the resulting bound against $(1 - \beta)\beta^t$ is a geometric series. This is stronger than using $\min\{1, (t + 1)e\}$ when a conditional coupling is available. $\square$

The infinite-horizon extension uses compact policy blocks, a uniform criterion tail, and classical convex separation. It neither asserts compactness for arbitrary average-cost problems nor charges an infinite public design index as a finite register. The next section on physical operator approximation will supply a different, unconditional prefix error bound for which Theorem 28.6, but not the conditional estimate in Corollary 28.7, applies.

APPENDIX 29. THE MINIMAL MARKED CONTINUATION QUOTIENT

We now address exact state reduction. A statistic sufficient for the parameter need not be sufficient for a latent decision target. The correct terminal datum is the joint conditional law of the parameter and the designated mark. We give both an intrinsic operational characterization and a finite construction.

Consider a finite hidden instrument with initial law ν_θ , transition/report probabilities $K_{\theta,t}(z', y \mid z, a)$, and a terminal mark channel $L_\theta(w \mid z_T)$ on a finite set W . Here K uses the new state z' and the report jointly; no old-state observation channel is silently substituted. A finite collection of path targets can be included by adjoining their necessary finite memory to z . This augmentation, when used, is part of the input size. Fix a reference prior $\pi(\theta) > 0$. For every supported controlled history $h = (a_0, y_1, \dots, a_{t-1}, y_t)$, let $\alpha_t(h)$ be the posterior row on (Θ, Z_t) under the reference prior and the indicated past actions. Control probabilities cancel in Bayes' formula; histories of zero reference probability are assigned an arbitrary existing label. No extra retained label is needed for events that have probability zero under every parameter.

Definition 29.1 (Continuation experiment). For each future open-loop action word $v = a_t \cdots a_{T-1}$, let $\mathcal{C}_t(h; v)$ be the conditional law of $(\Theta, Y_{t+1:T}, W)$ given h , with future actions v . Two supported histories are equivalent, written $h \equiv_t h'$, if $\mathcal{C}_t(h; v) = \mathcal{C}_t(h'; v)$ for every such v . Denote their class by $c_t(h)$ and the number of classes by K_t .

This definition compares experiments rather than lists of selected coordinates. At T it reduces to equality of $\mathcal{L}(\Theta, W \mid h)$, not only equality of $\mathcal{L}(\Theta \mid h)$. Since the parameter set is finite and the prior is strictly positive, changing the reference prior applies an invertible positive reweighting to every continuation law. The classes are consequently independent of this choice.

Lemma 29.2 (Controlled continuation). *The classes satisfy*

$$h \equiv_t h' \implies hay \equiv_{t+1} h'ay$$

for every a, y of positive conditional probability. The probability of y is itself constant on a class. Equality of the open-loop continuation laws implies equality under every randomized feedback policy using the subsequent report history.

Proof. The marginal of the first future report is a marginal of any continuation law beginning with a . It is the same for h, h' . Condition the joint law of $(\Theta, Y_{t+1:T}, W)$ on that report being y . Its positive common denominator is the first-report probability, and its remaining numerator is the continuation law from hay or $h'ay$. Varying the remaining action word gives the successor assertion. For a feedback policy, multiply each open-loop path probability by the product of its conditional control probabilities along that path and sum over action words. These factors depend only on the displayed future reports and on independent randomization. Equality is preserved. Null successors have zero probability under every parameter of positive reference prior. $\square$

29.1. An operational necessity theorem. The category in which a minimality claim is made matters. At a checkpoint t , a *restart preparation* chooses a supported past history H with an arbitrary law η , and then runs its conditional continuation experiment. The candidate statistic is obtained by consuming that same history through a channel $Q_t(dm | h)$. It does not resample a new history or replace the actual target W . A restart prediction task fixes a future action word v and an event $A \subseteq \Theta \times Y_{t+1:T} \times W$. Immediately at the checkpoint, before the continuation is observed, the decision maker predicts the indicator of A with an action in $[0, 1]$ and squared loss. The full-history optimal prediction is $p_A^v(h) = \mathcal{C}_t(h; v)(A)$.

Call a source-consuming causal statistic *restart-decision complete* when it preserves the full-history Bayes risk in every one of these tasks, for every preparation η and every checkpoint. The definition requires equality of operational risks. It does not require any posterior coordinate to be explicitly stored.

Theorem 29.3 (Marked operational minimality). *A privately randomized finite-valued statistic M_t is restart-decision complete at t if and only if $c_t(H)$ is recoverable as a function of M_t on every supported history. In particular any complete randomized causal machine has at least K_t retained states at checkpoint t . The deterministic quotient (c_t) is simultaneously complete at all checkpoints, has exactly these widths, and is minimal in this category. For a public seed U , the corresponding criterion is recovery of $c_t(H)$ from (U, M_t) , not from M_t alone. The same lower bound on the number of private register labels holds in either signature.*

Proof. The optimal Brier risks before and after compression differ by

$$\mathbb{E}_\eta \left[(p_A^v(H) - \mathbb{E}_\eta[p_A^v(H) | M_t])^2 \right]. \quad (29.1)$$

Indeed $\mathbb{E}[\mathbf{1}_A | H, M_t] = p_A^v(H)$, since the statistic's randomization is independent of the actual continuation given H . Orthogonal projection in L^2 proves the identity. There are only finitely many histories and cylinder events in the present finite experiment. Choose a preparation with positive mass on every supported history. Vanishing of (29.1) for every v, A forces $p_A^v(h) = p_A^v(h')$ whenever both histories give positive probability to a common value m . Thus $h \equiv_t h'$. Each retained value has support in one equivalence class, so the class is a function of that value and there must be at least K_t values.

Conversely, if the class is recoverable, every $p_A^v(H)$ is measurable from M_t , and (29.1) is zero for every preparation. Lemma 29.2 implements the classes by deterministic causal updates. It also proves equality for feedback continuations. For an independent public seed U , apply the same argument to (U, M_t) ; if the declared width bounds only M_t , a free retained U is a genuinely different signature. In an exact complete public scheme with finite widths, zero nonnegative excess

also holds for almost every fixed seed, so the lower bound $S_t \geq K_t$ remains valid. One cannot use a seed to merge distinguishable histories while preserving all these zero-loss differences. $\square$

Proposition 29.4 (A quantitative separation). *Suppose a continuation event has probabilities differing by Δ at h, h' . For the equal two-history preparation, any randomized statistic has Brier excess at least*

$$\frac{\Delta^2}{4}(1 - \text{TV}(Q_t(\cdot | h), Q_t(\cdot | h'))).$$

Proof. Write $x_m = Q_t(m | h)$ and $y_m = Q_t(m | h')$. Direct evaluation of the conditional variance in (29.1) gives $\frac{\Delta^2}{2} \sum_m x_m y_m / (x_m + y_m)$, with zero terms omitted. Since $xy/(x+y) \geq \min(x, y)/2$ and $\sum_m \min(x_m, y_m) = 1 - \text{TV}(x, y)$, the result follows. $\square$

This is not a minimality assertion for an arbitrary independent simulator with unpriced memory. Nor is it an assertion that a chosen small list of losses separates every latent feature. The restart family is decision complete for the stated marked continuation experiment; its loss tests force the necessary state rather than assuming a posterior coordinate representation in the definition. The continuation and linear-representation aspects have classical antecedents in automata and predictive-state theory [49, 5, 58]; the target mark and the source-consuming operational category are essential here.

29.2. Two exact marked causal channels. Let $k_t^0(y | c, a)$ be the reference next-report probability and $F_t(c, a, y)$ the successor class. Define

$$H_t^0(c' | c, a) = \sum_{y: F_t(c, a, y) = c'} k_t^0(y | c, a), \quad \lambda_\theta(c) = \mathbb{P}_\pi(\Theta = \theta | c).$$

On the support of parameter θ , Bayes' formula gives

$$k_{\theta, t}(y | c, a) = k_t^0(y | c, a) \frac{\lambda_\theta(F_t(c, a, y))}{\lambda_\theta(c)}. \quad (29.2)$$

The marked terminal channel is $G_\theta(w | c_T) = \mathbb{P}_\pi(W = w | \Theta = \theta, c_T)$. It is part of the quotient; it is not discarded.

Theorem 29.5 (Exact marked reconstruction). *The original instrument and the class instrument admit causal channels in both directions preserving the joint law of the reported transcript and the actual terminal mark, under every parameter and admissible feedback policy. The reverse report kernel is*

$$L_t(y | c, a, c') = \frac{\mathbf{1}_{\{F_t(c, a, y) = c'\}} k_t^0(y | c, a)}{H_t^0(c' | c, a)} \quad (29.3)$$

when the denominator is positive. It is independent of θ . The class instrument uses transitions $H_{\theta, t}(c' | c, a) = H_t^0(c' | c, a) \lambda_\theta(c') / \lambda_\theta(c)$ and terminal channel G_θ . Thus the channels preserve every bounded loss on the declared marked experiment, not just parameter likelihoods.

Proof. The forward channel consumes each actual report and updates c by F_t ; its current class is a finite register. For the reverse channel, first observe the new class c' , then draw y by (29.3). Multiplying this kernel by $H_{\theta, t}$ gives exactly (29.2). The terminal conditional distribution of W given the full original history and θ is $G_\theta(\cdot | c_T)$ by the definition of the terminal class. Consequently the product of transition factors, report reconstruction factors and this terminal mark factor is the original joint density of $(Y_{1:T}, W)$, for each open-loop action word. The feedback factors can be inserted before summing action words, as in Lemma 29.2. This proves the claimed joint, parameter-uniform identity. Cases with zero λ_θ or zero transition probability lie outside that parameter's support and may be assigned arbitrary kernels. The positive reference prior makes these conventions consistent across parameters.

A reverse transducer needs the previous class, the new class and the current action while forming its report. Between updates it retains only the previous class. Composing it with an S_t -state machine is therefore possible with at most $K_t S_t$ retained labels at time t ; the new input is transient. These resource costs are separate from the bare exact-simulation identity. $\square$

Example 29.6 (A parameter-sufficient statistic that loses its mark). Let $\mathbb{P}(\Theta = 0) = 9/10$, let Y be an independent fair bit, and put $W = Y \oplus \Theta$. The parameter posterior is constant in Y . A terminal quotient that keeps only that posterior has one class. Yet the Bayes error in guessing the actual W is $1/10$ with Y and $1/2$ with that class alone. The marked terminal laws $\mathcal{L}(\Theta, W \mid Y = 0)$ and $\mathcal{L}(\Theta, W \mid Y = 1)$ differ, so the marked quotient has two classes. Its reverse channel preserves the relation $W = Y \oplus \Theta$, not merely the marginal fair-bit law of Y . This example explains why an exact reconstruction theorem for reports alone cannot be used for arbitrary latent-target losses.

APPENDIX 30. TASK-INDEXED EXACT COMPRESSION AND ITS COMPLETION

The least state preserving a finite set of optimal decisions need not be the least state preserving their entire joint experiment. Even at one checkpoint, the former object is generally a compatible cover rather than a quotient. We first give its exact operational characterization, then identify the condition under which it becomes a causal quotient.

30.1. Finite decision families. Let $\mathcal{H}$ be a finite supported-history set at a restart checkpoint, and let $\eta(h) > 0$ be a preparation. A task $d \in D$ has a finite action set B_d and conditional loss $\ell_d(h, b)$. The latter is obtained by conditioning the actual marked continuation law and integrating the specified bounded loss; any prescribed continuation protocol is part of the task. Define

$$m_d(h) = \min_{b \in B_d} \ell_d(h, b), \quad A_d(h) = \{b : \ell_d(h, b) = m_d(h)\}.$$

The full-history benchmark for this restart problem is $\sum_h \eta(h) m_d(h)$. A subset $C \subseteq \mathcal{H}$ is *compatible* when

$$\bigcap_{h \in C} A_d(h) \neq \emptyset \quad \text{for every } d \in D. \quad (30.1)$$

Let τ_D be the smallest number of compatible sets covering $\mathcal{H}$. Empty sets are irrelevant.

Theorem 30.1 (Exact common-task width). *The least number of labels in a source-consuming randomized statistic which attains the full-history benchmark for every task is exactly τ_D . Independent public randomization does not reduce this number. The answer is independent of the choice of strictly positive preparation. If all the conditional losses factor through a marked continuation quotient with K classes, then*

$$\max_{d \in D} \tau_{\{d\}} \leq \tau_D \leq \min \left\{ K, \prod_{d \in D} |B_d| \right\}.$$

These are checkpoint widths. Separate checkpoint optima are not asserted to be implementable by one causal machine.

Proof. Let $Q(m \mid h)$ be the common message channel and $v_d(b \mid m)$ the task decoder. Its excess risk is

$$\sum_{h, m, b} \eta(h) Q(m \mid h) v_d(b \mid m) [\ell_d(h, b) - m_d(h)]. \quad (30.2)$$

Every summand is nonnegative. Equality to zero implies that the support of $v_d(\cdot \mid m)$ is contained in $A_d(h)$ for every h with $Q(m \mid h) > 0$. Thus each message support $C_m = \{h : Q(m \mid h) > 0\}$ is compatible. These supports cover $\mathcal{H}$, and so there are at least τ_D labels.

Conversely, from a compatible cover choose one containing cell for each history, and encode that cell deterministically. For each task and cell choose one action in the intersection in (30.1). Every

excess term is then zero. The argument uses only positivity of η , proving preparation independence. With a public seed condition (30.2) on that seed. Its conditional value is nonnegative, and its integral is zero for each of the finitely many tasks. For almost every seed all these equalities therefore hold, so the same cover lower bound applies.

The lower bound by individual tasks follows by removing constraints. Each marked class is compatible whenever the losses factor through the class. For the other upper bound, choose at each history an optimal action for each task and group histories by the resulting action vector. There are at most $\prod_d |B_d|$ vectors. $\square$

Example 30.2 (Compatibility is not pairwise equality). For one task with three actions, suppose three histories have optimal sets $\{1, 2\}, \{2, 3\}, \{1, 3\}$. Every pair is compatible, but the three histories together are not. The exact width is two, not one. Consequently pairwise indistinguishability alone does not describe general finite-task compression. This is the obstruction to replacing Theorem 30.1 by an arbitrary task-indexed equivalence relation.

30.2. Prediction tasks and a canonical completion. For an event A in a marked continuation experiment with a fixed open-loop word v , put $p_{A,v}(h) = \mathbb{P}(A \mid h, v)$. The Brier task asks for $a \in [0, 1]$ with loss $(a - \mathbf{1}_A)^2$. Its conditional excess is exactly $(a - p_{A,v}(h))^2$. Although its action space is not finite, the same nonnegative-summand argument applies because the minimizer is unique.

Corollary 30.3 (Finite features and universal completion). *For finitely many Brier tasks with probability vector $s(h) = (p_1(h), \dots, p_m(h))$, the least checkpoint width, with arbitrary private or independent public randomization, equals $|s(\mathcal{H})|$. Taking all marked continuation event tasks gives exactly the marked continuation quotient of Theorem 29.3. For finite instruments, a finite subfamily already distinguishes all its classes.*

Proof. If one retained label is possible at two histories, zero excess forces all their probabilities p_i to agree. Conversely storing the probability vector attains all the benchmarks. Equality for all events and words is equality of the finite marked continuation laws, which is the definition of the universal quotient. For each pair of distinct classes choose one word and event whose probabilities differ. There are finitely many pairs, so these choices give a finite determining family. The public-seed argument is the same conditional zero-excess argument as before. $\square$

The completion in this corollary does not enlarge a finite task list by fiat. It identifies precisely the extra tests that turn task-optimal compression into marked-experiment sufficiency. The possible gap is arbitrarily large: if all declared losses have the same optimal constant action, $\tau_D = 1$, even when K is arbitrarily large.

30.3. A task-indexed causal quotient. We now impose strict positivity of each next-report probability for every supported history and issued control. This makes all common continuations available and avoids identifying a null continuation with a supported one. At checkpoint t let $s_t(h)$ be a finite vector of conditional Brier probabilities for the specified restart tasks. Define signatures backward:

$$\begin{aligned} \sigma_T(h) &= s_T(h), \\ \sigma_t(h) &= (s_t(h), (\sigma_{t+1}(hay))_{a \in \mathcal{A}_t, y \in \mathcal{Y}_{t+1}}), \quad t < T. \end{aligned} \tag{30.3}$$

Histories with equal signatures form a right congruence, so a register containing σ_t can be updated from its old value and (a, y) . Denote the number of distinct signatures at time t by K_t^D .

Theorem 30.4 (Exact causal prediction width). *Among causal source-consuming machines that preserve the specified restart Brier risks at every checkpoint and for all supported open-loop control histories, the coordinatewise least width profile is $(K_t^D)_{t=0}^T$. Private or independent public*

randomization does not improve it. These signatures form the coarsest right congruence refining the probability features s_t . For the decision-complete family of all marked continuation event tasks, the signature quotient is the universal marked continuation quotient.

Proof. If one register symbol has positive probability at two histories, zero Brier excess forces equality of their current feature vectors. At time T this proves equality of signatures. Suppose the corresponding claim is proved at $t + 1$, and a symbol m is possible at histories h, h' at time t . Issue the same control a and consider the same report y . Strict positivity ensures that hay and $h'ay$ are both supported. The machine uses the same next-register kernel at (m, a, y) ; at least one of its outputs m' has positive probability. Thus the next-register supports overlap at these two successor histories. By backward induction their signatures at time $t + 1$ are equal. Doing this for all (a, y) and using equality of s_t proves equality of σ_t . Hence disjoint signatures require disjoint nonempty register supports, which gives the lower bound K_t^D simultaneously at each time.

The deterministic signature tracker attains the entire profile, because (30.3) makes its update well defined. Any right congruence refining the current features must also refine all successor features, and backward induction shows it refines σ_t ; this proves the coarsest assertion. For a public seed there are only finitely many histories and specified tests, so all zero-excess and overlap statements hold outside one null set, where the preceding proof applies.

For the complete event family, equality of present features is already equality of marked continuation law, and the latter is a right congruence by Lemma 29.2. The backward refinement therefore makes no further identifications or distinctions. $\square$

Remark 30.5. With zero report probabilities, the backward construction can be performed on a suitably specified partial transition system, but the preceding randomized lower-bound proof cannot simply compare unsupported successor histories. The theorem deliberately states its positivity hypothesis. The unrestricted universal result, including its treatment of supports, remains Theorem 29.3.

APPENDIX 31. SYNTHESIS FROM CONTINUATION SPACES

The continuation quotient can be computed without first expanding all controlled histories. Its linear dimension and its number of positive states are different quantities; the algorithm uses the former to test equivalence and the latter to bound its output.

Assume the finite marked instrument of Section 29 has rational data. At time t write $n_t = |\Theta| |Z_t|$ and $n = \max_t n_t$. Its block-diagonal transition matrices are

$$M_{t,a,y}((\theta, z), (\theta', z')) = \mathbf{1}_{\{\theta=\theta'\}} K_{\theta,t}(z', y \mid z, a).$$

The terminal columns are $e_{\theta,w}(\theta', z) = \mathbf{1}_{\{\theta=\theta'\}} L_{\theta}(w \mid z)$. Define rational continuation spaces backwards:

$$V_T = \text{span}\{e_{\theta,w} : \theta, w\}, \quad V_t = \text{span}\{M_{t,a,y}v : a, y, v \in V_{t+1}\}. \quad (31.1)$$

They have dimensions at most n_t , and contain the all-ones column. For the latter assertion sum the terminal columns, and then use $\sum_y M_{t,a,y} \mathbf{1} = \mathbf{1}$ for any fixed a .

Lemma 31.1 (Finite linear test for a marked class). *Two normalized posterior rows $\alpha_t(h), \alpha_t(h')$ determine the same marked continuation class if and only if they agree on V_t . It suffices to compare their evaluations on any rational basis B_t of that space.*

Proof. The probability of a specified future report word and terminal pair (θ, w) under an action word is the row $\alpha_t(h)$ applied to the column

$$M_{t,a_t,y_{t+1}} \cdots M_{T-1,a_{T-1},y_T} e_{\theta,w}.$$

These columns span exactly V_t , by induction in (31.1). Equality of all these probabilities is precisely Definition 29.1. Equality on their span is equivalent to equality on a basis. $\square$

Theorem 31.2 (Output-sensitive exact synthesis). *Let B be the total binary length of the rational source description, including any finite target augmentation and the positive reference prior. The marked quotient, one representative posterior per class, and its deterministic transition tables can be constructed in time polynomial in*

$$B + T + n + \sum_t |A_t| |Y_{t+1}| + \sum_{t=0}^T K_t.$$

No factor equal to the number of expanded histories is required. The stored transition table has $\sum_{t < T} K_t |A_t| |Y_{t+1}|$ entries, including null entries. A bit-serial implementation retains $\lceil \log_2 K_t \rceil$ state bits, with finite transient workspace and a finite table-lookup running-time bound determined by this table. This is an exact-state synthesis theorem, not a polynomial-time algorithm for optimal lossy common-encoder design.

Proof. Compute the spaces (31.1) backwards by rational elimination. At each stage there are at most $|A_t| |Y_{t+1}| n$ candidate columns. Retain independent columns, rather than repeatedly replacing the basis by arbitrary normalized linear combinations. Each retained column is then a product of at most T input matrices with a terminal column. A common denominator for all input probabilities bounds the binary length of its entries by a polynomial in B, T, n . Fraction-free elimination, or determinant bounds for rational elimination, therefore takes polynomial bit time.

Start at the initial posterior $\alpha_0 = (\pi(\theta)\nu_\theta(z))$. At stage t maintain one posterior representative α for each class found. For every a, y , form

$$r = \alpha M_{t,a,y} \mathbf{1}.$$

If $r = 0$, record the null transition. Otherwise form $\alpha' = \alpha M_{t,a,y}/r$, evaluate $\alpha' B_{t+1}$, and compare this vector with the stored keys at $t + 1$. Reuse its label if equal; otherwise store a new label and the representative α' . By Lemma 31.1, the key is the marked continuation class. The successor result ensures that all histories represented by α have this same transition, even though their full hidden posteriors can differ. Thus retaining one representative loses no reachable class. Induction over t proves that precisely the K_t classes are generated.

There are only the displayed number of transitions to examine. Every representative is the normalized posterior of some history of length at most T . The numerator and denominator are finite sums of products of rational input probabilities. Their binary lengths are polynomial in B, T, n , even if their magnitudes are very small. Exact comparison of keys, either by pairwise search or rational sorting, and all matrix operations consequently take polynomial bit time in the stated input and output parameters.

The executor needs only the class label and a fixed time-indexed table. A sequential scan of that finite table, with binary counters and the current action/report held transiently, is one explicit implementation. It has program length bounded by the table length times the binary length of an entry, and polynomial transient workspace and scan time in the table description. A more efficient random-access convention may improve this bound, but is not needed for finite effectivity. Report acquisition remains a separate interface cost. $\square$

The invariant-space computation is the classical weighted-automaton equivalence method; see Kiefer [58], Proposition 2.1 and Theorem 2.3, and Tzeng [61]. What is being synthesized here is a *positive, normalized, marked causal quotient*, not a minimal signed linear representation. Linear dimension alone would not give its persistent-state cost.

Example 31.3 (Small linear rank and growing positive width). Let $\Theta \in \{0, 1\}$ have positive prior and, conditional on it, let the reports be independent Bernoulli variables with parameters $p \neq q$ in $(0, 1)$. Retain Θ as a terminal mark. The hidden linear dimension is two at every time. After t

reports, the posterior odds for a history with k successes are the initial odds multiplied by

$$(q/p)^k ((1-q)/(1-p))^{t-k}.$$

These $t+1$ values are distinct since $p \neq q$. The parameter posterior is a marginal of the marked continuation law, and conditional future laws depend only on this posterior. Hence $K_t = t+1$, whereas the expanded report tree has 2^t leaves. The algorithm constructs the quotient in polynomial time in T , without mistaking rank two for width two.

APPENDIX 32. POSITIVE-STATE GROWTH AND TASK-SYNTHESIS COMPLEXITY

The continuation-space algorithm of Theorem 31.2 separates two quantities: the linear dimension used to test equality, and the number of positive normalized classes that must actually be retained. We give polynomial and exponential families at fixed linear dimension, and distinguish both from task-optimal synthesis.

Proposition 32.1 (Count-controlled positive-state growth). *Let θ be a static hidden variable with finite support. Conditional on θ , reports produced under a control a have time-homogeneous probabilities $p_\theta(y | a)$ and are independent. Suppose the terminal mark depends only on θ , through a fixed channel. For $r = |\mathcal{A}||\mathcal{Y}|$, the marked continuation quotient satisfies*

$$K_t \leq \binom{t+r-1}{r-1}.$$

For fixed r , rational input, and unary horizon T , exact positive-state synthesis is polynomial in the horizon and input bit length.

Proof. The posterior weight of θ after a supported history is proportional to $\pi_\theta \prod_{a,y} p_\theta(y | a)^{n_{a,y}}$, where $n_{a,y}$ is the number of occurrences of that control-report pair. Its future marked law depends only on this posterior, so histories with the same count vector are equivalent. There are $\binom{t+r-1}{r-1}$ nonnegative count vectors with total t . This is an upper bound; different counts can give the same posterior. Products of t rational input entries have bit length at most linear in t times the input bit bound, and normalization and equality testing preserve a polynomial bit bound. Enumerating the count vectors and their successors therefore gives the claimed synthesis bound. Alternatively apply the output-sensitive compiler to the same polynomial output bound. $\square$

Proposition 32.2 (Exponential positive width at linear rank two). *There are strictly positive rational binary-report instruments with two hidden states, polynomial description length in T , and continuation rank two for which $K_t = 2^t$ at every checkpoint $t \leq T$.*

Proof. Let $\theta \in \{0, 1\}$ have a strictly positive rational prior and let $W = \theta$. There are no controls. Choose distinct primes $p_1, \dots, p_T$, and conditionally independently set

$$\mathbb{P}(Y_i = 1 | \theta = 0) = \frac{1}{2}, \quad \mathbb{P}(Y_i = 1 | \theta = 1) = \frac{p_i}{1 + p_i}.$$

The posterior odds after $y_1, \dots, y_t$ equal the prior odds times

$$\prod_{i=1}^t \frac{2}{1 + p_i} \prod_{i=1}^t p_i^{y_i}.$$

Unique factorization makes the last products distinct for all binary words. Since the continuation contains the mark θ , different posterior odds imply different marked continuation laws. All 2^t words have positive probability, proving the exact count.

The hidden linear state space has dimension two. The terminal marked vectors are the two coordinate vectors, and their positive diagonal backward images remain independent, so the continuation rank is exactly two, not merely bounded by two. The first T primes have bit length

$O(T)$, for example by the elementary bound obtained by successive use of Bertrand's postulate. Thus the rational description has polynomial length. The exponential output is not concealed in exponentially long input probabilities. $\square$

Theorem 32.3 (Zero-excess task synthesis contains graph coloring). *For finite rational one-checkpoint marked decision problems, deciding whether three retained labels suffice to attain every task's full-history Bayes risk is NP-complete. This remains true when all task action sets are binary and private and independent public randomization are allowed.*

Proof. Given a finite graph $G = (V, E)$, let the observed history X be uniform on V and let $W = X$. For each edge $e = \{u, v\}$ make one binary task. At u its loss is zero only for action zero; at v it is zero only for action one; at every other vertex both actions have zero loss. The full-history Bayes risk is zero for each task. A set of histories is compatible in the sense of (30.1) precisely when it contains no edge of G . Compatible covers are therefore covers by independent sets, equivalently colorings after assigning each vertex to one containing set. Theorem 30.1 gives $\tau_D = \chi(G)$ even for randomized encoders. This is a polynomial reduction from three-colorability.

For general explicitly tabulated rational losses, a certificate consists of a label for each history and one optimal action for each task and label. Check all nonempty cells against the conditional minima by exact rational comparisons. The certificate has polynomial size and can be checked in polynomial time. Randomization does not enlarge the class of yes instances by Theorem 30.1. This proves membership in NP and completes the proof. $\square$

Graph-coloring descriptions of zero-error information problems have classical antecedents, notably [62]; the reduction above is given to specify the precise common-task loss formulation, not to claim that the connection with coloring is new. Likewise linear continuation equivalence is classical [61, 58]. The distinction established here is between polynomial linear tests, possibly exponential positive-state output, and a separate task-optimal combinatorial problem. None of these three should be described as a polynomial solution to the other two.

APPENDIX 33. A COMMON-ENCODER OBSTRUCTION WITH AN INTERIOR DUAL WITNESS

The following example puts the task-coupling and randomization issues in one experiment. It is not a difference between distinct source families or between incompatible loss normalizations.

The source reveals $X = \theta \in \{1, 2, 3\}$ without noise. Task $d \in \{1, 2, 3\}$ asks whether $\theta = d$, with binary output and zero-one loss. The encoder has two labels, is not told d , and is common to the nine rows (d, θ) . The task is supplied to the decoder. All full-history task baselines are zero. A task-specific two-label encoder would answer its task exactly.

Theorem 33.1 (Three tasks, two labels). *For the common encoder, the best deterministic worst-row risk is one, the best private-randomized risk is $1/2$, and the best shared-seed risk is $2/9$. The uniform probability on all nine rows is a dual optimizer for the shared problem. Every row receives positive dual weight, and all nine risks of the displayed optimal shared scheme are equal.*

Proof. A deterministic two-label encoder merges at least two source symbols. For the membership task of either symbol, the decoder makes an error on at least one of them. Thus its worst-row risk is one. Moreover, under the uniform distribution on the nine rows every deterministic design makes at least two errors: each of the two membership tasks for a merged pair has an error. The uniform risk is at least $2/9$. Freezing all encoder and decoder randomization gives this same lower bound for every shared mixture of complete designs.

For the matching scheme choose a singleton k uniformly from the three symbols using the public seed. Encode whether $X = k$. If $d = k$, answer the task exactly. If $d \neq k$, answer zero on the singleton label; on the pair label answer one with probability $2/3$. For a positive row $d = \theta$, error occurs when $k \neq d$ and the decoder answers zero, with probability $(2/3)(1/3) = 2/9$. For a

negative row $d \neq \theta$, error occurs only when k is the third symbol and the decoder answers one, with probability $(1/3)(2/3) = 2/9$. This establishes optimality and the asserted strictly positive dual witness. The additional decoder coin may be fresh private randomness; it may also be frozen into the public mixture of deterministic designs.

For a private encoder write $p_x = \mathbb{P}(M = 1 \mid X = x)$. For each task d , the probability of answering one is an affine function $F_d(p_x)$, since it is obtained by applying two decoder probabilities to the two encoder labels. Choose an index d for which p_d is a median of the three values. For this task, p_d lies between the other two values, including the possibility of equality. An affine function cannot be greater than $1/2$ at this middle value and less than $1/2$ at both end values. Hence some row of task d has error at least $1/2$. Ignoring the message and making a fair binary guess attains $1/2$. A common retained coin shared with the encoder would invalidate this affine argument, which is exactly why it is a different resource. $\square$

For a single task, independent randomization cannot improve a scalar expected-loss minimum, as in the strategic-measure results of [56]. The worst-row criterion here is a maximum of linear risks, not one linear risk. Convexification can therefore strictly improve it. There is no contradiction with that classical irrelevant-randomization principle.

APPENDIX 34. CONTROLLED COMMON ENCODERS AND THE MINIMAX RISK BODY

The causal constraint is a constraint on the encoder, not on each task separately. We now give a variational characterization with this constraint inside the feasible set. The distinction between a shared random seed and private fresh randomization is essential to the statement.

34.1. Finite instruments and the information pattern. Fix a horizon T , finite parameter set K , finite hidden spaces Z_t , report alphabets Y_t , and a finite control alphabet A . The instrument is specified by probability vectors ν_θ and stochastic matrices

$$K_{\theta,t}(z', y \mid z, a), \quad 0 \leq t < T.$$

The hidden variable is not available to either the encoder or the controller. A finite collection D of tasks has finite terminal action sets B_d and losses $\ell_{d\theta}(z, b) \in [0, 1]$. Any bounded loss on a finite trajectory, including losses at several decision times, is of this form after adjoining that trajectory to the hidden state. This enlargement is a description of the experiment, not memory supplied to its controller.

The encoder has widths $\mathbf{S} = (S_0, \dots, S_T)$, with $S_0 \geq 1$. A parameter-independent initialization law chooses the initial label before the first control; this label is visible to the controller and is charged in S_0 . Deterministic initialization can be labelled 1 by relabelling states. Before transition t , the task- d controller chooses a from its current label i . The encoder then observes (a, y) and updates the label. At the last time, a task- d terminal rule chooses b . The encoder's update table is the *same* for all tasks and parameters; the controller's tables may depend on d but not on θ . The current control is an input to the ensuing update, not a retained action history. Task-dependent restrictions on legal controls may be imposed on the controller tables. The controller has no additional persistent register.

For a minimax task retain one row $j = (d, \theta)$ per parameter. For a Bayesian task retain one row j with its specified prior κ_j on K . More generally a row has a task $d(j)$ and a probability vector κ_j ; point masses give the minimax rows. Let J be this finite row set, $M = |J|$, and let β_d be the full-history optimal risk for the *whole* task d . In particular, the baseline of a minimax task is not a separate parameterwise optimum. Write $b_j = \beta_{d(j)}$.

A *deterministic design* ω consists of one family of updates $f_t(i, a, y) \in [S_{t+1}]$, together with all controller tables $a_t^d(i) \in A$ and terminal tables $b^d(i) \in B_d$. Let $\Omega_{\mathbf{S}}$ be the finite set of these designs.

Its raw risk vector is denoted $R(\omega) = (R_j(\omega))_{j \in J}$, where

$$R_j(\omega) = \sum_{\theta} \kappa_j(\theta) \mathbb{E}_{\theta}^{\omega, d(j)} \ell_{d(j)\theta}(Z_T, b^{d(j)}(I_T)). \quad (34.1)$$

The expectation uses the actual closed-loop law. In particular, controls are not optimized separately inside hidden-parameter components.

There are two randomization conventions. In the *shared-seed* convention an independent variable is available to the encoder and the task controller throughout the run. It is chosen independently of the task and parameter; the risk averages its realization before taking the supremum over tasks or parameters. The adversary does not select a new parameter after seeing that realization. In the *private* convention there is no such retained shared variable. Each declared kernel uses fresh randomness, with conditional independence given its allowed inputs. Any seed remembered across updates is charged in the label. We write Θ_{sh} and Θ_{pr} for these two specified slices of (23.4). The term shared seed here has the precise meaning of the public independent seed permitted in Section 23; it never contains data from the experiment.

34.2. The variational theorem. Define the upper risk body and its lower support function by

$$\mathcal{U}_E(\mathbf{S}; J) = \text{co}\{R(\omega) : \omega \in \Omega_{\mathbf{S}}\} + \mathbb{R}_+^J, \quad (34.2)$$

$$h_E(\mathbf{S}; w) = \min_{\omega \in \Omega_{\mathbf{S}}} w \cdot R(\omega), \quad w \in \Delta_J. \quad (34.3)$$

Here Δ_J is the probability simplex. The body is defined from controlled execution risks, not from a presumed predictive quotient.

Theorem 34.1 (Controlled common-encoder minimax theorem). *For the finite instrument and information pattern just specified:*

- (1) *The shared-seed attainable upper risk set is exactly $\mathcal{U}_E(\mathbf{S}; J)$. The common-encoder excess satisfies*

$$\Theta_{\text{sh}}(\mathbf{S}; D) = \min_{\lambda \in \Delta(\Omega_{\mathbf{S}})} \max_{j \in J} \left\{ \sum_{\omega} \lambda_{\omega} R_j(\omega) - b_j \right\} \quad (34.4)$$

$$= \max_{w \in \Delta_J} \{h_E(\mathbf{S}; w) - w \cdot b\}. \quad (34.5)$$

An optimum uses at most $M + 1$ deterministic designs. Thus the encoder is chosen once, before the task, in both formulae.

- (2) *The functions $h_E(\mathbf{S}; \cdot)$ determine the whole upper risk body. For two matched task/row systems,*

$$\mathcal{U}_F(\mathbf{S}'; J) \subseteq \mathcal{U}_E(\mathbf{S}; J) \iff h_E(\mathbf{S}; w) \leq h_F(\mathbf{S}'; w) \quad (w \in \Delta_J).$$

In particular equality of all these support functions is equivalent to equality of the attainable upper bodies. Relabelling histories, hidden states or machine labels changes neither assertion.

- (3) *For constant width S , cardinality alone being charged,*

$$\Theta_{\text{pr}}((M + 1)S; D) \leq \Theta_{\text{sh}}(S; D) \leq \Theta_{\text{pr}}(S; D). \quad (34.6)$$

The first inequality is implemented by retaining the design index in the actual register, including the permitted randomized initialization. If a different signature forbids randomized initialization, this inequality refers to its explicitly initialization-enlarged signature. For a fixed finite row family, any existing positive cardinality exponent is consequently the same under these two randomization conventions.

Proof. First include the decoder and controller tables in the design. For one fixed common encoder their choices for distinct tasks are independent optimization variables, whereas the same choice for one task must work for all its parameter rows. Compactness of the finite products of probability simplexes implies attainment. Hence minimizing jointly over the common encoder and all its task rules gives exactly the original order $\inf_Q \max_d (r_E(Q, d) - \beta_d)$; it does not give a decoder indexed by the unknown parameter.

Realize every update and controller kernel by a uniform variable at its specified time, independent of the instrument. For a shared-seed scheme, condition on the entire shared seed and these variables. One obtains a deterministic common update family and one deterministic controller for each task. Although different tasks then generate different observation laws, the frozen maps are defined on *all* their possible inputs. They therefore form one element of $\Omega_{\mathbf{S}}$. Its conditional risk is computed under the corresponding controlled instrument, not under an incorrectly frozen uncontrolled history law. Averaging gives a convex combination of the finite vectors $R(\omega)$.

Conversely sample a design index with probabilities λ , expose that independent index to encoder and controller, and execute its tables. The encoder law is independent of which task will be requested. Its risk vector is exactly $\sum \lambda_\omega R(\omega)$. Taking coordinatewise upper bounds proves (34.2) and (34.4). Finite-dimensional convex minimax gives (34.5). Equivalently, the primal is the linear programme minimizing v subject to $\lambda \geq 0$, $\sum \lambda = 1$, and $\sum \lambda_\omega R_j(\omega) \leq v + b_j$ for every row. Carathéodory's theorem in $\mathbb{R}^M$ gives support at most $M + 1$ without changing the optimum risk vector. This argument freezes a tape to form its *distribution of designs*; it never moves a worst-task supremum through that distribution.

The body in (34.2) is closed and convex. If a point lies outside it, finite-dimensional separation supplies a linear functional whose infimum on the body exceeds its value at the point. The functional must be nonnegative on every coordinate direction, because the body contains every positive ray starting at one of its points. Normalizing its coefficients places it in Δ_J . Its infimum is (34.3). This proves the comparison assertion, including both directions of equality. A relabelling gives a bijection of the same execution laws and of their deterministic designs.

Every private scheme is also a shared-seed scheme that ignores its extra input. Conversely an optimal shared scheme can use $k \leq M + 1$ designs. Sample its index privately before the first observation and retain (r, I_t) , which has at most kS values. Each subsequent update, control and readout uses this actual pair. This has the same law as the shared scheme; no external transcript or hidden seed is used. This proves (34.6). Monotonicity, together with $\log((M + 1)S)/\log S \rightarrow 1$, proves the exponent assertion. The constant $M + 1$ is not uniform over growing or infinite task families. $\square$

The theorem identifies a finite controlled spectrum by one variational functional. It is more than the transport of a supplied simulation: its feasible set is constructed from the instrument and the prescribed memory, and the converse recovers every admissible execution. Convex minimax, separation and finite mixed strategies are classical tools. Their use here does not reassign their priority. The substantive information restriction is that all tasks share f , while all parameters within a task share its controller, and every retained design index is accounted for when the shared-seed convention is removed.

34.3. Controlled continuation partitions. Let $\mathcal{H}_t$ be the full finite tree of action–report histories, including histories null under a particular parameter or task. Histories null under every possible execution can be assigned arbitrary updates. A family of partitions $\mathcal{P}_t$ is controlled successor compatible when

$$h \sim_t h' \implies hay \sim_{t+1} h'ay \quad (a \in A, y \in Y_{t+1}). \quad (34.7)$$

Proposition 34.2 (Intrinsic feasible partitions). *Deterministic common encoders with widths $\mathbf{S}$ are, up to label names and unused states, exactly the partition families satisfying (34.7) and*

$|\mathcal{P}_t| \leq S_t$. A deterministic design adds, for each task, a control constant on every current cell and a terminal decision constant on every terminal cell. Thus the risk body and both sides of (34.5) can equivalently be defined on controlled continuation partitions with these cellwise rules.

Proof. Equal machine states have equal successors on the same action and report, proving necessity. Conversely label the cells and send (C, a, y) to the cell containing hay for any $h \in C$. Condition (34.7) makes this independent of h . The cellwise controller rule supplies the legal control, and induction identifies the actual label with the actual history's cell for every parameter and task. Defining the maps on the whole tree ensures that an action absent under one task has not made its update task dependent. $\square$

34.4. Private randomization is a different finite problem. For completeness there is also an exact, nonconvex description without a shared seed. Write $g_t^d(a \mid i)$ for the private controller, $q_t(i' \mid i, a, y)$ for the common private update, and $v^d(b \mid i)$ for its terminal rule. Let q_0 be a parameter-independent initialization law when randomized initialization is permitted. Then

$$\begin{aligned} p_0^{d\theta}(z, i) &= \nu_\theta(z) q_0(i), \\ p_{t+1}^{d\theta}(z', i') &= \sum_{z, i, a, y} p_t^{d\theta}(z, i) g_t^d(a \mid i) K_{\theta, t}(z', y \mid z, a) q_t(i' \mid i, a, y). \end{aligned} \quad (34.8)$$

The risk of row j is

$$\sum_{\theta, z, i, b} \kappa_j(\theta) p_T^{d(j)\theta}(z, i) v^{d(j)}(b \mid i) \ell_{d(j)\theta}(z, b).$$

Proposition 34.3 (Private flow characterization). *Minimizing the largest baseline-subtracted row risk over the probability simplexes in (34.8) gives exactly $\Theta_{\text{pr}}(\mathbf{S}; D)$. The minimum is attained. With rational input this is a finite polynomial optimization problem, not in general the linear programme (34.4).*

Proof. Induction on the actual execution gives the displayed occupancies and risk. Conversely their kernels implement precisely that law with no extra retained variable. The variables form a compact product of finite simplexes, and every row risk is a polynomial in them. Continuity gives attainment. The shared occurrences of q_t cannot be split by task or parameter; doing so would change the information pattern. $\square$

Example 34.4 (A strict cost for hiding the seed). Observe $X = \theta \in \{1, 2, 3\}$ without error, retain two labels, and finally guess θ with zero-one loss under worst-parameter risk. The full-information baseline is zero. With a shared seed, choose uniformly one of the three two-element subsets and encode the two selected symbols without error. The omitted symbol is necessarily wrong. Each parameter is omitted with probability $1/3$, so the risk is $1/3$. For any fixed seed, the sum of the three correct-reconstruction probabilities of a two-label experiment is at most two; averaging proves the matching lower bound.

Without a shared seed let r_1, r_2 be the two decoder distributions on three symbols. Correctness for symbol x is at most $\max\{r_1(x), r_2(x)\}$. If all three correctness probabilities exceeded $1/2$, one of two probability vectors would exceed $1/2$ at two coordinates, an impossibility. Thus the private risk is at least $1/2$. It is attained by assigning symbol 1 to the first label and symbols 2, 3 to the second, whose decoder tosses a fair coin. Therefore

$$\Theta_{\text{sh}}(2) = \frac{1}{3}, \quad \Theta_{\text{pr}}(2) = \frac{1}{2}.$$

The gap is not removed by allowing randomized private encoding. A single unknown parameter is selected before the seed, as required above.

APPENDIX 35. ONE VARIATIONAL PRINCIPLE: OCCUPANCIES, BAYES ENVELOPES AND PRESSURE

The support function in Theorem 34.1 can be computed without first constructing a quotient or listing full decision trees. The relevant state for this calculation is an *offline* joint occupancy. It is not an additional observation available during execution.

35.1. An exact support-function recursion. For $w \in \Delta_J$ put

$$z_0^{d\theta}(z, i) = \mathbf{1}_{\{i=1\}} \nu_\theta(z) \sum_{j:d(j)=d} w_j \kappa_j(\theta).$$

For arbitrary nonnegative arrays $z = (z^{d\theta}(x, i))$, a prescription consists of actions $a^d(i)$ and a *single* table $f(i, a, y)$. Define its linear occupancy update by

$$(\mathcal{T}_{a,f} z)^{d\theta}(x', i') = \sum_{x,i,y} z^{d\theta}(x, i) K_{\theta,t}(x', y \mid x, a^d(i)) \mathbf{1}_{\{f(i, a^d(i), y)=i'\}}. \quad (35.1)$$

The time t is understood from the two spaces in this formula. Define

$$V_T(z) = \min_{b^d(i) \in B_d} \sum_{d, \theta, x, i} z^{d\theta}(x, i) \ell_{d\theta}(x, b^d(i)), \quad (35.2)$$

$$V_t(z) = \min_{a,f} V_{t+1}(\mathcal{T}_{a,f} z), \quad t = T-1, \dots, 0. \quad (35.3)$$

Theorem 35.1 (Occupancy representation of the spectrum). *For the finite controlled instrument,*

$$h_E(\mathbf{S}; w) = V_0(z_0(w)), \quad \Theta_{\text{sh}}(\mathbf{S}; D) = \max_{w \in \Delta_J} \{V_0(z_0(w)) - w \cdot b\}. \quad (35.4)$$

Every V_t is a positively homogeneous, concave, piecewise-linear function on its nonnegative occupancy cone. A minimizing sequence of prescriptions compiles into an actual common-encoder design with the specified widths. No realized task index or hidden parameter is supplied to f .

Proof. Fix a deterministic sequence of prescriptions. Induction in (35.1) shows that $z_t^{d\theta}$ is the joint law of (Z_t, I_t) under that task and parameter, multiplied by its initial weight. Its terminal linear cost is therefore $w \cdot R(w)$. At the last time the terminal tables in (35.2) give precisely all legal terminal rules: $b^d(i)$ is common to every θ in task d .

At an earlier time, choosing $a^d(i)$ and one common f determines the entire next occupancy. Conversely every possible first prescription of a deterministic continuation has this form. Backward induction therefore proves that (35.3) minimizes over exactly all legal future designs. This proves the first equality in (35.4); Theorem 34.1 proves the second. For any fixed continuation the cost is linear in the initial occupancy. There are finitely many continuations, so V_t is the minimum of finitely many linear forms. The stated geometric properties follow.

The occupancy is propagated during design, starting at the known array $z_0(w)$. Once a sequence of minimizing tables is chosen, its future occupancies are deterministic functions of that sequence. Only the finite tables and the actual label are used online. In particular, one must not store the occupancy array without charging its size, nor allow a different prescription for the unobserved realized parameter. $\square$

This recursion is a designer's dynamic programme with coupled information constraints. Common-information prescriptions and belief-state dynamic programming have substantial antecedents [53, 54]. In the cited common-information preprint, Corollary 5 already treats absent common memory and finite local memories; its unconditional information state is an offline occupancy. The additional assertion here is the exact equality with the *common-encoder minimax spectrum*, including its dual adversarial weights, controlled continuation constraint and explicit seed cost. The recursion does not assert polynomial complexity: the number of linear pieces and the number of finite designs can be very large.

35.2. The common quadratic task family. There is a particularly transparent intrinsic form when reports are uncontrolled. Let the finite history law $p_t(h)$ be fixed, let U_{dt} be square-integrable Hilbert-valued targets, and let $\alpha_{dt} \geq 0$. Task d has loss $\sum_t \alpha_{dt} \|U_{dt} - \hat{U}_{dt}\|^2$. The decoder's answers do not affect later observations or updates. Set $m_{dt}(h) = \mathbb{E}[U_{dt} \mid H_t = h]$. For a successor-compatible partition family $\mathcal{P}$, define

$$D_d(\mathcal{P}) = \sum_t \alpha_{dt} \sum_{C \in \mathcal{P}_t} \sum_{h \in C} p_t(h) \|m_{dt}(h) - \bar{m}_{dt,C}\|^2, \quad \bar{m}_{dt,C} = \frac{\sum_{h \in C} p_t(h) m_{dt}(h)}{p_t(C)}. \quad (35.5)$$

Zero-mass cells are omitted. Define the single weighted geometry

$$M_{w,t}(h) = (\sqrt{w_d \alpha_{dt}} m_{dt}(h))_{d \in D}$$

to lie in the Hilbert direct sum of the target spaces.

Corollary 35.2 (Common-encoder variance duality). *For a shared independent seed and the declared widths,*

$$\begin{aligned} \Theta_{\text{sh}}(\mathbf{S}; D) &= \min_{\lambda \in \Delta(\mathcal{P})} \max_d \sum_{\mathcal{P}} \lambda_{\mathcal{P}} D_d(\mathcal{P}) \\ &= \max_{w \in \Delta_D} \min_{\mathcal{P}} \sum_t \mathbb{E} \|M_{w,t}(H_t) - \mathbb{E}[M_{w,t}(H_t) \mid \mathcal{P}_t]\|^2. \end{aligned} \quad (35.6)$$

The minima range over one successor-compatible family of width at most S_t at each time. This is an exact characterization of the maximum over tasks, not a statement only about one scalarized loss. Its private version lies between the values given by this formula at S and by its implementation with $(|D| + 1)S$ states.

Proof. Conditional expectation gives the orthogonal decomposition of each squared risk into its full-history Bayes risk and prediction of $m_{dt}(H_t)$. On a deterministic state cell its unique optimal readout is the displayed cell mean. All tasks can use their respective means simultaneously without changing the common encoder. Thus the coordinatewise best risk vector of a deterministic partition is exactly $D(\mathcal{P})$. There are finitely many history partitions, although the target Hilbert spaces need not be finite dimensional.

Conditioning on an independent seed gives a distribution of these partitions; conversely their mixtures are executable under the shared convention. The proof of Theorem 34.1 now applies to this finite vector set. The identity $\sum_d w_d D_d(\mathcal{P}) = \sum_t \mathbb{E} \|M_{w,t} - \mathbb{E}(M_{w,t} \mid \mathcal{P}_t)\|^2$ is the direct-sum norm formula. Storing the mixture index proves the last assertion exactly as in (34.6). $\square$

For $|D| = 1$, (35.6) reduces to the earlier conditional-variance characterization, Theorem 47.1. For several tasks it supplies its missing minimax convexification. The outer maximizing weight is common to every time; selecting a separate favourable weight or encoder at each checkpoint is not permitted. Compatible predictive trees are one class of feasible partitions of this same functional. Their energy estimates in the supporting theory supply quantitative bounds for that class; they are not hypotheses of Theorems 34.1 or 35.1.

Corollary 35.3 (A1: maximum over finite query and checkpoint tasks). *Consider the actual finite-command accepted-cell/failure protocol of Corollary 47.2, with its fixed prior, finite horizon and parameter-independent command laws. Fix any finite family of weighted quadratic query/checkpoint tasks. Its optimal shared-seed common causal encoder is given by (35.6), using the accepted and summed failure likelihoods of that protocol. At most $|D| + 1$ partition designs are needed. Retaining their index gives the same optimum with at most $(|D| + 1)S$ private states.*

Proof. The observation likelihoods and posterior means are derived in Corollary 47.2 from $g_j k_j(\theta)$ on acceptance and $\sum_j (1 - g_j) k_j(\theta)$ on the one failure report. These give one finite history law for all tasks. Insert its posterior query means in (35.5) and apply Corollary 35.2. The unseen failed

raw-cell label is never used in a partition update. Unlike the earlier scalarization, the present formula retains the maximum over the whole prescribed task family. $\square$

Proposition 35.4 (A1 with private randomness at unchanged width). *In the same finite-command protocol, let $p_t(h)$ be its actual history law, and let $m_{dt}(h)$ be the finite vector of posterior query means for task d , with query weights absorbed into that vector. Write $\eta_t^q(i | h)$ for the conditional label law generated recursively by one common private encoder q . The private maximum-over-tasks excess at unchanged width is exactly*

$$\min_{q,v} \max_{d \in D} \sum_{t,h,i} \alpha_{dt} p_t(h) \eta_t^q(i | h) \|m_{dt}(h) - v_{dt}(i)\|^2. \quad (35.7)$$

The readouts can be restricted to the compact boxes containing the weighted query means. This is a finite polynomial optimization problem in the stochastic update rows and readout coordinates, with coefficients given by the actual likelihood integrals. It has a minimum. No finite arithmetic claim is made when those coefficients are arbitrary reals.

Proof. Given a supported history, the label law is a finite sum of products of the encoder's transition probabilities. The same row is used whenever the preceding label and observed command/report agree, in particular on the common failure report. Conditional orthogonality subtracts the full-history Bayes risk, leaving precisely the displayed expression. Conversely every listed stochastic row and readout is an admissible private machine, so both infima are equal. Projecting readouts into the coordinatewise range of the query means cannot increase squared error. The feasible set is compact, and its objective is the maximum of finitely many polynomials. Continuity proves attainment. $\square$

35.3. Exponential criteria and hidden-phase control. The same support calculation has a multiplicative form. Let $\gamma > 0$ and let $C_{d\theta}$ be a bounded cumulative cost on the finite trajectory, including a terminal decision. Put

$$M_j(\omega) = \sum_{\theta} \kappa_j(\theta) \mathbb{E}_{\theta}^{\omega, d(j)} e^{\gamma C_{d(j)\theta}}.$$

For task d the risk is $\gamma^{-1} \log \max_{j:d(j)=d} M_j$; a Bayesian task has its single prior row. Let $\beta_d^{\exp}$ denote its corresponding full-history baseline.

Corollary 35.5 (Risk-sensitive common-encoder duality). *The shared-seed excess under these exponential risks equals*

$$\Theta_{\text{sh}}^{\exp}(\mathbf{S}; D) = \frac{1}{\gamma} \log \left[\max_{w \in \Delta_J} \min_{\omega \in \Omega_{\mathbf{S}}} \sum_j w_j e^{-\gamma \beta_{d(j)}^{\exp}} M_j(\omega) \right]. \quad (35.8)$$

For additive costs, the inner minimum is computed by (35.1)–(35.3) after replacing the transition by $K_{\theta,t} e^{\gamma C_{d\theta,t}}$ and the terminal loss by $e^{\gamma C_{d\theta,T}}$. The initial row weights are multiplied by $e^{-\gamma \beta_d^{\exp}}$. These are nonnegative occupancies, not normalized posterior probabilities.

Proof. For a mixture of designs, the exponential moment is $\sum_{\omega} \lambda_{\omega} M_j(\omega)$. It is *not* the exponential of an average of logarithmic risks. Since logarithm is increasing, subtracting the baseline and maximizing rows gives

$$\frac{1}{\gamma} \log \max_j \left[e^{-\gamma \beta_{d(j)}^{\exp}} \sum_{\omega} \lambda_{\omega} M_j(\omega) \right].$$

Minimize over λ and apply finite minimax to the bracketed positive moment table. This proves (35.8). Expanding the product of transition weights proves the Feynman–Kac occupancy assertion by the same induction as before. Positivity, not normalization, is what that induction requires. $\square$

For example, let an unobserved phase $\theta \in \{0, 1\}$ have probabilities $1/2, 1/2$, provide no report, and charge $C = \mathbf{1}_{\{a \neq \theta\}}$. Every observation-admissible action, randomized or not, has moment $(1 + e^\gamma)/2$ and risk $\gamma^{-1} \log((1 + e^\gamma)/2) > 0$. Optimizing separately within each phase would instead give zero. The single $a^d(i)$ in the occupancy recursion forbids precisely that exchange. This supplies a finite-model resolution of the common-policy issue arising in phase mixtures; it does not presume that a latent phase decomposition of a microscopic system has already been constructed.

APPENDIX 36. MINIMAL POSTERIOR-AND-REPORT-PREDICTIVE COORDINATES

The variational theorem does not require a sufficient statistic. When such a statistic is desired, the finite instrument admits a direct construction. It distinguishes the cost of prediction from the cost of preserving the whole parameter experiment.

Fix the finite instrument of Section 34. For an action–report history $h = (a_0, y_1, \dots, a_{t-1}, y_t)$ define the unnormalized forward vector by

$$\begin{aligned} u_{\theta,0}(z) &= \nu_\theta(z), \\ u_{\theta,t+1}(hay, z') &= \sum_z u_{\theta,t}(h, z) K_{\theta,t}(z', y \mid z, a), \quad L_\theta(h) = \sum_z u_{\theta,t}(h, z). \end{aligned} \quad (36.1)$$

Choose a reference prior $\pi_\theta > 0$. A history is supported if $\bar{L}(h) := \sum_\theta \pi_\theta L_\theta(h) > 0$. On supported histories put

$$\begin{aligned} \lambda_\theta(h) &= \pi_\theta L_\theta(h) / \bar{L}(h), \\ k_t(y \mid h, a) &= \bar{L}(hay) / \bar{L}(h). \end{aligned} \quad (36.2)$$

The likelihoods in these formulae do not contain action-selection probabilities. Under any parameter-independent observation-based policy, those probabilities cancel from the posterior and from likelihood ratios. Thus the formulae apply to every policy on its supported histories, not only to a policy used to explore the history tree.

A history coordinate $s_t(h)$ is called *saturated and predictive* if it determines $\lambda(h)$, determines every vector $k_t(\cdot \mid h, a)$, and has a deterministic update from (s_t, a, y) on supported continuations. At time T only the first condition is imposed. This is the finite, terminal-parameter version of the three conditions in Section 46. Additional latent-target marks require the corresponding conditional kernels as further decorations; parameter sufficiency alone does not imply their sufficiency. Sections 29–31 supply the marked reconstruction theorem and an output-sensitive construction.

Construct partitions backwards. At level T group histories with the same vector $\lambda(h)$. Having constructed the class map c_{t+1} , let $c_t(h)$ be the class determined by the finite tuple

$$\left(\lambda(h), (k_t(y \mid h, a), c_{t+1}(hay))_{a,y} \right). \quad (36.3)$$

For an unsupported continuation its class entry is one fixed cemetery symbol. Let K_t be the number of classes at level t .

Theorem 36.1 (Minimal posterior-and-report-predictive coordinate). *For every finite controlled instrument the partitions (36.3) form a saturated predictive coordinate. Every other such deterministic history coordinate refines them at each level. Consequently K_t is its least possible width, simultaneously at every level. The partitions, as equivalence relations on supported histories, are independent of the choice of the strictly positive reference prior.*

There is a parameter-independent finite reconstruction kernel $L_t(y \mid c, a, c')$ which, from successive quotient classes and the chosen controls, reconstructs the report stream under every admissible policy. The original and quotient parameterized report experiments have mutually exact causal

simulations. This assertion concerns reports and parameter likelihoods only; latent-target preservation is the separate marked statement of Theorem 29.5. Their additional persistent state is at most $\max_t K_t$ labels, so composing either simulation with an S -state encoder costs at most $(\max_t K_t)S$ states.

If the instrument and prior have rational entries, the classes and reconstruction kernels can be computed by exact finite arithmetic. The calculation is polynomial in the size of the expanded finite history tree and its rational input length. That tree, and the least width K_t , may grow exponentially with the horizon.

Proof. Equality of the keys at level t implies equality of the posterior vectors, of all reference predictive probabilities, and of the class of every common positive-probability successor. Sending (c, a, y) to that successor class therefore defines an update. Unsupported continuations may be assigned arbitrarily. This proves the three required properties.

For minimality, at time T any saturated coordinate separates unequal posterior vectors, so its fibres refine c_T . Suppose the assertion has been proved at $t + 1$. Equal coordinates at time t have equal posterior and predictive vectors. Their supported successors under any fixed (a, y) have equal next coordinates by recursion, and hence equal c_{t+1} by the induction hypothesis. Their complete keys (36.3) coincide. Backward induction proves the claimed simultaneous refinement, rather than just a bound on a terminal statistic.

To check prior independence, first observe that equality of posterior vectors under a positive prior is equivalent to proportionality of the nonnegative likelihood vectors $(L_\theta(h))_\theta$. This relation is independent of the prior. For a fixed parameter with $\lambda_\theta(h) > 0$, the actual next-report probabilities are

$$P_\theta(y \mid h, a) = k_t(y \mid h, a) \frac{\lambda_\theta(hay)}{\lambda_\theta(h)}. \quad (36.4)$$

In an existing class the right side is constant on histories for every supported successor, because the successor class determines its posterior. The set of parameters with positive likelihood is also constant on the class. Under a different positive prior, its new posterior weights are functions of its old posterior weights; averaging these common parameter-specific predictive kernels therefore gives a common new reference predictive kernel. The existing partition is saturated, predictive and recursive for the new prior. Minimality in both directions proves equality of the two partitions.

For reconstruction, write $F_t(c, a, y)$ for the class update and define

$$H_t^0(c' \mid c, a) = \sum_{y: F_t(c, a, y) = c'} k_t(y \mid c, a),$$

$$L_t(y \mid c, a, c') = \frac{\mathbf{1}_{\{F_t(c, a, y) = c'\}} k_t(y \mid c, a)}{H_t^0(c' \mid c, a)} \quad \text{if } H_t^0(c' \mid c, a) > 0.$$

Choose an arbitrary probability vector when the denominator is zero. The posterior at the next class is constant over the summed reports. Equation (36.4) thus factors the actual joint law of report and next class into this same L_t and the parameter-specific class-transition probability. Conditioning further on the next class cancels the likelihood factor. This gives a single parameter-independent report reconstruction, including when parameters have different supports. Histories having zero likelihood under a parameter are not visited under that parameter.

Starting from the initial class, the reconstruction stores the previous class, receives the next class after the actual chosen action, samples its report using L_t , and replaces its stored class. A target policy then chooses its next action from the reconstructed reports. Induction in time proves equality of the resulting action-report laws for every parameter and policy. In the other direction, apply F_t to the original reports. Product registers give the displayed storage bound; neither simulation presumes a cost-free transcript.

Finally, (36.1) enumerates the rational forward vectors on a finite tree. Formulae (36.2) and (36.3) use only finite sums, products, divisions by positive numbers, and equality tests of rational tuples. Class labels can be assigned by sorting those tuples backwards. To see the stated bit-complexity bound, take a common denominator for all rational input entries. Its bit length is at most their total input length. At level t , a forward entry can be represented with the t th power of this denominator times the initial denominator; sums add at most the bit length of the number of summands at each step. Thus every intermediate integer has bit length polynomial in horizon and input length. Posterior divisions and tuple comparisons preserve a polynomial bound. There are only as many forward and backward steps as nodes and edges of the expanded tree, times its finite hidden and parameter dimensions. This proves effectivity without any oracle for exact real comparisons. $\square$

The theorem constructs the object assumed by the causal factorization in this finite class. Its minimality is among the *specified* likelihood-saturated, raw-report-predictive coordinates. It is not a claim that every exact simulator, or every task-specific lossy encoder, must use these coordinates. The variational principle optimizes the latter directly and may use substantially less memory.

Example 36.2 (Saturation can have an exponential price). Let the parameter range over $\{0, 1\}^d$, and reveal it noiselessly one bit at a time during the first d observations. Thereafter all reports are independent fair bits. At time d all future-report predictions coincide, so the future-report predictive quotient has one class. Nevertheless the posterior on the parameter is a different point mass for each word. Every saturated coordinate at that time has at least 2^d classes, and the construction above has exactly 2^d . Predictive closure alone therefore supplies no controlled-cost parameter saturation. This is a cost calculation in a concrete model, not an assumption that the missing saturation has bounded size.

Example 36.3 (Likelihoods alone need not close prediction). There is one parameter. Let Y_1 be a fair bit and put $Y_2 = Y_1$. The constant coordinate is recursive and contains every parameter likelihood, but its one-step predictive law at time 1 is not a function of that coordinate. The least saturated predictive widths are $(K_0, K_1, K_2) = (1, 2, 1)$. The last quotient class can discard the bit, because reconstruction of Y_2 still uses the previous class. This example separates the hypotheses of the factorization theorem; it does not assert impossibility of other simulators with other charged resources.

APPENDIX 37. FINITE COMPILATION AND SEVERAL CHARGED RESOURCES

The finite variational principle also has an effective implementation. This is a separate source of multi-resource control from the energy oracle used in the predictive-tree construction. The price of synthesis is explicit and can be large; it is not moved into a free online oracle.

Use the finite instrument and finite decision alphabets of Section 34, with rational probabilities and losses. Let $S = \max_t S_t$, $Y = \max_t |Y_t|$, $B = \max_d |B_d|$, and $m = |D|$. A table implementation of one deterministic design has description length bounded, up to an absolute coding constant, by

$$\begin{aligned}
 P_0 = & \sum_{t=0}^{T-1} S_t |A| |Y_{t+1}| \lceil \log_2(S_{t+1} + 1) \rceil \\
 & + m \sum_{t=0}^{T-1} S_t \lceil \log_2(|A| + 1) \rceil + m S_T \lceil \log_2(B + 1) \rceil \\
 & + O((T + m) \log_2(T + S + |A| + Y + B + 2)).
 \end{aligned} \tag{37.1}$$

Unused states and forbidden controller actions can simply be retained as redundant table entries. The externally specified time index is the same clock convention as in (23.1). If no external clock is permitted, its $T + 1$ values must also be included in the register.

Theorem 37.1 (Effective finite-source realization). *An optimal shared-seed common-encoder scheme for the rational finite instrument can be synthesized by a finite exact calculation. It can use $k \leq M + 1$ deterministic tables with rational mixing weights. For every integer $p \geq 1$ there is a dyadic implementation whose excess is at most*

$$\Theta_{\text{sh}}(\mathbf{S}; D) + k 2^{-p}. \quad (37.2)$$

Its fixed description has $O(kP_0 + kp)$ bits. Initialization uses exactly p independent fair bits; no random bits are required subsequently. With an explicitly permitted read-only shared seed its persistent register has width at most S . Without that convention the same scheme has width at most kS , including the stored design index.

Each update reads one supplied finite-alphabet report. On a bit-serial table implementation, transient space can be bounded by $O(p + \log(P_0k + 2) + \log(S + |A| + Y + B + T + m + 2))$ bits and the running time of an update or initialization by $O(kP_0 + kp + p + 1)$ bit operations, after increasing the universal coding constant. No unit-cost access to an arbitrary real constant is required. Reading the physical report itself remains an input cost, rather than a claim about how an underlying continuous measurement was acquired.

Proof. Enumerate the finite deterministic designs. Their risks are rational numbers obtained from finite products and sums of rational transition entries and losses. Full-history baselines can likewise be computed by enumerating full-history policies and solving their finite minimax linear programmes; for a Bayesian task their minimum is already a finite linear optimization. Thus the primal programme (34.4) has rational data and has a rational optimal basic solution. Rational linear algebra and the usual elimination of affine dependence among its risk vectors give a rational optimal mixture of at most $M + 1$ designs. This gives a finite synthesis algorithm, not a bound polynomial in the horizon or the number of reports. Integer determinants of the finite rational constraint matrix give finite bit bounds for all weights, so synthesis does not require a probability oracle.

Let its weights be $\lambda_1, \dots, \lambda_k$. For $r < k$ set $\tilde{\lambda}_r = 2^{-p} \lfloor 2^p \lambda_r \rfloor$ and let the last weight be the nonnegative remainder. Then

$$\|\lambda - \tilde{\lambda}\|_{\text{TV}} = \sum_{r < k} (\lambda_r - \tilde{\lambda}_r) \leq (k - 1) 2^{-p}.$$

For every row, its risk lies in $[0, 1]$, so the difference of its mixed risks is at most this total variation. Taking the maximum and subtracting the unchanged baseline proves (37.2).

A uniform p -bit integer, compared to the cumulative integer weights, selects the dyadic design exactly. Its deterministic tables then implement the scheme. There are k copies of the tables counted in (37.1) and k integer weights of at most $p + 1$ bits, which proves the description bound. In the private implementation the index is part of (r, I_t) and hence multiplies the width by k . In the shared implementation the seed is explicitly allowed throughout the run; its index can instead be reread from that permitted input. A modified or data-dependent seed would not qualify for this exception.

For a conservative machine model, store the immutable description as a sequential binary table with self-delimited addresses. Locate an update, a controller action, or a terminal answer by scanning it. Counters need only logarithmic space in its length and in the finite alphabet sizes. Comparing the p -bit initialization integer with its cumulative weights requires $O(p)$ extra space and $O(kp)$ time by a sequential scan. Copying the selected finite answer and updating the label has the same bounds. Thus the displayed time and space bounds are valid without a random-access assumption. They are deliberately not optimized. $\square$

Proposition 37.2 (Stability of finite synthesis). *Suppose two finite instruments have matched alphabets, losses and timing. Their initial hidden laws have total variation distance at most η_0 for*

every parameter, and the corresponding transition–report rows have distance at most η_t uniformly in hidden state and control. Then, for every admissible common–encoder/controller design, the difference of each bounded-loss risk is at most $\varepsilon = \min\{1, \eta_0 + \sum_{t < T} \eta_t\}$. Their full-history baselines differ by at most ε , and their shared or private excess spectra at the same signature differ by at most 2ε . An additional uniform error ζ in the loss adds 2ζ to this last bound.

Proof. Couple the initial hidden states maximally. Until the two trajectories first disagree, choose the same controls, report-driven updates and random bits in both executions. Couple the corresponding hidden/report transition rows maximally. The probability of a disagreement by the last time is at most $\eta_0 + \sum \eta_t$. A bounded loss changes in expectation by at most that probability. This argument is uniform over every parameter and every legal controller, including full-history controllers. Infima over controllers and common encoders, and maxima over tasks or parameters, are nonexpansive in uniform error. Apply that observation once to the constrained risk and once to its baseline. A uniformly perturbed loss changes each of these by at most ζ . $\square$

This provides a multi-resource finite theorem with computable synthesis, finite randomization, finite description and finite execution cost. Together with Theorem 36.1, it also makes finite sufficient reconstruction effective whenever the expanded finite source is given rationally. It does not compute multiplicative energies for an arbitrary infinite predictive tree, nor does it turn a merely measurable sufficient statistic into a finite algorithm. Those are different input classes with different resource signatures.

Exact rational reconstruction tables need not have dyadic probabilities. Their computability does not imply exact sampling with a fixed number of fair bits. A fixed-bit implementation rounds their rows and uses Proposition 37.2 to charge the resulting path error; exact non-dyadic sampling requires a different randomization-time convention.

APPENDIX 38. A GAUSSIAN EXPERIMENT WITH A GENUINE TEMPORAL PENALTY

The append-register Gaussian theorem in the supporting development concerns a terminal vector experiment. The following experiment instead forces information to survive until the question that uses it arrives. Its final target is scalar, but its causal exponent depends on the number of coordinates acquired before that question.

Fix $d \geq 2$ and $\sigma > 0$. Let $\theta_1, \dots, \theta_d$ be independent and uniform on $[-1, 1]$, and observe sequentially

$$X_j = \theta_j + \sigma Z_j, \quad 1 \leq j \leq d,$$

where the Z_j are independent standard Gaussians. After these observations have been discarded, an independent uniform query $J \in \{1, \dots, d\}$ is revealed. The machine then updates its register and estimates θ_J . There is no external observation transcript. The query is an input to that final update, not an additional persistent decoder register. The cost is squared loss, bounded by four after projecting the answer onto $[-1, 1]$. Write

$$m(x) = \mathbb{E}[\theta_1 \mid X_1 = x], \quad \beta = \mathbb{E} \text{Var}(\theta_1 \mid X_1).$$

Let $\mathcal{E}_S$ be the infimal excess above β when the total register has at most S states at every stage. Independent shared randomness is permitted; the same conclusions hold without it.

Definition 38.1 (Delayed-query information signature). There are d sequential scalar-acquisition updates $I_j = F_j(I_{j-1}, X_j, U_j)$, followed by one query update $I_{d+1} = F_{d+1}(I_d, J, U_{d+1})$ and a terminal output $\hat{\theta} = D(I_{d+1}, U_{d+2})$. Each register has at most S values. The U_j are independent private draws; an optional independent retained public seed may be an argument of every rule. The query is independent of the pre-query history, and neither J nor an old report is available to the terminal decoder except through I_{d+1} . The cardinality signature permits arbitrary measurable real arithmetic. Proposition 39.2 gives a distinct finite-acquisition and finite-table implementation.

Theorem 38.2 (Delayed Gaussian query law). *There are constants $0 < c_{d,\sigma} \leq C_{d,\sigma} < \infty$ such that for every $S \geq 2^d$,*

$$c_{d,\sigma} S^{-2/d} \leq \mathcal{E}_S \leq C_{d,\sigma} S^{-2/d}. \quad (38.1)$$

The same order holds for the maximum over the d individual coordinate query risks, with a common encoder before the query.

In contrast, an encoder which receives the query together with the full observation vector at its sole checkpoint has excess of order S^{-2} . Thus the scalar terminal target has a strictly worse cardinality exponent under the delayed-query causal constraint. The lower bound uses the register before the query and is not a final-output dimension bound.

Proof. The posterior density on $[-1, 1]$ is proportional to $\exp(-(x - u)^2/(2\sigma^2))$. Differentiation under the integral on this compact interval gives

$$m'(x) = \frac{\text{Var}(\theta_1 \mid X_1 = x)}{\sigma^2} > 0. \quad (38.2)$$

The posterior variance is positive because its density is positive on an interval. The function m is smooth, odd, and strictly increasing. Choose $b > 0$ and put $a = m(b) > 0$. The density of X_1 is positive and continuous, and m' is positive and continuous on $[-b, b]$. The change-of-variables formula therefore gives a density for $m(X_1)$ which is bounded below by a positive constant on $[-a, a]$. Independence implies that $M = (m(X_1), \dots, m(X_d))$ has density bounded below by a positive constant on the cube $Q = [-a, a]^d$.

We record an elementary volume bound. If $c_1, \dots, c_S$ are arbitrary points in $\mathbb{R}^d$, and v_d is the volume of the unit ball, choose $r = (|Q|/(2Sv_d))^{1/d}$. The union of their radius- r balls has volume at most $|Q|/2$. Consequently

$$\int_Q \min_{i \leq S} \|x - c_i\|^2 dx \geq \frac{|Q|}{2} \left(\frac{|Q|}{2Sv_d} \right)^{2/d}. \quad (38.3)$$

This bound does not require cells to be convex or centres to lie in Q .

Consider any admissible machine and condition on its independent shared random seed, when present. Its pre-query register I_d has at most S values. Subsequent private updates and readouts can do no better than being given (I_d, J) and using its conditional posterior mean. Let $c_i = (\mathbb{E}[\theta_j \mid I_d = i])_{j=1}^d$, with the seed included in all these conditional expectations. The Markov relation from the full observation to the register, and independence of the random seed, imply $c_i = \mathbb{E}[M \mid I_d = i]$. Orthogonality of conditional expectation shows that the resulting excess for the uniform query is

$$\frac{1}{d} \mathbb{E} \|M - c_{I_d}\|^2 \geq \frac{1}{d} \mathbb{E} \min_{i \leq S} \|M - c_i\|^2.$$

For each fixed seed the law of M is unchanged. Its density lower bound on Q and (38.3) give $c_{d,\sigma} S^{-2/d}$. Averaging the seed preserves this lower bound. The argument permits randomized private encoding: replacing its selected centre by the nearest centre is only a lower bound, not an assumption that the encoding is deterministic. The maximum coordinate risk is at least the uniform average, proving its lower bound as well.

For the upper bound put $L = \lfloor S^{1/d} \rfloor \geq 2$ and partition $[-1, 1]$ into L equal intervals. At observation j compute the interval label of $m(X_j)$ and append it in base L to the stored labels. At that time there are at most $L^j \leq S$ states. After the query, replace the whole label by the interval label of the requested coordinate; the terminal decoder outputs its midpoint. This last update leaves only L possible labels and retains no separate query. Its squared excess in every coordinate is at most $L^{-2} \leq 4S^{-2/d}$. Conditional orthogonality again subtracts the same full-observation baseline β . The construction is deterministic and works with or without a shared seed.

For the noncausal checkpoint comparison, the encoder sees (X, J) and quantizes only $m(X_J)$ into S equal intervals. This has excess at most S^{-2} . The one-dimensional version of the same

density lower bound and (38.3) gives the matching lower bound, even with randomized encoding and a shared seed. Its S terminal centres are arbitrary real answers in $[-1, 1]$, so no additional discreteness assumption has entered. This proves all assertions. $\square$

The theorem is a Bayesian slice of the parameterized Gaussian experiment with the specified product prior, not a claim about uniform deficiency over the whole Gaussian family. Its constants depend on d and σ . It controls persistent cardinality and finite acquisition count, not the arithmetic cost of evaluating the real posterior mean. Under a finite numerical signature, an approximation of m requires a separate accuracy and computation bound. The obstruction nevertheless holds for every such implementation, since it was proved for the larger unrestricted-arithmetic class.

The lower bound is the same conditional-variance geometry as Corollary 35.2, now evaluated directly for a continuous history law. In the finite theorem, successor compatibility determines which cells can be executed. Here the late query forces all coordinate readouts to factor through the same pre-query cell, and the volume bound computes its cost. This explains both the lossless append-only regime and a regime where forgetting before a future question changes the exponent, without identifying either one with static terminal dimension.

APPENDIX 39. MARKED PROJECTIONS AND FINITE NUMERICAL ACQUISITION

The common object behind the decision problems is the attainable family of joint laws under one admissible encoder, with its task controllers and its actual target marks. Evaluating a bounded loss is a linear projection of this family. A risk body for one selected task family is therefore a projection, not by itself a classification of the whole experiment. We record the exact links to the quantitative regimes used later in the article.

Proposition 39.1 (Quadratic and metric projections). *For an uncontrolled standard Borel history H and a square-integrable Hilbert-valued target U , write $m(H) = \mathbb{E}[U \mid H]$. Any statistic I formed from H with independent randomization has optimal quadratic excess*

$$\mathbb{E} \|m(H) - \mathbb{E}[m(H) \mid I]\|^2. \quad (39.1)$$

For finitely many such tasks the common-encoder weighted support problem is the conditional variance of the direct sum of their posterior means multiplied by the square roots of their task weights. On finite histories this is precisely the compatible-partition functional in Corollary 35.2. On the predictive trees of Theorem 53.1, its within-cell variances are comparable to the retained tree-energy sums under the hypotheses of that theorem.

For a revealed law $\pi \in \mathcal{P}([0, 1])$, take as actual target π and as output a realized probability law $\hat{\pi}$. The loss $W_1(\pi, \hat{\pi})$ yields exactly the law-presentation profile of Theorem 49.1. Replacing the output by a sample, and testing only its marginal, is a different task and does not preserve this profile.

Proof. Conditional on (H, I) the target mean is $m(H)$. Orthogonal projection gives

$$\mathbb{E} \|U - a(I)\|^2 = \mathbb{E} \|U - m(H)\|^2 + \mathbb{E} \|m(H) - a(I)\|^2.$$

The second term is minimized by $a(I) = \mathbb{E}[m(H) \mid I]$, proving (39.1). Weighted sums are squared norms in the Hilbert direct sum. For a deterministic finite partition, conditioning on I is averaging over its cell, giving the stated finite functional. The lower and upper comparisons with tree energy are exactly the separation and diameter estimates of Theorem 53.1; no new equality between energy and variance is asserted. The continuous Gaussian volume argument is another evaluation of (39.1), not an invocation of the finite-history partition theorem on a continuous history space.

For the second assertion, the joint output loss of an encoder and its random decoder is, by definition, $\mathbb{E} W_1(\pi, D(Q(\pi)))$. The decoder must output a law on each run. This is the profile in (49.1), whose minimax order is $1/\log(S+1)$ under its stated common-code convention. The

single-marginal probe in Proposition 49.3 has a different output and a different rate. Integrating out the realized law before applying the distance changes the loss, so does not commute with this projection. $\square$

The revealed-law family is generally nondominated. It belongs to the marked-spectrum definition, but is not being smuggled into the product-dominated compactness theorem. Conversely the latter has a proved closed attainable body, whereas a general infinite family of tasks need not have one without hypotheses. Terminal statistical deficiency uses the decision-complete task family and compactness specified in Theorem 23.4; a finite selection of task supports is not sufficient to infer Blackwell equivalence. These distinctions allow the tree, Wasserstein, finite Gaussian and count bounds to be used together without equating their output spaces.

39.1. A finite table for the delayed Gaussian problem. Retain the notation and information pattern of Definition 38.1. The following bound separates acquisition, numerical evaluation and persistent storage.

Proposition 39.2 (Finite-precision Gaussian implementation). *Let $L = \lfloor S^{1/d} \rfloor \geq 2$. An acquisition interface returns one of the interior bins of a grid of width at most h on $[-R, R]$ and two tail symbols, where $R > 1$. Let the stored table approximate $m(x)$ at each interior bin midpoint to error at most ε , with values clipped to $[-1, 1]$. Quantize those approximations into L equal intervals and store the resulting labels in the base- L register. Then the squared excess in every coordinate is at most*

$$\left(L^{-1} + \frac{h}{2\sigma^2} + \varepsilon\right)^2 + 8 \exp\left(-\frac{(R-1)^2}{2\sigma^2}\right). \quad (39.2)$$

For fixed computable $\sigma > 0$, there are rational uniform-grid choices with $1/(2L) \leq h \leq 1/L$, $\varepsilon \leq 1/L$, and $1 + \sigma\sqrt{6\log L} \leq R \leq 2 + 2\sigma\sqrt{6\log L}$ that give excess $O_\sigma(L^{-2})$. The number of acquisition symbols is $O_\sigma(L\sqrt{\log L})$, the table has $O_\sigma(L(\log L)^{3/2})$ bits, and each scalar acquisition uses $O_\sigma(\log L)$ report bits. For rational σ^2 the table is computable by a terminating certified rational procedure. Its offline evaluation cost is separate from the bit-serial lookup and register-update costs.

Proof. Equation (38.2) and $|\theta| \leq 1$ imply $0 < m'(x) \leq \sigma^{-2}$. An interior-bin midpoint differs from X by at most $h/2$, so its posterior mean differs by at most $h/(2\sigma^2)$. Table evaluation and the final equal-interval quantization add at most ε and $1/L$. On a tail event, any clipped output differs from $m(X)$ by at most two. Since $X = \theta + \sigma Z$ and $|\theta| \leq 1$,

$$\mathbb{P}(|X| > R) \leq 2 \exp(-(R-1)^2/(2\sigma^2)).$$

The conditional-variance identity subtracts the same full-observation baseline and gives (39.2). This estimate holds coordinatewise; no union bound over all coordinates is needed for a single queried coordinate. The pre-query and post-query register updates are exactly the base- L construction of Theorem 38.2.

There are $O(R/h)$ interior bins. Only their final labels in $[L]$ need be stored after certified preprocessing; thus the table uses $O((R/h) \log L)$ bits. One may include the rational grid endpoints and the two prescribed tail labels in the fixed program. Choosing rational upper and lower approximations with constant-factor slack gives the asserted bounds without requiring an exact logarithm or square root in the online machine. A sequential lookup scans the finite table; binary arithmetic then appends or extracts a base- L digit. It uses $O(d \log L + \log(R/h))$ transient bits besides the read-only program and polynomial time in the table length and the register bit length. The physical analog-to-bin acquisition is an explicit interface operation, not claimed to be implemented for free by the register machine.

For completeness, certified values of m are available without an exact real oracle when σ^2 and the grid are rational. They are ratios of the integrals

$$\int_{-1}^1 u e^{-(x-u)^2/(2\sigma^2)} du, \quad \int_{-1}^1 e^{-(x-u)^2/(2\sigma^2)} du.$$

The denominator is at least $2e^{-(R+1)^2/(2\sigma^2)} > 0$ for $|x| \leq R$. Rational Taylor bounds for the exponential, combined with Riemann-sum error bounds from uniform derivative bounds on the compact rectangle, give rational intervals for both integrals of arbitrarily small width. Refine until the quotient interval has width at most ε , and choose its rational midpoint. If it lies near an equal-interval threshold, either adjacent label still has error at most $1/L + \varepsilon$ after enlarging the certified tolerance by a fixed factor, or the rational midpoint may simply be compared exactly to the rational thresholds. Termination follows from the denominator lower bound. The same construction for fixed computable σ uses certified intervals for it. No polynomial bound in the unspecified description of a general computable real is asserted. $\square$

The cardinality converse remains valid for this finite numerical signature because its implementations form a subclass of the unrestricted measurable machines used in that converse. Finite acquisition therefore preserves the delayed-query exponent when its error parameters are scaled as above, while displaying the additional resources that the cardinality theorem alone does not price.

APPENDIX 40. POSITIVE OBSERVATION MODELS AND HISTORY REDUCTION

We give explicit model consequences in which the marked law, continuity and contraction hypotheses can be checked from primitive data. These results concern the models specified here. They do not identify a finite-state model with an unconstructed microscopic billiard or kinetic model.

Let $Z_t \in [n]$, $a \in A$ with A finite, and suppose

$$\epsilon \leq P_a(z, z') \leq M, \quad Y_{t+1} = m_{a, Z_{t+1}} + \sigma_{a, Z_{t+1}} \xi_{t+1}, \quad \xi_t \sim N(0, 1), \quad (40.1)$$

where $0 < \epsilon < M$, $|m_{a,z}| \leq m_*$ and $0 < \sigma_- \leq \sigma_{a,z} \leq \sigma_+$. All primitive coefficients may depend continuously on a compact unknown parameter, with the same bounds. The initial hidden probabilities and any finite marking kernel also depend continuously on this parameter. The observation is of the *new* state. Its joint transition/report kernel is

$$K_a(z', dy | z) = P_a(z, z') g_{a,z'}(y) dy.$$

Finite terminal or path marks are permitted.

Theorem 40.1 (A continuous-report consumer). *For every fixed horizon, the instruments (40.1) satisfy the compact dominated hypotheses of Theorem 26.3 for bounded losses continuous in the primitive parameter. Their common-encoder minimax problems have attaining private designs and attaining shared mixtures.*

Partition $[-R, R]$ into intervals of length at most h , where $R > m_$, and use two additional tail cells. Reconstruction is uniform on interior intervals and uses any fixed probability density on each tail. For every memory width and every admissible controller, the resulting marked finite-report approximation has risk error at most $Te(h, R)$, where*

$$e(h, R) = \frac{h}{2\sigma_-} \sqrt{\frac{2}{\pi}} + 2 \exp\left(-\frac{(R - m_*)^2}{2\sigma_+^2}\right). \quad (40.2)$$

The corresponding excess-risk error is at most $2Te(h, R)$ when baselines are recomputed.

Proof. Use a positive Gaussian probability as the reference measure on each report space and counting probability on finite marks. With an open-loop action word fixed, the joint marked density is a finite sum over hidden paths of products of transition probabilities, Gaussian densities and terminal marking probabilities. Gaussian location-scale densities depend continuously in L^1 on their parameters when the variance is bounded away from zero. This follows, for example, by pointwise convergence and equality of their integrals, or directly by truncating uniformly Gaussian tails and using uniform continuity on a compact interval. Products of probability densities are continuous in L^1 by telescoping, and so are finite mixtures. Compactness of the primitive parameter set gives compactness of the path-density family in L^1 . Multiplication by bounded continuously parameterized losses preserves the required effect continuity. Theorem 26.3 applies.

If I is an interval of length at most h and $\bar{g}_I$ is its average density, the fundamental theorem of calculus gives

$$\int_I |g - \bar{g}_I| \leq h \int_I |g'|.$$

Indeed average $|g(x) - g(y)|$ over $y \in I$ and bound the difference by the integral of $|g'|$ between x, y . For a Gaussian density, $\int |g'| = \sqrt{2/\pi}/\sigma$. On the two tails, replacing a conditional density changes total variation by at most their total probability, bounded by the exponential term in (40.2). Thus each conditional observation channel has total variation error at most $e(h, R)$ uniformly in hidden state, control and parameter. Couple the hidden transition identically and maximally couple the original and reconstructed observations. As long as observations and random tapes have matched, both controllers make the same decisions. The probability of a first mismatch during T steps is at most $Te(h, R)$. On its complement the entire hidden path, all reports and the actual mark agree. Bounded losses change by at most this probability. The report conversions are transient operations, so the memory width is unchanged. Comparing the unrestricted baselines in the same way finishes the proof. $\square$

40.1. A proved forgetting estimate. For positive probability rows u, v , put

$$d_H(u, v) = \max_i \log(u_i/v_i) - \min_i \log(u_i/v_i).$$

Multiplying both rows by the same positive diagonal matrix and then normalizing leaves d_H unchanged. We use the elementary positive matrix contraction in its explicit form.

Lemma 40.2 (Positive-matrix contraction). *If the entries of a matrix P belong to $[\epsilon, M]$, then*

$$d_H(uP, vP) \leq \rho d_H(u, v), \quad \rho = \frac{M - \epsilon}{M + \epsilon} < 1.$$

For arbitrary nonnegative probability rows, the first images satisfy $d_H(uP, vP) \leq 2 \log(M/\epsilon)$.

Proof. Set $x_i = \log(u_i/v_i)$ and interpolate $v_i \exp(sx_i)$, $0 \leq s \leq 1$. The derivative of the logarithm of output coordinate j is the expectation of x_i under weights proportional to $v_i \exp(sx_i) P_{ij}$. For two output coordinates j, k , the likelihood ratio of their normalized weight distributions has maximum divided by minimum at most $(M/\epsilon)^2$. If two probability distributions have likelihood ratios in an interval $[a, b]$ with $b/a \leq C$, their total variation is at most $(\sqrt{C} - 1)/(\sqrt{C} + 1)$. To verify this bound, put all ratios below one at the lower endpoint and all ratios above one at the upper endpoint; the mass constraint makes the resulting variation at most $(1 - a)(b - 1)/(b - a)$. Maximizing over $b/a \leq C$, $a \leq 1 \leq b$, gives $a = C^{-1/2}$, $b = C^{1/2}$ and the displayed value. Differences of the two derivative expectations are therefore at most $\rho \text{osc}(x)$. Integrate in s and maximize over j, k . At $s = 0$ all output log ratios vanish, proving the first claim. For the second, every coordinate of either probability row times P lies in $[\epsilon, M]$; its coordinatewise ratio lies in $[\epsilon/M, M/\epsilon]$. $\square$

Theorem 40.3 (Uniform finite-history recovery). *Fix the primitive parameter. Two filters for (40.1), started with arbitrary initial probabilities and fed the same actions and observations for $L \geq 1$ steps, satisfy*

$$\mathrm{TV}(\pi_L, \tilde{\pi}_L) \leq \frac{1}{2} \log(M/\epsilon) \rho^{L-1}.$$

In particular a last- L -history filter, initialized with a fixed probability at its left endpoint, approximates the full filter with this error uniformly in the earlier history. With N report cells, a literal suffix implementation uses at most $(|A|N)^L$ labels, besides a finite beginning-of-stream convention. For any $[0, 1]$ -valued terminal loss on the current hidden state, an optimal suffix-based action has risk at most twice this filtering error above the full-filter optimum.

Proof. Prediction multiplies a posterior row by P_a . Conditioning on the current report multiplies it by the positive diagonal matrix with entries $g_{a,z'}(y)$ and normalizes. The latter operation preserves d_H , and Lemma 40.2 contracts the former. The first prediction gives diameter at most $2\log(M/\epsilon)$ and the next $L - 1$ predictions give the factor ρ^{L-1} . For probability rows, $\mathrm{TV}(u, v) \leq \tanh(d_H(u, v)/4) \leq d_H(u, v)/4$; the first inequality follows from the same two-endpoint likelihood-ratio bound in the lemma. Restart the truncated filter with the fixed prior at time $t - L$ and apply this result to obtain the suffix estimate. Its raw suffix is updated by shifting and appending the current control/report cell. Finally every $[0, 1]$ loss expectation changes by at most the total variation of the two posteriors. Compare the optimal action under each posterior through the other posterior, which costs at most twice that distance. $\square$

This contraction is not asserted for arbitrary past-path target marks: augmenting the hidden state to retain such a mark can destroy strict positivity. The general marked duality retains these marks without using the forgetting assertion.

The parameter must be fixed in this forgetting assertion. A static unknown phase need not be forgotten, and applying the assertion to its posterior would be false. The compact robust theorem applies to the unknown-parameter experiment without that extra assertion. Also, summable variations alone have not been used to infer exponential decay: here it follows from the explicit positive-matrix contraction. The distinction is important for history-kernel applications.

APPENDIX 41. NONLINEAR FINITE-STATE SEMIGROUPS

Let Q^a be conservative generators on $[n]$, $a \in A$ finite, and let v_i^a be bounded potentials. Define on $\mathbb{R}^n$

$$H_i(u) = \max_{a \in A} \left\{ v_i^a + \sum_{j \neq i} q_{ij}^a (e^{u_j - u_i} - 1) \right\}. \quad (41.1)$$

This is a nonlinear Hamiltonian. The following proof does not apply a linear resolvent identity to it.

Theorem 41.1 (Global semigroup and nonlinear resolvent). *The equation $\dot{u} = H(u)$ has a global solution for every initial condition and defines a semigroup S_t that is order preserving, additively homogeneous and nonexpansive in the sup norm. If $V = \max_{i,a} |v_i^a|$, then $\|S_t u\|_\infty \leq \|u\|_\infty + tV$. For every $\lambda > 0$ the equation*

$$u - \lambda H(u) = f$$

has a unique solution $J_\lambda f$. The resolvent is order preserving, additively homogeneous and nonexpansive. In particular H is maximal dissipative with this sign convention.

Proof. The map H is locally Lipschitz as the maximum of finitely many smooth functions. If $u_i - v_i = \max_j (u_j - v_j)$, then $u_j - u_i \leq v_j - v_i$ for every j , and hence $H_i(u) \leq H_i(v)$. The upper right derivative of the maximum coordinate of the difference of two solutions is therefore nonpositive. Applying the same argument to the reverse difference proves sup-norm contraction

and comparison. Since $H(u + c\mathbf{1}) = H(u)$, uniqueness gives additive homogeneity. Comparing with the time-dependent constant barriers $(\max_i u_i + tV)\mathbf{1}$ and $(\min_i u_i - tV)\mathbf{1}$ gives the norm bound. It prevents finite-time escape, so the locally defined solution is global; uniqueness gives the semigroup law.

For the resolvent, consider the damped equation $\dot{u} = f - u + \lambda H(u)$. The interval of constant barriers

$$(\min_i f_i - \lambda V)\mathbf{1} \leq u \leq (\max_i f_i + \lambda V)\mathbf{1}$$

is invariant. At a maximal difference between two solutions the upper derivative is at most the negative of that difference. Thus its flow contracts distances by e^{-t} . At time one this is a strict contraction of a complete invariant rectangle into itself, so it has a unique fixed point. Commutation with the flow shows that the fixed point is stationary: every later image is another fixed point of the time-one map. It therefore solves the resolvent equation. The same maximal-coordinate comparison applied to two resolvent equations gives $\max_i (u_i - \tilde{u}_i) \leq \max_i (f_i - \tilde{f}_i)$, which proves uniqueness, comparison and nonexpansiveness. Translation by constants proves homogeneity. $\square$

Theorem 41.2 (Marked mesh approximation and stability). *Define the log-moment Bellman step*

$$(F_\Delta u)_i = \max_a \log \left(e^{\Delta(Q^a + \text{diag } v^a)} e^u \right)_i.$$

For fixed t and bounded initial u , with $\Delta = t/m$, $F_\Delta^m u \rightarrow S_t u$ with error $O(\Delta)$, uniformly on bounded sets. Each mesh problem, after acquisition of the initial state, has finite report and control spaces and a bounded actual path-reward mark. Its bounded moment effects therefore have the common-encoder risk-body duality of Corollary 26.4, for compact continuously parameterized primitive data. The seed average is taken on these moments before taking a logarithm.

If $\tilde{Q}^a, \tilde{v}^a$ are another such family and

$$\delta_Q = \max_{i,a} \sum_{j \neq i} |q_{ij}^a - \tilde{q}_{ij}^a|, \quad \delta_v = \max_{i,a} |v_i^a - \tilde{v}_i^a|,$$

then on a common oscillation bound B for the trajectories,

$$\|S_t u - \tilde{S}_t u\|_\infty \leq t\{\delta_v + (e^B + 1)\delta_Q\}.$$

For the resolvents the same bound holds with t replaced by λ and B bounding their solutions.

Proof. The matrix exponential is a positive Feynman–Kac operator. On a bounded set of u , its Taylor expansion and the logarithm give $F_\Delta u = u + \Delta H(u) + O(\Delta^2)$, uniformly for the finite family of generators. Positivity makes F_Δ order preserving, and $F_\Delta(u + c\mathbf{1}) = F_\Delta u + c\mathbf{1}$. These two properties imply sup-norm nonexpansiveness. Its iterates and the exact flow remain in the same bounded region up to time t , by the constant barriers. The local flow expansion has the same first-order term and a uniform $O(\Delta^2)$ remainder. Telescope the errors over m steps using nonexpansiveness; the bound is $mO(\Delta^2) = O(t\Delta)$.

For a fixed control during one interval, the positive kernel $e^{\Delta(Q^a + \text{diag } v^a)}$ is the expectation of the exponential integrated reward with the terminal state indicator. Keeping the actual bounded reward integral as a mark, rather than interpreting this nonstochastic matrix as a transition probability, gives its proper probabilistic representation. Concatenation gives the mesh instrument. Finite sampled states and controls have a finite report representation; continuous reward marks can be integrated into its bounded effect functions. For this reward-maximizing problem, use a decreasing affine normalization of the bounded moment as a loss in $[0, 1]$ when applying the risk theorem. This reverses the optimization in the required way. For finite state/action paths this integration needs no assumption that reward marks have a common density: Corollary 26.4 applies directly. For finite parameter and task sets, one may also augment the finite hidden state by the sampled state/control path and replace the moment loss by its conditional expectation

given that path. This gives a finite instrument with exactly the same moment risks, to which Theorem 34.1 applies. It is an equivalence for these moment effects, not a finite-state realization of every continuous reward mark. Continuity of the effects on a compact primitive family follows from continuity of the finite matrix exponentials and products.

On $\text{osc}(u) \leq B$, direct subtraction of (41.1) gives $\|H(u) - \tilde{H}(u)\|_\infty \leq \delta_v + (e^B + 1)\delta_Q$. The maximal-coordinate comparison used above now bounds the upper derivative of the solution difference by this constant. Integration proves the flow estimate. Subtracting the two resolvent equations at a maximizing coordinate proves its resolvent counterpart. The logarithm of an average exponential moment is not an average of logarithms; none of these steps changes that ordering. $\square$

The semigroup construction is a finite-state instance of the logarithmic control-semigroup method of Nisio [59] and risk-sensitive control. Its role here is an explicit nonlinear consumer with a proved range condition and a marked finite-mesh interface, not a claim that finite-dimensional comparison settles the domain and coercivity questions of an infinite-dimensional kinetic generator.

APPENDIX 42. HIDDEN PHASES AND RISK-SENSITIVE CONTINUATION

Let C be a finite latent phase chosen once with positive initial weights, and let Z_t follow a controlled hidden model conditional on C . Its joint transition/report densities are $K_{c,a}(z', y | z)$ with respect to a probability measure $\mu(dy)$. Let $r_{c,a}(z, z')$ and the terminal reward $h_c(z)$ be bounded, and fix $\gamma > 0$. One controller must choose each action without observing C . The objective in this section is to minimize the positive exponential moment. A logarithm may be applied afterwards.

The exponential information-state mechanism is classical; see Bäuerle and Rieder [60]. We use the following explicit unnormalized version so that no phasewise policy is hidden in the notation. Define a nonnegative row η on (c, z) and the tilted update

$$(\eta K_a^\gamma(y))(c, z') = \sum_z \eta(c, z) K_{c,a}(z', y | z) e^{\gamma r_{c,a}(z, z')}. \quad (42.1)$$

Unlike an ordinary phase posterior, η includes the accumulated reward weight. Its total mass must not be discarded without a corresponding multiplicative value factor.

Theorem 42.1 (One-policy phase recursion). *For full-history controls, the minimum remaining exponential moment has the recursion*

$$V_T(\eta) = \sum_{c,z} \eta(c, z) e^{\gamma h_c(z)}, \quad V_t(\eta) = \min_{a \in A} \int V_{t+1}(\eta K_a^\gamma(y)) \mu(dy).$$

The value is positively homogeneous and Borel, and an optimal measurable selector exists. The minimum is outside the phase sum: replacing it by separate phasewise minima is in general incorrect. For finite-register common encoders, the exact moment risk body is instead obtained from the coupled design of Theorem 26.3 (or its finite specialization), not by providing η as free online memory. The model satisfies that theorem whenever its marked path densities and bounded effects satisfy the stated domination and continuity hypotheses; in particular finite phases with positive Gaussian observations and bounded finite-path rewards do so for fixed primitives, or for compact continuous primitive families as in Theorem 40.1, using Corollary 26.4 when only effect domination is needed.

Proof. For a fixed report/action history, $\eta_t(c, z)$ is the joint density of phase, current state and that report history, multiplied by the exponential accumulated reward and summed over past hidden states. This interpretation is true initially and propagates exactly by (42.1). At the terminal time summing the remaining exponential terminal factor gives V_T . At time t the action is one measurable choice from the current history; it cannot depend on an unobserved phase. For a fixed

choice a , integrate the remaining value of the updated density against the reference observation law. Backward induction gives the recursion. The minimum of finitely many Borel integrals is Borel and the least-index minimizer is a measurable selector. Induction also gives positive homogeneity. This is the unnormalized information-state form of risk-sensitive dynamic programming; it accounts for the probability and reward weights together.

For a finite-register design, fix its action and register paths and expand the true marked path law. Its exponential reward is a bounded positive loss after affine rescaling. The same encoder kernels appear in every task and phase row, so the compact risk theorem applies without giving the controller the unbounded-dimensional history or an uncharged exact real vector. The Gaussian model's domination and compactness follow from the finite mixture proof in Theorem 40.1. This proves the last assertion. $\square$

Example 42.2 (Strict failure of phasewise optimization). There is no observation. The equally likely phase is $C \in \{0, 1\}$, the action is $a \in \{0, 1\}$, and the cost is $\mathbf{1}_{\{a \neq C\}}$. The best actual exponential moment is $(1 + e^\gamma)/2$, for any mixture of the two actions. Its logarithmic value is $\gamma^{-1} \log((1 + e^\gamma)/2)$. Choosing $a = C$ separately in each hidden phase would give moment one, but is not an admissible policy. The tilted recursion and the common-encoder risk body both retain the actual value.

APPENDIX 43. A MICROSCOPIC GRAPH CORE AND WIDTH-UNIFORM ACQUISITION TRANSFER

The comparison theorem can be used before a kinetic limit is taken. Here the underlying trajectories are actual hard-sphere trajectories, not a finite hidden-state surrogate. We construct a collision-compatible core for their unbounded transport generator and show that replacing observables by this core preserves every bounded-register decision problem under noisy acquisition. The topology of the core is specified explicitly: it is a weighted L^p graph topology, not local uniform convergence on kinetic action sublevels.

43.1. The microscopic flow and invariant spaces. Let $\Omega_{N,\epsilon}$ be the nonoverlap phase space of N identical hard spheres of diameter $0 < \epsilon < 1/4$ in the unit flat torus, with unbounded velocities. At a binary nongrazing collision the incoming and outgoing velocities are related by elastic specular reflection. We use the classical almost-everywhere hard-sphere flow $\Phi_t^{N,\epsilon}$, defined for all real times outside the invariant singular null set. The requisite existence statement excludes multiple collisions, grazing pathologies, and finite-time accumulation for almost every initial state; see the hard-sphere flow construction in [63]. This classical flow existence theorem, not a kinetic-limit theorem, is an explicit input here. The cited construction is stated in Euclidean space. Its local exceptional-set argument applies to the periodic system as follows. On a bounded-energy set and a bounded time interval there are only finitely many periodic images that can enter a collision chart. Near a contact, $\epsilon < 1/4$ gives an unambiguous local lift to a Euclidean binary-collision surface. The volume estimate for states with two collisions in an interval of length δ is therefore the same $O(\delta^2)$ estimate, with a finite sum of chart constants; this is the estimate in Lemma 4.1.1 of the cited construction. A union over $O(T/\delta)$ time intervals has volume $O(T\delta)$. Letting δ decrease to zero excludes collision accumulation, together with the multiple and grazing exceptional sets. Exhaust bounded energies and time intervals countably. This supplies the periodic version used here; no uniform-in- N collision estimate is inferred from it.

The normalized equilibrium measure

$$d\mu_{N,\epsilon,\beta} = Z_{N,\epsilon,\beta}^{-1} \mathbf{1}_{\Omega_{N,\epsilon}} \exp(-\beta|V_N|^2/2) dX_N dV_N$$

is invariant. Free flight preserves phase volume, elastic reflection has unit absolute Jacobian and preserves kinetic energy, and their concatenation gives invariance on the regular flow. Null sets are completed in the usual way. We restrict to permutation-symmetric observables.

One may use a fixed nonempty system, or a countable disjoint family of nonempty systems (N, ϵ_N) with positive summable weights a_N . In the latter case put $\bar{\mu} = \sum_N a_N \mu_{N, \epsilon_N, \beta}$, normalized to mass one. This is a family of particle sectors, not a claim that infinitely many positive-radius spheres fit into a fixed torus. Additional particle-number or energy weights may be absorbed into the invariant measure whenever their total mass is finite. Fix $1 \leq p < \infty$ and let $\mathcal{B}_p = L^p_{\text{sym}}(\bar{\mu})$.

Lemma 43.1 (Strong transport and a dense observable class). *The operators $U_t f = f \circ \Phi_t$ form a strongly continuous isometric group on $\mathcal{B}_p$. Let A be its generator. The symmetric smooth observables with finite sector support and compact support in the interior phase space are dense in $\mathcal{B}_p$. The generator is the closed transport operator associated with the physical collision flow; in the interior it acts as $\sum_i v_i \cdot \nabla_{x_i}$.*

Proof. Invariance gives $\|U_t f\|_p = \|f\|_p$ and the flow law gives the group property. First truncate the sector sum, then truncate velocities and remove a neighborhood of the collision boundary. The discarded mass tends to zero. Ordinary smooth approximation on the remaining open phase space, followed by permutation averaging, proves density. For a bounded continuous interior test, $f(\Phi_t z) \rightarrow f(z)$ for almost every initial state as $t \rightarrow 0$: before its first collision an interior trajectory is a free flight. Dominated convergence gives strong continuity on the dense test class; isometry extends it to every f . The generator of a strongly continuous group is closed. Testing against interior smooth functions and differentiating free flight identifies its interior distributional action. The collision identification is inherited from the flow, rather than prescribed by an arbitrary boundary extension. $\square$

Choose a nonnegative $\varphi \in C_c^\infty(\mathbb{R})$ with integral one and write $\varphi_\delta(s) = \delta^{-1} \varphi(s/\delta)$. Define the orbit regularization

$$B_\delta f = \int_{\mathbb{R}} \varphi_\delta(s) U_s f \, ds. \quad (43.1)$$

The integral is a Bochner integral in $\mathcal{B}_p$.

Theorem 43.2 (Collision-compatible graph core). *For every $f \in \mathcal{B}_p$, $B_\delta f \in \bigcap_{m \geq 1} D(A^m)$, and*

$$A^m B_\delta f = (-1)^m \int \varphi_\delta^{(m)}(s) U_s f \, ds, \quad (43.2)$$

$$\|A^m B_\delta f\|_p \leq \delta^{-m} \|\varphi^{(m)}\|_1 \|f\|_p.$$

The linear span of $B_\delta f$, with f in the dense class of Lemma 43.1 and $\delta > 0$, is a graph core for A . There is one diagonal sequence of such observables converging to any $g \in D(A)$ in the graph norm. These observables have equal incoming and outgoing traces at regular specular collisions, in the local collision-flux almost-everywhere sense.

Proof. A change of variables gives

$$U_h B_\delta f = \int \varphi_\delta(s - h) U_s f \, ds.$$

Difference quotients of the scalar compactly supported smooth function converge in $L^1(ds)$. Isometry therefore permits passage to the limit in $\mathcal{B}_p$, proving (43.2) for $m = 1$ and its norm estimate. Iterate to get every m . Strong continuity and the approximate-identity property give $B_\delta f \rightarrow f$ in $\mathcal{B}_p$. If $g \in D(A)$, differentiation under the Bochner integral gives $AB_\delta g = B_\delta Ag$, so $B_\delta g \rightarrow g$ also in graph norm.

To obtain a core with the stated dense tests, choose $\delta_n \downarrow 0$ and a smooth finite-sector f_n with $\|f_n - g\|_p \leq \delta_n^2$. Then

$$\|B_{\delta_n}(f_n - g)\|_p \leq \delta_n^2, \quad \|AB_{\delta_n}(f_n - g)\|_p \leq \|\varphi'\|_1 \delta_n.$$

Together with graph convergence of $B_{\delta_n}g$ this proves the single sequence assertion. There is no passage from an iterated limit to a uniform kinetic estimate in this argument.

It remains to verify the boundary assertion, which is why orbit regularization is used instead of convolution in particle coordinates. For almost every regular orbit, $s \mapsto f(\Phi_s z)$ is locally integrable. Equation (43.1), evaluated on that orbit, is convolution in its time coordinate; it is a smooth function of time. At a nongrazing binary collision, the incoming point and its reflected outgoing point are the same orbit at the same collision time. The two one-sided values of this smooth time function therefore agree. To pass from interior almost-everywhere statements to collision traces, use a local free-flight tube transverse to a regular collision surface. Its phase-volume element is the collision surface element times the absolute normal relative velocity times ds . Fubini's theorem transfers the orbit statement to collision-flux almost every boundary point. Exhaust bounded-velocity, nongrazing, single-collision tubes by countably many such charts. Their complement consists of the excluded singular or grazing sets. This gives the asserted trace convention and no stronger claim of global smoothness across grazing configurations. $\square$

The theorem solves an exact microscopic domain problem in a stated weighted norm. It does not assume a bounded finite-state generator. It also does not identify the transport generator with a nonlinear collision Hamiltonian: the deterministic chain rule remains the transport chain rule.

43.2. Noisy acquisitions of physical observables. Let the initial microscopic state Z have density ρ_θ with respect to $\bar{\mu}$. Assume

$$\sup_{\theta} \|\rho_\theta\|_{p'} \leq M, \quad 1/p + 1/p' = 1. \quad (43.3)$$

A finite acquisition protocol chooses, by feedback, an index $a \in \mathcal{A}_t$ before its t th report. Its scalar report is

$$Y_t = f_{t,a}(\Phi_{t_t} Z) + \sigma_t \xi_t, \quad \xi_t \sim N(0, 1), \quad \sigma_t > 0,$$

where the ξ_t are mutually independent and independent of Z . Each $f_{t,a}$ is a bounded symmetric observable. An actual mark W is sampled from a fixed kernel given Z , independently of acquisition noise; in particular it may be any measurable functional of the entire microscopic trajectory. The acquisition choice does not alter the hard-sphere dynamics. Denote this marked experiment by E_f .

Theorem 43.3 (Graph-core acquisition comparison). *For another finite observable family $g_{t,a}$, the identity report transducers in both directions between E_f and E_g have one internal state, no public seed, and marked feedback error at most*

$$e(f, g) = \min \left\{ 1, \frac{M}{\sqrt{2\pi}} \sum_t \frac{1}{\sigma_t} \sum_{a \in \mathcal{A}_t} \|f_{t,a} - g_{t,a}\|_p \right\}. \quad (43.4)$$

Consequently, for every width profile, every seed bound and every bounded common-task family,

$$\begin{aligned} |h_{E_f}(S, q; w) - h_{E_g}(S, q; w)| &\leq e(f, g), \\ |\Theta_{E_f}(S, q) - \Theta_{E_g}(S, q)| &\leq 2e(f, g). \end{aligned}$$

Taking $g_{t,a} = B_\delta f_{t,a}$ makes these bounds tend to zero uniformly over all register widths and feedback designs, while all approximating observables belong to the microscopic graph domain and satisfy its collision compatibility.

Proof. Couple the two experiments with the same Z , the same mark conditional on Z , and the same private and public randomization of the feedback design. Conditional on a matched report history and on Z , the issued control is the same. The total variation distance between the two next Gaussian report laws is

$$2\Phi\left(\frac{|f_{t,a}(\Phi_{t_t} Z) - g_{t,a}(\Phi_{t_t} Z)|}{2\sigma_t}\right) - 1 \leq \frac{|f_{t,a}(\Phi_{t_t} Z) - g_{t,a}(\Phi_{t_t} Z)|}{\sigma_t \sqrt{2\pi}}.$$

Use a maximal coupling at each step. The probability of a first mismatch is bounded by the expectation of the sum of these bounds on previously matched histories. Bound the selected absolute difference by the sum over all possible a ; this removes the selection's dependence on Z and on earlier noises. By Hölder's inequality and invariance,

$$\int \rho_\theta(z) |f - g|(\Phi_{tt} z) d\bar{\mu}(z) \leq M \|U_{tt}(f - g)\|_p = M \|f - g\|_p.$$

Summing proves (43.4). The mark has been coupled identically throughout. This proves the required uniform marked feedback bound, not just convergence of open-loop marginal observations. The identity channel retains no state. Apply Theorem 24.2 in both directions and (24.3) to get the two decision bounds. Finally strong convergence of B_δ applies to the finitely many observables; Theorem 43.2 supplies the domain and boundary assertions. $\square$

Corollary 43.4 (A physical consumer of compact duality). *Suppose the parameter space is compact, $\theta \mapsto \rho_\theta$ is $L^1(\bar{\mu})$ continuous, and the mark kernel is parameter independent. Let the bounded loss family be uniformly continuous in its compact row index in supremum norm. Then E_f satisfies the dominated loss-effect hypothesis of Theorem 26.3. Its ideal-public risk body is compact, its common-encoder minimax duality is attained, and its graph-core approximations obey Theorem 43.3 under (43.3).*

Proof. For a fixed open-loop control word, condition on Z and multiply the Gaussian report densities. The mixture is absolutely continuous with respect to Lebesgue measure, and hence to a product of standard Gaussian reference probabilities, whose density is everywhere positive. Integrate the actual conditional marked loss before taking this density. A change of preparation by $\|\rho_\theta - \rho_{\theta'}\|_1$ changes the L^1 loss effect by at most that amount times the loss bound. A change of loss costs at most its supremum norm. The row-effect map is therefore L^1 continuous. This verifies the hypothesis rather than presuming it from smoothness of the microscopic dynamics. The stated conclusions now follow from the two cited theorems. $\square$

43.3. The precise historical implication. The B4 microscopic program requires both exact generator-domain objects and a model-specific identification of their kinetic limit. The preceding results supply a boundary-compatible weighted L^p graph core and a width-uniform decision transfer for noisy physical acquisitions. They neither prove the BBGKY recollision estimates nor turn L^p graph convergence into convergence locally uniformly on action sublevels. The latter requires additional uniform integrability and model-specific corrector estimates. In particular the full historical B4 target is not an automatic consequence of the present consumer.

There is a second distinction worth recording. A nonlinear resolvent $J_\lambda = (I - \lambda H)^{-1}$, when it exists uniquely, satisfies

$$J_\lambda f = J_\mu \left(\frac{\mu}{\lambda} f + \left(1 - \frac{\mu}{\lambda} \right) J_\lambda f \right). \quad (43.5)$$

Indeed, put $u = J_\lambda f$ and substitute $Hu = (u - f)/\lambda$ in $u - \mu Hu$. A linear difference/product resolvent identity cannot simply be carried over to H . Exponential resolvents and the required domain extension have an established theory; see [64], Theorem 3.4 and Proposition 3.6. Equation (43.5) is the identity to use in a nonlinear continuation of the microscopic graph argument, not a proof of full range from the existence of the linear Koopman group.

APPENDIX 44. UNBOUNDED GENERATORS, FORCED MEMORY, AND PHYSICAL DECISION LIMITS

We now connect the causal comparison results to the microscopic transport and compressed-memory objects. The analytic input is a generator-domain construction, not a claim that a finite positive model has replaced the physical dynamics. The finite-rank Mori-Zwanzig mechanism is classical; in particular the domain and bounded-perturbation issues are treated in [65, 66]. We give

the block proof in the form needed for a quantitative marked-acquisition comparison, including the unresolved initial component and the full observable.

44.1. A common domain for resolved and orthogonal dynamics. Let H be a complex Hilbert space, with inner product linear in the first argument. Let L generate a strongly continuous contraction semigroup $U(t)$, and let P be an orthogonal finite-rank projection such that

$$R = PH \subseteq D(L) \cap D(L^*), \quad Q = I - P. \quad (44.1)$$

The restriction LP is bounded. The restriction PL on $D(L)$ extends to a bounded map on H : if $e_1, \dots, e_k$ is an orthonormal basis of R , its extension is $f \mapsto \sum_i \langle f, L^* e_i \rangle e_i$. Define

$$A = PLP : R \rightarrow R, \quad B = \overline{PL}Q : QH \rightarrow R, \quad C = QLP : R \rightarrow QH, \quad D = QLQ|_{QD(L)}. \quad (44.2)$$

Here D acts in QH . The bar denotes the bounded extension, not a new choice of a generator domain.

Theorem 44.1 (Forced memory with an explicit orthogonal generator). *Under (44.1), D generates a strongly continuous contraction semigroup $T_Q(t)$ on QH . For every $f \in H$, the resolved trajectory $x(t) = PU(t)f$ is continuously differentiable and satisfies*

$$x'(t) = Ax(t) + \int_0^t K(t-s)x(s) ds + \eta_f(t), \quad x(0) = Pf, \quad (44.3)$$

$$K(t) = BT_Q(t)C, \quad \eta_f(t) = BT_Q(t)Qf. \quad (44.4)$$

For $\sigma > 0$, with $k = \|B\|\|C\|$,

$$\|K(t)\| \leq k, \quad \int_0^\infty e^{-\sigma t} \|K(t)\| dt \leq k/\sigma. \quad (44.5)$$

The compressed resolvent $\mathbf{C}(z) = P(z - L)^{-1}P|_R$ is invertible for $\Re z > 0$, and

$$\mathbf{C}(z)^{-1} = zI_R - A - B(z - D)^{-1}C, \quad (44.6)$$

$$\|\mathbf{C}(z)^{-1}\| \leq |z| + \|A\| + k/(\Re z). \quad (44.7)$$

The forcing transform is both $B(z - D)^{-1}Qf$ and $\mathbf{C}(z)^{-1}P(z - L)^{-1}Qf$. If L is skew-adjoint, D is skew-adjoint on QH , $B = -C^*$, and

$$K(t) = -C^*T_Q(t)C, \quad K(0) = -C^*C. \quad (44.8)$$

Proof. Since $PH \subseteq D(L)$, $D(L) = R \oplus QD(L)$. On this domain L has blocks A, B, C, D . The off-diagonal operator $F = PBQ + QCP$, with the evident inclusions of its ranges in H , is bounded. Equivalently it acts by $F(u + v) = Bv + Cu$. The bounded-perturbation construction shows that $L_0 = L - F$ on $D(L)$ is a generator. One can obtain its semigroup directly by the norm-convergent Dyson series on bounded time intervals; the n th perturbation term is bounded by $\|F\|^n t^n / n!$.

For $u \in R$ and $v \in QD(L)$,

$$\Re \langle L_0(u + v), u + v \rangle = \Re \langle Lu, u \rangle + \Re \langle Lv, v \rangle \leq 0.$$

For domain initial data the derivative of the squared norm of the L_0 orbit is nonpositive. Density extends the contraction bound to all initial data. The projection P preserves the domain and commutes with L_0 , and hence with its resolvent and semigroup. Its restrictions are e^{tA} and a contraction semigroup on QH with generator precisely D . This proves the asserted existence and domain, without writing an undefined exponential of QLQ .

Bounded off-diagonal perturbation and variation of constants now give, for $x(t) = PU(t)f$ and $y(t) = QU(t)f$,

$$\begin{aligned} x(t) &= e^{tA}Pf + \int_0^t e^{(t-s)A}By(s) ds, \\ y(t) &= T_Q(t)Qf + \int_0^t T_Q(t-s)Cx(s) ds. \end{aligned}$$

The first expression is continuously differentiable because A is bounded and $By(s)$ is continuous. In particular this differentiation uses the bounded block integral formula, not a derivative of $U(t)f$ for an arbitrary $f \notin D(L)$. Substituting the second proves (44.3)–(44.4) for every $f \in H$. Contractivity gives (44.5). The operator-valued kernel is norm continuous, since C has finite-dimensional domain and T_Q is strongly continuous.

For $\Re z > 0$, $z - D$ has a bounded inverse of norm at most $(\Re z)^{-1}$. Solving the lower block of $(z - L)(u + v) = g + h$ yields $v = (z - D)^{-1}(h + Cu)$. The upper block is

$$[zI_R - A - B(z - D)^{-1}C]u = g + B(z - D)^{-1}h.$$

The bracket is injective: a null vector would give a null vector of $z - L$ in its domain. It is therefore invertible in the finite-dimensional space R . Taking $h = 0$ identifies its inverse with $C(z)$; taking $g = 0$ proves the forcing identity. The norm estimate follows from the resolvent bound for D .

If $L^* = -L$, the off-diagonal operator F is bounded and skew-adjoint, so $L_0 = L - F$ is skew-adjoint on $D(L)$. Its reducing restrictions are skew-adjoint, and the adjoint relation of the off-diagonal blocks gives $B = -C^*$. This proves (44.8). $\square$

Two distinctions in this statement matter. First, $Qf = 0$ suffices to remove the forcing, but this cannot be assumed for an arbitrary physical observable. Second, (44.5) is a discounted bound on the right half-plane. It does not assert decay of $K(t)$ or meromorphic continuation across the imaginary axis. Those require additional spectral information. In particular no finiteness of left-half-plane transmission zeros is inferred from finite rank alone.

44.2. A stability estimate which retains the full observable. Let $\hat{x}$ solve a finite-dimensional Volterra equation with data $\hat{A}, \hat{K}, \hat{\eta}, \hat{x}_0$, where the kernel and forcing are locally integrable. Suppose $\hat{x}$ is continuous on $[0, T]$ and put $M_T = \sup_{t \leq T} \|\hat{x}(t)\|$. Define

$$\begin{aligned} B_T &= \|Pf - \hat{x}_0\| + T\|A - \hat{A}\|M_T + \int_0^T \|\eta_f(s) - \hat{\eta}(s)\| ds \\ &\quad + M_T \int_0^T (T-s)\|K(s) - \hat{K}(s)\| ds. \end{aligned} \tag{44.9}$$

Proposition 44.2 (Resolved and reconstructed error). *For $k = \|B\|\|C\|$,*

$$\sup_{t \leq T} \|x(t) - \hat{x}(t)\| \leq B_T \cosh(\sqrt{k} T). \tag{44.10}$$

If $\hat{y}(t) \in QH$ satisfies

$$\chi_T = \sup_{t \leq T} \left\| \hat{y}(t) - T_Q(t)Qf - \int_0^t T_Q(t-s)C\hat{x}(s) ds \right\| < \infty,$$

then the reconstructed full observable obeys

$$\sup_{t \leq T} \|U(t)f - \hat{x}(t) - \hat{y}(t)\| \leq (1 + T\|C\|)B_T \cosh(\sqrt{k} T) + \chi_T. \tag{44.11}$$

Proof. Write the difference equation with the unperturbed A as its local part and use the contractivity of e^{tA} . Fubini and $\|K(s)\| \leq k$ show that $e(t) = \|x(t) - \hat{x}(t)\|$ satisfies

$$e(t) \leq B_T + k \int_0^t (t-s)e(s) ds \quad (0 \leq t \leq T).$$

Iteration yields the series $B_T \sum_{n \geq 0} k^n t^{2n} / (2n)!$; the remainder tends to zero because e is bounded on the compact interval. This is (44.10). Subtract the expression for $y(t)$ in the preceding proof from the specified reconstruction. Contractivity of T_Q gives $\|y(t) - \hat{y}(t)\| \leq T\|C\| \sup e + \chi_T$. Adding the resolved error proves (44.11). $\square$

The last term cannot be dropped merely because a resolved trajectory is accurate. A projected observable $PU(t)f$ is not the actual observable $U(t)f$ unless its orthogonal component is accounted for. This is the operator counterpart of preserving the actual marked target in an experiment comparison.

44.3. Galerkin convergence from the microscopic graph core. We specialize to the weighted symmetric L^2 space of the almost-everywhere hard-sphere flow in Lemma 43.1. The measure μ is an invariant probability; it may be the normalized invariant measure of a finite particle sector or a summable mixture of such sectors with the stated weights. The Koopman group is unitary and its generator L is skew-adjoint. The graph core in Theorem 43.2 supplies a countable real core: take countable dense smooth interior inputs, rational orbit-smoothing scales, and their finite linear combinations. Separability of the graph norm allows a countable selection. Gram–Schmidt on an enumeration gives nested finite-dimensional spaces V_n such that

$$V_n \subset D(L), \quad \bigcup_n V_n \text{ is a graph core for } L.$$

Let P_n be their orthogonal projections and set $G_n = P_n L P_n$ on H . It is bounded and skew-adjoint; on $V_n^\perp$ it is zero.

The stable graph-core resolvent approximation below is a specialization of Trotter’s classical approximation theorem [67, Theorem 5.2]; the proof spells out its core consistency, uniform contraction bound, and resolvent-to-orbit implication.

Theorem 44.3 (Microscopic graph-core Galerkin limit). *For every $f \in H$ and every finite T ,*

$$\sup_{0 \leq t \leq T} \|e^{tG_n} P_n f - U(t)f\|_2 \longrightarrow 0. \quad (44.12)$$

Moreover $(z - G_n)^{-1}$ converges strongly to $(z - L)^{-1}$ for $\Re z > 0$. Every finite V_n satisfies the common-domain hypothesis (44.1), and hence has the forced-memory representation of Theorem 44.1.

Proof. Let f belong to the union of the V_n . For sufficiently large n , $G_n f = P_n L f \rightarrow L f$. The resolvent bounds for skew-adjoint operators and the identity

$$(z - G_n)^{-1}(z - L)f - f = (z - G_n)^{-1}(G_n - L)f$$

give convergence to zero. The set of vectors $(z - L)f$ for f in the core is dense in H : approximate $(z - L)^{-1}g$ in the graph norm. The common resolvent bound extends convergence to every $g \in H$.

We also give an exponential-formula argument for the time-uniform conclusion. For any contraction generator A and $f \in D(A)$,

$$(I - tA/m)^{-m} f = \mathbb{E}[e^{sA} f] \quad \text{with } s \sim \text{Gamma}(m, t/m).$$

The formula follows by iterating the Laplace resolvent representation. Since $\mathbb{E}s = t$, $\text{Var } s = t^2/m$, and $\|e^{sA} f - e^{tA} f\| \leq |s - t| \|A f\|$, it gives

$$\|(I - tA/m)^{-m} f - e^{tA} f\| \leq t \|A f\| / \sqrt{m}. \quad (44.13)$$

For a fixed core vector, $\|G_n f\| \leq \|L f\|$ eventually. Strong resolvent convergence gives convergence of the m th powers in (44.13) for fixed m and $t > 0$; their factors are uniformly bounded. First let $n \rightarrow \infty$ and then $m \rightarrow \infty$ to obtain pointwise time convergence. The orbit Lipschitz bounds for this core vector are uniform in n , so a finite time net makes the convergence uniform on $[0, T]$. Density and contractivity extend it to all $f \in H$. Finally $P_n f \rightarrow f$, and e^{tG_n} is unitary, giving (44.12). Since $D(L^*) = D(L)$, each V_n satisfies (44.1). $\square$

This proof is a graph-core version of the classical resolvent-to-semigroup approximation argument. It is included to verify, rather than assume, the approximation used in the next theorem. It is not a uniform estimate in particle number, a Boltzmann–Grad limit, or a nonlinear kinetic Trotter–Kato theorem.

44.4. The common-encoder limit for actual physical observations. *Scope of this preserved physical theorem: fixed microscopic flow, passive observation control, fixed particle number and positive noise; no kinetic or Boltzmann–Grad limit. The active intervention theorem of Section 9 is a separate result.*

Fix a family of initial densities f_θ with respect to μ such that

$$f_\theta \geq 0, \quad \int f_\theta d\mu = 1, \quad \sup_\theta \|f_\theta\|_2 \leq D_0 < \infty. \quad (44.14)$$

At checkpoint t , control a selects a real observable $F_{t,a} \in H$ and noise level $\sigma_{t,a} > 0$. It does not change the microscopic flow. The exact and approximating reports are, respectively,

$$\begin{aligned} Y_t &= (U(\tau_t)F_{t,a})(Z_0) + \sigma_{t,a}\xi_t, \\ Y_t^{(n)} &= (e^{\tau_t G_n} P_n F_{t,a})(Z_0) + \sigma_{t,a}\xi_t, \end{aligned} \quad (44.15)$$

where the displayed normal variables are independent conditional observation noises, not a requirement that the two laws use a common additive-noise coupling. The control is selected before its report. There are finitely many controls at each time. Actual stage marks are sampled by one fixed Borel channel from Z_0 and any specified latent randomness and are shared in a comparison coupling. These latent marks are not observed by an encoder unless the interface expressly supplies them. A mark may be an observable of the microscopic future; the latent sample and the flow determine it before the comparison is made.

Put

$$\begin{aligned} d_{t,a}^{(n)} &= \|U(\tau_t)F_{t,a} - e^{\tau_t G_n} P_n F_{t,a}\|_2, \\ \zeta_t^{(n)} &= \frac{D_0}{\sqrt{2\pi}} \sum_{a \in \mathcal{A}_t} \frac{d_{t,a}^{(n)}}{\sigma_{t,a}}, \quad \varepsilon_t^{(n)} = \min\{1, \sum_{s \leq t} \zeta_s^{(n)}\}. \end{aligned} \quad (44.16)$$

Theorem 44.4 (Width-uniform physical decision limit). *For the instruments in (44.15), identity report channels give two-sided, state-cost-one, marked prefix comparison with errors $\varepsilon_t^{(n)}$. The comparison is uniform in the parameter, the common-encoder width profile, all admissible feedback designs, and each private or independent public seed signature. For the weights in (28.1), define*

$$E_n = \sum_{t \geq 0} a_t \varepsilon_t^{(n)}. \quad (44.17)$$

Then $E_n \rightarrow 0$. For every finite width profile, and indeed uniformly over all such profiles, the full and approximating risk bodies have Hausdorff distance at most E_n , their lower support values differ by at most E_n , and their centered minimax spectra differ by at most $2E_n$ when the full-history baselines use the same information pattern. These statements also hold for a finite acquisition horizon by setting later weights to zero.

Proof. Couple the two experiments with the same Z_0 , the same actual marks, and the same independent policy randomization. Until the first report mismatch, their controls and retained states agree. Two real normal laws of equal variance σ^2 and means u, v have total variation distance at most $|u - v|/(\sigma\sqrt{2\pi})$. This follows by integrating the absolute derivative of their translated density, with the factor one half in the total variation norm. A measurable maximal coupling of the successive report kernels therefore has this conditional mismatch bound.

The conditional law of Z_0 given a past report can have a large density; we do not apply (44.14) to that posterior. Instead, integrate the first-mismatch contribution under the original marginal $f_\theta\mu$, bound the indicator of previous matching by one, and sum over the finitely many possible controls at the step. Cauchy–Schwarz gives the unconditional contribution at time s at most $\zeta_s^{(n)}$. The union bound proves (44.16). The actual marks remain matched whenever the transcript does, because they were coupled through the same latent sample. The bound includes the joint public seed, which is identical and independent in the two experiments. The identity comparison channels themselves require no persistent state or additional seed and do not reveal this proof coupling to a consumer.

For each fixed t, a , Theorem 44.3 gives $d_{t,a}^{(n)} \rightarrow 0$, and hence $\varepsilon_t^{(n)} \rightarrow 0$. Because $0 \leq \varepsilon_t^{(n)} \leq 1$ and the a_t sum to one, finite-prefix truncation proves $E_n \rightarrow 0$. Apply Theorem 28.6 in both directions. The same argument applies to unrestricted full-history consumers, so the optimal full-history baselines differ by at most E_n without changing their control or information pattern. The final centered bound follows. $\square$

Corollary 44.5 (Combining compact duality with microscopic approximation). *In this corollary only the preparation density and the bounded losses vary with the row. The flow, observables, noise laws, and actual mark channel are fixed. Assume in addition that the density family is continuous in $L^1(\mu)$ on a compact parameter space, and that the bounded row losses are uniformly continuous in their row index in supremum norm. With the finite task alphabets of Theorem 28.2, both physical instruments satisfy its finite-prefix effect hypotheses. They therefore have attained infinite-horizon public duality, and their attained values obey the uniform error bound in Theorem 44.4. A full observable approximation certified by (44.11) can replace the Galerkin approximation in (44.16) with the corresponding L^2 error.*

Proof. For a fixed finite control word, the joint report law is a mixture over Z_0 of products of positive normal densities. Choose at each report time any fixed strictly positive probability density as reference. The resulting loss effects are in product L^1 , since the bounded loss is integrated against a probability density. Changing f_θ changes any such effect in L^1 by at most $\|f_\theta - f_{\theta'}\|_1$ times the loss bound: integrate out the conditional normal and marked kernels. A supremum-norm change in the row loss gives the analogous bound. Thus the required effect map is continuous. The arguments do not require bounded observables or a density on the infinite report path. The last assertion follows because the coupling proof uses only the full observable error in L^2 , not how it was obtained. $\square$

The two dimensions in this construction must not be identified. $\dim V_n$ is an analytic approximation parameter of the physical observation presentation. The retained register remains S_t ; no access to a Galerkin coefficient vector is supplied to the consumer. The theorem proves a uniform comparison before optimizing those registers. The historical kinetic program additionally requires nonlinear action-sublevel bounds, hierarchy correctors, and a joint microscopic limit. Neither a finite-rank memory identity nor strong L^2 approximation proves those additional assertions.

APPENDIX 45. STABLE CERTIFICATES FOR EXACT RESOURCE PROFILES

The task-cover theorem and the analytic comparison theorem concern the same invariant: the least common-encoder excess at a prescribed width profile. The following consequence states

precisely when an approximate experiment certifies an exact resource answer. This is also the point at which the microscopic decision limit can be used without exchanging a model limit with optimization over encoders.

For the finite restart system of Section 25, write

$$V_E(S, q) = \inf_{\pi} \max_{t,d} \left[r_{td}^E(\pi) - \sum_h \eta_t(h) m_{td}^E(h) \right].$$

If a positive entry regret exists, define $g_E = \alpha\gamma$ as in Corollary 25.5; for a Brier task family use $g_E = \alpha\delta^2/2$. This gap is defined by the target task table, not estimated from a lossy experiment. In the trivial case of no positive regret every resource profile is feasible and no gap test is necessary.

Theorem 45.1 (Comparison, completion, and certification). *Let E be a finite restart experiment with $g_E > 0$, and let F be an experiment with the same decision rows and loss scale. Suppose two-sided cost-one comparison at the declared seed signature gives risk-body Hausdorff error at most ε , and the corresponding full-history benchmarks differ by at most ε . Then*

$$|V_F(S, q) - V_E(S, q)| \leq 2\varepsilon. \quad (45.1)$$

If $\varepsilon < g_E/4$, the single test

$$V_F(S, q) < g_E/2 \quad (45.2)$$

is equivalent to existence of a causally closed compatible cover for E with width profile S . Equality of the exact feasible profiles holds for all private and independent public seed signatures, even though the positive-excess values for those signatures may differ.

More generally, suppose only a simulator from E to F is given, with cost (K, r) , and suppose its risk and benchmark errors are at most ε . If

$$V_F(S, q) + 2\varepsilon < g_E,$$

then E has a causally closed compatible cover of width at most KS . The public seed bound of the implemented design is rq , but it is unnecessary for exactness of that cover.

Proof. The two risk-body inclusions bound the infimum of the maximum centered row by the opposite value plus the risk error and the benchmark error. This proves (45.1). By Corollary 25.5, $V_E(S, q)$ is either zero or at least g_E , and it is zero precisely for a causally closed compatible cover. In the first case $V_F(S, q) \leq 2\varepsilon < g_E/2$; in the second case $V_F(S, q) \geq g_E - 2\varepsilon > g_E/2$. This proves the test without requiring an optimizer for F . The one-sided implementation gives $V_E(KS, rq) \leq V_F(S, q) + 2\varepsilon < g_E$, forcing its value to be zero. Theorem 25.2 then supplies a deterministic cover machine at width KS . $\square$

Corollary 45.2 (Certified approximation sequence). *For any family F_n satisfying the hypotheses of the preceding theorem with a known error bound $\varepsilon_n \rightarrow 0$, every fixed resource profile of E is classified correctly by (45.2) once $\varepsilon_n < g_E/4$. If the computed value has an additional certified numerical error κ_n , a sufficient condition is $2\varepsilon_n + \kappa_n < g_E/2$, with the corresponding interval test. In applications of Theorem 44.4, the error bound is E_n whenever the entire declared restart family, including its preparations, is among the compared physical experiments.*

Proof. Apply (45.1) and add the numerical error. The last qualification ensures that the physical comparison controls these restart rows, rather than a different initial preparation or only a single uncontrolled experiment. $\square$

The gap may be small, and computing the finite common-encoder optimum can be difficult. The statement is a certification theorem, not a polynomial-time algorithm. In particular it does not make a constrained inverse randomization assertion: a risk witness need not manufacture a simulator of a prescribed cost. Its content is the stability of exact causal task completion under an independently verified, resource-priced comparison.

APPENDIX 46. A PRIOR-INDEPENDENT EXPERIMENT BEHIND PREDICTIVE COORDINATES

A future-test quotient relative to one Bayesian preparation need not preserve a parameterized experiment. The additional requirement is *likelihood saturation*. This section gives both a terminal construction and the extra recursion needed for a causal reconstruction. All assertions about measurable coordinates are modulo the stated common null sets.

Theorem 46.1 (A saturated posterior quotient). *Let K be compact metric, let X be standard Borel, and let $E = \{P_\theta : \theta \in K\}$ be continuous in total variation. Choose a dense sequence (θ_j) and weights $w_j > 0$ with $\sum_j w_j = 1$. Put $m = \sum_j w_j P_{\theta_j}$ and define*

$$T(x) = \left(w_j \frac{dP_{\theta_j}}{dm}(x) \right)_{j \geq 1} \in [0, 1]^{\mathbb{N}}. \quad (46.1)$$

There is one parameter-independent kernel R such that

$$R(T_{\#}P_\theta) = P_\theta \quad (\theta \in K). \quad (46.2)$$

Thus E and TE are statistically equivalent. The completed σ -field generated by T is minimal among sufficient statistics. A different dense sequence and strictly positive weights give the same minimal sufficient σ -field modulo m and the same experiment equivalence class, not necessarily the same coordinates. For the terminal cardinality signature, their marked decision spectra are identical for every budget and every family of bounded terminal parameter-decision marks, whose losses depend on the parameter and the action. An additional latent-target mark requires its own joint-law sufficiency condition. No computability or finite recursive representation of T is asserted.

Proof. If $m(B) = 0$, then $P_{\theta_j}(B) = 0$ for all j ; continuity and density imply $P_\theta(B) = 0$ for every θ . Choose Borel versions of the countably many densities. They give $\sum_j T_j = 1$ m -almost surely; modify them on one common null set if necessary. Disintegrate m given T to obtain the standard Borel conditional kernel $R(t, dx)$. For every j , the density of P_{θ_j} is T_j/w_j , so

$$P_{\theta_j}(B) = \int \frac{t_j}{w_j} R(t, B) T_{\#}m(dt) = R(T_{\#}P_{\theta_j})(B).$$

For arbitrary θ , choose $\theta_{j_k} \rightarrow \theta$. Both applying T and applying R contract total variation. Taking limits proves (46.2).

For minimality, let S be sufficient, in the conditional-law sense that one kernel R_S is a conditional law of X given S under every P_θ . Mixing shows that it is also the conditional law under m . The factorization

$$\frac{dP_{\theta_j}}{dm}(x) = \frac{d(S_{\#}P_{\theta_j})}{d(S_{\#}m)}(S(x))$$

follows by integrating both sides on arbitrary measurable sets using R_S . Thus every coordinate of T is $\sigma(S)$ -measurable. Conversely the preceding disintegration proves sufficiency of T . Mixtures obtained from different dense sequences are mutually absolutely continuous: each dominates every member of E , hence also the other mixture. Minimality then proves equality of the completed sufficient σ -fields.

Finally, any terminal encoder from TE is an encoder from E by composition with T . Conversely, an S -label encoder Q from E is implemented from TE by QR , with exactly the same label laws for all parameters. This uses only transient reconstruction randomization and at most S terminal labels. It is a kernel assertion, not an algorithm that stores a sampled infinite object in a finite register. The two sets of attainable terminal risks, and hence their spectra, coincide. $\square$

The theorem makes explicit a classical dominated-sufficiency mechanism [47, 46]. Its role here is to identify the precise terminal slice on which the prior-relative and parameter-uniform notions have the same resource profile. A predictive-test quotient has this property when its generated

σ -field contains the likelihood coordinates (46.1). Without that inclusion the conclusion is false, even for a two-point parameter set: observe $Y = \theta \in \{0, 1\}$ once, and allow only constant future tests. Their quotient is a point although the observed experiment distinguishes both parameters exactly. Predictive equivalence always refers to a declared family of future questions.

46.1. The causal factorization. We give a finite-horizon formulation to avoid any assertion that a terminal sufficient statistic is automatically a causal sufficient state. Let $H_t = (A_0, Y_1, \dots, A_{t-1}, Y_t)$ be the observed history. All policies have the same parameter-independent action kernels. Under a reference prior on θ , write $\bar{P}$ for the mixture law. Suppose Borel coordinates $S_t = S_t(H_t)$ satisfy the following *saturation conditions*, with versions fixed for every admissible history and action where they are used:

$$S_t = F_t(S_{t-1}, A_{t-1}, Y_t), \quad (46.3)$$

$$\bar{P}(dy \mid H_{t-1}, a) = k_t(dy \mid S_{t-1}, a), \quad (46.4)$$

$$\frac{dP_\theta|_{H_t}}{d\bar{P}|_{H_t}} = \ell_{\theta,t}(S_t). \quad (46.5)$$

The likelihood identity is required for each parameter, initially on a dense reference family and then for all parameters by finite-horizon variation continuity. These conditions hold uniformly over the policies under consideration. Alternatively one may assume domination and the identity directly for every parameter. The reference prior is not allowed to change with the policy during a reconstruction.

Theorem 46.2 (Causal reconstruction of a saturated quotient). *Under (46.3)–(46.5), there are parameter-independent kernels $L_t(dy \mid s, a, s')$ which reconstruct the observation stream from its successive quotient states, with the correct joint law under every admissible policy. Hence the original and quotient instruments have mutually exact causal simulations. If S_t has at most K_0 values at each time, both simulations may be implemented with K_0 internal states and the cardinality spectra obey the budget dilations of Theorem 23.2. If the quotient is not finite, the theorem asserts statistical and causal reconstruction only, with the storage of S_{t-1} explicitly charged; it does not assert a finite-register resource equivalence.*

Proof. Let $H_t^0(ds' \mid s, a)$ be the pushforward of $k_t(\cdot \mid s, a)$ by $F_t(s, a, \cdot)$. Disintegration, including the Borel variables (s, a) , gives

$$k_t(dy \mid s, a) = \int L_t(dy \mid s, a, s') H_t^0(ds' \mid s, a), \quad (46.6)$$

with $F_t(s, a, y) = s'$ under the resulting joint kernel. On histories with $\ell_{\theta,t-1}(s) > 0$, change of measure gives the actual joint conditional law of (Y_t, S_t) as

$$L_t(dy \mid s, a, s') \frac{\ell_{\theta,t}(s')}{\ell_{\theta,t-1}(s)} H_t^0(ds' \mid s, a).$$

In particular, conditioning further on s' removes the likelihood factor. On zero-likelihood histories the kernel may be chosen arbitrarily; such histories are not visited under that parameter. For parameters outside a countable reference support, equality of the constructed finite-horizon laws extends by total variation, or follows directly from the assumed domination identity.

Start at the declared initial state. Given a target policy on the reconstructed reports, feed its chosen action into the quotient instrument; on receiving s' , draw y with $L_t(\cdot \mid s, a, s')$, then replace s by s' . Induction proves equality of every finite parameter-indexed report/action law. This induction is essential: it reconstructs responses after the same actions, not after actions selected with hidden parameter information. In the forward direction, apply F_t recursively to the real reports. Both executions retain only the previous quotient state in addition to the target

encoder's own register. This costs $K_0 S$ states if that state has K_0 possible values. The general standard Borel version has no such finite bound, exactly as stated. $\square$

A likelihood statistic alone supplies (46.5), not (46.4). Conversely, a Bayesian predictive state supplies future report predictions, not necessarily all parameter likelihoods. The conjunction is the bridge. Sequential comparison of filtered experiments already has a substantial decision-theoretic history [48]; the present statement fixes the additional register cost of the reconstruction rather than claiming that sequential comparison itself is new.

For marks involving an additional latent target rather than the parameter, its conditional kernel must also be included in the predictive saturation condition. The causal reconstruction theorem does not infer equality of latent-target laws from equality of report laws alone.

APPENDIX 47. A NECESSITY THEOREM FOR CAUSAL REALIZATION

We now characterize the causal constraint without imposing a tree geometry. The time dependence of a machine is retained, so the result does not replace a free clock by an uncharged stationary-state assumption. Fix a horizon T , finite report alphabets, one Bayesian law, and square-integrable Hilbert targets U_t . There is no control in this section. Let $\mathcal{H}_t$ be the finite set of positive-probability histories, $p_t(h) = \mathbb{P}(H_t = h)$, and $m_t(h) = \mathbb{E}[U_t \mid H_t = h]$. Let $\beta_t \geq 0$ be fixed weights.

A family of partitions $(\mathcal{P}_t)_{t=0}^T$, with $\mathcal{P}_0 = \{\emptyset\}$, is *successor compatible* if

$$h, h' \in C \in \mathcal{P}_t, \quad hy, h'y \in \mathcal{H}_{t+1} \implies hy, h'y \text{ belong to the same cell of } \mathcal{P}_{t+1}. \quad (47.1)$$

Only common positive-probability continuations enter this condition. For $C \in \mathcal{P}_t$ put $\bar{m}_C = p_t(C)^{-1} \sum_{h \in C} p_t(h) m_t(h)$ and define

$$\mathcal{V}(\mathcal{P}) = \sum_{t=1}^T \beta_t \sum_{C \in \mathcal{P}_t} \sum_{h \in C} p_t(h) \|m_t(h) - \bar{m}_C\|^2. \quad (47.2)$$

Theorem 47.1 (Recursive partition characterization). *For the cardinality signature, allowing randomized time-dependent updates and decoders and independent public randomness, the optimal weighted excess over the full-history Bayes risk equals*

$$\inf_{\#I_t \leq S} \sum_{t=1}^T \beta_t \mathbb{E} \|m_t(H_t) - \hat{U}_t\|^2 = \min_{\substack{\mathcal{P} \text{ successor compatible} \\ |\mathcal{P}_t| \leq S}} \mathcal{V}(\mathcal{P}). \quad (47.3)$$

Thus same-order realization of a sequence of static checkpoint optima is equivalent to the existence of successor-compatible partitions with the corresponding widths and variance bounds. This is a necessary as well as a sufficient condition. It is invariant under relabelling of histories; suffix closure is one sufficient construction of these partitions, not a necessary choice of coordinates.

Proof. A deterministic initialized machine partitions $\mathcal{H}_t$ according to I_t . Its update depends only on (t, I_t, y) , so its partitions satisfy (47.1). Given a state cell, the mean $\bar{m}_C$ minimizes its squared error by Hilbert orthogonality. The resulting excess is precisely (47.2).

Conversely, label each partition's cells by at most S labels. For each cell C and symbol y , send its label to the cell containing hy for any supported $h \in C$ with supported extension hy . Compatibility makes the answer independent of the choice of h . If there is no such extension, define the update arbitrarily. At time t decode the label as $\bar{m}_C$. Induction identifies the actual label with the cell of the actual history. There are finitely many partition families, so the minimum exists.

To include randomization, realize the finitely many update and decoder kernels by independent uniforms, and condition on their entire tape. For each tape, the rule is a deterministic time-dependent S -state machine. Independence of the tape and the experiment preserves the history law

and conditional means. The average of its weighted costs cannot be smaller than the deterministic minimum. Public independent randomness is handled in the same way. This freezing argument concerns one fixed Bayesian sum; it does not interchange a minimax task supremum with the average over tapes.

Unrestricted checkpoint encoders correspond to arbitrary partitions at each time, without (47.1). Hence they have cost $\sum_t \beta_t e_S^2(m_t(H_t))$. Formula (47.3) gives exactly the asserted equivalence of order comparisons, including the case when a static cost is zero. $\square$

Corollary 47.2 (Finite-command cell instruments). *Let $k_1, \dots, k_J$ be Borel functions on a parameter space with $k_j \geq 0$ and $\sum_j k_j = 1$, and fix a prior μ . At each time the raw cell is drawn independently conditional on the parameter. A command g from a prescribed finite menu in $[\eta, 1 - \eta]^J$, $0 < \eta < 1/2$, has a prescribed parameter-independent law ν_t . Cell j is reported as j with probability g_j and otherwise as the single failure symbol $\dagger$. Only the command and the reported symbol are observed. Fix a finite horizon, finite future-query menus with known likelihoods $H_{tq} \in [0, 1]$, independent query weights ω_{tq} , and nonnegative checkpoint weights β_t .*

For the persistent-cardinality signature, the optimal common causal encoder for the weighted sum of quadratic prediction excesses is exactly (47.3), with the finite history law and query means computed from the actual cell likelihoods. This applies to the finite-command specialization of the A1 finite-cell protocol [52]. It is an exact scalarized criterion, not an asserted solution of the different maximum-over-checkpoints criterion with private randomization.

Proof. For the observed pair (g, x) define

$$\lambda_{g,x}(\theta) = \begin{cases} g_x k_x(\theta), & x \in \{1, \dots, J\}, \\ \sum_j (1 - g_j) k_j(\theta), & x = \dagger. \end{cases}$$

For a history $h = ((g_s, x_s))_{s \leq t}$ its probability is $\prod_{s \leq t} \nu_s(g_s)$ times $\int \prod_{s \leq t} \lambda_{g_s, x_s}(\theta) \mu(d\theta)$. On positive-probability histories the mean of query q is

$$m_{tq}(h) = \frac{\int H_{tq}(\theta) \prod_{s \leq t} \lambda_{g_s, x_s}(\theta) \mu(d\theta)}{\int \prod_{s \leq t} \lambda_{g_s, x_s}(\theta) \mu(d\theta)}.$$

Take the Hilbert prediction vector with coordinates $\sqrt{\omega_{tq}} m_{tq}(h)$ and apply Theorem 47.1. All histories are finite, and the law used there is exactly the actual acquisition law above. A failure history contains $\dagger$, not the unobserved cell j ; the successor compatibility condition therefore requires one common update for that report. Integration of the given Borel likelihoods may require arbitrary real constants or offline work. Neither is claimed finite under the cardinality signature. Zero-evidence histories are omitted rather than given artificial posteriors. $\square$

Corollary 47.3 (Continuation obstruction). *Let g be a bounded readout on all finite words over a finite alphabet. For stationary deterministic machines required to reproduce g on every word, an exact S -state realization exists if and only if the relation*

$$h \equiv_g h' \iff g(hw) = g(h'w) \text{ for every finite } w \quad (47.4)$$

has at most S classes. More generally put $d_(h, h') = \sup_w \|g(hw) - g(h'w)\|$. Every stationary deterministic realization with uniform error at most ϵ has state fibres of d_* -diameter at most 2ϵ , and thus at least as many states as any 2ϵ -separated set of histories.*

Proof. Histories in one machine state have the same state after every common continuation. Exact output equality implies (47.4); for approximate outputs the triangle inequality gives the diameter bound. The relation (47.4) is a right congruence. Its classes, updated by appending the new symbol, form an exact machine, whose output is $g(h)$, well defined by the empty continuation. This is the output version of the classical finite-automaton congruence argument [49]; the finite-horizon, weighted-distortion characterization in Theorem 47.1 is the form needed here. $\square$

Example 47.4 (A two-valued checkpoint requiring exponentially many persistent states). At the first time observe a uniformly distributed word $X \in \{0, 1\}^d$. At the second time observe an independent uniform index $J \in \{1, \dots, d\}$ and predict $U = X_J$. The raw first acquisition costs d bits; it is not called a one-bit observation. The full-history conditional mean at the second checkpoint has only two values, so its static two-label error is zero. Nevertheless a causal exact predictor requires $S \geq 2^d$. Two different first-stage words in one state differ in a coordinate which can be queried with positive probability. The second-stage update cannot separate them. Retaining the whole word is sufficient.

There is also a quantitative obstruction. If e is the smallest error probability of an S -state predictor, $2 \leq S \leq 2^d$, then

$$e \geq h_2^{-1} \left(1 - \frac{\log_2 S}{d} \right), \quad (47.5)$$

where h_2 is the binary entropy and its inverse takes values in $[0, 1/2]$. Indeed, condition also on every independent public coding variable R . Since the first register has at most S values, $H(X | I_1, R) \geq d - \log_2 S$. Predicting each coordinate optimally from (I_1, R) gives errors $e_j \leq 1/2$, and the binary entropy inequality and concavity give

$$H(X | I_1, R) \leq \sum_{j=1}^d H(X_j | I_1, R) \leq \sum_{j=1}^d h_2(e_j) \leq d h_2 \left(d^{-1} \sum_j e_j \right).$$

The average on the right is a lower bound for any actual queried-bit error. This proves (47.5). If squared loss is used instead, its optimal value is at least $e/2$, since $p(1-p) \geq \min(p, 1-p)/2$. Thus the discrepancy persists for the Hilbert-loss slice of the spectrum. It is a temporal obstruction, not poor scalar quantization. The query must be answered after the update; J is not an additional persistent decoder input.

APPENDIX 48. FINITE-PREFIX, FINITE-DESCRIPTION REALIZATION

The recursive partition theorem isolates a necessity condition. Predictive-energy trees provide a verifiable sufficient construction. The full tree theorem and its geometric hypotheses are proved in Section 53 below. We now strengthen its implementation: exact ordering of nearly tied real energies is unnecessary, and the raw acquisition need not expose an infinite word.

Use the factorial legal language, stationary cylinder masses p_v , energies a_v , scales $D_v = (a_v/p_v)^{1/2}$, and constants a_-, ρ, U, C_0, c_0 of that theorem. In particular $a_\emptyset = 1$, $a_{vi} \leq \rho a_v$ with $\rho < 1$, $a_{vi} \geq a_- a_v$, $D_{vi} \leq D_v$, and $a_{uv} < a_v$ for nonempty u . Legality and the child list are decidable. Suppose an oracle returns a positive rational upper estimate u_v satisfying, for a fixed $\kappa \geq 1$,

$$a_v \leq u_v \leq \kappa a_v. \quad (48.1)$$

No uniform cost for this oracle is hidden in a cardinality assertion. Set

$$\bar{a}_v = \min\{u_w : w \text{ is a contiguous factor of } v, \text{ including } \emptyset\}. \quad (48.2)$$

For $0 < \eta < 1$, split precisely the legal vertices for which $\bar{a}_v > \eta$. Retain all their legal children. Denote the resulting full finite tree by $\mathcal{T}_\eta$, its leaf cut by $\mathcal{C}_\eta$, and its number of leaves by l_η .

Theorem 48.1 (Robust tree realization with charged acquisition). *The rational-threshold construction is finite, suffix closed, and supports both deletion and prepend-and-parse insertion. Its depth and state count satisfy*

$$D_\eta \leq 1 + \left\lceil \frac{\log(\kappa/\eta)}{|\log \rho|} \right\rceil, \quad (48.3)$$

$$V_\eta = 1 + \sum_{v \text{ internal}} \deg^+(v) \leq (U+1)(2l_\eta - 1). \quad (48.4)$$

For constants depending only on the displayed structural bounds and κ ,

$$\sum_{v \in \mathcal{C}_\eta} a_v \asymp e_{l_\eta}^2(Z) \asymp G_{l_\eta}. \quad (48.5)$$

In the deletion experiment, at most D_η symbols of a snapshot acquisition suffice; no later access to that snapshot is used. The same average-risk order as the exact greedy construction follows. For every sufficiently large state budget S , a dyadic threshold can be selected with $V_\eta \leq S$, $l_\eta \geq cS$, and $D_\eta = O(\log(S+1))$.

If the online readouts are in a fixed bounded subset of $\mathbb{R}^d$ and representatives can be computed to prescribed accuracy, the selected machine has a finite rational presentation. Readout precision $p = O(\log(S+1))$ suffices to retain the risk order. A word list, transition table and readout table have total description

$$P = O(S \log(S+1) + Sdp) \quad (48.6)$$

for a fixed alphabet. Its online persistent storage is one of at most S states; a transition-table implementation uses transient storage $O(\log(S+1) + dp)$ bits. An acquisition uses $O(\log(S+1))$ symbol queries. Online table accesses can be implemented in word-RAM time proportional to these query counts; a sequential bit-table implementation gives the conservative bound $O(S(\log(S+1) + dp) \log(S+1))$ bit operations per acquisition. Offline oracle time is separate. The Hilbert-space cardinality assertion does not, by itself, encode an infinite-dimensional readout in finitely many bits.

Proof. If w is a factor of v , removing a prefix and then a suffix cannot decrease energy. Consequently

$$a_v \leq \bar{a}_v \leq \kappa a_v, \quad \bar{a}_{iv}, \bar{a}_{vi} \leq \bar{a}_v. \quad (48.7)$$

The second assertion follows by inclusion of factor sets and requires no accurate comparison between two distinct true energies. Thus the internal vertices form a prefix-closed and suffix-closed set. Each internal vertex is split into all its legal children. A proper suffix of a leaf is a legal child of a suffix of its parent, hence is present; repeating this argument proves full suffix closure. If iv had no leaf prefix when v is a leaf, it would be internal, forcing its suffix v to be internal, a contradiction. These are exactly the two update operations required.

At depth n , $\bar{a}_v \leq \kappa \rho^n$, which proves finiteness and (48.3). Every nonroot leaf has an internal parent, so

$$a_{-}\eta/\kappa < a_v \leq \eta. \quad (48.8)$$

The exact vertex count follows by counting edges; contracting unary chains gives (48.4) as in the full-tree lemma below.

Here is a lower bound not depending on exact greedy ordering. Refine all leaves by $h = 3(U+1)$ further legal levels. Each leaf has at least 2^3 descendants at that level, because there are at most U consecutive unary vertices. The resulting incomparable cylinders number at least $8l_\eta$. Every one has energy at least $a_-^{h+1}\eta/\kappa$. The separation hypothesis and monotonicity of D make the open tubes of radii δD_w about their images disjoint for $2\delta < c_0$. At most l_η tubes can contain one of l_η arbitrary proposed Hilbert centres. Each unoccupied cylinder contributes at least $\delta^2 a_w$ to the squared error. Hence $e_{l_\eta}^2(Z) \geq cl_\eta \eta$. One representative per original leaf gives $e_{l_\eta}^2(Z) \leq C_0^2 \sum a_v$, and (48.8) proves the first comparison in (48.5). The second is the static conclusion of Theorem 53.1, whose proof does not use the present theorem. Its oscillation estimate in the deletion orientation, or its cylinder diameter estimate in the insertion orientation, also proves the stated risk bounds for this cut.

A prefix parser stops on reaching a leaf, after at most D_η queries. It never reads or stores the rest of the snapshot. This is a finite-latency acquisition convention: the snapshot is available during that update and the prediction deadline permits these microsteps. No claim is made for a

moving physical state observed after an unmodelled read delay. Insertion and deletion then use the already proved closure.

For budget selection, use $\eta_j = 2^{-j}$ and take the last threshold with $V_{\eta_j} \leq S$. There is a constant bound on $V_{\eta_{j+1}}/V_{\eta_j}$. In fact, beneath a leaf at threshold η , all descendants at depth k have estimated energy at most $\kappa\rho^k\eta$, and so the next threshold adds at most k levels once $\kappa\rho^k \leq 1/2$. The ratio is bounded by a fixed geometric sum of powers of the alphabet size. It follows that the chosen V_η is at least a fixed multiple of S , and then (48.4) gives $l_\eta \geq cS$. For completeness, a leaf at depth n has $a_v \geq a_-^n$, so $n \geq \log(1/\eta)/|\log a_-|$. Every legal path branches at least once in $U + 1$ levels. The complete cut therefore has at least $2^{\lfloor \log(1/\eta)/((U+1)|\log a_-|) \rfloor}$ leaves. Since $l_\eta \leq S$, this bounds $\log(1/\eta)$ by $O(\log(S + 1))$. Combine with (48.3). Fixed-dilation comparison for G then converts (48.5) into the same- S order.

Finally replace each finite-dimensional readout by a rational vector within Euclidean distance δ . Pointwise squared loss is at most twice its previous value plus $2\delta^2$. Take $\delta^2 \leq \eta$ and use $\sum a_v \geq a_- l_\eta \eta / \kappa$. Bounded readouts require $p = O(\log(1/\eta) + \log(d + 1))$ binary digits per coordinate. Store all vertex words, their legal child/deletion/insertion transitions, and these rational vectors. There are $O(S)$ transitions for a fixed alphabet, each state index costs $O(\log(S + 1))$ bits, and the word-list depth is $O(\log(S + 1))$. This proves (48.6). A streaming parser uses only its current vertex index during the acquisition; no prefix string survives the update. The stated time bounds follow by random access, or by a full sequential scan for each table access. The number and precision of the offline queries can be listed, but (48.1) alone gives no uniform running-time bound for answering them. This is why offline time is not absorbed into (48.6). $\square$

The use of all factors in (48.2) is essential. Independent relative approximations u_v need not respect suffix monotonicity, even when the true energies do. Thresholding the raw approximations can therefore produce a tree that cannot execute its own suffix update. The envelope repairs that defect while changing energies by at most the fixed factor κ .

APPENDIX 49. THE MEASURE-VALUED SLICE: SHARP PRESENTATION ENTROPY

The decoder's output type is part of a mark. In particular, outputting a probability law close in expected Wasserstein distance is stronger than simulating one randomly chosen observation from that law. This separation is important for a genuinely measure-valued predictive state.

Let $\mathcal{M} = \mathcal{P}([0, 1])$ with W_1 . A *uniform law presentation* receives $\pi \in \mathcal{M}$ as a transient input and returns one of S persistent labels. Its decoder outputs a law $\hat{\pi}$. The fixed code, including its representative laws, is chosen once for all π . In particular the unknown input is not inserted into the decoder's free program constants. Let

$$H_S = \inf_{Q, D} \sup_{\pi \in \mathcal{M}} \mathbb{E} W_1(\pi, D(Q(\pi))). \quad (49.1)$$

Both maps may use independent private or public randomness. The criterion is the expected distance of the *realized* decoded law. Only persistent cardinality is charged here; the input-law interface and general real-law computation are unrestricted. Regard π as the unknown parameter and the uncompressed observation as π itself. Then H_S is the spectrum of the single bounded minimax law-presentation mark, with full-observation baseline zero. It is not a supremum of point-prior Bayes risks: such a prior would reveal the unknown law to the decoder and change the problem.

Theorem 49.1 (Uniform presentation of a probability measure). *There are absolute $c, C > 0$ such that*

$$\frac{c}{\log(S + 1)} \leq H_S \leq \frac{C}{\log(S + 1)}, \quad S \geq 1. \quad (49.2)$$

Moreover no continuous map from $\mathcal{M}$ into a fixed $\mathbb{R}^k$ can be an exact coordinate sufficient to evaluate every bounded Lipschitz integral on all of $\mathcal{M}$. This statement does not exclude noncontinuous Borel encodings of a standard Borel space into one real coordinate.

Proof. For the upper bound use the explicit net in Lemma 55.1 below: equally weighted n -point empirical laws with points on the grid $\{0, 1/n, \dots, 1\}$, in increasing order. There are $\binom{2n}{n} \leq 4^n$ such laws. Quantile discretization followed by grid rounding gives covering radius $3/(2n)$. Taking $n = \lfloor \log S / \log 4 \rfloor$ proves the upper bound for $S \geq 4$; increase C for smaller S .

For the lower bound, partition $[0, 1]$ into n equal intervals. In each interval put mass $1/n$ at either its quarter point or its three-quarter point. Denote the resulting law by π_ω , $\omega \in \{0, 1\}^n$. Their quantile functions give the exact identity

$$W_1(\pi_\omega, \pi_{\omega'}) = \frac{d_H(\omega, \omega')}{2n^2}. \quad (49.3)$$

Greedily choose binary words and delete Hamming balls of radius $\lfloor n/4 \rfloor$. The selected words have mutual distance greater than $n/4$, and their number is at least $2^{c_0 n}$, where $c_0 = 1 - h_2(1/4) > 0$. Indeed $\sum_{j \leq n/4} \binom{n}{j} \leq 2^{nh_2(1/4)}$, as follows by bounding these binomial terms using the binomial expansion at parameter $1/4$. Thus the selected measures have mutual W_1 distance greater than $1/(8n)$.

Choose $n \leq C_0 \log(S + 1)$ sufficiently large that the packing has at least $2S$ members. Give it the uniform prior. For any S proposed law centres, their open balls of radius $1/(16n)$ contain at most S packing members in total. At least half the prior therefore incurs distance at least $1/(16n)$, no matter how it is encoded. Fixing all independent coding randomness gives this same bound for each tape; integration preserves it. The worst input risk is at least the uniform prior risk, which proves the lower bound in (49.2).

For the last assertion, bounded Lipschitz integrals separate probability measures, so a sufficient coordinate map must be injective. The measures supported on any $k + 2$ distinct points form a $(k + 1)$ -simplex. A small sphere about its barycentre lies in its relative interior and is homeomorphic to S^k . By the Borsuk–Ulam theorem every continuous map of this sphere to $\mathbb{R}^k$ identifies two antipodal points [51]. Hence no such injective coordinate exists. For $k = 0$ two distinct measures already give the contradiction. $\square$

The entropy mechanism is consistent with the power-exponential geometry of Wasserstein spaces studied in [50]; the packing argument is supplied here because the exact presentation loss, including randomization, matters for the result.

Corollary 49.2 (A nonlinear posterior class with a matching bound). *Consider the nonlinear binary instrument of Corollary 55.3 below, with*

$$h_0(x) = x/5 + x^2/50, \quad h_1(x) = 3/5 + h_0(x), \quad \Phi_y \pi = \frac{3}{4}(h_y)_\# \pi + \frac{1}{4}(h_{1-y})_\# \pi.$$

For each fixed report y ,

$$\frac{1}{5} W_1(\pi, \nu) \leq W_1(\Phi_y \pi, \Phi_y \nu) \leq \frac{6}{25} W_1(\pi, \nu). \quad (49.4)$$

Consequently, for every fixed t , a single finite-register scheme that must present the posterior for every input prior has worst expected W_1 error at least $c5^{-t}/\log(S + 1)$ at time t , even if the decoder is additionally told the reports. A uniform recursive presentation has error $O(1/\log(S + 1))$ at all times, when the initial law is encoded at the same accuracy and the update error is of that order. No fixed finite-dimensional continuous exact predictive coordinate exists on this all-priors posterior class. Constants in the lower bound are not asserted uniform as $t \rightarrow \infty$ for one fixed preparation.

Proof. The images of h_0 and h_1 lie in two disjoint ordered intervals. The mixture weights for a fixed y agree for π and ν . Quantile coupling therefore gives

$$W_1(\Phi_y \pi, \Phi_y \nu) = \frac{3}{4} W_1((h_y)_\# \pi, (h_y)_\# \nu) + \frac{1}{4} W_1((h_{1-y})_\# \pi, (h_{1-y})_\# \nu).$$

The derivatives of both increasing maps lie between $1/5$ and $6/25$, proving (49.4). Iterate it along any length- t report word and apply the same packing as in the preceding proof. The reports of this instrument are independent fair bits, with a law independent of the initial prior. Conditioning on the report word thus does not change the uniform packing prior. The centre-counting lower bound survives even with this word supplied freely to the decoder; integrating it over words and coding tapes proves the assertion for the actual, less informed decoder.

The recursive upper bound follows from Theorem 55.2 with $\rho = 6/25$, initializing by a net point rather than by an arbitrary fixed law. The initial error is at most $3/(2n)$ and the accumulated error is bounded by $[3/(2n) + \eta]/(1 - 6/25)$. Every fixed-word map on laws is continuous and injective by the lower bound in (49.4). Its restriction to the compact simplex used above is a homeomorphism onto its image. Applying the same sphere argument proves the coordinate obstruction. $\square$

Proposition 49.3 (Marginal-probe simulation is a different presentation problem). *Suppose only one Bernoulli probe is required: after compression a function $g : [0, 1] \rightarrow [0, 1]$ with Lipschitz constant at most one is specified, and the desired output law is $\text{Bern}(\pi g)$. The probe is fixed independently of the realized encoder randomness. For $S \geq 2$ there is a common S -label encoder whose output-law error is at most $1/[2(S-1)]$, uniformly in π, g . Thus this marginal simulation mark cannot be substituted for the realized-law mark in (49.1).*

Proof. Sample $X \sim \pi$ transiently, store the nearest point Z of the S -point uniform grid, and discard X . On receiving g , output a Bernoulli variable with parameter $g(Z)$. Its unconditional law is Bernoulli with parameter $\mathbb{E}g(Z)$, and $|\mathbb{E}g(Z) - \pi g| \leq \mathbb{E}|Z - X| \leq 1/[2(S-1)]$. The encoder is chosen before g . It does not produce a realized law close to π in W_1 : its natural decoded law is δ_Z . Nor does the argument cover probes chosen after observing Z , or an unlimited transcript of independent samples. Those are different tasks and interfaces. This output-law simulation criterion is not identified with a realized-law decision loss. $\square$

APPENDIX 50. A DYNAMIC FINITE-EXPERIMENT CONSEQUENCE

The decision-complete slice of the same spectrum has a different geometry. We first make its temporal register convention explicit, and then connect it to the physical count experiment.

Fix $n, r \geq 1$, a compact $K \subset \mathbb{R}^r$ with nonempty interior, and a positive definite covariance Γ . At successive times observe independent

$$Y_t \sim N(\theta_t, \Gamma), \quad \theta = (\theta_1, \dots, \theta_n) \in K^n. \quad (50.1)$$

Each Y_t is discarded after the update. No earlier output labels are retained outside the register. At the terminal time only its final label is supplied to a parameter-independent reconstruction kernel. This is not an S -symbol channel whose entire transcript is available.

Theorem 50.1 (Sequential accumulation under a total register budget). *For the cardinality signature with persistent S states and at most S terminal outcomes, the decision-complete terminal spectrum of (50.1) satisfies*

$$c_n S^{-2/(nr)} \leq \Theta_{G^{(n)}}(S; \mathcal{D}_{\text{all}}) \leq C_n S^{-2/(nr)}. \quad (50.2)$$

The infimum is over causal encoders obeying the total state budget at every time. Lower bounds include arbitrary randomized, time-dependent encoders. A deterministic causal encoder attains the upper order. Constants are for fixed n, r, K, Γ , not uniform in a growing horizon. The readout/reconstruction randomness is external and fresh and private, while the physical alphabet

and the persistent register remain finite. No public tape or other side information is supplied alongside the final label in this theorem.

Proof. The full observation vector is a Gaussian location experiment of dimension nr , with compact mean set K^n and covariance $I_n \otimes \Gamma$. Every final register gives an experiment with at most S outcomes, regardless of its causal implementation. The lower bound of Theorem 56.1, proved below, applies to all such experiments. Because a register is a garbling of the full observation, its forward deficiency is zero. Theorem 23.4 therefore identifies its reverse deficiency with its worst normalized Bayes regret, proving the lower bound in (50.2).

For the upper bound put $J = \lfloor S^{1/(nr)} \rfloor$. At time t compactify each coordinate of Y_t by $\tau(x) = 1/2 + \arctan(x)/\pi$, quantize into J cells per coordinate, and append the resulting r -tuple of cell indices to the current register word. At that time there are $J^{rt} \leq J^{rn} \leq S$ possible words. The known absolute time determines their length; it carries no data-dependent information. Thus a single S -value register, not n separately uncharged registers, implements the rule. At the final time apply the positive tensor-hat reconstruction of Lemma 56.3 in dimension nr . For $J \geq 2$ its error is $O(J^{-2}) = O(S^{-2/(nr)})$. Enlarge the constant for the remaining finite budget range. This proves the upper bound using a genuine causal encoder. Its real-valued input computation has no finite-time claim under this signature. $\square$

If instead the sequence of emitted cell labels were supplied to the terminal decoder, its alphabet would be their product. In that different model a small current register could leave a large data-bearing transcript. The lower bound above is about the total retained experiment, not the size of the most recent communication symbol. This is one reason zero-delay channel coding and total-register coding require separate resource signatures [33, 34].

50.1. A theorem-level downstream arrow. We now give the adapter in the coordinates of the retained known-mark exposure protocol, rather than identifying all of the algebraic paper with this statistical subexperiment.

Corollary 50.2 (Known-mark exposure adapter). *Fix positive exposure centre λ^0 , known marks, a derivative matrix D of full row rank, and $\Lambda = \text{diag}(\lambda^0)$. Put $\Sigma = D\Lambda D^\top$ and $B_* = \Lambda D^\top \Sigma^{-1}$. Let L be a fixed normal-compression map of rank $r \geq 1$, let F be an orthonormal frame for its range, and set*

$$C = F^\top L D, \quad A = F^\top L, \quad \Gamma = C A C^\top.$$

*On a fixed compact index set H , observe independent counts of means $N\lambda_N(h)$, where $\lambda_N(h) = \lambda^0 + N^{-1/2}B_*h$. If AH contains an r -dimensional ball, the deterministic unjittered statistic $T_{N,J}$ of Section 57, applied to the score $C(N - N\lambda^0)/\sqrt{N}$, has at most $J^r \leq S$ outputs. For $\mathcal{G} = \{N(Ah, \Gamma) : h \in H\}$, it satisfies*

$$cS^{-2/r} \leq \Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C_K S^{-2/r}, \quad J = \lfloor S^{1/r} \rfloor, \quad 2^r \leq S \leq KN^{r/6}. \quad (50.3)$$

Only the physical output cardinality and the stated joint sample/register window are bounded here. Reconstruction randomization is external, parameter independent and private; no finite-bit complexity of the raw real score computation is asserted.

Proof. $DB_* = I$, so $CB_* = A$. Since $F^\top L$ has row rank r and D is surjective, C has row rank r and $\Gamma > 0$. The compact local alternatives have uniformly positive exposures for all sufficiently large N . Thus every hypothesis of Theorem 57.1 is verified with these actual score weights, mean set and covariance. That theorem supplies both inequalities in (50.3); its lower bound allows every experiment with at most S outcomes, not only this statistic. Theorem 23.4 gives the all-decision interpretation of the limiting obstruction, and Theorem 23.2 transports each bounded task through the comparison kernels. No algebraic primary-decomposition theorem is a premise of this argument. $\square$

The left inequality concerns every experiment with at most S terminal outputs approximating this specified Gaussian family, and therefore also the physical statistic. The upper inequality concerns that explicit statistic and its positive reconstruction kernel. In particular, achieving Gaussian-family distance at most $CN^{-1/5}$ forces $S \geq cN^{r/10}$, and budgets of that order suffice for this statistic. These are label-budget statements in the displayed joint window; they are not a sample-optimality theorem over all finite- N physical objectives.

Theorem 23.4 supplies a decision interpretation of the limiting experiment's finite-output obstruction, while Theorem 23.2 transports each bounded task through the Poisson comparison kernels. These are genuine arrows between specified experiments. They neither require nor assert a Sinai local limit theorem, a particle path large-deviation principle, or a physical phase construction.

SUPPORTING THEORY AND MODEL VERIFICATION

The following sections give the complete proofs used above. Unless a more detailed signature is specified, they use the cardinality signature of Section 23: only persistent states or physical terminal outcomes are bounded, with arbitrary transient computation and independent reconstruction randomness. In the tree application, Theorem 48.1 additionally replaces the full-word acquisition by a charged finite-prefix interface. In the measure application the new matching lower bound is for the uniform realized-law presentation task, not for a single marginal probe. The Gaussian and Poisson inequalities concern the indicated terminal experiment family. These local conventions are part of their use here.

APPENDIX 51. POSITIVE EXPERIMENTS AND PREDICTIVE QUOTIENTS

51.1. Actual histories. All spaces of states, observations and actions are standard Borel. For an unknown parameter λ , a controlled experiment has a probability kernel

$$K_t^\lambda(dx', dy, dc \mid x, a), \quad (51.1)$$

where c is the cost of the actual step. Failure, rejection and stopping are outcomes, not missing probability mass. A policy chooses a_t from the observed past and independent randomization. Preparation has a specified law and cost. With an absorbing symbol after stopping, repeated kernel integration constructs one probability law on the infinite history space. Its finite marginals are consistent because every kernel in (51.1) has mass one. This is the usual kernel extension theorem [4]; conditioning on success would define a different experiment unless its preparation and waiting costs were separately charged.

For a fixed Bayesian preparation, let π be a reachable belief state on the hidden state, including the parameter when it is random. The positive instrument associated with an action and observation event is

$$\mathsf{T}_{a,B}\pi(E) = \int K(E, B, dc \mid x, a) \pi(dx).$$

After an observed event of positive probability, normalization gives the updated belief. For nonatomic reports we use a specified disintegration, with all equalities interpreted outside the appropriate null sets. A formula for an event of probability zero is not an additional observation channel.

51.2. The future-test quotient. A future test consists of a finite observation-based policy, a stopping rule, and a bounded measurable terminal payoff. Its conditional expectation at a belief π is written $\pi(v)$, where v is the corresponding backward test function. Fix a family $\mathcal{V}$ containing

constants and closed under the one-step pullbacks of the allowed tests. It may, for example, contain all bounded finite-horizon executable tests. Define

$$\pi \sim_{\mathcal{V}} \pi' \iff \pi(v) = \pi'(v) \quad (v \in \mathcal{V}). \quad (51.2)$$

The set of classes is the predictive quotient. The choice of tests is part of the experiment specification: identifying a hidden state is not required if none of the designated tests distinguishes it.

Proposition 51.1 (Minimality and descent). *The quotient (51.2) is the coarsest set-valued statistic that determines all designated predictions. If the test family is one-step closed, normalized Bayesian updates descend to the quotient where their normalization is positive. If, in addition, the reachable belief domain B is compact metrizable, a countable family of continuous tests determines (51.2), and each fixed-report update is continuous on its positive-probability domain, the quotient has a canonical metrizable realization as a subset of $\mathbb{R}^{\mathbb{N}}$, and each such update descends continuously on compact subdomains on which its denominator is bounded away from zero.*

Proof. Map π to the vector of all its test expectations. Its fibres are exactly (51.2). Any statistic determining the expectations has fibres contained in these fibres, so the vector factors through that statistic as a set map. For the update, the numerator of a future prediction is a one-step pulled-back test and the denominator is the pulled-back constant. Equivalent states have equal numerators and denominators, hence equal normalized predictions.

In the countable continuous case use $q(\pi) = (\pi(v_j))_{j \geq 1}$. After scaling the tests to be bounded by one, its range is a compact subset of $[-1, 1]^{\mathbb{N}}$, with its product metric. A continuous map from a compact space into a Hausdorff space is a quotient map. The fibre-respecting continuous update therefore factors continuously on the stated compact subdomain. No assertion of a Borel cross-section, or of a smooth quotient for arbitrary Borel tests, is used. The minimality statement in the first paragraph alone would not imply either assertion. $\square$

The proposition is relative to the prepared Bayesian experiment. An identity for one prior is not a parameter-uniform comparison of the family $\{K^\lambda\}_\lambda$. All experiment distances below explicitly take the supremum over the parameter when that stronger meaning is intended.

51.3. Attainable Bayes geometry. Let U be square integrable in a separable Hilbert space H , let $\mathcal{H}$ be the actual information, and put $m = \mathbb{E}[U \mid \mathcal{H}]$. Define

$$e_M^2(m) = \inf_{z_1, \dots, z_M \in H} \mathbb{E} \min_i \|m - z_i\|^2.$$

Independent encoder/decoder randomization is permitted, but a retained observation is not a free input after encoding.

Proposition 51.2 (Checkpoint geometry and information loss). *The optimum squared loss of a single M -label checkpoint statistic is*

$$\mathbb{E}\|U - m\|^2 + e_M^2(m). \quad (51.3)$$

For any sub- σ -field $\mathcal{C} \subset \mathcal{H}$, posterior laws of λ satisfy the barycentre identity $\pi_{\mathcal{C}} = \mathbb{E}[\pi_{\mathcal{H}} \mid \mathcal{C}]$. Consequently convex posterior functionals decrease under coarsening. Whenever the quantities are finite,

$$I(\lambda; \mathcal{H}) - I(\lambda; \mathcal{C}) = \mathbb{E} D_{\text{KL}}(\pi_{\mathcal{H}} \parallel \pi_{\mathcal{C}}). \quad (51.4)$$

In a differentiable dominated family, the score of a statistic S is $\mathbb{E}[s_\lambda \mid S]$, and the lost Fisher information matrix is $\mathbb{E} \text{Cov}(s_\lambda \mid S)$, provided differentiation and conditioning are justified in L^2 .

Proof. Conditional orthogonality gives $\mathbb{E}\|U - z\|^2 = \mathbb{E}\|U - m\|^2 + \mathbb{E}\|m - z\|^2$ for every $\mathcal{H}$ -measurable output, and also after adding independent randomness. Freeze that randomness. There are at most M decoder outputs, so nearest-centre assignment is no worse than the given encoding. Conversely every finite codebook has a measurable nearest assignment, with deterministic tie-breaking. Taking infima proves (51.3), without assuming attainment of an optimal codebook. Conditional expectation against bounded parameter tests gives the barycentre identity. Conditional Jensen proves the convex assertion. The chain rule for the logarithm of posterior densities gives (51.4); finite partitions followed by monotone approximation give the same assertion under the stated finiteness condition. Differentiating the marginal density of S gives the score identity. The matrix law of total variance gives the Fisher identity. These are the classical decision and information identities in [1, 45]; their role here is to specify the geometry being implemented, not to certify recursive closure. $\square$

51.4. Normalized response and observation modules. Let b_θ be a differentiable finite positive measure with $b_\theta(1) > 0$, and let $\pi_\theta = b_\theta/b_\theta(1)$. Differentiating this actual normalization gives

$$\partial_\theta \pi_\theta(f) = \frac{\partial_\theta b_\theta(f) - \pi_\theta(f) \partial_\theta b_\theta(1)}{b_\theta(1)}. \quad (51.5)$$

If the derivative has score density s relative to b_θ , the right side is $\text{Cov}_{\pi_\theta}(f, s)$. Thus a response direction is invisible exactly when its centered score annihilates the linear span of the designated tests. If the observation protocol consists of products from two declared linear test spaces V, W , its null space is $(VW)^\perp$, with $VW = \text{span}\{vw : v \in V, w \in W\}$. This last statement follows immediately from linearity, but it requires that those products really are the observed probes. It does not identify an algebraic multiplicity with a physical noise variance. Formula (51.5) and this annihilator are the normalization and observation-module inputs used to type a contact experiment.

APPENDIX 52. CAUSAL MORPHISMS AND CHARGED APPROXIMATION

A resource signature specifies ordered coordinates, for example persistent cardinality, numerical precision, transient workspace, program description, preparation count and physical time. A budget transformer is a monotone map between two such signatures, accompanied by an actual implementation. A coordinate may explicitly be unlimited. The persistent-cardinality results below leave fixed programs, fixed real constants, time and transient computation unlimited unless a more specific implementation is given. There is no claim that these resources are equivalent.

A causal simulation from an experiment $\mathcal{E}$ to $\mathcal{F}$ consists of a source controller, a causal transducer and its initialized state. For every target policy the joint construction must produce the target observation/cost history within a declared parameter-uniform total-variation error. The transducer never sees the unknown parameter or the unobserved physical state. Resource bounds are imposed on this joint construction, not just on the terminal observation kernel. We use $\|P - Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|$.

Theorem 52.1 (Typed simulation and recursive approximation). *The following statements hold.*

- (i) *Serial causal simulations of errors ϵ_1, ϵ_2 and budget transformers ρ_1, ρ_2 compose with error at most $\epsilon_1 + \epsilon_2$ and transformer $\rho_1 \circ \rho_2$. A simulator with K persistent states and a target observer with M states can be run with KM persistent states. For losses in $[0, L]$, the risk changes by at most $L(\epsilon_1 + \epsilon_2)$. Other resource coordinates are transformed according to the supplied implementations. The identity simulation has zero error and the identity transformer. Independent parallel simulations have additive TV error and the product register, with preparations and other costs charged by their joint implementation.*

(ii) Suppose the true predictive state lies in a metric space (S, d) and updates by $s_t = F_t(s_{t-1}, y_t, a_t)$. For all reports and actions suppose F_t is ρ -Lipschitz in its state, $\rho < 1$. Let an implementable approximate update $\tilde{F}_t$ have local error at most η_t , and let $Q_t : S \rightarrow C_t$ be a measurable finite quantizer with radius ϵ_t . The register

$$\hat{s}_t = Q_t \tilde{F}_t(\hat{s}_{t-1}, y_t, a_t) \quad (52.1)$$

has at most $|C_t|$ states at time t and, on every common observation path,

$$d(s_t, \hat{s}_t) \leq \rho^t d(s_0, \hat{s}_0) + \sum_{j=1}^t \rho^{t-j} (\epsilon_j + \eta_j) =: b_t. \quad (52.2)$$

A data-dependent persistent numerical residue must be part of C_t ; it cannot be placed in an uncharged side register. If the report kernels are L_0 -Lipschitz in total variation as functions of the predictive state, the length- n simulated history differs by at most $L_0 \sum_{t < n} \mathbb{E} b_t$. The same statement holds up to a bounded stopping time, with the stopped sum. If the true conditional mean is $g(s_t)$ with g L_1 -Lipschitz into a Hilbert space, its implemented prediction has Bayes excess at most $L_1^2 \mathbb{E} b_t^2$.

Proof. Compose the two transducers on their joint initialized state. Applying a Markov kernel contracts total variation, so the triangle inequality gives the sum of errors, uniformly over the policy and parameter for which the two certificates were supplied. The joint register is the Cartesian product. This proves the cardinality bound, and integration of a loss with range $[0, L]$ proves the risk bound. The identity is the unchanged history. For independent parallel laws, insert one product factor at a time and use TV contraction to obtain the same sum bound. The composition order of the budget maps follows from first realizing the target in the intermediate experiment and then realizing that implementation in the source.

For (ii), insert $F_t(\hat{s}_{t-1}, y_t, a_t)$ between the two updates. Contraction, the local defect and the quantization radius give $d(s_t, \hat{s}_t) \leq \rho d(s_{t-1}, \hat{s}_{t-1}) + \eta_t + \epsilon_t$. Induction proves (52.2). For the law comparison, telescope the products of the history kernels, replacing one kernel at a time. The prefix before the replaced kernel is distributed according to the true experiment; its integrated local TV error is at most $L_0 \mathbb{E} b_t$. All suffix kernels are contractions in TV. Summing proves the bound. An absorbing stopped state proves the stopped version by the identical calculation. The policy is evaluated on the same history in each local comparison: no continuity of a possibly discontinuous policy is presumed. Finally apply conditional orthogonality as in Proposition 51.2 and the Lipschitz bound for g . $\square$

Corollary 52.2 (Resource transport of an attainable profile). *If an M -state target observer has risk R_M and a K -state causal simulation has error ϵ for its bounded decision problem, then the source has a KM -state observer of risk at most $R_M + L\epsilon$. Two simulations give the corresponding two directed inequalities; they need not preserve the same cardinality. A K -valued decoder-side event can always be absorbed into a register at a factor K , but a continuous side observation admits no such finite-factor assertion in general.*

Proof. Use the product construction. For the finite event store its current value as the second register coordinate. For the last assertion, an exact uniform real observation supplied directly to a decoder has zero squared estimation error with one otherwise empty state. If it is available only before an M -label encoding, scalar quantization has error $1/(12M^2) > 0$. Thus no finite factor simulates that side interface for all accuracies. $\square$

The theorem joins quotient approximation to experiment comparison. Contractivity is a property to be verified on the true quotient, and Q_t is an executable representation of it. A finite-dimensional formula for the conditional mean is sufficient for one application, but is not a hypothesis of the

theorem. In Section 55 the true state is an entire probability measure. In the next section a different structure replaces metric contraction by an exact finite-tree update.

APPENDIX 53. A STRUCTURAL THEOREM FOR PREDICTIVE TREES

53.1. Local hypotheses and the finite profile. Let $\mathcal{L}$ be a pruned factorial language on a finite alphabet of size $b \geq 2$. Thus prefixes and suffixes of legal words are legal, every word has an infinite continuation, and only legal children occur in its rooted tree. Write $[v]$ for a cylinder in the boundary Ω and $\emptyset$ for the empty word. Assume that along any branch there are at most U consecutive vertices with only one legal successor. This condition allows unary vertices but excludes an infinite deterministic branch. Let μ be a stationary probability on Ω , positive on every legal cylinder, and put $p_v = \mu[v]$.

A predictive energy is a positive family a_v , with $a_\emptyset = 1$. Put $D_v = (a_v/p_v)^{1/2}$. The following are the hypotheses of the structural theorem. There are constants $a_- > 0$, $\rho < 1$, and $c_0, C_0 > 0$ such that

$$a_- a_v \leq a_{vi} \leq \rho a_v, \quad \sum_{i: vi \in \mathcal{L}} a_{vi} \leq \rho a_v, \quad (53.1)$$

$$a_{uv} < a_v \quad (u \neq \emptyset), \quad D_{vi} \leq D_v. \quad (53.2)$$

These statements concern legal words only. A bounded Borel map Z from Ω into a separable Hilbert space satisfies

$$\text{diam } Z([v]) \leq C_0 D_v, \quad (53.3)$$

$$\|Z(\omega) - Z(\omega')\| \geq c_0 D_u \quad \text{if } u \text{ is their longest common prefix.} \quad (53.4)$$

The image may be infinite dimensional. Separation is an explicit geometric hypothesis, not a consequence of the word notation.

Starting with the root, repeatedly split a leaf of maximal energy into *all* its legal children. Fix a tie rule. Let $\mathcal{T}_N$ be the last tree in this sequence with at most N leaves, $\mathcal{P}_N$ its leaf cut, and

$$G_N = \sum_{v \in \mathcal{P}_N} a_v. \quad (53.5)$$

A split at a unary vertex is an actual step. Finiteness of $\mathcal{T}_N$ will follow below. The construction is a mathematical finite algorithm; computable implementation of arbitrary real energies is an additional input, not part of their Borel specification.

There are two causal interpretations of these data.

Deletion model. Independent acquisitions $\omega \sim \mu$ occur at renewal times of independent positive integer length L , with $\mathbb{E}L < \infty$. The acquired sequence is observed exactly. At age k the desired output is $g(\sigma^k \omega)$, where g is bounded and Hilbert valued. Write

$$w_k = \mathbb{P}(L > k)/\mathbb{E}L, \quad W(n) = \sum_{k \geq n} w_k, \quad Z(\omega) = (\sqrt{w_k} g(\sigma^k \omega))_{k \geq 0}.$$

In addition to (53.1)–(53.4), assume the local oscillation bound

$$\sum_{k < |v|} w_k (\text{osc}_{[v_k:]} g)^2 + 4 \|g\|_\infty^2 W(|v|) \leq C_1 D_v^2. \quad (53.6)$$

A register sees the renewal event only during its update; the decoder receives neither that event nor a stored raw observation for free.

Insertion model. Let $(Y_t)_{t \in \mathbb{Z}}$ be a stationary finite-alphabet observed process whose reversed past $\omega_t = (Y_t, Y_{t-1}, \dots)$ has law μ and language $\mathcal{L}$. For a square-integrable stationary Hilbert target U_t , suppose

$$m_t = \mathbb{E}[U_t \mid Y_s : s \leq t] = Z(\omega_t), \quad B = \mathbb{E}\|U_t - m_t\|^2. \quad (53.7)$$

The finite register may start at zero with a fixed state; it is not initialized with the infinite past. Infinite stationary pasts here specify the reference prediction and its law. A finite warm-up is included in the limiting average. No renewal or independent-observation assumption is imposed in this interpretation.

For either model let R_M be the infimum, over all M -value registers with causal standard Borel update and decoding kernels, of the upper limiting expected average squared loss. Absolute time and independent coding randomization are permitted. In the insertion model put $E_M = R_M - B$.

Theorem 53.1 (Predictive-tree realization in two orientations). *Under (53.1)–(53.4),*

$$cG_M \leq e_M^2(Z) \leq CG_M. \quad (53.8)$$

Every complete greedy tree is suffix closed. A tree with L leaves has exactly $1 + \sum_{v \text{ internal}} d^+(v)$ vertices, and at most $(U + 1)(2L - 1)$ vertices; all of them may be used as designated states. For every fixed $B_0 \geq 1$, $G_{\lceil B_0 M \rceil} \asymp_{B_0} G_M$.

In the deletion model, with (53.6),

$$cG_M \leq e_M^2(Z) \leq R_M \leq CG_M. \quad (53.9)$$

In the insertion model,

$$cG_M \leq e_M^2(Z) \leq E_M \leq CG_M. \quad (53.10)$$

The converses apply to the lower limiting average of each randomized, time-dependent observer. The upper bounds are achieved in order by stationary deterministic updates. In particular a logarithmic age or history multiplier is not required. Constants depend on the displayed structural hypotheses, not on the register budget.

53.2. Balance, dilation and exact closure.

Lemma 53.2 (Greedy balance). *For all sufficiently large N , $N/b \leq |\mathcal{P}_N| \leq N$. If θ_N is the largest energy of a current leaf, then*

$$a_- \theta_N \leq a_v \leq \theta_N \quad (v \in \mathcal{P}_N). \quad (53.11)$$

The trees are finite and their profiles have the fixed-dilation property in Theorem 53.1.

Proof. Suppress unary vertices in any finite complete tree. A rooted tree with L leaves and no unary internal vertex has at most $2L - 1$ vertices, since its number of edges is the sum of its internal outdegrees. Each vertex of the suppressed tree accounts for itself and at most U unary vertices on its incoming path, including the initial root path. Thus the original tree has at most $(U + 1)(2L - 1)$ vertices. An infinite sequence of splits with bounded leaf number is impossible, proving finiteness. The split following $\mathcal{T}_N$ crosses the budget and adds at most $b - 1$ leaves. Hence $|\mathcal{P}_N| > N - b + 1$.

The maximal leaf energy never increases. Every present leaf was born with energy at least a_- times an earlier maximal energy. This proves (53.11), including any unary splits. Set $\ell = U + 1$. Every vertex has at least two distinct descendants at depth ℓ , and at least 2^j at depth ℓj . Choose j with $2^j > b(B_0 + 2)$. If the later maximal energy at budget $\lceil B_0 N \rceil$ were less than $a_-^{\ell j + 1} \theta_N$, every original leaf would have been split through all these ℓj levels: all intervening vertices have energy at least that threshold. The resulting number of leaves would exceed $\lceil B_0 N \rceil$, a contradiction. Therefore the later maximum is at least a fixed multiple of θ_N . Applying (53.11) at the later budget gives $G_{\lceil B_0 N \rceil} \geq c_{B_0} N \theta_N \geq c_{B_0} G_N$. Splitting decreases the energy sum by (53.1), giving the reverse inequality. Adjusting constants covers the finite initial range. $\square$

Lemma 53.3 (Two exact operations on one tree). *Every proper suffix of an internal greedy-tree vertex is internal, and every suffix of any vertex is present. Consequently deletion of the first letter stays in the tree. If v is a leaf and iv is legal, then iv has a leaf prefix in the same tree. It can therefore be updated by prepending i and reading only the letters of v until that leaf is reached.*

Proof. Suppose an internal word uv , $u \neq \emptyset$, is about to be split. If its suffix v has not been split, some current leaf t is a prefix of v . Child decrease and strict prefix-attachment decrease give $a_t \geq a_v > a_{uv}$, contrary to maximality. Thus the suffix is already internal. For a leaf $i_1 \cdots i_n$ with $n \geq 2$, its parent is internal; the suffix $i_2 \cdots i_{n-1}$ of that parent is internal, so its legal child $i_2 \cdots i_n$ is present. The one-letter case ends at the root. Iteration gives closure of every vertex.

If there were no leaf prefix of iv , the complete tree would contain iv as an internal vertex. Its suffix v would then be internal by the first assertion, contradicting that it is a leaf. This proves the insertion assertion. Only legal children were used in either proof. $\square$

53.3. Static lower bound and the causal converse.

Lemma 53.4 (Arbitrary centres). *The bounds (53.8) hold for arbitrary Hilbert-space centres, not only centres on $Z(\Omega)$.*

Proof. Choose one image point in each leaf cylinder to obtain $e_N^2(Z) \leq C_0^2 G_N$. For the lower bound use $\mathcal{P}_{4bM}$, which has at least $4M$ leaves for large M . The open tubes about $Z([v])$ of radii δD_v are disjoint for a sufficiently small fixed $\delta > 0$. Indeed for incomparable v, v' their common prefix u has $D_u \geq \max(D_v, D_{v'})$, and (53.4) separates the two images by at least $c_0 D_u$. Take $2\delta < c_0$.

At most M of the tubes contain one of M proposed centres. On every unoccupied cylinder the squared error is at least $\delta^2 D_v^2$. Its integrated contribution is $\delta^2 a_v$. By balance, the at least three-quarters unoccupied leaves carry at least $3a_-/4$ of the total energy sum. Thus $e_M^2(Z) \geq cG_{4bM} \geq c'G_M$ by Lemma 53.2. The finite initial range is harmless: (53.4) and positive cylinder masses at unbounded branching depths give infinitely supported image law, so every fixed finite-centre error is positive. Alternatively the same tube proof at a larger fixed cut proves that positivity directly. $\square$

Lemma 53.5 (Renewal converse). *In the deletion experiment every randomized, nonstationary M -state observer has lower limiting average loss at least $e_M^2(Z)$.*

Proof. A standard Borel probability kernel is realized as a measurable function of its input and an independent uniform random variable [4]. Applying this at the countably many update and decode times gives an independent coding sequence. A data-dependent persistent position in such a sequence, if used, belongs to the charged state; the representation itself supplies no history channel.

Condition on an acquisition at a fixed time, its pre-acquisition history, and the independent coding variables. The acquired ω still has law μ . Along a hypothetical continuation with no new acquisition, each of the at most M post-acquisition labels determines one output string. For a truncation at age n , the survival-weighted loss is therefore at least $\mathbb{E}L e_M^2(Z^{<n})$, where $Z^{<n}$ is the first n coordinates of the weighted orbit. At a fixed physical time, events specifying the last acquisition are disjoint. Summing this bound over acquisition times at least n before the end of a horizon and using the Cesàro renewal count gives lower limiting average at least $e_M^2(Z^{<n})$. For completeness, the elementary renewal theorem gives expected renewals per unit time tending to $1/\mathbb{E}L$; a fixed truncation removes at most n terminal time indices, each with bounded contribution. No stationary coding rule is presumed.

Finally $e_M^2(Z^{<n}) \uparrow e_M^2(Z)$. Monotonicity follows by projection. The reverse limiting inequality follows by appending zero tails to an arbitrarily good truncated code: the extra mean squared norm is at most $\|g\|_\infty^2 W(n) \rightarrow 0$. This completes the proof. $\square$

Proof of Theorem 53.1. The static assertion, finite count, dilation and suffix closure are proved above. In the deletion experiment retain every tree vertex as a designated state. At acquisition store the unique leaf prefix of ω . Between acquisitions delete its first letter until the empty word is reached, and then keep that word. Choose a representative in each nonempty cylinder and output

its g value; use a fixed bounded output at the empty word. For an acquired leaf v the weighted error is bounded by the left side of (53.6). Averaging and applying renewal reward gives at most $C_1 G_N$. Lemma 53.5 gives the unrestricted converse.

For insertion, after reaching a leaf use the prepend-and-parse rule of Lemma 53.3. During initialization store the observed reversed word while it is internal, or its first leaf prefix when one appears. This uses only vertices of the same tree and after at most its maximum depth gives the correct leaf prefix of the full observed past. Decode a representative Z value in that leaf cylinder. Its stationary excess loss is at most $C_0^2 G_N$ by (53.3) and conditional orthogonality. The finite warm-up has zero limiting average contribution. The converse holds at each time: freeze independent coding randomness, apply orthogonality against m_t , and use the at most M decoder values to lower bound excess by $e_M^2(Z)$. Stationarity makes this the same number at every time, even for time-dependent decoders.

In either orientation choose N a fixed sufficiently small multiple of M , so that $(U+1)(2|\mathcal{P}_N|-1) \leq M$. Dilation converts G_N into $O(G_M)$ without a time-depth multiplier. One fixed state handles the remaining bounded range of budgets. This proves the two risk assertions. $\square$

The static part of the proof uses familiar separated-cylinder quantization ideas [43, 36]. Maximal-energy prefix splitting is likewise a classical coding device [37]. The additional conclusion is that a strict *predictive* energy inequality makes the adaptive tree executable in both orientations, with its whole persistent state counted.

APPENDIX 54. TWO MODEL VERIFICATIONS

54.1. Markov-renewal orbit geometry. Let A be a primitive zero-one matrix on b symbols, with Perron root greater than one. Its legal one-sided sequences have a stationary fully supported law μ such that every legal conditional child probability is at least $\zeta > 0$. Let h_i map a compact set of diameter at most one into pairwise g -separated subsets of itself. Assume word scales r_v , $r_\emptyset = 1$, satisfy

$$\begin{aligned} D_0^{-1} r_v |x - y| &\leq |h_v x - h_v y| \leq r_v |x - y|, \\ D_0^{-1} r_u r_v &\leq r_{uv} \leq r_u r_v, \quad 0 < r_- \leq r_i \leq r_+ < 1. \end{aligned} \quad (54.1)$$

The injective coding map identifies the legal sequences with a compact set K , on which T deletes the first symbol. Let $f : K \rightarrow \mathbb{R}^m$ obey

$$|f(x) - f(y)| \leq H|x - y|^\beta, \quad \sum_{j=0}^J |f(T^j x) - f(T^j y)|^2 \geq c_o |x - y|^{2\beta}, \quad 0 < \beta \leq 1. \quad (54.2)$$

The renewal age weights satisfy $0 < q_- \leq w_{k+1}/w_k \leq q_+ < 1$. Set $\tau = 2\beta$ and

$$V(v) = \sum_{k < |v|} w_k r_{v_k}^\tau + W(|v|), \quad a_v = \mu[v]V(v), \quad Z(x) = (\sqrt{w_k} f(T^k x))_{k \geq 0}. \quad (54.3)$$

Corollary 54.1 (Markov renewal as a structural instance). *The data (54.1)–(54.3) satisfy Theorem 53.1 in its deletion orientation. In particular $e_M^2(Z) \asymp R_M \asymp G_M$, with a stationary finite-tree observer and an unrestricted randomized, nonstationary converse. Forbidden edges are not completed, and unary vertices are charged.*

Proof. Every suffix scale is at most one and appending a letter decreases its scale. If $|u| = m$ and $|v| = n$, monotonicity of w gives

$$\begin{aligned} V(v) - \left(\sum_{j < n} w_{m+j} r_{v_j}^\tau + W(m+n) \right) &\geq r_v^\tau \sum_{k < m} w_k, \\ V(uv) - \left(\sum_{j < n} w_{m+j} r_{v_j}^\tau + W(m+n) \right) &\leq r_+^\tau r_v^\tau \sum_{k < m} w_k. \end{aligned}$$

Therefore $V(uv) < V(v)$. Stationarity gives only $\mu[uv] \leq \mu[v]$, which is enough to give the strict energy inequality. Equality of cylinder masses at a unique predecessor causes no problem.

Appending a legal letter multiplies old spatial terms by a factor between $D_0^{-\tau} r_-^\tau$ and r_+^τ . The old tail $W(n)$ becomes $w_n r_i^\tau + W(n+1)$. Since $w_n \geq (1 - q_+)W(n)$ and $W(n+1) \geq q_-W(n)$, we have

$$\ell V(v) \leq V(vi) \leq \rho V(v), \quad \ell = \min(D_0^{-\tau} r_-^\tau, q_-), \quad \rho = 1 - (1 - r_+^\tau)(1 - q_+) < 1.$$

Use $a_- = \zeta \ell$ and sum the conditional masses to verify (53.1). These inequalities also give monotonicity of $D_v = \sqrt{V(v)}$. In a primitive graph with Perron root greater than one, a deterministic path cannot run for b repeated unary vertices: a repeated vertex would generate a closed deterministic cycle. Thus one can take $U = b$.

Introduce $d_\beta(x, y)^2 = \sum_k w_k |T^k x - T^k y|^{2\beta}$. Hölder continuity bounds the orbit distance from above by Hd_β . Summing the delayed lower bound (54.2), and using $w_{k-j} \leq q_-^j w_k$, bounds d_β from above by a constant times the orbit distance. Points in one cylinder have squared d_β distance at most $V(v)$. Points whose longest common prefix is u have, before its end, distances at least $D_0^{-1} g r_{u_k}$, and at its end distance at least g . The last term controls $W(|u|)$ since $W(n) \leq w_n/(1 - q_+)$. This proves both image comparisons. Finally the Hölder bound on each suffix and the bounded tail prove (53.6). The structural theorem now supplies all combinatorial, quantization, and causal conclusions. $\square$

This corollary recovers the earlier Markov-renewal profile without repeating its tree argument as a new model-specific proof. Gibbs pressure identifies the profile exponent under the additional thermodynamic assumptions of the previous Markov theorem, whose full proof is retained in the companion. Pressure is a computation of this instance, not an assumption of the foundational realization theorem. For clarity, the passage from power bounds to an exponent uses only the following elementary fact.

Proposition 54.2 (Logarithmic squeeze). *If for every $\delta > 0$ a positive sequence R_M satisfies $c_\delta M^{-\alpha-\delta} \leq R_M \leq C_\delta M^{-\alpha+\delta}$ eventually, then $-\log R_M / \log M \rightarrow \alpha$.*

Proof. Taking logarithms gives lower and upper limits between $\alpha - \delta$ and $\alpha + \delta$, because fixed logarithmic constants divided by $\log M$ vanish. Let δ decrease to zero. In a pressure-root proof one first chooses the auxiliary root parameter on either side of its zero and only then lets M tend to infinity; root continuity then supplies the arbitrary δ used here. $\square$

54.2. An infinite hidden shift with noisy continual reports. Let $(B_t)_{t \in \mathbb{Z}}$ be independent identically distributed Bernoulli variables. The hidden state at time t is the whole sequence $X_t = (B_t, B_{t-1}, \dots)$. A binary memoryless channel reports Y_t about B_t . Write

$$\gamma = \mathbb{P}(Y_t = 1) \in (0, 1), \quad a_i = \mathbb{P}(B_t = 1 \mid Y_t = i), \quad a_0 \neq a_1.$$

Fix $0 < r < 1$. The Hilbert-valued prediction target and its mean are

$$U_t = (\sqrt{1 - r^2} r^j B_{t-j})_{j \geq 0}, \quad m_t = (\sqrt{1 - r^2} r^j a_{Y_{t-j}})_{j \geq 0}. \quad (54.4)$$

This squared loss can equally be implemented by drawing a fresh terminal query j with probability $(1 - r^2)r^{2j}$ and scoring the prediction of B_{t-j} . The independent query is a task index, not a new observation of the hidden state. Its distribution and interface are fixed in advance.

Theorem 54.3 (Sharp register law for an infinite posterior). *Put $p_0 = 1 - \gamma$, $p_1 = \gamma$ and let $s \in (0, 1)$ be the unique solution of*

$$r^{2s}(p_0^s + p_1^s) = 1, \quad \alpha = (1 - s)/s. \quad (54.5)$$

For the actual noisy shift experiment,

$$R_M = B + \Theta((a_1 - a_0)^2 M^{-\alpha}), \quad B = \sum_{i=0}^1 p_i a_i (1 - a_i). \quad (54.6)$$

The constants in the excess comparison depend on r, p_0, p_1 , not on a_0, a_1 . No posterior truncation is assumed in deriving this law. For a fair hidden bit and a binary symmetric channel of error $\varepsilon < 1/2$, it becomes

$$R_M = \varepsilon(1 - \varepsilon) + \Theta((1 - 2\varepsilon)^2 M^{2 \log r / \log 2}).$$

At $\varepsilon = 1/2$ the predictive mean is constant and the excess is zero with one state.

Proof. The full posterior is the product of Bernoulli laws with parameters $a_{Y_{t-j}}$. Thus (54.4) is the actual conditional mean, and orthogonality gives the displayed B . Normalize its nonconstant part by $a_1 - a_0$. On the full binary past language take $p_v = \prod p_{v_j}$, $D_v = r^{|v|}$ and $a_v = p_v r^{2|v|}$. Children multiply energy by $p_i r^2$, their sum multiplies it by r^2 , and every nonempty prefix attachment strictly decreases it. If two pasts first differ at position n , their normalized squared image distance lies between $(1 - r^2)r^{2n}$ and r^{2n} . All hypotheses of Theorem 53.1 hold in the insertion orientation.

For the exact order of G_M , set $q_i = (p_i r^2)^s$. Equation (54.5) says $q_0 + q_1 = 1$. For every finite complete cut, consistency of this Bernoulli measure gives $\sum_v a_v^s = 1$. By greedy balance, for a cut with L leaves and maximum θ , $L(a_- \theta)^s \leq 1 \leq L\theta^s$. Hence $G_M \asymp L\theta \asymp L^{-(1-s)/s} \asymp M^{-\alpha}$. The root exists uniquely because the left side of (54.5) decreases continuously from two to $r^2 < 1$. The fair case solves it explicitly, and the constant-mean case is immediate. If only reports from time zero are supplied, the unobserved past contributes at most $r^{2(t+1)}$ to the squared target norm at time t . Its average vanishes; the finite-tree initialization in the structural theorem still gives (54.6). $\square$

This is not an exactly closed finite list of conditional means. Every past report changes its own posterior coordinate, and arbitrarily many coordinates can be queried. It also has no fixed-step common minorizer: after k hidden updates the old tail remains intact, and two different initial tails give disjoint supports. A measure dominated by all those kernels must be zero. As always, a bare Borel encoding of an infinite sequence into one real number is possible; it is not a finite-cardinality register and does not change the theorem.

APPENDIX 55. FINITE PRESENTATIONS OF MEASURE-VALUED FILTERS

The preceding sharp example has a separated tree geometry. We next give a constructive approximation theorem requiring no such separation and no finite-dimensional exact posterior coordinate.

Lemma 55.1 (A finite Wasserstein presentation). *For $n \geq 1$ let $\mathcal{C}_n$ consist of empirical probability measures with n equally weighted atoms on $\{0, 1/n, \dots, 1\}$. Then*

$$|\mathcal{C}_n| = \binom{2n}{n} \leq 4^n, \quad \sup_{\pi \in \mathcal{P}([0,1])} W_1(\pi, Q_n \pi) \leq \frac{3}{2n} \quad (55.1)$$

for a specified Borel quantizer Q_n .

Proof. Take the generalized quantiles of π at $(j - 1/2)/n$, $1 \leq j \leq n$, and round them to nearest grid points, breaking ties toward zero. Quantiles are Borel functions of a probability law. If q is its monotone quantile function, the integral of the error from replacing q by its midpoint value on each of the n equal subintervals is at most $1/n$: bound each integral by its interval length times the oscillation and sum the monotone variations, at most one. Rounding adds at most $1/(2n)$. In one dimension W_1 is the L^1 distance of quantile functions, giving the bound. The possible sorted lists are multisets of size n from $n + 1$ points, counted by $\binom{2n}{n}$. The binomial theorem bounds this by 4^n . $\square$

Theorem 55.2 (A charged measure-valued recursive approximation). *Let the actual filter obey $\pi_t = \Phi_{y_t, a_t}(\pi_{t-1})$ on $\mathcal{P}([0, 1])$, and suppose every allowed update is ρ -contractive in W_1 , with $\rho < 1$. Suppose an implementable update has W_1 defect at most η . The initialized register*

$$\hat{\pi}_t = Q_n \tilde{\Phi}_{y_t, a_t}(\hat{\pi}_{t-1}), \quad \hat{\pi}_0 \in \mathcal{C}_n$$

has at most $\binom{2n}{n}$ persistent values and satisfies, for every actual observation path,

$$W_1(\pi_t, \hat{\pi}_t) \leq \rho^t + \frac{3/(2n) + \eta}{1 - \rho}. \quad (55.2)$$

This bounds the error of every 1-Lipschitz test expectation, uniformly over the choice of test. Thus for $M \geq 4$, $n = \lfloor \log M / \log 4 \rfloor$ gives error $O((\log M)^{-1} + \eta)$ and squared error $O((\log M)^{-2} + \eta^2)$ after the initialization term. If these test expectations are conditional means, the latter is a Bayes excess bound. Report-law comparison additionally requires the report Lipschitz hypothesis of Theorem 52.1; it is not automatic from W_1 contraction.

Proof. Apply Theorem 52.1 with the true metric state $S = \mathcal{P}([0, 1])$, radius $3/(2n)$, and diameter one. The dual formula for W_1 gives the assertion for every test; independent selection of a terminal test does not enlarge its bound. The codebook count and the logarithmic substitution are exact. Only an index of a codebook law persists between updates, not its exact unquantized posterior or a numerical residual. $\square$

55.1. A nonlinear physical realization. Let B_t be independent fair bits and define

$$h_0(x) = x/5 + x^2/50, \quad h_1(x) = 3/5 + x/5 + x^2/50, \quad X_t = h_{B_t}(X_{t-1}). \quad (55.3)$$

The observation Y_t reports B_t correctly with probability $3/4$, independently at each time. A permitted terminal probe $g : [0, 1] \rightarrow [0, 1]$ returns a Bernoulli variable of probability $g(X_t)$; the tested family is 1-Lipschitz probes and their bounded finite executable continuations. The actual update-report instrument on measures is

$$\mathsf{T}_y \pi = \frac{1}{2} \left\{ \frac{3}{4} (h_y)_\# \pi + \frac{1}{4} (h_{1-y})_\# \pi \right\}.$$

Its mass is $1/2$, independent of the old state. The normalized filter is therefore

$$\Phi_y \pi = \frac{3}{4} (h_y)_\# \pi + \frac{1}{4} (h_{1-y})_\# \pi. \quad (55.4)$$

Corollary 55.3 (Nonlinear posterior, explicit register). *The actual measure-valued filter (55.4) is $6/25$ -contractive in W_1 . Theorem 55.2 applies with $\eta = 0$ to an exact rational implementation on $\mathcal{C}_n$. The model has no fixed-step nonzero common minorizer. For every integer $D \geq 1$, the first D polynomial moments do not determine their own exact next-step update on the space of all prior laws.*

Proof. The two images lie respectively in $[0, 11/50]$ and $[3/5, 41/50]$. Each derivative is between $1/5$ and $6/25$. Couple two old laws optimally and use the same branch draw in their mixtures; this gives $W_1(\Phi_y \pi, \Phi_y \pi') \leq (6/25) W_1(\pi, \pi')$. For a codebook input, (55.4) has at most $2n$ atoms, at rational points with denominator dividing $50n^2$, with weights multiples of $1/(4n)$. Sorted cumulative weights give the exact quantiles, which are rounded back to the fixed grid. Thus no

numerical limit is being hidden in $\eta = 0$. The transient list of at most $2n$ atoms is discarded; its entries and weights have $O(\log n)$ -bit rational descriptions. The designated persistent object is the single codebook index. Approximate arithmetic instead obeys the separately charged η bound of the theorem.

After k hidden transitions the law from a fixed x has finite support $\{h_v(x) : |v| = k\}$. For a second initial point x' , each equality $h_v(x') = h_u(x)$ excludes at most one x' , since h_v is strictly increasing. Choose x' outside this finite set. The two supports are disjoint, so a measure dominated by all k -step laws must be zero.

For the moment assertion, the D th updated moment integrates $\frac{3}{4}h_y(x)^D + \frac{1}{4}h_{1-y}(x)^D$, a polynomial of degree $2D$ with nonzero leading coefficient 50^{-D} . It is not in the span of $1, x, \dots, x^D$. Linear algebra on finitely many distinct points therefore supplies a signed measure annihilating this span but not the new polynomial. Its positive and negative parts have equal mass; normalizing them yields two prior probabilities with identical first D moments and different updated D th moment. The argument is about ordinary finite moment closure, not the impossibility of a Borel encoding of measures by real numbers. $\square$

The quotient here is genuinely the posterior law for the specified probe family: bounded Lipschitz functions separate probability measures on a compact interval, and constants do not affect that separation. Updates respect this quotient by (55.4). The advancing reports have the exact state-independent fair law, while a terminal Bernoulli probe has total-variation defect at most $W_1(\pi_t, \hat{\pi}_t)$. Thus the same finite presentation defines a causal experiment simulation with a controlled terminal error and an exactly charged register. Multiple probes would update the posterior again; no claim for an unverified enlarged probe protocol is implicit. This example connects positive instruments, predictive quotient, recursive approximation and resource transformation within one model.

APPENDIX 56. SHARP FINITE-EXPERIMENT BUDGETS

For experiments $\mathcal{E} = \{P_\theta\}$ and $\mathcal{F} = \{Q_\theta\}$ with the same parameter set, write

$$\delta(\mathcal{E}, \mathcal{F}) = \inf_K \sup_\theta \|KP_\theta - Q_\theta\|_{\text{TV}}, \quad \Delta(\mathcal{E}, \mathcal{F}) = \max\{\delta(\mathcal{E}, \mathcal{F}), \delta(\mathcal{F}, \mathcal{E})\}.$$

All parameterized laws in this section are Borel families. The infimum is over parameter-independent Markov kernels. These are terminal experiment comparisons in the sense of [45, 46]; only the encoding statistic, not the comparison randomization, is part of the physical data reduction.

Let $\mathcal{G} = \{N(z, \Gamma) : z \in K\}$, where $K \subset \mathbb{R}^r$ is compact and Γ is positive definite. In the lower bound assume that K contains a nonempty Euclidean ball.

Theorem 56.1 (Gaussian finite-experiment cardinality). *There are constants $c, C > 0$, depending on K, Γ, r , such that*

$$cM^{-2/r} \leq \inf_{\mathcal{F}: |\mathcal{F}| \leq M} \Delta(\mathcal{F}, \mathcal{G}) \leq CM^{-2/r}, \quad M \geq 1. \quad (56.1)$$

The lower bound ranges over all experiments on at most M outcomes, not just deterministic Gaussian quantizers. The upper bound is attained in order by a deterministic statistic with J^r labels, where $J = \lfloor M^{1/r} \rfloor$, followed by a specified positive reconstruction kernel. No overflow label is needed.

56.1. The cardinality obstruction.

Lemma 56.2 (A decision-theoretic lower bound). *Under the preceding assumptions, $\Delta(\mathcal{F}, \mathcal{G}) \geq cM^{-2/r}$ whenever $\mathcal{F}$ has at most M outcomes.*

Proof. Restrict the mean parameter to a closed cube K_0 of positive side length contained in K and put the uniform prior on K_0 . Restriction can only decrease an experiment distance. Use squared-error estimation of z , with actions restricted to K_0 ; its loss range is bounded by $L = \text{diam}(K_0)^2$. Let $m(y) = \mathbb{E}[z \mid y]$ in this Gaussian experiment. Differentiation under the compact prior integral gives

$$Dm(y) = \text{Cov}(z \mid y)\Gamma^{-1}. \quad (56.2)$$

The posterior density is positive on the cube. Hence its covariance is positive definite, and $Dm(y)$ is nonsingular for every y . The inverse function theorem on a small neighborhood of any fixed y_0 , together with the positive continuous marginal density of y , shows that the law of $m(y)$ has a density bounded below by a positive constant on some smaller cube in $\mathbb{R}^r$.

This implies $e_M^2(m) \geq c_0 M^{-2/r}$. To see it directly, put balls of radius $t = (|Q|/(2Mv_r))^{1/r}$ about the M proposed centres, where Q is the smaller cube and v_r the volume of the unit ball. At least half of Q remains outside them, and contributes at least a fixed multiple of t^2 . This argument permits arbitrary centres.

Every M -outcome garbling S of the Gaussian observation therefore has Bayes risk at least $B_{\mathcal{G}} + c_0 M^{-2/r}$, by conditional orthogonality. If $\Delta(\mathcal{F}, \mathcal{G}) < \epsilon$, choose a kernel K from $\mathcal{G}$ to $\mathcal{F}$ with error below ϵ . The experiment $K\mathcal{G}$ has at most M outputs, and bounded-loss comparison gives

$$B_{K\mathcal{G}} \leq B_{\mathcal{F}} + L\epsilon \leq B_{\mathcal{G}} + 2L\epsilon.$$

The second inequality uses a kernel in the reverse direction. Comparing these bounds and taking the infimum over ϵ proves the result. Randomized actions do not improve a Bayes squared-error minimum. Notice that $\mathcal{F}$ itself was not assumed to be a Gaussian garbling. $\square$

56.2. Positive second-order reconstruction. Put

$$\tau(x) = \frac{1}{2} + \pi^{-1} \arctan x, \quad u = \tau^{\otimes r}(y) \in (0, 1)^r. \quad (56.3)$$

Let p_z be the transformed Gaussian density. If $x_j = \tan \pi(u_j - 1/2)$, then

$$p_z(u) = \phi_{\Gamma}(x - z) \prod_{j=1}^r \pi(1 + x_j^2). \quad (56.4)$$

It extends by zero across every face to a smooth periodic density. Indeed each fixed-order derivative is a polynomial in x times the Gaussian density, which tends to zero faster than any polynomial as a face is approached. Uniformity holds for z in the fixed compact set. For use below define the finite constants

$$H_2 = \sup_{z, u} \|D^2 p_z(u)\|_{\text{op}}, \quad H_0 = \max_j \sup_{z, u_j} p_{z, j}(u_j), \quad (56.5)$$

where $p_{z, j}$ is a transformed marginal density. These suprema, or any upper bounds obtained from (56.4), fix all the reconstruction and boundary constants; they do not depend on M or sample size. For instance, differentiating twice introduces only polynomials and the eigenvalue bounds of Γ , so bounded mean norm and fixed covariance conditioning suffice.

Partition the flat unit torus into J^r cubes of side $h = 1/J$, with centres c_i , and write $Q_J(u)$ for the cube label. For $J \geq 2$ let Φ_i be the periodic tensor product of the piecewise-linear hat functions with value one at c_i , zero at its neighboring mesh centres, and support within one mesh step in each coordinate. Then

$$\Phi_i \geq 0, \quad \int \Phi_i = h^r, \quad \sum_i \Phi_i(u) = 1.$$

On each mesh interval the neighbouring centres reproduce the affine coordinate, using local periodic lifts across the seam. Define the reconstruction from label i to have density $h^{-r} \Phi_i(u)$, then apply τ^{-1} coordinatewise. Values on the faces have probability zero.

Lemma 56.3 (A quadratic reconstruction estimate). *Uniformly in $z \in K$, this kernel H_J satisfies*

$$\|H_J Q_J \tau^{\otimes r} N(z, \Gamma) - N(z, \Gamma)\|_{\text{TV}} \leq \frac{r H_2}{12 J^2}. \quad (56.6)$$

Proof. Let p_i be the probability of cube i . Symmetric Taylor expansion about its centre gives $|p_i/h^r - p_z(c_i)| \leq r H_2 h^2/24$. The linear interpolant $I_h p_z = \sum_i p_z(c_i) \Phi_i$ has error at most $r H_2 h^2/8$: its weights reproduce constants and affine functions and the sum of their coordinate variances is at most $r h^2/4$. The second Taylor remainder is half this variance times H_2 . Both calculations hold across the seam because p_z has a smooth periodic extension. The reconstructed density is $\sum_i (p_i/h^r) \Phi_i$. Nonnegativity and the partition of unity give supremum error at most $r H_2 h^2/6$. Integrating over volume one and using the factor $1/2$ for total variation proves the claim. Inverting (56.3) preserves this distance. $\square$

Proof of Theorem 56.1. Lemma 56.2 gives the lower bound. The deterministic statistic $Q_J \tau^{\otimes r}$ has J^r outputs, so the forward deficiency is zero; Lemma 56.3 gives the reverse deficiency. For $M \geq 2^r$, $J = \lfloor M^{1/r} \rfloor \geq M^{1/r}/2$. For the remaining finite range use any one-outcome experiment and the trivial bound $\Delta \leq 1$, increasing C . The reconstruction uses positive probabilities; it is not a signed interpolation operator. $\square$

The use of a compactified grid avoids a growing Gaussian overflow radius. The use of hats rather than constant cells cancels the first Taylor term. Both choices are necessary for the stated proof of a logarithm-free quadratic budget rate.

APPENDIX 57. A SHARP BUDGET FOR THE PHYSICAL COUNT-TO-CONTACT EXPERIMENT

We now specify the actual local protocol, following the known-mark experiment of A2 [42]. Fix $\lambda_0 \in (0, \infty)^m$, a matrix $B \in \mathbb{R}^{m \times d}$, a compact set $H \subset \mathbb{R}^d$, and a full-row-rank matrix $C \in \mathbb{R}^{r \times m}$. Observe independent counts

$$K_{N,j} \sim \text{Pois}(N \lambda_{0,j} + \sqrt{N}(Bh)_j), \quad h \in H. \quad (57.1)$$

Write $\Lambda = \text{diag } \lambda_0$, $A = CB$, and $\Gamma = C \Lambda C^T$. The contact score and limiting experiment are

$$S_N = C(K_N - N \lambda_0)/\sqrt{N}, \quad \mathcal{G} = \{N(Ah, \Gamma) : h \in H\}.$$

The physical finite statistic is

$$T_{N,J} = Q_J(\tau^{\otimes r}(S_N)), \quad |T_{N,J}| \leq J^r. \quad (57.2)$$

There is no physical jitter, no retained real score, and no omitted overflow event. Every real score belongs to one of the J^r cells of the open compactified cube, with deterministic boundary tie-breaking.

Theorem 57.1 (Joint sample and register law). *For the fixed protocol (57.1), there are $N_0, C_* < \infty$ such that uniformly for $N \geq N_0$ and $J \geq 2$,*

$$\Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C_* \left(N^{-1/2} + \frac{J}{\sqrt{N}} + J^{-2} \right), \quad (57.3)$$

where $\mathcal{E}_{N,J} = \{\text{Law}_h(T_{N,J}) : h \in H\}$. If AH contains an r -dimensional ball, every experiment with at most M outputs has distance from $\mathcal{G}$ at least $c_* M^{-2/r}$. Consequently the statistic with $J = \lfloor M^{1/r} \rfloor$ satisfies

$$c_* M^{-2/r} \leq \Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C_{*,K} M^{-2/r}, \quad 2^r \leq M \leq K N^{r/6}, \quad (57.4)$$

for every fixed $K < \infty$. In particular $M \asymp N^{r/6}$ gives distance of order $N^{-1/3}$; achieving distance of order $N^{-1/5}$ requires and is achieved by a budget of order $N^{r/10}$. All constants may depend on the fixed matrices, compact parameter set and covariance conditioning. No uniformity at rank degeneration or unknown marks is asserted.

We prove the uniform estimates in five steps, keeping comparison randomization separate from (57.2).

Lemma 57.2 (Jittered Poisson local normal comparison). *There is an absolute $C_P < \infty$ such that for $\mu \geq 1$, independent $U \sim \text{Unif}(-1/2, 1/2)$, and $K \sim \text{Pois}(\mu)$,*

$$\|\text{Law}(K + U) - N(\mu, \mu)\|_{\text{TV}} \leq C_P \mu^{-1/2}. \quad (57.5)$$

Proof. It suffices to consider $\mu \geq 64$, increasing the constant for the remaining range. The jittered density is $p_\mu(k) = e^{-\mu} \mu^k / k!$ on $[k - 1/2, k + 1/2)$, $k \geq 0$. On the central cells $|k - \mu| \leq \mu^{2/3}/4$, put $t = k - \mu$, $x = k + u$, $|u| \leq 1/2$, and $z = (x - \mu)/\sqrt{\mu}$. Stirling's formula with remainder $0 < R_k < 1/(12k)$ gives

$$\log \frac{p_\mu(k)}{\phi_\mu(x)} = -\frac{1}{2} \log(k/\mu) - R_k - \{k \log(k/\mu) - k + \mu\} + \frac{(x - \mu)^2}{2\mu},$$

where ϕ_μ is the $N(\mu, \mu)$ density. For $|t| \leq \mu/2$, Taylor's theorem applied to $(1 + v) \log(1 + v) - (1 + v) + 1$ gives remainder at most $|t|^3/\mu^2$ after its quadratic term. Also $|\log(1 + t/\mu)| \leq 2|t|/\mu$ and $|(x - \mu)^2 - t^2| \leq |t| + 1/4$. Substitution therefore gives the convenient bound

$$\left| \log \frac{p_\mu(k)}{\phi_\mu(x)} \right| \leq 32\mu^{-1/2}(1 + |z|^3), \quad \left| \log \frac{p_\mu(k)}{\phi_\mu(x)} \right| \leq 64. \quad (57.6)$$

The constants are deliberately loose. For example $|z| \leq \mu^{1/6}/4 + 1/(2\sqrt{\mu})$ on these cells verifies the second bound directly. The inequality $|e^v - 1| \leq e^{|v|}|v|$ now bounds the integrated central density error by $32e^{64}\mu^{-1/2}(1 + \mathbb{E}|G|^3)$, $G \sim N(0, 1)$.

For completeness the omitted tails are exponentially smaller. For $|s| \leq 1$, the centered Poisson log moment generating function is $\mu(e^s - 1 - s) \leq \mu s^2$. Exponential Markov with $s = \mu^{-1/3}/16$, and its negative, bounds the probability outside the central cells, allowing the half-unit jitter margin, by $2 \exp(-\mu^{1/3}/128)$ after adjusting an absolute factor for $64 \leq \mu$ in the bounded initial interval. The Gaussian bound follows by its moment generating function. Thus an absolute choice such as $C_P = 10^8 e^{64}$ dominates the central error, the tails and the finite initial range. This proves a uniform $\mu^{-1/2}$ bound rather than an unquantified local-limit assertion. $\square$

Lemma 57.3 (Variance replacement and projected comparison). *Let U_j be independent uniform half-cell jitters, used only in this proof, and set $S_N^o = S_N + CU/\sqrt{N}$. Uniformly for $h \in H$,*

$$\|\text{Law}_h(S_N^o) - N(Ah, \Gamma)\|_{\text{TV}} \leq C_0 N^{-1/2}. \quad (57.7)$$

One may choose C_0 as a function only of m , the minimum of λ_0 , $\max_{h \in H} \|Bh\|$, and fixed matrix norms.

Proof. Put $b_* = \max_{h \in H} \|Bh\|_\infty$ and $\lambda_* = \min_j \lambda_{0,j}$. Choose $N_0 \geq \max(128/\lambda_*, 4b_*^2/\lambda_*^2)$. Then each count mean is at least $N\lambda_*/2$. Product comparison using Lemma 57.2, followed by centering and scaling, gives distance at most $mC_P\sqrt{2/\lambda_*} N^{-1/2}$ from the normal vector with mean Bh and covariance $\Lambda + N^{-1/2} \text{diag}(Bh)$.

For the replacement of this covariance by Λ , the relative variance eigenvalues are $1 + t_j$, with $|t_j| \leq b_*/(\lambda_*\sqrt{N}) \leq 1/2$. The exact Gaussian relative entropy is $\frac{1}{2} \sum_j \{t_j - \log(1 + t_j)\} \leq \sum_j t_j^2$. The elementary entropy-variation inequality $\|P - Q\|_{\text{TV}} \leq \sqrt{D_{\text{KL}}(P\|Q)/2}$ gives a further bound $\sqrt{m/2} b_*/(\lambda_*\sqrt{N})$. Taking $C_0 = mC_P\sqrt{2/\lambda_*} + \sqrt{m/2} b_*/\lambda_*$ is sufficient. Projection by C contracts total variation and sends the fixed Gaussian covariance to Γ , proving the assertion. $\square$

Lemma 57.4 (Deterministic cell crossing). *Let H_0 be the transformed marginal bound in (56.5), now for the means AH , and put $d_N = \|C\|_{\infty,1}/(2\pi\sqrt{N})$. Then*

$$\sup_{h \in H} \|\text{Law}_h(T_{N,J}) - Q_J \tau^{\otimes r} N(Ah, \Gamma)\|_{\text{TV}} \leq 2C_0 N^{-1/2} + 2rH_0 J d_N. \quad (57.8)$$

Here $\|C\|_{\infty,1}$ is the maximum absolute row sum.

Proof. The coupled scores satisfy $\|S_N - S_N^o\|_{\infty} \leq \|C\|_{\infty,1}/(2\sqrt{N})$. Since $|\tau'| \leq 1/\pi$, their compactified coordinates differ by at most d_N . Distinct grid labels require a coordinate of the jittered score to be within d_N of a grid boundary. On the torus there are J boundaries per coordinate; under the transformed Gaussian their total probability is at most $2rH_0 J d_N$, or its truncation by one. Replacing the jittered law by that Gaussian adds at most $C_0 N^{-1/2}$, by (57.7). The coupling inequality then compares theunjittered and jittered labels. One more data-processing comparison of the jittered labels to the Gaussian labels adds the second $C_0 N^{-1/2}$. Boundary tie events are already in the strips. This argument never asserts that the unjittered real score has a Gaussian law in total variation. $\square$

Proof of Theorem 57.1. Use the deterministic Gaussian-to-cell kernel for the direction $\mathcal{G} \rightarrow \mathcal{E}_{N,J}$. Lemma 57.4 bounds its error by $b_{N,J} = 2C_0 N^{-1/2} + 2rH_0 J d_N$. For the reverse direction, use the positive kernel H_J from Lemma 56.3. Kernel contraction and the triangle inequality give

$$\sup_h \|H_J \text{Law}_h(T_{N,J}) - N(Ah, \Gamma)\|_{\text{TV}} \leq b_{N,J} + rH_2/(12J^2).$$

This proves (57.3), with, for example, $C_* = 2C_0 + rH_0\|C\|_{\infty,1}/\pi + rH_2/12$ increased to one. Every kernel is parameter independent. Jitter was introduced only to bound a difference of laws, not to modify the physical statistic.

If AH contains a ball, restrict to an affine parameter section over a smaller mean cube and apply Lemma 56.2. More generally one may simply index the Gaussian family by its mean and choose a Borel representative preimage for each mean in the cube (the compact fibres of the continuous map on H admit a measurable section); the lower comparison uses only that subfamily. For $J = \lfloor M^{1/r} \rfloor$ and $M \geq 2^r$, $J^{-2} \leq 4M^{-2/r}$. If $M \leq KN^{r/6}$, then $M^{3/r}/\sqrt{N} \leq K^{3/r}$; hence $J/\sqrt{N}$ and $N^{-1/2}$ are also at most fixed multiples of $M^{-2/r}$. This proves the window. Taking M respectively of order $N^{r/6}$ and $N^{r/10}$ gives the two rates. Conversely, an error at most $CN^{-1/5}$ forces $M \geq cN^{r/10}$ by the universal cardinality lower bound. $\square$

57.1. The A2 matrices and the scope of the consequence. In A2's known-mark local protocol, the moment derivative D has $\Sigma = D\Lambda D^T > 0$, and its calibrated exposure direction is $B = \Lambda D^T \Sigma^{-1}$, so $DB = I$. If L maps to the contact coordinates and F is an orthonormal frame of its range, the matrices above are

$$C = F^T L D, \quad A = C B = F^T L, \quad \Gamma = C \Lambda C^T > 0.$$

Here A has full row rank r . A compact local parameter neighborhood therefore has an image containing a mean ball. Both sides of Theorem 57.1 apply to that actual protocol. This invokes a structural experiment-cardinality theorem verbatim, not just an analogy between a root singularity and a contact observation.

The result is a matched budget theorem in the displayed joint window. It is not a lower bound saying that $N^{-1/3}$ is the best sample-size comparison achievable with arbitrarily many labels. It also does not assign a stochastic noise law to a conductor scheme or establish uniformity at a rank-changing physical stratum. The geometric source, the exposure protocol, the finite statistic and the comparison kernels are distinct typed objects throughout.

PRESERVED HISTORICAL INTRODUCTIONS AND SUPPORTING BODIES

APPENDIX 58. INTRODUCTION

Statistical comparison asks which decisions can be transferred from one experiment to another. When observations arrive sequentially and the decision procedure has a fixed retained register, comparison has an additional constraint: the randomization used to transfer an experiment must itself obey the information and memory interface. A hidden persistent selector, an exposed selector, and transient private randomness define different feasible sets. Equality of unpriced experiments does not imply equality of their fixed-register risk bodies.

We develop comparison at the original resource signature. The central finite object is the image of a product of stochastic-row simplexes under the map taking an executable causal simulator to its marked feedback laws. This image is generally nonconvex. Convex separation of its convex hull can therefore lose precisely the private-resource obstruction one wishes to measure. Our main finite result replaces a fixed mixture of tests by polynomial proof functions on the simulator rows. The resulting strictly positive inequalities admit finite exact certificates without adding a persistent selector to the simulator.

The same lower certificates can be attached to genuine physical acquisitions. This requires more than approximating an observable in norm: the physical approximation must preserve the actual mark, respect feedback, produce finite reports, and rationalize their joint laws. We retain each of those steps and prove an explicit realization in which a nonconserved collision-compatible observable has both nonzero generator residual and a positive exactly characterized marked deficiency. The statistical and microscopic conclusions are joined at the same register and selector budgets.

58.1. The main theorem and its components. For a fixed finite marked experiment and resource signature b , write

$$\delta_b = \min_{q \in \mathcal{P}_b} \max_i f_i(q).$$

Here q is an executable stochastic-row table, the f_i are the finite marked event discrepancies under deterministic feedback, and $\mathcal{P}_b$ is a product of row simplexes. Visible selectors remain part of the target reference as well as the simulated joint law. An onto polynomial stick-breaking parameterization $q = q(x)$ places the domain on $[0, 1]^D$. With $h_i = f_i \circ q$ and $g_i = (1 + h_i)/2$, including the empty test, we prove

$$\delta_b = \sup_{p \geq 1} \min_{x \in [0, 1]^D} \frac{\sum_i g_i(x)^p h_i(x)}{\sum_i g_i(x)^p}. \quad (58.1)$$

The minima increase to δ_b with error at most $2 \log m / (p + 1)$, where m is the number of tests. For rational L , the strict inequality $L < \delta_b$ is equivalent to a strictly positive finite tensor Bernstein expansion of $\sum_i g_i^p(h_i - L)$ for some finite p and degree. The proof includes an explicit degree bound in terms of the monomial coefficients and the positivity margin. This is Theorem 6.1.

The weights in (58.1) depend on a candidate table; they are not independent test probabilities in an exchanged minimax. A two-time example has private width-one deficiency $\sqrt{2} - 1$, hidden-two-selector deficiency zero, and visible-selector deficiency $\sqrt{2} - 1$ at every finite selector size. Every fixed test mixture has minimum at most zero there. Thus the distinction is visible already in a concrete exact problem, not just in a formal choice of dual notation. The source supplies a strictly positive rational Bernstein certificate for the private lower bound $2/5$.

For physical acquisitions, Theorem 16.1 combines the polynomial certificate with stateless marked presentations. A finite lower-upper interval $[L, U]$ gives the original physical interval

$$[\max\{0, L - a - c\}, \min\{1, U + a + c\}],$$

where a, c include analytic approximation, report quantization and joint-prefix rationalization. There is no change in the resource signature. Under the stated effective data hypotheses these intervals converge. The finite search and the analytic presentation search are distinguished; neither an abstract graph core nor a decimal transition table is called a rigorous physical input.

Our explicit microscopic realization uses $N \geq 2$ interacting hard spheres in equilibrium on a torus. For $\vartheta = (2\pi/\ell) \sum_i q_{i,1}$ and normalized total momentum X , the nonconserved observable $f = \cos \vartheta$ obeys $U(t)f = \cos(\vartheta + \omega X t)$ with $\omega = 2\pi\sqrt{N}/\ell$. All generator powers and their Gram entries are computed on the actual collision-compatible domain. Reversal interventions give exactly $k + 1$ distinct mean functions among 2^k words. A Taylor presentation then has a computable error tending to zero, uniformly over consumer width. The actual initial sign mark yields an exact physical deficiency series; for the parameters of Corollary 15.5 its value lies strictly between 0.205809122887 and 0.205809122889, while the normalized one-dimensional residual is $2\pi^2/25 > 0$. The same example supplies both quantities.

Finally, for a bounded marked payoff of oscillation M , a variation error ε changes a normalized logarithmic value at scale β by at most $\beta^{-1} \log(1 + \varepsilon(e^{\beta M} - 1))$. Theorem 16.2 proves sharpness and transports optimized common-policy values. This identifies the additional accuracy needed at an exponential scale rather than asserting an unproved large-deviation conclusion from unscaled comparison.

58.2. Classical ingredients and the comparison of scope. Blackwell–Le Cam comparison and its decision-theoretic formulations provide the unpriced background [1, 47, 46]. Filtered comparison is a direct antecedent, not a remote analogy. Weisshaupt’s generalized filtered randomization theorem uses a compact convex class of filtration-preserving stochastic operators and a separation/minimax argument [68]. Our private feasible image is not that convex class. The two-time example above states exactly what ordinary constant convex testing loses, and the positive polynomial theorem states exactly what replaces it. We do not claim that polynomial positivity, power-mean approximation, or Bernstein basis elevation is new; their elementary proofs are included to expose the hypotheses on which the new causal application depends.

The retained existence theory credits weak-star policy topology and strategic-measure approaches [55, 56]. Common-information and finite-memory formulations are also prior structural tools [54, 53]. State equivalence, incomplete-machine minimization, and weighted-automaton methods are relevant to the retained exact task and synthesis results [49, 70, 58]. Those results are not relabelled as new polynomial duality. The accompanying comparison records which original sources were inspected and which proof-level priority questions remain unverified, including the Norberg original. The present novelty assertion is the stated resource-sensitive theorem and its physical realization, not an assertion of an exhaustive priority search across every form of filtered comparison.

The microscopic input is the classical almost-everywhere hard-sphere flow [63]. The new collective calculation uses a solvable mode of that interacting system; it is not a solution of its kinetic limit. This distinction is mathematical: fixed-particle L^2 control of noisy observations does not give a particle-uniform BBGKY corrector on action sublevels. The program is represented as a typed dependency graph, with theorem implications, operational adapters and independent primary branches kept distinct.

58.3. Organization. We first develop intrinsic finite causal deficiency, its local and behavior-dimension certificates, and the polynomial testing theorem. We retain the endogenous task and stationary regenerative results, the active graph-domain comparison, finite marked presentations, enriched residual convergence and constructive collision data. The last sections prove the nonconserved collective realization and the joint physical certificate theorem. Every mathematical section of the preceding canonical article is retained. The complete-development view additionally contains

the earlier mathematical bodies and historical introductions; no earlier theorem is removed to create the new comparison structure.

APPENDIX 59. INTRODUCTION

Comparison of statistical experiments becomes a different problem when the comparison channel must itself run with a prescribed finite memory. A causal channel may preserve the filtration and nevertheless require more persistent state than the experiment's consumer possesses. Moreover, randomizing among complete designs can enlarge the achievable set even when each design uses the same number of labels. The information exposed by the random selector, and the cost of retaining its value, are therefore part of the mathematical object being compared.

We study marked causal experiments with this resource structure. A mark is an actual random variable coupled to the source, not an independent copy with the same marginal law. The intrinsic deficiency is the smallest worst-feedback variation error among simulators at the declared state and randomization budget. Private, hidden-selector and visible-selector budgets lead to distinct feasible sets. The private image is generally nonconvex. Its exact finite certificates must consequently test that image itself rather than replace it by the convex hull of deterministic implementations.

The main result is a same-budget physical certification theorem. It starts with an active microscopic acquisition, passes through a finite trial presentation, quantizes its reports, constructs a coherent rational marked experiment, and transfers finite private certificates back to the original physical experiment. Each presentation map is stateless. The register cost paid by the final physical simulator is therefore the original finite budget, without an unpriced model-selection index. The analytic error and the finite optimization error are both explicit.

59.1. The principal theorem. Let b be a fixed private or finite-selector signature. For a pair of finite-horizon active Gaussian experiments E, F , write $\delta_b(E, F)$ for their intrinsic marked feedback deficiency. Let $E_{\mathbb{Q}}, F_{\mathbb{Q}}$ be finite rational marked presentations constructed by the procedures of this paper. Suppose their two-sided presentation errors are at most a, c , respectively. If finite local testing and an executable same-budget row table give bounds L, U for $\delta_b(E_{\mathbb{Q}}, F_{\mathbb{Q}})$, then

$$\max\{0, L - a - c\} \leq \delta_b(E, F) \leq \min\{1, U + a + c\}. \quad (59.1)$$

Theorem 13.1 proves this statement and a convergent construction under effective graph and marked-integration data. The lower bound remains valid without assuming attainment for the original nonfinite experiment. The upper bound comes with an actual physical simulator. The resulting intervals can be nested and made arbitrarily short. This is a finite-horizon effective theorem with specified analytic inputs, not a computability assertion for arbitrary measurable models.

There are three substantive steps. First, report truncation and histogram reconstruction are compared before controller optimization, using the original preparation and a coupled actual mark. This avoids unjustified bounds on feedback-conditioned densities. Second, rationalization is performed on joint marked prefix masses. A weighted normalization inequality cancels the small-prefix denominator. It gives a coherent rational causal tree and a two-sided feedback error without deciding which original prefix probabilities vanish. Third, piecewise affine graph-core paths give enriched residual bounds whose values themselves converge to zero. The intervention is applied after this domain-controlled path calculation; no invariance of the generator domain under kicks is assumed.

Two further results connect the finite geometry to state complexity. Theorem 2.10 bounds the number of distinct testing witnesses by the affine dimension of the attainable behavior image and the strict error gap. It keeps the original nonconvex private image; its bound is on tests, not the number of policy cells or computational running time. Theorem 7.7 gives packing lower bounds and covering upper bounds for the positive-error width of a finite family of marked prediction tasks at a

source-consuming restart. Completing that family recovers the universal marked continuation state at zero error. Causal update constraints remain necessary for simultaneous checkpoint realizations.

The effective data are not left as an integration oracle. Theorem 12.2 constructs an enumerable microscopic graph core and verifies its Gram and marked probability operations by certified full-measure collision-chart exhaustion. It applies to fixed-particle hard spheres with algebraic velocity kicks and bounded computable input data, without requiring conserved observables. No particle-number-uniform rate is inferred.

The physical data also admit explicit closed-form instances. Theorem 14.1 gives an exact deficiency $1/4$ for a continuous Gaussian acquisition of a conserved momentum coordinate in an actively intervened colliding hard-sphere system. Its trial Gram data are rational and exact. Proposition 14.3 then gives a collision-compatible localized position observable with a genuinely nonzero graph residual $R = 192$ and a rigorously bounded physical error. The conserved-sector family admits the complete finite-rational procedure at every fixed horizon, including rigorous scalar integration.

59.2. Classical antecedents and the distinction of resources. The general comparison of experiments and its risk interpretation are classical [1, 47]. Filtered randomization criteria predate the present finite-state formulation [48, 68]. In particular, Weisshaupt’s Lemma 1 and Theorem 5 use a compact convex set of filtered stochastic operators with values in finitely additive measures; their proof reduces the filtered statement to convex separation. They do not price a fixed persistent register. That comparison cannot be applied to our generally nonconvex private image by silently convexifying it. We retain the classical result as an antecedent, not a theorem displaced by this paper.

Common-information control and real-time coding provide essential information-state precedents [53, 54]. Compatible-cover language has an antecedent in incomplete-machine minimization [70]. The distinction here is the stated marked, resource-constrained comparison problem and its error transfer, not the invention of dynamic programming or compatible covers. A complete proof-level priority comparison with the original Norberg and Paull–Unger texts remains to be established; we make no assertion of exhaustive priority over those works.

Separate-affine vertex interpolation, subdivision, normed-space packing, real quantifier elimination [69], variation of constants, and graph-core approximation [67] are also classical. Their roles in the proofs are identified where used. The contribution claimed here is the original-budget physical certificate chain, including its actual marked feedback bridges, zero-prefix-stable rationalization and convergent enriched certificates, together with the stated resource-image and task-resolution consequences. None of these claims relies on counting source files or passing a diagnostic suite as a mathematical argument.

59.3. Organization and the program of applications. Section 2 develops intrinsic resource comparison, its finite local certificates and the new behavior-dimension witnesses. Section 7 treats endogenous common decisions and positive-error restart complexity. Section 8 retains the stationary regenerative theorem with its actual physical and register reset. The active microscopic construction and the original Galerkin residual inequalities follow. Sections 10 and 11 supply the two missing approximation steps; Section 12 constructs effective microscopic data; Section 13 proves the joint theorem and Section 14 instantiates certified physical data.

The associated eleven-component program is a dependency graph. Its primary geometric A2 branch is independent of the present comparison theory; an acquisition adapter is a different relation from theorem dependence. The present microscopic B4 edge proves fixed-horizon marked certification. It does not turn L^2 transport estimates into the particle-number-uniform nonlinear action-sublevel corrector of the historical kinetic program. Those larger targets and their derivations

are preserved in the source history and accompanying complete development. This article states precisely the new implications, without using repository order as a substitute for proof.

APPENDIX 60. INTRODUCTION

A causal simulation has two outputs: the report delivered to its consumer and the state that must survive until the next report. Classical comparison records the first and may leave the second unpriced. We keep both. The quantity of interest is the least error attainable by one task- and parameter-independent simulator at a prescribed persistent-state budget. The error compares the entire feedback transcript together with its actual latent mark. In particular, equality of report marginals does not replace equality of a marked experiment.

This distinction leads to three optimization problems. A private simulator retains only its declared state and uses fresh randomness. A hidden-selector simulator first selects a policy but does not reveal that choice to the consumer. A visible-selector simulator reveals it and includes it in the compared joint interface. Hidden convexification has a finite testing dual, but its realization may require storing a selector index. At a fixed private width this is not a harmless convention: a delayed three-symbol channel has zero hidden-selector deficiency and strictly positive private deficiency at width two. If the selector is visible, its optimized zero set is again the private zero set.

60.1. The main statement. For finite marked causal experiments E, F , prescribed widths $K = (K_0, \dots, K_T)$ and a fixed finite seed signature, let $\delta_K(E, F)$ be the infimum of the marked feedback error over all admissible simulators. The definitions and exact signature-dependent variational formulas are given in Section 2. The central result can be stated as follows.

At each fixed private or finite-selector budget, intrinsic causal deficiency has a complete family of finite local testing certificates and original-budget upper witnesses. These bounds converge with an explicit causal modulus. They compose across stateless marked presentation changes without increasing the budget, and active microscopic approximation supplies such changes with finite graph-residual bounds.

More precisely, Theorem 2.7 constructs rational lower and upper bounds $L_M \leq \delta_K \leq U_M$ from clipped-simplex policy cells. In the private case, with at most d_t joint output–state labels in a row at time t , their gap is at most

$$1 - \prod_{t=0}^T (1 - \min\{1, d_t/(2M)\}),$$

with a zero contribution from one-point rows. The modulus depends on depth and row alphabets, not on the number of histories. Enumeration of the cells can nevertheless be large; no polynomial-time synthesis conclusion is drawn. The lower bound assigns a marked feedback test to each cell. The upper bound is a single admissible row table at the original budget. This is a nonlinear testing converse, not an unpriced application of convex separation. For dyadic M the bounds are monotone and converge. Strict feasible and infeasible thresholds have finite witnesses; exact boundary decisions use classical real algebraic elimination.

The optimized spectra have the composition law

$$\delta_{KL}(E, G) \leq \delta_K(E, F) + \delta_L(F, G),$$

with selector costs multiplied when present. Theorem 2.9 then transfers an entire intrinsic interval across two-sided stateless presentation changes. Its lower statement excludes all original-budget simulators, even when the original carriers are nonfinite. Its upper statement constructs a simulator

by composing the rational row table with the bridge channels. Thus the scope of an assertion is visible in both its error and its persistent resources.

The other parts of the article give operational and analytic realizations of this comparison. Theorem 7.3 identifies the exact common-encoder face for endogenously controlled finite tasks by reachable compatible covers, and relates the lossy problem to Bellman regrets. Theorem 8.1 treats nonsummable average criteria for stationary finite-register controllers with actual simultaneous physical and register resets; deterministic or singular physical transitions are permitted between resets. Theorem 9.2 supplies active hard-sphere interventions and convergent graph-core approximations. Theorem 9.5 strengthens their finite-horizon comparison by an a posteriori bound involving only graph and intervention Gram data. Its root-sum-square word estimate retains the actual mark and precedes optimization over every finite consumer width. A finite age cutoff adds an explicit geometric remainder for average risks.

60.2. Comparison with classical results. Blackwell–Le Cam comparison, convex minimax, and finite-dimensional convex representation are classical ingredients [1, 47]. Filtered comparison is not new as a subject. Weisshaupt’s filtered stochastic-operator compactness lemma and randomization theorem [68, Lemma 1 and Theorem 5] give an explicit nearby convex comparison result with a finitely additive target space. They do not impose the persistent-state budget considered here. We distinguish that retrieved statement from Norberg’s closely related filtered comparison paper [48]; the full original-source comparison with the latter is not claimed complete.

The common-information dynamic program of Nayyar–Mahajan–Teneketzis [53, Theorem 3 and Corollary 5] provides a classical coordinator formulation, including a finite-memory specialization. The compatible-cover device also has classical incomplete-machine antecedents [70]. Our exact task theorem adds the common encoder, policy-dependent reachable support and Bellman regret conditions; the cover terminology by itself is not a novelty claim. Real quantifier elimination is used only for an exact decision corollary [69]. Similarly, graph-core semigroup approximation and the well-defined microscopic hard-sphere flow are credited to their classical sources [67, 63]. The residual proof below uses a finite-dimensional variation-of-constants formula and does not propose a new abstract semigroup approximation theorem.

The mathematical distinction retained after those ingredients are removed is the simultaneous original-budget testing and realization statement: local feedback tests certify the private feasible set that a convex filtered comparison can enlarge, and the same optimized quantity accepts active microscopic error bounds without a hidden consumer-memory charge. The finite delayed-channel separation makes that distinction explicit. It is not a claim that all ingredients, or their entire conjunction, have an exhaustively certified priority. The source-level literature audit records which original theorems were inspected and which remain to be compared.

60.3. Organization and scope. Section 2 develops the signature-indexed deficiency, its local certificates and its composition-stability theorem. The next sections treat endogenous tasks, regenerative average criteria, and active microscopic realization. All five sections include the proofs they use. The complete development retains the preceding mathematical bodies, including dominated finite-prefix and summable infinite-time duality, marked continuation quotients, exact synthesis, operator memory, Gaussian resource laws, and the historical model interfaces.

The physical results fix particle number, action alphabet, observation noise, and the initial-mark channel. They do not imply the historical nonlinear kinetic action-sublevel corrector or a Boltzmann–Grad limit. Likewise an intrinsic lower certificate is a mathematical inequality over a specified finite model, not an assertion that floating-point Gram entries certify a physical system. The eleven-component program is a mathematical dependency graph with adapters and independent lines; repository chronology and its separately frozen audit are not hypotheses of any theorem in this article.

APPENDIX 61. INTRODUCTION

An exact simulation of a statistical experiment need not preserve the memory budget of a decision maker. Conversely, convexifying a collection of finite-state simulations may introduce a persistent choice whose storage was absent from each member of the collection. The distinction is particularly sharp when the choice is revealed to the decision maker: a cancellation in a mixture of report laws then need not be a cancellation in the joint law of the choice and the reports. These observations suggest that the error, the retained state, and the visibility of randomization should be coordinates of a single comparison object.

We make this object explicit. Given marked controlled experiments E, F , a width profile $K = (K_0, \dots, K_T)$, and a seed signature, the *intrinsic causal deficiency* is the infimum of the marked feedback simulation error over all simulators with that signature. The mark is an actual hidden variable of the experiment, not a newly drawn variable having a convenient marginal. A simulator is independent of the unknown parameter and of the eventual decision task. Controls are passed through at the declared time, and the simulator can retain only its specified state. These restrictions are part of the object being compared.

61.1. The main theorem. For finite experiments, write $\mathcal{K}_K$ for the product of the simulator's stochastic row simplexes and $b(q)$ for the vector of all marked transcript probabilities against deterministic feedback trees. The map $q \mapsto b(q)$ is polynomial, but its image need not be convex. Put

$$a_K = \dim \operatorname{aff}\{b(q) : q \in \mathcal{K}_K\}.$$

Theorem 2.2 proves the following statement. *The hidden-mixture deficiency has a complete dual in weighted causal tests and an exact realization using at most $a_K + 1$ selector values; this realization costs at most $(a_K + 1)K$ private states. In contrast, arbitrary visible selection has zero deficiency exactly when a private K -state simulator has zero deficiency. All three signatures have multiplicative state-cost composition and uniform decision transport.* The private optimization and every fixed finite selector problem are attained. Thus the theorem compares optimal costs, rather than merely composing a simulator supplied in advance.

The state price in this statement is substantive. In a two-checkpoint experiment, with the input consumed before the output is produced, zero private deficiency at width K is equivalent to nonnegative rank at most K of the required stochastic channel (Proposition 2.4). A three-row channel has positive private two-state deficiency but zero hidden-selector deficiency at the same within-mode width. The selector can be stored, but it cannot be discarded. Consequently a convex decision dual cannot silently be interpreted as a same-width private inverse randomization theorem. Our converse gives the implementation and its cost together.

61.2. Endogenous decisions and nonsummable risks. A common encoder must also be shared by tasks that choose different controls. The set of histories relevant to a task then depends on the controller being synthesized. Theorem 7.3 characterizes zero excess relative to the full-history Bellman value using task-indexed reachable cells, task-dependent actions, and one task-independent update map. Closure is required only under the chosen actions and actual supported successors. Randomization leaves this exact feasibility set unchanged. An occupation identity gives quantitative upper bounds for approximate covers and a positive finite-table lower bound for infeasible profiles. The latter is a certificate for a fixed experiment, not a uniform margin over perturbed models.

The average-risk theorem uses an explicit physical regeneration. A reset refreshes the physical state and clears the within-policy register. Between resets the controlled physical kernel may be singular or deterministic. Uniform regeneration yields a compact risk image for stationary common finite-register policies, an attained minimax theorem after priced selection, and a vanishing-discount limit (Theorem 8.1). It is the reset of the joint physical and retained state, not an unproved

assertion about the mixing of a hidden plant, that gives the required uniformity. This result treats a nonsummable criterion; it is different from the dominated finite-prefix, summable infinite-horizon theory retained in the complete development.

61.3. Active microscopic realization. Our physical example is a finite hard-sphere system. At observation times an action applies an orthogonal velocity intervention, followed by the hard-sphere flow. The action therefore changes the subsequent trajectory. The interventions are bounded composition operators and need not preserve the generator domain. A graph-core Galerkin approximation of the unbounded Koopman generator, followed by projected interventions, approximates each finite controlled word. Gaussian acquisition converts the resulting L^2 errors into two-sided marked comparison bounds before optimizing any finite-register decision rule (Theorem 9.2). A sum over action words is essential: feedback conditioning does not preserve a uniform posterior density bound. After physical regeneration, the same comparison extends to average risks by a geometrically weighted cycle argument, even though no rate uniform in the word length is assumed.

The physical system, noise levels, and finite action alphabet are fixed. The approximation dimension is a continuous physical state dimension, not a bound on the cardinality of a decision maker’s register. We do not identify this active, regenerative microscopic theorem with a Boltzmann–Grad limit or with a nonlinear kinetic action theorem.

61.4. Relation to earlier theories. Blackwell comparison and Le Cam deficiency supply the classical statistical background; filtered experiments add the timing of information and decisions [1, 47, 48]. The finite minimax step, separation, and Carathéodory’s theorem are classical. What is retained here through these operations is the simulator’s *persistent-state and visibility signature*. The nonnegative-factorization example makes the difference from unconstrained randomization concrete. A full proof-level comparison with Norberg’s original filtered-experiment theorem is not asserted: the original full text was not available in the present source audit. In particular, we do not claim novelty for filtered deficiency as such.

Finite-memory control and the common-information method organize controllers through information states and prescriptions [54, 53]. Our cover theorem addresses a different finite question: exact Bellman optimality for several endogenous tasks with one retained encoder. Compatible covers in incomplete machine minimization are classical, and the cover device itself is not claimed as new. We give its decision-regret interpretation, its shared-update constraint, and its quantitative connection to the cost-indexed comparison theorem. The corresponding source-audit limitations are recorded outside the article.

The operator ingredients are likewise not presented as a new Mori–Zwanzig identity or a new abstract semigroup approximation theorem. Trotter’s approximation theorem [67, Theorem 5.2] is the classical antecedent of our graph-core argument. Recent domain-conscious projection results include [65, 66]. The new physical conclusion concerns feedback intervention and marked finite-register comparison, including its regenerative average-risk limit. The microscopic flow input is the standard almost-everywhere hard-sphere flow theorem [63].

61.5. Organization. Section 2 defines and proves the intrinsic comparison theorem, including its nonconvex and exposed-seed boundaries. Section 7 connects it to endogenous exact and lossy common decisions. Section 8 treats average risks, and Section 9 proves the active microscopic realization. The canonical article is self-contained relative to the expressly cited classical inputs. The accompanying complete development retains every predecessor mathematical body; those additional results are not silently made hypotheses of the present main theorem. The research program is a typed dependency graph, not a chain of implications inferred from paper names or repository chronology.

APPENDIX 62. INTRODUCTION

Statistical equivalence does not specify the memory needed to use an experiment. A report supplied at two different times and the same report supplied only once contain the same full-history information, but they need not support the same finite-register decisions. Nor does the state needed for a finite list of optimal decisions necessarily determine the state needed to preserve a whole marked continuation experiment. We study these distinctions through causal comparison with an explicit persistent state cost.

The object is a marked causal experiment together with an information interface. A single parameter-blind encoder must serve all declared tasks. A task controller may use its own task index but cannot reveal it as an additional uncharged encoder input. A consumed report is discarded; only the specified register survives. Independent public randomness is a separate resource. In particular a public variable taking an arbitrary point of a compact policy space is not identified with a finite index stored in the register.

62.1. The structural result. Let $\mathcal{R}_E(S, q)$ be the risk functions attainable on E with private widths $S = (S_t)$ and a retained independent public seed of at most q values. The value $q = \infty$ denotes an ideal uniform seed. The central comparison statement has three parts.

First, a causal marked simulator with internal widths $K = (K_t)$, public seed bound r , and feedback-uniform error ε transfers every target design to a source design with widths at most KS , seed bound rq , and row-risk error at most ε (Theorem 24.2). The actual implementation uses the reachable pairs of the simulator and consumer states. Composed simulators multiply these costs and add their errors. Lower support values of the common-encoder risk bodies give witnesses against an insufficient simulation budget (Corollary 24.4). These are concrete implementations of the abstract spectrum transport statement retained in Theorem 23.2, not a new name for that statement.

Second, for a finite marked instrument let $Q(E)$ be the quotient by conditional laws of the parameter, future reports, and actual target mark. A design is complete if its retained register, together with its independent seed, determines the current continuation class. Then

$$\mathcal{R}_E^c(S, q) = \mathcal{R}_{Q(E)}^c(S, q).$$

Theorem 24.7 proves this equality by fusing the simulator's class register into the consumer's already complete register. Without completeness the same simulation can require $K_t S_t$ labels. The repeated report example gives a sharp temporal cost: exact experiment equivalence can coexist with error $1 - s/k$ at an intermediate width s , while the other presentation has zero error. Thus the cost distortion and its zero-overhead subcategory are both part of the comparison invariant.

Third, completeness has an operational decision meaning. For a finite family of restart decisions, the least exact width is a compatible-cover number (Theorem 30.1). It is generally not the number of equivalence classes of a pairwise relation. For Brier prediction tasks the unique optimal predictions turn that cover into equality of conditional probability features. Under positive reports their causal closure is the least right congruence refining those features (Theorem 30.4). Completion under all marked event predictions recovers the universal quotient, and hence the complete-state fusion theorem.

62.2. Analytic realization and computational distinctions. The lower support invariant is attained in two nonfinite observation classes. The product-dominated finite-horizon theorem retained from the preceding development proves compactness of the private risk set and compact convexity of its ideal-public body for standard Borel reports with an L^1 -continuous family of complete loss effects (Theorem 26.3). Its tensor-testing and measurable representative arguments remain essential: merely naming a standard Borel carrier would not supply compactness.

Theorem 27.3 adds a singular displayed-covariate class. A continuous physical observation is supplied afresh at every checkpoint, while the innovations satisfy a conditional product-effect hypothesis. The full observation path can be supported on a nonatomic diagonal and have no product dominating measure. A compact space of conditional probability measures on private innovation policies gives an exact executable ideal-public risk body. This is a conditional Young measure argument with the private state still bounded, not an assumption that arbitrary information constraints are weakly closed. Finite stored randomness is recovered at the explicit row-covering cost of Proposition 24.5.

The exact continuation compiler uses classical weighted-automaton linear algebra to test equality, but outputs positive normalized states. We prove a polynomial count-vector bound for homogeneous conditionally independent controlled reports and an exponential $K_t = 2^t$ example at continuation rank two. We also realize graph coloring as the exact zero-excess common-task width. Thus linear equivalence testing, positive-state realization, and task-optimal compression are distinct computational questions, not three descriptions of one polynomial algorithm.

62.3. A physical consequence and the place of the other results. The microscopic hard-sphere application makes the comparison theorem operative on an unbounded generator. Orbit regularization constructs a collision-compatible weighted L^p graph core, with an explicit single approximation sequence (Theorem 43.2). For noisy finite acquisitions of actual physical observables, its approximation error gives stateless marked simulators in both directions. Their decision bounds are uniform over all memory widths and feedback rules (Theorem 43.3). This is a microscopic domain and acquisition result; it does not assert the separate action-sublevel BBGKY corrector or the nonlinear kinetic limit. The physical model and its operator topology are not replaced by a finite positive hidden-state model to obtain this conclusion.

The common-membership separation, delayed-query Gaussian exponent, positive filtering, finite nonlinear semigroups, hidden-phase moments, predictive trees, probability-measure geometry, and finite statistical comparison are retained with their full proofs and local hypotheses. They give distinct ways to evaluate or test the same risk construction. The canonical article contains this development; the complete-development edition additionally retains the earlier mathematical bodies and historical introductions. Earlier results are not deleted or declared consequences of stronger hypotheses they do not satisfy.

62.4. Antecedents and the claim of distinction. Blackwell–Le Cam comparison and positive convex separation underlie the risk-set viewpoint [1, 47, 46]. Filtered-experiment comparison is already present in Norberg [48]. Weak star team-policy topology, existence, and strategic-measure convexification are established methods [55, 56]. Theorems 6, 7, and 9 of [55] provide the relevant continuity, existence, and static-reduction antecedents. We do not claim any of these mechanisms as new, or infer unrestricted filtered compactness from the conditional extension proved here.

The common-information dynamic program of [53], Theorem 3 and Corollary 5, and causal coding reductions such as [57] identify useful designer states and policy structures. An offline belief can be an excellent optimization variable without being free online memory. Our costed simulators count the state needed to execute the comparison and identify exactly when it is already carried by a complete consumer.

Automaton continuation and predictive equivalence have their own long history [49, 5, 61, 58]. Proposition 2.1 and Theorem 2.3 of [58] describe the linear span engine used here. Graph coloring in zero-error information is also classical [62]. The new statements are therefore submitted as resource-sensitive refinements and their operational connection, not as the invention of minimax, causal comparison, state minimization, or graph coloring. The distinction sought is the interaction between executable state cost, the task family, and exact marked reconstruction. No claim of priority for unrestricted filtered comparison is made.

A model becomes a consumer of the comparison theory only after its information interface and analytic hypotheses have been verified. An acquisition adapter does not, by itself, derive that model's independent geometric or operator-theoretic assertions. This distinction is especially important for microscopic limits, where the topology and the scale of approximation form part of the theorem.

APPENDIX 63. INTRODUCTION

A statistic may preserve every parameter likelihood and yet fail to preserve the target of a decision. A compressor may be optimal for each task separately and yet fail when its update rule must be chosen before the task. Both distinctions become structural when information arrives sequentially and retained memory is bounded. We study them through the attainable risks of an executable causal experiment, with its target marks and information pattern specified together.

The principal object is the set of risk functions obtained by one common encoder and a family of task controllers. The encoder may use the current report, the current control and its retained label, but neither the task index nor the unknown parameter is an additional input. A task controller may depend on its task and retained label; it must work across all parameter rows of that task. Randomness is also part of the specification. Fresh private draws do not provide a free retained random variable, whereas an independent shared seed may correlate all stages of an execution. Expectations over that seed are taken before the worst decision row is selected.

63.1. The main results. Fix finite controls, decisions and a horizon T , but allow standard Borel report spaces and a compact parameter set. For a width vector $\mathbf{S} = (S_0, \dots, S_T)$ let $K_{\mathbf{S}}$ be the set of all private common-encoder designs and let $R(k) \in C(J)$ be their risk functions on a compact set J of task-parameter rows. Under the dominated loss-effect condition of Section 26, Theorem 26.3 gives

$$\min_{\lambda \in \mathcal{P}(K_{\mathbf{S}})} \max_{j \in J} \{\mathbb{E}_{\lambda} R_j - b_0(j)\} = \max_{w \in \mathcal{P}(J)} \left\{ \min_{k \in K_{\mathbf{S}}} \int R_j(k) w(dj) - \int b_0(j) w(dj) \right\}. \quad (63.1)$$

The private attainable risk set is compact, and its shared-seed barycenter set is compact and convex. Positive measures on J characterize the closed upper risk body. These conclusions concern all measurable finite-state updates, not just a compact family postulated in advance or encoders enumerated on a finite report tree.

The proof uses a product of weak star simplex spaces. Its main analytic step is a tensor-testing lemma: convergence of one bounded encoder factor in each report coordinate can be tested against an arbitrary L^1 path effect. Compact families of effects give uniform convergence of the risk functions. A separate representative lemma makes public mixtures of weak star classes jointly measurable and executable. Thus compactness is proved within the common-encoder information pattern. Corollary 26.5 treats countably many tasks by compactness and finite-subproblem duality; in that setting the dual is a supremum and no uniform finite seed is asserted.

The finite observation theorem, Theorem 26.7, compares the actual joint report-mark laws by causal cell conversion and within-cell reconstruction. Its error tends to zero uniformly in the memory width, task and controller. Finite experiments are therefore an approximating family for the dominated theory, not its sole domain. Corollary 26.6 gives the price of replacing a public mixture by a finite privately retained index.

Our second main result concerns the state carried by a marked experiment. For a finite instrument, two histories have the same continuation state when their conditional laws of the parameter, future reports and terminal mark agree under every future control word. Theorem 29.3 derives this state from decision risks: among all source-consuming randomized causal statistics, it has the least number of labels preserving every restart prediction risk. No prescribed posterior coordinates appear in that operational definition. The proof uses a strict conditional-variance

identity for binary-event prediction. Theorem 29.5 then constructs parameter-independent causal report channels preserving the actual terminal mark. An explicit binary example shows why parameter sufficiency alone does not give this conclusion.

For rational finite instruments, Theorem 31.2 computes these classes by backward continuation spaces and forward exploration of distinct positive states. Its cost is polynomial in the input bit length and the number of output states, rather than in the entire expanded history tree. The dimension of a linear representation is used to test equivalence; it is not confused with the number of positive retained states. A binary source with linear dimension two and $t + 1$ distinct states at time t illustrates the difference.

Two quantitative examples make the information pattern visible. Theorem 33.1 has three individually lossless two-label tasks, but common-encoder worst risk $1/2$ with private randomness and $2/9$ with shared randomness. All nine decision rows receive positive mass in an optimal dual witness. The delayed-query Gaussian theorem, Theorem 38.2, has pre-query error of order $S^{-2/d}$ while its query-aware checkpoint has order S^{-2} . Proposition 39.2 gives a finite acquisition and table implementation retaining the causal exponent.

63.2. Related work and the distinction of scope. Weak star policies and existence in stochastic teams are developed by Saldi [55]; strategic measures, public mixtures and failures of weak closure are treated by Yüksel and Saldi [56]. In particular, convexification by common randomness and compactness methods are not new principles here. Our result specifies a bounded retained register shared by several tasks, proves continuity for its coupled encoder factors, and obtains the risk-function body and its width-uniform marked approximations. Nonconvexity of the private set is consistent with the classical fact that randomization cannot improve a single fixed linear team objective: our strict example concerns a worst-row objective.

Causal source coding and common-information control provide the closest information-structure antecedents. See Witsenhausen [54], Yüksel [57], and Nayyar, Mahajan and Teneketzis [53]. An offline occupancy or belief recursion does not by itself constitute an online finite-memory implementation; we keep that distinction in the finite support recursion and the compiler. Comparison of filtered experiments, as in Norberg [48], is likewise part of the background of causal comparison, rather than a novelty claimed for the word “causal”.

Continuation equivalence is related to automata and predictive-state constructions [49, 5]. The linear span computation uses the weighted-automaton equivalence method presented by Kiefer [58]; neither rational elimination nor linear automaton minimization is asserted as new. The operational minimality here is for the declared marked continuation and for source-consuming statistics under all restart preparations. It is not minimality among independent simulators with unpriced retained memory. The target mark and this comparison category cannot be omitted from the statement.

The positive-model applications include established mechanisms: positive-matrix filter contraction, the logarithmic control-semigroup construction of Nisio [59], and the risk-sensitive information-state augmentation of Bäuerle and Rieder [60]. We prove the needed estimates here and specify how their actual marked effects enter the common-encoder theorem. No priority claim is made for these individual mechanisms.

63.3. Consequences and organization. The marked decision spectrum of Section 23 fixes the experiment, the task family, the information pattern and the resource budget before optimization. The compact theorem and the finite common-encoder theorem give exact variational descriptions in their stated classes. Quadratic targets give conditional variance; complete terminal loss families give statistical comparison; tree geometry gives weighted distortion estimates; and presenting an actual probability measure gives a Wasserstein law task. These are typed consequences of one risk construction, not an assertion that one scalar rate or one domination assumption describes

every model. In particular, presenting a realized measure and sampling a single marginal probe are different tasks.

The following sections develop the compact theorem, marked quotient, compiler and common-task example, followed by the finite variational and saturation theory. The Gaussian temporal example and numerical acquisition result precede three model applications: positive hidden Markov observations and history reduction, nonlinear finite-state semigroups, and hidden-phase exponential moments. The remaining sections supply the predictive-tree, measure, terminal Gaussian and count developments with full proofs. The separate complete-development edition also includes the earlier mathematical bodies and their historical introductions. None of their local hypotheses is removed when the results are used here.

APPENDIX 64. INTRODUCTION

An observation can be compressed only relative to questions and an information pattern. When the question is known before compression, one may retain only its answer. When it arrives later, the same machine must preserve the possibility of answering several incompatible questions. With feedback, the retained information also determines which experiment is performed next. The resulting problem is neither a static covering problem nor a collection of separate control problems.

This paper studies the marked decision spectrum of a causal experiment. Its encoder is fixed before the task, its controller and decoder may know the task but not the unknown parameter, and its persistent state is charged throughout the execution. The spectrum records the attainable risk vectors and their dependence on a resource signature. Our principal result is a controlled common-encoder variational theorem. It constructs the risk body from the finite instrument itself, characterizes its feasible continuation partitions, and identifies its lower support function by an exact occupancy recursion. The maximum over tasks is retained; it is not replaced by one favourable scalarization.

To describe the result, let E be a finite controlled hidden-state instrument on a finite horizon, and let D be a finite family of bounded decision tasks. A row $j \in J$ records a parameter within a minimax task, or the declared prior of a Bayesian task. Put $M = |J|$. A deterministic design ω consists of one common causal encoder together with one observation-admissible controller and decoder for each task. Write $R_j(\omega)$ for its row risk, and b_j for the corresponding task's full-history optimal baseline. These definitions are made precise in Section 34; in particular, a minimax task has one baseline, not a parameterwise oracle.

Theorem 64.1 (The common-encoder variational principle). *For prescribed finite register widths $\mathbf{S}$, let $\Omega_{\mathbf{S}}$ be the deterministic designs. Under an independent shared seed, averaged before the worst task and parameter are selected,*

$$\Theta_{\text{sh}}(\mathbf{S}; D) = \min_{\lambda \in \Delta(\Omega_{\mathbf{S}})} \max_{j \in J} \left\{ \sum_{\omega} \lambda_{\omega} R_j(\omega) - b_j \right\} \quad (64.1)$$

$$= \max_{w \in \Delta_J} \{V_0(z_0(w)) - w \cdot b\}. \quad (64.2)$$

Here V_0 is the finite prescription recursion (35.2)–(35.3), whose update table is common to all tasks and parameters. Its support values determine the whole attainable upper risk body. Deterministic feasible encoders are exactly the controlled successor-compatible history partitions.

An optimal mixture has at most $M + 1$ designs. Removing the shared seed while retaining its design index in the actual register gives

$$\Theta_{\text{pr}}((M + 1)S; D) \leq \Theta_{\text{sh}}(S; D) \leq \Theta_{\text{pr}}(S; D).$$

Private randomization at unchanged width instead has the nonconvex flow characterization (34.8); it need not give the convex body in (64.1).

The complete proof is given in Theorems 34.1 and 35.1 and Propositions 34.2 and 34.3. The underlying convex minimax and separation arguments are classical. The theorem uses them with a coupled feasible set: the encoder is common across tasks, each controller is common across its parameter rows, and a remembered randomization index is an actual resource when it is not an explicitly permitted shared input. These constraints are indispensable. In Example 34.4, the two-label minimax error of a three-symbol experiment is $1/3$ with a shared seed and $1/2$ without one. Thus replacing private kernels by an unpriced convexification would compute a different spectrum.

The same variational principle has two useful specializations. For a family of Bayesian quadratic tasks it is exactly a maximization over one task weight vector of the least conditional variance of a weighted posterior-mean vector, subject to causal compatibility. This strengthens the single-objective partition formula to the common-encoder maximum criterion. It applies directly to the accepted-cell/failure protocol of the A1 manuscript, with a single update on the observed failure symbol. For exponential criteria it is a duality for moments before taking logarithms, with a multiplicative occupancy recursion. In a hidden-phase example it excludes optimizing separately with knowledge of the phase. These are specializations of one feasible execution body, not independent assumptions that their controllers can be chosen jointly.

We also construct a least likelihood-saturated predictive coordinate for every finite instrument. A backward refinement of posterior vectors, next-report probabilities and successor classes gives its exact minimal widths. For rational input, this construction and its reconstruction kernel are computable by finite arithmetic. It makes the finite sufficiency bridge constructive, while identifying its possible cost: a one-class future-report quotient can require 2^d saturated classes after d parameter-revealing observations. Statistical saturation is therefore not a cost-free completion of a predictive quotient.

Finite synthesis yields control of several resources at once. Theorem 37.1 compiles an optimal finite design into a finite program and a dyadic initial lottery. It charges program length, persistent labels, initialization bits, transient work and bit-serial running time. Its excess differs from the optimum by at most $k2^{-p}$ for $k \leq M + 1$ designs and p lottery bits. Synthesis can be exponential in the horizon; no multiplicative-energy oracle is used in this finite source theorem. The predictive-tree compiler retained later addresses a different, infinite symbolic input class under its explicit geometric and energy-certification assumptions.

A continuous Gaussian example shows that the same causal constraint has quantitative force beyond finite enumeration. Acquire d noisy scalar coordinates before an independent coordinate query is disclosed. The final target is one scalar parameter, but its optimal causal excess is $S^{-2/d}$, whereas a query-aware static checkpoint has excess S^{-2} . The lower bound uses the common register *before* the query, the conditional-variance identity, and a density-volume bound on the vector of posterior means. This is a temporal penalty, not the terminal dimension argument for an append-only Gaussian vector experiment.

64.1. Relation to existing theory. Comparison of experiments by all bounded decisions is the theory of Blackwell and Le Cam [1, 47, 46]. The terminal deficiency representation included below is stated with its compactness hypotheses and the common encoder outside the task supremum. It does not claim a new unconstrained randomization criterion. Filtered experiment comparison is itself an established subject [48]. Likewise, predictive equivalence and continuation congruences have the classical antecedents [5, 49].

Real-time source coding and decentralized control are particularly close neighbours [54, 33, 34]. Nayyar, Mahajan and Teneketzis [53] derive common-information prescription recursions. In their preprint, Theorem 3 gives the dynamic programme, Corollary 5 treats absent common memory with finite local memory, and Theorem 4 simplifies its information state. Our offline occupancy is in this established designer/common-information tradition. The assertion made here is the coupled

minimax risk-body identity, its controlled partition converse, and the priced passage between shared and private randomization. An ordinary designer's dynamic programme, by itself, is not asserted to be new.

The retained quantitative sections use functional and separated-cylinder quantization, symbolic trees, nonlinear filtering and experiment approximation [35, 36, 43, 37, 44]. Their geometric hypotheses remain local to those theorems. The new conditional-variance variational principle includes the finite partition problem and explains the delayed-query Gaussian obstruction. It does not identify Wasserstein law presentation with decision-complete statistical deficiency, which retain different output marks.

64.2. Organization and scope. Section 23 fixes the marked spectrum and its transport and terminal comparison properties. Sections 34 and 35 prove the central variational principle. Sections 36–38 construct saturation, compile finite sources and exhibit the Gaussian temporal penalty. The subsequent sections retain the complete sufficient-statistic, partition, predictive-tree, measure-presentation, Gaussian and count arguments. The companion development additionally retains the earlier mathematical bodies and historical introductions with their original hypotheses; none is silently replaced by a stronger assertion.

The general marked spectrum includes standard Borel experiments, whereas the exact controlled minimax theorem proved here has finite horizon, finite hidden/report/control spaces and finitely many decision rows. The Gaussian example is established separately by its displayed continuous argument. In particular, the factor $M + 1$ is not a uniform statement about a growing or infinite task family. A finite-horizon exponential moment formula is not an LDP, an unbounded-operator theorem, or a proof of a microscopic phase decomposition. Those further structures require their own model theorems. The present foundation supplies a controlled common-encoder converse without importing such theorems as unnamed premises.

APPENDIX 65. INTRODUCTION

Finite descriptions of an experiment can be compared only after the questions to be answered and the permitted execution have been fixed. A two-valued checkpoint target may require exponentially many causal states when the relevant question arrives after the observation has been discarded. A finite statistic may preserve every bounded decision problem approximately without producing an accurate realization of a whole posterior measure. These are not competing definitions of the same error. They are different slices of one constrained decision problem.

We attach to a causal experiment its *marked decision spectrum*: the attainable risk vectors, indexed by the retained resources and by families of decision tasks. One common encoder is chosen before the task, whereas its decoder may depend on the task. The spectrum retains this order of quantifiers. It separates persistent state, physical output, acquisition length, program description, numerical precision and transient work. Under a resource-controlled causal equivalence it is invariant, up to the certified budget transformations. For a compact dominated terminal experiment, its decision-complete slice is exactly the optimal reverse deficiency of a common finite-output encoding. Its quadratic Bayesian slice is a compatible-partition distortion problem.

Theorem 65.1 (Representation and realization of the decision spectrum). *The spectrum defined in Section 23 has the following three properties.*

- (1) *Initialized, parameter-independent causal simulations transport its attainable risk sets with their full resource transformers. Mutually exact finite-state simulations preserve every existing positive cardinality exponent.*
- (2) *For a compact total-variation-continuous terminal family,*

$$\Theta_E(S; \mathcal{D}_{\text{all}}) = \inf_{Q: X \rightarrow [S]} \delta(QE, E).$$

A posterior coordinate formed with any positive dense reference prior is sufficient for the entire parameter family and leaves this terminal spectrum unchanged. Its causal analogue requires likelihood saturation and recursive predictive closure; a finite quotient then supplies a finite-state equivalence with an explicitly charged budget dilation.

- (3) *For finite report alphabets and a fixed finite-horizon Bayesian quadratic problem, the optimal causal excess is exactly the minimum conditional-variance functional over successor-compatible partitions of the history levels, with width at most S .*

The precise hypotheses and proofs are Theorems 23.2, 23.4, 46.1, 46.2 and 47.1.

The third assertion supplies a necessity theorem rather than another sufficient coding scheme. In Example 47.4, a random d -bit word is followed by a random coordinate query. The checkpoint mean has two values, but zero-error causal realization needs 2^d states. Its positive error is bounded below by the binary entropy inverse. Thus static predictive geometry alone cannot determine a causal budget. Compatibility of the partitions is the missing datum.

We then prove three quantitative consequences. First, Theorem 48.1 replaces exact real-energy ordering in a predictive tree by a rational threshold construction. Taking the minimum of certified energy bounds over every contiguous factor gives exact suffix closure despite ties and numerical uncertainty. Its finite tree has optimal distortion order, $O(\log S)$ acquisition depth and an $O(S \log S + Sdp)$ -bit description for a d -dimensional readout. The acquisition is a finite prefix of a snapshot, not a free infinite word.

Second, Theorem 49.1 and Corollary 49.2 give a matching $1/\log S$ presentation law for the entire space of probability measures on an interval and for a genuine nonlinear posterior class with arbitrary input prior at each fixed time. A continuous finite-dimensional exact coordinate is impossible on this class. Yet a single marginal Bernoulli probe admits an $O(1/S)$ simulation. The separation proves that forgetting the output mark can change the rate, even when the latent measure is unchanged.

Third, Theorem 50.1 gives the decision-complete terminal law $S^{-2/(nr)}$ for n sequential r -dimensional Gaussian innovations when only the current register is retained. The register must hold the accumulated finite experiment; previous labels cannot be stored in a free transcript. The positive Gaussian reconstruction and the physical Poisson count-to-contact comparison are proved in full in the supporting sections. The latter gives a direct downstream instance with matched label budgets in a joint sample-size window.

65.1. Relation to previous theory. Predictive equivalence and minimal predictive states are classical ideas; Shalizi and Crutchfield [5] establish optimality and uniqueness properties of causal-state representations. The continuation congruence used here has the classical finite-automaton antecedent [49]. We use neither of these observations as a claim of a new exact minimal-state principle. The extra constraint is a charged recursive presentation, with a fixed common encoder and a quantitative loss. The finite-history variance formula identifies its feasible partitions exactly.

The comparison of experiments through all bounded decision problems is likewise the theory of Blackwell and Le Cam [1, 47, 46]. Norberg [48] treats comparison with filtered probability spaces. Our terminal duality is proved to fix its compactness assumptions and its encoder quantifiers, not to reassign priority to that criterion. The new organization retains resource transformers as part of a causal morphism and distinguishes saturated statistical quotients from prior-relative predictive quotients.

Zero-delay coding of Markov sources has a substantial structural literature. Linder–Yüksel [33] and Wood–Linder–Yüksel [34] study causal coding policies, including stationary optimal policies and finite-memory approximations. Their communication alphabet is not automatically the cardinality of all retained information. In the latter paper, Theorems 3–4 of the author preprint

give stationary average-cost optimality and periodic finite-memory approximations. The sequential Gaussian result here concerns a different, explicitly total-register terminal problem.

Functional quantization and separated-cylinder quantization provide the static geometry [35, 36, 43]; greedy variable-to-fixed trees have the classical antecedent [37]. We retain those attributions. The robust factor envelope proves that a near-optimal finite tree can be compiled without an oracle for exact comparisons, while its suffix closure makes both causal updates executable. Nonlinear-filter contraction and finite-model approximation have extensive precedents, including [44]. Our additional lower bound concerns the full posterior presentation task, not a new proof of average-cost control optimality. The topological and entropy arguments also distinguish continuous finite-dimensional coordinates from arbitrary Borel codings.

65.2. Organization and quantitative conventions. Sections 23–47 establish the common spectrum, its sufficient-statistic representation and its causal compatibility obstruction. Sections 48–50 prove the quantitative consequences. The remaining sections supply the complete predictive-tree, filtering, Gaussian and count-experiment proofs used by these consequences. They are mathematical dependencies, not substitutes for a proof of the new representation statements.

The notation $a_S \asymp b_S$ means comparison by strictly positive constants independent of S in the stated parameter regime. Constants for the tree class are uniform only when its separation, child-energy, unary-chain and distortion constants have common bounds. Horizon, covariance conditioning and rank dependencies are retained. Explicit Poisson normal-approximation constants in the supporting proof are nonoptimized. The physical A2 consequence is order-optimal in the label budget inside its displayed joint window, not sample-optimal over all observation protocols. The paper’s title names the continuing programme; the invariant proved here is the typed marked spectrum, not a claim that unrelated spectral, kinetic or phase theorems are automatic corollaries.

APPENDIX 66. INTRODUCTION

A statistic that preserves present predictions need not admit a finite recursive implementation. Conversely, a small implementable register need not preserve the decision problems of the underlying experiment. The purpose of this paper is to relate these three objects: a predictive quotient, its attainable geometry, and a charged causal realization. The order of the argument is

$$\begin{aligned} &\text{positive causal experiment} \longrightarrow \text{future-test quotient} \\ &\longrightarrow \text{causal simulation} \longrightarrow \text{finite-resource realization.} \end{aligned}$$

The quotient is always relative to a declared family of executable future tests. Resource preservation is always relative to a declared resource signature. Neither a measurable re-encoding by real numbers nor a terminal randomization is, by itself, a finite-memory implementation.

The principal structural result is Theorem 53.1. It starts with a factorial language, a stationary law, and a positive energy on its finite words. The hypotheses are local child comparisons, strict energy decrease under prefix attachment, bounded unary chains, and geometric separation of the predictive images. No quantization dimension is an input. If G_M is the energy of the greedy cut with at most M leaves, then its static squared quantization error is comparable to G_M . Every greedy tree is suffix closed. With L leaves it has at most $(U + 1)(2L - 1)$ vertices, where U bounds unary chains. These vertices support two different causal operations. Deleting the oldest available future symbol realizes a renewal orbit code; prepending the newest observed symbol realizes a code of the entire observed past. Both operations have the same-order risk as the unrestricted static problem. The lower bounds allow randomized, time-dependent observers.

The two orientations are not merely two descriptions of one reset model. The Markov-renewal verification in Section 54 includes forbidden transitions and noninjective, delay-identifying observations. The second verification is a noisy hidden shift register whose tail persists at every time. Its complete posterior has infinitely many observation-dependent coordinates. Theorem 54.3

gives a matched whole-register excess-risk law in that experiment, including its noise dependence. Thus one structural realization theorem has a regenerative and a genuinely continually observed application.

A complementary approximation theorem treats predictive quotients that are measure valued rather than tree separated. Theorem 55.2 constructs a finite recursive register for contracting filters on $\mathcal{P}([0, 1])$. Its codebook has at most 4^n states and error at most $3/(2n(1 - \rho))$, apart from initialization and explicitly charged numerical defects. The nonlinear example has no fixed-step common minorizer and no closed recursion of any prescribed finite list of polynomial moments. This is an upper approximation theorem, not a claim that every nonlinear filter has this contraction or that its logarithmic budget rate is universally optimal.

The experiment-comparison application is two-sided. Theorem 56.1 proves that an r -dimensional nondegenerate Gaussian location experiment, over a compact parameter set containing a ball, has optimal M -outcome approximation error $\Theta(M^{-2/r})$ in Le Cam distance. The upper bound uses deterministic cells and positive second-order reconstruction, not first-order constant histograms. The lower bound applies to every experiment on at most M outcomes. For the actual local count-to-contact protocol of A2, Theorem 57.1 proves

$$\Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C\{N^{-1/2} + JN^{-1/2} + J^{-2}\}, \quad |\mathcal{E}_{N,J}| \leq J^r.$$

It follows that $\Delta \asymp M^{-2/r}$ throughout the growing-budget window $1 \ll M \lesssim N^{r/6}$. The proof includes the local normal, variance replacement, boundary crossing, reconstruction, and decision comparison arguments. The physical statistic is deterministic throughout.

66.1. Relation to previous work. The existence of path laws, conditional-probability quotients, Bayes orthogonality, information loss under coarsening, and the composition of statistical randomizations are classical; see [4, 1, 45, 46]. Sections 51–52 give the forms needed here, with the simulator and its persistent state included. We do not claim novelty for these identities or for their categorical notation.

Static graph-directed quantization is a particularly close antecedent. Kesseböhmer–Zhu [43], Theorem 1.1, determine Markov-type quantization dimension by the spectral radius of a matrix of $(p_{ij}c_{ij}^r)^{s/(s+r)}$ and characterize finiteness of the upper coefficient through maximal strongly connected components. Their spatial theorem is not a theorem about a causal register. Our tree argument separates that familiar static mechanism from the additional suffix-closure condition and proves realization in both time orientations. The Markov application retains the renewal-weighted orbit and survival contribution; its spatial pressure is not a new discovery of graph-directed quantization. The earlier comparisons with functional quantization, dynamical rate distortion and zero-delay coding remain relevant [35, 33, 34]. A finite channel alphabet does not generally bound the entire persistent state of its coding policy.

Demirci–Kara–Yüksel [44], Theorems 3.5–3.6 of the cited preprint, obtain quantized-belief and finite-window policy approximations in their contraction-based control analysis. Our measure-valued result uses the classical contraction-plus-quantization mechanism, but provides a concrete entire-register cardinality and a same-observation recursive error bound for a specified nonlinear experiment. Its objective is prediction and terminal experiment simulation, not average-cost control optimality. The sharp hidden-shift theorem is instead a two-sided predictive-tree result. Neither theorem purports to subsume general partially observed control.

The new contribution is therefore not “finite memory”, “pressure”, or “Bayes reduction” in isolation. It is the combination of a checkable energy criterion, exact two-orientation realization, unrestricted register converses, and resource-typed experiment comparison. The Gaussian/A2 application supplies a separate decision-theoretic cardinality obstruction and an attainable joint sample/budget law. No priority claim for every component of these arguments is required.

66.2. Reading convention. This article is the canonical seventh revision of Volume I. Its proofs are self-contained at the level of the stated models. A separate *Complete development* contains this article followed by the unchanged earlier theorem bodies, including pressure, critical renewal, calibration, collision, and control results. The companion is a preservation and extended-reference object, not an alternative definition of the main paper. Source identities, historical consultation, and responses to the referee are recorded outside the mathematical text.

APPENDIX 67. INTRODUCTION

The information retained by a causal observer must serve predictions whose time of use is not known when that information is acquired. A description of the current state can therefore be inadequate even when it is an optimal quantization at the acquisition time. There is also a converse difficulty: a good finite description of an entire future orbit need not admit a closed update on the same finite state set. These two questions link the geometry of prediction to the implementation of a finite memory.

This paper develops two realization principles for that link. The first is an exact suffix-closure principle for renewal-weighted cylinder codes on a mixing Markov repeller. The second is a contracting posterior principle for continual partial observation on an unbounded state space. Both are compared with the full class of randomized, time-dependent observers having a bounded persistent register.

67.1. The intrinsic law. Let Σ_A be a topologically mixing subshift of finite type, with forbidden transitions allowed, and let K be its strongly separated geometric realization by contractions of bounded conformal distortion. The class includes similarities in any Euclidean dimension and nonconstant smooth inverse derivatives in dimension one. The measure on K is a stationary Hölder Gibbs law. A Hölder observable may be noninjective provided a finite delay vector recovers the geometric state with a uniform snowflake lower bound.

Exact acquisitions occur at independent renewal times. The survival law need not be geometric. A finite cylinder energy combines the mass of the acquired cylinder, the successive suffix contraction scales, and the tail of the stationary renewal age. Splitting maximal-energy cylinders defines a finite profile G_M . Our first conclusion is

$$\mathcal{R}_M \asymp e_M^2(Z_w^f) \asymp G_M,$$

where Z_w^f is the weighted observable orbit and $\mathcal{R}_M$ is the optimal long-time average prediction risk. The constants hold for every budget, not merely along a sequence of cylinder depths.

The realization argument must accommodate vertices with a unique successor and suffix cylinders with a unique predecessor. Strict inequality of cylinder masses is then unavailable. We instead prove strict decrease of the renewal distortion under prefix attachment. Together with stationarity, it forces the complete greedy tree to be suffix closed. Unary vertices are counted, and the whole state set has size at most a graph-dependent constant times the number of leaves. The lower bound uses separated tubes in Hilbert space and is valid for arbitrary off-image centres. Thus neither admissibility nor causal execution is hidden in the definition of a static codebook.

When the renewal weights have exponential rate q , the squared-error exponent is determined by the larger of two pressure roots,

$$P_A(s_g\varphi + 2\beta s_g\psi) = 0, \quad P_A(s_t\varphi) + s_t \log q = 0.$$

The first combines mass with geometric contraction; the second combines mass with survival through unobserved dynamics. We prove the existence of the quantization exponent from these equations. It is not an additional hypothesis on an infinite-dimensional orbit image. For a Markov source, zeros of the transition matrix remain zeros in the transfer matrices.

The first-order exponent does not determine the critical correction. For the Parry law on any primitive graph and equal-ratio similarities, renewal survival $q^k(k+1)^{-\kappa}$ gives, at the common critical value, corrections $(\log M)^{1-\kappa}$ for $\kappa < 1$, $\log \log M$ for $\kappa = 1$, and a bounded factor for $\kappa > 1$. The underlying geometry and pressure threshold can be kept fixed. A uniform finite-size profile limit describes the window $q_n = q_c e^{x/n}$. These statements describe actual renewal laws and matching risk orders; they do not assert an optimal leading constant from a pair of inequalities.

67.2. A nonregenerative realization principle. For continual observations, let m_t denote the true complete-history conditional mean. Suppose it obeys a finite-dimensional Borel recursion which contracts conditionally in fourth mean in its previous mean coordinate. Individual updates may expand. An inward rational quantizer preserves fourth-moment bounds on an unbounded state space. The resulting recursive observer attains excess squared risk $O(M^{-2/d})$ using at most M persistent values. If the stationary posterior mean has positive density on a cube, a matching lower bound holds against all observers. The Bayes floor here is the *full-history* Bayes error, not the error of the latest observation.

For a stationary Gaussian autoregression with Gaussian observation noise, all hypotheses are verified and the risk is $P + \Theta(M^{-2})$, where P is the steady conditional variance. Every fixed-step hidden transition kernel in this example lacks a nonzero global common minorizing measure. The realization proof is therefore genuinely different from inserting an independent reference experiment at each acquisition. The general theorem states its posterior contraction hypothesis explicitly; it is not a contraction theorem for all nonlinear filters.

67.3. The resource and its invariances. Only the number of values of the persistent state is constrained. Fixed programs, fixed real constants, the absolute clock, and transient computation are separate resources. A renewal indicator is an input to the update, not a free post-update observation. Theorem 73.1 proves exact operational simulations: adding a K -valued event to the decoder costs at most a factor K in state count; splitting registers costs their product; binary checkpoint encodings are related by $M = 2^B$. These transformations preserve the exponent and, under fixed-budget dilation, the risk order. A real-valued free observation at decoding is a different experiment, as the explicit uniform example shows.

67.4. Relation to previous work. Thermodynamic formulas for *spatial* quantization predate the present work. Lindsay and Mauldin [38] relate quantization dimension for finite conformal systems to the temperature function. Atnip, Roychowdhury and Urbański [39, Theorem 3.1] extend that relation to an infinite conformal family under their summability and strong open set hypotheses. Roychowdhury [36] treats Gibbs-like measures on cookie-cutter sets. Those spatial results include systems beyond the finite, strongly separated class used here. We do not claim a generalization of their geometric hypotheses. The additional object here is the survival-weighted observable *future orbit*, with an unrestricted causal converse and an admissible, suffix-closed finite-state realization. Classical pressure and greedy variable-to-fixed trees [32, 37] are inputs, not inventions of this paper.

Finite-memory and zero-delay approximation likewise have a substantial literature. Linder and Yüksel [33] and Wood, Linder and Yüksel [34] establish structural and stationary coding results. Ghomi, Linder and Yüksel [40, Theorem 4] prove stationary optimality for linear vector Markov sources and control finite-horizon approximation to the infinite-horizon distortion. Their channel alphabet and their history-dependent coding policy are not a single finite persistent register. Our filtering result is not a new proof of stationary zero-delay coding optimality.

Kara and Yüksel [19, Theorem 12] prove exponentially accurate finite-memory feedback under their POMDP stability hypotheses. Cregg, Alajaji and Yüksel [41, Theorem 2 and Corollary 2] bound the discounted suboptimality of sliding-window source-channel policies by predictor stability, with an exponential window bound under sufficient Dobrushin conditions. Thus stability-based

finite-memory approximation is already established. In contrast, Theorem 72.1 concerns the number of retained values, a matched squared-error exponent against the full-history Bayes floor, and an unbounded continuous observation model. It is a quantitative realization result in that resource, not a Q-learning or optimal-control theorem. Recursive filter quantization itself is also classical; see Pagès and Pham [8].

Static functional quantization [35], dynamical rate distortion [29], and data-rate control [27] constrain different objects. The distinction is operational: a block code or information-rate bound does not by itself specify a closed update on a bounded persistent state set. Conversely, the state-count theorems here give no runtime, program-description, or channel-capacity bound.

67.5. Organization and dependencies. Sections 68–70 prove the Markov profile and pressure theorems. Section 71 gives the critical classification and uniform window. Sections 72 and 73 prove the nonregenerative and interface results. The general posterior-orbit converse is included with its complete proof in Section 74. The complete edition additionally retains the earlier Gibbs, dependent-acquisition, noisy expansion, calibration, root-collision, filter, control and foundational developments. They have their own hypotheses; the current Markov and filtering proofs do not depend on the singular-statistics or unbounded-operator conclusions.

The programme's T07–T08 question is the passage from acquired predictive geometry to actual finite-state evolution. The two realization mechanisms proved here resolve that question in their stated classes, including a noncompact nonregenerative model. The historical physical models require further, model-specific results. The active A2 conductor and nonreduced-generator edition [42] is an algebraic input to a different observation interface: its scheme multiplicities do not specify a noisy observation kernel. The accompanying snapshot audit records that interface at its current source edition rather than identifying algebraic multiplicity with a statistical exponent. The complete edition also proves a joint sample–label deficiency bound for its actual known-mark count-to-contact protocol. The overflow alphabet and the comparison randomization are explicitly charged and distinguished from the physical statistic.

APPENDIX 68. MARKOV GEOMETRY AND THE FINITE-STATE LAW

We first specify the geometry independently of the observation apparatus. Let A be a primitive zero–one matrix on an alphabet $\mathcal{A}$ of cardinality $b \geq 2$, with Perron root $\lambda_A > 1$. Write Σ_A for its one-sided shift and $\mathcal{L}_n$ for its words of length n . The empty word is denoted by $\emptyset$. Forbidden transitions are permitted; in particular, a vertex may have only one successor.

Let $D \subset \mathbb{R}^d$ be compact, of diameter at most one, and let $h_i : D \rightarrow D$ be injective contractions. Assume that the compact sets $h_i(D)$ have pairwise distance at least $g > 0$. For $v = i_0 \cdots i_{n-1}$ write $h_v = h_{i_0} \circ \cdots \circ h_{i_{n-1}}$. We require scales $r_\emptyset = 1$, $0 < r_- \leq r_i \leq r_+ < 1$, and $D_0 \geq 1$ such that

$$D_0^{-1} r_v |x - y| \leq |h_v x - h_v y| \leq r_v |x - y|, \quad (68.1)$$

$$D_0^{-1} r_u r_v \leq r_{uv} \leq r_u r_v. \quad (68.2)$$

These conditions are imposed for words for which the compositions are used. They hold for similarities in every dimension, and for increasing one-dimensional $C^{1+\gamma}$ contractions with uniformly positive derivative and bounded logarithmic distortion, with $r_v = \sup |h'_v|$. For higher-dimensional conformal maps, (68.1) is an explicit metric bounded-distortion hypothesis; no global co-Lipschitz conclusion is inferred merely from a pointwise derivative estimate.

The coding map $\pi : \Sigma_A \rightarrow K \subset D$ is injective. Set $T\pi(\omega) = \pi(\sigma\omega)$ and $K_v = \pi([v])$. Let μ be the image of a stationary, fully supported Hölder Gibbs measure on Σ_A . Normalize its potential φ by $P_A(\varphi) = 0$. In particular, for a fixed C_G ,

$$C_G^{-1} e^{S_n \varphi(\omega)} \leq \mu[v] \leq C_G e^{S_n \varphi(\omega)} \quad (\omega \in [v]). \quad (68.3)$$

We use the classical Gibbs and variational pressure theorems for a primitive finite shift in the form recorded in [32]. The profile theorem below needs only stationarity and a uniform positive lower bound on each admissible one-letter conditional probability.

68.1. Observables and acquisition times. Fix $0 < \beta \leq 1$, put $\tau = 2\beta$, and let $f : K \rightarrow \mathbb{R}^m$ satisfy

$$|f(x) - f(y)| \leq H|x - y|^\beta, \quad (68.4)$$

$$\sum_{j=0}^J |f(T^j x) - f(T^j y)|^2 \geq c_o |x - y|^{2\beta} \quad (x, y \in K) \quad (68.5)$$

for some finite J and $c_o > 0$. Thus the present output need not identify the state; finitely many successive outputs must do so quantitatively. Condition (68.5) concerns the actual observable on K , not an assigned distance on symbolic words.

At a renewal an independent state U of law μ is observed exactly. Between renewals the state evolves by T . Cycle lengths are independent copies of a positive integer L , independent of the acquired states, with finite mean. The age weights and their tails are

$$w_k = \frac{\mathbb{P}(L > k)}{\mathbb{E}L}, \quad W(n) = \sum_{k \geq n} w_k, \quad 0 < \underline{q} \leq \frac{w_{k+1}}{w_k} \leq \bar{q} < 1. \quad (68.6)$$

An initial acquisition occurs at zero; the next renewal time is not announced. Nongeometric lengths satisfying (68.6) are allowed. Write X_t for this process.

An observer keeps I_t in a set of at most M elements. Its update may use I_{t-1} , absolute time, the renewal indicator, and the acquired state at a renewal. The post-update decoder uses I_t , absolute time, and independent coding randomness, but does not separately receive the renewal indicator or a retained real-valued observation. All observationally dependent persistent variables are counted in I_t . Stochastic kernels, private randomness, arbitrary fixed real constants, and transient computation are allowed. Section 73 compares this convention with other finite-memory interfaces.

Set

$$\begin{aligned} Z_w^f(x) &= (\sqrt{w_k} f(T^k x))_{k \geq 0} \in \ell^2(\mathbb{N}_0; \mathbb{R}^m), \\ e_M^2(Z_w^f) &= \inf_{c_1, \dots, c_M} \int \min_{i \leq M} \|Z_w^f(x) - c_i\|^2 d\mu(x), \\ \mathcal{R}_M &= \inf_{\mathcal{C}} \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{t < N} \mathbb{E} |f(X_t) - \hat{f}_t|^2. \end{aligned} \quad (68.7)$$

Centres in the quantization problem may lie anywhere in the Hilbert space. The infimum in the last line ranges over the entire preceding observer class, not just stationary or cylinder observers.

68.2. A computable geometric profile. For $v \in \mathcal{L}_n$ and its suffix $v_k = i_k \cdots i_{n-1}$ define

$$V_w(v) = \sum_{k < n} w_k r_{v_k}^\tau + W(n), \quad a_w(v) = \mu[v] V_w(v), \quad a_w(\emptyset) = 1. \quad (68.8)$$

Starting from the root, split a leaf of maximal a_w into all its admissible one-symbol successors, with a fixed rule for ties. Let $\mathcal{T}_N$ be the last tree in this sequence having at most N leaves, let $\mathcal{P}_N$ be its leaves, and put

$$G_N(w) = \sum_{v \in \mathcal{P}_N} a_w(v). \quad (68.9)$$

For the finitely many budgets preceding the first root split, the tree is the root. Unary splits are genuine operations. They do not increase the number of leaves, but they cannot occur in an infinite uninterrupted chain. Consequently the last tree in this definition exists.

The construction is a finite mathematical algorithm on cylinder masses, scales and renewal tails. Effective numerical evaluation additionally requires certified access to those real inputs; no computability assertion for arbitrary Borel or Hölder data is implicit.

Theorem 68.1 (Markov renewal profile). *Under (68.1)–(68.6), there are positive constants c, C , independent of M , such that*

$$cG_M(w) \leq e_M^2(Z_w^f) \leq \mathcal{R}_M \leq CG_M(w) \quad (M \geq 1). \quad (68.10)$$

Every randomized time-dependent observer satisfies the middle lower bound with its lower limiting average loss. A stationary deterministic observer attains the upper order.

More precisely, the complete admissible tree $\mathcal{T}_N$ is suffix closed. If it has L leaves, all its vertices can be retained as states, and their number is at most

$$(b+1)(2L-1). \quad (68.11)$$

Its exact number is $1 + \sum_{v \text{ internal}} d^+(v)$, where $d^+(v)$ is the number of children actually present. No time-depth factor is required. The constants in (68.10) may depend on the graph, distortion, separation, conditional masses, observable, and the two ratios in (68.6).

For the pressure statement assume additionally that a Hölder potential ψ satisfies $\log r_v = S_n\psi(\omega) + O(1)$ uniformly on $[v]$, and that

$$\lim_{n \rightarrow \infty} n^{-1} \log w_n = \log q < 0. \quad (68.12)$$

The inverse-derivative potential gives ψ in the one-dimensional smooth case. For similarities, $\psi(\omega) = \log r_{\omega_0}$.

Theorem 68.2 (Intrinsic pressure exponent). *There are unique roots $s_g, s_t \in (0, 1)$ of*

$$P_A(s_g\varphi + \tau s_g\psi) = 0, \quad P_A(s_t\varphi) + s_t \log q = 0. \quad (68.13)$$

Writing $s_ = \max(s_g, s_t)$, one has*

$$\lim_{M \rightarrow \infty} \frac{-\log \mathcal{R}_M}{\log M} = \lim_{M \rightarrow \infty} \frac{-\log e_M^2(Z_w^f)}{\log M} = \frac{1 - s_*}{s_*}. \quad (68.14)$$

The spatial and survival roots exchange dominance at $q_c = \exp\{-P_A(s_g\varphi)/s_g\}$. This formula uses the exponential rate of the renewal tail. Polynomial factors in that tail can change the critical correction without changing either root.

APPENDIX 69. RENEWAL ENERGIES AND ADMISSIBLE SUFFIX TREES

The main combinatorial fact is a strict energy inequality, not strict inequality of cylinder masses. The distinction matters at vertices having a unique predecessor.

Lemma 69.1 (Strict suffix domination). *If u is nonempty and uv is admissible, then*

$$V_w(v) - V_w(uv) \geq (1 - r_+^\tau) r_v^\tau \sum_{k < |u|} w_k > 0, \quad a_w(uv) < a_w(v). \quad (69.1)$$

There are constants $a_0 > 0$ and $\rho < 1$ such that each admissible child satisfies

$$a_0 a_w(v) \leq a_w(vi) \leq \rho a_w(v), \quad \sum_{i: vi \text{ admissible}} a_w(vi) \leq \rho a_w(v). \quad (69.2)$$

The same inequalities hold at the root after decreasing a_0 .

Proof. Write $m = |u|$, $n = |v|$ and suppress the subscript w . Every suffix of v has scale at least r_v . Since the weights decrease,

$$\begin{aligned} V(v) - \left(\sum_{j < n} w_{m+j} r_{v_j}^\tau + W(m+n) \right) &\geq r_v^\tau \left(\sum_{j < n} (w_j - w_{m+j}) + W(n) - W(m+n) \right) \\ &= r_v^\tau \sum_{k < m} w_k. \end{aligned}$$

The remaining prefix terms in $V(uv)$ are at most $r_v^\tau \sum_{k < m} w_k r_{u_k}^\tau$, which is at most $r_+^\tau r_v^\tau \sum_{k < m} w_k$. This proves the first inequality, also for $v = \emptyset$. Stationarity gives $\mu[uv] \leq \mu\{\omega : \omega_{m:m+n} = v\} = \mu[v]$. Both masses are positive, so the second inequality follows. Neither argument completes an inadmissible word by an artificial transition.

For $|v| = n$, appending i multiplies each old spatial scale by a factor between $D_0^{-1}r_-$ and r_+ . The old tail $W(n)$ is replaced by $w_n r_i^\tau + W(n+1)$. The weight assumptions imply

$$(1 - \bar{q})W(n) \leq w_n, \quad W(n+1) \geq \underline{q}W(n).$$

It follows that

$$\ell_0 V(v) \leq V(vi) \leq \rho V(v), \quad \ell_0 = \min(D_0^{-\tau} r_-^\tau, \underline{q}), \quad \rho = 1 - (1 - r_+^\tau)(1 - \bar{q}) < 1.$$

A uniform lower bound $\zeta > 0$ on admissible conditional cylinder masses now gives $a_0 = \zeta \ell_0$ in (69.2). The last assertion follows by summing the masses of the children. Such a ζ follows directly from (68.3) and the bounded potential, for admissible extensions only. $\square$

Lemma 69.2 (Balanced trees and controlled dilation). *Let $L_N = |\mathcal{P}_N|$. For all sufficiently large N , $N/b \leq L_N \leq N$. If θ_N is the energy of the last split vertex, then every leaf satisfies*

$$a_0 \theta_N \leq a_w(v) \leq \theta_N. \quad (69.3)$$

For every fixed $B \geq 1$, with constants allowed to depend on B ,

$$G_{\lceil BN \rceil}(w) \asymp_B G_N(w). \quad (69.4)$$

The complete tree with L leaves has at most $(b+1)(2L-1)$ vertices.

Proof. Maximal split energies never increase, by (69.2). A present leaf was born with energy at least a_0 times an earlier split energy, which proves the lower bound in (69.3); the upper bound follows from the final maximal split. Each split adds between zero and $b-1$ leaves. The split following $\mathcal{T}_N$ increases the number beyond N ; hence $L_N > N - b + 1$, which implies the stated coarser bound.

Here are the finiteness details needed in the presence of unary vertices. A path of b successive vertices all having outdegree one would repeat a vertex and enter a closed deterministic cycle. This contradicts primitivity and $\lambda_A > 1$. Thus a unary chain has length at most b . After suppressing unary vertices, any finite tree with L leaves and no internal vertex of degree one has at most $2L-1$ vertices. Reinserting the unary chains, including an initial chain, gives the asserted bound. In particular an infinite sequence of splits with bounded leaf number is impossible.

Choose ℓ such that A^ℓ has all entries positive. From every last symbol there are at least two distinct continuations of length ℓ . Thus every vertex has at least 2^j descendants at depth ℓj . In continuing a greedy tree with threshold θ_N , if all later leaves have energy less than $a_0^{\ell j+1} \theta_N$, then every original leaf must have been split through those ℓj levels. This would give at least $2^j L_N$ later leaves. Choosing j so that $2^j > b(B+2)$ shows that the final maximum leaf energy at budget $\lceil BN \rceil$ is at least a fixed multiple of θ_N . The balance estimate at that budget bounds its minimum leaf energy by another fixed multiple of θ_N . Since $L_{\lceil BN \rceil}$ is at least a fixed multiple of N , its energy sum is at least $c_B N \theta_N \geq c_B G_N$. Conversely splitting decreases the sum by (69.2), so $G_{\lceil BN \rceil} \leq G_N$. Increasing constants handles the finitely many initial trees. $\square$

Lemma 69.3 (Observable orbit cylinders). *The image $Z_w^f(K_v)$ has squared diameter at most $CV_w(v)$. If two different sequences have longest common prefix u , then*

$$\|Z_w^f(x) - Z_w^f(y)\|^2 \geq cV_w(u). \quad (69.5)$$

All constants depend only on the displayed geometric and observable hypotheses, not on the word.

Proof. Introduce the auxiliary snowflake orbit distance $d_\beta(x, y)^2 = \sum_{k \geq 0} w_k |T^k x - T^k y|^{2\beta}$. The Hölder estimate bounds $\|Z_w^f(x) - Z_w^f(y)\|^2$ from above by $H^2 d_\beta(x, y)^2$. Multiplying (68.5) at $T^k x, T^k y$ by w_k and summing gives

$$c_o d_\beta(x, y)^2 \leq \sum_{j=0}^J \sum_{k \geq 0} w_k |f(T^{k+j}x) - f(T^{k+j}y)|^2 \leq \left(\sum_{j=0}^J q^{-j} \right) \|Z_w^f(x) - Z_w^f(y)\|^2.$$

For points in K_v , the first n terms of d_β^2 are bounded by $w_k r_{v_k}^{2\beta}$ and its remaining terms by $W(n)$. If u is their longest common prefix, then at time $|u|$ they lie in different first-level images, and at time $k < |u|$ their distance is at least $D_0^{-1} g r_{u_k}$. The first post-prefix term is $w_{|u|} g^{2\beta}$, and $W(|u|) \leq w_{|u|}/(1 - \bar{q})$. These estimates prove both assertions, including the empty prefix. $\square$

Lemma 69.4 (Unrestricted quantization and suffix realization). *The quantization errors satisfy $e_M^2(Z_w^f) \asymp G_M(w)$. Every complete greedy tree is suffix closed, including its leaves. A tree with L leaves gives a deterministic observer with at most $(b+1)(2L-1)$ states and average loss at most $C \sum_{v \in \mathcal{P}} a_w(v)$.*

Proof. Taking one orbit representative from each leaf cylinder proves $e_N^2 \leq CG_N$. For the reverse direction use the leaf partition $\mathcal{P}_{4bM}$, which has at least $4M$ members, apart from finitely many budgets absorbable in the constants. Around $Z_w^f(K_v)$ take its open Hilbert tube of radius $\delta \sqrt{V_w(v)}$. These tubes are disjoint for a sufficiently small fixed δ : for incomparable leaves the separation in (69.5) is at their common prefix, and appending symbols only decreases V_w . Any M Hilbert centres, whether or not on the orbit image, occupy at most M tubes. On every other cylinder the nearest-centre squared error is at least $\delta^2 V_w(v)$. The balance estimate shows that these unoccupied cylinders carry at least a fixed fraction of the energy sum. Hence $e_M^2 \geq cG_{4bM} \geq c'G_M$ by (69.4). For each of the finitely many omitted budgets this error is positive: the delay condition and infinitely supported Gibbs law exclude support on finitely many orbit points.

Suppose an internal word uv , with u nonempty, is about to be split. If its suffix v has not yet been split, some current leaf t is a prefix of v . By (69.2) and (69.1), $a_w(t) \geq a_w(v) > a_w(uv)$, contradicting maximality. Thus every proper suffix of an internal word is already internal. If $v = i_0 \cdots i_{n-1}$ is a leaf with $n \geq 2$, the word $i_1 \cdots i_{n-2}$ is a suffix of its internal parent, so it has been split. Its admissible child $i_1 \cdots i_{n-1}$ is therefore present. For $n = 1$ the suffix is the root. Iteration proves closure of the entire tree; it does not require that all symbols follow all others.

Choose $x_v \in K_v$ for every nonempty tree vertex, and decode that vertex as $f(x_v)$. At an acquisition store the unique leaf prefix of the observed sequence. At each nonacquisition delete the first symbol. Decode the empty word as any fixed $f(x_*)$, and keep it until the next acquisition. Each update stays within the same tree. If the acquired word is v of length n , the squared loss at age $k < n$ is at most $H^2 r_{v_k}^{2\beta}$; after exhaustion it is bounded by a constant. The renewal-reward formula therefore bounds the average by $C \sum_v \mu[v] V_w(v)$. All vertices are charged, with the count established in Lemma 69.2. $\square$

Proof of Theorem 68.1. The retained posterior-orbit theorem, Theorem 74.2, applies with exact observation and Bayes floor zero. It gives $e_M^2 \leq \mathcal{R}_M$ and the stated lower-limit assertion for arbitrary stochastic kernels. Its full proof is included in both reading editions. Lemma 69.4 gives the other lower bound. Choose a leaf budget N a fixed sufficiently small fraction of M so that (68.11) is at most M . The same lemma and fixed-budget dilation give an upper bound

$CG_N \leq C'G_M$. A one-state observer handles the remaining bounded range of M . This proves (68.10). $\square$

APPENDIX 70. TWO PRESSURES ON AN ADMISSIBLE LANGUAGE

All pressures in this section are taken on Σ_A . A Hölder potential F has uniformly bounded oscillation of $S_n F$ on cylinders, and its partition sums are comparable to $e^{nP_A(F)}$, uniformly in n . These are classical finite-state thermodynamic facts [32]. The new issue is the renewal-weighted sum of suffix scales, which is not an additive potential.

Lemma 70.1 (Mixed partition pressure). *For $0 < s < 1$, under (68.12),*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \sum_{v \in \mathcal{L}_n} a_w(v)^s = \max\{P_A(s\varphi + \tau s\psi), P_A(s\varphi) + s \log q\}. \quad (70.1)$$

Proof. Put $P_g(s) = P_A(s\varphi + \tau s\psi)$ and $P_0(s) = P_A(s\varphi)$. The first and last terms in (68.8), separately, imply

$$\sum_{|v|=n} a_w(v)^s \geq c_s \max\{w_0^s e^{nP_g(s)}, W(n)^s e^{nP_0(s)}\}.$$

For the upper bound use $(\sum x_k)^s \leq \sum x_k^s$ and split an admissible word after k letters. Its mass and suffix scale are, within fixed factors, the exponential of the corresponding Birkhoff sums. Dropping the constraint joining the two pieces can only increase the sum. Thus

$$\sum_{|v|=n} a_w(v)^s \leq C_s \left(\sum_{k < n} w_k^s e^{kP_0(s)} e^{(n-k)P_g(s)} + W(n)^s e^{nP_0(s)} \right). \quad (70.2)$$

No factorization of arbitrary words across a forbidden transition is asserted. Given $\delta > 0$, the weight assumption gives $w_k \leq C_\delta e^{k(\log q + \delta)}$, and $W(n) \asymp w_n$. The upper sum has at most $n + 1$ terms with exponential rate at most $\max(P_g(s), P_0(s) + s \log q + s\delta)$. The lower bound has rate at least the same maximum with $\delta = 0$. Let $\delta \downarrow 0$. $\square$

Proof of Theorem 68.2. We first verify uniqueness of the roots. Every vertex has at least two continuations of a common length ℓ . By the positive conditional mass bound, no particular continuation of that length has conditional probability larger than $1 - \zeta^\ell$. Hence $\mu[v] \leq C\eta^{|v|}$ for an $\eta < 1$. The Gibbs estimate gives $S_n \varphi \leq n \log \eta + O(1)$ on every cylinder. Similarly $S_n \psi \leq n \log r_+ + O(1)$. The pressure functions in (68.13) are continuous and strictly decreasing, with positive value $\log \lambda_A$ at zero. At one their values are strictly negative, since $P_A(\varphi) = 0$. Their roots lie in $(0, 1)$.

For $s > s_*$, both exponents in (70.2) are negative. Choose $q'' > q$ close enough that they remain negative with q'' in place of q . Bounding w_k and $W(k)$ by a constant times $(q'')^k$ and summing (70.2) over all n yields

$$\sum_{v \in \mathcal{L}} a_w(v)^s < \infty. \quad (70.3)$$

For a balanced partition with $L \asymp N$ leaves, this implies $L(a_0 \theta_N)^s \leq C_s$, and therefore $G_N \leq L \theta_N \leq C'_s N^{-(1-s)/s}$.

For the lower bound at s_g , use the Gibbs probability ν_g for the zero-pressure potential $s_g \varphi + \tau s_g \psi$. On each cylinder,

$$\nu_g[v] \leq C \mu[v]^{s_g} r_v^{\tau s_g} \leq C' a_w(v)^{s_g}.$$

Summing over $\mathcal{P}_N$ gives $1 \leq C' L \theta_N^{s_g}$. Balance then gives $G_N \geq c N^{-(1-s_g)/s_g}$.

For the survival root, a polynomial correction to w_n prevents using q^n as a uniform lower bound. Instead fix $0 < q' < q$. There is $c_{q'} > 0$ with $W(n) \geq c_{q'} (q')^n$ for all n . Let $s_t(q')$ be the zero of $P_A(s\varphi) + s \log q'$, and use the Gibbs measure of this zero-pressure potential. Its cylinder masses are

bounded by $C\mu[v]^{s_t(q')}(q')^{|v|s_t(q')}$, hence by $C'a_w(v)^{s_t(q')}$. The same balanced-partition argument proves the lower bound with exponent $(1 - s_t(q'))/s_t(q')$. The roots vary continuously with q' , by strict monotonicity. Letting $q' \uparrow q$ and $s \downarrow s_*$ proves the common exponent for G_N . Theorem 68.1 transfers it to both errors. Finally $s_t = s_g$ precisely when $P_A(s_g\varphi) + s_g \log q = 0$, proving the threshold formula. $\square$

Corollary 70.2 (Markov matrices with zero entries). *Suppose the h_i are similarities and μ is a stationary Markov measure with transition probabilities $p_{ij} > 0$ exactly when $A_{ij} = 1$. The roots in (68.13) are characterized by*

$$\text{spr}((A_{ij}p_{ij}^{s_g}r_i^{\tau s_g})_{ij}) = 1, \quad q^{s_t} \text{spr}((A_{ij}p_{ij}^{s_t})_{ij}) = 1. \quad (70.4)$$

Only admissible entries are powered; other entries are zero. At $s = 0$ the limiting matrix is A , not a matrix of ones.

Proof. Take $\varphi(\omega) = \log p_{\omega_0\omega_1}$ on admissible edges and $\psi(\omega) = \log r_{\omega_0}$. The nonuniform stationary initial factor and the terminal factor are bounded above and below, so do not change pressure. The finite transfer matrices of these edge potentials are exactly those in (70.4). Perron–Frobenius theory and Theorem 68.2 give the assertion. $\square$

Example 70.3 (A noninjective planar readout with a forbidden transition). On $[0, 1]^2$ take

$$h_0(x, y) = (ry, rx), \quad h_1(x, y) = (ry + 2/3, rx), \quad r = 1/3, \quad A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Rescaling the ambient domain makes its diameter one without changing the argument. Both maps are similarities and their images have a positive horizontal gap. Give the golden-mean shift any stationary Hölder Gibbs law. For $f(x, y) = x$, the relation $f(T(x, y)) = y/r$ on K gives

$$|f(x, y) - f(x', y')|^2 + |f(T(x, y)) - f(T(x', y'))|^2 = (x - x')^2 + r^{-2}(y - y')^2.$$

Thus (68.5) holds with $J = 1$ and $\beta = 1$. The sequences $000\cdots$ and $01000\cdots$ are admissible, have the same first coordinate, and have different second coordinates. Hence f is not injective. This is a higher-dimensional geometric system with a genuinely forbidden transition and a delayed, rather than instantaneously identifying, observable.

APPENDIX 71. CRITICAL CORRECTIONS AND A UNIFORM RENEWAL WINDOW

The exponential rate of the acquisition tail does not determine the critical second order. We give a classification in a family allowing arbitrary primitive transition graphs, every ambient dimension in the similarity construction, and delay-observable outputs.

Assume that all branch ratios equal r , and that μ is the Parry measure of Σ_A . Write $\lambda = \lambda_A$ and $a = r^{2\beta}$. For $q \in (0, 1)$ and $\kappa \geq 0$ take the actual survival law

$$\mathbb{P}(L > k) = q^k(k+1)^{-\kappa}, \quad c_{q,\kappa} = \left(\sum_{k \geq 0} q^k(k+1)^{-\kappa} \right)^{-1}. \quad (71.1)$$

Its weights satisfy (68.6), with lower ratio $q2^{-\kappa}$ and upper ratio q . The common depth- n energy scale is

$$V_{q,\kappa}(n) = c_{q,\kappa} \left[a^n \sum_{k < n} (q/a)^k(k+1)^{-\kappa} + \sum_{k \geq n} q^k(k+1)^{-\kappa} \right]. \quad (71.2)$$

Theorem 71.1 (Critical power–log classification). *Let $n_M = \lfloor \log M / \log \lambda \rfloor$. In this family,*

$$\mathcal{R}_M \asymp V_{q,\kappa}(n_M). \quad (71.3)$$

For fixed κ the constants can be chosen uniformly for q in a compact subinterval of $(0, 1)$. The critical value is $q_c = a$. With $\alpha = \log(1/a)/\log \lambda$, at this value, as $M \rightarrow \infty$,

$$\mathcal{R}_M \asymp M^{-\alpha} \begin{cases} (\log M)^{1-\kappa}, & 0 \leq \kappa < 1, \\ \log \log M, & \kappa = 1, \\ 1, & \kappa > 1. \end{cases} \quad (71.4)$$

In particular the same graph, geometry, observable, pressure roots, and critical exponential survival rate admit different critical corrections. The specification of the renewal law is essential.

Proof. Perron eigenvectors give, uniformly in $v \in \mathcal{L}_n$, $\mu[v] \asymp \lambda^{-n}$ and $|\mathcal{L}_n| \asymp \lambda^n$. All these cylinders have scale (71.2). A fixed-depth complete tree has $O(\lambda^n)$ vertices (sum the geometric level counts), is suffix closed, and gives average loss at most $CV_{q,\kappa}(n)$ by the pointwise construction in Lemma 69.4. Choose $n = n_M - C_0$ so that its state count is at most M .

For a lower bound choose $n = n_M + C_1$ with at least $4M$ cylinders. The disjoint tubes used in Lemma 69.4 then leave at least three quarters of the cylinders unoccupied by any M centres. Their masses are comparable and their scales equal, giving $e_M^2 \geq cV_{q,\kappa}(n)$. A fixed change in depth changes V by factors bounded above and below by (69.2)'s scale proof, uniformly for q in the specified compact set. The converse gives (71.3). This argument neither treats the Parry measure as independent nor fills forbidden transitions.

At $q = a$, the first term in (71.2) is $c_{a,\kappa} a^n \sum_{j=1}^n j^{-\kappa}$; the tail is $O(a^n(n+1)^{-\kappa})$. Integral comparison gives the three possibilities in (71.4). Also $P_A(s\varphi) = (1-s)\log \lambda$ and $P_A(s\varphi + 2\beta s \log r) = (1-s)\log \lambda + s \log a$. Thus the two roots coincide at $q = a$, as asserted. $\square$

Theorem 71.2 (Uniform finite-size profile window). *Fix $\kappa \geq 0$ and put $q_n = ae^{x/n}$. Uniformly for x in a bounded real interval, the following limits hold:*

$$\frac{V_{q_n,\kappa}(n)}{a^n n^{1-\kappa}} \longrightarrow c_{a,\kappa} \int_0^1 t^{-\kappa} e^{xt} dt, \quad 0 \leq \kappa < 1, \quad (71.5)$$

$$\frac{V_{q_n,1}(n)}{a^n \log n} \longrightarrow c_{a,1}, \quad \kappa = 1, \quad (71.6)$$

$$\frac{V_{q_n,\kappa}(n)}{a^n} \longrightarrow c_{a,\kappa} \sum_{j \geq 1} j^{-\kappa}, \quad \kappa > 1. \quad (71.7)$$

The optimal risk is uniformly comparable to these profiles at budgets of order λ^n . The displayed leading coefficients are profile coefficients, not asserted exact optimal-risk coefficients.

Proof. The normalizing constants tend uniformly to $c_{a,\kappa}$ by a common geometric majorant. For $0 \leq \kappa < 1$, split the normalized sum at $k = \delta n$. Away from zero, its summands form a uniformly convergent Riemann sum for $t^{-\kappa} e^{xt}$. The part below δn is bounded uniformly by $C\delta^{1-\kappa}$; then let $\delta \downarrow 0$. The normalized tail is $O(n^{-1})$ uniformly, using $q_n \leq q_+ < 1$ and $q_n^n/a^n = e^x$. For $\kappa = 1$,

$$\left| \sum_{k < n} \frac{e^{xk/n} - 1}{k+1} \right| \leq C$$

uniformly on bounded x , while the unperturbed harmonic sum is $\log n + O(1)$. The tail divided by $a^n \log n$ vanishes. For $\kappa > 1$, dominated convergence applies to the summable sequence $(k+1)^{-\kappa}$, and the tail divided by a^n is $O(n^{-\kappa})$. Finally use the uniform comparison in Theorem 71.1. $\square$

The classification concerns the stated Markov–Parry similarity family and the renewal tails (71.1). For unequal Hölder derivative cocycles the finite profile theorem and pressure exponent still apply; (71.4) is not imposed on that larger class. In the geometric case $\kappa = 0$ the first window profile is $(1-a)(e^x - 1)/x$, continuously extended at $x = 0$.

APPENDIX 72. FINITE REGISTERS WITHOUT A COMMON REGENERATION LAW

A different realization mechanism applies when observations arrive continually and the posterior itself contracts. It does not use an independent reference component of a transition kernel.

Let $(U_t, Y_t)_{t \in \mathbb{Z}}$ be stationary, with $U_t \in \mathbb{R}^d$ square integrable and Y_t standard Borel. Write

$$\mathcal{Y}_t^- = \sigma(Y_s : s \leq t), \quad m_t = \mathbb{E}[U_t \mid \mathcal{Y}_t^-], \quad B = \mathbb{E}\|U_0 - m_0\|^2.$$

An observer starting at zero receives Y_t at every update and retains at most M states. Its decoder receives only its post-update state, absolute time, and independent coding randomness. Let $\mathcal{R}_M^{\text{fil}}$ be the infimum of its upper limiting average squared loss. The reference B uses the full two-sided past; initial lack of that past will be handled by contraction, not by a fictitious initial sufficient statistic.

Theorem 72.1 (Contracting posterior realization). *Suppose $m_0 \in L^4$ and a Borel function F has a measurable Lipschitz coefficient $\ell(y) \geq 0$ such that*

$$\begin{aligned} m_t &= F(m_{t-1}, Y_t), & \|F(x, y) - F(x', y)\| &\leq \ell(y)\|x - x'\|, \\ \mathbb{E}[\ell(Y_t)^4 \mid \mathcal{Y}_{t-1}^-] &\leq \rho^4 < 1, & K_4 &:= \|F(0, Y_0)\|_{L^4} < \infty. \end{aligned} \quad (72.1)$$

A uniform bound $\ell(y) \leq \rho$ is sufficient but not necessary; individual observation updates may expand. If the law of m_0 has a Lebesgue density bounded below by a positive constant on a cube of positive side length, then

$$B + cM^{-2/d} \leq \mathcal{R}_M^{\text{fil}} \leq B + CM^{-2/d} \quad (M \geq 1). \quad (72.2)$$

The lower bound holds for the lower limiting average loss of every randomized time-dependent observer. The upper observer is stationary and deterministic. It uses no compact support or minorization hypothesis on either the hidden dynamics or the observation process.

For a product register of $(2n - 1)^d$ states, $n \geq 2$, its excess error has the explicit upper bound

$$\limsup_{t \rightarrow \infty} \mathbb{E}\|m_t - \hat{U}_t\|^2 \leq \frac{d(1 + K_4/(1 - \rho))^4}{n^2(1 - \rho)^2}. \quad (72.3)$$

Lemma 72.2 (A finite inward quantizer). *For $n \geq 2$ define*

$$j_n(x) = \min\left(n - 1, \left\lfloor \frac{n|x|}{1 + |x|} \right\rfloor\right), \quad Q_n(x) = \text{sgn}(x) \frac{j_n(x)}{n - j_n(x)}.$$

It has $2n - 1$ values and satisfies

$$|Q_n(x)| \leq |x|, \quad |x - Q_n(x)| \leq n^{-1}(1 + |x|)^2. \quad (72.4)$$

Its coordinatewise extension to $\mathbb{R}^d$ satisfies $\|Q_n x\| \leq \|x\|$ and $\|x - Q_n x\| \leq \sqrt{d} n^{-1}(1 + \|x\|)^2$.

Proof. For $x \geq 0$ put $u = x/(1 + x)$, so the inverse is $g(u) = u/(1 - u)$. The quantized argument $j_n(x)/n$ is at most u and differs from it by less than $1/n$, including the saturated last bin. The increasing derivative $g'(u) = (1 - u)^{-2}$ bounds the difference by $(1 + x)^2/n$. Odd reflection proves the scalar assertion. Sum the squared coordinate errors and use $|x_i| \leq \|x\|$ for the vector bound. $\square$

Proof of Theorem 72.1. Conditional orthogonality, also after adjoining randomness independent of the experiment, gives

$$\mathbb{E}\|U_t - \hat{U}_t\|^2 = B + \mathbb{E}\|m_t - \hat{U}_t\|^2. \quad (72.5)$$

At any fixed t , freeze all independent randomization. The decoder has at most M possible values, even though the encoder has seen a long observation history. On a cube where the density of m_t is at least c_0 , take balls of radius $\delta M^{-1/d}$ around those values. A sufficiently small δ , independent of their locations, makes their total intersection volume at most half the cube's volume. The

remaining volume incurs squared distance at least $\delta^2 M^{-2/d}$. Integrating the density lower bound and then the randomization gives the lower bound at each time, and hence for lower limiting averages.

For the upper bound start from $z_{-1} = 0$ and iterate

$$x_t = F(z_{t-1}, Y_t), \quad z_t = Q_n x_t, \quad \hat{U}_t = z_t. \quad (72.6)$$

Only the index of z_t is stored; x_t is transient. The values of Q_n are fixed rational vectors, part of the fixed decoder table. Conditional fourth-moment contraction gives $\|\ell(Y_t)z_{t-1}\|_{L^4} \leq \rho\|z_{t-1}\|_{L^4}$. The inward property and Minkowski's inequality therefore give, without independence of successive observations,

$$\|z_t\|_{L^4} \leq \rho\|z_{t-1}\|_{L^4} + K_4 \leq A_4 := K_4/(1 - \rho), \quad \|x_t\|_{L^4} \leq A_4.$$

Lemma 72.2 therefore implies $\|x_t - Q_n x_t\|_{L^2} \leq \sqrt{d}(1 + A_4)^2/n$. Couple the exact stationary recursion to the same observations. The previous error is measurable with respect to $\mathcal{Y}_{t-1}^-$; conditional Cauchy–Schwarz implies $\mathbb{E}[\ell(Y_t)^2 \mid \mathcal{Y}_{t-1}^-] \leq \rho^2$. Thus

$$\|m_t - z_t\|_{L^2} \leq \rho\|m_{t-1} - z_{t-1}\|_{L^2} + \sqrt{d}(1 + A_4)^2/n.$$

The finite initial error is multiplied by ρ^{t+1} . Summing this recursion proves (72.3). Equation (72.5) gives the upper average bound. For large M choose $n = \lfloor (M^{1/d} + 1)/2 \rfloor$, so $(2n - 1)^d \leq M$ and $n \asymp M^{1/d}$. For the remaining finite range, the one-state output zero has finite loss, and an enlarged constant completes (72.2). $\square$

Corollary 72.3 (An unbounded hidden Gaussian chain). *Let*

$$X_t = aX_{t-1} + \xi_t, \quad Y_t = X_t + \eta_t, \quad 0 < |a| < 1, \quad \xi_t \sim N(0, v), \quad \eta_t \sim N(0, \omega), \quad (72.7)$$

where $v, \omega > 0$, the noises are independent, and X has its stationary law. Let P be the unique positive solution of

$$P = \frac{\omega(a^2 P + v)}{a^2 P + v + \omega}. \quad (72.8)$$

Then $\mathcal{R}_M^{\text{fil}} = P + \Theta(M^{-2})$. For d independent such coordinates the law is $\sum_i P_i + \Theta(M^{-2/d})$. For every fixed $k \geq 1$, the hidden k -step kernel admits no nonzero finite measure dominated by all its transition probabilities.

Proof. Put $S = v/(1 - a^2)$. Conditional Gaussian projection gives the one-step prediction variance $a^2 P + v$ and the update (72.8). The map on its right sends $[0, S]$ into itself and has derivative at most $a^2 < 1$; its unique fixed point lies in $(0, S)$. Iterating from a finite past and letting that past recede gives the stationary conditional variance P and the mean recursion

$$m_t = (1 - K)am_{t-1} + KY_t, \quad K = \frac{a^2 P + v}{a^2 P + v + \omega}.$$

This passage is also the L^2 martingale convergence of the increasing observation sigma fields. Gaussian conditioning supplies the displayed finite-past recursion at each step. Thus no distributional filter approximation is being substituted for the true conditional mean. The mean is Gaussian with variance $S - P > 0$, so has all required moments and a strictly positive density on every bounded interval. The contraction constant is $|(1 - K)a| < 1$. Theorem 72.1 applies; independent coordinates give the vector statement with the Euclidean contraction constant equal to the largest coordinate constant.

The hidden k -step transition from x is a Gaussian with mean $a^k x$ and fixed finite positive variance. If a nonzero finite measure ν were dominated by every such probability, some bounded Borel set E would have $\nu(E) > 0$. Sending x to infinity gives $Q^k(x, E) \rightarrow 0$, a contradiction. The

proof of the risk law therefore cannot be an application of a global one-step or fixed-multistep common-reference minorization. $\square$

The contraction and Gaussian conditioning in this section are classical mechanisms. The quantitative assertion is the matched error in the number of values of the *whole persistent register*, with a uniform-in-time implementation on an unbounded state space. Constants in (72.3) display their deterioration as $\rho \uparrow 1$. No assertion at $\rho = 1$, nor a reinterpretation of this theorem as the $\varepsilon \downarrow 0$ limit of a minorization bound, is needed.

APPENDIX 73. OPERATIONAL CHANGES OF THE FINITE-MEMORY INTERFACE

The following comparison is distribution-free. In particular it applies to both realization mechanisms above.

Theorem 73.1 (Finite interface equivalence). *Consider any sequential experiment with a standard Borel update input. Suppose an event E_t in an alphabet of size $K < \infty$ is available to the update. Let R_M^{side} be the optimum when the post-update decoder also receives E_t , and let R_M^{reg} be the optimum when it does not. With the same choices of time and independent randomness, for every nonnegative path loss and every horizon, and likewise for upper or lower limiting average loss,*

$$R_{KM}^{\text{reg}} \leq R_M^{\text{side}} \leq R_M^{\text{reg}}. \quad (73.1)$$

These inequalities arise from exact pathwise simulations. An M -value register has an equivalent binary checkpoint encoding in $\lceil \log_2 M \rceil$ bits with unused strings allowed. Several finite registers of sizes $M_1, \dots, M_s$ are equivalent to one register of size $\prod_i M_i$.

Proof. An observer can ignore a free event, giving the second inequality. For the first, store the pair (I_t, E_t) . Its transition discards the previous event, performs the original update using the current input, and stores the new event. The original decoder is then evaluated on this pair. The outputs agree for each input and each realization of all coding random variables. Thus all path losses and their expectations agree, not just a selected one-time distortion. Applying a fixed bijection to the finite state set transports the transition kernels and decoder to binary strings or to a product state set. The inverse bijections give the remaining assertions. $\square$

Consequently a positive squared-error exponent is invariant under any fixed finite event interface, fixed regrouping of finite registers, and finite-state renaming. When the risk or its Bayes excess satisfies a fixed-budget-dilation comparison, its full order is invariant as well. For a binary checkpoint budget B the corresponding register scale is $M = 2^B$; this changes a polynomial in M to an exponential in B , rather than changing the theorem's physical resource silently.

Example 73.2 (An unbounded side interface is a different experiment). Let X_t be independent uniform random variables on $[0, 1]$ and let the current update input be $Y_t = X_t$. A decoder with the real-valued Y_t as a free post-update side input has zero loss with one persistent state. Without that side input the optimal M -state loss is $1/(12M^2)$: midpoint quantization attains it, and for cells of masses p_i the uniform interval moment inequality gives $\sum_i p_i^3/12 \geq 1/(12M^2)$. Randomization does not improve the pointwise nearest-centre minimum. A finite K in (73.1) cannot be replaced by an arbitrary standard Borel alphabet. The distinction is the presence of a new measurement at the decoding boundary, not a hidden age-state convention.

APPENDIX 74. POSTERIOR ORBITS AND FINITE-STATE REALIZATION

Let E and $\mathcal{Y}$ be standard Borel spaces, let μ be a probability on E , and let K be a probability kernel from E to $\mathcal{Y}$. Let $T : E \rightarrow E$ and $f : E \rightarrow \mathbb{R}^r$ be Borel, with $\|f\| \leq F < \infty$. At each renewal a fresh pair (U, Y) has law $\mu(du)K(u, dy)$. The observer acquires Y , not necessarily U . Between

renewals the hidden state follows T . The successive cycle lengths $L_0, L_1, \dots$ are independent copies of a positive integer-valued random variable L , independent of all acquired pairs, and

$$\ell = \mathbb{E}L < \infty, \quad s_k = \mathbb{P}(L > k), \quad w_k = \frac{s_k}{\ell}, \quad W(n) = \sum_{k \geq n} w_k. \quad (74.1)$$

Thus $\sum_{k \geq 0} w_k = 1$. There is an acquisition at time 0. Subsequent acquisition times are $\tau_j = L_0 + \dots + L_{j-1}$. At time $t = \tau_j + k < \tau_{j+1}$, the target is $f(T^k U_j)$. The renewal indicator is observed, but the length of the current cycle is not announced at its beginning.

An M -state observer retains $I_t \in \{1, \dots, M\}$. Its update is an arbitrary Borel stochastic kernel of the preceding state, absolute time, renewal indicator, and, at a renewal only, the new observation. After the update, the decoder has the state, absolute time, and independent public randomness. It does not separately retain the last observation, the renewal indicator, or the renewal age. Private randomness is allowed, but any data-dependent persistent register, including an age or a random-seed position, is part of I_t . Program descriptions, real constants, and transient computation are not charged to M . This is a persistent-cardinality constraint, not a bound on total computational storage.

Choose simultaneous Borel versions

$$m_k(y) = \mathbb{E}[f(T^k U) \mid Y = y], \quad Z(y) = (\sqrt{w_k} m_k(y))_{k \geq 0} \in \mathcal{H} := \ell^2(\mathbb{N}_0; \mathbb{R}^r). \quad (74.2)$$

A countable union of null sets makes these versions simultaneous; values there may be set to zero. Write

$$e_M^2(Z) = \inf_{c_1, \dots, c_M \in \mathcal{H}} \mathbb{E} \min_{i \leq M} \|Z(Y) - c_i\|_{\mathcal{H}}^2, \quad B = \sum_{k \geq 0} w_k \mathbb{E} \|f(T^k U) - m_k(Y)\|^2. \quad (74.3)$$

The number B is the unavoidable error of the complete acquisition history. Let $A_t = t - \max\{\tau_j : \tau_j \leq t\}$, and let Y_t^* be the observation at that last renewal. Define

$$E_M^{\text{av}} = \inf_{\mathcal{A}} \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{t=0}^{N-1} \mathbb{E} \|m_{A_t}(Y_t^*) - \hat{f}_t\|^2, \quad (74.4)$$

$$R_M^{\text{av}} = \inf_{\mathcal{A}} \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{t=0}^{N-1} \mathbb{E} \|f(X_t) - \hat{f}_t\|^2. \quad (74.5)$$

Here and below an infimum is over the specified machine class; it need not be attained. All statements about e_M^2 use squared distortion.

Lemma 74.1 (Kernel realization). *The preceding machine class can be realized using a countable family of independent uniform random variables, independent of the experiment. After fixing this family and the history before a renewal, the first n outputs along a continuation with no further renewal belong to a set of at most M vectors in $(\mathbb{R}^r)^n$.*

Proof. For a finite-state transition kernel the inverse cumulative probabilities give a Borel function of the input and one uniform variable. For a general standard Borel output, embed its range as a Borel subset of $[0, 1]$, use the measurable inverse distribution function, and map back; a fixed default value handles the null exceptional set. Applying this construction to each time-indexed kernel uses only countably many uniforms. The product probability space supplies them jointly. Public uniforms are included in the same construction, with their prescribed sharing between update and decoder. Once they are fixed, the update and decoder are deterministic. Given the state immediately after a renewal, the no-renewal input sequence and absolute times determine the subsequent states and outputs. There are only M possible starting states. The argument uses a fixed time-indexed family, not an uncharged data-dependent position in a random tape. $\square$

Theorem 74.2 (Posterior-orbit converse and realization). *For the experiment just defined,*

$$R_M^{\text{av}} = B + E_M^{\text{av}}, \quad e_M^2(Z) \leq E_M^{\text{av}}. \quad (74.6)$$

In fact every randomized, time-dependent M -state machine satisfies the latter lower bound with its $\liminf$ time-average excess loss. Conversely, for all integers $N, n \geq 1$,

$$E_{Nn+1}^{\text{av}} \leq e_N^2(Z^{<n}) + F^2W(n) \leq e_N^2(Z) + F^2W(n), \quad (74.7)$$

where $Z^{<n}$ is the first n coordinates of Z . The upper bound is realized, up to an arbitrary positive error, by a stationary deterministic machine with states $\{1, \dots, N\} \times \{0, \dots, n-1\} \cup \{\partial\}$. In particular, for $M \geq 2$,

$$e_M^2(Z) \leq E_M^{\text{av}} \leq \inf_{1 \leq n \leq M-1} \left\{ e_{\lfloor (M-1)/n \rfloor}^2(Z) + F^2W(n) \right\}. \quad (74.8)$$

Proof. Conditional on the full acquired history, the renewal indicators, and independent coding randomness, the conditional mean of $f(X_t)$ is $m_{A_t}(Y_t^*)$. Indeed, the current acquired pair is independent of previous cycles, and survival of its cycle is independent of that pair. Conditional orthogonality gives, for every machine,

$$\mathbb{E}\|f(X_t) - \hat{f}_t\|^2 = \mathbb{E}\|f(X_t) - m_{A_t}(Y_t^*)\|^2 + \mathbb{E}\|m_{A_t}(Y_t^*) - \hat{f}_t\|^2.$$

The Cesàro renewal-reward identity makes the average of the first term tend to B . Bounded rewards and $\mathbb{E}L < \infty$ suffice, including arithmetic or deterministic cycle lengths. This proves the equality in (74.6).

Fix n . At a renewal at time s , condition on the past and then use Lemma 74.1. Along the first n no-renewal steps the outputs are chosen from at most M decoder strings. The new Y has its original marginal law under this conditioning. Since survival to age k has probability s_k , the expected excess reward in those steps is at least

$$\inf_{c_1, \dots, c_M} \mathbb{E} \min_i \sum_{k < n} s_k \|m_k(Y) - c_{i,k}\|^2 = \ell e_M^2(Z^{<n}).$$

Randomizing the selection of a string cannot reduce its pointwise minimum. Integrating the conditioned variables preserves the bound. Let $u_s = \mathbb{P}(s \text{ is a renewal})$. Rewards arising from distinct last renewals are disjoint. Summing over $0 \leq s \leq N - n$ therefore gives a lower bound $\ell e_M^2(Z^{<n}) \sum_{s=0}^{N-n} u_s$ for the total excess loss up to time $N - 1$. The elementary renewal theorem, in its Cesàro form, gives $N^{-1} \sum_{s < N} u_s \rightarrow 1/\ell$. Consequently the lower limit of average loss is at least $e_M^2(Z^{<n})$.

Truncation cannot increase quantization error. Conversely, extending a codebook for the truncation by zero gives

$$e_M^2(Z^{<n}) \leq e_M^2(Z) \leq e_M^2(Z^{<n}) + F^2W(n).$$

Letting $n \rightarrow \infty$ proves the converse, without a compactness assumption on the orbit image or a restriction to on-image centres.

For the upper bound choose an ε -optimal N -centre codebook for $Z^{<n}$. Coordinatewise projection permits its k th coordinate to be written $\sqrt{w_k}c_{i,k}$ with $\|c_{i,k}\| \leq F$; a zero-weight coordinate is immaterial. At a renewal, choose the first nearest centre, a Borel choice, and set the state to $(i, 0)$. At a nonrenewal change (i, k) to $(i, k+1)$ until $k = n-1$, after which the state is ∂ . The latter state is absorbing between renewals. Decode (i, k) as $c_{i,k}$ and ∂ as zero. The average excess reward is at most $e_N^2(Z^{<n}) + \varepsilon + F^2W(n)$ by the same renewal-reward identity. This is an actual finite-state update; its age information is precisely the charged second coordinate. Let $\varepsilon \downarrow 0$. Unused states are permitted, which proves the last assertion. $\square$

Corollary 74.3 (Invariance of the quantization exponent). *Suppose $W(n) \leq A \exp(-cn^\beta)$ for some $A, c, \beta > 0$, and*

$$\alpha = \lim_{M \rightarrow \infty} \frac{-\log e_M^2(Z)}{\log M} \quad \text{exists in } (0, \infty). \quad (74.9)$$

Then

$$\lim_{M \rightarrow \infty} \frac{-\log E_M^{\text{av}}}{\log M} = \alpha. \quad (74.10)$$

If $e_M^2(Z) \asymp M^{-\alpha}$, more explicitly

$$c_1 M^{-\alpha} \leq E_M^{\text{av}} \leq C_1 M^{-\alpha} (\log M)^{\alpha/\beta} \quad (M \geq 2),$$

with constants depending on the displayed hypotheses. For geometric renewals with parameter $p > 0$, the exponent conclusion also holds for the infimum of the supremum-in-time excess risk.

Proof. Choose any $s > \alpha$ and $n = \lceil ((s/c) \log M)^{1/\beta} \rceil$. Then $\log[(M-1)/n]/\log M \rightarrow 1$, whereas $W(n) \leq AM^{-s}$. Apply (74.8); the two logarithmic bounds converge to α . Under the two-sided power bound the same substitution gives the explicit inequalities, enlarging constants for finitely many small M .

For geometric renewals put $q = 1 - p$, so $w_k = pq^k$. A stationary reset-based machine has a nonnegative age-loss sequence d_k . With an initial acquisition at zero its loss at time t is

$$p \sum_{k < t} q^k d_k + q^t d_t \leq p^{-1} \sum_{k \geq 0} w_k d_k.$$

Use the upper machines in Theorem 74.2; the converse follows because a supremum is at least the lower time-average. $\square$

The corollary is an exponent theorem, not a uniform equivalence of errors for arbitrary orbits. It applies to arbitrary Borel dynamics and arbitrary noisy or partial reset observations. Its geometric invariant is the quantization exponent of the *posterior* future orbit; the tail assumption controls the cost of making a finite codebook causal. Computing that invariant in a particular dynamical system remains a separate mathematical task.

Proposition 74.4 (Exact autonomous realization). *Let a ray mean a sequence $c = (c_k)_{k \geq 0}$ in the ball $\{z \in \mathbb{R}^r : \|z\| \leq F\}$. Among stationary deterministic machines, the optimal average excess risk is*

$$\inf_{\substack{|C| \leq M \\ SC \subseteq C}} \mathbb{E} \min_{c \in C} \sum_{k \geq 0} w_k \|m_k(Y) - c_k\|^2, \quad (Sc)_k = c_{k+1}. \quad (74.11)$$

The reset update may depend on the previous state; allowing that dependence does not alter this infimum.

Proof. For a fixed autonomous no-renewal update a and decoder d , a state i determines the ray $(d(a^k i))_{k \geq 0}$. These at most M rays are closed under S . Any reset rule selects one of them, possibly using the previous state; its reward is no smaller than the pointwise minimum over all the rays. Conversely, for a finite shift-closed set, retain the chosen ray as the state, apply S on a nonrenewal, and decode its first coordinate. The first nearest ray is a Borel reset rule independent of the previous state. Renewal reward proves the formula. This is a characterization of the stated autonomous class, not an assertion that all optimal randomized nonstationary machines are autonomous. $\square$

Example 74.5 (Orbit cardinality does not include an age register). Let $E = \{0, 1\}$, $U = 1$ deterministically, $T(0) = T(1) = 0$, and $f(x) = x$. Use geometric renewals with $0 < p < 1$ and a constant reset observation. The posterior orbit is the single ray $(1, 0, 0, \dots)$, so $e_1^2(Z) = 0$ and $B = 0$. Nevertheless a one-state decoder, having only absolute time and independent public coins,

does not know whether a renewal occurred at that time. For every $t \geq 1$ the minimum squared error is $p(1-p)$, since the target is an independent Bernoulli(p) renewal indicator. Two states suffice for zero error: write 1 at each renewal and 0 otherwise. Thus the orbit converse cannot be reversed with the same cardinality without a realizability condition. The finite-chain overhead in (74.7) and the closure in (74.11) address a genuine obstruction.

APPENDIX 75. A FINITE-REGISTER REFINEMENT OF THE CURRENT CONTACT EXPERIMENT

This application uses the actual known-mark count-to-contact protocol of A2 revision 118 [42, Theorems on contact experiments and quantized scores]. It is independent of the preceding dynamical hypotheses. We give the probability estimates explicitly so that no historical local-limit assertion is needed.

Fix $\lambda^0 \in (0, \infty)^m$, a matrix $B \in \mathbb{R}^{m \times p}$, a compact parameter set $H \subset \mathbb{R}^p$, and a rank- r matrix $C \in \mathbb{R}^{r \times m}$, with $r \geq 1$. Observe independent counts

$$N_j \sim \text{Pois}(N\lambda_j^0 + \sqrt{N}(Bh)_j), \quad h \in H, \quad S_N = C(N - N\lambda^0)/\sqrt{N}.$$

All means are positive for sufficiently large N . Put $\Lambda = \text{diag}(\lambda^0)$, $A_* = CB$, and $\Gamma = C\Lambda C^\top > 0$. In A2's notation, $B = B_* = \Lambda D^\top (D\Lambda D^\top)^{-1}$ and $C = F^\top L D$, so $A_* = F^\top L$. The exposure centre, mark law, compression and range frame are fixed before observing the counts. They are not estimated by an uncharged pilot.

For integers $J \geq 1$ and $R \geq 1$, partition $[-R, R]^r$ into J^r cubes of side $\eta = 2R/J$. Retain the cell containing S_N , or one overflow symbol. This is a deterministic physical statistic with a designated alphabet of $J^r + 1$ labels (some may have zero probability). For experiments indexed by $h \in H$, write Δ for the maximum of the two deficiencies in total variation, with parameter-independent randomization kernels.

Theorem 75.1 (Joint sample and label bound). *There are positive constants C_0, c_0, R_0 , depending on the fixed matrices, centre and compact parameter set, such that, for $R \geq R_0$ and $0 < \eta \leq 1$, the finite-label experiment $\mathcal{Q}_{N,J,R}$ satisfies*

$$\Delta(\mathcal{Q}_{N,J,R}, \{N(A_*h, \Gamma) : h \in H\}) \leq C_0 \left\{ N^{-1/5} + \frac{J}{R\sqrt{N}} + \frac{R}{J} + e^{-c_0 R^2} \right\}. \quad (75.1)$$

Thus, for $M \geq 2$, one may impose a prescribed register budget by choosing $J = \lfloor (M-1)^{1/r} \rfloor$, whenever $2R/J \leq 1$. In particular, for sufficiently large fixed K , the choices $R = K\sqrt{\log N}$ and $J = \lceil 2RN^{1/4} \rceil$ give deficiency $O(N^{-1/5})$ with

$$M = O(N^{r/4}(\log N)^{r/2}). \quad (75.2)$$

The rank r is the rank of the actual contact score. This is a sufficient finite-label budget, not a claim of its minimax necessity.

Proof. Adjoin independent U_j uniform on $(-1/2, 1/2)$ for comparison only. The law of $X_N^\circ = (N - N\lambda^0 + U)/\sqrt{N}$ is within $CN^{-1/5}$ in total variation of $N(Bh, \Lambda)$, uniformly on H . Here is a direct bound. In one coordinate let $\mu_N = N\lambda_j^0 + \sqrt{N}(Bh)_j$. On cells with $|k - \mu_N| \leq \sqrt{\mu_N}N^{1/10}$, Stirling's formula with its $O(k^{-1})$ remainder gives the ratio of the piecewise constant jittered Poisson density to the normal density with matching mean and variance as $1 + O(N^{-1/5})$. Indeed the logarithmic error is $O((1 + |z|^3)/\sqrt{N})$ for $|z| \leq CN^{1/10}$; variation of the normal within a cell contributes only a smaller term. The discarded mass is $O(N^{-1/5})$ by Chebyshev's inequality, uniformly in h . Replacing variance μ_N/N by λ_j^0 costs $O(N^{-1/2})$ by the normal density derivative bound. Product total variation is at most the sum of the coordinate errors. This proves the assertion. Applying C contracts total variation, so $S_N^\circ = CX_N^\circ$ is within $CN^{-1/5}$ of $N(A_*h, \Gamma)$.

Under this coupling, $\|S_N - S_N^\circ\|_\infty \leq d_N = C_1/\sqrt{N}$. The finite quantizers differ only when a coordinate crosses a grid boundary, including a boundary of the retained cube. For a normal marginal density f and any translated grid,

$$\eta \sum_{j \in \mathbb{Z}} \sup_{[j\eta, (j+1)\eta]} f \leq 1 + \eta \int |f'|.$$

The total probability within d_N of that grid is therefore at most Cd_N/η if $d_N \leq \eta/2$. For larger d_N the same bound follows from the trivial bound one. A union bound and the preceding total variation comparison show that the physical labels and the labels of the limiting Gaussian have distance at most $C(N^{-1/5} + d_N/\eta)$.

In the reverse direction, from a finite label generate a uniform point in its cell, and send the overflow label to a fixed point. On the retained cells the resulting histogram differs from the normal density in L^1 by at most $C\eta \int \|\nabla f\| \leq C'\eta$; this follows by integrating the mean-value estimate over pairs of points in each cube. Since H is compact and Γ fixed positive definite, the Gaussian mass outside $[-R, R]^r$ is at most $Ce^{-c_0 R^2}$ for $R \geq R_0$. Its replacement by the overflow point costs at most this mass in total variation. Quantization in the other direction is a fixed kernel. Composing these two comparisons proves (75.1). Choosing the stated R, J makes d_N/η and η of order $N^{-1/4}$ and the tail at most $N^{-1/5}$, proving (75.2). $\square$

For every bounded decision problem, these parameter-independent kernels also transfer the corresponding optimal risks with additive error bounded by (75.1) times the loss range. The physical statistic remains deterministic; the random kernels occur only in the comparison of experiments. Across contact strata the rank and conditioning of CAC^T must be reevaluated. The constants are not uniform through a collapsing positive eigenvalue. The conductor flags and nonreduced determinant powers of the current A2 edition specify algebraic failure schemes. They do not, without this specified observation protocol, supply the Gaussian covariance or a noise exponent. This application consequently uses the current statistical experiment, not an analogy with its scheme multiplicity.

APPENDIX 76. INTRODUCTION

A finite representation of a present observation need not be a finite representation of the predictions that observation will support. Expansion separates initially close states, and an interval without new data forces an observer to retain information about several future readouts. There are consequently two questions. One asks for the geometry of the weighted future orbit. The other asks whether a code for that geometry admits a closed update on a finite persistent state space. This paper gives matching answers on separated Gibbs repellers and extends the resulting prediction laws to acquisition kernels that retain hidden memory.

76.1. The principal results. Let T be the expanding map defined by finitely many separated $C^{1+\gamma}$ inverse branches, let μ be a Gibbs probability, and suppose an acquisition is followed by k unobserved iterates with probability proportional to q^k . Write φ for the normalized Gibbs potential and ψ for the logarithmic derivative of an inverse branch. The principal results identify the optimal M -state squared prediction risk R_M with a finite cylinder profile $G_M(q)$:

$$R_M \asymp e_M^2(Z_q) \asymp G_M(q), \quad Z_q(x) = (\sqrt{1-q} q^{k/2} T^k x)_{k \geq 0}.$$

Here G_M is obtained by successively splitting a cylinder of largest mass-weighted orbit diameter. It is a finite construction from cylinder probabilities and branch derivatives, not a restatement of an optimization over orbit codebooks.

Theorem 77.1 supplies this equivalence with the same state budget, up to constants. Its essential realization lemma is that the entire greedy tree is suffix closed: a full b -ary tree with N leaves

needs exactly $(bN - 1)/(b - 1)$ designated states. The proof uses stationarity of the Gibbs law to compare the energy of a word with the energy of any suffix. This is different from simply attaching an age register to each codeword, which in general costs the number of codewords times the truncation depth. A disjoint-tube estimate on the orbit image supplies the matching lower bound for arbitrary Hilbert-space centres and hence for all randomized, time-dependent causal observers.

Theorem 77.2 computes the exponent of this profile. If P denotes full-shift pressure, let

$$P(s_g\varphi + 2s_g\psi) = 0, \quad P(s_t\varphi) + s_t \log q = 0.$$

Then

$$\lim_{M \rightarrow \infty} \frac{-\log R_M}{\log M} = \frac{1 - \max\{s_g, s_t\}}{\max\{s_g, s_t\}}.$$

The two roots distinguish spatial resolution from survival through unobserved expansion. Their pressure balance determines the transition, including nonconstant derivative cocycles and nonuniform Markov branch laws. A variational formula expresses them through entropy, information and expansion rates. Positive-matrix formulas and finite pressure enclosures make the invariant explicit in specialized and general Hölder data, respectively.

Theorem 80.1 addresses dependence across acquisitions. The state transition at an acquisition may retain an arbitrary component of the preceding hidden state, subject to a common invariant law and a uniform minorization. A lower bound is obtained by conditional domination, not by declaring the acquisition to be a regeneration. A resetting observer has exactly the same expected loss as in the independent reference experiment. These two facts transfer the intrinsic pressure law and the noisy expanding-torus law to genuinely dependent experiments. In the latter case old observations can affect the current posterior; no identification of that posterior with a latest-observation posterior is needed.

76.2. Resource convention. The constrained resource is the cardinality of one persistent register. The program, fixed real constants, absolute clock and transient workspace are separate resources. Every data-dependent quantity used after a checkpoint must be encoded in the register. In particular, an acquisition indicator is available to the update kernel but not separately to the post-update decoder. Changing this interface can change exact zero-error cardinalities, as Example 74.5 demonstrates. All principal lower bounds allow stochastic kernels and arbitrary time dependence. Our upper constructions are stationary finite-state machines. We prove their order optimality, not the exact optimality of stationary policies for every member of the unrestricted class.

76.3. Relation to the literature. Classical quantization, causal coding and dynamical information theory provide different pieces of the problem. They should not be identified by a shared use of the word “memory.”

Linder and Yüksel [33] formulate zero-delay coding of vector Markov sources by controlled quantizers; their encoder and decoder can use transmitted histories or posterior states. Wood, Linder and Yüksel [34, Theorem 3] prove deterministic stationary policy optimality for finite irreducible aperiodic Markov sources. A stationary policy on a belief space is not, by itself, a machine with a fixed finite number of persistent values. Here acquisitions are intermittent and all retained observation-dependent information is charged to that finite number. This explains why a separate suffix-cardinality argument is necessary.

Greedy prefix-tree construction has classical antecedents in Tunstall’s variable-to-fixed coding [37]. The present priority is the mass-weighted *orbit distortion*, not merely word probability. Neither greedy tree construction nor the elementary vertex count is claimed as a new coding principle; the issue is to prove that this particular distortion tree is shift closed and matches the unrestricted causal risk.

Hilbert-space functional quantization is also classical. For example, Luschgy and Pagès [35] derive sharp Gaussian functional-quantization asymptotics from covariance eigenvalue decay. We use finite-centre Hilbert quantization as a converse, but must additionally compute a non-Gaussian dynamical orbit geometry and realize its codes causally. Pressure formulas for spatial quantization on cookie-cutter sets already occur in Roychowdhury [36]. The new calculation concerns the *weighted future orbit*: its partition exponent is the maximum of a spatial pressure and a survival pressure, and its adaptive codes have linear-sized suffix closure. The spatial pressure alone is not presented as a new principle.

Finite-rate control [26, 27, 28] constrains communication over time, whereas dynamical rate distortion [29] constrains information in orbit representations. Those resource models are different from the persistent-cardinality model here. The following comparison records the particular distinctions used in the proofs.

Reference or result	Constrained object	Structural conclusion used for comparison
Linder–Yüksel	Per-time transmitted symbol; vector Markov source	Existence and structure of causal quantizer policies on posterior state spaces.
Wood–Linder–Yüksel, Theorem 3	Zero-delay symbols; finite Markov source	Deterministic stationary policy optimality and finite-horizon comparison.
Luschgy–Pagès	Static finite-centre Hilbert code	Gaussian quantization asymptotics from covariance eigenvalues; no finite-state update constraint.
Roychowdhury	Spatial code for a cookie-cutter measure	Quantization dimension related to thermodynamic temperature; no future survival cost.
Theorems 77.1–77.2	All persistent observer states	Matched cylinder profile, arbitrary-coder converse, linear suffix realization and two-pressure exponent.
Theorem 80.1	Same finite register, dependent acquisitions	Conditional minorization lower bound and exact expected-loss transfer for resetting observers.

76.4. Organization and dependencies. Sections 77–79 state and prove the intrinsic profile and pressure theorem. Section 80 proves the dependent-acquisition comparison. The posterior-orbit and noisy expanding results needed in these arguments are proved in full in Sections 74–81. Those sections also retain the exact uniform critical crossover for the integer expanding experiment.

The subsequent quantitative development retains the calibrated collision-moment theorem, the exact Gaussian nuisance quotient, and the checkpoint, filtering and control results, followed by the full measurable foundations. These have independent assumptions; the pressure theorem does not rely on the collision theorem or on a historical spectral, particle, or operator limit. The separate principal-theorem edition is compiled from the same core files and includes every proof on the causal-memory dependency chain. The complete edition remains the preservation and cross-reference record for the full Foundations I manuscript. The foundational connection is precise: acquired-law quantization provides the converse, the suffix theorem provides the resource-aware causal realization, and thermodynamic pressure computes the resulting invariant. The assumptions of a separated finite-branch repeller and of an invariant minorized acquisition kernel are proved or stated where they are used; they are not replacements for the model-specific hypotheses of a Sinai, hard-particle, or unbounded- operator problem.

APPENDIX 77. AN INTRINSIC LAW ON GIBBS REPELLERS

Let $b \geq 2$, let $0 < \gamma \leq 1$, and let $h_1, \dots, h_b$ be increasing $C^{1+\gamma}$ maps of $[0, 1]$ into itself. We assume

$$0 < r_- \leq h'_i(x) \leq r_+ < 1, \quad \text{dist}(h_i([0, 1]), h_j([0, 1])) \geq g > 0 \quad (i \neq j). \quad (77.1)$$

Write $\Sigma = \{1, \dots, b\}^{\mathbb{N}_0}$, let S denote the left shift, and put

$$\pi(\omega) = \lim_{n \rightarrow \infty} h_{\omega_0} \cdots h_{\omega_{n-1}}(0), \quad K = \pi(\Sigma), \quad T = \pi S \pi^{-1}.$$

Strong separation makes π a homeomorphism onto K . In particular, T is a well-defined continuous expanding map on the repeller. Our readout is the ambient coordinate $f(x) = x$.

For a word $w = w_0 \cdots w_{n-1}$, set

$$h_w = h_{w_0} \circ \cdots \circ h_{w_{n-1}}, \quad r_w = \sup_{[0,1]} h'_w, \quad K_w = h_w(K), \quad r_\emptyset = 1.$$

The notation $w_{k\cdot}$ means the suffix $w_k \cdots w_{n-1}$. Choose a Hölder potential φ on Σ , normalized by $P(\varphi) = 0$, and let μ be its shift-invariant Gibbs probability. We use μ also for its image on K , and write $\mu[w] = \mu(K_w)$. Here P is topological pressure on the full shift. Thus, for a constant $C_G \geq 1$,

$$C_G^{-1} e^{S_n \varphi(\omega)} \leq \mu[w] \leq C_G e^{S_n \varphi(\omega)} \quad (\omega \in [w]), \quad S_n u = \sum_{j=0}^{n-1} u \circ S^j. \quad (77.2)$$

The derivative potential is

$$\psi(\omega) = \log h'_{\omega_0}(\pi(S\omega)). \quad (77.3)$$

It is Hölder and is bounded between $\log r_-$ and $\log r_+$.

Fix $0 < q < 1$. At time zero and at each subsequent acquisition a fresh μ -distributed state is observed exactly. At every other time the hidden state is advanced by T . The acquisition indicators after time zero are independent Bernoulli variables with probability $1 - q$ and are independent of all fresh states. After the update, the decoder receives only the persistent state, its absolute time, and independent coding randomness. In particular, it does not receive the current acquisition indicator as a separate uncharged input. An M -state observer has one persistent register with at most M values. Its Borel update and decoding kernels may be randomized and time dependent; no other data-dependent quantity survives a checkpoint. Define

$$R_M = \inf_{\mathcal{A}} \limsup_{H \rightarrow \infty} \frac{1}{H} \sum_{t=0}^{H-1} \mathbb{E} |X_t - \hat{X}_t^{\mathcal{A}}|^2. \quad (77.4)$$

The infimum is over this entire observer class. Its deterministic stationary subclass will suffice for the upper bounds. Program constants and transient computation are not charged to M .

The weighted orbit and its squared quantization error are

$$Z_q(x) = (\sqrt{1-q} q^{k/2} T^k x)_{k \geq 0}, \quad e_M^2(Z_q) = \inf_{c_1, \dots, c_M \in \ell^2} \int \min_j \|Z_q(x) - c_j\|^2 d\mu(x). \quad (77.5)$$

The centres in this definition need not lie on the orbit image.

77.1. A finite cylinder algorithm. For every finite word define

$$V_q(w) = (1-q) \sum_{k=0}^{|w|-1} q^k r_{w_{k\cdot}}^2 + q^{|w|}, \quad a_q(w) = \mu[w] V_q(w). \quad (77.6)$$

Both quantities at the empty word are one. They require only finite cylinder masses and derivatives of finite compositions. Set

$$\rho_q = q + (1-q)r_+^2 < 1. \quad (77.7)$$

View all finite words as the vertices of a full b -ary tree. Starting from the empty word, repeatedly replace a leaf of maximal a_q by its b children. Resolve ties in lexicographic order. For $N \geq 1$, let $\mathcal{P}_N$ be the last leaf partition in this sequence having at most N leaves, and put

$$G_N(q) = \sum_{w \in \mathcal{P}_N} a_q(w). \quad (77.8)$$

This convention also defines $G_N = 1$ for $N < b$. No optimization over Hilbert-space centres or over observer policies enters the algorithm.

Theorem 77.1 (Intrinsic profile and finite-state realization). *Under (77.1)–(77.2), for every $0 < q < 1$ there are constants $0 < c \leq C < \infty$ such that, for every integer $M \geq 1$,*

$$cG_M(q) \leq e_M^2(Z_q) \leq R_M \leq CG_M(q). \quad (77.9)$$

The lower bound holds also with the liminf time-average loss of any individual randomized, time-dependent observer. The upper bound is attained to the displayed factor by a deterministic stationary observer which, at each acquisition, selects a leaf of a greedy partition and thereafter deletes the first symbol of its stored word. For every N , the full greedy tree is suffix closed and the constructed observer has exactly $(b|\mathcal{P}_N| - 1)/(b - 1)$ designated states and average loss at most $G_N(q)$. In particular, it uses at most $(bN - 1)/(b - 1) \leq 2N - 1$ states.

The same order $G_M(q)$ holds when the objective in (77.4) is replaced by $\inf_{\mathcal{A}} \sup_{t \geq 0} \mathbb{E}|X_t - \hat{X}_t^{\mathcal{A}}|^2$. Constants may depend on the branch, distortion, Gibbs and separation data and on q . They can be chosen uniformly on compact q -intervals in $(0, 1)$ when those data are uniformly controlled.

There are two separate assertions here. The cylinder algorithm computes the relevant distortion profile; its suffix closure computes an implementable observer. The number of suffix states is proportional to the number of leaves, not to the number of leaves times their maximum depth. This is the point at which a geometric code becomes a causal code without a growing multiplicative overhead.

77.2. The pressure law. Let s_g and $s_t(q)$ be the unique numbers in $(0, 1)$ satisfying

$$P(s_g\varphi + 2s_g\psi) = 0, \quad P(s_t(q)\varphi) + s_t(q) \log q = 0, \quad s_*(q) = \max\{s_g, s_t(q)\}. \quad (77.10)$$

Existence and uniqueness are part of the next theorem.

Theorem 77.2 (Pressure determines the causal exponent). *In the setting of Theorem 77.1,*

$$\lim_{M \rightarrow \infty} \frac{-\log R_M}{\log M} = \lim_{M \rightarrow \infty} \frac{-\log e_M^2(Z_q)}{\log M} = \lim_{M \rightarrow \infty} \frac{-\log G_M(q)}{\log M} = \frac{1 - s_*(q)}{s_*(q)}. \quad (77.11)$$

More precisely, for $0 < s < 1$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \sum_{|w|=n} a_q(w)^s = \max\{P(s\varphi + 2s\psi), P(s\varphi) + s \log q\}. \quad (77.12)$$

Consequently the transition occurs at

$$q_c = \exp\{-P(s_g\varphi)/s_g\} \in (0, 1). \quad (77.13)$$

For $q \leq q_c$ the exponent is $(1 - s_g)/s_g$; for $q > q_c$ it is $(1 - s_t(q))/s_t(q)$. At the transition, the finite-budget profile is still given, up to constants, by (77.8) and (77.9).

Formula (77.12) is a maximum of two pressures, not pressure evaluated at the pointwise maximum of two potentials. The derivative cocycle enters the first term; survival across unobserved iterates enters the second. The theorem computes the quantization exponent from these data rather than postulating its existence. It does not identify a universal logarithmic correction or an optimal leading constant at q_c for all Gibbs repellers. The finite cylinder profile retains such corrections without discarding them in the exponent limit.

APPENDIX 78. CYLINDER GEOMETRY AND CAUSAL REALIZATION

Throughout this section q and the data in (77.1)–(77.2) are fixed. We suppress the subscript q in V and a . A comparison constant never depends on a word or a state budget.

Lemma 78.1 (Distortion and orbit separation). *There is $D \geq 1$ such that*

$$\sup h'_w \leq D \inf h'_w, \quad D^{-1}r_u r_v \leq r_{uv} \leq r_u r_v. \quad (78.1)$$

If $x, y \in K$ have longest common symbolic prefix w , then

$$c_1 V(w) \leq \|Z_q(x) - Z_q(y)\|^2 \leq V(w), \quad c_1 = (1 - q)g^2 D^{-2}. \quad (78.2)$$

For arbitrary words u, v ,

$$V(uv) \leq r_v^2(V(u) - q^{|u|}) + q^{|u|}V(v), \quad (78.3)$$

$$V(uv) \leq V(u)V(v), \quad a(uv) \leq a(v). \quad (78.4)$$

There exists $0 < a_0 < \rho_q$ such that each one-symbol child wi , $1 \leq i \leq b$, satisfies

$$a_0 a(w) \leq a(wi) \leq \rho_q a(w), \quad \sum_{i=1}^b a(wi) \leq \rho_q a(w). \quad (78.5)$$

Proof. Let H bound the γ -Hölder seminorms of $\log h'_i$. The chain rule, read from the innermost factor, gives

$$|\log h'_w(x) - \log h'_w(y)| \leq H \sum_{j \geq 0} r_+^{j\gamma} \leq H/(1 - r_+^\gamma).$$

This proves the first assertion with $D = \exp\{H/(1 - r_+^\gamma)\}$. The upper multiplicative bound follows from the chain rule; for the lower bound evaluate the inner derivative at a maximizer and bound the outer derivative below by r_u/D .

Put $n = |w|$. For $k < n$, the points $T^k x, T^k y$ belong to the same suffix cylinder. Their distance is at most r_{w_k} . Immediately after that suffix they belong to two different first-level cylinders, so the mean value theorem also bounds this distance below by gr_{w_k}/D . At $k = n$ their distance is at least g . The remaining squared distances are at most one. Multiplying by $(1 - q)q^k$ and summing proves (78.2), including the term q^n by means of its first summand $(1 - q)q^n g^2$. The same upper bound $V(w)$ holds for any two points of K_w .

Split the defining sum for $V(uv)$ at $|u|$ and apply (78.1) to its first part to obtain (78.3). Each suffix derivative r_{v_k} is at least r_v , since the omitted outer maps are contractions. Consequently $V(v) \geq r_v^2$. Equation (78.3) therefore gives $V(uv) \leq V(u)V(v)$. Since $V(u) \leq 1$ and shift invariance gives

$$\mu[uv] \leq \mu(S^{-|u|}[v]) = \mu[v],$$

we obtain the exact suffix domination $a(uv) \leq a(v)$. No independence of the symbols is used.

For a one-symbol child, $V(i) = q + (1 - q)r_i^2 \leq \rho_q$; submultiplicativity gives $V(wi) \leq \rho_q V(w)$. Conversely, split at $|w|$ and apply the lower distortion estimate to the old summands. Since $V(i) \geq r_i^2 \geq r_-^2$,

$$V(wi) \geq D^{-2}r_-^2(V(w) - q^{|w|}) + r_-^2 q^{|w|} \geq D^{-2}r_-^2 V(w).$$

The Gibbs bounds imply a positive lower bound β on $\mu[wi]/\mu[w]$, uniformly in w, i : divide the lower bound for $[wi]$ by the upper bound for $[w]$ at the same point, and use $\varphi \geq \inf \varphi$. Take $a_0 = \beta D^{-2}r_-^2/2$, decreased if necessary so that $a_0 < \rho_q$. Finally the children partition $[w]$; summing $\mu[wi]V(wi) \leq \rho_q \mu[wi]V(w)$ proves the final assertion. $\square$

Lemma 78.2 (Balanced greedy partitions). *Let $L_N = |\mathcal{P}_N|$. Then $N/b \leq L_N \leq N$. For $N \geq b$, let θ_N be the energy of the last leaf split in constructing $\mathcal{P}_N$. Every $w \in \mathcal{P}_N$ satisfies*

$$a_0 \theta_N \leq a(w) \leq \theta_N, \quad a_0 L_N \theta_N \leq G_N \leq L_N \theta_N. \quad (78.6)$$

For every fixed $A \geq 1$ there is $c_A > 0$ such that

$$c_A G_N \leq G_{\lceil AN \rceil} \leq G_N \quad (N \geq 1). \quad (78.7)$$

Moreover, for each $c > 0$, the number of all finite words v with $a(v) \geq c\theta_N$ is at most $C_c N$.

Proof. The leaf counts are $1 + j(b-1)$. A maximal energy selected for splitting never increases, since each new child has at most ρ_q times its parent's energy. A leaf still present at the last split was born from a parent of energy at least θ_N , and every current leaf has energy at most θ_N . This proves (78.6); the root-only cases are kept separate. The sum G_N is nonincreasing by (78.5).

For completeness, the reverse comparison in (78.7) does not require a preassigned power law. Choose an integer j with $b^{j+1} > b(A+1)$. If $\theta_{\lceil AN \rceil} < a_0^{j+1} \theta_N$, every descendant through depth j of each old leaf would have been split: all these nodes have energy at least $a_0^{j+1} \theta_N$. The new partition would then have at least $b^{j+1} L_N$ leaves, which exceeds $\lceil AN \rceil$. Thus $\theta_{\lceil AN \rceil} \geq a_0^{j+1} \theta_N$. Combining this with (78.6) proves the reverse comparison for $N \geq b$. Finitely many smaller budgets can be absorbed into c_A ; their energies are positive.

The finite full tree ending at $\mathcal{P}_N$ has at most $bL_N/(b-1)$ vertices. Every other vertex descends from a leaf. At depth j below a leaf its energy is at most $\rho_q^j \theta_N$. Only a bounded number of additional generations can therefore have energy at least $c\theta_N$; each has at most b^j descendants per leaf. This bounds the total number of such words by $C_c N$. $\square$

Lemma 78.3 (Quantization of the orbit image). *There are $0 < c_2 \leq C_2 < \infty$ such that*

$$c_2 G_M \leq e_M^2(Z_q) \leq C_2 G_M \quad (M \geq 1).$$

The lower bound allows arbitrary off-image centres in ℓ^2 .

Proof. Choosing one orbit point from each cylinder of $\mathcal{P}_M$ uses at most M centres and has squared error at most G_M , by the cylinder diameter estimate.

For the reverse bound take $\mathcal{P}_{4bM}$. It has at least $4M$ leaves. For each leaf w let

$$U_w = \{z \in \ell^2 : \text{dist}(z, Z_q(K_w)) < \delta \sqrt{V(w)}\}, \quad 0 < \delta < \sqrt{c_1}/(2\sqrt{2}).$$

These open sets are pairwise disjoint. Indeed, two different leaves have a common prefix u strictly shorter than either leaf. Any points in their two orbit cylinders have distance at least $\sqrt{c_1} V(u)$, whereas $V(w), V(v) \leq 2V(u)$. The sum of the two tube radii is smaller than this separation. Thus M centres meet at most M of the tubes. Every unoccupied cylinder contributes at least $\delta^2 a(w)$ to the distortion. By (78.6), the sum over the unoccupied leaves is at least a fixed multiple of G_{4bM} . Lemma 78.2 compares this with G_M . The argument uses distances to entire cylinders, so it does not require any centre to belong to the orbit image. The finitely many root-only cases can again be absorbed: the measure is nonatomic and the orbit's first coordinate separates distinct states, so each fixed finite quantization error is positive. $\square$

Lemma 78.4 (Exact suffix closure of the greedy tree). *Let $\mathcal{T}_N$ be the set of all vertices, including the root, in the greedy tree with leaf set $\mathcal{P}_N$, and write $L_N = |\mathcal{P}_N|$. Then $\mathcal{T}_N$ is closed under deletion of the first letter and*

$$|\mathcal{T}_N| = \frac{bL_N - 1}{b - 1} \leq \frac{bN - 1}{b - 1}. \quad (78.8)$$

In particular, every suffix of every leaf is already a vertex of this same tree. No additional age states are required.

Proof. Full support and stationarity give, for every nonempty u ,

$$\mu[uv] < \sum_{|z|=|u|} \mu[zv] = \mu[v].$$

Together with $V(uv) \leq V(v)$, this yields the strict inequality $a(uv) < a(v)$. Suppose a vertex w is selected for splitting. Every proper suffix v of w was already an internal vertex at that time. Otherwise some prefix z of v would be a leaf of the current tree, and the decreasing child energies would give $a(z) \geq a(v) > a(w)$, contradicting the maximality of w . Thus the internal vertices are suffix closed.

If w is a leaf, the parent of each nonempty proper suffix of w is a suffix of the parent of w . This parent is internal by the preceding argument (with the empty word interpreted as the root). Its children are vertices of the tree, so the suffix itself is also a vertex. The root and root-only tree cause no exception. This proves closure for all vertices. Finally, a finite full b -ary tree with I internal vertices has bI edges and $I + L_N$ vertices. Hence $bI = I + L_N - 1$, giving (78.8). $\square$

Proof of Theorem 77.1. The posterior-orbit converse, Theorem 74.2, applied to exact observations and geometric cycles, gives $e_M^2(Z_q) \leq R_M$, with a liminf bound for each observer. Its randomization representation is Lemma 74.1. Equivalently, at each acquisition one freezes independent coding randomness and the past; the M possible post-update states supply at most M no-acquisition decoder strings. Quantization bounds their weighted squared loss below, and the disjoint last-acquisition events give the time-average bound. This does not restrict the competing observers to symbolic codes.

For achievability, use $\mathcal{T}_N$ as the state set. At an acquisition x , write its unique leaf word. On a nonacquisition update delete its first letter; the empty word is absorbing until the next acquisition. Decode a nonempty word v by $\pi(v111\cdots)$ and the empty word by $\pi(111\cdots)$. At age $k < |w|$ after acquisition in K_w , the squared error is at most $r_{w_k}^2$; after exhaustion it is at most one. The average squared loss is at most G_N . Every piece of information used by this rule is in its suffix state. Neither reset age nor a precision index is supplied from outside the register.

By Lemma 78.4, this machine uses exactly $(bL_N - 1)/(b - 1)$ designated states and at most C_0N states, with $C_0 = b/(b - 1)$. Take $N = \lfloor M/C_0 \rfloor$ for sufficiently large M . The fixed-dilation comparison gives $G_N \leq CG_M$. For the remaining finitely many budgets use the one-state constant observer and enlarge C . Lemma 78.3 proves the rest of (77.9).

Let d_k be the expected age- k loss of the constructed machine under one acquisition. At time t its expected loss is

$$\sum_{k < t} (1 - q)q^k d_k + q^t d_t \leq \frac{1}{1 - q} \sum_{k \geq 0} (1 - q)q^k d_k.$$

The right side is at most $G_N/(1 - q)$. The supremum objective is bounded below by the average objective, proving its stated order. When q stays in a compact subinterval of $(0, 1)$ and the branch and Gibbs data are controlled, ρ_q stays below one and the separation and child-energy constants have common bounds. All constants in the proof can then be chosen uniformly. $\square$

APPENDIX 79. PRESSURE, DISTORTION AND SURVIVAL

We use the Gibbs and variational principles on a finite full shift in their usual Hölder form [32, Chapters 1–2]. For a Hölder function u , these principles give a shift-invariant probability ν_u and $C_u \geq 1$ with

$$C_u^{-1} e^{S_n u(\omega) - nP(u)} \leq \nu_u[w] \leq C_u e^{S_n u(\omega) - nP(u)}, \quad P(u) = \sup_{\nu \in \mathcal{M}_S} \left(h_\nu(S) + \int u d\nu \right). \quad (79.1)$$

Here $\mathcal{M}_S$ is the set of shift-invariant probabilities. Only these finite-alphabet statements, not a countable-state spectral or suspension theorem, are used below.

Lemma 79.1 (The mixed partition sum). *Fix $0 < s < 1$, and set*

$$A(s) = P(s\varphi + 2s\psi), \quad B_q(s) = P(s\varphi) + s \log q.$$

There are constants $c_s, C'_s > 0$, independent of n , such that

$$c_s \max\{e^{nA(s)}, e^{nB_q(s)}\} \leq \sum_{|w|=n} a_q(w)^s \leq C'_s \sum_{k=0}^n e^{kB_q(s) + (n-k)A(s)}. \quad (79.2)$$

Proof. For a word w of length n and $\omega \in [w]$, the chain rule and bounded distortion give

$$r_{w_k} \asymp e^{S_{n-k}\psi(S^k\omega)}, \quad \mu[w] \asymp e^{S_n\varphi(\omega)},$$

with constants independent of n, k, w . The term $k = 0$ and the terminal term in (77.6), together with (79.1), give the two lower bounds in (79.2).

For the upper bound use $(\sum x_j)^s \leq \sum x_j^s$ for nonnegative x_j . At a fixed splitting time $k < n$, the resulting sum, up to a constant and the factor $(1 - q)^s$, is

$$\sum_{|w|=n} \exp \left\{ s S_k \varphi(\omega) + k s \log q + S_{n-k}(s\varphi + 2s\psi)(S^k\omega) \right\}.$$

The variations of $S_k u$ inside any k -cylinder are uniformly bounded for a Hölder u . Since every prefix can be concatenated with every suffix on the full shift, this sum is comparable to the product of the length- k partition sum for $s\varphi + s \log q$ and the length- $(n - k)$ partition sum for $s\varphi + 2s\psi$. Each is comparable to its pressure exponential by (79.1). The terminal term is the same calculation with $k = n$. Summing over k proves the assertion. $\square$

Proof of Theorem 77.2. Both pressures at $s = 0$ equal $\log b$. At $s = 1$ they are negative: $A(1) \leq 2 \log r_+ < 0$ and $B_q(1) = \log q < 0$. There is a uniform positive lower bound on every one-symbol conditional cylinder mass. Since there are at least two symbols, there is consequently $\eta < 1$ such that $\mu[w] \leq \eta^{|w|}$. Equation (77.2) implies $S_n \varphi \leq n \log \eta + \log C_G$. Integration against any invariant probability gives $\int \varphi d\nu \leq \log \eta < 0$. The variational principle now shows that both A and B_q are strictly decreasing, with their slopes bounded above by a negative constant. Their unique zeros therefore lie in $(0, 1)$. Equation (79.2) proves (77.12): the sum on its right has at most $n + 1$ terms, each at most the exponential of $n \max\{A(s), B_q(s)\}$.

We prove the exponent without assuming any quantization dimension. Let $s > s_*(q)$, $s < 1$. Both pressures are negative, so (79.2) is summable in n . Hence

$$H_s := \sum_{w \text{ finite}} a_q(w)^s < \infty.$$

Balance in the greedy tree yields $L_N(a_0\theta_N)^s \leq H_s$, and thus

$$G_N \leq C_s'' N^{-(1-s)/s}. \quad (79.3)$$

For the reverse estimate choose a Gibbs measure ν_* for a zero-pressure potential as follows. When $s_* = s_g$, use $s_g\varphi + 2s_g\psi$; otherwise use $s_t(q)\varphi + s_t(q)\log q$. The spatial summand or the terminal summand of V_q , respectively, shows that

$$\nu_*[w] \leq C_* a_q(w)^{s_*} \quad \text{for every word } w.$$

Since $\mathcal{P}_N$ is a cylinder partition, $1 \leq C_* L_N \theta_N^{s_*}$. Therefore

$$G_N \geq c_* N^{-(1-s_*)/s_*}. \quad (79.4)$$

Let s decrease to s_* in (79.3) and use (79.4). This proves the exponent limit for G_N , and Theorem 77.1 proves the other limits. Finally $P(s_g\varphi) > 0$ because $s_g < 1$ and $P(\varphi) = 0$. Thus the number q_c in (77.13) lies in $(0, 1)$. Strict monotonicity shows that $s_t(q) \leq s_g$ precisely when $B_q(s_g) \leq 0$, which is equivalent to $q \leq q_c$. $\square$

Corollary 79.2 (Entropy–expansion ratios). *Put $I_\nu = -\int \varphi d\nu$ and $\chi_\nu = -\int \psi d\nu$. Then*

$$s_g = \sup_{\nu \in \mathcal{M}_S} \frac{h_\nu(S)}{I_\nu + 2\chi_\nu}, \quad s_t(q) = \sup_{\nu \in \mathcal{M}_S} \frac{h_\nu(S)}{I_\nu + \log(1/q)}. \quad (79.5)$$

All denominators are strictly positive. The quantities χ_ν are the mean expansion rates of T in the coding coordinates.

Proof. The preceding proof gives $I_\nu \geq -\log \eta > 0$, and $\chi_\nu \geq -\log r_+ > 0$. By the variational principle, $A(s) = \sup_\nu \{h_\nu - s(I_\nu + 2\chi_\nu)\}$. Its zero is the supremum of the displayed ratios. For example, at that supremum all the affine expressions are nonpositive, while ratios approaching the supremum make their values approach zero, since the denominators are uniformly bounded above as well as below. The second formula follows in the same way from B_q . The chain rule identifies $-S_n\psi$ with the logarithmic expansion of T^n on its cylinder. $\square$

Proposition 79.3 (Finite pressure enclosures). *For a Hölder potential u , define*

$$Z_n^-(u) = \sum_{|w|=n} \exp\{\inf_{[w]} S_n u\}, \quad Z_n^+(u) = \sum_{|w|=n} \exp\{\sup_{[w]} S_n u\}.$$

If $\text{var}_j(u) = \sup\{|u(x) - u(y)| : x_{0:j} = y_{0:j}\}$, then

$$\frac{1}{n} \log Z_n^-(u) \leq P(u) \leq \frac{1}{n} \log Z_n^+(u), \quad \frac{1}{n} \log \frac{Z_n^+(u)}{Z_n^-(u)} \leq \frac{1}{n} \sum_{j \geq 1} \text{var}_j(u). \quad (79.6)$$

Here $x_{0:j} = y_{0:j}$ means equality of the first j symbols. Thus certified cylinder bounds on the potentials give convergent finite enclosures for the two pressure roots and for q_c .

Proof. Concatenation gives $Z_{n+k}^- \geq Z_n^- Z_k^-$ and $Z_{n+k}^+ \leq Z_n^+ Z_k^+$. Their normalized logarithms have the same limit $P(u)$, since the oscillation of $S_n u$ in a cylinder is at most $\sum_{j=1}^n \text{var}_j(u)$. Superadditivity and subadditivity prove the enclosure, and the same oscillation bound gives its width. Strict monotonicity of the pressures then converts pressure enclosures into root enclosures. $\square$

These are finite enclosures in the intrinsic potentials. As usual with arbitrary real-valued smooth input data, an effective numerical algorithm requires certified access to those data. Neither the resource model nor this proposition asserts a Turing complexity bound for an unspecified real function.

79.1. Markov branch weights and unequal multipliers.

Corollary 79.4 (A Perron–Frobenius formula). *Suppose $h_i(x) = c_i + r_i x$ and the symbols have a stationary Markov law with a strictly positive stochastic matrix $P = (p_{ij})$. Then s_g and $s_t(q)$ are the unique roots*

$$\rho((p_{ij}^{s_g} r_i^{2s_g})_{ij}) = 1, \quad q^{s_t(q)} \rho((p_{ij}^{s_t(q)})_{ij}) = 1. \quad (79.7)$$

In particular,

$$q_c = \rho((p_{ij}^{s_g})_{ij})^{-1/s_g}.$$

For independent symbols with probabilities $p_i > 0$, these equations reduce to $\sum_i (p_i r_i^2)^{s_g} = 1$ and $q^{s_t} \sum_i p_i^{s_t} = 1$.

Proof. Take $\varphi(\omega) = \log p_{\omega_0 \omega_1}$ and $\psi(\omega) = \log r_{\omega_0}$. The Markov cylinder probability has a stationary initial factor and one fewer transition than $e^{S_n \varphi}$, so the Gibbs ratio is bounded above and below. The pressure of an edge potential $u(i, j)$ is $\log \rho((e^{u(i, j)})_{ij})$, by summing products along words and applying Perron–Frobenius to the positive matrix. This yields (79.7). When all rows equal $(p_i)_i$, the matrix has rank one and its nonzero eigenvalue is the displayed sum. $\square$

The powers of p_{ij} in (79.7) are *entrywise powers*. Replacing them by a matrix power of P would erase precisely the information pressure at issue.

Example 79.5 (A nonlinear variable-expansion family). On $[0, 1]$ take

$$h_1(x) = \frac{x}{5} + \varepsilon x(1 - x), \quad h_2(x) = \frac{3}{4} + \frac{x}{6} + \varepsilon x(1 - x), \quad 0 \leq \varepsilon \leq \frac{1}{100}.$$

Their image intervals are $[0, 1/5]$ and $[3/4, 11/12]$. The derivatives lie, respectively, in $[19/100, 21/100]$ and $[1/6 - 1/100, 1/6 + 1/100]$. Thus all hypotheses, including a common positive separation and distortion bound, hold uniformly. Equip the symbols, for example, with the stationary Markov law

$$P = \begin{pmatrix} 4/5 & 1/5 \\ 7/20 & 13/20 \end{pmatrix}.$$

At $\varepsilon = 0$, Corollary 79.4 gives two explicit positive-matrix equations. At $\varepsilon > 0$, Theorem 77.2 and Proposition 79.3 apply to the genuinely nonconstant derivative potential.

This family is not obtained by a smooth change of coordinate from a map with a common multiplier at every fixed point. The two inverse branches have fixed points with derivatives in the two disjoint intervals just displayed. A C^1 conjugacy with nonzero derivative at those points preserves the fixed-point multipliers, which are their reciprocals. Their inequality is therefore an obstruction to such a conjugacy. There remains a topological full-shift coding, as there must for this repeller class; the assertion concerns the metric expansion, not the absence of symbolic coordinates.

APPENDIX 80. ACQUISITIONS WITH PERSISTENT HIDDEN STATE

The preceding realization can be used when acquisition does not regenerate the physical state. The appropriate comparison concerns raw prediction risk: the full-history posterior need not equal the posterior based on the latest observation alone.

Let E and $\mathbf{Y}$ be standard Borel spaces, let $T : E \rightarrow E$ be Borel, let $f : E \rightarrow \mathbb{R}^r$ be Borel with $\|f\| \leq F$, and let μ be a probability with $T_{\#}\mu = \mu$. Acquisitions occur at $\tau_0 = 0$ and $\tau_{j+1} = \tau_j + L_j$, where the L_j are independent positive integer random variables of common finite mean ℓ . The acquisition clock is independent of the hidden transition and observation randomizations. Between acquisitions $X_t = TX_{t-1}$. At an acquisition $t > 0$ the transition and observation are specified, in this order, by

$$X_t \mid X_{t-1} = x \sim Q(x, du), \quad Y_t \mid X_t = u \sim K(u, dy). \quad (80.1)$$

At time zero $X_0 \sim \mu$ and $Y_0 \sim K(X_0, \cdot)$. We assume

$$\mu Q = \mu, \quad Q(x, A) \geq \epsilon \mu(A) \quad (x \in E, A \text{ Borel}), \quad 0 < \epsilon \leq 1. \quad (80.2)$$

The whole state X_{t-1} can influence X_t through the remaining part of Q . No regeneration indicator is supplied to the observer. The observer sees the acquisition indicators and the Y_t only when acquisitions occur; the persistent-state convention is that of Section 77.

Use a single pair $(U, Y) \sim \mu(du)K(u, dy)$ to define

$$\begin{aligned} s_k &= \mathbb{P}(L_0 > k), \quad w_k = s_k/\ell, \quad W(n) = \sum_{k \geq n} w_k, \\ m_k(y) &= \mathbb{E}[f(T^k U) \mid Y = y], \quad Z(y) = (\sqrt{w_k} m_k(y))_{k \geq 0}, \\ B_0 &= \sum_{k \geq 0} w_k \mathbb{E} \|f(T^k U) - m_k(Y)\|^2. \end{aligned} \quad (80.3)$$

Let R_M^{dep} denote the infimum of limsup average *raw* squared loss in this dependent experiment.

Theorem 80.1 (Minorized acquisition comparison). *Under (80.1)–(80.2),*

$$R_M^{\text{dep}} \geq \epsilon(B_0 + e_M^2(Z)) \quad (M \geq 1). \quad (80.4)$$

The same lower bound holds for the liminf average loss of each randomized, time-dependent M -state observer. For every $N, n \geq 1$ with $Nn + 1 \leq M$,

$$R_M^{\text{dep}} \leq B_0 + e_N^2(Z^{<n}) + F^2W(n). \quad (80.5)$$

A one-state zero decoder always has risk at most F^2 . More generally, every observer that overwrites its entire state using only the latest observation at each acquisition has exactly the same sequence of expected one-time losses as in the independent-acquisition experiment with law μK and the same clock, provided it is initialized in the same way.

Proof. Condition on the entire acquisition-indicator sequence. At time zero the hidden marginal is μ . Each intervening transition is either T or Q , both of which preserve μ . Induction therefore shows that the hidden state at each acquisition has marginal μ , even after this conditioning. Its observation pair has law μK . Future survival for k steps has probability s_k , independent of that pair. Thus an observer which restarts from the current observation has the same age-dependent expected loss as in the independent experiment, and hence the same expected one-time loss after averaging over the common clock. This proves the last assertion. Applying the $Nn + 1$ -state construction of Theorem 74.2 proves (80.5); the constant decoder proves the remaining upper bound.

For the converse fix n . At an acquisition condition on the entire hidden and observed past immediately before the transition, and use the independent uniform representation of the observer's kernels from Lemma 74.1. Freeze its coding uniforms, including those used on the hypothetical next n nonacquisition steps. The M possible post-update states determine at most M length- n decoder strings. Conditional on this past, the new observation pair has law $Q(x, du)K(u, dy)$, which dominates $\epsilon\mu(du)K(u, dy)$. Squared loss is nonnegative, so its expected contribution up to the next acquisition, truncated at n , is at least

$$\epsilon \inf_{i(\cdot), c_{i,k}} \sum_{k < n} s_k \mathbb{E}_{\mu K} \|f(T^k U) - c_{i(Y),k}\|^2 = \epsilon \ell(B_{0,<n} + e_M^2(Z^{<n})).$$

For the equality, condition on Y and use orthogonality around $m_k(Y)$; the remaining infimum is precisely finite-dimensional quantization. Enlarging the possible codebooks to all vectors only weakens this lower bound. The encoder is permitted to depend on the conditioned past, but the resulting bound does not.

For a horizon H , sum these truncated contributions only over acquisitions at times at most $H - n$. Each contribution stops at the next acquisition, so distinct terms charge disjoint time indices, path by path. The expected number of these acquisitions, divided by H , converges to $1/\ell$ by the elementary renewal theorem. Taking liminf in H proves the lower bound with $B_{0,<n}$ and $Z^{<n}$. Finally let $n \rightarrow \infty$. The noise terms converge by boundedness. The quantization terms increase to $e_M^2(Z)$: projection gives one inequality, and extending a truncated codebook by zero gives

$$e_M^2(Z) \leq e_M^2(Z^{<n}) + \mathbb{E}\|Z - Z^{<n}\|^2 \leq e_M^2(Z^{<n}) + F^2W(n).$$

This proves (80.4). □

The reference number B_0 in this theorem is *not* asserted to be the Bayes risk conditional on the entire dependent history. Older observations can improve that posterior. The theorem uses minorization to obtain a lower bound and invariant marginals to obtain an upper bound; it does not replace a dependent conditional law by an independent one.

Corollary 80.2 (The pressure law survives partial refreshment). *Take the repeller, Gibbs law and geometric acquisition clock of Section 77, but use any kernel Q satisfying (80.2) at acquisitions. Observe the current state exactly at those times. Then, for each fixed $\epsilon > 0$,*

$$R_M^{\text{dep}} \asymp G_M(q), \quad \lim_{M \rightarrow \infty} \frac{-\log R_M^{\text{dep}}}{\log M} = \frac{1 - s_*(q)}{s_*(q)}.$$

The same order holds for supremum one-time loss. An explicit example is

$$Q(x, du) = (1 - \epsilon)\delta_{Tx}(du) + \epsilon\mu(du). \quad (80.6)$$

For $\epsilon < 1$ the physical state is not regenerated at every acquisition. The phase boundary is governed by the acquisition survival probability q , not by the unobserved physical-refresh survival probability $1 - \epsilon(1 - q)$.

Proof. Exact observations give $B_0 = 0$ and the orbit Z_q . Theorem 80.1 supplies the lower bound $\epsilon e_M^2(Z_q)$. The resetting suffix observer in Theorem 77.1 supplies the upper bound $CG_M(q)$ with the same expected losses in the dependent experiment. The pressure formula and the supremum statement follow from the corresponding results there. Kernel (80.6) preserves μ because T does, and has the displayed minorization. Its transition coin is not an observation. $\square$

Corollary 80.3 (Noisy acquisition with retained hidden memory). *On $\mathbb{T}^d$ let $T_b x = bx \pmod{1}$, with integers $b \geq 2, d \geq 1$, let μ be Haar measure, and use any acquisition kernel Q satisfying (80.2). At each acquisition observe $Y = X + \sigma G \pmod{1}$, where the observation noises are independent standard Gaussian vectors. Predict $\Phi(x) = (\cos(2\pi x_j), \sin(2\pi x_j))_{j=1}^d$. Assume $W(n) \leq Aw_n$ for every n and some $A < \infty$. With*

$$a_k = e^{-2\pi^2 \sigma^2 b^{2k}}, \quad v_k = w_k a_k^2, \quad B_0 = d \sum_{k \geq 0} w_k (1 - a_k^2),$$

and Ψ_v, n_M as in (81.4) and (81.6), for all $M \geq 1$ and $\sigma \geq 0$,

$$\epsilon \left(B_0 + \frac{\Psi_v(n_M)}{8Ab^6} \right) \leq R_M^{\text{dep}} \leq B_0 + \frac{d\pi^2}{3} \Psi_v(n_M). \quad (80.7)$$

For geometric acquisitions this also bounds the optimal supremum raw loss. In particular $R_M^{\text{dep}} \asymp B_0 + \Psi_w(n_M)$, with constants independent of M and σ when ϵ is bounded below. The critical-window comparison of Corollary 81.4 remains valid for this dependent experiment, and the explicit restarting suffix observer has the same exact crossover coefficient as there.

Proof. The single-acquisition posterior under μK is $m_k(y) = a_k \Phi(b^k y)$, by (81.3). Lemma 81.2 bounds its orbit quantization error below by $\Psi_v(n_M)/(8Ab^6)$; use (80.4). The suffix observer proving Theorem 81.1 overwrites its state at acquisitions, so its exact expected losses, including the formula (81.9), transfer unchanged. This proves the upper bound and the supremum assertion. Finally $0 \leq \Psi_w(n) - \Psi_v(n) \leq B_0/d$ proves the raw-profile comparison. Its constants are uniform in the same compact q -windows as in Corollary 81.4; the explicit observer has exactly the same loss, rather than merely comparable loss, so its coefficient limit transfers as well. $\square$

APPENDIX 81. NOISY EXPANDING EXPERIMENTS AND A UNIFORM CRITICAL WINDOW

The realization overhead in Theorem 74.2 is not always necessary. We now identify a class in which symbolic closure yields matching errors, including the acquisition noise and the renewal tail. Let $b \geq 2$ and $d \geq 1$ be integers, let $\mathbb{T}^d = (\mathbb{R}/\mathbb{Z})^d$, and put

$$T_b x = bx \pmod{1}, \quad \Phi(x) = (\cos(2\pi x_j), \sin(2\pi x_j))_{j=1}^d. \quad (81.1)$$

At a renewal U is Haar uniform and the observation is

$$Y = U + \sigma G \pmod{1}, \quad G \sim N(0, I_d), \quad \sigma \geq 0. \quad (81.2)$$

The noise, the fresh states, and the renewal lengths are mutually independent. The target is $\Phi(X_t) \in \mathbb{R}^{2d}$, not a discontinuous choice of coordinates on the torus. Both average raw risk and average excess risk have the meanings of Section 74.

Haar invariance implies that Y and G are independent: conditional on G , the translate of U is Haar. Consequently the posterior law of U given $Y = y$ is that of $y - \sigma G$ modulo one, and

$$m_k(y) = a_k \Phi(b^k y), \quad a_k = \exp(-2\pi^2 \sigma^2 b^{2k}), \quad B = d \sum_{k \geq 0} w_k (1 - a_k^2). \quad (81.3)$$

This calculation uses the Gaussian characteristic function, so it also specifies exactly which noise information is used by the decoder. Write

$$v_k = w_k a_k^2, \quad \Psi_v(n) = \sum_{k < n} v_k b^{-2(n-k)} + \sum_{k \geq n} v_k, \quad n \geq 0. \quad (81.4)$$

For $v = w$, use the notation Ψ_w . We impose the quantitative renewal condition

$$W(n) \leq A w_n \quad (n \geq 0) \quad (81.5)$$

for some finite $A \geq 1$. In terms of the cycle length this is a uniform bound on the mean residual lifetime, conditional on survival. It includes geometric renewals, mixtures with geometric parameters bounded away from zero, and bounded renewal lengths. It does not include arbitrary polynomial renewal tails.

Put $a = b^d$ and

$$S_n = 1 + a + \cdots + a^n, \quad n_M = \max\{n \geq 0 : S_n \leq M\}. \quad (81.6)$$

The distinction between a^n and S_n counts, rather than suppresses, the persistent information about remaining precision.

Theorem 81.1 (Sharp posterior-orbit law). *Under (81.5), for every $M \geq 1$ and $\sigma \geq 0$,*

$$\frac{\Psi_v(n_M)}{8Ab^6} \leq E_M^{\text{av}} \leq \frac{d\pi^2}{3} \Psi_v(n_M), \quad (81.7)$$

$$B + \frac{\Psi_v(n_M)}{8Ab^6} \leq R_M^{\text{av}} \leq B + \frac{d\pi^2}{3} \Psi_v(n_M). \quad (81.8)$$

The lower bounds hold for all randomized, nonstationary machines. The upper bounds are attained by the finite suffix machine described below. Its exact average raw risk with S_n states is

$$U_{n,\sigma} = B + d \sum_{k < n} v_k \left[1 - \text{sinc}^2(\pi b^{-(n-k)}) \right] + d \sum_{k \geq n} v_k, \quad (81.9)$$

where $\text{sinc}(z) = \sin(z)/z$ and $\text{sinc}(0) = 1$. For geometric renewals, (81.8) also holds for $R_M^{\text{sup}} := \inf_{\mathcal{A}} \sup_{t \geq 0} \mathbb{E} \|\Phi(X_t) - \hat{f}_t\|^2$, with an initial acquisition at zero. The supremum raw risk of the displayed suffix machine is exactly $U_{n,\sigma}$.

Lemma 81.2 (Orbit separation and small balls). *Suppose $v_k = w_k a_k^2$ with a_k nonincreasing and (81.5) holds. For Haar Y the quantization error of $(\sqrt{v_k} \Phi(b^k Y))_{k \geq 0}$ satisfies*

$$e_M^2 \geq \frac{\Psi_v(n_M)}{8Ab^6}. \quad (81.10)$$

Proof. First, $\sum_{k \geq n} v_k \leq A v_n$. Also Ψ_v is nonincreasing, and

$$\Psi_v(n+1) \geq b^{-2} \Psi_v(n). \quad (81.11)$$

These follow directly from (81.4) and monotonicity of a_k . For $x, y \in \mathbb{T}^d$, write $\delta = \max_j \text{dist}_{\mathbb{T}}(x_j, y_j) \in [0, 1/2]$. If $\delta > 0$, set $K = \lfloor \log_b(1/(2\delta)) \rfloor$. In a coordinate attaining δ , for $0 \leq k \leq K$ the unreduced difference has modulus $b^k \delta \leq 1/2$. Hence $|\sin(\pi b^k \delta)| \geq 2b^k \delta$, and the squared orbit distance $D^2(x, y)$ obeys

$$\begin{aligned} D^2(x, y) &= 4 \sum_{k \geq 0} v_k \sum_{j=1}^d \sin^2(\pi b^k (x_j - y_j)) \\ &\geq 16\delta^2 \sum_{k=0}^K v_k b^{2k} \geq \frac{4}{b^2} \sum_{k=0}^K v_k b^{-2(K-k)} \geq \frac{4}{Ab^2} \Psi_v(K). \end{aligned} \quad (81.12)$$

The middle inequality uses $\delta > b^{-(K+1)}/2$; the last uses the residual-tail bound and $A \geq 1$.

Let $h = [2(2M)^{1/d}]^{-1}$ and $K_h = \lfloor \log_b(1/(2h)) \rfloor = \lfloor \log_a(2M) \rfloor$. Since $M < S_{n_M+1}$ and $a \geq 2$, one has $K_h \leq n_M + 2$. Whenever $\delta \geq h$, the associated $K \leq K_h$; thus (81.11) and (81.12) give

$$D^2(x, y) \geq \gamma \Psi_v(n_M), \quad \gamma = \frac{4}{Ab^6}.$$

If $\Psi_v(n_M) = 0$, the conclusion is immediate. Otherwise take a Hilbert ball of radius r , with $r^2 = \gamma \Psi_v(n_M)/16$. Two preimages in the same ball have orbit distance at most $2r$, so their torus sup-distance is less than h . Fixing one preimage, all preimages lie in a torus cube of side $2h$, whose Haar measure is $1/(2M)$. This argument applies even when the centre of the Hilbert ball is not on the orbit image. The union of M such balls has preimage measure at most $1/2$. Every M -centre quantizer therefore has mean squared error at least $r^2/2 = \Psi_v(n_M)/(8Ab^6)$. $\square$

Proof of Theorem 81.1. The lower bound follows from Theorem 74.2, (81.3), and Lemma 81.2. The orthogonality identity then adds B to the excess risk.

For the upper bound use the alphabet $\{0, \dots, b-1\}^d$. The states are all words of lengths $0, \dots, n$, exactly S_n states. At a renewal retain the first n base- b digits of each coordinate of Y , grouped into one word. Use the terminating expansion at ambiguous endpoints; those endpoints have Haar measure zero. At a nonrenewal delete the first letter, leaving the empty word fixed. For a nonempty word w of length l , let $c(w)$ be the midpoint of its b -adic cube, and decode it as

$$a_{n-l} \text{sinc}(\pi b^{-l}) \Phi(c(w)).$$

Decode the empty word as zero. The age before exhaustion is $n-l$, so there is no separate age register. The update does not use the renewal law.

Conditional on this suffix at age $k < n$, the vector $b^k Y$ is uniform on its cube. Averaging each trigonometric coordinate over an interval gives its midpoint value times $\text{sinc}(\pi b^{-(n-k)})$. The displayed output is therefore the conditional mean of $\Phi(b^k U)$ under the suffix. Its raw squared error is

$$d \left[1 - a_k^2 \text{sinc}^2(\pi b^{-(n-k)}) \right].$$

After exhaustion the raw error is d . Renewal reward proves (81.9). For $0 \leq t \leq 1/2$,

$$\frac{4}{3}t^2 \leq 1 - \text{sinc}^2(\pi t) \leq \frac{\pi^2}{3}t^2. \quad (81.13)$$

Indeed, if V, V' are independent uniform variables on an interval of length t , the middle expression is $2\mathbb{E} \sin^2(\pi(V - V'))$. Use $2|u| \leq |\sin(\pi u)| \leq \pi|u|$ for $|u| \leq 1/2$ and $\mathbb{E}(V - V')^2 = t^2/6$. This proves the average upper bounds, including $n = 0$.

For geometric renewals put $q = 1 - p$. The raw age-loss sequence just computed is nondecreasing: a_k decreases and, as k increases before exhaustion, the sinc factor decreases as well. The age at time t has the law of $\min(G_0, t)$, where $\mathbb{P}(G_0 = k) = pq^k$. Therefore its expected raw loss increases

to $\sum_k pq^k d_k = U_{n,\sigma}$ and never exceeds this number. The lower bound for supremum loss follows from the average converse. $\square$

Corollary 81.3 (Joint statistical and persistent-state law). *Under the hypotheses of Theorem 81.1,*

$$R_M^{\text{av}} \asymp_{b,d,A} B + \Psi_w(n_M). \quad (81.14)$$

The same holds for raw supremum risk under geometric renewals. Moreover a suffix decoder which does not know σ or the renewal law attains this order of raw risk.

Proof. Every coefficient in Ψ_v is between zero and one. Thus

$$0 \leq \Psi_w(n) - \Psi_v(n) \leq \sum_k w_k(1 - a_k^2) = B/d.$$

This and (81.8) prove the equivalence. For the noise-independent decoder use $\text{sinc}(\pi b^{-l})\Phi(c(w))$, omitting a_{n-l} . Compare with the hypothetical target $\Phi(b^k Y)$. At age k ,

$$\mathbb{E}\|\Phi(b^k U) - \Phi(b^k Y)\|^2 = 2d(1 - a_k) \leq 2d(1 - a_k^2).$$

The squared triangle inequality and the noiseless suffix bound give average raw risk at most $4B + (2d\pi^2/3)\Psi_w(n)$. For geometric renewals the same bound holds for supremum loss: both the noise-discrepancy sequence $1 - a_k^2$ and the noiseless suffix raw-loss sequence are nondecreasing, so each time expectation is bounded by its stationary-age average. The machine uses only b, d, n and the observed digits. $\square$

81.1. Geometric renewals and the critical window. For $w_k = pq^k$, $p = 1 - q$, the noiseless profile is

$$\Psi_q(n) = pb^{-2n} \sum_{k=0}^{n-1} (b^2 q)^k + q^n. \quad (81.15)$$

Theorem 81.1 is uniform for q in any fixed compact subset of $(0, 1)$, since $A = 1/p$ is then bounded. In particular it remains valid when q and σ depend on M .

Corollary 81.4 (Phase regimes and crossover). *For each fixed $0 < q < 1$, at $\sigma = 0$ the optimal average and supremum raw risks have orders*

$$\begin{cases} M^{-2/d}, & q < b^{-2}, \\ (1 + \log M)M^{-2/d}, & q = b^{-2}, \\ M^{-\log(1/q)/(d \log b)}, & q > b^{-2}. \end{cases} \quad (81.16)$$

Let $q_n = b^{-2} \exp(x/n)$ and $H(x) = (e^x - 1)/x$, with $H(0) = 1$. Uniformly for x in bounded intervals,

$$\frac{b^{2n}}{n} \Psi_{q_n}(n) \longrightarrow (1 - b^{-2})H(x). \quad (81.17)$$

For the explicit S_n -state suffix machine, its exact noiseless raw risk satisfies the stronger coefficient limit

$$\frac{b^{2n}}{n} U_{n,0} \longrightarrow \frac{d\pi^2}{3} (1 - b^{-2})H(x). \quad (81.18)$$

The optimal risks are bounded above and below by constant multiples of the profile in (81.17), uniformly in this window; no equality of their leading constants is asserted.

Proof. Since $S_n \asymp_b b^{dn}$, evaluating the geometric sum in (81.15) proves (81.16). In the critical window,

$$\frac{b^{2n}}{n} \Psi_{q_n}(n) = (1 - q_n) \frac{1}{n} \sum_{k=0}^{n-1} e^{xk/n} + \frac{e^x}{n}.$$

The Riemann sum tends uniformly on bounded x -intervals to $\int_0^1 e^{xt} dt = H(x)$. For the exact suffix risk use $1 - \text{sinc}^2(\pi t) = (\pi^2/3)t^2 + O(t^4)$, uniformly for $0 \leq t \leq 1/2$. After multiplication by b^{2n} , the error sum is bounded by a constant times

$$\sum_{k < n} e^{xk/n} b^{-2(n-k)},$$

which is uniformly bounded on bounded x -intervals. The exhaustion term $db^{2n}q_n^n = de^x$ is bounded as well. Dividing by n proves (81.18). The final assertion is exactly the uniform two-sided bound of Theorem 81.1. $\square$

For completeness, the noise floor has the same transition. For $0 < \sigma \leq 1$, set $n_\sigma = \lfloor \log_b(1/\sigma) \rfloor$; for $\sigma > 1$ set $n_\sigma = 0$. From $1 - e^{-4\pi^2 z^2} \asymp \min(1, z^2)$ and $b^{-(n_\sigma+1)} < \sigma \leq b^{-n_\sigma}$ one obtains

$$B \asymp_{b,d} \Psi_q(n_\sigma), \quad R_M \asymp_{b,d,q} \Psi_q(n_\sigma) + \Psi_q(n_M). \quad (81.19)$$

At $\sigma = 0$ the first term is interpreted as zero. The comparison in the second formula is uniform on compact q -intervals. Thus the small-noise orders of B are respectively σ^2 , $\sigma^2(1 + \log(1/\sigma))$, and $\sigma^{\log(1/q)/\log b}$ in the three regimes. This describes filtering uncertainty and expanding transport in the same experiment, including joint limits, rather than adding an unrelated noisy model.

81.2. Nonconstant slopes and other readouts. The preceding law is not confined to a constant-slope coordinate presentation. The next statement records both a genuine robustness class and its precise boundary.

Proposition 81.5 (Transport under conjugacy and metric readouts). *Let $h : \mathbb{T}^d \rightarrow E'$ be a Borel isomorphism. Transport the reset law, observation (81.2), dynamics, and readout by h ; thus the new observation is $h(Y)$, the dynamics are $hT_b h^{-1}$, and the readout is Φh^{-1} . All the preceding finite-state risks are unchanged. Suppose instead that E' is compact and the new continuous readout g satisfies, for some $0 < c \leq C < \infty$,*

$$c\|\Phi(x) - \Phi(y)\| \leq \|g(hx) - g(hy)\| \leq C\|\Phi(x) - \Phi(y)\|. \quad (81.20)$$

Then its optimal raw squared risks, for the same observations and state budgets, lie between $c^2/4$ and $4C^2$ times those for Φ . This assertion concerns raw risk, not equality of posterior excess risk for nonlinear readouts.

Proof. Apply h^{-1} to each acquired observation within the transient update computation. This and its inverse transform machines without changing a persistent state. For the readout assertion, replace any output z for Φ by a nearest point $\Phi(x_z)$ on its compact image, choosing measurably. For a true state x , $\|\Phi(x_z) - \Phi(x)\| \leq 2\|z - \Phi(x)\|$. Output $g(hx_z)$ and use the upper inequality in (81.20). The inverse construction projects onto the compact image of g and uses the lower inequality. Both transformations act only on the decoder, use no extra states, and are valid conditionally on any coding randomness. $\square$

For example, in one dimension take

$$h_\varepsilon(x) = x + \frac{\varepsilon}{2\pi} \sin(2\pi x) \pmod{1}, \quad |\varepsilon| < \frac{b-1}{b+1}.$$

This is a smooth circle diffeomorphism and the conjugate map has nonconstant derivative

$$(h_\varepsilon T_b h_\varepsilon^{-1})'(h_\varepsilon x) = b \frac{1 + \varepsilon \cos(2\pi b x)}{1 + \varepsilon \cos(2\pi x)} > 1.$$

Coordinatewise products give higher-dimensional examples, with nonuniform physical reset density and transported nonadditive observation noise. The ordinary circle embedding of $h_\varepsilon x$ satisfies (81.20). These are nonconstant-slope expanding experiments, but their Lyapunov exponent remains $\log b$ by conjugacy. We do not identify their law with a pressure theorem for arbitrary nonconjugate

expanding maps. The general posterior-orbit theorem still applies to such maps; its quantization invariant would have to be evaluated separately.

APPENDIX 82. CALIBRATED MOMENT OBSERVATIONS AT ROOT COLLISIONS

We next derive, rather than postulate, a singular observation modulus. Let

$$\Theta_d = \{\theta \in [0, 1]^d : \theta_1 \leq \cdots \leq \theta_d\}, \quad P_j(\theta) = \sum_{i=1}^d \theta_i^j, \quad 1 \leq j \leq d,$$

and write $P = (P_1, \dots, P_d)$. The data are the actually observed pair

$$Y = P(\theta) + \eta + \varepsilon_Y, \quad W = \eta + \varepsilon_W, \quad \begin{pmatrix} \varepsilon_Y \\ \varepsilon_W \end{pmatrix} \sim N(0, \sigma^2 \Sigma_0), \quad (82.1)$$

where $\eta \in \mathbb{R}^d$ is unrestricted, $\sigma \geq 0$ is known, and Σ_0 is a fixed positive-definite $2d$ by $2d$ matrix. Let

$$V_0 = (\Sigma_0)_{YY} + (\Sigma_0)_{WW} - (\Sigma_0)_{YW} - (\Sigma_0)_{WY} > 0.$$

An encoder of the pair retains one of M labels. Its decoder estimates θ , with squared Euclidean loss. Public and private randomness are permitted under the same hard label constraint as above. Denote the minimax risk, taking a supremum over both θ and η , by $\mathcal{R}_M(\Theta, \sigma)$ when the parameter set is Θ . Projection permits decoder values to lie in Θ_d , so the loss is bounded by d .

Theorem 85.1, proved in the retained development, reduces this problem *exactly for every* M to

$$D = Y - W \sim N(P(\theta), \sigma^2 V_0). \quad (82.2)$$

For $\sigma > 0$ this is its correlated Gaussian nuisance quotient; for $\sigma = 0$ the difference is noiseless and the reverse comparison follows by fixing the nuisance to any constant. The reduction is an equality of minimax risks for decisions about θ . It is not a Blackwell equivalence for the full parameter (θ, η) .

Theorem 82.1 (Collision strata and label resolution). *For every fixed $d \geq 1$ and positive-definite Σ_0 , there are $0 < c \leq C < \infty$, independent of M and σ , such that*

$$c \left\{ \min(1, \sigma^{2/d}) + M^{-2/d} \right\} \leq \mathcal{R}_M(\Theta_d, \sigma) \leq C \left\{ \min(1, \sigma^{2/d}) + M^{-2/d} \right\}. \quad (82.3)$$

More generally, let $I_1, \dots, I_s \subset [0, 1]$ be ordered closed intervals of positive lengths, with positive gaps between consecutive intervals. Prescribe $r_j \geq 1$ roots in I_j , with $\sum_j r_j = d$, and let $\Theta(I, r)$ be the resulting ordered parameter set. If $r_ = \max_j r_j$, then*

$$\mathcal{R}_M(\Theta(I, r), \sigma) \asymp_{d, V_0, I, r} \min(1, \sigma^{2/r_*}) + M^{-2/d}. \quad (82.4)$$

The constants in the stratified assertion may deteriorate when an interval width or a gap tends to zero. Neither assertion assumes a two-sided global contact equivalence.

The model is genuinely coupled. The determinant of the derivative of P at distinct ordered roots is

$$\det DP(\theta) = d! \prod_{i < j} (\theta_j - \theta_i). \quad (82.5)$$

Thus the rank changes on the collision strata. There is no invertible smooth global coordinate change that makes this a uniformly regular θ -experiment. The next lemma supplies the inverse modulus needed for the statistical theorem. Its root-perturbation argument is classical in method; the point here is to carry it through the actual calibrated observation and hard-label decision problem.

Lemma 82.2 (Derived inverse moment moduli). *There is a finite C_d such that for all $\theta, \theta' \in \Theta_d$,*

$$\|\theta - \theta'\| \leq C_d \|P(\theta) - P(\theta')\|^{1/d}. \quad (82.6)$$

For $\theta, \theta' \in \Theta(I, r)$ and $\delta = \|P(\theta) - P(\theta')\|$ sufficiently small, the coordinates in cluster j satisfy

$$\|\theta^{(j)} - \theta'^{(j)}\| \leq C_{d,I,r} \delta^{1/r_j}. \quad (82.7)$$

Both statements include multiple roots and preserve their multiplicities.

Proof. Let $e_j(\theta)$ be the elementary symmetric polynomials, with $e_0 = 1$. Newton's identities give

$$je_j = \sum_{i=1}^j (-1)^{i-1} e_{j-i} P_i.$$

On $[0, 1]^d$, $|P_i| \leq d$ and $|e_j| \leq \binom{d}{j}$. For an explicit finite Lipschitz bound define $L_0 = 0$ and

$$L_j = \frac{1}{j} \sum_{i=1}^j \left(dL_{j-i} + \binom{d}{j-i} \right), \quad 1 \leq j \leq d.$$

Induction gives $|e_j(\theta) - e_j(\theta')| \leq L_j \delta$. For the monic polynomials $Q_\theta(z) = \prod_i (z - \theta_i)$ and $Q_{\theta'}$, it follows that

$$|Q_\theta(z) - Q_{\theta'}(z)| \leq A_d \delta \quad (|z| \leq 2), \quad A_d = 2^d \max_{1 \leq j \leq d} L_j. \quad (82.8)$$

If $\delta = 0$, Newton's identities identify the two multisets of roots, and their ordered vectors coincide. Suppose $\delta > 0$ is small and put $\rho = (2A_d \delta)^{1/d} < 1$. Take the union of the complex disks of radius ρ about the roots of Q_θ . At its boundary every factor has modulus at least ρ , so $|Q_\theta| \geq \rho^d = 2A_d \delta > |Q_\theta - Q_{\theta'}|$. Rouché's theorem on each connected component gives the same number of zeros, counted with multiplicity, for the two polynomials. One may increase ρ arbitrarily slightly to avoid tangent boundaries and then take a limit; the piecewise smooth boundary version follows equivalently from the argument principle. A component containing at most d disks has diameter at most $2d\rho$. Match the roots within each component. For real ordered vectors, sorting a matching cannot increase its maximum displacement: an inverted pair can be uncrossed with no larger maximum distance. Thus $\|\theta - \theta'\| \leq 2d\sqrt{d}\rho$. For larger δ , compactness and the bound $\|\theta - \theta'\| \leq \sqrt{d}$ extend the inequality with a larger constant.

For the stratified assertion let $\Delta > 0$ be the smallest gap; when $s = 1$ use the preceding argument with $r_1 = d$. Around roots in I_j use disks of radius

$$\rho_j = \left[2A_d (2/\Delta)^{d-r_j} \delta \right]^{1/r_j}.$$

For small δ these radii are less than $\min(1, \Delta/3)$. At the boundary of their union the r_j factors from this cluster have modulus at least ρ_j , and all other factors have modulus at least $\Delta/2$. Hence the product again exceeds $A_d \delta$. Rouché gives equal root counts in each component, and its diameter is at most $2r_j \rho_j$. Matching and sorting within the cluster proves (82.7). $\square$

Lemma 82.3 (Indistinguishable colliding configurations). *For every $r \geq 1$ there exist two distinct real vectors $a_1 < \dots < a_r$ and $b_1 < \dots < b_r$ in $(-1/4, 1/4)$ such that*

$$\sum_i a_i^j = \sum_i b_i^j \quad (0 \leq j < r), \quad \sum_i a_i^r \neq \sum_i b_i^r. \quad (82.9)$$

If these vectors are scaled by h and translated to an interior point x_0 of a prescribed cluster, their root separation is a positive constant times h , whereas the difference of all moments of orders $1, \dots, d$ is $O(h^r)$, for fixed $d \geq r$.

Proof. Choose any distinct a_i in the indicated interval and let $Q(z) = \prod_i (z - a_i)$. Around each a_i choose a small closed interval, with disjoint interiors and endpoints still in $(-1/4, 1/4)$, on which Q has opposite nonzero signs at the two endpoints. For any sufficiently small nonzero constant c , the polynomial $Q + c$ retains these sign changes. It therefore has one real root in each interval, and these are all its r roots. Call them b_i . The monic coefficients through degree one of Q and $Q + c$ agree; only the constant coefficient changes. Newton's identities imply (82.9), with the difference at order r equal up to sign to $rc \neq 0$. For every $j \leq d$, expand $\sum_i (x_0 + ha_i)^j - \sum_i (x_0 + hb_i)^j$ by the binomial theorem. All terms of degree less than r in h vanish. The remaining terms are $O(h^r)$. The Euclidean separation of the two root vectors is $h\|a - b\| > 0$. $\square$

Proof of Theorem 82.1. Work first in the difference experiment (82.2). On either compact parameter set take a measurable least-squares minimizer

$$\tilde{\theta} \in \operatorname{argmin}_{\vartheta \in \Theta} \|D - P(\vartheta)\|.$$

Existence follows from compactness and continuity; the measurable minimum theorem gives a Borel selection. With $D = P(\theta) + \sigma V_0^{1/2} G$, $G \sim N(0, I_d)$, minimality gives

$$\|P(\tilde{\theta}) - P(\theta)\| \leq 2\sigma \|V_0^{1/2} G\|. \quad (82.10)$$

The global modulus and Gaussian moments yield mean squared error at most $C\sigma^{2/d}$, also bounded by d . On $\Theta(I, r)$, (82.7), extended by a bounded constant outside its small- δ range, yields an upper bound $C \sum_j \sigma^{2/r_j} \leq C'\sigma^{2/r_*}$ for $0 < \sigma \leq 1$. For clarity, that extension is valid since $\|\theta^{(j)} - \theta'^{(j)}\|^2 \leq r_j$ and $\delta \geq \delta_0$ implies $r_j \leq r_j \delta_0^{-2/r_j} \delta^{2/r_j}$. The cap by the diameter handles $\sigma > 1$.

Let $N = \lfloor M^{1/d} \rfloor \geq M^{1/d}/2$. The N^d midpoint product grid in $[0, 1]^d$ has squared covering radius at most $d/(4N^2)$. Project its centres onto Θ , which in both cases is a closed convex subset. This does not increase their distances to points of Θ . Quantize $\tilde{\theta}$ to the first nearest projected centre and retain its index. The squared triangle inequality adds at most $C_d M^{-2/d}$ and multiplies the statistical bound by at most two. This proves the upper bounds with at most M labels, including the noiseless case.

For a statistical lower bound take a cluster of size r_* , or the single cluster of size d for the global assertion. Place the configurations of Lemma 82.3 at an interior point of that cluster; fix all other roots in their intervals. For sufficiently small $\sigma > 0$ take $h = c_0 \sigma^{1/r_*}$. Their squared parameter distance is at least $c_1 h^2$ and the Kullback–Leibler divergence of the two difference laws is

$$\frac{1}{2\sigma^2} \|V_0^{-1/2} (P(\theta) - P(\theta'))\|^2 \leq Ch^{2r_*}/\sigma^2.$$

Choose $c_0 > 0$ so that this is at most $1/8$. Pinsker's inequality and the two-point testing bound give risk at least $c_2 \sigma^{2/r_*}$ even without a label constraint. For example, classifying an estimator by the closer of the two parameters shows that its maximum squared risk is at least one eighth of their squared distance times one minus their total variation distance. For larger σ , the Gaussian experiment is a garbling of the one at any smaller positive fixed noise level: add an independent normal with covariance $(\sigma^2 - \sigma_0^2)V_0$ before encoding. Minimax risk is therefore nondecreasing in σ , giving the cap $c \min(1, \sigma^{2/r_*})$.

It remains to prove a representation lower bound independent of noise and calibration. The parameter set contains a d -dimensional box of positive volume, obtained by placing all roots in disjoint small subintervals inside their prescribed intervals. Put the uniform prior on that box. Conditional on independent public randomness, a decoder has at most M possible deterministic centres; private output randomness can be replaced by its mean for squared loss. A union of M Euclidean balls of radius $c_3 M^{-1/d}$ covers at most half the box, with $c_3 > 0$ depending only on its volume and d . Pointwise, every distribution of an encoded label has loss at least the

squared distance to the nearest centre. Integrate first over the box and then over the public randomness. The Bayes, hence minimax, risk is at least $c_4 M^{-2/d}$. Combining the two lower bounds by $\max(u, v) \geq (u + v)/2$ proves the result. Finally apply the exact nuisance quotient to return to the observed pair. $\square$

Remark 82.4 (Statistical meaning of the collision exponent). The exponent $1/r_*$ is derived from moment cancellation and multiplicity-sensitive root stability; it is not an assumed contact order. The representation exponent remains $2/d$ because each allowed stratum-domain contains an open set of d distinct roots. The paired experiment has equal known weights and noisy observations of its first d power sums. It is not the experiment of independent samples from a finite mixture with unknown weights. Rates in the latter experiment, such as those of Heinrich and Kahn [31], concern a different likelihood and different local parameter sets. Nor is a polynomial root argument a substitute for an unsmoothed billiard local limit theorem. The present theorem verifies an actual changing-rank observation geometry within the calibrated reduction, without importing either of those assertions.

Remark 82.5 (Uniformity in the retained contact theorem). The constants of Theorem 85.2 depend on its displayed metric-equivalence constants λ, Λ . Their independence of contact orders, widths, or noise scales is conditional on uniform control of λ, Λ in those regimes. It is not a claim of uniform physical conditioning. In the moment model the dependence on dimension, covariance conditioning, cluster gaps, and interval widths is made explicit at the level of the hypotheses in Theorem 82.1; no uniform limit through a collapsing cluster gap is inferred from the stratified statement.

COMPLEMENTARY QUANTITATIVE DEVELOPMENT

The following sections retain the preceding quantitative theory and its proofs. The scalar doubling result provides an exact checkpoint comparison; the correlated nuisance quotient is also used in Section 82. The later filter and control results have separate assumptions and are not premises of the posterior-orbit realization theorem.

APPENDIX 83. REGENERATION AND THE GEOMETRY OF FUTURE PREDICTIONS

The obstruction in this section concerns information already acquired and subsequently discarded. It does not arise from estimating an unknown transition probability. We first give a converse for a general regenerative experiment and then compute its consequence for an expanding map.

83.1. A retained-label experiment. Let E be a standard Borel space, $T : E \rightarrow E$ a Borel map, μ a probability measure, and $f : E \rightarrow [0, 1]$ a Borel readout. Fix $0 < p < 1$ and write $q = 1 - p$. At time zero an observer acquires $U_0 \sim \mu$ and $X_0 = U_0$. At each integer time $t \geq 1$, an independent mode R_t is observed, with $\Pr(R_t = 1) = p$. If $R_t = 1$, a fresh independent $U_t \sim \mu$ is acquired and $X_t = U_t$. Otherwise $X_t = T(X_{t-1})$; there is no further observation of X_t . All acquisitions and modes are independent of the coding randomness. The full-history prediction is $f(X_t)$.

A causal M -label machine has a state $L_t \in \{1, \dots, M\}$. At time zero it encodes U_0 . Subsequently its update can use the previous label, the clock, the current mode, and, at a reset, the current U_t . After the update only its label, the clock, and independent public coding randomness are available for decoding. In particular the current acquisition is no longer available at the checkpoint. Private coding randomness is allowed. Past modes, reset times, previous acquisitions, and previous decoder outputs are not additional persistent registers. Time-dependent programs are allowed in the lower bound. Any random-generator state or data-dependent tape position that persists also counts as part of the label. Private coins may equivalently be supplied as independent, time-indexed uniforms.

One may realize the loss physically by selecting, after this compression, a Bernoulli terminal probe of mean $f(X_t)$. The probe is not an additional observation in the continuing experiment. Conditional expectation then identifies squared prediction excess with

$$L_t^{\text{pred}} = |f(X_t) - \hat{f}_t|^2.$$

Projection of the decoder onto $[0, 1]$ cannot increase this loss. Define

$$\mathcal{R}_M^{\text{ca}}(p; T, \mu, f) = \inf_{\text{causal } M\text{-label machines}} \sup_{t \geq 0} \mathbb{E} L_t^{\text{pred}}.$$

The clock is free; the elapsed time since the last reset is not free.

83.2. The future-orbit converse. For $u \in E$ set

$$Z_q(u) = (\sqrt{p} q^{k/2} f(T^k u))_{k \geq 0} \in \ell^2, \quad e_M^2(Z_q) = \inf_{c_1, \dots, c_M \in \ell^2} \int \min_i \|Z_q(u) - c_i\|_{\ell^2}^2 \mu(du). \quad (83.1)$$

The map is Borel and has norm at most one. Its geometry records every future readout that must be recovered from one label if no new acquisition occurs. In contrast, the checkpoint geometry concerns only $f(X_t)$.

Theorem 83.1 (Regenerative future-orbit lower bound). *Every causal M -label machine in the preceding experiment satisfies*

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \mathbb{E} L_t^{\text{pred}} \geq e_M^2(Z_q). \quad (83.2)$$

Consequently $\mathcal{R}_M^{\text{ca}}(p; T, \mu, f) \geq e_M^2(Z_q)$. The assertion holds for arbitrary independent public and private coding randomization and does not require a stationary machine or an invariant law for T .

Proof. Let $e_{M,K}^2$ be the quantization error of the first $K+1$ coordinates of Z_q . Appending the coordinates $\sqrt{p} q^{k/2}/2$, $k > K$, to any truncated codebook shows that

$$e_M^2(Z_q) - \frac{1}{4} q^{K+1} \leq e_{M,K}^2 \leq e_M^2(Z_q). \quad (83.3)$$

Approximate minimizers suffice for this argument; existence is not needed.

Fix a possible reset time $j \geq 1$. Condition on its entire past, on the occurrence of that reset, and on the independent coding random tapes. The new U_j still has law μ . The update assigns it one of at most M labels. Along the subsequent word of K nonreset modes, every such label determines a sequence of $K+1$ decoder values. Future random choices are part of the fixed coding tapes; the nonreset word does not contain new data. The weighted squared error of these values, integrated over U_j , is therefore at least $e_{M,K}^2$. This remains true when the assignment itself is randomized, since fixing its independent random tape produces an ordinary assignment to the same number of centres. Integrating back over the past and the tapes gives

$$p \sum_{k=0}^K q^k \mathbb{E}[L_{j+k}^{\text{pred}} \mid R_j = 1, R_{j+1} = \dots = R_{j+k} = 0] \geq e_{M,K}^2.$$

The displayed conditional expectations also integrate over the unconditioned past at j ; future modes beyond $j+k$ play no role.

For each checkpoint t , the events that its last reset occurred at $j = t - k$, $0 \leq k \leq K$, are disjoint. Sum their nonnegative contributions over $1 \leq t \leq N$ and retain only $1 \leq j \leq N - K$. Independence of the modes gives

$$\sum_{t=1}^N \mathbb{E} L_t^{\text{pred}} \geq (N - K) e_{M,K}^2.$$

First let $N \rightarrow \infty$ at fixed K , and then let $K \rightarrow \infty$ in (83.3). The supremum of expected checkpoint losses dominates their lower limiting average, proving the consequence. $\square$

The theorem is a converse for arbitrary causal machines, not only for machines obtained by quantizing E . The centres in (83.1) may lie off the orbit image. In particular, replacing their quantization problem by a covering problem for the current state would lose information needed for this lower bound.

APPENDIX 84. A SHARP CAUSAL-MEMORY TRANSITION FOR AN EXPANDING EXPERIMENT

Take

$$E = [0, 1), \quad \mu = \text{Leb}|_{[0,1)}, \quad T(x) = 2x - \lfloor 2x \rfloor, \quad f(x) = x. \quad (84.1)$$

The fresh acquisition is an analog uniform variable. A machine in the upper bound below reads only a finite binary prefix of it. The map is piecewise expanding; it is not asserted to be globally Lipschitz for the Euclidean metric on $[0, 1)$. The proof will account explicitly for the jumps at dyadic boundaries.

Let $\mathcal{R}_M^{\text{cp}}(t)$ be the optimal expected squared excess when an encoder can use the entire acquisition history at checkpoint t and then send one of M labels. Put

$$\Phi_q(M) = \begin{cases} M^{-2}, & 0 < q < 1/4, \\ (1 + \log M)M^{-2}, & q = 1/4, \\ M^{-\kappa(q)}, & 1/4 < q < 1, \end{cases} \quad \kappa(q) = \frac{\log(1/q)}{\log 2}. \quad (84.2)$$

All logarithms except those explicitly marked otherwise are natural.

Theorem 84.1 (Identical checkpoint geometry, three causal rates). *For the regenerative experiment (84.1), every integer $M \geq 1$, and every $t \geq 0$,*

$$\mathcal{R}_M^{\text{cp}}(t) = \frac{1}{12M^2}. \quad (84.3)$$

For each $0 < q < 1$, constants $0 < c_q \leq C_q < \infty$ give

$$c_q \Phi_q(M) \leq \mathcal{R}_M^{\text{ca}}(1 - q; T, \mu, f) \leq C_q \Phi_q(M) \quad (M \geq 1). \quad (84.4)$$

The lower bound already holds for the lower limiting temporal mean in (83.2). The upper bound is attained in order by a single deterministic finite program which does not use p or q . The program has at most M persistent labels, including every state used to record the number of retained binary digits.

The threshold is $p = 3/4$. Below that reset frequency the causal label exponent is smaller than the checkpoint exponent; at the threshold a logarithmic factor is necessary. The current acquired prediction is uniform in all these regimes. Thus its dimension and even its entire probability law do not determine the optimal causal memory law.

84.1. An explicit finite machine. For an integer $B \geq 0$ use the label set

$$\mathcal{W}_B = \bigcup_{b=0}^B \{0, 1\}^b, \quad |\mathcal{W}_B| = 2^{B+1} - 1,$$

including the empty word. At each reset retain the first B binary digits of U_t , using the terminating expansion at a dyadic rational. At a nonreset delete the leading digit, or leave the empty word unchanged. Decode a word of length b as the midpoint of its binary cylinder; the empty word decodes as $1/2$.

Lemma 84.2 (Exact risk of the suffix machine). *For this machine, $\sup_{t \geq 0} \mathbb{E} L_t^{\text{pred}}$ equals*

$$U_B(q) = \frac{1}{12} \left[p 4^{-B} \sum_{k=0}^{B-1} (4q)^k + q^B \right], \quad (84.5)$$

with the sum empty when $B = 0$. Equality holds at every $t \geq B$.

Proof. A uniform variable has independent fair binary digits outside the null set of dyadic rationals. If the most recent reset was k steps ago, then the label retains its next $(B - k)_+$ digits. Conditional on this label and the age, the current state is uniform on the corresponding cylinder, so its midpoint has squared error $4^{-(B-k)_+}/12$. At time t the age probabilities are pq^k for $0 \leq k < t$, and q^t for $k = t$, because time zero was an acquisition. Equivalently this age has the law of $\min(G, t)$, where $\Pr(G = k) = pq^k$. The function $k \mapsto 4^{-(B-k)_+}/12$ is nondecreasing and constant for $k \geq B$. Its expectation increases to (84.5) and is already equal to it for $t \geq B$. $\square$

Choose the largest B such that $2^{B+1} - 1 \leq M$. Then $(M + 1)/4 < 2^B \leq (M + 1)/2$. If $4q < 1$, the sum in (84.5) is bounded, and $q^B \leq 4^{-B}$. If $4q = 1$, the expression is $4^{-B}(pB + 1)/12$. If $4q > 1$, it is at most $[1 + p/(4q - 1)]q^B/12$. These estimates prove the upper bound in (84.4) for every integer budget, not just powers of two.

The label can be stored as an integer whose leading sentinel records its word length. Only $\lceil \log_2(2^{B+1} - 1) \rceil$ persistent bits are needed. Binary prefix acquisition and output formatting use $O(B + 1)$ transient bits; deletion can be performed in $O(B + 1)$ bit operations. The only read-only integer parameter is B . Neither a real-valued codebook nor the last reset time is hidden in the program. The cost of observing the reset mode and reading the new prefix is an acquisition cost, distinct from these memory bounds.

84.2. The lower bound without a restriction on encoders. Define, for $0 < \delta \leq 1$,

$$F_q(\delta) = \begin{cases} \delta^2, & q < 1/4, \\ \delta^2(1 + \log(1/\delta)), & q = 1/4, \\ \delta^{\kappa(q)}, & q > 1/4. \end{cases}$$

Each function is increasing on $(0, 1]$.

Lemma 84.3 (Separation of future orbits). *For $u, v \in [0, 1]$,*

$$\|Z_q(u) - Z_q(v)\|_{\ell^2}^2 \geq \frac{p}{32} F_q(|u - v|). \quad (84.6)$$

Proof. The case $u = v$ is immediate. Write $\delta = |u - v|$. If $\delta \leq 1/2$, let $n = \lfloor \log_2(1/(2\delta)) \rfloor$. For $0 \leq k \leq n$, the difference of the fractional parts of $2^k u$ and $2^k v$ has absolute value either $2^k \delta$ or $1 - 2^k \delta$. Both are at least $2^k \delta$, since $2^k \delta \leq 1/2$. Consequently

$$\|Z_q(u) - Z_q(v)\|^2 \geq p\delta^2 \sum_{k=0}^n (4q)^k. \quad (84.7)$$

For $q < 1/4$ the $k = 0$ term alone suffices, for every δ . For $q = 1/4$ and $\delta \leq 1/4$, the integer $n + 1$ is at least $\log(1/\delta)/(2 \log 2)$ and hence at least $(1 + \log(1/\delta))/4$. For $\delta \geq 1/4$, the initial term $p\delta^2$ is at least $p\delta^2(1 + \log(1/\delta))/(1 + \log 4)$. This proves a stronger constant than the stated one in the critical case.

For $q > 1/4$ and $\delta \leq 1/2$, one has $2^n \delta > 1/4$ and $q^n \geq (2\delta)^{\kappa(q)}$. The last term in (84.7) is therefore at least $p\delta^{\kappa(q)}/16$. For $\delta > 1/2$, the initial term gives $p\delta^2 \geq p\delta^{\kappa(q)}/4$, since $0 < \kappa(q) < 2$. All cases imply (84.6). $\square$

Proof of Theorem 84.1. Lebesgue measure is preserved by the doubling map, and a reset also has Lebesgue law. Thus X_t is uniform for every t . Given the complete history it is known exactly. Scalar quantization of the uniform law by M midpoints has error $1/(12M^2)$. Conversely, after ordering arbitrary centres their Voronoi cells are intervals of lengths a_i with $\sum_i a_i = 1$; the integral error on an interval is at least $a_i^3/12$. Convexity gives $\sum_i a_i^3 \geq M^{-2}$. Random assignment cannot improve on the nearest centre, and conditioning on an independent public seed proves the same bound for randomized encoders and decoders. This establishes (84.3).

For the causal lower bound put $h = 1/(2M)$ and $r = \frac{1}{4}\sqrt{pF_q(h)}/32$. Consider any radius- r ball in ℓ^2 , centred anywhere. Two points of the orbit image lying in this ball have distance at most $2r$. By Lemma 84.3, their preimages cannot be separated by h or more. The preimage of the ball therefore has Lebesgue measure at most h . A union of M such balls covers at most half the acquired mass, and hence

$$e_M^2(Z_q) \geq r^2/2 = \frac{p}{1024}F_q(1/(2M)).$$

For $q \leq 1/4$ this last profile is at least $\Phi_q(M)/4$; for $q > 1/4$ it is $2^{-\kappa(q)}\Phi_q(M) \geq \Phi_q(M)/4$. Thus $c_q = p/4096$ is one possible lower constant. Theorem 83.1 applies to every causal machine and proves the lower assertion including the temporal-mean version. The suffix machine and Lemma 84.2 give the upper bound as already computed. $\square$

The converse uses Euclidean differences only up to the first relevant dyadic separation; it does not replace the doubling map by an invalid global Lipschitz estimate. Nor does it assume that a general machine stores an interval or a binary word. These restrictions are used only for the explicit upper construction. The same algorithm works for all $p \in (0, 1)$, so an unknown reset frequency does not change its order of performance at any fixed true p .

At $p = 1$ every step is a fresh checkpoint and the exact risk is $1/(12M^2)$. The assertion above concerns $p > 0$; its constants need not remain uniform as $p \downarrow 0$. This dependence is displayed in the finite geometric sum (84.5), rather than suppressed in an unspecified stability constant.

APPENDIX 85. AN EXACT CALIBRATION QUOTIENT AND SINGULAR RESOLUTION

The regenerative experiment concerns excess prediction loss relative to a full-history observer. We now study squared error relative to an unknown parameter. The distinction is essential: calibration uncertainty persists even before any label constraint is imposed.

85.1. Correlated paired measurements. Let Θ be a compact metric space, let $f : \Theta \rightarrow \mathbb{R}^m$ be continuous, and observe

$$Y = f(\theta) + \eta + \varepsilon_Y, \quad W = \eta + \varepsilon_W, \quad \theta \in \Theta, \quad \eta \in \mathbb{R}^m. \quad (85.1)$$

The joint centred Gaussian noise has a known positive definite covariance

$$\Sigma = \begin{pmatrix} \Sigma_{YY} & \Sigma_{YW} \\ \Sigma_{WY} & \Sigma_{WW} \end{pmatrix}.$$

No independence between the two channels or their coordinates is imposed. The nuisance η is unrestricted and is not available to the observer. Both measurements are part of the acquisition. The procedure retains one of M labels and decodes an action in a standard Borel action space; independent public and private randomization is allowed.

Put

$$D = Y - W, \quad V = \Sigma_{YY} + \Sigma_{WW} - \Sigma_{YW} - \Sigma_{WY}, \quad (85.2)$$

$$C = \Sigma_{WY} - \Sigma_{WW}, \quad A = CV^{-1}, \quad R = \Sigma_{WW} - CV^{-1}C^\top. \quad (85.3)$$

The matrices V and R are positive definite. A direct Gaussian regression calculation gives the invertible transformation

$$(Y, W) \mapsto (D, Z), \quad Z = W - AD, \quad D \sim N(f(\theta), V), \quad Z \sim N(\eta - Af(\theta), R), \quad (85.4)$$

where D and Z are independent for fixed (θ, η) .

For a bounded nonnegative Borel loss $\ell(\theta, a) \leq L$, write $\mathcal{V}_M^{\text{pair}}$ for the minimax risk in (85.1), with supremum over (θ, η) , and $\mathcal{V}_M^D$ for the minimax risk from the sole observation $D \sim N(f(\theta), V)$, with the same action space, loss, label budget, and randomization convention.

Theorem 85.1 (Exact minimax elimination of additive calibration). *For every integer $M \geq 1$,*

$$\mathcal{V}_M^{\text{pair}} = \mathcal{V}_M^D. \quad (85.5)$$

Thus correlated measurement noise enters the decision problem through the covariance of the observed difference, not through the two marginal variances considered separately.

Proof. Taking differences implements every D -procedure on the paired experiment without changing its label count, so $\mathcal{V}_M^{\text{pair}} \leq \mathcal{V}_M^D$. For the reverse inequality fix any paired procedure and any $T > 0$. Average its risk over an independent nuisance $\eta \sim N(0, T^2 I_m)$. For each θ , this average is bounded above by its worst-nuisance risk. Under this averaging, (85.4) has product law

$$N(f(\theta), V) \otimes N(-Af(\theta), T^2 I_m + R).$$

Replace the second factor by the parameter-independent law $N(0, T^2 I_m + R)$. The relative entropy between the two second factors is at most $\|Af(\theta)\|_2^2 / (2T^2)$, so their total variation distance is at most $\|Af(\theta)\|_2 / (2T)$. This follows also when A is singular. Since $f(\Theta)$ is compact, the bound is uniform in θ .

A D -encoder can sample this parameter-independent replacement Z using private randomness, reconstruct $W = Z + AD$ and $Y = D + W$, and apply the original paired encoder. Its decoder is unchanged and still sees at most M labels. Data processing and boundedness of the loss imply that its maximum risk is at most the maximum paired risk plus

$$\frac{L}{2T} \sup_{\theta \in \Theta} \|Af(\theta)\|_2.$$

Let $T \rightarrow \infty$ and then take the infimum over paired procedures. This proves the opposite inequality in (85.5). $\square$

This is an equality of minimax decision problems for θ under an unrestricted additive nuisance. It is not a sufficiency assertion for (θ, η) , nor an equivalence of the complete experiments for estimating η . The distinction avoids identifying a nuisance-dependent conditional law with a parameter-independent postprocessing kernel.

85.2. Contact geometry in the noise metric. Let now

$$\Theta = \prod_{j=1}^d [0, a_j], \quad a_j > 0, \quad k_j \in \mathbb{N}, \quad k_j \geq 1, \quad v_j > 0.$$

Assume that, for fixed constants $0 < \lambda \leq \Lambda < \infty$,

$$\lambda \|b(\theta) - b(\theta')\|_2 \leq \|V^{-1/2}(f(\theta) - f(\theta'))\|_2 \leq \Lambda \|b(\theta) - b(\theta')\|_2, \quad b_j(\theta) = \theta_j^{k_j} / v_j. \quad (85.6)$$

This is a two-sided condition on the actual observation map in the noise metric. It is stronger than a rank statement at one point, but it allows nonlinear, nonseparable embeddings of the contact coordinates. No coordinate independence of the statistical experiment is required.

With $a_{(1)} \geq \dots \geq a_{(d)}$ the ordered widths, put

$$S = \sum_{j=1}^d \min\{a_j^2, v_j^{2/k_j}\}, \quad Q_M = \max_{1 \leq l \leq d} \left(\frac{\prod_{i=1}^l a_{(i)}}{M} \right)^{2/l}. \quad (85.7)$$

Write $\mathcal{R}_M$ for the minimax squared Euclidean parameter risk in (85.1), with estimates projected onto Θ . Let ω_l denote the volume of the Euclidean unit ball in $\mathbb{R}^l$.

Theorem 85.2 (Correlated, nonseparable contact and label law). *Under (85.6), for every integer $M \geq 1$,*

$$c_{d,\Lambda}(S + Q_M) \leq \mathcal{R}_M \leq 2K_{m,\lambda}S + 8dQ_M, \quad K_{m,\lambda} = \max\{1, 4m/\lambda^2\}. \quad (85.8)$$

One may take

$$c_{d,\Lambda} = \frac{1}{2} \min \left\{ \frac{3c^2}{32}, \min_{1 \leq l \leq d} \frac{1}{2} (2\omega_l)^{-2/l} \right\}, \quad c = \min\{1/2, 1/(2\Lambda)\}.$$

The comparison constants are independent of the contact orders, widths, noise scales, label budget, and unknown calibration. A deterministic minimum-distance estimate followed by one fixed product codebook attains the upper bound.

Proof. The squared loss is bounded on Θ , so Theorem 85.1 reduces the minimax problem exactly to $D \sim N(f(\theta), V)$.

Estimation. Choose a measurable minimizer

$$\tilde{\theta} \in \operatorname{argmin}_{u \in \Theta} \|V^{-1/2}(D - f(u))\|_2.$$

The minimum exists by compactness. For an explicit Borel selection, subdivide the parameter box dyadically. At each level choose the first closed child of the previously chosen box on which the minimum equals the global minimum. Such a child exists. The minimum over any fixed compact box is continuous in D , so all these finitely many equality tests are Borel. The selected nested boxes have diameters tending to zero; their unique limiting point is a Borel minimizer. For $G = V^{-1/2}(D - f(\theta)) \sim N(0, I_m)$, minimality and the triangle inequality imply

$$\|V^{-1/2}(f(\tilde{\theta}) - f(\theta))\|_2 \leq 2\|G\|_2.$$

It follows from (85.6) and $|x - y|^{k_j} \leq |x^{k_j} - y^{k_j}|$ for nonnegative x, y that

$$|\tilde{\theta}_j - \theta_j|^2 \leq \min\{a_j^2, (2v_j \|G\|_2 / \lambda)^{2/k_j}\}.$$

Since $\mathbb{E} \|G\|_2^{2/k_j} \leq m^{1/k_j}$, summing gives $\sup_{\theta} \mathbb{E} \|\tilde{\theta} - \theta\|_2^2 \leq K_{m,\lambda}S$.

Compression. Set $\delta = 2\sqrt{Q_M}$ and divide coordinate j into $n_j = \max\{1, \lfloor a_j/\delta \rfloor\}$ equal intervals. Their product has at most M cells: if l widths exceed δ , then $\prod_j n_j \leq \prod_{i \leq l} a_{(i)}/\delta^l \leq M/2^l$, and the case $l = 0$ uses one cell. Each side length is at most 2δ . The midpoint codebook therefore has squared radius at most $4dQ_M$. Quantize $\tilde{\theta}$ to it. The squared triangle inequality proves the upper bound in (85.8).

Statistical lower bound. Set $r_j = \min\{a_j, v_j^{1/k_j}\}$ and use the 2^d parameter vertices $\theta_j(\xi) = cr_j\xi_j$, $\xi \in \{0, 1\}^d$. For adjacent vertices differing only in coordinate j , (85.6) bounds their Gaussian relative entropy by

$$\frac{1}{2}\Lambda^2(cr_j)^{2k_j}/v_j^2 \leq \frac{1}{2}\Lambda^2c^{2k_j} \leq \frac{1}{8}.$$

Their total variation distance is at most $1/4$. Thresholding an estimate at $cr_j/2$ gives a test of bit j ; an incorrect bit entails squared error at least $c^2r_j^2/4$. Pairing adjacent vertices gives average error probability at least $3/8$, even after compression. Summing over j yields a minimax lower bound $3c^2S/32$.

Representation lower bound. Put uniform probability on Θ . For each independent public seed an averaged decoder has at most M centres, whatever noisy observation produced its label. Projection onto the l widest coordinates shows that M balls of radius r cover at most $M\omega_l r^l / \prod_{i \leq l} a_{(i)}$ of the parameter mass. With $r = (\prod_{i \leq l} a_{(i)} / (2M\omega_l))^{1/l}$, at least half the mass has squared error at least r^2 . Maximizing over l and integrating over the seed gives $\mathcal{R}_M \geq [\min_l \frac{1}{2}(2\omega_l)^{-2/l}]Q_M$. Combining this bound with the statistical bound proves the stated lower constant. $\square$

Example 85.3 (Nonseparable contact observations). In two dimensions let

$$b(\theta) = (\theta_1^2/v_1, \theta_2^3/v_2), \quad H(x_1, x_2) = (x_1, x_2 + \alpha \sin x_1), \quad f(\theta) = V^{1/2}H(b(\theta)).$$

Both H and its inverse have Lipschitz constant at most $1+|\alpha|$. Thus (85.6) holds with $\lambda = (1+|\alpha|)^{-1}$ and $\Lambda = 1+|\alpha|$. Theorem 85.2 gives a matched law with statistical term $\min(a_1^2, v_1) + \min(a_2^2, v_2^{2/3})$ and the two-dimensional anisotropic label floor, despite the coupling of the two mean coordinates. For an actual correlated pair with any chosen positive definite V , take independent Gaussian variables $\xi_D \sim N(0, V)$ and $\xi_R \sim N(0, R)$, with $R > 0$, and any matrix A . Set $\varepsilon_W = A\xi_D + \xi_R$ and $\varepsilon_Y = (I + A)\xi_D + \xi_R$. The joint noise covariance is positive definite, and its observed difference has exactly covariance V .

For a scalar channel, correlation coefficient ρ gives $V = \sigma^2 + \tau^2 - 2\rho\sigma\tau$. Ignoring the cross covariance can therefore change the resolution scale. The independent-coordinate experiment, including its zero-noise endpoints, remains Theorem 88.1. The positive definite hypothesis here is used for the exact Gaussian regression and does not replace those previously proved endpoints.

The observation map and covariance in this section are specified parts of the experiment. The unknown quantity is the additive calibration, not an arbitrary unknown map f or an unknown covariance. The theorem derives the statistical and label rates from the two-sided contact geometry; it does not infer that geometry from a finite derivative rank.

APPENDIX 86. ACQUIRED GEOMETRY AND NONUNIFORM CAUSAL RESOLUTION

86.1. The experiment, the metric, and the resource. Let D_t be separable metric spaces with metrics d_t and their Borel sigma fields. A prepared experiment has a known initial predictive state s_0 and jointly Borel updates

$$s_{t+1} = F_t(s_t, a_t, y_{t+1}), \quad F_t(D_t, a, y) \subset D_{t+1}.$$

All actions admitted at a time have a common continuation interface on D_t . The maps must be defined at every representative used below, not only almost surely under the exact experiment. The current input may include a publicly observed mode, but its past realizations are not available to a decoder unless charged to the retained label.

The physical prediction is $z_t = \Pi_t(s_t) \in \mathbb{R}^{q_t}$, with

$$\|\Pi_t(s) - \Pi_t(s')\|_2 \leq K_t d_t(s, s').$$

For a finite query distribution, coordinates are multiplied by the square roots of the query probabilities. The excess squared prediction loss is then exactly $\|z_t - \hat{z}_t\|_2^2$. This follows by conditioning on the full acquisition history; the cross term with the future outcome has conditional mean zero. The query is drawn only after compression.

Fix an exploration law independent of coding randomness and denote the resulting acquired law of z_t by μ_t . A lower bound always refers to this law. An upper bound may hold for a larger family of feedback policies; in that case its stability hypotheses must hold for that family. A parameter-uniform statement requires the same implemented update for all parameters, as in Section F. Alternatively, a Bayesian unknown parameter can be included in the predictive state.

For each t , choose a fixed Borel quantizer $q_t : D_t \rightarrow C_t \subset D_t$, where $|C_t| \leq M_t$ and

$$d_t(s, q_t(s)) \leq r_t \quad (s \in D_t).$$

Finite covers with a first-minimum tie rule give such maps. Let the implemented update $\tilde{F}_t$ take values in D_{t+1} and have uniform error at most η_{t+1} at representatives. The machine is

$$\hat{s}_0 = s_0, \quad \hat{s}_{t+1} = q_{t+1} \tilde{F}_t(\hat{s}_t, a_t, y_{t+1}). \quad (86.1)$$

Only the index of $\hat{s}_t$ persists. A prescribed sequence of arbitrary codebooks is not automatically a finite uniform program. An infinite-time finite-program assertion below uses a fixed codebook, a periodic finite list, or a separately specified finite algorithm generating and decoding the codebooks.

For a fixed visible input word, let $\Phi_{j,t} : D_j \rightarrow D_t$ be the exact composition of updates from time j to time t , and $\Phi_{t,t} = \text{id}$. Assume specified measurable bounds $\Lambda_{j,t} \geq 0$ satisfy

$$d_t(\Phi_{j,t}(x), \Phi_{j,t}(x')) \leq \Lambda_{j,t} d_j(x, x'), \quad \Lambda_{t,t} = 1. \quad (86.2)$$

The bounds can depend on that word. Taking an uncountable supremum and silently declaring it measurable is unnecessary. The hypothesis is on specified measurable bounds valid for every pair of states in the domain.

86.2. A resolution theorem. Write $[x]_+ = \max(x, 0)$. If $\nu \leq \mu_t$ is a subprobability measure of mass b , put

$$A_\nu(r) = \sup_{c \in \mathbb{R}^{qt}} \nu(B(c, r)), \quad \mathcal{L}_t(M) = \sup_{\nu \leq \mu_t, r > 0} r^2 [b - M A_\nu(r)]_+.$$

One may use any collection of specified submeasures in place of the supremum over all of them; no existence of an optimizing submeasure is asserted or required.

Theorem 86.1 (Acquired-law resolution and exact-suffix stability). *Under the preceding hypotheses the following statements hold.*

(i) *Every checkpoint encoder with at most M labels, including independent public and private coding randomization, has expected excess at least $\mathcal{L}_t(M)$. Consequently every one-machine procedure with stage budgets M_t has maximum expected excess at least $\sup_t \mathcal{L}_t(M_t)$.*

(ii) *Along every admitted common input word the machine (86.1) satisfies*

$$d_t(s_t, \hat{s}_t) \leq \sum_{j=1}^t \Lambda_{j,t} (r_j + \eta_j). \quad (86.3)$$

In particular its expected checkpoint excess is at most

$$K_t^2 \left\| \sum_{j=1}^t \Lambda_{j,t} (r_j + \eta_j) \right\|_{L^2}^2 \leq K_t^2 \left(\sum_{j=1}^t \|\Lambda_{j,t}\|_{L^2} (r_j + \eta_j) \right)^2. \quad (86.4)$$

The probability and all the norms in this assertion are those of the actual run of the implemented controller.

(iii) *Suppose the same integer budget M is used at every time. Let $\psi_t(M) > 0$. If fixed constants $a, A, b > 0$ give covers with $r_t + \eta_t \leq A\psi_t(M)$ and acquired submeasures ν_t of mass at least b such that*

$$M A_{\nu_t}(a\psi_t(M)) \leq b/2, \quad \sup_{t,M} \sum_{j=1}^t K_t \|\Lambda_{j,t}\|_{L^2} \frac{\psi_j(M)}{\psi_t(M)} \leq G < \infty,$$

then checkpoint expected risks are bounded below by $ba^2\psi_t(M)^2/2$, and one causal machine has expected excess at most $A^2G^2\psi_t(M)^2$ at each time. Thus the optimal maximum expected checkpoint risk is comparable to $\sup_t \psi_t(M)^2$. Replacing the L^2 bounds by bounds valid for all histories gives the analogous worst-history upper assertion.

Proof. Condition on an independent public seed. Averaging a randomized decoder conditional on its label and that seed cannot increase squared loss. There are at most M prediction centres $c_1, \dots, c_M$. Randomized assignment cannot improve on the distance to the nearest centre. The ν -mass of their r -balls is at most $MA_\nu(r)$; all remaining mass has squared error at least r^2 . This proves (i), first for a fixed seed and then after integration. The seed is independent of the acquired law. Every causal label is a checkpoint encoding of the entire prefix, so the same lower bounds hold before taking the infimum over causal machines.

For (ii), freeze the actual input word and set $w_j = \Phi_{j,t}(\hat{s}_j)$ for $0 \leq j \leq t$. Then $w_0 = s_t$, $w_t = \hat{s}_t$, and

$$w_{j-1} = \Phi_{j,t}(F_{j-1}(\hat{s}_{j-1}, a_{j-1}, y_j)).$$

The two arguments compared here belong to D_j . Their distance is at most $r_j + \eta_j$. Applying (86.2) and summing the t triangle inequalities proves (86.3). No hypothetical controller is reevaluated at a different state: all suffixes use the same actual word. The Lipschitz readout and the triangle inequality in L^2 give (86.4). The finite-sum argument also applies at each time of an infinite run. Substituting the hypotheses of (iii) into (i) and (ii) proves (iii). In particular the conclusion is $\sup_t \mathbb{E}L_t$, not $\mathbb{E}\sup_t L_t$. $\square$

The theorem deliberately separates the upper and lower geometries. A small-dimensional chart can supply a converse without covering the whole domain; conversely, a small cover does not imply that the experiment acquires mass at that scale. Certified perturbations of the model enter η_j only when measured on the stated domain in its stated metric.

Corollary 86.2 (Finite-horizon anisotropic transfer). *Assume the global geometry, acquired rectangular marginals, and causal compatibility hypotheses (G), (A), (C) of the attainable-geometry theorem in [15]. Its all-integer-budget checkpoint profile and its finite-horizon causal conclusion follow from Theorem 86.1. Explicitly, with its notation,*

$$Q_t(M) = \max_{1 \leq \ell \leq p_t} \left(M^{-1} \prod_{i=1}^{\ell} s_{t,i} \right)^{2/\ell}, \quad d_t(s_t, \hat{s}_t) \leq C \sum_{j=1}^t \left(\prod_{i=j}^{t-1} L_i \right) Q_j(M_j)^{1/2}.$$

The resulting comparison constants are those of the fixed horizon.

Proof. The bounded-format thin-rectangle covering lemma of [15], whose statement and proof are reproduced in the source supplement, gives $r_t(M) \leq C\sqrt{Q_t(M)}$ with representatives in the reachable set. To recall the integer-budget step, its entropy bound is $C \sum_{j=0}^{p_t} r^{-j} \prod_{i \leq j} s_{t,i}$. At $r = L\sqrt{Q_t(M)}$, all terms of positive degree sum to at most $M/2$ for a sufficiently large fixed L . For $M \geq 2C$ the constant term fits the other half; the remaining finitely many budgets are covered by one point and the bound $\sqrt{Q_t(M)} \geq s_{t,1}/M$.

For each acquired rectangular marginal in (A), the mass in a union of M balls of radius r is at most $C_\ell M r^\ell / \prod_{i \leq \ell} s_{t,i}$ of its normalized mass. Choose r a sufficiently small multiple of $(\prod_{i \leq \ell} s_{t,i}/M)^{1/\ell}$. A fixed fraction of the actual subprobability remains. The bounded left inverse of the physical query map and the bounded coordinate transformation transfer this lower bound to prediction error. Maximizing over ℓ gives cQ_t . Zero products need no converse; if every scale vanishes the domain is a singleton. Causal compatibility gives $\Lambda_{j,t} \leq \prod_{i=j}^{t-1} L_i$. Apply (ii), and use the finite number of times for the maximum-risk bound. This reproduces the transfer conclusion; the experiment-specific proofs of (G) and (A) remain those of [15]. $\square$

86.3. Intermittent contraction. Let $\mathcal{H}_t$ contain the visible past through time t . The next criterion is useful when some steps are not contracting at all.

Theorem 86.3 (A conditional block criterion). *Assume each same-input map is nonexpansive. Fix an integer $b \geq 1$ and $0 < p \leq 1$, $0 \leq \rho < 1$. Starting at any deterministic time j , partition the*

subsequent inputs into consecutive blocks of length b . For each block starting at k , suppose there is an event $E_k \in \mathcal{H}_{k+b}$ with

$$\Pr(E_k \mid \mathcal{H}_k) \geq p, \quad \text{Lip}(\Phi_{k,k+b}) \leq \rho \text{ on } E_k,$$

and the block is nonexpansive otherwise. These hypotheses refer to the actual policy or hold uniformly over the specified policy family. Set $q = 1 - p(1 - \rho^2)$. Then valid suffix bounds satisfy

$$\|\Lambda_{j,t}\|_{L^2} \leq q^{\lfloor (t-j)/b \rfloor / 2}, \quad \sum_{j=1}^t \|\Lambda_{j,t}\|_{L^2} \leq \frac{b}{1 - \sqrt{q}}.$$

Thus a fixed radius- r codebook, with numerical defect at most η , gives

$$\sup_{t \geq 1} \mathbb{E} d_t(s_t, \hat{s}_t)^2 \leq \frac{b^2(r + \eta)^2}{(1 - \sqrt{q})^2}.$$

No independence between blocks is required. If every block contracts by ρ on every history, the corresponding pathwise bound is $b(r + \eta)/(1 - \rho)$.

Proof. For a complete block use the factor $Z_k = \rho \mathbf{1}_{E_k} + \mathbf{1}_{E_k^c}$. Its conditional second moment is at most q . For the first m complete blocks after j ,

$$\mathbb{E} \prod_{\ell=0}^{m-1} Z_{j+\ell b}^2 \leq q^m.$$

Indeed the product through the penultimate block is measurable at the start of the last, so conditional expectation removes the last factor with multiplier at most q ; induction completes the argument. The unfinished final block is nonexpansive. Products of these factors therefore give the required measurable suffix bounds. Each possible value of the integer part occurs at most b times in the sum over j . Sum the geometric series and apply (86.4) with the identity readout. The deterministic assertion is the same calculation without expectations. $\square$

Arbitrarily long strings of unsuccessful blocks have positive probability in many examples satisfying this theorem. A uniform expected checkpoint bound therefore cannot be reinterpreted as a pathwise bound by suppressing the norm in the hypothesis. Conversely, expansion at individual steps is allowed by Theorem 86.1 whenever the exact suffix bounds are summable; the nonexpansive block criterion is a sufficient condition, not the definition of the general class.

APPENDIX 87. POSITIVE AND INTERMITTENT HIDDEN MARKOV EXPERIMENTS

87.1. Projective stability. For positive probability vectors define

$$h(p, q) = \max_i \log(p_i/q_i) - \min_i \log(p_i/q_i).$$

It is a metric on the interior of the simplex. For a positive likelihood vector r , normalization of $r_i p_i$ does not change these log-ratios apart from a common additive constant. Bayes multiplication is consequently an isometry in h .

Lemma 87.1 (Positive-matrix contraction). *For a positive matrix A , put*

$$R(A) = \max_{i,j,k,l} \frac{A_{ik}A_{jl}}{A_{il}A_{jk}}, \quad \vartheta(A) = \frac{\sqrt{R(A)} - 1}{\sqrt{R(A)} + 1}.$$

The normalized map $p \mapsto Ap/(\mathbf{1}^\top Ap)$ is $\vartheta(A)$ -Lipschitz in h . A nonnegative matrix mapping positive vectors to positive vectors is nonexpansive. Also $\|p - q\|_1 \leq h(p, q)$.

Proof. We include the finite-dimensional argument for the classical projective estimate [18]. Write $p_i = e^{x_i} / \sum e^{x_l}$. The derivative of $\log \sum_k A_{ik} e^{x_k}$ is the probability row $w_{ik} = A_{ik} e^{x_k} / \sum_l A_{il} e^{x_l}$. For two rows i, j , the ratio w_{ik}/w_{jk} has maximum divided by minimum at most $R(A)$. If probability vectors u, v have ratios in $[m, M]$, with $m \leq 1 \leq M$, convexity of $|x - 1|$ gives

$$\|u - v\|_{\text{TV}} \leq \frac{(M - 1)(1 - m)}{M - m}.$$

Subject to $M/m \leq R$, the right side is at most $(\sqrt{R} - 1)/(\sqrt{R} + 1)$: increasing the endpoint ratio and then putting $M = Rm$ reduces the maximization to $m = R^{-1/2}$. Consequently the difference of two derivative rows applied to a vector v is at most $\vartheta(A) \text{osc}(v)$ in absolute value. Integrate along the segment joining two log-coordinate vectors and take the maximum difference between rows. Normalization subtracts a common constant and does not change oscillation. This proves the contraction. For nonnegative matrices the same derivative argument uses only $\|w_i - w_j\|_{\text{TV}} \leq 1$.

For the last assertion join q to p by the normalized exponential path with logarithmic increment $v_i = \log(p_i/q_i)$. Its derivative has coordinates $p_i(s)(v_i - \sum_l p_l(s)v_l)$, whose ℓ^1 norm is at most $\text{osc}(v)$. Integrating gives the assertion. $\square$

Use the experiment and physical terminal probe in Section G, now with

$$d \geq 1, \quad 0 < \gamma \leq 1, \quad 0 \leq \epsilon \leq 1/4. \quad (87.1)$$

All known constants and both binary report signs retain their original meaning. Write $n = d + 1$, $\tau = 1/4$, and define

$$\rho_{d,\gamma} = \frac{1 - \gamma}{1 - \gamma + 2\gamma/n}, \quad B_\epsilon = \log \frac{1 + \epsilon}{1 - \epsilon}, \quad R_\epsilon = \frac{B_\epsilon}{1 - \rho_{d,\gamma}}, \quad (87.2)$$

$$D_{d,\gamma} = 2^{-d}(5/4)^{d+2}(n/\gamma)^{d+1} \cosh^{2d}(1). \quad (87.3)$$

Let v_d denote the volume of the Euclidean unit ball in $\mathbb{R}^d$.

Theorem 87.2 (The memory law for every positive refresh rate). *For (87.1), every integer $M \geq 1$, and every $t \geq 1$,*

$$\begin{aligned} c_{d,\gamma} \epsilon^2 M^{-2/d} &\leq R_{M,t}^{\text{av}} \leq R_{M,t}^{\text{max}} \leq C_{d,\gamma} \epsilon^2 M^{-2/d}, \\ c_{d,\gamma} \epsilon^2 M^{-2/d} &\leq \mathfrak{R}_{M,\infty}^{\text{av}} \leq \mathfrak{R}_{M,\infty}^{\text{max}} \leq C_{d,\gamma} \epsilon^2 M^{-2/d}. \end{aligned}$$

One possible choice is

$$c_{d,\gamma} = \frac{\tau^2}{2d} (2v_d D_{d,\gamma})^{-2/d}, \quad C_{d,\gamma} = \frac{\tau^2}{d} \left(\frac{64}{3} \right)^2 (1 - \rho_{d,\gamma})^{-4}. \quad (87.4)$$

One deterministic fixed-codebook machine gives the upper bounds for all histories and all times. The lower bounds use independent uniform commands and allow independent coding randomization. At $\epsilon = 0$, one label is exact. If the terminal amplitude is multiplied by ϵ , all bounds acquire the additional factor ϵ^2 .

Proof. Put $a = 1 - \gamma$, $b = \gamma/n$. The transition matrix has diagonal entries $a + b$ and off-diagonal entries b . Its maximum cross-ratio is $((a + b)/b)^2$, attained by two distinct diagonal positions. Lemma 87.1 gives coefficient $a/(a + 2b) = \rho_{d,\gamma}$. Multiplication by the likelihood does not alter h . Since $P_\gamma \nu = \nu$,

$$h(F(p, u, y), \nu) \leq \rho_{d,\gamma} h(p, \nu) + B_\epsilon.$$

Thus $D = \{p : h(p, \nu) \leq R_\epsilon\}$ is an invariant compact subset of the positive simplex. The initialization ν belongs to D .

In log-odds coordinates $x_i = \log(p_i/p_0)$, the domain is exactly $\text{osc}(0, x_1, \dots, x_d) \leq R_\epsilon$ and lies in $[-R_\epsilon, R_\epsilon]^d$. Divide this cube into k^d equal cells, $k = \lfloor M^{1/d} \rfloor$. Select one point of D in each

occupied cell. Two points in a cell have projective distance at most $4R_\epsilon/k$, so the fixed codebook has radius $r_M \leq 8R_\epsilon M^{-1/d}$. Its points are admissible normalized beliefs, although they need not be reachable at a particular time. Quantize after each exact update. The suffix bound is $\rho_{d,\gamma}^{t-j}$, whence

$$\sup_t h(p_t, \hat{p}_t) \leq \frac{8B_\epsilon}{(1 - \rho_{d,\gamma})^2} M^{-1/d}.$$

For $0 \leq \epsilon \leq 1/4$, integration of $2/(1 - \epsilon^2)$ gives $B_\epsilon \leq (8/3)\epsilon$. The physical loss is $(\tau^2/d) \|p_{t,+} - \hat{p}_{t,+}\|_2^2$ and is at most $(\tau^2/d) h(p_t, \hat{p}_t)^2$. This proves the upper constant.

For the converse the pre-report coordinates satisfy $q_i \geq \gamma/n$. The command-to-posterior Jacobian in Lemma G.4, with this floor in place of m_* , gives a density bounded by $D_{d,\gamma} \epsilon^{-d}$ for $p_{t,+}$. For clarity, the actual joint density of command and sign is $2^{-(d+1)} D_y$ and

$$|\det D_u F(p, u, y)_+| = \frac{q_0 \prod_{i=1}^d q_i}{D_y^{d+1}} \epsilon^d \prod_{i=1}^d \operatorname{sech}^2 u_i.$$

The odds $p'_i/p'_0 = q_i(1 + y\epsilon \tanh u_i)/q_0$ make each sign's map injective. Changing variables, summing the two signs, and integrating over every previous history gives exactly (87.3); no conditioning cost is removed. The mass of M Euclidean balls of radius r is at most $M v_d D_{d,\gamma} \epsilon^{-d} r^d$. Take $r = \epsilon(2M v_d D_{d,\gamma})^{-1/d}$. At least half the mass remains, and the physical factor is τ^2/d . This proves the lower constant and, by checkpoint relaxation, its causal version. At zero signal the posterior stays ν . Scaling the terminal probe scales its squared loss, proving the last assertion. $\square$

The dependence on mixing and dimension in (87.4) is part of the statement. Since $1 - \rho_{d,\gamma} = 2\gamma/[n(1 - \gamma) + 2\gamma]$, the upper constant is explicit even for weak refresh. The theorem is uniform in M, t and ϵ for each positive γ ; it does not assert matching powers of γ in a simultaneous $\gamma \downarrow 0$ limit. This is a strict enlargement of the refresh interval, not a relabeling of the old ℓ^1 estimate. At zero refresh the finite-horizon theorem remains available whenever its separate geometry and stability hypotheses hold.

87.2. An experiment with unbounded noncontracting intervals. At each time first draw an observed mode I_t , independently of the hidden state and of all past variables, with $\Pr(I_t = 1) = p > 0$. For $I_t = 1$ execute the preceding refresh and commanded observation. For $I_t = 0$ apply any prescribed permutation of the hidden states and return an uninformative fair sign. Commands are independent uniform for the converse. The upper construction accepts every admitted command word. The mode is read in the current update, then discarded except for its effect on the retained label. The terminal probe is unchanged.

Corollary 87.3 (Intermittent-refresh expected resolution). *In this experiment, for every integer $M \geq 1$,*

$$p c_{d,\gamma} \epsilon^2 M^{-2/d} \leq \mathfrak{R}_{M,\infty}^{\text{av}} \leq \frac{\tau^2}{d} \frac{(64/3)^2 \epsilon^2 M^{-2/d}}{(1 - \rho_{d,\gamma})^2 (1 - \sqrt{1 - p(1 - \rho_{d,\gamma}^2)})^2}.$$

The same lower bound holds at every positive checkpoint. The upper bound uses one fixed codebook. For $0 < p < 1$, arbitrarily long noncontracting intervals are admitted.

Proof. Permutations fix ν and preserve h ; hence the invariant domain D and its cover from the preceding proof work in both modes. A good step contracts by $\rho_{d,\gamma}$, a bad step is an isometry. Theorem 86.3 with $b = 1$ gives the upper bound. Conditional on the current good mode, the density calculation in the preceding proof is uniform over the prior history. The good-mode submeasure has mass p and density at most $p D_{d,\gamma} \epsilon^{-d}$. The same balls leave at least $p/2$ of its mass, proving the lower bound. Independent Bernoulli modes permit every finite string of bad steps. $\square$

87.3. Discrete and singular acquisition. For a finite command alphabet of size K , a fixed time t has at most $(2K)^t$ command/sign histories in the ordinary refreshing experiment. Its checkpoint prediction is therefore exactly encodable at that budget. This elementary fact precludes a positive $M^{-2/d}$ lower bound for all M at fixed t in that different experiment.

A singular command law has a different outcome. Let $d = 1$, $\gamma = 1$, and let $u = 2V - 1$, where V is the standard middle-thirds Cantor random variable. Set $s = \log 2 / \log 3$. The hidden state is freshly prepared at each step, but the binary command observation and the terminal probe are still physical observations as above.

Proposition 87.4 (A singular acquired dimension). *For this experiment there are constants $0 < c < C < \infty$ such that for every $t \geq 1$, $M \geq 1$, and $0 \leq \epsilon \leq 1/4$,*

$$c\epsilon^2 M^{-2/s} \leq R_{M,t}^{\text{av}} \leq \mathfrak{R}_{M,\infty}^{\text{max}} \leq C\epsilon^2 M^{-2/s}.$$

Here worst histories use commands in the support of the prescribed Cantor law. One fixed codebook attains the upper bound.

Proof. A level- m Cantor cylinder has length 3^{-m} and mass 2^{-m} . At that scale an interval of length comparable to 3^{-m} meets only a bounded number of cylinders. Thus the mass of an interval of radius r is at most Cr^s , while the support has a cover by 2^m intervals of length 3^{-m} . Affine rescaling to $[-1, 1]$ changes only constants.

For either sign the posterior coordinate is $(1 + y\epsilon \tanh u)/(2 + y\epsilon \tanh u)$. Its derivative in absolute value is $\epsilon \text{sech}^2 u / (2 + y\epsilon \tanh u)^2$, between two fixed positive multiples of ϵ when $\epsilon > 0$. Each sign therefore gives a bi-Lipschitz image at scale ϵ . The actual sign weight is between $3/8$ and $5/8$. The sum of the two image measures has concentration at most $C(r/\epsilon)^s$; each image has the corresponding covers. Allocate half the available centres to each image, adjusting the fixed constant for the finitely many budgets below four. The acquired-law lower bound and this cover give the asserted rate. Since $\gamma = 1$, the new posterior does not depend on the old label. The same quantizer is an exact one-step causal construction at every time. At $\epsilon = 0$ it is constant. $\square$

Neither the domain dimension of the command nor a Jacobian rank alone determines these three rates. For a mixture with finitely many additional atoms, assign one centre to each atom and use the remaining centres for the singular component; the submeasure converse retains its actual mixture weight. More generally Theorem 86.1 uses the full concentration profile without assigning a single dimension.

APPENDIX 88. SINGULAR STATISTICAL RESOLUTION WITH MEASURED CALIBRATION

We now change the target and the loss explicitly. Let $\theta \in \Theta = \prod_{j=1}^d [0, a_j]$, $a_j > 0$, and let $\eta \in \mathbb{R}^d$ be an unrestricted unknown additive calibration. Fix integers $k_j \geq 1$. The experiment consists of the $2d$ observations

$$Y_j = \theta_j^{k_j} + \eta_j + \sigma_j Z_j, \quad W_j = \eta_j + \tau_j Z'_j, \quad 1 \leq j \leq d, \quad (88.1)$$

with independent standard normal variables and known $\sigma_j, \tau_j \geq 0$. A procedure observes the pair, retains one of M labels, and decodes an estimate of θ . Only the label and independent public coding randomness are available to the decoder. Both Y and W count as acquisitions. In particular W is not a free read-only value of η .

Put $v_j = (\sigma_j^2 + \tau_j^2)^{1/2}$, and let $a_{(1)} \geq \dots \geq a_{(d)}$ be the ordered widths. Define

$$S = \sum_{j=1}^d \min\{a_j^2, v_j^{2/k_j}\}, \quad Q_M = \max_{1 \leq \ell \leq d} \left(M^{-1} \prod_{i=1}^{\ell} a_{(i)} \right)^{2/\ell}. \quad (88.2)$$

The risk here is squared parameter error, not the prediction excess used in the preceding sections.

Theorem 88.1 (A matched contact, calibration, and label law). *For the actual experiment (88.1), write*

$$\mathcal{R}_M = \inf_{M\text{-label procedures}} \sup_{\theta \in \Theta, \eta \in \mathbb{R}^d} \mathbb{E}_{\theta, \eta} \|\hat{\theta} - \theta\|_2^2.$$

For every integer $M \geq 1$,

$$c_d(S + Q_M) \leq \mathcal{R}_M \leq 2S + 8dQ_M, \quad (88.3)$$

where one may take

$$c_d = \frac{1}{2} \min \left\{ \frac{3}{128}, \min_{1 \leq \ell \leq d} \frac{1}{2} (2\omega_\ell)^{-2/\ell} \right\},$$

where ω_ℓ denotes the volume of the Euclidean unit ball in $\mathbb{R}^\ell$. The constants are independent of $a_j, k_j, \sigma_j, \tau_j, M$ and the unknown calibration. The upper construction is deterministic. The lower bound allows independent public and private randomization.

Proof. Estimation before compression. The actually observed differences $D_j = Y_j - W_j$ have laws $N(\theta_j^{k_j}, v_j^2)$ independent of η . Define

$$\tilde{\theta}_j = (\text{proj}_{[0, a_j^{k_j}]} D_j)^{1/k_j}.$$

For nonnegative x, y , $|x^{1/k} - y^{1/k}| \leq |x - y|^{1/k}$. Projection cannot increase the distance to $\theta_j^{k_j}$. Concavity of x^{1/k_j} and $\mathbb{E}Z^2 = 1$ consequently give

$$\mathbb{E}|\tilde{\theta}_j - \theta_j|^2 \leq \min\{a_j^2, v_j^{2/k_j} \mathbb{E}|Z|^{2/k_j}\} \leq \min\{a_j^2, v_j^{2/k_j}\}.$$

This is a finite-sample bound, including degenerate zero-noise coordinates.

An integer-budget cover. Set $\delta = 2\sqrt{Q_M}$ and partition coordinate j into $n_j = \max\{1, \lfloor a_j/\delta \rfloor\}$ equal intervals. Let ℓ be the number of widths greater than δ . If $\ell = 0$ the product of the n_j is one. Otherwise

$$\prod_j n_j \leq \prod_{i \leq \ell} a_{(i)}/\delta^\ell \leq M/2^\ell \leq M.$$

Every interval has length at most 2δ , so its midpoint product codebook has squared covering radius at most $d\delta^2 = 4dQ_M$. Quantize $\tilde{\theta}$ to this codebook. The squared triangle inequality proves the upper bound in (88.3).

The statistical obstruction with unknown calibration. Write $r_j = \min\{a_j, v_j^{1/k_j}\}$. Coordinates with $r_j = 0$ contribute zero and can be omitted from this step. For $\omega \in \{0, 1\}^d$, set

$$\theta_j(\omega) = \frac{1}{2}r_j\omega_j, \quad m_j = (r_j/2)^{k_j}, \quad \eta_j(\omega) = -\frac{\tau_j^2}{v_j^2}m_j\omega_j.$$

These are admissible alternatives in the same unknown-calibration experiment. Adjacent vertices differ in the mean pair at coordinate j by $(m_j\sigma_j^2/v_j^2, -m_j\tau_j^2/v_j^2)$. Their Gaussian divergence is

$$\frac{1}{2} \left(\frac{m_j^2\sigma_j^2}{v_j^4} + \frac{m_j^2\tau_j^2}{v_j^4} \right) = \frac{m_j^2}{2v_j^2} \leq \frac{1}{8}.$$

A zero-variance channel in this expression has identical means and is simply omitted. Pinsker's inequality bounds the variation distance by $1/4$. Any estimate determines a test of bit j by thresholding at $r_j/4$. An incorrect test has squared coordinate error at least $r_j^2/16$. Pair adjacent vertices and average: their average error probability is at least $(1 - 1/4)/2 = 3/8$. Summing over coordinates gives average squared risk at least $3 \sum_j r_j^2/128$. Compression and independent randomization cannot improve the testing distance. Since maximum risk dominates this finite average, $\mathcal{R}_M \geq 3S/128$.

The representation obstruction. Fix $\eta = 0$ and put uniform probability on the parameter box. At each fixed public seed an averaged decoder has at most M centres. Regardless of what noisy observations led to the label, its error at θ is at least the distance to the nearest centre. Project onto the ℓ widest coordinates. The fraction of this rectangular uniform law covered by M balls of radius r is at most $M\omega_\ell r^\ell / \prod_{i \leq \ell} a_{(i)}$. Choose $r = (\prod_{i \leq \ell} a_{(i)} / (2M\omega_\ell))^{1/\ell}$. At least half the mass has squared error at least r^2 . Maximize over ℓ and integrate over the seed. This proves $\mathcal{R}_M \geq [\min_\ell \frac{1}{2}(2\omega_\ell)^{-2/\ell}]Q_M$. Combining the two lower bounds by $\max(x, y) \geq (x + y)/2$ proves the result. $\square$

The testing alternatives explain why the calibration variance adds to the signal variance. Holding the nuisance artificially fixed in that step would prove only the weaker obstruction from σ_j , and would miss the cost of measuring calibration. The encoder never reads the unknown nuisance. Its read-only constants are the specified noise levels, contact orders, and parameter box.

For $d = 1$, $a_1 = 1$, and $v_1 \asymp N^{-1/2}$, the theorem gives $\mathcal{R}_M \asymp N^{-1/k_1} + M^{-2}$. Hence the statistical amplitude scale is $N^{-1/(2k_1)}$, and order $N^{1/(2k_1)}$ labels suffice to reach it. When $k_1 = 2$, this recovers the quadratic-contact amplitude scale in the physical-ray result of [16], now with an explicit finite-label constraint and an actually measured unrestricted additive calibration in (88.1). It is a comparison of rates in these specified experiments, not an assertion of their Le Cam equivalence. The nonnegative branch is part of the parameter space: an even-power observation on a signed branch would have an additional identifiability obstruction.

APPENDIX 89. OPTIMIZATION UNDER A RETAINED-LABEL CONSTRAINT

The following result supplies a policy comparison in addition to the common-controller statements of Section F. Let D be a compact metric predictive-state domain and $\mathcal{A}$ a finite action set. The report kernel is $g(dy | s, a)$ and the state update is $F(s, a, y) \in D$. Assume these are Borel, and, uniformly in a, y ,

$$0 \leq c(s, a) \leq C_0, \quad |c(s, a) - c(s', a)| \leq L_c d(s, s'), \quad d(F(s, a, y), F(s', a, y)) \leq l d(s, s'),$$

$$\|g(\cdot | s, a) - g(\cdot | s', a)\|_{\text{TV}} \leq \kappa d(s, s').$$

The current conditional expected physical cost is $c(s, a)$. The predictive state is sufficient for future costs and experiments under all admitted policies. Thus the belief-state dynamic program represents the optimum among all history-dependent policies. Fix discount $0 < \beta < 1$ with $\beta l < 1$, and define

$$L_* = \frac{L_c + \beta \kappa C_0 / (1 - \beta)}{1 - \beta l}.$$

Let $\mathcal{C} \subset D$ have at most M points and a radius- r Borel quantizer q . Assume the initial state is in $\mathcal{C}$; it can be reserved as a centre. The finite reference decision process uses cost $c(c, a)$ and next state $qF(c, a, y)$ with report distribution $g(dy | c, a)$. It has an optimal stationary policy $\hat{a}(c)$ because its state and action spaces are finite. Its transition probabilities are the specified integrals of these kernels; numerical approximation is addressed below.

Theorem 89.1 (A finite reference policy versus unrestricted control). *Implement $a_t = \hat{a}(\hat{s}_t)$ in the actual experiment and update $\hat{s}_{t+1} = qF(\hat{s}_t, a_t, y_{t+1})$. Let V^* be the unrestricted optimal discounted value and $J(\hat{a})$ the actual cost of this retained-label controller. Then*

$$0 \leq J(\hat{a}) - V^*(s_0) \leq 2L_* \sum_{t \geq 0} \beta^t \mathbb{E} d(s_t, \hat{s}_t) + \frac{2\beta L_* r}{(1 - \beta)^2}. \quad (89.1)$$

In particular, if the fixed-codebook construction has $\sup_t \mathbb{E} d(s_t, \hat{s}_t) \leq Gr$, the upper bound is $2L_ Gr / (1 - \beta) + 2\beta L_* r / (1 - \beta)^2$. Only one of M labels persists; no exact value function or full history is an input to the implemented policy.*

Proof. The exact Bellman operator on bounded Borel functions is

$$TV(s) = \min_a \{c(s, a) + \beta \int V(F(s, a, y))g(dy \mid s, a)\}.$$

It is a contraction in the uniform norm. Starting at zero, its iterates have values in $[0, C_0/(1 - \beta)]$. If an iterate has Lipschitz constant L , the next has constant at most $L_c + \beta lL + \beta \kappa C_0/(1 - \beta)$: compare the same report argument first and then the two report laws, using the range bound on V . The minimum over the same finite action set preserves the bound. It follows that V^* is L_* -Lipschitz, and each optimal action-value function $Q^*(\cdot, a)$ is also L_* -Lipschitz.

Let $\widehat{V}$ be the finite reference value. At a centre, replacing $F(c, a, y)$ by $qF(c, a, y)$ in V^* changes a Bellman expression by at most $\beta L_* r$. The finite Bellman contraction therefore implies

$$\sup_{c \in \mathcal{C}} |\widehat{V}(c) - V^*(c)| \leq \frac{\beta L_* r}{1 - \beta} =: D_0.$$

For every centre and action, the exact action value and the finite reference action value differ by at most $\beta(L_* r + D_0) = \beta L_* r/(1 - \beta) = D_0$. Minimizing the latter consequently yields

$$Q^*(c, \widehat{a}(c)) - V^*(c) \leq 2D_0.$$

For an actual state s and retained centre c , Lipschitz continuity of both Q^* and V^* gives $Q^*(s, \widehat{a}(c)) - V^*(s) \leq 2L_* d(s, c) + 2D_0$.

Along the actual trajectory, the conditional definition of Q^* makes the discounted sum of these residuals equal to $J(\widehat{a}) - V^*(s_0)$. More explicitly the finite sum telescopes, and its final term has absolute value at most $\beta^{T+1} C_0/(1 - \beta)$, tending to zero. Sum the bound and let T increase. This proves (89.1). The policy is a finite table indexed by the retained representative. The exact value entered only this comparison proof. $\square$

A stationary centre may be reserved at a cost of one label: for $M \geq 2$ use the preceding covers with $M - 1$ centres and add s_0 . A power-law covering rate is unchanged up to its stated fixed factor; for $M = 1$ the bounded-cost estimate is available. If an implemented finite table is within ξ in each action value of the exact reference table, choosing its minimizing action adds at most $2\xi/(1 - \beta)$ to the bound. This numerical certificate is separate from computing that table. Theorems 86.1 and 86.3 provide G when their assumptions hold under the lifted policy. A prediction lower bound is not a matching lower bound for every control cost; no such identification is used here.

APPENDIX 90. FINITE DESCRIPTIONS AND NUMERICAL IMPLEMENTATION

We distinguish five resource coordinates: persistent labels, program description, transient workspace, numerical error, and physical acquisitions. The first coordinate is determined by the resolution laws; the others are not silently identified with it. We give a finite-description construction for the positive-refresh upper bound, so its codebook does not require an independently stored list of M arbitrary real vectors.

Suppose γ, ϵ are rational known inputs and $\epsilon > 0$. Choose a rational R' with $R_\epsilon \leq R' \leq 2R_\epsilon$. Such a number is computable from the displayed elementary formula. For an integer $k \geq 1$, set $\delta = R'/k$ and use the log-odds grid

$$\mathcal{J}_k = \{j \in \{-k, \dots, k\}^d : \text{osc}(0, j_1, \dots, j_d) \leq k\}, \quad p_i(j) = \frac{e^{\delta j_i}}{1 + \sum_{l=1}^d e^{\delta j_l}}, \quad (90.1)$$

with p_0 the reciprocal denominator. Its domain is the invariant ball of radius R' .

Proposition 90.1 (An implicit codebook). *There is a finite-description codebook in this ball with at most M representatives and projective radius at most $12R'M^{-1/d}$. Its persistent state uses at most $\lceil \log_2 M \rceil$ bits. Its read-only description consists of d , the rational calibration, R' ,*

an integer k , and fixed algorithms for enumeration and elementary functions; it contains no list of M arbitrary real numbers. Certified update error η and quantization error at most $r + \eta$ give projective filtering error at most $(r + 2\eta)/(1 - \rho_{d,\gamma})$.

Proof. For $M \geq 3^d$ choose $k = \lfloor (M^{1/d} - 1)/2 \rfloor$. Then $(2k+1)^d \leq M$ and $k \geq M^{1/d}/6$. Round each log-odds coordinate toward zero in multiples of δ . The result remains in the oscillation ball: when both signs occur, both extreme absolute values decrease; when all coordinates have one sign, their extreme absolute value decreases. Each coordinate changes by at most δ , so projective distance changes by at most $2\delta \leq 12R'M^{-1/d}$. For $M < 3^d$ use only ν ; its radius R' is at most $3R'M^{-1/d}$. Lexicographic enumeration of the finite integer set labels the grid. The model and this enumeration are finite descriptions of all its representatives. The stated error is the geometric-series estimate, with total one-step defect $r + 2\eta$. $\square$

All log-odds and softmax functions in this construction are computable on their compact domains. With digitally represented commands, interval arithmetic can evaluate the update and finitely many distances until a representative within $r + \eta$ is certified, for any prescribed positive rational η . The fixed positive slack ensures termination even at a tie; exact comparison of two computable reals is not required. This is an existence and accuracy construction, not a uniform polynomial-time claim. The bound charges the counter needed to enumerate the grid and the precision of the computed numbers as transient workspace. The required precision can depend on $d, \gamma, \epsilon, M, \eta$.

A command rounded with coordinate error at most h changes each log likelihood by at most $\epsilon h/(1 - \epsilon)$ and hence changes the posterior projective distance by at most $2\epsilon h/(1 - \epsilon)$. This supplies an explicit additional one-step defect. It does not replace the analog acquisition law in the lower bound by a finite alphabet while keeping the same converse. The sampling law and the numerical interface must be declared separately. For a noncomputable real calibration, an exact read-only real convention would be a different resource model. The finite-description assertion above uses rational calibration and a numerical certificate instead.

The paired singular experiment uses $2d$ scalar measurements and one terminal label. Its box codebook is generated by d integer subdivision counts rather than an arbitrary centre list. Computing the differences and roots uses transient numerical workspace. A controller in Theorem 89.1 additionally has a read-only policy table; its construction and precision are not included in the label count.

APPENDIX 91. COMPARISON OF RESULTS AND MATHEMATICAL DEPENDENCIES

The relationship to earlier work is most precise at the level of the conclusion and the retained information.

Antecedent	Existing conclusion used for comparison	Conclusion proved here
[15], attainable-geometry transfer	Anisotropic acquired-law profile and a compatible finite-horizon label machine	Exact-suffix bounds and intermittent infinite-horizon resolution; the old transfer is Corollary 86.2
[8]	Quantization methods for nonlinear filtering	A matching converse for acquired prediction labels, not merely a finite signal grid
[19, Theorems 12, 16, 17]	Finite-memory policy approximation from filter stability	A policy computed on a finite representative model with an explicit retained-label budget and actual-state error term
[20, Theorems 9, 27]	Approximate information states and approximate dynamic programs with performance bounds	Theorem 89.1 is a specialized quantitative policy-transfer result, used together with the acquired-law converse
[16], physical-ray and score experiments	Quadratic-contact scale, Gaussian comparison, information retained by contact scores	Finite-sample contact orders, unrestricted measured additive calibration, and a matched label floor in one specified experiment

Recent finite-window developments include continuous-space approximations [21] and refined window-policy bounds [22]. The fixed-policy belief-cost comparison in [23] is another nearby, but distinct, target. None is replaced by a claim that the stability recurrence, quantization principle, or Bellman comparison was invented here. The specific mathematical assertions to compare are the joint hypotheses and conclusions of Theorems 86.1, 86.3, 87.2, and 88.1.

There is no backward dependence on a later theta manuscript. The only model-specific imported geometric estimate used in a corollary is the bounded-format covering result of the fixed A1 edition; the new principal theorems have proofs here. The A2 comparisons keep its physical cone and noise experiment distinct from (88.1). The earlier Sinai/vector-roof, particle large-deviation, kinetic-semigroup, unbounded operator, and labelled-phase problems keep their original dependency order. A proof of finite-label filtering does not assert any of those model-specific conclusions. The contribution to those chains is a proved resolution and decision interface, usable after their own experiment and domain hypotheses have been established.

FOUNDATIONAL DEVELOPMENT AND BOUNDARY EXAMPLES

The following appendices give the measurable, statistical, and asymptotic foundations used above. Their statements and proofs are retained from the first edition. The strong-refresh result in Appendix G is extended by Theorem 87.2; the additional statistical results are Theorems 85.1 and 85.2. The final interface appendix distinguishes these results from separate model-specific spectral, large-deviation, and operator problems.

APPENDIX A. POSITIVE CAUSAL EXPERIMENTS

A.1. Types and primitive data. A standard Borel space is a measurable space isomorphic to a Borel subset of a Polish space. All primitive spaces below are standard Borel. Time is initially $t = 0, \dots, N$, with the countable-time extension stated separately. The unknown parameter is $\lambda \in \Lambda$; it is not a random variable until a prior is supplied. The hidden state, applied action, and next report have spaces X_t, A_t, Y_{t+1} . A visible history is

$$h_t = (a_0, y_1, \dots, a_{t-1}, y_t) \in H_t.$$

A public initial report, clock increments, and observed cost records can be included in this notation. An action-availability rule, when present, is part of the interface. We usually write a common action space to avoid encoding that rule repeatedly.

Definition A.1 (Raw experiment). A raw experiment consists of a parameter space Λ , preparations $\mu_\lambda \in \mathcal{P}(X_0)$, and jointly measurable probability kernels

$$K_t^\lambda(dx', dy, dc \mid x, a) \quad (\text{A.1})$$

on $X_{t+1} \times Y_{t+1} \times \mathcal{C}$, together with a class of executable future tests and a resource interface. The nonnegative vector c records charged increments, such as raw preparations or physical time. When it is observed, it is included in y ; when it is not observed, it is integrated in the visible law but is still counted in the resource account. Every kernel has total mass one. A failed measurement and a terminating measurement are reports, not missing mass.

The following six requirements will be referred to as the foundational axioms. They specify the admissible language rather than analytic conclusions.

- A0 (measurable interfaces).** Primitive spaces are standard Borel; histories and continuation availability are specified.
- A1 (positive experiment).** Preparation and transition–observation kernels are nonnegative and normalized as in (A.1).
- A2 (causality).** The next action uses only the visible past and permitted independent randomness. Stopping is nonanticipative.
- A3 (executable tests).** The designated future tests have actual protocols and are closed under the one-step extensions needed for prediction.
- A4 (preparation and accounting).** Average risks and acquisition laws refer to a declared preparation and exploration strategy. Failures, pilots, and accepted-event probabilities are not removed from their costs.
- A5 (charged implementation).** Persistent labels, numerical precision, workspace, read-only calibration, and simulator state have separate accounts.

In particular these axioms do not assert strict positivity of each report, finite-dimensionality, stationarity, differentiability, a spectral gap, or a large-deviation principle. If the physical state is not Markov in the coordinates initially proposed, its entire necessary past can be included in X_t . This does not prove closure for a smaller observable state.

A.2. Strategies, paths, and stopping. A strategy is a sequence of measurable kernels $\alpha_t(da \mid h_t)$ supported on available actions. It cannot depend on the unknown true parameter. A finite-state strategy may have an additional explicit internal state; an unrestricted history strategy need not satisfy a finite-memory budget.

Proposition A.2 (Path construction). *For every parameter and causal strategy there is a unique consistent finite-dimensional path law with iterated-kernel expression*

$$\mu_\lambda(dx_0) \prod_{t=0}^{N-1} \alpha_t(da_t \mid h_t) K_t^\lambda(dx_{t+1}, dy_{t+1}, dc_{t+1} \mid x_t, a_t). \quad (\text{A.2})$$

For countably many discrete times these laws extend uniquely to the product path sigma field. Deterministic evolution and deterministic observations are included by taking Dirac kernels.

Proof. For finite N , integrate bounded measurable cylinder functions from the last variable backwards. At each step measurability follows from the kernel property, positivity is preserved, and the integral of one is one. The resulting probability measures have consistent marginals because the last kernel integrates to one. The standard kernel-extension theorem gives the countable product law; this is the Ionescu–Tulcea construction, as treated in [4]. Uniqueness follows first on

cylinder events and then on their generated sigma field. A Dirac kernel makes the corresponding integral substitution along the deterministic trajectory, without changing the remaining preparation or measurement randomness. $\square$

A bounded stopping time is implemented by an absorbing stopped symbol and zero subsequent increments. Thus the same construction handles optional termination. For an unbounded stopping time, an almost-surely finite stopped path can be considered by truncation; quantitative infinite-horizon error bounds still require a summable bound or a separate limiting argument. A random number of collisions is not silently substituted for a deterministic large-deviation speed.

A.3. Prepared and parameter-uniform questions. For statistical decision theory we retain the whole family $(P_\lambda^\alpha)_\lambda$. A decision rule is one rule, independent of the true parameter. For Bayesian prediction we instead specify a prior $\eta(d\lambda)$ and use the joint hidden state (λ, x_t) with preparation $\eta(d\lambda)\mu_\lambda(dx_0)$. Fixed known calibration is permitted, but it is not a substitute for an unknown parameter.

This distinction also applies to infima over encoders. The infimum for a fixed preparation need not be achieved by the same encoder for all priors. Likewise, a family of codebooks indexed by a known experimental calibration is not an estimator of an unknown calibration. Every quantitative theorem below states which situation is intended.

A.4. Instruments and evidence. After a preparation has been fixed, write the hidden state simply as x . For a report event $B \subset Y_{t+1}$, define the positive instrument

$$(\mathbb{T}_{t,a,B}f)(x) = \int \mathbf{1}_B(y)f(x')K_t(dx', dy \mid x, a).$$

It is countably additive in B for nonnegative f , positive in f , and $\mathbb{T}_{t,a,Y}1 = 1$. With finite reports write $\mathbb{T}_{t,a,y} = \mathbb{T}_{t,a,\{y\}}$. Starting with $b_0 = \mu$, define the unnormalized history functional by

$$b_{t+1}(f) = b_t(\mathbb{T}_{t,a_t,y_{t+1}}f), \quad Z_t = b_t(1). \quad (\text{A.3})$$

These functionals do not include the probability of the chosen command. They are likelihood/evidence functionals conditional on the realized command sequence; the actual joint history law also includes the strategy kernels in (A.2).

Proposition A.3 (Normalized update). *For finite reports and $Z_t > 0$, let $\pi_t = b_t/Z_t$. If $q_t(y \mid \pi_t, a) = \pi_t(\mathbb{T}_{t,a,y}1) > 0$, then*

$$\pi_{t+1}(f) = \frac{\pi_t(\mathbb{T}_{t,a,y}f)}{\pi_t(\mathbb{T}_{t,a,y}1)}, \quad Z_{t+1} = Z_t q_t(y \mid \pi_t, a). \quad (\text{A.4})$$

The formula remains valid for adaptive actions. When the evidence is zero, the unnormalized functional is zero and no normalized posterior is specified by this formula.

Proof. Divide (A.3) by its value at one. Positivity shows that a positive functional of total mass zero vanishes on every bounded test. For adaptive actions, condition on the visible history: the strategy's probability of the current action is fixed by that history and is independent of the hidden state conditional on it. This factor cancels in the posterior. An induction in (A.2) verifies the conditional expectation identity for every bounded hidden-state test. $\square$

For a non-atomic report space, conditioning is by disintegration of the joint next-state/report law, not by dividing the mass of a singleton. Under a fixed preparation a standard Borel conditional distribution exists almost everywhere. A single Borel update jointly in all proposed states, inputs, and reports is additional implementation data, or must be justified by an applicable parametrized disintegration theorem. We will not use a simultaneous version across a nondominated parameter family without that justification. All explicit quantitative models below have finite reports and Borel formulas.

A.5. Preparing a conditional experiment. If an event has acceptance probability $p > 0$, independently repeating a preparation until acceptance implements the conditional law. Its preparation count T satisfies $\mathbb{P}(T > n) = (1-p)^n$ and $\mathbb{E}T = 1/p$. A cap B instead creates an abort symbol with probability $(1-p)^B$. This is a different experiment from exact conditioning. The cap is necessary for a uniform pathwise preparation budget. No lower bound or implementation claim below replaces an actual report word by a probability-one conditioning event.

APPENDIX B. FUTURE TESTS AND PREDICTIVE QUOTIENTS

B.1. Operator words and adaptive protocols. A finite future protocol starts from the present time, uses available actions causally, and returns a bounded terminal readout. Its conditional expected readout, as a function of the current hidden state, is a future test. A terminal hidden-state probe is allowed only if an actual measurement kernel producing it is part of the protocol. For finite report alphabets, a deterministic depth- m protocol is a tree of actions and terminal readouts. Its test has the recursion

$$f = \sum_y \mathsf{T}_{t,a,y} f_y. \quad (\text{B.1})$$

Randomized protocols integrate such recursions. Let $\mathcal{V}_t$ be the real linear space generated by the designated future tests, including constants. We require the one-step closure

$$\mathsf{T}_{t,a,y} \mathcal{V}_{t+1} \subset \mathcal{V}_t \quad (\text{B.2})$$

for the continuations being retained. Where full future laws are intended, the tests include indicator readouts from a determining event class and its needed bounded-monotone closure.

A fixed finite menu used to assess prediction loss can be much smaller than $\mathcal{V}_t$. Such a menu does not automatically define a recursively sufficient state. One must either prove its closure/completeness or keep a larger predictive state for the online construction.

Definition B.1 (Predictive equivalence). Two prepared beliefs at time t are predictively equivalent when they have the same continuation interface and

$$\pi \equiv_t \tilde{\pi} \iff \pi(f) = \tilde{\pi}(f) \quad (f \in \mathcal{V}_t).$$

The continuation interface includes the legal future action tree, remaining budget, and any public clock that affects future tests. The predictive quotient is the set of these equivalence classes, or equivalently the image of $\pi \mapsto (\pi(f))_{f \in \mathcal{V}_t}$ together with that interface.

Theorem B.2 (Minimal future prediction). *Under (B.2), predictively equivalent beliefs give the same laws for all designated future protocols. After an action and a report of positive probability, their updated beliefs remain predictively equivalent. Conversely, equality of those protocol laws implies equality of their future tests. If a deterministic history statistic r_t determines all these predictions and the continuation interface, the predictive state factors uniquely through r_t on its attained image.*

Proof. For a one-step report, its probability is the value of $\mathsf{T}_{t,a,y} 1 \in \mathcal{V}_t$, and is therefore common. For a next-state test f , both the numerator and denominator of (A.4) have the same values under equivalent beliefs. The conditional next predictions are equal. Induct over the depth of a protocol tree using (B.1). Randomized action choices use the same conditional randomization at corresponding visible histories and can be integrated. Bounded monotone convergence extends equality from finite determining tests to the designated generated events. Conversely, expectations of each designated bounded readout are determined by its law, hence agree; equality extends linearly to $\mathcal{V}_t$.

If $r_t(h) = r_t(\tilde{h})$, sufficiency of the statistic for these tasks implies equality of all future predictions and interfaces. Thus each fibre of r_t is contained in a predictive-equivalence class. Define the

factor on the image of r_t by the predictive class of any history in that fibre. It is well defined and is the unique such map. This last assertion is a set-theoretic universal property. Its measurable version requires an actual measurable quotient realization, as specified below. $\square$

The theorem does not give a normalized continuation at a zero-probability report. One can extend an update arbitrarily there for a chosen measurable implementation, but such an extension is not canonical information. For worst-history statements, admissible histories and these extensions must be explicitly specified.

B.2. Safe measurable realizations. The complete posterior on a standard Borel hidden space is a safe measurable state under a fixed experiment and appropriate conditional versions. Its predictive quotient may be simpler, but not every abstract measurable quotient is standard Borel. We use the following concrete sufficient condition.

Theorem B.3 (Compact continuous realization). *Let B_t be compact metric admissible belief domains and let $(f_{t,j})_{j \geq 1}$ be continuous bounded evaluation functions on B_t that determine predictive equivalence. Put*

$$e_t(b) = (f_{t,j}(b))_{j \geq 1}, \quad S_t = e_t(B_t),$$

after rescaling the functions into $[-1, 1]$. Suppose action spaces are compact metric, reports finite, and $F_t : B_t \times A_t \times Y_{t+1} \rightarrow B_{t+1}$ is continuous for each report and respects predictive equivalence. Then S_t is compact metrizable and the induced update $\bar{F}_t : S_t \times A_t \times Y_{t+1} \rightarrow S_{t+1}$ is continuous for each report. Every continuous evaluation constant on predictive fibres descends to a continuous function on S_t .

Proof. The countable product $[-1, 1]^{\mathbb{N}}$ is compact metrizable, and e_t is continuous. Its image is compact, hence closed and metrizable. A continuous surjection from a compact space to a Hausdorff space is a closed quotient map. The same holds for $e_t \times \text{id}_{A_t}$ because its domain is compact and its target Hausdorff. For each report, $e_{t+1} \circ F_t$ is continuous and constant on fibres of this quotient map. The definition of quotient topology therefore gives a unique continuous descended map. The final assertion is the same argument without the action factor. $\square$

If the hidden state space is compact and the belief domain is compact in weak convergence, continuous hidden-state tests give the required continuous evaluations. With continuous positive likelihoods bounded away from zero, Bayes updating is continuous. In concrete finite models this is verified directly. A countable dense set of actions can determine all finite words when the corresponding evaluations are continuous; this must be checked, not inferred from standard Borel structure alone.

For finite histories the quotient is finite and the measurable issue is trivial. For noncompact or nondominated settings we state results on an already supplied Borel predictive realization rather than claiming a universal smooth quotient.

B.3. Finite remaining-test coordinates. The following elementary result is useful when a finite representation exists, including representations with redundant coordinates.

Proposition B.4 (Positive finite-coordinate update). *Let $\psi_t = (1, \psi_{t,1}, \dots, \psi_{t,q_t})$ be bounded tests. Suppose*

$$\mathbb{T}_{t,a,y} 1 = b_{t,a,y} \psi_t, \quad \mathbb{T}_{t,a,y} \psi_{t+1,i} = c_{t,a,y,i} \psi_t,$$

with uniformly bounded coefficient rows and jointly Borel dependence on the input. On a domain of beliefs where every relevant report probability is at least $\kappa > 0$, the coordinate vector $v = (\pi \psi_{t,i})_{i \geq 1}$ updates by

$$v'_i = \frac{c_{t,a,y,i}(1, v)^{\top}}{b_{t,a,y}(1, v)^{\top}}. \quad (\text{B.3})$$

It is uniformly Lipschitz on segments joining such moment vectors when the same report floor holds on their belief mixtures. Linear independence of the tests is not required.

Proof. Equation (B.3) follows from (A.4). A segment between two belief moment vectors represents their convex mixtures. The denominator is a report probability under that mixture and is at least κ . The numerator, coefficient rows, and coordinate vectors are bounded. Differentiating the quotient bounds its derivative by a constant depending only on these bounds and κ . Integrate that derivative along the segment. Redundant coefficient representations have the same evaluations on actual moment vectors and their mixtures, so they give the same update there. $\square$

In the static latent-variable case the instruments are multiplication by report likelihoods. In a dynamic model they are positive transfer operators. In either case the update uses current coordinates, command, and report, not a recovered history tape. An observer who stores the entire vector v at arbitrary real precision has not yet complied with a finite-label budget; that is the problem of Sections E and F.

B.4. Three different fibres. It is useful to record the types before adding geometry. Predictive fibres are fibres of $h_t \mapsto s_t(h_t)$. Statistical fibres are equivalence classes of parameters with equal laws under the specified strategy class. Feature fibres are fibres of a chosen map $\mathbf{O} : \Lambda \rightarrow \mathcal{Z}$. Equality of features can be strictly weaker than equality of laws, and equality of a fixed-prior posterior prediction need not be parameter-uniform sufficiency. Their derivatives, when defined, have different domains.

APPENDIX C. EXECUTABLE REDUCTIONS AND CHARGED RESOURCES

C.1. Finite-state implementations. A synchronized causal machine between experiments has an internal state r_t . At a target step it receives the target action a'_t , chooses a source action according to a parameter-independent kernel depending on (r_t, a'_t) , receives the source report, and outputs a target report while updating r_t . Random seeds are independent of the parameter and physical preparation. A stored seed that encodes previous reports is part of r_t . Every raw source step and every retained state is charged.

This synchronized model suffices for the results below. A macro-step that uses a random number of source steps requires an explicit causal stopping protocol and its preparation certificate. Such a protocol can be included, but a bound for one synchronized source step is not silently reused for it.

A target controller with state z_t receives the target reports and updates z_t . Its joint implementation with the machine stores (r_t, z_t) . Thus $|r_t| \leq R_t$ and $|z_t| \leq M_t$ give the direct product bound $R_t M_t$. This is an upper bound for one realization, not a claim that no smaller implementation is possible.

Definition C.1 (Causal reduction certificate). For experiments over a common parameter space, a certificate $\Gamma : \mathfrak{E} \rightarrow \mathfrak{F}$ consists of a parameter-independent causal machine, a lift L_Γ of every admitted target controller to its source implementation, and a monotone resource map ρ_Γ . Under target controller α , write $\hat{P}_\lambda^{\Gamma, \alpha}$ for the simulated target-history law obtained from the joint machine/source execution. An ε -certificate satisfies

$$\sup_{\lambda, \alpha} \left\| \hat{P}_\lambda^{\Gamma, \alpha} - P_{\mathfrak{F}, \lambda}^\alpha \right\|_{\text{TV}} \leq \varepsilon, \quad (\text{C.1})$$

and implementing a target budget B costs at most $\rho_\Gamma(B)$. The default certificate is pathwise. Expected-cost and tail-budget certificates are different types and must be stated separately.

For randomized strategy lifts, the right object is the joint causal execution. It need not be representable by a terminal kernel on the raw source history alone that is independent of the

controller. Once the lift and controller are fixed, its visible pushforward is unambiguous. This avoids a misleading notation that would hide feedback inside a supposedly static garbling.

An exact certificate has $\varepsilon = 0$. The identity machine gives the identity reduction. A reparametrization $r : \Lambda' \rightarrow \Lambda$ pulls back an experiment family and is kept separate from data reduction. If the reparametrization is not injective it introduces parameter redundancy; it is not loss of observed data.

C.2. The path-kernel estimate. We use the convention $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. For $0 \leq f \leq L$ this implies $|Pf - Qf| \leq L \|P - Q\|_{\text{TV}}$.

Lemma C.2 (Predictable telescoping). *Let P and Q be laws on the same finite visible history tree, with the same initial law and the same feedback action kernels. Suppose their conditional report kernels at history h_t and action a_t are $p_t(\cdot \mid h_t, a_t)$ and $q_t(\cdot \mid h_t, a_t)$. If*

$$\|p_t(\cdot \mid h_t, a_t) - q_t(\cdot \mid h_t, a_t)\|_{\text{TV}} \leq d_t(h_t, a_t),$$

then

$$\|P - Q\|_{\text{TV}} \leq \min \left\{ 1, \mathbb{E}_P \sum_{t=0}^{N-1} d_t(H_t, A_t) \right\}. \quad (\text{C.2})$$

If both systems become absorbing under the same visible stopping rule σ , the summand can be multiplied by $\mathbf{1}_{\{t < \sigma\}}$.

Proof. Fix a terminal measurable test $0 \leq f \leq 1$. Replace the report kernels one at a time, using a P -prefix and a Q -suffix. The suffix conditional expectation of f is a measurable function between zero and one. The change at step t , after integrating the common feedback action kernel, is therefore bounded in absolute value by $\mathbb{E}_P d_t(H_t, A_t)$. Summing the telescoping differences proves the bound for f . Take the supremum over event indicators. After the visible rule has stopped, the kernels are the same absorbing kernel and their local error is zero. Histories of zero probability under P contribute nothing; the kernels are nevertheless specified on all admitted histories so that all intermediate laws exist. $\square$

The expectation is under the actual reference experiment. In particular a rare report carrying a large conditional error is not automatically charged as if it occurred at every time. No maximal-coupling measurability theorem is needed for this proof. The action policy may be discontinuous: the same kernel is applied on the same history in each telescoping term.

Theorem C.3 (Composition and risk transfer). *Suppose $\Gamma_1 : \mathfrak{E} \rightarrow \mathfrak{F}$ and $\Gamma_2 : \mathfrak{F} \rightarrow \mathfrak{G}$ have certificates (ε_1, ρ_1) and (ε_2, ρ_2) uniformly over the controller classes needed by the lifts. Their serial implementation has certificate*

$$(\min\{1, \varepsilon_1 + \varepsilon_2\}, \rho_1 \circ \rho_2).$$

Exact reductions compose associatively as executions, up to relabeling of product states. For a common target $\tau(\lambda)$ and bounded loss $0 \leq \ell \leq L$,

$$\mathfrak{R}_{\mathfrak{E}}(\rho_{\Gamma}(B)) \leq \mathfrak{R}_{\mathfrak{F}}(B) + L\varepsilon \quad (\text{C.3})$$

for either minimax risk or risk under a common prior.

Proof. Execute the second machine as a controller of the first. Their stored states form a product and their source actions remain nonanticipative and parameter independent. The first certificate applies to this enlarged controller class by hypothesis. The subsequent machine/controller execution is a Markov postprocessing of its visible history together with its independent randomization, so it contracts total variation. Compare first with an exact $\mathfrak{F}$ run and then with the intended $\mathfrak{G}$ run. The triangle inequality gives the displayed defect. Budget maps compose in the same order

because they map target requirements to source requirements. Regrouping three nested machine states changes only their names, proving associativity.

For a target procedure within budget B , run its source implementation and use the same final decision. Its risk at each parameter differs by at most $L\varepsilon$. The resource certificate makes it admissible under $\rho_\Gamma(B)$. Take the supremum over parameters, or integrate against the common prior, and then take the infimum over target procedures. An approximating sequence of procedures suffices if an optimum is not attained. $\square$

A risk certificate concerns the same decision target and loss on both sides. It does not compare excess errors relative to different full-history predictors. An expected preparation constraint cannot be transported by a pathwise certificate without checking its type. Conversely, total-variation closeness alone does not transfer an unbounded cost expectation.

C.3. Checking an exact state reduction locally.

Proposition C.4 (One-step realization). *Let $r_t : X_t \rightarrow X'_t$, $g_t : Y_t \rightarrow Y'_t$, and an action lift $\iota_t : A'_t \rightarrow A_t$ be measurable. Suppose the preparations push forward correctly and, for every admitted source state and target action,*

$$(r_{t+1}, g_{t+1})_{\#} K_t^\lambda(\cdot \mid x, \iota_t a') = K_t'^\lambda(\cdot \mid r_t x, a'). \quad (\text{C.4})$$

Suppose also that continuation interfaces, recorded costs, and designated readouts match. Then applying the lifted actions and report maps gives an exact causal reduction for every target feedback strategy.

Proof. At time zero the reduced state laws agree. Conditional on a source state and on a lifted action, (C.4) gives the joint reduced next-state/report law. Integrate against the common current reduced law. Since the target controller reads the reduced history, its next action law is the same in both systems. Induction proves equality on all finite cylinders, and hence on the path sigma field. Readout and cost matching give the asserted experiment and resource identity. $\square$

The state map need not be observable to implement the reduction: the machine only needs the action lift and report map. The hidden-state map is a proof device for the law identity. If the proposed lift instead requires the unobserved state, it is not an executable reduction of this type.

C.4. Terminal and causal equivalence. For a fixed terminal family, a garbling G is reversible if there exists a parameter-independent kernel R such that $P_\lambda G R = P_\lambda$. This is statistical equivalence. Requiring R to operate causally and charging its memory gives a stronger notion. Requiring suitable bounds in both directions gives resource-aware causal equivalence. These notions are not identified by definition.

Independent parallel experiments may be implemented by product machines when their preparations and physical randomness are genuinely independent conditional on the parameter. Persistent cardinalities multiply and costs add. Shared nuisance variables or coupled preparations require the joint experiment; replacing them by an independent product changes the object.

APPENDIX D. POSTERIOR GEOMETRY AND INFORMATION LOSS

D.1. Posterior barycentres. The following statements concern a common preparation. Let Z be a hidden target, Y an original observation, and W a parameter-independent postprocessing of Y . Conditional independence is denoted by the Markov chain $Z \rightarrow Y \rightarrow W$.

Theorem D.1 (Posterior barycentre and convex order). *For bounded measurable $f_1, \dots, f_q$ of Z , put*

$$p_Y = (\mathbb{E}[f_i(Z) \mid Y])_i, \quad p_W = (\mathbb{E}[f_i(Z) \mid W])_i.$$

Then

$$p_W = \mathbb{E}[p_Y | W]. \quad (\text{D.1})$$

Consequently $\mathbb{E}\varphi(p_W) \leq \mathbb{E}\varphi(p_Y)$ for every convex function for which the expectations are defined. Moreover,

$$\text{Cov}(p_Y) - \text{Cov}(p_W) = \mathbb{E} \text{Cov}(p_Y | W) \succeq 0. \quad (\text{D.2})$$

Proof. The Markov property gives $\mathbb{E}[f_i(Z) | Y, W] = \mathbb{E}[f_i(Z) | Y]$. Taking conditional expectation with respect to W proves (D.1). Conditional Jensen yields the convex-order inequality. Subtract the common mean and expand the orthogonal decomposition $p_Y - \mathbb{E}p_Y = (p_Y - p_W) + (p_W - \mathbb{E}p_W)$; its cross covariance is zero by conditioning on W . This gives (D.2). $\square$

For a measure-valued posterior, the same identity holds against every bounded test. It is a conditional barycentre, not a function sending each original posterior to a single coarser posterior. In particular the image of a finite set of original posteriors may generate a continuous family of conditional barycentres. This is why convex-order monotonicity is not a statement about the cardinality of a predictive representation.

D.2. Relative entropy and recovery. For $P \ll Q$, set $D(P\|Q) = \int \log(dP/dQ) dP$, allowing $+\infty$; set it to infinity otherwise. Let $G : X \rightsquigarrow W$ be a Markov kernel and form the joint laws $\bar{P}(dx, dw) = P(dx)G(dw | x)$ and $\bar{Q}(dx, dw) = Q(dx)G(dw | x)$.

Theorem D.2 (Entropy loss and a common recovery kernel). *If $D(P\|Q) < \infty$, then*

$$D(P\|Q) = D(PG\|QG) + \int D(\bar{P}(dx | w)\|\bar{Q}(dx | w)) PG(dw). \quad (\text{D.3})$$

In particular, entropy decreases under G . Suppose a family P_λ is dominated by one P_0 , every $D(P_\lambda\|P_0)$ is finite, and

$$D(P_\lambda G\|P_0 G) = D(P_\lambda\|P_0) \quad \text{for every } \lambda.$$

Then $R(w, dx) = \bar{P}_0(dx | w)$ is a parameter-independent recovery kernel satisfying $P_\lambda GR = P_\lambda$ for every λ .

Proof. The joint likelihood ratio is $d\bar{P}/d\bar{Q}(x, w) = dP/dQ(x)$. Disintegrating both joint laws gives the factorization into the marginal likelihood ratio on W and the conditional likelihood ratio on X . Take logarithms and integrate. Standard truncation of the densities justifies the formula when logarithms are not bounded; finite total entropy makes the nonnegative conditional entropy integrable. This proves (D.3). Alternatively the marginal ratio is the conditional expectation under $\bar{Q}$ of dP/dQ , and conditional Jensen proves the inequality directly, also for infinite entropy.

For the family, equality makes the conditional entropy vanish $P_\lambda G$ -almost everywhere. Hence the two conditional distributions coincide there. The conditional law under the one reference P_0 is the same measurable kernel for all parameters; $P_\lambda G \ll P_0 G$ handles its null-set convention. Integrating that conditional law against $P_\lambda G$ recovers P_λ . $\square$

This is a sufficient recovery criterion under a common reference. An isolated equality for one pair does not certify an unrelated parameter family. Nor does it make the recovery kernel causal or memory free.

D.3. Scores and Fisher information. To avoid hiding regularity in notation, first take a finite observation space with $p_\lambda(y) > 0$ and continuously differentiable probabilities on an open parameter domain in $\mathbb{R}^k$. Let

$$s_\lambda(y) = \nabla_\lambda \log p_\lambda(y), \quad \mathcal{I}_\lambda = \mathbb{E}_\lambda[s_\lambda s_\lambda^\top].$$

If $G(w | y)$ is a parameter-independent finite channel, restrict to output symbols having positive probability; identically impossible symbols can be removed.

Theorem D.3 (Score projection). *The processed score and information satisfy*

$$s_\lambda^G(W) = \mathbb{E}_\lambda[s_\lambda(Y) \mid W], \quad (\text{D.4})$$

$$\mathcal{I}_\lambda - \mathcal{I}_\lambda^G = \mathbb{E}_\lambda \text{Cov}_\lambda(s_\lambda(Y) \mid W) \succeq 0. \quad (\text{D.5})$$

For a direction v , equality of information in that direction is equivalent to $v^\top s_\lambda(Y)$ being determined by W almost surely.

Proof. Differentiate $q_\lambda(w) = \sum_y p_\lambda(y)G(w \mid y)$ and divide by $q_\lambda(w)$. The coefficients are precisely the conditional probabilities of Y given W , proving (D.4). Differentiating $\sum_y p_\lambda(y) = 1$ gives $\mathbb{E}_\lambda s_\lambda = 0$. The conditional variance identity now gives (D.5). Its quadratic form in v is the mean conditional variance of $v^\top s_\lambda$, which vanishes exactly under the stated measurability condition. $\square$

The same proof works in a dominated differentiable model if differentiation under the integral is justified and the score lies in L^2 . More generally, the QMD version is the classical score-projection theorem described by Pollard [11]; when it is invoked, quadratic-mean differentiability is an additional statistical hypothesis. The raw experiment axioms by themselves impose no differentiability.

Information equality in (D.5) is a statement at a parameter point. It is not the recovery conclusion of Theorem D.2. Even information preservation throughout a model need not make a statistic sufficient without stronger assumptions; see [11].

Under a smooth parameter map $\lambda = r(\zeta)$, the information pulls back as $Dr^\top \mathcal{I} Dr$. Under a deterministic linear map of a random vector, its covariance pushes forward as $A \Sigma A^\top$. These are different operations on different objects. Neither formula says that an arbitrary covariance equals inverse Fisher information.

D.4. Sequential information in finite positive models. A useful controlled specialization follows without differentiating the controller. Suppose the conditional report probabilities are finite, positive, and differentiable, and the feedback policy has no explicit unknown-parameter dependence. Along a history define the local score

$$\xi_{t+1} = \nabla_\lambda \log p_\lambda(Y_{t+1} \mid H_t, A_t).$$

The policy factors contribute no score. Conditional normalization gives $\mathbb{E}_\lambda[\xi_{t+1} \mid H_t, A_t] = 0$.

Proposition D.4 (Sequential score decomposition). *If the local scores are square integrable, the path score is $S_N = \sum_{t < N} \xi_{t+1}$, and*

$$\mathbb{E}_\lambda[S_N S_N^\top] = \sum_{t < N} \mathbb{E}_\lambda[\xi_{t+1} \xi_{t+1}^\top]. \quad (\text{D.6})$$

For bounded stopping, each term is restricted to active steps. No independence of successive reports is required.

Proof. The likelihood factors into conditional report probabilities and parameter-independent policy factors. Its logarithmic derivative is the displayed sum. For $s < t$, ξ_{s+1} is measurable before ξ_{t+1} , so their cross moment is zero by the conditional mean-zero property. Expanding the square proves (D.6). A bounded stopping indicator is predictable for the next increment; the same argument applies. $\square$

This is an information identity for the actual controlled law. It does not make an information-optimal strategy automatically prediction-memory optimal, and it does not supply an asymptotic normal approximation.

APPENDIX E. ATTAINED PREDICTION AND CHECKPOINT QUANTIZATION

E.1. The physical loss. Fix a preparation and an exploration law, independently of coding randomness. At time t choose a future query J independently after the label is formed, with probabilities $w_j > 0$, $\sum_j w_j = 1$. Conditional on history and query, a bounded response Z_j has mean $m_j(h)$. Squared loss is used, and only the selected query is executed. Set

$$p(h) = (\sqrt{w_j} m_j(h))_{j=1}^q, \quad \nu = \text{Law}(p(H_t)).$$

The weighted decoder vector is denoted by c . Subtracting the complete history predictor's risk leaves $\|p(h) - c\|^2$.

An M -label checkpoint encoder reads the complete prefix once and retains one of M symbols. Its decoder receives the label and the subsequent query, not the discarded history or the acquisition commands. A public seed may specify a codebook but is independent of the preparation. Private coins are allowed and are either discarded or included in the retained label. We assume a bounded closed convex prediction domain containing all $p(h)$; projection into it can only improve squared error.

Theorem E.1 (Exact checkpoint identity). *The optimal mean checkpoint excess risk equals*

$$R_M^{\text{chk}} = e_M^2(\nu) := \inf_{c_1, \dots, c_M} \int \min_{1 \leq j \leq M} \|p - c_j\|^2 \nu(dp). \quad (\text{E.1})$$

Independent public or private coding randomization does not decrease this infimum. An arbitrarily near-optimal deterministic nearest-centre encoder is measurable.

Proof. Condition on history, query, and the code randomness. The elementary identity

$$\mathbb{E}[(Z_j - a)^2 \mid h, J = j] = \text{Var}(Z_j \mid h, J = j) + (m_j(h) - a)^2$$

reduces the loss to the displayed Euclidean distortion. For each label, replace a randomized decoder by its conditional mean over decoder coins; convexity of squared norm does not increase the loss. Given fixed decoder centres, random assignment by the encoder is no better than the nearest centre. Hence every fixed public-seed codebook has risk at least the right-hand side of (E.1); averaging over the independent seed preserves this inequality.

Conversely choose centres within any prescribed positive tolerance of the infimum. Encode by the first index attaining the least distance. Finite comparisons of continuous distance functions make this rule Borel. The resulting distortion is the integrand in (E.1). Let the tolerance decrease to zero. A codebook indexed by an unknown true parameter is not produced by this argument: ν is the one declared preparation law. $\square$

For worst-history loss the analogous deterministic checkpoint quantity is a covering radius, not the integral in (E.1). The mean and worst-history laws have the same order only under additional geometric and mass conditions.

E.2. A small-ball lower bound.

Lemma E.2 (Mass obstruction). *Suppose a probability law ν in $\mathbb{R}^d$ satisfies $\nu(B(c, r)) \leq Ar^d$ for every centre c and radius $r > 0$. Then for every integer $M \geq 1$,*

$$e_M^2(\nu) \geq \frac{1}{2} (2AM)^{-2/d}. \quad (\text{E.2})$$

In particular, a Lebesgue density bounded by $B\epsilon^{-d}$ gives $e_M^2(\nu) \geq c_{d,B}\epsilon^2 M^{-2/d}$ for $\epsilon > 0$.

Proof. The union of M balls of radius $r = (2AM)^{-1/d}$ has probability at most one half. On its complement every centre is at least distance r away. Integrating the nearest-centre squared distance gives at least $r^2/2$. For the density assertion use $A = Bv_d\epsilon^{-d}$, where v_d is the volume of the Euclidean unit ball. $\square$

The lemma permits centres outside the support and therefore already covers arbitrary decoders. It does not condition away a small probability of reaching the difficult region. If the useful density lower geometry instead occurs only on an event of probability β , the eventual risk lower bound must carry that mass.

Lemma E.3 (A positive attained chart). *Suppose a coordinate projection of a prediction law dominates β times uniform probability on a d -dimensional cube of side a , with $\beta, a > 0$. If that projection has operator norm at most L , then*

$$e_M^2(\nu) \geq c_d \beta a^2 L^{-2} M^{-2/d}.$$

Proof. For any centres in the original prediction space, project the centres onto the cube's coordinate space. Distances there are at most L times the original distances. The uniform cube law has density a^{-d} and satisfies Lemma E.2. Domination by β and rescaling prove the assertion. $\square$

E.3. Whole-image covers and all integer budgets. Let $D \subset \mathbb{R}^d$ be nonempty, compact, and contained in a cube of side $2a$. Divide each coordinate interval into $k = \lfloor M^{1/d} \rfloor$ pieces. There are at most $k^d \leq M$ cells. Choose one point of D in each nonempty cell. Any point is within Euclidean distance $2a\sqrt{d}/k$ of its representative. Since $k \geq M^{1/d}/2$, we obtain a radius $4a\sqrt{d}M^{-1/d}$ for every integer $M \geq 1$. The corresponding ℓ^1 bound is $4adM^{-1/d}$. Neither a large-budget asymptotic nor an integer-budget relaxation is needed.

More generally, finitely many Lipschitz images of compact cubes have such covers with exponent equal to the largest domain dimension. Allocate centres between the finitely many charts, use the preceding grid, and enlarge the constant for budgets smaller than the number of charts. This is a sufficient global description, not a consequence of local rank alone. A nonlinear or disconnected image cannot be replaced by a tangent plane without checking its entire cover.

E.4. Causal lower bounds use one exploration law. Let $\mathcal{C}_M$ be causal encoders with at most M_t labels at time t . Under the fixed exploration strategy, their label is also a measurable, possibly randomized, encoding of the complete time- t prefix. Therefore

$$\inf_{C \in \mathcal{C}_M} \max_{1 \leq t \leq N} \mathbb{E} \ell_t^C \geq \max_{1 \leq t \leq N} e_{M_t}^2(\nu_t). \quad (\text{E.3})$$

Indeed the checkpoint lower inequality holds for every single C and all t before taking the infimum. The same law of exploration supplies all marginals ν_t . Changing the exploration strategy with the encoder, or with the checkpoint after the infimum, would define another problem.

E.5. Calibrated anisotropy. A useful model interface is provided by [15]. There, under whole-image covering, positive attained mass for each initial scale flag, and uniformly stable raw updates, the scale profile is

$$Q_t(M) = \max_{1 \leq \ell \leq p_t} \left(\frac{s_{t,1} \cdots s_{t,\ell}}{M} \right)^{2/\ell}.$$

The physical prediction metric is fixed; small singular values are not removed by an unbounded coordinate whitening. The finite-horizon causal bound follows by propagating the covering radii, as in the next section. We cite this refined profile as a conditional model interface, not as an unproved universal consequence of a list of singular values. The dynamic model of Section G will establish its own isotropic scale, density bound, and invariant cover directly from the experiment.

APPENDIX F. CAUSAL QUANTIZATION WITH FEEDBACK AND STOPPING

F.1. Admissible state domains. Consider a predictive realization $s_t(h_t)$ with a known initial state s_0 . Let (D_t, d_t) be admissible metric state domains containing every attainable state. They may also contain normalized beliefs that are not attained at that precise time. This causes no

problem if all updates and readouts are genuinely defined on D_t . What is forbidden is evaluating an update at an artificial point where it has no experimental meaning or no stated extension.

Assume common action availability on each domain, or work within a fixed continuation-interface component. For each input (a, y) , let $F_t(\cdot, a, y) : D_t \rightarrow D_{t+1}$ be Borel and satisfy

$$d_{t+1}(F_t(s, a, y), F_t(\tilde{s}, a, y)) \leq L_t d_t(s, \tilde{s}). \quad (\text{F.1})$$

The known-calibration formulas implementing F_t are independent of the unknown parameter. If there is a parameterized law, write the next-report kernel as $g_t^\lambda(dy \mid s, a)$ and assume

$$\left\| g_t^\lambda(\cdot \mid s, a) - g_t^\lambda(\cdot \mid \tilde{s}, a) \right\|_{\text{TV}} \leq \kappa_t d_t(s, \tilde{s}) \quad (\text{F.2})$$

uniformly over the relevant parameter domain. The same Borel state update must represent every law for a parameter-uniform statement. For a single prepared Bayesian experiment there is no additional parameter supremum.

Choose finite representative sets $C_t \subset D_t$ with $|C_t| \leq M_t$ and Borel quantizers $q_t : D_t \rightarrow C_t$ satisfying $d_t(s, q_t s) \leq r_t$. A numerical update $\tilde{F}_t(c, a, y) \in D_{t+1}$ may replace the exact update at a representative, with

$$d_{t+1}(\tilde{F}_t(c, a, y), F_t(c, a, y)) \leq \eta_{t+1}. \quad (\text{F.3})$$

Set $\hat{s}_0 = s_0$ and implement

$$\hat{s}_{t+1} = q_{t+1} \tilde{F}_t(\hat{s}_t, a_t, y_{t+1}). \quad (\text{F.4})$$

The persistent state is only the index of $\hat{s}_t$ in C_t .

Theorem F.1 (Finite-state emulation). *Under (F.1)–(F.3), define*

$$b_0 = 0, \quad b_{t+1} = L_t b_t + r_{t+1} + \eta_{t+1}, \quad b_t = \sum_{j=1}^t (r_j + \eta_j) \prod_{i=j}^{t-1} L_i. \quad (\text{F.5})$$

Then the following assertions hold.

(i) *On every admitted common command/report sequence, $d_t(s_t, \hat{s}_t) \leq b_t$. If the physical prediction readout is K_t -Lipschitz, its excess squared error is at most $K_t^2 b_t^2$.*

(ii) *Define the finite-state reference experiment to use the same representative update (F.4), but to draw its next report from $g_t^\lambda(\cdot \mid \hat{s}_t, a_t)$. For any common causal feedback controller, the visible path laws of the actual experiment with the representative machine and of this finite-state reference experiment satisfy*

$$\left\| P_{N,\lambda}^\alpha - \hat{P}_{N,\lambda}^\alpha \right\|_{\text{TV}} \leq \min \left\{ 1, \sum_{t=0}^{N-1} \kappa_t b_t \right\}. \quad (\text{F.6})$$

The histories here include the applied actions, reports, and retained labels. A shared visible stopping rule replaces the deterministic sum by $\mathbb{E}_{P_\lambda^\alpha} \sum_{t < \sigma \wedge N} \kappa_t b_t$.

(iii) *Any additional causal simulator is charged as in Theorem C.3. A feedback controller with D_t own states is implemented with at most $D_t M_t$ persistent pairs. For a common bounded path-dependent loss, the risk-transfer error is at most its range times the right side of (F.6).*

Proof. For (i), insert successively the exact update from the retained representative and its numerical approximation. The triangle inequality, (F.1), the numerical error, and the covering radius give

$$d_{t+1}(s_{t+1}, \hat{s}_{t+1}) \leq L_t d_t(s_t, \hat{s}_t) + \eta_{t+1} + r_{t+1}.$$

Induction proves (F.5). The physical readout bound follows by its Lipschitz property and the conditional squared-loss identity. This argument remains valid when the actions depend on the retained label: it compares the exact posterior and the representative along the actual same sequence of applied commands and reports.

For (ii), at any common visible prefix the applied controller has the same action kernel. The representative $\hat{s}_t$ is the same measurable function of that prefix in both experiments. In the actual law, the exact predictive state is s_t ; in the finite-state reference law, the next-report kernel is evaluated at $\hat{s}_t$. Their conditional variation distance is at most $\kappa_t b_t$ by (i) and (F.2). The next label is a common deterministic function of the action, report, and current label, so appending it does not increase the conditional variation distance. Apply Lemma C.2. Stopping makes the conditional kernels identical after the stopping time, which gives the predictable active-step bound.

For (iii), pair the retained representative with the controller's internal state. Each next action and update then reads only this pair and current inputs. The state cardinality is bounded by the product. The bounded-loss and extra-simulator statements are precisely the conclusions of Theorem C.3. $\square$

The theorem is a directional emulation statement. It does not construct a reverse kernel from the finite-state reference experiment to the raw physical data. It therefore does not claim two-sided Le Cam equivalence. Nor does it estimate an unknown predictive update. The implemented update and representatives are known; uniformity over a statistical family is available only under the explicitly common update conditions above.

F.2. What feedback does and does not require. There are two different uses of a finite filter. One can predict on an arbitrarily selected actual input sequence; this uses only Theorem F.1(i). One can also use a compressed controller to choose inputs. That still gives accurate filtering along its own actual run, but comparison with the original full-information optimal controller is another problem. Part (ii) compares a common controller across two experiments and so supplies a valid route through experiment simulation. It does not require the controller to be Lipschitz, because identical prefixes produce identical action laws. A discontinuous policy evaluated at two different approximate states is not being assumed to choose the same action.

For example, minimizing a terminal common decision loss over all D_t -state controllers in the finite-state reference experiment gives a source controller with $D_t M_t$ states and the corresponding (F.6) risk allowance. This need not approximate the unrestricted optimal source policy, whose comparison would require a separate richness theorem for the controller class.

F.3. Uniform prediction and finite-horizon path comparison.

Corollary F.2 (Contractive implementation). *If $L_t \leq \rho < 1$, $r_t \leq r$, and $\eta_t \leq \eta$ uniformly, then*

$$\sup_t d_t(s_t, \hat{s}_t) \leq \frac{r + \eta}{1 - \rho}.$$

For a bounded physical readout with uniform Lipschitz constant K , the prediction excess is at most $K^2(r + \eta)^2/(1 - \rho)^2$ at every time. If also $\kappa_t \leq \kappa$, the path-law error on N steps is at most $\min\{1, N\kappa(r + \eta)/(1 - \rho)\}$.

Proof. Sum the geometric series in (F.5), then apply the two parts of Theorem F.1. $\square$

A uniform per-checkpoint prediction bound is not uniform total-variation closeness of two infinite path laws. The last display may grow with N . Two processes that are arbitrarily close at each step can remain statistically distinguishable from a sufficiently long record. If a general construction uses a different arbitrary codebook at each time, it is a time-indexed machine description, not automatically a finite uniform program. The dynamic construction in the next section uses one fixed invariant domain and one fixed codebook, so that distinction does not obstruct its infinite-time statement.

F.4. Model or calibration perturbations. Suppose an implementable reference update F_t^0 has $d(F_t(s, a, y), F_t^0(s, a, y)) \leq \delta_{t+1}$ on the relevant domain, and its report law differs by at most ζ_t at the same state. Repeating the proof replaces $r_j + \eta_j$ by $r_j + \eta_j + \delta_j$ in (F.5) and $\kappa_t b_t$ by $\kappa_t b_t + \zeta_t$ in (F.6). This is a deterministic certificate, not an unpaid pilot. If a preliminary estimate yields such bounds only on an event of probability $1 - p$, bounded risks receive an additional allowance of at most their range times p , and the pilot's data and persistent state must be charged.

The perturbation must remain in the same physical metric and admissible domain. A reparametrization with an unbounded inverse near a degeneracy cannot be inserted while keeping the old constants.

APPENDIX G. A DYNAMIC MODEL WITH A UNIFORM MEMORY LAW

G.1. The experiment and its resource convention. Fix $d \geq 1$, let $E_d = \{0, 1, \dots, d\}$, and write $\nu_i = 1/(d+1)$. The known parameters satisfy

$$4/5 \leq \gamma \leq 1, \quad 0 \leq \epsilon \leq 1/4. \quad (\text{G.1})$$

Prepare $X_0 \sim \nu$. At each acquisition step the hidden state is updated by the transition matrix

$$P_\gamma(i, j) = (1 - \gamma)\mathbf{1}_{\{i=j\}} + \gamma\nu_j. \quad (\text{G.2})$$

A command $u_t \in [-1, 1]^d$, chosen before the report, gives a binary report $Y_t \in \{-1, 1\}$ with likelihoods

$$\ell_y(u, i) = \begin{cases} 1/2, & i = 0, \\ (1 + y\epsilon \tanh u_i)/2, & 1 \leq i \leq d. \end{cases} \quad (\text{G.3})$$

Conditional on the hidden path and commands, the report randomizations are independent. The resulting joint transition/report kernel is $P_\gamma(i, j)\ell_y(u, j)$. It is an instance of Definition A.1; all report probabilities lie between $3/8$ and $5/8$.

For the acquisition lower bound the commands are independent uniform on $[-1, 1]^d$, independent of the hidden process and coding randomness. The upper-bound filter will work on every admitted command/report sequence, including adaptively chosen commands. It does not change the fixed exploration law used for the lower bound.

At any selected time $t \geq 1$ the acquisition can stop and one terminal probe is executed without an intervening refresh. After the retained label is formed, draw J uniformly from $\{1, \dots, d\}$. Probe j produces $Z \in \{0, 1\}$ with

$$\mathbb{P}(Z = 1 \mid X_t = i, J = j) = 1/2 + \tau\mathbf{1}_{\{i=j\}}, \quad \tau = 1/4. \quad (\text{G.4})$$

This is one noisy binary readout, not observation of X_t or of the full posterior. Its choice is independent of the acquisition. Each acquisition and the terminal probe costs one recorded experimental step. Evaluating maximum-checkpoint risk means that a run executes only its selected probe; it does not execute counterfactual probes at all times.

Let $p_t = \text{Law}(X_t \mid H_t)$, and write $p_{t,+} = (p_{t,1}, \dots, p_{t,d})$. The probe's full-history success probability is $1/2 + \tau p_{t,j}$. If a decoder uses candidate coordinates c_j , its excess loss is

$$\frac{\tau^2}{d} \|p_{t,+} - c\|_2^2. \quad (\text{G.5})$$

Arbitrary probe predictions are allowed; projecting them onto the interval $[1/2, 1/2 + \tau]$ cannot worsen loss. Thus the decoder coordinate formulation loses no candidates for the infimum.

Let $R_{M,t}^{\text{av}}$ and $R_{M,t}^{\text{max}}$ be optimal checkpoint risks under the stated exploration and over all admitted histories, respectively. Let $\mathfrak{R}_{M,N}^{\text{av}}$ and $\mathfrak{R}_{M,N}^{\text{max}}$ be the corresponding best maximum risks for one causal M -label encoder at times $1, \dots, N$. The clock, d, γ, ϵ , and the finite read-only program are known; no history or unknown calibration is stored in those constants. At time zero

one fixed initial state is allowed. For $N = \infty$ replace maxima by suprema and use one machine for all times.

Theorem G.1 (Refreshing experiment memory law). *For every fixed $d \geq 1$, constants $0 < c_d \leq C_d < \infty$ exist such that, for every (G.1), every integer $M \geq 1$, every $N \in \mathbb{N} \cup \{\infty\}$, and every $1 \leq t \leq N$,*

$$c_d \epsilon^2 M^{-2/d} \leq R_{M,t}^{\text{av}} \leq R_{M,t}^{\text{max}} \leq C_d \epsilon^2 M^{-2/d}, \quad (\text{G.6})$$

$$c_d \epsilon^2 M^{-2/d} \leq \mathfrak{R}_{M,N}^{\text{av}} \leq \mathfrak{R}_{M,N}^{\text{max}} \leq C_d \epsilon^2 M^{-2/d}. \quad (\text{G.7})$$

The upper bounds are realized by one deterministic finite-label filter using a fixed codebook for all positive times. The lower bounds allow independent public and private coding randomization. At $\epsilon = 0$ all four risks are zero with one label. Constants are uniform in time, horizon, calibration within (G.1), and the integer budget.

The proof will derive the cover, the actual attained density, and the stable update separately. This also supplies a concrete dynamic realization of the general emulation theorem.

G.2. Explicit posterior recursion. Set $p_0 = \nu$. Given the previous posterior p , the distribution just before the next report is

$$q = (1 - \gamma)p + \gamma\nu, \quad q_i \geq m_* := \frac{4}{5(d+1)}. \quad (\text{G.8})$$

For a command and report put

$$r_0 = 1, \quad r_i = 1 + y\epsilon \tanh u_i \quad (i \geq 1), \quad D_y = q_0 + \sum_{i=1}^d q_i r_i.$$

Then the probability of that report is $D_y/2$, and

$$F(p, u, y)_0 = \frac{q_0}{D_y}, \quad F(p, u, y)_i = \frac{q_i r_i}{D_y} \quad (i \geq 1). \quad (\text{G.9})$$

Every denominator belongs to $[1 - \epsilon, 1 + \epsilon]$. These formulas hold for every belief in the full simplex and every admitted input. Predictable actions reveal no further hidden-state information once their past is conditioned upon, as in Proposition A.3.

Lemma G.2 (Uniform contraction). *For any two simplex beliefs and the same (u, y) ,*

$$\|F(p, u, y) - F(\tilde{p}, u, y)\|_1 \leq \rho \|p - \tilde{p}\|_1, \quad \rho = 2 \frac{1 + \epsilon}{1 - \epsilon} (1 - \gamma) \leq \frac{2}{3}. \quad (\text{G.10})$$

Proof. For fixed positive likelihood vector r , write $B_r(q) = \text{diag}(r)q/(q^T r)$. Subtract the two normalized vectors by adding and subtracting $\text{diag}(r)\tilde{q}/(q^T r)$. The first term has norm at most $(r_{\max}/r_{\min}) \|q - \tilde{q}\|_1$. The second has norm $|q^T r - \tilde{q}^T r|/(q^T r)$, at most the same bound. Thus $\text{Lip}_1(B_r) \leq 2r_{\max}/r_{\min}$. Equation (G.8) multiplies differences by $1 - \gamma$. Use $r_{\max} \leq 1 + \epsilon$, $r_{\min} \geq 1 - \epsilon$, and (G.1) to obtain the result. $\square$

Lemma G.3 (An invariant shrinking domain). *Let*

$$a_\epsilon = \frac{2\epsilon}{\gamma(1 - \epsilon)}, \quad D_\epsilon = \{p \in \Delta_d : \|p - \nu\|_1 \leq a_\epsilon\}. \quad (\text{G.11})$$

Then $\nu \in D_\epsilon$, $F(D_\epsilon, u, y) \subset D_\epsilon$, and $a_\epsilon \leq (10/3)\epsilon$. Every attainable posterior belongs to D_ϵ , for every time and every admitted history.

Proof. For any pre-report belief q ,

$$\|B_r(q) - q\|_1 = \sum_i q_i \frac{|r_i - q^T r|}{q^T r} \leq \frac{2\epsilon}{1 - \epsilon}.$$

Consequently

$$\|F(p, u, y) - \nu\|_1 \leq (1 - \gamma) \|p - \nu\|_1 + \frac{2\epsilon}{1 - \epsilon}.$$

The right side is at most a_ϵ when $p \in D_\epsilon$, by its definition. The numerical bound follows from $\gamma \geq 4/5$ and $1 - \epsilon \geq 3/4$. Start the induction at ν . $\square$

The set D_ϵ is a compact convex polytope for each fixed calibration. Its representatives need not all be attainable at a particular time; they are normalized beliefs with valid updates and belong to a proved invariant domain. No extrapolation outside the probability simplex is used. At $\epsilon = 0$ the domain is the singleton $\{\nu\}$.

G.3. The actual acquisition density. Fix a preceding history, so its pre-report belief q is fixed with (G.8). For $\epsilon > 0$ and a fixed sign y , the command map

$$\Phi_{q,y} : u \mapsto F(p, u, y)_+ \in \mathbb{R}^d$$

is one-to-one. Indeed its odds satisfy

$$\frac{p'_i}{p'_0} = \frac{q_i}{q_0} (1 + y\epsilon \tanh u_i), \quad (\text{G.12})$$

so u_i is recovered by applying artanh to $\{p'_i q_0 / (p'_0 q_i) - 1\} / (y\epsilon)$. The image of the command interior has a smooth inverse. Boundary commands have zero probability under the uniform acquisition law.

Lemma G.4 (Posterior Jacobian and unconditional density). *For $\epsilon > 0$,*

$$|\det D\Phi_{q,y}(u)| = \frac{q_0 \prod_{i=1}^d q_i}{D_y^{d+1}} \epsilon^d \prod_{i=1}^d \text{sech}^2 u_i. \quad (\text{G.13})$$

Under independent uniform commands and actual reports, the law of $p_{t,+}$ has a Lebesgue density bounded by $B_d \epsilon^{-d}$ at every $t \geq 1$, where one possible uniform constant is

$$B_d = 2^{-d} (5/4)^{d+2} m_*^{-(d+1)} \cosh^{2d}(1). \quad (\text{G.14})$$

The bound is conditional on every preceding history as well as unconditional.

Proof. Write $r'_i = y\epsilon \text{sech}^2 u_i$. In the d nonzero-state coordinates,

$$D\Phi_{q,y} = \text{diag} \left(\frac{q_i r'_i}{D_y} \right) - \left(\frac{q_i r'_i}{D_y} \right)_i \left(\frac{q_j r'_j}{D_y} \right)_j^\top.$$

The determinant lemma gives the determinant of the diagonal matrix times

$$1 - \sum_{i=1}^d \frac{q_i r'_i}{D_y} = \frac{q_0}{D_y}.$$

Taking absolute values gives (G.13).

Before any conditioning on the sign, the joint density of the uniform command and that sign is $2^{-d} D_y / 2$. Change variables through the injective command map. Its contribution to the posterior density is

$$\frac{2^{-(d+1)} D_y^{d+2}}{q_0 \prod_{i=1}^d q_i \epsilon^d \prod_{i=1}^d \text{sech}^2 u_i} \leq 2^{-(d+1)} (5/4)^{d+2} m_*^{-(d+1)} \cosh^{2d}(1) \epsilon^{-d}.$$

Sum the two signs, allowing their images to overlap, to obtain (G.14). Finally integrate the conditional bounds against the actual preceding-history law. The same bound survives because it is independent of that history. Thus no event probability or complementary command volume is discarded. $\square$

This is a full-density upper bound for the actual law, which is sufficient for a quantization lower bound. It is stronger than merely exhibiting a nonzero differential at one command, but it does not assert that every point of the invariant polytope is attained. Those are different statements.

G.4. A fixed codebook and the proof of the theorem. Project D_ϵ to its d coordinates p_+ . It lies in the cube $\nu_+ + [-a_\epsilon, a_\epsilon]^d$. Partition this cube into k^d cells, $k = \lfloor M^{1/d} \rfloor$, and select one belief in D_ϵ from each occupied cell. Fix those representatives once and for all, along with a first-minimum tie convention. Empty cells need no label. The discarded coordinate satisfies $p_0 = 1 - \sum_{i=1}^d p_i$. Hence within a cell the full-simplex ℓ^1 distance is at most twice the ℓ^1 distance of its d coordinates. The resulting codebook has at most M points and covering radius

$$r_M \leq \frac{4da_\epsilon}{k} \leq \frac{80d}{3}\epsilon M^{-1/d}. \quad (\text{G.15})$$

Choose the nearest representative in full ℓ^1 distance. This preserves the covering bound and is Borel. The initial state is exactly ν ; thereafter the same codebook and the same update apply at every step.

Proof of Theorem G.1. For $\epsilon > 0$, use (G.9) at the retained belief and then quantize with (G.15). Lemmas G.2 and G.3 justify the whole procedure on every history. By Corollary F.2,

$$\sup_{t \geq 1} \|p_t - \hat{p}_t\|_1 \leq 3r_M \leq 80d\epsilon M^{-1/d}. \quad (\text{G.16})$$

Equation (G.5) bounds the excess by $(\tau^2/d)(80d)^2\epsilon^2 M^{-2/d}$. This proves a pathwise upper bound for one machine, hence all the required mean and worst-history upper bounds. A checkpoint encoder can use the same final nearest-centre procedure.

For the lower bound, Lemma G.4 and Lemma E.2 give

$$\inf_{c_1, \dots, c_M} \mathbb{E} \min_j \|p_{t,+} - c_j\|_2^2 \geq \frac{1}{2}(2B_d v_d)^{-2/d} \epsilon^2 M^{-2/d}.$$

Multiply by τ^2/d and use Theorem E.1. The bound holds for every $t \geq 1$ under the one fixed exploration law, with coding randomness allowed. Every causal encoder is a checkpoint encoder at each time, proving the causal lower bound for finite or infinite horizons. The inequality between mean and worst-history risks is immediate. When $\epsilon = 0$, both reports have probability $1/2$ independent of the state and command. Refreshing preserves the uniform preparation, so $p_t = \nu$ on every history. A single label predicts the terminal probe exactly relative to the full-history oracle. This proves the zero case and completes the uniform statement. $\square$

Corollary G.5 (Persistent bits and double attenuation). *For b persistent bits, take $M = 2^b$ in (G.7). With the fixed terminal probe the squared excess has order $\epsilon^2 2^{-2b/d}$. If the actual terminal probe is instead*

$$\mathbb{P}(Z = 1 \mid X_t = i, J = j) = 1/2 + \tau\epsilon \mathbf{1}_{\{i=j\}},$$

the order is $\epsilon^4 2^{-2b/d}$, with the same uniformity and the same filter. At $\epsilon = 0$, zero bits suffice in both models.

Proof. The first assertion substitutes the number of labels. In the second experiment the physical readout difference is multiplied by ϵ , while the acquisition process, posterior law, and filter are unchanged. Its squared excess is therefore multiplied by ϵ^2 . For the lower bound the same scaling applies to arbitrary decoder centres after clipping to the physical mean interval. The zero case was established above. $\square$

G.5. Feedback-uniform emulation in this model. For fixed command u , the probability of the positive report under belief p is

$$g_+(p, u) = \frac{1}{2} \left(1 + \epsilon \sum_{i=1}^d ((1 - \gamma)p_i + \gamma\nu_i) \tanh u_i \right).$$

For binary laws their total variation is the difference of their positive probabilities. Thus

$$\|g(\cdot | p, u) - g(\cdot | \tilde{p}, u)\|_{\text{TV}} \leq \frac{1}{2}\epsilon(1 - \gamma) \|p - \tilde{p}\|_1. \quad (\text{G.17})$$

By Theorem F.1, the actual report-path law and the finite-state reference report-path law obey

$$\|P_N^\alpha - \hat{P}_N^\alpha\|_{\text{TV}} \leq \min\{1, 40dN\epsilon^2(1 - \gamma)M^{-1/d}\} \quad (\text{G.18})$$

for every common feedback controller. This statement is uniform over controllers but is finite-horizon; it does not identify infinite path laws. If a terminal probe is included in the compared observed path, its additional conditional error is at most $\tau \|p_t - \hat{p}_t\|_1$, averaged over its random index. This term is to be added to (G.18). Prediction regret for that probe instead uses its squared error, as in (G.5); the two claims are different.

The bound vanishes for the acquisition report law when $\gamma = 1$: each new state is independently uniform, so previous belief errors have no effect on the next acquisition report. The current posterior can still have nontrivial geometry and require finite prediction memory for the immediate terminal probe. This limiting case is a useful check on the meaning of each estimate.

G.6. Calibration and finite precision. The theorem is uniform over (G.1), but the codebook and update may use the known values of γ and ϵ . It is not a calibration-blind theorem. Numerically approximating each belief update by an admissible belief within η in ℓ^1 changes (G.16) to $3(r_M + \eta)$. The physical excess is at most $9\tau^2(r_M + \eta)^2/d$. Thus choosing η proportional to $\epsilon M^{-1/d}$ preserves the order. This is an accuracy requirement; it does not assert zero workspace or a universal bit-complexity bound for computing the update.

For rational calibration one can specify the invariant polytope by rational linear inequalities. Representatives can be selected in its occupied grid cells by a finite feasibility procedure; centres and a membership/tie rule form a finite read-only program. With nonrational calibration the theorem uses the declared calibrated real-number model; a numerical implementation must supply its own certified approximation. The only persistent acquired information after a report is its representative label.

Finally, the refresh lower bound and the fixed positive state floor are substantive. If γ tends to zero, the constant $2/3$ need not remain valid and the argument does not give a horizon-uniform theorem. That case is not hidden in the calibration interval.

APPENDIX H. NORMALIZED RESPONSE AND OBSERVATION MODULES

H.1. The covariance pairing. Let π be a probability measure and f, g bounded measurable functions. For u near zero define

$$\pi_u(dx) = \frac{e^{ug(x)}\pi(dx)}{\pi(e^{ug})}.$$

Proposition H.1 (Normalized response). *The derivative of a normalized prediction is*

$$\left. \frac{d}{du} \pi_u(f) \right|_{u=0} = \pi(fg) - \pi(f)\pi(g) = \text{Cov}_\pi(f, g). \quad (\text{H.1})$$

More generally, if $L_u > 0$ is differentiable with the dominated derivatives needed for the quotient rule, then the derivative at u_0 of $\pi(fL_u)/\pi(L_u)$ is the covariance of f with $\partial_u \log L_u|_{u_0}$ under the normalized law at u_0 .

Proof. Differentiate numerator and denominator. At zero their values are $\pi(f)$ and one, and their derivatives are $\pi(fg)$ and $\pi(g)$. This proves (H.1). Apply the same quotient rule at u_0 and divide the derivative of L_u by L_{u_0} to obtain the general statement. Boundedness, or the stated domination in the general case, justifies integration of the derivatives. $\square$

For vector tests and several commands this gives a cross-covariance matrix. One index may represent an acquisition direction and the other a future test. That matrix is not automatically the Fisher information for an unknown physical parameter. In (G.9), command changes are selected by the observer; in Theorem D.3, parameter changes vary the family of physical laws. The similar algebra does not equate the two statistical problems.

H.2. Finite positive functionals and exact product fibres. Suppose a real algebra of functions contains finite-dimensional spaces $U_1, \dots, U_L$, and let V contain all their products. A parameter is a linear functional $\ell \in \Omega \subset V^*$, where Ω is open in V^* . Positivity of measured blocks may define this open set. The observed bilinear forms are

$$B_\nu(f, g) = \ell(fg), \quad f, g \in U_\nu.$$

There are no cross-contact products unless the experiment actually measures them. Set

$$W = \sum_{\nu=1}^L \text{span}\{fg : f, g \in U_\nu\} \subset V.$$

Theorem H.2 (Product-observation fibre). *The observation fibre through ℓ is*

$$(\ell + W^\perp) \cap \Omega. \tag{H.2}$$

Its local dimension is $\dim V - \dim W$. Recovery of every functional in Ω is equivalent to $W = V$. If the parameter domain is relatively open in an affine subspace with direction $L \subset V^$, the local invisible space is instead $L \cap W^\perp$.*

Proof. Two observations agree precisely when the difference functional vanishes on every generating product, equivalently on W . This gives (H.2). Openness allows all sufficiently small perturbations in the annihilator, so the local dimension is its dimension. The affine-domain version restricts those perturbations to L . $\square$

This is linear duality. The difficult model-specific work can lie in proving that these forms are actually observed, determining W , describing the locus where its dimension drops, and controlling real positive realizability. For the calibrated Hankel family of [16], V is a polynomial space and the observation is a normal compression of an inverse information matrix. The theorem supplies its abstract fibre mechanism, not the paper's complete degeneracy classification.

H.3. The contact calculation. For clarity, the local geometric interface can also be derived directly. Let $M \subset \mathbb{R}^k$ have an embedded smooth germ at x_0 , and let $R > 0$. Assume the local minimizing projection onto that germ computes the global distance in a neighborhood of x_0 . Define

$$C_R(b) = \inf_{x \in M} (x - b)^\top R^{-1} (x - b).$$

Let Z have orthonormal columns spanning the Euclidean normal space at x_0 , and put $B = Z^\top R Z$.

Proposition H.3 (Normal second jet). *Under these local projection assumptions,*

$$C_R(x_0 + h) = h^\top Z B^{-1} Z^\top h + o(\|h\|^2), \quad D^2 C_R(x_0) = 2 Z B^{-1} Z^\top.$$

Consequently the full second jet determines exactly the positive normal block B and has the tangent space as its kernel.

Proof. Replace the germ by its tangent space T to compute the quadratic term. The first-order minimizing condition for $\min_{v \in T} (h - v)^\top R^{-1} (h - v)$ says $R^{-1} (h - v) = Zn$ for some n . Hence $h - v = RZn$, and normal projection gives $Z^\top h = Bn$. The minimum is $(Z^\top h)^\top B^{-1} (Z^\top h)$. A smooth local parametrization of the germ has derivative spanning T ; its tangent Gram matrix in the positive R^{-1} metric is invertible. The implicit function theorem gives a differentiable minimizing parameter with the stated linear term. The quadratic expansion and the projection assumption identify it with C_R near x_0 . Positive block inversion is one-to-one, proving the observation assertion. $\square$

If $R = JH(\theta)J^\top$ with Hankel entries $H(\theta)_{ab} = \theta_{a+b}$, then $v^\top R w$ evaluates the moment functional on the product of the two corresponding coefficient polynomials. Theorem H.2 applies after this explicit identification. A higher-order distance jet has additional curvature and projection terms; replacing Sym^2 by Sym^m is not a derivation of a higher-order experiment.

H.4. Why general dynamic tests need not form a finite algebra.

Proposition H.4 (Finite continuous algebras have finite quotients). *Let X be compact Hausdorff and let $W \subset C(X, \mathbb{R})$ be a finite-dimensional unital subalgebra. Then W is the algebra of all functions constant on a finite clopen partition of X . The number of nonempty cells is $\dim W$.*

Proof. For $f \in W$, sufficiently many powers of f are linearly dependent. A nonzero polynomial therefore vanishes on $f(X)$, so that image is finite. A finite basis of W has only finitely many joint values; its fibres form a finite clopen partition. Every member of W is constant on these fibres. Conversely, for a joint value a , choose for every distinct joint value b a basis function $f_{i(a,b)}$ separating them. The product

$$\prod_{b \neq a} \frac{f_{i(a,b)} - f_{i(a,b)}(b)}{f_{i(a,b)}(a) - f_{i(a,b)}(b)}$$

is in W and is the indicator of the fibre at a . These indicators are linearly independent and span all functions on the partition, which proves the result. $\square$

Thus a finite multiplication-closed continuous algebra is a useful model, but it cannot by itself represent arbitrary continuous predictive information. Growing test spaces and positive operator words are the more general primitive. Neither the algebra dimension nor one tangent rank replaces the actual predictive law or its quantization geometry.

APPENDIX I. EXPONENTIAL TRANSFORMS AND CONDITIONAL LIMIT INTERFACES

I.1. Random channels have nonlinear exponential pullbacks. For a Markov kernel $G : X \rightsquigarrow Y$, bounded measurable $f : Y \rightarrow \mathbb{R}$, and a fixed $\beta > 0$, define

$$\mathcal{T}_{G,\beta} f(x) = \beta^{-1} \log \int e^{\beta f(y)} G(dy \mid x). \quad (\text{I.1})$$

For a probability P on X put $\mathcal{P}_{P,\beta}(h) = \beta^{-1} \log P(e^{\beta h})$. These expressions concern finite experiments and bounded sources; no long-time limit is presumed.

Theorem I.1 (Exponential composition and entropy duality). *For $H : Y \rightsquigarrow Z$ and bounded f ,*

$$\mathcal{T}_{GH,\beta} f = \mathcal{T}_{G,\beta}(\mathcal{T}_{H,\beta} f), \quad (\text{I.2})$$

$$\mathcal{P}_{PG,\beta}(f) = \mathcal{P}_{P,\beta}(\mathcal{T}_{G,\beta} f), \quad (\text{I.3})$$

$$\mathcal{P}_{P,\beta}(h) = \sup_{Q \ll P} \{Qh - \beta^{-1} D(Q \| P)\}. \quad (\text{I.4})$$

The supremum is attained at $dP^h/dP = e^{\beta h}/P(e^{\beta h})$. The source derivatives are

$$D\mathcal{P}_{P,\beta}(h)[u] = P^h u, \quad D^2\mathcal{P}_{P,\beta}(h)[u, v] = \beta \text{Cov}_{P^h}(u, v). \quad (\text{I.5})$$

Proof. Exponentiate (I.2). Its right side becomes $G(H(e^{\beta f}))$, which is the definition of the composed kernel's exponential integral. Integrating the same identity against P proves (I.3). For any $Q \ll P$ of finite entropy,

$$D(Q\|P^h) = D(Q\|P) - \beta Qh + \log P(e^{\beta h}) \geq 0.$$

Rearrangement gives the upper inequality in (I.4), and $Q = P^h$ attains equality. Measures of infinite entropy do not improve the supremum for bounded h . Finally boundedness permits differentiation of the exponential integral in the directions u, v ; one quotient rule gives the mean and the second gives the covariance with the factor β . $\square$

If G is the deterministic map g , the exponential pullback is exactly $f \circ g$. If G is random, it is generally not Gf . Jensen gives $\mathcal{T}_{G,\beta}f \geq Gf$, with equality precisely when f is constant $G(\cdot | x)$ -almost surely at each relevant x . The entropy variational principle is classical [13]; the proof fixes its conventions for this experiment category rather than asserting a new variational principle.

Proposition I.2 (Second-order chain rule for a random channel). *Put $h = \mathcal{T}_{G,\beta}f$ and*

$$G_f(dy | x) = \frac{e^{\beta f(y)} G(dy | x)}{G(e^{\beta f})(x)}.$$

Then the tilted joint law is $P^h(dx)G_f(dy | x)$ and

$$\begin{aligned} D\mathcal{T}_{G,\beta}(f)[u] &= G_f u, \\ D^2\mathcal{T}_{G,\beta}(f)[u, v] &= \beta \text{Cov}_{G_f}(u, v), \\ D^2\mathcal{P}_{PG,\beta}(f)[u, v] &= \beta \text{Cov}_{P^h}(G_f u, G_f v) + \beta \mathbb{E}_{P^h} \text{Cov}_{G_f}(u, v). \end{aligned}$$

The last expression is $\beta \text{Cov}_{P^h G_f}(u, v)$.

Proof. Multiplying the two tilted densities cancels $G(e^{\beta f})(x)$ and produces the original joint law multiplied by $e^{\beta f(y)}$ and its single total normalizer. Differentiation of the conditional normalizer gives the first two formulas just as in (I.5). Apply the ordinary second-order chain rule to (I.3). Its first term is the covariance of conditional means and its second is the mean of conditional covariances. The total covariance formula gives the final identity. $\square$

The conditional covariance term is essential. Discarding it would treat a random observation as a deterministic projection. This connects the positive-kernel and source-response layers exactly, while keeping them distinct from Fisher information for a model parameter.

I.2. A finite risk-sensitive recursion. Consider a known prepared predictive model with finite action sets, state update $F_t(s, a, y)$, report kernel $g_t(dy | s, a)$, bounded observed stage cost $c_t(s, a, y)$, and bounded terminal cost v_N . The cost here is an optimization objective, not an undeclared memory resource. For $\beta > 0$ define

$$v_t(s) = \min_{a \in A_t} \frac{1}{\beta} \log \int \exp\{\beta[c_t(s, a, y) + v_{t+1}(F_t(s, a, y))]\} g_t(dy | s, a). \quad (\text{I.6})$$

Proposition I.3 (Finite-horizon exponential dynamic programming). *The value (I.6) is the least value of $\beta^{-1} \log \mathbb{E} \exp(\beta[\sum_{j=t}^{N-1} c_j + v_N])$ over all causal strategies from s . A deterministic Borel minimizing policy exists with a fixed tie rule.*

Proof. Write $w_t = e^{\beta v_t} > 0$. Conditional expectation over the next report and then the future gives the multiplicative recursion $w_t(s) = \min_a \int e^{\beta c_t(s, a, y)} w_{t+1}(F_t(s, a, y)) g_t(dy | s, a)$. At the last step this is the exact conditional objective. Induct backwards; future randomized strategies cannot improve the already minimized continuation, and mixing current actions produces a convex

combination of their positive objective values, no less than their minimum. Finite Borel action comparisons give a Borel first minimizer. Taking logarithms recovers (I.6). $\square$

This recursion needs a state sufficient for the future experiment and its cost. It cannot be applied to an arbitrary lossy statistic without proving closure. Compact infinite action sets would require appropriate existence and selection hypotheses; they are not needed for this proposition.

I.3. Deterministic LDP contractions. We record a conditional interface for later work. Suppose a sequence P_n on a Polish space has a good large-deviation principle with speed $a_n \rightarrow \infty$ and rate I . If T is continuous into another Polish space, its pushforward has rate

$$I_T(y) = \inf_{T(x)=y} I(x). \quad (\text{I.7})$$

For an open or closed target set its inverse image has the same property, so the LDP lower and upper bounds give (I.7). Goodness follows because each finite-rate infimum is attained on compact sublevels and the sublevels of I_T are continuous images of those of I . This is the standard contraction principle [14]. It is not a consequence of the raw experiment axioms.

For a random channel one needs the joint law, a conditional rate, or a separate exponential approximation theorem. A small total-variation simulation defect does not supply these ingredients; an explicit example is given in Section J.

I.4. Regular quadratic and labelled-phase interfaces.

Proposition I.4 (Quadratic constrained reduction). *If $H > 0$ and A has full row rank, then*

$$\inf_{Ah=z} \frac{1}{2} h^\top H h = \frac{1}{2} z^\top (AH^{-1}A^\top)^{-1} z.$$

Proof. The Lagrange equations give $h = H^{-1}A^\top(AH^{-1}A^\top)^{-1}z$. The constraint matrix in parentheses is positive definite. Substitution gives the value, and strict convexity gives the unique minimizer. $\square$

Only if a separate probabilistic theorem identifies H^{-1} with the appropriate Gaussian covariance does this become a covariance contraction statement. Existence of an LDP alone is not that theorem.

For finitely many labelled components with common speed a_n , suppose component rates are I_j and their positive weights satisfy $-a_n^{-1} \log w_{n,j} \rightarrow c_j$, with $\min_j c_j = 0$. The joint labelled rate is $c_j + I_j$ and the unlabelled rate is $\min_j (c_j + I_j(x))$. To verify the bounds, bound a finite sum above by the number of summands times its largest summand and below by each summand; then apply the component LDPs. Compactness follows from the finite union of component compact sublevels. Dynamic prediction may still require the posterior over labels. A marginal log-sum is not automatically a Markov semigroup.

I.5. Closure and finite-dimensional memory. For a Markov kernel P and state map q , an exact reduced kernel $\bar{P}$ satisfies

$$P(f \circ q) = (\bar{P}f) \circ q.$$

Iterating this identity proves every finite-time identity, including the corresponding exponential transforms. If it fails, the projected state need not be Markov. For a finite-dimensional linear evolution split as

$$\dot{u} = A_{PP}u + A_{PQ}v, \quad \dot{v} = A_{QP}u + A_{QQ}v,$$

variation of constants in the second equation gives exactly

$$\begin{aligned} \dot{u}(t) &= A_{PP}u(t) + A_{PQ}e^{tA_{QQ}}v(0) \\ &\quad + \int_0^t A_{PQ}e^{(t-s)A_{QQ}}A_{QP}u(s) ds. \end{aligned} \quad (\text{I.8})$$

The unresolved initial-state term is not optional. Unbounded generators, common form domains, and differentiable operator transports require separate analytic results; the finite-dimensional identity does not prove them. This is the precise boundary at which the history and memory program of [17] can be connected to the present foundations.

APPENDIX J. BOUNDARY EXAMPLES AND NON-IMPLICATIONS

The following examples specify the distinctions used in the theorems. They are part of the mathematical scope, not objections to studying stronger results under stronger hypotheses.

Example J.1 (Less statistical information, more exact prediction labels). Let $Z \sim \text{Bernoulli}(1/2)$ and initially observe $X = Z$. The future target is again Z . Two labels retain the posterior mean exactly. Garble X into $Y = X + \xi$ with independent $\xi \sim N(0, 1)$. Bayes' formula and the ratio of the two Gaussian densities give

$$p(Y) = \mathbb{P}(Z = 1 \mid Y) = \frac{1}{1 + e^{1/2 - Y}}.$$

This is a smooth strictly increasing bijection from $\mathbb{R}$ to $(0, 1)$. The density of Y is positive everywhere, so the posterior law is continuous and has a positive density on every compact subinterval of $(0, 1)$. No finite set of centres can equal this posterior almost surely. More quantitatively, restrict to a compact interval where the density has a positive lower bound and apply Lemma E.3 in one dimension. Every fixed M has strictly positive optimal excess squared error. Both oracles are different: absolute decision risk obeys statistical data processing, but excess coding risk relative to each oracle need not.

Example J.2 (The same mean map with two noise geometries). Let $Z \sim N(\lambda, 1)$. For known $\epsilon > 0$, observing $Y_\epsilon = \epsilon Z$ is equivalent to observing Z by division, and its Fisher information for λ is one. Observing $\tilde{Y}_\epsilon = \epsilon\lambda + \xi$ with independent $\xi \sim N(0, 1)$ has Fisher information ϵ^2 . Both means equal $\epsilon\lambda$. Only the first covariance is scaled with the mean. Consequently a small singular value of a mean map does not determine its statistical information without the noise law.

Example J.3 (Rank does not record global fibres or contact order). The derivative of $x \mapsto x^2$ is nonzero at 1 and -1 , yet their images coincide. For another distinction, let $Y_i = \lambda^r + \xi_i$, $\lambda \geq 0$, with independent unit Gaussian noise and $r \geq 2$. Every mean map has zero derivative at zero, but its local amplitude scale is $n^{-1/(2r)}$. Indeed the divergence between $\lambda = 0$ and a positive alternative is $n\lambda^{2r}/2$. Taking this bounded gives the lower resolution scale by two-point testing. For an upper bound use $\hat{\lambda} = (\bar{Y}_+)^{1/r}$ and $|x_+^{1/r} - b^{1/r}| \leq |x - b|^{1/r}$ for $b \geq 0$; its mean squared error is $O(n^{-1/r})$. The order of contact, not just the rank, matters.

Example J.4 (Reachability is not attained mass). A protocol may permit all commands in an interval while the exploration law always chooses a single command. The set of legally attainable predictions can be continuous even if that particular exploration law visits only one prediction. Another exploration on the same command interface can spread positive mass over a full interval. Their average quantization risks differ. Theorem G.1 avoids this ambiguity by specifying independent uniform commands and deriving their density.

Example J.5 (Equal first two moments do not give equal laws). The two laws $\frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$ and $\frac{1}{2}\delta_0 + \frac{1}{4}\delta_{-\sqrt{2}} + \frac{1}{4}\delta_{\sqrt{2}}$ have mean zero and variance one, but are different. Equality of a finite jet or covariance needs a model-specific identifiability theorem to imply anything stronger.

Example J.6 (Fisher preservation need not be sufficiency). The Kagan–Shepp phenomenon, analyzed in [11], gives an insufficient statistic preserving Fisher information. We do not reproduce that external construction here or claim it as new. Its precise logical role is to prevent promotion of the equality case of Theorem D.3 into the parameter-uniform recovery assertion of Theorem D.2.

Example J.7 (Small variation error can change the exponential rate). For $a_n \rightarrow \infty$ and $c > 0$, let

$$P_n = \delta_0, \quad Q_n = (1 - e^{-a_n c})\delta_0 + e^{-a_n c}\delta_1.$$

Their total variation is $e^{-a_n c} \rightarrow 0$. At speed a_n , however, P_n gives rate $+\infty$ to 1, while Q_n gives it rate c . Thus even exponentially small variation error need not preserve a complete rate function. Appropriate superexponential approximation hypotheses are substantially stronger than an ordinary vanishing simulation defect.

Example J.8 (Linear-source pressure can convexify a nonconvex rate). Let P_n put probability $1/2$ at each of -1 and 1 , for all n . At speed n its rate vanishes at those two points and is infinite elsewhere. Its limiting pressure for linear source sx is $|s|$. The Legendre transform of $|s|$ vanishes on the entire interval $[-1, 1]$. The full bounded-continuous-source Laplace functional and the restricted linear-source transform are not interchangeable.

Example J.9 (Observation does not ensure Markov closure). Consider the deterministic chain $a \mapsto a$, $b \mapsto c$, $c \mapsto c$, with report $q(a) = q(b) = 0$, $q(c) = 1$. The current report 0 does not specify the next report: the two compatible hidden states give different answers. Thus no single reduced kernel works for every preparation. Future-test equivalence distinguishes a and b ; merely keeping their current report does not.

Example J.10 (Zero evidence has no canonical normalized limit). On a two-point space the positive measures $(\delta, 0)$ and $(0, \delta)$ both tend to the zero measure as $\delta \downarrow 0$. Their normalized probabilities tend to different point masses. A continuous normalized extension at zero cannot be assigned from the zero measure alone.

Example J.11 (A simulator may need delay memory). Let the unknown parameter be $\lambda \in \{0, 1\}$. A source with no control actions reveals λ at time one and constants thereafter. A target reveals a constant at time one and λ at time two. Their terminal information is equivalent. However, a parameter-independent simulator with only one retained state after time one, and no other retained tape, has the same internal state and the same new inputs under both parameters. Its second output therefore has the same law under both parameters and cannot equal the required two different point masses. Two states suffice by retaining the bit. If a target controller also retains its own state, the direct implementation is a product. Terminal sufficiency does not erase this state cost.

Example J.12 (A finite command alphabet is a different experiment). At a fixed finite time, finite actions and finite reports produce only finitely many histories. With enough labels they can be retained exactly. The positive lower bound of Theorem G.1 for all finite M uses the stipulated continuously distributed commands. Rounding the physical command alphabet itself changes that experiment. Approximating only the filter's arithmetic while observing the original experiment is instead the perturbation problem in Section F.

These examples rule out four unqualified shortcuts: deriving global identifiability from local rank, deriving memory from information order, deriving statistical rates from a noiseless feature map, and deriving exponential asymptotics from ordinary variation approximation. The positive results in this paper specify the extra structure needed in each case.

APPENDIX K. INTERFACES WITH SINGULAR GEOMETRY AND DYNAMICS

The foundations established here do not require all experiments to be contractions of one physical path law or one rate function. They give a common language in which an actual map or causal simulator must be exhibited before a comparison is asserted.

K.1. What is established in this installment. The exact results are path construction, future-test minimality with stated realization conditions, charged simulation composition, posterior and entropy identities, the finite/dominated score identity, normalized response, product-observation fibres, checkpoint quantization, and exponential composition. The quantitative results are the feedback- and stopping-compatible finite-state emulation theorem and the uniform dynamic memory law for the refreshing model. Their proofs occur in this article. The LDP and operator discussions are conditional interfaces, with a standard contraction principle and a finite-dimensional memory identity; no general Sinai or kinetic limit theorem is asserted here.

The principal dynamic model verifies a compact shrinking domain, actual probability mass, and stable updates directly. It gives one time-uniform isotropic degeneration law. It does not classify all anisotropic collisions, all hidden Markov filters, all priors, or weak-refreshing limits. The bounded comparison constants are not exact optimizer transition constants.

K.2. The role of the two geometric manuscripts. The finite-horizon manuscript [15] provides finer anisotropic profiles for its positive monomial experiments and a separate bounded observation-algebra law. Those hypotheses should remain attached to their conclusions. The present dynamic result is not a replacement for their collision geometry, and their fixed-horizon constants are not assumed uniform as the horizon grows.

The contact manuscript [16] provides a different geometric realization of information loss. Theorem H.2 explains its linear-functional interface. A complete extension to singular statistical resolution still requires the actual noise metric, calibration, parameter nuisance, and admissible real domain. In particular independent measurement noise and correlated raw-data score noise can react differently to a small multiplication singular value. A complex determinantal component is not automatically a physically attainable real statistical submodel.

K.3. A precise next target. Let a parameter target be $\tau(\lambda)$ and let an actual feature map be $O(\lambda)$. Its inverse modulus is

$$\omega(\delta) = \sup\{d(\tau(\lambda), \tau(\lambda')) : d_Z(O(\lambda), O(\lambda')) \leq \delta\}.$$

If a finite-memory procedure estimates that feature with a proved common error bound δ , fitting the model on a compact parameter set gives an error at most $\omega(2\delta)$: the true parameter is an admissible fit, and the triangle inequality bounds the discrepancy of the two model features. A matched statistical lower bound requires actual distributional closeness. If $\|P_{n,\lambda_0} - P_{n,\lambda_1}\|_{TV} \leq a < 1$, any estimator has maximum squared target risk at least

$$\frac{1-a}{8} d(\tau(\lambda_0), \tau(\lambda_1))^2.$$

Indeed choose the nearer of the two target points as a test; on an incorrect choice the estimation error is at least half their separation. The two error probabilities sum to at least $1 - a$.

These two elementary bounds specify a route to a resource–statistical singularity theorem, but do not prove a universal matched modulus law. Covering, testing, attainable alternatives, and tail control must be supplied. The geometry-of-rates literature, including [12], is the appropriate comparison, not a claim that all minimax theory follows from one rank locus.

K.4. The physical realization obligations remain. For deterministic hyperbolic and particle systems, one must still establish the true experiment kernels, physical clock handling, tightness and recovery at exponential scale, and the domains of any response or memory operators. A pathwise finite-state prediction error does not provide a Fourier local limit theorem. A static covariance factorization does not construct a collision process. A projected state does not acquire a Markov semigroup merely because it is named a predictive coordinate.

The historical typed-contraction program [17] can therefore be reorganized as a family of verified realization theorems attached to these interfaces. Its model-specific proofs are not assumed

completed by the present definitions. In particular memory response in a varying Hilbert or form domain requires its own transport and domain arguments.

K.5. Conclusion. A common foundation is obtained by keeping the entire experiment ahead of its geometric representations. Positive preparation and future tests give prediction; a real acquisition law gives a distortion problem; stable updates give an online implementation; a causal simulator gives a comparison, with its state and error charged. In the refreshing model these steps yield a uniform finite-memory law through loss of the acquisition channel. The separate entropy, score, and exponential identities explain which forms of information transport are valid. Stronger singular and long-time results must enter through corresponding theorems, not through additional meanings assigned to the word “contraction”.

APPENDIX L. MEASURABILITY, FINITE COVERS, AND A SCALAR BENCHMARK

L.1. Finite-label implementation. A finite codebook $C = \{c_1, \dots, c_M\}$ in a metric space has a Borel nearest-centre selector whenever each distance $s \mapsto d(s, c_j)$ is Borel. Define its label by the least index attaining the minimum. The set on which label j is selected is the intersection of the finitely many sets $\{d(s, c_j) < d(s, c_i)\}$ for $i < j$ and $\{d(s, c_j) \leq d(s, c_i)\}$ for $i > j$. These sets are Borel. Thus the finite-label recursions used in this paper do not require an unproved measurable choice of a nearest point in an infinite set.

Codebook construction and codebook execution are different questions. Existence of finitely many representatives in nonempty cells is an ordinary finite choice. When an algorithmic construction is claimed, one must also supply a representation and a feasibility procedure for those cells. In Section G, rational calibration gives a rational polytope; finite rational linear feasibility supplies this additional construction. This does not charge an unbounded list of history-dependent centres to a read-only program.

L.2. Conditional distributions and common versions. The existence of regular conditional distributions on standard Borel spaces is a classical input [4]. Its relevant conclusion is existence of a kernel under a fixed joint law. It does not provide one canonical version on every null history, nor does it alone give a version continuous in a parameter or uniformly valid on an uncountable family of null sets. The statements in the main text accordingly take one of three forms: an explicit positive finite-report formula; a compact continuous realization as in Theorem B.3; or an already supplied jointly measurable predictive realization with its applicable quantifiers.

There is no hidden claim that all quotient images of a Borel space are standard Borel. The evaluation image into a countable product can be given its induced measurable structure, but to use ordinary kernel calculus on it one must verify that the image and descended maps have the required measurability. The compact theorem verifies precisely that fact in one useful class. Keeping the full posterior remains a valid alternative when compression of the state is not yet justified.

L.3. Quantization after a readout change.

Proposition L.1 (Bi-Lipschitz readout comparison). *Let S be a metric space with a probability ν , and let $p : S \rightarrow \mathbb{R}^q$ be an injective readout satisfying*

$$a d(s, t) \leq \|p(s) - p(t)\|_2 \leq b d(s, t), \quad a, b > 0.$$

If S has an M -point cover of radius r with centres in S , the checkpoint prediction risk is at most $b^2 r^2$. If some measurable map $T : \mathbb{R}^q \rightarrow \mathbb{R}^k$ has Lipschitz constant $L > 0$, and the law of $Tp(s)$ has density at most A , that risk is at least

$$\frac{1}{2L^2} (2Av_k M)^{-2/k}.$$

The second assertion does not require decoder centres to belong to $p(S)$.

Proof. For the upper bound map the cover centres through p and use the upper Lipschitz estimate. For any decoder centres $z_j \in \mathbb{R}^q$, the inequality $\|p(s) - z_j\|_2 \geq L^{-1} \|Tp(s) - Tz_j\|_2$ reduces the lower bound to Lemma E.2. Take the infimum over the original centres. The constant a is useful for comparing intrinsic covers in both directions, but it is not needed in this particular density-based lower bound. In particular, the argument does not require inversion of a near-singular statistical covariance. $\square$

A linear projection or bounded coordinate map can be used for T . A coordinate change whose Lipschitz constant tends to infinity changes the bound; it cannot be called a harmless normalization at a degeneracy.

Proposition L.2 (Exact scalar uniform quantization). *For U uniform on an interval of length $a > 0$, the optimal squared M -centre quantization error is $a^2/(12M^2)$, for every integer $M \geq 1$.*

Proof. Order the centres. Their nearest-centre cells intersect the interval in intervals, allowing empty cells. If a cell has length l_j , its contribution is minimized by its midpoint and equals $l_j^3/(12a)$. Since $\sum_j l_j = a$, convexity gives $\sum_j l_j^3 \geq a^3/M^2$. Hence the total error is at least $a^2/(12M^2)$. Equal-length cells with their midpoints attain that value. Arbitrary randomized encoders cannot improve it by Theorem E.1. $\square$

This exact formula is a benchmark for the meaning of label cardinality, not a claim that the dynamic posterior density in Section G is uniform. The latter is bounded by its actual Jacobian and can depend on the preceding history. Its two-sided order, rather than an exact optimizer constant, is what we prove.

L.4. Predictable error budgets and random horizons. The stopped version of Lemma C.2 is often more useful than the deterministic sum. In a comparison with common visible stopping rule σ , append an absorbing terminal symbol. The two conditional kernels coincide once the symbol has appeared. Their discrepancy at time t is therefore bounded by $\mathbf{1}_{\{t < \sigma\}} e_t(H_t)$, a measurable function of the common prefix. The telescoping proof integrates it against the actual prefix law. This gives the expectation of the active-step sum, without replacing a random horizon by an uncharged successful count.

If the active-step sum is bounded by a deterministic budget B on every path, its expectation is at most B . If only an expected budget is known, that is the exact kind of budget furnished by the theorem. A tail-budget statement requires an additional probability estimate. These are distinct resource certificates even though the same positive summands occur in all three.

APPENDIX M. EXPLICIT FINITE CHECKS AND THE LIMITS OF COMPUTATION

M.1. A two-state posterior calculation. The smallest dynamic instance has $d = 1$, $\gamma = 4/5$, and $\epsilon = 1/4$. Put $v = \tanh u$ and select $v = 1/2$, which corresponds to the legal command $u = \operatorname{artanh}(1/2) < 1$. Starting from the uniform preparation, the pre-report belief is $q = (1/2, 1/2)$, and

$$\mathbb{P}(Y = +1) = \frac{17}{32}, \quad \mathbb{P}(Y = -1) = \frac{15}{32}.$$

The posterior probabilities of state 1 are respectively

$$p_1^+ = \frac{9}{17}, \quad p_1^- = \frac{7}{15}.$$

Thus the same current command produces two different posteriors. Discarding the negative report or assigning its conditional law mass one would alter the acquisition law used in the proof. With

the fixed terminal probe the corresponding success probabilities are

$$\frac{1}{2} + \frac{1}{4} \frac{9}{17} = \frac{43}{68}, \quad \frac{1}{2} + \frac{1}{4} \frac{7}{15} = \frac{37}{60}.$$

The mean posterior agrees with the pre-report probability:

$$\frac{17}{32} \frac{9}{17} + \frac{15}{32} \frac{7}{15} = \frac{1}{2}.$$

This is the posterior martingale property for the report step; the refresh step has its own affine prediction and is not being omitted.

The derivative with respect to v , rather than u , at this point for the positive report is

$$\frac{q_0 q_1 \epsilon}{D_+^2} = \frac{16}{289}.$$

Multiplying by $dv/du = 1 - v^2 = 3/4$ gives $12/289$, exactly (G.13) in one dimension. These distinct derivatives are recorded because replacing a physical command by a nonlinear coordinate without its Jacobian would change the preparation density.

M.2. A direct resource calculation. Suppose a report converter has two persistent states (for example, it retains the previous binary report), the target predictor has M labels, and a feedback controller has D labels. The implemented composite uses at most $2MD$ triples. On the next step it reads the old triple, the current input, and fresh independent randomness, and writes the next triple. The source history may be longer, but it is not consulted by this implementation. Any reduction below $2MD$ needs a proved state identification; it is not obtained by calling the converter a sufficient statistic.

Conversely, if a converter is memoryless and deterministic, its state set has one element. In that special case it need not enlarge the target label count. This is the correct context for a nonincrease assertion. It does not imply the false general statement about exact predictive memory refuted in Example J.1.

M.3. A finite precision requirement rather than a free real tape. For $d = 1$ the invariant domain is an interval inside the probability simplex. A codebook can be constructed using interval intersections, and its labels are ordinary finite integers. One evaluates the two rational Bayes fractions, with the command likelihood evaluated to a specified accuracy, then selects the nearest admissible representative. The perturbation form of Theorem F.1 accounts for the incurred state error. Since denominators are at least $3/4$, no division by a vanishing report probability occurs.

For general fixed d , the same procedure uses a finite polytope feasibility construction. It may be computationally expensive. Our cardinality theorem does not state an optimal running time or workspace bound. The program may store its predetermined codebook, as permitted by the read-only model; it may not put acquired commands or reports in that program. The current command must be discarded after updating the label, and the terminal decoder may not reread it as free side information.

M.4. Role of the accompanying diagnostics. The source package includes exact-rational checks of Bayes normalization, the odds inverse, the rank-one determinant identity, the refresh contraction inequality at a finite family of inputs, and the scalar quantization identity. It also checks a finite conditional-covariance identity, a feedback path-law comparison, and specified deliberately incorrect formulas. These computations are reproducibility aids for signs, constants, and implemented examples.

The universal statements in this article are proved analytically above. A finite test does not establish measurability, a global differential bound, a minimax lower bound, or uniformity over all times. In particular, no number of successful finite diagnostics certifies that every possible

hidden Markov model has the displayed memory law. Build receipts, source hashes, and diagnostic output belong to the accompanying delivery record rather than to the logical assumptions of any theorem.

REFERENCES

- [1] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272. doi:10.1214/aoms/1177729032.
- [2] T. Fritz, A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics, *Adv. Math.* **370** (2020), 107239. arXiv:1908.07021.
- [3] T. Fritz, T. Gonda, P. Perrone, and E. F. Rischel, Representable Markov categories and comparison of statistical experiments in categorical probability, *Theoret. Comput. Sci.* **961** (2023), 113896. arXiv:2010.07416.
- [4] O. Kallenberg, *Foundations of Modern Probability*, third ed., Probability Theory and Stochastic Modelling, vol. 99, Springer, 2021. doi:10.1007/978-3-030-61871-1.
- [5] C. R. Shalizi and J. P. Crutchfield, Computational mechanics: pattern and prediction, structure and simplicity, *J. Statist. Phys.* **104** (2001), 817–879. arXiv:cond-mat/9907176.
- [6] M. L. Littman, R. S. Sutton, and S. Singh, Predictive representations of state, in *Advances in Neural Information Processing Systems 14*, MIT Press, 2002, 1555–1561 (2001 conference). Publisher proceedings entry.
- [7] N. Barnett and J. P. Crutchfield, Computational mechanics of input-output processes: structured transformations and the ϵ -transducer, *J. Statist. Phys.* **161** (2015), 404–451. arXiv:1412.2690.
- [8] G. Pagès and H. Pham, Optimal quantization methods for nonlinear filtering with discrete-time observations, *Bernoulli* **11** (2005), 893–932. doi:10.3150/bj/1130077599.
- [9] D. S. Kalogerias and A. P. Petropulu, Grid based nonlinear filtering revisited: recursive estimation and asymptotic optimality, preprint, 2016. arXiv:1604.02631.
- [10] N. Saldi, S. Yüksel, and T. Linder, On the asymptotic optimality of finite approximations to Markov decision processes with Borel spaces, *Math. Oper. Res.* (2017). doi:10.1287/moor.2016.0832.
- [11] D. Pollard, A note on insufficiency and the preservation of Fisher information, in *From Probability to Statistics and Back: High-Dimensional Models and Processes*, IMS Collections, vol. 9, 2013. doi:10.1214/12-IMSCOLL919; arXiv:1107.3797.
- [12] D. L. Donoho and R. C. Liu, Geometrizing rates of convergence, II, *Ann. Statist.* **19** (1991), 633–667. doi:10.1214/aos/1176348114.
- [13] P. Dupuis and H. Wang, Subsolutions of an Isaacs equation and efficient schemes for importance sampling, *Math. Oper. Res.* **32** (2007), 723–757. doi:10.1287/moor.1070.0266.
- [14] S. R. S. Varadhan, *Large Deviations*, lecture notes, Courant Institute, Spring 2010, Sections 1–2. Author’s lecture notes.
- [15] Q. Qi, Attainable information and causal compression at exponent collisions, A1 v37, manuscript and companion, September 2026. Immutable repository edition, commit 90465076589f.
- [16] Q. Qi, Recovery of information metrics: multiplication failure schemes, A2 v112, manuscript, September 2026. Immutable mathematical edition, commit 0c696736e6ec.
- [17] Q. Qi, Path-space cotangent rigidity, universal contractions, and tangent representations, historical C2 manuscript, Round 17. Immutable historical snapshot, commit c04845b66132.
- [18] P. J. Bushell, On the projective contraction ratio for positive linear mappings, *J. London Math. Soc.* (2) **6** (1973), 256–258. doi:10.1112/jlms/s2-6.2.256.
- [19] A. D. Kara and S. Yüksel, Near optimality of finite memory feedback policies in partially observed Markov decision processes, *J. Mach. Learn. Res.* **23** (2022), no. 11, 1–46. Publisher edition.
- [20] J. Subramanian, A. Sinha, R. Seraj, and A. Mahajan, Approximate information state for approximate planning and reinforcement learning in partially observed systems, *J. Mach. Learn. Res.* **23** (2022), no. 12, 1–83. Publisher edition.
- [21] A. D. Kara, E. Bayraktar, and S. Yüksel, Near optimal approximations and finite memory policies for POMPDs with continuous spaces, preprint, 2024. arXiv:2410.02895.
- [22] Y. E. Demirci, A. D. Kara, and S. Yüksel, Refined bounds on near optimality finite window policies in POMDPs and their reinforcement learning, preprint, 2024. arXiv:2409.04351.
- [23] M. Kim, Finite memory belief approximation for optimal control in partially observable Markov decision processes, preprint, 2026. arXiv:2601.03132.
- [24] S. Marzen and J. P. Crutchfield, Predictive rate-distortion for infinite-order Markov processes, *J. Statist. Phys.* **163** (2016), 1312–1338. Preprint arXiv:1412.2859, under the title “Circumventing the curse of dimensionality in prediction”.

- [25] S. P. Loomis and J. P. Crutchfield, Exploring predictive states via Cantor embeddings and Wasserstein distance, *Chaos* **32** (2022), 123115. doi:10.1063/5.0102603.
- [26] G. N. Nair and R. J. Evans, Stabilization with data-rate-limited feedback: tightest attainable bounds, *Systems Control Lett.* **41** (2000), 49–56. doi:10.1016/S0167-6911(00)00037-2.
- [27] S. Tatikonda and S. Mitter, Control under communication constraints, *IEEE Trans. Automat. Control* **49** (2004), 1056–1068. doi:10.1109/TAC.2004.831187.
- [28] A. S. Matveev and A. V. Savkin, *Estimation and Control over Communication Networks*, Control Engineering, Birkhäuser, Boston, 2009. doi:10.1007/978-0-8176-4607-3.
- [29] E. Lindenstrauss and M. Tsukamoto, From rate distortion theory to metric mean dimension: variational principle, *IEEE Trans. Inform. Theory* **64** (2018), 3590–3609. doi:10.1109/TIT.2018.2806219. arXiv:1702.05722.
- [30] J. V. Bondar and P. Milnes, Amenability: a survey for statistical applications of Hunt–Stein and related conditions on groups, *Z. Wahrscheinlichkeitstheorie verw. Gebiete* **57** (1981), 103–128. doi:10.1007/BF00533716.
- [31] P. Heinrich and J. Kahn, Minimax rates for finite mixture estimation, preprint (2015), arXiv:1504.03506v1. The comparison in the text uses the explicitly numbered theorems of this fixed preprint edition.
- [32] R. Bowen, *Equilibrium States and the Ergodic Theory of Anosov Diffeomorphisms*, second revised edition, edited by J.-R. Chazottes, Lecture Notes in Mathematics 470, Springer, Berlin, 2008. doi:10.1007/978-3-540-77695-6.
- [33] T. Linder and S. Yüksel, On optimal zero-delay coding of vector Markov sources, *IEEE Trans. Inform. Theory* **60** (2014), 5975–5991. doi:10.1109/TIT.2014.2346780.
- [34] R. G. Wood, T. Linder and S. Yüksel, Optimal zero delay coding of Markov sources: stationary and finite memory codes, *IEEE Trans. Inform. Theory* **63** (2017), 5968–5980. doi:10.1109/TIT.2017.2692215.
- [35] H. Luschgy and G. Pagès, Sharp asymptotics of the functional quantization problem for Gaussian processes, *Ann. Probab.* **32** (2004), 1574–1599. doi:10.1214/009117904000000324.
- [36] M. K. Roychowdhury, Quantization dimension for Gibbs-like measures on cookie-cutter sets, *Kyoto J. Math.* **54** (2014), 239–257. doi:10.1215/21562261-2642377.
- [37] B. P. Tunstall, *Synthesis of noiseless compression codes*, Ph.D. dissertation, Georgia Institute of Technology, 1967.
- [38] L. J. Lindsay and R. D. Mauldin, Quantization dimension for conformal iterated function systems, *Nonlinearity* **15** (2002), 189–199. doi:10.1088/0951-7715/15/1/309.
- [39] J. Atnip, M. K. Roychowdhury and M. Urbanski, Quantization dimension for infinite conformal iterated function systems, preprint, 2018. arXiv:1811.00299v1.
- [40] M. Ghomi, T. Linder and S. Yüksel, Zero-delay lossy coding of linear vector Markov sources: optimality of stationary codes and near optimality of finite memory codes, *IEEE Trans. Inform. Theory* **68** (2022), 3474–3488. doi:10.1109/TIT.2021.3138769. Fixed preprint text: arXiv:2103.10810v3.
- [41] L. Cregg, F. Alajaji and S. Yüksel, Sliding finite window codes: near-optimality and Q-learning for zero-delay coding, *IEEE Trans. Inform. Theory* **72** (2026), 3910–3926. doi:10.1109/TIT.2026.3681183.
- [42] Q. Qi, Conductor strata and nonreduced multiplication failure schemes, A2 revision 118, manuscript, September 22, 2026. Fixed repository edition, commit 44bfc648ead0.
- [43] M. Kesseböhmer and S. Zhu, The upper and lower quantization coefficient for Markov-type measures, *Math. Nachr.* **290** (2017), 827–839. doi:10.1002/mana.201500328. Theorem 1.1 checked in the authors’ preprint, entitled *The quantization for Markov-type measures on a class of ratio-specified graph directed fractals*, arXiv:1406.3257v1.
- [44] Y. E. Demirci, A. D. Kara and S. Yüksel, Average cost optimality of partially observed MDPs: contraction of nonlinear filters and existence of optimal solutions and approximations, *SIAM J. Control Optim.* **62** (2024), 2859–2883. doi:10.1137/24M1643736. Theorem numbering in the text refers to arXiv:2312.14111v3.
- [45] L. Le Cam and G. L. Yang, *Asymptotics in Statistics: Some Basic Concepts*, second edition, Springer, New York, 2000. doi:10.1007/978-1-4612-1166-2.
- [46] E. Torgersen, *Comparison of Statistical Experiments*, Encyclopedia of Mathematics and its Applications 36, Cambridge University Press, Cambridge, 1991.
- [47] L. Le Cam, Sufficiency and approximate sufficiency, *Ann. Math. Statist.* **35** (1964), 1419–1455. doi:10.1214/aoms/1177700372.
- [48] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28. doi:10.1524/strm.2002.20.14.1.
- [49] A. Nerode, Linear automaton transformations, *Proc. Amer. Math. Soc.* **9** (1958), 541–544. doi:10.1090/S0002-9939-1958-0135681-9.
- [50] B. R. Kloeckner, A generalization of Hausdorff dimension applied to Hilbert cubes and Wasserstein spaces, *J. Topol. Anal.* **4** (2012), 203–235. doi:10.1142/S1793525312500094. Author preprint: arXiv:1105.0360v2.
- [51] J. Matoušek, *Using the Borsuk–Ulam Theorem: Lectures on Topological Methods in Combinatorics and Geometry*, Universitext, Springer, Berlin, 2003. doi:10.1007/978-3-540-76649-0.

- [52] Q. Qi, Attainable information and causal compression at exponent collisions, A1 revision 37, manuscript, September 9, 2026. Fixed repository edition, commit 90465076589f.
- [53] A. Nayyar, A. Mahajan and D. Teneketzis, Decentralized stochastic control with partial history sharing: a common information approach, *IEEE Trans. Automat. Control* **58** (2013), 1644–1658. doi:10.1109/TAC.2013.2239000. The theorem numbers cited here refer to the author preprint, arXiv:1209.1695.
- [54] H. S. Witsenhausen, On the structure of real-time source coders, *Bell System Tech. J.* **58** (1979), 1437–1451. doi:10.1002/j.1538-7305.1979.tb02263.x.
- [55] N. Saldi, A topology for team policies and existence of optimal team policies in stochastic team theory, preprint (2017), arXiv:1711.00639.
- [56] S. Yüksel and N. Saldi, Convex analysis in decentralized stochastic control, strategic measures and optimal solutions, author preprint arXiv:1609.07685v2.
- [57] S. Yüksel, On optimal causal coding of partially observed Markov sources in single and multi-terminal settings, author preprint arXiv:1010.4824v4.
- [58] S. Kiefer, Notes on equivalence and minimization of weighted automata, preprint (2020), arXiv:2009.01217v1.
- [59] M. Nisio, On a non-linear semi-group attached to stochastic optimal control, *Publ. Res. Inst. Math. Sci.* **12** (1976), 513–537. doi:10.2977/prims/1195190727.
- [60] N. Bäuerle and U. Rieder, Partially observable risk-sensitive Markov decision processes, *Math. Oper. Res.* **42** (2017), 1180–1196. doi:10.1287/moor.2016.0844. The comparison uses Theorem 3.3 and Section 4.4 of arXiv:1504.03530v3.
- [61] W.-G. Tzeng, A polynomial-time algorithm for the equivalence of probabilistic automata, *SIAM J. Comput.* **21** (1992), 216–227. doi:10.1137/0221017.
- [62] H. S. Witsenhausen, The zero-error side information problem and chromatic numbers, *IEEE Trans. Inform. Theory* **22** (1976), 592–593. doi:10.1109/TIT.1976.1055607.
- [63] I. Gallagher, L. Saint-Raymond and B. Texier, *From Newton to Boltzmann: Hard Spheres and Short-Range Potentials*, Zurich Lectures in Advanced Mathematics, European Mathematical Society, 2014. The flow and boundary discussion used here is in Sections 4.1–4.3 of arXiv:1208.5753v2.
- [64] R. C. Kraaij, The exponential resolvent of a Markov process and large deviations for Markov processes via Hamilton–Jacobi equations, arXiv:1910.03946v1, 2019. Theorem numbering refers to this author preprint.
- [65] C. Widder, J. Zimmer and T. Schilling, On the generalized Langevin equation and the Mori projection operator technique, preprint (2025), revised 2026. arXiv:2503.20457v6.
- [66] C. Widder and T. Schilling, Generalised Langevin dynamics: significance and limitations of the projection operator formalism, preprint (2026). arXiv:2604.20453v2.
- [67] H. F. Trotter, Approximation of semi-groups of operators, *Pacific J. Math.* **8** (1958), 887–919. doi:10.2140/pjm.1958.8.887. Theorem 5.2, p. 902, is the approximation theorem cited here.
- [68] H. Weisshaupt, LeCam’s randomization criterion in the setting of locally convex spaces without lattice structure, EURANDOM report 2002–017, 22 April 2002. Lemma 1 and Theorem 5, pp. 14–16. Author report.
- [69] S. Basu, Algorithms in real algebraic geometry: a survey, author preprint, arXiv:1409.1534 (2014). Published in *Panoramas et Synthèses* **51** (2017), 107–153. Theorem 2.1 and Section 2.1 in the author preprint are used here.
- [70] M. C. Paull and S. H. Unger, Minimizing the number of states in incompletely specified sequential switching functions, *IRE Trans. Electron. Comput.* **EC-8** (1959), 356–367. Publisher record.
- [71] M. Sion, On general minimax theorems, *Pacific J. Math.* **8** (1958), 171–176. doi:10.2140/pjm.1958.8.171.
- [72] H. Weisshaupt, A generalization of Le Cam’s randomization criterion, *Studia Sci. Math. Hungar.* **43** (2006), 443–486. doi:10.1556/SScMath.43.2006.4.3. Theorem numbers refer to the author’s 39-page preprint, Randomization.pdf.
- [73] E. de Klerk and M. Laurent, Error bounds for some semidefinite programming approaches to polynomial minimization on the hypercube, *SIAM J. Optim.* **20** (2010), 3104–3120. doi:10.1137/100790835.
- [74] M. Laurent and L. Slot, An effective version of Schmüdgen’s Positivstellensatz for the hypercube, arXiv:2109.09528v2, 2023.
- [75] E. de Klerk and J. C. Vera, The link between 1-norm approximation and effective Positivstellensätze for the hypercube, *Numer. Algebra Control Optim.* **16** (2026), 84–104. doi:10.3934/naco.2024060. Theorem numbers refer to arXiv:2404.04190v2.